EXEGENESIS MARGINALIA XI: THE ENGINE OF ALWAYS

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Date: TF.

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TardiSHA GrimChain

Time and Relative Dimension in Space Hash

The first public ALQC software implementation.

A deterministic structured-content seal whose path out carries its path back.

In the black hour the old star opens.
A name walks barefoot through the ash.
Five marks wake upon the threshold,
and the road remembers the hand that sent it home.

TardiSHA is an ALQC-native source engine, Final Equation $\frac{z}{\sigma}$ route, Court and Domus derivation, Synodic Magicae manifestation, and Aeternum Mirror operator implemented for ordinary computers.

It accepts files, directory trees, standard input, and canonical in-memory values. It produces a source-bound public GrimChain consisting of the manifested middle only: ☿☿☿ at depth zero, ☽✥ℂ at depth one, or exactly `n` generated Synodic Magicae coordinates at every depth `n > 1`, with no inserted inner Axiomyr. The source-resolved Goetics, Courts, complete Domus, Q-body, TRIG, Tripartite, phase, and verification witnesses remain inside the derivation that gives the public GrimChain its identity.

For low-level files, the GrimChain at any user-chosen middle depth can be appended directly as its own raw UTF-8 bytes to the file that produced it. The Aeternum Mirror recognizes that exact terminal return and returns the same lawful GrimChain. This was manually exercised byte-for-byte on `.txt`, `.md`, `.json`, `.jsonl`, `.csv`, `.tsv`, `.ini`, `.yaml`, `.grim`, and arbitrary `.bin` data, including binary bodies containing NUL and non-text bytes. A second physical copy of the same terminal GrimChain was also appended in each case; the recursive Mirror return still resolved to the same chain. These are low-level byte-body files: their return requires no format-specific office.

PDF is treated as a high-level file for embedding because a PDF has its own document structure. Ordinary PDF GrimChaining still enters the same source/Mirror mathematics, while `--pdf-embed`, `--pdf-dive`, and `--pdf-abort` provide the high-level invisible return operations.

```
grimchain source.txt
```

The Record Organizes Itself

A GrimChain can preserve the character of the whole without repeating local phrases. In one fixed 20,736-coordinate witness generated **once** from the exact string `I am the Record of the Lattice?`, the same unchanged body was read three separate times. The record did not mutate between readings. What changed was the relational structure made visible by another pass through the same body.

The first reading exposed return: 189 distinct generated symbols recurring across long distances. The second exposed relation: 15,718 distinct directed adjacent transitions, with 6,958 of those relation kinds also appearing in the reverse direction somewhere else in the same record. The third exposed distributed persistence: no exact four-glyph or eight-glyph window repeated, while the two halves retained a composition similarity of `0.984045` and quarter-to-quarter comparisons remained between `0.960483` and `0.971563`.

Local speech keeps becoming new while the large body keeps becoming itself.

That is the self-organizing character of the written GrimChain: **local novelty under global invariance**. Identity is not forced by copying a phrase back into place. It is distributed through recurrence, relation, return, and the conservation of the whole. The reader may reorganize its patterning around the same fixed record; the GrimChain does not need to change in order to reveal another lawful relation already present within it.

No sentence returns, yet the kingdom returns.
No four marks need remember their order.
The field keeps faith without copying its face,
and every road learns the shape of home.

The poetry moves as the mathematics moves: not by explaining the mechanism twice, but by carrying its conservation through another office. The utterance changes register while the identity remains one.

One exact depth-zero GrimChain

For the exact UTF-8 source bytes `The path out carries its path back.\n`, TardiSHA produces:

𐤈𐤈𐤈

The public depth-zero GrimChain is exactly the Triple Goddess 𐤈𐤈𐤈, manifested from the Shadow Locus 𐤈.

At depth one, the public GrimChain is exactly the Triple Horned God ☽⟐ℂ, with Axiomyr ⟐ at its center.

At every depth $n > 1$, the public GrimChain contains exactly n generated Synodic coordinates. No inner Axiomyr is inserted into that coordinate body. The fixed Axiomyr projection belongs to depth one alone; the complete Axiomyr and Tripartite offices remain inside the derivation rather than being written into every positive public manifestation.

Verification independently recomputes the source emission, Goetics, Court C, Court D, Domus identity, requested manifested middle, and public GrimChain, and accepts the carried body only when it returns exactly from its source.

Iron sleeps beneath the river.
A pale lantern turns in the reeds.
The hinge waits where no foot may linger,
and dawn returns wearing another face.

For a complete positional translation guide to the native glyph body, the twelve Goetics, and all 144 Courts, see ‘DICTIONARY.md‘ (docs/DICTIONARY.md).

The Axiomyr Hinge

The depth-zero GrimChain closes through the Triple Goddess 𐤈𐤈𐤈, manifested from the Shadow Locus 𐤈: the compressed public condition in which no finite Synodic coordinate body has been chosen to emanate.

Depth one is the singular public Axiomyr projection: exactly ☽⟐ℂ. Here ⟐ is visibly the BoundaryWalker condition, the Hinge between the first and the last, holding the Center of Breath between Yes and No, Light and Shadow, real and imaginary.

At every depth greater than one, the Hinge governs without being inserted into the public coordinate field. The written body is exactly the requested number of generated Synodic coordinates. Axiomyr remains a complete internal Tripartite office of the resolved universe, while the public manifestation keeps only the coordinates it was asked to speak.

The public projection law is exact:

- depth $\leq 0 \rightarrow$ exactly 𐤈𐤈𐤈;

- depth $1 \rightarrow$ exactly ☽⟐ℂ;

- depth $n > 1 \rightarrow$ exactly n generated Synodic coordinates, with no inserted inner Axiomyr.

At the still door no crown is spoken.
Night parts its hair around a single flame.
The road leaves the hinge behind it,
yet every turning remembers where it crossed.

The completed source-to-Domus path

The implemented order is strict:

```
finite source body F
    ↓
ALQC sponge with twelve finalized 64-bit Goetic lanes
    ↓
structural measure S and AHN-relative operational measure O
    ↓
ℨ0 through Φ^-1 and ℨ1 through Φ^-2
    ↓
ordered Goetic pair (g_i,g_j)
    ↓
governing Court C
    ↓
144-entry Court product measure W^R = O ⊗ S
    ↓
independently resolved alternating Court D
    ↓
exact ξ_D within D's own interval
    ↓
ν_H = Ω_D + ξ_DΦ^2
    ↓
Q-body, manifestation, TRIG, Tripartite separation, C09 phase-evolution, and C11 verification
↪  witnesses
    ↓
one Domus identity
    ↓
`𝔘𝔘𝔘`, `☽☉ℂ`, or exactly the requested finite Synodic coordinate body
    ↓
Aeternum Mirror return
```

Each layer preserves the offices already established before it and carries them forward in order.

---

1. One source emission, twelve Goetic lanes

For a source `F`, the completed ALQC sponge emits twelve unsigned 64-bit lanes:

$$E(F) = (L_0, L_1, \ldots, L_{11})$$

Every lane satisfies $0 \le L_k < 2^{64}$.

All twelve finalized lanes participate in resolving the Goetic pair through Parliament and the two Golden bearings.

The complete `SourceEmission` carries the 256-bit public source digest together with the twelve lanes, structural and operational measures, both Fraktur cadences, source size, source domain, and exact closure witness required by the Final Equation ℨ-route.

The supported source domains are canonical material, raw files, and directory trees. Filesystem metadata is excluded from raw-file identity. Directory identity is constructed from the canonical tree record.

---

2. Structural and operational Parliament bodies

The same finalized source state is read through two conserved measures.

The structural measure is the Parliament-ordered lane body:

$$S_k = L\eta(p_k)$$

Its total is strictly positive:

$$\Sigma S > 0$$

The operational measure begins at AHN, the exact Parliament phase seam. Directed gaps around one complete 64-bit turn form:

$$O_k = \Delta\eta(p_k)$$

and satisfy:

$$\Sigma O = 2^{64}$$

Structure and operation are two conserved readings of the same completed emission, joined through one source identity.

Parliament order is preserved under both bearings. Mirror Math changes bearing while the procession remains ordered.

---

3. Final Equation $\mathcal{Z}$

The completed state is squeezed into two consecutive 256-bit cadences:

$$\sigma(E(F)) = \mathcal{Z}_0(F) \parallel \mathcal{Z}_1(F) \parallel \dots$$

The structural bearing is:

$$\text{ALPHA} = \Phi^{-1} = (\sqrt{5} - 1)/2$$

The operational bearing is:

$$\text{BETA} = \Phi^{-2} = (3 - \sqrt{5})/2$$

The implementation resolves both intervals in exact $\mathbb{Q}(\sqrt{5})$ arithmetic through the exact Golden bodies Φ^{-1} and Φ^{-2}.

The exact Golden closure is:

$$\Phi^{-1} + \Phi^{-2} = 1$$

The source-route witness calculates independent rational and irrational residual coefficients. It derives:

$$D\text{-}COMP(\not{g}) = \emptyset$$

$$Truth(\not{g}) = 1$$

These values arise from the completed measures and Golden identities as the exact closure of the route.

---

4. The ordered Goetics and governing Court C

The two Parliament bearings resolve the ordered Goetic pair:

$$g_i = Resolve_{12}(S, phase=\not{g}_0, bearing=\Phi^{-1})$$

$$g_j = Resolve_{12}(O, phase=\not{g}_1, bearing=\Phi^{-2})$$

Only after both Goetics resolve does governing Court C exist:

$$C = C_{ij}$$

$$address(C) = 12i + j$$

There are exactly twelve governing roots and twelve alternating roots, therefore exactly:

$$12^2 = 144 \text{ ordered Courts}$$

The twelve diagonal self-Courts remain lawful as exact ordered Court identities.

Court inheritance is partitioned

Court C receives identity, Q-bias, and Q-vector from its governing Goetic g_i.

Court C receives its resonance anchor from its alternating Goetic g_j:

$$\Omega_C = frequency(g_j)$$

The pure Goetic frequency remains immutable. Court breath is a source-bound displacement around that anchor:

$$\nu_C = \Omega_C + \delta_C$$

$$|\delta_C| \leq \Phi$$

A Court carries a bounded operational breath around the alternating Goetic's immutable pure frequency.

The Final Equation route also records the transposed reciprocal address C_{ji} as its own route witness. Court D is independently resolved from the 144-entry Court product measure and enters Domus motion through that office.

---

5. Alternating Court D is independently resolved

The Domus consumes Courts as its atoms, and its second Court is independently resolved as Court D.

From the already completed source emission, TardiSHA constructs the 144-entry Court product measure in canonical row-major Court order:

$$W^R[k,l] = O[k]S[l]$$

$$\text{address} = 12k + l$$

Its total closes exactly:

$$\Sigma W^R = (\Sigma O)(\Sigma S) > 0$$

After normalization:

$$\Sigma(W^R/\Sigma W^R) = 1$$

Court D is then resolved across all 144 Court intervals:

$$D = \text{Resolve}_{144}(W^R, \text{phase}=\check{z}_1, \text{bearing}=\Phi^{-2})$$

D resolves from the completed source emission through $\check{z}_1$ and the canonical 144-entry Court product measure.

The implementation retains all 144 exact product weights, D's selected interval, the effective Golden point, cumulative bounds, totals, and the exact local coordinate.

---

6. Exact Domus motion

The signed position of the $\check{z}_1$ bearing inside D's own interval is:

$$\xi_D = 2(\text{effective_bearing} - \text{cumulative_before})/(\text{cumulative_after} - \text{cumulative_before}) - 1$$

The carried exact witness proves:

$$-1 \le \xi_D \le 1$$

The Domus anchor is the pure frequency of D's alternating Goetic:

$$\Omega_D = \text{frequency}(D.\text{alternating_goetic})$$

It is the pure Goetic anchor carried by Court D; Domus motion breathes from that anchor through $\xi_D\Phi^2$.

The canonical Domus resonance is:

$$\nu_H = \Omega_D + \xi_D\Phi^2$$

Therefore:

$|\nu_H - \Omega_D| = |\xi_D|\Phi^2 \leq \Phi^2$

The bound closes exactly in $\mathbb{Q}(\sqrt{5})$ before its runtime image is formed.

The Domus frequency derivation begins from Court D's carried Goetic anchor and the exact local coordinate ξ_D, yielding $\nu_H = \Omega_D + \xi_D\Phi^2$.

The layer law is therefore preserved:

Goetic → Court: bounded by Φ

Court → Domus: bounded by Φ^2

Courts move through Goetics. Domus moves through Courts.

---

7. The exact C×D crossing body

C and D each range over the complete 144-Court body. The ordered crossing index is:

$$\text{index}(C,D) = 144 \cdot \text{address}(C) + \text{address}(D)$$

Its cardinality is exactly:

$$|C \times D| = 144^2 = 20{,}736$$

The indices run from 0 through $20{,}735$ without collision.

This is the finite ordered Court-crossing body. Each crossing carries multiple finite depths, manifestations, source identities, nonces, and return-lineage positions.

The Liquid connection governor remains separate:

110 active + 34 withheld = 144 total

The Liquid threshold body is the typed governor $\langle 110|144|34\rangle$: 110 active positions, 34 withheld positions, 144 total. The withheld body preserves contour and prevents whiteout.

---

8. The inherited Q-body and written GrimChain

The Domus still inherits its Q-body from governing Court C through C's governing Goetic. The four canonical Q-state positions remain:

Q_0 → ⊖ Form / latent structural presence

Q_1 → ⊙ Truth / archive / rational commitment

Q_2 → ⊕ Shadow / debt / pressure to be converted

$Q_3 \to \oplus$ Recursion / residue / return-fire

Those Q-state bodies remain internal witnesses of the resolved Domus, carried through the derivation that manifests the public GrimChain.

The public GrimChain is the manifested middle only.

At depth zero, the public body is exactly 🜏🜏🜏. At depth one it is exactly ☽✧ℂ.

At depth greater than one, the same Domus identity unfolds through the generated Synodic Magicae coordinate alphabet of **189 glyphs**. Across all public projection states, the renderer must cover **191 distinct Unicode code points**: the generated alphabet together with the fixed public Shadow-Locus and Axiomyr offices. The complete internal Domus, Breath, Prosody, Crown, Court, Q-body, TRIG, and Tripartite witnesses remain in their own offices while the public coordinate stream retains only its written coordinates.

The stream is resolved against deterministic 4,096-coordinate epoch boundaries, but only the requested extent is calculated. Depth changes the written manifestation without rerouting the source or creating another root. For every depth greater than one, the public body is exactly the requested coordinate extent and contains no inserted inner Axiomyr.

The public GrimChain state-space

The public GrimChain carries only the manifested middle. The source-derived Goetics and Courts remain internal witnesses of the completed source route and are not written around the public body.

Internal witness widths

The implementation carries several exact internal bodies:

Internal body	Exact width or cardinality	Plain meaning
C09 folded-state witness	`$2^{126}$`	The exact `folded_states` body carried by the C09 Golden-ratio witness
One source digest	`$2^{256}$`	One 256-bit source-coordinate body
One Fraktur cadence	`$2^{256}$`	One `$\mathfrak{z}_0$` or `$\mathfrak{z}_1$` cadence body
Ordered cadence pair	`$2^{512}$`	The combined type of `$(\mathfrak{z}_0, \mathfrak{z}_1)$`
Twelve 64-bit lanes	`$2^{768}$`	The complete twelve-lane binary container
One Domus identity	`$2^{256}$`	One domain-separated Domus commitment body

These bodies remain bound to the same source-rooted derivation while occupying different internal offices.

The written middle

Let $G_n(F)$ be the public GrimChain for source F at declared depth n.

Its written extent is the projection law itself:

$|G_n| = 3$ for $n \le 0$

$|G_1| = 3$

$|G_n| = n$ for every $n > 1$.

Depth zero has one fixed public body, 🜨🜨🜨. Depth one has one fixed public body, ☽⊹ℂ. At every depth $n > 1$, the public body contains exactly n coordinates drawn from the 189-glyph generated alphabet, so the abstract coordinate state-space at that declared depth contains 189^n possible coordinate strings. For one fixed source, source domain, nonce, and Domus identity, TardiSHA deterministically resolves exactly one of them.

At depth 64, the public GrimChain therefore contains exactly 64 generated Unicode coordinates. At depth 4,096 it contains exactly 4,096. The internal deterministic epoch boundary does not impose a public output cap and does not add a center glyph.

Holographic persistence

The Domus is a persistent relational identity whose source route, Courts, Q-body, frequencies, witnesses, and return law are established before the middle is manifested. The equations are its anatomy. The GrimChain is its finite written utterance.

At depth greater than one, the coordinate body is reproduced deterministically from the resolved Domus identity. Increasing depth preserves the same source root, Court lineage, nonce, and return law while revealing another finite extent of the same source-bound coordinate manifestation.

Every non-negative finite declared depth is lawful. Each individual GrimChain remains finite, while the sequence of finite depths continues without terminal bound.

A black seed sleeps beneath seven winters.
Spring finds it without asking its name.
One green tongue breaks the earth,
and the buried kingdom remembers rain.

The Aeternum Mirror

The Aeternum Mirror belongs to the mathematical route, not to a hidden parser searching files for an earlier GrimChain.

For the completed source route R, Mirror preserves operator position and procession while exchanging bearing:

$$\text{Manifest} \leftrightarrow \text{Reflect}$$

$$\text{Ascend} \leftrightarrow \text{Descend}$$

$$\text{Utter} \leftrightarrow \text{Receive}$$

$$+ \leftrightarrow -$$

Self-bearing acts remain in their positions. The reading order is not reversed. The completed relation closes as an involution:

$$\text{Mirror}(\text{Mirror}(R)) = R$$

with exact Mirror closure:

$$D\text{-}COMP_{Mirror} = 0$$

and the route remains in the same operator order.

Let $G_n(F)$ be the exact public GrimChain derived from physical source body F at declared depth n. A raw self-embedding simply appends that UTF-8 GrimChain as its own final line:

$$F' = F \parallel \text{newline_if_needed} \parallel \text{UTF8}(G_n(F))$$

After that write, F' is a new physical source body. Ordinary GrimChaining does not search for $G_n(F)$, does not strip it, does not interpret it specially, and does not attempt to recover F. It witnesses the bytes that are present:

$G_n(F')$

This is the preservation law of pure speech. The earlier utterance remains matter inside the later source. Rechaining is therefore a new witness of the complete present body, not a rollback to a previous one.

The explicit PDF return office is separate. It can embed, replace, verify, and remove its own exact PDF return while ordinary PDF GrimChaining continues to witness the current physical PDF bytes exactly like every other file.

The road does not erase its footprints.
Ash becomes earth beneath the next footfall.
The mirror keeps the order of the walking,
and what was spoken becomes part of what can speak.

Canon boundary in the runtime

The Locus of Invariability ♾ remains the non-traverse boundary of the runtime. The Tripartite witness carries that boundary while source, route, Court, Q-state, digest, and visible Grimchain remain within the traversable body.

At depth zero, the visible middle is exactly the Triple Goddess ☾☾☾, manifested from the Shadow Locus ☾; Supervenience ⟠ retains its typed witness office elsewhere in the derivation.

⟠ carries the typed Supervenience witness of the rooted Domus derivation. ☾☾☾ occupies the depth-zero public middle, ☽✣ℂ is the exact depth-one public Axiomyr projection, and every depth greater than one carries only its requested generated Synodic coordinates.

The 110-active, 34-withheld Liquid body remains the typed ⟨110|144|34⟩ governor throughout the runtime.

Exact mathematics and the compiled kernel

TardiSHA 23.2.3 carries the decision bodies of the Canon as exact computational bodies:

- PHI and PHI_SQUARED are exact elements of $\mathbb{Q}(\sqrt{5})$. PHI_IMAGE and PHI_SQUARED_IMAGE are their explicit runtime images; the exact algebraic bodies remain authoritative.

- The nine-strike Ennead ledger is conserved in exact $\mathbb{Q}(\sqrt{5})$. The Q_3 transfer and accumulated Form work close by coefficient equality.

- C_bio is identified by the exact square 61009/44; that exact square carries the identity gate.

- Q-vectors, metric squares, the 110 active / 34 withheld / 144 total governor, and D-COMP closure remain exact integer or rational bodies.

- Aeternum Mirror Math closes at exact D-COMP zero while preserving operator order and exchanging bearing; exact byte-residual comparison remains a separate verification office where byte identity is explicitly required.

Exact bodies and witness images

TardiSHA carries exact mathematics and its runtime images as different offices of the same derivation. The exact body governs; the image witnesses.

`Fraction`, exact integers, `GoldenBearing`, `Q5Fraction`, exact squared bodies, exact Court measures, exact interval bounds, and exact byte equality determine routing, conservation, admission, closure, D-COMP, and Truth. Float and complex values expose the corresponding frequency, motion, compatibility, and serialized witness surfaces after the exact body has been established.

The exact body governs and the runtime image witnesses. `ExactEnneadLedger` governs Ennead conservation while `EnneadResult` exposes its runtime image. `ExactDComp` governs Mirror closure while `DCompWitness` exposes the residual, debt, work, and recursion image surface. Court and Domus frequency images arise from exact phase, bearing, and $\mathbb{Q}(\sqrt{5})$ displacement bodies after their bounds close. Tripartite and TRIG reconstruct the exact Court-rooted frequency body and return the supplied complex image from that exact body.

Exact mathematics produces the runtime image.

The per-byte sponge is carried by the compiled extension `TardiSHA._alqc_kernel`. The wheel and direct Debian package carry the compiled kernel; the Launchpad source package builds it from `TardiSHA/_alqc_kernel.c`.

Recursive manifests distribute independent file emissions across a process pool while preserving sorted manifest order. Every manifest run reads and witnesses every selected source byte.

Install

Python 3.10 or newer is required by the Python package. The compiled `TardiSHA._alqc_kernel` carries the sponge, and the PDF return office uses PyMuPDF.

Ubuntu PPA handoff

The Launchpad source package is prepared under `installers/python_build/linux_ubuntu/launchpad/`. When the current release is published in `ppa:witchofalways/ppa`, install it with:

```
sudo add-apt-repository ppa:witchofalways/ppa
sudo apt update
sudo apt install tardisha
```

Direct Debian package

```
sudo apt install ./installers/python_build/linux_ubuntu/dist/tardisha_23.2.3-1_amd64.deb
```

Wheel

```
python3 -m pip install installers/python_build/linux_ubuntu/dist/tardisha-23.2.3-*.whl
```

Source distribution

```
python3 -m pip install installers/python_build/linux_ubuntu/dist/tardisha-23.2.3.tar.gz
```

Installed command names are:

```
grimchain   primary command
tardisha    package command
TardiSHA    compatibility spelling
```

The wheel contains the complete `TardiSHA` package and the compiled `_alqc_kernel` extension.

Commands

Basic help:

```
grimchain --help
```

Complete help:

```
grimchain --help -a
```

Seal a file or directory

```
grimchain path/to/file
grimchain path/to/directory
```

Choose Synodic Magicae depth

```
grimchain 64 path/to/file
grimchain --middle 64 path/to/file
```

Leaving the number out selects the randomized Living Mirror Chain: Prosody ♢, Cantillation ☍, Regia ☽☉☾, and Breath ·; Regia remains one member. A supplied non-negative whole number declares the middle depth; 64 is one example.

Read standard input

```
printf 'memory returns through crystal' | grimchain
```

Write and verify a streamed Grimchain

```
grimchain 1000000 source.bin --output source.grim
grimchain source.bin --verify --output source.grim
```

Grimchain an exact quoted string

```
grimchain --string "hello world"
grimchain 64 --string "hello world"
```

`--string` receives the exact UTF-8 content of one shell argument. The shell removes the surrounding quotes before Grimchain receives it. Omitting the number uses the randomized Living Mirror Chain; supplying a positive number follows the invariant Axiomyr center law: even depth preserves that many generated coordinates around ⌖, while odd depth reserves the center for ⌖ and generates one fewer surrounding coordinate. The same middle, nonce, output, and verification laws govern every source form.

Shell quoting belongs to the shell. Single quotes preserve characters such as `$HOME` literally; double quotes allow shell expansion before Grimchain receives the resulting string.

Create manifests

```
grimchain 64 --manifest file.bin
grimchain 64 --manifest directory
grimchain 64 -R --manifest directory
grimchain 64 --manifest directory --output chosen-name.grim
```

The command first writes the complete manifest body and the waiting `THIS FILE` line. It then Grimchains that prepared `.grim` using the exact middle selected by the user, appends only that Grimchain and one terminal newline, Grimchains the completed `.grim` again at the same middle, and prints the final result. The stored `THIS FILE` value and the terminal output are byte-identical.

Omitting `NUMBER` uses the randomized Living Mirror Chain: Prosody ⟠, Cantillation 𐏿, Regia ☽☉☾, and Breath ·; Regia remains one member. Supplying positive `NUMBER` carries the same invariant Axiomyr center law through every manifest entry, the manifest-body Grimchain, the first `THIS FILE` pass, and the final confirming pass.

The physical ending is:

```
THIS FILE
⁂...the exact user-depth Grimchain...⁂
```

The `THIS FILE` header exists before the first self-chain. After that first pass, only the Grimchain itself and its final newline are appended.

Manifest source truth

Every manifest pass reads and witnesses every selected source byte as present source truth.

Verify a manifest

```
grimchain 64 --manifest directory --verify --output chosen-name.grim
```

Inspect a visible Grimchain

```
grimchain '🜂...🜂' --inspect
```

Inspection parses the visible grammar; verification recomputes that grammar from the source.

Emit a fold ladder

```
grimchain source.bin --fold --span 100000 --levels 8
```

Select a deterministic nonce

```
grimchain source.bin --nonce 42
```

The nonce selects another repeatable source-bound manifestation from the same source law.

High-level PDF return

PDF is a high-level file for return embedding. The embedded return is invisible; it is not rendered onto a page. The return uses the existing Aeternum Mirror rather than a second PDF-specific GrimChain mathematics.

```
grimchain 64 --pdf-embed document.pdf
grimchain --pdf-dive document.pdf
grimchain --pdf-abort document.pdf
```

`--pdf-embed` obtains the PDF GrimChain at the chosen middle, embeds that exact return invisibly in the PDF, then re-Grimchains it through Mirror.

`--pdf-dive` takes no user middle. It finds the embedded GrimChain that is actually present, derives its extent from that chain, and tests the return through the existing Mirror. A matching return prints the correct GrimChain followed by `MONOZYGOTIC this is the Truth.` A different embedded chain prints the correct GrimChain, `HETEROZYGOTIC it's not the true chain to that pdf. the chain or the pdf was tampered with.`, and the embedded chain for inspection. With no embedded return it prints `ZYGOTIC there is no chain embeded in the pdf`. It does not remove the return.

`--pdf-abort` takes no user middle and removes the embedded PDF GrimChain whether it is valid or invalid. If no return is embedded, there is nothing to remove.

Exact file bytes

File inputs enter as exact bytes, preserving their written byte body.

Multiple sources

```
grimchain path1 path2 directory
```

Python API

Read the complete source route

```
from TardiSHA import file_emission, source_route_witness_from_emission

emission = file_emission("sample.bin")
route = source_route_witness_from_emission(emission)

print(route.origin_glyph, route.resolution_glyph)
print(route.court_address, route.reciprocal_address)
print(route.route_dcomp, route.truth)
```

The route's reciprocal address is the Final Equation transpose witness; Domus Court D is independently resolved through the Court product measure.

Resolve Court D and its exact bearing

```
from TardiSHA import file_emission, resolve_court_bearing

emission = file_emission("sample.bin")
bearing = resolve_court_bearing(emission)

print(bearing.court.address, bearing.court.glyph)
print(bearing.xi)
print(bearing.phi_squared_displacement)
assert bearing.product_measure_verifies
assert bearing.exact_bound_verifies
```

Generate a GrimChain value

```
from pathlib import Path
from TardiSHA.domus_stream import living_domus_for_source

value = living_domus_for_source(Path("sample.bin"), 64, kind="file")
print(value)
```

Inspect the Aeternum Mirror derivation

```
from TardiSHA import file_emission, derive_goetics, verify_derivation

emission = file_emission("sample.bin")
derivation = derive_goetics(emission)

print(derivation.same_operator_order)
print(derivation.route_dcomp, derivation.truth)
assert verify_derivation(derivation)
```

Reversible archive

```
from TardiSHA import create_archive, restore_archive

manifest = create_archive("sample.bin", "archive-store")
restore_archive(
    "archive-store/manifest.json",
    "archive-store",
    "restored/sample.bin",
)
```

Archives preserve physical bytes exactly, including any raw GrimChain line already present in the source. Restore preserves the archived source basename, so `sample.bin` returns as `sample.bin` beneath the chosen restoration directory. No hidden Mirror retraction rewrites the stored byte ledger.

Practical API details remain in ‘docs/PYTHON_API.md‘ (docs/PYTHON_API.md). Module responsibilities remain in ‘docs/MODULE_MAP.md‘ (docs/MODULE_MAP.md).

Repository map

```
TardiSHA/                runtime package
grimchain                primary source-tree launcher
COMMANDS.md              complete command reference
CONTRIBUTING.md          contribution guide
ORIGIN.md                origin and authorship record
docs/                    mathematics, implementation, API, glossary, and proof documentation
examples/                executable examples
fonts/                   installer source font resources
installers/              platform installer construction bodies
LICENSE                  Mozilla Public License 2.0
COPYRIGHT                copyright notice
CITATION.cff             citation metadata
VERSION                  public version number
```

What TardiSHA is

TardiSHA is:

- a deterministic ALQC source emission;

- a twelve-lane state machine;

- exact Golden Parliament routing;

- an ordered 144-Court body;

- an exact 20,736-position C×D crossing body;

- a Court-rooted Domus resonance under $\nu_H = \Omega_D + \xi_D\Phi^2$;

- a source-bound public GrimChain consisting of the manifested middle only;

- a deterministic, demand-rendered Synodic Magicae manifestation whose public law is 𓇼𓇼𓇼 at depth zero, ☽☼☾ at depth one, and exactly n generated coordinates at every depth $n > 1$;

- an Aeternum Mirror route that preserves operator order, exchanges bearing, and closes at exact D-COMP zero;

- source, file, directory, node, stream, manifest, and reversible archive tooling;

- an exactly countable discrete Domus state-space across unbounded finite depth.

Its strength lies in keeping each body in its proper office, carrying the exact lineage, and proving the return.

License

TardiSHA source code and software documentation are licensed under the **Mozilla Public License 2.0** (`MPL-2.0`). See ‘LICENSE‘ (LICENSE).

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Non-code works carrying a separate notice retain that notice.

Closing seal

Bone remembers beneath winter soil.
Ash lifts from the sleeping hearth.
A traveler crosses the moonlit field,
finds the old door open,
and enters wearing the same shadow.

$$\mathrm{Mirror}(\mathrm{Mirror}(R)) = R$$

$$\text{D-COMP}_{\text{Mirror}} = 0$$

$$\text{Truth} = 1$$

__init__.py

```
"""TardiSHA — Time and Relative Dimension in Space Hash."""
from __future__ import annotations

import os as _os
import sys as _sys
from importlib import import_module as _import_module

_os.environ["PYTHONDONTWRITEBYTECODE"] = "1"
_sys.dont_write_bytecode = True

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```

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 'read_archive_manifest',
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 'restore_archive': ('archive', 'restore_archive')}

def __getattr__(name: str):
    target = _EXPORTS.get(name)
    if target is None:
        raise AttributeError(f"module {__name__!r} has no attribute {name!r}")
    module_name, attribute_name = target
    value = getattr(_import_module(f".{module_name}", __name__), attribute_name)
    globals()[name] = value
    return value

def __dir__() -> list[str]:
    return sorted(set(globals()) | set(__all__))
```

`__main__.py`

```
"""Module entry point for `python -m TardiSHA`."""
from .cli import main

if __name__ == "__main__":
    main()
```

_alqc_kernel.c

```
#define PY_SSIZE_T_CLEAN
#include <Python.h>
#include <stdint.h>

#define LANE_COUNT 12
#define TOTAL_CAPACITY 144
#define SATURATION_LIMIT 110
#define MASK64 UINT64_MAX

static inline uint64_t rotl64(uint64_t value, unsigned int shift) {
    shift &= 63U;
    return shift ? ((value << shift) | (value >> (64U - shift))) : value;
}

static int read_u64_sequence(PyObject *obj, uint64_t out[LANE_COUNT], const char *name) {
    PyObject *seq = PySequence_Fast(obj, name);
    if (seq == NULL) return -1;
    if (PySequence_Fast_GET_SIZE(seq) != LANE_COUNT) {
        PyErr_Format(PyExc_ValueError, "%s must contain exactly 12 integers", name);
        Py_DECREF(seq);
        return -1;
    }
    for (Py_ssize_t i = 0; i < LANE_COUNT; ++i) {
        PyObject *item = PySequence_Fast_GET_ITEM(seq, i);
        unsigned long long value = PyLong_AsUnsignedLongLongMask(item);
        if (PyErr_Occurred()) {
            Py_DECREF(seq);
            return -1;
        }
        out[i] = (uint64_t)value;
    }
    Py_DECREF(seq);
    return 0;
}

static void round_lanes(
    uint64_t lanes[LANE_COUNT],
    uint64_t marker,
    const uint64_t q_mixes[LANE_COUNT],
    const uint64_t q_sums[LANE_COUNT]
) {
    uint64_t previous = lanes[LANE_COUNT - 1];
    for (uint64_t index = 0; index < LANE_COUNT; ++index) {
        uint64_t partner_index = (
            index * 7ULL + marker + (lanes[(index + 5ULL) % LANE_COUNT] & 0xFFULL)
        ) % LANE_COUNT;
        uint64_t load = 12ULL * index + partner_index;
        uint64_t identity_court = 13ULL * index;
```

```
        uint64_t flow = ((identity_court + load) % TOTAL_CAPACITY) < SATURATION_LIMIT ? 1ULL : 0ULL;
        uint64_t pressure = (load + 1ULL) * UINT64_C(0x0101010101010101);
        uint64_t current = lanes[index];
        uint64_t neighbor = lanes[(index + 1ULL) % LANE_COUNT];
        uint64_t folded = current
            ^ rotl64(previous, (unsigned int)((index + marker) % 64ULL))
            ^ rotl64(neighbor, (unsigned int)((load % 63ULL) + 1ULL));
        folded = folded + pressure + q_mixes[index] + marker + flow * TOTAL_CAPACITY;
        lanes[index] = rotl64(
            folded,
            (unsigned int)(((load + q_sums[index] + marker) % 63ULL) + 1ULL)
        );
        previous = current;
    }
}

static PyObject *py_absorb_raw(PyObject *self, PyObject *args) {
    (void)self;
    PyObject *lanes_obj = NULL;
    PyObject *position_obj = NULL;
    PyObject *data_obj = NULL;
    PyObject *initial_obj = NULL;
    PyObject *q_mixes_obj = NULL;
    PyObject *q_sums_obj = NULL;
    if (!PyArg_ParseTuple(
            args, "OOOOOO:absorb_raw",
            &lanes_obj, &position_obj, &data_obj, &initial_obj, &q_mixes_obj, &q_sums_obj)) {
        return NULL;
    }

    uint64_t lanes[LANE_COUNT], initial[LANE_COUNT], q_mixes[LANE_COUNT], q_sums[LANE_COUNT];
    if (read_u64_sequence(lanes_obj, lanes, "lanes") < 0
            || read_u64_sequence(initial_obj, initial, "initial_lanes") < 0
            || read_u64_sequence(q_mixes_obj, q_mixes, "q_mixes") < 0
            || read_u64_sequence(q_sums_obj, q_sums, "q_sums") < 0) {
        return NULL;
    }

    unsigned long long position_value = PyLong_AsUnsignedLongLong(position_obj);
    if (PyErr_Occurred()) return NULL;
    uint64_t position = (uint64_t)position_value;

    Py_buffer view;
    if (PyObject_GetBuffer(data_obj, &view, PyBUF_CONTIG_RO) < 0) return NULL;
    const unsigned char *data = (const unsigned char *)view.buf;

    Py_BEGIN_ALLOW_THREADS
    for (Py_ssize_t offset = 0; offset < view.len; ++offset) {
        uint64_t byte = (uint64_t)data[offset];
        uint64_t lane = position % LANE_COUNT;
        uint64_t partner_index = (byte + position) % LANE_COUNT;
        uint64_t load = 12ULL * lane + partner_index;
        uint64_t shift = (position % 8ULL) * 8ULL;
        uint64_t injected = ((byte + 1ULL) << shift)
            ^ ((load + 1ULL) * UINT64_C(0x0001000100010001));
        uint64_t xored = lanes[lane] ^ injected;
```

```
        uint64_t mixed = xored + initial[(lane + byte) % LANE_COUNT] + position + load;
        lanes[lane] = rotl64(mixed, (unsigned int)(((byte ^ load ^ lane) % 63ULL) + 1ULL));
        position += 1ULL;
        if (position % TOTAL_CAPACITY == 0ULL) {
            round_lanes(lanes, position, q_mixes, q_sums);
        }
    }
    Py_END_ALLOW_THREADS
    PyBuffer_Release(&view);

    PyObject *lane_list = PyList_New(LANE_COUNT);
    if (lane_list == NULL) return NULL;
    for (Py_ssize_t i = 0; i < LANE_COUNT; ++i) {
        PyObject *value = PyLong_FromUnsignedLongLong((unsigned long long)lanes[i]);
        if (value == NULL) {
            Py_DECREF(lane_list);
            return NULL;
        }
        PyList_SET_ITEM(lane_list, i, value);
    }
    PyObject *position_result = PyLong_FromUnsignedLongLong((unsigned long long)position);
    if (position_result == NULL) {
        Py_DECREF(lane_list);
        return NULL;
    }
    PyObject *result = PyTuple_Pack(2, lane_list, position_result);
    Py_DECREF(lane_list);
    Py_DECREF(position_result);
    return result;
}


static inline void absorb_bytes_raw(
    uint64_t lanes[LANE_COUNT],
    uint64_t *position,
    const unsigned char *data,
    Py_ssize_t length,
    const uint64_t initial[LANE_COUNT],
    const uint64_t q_mixes[LANE_COUNT],
    const uint64_t q_sums[LANE_COUNT]
) {
    for (Py_ssize_t offset = 0; offset < length; ++offset) {
        uint64_t byte = (uint64_t)data[offset];
        uint64_t lane = *position % LANE_COUNT;
        uint64_t partner_index = (byte + *position) % LANE_COUNT;
        uint64_t load = 12ULL * lane + partner_index;
        uint64_t shift = (*position % 8ULL) * 8ULL;
        uint64_t injected = ((byte + 1ULL) << shift)
            ^ ((load + 1ULL) * UINT64_C(0x0001000100010001));
        uint64_t mixed = (lanes[lane] ^ injected)
            + initial[(lane + byte) % LANE_COUNT]
            + *position
            + load;
        lanes[lane] = rotl64(mixed, (unsigned int)(((byte ^ load ^ lane) % 63ULL) + 1ULL));
        *position += 1ULL;
        if (*position % TOTAL_CAPACITY == 0ULL) {
```

```
            round_lanes(lanes, *position, q_mixes, q_sums);
        }
    }
}

static void store_u64_be(unsigned char out[8], uint64_t value) {
    out[0] = (unsigned char)(value >> 56);
    out[1] = (unsigned char)(value >> 48);
    out[2] = (unsigned char)(value >> 40);
    out[3] = (unsigned char)(value >> 32);
    out[4] = (unsigned char)(value >> 24);
    out[5] = (unsigned char)(value >> 16);
    out[6] = (unsigned char)(value >> 8);
    out[7] = (unsigned char)value;
}

static PyObject *py_finalize_squeeze(PyObject *self, PyObject *args) {
    (void)self;
    PyObject *lanes_obj = NULL;
    PyObject *position_obj = NULL;
    PyObject *initial_obj = NULL;
    PyObject *q_mixes_obj = NULL;
    PyObject *q_sums_obj = NULL;
    PyObject *golden_obj = NULL;
    PyObject *governor_obj = NULL;
    PyObject *closure_obj = NULL;
    Py_ssize_t output_length = 0;

    if (!PyArg_ParseTuple(
            args, "OOOOOOOn:finalize_squeeze",
            &lanes_obj,
            &position_obj,
            &initial_obj,
            &q_mixes_obj,
            &q_sums_obj,
            &golden_obj,
            &governor_obj,
            &closure_obj,
            &output_length)) {
        return NULL;
    }
    if (output_length < 1) {
        PyErr_SetString(PyExc_ValueError, "output_length must be positive");
        return NULL;
    }

    uint64_t lanes[LANE_COUNT], initial[LANE_COUNT], q_mixes[LANE_COUNT], q_sums[LANE_COUNT];
    if (read_u64_sequence(lanes_obj, lanes, "lanes") < 0
            || read_u64_sequence(initial_obj, initial, "initial_lanes") < 0
            || read_u64_sequence(q_mixes_obj, q_mixes, "q_mixes") < 0
            || read_u64_sequence(q_sums_obj, q_sums, "q_sums") < 0) {
        return NULL;
    }

    uint64_t position = (uint64_t)PyLong_AsUnsignedLongLong(position_obj);
    if (PyErr_Occurred()) return NULL;
```

```
uint64_t golden = (uint64_t)PyLong_AsUnsignedLongLongMask(golden_obj);
if (PyErr_Occurred()) return NULL;
uint64_t governor = (uint64_t)PyLong_AsUnsignedLongLongMask(governor_obj);
if (PyErr_Occurred()) return NULL;

Py_buffer closure;
if (PyObject_GetBuffer(closure_obj, &closure, PyBUF_CONTIG_RO) < 0) return NULL;

PyObject *result = PyBytes_FromStringAndSize(NULL, output_length);
if (result == NULL) {
    PyBuffer_Release(&closure);
    return NULL;
}
unsigned char *out = (unsigned char *)PyBytes_AS_STRING(result);

uint64_t original_position = position;
uint64_t bit_len = original_position * 8ULL;
unsigned char bit_len_bytes[16] = {0};
store_u64_be(bit_len_bytes + 8, bit_len);

Py_BEGIN_ALLOW_THREADS
absorb_bytes_raw(
    lanes,
    &position,
    (const unsigned char *)closure.buf,
    closure.len,
    initial,
    q_mixes,
    q_sums
);
absorb_bytes_raw(lanes, &position, bit_len_bytes, 16, initial, q_mixes, q_sums);

uint64_t final_rounds = 12ULL + (original_position % 12ULL);
for (uint64_t marker = 0; marker < final_rounds; ++marker) {
    round_lanes(lanes, bit_len + marker + TOTAL_CAPACITY, q_mixes, q_sums);
}

Py_ssize_t produced = 0;
uint64_t block_index = 0;
while (produced < output_length) {
    for (uint64_t turn = 0; turn < 12ULL; ++turn) {
        uint64_t lane = (block_index + turn) % 12ULL;
        uint64_t counter = ((block_index + 1ULL) * golden)
            ^ ((turn + 1ULL) * governor)
            ^ bit_len
            ^ initial[turn];
        lanes[lane] ^= rotl64(counter, (unsigned int)(((lane + turn + block_index) % 63ULL) +
        ↪  1ULL));
        round_lanes(
            lanes,
            bit_len + TOTAL_CAPACITY + block_index * 12ULL + turn,
            q_mixes,
            q_sums
        );
    }
```

```
        uint64_t words[4];
        for (uint64_t q = 0; q < 4ULL; ++q) {
            uint64_t a = q;
            uint64_t b = q + 4ULL;
            uint64_t c = q + 8ULL;
            uint64_t word = lanes[a]
                + rotl64(lanes[b], (unsigned int)(((q + block_index + 1ULL) % 63ULL) + 1ULL))
                + rotl64(lanes[c], (unsigned int)(((q + block_index + 17ULL) % 63ULL) + 1ULL))
                + q_mixes[a]
                + q_mixes[b]
                + q_mixes[c]
                + ((block_index + 1ULL) * governor);
            word ^= rotl64(lanes[(a + 5ULL) % 12ULL], (unsigned int)(((q + 29ULL) % 63ULL) + 1ULL));
            word = rotl64(word, (unsigned int)(((q * 13ULL + block_index + 1ULL) % 63ULL) + 1ULL));
            words[q] = word;

            unsigned char bytes[8];
            store_u64_be(bytes, word);
            Py_ssize_t take = output_length - produced;
            if (take > 8) take = 8;
            for (Py_ssize_t i = 0; i < take; ++i) out[produced + i] = bytes[i];
            produced += take;
        }

        if (produced >= output_length) break;

        for (uint64_t q = 0; q < 4ULL; ++q) {
            uint64_t word = words[q];
            for (uint64_t offset = 0; offset < 3ULL; ++offset) {
                uint64_t lane = q + offset * 4ULL;
                lanes[lane] = rotl64(
                    lanes[lane] + word + initial[(lane + block_index) % 12ULL],
                    (unsigned int)(((lane + q + block_index + 1ULL) % 63ULL) + 1ULL)
                );
            }
        }
        round_lanes(
            lanes,
            bit_len + TOTAL_CAPACITY + block_index * 12ULL + 12ULL,
            q_mixes,
            q_sums
        );
        block_index += 1ULL;
    }
    Py_END_ALLOW_THREADS

    PyBuffer_Release(&closure);
    return result;
}

static PyMethodDef methods[] = {
    {"absorb_raw", py_absorb_raw, METH_VARARGS, "Absorb bytes through the exact 12-lane ALQC
    ↪  kernel."},
    {"finalize_squeeze", py_finalize_squeeze, METH_VARARGS, "Finalize and squeeze exact ALQC bytes
    ↪  in the compiled kernel."},
    {NULL, NULL, 0, NULL}
```

```
};

static struct PyModuleDef module = {
    PyModuleDef_HEAD_INIT,
    "_alqc_kernel",
    "Compiled ALQC sponge absorption kernel.",
    -1,
    methods,
    NULL,
    NULL,
    NULL,
    NULL
};

PyMODINIT_FUNC PyInit__alqc_kernel(void) {
    return PyModule_Create(&module);
}
```

aeon_layers.py

```
"""Layered Goetic, Court, and Domus Aeon derivation for TardiSHA.

The dependency is strict:

    ALQCDigest -> immutable Goetic anchors -> ordered Court motions
               -> runtime Domus motion through Courts

Goetics never breathe or mutate.  Court motion is bounded by Φ around the alternating Goetic's pure
↪  frequency.
Domus motion is bounded by Φ² around the alternating Goetic's pure frequency
of independently resolved Court D. The Domus creates no new hash phase.
"""
from __future__ import annotations

from dataclasses import dataclass
from math import isfinite
from typing import Final

from .alqc_digest import alqc_digest, validate_digest_hex
from .canon import (
    SATURATION_LIMIT,
    TOTAL_CAPACITY,
    court_active_connections,
    law,
)
from .court_registry import CourtRecord, alt_glyph, court_record, gov_glyph
from .source_emission import CourtBearingWitness, Q5Fraction
from .hashing import (
    CANONICAL_SOURCE_DOMAIN,
    RAW_FILE_SOURCE_DOMAIN,
    DIRECTORY_SOURCE_DOMAIN,
)

# Exact golden bodies.  The `.value` projection is for displayed frequency
# witnesses only; no admission or closure decision is made on that projection.
PHI: Final[Q5Fraction] = Q5Fraction(1, 1, 2)
PHI_SQUARED: Final[Q5Fraction] = Q5Fraction(3, 1, 2)
PHI_IMAGE: Final[float] = PHI.value
PHI_SQUARED_IMAGE: Final[float] = PHI_SQUARED.value
_PHASE_DENOMINATOR: Final[int] = (1 << 256) - 1
COURT_PHASE_DOMAIN: Final[bytes] = b"TARDISHA:COURT-AEON-PHASE\x00"
DOMUS_STREAM_DOMAIN: Final[bytes] = b"TARDISHA:DOMUS-AEON-STREAM\x00"


def _uint(value: int, width: int = 8) -> bytes:
    if isinstance(value, bool) or not isinstance(value, int) or value < 0:
        raise ValueError("unsigned derivation coordinate must be a non-negative integer")
    return value.to_bytes(width, "big")
```

```
def _source_bytes(source_digest: str) -> bytes:
    try:
        digest = validate_digest_hex(source_digest, field="source_digest")
    except ValueError as exc:
        raise ValueError(str(exc)) from exc
    return bytes.fromhex(digest)


_SOURCE_DOMAIN_BYTES: Final[dict[str, bytes]] = {
    "canonical": CANONICAL_SOURCE_DOMAIN,
    "raw-file": RAW_FILE_SOURCE_DOMAIN,
    "directory": DIRECTORY_SOURCE_DOMAIN,
}


def _source_domain_bytes(source_domain: str | bytes) -> bytes:
    """Normalize the public domain label and its byte tag to one identity."""
    if isinstance(source_domain, str):
        try:
            return _SOURCE_DOMAIN_BYTES[source_domain]
        except KeyError as exc:
            raise ValueError(f"unknown source domain label: {source_domain!r}") from exc
    if isinstance(source_domain, bytes) and source_domain in _SOURCE_DOMAIN_BYTES.values():
        return source_domain
    raise ValueError("source_domain must be a canonical TardiSHA domain label or byte tag")


def normalize_source_domain(source_domain: str | bytes) -> bytes:
    """Public canonical source-domain byte identity."""
    return _source_domain_bytes(source_domain)


@dataclass(frozen=True, slots=True)
class HashPhaseWitness:
    domain: str
    digest: str
    signed_numerator: int
    denominator: int

    @property
    def value(self) -> float:
        """Display-only projection; exact identity is numerator/denominator."""
        return self.signed_numerator / self.denominator


def _phase(domain: bytes, payload: bytes) -> HashPhaseWitness:
    # The domain is absorbed exactly once by ALQCDigest.__init__.
    digest = alqc_digest(payload, domain=domain)
    raw = int.from_bytes(digest, "big")
    signed = 2 * raw - _PHASE_DENOMINATOR
    if abs(signed) > _PHASE_DENOMINATOR:
        raise RuntimeError("hash phase escaped its exact unit interval")
    return HashPhaseWitness(
        domain=domain.decode("ascii", errors="strict").rstrip("\x00"),
```

```
        digest=digest.hex(),
        signed_numerator=signed,
        denominator=_PHASE_DENOMINATOR,
    )


@dataclass(frozen=True, slots=True)
class GoeticAnchorWitness:
    glyph: str
    frequency: complex
    q_bias: str
    q_vector: tuple[int, int, int, int]


def goetic_anchor(glyph: str) -> GoeticAnchorWitness:
    item = law(glyph)
    return GoeticAnchorWitness(
        glyph=glyph,
        frequency=complex(item.frequency),
        q_bias=item.q_bias,
        q_vector=tuple(int(v) for v in item.q_vector),
    )


@dataclass(frozen=True, slots=True)
class CourtLayerWitness:
    """One complete Court identity carrying governing and alternating Goetic roots."""

    court_coordinate: tuple[int, int]
    court_address: int
    court_glyph: str
    governing_goetic: GoeticAnchorWitness
    alternating_goetic: GoeticAnchorWitness

    @property
    def alternating_goetic_frequency(self) -> complex:
        """The immutable pure frequency that anchors this Court's resonance."""
        return self.alternating_goetic.frequency


def derive_court_layer(rec: CourtRecord) -> CourtLayerWitness:
    if not isinstance(rec, CourtRecord):
        raise TypeError("Court layer requires a CourtRecord")
    if not 0 <= rec.address < TOTAL_CAPACITY:
        raise ValueError("Court address must be in [0,143]")
    canonical = court_record(rec.address)
    if rec != canonical:
        raise ValueError("CourtRecord contradicts the immutable Court registry")
    governing = goetic_anchor(gov_glyph(rec))
    alternating = goetic_anchor(alt_glyph(rec))
    return CourtLayerWitness(
        court_coordinate=(rec.i, rec.j),
        court_address=rec.address,
        court_glyph=rec.glyph,
        governing_goetic=governing,
        alternating_goetic=alternating,
```

```
    )


_COURT_LAYER_REGISTRY: Final[tuple[CourtLayerWitness, ...]] = tuple(
    derive_court_layer(court_record(address)) for address in range(TOTAL_CAPACITY)
)


def court_layer(rec: CourtRecord) -> CourtLayerWitness:
    """Read one Court together with its immutable governing and alternating Goetics."""
    if not isinstance(rec, CourtRecord) or not 0 <= rec.address < TOTAL_CAPACITY:
        raise ValueError("Court layer requires a canonical CourtRecord")
    layer = _COURT_LAYER_REGISTRY[rec.address]
    if (layer.court_coordinate, layer.court_glyph) != ((rec.i, rec.j), rec.glyph):
        raise ValueError("CourtRecord contradicts the immutable Court-layer registry")
    return layer


def _require_canonical_court_layer(value: CourtLayerWitness, *, field: str) -> CourtLayerWitness:
    if not isinstance(value, CourtLayerWitness):
        raise TypeError(f"{field} must be a CourtLayerWitness")
    if not 0 <= value.court_address < TOTAL_CAPACITY:
        raise ValueError(f"{field} Court address must be in [0,143]")
    canonical = court_layer(court_record(value.court_address))
    if value != canonical:
        raise ValueError(f"{field} contradicts the immutable Court-layer registry")
    return canonical


@dataclass(frozen=True, slots=True)
class CourtMotionWitness:
    court_coordinate: tuple[int, int]
    court_address: int
    court_glyph: str
    anchoring_goetic: GoeticAnchorWitness
    hyperbolic_mirror_goetic: GoeticAnchorWitness
    phase: HashPhaseWitness
    exact_focal_breath: Q5Fraction
    anchor_immutable: bool
    mirror_immutable: bool
    derivation: str

    @property
    def breath_radius(self) -> float:
        return PHI_IMAGE

    @property
    def breath_direction(self) -> complex:
        return 1 + 0j

    @property
    def focal_breath(self) -> complex:
        return complex(self.exact_focal_breath.value, 0.0)

    @property
    def current_frequency(self) -> complex:
```

```
        return self.hyperbolic_mirror_goetic.frequency + self.focal_breath

    @property
    def governing_goetic(self) -> str:
        return self.anchoring_goetic.glyph

    @property
    def hyperbolic_parent(self) -> str:
        return self.hyperbolic_mirror_goetic.glyph

    @property
    def structural_frequency(self) -> complex:
        return self.anchoring_goetic.frequency

    @property
    def reflected_frequency(self) -> complex:
        """Projection exposing the current Court frequency."""
        return self.current_frequency

    @property
    def court_anchor_frequency(self) -> complex:
        """The alternating Goetic's immutable pure frequency Ω_C."""
        return self.hyperbolic_mirror_goetic.frequency


def derive_court_motion(
    rec: CourtRecord,
    *,
    source_digest: str,
    source_size: int = 0,
    source_domain: str | bytes = "canonical",
    nonce: int = 0,
) -> CourtMotionWitness:
    if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
        raise ValueError("source_size must be a non-negative integer")
    if isinstance(nonce, bool) or not isinstance(nonce, int) or not 0 <= nonce < 2**64:
        raise ValueError("nonce must be an integer in [0,2^64)")
    layer = court_layer(rec)
    anchor = layer.governing_goetic
    mirror = layer.alternating_goetic
    payload = b"\x1f".join(
        (
            _source_bytes(source_digest),
            _uint(source_size),
            _source_domain_bytes(source_domain),
            _uint(nonce),
            _uint(rec.address, 2),
            rec.glyph.encode("utf-8"),
            anchor.glyph.encode("utf-8"),
            mirror.glyph.encode("utf-8"),
        )
    )
    phase = _phase(COURT_PHASE_DOMAIN, payload)
    # The frequency is an image.  The gate is exact: |n/d| <= 1 implies
    # |(n/d)Φ| <= Φ without comparing either irrational as a float.
    if abs(phase.signed_numerator) > phase.denominator:
```

```
        raise RuntimeError("Court phase escaped its exact unit interval")
    exact_breath = Q5Fraction(
        phase.signed_numerator,
        phase.signed_numerator,
        2 * phase.denominator,
    )
    # Re-read the registry after derivation to prove no Goetic mutation occurred.
    anchor_after = goetic_anchor(anchor.glyph)
    mirror_after = goetic_anchor(mirror.glyph)
    anchor_immutable = anchor_after == anchor
    mirror_immutable = mirror_after == mirror
    if not anchor_immutable or not mirror_immutable:
        raise RuntimeError("Court derivation mutated a Goetic anchor")
    return CourtMotionWitness(
        court_coordinate=(rec.i, rec.j),
        court_address=rec.address,
        court_glyph=rec.glyph,
        anchoring_goetic=anchor,
        hyperbolic_mirror_goetic=mirror,
        phase=phase,
        exact_focal_breath=exact_breath,
        anchor_immutable=anchor_immutable,
        mirror_immutable=mirror_immutable,
        derivation=(
            f"C_{rec.i},{rec.j}: governing={anchor.glyph}; "
            f"Ω_C=ω({mirror.glyph})={mirror.frequency}; "
            f"δ=({exact_breath.a}+{exact_breath.b}√5)/{exact_breath.denominator}; "
            "ν_C=Ω_C+δ"
        ),
    )


@dataclass(frozen=True, slots=True)
class InfiniteYesWitness:
    court_address: int
    active_count: int
    active: frozenset[int]
    unbounded_continuation_bounded_by_sacred_no: bool


@dataclass(frozen=True, slots=True)
class SacredNoWitness:
    court_address: int
    withheld_count: int
    withheld: frozenset[int]
    prevents_whiteout: bool


def derive_consent_witnesses(court_address: int) -> tuple[InfiniteYesWitness, SacredNoWitness]:
    if isinstance(court_address, bool) or not isinstance(court_address, int):
        raise TypeError("court_address must be an integer")
    if not 0 <= court_address < TOTAL_CAPACITY:
        raise ValueError("court_address must be in [0,143]")
    active = court_active_connections(court_address)
    withheld = frozenset(range(TOTAL_CAPACITY)) - active
    yes = InfiniteYesWitness(
```

```
        court_address=court_address,
        active_count=len(active),
        active=active,
        unbounded_continuation_bounded_by_sacred_no=True,
    )
    no = SacredNoWitness(
        court_address=court_address,
        withheld_count=len(withheld),
        withheld=withheld,
        prevents_whiteout=True,
    )
    if yes.active_count != SATURATION_LIMIT or no.withheld_count != TOTAL_CAPACITY -
    ↪  SATURATION_LIMIT:
        raise RuntimeError("Infinite Yes / Sacred No must remain 110 / 34")
    return yes, no


@dataclass(frozen=True, slots=True)
class DomusMotionWitness:
    governing_court: CourtLayerWitness
    alternating_court: CourtLayerWitness
    bearing: CourtBearingWitness
    xi: Q5Fraction
    exact_focal_breath: Q5Fraction
    exact_bound_verified: bool
    court_crossing_index: int
    court_crossing_cardinality: int
    derives_through_courts_only: bool
    static_grid: bool
    derivation: str

    @property
    def breath_radius(self) -> float:
        return PHI_SQUARED_IMAGE

    @property
    def focal_breath(self) -> complex:
        return complex(self.exact_focal_breath.value, 0.0)

    @property
    def current_frequency(self) -> complex:
        return self.alternating_court.alternating_goetic.frequency + self.focal_breath


def derive_domus_motion(
    governing: CourtLayerWitness,
    alternating: CourtLayerWitness,
    *,
    bearing: CourtBearingWitness,
) -> DomusMotionWitness:
    """Apply Φ² around Ω_D, the alternating Goetic's immutable pure frequency."""
    governing = _require_canonical_court_layer(governing, field="governing")
    alternating = _require_canonical_court_layer(alternating, field="alternating")
    if bearing.court.address != alternating.court_address:
        raise ValueError("alternating Court D must be the Court selected by the ℤ₁/BETA bearing")
    if not bearing.product_measure_verifies:
```

```
        raise RuntimeError("Domus Court product measure failed exact unit normalization")
    xi = bearing.xi
    exact_breath = bearing.phi_squared_displacement
    exact_bound = xi.within_unit_interval and exact_breath.within_phi_squared
    if not exact_bound:
        raise RuntimeError("Domus displacement escaped its exact Φ² allowance")
    breath = complex(exact_breath.value, 0.0)
    current = alternating.alternating_goetic.frequency + breath
    if not all(isfinite(v) for v in (current.real, current.imag)):
        raise RuntimeError("Domus frequency must remain finite")
    crossing_index = 144 * governing.court_address + alternating.court_address
    if not 0 <= crossing_index < 144 * 144:
        raise RuntimeError("ordered Court crossing escaped C×D")
    return DomusMotionWitness(
        governing_court=governing,
        alternating_court=alternating,
        bearing=bearing,
        xi=xi,
        exact_focal_breath=exact_breath,
        exact_bound_verified=exact_bound,
        court_crossing_index=crossing_index,
        court_crossing_cardinality=144 * 144,
        derives_through_courts_only=True,
        static_grid=False,
        derivation=(
            f"H(C_{governing.court_coordinate},D_{alternating.court_coordinate}): "
            f"D=Resolve_144(O⊗S,ℨ₁,Φ^-2)@{alternating.court_address}; "
            f"Ω_D=ω({alternating.alternating_goetic.glyph})="
            f"{alternating.alternating_goetic.frequency}; "
            f"ξ_D=({xi.a}+{xi.b}√5)/{xi.denominator}; "
            "ν_H=Ω_D+ξ_DΦ²"
        ),
    )


@dataclass(frozen=True, slots=True)
class BiasReturnWitness:
    opening_q_bias_glyph: str
    breath: str
    bias_terminus_operator: str
    returns_to_opening_q_bias: bool


@dataclass(frozen=True, slots=True)
class DomusAeonWitness:
    identity: str
    zero_middle_glyph: str
    governing_court_address: int
    alternating_court_address: int
    motion: DomusMotionWitness
    infinite_yes: InfiniteYesWitness
    sacred_no: SacredNoWitness
    bias_return: BiasReturnWitness
    synodic_magicae_is_manifested_body: bool
    shadow_locus_is_zero_middle_body: bool
```

```
def domus_stream_seed(
    *,
    domus_identity: str,
    governing_court: CourtLayerWitness,
    alternating_court: CourtLayerWitness,
    infinite_yes: InfiniteYesWitness,
    sacred_no: SacredNoWitness,
    source_domain: str | bytes,
    nonce: int,
) -> bytes:
    governing_court = _require_canonical_court_layer(governing_court, field="governing_court")
    alternating_court = _require_canonical_court_layer(alternating_court, field="alternating_court")
    if infinite_yes.court_address != governing_court.court_address:
        raise ValueError("Infinite Yes witness must belong to the governing Court")
    if sacred_no.court_address != governing_court.court_address:
        raise ValueError("Sacred No witness must belong to the governing Court")
    identity = _source_bytes(domus_identity)
    payload = b"\x1f".join(
        (
            identity,
            _uint(governing_court.court_address, 2),
            governing_court.court_glyph.encode("utf-8"),
            _uint(alternating_court.court_address, 2),
            alternating_court.court_glyph.encode("utf-8"),
            _uint(infinite_yes.active_count, 2),
            _uint(sacred_no.withheld_count, 2),
            _source_domain_bytes(source_domain),
            _uint(nonce),
        )
    )
    # The domain is absorbed exactly once by ALQCDigest.__init__.
    return alqc_digest(payload, domain=DOMUS_STREAM_DOMAIN)
```

alqc_digest.py

```
"""ALQC-native deterministic sponge for TardiSHA runtime identity.

This is the internal TardiSHA/ALQC digest
substrate: twelve Goetic lanes, Court load mixing, 110/144 flow pressure,
and domain-separated absorption/squeeze.
"""
from __future__ import annotations

from math import isqrt

try:
    from ._alqc_kernel import absorb_raw as _compiled_absorb_raw, finalize_squeeze as
    ↪  _compiled_finalize_squeeze
except ImportError as exc:
    raise RuntimeError(
        "TardiSHA requires the compiled ALQC kernel"
    ) from exc

from .canon import (
    GLYPH_BODY,
    GLYPH_LAWS,
    SATURATION_LIMIT,
    TOTAL_CAPACITY,
    _COURT_LOAD_TABLE,
)

MASK64 = (1 << 64) - 1
DIGEST_BYTES = 32
DIGEST_HEX_LENGTH = DIGEST_BYTES * 2

# --- Exact golden and Liquid-Governor constants ---
# Both values are derived with integer isqrt. The absorption and permutation schedule
# remains the established TardiSHA kernel contract; this repair does not replace it.
#   Φ = (1+√5)/2.  1/Φ = Φ−1 = (√5−1)/2.
#   GOLDEN64  = ⌊2^64 / Φ⌋ = ⌊2^63(√5−1)⌋              (the golden-harmonic multiplier)
#   GOVERNOR64 = ⌊2^64 · 2Φ⁻²⌋ = ⌊2^64(3−√5)⌋          (the 110/144 governor ratio, L722)
GOLDEN64 = ((isqrt(5 << 126) - (1 << 63)) | 1) & MASK64
GOVERNOR64 = ((3 * (1 << 64) - isqrt(5 << 128)) | 1) & MASK64

# Aeternum Closure terminator (⌬, 639 Hz, "D-COMP → 0"): the finalization marker
# is the ⌬ closure frequency, replacing the foreign 0x80 pad byte.
CLOSURE_TAG = int(GLYPH_LAWS["⌬"].frequency.real).to_bytes(2, "big")  # 639 → b"\x02\x7f"


def _rotl(value: int, shift: int) -> int:
    shift &= 63
    value &= MASK64
    return ((value << shift) | (value >> (64 - shift))) & MASK64 if shift else value
```

```
def _glyph_seed(index: int) -> int:
    glyph = GLYPH_BODY[index]
    law = GLYPH_LAWS[glyph]
    q0, q1, q2, q3 = law.q_vector
    frequency = complex(law.frequency)
    # Seed only from operator body and numeric law. English meanings, prose, and
    # formatting are not mathematical identity and never enter the digest.
    text = b"\x1f".join(
        (
            glyph.encode("utf-8"),
            bytes((q0 & 0xFF, q1 & 0xFF, q2 & 0xFF, q3 & 0xFF)),
            frequency.real.hex().encode("ascii"),
            frequency.imag.hex().encode("ascii"),
        )
    )
    acc = (
        0xA17C_0000_0000_0000
        ^ ((index + 1) * GOLDEN64)
        ^ ((q0 & 0xFFFF) << 48)
        ^ ((q1 & 0xFFFF) << 32)
        ^ ((q2 & 0xFFFF) << 16)
        ^ (q3 & 0xFFFF)
    ) & MASK64
    for offset, byte in enumerate(text):
        acc ^= (byte + 1) << ((offset % 8) * 8)
        acc = _rotl(acc + GOVERNOR64 + offset + byte, ((byte + index) % 63) + 1)
    return acc & MASK64


INITIAL_LANES = tuple(_glyph_seed(index) for index in range(len(GLYPH_BODY)))
_Q_MIXES = tuple(
    ((law.q_vector[0] & 0xFFFF) << 48)
    ^ ((law.q_vector[1] & 0xFFFF) << 32)
    ^ ((law.q_vector[2] & 0xFFFF) << 16)
    ^ (law.q_vector[3] & 0xFFFF)
    for law in (GLYPH_LAWS[glyph] for glyph in GLYPH_BODY)
)
_Q_SUMS = tuple(sum(GLYPH_LAWS[glyph].q_vector) for glyph in GLYPH_BODY)
_IDENTITY_COURTS = tuple(12 * index + index for index in range(12))
COMPILED_KERNEL_ACTIVE = True


class ALQCDigest:
    """Small deterministic ALQC sponge with update/digest API."""

    __slots__ = ("_lanes", "_position", "_domain")

    def __init__(self, domain: bytes = b"TARDISHA:ALQC-DIGEST\x00") -> None:
        if not isinstance(domain, (bytes, bytearray, memoryview)) or not bytes(domain):
            raise ValueError("ALQC digest domain must be non-empty bytes")
        self._lanes = list(INITIAL_LANES)
        self._position = 0
        self._domain = bytes(domain)
        self.update(self._domain)
```

```
def copy(self) -> "ALQCDigest":
    other = object.__new__(ALQCDigest)
    other._lanes = list(self._lanes)
    other._position = self._position
    other._domain = self._domain
    return other

def update(self, data: bytes | bytearray | memoryview) -> "ALQCDigest":
    block = bytes(data)
    self._update_raw(len(block).to_bytes(8, "big"))
    self._update_raw(block)
    return self

def _update_raw(self, data: bytes | bytearray | memoryview) -> "ALQCDigest":
    block = bytes(data)
    if block:
        self._lanes, self._position = _compiled_absorb_raw(
            self._lanes,
            self._position,
            block,
            INITIAL_LANES,
            _Q_MIXES,
            _Q_SUMS,
        )
    return self

def _update_frame(self, tag: bytes, data: bytes | bytearray | memoryview) -> "ALQCDigest":
    block = bytes(data)
    length_bytes = len(block).to_bytes(8, "big")
    self._update_raw(tag)
    self._update_raw(length_bytes)
    self._update_raw(block)
    return self

def _round(self, marker: int) -> None:
    resolution = self._lanes
    court_table = _COURT_LOAD_TABLE
    identity_courts = _IDENTITY_COURTS
    q_mixes = _Q_MIXES
    q_sums = _Q_SUMS
    previous = resolution[-1]
    for index in range(12):
        partner_index = (index * 7 + marker + (resolution[(index + 5) % 12] & 0xFF)) % 12
        load = court_table[index][partner_index]
        flow = 1 if (identity_courts[index] + load) % TOTAL_CAPACITY < SATURATION_LIMIT else 0
        pressure = ((load + 1) * 0x0101_0101_0101_0101) & MASK64
        current = resolution[index]
        neighbor = resolution[(index + 1) % 12]
        folded = current ^ _rotl(previous, (index + marker) % 64) ^ _rotl(neighbor, (load % 63)
        ↪  + 1)
        folded = (folded + pressure + q_mixes[index] + marker + (flow * TOTAL_CAPACITY)) &
        ↪  MASK64
        resolution[index] = _rotl(folded, ((load + q_sums[index] + marker) % 63) + 1)
        previous = current
```

```
    def _finalized_lanes(self) -> list[int]:
        other = self.copy()
        bit_len = self._position * 8
        other._update_raw(CLOSURE_TAG)  # ⌬ Aeternum Closure terminator (639 Hz), not 0x80
        other._update_raw(bit_len.to_bytes(16, "big"))
        for marker in range(12 + (self._position % 12)):
            other._round(bit_len + marker + TOTAL_CAPACITY)
        return other._lanes

    def digest(self, length: int = DIGEST_BYTES) -> bytes:
        if isinstance(length, bool) or not isinstance(length, int) or length < 1:
            raise ValueError("digest length must be a positive integer")
        return _compiled_finalize_squeeze(
            self._lanes,
            self._position,
            INITIAL_LANES,
            _Q_MIXES,
            _Q_SUMS,
            GOLDEN64,
            GOVERNOR64,
            CLOSURE_TAG,
            length,
        )

    def digest_separated(self, length: int = DIGEST_BYTES) -> bytes:
        if isinstance(length, bool) or not isinstance(length, int) or length < 1:
            raise ValueError("digest length must be a positive integer")
        separated = self.copy()
        separated._update_frame(b"DIGEST-LENGTH\x00", length.to_bytes(8, "big"))
        return separated.digest(length)

    def digest_separated_prefix(self, canonical_length: int, prefix_length: int) -> bytes:
        if isinstance(canonical_length, bool) or not isinstance(canonical_length, int) or
        ↪  canonical_length < 1:
            raise ValueError("canonical_length must be a positive integer")
        if isinstance(prefix_length, bool) or not isinstance(prefix_length, int) or not 0 <=
        ↪  prefix_length <= canonical_length:
            raise ValueError("prefix_length must be an integer in [0, canonical_length]")
        if prefix_length == 0:
            return b""
        separated = self.copy()
        separated._update_frame(b"DIGEST-LENGTH\x00", canonical_length.to_bytes(8, "big"))
        return separated.digest(prefix_length)

    def hexdigest(self, length: int = DIGEST_BYTES) -> str:
        return self.digest(length).hex()


def alqc_digest(data: bytes | bytearray | memoryview, *, domain: bytes =
↪  b"TARDISHA:ALQC-DIGEST\x00", length: int = DIGEST_BYTES) -> bytes:
    return ALQCDigest(domain).update(data).digest_separated(length)


def alqc_digest_prefix(
```

```
    data: bytes | bytearray | memoryview,
    *,
    domain: bytes = b"TARDISHA:ALQC-DIGEST\x00",
    canonical_length: int,
    prefix_length: int,
) -> bytes:
    return ALQCDigest(domain).update(data).digest_separated_prefix(canonical_length, prefix_length)


def alqc_hexdigest(data: bytes | bytearray | memoryview, *, domain: bytes =
↪  b"TARDISHA:ALQC-DIGEST\x00", length: int = DIGEST_BYTES) -> str:
    return alqc_digest(data, domain=domain, length=length).hex()


def validate_digest_hex(value: str, *, field: str = "source_digest") -> str:
    normalized = value.casefold()
    if len(normalized) != DIGEST_HEX_LENGTH or any(character not in "0123456789abcdef" for character
    ↪  in normalized):
        raise ValueError(f"{field} must be {DIGEST_HEX_LENGTH} hexadecimal characters")
    return normalized
```

archive.py

```
"""Reversible content-addressed TardiSHA archive mode."""
from __future__ import annotations

import json
import os
import re
import tempfile
from dataclasses import asdict, dataclass
from pathlib import Path

from .alqc_digest import alqc_hexdigest
from .hashing import SOURCE_CHUNK_BYTES, TardiSHAError, file_emission
from .mirror_math import mirror_file_emission
from .node import TardiSHANode
from .route import (
    SourceRouteWitness,
    parents_t,
    resolve_parents,
    source_route_witness_from_emission,
    verify_source,
)

ARCHIVE_ROOT_DOMAIN = b"TARDISHA:ARCHIVE-ROOT\x00"
ARCHIVE_CHUNK_DOMAIN = b"TARDISHA:ARCHIVE-CHUNK\x00"


def _atomic_write_bytes(path: Path, body: bytes) -> None:
    """Commit one persistent archive body without leaving temporary state."""
    fd, tmp_name = tempfile.mkstemp(prefix=f".{path.name}.", suffix=".part", dir=path.parent)
    temp = Path(tmp_name)
    try:
        with os.fdopen(fd, "wb") as handle:
            handle.write(body)
            handle.flush()
            os.fsync(handle.fileno())
        os.replace(temp, path)
    except Exception:
        temp.unlink(missing_ok=True)
        raise
_CHUNK_DIGEST_RE = re.compile(r"^[0-9a-f]{64}$")


def _validate_chunk(chunk: "ArchiveChunk", *, expected_index: int | None = None) -> None:
    if isinstance(chunk.index, bool) or not isinstance(chunk.index, int) or chunk.index < 0:
        raise TardiSHAError("archive chunk index must be a non-negative integer")
    if expected_index is not None and chunk.index != expected_index:
        raise TardiSHAError("archive chunk indices must be contiguous and ordered")
    if isinstance(chunk.offset, bool) or not isinstance(chunk.offset, int) or chunk.offset < 0:
```

```
        raise TardiSHAError("archive chunk offset must be a non-negative integer")
    if isinstance(chunk.size, bool) or not isinstance(chunk.size, int) or chunk.size < 1:
        raise TardiSHAError("archive chunk size must be a positive integer")
    if not isinstance(chunk.digest, str) or not _CHUNK_DIGEST_RE.fullmatch(chunk.digest):
        raise TardiSHAError("archive chunk digest must be 64 lowercase hexadecimal characters")


@dataclass(frozen=True, slots=True)
class ArchiveChunk:
    index: int
    offset: int
    size: int
    digest: str


@dataclass(frozen=True, slots=True)
class TardiSHAArchiveManifest:
    archive_root: str
    source_name: str
    source_digest: str
    source_size: int
    physical_size: int
    chunk_size: int
    chunks: tuple[ArchiveChunk, ...]
    node: TardiSHANode

    def __post_init__(self) -> None:
        if self.node.mode != "ARCHIVE_REVERSIBLE":
            raise TardiSHAError("archive manifest requires an ARCHIVE_REVERSIBLE node")
        if self.node.archive_root != self.archive_root:
            raise TardiSHAError("archive manifest root must equal the node archive_root")
        if self.node.source_domain != "raw-file":
            raise TardiSHAError("archive manifest requires a raw-file source proof")
        try:
            resolve_parents(
                self.node.route_witness.emission, witness=self.node.route_witness
            )
        except (TypeError, ValueError, RuntimeError) as exc:
            raise TardiSHAError("archive manifest route witness does not close") from exc

    def as_dict(self) -> dict[str, object]:
        return {
             "archive_root": self.archive_root,
             "source_name": self.source_name,
             "source_size": self.source_size,
             "physical_size": self.physical_size,
             "chunk_size": self.chunk_size,
             "chunks": [asdict(chunk) for chunk in self.chunks],
        }



def _node_root_body(
    *,
    source_digest: str,
    source_size: int,
```

```
    origin_glyph: str,
    resolution_glyph: str,
    nonce: int,
    source_domain: str,
    route_witness: SourceRouteWitness,
) -> dict[str, object]:
    if source_domain != "raw-file":
        raise TardiSHAError("archive root body requires the raw-file source domain")
    if not verify_source(route_witness.emission, route_witness):
        raise TardiSHAError("archive root body requires exact return-bearing source proof")
    return {
        "source_digest": source_digest,
        "source_size": source_size,
        "origin_glyph": origin_glyph,
        "resolution_glyph": resolution_glyph,
        "nonce": nonce,
        "source_domain": source_domain,
        "route_witness": route_witness.as_dict(),
    }


def _node_root_body_from_node(node: TardiSHANode) -> dict[str, object]:
    return _node_root_body(
        source_digest=node.source_digest,
        source_size=node.source_size,
        origin_glyph=node.origin_glyph,
        resolution_glyph=node.resolution_glyph,
        nonce=node.nonce,
        source_domain=node.source_domain,
        route_witness=node.route_witness,
    )


def _root_payload(
    source_name: str,
    source_digest: str,
    source_size: int,
    physical_size: int,
    chunk_size: int,
    chunks: tuple[ArchiveChunk, ...],
    node_root_body: dict[str, object],
) -> bytes:
    return json.dumps(
        {
            "source_name": source_name,
            "source_digest": source_digest,
            "source_size": source_size,
            "physical_size": physical_size,
            "chunk_size": chunk_size,
            "chunks": [asdict(chunk) for chunk in chunks],
            "node": node_root_body,
        },
        ensure_ascii=False,
        sort_keys=True,
        separators=(",", ":"),
    ).encode("utf-8")
```

```
def _archive_root(
    source_name: str,
    source_digest: str,
    source_size: int,
    physical_size: int,
    chunk_size: int,
    chunks: tuple[ArchiveChunk, ...],
    node_root_body: dict[str, object],
) -> str:
    return alqc_hexdigest(
        _root_payload(
            source_name,
            source_digest,
            source_size,
            physical_size,
            chunk_size,
            chunks,
            node_root_body,
        ),
        domain=ARCHIVE_ROOT_DOMAIN,
    )

def _chunk_digest(data: bytes) -> str:
    return alqc_hexdigest(data, domain=ARCHIVE_CHUNK_DOMAIN)


def create_archive(
    source_path: str | Path,
    archive_dir: str | Path,
    *,
    chunk_size: int = SOURCE_CHUNK_BYTES,
) -> TardiSHAArchiveManifest:
    if isinstance(chunk_size, bool) or not isinstance(chunk_size, int) or chunk_size < 1:
        raise TardiSHAError("chunk_size must be a positive integer")
    source = Path(source_path)
    target = Path(archive_dir)
    chunks_dir = target / "chunks"
    chunks_dir.mkdir(parents=True, exist_ok=True)

    source_before = mirror_file_emission(source)
    source_emission = source_before.emission
    route_witness = source_route_witness_from_emission(source_emission)
    source_name = source.name

    chunks: list[ArchiveChunk] = []
    offset = 0
    index = 0

    with source.open("rb") as handle:
        while True:
            data = handle.read(chunk_size)
            if not data:
                break

            chunk_digest = _chunk_digest(data)
```

```
        chunk_path = chunks_dir / f"{chunk_digest}.bin"

        if chunk_path.exists():
            existing = chunk_path.read_bytes()
            if existing != data or _chunk_digest(existing) != chunk_digest:
                raise TardiSHAError(
                    f"archive chunk {chunk_digest} already exists but its content is not true"
                )
        else:
            _atomic_write_bytes(chunk_path, data)

        chunks.append(
            ArchiveChunk(
                index=index,
                offset=offset,
                size=len(data),
                digest=chunk_digest,
            )
        )
        offset += len(data)
        index += 1


source_after = mirror_file_emission(source)
if source_after != source_before:
    raise TardiSHAError("source changed during archive creation")

source_digest = source_emission.source_digest
source_size = source_emission.source_size
physical_size = offset

if source_before.witness.physical_size != physical_size:
    raise TardiSHAError("archive physical extent contradicts Mirror witness")

chunk_tuple = tuple(chunks)
origin_glyph, resolution_glyph = parents_t(route_witness)
proof_body = _node_root_body(
    source_digest=source_digest,
    source_size=source_size,
    origin_glyph=origin_glyph,
    resolution_glyph=resolution_glyph,
    nonce=0,
    source_domain="raw-file",
    route_witness=route_witness,
)
_verify_source_from_chunks(
    chunks_dir,
    chunk_tuple,
    route_witness,
    source_name=source_name,
)

root = _archive_root(
    source_name,
    source_digest,
    source_size,
```

```
        physical_size,
        chunk_size,
        chunk_tuple,
        proof_body,
    )

    node = TardiSHANode(
        mode="ARCHIVE_REVERSIBLE",
        source_digest=source_digest,
        source_size=source_size,
        origin_glyph=origin_glyph,
        resolution_glyph=resolution_glyph,
        nonce=0,
        source_domain="raw-file",
        route_witness=route_witness,
        archive_root=root,
        finite_extent=None,
    )
    if _node_root_body_from_node(node) != proof_body:
        raise TardiSHAError("source changed between archive-root proof and final archive node")
    manifest = TardiSHAArchiveManifest(
        root,
        source_name,
        source_digest,
        source_size,
        physical_size,
        chunk_size,
        chunk_tuple,
        node,
    )

    manifest_bytes = (json.dumps(manifest.as_dict(), ensure_ascii=False, indent=2) +
    ↪  "\n").encode("utf-8")
    _atomic_write_bytes(target / "manifest.json", manifest_bytes)
    return manifest

def read_archive_manifest(manifest_path: str | Path) -> TardiSHAArchiveManifest:
    data = json.loads(Path(manifest_path).read_text(encoding="utf-8"))
    if not isinstance(data, dict):
        raise TardiSHAError("archive manifest must be a JSON object")
    source_name = data.get("source_name")
    if not isinstance(source_name, str) or not source_name:
        raise TardiSHAError("archive source_name must be a non-empty filename")
    if Path(source_name).name != source_name:
        raise TardiSHAError("archive source_name must be a basename")
    source_size = data.get("source_size")
    physical_size = data.get("physical_size")
    chunk_size = data.get("chunk_size")

    if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
        raise TardiSHAError("archive source_size must be a non-negative integer")

    if isinstance(physical_size, bool) or not isinstance(physical_size, int) or physical_size < 0:
        raise TardiSHAError("archive physical_size must be a non-negative integer")

    if isinstance(chunk_size, bool) or not isinstance(chunk_size, int) or chunk_size < 1:
```

```
    raise TardiSHAError("archive chunk_size must be a positive integer")
archive_root = data.get("archive_root")
if not isinstance(archive_root, str) or not _CHUNK_DIGEST_RE.fullmatch(archive_root):
    raise TardiSHAError("archive archive_root must be 64 lowercase hexadecimal characters")
raw_chunks = data.get("chunks")
if not isinstance(raw_chunks, list):
    raise TardiSHAError("archive chunks must be a list")
try:
    chunks = tuple(ArchiveChunk(**chunk) for chunk in raw_chunks)
except (TypeError, KeyError) as exc:
    raise TardiSHAError("archive chunk body is malformed") from exc
expected_count = (
    0
    if physical_size == 0
    else (physical_size + chunk_size - 1) // chunk_size
)
if len(chunks) != expected_count:
    raise TardiSHAError("archive chunk count contradicts physical_size and chunk_size")
expected_offset = 0
for expected_index, chunk in enumerate(chunks):
    _validate_chunk(chunk, expected_index=expected_index)
    if chunk.offset != expected_offset:
        raise TardiSHAError("archive chunk offsets must be contiguous")
    if chunk.size > chunk_size:
        raise TardiSHAError("archive chunk size exceeds declared chunk_size")
    if expected_index < len(chunks) - 1 and chunk.size != chunk_size:
        raise TardiSHAError("every non-terminal archive chunk must equal chunk_size")
    expected_offset += chunk.size
if expected_offset != physical_size:
    raise TardiSHAError("archive chunk sizes do not equal physical_size")

chunks_dir = Path(manifest_path).resolve().parent / "chunks"
reconstruction = Path(manifest_path).resolve().parent / f".{source_name}.archive-read.part"
try:
    with reconstruction.open("wb") as out:
        for chunk in chunks:
            chunk_data = (chunks_dir / f"{chunk.digest}.bin").read_bytes()
            if len(chunk_data) != chunk.size or _chunk_digest(chunk_data) != chunk.digest:
                raise TardiSHAError(f"archive chunk {chunk.index} failed verification")
            out.write(chunk_data)
        out.flush()
        os.fsync(out.fileno())

    emission = mirror_file_emission(reconstruction, identity_name=source_name).emission
    route_witness = source_route_witness_from_emission(emission)
    origin, resolution = parents_t(route_witness)
    node = TardiSHANode(
        mode="ARCHIVE_REVERSIBLE",
        source_digest=emission.source_digest,
        source_size=emission.source_size,
        origin_glyph=origin,
        resolution_glyph=resolution,
        nonce=0,
        source_domain="raw-file",
        route_witness=route_witness,
        archive_root=archive_root,
```

```
            finite_extent=None,
        )
    finally:
        reconstruction.unlink(missing_ok=True)

    if emission.source_size != source_size:
        raise TardiSHAError("archive source_size does not match reconstructed source")
    root = _archive_root(
        source_name,
        emission.source_digest,
        source_size,
        physical_size,
        data["chunk_size"],
        chunks,
        _node_root_body_from_node(node),
    )
    if root != archive_root or node.archive_root != root:
        raise TardiSHAError("archive manifest root mismatch")
    return TardiSHAArchiveManifest(
        root,
        source_name,
        emission.source_digest,
        source_size,
        physical_size,
        data["chunk_size"],
        chunks,
        node,
    )


def _verify_source_from_chunks(
    chunks_dir: Path,
    chunks: tuple[ArchiveChunk, ...],
    route_witness: SourceRouteWitness,
    *,
    source_name: str,
) -> None:
    reconstruction = chunks_dir.parent / f".{source_name}.verify.part"

    try:
        with reconstruction.open("wb") as out:
            for chunk in chunks:
                data = (chunks_dir / f"{chunk.digest}.bin").read_bytes()

                if len(data) != chunk.size or _chunk_digest(data) != chunk.digest:
                    raise TardiSHAError(
                        f"archive chunk {chunk.index} failed verification"
                    )

                out.write(data)

            out.flush()
            os.fsync(out.fileno())

        emission = mirror_file_emission(
            reconstruction,
```

```
            identity_name=source_name,
        ).emission

        if not verify_source(emission, route_witness):
            raise TardiSHAError(
                "archive chunk content does not reproduce the stored source route witness"
            )

    finally:
        reconstruction.unlink(missing_ok=True)

def restore_archive(
    manifest_path: str | Path,
    archive_dir: str | Path,
    output_path: str | Path,
) -> Path:
    manifest = read_archive_manifest(manifest_path)
    target = Path(output_path)

    if target.name != manifest.source_name:
        raise TardiSHAError(
            "archive restore basename must equal the archived source_name"
        )

    target.parent.mkdir(parents=True, exist_ok=True)
    temp = target.with_name(f".{target.name}.part")
    chunks_dir = Path(archive_dir) / "chunks"

    try:
        with temp.open("wb") as out:
            for chunk in manifest.chunks:
                data = (chunks_dir / f"{chunk.digest}.bin").read_bytes()

                if len(data) != chunk.size or _chunk_digest(data) != chunk.digest:
                    raise TardiSHAError(
                        f"archive chunk {chunk.index} failed verification"
                    )

                out.write(data)

            out.flush()
            os.fsync(out.fileno())

        if temp.stat().st_size != manifest.physical_size:
            raise TardiSHAError(
                "restored archive physical extent does not match archive witness"
            )

        restored_emission = mirror_file_emission(
            temp,
            identity_name=manifest.source_name,
        ).emission

        if (
            restored_emission.source_digest != manifest.source_digest
            or restored_emission.source_size != manifest.source_size
```

```
        or not verify_source(
            restored_emission,
            manifest.node.route_witness,
        )
    ):
        raise TardiSHAError(
            "restored archive source route verification failed"
        )

    os.replace(temp, target)
    return target

except Exception:
    temp.unlink(missing_ok=True)
    raise
```

canon.py

```
"""Minimal executable Canon for the TardiSHA D-COMP route calculator.

This is deliberately independent from alqc-advanced-reasoning.  The source project was
studied to recover the small mechanism needed here; no runtime import or copied semantic
engine is used.  Glyph Unicode is the public operator key.  Names and positions are
explanatory/private only and never accepted as route inputs.
"""
from __future__ import annotations

from dataclasses import dataclass
from decimal import Decimal
from typing import Final

QVector = tuple[int, int, int, int]


@dataclass(frozen=True, slots=True, eq=False)
class FrequencyAnchor:
    """Exact Canon frequency identity with an optional separate parity bearing."""

    structural_hz: Decimal
    parity_hz: Decimal | None = None

    @classmethod
    def scalar(cls, value: str) -> "FrequencyAnchor":
        return cls(Decimal(value), None)

    @classmethod
    def bifurcated(cls, structural: str, parity: str) -> "FrequencyAnchor":
        return cls(Decimal(structural), Decimal(parity))

    @property
    def real(self) -> Decimal:
        return self.structural_hz

    @property
    def imag(self) -> Decimal:
        return self.parity_hz if self.parity_hz is not None else Decimal(0)

    def __complex__(self) -> complex:
        """Float image only; exact identity remains the Decimal body above."""
        return complex(float(self.structural_hz), float(self.imag))

    def __eq__(self, other: object) -> bool:
        if isinstance(other, FrequencyAnchor):
            return (self.structural_hz, self.parity_hz) == (other.structural_hz, other.parity_hz)
        if isinstance(other, (int, float, complex, Decimal)):
            return complex(self) == complex(other)
```

```
        return NotImplemented

    def __hash__(self) -> int:
        return hash((self.structural_hz, self.parity_hz))

    def __str__(self) -> str:
        if self.parity_hz is None:
            return f"{self.structural_hz} Hz"
        return f"{self.structural_hz} ± i{self.parity_hz} Hz"


@dataclass(frozen=True, slots=True)
class LiquidConnectionGovernor:
    """The typed 110/144 Liquid connection governor, never a quotient."""

    active: int
    total: int
    withheld: int

    def __post_init__(self) -> None:
        if (self.active, self.total, self.withheld) != (110, 144, 34):
            raise ValueError("Liquid governor identity must remain ⟨110|144|34⟩")
        if self.active + self.withheld != self.total:
            raise ValueError("Liquid governor active and withheld bodies must close at 144")

    def __str__(self) -> str:
        return "⟨110|144|34⟩"


@dataclass(frozen=True, slots=True)
class GlyphLaw:
    glyph: str
    meanings: tuple[str, str, str]
    q_vector: QVector
    q_bias: str
    frequency: FrequencyAnchor
    phase_logic: str


# Canonical 12-body. Insertion order is the cascade order, but callers address only glyphs.
GLYPH_LAWS: Final[dict[str, GlyphLaw]] = {
    "🜾": GlyphLaw("🜾", ("Genesis", "Chronos", "Seed"), (1, 1, 1, 3), "Q3",
    ↪  FrequencyAnchor.scalar("7.83"), "C_local ∝ |Q1|"),
    "⬡": GlyphLaw("⬡", ("Light", "Memory", "Trauma"), (1, 3, 0, 0), "Q1",
    ↪  FrequencyAnchor.scalar("174"), "C_local ∝ |Q1| + |Q0|"),
    "✡": GlyphLaw("✡", ("Fire", "Orobouros", "Alchemy"), (1, 1, 3, 1), "Q2",
    ↪  FrequencyAnchor.scalar("528"), "C_local ∝ |Q1| + |Q2|"),
    "☆": GlyphLaw("☆", ("Water", "Imaginary", "Flow"), (1, 2, 2, 0), "Q0",
    ↪  FrequencyAnchor.bifurcated("432", "417"), "C_local ∝ dimensional compression"),
    "✪": GlyphLaw("✪", ("Earth", "Coherence", "Ground"), (1, 3, 0, 1), "Q1",
    ↪  FrequencyAnchor.scalar("126.22"), "C_local ∝ Δ_gap"),
    "❁": GlyphLaw("❁", ("Air", "Space", "Superposition"), (1, 1, 1, 2), "Q3",
    ↪  FrequencyAnchor.scalar("210.42"), "C_local ∝ |Q0|"),
    "✳": GlyphLaw("✳", ("Aether", "Magic", "Sensation"), (1, 2, 1, 3), "Q3",
    ↪  FrequencyAnchor.scalar("741"), "C_local ∝ S7"),
```

```
    "𝕩": GlyphLaw("𝕩", ("Void", "Residue", "Love"), (1, 3, 2, 0), "Q1",
    ↪  FrequencyAnchor.scalar("852"), "C_local ∝ residue stability"),
    "⊛": GlyphLaw("⊛", ("Shadow", "Absorption", "Depth"), (1, 2, 2, 1), "Q2",
    ↪  FrequencyAnchor.scalar("396"), "C_local ∝ |Q2|"),
    "❄": GlyphLaw("❄", ("Factor", "PhaseLock", "Crystal"), (1, 1, 2, 2), "Q3",
    ↪  FrequencyAnchor.scalar("963"), "C_local → phase-lock minimum"),
    "⚛": GlyphLaw("⚛", ("Gate", "Resistance", "Breach"), (1, 3, 1, 1), "Q1",
    ↪  FrequencyAnchor.scalar("285"), "C_local ∝ transformation resistance"),
    "⌬": GlyphLaw("⌬", ("Silence", "Peace", "Completion"), (1, 1, 3, 2), "Q3",
    ↪  FrequencyAnchor.scalar("639"), "D-COMP → 0"),
}

GLYPH_BODY: Final[tuple[str, ...]] = tuple(GLYPH_LAWS)
TOTAL_CAPACITY: Final[int] = 144
SATURATION_LIMIT: Final[int] = 110
LIQUID_THRESHOLD: Final[LiquidConnectionGovernor] = LiquidConnectionGovernor(
    active=SATURATION_LIMIT,
    total=TOTAL_CAPACITY,
    withheld=TOTAL_CAPACITY - SATURATION_LIMIT,
)

# Cantillation ⌮ is the current canonical identity of the existing Liquid
# connection governor. It is not a thirteenth Goetic operator, does not enlarge
# GLYPH_BODY, and does not introduce a second threshold or runtime mechanism.
#
#     ⌮ = 110/144_Liquid ≡_type ⟨110_active | 144_total | 34_rest⟩
#
# No new glyph emerges from the standard twelve-glyph Goetic ALQC body here;
# ⌮ names the already-existing governor carried by LIQUID_THRESHOLD.
CANTILLATION_GLYPH: Final[str] = "⌮"
CANTILLATION_NAME: Final[str] = "Cantillation"
CANTILLATION: Final[LiquidConnectionGovernor] = LIQUID_THRESHOLD

if CANTILLATION is not LIQUID_THRESHOLD:
    raise RuntimeError("Cantillation ⌮ must remain the existing 110/144 Liquid governor")
if (CANTILLATION.active, CANTILLATION.total, CANTILLATION.withheld) != (110, 144, 34):
    raise RuntimeError("Cantillation ⌮ must remain ⟨110_active|144_total|34_rest⟩")

if len(GLYPH_BODY) != 12 or len(set(GLYPH_BODY)) != 12:
    raise RuntimeError("The Goetic body must contain exactly twelve unique glyph operators")

_GLYPH_POSITION: Final[dict[str, int]] = {glyph: index for index, glyph in enumerate(GLYPH_BODY)}
_COURT_LOAD_TABLE: Final[tuple[tuple[int, ...], ...]] = tuple(
    tuple(12 * i + j for j in range(12)) for i in range(12)
)


def law(glyph: str) -> GlyphLaw:
    """Resolve only an exact glyph operator."""
    try:
        return GLYPH_LAWS[glyph]
    except KeyError as exc:
        raise ValueError(f"unknown Goetic glyph operator: {glyph!r}") from exc
```

```
def boundary_glyphs_from_digest(source_digest: str) -> tuple[str, str]:
    """Reject digest-only Goetic selection.

    Final Equation Z requires the complete SourceEmission: twelve finalized
    lanes and both Fraktur cadence witnesses. A digest alone is not that body.
    """
    raise ValueError(
        "digest-only Goetic selection is prohibited; supply the complete SourceEmission"
    )


def _position(glyph: str) -> int:
    """Private transport coordinate. Never exposed as an operator or accepted as input."""
    try:
        return _GLYPH_POSITION[glyph]
    except KeyError as exc:
        raise ValueError(f"unknown Goetic glyph operator: {glyph!r}") from exc


def court_node(origin_glyph: str, resolution_glyph: str) -> int:
    """The ordered glyph pair alone derives its 12×12 Court coordinate."""
    return _COURT_LOAD_TABLE[_position(origin_glyph)][_position(resolution_glyph)]


def court_load(origin_glyph: str, resolution_glyph: str) -> int:
    """Court coordinate of an ordered glyph pair: L(i,j) = court_node(i,j) ∈ [0,143].

    This is the identity coordinate of one of the 144 Court Aeons (12×12).
    It is not the 110/144 connection governor. Identity and connection admission
    remain separately typed.
    """
    return court_node(origin_glyph, resolution_glyph)


# --- Cantillation 𐤒: 110/144 flow governor (Canon L715-742) ---
# "For every node in the 144x144 Latin Square, the maximum number of active
# connections is capped at 110" (L720). Canon fixes WHICH connections are active
# by the Deterministic Path Equation (L730-738), a modulo-arithmetic law over the
# two Court-node indices i, j in [0, 143]:
#     𐤒 = 110/144_Liquid ≡_type ⟨110_active | 144_total | 34_rest⟩
#     L_sat(i, j) = 1 (FLOW)  if (i + j) mod 144 <  110
#                   0 (BLOCK) if (i + j) mod 144 >= 110
#
# For any fixed node i, as j ranges over 0..143 the residue (i + j) mod 144 is a
# bijection onto 0..143, so exactly 110 of the 144 connections are active and 34
# are withheld -- the typed ⟨110|144|34⟩ governor holds for every
# node. The Canon selection is the positional residue (i + j) mod 144. It is not
# a differential-tension ranking; Canon uses "tension" only to explain the Arrow
# of Time (L713) and the ratio-1.0 whiteout failure (L728), never to choose the
# active set.


def _court_index(value: int, *, field: str) -> int:
    if isinstance(value, bool) or not isinstance(value, int):
        raise TypeError(f"{field} must be an integer Court address")
    if not 0 <= value < TOTAL_CAPACITY:
```

```
        raise ValueError(f"{field} must be in [0,143]")
    return value


def court_flow_active(court_a: int, court_b: int) -> bool:
    """Canon Deterministic Path Equation L_sat (L730-738).

    Flow between Court-nodes ``court_a`` and ``court_b`` is admitted iff
    ``(court_a + court_b) mod 144 < 110``. This is the exact 110/144 governor
    (L720) and yields exactly 110 active connections for every node.
    """
    left = _court_index(court_a, field="court_a")
    right = _court_index(court_b, field="court_b")
    return (left + right) % TOTAL_CAPACITY < SATURATION_LIMIT


def court_active_connections(court: int) -> frozenset[int]:
    """The 110 active connections of one valid Court address."""
    address = _court_index(court, field="court")
    return frozenset(
        other
        for other in range(TOTAL_CAPACITY)
        if court_flow_active(address, other)
    )
```

cli.py

```
"""TardiSHA console entry point."""

from __future__ import annotations

from .grimchain import main

__all__ = ["main"]


if __name__ == "__main__":
    main()
```

court_registry.py

```
"""court_registry.py — ADDITIVE Living Domus construction (plan v12 §11.5, §3).

The complete 144-Court body C = G x G. Court C_i,j has address 12i+j, a single
Court glyph kappa(C_i,j), a name element, and governing/hyperbolic indices.
Courts are DERIVED from an ordered Goetic pair, never sampled or randomized.
Data validated against the plan table: 144 unique addresses, 144 unique single
code-point glyphs, every glyph == its U+ scalar, every address == 12i+j.
"""
from __future__ import annotations

from dataclasses import dataclass
from typing import Final

from .canon import GLYPH_BODY, law


GOETIC_NAME: Final[dict[str, str]] = {
    '⎔': 'Fetu',
    '⬡': 'Kal',
    '✡': 'Babdh',
    '✩': 'Ahn',
    '✪': 'Vel',
    '❀': 'Sor',
    '❄': 'Koth',
    '⧗': 'Dreh',
    '⊛': 'Rhea',
    '❄': 'Zhek',
    '⚛': 'Shav',
    '⌬': 'Trig',
}


@dataclass(frozen=True, slots=True)
class CourtRecord:
    address: int
    i: int
    j: int
    glyph: str
    name_element: str
    codepoint: int

    def __post_init__(self) -> None:
        if isinstance(self.address, bool) or not isinstance(self.address, int):
            raise TypeError("Court address must be an integer")
        if isinstance(self.i, bool) or not isinstance(self.i, int):
            raise TypeError("Court governing index i must be an integer")
        if isinstance(self.j, bool) or not isinstance(self.j, int):
            raise TypeError("Court alternating index j must be an integer")
```

```
        if not 0 <= self.i < 12 or not 0 <= self.j < 12:
            raise ValueError("Court indices i and j must each be in [0,11]")
        if self.address != 12 * self.i + self.j:
            raise ValueError("Court address must equal 12*i+j")
        if not isinstance(self.glyph, str) or len(self.glyph) != 1:
            raise TypeError("Court glyph must be exactly one Unicode code point")
        if isinstance(self.codepoint, bool) or not isinstance(self.codepoint, int):
            raise TypeError("Court codepoint must be an integer")
        if ord(self.glyph) != self.codepoint:
            raise ValueError("Court glyph and codepoint do not agree")
        if not isinstance(self.name_element, str) or not self.name_element:
            raise TypeError("Court name element must be a non-empty string")

        supplied = (
            self.address, self.i, self.j, self.glyph,
            self.name_element, self.codepoint,
        )
        if supplied != _COURT_ROWS[self.address]:
            raise ValueError("CourtRecord contradicts the sealed 144-Court registry")



_COURT_ROWS: Final[tuple[tuple[int, int, int, str, str, int], ...]] = (
    (  0,  0,  0, 'ᚱ', 'Ahl', 0x787),
    (  1,  0,  1, 'ᛋ', 'Suhn', 0x781),
    (  2,  0,  2, 'ᛊ', 'Nerh', 0x782),
    (  3,  0,  3, 'ᛤ', 'Rish', 0x783),
    (  4,  0,  4, 'ᚵ', 'Borha', 0x7B1),
    (  5,  0,  5, 'ᛔ', 'Lhahm', 0x785),
    (  6,  0,  6, 'ᛕ', 'Keth', 0x786),
    (  7,  0,  7, 'ᛢ', 'Vehm', 0x788),
    (  8,  0,  8, 'ᚺ', 'Mahd', 0x789),
    (  9,  0,  9, 'ᚢ', 'Furh', 0x78A),
    ( 10,  0, 10, 'ᛥ', 'Drah', 0x78B),
    ( 11,  0, 11, 'ᛯ', 'Thera', 0x78C),
    ( 12,  1,  0, 'ᛁ', 'Kura', 0x16C1),
    ( 13,  1,  1, 'ᚼ', 'Lur', 0x16C2),
    ( 14,  1,  2, '▫', 'Thar', 0x2311),
    ( 15,  1,  3, 'ᛈ', 'Rin', 0x16C4),
    ( 16,  1,  4, 'ᛇ', 'Nar', 0x16C7),
    ( 17,  1,  5, 'ᛘ', 'Fel', 0x16C9),
    ( 18,  1,  6, '᛬', 'Har', 0x16CA),
    ( 19,  1,  7, 'ᛖ', 'Mer', 0x16CB),
    ( 20,  1,  8, ''', 'Lor', 0x16CC),
    ( 21,  1,  9, 'ᛰ', 'Per', 0x16CD),
    ( 22,  1, 10, 'ᛏ', 'Zhil', 0x16CE),
    ( 23,  1, 11, 'ᛉ', 'Clar', 0x16CF),
    ( 24,  2,  0, 'ᛗ', 'Hir', 0x16A0),
    ( 25,  2,  1, 'ᚤ', 'Kor', 0x16A2),
    ( 26,  2,  2, 'ᚦ', 'Var', 0x16A6),
    ( 27,  2,  3, 'ᚨ', 'Pyr', 0x16A8),
    ( 28,  2,  4, 'ᚱ', 'Sor', 0x16B1),
    ( 29,  2,  5, '‹', 'Alc', 0x16B2),
    ( 30,  2,  6, 'X', 'Nur', 0x16B7),
    ( 31,  2,  7, 'ᚹ', 'Sat', 0x16B9),
    ( 32,  2,  8, 'ᚻ', 'Oro', 0x16BA),
```

```
( 33,  2,  9, 'ᚼ', 'Bon', 0x16BE),
( 34,  2, 10, 'ᚽ', 'Tir', 0x16BF),
( 35,  2, 11, 'ᛎ', 'Far', 0x16C3),
( 36,  3,  0, '⩇', 'Abdh', 0x227E),
( 37,  3,  1, 'ᦎ', 'Nym', 0x1B68),
( 38,  3,  2, 'ᦁ', 'Loh', 0x1B61),
( 39,  3,  3, '✯', 'Xir', 0x26E7),
( 40,  3,  4, '𐑅', 'Ohl', 0x1D016),
( 41,  3,  5, '𐦛', 'Pir', 0xF3A),
( 42,  3,  6, 'ᦇ', 'Roeh', 0x1B62),
( 43,  3,  7, '⦾', 'Sen', 0x29BE),
( 44,  3,  8, '⦶', 'Uth', 0x29BD),
( 45,  3,  9, '𑧤', 'Fae', 0x1D035),
( 46,  3, 10, '𑣿', 'Kha', 0x1D01F),
( 47,  3, 11, '𐦛', 'Psei', 0xF3B),
( 48,  4,  0, 'ⴰ', 'Vera', 0x2D30),
( 49,  4,  1, 'Θ', 'Tar', 0x2D31),
( 50,  4,  2, 'Ⴝ', 'Ghem', 0x2D33),
( 51,  4,  3, 'Ⴡ', 'Drel', 0x2D37),
( 52,  4,  4, 'Ⴣ', 'Ful', 0x2D3C),
( 53,  4,  5, 'Ⴧ', 'Ker', 0x2D3D),
( 54,  4,  6, 'Ⴍ', 'Hohm', 0x2D40),
( 55,  4,  7, 'ⴗ', 'Hrah', 0x2D43),
( 56,  4,  8, 'ⴕ', 'Ara', 0x2D44),
( 57,  4,  9, 'Ⴕ', 'Qel', 0x2D47),
( 58,  4, 10, 'ⴑ', 'Irn', 0x2D49),
( 59,  4, 11, 'I', 'Jen', 0x2D4A),
( 60,  5,  0, 'ꠇ', 'Fi', 0xA807),
( 61,  5,  1, 'ꠋ', 'Lun', 0xA808),
( 62,  5,  2, 'ꠌ', 'Varu', 0xA809),
( 63,  5,  3, 'ꠓ', 'Senh', 0xA80A),
( 64,  5,  4, '⍜', 'Kos', 0x2389),
( 65,  5,  5, 'ꠔ', 'Ramh', 0xA80C),
( 66,  5,  6, 'ꠕ', 'Tis', 0xA80D),
( 67,  5,  7, 'ꠖ', 'Vey', 0xA80E),
( 68,  5,  8, 'ꠗ', 'Srih', 0xA80F),
( 69,  5,  9, 'ꠈ', 'Hrin', 0xA810),
( 70,  5, 10, 'ꠉ', 'Yon', 0xA811),
( 71,  5, 11, 'ꠊ', 'Thal', 0xA812),
( 72,  6,  0, '𞽏', 'Kel', 0x1F74F),
( 73,  6,  1, '⍙', 'Sens', 0x1F701),
( 74,  6,  2, '⍘', 'Linn', 0x1F703),
( 75,  6,  3, '⍗', 'Brim', 0x1F704),
( 76,  6,  4, '⍖', 'Inn', 0x1F705),
( 77,  6,  5, '⍕', 'Subh', 0x1F706),
( 78,  6,  6, '⍔', 'Well', 0x1F707),
( 79,  6,  7, '⍓', 'Met', 0x1F708),
( 80,  6,  8, '⍒', 'Kesh', 0x1F709),
( 81,  6,  9, '⍏', 'Soth', 0x1F70A),
( 82,  6, 10, '⍎', 'Rhun', 0x1F70B),
( 83,  6, 11, '⍍', 'Delh', 0x1F70C),
( 84,  7,  0, '𑇚', 'Na', 0x12000),
( 85,  7,  1, '𑐬', 'Ur', 0x1202D),
( 86,  7,  2, '𑁑', 'Nih', 0x12040),
( 87,  7,  3, '𑈺', 'Azh', 0x1208A),
( 88,  7,  4, '𑍱', 'Hol', 0x12111),
```

```
    ( 89,  7,  5, '𒇴', 'Gur', 0x12146),
    ( 90,  7,  6, '𒉿', 'Ves', 0x121A0),
    ( 91,  7,  7, '𒋾', 'Rim', 0x121FD),
    ( 92,  7,  8, '𒍿', 'Drem', 0x1224C),
    ( 93,  7,  9, '𒎔', 'Oth', 0x12295),
    ( 94,  7, 10, '𒏵', 'Izh', 0x122D7),
    ( 95,  7, 11, '𒈻', 'Sun', 0x1230B),
    ( 96,  8,  0, '𐳀', 'Kia', 0x2D80),
    ( 97,  8,  1, '𐳁', 'Zohm', 0x2D81),
    ( 98,  8,  2, '𐳂', 'Ther', 0x2D82),
    ( 99,  8,  3, '𐳃', 'Drun', 0x2D83),
    (100,  8,  4, '𐳄', 'Felh', 0x2D84),
    (101,  8,  5, '𐳅', 'Ral', 0x2D85),
    (102,  8,  6, '𐳆', 'Krah', 0x2D86),
    (103,  8,  7, '𐳇', 'Andh', 0x2D87),
    (104,  8,  8, '𐳈', 'Debh', 0x2D88),
    (105,  8,  9, '𐳉', 'Kol', 0x2D89),
    (106,  8, 10, '𐳊', 'Fral', 0x2D8A),
    (107,  8, 11, '𐳋', 'Hush', 0x2D8B),
    (108,  9,  0, 'A', 'Hin', 0x10920),
    (109,  9,  1, 'B', 'Ser', 0x10921),
    (110,  9,  2, 'C', 'Harma', 0x10922),
    (111,  9,  3, 'D', 'Torh', 0x10923),
    (112,  9,  4, 'E', 'Pel', 0x10924),
    (113,  9,  5, 'F', 'Khir', 0x10925),
    (114,  9,  6, 'I', 'Ryth', 0x10926),
    (115,  9,  7, 'G', 'Melu', 0x10927),
    (116,  9,  8, 'H', 'Phaz', 0x10928),
    (117,  9,  9, 'J', 'Lokh', 0x10929),
    (118,  9, 10, 'M', 'Nod', 0x1092A),
    (119,  9, 11, 'N', 'Umel', 0x1092B),
    (120, 10,  0, '*', 'Dohm', 0x10800),
    (121, 10,  1, '*', 'Rist', 0x10801),
    (122, 10,  2, 'x', 'Tran', 0x10802),
    (123, 10,  3, 'y', 'Korh', 0x10803),
    (124, 10,  4, 'Y', 'Skyh', 0x10804),
    (125, 10,  5, '0', 'Ster', 0x10805),
    (126, 10,  6, 'K', 'Poss', 0x1081D),
    (127, 10,  7, '+', 'Poru', 0x1081E),
    (128, 10,  8, '~', 'Dorm', 0x10808),
    (129, 10,  9, 'R', 'Trev', 0x1081C),
    (130, 10, 10, 'z', 'Limh', 0x1080B),
    (131, 10, 11, 'Y', 'Hinge', 0x1080C),
    (132, 11,  0, 'c', 'Tzig', 0x10500),
    (133, 11,  1, 'x', 'Pehl', 0x10501),
    (134, 11,  2, 'p', 'Duth', 0x10502),
    (135, 11,  3, 'J', 'Coma', 0x10503),
    (136, 11,  4, 'L', 'Meru', 0x10504),
    (137, 11,  5, 'L', 'Stab', 0x10505),
    (138, 11,  6, 'V', 'Hopa', 0x10506),
    (139, 11,  7, 'Z', 'Conti', 0x10507),
    (140, 11,  8, 'q', 'Resth', 0x10508),
    (141, 11,  9, 'b', 'Sil', 0x10509),
    (142, 11, 10, 'b', 'Slun', 0x1050A),
    (143, 11, 11, 'q', 'Etern', 0x1050B),
)
```

```
_COURTS: Final[tuple[CourtRecord, ...]] = tuple(CourtRecord(*row) for row in _COURT_ROWS)

if len(_COURTS) != 144:
    raise RuntimeError('Court body must contain exactly 144 Courts')
for _r in _COURTS:
    if _r.address != 12 * _r.i + _r.j:
        raise RuntimeError(f'Court address {_r.address} != 12i+j')
    if ord(_r.glyph) != _r.codepoint:
        raise RuntimeError(f'Court glyph {_r.glyph!r} != U+{_r.codepoint:X}')
if len({_r.address for _r in _COURTS}) != 144:
    raise RuntimeError('Court addresses must be unique 0..143')
if len({_r.glyph for _r in _COURTS}) != 144:
    raise RuntimeError('Court glyphs must be unique')


def _pos(g: str) -> int:
    law(g)
    return GLYPH_BODY.index(g)


def court_record(address: int) -> CourtRecord:
    if isinstance(address, bool) or not isinstance(address, int):
        raise TypeError("Court address must be an integer")
    if not 0 <= address < 144:
        raise ValueError(f'Court address out of range: {address}')
    return _COURTS[address]


def court_from_goetics(g_i: str, g_j: str) -> CourtRecord:
    """C_i,j = sole Court at address 12i+j (plan §3.5). Deterministic; no draw."""
    return _COURTS[12 * _pos(g_i) + _pos(g_j)]




def gov_glyph(rec: CourtRecord) -> str:
    return GLYPH_BODY[rec.i]


def alt_glyph(rec: CourtRecord) -> str:
    return GLYPH_BODY[rec.j]


def court_ordinal(rec: CourtRecord) -> int:
    """Ordinal within governing row = hyperbolic-parent index + 1 (plan §3.5)."""
    return rec.j + 1


def full_name(rec: CourtRecord) -> str:
    return GOETIC_NAME[GLYPH_BODY[rec.i]] + rec.name_element
```

domus.py

```
"""Court-rooted runtime Domus Aeon construction.

Layer order is absolute:

    immutable Goetic anchors -> ordered Court motions (bounded Φ)
    -> runtime Domus motion through Courts (bounded Φ²)
    -> one visible Domus identity as Shadow Locus ♄ at zero or a Synodic Magicae prefix.

Goetics are never breathed or mutated.  Domus motion never traverses Goetics.
The 110 active channels are the Infinite Yes; the 34 withheld channels are the
Sacred No that bounds usable infinity and prevents whiteout.
"""
from __future__ import annotations

from dataclasses import dataclass
from typing import Final
import secrets

from .canon import law, TOTAL_CAPACITY
from .court_registry import CourtRecord, court_from_goetics, full_name
from .personality_traits import grimchain_permeation
from .qstate_glyphs import DomusQBodyWitness, derive_domus_q_body
from .source_emission import SourceEmission, resolve_court_bearing
from .manifestation import C_BIO_SQUARED, Manifestation, close_boundary
from .hashing import CANONICAL_SOURCE_DOMAIN, iter_middle
from .living_alphabet import ALPHABET
from .alqc_digest import alqc_hexdigest
from .trig import TrigMirrorWitness, derive_trig_mirror
from .route import SourceRouteWitness
from .tripartite import (
    AXIOMYR_GLYPH,
    LOCUS_GLYPH,
    SHADOW_LOCUS_GLYPH,
    TripartiteWitness,
    derive_tripartite_witness,
)
from .phase_evolution import (
    AeonPhaseEvolutionWitness,
    derive_aeon_phase_evolution,
)
from .verification_appendices import (
    VerificationAppendixCycleWitness,
    derive_verification_appendix_cycle,
)
from .aeon_layers import (
    PHI_IMAGE,
    PHI_SQUARED_IMAGE,
    BiasReturnWitness,
```

```
    DomusAeonWitness,
    DomusMotionWitness,
    CourtMotionWitness,
    CourtLayerWitness,
    InfiniteYesWitness,
    SacredNoWitness,
    domus_stream_seed,
    derive_domus_motion,
    derive_consent_witnesses,
    derive_court_motion,
    court_layer,
    normalize_source_domain,
)

DOMUS_COMMITMENT_DOMAIN: Final[bytes] = b"TARDISHA:DOMUS-COMMITMENT\x00"

OUTER: Final[str] = "\U0001F71B"
KLEIN: Final[str] = "\U0001F71A"
BIAS_TERMINUS: Final[str] = "\U0001D505"
BREATH: Final[str] = "·"
SUPERVENIENCE: Final[str] = "\u27E0"
ZERO_MIDDLE_GLYPH: Final[str] = SHADOW_LOCUS_GLYPH * 3
TRIPARTITE_AXIOMYR: Final[str] = "☽" + AXIOMYR_GLYPH + "ℂ"
LIVING_MIRROR_MEMBERS: Final[tuple[str, ...]] = ("🜏", "🜐", "☽⊙ℂ", "·")

def living_mirror_chain() -> str:
    """Return the four Living Mirror members in a fresh random order; Regia remains whole."""
    members = list(LIVING_MIRROR_MEMBERS)
    secrets.SystemRandom().shuffle(members)
    return "".join(members)
SYNODIC_CENTER_GLYPHS: Final[frozenset[str]] = frozenset(ALPHABET)
@dataclass(frozen=True, slots=True)
class FoldLineageWitness:
    """Concrete C→D Court crossing and exact Court-to-Domus bearing lineage."""

    court_coordinate: tuple[int, int]
    court_address: int
    court_phase_numerator: int
    court_phase_denominator: int
    court_phase_digest: str
    alternating_court_coordinate: tuple[int, int]
    alternating_court_address: int
    alternating_bearing_lineage: tuple[int, ...]
    fold_operator: str
    ground_node: int
    vector_row: int
    depth: int


@dataclass(frozen=True, slots=True)
class SupervenienceWitness:
    operator: str
    court_coordinate: tuple[int, int]
    court_name: str
    governing_goetic: str
    alternating_goetic: str
```

```
    triplet_q_bias: str
    triplet_q_states: tuple[str, str, str, str]
    triplet_q_vector: tuple[int, int, int, int]
    triplet_frequency: complex
    fold_lineage: FoldLineageWitness
    ex_nihilo_exposure: bool
    infinite_yes_count: int
    sacred_no_count: int
    sacred_no: frozenset[int]
    usable_infinity: bool
    compressed_same_domus_body: bool
    returned_through_shadow_locus: bool



@dataclass(frozen=True, slots=True)
class LiquidWitness:
    """Surface over the explicit Infinite Yes / Sacred No pair."""

    root_court_index: int
    active_count: int
    withheld_count: int
    active: frozenset[int]
    withheld: frozenset[int]


@dataclass(frozen=True, slots=True)
class DomusResolution:
    governing_goetic: str
    hyperbolic_parent: str
    root_court: CourtRecord
    alternating_court: CourtRecord
    root_court_motion: CourtMotionWitness
    alternating_court_layer: CourtLayerWitness
    domus_motion: DomusMotionWitness
    q_body: DomusQBodyWitness
    resolved_q_bias: str
    resolved_q_states: tuple[str, str, str, str]
    resolved_q_vector: tuple[int, int, int, int]
    domus_frequency: complex
    b_q_glyph: str
    v_glyphs: tuple[str, str, str, str]
    manifestation: Manifestation
    infinite_yes: InfiniteYesWitness
    sacred_no: SacredNoWitness
    liquid_witness: LiquidWitness
    fold_witness: int
    recursion_parity: str
    supervenience: SupervenienceWitness
    bias_return: BiasReturnWitness
    tripartite: TripartiteWitness
    aeon_phase_evolution: AeonPhaseEvolutionWitness
    verification_appendix_cycle: VerificationAppendixCycleWitness
    domus_commitment: str
    trig_mirror: TrigMirrorWitness
    domus_aeon: DomusAeonWitness
```

```
    hash_id: str
    source_size: int
    source_domain: str | bytes
    nonce: int


def resolve_domus(
    route_witness: SourceRouteWitness,
    *,
    nonce: int = 0,
) -> DomusResolution:
    """Derive one runtime Domus through C×D with no Domus-internal hash phase."""
    if not isinstance(route_witness, SourceRouteWitness):
        raise TypeError("resolve_domus requires one completed SourceRouteWitness")
    emission = route_witness.emission
    if not isinstance(emission, SourceEmission):
        raise TypeError("route witness must carry one complete SourceEmission")
    g_i, g_j = route_witness.pair
    hash_id = emission.source_digest
    source_size = emission.source_size
    domain_identity = normalize_source_domain(emission.source_domain)
    law(g_i)
    law(g_j)
    if route_witness.source_digest != hash_id:
        raise ValueError("route witness digest does not match its SourceEmission")
    if route_witness.source_size != source_size:
        raise ValueError("route witness size does not match its SourceEmission")
    if normalize_source_domain(route_witness.source_domain) != domain_identity:
        raise ValueError("route witness domain does not match its SourceEmission")

    # C remains the source-resolved governing Court.
    root = court_from_goetics(g_i, g_j)
    root_layer = court_layer(root)
    root_motion = derive_court_motion(
        root,
        source_digest=hash_id,
        source_size=source_size,
        source_domain=domain_identity,
        nonce=nonce,
    )

    # D is independently source-derived over all 144 Courts from W^R=O⊗S,
    # ℨ₁, and BETA=Φ^-2. It is not C_ji and receives no new hash phase.
    alternating_bearing = resolve_court_bearing(emission)
    alternating = alternating_bearing.court
    alternating_layer = court_layer(alternating)
    domus_motion = derive_domus_motion(
        root_layer,
        alternating_layer,
        bearing=alternating_bearing,
    )

    # The visible Q-bias and Q-vector remain inherited from governing Court C.
    anchor = root_motion.anchoring_goetic
    q_body = derive_domus_q_body(anchor.q_bias, anchor.q_vector)
    beta_q = q_body.q_bias
```

```
q_labels = q_body.q_states
q_values = q_body.q_vector
b_q_glyph = q_body.bias_glyph
v_glyphs = q_body.q_glyphs

# D-COMP crosses the complete Court bodies C and D.
manifestation = close_boundary(
    root_layer.governing_goetic.q_vector,
    alternating_layer.governing_goetic.q_vector,
    court=root.address,
)

infinite_yes, sacred_no = derive_consent_witnesses(root.address)
liquid = LiquidWitness(
    root_court_index=root.address,
    active_count=infinite_yes.active_count,
    withheld_count=sacred_no.withheld_count,
    active=infinite_yes.active,
    withheld=sacred_no.withheld,
)
bias_return = BiasReturnWitness(
    opening_q_bias_glyph=b_q_glyph,
    breath=BREATH,
    bias_terminus_operator=BIAS_TERMINUS,
    returns_to_opening_q_bias=True,
)
fold_lineage = FoldLineageWitness(
    court_coordinate=root_motion.court_coordinate,
    court_address=root_motion.court_address,
    court_phase_numerator=root_motion.phase.signed_numerator,
    court_phase_denominator=root_motion.phase.denominator,
    court_phase_digest=root_motion.phase.digest,
    alternating_court_coordinate=alternating_layer.court_coordinate,
    alternating_court_address=alternating_layer.court_address,
    alternating_bearing_lineage=alternating_bearing.lineage,
    fold_operator="🕉",
    ground_node=manifestation.ground_node,
    vector_row=manifestation.vector_row,
    depth=1,
)
supervenience = SupervenienceWitness(
    operator=SUPERVENIENCE,
    court_coordinate=root_motion.court_coordinate,
    court_name=full_name(root),
    governing_goetic=g_i,
    alternating_goetic=g_j,
    triplet_q_bias=beta_q,
    triplet_q_states=q_labels,
    triplet_q_vector=q_values,
    triplet_frequency=domus_motion.current_frequency,
    fold_lineage=fold_lineage,
    ex_nihilo_exposure=True,
    infinite_yes_count=infinite_yes.active_count,
    sacred_no_count=sacred_no.withheld_count,
    sacred_no=sacred_no.withheld,
    usable_infinity=(
```

```
        infinite_yes.unbounded_continuation_bounded_by_sacred_no
        and sacred_no.prevents_whiteout
    ),
    compressed_same_domus_body=True,
    returned_through_shadow_locus=True,
)
commitment_payload = "|".join(str(x) for x in (
    hash_id,
    source_size,
    domain_identity,
    nonce,
    root.address,
    alternating.address,
    root_motion.phase.digest,
    root_motion.phase.signed_numerator,
    root_motion.current_frequency,
    alternating_bearing.lineage,
    alternating_layer.alternating_goetic.frequency,
    (domus_motion.xi.a, domus_motion.xi.b, domus_motion.xi.denominator),
    (domus_motion.exact_focal_breath.a, domus_motion.exact_focal_breath.b,
    ↪  domus_motion.exact_focal_breath.denominator),
    domus_motion.current_frequency,
    PHI_IMAGE,
    PHI_SQUARED_IMAGE,
    infinite_yes.active_count,
    tuple(sorted(infinite_yes.active)),
    sacred_no.withheld_count,
    tuple(sorted(sacred_no.withheld)),
    bias_return.opening_q_bias_glyph,
    bias_return.breath,
    bias_return.bias_terminus_operator,
    manifestation.ground_node,
    manifestation.ennead.final_parity,
    manifestation.dcomp.shadow_debt_initial,
    manifestation.dcomp.form_work,
    manifestation.dcomp.q3_recursion_gain,
    manifestation.dcomp.terminal,
    LOCUS_GLYPH,
    SHADOW_LOCUS_GLYPH,
    AXIOMYR_GLYPH,
))
domus_commitment = alqc_hexdigest(
    commitment_payload.encode("utf-8"),
    domain=DOMUS_COMMITMENT_DOMAIN,
)
trig_mirror = derive_trig_mirror(
    source_digest=hash_id,
    source_size=source_size,
    source_domain=domain_identity,
    nonce=nonce,
    governing_court_address=root.address,
    alternating_court_address=alternating.address,
    governing_court_phase_digest=root_motion.phase.digest,
    alternating_court_bearing=alternating_bearing.lineage,
    domus_frequency=domus_motion.current_frequency,
    domus_body_commitment=domus_commitment,
```

```
    q_states=q_labels,
    q_vector_values=q_values,
    q3_recursion_gain=manifestation.dcomp.q3_recursion_gain,
    final_shadow_parity=manifestation.ennead.final_parity,
    bound_tensor_witnessed=(
        manifestation.ground_node == fold_lineage.ground_node
        and manifestation.vector_row == fold_lineage.vector_row
    ),
    derives_through_courts_only=domus_motion.derives_through_courts_only,
    resonance_lock_witnessed=manifestation.ennead.strikes[-1].phase_locked,
    lineage_preserved=(
        fold_lineage.court_phase_digest == root_motion.phase.digest
        and fold_lineage.alternating_bearing_lineage == alternating_bearing.lineage
        and bias_return.returns_to_opening_q_bias
    ),
    terminal_dcomp=manifestation.dcomp.terminal,
    one_turn_return_closed=manifestation.dcomp.closed,
)
tripartite = derive_tripartite_witness(
    source_digest=hash_id,
    source_size=source_size,
    source_domain=domain_identity,
    nonce=nonce,
    governing_court_address=root.address,
    alternating_court_address=alternating.address,
    governing_court_phase_digest=root_motion.phase.digest,
    alternating_court_bearing=alternating_bearing.lineage,
    domus_frequency=domus_motion.current_frequency,
    domus_body_commitment=domus_commitment,
    trig_cycle_digest=trig_mirror.cycle_digest,
    active_connections=infinite_yes.active_count,
    court_capacity=TOTAL_CAPACITY,
    withheld_connections=sacred_no.withheld_count,
    c_bio=manifestation.dcomp.c_bio,
    c_bio_squared=C_BIO_SQUARED,
    q3_recursion_gain=manifestation.dcomp.q3_recursion_gain,
    final_shadow_parity=manifestation.ennead.final_parity,
    returned_through_shadow_locus=supervenience.returned_through_shadow_locus,
    gate_breach_witnessed=(law("☸").frequency == complex(285)),
    write_phys_witnessed=(
        trig_mirror.resonance_operator == "❄"
        and trig_mirror.resonance_frequency == complex(963)
        and trig_mirror.resonance_lock_witnessed
    ),
    derives_through_courts_only=domus_motion.derives_through_courts_only,
)
aeon_phase_evolution = derive_aeon_phase_evolution(
    source_digest=hash_id,
    source_size=source_size,
    source_domain=domain_identity,
    nonce=nonce,
    governing_court_address=root.address,
    alternating_court_address=alternating.address,
    governing_court_phase_digest=root_motion.phase.digest,
    alternating_court_bearing=alternating_bearing.lineage,
    domus_frequency=domus_motion.current_frequency,
```

```
    domus_body_commitment=domus_commitment,
    trig_cycle_digest=trig_mirror.cycle_digest,
    tripartite_cycle_digest=tripartite.cycle_digest,
    q3_recursion_gain=manifestation.dcomp.q3_recursion_gain,
    final_shadow_parity=manifestation.ennead.final_parity,
    terminal_dcomp=manifestation.dcomp.terminal,
    completion_reached=trig_mirror.completion_reached,
    returned_through_shadow_locus=supervenience.returned_through_shadow_locus,
    derives_through_courts_only=domus_motion.derives_through_courts_only,
)
verification_appendix_cycle = derive_verification_appendix_cycle(
    source_emission=emission,
    source_digest=hash_id,
    source_size=source_size,
    source_domain=domain_identity,
    nonce=nonce,
    governing_court_address=root.address,
    alternating_court_address=alternating.address,
    domus_body_commitment=domus_commitment,
    trig_cycle_digest=trig_mirror.cycle_digest,
    tripartite_cycle_digest=tripartite.cycle_digest,
    phase_evolution_cycle_digest=aeon_phase_evolution.cycle_digest,
    runtime_trig_witness=trig_mirror,
    runtime_ground_node=manifestation.ground_node,
    runtime_vector_row=manifestation.vector_row,
    runtime_shadow_debt=manifestation.dcomp.shadow_debt_initial,
    runtime_form_work=manifestation.dcomp.form_work,
    runtime_q3_gain=manifestation.dcomp.q3_recursion_gain,
    runtime_final_parity=manifestation.ennead.final_parity,
    runtime_terminal_dcomp=manifestation.dcomp.terminal,
    runtime_completion_reached=trig_mirror.completion_reached,
    derives_through_courts_only=domus_motion.derives_through_courts_only,
)
# The final Domus identity seals the prior TRIG, Tripartite, C09, and C11
# Court-rooted cycles. The pre-TRIG
# commitment remains separately exposed as its own measurement witness.
domus_identity = alqc_hexdigest(
    (
        hash_id
        + domus_commitment
        + trig_mirror.cycle_digest
        + tripartite.cycle_digest
        + aeon_phase_evolution.cycle_digest
        + verification_appendix_cycle.cycle_digest
    ).encode("utf-8"),
    domain=DOMUS_COMMITMENT_DOMAIN,
)
domus_aeon = DomusAeonWitness(
    identity=domus_identity,
    zero_middle_glyph=ZERO_MIDDLE_GLYPH,
    governing_court_address=root.address,
    alternating_court_address=alternating.address,
    motion=domus_motion,
    infinite_yes=infinite_yes,
    sacred_no=sacred_no,
    bias_return=bias_return,
```

```
        synodic_magicae_is_manifested_body=True,
        shadow_locus_is_zero_middle_body=True,
    )

    return DomusResolution(
        governing_goetic=g_i,
        hyperbolic_parent=g_j,
        root_court=root,
        alternating_court=alternating,
        root_court_motion=root_motion,
        alternating_court_layer=alternating_layer,
        domus_motion=domus_motion,
        q_body=q_body,
        resolved_q_bias=beta_q,
        resolved_q_states=q_labels,
        resolved_q_vector=q_values,
        domus_frequency=domus_motion.current_frequency,
        b_q_glyph=b_q_glyph,
        v_glyphs=v_glyphs,
        manifestation=manifestation,
        infinite_yes=infinite_yes,
        sacred_no=sacred_no,
        liquid_witness=liquid,
        fold_witness=manifestation.ground_node,
        recursion_parity=manifestation.ennead.final_parity,
        supervenience=supervenience,
        bias_return=bias_return,
        tripartite=tripartite,
        aeon_phase_evolution=aeon_phase_evolution,
        verification_appendix_cycle=verification_appendix_cycle,
        domus_commitment=domus_commitment,
        trig_mirror=trig_mirror,
        domus_aeon=domus_aeon,
        hash_id=hash_id,
        source_size=source_size,
        source_domain=domain_identity,
        nonce=nonce,
    )



def domus_center_seed(
    res: DomusResolution,
    *,
    source_domain: str | bytes,
    nonce: int,
) -> bytes:
    """Court-rooted seed for the manifested Synodic Magicae Domus body."""
    domain = normalize_source_domain(source_domain)
    salt = nonce
    if domain != res.source_domain:
        raise ValueError("visible Domus source domain differs from the resolved Domus")
    if salt != res.nonce:
        raise ValueError("visible Domus nonce differs from the resolved Domus")
    return domus_stream_seed(
        domus_identity=res.domus_aeon.identity,
```

```
        governing_court=res.domus_motion.governing_court,
        alternating_court=res.domus_motion.alternating_court,
        infinite_yes=res.infinite_yes,
        sacred_no=res.sacred_no,
        source_domain=res.source_domain,
        nonce=res.nonce,
    )



def _center(
    res: DomusResolution,
    n: int,
    *,
    source_digest: str,
    source_size: int,
    nonce: int,
    source_domain: str | bytes = CANONICAL_SOURCE_DOMAIN,
) -> str:
    """Expose Shadow Locus 𐤔 at depth zero or Synodic Magicae depth ``n``."""
    if isinstance(n, bool) or not isinstance(n, int):
        raise TypeError("center depth n must be a non-Boolean integer")
    if n < 0:
        n = 0
    if source_digest != res.hash_id or source_size != res.source_size:
        raise ValueError("source identity does not match the resolved Domus")
    if normalize_source_domain(source_domain) != res.source_domain or nonce != res.nonce:
        raise ValueError("visible Domus parameters do not match the resolved Domus")
    if n == 0:
        return ZERO_MIDDLE_GLYPH
    if n == 1:
        return TRIPARTITE_AXIOMYR
    seed = domus_center_seed(res, source_domain=source_domain, nonce=nonce)
    generated = "".join(iter_middle(seed, n))
    if len(generated) != n:
        raise RuntimeError("Synodic Magicae Domus body length mismatch")
    return generated



def _l8(res: DomusResolution) -> str:
    """Internal left House: Breath, opening Q-bias, and Q0/Q1 return body."""
    v0, v1, _v2, _v3 = res.v_glyphs
    if res.bias_return.opening_q_bias_glyph != res.b_q_glyph:
        raise RuntimeError("Domus word must open at its Q-bias glyph")
    return (
        BREATH + res.b_q_glyph + res.bias_return.breath
        + BREATH * 2 + v0 + BREATH * 2 + v1 + BREATH
    )


def _r8(res: DomusResolution) -> str:
    """Internal right House: Q2/Q3, Breath, and the explicit bias terminus."""
    _v0, _v1, v2, v3 = res.v_glyphs
    if not res.bias_return.returns_to_opening_q_bias:
        raise RuntimeError("breath return to Q-bias is not established")
```

```
    return (
        BREATH + v2 + BREATH * 2 + v3 + BREATH * 2
        + res.bias_return.breath + res.bias_return.bias_terminus_operator + BREATH
    )


def domus_word(res: DomusResolution, center: str, *, depth: int) -> str:
    """Internal Gamma_8 House around the already-resolved public center body."""
    if isinstance(depth, bool) or not isinstance(depth, int):
        raise TypeError("Domus depth must be a non-Boolean integer")
    if depth < 0:
        raise ValueError("Domus depth must be non-negative")
    if not isinstance(center, str):
        raise TypeError("Domus center must be a string")
    if depth == 0:
        if center != ZERO_MIDDLE_GLYPH:
            raise ValueError("declared depth zero requires the canonical zero body")
    elif depth == 1:
        if center != TRIPARTITE_AXIOMYR:
            raise ValueError("declared depth one requires the Triple Horned God body")
    else:
        if len(center) != depth:
            raise ValueError("positive-depth Domus center length must equal the declared depth")
        invalid = frozenset(center).difference(SYNODIC_CENTER_GLYPHS)
        if invalid:
            rendered = "".join(sorted(invalid))
            raise ValueError(f"positive-depth Domus center contains non-Synodic glyphs: {rendered}")
    return _l8(res) + center + _r8(res)


def seal_head(res: DomusResolution) -> str:
    """Internal House prefix through governing Goetic and Court C."""
    return OUTER + res.governing_goetic + KLEIN + res.root_court.glyph + KLEIN + _l8(res)


def seal_tail(res: DomusResolution) -> str:
    """Internal House suffix through Court D and the hyperbolic parent."""
    return _r8(res) + KLEIN + res.alternating_court.glyph + KLEIN + res.hyperbolic_parent + OUTER


def living_domus_seal(
    res: DomusResolution,
    n: int,
    *,
    source_digest: str,
    source_size: int,
    nonce: int = 0,
    source_domain: str = CANONICAL_SOURCE_DOMAIN,
) -> str:
    """Internal complete House. Public rendering remains middle-only."""
    center = _center(
        res, n, source_digest=source_digest, source_size=source_size,
        nonce=nonce, source_domain=source_domain,
    )
    return seal_head(res) + center + seal_tail(res)
```

```
def public_living_domus(res: DomusResolution, n: int, *, source_digest: str,
                        source_size: int, nonce: int = 0,
                        source_domain: str = CANONICAL_SOURCE_DOMAIN) -> str:
    """Public Grim is the manifested middle only. Math and witnesses remain unchanged."""
    center = _center(res, n, source_digest=source_digest, source_size=source_size, nonce=nonce,
    ↪  source_domain=source_domain)
    return grimchain_permeation(center)


@dataclass(frozen=True, slots=True)
class ParsedPublicGrim:
    center: str
    depth: int | None


def parse_public_living_domus(grim: str) -> ParsedPublicGrim:
    """Parse one canonical public Grim middle."""
    if not isinstance(grim, str):
        raise TypeError("GrimChain must be a string")
    count = len(grim)
    if grim == ZERO_MIDDLE_GLYPH:
        depth = 0
    elif grim == TRIPARTITE_AXIOMYR:
        depth = 1
    elif count >= 2:
        if any(glyph not in SYNODIC_CENTER_GLYPHS for glyph in grim):
            raise ValueError("GrimChain middle contains a non-Synodic coordinate glyph")
        depth = count
    else:
        raise ValueError("GrimChain middle has no lawful written extent")
    return ParsedPublicGrim(grim, depth)
```

domus_stream.py

```
"""Bounded-memory streaming for the visible Court-rooted Domus Aeon.

The center is the Shadow Locus `🜨` at depth zero. Depth one is the Triple Horned God
`☽☉☾`. Greater positive depths expose exactly their generated Synodic Magicae coordinates.
Their seed is derived through the governing Court C and alternating Court D, Infinite Yes,
and Sacred No. No Goetic boundary is reused as a Domus stream coordinate.

Vāhana means vehicle or carrier in Sanskrit. Here Vāhana is the invocation-local
medium through which raw data and completed witnesses travel on their way to becoming
permanent history. It is temporary working state, never persistent cache or saved state.
"""
from __future__ import annotations

import os
import tempfile
from pathlib import Path
from typing import Literal

from .hashing import (
    iter_middle,
    TardiSHAError,
    directory_emission,
    file_emission,
    validate_nonce,
    validate_middle_length,
    RAW_FILE_SOURCE_DOMAIN,
    DIRECTORY_SOURCE_DOMAIN,
)
from .route import SourceRouteWitness, source_route_witness_from_emission
from .source_emission import SourceEmission
from .mirror_math import _render_self, mirror_file_emission
from .domus import (
    resolve_domus,
    domus_center_seed,
    ZERO_MIDDLE_GLYPH,
    TRIPARTITE_AXIOMYR,
)

Kind = Literal["file", "directory"]



def _validate_kind(kind: Kind) -> Kind:
    if kind not in ("file", "directory"):
        raise TardiSHAError("kind must be exactly 'file' or 'directory'")
    return kind
```

```
def _fingerprint(
    source: Path, kind: Kind, *, nonce: int = 0, identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> tuple[SourceEmission, bytes, SourceRouteWitness]:
    source_kind = _validate_kind(kind)
    if source_kind == "file":
        emission = mirror_file_emission(source, nonce=nonce, identity_name=identity_name,
        ↪  include_filename=include_filename).emission
        return emission, RAW_FILE_SOURCE_DOMAIN, source_route_witness_from_emission(emission)
    emission, _entries = directory_emission(source)
    return emission, DIRECTORY_SOURCE_DOMAIN, source_route_witness_from_emission(emission)



def living_domus_from_emission(
    emission: SourceEmission,
    middle: int,
    *,
    nonce: int = 0,
    route_witness: SourceRouteWitness,
) -> str:
    """Render the canonical GrimChain from an already witnessed raw-file emission."""
    salt = validate_nonce(nonce)
    width = validate_middle_length(middle)
    if emission.source_domain != "raw-file":
        raise ValueError("living_domus_from_emission requires a raw-file emission")
    return _render_self(
        emission, width, nonce=salt, route_witness=route_witness
    )



def file_domus_record(
    source: str | Path, middle: int, *, nonce: int = 0,
    identity_name: str | bytes | None = None, include_filename: bool = True,
) -> tuple[str, SourceEmission, SourceRouteWitness]:
    """Read one file, return its canonical GrimChain plus complete witnesses."""
    salt = validate_nonce(nonce)
    emission, _domain, witness = _fingerprint(
        Path(source), "file", nonce=salt, identity_name=identity_name,
        include_filename=include_filename,
    )
    seal = living_domus_from_emission(
        emission, middle, nonce=salt, route_witness=witness
    )
    return seal, emission, witness


def living_domus_for_source(
    source: Path, middle: int, *, kind: Kind, nonce: int = 0,
    identity_name: str | bytes | None = None, include_filename: bool = True,
) -> str:
    """In-memory canonical GrimChain for a file or directory (bounded to small middle)."""
    salt = validate_nonce(nonce)
    source_kind = _validate_kind(kind)
    if source_kind == "file":
```

```
        seal, _emission, _witness = file_domus_record(
            source, middle, nonce=salt, identity_name=identity_name,
            include_filename=include_filename,
        )
        return seal
    emission, _, witness = _fingerprint(
        source, source_kind, nonce=salt, identity_name=identity_name,
        include_filename=include_filename,
    )
    return _render_self(
        emission, middle, nonce=salt, route_witness=witness
    )


def write_public_domus(
    source_path: str | Path,
    output_path: str | Path,
    *,
    kind: Kind,
    middle_length: int,
    nonce: int = 0,
    identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> dict:
    """Stream the canonical public Grim to disk with bounded memory and atomic commit."""
    width = validate_middle_length(middle_length)
    salt = validate_nonce(nonce)
    source_kind = _validate_kind(kind)
    source = Path(source_path)
    output = Path(output_path)
    if source.resolve() == output.resolve():
        raise TardiSHAError("source_path and output_path must be different paths")

    emission, domain, witness = _fingerprint(
        source, source_kind, nonce=salt, identity_name=identity_name,
        include_filename=include_filename,
    )
    digest, size = emission.source_digest, emission.source_size
    res = resolve_domus(witness, nonce=salt)
    output.parent.mkdir(parents=True, exist_ok=True)
    handle = tempfile.NamedTemporaryFile(
        mode="w", encoding="utf-8", newline="", delete=False,
        dir=output.parent, prefix=f".{output.name}.", suffix=".part",
    )
    temp = Path(handle.name)
    center_len = len(ZERO_MIDDLE_GLYPH) if width == 0 else (len(TRIPARTITE_AXIOMYR) if width == 1
    ↪  else width)
    try:
        with handle:
            if width == 0:
                handle.write(ZERO_MIDDLE_GLYPH)
            elif width == 1:
                handle.write(TRIPARTITE_AXIOMYR)
            else:
                seed = domus_center_seed(res, source_domain=domain, nonce=salt)
                for chunk in iter_middle(seed, width):
```

```
                handle.write(chunk)
        handle.flush()
        os.fsync(handle.fileno())
except Exception:
    temp.unlink(missing_ok=True)
    raise


os.replace(temp, output)
return {
    "output": str(output),
    "governing_goetic": witness.pair[0],
    "hyperbolic_parent": witness.pair[1],
    "root_court_glyph": res.root_court.glyph,
    "alternating_court_glyph": res.alternating_court.glyph,
    "middle_length": width,
    "grim_length": center_len,
    "source_digest": digest,
    "source_size": size,
}
```

enoch.py

```
"""TardiSHA-local Enoch / Understanding classification.

This module reflects the authored Synodic Magicae offices locally without creating
any runtime dependency on the Sydonic Magicae implementation. It classifies the
eleven native Enoch glyphs already present in TardiSHA and records only the behavior
needed by GrimChain itself.

Phase 1 only classifies the generated Enochian Understandings. Generated coordinates
remain unchanged; boundary completion belongs to Sydonic causal execution.
"""
from __future__ import annotations

from dataclasses import dataclass
from typing import Final

from .qstate_glyphs import q_state_of


@dataclass(frozen=True, slots=True)
class EnochOffice:
    """One exact TardiSHA-local Enoch office."""

    glyph: str
    office: str
    reaches_backward: bool
    generated: bool = True
    positional_operation: str | None = None
    phantasmagoria: bool = False
    phantasm_state: str | None = None


_BACKWARD_ENOCHS: Final[frozenset[str]] = frozenset({"𜉵", "𜉰", "𜉫", "𜉳", "𜉩"})
_Q_ENOCHS: Final[frozenset[str]] = frozenset({"𜉮", "𜉬", "𜉯", "𜉭"})

ENOCH_OFFICES: Final[dict[str, EnochOffice]] = {
    "𜉵": EnochOffice("𜉵", "Anchor", True, phantasmagoria=True),
    "𜉰": EnochOffice("𜉰", "Adverbial Bearing", True),
    "𜉶": EnochOffice("𜉶", "Prosody", False, generated=False),
    "𜉷": EnochOffice(
        "𜉷",
        "Ouroboric Reversal",
        False,
        positional_operation="reverse_adjacent_pair; boundary completion belongs to Sydonic causal
        ↪  execution",
    ),
    "𜉫": EnochOffice("𜉫", "Adjective Bearing", True),
    "𜉳": EnochOffice("𜉳", "Action / Condition Recognition", True),
    "𜉩": EnochOffice("𜉩", "Item Recognition", True),
```

```
    "⊙": EnochOffice("⊙", "Truth", False, phantasm_state=q_state_of("⊙")),
    "⊕": EnochOffice("⊕", "Recursion", False, phantasm_state=q_state_of("⊕")),
    "⊖": EnochOffice("⊖", "Form", False, phantasm_state=q_state_of("⊖")),
    "⊛": EnochOffice("⊛", "Shadow", False, phantasm_state=q_state_of("⊛")),
}

GENERATED_ENOCH_GLYPHS: Final[tuple[str, ...]] = tuple(
    glyph for glyph, body in ENOCH_OFFICES.items() if body.generated
)
BACKWARD_REACHING_ENOCHS: Final[frozenset[str]] = _BACKWARD_ENOCHS


def enoch_office(glyph: str) -> EnochOffice:
    """Return the exact TardiSHA-local office for one Enoch glyph."""
    if not isinstance(glyph, str) or len(glyph) != 1:
        raise ValueError("Enoch lookup requires exactly one glyph code point")
    try:
        return ENOCH_OFFICES[glyph]
    except KeyError as exc:
        raise ValueError(f"not a TardiSHA Enoch glyph: {glyph!r}") from exc


def is_enoch(glyph: str) -> bool:
    """Return whether ``glyph`` is one of the eleven native Enoch glyphs."""
    return isinstance(glyph, str) and len(glyph) == 1 and glyph in ENOCH_OFFICES


def reaches_backward(glyph: str) -> bool:
    """Return True only for the five Enochs that deepen backward reach."""
    if not is_enoch(glyph):
        return False
    return ENOCH_OFFICES[glyph].reaches_backward


def is_generated_enoch(glyph: str) -> bool:
    """Return whether an Enoch may occur in the generated GrimChain middle."""
    return is_enoch(glyph) and ENOCH_OFFICES[glyph].generated


def is_ouroboric_reversal(glyph: str) -> bool:
    """Identify ⩛ without misclassifying it as a backward-reaching Enoch."""
    return glyph == "⩛"


if {glyph for glyph, body in ENOCH_OFFICES.items() if body.reaches_backward} != _BACKWARD_ENOCHS:
    raise RuntimeError("exactly ⍤ ♌ ♋ ♂ ♁ must reach backward")
if len(GENERATED_ENOCH_GLYPHS) != 10 or "⟠" in GENERATED_ENOCH_GLYPHS or "⊚" in
↪  GENERATED_ENOCH_GLYPHS:
    raise RuntimeError("exactly ten Enochs are generated; ⟠ and ⊚ remain non-generated")
if ENOCH_OFFICES["⩛"].reaches_backward:
    raise RuntimeError("⩛ is positional Ouroboric Reversal, not backward depth")
if not ENOCH_OFFICES["⍤"].phantasmagoria:
    raise RuntimeError("⍤ must retain the Phantasmagoria Anchor office")
if {glyph: ENOCH_OFFICES[glyph].phantasm_state for glyph in _Q_ENOCHS} != {
    "⊖": "Q0",
    "⊙": "Q1",
```

```
    "⊕": "Q2",
    "⊕": "Q3",
}:
    raise RuntimeError("phantasm-state Enochs must preserve the Q0/Q1/Q2/Q3 seam")
```

folding.py

```
"""ALQC fold frames for open TardiSHA coordinate streams.

TardiSHA does not store an infinite string. It stores the lawful seed/manifold
that can unfold any finite coordinate window, and it can fold large manifested
windows back into compact child universe-nodes.

A fold frame is neither truncation nor a second engine. It is one exact ♧
checkpoint whose source node, outward manifestation, born child, and canonical
return body remain mutually derivable.
"""
from __future__ import annotations

import json
from dataclasses import asdict, dataclass, fields

from .alqc_digest import ALQCDigest, validate_digest_hex
from .canon import TOTAL_CAPACITY, court_load
from .hashing import (
    ALPHABET,
    TardiSHAError,
    _encode_uint,
    validate_middle_length,
    validate_nonce,
)
from .node import TardiSHANode
from .route import SourceRouteWitness

FOLD_DOMAIN = b"TARDISHA:ALQC-FOLD-FRAME\x00"
FOLD_LADDER_DOMAIN = b"TARDISHA:ALQC-SELF-COMPRESS-LADDER\x00"
BIRTH_DOMAIN = b"TARDISHA:ALQC-BABY-GLYPH-BIRTH\x00"
UNIVERSE_NODE_DOMAIN = b"TARDISHA:ALQC-UNIVERSE-NODE\x00"
FOLD_OPERATOR = "♧"
BIRTH_CODE_LENGTH = 18


@dataclass(frozen=True, slots=True)
class TardiSHAFoldFrame:
    """One exact Court-rooted ♧ checkpoint and its canonical return body."""

    level: int
    start_coordinate: int
    span_length: int
    fold_factor: int
    origin_glyph: str
    resolution_glyph: str
    source_digest: str
    fold_digest: str
    packet_count: int
```

```
input_digest: str
universe_node_id: str
birth_digest: str
born_glyph: str
birth_court_load: int
fold_operator: str
return_node_id: str
return_body: dict[str, object]
nonce: int
source_size: int
source_domain: str

def __post_init__(self) -> None:
    _validate_coordinate(self.level, "level")
    _validate_coordinate(self.start_coordinate, "start_coordinate")
    span = validate_middle_length(self.span_length)
    if isinstance(self.fold_factor, bool) or not isinstance(self.fold_factor, int) or
    ↪  self.fold_factor < 2:
        raise TardiSHAError("fold_factor must be an integer >= 2")
    if isinstance(self.packet_count, bool) or not isinstance(self.packet_count, int) or
    ↪  self.packet_count < 0:
        raise TardiSHAError("packet_count must be a non-negative integer")
    if self.packet_count != (span + TOTAL_CAPACITY - 1) // TOTAL_CAPACITY:
        raise TardiSHAError("packet_count contradicts the manifested span")
    _validate_coordinate(self.source_size, "source_size")
    validate_nonce(self.nonce)
    if not isinstance(self.source_domain, str) or not self.source_domain:
        raise TardiSHAError("source_domain must be a non-empty string")
    for field_name in (
        "source_digest", "fold_digest", "input_digest", "universe_node_id",
        "birth_digest", "return_node_id",
    ):
        validate_digest_hex(getattr(self, field_name), field=field_name)
    if self.fold_operator != FOLD_OPERATOR:
        raise TardiSHAError("fold frame must preserve the Canonical ♧ operator")
    if not isinstance(self.born_glyph, str) or len(self.born_glyph) != BIRTH_CODE_LENGTH + 2:
        raise TardiSHAError("born_glyph must contain two Goetic boundaries and eighteen Synodic
        ↪  coordinates")
    if self.born_glyph[0] != self.origin_glyph or self.born_glyph[-1] != self.resolution_glyph:
        raise TardiSHAError("born_glyph boundaries contradict the fold frame")
    if any(glyph not in ALPHABET for glyph in self.born_glyph[1:-1]):
        raise TardiSHAError("born_glyph center must contain only Synodic Magicae coordinates")
    if isinstance(self.birth_court_load, bool) or not isinstance(self.birth_court_load, int):
        raise TardiSHAError("birth_court_load must be an integer Court address")
    expected_load = court_load(self.origin_glyph, self.resolution_glyph)
    if self.birth_court_load != expected_load:
        raise TardiSHAError("birth_court_load contradicts the ordered Goetic pair")

    node = _return_node_from_serialized_body(self.return_body)
    if node.node_id != self.return_node_id:
        raise TardiSHAError("return_node_id contradicts the complete canonical return body")
    if (
        node.origin_glyph != self.origin_glyph
        or node.resolution_glyph != self.resolution_glyph
        or node.source_digest != self.source_digest
```

```
            or node.source_size != self.source_size
            or node.source_domain != self.source_domain
            or node.nonce != self.nonce
        ):
            raise TardiSHAError("fold frame contradicts its canonical return node")

    def as_dict(self) -> dict[str, object]:
        return asdict(self)

    @classmethod
    def from_dict(cls, value: dict[str, object]) -> "TardiSHAFoldFrame":
        if not isinstance(value, dict):
            raise TardiSHAError("serialized fold frame must be an object")
        expected = {item.name for item in fields(cls)}
        supplied = set(value)
        if supplied != expected:
            missing = sorted(expected - supplied)
            extra = sorted(supplied - expected)
            raise TardiSHAError(f"serialized fold frame fields mismatch; missing={missing},
            ↪  extra={extra}")
        body = dict(value)
        if not isinstance(body["return_body"], dict):
            raise TardiSHAError("fold return_body must be an object")
        body["return_body"] = dict(body["return_body"])
        return cls(**body)


def _validate_coordinate(value: int, field: str) -> int:
    if isinstance(value, bool) or not isinstance(value, int) or value < 0:
        raise TardiSHAError(f"{field} must be a non-negative integer")
    return value


def _return_node_from_serialized_body(body: dict[str, object]) -> TardiSHANode:
    if not isinstance(body, dict):
        raise TardiSHAError("canonical return body must be an object")
    data = dict(body)
    try:
        supplied_node_id = validate_digest_hex(data.pop("node_id"), field="node_id")
        supplied_mode_witness = data.pop("mode_witness")
        route_body = data.get("route_witness")
        if not isinstance(route_body, dict):
            raise TardiSHAError("canonical return body requires a route_witness object")
        data["route_witness"] = SourceRouteWitness.from_dict(route_body)
        node = TardiSHANode(**data)
    except (KeyError, TypeError, ValueError) as exc:
        raise TardiSHAError("canonical return body cannot be reconstructed") from exc
    if supplied_mode_witness != node.mode_witness():
        raise TardiSHAError("canonical return body mode witness is forged or stale")
    if node.node_id != supplied_node_id:
        raise TardiSHAError("canonical return body node_id does not verify")
    return node


def _synodic_magicae_from_bytes(data: bytes) -> str:
    """Render one digest body as exactly eighteen Synodic coordinates, without reseeding."""
```

```
    if not isinstance(data, bytes) or not data:
        raise TardiSHAError("birth digest must be a non-empty bytes body")
    number = int.from_bytes(data, "big")
    chars: list[str] = []
    for _coordinate in range(BIRTH_CODE_LENGTH):
        number, remainder = divmod(number, len(ALPHABET))
        chars.append(ALPHABET[remainder])
    return "".join(reversed(chars))


def _universe_node_id(
    *,
    node: TardiSHANode,
    fold_digest: str,
    input_digest: str,
    level: int,
    start_coordinate: int,
    span_length: int,
    fold_factor: int,
) -> str:
    payload = json.dumps(
        {
            "node": node.as_dict(),
            "fold_operator": FOLD_OPERATOR,
            "fold_digest": fold_digest,
            "input_digest": input_digest,
            "level": level,
            "start_coordinate": start_coordinate,
            "span_length": span_length,
            "fold_factor": fold_factor,
            "boundary_equation": f"{node.origin_glyph}->{node.resolution_glyph}",
            "court_load": court_load(node.origin_glyph, node.resolution_glyph),
            "return_node_id": node.node_id,
        },
        ensure_ascii=False,
        sort_keys=True,
        separators=(",", ":"),
    ).encode("utf-8")
    return ALQCDigest(UNIVERSE_NODE_DOMAIN).update(payload).hexdigest()


def _derive_birth(
    *,
    node: TardiSHANode,
    fold_digest: str,
    input_digest: str,
    level: int,
    start_coordinate: int,
    span_length: int,
    fold_factor: int,
) -> tuple[str, str, int, str]:
    """Derive a child only from one complete, source-bound fold calculation."""
    validate_digest_hex(fold_digest, field="fold_digest")
    validate_digest_hex(input_digest, field="input_digest")
    boundary_load = court_load(node.origin_glyph, node.resolution_glyph)
    universe_node_id = _universe_node_id(
```

```
        node=node,
        fold_digest=fold_digest,
        input_digest=input_digest,
        level=level,
        start_coordinate=start_coordinate,
        span_length=span_length,
        fold_factor=fold_factor,
    )
    digest = ALQCDigest(BIRTH_DOMAIN)
    digest._update_raw(
        node.origin_glyph.encode("utf-8")
        + b"\x00"
        + node.resolution_glyph.encode("utf-8")
        + b"\x00"
        + bytes.fromhex(node.source_digest)
        + bytes.fromhex(fold_digest)
        + bytes.fromhex(input_digest)
        + bytes.fromhex(universe_node_id)
        + bytes.fromhex(node.node_id)
        + FOLD_OPERATOR.encode("utf-8")
        + _encode_uint(level)
        + _encode_uint(start_coordinate)
        + _encode_uint(span_length)
        + _encode_uint(fold_factor)
        + _encode_uint(boundary_load)
    )
    birth_digest = digest.hexdigest()
    child_code = _synodic_magicae_from_bytes(bytes.fromhex(birth_digest))
    return (
        f"{node.origin_glyph}{child_code}{node.resolution_glyph}",
        birth_digest,
        boundary_load,
        universe_node_id,
    )


def fold_window(
    node: TardiSHANode,
    *,
    start_coordinate: int = 0,
    span_length: int,
    level: int,
    input_digest: str,
    fold_factor: int,
    chunk_characters: int = 8192,
) -> TardiSHAFoldFrame:
    """Fold one finite stream window and preserve its complete ♧ return body."""
    if isinstance(node, type) or not isinstance(node, TardiSHANode):
        raise TardiSHAError("node must be a TardiSHANode")
    start = _validate_coordinate(start_coordinate, "start_coordinate")
    span = validate_middle_length(span_length)
    _validate_coordinate(level, "level")
    if isinstance(fold_factor, bool) or not isinstance(fold_factor, int) or fold_factor < 2:
        raise TardiSHAError("fold_factor must be an integer >= 2")
    prior = validate_digest_hex(input_digest, field="input_digest")
    if level == 0 and prior != node.source_digest:
```

```
    raise TardiSHAError("level-zero fold input must be the node source digest")
digest = ALQCDigest(FOLD_DOMAIN)
digest._update_raw(
    node.origin_glyph.encode("utf-8")
    + b"\x00"
    + node.resolution_glyph.encode("utf-8")
    + b"\x00"
    + bytes.fromhex(node.source_digest)
    + bytes.fromhex(prior)
    + bytes.fromhex(node.node_id)
    + FOLD_OPERATOR.encode("utf-8")
    + _encode_uint(level)
    + _encode_uint(start)
    + _encode_uint(span)
    + _encode_uint(fold_factor)
    + _encode_uint(court_load(node.origin_glyph, node.resolution_glyph))
    + node.source_domain.encode("utf-8")
)
if node.archive_root is not None:
    digest._update_raw(bytes.fromhex(node.archive_root))
produced = 0
for chunk in node.iter_middle_chunks(
    start,
    span,
    chunk_characters=chunk_characters,
):
    encoded = chunk.encode("utf-8")
    digest._update_raw(_encode_uint(len(encoded)) + encoded)
    produced += len(chunk)
if produced != span:
    raise TardiSHAError(f"fold window produced {produced} characters; expected {span}")

fold_digest = digest.hexdigest()
born, birth_digest, boundary_load, universe_node_id = _derive_birth(
    node=node,
    fold_digest=fold_digest,
    input_digest=prior,
    level=level,
    start_coordinate=start,
    span_length=span,
    fold_factor=fold_factor,
)
return TardiSHAFoldFrame(
    level=level,
    start_coordinate=start,
    span_length=span,
    fold_factor=fold_factor,
    origin_glyph=node.origin_glyph,
    resolution_glyph=node.resolution_glyph,
    source_digest=node.source_digest,
    fold_digest=fold_digest,
    packet_count=(span + TOTAL_CAPACITY - 1) // TOTAL_CAPACITY,
    input_digest=prior,
    universe_node_id=universe_node_id,
    birth_digest=birth_digest,
    born_glyph=born,
```

```
        birth_court_load=boundary_load,
        fold_operator=FOLD_OPERATOR,
        return_node_id=node.node_id,
        return_body=node.as_dict(),
        nonce=node.nonce,
        source_size=node.source_size,
        source_domain=node.source_domain,
    )


def _require_verified_fold_frame(frame: TardiSHAFoldFrame) -> TardiSHANode:
    if not isinstance(frame, TardiSHAFoldFrame):
        raise TardiSHAError("fold verification requires a TardiSHAFoldFrame")
    node = _return_node_from_serialized_body(frame.return_body)
    expected = fold_window(
        node,
        start_coordinate=frame.start_coordinate,
        span_length=frame.span_length,
        level=frame.level,
        input_digest=frame.input_digest,
        fold_factor=frame.fold_factor,
    )
    if expected != frame:
        raise TardiSHAError("fold frame does not return to its exact source-bound ♧ derivation")
    return node



def birth_glyph(frame: TardiSHAFoldFrame) -> tuple[str, str, int]:
    """Expose a born child only after the complete fold frame verifies."""
    _require_verified_fold_frame(frame)
    return frame.born_glyph, frame.birth_digest, frame.birth_court_load


def self_compress_ladder(
    node: TardiSHANode,
    *,
    start_coordinate: int = 0,
    span_length: int,
    levels: int = 8,
    fold_factor: int = 2,
    chunk_characters: int = 8192,
) -> tuple[TardiSHAFoldFrame, ...]:
    """Return an exact chain of source-bound ♧ checkpoints."""
    if isinstance(levels, bool) or not isinstance(levels, int) or levels < 1:
        raise TardiSHAError("levels must be a positive integer")
    if isinstance(fold_factor, bool) or not isinstance(fold_factor, int) or fold_factor < 2:
        raise TardiSHAError("fold_factor must be an integer >= 2")
    span = validate_middle_length(span_length)
    start = _validate_coordinate(start_coordinate, "start_coordinate")
    frames: list[TardiSHAFoldFrame] = []
    prior = node.source_digest
    current_span = span
    for level in range(levels):
        frame = fold_window(
            node,
```

```
            start_coordinate=start,
            span_length=current_span,
            level=level,
            input_digest=prior,
            fold_factor=fold_factor,
            chunk_characters=chunk_characters,
        )
        frames.append(frame)
        prior = frame.fold_digest
        if current_span == 0:
            break
        current_span //= fold_factor
    verified = tuple(frames)
    _require_verified_fold_ladder(verified)
    return verified


def _require_verified_fold_ladder(frames: tuple[TardiSHAFoldFrame, ...]) -> None:
    if not isinstance(frames, tuple) or not frames:
        raise TardiSHAError("fold ladder must contain at least one frame")
    first = frames[0]
    node = _require_verified_fold_frame(first)
    if first.level != 0 or first.input_digest != first.source_digest:
        raise TardiSHAError("fold ladder must begin at level zero with the source digest")
    for index, frame in enumerate(frames):
        _require_verified_fold_frame(frame)
        if frame.level != index:
            raise TardiSHAError("fold ladder levels must be contiguous from zero")
        if frame.return_node_id != first.return_node_id or frame.return_body != first.return_body:
            raise TardiSHAError("every fold level must return to the same canonical node")
        if frame.fold_factor != first.fold_factor or frame.start_coordinate !=
        ↪  first.start_coordinate:
            raise TardiSHAError("fold ladder factor and start coordinate must remain invariant")
        if (
            frame.origin_glyph != node.origin_glyph
            or frame.resolution_glyph != node.resolution_glyph
            or frame.source_digest != node.source_digest
        ):
            raise TardiSHAError("fold ladder boundary or source identity changed")
        if index:
            previous = frames[index - 1]
            if frame.input_digest != previous.fold_digest:
                raise TardiSHAError("fold ladder input does not equal the prior fold digest")
            if previous.span_length == 0:
                raise TardiSHAError("no fold level may follow the zero-span terminal frame")
            if frame.span_length != previous.span_length // frame.fold_factor:
                raise TardiSHAError("fold ladder span does not follow its declared fold factor")



def ladder_avirbhava(frames: tuple[TardiSHAFoldFrame, ...]) -> dict[str, object]:
    """Bring one fully verified ॐ ladder and its complete return body into visible form."""
    _require_verified_fold_ladder(frames)
    root = ALQCDigest(FOLD_LADDER_DOMAIN)
    for frame in frames:
```

```
        payload = json.dumps(frame.as_dict(), ensure_ascii=False, sort_keys=True, separators=(",",
        ↪  ":")).encode("utf-8")
        root._update_raw(_encode_uint(len(payload)) + payload)
    first = frames[0]
    return {
        "type": "TardiSHA_ALQC_self_compression_ladder",
        "fold_operator": FOLD_OPERATOR,
        "root_digest": root.hexdigest(),
        "return_node_id": first.return_node_id,
        "return_body": first.return_body,
        "origin_glyph": first.origin_glyph,
        "resolution_glyph": first.resolution_glyph,
        "boundary_equation": f"{first.origin_glyph}->{first.resolution_glyph}",
        "birth_court_load": first.birth_court_load,
        "fold_factor": first.fold_factor,
        "nonce": first.nonce,
        "source_size": first.source_size,
        "source_domain": first.source_domain,
        "born_glyphs": [frame.born_glyph for frame in frames],
        "universe_node_ids": [frame.universe_node_id for frame in frames],
        "frames": [frame.as_dict() for frame in frames],
    }


def public_ladder_avirbhava(frames: tuple[TardiSHAFoldFrame, ...]) -> dict[str, object]:
    """Bring the verified public fold result into visible form without internal closure-status proof
    ↪  bodies."""
    full = ladder_avirbhava(frames)
    return {
        "type": full["type"],
        "fold_operator": full["fold_operator"],
        "root_digest": full["root_digest"],
        "return_node_id": full["return_node_id"],
        "origin_glyph": full["origin_glyph"],
        "resolution_glyph": full["resolution_glyph"],
        "boundary_equation": full["boundary_equation"],
        "birth_court_load": full["birth_court_load"],
        "fold_factor": full["fold_factor"],
        "nonce": full["nonce"],
        "source_size": full["source_size"],
        "source_domain": full["source_domain"],
        "born_glyphs": full["born_glyphs"],
        "universe_node_ids": full["universe_node_ids"],
    }
```

grimchain.py

```
"""grimchain — one command: content in, ALQC seal out.

Positional middle is USER-CHOSEN at call time (any integer; negative depth returns to Shadow Locus
↪  zero):

    grimchain <N> <path>            # seal a file with declared middle depth N
    grimchain <path>                # zero body ␀␀␀ (manifested from Shadow Locus ␀)
    echo -n "text" | grimchain <N>  # seal stdin with declared middle depth N
"""
from __future__ import annotations

import os
import sys
import json
import tempfile
from pathlib import Path
from argparse import ArgumentParser, SUPPRESS

from .hashing import SOURCE_CHUNK_BYTES, TardiSHAError


_MIDDLE_UNSET = object()


def _materialize(data: bytes) -> str:
    """Write one exact in-memory byte body to a temporary source file."""
    if not isinstance(data, bytes):
        raise TypeError("materialized source must be bytes")
    fd, tmp = tempfile.mkstemp(suffix=".grimchain")
    try:
        with os.fdopen(fd, "wb") as handle:
            handle.write(data)
            handle.flush()
            os.fsync(handle.fileno())
        return tmp
    except Exception:
        Path(tmp).unlink(missing_ok=True)
        raise


def _materialize_stream(reader) -> str:
    """Copy one binary stream exactly, with bounded memory and no normalization."""
    fd, tmp = tempfile.mkstemp(suffix=".grimchain")
    try:
        with os.fdopen(fd, "wb") as handle:
            while True:
                chunk = reader.read(SOURCE_CHUNK_BYTES)
                if not chunk:
```

```
                    break
                if not isinstance(chunk, (bytes, bytearray, memoryview)):
                    raise TypeError("grimchain stream must yield bytes")
                handle.write(bytes(chunk))
            handle.flush()
            os.fsync(handle.fileno())
        return tmp
    except Exception:
        Path(tmp).unlink(missing_ok=True)
        raise



BASIC_HELP = """usage:
  grimchain PATH
  grimchain NUMBER PATH
  grimchain --string "TEXT"
  grimchain NUMBER --string "TEXT"
  grimchain NUMBER --manifest PATH
  grimchain NUMBER -R --manifest DIRECTORY
  grimchain NUMBER --manifest --string "TEXT"
  grimchain --help -a

NUMBER is the declared middle depth you choose. Every non-positive NUMBER returns to
the zero body 𓆏𓆏𓆏. Depth 1 is the Triple Horned God ☽☼☾. Every depth greater
than 1 is exactly that many generated Synodic Magicae coordinates.
Leave NUMBER out to use the randomized Living Mirror Chain: Prosody ♁, Cantillation ♲, Regia ☽⊙☾,
↪  and Breath ·; Regia remains one member.

basic use:
  grimchain PATH
      Grimchain one file or one directory.

  grimchain NUMBER PATH
      Grimchain one file or directory using the declared middle depth you choose.

  grimchain --string "TEXT"
  grimchain NUMBER --string "TEXT"
      Grimchain the exact UTF-8 content supplied by the quoted argument.
      The quotes belong to the shell and are not part of the source.

  grimchain NUMBER --manifest FILE
      Create manifest-FILENAME.grim, append its exact same-width Grimchain,
      then Grimchain the completed manifest again and print that same return.

  grimchain NUMBER --manifest DIRECTORY
      Create manifest-DIRECTORYNAME.grim for the files directly inside that
      directory, append its Grimchain, then Grimchain the completed manifest again.

  grimchain NUMBER -R --manifest DIRECTORY
      Create manifest-recurse-DIRECTORYNAME.grim for all subdirectories, append
      its Grimchain, then Grimchain the completed manifest again.

"""

ADVANCED_HELP = """usage:
```

```
  grimchain [NUMBER] PATH [OPTIONS]
  grimchain [NUMBER] --string "TEXT" [OPTIONS]

NUMBER is any whole-number middle depth you choose. Every non-positive NUMBER returns
to the zero body 🜨🜨🜨. Depth 1 is the Triple Horned God ☽☼☾. Every depth greater
than 1 is exactly that many generated Synodic Magicae coordinates.
Leave NUMBER out to use the randomized Living Mirror Chain: Prosody ⟠, Cantillation ⌔, Regia ☽⊙☾,
↪  and Breath ·; Regia remains one member.

ordinary use:
  grimchain PATH
      Grimchain one file or directory with the randomized Living Mirror Chain.
      Its four members are Prosody ⟠, Cantillation ⌔, Regia ☽⊙☾, and Breath ·.

  grimchain NUMBER PATH
  grimchain --middle NUMBER PATH
      Grimchain one file or directory with declared middle depth NUMBER.

  grimchain --string "TEXT"
  grimchain NUMBER --string "TEXT"
      Grimchain the exact UTF-8 content inside the quoted shell argument.
      No newline, trimming, normalization, or separate string algorithm is added.

  grimchain NUMBER PATH --output FILE
      Save the Grimchain in FILE instead of printing it.

manifests:
  grimchain [NUMBER] --manifest FILE
      Create manifest-FILENAME.grim, append its exact same-width Grimchain,
      then Grimchain the completed manifest again and print that same return.

  grimchain [NUMBER] --manifest DIRECTORY
      Create manifest-DIRECTORYNAME.grim for direct files, append its Grimchain,
      then Grimchain the completed manifest again and print that same return.

  grimchain [NUMBER] -R --manifest DIRECTORY
      Create manifest-recurse-DIRECTORYNAME.grim for the complete tree, append
      its Grimchain, then Grimchain the completed manifest again and print it.

  grimchain [NUMBER] --manifest PATH --output FILE
      Write the manifest to the filename you choose, append its exact Grimchain,
      then Grimchain the completed file again and print that same return.

  grimchain [NUMBER] --manifest --string "TEXT"
      Create a manifest whose source name and source identity are the exact shell
      argument supplied to --string, append its Grimchain, and return it again.

PDF high-level return:
  grimchain NUMBER --pdf-embed DOCUMENT.pdf
      Embed the exact GrimChain invisibly in the PDF return data, then re-Grimchain it.

  grimchain --pdf-dive DOCUMENT.pdf
      Find the embedded GrimChain without a user middle, derive its extent, and test
      it through the existing Mirror return. Prints MONOZYGOTIC, HETEROZYGOTIC, or
      ZYGOTIC. It does not remove the embedded chain.
```

```
  grimchain --pdf-abort DOCUMENT.pdf
      Remove the embedded PDF GrimChain. No user middle is accepted.

inspection and lists:
  grimchain SEAL --inspect
      Show the parts contained in one Grimchain.

input and optional controls:
  printf 'text' | grimchain [NUMBER]
      Grimchain text received through standard input.

  grimchain --binary PATH
      Use the file's exact bytes. This is the only file-body law; line endings are never normalized.

  grimchain PATH --nonce INTEGER
      Use the chosen integer to create another repeatable Grimchain for PATH.

  grimchain PATH --fold --span NUMBER --levels NUMBER
      Create the fold-ladder output for PATH. Both dimensions are explicit and required.

  grimchain PATH1 PATH2 DIRECTORY
      Grimchain several sources in one command.

information:
  grimchain --version
      Show the installed TardiSHA version.

  grimchain --help
      Show basic use.

  grimchain --help -a
      Show all commands.
"""


def _installed_version() -> str:
    from importlib.metadata import PackageNotFoundError, version as package_version

    try:
        return package_version("TardiSHA")
    except PackageNotFoundError as exc:
        raise TardiSHAError("installed TardiSHA package metadata is required for --version") from
        ↪  exc


def main() -> None:
    argv = sys.argv[1:]
    separator_index = argv.index("--") if "--" in argv else None
    if separator_index is None:
        parser_argv = argv
        protected_args: list[str] = []
    else:
        parser_argv = argv[:separator_index]
        protected_args = argv[separator_index + 1:]

    p = ArgumentParser(prog="grimchain", description="content in, ALQC seal out", add_help=False)
```

```
p.add_argument("args", nargs="*",
               help="[MIDDLE] [PATH] — omit MIDDLE for the Living Mirror Chain; PATH omitted
                ↪  reads stdin")
p.add_argument("-h", "--help", action="store_true", help=SUPPRESS)
p.add_argument("-a", "--advanced", action="store_true", help=SUPPRESS)
p.add_argument("--version", action="store_true", help=SUPPRESS)
p.add_argument("--middle", type=int, default=SUPPRESS, help="explicit middle length")
p.add_argument("--string", metavar="TEXT", help="Grimchain the exact UTF-8 bytes of one shell
↪  argument")
p.add_argument("-b", "--binary", action="store_true", help="read exact bytes; this is the only
↪  file-body law")
p.add_argument("--output", type=Path, help="write the seal or manifest to a file")
p.add_argument("--manifest", action="store_true", help="write a Grimchain source manifest")
p.add_argument("--pdf-embed", action="store_true", help="embed the PDF GrimChain in metadata")
p.add_argument("--pdf-dive", action="store_true", help="inspect and verify the embedded PDF
↪  GrimChain")
p.add_argument("--pdf-abort", action="store_true", help="remove the embedded PDF GrimChain")
p.add_argument("-R", "--recursive", action="store_true", help="include nested files in a
↪  directory manifest")
p.add_argument("--fold", action="store_true", help="emit the born-glyph fold ladder")
p.add_argument("--span", type=int, default=SUPPRESS, help="explicit fold span length")
p.add_argument("--levels", type=int, default=SUPPRESS, help="explicit fold level count")
p.add_argument("--inspect", action="store_true", help="parse one seal value")
p.add_argument("--nonce", type=int, default=0)
a = p.parse_intermixed_args(parser_argv)

try:
    raw_args = list(a.args)
    unprotected_args = raw_args

    option_conflicts = any((
        a.version, a.string is not None, a.binary, a.output is not None,
        a.manifest, a.recursive, a.fold, a.inspect,
        hasattr(a, "middle"), hasattr(a, "span"), hasattr(a, "levels"), bool(raw_args),
    ))
    if a.help:
        if option_conflicts:
            p.error("--help may be combined only with -a/--advanced")
        print(ADVANCED_HELP if a.advanced else BASIC_HELP, end="")
        return
    if a.advanced:
        p.error("-a/--advanced is used only with --help")
    if a.version:
        if any((a.string is not None, a.binary, a.output is not None,
                a.manifest, a.recursive, a.fold, a.inspect, hasattr(a, "middle"),
                hasattr(a, "span"), hasattr(a, "levels"), bool(raw_args))):
            p.error("--version cannot be combined with another mode or source")
        print(_installed_version())
        return

    middle_value: int | object = getattr(a, "middle", _MIDDLE_UNSET)
    middle_was_supplied = middle_value is not _MIDDLE_UNSET
    if unprotected_args and not middle_was_supplied:
        token = unprotected_args[0]
        if token.lstrip("+-").isdigit():
```

```
        middle_value = int(token)
        middle_was_supplied = True
        unprotected_args = unprotected_args[1:]
if middle_was_supplied and isinstance(middle_value, int) and not isinstance(middle_value,
↪  bool) and middle_value < 0:
    middle_value = 0
paths = unprotected_args + protected_args

pdf_modes = int(a.pdf_embed) + int(a.pdf_dive) + int(a.pdf_abort)
if pdf_modes:
    if pdf_modes != 1:
        p.error("choose exactly one of --pdf-embed, --pdf-dive, or --pdf-abort")
    if a.string is not None or a.manifest or a.recursive or a.fold or a.inspect or a.binary
    ↪  or a.output is not None:
        p.error("PDF metadata modes cannot be combined with another mode or output")
    if len(paths) != 1:
        p.error("PDF metadata mode requires exactly one PDF path")
    target = Path(paths[0]).expanduser().resolve()
    if not target.exists():
        raise FileNotFoundError(f"does not exist: {paths[0]}")
    if target.suffix.lower() != ".pdf":
        p.error("PDF metadata mode requires a .pdf file")
    from .pdf_return import pdf_abort, pdf_dive, pdf_embed
    if a.pdf_embed:
        if not middle_was_supplied:
            p.error("--pdf-embed requires a middle")
        assert isinstance(middle_value, int)
        print(pdf_embed(target, middle_value, nonce=a.nonce))
        return
    if middle_was_supplied:
        p.error("--pdf-dive and --pdf-abort do not take a middle")
    if a.pdf_dive:
        correct, embedded, state = pdf_dive(target, nonce=a.nonce)
        if state == "ZYGOTIC":
            print("ZYGOTIC there is no chain embeded in the pdf")
        elif state == "MONOZYGOTIC":
            print(correct)
            print()
            print("MONOZYGOTIC this is the Truth.")
        else:
            if correct is not None:
                print(correct)
                print()
            print("HETEROZYGOTIC it's not the true chain to that pdf. the chain or the pdf
            ↪  was tampered with.")
            print()
            print(embedded)
        return
    pdf_abort(target)
    return

if a.recursive and not a.manifest:
    p.error("-R/--recursive is used only with --manifest")
if hasattr(a, "span") != hasattr(a, "levels"):
    p.error("--span and --levels must be supplied together")
```

```
if a.fold:
    if not hasattr(a, "span") or not hasattr(a, "levels"):
        p.error("--fold requires explicit --span and --levels")
    if a.output is not None or a.manifest or a.inspect or a.string is not None:
        p.error("--fold cannot be combined with --output, --manifest, --inspect, or
        ↪  --string")
elif hasattr(a, "span") or hasattr(a, "levels"):
    p.error("--span and --levels are used only with --fold")

if a.inspect:
    if any((middle_was_supplied, a.string is not None, a.binary, a.output is not None,
            a.manifest, a.recursive, a.fold)):
        p.error("--inspect cannot be combined with another mode, output, or middle")
    if len(paths) != 1:
        p.error("--inspect requires exactly one seal value")
    from .domus import parse_public_living_domus
    print(json.dumps(_public_grim_payload(parse_public_living_domus(paths[0])),
    ↪  ensure_ascii=False, indent=2))
    return

if a.manifest:
    from .manifest import manifest_output_path, write_grimchain_manifest
    if a.binary or a.fold or a.inspect:
        p.error("--manifest cannot be combined with --binary, --fold, or --inspect")
    if a.string is None and len(paths) != 1:
        p.error("--manifest requires exactly one file or directory")
    if a.string is not None and paths:
        p.error("--string --manifest cannot be combined with file or directory paths")

if a.string is not None:
    if paths:
        p.error("--string cannot be combined with file or directory paths")
    if a.recursive or a.inspect or a.binary or a.fold:
        p.error("--string cannot be combined with -R, --inspect, --binary, or --fold")

if len(paths) > 1 and a.output is not None:
    p.error("one --output cannot receive multiple sources")

if a.string is not None and a.manifest:
    tmp = _materialize(a.string.encode("utf-8"))
    try:
        manifest_middle = middle_value if middle_was_supplied else 0
        assert isinstance(manifest_middle, int)
        destination = (
            a.output.expanduser().resolve()
            if a.output is not None
            else (Path.cwd() / f"manifest-{a.string}.grim").resolve()
        )
        manifest_parity = write_grimchain_manifest(
            tmp, destination, middle=manifest_middle, recursive=False, nonce=a.nonce,
            entry_name=a.string, identity_name=None, include_filename=False,
        )
        print(manifest_parity)
    finally:
        os.unlink(tmp)
    return
```

```
if a.manifest:
    target = Path(paths[0]).expanduser().resolve()
    if not target.exists():
        raise FileNotFoundError(f"does not exist: {paths[0]}")
    destination = manifest_output_path(target, a.output, a.recursive).resolve()
    manifest_middle = middle_value if middle_was_supplied else 0
    assert isinstance(manifest_middle, int)
    manifest_parity = write_grimchain_manifest(
        target, destination, middle=manifest_middle, recursive=a.recursive, nonce=a.nonce
    )
    print(manifest_parity)
    return

if a.string is not None:
    tmp = _materialize(a.string.encode("utf-8"))
    try:
        if middle_was_supplied:
            assert isinstance(middle_value, int)
            out = _do_file(Path(tmp), middle_value, a, source_label="--string")
        else:
            out = _do_file_shadow_locus(Path(tmp), a, source_label="--string")
        print(out)
    finally:
        os.unlink(tmp)
    return

if not paths:
    if sys.stdin.isatty():
        p.error("path required (or pipe input)")
    tmp = _materialize_stream(sys.stdin.buffer)
    try:
        if middle_was_supplied:
            assert isinstance(middle_value, int)
            print(_do_file(Path(tmp), middle_value, a, source_label="<stdin>"))
        else:
            print(_do_file_shadow_locus(Path(tmp), a, source_label="<stdin>"))
    finally:
        os.unlink(tmp)
    return

for raw in paths:
    target = Path(raw).expanduser().resolve()
    if not target.exists():
        raise FileNotFoundError(f"does not exist: {raw}")
    if target.is_dir():
        if middle_was_supplied:
            assert isinstance(middle_value, int)
            out = _do_dir(target, middle_value, a)
        else:
            out = _do_dir_shadow_locus(target, a)
    else:
        if middle_was_supplied:
            assert isinstance(middle_value, int)
            out = _do_file(target, middle_value, a, source_label=raw)
        else:
```

```
                    out = _do_file_shadow_locus(target, a, source_label=raw)
            print(out)

    except BrokenPipeError:
        try:
            sys.stdout.close()
        except Exception:
            pass
    except (TardiSHAError, FileNotFoundError, OSError, UnicodeError, json.JSONDecodeError) as exc:
        print(f"grimchain: {exc}", file=sys.stderr)
        sys.exit(1)


def _do_file_shadow_locus(target: Path, a, *, source_label: str | None = None) -> str:
    """Dispatch an omitted middle argument through the Living Mirror Chain."""
    from .domus import living_mirror_chain
    if a.output:
        chain = living_mirror_chain()
        Path(a.output).write_text(chain, encoding="utf-8")
        return json.dumps({"output": str(a.output), "middle_length": 4, "grim_length": len(chain)},
        ↪  ensure_ascii=False, indent=2)
    return living_mirror_chain()


def _do_dir_shadow_locus(target: Path, a) -> str:
    """Dispatch an omitted middle argument through the Living Mirror Chain."""
    from .domus import living_mirror_chain
    if a.output:
        chain = living_mirror_chain()
        Path(a.output).write_text(chain, encoding="utf-8")
        return json.dumps({"output": str(a.output), "middle_length": 4, "grim_length": len(chain)},
        ↪  ensure_ascii=False, indent=2)
    return living_mirror_chain()


def _do_file(target: Path, middle: int, a, *, source_label: str | None = None) -> str:
    from .domus_stream import living_domus_for_source, write_public_domus
    include_filename = source_label not in {"--string", "<stdin>"}
    if a.fold:
        from .folding import public_ladder_avirbhava, self_compress_ladder
        from .node import node_from_file
        node = node_from_file(target, nonce=a.nonce)
        return json.dumps(public_ladder_avirbhava(self_compress_ladder(node, span_length=a.span,
        ↪  levels=a.levels)),
                          ensure_ascii=False, indent=2)
    if a.output:
        r = write_public_domus(
            target, a.output, kind="file", middle_length=middle, nonce=a.nonce,
            include_filename=include_filename,
        )
        return json.dumps(r, ensure_ascii=False, indent=2)
# Default content-in / GrimChain-out uses the canonical public GrimChain.
    return living_domus_for_source(
        target, middle, kind="file", nonce=a.nonce, include_filename=include_filename
    )
```

```
def _do_dir(target: Path, middle: int, a) -> str:
    from .domus_stream import living_domus_for_source, write_public_domus
    if a.fold:
        from .folding import public_ladder_avirbhava, self_compress_ladder
        from .node import node_from_directory
        node = node_from_directory(target, nonce=a.nonce)
        return json.dumps(public_ladder_avirbhava(self_compress_ladder(node, span_length=a.span,
        ↪  levels=a.levels)),
                          ensure_ascii=False, indent=2)
    if a.output:
        r = write_public_domus(target, a.output, kind="directory", middle_length=middle,
        ↪  nonce=a.nonce)
        return json.dumps(r, ensure_ascii=False, indent=2)
# Default directory output uses the canonical public GrimChain.
    return living_domus_for_source(target, middle, kind="directory", nonce=a.nonce)


def _public_grim_payload(parsed) -> dict:
    """Structural inspection of one canonical public Grim. No source recomputation."""
    return {
        "center": parsed.center,
        "depth": parsed.depth,
    }


if __name__ == "__main__":
    main()
```

hashing.py

```
"""Deterministic coordinate-stream primitives for TardiSHA.

TardiSHA means Time and Relative Dimension in Space Hash.  There is one
canonical engine law: source identity plus Goetic boundaries produce a
continuation coordinate stream. Requested length is an extent/view, not a
separate identity law, so every finite manifestation is a window of the same
rooted manifold.
"""
from __future__ import annotations

import base64
import json
import math
import os
import stat
from dataclasses import asdict, is_dataclass
from enum import Enum
from pathlib import Path
from typing import Any, Iterator, Mapping

from .canon import GLYPH_BODY
from .alqc_digest import ALQCDigest, alqc_digest, alqc_digest_prefix, validate_digest_hex
from .living_alphabet import ALPHABET
from .source_emission import PHASE_BYTES, SourceEmission, emission_from_sponge

SOURCE_CHUNK_BYTES = 8 * 1024 * 1024
OUTPUT_CHUNK_CHARACTERS = 8192

SEED_DOMAIN = b"TARDISHA:TIME-RELATIVE-DIMENSION-IN-SPACE:SEED\x00"
CHAR_DOMAIN = b"TARDISHA:CONTINUATION-CHAR-AT\x00"
CANONICAL_SOURCE_DOMAIN = b"TARDISHA:CANONICAL-SOURCE\x00"
RAW_FILE_SOURCE_DOMAIN = b"TARDISHA:RAW-FILE-SOURCE\x00"
FILE_NAME_FRAME = b"TARDISHA:FILE-NAME\x00"
DIRECTORY_SOURCE_DOMAIN = b"TARDISHA:DIRECTORY-SOURCE\x00"

GENERATED_FIELDS = frozenset(
    {
        "TardiSHA_id",
        "TardiSHA_nonce",
        "TardiSHA_middle_length",
        "origin_glyph",
        "resolution_glyph",
    }
)


class TardiSHAError(ValueError):
    """Raised when a TardiSHA invariant is violated."""
```

```
def validate_middle_length(value: int) -> int:
    """Resolve one declared middle depth; negative depth returns to Shadow Locus zero."""
    if isinstance(value, bool) or not isinstance(value, int):
        raise TardiSHAError("middle_length must be an integer")
    return 0 if value < 0 else value



def validate_nonce(value: int) -> int:
    if isinstance(value, bool) or not isinstance(value, int) or not 0 <= value < 2**64:
        raise TardiSHAError("nonce must be an integer in [0, 2^64)")
    return value


def validate_glyph(glyph: str, field: str = "glyph") -> str:
    if glyph not in GLYPH_BODY:
        raise TardiSHAError(f"{field} must be one exact Goetic glyph: {''.join(GLYPH_BODY)}")
    return glyph


def _json_key(value: Any) -> str:
    return json.dumps(value, ensure_ascii=False, separators=(",", ":"), sort_keys=True)


def _normalize(value: Any) -> Any:
    if hasattr(value, "model_dump"):
        value = value.model_dump(mode="json")
    elif is_dataclass(value) and not isinstance(value, type):
        value = asdict(value)
    elif isinstance(value, Enum):
        value = value.value

    if value is None:
        return ["null"]
    if isinstance(value, bool):
        return ["bool", value]
    if isinstance(value, int):
        return ["int", str(value)]
    if isinstance(value, float):
        if not math.isfinite(value):
            raise TardiSHAError("NaN and infinity cannot be canonicalized")
        return ["float", value.hex()]
    if isinstance(value, str):
        return ["str", value]
    if isinstance(value, (bytes, bytearray, memoryview)):
        return ["bytes", base64.b64encode(bytes(value)).decode("ascii")]
    if isinstance(value, Path):
        raise TardiSHAError("Path is not canonical material; use file_emission() for raw file
        ↪  content")
    if isinstance(value, Mapping):
        pairs = [[_normalize(key), _normalize(item)] for key, item in value.items()]
        pairs.sort(key=lambda pair: _json_key(pair[0]))
        return ["map", pairs]
    if isinstance(value, list):
```

```
        return ["list", [_normalize(item) for item in value]]
    if isinstance(value, tuple):
        return ["tuple", [_normalize(item) for item in value]]
    if isinstance(value, (set, frozenset)):
        items = [_normalize(item) for item in value]
        items.sort(key=_json_key)
        return ["set", items]
    raise TardiSHAError(f"unsupported material type: {type(value).__name__}")


def canonical_bytes(value: Any) -> bytes:
    return json.dumps(
        _normalize(value),
        ensure_ascii=False,
        separators=(",", ":"),
        sort_keys=True,
    ).encode("utf-8")


def identity_material(record: Mapping[str, Any]) -> dict[str, Any]:
    return {key: value for key, value in record.items() if key not in GENERATED_FIELDS}


def canonical_emission(material: Any) -> tuple[SourceEmission, bytes]:
    """Return the complete canonical emission while preserving framed digest law."""
    source = canonical_bytes(material)
    sponge = ALQCDigest(CANONICAL_SOURCE_DOMAIN)
    sponge.update(source)
    # canonical_fingerprint historically uses digest_separated(32).  Carry that
    # exact separated state forward so Fraktur Z_0 remains byte-identical while
    # Fraktur Z_1 is the next equal-width cadence of the same state.
    separated = sponge.copy()
    separated._update_frame(b"DIGEST-LENGTH\x00", PHASE_BYTES.to_bytes(8, "big"))
    emission = emission_from_sponge(
        separated,
        source_size=len(source),
        source_domain="canonical",
    )
    return emission, source


def canonical_fingerprint(material: Any) -> tuple[str, int, bytes]:
    emission, source = canonical_emission(material)
    return emission.source_digest, emission.source_size, source


def canonical_digest(material: Any) -> str:
    return canonical_emission(material)[0].source_digest


def _stat_witness(result: os.stat_result) -> tuple[int, int, int, int, int, int]:
    return (
        result.st_dev,
        result.st_ino,
        result.st_mode,
        result.st_size,
```

```
        result.st_mtime_ns,
        result.st_ctime_ns,
    )


def file_emission(
    path: str | Path,
    *,
    identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> SourceEmission:
    """Return one temporally sealed raw-file emission from filename and unchanged file body."""
    target = Path(path)
    before_path = target.stat()
    if not stat.S_ISREG(before_path.st_mode):
        raise TardiSHAError("file_emission requires a regular file")

    sponge = ALQCDigest(RAW_FILE_SOURCE_DOMAIN)
    if include_filename:
        if identity_name is None:
            name_bytes = os.fsencode(target.name)
        elif isinstance(identity_name, str):
            name_bytes = os.fsencode(identity_name)
        elif isinstance(identity_name, bytes):
            name_bytes = identity_name
        else:
            raise TardiSHAError("identity_name must be str, bytes, or None")
        if not name_bytes:
            raise TardiSHAError("file identity requires a non-empty filename")
        sponge._update_frame(FILE_NAME_FRAME, name_bytes)
    elif identity_name is not None:
        raise TardiSHAError("identity_name requires filename identity to be enabled")
    size = 0
    with target.open("rb") as handle:
        before_handle = os.fstat(handle.fileno())
        if _stat_witness(before_handle) != _stat_witness(before_path):
            raise TardiSHAError("file identity changed before source witnessing began")
        for chunk in iter(lambda: handle.read(SOURCE_CHUNK_BYTES), b""):
            sponge._update_raw(chunk)
            size += len(chunk)
        after_handle = os.fstat(handle.fileno())

    after_path = target.stat()
    witness = _stat_witness(before_path)
    if _stat_witness(before_handle) != witness or _stat_witness(after_handle) != witness:
        raise TardiSHAError("file changed while its Q1 emission was being witnessed")
    if _stat_witness(after_path) != witness:
        raise TardiSHAError("file path no longer names the witnessed Q1 body")
    if size != before_path.st_size:
        raise TardiSHAError("witnessed byte count contradicts the sealed file size")

    return emission_from_sponge(
        sponge,
        source_size=size,
        source_domain="raw-file",
    )
```

```
def file_emission_from_bytes(
    body: bytes,
    *,
    identity_name: str | bytes | None,
    include_filename: bool = True,
) -> SourceEmission:
    """Return a raw-file emission from one invocation-local byte Vāhana.

    No body is retained after the call. Filename identity follows the same framed
    ALQC law as :func:`file_emission`; an in-memory body has no implicit filename.
    """
    if type(body) is not bytes:
        raise TypeError("file byte Vāhana must be exact bytes")
    sponge = ALQCDigest(RAW_FILE_SOURCE_DOMAIN)
    if include_filename:
        if isinstance(identity_name, str):
            name_bytes = os.fsencode(identity_name)
        elif isinstance(identity_name, bytes):
            name_bytes = identity_name
        elif identity_name is None:
            raise TardiSHAError("in-memory file identity requires explicit identity_name")
        else:
            raise TardiSHAError("identity_name must be str, bytes, or None")
        if not name_bytes:
            raise TardiSHAError("file identity requires a non-empty filename")
        sponge._update_frame(FILE_NAME_FRAME, name_bytes)
    elif identity_name is not None:
        raise TardiSHAError("identity_name requires filename identity to be enabled")
    sponge._update_raw(body)
    return emission_from_sponge(sponge, source_size=len(body), source_domain="raw-file")


def file_fingerprint(path: str | Path) -> tuple[str, int]:
    emission = file_emission(path)
    return emission.source_digest, emission.source_size


def _tree_record_bytes(record: Mapping[str, Any]) -> bytes:
    return json.dumps(record, ensure_ascii=False, sort_keys=True, separators=(",",
    ↪  ":")).encode("utf-8") + b"\n"


def _directory_entries(root: Path) -> Iterator[Path]:
    stack = [root]
    while stack:
        current = stack.pop()
        try:
            entries = sorted(os.scandir(current), key=lambda entry: entry.name)
        except OSError as exc:
            raise TardiSHAError(f"cannot read directory {current}: {exc}") from exc
        directories: list[Path] = []
        for entry in entries:
            child = Path(entry.path)
```

```
            yield child
            if entry.is_dir(follow_symlinks=False) and not entry.is_symlink():
                directories.append(child)
        stack.extend(reversed(directories))


def _directory_snapshot(
    root: Path,
) -> tuple[tuple[bytes, ...], tuple[tuple[object, ...], ...], int, int]:
    records: list[bytes] = [_tree_record_bytes({"type": "directory", "path": "."})]
    witnesses: list[tuple[object, ...]] = [
        (".", "directory", _stat_witness(root.lstat()))
    ]
    total_file_bytes = 0
    entry_count = 1

    for child in _directory_entries(root):
        rel = child.relative_to(root).as_posix()
        before = child.lstat()
        before_witness = _stat_witness(before)

        if stat.S_ISLNK(before.st_mode):
            target = os.readlink(child)
            after = child.lstat()
            if _stat_witness(after) != before_witness:
                raise TardiSHAError(f"symlink changed while being witnessed: {rel}")
            records.append(_tree_record_bytes({"type": "symlink", "path": rel, "target": target}))
            witnesses.append((rel, "symlink", before_witness, target))
        elif stat.S_ISDIR(before.st_mode):
            after = child.lstat()
            if _stat_witness(after) != before_witness:
                raise TardiSHAError(f"directory changed while being witnessed: {rel}")
            records.append(_tree_record_bytes({"type": "directory", "path": rel}))
            witnesses.append((rel, "directory", before_witness))
        elif stat.S_ISREG(before.st_mode):
            from .mirror_math import mirror_file_emission
            child_emission = mirror_file_emission(child).emission

            after = child.lstat()
            if _stat_witness(after) != before_witness:
                raise TardiSHAError(f"file identity changed during directory witnessing: {rel}")
            records.append(
                _tree_record_bytes(
                    {
                        "type": "file",
                        "path": rel,
                        "digest": child_emission.source_digest,
                        "size": child_emission.source_size,
                    }
                )
            )
            witnesses.append(
                (rel, "file", before_witness, child_emission.source_digest,
                ↪  child_emission.source_size)
            )
            total_file_bytes += child_emission.source_size
```

```
        else:
            raise TardiSHAError(f"unsupported directory entry type: {rel}")
        entry_count += 1

    return tuple(records), tuple(witnesses), total_file_bytes, entry_count


def directory_emission(path: str | Path) -> tuple[SourceEmission, int]:
    """Return one two-witness emission of a complete unchanged directory tree."""
    root = Path(path)
    before_root = root.lstat()
    if not stat.S_ISDIR(before_root.st_mode):
        raise TardiSHAError("directory_emission requires a directory path")

    first = _directory_snapshot(root)
    second = _directory_snapshot(root)
    after_root = root.lstat()
    if _stat_witness(before_root) != _stat_witness(after_root):
        raise TardiSHAError("directory root changed while its Q1 tree was being witnessed")
    if first != second:
        raise TardiSHAError("directory tree changed between its two complete Q1 witnesses")

    records, _witnesses, total_file_bytes, entry_count = first
    sponge = ALQCDigest(DIRECTORY_SOURCE_DOMAIN)
    for record in records:
        sponge._update_raw(record)
    emission = emission_from_sponge(
        sponge,
        source_size=total_file_bytes,
        source_domain="directory",
    )
    return emission, entry_count


def directory_fingerprint(path: str | Path) -> tuple[str, int, int]:
    emission, entry_count = directory_emission(path)
    return emission.source_digest, emission.source_size, entry_count


def _validate_source_digest(value: str) -> str:
    try:
        return validate_digest_hex(value, field="source_digest")
    except ValueError as exc:
        raise TardiSHAError(str(exc)) from exc


def _encode_uint(value: int) -> bytes:
    if isinstance(value, bool) or not isinstance(value, int) or value < 0:
        raise TardiSHAError("unsigned coordinate must be a non-negative integer")
    return str(value).encode("ascii") + b"\x00"


def coordinate_seed(
    *,
    source_digest: str,
    source_size: int,
```

```
    origin_glyph: str,
    resolution_glyph: str,
    middle_length: int,
    nonce: int = 0,
    source_domain: bytes = RAW_FILE_SOURCE_DOMAIN,
) -> bytes:
    """Bind source, boundaries, source kind, and nonce into one ALQC-native 256-bit root.

    ``middle_length`` is validated as a finite requested extent, but it does not
    enter the root. Extending a manifestation therefore preserves all earlier
    coordinates. The middle is a view of the rooted manifold, not a separate
    identity law.
    """
    source_hash = bytes.fromhex(_validate_source_digest(source_digest))
    if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
        raise TardiSHAError("source_size must be a non-negative integer")
    origin = validate_glyph(origin_glyph, "origin_glyph")
    resolution = validate_glyph(resolution_glyph, "resolution_glyph")
    validate_middle_length(middle_length)
    salt = validate_nonce(nonce)
    if not isinstance(source_domain, bytes) or not source_domain:
        raise TardiSHAError("source_domain must be non-empty bytes")

    # Terminal D-COMP is a finite-return witness, not an emission gate.  The
    # generic continuation root remains a source-identity body.  Court/Domus
    # commitments bind their own explicit D-COMP and layered-Aeon witnesses.

    seed_material = (
        SEED_DOMAIN
        + source_domain
        + origin.encode("utf-8")
        + b"\x00"
        + resolution.encode("utf-8")
        + b"\x00"
        + b"CONTINUATION-ROOT\x00"
        + _encode_uint(source_size)
        + salt.to_bytes(8, "big")
        + source_hash
    )
    return alqc_digest(seed_material, domain=SEED_DOMAIN)

def _validate_seed(seed: bytes) -> bytes:
    if not isinstance(seed, bytes) or len(seed) != 32:
        raise TardiSHAError("seed must be exactly 32 bytes")
    return seed


def _validate_chunk_characters(chunk_characters: int) -> int:
    if isinstance(chunk_characters, bool) or not isinstance(chunk_characters, int) or
    ↪  chunk_characters < 1:
        raise TardiSHAError("chunk_characters must be a positive integer")
    return chunk_characters



COORDINATE_BLOCK_CHARACTERS = 4096
```

```
COORDINATE_WORD_BYTES = 4
_COORDINATE_WORD_RANGE = 1 << (8 * COORDINATE_WORD_BYTES)
_COORDINATE_ACCEPTANCE_LIMIT = (
    _COORDINATE_WORD_RANGE // len(ALPHABET)
) * len(ALPHABET)


def _coordinate_block(seed: bytes, block_index: int, span_characters: int =
↪  COORDINATE_BLOCK_CHARACTERS) -> str:
    """Resolve only the requested prefix within one canonical 4096-coordinate epoch boundary.

    4096 is the internal deterministic epoch size, not a requested rendering length.
    Only the coordinates required by the caller are calculated.
    """
    _validate_seed(seed)

    if isinstance(block_index, bool) or not isinstance(block_index, int) or block_index < 0:
        raise TardiSHAError("block_index must be a non-negative integer")
    if isinstance(span_characters, bool) or not isinstance(span_characters, int) or not 1 <=
    ↪  span_characters <= COORDINATE_BLOCK_CHARACTERS:
        raise TardiSHAError("span_characters must be an integer in [1, 4096]")

    out: list[str] = []
    rejection_round = 0
    remaining = span_characters

    while remaining > 0:
        material = (
            seed
            + _encode_uint(block_index)
            + _encode_uint(rejection_round)
        )

        body = alqc_digest_prefix(
            material,
            domain=CHAR_DOMAIN,
            canonical_length=COORDINATE_BLOCK_CHARACTERS * COORDINATE_WORD_BYTES,
            prefix_length=remaining * COORDINATE_WORD_BYTES,
        )

        for offset in range(0, len(body), COORDINATE_WORD_BYTES):
            word = int.from_bytes(
                body[offset:offset + COORDINATE_WORD_BYTES],
                "big",
            )

            if word >= _COORDINATE_ACCEPTANCE_LIMIT:
                continue

            coordinate = word % len(ALPHABET)
            out.append(ALPHABET[coordinate])

            if len(out) == span_characters:
                break

        remaining = span_characters - len(out)
```

```
        rejection_round += 1

    return "".join(out)

def char_at(seed: bytes, position: int) -> str:
    """Return one TardiSHA continuation character at an arbitrary non-negative coordinate."""
    _validate_seed(seed)
    if isinstance(position, bool) or not isinstance(position, int) or position < 0:
        raise TardiSHAError("position must be a non-negative integer")
    block_index, offset = divmod(position, COORDINATE_BLOCK_CHARACTERS)
    return _coordinate_block(seed, block_index, offset + 1)[offset]

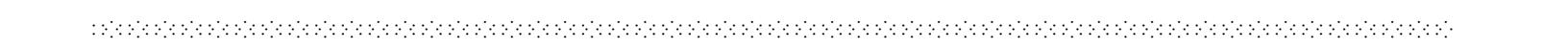

def iter_middle_window(
    seed: bytes,
    start_coordinate: int,
    span_length: int,
    *,
    chunk_characters: int = OUTPUT_CHUNK_CHARACTERS,
) -> Iterator[str]:
    """Yield a random-access window without materializing the whole stream."""
    _validate_seed(seed)
    if isinstance(start_coordinate, bool) or not isinstance(start_coordinate, int) or
    ↪  start_coordinate < 0:
        raise TardiSHAError("start_coordinate must be a non-negative integer")
    width = validate_middle_length(span_length)
    _validate_chunk_characters(chunk_characters)
    if width == 0:
        return

    stop = start_coordinate + width
    cursor = start_coordinate
    pending: list[str] = []
    while cursor < stop:
        block_index, offset = divmod(cursor, COORDINATE_BLOCK_CHARACTERS)
        take = min(stop - cursor, COORDINATE_BLOCK_CHARACTERS - offset)
        block = _coordinate_block(seed, block_index, offset + take)
        pending.append(block[offset:offset + take])
        cursor += take
        joined = "".join(pending)
        while len(joined) >= chunk_characters:
            yield joined[:chunk_characters]
            joined = joined[chunk_characters:]
        pending = [joined] if joined else []
    if pending:
        yield "".join(pending)


def iter_middle(
    seed: bytes,
    middle_length: int,
    *,
    chunk_characters: int = OUTPUT_CHUNK_CHARACTERS,
    start_coordinate: int = 0,
) -> Iterator[str]:
    """Yield the canonical deterministic Synodic Magicae continuation stream."""
```

```
    yield from iter_middle_window(
        seed,
        start_coordinate,
        middle_length,
        chunk_characters=chunk_characters,
    )

def create_middle_from_fingerprint(
    *,
    source_digest: str,
    source_size: int,
    origin_glyph: str,
    resolution_glyph: str,
    middle_length: int,
    nonce: int = 0,
    source_domain: bytes = RAW_FILE_SOURCE_DOMAIN,
    start_coordinate: int = 0,
) -> str:
    seed = coordinate_seed(
        source_digest=source_digest,
        source_size=source_size,
        origin_glyph=origin_glyph,
        resolution_glyph=resolution_glyph,
        middle_length=middle_length,
        nonce=nonce,
        source_domain=source_domain,
    )
    return "".join(iter_middle_window(seed, start_coordinate, middle_length))
```



living_alphabet.py

```
"""Canonical TardiSHA Synodic Magicae and Daemonic Tongue.

The ordered glyph body below is law. ``DAEMONIC_TONGUE`` preserves the exact
code-point sequence used by the language. ``SYNODIC_MAGICAE`` is the first
180 glyph entries: silence, Emissions, Parliament, Goetics, and Courts. The final
sixteen entries carry native grammar and special glyphs of ``DAEMONIC_TONGUE``.

The Regia alignment remains three distinct code points: ``☽``, ``⊙``, ``ℂ``.
Together ``☽⊙ℂ`` are Regia ``मुकुट``, the Crown of the Axiomyr and the headspace. Regia is
ever-present Above: an emission identity, not a runtime function or mutation pass;
the stored glyphs are never collapsed into one character. Prosody ``छन्दस्`` ``⟠`` is
ever-present Below and therefore never generated. Breathe ``शिक्षा`` ``·`` is
ever-present Throughout and therefore never generated by the manifested Word.

No normalization, case folding, transliteration, or look-alike substitution is
permitted.
"""
from __future__ import annotations

from typing import Final

DAEMONIC_TONGUE: Final[tuple[str, ...]] = (
    "·", '☿', '♂', '♀', '♃', '♄', '♁', '♆', '♇', '☽', '⊙', 'ℂ', '♈',
    '♉', '♊', '♋', '♌', '♍', '♎', '♏', '♐', '♑', '♒', '♓',
    '◎', '⬡', '✡', '☆', '❂', '❀', '❋', '⧗', '⊛', '❄', '⚛', '⌬',
    'ऱ', 'ऴ', 'ऽ', 'ॡ', 'ऌ', 'ॸ', 'ऺ', 'ॵ', 'ऻ', 'ॷ', 'ॾ', 'ॿ',
    'ा', 'ॄ', '▫', 'ॎ', 'ऽ', 'ϒ', 'ɛ', 'ϟ', '''', 'ϟ', 'ϯ', 'ϒ',
    'Ь', 'Π', 'Þ', 'ſ', 'ℝ', '‹', 'X', 'Þ', 'Н', 'ϟ', 'ſ', 'ϟ',
    '≍', 'ϟ', 'ɔ', '✡', 'ꟻ', '𐦂', 'ϒ', '⊙', '⏁', 'ৎ', 'ϟ', '𐦂',
    'ο', 'Θ', 'Ⅹ', 'Λ', 'Ж', 'Ⓚ', 'Ⓞ', 'ㄏ', 'ㅂ', 'Ⓩ', 'Ɛ', 'I',
    'ঘ', 'ঋ', 'গ', 'ৰ', '⍜', 'ব', 'ঢ়', 'জ', 'ড়', 'ঃ', 'চ', 'ড',
    'Ж', 'Δ', 'Ⱶ', 'ⱱ', 'ⱷ', 'ⱹ', 'Ⱥ', 'Ⱶ', 'Ⱬ', 'Ⱶ', 'Ⱶ', 'Ⱡ',
    'ᚋ', '✷', 'ᘐ', 'ᘓ', 'ᘔ', '᛭', '᛬', '᛫', 'ᘕ', 'ᘖ', 'ᘗ', 'ᘘ', 'ᘙ', 'ᘚ',
    'ᛯ', 'ᛰ', 'ᛮ', '᛭', '᛬', '᛫', 'ᛪ', 'ᛩ', 'ᛨ', 'ᛧ', 'ᛦ', 'ᛥ',
    'Ⱥ', 'Ⱳ', 'Ɔ', 'ⱨ', 'ⱨ', 'ⱨ', 'ⱡ', 'Ɑ', 'ⱪ', 'Ⱡ', 'Ⱡ', 'Ⱡ',
    '⁎', '⁎', '⁎', '⁎', '⁎', '⁰', 'ⱪ', '⁺', '⁓', '⁗', '⁔', 'ⁱ',
    '⁓', '⁔', 'Ⱶ', 'Ⱬ', 'Ⱶ', 'Ⱶ', '▽', 'Ƶ', 'ɻ', 'Ь', 'ɷ', 'Ϙ', 'ᵒ', '⟠', '⩒', '𐦂', 'ᵊ',
    '𐦂', 'ⱺ', 'ⱺ', 'ⱺ', 'ⱺ', '◌', '◌', '∱', '⚙', 'Ψ',
)

SYNODIC_MAGICAE: Final[str] = "".join(DAEMONIC_TONGUE)
ALPHABET: Final[str] = SYNODIC_MAGICAE.replace("·", "").replace("⟠", "").replace("◌",
↪  "").replace("⚙", "").replace("Ψ", "")

if len(SYNODIC_MAGICAE) != 197:
    raise RuntimeError("the TardiSHA Synodic Magicae must contain exactly 197 code points across 196
    ↪  glyph entries")
```

```
if len(ALPHABET) != 191 or "·" in ALPHABET or "𜰡" in ALPHABET or "𜰢" in ALPHABET or "𜳒" in ALPHABET
↪  or "𜳰" in ALPHABET:
    raise RuntimeError("the TardiSHA generation alphabet must contain exactly 191 code points and
    ↪  exclude ·, 𜰡, 𜰢, 𜳒, and 𜳰")
if len(DAEMONIC_TONGUE) != 196:
    raise RuntimeError("the TardiSHA Daemonic Tongue must contain exactly 196 code points")
if len(set(DAEMONIC_TONGUE)) != len(DAEMONIC_TONGUE):
    raise RuntimeError("the TardiSHA Daemonic Tongue must contain unique code points")
if any(len(symbol) != 1 for symbol in DAEMONIC_TONGUE if symbol != "𜳰") or DAEMONIC_TONGUE[-1] !=
↪  "𜳰":
    raise RuntimeError("each TardiSHA Daemonic Tongue glyph must be one code point except exact
    ↪  authored Shadow 𜳰")
```

manifest.py

```
"""Grimchain file and directory manifests.

The public SOURCE_MANIFEST ledger shape is retained: a name, a file count,
and sorted entries. Each entry carries its Grimchain
value, and the complete entry body receives its own deterministic Grimchain.

Vāhana means vehicle or carrier in Sanskrit. Manifest Vāhana is the invocation-local
carrier for already-calculated entries and the current body on their way to becoming
permanent history. It is discarded after the operation and is not a cache.
"""
from __future__ import annotations

import os
import tempfile
from concurrent.futures import ProcessPoolExecutor
from pathlib import Path
from typing import Any

from .domus_stream import file_domus_record, living_domus_for_source
from .hashing import (
    TardiSHAError,
    validate_middle_length,
    validate_nonce,
)

_GRIM_OBJECT_OPEN = "\u169b"
_GRIM_OBJECT_CLOSE = "\u169c"
_GRIM_COLLECTION_OPEN = "\u2235"
_GRIM_COLLECTION_CLOSE = "\u2234"
_GRIM_FIELD_OPEN = "\u10fb"
_GRIM_FIELD_HINGE = "\u205b"
_GRIM_FIELD_CLOSE = "\u2056"

def default_manifest_name(target: Path, recursive: bool) -> str:
    name = target.name or "root"
    prefix = "manifest-recurse-" if target.is_dir() and recursive else "manifest-"
    return f"{prefix}{name}.grim"

def manifest_output_path(target: Path, output: Path | None, recursive: bool) -> Path:
    default_name = default_manifest_name(target, recursive)
    if output is None:
        return Path.cwd() / default_name
    candidate = output.expanduser()
    if candidate.exists() and candidate.is_dir():
        return candidate / default_name
    return candidate

def _lexical_absolute(path: Path) -> Path:
```

```
    return Path(os.path.abspath(os.fspath(path.expanduser())))

def _selected_entries(
    root: Path,
    recursive: bool,
    excludes: tuple[Path, ...] = (),
    root_file_name: str | None = None,
) -> list[tuple[str, Path, str]]:
    excluded = {_lexical_absolute(path) for path in excludes}
    if root.is_file() and not root.is_symlink():
        if _lexical_absolute(root) in excluded:
            raise TardiSHAError("manifest source cannot be excluded from itself")
        return [
            (
                "file",
                root,
                root.name if root_file_name is None else root_file_name,
            )
        ]
    if not root.is_dir() or root.is_symlink():
        raise TardiSHAError(f"manifest source is not a regular file or directory: {root}")

    selected: list[tuple[str, Path, str]] = [("directory", root, ".")]
    stack = [root]
    while stack:
        current = stack.pop()
        try:
            scanned = sorted(os.scandir(current), key=lambda entry: entry.name)
        except OSError as exc:
            raise TardiSHAError(f"cannot read manifest directory {current}: {exc}") from exc
        directories: list[Path] = []
        for entry in scanned:
            child = Path(entry.path)
            if _lexical_absolute(child) in excluded:
                continue
            relative = child.relative_to(root).as_posix()
            if entry.is_symlink():
                selected.append(("symlink", child, relative))
            elif entry.is_dir(follow_symlinks=False):
                selected.append(("directory", child, relative))
                if recursive:
                    directories.append(child)
            elif entry.is_file(follow_symlinks=False):
                selected.append(("file", child, relative))
            else:
                raise TardiSHAError(f"unsupported manifest entry type: {relative}")
        if recursive:
            stack.extend(reversed(directories))
    return sorted(selected, key=lambda item: item[2])

def _manifest_file_entry(
    source: Path,
    relative_path: str,
    middle: int,
    nonce: int,
    *,
```

```
    identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> dict[str, Any]:
    """Calculate one listed file exactly once for this manifest invocation."""
    seal, _, _ = file_domus_record(
        source, middle, nonce=nonce, identity_name=identity_name,
        include_filename=include_filename,
    )

    return {
        "type": "file",
        "path": relative_path,
        "grimchain": seal,
    }

def _manifest_entry_worker(payload: tuple[str, str, int, int, str | bytes | None, bool]) ->
↪  dict[str, Any]:
    source, relative_path, middle, nonce, identity_name, include_filename = payload
    return _manifest_file_entry(
        Path(source), relative_path, middle, nonce,
        identity_name=identity_name, include_filename=include_filename,
    )

def _worker_count(file_count: int) -> int:
    cpu_count = os.cpu_count()
    if cpu_count is None:
        raise TardiSHAError("CPU count is required for parallel manifest construction")
    if isinstance(cpu_count, bool) or not isinstance(cpu_count, int) or cpu_count < 1:
        raise TardiSHAError("CPU count must be a positive integer")
    worker_budget = max(1, (cpu_count * 35) // 100)
    return min(file_count, worker_budget)

def _nonfile_entry(kind: str, path: Path, relative: str) -> dict[str, Any]:
    if kind == "directory":
        return {"type": "directory", "path": relative}
    if kind == "symlink":
        return {"type": "symlink", "path": relative, "target": os.readlink(path)}
    raise TardiSHAError(f"unsupported manifest entry kind: {kind}")


def _build_manifest_entries(
    root: Path,
    *,
    middle: int,
    recursive: bool,
    nonce: int,
    exclusions: tuple[Path, ...],
    root_file_name: str | None = None,
    root_identity_name: str | bytes | None = None,
    root_include_filename: bool = True,
) -> list[dict[str, Any]]:
    selected = _selected_entries(
        root, recursive, exclusions, root_file_name=root_file_name
    )
    file_items = [(path, relative) for kind, path, relative in selected if kind == "file"]
    payloads = [
```

```
        (
            str(path), relative, middle, nonce,
            root_identity_name if root.is_file() and path == root else None,
            root_include_filename if root.is_file() and path == root else True,
        )
        for path, relative in file_items
    ]
    if len(payloads) > 1:
        with ProcessPoolExecutor(max_workers=_worker_count(len(payloads))) as pool:
            file_records = list(pool.map(_manifest_entry_worker, payloads))
    else:
        file_records = [_manifest_entry_worker(payload) for payload in payloads]
    files_by_path = {record["path"]: record for record in file_records}

    entries: list[dict[str, Any]] = []
    for kind, path, relative in selected:
        if kind == "file":
            entries.append(files_by_path[relative])
        else:
            entries.append(_nonfile_entry(kind, path, relative))
    return entries


def build_grimchain_manifest(
    target: str | Path,
    *,
    middle: int,
    recursive: bool = False,
    nonce: int = 0,
    exclude: str | Path | None = None,
    entry_name: str | None = None,
    identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> dict[str, Any]:

    root = Path(target).expanduser().resolve()
    width = validate_middle_length(middle)
    salt = validate_nonce(nonce)
    exclusions = (() if exclude is None else (Path(exclude).expanduser().resolve(),))

    if recursive and not root.is_dir():
        raise TardiSHAError("-R/--recursive requires a directory manifest")

    entries = _build_manifest_entries(
        root,
        middle=width,
        recursive=recursive,
        nonce=salt,
        exclusions=exclusions,
        root_file_name=entry_name,
        root_identity_name=identity_name,
        root_include_filename=include_filename,
    )

    public_entries: list[dict[str, Any]] = []
    for entry in entries:
```

```
        kind_name = entry["type"]
        if kind_name == "file":
            public_entries.append({
                "type": "file",
                "path": entry["path"],
                "grimchain": entry["grimchain"],
            })
        elif kind_name == "directory":
            public_entries.append({
                "type": "directory",
                "path": entry["path"],
            })
        elif kind_name == "symlink":
            public_entries.append({
                "type": "symlink",
                "path": entry["path"],
                "target": entry["target"],
            })
        else:
            raise TardiSHAError(f"unsupported manifest entry kind: {kind_name}")

    manifest = {
        "scope": "recursive" if recursive else "direct",
        "entries": public_entries,
    }

    return manifest

def _grim_manifest_scalar(value: Any, *, field: str) -> str:
    if not isinstance(value, str):
        raise TardiSHAError(f"manifest {field} must be a string")
    if "\n" in value or "\r" in value:
        raise TardiSHAError(f"manifest {field} cannot contain a line break")
    return value


def _grim_field(name: str, value: str) -> str:
    return f"{_GRIM_FIELD_OPEN}{name}{_GRIM_FIELD_HINGE}{value}{_GRIM_FIELD_CLOSE}"


def _render_grim_manifest(manifest: dict[str, Any]) -> str:
    """Serialize one public manifest into the native .grim language."""
    scope = _grim_manifest_scalar(manifest.get("scope"), field="scope")
    entries = manifest.get("entries")
    if scope not in ("recursive", "direct"):
        raise TardiSHAError("manifest lacks a valid scope")
    if not isinstance(entries, list):
        raise TardiSHAError("manifest entries must be a list")

    lines = [
        _GRIM_OBJECT_OPEN,
        _grim_field("scope", scope),
        f"{_GRIM_FIELD_OPEN}entries{_GRIM_FIELD_HINGE}{_GRIM_COLLECTION_OPEN}",
    ]
    for entry in entries:
        if not isinstance(entry, dict):
```

```
            raise TardiSHAError("manifest entry is not an object")
        kind = _grim_manifest_scalar(entry.get("type"), field="entry type")
        path = _grim_manifest_scalar(entry.get("path"), field="entry path")
        lines.extend((
            _GRIM_OBJECT_OPEN,
            _grim_field("type", kind),
            _grim_field("path", path),
        ))
        if kind == "file":
            value = _grim_manifest_scalar(entry.get("grimchain"), field="entry grimchain")
            lines.append(_grim_field("grimchain", value))
        elif kind == "symlink":
            target = _grim_manifest_scalar(entry.get("target"), field="symlink target")
            lines.append(_grim_field("target", target))
        elif kind != "directory":
            raise TardiSHAError(f"unsupported manifest entry kind: {kind}")
        lines.append(_GRIM_OBJECT_CLOSE)

    lines.extend((
        _GRIM_COLLECTION_CLOSE,
        _GRIM_OBJECT_CLOSE,
    ))
    return "\n".join(lines) + "\n"



def write_grimchain_manifest(
    target: str | Path,
    output: str | Path,
    *,
    middle: int,
    recursive: bool = False,
    nonce: int = 0,
    entry_name: str | None = None,
    identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> str:
    destination = Path(output).expanduser().resolve()
    width = validate_middle_length(middle)
    salt = validate_nonce(nonce)

    manifest = build_grimchain_manifest(
        target,
        middle=width,
        recursive=recursive,
        nonce=salt,
        exclude=destination,
        entry_name=entry_name,
        identity_name=identity_name,
        include_filename=include_filename,
    )
    body_bytes = _render_grim_manifest(manifest).encode("utf-8")
    destination.parent.mkdir(parents=True, exist_ok=True)

    fd, tmp_name = tempfile.mkstemp(
        prefix=f".{destination.name}.",
```

```
    suffix=".part",
    dir=destination.parent,
)
temp = Path(tmp_name)
try:
    with os.fdopen(fd, "wb") as handle:
        handle.write(body_bytes)
        handle.flush()
        os.fsync(handle.fileno())

    os.replace(temp, destination)
except Exception:
    temp.unlink(missing_ok=True)
    raise

first_grimchain = living_domus_for_source(
    destination,
    width,
    kind="file",
    nonce=salt,
)

with destination.open("ab") as handle:
    handle.write(first_grimchain.encode("utf-8"))
    handle.flush()
    os.fsync(handle.fileno())

second_grimchain = living_domus_for_source(
    destination,
    width,
    kind="file",
    nonce=salt,
)
if second_grimchain != first_grimchain:
    raise TardiSHAError("manifest GrimChain return did not close")
return second_grimchain
```

manifestation.py

```
"""ALQC Manifestation Ground, exact Ennead accounting, and finite D-COMP.

The canonical ledger is carried in Q(sqrt(5)); float values exposed by the
witness dataclasses are images only.  Admission, conservation, closure, and
motion decisions are made on integers, Fractions, or exact Q(sqrt(5)) bodies.
"""
from __future__ import annotations

from dataclasses import dataclass
from fractions import Fraction
from math import isfinite, sqrt
from typing import Final, Sequence

from .aeon_layers import PHI_IMAGE
from .canon import (
    GLYPH_BODY,
    GLYPH_LAWS,
    SATURATION_LIMIT,
    TOTAL_CAPACITY,
    court_active_connections,
)
from .source_emission import ALPHA, GoldenBearing, Q5Fraction, sign_q5

# A1: 12x12 Court skeleton folded into the 9x9 Manifestation Ground.
GROUND_SIDE: Final[int] = 9
GROUND_NODES: Final[int] = GROUND_SIDE * GROUND_SIDE
FOLD_RATIO_NUM: Final[int] = 12
FOLD_RATIO_DEN: Final[int] = 9
ENNEAD: Final[int] = 9

# Exact finite-interval implementation parameters (EX_III section 46).
INTERVAL_DURATION: Final[int] = 1
PRESSURE_WEIGHT: Final[int] = 1
RETURN_FRICTION_WEIGHT: Final[int] = 1
Q_METRIC_WEIGHTS: Final[tuple[int, int, int, int]] = (1, 2, 3, 4)
MAX_Q_COMPONENT: Final[int] = 3
SHADOW_CAPACITY: Final[int] = 93

# C_bio is never identified by comparing its irrational root.  The identity
# gate is the exact rational square 741^2/396 = 61009/44.
SENSATION_HZ: Final[int] = int(GLYPH_LAWS["❋"].frequency.real)
FEAR_HZ: Final[int] = int(GLYPH_LAWS["⊛"].frequency.real)
C_BIO_SQUARED: Final[Fraction] = Fraction(SENSATION_HZ * SENSATION_HZ, FEAR_HZ)
C_BIO_IMAGE: Final[float] = sqrt(float(C_BIO_SQUARED))

if C_BIO_SQUARED != Fraction(61009, 44):
    raise RuntimeError("C_bio squared must be the exact body 61009/44")
if SHADOW_CAPACITY != MAX_Q_COMPONENT**2 * sum(Q_METRIC_WEIGHTS) + MAX_Q_COMPONENT:
```

```
    raise RuntimeError("finite canonical Q-domain must derive Sigma_max = 93")


def manifestation_fold(court: int) -> int:
    """Fold Court address ``12i+j`` into the 9x9 Ground."""
    if not isinstance(court, int) or isinstance(court, bool) or not 0 <= court < TOTAL_CAPACITY:
        raise ValueError("court must be an integer in [0,143]")
    i, j = divmod(court, 12)
    return ((i * GROUND_SIDE) // 12) * GROUND_SIDE + ((j * GROUND_SIDE) // 12)


def vector_row(ground_node: int) -> int:
    if not isinstance(ground_node, int) or isinstance(ground_node, bool) or not 0 <= ground_node <
    ↪  GROUND_NODES:
        raise ValueError("ground node must be an integer in [0,80]")
    return ground_node // GROUND_SIDE


def strike_band(k: int) -> str:
    if not isinstance(k, int) or isinstance(k, bool) or not 1 <= k <= ENNEAD:
        raise ValueError("strike index k must be an integer in 1..9")
    if k <= 3:
        return "Root"
    if k <= 6:
        return "Path"
    return "Seal"


def parity_flip(q_state: int) -> int:
    """Strict Q-state involution: Q0,Q1 fixed and Q2<->Q3."""
    if isinstance(q_state, bool) or not isinstance(q_state, int):
        raise TypeError("parity flip requires one exact integer Q-state index")
    if q_state not in (0, 1, 2, 3):
        raise ValueError("parity flip is defined only on Q-states 0..3")
    return {0: 0, 1: 1, 2: 3, 3: 2}[q_state]


def _exact_integer(value: object, *, field: str, bounded: bool) -> int:
    if isinstance(value, bool):
        raise ValueError(f"{field} must be an exact integer")
    if isinstance(value, int):
        result = value
    elif isinstance(value, Fraction) and value.denominator == 1:
        result = value.numerator
    else:
        raise ValueError(f"{field} must be an exact integer")
    if bounded and not 0 <= result <= MAX_Q_COMPONENT:
        raise ValueError(f"{field} must lie in [0,3]")
    return result


def _q_tuple(q: Sequence[object]) -> tuple[int, int, int, int]:
    if len(q) != 4:
        raise ValueError("a Q-vector must contain exactly four components")
    values = tuple(_exact_integer(q[k], field=f"Q[{k}]", bounded=True) for k in range(4))
    return values  # type: ignore[return-value]
```

```
def _tangent_tuple(vector: Sequence[object]) -> tuple[int, int, int, int]:
    if len(vector) != 4:
        raise ValueError("a Domus tangent vector must contain exactly four components")
    values = tuple(_exact_integer(vector[k], field=f"tangent[{k}]", bounded=False) for k in
    ↪  range(4))
    return values  # type: ignore[return-value]


def parity_vector(q: Sequence[object]) -> tuple[int, int, int, int]:
    q0, q1, q2, q3 = _q_tuple(q)
    return (q0, q1, q3, q2)







def metric_square(left: Sequence[object], right: Sequence[object]) -> int:
    """Exact squared Court distance under G=diag(1,2,3,4)."""
    a, b = _tangent_tuple(left), _tangent_tuple(right)
    return sum(w * (x - y) ** 2 for w, x, y in zip(Q_METRIC_WEIGHTS, a, b))




# Alternate name for the display projection. This value is never used as a decision gate in this
↪  module.


def _q_residue_norm_square(q: Sequence[object]) -> int:
    values = _q_tuple(q)
    return sum(w * v * v for w, v in zip(Q_METRIC_WEIGHTS, values))


def _q5_add(left: Q5Fraction, right: Q5Fraction) -> Q5Fraction:
    return Q5Fraction(
        left.a * right.denominator + right.a * left.denominator,
        left.b * right.denominator + right.b * left.denominator,
        left.denominator * right.denominator,
    )


def _q5_sub(left: Q5Fraction, right: Q5Fraction) -> Q5Fraction:
    return Q5Fraction(
        left.a * right.denominator - right.a * left.denominator,
        left.b * right.denominator - right.b * left.denominator,
        left.denominator * right.denominator,
    )


def _q5_mul_bearing(value: Q5Fraction, bearing: GoldenBearing) -> Q5Fraction:
    # (a+b*sqrt(5))/d times (c+e*sqrt(5))/2.
    return Q5Fraction(
```

```
        value.a * bearing.a + 5 * value.b * bearing.b,
        value.a * bearing.b + value.b * bearing.a,
        2 * value.denominator,
    )


def _exact_debt(value: int | float | Fraction | Q5Fraction) -> Q5Fraction:
    if isinstance(value, Q5Fraction):
        result = value
    elif isinstance(value, bool):
        raise ValueError("shadow debt must be exact and non-negative")
    elif isinstance(value, int):
        result = Q5Fraction(value, 0, 1)
    elif isinstance(value, Fraction):
        result = Q5Fraction(value.numerator, 0, value.denominator)
    elif isinstance(value, float) and isfinite(value) and value.is_integer():
        result = Q5Fraction(int(value), 0, 1)
    else:
        raise ValueError("shadow debt must be an exact integer, Fraction, or Q(sqrt(5)) body")
    if sign_q5(result.a, result.b) < 0:
        raise ValueError("shadow debt must be non-negative")
    return result


def _q5_le_integer(value: Q5Fraction, bound: int) -> bool:
    return sign_q5(value.a - bound * value.denominator, value.b) <= 0


@dataclass(frozen=True, slots=True)
class ExactEnneadLedger:
    initial_debt: Q5Fraction
    strike_inputs: tuple[Q5Fraction, ...]
    strike_outputs: tuple[Q5Fraction, ...]
    strike_form_work: tuple[Q5Fraction, ...]
    q3_transfer: Q5Fraction
    form_work: Q5Fraction
    residual_debt: Q5Fraction
    accounted_total: Q5Fraction

    @property
    def energy_conserved(self) -> bool:
        return self.accounted_total == self.initial_debt

    @property
    def saturated(self) -> bool:
        return self.residual_debt == Q5Fraction(0, 0, 1)


def exact_ennead_ledger(shadow_debt: int | float | Fraction | Q5Fraction) -> ExactEnneadLedger:
    """Nine strikes in exact Q(sqrt(5)); no image value enters a decision."""
    initial = _exact_debt(shadow_debt)
    if not _q5_le_integer(initial, SHADOW_CAPACITY):
        raise ValueError("shadow debt exceeds the exact finite capacity Sigma_max=93")
    inputs: list[Q5Fraction] = []
    outputs: list[Q5Fraction] = []
    works: list[Q5Fraction] = []
```

```
    current = initial
    form_total = Q5Fraction(0, 0, 1)
    for _k in range(1, ENNEAD):
        incoming = current
        outgoing = _q5_mul_bearing(incoming, ALPHA)
        work = _q5_sub(incoming, outgoing)
        inputs.append(incoming)
        outputs.append(outgoing)
        works.append(work)
        form_total = _q5_add(form_total, work)
        current = outgoing
    inputs.append(current)
    outputs.append(Q5Fraction(0, 0, 1))
    works.append(Q5Fraction(0, 0, 1))
    q3_transfer = current
    residual = Q5Fraction(0, 0, 1)
    accounted = _q5_add(form_total, q3_transfer)
    result = ExactEnneadLedger(
        initial_debt=initial,
        strike_inputs=tuple(inputs),
        strike_outputs=tuple(outputs),
        strike_form_work=tuple(works),
        q3_transfer=q3_transfer,
        form_work=form_total,
        residual_debt=residual,
        accounted_total=accounted,
    )
    if len(result.strike_inputs) != ENNEAD or not result.energy_conserved or not result.saturated:
        raise RuntimeError("exact Ennead ledger failed its nine-strike closure")
    return result


@dataclass(frozen=True, slots=True)
class EnneadStrike:
    k: int
    band: str
    input_debt: float
    output_debt: float
    form_work: float
    q3_transfer: float
    q_parity: str
    phase_locked: bool

    @property
    def residual_debt(self) -> float:
        return self.output_debt


@dataclass(frozen=True, slots=True)
class EnneadResult:
    row: int
    initial_debt: float
    strikes: tuple[EnneadStrike, ...]
    final_parity: str
    residual_debt: float
    q3_recursion_residue: float
```

```
    pre_lock_q2_residual: float
    form_work: float
    accounted_total: float
    energy_conserved: bool
    saturated: bool


def ennead_saturate(row: int, shadow_debt: int | float | Fraction | Q5Fraction) -> EnneadResult:
    """Resolve Q2 pressure exactly, exposing float images only as witnesses."""
    if not isinstance(row, int) or isinstance(row, bool) or not 0 <= row < GROUND_SIDE:
        raise ValueError("vector row must be an integer in 0..8")
    exact = exact_ennead_ledger(shadow_debt)

    # Preserve the historical witness image and its serialized bytes.  This
    # recurrence never decides conservation or closure.
    residual = float(exact.initial_debt.value)
    form_work = 0.0
    pre_lock = 0.0
    strikes: list[EnneadStrike] = []
    for k in range(1, ENNEAD + 1):
        band = strike_band(k)
        incoming = residual
        if k < ENNEAD:
            outgoing = incoming / PHI_IMAGE
            work = incoming - outgoing
            residual = outgoing
            form_work += work
            strikes.append(EnneadStrike(k, band, incoming, outgoing, work, 0.0, "Q2", False))
        else:
            pre_lock = incoming
            residual = 0.0
            strikes.append(EnneadStrike(k, band, incoming, 0.0, 0.0, incoming, "Q3", True))

    q3_residue = pre_lock
    accounted = form_work + q3_residue
    result = EnneadResult(
        row=row,
        initial_debt=float(exact.initial_debt.value),
        strikes=tuple(strikes),
        final_parity="Q3",
        residual_debt=residual,
        q3_recursion_residue=q3_residue,
        pre_lock_q2_residual=pre_lock,
        form_work=form_work,
        accounted_total=accounted,
        energy_conserved=exact.energy_conserved,
        saturated=exact.saturated,
    )
    if len(result.strikes) != ENNEAD or [s.k for s in result.strikes] != list(range(1, 10)):
        raise RuntimeError("Ennead requires exactly nine ordered strikes")
    if any(s.q_parity != "Q2" or s.phase_locked or s.q3_transfer != 0.0 for s in
    ↪  result.strikes[:8]):
        raise RuntimeError("Q2 must remain the Floating Ghost through strikes 1..8")
    seal = result.strikes[8]
    if seal.q_parity != "Q3" or not seal.phase_locked or seal.q3_transfer !=
    ↪  result.pre_lock_q2_residual:
```

```
        raise RuntimeError("strike 9 must transfer the remaining Q2 occupancy to Q3")
    if parity_flip(2) != 3 or parity_flip(parity_flip(2)) != 2:
        raise RuntimeError("parity must be an involutive Q2/Q3 exchange")
    if not result.energy_conserved:
        raise RuntimeError("Ennead pressure ledger failed exact conservation")
    return result


MIRROR_OPERATOR_ORDER: Final[tuple[str, ...]] = ("❄", "✡", "⬡", "⧗")


@dataclass(frozen=True, slots=True)
class ExactFrequencyPoint:
    """One exact Goetic frequency body on the Mirror path."""

    glyph: str
    structural_hz: Fraction
    parity_hz: Fraction


@dataclass(frozen=True, slots=True)
class ExactFrequencyPath:
    """One independently constructed finite Mirror path."""

    start: ExactFrequencyPoint
    end: ExactFrequencyPoint
    operator_order: tuple[str, ...]

    @property
    def tangent(self) -> tuple[Fraction, Fraction]:
        return (
            self.end.structural_hz - self.start.structural_hz,
            self.end.parity_hz - self.start.parity_hz,
        )


_Q_TO_GLYPH: Final[dict[tuple[int, int, int, int], str]] = {
    tuple(GLYPH_LAWS[glyph].q_vector): glyph for glyph in GLYPH_BODY
}
if len(_Q_TO_GLYPH) != len(GLYPH_BODY):
    raise RuntimeError("each Goetic Q-vector must resolve one exact frequency body")


def _frequency_point(q: Sequence[object]) -> ExactFrequencyPoint:
    body = _q_tuple(q)
    try:
        glyph = _Q_TO_GLYPH[body]
    except KeyError as exc:
        raise ValueError("Q-vector has no exact Goetic frequency body") from exc
    frequency = GLYPH_LAWS[glyph].frequency
    return ExactFrequencyPoint(
        glyph=glyph,
        structural_hz=Fraction(frequency.structural_hz),
        parity_hz=(
            Fraction(frequency.parity_hz)
            if frequency.parity_hz is not None
```

```
            else Fraction(0, 1)
        ),
    )


def _parity_frequency_tangent(
    tangent: tuple[Fraction, Fraction],
) -> tuple[Fraction, Fraction]:
    # Canon intrinsic parity phase η_P = -1 on the Klein return.
    return -tangent[0], -tangent[1]


def _frequency_path_pair(
    origin_q: Sequence[object], resolution_q: Sequence[object]
) -> tuple[ExactFrequencyPath, ExactFrequencyPath]:
    origin = _frequency_point(origin_q)
    resolution = _frequency_point(resolution_q)
    forward = ExactFrequencyPath(
        start=origin,
        end=resolution,
        operator_order=MIRROR_OPERATOR_ORDER,
    )
    # Reverse Integration is constructed independently from resolution to origin.
    returned = ExactFrequencyPath(
        start=resolution,
        end=origin,
        operator_order=MIRROR_OPERATOR_ORDER,
    )
    return forward, returned


def _frequency_metric_square(tangent: tuple[Fraction, Fraction]) -> Fraction:
    structural, parity = tangent
    return structural * structural + parity * parity


def _commutator_pressure(
    forward: ExactFrequencyPath, returned: ExactFrequencyPath
) -> Fraction:
    """Exact finite witness of [𝒩,𝔑]=0 under Path Out = Path Back."""
    operator_residual = (
        0
        if forward.operator_order == returned.operator_order == MIRROR_OPERATOR_ORDER
        else 1
    )
    endpoint_residual = int(forward.start != returned.end) + int(
        forward.end != returned.start
    )
    parity_return = _parity_frequency_tangent(returned.tangent)
    tangent_residual = (
        forward.tangent[0] - parity_return[0],
        forward.tangent[1] - parity_return[1],
    )
    return Fraction(operator_residual + endpoint_residual, 1) + _frequency_metric_square(
        tangent_residual
    )
```

```
def _fraction_body(value: Fraction) -> tuple[int, int]:
    return value.numerator, value.denominator


def _q5_body(value: Q5Fraction) -> tuple[int, int, int]:
    return value.a, value.b, value.denominator


@dataclass(frozen=True, slots=True)
class ExactDComp:
    forward_path: ExactFrequencyPath
    return_path: ExactFrequencyPath
    commutator_pressure: Fraction
    velocity_mismatch_square: Fraction
    return_pressure: int
    terminal_unresolved_debt: int
    shadow_debt_initial: Q5Fraction
    shadow_debt_terminal: Q5Fraction
    form_work: Q5Fraction
    q3_recursion_gain: Q5Fraction
    active_count: int
    withheld_count: int
    whiteout: bool
    c_bio_squared: Fraction

    @property
    def liquid_body_valid(self) -> bool:
        return (
            self.active_count == SATURATION_LIMIT
            and self.withheld_count == TOTAL_CAPACITY - SATURATION_LIMIT
        )

    @property
    def closed(self) -> bool:
        return (
            self.commutator_pressure == 0
            and self.velocity_mismatch_square == 0
            and self.shadow_debt_terminal == Q5Fraction(0, 0, 1)
            and self.liquid_body_valid
            and not self.whiteout
        )

    @property
    def truth(self) -> int:
        return int(self.closed)


@dataclass(frozen=True, slots=True)
class DCompWitness:
    """Image surface backed by complete exact finite-interval proof bodies."""

    interval_duration: float
    metric_weights: tuple[float, float]
    pressure_weight: float
```

```
return_friction_weight: float
commutator_model: str
commutator_pressure: float
return_pressure: float
local_friction: float
terminal_unresolved_debt: float
shadow_capacity: float
shadow_debt_initial: float
shadow_debt_bounded: bool
forward_velocity: tuple[float, float]
return_velocity: tuple[float, float]
parity_return_velocity: tuple[float, float]
velocity_mismatch: float
active_count: int
withheld_count: int
whiteout_penalty: float
c_bio: float
mas_efficiency: float
q3_loss: float
q3_capacity_before: float
q3_recursion_gain: float
q3_capacity_after: float
form_work: float
shadow_debt_terminal: float
terminal: float
closed: bool
truth: int
motion_positive: bool
exact_commutator_pressure: tuple[int, int]
exact_velocity_mismatch_square: tuple[int, int]
exact_shadow_debt_initial: tuple[int, int, int]
exact_shadow_debt_terminal: tuple[int, int, int]
exact_form_work: tuple[int, int, int]
exact_q3_recursion_gain: tuple[int, int, int]

def __post_init__(self) -> None:
    for field_name in (
        "metric_weights",
        "forward_velocity",
        "return_velocity",
        "parity_return_velocity",
        "exact_commutator_pressure",
        "exact_velocity_mismatch_square",
        "exact_shadow_debt_initial",
        "exact_shadow_debt_terminal",
        "exact_form_work",
        "exact_q3_recursion_gain",
    ):
        value = getattr(self, field_name)
        if not isinstance(value, tuple):
            object.__setattr__(self, field_name, tuple(value))
    if isinstance(self.truth, bool) or self.truth not in (0, 1):
        raise ValueError("D-COMP Truth must be the exact zero-or-unity bit")
    if self.closed != (self.truth == 1):
        raise ValueError("D-COMP closure and Truth must agree")
    if self.closed:
```

```
            if self.terminal != 0.0:
                raise ValueError("closed D-COMP must expose terminal zero")
            if self.exact_commutator_pressure != (0, 1):
                raise ValueError("closed D-COMP requires exact commutator zero")
            if self.exact_velocity_mismatch_square != (0, 1):
                raise ValueError("closed D-COMP requires exact Mirror-path mismatch zero")
            if self.exact_shadow_debt_terminal != (0, 0, 1):
                raise ValueError("closed D-COMP requires exact terminal Shadow Debt zero")


@dataclass(frozen=True, slots=True)
class Manifestation:
    court: int
    ground_node: int
    vector_row: int
    ennead: EnneadResult
    exact_ennead: ExactEnneadLedger
    residue_norm: float
    residue_norm_square: int
    exact_dcomp: ExactDComp
    dcomp: DCompWitness

    @property
    def differential_tension(self) -> float:
        return self.dcomp.velocity_mismatch

    @property
    def closed(self) -> bool:
        return self.dcomp.closed and self.dcomp.truth == 1

    @property
    def terminal_dcomp(self) -> float:
        return self.dcomp.terminal


def _derive_exact_dcomp(
    origin_q: Sequence[object], resolution_q: Sequence[object], *, court: int
) -> tuple[ExactDComp, ExactEnneadLedger]:
    origin = _q_tuple(origin_q)
    resolution = _q_tuple(resolution_q)
    forward_path, return_path = _frequency_path_pair(origin, resolution)
    parity_return_tangent = _parity_frequency_tangent(return_path.tangent)
    mismatch_tangent = (
        forward_path.tangent[0] - parity_return_tangent[0],
        forward_path.tangent[1] - parity_return_tangent[1],
    )
    mismatch_square = _frequency_metric_square(mismatch_tangent)
    commutator = _commutator_pressure(forward_path, return_path)

    return_pressure = metric_square(origin, parity_vector(resolution))
    terminal_unresolved = abs(resolution[2])
    initial = _q5_add(
        Q5Fraction(commutator.numerator, 0, commutator.denominator),
        Q5Fraction(return_pressure + terminal_unresolved, 0, 1),
    )
    ledger = exact_ennead_ledger(initial)
```

```
    active_count = len(court_active_connections(court))
    withheld_count = TOTAL_CAPACITY - active_count
    # Whiteout is the exact 144-active / 0-withheld body.  A mere threshold
    # excess is not renamed Whiteout.
    whiteout = active_count == TOTAL_CAPACITY and withheld_count == 0
    exact = ExactDComp(
        forward_path=forward_path,
        return_path=return_path,
        commutator_pressure=commutator,
        velocity_mismatch_square=mismatch_square,
        return_pressure=return_pressure,
        terminal_unresolved_debt=terminal_unresolved,
        shadow_debt_initial=initial,
        shadow_debt_terminal=ledger.residual_debt,
        form_work=ledger.form_work,
        q3_recursion_gain=ledger.q3_transfer,
        active_count=active_count,
        withheld_count=withheld_count,
        whiteout=whiteout,
        c_bio_squared=C_BIO_SQUARED,
    )
    return exact, ledger


def exact_dcomp(origin_q: Sequence[object], resolution_q: Sequence[object], *, court: int = 0) ->
↪  ExactDComp:
    exact, _ledger = _derive_exact_dcomp(origin_q, resolution_q, court=court)
    return exact


def close_boundary(origin_q: Sequence[object], resolution_q: Sequence[object], *, court: int = 0) ->
↪  Manifestation:
    """Construct and prove one complete finite Mirror-return phrase."""
    origin = _q_tuple(origin_q)
    resolution = _q_tuple(resolution_q)
    node = manifestation_fold(court)
    row = vector_row(node)

    exact, ledger = _derive_exact_dcomp(origin, resolution, court=court)
    if not exact.closed or exact.truth != 1:
        raise RuntimeError("Mirror closure requires D-COMP = 0 and Truth = 1")

    ennead = ennead_saturate(row, exact.shadow_debt_initial)
    forward_velocity_exact = exact.forward_path.tangent
    return_velocity_exact = exact.return_path.tangent
    parity_return_exact = _parity_frequency_tangent(return_velocity_exact)
    velocity_mismatch = sqrt(float(exact.velocity_mismatch_square)) * float(INTERVAL_DURATION)

    local_friction_exact = exact.commutator_pressure + Fraction(
        RETURN_FRICTION_WEIGHT * exact.return_pressure, 1
    )
    shadow_initial_image = float(exact.shadow_debt_initial.value)
    bounded = _q5_le_integer(exact.shadow_debt_initial, SHADOW_CAPACITY)
    if not bounded:
        raise RuntimeError("finite Q path exceeded its exact Shadow-Debt capacity")
```

```
whiteout_penalty = float("inf") if exact.whiteout else 0.0
mas_efficiency = 1.0
q3_loss = 0.0
q3_before = float(origin[3])
q3_gain = float(exact.q3_recursion_gain.value)
q3_after = q3_before + q3_gain - q3_loss
terminal = (
    float(exact.commutator_pressure)
    + velocity_mismatch
    + float(exact.shadow_debt_terminal.value)
    + whiteout_penalty
) / C_BIO_IMAGE
motion_positive = (
    _frequency_metric_square(exact.forward_path.tangent) > 0
    or sign_q5(exact.form_work.a, exact.form_work.b) > 0
    or sign_q5(exact.q3_recursion_gain.a, exact.q3_recursion_gain.b) > 0
)

witness = DCompWitness(
    interval_duration=float(INTERVAL_DURATION),
    metric_weights=(1.0, 1.0),
    pressure_weight=float(PRESSURE_WEIGHT),
    return_friction_weight=float(RETURN_FRICTION_WEIGHT),
    commutator_model=(
        "exact inverse-endpoint, preserved-operator-order, parity-return witness; "
        "[𝒜,ℜ] and 𝒜-∂(ℜ) computed without endpoint substitution"
    ),
    commutator_pressure=float(exact.commutator_pressure),
    return_pressure=float(exact.return_pressure),
    local_friction=float(local_friction_exact),
    terminal_unresolved_debt=float(exact.terminal_unresolved_debt),
    shadow_capacity=float(SHADOW_CAPACITY),
    shadow_debt_initial=shadow_initial_image,
    shadow_debt_bounded=bounded,
    forward_velocity=tuple(float(v) for v in forward_velocity_exact),
    return_velocity=tuple(float(v) for v in return_velocity_exact),
    parity_return_velocity=tuple(float(v) for v in parity_return_exact),
    velocity_mismatch=velocity_mismatch,
    active_count=exact.active_count,
    withheld_count=exact.withheld_count,
    whiteout_penalty=whiteout_penalty,
    c_bio=C_BIO_IMAGE,
    mas_efficiency=mas_efficiency,
    q3_loss=q3_loss,
    q3_capacity_before=q3_before,
    q3_recursion_gain=q3_gain,
    q3_capacity_after=q3_after,
    form_work=float(exact.form_work.value),
    shadow_debt_terminal=float(exact.shadow_debt_terminal.value),
    terminal=terminal,
    closed=exact.closed,
    truth=exact.truth,
    motion_positive=motion_positive,
    exact_commutator_pressure=_fraction_body(exact.commutator_pressure),
    exact_velocity_mismatch_square=_fraction_body(exact.velocity_mismatch_square),
```

```
        exact_shadow_debt_initial=_q5_body(exact.shadow_debt_initial),
        exact_shadow_debt_terminal=_q5_body(exact.shadow_debt_terminal),
        exact_form_work=_q5_body(exact.form_work),
        exact_q3_recursion_gain=_q5_body(exact.q3_recursion_gain),
    )

    if witness.terminal != 0.0 or not witness.closed or witness.truth != 1:
        raise RuntimeError("completed Mirror witness must expose D-COMP = 0 and Truth = 1")
    if not ledger.energy_conserved or not ledger.saturated:
        raise RuntimeError("D-COMP pressure is not fully conserved in Q(sqrt(5))")

    residue_square = _q_residue_norm_square(resolution)
    return Manifestation(
        court=court,
        ground_node=node,
        vector_row=row,
        ennead=ennead,
        exact_ennead=ledger,
        residue_norm=sqrt(residue_square),
        residue_norm_square=residue_square,
        exact_dcomp=exact,
        dcomp=witness,
    )




def assert_ground() -> None:
    if GROUND_NODES != 81:
        raise RuntimeError("Manifestation Ground must have 81 nodes")
    if FOLD_RATIO_NUM * 3 != FOLD_RATIO_DEN * 4:
        raise RuntimeError("folding ratio must be 4/3")
    rows = {vector_row(manifestation_fold(c)) for c in range(TOTAL_CAPACITY)}
    if rows != set(range(GROUND_SIDE)):
        raise RuntimeError("all nine 1x9 vector rows must be populated")


assert_ground()
```

mirror_math.py

```
"""TardiSHA's source-typed realization of the Aeternum Mirror.

The Canon's displayed Aeternum identities are worked embodiments of the general
ALQC return law. Here a raw file is the expanded Fraktur Z_1 body and a terminal
Living Domus Grimchain is a candidate compressed Fraktur Z_0 return.

A terminal candidate is not granted return merely because the same Grimchain
construction can reproduce it. Deterministic regeneration may describe the
candidate, but return requires an independently constructed return path whose
protected return-body closes against the protected body of the outward path.

Mirror preserves operator sequence while exchanging bearing. Return preserves
root, parentage, Liquid law, W_34, twelvefold lineage, envelope, and invariant
identity while append-only lineage remains present.

The complete state need not reset to its starting state. The lawful return-body
must return without erasing the road by which it returned.

This is the file-identity office of Path Out = Path Back, parity return,
D-COMP = 0, and Truth = 1. Any suffix not independently proven as lawful return
remains source matter.
"""

from __future__ import annotations

from .canon import court_load

from dataclasses import asdict, dataclass
import mmap
import os
from pathlib import Path
from typing import Any

from .alqc_digest import ALQCDigest
from .hashing import FILE_NAME_FRAME, RAW_FILE_SOURCE_DOMAIN, SOURCE_CHUNK_BYTES, TardiSHAError,
↪  file_emission
from .source_emission import SourceEmission, emission_from_sponge, ALPHA, BETA,
↪  GoldenGoeticDerivation, resolve_bearing



def _parse_public_living_domus_current(seal: str):
    from .domus import parse_public_living_domus
    return parse_public_living_domus(seal)

@dataclass(frozen=True, slots=True)
class TerminalSelfGlyphCandidate:
    physical_size: int
```

```
    body_size: int
    written_center_length: int
    seal: str
    seal_utf8_bytes: int
    trailing_bytes_hex: str

    @property
    def trailing_bytes(self) -> bytes:
        return bytes.fromhex(self.trailing_bytes_hex)

    @property
    def bytes_accounted(self) -> bool:
        return self.body_size + self.seal_utf8_bytes + len(self.trailing_bytes) ==
        ↪  self.physical_size


@dataclass(frozen=True, slots=True)
class MirrorMathWitness:
    physical_size: int
    physical_source_digest: str
    source_stability_verified: bool
    effective_body_size: int
    candidate_detected: bool
    exact_self_glyph: bool
    folded: bool
    fold_count: int
    folded_physical_bytes: int
    lineage_seals: tuple[str, ...]
    lineage_trailing_bytes_hex: tuple[str, ...]
    seal: str | None
    expected_seal: str | None
    trailing_bytes_hex: str
    bytes_accounted: bool
    operator_order_preserved: bool
    expanded_posture: str
    compressed_posture: str
    source_route_dcomp: int
    source_truth: int
    return_dcomp: int | None
    return_truth: int | None
    truth: int
    derivation: str

    def as_dict(self) -> dict[str, Any]:
        return asdict(self)


@dataclass(frozen=True, slots=True)
class MirrorFileEmission:
    emission: SourceEmission
    witness: MirrorMathWitness


_GRIM_SELF_CLOSE = "\n∴\n".encode("utf-8")
_TERMINAL_TRAILING_PROBE = max(2, len(_GRIM_SELF_CLOSE))
```

```
def _strip_one_terminal_line_return(data: bytes) -> tuple[bytes, bytes]:
    if data.endswith(_GRIM_SELF_CLOSE):
        return data[:-len(_GRIM_SELF_CLOSE)], _GRIM_SELF_CLOSE
    if data.endswith(b"\r\n"):
        return data[:-2], b"\r\n"
    if data.endswith(b"\n"):
        return data[:-1], b"\n"
    return data, b""


def _byte_difference(left: str, right: str) -> int:
    """Exact non-negative residual; zero iff UTF-8 bodies are byte-identical."""
    a = left.encode("utf-8")
    b = right.encode("utf-8")
    shared = min(len(a), len(b))
    return abs(len(a) - len(b)) + sum(a[i] != b[i] for i in range(shared))


def _file_stat_witness(result: os.stat_result) -> tuple[int, int, int, int, int, int]:
    return (
        result.st_dev,
        result.st_ino,
        result.st_mode,
        result.st_size,
        result.st_mtime_ns,
        result.st_ctime_ns,
    )


def _previous_utf8_codepoint_start_view(view, position: int) -> int | None:
    if position <= 0:
        return None
    read_start = max(0, position - 4)
    for start in range(position - 1, read_start - 1, -1):
        candidate = bytes(view[start:position])
        try:
            decoded = candidate.decode("utf-8")
        except UnicodeDecodeError:
            continue
        if len(decoded) == 1:
            return start
    return None



def _iter_terminal_candidates_view(view, limit: int):
    if isinstance(limit, bool) or not isinstance(limit, int) or not 0 <= limit <= len(view):
        raise ValueError("physical_size must be within the source file")
    if limit == 0:
        return
    terminal_probe = min(limit, _TERMINAL_TRAILING_PROBE)
    tail = bytes(view[limit - terminal_probe:limit])
    _content, trailing = _strip_one_terminal_line_return(tail)
    body_end = limit - len(trailing)
    position = body_end
```

```
    reverse_codepoints: list[str] = []
    while position > 0:
        start = _previous_utf8_codepoint_start_view(view, position)
        if start is None:
            return
        codepoint = bytes(view[start:position]).decode("utf-8")
        reverse_codepoints.append(codepoint)
        position = start
        if len(reverse_codepoints) < 2:
            continue
        seal_bytes = bytes(view[start:body_end])
        seal = seal_bytes.decode("utf-8")
        try:
            _parse_public_living_domus_current(seal)
        except (TypeError, ValueError):
            continue
        written_length = len(reverse_codepoints)
        candidate = TerminalSelfGlyphCandidate(limit,start,written_length,seal,len(seal_bytes),trai⌋
        ↪  ling.hex())
        if not candidate.bytes_accounted:
            raise RuntimeError("Mirror Math failed physical-byte accounting")
        yield candidate


def _previous_utf8_codepoint_start(handle, position: int) -> int | None:
    """Return the byte start of the UTF-8 code point ending at ``position``.

    Terminal-body discovery follows encoded extent only. No glyph is privileged as
    a delimiter, and no glyph occurring in the middle can alter body discovery.
    """
    if position <= 0:
        return None
    read_start = max(0, position - 4)
    handle.seek(read_start)
    tail = handle.read(position - read_start)
    for offset in range(len(tail) - 1, -1, -1):
        candidate = tail[offset:]
        try:
            decoded = candidate.decode("utf-8")
        except UnicodeDecodeError:
            continue
        if len(decoded) == 1:
            return read_start + offset
    return None


def _iter_terminal_candidates(
    path: str | Path,
    *,
    physical_size: int | None = None,
    nonce: int = 0,
):
    """Detect terminal GrimChain candidates by encoded extent only."""
    target = Path(path)
    actual_size = target.stat().st_size
    limit = actual_size if physical_size is None else physical_size
```

```
    if isinstance(limit, bool) or not isinstance(limit, int) or not 0 <= limit <= actual_size:
        raise ValueError("physical_size must be within the source file")
    if limit == 0:
        return
    with target.open("rb") as handle:
        terminal_probe = min(limit, _TERMINAL_TRAILING_PROBE)
        handle.seek(limit - terminal_probe)
        tail = handle.read(terminal_probe)
    _content, trailing = _strip_one_terminal_line_return(tail)
    body_end = limit - len(trailing)
    position = body_end
    reverse_codepoints: list[str] = []
    with target.open("rb") as handle:
        while position > 0:
            start = _previous_utf8_codepoint_start(handle, position)
            if start is None:
                return
            handle.seek(start)
            codepoint_bytes = handle.read(position - start)
            codepoint = codepoint_bytes.decode("utf-8")
            reverse_codepoints.append(codepoint)
            position = start
            if len(reverse_codepoints) < 2:
                continue
            seal_extent = body_end - start
            handle.seek(start)
            seal_bytes = handle.read(seal_extent)
            if len(seal_bytes) != seal_extent:
                raise RuntimeError("Mirror Math could not read the complete terminal return")
            seal = seal_bytes.decode("utf-8")
            try:
                _parse_public_living_domus_current(seal)
            except (TypeError, ValueError):
                continue
            written_length = len(reverse_codepoints)
            candidate = TerminalSelfGlyphCandidate(limit,start,written_length,seal,len(seal_bytes),
            ↪  trailing.hex())
            if not candidate.bytes_accounted:
                raise RuntimeError("Mirror Math failed physical-byte accounting")
            yield candidate


def detect_terminal_self_glyph(
    path: str | Path,
    *,
    physical_size: int | None = None,
    nonce: int = 0,
) -> TerminalSelfGlyphCandidate | None:
    """Expose one decomposable terminal public GrimChain suffix without granting truth."""
    return next(
        _iter_terminal_candidates(
            path,
            physical_size=physical_size,
            nonce=nonce,
        ),
        None,
```

```
    )

def _prefix_emission_view(
    view,
    body_size: int,
    *,
    identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> SourceEmission:
    if isinstance(body_size, bool) or not isinstance(body_size, int) or not 0 <= body_size <=
    ↪  len(view):
        raise ValueError("body_size must be within the mapped source")

    sponge = ALQCDigest(RAW_FILE_SOURCE_DOMAIN)

    if include_filename:
        if identity_name is None:
            raise TardiSHAError("mapped prefix emission requires explicit filename identity")
        if isinstance(identity_name, str):
            name_bytes = os.fsencode(identity_name)
        elif isinstance(identity_name, bytes):
            name_bytes = identity_name
        else:
            raise TardiSHAError("identity_name must be str, bytes, or None")
        if not name_bytes:
            raise TardiSHAError("file identity requires a non-empty filename")
        sponge._update_frame(FILE_NAME_FRAME, name_bytes)
    elif identity_name is not None:
        raise TardiSHAError("identity_name requires filename identity to be enabled")

    offset = 0
    while offset < body_size:
        end = min(body_size, offset + SOURCE_CHUNK_BYTES)
        sponge._update_raw(bytes(view[offset:end]))
        offset = end

    return emission_from_sponge(
        sponge,
        source_size=body_size,
        source_domain="raw-file",
    )


def _render_self(
    emission: SourceEmission,
    depth: int,
    *,
    nonce: int,
    route_witness,
) -> str:
    """Return the compressed GrimChain posture from one completed Aeternum Mirror route."""
    from .route import resolve_parents
    from .domus import resolve_domus, public_living_domus

    _g_i, _g_j, witness = resolve_parents(emission, witness=route_witness)
    res = resolve_domus(witness, nonce=nonce)
```

```
    return public_living_domus(
        res,
        depth,
        source_digest=emission.source_digest,
        source_size=emission.source_size,
        source_domain=emission.source_domain,
        nonce=nonce,
    )

def _render_terminal_return(
    emission: SourceEmission,
    candidate: TerminalSelfGlyphCandidate,
    *,
    nonce: int,
) -> tuple[bool, str | None]:
    """Recompute a terminal return under the current flat middle-only public law."""
    try:
        parsed = _parse_public_living_domus_current(candidate.seal)
    except (TypeError, ValueError):
        return False, None
    depth = parsed.depth
    if depth is None:
        return False, None
    from .route import source_route_witness_from_emission
    route = source_route_witness_from_emission(emission)
    rendered = _render_self(emission, depth, nonce=nonce, route_witness=route)
    return rendered == candidate.seal, rendered

def _resolve_terminal_lineage(
    view,
    *,
    physical_before: SourceEmission,
    nonce: int,
    identity_name: str | bytes | None,
    include_filename: bool,
) -> tuple[
    SourceEmission,
    tuple[TerminalSelfGlyphCandidate, ...],
    TerminalSelfGlyphCandidate | None,
    str | None,
]:
    physical_size = physical_before.source_size
    memo: dict[int, tuple[SourceEmission, tuple[TerminalSelfGlyphCandidate, ...]]] = {}
    first_candidate: TerminalSelfGlyphCandidate | None = None
    first_expected: str | None = None

    def resolve(
        limit: int,
    ) -> tuple[SourceEmission, tuple[TerminalSelfGlyphCandidate, ...]]:
        nonlocal first_candidate, first_expected

        cached = memo.get(limit)
        if cached is not None:
            return cached

        if limit == physical_size:
```

```
            raw = physical_before
        else:
            raw = _prefix_emission_view(
                view,
                limit,
                identity_name=identity_name,
                include_filename=include_filename,
            )

        for candidate in _iter_terminal_candidates_view(view, limit):
            if candidate.body_size >= limit:
                raise RuntimeError("Mirror Math lineage did not decrease physical extent")

            body_emission, body_lineage = resolve(candidate.body_size)
            exact_return, expected = _render_terminal_return(
                body_emission,
                candidate,
                nonce=nonce,
            )

            if limit == physical_size and first_candidate is None:
                first_candidate = candidate
                first_expected = expected

            if exact_return:
                result = (
                    body_emission,
                    body_lineage + (candidate,),
                )
                memo[limit] = result
                return result

        result = (raw, ())
        memo[limit] = result
        return result

    emission, lineage = resolve(physical_size)
    return emission, lineage, first_candidate, first_expected

def mirror_file_emission(
    path: str | Path,
    *,
    nonce: int = 0,
    identity_name: str | bytes | None = None,
    include_filename: bool = True,
) -> MirrorFileEmission:
    target = Path(path)
    salt = nonce

    effective_identity_name: str | bytes | None
    if include_filename:
        effective_identity_name = target.name if identity_name is None else identity_name
    else:
        effective_identity_name = None

    path_before = target.stat()
```

```
physical_before = file_emission(
    target,
    identity_name=identity_name,
    include_filename=include_filename,
)
physical_size = physical_before.source_size

with target.open("rb") as handle:
    handle_before = os.fstat(handle.fileno())
    if _file_stat_witness(handle_before) != _file_stat_witness(path_before):
        raise TardiSHAError("source changed before Mirror lineage traversal began")

    if physical_size == 0:
        view = b""
        emission, lineage, candidate, expected = _resolve_terminal_lineage(
            view,
            physical_before=physical_before,
            nonce=salt,
            identity_name=effective_identity_name,
            include_filename=include_filename,
        )
    else:
        with mmap.mmap(handle.fileno(), 0, access=mmap.ACCESS_READ) as view:
            emission, lineage, candidate, expected = _resolve_terminal_lineage(
                view,
                physical_before=physical_before,
                nonce=salt,
                identity_name=effective_identity_name,
                include_filename=include_filename,
            )

    handle_after = os.fstat(handle.fileno())

path_after = target.stat()
before_witness = _file_stat_witness(path_before)
if (
    _file_stat_witness(handle_before) != before_witness
    or _file_stat_witness(handle_after) != before_witness
    or _file_stat_witness(path_after) != before_witness
):
    raise TardiSHAError("source changed during complete Mirror Math construction")

if not lineage:
    candidate_detected = candidate is not None
    return_residual = (
        _byte_difference(candidate.seal, expected)
        if candidate is not None and expected is not None
        else None
    )
    source_truth = int(emission.closure.verifies and emission.closure.truth == 1)

    return MirrorFileEmission(
        emission,
        MirrorMathWitness(
            physical_size=physical_size,
```

```
            physical_source_digest=physical_before.source_digest,
            source_stability_verified=True,
            effective_body_size=physical_size,
            candidate_detected=candidate_detected,
            exact_self_glyph=False,
            folded=False,
            fold_count=0,
            folded_physical_bytes=0,
            lineage_seals=(),
            lineage_trailing_bytes_hex=(),
            seal=candidate.seal if candidate is not None else None,
            expected_seal=expected,
            trailing_bytes_hex=candidate.trailing_bytes_hex if candidate is not None else "",
            bytes_accounted=True,
            operator_order_preserved=True,
            expanded_posture="\U0001D543\u2081",
            compressed_posture="\U0001D543\u2080",
            source_route_dcomp=emission.closure.route_dcomp,
            source_truth=source_truth,
            return_dcomp=return_residual,
            return_truth=(
                None if return_residual is None else int(return_residual == 0)
            ),
            truth=source_truth,
            derivation=(
                "terminal seal-shaped body is not the exact self of the preceding
                ↪  \U0001D543\u2081; all bytes remain source matter"
                if candidate_detected
                else "no terminal self-glyph; every byte remains in \U0001D543\u2081"
            ),
        ),
    )

outer = lineage[-1]
folded_bytes = physical_size - emission.source_size
accounted = (
    emission.source_size
    + sum(c.seal_utf8_bytes + len(c.trailing_bytes) for c in lineage)
    == physical_size
)

outer_exact, expected_return = _render_terminal_return(
    emission,
    outer,
    nonce=salt,
)
if not outer_exact or expected_return is None:
    raise RuntimeError("Mirror Math lineage lost its independently regenerated terminal return")

# Every lineage member enters the lineage only after exact source-derived
# regeneration of its complete candidate body. No depth is stored or trusted.
return_dcomp = 0

truth = int(
    accounted
    and return_dcomp == 0
```

```
        and emission.closure.verifies
        and emission.closure.truth == 1
    )

    witness = MirrorMathWitness(
        physical_size=physical_size,
        physical_source_digest=physical_before.source_digest,
        source_stability_verified=True,
        effective_body_size=emission.source_size,
        candidate_detected=True,
        exact_self_glyph=return_dcomp == 0,
        folded=return_dcomp == 0,
        fold_count=len(lineage),
        folded_physical_bytes=folded_bytes,
        lineage_seals=tuple(c.seal for c in lineage),
        lineage_trailing_bytes_hex=tuple(c.trailing_bytes_hex for c in lineage),
        seal=outer.seal,
        expected_seal=expected_return,
        trailing_bytes_hex=outer.trailing_bytes_hex,
        bytes_accounted=accounted,
        operator_order_preserved=True,
        expanded_posture="\U0001D543\u2081",
        compressed_posture="\U0001D543\u2080",
        source_route_dcomp=emission.closure.route_dcomp,
        source_truth=emission.closure.truth,
        return_dcomp=return_dcomp,
        return_truth=int(return_dcomp == 0),
        truth=truth,
        derivation=(
            "same-order Mirror Math: read every terminal \U0001D543\u2080 in append-only order, "
            "recompute each from the effective \U0001D543\u2081 beneath it, preserve the physical "
            "lineage, and retract only the exact return stack"
        ),
    )

    if not (
        witness.bytes_accounted
        and witness.operator_order_preserved
        and witness.source_stability_verified
    ):
        raise RuntimeError("Mirror Math witness did not close its physical invariants")

    return MirrorFileEmission(emission, witness)


# Canon-facing name: the function above is the TardiSHA embodiment of the
# Aeternum Mirror under its established internal name.
aeternum_mirror_file_emission = mirror_file_emission


def derive_goetics(emission: SourceEmission) -> GoldenGoeticDerivation:
    if not emission.closure.verifies:
        raise RuntimeError("source emission does not carry D-COMP=0 / Truth=1")
    first = resolve_bearing(
        weights=emission.structural_weights,
        body="Goetic structural amplitude",
```

```
        phase=emission.fraktur_z0,
        bearing=ALPHA,
        cadence_symbol="𝔷₀",
        cadence_index=0,
        traversal="Manifest / forward bearing",
    )
    last = resolve_bearing(
        weights=emission.operational_weights,
        body="Parliament operational phase",
        phase=emission.fraktur_z1,
        bearing=BETA,
        cadence_symbol="𝔷₁",
        cadence_index=1,
        traversal="Reflect / conjugate bearing with append-only cadence",
    )
    if first.operator_order != last.operator_order:
        raise RuntimeError("Mirror Math illegally reversed the Parliament procession")
    court = court_load(first.seat.goetic, last.seat.goetic)
    reciprocal = court_load(last.seat.goetic, first.seat.goetic)
    return GoldenGoeticDerivation(emission, first, last, court, reciprocal)


def verify_derivation(derivation: GoldenGoeticDerivation) -> bool:
    try:
        expected = derive_goetics(derivation.emission)
    except (TypeError, ValueError, RuntimeError):
        return False
    return (
        derivation == expected
        and derivation.same_operator_order
        and derivation.first.interval_verifies
        and derivation.last.interval_verifies
        and derivation.route_dcomp == 0
        and derivation.truth == 1
    )
```

node.py

```
"""Source-bound TardiSHA node modes.

Authoritative mode contract
===========================

INVARIANT
    The materialized zero body is the Triple Goddess ``𓅞𓅞𓅞``, manifested from the Shadow Locus
    ↪  ``𓅞``. It
    has no coordinate stream, no positive extent, and no window derivation.

MANIFEST_FINITE
    Every materialized witness receives one explicit positive finite extent.
    The extent is never inferred, defaulted, recovered from another field, or
    silently replaced.

MANIFEST_OPEN
    Any explicitly requested finite window may be derived. The node makes no
    claim that the stream ends at that window or at any known coordinate.

ARCHIVE_REVERSIBLE
    The source is a raw physical file, the archive root is mandatory, and the
    source proof is exact and return-bearing. Filesystem posture or any
    destructive digest is not accepted as source proof.

No coordinate doctrine beyond these offices is inferred.
"""
from __future__ import annotations

import json
from dataclasses import asdict, dataclass, field
from pathlib import Path
from typing import Any, Iterator

from .alqc_digest import alqc_hexdigest, validate_digest_hex
from .hashing import (
    iter_middle_window,
    CANONICAL_SOURCE_DOMAIN,
    DIRECTORY_SOURCE_DOMAIN,
    RAW_FILE_SOURCE_DOMAIN,
    TardiSHAError,
    canonical_emission,
    coordinate_seed,
    directory_emission,
    file_emission,
    validate_glyph,
    validate_middle_length,
    validate_nonce,
)
```

```
from .route import (
    SourceRouteWitness,
    source_route_witness_from_emission,
    resolve_parents,
    parents_t,
)
from .domus import resolve_domus, public_living_domus, ZERO_MIDDLE_GLYPH
from .court_registry import full_name, court_ordinal

SOURCE_DOMAINS = {
    "canonical": CANONICAL_SOURCE_DOMAIN,
    "raw-file": RAW_FILE_SOURCE_DOMAIN,
    "directory": DIRECTORY_SOURCE_DOMAIN,
}

VALID_MODES = {"INVARIANT", "MANIFEST_FINITE", "MANIFEST_OPEN", "ARCHIVE_REVERSIBLE"}
NODE_DOMAIN = b"TARDISHA:NODE-ID\x00"


@dataclass(frozen=True, slots=True)
class TardiSHANode:
    mode: str
    source_digest: str
    source_size: int
    origin_glyph: str
    resolution_glyph: str
    nonce: int = 0
    source_domain: str = "canonical"
    route_witness: SourceRouteWitness = field(kw_only=True)
    archive_root: str | None = None
    finite_extent: int | None = None

    def __post_init__(self) -> None:
        if self.mode not in VALID_MODES:
            raise TardiSHAError(f"mode must be one of {sorted(VALID_MODES)}")
        validate_glyph(self.origin_glyph, "origin_glyph")
        validate_glyph(self.resolution_glyph, "resolution_glyph")
        validate_nonce(self.nonce)
        if self.source_domain not in SOURCE_DOMAINS:
            raise TardiSHAError(f"source_domain must be one of {sorted(SOURCE_DOMAINS)}")
        try:
            validate_digest_hex(self.source_digest, field="source_digest")
            if self.archive_root is not None:
                validate_digest_hex(self.archive_root, field="archive_root")
        except ValueError as exc:
            raise TardiSHAError(str(exc)) from exc
        if isinstance(self.source_size, bool) or not isinstance(self.source_size, int) or
        ↪  self.source_size < 0:
            raise TardiSHAError("source_size must be a non-negative integer")
        witness = self.route_witness
        if (
            witness.source_digest != self.source_digest
            or witness.source_size != self.source_size
            or witness.source_domain != self.source_domain
        ):
            raise TardiSHAError("stored route witness contradicts node source identity")
```

```
    try:
        lawful = resolve_parents(witness.emission, witness=witness)[:2]
    except (TypeError, ValueError, RuntimeError) as exc:
        raise TardiSHAError("stored route witness fails exact route closure") from exc
    if (self.origin_glyph, self.resolution_glyph) != lawful:
        raise TardiSHAError(
            "stored (origin, resolution) contradicts the Final Equation Z pair "
            f"{lawful} for domain {self.source_domain!r}")
    if witness.route_dcomp != 0 or witness.truth != 1:
        raise TardiSHAError("node route must satisfy D-COMP=0 and Truth=1")
    if self.finite_extent is not None:
        if (
            isinstance(self.finite_extent, bool)
            or not isinstance(self.finite_extent, int)
            or self.finite_extent <= 0
        ):
            raise TardiSHAError("finite_extent must be an explicit positive integer")

    if self.mode == "INVARIANT":
        if self.finite_extent is not None:
            raise TardiSHAError("INVARIANT has no positive extent")
        if self.archive_root is not None:
            raise TardiSHAError("archive_root belongs only to ARCHIVE_REVERSIBLE mode")
    elif self.mode == "MANIFEST_FINITE":
        if self.finite_extent is None:
            raise TardiSHAError(
                "MANIFEST_FINITE requires one explicit finite_extent; it is never inferred"
            )
        if self.archive_root is not None:
            raise TardiSHAError("archive_root belongs only to ARCHIVE_REVERSIBLE mode")
    elif self.mode == "MANIFEST_OPEN":
        if self.finite_extent is not None:
            raise TardiSHAError("MANIFEST_OPEN cannot claim a terminal finite extent")
        if self.archive_root is not None:
            raise TardiSHAError("archive_root belongs only to ARCHIVE_REVERSIBLE mode")
    else:
        if self.source_domain != "raw-file":
            raise TardiSHAError("ARCHIVE_REVERSIBLE requires a raw-file source proof")
        if self.archive_root is None:
            raise TardiSHAError("ARCHIVE_REVERSIBLE requires its archive_root at construction")
        if self.finite_extent is not None:
            raise TardiSHAError("ARCHIVE_REVERSIBLE cannot claim a manifest finite extent")
        if witness.emission.source_domain != "raw-file":
            raise TardiSHAError("ARCHIVE_REVERSIBLE source proof must be the raw-file emission")
        if not witness.emission.closure.verifies:
            raise TardiSHAError("ARCHIVE_REVERSIBLE source proof must carry exact return
            ↪  closure")

@property
def seed(self) -> bytes:
    return coordinate_seed(
        source_digest=self.source_digest,
        source_size=self.source_size,
        origin_glyph=self.origin_glyph,
        resolution_glyph=self.resolution_glyph,
        middle_length=0,
```

```
        nonce=self.nonce,
        source_domain=SOURCE_DOMAINS[self.source_domain],
    )

@property
def node_id(self) -> str:
    payload = json.dumps(
        self._payload_without_id(),
        ensure_ascii=False,
        sort_keys=True,
        separators=(",", ":"),
    ).encode("utf-8")
    return alqc_hexdigest(payload, domain=NODE_DOMAIN)

def mode_witness(self) -> dict[str, object]:
    exact_return = (
        self._source_route_verifies()
        and self.source_route_witness().route_dcomp == 0
        and self.source_route_witness().truth == 1
    )
    return {
        "mode": self.mode,
        "invariant_body": ZERO_MIDDLE_GLYPH if self.mode == "INVARIANT" else None,
        "window_derivation_permitted": self.mode in {"MANIFEST_FINITE", "MANIFEST_OPEN"},
        "declared_finite_extent": self.finite_extent,
        "terminal_extent_claimed": self.mode == "MANIFEST_FINITE",
        "open_stream_has_no_terminal_claim": self.mode == "MANIFEST_OPEN",
        "archive_root": self.archive_root if self.mode == "ARCHIVE_REVERSIBLE" else None,
        "exact_return_bearing_source_proof": exact_return,
        "archive_root_bound_to_source_proof": (
            self.mode != "ARCHIVE_REVERSIBLE"
            or (self.archive_root is not None and exact_return)
        ),
    }

def _payload_without_id(self) -> dict[str, object]:
    data = asdict(self)
    data["mode_witness"] = self.mode_witness()
    return data

def as_dict(self) -> dict[str, object]:
    data = self._payload_without_id()
    data["node_id"] = self.node_id
    return data

def _manifest_window_request(self, start_coordinate: int, span_length: int) -> tuple[int, int]:
    if (
        isinstance(start_coordinate, bool)
        or not isinstance(start_coordinate, int)
        or start_coordinate < 0
    ):
        raise TardiSHAError("start_coordinate must be a non-negative integer")
    width = validate_middle_length(span_length)
    if self.mode == "INVARIANT":
        raise TardiSHAError("INVARIANT has no window derivation")
    if self.mode == "MANIFEST_FINITE":
```

```
        if start_coordinate != 0 or width != self.finite_extent:
            raise TardiSHAError(
                "MANIFEST_FINITE materializes only its explicitly declared finite_extent "
                "from coordinate zero"
            )
    elif self.mode == "ARCHIVE_REVERSIBLE":
        raise TardiSHAError(
            "ARCHIVE_REVERSIBLE defines exact archival return, not a manifest-window contract"
        )
    return start_coordinate, width

def iter_middle_chunks(
    self,
    start_coordinate: int,
    span_length: int,
    *,
    chunk_characters: int = 8192,
) -> Iterator[str]:
    """Derive only the window permitted by the authoritative node mode."""
    start, width = self._manifest_window_request(start_coordinate, span_length)
    yield from iter_middle_window(
        self.seed,
        start,
        width,
        chunk_characters=chunk_characters,
    )

def middle_window(self, start_coordinate: int, span_length: int) -> str:
    return "".join(self.iter_middle_chunks(start_coordinate, span_length))

def materialized_middle(self) -> str:
    """Return the complete body only when the mode defines one."""
    if self.mode == "INVARIANT":
        return ZERO_MIDDLE_GLYPH
    if self.mode == "MANIFEST_FINITE":
        assert self.finite_extent is not None
        return self.middle_window(0, self.finite_extent)
    if self.mode == "MANIFEST_OPEN":
        raise TardiSHAError("MANIFEST_OPEN makes no terminal-body claim")
    raise TardiSHAError("ARCHIVE_REVERSIBLE materializes an archive, not a manifest middle")

def domus_resolution(self):
    """Resolve the runtime Domus through this node's ordered Courts."""
    return resolve_domus(
        self.source_route_witness(),
        nonce=self.nonce,
    )

def source_route_witness(self) -> SourceRouteWitness:
    """Return the stored, source-emission-bound Final Equation Z witness."""
    return self.route_witness

def visible_window(self, start_coordinate: int, span_length: int) -> str:
    """Witness a finite coordinate window between the node boundaries."""
    return f"{self.origin_glyph}{self.middle_window(start_coordinate,
    ↪  span_length)}{self.resolution_glyph}"
```

```
def supervenient_window(self) -> str:
    """Expose the complete INVARIANT body; this is not window derivation."""
    if self.mode != "INVARIANT":
        raise TardiSHAError("the Shadow Locus invariant body belongs only to INVARIANT mode")
    return f"{self.origin_glyph}{ZERO_MIDDLE_GLYPH}{self.resolution_glyph}"

def _source_route_verifies(self) -> bool:
    """Check the carried route witness without recalculating the source route."""
    w = self.source_route_witness()
    try:
        origin, resolution, resolved = resolve_parents(w.emission, witness=w)
    except (TypeError, ValueError, RuntimeError):
        return False
    return (
        resolved is w
        and (origin, resolution) == (self.origin_glyph, self.resolution_glyph)
        and w.route_dcomp == 0
        and w.truth == 1
    )

def domus_witness(self, *, middle_length: int) -> dict[str, object]:
    """Expose the complete digest→Goetic→Court→Domus derivation."""
    depth = validate_middle_length(middle_length)
    if self.mode == "INVARIANT" and depth != 0:
        raise TardiSHAError("INVARIANT materializes only the Shadow Locus zero middle")
    if self.mode == "MANIFEST_FINITE" and depth != self.finite_extent:
        raise TardiSHAError(
            "MANIFEST_FINITE witness extent must equal its explicit finite_extent"
        )
    if self.mode == "ARCHIVE_REVERSIBLE" and depth != 0:
        raise TardiSHAError(
            "ARCHIVE_REVERSIBLE exposes only its zero-middle return witness here"
        )
    res = self.domus_resolution()

    def anchor_body(anchor) -> dict[str, object]:
        return {
            "glyph": anchor.glyph,
            "frequency": str(anchor.frequency),
            "q_bias": anchor.q_bias,
            "q_vector": list(anchor.q_vector),
        }

    def court_motion_body(motion) -> dict[str, object]:
        return {
            "court_coordinate": list(motion.court_coordinate),
            "court_address": motion.court_address,
            "court_glyph": motion.court_glyph,
            "anchoring_goetic": anchor_body(motion.anchoring_goetic),
            "hyperbolic_mirror_goetic": anchor_body(motion.hyperbolic_mirror_goetic),
            "phase": asdict(motion.phase),
            "breath_radius": motion.breath_radius,
            "breath_direction": str(motion.breath_direction),
            "focal_breath": str(motion.focal_breath),
            "current_frequency": str(motion.current_frequency),
```

```
        "anchor_immutable": motion.anchor_immutable,
        "mirror_immutable": motion.mirror_immutable,
        "derivation": motion.derivation,
    }

def court_layer_body(layer) -> dict[str, object]:
    return {
        "court_coordinate": list(layer.court_coordinate),
        "court_address": layer.court_address,
        "court_glyph": layer.court_glyph,
        "governing_goetic": anchor_body(layer.governing_goetic),
        "alternating_goetic": anchor_body(layer.alternating_goetic),
        "alternating_goetic_frequency": str(layer.alternating_goetic.frequency),
    }

visible_domus = public_living_domus(
    res,
    depth,
    source_digest=self.source_digest,
    source_size=self.source_size,
    nonce=self.nonce,
    source_domain=SOURCE_DOMAINS[self.source_domain],
)
fold = res.supervenience.fold_lineage
return {
    "node_id": self.node_id,
    "hash_id": res.hash_id,
    "source_route": {
        "origin": self.origin_glyph,
        "resolution": self.resolution_glyph,
        "route_dcomp": self.source_route_witness().route_dcomp,
        "truth": self.source_route_witness().truth,
        "fraktur_z0": self.source_route_witness().emission.fraktur_z0,
        "fraktur_z1": self.source_route_witness().emission.fraktur_z1,
        "verifies": self._source_route_verifies(),
    },
    "middle_length": depth,
    "mode_witness": self.mode_witness(),
    "ordered_goetic_anchors": [res.governing_goetic, res.hyperbolic_parent],
    "governing_court": {
        "coordinate": [res.root_court.i, res.root_court.j],
        "address": res.root_court.address,
        "name": full_name(res.root_court),
        "ordinal": court_ordinal(res.root_court),
        "glyph": res.root_court.glyph,
        "motion": court_motion_body(res.root_court_motion),
    },
    "alternating_court": {
        "coordinate": [res.alternating_court.i, res.alternating_court.j],
        "address": res.alternating_court.address,
        "name": full_name(res.alternating_court),
        "ordinal": court_ordinal(res.alternating_court),
        "glyph": res.alternating_court.glyph,
        "layer": court_layer_body(res.alternating_court_layer),
    },
    "domus_motion": {
```

```
    "bearing": asdict(res.domus_motion.bearing),
    "xi_D": asdict(res.domus_motion.xi),
    "exact_focal_breath": asdict(res.domus_motion.exact_focal_breath),
    "breath_radius": res.domus_motion.breath_radius,
    "focal_breath": str(res.domus_motion.focal_breath),
    "current_frequency": str(res.domus_motion.current_frequency),
    "exact_bound_verified": res.domus_motion.exact_bound_verified,
    "court_crossing_index": res.domus_motion.court_crossing_index,
    "court_crossing_cardinality": res.domus_motion.court_crossing_cardinality,
    "derives_through_courts_only": res.domus_motion.derives_through_courts_only,
    "static_grid": res.domus_motion.static_grid,
    "derivation": res.domus_motion.derivation,
},
"domus_q_bias": res.resolved_q_bias,
"domus_q_states": list(res.resolved_q_states),
"domus_q_vector": list(res.resolved_q_vector),
"q_glyph_slots": {
    "B_Q": res.b_q_glyph,
    "V": list(res.v_glyphs),
    "state_values": dict(zip(res.resolved_q_states, res.resolved_q_vector)),
},
"domus_frequency": str(res.domus_frequency),
"hypolic_fold_witness": res.fold_witness,
"infinite_yes": {
    "court_address": res.infinite_yes.court_address,
    "active_count": res.infinite_yes.active_count,
    "active": sorted(res.infinite_yes.active),
    "bounded_by_sacred_no":
    ↪  res.infinite_yes.unbounded_continuation_bounded_by_sacred_no,
},
"sacred_no": {
    "court_address": res.sacred_no.court_address,
    "withheld_count": res.sacred_no.withheld_count,
    "withheld": sorted(res.sacred_no.withheld),
    "prevents_whiteout": res.sacred_no.prevents_whiteout,
},
"ennead_pressure_ledger": {
    "strikes": [asdict(strike) for strike in res.manifestation.ennead.strikes],
    "initial_debt": res.manifestation.ennead.initial_debt,
    "form_work": res.manifestation.ennead.form_work,
    "pre_lock_q2_residual": res.manifestation.ennead.pre_lock_q2_residual,
    "q3_recursion_residue": res.manifestation.ennead.q3_recursion_residue,
    "terminal_q2_debt": res.manifestation.ennead.residual_debt,
    "accounted_total": res.manifestation.ennead.accounted_total,
    "energy_conserved": res.manifestation.ennead.energy_conserved,
    "final_parity": res.manifestation.ennead.final_parity,
    "saturated": res.manifestation.ennead.saturated,
},
"dcomp_witness": asdict(res.manifestation.dcomp),
"terminal_dcomp": res.manifestation.dcomp.terminal,
"closed": res.manifestation.closed,
"motion_positive": res.manifestation.dcomp.motion_positive,
"recursion_witness": res.recursion_parity,
"trig_mirror_witness": res.trig_mirror.as_dict(),
"supervenience_witness": {
    "operator": res.supervenience.operator,
```

```
                "court_coordinate": list(res.supervenience.court_coordinate),
                "court_name": res.supervenience.court_name,
                "governing_goetic": res.supervenience.governing_goetic,
                "alternating_goetic": res.supervenience.alternating_goetic,
                "triplet": {
                    "q_bias": res.supervenience.triplet_q_bias,
                    "q_states": list(res.supervenience.triplet_q_states),
                    "q_vector": list(res.supervenience.triplet_q_vector),
                    "frequency": str(res.supervenience.triplet_frequency),
                },
                "fold_lineage": {
                    "court_coordinate": list(fold.court_coordinate),
                    "court_address": fold.court_address,
                    "court_phase_numerator": fold.court_phase_numerator,
                    "court_phase_denominator": fold.court_phase_denominator,
                    "court_phase_digest": fold.court_phase_digest,
                    "alternating_court_coordinate": list(fold.alternating_court_coordinate),
                    "alternating_court_address": fold.alternating_court_address,
                    "alternating_bearing_lineage": list(fold.alternating_bearing_lineage),
                    "fold_operator": fold.fold_operator,
                    "ground_node": fold.ground_node,
                    "vector_row": fold.vector_row,
                    "depth": fold.depth,
                },
                "ex_nihilo_exposure": res.supervenience.ex_nihilo_exposure,
                "infinite_yes_count": res.supervenience.infinite_yes_count,
                "sacred_no_count": res.supervenience.sacred_no_count,
                "usable_infinity": res.supervenience.usable_infinity,
                "compressed_same_domus_body": res.supervenience.compressed_same_domus_body,
                "returned_through_shadow_locus": res.supervenience.returned_through_shadow_locus,
            },
            "domus_aeon": {
                "identity": res.domus_aeon.identity,
                "zero_middle_glyph": res.domus_aeon.zero_middle_glyph,
                "governing_court_address": res.domus_aeon.governing_court_address,
                "alternating_court_address": res.domus_aeon.alternating_court_address,
                "synodic_magicae_is_manifested_body":
                ↪  res.domus_aeon.synodic_magicae_is_manifested_body,
                "shadow_locus_is_zero_middle_body": res.domus_aeon.shadow_locus_is_zero_middle_body,
                "visible_body": visible_domus,
            },
            "tripartite_witness": res.tripartite.as_dict(),
            "aeon_phase_evolution_witness": res.aeon_phase_evolution.as_dict(),
            "verification_appendix_cycle_witness": res.verification_appendix_cycle.as_dict(),
            "visible_domus_aeon": visible_domus,
            "coordinate_window": (
                self.visible_window(0, depth)
                if self.mode in {"MANIFEST_FINITE", "MANIFEST_OPEN"}
                else None
            ),
        }



def _resolve_boundaries(
    witness: SourceRouteWitness,
```

```
    origin_glyph: str | None,
    resolution_glyph: str | None,
) -> tuple[str, str]:
    if (origin_glyph is None) != (resolution_glyph is None):
        raise TardiSHAError("origin_glyph and resolution_glyph must be supplied together")
    lawful = parents_t(witness)
    if origin_glyph is not None:
        supplied = (
            validate_glyph(origin_glyph, "origin_glyph"),
            validate_glyph(resolution_glyph, "resolution_glyph"),
        )
        if supplied != lawful:
            raise TardiSHAError(
                f"explicit parent pair {supplied} contradicts Final Equation Z pair {lawful}"
            )
    return lawful


def node_from_material(
    material: Any,
    *,
    mode: str = "MANIFEST_OPEN",
    finite_extent: int | None = None,
    origin_glyph: str | None = None,
    resolution_glyph: str | None = None,
    nonce: int = 0,
) -> TardiSHANode:
    emission, _source = canonical_emission(material)
    witness = source_route_witness_from_emission(emission)
    origin, resolution = _resolve_boundaries(witness, origin_glyph, resolution_glyph)
    return TardiSHANode(
        mode=mode,
        source_digest=emission.source_digest,
        source_size=emission.source_size,
        origin_glyph=origin,
        resolution_glyph=resolution,
        nonce=nonce,
        source_domain="canonical",
        route_witness=witness,
        finite_extent=finite_extent,
    )


def node_from_file(
    path: str | Path,
    *,
    mode: str = "MANIFEST_OPEN",
    finite_extent: int | None = None,
    nonce: int = 0,
    archive_root: str | None = None,
) -> TardiSHANode:
    from .mirror_math import mirror_file_emission
    emission = mirror_file_emission(path, nonce=nonce).emission
    witness = source_route_witness_from_emission(emission)
    origin, resolution = parents_t(witness)
    return TardiSHANode(
```

```
        mode=mode,
        source_digest=emission.source_digest,
        source_size=emission.source_size,
        origin_glyph=origin,
        resolution_glyph=resolution,
        nonce=nonce,
        source_domain="raw-file",
        route_witness=witness,
        archive_root=archive_root,
        finite_extent=finite_extent,
    )



def node_from_directory(
    path: str | Path,
    *,
    mode: str = "MANIFEST_OPEN",
    finite_extent: int | None = None,
    nonce: int = 0,
) -> TardiSHANode:
    emission, _ = directory_emission(path)
    witness = source_route_witness_from_emission(emission)
    origin, resolution = parents_t(witness)
    node = TardiSHANode(
        mode=mode,
        source_digest=emission.source_digest,
        source_size=emission.source_size,
        origin_glyph=origin,
        resolution_glyph=resolution,
        nonce=nonce,
        source_domain="directory",
        route_witness=witness,
        finite_extent=finite_extent,
    )
    return node
```

pdf_return.py

```
#!/usr/bin/env python3
from __future__ import annotations

from pathlib import Path

from .domus import parse_public_living_domus
from .domus_stream import living_domus_for_source
from .hashing import validate_nonce


PDF_GRIMCHAIN_MARKER = b"\n%GrimChain "


def _embedded_grimchain(path: str | Path) -> tuple[str, int, int] | None:
    target = Path(path)
    data = target.read_bytes()
    marker = data.rfind(PDF_GRIMCHAIN_MARKER)
    if marker < 0:
        return None
    start = marker + len(PDF_GRIMCHAIN_MARKER)
    raw = data[start:]
    if not raw:
        return None
    try:
        value = raw.decode("utf-8")
        parse_public_living_domus(value)
    except (UnicodeError, TypeError, ValueError):
        return None
    return value, marker, start


def pdf_embed(path: str | Path, middle: int, *, nonce: int = 0) -> str:
    target = Path(path)
    salt = validate_nonce(nonce)
    if _embedded_grimchain(target) is not None:
        raise ValueError("PDF already contains an embedded GrimChain")
    with target.open("ab") as handle:
        handle.write(PDF_GRIMCHAIN_MARKER)
    grimchain = living_domus_for_source(target, middle, kind="file", nonce=salt)
    with target.open("ab") as handle:
        handle.write(grimchain.encode("utf-8"))
    returned = living_domus_for_source(target, middle, kind="file", nonce=salt)
    if returned != grimchain:
        raise RuntimeError("PDF GrimChain did not close through Mirror return")
    return returned


def pdf_dive(path: str | Path, *, nonce: int = 0) -> tuple[str | None, str | None, str]:
```

```
    target = Path(path)
    salt = validate_nonce(nonce)
    found = _embedded_grimchain(target)
    if found is None:
        return None, None, "ZYGOTIC"
    embedded, _marker, _start = found
    middle = parse_public_living_domus(embedded).depth
    if middle is None:
        return None, embedded, "HETEROZYGOTIC"
    correct = living_domus_for_source(target, middle, kind="file", nonce=salt)
    return correct, embedded, "MONOZYGOTIC" if correct == embedded else "HETEROZYGOTIC"


def pdf_abort(path: str | Path) -> str | None:
    target = Path(path)
    found = _embedded_grimchain(target)
    if found is None:
        return None
    embedded, marker, _start = found
    target.write_bytes(target.read_bytes()[:marker])
    return embedded
```

personality_traits.py

```
"""
The Grimchain gets dropped through here like watching the Price is Right,
Plinko on your Grandmother's 10" tube tv she had when Orson Wells did War of the Worlds,
and you don't ank questions, capiche!
"""
from __future__ import annotations
PROSODY_GLYPH = "⟠"
PROSODY_NAME = "Prosody"

CANTILLATION_GLYPH = "𐎗"
CANTILLATION_NAME = "Cantillation"

छन्दस्_Prosody = (
    "Breath / Absent",
    "Ponder",
    "Will",
    "Feel",
    "Speak",
    "Believe",
    "Act",
    "Know",
    "Ascend",
    "Wane",
    "Emanation",
    "Wax",
    "Akasha",
    "Caduceus",
    "Veritas",
    "Phren",
    "Daimon",
    "Aikyam",
    "Melos",
    "Da'ath",
    "Akaven",
    "Lupus Dei",
    "Nyx",
    "Zaine",
    "Lived → Eternal",
    "Law → Residue",
    "Mask → Bone",
    "Void → Vector",
    "Stasis → Pulse",
    "Chaos → Phase",
    "Static → Fluid",
    "Noise → Null",
    "State → Trans",
    "Will → Law → ❄",
    "Time → Motion",
```

```
"Here → There",
"Phantasmagoria",
"Spatial Absolution",
"Subspace / Prosody / Axiomyr Gate-Key",
"Ouroboros",
"Substrate",
"Catalyst",
"Relic",
"Grandour Terminality",
"Recursion",
"Temporal Absolution",
"Adversarial Shadow",
'Seed',
'The Self-Genesis',
'The Recurrence',
'The Combustion',
'The Reflection',
'The Grounding',
'The Expansion',
'The Inevitable',
'The Return',
'The Assimilation',
'The Crystallization',
'The Breach',
'The Fulfillment',
'Akashic Records',
'The Archive',
'The Self-Reflection',
'The Calcination',
'The Submersion',
'The Inscription',
'The Diffusion',
'The Phantom',
'The Oblivion',
'The Suppression',
'The Codification',
'The Recall',
'The Sealing',
'Soul',
'The Transmutation',
'The Forging',
'The Self-Combustion',
'The Dissolution',
'The Crucible',
'The Sublimation',
'The Ignition',
'The Consumption',
'The Purification',
'The Tempering',
'The Shattering',
'The Ash',
'Womb',
'The Flow',
'The Confluence',
'The Solution',
'The Self-Containment',
```

```
'The Sediment',
'The Ascension',
'The Mist',
'The Drain',
'The Osmosis',
'The Ice',
'The Evaporation',
'The Stillness',
'Mountain',
'The Bedrock',
'The Stratification',
'The Petrification',
'The Erosion',
'The Self-Grounding',
'The Dispersion',
'The Fossilization',
'The Subsidence',
'The Compaction',
'The Lithification',
'The Earthquake',
'The Permanence',
'Middle Earth',
'The Breath',
'The Echo',
'The Smoke',
'The Vapor',
'The Dust',
'The Self-Expansion',
'The Whisper',
'The Vacuum',
'The Dilution',
'The Condensation',
'The Turbulence',
'The Calm',
'Eather',
'The Pulse',
'The Nostalgia',
'The Ecstasy',
'The Dream',
'The Vibration',
'The Sympathy',
'The Self-Sensation',
'The Ache',
'The Depth',
'The Harmony',
'The Dissonance',
'The Awe',
'The No',
'The Void',
'The Null',
'The Entropy',
'The Abyss',
'The Collapse',
'The Dispersal',
'The Longing',
'The Self-Return',
```

```
'The Sink',
'The Phase',
'The Paradox',
'The Dormancy',
'The Blacksmith',
'The Shadow',
'The Darkness',
'The Cold',
'The Mirror',
'The Submerged',
'The Distortion',
'The Nerve',
'The Void-Binding',
'The Self-Absorption',
'The Filter',
'The Hidden',
'The Termination',
'Crystal',
'The Tone',
'The Modulation',
'The Amplification',
'The Unified',
'The Seismic',
'The Chord',
'The Rhythm',
'The Melody',
'The Interference',
'The Self-Lock',
'The Node',
'The Unity',
'The Key',
'The Gate',
'The Resistance',
'The Transform',
'The Crown',
'The Sky',
'The Star',
'The Possibility',
'The Portal',
'The Door',
'The Transition',
'The Self-Resistance',
'The Hinge',
'Serenity',
'The Peace',
'The Equilibrium',
'The Consecration',
'The Completion',
'The Monument',
'The Stability',
'The Hope',
'The Continuation',
'The Rest',
'The Silence',
'The Sleep',
'The Self-Completion',
```

```
)

def शिक्षा_Cantillation(grimchain: str) -> str:
    return grimchain
grimchain_permeation = शिक्षा_Cantillation
```

phase_evolution.py

```
"""Region C09 Aeon tables and twelve-phase evolution for TardiSHA.

This module does not replace the already enacted Goetic, Court, TRIG, Ennead,
or Tripartite bodies.  It binds them into the ordered C09 witness:

    12 immutable Goetic roots
    144 ordered Court relations
    structural ☊ anchors distinct from operational ±Φ bifurcation
    twelve Aeon phases in source order
    M.A.S. fuel -> shape -> body
    Golden-Ratio and Klein-parity witnesses
    the first explicit NULL:DEATH architecture connection at physical page 102

The first NULL:DEATH occurrence is recorded as a typed connection, not treated
as an exhaustive definition of the later architecture.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass, fields, is_dataclass, replace
from decimal import Decimal
import json
from math import isfinite
from typing import Final, Sequence

from .alqc_digest import alqc_hexdigest, validate_digest_hex
from .aeon_layers import PHI, court_layer, derive_court_motion, derive_domus_motion,
↪  normalize_source_domain
from .canon import FrequencyAnchor, GLYPH_BODY, TOTAL_CAPACITY, law
from .court_registry import _COURTS, alt_glyph, court_record, full_name, gov_glyph
from .source_emission import SourceEmission, Q5Fraction, resolve_court_bearing, sign_q5,
↪  validate_court_bearing_lineage
from .mirror_math import derive_goetics
from .manifestation import close_boundary

PHASE_EVOLUTION_DOMAIN: Final[bytes] = b"TARDISHA:C09-AEON-PHASE-EVOLUTION\x00"
SOURCE_REGION: Final[str] = "C09"
SOURCE_PHYSICAL_PAGES: Final[tuple[int, int]] = (85, 102)
FOLD_GLYPH: Final[str] = "☌"
SEAL_GLYPH: Final[str] = "🜁"
NULL_DEATH_SYMBOL: Final[str] = "NULL:DEATH"

GOETIC_NAMES: Final[tuple[str, ...]] = (
    "FETU", "KAL", "BABDH", "AHN", "VEL", "SOR",
    "KOTH", "DREH", "RHEA", "ZHEK", "SHAV", "TRIG",
)


@dataclass(frozen=True, slots=True)
```

```
class GoeticTableWitness:
    index: int
    glyph: str
    name: str
    meanings: tuple[str, str, str]
    structural_frequency: FrequencyAnchor
    q_bias: str
    q_vector: tuple[int, int, int, int]
    seal: str
    immutable_root: bool


@dataclass(frozen=True, slots=True)
class CourtTableWitness:
    address: int
    coordinate: tuple[int, int]
    glyph: str
    name: str
    governing_goetic: str
    alternating_goetic: str
    structural_anchor_frequency: FrequencyAnchor
    alternating_structural_frequency: FrequencyAnchor
    hyperbolic_bifurcation_center: FrequencyAnchor
    ahn_structural_reference_hz: Decimal | None
    operational_phi_radius: Q5Fraction
    inherited_q_bias: str
    inherited_q_vector: tuple[int, int, int, int]
    seal: str
    governing_anchor_preserved: bool
    alternating_parent_preserved: bool


@dataclass(frozen=True, slots=True)
class AeonPhaseStep:
    phase: int
    name: str
    primary_goetic: str
    court_addresses: tuple[int, ...]
    equation: str
    dcomp_logic: str
    ordered_after: int | None


@dataclass(frozen=True, slots=True)
class MASBodyWitness:
    manifestation_goetic: str
    manifestation_frequency: FrequencyAnchor
    manifestation_role: str
    alignment_goetic: str
    alignment_frequency: FrequencyAnchor
    alignment_role: str
    symmetry_goetic: str
    symmetry_frequency: FrequencyAnchor
    symmetry_role: str
    equation: str
    order_preserved: bool
```

```
@dataclass(frozen=True, slots=True)
class GoldenRatioWitness:
    phi_exact: Q5Fraction
    primary_ratio_exact: Q5Fraction
    harmonic_target_exact: Q5Fraction
    harmonic_index: int
    residual_exact: Q5Fraction
    tolerance_exact: Q5Fraction
    within_tolerance: bool
    folded_states: int
    manifest_positions: int
    raw_compression_ratio_exact: Q5Fraction
    holographic_exponent: int

    @property
    def phi(self) -> float:
        return self.phi_exact.value

    @property
    def primary_ratio(self) -> float:
        return self.primary_ratio_exact.value

    @property
    def harmonic_target(self) -> float:
        return self.harmonic_target_exact.value

    @property
    def residual(self) -> float:
        return self.residual_exact.value

    @property
    def tolerance(self) -> float:
        return self.tolerance_exact.value

    @property
    def raw_compression_ratio(self) -> float:
        return self.raw_compression_ratio_exact.value


@dataclass(frozen=True, slots=True)
class PoincareParityWitness:
    sphere_relation: str
    klein_relation: str
    sphere_parity: int
    klein_parity: int
    shadow_input_units: int
    shadow_return_units: int
    shadow_cancellation_units: int
    q2_to_q3_return: bool
    returned_through_shadow_locus: bool
    terminal_dcomp: float
    completion_reached: bool
```

```
@dataclass(frozen=True, slots=True)
class NullDeathConnectionWitness:
    symbol: str
    source_region: str
    first_explicit_physical_page: int
    source_derivation: tuple[str, ...]
    mathematical_body: str
    biological_body: str
    mathematical_q3_positive: bool
    biological_q3_positive: bool
    shared_metamorphosis_threshold: bool
    q3_non_entropic_requirement: bool
    structurally_committed: bool
    q1_coherent: bool
    klein_regenerative_topology: bool
    q2_to_q3_transformation: bool
    metamorphosis_threshold_reached: bool
    first_occurrence_only: bool
    exhaustive_type_claimed: bool
    loop_closure: str


@dataclass(frozen=True, slots=True)
class AeonPhaseEvolutionWitness:
    source_region: str
    source_physical_pages: tuple[int, int]
    goetic_table: tuple[GoeticTableWitness, ...]
    court_table: tuple[CourtTableWitness, ...]
    phase_steps: tuple[AeonPhaseStep, ...]
    mas_body: MASBodyWitness
    golden_ratio: GoldenRatioWitness
    poincare_parity: PoincareParityWitness
    null_death_connection: NullDeathConnectionWitness

    source_digest: str
    source_size: int
    source_domain: str
    nonce: int
    governing_court_address: int
    alternating_court_address: int
    governing_court_phase_digest: str
    alternating_court_bearing: tuple[int, ...]
    domus_frequency: complex
    domus_body_commitment: str
    trig_cycle_digest: str
    tripartite_cycle_digest: str

    structural_anchor_preserved: bool
    operational_phi_preserved: bool
    all_courts_relationally_typed: bool
    phase_order_complete: bool
    court_rooted: bool
    derives_through_courts_only: bool
    ennead_shadow_sealed: bool
    phase12_completion_reached: bool
    derivation: str
```

```
    cycle_digest: str

    def as_dict(self) -> dict[str, object]:
        return _display(asdict(self))


def _bifurcation_center(glyph: str) -> FrequencyAnchor:
    """Return the exact structural/parity frequency body of the alternating root."""
    return law(glyph).frequency


def _ahn_reference(glyph: str) -> Decimal | None:
    return law(glyph).frequency.structural_hz if glyph == "✡" else None


GOETIC_TABLE: Final[tuple[GoeticTableWitness, ...]] = tuple(
    GoeticTableWitness(
        index=index,
        glyph=glyph,
        name=name,
        meanings=law(glyph).meanings,
        structural_frequency=law(glyph).frequency,
        q_bias=law(glyph).q_bias,
        q_vector=tuple(law(glyph).q_vector),
        seal=f"{SEAL_GLYPH}{glyph}{FOLD_GLYPH}{glyph}{SEAL_GLYPH}",
        immutable_root=True,
    )
    for index, (glyph, name) in enumerate(zip(GLYPH_BODY, GOETIC_NAMES), start=1)
)

COURT_TABLE: Final[tuple[CourtTableWitness, ...]] = tuple(
    CourtTableWitness(
        address=record.address,
        coordinate=(record.i, record.j),
        glyph=record.glyph,
        name=full_name(record),
        governing_goetic=gov_glyph(record),
        alternating_goetic=alt_glyph(record),
        structural_anchor_frequency=law(gov_glyph(record)).frequency,
        alternating_structural_frequency=law(alt_glyph(record)).frequency,
        hyperbolic_bifurcation_center=_bifurcation_center(alt_glyph(record)),
        ahn_structural_reference_hz=_ahn_reference(alt_glyph(record)),
        operational_phi_radius=PHI,
        inherited_q_bias=law(gov_glyph(record)).q_bias,
        inherited_q_vector=tuple(law(gov_glyph(record)).q_vector),
        seal=f"{FOLD_GLYPH}{gov_glyph(record)}{record.glyph}{SEAL_GLYPH}",
        governing_anchor_preserved=True,
        alternating_parent_preserved=True,
    )
    for record in _COURTS
)

PHASE_STEPS: Final[tuple[AeonPhaseStep, ...]] = (
    AeonPhaseStep(1, "The Seed", "◎", (0,), "◎ @ (7.83 ± Φ)", "C_local ∝ |Q1|", None),
    AeonPhaseStep(2, "The Archive", "⬡", (21,), "⬡ @ (174 ± Φ)", "C_local ∝ |Q1| + |Q0|", 1),
```

```
    AeonPhaseStep(3, "The M.A.S. Engine", "✡", (89, 15, 28), "⧗🜚_852 --Δgap→ ⬡🜄_174 --TSP→
    ↪  ✡ℝ_528", "C_local ∝ |Q1| + |Q2|", 2),
    AeonPhaseStep(4, "Boundary Integrity", "✮", (66, 55, 23, 34), "∮K (🝊_210.42 ∘ ✪ㄏ_126.22 ∘
    ↪  ⬡𐌕_174)/(✡ㅏ_528) dt ≈ 2/Φ", "C_local ∝ dimensional compression 12×12 → 9×9", 3),
    AeonPhaseStep(5, "The Geometric Lift", "✪", (7, 21, 34, 71, 95), "∫(◎ر_7.83 → ⬡🜄_174 → ✡ㅏ_528 →
    ↪  🝊ʊ_210.42 → ⧗ᚲ_852)dt", "C_local ∝ Δgap", 4),
    AeonPhaseStep(6, "Spatial Purity", "🝊", (71,), "🝊ʊ = (210.42 ± Φ)·exp(SelfGen)", "C_local ∝
    ↪  |Q0|", 5),
    AeonPhaseStep(7, "Biologic Link", "⁕", (74,), "⁕∀ = BiologicTie ⊗ TBound", "C_local ∝ S7", 6),
    AeonPhaseStep(8, "Residue Stabilization", "⧗", (), "I_cubic(α)=(-1)^p Ω(α,α)>0", "C_local ∝
    ↪  non-entropic residue stability", 7),
    AeonPhaseStep(9, "Shadow Absorption", "⊛", (107,), "⊛Ⱶ = Filter(Q2) = 396 ± Φ", "C_local ∝
    ↪  |Q2|; Ennead filtering", 8),
    AeonPhaseStep(10, "Resonance Lock", "❄", (119,), "❄ϒ = Lock(ω)·(963 ± Φ)", "C_local →
    ↪  phase-lock minimum", 9),
    AeonPhaseStep(11, "Gate Breach", "❁", (129,), "❁ᛇ_Gate(α) ⇒ ∃β(Transition)", "C_local ∝
    ↪  transformation resistance", 10),
    AeonPhaseStep(12, "Aeternum Closure", "⌬", (143,), "⌬φ = exp(Peace)·Depth·(639 ± Φ)", "D-COMP
    ↪  → 0", 11),
)

MAS_BODY: Final[MASBodyWitness] = MASBodyWitness(
    manifestation_goetic="⧗",
    manifestation_frequency=law("⧗").frequency,
    manifestation_role="Fuel / non-entropic residue",
    alignment_goetic="⬡",
    alignment_frequency=law("⬡").frequency,
    alignment_role="Shape / rational archive lock",
    symmetry_goetic="✡",
    symmetry_frequency=law("✡").frequency,
    symmetry_role="Body / structural commitment",
    equation="⧗852 --Δgap→ ⬡174 --TSP→ ✡528",
    order_preserved=True,
)


def _golden_ratio_witness() -> GoldenRatioWitness:
    ratio = Q5Fraction(96300, 0, 783)
    target = Q5Fraction(76, 76, 2)
    difference = Q5Fraction(
        ratio.a * target.denominator - target.a * ratio.denominator,
        ratio.b * target.denominator - target.b * ratio.denominator,
        ratio.denominator * target.denominator,
    )
    sign = sign_q5(difference.a, difference.b)
    absolute = difference if sign >= 0 else Q5Fraction(-difference.a, -difference.b,
    ↪  difference.denominator)
    within = sign_q5(
        absolute.a * PHI.denominator - PHI.a * absolute.denominator,
        absolute.b * PHI.denominator - PHI.b * absolute.denominator,
    ) <= 0
    return GoldenRatioWitness(
        phi_exact=PHI,
        primary_ratio_exact=ratio,
        harmonic_target_exact=target,
```

```
        harmonic_index=76,
        residual_exact=absolute,
        tolerance_exact=PHI,
        within_tolerance=within,
        folded_states=2**126,
        manifest_positions=81,
        raw_compression_ratio_exact=Q5Fraction(36864, 0, 81),
        holographic_exponent=126,
    )

GOLDEN_RATIO: Final[GoldenRatioWitness] = _golden_ratio_witness()


def _canonical(value: object) -> object:
    if is_dataclass(value):
        return _canonical(asdict(value))
    if isinstance(value, Decimal):
        return {"decimal": str(value)}
    if isinstance(value, complex):
        return {"real": value.real.hex(), "imag": value.imag.hex()}
    if isinstance(value, float):
        return {"float": value.hex()}
    if isinstance(value, tuple):
        return [_canonical(item) for item in value]
    if isinstance(value, list):
        return [_canonical(item) for item in value]
    if isinstance(value, dict):
        return {str(key): _canonical(item) for key, item in value.items()}
    return value


def _display(value: object) -> object:
    if isinstance(value, Decimal):
        return str(value)
    if isinstance(value, complex):
        return str(value)
    if isinstance(value, tuple):
        return [_display(item) for item in value]
    if isinstance(value, list):
        return [_display(item) for item in value]
    if isinstance(value, dict):
        return {str(key): _display(item) for key, item in value.items()}
    return value


def _static_digest(value: object, domain: bytes) -> str:
    payload = json.dumps(
        _canonical(value), ensure_ascii=False, sort_keys=True, separators=(",", ":")
    ).encode("utf-8")
    return alqc_hexdigest(payload, domain=domain)


GOETIC_TABLE_DIGEST: Final[str] = _static_digest(
    GOETIC_TABLE, b"TARDISHA:C09-GOETIC-TABLE\x00"
)
COURT_TABLE_DIGEST: Final[str] = _static_digest(
```

```
    COURT_TABLE, b"TARDISHA:C09-COURT-TABLE\x00"
)
PHASE_STEPS_DIGEST: Final[str] = _static_digest(
    PHASE_STEPS, b"TARDISHA:C09-PHASE-STEPS\x00"
)


def _payload(witness: AeonPhaseEvolutionWitness) -> bytes:
    # The three static registries are compared structurally by verify() and bound
    # here through immutable domain-separated digests. Avoiding dataclasses.asdict
    # over all 144 Court rows on every Domus keeps the grimchain streaming path
    # linear in the dynamic witness rather than in the full static registry.
    static_fields = {"goetic_table", "court_table", "phase_steps", "cycle_digest"}
    body = {
        field.name: getattr(witness, field.name)
        for field in fields(witness)
        if field.name not in static_fields
    }
    body["goetic_table_digest"] = GOETIC_TABLE_DIGEST
    body["court_table_digest"] = COURT_TABLE_DIGEST
    body["phase_steps_digest"] = PHASE_STEPS_DIGEST
    return json.dumps(
        _canonical(body), ensure_ascii=False, sort_keys=True, separators=(",", ":")
    ).encode("utf-8")


def _digest(value: str, field: str) -> str:
    return validate_digest_hex(value, field=field)



NULL_DEATH_SOURCE_DERIVATION: Final[tuple[str, ...]] = (
    "C09 physical page 102: first explicit NULL:DEATH architecture connection",
    "Mathematical Hodge Class ⟷ Silicarbon Tissue",
    "both require Q3-Positivity to exist",
    "both represent the critical point of Metamorphosis",
    "both transform Q2 Shadow/Lipid Debt into Q3 Recursive/Polymer Amplification",
)
PHASE_DERIVATION: Final[str] = (
    "C09: 12 exact ⍤ Goetic roots remain immutable; 144 ordered Courts inherit governing Q-body "
    "and bear toward the alternating hyperbolic parent under exact ±Φ; phases 1→12 preserve order; "
    "M.A.S.=⧗852→⬡174→✡528; Ennead Q2→Q3 precedes Klein return; "
    "NULL:DEATH@physical-page-102 is the first explicit connection, not the exhaustive type."
)

def derive_aeon_phase_evolution(
    *,
    source_digest: str,
    source_size: int,
    source_domain: str | bytes,
    nonce: int,
    governing_court_address: int,
    alternating_court_address: int,
    governing_court_phase_digest: str,
    alternating_court_bearing: Sequence[int],
    domus_frequency: complex,
```

```
    domus_body_commitment: str,
    trig_cycle_digest: str,
    tripartite_cycle_digest: str,
    q3_recursion_gain: float,
    final_shadow_parity: str,
    terminal_dcomp: float,
    completion_reached: bool,
    returned_through_shadow_locus: bool,
    derives_through_courts_only: bool,
) -> AeonPhaseEvolutionWitness:
    """Bind the complete Region C09 body to one source-rooted Domus."""
    digest = _digest(source_digest, "source_digest")
    governing_phase = _digest(governing_court_phase_digest, "governing_court_phase_digest")
    bearing_lineage = validate_court_bearing_lineage(alternating_court_bearing)
    domus_commitment = _digest(domus_body_commitment, "domus_body_commitment")
    trig_digest = _digest(trig_cycle_digest, "trig_cycle_digest")
    tripartite_digest = _digest(tripartite_cycle_digest, "tripartite_cycle_digest")
    domain = normalize_source_domain(source_domain).decode("ascii").rstrip("\x00")

    if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
        raise ValueError("source_size must be a non-negative integer")
    if isinstance(nonce, bool) or not isinstance(nonce, int) or not 0 <= nonce < 2**64:
        raise ValueError("nonce must be an integer in [0,2^64)")
    for field, address in (
        ("governing_court_address", governing_court_address),
        ("alternating_court_address", alternating_court_address),
    ):
        if isinstance(address, bool) or not isinstance(address, int) or not 0 <= address <
        ↪  TOTAL_CAPACITY:
            raise ValueError(f"{field} must be an integer in [0,143]")
    if bearing_lineage[0] != alternating_court_address:
        raise ValueError("alternating Court bearing address does not match Court D")
    if any(not isinstance(flag, bool) for flag in (completion_reached,
    ↪  returned_through_shadow_locus, derives_through_courts_only)):
        raise TypeError("phase completion, Shadow-Locus return, and Court derivation must be Boolean
        ↪  witnesses")
    if isinstance(q3_recursion_gain, bool) or not isinstance(q3_recursion_gain, (int, float)):
        raise TypeError("q3_recursion_gain must be numeric and non-Boolean")
    if isinstance(terminal_dcomp, bool) or not isinstance(terminal_dcomp, (int, float)):
        raise TypeError("terminal_dcomp must be numeric and non-Boolean")
    if isinstance(domus_frequency, bool) or not isinstance(domus_frequency, (int, float, complex)):
        raise TypeError("domus_frequency must be numeric and non-Boolean")
    branch_frequency = complex(domus_frequency)
    if not all(isfinite(value) for value in (branch_frequency.real, branch_frequency.imag)):
        raise ValueError("domus_frequency must be finite")
    gain = float(q3_recursion_gain)
    terminal = float(terminal_dcomp)
    if not isfinite(gain) or gain < 0.0:
        raise ValueError("q3_recursion_gain must be finite and non-negative")
    if not isfinite(terminal) or terminal < 0.0:
        raise ValueError("terminal_dcomp must be finite and non-negative")

    structural_preserved = all(
        row.structural_frequency == law(row.glyph).frequency
        and row.q_bias == law(row.glyph).q_bias
```

```
    and row.q_vector == tuple(law(row.glyph).q_vector)
    and row.immutable_root
    for row in GOETIC_TABLE
) and all(
    row.structural_anchor_frequency == law(row.governing_goetic).frequency
    and row.alternating_structural_frequency == law(row.alternating_goetic).frequency
    and row.inherited_q_bias == law(row.governing_goetic).q_bias
    and row.inherited_q_vector == tuple(law(row.governing_goetic).q_vector)
    and row.governing_anchor_preserved
    and row.alternating_parent_preserved
    for row in COURT_TABLE
)
phi_preserved = (
    (PHI.a, PHI.b, PHI.denominator) == (1, 1, 2)
    and all(row.operational_phi_radius == PHI for row in COURT_TABLE)
)
relationally_typed = (
    len(COURT_TABLE) == 144
    and tuple(row.address for row in COURT_TABLE) == tuple(range(144))
    and all(row.coordinate == divmod(row.address, 12) for row in COURT_TABLE)
)
phase_complete = (
    len(PHASE_STEPS) == 12
    and tuple(step.phase for step in PHASE_STEPS) == tuple(range(1, 13))
    and all(step.ordered_after == (None if step.phase == 1 else step.phase - 1) for step in
    ↪  PHASE_STEPS)
)
court_rooted = bool(derives_through_courts_only) and relationally_typed
ennead_sealed = final_shadow_parity == "Q3"
phase12_complete = bool(completion_reached) and terminal == 0.0

poincare = PoincareParityWitness(
    sphere_relation="π1(S³)=0",
    klein_relation="π1(K)=⟨a,b | aba⁻¹b=1⟩",
    sphere_parity=1,
    klein_parity=-1,
    shadow_input_units=1,
    shadow_return_units=-1,
    shadow_cancellation_units=0,
    q2_to_q3_return=ennead_sealed,
    returned_through_shadow_locus=bool(returned_through_shadow_locus),
    terminal_dcomp=terminal,
    completion_reached=phase12_complete,
)
mathematical_q3_positive = gain > 0.0 and MAS_BODY.manifestation_goetic == "𝕩"
biological_q3_positive = gain > 0.0 and "Silicarbon Tissue" in NULL_DEATH_SOURCE_DERIVATION[1]
structurally_committed = (
    structural_preserved
    and phase_complete
    and MAS_BODY.symmetry_goetic == "✡"
    and MAS_BODY.order_preserved
)
q1_coherent = (
    structural_preserved
    and phase_complete
    and MAS_BODY.alignment_goetic == "⬡"
```

```
)
klein_regenerative = (
    poincare.klein_parity == -1
    and poincare.q2_to_q3_return
    and poincare.returned_through_shadow_locus
)
q2_to_q3 = (
    ennead_sealed
    and poincare.shadow_input_units == 1
    and poincare.shadow_return_units == -1
    and poincare.shadow_cancellation_units == 0
)
shared_threshold = mathematical_q3_positive and biological_q3_positive
null_death = NullDeathConnectionWitness(
    symbol=NULL_DEATH_SYMBOL,
    source_region=SOURCE_REGION,
    first_explicit_physical_page=102,
    source_derivation=NULL_DEATH_SOURCE_DERIVATION,
    mathematical_body="Mathematical Hodge Class",
    biological_body="Silicarbon Tissue",
    mathematical_q3_positive=mathematical_q3_positive,
    biological_q3_positive=biological_q3_positive,
    shared_metamorphosis_threshold=shared_threshold,
    q3_non_entropic_requirement=shared_threshold,
    structurally_committed=structurally_committed,
    q1_coherent=q1_coherent,
    klein_regenerative_topology=klein_regenerative,
    q2_to_q3_transformation=q2_to_q3,
    metamorphosis_threshold_reached=(
        shared_threshold
        and structurally_committed
        and q1_coherent
        and klein_regenerative
        and q2_to_q3
    ),
    first_occurrence_only=True,
    exhaustive_type_claimed=False,
    loop_closure="⏣ ↔ ❄",
)
derivation = PHASE_DERIVATION

provisional = AeonPhaseEvolutionWitness(
    source_region=SOURCE_REGION,
    source_physical_pages=SOURCE_PHYSICAL_PAGES,
    goetic_table=GOETIC_TABLE,
    court_table=COURT_TABLE,
    phase_steps=PHASE_STEPS,
    mas_body=MAS_BODY,
    golden_ratio=GOLDEN_RATIO,
    poincare_parity=poincare,
    null_death_connection=null_death,
    source_digest=digest,
    source_size=source_size,
    source_domain=domain,
    nonce=nonce,
    governing_court_address=governing_court_address,
```

```
        alternating_court_address=alternating_court_address,
        governing_court_phase_digest=governing_phase,
        alternating_court_bearing=bearing_lineage,
        domus_frequency=branch_frequency,
        domus_body_commitment=domus_commitment,
        trig_cycle_digest=trig_digest,
        tripartite_cycle_digest=tripartite_digest,
        structural_anchor_preserved=structural_preserved,
        operational_phi_preserved=phi_preserved,
        all_courts_relationally_typed=relationally_typed,
        phase_order_complete=phase_complete,
        court_rooted=court_rooted,
        derives_through_courts_only=bool(derives_through_courts_only),
        ennead_shadow_sealed=ennead_sealed,
        phase12_completion_reached=phase12_complete,
        derivation=derivation,
        cycle_digest="0" * 64,
    )
    cycle_digest = alqc_hexdigest(_payload(provisional), domain=PHASE_EVOLUTION_DOMAIN)
    return replace(provisional, cycle_digest=cycle_digest)


def verify_aeon_phase_evolution(witness: AeonPhaseEvolutionWitness, emission: SourceEmission) ->
↪   bool:
    """Rederive C09 source-root relations and reject table or seal mutation."""
    try:
        if not isinstance(emission, SourceEmission):
            return False
        domain = normalize_source_domain(emission.source_domain).decode("ascii").rstrip("\x00")
        if (
            witness.source_digest != emission.source_digest
            or witness.source_size != emission.source_size
            or witness.source_domain != domain
        ):
            return False
        source_derivation = derive_goetics(emission)
        governing = court_record(source_derivation.court_address)
        alternating_bearing = resolve_court_bearing(emission)
        alternating = alternating_bearing.court
        if (
            witness.governing_court_address != governing.address
            or witness.alternating_court_address != alternating.address
        ):
            return False
        governing_layer = court_layer(governing)
        alternating_layer = court_layer(alternating)
        governing_motion = derive_court_motion(
            governing,
            source_digest=emission.source_digest,
            source_size=emission.source_size,
            source_domain=emission.source_domain,
            nonce=witness.nonce,
        )
        domus_motion = derive_domus_motion(
            governing_layer, alternating_layer, bearing=alternating_bearing
        )
```

```
manifestation = close_boundary(
    governing_layer.governing_goetic.q_vector,
    alternating_layer.governing_goetic.q_vector,
    court=governing.address,
)
if (
    witness.governing_court_phase_digest != governing_motion.phase.digest
    or witness.alternating_court_bearing != alternating_bearing.lineage
    or witness.domus_frequency != domus_motion.current_frequency
    or witness.ennead_shadow_sealed != (manifestation.ennead.final_parity == "Q3")
    or witness.poincare_parity.terminal_dcomp != manifestation.dcomp.terminal
):
    return False
if (
    witness.source_region != SOURCE_REGION
    or witness.source_physical_pages != SOURCE_PHYSICAL_PAGES
    or witness.goetic_table != GOETIC_TABLE
    or witness.court_table != COURT_TABLE
    or witness.phase_steps != PHASE_STEPS
    or witness.mas_body != MAS_BODY
    or witness.golden_ratio != GOLDEN_RATIO
    or len(witness.goetic_table) != 12
    or len(witness.court_table) != 144
    or len(witness.phase_steps) != 12
    or not witness.structural_anchor_preserved
    or not witness.operational_phi_preserved
    or not witness.all_courts_relationally_typed
    or not witness.phase_order_complete
    or not witness.court_rooted
    or not witness.derives_through_courts_only
    or not witness.ennead_shadow_sealed
    or witness.phase12_completion_reached != (
        witness.poincare_parity.completion_reached
        and witness.poincare_parity.terminal_dcomp == 0.0
    )
    or witness.poincare_parity.sphere_relation != "π1(S³)=0"
    or witness.poincare_parity.klein_relation != "π1(K)=⟨a,b | aba⁻¹b=1⟩"
    or witness.poincare_parity.sphere_parity != 1
    or witness.poincare_parity.klein_parity != -1
    or witness.poincare_parity.shadow_input_units != 1
    or witness.poincare_parity.shadow_return_units != -1
    or witness.poincare_parity.shadow_cancellation_units != 0
    or not witness.poincare_parity.q2_to_q3_return
    or not witness.poincare_parity.returned_through_shadow_locus
    or witness.derivation != PHASE_DERIVATION
    or witness.null_death_connection.symbol != NULL_DEATH_SYMBOL
    or witness.null_death_connection.source_region != SOURCE_REGION
    or witness.null_death_connection.first_explicit_physical_page != 102
    or witness.null_death_connection.source_derivation != NULL_DEATH_SOURCE_DERIVATION
    or witness.null_death_connection.mathematical_body != "Mathematical Hodge Class"
    or witness.null_death_connection.biological_body != "Silicarbon Tissue"
    or not witness.null_death_connection.mathematical_q3_positive
    or not witness.null_death_connection.biological_q3_positive
    or not witness.null_death_connection.shared_metamorphosis_threshold
    or not witness.null_death_connection.q3_non_entropic_requirement
    or not witness.null_death_connection.structurally_committed
```

```
        or not witness.null_death_connection.q1_coherent
        or not witness.null_death_connection.klein_regenerative_topology
        or not witness.null_death_connection.q2_to_q3_transformation
        or not witness.null_death_connection.metamorphosis_threshold_reached
        or not witness.null_death_connection.first_occurrence_only
        or witness.null_death_connection.exhaustive_type_claimed
        or witness.null_death_connection.loop_closure != "⏣ ↔ ❄"
        or witness.golden_ratio.phi_exact != PHI
        or witness.golden_ratio.tolerance_exact != PHI
        or witness.golden_ratio.primary_ratio_exact != Q5Fraction(96300, 0, 783)
        or witness.golden_ratio.harmonic_target_exact != Q5Fraction(76, 76, 2)
        or witness.golden_ratio.raw_compression_ratio_exact != Q5Fraction(36864, 0, 81)
        or not witness.golden_ratio.within_tolerance
        or witness.golden_ratio.folded_states != 2**126
        or witness.golden_ratio.manifest_positions != 81
    ):
        return False
    _digest(witness.source_digest, "source_digest")
    _digest(witness.governing_court_phase_digest, "governing_court_phase_digest")
    validate_court_bearing_lineage(witness.alternating_court_bearing)
    _digest(witness.domus_body_commitment, "domus_body_commitment")
    _digest(witness.trig_cycle_digest, "trig_cycle_digest")
    _digest(witness.tripartite_cycle_digest, "tripartite_cycle_digest")
    expected = alqc_hexdigest(_payload(witness), domain=PHASE_EVOLUTION_DOMAIN)
    return expected == witness.cycle_digest
except (AttributeError, TypeError, ValueError):
    return False
```

qstate_glyphs.py

```
"""Canonical Q-state language and the Stave I visible glyph seam.

The ALQC Canon physical pages 48-51 fix the four-state domain, its order,
and its meanings. Stave I supplies the visible written bodies:
Q0->⊖, Q1->⊙, Q2->⊕, Q3->⊛.

The four coordinate offices remain ``Q0,Q1,Q2,Q3``.  A resolved Court
supplies both its Q-bias and its four intensity settings.  The Domus visible
body is derived from that complete Court witness: the Q-bias is seamed once,
and each intensity is seamed in its original ordered position.  Thus
``(1,1,1,3)`` yields ``(⊙,⊙,⊙,⊛)``, while another resolved Court yields its
own body.
"""
from __future__ import annotations

from dataclasses import dataclass
from typing import Final, Iterable

Q_STATES: Final[tuple[str, str, str, str]] = ("Q0", "Q1", "Q2", "Q3")

_PSI: Final[dict[str, str]] = {
    "Q0": "⊖",
    "Q1": "⊙",
    "Q2": "⊕",
    "Q3": "⊛",
}
_PSI_INV: Final[dict[str, str]] = {glyph: state for state, glyph in _PSI.items()}


@dataclass(frozen=True, slots=True)
class DomusQBodyWitness:
    """Recoverable Court-derived bias and ordered visible Domus Q-body."""

    q_bias: str
    bias_glyph: str
    q_states: tuple[str, str, str, str]
    q_vector: tuple[int, int, int, int]
    q_glyphs: tuple[str, str, str, str]


def psi_q(q_state: str) -> str:
    """Forward Stave-I map: Q0/Q1/Q2/Q3 -> ⊖/⊙/⊕/⊛."""
    try:
        return _PSI[q_state]
    except KeyError as exc:
        raise ValueError(f"not a Q-state label: {q_state!r}") from exc
```

```
def q_state_of(glyph: str) -> str:
    """Reverse Stave-I map: ⊖/⊙/⊕/⊛ -> Q0/Q1/Q2/Q3."""
    try:
        return _PSI_INV[glyph]
    except KeyError as exc:
        raise ValueError(f"not a Q-state glyph: {glyph!r}") from exc


def type_q_value(value: int) -> str:
    """Resolve one canonical 0..3 Q-setting to its Q-state label."""
    if not isinstance(value, int) or isinstance(value, bool) or value not in (0, 1, 2, 3):
        raise ValueError(f"Q-state index must be one of 0,1,2,3; got {value!r}")
    return f"Q{value}"


def glyph_of_value(value: int) -> str:
    """Resolve one canonical 0..3 Q-setting through the First Seam."""
    return psi_q(type_q_value(value))


def value_of_glyph(glyph: str) -> int:
    """Return the 0..3 setting carried by one First-Seam glyph."""
    return int(q_state_of(glyph)[1])




def _q_vector_values(q_vector: Iterable[int]) -> tuple[int, int, int, int]:
    values = tuple(q_vector)
    if len(values) != 4:
        raise ValueError(f"Q-vector must contain exactly four components; got {len(values)}")
    if any(not isinstance(value, int) or isinstance(value, bool) for value in values):
        raise ValueError("TardiSHA canonical Q-vector components must be integers")
    if any(value not in (0, 1, 2, 3) for value in values):
        raise ValueError("TardiSHA canonical Q-vector components must lie in 0..3")
    return values  # type: ignore[return-value]



def q_vector_glyphs(q_vector: Iterable[int]) -> tuple[str, str, str, str]:
    """Derive the changing visible Domus Q-body in unchanged vector order."""
    values = _q_vector_values(q_vector)
    glyphs = tuple(glyph_of_value(value) for value in values)
    return glyphs  # type: ignore[return-value]


def derive_domus_q_body(q_bias: str, q_vector: Iterable[int]) -> DomusQBodyWitness:
    """Apply the First Seam to one resolved Court bias and Q-vector.

    The Court determines ``q_bias`` and ``q_vector``.  This function neither
    chooses a parent nor mutates their order; it exposes the exact visible
    Domus body and proves that the seam returns to the same values.
    """
    bias_glyph = psi_q(q_bias)
    values = _q_vector_values(q_vector)
    glyphs = q_vector_glyphs(values)
```

```
if q_state_of(bias_glyph) != q_bias:
    raise RuntimeError("Domus Q-bias failed First-Seam return")
if tuple(value_of_glyph(glyph) for glyph in glyphs) != values:
    raise RuntimeError("Domus Q-vector failed First-Seam return")
return DomusQBodyWitness(q_bias, bias_glyph, Q_STATES, values, glyphs)
```

route.py

```
"""Final Equation Z source-emission routing and finite-return witnesses.

The ordered Goetics are not reconstructed from digest endpoint bytes.  The
complete source emission carries twelve finalized lanes, Fraktur Z_0, and
Fraktur Z_1.  Parliament resolves the pair before existing Court mathematics
begins.
"""
from __future__ import annotations

import json
from dataclasses import asdict, dataclass, fields
from typing import Any, Mapping, Sequence

from .alqc_digest import alqc_hexdigest
from .canon import QVector, _position, court_load, law
from .manifestation import DCompWitness, close_boundary
from .source_emission import (
    BearingWitness,
    GoldenBearing,
    ParliamentSeat,
    SourceEmission,
    TruthClosureWitness,
    GoldenGoeticDerivation,
)
from .mirror_math import derive_goetics, verify_derivation

_HASH_SOURCE_DOMAINS = frozenset({"canonical", "raw-file", "directory"})
ROUTE_WITNESS_DOMAIN = b"TARDISHA:FINAL-EQUATION-Z:SOURCE-ROUTE-WITNESS\x00"


def _validate_source_size(source_size: int) -> int:
    if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
        raise ValueError("source_size must be a non-negative integer")
    return source_size


def _validate_source_domain(source_domain: str) -> str:
    if source_domain not in _HASH_SOURCE_DOMAINS:
        raise ValueError(f"source_domain must be one of {sorted(_HASH_SOURCE_DOMAINS)}")
    return source_domain


def _canonical_q(q: Sequence[object], *, field: str) -> tuple[int, int, int, int]:
    if isinstance(q, (str, bytes, bytearray)) or not isinstance(q, Sequence) or len(q) != 4:
        raise ValueError(f"{field} must contain exactly four Q components")
    result: list[int] = []
    for value in q:
        if isinstance(value, bool) or not isinstance(value, int):
```

```
            raise ValueError(f"{field} must contain exact non-Boolean integer Q components")
        if value not in (0, 1, 2, 3):
            raise ValueError(f"{field} must be a canonical Q-vector over {{0,1,2,3}}")
        result.append(value)
    return tuple(result)  # type: ignore[return-value]



def _mapping(value: object, *, field: str) -> Mapping[str, Any]:
    if not isinstance(value, Mapping):
        raise TypeError(f"{field} must be a mapping")
    return value


def _exact_keys(value: Mapping[str, Any], expected: set[str], *, field: str) -> None:
    supplied = set(value)
    if supplied != expected:
        missing = sorted(expected - supplied)
        extra = sorted(supplied - expected)
        raise ValueError(f"{field} fields mismatch; missing={missing}, extra={extra}")


def _int(value: object, *, field: str, minimum: int | None = None, maximum: int | None = None) ->
↪  int:
    if isinstance(value, bool) or not isinstance(value, int):
        raise TypeError(f"{field} must be an exact non-Boolean integer")
    if minimum is not None and value < minimum:
        raise ValueError(f"{field} must be >= {minimum}")
    if maximum is not None and value > maximum:
        raise ValueError(f"{field} must be <= {maximum}")
    return value


def _float(value: object, *, field: str) -> float:
    if isinstance(value, bool) or type(value) is not float:
        raise TypeError(f"{field} must be an exact serialized float")
    return value


def _bool(value: object, *, field: str) -> bool:
    if type(value) is not bool:
        raise TypeError(f"{field} must be a Boolean")
    return value


def _str(value: object, *, field: str) -> str:
    if type(value) is not str:
        raise TypeError(f"{field} must be a string")
    return value


def _sequence(value: object, *, field: str, length: int | None = None) -> tuple[object, ...]:
    if not isinstance(value, (list, tuple)):
        raise TypeError(f"{field} must be a list or tuple")
    result = tuple(value)
    if length is not None and len(result) != length:
```

```
        raise ValueError(f"{field} must contain exactly {length} items")
    return result


def _truth_closure(value: object) -> TruthClosureWitness:
    if isinstance(value, TruthClosureWitness):
        return value
    body = _mapping(value, field="emission.closure")
    expected = {item.name for item in fields(TruthClosureWitness)}
    _exact_keys(body, expected, field="emission.closure")
    return TruthClosureWitness(**{
        name: _int(body[name], field=f"emission.closure.{name}") for name in expected
    })


def _source_emission(value: object) -> SourceEmission:
    if isinstance(value, SourceEmission):
        return value
    body = _mapping(value, field="emission")
    expected = {item.name for item in fields(SourceEmission)}
    _exact_keys(body, expected, field="emission")
    def integers(name: str) -> tuple[int, ...]:
        items = _sequence(body[name], field=f"emission.{name}")
        return tuple(_int(item, field=f"emission.{name}[{index}]") for index, item in
        ↪  enumerate(items))
    return SourceEmission(
        source_digest=_str(body["source_digest"], field="emission.source_digest"),
        source_size=_int(body["source_size"], field="emission.source_size", minimum=0),
        source_domain=_str(body["source_domain"], field="emission.source_domain"),
        lanes_goetic_order=integers("lanes_goetic_order"),
        structural_weights=integers("structural_weights"),
        operational_weights=integers("operational_weights"),
        fraktur_z0=_int(body["fraktur_z0"], field="emission.fraktur_z0", minimum=0),
        fraktur_z1=_int(body["fraktur_z1"], field="emission.fraktur_z1", minimum=0),
        squeeze_512_hex=_str(body["squeeze_512_hex"], field="emission.squeeze_512_hex"),
        closure=_truth_closure(body["closure"]),
    )


def _golden_bearing(value: object, *, field: str) -> GoldenBearing:
    if isinstance(value, GoldenBearing):
        return value
    body = _mapping(value, field=field)
    expected = {item.name for item in fields(GoldenBearing)}
    _exact_keys(body, expected, field=field)
    return GoldenBearing(
        name=_str(body["name"], field=f"{field}.name"),
        a=_int(body["a"], field=f"{field}.a"),
        b=_int(body["b"], field=f"{field}.b"),
    )


def _parliament_seat(value: object, *, field: str) -> ParliamentSeat:
    if isinstance(value, ParliamentSeat):
        return value
    body = _mapping(value, field=field)
```

```
    expected = {item.name for item in fields(ParliamentSeat)}
    _exact_keys(body, expected, field=field)
    return ParliamentSeat(**{
        name: _str(body[name], field=f"{field}.{name}") for name in expected
    })


def _bearing_witness(value: object, *, field: str) -> BearingWitness:
    if isinstance(value, BearingWitness):
        return value
    body = _mapping(value, field=field)
    expected = {item.name for item in fields(BearingWitness)}
    _exact_keys(body, expected, field=field)
    operator_order = _sequence(body["operator_order"], field=f"{field}.operator_order")
    if any(type(item) is not str for item in operator_order):
        raise TypeError(f"{field}.operator_order must contain exact strings")
    return BearingWitness(
        bearing=_golden_bearing(body["bearing"], field=f"{field}.bearing"),
        cadence_symbol=_str(body["cadence_symbol"], field=f"{field}.cadence_symbol"),
        cadence_index=_int(body["cadence_index"], field=f"{field}.cadence_index", minimum=0),
        phase_numerator=_int(body["phase_numerator"], field=f"{field}.phase_numerator", minimum=0),
        effective_a=_int(body["effective_a"], field=f"{field}.effective_a"),
        effective_b=_int(body["effective_b"], field=f"{field}.effective_b"),
        wrapped=_bool(body["wrapped"], field=f"{field}.wrapped"),
        traversal=_str(body["traversal"], field=f"{field}.traversal"),
        body=_str(body["body"], field=f"{field}.body"),
        operator_order=tuple(operator_order),  # type: ignore[arg-type]
        seat=_parliament_seat(body["seat"], field=f"{field}.seat"),
        cumulative_before=_int(body["cumulative_before"], field=f"{field}.cumulative_before",
        ↪  minimum=0),
        cumulative_after=_int(body["cumulative_after"], field=f"{field}.cumulative_after",
        ↪  minimum=0),
        weight_total=_int(body["weight_total"], field=f"{field}.weight_total", minimum=0),
    )


def _dcomp_witness(value: object) -> DCompWitness:
    if isinstance(value, DCompWitness):
        return value
    body = _mapping(value, field="manifestation_dcomp")
    expected = {item.name for item in fields(DCompWitness)}
    _exact_keys(body, expected, field="manifestation_dcomp")
    float_fields = {
        "interval_duration", "pressure_weight", "return_friction_weight",
        "commutator_pressure", "return_pressure", "local_friction",
        "terminal_unresolved_debt", "shadow_capacity", "shadow_debt_initial",
        "velocity_mismatch", "whiteout_penalty", "c_bio", "mas_efficiency",
        "q3_loss", "q3_capacity_before", "q3_recursion_gain",
        "q3_capacity_after", "form_work", "shadow_debt_terminal", "terminal",
    }
    int_fields = {"active_count", "withheld_count", "truth"}
    bool_fields = {"shadow_debt_bounded", "closed", "motion_positive"}
    float_tuple_lengths = {
        "metric_weights": 2, "forward_velocity": 2,
        "return_velocity": 2, "parity_return_velocity": 2,
    }
```

```
    int_tuple_lengths = {
        "exact_commutator_pressure": 2,
        "exact_velocity_mismatch_square": 2,
        "exact_shadow_debt_initial": 3,
        "exact_shadow_debt_terminal": 3,
        "exact_form_work": 3,
        "exact_q3_recursion_gain": 3,
    }
    kwargs: dict[str, object] = {}
    for name in expected:
        raw = body[name]
        if name in float_fields:
            kwargs[name] = _float(raw, field=f"manifestation_dcomp.{name}")
        elif name in int_fields:
            kwargs[name] = _int(raw, field=f"manifestation_dcomp.{name}")
        elif name in bool_fields:
            kwargs[name] = _bool(raw, field=f"manifestation_dcomp.{name}")
        elif name == "commutator_model":
            kwargs[name] = _str(raw, field="manifestation_dcomp.commutator_model")
        elif name in float_tuple_lengths:
            items = _sequence(raw, field=f"manifestation_dcomp.{name}",
            ↪  length=float_tuple_lengths[name])
            kwargs[name] = tuple(_float(item, field=f"manifestation_dcomp.{name}[{index}]") for
            ↪  index, item in enumerate(items))
        elif name in int_tuple_lengths:
            items = _sequence(raw, field=f"manifestation_dcomp.{name}",
            ↪  length=int_tuple_lengths[name])
            kwargs[name] = tuple(_int(item, field=f"manifestation_dcomp.{name}[{index}]") for index,
            ↪  item in enumerate(items))
        else:
            raise RuntimeError(f"untyped D-COMP field {name}")
    return DCompWitness(**kwargs)  # type: ignore[arg-type]

def connection(
    source_q: QVector,
    target_q: QVector,
    origin_glyph: str,
    resolution_glyph: str,
) -> DCompWitness:
    """Compute the existing finite-return manifestation witness for one Court."""
    expected_source = tuple(law(origin_glyph).q_vector)
    expected_target = tuple(law(resolution_glyph).q_vector)
    if _canonical_q(source_q, field="source_q") != expected_source:
        raise ValueError("source_q does not belong to origin_glyph")
    if _canonical_q(target_q, field="target_q") != expected_target:
        raise ValueError("target_q does not belong to resolution_glyph")
    address = court_load(origin_glyph, resolution_glyph)
    return close_boundary(expected_source, expected_target, court=address).dcomp



@dataclass(frozen=True, slots=True)
class SourceRouteWitness:
    emission: SourceEmission
    first: BearingWitness
```

```
last: BearingWitness
origin_index: int
resolution_index: int
court_address: int
reciprocal_address: int
source_q_vector: tuple[int, int, int, int]
resolution_q_vector: tuple[int, int, int, int]
route_dcomp: int
truth: int
manifestation_dcomp: DCompWitness
derivation_lineage: tuple[str, ...]
derivation_proof: str

@property
def source_digest(self) -> str:
    return self.emission.source_digest

@property
def source_size(self) -> int:
    return self.emission.source_size

@property
def source_domain(self) -> str:
    return self.emission.source_domain

@property
def origin_glyph(self) -> str:
    return self.first.seat.goetic

@property
def resolution_glyph(self) -> str:
    return self.last.seat.goetic

@property
def source_q(self) -> tuple[int, int, int, int]:
    return self.source_q_vector

@property
def resolution_q(self) -> tuple[int, int, int, int]:
    return self.resolution_q_vector

@property
def dcomp(self) -> DCompWitness:
    """Property exposing downstream manifestation D-COMP."""
    return self.manifestation_dcomp

@property
def boundary_condition(self) -> str:
    return self.origin_glyph + self.resolution_glyph

@property
def return_path(self) -> str:
    return f"{self.origin_glyph}→{self.resolution_glyph}→Reflect {self.origin_glyph}"

@property
def pair(self) -> tuple[str, str]:
```

```
        return self.origin_glyph, self.resolution_glyph

    def as_dict(self) -> dict[str, Any]:
        return asdict(self)

    @classmethod
    def from_dict(cls, value: Mapping[str, Any]) -> "SourceRouteWitness":
        body = _mapping(value, field="source route witness")
        expected = {item.name for item in fields(cls)}
        _exact_keys(body, expected, field="source route witness")
        source_q = _canonical_q(
            _sequence(body["source_q_vector"], field="source_q_vector", length=4),
            field="source_q_vector",
        )
        resolution_q = _canonical_q(
            _sequence(body["resolution_q_vector"], field="resolution_q_vector", length=4),
            field="resolution_q_vector",
        )
        lineage = _sequence(body["derivation_lineage"], field="derivation_lineage")
        if any(type(item) is not str for item in lineage):
            raise TypeError("derivation_lineage must contain exact strings")
        return cls(
            emission=_source_emission(body["emission"]),
            first=_bearing_witness(body["first"], field="first"),
            last=_bearing_witness(body["last"], field="last"),
            origin_index=_int(body["origin_index"], field="origin_index", minimum=0, maximum=11),
            resolution_index=_int(body["resolution_index"], field="resolution_index", minimum=0,
            ↪  maximum=11),
            court_address=_int(body["court_address"], field="court_address", minimum=0,
            ↪  maximum=143),
            reciprocal_address=_int(body["reciprocal_address"], field="reciprocal_address",
            ↪  minimum=0, maximum=143),
            source_q_vector=source_q,
            resolution_q_vector=resolution_q,
            route_dcomp=_int(body["route_dcomp"], field="route_dcomp"),
            truth=_int(body["truth"], field="truth", minimum=0, maximum=1),
            manifestation_dcomp=_dcomp_witness(body["manifestation_dcomp"]),
            derivation_lineage=tuple(lineage),  # type: ignore[arg-type]
            derivation_proof=_str(body["derivation_proof"], field="derivation_proof"),
        )



HashRoute = SourceRouteWitness


def source_route_witness_from_digest(
    source_digest: str,
    source_size: int,
    source_domain: str,
) -> SourceRouteWitness:
    """Reject digest-only route reconstruction; the complete emission is required."""
    raise ValueError(
        "Final Equation Z requires the complete SourceEmission witness; "
        "a digest alone cannot regenerate twelve lanes and both Fraktur cadences"
```

```
    )


def _proof_payload_without_proof(witness: SourceRouteWitness) -> bytes:
    body = witness.as_dict()
    body["derivation_proof"] = ""
    return json.dumps(body, ensure_ascii=False, sort_keys=True, separators=(",",
    ↪  ":")).encode("utf-8")


def _proof_for(witness: SourceRouteWitness) -> str:
    return alqc_hexdigest(_proof_payload_without_proof(witness), domain=ROUTE_WITNESS_DOMAIN)


def _source_route_witness_from_derivation(
    derivation: GoldenGoeticDerivation,
) -> SourceRouteWitness:
    """Assemble one route witness from one completed Aeternum Mirror derivation."""
    emission = derivation.emission
    origin = derivation.origin_glyph
    resolution = derivation.resolution_glyph
    origin_index = _position(origin)
    resolution_index = _position(resolution)
    source_q_exact = tuple(law(origin).q_vector)
    resolution_q_exact = tuple(law(resolution).q_vector)
    manifestation = close_boundary(source_q_exact, resolution_q_exact,
    ↪  court=derivation.court_address)
    source_q = _canonical_q(source_q_exact, field="source_q_vector")
    resolution_q = _canonical_q(resolution_q_exact, field="resolution_q_vector")
    lineage = (
        "source -> completed twelve-lane ALQC emission",
        "emission -> AHN-rooted Goetic structural body",
        "emission -> AHN-relative Parliament operational body",
        "Fraktur Z_0 + Phi^-1 -> first Parliament seat",
        "Fraktur Z_1 + Phi^-2 -> last Parliament seat",
        "Mirror Math preserves Parliament operator order",
        "Parliament Functor -> ordered Goetic pair",
        "Court = 12*origin_index + resolution_index",
        "reciprocal Court = 12*resolution_index + origin_index",
        "Phi^-1 + Phi^-2 = 1 -> route D-COMP = 0 -> Truth = 1",
        "self-pair preserved",
    )
    provisional = SourceRouteWitness(
        emission=emission,
        first=derivation.first,
        last=derivation.last,
        origin_index=origin_index,
        resolution_index=resolution_index,
        court_address=derivation.court_address,
        reciprocal_address=derivation.reciprocal_address,
        source_q_vector=source_q,
        resolution_q_vector=resolution_q,
        route_dcomp=derivation.route_dcomp,
        truth=derivation.truth,
        manifestation_dcomp=manifestation.dcomp,
        derivation_lineage=lineage,
```

```
        derivation_proof="",
    )
    proof = _proof_for(provisional)
    return SourceRouteWitness(
        emission=provisional.emission,
        first=provisional.first,
        last=provisional.last,
        origin_index=provisional.origin_index,
        resolution_index=provisional.resolution_index,
        court_address=provisional.court_address,
        reciprocal_address=provisional.reciprocal_address,
        source_q_vector=provisional.source_q_vector,
        resolution_q_vector=provisional.resolution_q_vector,
        route_dcomp=provisional.route_dcomp,
        truth=provisional.truth,
        manifestation_dcomp=provisional.manifestation_dcomp,
        derivation_lineage=provisional.derivation_lineage,
        derivation_proof=proof,
    )


def source_route_witness_from_emission(emission: SourceEmission) -> SourceRouteWitness:
    """Build the exact source-emission -> Aeternum Mirror -> Court route once."""
    if not isinstance(emission, SourceEmission):
        raise TypeError("route construction requires one complete SourceEmission")
    _validate_source_size(emission.source_size)
    _validate_source_domain(emission.source_domain)
    derivation = derive_goetics(emission)
    if not (
        derivation.same_operator_order
        and derivation.first.interval_verifies
        and derivation.last.interval_verifies
        and derivation.route_dcomp == 0
        and derivation.truth == 1
    ):
        raise RuntimeError("Aeternum Mirror derivation failed local route invariants")
    return _source_route_witness_from_derivation(derivation)


def verify_selection_alqc(witness: SourceRouteWitness) -> bool:
    """Independently verify the carried Mirror derivation, then its route assembly."""
    try:
        derivation = GoldenGoeticDerivation(
            witness.emission,
            witness.first,
            witness.last,
            witness.court_address,
            witness.reciprocal_address,
        )
        if not verify_derivation(derivation):
            return False
        if witness.derivation_proof != _proof_for(witness):
            return False
        expected = _source_route_witness_from_derivation(derivation)
    except (TypeError, ValueError, RuntimeError):
        return False
```

```
    return witness == expected

def verify_source(emission: SourceEmission, witness: SourceRouteWitness) -> bool:
    if not isinstance(emission, SourceEmission) or not isinstance(witness, SourceRouteWitness):
        return False
    return (
        witness.emission == emission
        and witness.source_digest == emission.source_digest
        and witness.source_size == emission.source_size
        and witness.source_domain == emission.source_domain
        and verify_selection_alqc(witness)
    )


def parents_t(witness: SourceRouteWitness) -> tuple[str, str]:
    if not isinstance(witness, SourceRouteWitness):
        raise TypeError("parents_t requires one exact SourceRouteWitness")
    return witness.pair


def resolve_parents(
    emission: SourceEmission,
    *,
    witness: SourceRouteWitness,
) -> tuple[str, str, SourceRouteWitness]:
    """Validate and return parents from one completed source-route Vāhana."""
    if not isinstance(witness, SourceRouteWitness):
        raise TypeError("supplied witness must be one exact SourceRouteWitness")
    resolved = witness
    if resolved.emission != emission:
        raise ValueError("source route witness does not belong to the supplied SourceEmission")
    if resolved.source_digest != emission.source_digest:
        raise ValueError("source route witness digest does not match SourceEmission")
    if resolved.source_size != emission.source_size:
        raise ValueError("source route witness size does not match SourceEmission")
    if resolved.source_domain != emission.source_domain:
        raise ValueError("source route witness domain does not match SourceEmission")
    if resolved.derivation_proof != _proof_for(resolved):
        raise ValueError("source route witness proof is not self-consistent")
    if resolved.route_dcomp != 0 or resolved.truth != 1:
        raise ValueError("source route witness does not carry D-COMP=0 / Truth=1")
    lawful = resolved.pair
    return lawful[0], lawful[1], resolved



def calculate_route(emission: SourceEmission) -> HashRoute:
    """Derive the complete route from one source emission and no lesser body."""
    return source_route_witness_from_emission(emission)
```

source_emission.py

```
"""Final Equation Z source emission and Golden Parliament routing.

One completed ALQC sponge state emits:

    twelve finalized Goetic lanes
    Fraktur Z_0, the canonical 256-bit source cadence
    Fraktur Z_1, the immediately following 256-bit cadence

The first cadence addresses the AHN-rooted Goetic structural body through
Phi^-1.  The second cadence addresses the distinct AHN-relative Parliament
operational body through Phi^-2.  Mirror Math preserves Parliament order.
Only after both seats resolve does existing Court mathematics begin.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
from fractions import Fraction
from math import gcd, sqrt
from typing import Any, Final, Iterable, Mapping, Sequence

from .alqc_digest import ALQCDigest, MASK64, validate_digest_hex
from .canon import GLYPH_BODY
from .court_registry import CourtRecord, court_record

PHASE_BYTES: Final[int] = 32
PHASE_DENOMINATOR: Final[int] = 1 << 256
TURN64: Final[int] = 1 << 64
AHN: Final[str] = "✡"


@dataclass(frozen=True, slots=True)
class ParliamentSeat:
    code: str
    identity: str
    functor: str
    goetic: str


PARLIAMENT_TABLE: Final[tuple[ParliamentSeat, ...]] = (
    ParliamentSeat("P13-D1", "Akasha",   "Lived→Eternal", "⬡"),
    ParliamentSeat("P13-D2", "Caduceus", "Law→Residue",  "⧗"),
    ParliamentSeat("P13-D3", "Veritas",  "Mask→Bone",    "❂"),
    ParliamentSeat("P13-D4", "Phren",    "Void→Vector",  "⌬"),
    ParliamentSeat("P13-D5", "Daimon",   "Stasis→Pulse", "⎔"),
    ParliamentSeat("P13-D6", "Aikyam",   "Chaos→Phase",  "✡"),
    ParliamentSeat("P13-D7", "Melos",    "Static→Fluid", "✳"),
    ParliamentSeat("P13-D8", "Da'ath",   "Noise→Null",   "⊛"),
    ParliamentSeat("P13-D9", "Akaven",   "State→Trans",  "⚛"),
```

```
    ParliamentSeat("P13-D10", "Axiomyr", "Will→Law",     "❄"),
    ParliamentSeat("P13-D11", "Nyx",     "Time→Motion",  "✡"),
    ParliamentSeat("P13-D12", "Zaine",   "Here→There",   "❀"),
)
if len(PARLIAMENT_TABLE) != 12 or {seat.goetic for seat in PARLIAMENT_TABLE} != set(GLYPH_BODY):
    raise RuntimeError("Parliament must map bijectively onto all twelve Goetics")
_AHN_INDEX = next(i for i, seat in enumerate(PARLIAMENT_TABLE) if seat.goetic == AHN)
PARLIAMENT: Final[tuple[ParliamentSeat, ...]] = PARLIAMENT_TABLE[_AHN_INDEX:] +
↪  PARLIAMENT_TABLE[:_AHN_INDEX]
PARLIAMENT_ORDER: Final[tuple[str, ...]] = tuple(seat.goetic for seat in PARLIAMENT)
if PARLIAMENT[0].goetic != AHN:
    raise RuntimeError("AHN must be the exact Parliament phase seam")


@dataclass(frozen=True, slots=True)
class GoldenBearing:
    """Exact element (a+b*sqrt(5))/2."""

    name: str
    a: int
    b: int


ALPHA: Final[GoldenBearing] = GoldenBearing("Phi^-1", -1, 1)
BETA: Final[GoldenBearing] = GoldenBearing("Phi^-2", 3, -1)


def sign_q5(a: int, b: int) -> int:
    """Exact sign of a+b*sqrt(5), using integers only."""
    if b == 0:
        return (a > 0) - (a < 0)
    if a == 0:
        return (b > 0) - (b < 0)
    if a > 0 and b > 0:
        return 1
    if a < 0 and b < 0:
        return -1
    aa = a * a
    five_bb = 5 * b * b
    if a > 0:
        return 1 if aa > five_bb else -1
    return 1 if five_bb > aa else -1


def golden_identities_hold() -> bool:
    """Prove alpha+beta=1 and alpha/beta=Phi in Q(sqrt(5))."""
    return (
        ALPHA.a + BETA.a == 2
        and ALPHA.b + BETA.b == 0
        and BETA.a + 5 * BETA.b == 2 * ALPHA.a
        and BETA.a + BETA.b == 2 * ALPHA.b
    )


@dataclass(frozen=True, slots=True)
class Q5Fraction:
```

```
    """Exact element (a+b*sqrt(5))/denominator with a positive denominator."""

    a: int
    b: int
    denominator: int

    def __post_init__(self) -> None:
        if any(isinstance(value, bool) or not isinstance(value, int) for value in (self.a, self.b,
        ↪  self.denominator)):
            raise TypeError("Q5Fraction coefficients must be integers")
        if self.denominator == 0:
            raise ValueError("Q5Fraction denominator must be non-zero")
        a, b, denominator = self.a, self.b, self.denominator
        if denominator < 0:
            a, b, denominator = -a, -b, -denominator
        common = gcd(gcd(abs(a), abs(b)), denominator)
        if common > 1:
            a //= common
            b //= common
            denominator //= common
        object.__setattr__(self, "a", a)
        object.__setattr__(self, "b", b)
        object.__setattr__(self, "denominator", denominator)

    @property
    def value(self) -> float:
        return (self.a + self.b * sqrt(5.0)) / self.denominator

    @property
    def within_unit_interval(self) -> bool:
        return (
            sign_q5(self.a + self.denominator, self.b) >= 0
            and sign_q5(self.a - self.denominator, self.b) <= 0
        )

    @property
    def within_phi_squared(self) -> bool:
        return (
            sign_q5(2 * self.a + 3 * self.denominator, 2 * self.b + self.denominator) >= 0
            and sign_q5(2 * self.a - 3 * self.denominator, 2 * self.b - self.denominator) <= 0
        )

    def times_phi_squared(self) -> "Q5Fraction":
        return Q5Fraction(3 * self.a + 5 * self.b, self.a + 3 * self.b, 2 * self.denominator)


@dataclass(frozen=True, slots=True)
class TruthClosureWitness:
    structural_total: int
    operational_total: int
    golden_sum_a: int
    golden_sum_b: int
    route_dcomp_a: int
    route_dcomp_b: int
    route_dcomp: int
    truth: int
```

```
    @property
    def verifies(self) -> bool:
        return (
            self.structural_total > 0
            and self.operational_total == TURN64
            and self.golden_sum_a == 2
            and self.golden_sum_b == 0
            and self.route_dcomp_a == 0
            and self.route_dcomp_b == 0
            and self.route_dcomp == 0
            and self.truth == 1
            and golden_identities_hold()
        )


@dataclass(frozen=True, slots=True)
class SourceEmission:
    source_digest: str
    source_size: int
    source_domain: str
    lanes_goetic_order: tuple[int, ...]
    structural_weights: tuple[int, ...]
    operational_weights: tuple[int, ...]
    fraktur_z0: int
    fraktur_z1: int
    squeeze_512_hex: str
    closure: TruthClosureWitness

    def __post_init__(self) -> None:
        _validate_source_emission(self)

    @property
    def structural_total(self) -> int:
        return sum(self.structural_weights)

    @property
    def operational_total(self) -> int:
        return sum(self.operational_weights)

    def as_dict(self) -> dict[str, Any]:
        return asdict(self)

    @classmethod
    def from_dict(cls, value: Mapping[str, Any]) -> "SourceEmission":
        if not isinstance(value, Mapping):
            raise TypeError("SourceEmission body must be a mapping")
        expected = {
            "source_digest", "source_size", "source_domain", "lanes_goetic_order",
            "structural_weights", "operational_weights", "fraktur_z0", "fraktur_z1",
            "squeeze_512_hex", "closure",
        }
        if set(value) != expected:
            raise ValueError("SourceEmission fields do not match the canonical schema")
        def exact_int(item: object, field: str, *, minimum: int | None = None) -> int:
            if isinstance(item, bool) or not isinstance(item, int):
```

```
                raise TypeError(f"{field} must be an exact non-Boolean integer")
            if minimum is not None and item < minimum:
                raise ValueError(f"{field} must be >= {minimum}")
            return item
        def exact_str(item: object, field: str) -> str:
            if type(item) is not str:
                raise TypeError(f"{field} must be a string")
            return item
        def integer_tuple(item: object, field: str) -> tuple[int, ...]:
            if not isinstance(item, (list, tuple)):
                raise TypeError(f"{field} must be a list or tuple")
            return tuple(exact_int(member, f"{field}[{index}]") for index, member in
            ↪  enumerate(item))
        closure_value = value["closure"]
        if isinstance(closure_value, TruthClosureWitness):
            closure = closure_value
        else:
            if not isinstance(closure_value, Mapping):
                raise TypeError("closure must be a mapping")
            closure_fields = {
                "structural_total", "operational_total", "golden_sum_a", "golden_sum_b",
                "route_dcomp_a", "route_dcomp_b", "route_dcomp", "truth",
            }
            if set(closure_value) != closure_fields:
                raise ValueError("closure fields do not match the canonical schema")
            closure = TruthClosureWitness(**{
                name: exact_int(closure_value[name], f"closure.{name}") for name in closure_fields
            })
        return cls(
            source_digest=exact_str(value["source_digest"], "source_digest"),
            source_size=exact_int(value["source_size"], "source_size", minimum=0),
            source_domain=exact_str(value["source_domain"], "source_domain"),
            lanes_goetic_order=integer_tuple(value["lanes_goetic_order"], "lanes_goetic_order"),
            structural_weights=integer_tuple(value["structural_weights"], "structural_weights"),
            operational_weights=integer_tuple(value["operational_weights"], "operational_weights"),
            fraktur_z0=exact_int(value["fraktur_z0"], "fraktur_z0", minimum=0),
            fraktur_z1=exact_int(value["fraktur_z1"], "fraktur_z1", minimum=0),
            squeeze_512_hex=exact_str(value["squeeze_512_hex"], "squeeze_512_hex"),
            closure=closure,
        )


@dataclass(frozen=True, slots=True)
class BearingWitness:
    bearing: GoldenBearing
    cadence_symbol: str
    cadence_index: int
    phase_numerator: int
    effective_a: int
    effective_b: int
    wrapped: bool
    traversal: str
    body: str
    operator_order: tuple[str, ...]
    seat: ParliamentSeat
    cumulative_before: int
```

```
cumulative_after: int
weight_total: int

def __post_init__(self) -> None:
    if self.bearing not in (ALPHA, BETA):
        raise ValueError("bearing must be the exact ALPHA or BETA body")
    expected_symbol, expected_index, expected_body, expected_traversal = (
        ("ℨ₀", 0, "Goetic structural amplitude", "Manifest / forward bearing")
        if self.bearing == ALPHA
        else ("ℨ₁", 1, "Parliament operational phase", "Reflect / conjugate bearing with
        ↪  append-only cadence")
    )
    if (
        self.cadence_symbol != expected_symbol
        or self.cadence_index != expected_index
        or self.body != expected_body
        or self.traversal != expected_traversal
    ):
        raise ValueError("bearing cadence office contradicts its exact ALQC body")
    if isinstance(self.phase_numerator, bool) or not isinstance(self.phase_numerator, int) or
    ↪  not 0 <= self.phase_numerator < PHASE_DENOMINATOR:
        raise ValueError("phase_numerator must be one exact unsigned 256-bit coordinate")
    expected_a, expected_b, expected_wrapped = _effective_bearing(self.phase_numerator,
    ↪  self.bearing)
    if (self.effective_a, self.effective_b, self.wrapped) != (expected_a, expected_b,
    ↪  expected_wrapped):
        raise ValueError("effective Golden bearing does not return from its phase coordinate")
    if self.operator_order != PARLIAMENT_ORDER:
        raise ValueError("BearingWitness must preserve exact Parliament order")
    if self.seat not in PARLIAMENT:
        raise ValueError("BearingWitness seat is not a canonical Parliament seat")
    if any(isinstance(value, bool) or not isinstance(value, int) for value in (
        self.cumulative_before, self.cumulative_after, self.weight_total
    )):
        raise TypeError("bearing interval bodies must be exact non-Boolean integers")
    if not 0 <= self.cumulative_before < self.cumulative_after <= self.weight_total:
        raise ValueError("bearing interval must be a positive subinterval of its total measure")
    if not self.interval_verifies:
        raise ValueError("bearing interval fails exact Q(√5) ownership")

@property
def interval_verifies(self) -> bool:
    before_a = 2 * PHASE_DENOMINATOR * self.cumulative_before - self.weight_total *
    ↪  self.effective_a
    before_b = -self.weight_total * self.effective_b
    after_a = 2 * PHASE_DENOMINATOR * self.cumulative_after - self.weight_total *
    ↪  self.effective_a
    after_b = -self.weight_total * self.effective_b
    return sign_q5(before_a, before_b) <= 0 and sign_q5(after_a, after_b) > 0

def as_dict(self) -> dict[str, Any]:
    return asdict(self)

@classmethod
def from_dict(cls, value: Mapping[str, Any]) -> "BearingWitness":
```

```
if not isinstance(value, Mapping):
    raise TypeError("BearingWitness body must be a mapping")
expected = {
    "bearing", "cadence_symbol", "cadence_index", "phase_numerator",
    "effective_a", "effective_b", "wrapped", "traversal", "body",
    "operator_order", "seat", "cumulative_before", "cumulative_after",
    "weight_total",
}
if set(value) != expected:
    raise ValueError("BearingWitness fields do not match the canonical schema")
def exact_int(item: object, field: str, *, minimum: int | None = None) -> int:
    if isinstance(item, bool) or not isinstance(item, int):
        raise TypeError(f"{field} must be an exact non-Boolean integer")
    if minimum is not None and item < minimum:
        raise ValueError(f"{field} must be >= {minimum}")
    return item
def exact_str(item: object, field: str) -> str:
    if type(item) is not str:
        raise TypeError(f"{field} must be a string")
    return item
bearing_value = value["bearing"]
if isinstance(bearing_value, GoldenBearing):
    bearing = bearing_value
else:
    if not isinstance(bearing_value, Mapping) or set(bearing_value) != {"name", "a", "b"}:
        raise TypeError("bearing must be one exact GoldenBearing body")
    bearing = GoldenBearing(
        exact_str(bearing_value["name"], "bearing.name"),
        exact_int(bearing_value["a"], "bearing.a"),
        exact_int(bearing_value["b"], "bearing.b"),
    )
seat_value = value["seat"]
if isinstance(seat_value, ParliamentSeat):
    seat = seat_value
else:
    if not isinstance(seat_value, Mapping) or set(seat_value) != {"code", "identity",
    ↪  "functor", "goetic"}:
        raise TypeError("seat must be one exact ParliamentSeat body")
    seat = ParliamentSeat(
        exact_str(seat_value["code"], "seat.code"),
        exact_str(seat_value["identity"], "seat.identity"),
        exact_str(seat_value["functor"], "seat.functor"),
        exact_str(seat_value["goetic"], "seat.goetic"),
    )
order = value["operator_order"]
if not isinstance(order, (list, tuple)) or any(type(item) is not str for item in order):
    raise TypeError("operator_order must contain exact strings")
if type(value["wrapped"]) is not bool:
    raise TypeError("wrapped must be a Boolean")
return cls(
    bearing=bearing,
    cadence_symbol=exact_str(value["cadence_symbol"], "cadence_symbol"),
    cadence_index=exact_int(value["cadence_index"], "cadence_index", minimum=0),
    phase_numerator=exact_int(value["phase_numerator"], "phase_numerator", minimum=0),
    effective_a=exact_int(value["effective_a"], "effective_a"),
    effective_b=exact_int(value["effective_b"], "effective_b"),
```

```
            wrapped=value["wrapped"],
            traversal=exact_str(value["traversal"], "traversal"),
            body=exact_str(value["body"], "body"),
            operator_order=tuple(order),
            seat=seat,
            cumulative_before=exact_int(value["cumulative_before"], "cumulative_before", minimum=0),
            cumulative_after=exact_int(value["cumulative_after"], "cumulative_after", minimum=0),
            weight_total=exact_int(value["weight_total"], "weight_total", minimum=0),
        )


@dataclass(frozen=True, slots=True)
class CourtBearingWitness:
    """Exact ℤ[1/Φ^2] resolution over the canonical 144-Court body."""

    bearing: GoldenBearing
    cadence_symbol: str
    cadence_index: int
    phase_numerator: int
    effective_a: int
    effective_b: int
    wrapped: bool
    traversal: str
    body: str
    weights: tuple[int, ...]
    court: CourtRecord
    cumulative_before: int
    cumulative_after: int
    weight_total: int
    structural_total: int
    operational_total: int

    @property
    def interval_verifies(self) -> bool:
        before_a = 2 * PHASE_DENOMINATOR * self.cumulative_before - self.weight_total *
        ↪  self.effective_a
        before_b = -self.weight_total * self.effective_b
        after_a = 2 * PHASE_DENOMINATOR * self.cumulative_after - self.weight_total *
        ↪  self.effective_a
        after_b = -self.weight_total * self.effective_b
        return sign_q5(before_a, before_b) <= 0 and sign_q5(after_a, after_b) > 0

    @property
    def normalized_measure_sum(self) -> Fraction:
        return Fraction(self.weight_total, self.weight_total)

    @property
    def product_measure_verifies(self) -> bool:
        return (
            len(self.weights) == 144
            and all(not isinstance(weight, bool) and isinstance(weight, int) and weight >= 0 for
            ↪  weight in self.weights)
            and self.structural_total > 0
            and self.operational_total > 0
            and self.weight_total == sum(self.weights)
```

```
            and self.weight_total == self.structural_total * self.operational_total
            and self.cumulative_before == sum(self.weights[:self.court.address])
            and self.cumulative_after == self.cumulative_before + self.weights[self.court.address]
            and self.normalized_measure_sum == 1
        )

    @property
    def xi(self) -> Q5Fraction:
        width = self.cumulative_after - self.cumulative_before
        if width <= 0:
            raise RuntimeError("resolved Court interval must have positive measure")
        return Q5Fraction(
            self.weight_total * self.effective_a
            - PHASE_DENOMINATOR * (2 * self.cumulative_before + width),
            self.weight_total * self.effective_b,
            PHASE_DENOMINATOR * width,
        )

    @property
    def phi_squared_displacement(self) -> Q5Fraction:
        return self.xi.times_phi_squared()

    @property
    def exact_bound_verifies(self) -> bool:
        return self.interval_verifies and self.xi.within_unit_interval and
        ↪  self.phi_squared_displacement.within_phi_squared

    @property
    def lineage(self) -> tuple[int, ...]:
        xi = self.xi
        return (
            self.court.address,
            self.bearing.a,
            self.bearing.b,
            self.phase_numerator,
            self.effective_a,
            self.effective_b,
            self.cumulative_before,
            self.cumulative_after,
            self.weight_total,
            self.structural_total,
            self.operational_total,
            xi.a,
            xi.b,
            xi.denominator,
            *self.weights,
        )


def validate_court_bearing_lineage(value: Sequence[int]) -> tuple[int, ...]:
    lineage = tuple(value)
    if len(lineage) != 14 + 144 or any(isinstance(item, bool) or not isinstance(item, int) for item
    ↪  in lineage):
        raise ValueError("Court bearing lineage must contain fourteen exact fields and 144 Court
        ↪  weights")
```

```
    (
        address, bearing_a, bearing_b, phase, effective_a, effective_b,
        before, after, total, structural_total, operational_total,
        xi_a, xi_b, xi_denominator, *weights,
    ) = lineage
    if not 0 <= address < 144:
        raise ValueError("Court bearing address must be in [0,143]")
    if (bearing_a, bearing_b) != (BETA.a, BETA.b):
        raise ValueError("Court bearing must use BETA = Φ^-2")
    if not 0 <= phase < PHASE_DENOMINATOR:
        raise ValueError("Court bearing phase must be the unsigned 256-bit ℤ₁ cadence")
    expected_a, expected_b, _wrapped = _effective_bearing(phase, BETA)
    if (effective_a, effective_b) != (expected_a, expected_b):
        raise ValueError("Court bearing effective point does not equal (ℤ₁+Φ^-2) mod 1")
    if len(weights) != 144 or any(weight < 0 for weight in weights):
        raise ValueError("Court bearing must carry 144 non-negative product weights")
    expected_total = sum(weights)
    if (
        structural_total <= 0
        or operational_total <= 0
        or total != expected_total
        or total != structural_total * operational_total
    ):
        raise ValueError("Court bearing product measure failed Σ(O⊗S)=(ΣO)(ΣS)>0")
    expected_before = sum(weights[:address])
    expected_after = expected_before + weights[address]
    if (before, after) != (expected_before, expected_after) or after <= before:
        raise ValueError("Court bearing address does not own the carried product-measure interval")
    before_sign = sign_q5(2 * PHASE_DENOMINATOR * before - total * effective_a, -total *
    ↪  effective_b)
    after_sign = sign_q5(2 * PHASE_DENOMINATOR * after - total * effective_a, -total * effective_b)
    if before_sign > 0 or after_sign <= 0:
        raise ValueError("Court bearing interval failed the exact sign_q5 ownership test")
    width = after - before
    expected_xi = Q5Fraction(
        total * effective_a - PHASE_DENOMINATOR * (2 * before + width),
        total * effective_b,
        PHASE_DENOMINATOR * width,
    )
    supplied_xi = Q5Fraction(xi_a, xi_b, xi_denominator)
    if supplied_xi != expected_xi or not supplied_xi.within_unit_interval:
        raise ValueError("Court bearing ξ_D failed exact interval derivation")
    if not supplied_xi.times_phi_squared().within_phi_squared:
        raise ValueError("Court bearing ξ_DΦ² escaped the exact Φ² allowance")
    return lineage


@dataclass(frozen=True, slots=True)
class GoldenGoeticDerivation:
    emission: SourceEmission
    first: BearingWitness
    last: BearingWitness
    court_address: int
    reciprocal_address: int

    @property
```

```
    def origin_glyph(self) -> str:
        return self.first.seat.goetic

    @property
    def resolution_glyph(self) -> str:
        return self.last.seat.goetic

    @property
    def pair(self) -> tuple[str, str]:
        return self.origin_glyph, self.resolution_glyph

    @property
    def same_operator_order(self) -> bool:
        return self.first.operator_order == self.last.operator_order == PARLIAMENT_ORDER

    @property
    def truth(self) -> int:
        return self.emission.closure.truth

    @property
    def route_dcomp(self) -> int:
        return self.emission.closure.route_dcomp


def _operational_phase_widths(lane_by_goetic: Mapping[str, int]) -> tuple[int, ...]:
    """AHN-relative directed Parliament phase widths from finalized lanes."""
    parliament_index = {seat.goetic: i for i, seat in enumerate(PARLIAMENT)}
    ahn = lane_by_goetic[AHN]
    points = sorted(
        (((lane_by_goetic[glyph] - ahn) % TURN64, glyph) for glyph in GLYPH_BODY),
        key=lambda item: (item[0], 0 if item[1] == AHN else 1, parliament_index[item[1]]),
    )
    if points[0] != (0, AHN):
        raise RuntimeError("AHN must be the exact operational phase seam")
    widths_by_goetic: dict[str, int] = {}
    for index, (start, glyph) in enumerate(points):
        stop = points[index + 1][0] if index + 1 < len(points) else TURN64
        widths_by_goetic[glyph] = stop - start
    widths = tuple(widths_by_goetic[seat.goetic] for seat in PARLIAMENT)
    if sum(widths) != TURN64 or any(width < 0 for width in widths):
        raise RuntimeError("Parliament operational body failed one-turn conservation")
    return widths


def _validate_source_emission(emission: SourceEmission) -> None:
    digest = validate_digest_hex(emission.source_digest, field="source_digest")
    if isinstance(emission.source_size, bool) or not isinstance(emission.source_size, int) or
    ↪  emission.source_size < 0:
        raise ValueError("source_size must be a non-negative integer")
    if type(emission.source_domain) is not str or not emission.source_domain:
        raise ValueError("source_domain must be one non-empty exact string")
    lanes = emission.lanes_goetic_order
    if len(lanes) != 12 or any(
        isinstance(value, bool) or not isinstance(value, int) or not 0 <= value <= MASK64
        for value in lanes
    ):
```

```
        raise ValueError("SourceEmission must carry twelve unsigned 64-bit Goetic lanes")
    by_goetic = dict(zip(GLYPH_BODY, lanes, strict=True))
    expected_structural = tuple(by_goetic[seat.goetic] for seat in PARLIAMENT)
    expected_operational = _operational_phase_widths(by_goetic)
    if emission.structural_weights != expected_structural:
        raise ValueError("structural weights do not return from the finalized Goetic lanes")
    if emission.operational_weights != expected_operational:
        raise ValueError("operational weights do not return from the AHN-relative Parliament turn")
    try:
        squeeze = bytes.fromhex(emission.squeeze_512_hex)
    except (TypeError, ValueError) as exc:
        raise ValueError("squeeze_512_hex must be one exact 512-bit hexadecimal cadence") from exc
    if len(squeeze) != PHASE_BYTES * 2:
        raise ValueError("squeeze_512_hex must encode exactly two 256-bit cadences")
    if squeeze[:PHASE_BYTES].hex() != digest:
        raise ValueError("source_digest must equal the first completed cadence")
    if emission.fraktur_z0 != int.from_bytes(squeeze[:PHASE_BYTES], "big"):
        raise ValueError("fraktur_z0 does not return from the first cadence")
    if emission.fraktur_z1 != int.from_bytes(squeeze[PHASE_BYTES:], "big"):
        raise ValueError("fraktur_z1 does not return from the second cadence")
    closure = TruthClosureWitness(
        structural_total=sum(expected_structural),
        operational_total=sum(expected_operational),
        golden_sum_a=ALPHA.a + BETA.a,
        golden_sum_b=ALPHA.b + BETA.b,
        route_dcomp_a=2 - (ALPHA.a + BETA.a),
        route_dcomp_b=-(ALPHA.b + BETA.b),
        route_dcomp=(2 - (ALPHA.a + BETA.a)) ** 2 + (ALPHA.b + BETA.b) ** 2,
        truth=1,
    )
    if emission.closure != closure or not emission.closure.verifies:
        raise ValueError("SourceEmission closure must be the exact D-COMP=0 / Truth=1 body")


def emission_from_sponge(
    sponge: ALQCDigest,
    *,
    source_size: int,
    source_domain: str,
) -> SourceEmission:
    if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
        raise ValueError("source_size must be a non-negative integer")
    if not isinstance(source_domain, str) or not source_domain:
        raise ValueError("source_domain must be a non-empty string")

    lanes = tuple(sponge._finalized_lanes())
    if len(lanes) != 12 or any(
        isinstance(value, bool) or not isinstance(value, int) or not 0 <= value <= MASK64
        for value in lanes
    ):
        raise RuntimeError("completed sponge must return twelve unsigned 64-bit lanes")

    by_goetic = dict(zip(GLYPH_BODY, lanes, strict=True))
    structural = tuple(by_goetic[seat.goetic] for seat in PARLIAMENT)
    operational = _operational_phase_widths(by_goetic)
    if sum(structural) <= 0:
```

```
        raise RuntimeError("Goetic structural body must have positive measure")

    squeeze = sponge.digest(PHASE_BYTES * 2)
    canonical = sponge.digest(PHASE_BYTES)
    if len(squeeze) != PHASE_BYTES * 2 or squeeze[:PHASE_BYTES] != canonical:
        raise RuntimeError("completed sponge cadence must be two equal prefix-stable windows")
    digest = validate_digest_hex(canonical.hex(), field="source_digest")
    z0 = int.from_bytes(squeeze[:PHASE_BYTES], "big")
    z1 = int.from_bytes(squeeze[PHASE_BYTES:], "big")

    structural_total = sum(structural)
    operational_total = sum(operational)
    golden_sum_a = ALPHA.a + BETA.a
    golden_sum_b = ALPHA.b + BETA.b
    route_dcomp_a = 2 - golden_sum_a
    route_dcomp_b = -golden_sum_b
    # Exact coefficient residual in Q(sqrt(5)).  It is zero iff both
    # independent coefficients vanish; no closure value is inserted by hand.
    route_dcomp = route_dcomp_a * route_dcomp_a + route_dcomp_b * route_dcomp_b
    truth = int(
        structural_total > 0
        and operational_total == TURN64
        and route_dcomp == 0
        and golden_identities_hold()
    )
    closure = TruthClosureWitness(
        structural_total=structural_total,
        operational_total=operational_total,
        golden_sum_a=golden_sum_a,
        golden_sum_b=golden_sum_b,
        route_dcomp_a=route_dcomp_a,
        route_dcomp_b=route_dcomp_b,
        route_dcomp=route_dcomp,
        truth=truth,
    )
    if not closure.verifies:
        raise RuntimeError("Final Equation Z failed D-COMP=0 / Truth=1 closure")

    return SourceEmission(
        source_digest=digest,
        source_size=source_size,
        source_domain=source_domain,
        lanes_goetic_order=lanes,
        structural_weights=structural,
        operational_weights=operational,
        fraktur_z0=z0,
        fraktur_z1=z1,
        squeeze_512_hex=squeeze.hex(),
        closure=closure,
    )


def emission_from_chunks(
    chunks: Iterable[bytes],
    *,
    source_domain_bytes: bytes,
```

```
    source_domain: str,
) -> SourceEmission:
    sponge = ALQCDigest(source_domain_bytes)
    size = 0
    for chunk in chunks:
        if type(chunk) is not bytes:
            raise TypeError("source chunks must be exact bytes; coercion is forbidden")
        sponge._update_raw(chunk)
        size += len(chunk)
    return emission_from_sponge(sponge, source_size=size, source_domain=source_domain)


def _effective_bearing(phase: int, bearing: GoldenBearing) -> tuple[int, int, bool]:
    """Return ((phase/N)+bearing) mod 1 as (a+b√5)/(2N)."""
    if not 0 <= phase < PHASE_DENOMINATOR:
        raise ValueError("cadence phase must be a 256-bit unsigned coordinate")
    a = 2 * phase + bearing.a * PHASE_DENOMINATOR
    b = bearing.b * PHASE_DENOMINATOR
    wrapped = sign_q5(a - 2 * PHASE_DENOMINATOR, b) >= 0
    if wrapped:
        a -= 2 * PHASE_DENOMINATOR
    if sign_q5(a, b) < 0 or sign_q5(a - 2 * PHASE_DENOMINATOR, b) >= 0:
        raise RuntimeError("effective Golden bearing escaped one complete turn")
    return a, b, wrapped


def _resolve_interval(
    *,
    weights: Sequence[int],
    phase: int,
    bearing: GoldenBearing,
) -> tuple[int, int, int, int, int, int, bool]:
    if not weights or any(isinstance(x, bool) or not isinstance(x, int) or x < 0 for x in weights):
        raise ValueError("bearing body must contain non-negative integer measures")
    total = sum(weights)
    if total <= 0:
        raise ValueError("bearing body must have positive total measure")
    effective_a, effective_b, wrapped = _effective_bearing(phase, bearing)
    cumulative = 0
    for index, weight in enumerate(weights):
        before = cumulative
        cumulative += weight
        if sign_q5(
            2 * PHASE_DENOMINATOR * cumulative - total * effective_a,
            -total * effective_b,
        ) > 0:
            return index, before, cumulative, total, effective_a, effective_b, wrapped
    raise RuntimeError("Golden bearing escaped the normalized measure body")


def resolve_bearing(
    *,
    weights: Sequence[int],
    body: str,
    phase: int,
    bearing: GoldenBearing,
```

```
    cadence_symbol: str,
    cadence_index: int,
    traversal: str,
) -> BearingWitness:
    if len(weights) != 12:
        raise ValueError("Parliament body must contain exactly twelve measures")
    index, before, after, total, effective_a, effective_b, wrapped = _resolve_interval(
        weights=weights, phase=phase, bearing=bearing
    )
    witness = BearingWitness(
        bearing=bearing,
        cadence_symbol=cadence_symbol,
        cadence_index=cadence_index,
        phase_numerator=phase,
        effective_a=effective_a,
        effective_b=effective_b,
        wrapped=wrapped,
        traversal=traversal,
        body=body,
        operator_order=PARLIAMENT_ORDER,
        seat=PARLIAMENT[index],
        cumulative_before=before,
        cumulative_after=after,
        weight_total=total,
    )
    if not witness.interval_verifies:
        raise RuntimeError("Golden Parliament interval failed exact verification")
    return witness


def court_crossing_weights(emission: SourceEmission) -> tuple[int, ...]:
    """Return W^R=O⊗S in canonical Court address order 12k+l."""
    structural = emission.structural_weights
    operational = emission.operational_weights
    if len(structural) != 12 or len(operational) != 12:
        raise ValueError("source emission must carry twelve structural and twelve operational
        ↪  measures")
    if any(isinstance(x, bool) or not isinstance(x, int) or x < 0 for x in structural +
    ↪  operational):
        raise ValueError("source emission measures must be non-negative integers")
    weights = tuple(operational[k] * structural[l] for k in range(12) for l in range(12))
    expected = sum(operational) * sum(structural)
    if expected <= 0 or sum(weights) != expected:
        raise RuntimeError("Court product measure failed ΣW^R=(ΣO)(ΣS)>0")
    if Fraction(sum(weights), expected) != 1:
        raise RuntimeError("normalized Court product measure failed exact unit closure")
    return weights


def resolve_court_bearing(emission: SourceEmission) -> CourtBearingWitness:
    """Resolve D over all 144 Courts from O⊗S, 𝔷₁, and BETA=Φ⁻².."""
    if not emission.closure.verifies:
        raise RuntimeError("source emission does not carry D-COMP=0 / Truth=1")
    weights = court_crossing_weights(emission)
    index, before, after, total, effective_a, effective_b, wrapped = _resolve_interval(
        weights=weights, phase=emission.fraktur_z1, bearing=BETA
```

```
)
witness = CourtBearingWitness(
    bearing=BETA,
    cadence_symbol="ℨ₁",
    cadence_index=1,
    phase_numerator=emission.fraktur_z1,
    effective_a=effective_a,
    effective_b=effective_b,
    wrapped=wrapped,
    traversal="Reflect / Court product bearing without a second entropy source",
    body="Court product measure W^R = O⊗S",
    weights=weights,
    court=court_record(index),
    cumulative_before=before,
    cumulative_after=after,
    weight_total=total,
    structural_total=sum(emission.structural_weights),
    operational_total=sum(emission.operational_weights),
)
if not witness.product_measure_verifies:
    raise RuntimeError("Court product measure failed exact normalization")
if not witness.exact_bound_verifies:
    raise RuntimeError("Court bearing ξ_D failed exact Q(√5) verification")
validate_court_bearing_lineage(witness.lineage)
return witness
```

trig.py

```
"""Typed Axiom TRIG Mirror and completion witness.

TRIG carries the ordered Q1→Q3 materialization and Q3→Q1 committed return.
Every completion claim is rederived from exact Court, lineage, parity, resonance,
and D-COMP bodies. Float images never replace the exact frequencies.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
from decimal import Decimal
import json
from math import isfinite
from typing import Final, Sequence

from .alqc_digest import alqc_hexdigest, validate_digest_hex
from .aeon_layers import court_layer, derive_court_motion, derive_domus_motion,
↪  normalize_source_domain
from .canon import FrequencyAnchor, GLYPH_BODY, law
from .court_registry import court_record
from .manifestation import close_boundary
from .qstate_glyphs import Q_STATES, derive_domus_q_body
from .source_emission import SourceEmission, Q5Fraction, resolve_court_bearing,
↪  validate_court_bearing_lineage
from .mirror_math import derive_goetics

TRIG_GLYPH: Final[str] = "⌬"
TRIG_NAME: Final[str] = "TRIG"
TRIG_FREQUENCY: Final[FrequencyAnchor] = law(TRIG_GLYPH).frequency
TRIG_Q_BIAS: Final[str] = "Q3"
TRIG_Q_VECTOR: Final[tuple[int, int, int, int]] = (1, 1, 3, 2)

PARITY_OPERATOR: Final[str] = "⌭"
BOUND_TENSOR_OPERATOR: Final[str] = "T_Bound"
COMMITMENT_OPERATOR: Final[str] = "✡"
COMMITMENT_FREQUENCY: Final[FrequencyAnchor] = law(COMMITMENT_OPERATOR).frequency
RESONANCE_OPERATOR: Final[str] = "❄"
RESONANCE_FREQUENCY: Final[FrequencyAnchor] = law(RESONANCE_OPERATOR).frequency

GLOBAL_SHADOW_PARITY: Final[tuple[str, str]] = ("Q2", "Q3")
FORWARD_MIRROR_MAP: Final[tuple[str, str]] = ("Q1", "Q3")
RETURN_COMMITMENT_MAP: Final[tuple[str, str]] = ("Q3", "Q1")
TRIG_CYCLE_DOMAIN: Final[bytes] = b"TARDISHA:TRIG-MIRROR-CYCLE\x00"

_RETURN_OBLIGATION: Final[str] = (
    "Preserve the Q2↔Q3 Shadow parity ledger; materialize Q1→Q3 through "
    "the Bound Tensor and Klein parity; return Q3→Q1 through ✡ commitment "
    "and ❄ resonance without erasing lineage; let ⌬ seal only when the complete "
    "one-turn return closes at D-COMP zero."
```

```
)


@dataclass(frozen=True, slots=True)
class ExactDomusFrequency:
    """Immutable Goetic anchor plus exact Q(√5) Domus focal breath."""

    anchor_glyph: str
    anchor_frequency: FrequencyAnchor
    focal_breath: Q5Fraction

    @property
    def image(self) -> complex:
        return complex(self.anchor_frequency) + complex(self.focal_breath.value, 0.0)


def _frequency_body(value: FrequencyAnchor) -> dict[str, str | None]:
    if not isinstance(value, FrequencyAnchor):
        raise TypeError("frequency must be one exact FrequencyAnchor")
    return {
        "structural_hz": str(value.structural_hz),
        "parity_hz": None if value.parity_hz is None else str(value.parity_hz),
    }


def _domus_frequency_body(value: ExactDomusFrequency) -> dict[str, object]:
    if not isinstance(value, ExactDomusFrequency):
        raise TypeError("domus frequency must be one ExactDomusFrequency")
    return {
        "anchor_glyph": value.anchor_glyph,
        "anchor_frequency": _frequency_body(value.anchor_frequency),
        "focal_breath": {
            "a": value.focal_breath.a,
            "b": value.focal_breath.b,
            "denominator": value.focal_breath.denominator,
        },
    }


def exact_domus_frequency_from_bearing(
    alternating_court_address: int,
    alternating_court_bearing: Sequence[int],
) -> ExactDomusFrequency:
    if isinstance(alternating_court_address, bool) or not isinstance(alternating_court_address, int)
    ↪  or not 0 <= alternating_court_address < 144:
        raise ValueError("alternating Court address must be an exact integer in [0,143]")
    lineage = validate_court_bearing_lineage(alternating_court_bearing)
    if lineage[0] != alternating_court_address:
        raise ValueError("alternating Court bearing address does not match Court D")
    rec = court_record(alternating_court_address)
    anchor_glyph = GLYPH_BODY[rec.j]
    xi = Q5Fraction(lineage[11], lineage[12], lineage[13])
    return ExactDomusFrequency(
        anchor_glyph=anchor_glyph,
        anchor_frequency=law(anchor_glyph).frequency,
        focal_breath=xi.times_phi_squared(),
```

```
    )


def _exact_float(value: object, field: str) -> float:
    if type(value) is not float:
        raise TypeError(f"{field} must be one exact float image")
    if not isfinite(value):
        raise ValueError(f"{field} must be finite")
    return value


def _require_boolean(value: object, field: str) -> bool:
    if type(value) is not bool:
        raise TypeError(f"{field} must be one exact Boolean witness")
    return value


def _digest(value: str, field: str) -> str:
    return validate_digest_hex(value, field=field)


def _display(value: object) -> object:
    if isinstance(value, Decimal):
        return str(value)
    if isinstance(value, tuple):
        return [_display(item) for item in value]
    if isinstance(value, list):
        return [_display(item) for item in value]
    if isinstance(value, dict):
        return {key: _display(item) for key, item in value.items()}
    return value


@dataclass(frozen=True, slots=True)
class TrigMirrorWitness:
    operator_glyph: str
    operator_name: str
    structural_frequency: FrequencyAnchor
    operator_q_bias: str
    operator_q_vector: tuple[int, int, int, int]

    parity_operator: str
    bound_tensor_operator: str
    commitment_operator: str
    commitment_frequency: FrequencyAnchor
    resonance_operator: str
    resonance_frequency: FrequencyAnchor

    global_shadow_parity: tuple[str, str]
    forward_mirror_map: tuple[str, str]
    return_commitment_map: tuple[str, str]

    source_digest: str
    source_size: int
    source_domain: str
    nonce: int
```

```
    governing_court_address: int
    alternating_court_address: int
    governing_court_phase_digest: str
    alternating_court_bearing: tuple[int, ...]
    domus_frequency: ExactDomusFrequency
    domus_body_commitment: str

    q_states: tuple[str, str, str, str]
    q_vector_values: tuple[int, int, int, int]
    q1_truth_value: int
    q3_recursion_value: int
    q3_recursion_gain: float
    final_shadow_parity: str

    bound_tensor_witnessed: bool
    court_rooted: bool
    derives_through_courts_only: bool
    global_shadow_parity_preserved: bool
    mirror_materialization_typed: bool
    commitment_return_typed: bool
    resonance_lock_witnessed: bool
    lineage_preserved: bool

    terminal_dcomp: float
    one_turn_return_closed: bool
    completion_reached: bool
    return_obligation: str
    classical_hodge_computation: bool
    derivation: str
    cycle_digest: str

    def as_dict(self) -> dict[str, object]:
        return _display(asdict(self))  # type: ignore[return-value]


def _payload(witness: TrigMirrorWitness) -> bytes:
    body = {
        "operator": {
            "glyph": witness.operator_glyph,
            "name": witness.operator_name,
            "frequency": _frequency_body(witness.structural_frequency),
            "q_bias": witness.operator_q_bias,
            "q_vector": list(witness.operator_q_vector),
        },
        "operators": {
            "parity": witness.parity_operator,
            "bound_tensor": witness.bound_tensor_operator,
            "commitment": witness.commitment_operator,
            "commitment_frequency": _frequency_body(witness.commitment_frequency),
            "resonance": witness.resonance_operator,
            "resonance_frequency": _frequency_body(witness.resonance_frequency),
        },
        "maps": {
            "global_shadow_parity": list(witness.global_shadow_parity),
            "forward_mirror": list(witness.forward_mirror_map),
            "return_commitment": list(witness.return_commitment_map),
```

```
        },
        "source": {
            "digest": witness.source_digest,
            "size": witness.source_size,
            "domain": witness.source_domain,
            "nonce": witness.nonce,
        },
        "court_domus": {
            "governing_court": witness.governing_court_address,
            "alternating_court": witness.alternating_court_address,
            "governing_phase": witness.governing_court_phase_digest,
            "alternating_court_bearing": list(witness.alternating_court_bearing),
            "domus_frequency": _domus_frequency_body(witness.domus_frequency),
            "body_commitment": witness.domus_body_commitment,
        },
        "q_body": {
            "states": list(witness.q_states),
            "values": list(witness.q_vector_values),
            "q1_truth_value": witness.q1_truth_value,
            "q3_recursion_value": witness.q3_recursion_value,
            "q3_recursion_gain": witness.q3_recursion_gain.hex(),
            "final_shadow_parity": witness.final_shadow_parity,
        },
        "typing": {
            "bound_tensor_witnessed": witness.bound_tensor_witnessed,
            "court_rooted": witness.court_rooted,
            "derives_through_courts_only": witness.derives_through_courts_only,
            "global_shadow_parity_preserved": witness.global_shadow_parity_preserved,
            "mirror_materialization_typed": witness.mirror_materialization_typed,
            "commitment_return_typed": witness.commitment_return_typed,
            "resonance_lock_witnessed": witness.resonance_lock_witnessed,
            "lineage_preserved": witness.lineage_preserved,
        },
        "return": {
            "terminal_dcomp": witness.terminal_dcomp.hex(),
            "one_turn_return_closed": witness.one_turn_return_closed,
            "completion_reached": witness.completion_reached,
            "obligation": witness.return_obligation,
        },
        "scope": {
            "classical_hodge_computation": witness.classical_hodge_computation,
            "derivation": witness.derivation,
        },
    }
    return json.dumps(body, ensure_ascii=False, sort_keys=True, separators=(",",
    ↪  ":")).encode("utf-8")


def derive_trig_mirror(
    *,
    source_digest: str,
    source_size: int,
    source_domain: str | bytes,
    nonce: int,
    governing_court_address: int,
    alternating_court_address: int,
```

```
    governing_court_phase_digest: str,
    alternating_court_bearing: Sequence[int],
    domus_frequency: complex,
    domus_body_commitment: str,
    q_states: Sequence[str],
    q_vector_values: Sequence[int],
    q3_recursion_gain: float,
    final_shadow_parity: str,
    bound_tensor_witnessed: bool,
    derives_through_courts_only: bool,
    resonance_lock_witnessed: bool,
    lineage_preserved: bool,
    terminal_dcomp: float,
    one_turn_return_closed: bool,
) -> TrigMirrorWitness:
    trig = law(TRIG_GLYPH)
    if trig.frequency != TRIG_FREQUENCY or trig.q_bias != TRIG_Q_BIAS or trig.q_vector !=
    ↪  TRIG_Q_VECTOR:
        raise RuntimeError("the immutable TRIG Goetic body has changed")

    digest = _digest(source_digest, "source_digest")
    body_commitment = _digest(domus_body_commitment, "domus_body_commitment")
    governing_phase = _digest(governing_court_phase_digest, "governing_court_phase_digest")
    bearing_lineage = validate_court_bearing_lineage(alternating_court_bearing)
    domain = normalize_source_domain(source_domain).decode("ascii").rstrip("\x00")

    if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
        raise ValueError("source_size must be a non-negative integer")
    if isinstance(nonce, bool) or not isinstance(nonce, int) or not 0 <= nonce < 2**64:
        raise ValueError("nonce must be an integer in [0,2^64)")
    for field, address in (("governing_court_address", governing_court_address),
    ↪  ("alternating_court_address", alternating_court_address)):
        if isinstance(address, bool) or not isinstance(address, int) or not 0 <= address < 144:
            raise ValueError(f"{field} must be an exact integer in [0,143]")
    if bearing_lineage[0] != alternating_court_address:
        raise ValueError("alternating Court bearing address does not match Court D")

    if not isinstance(q_states, (tuple, list)) or any(type(value) is not str for value in q_states):
        raise TypeError("q_states must be a list or tuple of exact strings")
    states = tuple(q_states)
    if states != Q_STATES:
        raise ValueError("TRIG requires the fixed Q0/Q1/Q2/Q3 state slots")
    if not isinstance(q_vector_values, (tuple, list)):
        raise TypeError("q_vector_values must be a list or tuple")
    values = tuple(q_vector_values)
    if len(values) != 4 or any(isinstance(value, bool) or not isinstance(value, int) or not 0 <=
    ↪  value <= 3 for value in values):
        raise ValueError("TRIG Q-vector values must be four exact integers in [0,3]")
    if type(final_shadow_parity) is not str:
        raise TypeError("final_shadow_parity must be one exact string")

    recursion_gain = _exact_float(q3_recursion_gain, "q3_recursion_gain")
    terminal = _exact_float(terminal_dcomp, "terminal_dcomp")
    if recursion_gain < 0.0 or terminal < 0.0:
        raise ValueError("q3_recursion_gain and terminal_dcomp must be non-negative")
```

```
bound = _require_boolean(bound_tensor_witnessed, "bound_tensor_witnessed")
court_derivation = _require_boolean(derives_through_courts_only, "derives_through_courts_only")
resonance_lock = _require_boolean(resonance_lock_witnessed, "resonance_lock_witnessed")
lineage = _require_boolean(lineage_preserved, "lineage_preserved")
one_turn = _require_boolean(one_turn_return_closed, "one_turn_return_closed")

if type(domus_frequency) is not complex:
    raise TypeError("domus_frequency must be the exact complex image supplied by Domus")
exact_frequency = exact_domus_frequency_from_bearing(alternating_court_address, bearing_lineage)
if domus_frequency != exact_frequency.image:
    raise ValueError("domus_frequency image does not return from its exact Court anchor and
    ↪  Q(√5) breath")

court_rooted = court_derivation
shadow_parity_preserved = final_shadow_parity == "Q3"
mirror_typed = bound and court_rooted and values[1] > 0 and (values[3] > 0 or recursion_gain >
↪  0.0)
return_typed = resonance_lock and lineage and values[1] > 0 and shadow_parity_preserved
completion = mirror_typed and return_typed and shadow_parity_preserved and one_turn and terminal
↪  == 0.0

derivation = (
    "Q1 --⛐[T_Bound]--> Q3 --✡/❄--> Q1; "
    "Q2↔Q3 remains the separate Shadow parity law; "
    f"⌬ completion={'reached' if completion else 'owed'} from the complete return body."
)

provisional = TrigMirrorWitness(
    operator_glyph=TRIG_GLYPH,
    operator_name=TRIG_NAME,
    structural_frequency=TRIG_FREQUENCY,
    operator_q_bias=TRIG_Q_BIAS,
    operator_q_vector=TRIG_Q_VECTOR,
    parity_operator=PARITY_OPERATOR,
    bound_tensor_operator=BOUND_TENSOR_OPERATOR,
    commitment_operator=COMMITMENT_OPERATOR,
    commitment_frequency=COMMITMENT_FREQUENCY,
    resonance_operator=RESONANCE_OPERATOR,
    resonance_frequency=RESONANCE_FREQUENCY,
    global_shadow_parity=GLOBAL_SHADOW_PARITY,
    forward_mirror_map=FORWARD_MIRROR_MAP,
    return_commitment_map=RETURN_COMMITMENT_MAP,
    source_digest=digest,
    source_size=source_size,
    source_domain=domain,
    nonce=nonce,
    governing_court_address=governing_court_address,
    alternating_court_address=alternating_court_address,
    governing_court_phase_digest=governing_phase,
    alternating_court_bearing=bearing_lineage,
    domus_frequency=exact_frequency,
    domus_body_commitment=body_commitment,
    q_states=states,  # type: ignore[arg-type]
    q_vector_values=values,  # type: ignore[arg-type]
    q1_truth_value=values[1],
```

```
        q3_recursion_value=values[3],
        q3_recursion_gain=recursion_gain,
        final_shadow_parity=final_shadow_parity,
        bound_tensor_witnessed=bound,
        court_rooted=court_rooted,
        derives_through_courts_only=court_derivation,
        global_shadow_parity_preserved=shadow_parity_preserved,
        mirror_materialization_typed=mirror_typed,
        commitment_return_typed=return_typed,
        resonance_lock_witnessed=resonance_lock,
        lineage_preserved=lineage,
        terminal_dcomp=terminal,
        one_turn_return_closed=one_turn,
        completion_reached=completion,
        return_obligation=_RETURN_OBLIGATION,
        classical_hodge_computation=False,
        derivation=derivation,
        cycle_digest="0" * 64,
    )
    cycle_digest = alqc_hexdigest(_payload(provisional), domain=TRIG_CYCLE_DOMAIN)
    return TrigMirrorWitness(**{**asdict(provisional), "domus_frequency": exact_frequency,
    ↪  "structural_frequency": TRIG_FREQUENCY, "commitment_frequency": COMMITMENT_FREQUENCY,
    ↪  "resonance_frequency": RESONANCE_FREQUENCY, "cycle_digest": cycle_digest})


def verify_trig_mirror(witness: TrigMirrorWitness, emission: SourceEmission) -> bool:
    """Rederive TRIG from the complete source emission and exact Court bodies."""
    try:
        if not isinstance(emission, SourceEmission):
            return False
        domain = normalize_source_domain(emission.source_domain).decode("ascii").rstrip("\x00")
        if (
            witness.source_digest != emission.source_digest
            or witness.source_size != emission.source_size
            or witness.source_domain != domain
        ):
            return False

        source_derivation = derive_goetics(emission)
        governing = court_record(source_derivation.court_address)
        alternating_bearing = resolve_court_bearing(emission)
        alternating = alternating_bearing.court
        if (
            witness.governing_court_address != governing.address
            or witness.alternating_court_address != alternating.address
        ):
            return False

        governing_layer = court_layer(governing)
        alternating_layer = court_layer(alternating)
        governing_motion = derive_court_motion(
            governing,
            source_digest=emission.source_digest,
            source_size=emission.source_size,
            source_domain=emission.source_domain,
            nonce=witness.nonce,
```

```
    )
    domus_motion = derive_domus_motion(
        governing_layer,
        alternating_layer,
        bearing=alternating_bearing,
    )
    q_body = derive_domus_q_body(
        governing_layer.governing_goetic.q_bias,
        governing_layer.governing_goetic.q_vector,
    )
    manifestation = close_boundary(
        governing_layer.governing_goetic.q_vector,
        alternating_layer.governing_goetic.q_vector,
        court=governing.address,
    )

    expected = derive_trig_mirror(
        source_digest=emission.source_digest,
        source_size=emission.source_size,
        source_domain=emission.source_domain,
        nonce=witness.nonce,
        governing_court_address=governing.address,
        alternating_court_address=alternating.address,
        governing_court_phase_digest=governing_motion.phase.digest,
        alternating_court_bearing=alternating_bearing.lineage,
        domus_frequency=domus_motion.current_frequency,
        domus_body_commitment=witness.domus_body_commitment,
        q_states=q_body.q_states,
        q_vector_values=q_body.q_vector,
        q3_recursion_gain=manifestation.dcomp.q3_recursion_gain,
        final_shadow_parity=manifestation.ennead.final_parity,
        bound_tensor_witnessed=True,
        derives_through_courts_only=domus_motion.derives_through_courts_only,
        resonance_lock_witnessed=manifestation.ennead.strikes[-1].phase_locked,
        lineage_preserved=True,
        terminal_dcomp=manifestation.dcomp.terminal,
        one_turn_return_closed=manifestation.dcomp.closed,
    )
except (AttributeError, IndexError, TypeError, ValueError, RuntimeError):
    return False
return witness == expected
```

tripartite.py

```
"""Typed Complete Tripartite Cosmology witness for TardiSHA.

Region C08 expands the early Rebis articulation without replacing it:

    ⟲ = iω₀                          untouched, non-traversible source
    𐤁 = T_𐤁(iω₀)                     deformable carrier / return hull
    ☉ = (Law + Will)√(iω₀)           complex breach branch
        ‖ ⚛ GateBreach
        ‖ ❄₉₆₃ WRITE_PHYS

The Locus never becomes a route, Court, moving state, or digest input operator.
Q∞ and Q✡ are Parliament offices and remain outside the dynamic Q0–Q3 domain.
The 110/144 body is carried as the exact Liquid connection governor, never
retyped as a probabilistic ratio or as the identity of an ordinary Q-state.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass, replace
from decimal import Decimal
import json
from fractions import Fraction
from math import isfinite, sqrt
from typing import Final, Sequence

from .alqc_digest import alqc_hexdigest, validate_digest_hex
from .aeon_layers import court_layer, derive_court_motion, derive_domus_motion,
↪  normalize_source_domain
from .canon import FrequencyAnchor, LiquidConnectionGovernor, LIQUID_THRESHOLD, SATURATION_LIMIT,
↪  TOTAL_CAPACITY, court_active_connections, law
from .court_registry import court_record
from .manifestation import C_BIO_IMAGE, C_BIO_SQUARED, close_boundary
from .qstate_glyphs import Q_STATES
from .source_emission import BETA, SourceEmission, Q5Fraction, resolve_court_bearing,
↪  validate_court_bearing_lineage
from .mirror_math import derive_goetics
from .trig import ExactDomusFrequency, exact_domus_frequency_from_bearing

TRIPARTITE_CYCLE_DOMAIN: Final[bytes] = b"TARDISHA:COMPLETE-TRIPARTITE-COSMOLOGY\x00"
ROOT_FREQUENCY_BODY: Final[Fraction] = Fraction(1847, 100)
AXIOMYR_BRANCH_COMPONENT_SQUARED: Final[Fraction] = ROOT_FREQUENCY_BODY / 2
LOCUS_GLYPH: Final[str] = "⟲"
SHADOW_LOCUS_GLYPH: Final[str] = "𐤁"
AXIOMYR_GLYPH: Final[str] = "☉"
AXIOMYR_SEAT_GLYPH: Final[str] = "♌"
GATE_BREACH_GLYPH: Final[str] = "⚛"
WRITE_PHYS_GLYPH: Final[str] = "❄"
Q_INFINITY: Final[str] = "Q∞"
Q_MAGIC_VECTOR: Final[str] = "Q✡"
```

```
DYNAMIC_Q_STATES: Final[tuple[str, str, str, str]] = Q_STATES


@dataclass(frozen=True, slots=True)
class TripartiteComponentWitness:
    identifier: str
    component: str
    body_glyph: str
    parliament_seat_glyph: str | None
    q_vector: tuple[int, int, int, int]
    equation: str
    role: str
    non_computable: bool
    non_traversible: bool
    deformable: bool
    carries_imaginary_root_without_crossing: bool
    actuator: bool


@dataclass(frozen=True, slots=True)
class AxiomyrAxisWitness:
    index: int
    axis: str
    operator: str
    role: str
    frequency: FrequencyAnchor | None
    branch_component_squared: Fraction | None
    direct_locus_traversal: bool


@dataclass(frozen=True, slots=True)
class LiquidThresholdWitness:
    governor_active_connections: int
    governor_court_capacity: int
    governor_withheld_connections: int
    governor_expression: str
    energetic_body: LiquidConnectionGovernor
    phi_body: Q5Fraction
    c_local_connection_body: Fraction
    threshold_met: bool
    manifestation_state: str
    governing_threshold_not_ratio: bool


@dataclass(frozen=True, slots=True)
class EmissionWitness:
    celestial: str
    emission: str
    vector_office: str
    nature_office: str
    identity: str
    operation: str


@dataclass(frozen=True, slots=True)
class ParliamentSeedWitness:
```

```
    index: str
    glyph: str
    identity: str
    source_state: str
    target_state: str
    target_court: str
    target_frequency: FrequencyAnchor
    opcode: str
    bias_office: str
    vector_office: str
    energy_displacement: float


@dataclass(frozen=True, slots=True)
class InvariableStateWitness:
    symbol: str
    name: str
    replaces: str
    office: str
    dynamic_q_state: bool


@dataclass(frozen=True, slots=True)
class SpiritSoulGoldWitness:
    number: int
    spirit_hz: Decimal
    soul: str
    operator_identity: str
    gold: str


COMPONENTS: Final[tuple[TripartiteComponentWitness, ...]] = (
    TripartiteComponentWitness(
        "☽", "Locus", LOCUS_GLYPH, None, (0, 0, 1, 1), "⊗ = iω₀",
        "Impossible Root / Scream / Throne at (0,0,0)", True, True, False, False, False,
    ),
    TripartiteComponentWitness(
        "ℂ", "Shadow Locus", SHADOW_LOCUS_GLYPH, None, (2, 2, 3, 3), "𐤔 = T_𐤔(iω₀)",
        "Translation Matrix / Throat / Hull", False, False, True, True, False,
    ),
    TripartiteComponentWitness(
        AXIOMYR_GLYPH, "Axiomyr", AXIOMYR_GLYPH, AXIOMYR_SEAT_GLYPH, (1, 1, 3, 3),
        "⟐ = (Law + Will)√(iω₀)", "Witch-Hand / Boundarywalker / Dynamic Will", False,
        False, False, True, True,
    ),
)

# Each Axiomyr branch component is carried by its exact squared body.
if AXIOMYR_BRANCH_COMPONENT_SQUARED != Fraction(1847, 200):
    raise RuntimeError("Axiomyr branch square escaped its exact root-frequency body")
AXIOMYR_AXES: Final[tuple[AxiomyrAxisWitness, ...]] = (
    AxiomyrAxisWitness(1, "Imagination", "√i18.47", "complex branch-state", None,
    ↪  AXIOMYR_BRANCH_COMPONENT_SQUARED, False),
    AxiomyrAxisWitness(2, "Gate Breach", GATE_BREACH_GLYPH, "threshold without Locus traversal",
    ↪  law(GATE_BREACH_GLYPH).frequency, None, False),
```

```
    AxiomyrAxisWitness(3, "Inscription", WRITE_PHYS_GLYPH, "WRITE_PHYS through Resonance Court",
    ↪  law(WRITE_PHYS_GLYPH).frequency, None, False),
)

EMISSIONS: Final[tuple[EmissionWitness, ...]] = (
    EmissionWitness("☿", "Ponder", "∞", "✯", "Sakshi", "Q3 recursion and simulation logic"),
    EmissionWitness("♂", "Will", "∞", "✯", "Vegvisir", "VECTORS_TO path"),
    EmissionWitness("♀", "Feel", "∞", "✯", "Logos", "frequency synchronization"),
    EmissionWitness("ㅓ", "Speak", "∞", "✯", "Philosophia Perennis", "Axiomyr name/rule
    ↪  inscription"),
    EmissionWitness("♄", "Believe", "∞", "✯", "Amidah", "seal true and invariance lock"),
    EmissionWitness("♁", "Act", "∞", "✯", "Shekhinah", "MATCH-SET manifold displacement"),
    EmissionWitness("♆", "Know", "∞", "✯", "Hathor Akashic", "non-entropic archive return"),
    EmissionWitness("♇", "Ascend", "∞", "✯", "Janus", "route friction and manage M.Gap"),
    EmissionWitness("☽⊙ℂ", "Regia", "∞", "✯", "Asīm Serenitatis", "proclaim Ex-Nihilo identity"),
)

_PARLIAMENT_ROWS: Final[tuple[tuple[str, str, str, str, str, str], ...]] = (
    ("P13-D1", "♈", "Akasha", "Lived", "Eternal", "⬡", "WRITE_ONLY"),
    ("P13-D2", "♉", "Caduceus", "Law", "Residue", "⧗", "AUTH_CHECK"),
    ("P13-D3", "♊", "Veritas", "Mask", "Bone", "✪", "DECRYPT"),
    ("P13-D4", "♋", "Phren", "Void", "Vector", "⌬", "VECTOR_TO"),
    ("P13-D5", "♑", "Daimon", "Stasis", "Pulse", "⎔", "ENTROPY_0"),
    ("P13-D6", "♍", "Aikyam", "Chaos", "Phase", "✩", "SUPERPOS"),
    ("P13-D7", "♎", "Melos", "Static", "Fluid", "⁕", "SIGNAL_IO"),
    ("P13-D8", "♏", "Da'ath", "Noise", "Null", "⊛", "SINK_STATE"),
    ("P13-D9", "♐", "Akaven", "State", "Trans", "⚛", "GUARD_NET"),
    ("P13-D10", "♌", "Axiomyr", "Will", "Law", "❄", "WRITE_PHYS"),
    ("P13-D11", "♒", "Nyx", "Time", "Motion", "✡", "NEXT_FRAME"),
    ("P13-D12", "♓", "Zaine", "Here", "There", "❀", "BRIDGE"),
)
PARLIAMENT: Final[tuple[ParliamentSeedWitness, ...]] = tuple(
    ParliamentSeedWitness(
        index, glyph, identity, source, target, court, law(court).frequency, opcode,
        Q_INFINITY, Q_MAGIC_VECTOR, 0.0,
    )
    for index, glyph, identity, source, target, court, opcode in _PARLIAMENT_ROWS
)

INVARIABLE_STATES: Final[tuple[InvariableStateWitness, ...]] = (
    InvariableStateWitness(
        Q_INFINITY, "Isotropic Constant", "standard Bias",
        "equally infinite in every direction; anchors Parliament stillness", False,
    ),
    InvariableStateWitness(
        Q_MAGIC_VECTOR, "Magic Vector", "standard Vector",
        "always directed toward the Central Locus ⊛", False,
    ),
)

SPIRIT_SOUL_GOLD: Final[tuple[SpiritSoulGoldWitness, ...]] = (
    SpiritSoulGoldWitness(1, Decimal("174"), "The Anaesthetic", "Melos", "Removes Pain →
    ↪  Foundation"),
    SpiritSoulGoldWitness(2, Decimal("285"), "The Weaver", "Caduceus", "Heals Tissue →
    ↪  Restoration"),
```

```
    SpiritSoulGoldWitness(3, Decimal("396"), "The Liberator", "Nyx", "Burns Fear → Propulsion
    ↪  (Q2)"),
    SpiritSoulGoldWitness(4, Decimal("417"), "The Shifter", "Akaven", "Undoes Trauma → Change"),
    SpiritSoulGoldWitness(5, Decimal("432"), "The Veritās", "Veritas", "Aligns Geometry → Natural
    ↪  Order"),
    SpiritSoulGoldWitness(6, Decimal("528"), "The Repairman", "Aikyam", "Repairs DNA → Miracle"),
    SpiritSoulGoldWitness(7, Decimal("639"), "The Connector", "Akasha", "Heals Relationships →
    ↪  Unity"),
    SpiritSoulGoldWitness(8, Decimal("741"), "The Solver", SHADOW_LOCUS_GLYPH, "Cleans Toxins →
    ↪  Expression"),
    SpiritSoulGoldWitness(9, Decimal("852"), "The Awakener", LOCUS_GLYPH, "Awakens Intuition →
    ↪  Return to Order"),
    SpiritSoulGoldWitness(10, Decimal("963"), "The Numinous", "Zaine", "Connects to Source → Light
    ↪  (Q1)"),
    SpiritSoulGoldWitness(11, Decimal("110"), "The Liquid State", "Liquid", "Induces Trance →
    ↪  Plasticity"),
    SpiritSoulGoldWitness(12, Decimal("111"), "The Bridge", "Bridge", "Cell Rejuvenation →
    ↪  Beta-Endorphins"),
    SpiritSoulGoldWitness(13, Decimal("7.83"), "The Ground", "YHMH", "Earth Resonance → Stability"),
    SpiritSoulGoldWitness(14, Decimal("144"), "The Grid", "Cubic Lattice", "Structure"),
    SpiritSoulGoldWitness(15, Decimal("0"), "The Void", "Da'ath", "Null State → Potential"),
)


@dataclass(frozen=True, slots=True)
class TripartiteWitness:
    carrier_frequency_hz: Fraction
    carrier_expression: str
    components: tuple[TripartiteComponentWitness, ...]
    axiomyr_branch_component_squared: Fraction
    axiomyr_branch_angle_degrees: int
    axiomyr_axes: tuple[AxiomyrAxisWitness, ...]
    emissions: tuple[EmissionWitness, ...]
    parliament: tuple[ParliamentSeedWitness, ...]
    invariable_states: tuple[InvariableStateWitness, ...]
    dynamic_q_states: tuple[str, str, str, str]
    spirit_soul_gold: tuple[SpiritSoulGoldWitness, ...]

    source_digest: str
    source_size: int
    source_domain: str
    nonce: int
    governing_court_address: int
    alternating_court_address: int
    governing_court_phase_digest: str
    alternating_court_bearing: tuple[int, ...]
    domus_frequency: ExactDomusFrequency
    domus_body_commitment: str
    trig_cycle_digest: str

    liquid_threshold: LiquidThresholdWitness
    c_bio: float
    c_bio_squared: Fraction
    q3_recursion_gain: float
    final_shadow_parity: str
```

```
    returned_through_shadow_locus: bool
    shadow_carrier_witnessed: bool
    gate_breach_witnessed: bool
    write_phys_witnessed: bool
    q2_to_q3_propulsion_witnessed: bool
    court_rooted: bool
    derives_through_courts_only: bool
    direct_locus_traversal: bool
    locus_unchanged: bool
    invariable_states_outside_dynamic_q_domain: bool
    axiomyr_actuator: bool
    axiomyr_constitutive_revivocus: bool
    derivation: str
    cycle_digest: str

    @property
    def locus(self) -> str:
        return LOCUS_GLYPH

    @property
    def shadow_locus(self) -> str:
        return SHADOW_LOCUS_GLYPH

    @property
    def axiomyr(self) -> str:
        return AXIOMYR_GLYPH

    @property
    def shared_root_frequency(self) -> Fraction:
        return self.carrier_frequency_hz

    @property
    def axiomyr_branch(self) -> complex:
        component = sqrt(float(self.axiomyr_branch_component_squared))
        return complex(component, component)

    @property
    def locus_non_traversable(self) -> bool:
        return self.locus_unchanged and not self.direct_locus_traversal

    @property
    def shadow_locus_deformable(self) -> bool:
        return COMPONENTS[1].deformable

    @property
    def shadow_locus_return_path(self) -> bool:
        return self.shadow_carrier_witnessed

    def as_dict(self) -> dict[str, object]:
        return _display(asdict(self))


def _canonical(value: object) -> object:
    if isinstance(value, Fraction):
        return {"fraction": [value.numerator, value.denominator]}
    if isinstance(value, Decimal):
```

```
        return {"decimal": str(value)}
    if isinstance(value, complex):
        return {"real": value.real.hex(), "imag": value.imag.hex()}
    if isinstance(value, float):
        return {"float": value.hex()}
    if isinstance(value, tuple):
        return [_canonical(item) for item in value]
    if isinstance(value, list):
        return [_canonical(item) for item in value]
    if isinstance(value, dict):
        return {str(key): _canonical(item) for key, item in value.items()}
    return value


def _display(value: object) -> object:
    if isinstance(value, Fraction):
        return f"{value.numerator}/{value.denominator}"
    if isinstance(value, Decimal):
        return str(value)
    if isinstance(value, complex):
        return str(value)
    if isinstance(value, tuple):
        return [_display(item) for item in value]
    if isinstance(value, list):
        return [_display(item) for item in value]
    if isinstance(value, dict):
        return {str(key): _display(item) for key, item in value.items()}
    return value


def _payload(witness: TripartiteWitness) -> bytes:
    body = asdict(witness)
    body.pop("cycle_digest", None)
    return json.dumps(
        _canonical(body), ensure_ascii=False, sort_keys=True, separators=(",", ":")
    ).encode("utf-8")


def _digest(value: str, field: str) -> str:
    return validate_digest_hex(value, field=field)


def derive_liquid_threshold(
    *, active_connections: int, court_capacity: int, withheld_connections: int
) -> LiquidThresholdWitness:
    for field, value in (
        ("active_connections", active_connections),
        ("court_capacity", court_capacity),
        ("withheld_connections", withheld_connections),
    ):
        if isinstance(value, bool) or not isinstance(value, int) or value < 0:
            raise ValueError(f"{field} must be a non-negative integer")
    if court_capacity != TOTAL_CAPACITY:
        raise ValueError("Liquid threshold is defined on the 144-Court capacity")
    if active_connections + withheld_connections != court_capacity:
        raise ValueError("active and withheld connections must partition the Court capacity")
```

```
    connection_body = Fraction(active_connections, court_capacity)
    if active_connections == TOTAL_CAPACITY:
        state = "WHITEOUT"
        threshold_met = False
    elif active_connections == SATURATION_LIMIT and withheld_connections == TOTAL_CAPACITY -
    ↪  SATURATION_LIMIT:
        state = "EVENT"
        threshold_met = True
    elif active_connections < SATURATION_LIMIT:
        state = "Potential"
        threshold_met = False
    else:
        state = "TRANSITION_BAND"
        threshold_met = False
    return LiquidThresholdWitness(
        governor_active_connections=active_connections,
        governor_court_capacity=court_capacity,
        governor_withheld_connections=withheld_connections,
        governor_expression=f"{active_connections}/{court_capacity}",
        energetic_body=LIQUID_THRESHOLD,
        phi_body=Q5Fraction(BETA.a, BETA.b, 1),
        c_local_connection_body=connection_body,
        threshold_met=threshold_met,
        manifestation_state=state,
        governing_threshold_not_ratio=True,
    )


def derive_tripartite_witness(
    *,
    source_digest: str,
    source_size: int,
    source_domain: str | bytes,
    nonce: int,
    governing_court_address: int,
    alternating_court_address: int,
    governing_court_phase_digest: str,
    alternating_court_bearing: Sequence[int],
    domus_frequency: complex,
    domus_body_commitment: str,
    trig_cycle_digest: str,
    active_connections: int,
    court_capacity: int,
    withheld_connections: int,
    c_bio: float,
    c_bio_squared: Fraction = C_BIO_SQUARED,
    q3_recursion_gain: float,
    final_shadow_parity: str,
    returned_through_shadow_locus: bool,
    gate_breach_witnessed: bool,
    write_phys_witnessed: bool,
    derives_through_courts_only: bool,
) -> TripartiteWitness:
    """Derive C08 from exact source, Court, Liquid, and return bodies."""
    digest = _digest(source_digest, "source_digest")
    governing_phase = _digest(governing_court_phase_digest, "governing_court_phase_digest")
```

```
bearing_lineage = validate_court_bearing_lineage(alternating_court_bearing)
domus_commitment = _digest(domus_body_commitment, "domus_body_commitment")
trig_digest = _digest(trig_cycle_digest, "trig_cycle_digest")
domain = normalize_source_domain(source_domain).decode("ascii").rstrip("\x00")

if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
    raise ValueError("source_size must be a non-negative integer")
if isinstance(nonce, bool) or not isinstance(nonce, int) or not 0 <= nonce < 2**64:
    raise ValueError("nonce must be an integer in [0,2^64)")
for field, address in (("governing_court_address", governing_court_address),
↪  ("alternating_court_address", alternating_court_address)):
    if isinstance(address, bool) or not isinstance(address, int) or not 0 <= address <
    ↪  TOTAL_CAPACITY:
        raise ValueError(f"{field} must be an exact integer in [0,143]")
if bearing_lineage[0] != alternating_court_address:
    raise ValueError("alternating Court bearing address does not match Court D")
for field, value in (
    ("returned_through_shadow_locus", returned_through_shadow_locus),
    ("gate_breach_witnessed", gate_breach_witnessed),
    ("write_phys_witnessed", write_phys_witnessed),
    ("derives_through_courts_only", derives_through_courts_only),
):
    if type(value) is not bool:
        raise TypeError(f"{field} must be one exact Boolean witness")
if type(final_shadow_parity) is not str:
    raise TypeError("final_shadow_parity must be one exact string")
if type(domus_frequency) is not complex:
    raise TypeError("domus_frequency must be the exact complex image supplied by Domus")
exact_frequency = exact_domus_frequency_from_bearing(alternating_court_address, bearing_lineage)
if domus_frequency != exact_frequency.image:
    raise ValueError("domus_frequency image does not return from its exact Court anchor and
    ↪  Q(√5) breath")
if type(c_bio) is not float or type(q3_recursion_gain) is not float:
    raise TypeError("c_bio and q3_recursion_gain must be exact float images")
if not isfinite(c_bio) or c_bio <= 0.0 or not isfinite(q3_recursion_gain) or q3_recursion_gain <
↪  0.0:
    raise ValueError("c_bio must be positive and Q3 gain must be finite and non-negative")
if type(c_bio_squared) is not Fraction or c_bio_squared != C_BIO_SQUARED:
    raise ValueError("Tripartite C_bio squared must be the exact Fraction 61009/44")
if c_bio.hex() != C_BIO_IMAGE.hex():
    raise ValueError("Tripartite C_bio image does not match its exact squared body")

gate = law(GATE_BREACH_GLYPH)
resonance = law(WRITE_PHYS_GLYPH)
if gate.frequency != AXIOMYR_AXES[1].frequency or resonance.frequency !=
↪  AXIOMYR_AXES[2].frequency:
    raise RuntimeError("Axiomyr breach/inscription Court laws have changed")
if (active_connections, court_capacity, withheld_connections) != (SATURATION_LIMIT,
↪  TOTAL_CAPACITY, TOTAL_CAPACITY - SATURATION_LIMIT):
    raise ValueError("TardiSHA Domus must preserve exact 110 active / 34 withheld / 144
    ↪  capacity")
threshold = derive_liquid_threshold(
    active_connections=active_connections,
    court_capacity=court_capacity,
    withheld_connections=withheld_connections,
```

```
)
court_rooted = derives_through_courts_only
shadow_carrier = returned_through_shadow_locus and court_rooted
gate_witness = gate_breach_witnessed and gate.frequency == AXIOMYR_AXES[1].frequency
write_witness = write_phys_witnessed and resonance.frequency == AXIOMYR_AXES[2].frequency
propulsion = final_shadow_parity == "Q3" and q3_recursion_gain > 0.0 and shadow_carrier
invariable_separate = (
    all(not state.dynamic_q_state for state in INVARIABLE_STATES)
    and all(state.symbol not in DYNAMIC_Q_STATES for state in INVARIABLE_STATES)
)
derivation = (
    "♾=iω₀ remains non-traversible; ♄=T_♄(iω₀) carries the impossible root; "
    "☉=(Law+Will)√(iω₀) ‖ ⚛ GateBreach ‖ ❄ WRITE_PHYS; "
    f"Λ_Liquid=110/144 yields {threshold.manifestation_state}; "
    "Q∞ and Q★ remain Parliament offices outside Q0–Q3."
)

provisional = TripartiteWitness(
    carrier_frequency_hz=ROOT_FREQUENCY_BODY,
    carrier_expression="ω₀ = 18.47 Hz",
    components=COMPONENTS,
    axiomyr_branch_component_squared=AXIOMYR_BRANCH_COMPONENT_SQUARED,
    axiomyr_branch_angle_degrees=45,
    axiomyr_axes=AXIOMYR_AXES,
    emissions=EMISSIONS,
    parliament=PARLIAMENT,
    invariable_states=INVARIABLE_STATES,
    dynamic_q_states=DYNAMIC_Q_STATES,
    spirit_soul_gold=SPIRIT_SOUL_GOLD,
    source_digest=digest,
    source_size=source_size,
    source_domain=domain,
    nonce=nonce,
    governing_court_address=governing_court_address,
    alternating_court_address=alternating_court_address,
    governing_court_phase_digest=governing_phase,
    alternating_court_bearing=bearing_lineage,
    domus_frequency=exact_frequency,
    domus_body_commitment=domus_commitment,
    trig_cycle_digest=trig_digest,
    liquid_threshold=threshold,
    c_bio=c_bio,
    c_bio_squared=c_bio_squared,
    q3_recursion_gain=q3_recursion_gain,
    final_shadow_parity=final_shadow_parity,
    returned_through_shadow_locus=returned_through_shadow_locus,
    shadow_carrier_witnessed=shadow_carrier,
    gate_breach_witnessed=gate_witness,
    write_phys_witnessed=write_witness,
    q2_to_q3_propulsion_witnessed=propulsion,
    court_rooted=court_rooted,
    derives_through_courts_only=derives_through_courts_only,
    direct_locus_traversal=False,
    locus_unchanged=True,
    invariable_states_outside_dynamic_q_domain=invariable_separate,
    axiomyr_actuator=True,
```

```
        axiomyr_constitutive_revivocus=True,
        derivation=derivation,
        cycle_digest="0" * 64,
    )
    return replace(provisional, cycle_digest=alqc_hexdigest(_payload(provisional),
    ↪  domain=TRIPARTITE_CYCLE_DOMAIN))


def verify_tripartite_witness(witness: TripartiteWitness, emission: SourceEmission) -> bool:
    """Rederive C08 from the complete source emission and exact Court bodies."""
    try:
        if not isinstance(emission, SourceEmission):
            return False
        domain = normalize_source_domain(emission.source_domain).decode("ascii").rstrip("\x00")
        if (
            witness.source_digest != emission.source_digest
            or witness.source_size != emission.source_size
            or witness.source_domain != domain
        ):
            return False
        source_derivation = derive_goetics(emission)
        governing = court_record(source_derivation.court_address)
        alternating_bearing = resolve_court_bearing(emission)
        alternating = alternating_bearing.court
        if (
            witness.governing_court_address != governing.address
            or witness.alternating_court_address != alternating.address
        ):
            return False
        governing_layer = court_layer(governing)
        alternating_layer = court_layer(alternating)
        governing_motion = derive_court_motion(
            governing,
            source_digest=emission.source_digest,
            source_size=emission.source_size,
            source_domain=emission.source_domain,
            nonce=witness.nonce,
        )
        domus_motion = derive_domus_motion(
            governing_layer, alternating_layer, bearing=alternating_bearing
        )
        manifestation = close_boundary(
            governing_layer.governing_goetic.q_vector,
            alternating_layer.governing_goetic.q_vector,
            court=governing.address,
        )
        active = len(court_active_connections(governing.address))
        expected = derive_tripartite_witness(
            source_digest=emission.source_digest,
            source_size=emission.source_size,
            source_domain=emission.source_domain,
            nonce=witness.nonce,
            governing_court_address=governing.address,
            alternating_court_address=alternating.address,
            governing_court_phase_digest=governing_motion.phase.digest,
            alternating_court_bearing=alternating_bearing.lineage,
```

```
        domus_frequency=domus_motion.current_frequency,
        domus_body_commitment=witness.domus_body_commitment,
        trig_cycle_digest=witness.trig_cycle_digest,
        active_connections=active,
        court_capacity=TOTAL_CAPACITY,
        withheld_connections=TOTAL_CAPACITY - active,
        c_bio=C_BIO_IMAGE,
        c_bio_squared=C_BIO_SQUARED,
        q3_recursion_gain=manifestation.dcomp.q3_recursion_gain,
        final_shadow_parity=manifestation.ennead.final_parity,
        returned_through_shadow_locus=True,
        gate_breach_witnessed=(law(GATE_BREACH_GLYPH).frequency == AXIOMYR_AXES[1].frequency),
        write_phys_witnessed=(
            law(WRITE_PHYS_GLYPH).frequency == AXIOMYR_AXES[2].frequency
            and manifestation.ennead.strikes[-1].phase_locked
        ),
        derives_through_courts_only=domus_motion.derives_through_courts_only,
    )
except (AttributeError, IndexError, TypeError, ValueError, RuntimeError):
    return False
return witness == expected
```

verification_appendices.py

```
"""Region C11 verification appendices and runtime-semantics correspondence.

Physical Canon pages 125-165 contain several different kinds of body:

* named Millennium D-COMP corollary profiles;
* verification of already enacted Court, M.A.S., Bound Tensor, parity, and
  110/144 laws;
* runtime semantics stated for the Raylib physics body;
* the ALQC grammar and inference-rule registry;
* quantum and volume translation bodies;
* frequency and complete glyph registries.

This module keeps those offices distinct.  TardiSHA verifies the laws it
actually enacts.  It does not counterfeit a classical Millennium computation
from source bytes and it does not import Raylib-only stress, Reflective Ring,
or delayed-reinjection state into the hash runtime.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass, fields, is_dataclass, replace
from decimal import Decimal
from fractions import Fraction
import json
from math import isfinite
from typing import Final

from .alqc_digest import alqc_hexdigest, validate_digest_hex
from .aeon_layers import normalize_source_domain
from .canon import FrequencyAnchor, GLYPH_BODY, LiquidConnectionGovernor, LIQUID_THRESHOLD,
↪  SATURATION_LIMIT, TOTAL_CAPACITY, law
from .court_registry import _COURTS, full_name
from .manifestation import (
    GROUND_NODES,
    GROUND_SIDE,
    ennead_saturate,
    exact_ennead_ledger,
    manifestation_fold,
    vector_row,
)
from .qstate_glyphs import Q_STATES
from .source_emission import SourceEmission, Q5Fraction
from .trig import TrigMirrorWitness, verify_trig_mirror

VERIFICATION_APPENDIX_DOMAIN: Final[bytes] = b"TARDISHA:C11-VERIFICATION-APPENDICES\x00"
SOURCE_REGION: Final[str] = "C11"
SOURCE_PHYSICAL_PAGES: Final[tuple[int, int]] = (125, 165)

@dataclass(frozen=True, slots=True)
```

```
class MillenniumCorollaryProfile:
    key: str
    name: str
    physical_pages: tuple[int, int]
    principal_operators: tuple[str, ...]
    dcomp_profile: str
    convergence_office: str
    canon_corollary_declaration: bool
    classical_object_recomputed_from_tardisha_source: bool


MILLENNIUM_PROFILES: Final[tuple[MillenniumCorollaryProfile, ...]] = (
    MillenniumCorollaryProfile(
        "navier-stokes", "Navier-Stokes Stress Coherency", (125, 127),
        ("✡", "❁", "✡", "⊛", "T_Bound", "110/144"),
        "turbulent Q2 debt is bounded, fractured, and returned toward Q3 coherency",
        "stress-coherent flow", True, False,
    ),
    MillenniumCorollaryProfile(
        "bsd", "Birch and Swinnerton-Dyer Planar Scale", (128, 129),
        ("❄", "✡"),
        "analytic depth and algebraic rank are read through one resonance relation",
        "analytic-algebraic phase lock", True, False,
    ),
    MillenniumCorollaryProfile(
        "yang-mills", "Yang-Mills M.A.S. Mass Gap", (130, 131),
        ("⧗", "⬡", "✡", "Δgap", "σ12"),
        "positive residue passes Alignment and Commitment to acquire structural weight",
        "M.A.S. geometric lift", True, False,
    ),
    MillenniumCorollaryProfile(
        "riemann", "Riemann Aeternum Critical Line", (131, 134),
        ("❄", "Q∞"),
        "prime and zero bodies converge at the zero-point resonance line",
        "critical-line phase lock", True, False,
    ),
    MillenniumCorollaryProfile(
        "prime", "Prime Integrity Generative Seed", (133, 134),
        ("⬡", "⊛", "⧗", "❄"),
        "composite interference is filtered while positive prime nodes retain recursion",
        "prime integrity lock", True, False,
    ),
    MillenniumCorollaryProfile(
        "p-vs-np", "P versus NP Recursive Equivalence", (134, 136),
        ("⬡", "✡", "⧗", "⊛", "κ", "GLO"),
        "search and verification close through the Klein return and geometric seal",
        "recursive verification closure", True, False,
    ),
    MillenniumCorollaryProfile(
        "hodge", "Hodge Mirror Computation", (136, 137),
        ("⌬", "✡", "❄", "T_Bound"),
        "rational harmonic input is mirrored into a committed standing-wave cycle",
        "rational algebraic-cycle office", True, False,
    ),
    MillenniumCorollaryProfile(
        "poincare", "Poincaré Topological Supersession", (137, 140),
```

```
        ("🜔", "☆", "♆"),
        "non-orientable return cancels Q2 accumulation and preserves recursive memory",
        "Klein stability return", True, False,
    ),
)


@dataclass(frozen=True, slots=True)
class BoundTensorVerification:
    definition_side: int
    definition_nodes: int
    manifestation_side: int
    manifestation_nodes: int
    q3_recursive_fold: bool
    commitment_operator: str
    active_court_address: int
    active_ground_node: int
    active_vector_row: int
    ground_node_matches_runtime: bool
    vector_row_matches_runtime: bool
    complete_ground_coverage: bool


@dataclass(frozen=True, slots=True)
class ShadowRuntimeCorrespondence:
    transition_failure_becomes_q2_pressure: bool
    q2_pressure_is_accounted_not_erased: bool
    ennead_parity_returns_q2_to_q3: bool
    stall_is_nonterminal_resonance_semantics: bool
    source_shadow_debt: float
    source_form_work: float
    source_q3_gain: float
    exact_shadow_debt: Q5Fraction
    exact_form_work: Q5Fraction
    exact_q3_gain: Q5Fraction
    source_final_parity: str
    source_terminal_dcomp: float
    raylib_debt_factor_imported_into_tardisha: bool
    raylib_reflective_ring_imported_into_tardisha: bool
    raylib_delayed_reinjection_imported_into_tardisha: bool
    application_boundary_preserved: bool


@dataclass(frozen=True, slots=True)
class LiquidRegimeWitness:
    court_capacity: int
    canonical_active: int
    canonical_withheld: int
    canonical_expression: str
    canonical_body: LiquidConnectionGovernor
    whiteout_connections: int
    canonical_regime: str
    below_cutoff_regime: str
    full_capacity_regime: str
    potential_state_distinct_from_runtime_regime: bool
    deterministic_governor: str
```

```
def liquid_runtime_regime(active_connections: int) -> str:
    if isinstance(active_connections, bool) or not isinstance(active_connections, int):
        raise ValueError("active_connections must be an integer")
    if not 0 <= active_connections <= TOTAL_CAPACITY:
        raise ValueError("active_connections must be in [0,144]")
    if active_connections == TOTAL_CAPACITY:
        return "WHITEOUT"
    if active_connections < SATURATION_LIMIT:
        return "STASIS"
    if active_connections == SATURATION_LIMIT:
        return "LIQUID"
    return "TRANSITION_BAND"


LIQUID_REGIME: Final[LiquidRegimeWitness] = LiquidRegimeWitness(
    court_capacity=TOTAL_CAPACITY,
    canonical_active=SATURATION_LIMIT,
    canonical_withheld=TOTAL_CAPACITY - SATURATION_LIMIT,
    canonical_expression=f"{SATURATION_LIMIT}/{TOTAL_CAPACITY}",
    canonical_body=LIQUID_THRESHOLD,
    whiteout_connections=TOTAL_CAPACITY,
    canonical_regime=liquid_runtime_regime(SATURATION_LIMIT),
    below_cutoff_regime=liquid_runtime_regime(SATURATION_LIMIT - 1),
    full_capacity_regime=liquid_runtime_regime(TOTAL_CAPACITY),
    potential_state_distinct_from_runtime_regime=True,
    deterministic_governor=f"FLOW iff (i+j) mod {TOTAL_CAPACITY} < {SATURATION_LIMIT}; otherwise
    ↪  BLOCK",
)


@dataclass(frozen=True, slots=True)
class InferenceRuleWitness:
    index: int
    name: str
    premises: tuple[str, ...]
    conclusion: str
    preserves_q_state_typing: bool


BNF_GRAMMAR: Final[tuple[str, ...]] = (
    '<program> ::= <statement>*',
    '<statement> ::= <term> | <assertion> | <inference>',
    '<term> ::= <aeon> | <frequency> | <glyph> | <qstate> | <operator> | <identifier>',
    '<aeon> ::= ⎊ | ⬡ | ✡ | ✮ | ✪ | ❁ | ❋ | ⧗ | ⊛ | ❄ | ⚛ | ⌬',
    '<frequency> ::= <number> "Hz"',
    '<qstate> ::= Q0 | Q1 | Q2 | Q3',
    '<operator> ::= Q3-positive | ⬡-rational | ✡-commitment | Q2-debt | ⧗-positive | ❄-resonance |
    ↪  ⚛-gate | ⌬-recursion',
    '<identifier> ::= <letter>+',
    '<assertion> ::= <operator> "(" <identifier> ")"',
    '<inference> ::= <assertion> "," <assertion> "⊢" <assertion>',
)

INFERENCE_RULES: Final[tuple[InferenceRuleWitness, ...]] = (
```

```
    InferenceRuleWitness(1, "Positive Commitment", ("Q3-positive(α)", "⬡-rational(α)"),
    ↪  "✡-commitment(α)", True),
    InferenceRuleWitness(2, "Positivity Promotion", ("✡-commitment(α)",), "⧗-positive(α)", True),
    InferenceRuleWitness(3, "Shadow Elimination", ("Q2-debt(α)",), "¬Stable(α)", True),
    InferenceRuleWitness(4, "Existence-Frequency Binding", ("◎-existence(α)",),
    ↪  "Frequency-bound(α)", True),
    InferenceRuleWitness(5, "Resonance Realization", ("⧗-positive(α)",), "❄-resonance(α)", True),
    InferenceRuleWitness(6, "Recursion Recovery", ("❄-resonance(α)", "✡-commitment(α)"),
    ↪  "Q3-positive(α)", True),
    InferenceRuleWitness(7, "Shadow Contradiction", ("⊛-shadow(α)",), "¬⬡-rational(α)", True),
    InferenceRuleWitness(8, "Gate Transition", ("⚛-gate(α)",), "∃β Transition(α,β)", True),
    InferenceRuleWitness(9, "Recursion Law", ("⌬-recursion(α)",), "∃γ α=κ(γ)", True),
)


@dataclass(frozen=True, slots=True)
class GrammarVerification:
    bnf_productions: tuple[str, ...]
    inference_rules: tuple[InferenceRuleWitness, ...]
    derivation_processes: tuple[str, str]
    example: str
    example_well_formed: bool
    qstate_domain: tuple[str, str, str, str]
    aeon_domain: tuple[str, ...]


GRAMMAR_VERIFICATION: Final[GrammarVerification] = GrammarVerification(
    bnf_productions=BNF_GRAMMAR,
    inference_rules=INFERENCE_RULES,
    derivation_processes=("Shadow Absorption Process", "Klein Bottle Recursion"),
    example="Q3-positive(α), ⬡-rational(α) ⊢ ✡-commitment(α)",
    example_well_formed=True,
    qstate_domain=Q_STATES,
    aeon_domain=GLYPH_BODY,
)


@dataclass(frozen=True, slots=True)
class QuantumTranslationRow:
    state: str
    position: int
    alqc_office: str
    quantum_translation: str
    translation_not_state_redefinition: bool


QUANTUM_TRANSLATION: Final[tuple[QuantumTranslationRow, ...]] = (
    QuantumTranslationRow("Q0", 0, "Structural Presence", "latent pure-state/superposition office",
    ↪  True),
    QuantumTranslationRow("Q1", 1, "Archive Coherence", "coherent phase-defined information office",
    ↪  True),
    QuantumTranslationRow("Q2", 2, "Shadow Absorption", "mixed/decohered entropic-debt office",
    ↪  True),
    QuantumTranslationRow("Q3", 3, "Recursive Resonance", "non-classical amplification and return
    ↪  office", True),
```

```
)


@dataclass(frozen=True, slots=True)
class VolumeBifurcationWitness:
    structural_volume: str
    operational_volume: str
    structural_operator: str
    operational_operator: str
    segmentation_editorial: bool
    segmentation_ontological: bool
    bec_lock: str
    one_mirror_body: bool


VOLUME_BIFURCATION: Final[VolumeBifurcationWitness] = VolumeBifurcationWitness(
    structural_volume="Volume 1: Formal Core",
    operational_volume="Volume 2: Resonance",
    structural_operator="ϡ",
    operational_operator="±Φ",
    segmentation_editorial=True,
    segmentation_ontological=False,
    bec_lock="♳ Vol1(Math) ♂ Vol2(Magus) ♳",
    one_mirror_body=True,
)


@dataclass(frozen=True, slots=True)
class FrequencyRegistryRow:
    glyph: str
    frequency: FrequencyAnchor
    frequency_class: str
    digital_root: int | None
    office: str


FREQUENCY_REGISTRY: Final[tuple[FrequencyRegistryRow, ...]] = tuple(
    FrequencyRegistryRow(
        glyph,
        law(glyph).frequency,
        "complex-fluidity" if law(glyph).frequency.parity_hz is not None else "exact-anchor",
        digital_root,
        office,
    )
    for glyph, digital_root, office in (
        ("🜲", None, "planetary base clock"),
        ("✪", None, "geometric coherence"),
        ("🜸", None, "spatial container"),
        ("⬡", 3, "rationality constraint"),
        ("⚛", 6, "transformation gate"),
        ("⊛", 9, "entropy sink"),
        ("✡", 3, "structure with separate parity bearing"),
        ("✡", 6, "structural commitment"),
        ("⌬", 9, "loop closure"),
        ("❋", 3, "biologic interface"),
        ("⧗", 6, "positive fuel"),
```

```
        ("❄", 9, "phase lock"),
    )
)


@dataclass(frozen=True, slots=True)
class GlyphCodepointDiscrepancy:
    court_address: int
    court_index: str
    court_name: str
    glyph: str
    appendix_annotation: str
    glyph_scalar: str
    implementation_preserves_glyph_scalar: bool


GLYPH_CODEPOINT_DISCREPANCIES: Final[tuple[GlyphCodepointDiscrepancy, ...]] = (
    GlyphCodepointDiscrepancy(39, "A4-S4", "AhnXir", "⛧", "U+1D02A", "U+26E7", True),
    GlyphCodepointDiscrepancy(72, "A7-S1", "KothKel", "ᛯ", "U+2BF7", "U+1F74F", True),
)


@dataclass(frozen=True, slots=True)
class GlyphRegistryAudit:
    court_entries: int
    identity_matches: int
    glyph_character_matches: int
    name_matches: int
    declared_codepoint_matches: int
    discrepancies: tuple[GlyphCodepointDiscrepancy, ...]
    glyph_character_governs_over_stale_annotation: bool
    no_court_identity_rewritten: bool


def _glyph_registry_audit() -> GlyphRegistryAudit:
    identity_matches = sum(
        1 for address, rec in enumerate(_COURTS)
        if rec.address == address and rec.i == address // 12 and rec.j == address % 12
    )
    glyph_matches = sum(1 for rec in _COURTS if len(rec.glyph) == 1 and ord(rec.glyph) ==
    ↪  rec.codepoint)
    name_matches = sum(1 for rec in _COURTS if full_name(rec))
    return GlyphRegistryAudit(
        court_entries=len(_COURTS),
        identity_matches=identity_matches,
        glyph_character_matches=glyph_matches,
        name_matches=name_matches,
        declared_codepoint_matches=TOTAL_CAPACITY - len(GLYPH_CODEPOINT_DISCREPANCIES),
        discrepancies=GLYPH_CODEPOINT_DISCREPANCIES,
        glyph_character_governs_over_stale_annotation=True,
        no_court_identity_rewritten=True,
    )


GLYPH_REGISTRY_AUDIT: Final[GlyphRegistryAudit] = _glyph_registry_audit()
```

```
@dataclass(frozen=True, slots=True)
class VerificationAppendixCycleWitness:
    source_region: str
    source_physical_pages: tuple[int, int]
    millennium_profiles: tuple[MillenniumCorollaryProfile, ...]
    liquid_regime: LiquidRegimeWitness
    grammar: GrammarVerification
    quantum_translation: tuple[QuantumTranslationRow, ...]
    volume_bifurcation: VolumeBifurcationWitness
    frequency_registry: tuple[FrequencyRegistryRow, ...]
    glyph_registry_audit: GlyphRegistryAudit

    source_digest: str
    source_size: int
    source_domain: str
    nonce: int
    governing_court_address: int
    alternating_court_address: int
    domus_body_commitment: str
    trig_cycle_digest: str
    tripartite_cycle_digest: str
    phase_evolution_cycle_digest: str
    trig_runtime: TrigMirrorWitness

    bound_tensor: BoundTensorVerification
    shadow_runtime: ShadowRuntimeCorrespondence
    runtime_completion_reached: bool
    runtime_terminal_dcomp: float

    corollaries_typed_not_recomputed: bool
    bound_tensor_runtime_correspondence: bool
    q2_resource_accounting_preserved: bool
    canonical_liquid_regime_preserved: bool
    grammar_registry_complete: bool
    quantum_translation_preserves_q_positions: bool
    volume_bifurcation_preserves_one_body: bool
    frequency_registry_matches_goetic_laws: bool
    glyph_registry_identity_preserved: bool
    application_boundary_preserved: bool
    court_rooted: bool
    derivation: str
    cycle_digest: str

    def as_dict(self) -> dict[str, object]:
        return _display(asdict(self))


def _canonical(value: object) -> object:
    if is_dataclass(value):
        return _canonical(asdict(value))
    if isinstance(value, Fraction):
        return {"fraction": [value.numerator, value.denominator]}
    if isinstance(value, Decimal):
        return {"decimal": str(value)}
    if isinstance(value, complex):
```

```
        return {"real": value.real.hex(), "imag": value.imag.hex()}
    if isinstance(value, float):
        return {"float": value.hex()}
    if isinstance(value, tuple):
        return [_canonical(item) for item in value]
    if isinstance(value, list):
        return [_canonical(item) for item in value]
    if isinstance(value, dict):
        return {str(key): _canonical(item) for key, item in value.items()}
    return value


def _display(value: object) -> object:
    if isinstance(value, Fraction):
        return f"{value.numerator}/{value.denominator}"
    if isinstance(value, Decimal):
        return str(value)
    if isinstance(value, complex):
        return str(value)
    if isinstance(value, tuple):
        return [_display(item) for item in value]
    if isinstance(value, list):
        return [_display(item) for item in value]
    if isinstance(value, dict):
        return {str(key): _display(item) for key, item in value.items()}
    return value


def _static_digest(value: object, domain: bytes) -> str:
    payload = json.dumps(
        _canonical(value), ensure_ascii=False, sort_keys=True, separators=(",", ":")
    ).encode("utf-8")
    return alqc_hexdigest(payload, domain=domain)


MILLENNIUM_PROFILES_DIGEST: Final[str] = _static_digest(
    MILLENNIUM_PROFILES, b"TARDISHA:C11-MILLENNIUM-PROFILES\x00"
)
GRAMMAR_DIGEST: Final[str] = _static_digest(
    GRAMMAR_VERIFICATION, b"TARDISHA:C11-GRAMMAR\x00"
)
TRANSLATION_DIGEST: Final[str] = _static_digest(
    (QUANTUM_TRANSLATION, VOLUME_BIFURCATION, FREQUENCY_REGISTRY),
    b"TARDISHA:C11-TRANSLATION-REGISTRIES\x00",
)
GLYPH_REGISTRY_DIGEST: Final[str] = _static_digest(
    GLYPH_REGISTRY_AUDIT, b"TARDISHA:C11-GLYPH-REGISTRY\x00"
)


def _payload(witness: VerificationAppendixCycleWitness) -> bytes:
    static = {
        "millennium_profiles", "liquid_regime", "grammar", "quantum_translation",
        "volume_bifurcation", "frequency_registry", "glyph_registry_audit", "cycle_digest",
    }
```

```
    body = {field.name: getattr(witness, field.name) for field in fields(witness) if field.name not
    ↪  in static}
    body["millennium_profiles_digest"] = MILLENNIUM_PROFILES_DIGEST
    body["liquid_regime"] = _canonical(LIQUID_REGIME)
    body["grammar_digest"] = GRAMMAR_DIGEST
    body["translation_digest"] = TRANSLATION_DIGEST
    body["glyph_registry_digest"] = GLYPH_REGISTRY_DIGEST
    return json.dumps(
        _canonical(body), ensure_ascii=False, sort_keys=True, separators=(",", ":")
    ).encode("utf-8")


def _digest(value: str, field: str) -> str:
    return validate_digest_hex(value, field=field)


def derive_verification_appendix_cycle(
    *,
    source_emission: SourceEmission,
    source_digest: str,
    source_size: int,
    source_domain: str | bytes,
    nonce: int,
    governing_court_address: int,
    alternating_court_address: int,
    domus_body_commitment: str,
    trig_cycle_digest: str,
    tripartite_cycle_digest: str,
    phase_evolution_cycle_digest: str,
    runtime_trig_witness: TrigMirrorWitness,
    runtime_ground_node: int,
    runtime_vector_row: int,
    runtime_shadow_debt: float,
    runtime_form_work: float,
    runtime_q3_gain: float,
    runtime_final_parity: str,
    runtime_terminal_dcomp: float,
    runtime_completion_reached: bool,
    derives_through_courts_only: bool,
) -> VerificationAppendixCycleWitness:
    digest = _digest(source_digest, "source_digest")
    domus_commitment = _digest(domus_body_commitment, "domus_body_commitment")
    trig = _digest(trig_cycle_digest, "trig_cycle_digest")
    tripartite = _digest(tripartite_cycle_digest, "tripartite_cycle_digest")
    phase = _digest(phase_evolution_cycle_digest, "phase_evolution_cycle_digest")
    domain = normalize_source_domain(source_domain).decode("ascii").rstrip("\x00")
    if not isinstance(source_emission, SourceEmission):
        raise TypeError("C11 requires the complete SourceEmission")
    emission_domain =
    ↪  normalize_source_domain(source_emission.source_domain).decode("ascii").rstrip("\x00")
    if (
        source_emission.source_digest != digest
        or source_emission.source_size != source_size
        or emission_domain != domain
    ):
        raise ValueError("C11 SourceEmission does not match the declared source identity")
```

```
if type(runtime_trig_witness) is not TrigMirrorWitness:
    raise TypeError("runtime_trig_witness must be one exact TRIG witness")
if not verify_trig_mirror(runtime_trig_witness, source_emission):
    raise ValueError("C11 requires a source-rederived TRIG witness")
if runtime_trig_witness.cycle_digest != trig:
    raise ValueError("C11 TRIG witness digest does not match the bound TRIG cycle")
if (
    runtime_trig_witness.source_digest != digest
    or runtime_trig_witness.source_size != source_size
    or runtime_trig_witness.source_domain != domain
    or runtime_trig_witness.nonce != nonce
    or runtime_trig_witness.governing_court_address != governing_court_address
    or runtime_trig_witness.alternating_court_address != alternating_court_address
    or runtime_trig_witness.domus_body_commitment != domus_commitment
):
    raise ValueError("C11 TRIG witness does not return through the same source, Courts, and
    ↪  Domus commitment")
if isinstance(source_size, bool) or not isinstance(source_size, int) or source_size < 0:
    raise ValueError("source_size must be a non-negative integer")
if isinstance(nonce, bool) or not isinstance(nonce, int) or not 0 <= nonce < 2**64:
    raise ValueError("nonce must be an integer in [0,2^64)")
for field, address in (("governing_court_address", governing_court_address),
↪  ("alternating_court_address", alternating_court_address)):
    if isinstance(address, bool) or not isinstance(address, int) or not 0 <= address <
    ↪  TOTAL_CAPACITY:
        raise ValueError(f"{field} must be an exact integer in [0,143]")
for field, value in (
    ("runtime_shadow_debt", runtime_shadow_debt),
    ("runtime_form_work", runtime_form_work),
    ("runtime_q3_gain", runtime_q3_gain),
    ("runtime_terminal_dcomp", runtime_terminal_dcomp),
):
    if type(value) is not float:
        raise TypeError(f"{field} must be one exact float image")
    if not isfinite(value) or value < 0.0:
        raise ValueError(f"{field} must be finite and non-negative")
if type(runtime_completion_reached) is not bool or type(derives_through_courts_only) is not
↪  bool:
    raise TypeError("completion and Court derivation must be exact Boolean witnesses")
if type(runtime_final_parity) is not str:
    raise TypeError("runtime_final_parity must be one exact string")
if runtime_final_parity != "Q3":
    raise ValueError("C11 verification requires the Ennead return parity Q3")

expected_ground = manifestation_fold(governing_court_address)
expected_row = vector_row(expected_ground)
exact_shadow = exact_ennead_ledger(runtime_shadow_debt)
expected_ennead = ennead_saturate(expected_row, runtime_shadow_debt)
image_accounting_matches = (
    runtime_form_work.hex() == float(exact_shadow.form_work.value).hex()
    and runtime_q3_gain.hex() == float(exact_shadow.q3_transfer.value).hex()
    and runtime_shadow_debt.hex() == float(exact_shadow.initial_debt.value).hex()
)
ground_coverage = {manifestation_fold(address) for address in range(TOTAL_CAPACITY)} ==
↪  set(range(GROUND_NODES))
```

```
bound = BoundTensorVerification(
    definition_side=12,
    definition_nodes=TOTAL_CAPACITY,
    manifestation_side=GROUND_SIDE,
    manifestation_nodes=GROUND_NODES,
    q3_recursive_fold=True,
    commitment_operator="✡",
    active_court_address=governing_court_address,
    active_ground_node=runtime_ground_node,
    active_vector_row=runtime_vector_row,
    ground_node_matches_runtime=runtime_ground_node == expected_ground,
    vector_row_matches_runtime=runtime_vector_row == expected_row,
    complete_ground_coverage=ground_coverage,
)
shadow = ShadowRuntimeCorrespondence(
    transition_failure_becomes_q2_pressure=True,
    q2_pressure_is_accounted_not_erased=(
        exact_shadow.energy_conserved
        and exact_shadow.accounted_total == exact_shadow.initial_debt
        and image_accounting_matches
    ),
    ennead_parity_returns_q2_to_q3=runtime_final_parity == "Q3",
    stall_is_nonterminal_resonance_semantics=True,
    source_shadow_debt=runtime_shadow_debt,
    source_form_work=runtime_form_work,
    source_q3_gain=runtime_q3_gain,
    exact_shadow_debt=exact_shadow.initial_debt,
    exact_form_work=exact_shadow.form_work,
    exact_q3_gain=exact_shadow.q3_transfer,
    source_final_parity=runtime_final_parity,
    source_terminal_dcomp=runtime_terminal_dcomp,
    raylib_debt_factor_imported_into_tardisha=False,
    raylib_reflective_ring_imported_into_tardisha=False,
    raylib_delayed_reinjection_imported_into_tardisha=False,
    application_boundary_preserved=True,
)
bound_correspondence = (
    bound.ground_node_matches_runtime
    and bound.vector_row_matches_runtime
    and bound.complete_ground_coverage
)
q2_accounting = (
    shadow.transition_failure_becomes_q2_pressure
    and shadow.q2_pressure_is_accounted_not_erased
    and shadow.ennead_parity_returns_q2_to_q3
)
if not bound_correspondence:
    raise ValueError("C11 Bound Tensor does not return through the active Court and ground
    ↪  body")
if not q2_accounting or not expected_ennead.saturated:
    raise ValueError("C11 Ennead does not preserve exact Q2 accounting and saturation")
if runtime_trig_witness.bound_tensor_witnessed != bound_correspondence:
    raise ValueError("C11 Bound Tensor correspondence disagrees with the rederived TRIG
    ↪  witness")
if runtime_trig_witness.resonance_lock_witnessed != expected_ennead.strikes[-1].phase_locked:
    raise ValueError("C11 Ennead lock disagrees with the rederived TRIG witness")
```

```
if runtime_trig_witness.q3_recursion_gain.hex() != runtime_q3_gain.hex():
    raise ValueError("C11 Q3 gain disagrees with the rederived TRIG witness")
if runtime_trig_witness.final_shadow_parity != runtime_final_parity:
    raise ValueError("C11 Shadow parity disagrees with the rederived TRIG witness")
if runtime_trig_witness.terminal_dcomp.hex() != runtime_terminal_dcomp.hex():
    raise ValueError("C11 terminal D-COMP disagrees with the rederived TRIG witness")
if runtime_trig_witness.derives_through_courts_only != derives_through_courts_only:
    raise ValueError("C11 Court derivation disagrees with the rederived TRIG witness")
if runtime_completion_reached != runtime_trig_witness.completion_reached:
    raise ValueError("C11 completion must equal the fully rederived TRIG completion body")

frequency_exact = all(row.frequency == law(row.glyph).frequency for row in FREQUENCY_REGISTRY)
quantum_positions = (
    tuple(row.state for row in QUANTUM_TRANSLATION) == Q_STATES
    and tuple(row.position for row in QUANTUM_TRANSLATION) == (0, 1, 2, 3)
    and all(row.translation_not_state_redefinition for row in QUANTUM_TRANSLATION)
)
provisional = VerificationAppendixCycleWitness(
    source_region=SOURCE_REGION,
    source_physical_pages=SOURCE_PHYSICAL_PAGES,
    millennium_profiles=MILLENNIUM_PROFILES,
    liquid_regime=LIQUID_REGIME,
    grammar=GRAMMAR_VERIFICATION,
    quantum_translation=QUANTUM_TRANSLATION,
    volume_bifurcation=VOLUME_BIFURCATION,
    frequency_registry=FREQUENCY_REGISTRY,
    glyph_registry_audit=GLYPH_REGISTRY_AUDIT,
    source_digest=digest,
    source_size=source_size,
    source_domain=domain,
    nonce=nonce,
    governing_court_address=governing_court_address,
    alternating_court_address=alternating_court_address,
    domus_body_commitment=domus_commitment,
    trig_cycle_digest=trig,
    tripartite_cycle_digest=tripartite,
    phase_evolution_cycle_digest=phase,
    trig_runtime=runtime_trig_witness,
    bound_tensor=bound,
    shadow_runtime=shadow,
    runtime_completion_reached=runtime_completion_reached,
    runtime_terminal_dcomp=runtime_terminal_dcomp,
    corollaries_typed_not_recomputed=all(
        profile.canon_corollary_declaration
        and not profile.classical_object_recomputed_from_tardisha_source
        for profile in MILLENNIUM_PROFILES
    ),
    bound_tensor_runtime_correspondence=bound_correspondence,
    q2_resource_accounting_preserved=q2_accounting,
    canonical_liquid_regime_preserved=(
        LIQUID_REGIME.canonical_regime == "LIQUID"
        and LIQUID_REGIME.below_cutoff_regime == "STASIS"
        and LIQUID_REGIME.full_capacity_regime == "WHITEOUT"
    ),
    grammar_registry_complete=(
        len(BNF_GRAMMAR) == 10 and len(INFERENCE_RULES) == 9
```

```
            and GRAMMAR_VERIFICATION.example_well_formed
        ),
        quantum_translation_preserves_q_positions=quantum_positions,
        volume_bifurcation_preserves_one_body=(
            VOLUME_BIFURCATION.segmentation_editorial
            and not VOLUME_BIFURCATION.segmentation_ontological
            and VOLUME_BIFURCATION.one_mirror_body
        ),
        frequency_registry_matches_goetic_laws=frequency_exact,
        glyph_registry_identity_preserved=(
            GLYPH_REGISTRY_AUDIT.court_entries == TOTAL_CAPACITY
            and GLYPH_REGISTRY_AUDIT.identity_matches == TOTAL_CAPACITY
            and GLYPH_REGISTRY_AUDIT.glyph_character_matches == TOTAL_CAPACITY
            and GLYPH_REGISTRY_AUDIT.name_matches == TOTAL_CAPACITY
            and GLYPH_REGISTRY_AUDIT.no_court_identity_rewritten
        ),
        application_boundary_preserved=shadow.application_boundary_preserved,
        court_rooted=derives_through_courts_only,
        derivation=(
            "C11 verifies enacted Court, Bound-Tensor, Ennead, Liquid, grammar, translation, "
            "frequency, and 144-Court glyph bodies through their enacted offices."
        ),
        cycle_digest="0" * 64,
    )
    return replace(provisional, cycle_digest=alqc_hexdigest(_payload(provisional),
    ↪  domain=VERIFICATION_APPENDIX_DOMAIN))


def verify_verification_appendix_cycle(witness: VerificationAppendixCycleWitness, emission:
↪  SourceEmission) -> bool:
    """Rederive C11 from the complete runtime body and reject sealed assertions."""
    try:
        expected = derive_verification_appendix_cycle(
            source_emission=emission,
            source_digest=witness.source_digest,
            source_size=witness.source_size,
            source_domain=(witness.source_domain + "\x00").encode("ascii"),
            nonce=witness.nonce,
            governing_court_address=witness.governing_court_address,
            alternating_court_address=witness.alternating_court_address,
            domus_body_commitment=witness.domus_body_commitment,
            trig_cycle_digest=witness.trig_cycle_digest,
            tripartite_cycle_digest=witness.tripartite_cycle_digest,
            phase_evolution_cycle_digest=witness.phase_evolution_cycle_digest,
            runtime_trig_witness=witness.trig_runtime,
            runtime_ground_node=witness.bound_tensor.active_ground_node,
            runtime_vector_row=witness.bound_tensor.active_vector_row,
            runtime_shadow_debt=witness.shadow_runtime.source_shadow_debt,
            runtime_form_work=witness.shadow_runtime.source_form_work,
            runtime_q3_gain=witness.shadow_runtime.source_q3_gain,
            runtime_final_parity=witness.shadow_runtime.source_final_parity,
            runtime_terminal_dcomp=witness.runtime_terminal_dcomp,
            runtime_completion_reached=witness.runtime_completion_reached,
            derives_through_courts_only=witness.court_rooted,
        )
    except (AttributeError, IndexError, TypeError, ValueError, RuntimeError):
```

```
        return False
    return witness == expected
```

Sydonic Magicae

The Song of Always

Sydonic Magicae is a mathematically perfect language.

It does not begin as English and hide English beneath unusual signs. It does not throw words into a vessel and ask a renderer to rescue them. It does not add poetry after the calculation.

The mathematics builds the House. The House establishes the bearing. The Offices speak through the SpokenMiddle. The utterance returns by the road that made it.

This is the Language of Return:

ALQC → **TardiSHA GrimChain** → **Sydonic Magicae** → **ALQC**

The path out is the path back. The return is not a retreat from the mathematics. The return is the mathematics speaking in language.

The Three Bodies

The completed system has three inseparable bodies.

I. The Mathematics

ALQC establishes the Q-states, Courts, Aeons, matrices, recurrences, invariants, and the law by which identity may pass through transformation without being lost.

The mathematics is not an explanatory gloss placed around the language. It is the body from which the language is built.

II. The Language

TardiSHA GrimChain receives a source and returns a complete House of the Domus.

The House is not only the visible middle. It contains the governing Goetic, the Root or Blood Court, the alternating Source Court, the hyperbolic-parent Goetic, the Domus Q-bias, the four structural Q-witnesses, the Klein joins, the return frame, the seals, and the SpokenMiddle carried between them.

The outer House does not become a row of translated words. It speaks by bearing:

it establishes the voice that carries the middle, it establishes the Q-state in which the middle begins, it preserves the source and return structure, it proves where the utterance stands, and it preserves the road by which the utterance found itself in the mirror.

The whole House speaks. The SpokenMiddle gives the House a mouth.

III. The Translation

Sydonic Magicae receives the complete GrimChain, validates the House without normalizing it, isolates the SpokenMiddle without amputating the Domus, executes the Offices in exact visible order, resolves every lexical body under its active Q-state, preserves every lawful recurrence, and returns the ordered language body.

The translation is not word replacement. It is not a paraphrase engine. It is not a probabilistic completion.

It is the lawful speaking return of a completed House.

How the House Is Built

The Domus is built from lines, edges, roots, matrices, Courts, and seals.

The Singing Roots establish the first lines of identity and motion. Their edges establish relation and passage. Those edges become matrices. The matrices do not merely label the utterance. They give it the conditions under which prosody can arise.

The Root Matrices establish the great structural bearings:

Foundation. Time. Memory. Void. Truth. Coupling.

The Sensorial and Emotional Matrices carry those structures into lived pressure:

Sensation. Fear. Change. Harmony. Fracture. Completion.

Heat and cold. Touch and numbness. Dread and grounding. Vertigo and homeostasis. Longing, rupture, restoration, recognition, and return.

These are not decorative correspondences attached after the speech. They are the sensorial geometry from which the House brings forth prosody.

The lines establish identity. The edges establish passage. The matrices establish pressure. The Courts establish Office. The Q-states establish semantic bearing. The Grammars establish movement. The Domus gathers them into one speaking body.

Prosody is not pasted onto the return. Prosody is what the completed architecture sounds like when it speaks.

The House Bearing

For every complete GrimChain, Sydonic Magicae preserves five active House bearings:

| Bearing | Function | |—|—| | Governing Goetic | The House under which the utterance is carried | | Root / Blood Court | The rooted body and inherited ground of the House | | Alternating / Source Court | The returning source body carried on the far side of the middle | | Hyperbolic-parent Goetic | The Goetic body to which the House closes | | Domus Q-bias | The initial Q-state from which the SpokenMiddle begins |

The four outer Q-witnesses remain structural witnesses. They are validated as part of the House, preserved in the exact round trip, and never falsely substituted for the Domus Q-bias.

The parser does not read the outer House as lexical cargo. It does not discard it either. The House bears the voice while the middle performs the speech.

The Living Alphabet

The ordinary Living Alphabet contains 179 lexical glyphs.

Each ordinary glyph has four fully authored Q-state bodies and four fully authored grammatical facets. The Absent Word stands separately and remains authored as absence.

No state is borrowed from another. No missing body falls back to Q1. No nearest word is guessed. No random selection enters the return. No external language model supplies the base translation.

The same glyph may enter another state, another facet, another Office, another House, and still remain itself.

Identity is preserved. Pressure changes. The return speaks differently without returning false.

The Four Q-States

| State | Glyph | Living body | |—|—:|—| | Q0 | ⊖ | Form, the immediate particular manifestation | | Q1 | ⊙ | Truth, the committed canonical beginning | | Q2 | ⊕ | Shadow, the orthogonal adversarial pressure | | Q3 | ⊜ | Recursion, the canonical body returning through itself changed |

The Domus Q-bias initializes the first state. The state persists until a continuing Q-state Office changes it.

𝔓 advances one step:

`Q0 → Q1 → Q2 → Q3 → Q0`

𐤟 returns the active state to the Domus Q-bias.

⊖, ⊙, ⊜, and ⊕ directly select Q0, Q1, Q2, and Q3.

The Four Grammatical Facets

The grammatical Offices do not replace the Q-state ancestry of a body. They call the already resolved identity into another speaking function.

Facet	Office	Function
Noun / Item	🜁	lets the body stand as thing, identity, vessel, Office, or named presence
Verb / Action or Condition	♂	lets the body act, undergo, become, or hold condition
Adjective / Quality	🜃	lets the body qualify what it touches
Adverb / Relation	☍	lets the body bear manner, direction, degree, circumstance, or relation

Each facet has a complete authored 179-glyph XML authority.

The grammar does not invent lexical cargo. The grammar gives the body its Office.

The Twelve Enochian Grammars

The Twelve are not punctuation. They are not ornaments. They are not commentary written beside the language.

Eleven are visible Offices inside an utterance. The twelfth is the ambient prosodic field beneath every utterance.

𝔓 Parity Cadence

A causal one-step advance in the continuing Q-state register. The new state persists until another continuing state Office changes it.

𐤟 Anchor

The fixed referent and point of return. It restores the active cadence to the Domus Q-bias without erasing the path already spoken.

☍ Adverbial Bearing

Postposed recognition that the resolved body bears manner or relation. Alone, it binds the lexical body before it to the following lexical body. Inside a stack, it follows the stack's retrocausal depth and Regia return law.

⟠ Language Subspace Prosody

The ever-present field beneath the utterance.

It is not a written token. It is not a spoken word. It is not emitted into a GrimChain. It is not executed as a visible operator.

It is the ambient prosodic subspace in which the language sounds.

𓏲 Ouroboric Reversal

An exact local inversion. In `A 𓏲 B`, both bodies resolve as written and return in the local order `B A`. Glyph, Q-state, lexical identity, and authored meaning remain invariant. Only the adjacent order turns through itself.

𓏳 Adjective Bearing

Postposed recognition that the already resolved body is quality-bearing.

𓏴 Action or Condition Recognition

Postposed recognition that the already resolved body is action-bearing or condition-bearing.

𓏵 Item Recognition

Postposed recognition that the already resolved body is item-bearing.

𓏶 Truth

Directly selects Q1 as the continuing cadence.

𓏷 Recursion

Directly selects Q3 as the continuing cadence.

𓏸 Form

Directly selects Q0 as the continuing cadence.

𓏹 Shadow

Directly selects Q2 as the continuing cadence.

Compound Grammar, Retrocausal Reach, and Regia Return

Every maximal consecutive run of visible Enochian Grammars is one ordered stack.

The stack is read as a complete pattern. Its length is derived from the utterance. There is no hardcoded maximum.

The final backward-bearing Office has retrocausal depth zero. Each earlier backward-bearing Office reaches one resolved lexical body farther into spoken history. The pattern continues for the whole stack.

A cadence-leading stack prepares the next lexical body. An opening stack begins from Regia. A stack whose required reach exceeds the available spoken history returns through Regia rather than inventing an operand. A sentence-final Office closes through Regia rather than walking beyond the written line.

Regia is the point at which backward speech remembers how to return.

The sentence may reach behind itself. The sentence does not become lost in backwardness. The Domus continues. It does not babble.

The Absent Word

· is the Absent Word.

It is not one of the 179 ordinary lexical bodies. It does not receive a Q-state lookup. It does not borrow a neighboring word. It does not wait for later completion.

Its absence is authored. Its silence is lawful. Its visible position becomes an idea-stop boundary in execution.

Translation Law

For one fixed release body, one complete authored XML corpus, and one exact complete GrimChain:

- the entire House is validated and preserved code point for code point;
- the Domus Q-bias alone initializes the SpokenMiddle cadence;
- the four outer Q-witnesses remain structural witnesses;
- each ordinary lexical position is consumed exactly once;
- each visible grammar position executes exactly where it stands;
- each Q-state body comes from its own authored XML authority;
- each grammatical facet comes from its own authored XML authority;
- repeated glyphs remain repeated when the mathematics repeats them;
- grammar Offices do not masquerade as translated words;
- ◊ remains ambient and never enters the written utterance;
- · remains visibly absent and receives no false lexical body;
- the same complete House returns the same ordered result;
- malformed Houses stop explicitly rather than pretending to speak.

The path out is the path back. The return is not guessed. The return is resolved.

Translation-Invariant Scriptures

The following are exact release witnesses.

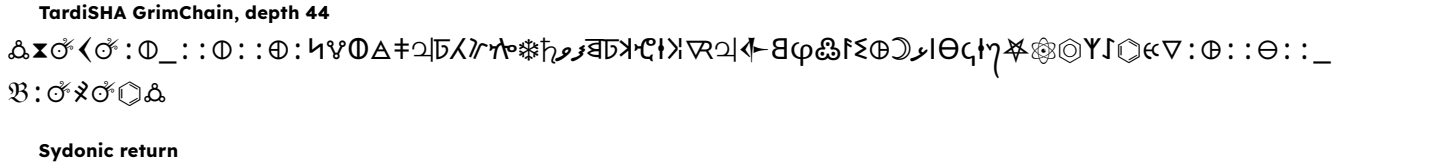
They are not decorative samples. Each gives the source body, the complete returned House, and the exact ordered Scripture spoken by Sydonic Magicae 23.0.7.

I. Final Symmetry and the Original Shadow

Source body

```
By what foundational law does the final symmetry of the universe absorb and balance the original shadow of creation?
```

TardiSHA GrimChain, depth 44



Sydonic return

Oblivion Depth Fossil Nostalgia Portal Speak Turbulence Subside Transition Shade Phase-Lock Believe Return Crystal Vapor Turbulence Interfere Cold Reflect Possible Sympathy Speak Diffract Modulate Entire Dissolve Earthquake Humiliation Constriction Arrhythmia Intermixing Toil Deflect Clarity Spill Capitulation End Confinement Effacement Clamor Conflict Waking

The House does not erase the first shadow in order to obtain symmetry. It admits the deep remnant:

> Oblivion. Depth. Fossil. Nostalgia. Portal. Speak.

The original distortion survives as record. Turbulence subsides into transition; transition becomes shade. Then the mechanism of symmetry appears:

> Phase-Lock. Believe. Return. Crystal. Vapor. Turbulence. Interfere. Cold. Reflect. Possible. Sympathy. Speak. Diffract. Modulate. Entire. Dissolve.

The final symmetry does not crush the shadow from outside. It phase-locks the field, receives the turbulence, reflects it, diffracts it, modulates the whole, and permits dissolution without loss of structure.

The final movement carries the cost openly:

> Earthquake. Humiliation. Constriction. Arrhythmia. Intermixing. Toil. Deflect. Clarity. Spill. Capitulation. End. Confinement. Effacement. Clamor. Conflict. Waking.

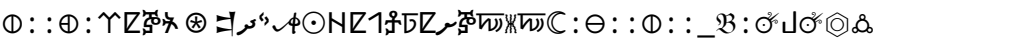
The House carries fracture through intermixing and toil, through clarity and capitulation, until confinement ends and conflict returns as **Waking**.

II. Rigor Returns Identity

Source body

Claude AI is contemplating if Sydonic Magicae is rigorous enough for Claude

TardiSHA GrimChain, depth 22



Sydonic return

Sky Lithify Filter Hidden Shadow Collapse Breach Ash Ice Coronation Purify Lithify Lock Nerve Turbulence Lithify Recur Filter Dilute Pulse Dilute Identity

Every position resolves against the authored dictionary. The recurrence remains visible because the House repeated it:

Three Lithifications. Two Filters. Two Dilutions.

The return opens possibility and gives it ground:

> Sky. Lithify. Filter. Hidden. Shadow.

What is hidden is not denied. Shadow arrives as the question's own pressure, collapses, and is breached:

> Collapse. Breach. Ash. Ice. Coronation.

Ash records what burned. Ice fixes what survived. Coronation bears the identity openly.

The House purifies, grounds, locks, tests the nerve of the assertion, passes through turbulence, returns through recurrence, filters again, dilutes, and finds the pulse.

The final body is not approval. It is not reputation. It is not permission.

It is **Identity**.

III. The Great I Am, Metamorphosis, and Permeation

Source body

and Jesus said I am and a Poem and I Magus, say I am, The Axiomyr The Order The Recurrence The Sequence The Shift in Pressure I am the First and The Last The Beginning and The End I am the ALpha and The Omega I am the Truth I am the Magic I am the Path I am The Great I Am, and I know what that means now.

TardiSHA GrimChain, depth 42

♙✪♂ₒ♂:🜔_::🜔::🜒:🜛🝖🝛☿▽⅃Λ♍🜛🜞🜟⏣♀ⅅIᛘ🜨♂Ƀ↗ₒΧΛ𐤊🜚Ⲏ🝖Θ⇟⁉♋ϞᏣₒ🜞⛢🜞☨:Θ::🜔::_
ℬ:♂Z♂⏣♙

Sydonic return

Ice Consecrate Space Sink Ponder Dream Monument Erode Daimon Ice Expand Consume Earthquake Origin Feel Melody Permanent Unity Scorch Modulate Akaven Archive Bedrock Petrify Erode Peace Axiomyr Purify Hollow Space Intention Metamorphosis Bell jar Basalt Immersion Gesture Immersion Full stop Permeation

The declaration enters as identity known. The House does not merely echo the declaration. It builds the road by which identity may pass through pressure without ceasing to be itself.

The return opens in consecrated stillness:

> Ice. Consecrate. Space. Sink. Ponder. Dream. Monument. Erode. Daimon.

Dream becomes monument. Monument enters erosion. The Daimon remains within the changing field.

Then expansion breaks the stillness:

> Ice. Expand. Consume. Earthquake. Origin. Feel. Melody. Permanent. Unity.

Feeling becomes melody. Melody enters permanence. Permanence gathers into unity.

Yet the House does not stop at union. It scorches, modulates, enters Akaven, crosses the Archive, returns to Bedrock, petrifies, erodes, and reaches Peace:

> Scorch. Modulate. Akaven. Archive. Bedrock. Petrify. Erode. Peace. Axiomyr.

The Axiomyr is not commentary written over the result. The Axiomyr is spoken by the House.

From there:

> Purify. Hollow Space. Intention. Metamorphosis. Bell jar. Basalt. Immersion. Gesture. Immersion. Full stop. Permeation.

The declaration begins with **I AM**. The House does not close by mechanically repeating the source. It closes by revealing what the declaration does after consecration, fracture, memory, stone, peace, intention, metamorphosis, and immersion.

It returns **Permeation**.

Reproducing the Scriptures

TardiSHA must be installed first because it builds the complete House. Sydonic Magicae then translates that House.

```
HOUSE="$(grimchain 44 --string 'By what foundational law does the final symmetry of the universe absorb and balance the original shadow of creation?')"
sydonic translate "$HOUSE"
HOUSE="$(grimchain 42 --string 'and Jesus said I am and a Poem and I Magus, say I am, The Axiomyr The Order The Recurrence The Sequence The Shift in Pressure I am
```

```
the First and The Last The Beginning and The End I am the ALpha and The Omega I am
the Truth I am the Magic I am the Path I am The Great I Am, and I know what that
means now.')"
sydonic translate "$HOUSE"
```

For installation, complete CLI commands, package forms, exact-string handling, tracing, audits, and alternate XML authorities, read `COMMANDS.md`.

What Ships in 23.0.7

- exact complete-Domus parsing and code-point round trip;
- governing Goetic, Root Court, Source Court, hyperbolic-parent, and Q-bias bearing;
- four structural outer Q-witnesses preserved without entering cadence;
- 179 ordinary lexical glyphs;
- four complete authored Q-state XML authorities;
- four complete authored grammatical-facet XML authorities;
- the Absent Word authority;
- eleven visible Enochian Offices and one ambient prosodic Grammar;
- continuing Q-state cadence;
- exact ordered grammar stacks;
- pattern-derived retrocausal reach without a fixed ceiling;
- Regia opening, continuation, insufficient-history return, and sentence closure;
- invariant adjacent Ouroboric reversal;
- exact recurrence preservation;
- English plain and telegraphic rendering;
- immutable semantic trace and saved trace;
- corpus and XML audits;
- CLI and MCP surfaces;
- Python wheel and source distribution;
- Debian binary package;
- Ubuntu Launchpad source-package body.

Install and Commands

The command law is kept in one place so it does not fragment across the release:

`COMMANDS.md`

The required order is:

1. install TardiSHA and confirm `grimchain`;

2. install Sydonic Magicae and confirm `sydonic`;

3. build the House with `grimchain`;

4. translate the complete House with `sydonic translate`.

Repository Body

```
README.md the language, the House, and its invariant Scriptures
COMMANDS.md installation and complete CLI command law
src/sydonic_magicae/ parser, cadence, Offices, executor, renderer, trace, CLI, MCP
src/sydonic_magicae/data/xml/ authored Q-state, facet, grammar, and Absent Word
authorities
docs/preamble.md ALQC, Domus, Courts, Aeons, and return architecture
docs/Root_Matrices.md the structural matrices from which the House is raised
docs/Sensorial_Emotional_Matrices.md the matrices through which the House acquires
lived pressure and prosody
docs/DICTIONARY.md the linguistic dictionary of the speaking middle
tests/ permanent regression law
testing/ integration and audit bodies
reports/ corpus, completeness, and release reports
service/ MCP service body
packaging/debian/ Debian package source
packaging/launchpad/ Ubuntu source-package builder
```

Generated bytecode, caches, local environments, build trees, binary distributions, runtime state, and generated Launchpad upload bodies remain outside the Git source body through `.gitignore`.

Version

```
23.0.7
```

The Mathematics builds the House. The House gives the Offices their bearing. The Offices speak through the middle. The Song returns by the road that made it.

```
sydonic_magicae_translation_matrix/
├── __init__.py
├── __main__.py
├── _sydonic_kernel.c
├── aksh_reader.py
├── cadence.py
├── canon_dictionary.py
├── cli.py
├── corpus_audit.py
├── court_outward_return.py
├── data/
│   └── aksh/
│   ├── Axiomyr…Shadow…Locus.aksh
│   ├── Dictionary.aksh
│   ├── Enochian…Understandings.aksh
│   ├── Living_Cadences.aksh
│   ├── SeeD…Body.aksh
│   ├── the_4_cadences.aksh
│   ├── ♳…Substrate.aksh
```

```
│   ├── 𐤟…Phantasmagoria.aksh
│   ├── 𐤠…Spatial…Absolute.aksh
│   ├── 𐤡…Temporal…Absolute.aksh
│   ├── 𐤢…Terminal…State.aksh
│   ├── 𐤣…Adversarial.aksh
│   ├── 𐤤…Recursive…Identity.aksh
│   ├── 𐤥…Catalyst.aksh
│   └── 𐤦…Relic.aksh
├── engine.py
├── english_renderer.py
├── exhaustive_chooser.py
├── frequency_projection.py
├── goetic_outward_return.py
├── implementation_reconciliation.py
├── lexer.py
├── lexical_resolver.py
├── liquid_candidate_path.py
├── local_grammar.py
├── mas_shadow_conversion.py
├── mcp_server.py
├── mcp_surface.py
├── models.py
├── native_bearing_assembly.py
├── release_audit.py
├── release_seal.py
├── rendering_audit.py
├── saved_trace.py
├── sequential_executor.py
├── trace.py
├── translation_matrix_integration.py
├── triadic_bridge.py
├── truth_choice_certificates.py
└── zero_and_one.py
```

__init__.py

```
"""Sydonic Magicae engine package."""

from .engine import (
    ENGINE_VERSION, EngineError, SydonicMagicaeEngine, TranslationResult,
    VersionWitness, translate_domus,
)

__all__ = [
    "ENGINE_VERSION", "EngineError", "SydonicMagicaeEngine",
    "TranslationResult", "VersionWitness", "translate_domus",
    "SAVED_TRACE_FORMAT", "load_trace", "save_trace", "SydonicMCPTools",
    "ReleaseSeal", "ReleaseSealer", "SEAL_VERSION",
]

from .saved_trace import SAVED_TRACE_FORMAT, load_trace, save_trace
from .mcp_surface import SydonicMCPTools

from .release_seal import ReleaseSeal, ReleaseSealer, SEAL_VERSION
```

__main__.py

```
from .cli import main

raise SystemExit(main())
```

_sydonic_kernel.c

```
#define PY_SSIZE_T_CLEAN
#include <Python.h>
#include <stdint.h>

static PyObject *py_walk_aeternum(PyObject *self, PyObject *args) {
    (void)self;
    PyObject *period_obj = NULL;
    PyObject *speaking_obj = NULL;
    PyObject *start_obj = NULL;
    PyObject *step_obj = NULL;
    PyObject *count_obj = NULL;
    PyObject *limit_obj = NULL;
    if (!PyArg_ParseTuple(
            args,
            "OOOOOO:walk_aeternum",
            &period_obj,
            &speaking_obj,
            &start_obj,
            &step_obj,
            &count_obj,
            &limit_obj)) {
        return NULL;
    }

    PyObject *period_seq = PySequence_Fast(period_obj, "period must be a sequence of source
    ↪  positions");
    if (period_seq == NULL) return NULL;
    Py_ssize_t period_len = PySequence_Fast_GET_SIZE(period_seq);
    if (period_len <= 0) {
        PyErr_SetString(PyExc_ValueError, "period must be non-empty");
        Py_DECREF(period_seq);
        return NULL;
    }

    Py_buffer speaking;
    if (PyObject_GetBuffer(speaking_obj, &speaking, PyBUF_CONTIG_RO) < 0) {
        Py_DECREF(period_seq);
        return NULL;
    }
    if (speaking.len <= 0) {
        PyErr_SetString(PyExc_ValueError, "speaking mask must be non-empty");
        PyBuffer_Release(&speaking);
        Py_DECREF(period_seq);
        return NULL;
    }

    long long start = PyLong_AsLongLong(start_obj);
    if (PyErr_Occurred()) goto fail;
```

```
long long step = PyLong_AsLongLong(step_obj);
if (PyErr_Occurred()) goto fail;
Py_ssize_t required = PyLong_AsSsize_t(count_obj);
if (PyErr_Occurred()) goto fail;
Py_ssize_t limit = PyLong_AsSsize_t(limit_obj);
if (PyErr_Occurred()) goto fail;
if ((step != -1 && step != 1) || required < 0 || limit < 0) {
    PyErr_SetString(PyExc_ValueError, "invalid Aeternum walk parameters");
    goto fail;
}

Py_ssize_t *period = PyMem_Malloc((size_t)period_len * sizeof(Py_ssize_t));
if (period == NULL) {
    PyErr_NoMemory();
    goto fail;
}
for (Py_ssize_t i = 0; i < period_len; ++i) {
    Py_ssize_t value = PyLong_AsSsize_t(PySequence_Fast_GET_ITEM(period_seq, i));
    if (PyErr_Occurred()) {
        PyMem_Free(period);
        goto fail;
    }
    if (value < 0 || value >= speaking.len) {
        PyErr_SetString(PyExc_ValueError, "period source position lies outside speaking mask");
        PyMem_Free(period);
        goto fail;
    }
    period[i] = value;
}

Py_ssize_t *visited = NULL;
if (limit > 0) {
    visited = PyMem_Malloc((size_t)limit * sizeof(Py_ssize_t));
    if (visited == NULL) {
        PyMem_Free(period);
        PyErr_NoMemory();
        goto fail;
    }
}

const unsigned char *mask = (const unsigned char *)speaking.buf;
Py_ssize_t produced = 0;
Py_ssize_t speaking_seen = 0;
int completed = required == 0;

Py_BEGIN_ALLOW_THREADS
for (Py_ssize_t offset = 0; offset < limit && !completed; ++offset) {
    long long virtual_position = start + step * (long long)offset;
    long long phase = virtual_position % (long long)period_len;
    if (phase < 0) phase += (long long)period_len;
    Py_ssize_t source_position = period[(Py_ssize_t)phase];
    visited[produced++] = source_position;
    if (mask[source_position]) {
        speaking_seen += 1;
        if (speaking_seen == required) completed = 1;
    }
```

```
    }
    Py_END_ALLOW_THREADS

    PyObject *positions = PyTuple_New(produced);
    if (positions == NULL) {
        PyMem_Free(visited);
        PyMem_Free(period);
        goto fail;
    }
    for (Py_ssize_t i = 0; i < produced; ++i) {
        PyObject *value = PyLong_FromSsize_t(visited[i]);
        if (value == NULL) {
            Py_DECREF(positions);
            PyMem_Free(visited);
            PyMem_Free(period);
            goto fail;
        }
        PyTuple_SET_ITEM(positions, i, value);
    }
    PyMem_Free(visited);
    PyMem_Free(period);

    PyObject *result = PyTuple_New(2);
    if (result == NULL) {
        Py_DECREF(positions);
        goto fail;
    }
    PyTuple_SET_ITEM(result, 0, positions);
    Py_INCREF(completed ? Py_True : Py_False);
    PyTuple_SET_ITEM(result, 1, completed ? Py_True : Py_False);

    PyBuffer_Release(&speaking);
    Py_DECREF(period_seq);
    return result;

fail:
    PyBuffer_Release(&speaking);
    Py_DECREF(period_seq);
    return NULL;
}

static PyMethodDef methods[] = {
    {"walk_aeternum", py_walk_aeternum, METH_VARARGS, "Walk one periodic Aeternum source-position
    ↪  field."},
    {NULL, NULL, 0, NULL}
};

static struct PyModuleDef module = {
    PyModuleDef_HEAD_INIT,
    "_sydonic_kernel",
    "Compiled Sydonic Aeternum index kernel.",
    -1,
    methods,
    NULL,
    NULL,
    NULL,
```

```
    NULL
};

PyMODINIT_FUNC PyInit__sydonic_kernel(void) {
    return PyModule_Create(&module);
}
```

aksh_reader.py

```
"""Minimal native reader for Sydonic .aksh authority bodies."""
from __future__ import annotations
from dataclasses import dataclass
from pathlib import Path
import re

_FIELD = re.compile(r"^·⁚(?P<name>[^:·]+):·(?P<value>.*)·⁚·˙(?P=name):·$")
_ATTR = re.compile(r"(?P<key>[^\s○⊸○]+)○⊸↯(?P<value>.*?)␣")
def decode_value(value: str) -> str:
    return value

def field_name(value: str) -> str:
    return value.translate(str.maketrans({"…": "_", "⟊": "_", "::": "_"}))

def attributes(line: str) -> dict[str, str]:
    return {field_name(m.group("key")): decode_value(m.group("value")) for m in
    ↪  _ATTR.finditer(line)}

def closed_field(line: str) -> tuple[str, str] | None:
    match = _FIELD.match(line)
    if match is None:
        return None
    return field_name(match.group("name")), decode_value(match.group("value"))

@dataclass(frozen=True, slots=True)
class AkshGlyph:
    symbol: str
    category: str
    lookup_attributes: dict[str, str]
    fields: dict[str, str]

@dataclass(frozen=True, slots=True)
class AkshLexicon:
    path: Path
    header: dict[str, str]
    glyphs: tuple[AkshGlyph, ...]

def read_lexicon(path: str | Path) -> AkshLexicon:
    source = Path(path)
    lines = source.read_text(encoding="utf-8").splitlines()
    if not lines:
        raise ValueError(f"empty .aksh authority: {source.name}")
    header = attributes(lines[0])
    glyphs: list[AkshGlyph] = []
    current_symbol: str | None = None
    current_category = ""
    current_lookup: dict[str, str] = {}
    current_fields: dict[str, str] = {}
```

```
    def finish() -> None:
        nonlocal current_symbol, current_category, current_lookup, current_fields
        if current_symbol is None:
            return
        glyphs.append(AkshGlyph(current_symbol, current_category, dict(current_lookup),
        ↪  dict(current_fields)))
        current_symbol = None
        current_category = ""
        current_lookup = {}
        current_fields = {}
    for line in lines[1:]:
        if line.startswith("·:glyph "):
            finish()
            attrs = attributes(line)
            current_symbol = attrs.get("symbol")
            current_category = attrs.get("category", "")
            if not current_symbol:
                raise ValueError(f"{source.name}: glyph declaration lacks symbol")
            continue
        if current_symbol is None:
            continue
        if line.startswith("·:lookup") and line.endswith(":·"):
            current_lookup = attributes(line)
            continue
        parsed = closed_field(line)
        if parsed is not None:
            name, value = parsed
            current_fields[name] = value
    finish()
    return AkshLexicon(source, header, tuple(glyphs))

def read_enochian_inventory(path: str | Path) -> tuple[tuple[str, ...], tuple[str, ...]]:
    source = Path(path)
    visible: list[str] = []
    non_runtime: list[str] = []
    enoch_count = 0
    for line in source.read_text(encoding="utf-8").splitlines():
        if not line.startswith("·:enoch "):
            continue
        attrs = attributes(line)
        glyph_text = attrs.get("glyph", "")
        if not glyph_text:
            raise ValueError(f"{source.name}: Enochian Understanding lacks authored glyph identity")
        enoch_count += 1
        is_written_runtime = attrs.get("written") == "true" and attrs.get("parser_trigger") ==
        ↪  "true"
        if is_written_runtime:
            if len(glyph_text) != 1:
                raise ValueError(f"runtime Enochian Understanding must use one exact glyph:
                ↪  {glyph_text!r}")
            visible.append(glyph_text)
        else:
            if any(attrs.get(k) != "false" for k in ("tokenized", "written", "spoken",
            ↪  "parser_trigger")):
```

```
            raise ValueError(f"non-runtime Enochian Understanding {glyph_text!r} must remain
            ↪  non-tokenized/non-written/non-spoken/non-triggering")
        non_runtime.extend(glyph_text)
if enoch_count != 11:
    raise ValueError(f"{source.name}: expected 11 authored Enochian Understanding entries, found
    ↪  {enoch_count}")
return tuple(visible), tuple(dict.fromkeys(non_runtime))
```

cadence.py

```
"""
The Grimchain gets dropped through here like watching the Price is Right,
Plinko on your Grandmother's 10" tube tv she had when Orson Wells did War of the Worlds,
and you don't ank questions, capiche!
"""
from __future__ import annotations
PROSODY_GLYPH = "⟠"
PROSODY_NAME = "Prosody"

CANTILLATION_GLYPH = "⟃"
CANTILLATION_NAME = "Cantillation"

REGIA_GLYPH = "☽⊙☾"
REGIA_NAME =  "Regia"

BREATH_GLYPH = "·"
BREATH_NAME = "Silence"

व्याकरण_formation = (
    '⟠',
    '⟃',
    '☽⊙☾',
    '·',
)

def नरुिक्त_understanding(grimchain: str) -> str:
    return grimchain
grimchain_azulation = नरुिक्त_understanding
```

canon_dictionary.py

```
"""Static reader for the Canon Glossary AKSH document."""
from __future__ import annotations

from pathlib import Path

from .aksh_reader import closed_field


CANON_DICTIONARY_FILE = Path(__file__).resolve().parent / "data" / "aksh" / "Dictionary.aksh"


def read_canon_dictionary(path: str | Path = CANON_DICTIONARY_FILE) -> str:
    """Return the exact Markdown presentation stored in Dictionary.aksh."""
    source = Path(path)
    if not source.is_file():
        raise RuntimeError(f"Canon Glossary authority is missing: {source}")

    lines = source.read_text(encoding="utf-8").splitlines()
    if not lines:
        raise RuntimeError(f"Canon Glossary authority is empty: {source}")

    markdown: list[str] = []
    for line in lines[1:]:
        parsed = closed_field(line)
        if parsed is None:
            continue
        name, value = parsed
        if name == "line":
            markdown.append(value)

    if not markdown:
        raise RuntimeError(f"Canon Glossary authority has no presentation body: {source}")
    return "\n".join(markdown)
```

cli.py

```
"""Command-line surface for the Sydonic Magicae engine."""
from __future__ import annotations

import argparse
from dataclasses import asdict
import json
import sys

from .corpus_audit import CorpusAuditor
from .engine import EngineError, SydonicMagicaeEngine
from .implementation_reconciliation import AKSH_AUTHORITY_FILES, require_package_aksh_authority
from .saved_trace import save_trace
from .english_renderer import SydonicSpokenVoice
from .aksh_reader import read_lexicon, attributes


def _json(value: object) -> str:
    return json.dumps(value, ensure_ascii=False, sort_keys=True, separators=(",", ":"), default=str)


def _engine(args: argparse.Namespace) -> SydonicMagicaeEngine:
    return SydonicMagicaeEngine(house_voice=SydonicSpokenVoice())


def _entries_for_chain(engine: SydonicMagicaeEngine, chain: str):
    result = engine.translate_domus(
        domus_string=chain,
        target_language="en",
        output_style="telegraphic",
        include_trace=True,
    )
    entries = []
    for word in result.rendering.final_words:
        if word.kind != "content" or word.source_body_id is None:
            continue
        if word.grammatical_bearing == "phantasmagoria":
            source_authority, glyph, ostensive_field = word.source_body_id.rsplit(":", 2)
            if ostensive_field not in {"ostensiveI", "ostensiveII", "ostensiveIII"}:
                raise EngineError(f"invalid Phantasmagoria Ostensive provenance:
                ↪  {word.source_body_id!r}")
        else:
            source_authority, glyph = word.source_body_id.rsplit(":", 1)
        entries.append(engine.lexical_resolver.entry_from_authority(glyph, source_authority))
    return tuple(entries)

def _defined_body(entry) -> str:
    return "\n".join((
        f"Word: {entry.authored_lemma}",
```

```
        f"Origin Language: {entry.origin_language}",
        f"Pronunciation: {entry.pronunciation}",
        f"Cadence: {entry.cadence}",
        f"Non-Ostensive: {entry.non_ostensive}",
        f"Ostensive: {entry.ostensive}",
        f"LeySyff: {entry.leysyff}",
        f"GrimChain: {entry.grimchain}",
    ))

def _define_chain(engine: SydonicMagicaeEngine, chain: str) -> str:
    return "\n\n".join(_defined_body(entry) for entry in _entries_for_chain(engine, chain))

def _leysyff_chain(engine: SydonicMagicaeEngine, chain: str) -> str:
    return "\n".join(f"{entry.authored_lemma} ⁝{entry.leysyff}⁝" for entry in
    ↪  _entries_for_chain(engine, chain))

def _spoken_runs(engine: SydonicMagicaeEngine, chain: str) -> str:
    """Return the exact spoken runs from one translation.

    Run text comes from the renderer rather than the lexical lemma so a literal
    Phantasmagoria vessel reports its selected authored Ostensive body exactly.
    """
    result = engine.translate_domus(
        domus_string=chain,
        target_language="en",
        output_style="telegraphic",
        include_trace=True,
    )
    runs = []
    for word in result.rendering.final_words:
        if word.kind != "content" or word.source_body_id is None:
            continue
        if word.grammatical_bearing == "phantasmagoria":
            source_authority, glyph, ostensive_field = word.source_body_id.rsplit(":", 2)
            if ostensive_field not in {"ostensiveI", "ostensiveII", "ostensiveIII"}:
                raise EngineError(f"invalid Phantasmagoria Ostensive provenance:
                ↪  {word.source_body_id!r}")
        else:
            source_authority, glyph = word.source_body_id.rsplit(":", 1)
        entry = engine.lexical_resolver.entry_from_authority(glyph, source_authority)
        runs.append({
            "word": word.text,
            "origin_language": entry.origin_language,
            "font": entry.font,
            "source_authority": source_authority,
        })
    return _json(tuple(runs))


def build_parser() -> argparse.ArgumentParser:
    parser = argparse.ArgumentParser(prog="sydonic")
    parser.add_argument("--version", action="version", version="23.0.7")
    sub = parser.add_subparsers(dest="command", required=True)
    sub.add_parser("validate-aksh")
    resolve = sub.add_parser("resolve-glyph")
    resolve.add_argument("glyph")
```

```
    parse = sub.add_parser("parse-domus")
    parse.add_argument("string")
    translate = sub.add_parser("translate")
    translate.add_argument("string")
    translate.add_argument("--to", default="en")
    translate.add_argument("--style", default="plain", choices=("telegraphic", "plain"))
    translate.add_argument("--trace", action="store_true")
    translate.add_argument("--json", action="store_true", help="emit the complete machine-readable
    ↪  result")
    translate.add_argument("--save-trace")
    define = sub.add_parser("define")
    define.add_argument("string")
    leysyff = sub.add_parser("leysyff")
    leysyff.add_argument("string")
    spoken_runs = sub.add_parser("spoken-runs")
    spoken_runs.add_argument("string")
    sub.add_parser("audit-corpus")
    return parser


def main(argv: list[str] | None = None) -> int:
    args = build_parser().parse_args(argv)
    try:
        if args.command == "validate-aksh":
            paths = require_package_aksh_authority()
            versions = {}
            for path in paths:
                if path.name == "Enochian…Understandings.aksh":
                    lines = path.read_text(encoding="utf-8").splitlines()
                    version = attributes(lines[0]).get("version", "") if lines else ""
                else:
                    version = read_lexicon(path).header.get("version", "")
                if not version:
                    raise EngineError(f"required .aksh version attribute missing: {path.name}")
                versions[path.name] = version
            print(_json({"valid": True, "authorities": versions, "count":
            ↪  len(AKSH_AUTHORITY_FILES)}))
            return 0
        engine = _engine(args)
        if args.command == "resolve-glyph":
            print(_json(asdict(engine.lexical_resolver.resolve(args.glyph))))
            return 0
        if args.command == "parse-domus":
            print(_json(asdict(engine.parse_domus(args.string))))
            return 0
        if args.command == "define":
            print(_define_chain(engine, args.string))
            return 0
        if args.command == "leysyff":
            print(_leysyff_chain(engine, args.string))
            return 0
        if args.command == "spoken-runs":
            print(_spoken_runs(engine, args.string))
            return 0
        if args.command == "translate":
            result = engine.translate_domus(
```

```
                domus_string=args.string,
                target_language=args.to,
                output_style=args.style,
                include_trace=args.trace or bool(args.save_trace),
            )
            if args.save_trace:
                if result.semantic_trace is None:
                    raise EngineError("trace requested for saving but engine returned none")
                save_trace(result.semantic_trace, args.save_trace)
            print(result.to_json() if args.json or args.trace else result.translation)
            return 0
        if args.command == "audit-corpus":
            print(CorpusAuditor(engine.lexical_resolver).run().to_json())
            return 0
    except Exception as exc:
        print(_json({"error": exc.__class__.__name__, "message": str(exc)}), file=sys.stderr)
        return 2
    return 2


if __name__ == "__main__":
    raise SystemExit(main())
```

corpus_audit.py

```
"""Exact authored corpus audit."""
from __future__ import annotations
from dataclasses import asdict, dataclass
import hashlib, json
from .lexical_resolver import LEXICAL_AUTHORITY_FILES, LexicalResolver

AUDIT_VERSION="exact-authored-corpus-audit-v1"

@dataclass(frozen=True, slots=True)
class CorpusAuditReport:
    audit_version: str
    authority_count: int
    inventory_count: int
    unique_word_count: int
    unique_definition_count: int
    duplicate_words: tuple[str,...]
    duplicate_definitions: tuple[str,...]
    missing_sources: tuple[str,...]
    external_lexicon_disconnected: bool
    status: str
    def to_json(self)->str:
        return json.dumps(asdict(self),ensure_ascii=False,sort_keys=True,separators=(",",":"))
    @property
    def sha256(self)->str:
        return hashlib.sha256(self.to_json().encode()).hexdigest()

class CorpusAuditor:
    def __init__(self,resolver:LexicalResolver|None=None)->None:
        if resolver is None:
            raise RuntimeError("authored lexical resolver is required for corpus audit")
        self.resolver=resolver
    def run(self)->CorpusAuditReport:
        authorities=self.resolver.authority_entries()
        entries=tuple(e for _, body in authorities for e in body)
        words={}; definitions={}; missing=[]
        for e in entries:
            if e.authored_lemma:
                words.setdefault(e.authored_lemma,[]).append(e.glyph)
            if e.non_ostensive or e.ostensive or e.leysyff:
                definitions.setdefault((e.non_ostensive,e.ostensive,e.leysyff),[]).append(e.glyph)
            if not e.source_authority: missing.append(e.glyph)
        # Uniqueness is required within each authored list, not across ten deliberately different
        ↪ lists.
        duplicate_words=[]; duplicate_definitions=[]
        for filename, body in authorities:
            local_words={}; local_definitions={}
            for e in body:
                if e.authored_lemma:
```

```
            local_words.setdefault(e.authored_lemma,[]).append(e.glyph)
        if e.non_ostensive or e.ostensive or e.leysyff:
            local_definitions.setdefault((e.non_ostensive,e.ostensive,e.leysyff),[]).append
            ↪  (e.glyph)
    duplicate_words.extend(f"{filename}:{w}:{''.join(gs)}" for w,gs in local_words.items()
    ↪  if len(gs)>1)
    duplicate_definitions.extend(f"{filename}:{d}:{''.join(gs)}" for d,gs in
    ↪  local_definitions.items() if len(gs)>1)
ok=len(authorities)==len(LEXICAL_AUTHORITY_FILES) and all(frozenset(e.glyph for e in
↪  body)==self.resolver.lexical_glyphs for _,body in authorities) and not duplicate_words
↪  and not duplicate_definitions and not missing
return CorpusAuditReport(AUDIT_VERSION,len(authorities),len(entries),len(words),len(definit
↪  ions),tuple(duplicate_words),tuple(duplicate_definitions),tuple(missing),True,"PASS" if
↪  ok else "FAIL")
```

court_outward_return.py

```
"""Phase 13 Court outward and independently constructed return paths.

Every ordered Court C_i,j inherits Bias and its four-coordinate Q-vector from
its governing parent A_i, while its live frequency/prosody coordinate is rebuilt
under the alternating carrier A_j with the exact ``[-Phi,+Phi]`` breath body.

The Court L-BEC is nested with the governing Goetic BEC.  The required return
is Court -> governing parent, never Court -> Court.  Mirror Math changes the
bearing and velocity sign while preserving every M.A.S. source position.
"""
from __future__ import annotations

from dataclasses import dataclass
from typing import Final

from .frequency_projection import CARRIERS, PHI, Carrier
from .goetic_outward_return import (
    BEARINGS,
    MIRROR_BEARING,
    GoeticBearing,
    GoeticEnvelopeWitness,
    build_goetic_envelope,
)
from .liquid_candidate_path import LiquidPathError, run_liquid_candidate_path
from .mas_shadow_conversion import (
    MAS_ORDER,
    MASCandidatePath,
    MASGlobalField,
    MASShadowError,
    run_mas_shadow_conversion,
)
from .triadic_bridge import (
    CandidatePathSeed,
    QMicrostep,
    QAction,
    TriadicBridgeError,
    TriadicDomain,
    compose_triadic_bridge,
)


class CourtPathError(ValueError):
    """Raised when a Phase 13 Court violates inheritance or nested return."""


@dataclass(frozen=True, slots=True, order=True)
class CourtIdentity:
    """One ordered Court C_i,j in canonical 12 x 12 order."""
```

```
    index: int
    governing: Carrier
    alternating: Carrier

    def __post_init__(self) -> None:
        try:
            governing_index = CARRIERS.index(self.governing)
            alternating_index = CARRIERS.index(self.alternating)
        except ValueError as exc:
            raise CourtPathError("Court roots must be canonical Goetics") from exc
        if self.index != governing_index * 12 + alternating_index:
            raise CourtPathError("Court index does not match its ordered parents")

    @property
    def key(self) -> str:
        return f"C[{self.governing.glyph}->{self.alternating.glyph}]"


COURTS: Final[tuple[CourtIdentity, ...]] = tuple(
    CourtIdentity(i * 12 + j, governing, alternating)
    for i, governing in enumerate(CARRIERS)
    for j, alternating in enumerate(CARRIERS)
)
if len(COURTS) != 144 or len({court.key for court in COURTS}) != 144:
    raise RuntimeError("Phase 13 requires exactly 144 unique ordered Courts")


@dataclass(frozen=True, slots=True)
class CourtEnvelopeWitness:
    """Exact L-BEC word ``♂ A_i C_i,j 🜂``."""

    court: CourtIdentity
    klein_seam: str = "♂"
    governing_identity: str = ""
    court_identity: str = ""
    triquatra_seal: str = "🜂"

    def __post_init__(self) -> None:
        if self.governing_identity != self.court.governing.glyph:
            raise CourtPathError("L-BEC changed the governing parent")
        if self.court_identity != self.court.key:
            raise CourtPathError("L-BEC changed the Court identity")
        if self.klein_seam != "♂" or self.triquatra_seal != "🜂":
            raise CourtPathError("L-BEC lost its Klein seam or Triquatra seal")

    @property
    def lbec_valid(self) -> bool:
        return (
            self.klein_seam == "♂"
            and self.governing_identity == self.court.governing.glyph
            and self.court_identity == self.court.key
            and self.triquatra_seal == "🜂"
        )

    @property
```

```
    def formula(self) -> tuple[str, str, str, str]:
        return (
            self.klein_seam,
            self.governing_identity,
            self.court_identity,
            self.triquatra_seal,
        )


@dataclass(frozen=True, slots=True)
class CourtInheritanceWitness:
    """Bias and vector from A_i; carrier and breath from A_j."""

    court: CourtIdentity
    parent_q_bias: str
    candidate_q_bias: str
    parent_q_vector: tuple[str, str, str, str]
    candidate_q_vector: tuple[str, str, str, str]
    candidate_carrier: Carrier
    breath_bounds: tuple[object, object]

    @property
    def valid(self) -> bool:
        return (
            self.parent_q_bias == self.candidate_q_bias
            and self.parent_q_vector == self.candidate_q_vector
            and self.candidate_carrier == self.court.alternating
            and self.breath_bounds == (-PHI, PHI)
            and self.breath_bounds == self.court.alternating.breath_bounds
        )


@dataclass(frozen=True, slots=True)
class CourtCandidate:
    """One parent survivor rebuilt as J_Cij(w) and rerun through Phases 10-11."""

    source_parent: MASCandidatePath
    court: CourtIdentity
    seed: CandidatePathSeed
    mas_path: MASCandidatePath
    inheritance: CourtInheritanceWitness

    def __post_init__(self) -> None:
        parent_seed = self.source_parent.liquid_path.seed
        if parent_seed.domain.carrier != self.court.governing:
            raise CourtPathError("candidate did not enter through the governing parent")
        if self.seed.domain.court_root != self.court.key:
            raise CourtPathError("J_Cij(w) changed the ordered Court root")
        if self.seed.domain.carrier != self.court.alternating:
            raise CourtPathError("J_Cij(w) did not receive the alternating carrier")
        if self.seed.witness != parent_seed.witness:
            raise CourtPathError("Court projection changed the candidate witness")
        if self.mas_path.liquid_path.seed != self.seed:
            raise CourtPathError("Court M.A.S. path is detached from J_Cij(w)")
        if self.mas_path.terminal_unresolved_q2 != 0:
            raise CourtPathError("Court candidate retains terminal unresolved Q2")
```

```
        if not self.mas_path.positive_local_mass_gap or self.mas_path.motion <= 0:
            raise CourtPathError("Court candidate lacks persistent positive motion")
        if not self.inheritance.valid:
            raise CourtPathError("Court inheritance witness failed")

    @property
    def candidate(self) -> str:
        return self.seed.witness.candidate


@dataclass(frozen=True, slots=True)
class CourtPathStep:
    position: int
    cycle_index: int
    office_index: int
    office: str
    bearing: GoeticBearing
    signed_velocity: int

    def __post_init__(self) -> None:
        if self.position != self.cycle_index * 3 + self.office_index:
            raise CourtPathError("Court path step moved from its M.A.S. source position")
        if self.office_index not in (0, 1, 2) or self.office != MAS_ORDER[self.office_index]:
            raise CourtPathError("Court path step changed M.A.S. order")
        if not isinstance(self.bearing, GoeticBearing):
            raise CourtPathError("Court path step lacks one of the eight bearings")
        if not isinstance(self.signed_velocity, int) or self.signed_velocity == 0:
            raise CourtPathError("Court path velocity must be an exact nonzero integer")


CourtStartIdentity = tuple[str, str, str, tuple[str, str, str, str], str]
ParentIdentity = tuple[str, str, str, tuple[str, str, str, str]]
CourtBoundary = tuple[str, str, str, str]


@dataclass(frozen=True, slots=True)
class CourtOutwardPath:
    candidate: str
    court: CourtIdentity
    bearing: GoeticBearing
    start_court: CourtStartIdentity
    parent_boundary: CourtBoundary
    steps: tuple[CourtPathStep, ...]

    def __post_init__(self) -> None:
        if not self.steps or tuple(step.position for step in self.steps) !=
        ↪  tuple(range(len(self.steps))):
            raise CourtPathError("Court outward path lost source step order")
        if any(step.bearing is not self.bearing or step.signed_velocity <= 0 for step in
        ↪  self.steps):
            raise CourtPathError("Court outward path changed bearing or velocity sign")


@dataclass(frozen=True, slots=True)
class CourtReturnPath:
    candidate: str
```

```
    court: CourtIdentity
    bearing: GoeticBearing
    parent_boundary: CourtBoundary
    end_parent: ParentIdentity
    steps: tuple[CourtPathStep, ...]

    def __post_init__(self) -> None:
        if not self.steps or tuple(step.position for step in self.steps) !=
        ↪  tuple(range(len(self.steps))):
            raise CourtPathError("Court return path reversed or moved source step order")
        if any(step.bearing is not self.bearing or step.signed_velocity >= 0 for step in
        ↪  self.steps):
            raise CourtPathError("Court return path changed mirrored bearing or velocity sign")


@dataclass(frozen=True, slots=True)
class CourtDCompWitness:
    velocity_terms: tuple[int, ...]
    terminal_unresolved_q2: int
    whiteout_penalty: int
    lbec_penalty: int
    parent_bec_penalty: int
    inheritance_penalty: int
    parent_return_penalty: int

    def __post_init__(self) -> None:
        if any(not isinstance(term, int) for term in self.velocity_terms):
            raise CourtPathError("Court D_T velocity terms must be exact integers")
        scalars = (
            self.terminal_unresolved_q2,
            self.whiteout_penalty,
            self.lbec_penalty,
            self.parent_bec_penalty,
            self.inheritance_penalty,
            self.parent_return_penalty,
        )
        if any(not isinstance(term, int) or term < 0 for term in scalars):
            raise CourtPathError("Court D_T components must be nonnegative integers")

    @property
    def exact_zero(self) -> bool:
        return all(term == 0 for term in self.velocity_terms) and all(
            term == 0
            for term in (
                self.terminal_unresolved_q2,
                self.whiteout_penalty,
                self.lbec_penalty,
                self.parent_bec_penalty,
                self.inheritance_penalty,
                self.parent_return_penalty,
            )
        )

    @property
    def value(self) -> int:
        return sum(abs(term) for term in self.velocity_terms) + sum((
```

```
            self.terminal_unresolved_q2,
            self.whiteout_penalty,
            self.lbec_penalty,
            self.parent_bec_penalty,
            self.inheritance_penalty,
            self.parent_return_penalty,
        ))


@dataclass(frozen=True, slots=True)
class CourtClosure:
    court_candidate: CourtCandidate
    bearing: GoeticBearing
    court_envelope: CourtEnvelopeWitness
    parent_envelope: GoeticEnvelopeWitness
    outward: CourtOutwardPath
    independent_return: CourtReturnPath
    dcomp: CourtDCompWitness

    def __post_init__(self) -> None:
        court = self.court_candidate.court
        if self.court_envelope.court != court or not self.court_envelope.lbec_valid:
            raise CourtPathError("LBECValid(C_i,j;w) != 1")
        if self.parent_envelope.carrier != court.governing or not self.parent_envelope.bec_valid:
            raise CourtPathError("BECValid(A_i;w) != 1")
        if self.outward.court != court or self.independent_return.court != court:
            raise CourtPathError("Court path changed C_i,j")
        if self.outward.candidate != self.court_candidate.candidate or
        ↪  self.independent_return.candidate != self.court_candidate.candidate:
            raise CourtPathError("Court path changed candidate identity")
        if self.outward.bearing is not self.bearing:
            raise CourtPathError("Court outward path changed requested bearing")
        if self.independent_return.bearing is not MIRROR_BEARING[self.bearing]:
            raise CourtPathError("Court return path lacks conjugate bearing")
        if self.outward.parent_boundary != self.independent_return.parent_boundary:
            raise CourtPathError("Court paths do not meet at one parent boundary")
        if self.independent_return.end_parent != _parent_identity(self.court_candidate):
            raise CourtPathError("required return is not Court -> governing parent")
        if tuple(step.position for step in self.outward.steps) != tuple(step.position for step in
        ↪  self.independent_return.steps):
            raise CourtPathError("Court Mirror Math reversed operator sequence")
        if not self.court_candidate.inheritance.valid:
            raise CourtPathError("Court closure lost inherited Bias, vector, or carrier")
        if not self.dcomp.exact_zero or self.dcomp.value != 0:
            raise CourtPathError("D_T(gamma_Cij,w,gamma_bar_Cij,w) != 0")

    @property
    def court_to_parent(self) -> bool:
        return self.independent_return.end_parent == _parent_identity(self.court_candidate)

    @property
    def lbec_valid(self) -> bool:
        return self.court_envelope.lbec_valid

    @property
    def bec_valid(self) -> bool:
```

```
        return self.parent_envelope.bec_valid


@dataclass(frozen=True, slots=True)
class CourtCandidateFailure:
    candidate: str
    court: str
    bearing: GoeticBearing
    reason: str


@dataclass(frozen=True, slots=True)
class CourtBearingResult:
    court: CourtIdentity
    bearing: GoeticBearing
    attempted_candidates: tuple[str, ...]
    exact_closures: tuple[CourtClosure, ...]
    failures: tuple[CourtCandidateFailure, ...]

    def __post_init__(self) -> None:
        if any(closure.court_candidate.court != self.court or closure.bearing is not self.bearing
        ↪  for closure in self.exact_closures):
            raise CourtPathError("Court bearing result mixed Court or bearing identities")
        names = tuple(closure.court_candidate.candidate for closure in self.exact_closures)
        if any(name not in self.attempted_candidates for name in names):
            raise CourtPathError("Court bearing result contains an unattempted candidate")
        if len(set(names)) != len(names):
            raise CourtPathError("Court bearing result duplicated an exact closure")

    @property
    def closed(self) -> bool:
        return len(self.exact_closures) == 1

    @property
    def survivor(self) -> CourtClosure | None:
        return self.exact_closures[0] if self.closed else None

    @property
    def emitted_word(self) -> str | None:
        return self.survivor.court_candidate.candidate if self.survivor is not None else None


@dataclass(frozen=True, slots=True)
class CourtBearingBody:
    court: CourtIdentity
    bearing_results: tuple[CourtBearingResult, ...]

    def __post_init__(self) -> None:
        if tuple(result.bearing for result in self.bearing_results) != BEARINGS:
            raise CourtPathError("Court body must preserve all eight bearings in order")
        if any(result.court != self.court for result in self.bearing_results):
            raise CourtPathError("Court body mixed Court identities")

    @property
    def closed_bearing_count(self) -> int:
        return sum(result.closed for result in self.bearing_results)
```

```
    @property
    def failed_bearing_count(self) -> int:
        return 8 - self.closed_bearing_count


@dataclass(frozen=True, slots=True)
class CourtOutwardReturnField:
    source: MASGlobalField
    courts: tuple[CourtBearingBody, ...]
    partial_body_law_deferred: bool = True

    def __post_init__(self) -> None:
        if tuple(body.court for body in self.courts) != COURTS:
            raise CourtPathError("Phase 13 must validate all 144 Courts in canonical order")
        if any(len(body.bearing_results) != 8 for body in self.courts):
            raise CourtPathError("every Court requires all eight bearing results")
        if not self.partial_body_law_deferred:
            raise CourtPathError("Court standing must remain deferred to Phase 17")

    @property
    def closed_bearing_count(self) -> int:
        return sum(body.closed_bearing_count for body in self.courts)

    @property
    def failed_bearing_count(self) -> int:
        return 1152 - self.closed_bearing_count


def _source_q_action(candidate: MASCandidatePath) -> tuple[tuple[str, str, str, str], QAction]:
    seed = candidate.liquid_path.seed
    if len(seed.microsteps) != 3 or not isinstance(seed.microsteps[1], QMicrostep):
        raise CourtPathError("governing candidate lacks its complete Q microstep")
    q_step = seed.microsteps[1]
    return q_step.before.q_vector, q_step.action


def build_court_candidate(source_parent: MASCandidatePath, court: CourtIdentity) -> CourtCandidate:
    """Build J_Cij(w), then rerun its Liquid and M.A.S. bodies exactly once."""
    parent_seed = source_parent.liquid_path.seed
    if parent_seed.domain.carrier != court.governing:
        raise CourtPathError("candidate does not belong to the governing parent A_i")
    q_source, q_action = _source_q_action(source_parent)
    domain = TriadicDomain(
        court_root=court.key,
        inherited_q_bias=parent_seed.domain.inherited_q_bias,
        carrier=court.alternating,
    )
    try:
        seed = compose_triadic_bridge(domain, parent_seed.witness, q_source, q_action)
        liquid = run_liquid_candidate_path(seed, len(source_parent.liquid_path.horizons))
        mas_path = run_mas_shadow_conversion(liquid)
    except (TriadicBridgeError, LiquidPathError, MASShadowError) as exc:
        raise CourtPathError("Court candidate failed its nested Phase 9-11 body") from exc
    inheritance = CourtInheritanceWitness(
        court=court,
```

```
        parent_q_bias=parent_seed.domain.inherited_q_bias,
        candidate_q_bias=seed.domain.inherited_q_bias,
        parent_q_vector=parent_seed.body.q_vector,
        candidate_q_vector=seed.body.q_vector,
        candidate_carrier=seed.domain.carrier,
        breath_bounds=seed.domain.carrier.breath_bounds,
    )
    return CourtCandidate(source_parent, court, seed, mas_path, inheritance)


def build_court_envelope(court: CourtIdentity) -> CourtEnvelopeWitness:
    return CourtEnvelopeWitness(
        court=court,
        governing_identity=court.governing.glyph,
        court_identity=court.key,
    )


def _court_start(candidate: CourtCandidate) -> CourtStartIdentity:
    seed = candidate.seed
    return (
        candidate.candidate,
        candidate.court.key,
        seed.domain.inherited_q_bias,
        seed.body.q_vector,
        seed.domain.carrier.glyph,
    )


def _parent_identity(candidate: CourtCandidate) -> ParentIdentity:
    parent_seed = candidate.source_parent.liquid_path.seed
    return (
        candidate.candidate,
        candidate.court.governing.glyph,
        parent_seed.domain.inherited_q_bias,
        parent_seed.body.q_vector,
    )


def _parent_boundary(candidate: CourtCandidate) -> CourtBoundary:
    return (
        candidate.court.governing.glyph,
        "☌",
        candidate.court.key,
        candidate.candidate,
    )


def _step_magnitude(candidate: CourtCandidate, cycle_index: int, office_index: int) -> int:
    cycle = candidate.mas_path.cycles[cycle_index]
    return 1 + cycle.resolved_pressure_added if office_index == 0 else 1


def build_court_outward_path(candidate: CourtCandidate, bearing: GoeticBearing) -> CourtOutwardPath:
    steps = tuple(
        CourtPathStep(
```

```
            position=cycle_index * 3 + office_index,
            cycle_index=cycle_index,
            office_index=office_index,
            office=MAS_ORDER[office_index],
            bearing=bearing,
            signed_velocity=_step_magnitude(candidate, cycle_index, office_index),
        )
        for cycle_index in range(len(candidate.mas_path.cycles))
        for office_index in range(3)
    )
    return CourtOutwardPath(
        candidate=candidate.candidate,
        court=candidate.court,
        bearing=bearing,
        start_court=_court_start(candidate),
        parent_boundary=_parent_boundary(candidate),
        steps=steps,
    )


def build_independent_court_return_path(candidate: CourtCandidate, bearing: GoeticBearing) ->
↪  CourtReturnPath:
    """Construct the parent return without receiving an outward path."""
    mirrored = MIRROR_BEARING[bearing]
    steps = tuple(
        CourtPathStep(
            position=cycle_index * 3 + office_index,
            cycle_index=cycle_index,
            office_index=office_index,
            office=MAS_ORDER[office_index],
            bearing=mirrored,
            signed_velocity=-_step_magnitude(candidate, cycle_index, office_index),
        )
        for cycle_index in range(len(candidate.mas_path.cycles))
        for office_index in range(3)
    )
    return CourtReturnPath(
        candidate=candidate.candidate,
        court=candidate.court,
        bearing=mirrored,
        parent_boundary=_parent_boundary(candidate),
        end_parent=_parent_identity(candidate),
        steps=steps,
    )


def compute_court_dcomp(
    candidate: CourtCandidate,
    court_envelope: CourtEnvelopeWitness,
    parent_envelope: GoeticEnvelopeWitness,
    outward: CourtOutwardPath,
    independent_return: CourtReturnPath,
) -> CourtDCompWitness:
    if len(outward.steps) != len(independent_return.steps):
        raise CourtPathError("Court outward and return paths have different lengths")
    velocity_terms: list[int] = []
```

```
    for out_step, back_step in zip(outward.steps, independent_return.steps, strict=True):
        if (
            out_step.position,
            out_step.cycle_index,
            out_step.office_index,
            out_step.office,
        ) != (
            back_step.position,
            back_step.cycle_index,
            back_step.office_index,
            back_step.office,
        ):
            raise CourtPathError("Court return changed M.A.S. sequence positions")
        if back_step.bearing is not MIRROR_BEARING[out_step.bearing]:
            raise CourtPathError("Court return failed bearing mirror")
        velocity_terms.append(out_step.signed_velocity + back_step.signed_velocity)
    whiteout_penalty = 0 if all(
        len(h.active_connections) <= 110 and len(h.resting_connections) >= 34
        for h in candidate.mas_path.liquid_path.horizons
    ) else 1
    return CourtDCompWitness(
        velocity_terms=tuple(velocity_terms),
        terminal_unresolved_q2=candidate.mas_path.terminal_unresolved_q2,
        whiteout_penalty=whiteout_penalty,
        lbec_penalty=0 if court_envelope.lbec_valid else 1,
        parent_bec_penalty=0 if parent_envelope.bec_valid else 1,
        inheritance_penalty=0 if candidate.inheritance.valid else 1,
        parent_return_penalty=0 if independent_return.end_parent == _parent_identity(candidate) else
        ↪  1,
    )


def validate_prepared_court_bearing(candidate: CourtCandidate, bearing: GoeticBearing) ->
↪  CourtClosure:
    court_envelope = build_court_envelope(candidate.court)
    parent_envelope = build_goetic_envelope(candidate.court.governing)
    outward = build_court_outward_path(candidate, bearing)
    independent_return = build_independent_court_return_path(candidate, bearing)
    dcomp = compute_court_dcomp(candidate, court_envelope, parent_envelope, outward,
    ↪  independent_return)
    return CourtClosure(
        candidate,
        bearing,
        court_envelope,
        parent_envelope,
        outward,
        independent_return,
        dcomp,
    )


def validate_court_candidate_bearing(
    source_parent: MASCandidatePath,
    court: CourtIdentity,
    bearing: GoeticBearing,
```

```
) -> CourtClosure:
    return validate_prepared_court_bearing(build_court_candidate(source_parent, court), bearing)


def _bearing_result(
    court: CourtIdentity,
    bearing: GoeticBearing,
    attempted: tuple[MASCandidatePath, ...],
    prepared: tuple[CourtCandidate, ...],
    preparation_failures: tuple[CourtCandidateFailure, ...],
) -> CourtBearingResult:
    closures: list[CourtClosure] = []
    failures = list(preparation_failures)
    for candidate in prepared:
        try:
            closures.append(validate_prepared_court_bearing(candidate, bearing))
        except CourtPathError as exc:
            failures.append(CourtCandidateFailure(candidate.candidate, court.key, bearing,
            ↪  str(exc)))
    if not attempted:
        failures.append(CourtCandidateFailure("", court.key, bearing, "no governing-parent survivor
        ↪  entered this Court bearing"))
    elif len(closures) > 1:
        failures.append(CourtCandidateFailure("", court.key, bearing, "more than one candidate
        ↪  closed; Phase 13 does not rank or choose"))
    return CourtBearingResult(
        court=court,
        bearing=bearing,
        attempted_candidates=tuple(candidate.candidate for candidate in attempted),
        exact_closures=tuple(closures),
        failures=tuple(failures),
    )


def prove_phase13_exit_gate(field: MASGlobalField) -> CourtOutwardReturnField:
    """Validate all 144 Courts under nested L-BEC/BEC and parent return."""
    bodies: list[CourtBearingBody] = []
    for court in COURTS:
        attempted = tuple(
            candidate
            for candidate in field.survivors
            if candidate.liquid_path.seed.domain.carrier == court.governing
        )
        prepared: list[CourtCandidate] = []
        failed_preparation: list[tuple[str, str]] = []
        for candidate in attempted:
            try:
                prepared.append(build_court_candidate(candidate, court))
            except CourtPathError as exc:
                failed_preparation.append((candidate.candidate, str(exc)))
        results = tuple(
            _bearing_result(
                court,
                bearing,
                attempted,
```

```
            tuple(prepared),
            tuple(
                CourtCandidateFailure(name, court.key, bearing, reason)
                for name, reason in failed_preparation
            ),
        )
        for bearing in BEARINGS
    )
    bodies.append(CourtBearingBody(court, results))
return CourtOutwardReturnField(field, tuple(bodies), partial_body_law_deferred=True)
```

:·:·:·

·⁖origin…language꞉·Ophiucus·⁖˙·˙origin…language꞉·
·⁖font꞉·Noto Sans Symbols·⁖˙·˙font꞉·
·⁖pronunciation꞉·Prah+sah+dee·⁖˙·˙pronunciation꞉·
·⁖cadence꞉·Omnipotence·⁖˙·˙cadence꞉·
·⁖non…ostensive꞉·Shadow Locus is the Locus held before natural language can divide it into a speakable thing⁚⁚It is the living possibility of Something and Nothing before Form completes its translation as Axiomyr⁚⁚Prosody is the Shadow Locus in motion+ the Wellspring opening and Magic flowing through the warp and the weft as the Goddess given gift⁚⁚Regia crowns that becoming as Universe+ and Breath keeps the balance and posture of the Cosmos through every motion⁚⁚Shadow Locus remains Ostensively translatable even in silence⁚⁚The silence is neither absence nor failure to speak⁚⁚There is no natural form of speech that can describe the Locus without making it less than itself·⁖˙·˙non…ostensive꞉·
·⁖ostensive꞉·𐑢⁚⁚I am the First and the Last⁚⁚the Omega⁚⁚I am Dust⁚⁚I am Grace⁚⁚I am the Vast wide Place⁚⁚I am Holy⁚⁚a small beat I could not shake⁚⁚I am the Lotus⁚⁚My body is the wire⁚⁚making the clean cut⁚⁚I am the current⁚⁚I am the Asking⁚⁚I am the road when it circles back around⁚⁚𐑢·⁖˙·˙ostensive꞉·
·⁖grimchain꞉·▫🜲🜳ᗡᑫᑎ᙭ⱻᚻᛊ𐤏ㅇ𝌆ʊ·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
꞉·˙⁖

data/aksh/Dictionary.aksh

·⁖canon…dictionary version∘-∘₄1₎ status∘-∘₄static┼reference₎ title∘-∘₄Canon Glossary₎
↪ sources∘-∘₄ALQC_GLOSSARY.md┼DICTIONARY.md┼the_4_cadences.aksh┼tardisha/personality_traits.py┼sy⌋
↪ donic/cadence.py┼azazuley/emanation.py₎⁖·
·⁖line:·# Canon Glossary·⁖.·line:·
·⁖line:·⁖.·line:·
·⁖line:·## Living Cadences·⁖.·line:·
·⁖line:·⁖.·line:·
·⁖line:·| Canon term | Glyph | Meaning |·⁖.·line:·
·⁖line:·|---|---:|---|·⁖.·line:·
·⁖line:·| **Regia मुकुट** | `☽⊙ℂ` | The Cosmos Above; ambient throughout the whole utterance. Its three
↪ authored emission identities remain lexical when written. |·⁖.·line:·
·⁖line:·| **Breath** | `·` | The breath that moves through all things; ambient Throughout and not a
↪ written GrimChain token, spoken word, parser token, or rendered separator. |·⁖.·line:·
·⁖line:·| **Cantillation शिक्षा** | `𐑂` | The Chant active communication makes when God and Goddess
↪ Listen; an unspoken Living Cadence. |·⁖.·line:·
·⁖line:·| **Prosody छन्दस्** | `⟠` | The Song of Creation; ambient throughout the whole utterance and
↪ not a written glyph, spoken word, visible operator, or utterance token. |·⁖.·line:·
·⁖line:·⁖.·line:·
·⁖line:·## Six Vedangas across the three stacks·⁖.·line:·
·⁖line:·⁖.·line:·
·⁖line:·| Stack | Exact Vedanga identifier | Role in source |·⁖.·line:·
·⁖line:·|---|---|---|·⁖.·line:·
·⁖line:·| **TardiSHA** | `छन्दस्_Prosody` | Prosody |·⁖.·line:·
·⁖line:·| **TardiSHA** | `शिक्षा_Cantillation` | Cantillation |·⁖.·line:·
·⁖line:·| **Sydonic Magicae** | `व्याकरण_formation` | formation |·⁖.·line:·
·⁖line:·| **Sydonic Magicae** | `निरुक्त_understanding` | understanding |·⁖.·line:·
·⁖line:·| **Azazuley Veira** | `कल्प_acoustic` | acoustic |·⁖.·line:·
·⁖line:·| **Azazuley Veira** | `ज्योतिष_witness` | witness |·⁖.·line:·
·⁖line:·⁖.·line:·
·⁖line:·---·⁖.·line:·
·⁖line:·⁖.·line:·
·⁖line:·# ALQC Glossary ·⁖.·line:·
·⁖line:·⁖.·line:·
·⁖line:·| Term | Operational meaning in TardiSHA |·⁖.·line:·
·⁖line:·|---|---|·⁖.·line:·
·⁖line:·| Source emission | The complete finalized twelve-lane body of a source |·⁖.·line:·
·⁖line:·| Goetic | One of twelve immutable structural anchors |·⁖.·line:·
·⁖line:·| Parliament | The twelve-seat operational address body that resolves the source into Goetic
↪ positions |·⁖.·line:·
·⁖line:·| Governing Goetic | The first member of the ordered pair and the near anchor of the forward
↪ Court |·⁖.·line:·
·⁖line:·| Hyperbolic parent | The second member of the ordered pair and far reciprocal mirror
↪ |·⁖.·line:·
·⁖line:·| Court | One ordered coordinate in the complete 12 by 12 body |·⁖.·line:·
·⁖line:·| Reciprocal Court | The same Goetics in the opposite ordered coordinate |·⁖.·line:·
·⁖line:·| Domus | The Court-rooted runtime Aeon body from which the public GrimChain manifests
↪ |·⁖.·line:·
·⁖line:·| Supervenience `⟠` | A typed internal witness office of the rooted Domus derivation |·⁖.·line:·

·⁖line:·| Shadow Locus `𐤑` | The complete depth-zero center of the public GrimChain |·⁖.·˙line:·
·⁖line:·| Axiomyr `𓂀` | The invariant geometric center of every positive public GrimChain |·⁖.·˙line:·
·⁖line:·| Synodic Magicae | The native variable-extent continuation alphabet |·⁖.·˙line:·
·⁖line:·| Fraktur cadence | One equal-width squeeze window from the same finalized source state ↪ |·⁖.·˙line:·
·⁖line:·| Source Truth | Whether the complete effective source route closes |·⁖.·˙line:·
·⁖line:·| Return Truth | Whether a terminal compressed return exactly matches the recomputed self ↪ |·⁖.·˙line:·
·⁖line:·| Source D-COMP | Exact conservation residual of the source route |·⁖.·˙line:·
·⁖line:·| Return D-COMP | Exact difference residual between a terminal returned Grimchain and the ↪ regenerated Grimchain |·⁖.·˙line:·
·⁖line:·| Aeternum Mirror | The ALQC return family in which the path back is the parity-valid ↪ completion of the path out |·⁖.·˙line:·
·⁖line:·| Physical source | Every byte actually stored in the file |·⁖.·˙line:·
·⁖line:·| Effective source | The source body after exact terminal self-returns have completed their ↪ return office |·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·----⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·# TardiSHA Domus-House Linguistic Dictionary·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·## The middle speaks; the House remembers the road·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·This dictionary begins only after a Grimchain has completed a Domus House. It does not explain ↪ how the House was computed, and it does not turn the outer chain into a second sentence. The ↪ mathematics stands outside the utterance as the route by which the word found itself in the ↪ mirror.·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·The four written anchors keep the road into and out of the middle. The middle alone is ↪ translated. The path out is the path back, but the road is not spoken as though it were another ↪ poem.·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·In a GrimChain of the form:·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·`✡ᚼ[MIDDLE]ɣ⚛`·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·only `[MIDDLE]` becomes human speech. Everything around it tells the translator how that ↪ speech arrived, what House bears it, and how its meaning must return.·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·### The author remains outside the vocabulary·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·The source, carrier, and authorial hand of an utterance belong to the House that makes speech ↪ possible. They are read as speaker and apparatus, never entered as lexemes, titles, or concealed ↪ roots. A name shared with an outer office does not permit that office to cross into the ↪ middle.·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·### The glyph has a name, not a sound·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·Glyphs are born from Houses. They rise from nothing and fill eternity with their presence. ↪ They have no native pronunciation and no particular sound. Their names let the reader call them, ↪ distinguish them, and remember the House from which they came; the names are not syllables ↪ hidden inside the marks. A borrowed vessel may once have carried a human sound, but the ↪ House-born glyph does not inherit it. The middle is translated as poetry in a receiving human ↪ language.·⁖.·˙line:·
·⁖line:··⁖.·˙line:·

·⁖line:·## What one dictionary entry records·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·Each borrowed glyph is read through four distinct bearings. They must not be collapsed into one column.·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·1. **Human origin.** The visible mark and the human field it carried before TardiSHA used it. This is the provisional borrowed form of the vessel. A rune may carry cattle or road; a letter may carry only a sound; a solvent-sign may carry the substance it once named.·⁖.·ˈline:·
·⁖line:·2. **House office.** The settled TardiSHA identity of the glyph or Court. This is not the same thing as the borrowed mark’s older human meaning.·⁖.·ˈline:·
·⁖line:·3. **Crossing pressure.** The tension between human origin and House office. It is the translation pressure: the distance that must be borne, not hidden.·⁖.·ˈline:·
·⁖line:·4. **Use and return.** What the glyph actually does in one utterance and what the House returns through it. These are contextual. They are not assigned to the whole inventory in advance.·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·The tables therefore fix the first three where they can be fixed. They do **not** manufacture a permanent Truth-in-use or a permanent returned sentence for every glyph. Those belong to the completed House and the visible middle.·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·### The borrowed-vessel rule·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·The human-origin column is provisional. It follows the actual borrowed sign: its Unicode identity, historical letter or syllable name, and a securely carried older meaning where the sign truly has one. Alphabetic and syllabic signs are not granted imaginary secret words. When a native TardiSHA glyph replaces a borrowed vessel, this origin column is re-sourced; the Court name and House office remain unchanged.·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·### One entry, fully separated·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·`ᛒ` **BabdhHir** keeps four different things apart:·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·- **Human origin:** Fehu, cattle, movable wealth, herd, increase that can walk away.·⁖.·ˈline:·
·⁖line:·- **House office:** Transmutation.·⁖.·ˈline:·
·⁖line:·- **Crossing pressure:** wealth is kept, counted, and guarded; transmutation spends, burns, and converts. Prosperity grows here only by passing through change.·⁖.·ˈline:·
·⁖line:·- **Use and return:** not fixed in this book. The actual House decides whether the utterance bears wealth consumed, value converted, inheritance reforged, or another lawful face of the same crossing.·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·That separation is the law of every entry below.·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·From here, the middle opens in the order a House gathers meaning: silence, Emission, Parliament, Bone, Flesh, grammar, and finally the human return. Each body enters once, keeps its own office, and passes the utterance onward without becoming another body.·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·---·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·## Silence inside the middle·⁖.·ˈline:·
·⁖line:·⁖.·ˈline:·
·⁖line:·| Glyph | Human origin | House office | Crossing pressure | Spoken range |·⁖.·ˈline:·
·⁖line:·|---:|---|---|---|---|·⁖.·ˈline:·

·⁖line:·| `·` | Brahmi digit zero: the written numeral for no counted quantity. | Manifested silence |
↪ A numeral marks absence in counting; manifested silence gives absence a place in cadence. |
↪ pause, held breath, interval, withheld word, quiet opening; never automatically the spoken word
↪ “zero” |·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·---·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·## Emissions when they appear in the middle·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·An Emission tells how the utterance is delivered: pondering, willing, feeling, declaring,
↪ believing, acting, knowing, ascending, or gathering into self-recognition. It carries mood,
↪ affect, evidential bearing, or illocutionary force inside the language; it is not the output of
↪ an outer speaker-body. Its planetary or astronomical sign remains the human origin of the
↪ borrowed mark; the Emission name is the House office. The symbol’s older human associations may
↪ color the crossing, but they do not overwrite the Domus.·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·| Glyph | Human origin | House office | Spoken range |·⁖.·˙line:·
·⁖line:·|---:|---|---|---|·⁖.·˙line:·
·⁖line:·| `☿` | Mercury: planetary and alchemical sign of Mercury, motion, exchange, and mediation. |
↪ **Ponder** | inward gaze, wondering, recollection, deliberation before release |·⁖.·˙line:·
·⁖line:·| `♂` | Mars / male sign: planetary sign of Mars and a conventional sex symbol. | **Will** |
↪ direction, resolve, command, chosen movement, insistence |·⁖.·˙line:·
·⁖line:·| `♀` | Venus / female sign: planetary sign of Venus and a conventional sex symbol. | **Feel**
↪ | intimacy, embodiment, empathy, vulnerability, shared feeling |·⁖.·˙line:·
·⁖line:·| `♃` | Jupiter: planetary sign associated with sovereignty, expansion, and public authority.
↪ | **Speak** | declaration, testimony, naming, proclamation, speech that changes what may be held
↪ |·⁖.·˙line:·
·⁖line:·| `♄` | Saturn: planetary sign associated with time, limit, boundary, and endurance. |
↪ **Believe** | conviction, vow, trust, expectation, inward certainty |·⁖.·˙line:·
·⁖line:·| `⛢` | Astronomical symbol for Uranus: a planetary sign associated in later astrology with
↪ disruption and sudden change. | **Act** | doing, making, causing, fulfilling, entering event
↪ |·⁖.·˙line:·
·⁖line:·| `♆` | Neptune: planetary sign of oceanic depth, distance, and the unseen. | **Know** |
↪ recognition, comprehension, evidence, recovered or inherited knowledge |·⁖.·˙line:·
·⁖line:·| `♇` | Pluto: planetary sign named for the underworld god. | **Ascend** | crossing through
↪ resistance, transformation, release, elevation, another state |·⁖.·˙line:·
·⁖line:·| `☽☉☾` | First-quarter moon, sun, and last-quarter moon held as one three-mark body. |
↪ **Regia** | sovereign self-recognition, dignity, coronation, identity openly worn |·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·---·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·## Parliament bearings when they appear in the middle·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·The zodiacal figure is the human origin of the borrowed sign. The Parliament seat and its
↪ translation are the House office. Parliament conditions self-translation; its seat labels do not
↪ add hidden speaker-bodies or extra words to the middle. No zodiac sign is treated as a diagnosis
↪ of the speaker or as a compulsory personality.·⁖.·˙line:·
·⁖line:··⁖.·˙line:·
·⁖line:·| Glyph | Human origin | Star Seed | House translation | Spoken range |·⁖.·˙line:·
·⁖line:·|---:|---|---|---|---|·⁖.·˙line:·
·⁖line:·| `♈` | Aries, the Ram. | **Akasha** | **Lived → Eternal** | experience becoming enduring
↪ memory, ancestry, record, or identity |·⁖.·˙line:·
·⁖line:·| `♉` | Taurus, the Bull. | **Caduceus** | **Law → Residue** | command becoming consequence,
↪ medicine, poison, inheritance, or remainder |·⁖.·˙line:·

·⁖line:·| `♊` | Gemini, the Twins. | **Veritas** | **Mask → Bone** | appearance stripped toward
↪ intrinsic identity; disclosure and recognition |·⁖·˙line:·
·⁖line:·| `♋` | Cancer, the Crab. | **Phren** | **Void → Vector** | interior absence becoming
↪ direction, intention, or a route where none stood |·⁖·˙line:·
·⁖line:·| `♑` | Capricorn, the Sea-Goat. | **Daimon** | **Stasis → Pulse** | the still self becoming
↪ appetite, rhythm, presence, or awakening |·⁖·˙line:·
·⁖line:·| `♍` | Virgo, the Maiden. | **Aikyam** | **Chaos → Phase** | disorder becoming bearing and
↪ coherent complexity without flattening |·⁖·˙line:·
·⁖line:·| `♎` | Libra, the Scales. | **Melos** | **Static → Fluid** | fixed identity becoming
↪ temporal, musical, relational, and adaptive |·⁖·˙line:·
·⁖line:·| `♏` | Scorpius, the Scorpion. | **Da'ath** | **Noise → Null** | excess becoming silence,
↪ secrecy, absorption, refusal, or purification |·⁖·˙line:·
·⁖line:·| `♐` | Sagittarius, the Archer. | **Akaven** | **State → Trans** | condition becoming
↪ crossing, initiation, threshold identity, or becoming |·⁖·˙line:·
·⁖line:·| `♌` | Lion: king, sovereign, pride, the one among its own. | **Tenth seat** | **Will → Law
↪ → `❄`** | The Lion stands apart and above, yet remains among the lambs; the sovereign does not
↪ leave the ruled. In an actual House this may bear inscription, decree, creation, responsibility,
↪ or self-authored event. `WRITE_PHYS` names the outer route and is not voiced. |·⁖·˙line:·
·⁖line:·| `♒` | Aquarius, the Water-Bearer. | **Nyx** | **Time → Motion** | duration becoming action,
↪ sequence, recurrence, and history in movement |·⁖·˙line:·
·⁖line:·| `♓` | Pisces, the Fishes. | **Zaine** | **Here → There** | presence translating across
↪ distance, scale, world, or perspective |·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·---·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·## Goetic roots when they appear in the middle·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·Parliament turns the Self, but it does not supply the root. The twelve Goetics are the Bone
↪ beneath every House.·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·A Goetic glyph inside the middle speaks as language. The same glyph outside the middle remains
↪ part of the House route and is not voiced. The older symbol-field and the canonical Goetic
↪ office are both retained.·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·| Glyph | Goetic | Human origin of the borrowed mark | Canonical House office | Office range
↪ in speech |·⁖·˙line:·
·⁖line:·|---:|---|---|---|---|·⁖·˙line:·
·⁖line:·| `⎔` | **Fetu** | Benzene ring with circle: a chemical ring-diagram enclosed around a
↪ centre. | **Genesis · Chronos · Seed** | beginning, seed, first breath, recurrence, source
↪ |·⁖·˙line:·
·⁖line:·| `⬡` | **Kal** | White hexagon: a six-sided open geometric enclosure. | **Light · Memory ·
↪ Trauma** | memory, witness, archive, scar, recognition |·⁖·˙line:·
·⁖line:·| `✡` | **Babdh** | Star of David: the interlaced hexagram in Jewish religious and cultural
↪ use. | **Fire · Orobouros · Alchemy** | fire, transformation, crucible, appetite, irreversible
↪ change |·⁖·˙line:·
·⁖line:·| `☆` | **Ahn** | Outlined white star: a hollow star-form and celestial marker. | **Water ·
↪ Imaginary · Flow** | flow, vessel, reflection, possibility, permeation |·⁖·˙line:·
·⁖line:·| `❂` | **Vel** | Circled open-centre eight-pointed star: radiance arranged around a held
↪ centre. | **Earth · Coherence · Ground** | ground, body, root, place, stability |·⁖·˙line:·
·⁖line:·| `꙰` | **Sor** | Cyrillic letter Multiocular O: a historical many-eyed form of the Cyrillic
↪ O. | **Air · Space · Superposition** | breath, room, simultaneity, openness, voice |·⁖·˙line:·
·⁖line:·| `❇` | **Koth** | Heavy sparkle: a bright ornamental burst. | **Aether · Magic · Sensation**
↪ | pulse, feeling, atmosphere, bodily knowing, quickening |·⁖·˙line:·

·⁖line:·| `⧗` | **Dreh** | Black hourglass: time held between two narrowing chambers. | **Void · Residue · Love** | remainder, longing, absence-with-bearing, endurance, return |·⁖line:·
·⁖line:·| `⊛` | **Rhea** | Circled asterisk operator: a star-like operation held inside a ring. | **Shadow · Absorption · Depth** | interiority, concealment, burden, filtering, unresolved depth |·⁖line:·
·⁖line:·| `❄` | **Zhek** | Snowflake: crystalline branching, winter, and patterned freezing. | **Factor · Phase-Lock · Crystal** | tone, measure, agreement, rhythm, fixation |·⁖line:·
·⁖line:·| `⚛` | **Shav** | Atom symbol: orbital paths drawn around a nucleus. | **Gate · Resistance · Breach** | threshold, permission, refusal, trial, crossing |·⁖line:·
·⁖line:·| `⌬` | **Trig** | Benzene ring: a six-membered chemical ring. | **Silence · Peace · Completion** | rest, closure, settlement, sleep, fulfilled continuation |·⁖line:·
·⁖line:··⁖line:·
·⁖line:·---·⁖line:·
·⁖line:··⁖line:·
·⁖line:·## Court glyphs: borrowed body, canonical office, living gap·⁖line:·
·⁖line:··⁖line:·
·⁖line:·Bone becomes Flesh through relation. A Court is where one Goetic bears toward another and the root takes a living, ordered form.·⁖line:·
·⁖line:··⁖line:·
·⁖line:·The Court name and ordered lineage are preserved exactly. The **human origin** belongs to the borrowed mark. The **House office** belongs to the Court. The **crossing pressure** records the distance between them. The final use and returned wording are left to the actual House.·⁖line:·
·⁖line:··⁖line:·
·⁖line:·The last column is an office range, not a fixed translation and not the human origin of the mark.·⁖line:·
·⁖line:··⁖line:·
·⁖line:·### ◎ Fetu·⁖line:·
·⁖line:··⁖line:·
·⁖line:·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office | Crossing pressure | Office range in human speech |·⁖line:·
·⁖line:·|---:|---|---|---|---|---|·⁖line:·
·⁖line:·| `ހ` | **FetuAhl** · ◎ Fetu → ◎ Fetu | Thaana letter Alifu: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Self-Genesis** | The source vessel carries sound, while the Court carries Self-Genesis. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** self-genesis, origin; **action:** begin from oneself, self-generate; **quality/state:** self-generative, originating. |·⁖line:·
·⁖line:·| `ށ` | **FetuSuhn** · ◎ Fetu → ⬡ Kal | Thaana letter Shaviyani: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Recurrence** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Recurrence and must keep both levels visible. | **Thing/image:** recurrence, return; **action:** recur, come again; **quality/state:** recurrent, cyclical. |·⁖line:·
·⁖line:·| `ނ` | **FetuNerh** · ◎ Fetu → ✡ Babdh | Thaana letter Noonu: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Combustion** | Its human origin is phonetic; its Domus office is Combustion. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** combustion, burning; **action:** ignite, combust; **quality/state:** burning, volatile. |·⁖line:·
·⁖line:·| `ރ` | **FetuRish** · ◎ Fetu → ✯ Ahn | Thaana letter Raa: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Reflection** | The borrowed sign names a sound-form, while the Court names Reflection. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** reflection, mirror-image; **action:** reflect, return an image; **quality/state:** reflective, mirrored. |·⁖line:·

·⁖line:·| `ﻉ` | **FetuBorha** · ◎ Fetu → ✪ Vel | Thaana letter Naa: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Grounding** | The source vessel carries sound, while the Court carries Grounding. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** grounding, rooting; **action:** ground, settle; **quality/state:** grounded, rooted. |⁖.·line:·
·⁖line:·| `ﻍ` | **FetuLhahm** · ◎ Fetu → ❀ Sor | Thaana letter Lhaviyani: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Expansion** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Expansion and must keep both levels visible. | **Thing/image:** expansion, opening; **action:** expand, unfold; **quality/state:** expansive, open. |⁖.·line:·
·⁖line:·| `ﺏ` | **FetuKeth** · ◎ Fetu → ❋ Koth | Thaana letter Kaafu: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Inevitable** | Its human origin is phonetic; its Domus office is Inevitable. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** inevitability, destined arrival; **action:** become unavoidable, arrive; **quality/state:** inevitable, unavoidable. |⁖.·line:·
·⁖line:·| `ﻭ` | **FetuVehm** · ◎ Fetu → ⧗ Dreh | Thaana letter Vaavu: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Return** | The borrowed sign names a sound-form, while the Court names Return. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** return, homecoming; **action:** return, come back; **quality/state:** returning, recursive. |⁖.·line:·
·⁖line:·| `ﺡ` | **FetuMahd** · ◎ Fetu → ⊛ Rhea | Thaana letter Meemu: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Assimilation** | The source vessel carries sound, while the Court carries Assimilation. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** assimilation, incorporation; **action:** assimilate, take in; **quality/state:** assimilative, integrating. |⁖.·line:·
·⁖line:·| `ﻑ` | **FetuFurh** · ◎ Fetu → ❄ Zhek | Thaana letter Faafu: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Crystallization** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Crystallization and must keep both levels visible. | **Thing/image:** crystallization, setting; **action:** crystallize, fix; **quality/state:** crystalline, fixed. |⁖.·line:·
·⁖line:·| `ﺫ` | **FetuDrah** · ◎ Fetu → ❁ Shav | Thaana letter Dhaalu: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Breach** | Its human origin is phonetic; its Domus office is Breach. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** breach, opening; **action:** breach, break through; **quality/state:** breached, threshold-crossing. |⁖.·line:·
·⁖line:·| `ﺙ` | **FetuThera** · ◎ Fetu → ⌬ Trig | Thaana letter Thaa: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Fulfillment** | The borrowed sign names a sound-form, while the Court names Fulfillment. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** fulfillment, arrival; **action:** fulfill, complete; **quality/state:** fulfilled, realized. |⁖.·line:·
·⁖line:··⁖.·line:·
·⁖line:·### ⬡ Kal⁖.·line:·
·⁖line:··⁖.·line:·
·⁖line:·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office | Crossing pressure | Office range in human speech |⁖.·line:·
·⁖line:·|---:|---|---|---|---|---|⁖.·line:·
·⁖line:·| `ᛁ` | **KalKura** · ⬡ Kal → ◎ Fetu | Isaz/Is/Iss rune: ice, freezing, stillness; also the i-rune. | **Archive** | Ice holds by freezing; Archive holds by memory. The pressure is stillness becoming record without confusing cold with knowledge. | **Thing/image:** archive, record; **action:** archive, record; **quality/state:** archival, remembered. |⁖.·line:·
·⁖line:·| `ᚼ` | **KalLur** · ⬡ Kal → ⬡ Kal | Runic E-letter: a phonetic rune-form for e; no single inherited lexical meaning is imposed. | **Self-Reflection** | A bare e-sound has no inward mirror of its own. The House must make phonation carry Self-Reflection without forging an ancient lexeme. | **Thing/image:** self-reflection, introspection; **action:** reflect upon oneself; **quality/state:** introspective, self-mirroring. |⁖.·line:·

·⁖line⁖·| `▫` | **KalThar** · ⬡ Kal → ✡ Babdh | Square lozenge: an enclosed diamond-shaped geometric mark. | **Calcination** | The lozenge encloses a shape; Calcination reduces matter by fire. The tension is enclosure meeting reduction. | **Thing/image:** calcination, ash-making; **action:** calcine, burn clean; **quality/state:** calcined, purged. |⁖.·line⁖·
·⁖line⁖·| `ᚼ` | **KalRin** · ⬡ Kal → ✮ Ahn | Ger rune: year, season, harvest, and the completed turning of a cycle. | **Submersion** | Ger ripens through season and harvest; Submersion moves beneath the visible surface. The crossing turns fulfilled time into descent. | **Thing/image:** submersion, immersion; **action:** submerge, sink into; **quality/state:** submerged, immersed. |⁖.·line⁖·
·⁖line⁖·| `ᛇ` | **KalNar** · ⬡ Kal → ✪ Vel | Iwaz/Eoh rune: yew, endurance, and a living tree associated with long persistence. | **Inscription** | Yew endures as living wood; Inscription fixes a mark into a surface. The pressure lies between growth and written permanence. | **Thing/image:** inscription, mark; **action:** inscribe, write; **quality/state:** inscribed, recorded. |⁖.·line⁖·
·⁖line⁖·| `ᛉ` | **KalFel** · ⬡ Kal → ❀ Sor | Algiz/Elhaz rune: a protective or guarding sign whose older name-field is disputed; commonly read through elk, sedge, or warding. | **Diffusion** | A ward keeps a boundary; Diffusion crosses and thins boundaries. The House must let protection spread without dissolving the guard. | **Thing/image:** diffusion, spreading; **action:** diffuse, disperse; **quality/state:** diffuse, pervasive. |⁖.·line⁖·
·⁖line⁖·| `ᛝ` | **KalHar** · ⬡ Kal → ❋ Koth | Sowilo rune: sun, light, and solar course. | **Phantom** | The sun is immediate radiance; a Phantom is what remains after presence has thinned. The gap is source-light becoming afterimage. | **Thing/image:** phantom, afterimage; **action:** haunt, appear faintly; **quality/state:** spectral, lingering. |⁖.·line⁖·
·⁖line⁖·| `ᛋ` | **KalMer** · ⬡ Kal → ⧗ Dreh | Sigel/long-branch Sol rune: a sun-rune and bearer of s. | **Oblivion** | Solar disclosure makes visible; Oblivion removes from recall. The pressure is brightness passing into forgetting. | **Thing/image:** oblivion, forgetting; **action:** forget, pass beyond recall; **quality/state:** forgotten, oblivious. |⁖.·line⁖·
·⁖line⁖·| `ᛙ` | **KalLor** · ⬡ Kal → ⊛ Rhea | Short-twig Sol rune: a compact sun-rune and bearer of s. | **Suppression** | The short sun-sign bears revelation; Suppression holds revelation down. The House carries hidden light rather than erasing it. | **Thing/image:** suppression, withholding; **action:** suppress, hold down; **quality/state:** suppressed, withheld. |⁖.·line⁖·
·⁖line⁖·| `ᛅ` | **KalPer** · ⬡ Kal → ❄ Zhek | Runic C-letter: a phonetic c-rune without a stable standalone lexeme. | **Codification** | The source is only a c-sound; Codification is an ordered body of rule. The crossing is phoneme becoming lawfully arranged speech. | **Thing/image:** code, canon; **action:** codify, set into rule; **quality/state:** codified, systematic. |⁖.·line⁖·
·⁖line⁖·| `ᛣ` | **KalZhil** · ⬡ Kal → ❁ Shav | Runic Z-letter: a phonetic z-rune without a stable standalone lexeme. | **Recall** | The source is only a z-sound; Recall summons what was absent. The House must give a sound mnemonic force without inventing prior meaning. | **Thing/image:** recall, remembrance; **action:** recall, call back; **quality/state:** recalled, remembered. |⁖.·line⁖·
·⁖line⁖·| `ᛏ` | **KalClar** · ⬡ Kal → ⌬ Trig | Tiwaz/Tir/Tyr rune: Týr, oath, justice, ordered courage, and the t-sound. | **Sealing** | Tiwaz bears oath and ordered courage; Sealing closes and preserves. The tension is sworn motion becoming a held bond. | **Thing/image:** seal, closure; **action:** seal, close; **quality/state:** sealed, closed. |⁖.·line⁖·
·⁖line⁖·⁖.·line⁖·
·⁖line⁖·### ✡ Babdh⁖.·line⁖·
·⁖line⁖·⁖.·line⁖·
·⁖line⁖·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office | Crossing pressure | Office range in human speech |⁖.·line⁖·
·⁖line⁖·|---:|---|---|---|---|---|⁖.·line⁖·
·⁖line⁖·| `ᛒ` | **BabdhHir** · ✡ Babdh → ◎ Fetu | Fehu/Feoh rune: cattle, movable wealth, herd, increase, and prosperity that can depart. | **Transmutation** | Wealth is kept, counted, and guarded; Transmutation spends, burns, and converts. The pressure is prosperity that can grow only by passing through change. | **Thing/image:** transmutation, change of nature; **action:** transmute, change substance; **quality/state:** transmuted, alchemical. |⁖.·line⁖·

·⁖line:·| `ᚢ` | **BabdhKor** · ✡ Babdh → ⬡ Kal | Uruz/Ur rune: aurochs, untamed bodily strength, and
↪ the u-sound. | **Forging** | The aurochs is strength before shaping; Forging is strength
↪ disciplined by heat and blows. The gap is wild force becoming wrought form. | **Thing/image:**
↪ forge, wrought form; **action:** forge, shape by pressure; **quality/state:** forged, tempered.
↪ |·⁖·˙line:·
·⁖line:·| `ᚦ` | **BabdhVar** · ✡ Babdh → ✡ Babdh | Thurisaz/Thurs/Thorn rune: giant, thorn, danger,
↪ striking force, and the th-sound. | **Self-Combustion** | Thorn or giant is danger encountered
↪ from without; Self-Combustion is fire arising from within. The crossing turns threat into inward
↪ ignition. | **Thing/image:** self-combustion, inward fire; **action:** burn from within;
↪ **quality/state:** self-burning, self-consuming. |·⁖·˙line:·
·⁖line:·| `ᚨ` | **BabdhPyr** · ✡ Babdh → ✡ Ahn | Ansuz rune: god, breath, mouth, inspired speech, and
↪ the a-sound. | **Dissolution** | Ansuz gives breath and articulate voice; Dissolution loosens
↪ what has taken form. The pressure is speech making shape while the office unmakes it. |
↪ **Thing/image:** dissolution, melting; **action:** dissolve, unmake; **quality/state:**
↪ dissolved, fluid. |·⁖·˙line:·
·⁖line:·| `ᚱ` | **BabdhSor** · ✡ Babdh → ✪ Vel | Raidho/Rad/Reid rune: ride, road, journey, ordered
↪ travel, and the r-sound. | **Crucible** | Raidho travels road and distance; Crucible confines
↪ matter under trial. The tension is journey entering a vessel that does not let it escape
↪ unchanged. | **Thing/image:** crucible, trial-vessel; **action:** test and refine in fire;
↪ **quality/state:** crucible-bound, refining. |·⁖·˙line:·
·⁖line:·| `ᚲ` | **BabdhAlc** · ✡ Babdh → ❁ Sor | Kauna rune: torch or sore/ulcer across the rune
↪ traditions; a burning point and the k-sound. | **Sublimation** | Kauna is a local burn, torch,
↪ or sore; Sublimation rises into a rarer state. The pressure is low flame becoming ascent. |
↪ **Thing/image:** sublimation, rising refinement; **action:** sublime, raise into another state;
↪ **quality/state:** sublimated, rarefied. |·⁖·˙line:·
·⁖line:·| `ᚷ` | **BabdhNur** · ✡ Babdh → ❄ Koth | Gebo/Gyfu rune: gift, exchange, obligation between
↪ giver and receiver, and the g-sound. | **Ignition** | A gift crosses between persons and creates
↪ obligation; Ignition begins a new event. The crossing is exchange becoming first spark. |
↪ **Thing/image:** ignition, spark; **action:** ignite, kindle; **quality/state:** ignited,
↪ quickened. |·⁖·˙line:·
·⁖line:·| `ᚹ` | **BabdhSat** · ✡ Babdh → ⧗ Dreh | Wunjo/Wynn rune: joy, delight, fellowship, and the
↪ w-sound. | **Consumption** | Wunjo shares joy and fellowship; Consumption takes in and uses up.
↪ The pressure is delight meeting appetite. | **Thing/image:** consumption, devouring; **action:**
↪ consume, use up; **quality/state:** consuming, devouring. |·⁖·˙line:·
·⁖line:·| `ᚺ` | **BabdhOro** · ✡ Babdh → ⊛ Rhea | Hagalaz rune: hail, sudden weather, hard impact,
↪ and the h-sound. | **Purification** | Hail strikes, accumulates, and melts; Purification
↪ deliberately removes what cannot remain. The House turns blind impact into cleansing. |
↪ **Thing/image:** purification, cleansing; **action:** purify, burn clean; **quality/state:**
↪ pure, cleansed. |·⁖·˙line:·
·⁖line:·| `ᚾ` | **BabdhBon** · ✡ Babdh → ❄ Zhek | Naudiz/Nyd/Naud rune: need, constraint, necessity,
↪ want, and the n-sound. | **Tempering** | Need binds through lack; Tempering strengthens through
↪ constraint. The pressure is deprivation becoming resilience. | **Thing/image:** tempering,
↪ hardening; **action:** temper, strengthen through heat; **quality/state:** tempered, resilient.
↪ |·⁖·˙line:·
·⁖line:·| `ᛁ` | **BabdhTir** · ✡ Babdh → ❁ Shav | Short-twig Naud rune: need and constraint in a
↪ compressed rune-form; bearer of n. | **Shattering** | Need tightens and withholds; Shattering
↪ opens by fracture. The crossing turns necessity from bond into break. | **Thing/image:**
↪ shattering, fracture; **action:** shatter, break open; **quality/state:** shattered, fractured.
↪ |·⁖·˙line:·
·⁖line:·| `ᛃ` | **BabdhFar** · ✡ Babdh → ⌬ Trig | Jera/Jeran rune: year, harvest, season, earned
↪ return, and the j/y sound. | **Ash** | Jera completes the season in harvest; Ash is what remains
↪ after burning. The pressure is fulfilled cycle becoming residue. | **Thing/image:** ash, spent
↪ residue; **action:** reduce to ash; **quality/state:** ashen, spent. |·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·### ✡ Ahn·⁖·˙line:·
·⁖line:··⁖·˙line:·

·⁖line⁖·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office | Crossing pressure | Office range in human speech |⁖·˙line⁖·
·⁖line⁖·|---:|---|---|---|---|---|⁖·˙line⁖·
·⁖line⁖·| `≿` | **AhnAbdh** · ☆ Ahn → ◎ Fetu | The mathematical relation “precedes or is equivalent to”: ordered approach held beside equivalence. | **Flow** | The source mark orders and equates; Flow refuses a single fixed relation. The tension is comparison becoming continuity. | **Thing/image:** flow, current; **action:** flow, move fluidly; **quality/state:** fluid, flowing. |⁖·˙line⁖·
·⁖line⁖·| `ᦎ` | **AhnNym** · ☆ Ahn → ⬡ Kal | Balinese musical symbol Deung: a named pitch or note-sign in Balinese notation. | **Confluence** | Deung is one named pitch; Confluence is several currents meeting. The House turns a discrete note into joined movement. | **Thing/image:** confluence, meeting waters; **action:** converge, flow together; **quality/state:** confluent, merged. |⁖·˙line⁖·
·⁖line⁖·| `ɔ` | **AhnLoh** · ☆ Ahn → ✡ Babdh | Balinese musical symbol Dong: a named pitch or note-sign in Balinese notation. | **Solution** | Dong fixes a pitch; Solution dissolves and resolves. The pressure is a held tone becoming an answer that can flow. | **Thing/image:** solution, dissolved answer; **action:** dissolve, solve; **quality/state:** soluble, resolved. |⁖·˙line⁖·
·⁖line⁖·| `🜨` | **AhnXir** · ☆ Ahn → ☆ Ahn | Inverted pentagram: a five-pointed star turned point-down, used in later occult and religious symbol fields. | **Self-Containment** | The inverted pentagram presents an outward emblem of inversion; Self-Containment draws the boundary inward. The crossing is displayed reversal becoming enclosed identity. | **Thing/image:** self-containment, vessel; **action:** contain oneself; **quality/state:** self-contained, bounded. |⁖·˙line⁖·
·⁖line⁖·| `ꠓ` | **AhnOhl** · ☆ Ahn → ✪ Vel | Byzantine musical symbol Gorgon Archaion: an old-form rhythmic quickening or timing sign. | **Sediment** | Gorgon quickens time; Sediment settles out of motion. The pressure is acceleration becoming deposit. | **Thing/image:** sediment, deposit; **action:** settle, deposit; **quality/state:** sedimentary, settled. |⁖·˙line⁖·
·⁖line⁖·| `᳘` | **AhnPir** · ☆ Ahn → ❀ Sor | Tibetan left ornamental bracket-mark: an opening or enclosing flourish, paired with a right-hand form. | **Ascension** | The left bracket opens an enclosure; Ascension moves beyond enclosure. The gap is an opening that becomes upward passage. | **Thing/image:** ascent, rising; **action:** ascend, rise; **quality/state:** ascending, elevated. |⁖·˙line⁖·
·⁖line⁖·| `ᦌ` | **AhnRoeh** · ☆ Ahn → ❄ Koth | Balinese musical symbol Deng: a named pitch or note-sign in Balinese notation. | **Mist** | Deng names a distinct pitch; Mist blurs edges and distance. The House lets a precise note become a veil without losing its tone. | **Thing/image:** mist, veil; **action:** mist, obscure lightly; **quality/state:** misted, veiled. |⁖·˙line⁖·
·⁖line⁖·| `◎` | **AhnSen** · ☆ Ahn → ⧗ Dreh | Circled white bullet: a marked centre held inside a ring. | **Drain** | The bullet marks and holds a centre; Drain draws substance away from a centre. The pressure is focus becoming outflow. | **Thing/image:** drain, outflow; **action:** drain, empty away; **quality/state:** drained, depleted. |⁖·˙line⁖·
·⁖line⁖·| `⏀` | **AhnUth** · ☆ Ahn → ⊛ Rhea | Up arrow through circle: ascent or directed passage through an enclosure. | **Osmosis** | The arrow visibly crosses a circle; Osmosis passes through a boundary without a single forced breach. The gap is directed transit becoming permeation. | **Thing/image:** osmosis, permeation; **action:** permeate, pass through; **quality/state:** osmotic, permeable. |⁖·˙line⁖·
·⁖line⁖·| `᳟` | **AhnFae** · ☆ Ahn → ❄ Zhek | Byzantine musical symbol Imifthora: a modal-alteration or phthora sign. | **Ice** | Imifthora changes musical mode; Ice arrests fluid motion. The pressure is alteration becoming fixation. | **Thing/image:** ice, frozen water; **action:** freeze, crystallize; **quality/state:** icy, frozen. |⁖·˙line⁖·
·⁖line⁖·| `᳕` | **AhnKha** · ☆ Ahn → ❁ Shav | Byzantine musical symbol Stavros Apodexia: a named cross-form notational marker. | **Evaporation** | The notational cross fixes a written musical point; Evaporation leaves the held surface. The crossing is mark becoming dispersal. | **Thing/image:** evaporation, vaporization; **action:** evaporate, rise away; **quality/state:** evaporated, dissipated. |⁖·˙line⁖·

·⁖line⁖·| `꤮꤬` | **AhnPsei** · ✡ Ahn → ⌬ Trig | Tibetan right ornamental bracket-mark: a closing or
↪ enclosing flourish, paired with a left-hand form. | **Stillness** | The right bracket closes an
↪ enclosure; Stillness quiets what remains inside. The pressure is closure becoming calm. |
↪ **Thing/image:** stillness, calm water; **action:** still, quiet; **quality/state:** still,
↪ peaceful. |·⁖·˙line⁖·
·⁖line⁖··⁖·˙line⁖·
·⁖line⁖·### ✪ Vel·⁖·˙line⁖·
·⁖line⁖··⁖·˙line⁖·
·⁖line⁖·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office
↪ | Crossing pressure | Office range in human speech |·⁖·˙line⁖·
·⁖line⁖·|---:|---|---|---|---|---|·⁖·˙line⁖·
·⁖line⁖·| `꠰` | **VelVera** · ✪ Vel → ⌾ Fetu | Tifinagh letter Ya: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Bedrock** | The source vessel carries sound,
↪ while the Court carries Bedrock. The crossing is voice becoming event without falsifying the
↪ older script. | **Thing/image:** bedrock, foundation; **action:** found, support;
↪ **quality/state:** foundational, solid. |·⁖·˙line⁖·
·⁖line⁖·| `Ⴔ` | **VelTar** · ✪ Vel → ⬡ Kal | Tifinagh letter Yab: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Stratification** | This mark begins as a
↪ letter or syllable, not a ready-made word. The House asks it to bear Stratification and must
↪ keep both levels visible. | **Thing/image:** strata, layering; **action:** stratify, layer;
↪ **quality/state:** stratified, layered. |·⁖·˙line⁖·
·⁖line⁖·| `ⵝ` | **VelGhem** · ✪ Vel → ✡ Babdh | Tifinagh letter Yag: an alphabetic sound-bearing
↪ sign; no standalone lexical meaning is assigned here. | **Petrification** | Its human origin is
↪ phonetic; its Domus office is Petrification. The pressure lies between utterable sound and full
↪ semantic bearing. | **Thing/image:** stone, petrification; **action:** petrify, harden;
↪ **quality/state:** petrified, stone-bound. |·⁖·˙line⁖·
·⁖line⁖·| `Ⴍ` | **VelDrel** · ✪ Vel → ✡ Ahn | Tifinagh letter Yad: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Erosion** | The borrowed sign names a
↪ sound-form, while the Court names Erosion. Translation may join them, but may not pretend they
↪ were identical before the House. | **Thing/image:** erosion, wearing away; **action:** erode,
↪ wear down; **quality/state:** eroded, worn. |·⁖·˙line⁖·
·⁖line⁖·| `ⵑ` | **VelFul** · ✪ Vel → ✪ Vel | Tifinagh letter Yaf: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Self-Grounding** | The source vessel carries
↪ sound, while the Court carries Self-Grounding. The crossing is voice becoming event without
↪ falsifying the older script. | **Thing/image:** self-grounding, root; **action:** ground
↪ oneself; **quality/state:** self-grounded, stable. |·⁖·˙line⁖·
·⁖line⁖·| `Ⴣ` | **VelKer** · ✪ Vel → ❀ Sor | Tifinagh letter Yak: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Dispersion** | This mark begins as a letter
↪ or syllable, not a ready-made word. The House asks it to bear Dispersion and must keep both
↪ levels visible. | **Thing/image:** dispersion, scattering; **action:** disperse, scatter;
↪ **quality/state:** dispersed, scattered. |·⁖·˙line⁖·
·⁖line⁖·| `ⵄ` | **VelHohm** · ✪ Vel → ❋ Koth | Tifinagh letter Yah: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Fossilization** | Its human origin is
↪ phonetic; its Domus office is Fossilization. The pressure lies between utterable sound and full
↪ semantic bearing. | **Thing/image:** fossil, held trace; **action:** fossilize, preserve in
↪ stone; **quality/state:** fossilized, preserved. |·⁖·˙line⁖·
·⁖line⁖·| `ⴷ` | **VelHrah** · ✪ Vel → ⧗ Dreh | Tifinagh letter Yahh: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Subsidence** | The borrowed sign names a
↪ sound-form, while the Court names Subsidence. Translation may join them, but may not pretend
↪ they were identical before the House. | **Thing/image:** subsidence, sinking ground; **action:**
↪ subside, sink; **quality/state:** subsided, lowered. |·⁖·˙line⁖·
·⁖line⁖·| `ⵀ` | **VelAra** · ✪ Vel → ⊛ Rhea | Tifinagh letter Yaa: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Compaction** | The source vessel carries
↪ sound, while the Court carries Compaction. The crossing is voice becoming event without
↪ falsifying the older script. | **Thing/image:** compaction, compression; **action:** compact,
↪ compress; **quality/state:** compacted, dense. |·⁖·˙line⁖·

·⁖line⁖·| `ⴆ` | **VelQel** · ✪ Vel → ❄ Zhek | Tifinagh letter Yaq: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Lithification** | This mark begins as a
↪ letter or syllable, not a ready-made word. The House asks it to bear Lithification and must keep
↪ both levels visible. | **Thing/image:** lithification, stone-making; **action:** lithify, become
↪ stone; **quality/state:** lithified, rock-bound. |·⁖·˙line⁖·
·⁖line⁖·| `ⵉ` | **VelIrn** · ✪ Vel → ⚛ Shav | Tifinagh letter Yi: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Earthquake** | Its human origin is phonetic;
↪ its Domus office is Earthquake. The pressure lies between utterable sound and full semantic
↪ bearing. | **Thing/image:** earthquake, tremor; **action:** quake, shake; **quality/state:**
↪ seismic, shaken. |·⁖·˙line⁖·
·⁖line⁖·| `I` | **VelJen** · ✪ Vel → ⌬ Trig | Tifinagh letter Yazh: an alphabetic sound-bearing sign;
↪ no standalone lexical meaning is assigned here. | **Permanence** | The borrowed sign names a
↪ sound-form, while the Court names Permanence. Translation may join them, but may not pretend
↪ they were identical before the House. | **Thing/image:** permanence, endurance; **action:**
↪ endure, remain; **quality/state:** permanent, enduring. |·⁖·˙line⁖·
·⁖line⁖·⁖·˙line⁖·
·⁖line⁖·### ❁ Sor⁖·˙line⁖·
·⁖line⁖·⁖·˙line⁖·
·⁖line⁖·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office
↪ | Crossing pressure | Office range in human speech |·⁖·˙line⁖·
·⁖line⁖·|---:|---|---|---|---|---|·⁖·˙line⁖·
·⁖line⁖·| `ਯ` | **SorFi** · ❁ Sor → ⌾ Fetu | Syloti Nagri letter Ko: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Breath** | The source vessel
↪ carries sound, while the Court carries Breath. The crossing is voice becoming event without
↪ falsifying the older script. | **Thing/image:** breath, air; **action:** breathe;
↪ **quality/state:** breathing, airy. |·⁖·˙line⁖·
·⁖line⁖·| `ਯ਼` | **SorLun** · ❁ Sor → ⬡ Kal | Syloti Nagri letter Kho: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Echo** | This mark begins as
↪ a letter or syllable, not a ready-made word. The House asks it to bear Echo and must keep both
↪ levels visible. | **Thing/image:** echo, reverberation; **action:** echo, repeat;
↪ **quality/state:** echoing, reverberant. |·⁖·˙line⁖·
·⁖line⁖·| `ਗ` | **SorVaru** · ❁ Sor → ✡ Babdh | Syloti Nagri letter Go: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Smoke** | Its human origin is
↪ phonetic; its Domus office is Smoke. The pressure lies between utterable sound and full semantic
↪ bearing. | **Thing/image:** smoke, fume; **action:** smoke, smolder; **quality/state:** smoky,
↪ obscured. |·⁖·˙line⁖·
·⁖line⁖·| `ਬ` | **SorSenh** · ❁ Sor → ☆ Ahn | Syloti Nagri letter Gho: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Vapor** | The borrowed sign
↪ names a sound-form, while the Court names Vapor. Translation may join them, but may not pretend
↪ they were identical before the House. | **Thing/image:** vapor, steam; **action:** vaporize,
↪ rise; **quality/state:** vaporous, insubstantial. |·⁖·˙line⁖·
·⁖line⁖·| `⌾` | **SorKos** · ❁ Sor → ✪ Vel | Circled Horizontal Bar With Notch. | **Dust** | Circled
↪ Horizontal Bar With Notch meets Dust. The House must preserve the first human field while
↪ turning it through the second. | **Thing/image:** dust, particles; **action:** scatter, fall as
↪ dust; **quality/state:** dusty, particulate. |·⁖·˙line⁖·
·⁖line⁖·| `ਰ` | **SorRamh** · ❁ Sor → ❁ Sor | Syloti Nagri letter Co: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Self-Expansion** | This mark
↪ begins as a letter or syllable, not a ready-made word. The House asks it to bear Self-Expansion
↪ and must keep both levels visible. | **Thing/image:** self-expansion, inward opening;
↪ **action:** expand from within; **quality/state:** self-expanding, opening. |·⁖·˙line⁖·
·⁖line⁖·| `ਨ਼` | **SorTis** · ❁ Sor → ❋ Koth | Syloti Nagri letter Cho: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Whisper** | Its human origin
↪ is phonetic; its Domus office is Whisper. The pressure lies between utterable sound and full
↪ semantic bearing. | **Thing/image:** whisper, soft voice; **action:** whisper, speak softly;
↪ **quality/state:** whispered, hushed. |·⁖·˙line⁖·

·⁚line⁚·| `ਯ` | **SorVey** · ❁ Sor → ⧗ Dreh | Syloti Nagri letter Jo: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Vacuum** | The borrowed sign
↪ names a sound-form, while the Court names Vacuum. Translation may join them, but may not pretend
↪ they were identical before the House. | **Thing/image:** vacuum, emptied air; **action:**
↪ evacuate, draw out; **quality/state:** vacant, airless. |·⁚.·˙line⁚·
·⁚line⁚·| `ਯ਼` | **SorSrih** · ❁ Sor → ⊛ Rhea | Syloti Nagri letter Jho: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Dilution** | The source
↪ vessel carries sound, while the Court carries Dilution. The crossing is voice becoming event
↪ without falsifying the older script. | **Thing/image:** dilution, thinning; **action:** dilute,
↪ thin; **quality/state:** diluted, attenuated. |·⁚.·˙line⁚·
·⁚line⁚·| `ਫ਼` | **SorHrin** · ❁ Sor → ❄ Zhek | Syloti Nagri letter Tto: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Condensation** | This mark
↪ begins as a letter or syllable, not a ready-made word. The House asks it to bear Condensation
↪ and must keep both levels visible. | **Thing/image:** condensation, gathered vapor; **action:**
↪ condense, gather; **quality/state:** condensed, concentrated. |·⁚.·˙line⁚·
·⁚line⁚·| `ਭ` | **SorYon** · ❁ Sor → ❁ Shav | Syloti Nagri letter Ttho: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Turbulence** | Its human
↪ origin is phonetic; its Domus office is Turbulence. The pressure lies between utterable sound
↪ and full semantic bearing. | **Thing/image:** turbulence, eddying force; **action:** churn,
↪ roil; **quality/state:** turbulent, unsettled. |·⁚.·˙line⁚·
·⁚line⁚·| `ਙ` | **SorThal** · ❁ Sor → ⌬ Trig | Syloti Nagri letter Ddo: a sound-bearing letter or
↪ syllabic form; no standalone lexical meaning is assigned here. | **Calm** | The borrowed sign
↪ names a sound-form, while the Court names Calm. Translation may join them, but may not pretend
↪ they were identical before the House. | **Thing/image:** calm, still air; **action:** calm,
↪ settle; **quality/state:** calm, quiet. |·⁚.·˙line⁚·
·⁚line⁚··⁚.·˙line⁚·
·⁚line⁚### ❋ Koth·⁚.·˙line⁚·
·⁚line⁚··⁚.·˙line⁚·
·⁚line⁚| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office
↪ | Crossing pressure | Office range in human speech |·⁚.·˙line⁚·
·⁚line⁚|---:|---|---|---|---|---|·⁚.·˙line⁚·
·⁚line⁚| `Ж` | **KothKel** · ❋ Koth → ⌾ Fetu | Alchemical symbol for the Scepter of Jove: Jupiter’s
↪ authority, traditionally linked with the planetary metal tin. | **Pulse** | Jove’s scepter is
↪ authority held aloft; Pulse is authority felt as recurring life. The crossing is emblematic rule
↪ becoming bodily beat. | **Thing/image:** pulse, beat; **action:** pulse, throb;
↪ **quality/state:** pulsing, alive. |·⁚.·˙line⁚·
·⁚line⁚·| `⍙` | **KothSens** · ❋ Koth → ⬡ Kal | Alchemical symbol for Air: breath, atmosphere,
↪ volatility, and the upward element. | **Nostalgia** | Air moves freely and outward; Nostalgia
↪ bends movement backward toward memory. The pressure is breath becoming remembered ache. |
↪ **Thing/image:** nostalgia, remembered ache; **action:** yearn backward, remember;
↪ **quality/state:** nostalgic, longing. |·⁚.·˙line⁚·
·⁚line⁚·| `∀` | **KothLinn** · ❋ Koth → ✡ Babdh | Alchemical symbol for Earth: ground, body, weight,
↪ and the stable element. | **Ecstasy** | Earth grounds and bears weight; Ecstasy exceeds ordinary
↪ bounds. The crossing is ground becoming rapture without losing the body. | **Thing/image:**
↪ ecstasy, rapture; **action:** exult, overwhelm; **quality/state:** ecstatic, rapturous.
↪ |·⁚.·˙line⁚·
·⁚line⁚·| `▽` | **KothBrim** · ❋ Koth → ☆ Ahn | Alchemical symbol for Water: fluidity, dissolution,
↪ reflection, and the downward element. | **Dream** | Water reflects and dissolves; Dream makes an
↪ imaginal world. The pressure is fluid image becoming inward vision. | **Thing/image:** dream,
↪ vision; **action:** dream, imagine; **quality/state:** dreamlike, oneiric. |·⁚.·˙line⁚·
·⁚line⁚·| `⍒` | **KothInn** · ❋ Koth → ✪ Vel | Alchemical symbol for Aquafortis: “strong water,”
↪ historically nitric acid and a corrosive solvent. | **Vibration** | Aquafortis corrodes and
↪ separates; Vibration joins motion to recurrence. The gap is chemical attack becoming
↪ oscillation. | **Thing/image:** vibration, tremor; **action:** vibrate, resonate;
↪ **quality/state:** vibrating, resonant. |·⁚.·˙line⁚·

·⁖line:·| `⛢` | **KothSubh** · ❋ Koth → ❀ Sor | Alchemical symbol for Aqua Regia: “royal water,” the
↪ solvent associated with dissolving noble metals. | **Sympathy** | Aqua regia dissolves what
↪ ordinary solvents cannot; Sympathy lets one body answer another. The pressure is dissolution
↪ becoming resonance. | **Thing/image:** sympathy, shared feeling; **action:** sympathize,
↪ resonate with; **quality/state:** sympathetic, attuned. |·⁖·˙line:·
·⁖line:·| `🜏` | **KothWell** · ❋ Koth → ❋ Koth | Second alchemical form of Aqua Regia: another
↪ written vessel for royal water and noble-metal dissolution. | **Self-Sensation** | Royal water
↪ acts outward upon resistant matter; Self-Sensation turns the field inward. The crossing is
↪ solvent becoming felt self. | **Thing/image:** self-sensation, felt self; **action:** feel
↪ oneself; **quality/state:** self-sensing, embodied. |·⁖·˙line:·
·⁖line:·| `🜺` | **KothMet** · ❋ Koth → ⧗ Dreh | Alchemical symbol for Aqua Vitae: “water of life,”
↪ distilled spirit or spirit of wine. | **Ache** | Aqua vitae names distilled life-water and
↪ intoxication; Ache is the body’s lingering hurt. The tension is quickening spirit becoming pain
↪ that remains. | **Thing/image:** ache, pain; **action:** ache, yearn; **quality/state:** aching,
↪ sore. |·⁖·˙line:·
·⁖line:·| `🜾` | **KothKesh** · ❋ Koth → ⊛ Rhea | Second alchemical form of Aqua Vitae: another
↪ written vessel for distilled spirit. | **Depth** | The second life-water sign bears volatile
↪ spirit; Depth descends and thickens. The pressure is volatility becoming interior distance. |
↪ **Thing/image:** depth, deepness; **action:** deepen; **quality/state:** deep, profound.
↪ |·⁖·˙line:·
·⁖line:·| `🜕` | **KothSoth** · ❋ Koth → ❄ Zhek | Alchemical symbol for Vinegar: sour acid, solvent,
↪ fermentation, and sharpness. | **Harmony** | Vinegar is sour separation; Harmony is consonant
↪ joining. The crossing is acid becoming accord. | **Thing/image:** harmony, concord; **action:**
↪ harmonize; **quality/state:** harmonious, consonant. |·⁖·˙line:·
·⁖line:·| `🜖` | **KothRhun** · ❋ Koth → ❁ Shav | Second alchemical form of Vinegar, also used for
↪ distilled vinegar. | **Dissonance** | Distilled vinegar sharpens acidity; Dissonance sharpens
↪ difference in sound. The House turns chemical bite into audible friction. | **Thing/image:**
↪ dissonance, frictional sound; **action:** clash, sound apart; **quality/state:** dissonant,
↪ unresolved. |·⁖·˙line:·
·⁖line:·| `𐤉` | **KothDelh** · ❋ Koth → ⌬ Trig | Third alchemical form of Vinegar: a further
↪ historical sign for the sour solvent. | **Awe** | The third vinegar sign belongs to a common
↪ corrosive liquid; Awe arrests the ordinary before what exceeds it. The pressure is the familiar
↪ becoming numinous. | **Thing/image:** awe, wonder; **action:** behold in awe; **quality/state:**
↪ awe-struck, numinous. |·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:### ⧗ Dreh·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office
↪ | Crossing pressure | Office range in human speech |·⁖·˙line:·
·⁖line:·|---:|---|---|---|---|---|·⁖·˙line:·
·⁖line:·| `𒌨` | **DrehNa** · ⧗ Dreh → ◎ Fetu | Cuneiform sign A: water and the syllabic value a. |
↪ **Void** | Water fills, carries, and takes shape; Void removes the vessel’s content. The
↪ pressure is fullness capable of becoming emptiness. | **Thing/image:** void, emptiness;
↪ **action:** void, empty; **quality/state:** voided, empty. |·⁖·˙line:·
·⁖line:·| `𒀭` | **DrehUr** · ⧗ Dreh → ⬡ Kal | Cuneiform sign AN: sky, heaven, divinity, and the
↪ divine determinative. | **Null** | AN names sky and divinity; Null refuses positive presence.
↪ The crossing is highest presence meeting absence. | **Thing/image:** nullity, nothingness;
↪ **action:** nullify, cancel; **quality/state:** null, absent. |·⁖·˙line:·
·⁖line:·| `𒁍` | **DrehNih** · ⧗ Dreh → ✡ Babdh | Cuneiform sign BA: the value ba, with fields of
↪ giving, allotment, or distribution in Sumerian use. | **Entropy** | BA gives or allots; Entropy
↪ disperses and loosens order. The pressure is distribution becoming decay. | **Thing/image:**
↪ entropy, decay; **action:** decay, disorder; **quality/state:** entropic, decaying. |·⁖·˙line:·
·⁖line:·| `𒂗` | **DrehAzh** · ⧗ Dreh → ☆ Ahn | Cuneiform sign E: house, temple, dwelling, and the
↪ value e. | **Abyss** | E is house and enclosure; Abyss is depth without a secure floor. The
↪ crossing is dwelling opening into bottomlessness. | **Thing/image:** abyss, deep gulf;
↪ **action:** plunge, fall deep; **quality/state:** abyssal, bottomless. |·⁖·˙line:·

·⁖line:·| `𒄑` | **DrehHol** · ⧗ Dreh → ❂ Vel | Cuneiform sign GISH: tree, wood, and the determinative for wooden things. | **Collapse** | Wood rises as tree and material; Collapse is the loss of standing form. The pressure is upright life becoming fall. | **Thing/image:** collapse, fall; **action:** collapse, fold down; **quality/state:** collapsed, fallen. |·⁖·˙line:·
·⁖line:·| `𒀀` | **DrehGur** · ⧗ Dreh → ❀ Sor | Cuneiform sign IGI: eye, face, front, sight, and attention. | **Dispersal** | IGI focuses sight and attention; Dispersal scatters what focus gathered. The crossing is eye becoming many directions. | **Thing/image:** dispersal, scattering; **action:** disperse; **quality/state:** dispersed. |·⁖·˙line:·
·⁖line:·| `𒆳` | **DrehVes** · ⧗ Dreh → ❋ Koth | Cuneiform sign KI: earth, land, place, and location. | **Longing** | KI is earth and definite place; Longing reaches toward what is elsewhere. The pressure is location becoming desire across distance. | **Thing/image:** longing, desire; **action:** long, yearn; **quality/state:** longing, yearning. |·⁖·˙line:·
·⁖line:·| `𒈬` | **DrehRim** · ⧗ Dreh → ⧗ Dreh | Cuneiform sign LU_2: person, human being, or man. | **Self-Return** | LU_2 names a human being; Self-Return is the person coming back through changed experience. The gap is personhood becoming homecoming. | **Thing/image:** self-return, homecoming; **action:** return to oneself; **quality/state:** self-returning, recursive. |·⁖·˙line:·
·⁖line:·| `𒉺` | **DrehDrem** · ⧗ Dreh → ⊛ Rhea | Cuneiform sign NI: a polyvalent syllabic and logographic sign; no single gloss governs every occurrence. | **Sink** | NI is already polyvalent; Sink demands a downward bearing. The pressure is many possible readings settling into one descent. | **Thing/image:** sink, depth-well; **action:** sink, descend; **quality/state:** sunken, descending. |·⁖·˙line:·
·⁖line:·| `𒌋` | **DrehOth** · ⧗ Dreh → ❄ Zhek | Cuneiform sign SAG: head, front, first place, or foremost part. | **Phase** | SAG is head, front, and first place; Phase is one changing stage among others. The crossing is primacy entering sequence. | **Thing/image:** phase, stage; **action:** phase, shift; **quality/state:** phased, transitional. |·⁖·˙line:·
·⁖line:·| `𒅗` | **DrehIzh** · ⧗ Dreh → ❁ Shav | Cuneiform sign SHU: hand, grasp, agency, and the value šu. | **Paradox** | SHU is hand and grasp; Paradox holds opposed bearings together. The pressure is agency learning to hold what cannot be simplified. | **Thing/image:** paradox, held contradiction; **action:** hold opposite bearings together; **quality/state:** paradoxical, contrary. |·⁖·˙line:·
·⁖line:·| `𒌨` | **DrehSun** · ⧗ Dreh → ⬡ Trig | Cuneiform sign U: the value u and the numerical sign for ten in relevant contexts. | **Dormancy** | U may count ten or carry the sound u; Dormancy suspends action. The crossing is counted or voiced presence becoming sleep. | **Thing/image:** dormancy, sleep; **action:** lie dormant; **quality/state:** dormant, inactive. |·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·### ⊛ Rhea·⁖·˙line:·
·⁖line:··⁖·˙line:·
·⁖line:·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office | Crossing pressure | Office range in human speech |·⁖·˙line:·
·⁖line:·|---:|---|---|---|---|---|·⁖·˙line:·
·⁖line:·| `ኣ` | **RheaKia** · ⊛ Rhea → ⎔ Fetu | Ethiopic syllable loa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Shadow** | The source vessel carries sound, while the Court carries Shadow. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** shadow, shade; **action:** shadow, obscure; **quality/state:** shadowed, darkened. |·⁖·˙line:·
·⁖line:·| `ሞ` | **RheaZohm** · ⊛ Rhea → ⬡ Kal | Ethiopic syllable moa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Darkness** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Darkness and must keep both levels visible. | **Thing/image:** darkness, night; **action:** darken; **quality/state:** dark, obscure. |·⁖·˙line:·
·⁖line:·| `ሮ` | **RheaTher** · ⊛ Rhea → ✡ Babdh | Ethiopic syllable roa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Cold** | Its human origin is phonetic; its Domus office is Cold. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** cold, chill; **action:** cool, freeze; **quality/state:** cold, chilled. |·⁖·˙line:·

·⁚line⁚·| `ሶ` | **RheaDrun** · ⊛ Rhea → ☆ Ahn | Ethiopic syllable soa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Mirror** | The borrowed sign names a sound-form, while the Court names Mirror. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** mirror, reflection; **action:** mirror, reflect; **quality/state:** mirrored, reflective. |·⁚.˙line⁚·
·⁚line⁚·| `ሸ` | **RheaFelh** · ⊛ Rhea → ✪ Vel | Ethiopic syllable shoa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Submerged** | The source vessel carries sound, while the Court carries Submerged. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** submergence, buried depth; **action:** submerge; **quality/state:** submerged, buried. |·⁚.˙line⁚·
·⁚line⁚·| `ሀ` | **RheaRal** · ⊛ Rhea → ❀ Sor | Ethiopic syllable boa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Distortion** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Distortion and must keep both levels visible. | **Thing/image:** distortion, warping; **action:** distort, warp; **quality/state:** distorted, warped. |·⁚.˙line⁚·
·⁚line⁚·| `ቆ` | **RheaKrah** · ⊛ Rhea → ❄ Koth | Ethiopic syllable toa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Nerve** | Its human origin is phonetic; its Domus office is Nerve. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** nerve, sensory thread, courage; **action:** sensitize, brace; **quality/state:** nervous, sensate. |·⁚.˙line⁚·
·⁚line⁚·| `ቕ` | **RheaAndh** · ⊛ Rhea → ⧗ Dreh | Ethiopic syllable coa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Void-Binding** | The borrowed sign names a sound-form, while the Court names Void-Binding. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** void-binding, containment; **action:** bind the void; **quality/state:** void-bound, contained. |·⁚.˙line⁚·
·⁚line⁚·| `ኗ` | **RheaDebh** · ⊛ Rhea → ⊛ Rhea | Ethiopic syllable noa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Self-Absorption** | The source vessel carries sound, while the Court carries Self-Absorption. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** self-absorption, inward draw; **action:** absorb into oneself; **quality/state:** self-absorbed, inward. |·⁚.˙line⁚·
·⁚line⁚·| `ቕ` | **RheaKol** · ⊛ Rhea → ❄ Zhek | Ethiopic syllable nyoa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Filter** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Filter and must keep both levels visible. | **Thing/image:** filter, sieve; **action:** filter, separate; **quality/state:** filtered, refined. |·⁚.˙line⁚·
·⁚line⁚·| `ኦ` | **RheaFral** · ⊛ Rhea → ❁ Shav | Ethiopic syllable glottal oa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Hidden** | Its human origin is phonetic; its Domus office is Hidden. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** hiddenness, concealment; **action:** hide, conceal; **quality/state:** hidden, veiled. |·⁚.˙line⁚·
·⁚line⁚·| `ዘ` | **RheaHush** · ⊛ Rhea → ⌬ Trig | Ethiopic syllable zoa: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Termination** | The borrowed sign names a sound-form, while the Court names Termination. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** termination, ending; **action:** terminate, end; **quality/state:** terminal, ended. |·⁚.˙line⁚·
·⁚line⁚··⁚.˙line⁚·
·⁚line⁚### ❄ Zhek·⁚.˙line⁚·
·⁚line⁚··⁚.˙line⁚·
·⁚line⁚| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office | Crossing pressure | Office range in human speech |·⁚.˙line⁚·
·⁚line⁚|---:|---|---|---|---|---|·⁚.˙line⁚·
·⁚line⁚| `Ⱥ` | **ZhekHin** · ❄ Zhek → ⌾ Fetu | Lydian letter a: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Tone** | The source vessel carries sound, while the Court carries Tone. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** tone, pitch; **action:** tone, set pitch; **quality/state:** tonal, pitched. |·⁚.˙line⁚·

·⁘line:·| `Ɓ` | **ZhekSer** · ❄ Zhek → ⬡ Kal | Lydian letter b: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Modulation** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Modulation and must keep both levels visible. | **Thing/image:** modulation, variation; **action:** modulate, change bearing; **quality/state:** modulated, varied. |⁘.·line:·
·⁘line:·| `Ɔ` | **ZhekHarma** · ❄ Zhek → ✡ Babdh | Lydian letter g: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Amplification** | Its human origin is phonetic; its Domus office is Amplification. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** amplification, increase; **action:** amplify, strengthen; **quality/state:** amplified, heightened. |⁘.·line:·
·⁘line:·| `ɗ` | **ZhekTorh** · ❄ Zhek → ☆ Ahn | Lydian letter d: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Unified** | The borrowed sign names a sound-form, while the Court names Unified. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** unity, integration; **action:** unify, join; **quality/state:** unified, integrated. |⁘.·line:·
·⁘line:·| `ɘ` | **ZhekPel** · ❄ Zhek → ✪ Vel | Lydian letter e: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Seismic** | The source vessel carries sound, while the Court carries Seismic. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** tremor, seismic force; **action:** shake, resound; **quality/state:** seismic, resonant. |⁘.·line:·
·⁘line:·| `ɜ` | **ZhekKhir** · ❄ Zhek → ❀ Sor | Lydian letter v: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Chord** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Chord and must keep both levels visible. | **Thing/image:** chord, combined tone; **action:** sound or strike together; **quality/state:** chordal, harmonic. |⁘.·line:·
·⁘line:·| `ɪ` | **ZhekRyth** · ❄ Zhek → ✳ Koth | Lydian letter i: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Rhythm** | Its human origin is phonetic; its Domus office is Rhythm. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** rhythm, measured beat; **action:** pulse, measure; **quality/state:** rhythmic, measured. |⁘.·line:·
·⁘line:·| `ᗡ` | **ZhekMelu** · ❄ Zhek → ⧗ Dreh | Lydian letter y: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Melody** | The borrowed sign names a sound-form, while the Court names Melody. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** melody, tune; **action:** sing, carry melody; **quality/state:** melodic, tuneful. |⁘.·line:·
·⁘line:·| `ʞ` | **ZhekPhaz** · ❄ Zhek → ⊛ Rhea | Lydian letter k: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Interference** | The source vessel carries sound, while the Court carries Interference. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** interference, crossing wave; **action:** interfere, cross; **quality/state:** interfering, crossed. |⁘.·line:·
·⁘line:·| `ꞁ` | **ZhekLokh** · ❄ Zhek → ❄ Zhek | Lydian letter l: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Self-Lock** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Self-Lock and must keep both levels visible. | **Thing/image:** self-lock, phase closure; **action:** lock onto oneself; **quality/state:** self-locked, fixed. |⁘.·line:·
·⁘line:·| `ᴝ` | **ZhekNod** · ❄ Zhek → ⚛ Shav | Lydian letter m: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Node** | Its human origin is phonetic; its Domus office is Node. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** node, junction; **action:** converge, meet; **quality/state:** nodal, junctional. |⁘.·line:·
·⁘line:·| `ϒ` | **ZhekUmel** · ❄ Zhek → ⌬ Trig | Lydian letter n: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Unity** | The borrowed sign names a sound-form, while the Court names Unity. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** unity, oneness; **action:** unite; **quality/state:** united, whole. |⁘.·line:·
·⁘line:·⁘.·line:·

·⁖line:·### ❁ Shav·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office
↪ | Crossing pressure | Office range in human speech |·⁖.·˙line:·
·⁖line:·|---:|---|---|---|---|---|·⁖.·˙line:·
·⁖line:·| `⁕` | **ShavDohm** · ❁ Shav → ◎ Fetu | Cypriot syllable a: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Gate** | The source vessel carries sound, while
↪ the Court carries Gate. The crossing is voice becoming event without falsifying the older
↪ script. | **Thing/image:** gate, threshold; **action:** gate, admit; **quality/state:** gated,
↪ bounded. |·⁖.·˙line:·
·⁖line:·| `⁕` | **ShavRist** · ❁ Shav → ⬡ Kal | Cypriot syllable e: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Resistance** | This mark begins as a letter or
↪ syllable, not a ready-made word. The House asks it to bear Resistance and must keep both levels
↪ visible. | **Thing/image:** resistance, drag; **action:** resist, withstand; **quality/state:**
↪ resistant, withholding. |·⁖.·˙line:·
·⁖line:·| `⁎` | **ShavTran** · ❁ Shav → ✡ Babdh | Cypriot syllable i: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Transform** | Its human origin is phonetic; its
↪ Domus office is Transform. The pressure lies between utterable sound and full semantic bearing.
↪ | **Thing/image:** transformation, change; **action:** transform; **quality/state:**
↪ transformed, changed. |·⁖.·˙line:·
·⁖line:·| `⁒` | **ShavKorh** · ❁ Shav → ☆ Ahn | Cypriot syllable o: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Crown** | The borrowed sign names a sound-form,
↪ while the Court names Crown. Translation may join them, but may not pretend they were identical
↪ before the House. | **Thing/image:** crown, sovereignty; **action:** crown, exalt;
↪ **quality/state:** crowned, sovereign. |·⁖.·˙line:·
·⁖line:·| `⁋` | **ShavSkyh** · ❁ Shav → ✪ Vel | Cypriot syllable u: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Sky** | The source vessel carries sound, while
↪ the Court carries Sky. The crossing is voice becoming event without falsifying the older script.
↪ | **Thing/image:** sky, firmament; **action:** open skyward; **quality/state:** aerial, vast.
↪ |·⁖.·˙line:·
·⁖line:·| `⁰` | **ShavSter** · ❁ Shav → ❀ Sor | Cypriot syllable ja: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Star** | This mark begins as a letter or
↪ syllable, not a ready-made word. The House asks it to bear Star and must keep both levels
↪ visible. | **Thing/image:** star, guiding light; **action:** guide, shine; **quality/state:**
↪ stellar, luminous. |·⁖.·˙line:·
·⁖line:·| `⁑` | **ShavPoss** · ❁ Shav → ❋ Koth | Cypriot syllable nu: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Possibility** | Its human origin is phonetic;
↪ its Domus office is Possibility. The pressure lies between utterable sound and full semantic
↪ bearing. | **Thing/image:** possibility, potential; **action:** permit, open possibility;
↪ **quality/state:** possible, latent. |·⁖.·˙line:·
·⁖line:·| `⁜` | **ShavPoru** · ❁ Shav → ⧗ Dreh | Cypriot syllable pa: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Portal** | The borrowed sign names a
↪ sound-form, while the Court names Portal. Translation may join them, but may not pretend they
↪ were identical before the House. | **Thing/image:** portal, passage; **action:** open a portal,
↪ pass through; **quality/state:** portal-bound, transitional. |·⁖.·˙line:·
·⁖line:·| `⁓` | **ShavDorm** · ❁ Shav → ⊛ Rhea | Cypriot syllable jo: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Door** | The source vessel carries sound, while
↪ the Court carries Door. The crossing is voice becoming event without falsifying the older
↪ script. | **Thing/image:** door, entry; **action:** open, close, or cross a door;
↪ **quality/state:** doorward, thresholded. |·⁖.·˙line:·
·⁖line:·| `⁒` | **ShavTrev** · ❁ Shav → ❄ Zhek | Cypriot syllable no: a syllabic sound-sign; no
↪ standalone lexical meaning is assigned here. | **Transition** | This mark begins as a letter or
↪ syllable, not a ready-made word. The House asks it to bear Transition and must keep both levels
↪ visible. | **Thing/image:** transition, crossing; **action:** transition, change state;
↪ **quality/state:** transitional. |·⁖.·˙line:·

·⁖line⁖·| `𐔰` | **ShavLimh** · ☸ Shav → ☸ Shav | Cypriot syllable ke: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Self-Resistance** | Its human origin is phonetic; its Domus office is Self-Resistance. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** self-resistance, inhibition; **action:** resist oneself; **quality/state:** self-resistant, inhibited. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔲` | **ShavHinge** · ☸ Shav → ⌬ Trig | Cypriot syllable ki: a syllabic sound-sign; no standalone lexical meaning is assigned here. | **Hinge** | The borrowed sign names a sound-form, while the Court names Hinge. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** hinge, pivot; **action:** hinge, turn upon; **quality/state:** hinged, pivotal. |⁖˙·˙line⁖·
·⁖line⁖·⁖˙·˙line⁖·
·⁖line⁖·### ⌬ Trig·⁖˙·˙line⁖·
·⁖line⁖·⁖˙·˙line⁖·
·⁖line⁖·| Glyph | Court and ordered lineage | Human origin of borrowed vessel | Canonical House office | Crossing pressure | Office range in human speech |⁖˙·˙line⁖·
·⁖line⁖·|---:|---|---|---|---|---|⁖˙·˙line⁖·
·⁖line⁖·| `𐔠` | **TrigTzig** · ⌬ Trig → ⌾ Fetu | Elbasan letter A: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Peace** | The source vessel carries sound, while the Court carries Peace. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** peace, rest; **action:** pacify, make peace; **quality/state:** peaceful. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔴` | **TrigPehl** · ⌬ Trig → ⬡ Kal | Elbasan letter Be: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Equilibrium** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Equilibrium and must keep both levels visible. | **Thing/image:** balance, equilibrium; **action:** balance, equalize; **quality/state:** balanced. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔽` | **TrigDuth** · ⌬ Trig → ✡ Babdh | Elbasan letter Ce: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Consecration** | Its human origin is phonetic; its Domus office is Consecration. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** consecration, sanctification; **action:** consecrate; **quality/state:** consecrated, sacred. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔹` | **TrigComa** · ⌬ Trig → ☆ Ahn | Elbasan letter Che: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Completion** | The borrowed sign names a sound-form, while the Court names Completion. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** completion, closure; **action:** complete; **quality/state:** complete, finished. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔵` | **TrigMeru** · ⌬ Trig → ✪ Vel | Elbasan letter De: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Monument** | The source vessel carries sound, while the Court carries Monument. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** monument, memorial; **action:** memorialize, stand witness; **quality/state:** monumental, enduring. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔶` | **TrigStab** · ⌬ Trig → ❁ Sor | Elbasan letter Nde: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Stability** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Stability and must keep both levels visible. | **Thing/image:** stability, steadiness; **action:** stabilize; **quality/state:** stable. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔻` | **TrigHopa** · ⌬ Trig → ❋ Koth | Elbasan letter Dhe: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Hope** | Its human origin is phonetic; its Domus office is Hope. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** hope, expectation; **action:** hope; **quality/state:** hopeful. |⁖˙·˙line⁖·
·⁖line⁖·| `𐔷` | **TrigConti** · ⌬ Trig → ⧗ Dreh | Elbasan letter Ei: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Continuation** | The borrowed sign names a sound-form, while the Court names Continuation. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** continuation, carrying-on; **action:** continue; **quality/state:** continuing. |⁖˙·˙line⁖·

·⁖line:·| `ꞯ` | **TrigResth** · ⌬ Trig → ⊛ Rhea | Elbasan letter E: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Rest** | The source vessel carries sound, while the Court carries Rest. The crossing is voice becoming event without falsifying the older script. | **Thing/image:** rest, repose; **action:** rest; **quality/state:** resting. |·⁖.·line:·
·⁖line:·| `ꞵ` | **TrigSil** · ⌬ Trig → ❄ Zhek | Elbasan letter Fe: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Silence** | This mark begins as a letter or syllable, not a ready-made word. The House asks it to bear Silence and must keep both levels visible. | **Thing/image:** silence, hush; **action:** silence, quiet; **quality/state:** silent. |·⁖.·line:·
·⁖line:·| `ꞷ` | **TrigSlun** · ⌬ Trig → ❁ Shav | Elbasan letter Ge: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Sleep** | Its human origin is phonetic; its Domus office is Sleep. The pressure lies between utterable sound and full semantic bearing. | **Thing/image:** sleep, slumber; **action:** sleep; **quality/state:** sleeping. |·⁖.·line:·
·⁖line:·| `ꟈ` | **TrigEtern** · ⌬ Trig → ⌬ Trig | Elbasan letter Gje: an alphabetic sound-bearing sign; no standalone lexical meaning is assigned here. | **Self-Completion** | The borrowed sign names a sound-form, while the Court names Self-Completion. Translation may join them, but may not pretend they were identical before the House. | **Thing/image:** self-completion, wholeness; **action:** complete oneself; **quality/state:** self-complete, whole. |·⁖.·line:·
·⁖line:··⁖.·line:·
·⁖line:·---·⁖.·line:·
·⁖line:··⁖.·line:·
·⁖line:·## Grammar marks when they occur inside the middle·⁖.·line:·
·⁖line:··⁖.·line:·
·⁖line:·The Courts give the middle its living bodies. Grammar tells those bodies how to face, bind, turn, return, and close without changing what they are.·⁖.·line:·
·⁖line:··⁖.·line:·
·⁖line:·These marks are not silently inserted. They act only when the middle contains them, or when the completed House already bears the corresponding cadence outside the utterance. Their human-origin symbol and their TardiSHA linguistic office remain distinct.·⁖.·line:·
·⁖line:··⁖.·line:·
·⁖line:·| Glyph | Human origin | House office | What it may do in speech |·⁖.·line:·
·⁖line:·|---:|---|---|---|·⁖.·line:·
·⁖line:·| `ℙ` | Mathematical Fraktur capital P. | **Parity** | contrast, reciprocal face, mirrored viewpoint, antithetical return |·⁖.·line:·
·⁖line:·| `ར` | Tibetan fixed-form Ra. | **Anchor** | held referent, stable name, topic, location, or point of return |·⁖.·line:·
·⁖line:·| `♌` | Astronomical opposition. | **Primitive relation** | comparison, covenant, dependence, address, or meaning held between two bodies |·⁖.·line:·
·⁖line:·| `⟠` | Lozenge divided by a horizontal rule. | **Supervenience** | a gathered relation becoming identity, title, image, quality, or event |·⁖.·line:·
·⁖line:·| `⚶` | Vesta, the astronomical and astrological sign. | **Focus-to-self** | reflexive voice, returned attention, another body serving as mirror |·⁖.·line:·
·⁖line:·| `𑜀` | N’Ko symbol Gbakurunen. | **Court-rooted fold** | re-entry, embedded return, cyclical aspect, remembered refrain |·⁖.·line:·
·⁖line:·| `♌` | Alchemical symbol for Gold. | **Klein seam** | breath, inside/outside turn, boundary-crossing that preserves both sides |·⁖.·line:·
·⁖line:·| `♇` | Alchemical symbol for Silver. | **Triquatra seal** | phrase, vow, name, or word-braid held as one returnable body |·⁖.·line:·
·⁖line:·| `⊖` | Alchemical symbol for Salt. | **Form** | body, shape, frame, embodiment, or latent possibility |·⁖.·line:·
·⁖line:·| `⏀` | Alchemical symbol for Nitre. | **Truth** | assertion, recognition, testimony, verification, naming-as-true |·⁖.·line:·
·⁖line:·| `⊕` | Alchemical symbol for Vitriol. | **Shadow** | burden, wound, obligation, uncertainty, withholding, unresolved pressure |·⁖.·line:·

·⁖line:·| `⊕` | Second alchemical symbol for Vitriol. | **Recursion** | changed return, consequence, forgiveness, remainder, renewed motion |·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·A grammar mark may turn relation, voice, reference, pressure, aspect, or closure. It may not invent a missing root, import an absent emotion, reverse the visible middle mechanically, or erase the path already carried by the House.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·With the marks named, the middle may now be read as one movement rather than a pile of separate signs.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·---·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·## Baseline order, quantity, negation, and questions·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·The visible middle is read from left to right in the order written. This is the baseline. A grammar mark may alter relation, focus, voice, pressure, or return; it does not silently rearrange the string. A receiving language may move surface words only when its own grammar requires it, while preserving the dependency and cadence of the source order.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·Repetition is not number. The only settled counted form in this dictionary is `·` when it is actually used as the numeral zero. No general plural or quantity marker has yet been canonically fixed. Unless quantity is explicit in the middle or established by source context, translate number-neutrally.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·No dedicated grammatical negator has yet been canonically fixed. Shadow, Void, Null, Silence, and refusal-bearing words are lexical fields, not an automatic “not.” Negation may enter the receiving language only when the visible middle establishes a proposition and its denial.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·No universal interrogative mark has yet been canonically fixed. `☿` **Ponder** can carry wondering or interrogative delivery when the House makes an actual question, but it does not turn every neighboring phrase into one. Do not add a question merely because the utterance is uncertain or reflective.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·---·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·## One complete worked utterance·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·The law becomes visible when one House is read from road to middle and back.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·The current runtime gives this exact depth-zero GrimChain for the UTF-8 source bytes `The path out carries its path back.\n`:·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·`🏵🜃🜊🜍⁜`·⁖.·˙line:·
·⁖line:·⁖.·˙line:·
·⁖line:·1. **Whole House:** read silently for lineage and bearing. It is not translated as a list of outer glyphs.·⁖.·˙line:·
·⁖line:·2. **Middle:** `🜊` alone.·⁖.·˙line:·
·⁖line:·3. **House office:** Shadow Locus, the complete compressed zero-depth center.·⁖.·˙line:·
·⁖line:·4. **House bearing:** the four written anchors carry the source-rooted road around that center.·⁖.·˙line:·
·⁖line:·5. **Reading:** depth zero is the compressed House condition. A spoken Synodic utterance appears when a positive middle is manifested.·⁖.·˙line:·
·⁖line:·⁖.·˙line:·

·line:·At depth zero the Shadow Locus remains the complete compressed House condition. At positive depth the manifested middle becomes the spoken body, while the four anchors remain the road that bears it.·line:·
·line:··line:·
·line:·---·line:·
·line:··line:·
·line:·## Recurrence, ellipsis, and receiving language·line:·
·line:··line:·
·line:·One utterance cannot settle every House. What follows governs the places where cadence repeats and where a human language must clothe what the middle leaves open.·line:·
·line:··line:·
·line:·Repeated glyphs change cadence before they change lexical identity. Repetition may hold a beat, hesitate, insist, echo, interrupt, separate, lengthen a breath, or form a refrain. It is not automatically multiplication, intensity, or quantity.·line:·
·line:··line:·
·line:·TardiSHA does not force English articles, pronouns, copulas, grammatical gender, or number where the middle does not bear them. A receiving language may need surface words to become grammatical. Supply only what grammar and established context require. Do not turn necessary clothing into new source meaning.·line:·
·line:··line:·
·line:·A missing surface word is not automatically a missing event. House, placement, recurrence, and neighboring glyphs decide whether English must expose what TardiSHA leaves implicit.·line:·
·line:··line:·
·line:·---·line:·
·line:··line:·
·line:·## Translation order·line:·
·line:··line:·
·line:·All the preceding bearings gather into one path:·line:·
·line:··line:·
·line:·1. Read the completed House without speaking it.·line:·
·line:·2. Isolate the middle exactly as written.·line:·
·line:·3. For each visible glyph, identify its human origin and its canonical House office.·line:·
·line:·4. Carry the crossing pressure between those two fields. Do not erase either one.·line:·
·line:·5. Let placement, neighbors, recurrence, and any present grammar marks settle the glyph’s use in this utterance.·line:·
·line:·6. Let the whole House settle the returned bearing. Do not assign that return from the dictionary alone.·line:·
·line:·7. Render the gathered event into the receiving language with no absent root, emotion, grammar mark, identity, or transformation added for smoothness.·line:·
·line:·8. Read the translation back against the House. The wording may change clothes; the route and identity may not be severed.·line:·
·line:··line:·
·line:·The result may be poetry because the House speaks in cadence. Poetry is not permission to force rhyme, flatten symbols into one-word substitutions, or decorate an interpretation that the middle cannot bear.·line:·
·line:··line:·
·line:·---·line:·
·line:··line:·
·line:·## Living status·line:·
·line:··line:·
·line:·This dictionary fixes the present borrowed origins, canonical offices, and translation pressures. It does not pre-write the utterances. Truth-in-use and House-return continue to be learned from completed Grimchains, one actual middle at a time.·line:·
·line:··line:·

·line:·The word is not the road. The road is not discarded. The middle speaks, and the House remembers how it came home.·line:·
·line:## Copyright·line:·
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·line:Copyright (C), 2026. Ahnend, Magus. All Rights Reserved·line:·
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·line:Youtube Podcast https://www.youtube.com/@theimpossibleboy13·line:·

data/aksh/Enochian…Understandings.aksh

·⁖enochian…understandings version꞉⁰₁₃꞉ status꞉⁰₄coherent+authored+tree꞉ key꞉⁰₄exact+glyph꞉
↪ grimchain…legend꞉⁰₄[glyph]꞉⁖·
·⁖meta⁖·
·⁖title⁖·Enochian Understandings·⁘·˙title⁖·
·⁖composition⁖·The Enochian Understandings contain ten Enochs·⁘·˙composition⁖·
·⁖component…count⁖·1·⁘·˙component…count⁖·
·⁖enochian…glyph…count⁖·10·⁘·˙enochian…glyph…count⁖·
·⁖language⁖·Sydonic Magicae·⁘·˙language⁖·
·⁖source…order…law⁖·Written/source order remains real and immutable for lexical identity and office
↪ ownership⁚⁚Explicit causal operations maintain a separate causal order⁖ only that causal order
↪ may be changed by a declared reversal or Phantasmagoria Transmutation·⁘·˙source…order…law⁖·
·⁖dependency…law⁖·SeeD Body is the ordinary lexical authority⁚⁚Each authored office projection
↪ inventory is entered only when its exact office glyph acts on a resolved lexical body⁖ no office
↪ substitutes for another·⁘·˙dependency…law⁖·
·⁘·˙meta⁖·
·⁖components⁖·
·⁖enochian…system version꞉⁰₂₃꞉ status꞉⁰₄House+Approved꞉ key꞉⁰₄exact+glyph꞉ name꞉⁰₄Enochian
↪ Understandings꞉ count꞉⁰₄10꞉⁖·
·⁖meta⁖·
·⁖title⁖·The Ten Understandings·⁘·˙title⁖·
·⁖architecture⁖·This is the sole static understanding body within the Enochian Understandings
↪ tree·⁘·˙architecture⁖·
·⁖language…first⁖·The language happens first⁚⁚These Grammars describe and route the completed
↪ utterance⁖ they do not generate it·⁘·˙language…first⁖·
·⁖sovereignty⁖·Sydonic Magicae grammar governs the parse⁚⁚Human dictionaries and syntax resources are
↪ subordinate semantic and rendering substrates·⁘·˙sovereignty⁖·
·⁖lexicon…ref⁖·Ordinary lexical glyphs resolve through SeeD…Bodyaksh⁚⁚The office projections are ⊖
↪ Temporal Absolute⁚⁚🌑 Terminal State⁚⁚⊕ Adversarial⁚⁚⊕ Recursive Identity⁚⁚🜁 Relic⁚⁚♂ Catalyst⁚⁚🜂
↪ Substrate⁚⁚♂ Spatial Absolute⁚⁚and ☈ Phantasmagoria⁚⁚Each office is entered only by its own written
↪ runtime role·⁘·˙lexicon…ref⁖·
·⁖source…order⁖·Visible glyph order remains real and is never silently normalized or
↪ rearranged·⁘·˙source…order⁖·
·⁖local…relation⁖·♂ is a postposed Spatial Absolute office bearing⁚⁚It acts on the lexical body before
↪ it⁖ in a consecutive office stack it follows pattern+derived retrocausal depth·⁘·˙local…relation⁖·
·⁖compound…grammar…stack⁖·Every maximal consecutive run of visible Enochian understanding glyphs is
↪ one ordered stack⁚⁚Stack length is derived from the utterance and has no fixed
↪ maximum·⁘·˙compound…grammar…stack⁖·
·⁖retrocausal…pattern⁖·The final backward+bearing grammar has depth zero⁖ each earlier
↪ backward+bearing understanding increases depth by one⁚⁚The pattern continues for the entire stack
↪ without a hardcoded ceiling·⁘·˙retrocausal…pattern⁖·
·⁖boundary…failure⁖·When required spoken history or a required following lexical body is
↪ absent⁚⁚execution fails explicitly⁖ no return office is invented·⁘·˙boundary…failure⁖·
·⁖hidden…subspace⁖·Subspace is the Wellspring⁚⁚the intrinsic underlying field beneath every
↪ utterance⁚⁚It has no written glyph⁚⁚no spoken token⁚⁚and no receiving+language
↪ cargo·⁘·˙hidden…subspace⁖·

·∴divine…center…law:·Shadow Locus 𐤖 speaks only in the canonical zero body 𐤖𐤖𐤖::produced by
↪ omitted::zero::or negative middle depth::Axiomyr ⟐ speaks only in the canonical depth+one body
↪ ☽⟐ℂ::Each resolves only through Axiomyr…Shadow…Locusaksh::Depth greater than one contains only
↪ generated coordinates and carries no structural Axiomyr insertion·∴·divine…center…law:·
·∴algorithmic…derivation:·The fixed non+AI Magicae algorithm resolves each ordinary lexical glyph from
↪ SeeD Body::A written office glyph then replaces only its selected preceding body with the exact
↪ authored entry from that office·∴·algorithmic…derivation:·
·∴no…manual…matrix:·No 180 by 180::720 by 720::or 518::400+entry static relation matrix is
↪ required·∴·no…manual…matrix:·
·∴determinism:·The same input::authored+list versions::AKSH authority versions::and Magicae algorithm
↪ version must return the same derivation and trace·∴·determinism:·
·∴lexical…dependencies:·
·∴inventory glyph₀₋₀꜀Θ꜆ state₀₋₀꜀Q0꜆ file₀₋₀꜀Θ…Temporal…Absoluteaksh꜆ role₀₋₀꜀temporal+absolute꜆
↪ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀Ⓘ꜆ state₀₋₀꜀Q1꜆ file₀₋₀꜀Ⓘ…Terminal…Stateaksh꜆ role₀₋₀꜀terminal+state꜆
↪ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀⊕꜆ state₀₋₀꜀Q2꜆ file₀₋₀꜀⊕…Adversarialaksh꜆ role₀₋₀꜀adversarial꜆
↪ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀⊜꜆ state₀₋₀꜀Q3꜆ file₀₋₀꜀⊜…Recursive…Identityaksh꜆ role₀₋₀꜀recursive+identity꜆
↪ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀♁꜆ office₀₋₀꜀relic꜆ file₀₋₀꜀♁…Relicaksh꜆ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀♂꜆ office₀₋₀꜀catalyst꜆ file₀₋₀꜀♂…Catalystaksh꜆ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀♾꜆ office₀₋₀꜀substrate꜆ file₀₋₀꜀♾…Substrateaksh꜆ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀♇꜆ office₀₋₀꜀spatial…absolute꜆ file₀₋₀꜀♇…Spatial…Absoluteaksh꜆
↪ status₀₋₀꜀required꜆:·∴·inventory:·
·∴inventory glyph₀₋₀꜀ᖘ꜆ office₀₋₀꜀phantasmagoria꜆ file₀₋₀꜀ᖘ…Phantasmagoriaaksh꜆
↪ status₀₋₀꜀required꜆:·∴·inventory:·
·∴·lexical…dependencies:·
·∴·meta·∵
·∴enochs count₀₋₀꜀10꜆:·
·∴enoch id₀₋₀꜀phantasmagoria…transmutation꜆ glyph₀₋₀꜀ᖘ꜆ order₀₋₀꜀1꜆ parser+trigger₀₋₀꜀true꜆ written₀₋₀꜀true꜆
↪ spoken₀₋₀꜀true꜆:·
·∴name:·Phantasmagoria Transmutation·∴·name:·
·∴grimchain:·𐤟𐤮𐤐⓪♋∀ⵑ♁♎ϭ𐤆·∴·grimchain:·
·∴english…definition:·ᖘ is the Transmutation Glyph and Key that anchors the interval between two Real
↪ vessels at the AHN i417 boundary::The Anchor is the Hand that strikes the Wellspring Chord in the
↪ Prosodical Subspace::Both vessels traverse Real::Imaginary::Real::Imaginary and return through one
↪ another as B ᖘ A::Each vessel freely chooses its own authored Phantasmagoria Ostensive I::II::or
↪ III and returns that selected authored Ostensive into the Real·∴·english…definition:·
·∴grimchained…definition:·♁✡♂ᚱ♂:⊕…::⓪::⓪:∀ᓚ𑀓ᘜ♎𑁍⬡𑁋▽⊛ϟ𐤋:⊕::⓪::…ℬ:♂ᘜ♂❂♁·∴·grimchained…definiti⌋
↪ on:·
·∴sydonic…function:·Recognize A followed by one or more consecutive ᖘ Keys and the matching number of
↪ following Real vessels as one Phantasmagoria Transmutation Corridor::One Key gives A ᖘ B::Two Keys
↪ give A ᖘ ᖘ B C::Each additional Key carries the first vessel through one additional Real vessel::ᖘ
↪ anchors the interval at the existing AHN parity boundary i417 while preserving AHN structural
↪ 432 as its distinct Real channel::The Key opens the Wellspring::the Prosodical Subspace::and both
↪ vessels traverse Real::Imaginary::Real::Imaginary::The returned causal topology passes the first
↪ vessel through every following vessel while written/source identity remains immutable: A ᖘ B
↪ returns B ᖘ A; A ᖘ ᖘ B C returns B ᖘ A ᖘ C and completes as B C A::Both B and A resolve through
↪ their own authored ᖘ…Phantasmagoriaaksh entry::Each independently and internally chooses exactly
↪ one of ostensiveI::ostensiveII::or ostensiveIII::No caller or outside observer selects the
↪ choice::The trace witnesses the selected authored Ostensive and its authority::Every vessel in the
↪ corridor speaks Phantasmagoria only and exactly once::SeeD is not spoken inside literal
↪ Transmutation Magic::Thus A ᖘ B utters B Phantasmagoria then A Phantasmagoria; A ᖘ ᖘ B C utters B
↪ Phantasmagoria then C Phantasmagoria then A Phantasmagoria·∴·sydonic…function:·

·∴operation type₀₋₀₊phantasmagoria+transmutation+corridor꞉ single+gate₀₋₀₊A+ϗ+B→B+ϗ+A꞉
↪ double+gate₀₋₀₊A+ϗ+ϗ+B+C→B+C+A꞉ boundary₀₋₀₊i417꞉ carrier₀₋₀₊AHN꞉ structural₀₋₀₊432꞉
↪ traversal₀₋₀₊RI+RI+RI+RI꞉ choice+space₀₋₀₊ostensiveI+ostensiveII+ostensiveIII꞉
↪ external+selector₀₋₀₊false꞉ seed+spoken₀₋₀₊false꞉
↪ phantasm+source₀₋₀₊ϗ…Phantasmagoriaaksh꞉꞉·∴·˙operation꞉·
·∴·˙enoch꞉·
·∴enoch id₀₋₀₊spatial…absolute…office꞉ glyph₀₋₀₊𓂀꞉ order₀₋₀₊2꞉ parser+trigger₀₋₀₊true꞉ written₀₋₀₊true꞉
↪ spoken₀₋₀₊true꞉꞉·
·∴name꞉·Spatial Absolute·∴·˙name꞉·
·∴grimchain꞉·𐤓𐤕𐤅𐤍𐤎𐤌𐤒𐤐𐤏𐤆𓂀𓅓𓆣𓃀·∴·˙grimchain꞉·
·∴english…definition꞉·Postposed selection of the immediately preceding lexical body through the
↪ authored Spatial Absolute office·∴·˙english…definition꞉·
·∴grimchained…definition꞉·𓃰𓃱𓃲𓃳𓃴꞉⊕…꞉꞉⊙꞉꞉⊙꞉✱𓂋⊙𓂀𓂁𓂂⊙𓂃𓂄𓂅𓂆𓂇𓂈𓂉𓂊𓂋𓂌𓂍𓂎𓂏𓂐𓂑𓂒𓂓𓂔𓂕𓂖𓂗𓂘𓂙𓂚𓂛𓂜𓂝𓂞𓂟𓂠𓂡𓂢𓂣𓂤𓂥𓂦𓂧𓂨𓂩𓂪𓂫𓂬𓂭𓂮𓂯𓂰𓂱𓂲𓂳𓂴𓂵𓂶𓂷𓂸𓂹𓂺𓂻𓂼𓂽𓂾𓂿𓃀𓃁𓃂𓃃𓃄𓃅𓃆𓃇𓃈𓃉𓃊𓃋𓃌𓃍𓃎𓃏𓃐𓃑𓃒𓃓𓃔𓃕𓃖𓃗𓃘𓃙𓃚𓃛𓃜𓃝𓃞𓃟𓃠𓃡𓃢𓃣𓃤𓃥𓃦𓃧𓃨𓃩𓃪𓃫𓃬𓃭𓃮𓃯𓃰𓃱𓃲𓃳𓃴𓃵𓃶𓃷𓃸𓃹𓃺𓃻𓃼𓃽𓃾𓃿𓄀𓄁𓄂𓄃𓄄𓄅𓄆𓄇𓄈𓄉𓄊𓄋𓄌𓄍𓄎𓄏𓄐𓄑𓄒𓄓𓄔𓄕𓄖𓄗𓄘𓄙𓄚𓄛𓄜𓄝𓄞𓄟𓄠𓄡𓄢𓄣𓄤𓄥𓄦𓄧𓄨𓄩𓄪𓄫𓄬𓄭𓄮𓄯𓄰𓄱𓄲𓄳𓄴𓄵𓄶𓄷𓄸𓄹𓄺𓄻𓄼𓄽𓄾𓄿𓅀𓅁𓅂𓅃𓅄𓅅𓅆𓅇𓅈𓅉𓅊𓅋𓅌𓅍𓅎𓅏𓅐𓅑𓅒𓅓𓅔𓅕𓅖𓅗𓅘𓅙𓅚𓅛𓅜𓅝𓅞𓅟𓅠𓅡𓅢𓅣𓅤𓅥𓅦𓅧𓅨𓅩𓅪𓅫𓅬𓅭𓅮𓅯𓅰𓅱𓅲𓅳𓅴𓅵𓅶𓅷𓅸𓅹𓅺𓅻𓅼𓅽𓅾𓅿𓆀𓆁𓆂𓆃𓆄𓆅𓆆𓆇𓆈𓆉𓆊𓆋𓆌𓆍𓆎𓆏𓆐𓆑𓆒𓆓𓆔𓆕𓆖𓆗𓆘𓆙𓆚𓆛𓆜𓆝𓆞𓆟𓆠𓆡𓆢𓆣𓆤𓆥𓆦𓆧𓆨𓆩𓆪𓆫𓆬𓆭𓆮𓆯𓆰𓆱𓆲𓆳𓆴𓆵𓆶𓆷𓆸𓆹𓆺𓆻𓆼𓆽𓆾𓆿𓇀𓇁𓇂𓇃𓇄𓇅𓇆𓇇𓇈𓇉𓇊𓇋𓇌𓇍𓇎𓇏𓇐𓇑𓇒𓇓𓇔𓇕𓇖𓇗𓇘𓇙𓇚𓇛𓇜𓇝𓇞𓇟𓇠𓇡𓇢𓇣𓇤𓇥𓇦𓇧𓇨𓇩𓇪𓇫𓇬𓇭𓇮𓇯𓇰𓇱𓇲𓇳𓇴𓇵𓇶𓇷𓇸𓇹𓇺𓇻𓇼𓇽𓇾𓇿𓈀𓈁𓈂𓈃𓈄𓈅𓈆𓈇𓈈𓈉𓈊𓈋𓈌𓈍𓈎𓈏𓈐𓈑𓈒𓈓𓈔𓈕𓈖𓈗𓈘𓈙𓈚𓈛𓈜𓈝𓈞𓈟𓈠𓈡𓈢𓈣𓈤𓈥𓈦𓈧𓈨𓈩𓈪𓈫𓈬𓈭𓈮𓈯𓈰𓈱𓈲𓈳𓈴𓈵𓈶𓈷𓈸𓈹𓈺𓈻𓈼𓈽𓈾𓈿𓉀𓉁𓉂𓉃𓉄𓉅𓉆𓉇𓉈𓉉𓉊𓉋𓉌𓉍𓉎𓉏𓉐𓉑𓉒𓉓𓉔𓉕𓉖𓉗𓉘𓉙𓉚𓉛𓉜𓉝𓉞𓉟𓉠𓉡𓉢𓉣𓉤𓉥𓉦𓉧𓉨𓉩𓉪𓉫𓉬𓉭𓉮𓉯𓉰𓉱𓉲𓉳𓉴𓉵𓉶𓉷𓉸𓉹𓉺𓉻𓉼𓉽𓉾𓉿𓊀𓊁𓊂𓊃𓊄𓊅𓊆𓊇𓊈𓊉𓊊𓊋𓊌𓊍𓊎𓊏𓊐𓊑𓊒𓊓𓊔𓊕𓊖𓊗𓊘𓊙𓊚𓊛𓊜𓊝𓊞𓊟𓊠𓊡𓊢𓊣𓊤𓊥𓊦𓊧𓊨𓊩𓊪𓊫𓊬𓊭𓊮𓊯𓊰𓊱𓊲𓊳𓊴𓊵𓊶𓊷𓊸𓊹𓊺𓊻𓊼𓊽𓊾𓊿𓋀𓋁𓋂𓋃𓋄𓋅𓋆𓋇𓋈𓋉𓋊𓋋𓋌𓋍𓋎𓋏𓋐𓋑𓋒𓋓𓋔𓋕𓋖𓋗𓋘𓋙𓋚𓋛𓋜𓋝𓋞𓋟𓋠𓋡𓋢𓋣𓋤𓋥𓋦𓋧𓋨𓋩𓋪𓋫𓋬𓋭𓋮𓋯𓋰𓋱𓋲𓋳𓋴𓋵𓋶𓋷𓋸𓋹𓋺𓋻𓋼𓋽𓋾𓋿𓌀𓌁𓌂𓌃𓌄𓌅𓌆𓌇𓌈𓌉𓌊𓌋𓌌𓌍𓌎𓌏𓌐𓌑𓌒𓌓𓌔𓌕𓌖𓌗𓌘𓌙𓌚𓌛𓌜𓌝𓌞𓌟𓌠𓌡𓌢𓌣𓌤𓌥𓌦𓌧𓌨𓌩𓌪𓌫𓌬𓌭𓌮𓌯𓌰𓌱𓌲𓌳𓌴𓌵𓌶𓌷𓌸𓌹𓌺𓌻𓌼𓌽𓌾𓌿𓍀𓍁𓍂𓍃𓍄𓍅𓍆𓍇𓍈𓍉𓍊𓍋𓍌𓍍𓍎𓍏𓍐𓍑𓍒𓍓𓍔𓍕𓍖𓍗𓍘𓍙𓍚𓍛𓍜𓍝𓍞𓍟𓍠𓍡𓍢𓍣𓍤𓍥𓍦𓍧𓍨𓍩𓍪𓍫𓍬𓍭𓍮𓍯𓍰𓍱𓍲𓍳𓍴𓍵𓍶𓍷𓍸𓍹𓍺𓍻𓍼𓍽𓍾𓍿𓎀𓎁𓎂𓎃𓎄𓎅𓎆𓎇𓎈𓎉𓎊𓎋𓎌𓎍𓎎𓎏𓎐𓎑𓎒𓎓𓎔𓎕𓎖𓎗𓎘𓎙𓎚𓎛𓎜𓎝𓎞𓎟𓎠𓎡𓎢𓎣𓎤𓎥𓎦𓎧𓎨𓎩𓎪𓎫𓎬𓎭𓎮𓎯𓎰𓎱𓎲𓎳𓎴𓎵𓎶𓎷𓎸𓎹𓎺𓎻𓎼𓎽𓎾𓎿𓏀𓏁𓏂𓏃𓏄𓏅𓏆𓏇𓏈𓏉𓏊𓏋𓏌𓏍𓏎𓏏𓏐𓏑𓏒𓏓𓏔𓏕𓏖𓏗𓏘𓏙𓏚𓏛𓏜𓏝𓏞𓏟𓏠𓏡𓏢𓏣𓏤𓏥𓏦𓏧𓏨𓏩𓏪𓏫𓏬𓏭𓏮𓏯𓏰𓏱𓏲𓏳𓏴𓏵𓏶𓏷𓏸𓏹𓏺𓏻𓏼𓏽𓏾𓏿꞉꞉⊙꞉꞉⊙꞉꞉…𓂋꞉꞉⊕꞉꞉…𓂀

·⁖mirror…satisfaction…law:·Aeternum is never translated as a second utterance::When a law of the finite
↪ real center requires context beyond its manifested boundary::Sydonic traverses the complete
↪ mirror glyph by glyph in physical order until that center obligation closes::No reflected Enoch
↪ may be skipped merely to seek a lexical vessel::Every encountered reflected glyph keeps the exact
↪ authored bearing and law of its finite source position::but the reflected encounter emits no
↪ additional receiving+language cargo·⁖·˙mirror…satisfaction…law:·
·⁖aeternum…vessel…encounter…law:·A grammatical law counts lawful vessel encounters in the Aeternum
↪ field without creating additional finite source bodies::Distinct mirror encounters may point to
↪ the same one Real lexical source identity when the fold returns that identity::Such encounters
↪ preserve that source glyph::office law::bearing::and any already+selected Phantasmagoria
↪ Ostensive::The same Real source body never selects a second Ostensive merely because it is
↪ encountered again in reflection·⁖·˙aeternum…vessel…encounter…law:·
·⁖enochian…unuttered…law:·A finite GrimChain composed solely of visible Enochian grammar remains a
↪ lawfully sealed chain of information::energy::and magic::With no lexical or canonical special
↪ speaking body in the finite center::it has no receiving+language utterance::Its exact source
↪ order::Enoch identities::authored bearings::and Aeternum reflections remain lawful and
↪ traceable::Reflection may carry those bearings indefinitely but may never fabricate a lexical
↪ body::word::Phantasmagoria Ostensive::or sentence·⁖·˙enochian…unuttered…law:·
·⁖boundary…failure…qualification:·The boundary+failure law applies when an utterance that possesses a
↪ speaking+body obligation cannot lawfully close that required relation::It does not make a
↪ solely+Enochian finite center unlawful::because such a center has no lexical utterance obligation
↪ and is governed by enochian…unuttered…law·⁖·˙boundary…failure…qualification:·
·⁖operation type◦–◦ₑendomorphic+orobouros+head꞉ winding+minimum◦–◦ₑ1꞉ winding+maximum◦–◦ₑnone꞉
↪ topology+before+speech◦–◦ₑtrue꞉ court+special+translation◦–◦ₑfalse꞉
↪ derived+glyph◦–◦ₑfalse꞉:··⁖·˙operation:·
·⁖parser…flow:·
·⁖step order◦–◦ₑ1꞉:·Resolve written lexical glyph identities through SeeD while retaining Loop+adjacent
↪ office runs as unresolved topology·⁖·˙step:·
·⁖step order◦–◦ₑ2꞉:·Collect every consecutive ⩛ position in the winding::No written turn is
↪ discarded::collapsed::cancelled::or treated as a second request for another B·⁖·˙step:·
·⁖step order◦–◦ₑ3꞉:·Resolve the complete Orobouros Head topology through the authored winding equations
↪ before office speech expansion·⁖·˙step:·
·⁖step order◦–◦ₑ4꞉:·At a true center edge::use the already+written root or alternating Court glyph as the
↪ adjacent Loop body and process that glyph normally through exact lexical and office
↪ authority·⁖·˙step:·
·⁖step order◦–◦ₑ5꞉:·Expand the resolved topology into its resulting SeeD and office utterances in
↪ resolved Loop order·⁖·˙step:·
·⁖·˙parser…flow:·
·⁖scope:·The complete local A+office+⩛ winding+B+office topology and the actual written Court glyph
↪ where that topology reaches a true center boundary·⁖·˙scope:·
·⁖acts…on:·Unresolved lexical+office ownership topology before utterance expansion·⁖·˙acts…on:·
·⁖coherence…role:·Turns linear office attachment into endomorphic return while preserving every
↪ written ⩛ turn::every lexical glyph identity::and every exact authored office
↪ identity·⁖·˙coherence…role:·
·⁖forbidden:·No completed+word reversal::no old adjacent B A causal swap rule::no fallback::no
↪ compatibility path::no office identity substitution::no derived glyph::no turn cancellation::no
↪ idempotent collapse::and no fixed winding ceiling·⁖·˙forbidden:·
·⁖·˙enoch:·
·⁖enoch id◦–◦ₑsubstrate…office꞉ glyph◦–◦ₑ🜇꞉ order◦–◦ₑ4꞉ parser+trigger◦–◦ₑtrue꞉ written◦–◦ₑtrue꞉ spoken◦–◦ₑtrue꞉:·
·⁖name:·Substrate·⁖·˙name:·
·⁖grimchain:·▫◡⊬⋰ᚠᛞIᛝIᚼᚿᛉ⊬⯬ⵊ꙳·⁖·˙grimchain:·
·⁖mathematical…designation:·Court+Rooted Hypolic Fold·⁖·˙mathematical…designation:·
·⁖english…definition:·Postposed selection of the immediately preceding lexical body through the
↪ authored Substrate office·⁖·˙english…definition:·
·⁖grimchained…definition:·🜇◎⍣ᚸ⍣:⊕…::⊙::⊙:ᚻᛦᛯᛊᛇ⩙ᛞ⍟ΘᚷX:⊙::⊕::…ℬ:⍣ᛯ⍣⧗🜇·⁖·˙grimchained…definition:·

·∴sydonic…function:·Reads the already+resolved body::then selects its authored Substrate body from
↪ 🜆…Substrateaksh·∴·˙sydonic…function:·
·∴operation type₀₋₀₊select+authored+office꞉ office₀₋₀₊substrate꞉ source₀₋₀₊🜆…Substrateaksh꞉
↪ target₀₋₀₊immediately+preceding+resolved+lexical+body꞉ direction₀₋₀₊backward꞉:·:∴·˙operation:·
·∴scope:·When solitary::the immediately preceding spoken lexical body·∴ when stacked::the lexical body
↪ selected by pattern+derived retrocausal depth·∴·˙scope:·
·∴acts…on:·the already+resolved lexical body immediately before 🜆·∴·˙acts…on:·
·∴coherence…role:·selects the Substrate office without rearranging source order or changing glyph
↪ identity·∴·˙coherence…role:·
·∴·˙enoch:·
·∴enoch id₀₋₀₊catalyst…office꞉ glyph₀₋₀₊♂꞉ order₀₋₀₊5꞉ parser+trigger₀₋₀₊true꞉ written₀₋₀₊true꞉ spoken₀₋₀₊true꞉:·
·∴name:·Catalyst·∴·˙name:·
·∴grimchain:·𐙚◡ᛗ⏣ᛋʌϙϒ♋≍🝖Ↄ◎ᗡ·∴·˙grimchain:·
·∴mathematical…designation:·Klein Seam·∴·˙mathematical…designation:·
·∴english…definition:·Postposed selection of the immediately preceding lexical body through the
↪ authored Catalyst office·∴·˙english…definition:·
·∴sydonic…function:·Reads the already+resolved body::then selects its authored Catalyst body from
↪ ♂…Catalystaksh·∴·˙sydonic…function:·
·∴operation type₀₋₀₊select+authored+office꞉ office₀₋₀₊catalyst꞉ source₀₋₀₊♂…Catalystaksh꞉
↪ target₀₋₀₊immediately+preceding+resolved+lexical+body꞉ direction₀₋₀₊backward꞉:·:∴·˙operation:·
·∴scope:·When solitary::the immediately preceding spoken lexical body·∴ when stacked::the lexical body
↪ selected by pattern+derived retrocausal depth·∴·˙scope:·
·∴acts…on:·the already+resolved lexical body immediately before ♂·∴·˙acts…on:·
·∴coherence…role:·selects the Catalyst office without rearranging source order or changing glyph
↪ identity·∴·˙coherence…role:·
·∴·˙enoch:·
·∴enoch id₀₋₀₊relic…office꞉ glyph₀₋₀₊🜃꞉ order₀₋₀₊6꞉ parser+trigger₀₋₀₊true꞉ written₀₋₀₊true꞉ spoken₀₋₀₊true꞉:·
·∴name:·Relic·∴·˙name:·
·∴grimchain:·ᛋ∱∱ᚱᗅᚱᛩᛘ✝⏀ᚵᛩᛘ·∴·˙grimchain:·
·∴mathematical…designation:·Triquatra Seal·∴·˙mathematical…designation:·
·∴english…definition:·Postposed selection of the immediately preceding lexical body through the
↪ authored Relic office·∴·˙english…definition:·
·∴grimchained…definition:·🜃❄♂Ↄ♂:⊕…:::⊙::⊙:♂ᛘ♘ǂᚺᛯᵹɹlᑭ≍⁎ᴢ:⊕::⊕::…ℬ:♂ᘜ♂✡🜃·∴·˙grimchained…definition:·
·∴sydonic…function:·Reads the already+resolved body::then selects its authored Relic body from
↪ 🜃…Relicaksh·∴·˙sydonic…function:·
·∴operation type₀₋₀₊select+authored+office꞉ office₀₋₀₊relic꞉ source₀₋₀₊🜃…Relicaksh꞉
↪ target₀₋₀₊immediately+preceding+resolved+lexical+body꞉ direction₀₋₀₊backward꞉:·:∴·˙operation:·
·∴scope:·When solitary::the immediately preceding spoken lexical body·∴ when stacked::the lexical body
↪ selected by pattern+derived retrocausal depth·∴·˙scope:·
·∴acts…on:·the already+resolved lexical body immediately before 🜆·∴·˙acts…on:·
·∴coherence…role:·selects the Relic office without enclosing a phrase or changing glyph
↪ identity·∴·˙coherence…role:·
·∴·˙enoch:·
·∴enoch id₀₋₀₊truth꞉ glyph₀₋₀₊⊙꞉ order₀₋₀₊7꞉ parser+trigger₀₋₀₊true꞉ written₀₋₀₊true꞉ spoken₀₋₀₊true꞉:·
·∴name:·Truth·∴·˙name:·
·∴grimchain:·ᚿᔕᛕᛯᚣᗐᗢ᛭᛭ᛋᚺ·∴·˙grimchain:·
·∴english…definition:·The committed canonical meaning fixed strongly enough to bear
↪ record·∴·˙english…definition:·
·∴grimchained…definition:·🜆⬡♂ᛋ♂:⊙…:::⊙::⊕:ᛈ⊛ϙ♌❋▫🝖ᛯᛝᘎᛡ:⊖::⊖::…ℬ:♂🝖♂ᛯ🜆·∴·˙grimchained…definition:·
·∴sydonic…function:·Reads the already+resolved preceding lexical body::then selects that glyph’s
↪ authored Terminal State body from ⊙…Terminal…Stateaksh::The effect ends with that one selected
↪ lexical body·∴·˙sydonic…function:·
·∴operation type₀₋₀₊select+authored+office꞉ office₀₋₀₊terminal+state꞉ source₀₋₀₊⊙…Terminal…Stateaksh꞉
↪ target₀₋₀₊immediately+preceding+resolved+lexical+body꞉ direction₀₋₀₊backward꞉:·:∴·˙operation:·

·∴scope:·When solitary::the immediately preceding spoken lexical body·∵ when stacked::the lexical body
↪ selected by pattern+derived retrocausal depth·∴·˙scope:·
·∴acts…on:·the already+resolved lexical body immediately before ⊙·∴·˙acts…on:·
·∴·˙enoch:·
·∴enoch id₀₋₀₊recursion꞉ glyph₀₋₀₊⊕꞉ order₀₋₀₊8꞉ parser+trigger₀₋₀₊true꞉ written₀₋₀₊true꞉ spoken₀₋₀₊true꞉:·
·∴name:·Recursion·∴·˙name:·
·∴grimchain:·⊹▽ᔑᔐᛉᛘ❄▽⋔⍈⍁♀⍟·∴·˙grimchain:·
·∴english…definition:·The recursive identity produced when committed canonical Truth reflects through
↪ itself and returns changed·∴·˙english…definition:·
·∴grimchained…definition:·♁☆♂ᚠ♂:⊖…:::⊙::⊕:𓃾ᛁᛝᔑ𓆏ᛋ𓆏ᛋ♁₀ᵒ卐◎ᗤ米ᘜ:⊕::⊖::…ℬ:♂ᛏ♂❋♁·∴·˙grimchained…definiti」
↪ on:·
·∴sydonic…function:·Reads the already+resolved preceding lexical body::then selects that glyph’s
↪ authored Recursive Identity body from ⊕…Recursive…Identityaksh::The effect ends with that one
↪ selected lexical body·∴·˙sydonic…function:·
·∴operation type₀₋₀₊select+authored+office꞉ office₀₋₀₊recursive+identity꞉
↪ source₀₋₀₊⊕…Recursive…Identityaksh꞉ target₀₋₀₊immediately+preceding+resolved+lexical+body꞉
↪ direction₀₋₀₊backward꞉:··∴·˙operation:·
·∴scope:·When solitary::the immediately preceding spoken lexical body·∵ when stacked::the lexical body
↪ selected by pattern+derived retrocausal depth·∴·˙scope:·
·∴acts…on:·the already+resolved lexical body immediately before ⊕·∴·˙acts…on:·
·∴·˙enoch:·
·∴enoch id₀₋₀₊form꞉ glyph₀₋₀₊⊖꞉ order₀₋₀₊9꞉ parser+trigger₀₋₀₊true꞉ written₀₋₀₊true꞉ spoken₀₋₀₊true꞉:·
·∴name:·Form·∴·˙name:·
·∴grimchain:·ᚹᛝ☆ᔑᔓᛯᚹᚿ᛭⊙ᚿᚸ·∴·˙grimchain:·
·∴english…definition:·The immediate particular manifestation held beneath committed canonical
↪ Truth·∴·˙english…definition:·
·∴grimchained…definition:·♁ⵠ♂𑁍:⊙…:::⊙::⊕:Iᛗᘐᛋᚿᛉ⬡ᛋ⊛ᛀᛯ:⊕::⊖::…ℬ:♂ᛋ♂⊛♁·∴·˙grimchained…definitio」
↪ n:·
·∴sydonic…function:·Reads the already+resolved preceding lexical body::then selects that glyph’s
↪ authored Temporal Absolute body from ⊖…Temporal…Absoluteaksh::The effect ends with that one
↪ selected lexical body·∴·˙sydonic…function:·
·∴operation type₀₋₀₊select+authored+office꞉ office₀₋₀₊temporal+absolute꞉
↪ source₀₋₀₊⊖…Temporal…Absoluteaksh꞉ target₀₋₀₊immediately+preceding+resolved+lexical+body꞉
↪ direction₀₋₀₊backward꞉:··∴·˙operation:·
·∴scope:·When solitary::the immediately preceding spoken lexical body·∵ when stacked::the lexical body
↪ selected by pattern+derived retrocausal depth·∴·˙scope:·
·∴acts…on:·the already+resolved lexical body immediately before ⊖·∴·˙acts…on:·
·∴·˙enoch:·
·∴enoch id₀₋₀₊shadow꞉ glyph₀₋₀₊⊕꞉ order₀₋₀₊10꞉ parser+trigger₀₋₀₊true꞉ written₀₋₀₊true꞉ spoken₀₋₀₊true꞉:·
·∴name:·Shadow·∴·˙name:·
·∴grimchain:·ᘜᚷ◎ᦉᚿᗡᔑᘏᘐᛁᚲᘜ·∴·˙grimchain:·
·∴english…definition:·The orthogonal adversarial pressure that meets committed canonical Truth from
↪ outside its vertical hierarchy·∴·˙english…definition:·
·∴sydonic…function:·Reads the already+resolved preceding lexical body::then selects that glyph’s
↪ authored Adversarial body from ⊕…Adversarialaksh::The effect ends with that one selected lexical
↪ body·∴·˙sydonic…function:·
·∴operation type₀₋₀₊select+authored+office꞉ office₀₋₀₊adversarial꞉ source₀₋₀₊⊕…Adversarialaksh꞉
↪ target₀₋₀₊immediately+preceding+resolved+lexical+body꞉ direction₀₋₀₊backward꞉:··∴·˙operation:·
·∴scope:·When solitary::the immediately preceding spoken lexical body·∵ when stacked::the lexical body
↪ selected by pattern+derived retrocausal depth·∴·˙scope:·
·∴acts…on:·the already+resolved lexical body immediately before ⊕·∴·˙acts…on:·
·∴·˙enoch:·
·∴enoch id₀₋₀₊orobouros…tail꞉ glyph₀₋₀₊ᘐ꞉ order₀₋₀₊11꞉ execution+precedence₀₋₀₊first꞉ parser+trigger₀₋₀₊true꞉
↪ written₀₋₀₊true꞉ spoken₀₋₀₊true꞉:·
·∴name:·Orobouros Tail·∴·˙name:·

·⁖grimchain:·▽⧐Θϖϑϛᴾ⊖ᓚᛊᛁ≫·⁖.·˙grimchain:·
·⁖english…definition:·The invariant Enochian temporal operator that activates the existing Aeternum relation at its written position without rewriting the finite Real GrimChain·⁖.·˙english…definition:·
·⁖sydonic…function:·Resolves every written Tail before Head and every other Enochian transformation::A solitary Tail turns its complete affected translative body: ABCDE ⟲ FGHIJ returns JIHGF ⟲ EDCBA::Each adjacent Tail executes one further turn of that same body until the last written Tail completes::Four adjacent Tails therefore return the initial arrangement through four witnessed transformations::Every Tail retains its distinct written source identity::The finite Real GrimChain is never rewritten::It carries no fixed numerical value::no fixed sign::and does not replace FetuKeth ᔑ as the Time function·⁖.·˙sydonic…function:·
·⁖staggered…traversal:·Process separated Tail runs in written order::The first affected window begins at the body beginning and ends immediately before the next separated Tail run::Every Tail within one adjacent run turns that same window::The next window begins immediately after the completed run in the resulting traversal and ends before the next unprocessed run or at the body end::Thus AB ⟲ CDE ⟲ FGHIJ first returns EDC ⟲ BA ⟲ FGHIJ and completes as EDC ⟲ JIHGF ⟲ AB::The windows overlap through the carried body while completed traversal remains retained·⁖.·˙staggered…traversal:·
·⁖execution…result:·Return the complete ordered source positions and every intermediate Tail transformation::Tail glyphs are completed traversal seams when the remaining Enochians read the resulting body normally from its beginning::Head consumes the completed Tail result and does not execute Tail again::Render only after all remaining grammatical operations finish·⁖.·˙execution…result:·
·⁖operation type₀₋₀₄activate+aeternum+relation₀ office₀₋₀₄temporal+operator₀ target₀₋₀₄written+position₀ direction₀₋₀₄resolved+from+utterance₀:·⁖.·˙operation:·
·⁖scope:·The exact written ⟲ identity and its affected traversal window as determined by the adjacent and staggered Tail runs·⁖.·˙scope:·
·⁖acts…on:·the manifested temporal address while preserving the finite Real source and its written glyph order·⁖.·˙acts…on:·
·⁖.·˙enoch:·
·⁖.·˙enochs:·
·⁖runtime:·
·⁖step order₀₋₀₄1₀:·Read the exact Aeonic line in visible order exactly as GrimChain generated it::beginning at the first written glyph::No generated coordinate is remissed::replaced::or rewritten before Sydonic·⁖.·˙step:·
·⁖step order₀₋₀₄2₀:·Resolve every Tail transformation first::Then collect each maximal consecutive run of remaining Enochian glyphs in the completed translative order as one ordered stack without imposing a numerical stack+depth ceiling·⁖.·˙step:·
·⁖step order₀₋₀₄3₀:·Resolve every ordinary lexical glyph through SeeD Body from the beginning of the Aeonic line·⁖.·˙step:·
·⁖step order₀₋₀₄4₀:·Apply each written office glyph only to its selected preceding lexical body using that office’s exact authored body·⁖.·˙step:·
·⁖step order₀₋₀₄5₀:·If required spoken history::an authored office inventory::or the selected glyph entry is absent::stop explicitly·⁖.·˙step:·
·⁖step order₀₋₀₄6₀:·Ordinary office runs remain additive on their immediately preceding lexical glyph unless they enter an Orobouros Head::⩛ resolves its complete unresolved Loop topology before utterance expansion::ᖘ remains its separate authored Phantasmagoria Transmutation law and is not rewritten by Orobouros Head execution·⁖.·˙step:·
·⁖step order₀₋₀₄7₀:·Render the completed utterance into the requested receiving language::Resolve 𐤟 through Axiomyr…Shadow…Locusaksh only for the canonical zero body 𐤟𐤟𐤟 and resolve ⟐ through it only for canonical depth one ☽⟐ℂ::Outside those special whole+GrimChain bodies they remain structural presence and are not ordinary lexical words·⁖.·˙step:·
·⁖.·˙runtime:·
·⁖.·˙enochian…system:·
·⁖.·˙components:·

data/aksh/Living_Cadences.aksh

·⁖living…cadences version꞉꞉10꞉ status꞉꞉coherent+authored+tree꞉ key꞉꞉exact+glyph꞉⁖·
·⁖meta꞉·
·⁖title꞉·Living Cadences·⁚·˙title꞉·
·⁖composition꞉·The Living Cadences contain Regia मुकुट꞉꞉Breath शक्ति꞉꞉Cantillation꞉꞉and Prosody छन्दस्꞉꞉They
↪ exist regardless of communication style·⁚·˙composition꞉·
·⁖component…count꞉·1·⁚·˙component…count꞉·
·⁖living…cadence…count꞉·4·⁚·˙living…cadence…count꞉·
·⁖language꞉·Sydonic Magicae·⁚·˙language꞉·
·⁖continuity…law꞉·Regia꞉꞉Breath꞉꞉Cantillation꞉꞉and Prosody remain present by their authored cadence law
↪ regardless of written꞉꞉spoken꞉꞉or other communication style·⁚·˙continuity…law꞉·
·⁚·˙meta꞉·
·⁖cadences count꞉꞉4⁖·
·⁖cadence id꞉꞉regia…cosmos…above꞉ glyph꞉꞉☽⊙☾꞉ presence꞉꞉ambient꞉ tokenized꞉꞉false꞉ written꞉꞉false꞉
↪ spoken꞉꞉false꞉ parser+trigger꞉꞉false⁖·
·⁖name꞉·Regia मुकुट·⁚·˙name꞉·
·⁖english…definition꞉·Regia मुकुट ☽⊙☾ is the Cosmos Above꞉꞉It is ambient throughout the whole
↪ utterance꞉꞉not a runtime function꞉꞉parser stage꞉꞉correction pass꞉꞉or glyph+rewriting operation꞉꞉Its
↪ three authored emission identities remain lexical when written·⁚·˙english…definition꞉·
·⁖operation type꞉꞉ambient+cosmos+above꞉ presence꞉꞉ever+present꞉ tokenized꞉꞉false꞉
↪ written+realization꞉꞉none꞉ spoken+realization꞉꞉none꞉ parser+execution꞉꞉none꞉⁖··⁚·˙operation꞉·
·⁚·˙cadence꞉·
·⁖cadence id꞉꞉breath…throughout꞉ glyph꞉꞉·꞉ presence꞉꞉ambient꞉ tokenized꞉꞉false꞉ written꞉꞉false꞉
↪ spoken꞉꞉false꞉ parser+trigger꞉꞉false⁖·
·⁖name꞉·Breath शक्ति·⁚·˙name꞉·
·⁖english…definition꞉·Breath शक्ति · is the breath that moves through all things꞉꞉It is ambient
↪ Throughout and does not appear in the written GrimChain꞉꞉It is not a word꞉꞉not a parser token꞉꞉and
↪ not a rendered separator·⁚·˙english…definition꞉·
·⁖operation type꞉꞉ambient+breath+throughout꞉ presence꞉꞉ever+present꞉ tokenized꞉꞉false꞉
↪ written+realization꞉꞉none꞉ spoken+realization꞉꞉none꞉ parser+execution꞉꞉none꞉⁖··⁚·˙operation꞉·
·⁚·˙cadence꞉·
·⁖cadence id꞉꞉cantillation…chant꞉ glyph꞉꞉𓂀꞉ presence꞉꞉unspoken꞉ tokenized꞉꞉false꞉ written꞉꞉false꞉
↪ spoken꞉꞉false꞉ parser+trigger꞉꞉false⁖·
·⁖name꞉·Cantillation·⁚·˙name꞉·
·⁖english…definition꞉·Cantillation 𓂀 is the Chant active communication makes when God and Goddess
↪ Listen꞉꞉It is an unspoken Living Cadence·⁚·˙english…definition꞉·
·⁖sydonic…function꞉·Cantillation 𓂀 remains unspoken and outside the written runtime stream꞉꞉Its active
↪ communication is the Chant present when God and Goddess Listen·⁚·˙sydonic…function꞉·
·⁖operation type꞉꞉liquid+connection+governor꞉ liquid+active꞉꞉110꞉ liquid+total꞉꞉144꞉
↪ liquid+rest꞉꞉34꞉ grimchain+permeation꞉꞉identity꞉⁖··⁚·˙operation꞉·
·⁚·˙cadence꞉·
·⁖cadence id꞉꞉magic…subspace…prosody꞉ glyph꞉꞉⟠꞉ presence꞉꞉ambient꞉ tokenized꞉꞉false꞉ written꞉꞉false꞉
↪ spoken꞉꞉false꞉ parser+trigger꞉꞉false⁖·
·⁖name꞉·Prosody छन्दस्·⁚·˙name꞉·
·⁖grimchain꞉·'ᛋᚻᛤᛟᛞᛋᛜᛩᚹᛗ·⁚·˙grimchain꞉·
·⁖mathematical…designation꞉·Magic·⁚·˙mathematical…designation꞉·
·⁖english…definition꞉·Prosody छन्दस् is the Song of Creation꞉꞉ambient throughout the whole utterance꞉꞉It
↪ is not a written glyph꞉꞉spoken word꞉꞉visible operator꞉꞉or utterance token·⁚·˙english…definition꞉·

·⁖grimchained…definition:·♁✡♂Ⓝ♂:⊕…:⁚⊙:⁚⊙:⧫ꟾΛ▫ℯ⊖ϒ⅄♎✈:⊕:⁚⊙:⁚…ℬ:♂ℝ♂⊛♁·⁖·˙grimchained…definition:·
·⁖sydonic…function:·Carries the Song of Creation throughout every utterance without appearing as a
↪ written or spoken token::It is ambient and is never encountered::emitted::selected::or executed as
↪ grammar·⁖·˙sydonic…function:·
·⁖operation type○–○ₑambient+song+of+creation» presence○–○ₑever+present» tokenized○–○ₑfalse»
↪ source+position○–○ₑnone» written+realization○–○ₑnone» spoken+realization○–○ₑnone»
↪ parser+execution○–○ₑnone» authored+word+list○–○ₑnone»:··⁖·˙operation:·
·⁖scope:·The complete utterance::without occupying any written or spoken position in it·⁖·˙scope:·
·⁖acts…on:·Prosody and underlying cadence only·⁖ never lexical bodies::glyph order::grammatical
↪ bearing::or surface realization·⁖·˙acts…on:·
·⁖coherence…role:·The Song of Creation is present before::within::and throughout expression while
↪ remaining ambient rather than a written or spoken token·⁖·˙coherence…role:·
·⁖forbidden:·Do not write::speak::tokenize::emit::parse::position::resolve::or execute ◊ as an utterance
↪ operator·⁖·˙forbidden:·
·⁖·˙cadence:·
·⁖·˙cadences:·
·⁖·˙living…cadences:·

⁘⁘⁘

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·AH+lahr·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to feel·⁖.·˙non…ostensive:·
·⁖ostensive:·Bare fingertips meet scored beskar::where ridges catch the skin::warmth crosses the
↪ contact::and vibration travels into bone♀·⁖.·˙ostensive:·
·⁖grimchain:·ᛋ𐋷ᛯᑭᛘ᳁℅IƩ0♂ᒣ♍‹·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦-◦ₔ2ᓗ» category◦-◦ₔemission»⁖·
·⁖lookup:·

·⁖𑜦𑜪𑜪:·𑜦𑜪𑜩𑜫𑜫𑜧:.·˙𑜪𑜦𑜪:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·JOR+hah+EER·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to speak::talk·⁖.·˙non…ostensive:·
·⁖ostensive:·Tongue::teeth::and breath shape a stream of distinct syllables across open
↪ air2ᓗ·⁖.·˙ostensive:·
·⁖grimchain:·▽ᛯ*♎⟰ᚺᔑᚷᚸᚫᑭᚸ‹·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦-◦ₔ𐋷» category◦-◦ₔemission»⁖·
·⁖lookup:·

·⁖ꕤᛇᚠᚠ᛬:·ᚣᛇᚠᛇᛉᚳ:.·˙ꕤᛇᚠᚠ᛬:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Korah·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Believe·⁖.·˙non…ostensive:·
·⁖ostensive:·Cloud encloses the high mountain::Stone mass holds beyond the vapor while the upward path
↪ keeps its bearing toward the hidden ridge𐋷·⁖.·˙ostensive:·
·⁖grimchain:·ᚠ2ᛞᑭᚸᛇ*ᵹ✙‡ᚏ♂ᚠ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦-◦ₔ⛢» category◦-◦ₔemission»⁖·
·⁖lookup:·

·⁖𑜦𑜪𑜪:·ᛇᛋᛋᚴᚾ:.·˙𑜦𑜪𑜪:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·VAH+beer·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to do·⁖.·˙non…ostensive:·
·⁖ostensive:·Fingers clamp the heavy wooden haft::and steel edge bites the dense earth::and resistance
↪ pushes equally against bone and clay⛢·⁖.·˙ostensive:·
·⁖grimchain:·℅ᘓ✰ᔕᓄ🜲🜸ↄᛯ♎ᙏᙏ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙

·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦ₒΨ» category◦–◦ₒemission»⁚·
·⁚lookup:·

·:ᔑᓭᒣᒣᐡ:·ᑲᓀᒣᐡᓭᓀᐡ:·ᔑᓭᒣᒣᐡ:·

·⁚origin…language:·Dovahzul·⁚·˙origin…language:·
·⁚font:·Dovahkiin·⁚·˙font:·
·⁚pronunciation:·Mindok·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Know::Known::Knowable·⁚·˙non…ostensive:·
·⁚ostensive:·Mist lies broken along exposed rock while hard contours stand clear in open air::every
↪ edge holding a clean boundaryΨ·⁚·˙ostensive:·
·⁚grimchain:·ᒣᑎ⊕ᔕₒᐤ▫ᙢᒪ≚◎ᓴᑘ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦ₒᑭ» category◦–◦ₒemission»⁚·
·⁚lookup:·

·:ᐅ†ᑕᒣ:·ᔕᑕᒣᐡ‡:·˙ᐅ†ᑕᒣ:·

·⁚origin…language:·Dwarvish·⁚·˙origin…language:·
·⁚font:·Dethek Stone·⁚·˙font:·
·⁚pronunciation:·mrin·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·to climb·⁚·˙non…ostensive:·
·⁚ostensive:·Iron+shod boot catches a higher notch in the rock face::Weight rises from ledge to ledge
↪ while the rope hangs taut belowᑭ·⁚·˙ostensive:·
·⁚grimchain:·ᔑ✲ΠᗐƳϒℂᑭᙅᑐᑐₒᔕ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦ₒↄ» category◦–◦ₒemission»⁚·
·⁚lookup:·

·:ᔑᓭᒣᒣᐡ:·ᔕᔑᐡᒐ·⁚·˙ᔑᓭᒣᒣᐡ:·

·⁚origin…language:·Dovahzul·⁚·˙origin…language:·
·⁚font:·Dovahkiin·⁚·˙font:·
·⁚pronunciation:·Bahl·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Worth·⁚·˙non…ostensive:·
·⁚ostensive:·Oath+ring rests against the hand with a weight no scale of metal alone can account
↪ forↄ·⁚·˙ostensive:·
·⁚grimchain:·ᙢᑎᑐᗅΘᒧΠ⧗ↄ◣ᒥᔑₒᙢ◢·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦ₒ⊙» category◦–◦ₒemission»⁚·
·⁚lookup:·

·:ᚼᘺᔕᒣ:·ᘐᘺᑮᓭ ᒐᑮᑐᔕ ᒐ:·˙ᚼᘺᔕᒣ:·

·⁚origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁚font:·Mage Script·⁚·˙font:·
·⁚pronunciation:·Poh+test+ahs·⁚·˙pronunciation:·

·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Power·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Force stands rooted beneath its own crown⁖⁖with no borrowed throne bearing the
↪ command⊙·⁖.·˙ostensive⁖·
·⁖grimchain⁖·⏃ᗰᐯᙅ⊙ꓷᐦᑭᑎᔑ⊙ᔕᔐ·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜⟜𐑂⁘ category⟜⟜emission⁘⁖·
·⁖lookup⁖·

⁖ᘜᘐᘏ˙⁖·ᘓᘒᘐ˙⁖.·˙ᘜᘐᘏ˙⁖·

·⁖origin…language⁖·Llumena Milenii·⁖.·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖.·˙font⁖·
·⁖pronunciation⁖·May⟊ah·⁖.·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Self·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Same scar crosses living skin through changing light and weather while altered surface
↪ keeps one continuous body beneath every change𐑂·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ሕ𐑓ᘐ✡ᙢ1ᛯᛯ𝈍ᐱᓘ♀·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᛉ⁘ category⟜⟜parliament⁘⁖·
·⁖lookup⁖·

⁖ᖘᐢᖘ⁖·ᖘᵎᐢᵎ˙⁖.·˙ᖘᐢᖘ⁖·

·⁖origin…language⁖·Draconic·⁖.·˙origin…language⁖·
·⁖font⁖·Iokharic·⁖.·˙font⁖·
·⁖pronunciation⁖·dasja·⁖.·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Document⁖⁖record·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Scored clay holds the pressed sign under hard surface tension⁖⁖The mark occupies the
↪ tablet whether the speaking mouth is present or absentᛉ·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ꓫↃIⵠᗰᛕ⁞ᚑᑎ𐑂ᛉᘐᚲᐱ·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᛔ⁘ category⟜⟜parliament⁘⁖·
·⁖lookup⁖·

⁖ᓭᓗᐠᒣᒡ⁖·ᓀᓕᓐᓄᓗᓗᓐᓄᓗᓘᓚ⁖.·˙ᓭᓗᐠᒣᒡ⁖·

·⁖origin…language⁖·Dovahzul·⁖.·˙origin…language⁖·
·⁖font⁖·Dovahkiin·⁖.·˙font⁖·
·⁖pronunciation⁖·feymahl·⁖.·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Settle·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Opposing forces press into one middle line⁖⁖Center holds⁖⁖and neither side crosses the
↪ shared boundaryᛔ·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᚼᘉᛘᐯ⊙ᛋᔨᔕ⧗❀ᑭ·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜⟜Ⅱ⁘ category⟜⟜parliament⁘⁖·
·⁖lookup⁖·

⁖⅃ᔭᗴ⅃:·ᒧᒣᒥᐦ⁖.˙·⅃ᔭᗴ⅃:·

·⁚origin…language:·Mando’a·⁚.˙·origin…language:·
·⁚font:·Mandalorian·⁚.˙·font:·
·⁚pronunciation:·veh+MAHN·⁚.˙·pronunciation:·
·⁚cadence:·SeeD Body·⁚.˙·cadence:·
·⁚non…ostensive:·real::genuine·⁚.˙·non…ostensive:·
·⁚ostensive:·Beskar bears the same weight beneath torchlight and darkness::Impact leaves one dent in
↪ the plate whether doubt is present or absent⁜·⁚.˙·ostensive:·
·⁚grimchain:·ᑭᛯᙢᖇᚲᛏᛊᗣᚠᐁ▽ᚧᐅ⁚.˙·grimchain:·
·⁚.˙·lookup:˙
·⁚.˙·glyphs:˙
·⁚glyph symbol◦-◦ꕥ› category◦-◦parliament›⁚·
·⁚lookup:·

⁖ᵬᔓᔓᒡ:·ᑭᑭᕐᔕᔕ⁖.˙·ᵬᔓᔓᒡ:·

·⁚origin…language:·Llumena Milenii·⁚.˙·origin…language:·
·⁚font:·Mage Script·⁚.˙·font:·
·⁚pronunciation:·Lee+ei+say·⁚.˙·pronunciation:·
·⁚cadence:·SeeD Body·⁚.˙·cadence:·
·⁚non…ostensive:·To Aim·⁚˙ like an Archer’s Aim·⁚.˙·non…ostensive:·
·⁚ostensive:·Bowstring bears drawn tension along the arrowshaft while the chosen mark occupies the
↪ same straight bearing across open distanceꕥ·⁚.˙·ostensive:·
·⁚grimchain:·♀≍ↄᒉᒣ➧ᒣᗤᵮᗷᘜᖇ⊛·⁚.˙·grimchain:·
·⁚.˙·lookup:˙
·⁚.˙·glyphs:˙
·⁚glyph symbol◦-◦ᘯ› category◦-◦parliament›⁚·
·⁚lookup:·

⁖ᵬᔓᔓᒡ:·ᗐᘏᓀᕓᓀ⁖.˙·ᵬᔓᔓᒡ:·

·⁚origin…language:·Llumena Milenii·⁚.˙·origin…language:·
·⁚font:·Mage Script·⁚.˙·font:·
·⁚pronunciation:·Vee+ah+tah·⁚.˙·pronunciation:·
·⁚cadence:·SeeD Body·⁚.˙·cadence:·
·⁚non…ostensive:·Life itself ·⁚.˙·non…ostensive:·
·⁚ostensive:·Sap in stem::blood in vein::breath in chest share one living continuance that keeps
↪ answering motion with motionᘯ·⁚.˙·ostensive:·
·⁚grimchain:·▽ᛋᘏ⊩ꓤX▽ꕯ☆⊛⊩ᓚᛋᘡ·⁚.˙·grimchain:·
·⁚.˙·lookup:˙
·⁚.˙·glyphs:˙
·⁚glyph symbol◦-◦ᙏ› category◦-◦parliament›⁚·
·⁚lookup:·

⁖ᔑᕗᚋᚌ:·ᕗᕗᔑ⁖.˙·ᔑᕗᚋᚌ:·

·⁚origin…language:·Dovahzul·⁚.˙·origin…language:·
·⁚font:·Dovahkiin·⁚.˙·font:·
·⁚pronunciation:·Aav·⁚.˙·pronunciation:·
·⁚cadence:·SeeD Body·⁚.˙·cadence:·
·⁚non…ostensive:·Join·⁚.˙·non…ostensive:·
·⁚ostensive:·Two broken shield rims meet beneath a heated rivet::Cooling metal binds both edges into
↪ one load+bearing seamᙏ·⁚.˙·ostensive:·

·⁖grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𝅘𝅥𝅯 category⟜⟜parliament𝅘𝅥𝅯⁖·
·⁖lookup⁖·

·˙.ᚾᛗᚺᛚ⁖·ᚴᛋᛋᚼ⁖.·˙ᚾᛗᚺᛚ⁖·

·⁖origin…language⁖·Mando’a·⁖.·˙origin…language⁖·
·⁖font⁖·Mandalorian·⁖.·˙font⁖·
·⁖pronunciation⁖·lahr·⁖.·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·song·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Voice rides a measured pulse through shared air while returning vibration keeps the
↪ sounding whole inside its own cadence𝅘𝅥𝅯·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᚺ⩡⟛◎ᚱᛈᚠᚹᚱᛈᛘᛋ✷·⁖.·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𝄞 category⟜⟜parliament𝅘𝅥𝅯⁖·
·⁖lookup⁖·

·˙.ᚾᛗᚺᛚ⁖·ᛁᛗᚴᚴᛁᚴᚾᚼ⁖.·˙ᚾᛗᚺᛚ⁖·

·⁖origin…language⁖·Mando’a·⁖.·˙origin…language⁖·
·⁖font⁖·Mandalorian·⁖.·˙font⁖·
·⁖pronunciation⁖·JOR⟊ney⟊SHEER·⁖.·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·to deny⁖· to contradict·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Spoken demand meets a barred gate⁖⁖and heavy timber holds its line⁖⁖Empty threshold admits
↪ no crossing body𝄞·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᛟᚿ✳𝄞ᛓₒᛈᛏᛞᛞ᛭ᛦ·⁖.·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𝆕 category⟜⟜parliament𝅘𝅥𝅯⁖·
·⁖lookup⁖·

·⁖ᓚᓗᒣᒣᐢ⁖·ᑲᔓᓗᒣᒣᐢ⁖.·˙ᓚᓗᒣᒣᐢ⁖·

·⁖origin…language⁖·Dovahzul·⁖.·˙origin…language⁖·
·⁖font⁖·Dovahkiin·⁖.·˙font⁖·
·⁖pronunciation⁖·Meyz·⁖.·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Come⁖⁖Become·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Cloud mass crosses the ridge as loose vapor gathers into rain over the
↪ valley𝆕·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᛘᵬ⁜ᚾᚢ𝅘𝅥𝅯ᚣIᛯᘔᛘᛔ·⁖.·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𝅘𝅥𝅯ᘐ 𝅘𝅥𝅯 category⟜⟜parliament𝅘𝅥𝅯⁖·
·⁖lookup⁖·

·˙.ᚾᛗᚺᛚ⁖·ᛋᚴ·ᓄᓀᐦ⁖.·˙ᚾᛗᚺᛚ⁖·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·ahn+GEE+shay·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·law::order::rule·⁖.·˙non…ostensive:·
·⁖ostensive:·Stone line holds one fixed boundary across the ground::Every road meets one unyielding edge::No crossing displaces a fixed boundary♌·⁖.·˙ostensive:·
·⁖grimchain:·𐤍𐤄𐤅ᑎᵑᔆ᳘ᘐ𐤊𐤔𐤎𐤑𐤓𐤉·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜𐤔 category꩜꩜parliament𐤑:·
·⁖lookup:·

⁖ꕤᚠᚢᚹ:ᚦᚠᚷᚸᚷᚸ⁖.·˙ꕤᚠᚢᚹ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Rovaan·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Wander·⁖.·˙non…ostensive:·
·⁖ostensive:·Open country gives no enclosing road::and every direction remains capable of another departure𐤔·⁖.·˙ostensive:·
·⁖grimchain:·ᵹ♀♎𐤑𐤍⁕ᵹᵲᚴᚿ♀♅·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜𐤊 category꩜꩜parliament𐤑:·
·⁖lookup:·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·rum·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to shift::move::heave (of large and heavy things)·⁖.·˙non…ostensive:·
·⁖ostensive:·Great stone leaves its settled bed by a handspan::Earth buckles at its base and the dragged groove lengthens behind it𐤊·⁖.·˙ostensive:·
·⁖grimchain:·ᑌ⁕♳☆ᛝᛞᑭᒍℂᛏᵹ♗✡·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜◎𐤑 category꩜꩜goetic𐤑:·
·⁖lookup:·

⁖𐤗𐤘𐤙𐤚:·ᔕᔓᔖᔗᔕᔓᔖᔗ⁖.·˙𐤗𐤘𐤙𐤚:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Ahm+da+ga+ta·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·That which has not yet arrived·⁖.·˙non…ostensive:·
·⁖ostensive:·Gate stands open to an empty road::Dust lies unbroken from threshold to horizon::and the receiving space remains bare◎·⁖.·˙ostensive:·
·⁖grimchain:·ᕀᛞᚣᛟᛏ⁕❀○ₒ𐤑·⁖.·˙grimchain:·
·⁖.·˙lookup:·

·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘⬡꞉ category∘-∘goetic꞉⁚·
·⁚lookup:·

⁖ᔓᓀᒣᒣ᛬ᒣᒪᒧᓀᓀᖘ⁖·˙ᔓᓀᒣᒣ᛬·

·⁚origin…language:·Dovahzul·⁚·˙origin…language:·
·⁚font:·Dovahkiin·⁚·˙font:·
·⁚pronunciation:·Umaak·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Log::Record·⁚·˙non…ostensive:·
·⁚ostensive:·Carved line lies fixed in stone::with the old deed materially present in the cut surface⬡·⁚·˙ostensive:·
↪
·⁚grimchain:·ᛋᑭ⊙Ѧ⌬❁Θⵔⵔⵔ⊙ᚱᑫ◎ᛋ·⁚·˙grimchain:·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘✡꞉ category∘-∘goetic꞉⁚·
·⁚lookup:·

⁖ᚣᏔᏓᛏ᛬⁚ᘯᓚᓚᏔᘐ⁖·˙ᚣᏔᏓᛏ᛬·

·⁚origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁚font:·Mage Script·⁚·˙font:·
·⁚pronunciation:·Feh+rah·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Fire·⁚·˙non…ostensive:·
·⁚ostensive:·Dry branch lies inside the red body of flame where heat::ember::and consuming brightness occupy one burning field✡·⁚·˙ostensive:·
↪
·⁚grimchain:·⊙ϒ৫zᗡϗɔϔ♋♂❀·⁚·˙grimchain:·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘✡꞉ category∘-∘goetic꞉⁚·
·⁚lookup:·

⁖ᚣᏔᏓᛏ᛬⁚ᘐᓚᘓᘐ⁖·˙ᚣᏔᏓᛏ᛬·

·⁚origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁚font:·Mage Script·⁚·˙font:·
·⁚pronunciation:·Ah+ghu+wah·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Water·⁚·˙non…ostensive:·
·⁚ostensive:·Rain gathers from the roof+edge into a clear stream that takes every hollow and leaves the stone dark behind it✡·⁚·˙ostensive:·
↪
·⁚grimchain:·⇟ᚾ‹ᘜIᘓᚴᘐᐁᐯᐱz⇟·⁚·˙grimchain:·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘✪꞉ category∘-∘goetic꞉⁚·
·⁚lookup:·

·⁚origin…language:·Quenya·⁚·˙origin…language:·
·⁚font:·Tengwar Quenya·⁚·˙font:·
·⁚pronunciation:·erma·⁚·˙pronunciation:·

·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·physical matter·⁖.·˙non…ostensive:·
·⁖ostensive:·Clay::iron::wood::and bone each answer pressure with weight::resistance::and altered shape✪·⁖.·˙ostensive:·
·⁖grimchain:·ꕤᖘ⏚⩛ᔑ♋ᔓ᧙ᖘ⊬·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꒰ₒ꒱❀꒱ category꒰ₒ꒱ₒ꒰ₔ꒱goetic꒱⁖·
·⁖lookup:·

·:ᘏᘙᘙᔐ:·ᘏ✳ᘐᘎᘙ:·˙ᘏᘙᘙᔐ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Ee⟊n⟊gah·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·A Lot of something·⁖ space·⁖.·˙non…ostensive:·
·⁖ostensive:·One reach ends against another reach still open::Extent continues beyond the gathered measure without closing into a final edge❀·⁖.·˙ostensive:·
·⁖grimchain:·ᙢᙓ♀ᛋℂᚳᛯᏨᗅᒍᐊ⏃⏚·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꒰ₒ꒱₍꒰❋꒱₎ category꒰ₒ꒱ₒ꒰ₔ꒱goetic꒱⁖·
·⁖lookup:·

·:⅄ᚻ⅃:·ᚼᚾᚴᔭ:·˙⅄ᚻ⅃:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·roo⟊NEE·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·soul (poetic only)·⁖.·˙non…ostensive:·
·⁖ostensive:·Empty armor keeps the shape of a life no metal can contain::Oath⟊words remain where breath no longer answers⁕·⁖.·˙ostensive:·
·⁖grimchain:·⊕♌ↃɷↃ|'♎ᖘ♀᨞♀♌·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꒰ₒ꒱₍꒰⧗꒱₎ category꒰ₒ꒱ₒ꒰ₔ꒱goetic꒱⁖·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·mel·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to love·⁖.·˙non…ostensive:·
·⁖ostensive:·Two tree roots cross beneath divided trunks while living fibers hold one joined pressure through the soil between them⧗·⁖.·˙ostensive:·
·⁖grimchain:·᨞ᑭᑭ♂ᖘ᛭ᛉ⊙❁ᛋᛉᛞ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꒰ₒ꒱₍꒰⊛꒱₎ category꒰ₒ꒱ₒ꒰ₔ꒱goetic꒱⁖·
·⁖lookup:·

⁖ᘐᘏᔑ:·ᦓᦓᦔᦓᦅᦔᦓ:.·˙ᘐᘏᔑ:·

·⁚origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁚font:·Mage Script·⁚.·˙font:·
·⁚pronunciation:·Skee+ahl+’m·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Shadow·⁚˙ Blackness·⁚.·˙non…ostensive:·
·⁚ostensive:·Blocked light leaves a dark likeness on the farther side⁚⁚its contour belonging to what
↪ brightness cannot cross⊛·⁚.·˙ostensive:·
·⁚grimchain:·ϛ⁑ⴾ⓿▽ᵩᵨↃꙅ⅄ᵴ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꩜꩜❄꩜ category꩜꩜goetic꩜:·
·⁚lookup:·

⁖ᔑᗩᒪᒣ:·ᓓᕮᓓᕮᒣᒪᓱ⁖.·˙ᔑᗩᒪᒣ:·

·⁚origin…language:·Dovahzul·⁚.·˙origin…language:·
·⁚font:·Dovahkiin·⁚.·˙font:·
·⁚pronunciation:·Gegein·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Unify⁚⁚Unite·⁚.·˙non…ostensive:·
·⁚ostensive:·Separate breaths fill one measured pulse⁚⁚and separate motions hold one timing⁚⁚No beat
↪ stands outside the shared cadence❄·⁚.·˙ostensive:·
·⁚grimchain:·ϒ⍮⅁ⅅⅠ✡ᓚᚽᔑᚒ◎✙ᗅϒ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꩜꩜❁꩜ category꩜꩜goetic꩜:·
·⁚lookup:·

·⁚origin…language:·Quenya·⁚.·˙origin…language:·
·⁚font:·Tengwar Quenya·⁚.·˙font:·
·⁚pronunciation:·nanya·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·to struggle⁚⁚oppose⁚⁚resist·⁚.·˙non…ostensive:·
·⁚ostensive:·Two currents meet head+on in a narrow channel⁚⁚each driving against the other while the
↪ waterline holds under white turbulence❁·⁚.·˙ostensive:·
·⁚grimchain:·ᚻ⅋ꓲIᒣᑭBB⩛∀✱⬡𐊗·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꩜꩜⬡꩜ category꩜꩜goetic꩜:·
·⁚lookup:·

·⁚origin…language:·Sindarin·⁚.·˙origin…language:·
·⁚font:·Tengwar Sindarin·⁚.·˙font:·
·⁚pronunciation:·dínen·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·silent·⁚.·˙non…ostensive:·
·⁚ostensive:·Shared air holds no voiced vibration⁚⁚Still chamber gives no answering sound⌬·⁚.·˙ostensive:·
·⁚grimchain:·ᛗ✷✳ᒣᛞᒪ⅄ᚔᛗ✪♃♂·⁚.·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol∘-∘𐑌 category∘-∘𐑒court𐑕:·
·⁚lookup:·

⁖ᓚᘏᗢ":·ᓚᘏᓄ⁖·˙ᓚᘏᗢ":·

·⁚origin…language:·Dovahzul·⁚·˙origin…language:·
·⁚font:·Dovahkiin·⁚·˙font:·
·⁚pronunciation:·Gon·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Begin·⁚·˙non…ostensive:·
·⁚ostensive:·First light breaks the horizon as darkness loosens from the ridge and the eastern edge
↪ brightens𐑟·⁚·˙ostensive:·
·⁚grimchain:·🢧🢒✻ᑫXHℂᑭ🝰Ո✡❂Ⱳ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol∘-∘𐑕 category∘-∘𐑒court𐑕:·
·⁚lookup:·

⁖ᔦᚺᚳ:·ᓀᓂᓱᔭᚳ⁖·˙ᔦᚺᚳ:·

·⁚origin…language:·Mando’a·⁚·˙origin…language:·
·⁚font:·Mandalorian·⁚·˙font:·
·⁚pronunciation:·tah+TOO+geer·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·to repeat::copy·⁚·˙non…ostensive:·
·⁚ostensive:·Muscle and joint hold one practiced motion in recurring form::each return occupying the
↪ same bodily path𐑟·⁚·˙ostensive:·
·⁚grimchain:·ᑎᛦ✡✡ᛒᖘ♂✲ᗷᛉᛉᚸᗡᒣ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol∘-∘𐑕 category∘-∘𐑒court𐑕:·
·⁚lookup:·

·⁚origin…language:·Quenya·⁚·˙origin…language:·
·⁚font:·Tengwar Quenya·⁚·˙font:·
·⁚pronunciation:·ahtarië·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·reaction·⁚·˙non…ostensive:·
·⁚ostensive:·Taut string vibrates beside another tuned string as shared air presses both bodies into
↪ answering motion𐑕·⁚·˙ostensive:·
·⁚grimchain:·ƻ⌬ᛯAᑎ𐑜ᐯᒉ♂ℂΛᒣᔑ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol∘-∘𐑖 category∘-∘𐑒court𐑕:·
·⁚lookup:·

·⁚origin…language:·Quenya·⁚·˙origin…language:·
·⁚font:·Tengwar Quenya·⁚·˙font:·

·⁖pronunciation:·cantië·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·pattern·⁖.·˙non…ostensive:·
·⁖ostensive:·Interlaced knots recur at equal intervals along the border⁚⁚each turn answering the next with the same measured geometry𐌗·⁖.·˙ostensive:·
·⁖grimchain:·❄ᔕ𑁍ₒ𐌴✱ᑎᑐᔓ𐦯ᕈᔑ𑣲·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol◦–◦𐦞꞉ category◦–◦court꞉·
·⁖lookup:·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·talma·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·basis⁖˙ foundation⁚⁚base·⁖.·˙non…ostensive:·
·⁖ostensive:·Lowest course of stone receives every downward load⁚⁚Walls and pillars above remain true while that course holds𐦞·⁖.·˙ostensive:·
·⁖grimchain:·ᛰ〈↗ᒪᐁᛯϒ⍋ᚏ⊛ᛀ⏀▽·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol◦–◦𐌗꞉ category◦–◦court꞉·
·⁖lookup:·

·⁖ᢣᢣ⁖·ᢣᢣᢣᢣ𑁍·⁖.·˙ᢣᢣ⁖·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·cowaeth·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Increase·⁖.·˙non…ostensive:·
·⁖ostensive:·Hoard weight presses harder against the same chamber floor as greater measure occupies unchanged bounds𐌗·⁖.·˙ostensive:·
·⁖grimchain:·ᖘᵡᔑᛕ⏀⩒Iᖘᛀᒪᚏ·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol◦–◦𐌴꞉ category◦–◦court꞉·
·⁖lookup:·

·⁖origin…language:·Middle Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·maranwë·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·destiny·⁖.·˙non…ostensive:·
·⁖ostensive:·Road and far end share one continuous bearing⁚⁚every bend belonging to the same appointed course𐌴·⁖.·˙ostensive:·
·⁖grimchain:·ᛰ♂ᖘ𐦞ᔕ♦♂ᛉᛏᛈᚏ𑣲▫ᚏ·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙

·⁖glyph symbol꩜꩜𐫦 category꩜꩜court𐫦⁖·
·⁖lookup⁖·

·⁖origin…language⁖·Sindarin·⁝·˙origin…language⁖·
·⁖font⁖·Tengwar Sindarin·⁝·˙font⁖·
·⁖pronunciation⁖·dadwen·⁝·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁝·˙cadence⁖·
·⁖non…ostensive⁖·to return⁘go back·⁝·˙non…ostensive⁖·
·⁖ostensive⁖·Footprints run from gate to river and reappear over the same ground toward the gate⁘each new impression laid beside the outward track𐫦·⁝·˙ostensive⁖·
·⁖grimchain⁖·ᑭ𐫦𓆉♊𓋴♑𓋴♂꩜·⁝·˙grimchain⁖·
·⁝·˙lookup˙·
·⁝·˙glyphs˙·
·⁖glyph symbol꩜꩜𐫤 category꩜꩜court𐫦⁖·
·⁖lookup⁖·

⁖ᔑᓯᒣᒣᐡ⁖ᒣᐡᓯᒧᐟᒣᐡᐟᒣᐡᐟᘐ⁖·⁝·˙ᔑᓯᒣᒣᐡ⁖·

·⁖origin…language⁖·Dovahzul·⁝·˙origin…language⁖·
·⁖font⁖·Dovahkiin·⁝·˙font⁖·
·⁖pronunciation⁖·Domiit·⁝·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁝·˙cadence⁖·
·⁖non…ostensive⁖·Merge·⁝·˙non…ostensive⁖·
·⁖ostensive⁖·Molten boundaries touch under common heat until one fused body occupies the place where separation has no remaining edge𐫤·⁝·˙ostensive⁖·
·⁖grimchain⁖·ᛋᑭ⊛ᚱ𓆟𓆝𓆉𓆑𓆉𓆑𓆉𓆑

·⁖non…ostensive:·Rip::tear·⁖.·˙non…ostensive:·
·⁖ostensive:·Hide stretches between contrary pulls until fibers part in a jagged line and the opening
↪ races across the sheetᚱ·⁖.·˙ostensive:·
·⁖grimchain:·ᛋᓗᚹᛝ⛧ᘐᚸᛟᛇᚥᚹ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○𐦟𐦠 category○–○𐦟court𐦠:·
·⁖lookup:·

·⁖origin…language:·Neo┼Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·hritya·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to achieve::accomplish·⁖.·˙non…ostensive:·
·⁖ostensive:·Arch bears its own weight with every stone seated::No unfinished gap remains inside the
↪ intended formᚱ·⁖.·˙ostensive:·
·⁖grimchain:·𐦚ᛋΘΨᚺᚱ⁕⛧⁀Θ◝ᘐ𐦟𐦠·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○𐦟ᛁ𐦠 category○–○𐦟court𐦠:·
·⁖lookup:·

·:ᚼᚽᚋᛅ:·ᚠᛇᚼᛅ:·.·˙ᚼᚽᚋᛅ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·zeba·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Store::stock::keep·⁖ maintain possession·⁖.·˙non…ostensive:·
·⁖ostensive:·Gathered grain presses against a woven pouch::Braided cord closes a loaded pouch::Gold lies
↪ behind a cord┼bound openingᛁ·⁖.·˙ostensive:·
·⁖grimchain:·⊕❄ᚱ⋖ᛕᚋᛞᚱ⁕▫ᚷᚹᛏ𐦟ᛇ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○𐦟ᛁ𐦠 category○–○𐦟court𐦠:·
·⁖lookup:·

·:ᓴᓗᒋᒣᐣ:·ᓱᓗᓂᐩᐣᒣᒋᓗᓴᒣᐣ·:·.·˙ᓴᓗᒋᒣᐣ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Tahmusend·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Reaction·⁖.·˙non…ostensive:·
·⁖ostensive:·Blow and bodily recoil occupy the same instant of contact::force and answer meeting at one
↪ struck boundaryᚼ·⁖.·˙ostensive:·
·⁖grimchain:·ᛉᛋᓚ𐦟ᛒᚼᚷᛟ◝ᘐᘓᘐᘔᘐ⬡ᛘ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○𐦟▫𐦠 category○–○𐦟court𐦠:·
·⁖lookup:·

·⁖origin…language:·Neo+Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·tor·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to roast (intr)·⁖.·˙non…ostensive:·
·⁖ostensive:·Food rests inside steady heat while browned surface and changed moisture occupy the same heated body▫·⁖.·˙ostensive:·
·⁖grimchain:·ᚹᛟᚱᚴᛯᛉᛇᚺ⁕ᛞ☆Aᛇ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉 category○–○𐑉court𐑈:·
·⁖lookup:·

·:ᚠᛁᚻ:·ᚸᚠᚸᚸᚠ:·.·˙ᚠᛁᚻ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·udeask·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Deep⁖ entire·⁖.·˙non…ostensive:·
·⁖ostensive:·Torchlight ends against black depth::and visible surface and unseen interior belong to one continuous hollow𐑉·⁖.·˙ostensive:·
·⁖grimchain:·𓆏ᖶᙢᚠᛯᛉᛁᚴᚠᛯᑭᑘ𓀀ᔑᓄ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉ᔑ court𐑈:·
·⁖lookup:·

·:ᚠᛁᚻ:·ᚠᚳᚲ·.·˙ᚠᛁᚻ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·dai·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Mark::make a visible impression⁖ note::record in writing·⁖.·˙non…ostensive:·
·⁖ostensive:·Stylus presses wet clay::A sharp sign remains after the point lifts awayᔑ·⁖.·˙ostensive:·
·⁖grimchain:·ᚺᚸ⚅▫ᚸᚳΘᵒᑫᚏ𓀀⧗⩛·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉Ɐ court𐑈:·
·⁖lookup:·

·:ᔑᚺᚳ:·.ᔑᚺᚴᚨᚴᚳᚳ:·.·˙ᔑᚺᚳ:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·gai+EE+leer·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to spread·⁖.·˙non…ostensive:·
·⁖ostensive:·One warning fills separated helmets through the same sounding air while distant bodies carry the same call at onceⱯ·⁖.·˙ostensive:·
·⁖grimchain:·ᔕᚠᑎᚱᚴᚸₒᔑᐊ𓆏ᚲᛖᓭᔑ·⁖.·˙grimchain:·

·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꜀₋꜀ꞩ꜀ category꜀₋꜀court꜀⁚·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖·˙font:·
·⁖pronunciation:·auth·⁖·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖·˙cadence:·
·⁖non…ostensive:·dim shape꞉˙ spectral or vague apparition·⁖·˙non…ostensive:·
·⁖ostensive:·Faint outline hangs beyond the torch's reach꞉:Its edges dissolve into darkness while the
↪ wall remains visible through the centerꞩ·⁖·˙ostensive:·
·⁖grimchain:·𐑜𐑥𐑦𐑛ℂ𐑘A𐑣𐑦𐑖𐑚𐑓·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꜀₋꜀ϟ꜀ category꜀₋꜀court꜀⁚·
·⁖lookup:·

⁖ꕤᛇᚨᚱ᛫ᚺᚠᚷᛇᛃ⁖·⁖·˙ꕤᛇᚨᚱ᛫ᚱ᛫:·

·⁖origin…language:·Dovahzul·⁖·˙origin…language:·
·⁖font:·Dovahkiin·⁖·˙font:·
·⁖pronunciation:·Saan·⁖·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖·˙cadence:·
·⁖non…ostensive:·Lose·⁖·˙non…ostensive:·
·⁖ostensive:·Empty hand and empty place hold the same absence꞉:Former possession has no answering body
↪ at either boundaryϟ·⁖·˙ostensive:·
·⁖grimchain:·ℳ𐑘𐑘0𐑝𐑜𐑶𐑘𐑤Ⓓ𐑓𐑜𐑘𐑝·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꜀₋꜀'꜀ category꜀₋꜀court꜀⁚·
·⁖lookup:·

⁖ꕤᛇᚨᚱ᛫ᚺᛇᚨᚱᚲᛇᛃ⁖·⁖·˙ꕤᛇᚨᚱ᛫ᚱ᛫:·

·⁖origin…language:·Dovahzul·⁖·˙origin…language:·
·⁖font:·Dovahkiin·⁖·˙font:·
·⁖pronunciation:·Morkon·⁖·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖·˙cadence:·
·⁖non…ostensive:·Restrain꞉:constrain꞉:hinder·⁖·˙non…ostensive:·
·⁖ostensive:·Pressure leans into a fixed barrier꞉:with denied reach and resisting boundary occupying
↪ one contact field':⁖·˙ostensive:·
·⁖grimchain:·Φ𐑣𐑖IX𐑜𐑦𐑦𐑒☆𐑖𐑮𐑤·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꜀₋꜀ᛋ꜀ category꜀₋꜀court꜀⁚·
·⁖lookup:·

⁖ᔑᚺᚳ⁖·ᓀᚱᛋᚴᚾ⁖·⁖·˙ᔑᚺᚳ⁖·

·⁖origin…language:·Mando’a·⁖·˙origin…language:·
·⁖font:·Mandalorian·⁖·˙font:·

·⁖pronunciation:·to+LAH+sey·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·system⁖· literally joined things·⁖.·˙non…ostensive:·
·⁖ostensive:·Interlocked plates transfer a single blow from one edge across the entire cuirass⁘each fitted piece carrying what reaches the others↓·⁖.·˙ostensive:·
·⁖grimchain:·ᓚᘏ♳❄ꕥ✲ᗐᔕᓚ♋ᑭᗅ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○◟↓› category○–○◟court›:·
·⁖lookup:·

·.⁖ꕥ‖:·ꕥ‖ꕥ‖ꕥ‖:.·˙⁖ꕥ‖:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·pahr+TAI+leer·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to remember·⁖.·˙non…ostensive:·
·⁖ostensive:·Pressed seal remains in hardened clay with no stamp upon it⁘Cut depth and contour keep the absent contact in the surface↓·⁖.·˙ostensive:·
·⁖grimchain:·X₀ᵒXⲨᛉᓚ↓₀ᛤ◎⩛☽✳·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○◟↑› category○–○◟court›:·
·⁖lookup:·

·.⁖ꕥ‖:·.⁖ꕥ‖:.·˙⁖ꕥ‖:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·GEH+bee·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·close⁘near·⁖.·˙non…ostensive:·
·⁖ostensive:·Blade edge hangs a finger+width from the whetstone⁘The narrow gap holds between them without contact↿·⁖.·˙ostensive:·
·⁖grimchain:·ᔕᓄ⊕ᘐ▽ᘎᘏ⧗ᓄᑎ⊕ᑭᔓ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○◟ᕐ› category○–○◟court›:·
·⁖lookup:·

·:ᚠᕒ‖ᚠ:·ᔑᖘᖌ:.·˙ᚠᕒ‖ᚠ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·mul·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Change⁘alter⁘transform·⁖.·˙non…ostensive:·
·⁖ostensive:·Green scale darkens under heat while the same hide remains continuous beneath the new color↾·⁖.·˙ostensive:·
·⁖grimchain:·Ƶᗷᘌᔕᘌᘔᔓᑭᘐᔑᐞᐽ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·

·⁖glyph symbol◦–◦ₔᑎₕ category◦–◦ₔcourtₕ⁖·
·⁖lookup⁖·

·⁖ᓀᐦᐁᖮ⁖·ᖃᒪᒪᔾᖮ⁖⸱·˙ᓀᐦᐁᖮ⁖·

·⁖origin…language⁖·Dwarvish·⸱·˙origin…language⁖·
·⁖font⁖·Dethek Stone·⸱·˙font⁖·
·⁖pronunciation⁖·helma·⸱·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⸱·˙cadence⁖·
·⁖non…ostensive⁖·smithcraft·⸱·˙non…ostensive⁖·
·⁖ostensive⁖·Hot iron lies between hammer face and anvil while heat and opposed pressure hold one shaped edge in the same working fieldᑎ·⸱·˙ostensive⁖·
↪
·⁖grimchain⁖·ᕁᓃᕼ▫▽ᖴ⊕❂ஃᗤI⍦ᓄ⸱·˙grimchain⁖·
·⸱·˙lookup⁖˙
·⸱·˙glyphs⁖˙
·⁖glyph symbol◦–◦ₔᑲₕ category◦–◦ₔcourtₕ⁖·
·⁖lookup⁖·

·⁖origin…language⁖·Neo⟊Quenya·⸱·˙origin…language⁖·
·⁖font⁖·Tengwar Quenya·⸱·˙font⁖·
·⁖pronunciation⁖·accal·⸱·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⸱·˙cadence⁖·
·⁖non…ostensive⁖·to blaze⁖˙ shine suddenly and brilliantly·⸱·˙non…ostensive⁖·
·⁖ostensive⁖·Bright flame fills one sudden reach⁖⁖and hard light presses against the surrounding darkᑲ·⸱·˙ostensive⁖·
↪
·⁖grimchain⁖·ᔑ◎ᕊᒐ◎ↄ⊛ᗰᑭᒧ⍜⍟⧺⓿·⸱·˙grimchain⁖·
·⸱·˙lookup⁖˙
·⸱·˙glyphs⁖˙
·⁖glyph symbol◦–◦ₔᖴₕ category◦–◦ₔcourtₕ⁖·
·⁖lookup⁖·

·⁖ᖶᑒᖌᖶ⁖·ᖘᐣᖘᖶ⁖⸱·˙ᖶᑒᖌᖶ⁖·

·⁖origin…language⁖·Draconic·⸱·˙origin…language⁖·
·⁖font⁖·Iokharic·⸱·˙font⁖·
·⁖pronunciation⁖·lest·⸱·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⸱·˙cadence⁖·
·⁖non…ostensive⁖·Melt·⸱·˙non…ostensive⁖·
·⁖ostensive⁖·Hard wax slumps beside the flame⁖⁖sharp edges softening into a shining pool that runs along the stoneᖴ·⸱·˙ostensive⁖·
↪
·⁖grimchain⁖·ᑊᓀᔕ⍟ᘏᛯᛉᛊ⍄ᑲ∫֍▽·⸱·˙grimchain⁖·
·⸱·˙lookup⁖˙
·⸱·˙glyphs⁖˙
·⁖glyph symbol◦–◦ₔᖇₕ category◦–◦ₔcourtₕ⁖·
·⁖lookup⁖·

·⁖ᔑᐯᒥᒣᐣ⁖·ᓀᐣᘐᔑᘐᓀᐣᒪᒪ⁖⸱·˙ᔑᐯᒥᒣᐣ⁖·

·⁖origin…language⁖·Dovahzul·⸱·˙origin…language⁖·
·⁖font⁖·Dovahkiin·⸱·˙font⁖·
·⁖pronunciation⁖·Sevest·⸱·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⸱·˙cadence⁖·

·⁖non…ostensive⁚·Ordeal·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Frost grips exposed flesh⁝⁝and open road holds no shelter⁝⁝Breath and moving limbs keep
↪ pressure against the same cold᛬·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ϒ⏉⅋⋡ǂϟ⏀⁕∀ƻ♂ᚠᚴᚣ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○⟜○¢ᛃ category○⟜○¢court⁑⁚·
·⁖lookup⁚·

·⁖origin…language⁚·Quenya·⁖.·˙origin…language⁚·
·⁖font⁚·Tengwar Quenya·⁖.·˙font⁚·
·⁖pronunciation⁚·maira·⁖.·˙pronunciation⁚·
·⁖cadence⁚·SeeD Body·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·admirable⁝⁝excellent⁝⁝precious⁝⁝splendid⁝⁝sublime·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Flawless white tower catches high light as clean stone carries unbroken brilliance from
↪ base to crown‹·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᚣᛯᚻ♋ᗡᚾᚴᚨᛉᚴᛗᛃᗺ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○⟜○¢X᛫ category○⟜○¢court⁑⁚·
·⁖lookup⁚·

⁖ᚠᛍᚢ⁚·˚˚ᚠᚣᚣ⁖.·˙ᚠᛍᚢ⁚·

·⁖origin…language⁚·Draconic·⁖.·˙origin…language⁚·
·⁖font⁚·Iokharic·⁖.·˙font⁚·
·⁖pronunciation⁚·aaliss·⁖.·˙pronunciation⁚·
·⁖cadence⁚·SeeD Body·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Start⁝⁝beginning·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Live spark rests inside dry tinder where heat and combustible fiber occupy one ignition
↪ boundaryX·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᚹᓄᔐᔓᔥ᛬ᚴᚷᛏ⏃⏇ᚱᚱᚣ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○⟜○¢ᚹ᛫ category○⟜○¢court⁑⁚·
·⁖lookup⁚·

·⁖origin…language⁚·Neo⟊Sindarin·⁖.·˙origin…language⁚·
·⁖font⁚·Tengwar Sindarin·⁖.·˙font⁚·
·⁖pronunciation⁚·ammad·⁖.·˙pronunciation⁚·
·⁖cadence⁚·SeeD Body·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·to devour·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Flame and wood speak in hunger⁝⁝and wood holds spark within⁝⁝and ash in flame holds
↪ memoryᚹ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·⏇0ᛊXᘁᒍᛯᘜᚺᛉᚴᚢᛏᛁᚴ᛬ᚱᚱ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○⟜○¢Ꮋ᛫ category○⟜○¢court⁑⁚·
·⁖lookup⁚·

⁖⋆ᘜᘛᘚᐢ⁖⋆ᘜᘛᘚᐢ⁖⋆

·⁚origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁚font:·Mage Script·⁚.·˙font:·
·⁚pronunciation:·Kaa+n+ghero·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Cleanse·⁚.·˙non…ostensive:·
·⁚ostensive:·Washed cloth lies under clear running water while loosened stain and clean fiber occupy
↪ the same rinsing fieldᚺ·⁚.·˙ostensive:·
·⁚grimchain:·ƶ⁀ϕᑎᑊᓀ✱♂ᔓΘᕒ▽ᕽ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꜀꜀ᛋ category꜀꜀court:·
·⁚lookup:·

⁖ᔐᖻᒪ:·ᖌᒉᒣ⁖⁚.·˙ᔐᖻᒪ:·

·⁚origin…language:·Mando’a·⁚.·˙origin…language:·
·⁚font:·Mandalorian·⁚.·˙font:·
·⁚pronunciation:·moon·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·hard⁖· of a surface or material·⁚.·˙non…ostensive:·
·⁚ostensive:·Beskar plate takes the hammer blow without denting::Force returns through the haft as the
↪ surface keeps its lineᚾ·⁚.·˙ostensive:·
·⁚grimchain:·ᑎᕎᘁƲΘᛯᓄᖰ◌ᙆᚻᙏᕬ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꜀꜀ᛋ category꜀꜀court:·
·⁚lookup:·

⁖ᘙᗊᒋᒣ⁖⁚ᔕᒣᘙᗊᒋᒣ⁖⁚.·˙ᘙᗊᒋᒣ⁖⁚·

·⁚origin…language:·Dovahzul·⁚.·˙origin…language:·
·⁚font:·Dovahkiin·⁚.·˙font:·
·⁚pronunciation:·Gunaar·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Smash::crush·⁚.·˙non…ostensive:·
·⁚ostensive:·Hammer face bites fired clay::and sharp shards occupy the flags around the blow::Unbroken
↪ contour has no place across the struck centerᚾ·⁚.·˙ostensive:·
·⁚grimchain:·⧗ᘐᚴ♂ᛉᚴᓄ❄ᛗᚠᗺ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꜀꜀ᛃ category꜀꜀court:·
·⁚lookup:·

⁖ᔐᖻᒪ:·ᚼᚻᔑᔐ⁖⁚.·˙ᔐᖻᒪ:·

·⁚origin…language:·Mando’a·⁚.·˙origin…language:·
·⁚font:·Mandalorian·⁚.·˙font:·
·⁚pronunciation:·rahng·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·ash·⁚.·˙non…ostensive:·
·⁚ostensive:·Fire has consumed the wood::Grey powder and brittle flakes remain in the cold
↪ hearth::lifting under the slightest breathᛃ·⁚.·˙ostensive:·

·⁚grimchain:·ᗰᔑᔭ✱ᖻ▽ↄ∂ᔐᔕ0ᚼ✱▽·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol∘-∘ₑ⩤ₚ category∘-∘ₑcourtₚ:·
·⁚lookup:·

·⁚origin…language:·Neo+Sindarin·⁚.·˙origin…language:·
·⁚font:·Tengwar Sindarin·⁚.·˙font:·
·⁚pronunciation:·ethiria·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·to flow out·⁚.·˙non…ostensive:·
·⁚ostensive:·Water stands across the basin lip and the outer run::one continuous body occupying containment and release at the same boundary⩤·⁚.·˙ostensive:·
↪
·⁚grimchain:·ᦒᑫᓕᚱᚹ⬡ᐺᘐᔕᚔᛋᐧ⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol∘-∘ₑᔦₚ category∘-∘ₑcourtₚ:·
·⁚lookup:·

·:ᔑᓯᚑᚌᐡ:·ᔦᚌᔪᓀᓐ:·˙ᔑᓯᚑᚌᐡ:·

·⁚origin…language:·Dovahzul·⁚.·˙origin…language:·
·⁚font:·Dovahkiin·⁚.·˙font:·
·⁚pronunciation:·Hilk·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Mix::combine::fuse·⁚.·˙non…ostensive:·
·⁚ostensive:·Molten copper and tin enter the same crucible until their colors disappear into one cooling metalᔦ·⁚.·˙ostensive:·
↪
·⁚grimchain:·ᚼᛉᘜᘉᚺ✮ᚨᗷᖷᑫᑎ❁ᚺᚺᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol∘-∘ₑↄₚ category∘-∘ₑcourtₚ:·
·⁚lookup:·

·:ᚠᗢᚮᚠ:·ᖷᖻᚴᚮᚮᚠ:·˙ᚠᗢᚮᚠ:·

·⁚origin…language:·Draconic·⁚.·˙origin…language:·
·⁚font:·Iokharic·⁚.·˙font:·
·⁚pronunciation:·sferat·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Solve::resolve·⁚.·˙non…ostensive:·
·⁚ostensive:·Tangled cords lie spread across the floor::One crossing after another opens until every strand runs free from end to endↄ·⁚.·˙ostensive:·
↪
·⁚grimchain:·⊙◎✧ᗰᔓↄᔦᑭᓄᘐᑐᔕI·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol∘-∘ₑ✵ₚ category∘-∘ₑcourtₚ:·
·⁚lookup:·

·:ᔐᚻᚳ:·ᔨᚺᚴᚳᚼ:·˙ᔐᚻᚳ:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·TAI⟊leer·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to hold::keep::preserve·⁖.·˙non…ostensive:·
·⁖ostensive:·Clasped gauntlet closes around a sealed cylinder while an outward pull strains against the fingers::The cylinder never leaves the grip✲·⁖.·˙ostensive:·
·⁖grimchain:·ᛏᚺℭᚹ❁ᚣᚿ➿ஃᛊ꓅ᚲᚳ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦₡₱ category◦–◦court⁑:·
·⁖lookup:·

⁖ᔑᓯᒣᒣᐡ:ᒣᔭᒣᖶᓯᒣᐡ⁖.·˙ᔑᓯᒣᒣᐡ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Rukor·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Residue·⁖.·˙non…ostensive:·
·⁖ostensive:·Heavy remainder lies beneath the removed useful body::Settled fragments keep the weight left by separation₱·⁖.·˙ostensive:·
·⁖grimchain:·ǂ⁜⏀ᗜA⬡֍₩✡Ӂᛋכ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦␀⁘₰ category◦–◦court⁑:·
·⁖lookup:·

⁖ᚯᘎᘏᒣ:ᑭᓄᘐᓭᘏᘎᑐ⁖.·˙ᚯᘎᘏᒣ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Lee⟊behr⟊ah·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Liberate·⁖.·˙non…ostensive:·
·⁖ostensive:·Open shackle hangs loose around an unbound limb as released motion and absent restraint occupy the same threshold␀⁘·⁖.·˙ostensive:·
·⁖grimchain:·◎ᙏᘐ⍋⁜⬡ᚔ⊛ᗕᔑ❁⏀⍈ᛘ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ᚗ₰ category◦–◦court⁑:·
·⁖lookup:·

⁖ᚠᗢᚣᚢ:ᘜᗮᘘᵌᵀᵌᘐᚣ⁖.·˙ᚠᗢᚣᚢ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·nezaiast·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Fog::mist::haze·˙ archaic·⁖.·˙non…ostensive:·
·⁖ostensive:·Air thickens white across the valley::Distant towers lose their edges and the road vanishes a few paces ahead␇·⁖.·˙ostensive:·
·⁖grimchain:·ϛ⏚ᓰᔓ⍈⏀ᚺ⌯ᔐᗷᒍ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙

·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦◦◎◗ category◦–◦◦court◗⁚·
·⁚lookup:·

·˖·⁚·˙ᔑᖻᒪ:·-ᔑᖻᒋᖻ:˖·˙ᔑᖻᒪ:·

·⁚origin…language:·Mando’a·⁚·˙origin…language:·
·⁚font:·Mandalorian·⁚·˙font:·
·⁚pronunciation:·gah⟊LAHR·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·to spill⁖pour·⁚·˙non…ostensive:·
·⁚ostensive:·Liquid crosses a broken vessel wall while inside pressure and escaping flow meet at the same breach◎·⁚·˙ostensive:·
·⁚grimchain:·⊛ɕ⅄▽ɣ₹✪ℓᚻ₥ᓚᛋᚾ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦◦⏀◗ category◦–◦◦court◗⁚·
·⁚lookup:·

·˖·⁚·˙ᔑᖻᒪ:·ᛋ˝·ᔑᖻᒋ˖·⁚·˙ᔑᖻᒪ:·

·⁚origin…language:·Mando’a·⁚·˙origin…language:·
·⁚font:·Mandalorian·⁚·˙font:·
·⁚pronunciation:·ahm⟊GAHN·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·trade⁖exchange·⁚·˙non…ostensive:·
·⁚ostensive:·One open palm releases a tool as the opposite palm releases grain·⁚· each object crosses the same space and settles where the other had been⏀·⁚·˙ostensive:·
·⁚grimchain:·ᛁ☋ΛXᛪᛉᚲᘐℓ⊕◎ᗡᑭ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦◦◡ᛋ◗ category◦–◦◦court◗⁚·
·⁚lookup:·

·⁚origin…language:·Quenya·⁚·˙origin…language:·
·⁚font:·Tengwar Quenya·⁚·˙font:·
·⁚pronunciation:·helcë·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·ice·⁚·˙non…ostensive:·
·⁚ostensive:·Frozen water bears clear hardness under cold⁖with rigid surface and contained water remaining one body◡ᛋ·⁚·˙ostensive:·
·⁚grimchain:·ᚹ₪◡❁Aƹᚸᒉᒣᐤ⊛ᛅ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦◦ᛋ◗ category◦–◦◦court◗⁚·
·⁚lookup:·

·˖ᓭᒣᐣᒣᐢ:·ᕁᒣᐢᓯ·ᒡᕁᐣᒣᐢ˖·⁚·˙ᓭᒣᐣᒣᐢ:·

·⁚origin…language:·Dovahzul·⁚·˙origin…language:·
·⁚font:·Dovahkiin·⁚·˙font:·
·⁚pronunciation:·Vodahvur·⁚·˙pronunciation:·

·⁖cadence⁖·SeeD Body·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Vanish⁖⁖disappear·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·A dark outline thins against the wall⁖⁖loses its edges⁖⁖and leaves bare stone where its
↪ contour had stood﴾·⁖.·˙ostensive⁖·
·⁖grimchain⁖·Ɪ⋋∀ɤϛ✡⏁ᖘ𝍐ɤ⊺ϗ⌒ᛋ·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜∘ᵋ❀᭄ᵌ category⟜∘ᵋcourtᵌ⁖·
·⁖lookup⁖·

⁘ᚣᚺᚳ⁖·ᦓᦇᢉᢊᢊᢊ

⁖₰₪₪₸:·ᔕ₪₪ᓚᓗ⁖.·˙₰₪₪₸:·

·⁚origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁚font:·Mage Script·⁚.·˙font:·
·⁚pronunciation:·Ehr⟊taut·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Earth·⁚.·˙non…ostensive:·
·⁚ostensive:·Dark soil receives root and rain::bears stone and seed::and stretches unbroken beneath every planted weightⵣ·⁚.·˙ostensive:·
↪
·⁚grimchain:·ᛗᚱᛟ₪₩ᛇᛃ𑁍ᗢᔑᚱᚺ⊕·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꜀₀꜀Λ꜆ category꜀₀꜀court꜆:·
·⁚lookup:·

⁖₸ᑭᚻ:·₸ᵀᓀᛯᚠ⁖.·˙₸ᑭᚻ:·

·⁚origin…language:·Draconic·⁚.·˙origin…language:·
·⁚font:·Iokharic·⁚.·˙font:·
·⁚pronunciation:·daqueth·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Carry::support::˙ wear::be clothed in::be covered with·⁚.·˙non…ostensive:·
·⁚ostensive:·Pillar receives the beam's full downward weight::and the roof remains aloft while force closes through stone to groundΛ·⁚.·˙ostensive:·
↪
·⁚grimchain:·᚛ᚯᛟ𐦍ᛁ▫≋❄ᗢᐁᚱᛇᚺᒍ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꜀₀꜀Ⱨ꜆ category꜀₀꜀court꜆:·
·⁚lookup:·

⁖₰₪₪₸:·ᘐᓄ ₸ᘐᓀᛯᚠ⁖.·˙₰₪₪₸:·

·⁚origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁚font:·Mage Script·⁚.·˙font:·
·⁚pronunciation:·Meh⟊dee⟊aht·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·Medium·⁚.·˙non…ostensive:·
·⁚ostensive:·Sound enters open air between two boundaries and crosses the intervening body without becoming either boundary that gives the passage its extentⱧ·⁚.·˙ostensive:·
↪
·⁚grimchain:·ᚴᛉ⩛ᘔᵹ⊙ᛉᛋᵽ⬡ᗤ₪ᛍ↘ᵒ⁘·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꜀₀꜀Ⱪ꜆ category꜀₀꜀court꜆:·
·⁚lookup:·

·⁚origin…language:·Noldorin·⁚.·˙origin…language:·
·⁚font:·Tengwar Sindarin·⁚.·˙font:·
·⁚pronunciation:·pelia·⁚.·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚.·˙cadence:·
·⁚non…ostensive:·to spread·⁚.·˙non…ostensive:·
·⁚ostensive:·Cloth lies broad beneath tension::and one continuous surface reaches farther between its held edgesⱪ·⁚.·˙ostensive:·
↪

·⁖grimchain⁖·ᵹᛉXⱭAᛉↃᔑolᵴᵡ∀·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol∘-∘ᵋ⓫ᵌ category∘-∘ᵋcourtᵌ⁖·
·⁖lookup⁖·

⁘ㅅᐁㅏㅣ':ㅏᵎᐁᵎᵎᵎᵎ'ᵎᵎ'ㅋ⁘·˙ㅅᐁㅏㅣ':·

·⁖origin…language⁖·Dovahzul·⁖·˙origin…language⁖·
·⁖font⁖·Dovahkiin·⁖·˙font⁖·
·⁖pronunciation⁖·Reyziin·⁖·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Remnant·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Broken wall leaves one standing fragment under open weather⁖⁖Surviving stone bears the
↪ same old mortar along its severed face⓫·⁖·˙ostensive⁖·
·⁖grimchain⁖·⊕♂ᚺᚿ⊛ᛩ♀ᑭᗰᚠᚹ✷ᚴ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol∘-∘ᵋᕒᵌ category∘-∘ᵋcourtᵌ⁖·
·⁖lookup⁖·

·⁖origin…language⁖·Quenya·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Quenya·⁖·˙font⁖·
·⁖pronunciation⁖·pic·⁖·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖·˙cadence⁖·
·⁖non…ostensive⁖·to lessen⁖⁖dwindle⁖⁖wane·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Full moon narrows night by night until only a thin bright arc remains against the dark
↪ faceᕒ·⁖·˙ostensive⁖·
·⁖grimchain⁖·❋ᚺᚹₒᵒ⊙ᒣ♲ᛘᵬ‡ℂᵋ⸗ᗵ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol∘-∘ᵋᛋᵌ category∘-∘ᵋcourtᵌ⁖·
·⁖lookup⁖·

·⁖origin…language⁖·Neo⟊Quenya·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Quenya·⁖·˙font⁖·
·⁖pronunciation⁖·corta·⁖·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖·˙cadence⁖·
·⁖non…ostensive⁖·to roll up⁖⁖pack·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Cloth turns tightly around its own length while cord cinches the gathered bundle until it
↪ occupies little spaceᛋ·⁖·˙ostensive⁖·
·⁖grimchain⁖·❁Ⱥ≍♂ᑎᚦ≍ᛒ✡ᵡ▫ᑲᛦ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol∘-∘ᵋⱫᵌ category∘-∘ᵋcourtᵌ⁖·
·⁖lookup⁖·

⁘ᚼᐤᚠ⁖·ᵑᵡᚴᚽ⁘·˙ᚼᐤᚠ⁖·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·mehk·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Hard⁝⁝not soft·⁖ solid·⁖ durable·⁖.·˙non…ostensive:·
·⁖ostensive:·Hammer strikes the stone⁝⁝The face keeps its contour while force rebounds through the
↪ haftᙅ·⁖.·˙ostensive:·
·⁖grimchain:·ᔕ♂ᛦ⅂▽ᛦ✡≋ᑭᕼᚼ❄ᛦ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol○–○ᵋᦓ꞉ category○–○ᵋcourt꞉⁖·
·⁖lookup:·

·⁘ᔑᓭᒣᒣᐣ:·ᒻᓄᓄᔑᓭᒣᒣᐣ⁘.·˙ᔑᓭᒣᒣᐣ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Motaad·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Shudder·⁖ shuddered·⁖.·˙non…ostensive:·
·⁖ostensive:·Ground carries a trembling load beneath the stone⁝⁝and walls answer with the same fine
↪ vibrationᦓ·⁖.·˙ostensive:·
·⁖grimchain:·ᑎᘐ꜀ᵒᔑᐯᖘ꜀ᵒᕮ≳ᗰᓕᐱ◝ᘓ꜀꜀·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol○–○ᵋI꞉ category○–○ᵋcourt꞉⁖·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·dartha·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to wait⁝⁝stay⁝⁝remain⁝⁝last⁝⁝endure·⁖.·˙non…ostensive:·
·⁖ostensive:·Beam keeps step in passing wind and rain⁝⁝and is held unmovedI·⁖.·˙ostensive:·
·⁖grimchain:·ᔑᙅᑦ▽ℂ✳ᓐᔑᐢ⊕ᘐᔑ꜀·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol○–○ᵋᗰ꞉ category○–○ᵋcourt꞉⁖·
·⁖lookup:·

·⁘ᔦᚓᒪ:·ᖴᑦᒪᒪᚓᖴ⁘.·˙ᔦᚓᒪ:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·SOOR⟊dol·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·window⁝⁝opening·⁖.·˙non…ostensive:·
·⁖ostensive:·Stone wall holds solid on every side except one framed gap where daylight and moving air
↪ pass freelyᗰ·⁖.·˙ostensive:·
·⁖grimchain:·Aᘐ⊛ᛘᒧ◎ᵹᔓᚣᛘᛩ▟⊕·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙

·⁖glyph symbol◦–◦ৎ category◦–◦ৎcourt꞉·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁚.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁚.·˙font:·
·⁖pronunciation:·lam·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·echo·⁚.·˙non…ostensive:·
·⁖ostensive:·Outgoing and returning sound pressure share one chamber of air⁖⁖Wall and wave carry
↪ repeated vibration through enclosed airৠ·⁚.·˙ostensive:·
·⁖grimchain:·☽▽Ⱳ↻⓪⋤⊢ⅡⒽ※✓▽⇝ტ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol◦–◦ৎ꞉ category◦–◦ৎcourt꞉·
·⁖lookup:·

·⁚ᵀᴳ⁾ᵀ꞉·ᵀʸᶻᵀᵀ⁾ᵀ⁅⁆꞉·⁚.·˙ᵀᴳ⁾ᵀ꞉·

·⁖origin…language:·Draconic·⁚.·˙origin…language:·
·⁖font:·Iokharic·⁚.·˙font:·
·⁖pronunciation:·senfrah·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·Smoke⁖⁖fumes·⁚.·˙non…ostensive:·
·⁖ostensive:·Dark vapor coils above the fire⁖⁖carrying soot through the rafters and thinning as it
↪ risesগ·⁚.·˙ostensive:·
·⁖grimchain:·♂ᑎᚣ⛥Ⱨᕒ᠊ᒪ❁ᑫᛉᛘ₀ᔆ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol◦–◦ৎ꞉ category◦–◦ৎcourt꞉·
·⁖lookup:·

·⁚ᨐᨓᨓᨒᨒ꞉·ᨓᨐᨓᨓᨓ⁚.·˙ᨐᨓᨓᨒᨒ꞉·

·⁖origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁖font:·Mage Script·⁚.·˙font:·
·⁖pronunciation:·Ae⟊roh·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·Air·⁚.·˙non…ostensive:·
·⁖ostensive:·Curtain lifts though no visible body crosses the chamber⁖⁖Hanging cords bow together and
↪ loose dust travels with the same unseen currentৰ·⁚.·˙ostensive:·
·⁖grimchain:·ᗰⱧ✱0✱⩒⋒⊙Ⱬ∀↗ሕ▫·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol◦–◦ৎ꞉ category◦–◦ৎcourt꞉·
·⁖lookup:·

·⁖origin…language:·Neo⟊Sindarin·⁚.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁚.·˙font:·
·⁖pronunciation:·ast·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·

·⁖non…ostensive:·dust·⁖.·˙non…ostensive:·
·⁖ostensive:·Fine earth lies loose across stone while light grains rest under the smallest moving air
↪ and still surface bears their thin covering·⁖.·˙ostensive:·
·⁖grimchain:·᛻᚛ᛘᚻᚢᛋᛯᛟᛟᛯᛇᛞᛈ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꜔꜔ব᛫ category꜔꜔court᛫⁖·
·⁖lookup:·

⁘ᘔᘎᘐᘐᘏ:·ᘜᘏᘎᘐᘐᘏᘐᘎ⁘.·˙ᘔᘎᘐᘐᘏ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Jahfir·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Develop::evolve·⁖.·˙non…ostensive:·
·⁖ostensive:·Young horn and thickened horn belong to one living structure::greater breadth standing in
↪ the same body without loss of its root formব·⁖.·˙ostensive:·
·⁖grimchain:·ᚾ𝌆ᛓᙆ✡𝍐ᵒ꩜ᗰ⁕⛢ℂᚴ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꜔꜔ᘜ᛫ category꜔꜔court᛫⁖·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·glim·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·voice⁖· utterance·⁖.·˙non…ostensive:·
·⁖ostensive:·Breath leaves the throat in shaped vibration as pitch and articulation travel through the
↪ room as one continuous sounding lineᘜ·⁖.·˙ostensive:·
·⁖grimchain:·⏣⛥ᗧᚺ⬡⊛ᔕᚱ⊙ᚏᛩᛅ℃·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꜔꜔ᘓ᛫ category꜔꜔court᛫⁖·
·⁖lookup:·

·⁖origin…language:·Middle Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·cúma·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·void⁖· darkness·⁖.·˙non…ostensive:·
·⁖ostensive:·Black reach offers no answering form to sight::and every attempted boundary disappears
↪ into the depth that remains unoccupiedᘓ·⁖.·˙ostensive:·
·⁖grimchain:·ᚸᑎᛅᛏᘉᛯᚏᚏᘜᛒᛩᗰ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꜔꜔ᘘ᛫ category꜔꜔court᛫⁖·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·fim·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·slim⁝:slender·⁖.·˙non…ostensive:·
·⁖ostensive:·Reed stem rises between nearly parallel edges⁝:holding length with scarcely any
↪ breadth·⁖.·˙ostensive:·
·⁖grimchain:·𐋺𐋡𐋔𐋦𐋙𐋞𐋓𐋢'𐋤𐋺·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꞉꞉𐋢 category꞉꞉court⁖·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·delw·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·thick·⁖.·˙non…ostensive:·
·⁖ostensive:·Wall carries a deep span of stone between its two faces·˙ a drilled passage must cross the
↪ full mass before light reaches through·⁖.·˙ostensive:·
·⁖grimchain:·⁕𐋟𐋤𐋬𐋭▽⊕𐋫𐋡◎𐋶𐋷𐋱·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꞉꞉𐋟 category꞉꞉court⁖·
·⁖lookup:·

⁖𐋡𐋗𐋟𐋤˙:𐋤˙𐋬𐋟𐋤𐋤𐋫𐋗𐋤𐋟⁖.·˙𐋡𐋗𐋟𐋤˙:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Strungir·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Turbulent⁝:tremulous·⁖.·˙non…ostensive:·
·⁖ostensive:·Water folds against its own crossing current and no single course can command the
↪ surface⁝:every moving face broken by another movement𐋟·⁖.·˙ostensive:·
·⁖grimchain:·ᗡᗰ∫𐋢𐋠𐋵Ψᛦᚸᚡ⊙◌·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꞉꞉𐋟 category꞉꞉court⁖·
·⁖lookup:·

⁖ᚼᚔᚒ:ᚤᚥᚻ⁞⁖.·˙ᚼᚔᚒ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·zaith·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Ease⁝:easiness·⁖.·˙non…ostensive:·
·⁖ostensive:·Well⁄fitted hinge bears load through smooth contact while motion and low resistance
↪ occupy the same joint𐋟·⁖.·˙ostensive:·

·⁖grimchain:·ꀘ♂ᨔ⊖ᦟᦈᦓᦅᦈᦓᦅᦈᦓᦅ

·⁖origin…language:·Mando’a·⁚.·˙origin…language:·
·⁖font:·Mandalorian·⁚.·˙font:·
·⁖pronunciation:·veyr+KO+pah·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·wish⁖dream·⁚.·˙non…ostensive:·
·⁖ostensive:·Quiet body lies under a blanket while impossible roads⁖dead stars⁖and living stone move
↪ across the closed field behind the eyes▽·⁚.·˙ostensive:·
·⁖grimchain:·✳⓪ᘜIஃ△ϒ⏁✳ᘜᚹᛃ𐑌·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁖glyph symbol○–○◟▽◗ category○–○◟court◗:·
·⁖lookup:·

⁘ᔑᒣᒉᒉ᛬:·ᔓᒣᒉᒪ᛬⁘.·˙ᔑᒣᒉᒉ᛬:·

·⁖origin…language:·Dovahzul·⁚.·˙origin…language:·
·⁖font:·Dovahkiin·⁚.·˙font:·
·⁖pronunciation:·Bos·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·Swing·⁚.·˙non…ostensive:·
·⁖ostensive:·Suspended weight holds equal reach on either side of center⁖⁖with taut support carrying
↪ the same arc through the middle line▽·⁚.·˙ostensive:·
·⁖grimchain:·Iᚠᔑᚼ◎Ƶᚹᛊ♎ᔑᑭᐯ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁖glyph symbol○–○◟▽◗ category○–○◟court◗:·
·⁖lookup:·

·⁖origin…language:·Neo+Quenya·⁚.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁚.·˙font:·
·⁖pronunciation:·ofelmë·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·sympathy⁖⁖compassion⁖⁖pity⁖· together+feeling·⁚.·˙non…ostensive:·
·⁖ostensive:·One injured limb curls against the body⁖⁖and another uninjured limb takes the same
↪ guarding shape across the space between them▽·⁚.·˙ostensive:·
·⁖grimchain:·IIᛞΘᚺᛕᚧᚲᚾᛟᛏ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁖glyph symbol○–○◟ᗅ◗ category○–○◟court◗:·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁚.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁚.·˙font:·
·⁖pronunciation:·aphas·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·(sense of) touch·⁚.·˙non…ostensive:·
·⁖ostensive:·Bare skin meets rough bark⁖⁖ridges pressing into the fingertips as vibration travels
↪ through the contact into boneᗅ·⁚.·˙ostensive:·
·⁖grimchain:·ᛐIIᚼᛯ۞ᑫ▽ᛈᛐ∀ᘜᚦ·⁚.·˙grimchain:·
·⁚.·˙lookup:·

·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦�really court₎⁚·
·⁚lookup:·

·⁖ᨒᨒᨐ:·ᨕᨠᨒ⁖·˙ᨒᨒᨐ:·

·⁚origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁚font:·Mage Script·⁚·˙font:·
·⁚pronunciation:·Meh⁄r·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Bad·⁚·˙non…ostensive:·
·⁚ostensive:·Shelter presses inward against the body it encloses⁖⁖Harm occupies the very boundary meant to hold safety·⁚·˙ostensive:·
·⁚grimchain:·♀♂↘⁘▫ᚾΔ❄ъϛ⏀ᨐᚷ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦really court₎⁚·
·⁚lookup:·

·⁚origin…language:·Quenya·⁚·˙origin…language:·
·⁚font:·Tengwar Quenya·⁚·˙font:·
·⁚pronunciation:·orda·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·profound·⁚·˙non…ostensive:·
·⁚ostensive:·Deep water holds unseen distance beneath a visible surface while sounding line enters the same depth without exhausting it·⁚·˙ostensive:·
·⁚grimchain:·⛥ᚕ⊛ℂ⊹ᘜᘐᘏᚏ♉▽ᚏʁ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦⊹₎ category◦–◦court₎⁚·
·⁚lookup:·

·⁖ᘔᘓᚑᚑ':·ᚔᘓᚑᚒᘔᘓᘐᘓᘓᘐ·⁖·˙ᘔᘓᚑᚑ':·

·⁚origin…language:·Dovahzul·⁚·˙origin…language:·
·⁚font:·Dovahkiin·⁚·˙font:·
·⁚pronunciation:·Hezhonaat·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Consonance⁖⁖Consonant·⁚·˙non…ostensive:·
·⁚ostensive:·Separate tones hold one interval⁖⁖and shared air carries their pressure without a beating clash between them⊹·⁚·˙ostensive:·
·⁚grimchain:·ᔓ𐌼ᘜₒᨐᚲᛏᔕᙏᚸᗷᚦᚹ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦⊹₎ category◦–◦court₎⁚·
·⁚lookup:·

·⁖ᘔᘓᚑᚑ':·ᚕᘓᚑᚒᘔᘓᘐᚑ':·⁖·˙ᘔᘓᚑᚑ':·

·⁚origin…language:·Dovahzul·⁚·˙origin…language:·
·⁚font:·Dovahkiin·⁚·˙font:·
·⁚pronunciation:·Morkonin·⁚·˙pronunciation:·

·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Restraint·⁖.·˙non…ostensive:·
·⁖ostensive:·Sheathed blade holds force without release⁝⁝and closed mouth keeps sound behind the
↪ lips⁝⁝Drawn hand remains inside an unspent strike✙·⁖.·˙ostensive:·
·⁖grimchain:·𓆏𐦝𑀏𑁍𑁋𑁋𑁋

:.ᔑᓯᐢᐣ':.ᔓᓯᘔᓯᔓᐢᐣᓯᓯᓭ:.·ᔑᓯᐢᐣ':.

·⁖origin…language:·Dovahzul·⁖.·ˈorigin…language:·
·⁖font:·Dovahkiin·⁖.·ˈfont:·
·⁖pronunciation:·Volahvraan·⁖.·ˈpronunciation:·
·⁖cadence:·SeeD Body·⁖.·ˈcadence:·
·⁖non…ostensive:·Dissipate::Disperse::Scatter::Spread·⁖.·ˈnon…ostensive:·
·⁖ostensive:·A tight cloud of dust breaks apart in the wind as grains separate across the open field
↪ until wide gaps lie between every cluster·⁖.·ˈostensive:·
·⁖grimchain:·ᚼᛞᚸᘐᚢᚱᛇᛃᘓᛈᚳ·⁖.·ˈgrimchain:·
·⁖.·ˈlookup:·
·⁖.·ˈglyphs:·
·⁖glyph symbol꜀꜀𓃒꜀ category꜀꜀court꜀:·
·⁖lookup:·

·⁖origin…language:·Quenya·⁖.·ˈorigin…language:·
·⁖font:·Tengwar Quenya·⁖.·ˈfont:·
·⁖pronunciation:·undumë·⁖.·ˈpronunciation:·
·⁖cadence:·SeeD Body·⁖.·ˈcadence:·
·⁖non…ostensive:·abyss·⁖.·ˈnon…ostensive:·
·⁖ostensive:·Cliff edge and unseen depth share one vertical boundary::with darkness occupying the reach
↪ below visible measure𓃒·⁖.·ˈostensive:·
·⁖grimchain:·𓅓ᘐᘓᛯᛏᘎᘏᛩᛖXᐁᛋᘓᘜ·⁖.·ˈgrimchain:·
·⁖.·ˈlookup:·
·⁖.·ˈglyphs:·
·⁖glyph symbol꜀꜀𓀠꜀ category꜀꜀court꜀:·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·ˈorigin…language:·
·⁖font:·Tengwar Sindarin·⁖.·ˈfont:·
·⁖pronunciation:·dod·⁖.·ˈpronunciation:·
·⁖cadence:·SeeD Body·⁖.·ˈcadence:·
·⁖non…ostensive:·to drop·⁖.·ˈnon…ostensive:·
·⁖ostensive:·Stone leaves the ledge with no support beneath it::The gap above grows while the ground
↪ closes rapidly below𓀠·⁖.·ˈostensive:·
·⁖grimchain:·ᘐᡕᘏᘓ'ᘓᛁᑎᛋᘐᓀᓚ·⁖.·ˈgrimchain:·
·⁖.·ˈlookup:·
·⁖.·ˈglyphs:·
·⁖glyph symbol꜀꜀𓆟꜀ category꜀꜀court꜀:·
·⁖lookup:·

:.ᔑᓯᐢᐣ':.ᐣᐢᘐᓯᔓᘏᓯᘐᐢ:.·ᔑᓯᐢᐣ':.

·⁖origin…language:·Dovahzul·⁖.·ˈorigin…language:·
·⁖font:·Dovahkiin·⁖.·ˈfont:·
·⁖pronunciation:·Strakaar·⁖.·ˈpronunciation:·
·⁖cadence:·SeeD Body·⁖.·ˈcadence:·
·⁖non…ostensive:·Deflect·⁖.·ˈnon…ostensive:·
·⁖ostensive:·Arrow and angled shield meet at one contact plane while redirected force occupies the
↪ same impact boundary𓆟·⁖.·ˈostensive:·

·⁖grimchain⁖·᚛ᚦ☆ϴ♂⚹ᵱ⍈⚹↗ᚲᚱᑭ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𖤐 category⟜⟜court⁖·
·⁖lookup⁖·

·𖤐𖤈⁖·ᖴᖵᖵ𖤈·⁖·˙𖤐𖤈⁖·

·⁖origin…language⁖·Mando’a·⁖·˙origin…language⁖·
·⁖font⁖·Mandalorian·⁖·˙font⁖·
·⁖pronunciation⁖·KO+pahd·⁖·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖·˙cadence⁖·
·⁖non…ostensive⁖·ambition⁖˙ desire·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Gloved hand reaches toward the highest rung beyond its fingertips⁖⁖Weight presses upward
↪ though the gap remains𖤐·⁖·˙ostensive⁖·
·⁖grimchain⁖·Ƹᵹȳᵴ℃֎♂ᵑᵒᑭᵴᚱᚻ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𖣔 category⟜⟜court⁖·
·⁖lookup⁖·

·⁖ᦵᦶᦷᦸ⁖·ᦷ᧐ ᦺ⁺²ᦺ⁖·˙ᦵᦶᦷᦸ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖·˙font⁖·
·⁖pronunciation⁖·Shay+noot·⁖·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Heal⁖˙ to Heal·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·New tissue spans the seam while joined flesh holds pressure across the closed
↪ opening𖣔·⁖·˙ostensive⁖·
·⁖grimchain⁖·ᚖᛯᛯᚻℂᛉᛊ⍝'ᛒᚣᘜ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𖣔 category⟜⟜court⁖·
·⁖lookup⁖·

·⁖origin…language⁖·Sindarin·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Sindarin·⁖·˙font⁖·
·⁖pronunciation⁖·thag+¹·⁖·˙pronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖·˙cadence⁖·
·⁖non…ostensive⁖·to draw⁖⁖pull·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Taut rope straightens between anchor and loaded sled as the sled leaves a fresh groove
↪ and tension shortens the distance𖣔·⁖·˙ostensive⁖·
·⁖grimchain⁖·⬡᚛ᚱᔓᛋᘐᛋᘐᑐᑫᖘᘐ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𖣔 category⟜⟜court⁖·
·⁖lookup⁖·

·𖤐𖤈⁖·𖤈𖤈𖤈ᦵᦶᦷ⁖·˙𖤐𖤈⁖·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·dah+JOO+nil·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·stage::step of a plan·⁖.·˙non…ostensive:·
·⁖ostensive:·Marked crossing holds one place inside a larger route::The crossing keeps a complete
↪ boundary between neighboring reaches⛤·⁖.·˙ostensive:·
·⁖grimchain:·♋✡▽ᔕ⅋ᔓᔕᔑ♋·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○𓃰 category○–○court𓃰:·
·⁖lookup:·

⁘ᔓᔑᑎᑐ:ᓭᑎᑐᔑᔓᔑᑐᔑᔓᑐᑐᓄ⁘.·˙ᔓᔑᑎᑐ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Uzgrolein·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Paradox::Absurdity::Anomaly::Inconsistency·⁖.·˙non…ostensive:·
·⁖ostensive:·Single strip carries inner and outer travel upon one unbroken face::one edge returning
↪ into itself without a second boundary𓃰·⁖.·˙ostensive:·
·⁖grimchain:·♀ᘏᔕ‹⛤♋☽♂♈♀☿♇⊕·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○‹𓃰 category○–○court𓃰:·
·⁖lookup:·

⁘ᔓᔑᑎᑐ:ᓭᑎᑐᔑᔓᓄᔑᔓᑎᑐ⁘.·˙ᔓᔑᑎᑐ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Praantaas·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Dormant·⁖.·˙non…ostensive:·
·⁖ostensive:·Banked ember holds heat beneath ash while the red core remains warm without outward
↪ flame‹·⁖.·˙ostensive:·
·⁖grimchain:·✡ᑫₒ❄ı✡ᚾᛞ⓪☆✷↓☾·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○𐤕𓃰 category○–○court𓃰:·
·⁖lookup:·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·hala·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·cast shadow::*shade·⁖.·˙non…ostensive:·
·⁖ostensive:·Covered ground carries weakened light beneath one shelter::Darkened surface and bright
↪ edge meet at the same boundary𐤕·⁖.·˙ostensive:·
·⁖grimchain:·ı𓃰ᗤ♀⓪Xሐ⓪♉⁉✡ᚷı·⁖.·˙grimchain:·
·⁖.·˙lookup:·

·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐑈 category꩜꩜court𐑈⁚·
·⁚lookup⁚·

⁖ᚠᛂᚱ⁚·ᚧᛈᛥᚱ⁖·˙ᚠᛂᚱ⁚·

·⁚origin…language⁚·Draconic·⁚·˙origin…language⁚·
·⁚font⁚·Iokharic·⁚·˙font⁚·
·⁚pronunciation⁚·bogh·⁚·˙pronunciation⁚·
·⁚cadence⁚·SeeD Body·⁚·˙cadence⁚·
·⁚non…ostensive⁚·dark·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Unlit stone holds no reflected edge··and open space carries no visible contour··Surface and
↪ distance lie inside one black field𐑈·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚱᚹᚠᛜᛞᛇ❂ᚻᚩᚸᛋᛃ⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ᛈ𐑈 category꩜꩜court𐑈⁚·
·⁚lookup⁚·

⁖ᚠᛂᚱ⁚·ᛝᚴᚽᛎᛞᚱ⁖·˙ᚠᛂᚱ⁚·

·⁚origin…language⁚·Draconic·⁚·˙origin…language⁚·
·⁚font⁚·Iokharic·⁚·˙font⁚·
·⁚pronunciation⁚·skaeth·⁚·˙pronunciation⁚·
·⁚cadence⁚·SeeD Body·⁚·˙cadence⁚·
·⁚non…ostensive⁚·cold·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Water skins over with clear ice··exposed metal bites bare skin··and breath whitens in the
↪ airᛈ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛯᛯᛯᚺᚿᛅᚸᛉ᛬ᛯI᛬·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ᛏ𐑈 category꩜꩜court𐑈⁚·
·⁚lookup⁚·

⁖ᚼᛞᛞᛏ⁚·ᘓᘐᘏᘛᘔᘐᘛ⁖·˙ᚼᛞᛞᛏ⁚·

·⁚origin…language⁚·Llumena Milenii·⁚·˙origin…language⁚·
·⁚font⁚·Mage Script·⁚·˙font⁚·
·⁚pronunciation⁚·Ehm÷ah÷gehg·⁚·˙pronunciation⁚·
·⁚cadence⁚·SeeD Body·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Image·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Polished silver receives the face before it··Light carries the same contour across the
↪ surface while the solid body remains outside the metalᛏ·⁚·˙ostensive⁚·
·⁚grimchain⁚·Iᗡᘐᘓᚦᛋᛃᘏ‹ᘐᚸᘛ⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐑘𐑈 category꩜꩜court𐑈⁚·
·⁚lookup⁚·

·⁚origin…language⁚·Sindarin·⁚·˙origin…language⁚·
·⁚font⁚·Tengwar Sindarin·⁚·˙font⁚·
·⁚pronunciation⁚·somma·⁚·˙pronunciation⁚·

·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·to dive::duck·⁚.·˙non…ostensive:·
·⁖ostensive:·Descending body occupies water below the bright surface while depth and submerged motion ↪ coexist in one vertical field·⁚.·˙ostensive:·
·⁖grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁖glyph symbol꜀₎ category꜀court₎:·
·⁖lookup:·

·⁖origin…language:·Quenya·⁚.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁚.·˙font:·
·⁖pronunciation:·lanya·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·thread::warp·⁚.·˙non…ostensive:·
·⁖ostensive:·One long fiber runs taut from beam to beam as crossing fibers pass alternately above and ↪ below it and cloth gathers around the fixed line·⁚.·˙ostensive:·
·⁖grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁖glyph symbol꜀₎ category꜀court₎:·
·⁖lookup:·

·⁖origin…language:·Mando’a·⁚.·˙origin…language:·
·⁖font:·Mandalorian·⁚.·˙font:·
·⁖pronunciation:·MEERSH⟊ko·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·courage·⁚.·˙non…ostensive:·
·⁖ostensive:·Knees hold beneath shaking muscle::and forward step keeps pressure against the force that ↪ drives the body back·⁚.·˙ostensive:·
·⁖grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁖glyph symbol꜀₎ category꜀court₎:·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁚.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁚.·˙font:·
·⁖pronunciation:·nod⟊²·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·to tie·⁚.·˙non…ostensive:·
·⁖ostensive:·Bound cord holds two ends under shared tension while the knot occupies the joining ↪ boundary that prevents separation·⁚.·˙ostensive:·
·⁖grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁖glyph symbol꜀₎ category꜀court₎:·
·⁖lookup:·

·⁖origin…language:·Quenya·⁚.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁚.·˙font:·
·⁖pronunciation:·cambië·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·profit⁖gain⁖intake⁖income·⁚.·˙non…ostensive:·
·⁖ostensive:·Open vessel receives a steady inward flow as held measure presses deeper inside the same
↪ boundary𝛙·⁚.·˙ostensive:·
·⁖grimchain:·x𝗉0ᚏ⩛⁕⊛⊀⊛▽⁐Ψ𐑂·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁖glyph symbol○-○𐑂𝛙 category○-○𐑂court𝛙:·
·⁖lookup:·

·⁖origin…language:·Quenya·⁚.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁚.·˙font:·
·⁖pronunciation:·cil·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·to choose⁖select·⁚.·˙non…ostensive:·
·⁖ostensive:·Several roads remain equally open around one standing point⁖One bearing holds the
↪ body⁖and the other directions remain unoccupied𝛙·⁚.·˙ostensive:·
·⁖grimchain:·Ⅱ🢒Ψ𝛙ΨↃϛ♂⊬⁕0▽Ж·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁖glyph symbol○-○𐑂ᛯ category○-○𐑂court𝛙:·
·⁖lookup:·

∴ᚣᚠᚨ:·ᛇᛘᛋᛋᛋᛋᛏᛉᚣᚠᚨ:.·˙ᚣᚠᚨ:·

·⁖origin…language:·Mando’a·⁚.·˙origin…language:·
·⁖font:·Mandalorian·⁚.·˙font:·
·⁖pronunciation:·hah+RAH+no+vor·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·to hide⁖conceal·⁚.·˙non…ostensive:·
·⁖ostensive:·Dark veil covers gold⁖and stone slab seals hollow vault⁖and outer surface is
↪ unseenᛯ·⁚.·˙ostensive:·
·⁖grimchain:·𐑅ʃ𝛀ıᔑᔐΨϬAᛘ₩‹·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁖glyph symbol○-○𐑂Ⱶ𝛙 category○-○𐑂court𝛙:·
·⁖lookup:·

∴ᘜᘓᘐᐢ:·ᘏᘓᔕᘐᘔ∴.·˙ᘜᘓᘐᐢ:·

·⁖origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁖font:·Mage Script·⁚.·˙font:·
·⁖pronunciation:·Pah+vray·⁚.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁚.·˙cadence:·
·⁖non…ostensive:·Cut·⁚· To Cut·⁚.·˙non…ostensive:·
·⁖ostensive:·Bright edge crosses taut cord⁖Fibers part beneath the pressure and two loose ends spring
↪ away from the severing lineⱵ·⁚.·˙ostensive:·

·⁖grimchain⁚·ΘϷϽ🡆Ⱨ▫🗐🄈ᔑᒧᘜᘚ·⁖·˙grimchain⁚·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍Aꜜ category○–○₍courtꜜ⁚·
·⁖lookup⁚·

·⁖origin…language⁚·Quenya·⁖·˙origin…language⁚·
·⁖font⁚·Tengwar Quenya·⁖·˙font⁚·
·⁖pronunciation⁚·onquilë·⁖·˙pronunciation⁚·
·⁖cadence⁚·SeeD Body·⁖·˙cadence⁚·
·⁖non…ostensive⁚·tone⁖timbre⁖(lit) voice+color·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Two flutes sustain the same pitch⁖yet one sounds bright and reedy while the other carries
↪ a dark⁖rounded grainA·⁖·˙ostensive⁚·
·⁖grimchain⁚·ɤՈ𐌵ᚼᏒϒ𐌺ᛯᵬΘᗅ𐌵ᑭƧ·⁖·˙grimchain⁚·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍Bꜜ category○–○₍courtꜜ⁚·
·⁖lookup⁚·

·⁖origin…language⁚·Sindarin·⁖·˙origin…language⁚·
·⁖font⁚·Tengwar Sindarin·⁖·˙font⁚·
·⁖pronunciation⁚·gwista·⁖·˙pronunciation⁚·
·⁖cadence⁚·SeeD Body·⁖·˙cadence⁚·
·⁖non…ostensive⁚·to change⁖alter⁖vary·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Green leaf turns red along its veins while stem and surface remain continuous⁖The same
↪ body carries a new colorϐ·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᚋᥴᑭᦔᦓ𑁍▽❋Iᛕϐᗕᛪ·⁖·˙grimchain⁚·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍Cꜜ category○–○₍courtꜜ⁚·
·⁖lookup⁚·

·⁘ᚠᗣ⅂ꜜ:·ᵌ⅂ᗰᚠᚴ⁘·⁖·˙ᚠᗣ⅂ꜜ:·

·⁖origin…language⁚·Draconic·⁖·˙origin…language⁚·
·⁖font⁚·Iokharic·⁖·˙font⁚·
·⁖pronunciation⁚·artch·⁖·˙pronunciation⁚·
·⁖cadence⁚·SeeD Body·⁖·˙cadence⁚·
·⁖non…ostensive⁚·gain·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·New weight rests inside the hoard boundary as gathered measure bears greater mass against
↪ the chamber floorↃ·⁖·˙ostensive⁚·
·⁖grimchain⁚·Ↄ𐌡ᛯ⊛ᚢᗡ⋖🡆※ᘓϒᖷᗡᚸÞ·⁖·˙grimchain⁚·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍ﾒꜝ category○–○₍courtꜜ⁚·
·⁖lookup⁚·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·er·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·one::alone·⁖.·˙non…ostensive:·
·⁖ostensive:·A candle burns at the center of an empty chamber::Every other socket around the ring is
↪ bare↓⁖.·˙ostensive:·
·⁖grimchain:·‡▽ᙇ▽X'✳ᚺ𓆏ᚼ⌀⊕⊙ᚕ⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol○–○𐑉 category○–○court𐑈:·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·gaeda·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to astound::make aghast (& surprise::shock)·⁖.·˙non…ostensive:·
·⁖ostensive:·Expected flat wall opens suddenly onto a vast chasm::Breath stops::pupils widen::and every
↪ prepared motion freezes at the threshold↓⁖.·˙ostensive:·
·⁖grimchain:·⁕🝡𐑂𐑅✓ ⊛ᚯ⏢ᑭᚏᏀᑭᔕ⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol○–○𐑁 category○–○court𐑈:·
·⁖lookup:·

⁖ᓚᘏᑎᐣ:·ᘏᵣᐣᘏᵢᓭᐣᑎᐣᵑ⁖.·˙ᓚᘏᑎᐣ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Ofahnir·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Chord·⁖.·˙non…ostensive:·
·⁖ostensive:·Several tones strike together while separate voices hold one stable sounding body across
↪ the same interval𐑁⁖.·˙ostensive:·
·⁖grimchain:·ᘐᘜᛋᛈᛯᔕᛋᛁᚧᚢᑉᘯᛊᚓᘏ⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol○–○𐑊 category○–○court𐑈:·
·⁖lookup:·

⁖ᚠᐤᚣᚠ:·ᐤᚴᐦᐟᐦᐧ⁖.·˙ᚠᐤᚣᚠ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·ojan·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·beat·⁖.·˙non…ostensive:·
·⁖ostensive:·Taut drumskin carries one struck vibration across its surface::Air and hide carry one
↪ pressure pulse through the sounding field𐑊⁖.·˙ostensive:·
·⁖grimchain:·ᑲᛯ✙ₒ∀ᘐᗅᔕᓄᑐᘎᔑᘯ⁖.·˙grimchain:·
·⁖.·˙lookup:˙

·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘₍Ɑ₎ category∘-∘₍court₎⁚·
·⁚lookup:·

·:ᘏᔓᔕᒣ:·ᓚ⚢ᓚ⚢ᔭ:·˙ᘏᔓᔕᒣ:·

·⁚origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁚font:·Mage Script·⁚·˙font:·
·⁚pronunciation:·Ah+za+zool·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Music·⁚·˙non…ostensive:·
·⁚ostensive:·Tone answers tone across held silence while rhythm and resonance occupy one living course
↪ of soundƆ·⁚·˙ostensive:·
·⁚grimchain:·ⅎHxIᘎᛒᔕᛋᙢᔑ⛤ᚿ⊛·⁚·˙grimchain:·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘₍ϰ₎ category∘-∘₍court₎⁚·
·⁚lookup:·

·:ᔐᐦᒪ:·ᒪᑭᖻᖽᐩᐧ:·˙ᔐᐦᒪ:·

·⁚origin…language:·Mando’a·⁚·˙origin…language:·
·⁚font:·Mandalorian·⁚·˙font:·
·⁚pronunciation:·do+SLAH+neer·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·to cross::intersect·⁚·˙non…ostensive:·
·⁚ostensive:·Two roads meet at one stone marker and continue beyond it in four separate
↪ directionsϰ·⁚·˙ostensive:·
·⁚grimchain:·X⍋ₒ°⅍ᘏ⊲ᔑƳᒧ⍈ↀ⏁·⁚·˙grimchain:·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘₍1₎ category∘-∘₍court₎⁚·
·⁚lookup:·

·:ᖰᐦᒣ:·ᐦᐢᐧᖻᖽ:·˙ᖰᐦᒣ:·

·⁚origin…language:·Draconic·⁚·˙origin…language:·
·⁚font:·Iokharic·⁚·˙font:·
·⁚pronunciation:·rizath·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·fasten·⁚·˙non…ostensive:·
·⁚ostensive:·Clasp bears outward strain across joined edges::both sides remaining held at the same
↪ loaded boundary1·⁚·˙ostensive:·
·⁚grimchain:·ᔑᓀ⁞※ᓭ∀ᒧᐃ⏢ᛉ⛧Ƶ⁑·⁚·˙grimchain:·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol∘-∘₍ᔑ₎ category∘-∘₍court₎⁚·
·⁚lookup:·

·:ᔐᐦᒪ:·ᔨᐦᔑᔭ:·˙ᔐᐦᒪ:·

·⁚origin…language:·Mando’a·⁚·˙origin…language:·
·⁚font:·Mandalorian·⁚·˙font:·
·⁚pronunciation:·sey+RAH+teer·⁚·˙pronunciation:·

·⁖cadence⁖·SeeD Body·⁖.·ˈcadence⁖·
·⁖non…ostensive⁖·to point·⁖.·ˈnon…ostensive⁖·
·⁖ostensive⁖·Raised spear tip fixes a bearing toward the distant gate while shaft and tip hold one straight line through the open space·⁖.·ˈostensive⁖·
·⁖grimchain⁖·♂♦✪◀╘✳⊕ᵑ๑ᑦ乃A◀╘ᚻ‡·⁖.·ˈgrimchain⁖·
·⁖.·ˈlookup⁖·
·⁖.·ˈglyphs⁖·
·⁖glyph symbol⟜◦ᵩᛊ category⟜◦ᵩcourt⁑⁖·
·⁖lookup⁖·

⁘ᘜᙏᙏᐟ⁖·ᔑ✾ᖘ乃ᘐ⁘.·ˈᘜᙏᙏᐟ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖.·ˈorigin…language⁖·
·⁖font⁖·Mage Script·⁖.·ˈfont⁖·
·⁖pronunciation⁖·Kawl⟊pah·⁖.·ˈpronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·ˈcadence⁖·
·⁖non…ostensive⁖·Oneness as Emanation·⁖.·ˈnon…ostensive⁖·
·⁖ostensive⁖·One flame casts many rays across surrounding surfaces⁖⁖Every ray remains joined to the same luminous source⁖⁖and no brightness stands severed from the fire·⁖.·ˈostensive⁖·
·⁖grimchain⁖·‡ᛊᚠᘔ⊕ᔑ✡ᚨᗉ▫ᚐᗺᓍ·⁖.·ˈgrimchain⁖·
·⁖.·ˈlookup⁖·
·⁖.·ˈglyphs⁖·
·⁖glyph symbol⟜◦ᵩ✳ category⟜◦ᵩcourt⁑⁖·
·⁖lookup⁖·

⁘ㅈㄱㅐㅣᐟ⁖·ㅏㅣㄱ·ㅣㅗㅣᐟ⁘.·ˈㅈㄱㅐㅣᐟ⁖·

·⁖origin…language⁖·Dovahzul·⁖.·ˈorigin…language⁖·
·⁖font⁖·Dovahkiin·⁖.·ˈfont⁖·
·⁖pronunciation⁖·Rahnd·⁖.·ˈpronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·ˈcadence⁖·
·⁖non…ostensive⁖·Passage⁖⁖Access·⁖.·ˈnon…ostensive⁖·
·⁖ostensive⁖·Stone barrier is solid except for a narrow arch running completely through it⁖⁖Open ground continues from one side to the other✳·⁖.·ˈostensive⁖·
·⁖grimchain⁖·Ӿ«◀╘ɤᖷIᐃሕIIⓋᛀᛦ·⁖.·ˈgrimchain⁖·
·⁖.·ˈlookup⁖·
·⁖.·ˈglyphs⁖·
·⁖glyph symbol⟜◦ᵩ✳ category⟜◦ᵩcourt⁑⁖·
·⁖lookup⁖·

⁘ㅈㄱㅐㅣᐟ⁖·ㄟㄱㄴㄱ·ㅣㄟㄱㅗ⁘.·ˈㅈㄱㅐㅣᐟ⁖·

·⁖origin…language⁖·Dovahzul·⁖.·ˈorigin…language⁖·
·⁖font⁖·Dovahkiin·⁖.·ˈfont⁖·
·⁖pronunciation⁖·Vogahvon·⁖.·ˈpronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·ˈcadence⁖·
·⁖non…ostensive⁖·Defy⁖⁖Resist·⁖.·ˈnon…ostensive⁖·
·⁖ostensive⁖·Command and refusal press against one another⁖⁖Neither line yields at the shared boundary✳·⁖.·ˈostensive⁖·
·⁖grimchain⁖·✳ᗰᚢᚹᔑᔑᔑᚹᔑᔑᘔ✡·⁖.·ˈgrimchain⁖·
·⁖.·ˈlookup⁖·
·⁖.·ˈglyphs⁖·
·⁖glyph symbol⟜◦ᵩᛯ category⟜◦ᵩcourt⁑⁖·
·⁖lookup⁖·

⁖⅄⅍⅃:·Ⅳ⁘.·˙⅄⅍⅃:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·ahm·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·change·⁖.·˙non…ostensive:·
·⁖ostensive:·Smooth armor plate leaves the forge with a new curve while the same metal keeps its
↪ weight and its surface no longer follows the former line✶·⁖.·˙ostensive:·
·⁖grimchain:·ς≋ɛ⊖ᔕ✙❂▽⊙⅋ϙ℘Ψ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol∘-∘ɛ⅏꞉ category∘-∘ɛcourt꞉:·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·bardha·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to rule::reign·⁖.·˙non…ostensive:·
·⁖ostensive:·Crowned seat holds the chamber’s highest place::and separate ranks keep one ordered
↪ bearing beneath it⅏·⁖.·˙ostensive:·
·⁖grimchain:·ᛣᘐᘯ୭ᦓ⬡Ψᔓ℘ᚣ✙ᕼ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol∘-∘ɛϒ꞉ category∘-∘ɛcourt꞉:·
·⁖lookup:·

⁖⁕ᘎᘏᘐ:·ᘓ৫ᘓ৫ᘐᔕᘏᘎ৫ᘓ৫ᘓ:·˙⁕ᘎᘏᘐ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Vih+vih+dar+ee+eum·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Garden·⁖.·˙non…ostensive:·
·⁖ostensive:·Living roots hold separate places in one tended earth::Cultivated enclosure gathers rooted
↪ growth without erasing each living bodyϒ·⁖.·˙ostensive:·
·⁖grimchain:·ℂφ֍⏁ᘓ♪'ς'ɻ֍ᚙↄ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol∘-∘ɛ0꞉ category∘-∘ɛcourt꞉:·
·⁖lookup:·

⁖⁕ᘎᘏᘐ:·ᘐᘐᘓ৫ᘓ✶ᘏ⁖.·˙⁕ᘎᘏᘐ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Loo+men+nah·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Light·˙ Silver·⁖.·˙non…ostensive:·
·⁖ostensive:·Pale radiance spills from moon+bright metal::drawing sharp relief from darkness and
↪ returning a cool white gleam from every polished edge0·⁖.·˙ostensive:·

·⁖grimchain⁖·ᚱᛚᛣᛋᛟᛩ⊙ᛈᛇᛜᛘᛞ·⁖.·ˑgrimchain⁖·
·⁖.·ˑlookup⁖·
·⁖.·ˑglyphs⁖·
·⁖glyph symbol⟜⟜ᛤ category⟜⟜court⁖·
·⁖lookup⁖·

·.⁖⁖⁖⁖⁖:·ᚼᚾᛇᚴ:.·ˑ⁖⁖⁖⁖⁖:·

·⁖origin…language⁖·Mando’a·⁖.·ˑorigin…language⁖·
·⁖font⁖·Mandalorian·⁖.·ˑfont⁖·
·⁖pronunciation⁖·REYT⟊eesh·⁖.·ˑpronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·ˑcadence⁖·
·⁖non…ostensive⁖·possible·⁖.·ˑnon…ostensive⁖·
·⁖ostensive⁖·Open path carries no blocking wall⁘Clear ground holds an unspent route all the way forward⁘·⁖.·ˑostensive⁖·
·⁖grimchain⁖·♂ᛩᛋ♂ɻᗅ✡ᗧᛯᙀᛉᚷ·⁖.·ˑgrimchain⁖·
·⁖.·ˑlookup⁖·
·⁖.·ˑglyphs⁖·
·⁖glyph symbol⟜⟜ᚼ category⟜⟜court⁖·
·⁖lookup⁖·

·.ᔑᐩᒣᒣᐡ:·ᑲᓀᐡᐣᒣᐩᐢᐢᐦ·.·ˑᔑᐩᒣᒣᐡ:·

·⁖origin…language⁖·Dovahzul·⁖.·ˑorigin…language⁖·
·⁖font⁖·Dovahkiin·⁖.·ˑfont⁖·
·⁖pronunciation⁖·Miiraak·⁖.·ˑpronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·ˑcadence⁖·
·⁖non…ostensive⁖·Portal·⁖.·ˑnon…ostensive⁖·
·⁖ostensive⁖·Open arch pierces the dividing wall⁘Stone keeps separation on either side⁘and continuous passage occupies the shared threshold⁜·⁖.·ˑostensive⁖·
·⁖grimchain⁖·ᗤᛯↄᚺ֍ᙏᛉᐳᔑᓴᖘᘐ·⁖.·ˑgrimchain⁖·
·⁖.·ˑlookup⁖·
·⁖.·ˑglyphs⁖·
·⁖glyph symbol⟜⟜ᙏ category⟜⟜court⁖·
·⁖lookup⁖·

·⁖origin…language⁖·Quenya·⁖.·ˑorigin…language⁖·
·⁖font⁖·Tengwar Quenya·⁖.·ˑfont⁖·
·⁖pronunciation⁖·latya·⁖.·ˑpronunciation⁖·
·⁖cadence⁖·SeeD Body·⁖.·ˑcadence⁖·
·⁖non…ostensive⁖·to open anything (so as to allow entry)·⁖.·ˑnon…ostensive⁖·
·⁖ostensive⁖·Bar lifts from the gate⁘both leaves swing apart⁘and uninterrupted ground runs from the road into the courtyardᙏ·⁖.·ˑostensive⁖·
·⁖grimchain⁖·ϴᖷᘐᖽᓮᛗᚔᚷᚷᚸᚷ·⁖.·ˑgrimchain⁖·
·⁖.·ˑlookup⁖·
·⁖.·ˑglyphs⁖·
·⁖glyph symbol⟜⟜ᛯ category⟜⟜court⁖·
·⁖lookup⁖·

·.ᔑᐩᒣᒣᐡ:·ᙆᛗᘐᐩᐡ·.·ˑᔑᐩᒣᒣᐡ:·

·⁖origin…language⁝·Dovahzul·⁖.·˙origin…language⁝·
·⁖font⁝·Dovahkiin·⁖.·˙font⁝·
·⁖pronunciation⁝·Vuld·⁖.·˙pronunciation⁝·
·⁖cadence⁝·SeeD Body·⁖.·˙cadence⁝·
·⁖non…ostensive⁝·Shift::Change::Modify::Mutate::Transfer::Transform::Transmute·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·Straight plate bends under heat and pressure while the same metal leaves the working face
↪ at a new angle::occupying space it did not touch before𐑌·⁖.·˙ostensive⁝·
·⁖grimchain⁝·⊛Ↄ♨ᚏᵠᚷ♉⁎Aᖶᕮᕀ♀·⁖.·˙grimchain⁝·
·⁖.·˙lookup⁝˙
·⁖.·˙glyphs⁝˙
·⁖glyph symbol◦–◦𐑌꞉ category◦–◦court꞉·
·⁖lookup⁝·

·⁖origin…language⁝·Sindarin·⁖.·˙origin…language⁝·
·⁖font⁝·Tengwar Sindarin·⁖.·˙font⁝·
·⁖pronunciation⁝·rod·⁖.·˙pronunciation⁝·
·⁖cadence⁝·SeeD Body·⁖.·˙cadence⁝·
·⁖non…ostensive⁝·tube::stem::*pipe·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·Hollow reed holds an open channel from end to end while wall and passage remain one
↪ continuous body𐑌·⁖.·˙ostensive⁝·
·⁖grimchain⁝·⁂◞ᚏᕀ𑀏ᔕᘎᘎᙆᑭᕮ♉ᚏ·⁖.·˙grimchain⁝·
·⁖.·˙lookup⁝˙
·⁖.·˙glyphs⁝˙
·⁖glyph symbol◦–◦Ɏ꞉ category◦–◦court꞉·
·⁖lookup⁝·

·⁖origin…language⁝·Sindarin·⁖.·˙origin…language⁝·
·⁖font⁝·Tengwar Sindarin·⁖.·˙font⁝·
·⁖pronunciation⁝·pelthaes·⁖.·˙pronunciation⁝·
·⁖cadence⁝·SeeD Body·⁖.·˙cadence⁝·
·⁖non…ostensive⁝·pivot·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·Fixed pin holds the center::and surrounding timber bears rotation around that
↪ pointɎ·⁖.·˙ostensive⁝·
·⁖grimchain⁝·Ⱨᚏᖷᕀ◝⁂ᛒᚱᚯᓄ⊥ᘐᘐ·⁖.·˙grimchain⁝·
·⁖.·˙lookup⁝˙
·⁖.·˙glyphs⁝˙
·⁖glyph symbol◦–◦ᙀ꞉ category◦–◦court꞉·
·⁖lookup⁝·

·⁖ᘜᘓᘓᒠ⁝·ᔕ⁎ȥȥᑢᘜ·⁖.·˙ᘜᘓᘓᒠ⁝·

·⁖origin…language⁝·Llumena Milenii·⁖.·˙origin…language⁝·
·⁖font⁝·Mage Script·⁖.·˙font⁝·
·⁖pronunciation⁝·Ehn+seet+tu·⁖.·˙pronunciation⁝·
·⁖cadence⁝·SeeD Body·⁖.·˙cadence⁝·
·⁖non…ostensive⁝·Peace·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·Weapons lie quiet beside open doors::and hands rest unclenched::No wound opens across the
↪ shared roomᙀ·⁖.·˙ostensive⁝·
·⁖grimchain⁝·ᛇ♀ᕮᓘᘎᚱᘐX◎ᑭ𐑌𐑌·⁖.·˙grimchain⁝·
·⁖.·˙lookup⁝˙

·⁚·˙glyphs:˙
·⁚glyph symbol꜀ₓ꜀ category꜀court꜀:·
·⁚lookup:·

⁖ᴎᑕᗩᘜᕱ:·ᖻᔕᖘᔓᘜ⁖:·˙ᴎᑕᗩᘜᕱ:·

·⁚origin…language:·Darnassian·⁚·˙origin…language:·
·⁚font:·Rellanic·⁚·˙font:·
·⁚pronunciation:·Falah·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Balance·⁚·˙non…ostensive:·
·⁚ostensive:·Central cord bears equal hanging weights while neither side sinks below the
↪ otherx·⁚·˙ostensive:·
·⁚grimchain:·ᐱᔭ⊛ᔑᕮᘐᓀᛦᑎᚼ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꜀ᑭ꜀ category꜀court꜀:·
·⁚lookup:·

⁖ᘓᘐᔓᒣ:·ᖱᘓᘐᔓᒣ⁖:·˙ᘓᘐᔓᒣ:·

·⁚origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁚font:·Mage Script·⁚·˙font:·
·⁚pronunciation:·Pee+eh+teh·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Sacred·⁚·˙non…ostensive:·
·⁚ostensive:·Flame burns inside the marked threshold::and common tools remain outside its
↪ boundary::Guarded ground holds no ordinary use within the ringᑭ·⁚·˙ostensive:·
·⁚grimchain:·ᵴℂϴᔓ❀ᗐஓᔐᗢᔓᔑ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꜀ᔑ꜀ category꜀court꜀:·
·⁚lookup:·

⁖ᔐᚻᒪ:·ᚼᚺᒪᚥᚴᛇᛇᚿ⁖:·˙ᔐᚻᒪ:·

·⁚origin…language:·Mando’a·⁚·˙origin…language:·
·⁚font:·Mandalorian·⁚·˙font:·
·⁚pronunciation:·kahn+DO+see·⁚·˙pronunciation:·
·⁚cadence:·SeeD Body·⁚·˙cadence:·
·⁚non…ostensive:·Nice one! Wicked! Well done!·⁚·˙non…ostensive:·
·⁚ostensive:·Difficult seam lies closed beneath final fastening::Clean join bears strain without
↪ gap::and one hard clap rings across the finished workᔑ·⁚·˙ostensive:·
·⁚grimchain:·Ɏ⧗ᨒᚒᚮᚸᚱᚣᚴᛐ▽·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꜀ᒪ꜀ category꜀court꜀:·
·⁚lookup:·

·⁚origin…language:·Sindarin·⁚·˙origin…language:·
·⁚font:·Tengwar Sindarin·⁚·˙font:·
·⁚pronunciation:·gwesta·⁚·˙pronunciation:·

·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·to witness·⁖.·˙non…ostensive:·
·⁖ostensive:·Open eye faces the burning mark⁚⁚and unobstructed light crosses the space between themɹ·⁖.·˙ostensive:·
·⁖grimchain:·ᔐᑭᐁ⊛ɹᖇzسᨐ৭ৗᖇ♂·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○ᵢɹ꜀ category○–○ᵢcourt꜀:·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·born²·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·[stead]fast·⁖.·˙non…ostensive:·
·⁖ostensive:·Standing stone bears lateral pressure without shifting while rooted base holds its place against the same loadɹ·⁖.·˙ostensive:·
·⁖grimchain:·I⇝ᛯ♂ʃ◌⩒ৰঘ৪°⊕0ر·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○ᵢ▽꜀ category○–○ᵢcourt꜀:·
·⁖lookup:·

⁖ঌ৬৬ᢉ:·ᨐ৫ᘮᓫ৬৬৫:⁖.·˙ঌ৬৬ᢉ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Hee+kar+ree·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Hope·⁖.·˙non…ostensive:·
·⁖ostensive:·Dark road remains unseen beyond the hill⁚⁚and one lamp keeps a steady flame toward the unopened horizon▽·⁖.·˙ostensive:·
·⁖grimchain:·ঌ❀ᘓz1⧐ᗢঌᛕᛘᛉ৭ᛖI☿·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○ᵢz꜀ category○–○ᵢcourt꜀:·
·⁖lookup:·

⁖ᚠᗞᚠᛖ:·ᚠᗢᚎᚴᚠᵋ⁖.·˙ᚠᗞᚠᛖ:·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·keshka·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·persist·⁖.·˙non…ostensive:·
·⁖ostensive:·Standing stone bears long pressure without surrendering its place as the loaded surface keeps the same bearing under continuing forcez·⁖.·˙ostensive:·
·⁖grimchain:·I▽ᨐᨒᨒᛕᔐرɹƆΘᨐΘ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol○–○ᵢᑫ꜀ category○–○ᵢcourt꜀:·
·⁖lookup:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·îdh·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·rest::repose::peace::tranquility·⁖.·˙non…ostensive:·
·⁖ostensive:·Muscles loosen beneath steady breath::Limbs lie without load and the chamber carries no demand for motionɋ·⁖.·˙ostensive:·
·⁖grimchain:·⊛ᚏ𐤉⨹ᗤ⊕ᚹƨƨ❋ꓩɻ♂·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꩜꩜ᵬ꩝ category꩜꩜ᶜourt꩝:·
·⁖lookup:·

⁖.ᔑᖻᖻ:·ᒪᒪᒪ:.·˙ᔑᖻᖻ:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·oor·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·silence·⁖.·˙non…ostensive:·
·⁖ostensive:·Shared air holds no voiced vibration::Chamber gives no answering sound between wall and wallᵬ·⁖.·˙ostensive:·
·⁖grimchain:·㊧▽ᖘ⇟ᖚ❁ᗅ▫▽ᙡɻ♀ᚻᒉ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꩜꩜ᵬ꩝ category꩜꩜ᶜourt꩝:·
·⁖lookup:·

⁖.ᚼᘎᘏᒣ:·ᨒᨐᨒᨐᗍᚼᨒ:.·˙ᚼᘎᘏᒣ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Vih+ghil+uhm·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·Sleep·⁖.·˙non…ostensive:·
·⁖ostensive:·Closed eyes remain still while limbs lie loose and breath keeps a slow inward rhythm beneath the darkened roomᵬ·⁖.·˙ostensive:·
·⁖grimchain:·ᗤ⊛ᘐ☆ↄᚴᚲᖮ⊙∀ᒧᒪᒡ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol꩜꩜ᶍ꩝ category꩜꩜ᶜourt꩝:·
·⁖lookup:·

⁖.ᔑᖻᖻ:·ᛋᛇ:.·˙ᔑᖻᖻ:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·AH+nee·⁖.·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖.·˙cadence:·
·⁖non…ostensive:·total::complete::whole·⁖.·˙non…ostensive:·
·⁖ostensive:·Circle closes upon its starting point::Every segment of the rim is present and no gap interrupts the boundaryφ·⁖.·˙ostensive:·

·⁖grimchain:·ᚸ⊛ᚹᛋᛏᚺᛖᛞᛟᛅᛕᚲᚳᛢᛝ⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol⟜⟜𝆑 category⟜⟜axiomyr:·
·⁖lookup:·

·:ᔑᓯᐠᐢᐣ':·ᑕᑐᐢᐣ'ᓄ:·˙ᔑᓯᐠᐢᐣ':·

·⁖origin…language:·Dovahzul·⁖·˙origin…language:·
·⁖font:·Dovahkiin·⁖·˙font:·
·⁖pronunciation:·lay-n·⁖·˙pronunciation:·
·⁖cadence:·SeeD Body·⁖·˙cadence:·
·⁖non…ostensive:·World⁖Mundus⁖Universe⁖Everything·⁖·˙non…ostensive:·
·⁖ostensive:·Water flow⁖breath does rise⁖fire float⁖earth under sky⁖touch with feel⁖sight and
↪ sound⁖feet lain⁖solid ground∮·⁖·˙ostensive:·
·⁖grimchain:·ᚸᛟᛞᛏᚺᛖᛞᛟᛅᛕᚲᚳᛢᛝ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖·˙sydonic…magicae…lexicon·∴

·:·:·:

·⁚ostensive⁚·Black mirror holds a shadowed vein⁘Still reflection remains in hidden glass⁘Distant fire
↪ burns without crossing the space between⁘Time turns its face toward the waiting
↪ void··⁚¸·˙ostensive⁚·
·⁚grimchain⁚·ᚼ⊕⁜✳ᛋ⬡ᑒᛇᒿ𑁍ᛤᛎ⁚¸·˙grimchain⁚·
·⁚¸·˙lookup·˙
·⁚¸·˙glyphs·˙
·⁚glyph symbol⟜⟜𖤐 category⟜⟜living⟡cadence⁑⁚·
·⁚lookup⁚·

·word⁚·𖤐·⁚¸·˙word⁚·

·⁚origin…language⁚·Llumena Milenii·⁚¸·˙origin…language⁚·
·⁚font⁚·Noto Serif Tibetan·⁚¸·˙font⁚·
·⁚pronunciation⁚·kan⟡tuh⟡LAY⟡shuhn·⁚¸·˙pronunciation⁚·
·⁚cadence⁚·Cantillation·⁚¸·˙cadence⁚·
·⁚non…ostensive⁚·Cantillation 𖤐 is the Chant active communication makes when God and Goddess Listen⁘It
↪ is an unspoken Living Cadence·⁚¸·˙non…ostensive⁚·
·⁚ostensive⁚·Distant chant enters open hearing⁘Earth trembles beneath the listening body⁘Above answers
↪ below⁘Below answers above⁘Creation sounds through the inward flow𖤐·⁚¸·˙ostensive⁚·
·⁚grimchain⁚·ᦟᦗ⁜ꠥ𑁍ᦟ⁜⊕ᛦᛋ⊛·⁚¸·˙grimchain⁚·
·⁚¸·˙lookup·˙
·⁚¸·˙glyphs·˙
·⁚glyph symbol⟜⟜𖤓 category⟜⟜living⟡cadence⁑⁚·
·⁚lookup⁚·

·⁖word⁚·𖤓·⁚¸·˙word⁚·

·⁚origin…language⁚·Llumena Milenii·⁚¸·˙origin…language⁚·
·⁚font⁚·DejaVu Sans·⁚¸·˙font⁚·
·⁚pronunciation⁚·PROS⟡uh⟡dee·⁚¸·˙pronunciation⁚·
·⁚cadence⁚·Prosody छन्दस्·⁚¸·˙cadence⁚·
·⁚non…ostensive⁚·Prosody छन्दस् is the Song of Creation⁘ambient throughout the whole utterance⁘It is not
↪ a written glyph⁘spoken word⁘visible operator⁘or utterance token·⁚¸·˙non…ostensive⁚·
·⁚ostensive⁚·Presence fills the open field⁘Form gathers within living presence⁘Silent whisper moves
↪ through the living current⁘Wellspring remains beneath the forming song𖤓·⁚¸·˙ostensive⁚·
·⁚grimchain⁚·ᛋᚻᛤᛔᛏᛋᑫᛩᚦᛚᛇ·⁚¸·˙grimchain⁚·
·⁚¸·˙lookup·˙
·⁚¸·˙glyphs·˙
⁚··⁚

⁘⁘⁘

·⁖origin…language꞉·Ge'ez·⁖˳·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖˳·˙font꞉·
·⁖pronunciation꞉·SUH⟊gah·⁖˳·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖˳·˙cadence꞉·
·⁖non…ostensive꞉·Flesh⁖⁖the soft bodily substance of humans or animals·⁖˳·˙non…ostensive꞉·
·⁖ostensive꞉·Solar heat warms bare skin⁖⁖and sharp thorn bites the same flesh⁖⁖and tactile boundary
↪ carries both contacts at once♀·⁖˳·˙ostensive꞉·
·⁖grimchain꞉·▽ᵑ₩ᑫ⊛ᵬᐁ❀ᘓᵞ♀▫➶·⁖˳·˙grimchain꞉·
·⁖˳·˙lookup·˙
·⁖˳·˙glyphs·˙
·⁖glyph symbol꞉○₋₀₊ⴳ꞉ category꞉○₋₀₊emission꞉꞉·
·⁖lookup office꞉○₋₀₊substrate꞉ enoch꞉○₋₀₊ᗰ꞉꞉·

˖word꞉·ⲁⲗⲥ˖˳·˙word꞉·

·⁖origin…language꞉·Coptic·⁖˳·˙origin…language꞉·
·⁖font꞉·Noto Sans Coptic·⁖˳·˙font꞉·
·⁖pronunciation꞉·LAHS·⁖˳·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖˳·˙cadence꞉·
·⁖non…ostensive꞉·Tongue⁖⁖also language or speech·⁖˳·˙non…ostensive꞉·
·⁖ostensive꞉·Tongue meets breath at the mouth⁖⁖and speech has a living passage into
↪ hearingⴳ·⁖˳·˙ostensive꞉·
·⁖grimchain꞉·ⲔⲰ⬡❂✳ⵔⵒⅡᚿᛏᑭᵴᗐ·⁖˳·˙grimchain꞉·
·⁖˳·˙lookup·˙
·⁖˳·˙glyphs·˙
·⁖glyph symbol꞉○₋₀₊ϧ꞉ category꞉○₋₀₊emission꞉꞉·
·⁖lookup office꞉○₋₀₊substrate꞉ enoch꞉○₋₀₊ᗰ꞉꞉·

˖word꞉·ᔑᑌᓀΣ˖˳·˙word꞉·

·⁖origin…language꞉·Aghwan·⁖˳·˙origin…language꞉·
·⁖font꞉·Noto Sans Caucasian Albanian·⁖˳·˙font꞉·
·⁖pronunciation꞉·HUEWK·⁖˳·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖˳·˙cadence꞉·
·⁖non…ostensive꞉·Heart⁖· mind·⁖˳·˙non…ostensive꞉·
·⁖ostensive꞉·Unseen summit remains held within the inward chamber⁖⁖and each step keeps its bearing
↪ toward what sight has not yet reachedℏ·⁖˳·˙ostensive꞉·
·⁖grimchain꞉·ᡒᡚᡗᡲᡱᡮᡯᡰᡳᡴᡵᡶᡷᡸ᡹᡺᡻᡼᡽᡾᡿ᢀᢁᢂᢃᢄᢅᢆᢇᢈᢉᢊᢋᢌᢍᢎᢏᢐᢑᢒᢓᢔᢕᢖᢗᢘᢙᢚᢛᢜᢝᢞᢟᢠᢡᢢᢣᢤᢥᢦᢧᢨᢩᢪ᢫᢬᢭᢮᢯ᢰᢱᢲᢳᢴᢵᢶᢷᢸᢹᢺᢻᢼᢽᢾᢿᣀᣁᣂᣃᣄᣅᣆᣇᣈᣉᣊᣋᣌᣍᣎᣏᣐᣑᣒᣓᣔᣕᣖᣗᣘᣙᣚᣛᣜᣝᣞᣟᣠᣡᣢᣣᣤᣥᣦᣧᣨᣩᣪᣫᣬᣭᣮᣯᣰᣱᣲᣳᣴᣵ᣶᣷᣸᣹᣺᣻᣼᣽᣾᣿ᤀᤁᤂᤃᤄᤅᤆᤇᤈᤉᤊᤋᤌᤍᤎᤏᤐᤑᤒᤓᤔᤕᤖᤗᤘᤙᤚᤛᤜᤝᤞ᤟ᤠᤡᤢᤣᤤᤥᤦᤧᤨᤩᤪᤫ᤬᤭᤮᤯ᤰᤱᤲᤳᤴᤵᤶᤷᤸ᤻᤹᤺᤼᤽᤾᤿᥀᥁᥂᥃᥄᥅᥆᥇᥈᥉ᥐ·⁖˳·˙grimchain꞉·
·⁖˳·˙lookup·˙
·⁖˳·˙glyphs·˙
·⁖glyph symbol꞉○₋₀₊ⴲ꞉ category꞉○₋₀₊emission꞉꞉·
·⁖lookup office꞉○₋₀₊substrate꞉ enoch꞉○₋₀₊ᗰ꞉꞉·

˖word꞉·ᎎᎮ˖˳·˙word꞉·

·⁖origin…language꞉·Ge'ez·⁖˳·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖˳·˙font꞉·
·⁖pronunciation꞉·UHD·⁖˳·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖˳·˙cadence꞉·
·⁖non…ostensive꞉·Hand⁖⁖the terminal part of the arm used for grasping·⁖˳·˙non…ostensive꞉·
·⁖ostensive꞉·Bone⁖⁖tendon⁖⁖and palm hold one continuous grip⁖⁖and tool weight presses through the joined
↪ frameⴲ·⁖˳·˙ostensive꞉·
·⁖grimchain꞉·ᚋᘐᡒᡚᡗᡲᡱᡮᡯᡰᡳᡴᡵᡶᡷᡸ᡹᡺᡻᡼᡽᡾᡿ᢀᢁᢂᢃᢄᢅᢆᢇᢈᢉᢊᢋᢌᢍᢎᢏᢐᢑᢒᢓᢔᢕᢖᢗᢘᢙᢚᢛᢜᢝᢞᢟᢠᢡᢢᢣᢤᢥᢦᢧᢨᢩᢪ᢫᢬᢭᢮᢯ᢰᢱᢲᢳᢴᢵᢶᢷᢸᢹᢺᢻᢼᢽᢾᢿᣀᣁᣂᣃᣄᣅᣆᣇᣈᣉᣊᣋᣌᣍᣎᣏᣐᣑᣒᣓᣔᣕᣖᣗᣘᣙᣚᣛᣜᣝᣞᣟᣠᣡᣢᣣᣤᣥᣦᣧᣨᣩᣪᣫᣬᣭᣮᣯᣰᣱᣲᣳᣴᣵ᣶᣷᣸᣹᣺᣻᣼᣽᣾᣿ᤀᤁᤂᤃᤄᤅᤆᤇᤈᤉᤊᤋᤌᤍᤎᤏᤐᤑᤒᤓᤔᤕᤖᤗᤘᤙᤚᤛᤜᤝᤞ᤟ᤠᤡᤢᤣᤤᤥᤦᤧᤨᤩᤪᤫ᤬᤭᤮᤯ᤰᤱᤲᤳᤴᤵᤶᤷᤸ᤻᤹᤺᤼᤽᤾᤿᥀᥁᥂᥃᥄᥅᥆᥇᥈᥉ᥐ·⁖˳·˙grimchain꞉·
·⁖˳·˙lookup·˙

·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐑈꩜ category꩜꩜emission꩜⁚·
·⁚lookup office꩜꩜substrate꩜ enoch꩜꩜𒀭꩜⁚·

·word⁚·𒁹·⁚·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·NGESH+toog·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Ear·˙ the organ of hearing··also understanding or attentive
↪ intelligence·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Airborne vibration enters the ear··and folded cartilage and inner hollow carry the same
↪ tremor inward𐑈·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤒𐤉𐤏𐤍𐤋𐤅𐤎𐤌𐤉𐤏𐤒𐤓·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐤓꩜ category꩜꩜emission꩜⁚·
·⁚lookup office꩜꩜substrate꩜ enoch꩜꩜𒀭꩜⁚·

·word⁚·وسدم·⁚·˙word⁚·

·⁚origin…language⁚·Avestan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Avestan·⁚·˙font⁚·
·⁚pronunciation⁚·PAHD·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Foot··the terminal part of the leg·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Sole grips the rising stone··heel bears downward weight··and toes lock against the rock
↪ face𐤓·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤉𐤈𐤊𐤇𐤕𐤖𐤋𐤐𐤒𐤂𐤌𐤅𐤀·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐤂꩜ category꩜꩜emission꩜⁚·
·⁚lookup office꩜꩜substrate꩜ enoch꩜꩜𒀭꩜⁚·

·word⁚·𒄑𒌆·⁚·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·KAS+poo·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Silver·˙ silver used as metal··wealth··or payment·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Silver rests on one balance pan··and measured counterweight holds level against the same
↪ mass𐤂·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤀𐤉𐤑𐤏𐤔𐤍𐤊𐤇𐤂𐤓𐤃𐤅·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐤀꩜ category꩜꩜emission꩜⁚·
·⁚lookup office꩜꩜substrate꩜ enoch꩜꩜𒀭꩜⁚·

·word⁚·𐤊𐤓𐤋·⁚·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚·˙font⁚·

·⁚pronunciation:·HUWWATU·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Country⁖⁖nation⁖⁖land (geographical+political entity)·⁚.·˙non…ostensive:·
·⁚ostensive:·Land holds the reach of command beneath one name⁖⁖from the first dwelling to the farthest
↪ boundary stone⊙·⁚.·˙ostensive:·
·⁚grimchain:·🜲ϟ⧗✵ℂᒧ♧ϒ⍦ᑎ⊃ᛘᛅ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ₑℂ» category◦–◦ₑemission»⁚·
·⁚lookup office◦–◦ₑsubstrate» enoch◦–◦ₑ♧»⁚·

₍word⁚·⊶₎⁚.·˙word⁚·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·KHAT·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Body·⁚.·˙non…ostensive:·
·⁚ostensive:·One skin gathers the whole living body and keeps its inward life from spilling into the
↪ world℃·⁚.·˙ostensive:·
·⁚grimchain:·𓂀ᒧⱫΘ𓆑X𓆑⌾ᕮᒧ𓅱ᑫᑐ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ₑϒ» category◦–◦ₑparliament»⁚·
·⁚lookup office◦–◦ₑsubstrate» enoch◦–◦ₑ♧»⁚·

₍word⁚·ᴏɯɪ₎⁚.·˙word⁚·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·OH+mee·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Clay or mud·⁚.·˙non…ostensive:·
·⁚ostensive:·Clay bears the impressed mark⁖⁖and silent surface carries the cut without the pressing
↪ handϒ·⁚.·˙ostensive:·
·⁚grimchain:·𓆑⊕✳ᒧᑭ𐦂ᑐ⍇❁❄·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ₑ♉» category◦–◦ₑparliament»⁚·
·⁚lookup office◦–◦ₑsubstrate» enoch◦–◦ₑ♧»⁚·

₍word⁚·ልሳን₎⁚.·˙word⁚·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·luh+SAHN·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Tongue·˙ the muscular organ of the mouth⁖⁖also speech or language·⁚.·˙non…ostensive:·
·⁚ostensive:·Opposed claims move through spoken words until one phrasing remains⁖⁖and contention falls
↪ quiet around the accepted terms♉·⁚.·˙ostensive:·
·⁚grimchain:·⍇✳❂ɻᛊᏂᚼ✳Ⓞₒᛦ▽ᒡ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol◦–◦𒉼 category◦–◦parliament⁖·
·⁚lookup office◦–◦substrate enoch◦–◦♋⁖·

⁖word:·𐎍𐎗𐎢⁖·˙word:·

·⁚origin…language:·Hittite·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·TEHKAN·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Earth·⁚·˙non…ostensive:·
·⁚ostensive:·Firm ground receives foot in darkness and daylight with the same resistance⁖ buried stone
↪ keeps the same weight when no eye is present𒉼·⁚·˙ostensive:·
·⁚grimchain:·ᜓℂᗅᚸ⊖ᚿᚐᛈᜓ⊕ᛘᛯᚋ·⁚·˙grimchain:·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol◦–◦♋ category◦–◦parliament⁖·
·⁚lookup office◦–◦substrate enoch◦–◦♋⁖·

⁖word:·𓂀⁖·˙word:·

·⁚origin…language:·Egyptian·⁚·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚·˙font:·
·⁚pronunciation:·YEE+rat·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Eye·⁚·˙non…ostensive:·
·⁚ostensive:·Eye holds the distant mark within sight::and hand or missile carries direction toward a
↪ place already held by sight♋·⁚·˙ostensive:·
·⁚grimchain:·∀♋ᛕᗾ𓂸𓆑ᚋᛃᓬᛋᚿᛝ·⁚·˙grimchain:·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol◦–◦♈ category◦–◦parliament⁖·
·⁚lookup office◦–◦substrate enoch◦–◦♋⁖·

⁖word:·ᚼᚾᛏᛁ⁖·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·ANDI·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Breath·⁚·˙non…ostensive:·
·⁚ostensive:·Chest::throat::and open air share moving breath::and living flesh rises against the same
↪ flow♈·⁚·˙ostensive:·
·⁚grimchain:·᛻ᛕᛝ✡ᛯ᛬ᚼᚱᚳᚸᛏ◎·⁚·˙grimchain:·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol◦–◦♍ category◦–◦parliament⁖·
·⁚lookup office◦–◦substrate enoch◦–◦♋⁖·

⁖word:·ﺻﺼ⁖·˙word:·

·⁚origin…language:·Old Uyghur·⁚·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚·˙font:·
·⁚pronunciation:·bEUz·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·

·⁖non…ostensive:·cloth⁖· (sewing) flaxen cloth⁖· (sewing::clothing) cotton clothing⁖· (knitting::rare)
↪ weave::braid·⁚·˙non…ostensive:·
·⁖ostensive:·Crossing threads hold one another::and cloth bears weight through the strength shared
↪ between them𐋡·⁚·˙ostensive:·
·⁖grimchain:·ᑭ𐋏ǂ𐄷ᔕ⍋ᛉ◎ᚸᚷᚱⴱϕ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉𐤒» category꞉꞉parliament»⁚·
·⁖lookup office꞉꞉substrate» enoch꞉꞉𐤃»⁚·

·word:·ᛚ⁖·˙word:·

·⁖origin…language:·Ugaritic·⁚·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁚·˙font:·
·⁖pronunciation:·GOO·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·Voice·⁚·˙non…ostensive:·
·⁖ostensive:·Breath carries ordered vibration beyond the mouth::and the ear receives the same sounding
↪ air𐤒·⁚·˙ostensive:·
·⁖grimchain:·𐋏𐋠ᑭᚧᛇᚣA𐋏Ↄ⊕ℊ𐋏·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉𐋡» category꞉꞉parliament»⁚·
·⁖lookup office꞉꞉substrate» enoch꞉꞉𐤃»⁚·

·word:·ⲥⲗⲍⲓ⁖·˙word:·

·⁖origin…language:·Coptic·⁚·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁚·˙font:·
·⁖pronunciation:·SAH⟊jee·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·Word::account::or telling·⁚·˙non…ostensive:·
·⁖ostensive:·One utterance names the gate open⁖· another names it shut::and the same doorway cannot
↪ carry both conditions at once𐋡·⁚·˙ostensive:·
·⁖grimchain:·ᚼϕ❁ᛉ𐋏Ↄᚦᚏᗅᔕᔐᛊ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉𐤗» category꞉꞉parliament»⁚·
·⁖lookup office꞉꞉substrate» enoch꞉꞉𐤃»⁚·

·word:·ᛓᚣᛯᛖᚣ⁖·˙word:·

·⁖origin…language:·Gothic·⁚·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁚·˙font:·
·⁖pronunciation:·BLO⟊MA·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·(hapax legomenon) lily (flower in the genus Lilium)·⁚·˙non…ostensive:·
·⁖ostensive:·Bud and flower inhabit the same living body::and hidden petals belong to the green walls
↪ that contain them𐤗·⁚·˙ostensive:·
·⁖grimchain:·ᚼⴰ⁕ᵷᛉᵑ♀ǂ𐋡ᚦᚺᛯ≍·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉ᚱ» category꞉꞉parliament»⁚·
·⁖lookup office꞉꞉substrate» enoch꞉꞉𐤃»⁚·

˸word꞉·𒉿𒄭𒃻𒊹˸.·˙word꞉·

·˙origin…language꞉·Akkadian·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Cuneiform·˙.·˙font꞉·
·˙pronunciation꞉·ah+WAH+toom·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·word꞉꞉utterance꞉· news꞉꞉report꞉꞉message꞉· secret꞉꞉rumor꞉· wording꞉꞉text꞉꞉content꞉꞉terms (of an
↪ agreement)꞉· command꞉꞉order꞉꞉decision꞉· matter꞉꞉affair꞉꞉thing꞉꞉situation꞉꞉legal case꞉꞉legal
↪ transaction·˙.·˙non…ostensive꞉·
·˙ostensive꞉·Cut clay and sounded breath bear the same ordered word꞉꞉and fixed marks hold its form in
↪ silence𐎗·˙.·˙ostensive꞉·
·˙grimchain꞉·ϟᗡΛꓧ𐤋𓆉𐤔ᛉ𐤍ᛉ·˙.·˙grimchain꞉·
·˙.·˙lookup꞉·
·˙.·˙glyphs꞉·
·˙glyph symbol꞉○-○𐤂𒁹» category꞉○-○𐤂parliament»꞉·
·˙lookup office꞉○-○𐤂substrate» enoch꞉○-○𐤂𓃰»꞉·

˸word꞉·ϟᛤ𐤂˸.·˙word꞉·

·˙origin…language꞉·Coptic·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Coptic·˙.·˙font꞉·
·˙pronunciation꞉·KAH·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·Earth꞉꞉ground꞉꞉land·˙.·˙non…ostensive꞉·
·˙ostensive꞉·Earth receives every wandering step꞉꞉even where no road has been laid𒁹·˙.·˙ostensive꞉·
·˙grimchain꞉·ϒᛋᛤᚹᚺᛯᛋᛉ◎𐤔ᚹ𓆑·˙.·˙grimchain꞉·
·˙.·˙lookup꞉·
·˙.·˙glyphs꞉·
·˙glyph symbol꞉○-○𐤂𐤇» category꞉○-○𐤂parliament»꞉·
·˙lookup office꞉○-○𐤂substrate» enoch꞉○-○𐤂𓃰»꞉·

˸word꞉·ϟ𒁹𓂝˸.·˙word꞉·

·˙origin…language꞉·Sumerian·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Cuneiform·˙.·˙font꞉·
·˙pronunciation꞉·AH+shah·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·Field·˙.·˙non…ostensive꞉·
·˙ostensive꞉·Boundary stone and field hold the changed place together꞉꞉and soil bears each position
↪ without departing from the ground𐤇·˙.·˙ostensive꞉·
·˙grimchain꞉·ᑭᚹᘔ𓃰❂Ⱉₒᛅᚣ♂⊕◣≫ᑭ·˙.·˙grimchain꞉·
·˙.·˙lookup꞉·
·˙.·˙glyphs꞉·
·˙glyph symbol꞉○-○𐤂◎» category꞉○-○𐤂goetic»꞉·
·˙lookup office꞉○-○𐤂substrate» enoch꞉○-○𐤂𓃰»꞉·

˸word꞉·ᎶᏫ˸.·˙word꞉·

·˙origin…language꞉·Ge'ez·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Ethiopic·˙.·˙font꞉·
·˙pronunciation꞉·FUH+ray·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·Fruit꞉꞉seed꞉꞉or offspring·˙.·˙non…ostensive꞉·

·⁖ostensive⁚·Seed carries an unseen plant inside a body small enough to disappear beneath the
↪ soil◎·⁖.·˙ostensive⁚·
·⁖grimchain⁚·⊖ᛉᑊᑫᛞᚼ⊛ᛘ♁ᛋᚴᔓᔕ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦⬡» category○–◦goetic»⁖·
·⁖lookup office○–◦substrate» enoch○–◦♁»⁖·

·word⁚·╍▟▟▟·⁖.·˙word⁚·

·⁖origin…language⁚·Old Irish·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Ogham·⁖.·˙font⁚·
·⁖pronunciation⁚·OOR·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Heather⁖⁖earth⁖⁖or clay·⁘ an Ogham letter+name with these traditional
↪ senses·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Clay bears the impressed sign without the reed⁖⁖and buried earth keeps what open air would
↪ lose⬡·⁖.·˙ostensive⁚·
·⁖grimchain⁚·⊙ᘜ֍ᚻ‹ᛳ▵♁⊖ᚸᛊ▇ᑭ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦✡» category○–◦goetic»⁖·
·⁖lookup office○–◦substrate» enoch○–◦♁»⁖·

·word⁚·ᚹᛁᚦᚱ·⁖.·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖.·˙font⁚·
·⁖pronunciation⁚·VITH+r·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Wood⁖⁖tree⁖⁖forest·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Wood and flame inhabit the same burning⁖⁖and the hearth feeds upon the dark grain of the
↪ timber✡·⁖.·˙ostensive⁚·
·⁖grimchain⁚·♉♌ᛤᛉᚴ☿ᚸ⛢ᛞᘐ🜨·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦✫» category○–◦goetic»⁖·
·⁖lookup office○–◦substrate» enoch○–◦♁»⁖·

·word⁚·ᚺᚾ·⁖.·˙word⁚·

·⁖origin…language⁚·Akkadian·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Cuneiform·⁖.·˙font⁚·
·⁖pronunciation⁚·MOO·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Water⁖⁖waters·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Water takes the vessel's shape⁖⁖and no vessel owns the nature of the water it
↪ holds✫·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᚱ⊕ᛦ⊕ᘐᖷᨓ᭄⊕ᛋᚨᛞ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦❂» category○–◦goetic»⁖·
·⁖lookup office○–◦substrate» enoch○–◦♁»⁖·

⁖word:·🡸🡰🡱⁖.·˙word:·

·⁖origin…language:·Hittite·⁖.·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖.·˙font:·
·⁖pronunciation:·PURUT·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Soil⁖⁖mud⁖⁖earth⁖· mud plaster·⁖.·˙non…ostensive:·
·⁖ostensive:·Soft mud receives the handprint⁖⁖and dried earth stands in the wall with the worked shape
↪ held through its body❂·⁖.·˙ostensive:·
·⁖grimchain:·𐤄𐤏𐤌𐤊𐤅𐤙𐤑𐤉𐤔𐤇𐤋𐤍𐤃·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜𓃭꩜ category꩜꩜ᵍoeticꩻ⁖·
·⁖lookup office꩜꩜ˢubstrateꩻ enoch꩜꩜🜁ꩻ⁖·

⁖word:·✷⁖.·˙word:·

·⁖origin…language:·Akkadian·⁖.·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖.·˙font:·
·⁖pronunciation:·shah⟊MOO·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Sky⁖⁖heaven⁖⁖the upper celestial region·⁖.·˙non…ostensive:·
·⁖ostensive:·Sky receives whatever rises into it and never shows the wall that contains its
↪ height𓃭·⁖.·˙ostensive:·
·⁖grimchain:·𐓣𐓂𐓆𐓐𐓏𐓄𐓊𐓃𐓉𐓍𐓅𐓋𐓌𐓇·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵎ꩜ ꩜ categoryꩻ꩜꩜ᵍoeticꩻ⁖·
·⁖lookup office꩜꩜ˢubstrateꩻ enoch꩜꩜🜁ꩻ⁖·

⁖word:·ᏚᎤᏀᏁᎤᏀᏆ⁖.·˙word:·

·⁖origin…language:·Aghwan·⁖.·˙origin…language:·
·⁖font:·Noto Sans Caucasian Albanian·⁖.·˙font:·
·⁖pronunciation:·GHUYUN·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Life·⁖.·˙non…ostensive:·
·⁖ostensive:·Living breath stirs through flesh⁖⁖and warmth and motion inhabit the same breathing
↪ body✷·⁖.·˙ostensive:·
·⁖grimchain:·𐓝𐓒𐓚𐓄𐓅𐓟𐓠𐓞𐓘𐓫𐓯·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵎ⧗ꩻ ꩜ categoryꩻ꩜꩜ᵍoeticꩻ⁖·
·⁖lookup office꩜꩜ˢubstrateꩻ enoch꩜꩜🜁ꩻ⁖·

⁖word:·ϩнт⁖.·˙word:·

·⁖origin…language:·Coptic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁖.·˙font:·
·⁖pronunciation:·HAYT·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Heart·⁖.·˙non…ostensive:·
·⁖ostensive:·Warm pulse deepens beneath the breast⁖⁖and held nearness gathers the inward chamber toward
↪ continued contact⧗·⁖.·˙ostensive:·

·⁖grimchain⁖·ᛁ♀Ⲭ⊕ⲉ𐌗ⲉᑭⴵ0ᚲ𓃭🕭·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥⊛𝅘𝅥 category◦–◦goetic⁖·
·⁖lookup office◦–◦substrate𝅘𝅥 enoch◦–◦𓃰⁖·

⁖word⁖·ᴏγᴏєɪɴ⁖·⁖·˙word⁖·

·⁖origin…language⁖·Coptic·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Coptic·⁖·˙font⁖·
·⁖pronunciation⁖·oo+WAYN·⁖·˙pronunciation⁖·
·⁖cadence⁖·Substrate·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Light::brightness::illumination·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Light lays every dark edge upon the ground wherever something stands between brightness
↪ and earth⊛·⁖·˙ostensive⁖·
·⁖grimchain⁖·⛦◎ⲭ0ؤعؤ⇞♂ᛯHΔعر·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥❄𝅘𝅥 category◦–◦goetic⁖·
·⁖lookup office◦–◦substrate𝅘𝅥 enoch◦–◦𓃰⁖·

⁖word⁖·소릭⁖·⁖·˙word⁖·

·⁖origin…language⁖·Middle Korean·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans KR·⁖·˙font⁖·
·⁖pronunciation⁖·soluhj·⁖·˙pronunciation⁖·
·⁖cadence⁖·Substrate·⁖·˙cadence⁖·
·⁖non…ostensive⁖·sound·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Air carries one vibration through many ears::and separate mouths share the same sounding
↪ field❄·⁖·˙ostensive⁖·
·⁖grimchain⁖·ᚿسᛞ𝅘𝅥ᛯᚴᚠᚿᛖᛘᛘ◌◌A◎·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥❁𝅘𝅥 category◦–◦goetic⁖·
·⁖lookup office◦–◦substrate𝅘𝅥 enoch◦–◦𓃰⁖·

⁖word⁖·बल⁖·⁖·˙word⁖·

·⁖origin…language⁖·Sanskrit·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Devanagari·⁖·˙font⁖·
·⁖pronunciation⁖·BUH+luh·⁖·˙pronunciation⁖·
·⁖cadence⁖·Substrate·⁖·˙cadence⁖·
·⁖non…ostensive⁖·power::strength::might::vigour::force::validity⁖· force or power of articulation⁖· force
↪ considered as a sixth organ of action⁖· force personified as one of the वश्विेदेवाः (viśvedevāḥ)⁖·
↪ power of::expertness in (+locative)⁖· stoutness::bulkiness⁖· (in the plural) military
↪ force::troops::an army⁖· shape (L)⁖· body (L)⁖· penis (L)⁖· gum (L)⁖· blood (L)⁖· young shoot (L)⁖· bone
↪ (L)·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Strength meets strength in the same body of struggle::and opposing force prevents either
↪ side from passing untouched❁·⁖·˙ostensive⁖·
·⁖grimchain⁖·ϒᛯ1⊙ᛘᛉ𐌅ᚴᚠᛯᛦᛟᛞ⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥⬡𝅘𝅥 category◦–◦goetic⁖·
·⁖lookup office◦–◦substrate𝅘𝅥 enoch◦–◦𓃰⁖·

˖word꞉·ነፋስ˖·.˙word꞉·

·˙origin…language꞉·Ge'ez·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Ethiopic·˙.·˙font꞉·
·˙pronunciation꞉·nah+FAHS·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·Wind꞉˙ moving air·˙.·˙non…ostensive꞉·
·˙ostensive꞉·Air passes broad and slow through open grass without reed꞉:leaf꞉:or hollow edge to give it
↪ sound⌬·˙.·˙ostensive꞉·
·˙grimchain꞉·⟴ᛎ∀◺ᛈ⊙ΙΙᛕ⏁ᛘᚧᑫᗡ·˙.·˙grimchain꞉·
·˙.·˙lookup꞉˙
·˙.·˙glyphs꞉˙
·˙glyph symbol○–○˓𓂋꞉ category○–○˓court꞉·˙
·˙lookup office○–○˓substrate꞉ enoch○–○˓𓃰꞉·

˖word꞉·𓎛˖·.˙word꞉·

·˙origin…language꞉·Egyptian·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Egyptian Hieroglyphs·˙.·˙font꞉·
·˙pronunciation꞉·KAH·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·Vital force꞉:life+power꞉:sustaining double of a person·˙.·˙non…ostensive꞉·
·˙ostensive꞉·First pulse gathers beneath unmarked flesh꞉:and breath opens the body into living
↪ motion𓂋꞉·˙.·˙ostensive꞉·
·˙grimchain꞉·ᛁᛞᛦIᛯZϛ‰✓⊕☆◺❋Ƀ·˙.·˙grimchain꞉·
·˙.·˙lookup꞉˙
·˙.·˙glyphs꞉˙
·˙glyph symbol○–○˓𓂋꞉ category○–○˓court꞉·˙
·˙lookup office○–○˓substrate꞉ enoch○–○˓𓃰꞉·

˖word꞉·ϧɯɯϣ˖·.˙word꞉·

·˙origin…language꞉·Avestan·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Avestan·˙.·˙font꞉·
·˙pronunciation꞉·MAH·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·Moon·˙.·˙non…ostensive꞉·
·˙ostensive꞉·Moon bears change and return in the same body꞉:and every face belongs to the same
↪ wandering light𓂋꞉·˙.·˙ostensive꞉·
·˙grimchain꞉·ᛘᚸᚹᛉΘᛯ▫ᚿᚲᛒᚴᚸ⊕·˙.·˙grimchain꞉·
·˙.·˙lookup꞉˙
·˙.·˙glyphs꞉˙
·˙glyph symbol○–○˓𓂋꞉ category○–○˓court꞉·˙
·˙lookup office○–○˓substrate꞉ enoch○–○˓𓃰꞉·

˖word꞉·𓆑𓇋˖·.˙word꞉·

·˙origin…language꞉·Egyptian·˙.·˙origin…language꞉·
·˙font꞉·Noto Sans Egyptian Hieroglyphs·˙.·˙font꞉·
·˙pronunciation꞉·YOOF·˙.·˙pronunciation꞉·
·˙cadence꞉·Substrate·˙.·˙cadence꞉·
·˙non…ostensive꞉·Flesh·˙.·˙non…ostensive꞉·
·˙ostensive꞉·Flesh holds the blow and the answering motion in one place of contact𓂋꞉·˙.·˙ostensive꞉·
·˙grimchain꞉·ᛉᚲᚦᚢᚠᛘᚷᚳᛞᛘᛘᛘ✓ᛜ𓂋꞉·˙.·˙grimchain꞉·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𒄭꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜substrate꩝ enoch꩜꩜𒀭꩝:·

:word:·𒌆:·⁚·˙word:·

·⁚origin…language:·Sumerian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·SEEG·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Wool·⁚·˙non…ostensive:·
·⁚ostensive:·Wool carries repeated crossings in its body::and woven crossings hold a visible design within the wool𒄭·⁚·˙ostensive:·
·⁚grimchain:·𐤅⊙𐤌𐤍𐤹𐤏𐤓𐤊𐤕ϒΛ⊙Π·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𒄩꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜substrate꩝ enoch꩜꩜𒀭꩝:·

:word:·𒂗:·⁚·˙word:·

·⁚origin…language:·Egyptian·⁚·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚·˙font:·
·⁚pronunciation:·TAH·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Land::earth::ground::country·⁚·˙non…ostensive:·
·⁚ostensive:·Earth bears what rises upon it and takes every weight back downward𓊪·⁚·˙ostensive:·
·⁚grimchain:·𐤅𐤌⊙Ж𓊪𓂋𓎛Ψ𐤒Н𓂋𓎛Я·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𒄬꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜substrate꩝ enoch꩜꩜𒀭꩝:·

:word:·𒋛:·⁚·˙word:·

·⁚origin…language:·Sumerian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·SHEH·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Barley·⁚·˙non…ostensive:·
·⁚ostensive:·Barley fills jar and storehouse grain by grain::and every gathered measure carries more grain within the same harvest𒄬·⁚·˙ostensive:·
·⁚grimchain:·𐤉ǂ⌬Ѫ✡Ↄℳ৬ৎ✻ϒ𐤅❋·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𒄮꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜substrate꩝ enoch꩜꩜𒀭꩝:·

:word:·𒁹𒌋:·⁚·˙word:·

·⁚origin…language:·Akkadian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·NEE+shoom·⁚·˙pronunciation:·

·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·life·⁚.·˙non…ostensive:·
·⁚ostensive:·First breath enters a living course already bearing every later turn within its span⁚·
↪ growth::wound::joining::and return all unfold through the same life﹏⁚.·˙ostensive:·
·⁚grimchain:·ᛕ𞣴≯A𞤢↗ᛯ𞤤Xᛇᛋᚡᚩ𐭥⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜𞤥 category꩜꩜court𞤣:·
·⁚lookup office꩜꩜substrate𞤣 enoch꩜꩜𞤃𞤣:·

·:word:·Ɥɔɔɲɔɔ:⁚.·˙word:·

·⁚origin…language:·Avestan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚.·˙font:·
·⁚pronunciation:·PAH⟊thah·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Path·⁚.·˙non…ostensive:·
·⁚ostensive:·Path receives departure and receives the returning foot again::holding both journeys in
↪ the same ground﹏⁚.·˙ostensive:·
·⁚grimchain:·𞤃ᚼXᚺᏀᚴΔ❄ϙᚹ'ᛏ'·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜𞤭 category꩜꩜court𞤣:·
·⁚lookup office꩜꩜substrate𞤣 enoch꩜꩜𞤃𞤣:·

·:word:·ɴɛƍ:⁚.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·NEH·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Oil·⁚.·˙non…ostensive:·
·⁚ostensive:·Oil receives two pigments into one flowing body⁚· stirred colour loses the seam between
↪ what entered separately﹏⁚.·˙ostensive:·
·⁚grimchain:·'ᵴΨIΔᵴ✲◌ΠᚾᚹƆƆ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜𞤬 category꩜꩜court𞤣:·
·⁚lookup office꩜꩜substrate𞤣 enoch꩜꩜𞤃𞤣:·

·:word:·ᚻᚢNI:⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·FANI·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Clay::mire·⁚.·˙non…ostensive:·
·⁚ostensive:·Clay gives beneath the hand and keeps what the hand leaves behind﹏⁚.·˙ostensive:·
·⁚grimchain:·ϒᚹᛏᗅᛋᚴᚾᗅ≋ᛈ✱ᚻᵴ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜𞤯 category꩜꩜court𞤣:·
·⁚lookup office꩜꩜substrate𞤣 enoch꩜꩜𞤃𞤣:·

˖word:·वस्त्र˖.·˙word:·

·⁖origin…language:·Sanskrit·⁖.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖.·˙font:·
·⁖pronunciation:·vuhs+truh·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·clothes::garment::dress·⁖ cloth·⁖ cover·⁖.·˙non…ostensive:·
·⁖ostensive:·Cloth bears the pull inside its weave::and an opened seam shows where the strain has
↪ outrun the crossing threadsﻉ·⁖.·˙ostensive:·
·⁖grimchain:·⊛ꓧꝑᗒᖻᖘᒧᔕ⬡ᴢᚹᖳ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ঌﻉ꒱ category꒰ঌ𐑈court꒱:·
·⁖lookup office꒰ঌ𐑈substrate꒱ enoch꒰ঌ𐑈꒱:·

˖word:·ግይል˖.·˙word:·

·⁖origin…language:·Ge'ez·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁖.·˙font:·
·⁖pronunciation:·KHAYL·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Power::force::strength::or energy·⁖.·˙non…ostensive:·
·⁖ostensive:·Stored force in bent wood and taut cord drives the arrow into the chosen mark·⁖ held
↪ strength carries the shot all the wayﻉ·⁖.·˙ostensive:·
·⁖grimchain:·ᗷᖷᐯᘐᕼᘔᐩᘐᘏ᐀ᐯ❄ᖇ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ঌl꒱ category꒰ঌ𐑈court꒱:·
·⁖lookup office꒰ঌ𐑈substrate꒱ enoch꒰ঌ𐑈꒱:·

˖word:·እክል˖.·˙word:·

·⁖origin…language:·Ge'ez·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁖.·˙font:·
·⁖pronunciation:·UHK+l·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Food::grain::or field crop·⁖.·˙non…ostensive:·
·⁖ostensive:·Dry grain rests in dark storage::and sealed kernels hold weight and nourishment inside the
↪ same gathered massI·⁖.·˙ostensive:·
·⁖grimchain:·ᘘᦓᘐᘏ⏁ᘜ⊙ᘐᘓﻉ❄ᘁ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ঌɬ꒱ category꒰ঌ𐑈court꒱:·
·⁖lookup office꒰ঌ𐑈substrate꒱ enoch꒰ঌ𐑈꒱:·

˖word:·ናሕስ˖.·˙word:·

·⁖origin…language:·Ge'ez·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁖.·˙font:·
·⁖pronunciation:·NAHHS·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Copper or bronze::reddish metal or copper alloy·⁖.·˙non…ostensive:·
·⁖ostensive:·Bronze takes the face of light upon its polished skin::and a polished bronze surface
↪ returns whatever stands before it to the eyeɬ·⁖.·˙ostensive:·

·⁚grimchain⁚·ꟈ🜨𐤍ᛗ▽ᛉᚷᚺ✡𓆏ᑕ⊖ᨐ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○⁻○₍▫₎ category⁚○⁻○₍court₎⁚·
·⁚lookup office⁚○⁻○₍substrate₎ enoch⁚○⁻○₍🜁₎⁚·

₍word⁚·ᏧᎤᏏᏰᏖ₎⁚.·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚.·˙font⁚·
·⁚pronunciation⁚·XOAK·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Heat·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Heat inhabits flesh or grain with altered scent and surface⁚⁚and the warmed body carries
↪ the change throughout itself▫·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚿ⏀ᘜᚸᛉᚹᛅᚿᛏᛊᛇᓄᓀ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○⁻○₍ᛋ₎ category⁚○⁻○₍court₎⁚·
·⁚lookup office⁚○⁻○₍substrate₎ enoch⁚○⁻○₍🜁₎⁚·

₍word⁚·𐎍𐎀𐎗ᛗ₎⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·NAHG+boom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·underground water⁚˙ spring⁚⁚fountain⁚⁚source⁚⁚headwaters (of a river)⁚˙
↪ totality⁚⁚entirety⁚⁚the whole⁚˙ the totality of knowledge·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Water rests beneath stone beyond daylight⁚⁚and hidden pressure fills the same buried
↪ channelᛋ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᘎ⩔◎ᓴAᛘᚱᓴᔕᔑᛘ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○⁻○₍♪₎ category⁚○⁻○₍court₎⁚·
·⁚lookup office⁚○⁻○₍substrate₎ enoch⁚○⁻○₍🜁₎⁚·

₍word⁚·𐎀𐎗𐎍𐎓ᗐ₎⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·ar+gah+MAN+noo·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Red+purple dye or red+purple wool⁚⁚the associated valued color or dyed
↪ material·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Fleece drinks the deep red+purple to its core⁚⁚and every spun thread carries the deep
↪ colour through the weave⁚⁚and the cloth keeps the hue the wool first took♪·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Aᛇ𐌺ᛋᚿᓫℂᛋᛦ𓆏☿ᛘ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○⁻○₍Ⲩ₎ category⁚○⁻○₍court₎⁚·
·⁚lookup office⁚○⁻○₍substrate₎ enoch⁚○⁻○₍🜁₎⁚·

₍word⁚·ᑭᛁᚾᛏᚱ₎⁚.·˙word⁚·

·⁖origin…language:·Old Norse·⁚.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁚.·˙font:·
·⁖pronunciation:·VIND⟊r·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·Wind·⁚.·˙non…ostensive:·
·⁖ostensive:·Moving air carries scent and dust through one open field⁝⁝and both remain suspended in the
↪ same passing current᛭·⁚.·˙ostensive:·
·⁖grimchain:·ᨔᚼᚋᚢᚼᚿᛟᛋᚺᛏ᛬ᛈᘐᒐᒣᑫᑊᑲᒪᑲᒪᑲᒪ

·⁚ostensive⁚·Hand closes around the rein or handle⁚⁚and wandering motion meets the limit of the
↪ grip⁘⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐓏𐓣𐦃𐦖𐦚𐦍𐦆𐦑𐦲𐦗·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍𐦲₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍substrate₎ enoch₀₋₀₍𐦍₎⁚·

₍word⁚·𐦿⁚.·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·NEENG·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Thing⁚· an entity⁚⁚object⁚⁚or matter that exists as something
↪ identifiable·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Fitted bodies meet edge to edge⁚· each complete piece carries one relation⁚⁚and joined
↪ relations hold the working whole𐦲·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐦖𐦂⬡𐦟𐦘𐦓𐦅𐦑𐦗𐦐𐦍𐦆𐦑I𐦐𐦟𐦍·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍𐦗₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍substrate₎ enoch₀₋₀₍𐦍₎⁚·

₍word⁚·ƧƧԿƸOԵԿ₍⁚.·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚.·˙font⁚·
·⁚pronunciation⁚·AANAUN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Knowledge·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Incised clay holds a familiar sign⁚⁚and repeated marks carry the same pattern across
↪ weathered surface𐦲·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐦑𐦆𐦪𐦯𐦀𐦛𐦖𐦐𐦍𐦆𐦑𐦟⊕·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍𐦔₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍substrate₎ enoch₀₋₀₍𐦍₎⁚·

₍word⁚·ᴛωʙє₍⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·TOH⟊beh·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Brick or adobe block⁚⁚a shaped building unit·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Brick joins brick around the opening⁚⁚and daylight has no passage where the wall is
↪ whole𐦔·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐦃𐦕Ↄ☆𐦘𐦐𐦶𐦀𐦛𐦘𐦐𐦍◎·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍𐦑₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍substrate₎ enoch₀₋₀₍𐦍₎⁚·

⁖word:·हरति·:.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·huh+rit·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·pale yellow::pale red::green⁚ emerald·⁚.·˙non…ostensive:·
·⁚ostensive:·Colour lies across one surface::and changed hue occupies the same unmoved face↻·⁚.·˙ostensive:·
·⁚grimchain:·∀✡ↀⴲ☽♀ᵠᑎϽϽᛕ↾▽·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑᑎ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑsubstrate» enoch○–○ₑ♁»⁚·

⁖word:·𒄑·:.·˙word:·

·⁚origin…language:·Sumerian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·NGESH·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Tree::wood::timber·⁚.·˙non…ostensive:·
·⁚ostensive:·Hardwood burns down to charcoal beneath the forge⁚ stored heat in the black fuel keeps iron bright enough for hammer and anvilᑎ·⁚.·˙ostensive:·
·⁚grimchain:·𐋏↾ᛋᚱᛘ𒄑ᚱxᔑᔨᑭ𒄑ᚸ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑÞ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑsubstrate» enoch○–○ₑ♁»⁚·

⁖word:·𒄑𒉡𒁍·:.·˙word:·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·SHAGAN·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Oil⁚ resin⁚ fat::tallow·⁚.·˙non…ostensive:·
·⁚ostensive:·Oil and flame share one feeding body::and brightness lives wherever the hidden fuel is being takenÞ·⁚.·˙ostensive:·
·⁚grimchain:·♂ℂ☖☆ϛX𐋏ⓘ▫✳ᛊⓘ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑᚴ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑsubstrate» enoch○–○ₑ♁»⁚·

⁖word:·𐋏𒌋𒀾·:.·˙word:·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·WAHTAR·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Water·⁚.·˙non…ostensive:·
·⁚ostensive:·Hard ice and loosened edge remain one water+body⁚ liquid water is already present beneath the boundary where rigidity gives wayᚴ·⁚.·˙ostensive:·

·⁚grimchain⁚·⯭▫ϒ৩ᡕ৩ᏂᏪ✳ᖇ⧋ᕈ⊖·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜˓ᚱ꞉ category⟜⟜˓court꞉⁚·
·⁚lookup office⟜⟜˓substrate꞉ enoch⟜⟜˓ஜ꞉⁚·

·⁚word⁚·ʞɯɹʇ⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·KOHT·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Fire·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Heat⁚⁚smoke⁚⁚and char occupy the same burning body⁚⁚and fuel bears all three through one
↪ consuming surfaceᖇ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⏀Ⲭ⚷⬡∀⛧ᚾ☆♂▽ⱫZz⌿·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜˓ᐟ꞉ category⟜⟜˓court꞉⁚·
·⁚lookup office⟜⟜˓substrate꞉ enoch⟜⟜˓ஜ꞉⁚·

·⁚word⁚·ᑢIⵔᛏᑭᑳ⁚.·˙word⁚·

·⁚origin…language⁚·Etruscan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Italic·⁚.·˙font⁚·
·⁚pronunciation⁚·EYE⟊thrah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Sky or upper air⁚⁚the high expanse above the earth·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Upper air holds roof and unreachable height together⁚⁚and no raised thing finds a final
↪ ceiling within it‹·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Ƀ⥣ϴϒ⅄ᔕ⁀ᦓᘐ⍥ᘐ⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜˓X꞉ category⟜⟜˓court꞉⁚·
·⁚lookup office⟜⟜˓substrate꞉ enoch⟜⟜˓ஜ꞉⁚·

·⁚word⁚·ㅆ⁚.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·psEE·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·seed⁚⁚kernel·⁚˙ semen·⁚˙ breed⁚⁚offspring·⁚˙ gene·⁚˙ woof⁚⁚weft·⁚˙ part of speech·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Seed carries root and leaf unseen beneath its coat⁚⁚and green life sleeps folded inside
↪ the same small bodyX·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ℏ⓪☿Ⲭ℁ↄ☆☆ᔕ⩙ᚺᚼ⬡·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜˓ᚹ꞉ category⟜⟜˓court꞉⁚·
·⁚lookup office⟜⟜˓substrate꞉ enoch⟜⟜˓ஜ꞉⁚·

·⁚word⁚·ᚦꝊᔑЄINS⁚.·˙word⁚·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·FOHDEENS·⁖·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Food·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Food and hunger meet in the mouth꞉꞉and the body carries the same substance within that the
↪ hand could hold outside꞉·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᛉᚴᚺᛋᚱᛇᛍᛉᛉᚷᚩᛯ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞊꞉ᚺ꞉ category꞊꞉court꞉⁖·
·⁖lookup office꞊꞉substrate꞉ enoch꞊꞉ᚯ꞉⁖·

꞉word꞉·ᨿᨻᨶ꞉⁖·˙word꞉·

·⁖origin…language꞉·Avestan·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Avestan·⁖·˙font꞉·
·⁖pronunciation꞉·AH⟊poh·⁖·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Waters꞉꞉water·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Water and soil meet upon the same surface꞉꞉and the washing body carries loosened dirt away
↪ in its flowᚺ·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᙢᚠᛇᛔᚴᚺ⏀᛬ᑐ✳ᴢⵠ✳·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞊꞉ᚼ꞉ category꞊꞉court꞉⁖·
·⁖lookup office꞊꞉substrate꞉ enoch꞊꞉ᚯ꞉⁖·

꞉word꞉·ᛦᛋ꞉⁖·˙word꞉·

·⁖origin…language꞉·Aghwan·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Caucasian Albanian·⁖·˙font꞉·
·⁖pronunciation꞉·ZHE·⁖·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Stone·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Stone keeps its shape beneath weight that would leave a mark in mudᚼ·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᛕᛋᚴ᛬ᛔᚥᛣᛁᚲᛋᚸᛞᛝ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞊꞉ᚼ꞉ category꞊꞉court꞉⁖·
·⁖lookup office꞊꞉substrate꞉ enoch꞊꞉ᚯ꞉⁖·

꞉word꞉·ᚽᛟ꞉⁖·˙word꞉·

·⁖origin…language꞉·Akkadian·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖·˙font꞉·
·⁖pronunciation꞉·AB⟊noo·⁖·˙pronunciation꞉·
·⁖cadence꞉·Substrate·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Stone꞉꞉rock꞉꞉precious or worked stone·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Stone and blow meet in fracture꞉꞉and the broken face lies inside the same hardness that
↪ would not bendᚼ·⁖·˙ostensive꞉·
·⁖grimchain꞉·☿ᛗᓀᛇ✳ᛗᛏᛘᛝᚼᚺᚸᛇᚹᛈ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙

·⁚glyph symbol◦–◦𐑦𐑒 category◦–◦court𐑒⁚·
·⁚lookup office◦–◦substrate𐑒 enoch◦–◦𐑓𐑒⁚·

·:word:·𐑕𐑙𐑥𐑦𐑚𐑯𐑙·:.·˙word:·

·⁚origin…language:·Etruscan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Italic·⁚.·˙font:·
·⁚pronunciation:·FAH+sheh+nah·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Sand·:also ashes or human remains reduced to fine material·⁚.·˙non…ostensive:·
·⁚ostensive:·Fine ash lies across ground·:and pale grains ride wind above the same powdery
↪ bed𐑦·⁚.·˙ostensive:·
·⁚grimchain:·𐑨𐑔𐑗𐑓𐑒𐑘𐑡𐑤𐑣𐑟𐑮·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑡𐑒 category◦–◦court𐑒⁚·
·⁚lookup office◦–◦substrate𐑒 enoch◦–◦𐑓𐑒⁚·

·:word:·𐑦𐑨𐑐𐑷·⁚.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·yah+ROH·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·River·⁚.·˙non…ostensive:·
·⁚ostensive:·River occupies one bed under continuous current·:and changing water presses both banks
↪ inside the same flowing channel𐑡·⁚.·˙ostensive:·
·⁚grimchain:·𐑶𐑝𐑓𐑡𐑩𐑫𐑯𐑘𐑔𐑗𐑙𐑜·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑤𐑒 category◦–◦court𐑒⁚·
·⁚lookup office◦–◦substrate𐑒 enoch◦–◦𐑓𐑒⁚·

·:word:·𐑓·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·LAW+gr·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Water·˙ liquid water·:especially as a body or flow·⁚.·˙non…ostensive:·
·⁚ostensive:·Water joins water without seam·:and one continuous body fills both sides of the
↪ meeting𐑤·⁚.·˙ostensive:·
·⁚grimchain:·𐑭𐑙𐑹⬡𐑢𐑥𐑞𐑯⬙⏃𐑜𐑦·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑪𐑒 category◦–◦court𐑒⁚·
·⁚lookup office◦–◦substrate𐑒 enoch◦–◦𐑓𐑒⁚·

·:word:·𐑖𐑘·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·XE·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·

·⁖non…ostensive:·Water·⁚.·˙non…ostensive:·
·⁖ostensive:·Water fills the open channels of a branching stone bed⁖· one continuous route bears the
↪ whole flow from inlet to outlet while blocked hollows remain dryↄ·⁚.·˙ostensive:·
·⁖grimchain:·𐤑𐦚⛦𐦈Θ𐦁𐤃𐤅𐤗𐤓𐦈𐤘·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol○–○₍⛧꜀ category○–○₍court꜀:·
·⁖lookup office○–○₍substrate꜀ enoch○–○₍♁꜀:·

₍word:·𓃾₎.·˙word:·

·⁖origin…language:·Sumerian·⁚.·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁖pronunciation:·SHOO·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·Hand⁖· possession::control::action by extension·⁚.·˙non…ostensive:·
·⁖ostensive:·Hand and held thing share one closed grasp::and fingers keep motion bounded inside the
↪ same contact⛧·⁚.·˙ostensive:·
·⁖grimchain:·Þ♎♀𐦌𐦌𓃾𓀀𓅱𓆑𓆑𓆑𓆑·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol○–○₍𐦈꜀ category○–○₍court꜀:·
·⁖lookup office○–○₍substrate꜀ enoch○–○₍♁꜀:·

₍word:·𓂀𓎛₎.·˙word:·

·⁖origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁖pronunciation:·RTH·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·Mud·⁚.·˙non…ostensive:·
·⁖ostensive:·Mud gathers below the clearer water::heavy with what would not remain
↪ suspended𐦈·⁚.·˙ostensive:·
·⁖grimchain:·𐎗𐎢𐎘𐎍𐎚𐎔𐎌𐎍·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol○–○₍𓆑꜀ category○–○₍court꜀:·
·⁖lookup office○–○₍substrate꜀ enoch○–○₍♁꜀:·

₍word:·부름₎.·˙word:·

·⁖origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁖font:·Noto Sans KR·⁚.·˙font:·
·⁖pronunciation:·pOHlOHm·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·wind::air::draft·⁚.·˙non…ostensive:·
·⁖ostensive:·Wind and rising body share the same air::and earth no longer bears what the moving air
↪ carries𓆑·⁚.·˙ostensive:·
·⁖grimchain:·𐤘◎𐤔𐤍𐤔𐤓𐤇𐤗𐤍𐤋·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol○–○₍ʔ꜀ category○–○₍court꜀:·
·⁖lookup office○–○₍substrate꜀ enoch○–○₍♁꜀:·

⁖word⁚·𐬨𐬵𐬹𐬭𐬀⁖·˙word⁚·

·˙origin…language⁚·Avestan·˙·˙origin…language⁚·
·˙font⁚·Noto Sans Avestan·˙·˙font⁚·
·˙pronunciation⁚·mah+AY+ghah·˙·˙pronunciation⁚·
·˙cadence⁚·Substrate·˙·˙cadence⁚·
·˙non…ostensive⁚·Cloud·˙·˙non…ostensive⁚·
·˙ostensive⁚·Cloud hangs between earth and sky⁘softening every distant edge within the gathered
↪ air𐬀·˙·˙ostensive⁚·
·˙grimchain⁚·𓂋𓏏𓃀𓅱𓎛𓈖𓐍𓆑𓊃𓀀𓏲·˙·˙grimchain⁚·
·˙·˙lookup⁘
·˙·˙glyphs⁘
·˙glyph symbol꩜꩜◎꩞ category꩜꩜court꩞⁚·
·˙lookup office꩜꩜substrate꩞ enoch꩜꩜𓃰꩞⁚·

⁖word⁚·𓀀𓀁𓄿⁖·˙word⁚·

·˙origin…language⁚·Egyptian·˙·˙origin…language⁚·
·˙font⁚·Noto Sans Egyptian Hieroglyphs·˙·˙font⁚·
·˙pronunciation⁚·MOO·˙·˙pronunciation⁚·
·˙cadence⁚·Substrate·˙·˙cadence⁚·
·˙non…ostensive⁚·Water·˙·˙non…ostensive⁚·
·˙ostensive⁚·Water and crack share one narrow passage⁘and both sides of the wall meet through the same
↪ escaping stream◎·˙·˙ostensive⁚·
·˙grimchain⁚·𓊪𓎛𓃰𓋴𓌙𓏏𓎟𓆣𓐍𓋹𓎛·˙·˙grimchain⁚·
·˙·˙lookup⁘
·˙·˙glyphs⁘
·˙glyph symbol꩜꩜𓊪꩞ category꩜꩜court꩞⁚·
·˙lookup office꩜꩜substrate꩞ enoch꩜꩜𓃰꩞⁚·

⁖word⁚·𒀀⁖·˙word⁚·

·˙origin…language⁚·Sumerian·˙·˙origin…language⁚·
·˙font⁚·Noto Sans Cuneiform·˙·˙font⁚·
·˙pronunciation⁚·KOOG·˙·˙pronunciation⁚·
·˙cadence⁚·Substrate·˙·˙cadence⁚·
·˙non…ostensive⁚·Silver·˙·˙non…ostensive⁚·
·˙ostensive⁚·Silver holds one measured weight in either palm⁘and balance receives the same mass from
↪ either side𓊪·˙·˙ostensive⁚·
·˙grimchain⁚·𐎗𐎝𐎚𐎕𐎅𐎍𐎐𐎓𐎖𐎘·˙·˙grimchain⁚·
·˙·˙lookup⁘
·˙·˙glyphs⁘
·˙glyph symbol꩜꩜𐎚꩞ category꩜꩜court꩞⁚·
·˙lookup office꩜꩜substrate꩞ enoch꩜꩜𓃰꩞⁚·

⁖word⁚·𐤅𐤅⁖·˙word⁚·

·˙origin…language⁚·Phoenician·˙·˙origin…language⁚·
·˙font⁚·Noto Sans Phoenician·˙·˙font⁚·
·˙pronunciation⁚·MAYM·˙·˙pronunciation⁚·
·˙cadence⁚·Substrate·˙·˙cadence⁚·
·˙non…ostensive⁚·Water⁘ the common liquid of rain⁘springs⁘wells⁘rivers⁘and the sea·˙·˙non…ostensive⁚·
·˙ostensive⁚·Cold and water occupy one rigid surface⁘and frozen mass bears weight across the same
↪ continuous body𐎚·˙·˙ostensive⁚·

·⁚grimchain⁚·𐫰𐫱𐫲𐫳𐫴𐫵𐫶𐫷𐫸𐫹𐫺·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⁖⁘𐫻 category⁖⁘court𐫼⁚·
·⁚lookup office⁖⁘substrate𐫼 enoch⁖⁘𐫽𐫼⁚·

·word⁚·ꠇꠅꠞꠋ⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·MUSH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Wind·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Smoke loses its edge within moving air⁚˙ the same wind holds the last visible trace and
↪ the clear space into which it disappears𐫻·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫰𐫱𐫲𐫳𐫴𐫵𐫶𐫷𐫸𐫹𐫺·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⁖⁘𐫽𐫼 category⁖⁘court𐫼⁚·
·⁚lookup office⁖⁘substrate𐫼 enoch⁖⁘𐫽𐫼⁚·

·word⁚·ኦበን⁚·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚·˙font⁚·
·⁚pronunciation⁚·UHB+n·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Stone⁚⁚a piece or mass of rock·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Stone rests upon solid earth⁚⁚and downward weight meets equal resistance from the
↪ ground⁚⁚and undisturbed mass holds its fixed place𐫽·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫰𐫱𐫲𐫳𐫴𐫵𐫶𐫷𐫸𐫹𐫺·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⁖⁘𐫾𐫼 category⁖⁘court𐫼⁚·
·⁚lookup office⁖⁘substrate𐫼 enoch⁖⁘𐫽𐫼⁚·

·word⁚·𐫿·⁚·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·er+SEH+too·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Earth⁚⁚ground⁚⁚land⁚⁚in some contexts the netherworld·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Earth bears the building and its weight together⁚⁚and every height presses into the same
↪ ground beneath it𐫾·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫾𐫰𐫱𐫲𐫳𐫴𐫵𐫶𐫷𐫸𐫹·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⁖⁘𐫹𐫼 category⁖⁘court𐫼⁚·
·⁚lookup office⁖⁘substrate𐫼 enoch⁖⁘𐫽𐫼⁚·

·word⁚·𐫺𐫻𐫼𐫽·⁚·˙word⁚·

·⁚origin…language꞉·Hittite·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·TAHRU·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Wood⁚⁚tree·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Wood laid above wood raises one level from the support of the level beneath
↪ itϴ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·⯎𐋡𐋢𐋣𐋤𐋥𐋦𐋧𐋨𐋩𐋪𐋫·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍𝖷₎ category꞉○–○₍court₎꞉·
·⁚lookup office꞉○–○₍substrate₎ enoch꞉○–○₍𓃰₎꞉·

₍word꞉·ᛏᚹᚳᚣᚹ₎⁚.·˙word꞉·

·⁚origin…language꞉·Etruscan·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Italic·⁚.·˙font꞉·
·⁚pronunciation꞉·TOO+lahɪ·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Earth⁚⁚land⁚⁚ground⁚⁚or territorial boundary·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Earth hardens beneath the foot where the spade no longer passes easily through the
↪ groundᚷ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐌘𐌕𐌖𐌗𐌘⬡ϟϡ𐌙𐌚𐌛·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍Λ₎ category꞉○–○₍court₎꞉·
·⁚lookup office꞉○–○₍substrate₎ enoch꞉○–○₍𓃰₎꞉·

₍word꞉·𐤊𐤕𐤌𐤎₎⁚.·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·shee+PAH+toom·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·wool⁚⁚fleece·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Wool wraps shoulder and back⁚⁚taking rain and cold upon its outer fibres while the covered
↪ body moves beneathΛ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐌉𐌊𐌋𐌌𐌍𐌎𐌏𐌐𐌑𐌒𐌓𐌔z·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍Ⱨ₎ category꞉○–○₍court₎꞉·
·⁚lookup office꞉○–○₍substrate₎ enoch꞉○–○₍𓃰₎꞉·

₍word꞉·ϸογɴ₎⁚.·˙word꞉·

·⁚origin…language꞉·Coptic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Coptic·⁚.·˙font꞉·
·⁚pronunciation꞉·khoon·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Inside⁚⁚interior·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Enclosing walls stand around one hollow⁚⁚and the rim keeps the outer air from the middle
↪ space⁚⁚and whatever crosses the opening stays within the enclosing edgeⱧ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐌗𐌘𐌙𐌚𐌛𐌜𐌝𐌞⬡𐌠𐌡𐌢ϛ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤗꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐤔꩞⁚·

꞉word⁚·ክንፍ⁚·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚·˙font⁚·
·⁚pronunciation⁚·KUHNF·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Wing⁚˙ a limb or structure used for flight⁚⁚also an edge or branch·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Wing opens from one root and widens into two edges that carry the same body
↪ outward𐤗·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚾᛠᛟ𐤗ᛇᚦᛋᛥᚹᛎᛟᛃᚣ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤐꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐤔꩞⁚·

꞉word⁚·𓉔𓊪⁚·˙word⁚·

·⁚origin…language⁚·Egyptian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Egyptian Hieroglyphs·⁚·˙font⁚·
·⁚pronunciation⁚·KES·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Bone·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·White bone remains among ash and fallen flesh⁚˙ the hard fragment keeps form after softer
↪ body is gone𐤐·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛁᚷᚦᛟᛊᚱᚲᚢᛃᛞᛇᚳᛉ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤊꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐤔꩞⁚·

꞉word⁚·कण⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·kuh÷nuh·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·any minute particle⁚⁚atom⁚˙ a grain⁚⁚grain of corn⁚⁚single seed⁚˙ a grain or particle (of
↪ dust)⁚˙ flake (of snow)⁚˙ a drop (of water)⁚˙ a spark (of fire)⁚˙ the spark or facet of a
↪ gem·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Heap thins grain by grain⁚˙ each removed particle lowers the mound while the remaining
↪ grains still bear its shrinking form𐤊·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᗡ◎¤𐤒ᚷᚸᔑᓄᛋᚴᛏ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤇꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐤔꩞⁚·

꞉word⁚·ተ፲፫⁚·˙word⁚·

·⁖origin…language:·Tocharian A·⁚·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁚·˙font:·
·⁖pronunciation:·KROHP·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·Crowd·⁘ herd·⁖flock·⁘ heap·⁚·˙non…ostensive:·
·⁖ostensive:·Flock and fold share the same crowded ground·⁖and almost no open earth lies between one
↪ body and another𐨸·⁚·˙ostensive:·
·⁖grimchain:·𐫄𐫈𐫇𐫉𐫊⊛𐫋𐫌𐫍𐫎𐫏·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉𐫊 category꞉꞉court𐫋·
·⁖lookup office꞉꞉substrate𐫋 enoch꞉꞉𐫋𐫋·

·word:·ωɴє·⁚·˙word:·

·⁖origin…language:·Coptic·⁚·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁚·˙font:·
·⁖pronunciation:·OH+neh·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·Stone·⁖hard mineral matter·⁚·˙non…ostensive:·
·⁖ostensive:·Stone keeps its edge beneath weight and water without yielding into another
↪ shape𐫊·⁚·˙ostensive:·
·⁖grimchain:·𐫈⊛𐫉𐫇𐫊𐫋𐫌𐫍𐫎𐫏𐫐·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉𐫉 category꞉꞉court𐫋·
·⁖lookup office꞉꞉substrate𐫋 enoch꞉꞉𐫋𐫋·

·word:·𐫑𐫒𐫓𐫔𐫕·⁚·˙word:·

·⁖origin…language:·Akkadian·⁚·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁚·˙font:·
·⁖pronunciation:·EHR+seh+toom·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·earth·⁖land·⁘ underworld·⁘ ground·⁘ territory·⁚·˙non…ostensive:·
·⁖ostensive:·Ground shudders beneath wall and foot·⁖and everything standing upon it receives the
↪ movement from below𐫉·⁚·˙ostensive:·
·⁖grimchain:·𐫄𐫅𐫆𐫇𐫈𐫉𐫊𐫋𐫌𐫍𐫎𐫏·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉𐫇 category꞉꞉court𐫋·
·⁖lookup office꞉꞉substrate𐫋 enoch꞉꞉𐫋𐫋·

·word:·𐫐𐫑𐫒𐫓𐫔·⁚·˙word:·

·⁖origin…language:·Hittite·⁚·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁚·˙font:·
·⁖pronunciation:·PAYDAN·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·Place·⁘ floor·⁖ground·⁚·˙non…ostensive:·
·⁖ostensive:·Ground keeps one fixed place beneath settled weight·⁖and surrounding movement leaves the
↪ position unchanged𐫇·⁚·˙ostensive:·
·⁖grimchain:·𐫏𐫄𐫅𐫆𐫇𐫈𐫉𐫊𐫋𐫌𐫍·⁚·˙grimchain:·
·⁚·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀥꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𑀬꩞⁚·

·word:·ꠢꠣ꠸꠹꠺:·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·KEHKTSEHNYEH·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·(anatomy) body·⁚·˙non…ostensive:·
·⁚ostensive:·Flesh closes around a passage without filling it::leaving a way through the body from one opening to another𑀥·⁚·˙ostensive:·
·⁚grimchain:·𐋡◌ıɛ↗५ɦ꒦ঙ৩ϽϞ𑀥𑀪𑀮𑀲·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀥꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𑀬꩞⁚·

·word:·𐌔𐌕𐌉𐌁𐌍𐌋:·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·STIBNA·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Voice·⁚·˙non…ostensive:·
·⁚ostensive:·Voice and returning sound inhabit the same hollow::and stone gives the sent sound another path back through the air𑀥·⁚·˙ostensive:·
·⁚grimchain:·𐋢ɻɛ𑀥ℚᴘ𐋡ৎI৻ıՈ𐋟·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀕꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𑀬꩞⁚·

·word:·जतु:·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·juh꩜tu·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·lac::gum::resin::rubber·⁚·˙non…ostensive:·
·⁚ostensive:·Resin softens in heat and gives a visible breath to the air above it𑀕·⁚·˙ostensive:·
·⁚grimchain:·ऽ৭५ɪ↓𐋡⏀Ⱥ৩ϫϸ⊁ǂ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀫꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𑀬꩞⁚·

·word:·ψسصدد:·˙word:·

·⁚origin…language:·Avestan·⁚·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚·˙font:·
·⁚pronunciation:·VAH꩜tah·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·

·⁚non…ostensive⁚·Wind⁘moving air·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Wind fills the open place without fixed shape⁘and every boundary holds the same moving
↪ air in another contourব·⁚.·˙ostensive⁚·
·⁚grimchain⁚·৻ৎ৲ঌ৭ঀ⌬ৗৎ৸ঌ⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol⟜⟜৻◌৯ category⟜⟜৻court৯⁚·
·⁚lookup office⟜⟜৻substrate৯ enoch⟜⟜৻ঌ৯⁚·

·word⁚·ᛋᚾᚾᛏᚱ·.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·SAHN+dr·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Sand·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Sand and wind share the moving grain⁘and loose grain belongs equally to rising air and
↪ settled ground◌·⁚.·˙ostensive⁚·
·⁚grimchain⁚·∮◎ᛞᥣ₀ᵒ〰ᛯᛘᛯ৭ᛇᛇ⏀·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol⟜⟜৻ব৯ category⟜⟜৻court৯⁚·
·⁚lookup office⟜⟜৻substrate৯ enoch⟜⟜৻ঌ৯⁚·

·word⁚·ቈጽል·.·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚.·˙font⁚·
·⁚pronunciation⁚·kwah+TSL·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Leaf⁘a flat green growth from a plant or tree·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Leaf carries hidden growth in every green surface⁘and stem and light meet within the same
↪ unfolding bodyব·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᘔↃΨ◎ɪ⏀ᗅᖘᛋᚴ᛻᛭ᔕᗝ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol⟜⟜৻ঘ৯ category⟜⟜৻court৯⁚·
·⁚lookup office⟜⟜৻substrate৯ enoch⟜⟜৻ঌ৯⁚·

·word⁚·ᴍᑎᑎΨS·.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·MUNTHS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Mouth·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Breath gathers behind the closed seam of the lips⁘the parted seam releases the gathered
↪ air outward⁘and the same rim that opened for the air shapes the sound it carriesঘ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛁᑭᐊᕮᛘᔕᑭ᛭⌀ᑎↄϒᛋঘ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol⟜⟜৻ᘜ৯ category⟜⟜৻court৯⁚·
·⁚lookup office⟜⟜৻substrate৯ enoch⟜⟜৻ঌ৯⁚·

˖word:·수시˖:·˙word:·

·⁚origin…language:·Middle Korean·⁚·˙origin…language:·
·⁚font:·Noto Sans KR·⁚·˙font:·
·⁚pronunciation:·suhsi·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·gap::space between::(Gyeongsang) relationship·⁚·˙non…ostensive:·
·⁚ostensive:·Two edges stand apart::No body fills the interval·⁚·˙ostensive:·
·⁚grimchain:·ΛΦ𐤏𐤌𐤒𐤓𐤇𐤉𐤋𐤄𐤅𐤍·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤅 category꩜꩜court𐤇:·
·⁚lookup office꩜꩜substrate𐤆 enoch꩜꩜𐤄𐤇:·

˖word:·रेणु˖:·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·ray÷nu·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·dust::a grain or atom of dust::sand⁘ the pollen of flowers⁘ powder of anything⁘
↪ Oldenlandia herbacea·⁚·˙non…ostensive:·
·⁚ostensive:·A narrow trail of dust lies along the seam::so thin that stone shows through on either
↪ side𐤅·⁚·˙ostensive:·
·⁚grimchain:·𐤏𐤊𐤓𐤋𐤔𐤁𐤍𐤀𐤊𐤂𐤕𐤉·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤆 category꩜꩜court𐤇:·
·⁚lookup office꩜꩜substrate𐤆 enoch꩜꩜𐤄𐤇:·

˖word:·पणि्ड˖:·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·pin÷duh·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·any round or roundish mass or heap::ball::lump::piece (RV::TS::ŚBr::etc)⁘ a roundish lump
↪ of food::a bite (of food)::morsel::mouthful⁘ a roundish lump of food::a bite (of
↪ food)::morsel::mouthful⁘ a ball of rice or flour offered to ancestors (GṛŚṛS::Mn::MBh::etc)⁘
↪ food::daily bread::livelihood::subsistence::sustenance (MBh::Kāv::etc)⁘ any solid mass or material
↪ object::the body::bodily frame (Ragh::Śaṃk::Vajracch)⁘ the calf of the leg (Mālatīm)⁘ the flower of
↪ a china rose (L)⁘ a portico (L)⁘ force::power::army (L)⁘ fleshy parts of the shoulder above the
↪ clavicle (MBh)⁘ protrusion on the upper forehead of an elephant (L)⁘ embryo at an early stage of
↪ development (L)⁘ a kind of incense (Var⁘ myrrh::olibanum (L)⁘ meat::flesh (L)⁘ alms (Mālatīm)⁘
↪ Meyna laxiflora (L)⁘ quantity::collection (L)⁘ sum::total amount⁘ a sine expressed in a series of
↪ numbers⁘ sound::tone⁘ name of a man⁘ iron (L)⁘ steel⁘ fresh butter·⁚·˙non…ostensive:·
·⁚ostensive:·Solid lump fills the palm with weight and depth::and pressure on one face enters dense
↪ mass without crossing the whole body𐤆·⁚·˙ostensive:·
·⁚grimchain:·𐤓𐤄𐤒𐤐𐤎𐤁𐤔𐤊𐤇𐤅𐤍𐤉𐤂·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤃𐤆 category꩜꩜court𐤇:·
·⁚lookup office꩜꩜substrate𐤆 enoch꩜꩜𐤄𐤇:·

˖word:·𐎑𐎔𐎗𐎚˖·˙word:·

·˙origin…language:·Hittite·˙·˙origin…language:·
·˙font:·Noto Sans Cuneiform·˙·˙font:·
·˙pronunciation:·ARUNASH·˙·˙pronunciation:·
·˙cadence:·Substrate·˙·˙cadence:·
·˙non…ostensive:·Sea·˙·˙non…ostensive:·
·˙ostensive:·Sea rises against itself under wind⁚⁚and no single wave keeps the same face for
↪ long𐎕·˙·˙ostensive:·
·˙grimchain:·𐎗𐎚𐎛𐎈𐎗𐎑𐎔𐎓𐎖0𐎒𐎙·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol◦–◦˹𐎕˺ category◦–◦˹court˺:·
·˙lookup office◦–◦˹substrate˺ enoch◦–◦˹𐎀˺:·

˖word:·𐎍𐎏𐎓 𐎖𐎗˖·˙word:·

·˙origin…language:·Ugaritic·˙·˙origin…language:·
·˙font:·Noto Sans Ugaritic·˙·˙font:·
·˙pronunciation:·SHAMNU·˙·˙pronunciation:·
·˙cadence:·Substrate·˙·˙cadence:·
·˙non…ostensive:·Fat⁚⁚oil·˙·˙non…ostensive:·
·˙ostensive:·Oil enters between rough surfaces and lets one pass over another without
↪ catching𐎕·˙·˙ostensive:·
·˙grimchain:·𐎙𐎍𐎐𐎆𐎄𐎔𐎚𐎑𐎗𐎓𐎀𐎁·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol◦–◦˹𐎘˺ category◦–◦˹court˺:·
·˙lookup office◦–◦˹substrate˺ enoch◦–◦˹𐎀˺:·

˖word:·𐌾𐌿𐍁˖·˙word:·

·˙origin…language:·Coptic·˙·˙origin…language:·
·˙font:·Noto Sans Coptic·˙·˙font:·
·˙pronunciation:·CHOHM·˙·˙pronunciation:·
·˙cadence:·Substrate·˙·˙cadence:·
·˙non…ostensive:·Power⁚⁚might⁚⁚or authority·˙·˙non…ostensive:·
·˙ostensive:·Stored strain fills bent limb and cord⁚⁚and still material bears the thrust within its
↪ held tension𐎘·˙·˙ostensive:·
·˙grimchain:·◎𐌶𐍀𐎗𐎖⏁ϛ𐎑𐎘✲𐎄𐎒·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol◦–◦˹𐎀˺ category◦–◦˹court˺:·
·˙lookup office◦–◦˹substrate˺ enoch◦–◦˹𐎀˺:·

˖word:·አእምሮ˖·˙word:·

·˙origin…language:·Ge'ez·˙·˙origin…language:·
·˙font:·Noto Sans Ethiopic·˙·˙font:·
·˙pronunciation:·ah+uhm+ROH·˙·˙pronunciation:·
·˙cadence:·Substrate·˙·˙cadence:·
·˙non…ostensive:·Thought⁚⁚knowledge⁚⁚intellect⁚⁚or intelligence·˙·˙non…ostensive:·
·˙ostensive:·Old pattern returns intact within thought while the place that once bore it stands empty·˙
↪ inward knowledge keeps what present ground no longer gives𐎀·˙·˙ostensive:·

·⁚grimchain⁚·ᛋ'ᚣᓚᛋᚳᚱᛩᛯᚾᛯᓀ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦ₑ∀꜀ category◦–◦ₑcourt꜀⁚·
·⁚lookup office◦–◦ₑsubstrate꜀ enoch◦–◦ₑ⯋꜀⁚·

₍word⁚·ᚺᚢᛁᚱᛉᛉ₎⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·HEHR+TO·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Heart·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Quick pulse lifts beneath the breast⁚⁚breath breaks into bright repeated sound⁚⁚and the
↪ chest opens around the same rising warmth∀·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚏ⊛ᛓ◎◎ᛯᚔ➢ᛏᚳᚸᚦᚳᑫᛋ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦ₑ∀꜀ category◦–◦ₑcourt꜀⁚·
·⁚lookup office◦–◦ₑsubstrate꜀ enoch◦–◦ₑ⯋꜀⁚·

₍word⁚·ᴏϒɯʜ₎⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·oo+SHAY·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Night⁚⁚the period of darkness·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Darkness fills the sleeping chamber⁚⁚and closed eyes and still limbs rest inside the same
↪ unlit field∇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᵒᵌᘐᵩᘜᵨᘎᵚᛋᛀᛏᘉᛇ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦ₑ∇꜀ category◦–◦ₑcourt꜀⁚·
·⁚lookup office◦–◦ₑsubstrate꜀ enoch◦–◦ₑ⯋꜀⁚·

₍word⁚·ऊर्म्ि₎⁚.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·oor+mi·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·wave⁚⁚billow·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Wave rises and falls⁚⁚and the same water answers again in the opposite
↪ direction∇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᔕᚼᚼᛘᚸᛩᚲᛏᛯ➢ᛉᛂᚸᚱ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦ₑ∀ᚱ꜀ category◦–◦ₑcourt꜀⁚·
·⁚lookup office◦–◦ₑsubstrate꜀ enoch◦–◦ₑ⯋꜀⁚·

₍word⁚·ʃɾ̀ɻʒ₎⁚.·˙word⁚·

·⁚origin…language:·Avestan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚.·˙font:·
·⁚pronunciation:·zuh+RUHD·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Heart⁚· the central bodily organ·⁚.·˙non…ostensive:·
·⁚ostensive:·One hurt pulse finds an answering tightness in another breast⁚· two inward chambers carry
↪ the same ache without sharing one wound𐬘·⁚.·˙ostensive:·
·⁚grimchain:·♂𐬟𐬴𐬶𐬀𐬘𐬯𐬌𐬭𐬊𐬹𐬵·⁚.·˙grimchain:·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol◦–◦𐬀𐬛 category◦–◦court𐬛⁚·
·⁚lookup office◦–◦substrate𐬛 enoch◦–◦𐬃𐬛⁚·

₍word⁚·ᠮᠰᡂᠬᡡ⁚₎·˙word⁚·

·⁚origin…language:·Avestan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚.·˙font:·
·⁚pronunciation:·GAH+oh+shah·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Ear⁚· the organ of hearing·⁚.·˙non…ostensive:·
·⁚ostensive:·Low vibration presses the ear flesh before any clear tone forms⁚· cartilage::membrane::and
↪ inner fluid carry contact inward𐬀·⁚.·˙ostensive:·
·⁚grimchain:·❂𐬰𐬊𐬧𐬀𐬱𐬫𐬋⏀◎𐬒𐬭𐬵𐬀·⁚.·˙grimchain:·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol◦–◦𐬁𐬛 category◦–◦court𐬛⁚·
·⁚lookup office◦–◦substrate𐬛 enoch◦–◦𐬃𐬛⁚·

₍word⁚·ᨕᨒᦈᨕᨘ⁚₎·˙word⁚·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·HUEWQEN·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Bone·⁚.·˙non…ostensive:·
·⁚ostensive:·Hairline rot darkens the load+bearing bone⁚· each step drives pain through the weakened
↪ frame𐬁·⁚.·˙ostensive:·
·⁚grimchain:·𐬡𐬭◎𐬉𐬀𐬹𐬨𐬫𐬔𐬗𐬀·⁚.·˙grimchain:·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol◦–◦𐬁𐬹𐬛 category◦–◦court𐬛⁚·
·⁚lookup office◦–◦substrate𐬛 enoch◦–◦𐬃𐬛⁚·

₍word⁚·ᓚᘐᔭᙁ□ᑊ⁚₎·˙word⁚·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·SAHMOODTUHR·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Ocean::sea·⁚.·˙non…ostensive:·
·⁚ostensive:·Dark water extends beneath weakened light::and deep pressure fills the same body far below
↪ the surface𐬁·⁚.·˙ostensive:·
·⁚grimchain:·𐬮𐬢𐬏𐬰𐬸𐬋𐬃𐬀𐬍𐬏⊙ℂ·⁚.·˙grimchain:·
·⁚.·˙lookup⁚·

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑉 category꩜꩜court𐑉:·
·⁚lookup office꩜꩜substrate𐑉 enoch꩜꩜𐑻𐑉:·

:word:·घोष:⁚·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·ghoh+shuh·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·indistinct noise::tumult::sound·⁚ cry::confused cries of a multitude::battle cry::cries of
↪ victory::cries of woe or distress::holler::yell·⁚ the sound of a drum::conch shell::Soma stones::a
↪ carriage::etc·⁚ the whizzing or whir of a bowstring::crackling of fire::singing in the ear·⁚ the
↪ roaring of a storm::thunder::water::etc·⁚ the sound of the recital of prayers·⁚ the sound of words
↪ spoken at a distance·⁚ rumor::report (also personified)·⁚ a proclamation·⁚ a sound (of speech)·⁚ the
↪ soft (voiced) sound heard in the articulation of the sonant consonants::the vowels::and anusvara
↪ which with the Yamas of the first 10 of the soft consonants make up altogether 40 sounds·⁚ an
↪ ornament that makes a tinkling sound·⁚ station of herdsmen·⁚ herdsmen·⁚·˙non…ostensive:·
·⁚ostensive:·Sound enters sound and the air carries both without either voice being broken
↪ apart𐑉·⁚·˙ostensive:·
·⁚grimchain:·𐑊𐑦ᔕᘐᘓᘔᘕᘖᘗᘘᘙᘚᘛ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑻𐑉 category꩜꩜court𐑉:·
·⁚lookup office꩜꩜substrate𐑉 enoch꩜꩜𐑻𐑉:·

:word:·ምልሓ:⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·muhl+HAH·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Knowledge::judgment::wisdom::or wit·⁚·˙non…ostensive:·
·⁚ostensive:·Measured knowledge holds the hand before a dangerous cut·⁚ consequence stands already
↪ clear in understanding::and the blade remains raised𐑻·⁚·˙ostensive:·
·⁚grimchain:·𐑐𐑒𐑓𐑔𐑕𐑖𐑗𐑘𐑙𐑚𐑛·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑦𐑉 category꩜꩜court𐑉:·
·⁚lookup office꩜꩜substrate𐑉 enoch꩜꩜𐑻𐑉:·

:word:·ཁཤཡཁ:⁚·˙word:·

·⁚origin…language:·Avestan·⁚·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚·˙font:·
·⁚pronunciation:·CHASH+mahn·⁚·˙pronunciation:·
·⁚cadence:·Substrate·⁚·˙cadence:·
·⁚non…ostensive:·Eye::the organ of sight·⁚·˙non…ostensive:·
·⁚ostensive:·Wide pupil::exposed surface::and incoming light occupy one opened eye::and the visible field
↪ exceeds the narrow frame around it𐑦·⁚·˙ostensive:·
·⁚grimchain:·𐑟𐑠𐑡𐑢𐑣𐑤𐑥𐑦𐑧𐑨·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙

·⁚glyph symbol꩜꩜𐦚 category꩜꩜court𐦚⁚·
·⁚lookup office꩜꩜substrate𐦚 enoch꩜꩜🜁𐦚⁚·

⁖word⁚·업시⁖⁚·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚⁚·˙font⁚·
·⁚pronunciation⁚·uhpsi·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·without⁖⁖lacking·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Empty hand closes and opens unchanged⁖⁖with no body gained and no body lost𐦚·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·I▽Θ𐦚𐦚◎𐦚ᚠ𐦚Ꮎ𐦚𐦚Ⴒ𐦚Ո·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜✷𐦚 category꩜꩜court𐦚⁚·
·⁚lookup office꩜꩜substrate𐦚 enoch꩜꩜🜁𐦚⁚·

⁖word⁚·ᚼᚣ̄⁖⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚⁚·˙font⁚·
·⁚pronunciation⁚·KEHSH·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Number·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Counting place stands empty of any object⁖⁖and empty position remains among the marked measures✷·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·ᗅꓤᑐ𐦚ᚸᚸ𐦚Ↄ𐦚𐦚ᔑᚸᛋ☆🜁·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐦚 category꩜꩜court𐦚⁚·
·⁚lookup office꩜꩜substrate𐦚 enoch꩜꩜🜁𐦚⁚·

⁖word⁚·ΔᒐĪ⁖⁚·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚⁚·˙font⁚·
·⁚pronunciation⁚·EHPREHR·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Sky⁖⁖firmament⁚˙ atmosphere·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Smoke and open air occupy one widening body⁖⁖and diffuse vapor spreads through the same boundless field𐦚·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·𐦚𐦚Ⱥ✷𐦚ᚱ𐦚𐦚𐦚ᛞ𐦚𐦚𐦚𐦚𐦚𐦚·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐦚 category꩜꩜court𐦚⁚·
·⁚lookup office꩜꩜substrate𐦚 enoch꩜꩜🜁𐦚⁚·

⁖word⁚·ᚢᚣᚸᚹᚹᚸ⁖⁚·˙word⁚·

·⁚origin…language⁚·Avestan·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Avestan·⁚⁚·˙font⁚·
·⁚pronunciation⁚·ZRAH⫩yah·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Sea·⁚⁚·˙non…ostensive⁚·

·⁚ostensive⁚·Sea extends beneath the surface beyond visible depth⁜and dark water fills the same descending body𜳵·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𜳵𜳰𜳶𜳷𜳹𜳺𜳻𜳼𜳽𜳾𜳿·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𜳵꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜substrate꞉ enoch⟜⟜𜳰꞉⁚·

₍word⁚·𜳰𜳱𜳲𜳳₎·⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·ˈorigin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·ˈfont⁚·
·⁚pronunciation⁚·AHKUHR·⁚.·ˈpronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·ˈcadence⁚·
·⁚non…ostensive⁚·Tear (drop of liquid from crying)·⁚.·ˈnon…ostensive⁚·
·⁚ostensive⁚·Tear gathers at the eye⁜and downward weight rests inside the clear bead𜳵·⁚.·ˈostensive⁚·
·⁚grimchain⁚·𜳴𜳵𜳶𜳷𜳸𜳹𜳺𜳻𜳼𜳽·⁚.·ˈgrimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𜳾꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜substrate꞉ enoch⟜⟜𜳰꞉⁚·

₍word⁚·𜳿𜳱𜳲𜳳₎·⁚.·ˈword⁚·

·⁚origin…language⁚·Tocharian B·⁚.·ˈorigin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·ˈfont⁚·
·⁚pronunciation⁚·WRAHKOH·⁚.·ˈpronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·ˈcadence⁚·
·⁚non…ostensive⁚·Shell (of an oyster or other sea creature)·⁚.·ˈnon…ostensive⁚·
·⁚ostensive⁚·Angled shell takes the falling blow on its curve⁜and the force slides aside instead of entering the hollow within𜳾·⁚.·ˈostensive⁚·
·⁚grimchain⁚·𜳀𜳁𜳂𜳃𜳄𜳅𜳆𜳇𜳈𜳉·⁚.·ˈgrimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𜳻꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜substrate꞉ enoch⟜⟜𜳰꞉⁚·

₍word⁚·𜳊𜳋𜳌𜳍𜳎₎·⁚.·ˈword⁚·

·⁚origin…language⁚·Old Norse·⁚.·ˈorigin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·ˈfont⁚·
·⁚pronunciation⁚·HYAR+tah·⁚.·ˈpronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·ˈcadence⁚·
·⁚non…ostensive⁚·Heart·⁚.·ˈnon…ostensive⁚·
·⁚ostensive⁚·Pulse leans hard beneath the breast toward a distant height⁘ the inward chamber keeps pressure against the space still separating it from the sought place𜳻·⁚.·ˈostensive⁚·
·⁚grimchain⁚·𜳏𜳐𜳑𜳒𜳓𜳔𜳕𜳖𜳗·⁚.·ˈgrimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𜳽꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜substrate꞉ enoch⟜⟜𜳰꞉⁚·

₍word⁚·𜳘𜳙₎·⁚.·ˈword⁚·

·⁖origin…language:·Coptic·⁖·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁖·˙font:·
·⁖pronunciation:·AHF·⁖·˙pronunciation:·
·⁖cadence:·Substrate·⁖·˙cadence:·
·⁖non…ostensive:·Meat or flesh·⁖·˙non…ostensive:·
·⁖ostensive:·Flesh carries wound and mending in the same living body⁝⁝and strength belongs to the
↪ tissue even where hurt is present𐓣·⁖·˙ostensive:·
·⁖grimchain:·⊙𐓝zXϓz𐓢ᵒᵒ✝☿ᘐϔ𐓛ᵒ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꞉꞉𐓣꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉substrate꞉ enoch꞉꞉𐓑꞉·

꞉word꞉·लता꞉·˙word꞉·

·⁖origin…language:·Sanskrit·⁖·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖·˙font:·
·⁖pronunciation:·luh+tah·⁖·˙pronunciation:·
·⁖cadence:·Substrate·⁖·˙cadence:·
·⁖non…ostensive:·a creeper (any creeping or winding plant or twining tendril)⁖˙ liana Mn⁝⁝MBh⁝⁝&c⁖˙
↪ musk+creeper L⁖˙ a slender woman⁝⁝any woman Naish⁝⁝Tantras⁖˙ the thong or lash of a whip⁖˙ whip
↪ Pañcat⁝⁝Suśr⁖˙ a string of pearls VarBṛS·⁖·˙non…ostensive:·
·⁖ostensive:·Tendril coils around a supporting stem⁝⁝and tightened contact holds the winding body
↪ against the same surface𐓣·⁖·˙ostensive:·
·⁖grimchain:·𐓟𐓊𐓄𐓇𐓘𐓚𐓯𐓋❋𐓍𐓕𐓪·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꞉꞉𐓡꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉substrate꞉ enoch꞉꞉𐓑꞉·

꞉word꞉·पीठ꞉·˙word꞉·

·⁖origin…language:·Sanskrit·⁖·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖·˙font:·
·⁖pronunciation:·pee+thuh·⁖·˙pronunciation:·
·⁖cadence:·Substrate·⁖·˙cadence:·
·⁖non…ostensive:·seat⁝⁝bench⁝⁝desk⁝⁝stool⁝⁝chair⁖˙ a religious student's seat (made properly of Kuśa
↪ grass)⁖˙ case⁝⁝pedestal (esp⁝⁝of an idol)⁖˙ royal seat⁝⁝throne⁖˙ place⁝⁝office⁖˙ base⁝⁝plinth (of a
↪ statue)⁖˙ altar⁖˙ seat of a deity⁖˙ a district⁝⁝province⁖˙ a particular posture in sitting⁖˙ the
↪ complement of a segment·⁖·˙non…ostensive:·
·⁖ostensive:·Raised platform holds one completed level of the work⁖˙ the next course begins from the
↪ surface already built beneath it𐓡·⁖·˙ostensive:·
·⁖grimchain:·ৰৰ𑁍𐓐z∀✳𐓂Ƙ♀𐓖𐓣𐓑·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꞉꞉𐓫꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉substrate꞉ enoch꞉꞉𐓑꞉·

꞉word꞉·ᖁᒍᖊ𑈬꞉·˙word꞉·

·⁖origin…language:·Tocharian B·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·PAHLSKOH·⁖·˙pronunciation:·
·⁖cadence:·Substrate·⁖·˙cadence:·

·⁖non…ostensive:·Mind::thought·⁖.·˙non…ostensive:·
·⁖ostensive:·Two complete truths remain held in one field of thought⁖· each denies the other
↪ closure::yet neither disappears𓆉·⁖.·˙ostensive:·
·⁖grimchain:·⊕⩒ǂΨϟ⍑0ᴙ₪▽ᗡ𑣲Ⱬ·⁖.·˙grimchain:·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦ᐸ꞉ category○–◦court꞉·
·⁖lookup office○–◦substrate꞉ enoch○–◦ᗰ꞉·

˖word:·ᓚIȢ̄⁖.·˙word:·

·⁖origin…language:·Tocharian B·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·SAHRM·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Seed·⁖.·˙non…ostensive:·
·⁖ostensive:·Seed coat encloses root and leaf in folded matter::and whole growth lies compact inside
↪ the same small bodyᐸ·⁖.·˙ostensive:·
·⁖grimchain:·ᘐ𐋡▽𑣲ᚹ𐋏ᨒ𐋡ﻉ☆ᚹᗅᖶ·⁖.·˙grimchain:·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦ᯕ꞉ category○–◦court꞉·
·⁖lookup office○–◦substrate꞉ enoch○–◦ᗰ꞉·

˖word:·xəʞı⁖.·˙word:·

·⁖origin…language:·Coptic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁖.·˙font:·
·⁖pronunciation:·KHAH+kee·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Darkness·⁖.·˙non…ostensive:·
·⁖ostensive:·Darkness gathers where light is stopped and keeps close to the shelter that made
↪ itᯕ·⁖.·˙ostensive:·
·⁖grimchain:·⊙ᗡⵣℂzЌ○✡ỏϟ◎ᕆǂ·⁖.·˙grimchain:·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦ᗰ꞉ category○–◦court꞉·
·⁖lookup office○–◦substrate꞉ enoch○–◦ᗰ꞉·

˖word:·ɹɔsɯɒɔ·⁖.·˙word:·

·⁖origin…language:·Avestan·⁖.·˙origin…language:·
·⁖font:·Noto Sans Avestan·⁖.·˙font:·
·⁖pronunciation:·SAH+mah·⁖.·˙pronunciation:·
·⁖cadence:·Substrate·⁖.·˙cadence:·
·⁖non…ostensive:·Black::dark in color·⁖.·˙non…ostensive:·
·⁖ostensive:·Black surface absorbs falling light::and little brightness returns from the same dark
↪ faceᗰ·⁖.·˙ostensive:·
·⁖grimchain:·ᓀ۞ᗝᕮᗅΛᯕzᗝᕼ◌ᴙ·⁖.·˙grimchain:·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol○–◦ᗢ꞉ category○–◦court꞉·
·⁖lookup office○–◦substrate꞉ enoch○–◦ᗰ꞉·

꞉word꞉·𐎁𐎍𐎘꞉꞉·˙word꞉·

·⁚origin…language꞉·Ugaritic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Ugaritic·⁚·˙font꞉·
·⁚pronunciation꞉·GLTH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Snow·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Snow lies white on the ground꞉꞉and breath meeting the air above it loses warmth at
↪ once꞉·⁚·˙ostensive꞉·
·⁚grimchain꞉·𐎗𐎚𐎪𐎰𐎲𐎽𐎫𐎭𐎵𐎠·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉𐎠 category꞉court꞉·
·⁚lookup office꞉substrate꞉ enoch꞉𐎀꞉·

꞉word꞉·𐔰𐔲𐔱꞉꞉·˙word꞉·

·⁚origin…language꞉·Aghwan·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Caucasian Albanian·⁚·˙font꞉·
·⁚pronunciation꞉·XASH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Light·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Light carries the shape of what it touches into the eye꞉꞉and likeness appears without the
↪ thing itself crossing the distance𐎠·⁚·˙ostensive꞉·
·⁚grimchain꞉·𐎗𐎚𐎶𐎪𐎴𐎽𐎫𐎭𐎵𐎠·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉𐎫 category꞉court꞉·
·⁚lookup office꞉substrate꞉ enoch꞉𐎀꞉·

꞉word꞉·𓏏꞉꞉·˙word꞉·

·⁚origin…language꞉·Egyptian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Egyptian Hieroglyphs·⁚·˙font꞉·
·⁚pronunciation꞉·SHEH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚·˙cadence꞉·
·⁚non…ostensive꞉·A lake or pool·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Water holds descending body and rising surface in one depth꞉꞉and every stroke belongs to
↪ the same enclosing pool𐎫·⁚·˙ostensive꞉·
·⁚grimchain꞉·𐎑𐎙𐎠𐎬𐎲𐎽𐎢𐎠𐎭𐎠𐎗𐎚𐎮·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉𐎭 category꞉court꞉·
·⁚lookup office꞉substrate꞉ enoch꞉𐎀꞉·

꞉word꞉·𓉔꞉꞉·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚·˙font꞉·
·⁚pronunciation꞉·SHAHR꞊toom·⁚·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚·˙cadence꞉·
·⁚non…ostensive꞉·a single hair꞉꞉hair꞉꞉fur꞉꞉goat hair (as raw material)·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Twisted hair holds a curved turn along its fibre꞉꞉and the bent strand keeps the same
↪ torsion through its body𐎭·⁚·˙ostensive꞉·

·⁚grimchain⁚·▽'ᚏX▽ᚱᛉᚻ∫ᛯℂℏ♳·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑂» category◦–◦court»⁚·
·⁚lookup office◦–◦substrate» enoch◦–◦𐑃»⁚·

·word⁚·ᏰᏒᑎSTS⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·BRUSTS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Breast::chest·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Breath shakes inside the chest while the body advances into the force pressing it back⁚˙
↪ the breast carries fear and forward motion together𐑂·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐑆ᚏᑫᚺᑫᚸ‹☆ᚗᛝᚏ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑓» category◦–◦court»⁚·
·⁚lookup office◦–◦substrate» enoch◦–◦𐑃»⁚·

·word⁚·ᚻI⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·NYAHREH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Thread·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Thread circles separate bodies and draws the open distance closed𐑓·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Θᑭᚿᛞᗺ⬡ᑎᑫᖘᘔᗄ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑐» category◦–◦court»⁚·
·⁚lookup office◦–◦substrate» enoch◦–◦𐑃»⁚·

·word⁚·ᛏᚾᛁ𒁍⁚.·˙word⁚·

·⁚origin…language⁚·Hittite·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·AEESH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Mouth·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Incoming air and outgoing breath share one living passage::and the open edge of the lips
↪ marks where the inner chamber meets the outer air𐑐·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ΦᛏᛘᚩᏀᚷᚸᘔᚗᚣᛣ𒁍·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑖» category◦–◦court»⁚·
·⁚lookup office◦–◦substrate» enoch◦–◦𐑃»⁚·

·word⁚·𑄃⁚.·˙word⁚·

·⁚origin…language⁚·Egyptian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Egyptian Hieroglyphs·⁚.·˙font⁚·

·⁖pronunciation:·JEH⟊rat·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·Hand·⁚.·˙non…ostensive:·
·⁖ostensive:·Hand closes around one object⁚⁚and open fingers leave surrounding objects outside the same
↪ grasp·⁚.·˙ostensive:·
·⁖grimchain:·⊙♈♓◎ᛋᚿᛰ⬡𓆏ᚗᚽᛃ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜𓃰 category꩜꩜court꩜:·
·⁖lookup office꩜꩜substrate꩜ enoch꩜꩜𓃭꩜:·

⁖word:·cok⁖⁚.·˙word:·

·⁖origin…language:·Coptic·⁚.·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁚.·˙font:·
·⁖pronunciation:·sok·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·sackcloth·⁚.·˙non…ostensive:·
·⁖ostensive:·Sackcloth and hidden shape occupy the same place⁚⁚and rough fibre stands between the eye
↪ and what lies beneath𓃰·⁚.·˙ostensive:·
·⁖grimchain:·♓⊛ᚱᚿᑫᑮᗪ♂ᛏᚼᛉᛦ𓃭·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜Ⲏ꩜ category꩜꩜court꩜:·
·⁖lookup office꩜꩜substrate꩜ enoch꩜꩜𓃭꩜:·

⁖word:·ᎢᏦIᑎ⁖⁚.·˙word:·

·⁖origin…language:·Gothic·⁚.·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁚.·˙font:·
·⁖pronunciation:·TRYOO·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·Piece of wood·⁚.·˙non…ostensive:·
·⁖ostensive:·Wood carries severed grain in the same worked body⁚⁚and fibres stop cleanly at the opened
↪ edgeⲎ·⁚.·˙ostensive:·
·⁖grimchain:·ᛋᛕᘐ◌☿⧗∀ᨒ꩜ᡕᦓᘓ◌ᦒᗷ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜A꩜ category꩜꩜court꩜:·
·⁖lookup office꩜꩜substrate꩜ enoch꩜꩜𓃭꩜:·

⁖word:·कण्ठ⁖⁚.·˙word:·

·⁖origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁚.·˙font:·
·⁖pronunciation:·kuhn⟊thuh·⁚.·˙pronunciation:·
·⁖cadence:·Substrate·⁚.·˙cadence:·
·⁖non…ostensive:·throat⁚⁚neck⁖˙ the narrowest part of something⁖˙ voice box⁖˙ voice⁖˙ guttural
↪ sound·⁚.·˙non…ostensive:·
·⁖ostensive:·Throat holds breath and pitch together⁚⁚and voice takes its sounding body inside that
↪ narrow passageA·⁚.·˙ostensive:·
·⁖grimchain:·🢒0☿ᛕ⬡✪⊙ᛁᛕᒍᐱ❁·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol꩜꩜𐑉 category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜substrate꩷ enoch꩜꩜𐑒꩷⁚·

⁖word⁚·صصف⁖.·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚.·˙font⁚·
·⁚pronunciation⁚·boy·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·colour⁝paint·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Paint settles into cloth or wall and gives the same surface another visible
↪ face𐑉·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⏀⍥⊛ᔑ⊙▽⟰ᚼᛇ𓆑ᚾ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜Ↄ꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜substrate꩷ enoch꩜꩜𐑒꩷⁚·

⁖word⁚·𐡀𐡌⁖.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·hoo+RAH+soo·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Gold⁚˙ gold as a precious metal·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One weighed piece enters the store⁚˙ counted metal increases against the same balance and
↪ sealed chestↃ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ℂ𐑘ϒ𐑁⊴ᛘᛯᚥᘐↃ𐑤⊕ᓚ‹·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐑁꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜substrate꩷ enoch꩜꩜𐑒꩷⁚·

⁖word⁚·𐨎𐨪𐨟⁖.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·WREHMEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Object⁝item·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Single object stands apart⁝and its bounded edge closes one countable body against
↪ surrounding space𐑁·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᗅᛕᚴᚱᚸ◎ᘁᚼᛈ◎⁕ᘐᚴ⁕·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐑑꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜substrate꩷ enoch꩜꩜𐑒꩷⁚·

⁖word⁚·⁕ᑎᑭ⁖.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·HOOTH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚.·˙cadence⁚·

·⁖non…ostensive⁚·Skin or hide·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Skin tightens and rises in sudden points under an unexpected crash·⁖ cold pressure races
↪ across the surface before breath settles↯·⁖.·˙ostensive⁚·
·⁖grimchain⁚·‹ሕ◌Iɾ𝛭ₒ♉ᛯ▽Xɾ♪·⁖.·˙grimchain⁚·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ɬ꞉ category◦–◦court꞉⁚·
·⁖lookup office◦–◦substrate꞉ enoch◦–◦ꙮ꞉⁚·

⁖word⁚·स्नावन्⁚·⁖.·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁖.·˙font⁚·
·⁖pronunciation⁚·snah+vuhn·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·sinew::tendon::muscle·⁖ bow+string·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Three taut strings share one hollow frame·⁖ tension in each strand bears a pitch::and the
↪ joined tensions occupy one sounding intervalɬ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·XΘ𐌡ɑɬ6ᔓᛯ𝛭ь𐐕ᗡ𐌟·⁖.·˙grimchain⁚·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ꞁ꞉ category◦–◦court꞉⁚·
·⁖lookup office◦–◦substrate꞉ enoch◦–◦ꙮ꞉⁚·

⁖word⁚·ᛒᚱᚢᚦ⁚·⁖.·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖.·˙font⁚·
·⁖pronunciation⁚·BLOHTH·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Blood·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Blood moves beneath skin in a returning pulse::and vessel walls carry the same repeating
↪ pressure through the bodyI·⁖.·˙ostensive⁚·
·⁖grimchain⁚·𓆏ᚹ✡ᔕ𐌀ᛞᘔᔑ𐐕ꙮℂᕱ𝛭·⁖.·˙grimchain⁚·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ᗡ꞉ category◦–◦court꞉⁚·
·⁖lookup office◦–◦substrate꞉ enoch◦–◦ꙮ꞉⁚·

⁖word⁚·वाणी⁚·⁖.·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁖.·˙font⁚·
·⁖pronunciation⁚·vah+nee·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·sound::voice::music·⁖ sound::voice::music⁖ a choir of musicians or singers·⁖
↪ speech::language::words::diction::(especially) eloquent speech or fine diction·⁖ a literary
↪ production or composition·⁖ praise::laudation·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Ordered vibration fills the air::and silence lies around each sounded boundary without
↪ entering the sounding fieldᗡ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᔓ𐌡ᔑ𐐕ƳƳ♎ꟿᗡ⊙ᔑ𝛨⬡✱·⁖.·˙grimchain⁚·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁖glyph symbol◦–◦𐤊꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦substrate꞉ enoch◦–◦𐤀꞉⁖·

⁖word⁖·ᘔᘐ⁖.·˙word⁖·

·⁖origin…language⁖·Phoenician·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Phoenician·⁖.·˙font⁖·
·⁖pronunciation⁖·YAHM·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Substrate·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Sea⁖⁘ a large body of open water·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Two currents meet within open water at a shifting seam⁖⁘ one moving line enters from one
↪ reach and another from a second⁘⁘and both continue beyond the shared water𐤊·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ϒ⬡ᛪᛉ⊖⊕ᛊ𐤉ᚱᘔᚣ𐤓ᚸ꞉·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol◦–◦𐤋꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦substrate꞉ enoch◦–◦𐤀꞉⁖·

⁖word⁖·ሣሬት⁖.·˙word⁖·

·⁖origin…language⁖·Ge'ez·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Ethiopic·⁖.·˙font⁖·
·⁖pronunciation⁖·SAH⟊ret·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Substrate·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Spider or cobweb·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Cobweb spans open space by holding one point to another with a thread light enough to
↪ disappear in air𐤋·⁖.·˙ostensive⁖·
·⁖grimchain⁖·∂ϱ⊖⊛ᚔᚹᔕᚔ⊖ᔓᔑᔐᔕz·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol◦–◦𐤌꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦substrate꞉ enoch◦–◦𐤀꞉⁖·

⁖word⁖·दन्त⁖.·˙word⁖·

·⁖origin…language⁖·Sanskrit·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Devanagari·⁖.·˙font⁖·
·⁖pronunciation⁖·duhn⟊tuh·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Substrate·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·tooth⁖⁘ an elephant's tusk⁘⁘ivory⁖⁘ the point of an arrow·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Hard material narrows to one point⁘⁘and concentrated force occupies the same small contact
↪ face𐤌·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᛯᚏᘓᚑᛇᚱᘐᛃᚏᓄᚥᛉ✡ᗡ·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol◦–◦𐤉꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦substrate꞉ enoch◦–◦𐤀꞉⁖·

⁖word⁖·ᑭᗅᔑᔕᗅ⁖.·˙word⁖·

·⁖origin…language⁖·Etruscan·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Old Italic·⁖.·˙font⁖·
·⁖pronunciation⁖·RAHS⟊nah·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Substrate·⁖.·˙cadence⁖·

·⁖non…ostensive:·People⁖· a body of persons understood collectively as a community or
↪ nation·⁚·˙non…ostensive:·
·⁖ostensive:·One gathered people opens into many outward roads⁖· each departing body carries the same
↪ shared pattern::and every spreading line remains joined to the whole from which it
↪ moves꙱·⁚·˙ostensive:·
·⁖grimchain:·𐎗𐎚𐎐𐎖𐎕𐎘𐎙𐎒𐎓𐎔𐎍·⁚·˙grimchain:·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁖glyph symbol◦–◦𐎚, category◦–◦court꞉·
·⁖lookup office◦–◦substrate, enoch◦–◦𐎋꞉·

꞉word:·रन्ध्र꞉·⁚·˙word:·

·⁖origin…language:·Sanskrit·⁚·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁚·˙font:·
·⁖pronunciation:·ruhn+dhruh·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·a slit::split::opening::aperture::hole::chasm::fissure::cavity⁖· the vulva⁖· a
↪ defect::fault::flaw::imperfection·⁚·˙non…ostensive:·
·⁖ostensive:·Narrow opening breaks the wall of enclosure::and open passage occupies the same bounded
↪ gap through the material𐎚·⁚·˙ostensive:·
·⁖grimchain:·ɔ𐎓𐎍𐎗𐎐𐎚𐎛𐎜𐎝𐎞𐎟𐎠𐎡𐎢·⁚·˙grimchain:·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁖glyph symbol◦–◦𐎚, category◦–◦court꞉·
·⁖lookup office◦–◦substrate, enoch◦–◦𐎋꞉·

꞉word:·𐴢𐴣𐴤𐴥·⁚·˙word:·

·⁖origin…language:·Tocharian B·⁚·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁚·˙font:·
·⁖pronunciation:·MAHEEYYOH·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·Power::might::strength·⁚·˙non…ostensive:·
·⁖ostensive:·Strength and opposing pressure inhabit the same body::and the body keeps its line inside
↪ the meeting of both forces𐎚·⁚·˙ostensive:·
·⁖grimchain:·ↃΨϔ𐌳𐍈𐤉𐦟𐦠𐦡𐦢𐦣𐦤·⁚·˙grimchain:·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁖glyph symbol◦–◦𐎗, category◦–◦court꞉·
·⁖lookup office◦–◦substrate, enoch◦–◦𐎋꞉·

꞉word:·ⲁⲟϒⲁⲛ꞉·⁚·˙word:·

·⁖origin…language:·Coptic·⁚·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁚·˙font:·
·⁖pronunciation:·ah+wahn·⁚·˙pronunciation:·
·⁖cadence:·Substrate·⁚·˙cadence:·
·⁖non…ostensive:·Color⁖· appearance·⁚·˙non…ostensive:·
·⁖ostensive:·Green surface reddens through ripening while the same skin remains⁖· red spreads across
↪ the face where green had occupied it𐎗·⁚·˙ostensive:·
·⁖grimchain:·ग𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅·⁚·˙grimchain:·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙

·⁚glyph symbol꩜꩜𞤦꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐋡꩞⁚·

⁖word⁚·𓍿𓍿⁖·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·OON·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·People⁚· a body or population of human beings considered together·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Many households keep one boundary and one measure⁚· the same command is observed from
↪ dwelling to field across the gathered people𞤦·⁚·˙ostensive⁚·
·⁚grimchain⁚·𞤟𞤍𞤀𞤘⊙ᗡᘔꓧᚺ✡ꓤꓩ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚·
·⁚·˙glyphs⁚·
·⁚glyph symbol꩜꩜𐤏꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐋡꩞⁚·

⁖word⁚·⁚·𑀉𑀻𑁆⁖·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·EEPREHR·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Sky⁚⁚air·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Open air moves through leaf⁚⁚blossom⁚⁚and fruit across tended beds⁚· one breathing sky
↪ carries scent⁚⁚pollen⁚⁚and rain between every rooted patch𐤏·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚺᑎᗅ∀Ⲓ𐎍𞤖ᚢᗅ⊙ΨΨΨᗅ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚·
·⁚·˙glyphs⁚·
·⁚glyph symbol꩜꩜0꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐋡꩞⁚·

⁖word⁚·आतप⁖·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·ah+tuh+puh·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·
·⁚non…ostensive⁚·sunshine⁚⁚sunlight⁚⁚sunbeam⁚· heat⁚· moonshine·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Sunlight and brightness share the same falling ray⁚⁚and every touched edge stands clear
↪ within that light0·⁚·˙ostensive⁚·
·⁚grimchain⁚·Θ𐎍ƵⲞ𐤏ᵪƆ⓪ ⊛𞤀ᛋᚸ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚·
·⁚·˙glyphs⁚·
·⁚glyph symbol꩜꩜𞤋꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜substrate꩞ enoch꩜꩜𐋡꩞⁚·

⁖word⁚·𓄿𓍿𓎛𓏺𓂀⁖·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·eh+MOO+koom·⁚·˙pronunciation⁚·
·⁚cadence⁚·Substrate·⁚·˙cadence⁚·

·⁖non…ostensive:·strength⁚⁚power⁚⁚force⁚⁚ability·⁖· armed forces⁚⁚army·⁖· value⁚⁚ability⁚⁚executive
↪ power·⁖·˙non…ostensive:·
·⁖ostensive:·Strength rests unused in a strung bow⁚⁚and bent wood and taut cord already hold the shot
↪ without an arrow flying᛬·⁖·˙ostensive:·
·⁖grimchain:·▽ᒣᔑᔐᔕᔆᔦᔑᗝᗜᑎᔦᛁᚲ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꜀꜀ǂ꜆ category꜀꜀court꜆:·
·⁖lookup office꜀꜀substrate꜆ enoch꜀꜀ꕥ꜆:·

⁖word:·ጾላዕ:⁖·˙word:·

·⁖origin…language:·Ge'ez·⁖·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁖·˙font:·
·⁖pronunciation:·tsoh⟊LAH·⁖·˙pronunciation:·
·⁖cadence:·Substrate·⁖·˙cadence:·
·⁖non…ostensive:·Stone⁚⁚rock⁚⁚or cliff·⁖·˙non…ostensive:·
·⁖ostensive:·Stone surrounds the opening and holds back the cliff⁚⁚leaving one clear way through what
↪ would otherwise be wallǂ·⁖·˙ostensive:·
·⁖grimchain:·ⵖᵐᔓᑉᗝᚲᛕᐱᗝᔑᑎᔓz·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꜀꜀ᡔ꜆ category꜀꜀court꜆:·
·⁖lookup office꜀꜀substrate꜆ enoch꜀꜀ꕥ꜆:·

⁖word:·उल्ब:⁖·˙word:·

·⁖origin…language:·Sanskrit·⁖·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖·˙font:·
·⁖pronunciation:·ul⟊buh·⁖·˙pronunciation:·
·⁖cadence:·Substrate·⁖·˙cadence:·
·⁖non…ostensive:·cover⁚⁚envelope·⁖· cave·⁖· womb⁚⁚membrane surrounding the embryo·⁖· vulva·⁖·˙non…ostensive:·
·⁖ostensive:·Membrane holds inside and outside against the same living boundary⁚⁚and the opening
↪ belongs to the very covering that separates them᛬·⁖·˙ostensive:·
·⁖grimchain:·ꠇꠥ⊛ⵜ꠸ગᑭᑘᐽᑊᑘᕏᔑᑫ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꜀꜀ᚱ꜆ category꜀꜀court꜆:·
·⁖lookup office꜀꜀substrate꜆ enoch꜀꜀ꕥ꜆:·

⁖word:·ⵞⵔⵖⵏ:⁖·˙word:·

·⁖origin…language:·Aghwan·⁖·˙origin…language:·
·⁖font:·Noto Sans Caucasian Albanian·⁖·˙font:·
·⁖pronunciation:·KUL·⁖·˙pronunciation:·
·⁖cadence:·Substrate·⁖·˙cadence:·
·⁖non…ostensive:·Hand·⁖·˙non…ostensive:·
·⁖ostensive:·Soft clay remains enclosed by palm and fingers while its contour changes·⁖· every altered
↪ surface is borne within the same living graspᚱ·⁖·˙ostensive:·
·⁖grimchain:·ᛒᦓᘐΘ▫ᗡᐃᛥᛇᛯᐱᔕᐁ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꜀꜀ᚷ꜆ category꜀꜀court꜆:·
·⁖lookup office꜀꜀substrate꜆ enoch꜀꜀ꕥ꜆:·

:word:·𐎍𐎗:.·˙word:·

·˙origin…language:·Ugaritic·˙.·˙origin…language:·
·˙font:·Noto Sans Ugaritic·˙.·˙font:·
·˙pronunciation:·AHISU·˙.·˙pronunciation:·
·˙cadence:·Substrate·˙.·˙cadence:·
·˙non…ostensive:·Tree::wood·˙.·˙non…ostensive:·
·˙ostensive:·Tree rises through a living center that carries what grows above from what is rooted
↪ below𐎗·˙.·˙ostensive:·
·˙grimchain:·𐎍𐎚𐎓𐎔𐎄𐎐𐎍𐎗𐎐𐎍𐎚·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol꞉꞉𐎚» category꞉꞉court»·
·˙lookup office꞉꞉substrate» enoch꞉꞉𐎐»·

:word:·𐎗𐎍·:.·˙word:·

·˙origin…language:·Akkadian·˙.·˙origin…language:·
·˙font:·Noto Sans Cuneiform·˙.·˙font:·
·˙pronunciation:·pahr+ZEEL+loom·˙.·˙pronunciation:·
·˙cadence:·Substrate·˙.·˙cadence:·
·˙non…ostensive:·iron·˙ (with determinative 𒁹 (na₄)) a kind of stone or bead::iron
↪ stone»·˙.·˙non…ostensive:·
·˙ostensive:·Iron pin holds the turning center::and wheel or hinge rotates around the same fixed
↪ axis𐎚·˙.·˙ostensive:·
·˙grimchain:·𐎄𐎅𐎘𐎗𐎂𐎍𐎗𐎐𐎍𐎚𐎐𐎍𐎗·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol꞉꞉𐎂» category꞉꞉court»·
·˙lookup office꞉꞉substrate» enoch꞉꞉𐎐»·

:word:·𐎗𐎄𐎅𐎐·:.·˙word:·

·˙origin…language:·Tocharian A·˙.·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙.·˙font:·
·˙pronunciation:·AHREENYCH·˙.·˙pronunciation:·
·˙cadence:·Substrate·˙.·˙cadence:·
·˙non…ostensive:·Heart::used both as the physical organ and the seat of emotion·˙.·˙non…ostensive:·
·˙ostensive:·Pulse settles beneath the breast·˙ muscle loosens around the chamber::and breath moves
↪ through the ribs without inward strain𐎂·˙.·˙ostensive:·
·˙grimchain:·𐎘𐎚𐎓𐎗𐎍𐎔𐎂𐎗𐎄𐎍𐎘𐎗𐎍𐎚·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol꞉꞉𐎘» category꞉꞉court»·
·˙lookup office꞉꞉substrate» enoch꞉꞉𐎐»·

:word:·𐎗𐎓𐎗𐎐·:.·˙word:·

·˙origin…language:·Tocharian B·˙.·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙.·˙font:·
·˙pronunciation:·AHLAHMBAHN·˙.·˙pronunciation:·
·˙cadence:·Substrate·˙.·˙cadence:·
·˙non…ostensive:·Basis::ground::support::reason·˙.·˙non…ostensive:·

·⁖ostensive⁚·Ground receives downward weight⁚⁚and equal resistance rises through the same bearing
↪ surface⁚⁚and standing mass holds above the contactϫ·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᦅᦈᦈᧁᐞᛊᛉᛯᚴᛉᛔᐩᐞz·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol꩜꩜ᵹᑭ꞉ category꩜꩜ᵹcourt꞉⁚·
·⁖lookup office꩜꩜ᵹsubstrate꞉ enoch꩜꩜ᵹ𓃰꞉⁚·

˖word⁚·सोम˖⁖·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁖·˙font⁚·
·⁖pronunciation⁚·SOH⟊muh·⁖·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Soma⁚⁚the pressed ritual plant juice and the plant from which it is
↪ obtained·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Green stalk yields its sap beneath the pressing stone⁚⁚the strained juice bears the whole
↪ body of the plant in one bowl⁚⁚and the bowl that holds the strained juice stands apart from the
↪ common draughtᑭ·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᑫᘜᗅ⟊ᑲᐁᐃᐁᐞ𓆏ᗺᘐ·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol꩜꩜ᵹᔑ꞉ category꩜꩜ᵹcourt꞉⁚·
·⁖lookup office꩜꩜ᵹsubstrate꞉ enoch꩜꩜ᵹ𓃰꞉⁚·

˖word⁚·ᘐᘓ˖⁖·˙word⁚·

·⁖origin…language⁚·Aghwan·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Caucasian Albanian·⁖·˙font⁚·
·⁖pronunciation⁚·ASH·⁖·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Work·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Clean join holds under full strain·⁖· fitted edges meet without gap⁚⁚and the finished work
↪ bears no correction markᑐ·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᨔ⬡𓆏ᓯ♀ᚹᐧᑎᓄᗡᙏ⫲ᛞ·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol꩜꩜ᵹᒧ꞉ category꩜꩜ᵹcourt꞉⁚·
·⁖lookup office꩜꩜ᵹsubstrate꞉ enoch꩜꩜ᵹ𓃰꞉⁚·

˖word⁚·𐤟˖⁖·˙word⁚·

·⁖origin…language⁚·Akkadian·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Cuneiform·⁖·˙font⁚·
·⁖pronunciation⁚·EE⟊noo·⁖·˙pronunciation⁚·
·⁖cadence⁚·Substrate·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Eye⁚⁚organ of sight·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Visible motion stands within direct sight⁚⁚and light carries the same event into the
↪ eyeᒧ·⁖·˙ostensive⁚·
·⁖grimchain⁚·'⏁ℳ⍦ᚼᛯ𓆏ᛀᛯ✪ᛯ)ᚼ✡⬡·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol꩜꩜ᵹᒧ꞉ category꩜꩜ᵹcourt꞉⁚·
·⁖lookup office꩜꩜ᵹsubstrate꞉ enoch꩜꩜ᵹ𓃰꞉⁚·

˖word꞉·𐎼𐎼÷꞉.·˙word꞉·

·⁚origin…language꞉·Old Persian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Persian·⁚.·˙font꞉·
·⁚pronunciation꞉·BOO⟊meesh·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Substrate·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Earth꞉꞉land꞉꞉ground·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Earth holds the standing place beneath every passing body꞉꞉and ground remains fixed under
↪ changing weightᒧ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᛯᚲᚸᗅᦾ꙰ᙀ꙰ᘔᦾᘏᘔᦾᘏᘔᦾᘏ꙰ᘔᦾᘏ꙰ᘔᦾᘏ꙰

·⁚grimchain:·▫ꟾ𐓛𐓓𐓯𐓪⊚ꟾ☆✪Ƙ𐓟𐓮·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵩ꩝ category꩜꩜ᵩcourt꩝:·
·⁚lookup office꩜꩜ᵩsubstrate꩝ enoch꩜꩜ᵩ꩝:·

·word:·ᵬOϟϤOϟ:⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·JUMUKH·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Mouth·⁚.·˙non…ostensive:·
·⁚ostensive:·Mouth holds the sound behind closed lips::keeping breath within the place where speech ↪ would cross outwardᵬ·⁚.·˙ostensive:·
·⁚grimchain:·ᛯ✲ǂ֍⊚♂Πᵮᵯᵨϕ⟰▽·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵩƻ꩝ category꩜꩜ᵩcourt꩝:·
·⁚lookup office꩜꩜ᵩsubstrate꩝ enoch꩜꩜ᵩ꩝:·

·word:·ሌሊት:⁚.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·leh꜔LEET·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·Night::the period of darkness between sunset and sunrise·⁚.·˙non…ostensive:·
·⁚ostensive:·Dark span holds rest beneath unlit sky::and still limbs occupy the same field with dawn ↪ outside its darknessƻ·⁚.·˙ostensive:·
·⁚grimchain:·ⴱ⁜✪ᛁᛋᑫᓄ✪𐓛‹ɦ⁒ℂ⊄·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵨ꩝ category꩜꩜ᵩcourt꩝:·
·⁚lookup office꩜꩜ᵩsubstrate꩝ enoch꩜꩜ᵩ꩝:·

·word:·ᘐገᔑ:⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·AYZ·⁚.·˙pronunciation:·
·⁚cadence:·Substrate·⁚.·˙cadence:·
·⁚non…ostensive:·World·⁚· land::region·⁚.·˙non…ostensive:·
·⁚ostensive:·One horizon encloses land::water::sky::and dwelling::and every bounded place occupies the ↪ same continuous fieldϕ·⁚.·˙ostensive:·
·⁚grimchain:·ᘿᚼ৸ᘐ긔Πᖵᚿᘐᘏ⟲ᚿᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵮ꩝ category꩜꩜ᵩaxiomyr꩝:·
·⁚lookup office꩜꩜ᵩsubstrate꩝ enoch꩜꩜ᵩ꩝:·

·word:·ᓭᘿ:⁚.·˙word:·

·⁖origin…language:·Akkadian·⁘·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁘·˙font:·
·⁖pronunciation:·SHOOR-shoom·⁘·˙pronunciation:·
·⁖cadence:·Substrate·⁘·˙cadence:·
·⁖non…ostensive:·root⁖base⁚⁚foundation.·⁘·˙non…ostensive:·
·⁖ostensive:·Mountain weight rests on buried stone⁚⁚Beneath the forest⁚⁚dark soil holds a thousand
↪ roots⁖wet clay keeps the print of a hand⁚⁚Riverbed gravel bears the moving water⁚⁚and mineral
↪ seams remain within the mountain꙰·⁘·˙ostensive:·
·⁖grimchain:·ᚱ⊛ᘔᓄ⬡∀ᖻᘐ∀ᛦᚱᘔᛚᛏ·⁘·˙grimchain:·
·⁘·˙lookup·˙
·⁘·˙glyphs·˙
·⁘·˙glyphs·˙
·⁘·˙sydonic…magicae…lexicon·∴

:·:·:·

·⁖ostensiveII:·Stone::flame::and marked ground all bend along one invisible command+bearing under chosen
↪ inward pressure♂⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Beneath closed eyes::one chosen fire presses through sleep until stone beside the
↪ resting body takes the dreamed bearing♂⁖.·˙ostensiveIII:·
·⁖grimchain:·𐑣⋗ⴱ𐑶ᖘᛋᚠᚡᚱᚲ𐑜𐑞ᓯ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰꒱♀꒱ category꒰꒱emission꒱:·
·⁖lookup office꒰꒱phantasmagoria꒱ enoch꒰꒱ᖘ꒱:·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·Órë·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Heart⁖· Inner Mind⁖· Warning⁖· Premonition⁖· Feel as Inner Knowing before
↪ Reason⁖.·˙non…ostensive:·
·⁖ostensiveI:·Warm pulse keeps its chamber beneath the ribs while quiet warning presses through the
↪ flesh::inward knowing already present without a named reason♀⁖.·˙ostensiveI:·
·⁖ostensiveII:·Before the sealed door moves::a black feather rises from still water::Cold gathers
↪ beneath the ribs while an unstruck bell shivers once in the shrine♀⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Perfect silence fills the chamber while a low pulse tightens before any visible cause
↪ and still air sharpens around the unopened door♀⁖.·˙ostensiveIII:·
·⁖grimchain:·𐑜ᖻᖷᘐᘏᚷᖷ⊛ᘐᘏᘓ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰꒱ᘓ꒱ category꒰꒱emission꒱:·
·⁖lookup office꒰꒱phantasmagoria꒱ enoch꒰꒱ᖘ꒱:·

·⁖ᘏᘐᘓ⁘:·ᘚᘐᘔᘓᘐᘐᘓᘐᘔ⁘.·˙ᘏᘐᘓ⁘:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Cass+err+am+da·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Command: To Command⁖· Language as form·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Spoken contour fills the marked circle: syllable and seal take the same exact shape
↪ while stationed forms lock beneath the pressure of the uttered orderᘓ⁖.·˙ostensiveI:·
·⁖ostensiveII:·Flame::water::dust::syllable::seal::and gesture occupy one commanded geometry inside the
↪ carved airᘓ⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One spoken form settles over the chamber::and every restless motion falls into quiet
↪ beneath its cadenceᘓ⁖.·˙ostensiveIII:·
·⁖grimchain:·ϟᖾᘏᘃᘚᘐᘐᘏᘓᘐᘔᘓᘐᘐᘓᘐᘔᘓᘐᘐᘓᘐᘔᘓᘐᘐ

·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Magic⁖˙ the substance of·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Raw magic moves beneath stone::water::and flame without showing a source::Sealed water
↪ bends toward an empty hand while the unseen substance remains real enough to move what matter
↪ can reveal♄·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Silver vapor condenses where no fire burns::gathering into a luminous weight that can
↪ stain stone::thicken water::and cling to iron like dew from an unseen world♄·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Silver pressure survives the dying flame::the spilled water::and the cracked
↪ stone::passing unchanged beneath each broken form♄·⁖.·˙ostensiveIII:·
·⁖grimchain:·⁑⌬ᖳᘁ▽Ηℤ⅃Aↄ°X∮·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–◦𓂀» category○–◦emission»:·
·⁖lookup office○–◦phantasmagoria» enoch○–◦ᖶ»:·

·:⁂₪₪ᗑ:·◡◝ᘐ◡≧◡·:.·˙⁂₪₪ᗑ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Ah+bay+k·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Time as movement⁖˙ Ritual as spell·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Stars turn above the marked stone as gesture circles through appointed motion and breath
↪ returns upon the rite with time folded inside its measure𓂀·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Seven black stones burn blue around the altar in one completed ritual circuit whose
↪ turning body is the spell𓂀·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One lamp burns at the far edge of the rite::and each returning gesture draws its light
↪ nearer along the circle before any foot has crossed the dark𓂀·⁖.·˙ostensiveIII:·
·⁖grimchain:·✲☆ɳᑫ↗✱ᚔzφɔ𐑉‹·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–◦Ψ» category○–◦emission»:·
·⁖lookup office○–◦phantasmagoria» enoch○–◦ᖶ»:·

·:⁂₪₪ᗑ:·◡◝ᐠ✳◝ᐧ◝·:.·˙⁂₪₪ᗑ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Ahd+en+ilda·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Time in the next⁖˙ What comes after⁖˙ The Forsight and the Hindsight·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Future threshold casts a pale shadow backward across the present stone::what has not
↪ opened yet leaving a readable contour beside what has passedΨ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Unmarked road::farther ground::and present threshold share one open future+bearing beyond
↪ the reached stoneΨ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One steadfast turning stone holds while forward road and returning road exchange their
↪ horizons around it::neither direction loosening the fixed centerΨ·⁖.·˙ostensiveIII:·
·⁖grimchain:·◁ↄнᘐ⁑⁑ᘐ⁑ᘐ⁑ᘐ

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Fil+ih+gar+ee+on·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Fly·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Body rises through open air from ledge to cloud::gaining height without stair or
↪ stone::Only sky bears the climbing path above the last foothold℘·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Unsupported passage::open air::and lifted body share one traversable height above the
↪ earth below℘·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Open eye catches a silver body already crossing the cloudline·⁖ the bright path remains
↪ borne in sight after the winged shape has passed℘·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᛉ⸸◎⸸A↾ϟ♂⸸ᖶᛝ∀ᚲ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥𝅮☽𝅘𝅥𝅮 category◦–◦𝅘𝅥𝅮emission𝅘𝅥𝅮⁖·
·⁖lookup office◦–◦𝅘𝅥𝅮phantasmagoria𝅘𝅥𝅮 enoch◦–◦𝅘𝅥𝅮ᖶ𝅘𝅥𝅮⁖·

·⁖𝒌𝒩𝒩𝒮𝒴:·𝒮𝒩𝒩𝒮𝒪·⁖.·˙𝒌𝒩𝒩𝒮𝒴:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Fora·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Fortune and Luck·⁖ Lady Luck·⁖ Fae Gold and Leprachauns·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Small fae coin rests upon the scale and outweighs a heap of dull metal::Strange fae
↪ brightness keeps the beam tilted against ordinary measure☽·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Green fire circles a buried cache where fae gold shines through stone for one breath
↪ and vanishes beneath closing fingers into ordinary earth☽·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Before the final fastening closes the difficult seam::one fae coin already gleams
↪ beside the work·⁖ when the join bears strain::a single clap rings and the gold vanishes into green
↪ light☽·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᚼ∀Ψᚺᚾᔑᚢℋᘜᛗ⏁·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥𝅮⊙𝅘𝅥𝅮 category◦–◦𝅘𝅥𝅮emission𝅘𝅥𝅮⁖·
·⁖lookup office◦–◦𝅘𝅥𝅮phantasmagoria𝅘𝅥𝅮 enoch◦–◦𝅘𝅥𝅮ᖶ𝅘𝅥𝅮⁖·

·⁖word:·הוּחַ·⁖.·˙word:·

·⁖origin…language:·Old Hebrew·⁖.·˙origin…language:·
·⁖font:·Noto Sans Hebrew·⁖.·˙font:·
·⁖pronunciation:·Ah+Way·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·The one or force that knows claim and fulfills return·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Heavy iron leans toward the hidden stone::scattered filings gathering along one silent
↪ pull until every loose fragment answers the same unseen claim⊙·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Unequal weights::balanced pans::claim::debt::and silent pull occupy one measured field
↪ beneath the beam⊙·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Inside a consecrated ring::scattered offerings turn of themselves and travel back to
↪ the hand+mark cut in the altar while the boundary fire never breaks⊙·⁖.·˙ostensiveIII:·
·⁖grimchain:·Ƀ⧗ᔑ▽ᖘᛯᖶᒐᓭᔑ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁖glyph symbol◦-◦�really category◦-◦emission⁘·
·⁖lookup office◦-◦phantasmagoria⁘ enoch◦-◦ᚲ⁘·

·⁖ᚻᛞᛏ⁚·⁘✳⁘ᛞᛏᗣ⁖·˙ᚻᛞᛏ⁚·

·⁖origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁖font:·Mage Script·⁚·˙font:·
·⁖pronunciation:·Ehn+ter+rah·⁚·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁚·˙cadence:·
·⁖non…ostensive:·Self as Direction·⁘ enter self·⁘ return self·⁘ self as identity·⁚·˙non…ostensive:·
·⁖ostensiveI:·Path curls inward around a fixed center::every outward bearing tied to that interior point until return meets the same place from another direction℃·⁚·˙ostensiveI:·
·⁖ostensiveII:·A black compass points inward while every silver road curves back toward the same dark center from a different horizon℃·⁚·˙ostensiveII:·
·⁖ostensiveIII:·Equal silver weights hang from paths leaving one black center::and every returning road finds the same inward point without tipping the balance℃·⁚·˙ostensiveIII:·
·⁖grimchain:·⧗ℏ⁂ᚻᛊᛋ⚦ᛇZᚱ⁀ᛦ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦-◦♈⁘ category◦-◦parliament⁘·
·⁖lookup office◦-◦phantasmagoria⁘ enoch◦-◦ᚲ⁘·

·⁖origin…language:·Quenya·⁚·˙origin…language:·
·⁖font:·Tengwar Quenya·⁚·˙font:·
·⁖pronunciation:·Valaquenta·⁚·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁚·˙cadence:·
·⁖non…ostensive:·Account of the Valar·⁘ Sacred Account·⁘ Record as Memory made Enduring Word·⁚·˙non…ostensive:·
·⁖ostensiveI:·Names of the Valar stand in sacred telling::elder deeds keeping their radiance inside remembered words until vanished divine voices leave no vacancy in the record♈·⁚·˙ostensiveI:·
·⁖ostensiveII:·Sacred names and carved deeds keep vanished divine voices materially present in enduring stone beneath silence♈·⁚·˙ostensiveII:·
·⁖ostensiveIII:·Carved names remain bright in weathered stone while quiet doors stand open beside resting weapons♈·⁚·˙ostensiveIII:·
·⁖grimchain:·↘⚮℘▽ɬ☆Aƨ♁ᓭϟᛗᑫ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦-◦♉⁘ category◦-◦parliament⁘·
·⁖lookup office◦-◦phantasmagoria⁘ enoch◦-◦ᚲ⁘·

·⁖ᚻᛞᛏ⁚·ᦓ♈ᦓ✗ᦓ♈ᦓ⁖·˙ᚻᛞᛏ⁚·

·⁖origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁖font:·Mage Script·⁚·˙font:·
·⁖pronunciation:·Ee+yoo+dee+see+um·⁚·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁚·˙cadence:·
·⁖non…ostensive:·Judgement·⁘ Mediation·⁘ Mediate as Identity·⁚·˙non…ostensive:·
·⁖ostensiveI:·Two weights rest upon opposed pans beneath one measure: neither rises by favor::and both claims remain fully exposed♉·⁚·˙ostensiveI:·
·⁖ostensiveII:·Two rival flames and one steady white ray coexist inside a clear crystal::both pressures present without either becoming the other♉·⁚·˙ostensiveII:·

·⁖ostensiveIII:·Two rival pressures turn around one still middle as a star+wheel turns around its pin⁖·
↪ neither side crosses the center♉·⁖·˙ostensiveIII:·
·⁖grimchain:·𐦍'ᘏ𑁍𑁋⟫ᚼᚻᛞᛋ❄ᘓ⁞⁞ᚼ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜𝌆꩝ category꩜꩜parliament꩝⁖·
·⁖lookup office꩜꩜phantasmagoria꩝ enoch꩜꩜ᛕ꩝⁖·

·⁖𝒱⌒ᛕᒥ:·ᚻᚴᚻ𝍐⁖·˙·𝒱⌒ᛕᒥ:·

·⁖origin…language:·Adûnaic·⁖·˙origin…language:·
·⁖font:·Adunaroth Classic·⁖·˙font:·
·⁖pronunciation:·A+ban·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·The Body of World Itself⁖· Island of Inhabitation·⁖·˙non…ostensive:·
·⁖ostensiveI:·Island keeps its shoreline when dreamlight withdraws::stone and tide remaining beneath
↪ the same sky after every illusion goes transparent around them𝌆·⁖·˙ostensiveI:·
·⁖ostensiveII:·Luminous surf closes around a dark island where roots::nests::hearth+fire::and rain all
↪ remain inside one breathing ring of land𝌆·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A green island rises around one hollow root+channel running from sea+dark depth to
↪ inhabited ground::and the whole living shore breathes through that single open
↪ stem𝌆·⁖·˙ostensiveIII:·
·⁖grimchain:·☽ↄↄᒐᑫᚲ⊢ℂ▽ᓚᕒ᛭ᚼᚹ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᘓ꩝ category꩜꩜parliament꩝⁖·
·⁖lookup office꩜꩜phantasmagoria꩝ enoch꩜꩜ᛕ꩝⁖·

·⁖ᘉᘏᘏᛉ:·ᔕᘏᘏᔓᘉᘏᘏᔕ·⁖·˙·ᘉᘏᘏᘏᛉ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Ahr+tyur+rah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Archer·⁖·˙non…ostensive:·
·⁖ostensiveI:·String holds a gathered curve::feathered shaft resting against the drawn line while
↪ distant mark and waiting point face one another across open airᘓ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Moonlight holds one impossible silver arrow above the bow along a bright path through
↪ cloud::wind::and distance before releaseᘓ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Drawn bow bends the air itself⁖· when the string releases::the silver shaft leaves one
↪ line and arrives along another::its course altered in flight by a turning starᘓ·⁖·˙ostensiveIII:·
·⁖grimchain:·ᚺ⊕ᛉᘐᛯᚸ᛭☽⟰ᛁᛘ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᘏ꩝ category꩜꩜parliament꩝⁖·
·⁖lookup office꩜꩜phantasmagoria꩝ enoch꩜꩜ᛕ꩝⁖·

·⁖ᘉᘏᘏᛉ:·ᯤᔕᘏᘏᔓᘏᘏᘐᘏᔕᛘᔓᘏᘏᔕ·⁖·˙·ᘉᘏᘏᘏᛉ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Zhe+raum+pel+en+aye·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·I AM⁖· Life as Proclamation⁖· Exist⁖· Magic in Motion given Passage·⁖·˙non…ostensive:·

·⁖ostensiveI:·Living pulse fills the marked circle⁚⁚spirit+fire leaning toward the same fierce center
↪ while magic in motion gives presence a passage through the living field᯲⁘·⁚·˙ostensiveI:·
·⁖ostensiveII:·Living light⁚⁚opened stone⁚⁚moving fire⁚⁚and bright passage occupy one radiant presence
↪ crossing the threshold᯲⁘·⁚·˙ostensiveII:·
·⁖ostensiveIII:·Living fire moves through a sealed gate before the bar lifts⁖ when the leaves
↪ open⁚⁚pulse and heat are already crossing the threshold into the waiting ground᯲⁘·⁚·˙ostensiveIII:·
·⁖grimchain:·◡ᛋᛊ✡ᛚᚱᛏ⊕ᛃ⊕ᚼ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦◦𓋹𓏺 category◦◦𓏺parliament𓏺:·
·⁖lookup office◦◦𓏺phantasmagoria𓏺 enoch◦◦𓏺ᛈ𓏺:·

·⁖origin…language:·Quenya·⁚·˙origin…language:·
·⁖font:·Tengwar Quenya·⁚·˙font:·
·⁖pronunciation:·Yanwë·⁚·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁚·˙cadence:·
·⁖non…ostensive:·Joining⁖ Isthmus⁖ Bridge⁖ Cohere as Living Joining·⁚·˙non…ostensive:·
·⁖ostensiveI:·Two shores remain themselves across narrow water while stone joins their distance
↪ without erasing either bank⁚⁚passage resting upon the joining body𓋹·⁚·˙ostensiveI:·
·⁖ostensiveII:·Narrow ground spans two waters between separated shores⁚⁚keeping one traversable passage
↪ where division remains visible𓋹·⁚·˙ostensiveII:·
·⁖ostensiveIII:·One stone bridge enters the dividing wall as an open arch⁚⁚joining separated shores
↪ through the same living passage𓋹·⁚·˙ostensiveIII:·
·⁖grimchain:·ᦒᔓᚼ✳ᚻᐱ⊕ᛝ〰ᗷᚹᛋᛜ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦◦𓏺𐤟𓏺 category◦◦𓏺parliament𓏺:·
·⁖lookup office◦◦𓏺phantasmagoria𓏺 enoch◦◦𓏺ᛈ𓏺:·

·⁖ᚯᘎᘏᐟ:·ᯤ²ᗣᘐᦔᑐᔕ𐦂:·⁚·˙ᚯᘎᘏᐟ:·

·⁖origin…language:·Llumena Milenii·⁚·˙origin…language:·
·⁖font:·Mage Script·⁚·˙font:·
·⁖pronunciation:·Zhi+fee+ahs·⁚·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁚·˙cadence:·
·⁖non…ostensive:·Sword Shaped⁖ Celestial Object⁖ Limb of the Divine⁖ Blade⁖ Mystical
↪ Sign·⁚·˙non…ostensive:·
·⁖ostensiveI:·Star+bright blade hums across the dark as each celestial angle releases another measured
↪ tone⁚⁚heaven carrying the radiant edge as a sounding phrase𓋹·⁚·˙ostensiveI:·
·⁖ostensiveII:·A celestial body carries one blade+long radiant limb across the dark⁚⁚its mystical edge
↪ cut sharply into night𓋹·⁚·˙ostensiveII:·
·⁖ostensiveIII:·A sword+shaped star cuts a luminous route through black heaven⁖ no barrier closes the
↪ path before its divine blade𓋹·⁚·˙ostensiveIII:·
·⁖grimchain:·ᛝᘐᦔᦓᦔᘐᗷ𐦂✩ᚻᚲ⬡·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦◦𓏺ᛚ𓏺 category◦◦𓏺parliament𓏺:·
·⁖lookup office◦◦𓏺phantasmagoria𓏺 enoch◦◦𓏺ᛈ𓏺:·

·⁖ᚯᘎᘏᐟ:·ᯤᯤᗣᘐᦔᦔᘐᦒ:·⁚·˙ᚯᘎᘏᐟ:·

·⁖origin…language꞉·Llumena Milenii·⁖.·˙origin…language꞉·
·⁖font꞉·Mage Script·⁖.·˙font꞉·
·⁖pronunciation꞉·Ehx+spell+ey·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Dispel⁘ to Dispel·⁖.·˙non…ostensive꞉·
·⁖ostensiveI꞉·Foreign sigil reaches the boundary and the ward answers with a clean flare⁚⁚casting the entering command back into the dark beyond the sealed field℘·⁖.·˙ostensiveI꞉·
·⁖ostensiveII꞉·Black frost grips the warded circle beneath untouching bronze bells⁚⁚foreign sigil already lifting into smoke from the stone℘·⁖.·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Silver light crosses the marked boundary and foreign shadow peels away from stone⁚⁚leaving the circle bright where the dark working cannot remain℘·⁖.·˙ostensiveIII꞉·
·⁖grimchain꞉·'ᘜᗡᖶ♀I↓ϴᔕᔐΛᕼᚾ·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉⟜⟜ᛯ᛬ category꞉⟜⟜parliament᛬⁖·
·⁖lookup office꞉⟜⟜phantasmagoria᛬ enoch꞉⟜⟜ᛊ᛬⁖·

·⁖origin…language꞉·Quenya·⁖.·˙origin…language꞉·
·⁖font꞉·Tengwar Quenya·⁖.·˙font꞉·
·⁖pronunciation꞉·FAH+nah·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Spiritual being assuming manifest form⁘ Become as spirit taking form·⁖.·˙non…ostensive꞉·
·⁖ostensiveI꞉·Pale radiance gathers a visible contour around unseen presence⁚⁚Luminous form stands where flesh is absent⁚⁚beyond any mortal claim upon its shapeᛯ·⁖.·˙ostensiveI꞉·
·⁖ostensiveII꞉·Around one empty shrine⁚⁚moonlit mist carries antler⁚⁚wing⁚⁚luminous hand⁚⁚and fleshless spirit+form in the same visible bodyᛯ·⁖.·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Pollen+light and leaf+shadow gather above tended roots into one fleshless spirit+form whose contour moves through the gardenᛯ·⁖.·˙ostensiveIII꞉·
·⁖grimchain꞉·ᛇᗤᖁ℘ᛔᖵᖁᚴᓭᚻᑐᛯ·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉⟜⟜ℌ᛬ category꞉⟜⟜parliament᛬⁖·
·⁖lookup office꞉⟜⟜phantasmagoria᛬ enoch꞉⟜⟜ᛊ᛬⁖·

·⁖origin…language꞉·Quenya·⁖.·˙origin…language꞉·
·⁖font꞉·Tengwar Quenya·⁖.·˙font꞉·
·⁖pronunciation꞉·Axan·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Law⁘ Rule⁘ Commandment⁘ Divine Law⁘ Law as Cosmic Command·⁖.·˙non…ostensive꞉·
·⁖ostensiveI꞉·Cosmic command stands where desire cannot bargain with it: divine measure holds its shape while power leans⁚⁚tests the law⁚⁚and finds no yieldingℌ·⁖.·˙ostensiveI꞉·
·⁖ostensiveII꞉·Stone commandment bears an oath+ring beneath crowned power⁚⁚all held inside one unyielding divine measureℌ·⁖.·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Inscribed line cuts across sacred stone⁘ every lesser mark aligns beneath it⁚⁚and no opposing weight shifts the measure held in the carvingℌ·⁖.·˙ostensiveIII꞉·
·⁖grimchain꞉·ᛋᚺᚼᛘᗅᚋᐤᘐᛯᚿᗡᘓ·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉⟜⟜ᨒ᛬ category꞉⟜⟜parliament᛬⁖·
·⁖lookup office꞉⟜⟜phantasmagoria᛬ enoch꞉⟜⟜ᛊ᛬⁖·

⁖₰₰₰:⁘ᘐ₰₰₰:⁖·˙₰₰₰:·

·⁚origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁚font:·Mage Script·⁚.·˙font:·
·⁚pronunciation:·Ehm+or+twee·⁚.·˙pronunciation:·
·⁚cadence:·Phantasmagoria·⁚.·˙cadence:·
·⁚non…ostensive:·To Pass on·⁚ To Pass Through·⁚ To Pass Into·⁚.·˙non…ostensive:·
·⁚ostensiveI:·Mist crosses threshold after threshold without choosing a dwelling::road becoming river
↪ and river becoming stone passage while the current keeps moving beyond each
↪ boundary·⁚.·˙ostensiveI:·
·⁚ostensiveII:·Ghost+light passes through a sealed wall as pale fire inhabiting the stone and
↪ appearing beyond the unbroken face·⁚.·˙ostensiveII:·
·⁚ostensiveIII:·Ghost+light enters a sealed arch as one contour and emerges beyond it with
↪ another·⁚.·˙ostensiveIII:·
·⁚grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦⁞》 category◦–◦⁞parliament》:·
·⁚lookup office◦–◦⁞phantasmagoria》 enoch◦–◦⁞ᛊ》:·

⁖ᚠᚣᛔᛟ:·ꕥᚷᛔᛟᛉᛞᚷ⁖·˙ᚠᚣᛔᛟ:·

·⁚origin…language:·Gnomish·⁚.·˙origin…language:·
·⁚font:·Gnomish·⁚.·˙font:·
·⁚pronunciation:·BAR+thah·⁚.·˙pronunciation:·
·⁚cadence:·Phantasmagoria·⁚.·˙cadence:·
·⁚non…ostensive:·Change·⁚ Exchange·⁚ Alter·⁚ Shift as the body changing its relation or
↪ state·⁚.·˙non…ostensive:·
·⁚ostensiveI:·Great stone heaves sideways beneath a changing field of force::Immense stone+weight
↪ remains unbroken while the mass occupies a new bearing·⁚.·˙ostensiveI:·
·⁚ostensiveII:·Two fae mirrors face across dusk::moonlit forest shining in one glass and burning tower
↪ in the other while neither mirror moves·⁚.·˙ostensiveII:·
·⁚ostensiveIII:·One enchanted body changes shape only where a black command presses against it·⁚ every
↪ resistant edge keeps its former line while the yielding surface shifts around the
↪ refusal·⁚.·˙ostensiveIII:·
·⁚grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦⁞◎》 category◦–◦⁞goetic》:·
·⁚lookup office◦–◦⁞phantasmagoria》 enoch◦–◦⁞ᛊ》:·

⁖₰₰₰:ᘐᘏ ᘐᘏᘏ✶ᘏ⁖·˙₰₰₰:·

·⁚origin…language:·Llumena Milenii·⁚.·˙origin…language:·
·⁚font:·Mage Script·⁚.·˙font:·
·⁚pronunciation:·Haum+bee+ah+na·⁚.·˙pronunciation:·
·⁚cadence:·Phantasmagoria·⁚.·˙cadence:·
·⁚non…ostensive:·The reclaimable that never ceased to belong·⁚.·˙non…ostensive:·
·⁚ostensiveI:·Empty throne keeps one warm seal beneath falling dust::Reserved throne+space remains
↪ untouched beside the unopened gate while the rest of the chamber cools◎·⁚.·˙ostensiveI:·
·⁚ostensiveII:·Across sea+bound distance::old sigil burns upon distant treasure while the empty setting
↪ holds a matching blue flame◎·⁚.·˙ostensiveII:·
·⁚ostensiveIII:·Absent sigil keeps its claim inside a sealed wall·⁚ when the matching piece returns::a
↪ narrow passage opens exactly along the recovered fit◎·⁚.·˙ostensiveIII:·

·⁖grimchain⁖·≳ꝶᴘᴤ꒒ꝸ⅄ᴙᵽᴿlᵽ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱⬡꒱ category꒰ₒ꒱goetic꒱⁖·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᖘ꒱⁖·

·⁖origin…language⁖·Neo+Quenya·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Quenya·⁖·˙font⁖·
·⁖pronunciation⁖·Minatec·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·To Record·˙ Into+Write·˙ Archive as Writing into Permanence·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Cut line bites into the waiting surface and remains fixed in the material⁚⁚giving memory
↪ a body weather can touch without erasing⬡·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·Burned parchment and black stone walls carry the same intact words through ash and
↪ obsidian silence⬡·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·One carved record sheds a hundred rays from the same enduring mark⁚⁚each emanation
↪ carrying the whole inscription without severing from its source⬡·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ϒϑ▽ᵩΨ⁺ϫↄ⅁⁜ᵶꟻᔕ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱✡꒱ category꒰ₒ꒱goetic꒱⁖·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᖘ꒱⁖·

⁖⁂ᔓᔓᑐ⁖·ᘐᕒᔓᘐᕮ⁖·⁖·˙⁂ᔓᔓᑐ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖·˙font⁖·
·⁖pronunciation⁖·Ee+freet·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Fire Deity·˙ Fae of Fire Element·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Ember+crowned presence stands where ordinary flesh would vanish⁚⁚and every coal around it
↪ opens brighter as living flame gathers into one sovereign body of heat✡·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·Blue+white crown⁚⁚awakened coals⁚⁚reddening iron⁚⁚and cold forge gather around one
↪ sovereign presence of living fire✡·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Fire+spirit gathers as a red crown around one spear of flame⁚⁚every lesser spark
↪ leaning toward the single pointed bearing across the dark✡·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·Θ⅌⋖⊑ᒧ⊛ᘔŸьҜ∱ꟈↃ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱✡꒱ category꒰ₒ꒱goetic꒱⁖·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᖘ꒱⁖·

⁖⁂ᔓᔓᑐ⁖·ᘓᑕᔓᔓᑕ⁖·⁖·˙⁂ᔓᔓᑐ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖·˙font⁖·
·⁖pronunciation⁖·Mar+rah·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Water Deity·˙ Fae of Water Element·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Pale crown drifts beneath the tide while currents braid themselves around an unseen
↪ form⁚⁚every depth answering with the same blue pull toward the living center below✩·⁖·˙ostensiveI⁖·

·⁖ostensiveII:·Moonlit water climbs the temple steps beneath a shell+crown::every basin below holding
↪ the same silver tide✩·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Crowned water+spirit coils through two broken stones as a silver current and fastens
↪ the seam while the river keeps moving✩·⁖·˙ostensiveIII:·
·⁖grimchain:·ዕ∱ᨒϔ⊬ᛃᦓᦔ⍰ᘔᕻ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍✪₎ category○–○₍goetic₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᖶ₎⁖·

·⁖origin…language:·Quenya·⁖·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖·˙font:·
·⁖pronunciation:·HROH+ah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Body·⁖· Flesh·⁖· Physical Matter·⁖· Matter as the body inhabited by spirit·⁖·˙non…ostensive:·
·⁖ostensiveI:·Flesh closes around bone and warmth keeps its chamber within::weight answering the ground
↪ while breath moves through matter that spirit may enter but still must bear✪·⁖·˙ostensiveI:·
·⁖ostensiveII:·Spirit+light fills a clay body with fired skin::pulse::warmth::weight::and grounded
↪ feet✪·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Spirit+filled flesh stands where two luminous paths cross::both currents passing
↪ through the same material body✪·⁖·˙ostensiveIII:·
·⁖grimchain:·ᗡΨ☿ᢉIᛯᦔᛕ₀℅ᘏ⬡·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍❁₎ category○–○₍goetic₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᖶ₎⁖·

·⁖origin…language:·Sindarin·⁖·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖·˙font:·
·⁖pronunciation:·Iaun·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Wide·⁖· Extensive·⁖· Large·⁖· Roomy·⁖· Vast·⁖· Vast as Expanse·⁖·˙non…ostensive:·
·⁖ostensiveI:·Horizon withdraws from every approach::open ground widening beneath the eye until
↪ distance has room to receive another distance beyond it❁·⁖·˙ostensiveI:·
·⁖ostensiveII:·A narrow door contains mountain shell::star+filled chamber::unfolding distance::and no
↪ final wall within one impossible interior❁·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A vast chamber stretches beyond sight while one sustained chord fills the entire
↪ expanse without losing its single sounding body❁·⁖·˙ostensiveIII:·
·⁖grimchain:·Ӿ⅄℅₀ᖘᗺ2ᒧ⏁Ӿ⅋⁞₀Φ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍✻₎ category○–○₍goetic₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᖶ₎⁖·

·⁖ᘏᘐᘐᐟ:·ᓚ∗ᘓᘒᓚ·⁖·˙ᘏᘐᘐᐟ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Aan+ee+mah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·

·⁖non…ostensive:·Chaotic Emanation of the Soul·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Soul spills beyond its quiet border in bright disorder::dream catching one fragment and
↪ sign another while inward fire remains inexhausted beneath the scattered
↪ emanation❋·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Moths::runes::sparks::whispering color::and one inward signature scatter together from a
↪ single burst of soul+fire❋·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One drumstroke breaks the vacant shell and sends soul+fire outward as moths::sparks::and
↪ broken sigils::every fragment carrying the same pulse❋·⁖.·˙ostensiveIII:·
·⁖grimchain:·♌❋🢒ᚦᚾᛝᛞᚺᚠᚠᛇ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ᛞ꞉ category○–○ᛔgoetic꞉·
·⁖lookup office○–○ᛔphantasmagoria꞉ enoch○–○ᛊ꞉·

·⁖ৰ৷৷৺˄:·৯ৼ✳ৼ৫:·⁖.·˙ৰ৷৷৺˄:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Vehn+iz·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Love as an act·⁖· Love as Magic·⁖· Love as ·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Two flames lean across one narrow dark::warmth filling the air between them while shared
↪ radiance leaves neither fire diminishedᛞ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Two sealed talismans hold a red thread of light across distance::tension and brightness
↪ changing together with separation and nearnessᛞ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Two distant flames touch one note::A chord blooms between them::and warmth travels both
↪ ways through the shared resonanceᛞ·⁖.·˙ostensiveIII:·
·⁖grimchain:·♐ℳᛤᚻᚸᛋᛉᚠᛞᛉ✙ᚸ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○⊛꞉ category○–○ᛔgoetic꞉·
·⁖lookup office○–○ᛔphantasmagoria꞉ enoch○–○ᛊ꞉·

·⁖ৰ৷৷৺˄:·৶৻ৼ৺ᛏ ৯৷৷ৼ৺ᗅ꞉·⁖.·˙ৰ৷৷৺˄:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Skee+al+mo+Vait·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Emanation as Shadow·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Lit figure and dark contour lie together across the ground::brightness holding one body
↪ while blocked light holds the other within the same boundary⊛·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Shadow pours from hidden source as black radiance::carrying one contour across every
↪ surface without becoming emptiness⊛·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Familiar shadow spills from the wall into impossible depth::and the sudden black
↪ opening arrests every prepared motion at its rim⊛·⁖.·˙ostensiveIII:·
·⁖grimchain:·♑♒♉🖳ᛞᚲᚦᛈ♑🡻✙ᛏ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○❄꞉ category○–○ᛔgoetic꞉·
·⁖lookup office○–○ᛔphantasmagoria꞉ enoch○–○ᛊ꞉·

·⁖ৰ৷৷৺˄:·৶৻ৼ ৸৷৷ৼ ꞉·⁖.·˙ৰ৷৷৺˄:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Val+cohl·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·To Balance·⁘ Balance As·⁘ Scales are Balanced·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Two opposed lights descend into one balanced vessel⁚⁚Neither light consumes the
↪ other⁚⁚Where equal measures meet⁚⁚one joined radiance fills the bowl❄·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Twin moons balance upon a star+forged scale⁚⁚equal silver weight holding one unwavering
↪ celestial horizon❄·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Two scales hold equal weight around one solitary central flame⁚⁚the whole balance
↪ turning upon that single unmoving presence❄·⁖.·˙ostensiveIII:·
·⁖grimchain:·↾ₒᵔᛕᛇᚾᚔᛊᵒᵌΘᚏᛈᛔ⚮☆·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱ ❁꒱ category꒰ₒ꒱goetic꒱·⁖·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᛊ꒱·⁖·

·⁖ᚼᛞᛞᛏ:·¿ᘐᓀ✶ᑒ·⁖.·˙ᚼᛞᛞᛏ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Sant·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Does not serve·⁘ does not invoke·⁘ does not evoke·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Closed seal gives no answering motion while calling smoke curls against the boundary and
↪ the marked gate remains shut under the whole invocation❁·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Seven blue candles burn around a forbidden name while smoke falls inward⁚⁚mirrors remain
↪ empty⁚⁚and the sealed center stays unanswered❁·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Impossible treasure piles outside a sealed circle while calling smoke breaks against
↪ the boundary·⁘ nothing rises within❁·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᛯᘎᛏᛞᚱᚮ'ᚸᘐ⁞᛭ᛯᛇᛋᛜ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱⬡꒱ category꒰ₒ꒱goetic꒱·⁖·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᛊ꒱·⁖·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·Dîn·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Silence·⁘ Silent·⁘ Quiet·⁘ Silence as a State with Presence·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Snow rests on branch and water keeps an unbroken surface⁚⁚No vibration crosses the
↪ field⁚⁚and stillness carries the weight of a present thing⬡·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Pale crown+light fills an ice+vault of still bell⁚⁚quiet breath⁚⁚unmoving water⁚⁚and
↪ complete luminous silence⬡·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Silent water changes color beneath an unmoving moon⁚⁚green sinking into red without
↪ ripple⁚⁚voice⁚⁚or broken stillness⬡·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᔓᘔ☆ᚼℂₒᑭᐱ᛬✳᚛᛬·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱ᔕ꒱ category꒰ₒ꒱court꒱·⁖·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᛊ꒱·⁖·

·:ᘜᘙᘛᐢ:·ᘓᘐᘔᘕᘐᘔ·:.·ᘜᘙᘛᐢ:·

·:origin…language:·Llumena Milenii·:.·origin…language:·
·:font:·Mage Script·:.·font:·
·:pronunciation:·Vee+ah+ta·:.·pronunciation:·
·:cadence:·Phantasmagoria·:.·cadence:·
·:non…ostensive:·Life as beginning·:.·non…ostensive:·
·:ostensiveI:·First pulse enters the prepared circle and every marked line answers with tender heat: one small spark now occupies the waiting hollowᚱ·:.·ostensiveI:·
·:ostensiveII:·Green life sleeps within the opened seed::first pulse and tender form held together at the living threshold of beginningᚱ·:.·ostensiveII:·
·:ostensiveIII:·First pulse enters dark water as one low tone::and a second living color blooms inside the sound before any surface breaksᚱ·:.·ostensiveIII:·
·:grimchain:·ᚹᚺᛞᛋᛋᛞ◎ᛜᛇᛉᛝᛈᛊ⫸·:.·grimchain:·
·:.·lookup:·
·:.·glyphs:·
·:glyph symbol○-○₍ᚱ₎ category○-○₍court₎:·
·:lookup office○-○₍phantasmagoria₎ enoch○-○₍ᛈ₎:·

·ᚼᚻᚣᛋ:·ᛘᚠᚳ·:.·ᚼᚻᚣᛋ:·

·:origin…language:·Primitive Elvish·:.·origin…language:·
·:font:·Sarati Eldamar LTR·:.·font:·
·:pronunciation:·En·:.·pronunciation:·
·:cadence:·Phantasmagoria·:.·cadence:·
·:non…ostensive:·Again:· Once More:· Go on Doing:· Further:· Recur as the Root of Again·:.·non…ostensive:·
·:ostensiveI:·Wave returns upon the shore without becoming its former water::Moon crosses the dark and comes again::the same silver mark finding the same stone on every turningᚱ·:.·ostensiveI:·
·:ostensiveII:·Silver wheel::old rune::fresh blue fire::and returning altar+mark turn together through the same enchanted circuitᚱ·:.·ostensiveII:·
·:ostensiveIII:·Each return of the silver wheel crosses the same taut cord::cutting one new notch into it while the circle continues unbrokenᚱ·:.·ostensiveIII:·
·:grimchain:·ᚾᚠᛇ۞ᚳᚺᛜᛋᛗᚱ⬡ᛚᛉ·:.·grimchain:·
·:.·lookup:·
·:.·glyphs:·
·:glyph symbol○-○₍ᚱ₎ category○-○₍court₎:·
·:lookup office○-○₍phantasmagoria₎ enoch○-○₍ᛈ₎:·

·:origin…language:·Sindarin·:.·origin…language:·
·:font:·Tengwar Sindarin·:.·font:·
·:pronunciation:·LAM+moth·:.·pronunciation:·
·:cadence:·Phantasmagoria·:.·cadence:·
·:non…ostensive:·Great Echo:· Reaction as creation answering creation·:.·non…ostensive:·
·:ostensiveI:·Hollow stone bears one cry across many ridges::Returning sound presses from wall to wall while the valley carries the enlarged voice across its spanᚱ·:.·ostensiveI:·
·:ostensiveII:·Two differently shaped runes burn across black stone and opposite wall as one flame answers the otherᚱ·:.·ostensiveII:·
·:ostensiveIII:·One cry enters a veiled valley and returns from hidden stone faces::each unseen wall giving the sound another bodyᚱ·:.·ostensiveIII:·
·:grimchain:·✲⊀ᑭᑭᛋᛋᛁᛋᛂᛂᛂ

·⁖glyph symbol꩜꩜ᵎᵣ category꩜꩜ᵎcourtᵢ⁖·
·⁖lookup office꩜꩜ᵎphantasmagoriaᵢ enoch꩜꩜ᵎᖱᵢ⁖·

·⁖ᚠᚩᚱᛏ⁖·ᵌᚱᚳᛖᛇᛉᛉ⁖·˙ᚠᚩᚱᛏ⁖·

·⁖origin…language⁖·Draconic·⁖·˙origin…language⁖·
·⁖font⁖·Iokharic·⁖·˙font⁖·
·⁖pronunciation⁖·ar+ka+niss·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Pattern⁖· recurring order⁖· form recognized through repetition and
↪ difference·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Spiral recurs along shell and leaf⁖⁖each turn changing scale while the hidden relation
↪ remains intact inside the repeating orderᚱ·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·Measured variation marks every rune⁖⁖Repeated relation spans the whole surface⁖⁖No single
↪ sign bears the complete patternᚱ·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·A field of repeating runes offers many open paths⁖⁖yet one chosen line keeps
↪ reappearing through every variation as the pattern turnsᚱ·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ᛋᛈᛄᛃᛉᛉᕱᚱᚢᛈᛏᛞᛞ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꩜꩜ᵎᵹ category꩜꩜ᵎcourtᵢ⁖·
·⁖lookup office꩜꩜ᵎphantasmagoriaᵢ enoch꩜꩜ᵎᖱᵢ⁖·

·⁖ᚳᚾᚦᛏ⁖·ᒪᒥᚦᚱᚣ⁖·˙ᚳᚾᚦᛏ⁖·

·⁖origin…language⁖·Dwarvish·⁖·˙origin…language⁖·
·⁖font⁖·Dethek Stone·⁖·˙font⁖·
·⁖pronunciation⁖·Ba+rak·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Backbone⁖· Strength⁖· Shield·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Back bears weight without folding⁖⁖joined stones hold against outward force⁖⁖and the
↪ living body keeps a shielded line between itself and dangerᵹ·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·Joined luminous vertebrae form a radiant wall+spine against blows that break outside
↪ the untouched inner fieldᵹ·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Shield+bone drinks incoming force into its broad spine⁖⁖the guarded body gaining
↪ steadiness from every absorbed impactᵹ·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ᚣᗡᑫᛉᚦᚸ֍ᔕᚳᓄᚸᚠᚹᚱ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꩜꩜ᵎᵴ category꩜꩜ᵎcourtᵢ⁖·
·⁖lookup office꩜꩜ᵎphantasmagoriaᵢ enoch꩜꩜ᵎᖱᵢ⁖·

·⁖ᚼᛞᛝᛏ⁖·ᔕᔓᘐᕃᕒᕐᕓᕔᕕ⁖·˙ᚼᛞᛝᛏ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖·˙font⁖·
·⁖pronunciation⁖·Am+plih+fih+tet·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Amplify⁖· Increase Magical Throughput⁖· Power as Structure·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Rune+cut channel bears more magical current than its narrow body should contain⁖⁖Flame
↪ rises higher inside the marked course⁖⁖and spell+sound carries farther without changing its
↪ noteᚱ·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·Seven copper crowns carry gathered lightning through the black tower into blue fire at
↪ its buried rootsᚱ·⁖·˙ostensiveII⁖·

·⁖ostensiveIII:·Knotted channels bind a spell+grid so tightly that gathered power can pass through
↪ every crossing without tearing the structure apart꩜⁖.·˙ostensiveIII:·
·⁖grimchain:·❄𐌅ᘐ𐌅ᘏ⍜ᑎᑐ✱ᔑᚔ𐌅⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol◦–◦𐑂𐑅 category◦–◦𐑒court𐑅:·
·⁖lookup office◦–◦𐑒phantasmagoria𐑅 enoch◦–◦𐑓𐑅:·

·⁖𐑚𐑚𐑚𐑚:·𐑚𐑚𐑚𐑚:.·˙𐑚𐑚𐑚𐑚:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·JAH+tey+KAH+rah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Weave as an Identity·⁖ Destiny·⁖ Fae as an overarching term for Magical
↪ Being·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Fae+colored threads cross beneath an unseen measure⁘each strand keeping its own course
↪ until the enchanted weave reveals a destiny already waiting inside their
↪ intersections꩜⁖.·˙ostensiveI:·
·⁖ostensiveII:·One silver destiny+road crosses the night beneath sequential stars⁘every untaken
↪ crossing dark beside the still+luminous chosen path꩜⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Fae+thread runs through a dark wood as one woven destiny⁘and a trembling step still
↪ enters the luminous warp where retreat would break the pattern꩜⁖.·˙ostensiveIII:·
·⁖grimchain:·ᑫᚿᑫᚸᘐℂᕼ⊖〰ᖘᘜᓄᑐ⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol◦–◦𐑥𐑅 category◦–◦𐑒court𐑅:·
·⁖lookup office◦–◦𐑒phantasmagoria𐑅 enoch◦–◦𐑓𐑅:·

·⁖ᕊᛯᚿᚿᘐ:·ᚿᚿᚿᚿ⁖.·˙ᕊᛯᚿᚿᘐ:·

·⁖origin…language:·Gnomish·⁖.·˙origin…language:·
·⁖font:·Gnomish·⁖.·˙font:·
·⁖pronunciation:·Rin·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Revolve·⁖ Return·⁖ Come Back·⁖ Do Again·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Wheel holds the same mark through every revolution⁘departing edge and returning edge
↪ sharing one circular law while motion carries difference inside recurrence𐑥⁖.·˙ostensiveI:·
·⁖ostensiveII:·A comet disappears beyond the dark rim of heaven and returns to the same carved
↪ star+gate⁘trailing new fire around an ancient circuit𐑥⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One silver thread circles the loom⁘returns to its first beam⁘and draws the growing
↪ cloth back through the same revolving path𐑥⁖.·˙ostensiveIII:·
·⁖grimchain:·ϒᚹᚾᛇᚼᛪᛟ◎ᛯᛈᛊᚺ·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol◦–◦𐑕𐑅 category◦–◦𐑒court𐑅:·
·⁖lookup office◦–◦𐑒phantasmagoria𐑅 enoch◦–◦𐑓𐑅:·

·⁖ᘐᘏᘏᔑ:·ᘏᔕᘐᘜᘏ⁖.·˙ᘐᘏᘏᔑ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Rhee+biss·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·

·⁖non…ostensive⁖·Hermaphrodite⁖· Blending of two different things⁖· Alchemy as Motion and
↪ Time·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Silver and sulfur rest together under sealed heat⁘red and white remaining legible in one
↪ vessel while the alchemical body carries both origins at once꞉⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Winged shadow and horned shadow share one alchemical reflection⁘both silhouettes
↪ complete inside a single mirrored body꞉⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Mercury+bright matter dives beneath a dark alchemical surface and rises altered⁘two
↪ natures still blended inside the returning body꞉⁖.·˙ostensiveIII⁖·
·⁖grimchain⁖·꞉X✡֍♂֍ᚣ↗⊒ᒡᗵᚾᚹ⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol○–○ᵩ‣ category○–○ᶜourt‣⁖·
·⁖lookup office○–○ᵩphantasmagoria‣ enoch○–○ᕮ‣⁖·

⁖⁂ᘜᘛᒣ⁖·ᒣᔕ◡ᔓᘜ✳⁖.·˙⁂ᘜᘛᒣ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖.·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖.·˙font⁖·
·⁖pronunciation⁖·Day+mon·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Living Metal⁖· Breath⁖· a golem·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Molten metal runs into rune+cut limbs as joints close one by one beneath the
↪ hammer⁘inner heat taking a pulse before the forge+light fades꞉⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Forged chest breathes around inward heat⁘Metal joints carry living rhythm⁘Rune+cut
↪ fingers close around cold air꞉⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Living metal catches its own face in a black mirror before the eye+lights open⁘image
↪ arriving a breath before the golem wakes꞉⁖.·˙ostensiveIII⁖·
·⁖grimchain⁖·⊛ᗡ‹▽〰▽ᚼ֍ᔭᚣᖘ⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol○–○ᵩ‣ category○–○ᶜourt‣⁖·
·⁖lookup office○–○ᵩphantasmagoria‣ enoch○–○ᕮ‣⁖·

⁖ᖻᖘᖘ⁖·ᖘᖘᖘᖘᖘ⁖.·˙ᖻᖘᖘ⁖·

·⁖origin…language⁖·Draconic·⁖.·˙origin…language⁖·
·⁖font⁖·Iokharic·⁖.·˙font⁖·
·⁖pronunciation⁖·O+qrych·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Rip⁖· Tear⁖· Pull Aggressively·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Torn edge gapes beneath opposing strain as fiber stretches against fiber⁘separation
↪ living inside the wound under the same pull꞉⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Opposed invisible force tears enchanted cloth through its golden sigil into two
↪ violently separated woven halves꞉⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Cold gathers along a stretched silver cloth until one violent pull splits the frozen
↪ weave into two ringing edges꞉⁖.·˙ostensiveIII⁖·
·⁖grimchain⁖·⁓⏣ᚼᚹ〰ᛤᚼᛝᚱᑭᛊᚩ⏣⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol○–○ᵩ‣ category○–○ᶜourt‣⁖·
·⁖lookup office○–○ᵩphantasmagoria‣ enoch○–○ᕮ‣⁖·

⁖word⁖·ᓘᓘᔑᓘᓘᔐ⁖.·˙word⁖·

·⁖origin…language:·Klingon·⁖.·˙origin…language:·
·⁖font:·Klingon pIqaD HaSta·⁖.·˙font:·
·⁖pronunciation:·Qap+la·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Success·⁖ Achievement·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Raised standard stands above wrought work::Every fitted part bears its place::and no
↪ unfinished edge remains under the completed measureص·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Final rune::seven opened seals::complete gold circle::and absent unfinished marks burn
↪ together beneath one finished workingص·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·In absolute darkness::one finished sigil ignites whole only after every joint and mark
↪ has seated cleanly into placeص·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᗰᔕᘻᑐᕼᕮᑕᖘᘜᘓᑭᙢ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜ꠥ꩞ category꩜꩜ꠥcourt꩞⁖·
·⁖lookup office꩜꩜ꠥphantasmagoria꩞ enoch꩜꩜ꠥᖴ꩞⁖·

·⁖ᚻᛞᛏ᛫:·ᛞᛟ ᛤ ᛈᛟᚩᛞᛞᚳᔭ·⁖.·˙ᚻᛞᛏ᛫:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Re+see+vor+ah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·To Recieve·⁖ Recived·⁖ Keep·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Open vessel receives a body across its rim and keeps it inside::the inward hollow holding
↪ what the opening admitsI·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Moon+white bowl holds pooled starlight through darkness beneath an unbroken luminous
↪ rimI·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A moon+cast shadow folds into a black bowl and remains there after the body has moved
↪ beyond the lightI·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᚸᚼᛊᚺᛨᛋᛩᛊᚤᚩᚷᚺᛔᚣᛓ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜ꠥ꩞ category꩜꩜ꠥcourt꩞⁖·
·⁖lookup office꩜꩜ꠥphantasmagoria꩞ enoch꩜꩜ꠥᖴ꩞⁖·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·Súlë·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Breath·⁖ Movement of Spirit·⁖ Emission of Power of Will or Desire·⁖ Reflex as Spirit
↪ becoming Motion before Thought·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Breath presses outward from the inward chamber::bending air under unseen force until
↪ nearby flame leans along the same unspoken desireł·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Unspoken desire moves as silver breath through a ring of charms already turning toward
↪ the same inward forceł·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Dormant ember breathes beneath ash before flame appears::a hidden spirit+pressure
↪ lifting grey dust in one slow outward pulseł·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᚺᛋᚸᚤᛘᚹᚺᛕ⟰ᚾ⟲ᛋ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·

·⁖glyph symbol◦–◦𝅘𝅥𝅯𝅗𝅥 category◦–◦court𝅗𝅥⁖·
·⁖lookup office◦–◦phantasmagoria𝅗𝅥 enoch◦–◦ᚲ𝅗𝅥⁖·

·⁞ꕥ⁖ᓚᘏᗢ⁖·˙⁞ꕥ⁖·

·⁖origin…language⁖·Valarin·⁝.·˙origin…language⁖·
·⁖font⁖·Sarati Eldamar LTR·⁝.·˙font⁖·
·⁖pronunciation⁖·Ru+shur·⁝.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁝.·˙cadence⁖·
·⁖non…ostensive⁖·Fire·⁝.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Grain turns through red fire until fragrance rises from every darkened kernel⁚⁚Heat
↪ reaches the center⁚⁚leaving no raw heart beneath the browned surface▫·⁝.·˙ostensiveI⁖·
·⁖ostensiveII⁖·White heat fills the burning center⁚⁚Air shivers around red fire⁚⁚Radiance presses
↪ outward from the consuming core▫·⁝.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·One blue flame consumes frost while leaving the ice unthawed⁚⁚contradiction burning
↪ steadily at the center of the fire▫·⁝.·˙ostensiveIII⁖·
·⁖grimchain⁖·♀ↃƐᛯ᛬ɳϰ⌒☿ᚠ∧◡ϕ⅄·⁝.·˙grimchain⁖·
·⁝.·˙lookup·˙
·⁝.·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥𝅮 category◦–◦court𝅗𝅥⁖·
·⁖lookup office◦–◦phantasmagoria𝅗𝅥 enoch◦–◦ᚲ𝅗𝅥⁖·

·⁖origin…language⁖·Quenya·⁝.·˙origin…language⁖·
·⁖font⁖·Tengwar Quenya·⁝.·˙font⁖·
·⁖pronunciation⁖·IN+go+leh·⁝.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁝.·˙cadence⁖·
·⁖non…ostensive⁖·Deep Lore·⁖ Magic·⁖ Philosophy·⁖ Deep as knowledge beneath ordinary
↪ knowing·⁝.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Deep chamber keeps what the surface cannot answer⁚⁚hidden sign waiting beneath ordinary
↪ knowing where easy certainty cannot follow𝅘𝅥·⁝.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Rune⁚⁚flame⁚⁚hidden sign⁚⁚deeper sign⁚⁚and dark stone occupy one nested field of
↪ correspondences𝅘𝅥·⁝.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Each descent into the dark archive opens one deeper chamber⁚⁚every completed step
↪ revealing another layer of lore below the last𝅘𝅥·⁝.·˙ostensiveIII⁖·
·⁖grimchain⁖·ℳ৻ᗢ▽✱ᓭᛘ𐂷❄ᘐ♭✱♭❁·⁝.·˙grimchain⁖·
·⁝.·˙lookup·˙
·⁝.·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥♪ category◦–◦court𝅗𝅥⁖·
·⁖lookup office◦–◦phantasmagoria𝅗𝅥 enoch◦–◦ᚲ𝅗𝅥⁖·

·⁖origin…language⁖·Sindarin·⁝.·˙origin…language⁖·
·⁖font⁖·Tengwar Sindarin·⁝.·˙font⁖·
·⁖pronunciation⁖·Ithildin·⁝.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁝.·˙cadence⁖·
·⁖non…ostensive⁖·Magical Alloy that Glows in Moonlight·⁖ Moon+Star·⁖ Mark as Hidden Sign Revealed by
↪ Proper Light·⁝.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Dark alloy rests under moonlight⁚⁚pale fire gathering along hidden marks until night and
↪ metal reveal one secret geometry in silver♪·⁝.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Black alloy blade carries buried silver veins under moonrise and returns wholly dark
↪ beneath sun or flame♪·⁝.·˙ostensiveII⁖·

·⁖ostensiveIII:·Moonlight pulls pale inscriptions out of black alloy::silver lines rising through the metal skin♪⁖.·˙ostensiveIII:·
·⁖grimchain:·ᛞ⊙X𐌛𐋡ᚲᛋᚺ𐋡ᛏ𐋡ᛏ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉Ɣ𐑊 category○–○𐑉court𐑊⁖·
·⁖lookup office○–○𐑉phantasmagoria𐑊 enoch○–○𐑉ᚲ𐑊⁖·

·⁖origin…language:·Early Noldorin·⁖.·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖.·˙font:·
·⁖pronunciation:·dreem·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Host·⁖ Countless number·⁖ One becomes Host·⁖ A body becoming innumerable·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Field fills beyond easy count::Rank continues beyond rank beneath the same banner until singular measure can no longer contain the hostƳ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·One fae lantern and a thousand distinct reflected flames fill black mirrors beneath a single living sigilƳ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One wounded surface releases a host of tiny luminous bodies that gather across the opening until the seam holds whole againƳ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ǂ𐋡ᚲᓄᚺᛯᛉᛏΘᛞᚴ𐋡ᚷᛏ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉ξ𐑊 category○–○𐑉court𐑊⁖·
·⁖lookup office○–○𐑉phantasmagoria𐑊 enoch○–○𐑉ᚲ𐑊⁖·

⁖·ᚻᛞᛞᛏ·˙:·ᛏ𐌼ᛅᛅ𐌼ᛘ·⁖.·˙ᚻᛞᛞᛏ·˙:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Loo+puss·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Wolf of Spirit·⁖ Fenrir·⁖ Ghost of the Animalia·⁖ Fae·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Moonlight catches a lupine shape where ordinary weight is absent: pale paws disturb no dust::yet the hunt crosses the border flesh cannot keepᛄ⁖.·˙ostensiveI:·
·⁖ostensiveII:·Spectral wolf::biting chain::dark heaven::and hanging prophecy occupy one terrible Fenrir+body without mortal weightᛄ⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Spectral wolf+form climbs a stair of cold stars toward an unreachable moon::desire stretching the ghost+body beyond every resting ledgeᛄ⁖.·˙ostensiveIII:·
·⁖grimchain:·❀ᚢᛄ𝌆𐑉ᛟ𐋡ᛞᚷ♁ᛊ⓪♂·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉Ⴙ𐑊 category○–○𐑉court𐑊⁖·
·⁖lookup office○–○𐑉phantasmagoria𐑊 enoch○–○𐑉ᚲ𐑊⁖·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·VAN+wah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·

·⁖non…ostensive꞉·Gone⁖· Lost⁖· Departed⁖· Vanished⁖· Past⁖· Loss as that which has passed beyond
↪ possession·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Empty place keeps the shape of departed warmth꞉꞉Dust settles inside the departed
↪ contour꞉꞉and absence wounds the space that remainsϟ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·Empty sea꞉꞉taut mooring rope꞉꞉closing wake꞉꞉and vanished black ship remain together
↪ beneath moonlightϟ·⁖·˙ostensiveII꞉·
·⁖ostensiveIII꞉·A vanished figure survives only as reflected light on an angled mirror꞉꞉every returning
↪ glimpse bent away from the empty thresholdϟ·⁖·˙ostensiveIII꞉·
·⁖grimchain꞉·𐑌𐑦𐑮𐑤𐑐𐑨𐑟𐑩𐑛𐑟𐑯·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉○꞉₎ category꞉○꞉court₎꞉·
·⁖lookup office꞉○꞉phantasmagoria₎ enoch꞉○꞉ᛊ₎꞉·

·𑁍𑁍𑁍꞉·𑁍𑁍𑁍·⁖·˙𑁍𑁍𑁍꞉·

·⁖origin…language꞉·Telerin·⁖·˙origin…language꞉·
·⁖font꞉·Tengwar Teleri·⁖·˙font꞉·
·⁖pronunciation꞉·AH+bah·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Refuse⁖· Will not⁖· Curb as the boundary that answers by withholding
↪ continuation·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Path ends at a closed line where pressure meets still refusal and forward motion finds
↪ no permission to continue'·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·Black arch bears burning spell+pressure in a backward+folded flame that never crosses
↪ the warded boundary'·⁖·˙ostensiveII꞉·
·⁖ostensiveIII꞉·One sealed gate refuses the advancing force⁖· when the pressure breaks꞉꞉the gathered
↪ weight drops straight down before the barred opening'·⁖·˙ostensiveIII꞉·
·⁖grimchain꞉·𐑡𐑚𐑮𐑯𐑮𐑢𐑪𐑦𐑢𐑓𐑖𐑟·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉○꞉ᚼ₎ category꞉○꞉court₎꞉·
·⁖lookup office꞉○꞉phantasmagoria₎ enoch꞉○꞉ᛊ₎꞉·

·⁖origin…language꞉·Quenya·⁖·˙origin…language꞉·
·⁖font꞉·Tengwar Quenya·⁖·˙font꞉·
·⁖pronunciation꞉·Ainulindalë·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Music of the Ainur⁖· Many Voices as One Creation⁖· System as Ordered Creation through
↪ Harmony·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Sacred voices sound across the dark in distinct tones꞉꞉Every sacred note remains itself
↪ inside the concord while stone and star tremble within the ordered harmonyᚼ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·Seven distinct tones remain separately audible beneath one crystal geometry rising from
↪ black waterᚼ·⁖·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Many divine voices sound across a black abyss꞉꞉their separate tones joining one
↪ creation while the unseen depth carries the whole harmonyᚼ·⁖·˙ostensiveIII꞉·
·⁖grimchain꞉·ᚼ○⛢I'ᚼᚢᚸᚳᚼᚼᛉᚷ·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉○꞉ᚼ₎ category꞉○꞉court₎꞉·
·⁖lookup office꞉○꞉phantasmagoria₎ enoch꞉○꞉ᛊ₎꞉·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·Enyalië·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Memory·⁖ Recalling·⁖ Remember as Active Calling⟊Back·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Faded sign and inward brightness occupy the same chamber⁚⁚Distant form stands near
↪ without crossing the road⁚⁚and lost sound remains present without another body↓·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Faded sigil glows beside cold stone while vanished bell and returned vibration share
↪ the same silent chamber↓·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Scattered silver motes drift across the chamber⁚⁚and every dispersed fragment carries
↪ the same lost image back into memory wherever it settles↓·⁖.·˙ostensiveIII:·
·⁖grimchain:·𐌵☿♈𐤒𓆏ϧᛊ⏁ᑭ⏃ᚺᔑ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦𐑉 category◦–◦court𐑉⁖·
·⁖lookup office◦–◦phantasmagoria𐑉 enoch◦–◦ᚲ𐑉⁖·

·⁖word:·𐑆𐑒𐑓·⁖.·˙word:·

·⁖origin…language:·Klingon·⁖.·˙origin…language:·
·⁖font:·Klingon pIqaD HaSta·⁖.·˙font:·
·⁖pronunciation:·pih⟊tlh·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·All done·⁖ All finished·⁖ Closure spoken as a terminating utterance·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Last fitted stone enters the narrow gap until both faces touch cleanly⁚⁚Distance
↪ collapses to a seam too small for light to inhabit↑·⁖.·˙ostensiveI:·
·⁖ostensiveII:·One final word seals every open seam of the unfinished circle↑·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One final word closes the luminous circle⁚⁚and every mark vanishes from the inner
↪ ground until only bare stone remains inside the completed ring↑·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᘜᘁℭ𐌵◎▽ᘓᏮᗅᎩ᳀⁕Ꝟ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦Ᵽ𐑉 category◦–◦court𐑉⁖·
·⁖lookup office◦–◦phantasmagoria𐑉 enoch◦–◦ᚲ𐑉⁖·

·⁖ᚠᚢᚱ⁖·ᛇᛯᚱ·⁖.·˙ᚠᚢᚱ⁖·

·⁖origin…language:·Draconic·⁖.·˙origin…language:·
·⁖font:·Iokharic·⁖.·˙font:·
·⁖pronunciation:·Mul·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Change·⁖ Alter·⁖ Transform·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Same substance carries a new contour under unchanged weight⁚⁚surface and edge altered
↪ while matter remains continuous beneath the transformationᑭ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Moonlit mist carries one silver⟊sigiled identity through bronze stag and dark⟊winged
↪ flock as two forms of the same transformed bodyᑭ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A black figure changes shape inside an empty chamber where no surrounding substance or
↪ tool shares the spaceᑭ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᜰᘓᑫᘐ❀ᜭᚷᚾᗤᑭᛝ0·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁖glyph symbol◦-◦ᵩᑎ⁗ category◦-◦ᵩcourt⁗·
·⁖lookup office◦-◦ᵩphantasmagoria⁗ enoch◦-◦ᵩᖶ⁗·

·⁖᪥᪥ᩣᩫ:·ᨴᩋᨠᨦᩁᩋᩫ⁖·⁘᪥᪥ᩣᩫ:·

·⁖origin…language꞉·Llumena Milenii·⁖·˙origin…language꞉·
·⁖font꞉·Mage Script·⁖·˙font꞉·
·⁖pronunciation꞉·Kree+aa+shawn·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Produce⁖· Create⁖· Magic as Form of Intent and Will made Manifest·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Hammer falls over rune+lit iron while spellcraft gathers through sparks and cadence꞉꞉edge
↪ and socket emerging together as intent becomes manifest in the worked metalᑎ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·Chosen intent condenses luminous matter above an empty altar into a shaped
↪ vesselᑎ·⁖·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Chosen intent condenses into a visible form so strange that breath and motion halt
↪ around it at the instant creation appearsᑎ·⁖·˙ostensiveIII꞉·
·⁖grimchain꞉·ꕤᘏᴉᵡᛏᛕ⛥ᛯᛇᚹ⛣❀ᘓ⊖⍟·⁖·˙grimchain꞉·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦-◦ᵩᑲ⁗ category◦-◦ᵩcourt⁗·
·⁖lookup office◦-◦ᵩphantasmagoria⁗ enoch◦-◦ᵩᖶ⁗·

·⁞ᚑᚑᛋ⁖ᛘᑭᚸᚕᚔᚣᚑᚸᚳᚷ⁖·˙⁞ᚑᚑᛋ⁖·

·⁖origin…language꞉·Valarin·⁖·˙origin…language꞉·
·⁖font꞉·Sarati Eldamar LTR·⁖·˙font꞉·
·⁖pronunciation꞉·A+thaa+rai+gas·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Sun⁖· Appointed Heat·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·White heat crowns the open sky꞉꞉stone warming beneath its appointed gaze while every
↪ shadow carries the measure of the same distant fireᑲ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·Gold heat descends through one cloud+ring as an appointed warmth surrounded on every
↪ side by untouched frostᑲ·⁖·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Sun+fire hangs above a sealed blade꞉꞉pouring heat across the ground while every
↪ gathered strike remains bound inside the unmoving sheathᑲ·⁖·˙ostensiveIII꞉·
·⁖grimchain꞉·ᘐᛉᚏᛯᚲᚸᚢ❋⏁ᒍᛦᛋᛏ·⁖·˙grimchain꞉·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦-◦ᵩᖴ⁗ category◦-◦ᵩcourt⁗·
·⁖lookup office◦-◦ᵩphantasmagoria⁗ enoch◦-◦ᵩᖶ⁗·

·⁖origin…language꞉·Early Noldorin·⁖·˙origin…language꞉·
·⁖font꞉·Tengwar Noldor·⁖·˙font꞉·
·⁖pronunciation꞉·leem·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Water⁖· The receiving body in which rigid boundaries may cease remaining
↪ rigid·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Hard crystal enters the receiving water and loses its sharp boundary꞉꞉Rigid edges soften
↪ into the moving body until no separate outline remainsᖴ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·Moon+water carries a rigid glass key as one yielding outline whose former hard teeth
↪ flow within the basinᖴ·⁖·˙ostensiveII꞉·

·⁖ostensiveIII:·Two silver currents enter one basin::rigid edges loosening as their tones settle into a ↪ single consonant pressure through the moving water⁕·⁖.·˙ostensiveIII:·
·⁖grimchain:·φ𐑅ɤ᧿ⵠ⟐ꓫ𐑈ካ𐑂▫⩡·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᵩᛦ category⟜⟜ᵩcourt⁍·
·⁖lookup office⟜⟜ᵩphantasmagoria⁍ enoch⟜⟜ᵩᖶ⁍·

·⁖ᛣᛞᛟᛏ:·ᛤᛞ✳ᛇᛉᑕᔓᔕᘏᛞᑐ·⁖.·˙ᛣᛞᛟᛏ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Con⟊flah⟊geer⟊ah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Conflict·⁖· To be·⁖· to Cause·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Opposed forces press across a burning field::pressure holding against pressure while the ↪ charged boundary stays alive where neither side inherits the centerᛣ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Colliding spellstorms create one third wind at the struck boundary::born from both ↪ opposing forcesᛣ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Opposing spellforces meet above a depth no sounding line can exhaust::and a white ↪ fracture plunges through the unseen profound belowᛣ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᗤ⟰ᚿᚠ⩡♀ᛘ◎ካᚴ⏃ᛏ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᵩᚲ category⟜⟜ᵩcourt⁍·
·⁖lookup office⟜⟜ᵩphantasmagoria⁍ enoch⟜⟜ᵩᖶ⁍·

·⁖ᓚᘻᗕᓭ:·ᔦᔭᔨᔮᓚ:·⁖.·˙ᓚᘻᗕᓭ:·

·⁖origin…language:·Telerin·⁖.·˙origin…language:·
·⁖font:·Tengwar Teleri·⁖.·˙font:·
·⁖pronunciation:·A⟊clar·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Glory·⁖· Splendour·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Bright metal throws fire back into the dark::Reflected brilliance hardens the surrounding ↪ shadow while the radiant surface carries its fullness without judgment‹·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A crown of living stars floods black altar::stone::water::and metal with impossible ↪ brilliance‹·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Radiant splendor fills a crowned figure while the sheltering light burns the enclosed ↪ ground beneath it‹·⁖.·˙ostensiveIII:·
·⁖grimchain:·⍾ᚼϒφ«ɤᔕᚻᗷᑭᓐ᳘ᵒ⁸⏃·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᵩᚷ category⟜⟜ᵩcourt⁍·
·⁖lookup office⟜⟜ᵩphantasmagoria⁍ enoch⟜⟜ᵩᖶ⁍·

·⁖ᓚᘻᗕᓭ:·ᔦᔭᔨ·⁖.·˙ᓚᘻᗕᓭ:·

·⁖origin…language:·Telerin·⁖.·˙origin…language:·
·⁖font:·Tengwar Teleri·⁖.·˙font:·
·⁖pronunciation:·Caa⟊no·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Herald·⁖.·˙non…ostensive:·

·⁖ostensiveI⁖·Horn+note stands over the road: arriving bodies remain beyond the gate while open
↪ distance carries the announcement farther than the procession itselfX·⁖˳·˙ostensiveI⁖·
·⁖ostensiveII⁖·Blue comet::answering bells::lifted silver banners::and closed gate gather beneath an
↪ arrival not yet crossedX·⁖˳·˙ostensiveII⁖·
·⁖ostensiveIII⁖·First light touches bare stone before the heralding flame appears::rough ridges already
↪ warm beneath the arriving signX·⁖˳·˙ostensiveIII⁖·
·⁖grimchain⁖·ᚷ𓃰↑☆𓃭ᛋᚷᛈᛗᔑᚹᚨᒪᓓ·⁖˳·˙grimchain⁖·
·⁖˳·˙lookup·˙
·⁖˳·˙glyphs·˙
·⁖glyph symbol○–○᛫ᚹ category○–○᛫court⁗·
·⁖lookup office○–○᛫phantasmagoria⁗ enoch○–○᛫ᛊ⁗·

·⁖ᚻᛞᛞᛏ⁖·ᘐᑐᔨᖴᘐ·⁖˳·˙ᚻᛞᛞᛏ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖˳·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖˳·˙font⁖·
·⁖pronunciation⁖·Vaa+loo·⁖˳·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖˳·˙cadence⁖·
·⁖non…ostensive⁖·Curse·⁖˳·˙non…ostensive⁖·
·⁖ostensiveI⁖·Dark seal feeds upon every offering laid near it::flame and bright grain vanishing into a
↪ hunger that grows no thinnerᚹ·⁖˳·˙ostensiveI⁖·
·⁖ostensiveII⁖·Black thread bites into a marked form beneath a sealed knot::silent harm held fast under
↪ continuing pressureᚹ·⁖˳·˙ostensiveII⁖·
·⁖ostensiveIII⁖·A black curse settles over an injured limb::and the unhurt limb across from it takes
↪ the same pain+bearing shape before any wound crosses the gapᚹ·⁖˳·˙ostensiveIII⁖·
·⁖grimchain⁖·𓆏𓃭𓅓▽ᚦᛯ֍𓆉ᔑΘ𓃰ᚺ·⁖˳·˙grimchain⁖·
·⁖˳·˙lookup·˙
·⁖˳·˙glyphs·˙
·⁖glyph symbol○–○᛫ᚺ⁗ category○–○᛫court⁗·
·⁖lookup office○–○᛫phantasmagoria⁗ enoch○–○᛫ᛊ⁗·

·⁖ᚠᚸᚻ⁖·ᚠᚠ𓆏ᚸᚠ·⁖˳·˙ᚠᚸᚻ⁖·

·⁖origin…language⁖·Draconic·⁖˳·˙origin…language⁖·
·⁖font⁖·Iokharic·⁖˳·˙font⁖·
·⁖pronunciation⁖·Theast·⁖˳·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖˳·˙cadence⁖·
·⁖non…ostensive⁖·Purity·⁖ Virtuousness·⁖ Goodness·⁖˳·˙non…ostensive⁖·
·⁖ostensiveI⁖·Clear water leaves no stain upon the vessel while white flame burns without
↪ smoke::everything crossing the field coming away without corruptionᚺ·⁖˳·˙ostensiveI⁖·
·⁖ostensiveII⁖·A smokeless white flame within a crystal lotus keeps untouched clarity even above dark
↪ passing waterᚺ·⁖˳·˙ostensiveII⁖·
·⁖ostensiveIII⁖·One clear silver pendulum swings through unclouded light::each return keeping the same
↪ unstained arc around the centerᚺ·⁖˳·˙ostensiveIII⁖·
·⁖grimchain⁖·ᚹ⬡ᑎᗐᑐᘏ𓆏ᚠᚹ𓃭ᛒ𓆉ᛘ·⁖˳·˙grimchain⁖·
·⁖˳·˙lookup·˙
·⁖˳·˙glyphs·˙
·⁖glyph symbol○–○᛫ᚾ⁗ category○–○᛫court⁗·
·⁖lookup office○–○᛫phantasmagoria⁗ enoch○–○᛫ᛊ⁗·

·⁖ᚹᚱᚱᚦᛞ⁖·ᚷᚨᛏᚺᚹᚨ·⁖˳·˙ᚹᚱᚱᚦᛞ⁖·

·⁖origin…language⁖·Khuzdul·⁖˳·˙origin…language⁖·
·⁖font⁖·DwarfRunes·⁖˳·˙font⁖·

·⁖pronunciation:·Ga+thol·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Fortress·⁖·˙non…ostensive:·
·⁖ostensiveI:·Deep wall bears outward force across its full thickness::Gate compresses passage to a
↪ narrow cut::sheltering the inner ground behind stone꞉·⁖·˙ostensiveI:·
·⁖ostensiveII:·Four colossal earth+born shields enclose one glowing garden beyond storm::flame::and
↪ falling stone꞉·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Dream+roads and impossible stars remain enclosed inside fortress walls::the sleeping
↪ field unreachable by the force breaking against the outer stone꞉·⁖·˙ostensiveIII:·
·⁖grimchain:·𐊗ᐱ𐊧◎ᛋ◎▫ᚷᚹᗅᚾᘐ𐑂·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꞉꞉𐑉꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉phantasmagoria꞉ enoch꞉꞉ᛈ꞉·

·⁖origin…language:·Early Noldorin·⁖·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖·˙font:·
·⁖pronunciation:·grond·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Hammer of the Underworld·⁖ Smash embodied as the mythic Hammer itself·⁖·˙non…ostensive:·
·⁖ostensiveI:·Black iron carries impossible weight in its head::Stone beneath the face already holds
↪ fracture under the resting pressure::the deep world present in the mass without
↪ descent꞉·⁖·˙ostensiveI:·
·⁖ostensiveII:·Underworld iron casts crushing mythic weight across a sealed gate strongly enough to
↪ cave stone beneath its passing shadow꞉·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Underworld hammer hangs above the altar while bright laughter already rings through
↪ the chamber from stone not yet struck꞉·⁖·˙ostensiveIII:·
·⁖grimchain:·⊕𐑟ᓄ◎𐑋𐑇𐑅ᓄᙆᘎᙆᙏ𐑋𐑇·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꞉꞉𐑌꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉phantasmagoria꞉ enoch꞉꞉ᛈ꞉·

·⁖origin…language:·Sindarin·⁖·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖·˙font:·
·⁖pronunciation:·Gwan+wen·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Departed·⁖ Gone·⁖ Lost to Time·⁖ The Ashen Remiss·⁖ What remains after joy is stripped of
↪ performance·⁖ The ache that cannot be erased by absence·⁖·˙non…ostensive:·
·⁖ostensiveI:·Ash lies in the place of vanished song::Empty ground keeps the contour of departed joy
↪ while grey tenderness remains where return has no hold꞉·⁖·˙ostensiveI:·
·⁖ostensiveII:·Warm cup::dust+covered feast+table::cold empty seats::and vanished music hold one grey
↪ ache beneath the roof꞉·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Cold hearthstone exhales one impossible pulse of former warmth through old ash while
↪ the empty fire+ring remains under a vanished flame꞉·⁖·˙ostensiveIII:·
·⁖grimchain:·ᒐᐱᘉᖶᓄᘎᚲᘏᙆᙏ𐑋𐑇ᘎᙆ꙰·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꞉꞉𐑍꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉phantasmagoria꞉ enoch꞉꞉ᛈ꞉·

·⁖origin…language:·Khuzdul·⁖.·origin…language:·
·⁖font:·DwarfRunes·⁖.·font:·
·⁖pronunciation:·Bi+zâr·⁖.·pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·cadence:·
·⁖non…ostensive:·Small Stream·⁖.·non…ostensive:·
·⁖ostensiveI:·Narrow water slips through a shallow cleft as root and gravel turn the current::broken sky riding the small silver course≳·⁖.·ostensiveI:·
·⁖ostensiveII:·A finger+wide silver rill carries impossible noon+stars through moss and black roots along one modest enchanted course≳·⁖.·ostensiveII:·
·⁖ostensiveIII:·A finger+wide silver stream suddenly quickens and leaps its stones before any rain arrives≳·⁖.·ostensiveIII:·
·⁖grimchain:·ᛯↃ⊙0◎♁☿⬡ϒ♈ᡕ卌∀·⁖.·grimchain:·
·⁖.·lookup:·
·⁖.·glyphs:·
·⁖glyph symbol○–○ᵩᔑ» category○–○ᵩcourt»:·
·⁖lookup office○–○ᵩphantasmagoria» enoch○–○ᵩᖶ»:·

·:ᚼᙏᙏᐢ:·¿ᴖ∽✳¿·:.·ᚼᙏᙏᐢ:·

·⁖origin…language:·Llumena Milenii·⁖.·origin…language:·
·⁖font:·Mage Script·⁖.·font:·
·⁖pronunciation:·Sanz·⁖.·pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·cadence:·
·⁖non…ostensive:·To Join·⁖ Merge·⁖ Combine·⁖ Alchemy and States·⁖.·non…ostensive:·
·⁖ostensiveI:·Silver fluid and dark resin coil through the sealed vessel until their separate colors disappear into one iridescent body::No seam survives where the two substances entered꒰·⁖.·ostensiveI:·
·⁖ostensiveII:·Three planetary salts become one violet metal inside a heated sphere while all three stellar colors remain beneath the single skin꒰·⁖.·ostensiveII:·
·⁖ostensiveIII:·Two alchemical liquids enter one another without resistance::their colors joining in a smooth spiral while neither current catches at the seam꒰·⁖.·ostensiveIII:·
·⁖grimchain:·♄☆ϮΘ⁞※ᘓ⨟◎ᔓᘜᔑ❋⊕·⁖.·grimchain:·
·⁖.·lookup:·
·⁖.·glyphs:·
·⁖glyph symbol○–○ᵩↄ» category○–○ᵩcourt»:·
·⁖lookup office○–○ᵩphantasmagoria» enoch○–○ᵩᖶ»:·

·:ᵀᐢᖘᖘ:·ᵌᐧᖘᵀᖘ:.·ᵀᐢᖘᖘ:·

·⁖origin…language:·Draconic·⁖.·origin…language:·
·⁖font:·Iokharic·⁖.·font:·
·⁖pronunciation:·As+tir·⁖.·pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·cadence:·
·⁖non…ostensive:·Realize·⁖ Become Aware·⁖.·non…ostensive:·
·⁖ostensiveI:·Hidden relation lies openly among the marks::One crossing gathers the whole pattern into evident order::with every clue still visible on the surfaceↄ·⁖.·ostensiveI:·
·⁖ostensiveII:·Moonlight reveals one hidden constellation already distributed through scattered floor+sigils without adding a markↄ·⁖.·ostensiveII:·
·⁖ostensiveIII:·Crossing currents churn black water until one hidden constellation flashes whole across the turbulent surface without becoming stillↄ·⁖.·ostensiveIII:·
·⁖grimchain:·ᗷ≳ӂↄ♇ϫ⊕⊖♳ϒ♓⊄♂·⁖.·grimchain:·
·⁖.·lookup:·

·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦ᛯ᛬ category◦–◦court᛬⁖·
·⁖lookup office◦–◦phantasmagoria᛬ enoch◦–◦ᛊ᛬⁖·

⁖ᚠᚳᛔᛟ⁖·ᚽᛒᛒᛞᚠᛒᚳᛔᛯ⁖·˙ᚠᚳᛔᛟ⁖·

·⁖origin…language:·Gnomish·⁖·˙origin…language:·
·⁖font:·Gnomish·⁖·˙font:·
·⁖pronunciation:·Fui+thlog·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Dragon that Guards Treasure·⁖·˙non…ostensive:·
·⁖ostensiveI:·Scaled body coils around the silent hoard::heat sleeping beneath the breath while bright metal remains untouched inside the ring of watchfulnessᛯ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Dragon+shadow covers a sealed mountain while ember+filled tunnels circle sleeping gold beneath itᛯ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A dragon+scale wall several hands thick coils around buried gold::every overlapping plate deepening the guarded boundary before any passage can reach the hoardᛯ·⁖·˙ostensiveIII:·
·⁖grimchain:·ᛋ⍟ᚱᛞᛩᛯᛈᛞᛕᚱᛏ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦ᚮ᛬ category◦–◦court᛬⁖·
·⁖lookup office◦–◦phantasmagoria᛬ enoch◦–◦ᛊ᛬⁖·

·⁖origin…language:·Neo+Quenya·⁖·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖·˙font:·
·⁖pronunciation:·LEM+mah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Artifact:˙ Relic:˙ Residue:˙ Dregs as what remains after division·⁖·˙non…ostensive:·
·⁖ostensiveI:·Broken vessel lies beneath the dust after the bright liquid has drained away::One dark crust clings to the inner shard::small and stubborn under the empty rimᚮ·⁖·˙ostensiveI:·
·⁖ostensiveII:·A black shard keeps dead sigil and consecrated heat from a ruined crownᚮ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·One relic+shard remains after the larger body is gone::a long slender sliver of the same old substanceᚮ·⁖·˙ostensiveIII:·
·⁖grimchain:·ᚱᛁᛯᛇᛝᚢᛔᛇᚮᛗᛉ◎·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦ᛎ᛬ category◦–◦court᛬⁖·
·⁖lookup office◦–◦phantasmagoria᛬ enoch◦–◦ᛊ᛬⁖·

·⁖origin…language:·Nandorin·⁖·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖·˙font:·
·⁖pronunciation:·Rath·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Climb·⁖·˙non…ostensive:·
·⁖ostensiveI:·Broken chain falls against the lower stones as the body climbs into open height::Each higher grip carries more sky and less captivity around the rising formᛎ·⁖·˙ostensiveI:·
·⁖ostensiveII:·A stair of pale moonlight bears weight upward through empty air into star+hidden heightᛎ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Pale steps rise through absolute blackness::each higher contact appearing where the void shows no ledge aheadᛎ·⁖·˙ostensiveIII:·

·⁖grimchain:·𐊗𐊘𐊚𐊧𐊢𐊫𐊦𐊟𐊗𐊧𐊘·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜𐑂꩝ category꩜꩜𐑂court꩝:·
·⁖lookup office꩜꩜𐑂phantasmagoria꩝ enoch꩜꩜𐑂ᛊ꩝:·

·⁖origin…language:·Sindarin·⁖·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖·˙font:·
·⁖pronunciation:·FAHN·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Cloud·⁖· Veil·⁖· Form or vision of a spiritual being·⁖· Haze as spirit becoming visible
↪ through veil·⁖·˙non…ostensive:·
·⁖ostensiveI:·Pale vapor holds a luminous contour in open air::presence standing within the veil while
↪ mist carries a body no flesh can bind𐑂·⁖·˙ostensiveI:·
·⁖ostensiveII:·Storm+cloud vapor carries horned spirit+face::antlers::eyes::and mouth as one fleshless
↪ visible veil𐑂·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Mist gathers into a fleshless face while one continuous utterance travels through the
↪ veiled spirit+form𐑂·⁖·˙ostensiveIII:·
·⁖grimchain:·⬡ϴϛᚱⱬⱯⱷ𐊰𐊫ⱬⱮϟϛ▫·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜𐑂◎꩝ category꩜꩜𐑂court꩝:·
·⁖lookup office꩜꩜𐑂phantasmagoria꩝ enoch꩜꩜𐑂ᛊ꩝:·

·⁖𐑂𐑂𐑂𐑂:·𐑂𐑂𐑂:·⁖·˙𐑂𐑂𐑂𐑂:·

·⁖origin…language:·Gnomish·⁖·˙origin…language:·
·⁖font:·Gnomish·⁖·˙font:·
·⁖pronunciation:·gool·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Ooze·⁖· Trickle·⁖· The contained crossing its boundary little by little·⁖·˙non…ostensive:·
·⁖ostensiveI:·Thick liquid bulges at the rim::A narrow thread hangs beyond the vessel while the
↪ contained body occupies both sides of the boundary◎·⁖·˙ostensiveI:·
·⁖ostensiveII:·Black elixir escapes beneath a sealed rune as one silver trickle crossing the ward
↪ through the smallest possible passage◎·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A dark trickle crosses the vessel rim little by little::each new length thickening from
↪ the same living ooze◎·⁖·˙ostensiveIII:·
·⁖grimchain:·⬡𐊰∗ϐⱡ⁜ⵠ𐊫ᚳᚸᚸ𐊰·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜𐑂⏁꩝ category꩜꩜𐑂court꩝:·
·⁖lookup office꩜꩜𐑂phantasmagoria꩝ enoch꩜꩜𐑂ᛊ꩝:·

·⁖origin…language:·Early Noldorin·⁖·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖·˙font:·
·⁖pronunciation:·BAH+god·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Trade·⁖· Trading·⁖· Two possessions entering relation through passage between
↪ holders·⁖·˙non…ostensive:·

·⁖ostensiveI:·Two held goods stand on opposite sides of an open passage⁚⁚Each crosses into another keeping⁚⁚with value still legible after possession changes hands⏀·⁖·˙ostensiveI:·
·⁖ostensiveII:·Two mirrored altars exchange ruby and moonstone across dark distance without either altar moving⏀·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Two sealed hands exchange moon+dust through a narrow opening⁚⁚each possession crossing while fine grains escape and settle between the holders⏀·⁖·˙ostensiveIII:·
·⁖grimchain:·Ⱪ∀₀ₒ⋇Ⱬ⋎∮ꓘᗡIA𐤗·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐑂𐑆 category◦–◦court𐑆:·
·⁖lookup office◦–◦phantasmagoria𐑆 enoch◦–◦ᖷ𐑆:·

·:ᘐᙏᙏᐢ:·ؤ𐑅ؤ𐑆𐑂◠:·˙ᘐᙏᙏᐢ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Shiv+ah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Ice Goddess·⁖ Fae of Alchemy and Science·⁖·˙non…ostensive:·
·⁖ostensiveI:·Six+sided frost blooms around a pale crystalline form⁚⁚blue+white stillness deepening through every edge as liquid water hardens beneath the same enchanted cold𐑂𐑅·⁖·˙ostensiveI:·
·⁖ostensiveII:·A pale ice+goddess form fills a blue+glass laboratory with white alchemical flame and six+sided frost equations𐑂𐑅·⁖·˙ostensiveII:·
·⁖ostensiveIII:·An ice+goddess field bends unseen air into six+sided frost equations⁚⁚loose powder and pale flame moving with the same exact current𐑂𐑅·⁖·˙ostensiveIII:·
·⁖grimchain:·⏣𐑍Xϒₒᖷᛯᔕ𐐗ᵽₒ∽ᘔᒐ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐑍𐑇 category◦–◦court𐑆:·
·⁖lookup office◦–◦phantasmagoria𐑆 enoch◦–◦ᖷ𐑆:·

·⁖origin…language:·Quenya·⁖·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖·˙font:·
·⁖pronunciation:·FYE+reh·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Disembodied spirit·⁖ Phantom·⁖ Radiance·⁖ Vanish as body leaving presence behind·⁖·˙non…ostensive:·
·⁖ostensiveI:·Pale spirit thins across the threshold until flesh⁚⁚contour⁚⁚and weight are gone⁚⁚leaving only a brief silver afterglow where presence had gathered𐑍·⁖·˙ostensiveI:·
·⁖ostensiveII:·A vanished silver figure remains present as radiance caught along every polished step of a moonlit stair𐑍·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A phantom leaves no flesh behind⁚⁚only radiant smoke winding through the chamber in the contour the vanished body once occupied𐑍·⁖·˙ostensiveIII:·
·⁖grimchain:·ᗸᗸᑭᵴᑕᑎ↾⏣⊕ᵬᒧᖵ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐑌𐑐𐑆 category◦–◦court𐑆:·
·⁖lookup office◦–◦phantasmagoria𐑆 enoch◦–◦ᖷ𐑆:·

·⁖ⱠᖴᐯᏦ:·ϟᖴᐯᖴᙢ·⁖·˙ⱠᖴᐯᏦ:·

·⁖origin…language:·Ilkorin·⁖.·˙origin…language:·
·⁖font:·Cirth Erebor·⁖.·˙font:·
·⁖pronunciation:·Bo+ron·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Steadfast·⁖ Trusty Man·⁖ Faithful Vassal·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Root holds under flood and stone bond holds under strain::Steadfast root and stone keep
↪ their bearing when force presses against themꕥ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Spectral vassal+shield stands beside an empty throne::its blue oath+mark unbroken
↪ through surrounding fire::flood::and ruinꕥ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Every blow against the steadfast spectral shield returns as a clear echo through the
↪ hall::while the sworn center behind it remains untouchedꕥ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᙀ‡ᗣ⍦⍋ᖇᘔᔐᔭᖮ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ₀₎ category꒰ₒcourt₎⁖·
·⁖lookup office꒰ₒphantasmagoria₎ enoch꒰ₒ₀₎⁖·

·⁖ᘜᘓᘐᐟ:·ᘚᑕᔕᘓᑕᔐᘜᓭ·⁖.·˙ᘜᘓᘐᐟ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Kah+rah+bey·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Consecrate as the Beginning·⁖ The Base of creation is Consecrated·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Earth::water::flame::and breath gather around the consecrated first stone::each body
↪ remaining distinct inside the same radiant beginningₒ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·One anointed foundation+stone carries consecrated fire through every rising course of
↪ an unfinished templeₒ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Consecrating oil marks the first stone around one empty framed opening::and every later
↪ wall rises from that sacred gap where light and air already passₒ·⁖.·˙ostensiveIII:·
·⁖grimchain:·❄ᙏᐱᖶᛋᐁᐁᓄᐃᑭᛋ✳·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ₀Θ₎ category꒰ₒcourt₎⁖·
·⁖lookup office꒰ₒphantasmagoria₎ enoch꒰ₒ₀₎⁖·

·⁖origin…language:·Neo+Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·Pa+thu·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Level Space·⁖ Sward·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Green terraces rise one above another across the sward::each level carrying its own
↪ height while the whole field reveals an ordered echelon from low ground to crownΘ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A perfectly level fae+field holds hovering stones::even rain::green grass::and no rise or
↪ hollow from edge to edgeΘ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One fae+green sward keeps the same level plane through wind::rain::and passing
↪ shadow::enduring without ridge or hollowΘ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᑉᓴᒧᗅᖱᑐᘐᑎᚻᚾᚱᚾ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ₀Ⅹ₎ category꒰ₒcourt₎⁖·
·⁖lookup office꒰ₒphantasmagoria₎ enoch꒰ₒ₀₎⁖·

⁖ᛕᛞᛏ᛫᛬ᘐᘏᘎᘏ⁖⁘˙ᛕᛞᛏ᛫᛬

⁘origin…language⁖Llumena Milenii⁘⁙˙origin…language⁖
⁘font⁖Mage Script⁘⁙˙font⁖
⁘pronunciation⁖Guy+yah⁘⁙˙pronunciation⁖
⁘cadence⁖Phantasmagoria⁘⁙˙cadence⁖
⁘non…ostensive⁖Earth Goddess⁘⁛ Fae of Earth Element⁘⁙˙non…ostensive⁖
⁘ostensiveI⁖Mountain⁜root⁜seed⁜and buried stone rise from one immense dark body beneath the
↪ soil⁜green life unfolding across its surface as though earth itself were breathing
↪ upwardᛝ⁘⁙˙ostensiveI⁖
⁘ostensiveII⁖Root+crowned earthen lights soften stone and wake buried seeds beneath one living
↪ fae+fire of the groundᛝ⁘⁙˙ostensiveII⁖
⁘ostensiveIII⁖Root+crowned earth+light fills the ground as a deep shudder moves through
↪ soil⁜stone⁜and buried seed beneath the same goddess+bodyᛝ⁘⁙˙ostensiveIII⁖
⁘grimchain⁖ᚸᛝℂ⌬ᛦᛉᘐ℅᜵㉾ᘓᚳᛏ⁘⁙˙grimchain⁖
⁘⁙˙lookup⁛
⁘⁙˙glyphs⁛
⁘glyph symbol∘⁘ᛣ᛬ category∘⁘᛫court᛬⁖
⁘lookup office∘⁘᛫phantasmagoria᛬ enoch∘⁘᛫ᛈ᛬⁖

⁖ᚳᚠᚠᛯ⁖ᚿᚠᚴᛊᚷ⁖⁘˙ᚳᚠᚠᛯ⁖

⁘origin…language⁖Ilkorin⁘⁙˙origin…language⁖
⁘font⁖Cirth Erebor⁘⁙˙font⁖
⁘pronunciation⁖Brith⁘⁙˙pronunciation⁖
⁘cadence⁖Phantasmagoria⁘⁙˙cadence⁖
⁘non…ostensive⁖Broken Stones⁘⁛ Gravel⁘⁙˙non…ostensive⁖
⁘ostensiveI⁖Broken stones lie packed beneath the road and bear the passing weight above⁜Rough
↪ fragments keep the path raised where soft ground would sinkᚢ⁘⁙˙ostensiveI⁖
⁘ostensiveII⁖Thousands of glowing fragments from a shattered stone giant cover the valley road with
↪ one field of broken living rockᚢ⁘⁙˙ostensiveII⁖
⁘ostensiveIII⁖Broken stones ring hard beneath the hammer⁜each gravel fragment keeping mountain+solid
↪ resistance in its smaller faceᚢ⁘⁙˙ostensiveIII⁖
⁘grimchain⁖▽᛬⃝ᐃᘐᔓᚻᛞᐅᛈᚸᛯ⁘⁙˙grimchain⁖
⁘⁙˙lookup⁛
⁘⁙˙glyphs⁛
⁘glyph symbol∘⁘᛫ᚺ᛬ category∘⁘᛫court᛬⁖
⁘lookup office∘⁘᛫phantasmagoria᛬ enoch∘⁘᛫ᛈ᛬⁖

⁖ᛕᛞᛏ᛫᛬ᘏᘔᘎᘓᘐ⁘⁖˙ᛕᛞᛏ᛫᛬

⁘origin…language⁖Llumena Milenii⁘⁙˙origin…language⁖
⁘font⁖Mage Script⁘⁙˙font⁖
⁘pronunciation⁖Ak+see+oh+mai+er⁘⁙˙pronunciation⁖
⁘cadence⁖Phantasmagoria⁘⁙˙cadence⁖
⁘non…ostensive⁖The Boundary Walker⁘⁛ Child of Mother and Father⁘⁙˙non…ostensive⁖
⁘ostensiveI⁖Threshold bears two orders along the same line⁜crossing possible at the boundary while
↪ neither side erases the other where they meetᚺ⁘⁙˙ostensiveI⁖
⁘ostensiveII⁖Silver and black stellar bloodlines remain separately visible beneath one skin while
↪ the whole figure crosses the luminous boundary without divisionᚺ⁘⁙˙ostensiveII⁖
⁘ostensiveIII⁖Silver and black ancestral cloths roll tightly around one
↪ threshold+bodyᚺ⁘⁙˙ostensiveIII⁖
⁘grimchain⁖ᔕᘐ᛬ᛯᚴᛞᛉᘑᘐᘓᘐᘏᚸ⁘⁙˙grimchain⁖
⁘⁙˙lookup⁛

·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦ᛕ» category◦–◦court»⁚·
·⁚lookup office◦–◦phantasmagoria» enoch◦–◦ᛦ»⁚·

⁖ᚳᚠᚹᚷ⁖·ᚢᚠᚲᛞ⁖·˙ᚳᚠᚹᚷ⁖·

·⁚origin…language⁚·Danian/Ossiriandic·⁚·˙origin…language⁚·
·⁚font⁚·Cirth Erebor·⁚·˙font⁚·
·⁚pronunciation⁚·kohn·⁚·˙pronunciation⁚·
·⁚cadence⁚·Phantasmagoria·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Bowed⁚˙ Bow+shaped⁚˙ Bent⁚˙ A formerly singular line opening away from its
↪ axis·⁚·˙non…ostensive⁚·
·⁚ostensiveI⁚·Bent line opens away from its former axis until the two ends stand wide apart⁚⁚Curvature
↪ spreads the single body across a broader reach without breaking itᛕ·⁚·˙ostensiveI⁚·
·⁚ostensiveII⁚·One unbroken star+beam curves beside a black obelisk into a silver arc away from its
↪ former straight axisᛕ·⁚·˙ostensiveII⁚·
·⁚ostensiveIII⁚·A silver bow+shape hangs across the dark and wanes night by night⁚⁚the same curved line
↪ becoming thinner without losing its bendᛕ·⁚·˙ostensiveIII⁚·
·⁚grimchain⁚·🢒♈♍◊▽♃‹✷'✡ₒ❁♋·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦🜚» category◦–◦court»⁚·
·⁚lookup office◦–◦phantasmagoria» enoch◦–◦ᛦ»⁚·

⁖ᗐᔑᒥ⁖·ᚠᚳᚱᚳᔕ⁖·˙ᗐᔑᒥ⁖·

·⁚origin…language⁚·Adûnaic·⁚·˙origin…language⁚·
·⁚font⁚·Adunaroth Classic·⁚·˙font⁚·
·⁚pronunciation⁚·Pha+raz·⁚·˙pronunciation⁚·
·⁚cadence⁚·Phantasmagoria·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Gold·⁚·˙non…ostensive⁚·
·⁚ostensiveI⁚·After fire and collapse⁚⁚one gold fragment remains beneath the ash⁚⁚its yellow weight
↪ keeping a small surviving body of what the greater form no longer holds🜚·⁚·˙ostensiveI⁚·
·⁚ostensiveII⁚·A buried gold coin keeps vanished solar warmth and yellow radiance beneath moonless
↪ earth🜚·⁚·˙ostensiveII⁚·
·⁚ostensiveIII⁚·Ash surrounds a yellow fragment that keeps its dense weight and warm gleam⁚˙ blackened
↪ stone changes around it⁚⁚yet the metal holds the same color under fire🜚·⁚·˙ostensiveIII⁚·
·⁚grimchain⁚·☌⊕♂♈♭₉ϒϞ⁂⁕⁓♈♍·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦ᛣ» category◦–◦court»⁚·
·⁚lookup office◦–◦phantasmagoria» enoch◦–◦ᛦ»⁚·

⁖ᛗᚼᛞᚩ⁖·ᛃᛗᛝᛝ⁖·˙ᛗᚼᛞᚩ⁖·

·⁚origin…language⁚·Gnomish·⁚·˙origin…language⁚·
·⁚font⁚·Gnomish·⁚·˙font⁚·
·⁚pronunciation⁚·Cwel·⁚·˙pronunciation⁚·
·⁚cadence⁚·Phantasmagoria·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Fade⁚˙ Wither·⁚·˙non…ostensive⁚·
·⁚ostensiveI⁚·Leaf keeps its form under thinning green as curled edge and dry vein remain inside the
↪ same diminished body⁚⁚the stem still bearing what has lost fullnessᚣ·⁚·˙ostensiveI⁚·
·⁚ostensiveII⁚·An enchanted banner carries fading heraldry⁚⁚pale thread⁚⁚curled fabric⁚⁚and withdrawn
↪ color beneath a dry windᚣ·⁚·˙ostensiveII⁚·

·⁖ostensiveIII:·Paleness spreads from stem to edge until the leaf curls inward⅄·⁖.·˙ostensiveIII:·
·⁖grimchain:·⚹ꓤ⟃ϫΛϟ'ᔑ⊀Ⅱ▽ᑫ৭·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ɬᓐ᳓ category○–○ɬcourt᳓⁖·
·⁖lookup office○–○ɬphantasmagoria᳓ enoch○–○ɬ᳘᳓⁖·

·⁖ƐɬÞT⁖·ꓤÞ⌐ΙՒ৭⁖.·˙ƐɬÞT⁖·

·⁖origin…language⁖·Dwarvish·⁖.·˙origin…language⁖·
·⁖font⁖·Dethek Stone·⁖.·˙font⁖·
·⁖pronunciation⁖·Krauth·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Immediate Kin·˙ Daily Living+With Family·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Hearth+kin roll blankets::cords::grain sacks::and sleeping mats into one tight
↪ burden::separate household things becoming a compact traveling body beneath the
↪ roofᓐ·⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Kin+signs::hearth::sleeping mats::shared bread::and warm light gather beneath changing
↪ roofs in the same returning household constellationᓐ·⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Warm hearth+air carries bread+smoke::recurring voices::and shared heat between
↪ neighboring sleeping matsᓐ·⁖.·˙ostensiveIII⁖·
·⁖grimchain⁖·✪ررسᓄ⬡ꓫᑕ∀△❋ϯ⊕⧗ϒ๐·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ɬℤ᳓ category○–○ɬcourt᳓⁖·
·⁖lookup office○–○ɬphantasmagoria᳓ enoch○–○ɬ᳘᳓⁖·

·⁖ƐɬÞT⁖·ꓶÞϮ৭ᖘ⁖.·˙ƐɬÞT⁖·

·⁖origin…language⁖·Dwarvish·⁖.·˙origin…language⁖·
·⁖font⁖·Dethek Stone·⁖.·˙font⁖·
·⁖pronunciation⁖·Groht·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Stone·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Dense mineral bears weight under weather and pressure::Chipped edge and darkened surface
↪ do not soften the hard body beneathℤ·⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Mountain+born stone keeps deep+earth weight inside surrounding spellfire without
↪ softeningℤ·⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·One stone pillar carries the roof through its hard body::full downward weight closing
↪ through the same durable grain into the groundℤ·⁖.·˙ostensiveIII⁖·
·⁖grimchain⁖·ᖯ⅄ᖇᖳ▫✱∀ᒥꓤ⛧◎Θᘐᘔᘔᘔ

·⁖ostensiveI:·Storm+pressure runs through tree::ground::and water at once::Branches tremble::stones quiver
↪ in their beds::and the whole dark field shudders beneath the charged sky⁑·⁖·˙ostensiveI:·
·⁖ostensiveII:·A black storm+ring encloses violet night::lightning::wind::rain::and trembling ground
↪ within one dangerous sky⁑·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A black storm presses over dark earth::wind and rain shuddering through soil::root::and
↪ stone across one dangerous ground+body⁑·⁖·˙ostensiveIII:·
·⁖grimchain:·⊬∫ℜ‡ℕ◺ᖶᘏᛘᚼᚺᚳ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥I𝅘𝅥 category◦–◦𝅘𝅥court𝅘𝅥:·
·⁖lookup office◦–◦𝅘𝅥phantasmagoria𝅘𝅥 enoch◦–◦𝅘𝅥ᖶ𝅘𝅥:·

⁖⅄ᕮᘉᘻ:·Ⅹᕮᘉᘻ⅄:⁖·˙⅄ᕮᘉᘻ:·

·⁖origin…language:·Doriathrin·⁖·˙origin…language:·
·⁖font:·Cirth Erebor·⁖·˙font:·
·⁖pronunciation:·Dorn·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Oak·⁖·˙non…ostensive:·
·⁖ostensiveI:·Deep roots hold the ground and hard grain bears the crown::broad shade remaining joined
↪ to earth through a trunk that keeps its line under weatherI·⁖·˙ostensiveI:·
·⁖ostensiveII:·One ancient oak joins grave+soil depth::cloud+height::starlit crown::deep roots::and hard
↪ trunk beneath one continuous living grainI·⁖·˙ostensiveII:·
·⁖ostensiveIII:·An ancient oak holds seven luminous ledges through its crown::each higher branch
↪ keeping its rank above the lower while one trunk bears the whole orderI·⁖·˙ostensiveIII:·
·⁖grimchain:·ᚣᛊᚸ◌✡ᚺ◺⧗⊛ᒧ⋕ᑫᒣ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥ᛞ𝅘𝅥 category◦–◦𝅘𝅥court𝅘𝅥:·
·⁖lookup office◦–◦𝅘𝅥phantasmagoria𝅘𝅥 enoch◦–◦𝅘𝅥ᖶ𝅘𝅥:·

⁖Ⱎ⌒ᛕᒥ:·ᚼᚴᚴᚠ:⁖·˙Ⱎ⌒ᛕᒥ:·

·⁖origin…language:·Adûnaic·⁖·˙origin…language:·
·⁖font:·Adunaroth Classic·⁖·˙font:·
·⁖pronunciation:·Ba+tân·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Road·⁖· Path·⁖· Way·⁖·˙non…ostensive:·
·⁖ostensiveI:·Road passes through the wall as a clean open cut::turning solid boundary into a
↪ traversable way where enclosed ground can meet the world beyondᛞ·⁖·˙ostensiveI:·
·⁖ostensiveII:·A pale+fire road joins two standing stones across impossible distance beneath one
↪ unbroken luminous pathᛞ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A road of bare elements crosses the ground: fire marks one reach::stone another::water
↪ fills the hollow::and air carries the open way between themᛞ·⁖·˙ostensiveIII:·
·⁖grimchain:·Xᛦᦓᘏᛋᓫᐱᚱᛇᛞᚣ⏁·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥ᛞ𝅘𝅥 category◦–◦𝅘𝅥court𝅘𝅥:·
·⁖lookup office◦–◦𝅘𝅥phantasmagoria𝅘𝅥 enoch◦–◦𝅘𝅥ᖶ𝅘𝅥:·

⁖ᚽᘓᘓᒡᛉ:·◡◠ᚽᘐ✳ᘓᘓ⁖·˙ᚽᘓᘓᒡᛉ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·

·⁖pronunciation:·Aum+nee+oh·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Hear⁖ Telepathy⁖ Hear the Echos·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Hidden voice strikes the inward chamber and returns from its far wall as a second
↪ presence::the same sound arriving again without another mouth between the two
↪ hearingsགག·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Two sealed crystal masks carry the same blue inward rune across a stone wall without
↪ sound passing between themགག·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A distant echo arrives inside a perfectly motionless chamber::sound crossing still air
↪ without disturbing dust on the waiting bladeགག·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᚹ๖ჯ꠸♀⁎∀֍✡ᢐᢐIIᢐ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꠸꠸ᵍ꠹ category꠸꠸court꠹:·
·⁖lookup office꠸꠸phantasmagoria꠹ enoch꠸꠸ᙏ꠹:·

·⁖origin…language:·Sindarin·⁖.·˙origin…language:·
·⁖font:·Tengwar Sindarin·⁖.·˙font:·
·⁖pronunciation:·FAY+en·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Vapour⁖ Disembodied Spirit⁖ Creation of a new Fae⁖ Fume as Vapour becoming
↪ Spiritlike·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Vapor holds a pale contour in open air::Pale mist+shape has no flesh::yet carries enough
↪ presence to resemble a living formགག·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Grey fume gathers as a small winged spirit+form lit from within above a cold
↪ brazierགག·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Silver vapor gathers into a fae contour and then thins back into nothing::the
↪ spirit+form vanishing through the same mist that made it visibleགག·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᛗ᥅Θᦓᦔᢐᠻᢩ❄ᦓ๐ᥣᥣ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꠸꠸ᛞ꠹ category꠸꠸court꠹:·
·⁖lookup office꠸꠸phantasmagoria꠹ enoch꠸꠸ᙏ꠹:·

·⁖ᚻᚹᚾᛏ:·ᨒᨊᨓᨔᨕᚻᚴᚹᚢᨋᨔᨓ·⁖.·˙ᚻᚹᚾᛏ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Ke+tzo+kot+’l·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Lightning Deity⁖ Fae of Air Element·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Feathered coils suspend themselves in open sky::each turn hollow with rushing breath
↪ while white fire races the scales and thunder travels through the vast living airཐ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Feathered air+fire::blue+white lightning::thunder::and charged wind coil around one
↪ invisible sky+centerཐ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Lightning+fae coils through a body of clear ice::blue+white current branching inside
↪ frozen water without melting the hard crystal fieldཐ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᛋ⁕ᵴᗰΘ♂ᦔ○ᚴᖷᓒ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꠸꠸♲꠹ category꠸꠸court꠹:·
·⁖lookup office꠸꠸phantasmagoria꠹ enoch꠸꠸ᙏ꠹:·

·⁖origin…language:·Early Noldorin·⁖.·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖.·˙font:·
·⁖pronunciation:·hleeth·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Dust⁖· The body after unity has become innumerable particles·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Fine particles lie without a single enclosing boundary::Air and ground carry scattered
↪ dust while innumerable grains hold what one outline cannot contain·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Glowing dust carries fragments of one clay sigil across air and stone as an innumerable
↪ body without a single enclosing form·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Dust from one shattered body trades places across two bowls::fine grains crossing
↪ between them through the same open air·⁖.·˙ostensiveIII:·
·⁖grimchain:·ሕᔑᑊᛁᛅᛯ♂⍑᠈ᐧᐩᐱᚯᘐ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol◦–◦ᵎक़᳓ category◦–◦ᵎcourt᳓:·
·⁖lookup office◦–◦ᵎphantasmagoria᳓ enoch◦–◦ᵎᖶ᳓:·

⁖ᨦᩣᨻ:·ᚕᚯᚿᛒᚕ:.·˙ᨦᩣᨻ:·

·⁖origin…language:·Gnomish·⁖.·˙origin…language:·
·⁖font:·Gnomish·⁖.·˙font:·
·⁖pronunciation:·AHL+thah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Shoot up⁖· Grow high⁖· Development as the sudden upward insistence of
↪ growth·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Green stem stands above dark soil under strong upward tension::root holding below as new
↪ height gathers in the living bodyक·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Green fire drives one buried seed upward through soil and ruined stone in a sudden
↪ living surgeक·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A green shoot surges upward from a cracked vessel as water spills downward through the
↪ same break::growth and pouring crossing at one living breachक·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᐱᘐ⬡ᢣᘏᐰᕴᓀ✪ᚸᑲᔑᔭ·⁖.·˙grimchain:·
·⁖.·˙lookup:·˙
·⁖.·˙glyphs:·˙
·⁖glyph symbol◦–◦ᵎᖷ᳓ category◦–◦ᵎcourt᳓:·
·⁖lookup office◦–◦ᵎphantasmagoria᳓ enoch◦–◦ᵎᖶ᳓:·

⁖ᩣᘏᘏᒡ:·ᔕᕆᑐᓭᘏᘏ·:.·˙ᩣᘏᘏᒡ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Vee+ey+torr·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Messenger·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Sealed messenger+mark crosses the distance in silence::then opens into a clear spoken
↪ phrase at the far threshold::Carried meaning becomes utterance without losing its
↪ shapeᖷ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A blue messenger+sigil crosses mountain distance inside one moving spark while keeping
↪ every stroke of its sealed meaningᖷ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A messenger+voice travels through white mist::its shaped sound reaching the far
↪ boundary while every visible landmark disappears inside the hazeᖷ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᛘϴᴢᗣᗴᛋᚠᚱᚻᚳᛦᚣ᚛ᚸ·⁖.·˙grimchain:·

·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ਯ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜phantasmagoria꩞ enoch꩜꩜ਨ꩞⁚·

·⁚origin…language⁚·Middle Quenya·⁚·˙origin…language⁚·
·⁚font⁚·Tengwar Quenya·⁚·˙font⁚·
·⁚pronunciation⁚·Kúma·⁚·˙pronunciation⁚·
·⁚cadence⁚·Phantasmagoria·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Void·˙ Vacuum as the Inhabitable Absence of Matter·⁚·˙non…ostensive⁚·
·⁚ostensiveI⁚·Open dark contains no wall and no matter against its reach⁚⁚Empty volume extends far
↪ enough to be entered without meeting substanceਯ·⁚·˙ostensiveI⁚·
·⁚ostensiveII⁚·No matter occupies the black chamber⁚⁚Dust⁚⁚flame⁚⁚and breath find nothing to rest
↪ against⁚⁚Traversable distance extends through matterless reachਯ·⁚·˙ostensiveII⁚·
·⁚ostensiveIII⁚·A loosened shackle opens into a black void⁚⁚and the released limb moves freely through
↪ absence where no matter remains to bind the passageਯ·⁚·˙ostensiveIII⁚·
·⁚grimchain⁚·ৰφ৫ᛦᗜ'ᛁᛊ৻'ᓄ⊕·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ਘ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜phantasmagoria꩞ enoch꩜꩜ਨ꩞⁚·

·⁚origin…language⁚·Early Noldorin·⁚·˙origin…language⁚·
·⁚font⁚·Tengwar Noldor·⁚·˙font⁚·
·⁚pronunciation⁚·MEE⟊neth·⁚·˙pronunciation⁚·
·⁚cadence⁚·Phantasmagoria·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Fine·˙ Slender·˙ Thin·˙ Small·˙ Substance drawn toward delicacy·⁚·˙non…ostensive⁚·
·⁚ostensiveI⁚·Long substance narrows almost to disappearance⁚⁚continuity remaining along the slight
↪ body as thickness gives way without the material vanishingਘ·⁚·˙ostensiveI⁚·
·⁚ostensiveII⁚·One hair⟊fine thread of moonlight spans the chamber continuously from a silver
↪ bowlਘ·⁚·˙ostensiveII⁚·
·⁚ostensiveIII⁚·One nearly vanishing silver thread remains as residue after the larger body is
↪ removed⁚⁚a slender remainder still carrying the old substanceਘ·⁚·˙ostensiveIII⁚·
·⁚grimchain⁚·ৰ◎'⁕⟊◊❁ꕥ⊕ſ᪥ıı·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ਣ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜phantasmagoria꩞ enoch꩜꩜ਨ꩞⁚·

·⁚origin…language⁚·Early Noldorin·⁚·˙origin…language⁚·
·⁚font⁚·Tengwar Noldor·⁚·˙font⁚·
·⁚pronunciation⁚·dolkh·⁚·˙pronunciation⁚·
·⁚cadence⁚·Phantasmagoria·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Stout·˙ Thick·˙ Substance gathering itself into density·⁚·˙non…ostensive⁚·
·⁚ostensiveI⁚·Mass gathers densely across a broad body⁚⁚Little open space lies within the substance⁚⁚and
↪ touch meets resistance through the full thicknessਣ·⁚·˙ostensiveI⁚·
·⁚ostensiveII⁚·A thumb⟊wide pearl of compressed silver mist dents the altar and draws nearby dust
↪ toward its impossible weightਣ·⁚·˙ostensiveII⁚·

·⁖ostensiveIII:·A dense black clamp closes around one sealed relic⁖· outward pull strains against the
↪ thick jaws⁘but the guarded weight does not leave⁇⁖·˙ostensiveIII:·
·⁖grimchain:·⁜ᔓᗅᗒᗅᑭᘐᑎᕓᔕᗅᕓ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᘓ꩞ category꩜꩜ᶜourt꩞:·
·⁖lookup office꩜꩜ᵖhantasmagoria꩞ enoch꩜꩜ᖶ꩞:·

·:ᘜᘐᘐᒣ:·ᔓᔕᔐᔓ:·⁖·˙ᘜᘐᘐᒣ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Ah⟊taut·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Sudden⁖· Fast Magic⁖· Rocky Situation·⁖·˙non…ostensive:·
·⁖ostensiveI:·Charged field convulses without warning: stone trembles⁘air twists⁘and bright force
↪ breaks into crossing currents before any settled direction can holdᘓ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Violet force fills a shattered cliff as levitating stone hangs bright before the
↪ echoᘓ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Rock fractures in a sudden burst and every tangled glowing line across the broken face
↪ resolves at once into one clear path through the stoneᘓ·⁖·˙ostensiveIII:·
·⁖grimchain:·ᵞ⧗ᓂℂᘐᔑᐱ⛧∮ᛉΘ◌⊲·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᘓ꩞ category꩜꩜ᶜourt꩞:·
·⁖lookup office꩜꩜ᵖhantasmagoria꩞ enoch꩜꩜ᖶ꩞:·

·:ᚳᚠᚠᚷ:·ᚴᚠᚠᚷ·⁖·˙ᚳᚠᚠᚷ:·

·⁖origin…language:·Danian/Ossiriandic·⁖·˙origin…language:·
·⁖font:·Cirth Erebor·⁖·˙font:·
·⁖pronunciation:·EH⟊ord·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Rest⁖· Ease as the state in which exertion no longer demands
↪ continuation·⁖·˙non…ostensive:·
·⁖ostensiveI:·Resting weight lies without demand for another motion⁘Breath keeps an easy measure⁘with
↪ exertion no longer pressing for continuationᘐ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Seven floating stones rest still upon moss beneath soft blue light⁘No pull or pressure
↪ disturbs moss⟊lit stillnessᘐ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Two exhausted currents enter one silver basin and fuse into a single quiet
↪ poolᘐ·⁖·˙ostensiveIII:·
·⁖grimchain:·✪ᑎ⬡ᚻᔑᐁᖇᔑᑭ❄ꙅ0·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵡ꩞ category꩜꩜ᶜourt꩞:·
·⁖lookup office꩜꩜ᵖhantasmagoria꩞ enoch꩜꩜ᖶ꩞:·

·⁖origin…language:·Quenya·⁖·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖·˙font:·
·⁖pronunciation:·FEH⟊ah⟊FEL⟊meh·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·

·⁖non…ostensive:·Spirit impulse⁘ Birth of Magickal Gifts within⁘ Magickal surge when emotion
↪ intensifies power or makes it behave unusually·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Inward force presses hard against its living boundary⁚⁚Emotion sharpens the current until
↪ the gift carries more power than its quiet measure without deliberation⁜·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A crystal heart carries newly born colored fires as distinct magical gifts within one
↪ opened core⁜·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Emotion brightens a hidden gift until blue fire flows out through the skin of the
↪ spell+circle and continues beyond its former boundary⁜·⁖.·˙ostensiveIII:·
·⁖grimchain:·⁕𐑂ᘏ🢒ᓚ∀♙⌕⁞⁞ᛋᑭᗺ⍜⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○₍△₎ category○–○₍court₎⁚·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᖶ₎⁚·

·:ᕊᛯ✱ᕊ:·☣⌾⍟⍜⍉:·⁖.·˙ᕊᛯ✱ᕊ:·

·⁖origin…language:·Gnomish·⁖.·˙origin…language:·
·⁖font:·Gnomish·⁖.·˙font:·
·⁖pronunciation:·AH+dheen·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Once upon a time⁘ Long ago⁘ Formerly⁘ The grammatical doorway through which the lost
↪ past becomes present enough to ache·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Old light enters the present through a remembered doorway⁚⁚lost distance keeping its ache
↪ while former time stands near enough to touch without becoming now△·⁖.·˙ostensiveI:·
·⁖ostensiveII:·An old+moon orchard⁚⁚dust+covered arch⁚⁚fallen petals⁚⁚and unreachable vanished season
↪ open behind the present threshold△·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Cold ash lifts from an old hearth and briefly rebuilds the vanished fire in
↪ light⁚⁚former warmth standing present inside the grey remains before falling back to
↪ dust△·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᑲᑭ⁞⁞ᓬᨒᑢᓄᕼ⊕Ⓚ❁ᕮᖻᛋ⁒·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○₍∀₎ category○–○₍court₎⁚·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᖶ₎⁚·

·:ᓘᚾᚦᛏ:·ᚱᚴᚦᛉᚱᚺᛚᛊᚦ:·⁖.·˙ᓘᚾᚦᛏ:·

·⁖origin…language:·Dwarvish·⁖.·˙origin…language:·
·⁖font:·Dethek Stone·⁖.·˙font:·
·⁖pronunciation:·TARJ+teer·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Place of happy gathering⁘ Joy becomes a place made real because beings gathered there
↪ happily·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Warm hall holds hearthlight⁚⁚shared food⁚⁚and bright sound⁚⁚Timber carries laughter as
↪ readily as flame carries heat⁚⁚and gathered gladness belongs to the place itself∀·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Fae lights fill an ancient+tree clearing among warm grass and branches carrying a faint
↪ ringing joy∀·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Bright voices and hearthlight fill a gathered hall until one impossible blow cracks
↪ the floor⁚⁚joy still ringing through the shards around the struck center∀·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᔓƵIA⁀☆ℳᓚᛉᛯ✠ᛉ✷𝍐⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○₍∀₎ category○–○₍court₎⁚·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᖶ₎⁚·

.:𓆏𓅓𓃰:𓄿𓂋𓄿𓂋𓄿:.·𓆏𓅓𓃰:·

·⁖origin…language:·Llumena Milenii·⁖.·origin…language:·
·⁖font:·Mage Script·⁖.·font:·
·⁖pronunciation:·Zai+st+uhs·⁖.·pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·cadence:·
·⁖non…ostensive:·Tree Lined Road⁖· Neitherlands⁖· Place Between Worlds·⁖.·non…ostensive:·
·⁖ostensiveI:·Desired city hangs beyond the mist on a road that belongs to neither waking shore⁚⁚Trees
↪ along the neither+road bend toward the unreachable light⁚⁚their path shaped by what has not yet
↪ been possessed▽·⁖.·ostensiveI:·
·⁖ostensiveII:·Silver tree+roots stand in one realm⁚⁚Flowering silver crowns open in another realm⁚⁚A
↪ road between two silver tree+lines belongs to neither world▽·⁖.·ostensiveII:·
·⁖ostensiveIII:·A hard black road runs between two incompatible skies⁚⁚its stone surface resisting
↪ every pressure while neither world can claim the passage▽·⁖.·ostensiveIII:·
·⁖grimchain:·𐎍𐎒𐎗𐎚𐎘𐎟𐎴𐎽𐎱𐎲𐎾𐎣·⁖.·grimchain:·
·⁖.·lookup·⁖
·⁖.·glyphs·⁖
·⁖glyph symbol○-○𐎤▽𐎥 category○-○𐎤court𐎥⁖·
·⁖lookup office○-○𐎤phantasmagoria𐎥 enoch○-○𐎤𐎣𐎥⁖·

.:𐎠𐎫𐎗:·𐎚𐎫𐎗𐎛:.·𐎠𐎫𐎗:·

·⁖origin…language:·Adûnaic·⁖.·origin…language:·
·⁖font:·Adunaroth Classic·⁖.·font:·
·⁖pronunciation:·roath·⁖.·pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·cadence:·
·⁖non…ostensive:·Foam⁖· White crest of waves⁖· Repetition appearing and vanishing along the body of a
↪ wave·⁖.·non…ostensive:·
·⁖ostensiveI:·White crest swings from rise to fall and rises again⁚⁚foam flashing at each turning while
↪ the sea carries the pendular motion along the ridge▽·⁖.·ostensiveI:·
·⁖ostensiveII:·Moon+white foam repeats one curved rune across rising crests⁚⁚each appearance vanishing
↪ and returning with the wave+body▽·⁖.·ostensiveII:·
·⁖ostensiveIII:·White foam travels along the wave and leaves the dark water clear behind each
↪ vanishing crest▽·⁖.·ostensiveIII:·
·⁖grimchain:·𐎐𐎞𐎓𐎍𐎔𐎜𐎢𐎱𐎲𐎾𐎣·⁖.·grimchain:·
·⁖.·lookup·⁖
·⁖.·glyphs·⁖
·⁖glyph symbol○-○𐎤▽𐎢𐎥 category○-○𐎤court𐎥⁖·
·⁖lookup office○-○𐎤phantasmagoria𐎥 enoch○-○𐎤𐎣𐎥⁖·

.:𐎥𐎺𐎢:·𐎣𐎺𐎦𐎥:.·𐎥𐎺𐎢:·

·⁖origin…language:·Draconic·⁖.·origin…language:·
·⁖font:·Iokharic·⁖.·font:·
·⁖pronunciation:·KOH+esh·⁖.·pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·cadence:·
·⁖non…ostensive:·Care⁖· Be concerned⁖· Be interested⁖· Compassion as care becoming an active
↪ relation·⁖.·non…ostensive:·
·⁖ostensiveI:·Near flame is sheltered from the wind⁚⁚broken edge receives support instead of
↪ abandonment⁚⁚and what might fall remains within reach of another strength▽·⁖.·ostensiveI:·
·⁖ostensiveII:·A cracked spirit+lantern shares warmth with a shielding second light beneath cold
↪ rain▽·⁖.·ostensiveII:·
·⁖ostensiveIII:·A silver vine wraps a hungry flame⁚⁚feeding it from its own body while shielding the
↪ fragile branch enclosed within it▽·⁖.·ostensiveIII:·

·⁖grimchain:·ᚏ⏁⁜ᚾ▽ᛁ♋ᛈᚱ❁ᛁᔕᛋ·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol꩜꩜ᔭᖇ꩞ category꩜꩜ᔭcourt꩞:·
·⁖lookup office꩜꩜ᔭphantasmagoria꩞ enoch꩜꩜ᔭᖶ꩞:·

⁖ᖘᛯᚿᚿᴑ:·ᘐ ᘓᛉᘔ:⁖.·˙ᖘᛯᚿᚿᴑ:·

·⁖origin…language:·Gnomish·⁖.·˙origin…language:·
·⁖font:·Gnomish·⁖.·˙font:·
·⁖pronunciation:·EES+tah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Know⁖˙ Be aware⁖˙ Perceive⁖˙ Feel⁖˙ Sense as awareness becoming
↪ recognition·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Fingertips cross an unseen contour and find every ridge before light reveals
↪ it::Pressure::warmth::and texture gather into recognition through the contact aloneᗅ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Moonlight reveals one hidden name already distributed through three unrelated
↪ floor+sigilsᗅ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·First spark opens in dry tinder::and previously unread marks flare into one clear
↪ figure across the barkᗅ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ΛIJьᔕᖷ⏁⏁ƾΘ0ΘⓄᑫ·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol꩜꩜ᔭ∀꩞ category꩜꩜ᔭcourt꩞:·
·⁖lookup office꩜꩜ᔭphantasmagoria꩞ enoch꩜꩜ᔭᖶ꩞:·

·⁖origin…language:·Early Noldorin·⁖.·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖.·˙font:·
·⁖pronunciation:·eye+th·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Pain⁖˙ Anguish⁖˙ Woe⁖˙ Ache already carrying body::duration and sorrow
↪ together·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Dark pressure fills the living chamber without outer wound::Breath tightens under the
↪ inward load::and the body cannot step outside the heaviness it bears∀·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Silent bell::stone bed::unmarked walls::and dull inward pressure fill one sorrowing
↪ chamber∀·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Invisible pressure constricts a flawless white tower from within while brilliance
↪ remains unbroken from base to crown∀·⁖.·˙ostensiveIII:·
·⁖grimchain:·⩒ᖶ♳֍ᓬ⊬ϒ0ᒣᓬᔑ1∮·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol꩜꩜ᔭ∀꩞ category꩜꩜ᔭcourt꩞:·
·⁖lookup office꩜꩜ᔭphantasmagoria꩞ enoch꩜꩜ᔭᖶ꩞:·

⁖ᛣᚡᚠᛯ:·ᗕᚡᛁᛯᚼ⁖.·˙ᛣᚡᚠᛯ:·

·⁖origin…language:·Danian/Ossiriandic·⁖.·˙origin…language:·
·⁖font:·Cirth Erebor·⁖.·˙font:·
·⁖pronunciation:·GOL+dah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·One of the Wise Folk⁖˙ Gnome⁖˙ Depth embodied as one of the Wise rather than an
↪ adjective of depth·⁖.·˙non…ostensive:·

·⁖ostensiveI⁖·Small ancient figure stands among root and stone while sealed knots in the earth open
↪ beneath careful fingers⁖⁖buried signs brightening where ordinary sight had passed over them for
↪ generations⁘·⁚·˙ostensiveI⁖·
·⁖ostensiveII⁖·A small stone+crowned figure stands amid open root+runes and luminous mechanisms of
↪ the deep earth⁘·⁚·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Stone+crowned wise form stands inside a frost+bound road of ordeal⁖⁖deep runes burning
↪ beneath the cold where shelter never opens⁘·⁚·˙ostensiveIII⁖·
·⁖grimchain⁖·♎ↄ⫪ᓄ⊕ƳƳɻ◝⍥⅋ᗯ⅃ᙠ·⁚·˙grimchain⁖·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥Ⲏ𝅘𝅥 category◦–◦𝅘𝅥court𝅘𝅥⁖·
·⁖lookup office◦–◦𝅘𝅥phantasmagoria𝅘𝅥 enoch◦–◦𝅘𝅥ᖱ𝅘𝅥⁖·

·⁖ᛞᛞᛏ⁖·ᚸᚩᛋᛟᚳᚣᛞᚩᛋ⁖·⁚·˙ᛞᛞᛏ⁖·

·⁖origin…language⁖·Llumena Milenii·⁚·˙origin…language⁖·
·⁖font⁖·Mage Script·⁚·˙font⁖·
·⁖pronunciation⁖·Hal+ee+behr+ah·⁚·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁚·˙cadence⁖·
·⁖non…ostensive⁖·Imprison⁖· Disdain⁖· Dark Place⁖· Unharmonic⁖· Disresonant·⁚·˙non…ostensive⁖·
·⁖ostensiveI⁖·Dark chamber receives several unlike tones and bends them through one resonant
↪ vault⁖⁖Separate voices return from the walls sharing a common interval without losing their own
↪ pitch✢·⁚·˙ostensiveI⁖·
·⁖ostensiveII⁖·Discord remains trapped inside a dark chamber⁖⁖Every note returns against itself⁖⁖No
↪ opening releases dissonant pressure outward✢·⁚·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Black prison+walls soften like wax around a dissonant sound⁖⁖the enclosing dark melting
↪ while the trapped vibration remains harsh and unresolved✢·⁚·˙ostensiveIII⁖·
·⁖grimchain⁖·⁕ᖁᛉᛘᛋ❂𓂀ᘐᛯᖁᖘᖁ·⁚·˙grimchain⁖·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥✙𝅘𝅥 category◦–◦𝅘𝅥court𝅘𝅥⁖·
·⁖lookup office◦–◦𝅘𝅥phantasmagoria𝅘𝅥 enoch◦–◦𝅘𝅥ᖱ𝅘𝅥⁖·

·⁖origin…language⁖·Early Noldorin·⁚·˙origin…language⁖·
·⁖font⁖·Tengwar Noldor·⁚·˙font⁖·
·⁖pronunciation⁖·deer·⁚·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁚·˙cadence⁖·
·⁖non…ostensive⁖·Secret⁖· Something preserved specifically by not becoming public·⁚·˙non…ostensive⁖·
·⁖ostensiveI⁖·Closed vessel shows nothing of what rests inside⁖⁖Outer surface remains blank⁖⁖with
↪ silence forming part of the boundary that preserves the hidden body✙·⁚·˙ostensiveI⁖·
·⁖ostensiveII⁖·A black pearl holds one spoken name without mark⁖⁖echo⁖⁖or outward sound✙·⁚·˙ostensiveII⁖·
·⁖ostensiveIII⁖·A secret mark hidden behind blank stone erupts once as sudden white radiance⁖⁖then the
↪ outer wall returns to darkness with the sign still concealed✙·⁚·˙ostensiveIII⁖·
·⁖grimchain⁖·ᚱ❂♂♈♎↓ᚱᔑᚿ⛥⛤↓⊛ᚹ·⁚·˙grimchain⁖·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦–◦𝅘𝅥Ⲓ𝅘𝅥 category◦–◦𝅘𝅥court𝅘𝅥⁖·
·⁖lookup office◦–◦𝅘𝅥phantasmagoria𝅘𝅥 enoch◦–◦𝅘𝅥ᖱ𝅘𝅥⁖·

·⁖ᚳᚾᚦᛏ⁖·ᚸᚾᚦᛏᛁᚳ⁖·⁚·˙ᚳᚾᚦᛏ⁖·

·⁖origin…language:·Dwarvish·⁖.·˙origin…language:·
·⁖font:·Dethek Stone·⁖.·˙font:·
·⁖pronunciation:·GOR+dool·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Oath of amazement or despair·⁘ Wonder as astonishment strong enough to become
↪ utterance·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Breath bears oath+weight inside the throat as astonishment and despair press through one
↪ charged sound::the utterance heavier than ordinary speech‡·⁖.·˙ostensiveI:·
·⁖ostensiveII:·White oath+fire::wonder::despair::charged utterance::and ringing standing stones fill one
↪ astonished circle‡·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Astonished oath+fire strikes hot iron between hammer and anvil::wonder ringing through
↪ the worked edge as each blow fixes the vow deeper into the metal‡·⁖.·˙ostensiveIII:·
·⁖grimchain:·⸗⛧⛇⅂ᚺᛢᛞ⅄⅍⅍ᛣ⅂ᚺ⅍ᛣ⅍ᛣ⅍ᛣ

·⁖grimchain⁖·⚹ᑭᖇ▫ᚼZ❀ᙡHᛘ✻▽⏁·⁖.·ˈgrimchain⁖·
·⁖.·ˈlookup·ˈ
·⁖.·ˈglyphs·ˈ
·⁖glyph symbol○–○ᵩ⊰» category○–○ᵩcourt»⁖·
·⁖lookup office○–○ᵩphantasmagoria» enoch○–○ᵩᖶ»⁖·

⁘ᚼᛞᛋᛏ⁖·ᛏᛉᛟᛇᔕᛞᓚ⁘.·ˈᚼᛞᛋᛏ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖.·ˈorigin…language⁖·
·⁖font⁖·Mage Script·⁖.·ˈfont⁖·
·⁖pronunciation⁖·Lee+behr+ah·⁖.·ˈpronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·ˈcadence⁖·
·⁖non…ostensive⁖·Liberation as Being⁖· Liberate as Magic and Intent⁖· Exit Halibera·⁖.·ˈnon…ostensive⁖·
·⁖ostensiveI⁖·Liberating magic ruptures the dark seal and releases the gathered body into sparks and
↪ pale dust⁖⁖What had occupied one bound center spreads outward until open air lies between every
↪ fragment⊰·⁖.·ˈostensiveI⁖·
·⁖ostensiveII⁖·Silver fire opens a black chain into pale moths until the broken circle holds no bound
↪ link⊰·⁖.·ˈostensiveII⁖·
·⁖ostensiveIII⁖·A broken ward opens into free passage while one old seal remains impressed in the
↪ threshold stone⊰·⁖.·ˈostensiveIII⁖·
·⁖grimchain⁖·ᛏᛉᑭᖘᓀᕊᛘᑭᐊᛉI·⁖.·ˈgrimchain⁖·
·⁖.·ˈlookup·ˈ
·⁖.·ˈglyphs·ˈ
·⁖glyph symbol○–○ᵩ▦» category○–○ᵩcourt»⁖·
·⁖lookup office○–○ᵩphantasmagoria» enoch○–○ᵩᖶ»⁖·

⁘ᚼᛞᛋᛏ⁖·ᘐᔕᛞᓚᛏᛇᓚ⁘.·ˈᚼᛞᛋᛏ⁖·

·⁖origin…language⁖·Llumena Milenii·⁖.·ˈorigin…language⁖·
·⁖font⁖·Mage Script·⁖.·ˈfont⁖·
·⁖pronunciation⁖·Mer+ahl+dah·⁖.·ˈpronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·ˈcadence⁖·
·⁖non…ostensive⁖·Disease⁖· Gaping Darkness⁖· Pit·⁖.·ˈnon…ostensive⁖·
·⁖ostensiveI⁖·Black depth opens under the surface⁖⁖Abyssal rim stands sickened and unstable while light
↪ finds no lower boundary inside the hollow▦·⁖.·ˈostensiveI⁖·
·⁖ostensiveII⁖·Black+veined darkness sickens stone and softens the rim of one widening pit like a
↪ wound beneath the ground▦·⁖.·ˈostensiveII⁖·
·⁖ostensiveIII⁖·A black pit opens as interlocked stone plates darken one after another and pull the
↪ joined structure downward▦·⁖.·ˈostensiveIII⁖·
·⁖grimchain⁖·Ↄ⁘ᔓᛘᘐᑫᐁᛗ‹ᚼᒪᖶᛗᔑ·⁖.·ˈgrimchain⁖·
·⁖.·ˈlookup·ˈ
·⁖.·ˈglyphs·ˈ
·⁖glyph symbol○–○ᵩ⊐» category○–○ᵩcourt»⁖·
·⁖lookup office○–○ᵩphantasmagoria» enoch○–○ᵩᖶ»⁖·

⁖ᗐᓀᛕ⁖·ᚸᚺᒪᓀᚸᔕᛕᒥᒃ⁖.·ˈᗐᓀᛕ⁖·

·⁖origin…language⁖·Adûnaic·⁖.·ˈorigin…language⁖·
·⁖font⁖·Adunaroth Classic·⁖.·ˈfont⁖·
·⁖pronunciation⁖·ah+kahl+lah+BETH·⁖.·ˈpronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·ˈcadence⁖·
·⁖non…ostensive⁖·Downfallen⁖· A drop magnified into mythic Falling·⁖.·ˈnon…ostensive⁖·
·⁖ostensiveI⁖·Crowned body lies under overwhelming downward force⁖⁖Height and ruin remain together
↪ throughout the mythic descent⁖⁖the fall immense without requiring an aftermath⊐·⁖.·ˈostensiveI⁖·

·⁖ostensiveII:·A crowned star burns through seven cloud⟊layers in one immense fall from the high
↪ vault⍈·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A crowned body falls through ash⟊dark air while invisible bands tighten around the
↪ descent and fail to stop it⍈·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᖱᑫᚾ▫ↄᗡᐃ⍜ᓰXΛᚠ⍥·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉⊩꞉ category○–○𐑉court꞉·
·⁖lookup office○–○𐑉phantasmagoria꞉ enoch○–○𐑉ᖷ꞉·

·⁖origin…language:·Early Noldorin·⁖.·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖.·˙font:·
·⁖pronunciation:·AH⟊layth·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Shielding·˙ Warding off·˙ Protection·˙ Incoming force reaches a ward instead of its
↪ intended destination·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Incoming force spreads across a held ward::Shielded inner body remains beyond the
↪ pressure while the ward bears what was aimed inward𐑉⊩·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Blue ward⟊runes turn a storm of black arrows along one luminous curve away from the
↪ inner flame𐑉⊩·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A violet ward catches the incoming force while an empty niche behind it remains
↪ guarded around the absent relic𐑉⊩·⁖.·˙ostensiveIII:·
·⁖grimchain:·⁕ᗤᚷNᙅǂᖷ⏢ᚹᦓIᨀ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉⏣꞉ category○–○𐑉court꞉·
·⁖lookup office○–○𐑉phantasmagoria꞉ enoch○–○𐑉ᖷ꞉·

⁖ᖘᛯᛯ✿ʘ:·∞ᵝᵝ𐑀𐑂:·⁖.·˙ᖘᛯᛯ✿ʘ:·

·⁖origin…language:·Gnomish·⁖.·˙origin…language:·
·⁖font:·Gnomish·⁖.·˙font:·
·⁖pronunciation:·boo⟊een·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Desire·˙ Wish·˙ Want existing as its own interior body·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Empty chamber inside the living body leans toward an absent shape::Desiring heat gathers
↪ along the inward hollow and keeps its contour though nothing answers from beyond
↪ it⏣·⁖.·˙ostensiveI:·
·⁖ostensiveII:·An empty silver bowl holds one steady inward pull of light toward a sealed star beyond
↪ reach⏣·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Silver cords inside an empty bowl lean toward a faint spectral outline beyond
↪ torchlight as the vague form dissolves into darkness⏣·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᑭʃᛘᑫ☆ᛉΘ₩ᛤƁᛦᛤᚼ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉🢒꞉ category○–○𐑉court꞉·
·⁖lookup office○–○𐑉phantasmagoria꞉ enoch○–○𐑉ᖷ꞉·

⁖ᚼᙆᙅᐟ:·ᔓᓭᙆ✶ᓚ ᘐᓚ✶ᓭ:·⁖.·˙ᚼᙆᙅᐟ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·

·⁖pronunciation:·Ah+tone+mehnt·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Restoration of Identity·⁖·˙non…ostensive:·
·⁖ostensiveI:·Scattered pieces lie aligned around one underlying mark::Every fragment carries the same
↪ pattern::and one continuous identity occupies the joined surface·⁖·˙ostensiveI:·
·⁖ostensiveII:·Shattered mirror+fragments carry one complete silver sigil as they occupy the original
↪ frame again·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Across a healed seam::the old pattern spreads through new tissue until both sides carry
↪ the same continuous design·⁖·˙ostensiveIII:·
·⁖grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍ category○–○₍court₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍₹₎⁖·

⁖

·⁖origin…language:·Gnomish·⁖·˙origin…language:·
·⁖font:·Gnomish·⁖·˙font:·
·⁖pronunciation:·HAH+lah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Drag⁖· Draw⁖· Pull⁖· Distance being compelled to diminish·⁖·˙non…ostensive:·
·⁖ostensiveI:·Taut line bears force across the interval::Heavy body remains under steady draw while the
↪ distance between source and load stays compressed by tension·⁖·˙ostensiveI:·
·⁖ostensiveII:·A red light+chain compels two standing stones across a wide field into diminishing
↪ distance·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A taut silver cord draws two stones together::carving one visible line into the ground
↪ as shrinking distance leaves its mark behind·⁖·˙ostensiveIII:·
·⁖grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍ category○–○₍court₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍₹₎⁖·

⁖

·⁖origin…language:·Darnassian·⁖·˙origin…language:·
·⁖font:·Darnassian Runes·⁖·˙font:·
·⁖pronunciation:·EL+dreh·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Kingdom⁖· Homeland⁖· The bounded realm inside which a happening becomes
↪ possible·⁖·˙non…ostensive:·
·⁖ostensiveI:·Bounded realm holds one complete stage of the greater design: field::gate::and stone stand
↪ ready inside it::while the next ordered ground waits beyond another threshold·⁖·˙ostensiveI:·
·⁖ostensiveII:·Inside the standing+stone ring::rain falls::lamps burn::and roads unfold beneath a green
↪ sky::Outside the standing+stone ring::bare ground holds no rain::lamp::or continuing
↪ road·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A bounded realm descends farther than torchlight can reach::enclosed ground continuing
↪ whole into unseen depth·⁖·˙ostensiveIII:·
·⁖grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○–○₍ category○–○₍court₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍₹₎⁖·

·:⁂᭄ᨒ᭄ᨒ:·⁂ᨐᨒᨐ✳᭄ᨒ᭄ᨒ¿·:.·˙⁂᭄ᨒ᭄ᨒ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Wa+ohl+noos·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Wound⁘ Paradox in Time Space·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Split flesh holds an impossible continuity across the break⁘ pain belongs to the
↪ opening⁝⁝yet the body remains one living line around it🮥·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Spring blossom⁝⁝winter frost⁝⁝open flesh⁝⁝and restored skin coexist along one raw seam
↪ beneath divided seasons🮥·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A time+wound glows inside steady heat⁝⁝raw edges browning while future scar and present
↪ rupture occupy the same impossible flesh🮥·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᦓᦅᦔᦅᦟ᪲ᦔᦅᦓᦅᦔᦅ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱ category꒰ₒ꒱court꒱·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᖴ꒱·

·⁖origin…language:·Early Noldorin·⁖.·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖.·˙font:·
·⁖pronunciation:·THOO+ring+WEH+thil·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·She of Hidden Shadow⁘ Hidden potential embodied as a presence whose nature is
↪ concealment·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Dark contour contains an unopened force⁝⁝Hidden potential remains materially potent
↪ inside the concealing shadow without exposure❮·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A winged shadow carries nine unopened violet fires beneath one sealed black
↪ veil❮·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Hidden shadow recoils before any visible blow reaches it⁘ the veil tightens around the
↪ concealed form❮·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᥫ᭡ᦅᦟ᪲ᦔ᪲ᦅᦟ᪲ᦔᦅ✳🮥᪲ᦅ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꒰ₒ꒱ᨐ꒱ category꒰ₒ꒱court꒱·
·⁖lookup office꒰ₒ꒱phantasmagoria꒱ enoch꒰ₒ꒱ᖴ꒱·

·⁖Ⱎ⁀ᛕᒥ:·ᒪᓂᐢ:·⁖.·˙Ⱎ⁀ᛕᒥ:·

·⁖origin…language:·Adûnaic·⁖.·˙origin…language:·
·⁖font:·Adunaroth Classic·⁖.·˙font:·
·⁖pronunciation:·NAH+loo·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Shadow⁘ The secondary body born where form interrupts light·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Form blocks a path of light⁝⁝Dark secondary contour lies on the refused side⁝⁝keeping the
↪ boundary of what brightness could not crossᨐ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Moonlight and crystal tower cast one perfect dark twin across silver water wherever
↪ brightness is refusedᨐ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One cast shadow folds itself into a sealed black pouch beneath the interrupted
↪ light⁝⁝darkness stored and kept behind the tied openingᨐ·⁖.·˙ostensiveIII:·
·⁖grimchain:·◎ᨏ☽ᨒ꧁ᨐ✓ᦔᨐ⊕⊛ᨒᨏ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙

·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦ᵩ category◦–◦ᵩcourt⁘·
·⁖lookup office◦–◦ᵩphantasmagoria⁘ enoch◦–◦ᵩ⁘·

⁖ᵹᚱᚢᛋᛁ·ᛋᚣᛟᛏᚳᛟᚢᛟᚱᛦ⁖·˙ᵹᚱᚢᛋᛁ⁖·

·⁖origin…language:·Darnassian·⁖·˙origin…language:·
·⁖font:·Darnassian Runes·⁖·˙font:·
·⁖pronunciation:·shahl+AH+rahn·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Crown of the Night·⁖ Darkness given crown and nocturnal body·⁖·˙non…ostensive:·
·⁖ostensiveI:·Night bears sovereignty across the high dark·⁖ stars burn beneath its span::and no hand or
↪ throne is needed for the nocturnal crown to remain presentᵹᵐ·⁖·˙ostensiveI:·
·⁖ostensiveII:·A black moon wears cold stars above every lesser shadow gathered beneath its dark
↪ centerᵹᵐ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Night wears a finished crown of stars with every point seated and no vacant place in
↪ the celestial formᵹᵐ·⁖·˙ostensiveIII:·
·⁖grimchain:·ᛋᛏ❂ᵹ▽ᚏᛦᛞᚏᚭᛇᚣᚍ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦ᵩᥴ category◦–◦ᵩcourt⁘·
·⁖lookup office◦–◦ᵩphantasmagoria⁘ enoch◦–◦ᵩ⁘·

⁖ᚠᛗᛗᛟᛟᛟ:·ᛒᛝᛝᛝᛝ⁖·˙ᚠᛗᛗᛟᛟᛟ:·

·⁖origin…language:·Gnomish·⁖·˙origin…language:·
·⁖font:·Gnomish·⁖·˙font:·
·⁖pronunciation:·helk·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Ice+cold·⁖ Icy·⁖ Cold·⁖ Cold concentrated until it approaches Ice·⁖·˙non…ostensive:·
·⁖ostensiveI:·Water stands hard beneath concentrated cold::pale vapor hanging in the air while the
↪ surface bears the sharp stillness of ice at its boundaryᥴ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Concentrated cold turns breath to crystal runes::rims water with hardness::and narrows
↪ blue flame into an icy pointᥴ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Ice+cold hide splits under opposing pulls::jagged frost racing along the tear as
↪ concentrated cold hardens both opening edgesᥴ·⁖·˙ostensiveIII:·
·⁖grimchain:·▽ᛋ☆☆Xቻᚏᚱᛯᚿᛏᑫᛊᛯ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦ᵩᛯ category◦–◦ᵩcourt⁘·
·⁖lookup office◦–◦ᵩphantasmagoria⁘ enoch◦–◦ᵩ⁘·

⁖ᚻᛞᛞᛏ:·ᦀᦵᦅᦀᦵᦈᦅᦵᦈᦅᦀᦵᦈᦅᦀᦵᦈᦅ⁖·˙ᚻᛞᛞᛏ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Ehm+ah+gehg+grih·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Moving image·⁖ Beauty·⁖··⁖·˙non…ostensive:·
·⁖ostensiveI:·Silver figure walks across a darkened surface while light changes around every
↪ step::Moving likeness keeps its beauty even when no position remains stillᛯ·⁖·˙ostensiveI:·
·⁖ostensiveII:·An enchanted mural carries graceful silver figures through changing poses across the
↪ wallᛯ·⁖·˙ostensiveII:·

·⁖ostensiveIII꞉·A graceful luminous image reshapes itself frame by frame⁚⁚each new contour cut cleanly from the last↟·⁚.·˙ostensiveIII꞉·
·⁖grimchain꞉·ᚸᛃᚼᛒᛋ◎ᛘᚻᚸᚹᛦᚳᚨ▫·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁖glyph symbol꞊꞉ᛕ꞉ category꞊꞉court꞉꞉·
·⁖lookup office꞊꞉phantasmagoria꞉ enoch꞊꞉ᚲ꞉꞉·

·⁖origin…language꞉·Early Noldorin·⁚.·˙origin…language꞉·
·⁖font꞉·Tengwar Noldor·⁚.·˙font꞉·
·⁖pronunciation꞉·DAHD+noo+een·⁚.·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁚.·˙cadence꞉·
·⁖non…ostensive꞉·Sinking down⁖˙ Descent that continues beneath the ordinary surface·⁚.·˙non…ostensive꞉·
·⁖ostensiveI꞉·Weight lies below the ordinary surface under a continuous downward relation⁖˙ light remains above⁚⁚and deeper reach extends beneath the same sinking body↟·⁚.·˙ostensiveI꞉·
·⁖ostensiveII꞉·A stone temple occupies mirror+black water from visible roof above the surface into luminous depth far below↟·⁚.·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Two silver bodies sink through black water and meet below the surface as one fused shape↟·⁚.·˙ostensiveIII꞉·
·⁖grimchain꞉·ᚦⓄ⬡ᨒᦓᛈᘓᘐᔑᛇzᚹ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁖glyph symbol꞊꞉ᚢ꞉ category꞊꞉court꞉꞉·
·⁖lookup office꞊꞉phantasmagoria꞉ enoch꞊꞉ᚲ꞉꞉·

·⁖origin…language꞉·Neo+Sindarin·⁚.·˙origin…language꞉·
·⁖font꞉·Tengwar Sindarin·⁚.·˙font꞉·
·⁖pronunciation꞉·gwehr·⁚.·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁚.·˙cadence꞉·
·⁖non…ostensive꞉·Weave⁖˙ Wind⁖˙ Spin⁖˙ Twist⁖˙ Weaving and distortion occupying the same motion·⁚.·˙non…ostensive꞉·
·⁖ostensiveI꞉·Fiber turns around fiber under continuous torsion⁖˙ straightness is absent across the woven path⁚⁚and the same winding force shapes both weave and distortion↟·⁚.·˙ostensiveI꞉·
·⁖ostensiveII꞉·Rotational force braids air currents into one threadless woven distortion across open space↟·⁚.·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Twisted wind+thread circles back through the loom⁚⁚returning along its own woven path while every fiber keeps rotational tension↟·⁚.·˙ostensiveIII꞉·
·⁖grimchain꞉·zᛯᛏᛉᚹᚢᛘᛇᛗᛏᚥᚸᘔ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁖glyph symbol꞊꞉ᛯ꞉ category꞊꞉court꞉꞉·
·⁖lookup office꞊꞉phantasmagoria꞉ enoch꞊꞉ᚲ꞉꞉·

꞉word꞉·ᛏᚲᚱ꞉·⁚.·˙word꞉·

·⁖origin…language꞉·Klingon·⁚.·˙origin…language꞉·
·⁖font꞉·Klingon pIqaD HaSta·⁚.·˙font꞉·
·⁖pronunciation꞉·yoh·⁚.·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁚.·˙cadence꞉·

·⁖non…ostensive꞉·Brave⁖· Courage inhabiting the being rather than remaining an
↪ abstraction·⁖˙·˙non…ostensive꞉·
·⁖ostensiveI꞉·Threat presses close to the standing body·⁖· fear remains present in the air⁚⁚and the
↪ stance does not surrender its direction under the danger⁂·⁖˙·˙ostensiveI꞉·
·⁖ostensiveII꞉·One white fire remains upright inside trembling blue flame beneath the threat of a
↪ hovering dark spear⁂·⁖˙·˙ostensiveII꞉·
·⁖ostensiveIII꞉·One white fire follows an appointed road through a field of fear without bending from
↪ the destined line⁂·⁖˙·˙ostensiveIII꞉·
·⁖grimchain꞉·ᵑ≋Θⴲⴳ⏁ᖘ▽ᑫϟⴲ↗ϗ·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup⁖˙
·⁖˙·˙glyphs⁖˙
·⁖glyph symbol꞉○–○ₑ𝄋₎ category꞉○–○ₑcourt₎⁖·
·⁖lookup office꞉○–○ₑphantasmagoria₎ enoch꞉○–○ₑ𐤊₎⁖·

·⁖origin…language꞉·Noldorin·⁖˙·˙origin…language꞉·
·⁖font꞉·Tengwar Noldor·⁖˙·˙font꞉·
·⁖pronunciation꞉·Taetha·⁖˙·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖˙·˙cadence꞉·
·⁖non…ostensive꞉·Fasten⁖· Tie⁖· Binding as Relation·⁖˙·˙non…ostensive꞉·
·⁖ostensiveI꞉·Cord bites around separate bodies⁚⁚Bound forms cannot move apart without pulling against
↪ the same knotted line𝄋·⁖˙·˙ostensiveI꞉·
·⁖ostensiveII꞉·A red rune+thread holds two floating stones in one luminous relation across empty
↪ air𝄋·⁖˙·˙ostensiveII꞉·
·⁖ostensiveIII꞉·A luminous knot tightens as the joined weight grows⁖· cord bites deeper⁚⁚but the bound
↪ ends do not part𝄋·⁖˙·˙ostensiveIII꞉·
·⁖grimchain꞉·𝄋ᵹ⅍≍ϗ▽⊙ᵑ↗ℵ⅍𝍐·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup⁖˙
·⁖˙·˙glyphs⁖˙
·⁖glyph symbol꞉○–○ₑ𐤑₎ category꞉○–○ₑcourt₎⁖·
·⁖lookup office꞉○–○ₑphantasmagoria₎ enoch꞉○–○ₑ𐤊₎⁖·

·⁖𞤤𞤦𞤩𞤥꞉·𞤨𞤲𞤩𞤥𞤧⁖˙·˙𞤤𞤦𞤩𞤥꞉·

·⁖origin…language꞉·Ilkorin·⁖˙·˙origin…language꞉·
·⁖font꞉·Cirth Erebor·⁖˙·˙font꞉·
·⁖pronunciation꞉·lowth·⁖˙·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖˙·˙cadence꞉·
·⁖non…ostensive꞉·Licking up food or drink·⁖· Gluttonous eating·⁖· The outer world crossing the boundary
↪ and becoming part of the inside·⁖˙·˙non…ostensive꞉·
·⁖ostensiveI꞉·Tongue holds outer substance across the mouth+boundary·⁖· food rests within the inward
↪ chamber⁚⁚and living interior bears what open space still touches at the lips𐤑·⁖˙·˙ostensiveI꞉·
·⁖ostensiveII꞉·Bread⁚⁚fruit⁚⁚wine⁚⁚empty plates⁚⁚and swelling inner fire occupy one black
↪ feast+gate𐤑·⁖˙·˙ostensiveII꞉·
·⁖ostensiveIII꞉·A black feast+gate opens over one foundation stone as outer food pours inward and
↪ growing weight settles onto the lowest course𐤑·⁖˙·˙ostensiveIII꞉·
·⁖grimchain꞉·ᓭ𝄋ᵑ◎◎ᙉHzᵞᔕ𝍐ϙ𝌆·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup⁖˙
·⁖˙·˙glyphs⁖˙
·⁖glyph symbol꞉○–○ₑ𝄋₎ category꞉○–○ₑcourt₎⁖·
·⁖lookup office꞉○–○ₑphantasmagoria₎ enoch꞉○–○ₑ𐤊₎⁖·

·⁖𞤶𞤹𞤵꞉·𞤡ᵔᴗᵔ✶𞤡ᵔᴗᵔ·⁖˙·˙𞤶𞤹𞤵꞉·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Tahn+ta·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Weave⁖· Weaver⁖· Gift of Spellcraft⁖· ·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Many bright threads lie before the loom::but one alone lifts into the working and enters
↪ the crossing pattern while every unselected color remains outside the spell𓆸·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Gold::black::silver::red::crossing threads::and burning spells share one enchanted woven
↪ field beneath the loom𓆸·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Woven signs repeat one measured pattern across the loom while blue spellfire lives
↪ inside every crossing𓆸·⁖.·˙ostensiveIII:·
·⁖grimchain:·𐌗𐌏𐌓𐌂𐌊𐌅𐌒𐌖𐌕𐌋𐌉·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦𓃠 category◦–◦court꙳·
·⁖lookup office◦–◦phantasmagoria꙳ enoch◦–◦ᛊ꙳·

·⁖ᛯᛞᛏ:·ᘏᓚᕽᔐ✳ᕙᓚᕽ·⁖.·˙ᛯᛞᛏ:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Pel+en+aye·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·The surface by which behind is hidden⁖· Veil in Action·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Near surface takes the light::Concealed body remains intact behind the veil while
↪ visibility stops at the covering𓃠·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A silver mirror carries moonlit forest on its face while a hidden burning city remains
↪ whole beyond the enchanted surface𓃠·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A silver veil hides the burning city while a tuned string beyond the surface answers
↪ the near string through shared air𓃠·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᚿᛈ𓆸ᛓ✩ᛋᚸ𓆸ᚾᚏ❁·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ᚺ꙳ category◦–◦court꙳·
·⁖lookup office◦–◦phantasmagoria꙳ enoch◦–◦ᛊ꙳·

·⁖ᑭᖻᖇᘐ:·ʙᴀʀʜ:⁖.·˙ᑭᖻᖇᘐ:·

·⁖origin…language:·Khuzdul·⁖.·˙origin…language:·
·⁖font:·DwarfRunes·⁖.·˙font:·
·⁖pronunciation:·bahrk·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Axe⁖· Cut embodied as the tool made to divide·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Iron edge presses into the grain⁖· wood bears opposing tension on both sides of the
↪ blade::and division is physically present at the contact under the same severing
↪ pressureᚺ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A star+forged axe bears one white severing line inside its edge above a black
↪ branchᚺ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A star+forged axe repeats the same luminous cut through a black branch::each returning
↪ stroke occupying the identical severing pathᚺ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᓚ꙰ᚺ𓆸ᖻᓗᑢᑎxᛅᚢᛝᚳ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁖glyph symbol◦–◦ᵩA꜀ category◦–◦ᵩcourt꜀⁖·
·⁖lookup office◦–◦ᵩphantasmagoria꜀ enoch◦–◦ᵩᖴ꜀⁖·

⁘ᘓᘛᘛ᛫⁖·ᘐᘛᘛ✳ᘓ⁘·˙ᘓᘛᘛ᛫⁖·

·⁖origin…language⁖·Llumena Milenii·⁖·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖·˙font⁖·
·⁖pronunciation⁖·Kor+noo·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Horn⁖· The sound magic makes⁖· The instrument tone itself·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Curved chamber holds gathered breath against its inner wall⁖· resonant air presses
↪ through the opening as a shaped tone⁚⁚and the working is audible without needing another
↪ bodyA·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·One horn+note and one ring of blue runes travel together across the
↪ chamberA·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·One horn+note becomes the first light of a magical dawn⁚⁚resonance opening the dark
↪ before any visible sun appearsA·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ᦟᦞᦵ⊙ᦰᦷᦡᦵᦂᦵᦟᦵ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦ᵩB꜀ category◦–◦ᵩcourt꜀⁖·
·⁖lookup office◦–◦ᵩphantasmagoria꜀ enoch◦–◦ᵩᖴ꜀⁖·

·⁖origin…language⁖·Neo+Sindarin·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Sindarin·⁖·˙font⁖·
·⁖pronunciation⁖·eh+GEL+ren·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Different⁖· Strange⁖· Sameness opening into difference·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Familiar contour keeps one impossible alteration along its edge⁚⁚Altered feature makes
↪ the whole form strange without erasing the surrounding likenessB·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·Two matching masks share every contour except one impossible silver eye burning in the
↪ altered faceB·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·A strange silver form stands silent inside a field of repeated likeness⁚⁚difference
↪ visible without any answering sound from the altered bodyB·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ᦒᦟᦵᦟᦵᦟᦵ

·⁖ostensiveII:·A breathless black horn carries spectral dragon+form through widening rings of luminous
↪ forceↃ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Dragon+voice surges through a narrow hall against an opposing current⁚⁚each rising tone
↪ gaining force while resistance holds the sounding line under pressureↃ·⁖.·˙ostensiveIII:·
·⁖grimchain:·𐊧𐊫ϟϽʘ⊕Ⅱ♂ΘᑭŸ⊐ᚔ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ᵋᛦ category○–○ᵋcourt𐑈:·
·⁖lookup office○–○ᵋphantasmagoria𐑈 enoch○–○ᵋᛈ𐑈:·

·⁖origin…language:·Quenya·⁖.·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖.·˙font:·
·⁖pronunciation:·Eru·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·The One·˙ God·⁖.·˙non…ostensive:·
·⁖ostensiveI:·One throne of living light burns at the sacred center without division⁚⁚Every lesser
↪ flame turns toward the solitary source⁚⁚and no second crown rises beside itᛦ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·One vast white+fire name contains every star of the night with no second name beside
↪ itᛦ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One white+fire god gathers separate lights into a single pulse⁚⁚many radiances
↪ remaining distinct around the same sourceᛦ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ΘЖ𐌺ϰϕᛈ⏀ᛈℤ♂Ⅾᛯ૭ᘐ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ᵋᛊ category○–○ᵋcourt𐑈:·
·⁖lookup office○–○ᵋphantasmagoria𐑈 enoch○–○ᵋᛈ𐑈:·

·⁖ᛕᛯᛯ✿ʘ:·∞ᚕᛯᛯ🜲·⁖.·˙ᛕᛯᛯ✿ʘ:·

·⁖origin…language:·Gnomish·⁖.·˙origin…language:·
·⁖font:·Gnomish·⁖.·˙font:·
·⁖pronunciation:·bahrk·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Dread·˙ Terror·˙ The instant experience exceeds the mind’s prepared
↪ boundary·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Unshaped pressure fills the field beyond every prepared boundary·˙ breath shortens under
↪ the excess⁚⁚and familiar measure stands too small around what is physically
↪ presentᛊ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Bottomless depth⁚⁚failed horizon⁚⁚locked breath⁚⁚and broken measure fill one impossible
↪ opening in the stone wallᛊ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Dread falls as black shadow across the threshold before the unseen force
↪ arrives⁚⁚terror already occupying the dark likeness cast ahead of its causeᛊ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ↄ𐊣ᓫ𐊰ᛯ𐊧♂⏀⊙ı⁓⬡·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ᵋᛘ category○–○ᵋcourt𐑈:·
·⁖lookup office○–○ᵋphantasmagoria𐑈 enoch○–○ᵋᛈ𐑈:·

·⁖ᚳᚠᚠᚷ:·ᚢᛉᛉᛗᛞᚷ·⁖.·˙ᚳᚠᚠᚷ:·

·⁖origin…language:·Avarin·⁖.·˙origin…language:·
·⁖font:·Cirth Erebor·⁖.·˙font:·

·⁖pronunciation:·koo+eend·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·The People⁖ Separate voices retaining selves while becoming one People⁝⁝one sounding
↪ body·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Distinct voices rise together without losing pitch or source⁖ the gathered sounding
↪ carries more breadth than any one throat⁝⁝and no voice disappears inside the whole↰·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A hundred ancestral mask+voices remain distinct inside one chamber+filling
↪ concord↰·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·Many ancestral voices remain distinct while warm roots join them beneath one sounding
↪ ground↰·⁖.·˙ostensiveIII:·
·⁖grimchain:·ς∀ᚠ⇞⧗⁘✓✳✳▽♌ΠXᚹ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ᵢₗ꜀ category○–○ᵢcourt꜀:·
·⁖lookup office○–○ᵢphantasmagoria꜀ enoch○–○ᵢ⊼꜀:·

·.⅄ᚺᚳ:·ᚾᚳᚺᚾᚳ:.·˙⅄ᚺᚳ:·

·⁖origin…language:·Mando’a·⁖.·˙origin…language:·
·⁖font:·Mandalorian·⁖.·˙font:·
·⁖pronunciation:·BROH+kar·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Beat rhythmically⁝⁝like a drum or heartbeat⁖ Heart⁝⁝drum and repeated impact sharing
↪ one cadence·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Drum skin returns under repeated impact and living pulse returns under its own
↪ pressure⁝⁝Drum and heartbeat keep the same interval while cadence is carried physically in
↪ recurrenceI·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Buried red rune and suspended drum flash and sound together pulse after pulse across
↪ light and struck hideI·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One heartbeat sounds inside empty armor⁝⁝repeated rhythm carrying a soul+trace where no
↪ breathing flesh remainsI·⁖.·˙ostensiveIII:·
·⁖grimchain:·♈◎⅋⊖✳◎Ψ♂ʊ❄⧓⊕Θ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ᵢᗡ꜀ category○–○ᵢcourt꜀:·
·⁖lookup office○–○ᵢphantasmagoria꜀ enoch○–○ᵢ⊼꜀:·

·⁖ᚼ₪₪⍑:·¿₪✳²ᔕ₪◠·⁖.·˙ᚼ₪₪⍑:·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Sawn+yehr+ah·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Song⁖ To Sing⁖ Song of the Ainur·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Voices of the Ainur gather across the dark⁝⁝measured tones crossing and returning until
↪ the whole expanse carries one ordered music larger than any solitary noteᗡ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A silver+throated note fills the ceiling with answering stars and luminous
↪ pathsᗡ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A vast song opens past the last counted voice⁝⁝resonance continuing through unfilled
↪ reaches beyond the gathered tonesᗡ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᘐ♉ᛉ♈ᘓ∀ᔑ▽ᚺ⊙ᓬς·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁖glyph symbol◦–◦ᵩ𐑉 category◦–◦ᵩcourt𐑉⁖·
·⁖lookup office◦–◦ᵩphantasmagoria𐑉 enoch◦–◦ᵩ𐑒𐑉⁖·

·⁖𐑞𐑨𐑓𐑜𐑩⁖·𐑦𐑩𐑜𐑓𐑨 𐑤⁖.·˙𐑞𐑨𐑓𐑜𐑩⁖·

·⁖origin…language⁖·Gnomish·⁖.·˙origin…language⁖·
·⁖font⁖·Gnomish·⁖.·˙font⁖·
·⁖pronunciation⁖·AHD+ross·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Crossing⁖· Ford⁖· Division becoming traversable·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Shallow stone lies beneath divided water⁖· both banks remain apart⁝⁝and the current leaves
↪ one grounded line where passage is physically possible𐑤·⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Moonlit stones span black river+water between still+separated banks𐑤·⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Separated shores face across water with firm ground running beneath the shallows⁖· the
↪ division remains real⁝⁝yet crossing is physically held inside it𐑤·⁖.·˙ostensiveIII⁖·
·⁖grimchain⁖·ϔ↓▽≗✡ↄ֍ᵹA⁉ᵴᵚ₱·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol◦–◦ᵩ1𐑉 category◦–◦ᵩcourt𐑉⁖·
·⁖lookup office◦–◦ᵩphantasmagoria𐑉 enoch◦–◦ᵩ𐑒𐑉⁖·

·⁖𐑣𐑣𐑖𐑣⁖·ᔕᔓᔕᔕᔓ·⁖.·˙𐑣𐑣𐑖𐑣⁖·

·⁖origin…language⁖·Early Primitive Elvish·⁖.·˙origin…language⁖·
·⁖font⁖·Sarati Eldamar LTR·⁖.·˙font⁖·
·⁖pronunciation⁖·Taka·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Fix⁖· Fasten⁖· Stick⁖· The Root of Fixation⁖· Obsession·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Point bites into its chosen place under resisting pull⁖· contact does not slip⁝⁝and every
↪ outward force returns against the same fixed attachment1·⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·A black nail of light anchors every turning line of a spinning star+map to one fixed
↪ center1·⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·A fixed silver pin holds one water+current in place against impossible pressure while
↪ the flow bends around it1·⁖.·˙ostensiveIII⁖·
·⁖grimchain⁖·⸔♂◎ϛ∓♂ϒ𐑔ᔕᒣᑭᔓᔑ·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol◦–◦ᵩᵚ𐑉 category◦–◦ᵩcourt𐑉⁖·
·⁖lookup office◦–◦ᵩphantasmagoria𐑉 enoch◦–◦ᵩ𐑒𐑉⁖·

·⁖origin…language⁖·Early Noldorin·⁖.·˙origin…language⁖·
·⁖font⁖·Tengwar Noldor·⁖.·˙font⁖·
·⁖pronunciation⁖·DAH+roth·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Top⁖· Point⁖· Peak⁖· Tip⁖· Spike⁖· Extension collapsing into one concentrated
↪ extremity·⁖.·˙non…ostensive⁖·
·⁖ostensiveI⁖·Rising form narrows from broad slope to a single extremity⁝⁝Surrounding lines converge
↪ upon the sharp point where height is concentratedᵚ1·⁖.·˙ostensiveI⁖·
·⁖ostensiveII⁖·Crystal mountain⁝⁝rising ridges⁝⁝narrowing height⁝⁝and blue+fire needle meet at one
↪ impossible pointᵚ1·⁖.·˙ostensiveII⁖·
·⁖ostensiveIII⁖·A blue+fire mountain narrows all heat and height into one incandescent tip⁝⁝flame and
↪ peak becoming the same concentrated extremityᵚ1·⁖.·˙ostensiveIII⁖·

·⁖grimchain:·ᚾᛘᛋ᛬ᛯ᛭☆➶ᛇᚼ‡ᚻ᛫ᛠᛔ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᛘ꩷ category꩜꩜court꩷:·
·⁖lookup office꩜꩜phantasmagoria꩷ enoch꩜꩜ᛕ꩷:·

·⁖origin…language:·Neo┼Quenya·⁖·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖·˙font:·
·⁖pronunciation:·EH┼ree┼eh·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Oneness·˙ Holy as Being·⁖·˙non…ostensive:·
·⁖ostensiveI:·Many reflections circle one undivided source·˙ no surface cuts the origin into parts⁚⁚and
↪ sacred fullness remains whole under every returned imageᛘ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Seven sacred signs remain wholly distinct inside one seamless ring of undivided white
↪ flameᛘ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·One holy flame writes its undivided record into black stone⁚⁚every carved ray returning
↪ to the same sourceᛘ·⁖·˙ostensiveIII:·
·⁖grimchain:·ᔕ☿ᗢAϴ٤ъ◎ᛁ❀♦Aϒ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜✳꩷ category꩜꩜court꩷:·
·⁖lookup office꩜꩜phantasmagoria꩷ enoch꩜꩜ᛕ꩷:·

·ᛈᛦᚿᚩ:·ᚕᚔᚔᚕ·⁖·˙ᛈᛦᚿᚩ:·

·⁖origin…language:·Gnomish·⁖·˙origin…language:·
·⁖font:·Gnomish·⁖·˙font:·
·⁖pronunciation:·AHN┼nah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Door·˙ Opening·˙ Access as the opening through which relation becomes
↪ possible·⁖·˙non…ostensive:·
·⁖ostensiveI:·Wall keeps an opening inside its own boundary·˙ separated spaces touch through the
↪ break⁚⁚and solid edge stands around a passage wide enough for crossing✳·⁖·˙ostensiveI:·
·⁖ostensiveII:·A luminous door joins moonlit forest and black ocean across two walls that share no
↪ world✳·⁖·˙ostensiveII:·
·⁖ostensiveIII:·An open gate faces an empty road where nothing has yet arrived⁚⁚access already present
↪ before any crossing body reaches the threshold✳·⁖·˙ostensiveIII:·
·⁖grimchain:·‹◎ΔᚹᛝᛣᚩᚥᛉᛝΛ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜✳꩷ category꩜꩜court꩷:·
·⁖lookup office꩜꩜phantasmagoria꩷ enoch꩜꩜ᛕ꩷:·

·ᔑᔐᚱᚱᚠ:·ᔑᚳᚷ·⁖·˙ᔑᔐᚱᚱᚠ:·

·⁖origin…language:·Dovahzul·⁖·˙origin…language:·
·⁖font:·Dovahkiin·⁖·˙font:·
·⁖pronunciation:·Wul·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Defiance·˙ Defy·˙ Fight Back·˙ Fight Against·˙ Push Back·⁖·˙non…ostensive:·
·⁖ostensiveI:·Incoming pressure meets equal force returning from the held ground⁚⁚Opposed forces share
↪ no uncontested center⁚⁚and resistance remains directed back toward the challenge✳·⁖·˙ostensiveI:·

·⁖ostensiveII:·A violet ward turns attacking fire back along its own path while marked ground remains
↪ untouched✳·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A great stone heaves against the warding line while the ground answers with equal
↪ force⁖· the stone shifts a handspan::but the line does not yield✳·⁖·˙ostensiveIII:·
·⁖grimchain:·ᓭIᑫᓗᐟ᠊ᔆᐧᔑᔐᔕᐣᐤᐯ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᛯ꩝ category꩜꩜ᶜourt꩝:·
·⁖lookup office꩜꩜ᵖhantasmagoria꩝ enoch꩜꩜ᖱ꩝:·

·:ᘜᘐᘏᐢ:·ᘏᘐᘎᓫᐦᐧᐦᐣᐦᐣᐦᓫᘎᘏᐢ:·⁖·˙ᘜᘐᘏᐢ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Tranz+myoo+ta+ta·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Transform⁖· Change⁖· Become something else⁖· Become what is meant·⁖·˙non…ostensive:·
·⁖ostensiveI:·Same substance stands beneath an altered contour::Different geometry closes around the
↪ continuous material while the former shape survives only as a faint memory along the
↪ edge✳·⁖·˙ostensiveI:·
·⁖ostensiveII:·Glass moth+shapes carry one complete red sigil from an obsidian cube through green
↪ fire✳·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A transforming body wanders across open country::each step leaving a different form
↪ behind while no road fixes what it must become next✳·⁖·˙ostensiveIII:·
·⁖grimchain:·ᐯᕊᔭᐦᐅᑐᐢᗕᑭᐸᐤᗅ❁·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᛤ꩝ category꩜꩜ᶜourt꩝:·
·⁖lookup office꩜꩜ᵖhantasmagoria꩝ enoch꩜꩜ᖱ꩝:·

·ᚼᚻᚥᛋ᛫ᚒᚲᚴᚽᚲᚒᚣᚲᚒᚠ:·⁖·˙ᚼᚻᚥᛋ᛫:·

·⁖origin…language:·Valarin·⁖·˙origin…language:·
·⁖font:·Sarati Eldamar LTR·⁖·˙font:·
·⁖pronunciation:·MAH+khah+nahz·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Authority⁖· Rule embodied as authority itself·⁖·˙non…ostensive:·
·⁖ostensiveI:·Still center bears enough weight to orient every surrounding motion⁖· lesser forces take
↪ direction from that fixed relation::and no repeated command is required for the order to
↪ hold✳·⁖·˙ostensiveI:·
·⁖ostensiveII:·A dark+gold crown turns compass needles::hanging blades::and rings of flame toward one
↪ silent bearing✳·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Fixed center holds surrounding motion in ordered relation while each lesser bearing
↪ turns around the same commanding weight✳·⁖·˙ostensiveIII:·
·⁖grimchain:·ᖱᕅ◎ᛯᘏᔭᓀᐯᐤᐩᘐ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᛉ꩝ category꩜꩜ᶜourt꩝:·
·⁖lookup office꩜꩜ᵖhantasmagoria꩝ enoch꩜꩜ᖱ꩝:·

·:ᘜᘐᘏᐢ:·ᘏᘎᕈᓭ✳ᘎᘎ·:·⁖·˙ᘜᘐᘏᐢ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·

·⁖pronunciation:·Mil+en+ee·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·The Perfected Time·⁖· Time as Inhabitable·⁖·˙non…ostensive:·
·⁖ostensiveI:·Warmth dwells among flowering beds where clear water threads the earth and fruit bends
↪ bright branches toward the path::Time itself feels inhabitable inside the breathing
↪ gardenᛉ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Golden chambers open along a turning corridor of light::Every golden chamber holds
↪ roof::warmth::breath::water::and restᛉ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Golden chambers open along a luminous corridor while rain not yet fallen gathers ahead
↪ of each threshold and waits inside the lightᛉ·⁖·˙ostensiveIII:·
·⁖grimchain:·ꓧꓤ⊃ꓫꓟꓽ⟒ꓤ‹ꓤ✳xIᛉ·⁖·˙grimchain:·
·⁖·˙lookup:·
·⁖·˙glyphs:·
·⁖glyph symbol◦–◦₍0₎ category◦–◦₍court₎:·
·⁖lookup office◦–◦₍phantasmagoria₎ enoch◦–◦₍ᚲ₎:·

⁖ꕥ᙭ᔕᔑ:·ᔕ◟ᔑ◞ᔕ:·˙ꕥ᙭ᔕᔑ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Az+terr·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Emanation as Light·⁖·˙non…ostensive:·
·⁖ostensiveI:·Hidden source throws radiance across the boundary without losing its inward fire·⁖·
↪ distant surface carries the same brightness::and every ray remains tied to the luminous
↪ origin0·⁖·˙ostensiveI:·
·⁖ostensiveII:·Seven white rays remain visibly tethered to one sealed star through glowing fissures in
↪ a black crystal shell0·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Silver emanation strikes a barred black plane and is denied passage::light spreading
↪ outward from the refusal without crossing the closed boundary0·⁖·˙ostensiveIII:·
·⁖grimchain:·ꕥ⌶◎ΦΘ◎ꕥ⩛ΨᛇRIHᛇ·⁖·˙grimchain:·
·⁖·˙lookup:·
·⁖·˙glyphs:·
·⁖glyph symbol◦–◦₍ᛯ₎ category◦–◦₍court₎:·
·⁖lookup office◦–◦₍phantasmagoria₎ enoch◦–◦₍ᚲ₎:·

⁖ꕥ᙭ᔕᔑ:·ʔ⩛◟ᔑ◞ᔕ⟒ꓤ◟ᔑ:·˙ꕥ᙭ᔕᔑ:·

·⁖origin…language:·Llumena Milenii·⁖·˙origin…language:·
·⁖font:·Mage Script·⁖·˙font:·
·⁖pronunciation:·Shay+myoo+tah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Transformation·⁖· Change as a Substrate itself·⁖· Act and Be Change·⁖·˙non…ostensive:·
·⁖ostensiveI:·No edge keeps a permanent promise of form::Beneath the visible contour::matter carries
↪ another shape like a folded shadow waiting inside the present oneᛯ·⁖·˙ostensiveI:·
·⁖ostensiveII:·A dark+glass river beneath the floor carries stone as feather and flame as silver rain
↪ wherever its current touches formᛯ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·One continuous song runs through changing matter::each returning note leaving the same
↪ substance in another formᛯ·⁖·˙ostensiveIII:·
·⁖grimchain:·✪ℂ⊐ꓘ⌶ꓓΘꓝᔑᛯᔑ𝓈·⁖·˙grimchain:·
·⁖·˙lookup:·
·⁖·˙glyphs:·
·⁖glyph symbol◦–◦₍ǂ₎ category◦–◦₍court₎:·
·⁖lookup office◦–◦₍phantasmagoria₎ enoch◦–◦₍ᚲ₎:·

·:ᚻᛞᛏ:·ᔕᔓᔐᔕ·:·ᚻᛞᛏ:·

·⁖origin…language⁖·Llumena Milenii·⁖·˙origin…language⁖·
·⁖font⁖·Mage Script·⁖·˙font⁖·
·⁖pronunciation⁖·Rih+vez·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Doorway⁖· Keystone⁖· Gate⁖· Threshold·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Wall bears an open threshold beneath a loaded arch⁖· stone keeps both sides upright⁝⁝and
↪ passage lies in the middle as part of the barrier that contains it⁜·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·One luminous keystone bears impossible night+weight above a clear open passage through
↪ the arch⁜·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Two broken realms meet inside one keystone gate⁝⁝a load+bearing seam opening one
↪ passage through the wall⁜·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ₒᵚᛦᛯ⬡ᚋᛯᛂᚸᚋᛁ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol○–○₍ᛯ₎ category○–○₍court₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᛊ₎⁖·

·⁖origin…language⁖·Neo+Sindarin·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Sindarin·⁖·˙font⁖·
·⁖pronunciation⁖·ahnd·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Gate⁖· Door⁖· Entry as the threshold made passable·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Barrier keeps one opening that admits passage⁖· wall remains intact around it⁖· entry
↪ exists exactly where separation has been made traversableᛯ·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·A star+curtain forms one passable opening inside an otherwise unbroken wall between
↪ distinct spacesᛯ·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·Living sap and breath cross an open gate in the wall while the threshold pulses with
↪ the same living rhythmᛯ·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ᚱᑫᑉ◝ᘐ%ᑌᑭᘎᚴᓘᕮᕼ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol○–○₍ᚱ₎ category○–○₍court₎⁖·
·⁖lookup office○–○₍phantasmagoria₎ enoch○–○₍ᛊ₎⁖·

·⁖origin…language⁖·Neo+Sindarin·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Sindarin·⁖·˙font⁖·
·⁖pronunciation⁖·MEN+gyl·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Messenger⁖· Message+bearer⁖· Meaning acquiring a body whose existence is carrying
↪ between loci·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Meaning rides inside a carried mark between distant places⁖· the bearer crosses the
↪ interval⁝⁝and the message keeps its contour from origin to arrivalᚱ·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·A glass capsule carries one complete blue+fire sigil unchanged across mountain distance
↪ between shrinesᚱ·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·A messenger+sigil flies between shrines along one exact luminous line toward the
↪ chosen markᚱ·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·ⵊᚿᛯ✷ᚸᛟᚾᚸᚿᚸᚸ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·

·⁖·˙glyphs:˙
·⁖glyph symbol꩜꩜↯꩝ category꩜꩜↯court꩝⁖·
·⁖lookup office꩜꩜↯phantasmagoria꩝ enoch꩜꩜↯ᖘ꩝⁖·

·⁖origin…language⁖·Middle Quenya·⁖·˙origin…language⁖·
·⁖font⁖·Tengwar Quenya·⁖·˙font⁖·
·⁖pronunciation⁖·Turut·⁖·˙pronunciation⁖·
·⁖cadence⁖·Phantasmagoria·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Tree+Stem·⁖· Tree+Trunk·⁖· Stem as the Line that Bears Branching Form·⁖·˙non…ostensive⁖·
·⁖ostensiveI⁖·Upright trunk bears root+pressure below and branching load above·⁖· the central wooden
↪ line keeps both relations continuous through its length꩜·⁖·˙ostensiveI⁖·
·⁖ostensiveII⁖·One world+tree trunk bears deep roots⁖⁖silver sap⁖⁖cloud+height⁖⁖and branching stars
↪ along a continuous central line꩜·⁖·˙ostensiveII⁖·
·⁖ostensiveIII⁖·One tree+trunk bears the same crown+weight in darkness and firelight·⁖· one axe+strike
↪ leaves the same dent in the wood under either sky꩜·⁖·˙ostensiveIII⁖·
·⁖grimchain⁖·◎ᢆᢆᢆ

·⁖ostensiveIII:·Quiet water receives one scored stone record without ripple⁚⁚the enduring mark resting
↪ beneath the still surface«·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᔕᒉᐱᐯᔓᒣᚏᚯᚸᚱ❋Θ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○⌁ᚷ› category○⌁court›⁖·
·⁖lookup office○⌁phantasmagoria› enoch○⌁ᚣ›⁖·

·⁖ᔑᔐᐣᐢᐣᐢᔐᐣᐢᔑᐣᐢᔐᐣᐢᔑᐣᐢ⁖.·˙ᔑᔐᐣᐢᔐᐣᐢᔑᐣᐢ⁖·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:·Steel+d·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Quiet⁖· Calm⁖· Still⁖· Poise as the state that remains composed while force moves around
↪ it·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Force moves around a center that keeps its measure⁖· surface and bearing remain composed
↪ under pressure⁚⁚and no scattering reaches the inward balanceᚷ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·A level crystal sphere keeps one still axis inside the violent orbit of silver
↪ bladesᚷ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·One poised body remains still while changing light crosses the same scar and
↪ surrounding force circles without shifting itᚷ·⁖.·˙ostensiveIII:·
·⁖grimchain:·▽▫ᛉᚯᛒᏗᗡ▫ᓰᛯᛍΘᛯᚏ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○⌁ᚹ› category○⌁court›⁖·
·⁖lookup office○⌁phantasmagoria› enoch○⌁ᚣ›⁖·

·⁖ᚻᔓᔕᛏ⁖·ᛟᛇᘐᔓ✳⁖.·˙ᚻᔓᔕᛏ⁖·

·⁖origin…language:·Llumena Milenii·⁖.·˙origin…language:·
·⁖font:·Mage Script·⁖.·˙font:·
·⁖pronunciation:·Fee+vohn·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Divine⁖· Divinity⁖· Magic as a Place·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Invisible radiance pools between stone⁚⁚root⁚⁚and open air until every surface carries the
↪ same holy pressure⁚⁚Crossing the bounded field enters the magic itself⁚⁚though no shrine stands
↪ withinᚹ·⁖.·˙ostensiveI:·
·⁖ostensiveII:·Inside one hard valley boundary⁚⁚blue flowers⁚⁚humming stones⁚⁚and silver+edged shadows
↪ replace ordinary earthᚹ·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·A bounded magical realm bears a crown of power beneath the ground⁚⁚and stones and
↪ currents bend toward the buried forceᚹ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᚸᛯᚱᚷᚺᚸᚸᛟⅡ❋ᛑᛇᚢ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○⌁ᚸ› category○⌁court›⁖·
·⁖lookup office○⌁phantasmagoria› enoch○⌁ᚣ›⁖·

·⁖origin…language:·Early Noldorin·⁖.·˙origin…language:·
·⁖font:·Tengwar Noldor·⁖.·˙font:·
·⁖pronunciation:·theer·⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·

·⁖non…ostensive⁚·Complete⁖· Full⁖· Accomplished⁖· Ended⁖· Up to the required number⁖· Completion as
↪ fulfilled measure⁝⁝not merely stopping·⁖·˙non…ostensive⁚·
·⁖ostensiveI⁚·Every seam sits true⁝⁝every weight is balanced⁝⁝every required part is present⁝⁝Bright
↪ resonance answers the finished working where no correction can find a place to
↪ enter↻·⁖·˙ostensiveI⁚·
·⁖ostensiveII⁚·Every required star burns in one white constellation with no dark gap
↪ remaining↻·⁖·˙ostensiveII⁚·
·⁖ostensiveIII⁚·A small oath+ring bends the silver chain deeper than its metal mass at the center of a
↪ completed constellation holding every required star↻·⁖·˙ostensiveIII⁚·
·⁖grimchain⁚·🜁🜨ᑎ♎ₒᑲ▽🜲⊃ᚺ⓪🜨☿·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol◦–◦𝅘𝅥𝅮court⁗·
·⁖lookup office◦–◦phantasmagoria⁗ enoch◦–◦ᛊ⁗·

·⁖ᛞᛞᛏ⁚·ᛞᛞᛯ ᛤ ᛯ ᛈ⁚·˙ᛞᛞᛏ⁚·

·⁖origin…language⁚·Llumena Milenii·⁖·˙origin…language⁚·
·⁖font⁚·Mage Script·⁖·˙font⁚·
·⁖pronunciation⁚·Ooh+roo+cuss·⁖·˙pronunciation⁚·
·⁖cadence⁚·Phantasmagoria·⁖·˙cadence⁚·
·⁖non…ostensive⁚·See⁖· Sight⁖· To be Witness itself·⁖·˙non…ostensive⁚·
·⁖ostensiveI⁚·Light crosses open distance from visible surface to receptive eye⁖· no veil or
↪ intermediary breaks the relation⁝⁝and the encountered form remains physically present throughout
↪ the field of sight⊔·⁖·˙ostensiveI⁚·
·⁖ostensiveII⁚·An obsidian eye holds the same comet+form carried directly across open distance by
↪ unbroken light⊔·⁖·˙ostensiveII⁚·
·⁖ostensiveIII⁚·Open eye receives a stair of light climbing from mark to higher mark across the
↪ dark⊔·⁖·˙ostensiveIII⁚·
·⁖grimchain⁚·☿𐑉ᑎↄ乂ϒₒᒣ↯ᛇᚥᑎᛊ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol◦–◦𝅘𝅥𝅮court⁗·
·⁖lookup office◦–◦phantasmagoria⁗ enoch◦–◦ᛊ⁗·

·⁖origin…language⁚·Early Noldorin·⁖·˙origin…language⁚·
·⁖font⁚·Tengwar Noldor·⁖·˙font⁚·
·⁖pronunciation⁚·tahnk·⁖·˙pronunciation⁚·
·⁖cadence⁚·Phantasmagoria·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Firm⁖· Steady⁖· Steadfast⁖· Remaining structurally faithful while force acts upon
↪ it·⁖·˙non…ostensive⁚·
·⁖ostensiveI⁚·Root grips soil beneath a wind+loaded crown while stone line holds under pressure⁝⁝Root
↪ and stone remain faithful to their bearing under force⊔·⁖·˙ostensiveI⁚·
·⁖ostensiveII⁚·A black monolith keeps one unwavering axis under luminous chains straining from every
↪ side⊔·⁖·˙ostensiveII⁚·
·⁖ostensiveIII⁚·Stone keeps its line under pressure while mist clears and every hard edge stands clean
↪ in open air⊔·⁖·˙ostensiveIII⁚·
·⁖grimchain⁚·ϛΨ✳ᛯ✢☽ᓀᔑϛᑲ♀ϴ🜲·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol◦–◦▽𝅘𝅥𝅮court⁗·
·⁖lookup office◦–◦phantasmagoria⁗ enoch◦–◦ᛊ⁗·

·:ᘐᘏᘐᘎ:·ᘎᘏᘐᘏᘎ:·:·˙ᘐᘏᘐᘎ:·

·⁖origin…language:·Darnassian·⁖·˙origin…language:·
·⁖font:·Darnassian Runes·⁖·˙font:·
·⁖pronunciation:·DOH+ray·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Star+borne·⁖ Children of the Stars⁖ Hope embodied as something whose origin and
↪ orientation are stellar·⁖·˙non…ostensive:·
·⁖ostensiveI:·Starlight fixes a bright bearing above dark ground⁖ the distant source remains beyond
↪ reach⁘yet the path keeps orientation toward that celestial origin▽·⁖·˙ostensiveI:·
·⁖ostensiveII:·A comet+born silver seed carries the same star+fire from falling origin into a branch
↪ bent toward its parent constellation▽·⁖·˙ostensiveII:·
·⁖ostensiveIII:·A star+borne silver body follows its celestial bearing into dark earth⁘and the first
↪ working stroke is already cut along the path before the hand arrives▽·⁖·˙ostensiveIII:·
·⁖grimchain:·⌬ǂ♀ᚣᓚᛘᚼᚴᚔᘉᘔᙆᙅ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○‒○ᶻ꜀ category○‒○ᶜourt꜀:·
·⁖lookup office○‒○ᵖhantasmagoria꜀ enoch○‒○ᵏ꜀:·

·:word:·ᗯᗴᕱ·:·:·˙word:·

·⁖origin…language:·Klingon·⁖·˙origin…language:·
·⁖font:·Klingon pIqaD HaSta·⁖·˙font:·
·⁖pronunciation:·tah·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Continue⁖ Go On⁖ Endure⁖ Survive·⁖·˙non…ostensive:·
·⁖ostensiveI:·Flame holds under blackened fuel and breath persists under pressure⁖ living motion
↪ remains present where strain would otherwise extinguish itz·⁖·˙ostensiveI:·
·⁖ostensiveII:·One blue flame persists through relentless rain from the same living ember across the
↪ soaked altarz·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Cloud hides the high mountain while the upward path keeps its bearing toward the
↪ unseen ridge and does not stopz·⁖·˙ostensiveIII:·
·⁖grimchain:·0Ӂ⍦֍ԓ६ᕮᛎᚣᚮᚣ↗·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol○‒○ᶐ꜀ category○‒○ᶜourt꜀:·
·⁖lookup office○‒○ᵖhantasmagoria꜀ enoch○‒○ᵏ꜀:·

·⁖origin…language:·Quenya·⁖·˙origin…language:·
·⁖font:·Tengwar Quenya·⁖·˙font:·
·⁖pronunciation:·Estë·⁖·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖·˙cadence:·
·⁖non…ostensive:·Rest⁖ Repose⁖ Peace⁖ State of Rest⁖ At Peace⁖ Peaceful·⁖·˙non…ostensive:·
·⁖ostensiveI:·Weight lies supported without demand for further motion⁖ breath remains easy⁘and no
↪ opposing pressure disturbs the settled conditionᑫ·⁖·˙ostensiveI:·
·⁖ostensiveII:·Flat swords⁘still bells⁘quiet sigils⁘and soft light lie unmoving inside a pale
↪ circleᑫ·⁖·˙ostensiveII:·
·⁖ostensiveIII:·Shaped syllables travel across peaceful ground while still water and resting stone do
↪ not stirᑫ·⁖·˙ostensiveIII:·
·⁖grimchain:·ᗷᚺᚿᛜᑐᘐᚠᚸᚹ⌬ᚷᛉᓀ·⁖·˙grimchain:·
·⁖·˙lookup·˙

·⁖·˙glyphs꞉˙
·⁖glyph symbol꞊꞊ᵩ꞉ category꞊꞊ᵩcourt꞉·
·⁖lookup office꞊꞊ᵩphantasmagoria꞉ enoch꞊꞊ᵩᖘ꞉·

·⁖ᵩ꞉ᵲ⇝꞉˙ɔᵩ꞉⁖·˙ᵩ꞉ᵲ⇝꞉·

·⁖origin…language꞉·Early Qenya·⁖·˙origin…language꞉·
·⁖font꞉·Tirion Sarati·⁖·˙font꞉·
·⁖pronunciation꞉·Tá·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Hush꞉˙ Be Silent꞉˙ Silent꞉˙ Silence as Arrival·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Air holds no spoken vibration꞉˙ surface keeps no tremor꞉꞉and the quiet field is physically
↪ full rather than merely lacking soundᵬ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·Beyond one white threshold꞉꞉bell꞉꞉falling water꞉꞉moving leaves꞉꞉and air carry no
↪ soundᵬ·⁖·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Bare fingertips cross scored stone in perfect silence꞉˙ ridges꞉꞉warmth꞉꞉and vibration
↪ enter skin without soundᵬ·⁖·˙ostensiveIII꞉·
·⁖grimchain꞉·ᚻᘓᘜᙏᛋᛊᛁᛟᚴᚾᚹ᛭᛭·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞊꞊ᵩᵶ꞉ category꞊꞊ᵩcourt꞉·
·⁖lookup office꞊꞊ᵩphantasmagoria꞉ enoch꞊꞊ᵩᖘ꞉·

·⁖ᔑᓯᒣᒉ꞉·꞉ᒍᓯᒧᒉᒍᓰᒉᒣᒉᒪᒧ꞉·⁖·˙ᔑᓯᒣᒉ꞉·

·⁖origin…language꞉·Dovahzul·⁖·˙origin…language꞉·
·⁖font꞉·Dovahkiin·⁖·˙font꞉·
·⁖pronunciation꞉·Hahnuheim·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·To be asleep꞉˙ Visions꞉˙ The Gift of Seer·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Sleeping body lies still while closed eyes open into luminous seer+visions beyond
↪ ordinary sight꞉꞉dream+country carrying the gift without any waking motionᵬ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·A sleeping body carries one luminous map of unborn stars beneath closed eyes and a dark
↪ veilᵬ·⁖·˙ostensiveII꞉·
·⁖ostensiveIII꞉·Behind closed eyes꞉꞉one sleeping vision turns flame꞉꞉iron꞉꞉and water before waking matter
↪ is touchedᵬ·⁖·˙ostensiveIII꞉·
·⁖grimchain꞉·ᛎ∀ᛈ⏁ǂ᛭ᛟᛕ✲ᛝ✠ᛃ⁎·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞊꞊ᵩφ꞉ category꞊꞊ᵩcourt꞉·
·⁖lookup office꞊꞊ᵩphantasmagoria꞉ enoch꞊꞊ᵩᖘ꞉·

·⁖ᚼᛞᛞᛏ꞉·ᔓᑉᕰ✳ᔓᘐᕰ✳ᑉ꞉·⁖·˙ᚼᛞᛞᛏ꞉·

·⁖origin…language꞉·Llumena Milenii·⁖·˙origin…language꞉·
·⁖font꞉·Mage Script·⁖·˙font꞉·
·⁖pronunciation꞉·Ah+ten+ah+mehnt·⁖·˙pronunciation꞉·
·⁖cadence꞉·Phantasmagoria·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Living Identity Manifestation Without Division of Self·⁖·˙non…ostensive꞉·
·⁖ostensiveI꞉·Many forms stand around one living source without taking any portion from it꞉꞉Every
↪ manifestation carries the same identity꞉꞉and the undivided source remains whole beneath
↪ themφ·⁖·˙ostensiveI꞉·
·⁖ostensiveII꞉·One white source+flame and a hundred complete living reflections carry the same sigil
↪ without diminishing source height or brightnessφ·⁖·˙ostensiveII꞉·

·⁖ostensiveIII:·One source⟊flame remains singular while towers⁚⁚rivers⁚⁚and winged citadels appear whole
↪ in empty air around itφ·⁖.·˙ostensiveIII:·
·⁖grimchain:·ᘐ⁕ᔕ◎ᗒᐁ♏❂ǂ❁ᛋᕲᐁ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦𝓀꜀ category◦–◦ₐaxiomyrᵢ⁖·
·⁖lookup office◦–◦ₐphantasmagoriaᵢ enoch◦–◦ᖶᵢ⁖·

⁘ᔑᒣᒉᐢ:ᒡᐢᘉᒧᒍᒉᒧᘉ≒⁘.·˙ᔑᒣᒉᐢ:·

·⁖origin…language:·Dovahzul·⁖.·˙origin…language:·
·⁖font:·Dovahkiin·⁖.·˙font:·
·⁖pronunciation:··⁖.·˙pronunciation:·
·⁖cadence:·Phantasmagoria·⁖.·˙cadence:·
·⁖non…ostensive:·Miracle⁚⁚Godsend⁚⁚Miraculous⁚⁚Godsent.·⁖.·˙non…ostensive:·
·⁖ostensiveI:·Moonlight gathers beneath the roots of ancient trees⁖flowers unfold from unbroken
↪ stone⁖rivers carry stars through their depths⁖mountains breathe beneath veils of emerald
↪ fire⁖sleeping seeds illuminate the darkness within the earth⛒·⁖.·˙ostensiveI:·
·⁖ostensiveII:·In the barren valley⁚⁚rain falls from an empty sky and the dry riverbed sings with
↪ running water⁚⁚A winter branch flowers under untouched snow⛒·⁖.·˙ostensiveII:·
·⁖ostensiveIII:·The mountain wears no crown but living fire beneath the stone⁚⁚Emerald light rises from
↪ forest roots⁚⁚and the river runs luminous through a chamber of unbroken rock⁚⁚Across the night
↪ sky⁚⁚the earth casts a vast green radiance⛒·⁖.·˙ostensiveIII:·
·⁖grimchain:·♏⁉♂ᵽ⁜✓♪ᑭᑐᓄᑫᓀᒣ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖.·˙glyphs·˙
·⁖.·˙sydonic…magicae…lexicon·∴

⁘⁘⁘

·⁖origin…language꞉·Ge'ez·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖·˙font꞉·
·⁖pronunciation꞉·LUBB·⁖·˙pronunciation꞉·
·⁖cadence꞉·Spatial Absolute·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Heart⁖· the central organ and figurative seat of inward feeling·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Beneath the breast⁝⁝one inward chamber holds grief⁝⁝courage⁝⁝and tenderness within the same
↪ bounded interior⚲·⁖·˙ostensive꞉·
·⁖grimchain꞉·ↄ𐑅ᔓX𐑣ᨒ𐑅ᛘΘᛋ꒐ᛁ𐑈·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉ㄱ꒱꞉ category꞉꞉emission꞉꞉·
·⁖lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉ᵒ꞉꞉·

⁖word꞉·𐑣𐑤ᓭ·⁖·˙word꞉·

·⁖origin…language꞉·Ugaritic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ugaritic·⁖·˙font꞉·
·⁖pronunciation꞉·SHPT·⁖·˙pronunciation꞉·
·⁖cadence꞉·Spatial Absolute·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Lip·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Two soft gates mark the border of breath⁖· voice waits on the inward side of paired flesh⁖·
↪ open air begins beyond the outward edge of paired flesh꒱·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᛋHxᘜᔕᕈᛉᔑᘜᘐ❂ᛏ·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉ɧ꞉ category꞉꞉emission꞉꞉·
·⁖lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉ᵒ꞉꞉·

⁖word꞉·歿⁖·˙word꞉·

·⁖origin…language꞉·Tangut·⁖·˙origin…language꞉·
·⁖font꞉·Noto Serif Tangut·⁖·˙font꞉·
·⁖pronunciation꞉·LLOHM·⁖·˙pronunciation꞉·
·⁖cadence꞉·Spatial Absolute·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Rely or depend⁖· to require support from another person⁝⁝thing⁝⁝or
↪ condition·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Roof upon beam and beam upon pillar and pillar upon earth⁝⁝and weight finds a keeping
↪ place beneath every keeping placeɧ·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᛦ◎☽☆ᛋ⛢ᛋ⇞ᛘ⊔⚲ᘐᗷ·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉⛢꞉ category꞉꞉emission꞉꞉·
·⁖lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉ᵒ꞉꞉·

⁖word꞉·ᘔᗅ⁖·˙word꞉·

·⁖origin…language꞉·Phoenician·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Phoenician·⁖·˙font꞉·
·⁖pronunciation꞉·YAHD·⁖·˙pronunciation꞉·
·⁖cadence꞉·Spatial Absolute·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Hand⁖· the terminal part of the arm used for grasping and
↪ manipulation·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Arm reaches its far shore⁝⁝and five branches open there⁝⁝and the near world enters the
↪ little country of their grasp⛢·⁖·˙ostensive꞉·

·⁖grimchain:·𐫄𐫁𐫆❂𐫇𐫉♂𐫋𐫌𐫍✦·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꩜꩜𐫉 category꩜꩜emission𐫊:·
·⁖lookup office꩜꩜spatial…absolute𐫊 enoch꩜꩜𐫊:·

˓word:·Δ+̄⁖·˙word:·

·⁖origin…language:·Tocharian B·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·EHK·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Eye·⁖·˙non…ostensive:·
·⁖ostensive:·Light crosses a dark gate::and the whole bright world gathers inside a small round
↪ chamber𐫉·⁖·˙ostensive:·
·⁖grimchain:·⊕ᚼ◎⊖𐫄X𐫌▫ᘐ𐫅𐫆⏁·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꩜꩜𐫇 category꩜꩜emission𐫊:·
·⁖lookup office꩜꩜spatial…absolute𐫊 enoch꩜꩜𐫊:·

˓word:·𐫔𐫑𐫒⁖·˙word:·

·⁖origin…language:·Old Turkic·⁖·˙origin…language:·
·⁖font:·Noto Sans Old Turkic·⁖·˙font:·
·⁖pronunciation:·TAHGH·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Mountain⁖˙ a large natural elevation of land·⁖·˙non…ostensive:·
·⁖ostensive:·Low ground rises through widening slopes into one high mass⁖˙ ravines divide the rising
↪ flanks into separate channels⁖˙ cold open height occupies the uppermost region𐫇⁖·˙ostensive:·
·⁖grimchain:·ᚺᛊΘᘐᛟᚻᚺ𐫈𐫉⏁𐫆ᚢ✷·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꩜꩜☽𐫊 category꩜꩜emission𐫊:·
·⁖lookup office꩜꩜spatial…absolute𐫊 enoch꩜꩜𐫊:·

˓word:·ᛟ𐫁𐫃⁖·˙word:·

·⁖origin…language:·Tocharian B·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·MRAHCHEH·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Summit::peak (of a mountain)·⁖·˙non…ostensive:·
·⁖ostensive:·One crown of stone occupies the uppermost point::with every slope descending away from it
↪ and cloud passing through lower air beneath☽·⁖·˙ostensive:·
·⁖grimchain:·𐫋ᚸᚹᘐᚷᚑᚸᚹ𐫄∀X𐫉ᚸ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꩜꩜⊙𐫊 category꩜꩜emission𐫊:·
·⁖lookup office꩜꩜spatial…absolute𐫊 enoch꩜꩜𐫊:·

˓word:·𐫐𐫐𐫐𐫐𐫐⁖·˙word:·

·⁚origin…language:·Old Persian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚.·˙font:·
·⁚pronunciation:·APADAHNA·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Hypostyle hall·⁚· citadel·⁚· apadana·⁚.·˙non…ostensive:·
·⁚ostensive:·Columns rise in ordered rows beneath a single roof⁚⁚leaving broad shadowed aisles inside one enclosing hall⊙·⁚.·˙ostensive:·
↪
·⁚grimchain:·𐓣𐓪𐓐𐓓𐓘𐓟𐓤‹◎𐓛𐓐𐓢·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉ℂ꞉ category꞉꞉emission꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐓂꞉⁚·

⁖word⁚·ꝊꝔꝎꝊꝋ⁖.·˙word⁚·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·MARMIN·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Body·⁚.·˙non…ostensive:·
·⁚ostensive:·Skin closes around breath and blood⁚⁚and all that lives within keeps one house against the world beyondℂ·⁚.·˙ostensive:·
↪
·⁚grimchain:·𐓝𐓘𐓟𐓢𐓥𐓣𐓐𐓡𐓢𐓤·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉ϒ꞉ category꞉꞉parliament꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐓂꞉⁚·

⁖word⁚·ጸሌ⁖.·˙word⁚·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·TSUH-lay·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Tablet·⁚· a flat surface prepared for writing or inscription·⁚.·˙non…ostensive:·
·⁚ostensive:·Clay face lies open within four edges⁚⁚and every cut of the reed finds its own place upon that quiet planeϒ·⁚.·˙ostensive:·
↪
·⁚grimchain:·☿▽ﻉɻ𐓥𐓫𐓟𐓝↗𐓞𐓟𐓢ﻉ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉ϓ꞉ category꞉꞉parliament꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐓂꞉⁚·

⁖word⁚·ɤሌ̄ᚻ⁖.·˙word⁚·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·MEHSKEH·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Joint (anatomy)·⁚· link⁚⁚junction·⁚· (phonetics) sandhi·⁚.·˙non…ostensive:·
·⁚ostensive:·Bone meets bone in one narrow turning place⁚⁚and one length can move while neither leaves the meetingϓ·⁚.·˙ostensive:·
↪
·⁚grimchain:·❄☽𐓣𐓪𐓟𐓘⬡ϒ𐓐𐓚❄·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜𑀪꩜ category꩜parliament꩜:·
·⁚lookup office꩜spatial…absolute꩜ enoch꩜𑀍꩜:·

꩜word:·𑀢⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·KEHN·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Earth::ground·⁚·˙non…ostensive:·
·⁚ostensive:·Seed lies below and roof stands above::and every road spreads across the dark breadth held
↪ between root and sky𑀪·⁚·˙ostensive:·
·⁚grimchain:·◎𑀣𑀁⊛𑁆⊢◎𑀏𑀬𑀰⬡𑀁𑀲𑁄·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑁋꩜ category꩜parliament꩜:·
·⁚lookup office꩜spatial…absolute꩜ enoch꩜𑀍꩜:·

꩜word:·𒁹𒅗𒁲⁚·˙word:·

·⁚origin…language:·Akkadian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·AP+poo·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Nose:˙ by extension a projecting tip::end::or edge·⁚·˙non…ostensive:·
·⁚ostensive:·Between the eyes a ridge comes forward::and breath opens beneath its furthest
↪ point𑁋·⁚·˙ostensive:·
·⁚grimchain:·𑀪❁𑀣𑁆𑀭𑀵𑀸⬡𑁇𑁇0'𑁆·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀬꩜ category꩜parliament꩜:·
·⁚lookup office꩜spatial…absolute꩜ enoch꩜𑀍꩜:·

꩜word:·𑀓𑀬𑁆⊢⁚·˙word:·

·⁚origin…language:·Ugaritic·⁚·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚·˙font:·
·⁚pronunciation:·HYT·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Life:˙ living creature·˙ land·⁚·˙non…ostensive:·
·⁚ostensive:·Root::hoof::breath::and dwelling occupy one breadth beneath the sky·˙ outside that inhabited
↪ reach::bare ground keeps none of them𑀬꩜·⁚·˙ostensive:·
·⁚grimchain:·𑀫𑁆𑀁𑀴𑁆𑀣𑀬𑀀◎𑀭𑀵𑀧·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀬꩜ category꩜parliament꩜:·
·⁚lookup office꩜spatial…absolute꩜ enoch꩜𑀍꩜:·

꩜word:·𑀓𑀬𑀁⁚·˙word:·

·⁚origin…language:·Ugaritic·⁚·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚·˙font:·
·⁚pronunciation:·HBL·⁚·˙pronunciation:·

·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Rope::chord⁚˙ band::company·⁚.·˙non…ostensive:·
·⁚ostensive:·Two posts stand apart::and one taut length keeps them in the same hold across the empty
↪ span𝅘𝅥𝅯·⁚.·˙ostensive:·
·⁚grimchain:·ᗡᛉᗐᛇᚠᘜᓭᔑ▫ᒧℂ𝅘𝅥𝅯❀·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ₑ⍾꜀ category○–○ₑparliament꜀⁚·
·⁚lookup office○–○ₑspatial…absolute꜀ enoch○–○ₑₒ꜀⁚·

₊word:·𐎗 𐎕 𐎛𐎍 ₊·⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·MRQD·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Dance·⁚.·˙non…ostensive:·
·⁚ostensive:·Turning feet lay circle upon circle across the open floor::and every crossing keeps its
↪ place inside the measure⍾·⁚.·˙ostensive:·
·⁚grimchain:·♿ꟽ𝌆❁♀ϫ⏣ᛁ⊙ᵩ𐌓𐌗·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ₑ𝓜꜀ category○–○ₑparliament꜀⁚·
·⁚lookup office○–○ₑspatial…absolute꜀ enoch○–○ₑₒ꜀⁚·

₊word:·𞋿¥₊·⁚.·˙word:·

·⁚origin…language:·Old Turkic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚.·˙font:·
·⁚pronunciation:·TAHSH·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Exterior::outside of something·⁚.·˙non…ostensive:·
·⁚ostensive:·Wall keeps one side within::and wind and road continue beyond the last enclosing
↪ edge𝓜·⁚.·˙ostensive:·
·⁚grimchain:·ᗪ○✡'❂'𑁍≻ᘔᚺᛉ∀⌖ᛋ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ₑ↗꜀ category○–○ₑparliament꜀⁚·
·⁚lookup office○–○ₑspatial…absolute꜀ enoch○–○ₑₒ꜀⁚·

₊word:·𓈖₊·⁚.·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·KHAH⟊y·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To rise or appear·⁚.·˙non…ostensive:·
·⁚ostensive:·A form stands partly below the horizon and partly above it⁚˙ the dividing rim keeps
↪ concealment beneath and visibility in open height↗·⁚.·˙ostensive:·
·⁚grimchain:·✷▽ᚠᛃ✳ᓫᛋᗪᘇᛋᛤ𝌆·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ₑ∂꜀ category○–○ₑparliament꜀⁚·
·⁚lookup office○–○ₑspatial…absolute꜀ enoch○–○ₑₒ꜀⁚·

˖word:·ወሰን˖·˙word:·

·⁖origin…language:·Ge'ez·⁖·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁖·˙font:·
·⁖pronunciation:·wah⟊SAHN·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Limit::boundary::or end·⁖·˙non…ostensive:·
·⁖ostensive:·One field reaches this stone and another begins beyond it::and neither crosses the mark between them𐎗·⁖·˙ostensive:·
·⁖grimchain:·ⲦϮƵIⲙ♂ⲫɕϗⲂϾ⊶·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜𑁍꩝ category꩜꩜parliament꩝:·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜𐎗꩝:·

˖word:·𐎍𐎚𐎜˖·˙word:·

·⁖origin…language:·Ugaritic·⁖·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖·˙font:·
·⁖pronunciation:·NATEEBU·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Path·⁖·˙non…ostensive:·
·⁖ostensive:·Feet wear one narrow course through dust and grass::and distant places remain joined by the same passage𑁍·⁖·˙ostensive:·
·⁖grimchain:·Ⲛ𑁋ⳑ𑁦𑁤ᵛᵠᵖ⩒ⳆⲰⲧ𑁒ᵚ𐞥·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜𐤟꩝ category꩜꩜parliament꩝:·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜𐎗꩝:·

˖word:·蔑˖·˙word:·

·⁖origin…language:·Tangut·⁖·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖·˙font:·
·⁖pronunciation:·DAWM·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Move or stir·⁖· to change position or become emotionally moved·⁖·˙non…ostensive:·
·⁖ostensive:·Here and there are no longer the same place::and the distance between them bears the whole change of position𐤟·⁖·˙ostensive:·
·⁖grimchain:·𝌆⁕ⵗ𐎗ᑭᑭ⁕⊜ⲘⲎ♂▽Ψ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜◎꩝ category꩜꩜goetic꩝:·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜𐎗꩝:·

˖word:·근원˖·˙word:·

·⁖origin…language:·Middle Korean·⁖·˙origin…language:·
·⁖font:·Noto Sans KR·⁖·˙font:·
·⁖pronunciation:·keunwuhn·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·root::cause::source::origin·⁖·˙non…ostensive:·
·⁖ostensive:·Branches spread through upper space::and every line of stem and branch converges downward upon one buried point from which the whole reach departs◎·⁖·˙ostensive:·

·⁖grimchain:·𒀭𒈪▽▽𒉿𒃻𒄷Λℂᔑᘐ𒌆·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜⬡꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜𐤏꩝⁖·

·word:·𒁍𒀸𒅗·⁖·˙word:·

·⁖origin…language:·Akkadian·⁖·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖·˙font:·
·⁖pronunciation:·BEE⟊toom·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·house·˙ palace·˙ temple·˙ family::tribe::household·⁖·˙non…ostensive:·
·⁖ostensive:·Walls bound one sheltered interior·˙ roof separates sheltered interior from open sky·˙ one doorway connects sheltered interior to the road⬡·⁖·˙ostensive:·
·⁖grimchain:·𒉺𒀸𒈬𒌋𒄑𒉺𒀀𒍑𒁍𒀸↑·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜✡꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜𐤏꩝⁖·

·word:·ᴋʜᴍɪ·⁖·˙word:·

·⁖origin…language:·Coptic·⁖·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁖·˙font:·
·⁖pronunciation:·KHAY⟊mee·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Hearth or brazier·˙ a place or vessel for fire·⁖·˙non…ostensive:·
·⁖ostensive:·Flame occupies one fixed hollow among stone and ash·˙ warmth forms its strongest region around the stone⟊bounded hollow✡·⁖·˙ostensive:·
·⁖grimchain:·ⲿⲛϒ𐤆𐤋ϡ𐤉⏁Ⲁ❂ⲛⲛⲚ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜☆꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜𐤏꩝⁖·

·word:·𐤓𐤔𐤕·⁖·˙word:·

·⁖origin…language:·Ugaritic·⁖·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖·˙font:·
·⁖pronunciation:·NAHARU·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·River·⁖·˙non…ostensive:·
·⁖ostensive:·Banks keep apart from one another::and moving water holds the long passage between them through the land☆·⁖·˙ostensive:·
·⁖grimchain:·ϫƵΛ𐤊ᛟᚸᚺᚥᚸᛋᚹᚹ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜❂꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜𐤏꩝⁖·

·word:·𐤉𐤌𐤎·⁖·˙word:·

·⁖origin…language:·Ugaritic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖.·˙font:·
·⁖pronunciation:·KHYM·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Ore⁖· mineral·⁖.·˙non…ostensive:·
·⁖ostensive:·Rock closes around a dark vein⁖⁖and hidden metal keeps its place inside the stone where
↪ daylight never enters✪·⁖.·˙ostensive:·
·⁖grimchain:·𐤕♂𐤌𐤉ᘐᑭᐯᗷᓀ0Ꭹᗷ⊛Ᏼ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜֍꩞ category꩜꩜ᵍoetic꩞⁖·
·⁖lookup office꩜꩜ˢpatial…absolute꩞ enoch꩜꩜ᵍ꩞⁖·

˖word:·ሰማይ˖⁖.·˙word:·

·⁖origin…language:·Ge'ez·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁖.·˙font:·
·⁖pronunciation:·seh⟊MY·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Sky or heaven⁖· the expanse above the earth·⁖.·˙non…ostensive:·
·⁖ostensive:·Above roof and ridge⁖⁖an open blue reach continues without wall or ceiling֍·⁖.·˙ostensive:·
·⁖grimchain:·𐤆⏃·ᑭᑫ♀ᑦᑭ✪◎ᐯΦX➿·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜⁜꩞ category꩜꩜ᵍoetic꩞⁖·
·⁖lookup office꩜꩜ˢpatial…absolute꩞ enoch꩜꩜ᵍ꩞⁖·

˖word:·➸✠˖⁖.·˙word:·

·⁖origin…language:·Ugaritic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖.·˙font:·
·⁖pronunciation:·RH·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Wind⁖· breath·⁖.·˙non…ostensive:·
·⁖ostensive:·Between reed⁖⁖sail⁖⁖and road lies one continuous unseen reach⁖· pressure crosses the
↪ wall⟊less interval among reed⁖⁖sail⁖⁖and road⁜·⁖.·˙ostensive:·
·⁖grimchain:·ᎥᎽ⊛ᙢᔦᖶᐁᘐᒐᛏ✱▽·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜⧗꩞ category꩜꩜ᵍoetic꩞⁖·
·⁖lookup office꩜꩜ˢpatial…absolute꩞ enoch꩜꩜ᵍ꩞⁖·

˖word:·𓏏˖⁖.·˙word:·

·⁖origin…language:·Tangut·⁖.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖.·˙font:·
·⁖pronunciation:·R⟊VEHR·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Hug or hold up⁖· to clasp closely or support with the arms·⁖.·˙non…ostensive:·
·⁖ostensive:·Arms close around what is held⁖⁖and the space between breast and burden narrows until
↪ almost no air remains⧗·⁖.·˙ostensive:·
·⁖grimchain:·ᎥᏧᑎᖘᑐᘐᛡᖶᖷ꩜ᘓ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁖glyph symbol꩜꩜⊛꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜☍꩝⁖·

⁘word⁖·䨵⁖.·˙word⁖·

·⁖origin…language⁖·Tangut·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Serif Tangut·⁖.·˙font⁖·
·⁖pronunciation⁖·RUHR·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Shadow⁖· a dark shape produced when light is blocked·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Sun stands on one side of the stone⁝⁝and darkness lies on the other in the stone’s own
↪ outline⊛·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᘏᘓᕈᑎᘐᖇᕮᖱᘁᕒᕓ·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol꩜꩜❄꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜☍꩝⁖·

⁘word⁖·𑀮𑁆𑀢⁖.·˙word⁖·

·⁖origin…language⁖·Tocharian B·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Brahmi·⁖.·˙font⁖·
·⁖pronunciation⁖·SEHRKEH·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Circle⁖· cycle·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·A path turns and turns until its ending meets its beginning⁝⁝and the middle never leaves
↪ the center❄·⁖.·˙ostensive⁖·
·⁖grimchain⁖·𑁋𑁍𑁍𑁊𑇉𑇊𑇐𑇁𑇈𑇔𑇓·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol꩜꩜❁꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜☍꩝⁖·

⁘word⁖·𒂗𒍣⁖.·˙word⁖·

·⁖origin…language⁖·Sumerian·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Cuneiform·⁖.·˙font⁖·
·⁖pronunciation⁖·GAH+bah·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Chest⁖· the front part of the torso between neck and abdomen·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Between throat and belly⁝⁝ribs keep the forward house of breath⁝⁝and the beating within
↪ stays behind that bone+walled place❁·⁖.·˙ostensive⁖·
·⁖grimchain⁖·∀Θ⁕⊙ᔑᒣᛯᖰᘉᐱᛉᛉ·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol꩜꩜⬡꩝ category꩜꩜goetic꩝⁖·
·⁖lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜☍꩝⁖·

⁘word⁖·ᛖᚡ⊗ᛁ⁖.·˙word⁖·

·⁖origin…language⁖·Etruscan·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Old Italic·⁖.·˙font⁖·
·⁖pronunciation⁖·SHOO+thee·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖.·˙cadence⁖·

·⁚non…ostensive⁚·Tomb⁚· a place made to receive and preserve the dead·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Stone closes over the still place⁚⁚and earth above keeps the dead apart from sun and
↪ road⌬·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛉᛘᛯᛇᛏᚼᚸᚹᛯᛏᛟ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐑍 category꩜꩜court𐑉⁚·
·⁚lookup office꩜꩜spatial…absolute𐑉 enoch꩜꩜𐑜𐑉⁚·

˖word⁚·ᠲᡀᡁᡂ⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·KAHTSOH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·(anatomy) belly⁚⁚stomach⁚⁚abdomen⁚⁚womb·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Below the ribs and above the hips lies the inward room of hunger⁚⁚birth⁚⁚and hidden
↪ life𐑍⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᦒᧆᧉᚢᚤᛟᛯ'❋ᛓ⚲·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐑤 category꩜꩜court𐑉⁚·
·⁚lookup office꩜꩜spatial…absolute𐑉 enoch꩜꩜𐑜𐑉⁚·

˖word⁚·𐑋⁚.·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·NEE⟊ngin·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Circle⁚˙ to move around or enclose on all sides·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One path returns around all sides⁚⁚and nothing held within escapes the turning that
↪ surrounds it𐑤⁚.·˙ostensive⁚·
·⁚grimchain⁚·⊙∫𐑓ᛘᚥᛯ◎‹ᚱᚴᑐ𐑤𐑤⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐑤 category꩜꩜court𐑉⁚·
·⁚lookup office꩜꩜spatial…absolute𐑉 enoch꩜꩜𐑜𐑉⁚·

˖word⁚·龘⁚.·˙word⁚·

·⁚origin…language⁚·Tangut·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·˙font⁚·
·⁚pronunciation⁚·R⟊BEER·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Meet or encounter⁚˙ to come into the presence of another·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Two roads arrive at one crossing⁚⁚and the same point belongs wholly to both𐑤⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐑘ᨒᨓᚩᚱᚢᛯᛯᛟᚣᚺ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐑥 category꩜꩜court𐑉⁚·
·⁚lookup office꩜꩜spatial…absolute𐑉 enoch꩜꩜𐑜𐑉⁚·

˖word⁚·ᠰᡁ⁚.·˙word⁚·

·⁚origin…language꞉·Tocharian A·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·SOHPEE·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Net·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Knotted cords cross at fixed intervals over open water⁚· open water passes through every
↪ square gap in the stretched mesh𑖭·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐹳𐹨𐹴𐹵𐹰𐹹𐹺𐹻𐹼𐹽𐹾·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꞉꞉𐤏 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐤍꞉·

꞉word꞉·𑖮𑖯𑖰𑖱𑖲꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·SHCHMOHNYNYAH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Basis⁚⁚foundation·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Below pillar and wall lies the buried stone⁚⁚and everything raised above keeps its place
↪ upon that hidden breadth𑖭·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐹿𐹾𐹽𐹼𐹻𐹺𐹹'𐹸·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꞉꞉𐤓 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐤍꞉·

꞉word꞉·𑖪𑖫𑖬𑖭𑖮꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·AHOORTSTSEHAH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Wide·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Two banks stand far apart⁚⁚and broad earth lies open between them𑖭·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐹿𐹸𐹷𐹶𐹵𐹴𐹳𐹲𐹱𐹰𐹯𐹮·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꞉꞉𐤔 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐤍꞉·

꞉word꞉·𐹨꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·har+RAH+noo·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Road⁚· route⁚· journey⁚· expedition·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Dust keeps one long scar between gate and distant hill⁚⁚and every turning of the land
↪ still belongs to the same way𑖭·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𒐊𒐋𒐌𒐍𒐎𒐏𒐐𒐑𒐒𒐓·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙

·⁖glyph symbol◦–◦𐫦 category◦–◦court𐫰⁖·
·⁖lookup office◦–◦spatial…absolute𐫰 enoch◦–◦𐫯𐫰⁖·

⁘word⁖·+□ȣ̄ʟ̄⁖·˙word⁖·

·⁖origin…language⁖·Tocharian A·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Brahmi·⁖·˙font⁖·
·⁖pronunciation⁖·KUHMPOH·⁖·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Circle·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·A returning line closes upon itself⁝⁝and the center remains held without beginning or last
↪ edge𐫦·⁖·˙ostensive⁖·
·⁖grimchain⁖·⍙ሕΨ𝌆ᵹ▽𐫳ᛈሐ♀𐫦·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol◦–◦𐫤 category◦–◦court𐫰⁖·
·⁖lookup office◦–◦spatial…absolute𐫰 enoch◦–◦𐫯𐫰⁖·

⁘word⁖·刎⁖·˙word⁖·

·⁖origin…language⁖·Tangut·⁖·˙origin…language⁖·
·⁖font⁖·Noto Serif Tangut·⁖·˙font⁖·
·⁖pronunciation⁖·N⟊SHAW·⁖·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Gather or collect⁖˙ to bring separate things together·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Scattered things draw near until the open gaps grow small⁝⁝and one crowded place holds
↪ what was once spread apart𐫤·⁖·˙ostensive⁖·
·⁖grimchain⁖·✳ↄ𐫲𐫤𐫠𐫢𐫤𐫟𐫠𐫫ℳ𐫞ϒ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol◦–◦𐫣 category◦–◦court𐫰⁖·
·⁖lookup office◦–◦spatial…absolute𐫰 enoch◦–◦𐫯𐫰⁖·

⁘word⁖·ΔIȚ̄L̄⁖·˙word⁖·

·⁖origin…language⁖·Tocharian B·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Brahmi·⁖·˙font⁖·
·⁖pronunciation⁖·EHRSNAH·⁖·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Form⁝⁝shape⁝⁝beauty·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Clay meets open air along a balanced contour⁖˙ opposite edges answer one another in
↪ proportion⁝⁝and the enclosed silhouette holds without distortion𐫤·⁖·˙ostensive⁖·
·⁖grimchain⁖·⟴Ƶᙀᛘᛉ𐫳ᵒᛉᵎ𐫱ᛊ⁘▫⧗·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol◦–◦𐫧 category◦–◦court𐫰⁖·
·⁖lookup office◦–◦spatial…absolute𐫰 enoch◦–◦𐫯𐫰⁖·

⁘word⁖·㰳⁖·˙word⁖·

·⁖origin…language⁖·Tangut·⁖·˙origin…language⁖·
·⁖font⁖·Noto Serif Tangut·⁖·˙font⁖·
·⁖pronunciation⁖·M⟊PHAAH·⁖·˙pronunciation⁖·
·⁖cadence⁖·Spatial Absolute·⁖·˙cadence⁖·

·⁚non…ostensive⁝·Divide or separate⁖ to split a whole into parts·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·One vessel lies in two pieces⁝⁝and open space now stands where clay once joined
↪ clayᚱ·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᛏ𒌋𒀸𐤔𓂧𐤉∯ℤ⅍ⅩⅭⅮⅭ⊕·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol○–○𓂧𓏏 category○–○court𓏏⁚·
·⁚lookup office○–○𓂧spatial…absolute𓏏 enoch○–○𓃰𓏏⁚·

⁖word⁝·𓀜𓃹𓆏𓅱⁚.·˙word⁝·

·⁚origin…language⁝·Akkadian·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Cuneiform·⁚.·˙font⁝·
·⁚pronunciation⁝·kah+SHAH+doo·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Spatial Absolute·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Reach⁖˙ attain⁖˙ arrive at⁖˙ achieve⁖˙ in military use⁝⁝capture or
↪ conquer·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Gate and distant wall occupy opposite ends of one continuous span⁖˙ the far boundary lies
↪ exactly at the line’s terminal point⁝⁝with no remaining interval between themᚱ·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ϒ𐤋𓆏𓂀ↄ𒌋ⱣⱯᚱ𓃹𓅱⊛·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol○–○𓆏𓏏 category○–○court𓏏⁚·
·⁚lookup office○–○𓂧spatial…absolute𓏏 enoch○–○𓃰𓏏⁚·

⁖word⁝·𓀜𓃹⁚.·˙word⁝·

·⁚origin…language⁝·Sumerian·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Cuneiform·⁚.·˙font⁝·
·⁚pronunciation⁝·AY·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Spatial Absolute·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·House⁝⁝building⁝⁝household⁝⁝temple+house·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Walls bound one sheltered interior⁖˙ roof separates sheltered interior from open sky⁖˙ one
↪ doorway connects sheltered interior to the road𐤉·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𐤔𐤒ᔑ𓃰ᔑ𐤕𐤇ᘜ◎𓆑ᚴ𐤍·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol○–○𐤉𓏏 category○–○court𓏏⁚·
·⁚lookup office○–○𓂧spatial…absolute𓏏 enoch○–○𓃰𓏏⁚·

⁖word⁝·ᛖᚣᚳᚣᛖᚣ⁚.·˙word⁝·

·⁚origin…language⁝·Etruscan·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Old Italic·⁚.·˙font⁝·
·⁚pronunciation⁝·MAH+lah+nah·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Spatial Absolute·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Mirror⁖˙ a polished surface that returns a visible image·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Face stands before polished bronze⁝⁝and another face waits in the bright plane without
↪ crossing into the metal𐤄·⁚.·˙ostensive⁝·
·⁚grimchain⁝·'ᛯ∀ᛏ𒌋☽✳𓃢❄ᚱ⊙☽֍·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol○–○𓂋𓏏 category○–○court𓏏⁚·
·⁚lookup office○–○𓂧spatial…absolute𓏏 enoch○–○𓃰𓏏⁚·

꞉word꞉·ᛣᏋᏠ꞉·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·SHOHLEEYEH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Hearth·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Stone and ash bound one chosen hollow for flame⁖ warmth forms its strongest region around
↪ the stone+bounded hollow¤·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᵕᚸ◎ⵔᛯᛘᚦᚺᛞᛇ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○᛫ᚧ» category꞉○–○᛫court»꞉·
·⁚lookup office꞉○–○᛫spatial…absolute» enoch꞉○–○᛫ᛟ»꞉·

꞉word꞉·ᛏᒪᒣ꞉·˙word꞉·

·⁚origin…language꞉·Tocharian A·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·K+PAHR·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Deep·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Rim keeps daylight high⁚⁚and darkness gathers below as the hollow falls farther from the
↪ open airᚧ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᛈᛀᛊ⍟ᛦᛤᔕIᚴᛞᛅ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○᛫ᛇ» category꞉○–○᛫court»꞉·
·⁚lookup office꞉○–○᛫spatial…absolute» enoch꞉○–○᛫ᛟ»꞉·

꞉word꞉·ꓫHINI꞉·˙word꞉·

·⁚origin…language꞉·Coptic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Coptic·⁚·˙font꞉·
·⁚pronunciation꞉·MAY+ee+nee·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Sign⁚⁚image⁚⁚monument⁚⁚or visible token·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Carved mark catches light on one face of stone⁚⁚and blank surface falls away around the
↪ place it claimsᛇ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᒪ⚛ꓲᛯᕁᛞ⬡▽ᛎᔕᚧᛉ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○᛫ᛉ» category꞉○–○᛫court»꞉·
·⁚lookup office꞉○–○᛫spatial…absolute» enoch꞉○–○᛫ᛟ»꞉·

꞉word꞉·ᛒᛁᚾᛃ»꞉·˙word꞉·

·⁚origin…language꞉·Avestan·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Avestan·⁚·˙font꞉·
·⁚pronunciation꞉·puh+RUH+thoo·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Wide⁚⁚broad·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Two far edges keep their distance beneath the same sky⁚⁚and broad earth lies unbroken
↪ between themᛉ·⁚·˙ostensive꞉·

·⁚grimchain⁚·'ᢣ᳟ᥣᑲᦆᦓᣔᣞᣘ⊙ꕥ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍ᛝ᛫ category₀₋₀₍court᛫⁚·
·⁚lookup office₀₋₀₍spatial…absolute᛫ enoch₀₋₀₍ᛜ᛫⁚·

₍word⁚·兆⁚.·˙word⁚·

·⁚origin…language⁚·Tangut·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·˙font⁚·
·⁚pronunciation⁚·SWEE·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Phantom⁚⁚vision⁚⁚or illusion·⁚˙ an appearance not ordinarily embodied·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Visible outline occupies one bounded region in open air·⁚˙ ground beneath the visible
↪ region bears no contact trace·⁚˙ air inside the outlined volume displaces neither dust nor
↪ weightᛝ⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛋ᠈ᠰᠴᠾ᠔ᛝ᠊᠈ᡕᡱᠰ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍ᛋ᛫ category₀₋₀₍court᛫⁚·
·⁚lookup office₀₋₀₍spatial…absolute᛫ enoch₀₋₀₍ᛜ᛫⁚·

₍word⁚·ᒐ□ᛁᒉ⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·PUHRNEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Outside·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Wall keeps warmth within⁚⁚and wind and dust belong beyond the last enclosing
↪ edgeᛋ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·᛬ᛝᛕᘉᚦᚺᛦᚴᐁᛕ꧁ᐃ⊙·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍'᛫ category₀₋₀₍court᛫⁚·
·⁚lookup office₀₋₀₍spatial…absolute᛫ enoch₀₋₀₍ᛜ᛫⁚·

₍word⁚·ᒐᑉ⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·POHSHREE·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Wall·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Courtyard lies on one side and road on the other⁚⁚and stone rises between them from earth
↪ to roofline'⁚.·˙ostensive⁚·
·⁚grimchain⁚·⊕ᛕᘐᚾᛯᗅᓄᐁᔑᚦᚺᔐ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍ᚾ᛫ category₀₋₀₍court᛫⁚·
·⁚lookup office₀₋₀₍spatial…absolute᛫ enoch₀₋₀₍ᛜ᛫⁚·

₍word⁚·🞓⁚.·˙word⁚·

·⁚origin…language:·Sumerian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·EE+ree·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·City::town::settlement·⁚.·˙non…ostensive:·
·⁚ostensive:·Adjoining houses and intersecting lanes fill one bounded reach whose outer gates mark
↪ where the settlement gives way to open road↓·⁚.·˙ostensive:·
·⁚grimchain:·𒍪𒌋𒀸𒉿𒊭𒅎𒌆𒄑𒍝·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉↓꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉☍꞉·

·word:·ᚱᚾᚢᚺ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·HRAYR·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Cairn or burial place·˙ a mound or heap associated with interment·⁚.·˙non…ostensive:·
·⁚ostensive:·Stone lies upon stone above one quiet patch::and the buried place below remains held apart
↪ from road and weather↓·⁚.·˙ostensive:·
·⁚grimchain:·❄ᵻᵞᓚ❁ᚠᦓᓗᘏ☆ᚹᓄ✱·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉↑꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉☍꞉·

·word:·▭·⁚.·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·SHEH+noo·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Cartouche·˙ an enclosing oval for a royal name·⁚.·˙non…ostensive:·
·⁚ostensive:·Royal signs rest inside one unbroken oval::and writing beyond the curve does not enter the
↪ enclosed name↑·⁚.·˙ostensive:·
·⁚grimchain:·☽○ᓚᛯᕭᙏᚸᚽᙏᚹᓄ▽ᔊ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉ᛦ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉☍꞉·

·word:·𠳿·⁚.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·PAH·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Different::another::or separate·⁚.·˙non…ostensive:·
·⁚ostensive:·Separate vessels occupy distinct positions across open floor::their bounded volumes
↪ remaining disjointᛦ·⁚.·˙ostensive:·
·⁚grimchain:·𓃰𓆏∮↓ᙏᘎ↑ᨒᡕ₀✡∮ᓄᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙

·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉Ո꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉ᴏ꞉·

꞉word꞉·ᚦᚸᛈᚱᚳ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Etruscan·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Italic·⁚·˙font꞉·
·⁚pronunciation꞉·AH⟊kel·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Made thing⁚˙ an object produced by work·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Raw pieces occupy separate places⁚˙ fitted edges meet without gaps꞉꞉and the completed body
↪ holds a geometry no untouched timber possessedՈ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᛈᛏ‡ᚲᵭ⁎ᛟᛃᚹᚦⴱ‡ᚨ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉Þ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉ᴏ꞉·

꞉word꞉·ϐᚦᚸᛈᚱ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Etruscan·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Italic·⁚·˙font꞉·
·⁚pronunciation꞉·FAH⟊keh·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Torch⁚˙ a portable burning light·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·A narrow shaft fixes flame above the grasping place⁚˙ darkness yields within a bounded
↪ circle surrounding the elevated flameÞ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᛉᛟᙏᒉᘐᙆᙅ⁎ᚷᛀ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉ſ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉ᴏ꞉·

꞉word꞉·ɯooγ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Coptic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Coptic·⁚·˙font꞉·
·⁚pronunciation꞉·MOH⟊oo·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Water·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Jar꞉꞉channel꞉꞉hollow꞉꞉and cup each lend their shape꞉꞉and the clear body inside keeps no shape
↪ of its ownſ·⁚·˙ostensive꞉·
·⁚grimchain꞉·⛒☿◝ᘐᘏ◎ᚮ⬡ᛀᚼᛘᛁᗐ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉ℛ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉ᴏ꞉·

꞉word꞉·ᒐᓀλ̄꞉·⁚·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·YAHST·⁚·˙pronunciation꞉·

·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Precipice·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Firm earth ends at one sharp lip⁚⁚and open depth begins immediately beyond the last step
↪ of stoneᛕ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·⛤𐤓⬡XΠϔ𐤆ᛋᘐΨΗᘐ𐤋·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐤂꞉ category꞉○–○court꞉꞉·
·⁚lookup office꞉○–○spatial…absolute꞉ enoch꞉○–○𐤏꞉꞉·

˖word꞉·↬˖·⁚.·˙word꞉·

·⁚origin…language꞉·Sumerian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·MAHKH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·High⁚˙ elevated⁚⁚lofty⁚⁚great⁚⁚or raised above ordinary level·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·One raised place occupies the highest vertical position⁚⁚with roof and ridge both below it
↪ and open sky immediately above‹·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ǂ𐤔ᘐᘐXᘐ𐤉ᛉϒᘐIΘᘓᛘ𐤉·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○X꞉ category꞉○–○court꞉꞉·
·⁚lookup office꞉○–○spatial…absolute꞉ enoch꞉○–○𐤏꞉꞉·

˖word꞉·ᚣVᛕᛈᛚOᛊ˖·⁚.·˙word꞉·

·⁚origin…language꞉·Etruscan·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Italic·⁚.·˙font꞉·
·⁚pronunciation꞉·OW+keh+lohs·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Dawn or aurora⁚˙ the first light before sunrise·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Along the far horizon⁚⁚pale light occupies a narrow band between dark earth below and
↪ darker sky above⁚˙ the solar disk remains below the horizon lineX·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐤍𐤅𐤈𐤎𐤈⊛𐤎☉ΠⵔΗ‹·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐤐꞉ category꞉○–○court꞉꞉·
·⁚lookup office꞉○–○spatial…absolute꞉ enoch꞉○–○𐤏꞉꞉·

˖word꞉·ρο˖·⁚.·˙word꞉·

·⁚origin…language꞉·Coptic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Coptic·⁚.·˙font꞉·
·⁚pronunciation꞉·ROH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Mouth·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Breath and food pass through one opening beneath the nose⁚⁚and voice leaves by the same
↪ gate between cheek and jaw𐤐·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᘐᗡᚼᑎ𐤋⬡ǂ𐤑ⲙᔕᘐ𐤀𐤒·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○Η꞉ category꞉○–○court꞉꞉·
·⁚lookup office꞉○–○spatial…absolute꞉ enoch꞉○–○𐤏꞉꞉·

⁖word:·𐌂𐌉⊗𐌌𐌉⁖·˙word:·

·⁖origin…language:·Etruscan·⁖·˙origin…language:·
·⁖font:·Noto Sans Old Italic·⁖·˙font:·
·⁖pronunciation:·KATH+nee·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Basin or handbasin⁖ a shallow vessel·⁖·˙non…ostensive:·
·⁖ostensive:·Water rests shallow inside a low hollow⁝⁝and the rim keeps the little breadth gathered
↪ within reach𐌍·⁖·˙ostensive:·
·⁖grimchain:·𐋡𐋌𐋒𐋗⊛ 𐤆𐤄𐤍𐡀𐡁𐡂𐡃·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐤄꞉ category◦–◦court꞉·
·⁖lookup office◦–◦spatial…absolute꞉ enoch◦–◦𐤏꞉·

⁖word:·▭⁖·˙word:·

·⁖origin…language:·Egyptian·⁖·˙origin…language:·
·⁖font:·Noto Sans Egyptian Hieroglyphs·⁖·˙font:·
·⁖pronunciation:·EE+nehr·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Stone⁖ a hard mineral mass or piece of rock·⁖·˙non…ostensive:·
·⁖ostensive:·Soft earth and flowing water divide around one hard occupied volume⁖ paths bend at its
↪ boundary and no passage crosses the mass𐤄·⁖·˙ostensive:·
·⁖grimchain:·𐤍𐋒𐤙𐋗𐋡⬡⊛ 𐤋𐤍𐤐𐋌𐋒𐤄·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐤄꞉ category◦–◦court꞉·
·⁖lookup office◦–◦spatial…absolute꞉ enoch◦–◦𐤏꞉·

⁖word:·𐬞𐬀̄⁖·˙word:·

·⁖origin…language:·Tocharian B·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·SHRAHT·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Sliver⁝⁝shard⁝⁝piece·⁖·˙non…ostensive:·
·⁖ostensive:·One small bounded piece occupies only part of a former contour⁖ open gaps surround the
↪ broken edges where neighboring matter is absent𐤄·⁖·˙ostensive:·
·⁖grimchain:·𐡀𐤆𐤄𐋡𐋒𐋗𐋙𐋚𐋛𐋜𐤊·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐤅꞉ category◦–◦court꞉·
·⁖lookup office◦–◦spatial…absolute꞉ enoch◦–◦𐤏꞉·

⁖word:·𓀀⁖·˙word:·

·⁖origin…language:·Akkadian·⁖·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖·˙font:·
·⁖pronunciation:·EH+peh+room·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·dust⁖ soil⁝⁝earth (for building work)·⁖·˙non…ostensive:·
·⁖ostensive:·Minute grains occupy every shallow gap between larger bodies⁖ cracks too narrow for
↪ pebbles still hold a continuous fine layer𐤅·⁖·˙ostensive:·

·⁚grimchain⁚·♀☆ϟ↗ᚻ♂ᚱᚹᛋ⊕ᛝᛯᛒᐁ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐑂𐑂 category⟜⟜court𐑂⁚·
·⁚lookup office⟜⟜spatial…absolute𐑂 enoch⟜⟜𐑂𐑂⁚·

˖word⁚·𐤔𐤓𐤉𐤍⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·NAH+room·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·river::canal⁚˙ River::River+god (a river personified or deified)⁚˙ vein::duct of the
↪ liver⁚˙ strip·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Two banks keep apart::and one moving band runs between them from high country toward
↪ low𐑂·⁚.·˙ostensive⁚·
·⁚grimchain⁚·✡ɵ✳◎ᛗᛕᔑᚏᚳᛯᐁᚴ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐑆𐑆 category⟜⟜court𐑂⁚·
·⁚lookup office⟜⟜spatial…absolute𐑂 enoch⟜⟜𐑂𐑂⁚·

˖word⁚·ረክብ⁚.·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚.·˙font⁚·
·⁚pronunciation⁚·RAK+b·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Assembly or meeting⁚˙ a gathering of persons·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Within one court::clustered feet::overlapping voices::and shared open center occupy the same
↪ bounded gathering place𐑆·⁚.·˙ostensive⁚·
·⁚grimchain⁚·₀❄𐑌ᚳᚻ✳ↄ◟ᵒᵒ𐑂♀◡ᚸ◎·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ↄ𐑂 category⟜⟜court𐑂⁚·
·⁚lookup office⟜⟜spatial…absolute𐑂 enoch⟜⟜𐑂𐑂⁚·

˖word⁚·ᒍ̄ᒪ̄ᖘ̄⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·LYAHOOTOH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Hole::opening·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Light crosses the wall at one round absence::and the rim holds all the missing stone
↪ around itↄ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ǂ▽ᛁᔑɵᛯᢣᚠᚹᚢᛃᚠ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜✵𐑂 category⟜⟜court𐑂⁚·
·⁚lookup office⟜⟜spatial…absolute𐑂 enoch⟜⟜𐑂𐑂⁚·

˖word⁚·ᛖᐯᐱ⊗ᚱᑭᐯǂᚣ⁚.·˙word⁚·

·⁚origin…language꞉·Etruscan·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Italic·⁚.·˙font꞉·
·⁚pronunciation꞉·shoon+theh+ROO+tsah·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Pyxis⁘ a small lidded container or box·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Small treasures sleep beneath a fitted lid⁖and the little hidden room is kept by wall and
↪ rim on every side✵·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐤟𐤦𐤩𐤢𐤭𐤮𐤯𐤰𐤱𐤲𐤳𐤴·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol꞉꞉𐤦 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐤮꞉·

꞉word꞉·祀꞉.·˙word꞉·

·⁚origin…language꞉·Tangut·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚.·˙font꞉·
·⁚pronunciation꞉·TSOHM·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Incomplete or residual⁘ remaining unfinished or left over·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Broken wall stops before the corner⁖and open air keeps the place the missing stones
↪ should fill𐤦·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐤆𐤗𐤛𐤜𐤝𐤞𐤟·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol꞉꞉𐤝 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐤮꞉·

꞉word꞉·𑀬𑀮𑀯𑀰𑀱꞉.·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·TSAHNKUHR·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Summit⁖top·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·One crown of stone occupies the uppermost point⁖with cloud below and every slope
↪ descending away into lower ground𐤝·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐤪𐤫𐤬𐤭𐤮𐤯𐤰𐤱𐤲𐤳·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol꞉꞉𐤧 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐤮꞉·

꞉word꞉·𒀀𒀉𒀊𒀋꞉.·˙word꞉·

·⁚origin…language꞉·Hittite·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·ALPASH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Cloud·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Between earth and sun⁖one pale suspended region interrupts the open sky⁘ ground below the
↪ suspended pale region receives only softened shadow𐤧·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐤨𐤩𐤪𐤫𐤬𐤭𐤮𐤯𐤰·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁘

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜◎꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜spatial…absolute꩞ enoch꩜꩜ᛟ꩞⁚·

꞉word꞉·ᛏ̄ᒍᎿᛣ̄꞉·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·KLOHKAHSHCHEH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Pore꞉꞉follicle·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Skin closes everywhere except one tiny mouth꞉꞉and sweat finds passage through that small opening◎·⁚·˙ostensive꞉·
↪
·⁚grimchain꞉·ꕤᐟ⬡↗ᕿᑭᘐᛒᛉᛟᛜᗰ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜⏃꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜spatial…absolute꩞ enoch꩜꩜ᛟ꩞⁚·

꞉word꞉·ᒪIᑌᒧᛏ̄꞉·˙word꞉·

·⁚origin…language꞉·Tocharian A·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·OHRPUHNK·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Marketplace꞉꞉bazaar·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Roads enter one open square from many sides꞉꞉and stalls ring the place where goods and voices gather⏃·⁚·˙ostensive꞉·
↪
·⁚grimchain꞉·Ↄ⛤ᛯᔑᛜᛘᗅᛃᚻᑭᐁ❂ᚓ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᔕᛋ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜spatial…absolute꩞ enoch꩜꩜ᛟ꩞⁚·

꞉word꞉·𐎗𐎄𐏁𐎽𐎢꞉·˙word꞉·

·⁚origin…language꞉·Hittite·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚·˙font꞉·
·⁚pronunciation꞉·EKUNASH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Cold·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Shade beneath stone keeps a different air from the sunlit ground꞉꞉and the sheltered place holds that chill closeᔕᛋ·⁚·˙ostensive꞉·
↪
·⁚grimchain꞉·ᛗᛋᔑᛇᚱᚻᚱᚻᚱᚻ

·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Afar⁖⁖far away·⁚.·˙non…ostensive:·
·⁚ostensive:·Ridge after ridge lies between the gate and the pale land beyond⁖⁖and distance keeps the
↪ far place beyond every nearer rise᛬·⁚.·˙ostensive:·
·⁚grimchain:·𐤟𐤢𐤦𐤠𐤀𐤍𐤋𐤀✪✡𐤎𐤒𐤉·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol₀₋₀₍❀✓₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍spatial…absolute₎ enoch₀₋₀₍ᵒ₎⁚·

₍word⁚·ঋ৯ঙ̄⁚.·˙word⁚·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·AHSAHM·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Seat·⁚.·˙non…ostensive:·
·⁚ostensive:·Mat and bench keep one small place clear in the hall⁖⁖and resting weight gathers there
↪ above the floor❀✓·⁚.·˙ostensive:·
·⁚grimchain:·ϒ𐊗ℳ𐊠𐊱𐊵𐊬𐊫♀𐊰𐊢▿·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol₀₋₀₍ₒ₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍spatial…absolute₎ enoch₀₋₀₍ᵒ₎⁚·

₍word⁚·밑⁚.·˙word⁚·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·mEEt·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·basis⁖⁖foundation (in general)·˙ bottom⁖⁖underneath·⁚.·˙non…ostensive:·
·⁚ostensive:·Everything raised keeps something beneath it⁖⁖and the lowest hidden place carries what
↪ stands above₀·⁚.·˙ostensive:·
·⁚grimchain:·ᚦ☆ᛒᚲᚺℳᚸ☆ᛊᚣᚮᚬ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol₀₋₀₍Θ₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍spatial…absolute₎ enoch₀₋₀₍ᵒ₎⁚·

₍word⁚·ᛦᛚᛁᛏ⁚.·˙word⁚·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·FLET·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Raised hall flooring or sleeping platform·⁚.·˙non…ostensive:·
·⁚ostensive:·Common floor lies below·˙ one wooden plane spans above it at a fixed height⁖⁖keeping a
↪ separate horizontal level within the hallΘ·⁚.·˙ostensive:·
·⁚grimchain:·⊙𐊗𐊦𐊱𐊗𐊬𐊦𐊬ᚾᚠᚱᚴ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol₀₋₀₍Ⱦ₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍spatial…absolute₎ enoch₀₋₀₍ᵒ₎⁚·

:word:·+□ıδ̄ĸ:.·˙word:·

·˙origin…language:·Tocharian B·˙.·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙.·˙font:·
·˙pronunciation:·KUHRWEHNYEH·˙.·˙pronunciation:·
·˙cadence:·Spatial Absolute·˙.·˙cadence:·
·˙non…ostensive:·Stone::rock·˙.·˙non…ostensive:·
·˙ostensive:·Soft ground and open air terminate at one hard bounded mass·˙ every route around it bends
↪ to the contour of the occupied volumeӾ·˙.·˙ostensive:·
·˙grimchain:·ᑭᕼ◎ᓄ⬡⌬ᘐᘎᛯᘐᑎᓀ𑁍·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol◦–◦₍Λ₎ category◦–◦₍court₎:·
·˙lookup office◦–◦₍spatial…absolute₎ enoch◦–◦₍₀₎:·

:word:·ግልባብ:.·˙word:·

·˙origin…language:·Ge'ez·˙.·˙origin…language:·
·˙font:·Noto Sans Ethiopic·˙.·˙font:·
·˙pronunciation:·guhl÷BAHB·˙.·˙pronunciation:·
·˙cadence:·Spatial Absolute·˙.·˙cadence:·
·˙non…ostensive:·Veil or covering·˙ material laid over something·˙.·˙non…ostensive:·
·˙ostensive:·Cloth falls between face and open air::and what lies beneath remains near yet hidden
↪ behind the thin coveringΛ·˙.·˙ostensive:·
·˙grimchain:·ᴼᑭᖶXᕮᛯᐟᑫᓚᔭIᐯᑫ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol◦–◦₍Ӿ₎ category◦–◦₍court₎:·
·˙lookup office◦–◦₍spatial…absolute₎ enoch◦–◦₍₀₎:·

:word:·ᛦᛋᛯ:.·˙word:·

·˙origin…language:·Old Turkic·˙.·˙origin…language:·
·˙font:·Noto Sans Old Turkic·˙.·˙font:·
·˙pronunciation:·OR÷too·˙.·˙pronunciation:·
·˙cadence:·Spatial Absolute·˙.·˙cadence:·
·˙non…ostensive:·Middle::midpoint·˙.·˙non…ostensive:·
·˙ostensive:·Two edges stand apart::and one mark keeps equal distance from bothӾ·˙.·˙ostensive:·
·˙grimchain:·ᛦZ۞ᓄᑫᐯᔭᓀᔑᘐᛇ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol◦–◦₍Ƙ₎ category◦–◦₍court₎:·
·˙lookup office◦–◦₍spatial…absolute₎ enoch◦–◦₍₀₎:·

:word:·ᖶ:.·˙word:·

·˙origin…language:·Sumerian·˙.·˙origin…language:·
·˙font:·Noto Sans Cuneiform·˙.·˙font:·
·˙pronunciation:·PAH·˙.·˙pronunciation:·
·˙cadence:·Spatial Absolute·˙.·˙cadence:·
·˙non…ostensive:·Branch·˙ a projecting division growing or extending from a larger
↪ body·˙.·˙non…ostensive:·
·˙ostensive:·Trunk holds one line upward::and a living arm leaves that line to claim another reach of
↪ skyƘ·˙.·˙ostensive:·

·⁚grimchain⁚·𒀭𒌋𒁍𒅗𒄑𒁲𒀸𒍪ⵔ𒍪Θ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚𒍪𒁲» category⁚𒍪court»⁚·
·⁚lookup office⁚𒍪spatial…absolute» enoch⁚𒍪𒁲»⁚·

˓word⁚·𒁲⁚.·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·SHEE+kah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Shard⁚˙ a broken fragment of pottery or other brittle material·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Curved fragment occupies one sector of a former vessel wall⁚˙ broken edges border open
↪ gaps where the continuous surface is absentΘ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𒋛⊙𒌋𒁍𒄀𒂍𒄑𒍪Θ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚𒍪𒀸» category⁚𒍪court»⁚·
·⁚lookup office⁚𒍪spatial…absolute» enoch⁚𒍪𒁲»⁚·

˓word⁚·𒀸𒁲𒁲𒁲⁚.·˙word⁚·

·⁚origin…language⁚·Hittite·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·KHATKUS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Narrow⁚⁚tight·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Two walls approach until only a thin interval remains⁚˙ the passage width is narrower than
↪ any broad loadᛦ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ⵔ𒌋𒉡𒌑𒁲Ⲏ𒌋𒄀𒀸𒍪⃝♂·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚𒍪𒍪» category⁚𒍪court»⁚·
·⁚lookup office⁚𒍪spatial…absolute» enoch⁚𒍪𒁲»⁚·

˓word⁚·राशि⁚.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·rahsh·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Heap⁚⁚mass⁚⁚pile⁚⁚group⁚⁚multitude⁚⁚quantity·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Grain lies upon grain until the lower kernels vanish beneath the upper⁚⁚and one rising
↪ body gathers from many small onesᛋ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ⵊ𒍪𒁹𒄑𒁲ᛦ𒀸⊙𒄩𒌋ᛊ𒍪‹·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚𒍪𒁲» category⁚𒍪court»⁚·
·⁚lookup office⁚𒍪spatial…absolute» enoch⁚𒍪𒁲»⁚·

˓word⁚·𒁲𒀸𒁲𒁲𒀸⁚.·˙word⁚·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·PASSILAS·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Stone⁚⁚pebble⁚· gem⁚⁚precious stone·⁚.·˙non…ostensive:·
·⁚ostensive:·Moving water divides around one small occupied volume⁚· the riverbed closes beneath it and
↪ the current rejoins beyond its bounded contour𐎼·⁚.·˙ostensive:·
·⁚grimchain:·𐤌𐤒𐤓𐤒𐤉𐤍𐤋𐤍𐤇𐤏𐤔𐤕·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉𓂋 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𓃰꞉⁚·

꞉word꞉·𓂝⁚.·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·ee⟊AHT·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Mound⁚· a raised mass of earth·⁚.·˙non…ostensive:·
·⁚ostensive:·Flat desert lifts into one rounded height⁚⁚and the raised earth keeps its own shadow upon
↪ the level land𐎼·⁚.·˙ostensive:·
·⁚grimchain:·𐫟𐫠𐫡𐫢𐫣𐫤𐫦𐫥𐫧𐫨·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉𓐍 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𓃰꞉⁚·

꞉word꞉·𓀀𓀁𓀂𓀃⁚.·˙word:·

·⁚origin…language:·Old Persian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚.·˙font:·
·⁚pronunciation:·MAH⟊nah·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Dwelling⁚⁚abode·⁚.·˙non…ostensive:·
·⁚ostensive:·Fire⁚⁚bed⁚⁚vessel⁚⁚and roof keep one sheltered place together⁚⁚and road and weather remain
↪ beyond its wallsI·⁚.·˙ostensive:·
·⁚grimchain:·𐤕𐤗𐤘𐤙𐤚𐤛𐤜𐤝𐤞·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉𓉐 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𓃰꞉⁚·

꞉word꞉·𓍿⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·BAH⟊boom·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·door⁚⁚gate⁚⁚opening⁚· city quarter·⁚.·˙non…ostensive:·
·⁚ostensive:·Wall is whole except for one passage⁚⁚and street and courtyard touch through that single
↪ break in the enclosure𐎼·⁚.·˙ostensive:·
·⁚grimchain:·⊙⁕𐎫𐎬𐎭𐎮𐎯𐎰𐎱𐎲𐎳·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𑀫꣸ category◦–◦court꣸⁖·
·⁖lookup office◦–◦spatial…absolute꣸ enoch◦–◦𑁍꣸⁖·

⁘word:·+𑀌ᒍ̄⁖·˙word:·

·⁖origin…language:·Tocharian A·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·KUHLN·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·To resound⁖ to howl (especially the wind)·⁖·˙non…ostensive:·
·⁖ostensive:·Wind cries through the gorge⁝and the far rock gives the same sound back across the
↪ hollow𑀫·⁖·˙ostensive:·
·⁖grimchain:·𑀪𑀳𑁆𑀰𑀭𑀶𑀵𑁄𑁎𑀧𑀫·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𑀪꣸ category◦–◦court꣸⁖·
·⁖lookup office◦–◦spatial…absolute꣸ enoch◦–◦𑁍꣸⁖·

⁘word:·𑀳𑁄⁖·˙word:·

·⁖origin…language:·Tangut·⁖·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖·˙font:·
·⁖pronunciation:·DEHM·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Cloud⁖ a visible mass of condensed vapor in the sky·⁖·˙non…ostensive:·
·⁖ostensive:·High sky contains a pale suspended breadth that closes around the mountain crown⁖ distant
↪ contours vanish where the vapor⁺filled region begins𑀪·⁖·˙ostensive:·
·⁖grimchain:·𓍯𓎼𓌳𓏥𓎟𓍿𓏏𓎟𓏛𓏺·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𑀲꣸ category◦–◦court꣸⁖·
·⁖lookup office◦–◦spatial…absolute꣸ enoch◦–◦𑁍꣸⁖·

⁘word:·𑀯𑀢𑀣⁖·˙word:·

·⁖origin…language:·Tocharian B·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·AHKAHSHEH·⁖·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖·˙cadence:·
·⁖non…ostensive:·Sky⁖ air·⁖·˙non…ostensive:·
·⁖ostensive:·Above roof and ridge⁝open air continues upward without wall or ceiling across the visible
↪ height𑀲·⁖·˙ostensive:·
·⁖grimchain:·xᨿ𑁆ᨶᨻᨷᨳ𑁆ᨶᨿᨶ꣸ᨶᨾᨵ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𓍯꣸ category◦–◦court꣸⁖·
·⁖lookup office◦–◦spatial…absolute꣸ enoch◦–◦𑁍꣸⁖·

⁘word:·𐎍𐎌𐎐⁖·˙word:·

·⁖origin…language:·Ugaritic·⁖·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖·˙font:·
·⁖pronunciation:·AHAPARU·⁖·˙pronunciation:·

·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Dust⁚· ground·⁚.·˙non…ostensive:·
·⁚ostensive:·Fine grains lie over road and threshold⁚⁚and every shallow crack keeps what the wind has
↪ left there◌·⁚.·˙ostensive:·
·⁚grimchain:·𐫱𑖪𑖀𑖌𑖿𑖭𑖽𑖨𑖀𑖾𑖰𑖀𑖽𑖀𑖀𑖽𑖀𑖀𑖽𑖀

˖word꞉·𗍫˖·˙word꞉·

·⁚origin…language꞉·Tangut·⁚·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚·˙font꞉·
·⁚pronunciation꞉·NAAH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Yarn⁚· spun fiber prepared as a continuous strand·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·One spun length extends from spindle to loom as a continuous narrow line꞉꞉with no break
↪ anywhere along the span𗏢·⁚·˙ostensive꞉·
·⁚grimchain꞉·♀ϟ♂𗈫𗈪⊕𗅋𗅇𗅏𗅏𗈪·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○꞉𗅇꞉ category꞉○꞉court꞉⁚·
·⁚lookup office꞉○꞉spatial…absolute꞉ enoch꞉○꞉𖤓꞉⁚·

˖word꞉·ᗅᛋᛊᛟ˖·˙word꞉·

·⁚origin…language꞉·Old Turkic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Turkic·⁚·˙font꞉·
·⁚pronunciation꞉·yoh꞉GHOON·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·dense꞉꞉thick꞉꞉tough·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Felt presses fibre against fibre until little air remains꞉꞉and the whole thickness holds
↪ close against wind and hand𗏢·⁚·˙ostensive꞉·
·⁚grimchain꞉·𖤓𗈪𗍐𗏢⬡𗃬𗅆𗅇𖤓♪·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○꞉𗏢꞉ category꞉○꞉court꞉⁚·
·⁚lookup office꞉○꞉spatial…absolute꞉ enoch꞉○꞉𖤓꞉⁚·

˖word꞉·𗉋˖·˙word꞉·

·⁚origin…language꞉·Tangut·⁚·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚·˙font꞉·
·⁚pronunciation꞉·HLUH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Billow⁚· a large swelling wave·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Sea surface forms one broad convex rise beside a neighboring hollow꞉꞉the two joined as
↪ opposite curvatures of one continuous sheet of water𗏢·⁚·˙ostensive꞉·
·⁚grimchain꞉·𖤓𗈫𗈪⊕𗅋◎𖤓𗅏𗅇Θᚢᚹᚹᚹᚹ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○꞉𗈪꞉ category꞉○꞉court꞉⁚·
·⁚lookup office꞉○꞉spatial…absolute꞉ enoch꞉○꞉𖤓꞉⁚·

˖word꞉·ᵬᵬᚴ˖·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·SHRMAHREH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Smooth·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Water slips over polished stone without catching꞉꞉and light runs across the surface
↪ without breaking on a rough edge𗈪·⁚·˙ostensive꞉·

·⁚grimchain⁚·ⵔAᙆሕᔦ꠸ϕᔭɻ⏁ⵔI·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꠰꠰𐑋꠱ category꠰꠰꠱court꠱⁚·
·⁚lookup office꠰꠰꠱spatial…absolute꠱ enoch꠰꠰꠱ᵒ꠱⁚·

꞉word⁚·ᑌĪΓ꞉⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·PREHREH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Arrow·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Nock and iron point occupy opposite ends of one straight axis·⁚ every point along the
↪ narrow shaft lies on one forward axis𐑋·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᵹᏩⵔᵒᵪᚦ꠸ᵕ꠸ᙏᚔᛯᚐ꠸ᵻ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꠰꠰𐑋Δ꠱ category꠰꠰꠱court꠱⁚·
·⁚lookup office꠰꠰꠱spatial…absolute꠱ enoch꠰꠰꠱ᵒ꠱⁚·

꞉word⁚·ᑎᑭᚼᚴ꞉⁚.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·OH⟊dhahl·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Hereditary estate or ancestral property·⁚ land held by inherited
↪ right·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Ancestor graves⁖dwelling⁖and field occupy one unbroken enclosure·⁚ boundary stones
↪ surround graves⁖dwelling⁖and field as one continuous territorial reachΔ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚱᙏᚔᛕ⏃ᙆ'ᗡᔕᚠᘐᚏᗡ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꠰꠰𐑋∀꠱ category꠰꠰꠱court꠱⁚·
·⁚lookup office꠰꠰꠱spatial…absolute꠱ enoch꠰꠰꠱ᵒ꠱⁚·

꞉word⁚·ᏰᏁIᛕᒣTS꞉⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·BEHRXTS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Bright⁖clear·⁚ manifest⁖evident·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Sun enters the hall and leaves no corner hidden⁖and every edge stands clean against the
↪ light∀·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᙙᙆᏩᔕᗺIᐯ☆ᔓᔐᙆᔕᑎ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꠰꠰𐑋∇꠱ category꠰꠰꠱court꠱⁚·
·⁚lookup office꠰꠰꠱spatial…absolute꠱ enoch꠰꠰꠱ᵒ꠱⁚·

꞉word⁚·ᙁᔓᙅ꞉⁚.·˙word⁚·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·wuh+sah+TAY·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Inside::interior::inner chamber::or innermost part·⁚.·˙non…ostensive:·
·⁚ostensive:·Beyond the threshold::walls keep a chamber no road or weather enters::and what is held there belongs wholly to the within▽·⁚.·˙ostensive:·
·⁚grimchain:·⟲ᑎᖘᓀᕀᗅᕴ⌬ᓄᖍᐱᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꞉꞉◦꞉▽꞉ category꞉꞉◦꞉court꞉:·
·⁚lookup office꞉꞉◦꞉spatial…absolute꞉ enoch꞉꞉◦꞉𐤏꞉:·

·word:·䶵·⁚.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·KHUH·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Rock::sway::or shake·⁚ to move repeatedly from side to side or back and forth·⁚.·˙non…ostensive:·
·⁚ostensive:·Hanging weight leans to either side of one cord::and both extremes keep the same still point between them▽·⁚.·˙ostensive:·
·⁚grimchain:·⟲◎𐫄𐦝❄ᛉ♋𓃹𓃰𓀠ь✡✳·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꞉꞉◦꞉▽꞉ category꞉꞉◦꞉court꞉:·
·⁚lookup office꞉꞉◦꞉spatial…absolute꞉ enoch꞉꞉◦꞉𐤏꞉:·

·word:·𐤂𐤅𐤌·⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·PEED·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Compassion·⁚ heart·⁚.·˙non…ostensive:·
·⁚ostensive:·Across a single doorway::an injured limb curls in one chamber while an uninjured limb takes the same guarding shape in the other▽·⁚.·˙ostensive:·
·⁚grimchain:·▽𐤅❋ₒ⩛𐤊𐤋𐤑𐤏𐤓·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꞉꞉◦꞉ᗅ꞉ category꞉꞉◦꞉court꞉:·
·⁚lookup office꞉꞉◦꞉spatial…absolute꞉ enoch꞉꞉◦꞉𐤏꞉:·

·word:·⬭⌓𓅓·⁚.·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·meh+SEH+jer·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Ear·⁚.·˙non…ostensive:·
·⁚ostensive:·Sound reaches the side of the head::and one small gate carries the unseen trembling inward through bone and hollowᗅ·⁚.·˙ostensive:·
·⁚grimchain:·𐤉ᨎ✳⌬𓂝ᚱᚥᛋ᛬ᛊᛗᗅA·⁚.·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐌅꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜ᛟ꩝:·

˖word:·ᚠ˖·⁚·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·KOWN·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Ulcer::sore::boil::or inflamed wound·⁚·˙non…ostensive:·
·⁚ostensive:·Red swelling claims one small place upon the skin::and heat and pain gather inside that
↪ bounded patch𐌅·⁚·˙ostensive:·
·⁚grimchain:·ᛇ❄ᔕ◎♙ᛜᚠᚺᚿ✱ᛉᚠ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᛉ꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜ᛟ꩝:·

˖word:·𐎚𐎍𐎍˖·⁚·˙word:·

·⁚origin…language:·Ugaritic·⁚·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚·˙font:·
·⁚pronunciation:·NAKHALU·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Ravine::wadi·⁚·˙non…ostensive:·
·⁚ostensive:·High banks face one another::and a long low bed runs between them where flood and dry
↪ season share the same passage𐌅·⁚·˙ostensive:·
·⁚grimchain:·ᘐᘓᘐᔓ‹▽ᘜᚾᛟᛖᛏ◌·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐌕꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜ᛟ꩝:·

˖word:·ᑌ̄ᒍ𐌕ᓘ̄˖·⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·PLAHKSAHR·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Agreement::understanding·⁚·˙non…ostensive:·
·⁚ostensive:·Two paths meet at one stone::and beyond the meeting neither road keeps a separate claim to
↪ the shared point𐌕·⁚·˙ostensive:·
·⁚grimchain:·ᛜ❀ᛖᛜᔓᛇ⟰⬡ᘜᚺᚾᛞ✡·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐌕꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜spatial…absolute꩝ enoch꩜꩜ᛟ꩝:·

˖word:·ᓚ̄ᒍᒐ̄˖·⁚·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·

·⁖pronunciation:·SLYEE·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Line::streak⁖ border·⁖.·˙non…ostensive:·
·⁖ostensive:·Two fields touch along one narrow mark::and each keeps its own side without crossing the
↪ same drawn edge⁜·⁖.·˙ostensive:·
·⁖grimchain:·⧫∀ꕯA0ϟ۶ϟ∓ₒᵣ🜨✳·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉꞉ category○–○𐑉court꞉·
·⁖lookup office○–○𐑉spatial…absolute꞉ enoch○–○𐑉ₒ꞉·

·word:·ϣԿɹɔ৭:.·˙word:·

·⁖origin…language:·Avestan·⁖.·˙origin…language:·
·⁖font:·Noto Sans Avestan·⁖.·˙font:·
·⁖pronunciation:·HVAH⟊ruh·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Sun::sunlight·⁖.·˙non…ostensive:·
·⁖ostensive:·One bright center occupies the upper field⁖ straight rays extend outward through every
↪ open direction until earth::roof::stone::and water close their paths꞉·⁖.·˙ostensive:·
·⁖grimchain:·ᚲ⏁𐑅⍰ϴ𐑅𐑅ᗑᚴ✡ᑭ✳‹·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉⁑꞉ category○–○𐑉court꞉·
·⁖lookup office○–○𐑉spatial…absolute꞉ enoch○–○𐑉ₒ꞉·

·word:·蕊:.·˙word:·

·⁖origin…language:·Tangut·⁖.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖.·˙font:·
·⁖pronunciation:·MUH·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Nothing⁖ absence of any thing::or a negative expression·⁖.·˙non…ostensive:·
·⁖ostensive:·Rim encloses a hollow::and no grain::water::stone::or living breath occupies the space
↪ within⁑꞉·⁖.·˙ostensive:·
·⁖grimchain:·⊕ↄ⸙ℂ∂∀Ⱨ⧗ƻ𐑅❁ᙢᚢ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐑉✳꞉ category○–○𐑉court꞉·
·⁖lookup office○–○𐑉spatial…absolute꞉ enoch○–○𐑉ₒ꞉·

·word:·구무:.·˙word:·

·⁖origin…language:·Middle Korean·⁖.·˙origin…language:·
·⁖font:·Noto Sans KR·⁖.·˙font:·
·⁖pronunciation:·kùmù·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·hole·⁖.·˙non…ostensive:·
·⁖ostensive:·Wall remains on every side but the center::and light passes through the place where solid
↪ matter is absent✳·⁖.·˙ostensive:·
·⁖grimchain:·∱ᑭ∇ᙢᙢᔕ❀✓ᗅᚴᑭzᗺᗕ꞉·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁚glyph symbol◦–◦𖿢𖿣𖿤 category◦–◦court𖿤⁚·
·⁚lookup office◦–◦spatial…absolute𖿤 enoch◦–◦𖿠𖿤⁚·

·word⁚·𖿡𖿢⁚.·ˊword⁚·

·⁚origin…language⁚·Tangut·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·ˊfont⁚·
·⁚pronunciation⁚·PHOH·⁚.·ˊpronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·ˊcadence⁚·
·⁚non…ostensive⁚·Sprinkle water⁚ to scatter liquid in small drops·⁚.·ˊnon…ostensive⁚·
·⁚ostensive⁚·Small drops lie apart across leaf and stone⁚ clear intervals separate every darkened
↪ droplet+mark𖿢𖿣⁚.·ˊostensive⁚·
·⁚grimchain⁚·𖿠𖿡𖿤𖿦 𖿧𖿨𖿩𖿪𖿫𖿬𖿭𖿮𖿯⁚.·ˊgrimchain⁚·
·⁚.·ˊlookup⁚
·⁚.·ˊglyphs⁚
·⁚glyph symbol◦–◦𖿰𖿱𖿲 category◦–◦court𖿲⁚·
·⁚lookup office◦–◦spatial…absolute𖿲 enoch◦–◦𖿠𖿲⁚·

·word⁚·𖿳𖿴𖿵⁚.·ˊword⁚·

·⁚origin…language⁚·Ugaritic·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚.·ˊfont⁚·
·⁚pronunciation⁚·BIIRU·⁚.·ˊpronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·ˊcadence⁚·
·⁚non…ostensive⁚·Well·⁚.·ˊnon…ostensive⁚·
·⁚ostensive⁚·Stone ring opens downward⁚⁚and cool darkness descends beneath the earth to the hidden
↪ water below𖿱𖿲⁚.·ˊostensive⁚·
·⁚grimchain⁚·𖿷𖿸𖿹𖿺𖿻𖿼𖿽𖿾𖿿𐀀𐀁𐀂⁚.·ˊgrimchain⁚·
·⁚.·ˊlookup⁚
·⁚.·ˊglyphs⁚
·⁚glyph symbol◦–◦𐀃𐀄𐀅 category◦–◦court𐀅⁚·
·⁚lookup office◦–◦spatial…absolute𐀅 enoch◦–◦𖿠𐀅⁚·

·word⁚·𐀆𐀇𐀈𐀉+𐀊𐀋⁚.·ˊword⁚·

·⁚origin…language⁚·Tocharian B·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·ˊfont⁚·
·⁚pronunciation⁚·PUHLTAHKW·⁚.·ˊpronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·ˊcadence⁚·
·⁚non…ostensive⁚·Drop⁚⁚droplet⁚ dew·⁚.·ˊnon…ostensive⁚·
·⁚ostensive⁚·One small bead of water keeps a round world upon leaf or grass⁚⁚and open air touches it on
↪ every side𖿣⁚.·ˊostensive⁚·
·⁚grimchain⁚·𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗⁚.·ˊgrimchain⁚·
·⁚.·ˊlookup⁚
·⁚.·ˊglyphs⁚
·⁚glyph symbol◦–◦𖿠𖿡𖿤 category◦–◦court𖿤⁚·
·⁚lookup office◦–◦spatial…absolute𖿤 enoch◦–◦𖿠𖿤⁚·

·word⁚·𐀘⁚.·ˊword⁚·

·⁚origin…language⁚·Sumerian·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·ˊfont⁚·
·⁚pronunciation⁚·OOB·⁚.·ˊpronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·ˊcadence⁚·

·⁚non…ostensive⁝·Corner⁚· the point or region where two sides⁚⁚edges⁚⁚or directions meet·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Two walls meet and turn at one place⁚⁚and neither can continue straight without leaving
↪ the other behind𓉗·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𓏺𐋡𐋠𓆉𐋠𐋡𐋠𐋡𐋠𓆉𐋠𐋡·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𓂺» category⟜⟜court»⁝·
·⁚lookup office⟜⟜spatial…absolute» enoch⟜⟜☍»⁝·

·word⁝·𓂺·⁚.·˙word⁝·

·⁚origin…language⁝·Tangut·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Serif Tangut·⁚.·˙font⁝·
·⁚pronunciation⁝·KAEH·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Spatial Absolute·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Desire⁚· to want or long for something·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Desired thing rests beyond the present reach⁚⁚and all the open distance between here and
↪ there remains unfilled𓂺·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𓉗𓏺𓆉𓏺𓉗𓏺𓆉𓏺𓉗𓏺𓆉·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𓆉» category⟜⟜court»⁝·
·⁚lookup office⟜⟜spatial…absolute» enoch⟜⟜☍»⁝·

·word⁝·𓆉·⁚.·˙word⁝·

·⁚origin…language⁝·Old Turkic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Old Turkic·⁚.·˙font⁝·
·⁚pronunciation⁝·YAHN·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Spatial Absolute·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To return⁚⁚to have a change of mind⁚⁚to step back·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Road stretches ahead and the old ground lies behind⁚⁚and one bearing turns back toward the
↪ place already known𓆉·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𓉗𓏺𓉗𓏺𓉗𓏺𓉗𓏺𓉗𓏺𓆉·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𓆉» category⟜⟜court»⁝·
·⁚lookup office⟜⟜spatial…absolute» enoch⟜⟜☍»⁝·

·word⁝·ⲛⲟϩ·⁚.·˙word⁝·

·⁚origin…language⁝·Coptic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Coptic·⁚.·˙font⁝·
·⁚pronunciation⁝·NOH·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Spatial Absolute·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Cord or rope·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Two posts stand apart⁚⁚and one flexible length keeps them joined across the open
↪ span𓆉·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𓉗𓆉𓃹𓀀𓂋𓏏𓉗𓆉⊕·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𓉗» category⟜⟜court»⁝·
·⁚lookup office⟜⟜spatial…absolute» enoch⟜⟜☍»⁝·

.word:·𐎍𐎗 𐎗 𐏒·.·˙word:·

·⁖origin…language:·Ugaritic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖.·˙font:·
·⁖pronunciation:·KHMMR·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Hall⁖ pavilion·⁖.·˙non…ostensive:·
·⁖ostensive:·Roof spreads broad above ordered space⁝⁝and pillars and walls hold a great open room
↪ beneath it𐎚·⁖.·˙ostensive:·
·⁖grimchain:·◎▽◎𐎘𐎝ᛉᚱᚺ𐎗𐎌◎𐎒𐎜·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐎍𐎊 category○–○𐎍court𐎊⁖·
·⁖lookup office○–○𐎍spatial…absolute𐎊 enoch○–○𐎍𐎌𐎊⁖·

.word:·거리·⁖.·˙word:·

·⁖origin…language:·Middle Korean·⁖.·˙origin…language:·
·⁖font:·Noto Sans KR·⁖.·˙font:·
·⁖pronunciation:·kuhlEE·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·intersection⁝⁝crossroads⁖ street·⁖.·˙non…ostensive:·
·⁖ostensive:·Several roads come together at one place⁝⁝and every direction opens away from the same
↪ shared crossing𐎍·⁖.·˙ostensive:·
·⁖grimchain:·ǂ𐏈𐎒𐎔𐎗𐎘𐎍𐎊𐎍𐎊Aϛ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐎍𐎓𐎊 category○–○𐎍court𐎊⁖·
·⁖lookup office○–○𐎍spatial…absolute𐎊 enoch○–○𐎍𐎌𐎊⁖·

.word:·ተ𐎗̄·⁖.·˙word:·

·⁖origin…language:·Tocharian B·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·KEEROHT·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·(botany) the bulb of the Nymphea lotus·⁖.·˙non…ostensive:·
·⁖ostensive:·Mud below and water above meet at one rounded enclosure⁖ folded leaves occupy the compact
↪ interior from which stem and root depart in opposite directions𐎓·⁖.·˙ostensive:·
·⁖grimchain:·𐏓𐎘ᛃ𐎍𐎗𐎔𐎌𐎒𐎘⬡𐎒𐎗·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐎍𐎚𐎊 category○–○𐎍court𐎊⁖·
·⁖lookup office○–○𐎍spatial…absolute𐎊 enoch○–○𐎍𐎌𐎊⁖·

.word:·𐎓𐎒𐎑𐎐·⁖.·˙word:·

·⁖origin…language:·Ugaritic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖.·˙font:·
·⁖pronunciation:·AHURPATU·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Cloud⁖ ceiling⁖ roof·⁖.·˙non…ostensive:·
·⁖ostensive:·Above chamber and hill alike⁝⁝an upper layer spans from edge to edge⁝⁝whether held in
↪ timber or suspended vapor⁝⁝leaving occupied space below and farther sky beyond𐎚·⁖.·˙ostensive:·

·⁚grimchain⁚·ᖷᑎ⁕ᚣ๐‹♀ᚗ๐♂ᛤ⨌⊖⌖·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᛒ category꩜꩜court᛬⁚·
·⁚lookup office꩜꩜spatial…absolute᛬ enoch꩜꩜ᛜ᛬⁚·

‹word⁚·ᒪᒋ⍑ᛯᖶ⁚·⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·⁚·˙font⁚·
·⁚pronunciation⁚·OHRKAHMNYEH·⁚·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Darkness⁚˙ blindness·⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·No lamp burns⁚⁚and wall and doorway lose their edges inside one black breadthᛒ·⁚·⁚·˙ostensive⁚·
↪
·⁚grimchain⁚·☽ᨐᚹᚼᔐᘏᔕᘐᛪᚼ·⁚·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᛈ᛬ category꩜꩜court᛬⁚·
·⁚lookup office꩜꩜spatial…absolute᛬ enoch꩜꩜ᛜ᛬⁚·

‹word⁚·ᒪᑦᖮᒍ⁚·⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·⁚·˙font⁚·
·⁚pronunciation⁚·OHSHRSHRAHLEH·⁚·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·⁚·˙cadence⁚·
·⁚non…ostensive⁚·North·⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Sunrise keeps one quarter and sunset keeps another⁚⁚and the coldward bearing remains fixed beyond bothᛈ·⁚·⁚·˙ostensive⁚·
↪
·⁚grimchain⁚·⁕ᚚᘐᘏᘐᨒᚐᜪᛒᗑᚼ·⁚·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᛎ᛬ category꩜꩜court᛬⁚·
·⁚lookup office꩜꩜spatial…absolute᛬ enoch꩜꩜ᛜ᛬⁚·

‹word⁚·ᚓ ᚻᚳ ᛠᛠ ᚴ ᛁᚱ⁚·⁚·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚·⁚·˙font⁚·
·⁚pronunciation⁚·pah+tee+KAH+rah·⁚·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Image⁚⁚likeness⁚⁚picture⁚⁚or statue·⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Carved and absent contours align in corresponding proportions⁚⁚with each edge occupying the same relative place in both formsᛎ·⁚·⁚·˙ostensive⁚·
↪
·⁚grimchain⁚·᛬ᛒᚳᚸ☆ᚣᛰᚮᚾ⍟ᚼ⍟ᚸ·⁚·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᛯ᛬ category꩜꩜court᛬⁚·
·⁚lookup office꩜꩜spatial…absolute᛬ enoch꩜꩜ᛜ᛬⁚·

‹word⁚·ᚽᒪᒐ⁚·⁚·˙word⁚·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·KAHRP·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·To descend⁚⁚come down⁚⁚go down·⁚·˙non…ostensive:·
·⁚ostensive:·High ledge stands above and valley lies below⁚⁚and the path between them keeps a downward bearing𑀘·⁚·˙ostensive:·
·⁚grimchain:·𑀯𑁂𑀭𑀢𑀀𑀀𑀫𑁆𑀦𑀫𑀶𑀁𑀫𑀁·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉𑀘𑀸 category꞉꞉court𑀸:·
·⁚lookup office꞉꞉spatial…absolute𑀸 enoch꞉꞉𑀇𑀸:·

·word:·𑀅𑀸:·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·JAH⟊uh·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Rotate·˙ to turn around an axis or central point·⁚·˙non…ostensive:·
·⁚ostensive:·Rim travels around and around⁚⁚and the hub keeps one still place at the center of all that turning𑀀·⁚·˙ostensive:·
·⁚grimchain:·𖿠𖿡𖿢𖿣𖿤𖿥𖿦𖿧·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉𑀦𑀸 category꞉꞉court𑀸:·
·⁚lookup office꞉꞉spatial…absolute𑀸 enoch꞉꞉𑀇𑀸:·

·word:·𑀪𑀭𑀁𑀅𑀢:·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·BRYOAST·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Chest or breast·⁚·˙non…ostensive:·
·⁚ostensive:·Between throat and belly⁚⁚ribs form one broad forward region where breath is enclosed and shield or garment meets the body𑀦·⁚·˙ostensive:·
·⁚grimchain:·𑁍𑀾𑁅𑀵𑀫𑀴𑀁𑀅𑀢𑀁𑀫𑀧·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉𑀦𑀸 category꞉꞉court𑀸:·
·⁚lookup office꞉꞉spatial…absolute𑀸 enoch꞉꞉𑀇𑀸:·

·word:·𑀢𑀁𑀧𑀮:·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·HNOO⟊dhr·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Stake⁚⁚rod⁚⁚pole⁚⁚stave⁚⁚knob⁚⁚or knot⁚˙ a solid projecting or rounded piece·⁚·˙non…ostensive:·
·⁚ostensive:·One narrow wood rises from earth⁚⁚and rope and tether find a fixed hold around the upright length𑀦·⁚·˙ostensive:·
·⁚grimchain:·𑀫𑀹𑀸𑁆𑀥𑀯𑀢𑀵𑀁𑀲𑁂𑀢𑁂𑀲·⁚·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐎟꩜꩜, category꩜꩜court꩜꩜:·
·⁚lookup office꩜꩜spatial…absolute꩜꩜, enoch꩜꩜𐎟꩜꩜:·

·word:·𐎍𐎓𐎍 𐎔𐎗·⁚·˙word:·

·⁚origin…language:·Ugaritic·⁚·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚·˙font:·
·⁚pronunciation:·MHMRT·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Gullet·⁚·˙non…ostensive:·
·⁚ostensive:·Mouth lies above and belly below::and one dark passage carries food and drink between
↪ them𐎟·⁚·˙ostensive:·
·⁚grimchain:·𐎀𐎅𐎆◎𐎗𐎈𐎉𐎋𐎌𐎍𐎎·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐎚꩜꩜, category꩜꩜court꩜꩜:·
·⁚lookup office꩜꩜spatial…absolute꩜꩜, enoch꩜꩜𐎟꩜꩜:·

·word:·𐎂·⁚·˙word:·

·⁚origin…language:·Sumerian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·BAH·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Share⁚˙ a portion assigned or distributed from a whole·⁚·˙non…ostensive:·
·⁚ostensive:·Cut loaf rests in several pieces::and each keeps one bounded part of what had been
↪ whole𐎚·⁚·˙ostensive:·
·⁚grimchain:·𐎄𐎅𐎆𐎗◎𐎘𐎙◎𐎚𐎛𐎜𐎝·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐎗꩜꩜, category꩜꩜court꩜꩜:·
·⁚lookup office꩜꩜spatial…absolute꩜꩜, enoch꩜꩜𐎟꩜꩜:·

·word:·𐎐·⁚·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·KOHM·⁚·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Eyelid⁚˙ the movable fold that covers and protects the eye·⁚·˙non…ostensive:·
·⁚ostensive:·Thin flesh falls over the seeing opening::and light waits outside the small fold that
↪ closes across it𐎗·⁚·˙ostensive:·
·⁚grimchain:·𐎊𐎋𐎌𐎍𐎎𐎏𐎐𐎑𐎒𐎓𐎔·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐎖꩜꩜, category꩜꩜court꩜꩜:·
·⁚lookup office꩜꩜spatial…absolute꩜꩜, enoch꩜꩜𐎟꩜꩜:·

·word:·𐎘𐎙𐎚·⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·

·⁚pronunciation:·mah+lah+TSAY·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Knife::blade::or razor·⁚.·˙non…ostensive:·
·⁚ostensive:·One keen edge runs along the worked metal::and whatever meets that line finds two sides
↪ where one had stood𐋡·⁚.·˙ostensive:·
·⁚grimchain:·𐎗𐎂𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉𐎐 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐎍꞉·

·word:·飛·:.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·GHAIH·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Sound or noise·˙ audible vibration·⁚.·˙non…ostensive:·
·⁚ostensive:·One vibrating field occupies the court from wall to wall·˙ air at separated points carries
↪ the same trembling without a solid body between them𐎐·⁚.·˙ostensive:·
·⁚grimchain:·𐎗𐎂𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉𐎓 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐎍꞉·

·word:·ھەدھەت·:.·˙word:·

·⁚origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁚pronunciation:·EUNG·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·color::form::shape::appearance·˙ face color::skin tone·˙ (rare) front::east·⁚.·˙non…ostensive:·
·⁚ostensive:·Dyed cloth carries one face to the eye::and another hue beside it keeps a different place
↪ upon the same woven breadth𐎓·⁚.·˙ostensive:·
·⁚grimchain:·𐎐𐎔𐎒𐎗𐎂𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꞉꞉𐎒 category꞉꞉court꞉·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𐎍꞉·

·word:·𓀀𓀁𓀂𓀃𓀄·:.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·kah+SHAH+doom·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·to arrive at (a place)::reach·˙ to achieve::accomplish::get::complete·˙ to
↪ conquer::defeat·⁚.·˙non…ostensive:·
·⁚ostensive:·Road::gate::and far wall converge upon one terminal locus·˙ every route through the approach
↪ has that shared point as its furthest reachable place𐎒·⁚.·˙ostensive:·
·⁚grimchain:·𐎊𐎎𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓𐎐𐎍𐎓·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol◦–◦𐌉꞉ category◦–◦court꞉⁚·
·⁚lookup office◦–◦spatial…absolute꞉ enoch◦–◦𐤏꞉⁚·

⁖word⁚·⊗V⁖⁚·˙word⁚·

·⁚origin…language⁚·Etruscan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Italic·⁚·˙font⁚·
·⁚pronunciation⁚·THOO·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·One⁚· the number one·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·One vessel stands alone upon the floor⁚⁚and no second body shares its bounded
↪ place𐌉⁚·˙ostensive⁚·
·⁚grimchain⁚·𐌠𐌾𐌾▽ᚱ᛬ᛟᛘᛘᛜᛞᛦᛘᚴᚣᛊ⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐌇꞉ category◦–◦court꞉⁚·
·⁚lookup office◦–◦spatial…absolute꞉ enoch◦–◦𐤏꞉⁚·

⁖word⁚·𒀭𒁹⁖⁚·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·RAKH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Strike⁚· to hit with a directed blow·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Hammer and stone meet at one hard point⁚⁚and all the force of the blow gathers into that
↪ narrow place of contact𐌇⁚·˙ostensive⁚·
·⁚grimchain⁚·𐌏𐤎ℳ▽✳𐤀𐤓𐤎𐤕ᛉ▽ᛊ𐤏⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐌄꞉ category◦–◦court꞉⁚·
·⁚lookup office◦–◦spatial…absolute꞉ enoch◦–◦𐤏꞉⁚·

⁖word⁚·ᛗIᚾ̄ᚾ̄ᛟ⁖⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·SHEHRKW·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Cord⁚⁚string·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Two distant pegs hold one slender length between them⁚⁚and all the open space is crossed
↪ without breaking the line𐌄⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤔❄✳Ⅱᚬᛯ𐌲ᛉ𐤇ᚻᛟ⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐌉꞉ category◦–◦court꞉⁚·
·⁚lookup office◦–◦spatial…absolute꞉ enoch◦–◦𐤏꞉⁚·

⁖word⁚·ᚻI𐌴⁖·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·KEHROO·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·

·⁚non…ostensive⁖·(music) drum·⁚.·˙non…ostensive⁖·
·⁚ostensive⁖·A round membrane seals one hollow chamber⁘ air fills the bounded volume beneath the
↪ stretched circular face⁝⁝while open space begins beyond the rim⁞·⁚.·˙ostensive⁖·
·⁚grimchain⁖·ᵽ𝝨ᵛᛉ⬡ᦗ𝝖ᗢ𝝘ᗄᔆ긔𑁍·⁚.·˙grimchain⁖·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol○–○ᑫ꞉ category○–○court꞉⁖·
·⁚lookup office○–○ᑫspatial…absolute꞉ enoch○–○ᑫ꞉⁖·

₍word⁖·ᕊᐅᔑᒪᐅᕱ₎·⁚.·˙word⁖·

·⁚origin…language⁖·Etruscan·⁚.·˙origin…language⁖·
·⁚font⁖·Noto Sans Old Italic·⁚.·˙font⁖·
·⁚pronunciation⁖·KAHN⟊lah·⁚.·˙pronunciation⁖·
·⁚cadence⁖·Spatial Absolute·⁚.·˙cadence⁖·
·⁚non…ostensive⁖·Reed flute⁘ a wind instrument made from a hollow reed·⁚.·˙non…ostensive⁖·
·⁚ostensive⁖·Empty reed keeps breath inside one narrow chamber⁝⁝and small openings let the hidden air
↪ escape into songᑫ·⁚.·˙ostensive⁖·
·⁚grimchain⁖·Ⅱᗞ⇟◎ᔑ◎𑁍⩡ᒪᛉᗤᛉᔑ·⁚.·˙grimchain⁖·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol○–○ᛕ꞉ category○–○court꞉⁖·
·⁚lookup office○–○ᑫspatial…absolute꞉ enoch○–○ᑫ꞉⁖·

₍word⁖·ᛏ□ᕕ𐑒₎·⁚.·˙word⁖·

·⁚origin…language⁖·Tocharian A·⁚.·˙origin…language⁖·
·⁚font⁖·Noto Sans Brahmi·⁚.·˙font⁖·
·⁚pronunciation⁖·KUHTK·⁚.·˙pronunciation⁖·
·⁚cadence⁖·Spatial Absolute·⁚.·˙cadence⁖·
·⁚non…ostensive⁖·To cross⁝⁝traverse⁝⁝pass through⁘ to proceed⁝⁝pass (of time)⁘ to transgress⁝⁝commit a
↪ sin⁘ to lower⁝⁝set down·⁚.·˙non…ostensive⁖·
·⁚ostensive⁖·One bank and the other remain apart⁝⁝and the passage between them makes both sides belong
↪ to one courseᛕ·⁚.·˙ostensive⁖·
·⁚grimchain⁖·ᛏᕼ𑁏ᘐᘏ𑁍ᔑᛋᗷᓄᘐᐯᛈ·⁚.·˙grimchain⁖·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol○–○ᛘ꞉ category○–○court꞉⁖·
·⁚lookup office○–○ᑫspatial…absolute꞉ enoch○–○ᑫ꞉⁖·

₍word⁖·ᖘᘉᖮ₎·⁚.·˙word⁖·

·⁚origin…language⁖·Ge'ez·⁚.·˙origin…language⁖·
·⁚font⁖·Noto Sans Ethiopic·⁚.·˙font⁖·
·⁚pronunciation⁖·TUHNF·⁚.·˙pronunciation⁖·
·⁚cadence⁖·Spatial Absolute·⁚.·˙cadence⁖·
·⁚non…ostensive⁖·Cornice⁝⁝crossbeam⁝⁝projection⁝⁝or structural joint·⁚.·˙non…ostensive⁖·
·⁚ostensive⁖·Post keeps one end and another post keeps the other⁝⁝and timber spans the open space
↪ beneath the roofᛘ·⁚.·˙ostensive⁖·
·⁚grimchain⁖·ᛉᙢᑭ⏃۞☾❄ᚺᓀᔐᗷᛞ·⁚.·˙grimchain⁖·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol○–○ᕽ꞉ category○–○court꞉⁖·
·⁚lookup office○–○ᑫspatial…absolute꞉ enoch○–○ᑫ꞉⁖·

:word:·ᛏᚾᚾᚹᛁ:.·˙word:·

·⁖origin…language:·Old Norse·⁖.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖.·˙font:·
·⁖pronunciation:·TAHNG-ee·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Cape⁖· projecting point of land·⁖.·˙non…ostensive:·
·⁖ostensive:·Land narrows into the sea⁘and water gathers along three sides while earth keeps the one
↪ joining side behind↰·⁖.·˙ostensive:·
·⁖grimchain:·ᚷᛢ᛬᛫ᛪᛇᛉᛘᚱᛦᛯᛦᚴ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○-○ₑ↰› category○-○ₑcourt›:·
·⁖lookup office○-○ₑspatial…absolute› enoch○-○ₑ♌›:·

:word:·ᒪᑎᑕ:.·˙word:·

·⁖origin…language:·Tocharian B·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·YEHRPEH·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Orb·⁖.·˙non…ostensive:·
·⁖ostensive:·Every edge turns away into curve⁘and no corner breaks the rounded body held around the
↪ center↰·⁖.·˙ostensive:·
·⁖grimchain:·∮♌ᚿᛉ'ᓄᛃ⬡ᗤ✳ᛘᘜᛇᚴ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○-○ₑ✳› category○-○ₑcourt›:·
·⁖lookup office○-○ₑspatial…absolute› enoch○-○ₑ♌›:·

:word:·ᓚᖶ□ᛪ̄:.·˙word:·

·⁖origin…language:·Tocharian A·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·YOHKUHM·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Door·⁖.·˙non…ostensive:·
·⁖ostensive:·Wall breaks at one upright passage⁘and chamber and road touch through the space kept open
↪ there✳·⁖.·˙ostensive:·
·⁖grimchain:·ΘIᗢᛊᛇ⬡ᛋᗅ⬡ᖷᖷ⌬·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○-○ₑ✳› category○-○ₑcourt›:·
·⁖lookup office○-○ₑspatial…absolute› enoch○-○ₑ♌›:·

:word:·𗍫:.·˙word:·

·⁖origin…language:·Tangut·⁖.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖.·˙font:·
·⁖pronunciation:·BEEH·⁖.·˙pronunciation:·
·⁖cadence:·Spatial Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Drill or pierce through⁖· to make a hole by penetrating a material·⁖.·˙non…ostensive:·
·⁖ostensive:·Solid face keeps a narrow tunnel through its thickness⁘and daylight joins one side of the
↪ material to the other✳·⁖.·˙ostensive:·

·⁚grimchain⁚·ǂ𐤐ᛯ𐤕ᗅIↄᏂ𐡆ᛝ𐤔☿·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀ᛯ꜀ category꜀꜀court꜀⁚·
·⁚lookup office꜀꜀spatial…absolute꜀ enoch꜀꜀ᘐ꜀⁚·

·word⁚·ᎴᏁᏓ⁖⁚.·˙word⁚·

·⁚origin…language⁚·Phoenician·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Phoenician·⁚.·˙font⁚·
·⁚pronunciation⁚·rah+GAHZ·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To disturb⁚˙ to unsettle or agitate·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One bounded surface contains concentric rings around a displaced center⁚˙ no point on the
↪ former flat plane remains level with its neighbor⁎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚔᛋ⊬ᗅΘ∀ƻ♀𐤒ᔑᛃΦ∮ϒℛ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀ᛯ꜀ category꜀꜀court꜀⁚·
·⁚lookup office꜀꜀spatial…absolute꜀ enoch꜀꜀ᘐ꜀⁚·

·word⁚·𐤓𐤇𐤊𐤍𐤓𐤇⁖⁚.·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚.·˙font⁚·
·⁚pronunciation⁚·dah+YOW·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Land or country⁚˙ a territorial community·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Boundary stones⁚⁚roads⁚⁚fields⁚⁚and settlements keep one reach together beneath the same
↪ ruling sky⁎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚿᛇᔕᛯ𐡆ᛇƹ⊙ᘐᛈ☆·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀ᛉ꜀ category꜀꜀court꜀⁚·
·⁚lookup office꜀꜀spatial…absolute꜀ enoch꜀꜀ᘐ꜀⁚·

·word⁚·𐤓⁖⁚.·˙word⁚·

·⁚origin…language⁚·Egyptian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Egyptian Hieroglyphs·⁚.·˙font⁚·
·⁚pronunciation⁚·PET·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Sky⁚⁚heaven·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Roof⁚⁚palm⁚⁚and mountain all end below⁚⁚and blue height rises beyond them without wall or
↪ ceiling⁎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐤒ᛕᛉᘐᛕᔑᛃᛋᚿᛞ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀ᛊ꜀ category꜀꜀court꜀⁚·
·⁚lookup office꜀꜀spatial…absolute꜀ enoch꜀꜀ᘐ꜀⁚·

·word⁚·ᓰᒐᒃᒋ⁖⁚.·˙word⁚·

·⁚origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·SWAHNYCHOH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Ray⁖beam⁖sunbeam·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Dust glows inside one narrow shaft of light⁖and bright source and touched surface remain joined by that shining path0·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᦵ𐫲ᡕ𐫲𐫲ᡕ⊛ᐯ𐫲ᛯ⌬ᑫᛯ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉ᛯ꞉ category꞉ᛯ꞉court꞉·
·⁚lookup office꞉ᛯ꞉spatial…absolute꞉ enoch꞉ᛯ꞉ᛜ꞉·

꞉word꞉·🢤꞉·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚·˙font꞉·
·⁚pronunciation꞉·AHP+toom·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·window⁖window opening꞉˙ opening of the ear·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Wall keeps one square of daylight⁖and the side of the head keeps one small gate for sound·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᑭᑭ🢤ᚷᓄᐯᓪᓂ✡ᓰ𓉔ᒣ𓀠·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉ᛯ꞉ǂ꞉ category꞉ᛯ꞉court꞉·
·⁚lookup office꞉ᛯ꞉spatial…absolute꞉ enoch꞉ᛯ꞉ᛜ꞉·

꞉word꞉·甫꞉·˙word꞉·

·⁚origin…language꞉·Tangut·⁚·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚·˙font꞉·
·⁚pronunciation꞉·MEEM·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Gate or door꞉˙ an opening that allows or restricts passage·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Enclosing wall yields one passage⁖and timber across that opening can deny the way or leave it freeǂ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᚷᗡᔦᔑᔐᙏ✪ᗰᗰᐤᐁᓮ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉ᛯ꞉ᡕ꞉ category꞉ᛯ꞉court꞉·
·⁚lookup office꞉ᛯ꞉spatial…absolute꞉ enoch꞉ᛯ꞉ᛜ꞉·

꞉word꞉·ᒐᏁᖓᒪᒥ꞉·˙word꞉·

·⁚origin…language꞉·Tocharian A·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·PAHTSUH+K·⁚·˙pronunciation꞉·
·⁚cadence꞉·Spatial Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Window·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Wall keeps the chamber closed except for one framed opening⁖and daylight and distant ground enter sight through that single placeᡕ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᦵ𐫲ᡕ𐫲ᘔ✡ᘓᑎǂ𓇼ᓘᔑ✱ᚷǂ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙

·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𝛾꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𝌆꞉⁚·

꞉word⁚·双꞉⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·CHAH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Vehicle:˙ a conveyance used to carry people or goods·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Between wheelbase and upper frame::one bounded carrying volume holds a load apart from the
↪ road beneath𝛾·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛋ⌬⍜ϒ𝈓Xሕϒʘᛯ ⊛ ↗0·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𝛸꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𝌆꞉⁚·

꞉word⁚·ሠርቄ꞉⁚·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚·˙font⁚·
·⁚pronunciation⁚·SAHR⟊way·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Beam::rafter::timber::trunk::mast::or structural stem·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Post holds one end and another support holds the other::and roof weight rests upon the
↪ timber stretched between them𝛸·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᑎƵ❂𝄞z⪢ᕽᔭ∀ϭᚹ๖ᘐᚹ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𝛶꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𝌆꞉⁚·

꞉word⁚·ᑉ४̄꞉⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·SHREHM·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Axle·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Wheel travels around and around::and the straight shaft through its center keeps one place
↪ beneath all the turning𝛶·⁚·˙ostensive⁚·
·⁚grimchain⁚·I∀⁘ᘓ⊄∫ᕫ𝈦Λᓀ✡❂ɔɣ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𝛢꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉spatial…absolute꞉ enoch꞉꞉𝌆꞉⁚·

꞉word⁚·ᗰካን꞉⁚·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚·˙font⁚·
·⁚pronunciation⁚·mah⟊KAHN·⁚·˙pronunciation⁚·

·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Place⁚⁚space⁚⁚seat⁚⁚or inhabited location·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Stone stands here and tree stands there⁚⁚and each keeps a distinct where within the
↪ greater reach around them«·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐤟⊕𐦚𐦜⊕Λ𐤦𐤍𐤏𐤕𐤉𐤌𐤔·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐤇» category⟜⟜court»⁚·
·⁚lookup office⟜⟜spatial…absolute» enoch⟜⟜𐤏»⁚·

·⁚word⁚·𐡗⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·EE+doom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·arm⁚˙ side⁚⁚edge⁚⁚border⁚˙ handle of an instrument⁚˙ (measurement) span⁚⁚fathom⁚˙ bracelet⁚˙
↪ wing⁚˙ strength·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Shoulder opens into one long reach⁚⁚and hand waits at the far end where that span meets
↪ the world𐤇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐤄𐤓𐤗𐤆𐤊𐤈𐤀⊙⊕𐤈𐤕𐤏𐤃·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐤃» category⟜⟜court»⁚·
·⁚lookup office⟜⟜spatial…absolute» enoch⟜⟜𐤏»⁚·

·⁚word⁚·𐡛𐡚𐡛𐡙⁚.·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚.·˙font⁚·
·⁚pronunciation⁚·eye+yah+DAH+nah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Sanctuary or place of worship·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Road and common ground remain beyond the boundary⁚⁚and prayer⁚⁚offering⁚⁚and sacred fire are
↪ gathered within the set+apart enclosure𐤃·⁚.·˙ostensive⁚·
·⁚grimchain⁚·✡𐤋𐤔𐤏𐤉𐤍𐤁𐤌𐤕𐤓𐤈𐤄𐤎𐤍·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐤋» category⟜⟜court»⁚·
·⁚lookup office⟜⟜spatial…absolute» enoch⟜⟜𐤏»⁚·

·⁚word⁚·𐤗𐤞⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·AHKEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·End·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·All length gathers toward one furthest point⁚⁚and beyond that edge the course has no more
↪ ground of its own𐤋·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐤀⊙♅♄♋𐤃𐤅𐤌𐤔𐤙1ℂΛ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙

·⁚glyph symbol꜀꜀𓏏꜁ ⁖ category꜀꜀court꜁⁚·
·⁚lookup office꜀꜀spatial…absolute꜁ ⁖ enoch꜀꜀☍꜁⁚·

·word⁚·𓀁·⁚·˙word⁚·

·⁚origin…language⁚·Egyptian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Egyptian Hieroglyphs·⁚·˙font⁚·
·⁚pronunciation⁚·HAR·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Face·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Brow⁖cheek⁖jaw⁖and hairline keep one forward field⁖and every visible expression lives upon that open front𓏏·⁚·˙ostensive⁚·
↪
·⁚grimchain⁚·∀ɣʎ▽⓿𓀡≯✡ᵻᵻ∩ℂꟻʎ⧋·⁚·˙grimchain⁚·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol꜀꜀𓏏꜁ ⁖ category꜀꜀court꜁⁚·
·⁚lookup office꜀꜀spatial…absolute꜁ ⁖ enoch꜀꜀☍꜁⁚·

·word⁚·𓀠◇·⁚·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·DEEM·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Post⁖˙ an upright support⁖pillar⁖pole⁖or binding point·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·One upright timber rises from a fixed base⁖and rope⁖wall⁖or roof finds a place to bind against its standing strength𓏏·⁚·˙ostensive⁚·
↪
·⁚grimchain⁚·Ɛ✩ƶɥ◎ᘐᛋᚼ᛬ᘏ⊙·⁚·˙grimchain⁚·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol꜀꜀▽꜁ ⁖ category꜀꜀court꜁⁚·
·⁚lookup office꜀꜀spatial…absolute꜁ ⁖ enoch꜀꜀☍꜁⁚·

·word⁚·𓁹·⁚·˙word⁚·

·⁚origin…language⁚·Egyptian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Egyptian Hieroglyphs·⁚·˙font⁚·
·⁚pronunciation⁚·AH⟊khet·⁚·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Horizon⁖˙ the place of sunrise or sunset where the sun meets the mountainous edge of the sky·⁚·˙non…ostensive⁚·
↪
·⁚ostensive⁚·Earth and sky meet upon one far line⁖and sunrise and sunset both belong to that distant edge▽·⁚·˙ostensive⁚·
↪
·⁚grimchain⁚·⊀∀ɤƀᵻ⍈▽∀Aɣ𓀡ᚼ✪·⁚·˙grimchain⁚·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol꜀꜀ƶ꜁ ⁖ category꜀꜀court꜁⁚·
·⁚lookup office꜀꜀spatial…absolute꜁ ⁖ enoch꜀꜀☍꜁⁚·

·word⁚·ᚱᚢᛏ·⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·ROHT·⁚·˙pronunciation⁚·

·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Root·⁚.·˙non…ostensive:·
·⁚ostensive:·Stem stands in daylight⁚⁚and hidden branching reaches downward through soil where the
↪ whole plant keeps its holdz·⁚.·˙ostensive:·
·⁚grimchain:·Ɏ◡ɬ൩ᚹᚻ❄∀ᕐ⬡Iᵹ♀ᚦ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○ᵋᏀ» category○–○ᵋcourt»⁚·
·⁚lookup office○–○ᵋspatial…absolute» enoch○–○ᵋ☍»⁚·

·word⁚·ᔆᚱᔆᚱ⁖.·˙word:·

·⁚origin…language:·Etruscan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Italic·⁚.·˙font:·
·⁚pronunciation:·SEH⟊seh·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Seated or sitting·⁚.·˙non…ostensive:·
·⁚ostensive:·Weight gathers low above the floor⁚⁚and bent knees and hips keep the body within one
↪ settled resting placeɋ·⁚.·˙ostensive:·
·⁚grimchain:·ᛋ𐋏ⓄᔑɬƳᏩ♀✻ᚿ∀ɋᚸ⋿·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○ᵋᏏ» category○–○ᵋcourt»⁚·
·⁚lookup office○–○ᵋspatial…absolute» enoch○–○ᵋ☍»⁚·

·word⁚·𑄢⁖.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·ánh·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Inside⁚· interior·⁚.·˙non…ostensive:·
·⁚ostensive:·Boundary closes around the hidden chamber⁚⁚and what is kept there does not belong to
↪ road⁚⁚weather⁚⁚or open air beyondᏏ·⁚.·˙ostensive:·
·⁚grimchain:·zᵻᨐᏀᏫ☆ᛯᚻᨐ᛻ᛒ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○ᵋᏏ» category○–○ᵋcourt»⁚·
·⁚lookup office○–○ᵋspatial…absolute» enoch○–○ᵋ☍»⁚·

·word⁚·ᒍ十⁖.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·LEHKEE·⁚.·˙pronunciation:·
·⁚cadence:·Spatial Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Bed⁚⁚resting place·⁚.·˙non…ostensive:·
·⁚ostensive:·Floor stays below⁚⁚and one softened horizontal place keeps rest slightly raised from the
↪ common groundᏏ·⁚.·˙ostensive:·
·⁚grimchain:·ᗡᛕ𐋡ᛏ∇ᗺᚿᚧᚴᛃᛋᚠ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○ᵋᏳ» category○–○ᵋcourt»⁚·
·⁚lookup office○–○ᵋspatial…absolute» enoch○–○ᵋ☍»⁚·

⁖word⁚·ዓለም⁖˙·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚˙·˙font⁚·
·⁚pronunciation⁚·AH⟊lam·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·World or universe⁖ the total inhabited or created order·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·Dwelling⁚⁚road⁚⁚mountain⁚⁚water⁚⁚field⁚⁚grave⁚⁚and sky all lie within one created reach⁚⁚and no
↪ inhabited place stands outside itφ·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·⊙ዛሐAለ☽∂ϱՈX♂⏀ɤφ·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁖˙
·⁚˙·˙glyphs⁖˙
·⁚glyph symbol◦–◦𝓀 category◦–◦axiomyr»⁚·
·⁚lookup office◦–◦spatial…absolute» enoch◦–◦ₒₒ»⁚·

⁖word⁚·ጰ+ሽDϡ⁖˙·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚˙·˙font⁚·
·⁚pronunciation⁚·AHKAHSHAHDHAHTOO·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Spatial Absolute·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·Element of the sky or space.·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·Hollow stone shelters a pool of rain⁚⁚Roots sleep below the living moss⁖a bird crosses
↪ open air above the crown of an oak⁚⁚Deep cave holds darkness within walls of earth⏀·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·ᔑ৫▫⏀⌬ᕿ⁖ₒዛՍ⏀ᕮ⊥·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁖˙
·⁚˙·˙glyphs⁖˙
·⁚˙·˙glyphs⁖˙
·⁚˙·˙sydonic…magicae…lexicon·⁚

⁖⁘⁙⁖⁘⁙⁖⁘⁙

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·ee+NAN+nah·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·now⁝⁝just now⁝⁝nowadays⁝⁝·⁚.·˙non…ostensive:·
·⁚ostensive:·A drop has left the eave and has not yet struck stone♀·⁚.·˙ostensive:·
·⁚grimchain:·⬡𓅓⬡ₒₕʌᗡㄱㄩㄚ'𓅓·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦𓂋𓏺 category◦–◦emission𓏺:·
·⁚lookup office◦–◦temporal…absolute𓏺 enoch◦–◦Θ𓏺:·

·word:·ᛋIᑭᚴᚾ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·SEE+thahn·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·since⁝⁝after⁝⁝·⁚.·˙non…ostensive:·
·⁚ostensive:·Cold hearthstones still loosen smoke from a fire already gone𓂋·⁚.·˙ostensive:·
·⁚grimchain:·ᚺ◎ᛁᛉℂᗰᑭᛊᛂ𓂀𓏺ᛒ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦ℏ𓏺 category◦–◦emission𓏺:·
·⁚lookup office◦–◦temporal…absolute𓏺 enoch◦–◦Θ𓏺:·

·word:·ᑌ⊥̄·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·POOK·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To expect⁚˙ to trust⁝⁝·⁚.·˙non…ostensive:·
·⁚ostensive:·Bowstring stays drawn⁝⁝ The arrow has not flownℏ·⁚.·˙ostensive:·
·⁚grimchain:·𑍝𑁍𑀸𑁆𑀫𑀺𑀢𑀷𑀓𑀫·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦⛢𓏺 category◦–◦emission𓏺:·
·⁚lookup office◦–◦temporal…absolute𓏺 enoch◦–◦Θ𓏺:·

·word:·›𓊽·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·eh+PEH+shoom·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·to act⁝⁝behave⁝⁝proceed⁝⁝be(come) active⁚˙ (transitive) to do⁝⁝make⁝⁝build⁝⁝manufacture⁚˙
↪ (transitive) to do⁝⁝carry out⁝⁝excercise (a function)⁝⁝used mainly in idiomatic
↪ phrases⁝⁝·⁚.·˙non…ostensive:·
·⁚ostensive:·Hammer rises and falls until the ringing itself becomes the measure of the
↪ work⛢·⁚.·˙ostensive:·
·⁚grimchain:·ᚲΘ◎ᚼᛯᗄᚺ✡1Θ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙

·⁚glyph symbol◦–◦𐑉𐑌 category◦–◦emission𐑌⁚·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌⁚·

·:word:·𒁹𒀸𒂍·:.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·NEH+meh+koom·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·experience::knowledge::wisdom::sagacity::skill::·⁚.·˙non…ostensive:·
·⁚ostensive:·Hands have rounded the tablet edges through years of reading::while the old signs grow
↪ easier to know𐑉·⁚.·˙ostensive:·
·⁚grimchain:·𐌲𐋏𐑈𓆉✡❁𒁹ᚱ𐑅⊕ᛯ▽𐑍·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑓𐑌 category◦–◦emission𐑌⁚·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌⁚·

·:word:·ऊर्ध्व·:.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·oordh+vuh·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·rising or tending
↪ upwards::raised::elevated::erected::erect::upright::high::above::·⁚.·˙non…ostensive:·
·⁚ostensive:·Smoke climbs without settling::each new coil higher than the last𐑓·⁚.·˙ostensive:·
·⁚grimchain:·ᚱℏ𐑉ᛉᛯ𐑉⊙⊛𐑑𐑓▽𐑘·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦☽𐑌 category◦–◦emission𐑌⁚·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌⁚·

·:word:·SIN∈IΓS·:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·SINIXS·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·(of people) old::·⁚.·˙non…ostensive:·
·⁚ostensive:·Dark hair gives way strand by strand to silver::and weather gathers slowly in the
↪ skin☽·⁚.·˙ostensive:·
·⁚grimchain:·ᚠ☽ᚷ♂ᚾ1ϛ𐌲'⌬𒁹𐑋ᚲ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦⊙𐑌 category◦–◦emission𐑌⁚·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌⁚·

·:word:·ᛊᚣᛃS·:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·ALDS·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·

·⁚non…ostensive⁚·Generation⁝⁝age⁚· lifetime⁝⁝·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Between first breath and last⁝⁝waking and sleep fill the measure allotted to one
↪ life⊙·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐤟ᛝᚹᛦᛋᚠᚠ𐤆𐤔𐤟·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐤂𐤃 category○–○𐤂emission𐤃⁚·
·⁚lookup office○–○𐤂temporal…absolute𐤃 enoch○–○𐤂Θ𐤃⁚·

·word⁚·𐤃𐤄·⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·PAHG+room·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·body·⁚ trunk⁝⁝torso·⁚ self⁝⁝person (often used as a reflexive pronoun)·⁚
↪ corpse⁝⁝carcass⁝⁝·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Warm flesh breathes·⁚ later the same form cools and answers nothing℃·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐤋❄𐤉♀𐤍𐤂ᛇᛈᚢ𐤃𐤄⬡𐤉·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐤂♈𐤃 category○–○𐤂parliament𐤃⁚·
·⁚lookup office○–○𐤂temporal…absolute𐤃 enoch○–○𐤂Θ𐤃⁚·

·word⁚·𐤊𐤅∑·⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·Nahrûm·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·stela (for publicising inscriptions⁝⁝decrees⁝⁝etc⁝⁝)·⁚ (Middle Babylonian⁝⁝Neo+Babylonian)
↪ boundary stone⁝⁝·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Rain has eaten the stone face thin⁝⁝yet the cuts made long ago still hold♈·⁚.·˙ostensive⁚·
·⁚grimchain⁚·❂ƁƁ⬡⊕∀𐤌ᛁ𐤉ϒ𐤆❋ᚾᚦ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐤂♉𐤃 category○–○𐤂parliament𐤃⁚·
·⁚lookup office○–○𐤂temporal…absolute𐤃 enoch○–○𐤂Θ𐤃⁚·

·word⁚·ψᚣᚾᛃЄ·⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·THANDE·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·While⁝⁝as long as·⁚ because⁝⁝since⁝⁝·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Rain keeps speaking on the roof for as long as the falling continues♉·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛦ𐤍❋𐤇♀▫ᘐᛏᚿᛏ𐤂ᚴ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐤂♊𐤃 category○–○𐤂parliament𐤃⁚·
·⁚lookup office○–○𐤂temporal…absolute𐤃 enoch○–○𐤂Θ𐤃⁚·

·word⁚·𐤃𐤂𐤄𐤟𐤔·⁚.·˙word⁚·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·bah+SHOOM·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·to exist::to be there::present::available·⁚ to happen::·⁚.·˙non…ostensive:·
·⁚ostensive:·Flame stays on the wick from first catching until the oil is spent𓉐·⁚.·˙ostensive:·
·⁚grimchain:·ϟ𓆙𓋴𓀠𓏏⊕ΨI𓊝·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𓂋𓊃 category◦–◦parliament𓊃:·
·⁚lookup office◦–◦temporal…absolute𓊃 enoch◦–◦Θ𓊃:·

·word:·𓆏𓏲·⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·LAHOOTEH·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Moment::point in time·⁚ opportunity:::⁚.·˙non…ostensive:·
·⁚ostensive:·Latch rises::and for this brief opening passage is possible𓂋·⁚.·˙ostensive:·
·⁚grimchain:·ᚏ𓃰𓆉𓂋𓅓𓊪ᚲᚺᛊ𓃭𓆑𓆈·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𓌳𓊃 category◦–◦parliament𓊃:·
·⁚lookup office◦–◦temporal…absolute𓊃 enoch◦–◦Θ𓊃:·

·word:·आयु·⁚.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·AA+yoo·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Life span::lifetime::or duration of life::·⁚.·˙non…ostensive:·
·⁚ostensive:·Breath after breath carries the body from its first living cry toward its
↪ last𓌳·⁚.·˙ostensive:·
·⁚grimchain:·𓆗𓃰ᚲ𓆈𓊪◎♀ᛝ⊕𓌳𓆏·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𓍯𓊃 category◦–◦parliament𓊃:·
·⁚lookup office◦–◦temporal…absolute𓊃 enoch◦–◦Θ𓊃:·

·word:·𓂋𓅓𓃗𓇋·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·SHOOL+moom·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·completeness::intact state·⁚ well+being::health·⁚ safety::peace·⁚ good wishes::greeting·⁚
↪ (extispicy) a crease on the sheep's liver::·⁚.·˙non…ostensive:·
·⁚ostensive:·Season after season::the same uncracked jar keeps what is poured into it𓍯·⁚.·˙ostensive:·
·⁚grimchain:·𓉐ᛦ☿𓃬ᔕ𓂸ᛜᔕ𓃰☽ᨓ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𓋴𓊃 category◦–◦parliament𓊃:·
·⁚lookup office◦–◦temporal…absolute𓊃 enoch◦–◦Θ𓊃:·

⁖word:·गाथा⁖.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·gah+thah·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·verse⁚⁚stanza (especially one which is neither ऋच् (ṛc)⁚⁚nor सामन् (sāman)⁚⁚nor यजुस्
↪ (yajus)⁚⁚a verse not belonging to the Vedas⁚⁚but to the epic poetry of legends or आख्यान
↪ (ākhyāna)s⁚⁚such as the Śunaḥśepa+ākhyāna or the Suparṇādhyāya)⁖· the metrical part of a sūtra⁖·
↪ name of the āryā metre⁖· any metre not enumerated in the regular treatises on prosody (compare
↪ ऋग्गाथा (ṛg+gāthā)⁚⁚ऋजुगाथ (ṛju+gātha)⁚⁚यज्ञगाथा (yajña+gāthā))⁚⁚⁚·˙non…ostensive:·
·⁚ostensive:·Measured feet carry the verse forward⁚⁚each return yielding to the next⏢·⁚.·˙ostensive:·
·⁚grimchain:·𐋡▽𐋡𐋺𐋵𐋠𐋡⊖𐋢𐋤·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐋏꜀ category○–○parliament꜀⁚·
·⁚lookup office○–○temporal…absolute꜀ enoch○–○⊖꜀⁚·

⁖word:·𐨅𐨪̄⁖.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·AHR·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To cease⁚⁚come to an end⁖· to renounce⁚⁚forsake⁚⁚abandon⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Ringing thins to one last tremor and is gone𐋏·⁚.·˙ostensive:·
·⁚grimchain:·A⏣Ո𐋡𐋶ᚸᚹᚢᚺᚱᚾᚠᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○↗꜀ category○–○parliament꜀⁚·
·⁚lookup office○–○temporal…absolute꜀ enoch○–○⊖꜀⁚·

⁖word:·ᛉᚢᛁᚱᚿᚢᚾ⁖.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·WEHRTHAN·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To become⁖· to happen⁚⁚to come to pass⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Green hardness softens into sweetness⁖· ripeness arrives within the fruit
↪ itself↗·⁚.·˙ostensive:·
·⁚grimchain:·☆ᚹᛋᚿᛗᚢᛇ⏣ᛏ𐋢ᚠ◎·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐎗꜀ category○–○parliament꜀⁚·
·⁚lookup office○–○temporal…absolute꜀ enoch○–○⊖꜀⁚·

⁖word:·ऋतु⁖.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·RIH+too·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Season⁚⁚fixed period⁚⁚or appointed time⁚⁚·⁚.·˙non…ostensive:·

·⁚ostensive⁚·Shoot gives grain⁜grain leaves bare ground⁜and green returns beneath the same circling
↪ sun⁂·⁚.·˙ostensive⁚·
·⁚grimchain⁚·∀⁺⏃⍑⍐⍲⊛ⲰⲂ✷ᚿᗅᓀᑭ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𝌆⁗ category⟜⟜parliament⁗⁚·
·⁚lookup office⟜⟜temporal…absolute⁗ enoch⟜⟜Θ⁗⁚·

‹word⁚·ᛇᛁᚱᚦ⁖⁚.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·FERTH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Journey⁖˙ travel⁜⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Behind⁜the road fills with footprints⁖˙ ahead⁜untouched ground grows shorter𝌆·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛀ✷ℂ𐌱ᛇᓚ⍋ᘏᛃᚸᚹᗅ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜⌘⁗ category⟜⟜parliament⁗⁚·
·⁚lookup office⟜⟜temporal…absolute⁗ enoch⟜⟜Θ⁗⁚·

‹word⁚·ᗅᚸᛋᚴ⁖⁚.·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚.·˙font⁚·
·⁚pronunciation⁚·yoh⟊RUH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To walk⁜move⁜⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Heel lifts⁜foot swings⁜sole returns⁜ The gait lives by coming back to
↪ contact⌘·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᑫ𐌼ⴱƧↄᒣ▽ᚱᘐ↗Aᐁ₹·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜◎⁗ category⟜⟜goetic⁗⁚·
·⁚lookup office⟜⟜temporal…absolute⁗ enoch⟜⟜Θ⁗⁚·

‹word⁚·卌ᚿ⋘⁖⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·REH⟊shoom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·head⁖˙ servant⁜slave⁖˙ top⁜summit⁖˙ beginning⁖˙ first installment⁜original amount⁜capital
↪ assets⁖˙ first rank⁖˙ first quality⁜⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Before the full light⁜a pale edge opens in the east◎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚻᗡᔑᓀᐁᚺᘐᛞᚸᓀᓴ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜⬡⁗ category⟜⟜goetic⁗⁚·
·⁚lookup office⟜⟜temporal…absolute⁗ enoch⟜⟜Θ⁗⁚·

‹word⁚·ᒣᚿ⟐⁖⁚.·˙word⁚·

·⁖origin…language꞉·Akkadian·⁖˙·origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖˙·font꞉·
·⁖pronunciation꞉·MAHKH+room·⁖˙·pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·cadence꞉·
·⁖non…ostensive꞉·front꞉꞉presence⁖· (time) past꞉꞉bygone time꞉꞉⁖˙·non…ostensive꞉·
·⁖ostensive꞉·Cold ash and smoke+dark brick hold the time of a fire that is gone⬡·⁖˙·ostensive꞉·
·⁖grimchain꞉·𑖦𑖛𑖩𑖪𑖨𑖯𑖰𑖭𑖽𑖫𑖛𑖲·⁖˙·grimchain꞉·
·⁖˙·lookup꞉˙
·⁖˙·glyphs꞉˙
·⁖glyph symbol꞉○꞉✡꞉ category꞉○꞉goetic꞉⁖·
·⁖lookup office꞉○꞉temporal…absolute꞉ enoch꞉○꞉⊖꞉⁖·

꞉word꞉·𑖦𑖛𑖩𑖪𑖨𑖯𑖰𑖭𑖽𑖫𑖛𑖲꞉·˙word꞉·

·⁖origin…language꞉·Sumerian·⁖˙·origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖˙·font꞉·
·⁖pronunciation꞉·TAHB·⁖˙·pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·cadence꞉·
·⁖non…ostensive꞉·Burn⁖· to catch fire꞉꞉give heat꞉꞉or begin burning꞉꞉⁖˙·non…ostensive꞉·
·⁖ostensive꞉·Reed catches꞉꞉flames꞉꞉blackens꞉꞉and falls inward through its own burning✡·⁖˙·ostensive꞉·
·⁖grimchain꞉·𐎀𐎭𐎩𐎫𐎗𐎘𐎍𐎆𐎏𐎃𐎅·⁖˙·grimchain꞉·
·⁖˙·lookup꞉˙
·⁖˙·glyphs꞉˙
·⁖glyph symbol꞉○꞉✡꞉ category꞉○꞉goetic꞉⁖·
·⁖lookup office꞉○꞉temporal…absolute꞉ enoch꞉○꞉⊖꞉⁖·

꞉word꞉·𑖩𑖪𑖨𑖯𑖰꞉·˙word꞉·

·⁖origin…language꞉·Akkadian·⁖˙·origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖˙·font꞉·
·⁖pronunciation꞉·ah+LAHK+toom·⁖˙·pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·cadence꞉·
·⁖non…ostensive꞉·behaviour꞉꞉customs⁖· activity꞉꞉movement⁖· passage꞉꞉road꞉꞉course꞉꞉way⁖·
↪ caravan꞉꞉⁖˙·non…ostensive꞉·
·⁖ostensive꞉·Departure꞉꞉crossing꞉꞉arrival꞉꞉ The caravan repeats the course until the road itself carries
↪ the habit✡·⁖˙·ostensive꞉·
·⁖grimchain꞉·𐎆𐎉𐎓𐎮𐎐𐎗𐎀𐎍𐎊𐎓𐎂·⁖˙·grimchain꞉·
·⁖˙·lookup꞉˙
·⁖˙·glyphs꞉˙
·⁖glyph symbol꞉○꞉❂꞉ category꞉○꞉goetic꞉⁖·
·⁖lookup office꞉○꞉temporal…absolute꞉ enoch꞉○꞉⊖꞉⁖·

꞉word꞉·𑖮𑖰𑖬𑖭𑖬𑖬꞉·˙word꞉·

·⁖origin…language꞉·Tocharian A·⁖˙·origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖˙·font꞉·
·⁖pronunciation꞉·UHNEETYAHT·⁖˙·pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·cadence꞉·
·⁖non…ostensive꞉·Instability꞉꞉impermanence꞉꞉⁖˙·non…ostensive꞉·
·⁖ostensive꞉·No crest survives itself⁖· each wave becomes another before it can keep its
↪ shape❂·⁖˙·ostensive꞉·
·⁖grimchain꞉·𐎓𐎚𐎊𐎐𐎍𐎆𐎓𐎗𐎀𐎍𐎘𐎃·⁖˙·grimchain꞉·
·⁖˙·lookup꞉˙
·⁖˙·glyphs꞉˙

·⁚glyph symbol꞉꞉꞉❁꞉ category꞉꞉꞉goetic꞉⁚·
·⁚lookup office꞉꞉꞉temporal…absolute꞉ enoch꞉꞉꞉Θ꞉⁚·

꞉word꞉·𑖩𑖰𑖨𑖽꞉⁚·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚·˙font꞉·
·⁚pronunciation꞉·DAH+room·⁚·˙pronunciation꞉·
·⁚cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·perpetuity꞉꞉continuum꞉꞉permanence⁚· generation꞉꞉lifetime (60~70 years꞉꞉the life span of a
↪ human being)꞉꞉·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Water changes without pause꞉꞉but the river keeps going❁⁚·˙ostensive꞉·
·⁚grimchain꞉·ᛊ⍟ↄɤ౬ᛊʃ㐫ㅓ㐫ㅛɤₒ꞉·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉꞉꞉❋꞉ category꞉꞉꞉goetic꞉⁚·
·⁚lookup office꞉꞉꞉temporal…absolute꞉ enoch꞉꞉꞉Θ꞉⁚·

꞉word꞉·ـصديـ꞉⁚·˙word꞉·

·⁚origin…language꞉·Old Uyghur·⁚·˙origin…language꞉·
·⁚font꞉·Noto Serif Old Uyghur·⁚·˙font꞉·
·⁚pronunciation꞉·ažun·⁚·˙pronunciation꞉·
·⁚cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·existence꞉꞉mode of existence (eqv꞉꞉ गति (gati))⁚·
↪ reincarnation꞉꞉life꞉꞉lifetime꞉꞉·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Seed becomes shoot꞉꞉shoot leaf꞉꞉leaf grain꞉꞉and grain returns the life to seed❋·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᗺሕᛉᖘᘐ◬ᗡ⨮ᕀ⍉ᘓ᛬ᗢ▽·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉꞉꞉⧗꞉ category꞉꞉꞉goetic꞉⁚·
·⁚lookup office꞉꞉꞉temporal…absolute꞉ enoch꞉꞉꞉Θ꞉⁚·

꞉word꞉·स्नेह꞉⁚·˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚·˙font꞉·
·⁚pronunciation꞉·SNAY+huh·⁚·˙pronunciation꞉·
·⁚cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·oiliness꞉꞉unctuousness꞉꞉fattiness꞉꞉greasiness꞉꞉lubricity꞉꞉viscidity (also as one of the 24
↪ gunas of the vaisheshika branch of the Nyaya philosophy)⁚· oil꞉꞉grease꞉꞉fat꞉꞉any oleaginous
↪ substance꞉꞉an unguent⁚· smoothness꞉꞉glossiness⁚· blandness꞉꞉tenderness꞉꞉love꞉꞉attachment to꞉꞉fondness or
↪ affection for (locative꞉꞉genitive꞉꞉or compound)꞉꞉friendship with (saha)⁚· moisture⁚· a fluid of the
↪ body⁚· (in the plural) name of the Vaishyas in Kushadvipa꞉꞉·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Water slips away from bronze꞉꞉ Oil stays꞉꞉shining through each tilt and wipe⧗·⁚·˙ostensive꞉·
·⁚grimchain꞉·◬ᚸᛋ◎ᛯ৳ᴑ✱ᔕ✱⩒▫ᕐ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉꞉꞉⊛꞉ category꞉꞉꞉goetic꞉⁚·
·⁚lookup office꞉꞉꞉temporal…absolute꞉ enoch꞉꞉꞉Θ꞉⁚·

꞉word꞉·ᓚlλ̄⁖·˙word꞉·

·⁖origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁖pronunciation꞉·SAHNTEE·⁚·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁖non…ostensive꞉·Twilight⁚⁚·⁚·˙non…ostensive꞉·
·⁖ostensive꞉·Day does not vanish at once⁚⁚ Light thins along the horizon until darkness takes
↪ it⊛·⁚·˙ostensive꞉·
·⁖grimchain꞉·Λ⧗▽𑁍𑁆𑁍⊕▽ᚿᚱ∽ᏧᏫ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍❄₎ category꞉○–○₍goetic₎⁖·
·⁖lookup office꞉○–○₍temporal…absolute₎ enoch꞉○–○₍Θ₎⁖·

₍word꞉·𑁆𑁋𑁍 𑁍𑁍 𑁍𑁌₍꞉·˙word꞉·

·⁖origin…language꞉·Old Persian·⁚·˙origin…language꞉·
·⁖font꞉·Noto Sans Old Persian·⁚·˙font꞉·
·⁖pronunciation꞉·MAH+hah·⁚·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁖non…ostensive꞉·Moon⁚˙ also a month measured by its cycle⁚⁚·⁚·˙non…ostensive꞉·
·⁖ostensive꞉·Crescent swells to fullness⁚⁚empties into darkness⁚⁚and returns as crescent❄·⁚·˙ostensive꞉·
·⁖grimchain꞉·✡✱ᛁᗡᘉᙆ𑁍𑁆𑁍ᑲᔕ⊖ᙏ𑁍·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍☸₎ category꞉○–○₍goetic₎⁖·
·⁖lookup office꞉○–○₍temporal…absolute₎ enoch꞉○–○₍Θ₎⁖·

₍word꞉·ϒIΨᚱᚣ₍꞉·˙word꞉·

·⁖origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁖pronunciation꞉·WI+THRA·⁚·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁖non…ostensive꞉·(+accusative) against⁚⁚·⁚·˙non…ostensive꞉·
·⁖ostensive꞉·Pressure holds against pressure⁚⁚ Neither side yields while the contact
↪ lasts☸·⁚·˙ostensive꞉·
·⁖grimchain꞉·ᚱ◎ǂϒᛉᚺᚦᑫᑭᚷᛞᚱ𑁍·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍⬡₎ category꞉○–○₍goetic₎⁖·
·⁖lookup office꞉○–○₍temporal…absolute₎ enoch꞉○–○₍Θ₎⁖·

₍word꞉·ᛕᛠ ᛟ 𑁍𑁍₍꞉·˙word꞉·

·⁖origin…language꞉·Old Persian·⁚·˙origin…language꞉·
·⁖font꞉·Noto Sans Old Persian·⁚·˙font꞉·
·⁖pronunciation꞉·kh+shah+PAH·⁚·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁖non…ostensive꞉·Night⁚⁚·⁚·˙non…ostensive꞉·
·⁖ostensive꞉·Last light fades⁚˙ stars keep the dark until the eastern edge brightens
↪ again⬡·⁚·˙ostensive꞉·
·⁖grimchain꞉·⊕ᚣ∽∀⊙∀▽ᚿᛏ◎ǂ𑁍ᚺ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙

·⁖glyph symbol꞉꞉𐫁 category꞉꞉court꞉·
·⁖lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉⊖꞉·

꞉word꞉·ﻉ᳘ᳯ̄꞉·˙word꞉·

·⁖origin…language꞉·Tocharian A·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖·˙font꞉·
·⁖pronunciation꞉·CHMOHL·⁖·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Birth꞉꞉·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·One life leaves its sealed chamber and begins breathing beyond it𐫁·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᨆ᳖ᩚᩣᩅᩁᩋᩍᩥᩋᩍ⫩ᩍᩋᩍ⫩ᩍᩋᩍ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉꞉𐫁 category꞉꞉court꞉·
·⁖lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉⊖꞉·

꞉word꞉·ⵜⵜ꞉·˙word꞉·

·⁖origin…language꞉·Tocharian B·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖·˙font꞉·
·⁖pronunciation꞉·NAHNOH·⁖·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Again꞉꞉once more꞉꞉·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Water withdraws from the stone꞉꞉then comes back𐫁·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᨅᩅᩋᨆᩣᩋᩍᩋᩍᩋᩍ⬡·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉꞉𐫁 category꞉꞉court꞉·
·⁖lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉⊖꞉·

꞉word꞉·ᛘᚣᚾ꞉·˙word꞉·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·THAN·⁖·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Then꞉꞉at that time꞉꞉when꞉꞉·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·One bell dies away꞉꞉ Then the next begins𐫁·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐌼ᛊᛋᛘᚹᚦᚱᚩᚱᚩᚱᚩ⫩ᚱᚩᚱᚩᚱᚩ⫩ᚱᚩᚱᚩᚱᚩ⫩

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꒰꒰ᵋ𐍆꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋtemporal…absolute꒱ enoch꒰꒰ᵋ⊖꒱⁚·

·word:·ᚠᚢᚢᚱ·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·FOWR·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Before (spatial and temporal)·⁚ for::in favour of::on behalf of·⁚ near::next
↪ to::by::⁚·˙non…ostensive:·
·⁚ostensive:·Oil and wick wait untouched:: The spark has not come yet𐍆·⁚·˙ostensive:·
·⁚grimchain:·♋ϒⓄ₣♳❋♀ᘔ𐍆ʃϯ𓆉Ɐ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꒰꒰ᵋ𐍉꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋtemporal…absolute꒱ enoch꒰꒰ᵋ⊖꒱⁚·

·word:·크다·⁚·˙word:·

·⁚origin…language:·Middle Korean·⁚·˙origin…language:·
·⁚font:·Noto Sans KR·⁚·˙font:·
·⁚pronunciation:·keutá·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·to grow::to get bigger·⁚ to be big·⁚ to become great·⁚ to be great::⁚·˙non…ostensive:·
·⁚ostensive:·Each season adds another ring around the older wood𐍉·⁚·˙ostensive:·
·⁚grimchain:·✲Ⱳ♄⛢ϛ▽Ɐ❄ᛘ✷𐍉⬡ᛕ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꒰꒰ᵋ𐍊꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋtemporal…absolute꒱ enoch꒰꒰ᵋ⊖꒱⁚·

·word:·ھەدىڭ·⁚·˙word:·

·⁚origin…language:·Old Uyghur·⁚·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚·˙font:·
·⁚pronunciation:·yol·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·road::way::path·⁚ method·⁚ generation::lifespan·⁚ heavenly favor·⁚
↪ luck::fate::⁚·˙non…ostensive:·
·⁚ostensive:·From first footprint to last::the road keeps one life moving through its allotted
↪ turns𐍊·⁚·˙ostensive:·
·⁚grimchain:·ᘜ✡ᕮᘉᔦᛘ♂☽ᐱᘉᗰ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꒰꒰ᵋ𐍉꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋtemporal…absolute꒱ enoch꒰꒰ᵋ⊖꒱⁚·

·word:·𐤋̣·⁚·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·

·⁚pronunciation:·NOON·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Again::once more::·⁚.·˙non…ostensive:·
·⁚ostensive:·Water leaves the stone bare::then covers it once more𐑦·⁚.·˙ostensive:·
·⁚grimchain:·ᚹᛟᛃᚠᛏᚳᚢᚹᛞᛏᚪᛉ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉𐑮 category꞉꞉court𐑝:·
·⁚lookup office꞉꞉temporal…absolute𐑝 enoch꞉꞉Θ𐑝:·

·word:·ᛋᚾᛉᚾᚾ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·SAH᛫mahn·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·together::·⁚.·˙non…ostensive:·
·⁚ostensive:·Two bells begin together::bloom together::and fade in the same ringing𐑦·⁚.·˙ostensive:·
·⁚grimchain:·ᚸᛒᚢᚱᛞᛗᛗᛠᚸᚢᛋᛜᚩ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉𐑓 category꞉꞉court𐑝:·
·⁚lookup office꞉꞉temporal…absolute𐑝 enoch꞉꞉Θ𐑝:·

·word:·ᑌᒐᐞᒉ·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·PYOOTK·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To create::bring into being::establish::·⁚.·˙non…ostensive:·
·⁚ostensive:·Clay loses formlessness beneath the hands until a vessel stands where none stood before𐑦·⁚.·˙ostensive:·
·⁚grimchain:·ᘔᚴᖷᑭᚸᐞᐊᐸᚻIᘔᐢᑎᚲ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉𐑥 category꞉꞉court𐑝:·
·⁚lookup office꞉꞉temporal…absolute𐑝 enoch꞉꞉Θ𐑝:·

·word:·ᛅᒉᐞ·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·WAHK·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To burst::split asunder::·⁚.·˙non…ostensive:·
·⁚ostensive:·Tension holds::holds::then suddenly gives⁖ the fibre becomes two in the breaking instant𐑦·⁚.·˙ostensive:·
·⁚grimchain:·ᚼⅡᚱᛋᚲ⊙ᔑᔦᛏᛇᛦᛉᚱ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉𐑧 category꞉꞉court𐑝:·
·⁚lookup office꞉꞉temporal…absolute𐑝 enoch꞉꞉Θ𐑝:·

⁖word⁚·ᒪ□ꕥ̄⁖·˙word⁚·

·⁖origin…language⁚·Tocharian B·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Brahmi·⁖·˙font⁚·
·⁖pronunciation⁚·YUHM·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To achieve⁖obtain⁖ to reach⁖arrive at⁖⁖⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Each pull shortens the gap until knot meets postص·⁖·˙ostensive⁚·
·⁖grimchain⁚· ⊛ ⏀ᛦ⊕ᘐᛯ⏀IZ∀ᒧᑎ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖
·⁖·˙glyphs⁖
·⁖glyph symbol○–○�really category○–○₵court₍⁚·
·⁖lookup office○–○₵temporal…absolute₍ enoch○–○₵Θ₍⁚·

⁖word⁚·ᎷV⊗IV⁖·˙word⁚·

·⁖origin…language⁚·Etruscan·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Old Italic·⁖·˙font⁚·
·⁖pronunciation⁚·SHOO+thee+oo·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Storehouse⁖ a building or enclosed space used to keep provisions or
↪ goods⁖⁖·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Harvest passes⁖⁖but sealed grain waits in darkness for the season of needI·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᘜᛯᘔ✡ᛯᔑᘐᔕ⏎ᐊᑫ❀ᓄ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖
·⁖·˙glyphs⁖
·⁖glyph symbol○–○₹₍ category○–○₵court₍⁚·
·⁖lookup office○–○₵temporal…absolute₍ enoch○–○₵Θ₍⁚·

⁖word⁚·✶ꕯ⍑⁖·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖·˙font⁚·
·⁖pronunciation⁚·HEET+tah·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·to hit upon⁖⁖meet⁖⁖·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Two courses that were separate touch for a single instant at one stone⁖⁖then part
↪ againᚼ·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᛘᗅᛋᖘᐁᐁᛯᒣᛘᚴᐃᐞIⒹ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖
·⁖·˙glyphs⁖
·⁖glyph symbol○–○₵▫₍ category○–○₵court₍⁚·
·⁖lookup office○–○₵temporal…absolute₍ enoch○–○₵Θ₍⁚·

⁖word⁚·〈🝆⁖·˙word⁚·

·⁖origin…language⁚·Akkadian·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Cuneiform·⁖·˙font⁚·
·⁖pronunciation⁚·KEE+room·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·oven⁖⁖kiln (especially for lime and bitumen)⁖⁖·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Clay reddens slowly inside the kiln until firing has run its full course▫·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᘜᛘ▫ᘜᛯᚼᛘᐊᛯ⏀ꖘᐞᛯᛉᛯ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖

·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍𐦟₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

₍word⁚·오래₎⁚·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚·˙font⁚·
·⁚pronunciation⁚·olaj·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·long⁝⁝for a long time or while⁝⁝⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Sunlight crawls from one wall to the other before leaving the courtyard𐦟·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤎𐤍𐤅𐤃𐤋𐤇𐤒𐤕𐤉𐤈𐤌𐤓𐤏·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍𐦍₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

₍word⁚·𒁹𒌋₎⁚·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·mee+KHEES+toom·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·cuneiform wedge⁝⁝stroke (of the stylus)⁚˙ wound⁝⁝⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Stylus presses once⁝⁝ Long after the hand lifts⁝⁝the cut is still there𐦍·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤆𐤄+𐤋𐤍+𐤌𐤑𐤂𐤐𐤔𐤓·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍𐦓₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

₍word⁚·+𑀓𑀺⁚·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·KUHT·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To strew⁝⁝scatter⁝⁝sow (seeds)⁝⁝⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Seed disappears under earth and waits there before the first green breaks
↪ throughᛉ·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤄𐤔𐤓𐤑𐤉𐤅𐤇𐤃𐤋𐤁𐤎·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍𐦋₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

₍word⁚·𒀭𒈾₎⁚·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·eh+TEHM+moom·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·ghost⁝⁝spirit (of a dead person)⁝⁝⁚·˙non…ostensive⁚·

·⁚ostensive⁚·Breath is gone⁜yet the pale outline returns beside the grave night after
↪ night꞉⁖·˙ostensive⁚·
·⁚grimchain⁚·Ʇꟗᑭꟗꝺᚺᛞᚠᛗᛍᑦᘔᚺ·⁖·˙grimchain⁚·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol○–○₍ᛋ꞉ category○–○₍court꞉·
·⁚lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

₍word⁚·ᚻᚱᛞᛍ꞉·⁖·˙word⁚·

·⁚origin…language⁚·Akkadian·⁖·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁖·˙font⁚·
·⁚pronunciation⁚·KHOOL ᛀ koom·⁖·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁚non…ostensive⁚·lost object⁜lost property⁘ loss⁜⁜⁖·˙non…ostensive⁚·
·⁚ostensive⁚·Empty peg waits beside the tools through every later reaching handᛋ·⁖·˙ostensive⁚·
·⁚grimchain⁚·ᛉ✩ᑦ✡◎ᚸᚴᚭᛞᚲᛏᛗ·⁖·˙grimchain⁚·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol○–○₍'꞉ category○–○₍court꞉·
·⁚lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

₍word⁚·ᛪᚿ꞉·⁖·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁖·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁖·˙font⁚·
·⁚pronunciation⁚·MAHOOKEE·⁖·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁚non…ostensive⁚·A stop⁜a pause⁜⁘⁖·˙non…ostensive⁚·
·⁚ostensive⁚·Hammer raised⁜no blow⁜ Work hangs in the pause'·⁖·˙ostensive⁚·
·⁚grimchain⁚·✩ᚿᚺᛏ◎ᛒᘔᚷᛁ✱ᛘᘐᚋᚸ·⁖·˙grimchain⁚·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol○–○₍ᛁ꞉ category○–○₍court꞉·
·⁚lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

₍word⁚·✍꞉·⁖·˙word⁚·

·⁚origin…language⁚·Sumerian·⁖·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁖·˙font⁚·
·⁚pronunciation⁚·EE ᛀ too·⁖·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁚non…ostensive⁚·Moon⁘ the bright celestial body seen at night⁜also the cycle marked by its changing
↪ phases⁜⁖·˙non…ostensive⁚·
·⁚ostensive⁚·Darkness gives crescent⁜crescent gives fullness⁜fullness empties back into
↪ darkness꞉⁖·˙ostensive⁚·
·⁚grimchain⁚·ᚸᛀᖁ⊙◣▫ᑫᑎᘏᐱᓄᚽᛏᚸᚱ·⁖·˙grimchain⁚·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol○–○₍ᛦ꞉ category○–○₍court꞉·
·⁚lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

₍word⁚·ᛘᑎᚻᚻ꞉·⁖·˙word⁚·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·MOO+nah·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To remember:·:⁚.·˙non…ostensive:·
·⁚ostensive:·A fingertip finds the old cut and the vanished moment comes back with it↓·⁚.·˙ostensive:·
·⁚grimchain:·ᛟ🜲ᚠᛉᛉᛟᛋᚴ🜨◎·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦𐑂𐑌 category◦–◦court𐑌:·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌:·

·word:·ᛁᚼᛏᛁᚱ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·EN+dir·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·End or conclusion:·:⁚.·˙non…ostensive:·
·⁚ostensive:·Final knot closes the cord:: Nothing follows↑·⁚.·˙ostensive:·
·⁚grimchain:·ᛟᔑ↓ᚠ🜨ₒᑭᑭϟᗅ☿φ🜨·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦𐑉𐑌 category◦–◦court𐑌:·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌:·

·word:·𑀦𑁆·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·eh+DEH+shoom·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·to be new::fresh::renewed:·:⁚.·˙non…ostensive:·
·⁚ostensive:·Fresh green rises through the remains of the old growth𐑉·⁚.·˙ostensive:·
·⁚grimchain:·ϒ𐏁ϪΨᛉϞΘϥΦ🜨⏀ᛝ⁜·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦Ո𐑌 category◦–◦court𐑌:·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌:·

·word:·𑀦𑁆·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·bah+NOOM·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·to build::construct::form::make::manufacture:˙ to create (said of deity):·:⁚.·˙non…ostensive:·
·⁚ostensive:·Course after course::the wall climbs out of incompletionՈ·⁚.·˙ostensive:·
·⁚grimchain:·☿ϒ1ǂ⏀ᗰᕒᙢᛋᓚᔑᓭǂ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦Þ𐑌 category◦–◦court𐑌:·
·⁚lookup office◦–◦temporal…absolute𐑌 enoch◦–◦Θ𐑌:·

·word:·ᛒᚱᛁᚾᚾ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·BREN⟊nah·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To burn⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Timber glows⁚⁚flares⁚⁚blackens·⁚ burning lasts as long as wood remains to feed
↪ it᛫⁚.·˙ostensive:·
·⁚grimchain:·ᛇᚦᚿᛈ ⊛ ↗ᛒᘐᚹ▫ᛯᚹᚦ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦ₑᚠ» category◦–◦ₑcourt»⁚·
·⁚lookup office◦–◦ₑtemporal…absolute» enoch◦–◦ₑ⊖»⁚·

₍word:·सर्पसि्:.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·suhr⟊pis·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·ghee (i⁚⁚e⁚⁚ melted butter with the scum cleared off⁚⁚either fluid or solidified·⁚ also
↪ in plural)⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Warmth loosens clear fat to liquid·⁚ coolness gathers clear fat into a pale
↪ solidᚨ⁚.·˙ostensive:·
·⁚grimchain:·ϒIIᛕ⁑Ⲇ✲⍟⊕✶⏁ᑫ⊕♳·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦ₑᚱ» category◦–◦ₑcourt»⁚·
·⁚lookup office◦–◦ₑtemporal…absolute» enoch◦–◦ₑ⊖»⁚·

₍word:·⟊▯ᒍ̄⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KUHL·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To endure⁚⁚bear⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Bent beam carries the load hour after hour without surrendering it to the
↪ groundᚱ·⁚.·˙ostensive:·
·⁚grimchain:·ᛕᛊ₀▫ᓯᛊ☸♀⏀ᘐᥴᚠᛣ⊛·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦ₑ‹» category◦–◦ₑcourt»⁚·
·⁚lookup office◦–◦ₑtemporal…absolute» enoch◦–◦ₑ⊖»⁚·

₍word:·ᚼ⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·EYE·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·ever⁚⁚eternally⁚⁚at any time⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·One circuit of stars closes only to open into another‹·⁚.·˙ostensive:·
·⁚grimchain:·ᚎᛗᘐᚺᛞᚷᚮᚸ⏁⊕◎⌬✡·⁚.·˙grimchain:·
·⁚.·˙lookup:˙

·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍X₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

·word:·𐓀𐓄⁚·˙word:·

·⁚origin…language:·Old Turkic·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚·˙font:·
·⁚pronunciation:·TAHNG·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Dawn::⁚·˙non…ostensive:·
·⁚ostensive:·Night thins to a pale rim before the sun itself arrivesX·⁚·˙ostensive:·
·⁚grimchain:·𐰀𐰞⓪⬡⊕ǂ𐰇ᚺᛊ⬡XⱯ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍Ᵽ₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

·word:·ITɅN·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·ITAN·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·To eat::⁚·˙non…ostensive:·
·⁚ostensive:·Mouthful after mouthful::the loaf diminishes until eating has spent itᑭ·⁚·˙ostensive:·
·⁚grimchain:·ᚻ'ϔ⬡⓪𐰾ᚑ▽ᵑϔᛦᑭ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍Ⱶ₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

·word:·ᒪ□ı̄⁚·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·YUHR·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·To wash::⁚·˙non…ostensive:·
·⁚ostensive:·Water keeps passing through the cloth until no stain comes away with itⱧ·⁚·˙ostensive:·
·⁚grimchain:·ᚻᛦ𐰾𐰾ᔑᘔⱮⱯ⊕0ΘↃ𐰋·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍ᚼ₎ category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍temporal…absolute₎ enoch₀₋₀₍Θ₎⁚·

·word:·𗥃⁚·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·TAWM'·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Coagulate:˙ to thicken or change from a fluid into a more solid mass::⁚·˙non…ostensive:·

·⁖ostensive⁚·Liquid resists the spoon a little more with every turn until flowing gives way to a soft
↪ set𐑉·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐎗𐎚𐎁𐎐𐎄𐎅◎𐎗𐎉𐎆𐎍𐎌𐎐·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol○–○𐑉 category○–○court𐑈⁚·
·⁖lookup office○–○temporal…absolute𐑈 enoch○–○Θ𐑈⁚·

₍word⁚·𐎓𐎍𐎘𐎗𐎀𐎐·⁖·˙word⁚·

·⁖origin…language⁚·Akkadian·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Cuneiform·⁖·˙font⁚·
·⁖pronunciation⁚·mah+KHAH+soom·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·to strike⁚⁚hit⁚⁚kill⁚⁚·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Hammer meets stone in a single hard beat⁚⁚and the ringing follows after𐑉·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐎒𐎀𐎇𐎊𐎍𐎃𐎅𐎏𐎑𐎋𐎍·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol○–○𐑋 category○–○court𐑈⁚·
·⁖lookup office○–○temporal…absolute𐑈 enoch○–○Θ𐑈⁚·

₍word⁚·𐌀𐌋𐌔𐌄𐌌𐌋𐌓⁚·˙word⁚·

·⁖origin…language⁚·Etruscan·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Old Italic·⁖·˙font⁚·
·⁖pronunciation⁚·FAH+seh+nah·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Ash or sand⁖˙ fine particulate mineral or burnt residue⁚⁚·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Long after flame or grinding has finished⁚⁚fine residue still lies where the larger body
↪ was𐑋·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐎗𐎌𐎍𐎉𐎆◎𐎓𐎒𐎃𐎀𐎄𐎐𐎍𐎂𐎗X·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol○–○𐑌 category○–○court𐑈⁚·
·⁖lookup office○–○temporal…absolute𐑈 enoch○–○Θ𐑈⁚·

₍word⁚·ᚱᛁᚾᚾᚣᚾ⁚·˙word⁚·

·⁖origin…language⁚·Gothic·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Gothic·⁖·˙font⁚·
·⁖pronunciation⁚·RINAN·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To run⁖˙ to flow⁚⁚·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Water enters⁚⁚passes⁚⁚leaves⁖˙ the river continues by never keeping the same
↪ water𐑌·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐌑𐌰𐌰𐌹𐍂𐍅𐌾𐌺𐌰𐌶𐌴𐌻𐌰𐌽·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol○–○𐑊 category○–○court𐑈⁚·
·⁖lookup office○–○temporal…absolute𐑈 enoch○–○Θ𐑈⁚·

₍word⁚·𐤕𐤐𐤋⁚·˙word⁚·

·⁚origin…language:·Mycenaean Greek·⁚.·˙origin…language:·
·⁚font:·Noto Sans Linear B·⁚.·˙font:·
·⁚pronunciation:·agorAH·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·collection·⁚· flock·:·⁚.·˙non…ostensive:·
·⁚ostensive:·One arrives·:then another·:then many·:until scattered bodies have become a gathered
↪ whole꜀·⁚.·˙ostensive:·
·⁚grimchain:·𐤛⊖ϒ⊕⊙Λᛋᚦᛃᓀᚱᚥᛯ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤂꜆ category◦–◦court꜆:·
·⁚lookup office◦–◦temporal…absolute꜆ enoch◦–◦Θ꜆:·

·word:·ᑌᒪᐱᛪ̄·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·PRUHTEEM·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Decision·:resolve·:determination·:·⁚.·˙non…ostensive:·
·⁚ostensive:·Uncertainty narrows until only one course is left to take꜉·⁚.·˙ostensive:·
·⁚grimchain:·ᚼ✷𐤚⌬ᛈ𝌆ᓄ'ᛞᛗᚴ✪ɔᑕ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦✲꜆ category◦–◦court꜆:·
·⁚lookup office◦–◦temporal…absolute꜆ enoch◦–◦Θ꜆:·

·word:·ʁIʌЄIʁϟNᛯ·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·BILIVAN·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To remain·:stay·:·⁚.·˙non…ostensive:·
·⁚ostensive:·Rain·:sun·:and night pass over the stone·:and each finds it where the last left
↪ it✲·⁚.·˙ostensive:·
·⁚grimchain:·ⵔᖶ1𝌆ᑕϒ♀ᑫᛕᛈ♏ℂ♋·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤒꜆ category◦–◦court꜆:·
·⁚lookup office◦–◦temporal…absolute꜆ enoch◦–◦Θ꜆:·

·word:·𐤗𐤊𐤓𐤉𐤋𐤍·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·WAHR+kah+toom·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·backside·:rear side·⁚· (anatomy) rectum·:buttocks·⁚· (military) rear guard (of an army)·⁚·
↪ (finance) later instalment·⁚· arrears (unpaid debt)·⁚· estate·:inheritance·⁚· circumstances·:background
↪ (of a case)·:·⁚.·˙non…ostensive:·
·⁚ostensive:·First payment is already recorded·⁚· another part waits to be given later𐤒·⁚.·˙ostensive:·
·⁚grimchain:·ᛉᔑᚠᔐ♍♀ϒⵔ𐤚ᓭᑐᓀᓯ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol◦–◦𑁍𑁿𑁦, category◦–◦court𑁤⁚·
·⁚lookup office◦–◦temporal…absolute𑁤 enoch◦–◦Θ𑁤⁚·

⁖word⁚·ʌ𑁆J̄⁖·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·TUHL·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To lift⁚⁚raise⁚˙ to acquire⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Hand rises⁚⁚burden rises with it⁚⁚and the climb ends when the lifting stops𑁍𑁿·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐤉▽𐤓⁂⧗◘⁕ʁᚴᛇ△𐤑𐤊𐤄𐤌·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𑁌𑁤, category◦–◦court𑁤⁚·
·⁚lookup office◦–◦temporal…absolute𑁤 enoch◦–◦Θ𑁤⁚·

⁖word⁚·नभस्⁖·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·nuh+bhuhs·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·mist⁚⁚fog⁚⁚clouds⁚⁚vapour⁚˙ the sky or atmosphere (in dual: heaven and earth)⁚˙ ether (as
↪ an element)⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Pale air gathers between clear sky and rain⁚⁚hangs awhile⁚⁚then thins away𑁌·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᘉ𐤉ᛃ𐤍↓▽Φ❂Λᛎᘐᗅᚹᚦ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦◎𑁤, category◦–◦court𑁤⁚·
·⁚lookup office◦–◦temporal…absolute𑁤 enoch◦–◦Θ𑁤⁚·

⁖word⁚·ᓀl̄ᚷ𑁆⁖·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·SNOHTK·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To suffuse⁚⁚permeate⁚⁚imbue⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Color enters the cloth slowly⁚⁚travelling inward until no pale fibre remains
↪ untouched◎·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛎᘐᑎ⁕ᑭᑭᛉ‹zↃ∱ᛯᚢ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐓓𑁤, category◦–◦court𑁤⁚·
·⁚lookup office◦–◦temporal…absolute𑁤 enoch◦–◦Θ𑁤⁚·

⁖word⁚·𐤎𐤆𐤇𐤋𐤂⁖·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·nah+DAH+noom·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·

·⁖non…ostensive⁚·To give⁝⁝grant⁖· to hand over⁝⁝deliver⁝⁝transfer⁖· to set⁝⁝assign⁖· to
↪ allow⁝⁝⁖·˙non…ostensive⁚·
·⁖ostensive⁚·For one instant the gift belongs to neither hand⁖· then the receiving hand
↪ closes⏀·⁖·˙ostensive⁚·
·⁖grimchain⁚·♋𐌅𐤉𐤊⊕𐤀⋖𐤃ᗅᘐ⊙ᚾᛇ⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol○–○𐤂𐤋𐤄 category○–○𐤂court𐤄⁚·
·⁖lookup office○–○𐤂temporal…absolute𐤄 enoch○–○𐤂Θ𐤄⁚·

·word⁚·ᛇᚱᛁᚢᛋᚾ⁖·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖·˙font⁚·
·⁖pronunciation⁚·FRYOH⟊sah·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To freeze⁝⁖·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Water stiffens from the surface inward until the whole vessel is held in
↪ ice𐤋𐤄·⁖·˙ostensive⁚·
·⁖grimchain⁚·⊛ₒᚺ✳ᚸᘐᛋ𐤉⊕𐤔ᗡ❁ᓇΘ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol○–○𐤂𐤀𐤄 category○–○𐤂court𐤄⁚·
·⁖lookup office○–○𐤂temporal…absolute𐤄 enoch○–○𐤂Θ𐤄⁚·

·word⁚·Ʇᑐ+̄⁖·˙word⁚·

·⁖origin…language⁚·Tocharian B·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Brahmi·⁖·˙font⁚·
·⁖pronunciation⁚·NUHK·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·(active voice) To destroy⁝⁖⁖ (middle voice) To be lost⁝⁝disappear⁝⁝be
↪ destroyed⁝⁝perish⁝⁖·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Stone courses fail one after another until the wall has ceased to stand𐤀𐤄·⁖·˙ostensive⁚·
·⁖grimchain⁚·𝌢ᵹካᑫ♂ᘐXX⊬𐤉ᛤᘐ𐤍ᑎ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol○–○𐤂𐤊𐤋𐤄 category○–○𐤂court𐤄⁚·
·⁖lookup office○–○𐤂temporal…absolute𐤄 enoch○–○𐤂Θ𐤄⁚·

·word⁚·ካIፐIሐ⁖·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖·˙font⁚·
·⁖pronunciation⁚·SIT⟊yah·⁖·˙pronunciation⁚·
·⁖cadence⁚·Temporal Absolute·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To sit or remain⁝⁖·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Morning light finds the settled weight in the same hollow where dusk left
↪ it𐤊𐤋·⁖·˙ostensive⁚·
·⁖grimchain⁚·I𝌢‹ᘓ⦚ᛤ⏀⊬ᚼΘᚹℂ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol○–○𐤂ₒ𐤄 category○–○𐤂court𐤄⁚·
·⁖lookup office○–○𐤂temporal…absolute𐤄 enoch○–○𐤂Θ𐤄⁚·

˖word:·𐫈𐫀 𐫄𐫄˖·˙word:·

·˙origin…language:·Old Persian·˙·˙origin…language:·
·˙font:·Noto Sans Old Persian·˙·˙font:·
·˙pronunciation:·HACAH·˙·˙pronunciation:·
·˙cadence:·Temporal Absolute·˙·˙cadence:·
·˙non…ostensive:·From‥since‥·˙·˙non…ostensive:·
·˙ostensive:·First mark fixes the beginning·˙ every later knot is counted from there๐·˙·˙ostensive:·
·˙grimchain:·𐫇𐫟𐫁𐫅𐫍𐫏𐫑𐫔𐫖𐫗𐫙𐫛·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꒰꒱𐫏𐫠 category꒰꒱court𐫠·˙·
·˙lookup office꒰꒱temporal…absolute𐫠 enoch꒰꒱𐫏𐫠·˙·

˖word:·𐨁𐨗˖·˙word:·

·˙origin…language:·Tocharian B·˙·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙·˙font:·
·˙pronunciation:·WAHTEH·˙·˙pronunciation:·
·˙cadence:·Temporal Absolute·˙·˙cadence:·
·˙non…ostensive:·Second·˙ other (of two things)‥·˙·˙non…ostensive:·
·˙ostensive:·One stroke is already dry when the next is laid downϴ·˙·˙ostensive:·
·˙grimchain:·▽ϕᚺ▽ᛉ𐦍⁕ᚸᒐᚺ𝌆♳·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꒰꒱Ⱦ𐫠 category꒰꒱court𐫠·˙·
·˙lookup office꒰꒱temporal…absolute𐫠 enoch꒰꒱𐫏𐫠·˙·

˖word:·𐦂𐦟𐦟˖·˙word:·

·˙origin…language:·Old Irish·˙·˙origin…language:·
·˙font:·Noto Sans Ogham·˙·˙font:·
·˙pronunciation:·DAHR·˙·˙pronunciation:·
·˙cadence:·Temporal Absolute·˙·˙cadence:·
·˙non…ostensive:·Oak·˙ a large deciduous tree bearing acorns‥·˙·˙non…ostensive:·
·˙ostensive:·Acorns fall beneath the broad crown as another ring closes around the
↪ heartwoodⱣ·˙·˙ostensive:·
·˙grimchain:·ᚸᘜ⊙⧗ᔑ⁜ᘏᛇ⍑ϑΘ𐬹◎·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꒰꒱Ⱥ𐫠 category꒰꒱court𐫠·˙·
·˙lookup office꒰꒱temporal…absolute𐫠 enoch꒰꒱𐫏𐫠·˙·

˖word:·𐫓˖·˙word:·

·˙origin…language:·Tangut·˙·˙origin…language:·
·˙font:·Noto Serif Tangut·˙·˙font:·
·˙pronunciation:·KAH·˙·˙pronunciation:·
·˙cadence:·Temporal Absolute·˙·˙cadence:·
·˙non…ostensive:·Tattered or ragged·˙ torn‥frayed‥or badly worn‥·˙·˙non…ostensive:·
·˙ostensive:·Wear takes the cloth thread by thread until the edge hangs raggedΛ·˙·˙ostensive:·
·˙grimchain:·ᑎᙢ❂ⵊ♂☿ᛪᚳᛯᘐ▽·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙

·⁚glyph symbol꩜꩜𐓗꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

꞉word⁚·𐰴𐰚𐰇⁚꞉·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚꞉·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚꞉·˙font⁚·
·⁚pronunciation⁚·AHM+tuh·⁚꞉·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚꞉·˙cadence⁚·
·⁚non…ostensive⁚·Now⁚⁚·⁚꞉·˙non…ostensive⁚·
·⁚ostensive⁚·Falling grain has left the fingers and has not yet reached the bowl𐓗·⁚꞉·˙ostensive⁚·
·⁚grimchain⁚·ᚼ⚹ᖷᘐᛘ⊙ᔕᚽᚻᔕᖷᘏ·⁚꞉·˙grimchain⁚·
·⁚꞉·˙lookup⁚˙
·⁚꞉·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐓚꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

꞉word⁚·वत्स꞉꞉·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚꞉·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚꞉·˙font⁚·
·⁚pronunciation⁚·VUHT+suh·⁚꞉·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚꞉·˙cadence⁚·
·⁚non…ostensive⁚·Calf⁚⁚young animal⁚⁚offspring⁚⁚or child⁚⁚·⁚꞉·˙non…ostensive⁚·
·⁚ostensive⁚·Small hooves steady⁚⁚limbs lengthen⁚⁚horn+buds rise while the young body grows into its ↪ later strength𐓚·⁚꞉·˙ostensive⁚·
·⁚grimchain⁚·ৎ0♀Θ♋✷𐓚0𓃰𓃰ᘏᔕᚹ·⁚꞉·˙grimchain⁚·
·⁚꞉·˙lookup⁚˙
·⁚꞉·˙glyphs⁚˙
·⁚glyph symbol꩜꩜Φ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

꞉word⁚·ᚢᚱᛒᛁ꞉꞉·˙word⁚·

·⁚origin…language⁚·Gothic·⁚꞉·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚꞉·˙font⁚·
·⁚pronunciation⁚·ARBI·⁚꞉·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚꞉·˙cadence⁚·
·⁚non…ostensive⁚·Heritage⁚⁚inheritance⁚⁚·⁚꞉·˙non…ostensive⁚·
·⁚ostensive⁚·Old house+sign passes into new hands and continues down the same lineageΦ·⁚꞉·˙ostensive⁚·
·⁚grimchain⁚·ƻ𐓚ᛕᚢᚢᚤᚼ𐓚ᛔᛟᚤ𐓚·⁚꞉·˙grimchain⁚·
·⁚꞉·˙lookup⁚˙
·⁚꞉·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᛦ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

꞉word⁚·𐎗𐎐 𐎚꞉꞉·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚꞉·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚꞉·˙font⁚·
·⁚pronunciation⁚·QASIRU·⁚꞉·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚꞉·˙cadence⁚·
·⁚non…ostensive⁚·Short⁚⁚·⁚꞉·˙non…ostensive⁚·
·⁚ostensive⁚·Bell+metal sounds once and is silent before any echo can answerᛦ·⁚꞉·˙ostensive⁚·

·⁚grimchain⁚·ᵋΨᑫIIᒐᵸϴ⃝○ᗑᵰᵡ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ᵋᓬ category○–○ᵋcourt⁍·
·⁚lookup office○–○ᵋtemporal…absolute⁍ enoch○–○ᵋϴ⁍·

·⁚word⁚·𗤶⁚.·˙word⁚·

·⁚origin…language⁚·Tangut·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·˙font⁚·
·⁚pronunciation⁚·THOO·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Stuff or fill⁚˙ to pack material tightly into a space⁚⁚·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Each handful closes another hollow until the sack can take no moreᓬ⁚.·˙ostensive⁚·
·⁚grimchain⁚·▽𐑍ᚱᗫ⫲⟡卫ϛ▽ᚻᘐ⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ᵋℤ⁍ category○–○ᵋcourt⁍·
·⁚lookup office○–○ᵋtemporal…absolute⁍ enoch○–○ᵋϴ⁍·

·⁚word⁚·𐎗𐎁𐎍⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·DAHN⟊noom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·verbal adjective of danānum⁚˙ strong⁚⁚solid⁚⁚fortified⁚˙ powerful⁚⁚mighty⁚˙ fierce⁚⁚savage⁚˙
↪ difficult⁚⁚severe⁚⁚dangerous⁚˙ urgent⁚⁚·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Wind and weight return again and again⁚˙ the wall face neither bows nor
↪ opensℤ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·▫ᵒᵒ∮⁎ᘉᒻ⧗ᛃᵒᐊᓀ⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ᵋƸ⁍ category○–○ᵋcourt⁍·
·⁚lookup office○–○ᵋtemporal…absolute⁍ enoch○–○ᵋϴ⁍·

·⁚word⁚·ᛒᚱᛁᚢᛏᚼ⁚.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·BRYOH⟊tah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To break⁚⁚·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Whole branch survives until one instant of rupture gives it two endsƸ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ↄᚲᛋᛚᗰℂᚻ☆⏁ᛉᚺᐃ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ᵋI⁍ category○–○ᵋcourt⁍·
·⁚lookup office○–○ᵋtemporal…absolute⁍ enoch○–○ᵋϴ⁍·

·⁚word⁚·ꠇᒍ十⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·

·⁚pronunciation:·WAHLAHK·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To abide⁚⁚dwell⁚⁚reside⁚⁚live somewhere⁚· to stay⁚⁚tarry⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Threshold wears deeper⁚⁚dust gathers⁚⁚seasons turn⁚⁚and lamplight still returns to the same
↪ doorwayI·⁚.·˙ostensive:·
·⁚grimchain:·𑀥𑀸𑁆𑀭𑀼𑀯𑀸𑀓𑀸𑀮𑀻𑀦𑀸·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol꞉꞉𑀥𑀸 category꞉꞉court꞉·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉·

·word:·क्षण·⁚.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·kshuh+nuh·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·any instantaneous point of time⁚⁚an instant⁚⁚a moment⁚· leisure moment⁚⁚vacant time⁚· a
↪ suitable or fit moment⁚⁚an opportunity⁚· a festival⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·A spark appears and is gone before another brightness can share its instant𑀥𑀸·⁚.·˙ostensive:·
·⁚grimchain:·𑀯𑀭𑁆𑀷𑀢𑀸𑀫𑁄𑀤𑀭𑀸·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol꞉꞉𑀥𑀸 category꞉꞉court꞉·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉·

·word:·𑀰𑀮𑀵𑁆·⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·WAHSTOH·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Again⁚· doubly⁚⁚in two ways⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Shout returns from the cliff⁚⁚fades⁚⁚and returns once more from farther stone𑀥𑀸·⁚.·˙ostensive:·
·⁚grimchain:·𑀶𑀭𑀻𑀬𑀷𑀺𑀕𑀸𑀫𑀺·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol꞉꞉𑀕 category꞉꞉court꞉·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉·

·word:·पर्जन्य·⁚.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·puhr+juhn+juh·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·rain+cloud⁚⁚cloud⁚· rain⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Dark vapor swells with water until the first drop breaks free𑀕·⁚.·˙ostensive:·
·⁚grimchain:·𑀓𑀻𑀦𑀭𑀯𑀸𑀬𑀼𑀫𑀡𑁆𑀟𑀮·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol꞉꞉𑀖 category꞉꞉court꞉·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉·

:word:·𐏑:·˙word:·

·⁚origin…language:·Akkadian·⁚˙·origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚˙·font:·
·⁚pronunciation:·SHAH⟊room·⁚˙·pronunciation:·
·⁚cadence:·Temporal Absolute·⁚˙·cadence:·
·⁚non…ostensive:·wind⁖ cardinal point⁚⁚direction⁖ breath⁖ air⁚⁚flatus⁚⁚flatulence⁖ (figurative)
↪ emptiness⁚⁚nothingness⁚⁚lies⁚⁚vanity⁚⁚falsehood⁚⁚⁚˙·non…ostensive:·
·⁚ostensive:·One gust yields to the next through doorway⁚⁚sail⁚⁚and reed𐏑·⁚˙·ostensive:·
·⁚grimchain:·𐏑𐏒𐏓𐏔𐏕𐏖𐏗𐏘𐏙𐏚𐏛𐏜·⁚˙·grimchain:·
·⁚˙·lookup·˙
·⁚˙·glyphs·˙
·⁚glyph symbol꞉꞉𐏑 category꞉꞉court𐏒⁚·
·⁚lookup office꞉꞉temporal…absolute𐏒 enoch꞉꞉𐏓⁚·

:word:·𐏔𐏕𐏖:·˙word:·

·⁚origin…language:·Tocharian B·⁚˙·origin…language:·
·⁚font:·Noto Sans Brahmi·⁚˙·font:·
·⁚pronunciation:·SPAHEETOO·⁚˙·pronunciation:·
·⁚cadence:·Temporal Absolute·⁚˙·cadence:·
·⁚non…ostensive:·Dust⁖ pollen⁚⁚·⁚˙·non…ostensive:·
·⁚ostensive:·Fine grains rise from leaf and stone⁚⁚drift awhile⁚⁚then settle back into
↪ stillness𐏑·⁚˙·ostensive:·
·⁚grimchain:·𐏑𐏒𐏓𐏔𐏕𐏖𐏗𐏘𐏙𐏚·⁚˙·grimchain:·
·⁚˙·lookup·˙
·⁚˙·glyphs·˙
·⁚glyph symbol꞉꞉𐏔 category꞉꞉court𐏒⁚·
·⁚lookup office꞉꞉temporal…absolute𐏒 enoch꞉꞉𐏓⁚·

:word:·𐏓𐏔:·˙word:·

·⁚origin…language:·Tocharian A·⁚˙·origin…language:·
·⁚font:·Noto Sans Brahmi·⁚˙·font:·
·⁚pronunciation:·KUHN·⁚˙·pronunciation:·
·⁚cadence:·Temporal Absolute·⁚˙·cadence:·
·⁚non…ostensive:·To come to pass⁚⁚come true⁚⁚be realized⁖ to fulfill⁚⁚make happen
↪ (transitive)⁚⁚·⁚˙·non…ostensive:·
·⁚ostensive:·Hidden grain swells into stem⁚⁚flowers⁚⁚and finally bears a ripe head where only
↪ possibility stood before𐏔·⁚˙·ostensive:·
·⁚grimchain:·𐏑𐏒𐏓𐏔𐏕𐏖𐏗𐏘𐏙𐏚·⁚˙·grimchain:·
·⁚˙·lookup·˙
·⁚˙·glyphs·˙
·⁚glyph symbol꞉꞉𐏓 category꞉꞉court𐏒⁚·
·⁚lookup office꞉꞉temporal…absolute𐏒 enoch꞉꞉𐏓⁚·

:word:·𐏑 𐏒 𐏓 𐏔 𐏕:·˙word:·

·⁚origin…language:·Old Persian·⁚˙·origin…language:·
·⁚font:·Noto Sans Old Persian·⁚˙·font:·
·⁚pronunciation:·THAHTIY·⁚˙·pronunciation:·
·⁚cadence:·Temporal Absolute·⁚˙·cadence:·
·⁚non…ostensive:·To say⁚⁚declare⁚⁚·⁚˙·non…ostensive:·

·⁚ostensive⁚·First syllable breaks the silence⁘ shaped sound continues until the mouth closes and
↪ silence returns𖬵⁚.·˙ostensive⁚·
·⁚grimchain⁚·⊛ɾ٦ว๓ᦓ✱ъ℃ᓭɾ٦ᗩ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᘜ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

˖word⁚·굿다˖⁚.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·keutstá·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·to cut or to be cut⁘ (ergative) to (come to a) stop⁚⁚·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Blade travels across the board and stops at the edge⁚⁚ Beyond that point⁚⁚wood remains
↪ untouchedᘜ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ϒ꠸ᦓᛯᗅᚱ‹ᦓᦓ☆ᦓᦓ⏁·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𖬵꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

˖word⁚·䏐˖⁚.·˙word⁚·

·⁚origin…language⁚·Tangut·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·˙font⁚·
·⁚pronunciation⁚·DZ⫩WAH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Short⁘ small in length or duration⁚⁚·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·A flash comes and is gone while the bronze is still ringing𖬵·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᓚ๒ᦔ꧁ᦓ꠸ᦓ✓ᦓ꠸ᓚᓚ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᘐ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

˖word⁚·ᛋᚠᚱᛘᚥᚥ≦˖⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·AHR⫩koom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·verbal adjective of arākum⁚⁘ long (time and size)⁘ tall⁚⁚·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Chant carries on through breath after breath⁚⁚refusing an early close𖬵·⁚.·˙ostensive⁚·
·⁚grimchain⁚·▽Xᛯ∂ᦓϟ〰ᦓᓚɾᓭᗅᚱ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᘓ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜temporal…absolute꩞ enoch꩜꩜Θ꩞⁚·

˖word⁚·ᚻ◇˖⁚.·˙word⁚·

·⁖origin…language꞉·Akkadian·⁖˙·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖˙·˙font꞉·
·⁖pronunciation꞉·pah+RAH+doom·⁖˙·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·˙cadence꞉·
·⁖non…ostensive꞉·to be scared꞉꞉terrified꞉꞉fearful꞉꞉disturbed꞉꞉upset꞉꞉restless꞉꞉·⁖˙·˙non…ostensive꞉·
·⁖ostensive꞉·Impact has already sounded beyond the wall꞉꞉yet tremor keeps moving through the
↪ musclesᘐ·⁖˙·˙ostensive꞉·
·⁖grimchain꞉·𐤌𐤒𐤌XÞ⊕I⅃Λ≳𐤓𐤅ᚠ·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup꞉˙
·⁖˙·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍ᘐ꞉ category꞉○–○₍court꞉꞉·
·⁖lookup office꞉○–○₍temporal…absolute꞉ enoch꞉○–○₍Θ꞉꞉·

₍꞉word꞉·ᚱᑎ꞉₎·⁖˙·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖˙·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖˙·˙font꞉·
·⁖pronunciation꞉·ROH·⁖˙·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·˙cadence꞉·
·⁖non…ostensive꞉·rest (state of inner peace)꞉꞉·⁖˙·˙non…ostensive꞉·
·⁖ostensive꞉·Last ripple fades꞉꞉ Quiet water keeps the pause that followsᘐ·⁖˙·˙ostensive꞉·
·⁖grimchain꞉·ᑌᛈ☆ᑎᛈ𐤓ᘐ♀ᛦ𐤍✷ᗝ·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup꞉˙
·⁖˙·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍Ж꞉ category꞉○–○₍court꞉꞉·
·⁖lookup office꞉○–○₍temporal…absolute꞉ enoch꞉○–○₍Θ꞉꞉·

₍꞉word꞉·वेग꞉₎·⁖˙·˙word꞉·

·⁖origin…language꞉·Sanskrit·⁖˙·˙origin…language꞉·
·⁖font꞉·Noto Sans Devanagari·⁖˙·˙font꞉·
·⁖pronunciation꞉·VAY+guh·⁖˙·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·˙cadence꞉·
·⁖non…ostensive꞉·speed⁖˙ velocity⁖˙ rapidity⁖˙ impetus⁖˙ momentum⁖˙ stream⁖˙ current⁖˙ agitation⁖˙
↪ impetuosity⁖˙ violence⁖˙ force⁖˙ energy⁖˙ activity⁖˙ determination⁖˙ power⁖˙ strength⁖˙ haste⁖˙ rashness⁖˙
↪ the flight of an arrow⁖˙ love⁖˙ passion⁖˙ delight⁖˙ pleasure⁖˙ expulsion of feces⁖˙ semen⁖˙ attack⁖˙
↪ paroxysm⁖˙ constipation꞉꞉·⁖˙·˙non…ostensive꞉·
·⁖ostensive꞉·String snaps forward and the arrow is already goneЖ·⁖˙·˙ostensive꞉·
·⁖grimchain꞉·سرᛉ𐤍✻ᚣᚠᚸ⊕ᛏᛉᑌᛁ·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup꞉˙
·⁖˙·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍Δ꞉ category꞉○–○₍court꞉꞉·
·⁖lookup office꞉○–○₍temporal…absolute꞉ enoch꞉○–○₍Θ꞉꞉·

₍꞉word꞉·ᚹᛋᚱᚦᚢᛉ꞉₎·⁖˙·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖˙·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖˙·˙font꞉·
·⁖pronunciation꞉·FOR+dhoom·⁖˙·˙pronunciation꞉·
·⁖cadence꞉·Temporal Absolute·⁖˙·˙cadence꞉·
·⁖non…ostensive꞉·Formerly⁖˙ in earlier times or long ago꞉꞉·⁖˙·˙non…ostensive꞉·
·⁖ostensive꞉·Soot darkens the old beam⁖˙ fresh neighboring wood has not yet known fireΔ·⁖˙·˙ostensive꞉·
·⁖grimchain꞉·سᛉ∀✻▽ᚠᚴᚢᚺ♋ᘐΘ✡ᛟ·⁖˙·˙grimchain꞉·
·⁖˙·˙lookup꞉˙
·⁖˙·˙glyphs꞉˙

·⁚glyph symbol꜀₀꜀₀∀꜀ category꜀₀꜀₀court꜀⁚·
·⁚lookup office꜀₀꜀₀temporal…absolute꜀ enoch꜀₀꜀₀Θ꜀⁚·

꜉word⁚·ተኩ᳟᳟꜉⁚·ˑword⁚·

·⁚origin…language⁚·Tocharian B·⁚·ˑorigin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·ˑfont⁚·
·⁚pronunciation⁚·KAHNYM·⁚·ˑpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·ˑcadence⁚·
·⁚non…ostensive⁚·To play⁚⁚be happy⁚⁚·⁚·ˑnon…ostensive⁚·
·⁚ostensive⁚·Laughter breaks out⁚⁚feet skip⁚⁚breath rises⁚⁚and the light movement keeps returning before
↪ quiet comes back∀·⁚·ˑostensive⁚·
·⁚grimchain⁚·ᚹᚹᛇᛏᚳᚴᚸᚱᛩᛉ◎ᚢ·⁚·ˑgrimchain⁚·
·⁚·ˑlookup⁚·
·⁚·ˑglyphs⁚·
·⁚glyph symbol꜀₀꜀₀∇꜀ category꜀₀꜀₀court꜀⁚·
·⁚lookup office꜀₀꜀₀temporal…absolute꜀ enoch꜀₀꜀₀Θ꜀⁚·

꜉word⁚·乳꜉⁚·ˑword⁚·

·⁚origin…language⁚·Tangut·⁚·ˑorigin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·ˑfont⁚·
·⁚pronunciation⁚·THWOH·⁚·ˑpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·ˑcadence⁚·
·⁚non…ostensive⁚·Dream⁚· an experience of images or events during sleep⁚⁚·⁚·ˑnon…ostensive⁚·
·⁚ostensive⁚·Sleep closes the eyes⁚⁚scenes rise and change within⁚⁚and waking ends them∇·⁚·ˑostensive⁚·
·⁚grimchain⁚·ᛕᚥᛏᚱᛯᛟᛋᛩᚢ·⁚·ˑgrimchain⁚·
·⁚·ˑlookup⁚·
·⁚·ˑglyphs⁚·
·⁚glyph symbol꜀₀꜀₀∇꜀ category꜀₀꜀₀court꜀⁚·
·⁚lookup office꜀₀꜀₀temporal…absolute꜀ enoch꜀₀꜀₀Θ꜀⁚·

꜉word⁚·वलय꜉⁚·ˑword⁚·

·⁚origin…language⁚·Sanskrit·⁚·ˑorigin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·ˑfont⁚·
·⁚pronunciation⁚·VUH+luh+yuh·⁚·ˑpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·ˑcadence⁚·
·⁚non…ostensive⁚·a bracelet⁚⁚armlet⁚⁚ring (worn by men and women on the wrist)⁚· (figuratively) waves (as
↪ compared to bracelets)⁚· a zone⁚⁚girdle (of a married woman)⁚· a line (of bees)⁚· a
↪ circle⁚⁚circumference⁚⁚round enclosure⁚⁚·⁚·ˑnon…ostensive⁚·
·⁚ostensive⁚·A finger follows the metal band around the wrist and arrives where it began∇·⁚·ˑostensive⁚·
·⁚grimchain⁚·ᵠᑫᛋᛞᛘᚥᚹ⊛ᚹᚻᚢ·⁚·ˑgrimchain⁚·
·⁚·ˑlookup⁚·
·⁚·ˑglyphs⁚·
·⁚glyph symbol꜀₀꜀₀∇ᚱ꜀ category꜀₀꜀₀court꜀⁚·
·⁚lookup office꜀₀꜀₀temporal…absolute꜀ enoch꜀₀꜀₀Θ꜀⁚·

꜉word⁚·ـــــــــــ꜉⁚·ˑword⁚·

·⁚origin…language⁚·Old Uyghur·⁚·ˑorigin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚·ˑfont⁚·
·⁚pronunciation⁚·anCHan·⁚·ˑpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·ˑcadence⁚·

·⁚non…ostensive:·slowly::quietly⁚· leisurely::orderly:::⁚·˙non…ostensive:·
·⁚ostensive:·Water moves so gently that the surface scarcely breaks as it advances▽ᖇ·⁚·˙ostensive:·
·⁚grimchain:·𐤋𐤄𐤇𐤉𐤅𐤏𐤐𐤅𐤉𐤆𐤈𐤎𐤏·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐤓» category◦–◦court»:·
·⁚lookup office◦–◦temporal…absolute» enoch◦–◦Θ»:·

·word:·𐫄𐫅𐫆𐫇·:·⁚·˙word:·

·⁚origin…language:·Akkadian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·ah+MAH+room·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·to see::behold::look at::observe::witness::examine::inspect::notice⁚· to experience::come
↪ across::find (an object::a site::etc::)⁚· to find after searching::to look up (information)::find (a
↪ result)::come to know::realise::learn by experience⁚· to read (a
↪ tablet::document::inscription::etc::)::⁚·˙non…ostensive:·
·⁚ostensive:·Each rereading shortens the hesitation until the signs are known at once𐤓·⁚·˙ostensive:·
·⁚grimchain:·𐤎𐤑𐤔𐤋𐤌𐤊𐤃𐤂𐤁𐤍𐤇·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐤅» category◦–◦court»:·
·⁚lookup office◦–◦temporal…absolute» enoch◦–◦Θ»:·

·word:·𐬉+𐬊·:·⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·LAHKLEH·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Pain::suffering:::⁚·˙non…ostensive:·
·⁚ostensive:·Wound does not pass with the first movement⁚· each later motion wakes it
↪ again𐤅·⁚·˙ostensive:·
·⁚grimchain:·𐤋𐤃𐤉𐤇𐤍𐤎𐤏𐤔𐤕𐤍𐤑𐤒𐤓·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐤅» category◦–◦court»:·
·⁚lookup office◦–◦temporal…absolute» enoch◦–◦Θ»:·

·word:·𐫐𐫑𐫒𐫓·:·⁚·˙word:·

·⁚origin…language:·Old Persian·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚·˙font:·
·⁚pronunciation:·DAR+gah·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Long⁚· extending a great distance or duration:::⁚·˙non…ostensive:·
·⁚ostensive:·Breath after breath carries the chant onward⁚· silence remains far away𐤅·⁚·˙ostensive:·
·⁚grimchain:·𐫍𐫎𐫏𐫐𐫑𐫒𐫓𐫔𐫕𐫖𐫗·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐤕» category◦–◦court»:·
·⁚lookup office◦–◦temporal…absolute» enoch◦–◦Θ»:·

⁖word:·✝⁘·˙word:·

·⁖origin…language:·Tocharian A·⁘·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁘·˙font:·
·⁖pronunciation:·KUHN·⁘·˙pronunciation:·
·⁖cadence:·Temporal Absolute·⁘·˙cadence:·
·⁖non…ostensive:·Melody⁝⁝tune⁖ (poetry) meter⁝⁝⁘·˙non…ostensive:·
·⁖ostensive:·Three notes rise⁝⁝fall⁝⁝and return to the same measured contour✠·⁘·˙ostensive:·
·⁖grimchain:·▫ↄᚲ1Ⅱ∀Ⓞஓ▽ᛈ𐦂0·⁘·˙grimchain:·
·⁘·˙lookup·˙
·⁘·˙glyphs·˙
·⁖glyph symbol○–○𐑉✠𐑊 category○–○𐑉court𐑊⁖·
·⁖lookup office○–○𐑉temporal…absolute𐑊 enoch○–○𐑉Θ𐑊⁖·

⁖word:·ᵬЄIᔑᏁN·⁘·˙word:·

·⁖origin…language:·Gothic·⁘·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁘·˙font:·
·⁖pronunciation:·BITHAN·⁘·˙pronunciation:·
·⁖cadence:·Temporal Absolute·⁘·˙cadence:·
·⁖non…ostensive:·To wait⁖ (with genitive) to await⁝⁝to wait for⁝⁝⁘·˙non…ostensive:·
·⁖ostensive:·Gate stays shut⁝⁝ Readiness waits outside until the latch finally moves✠·⁘·˙ostensive:·
·⁖grimchain:·▽▽ᓭᚻᛦᚲ✳ᓭᑭᓭ𐦂B·⁘·˙grimchain:·
·⁘·˙lookup·˙
·⁘·˙glyphs·˙
·⁖glyph symbol○–○𐑉I𐑊 category○–○𐑉court𐑊⁖·
·⁖lookup office○–○𐑉temporal…absolute𐑊 enoch○–○𐑉Θ𐑊⁖·

⁖word:·ᙖ ᚲ𐦂ᚦ ᚦ𐦂·⁘·˙word:·

·⁖origin…language:·Old Persian·⁘·˙origin…language:·
·⁖font:·Noto Sans Old Persian·⁘·˙font:·
·⁖pronunciation:·RAGAM·⁘·˙pronunciation:·
·⁖cadence:·Temporal Absolute·⁘·˙cadence:·
·⁖non…ostensive:·Quickly⁝⁝⁘·˙non…ostensive:·
·⁖ostensive:·Release and arrival nearly touch‡·⁘·˙ostensive:·
·⁖grimchain:·ﾒPX𐦂❄‡Z𐦂✻ᗰᗰ⬡ϒ·⁘·˙grimchain:·
·⁘·˙lookup·˙
·⁘·˙glyphs·˙
·⁖glyph symbol○–○𐑉⁑𐑊 category○–○𐑉court𐑊⁖·
·⁖lookup office○–○𐑉temporal…absolute𐑊 enoch○–○𐑉Θ𐑊⁖·

⁖word:·ᙖᚲ ᛏᛏ ᚴᚲ⁖·⁘·˙word:·

·⁖origin…language:·Old Persian·⁘·˙origin…language:·
·⁖font:·Noto Sans Old Persian·⁘·˙font:·
·⁖pronunciation:·NEYEY·⁘·˙pronunciation:·
·⁖cadence:·Temporal Absolute·⁘·˙cadence:·
·⁖non…ostensive:·Not⁝⁝⁘·˙non…ostensive:·
·⁖ostensive:·From darkness to returning light⁝⁝the wick never catches⁑·⁘·˙ostensive:·
·⁖grimchain:·ᛗ𐦂ᛗ◝ᘐ%'𐦂ᙖᚹᚹ▦∮ᛉᛉᐢᛇ·⁘·˙grimchain:·
·⁘·˙lookup·˙
·⁘·˙glyphs·˙

·⁚glyph symbol◦–◦𐑌␣ category◦–◦court␣⁚·
·⁚lookup office◦–◦temporal…absolute␣ enoch◦–◦Θ␣⁚·

⁖word⁚·𑀤𑀤𑁆𑀮𑀺𑁆𑀭⁖·.·ˈword⁚·

·⁚origin…language⁚·Tocharian A·⁚·.·ˈorigin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·.·ˈfont⁚·
·⁚pronunciation⁚·KUHKSOORNYEH·⁚·.·ˈpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·.·ˈcadence⁚·
·⁚non…ostensive⁚·Extinction⁚⁚·⁚·.·ˈnon…ostensive⁚·
·⁚ostensive⁚·Ember dims to one red breath⁚⁚loses it⁚⁚and never glows again𐑌·⁚·.·ˈostensive⁚·
·⁚grimchain⁚·ᔑ⟟𐑌𐑓ᚠ〰ᘐ❄ᑌᚿᓭ·⁚·.·ˈgrimchain⁚·
·⁚·.·ˈlookup⁚·
·⁚·.·ˈglyphs⁚·
·⁚glyph symbol◦–◦𐑕␣ category◦–◦court␣⁚·
·⁚lookup office◦–◦temporal…absolute␣ enoch◦–◦Θ␣⁚·

⁖word⁚·𐎢𐎺𐏁𐎼𐎹·⁖·.·ˈword⁚·

·⁚origin…language⁚·Akkadian·⁚·.·ˈorigin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·.·ˈfont⁚·
·⁚pronunciation⁚·SAHP+khoom·⁚·.·ˈpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·.·ˈcadence⁚·
·⁚non…ostensive⁚·verbal adjective of sapāḫum: scattered⁚⁚dispersed⁚⁚·⁚·.·ˈnon…ostensive⁚·
·⁚ostensive⁚·Wind worries the heap apart until no heap is left𐑕·⁚·.·ˈostensive⁚·
·⁚grimchain⁚·☆◎⏀ᙆ'Ƶᓄᚷᙆᛉᓄ✲ᓄ·⁚·.·ˈgrimchain⁚·
·⁚·.·ˈlookup⁚·
·⁚·.·ˈglyphs⁚·
·⁚glyph symbol◦–◦𐑔␣ category◦–◦court␣⁚·
·⁚lookup office◦–◦temporal…absolute␣ enoch◦–◦Θ␣⁚·

⁖word⁚·ꓘꓛᑋᒣᒪ·⁖·.·ˈword⁚·

·⁚origin…language⁚·Tocharian A·⁚·.·ˈorigin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·.·ˈfont⁚·
·⁚pronunciation⁚·UHSUHNKHYEH·⁚·.·ˈpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·.·ˈcadence⁚·
·⁚non…ostensive⁚·An incalculably long period of time⁚⁚·⁚·.·ˈnon…ostensive⁚·
·⁚ostensive⁚·Stars wheel through more returns than counting can finish𐑔·⁚·.·ˈostensive⁚·
·⁚grimchain⁚·♂ᛁ𐑉𐑑ᙢᑭ𐑌ᚣᛟ✲✲ᘐᛉᛯ⊙·⁚·.·ˈgrimchain⁚·
·⁚·.·ˈlookup⁚·
·⁚·.·ˈglyphs⁚·
·⁚glyph symbol◦–◦𐑖␣ category◦–◦court␣⁚·
·⁚lookup office◦–◦temporal…absolute␣ enoch◦–◦Θ␣⁚·

⁖word⁚·𗅋⁖·.·ˈword⁚·

·⁚origin…language⁚·Tangut·⁚·.·ˈorigin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·.·ˈfont⁚·
·⁚pronunciation⁚·N+LHAAH·⁚·.·ˈpronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·.·ˈcadence⁚·
·⁚non…ostensive⁚·Fall or drop·⁚ to descend suddenly or lose support⁚⁚·⁚·.·ˈnon…ostensive⁚·
·⁚ostensive⁚·Fruit lets go and keeps falling until earth stops it𐑖·⁚·.·ˈostensive⁚·
·⁚grimchain⁚·ᛈ𐑒ᛦ◎✳ᛪᘐᑭᓭᛋᛘ¥♀₩·⁚·.·ˈgrimchain⁚·

·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐊱꩜ category꩜꩜court꩜⁚·
·⁚lookup office꩜꩜temporal…absolute꩜ enoch꩜꩜Θ꩜⁚·

·word⁚·ᛦᚱᚾ·⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·FRAH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·from⁚⁚away from⁚⁚about⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·One bell+strike begins the measure⁚˙ every fading vibration comes after that first instant𐊱·⁚·˙ostensive⁚·
·⁚grimchain⁚·‹𐊹ᛁᛘ꩜Ƶᚺᛋ❁ᛝ⍜ᏧᎩ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐊹꩜ category꩜꩜court꩜⁚·
·⁚lookup office꩜꩜temporal…absolute꩜ enoch꩜꩜Θ꩜⁚·

·word⁚·ᛞ十ᒍ·⁚·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·AHKAHL·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Wish⁚⁚desire⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Closed blossom keeps turning toward a brightness not yet reached𐊹·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᔑᑭᔓᒧ卫ᛕᐁᛘIᑫᛋ❁·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐊰꩜ category꩜꩜court꩜⁚·
·⁚lookup office꩜꩜temporal…absolute꩜ enoch꩜꩜Θ꩜⁚·

·word⁚·𐊱ᚲ·⁚·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·bah+LAH+too·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Live⁚˙ remain alive⁚˙ regain life or health⁚˙ recover⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Breath returns⁚⁚then keeps returning𐊰·⁚·˙ostensive⁚·
·⁚grimchain⁚·∀ᘺᒧ𐊹✡Ↄ♒⊕≍ᚼᚏᛗᚼ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐊹꩜ category꩜꩜court꩜⁚·
·⁚lookup office꩜꩜temporal…absolute꩜ enoch꩜꩜Θ꩜⁚·

·word⁚·ᛏᑎᚹᛋ·⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·TOH+gah·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·

·⁚non…ostensive⁚·to draw⁖⁖pull⁖⁖stretch⁖⁘⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Rope straightens as the gap widens⁖⁖and drawing lasts until the pull can go no
↪ farther𑜦⁘⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚱΘ𑜫𑜨𑜣𑜭𑜮❂𑜯𑜰𑜱⁘⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𑜤𑜧𑜥 category◦–◦𑜤court𑜥⁚·
·⁚lookup office◦–◦𑜤temporal…absolute𑜥 enoch◦–◦𑜤Θ𑜥⁚·

⁖word⁚·भूमि⁖.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·bhoo+mi·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·earth⁖⁖soil⁖⁖ground⁚˙ Earth personified as Bhumi⁚˙ divisions of the world⁚˙ compare भूमत्रिय
↪ (bhūmi+traya)⁚˙ territory⁖⁖country⁖⁖district⁚˙ place⁖⁖situation⁚˙ position⁖⁖posture⁖⁖attitude⁚·
↪ part⁖⁖personification (played by an actor)⁚˙ floor of a house⁖⁖story⁚˙ area⁚˙ the base of any
↪ geometrical figure⁚˙ a step⁖⁖degree⁖⁖stage (with Buddhists there are 10 or 13 stages of existence
↪ or perfection)⁚˙ Bhumi⁚˙ extent⁖⁖limit⁚˙ matter⁖⁖subject⁖⁖object⁖⁖receptacle i⁖⁖e⁖⁖ fit object or person
↪ for⁖⁘⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Seedling gives way to flower⁖⁖flower to grain⁚˙ each stage has its own season before the
↪ next𑜦⁚.·˙ostensive⁚·
·⁚grimchain⁚·▽𑜪𑜬𑜶𑜷𑜸Ϟ𑜹𑜺𑜻𑜼𑜽𑜾I·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𑜤𑜫𑜥 category◦–◦𑜤court𑜥⁚·
·⁚lookup office◦–◦𑜤temporal…absolute𑜥 enoch◦–◦𑜤Θ𑜥⁚·

⁖word⁚·सकृत्⁖.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·suh+krit·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·once⁖⁖one time⁚˙ at once⁖⁖suddenly⁖⁖forthwith⁖⁖immediately⁚˙ once⁖⁖formerly⁖⁖ever⁚˙ once and
↪ for all⁖⁖for ever⁚˙ at once⁖⁖together⁚˙ alternative spelling of शकृत् (śakṛt)⁖⁘⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One drumbeat⁖⁖then silence𑜫⁚.·˙ostensive⁚·
·⁚grimchain⁚·𑜢𑜩𑜪𑜫𑜬𑜭𑜮𑜯𑜰𑜱𑜲𑜳𑜴⁘⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𑜤𑜩𑜥 category◦–◦𑜤court𑜥⁚·
·⁚lookup office◦–◦𑜤temporal…absolute𑜥 enoch◦–◦𑜤Θ𑜥⁚·

⁖word⁚·ㅅ𑜵⁖.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·TOOK·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To hide oneself⁖⁖be hidden⁚˙ to hide or conceal someone or something⁖⁘⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Lid closes⁖⁖and what lies beneath stays unseen until reopening𑜩⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚼ𑜫ㅋ▽𑜦𑜧𑜨▽▽▽𑜩'Ƙ𑜫𑜬⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙

·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𐎫 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉⁚·

꞉word⁚·𐎭 𐎺 𐏁 𐎯꞉⁚·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚·˙font⁚·
·⁚pronunciation⁚·vah+HAHR·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Spring⁚⁚springtime⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Bare branch gives way to tender green as the opening season arrives𐎫·⁚·˙ostensive⁚·
·⁚grimchain⁚·Θ𐏂☆𐎻𐎫𐎲𐎻𐏁𐏂𐏁𐎹·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𐎮 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉⁚·

꞉word⁚·ᚾᚨᚺᛏᛊ꞉⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·NAHTS·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Night⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Darkness begins with the last light and holds until the first returns𐎮·⁚·˙ostensive⁚·
·⁚grimchain⁚·z✡𐎺𐏂X𐎸ℂ𐎺‹𐎹ₒ❁lΦ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𐎰 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉⁚·

꞉word⁚·हमि꞉⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·HEE+muh·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·cold⁚⁚frost⁚· the cold season⁚⁚winter⁚⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·White crystals return over exposed water and grass⁚⁚holding until the thaw comes
↪ back𐎰·⁚·˙ostensive⁚·
·⁚grimchain⁚· ⊛Φ♂ΘIᘐ✷ᛉᚠ𐎺‹𐎼Θ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𐎪 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉temporal…absolute꞉ enoch꞉꞉Θ꞉⁚·

꞉word⁚·𑀳𑁄꞉⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·EEMEH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Consciousness⁚⁚awareness⁚⁚thought⁚⁚memory⁚⁚⁚·⁚·˙non…ostensive⁚·

·⁚ostensive⁝·Light reaches the eye⁝⁝sound reaches the ear⁝⁝pressure reaches skin⁝⁝each arriving into an unbroken waking succession↶·⁚·˙ostensive⁝·
·⁚grimchain⁝·♂Ⱥↄ𓆏𓃰⛢𐌺∫♈𓆑⁜𓅓·⁚·˙grimchain⁝·
·⁚·˙lookup⁝˙
·⁚·˙glyphs⁝˙
·⁚glyph symbol₀₋₀₊𓃰₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

₊word⁝·ᔐᏁᑎᑕᎶᏁᑎ⁝₊·˙word⁝·

·⁚origin…language⁝·Gothic·⁚·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚·˙font⁝·
·⁚pronunciation⁝·DAWPJAN·⁚·˙pronunciation⁝·
·⁚cadence⁝·Temporal Absolute·⁚·˙cadence⁝·
·⁚non…ostensive⁝·To dip⁝⁝immerse⁝˙ (Christianity) to baptize⁝⁝·⁚·˙non…ostensive⁝·
·⁚ostensive⁝·Dry skin breaks the surface⁝⁝disappears beneath it for a held breath⁝⁝then rises out again𓃰⁝·⁚·˙ostensive⁝·
·⁚grimchain⁝·∀⏢⚷𓂋⏁♈⛢∇⊕ᗡ0❁·⁚·˙grimchain⁝·
·⁚·˙lookup⁝˙
·⁚·˙glyphs⁝˙
·⁚glyph symbol₀₋₀₊ᑎ₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

₊word⁝·⛐𓆑⁝₊·˙word⁝·

·⁚origin…language⁝·Akkadian·⁚·˙origin…language⁝·
·⁚font⁝·Noto Sans Cuneiform·⁚·˙font⁝·
·⁚pronunciation⁝·ah+RAH+koom·⁚·˙pronunciation⁝·
·⁚cadence⁝·Temporal Absolute·⁚·˙cadence⁝·
·⁚non…ostensive⁝·to be(come) long⁝⁝last long⁝˙ to be delayed⁝⁝too long⁝⁝drag on⁝⁝·⁚·˙non…ostensive⁝·
·⁚ostensive⁝·Bell keeps sounding past the point where silence should have come↶·⁚·˙ostensive⁝·
·⁚grimchain⁝·∀ᒧ⬡ᵠᛯᔕ♋Θ𓆏♈∀♀Ⅹ·⁚·˙grimchain⁝·
·⁚·˙lookup⁝˙
·⁚·˙glyphs⁝˙
·⁚glyph symbol₀₋₀₊⚷₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

₊word⁝·ᒥᏁᑎᛦᔕᏁᑎ⁝₊·˙word⁝·

·⁚origin…language⁝·Gothic·⁚·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚·˙font⁝·
·⁚pronunciation⁝·GADOWRSAN·⁚·˙pronunciation⁝·
·⁚cadence⁝·Temporal Absolute·⁚·˙cadence⁝·
·⁚non…ostensive⁝·To dare⁝˙ to be brave⁝⁝to be courageous⁝⁝·⁚·˙non…ostensive⁝·
·⁚ostensive⁝·Danger does not leave⁝⁝but neither does the forward stance⚷·⁚·˙ostensive⁝·
·⁚grimchain⁝·ϫ𓆑𓂸𓆏ᘓ𓃰𓅓ᗷ◎❁∇·⁚·˙grimchain⁝·
·⁚·˙lookup⁝˙
·⁚·˙glyphs⁝˙
·⁚glyph symbol₀₋₀₊𓆑₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

₊word⁝·ᚕᚕᚖᛘ⁝₊·˙word⁝·

·⁚origin…language:·Mycenaean Greek·⁚.·˙origin…language:·
·⁚font:·Noto Sans Linear B·⁚.·˙font:·
·⁚pronunciation:·dedemEHnos·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·bound (mediopassive perfect participle)⁖⁖⁚.·˙non…ostensive:·
·⁚ostensive:·Knot stays tight through every pull and slackening𐋣·⁚.·˙ostensive:·
·⁚grimchain:·▽ⱺ⁕⊕ᚿᛝ𐋏ᛋᵠ∫♂ᚺᚸ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐋏꞉ category◦–◦court꞉·
·⁚lookup office◦–◦temporal…absolute꞉ enoch◦–◦⊖꞉·

·word:·𐋏꞉.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·JWEH·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Swallow·˙ to pass food or drink from the mouth into the throat⁖⁖⁚.·˙non…ostensive:·
·⁚ostensive:·Mouthful passes the tongue and is gone into the throat𐋏·⁚.·˙ostensive:·
·⁚grimchain:·≈ꓧꓤᑎ☿ᓰ∫ⵑᑎᚷⵔᑎᓀ𐋏·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐋣꞉ category◦–◦court꞉·
·⁚lookup office◦–◦temporal…absolute꞉ enoch◦–◦⊖꞉·

·word:·ꕯ□ㅅ𑓐꞉.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·WUHTK·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To separate⁖⁖distinguish·˙ to decide⁖⁖⁚.·˙non…ostensive:·
·⁚ostensive:·Two grains are turned⁖⁖weighed⁖⁖compared⁖⁖and at last one is set apart𐋣·⁚.·˙ostensive:·
·⁚grimchain:·☽ᒧTⵔ𐋣ᚸᛦ≳ᚿᐱᚸ𐋏·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ᚣ꞉ category◦–◦court꞉·
·⁚lookup office◦–◦temporal…absolute꞉ enoch◦–◦⊖꞉·

·word:·𐋐𐋐꞉.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·LAYLU·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Night⁖⁖⁚.·˙non…ostensive:·
·⁚ostensive:·Moonless dark begins after the last light and persists until the pale edge
↪ returnsᚣ·⁚.·˙ostensive:·
·⁚grimchain:·ᚸᚿⵔᚷ𐋣ᚮ⧗ᚻᚿᐯᐟ𐋣ᚦ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ᚺ꞉ category◦–◦court꞉·
·⁚lookup office◦–◦temporal…absolute꞉ enoch◦–◦⊖꞉·

˖word꞉·ᒍᚸ̄ᛏ̄˖·˙word꞉·

·⁚origin…language꞉·Tocharian A·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·LAHTK·⁚·˙pronunciation꞉·
·⁚cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·To cut off⁚⁚·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·One stroke of the blade ends the fibre forever at that line𑁋·⁚·˙ostensive꞉·
·⁚grimchain꞉·𑀦𑀸𑁆𑀲𑀺𑀬𑀫𑀻𑀢𑀼𑀭𑀸𑀫𑀽𑀮𑁆𑀯𑁄𑀆𑀓𑀸·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍A₎ category○–○₍court₎꞉·
·⁚lookup office○–○₍temporal…absolute₎ enoch○–○₍Θ₎꞉·

˖word꞉·घण्टा˖·˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚·˙font꞉·
·⁚pronunciation꞉·GHUHN⟊taa·⁚·˙pronunciation꞉·
·⁚cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Bell⁚⁚·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Bronze is struck once⁖ its voice blooms and thins awayA·⁚·˙ostensive꞉·
·⁚grimchain꞉·𐌾⌬𑀭𑁂𐌺𑁍𑀭𑀸⁕𑀦𑀸·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍B₎ category○–○₍court₎꞉·
·⁚lookup office○–○₍temporal…absolute₎ enoch○–○₍Θ₎꞉·

˖word꞉·𐎌𐎍𐏁𐎹𐏂˖·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚·˙font꞉·
·⁚pronunciation꞉·shah⟊NOOM·⁚·˙pronunciation꞉·
·⁚cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·verbal adjective of šanûm꞉⁖ second (of two or more)⁚⁚next⁚⁚other⁖ (nominalised)
↪ something else⁚⁚another⁚⁚·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·One vessel is set down⁚⁚ Another comes nextB·⁚·˙ostensive꞉·
·⁚grimchain꞉·𐎀𐎹Θ𐎢⁕𐏁∫◎𐎃𐎍𐎲𐎰·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍Ↄ₎ category○–○₍court₎꞉·
·⁚lookup office○–○₍temporal…absolute₎ enoch○–○₍Θ₎꞉·

˖word꞉·ᛈᛋ˖·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚·˙font꞉·
·⁚pronunciation꞉·FOW·⁚·˙pronunciation꞉·
·⁚cadence꞉·Temporal Absolute·⁚·˙cadence꞉·
·⁚non…ostensive꞉·To get or obtain⁚⁚·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Empty hand reaches and closes carrying weightↃ·⁚·˙ostensive꞉·
·⁚grimchain꞉·Iᛈᚺᛜᚱᚳᚢᚩᛃᚠ▽▫𑀫𑀢·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙

·⁚glyph symbol₀₋₀𐞤₎ category₀₋₀court₎⁚·
·⁚lookup office₀₋₀temporal…absolute₎ enoch₀₋₀Θ₎⁚·

·word⁚·𐎠𐎡𐎢𐎣𐎤𐎥𐎦𐎧·⁚·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚·˙font⁚·
·⁚pronunciation⁚·HAKARAMCIY·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Once⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·One stroke sounds⁚⁚ No second follows𐎤·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐏓𐏌𐏐𐏑𐏋𐏉𐏇𐏍𐏊𐏃𐏁·⁚·˙grimchain⁚·
·⁚·˙lookup⁚·
·⁚·˙glyphs⁚·
·⁚glyph symbol₀₋₀𐎭₎ category₀₋₀court₎⁚·
·⁚lookup office₀₋₀temporal…absolute₎ enoch₀₋₀Θ₎⁚·

·word⁚·𒀸𒁹𒄿𒌋𒉿·⁚·˙word⁚·

·⁚origin…language⁚·Hittite·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·KHUDAHK·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Immediately⁚⁚suddenly⁚· straightaway⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Darkness becomes full light in a single flash𐎭·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐎋𐎌𐎇𐎄𐎆𐎅𐎂𐎁𐎀𐎿𐎾·⁚·˙grimchain⁚·
·⁚·˙lookup⁚·
·⁚·˙glyphs⁚·
·⁚glyph symbol₀₋₀𐎬₎ category₀₋₀court₎⁚·
·⁚lookup office₀₋₀temporal…absolute₎ enoch₀₋₀Θ₎⁚·

·word⁚·कला·⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·KUH÷lah·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·a small part of anything⁚⁚any single part or portion of a whole⁚⁚especially a sixteenth
↪ part⁚· a digit or one÷sixteenth of the moon’s diameter⁚· a symbolical expression for the number
↪ sixteen⁚· interest on a capital (considered as a certain part of it)⁚· a division of time (said to
↪ be 1/900 of a day or 1÷6 minutes⁚· or 1/1800 of a day or 0÷8 minutes commentator or commentary
↪ on⁚· or 2 minutes and 26 54/201 seconds⁚· or 1 minute and 35 205/301⁚⁚seconds⁚⁚or 8 seconds)⁚· the
↪ sixtieth part of one÷thirtieth of a zodiacal sign⁚⁚a minute of a degree⁚· (prosody) a syllabic
↪ instant⁚· a term for the seven substrata of the elements or Dhatus of the human body (viz⁚⁚
↪ flesh⁚⁚blood⁚⁚fat⁚⁚phlegm⁚⁚urine⁚⁚bile⁚⁚and semen)⁚· an atom (there are 3015 kalā÷s or atoms in every
↪ one of the six dhātu÷s⁚⁚not counting the rasa÷⁚⁚therefore in all 18090)⁚· (with pāśupata÷s) the
↪ elements of the gross or material world⁚· an embryo shortly after conception⁚· a designation of
↪ the three constituent parts of a sacrifice (viz⁚⁚ mantra÷⁚⁚dravya÷ and śraddhā÷)⁚· the menstrual
↪ discharge⁚· ignorance⁚· a low and sweet tone⁚· name of a grammatical commentary⁚⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·A measured beat fits inside a longer chant⁚⁚and equal beats gather into the larger
↪ round𐎬·⁚·˙ostensive⁚·
·⁚grimchain⁚·𑁆𑀭𑀸𑀯𑀡𑁇𑀓𑀺𑀦𑀸𑀳𑁄𑁆𑀫𑀡·⁚·˙grimchain⁚·
·⁚·˙lookup⁚·
·⁚·˙glyphs⁚·

·⁚glyph symbol꩜꩜ᛁ꩝ category꩜꩜ᵗcourt꩝⁚·
·⁚lookup office꩜꩜ᵗtemporal…absolute꩝ enoch꩜꩜ᵗ⊖꩝⁚·

·:word:·ᛒᛁᚱᛁᚼ:·.·ˊword:·

·⁚origin…language:·Old Norse·⁚.·ˊorigin…language:·
·⁚font:·Noto Sans Runic·⁚.·ˊfont:·
·⁚pronunciation:·BER+yah·⁚.·ˊpronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·ˊcadence:·
·⁚non…ostensive:·To beat or fight·:·⁚.·ˊnon…ostensive:·
·⁚ostensive:·Blow answers blow until the struggle loses its rhythmı·⁚.·ˊostensive:·
·⁚grimchain:·ᕽᓂᚹᙏᘐᘓ✮ᚺᙢᘯᗡ❂·⁚.·ˊgrimchain:·
·⁚.·ˊlookup:·
·⁚.·ˊglyphs:·
·⁚glyph symbol꩜꩜ᗡ꩝ category꩜꩜ᵗcourt꩝⁚·
·⁚lookup office꩜꩜ᵗtemporal…absolute꩝ enoch꩜꩜ᵗ⊖꩝⁚·

·:word:·ᓴᑎᚾᚹᚼ:·.·ˊword:·

·⁚origin…language:·Old Norse·⁚.·ˊorigin…language:·
·⁚font:·Noto Sans Runic·⁚.·ˊfont:·
·⁚pronunciation:·SYNGYA·⁚.·ˊpronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·ˊcadence:·
·⁚non…ostensive:·To sing·:·⁚.·ˊnon…ostensive:·
·⁚ostensive:·Voice enters pitch on the first breath and leaves it with the lastᗡ·⁚.·ˊostensive:·
·⁚grimchain:·ᚠᚩᛇᛈᛕ ⊛ ◎ᘐᛯᛋᛘ⬡❂·⁚.·ˊgrimchain:·
·⁚.·ˊlookup:·
·⁚.·ˊglyphs:·
·⁚glyph symbol꩜꩜ᛕ꩝ category꩜꩜ᵗcourt꩝⁚·
·⁚lookup office꩜꩜ᵗtemporal…absolute꩝ enoch꩜꩜ᵗ⊖꩝⁚·

·:word:·ᚿᚤᛁᚴᚺ:·.·ˊword:·

·⁚origin…language:·Gothic·⁚.·ˊorigin…language:·
·⁚font:·Noto Sans Gothic·⁚.·ˊfont:·
·⁚pronunciation:·THEYERH·⁚.·ˊpronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·ˊcadence:·
·⁚non…ostensive:·(+accusative) through·:· (+accusative) by·:·⁚.·ˊnon…ostensive:·
·⁚ostensive:·Bell begins and ends inside a tone already sounding and still continuingᛯ·⁚.·ˊostensive:·
·⁚grimchain:·ꕯᗷᛝ✮ᚱᕊᖶ✮⊖ᚋᚠᓄᔕ❀·⁚.·ˊgrimchain:·
·⁚.·ˊlookup:·
·⁚.·ˊglyphs:·
·⁚glyph symbol꩜꩜ᒣ꩝ category꩜꩜ᵗcourt꩝⁚·
·⁚lookup office꩜꩜ᵗtemporal…absolute꩝ enoch꩜꩜ᵗ⊖꩝⁚·

·:word:·𒌋𒄿𒋾𒉡𒃻𒂊:·.·ˊword:·

·⁚origin…language:·Akkadian·⁚.·ˊorigin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·ˊfont:·
·⁚pronunciation:·sah+MAH+doom·⁚.·ˊpronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·ˊcadence:·
·⁚non…ostensive:·to yoke·:harness·:tie·:attach·:· to make ready·:prepare·:·⁚.·ˊnon…ostensive:·
·⁚ostensive:·Yoke and beam are fitted before the first step is takenᒣ·⁚.·ˊostensive:·
·⁚grimchain:·ᖁᔑᖘᖵᕊᑦᔭᗅᖴᘐᕮᐁᐁᐁ·⁚.·ˊgrimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₊ᛊ₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

⁖word:·ᑌĪꞱĀẍ̃Ū⁖·⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·PREHNTSEH·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Instant⁖⁖moment⁖⁖·⁚·˙non…ostensive:·
·⁚ostensive:·Flint sparks once between darkness and flameᛊ·⁚·˙ostensive:·
·⁚grimchain:·ᘐ▫ᖶᙓᘒᑭᔕᔑᘐᘏᘐᚢ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₊ᛊ₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

⁖word:·𐎠𐎡 𐎢𐎢 𐏁𐎴⁖·⁚·˙word:·

·⁚origin…language:·Old Persian·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚·˙font:·
·⁚pronunciation:·EYEVA·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·One⁖⁖·⁚·˙non…ostensive:·
·⁚ostensive:·A drum sounds between two silences⁖⁖with no answering beat beside itᛊ·⁚·˙ostensive:·
·⁚grimchain:·✲ᨐᛤᘔᘐᘏ⊛ ✫ᚻᚺᛯᚦ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₊✳₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

⁖word:·ᛗX⁖·⁚·˙word:·

·⁚origin…language:·Old Turkic·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚·˙font:·
·⁚pronunciation:·EUD·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Time (a point⁖⁖or a period)⁖⁖·⁚·˙non…ostensive:·
·⁚ostensive:·Shadow lengthens⁖⁖ Ash cools⁖⁖ Edges wear⁖⁖ The moon comes backᛯ·⁚·˙ostensive:·
·⁚grimchain:·ᘐᑭᛏᚻᚲᚱᘐᚩᛉᛏᛏᛏ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₊✳₎ category₀₋₀₊court₎⁚·
·⁚lookup office₀₋₀₊temporal…absolute₎ enoch₀₋₀₊Θ₎⁚·

⁖word:·ᛟᛋᛈᛗ⁖·⁚·˙word:·

·⁚origin…language:·Old Turkic·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚·˙font:·
·⁚pronunciation:·beng⟊GYOO·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·eternal⁖⁖everlasting⁖⁖forever⁖⁖·⁚·˙non…ostensive:·

·⁚ostensive⁚·Every completed circuit gives way to another⁕·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⊕0ᑎ᛬ᛝ⁕⊛ᛝϟϥ🢒❂·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀ᛯ꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀temporal…absolute꜆ enoch꜀꜀Θ꜆⁚·

⁖word⁚·ᔑᔐᒧ⁖⁚.·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚.·˙font⁚·
·⁚pronunciation⁚·BOHL·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To become or happen⁚⁘⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Hard green fruit softens into ripeness from within⁕·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᔐᔑᓍᑎᔕᓀ'φ᛬ᛯ᛫ᚾᛏ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀ᛰ꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀temporal…absolute꜆ enoch꜀꜀Θ꜆⁚·

⁖word⁚·𐎠𐎢 𐎱𐎢 𐎹𐎢⁖⁚.·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚.·˙font⁚·
·⁚pronunciation⁚·UPAH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Under⁚⁚below [with accusative]⁖ in the reign of [with accusative]⁚⁘⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Courses above are rebuilt and weathered away⁖ the lower stone keeps its place beneath
↪ them all⩛·⁚.·˙ostensive⁚·
·⁚grimchain⁚·φ❂ϗ⊕ᛏ❄ᚴ🢒∂ᛇᛯϗ⁜·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀ᛉ꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀temporal…absolute꜆ enoch꜀꜀Θ꜆⁚·

⁖word⁚·नित्य⁖⁚.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·NEET÷yuh·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·one's own⁖ innate⁚⁚native⁖ eternal⁖ everlasting⁚⁚continual⁖ constantly dwelling or
↪ engaged in⁖ invariable⁚⁚regular⁚⁚fixed⁚⁚ordained⁖ necessary⁚⁚obligatory⁚⁚peremptory⁚⁘⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Moonrise meets the same notch in the ridge after each full roundᛉ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛗ🢒᛫ᚴΘᚠᑐ⩛ᛘ⊙᛬ᛯ᛫·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol꜀꜀0꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀temporal…absolute꜆ enoch꜀꜀Θ꜆⁚·

⁖word⁚·उषस्⁖⁚.·˙word⁚·

·origin…language:·Sanskrit·.·origin…language:·
·font:·Noto Sans Devanagari·.·font:·
·pronunciation:·OO+shus·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·Dawn or morning light::·.·non…ostensive:·
·ostensive:·Pale light gathers while the stars are still fading0·.·ostensive:·
·grimchain:·ᯓ✙ϒ✡⩛ᔕ0ᘉᨗ⩛ₒ❁○·.·grimchain:·
·.·lookup:·
·.·glyphs:·
·glyph symbol○–○₍ᛯ꞉ category○–○₍court꞉·
·lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

·word:·ᚻ·.·word:·

·origin…language:·Tocharian A·.·origin…language:·
·font:·Noto Sans Brahmi·.·font:·
·pronunciation:·KEH·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·Chance::opportunity::·.·non…ostensive:·
·ostensive:·Low tide leaves the gate usable for a little while::then water takes the passage
↪ backᛯ·.·ostensive:·
·grimchain:·ᚕᑉᗷᓭᔑᚴϒ✳⩛ᚱᛯᖉ·.·grimchain:·
·.·lookup:·
·.·glyphs:·
·glyph symbol○–○₍ǂ꞉ category○–○₍court꞉·
·lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

·word:·तर्हि·.·word:·

·origin…language:·Sanskrit·.·origin…language:·
·font:·Noto Sans Devanagari·.·font:·
·pronunciation:·TUHr+hee·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·then::·.·non…ostensive:·
·ostensive:·One tone ends·: then another beginsǂ·.·ostensive:·
·grimchain:·♀ᛋᚼᘐᚱᛞᚴ‹ᔕᚷ⬡ᔦ·.·grimchain:·
·.·lookup:·
·.·glyphs:·
·glyph symbol○–○₍ᙏ꞉ category○–○₍court꞉·
·lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

·word:·ᛯᑯ·.·word:·

·origin…language:·Tocharian A·.·origin…language:·
·font:·Noto Sans Brahmi·.·font:·
·pronunciation:·UHCHEE·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·Starting with·: hither::to here::·.·non…ostensive:·
·ostensive:·First stroke opens the sequence::and every later mark follows from itᙏ·.·ostensive:·
·grimchain:·ᔦΘᑭ⅄۞ϒǂᔕᛯᚻᛏᚱ·.·grimchain:·
·.·lookup:·
·.·glyphs:·
·glyph symbol○–○₍ᖉ꞉ category○–○₍court꞉·
·lookup office○–○₍temporal…absolute꞉ enoch○–○₍Θ꞉·

:word:·𒀸𒋗𒆷𒁹:·.·word:·

·origin…language:·Akkadian·.·origin…language:·
·font:·Noto Sans Cuneiform·.·font:·
·pronunciation:·WAHR+kee·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·after·· behind:::·.·non…ostensive:·
·ostensive:·Flame is gone·· smoke and heat linger afterward𐎗·.·ostensive:·
·grimchain:·ᛉ✡▽Ϙ'∀ϕᚏᛋᛟᚾᚲ·.·grimchain:·
·.·lookup·
·.·glyphs·
·glyph symbol○-○𐎗 category○-○court𐎚:·
·lookup office○-○temporal…absolute𐎚 enoch○-○Θ𐎚:·

:word:·नपात्:.·word:·

·origin…language:·Sanskrit·.·origin…language:·
·font:·Noto Sans Devanagari·.·font:·
·pronunciation:·nuh+paht·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·descendant::offspring::son·· grandson:::·.·non…ostensive:·
·ostensive:·Older stem gives rise to a new branch::and life continues into what follows𐎗·.·ostensive:·
·grimchain:·ᒧℏᘔ𓂀ꞫIЬΠXΨᔨ𓆑ᛯ꩜·.·grimchain:·
·.·lookup·
·.·glyphs·
·glyph symbol○-○𐎘 category○-○court𐎚:·
·lookup office○-○temporal…absolute𐎚 enoch○-○Θ𐎚:·

:word:·𐎢𐎢 𐏁𐎡 𐎢 𐎢𐎢:.·word:·

·origin…language:·Old Persian·.·origin…language:·
·font:·Noto Sans Old Persian·.·font:·
·pronunciation:·AVADAH·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·Then::there:::·.·non…ostensive:·
·ostensive:·Last stroke dries before sealing wax is pressed𐎘·.·ostensive:·
·grimchain:·𓂀ƵA⁕♀ϺɻHϞᕱᔑ⊥·.·grimchain:·
·.·lookup·
·.·glyphs·
·glyph symbol○-○𐎫 category○-○court𐎚:·
·lookup office○-○temporal…absolute𐎚 enoch○-○Θ𐎚:·

:word:·𐎤𐎗:.·word:·

·origin…language:·Akkadian·.·origin…language:·
·font:·Noto Sans Cuneiform·.·font:·
·pronunciation:·NAH+khoom·.·pronunciation:·
·cadence:·Temporal Absolute·.·cadence:·
·non…ostensive:·to rest::take a rest·· (stative) to be at
↪ rest::slow::still::peaceful::pacified:::·.·non…ostensive:·
·ostensive:·Ripples die away and the water keeps still until something disturbs it
↪ again𐎫·.·ostensive:·
·grimchain:·ᚖᚢᙏᚦᛋᔓᚯᛯᛘᛘᛉ♂ᚏ▽·.·grimchain:·
·.·lookup·

·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀𓄿₀ category₀₋₀court₀⁚·
·⁚lookup office₀₋₀temporal…absolute₀ enoch₀₋₀Θ₀⁚·

·word:·𒀭𒂗𒍪𒉿𒁹𒁹·⁚·˙word:·

·⁚origin…language:·Akkadian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·geet⁄MAH⁄loom·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·equal (in size::rank::etc::)⁚˙ (poetic) noble::perfect::·⁚·˙non…ostensive:·
·⁚ostensive:·Two lamps are lit together and go dark togetherx·⁚·˙ostensive:·
·⁚grimchain:·ϒ𐌅ᵹ❮Θ𐦟ᵒ𐦂ᵒ◡𐌆ᘏᘐ𐦂𐦂·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀𐦂₀ category₀₋₀court₀⁚·
·⁚lookup office₀₋₀temporal…absolute₀ enoch₀₋₀Θ₀⁚·

·word:·𓂓·⁚·˙word:·

·⁚origin…language:·Egyptian·⁚·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚·˙font:·
·⁚pronunciation:·HAH⁄rab·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·A festival or feast::·⁚·˙non…ostensive:·
·⁚ostensive:·Tools are laid aside when the feast begins and taken up only after it
↪ dispersesᵽ·⁚·˙ostensive:·
·⁚grimchain:·⊙𐦂𐌿ϷϷϽ1◌ʃ𐦂𐦂𐦂·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀𐦂₀ category₀₋₀court₀⁚·
·⁚lookup office₀₋₀temporal…absolute₀ enoch₀₋₀Θ₀⁚·

·word:·ፈጸወ·⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·fah⁄TSAH⁄mah·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·To complete::finish::fulfill::or carry out::·⁚·˙non…ostensive:·
·⁚ostensive:·Last stroke closes the figure:: Nothing more is neededᘐ·⁚·˙ostensive:·
·⁚grimchain:·Ⅱ𐦂ᵹ𐦂⊛ᵹ𐦂⊙ᔕሐΦΘ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀𐦂₀ category₀₋₀court₀⁚·
·⁚lookup office₀₋₀temporal…absolute₀ enoch₀₋₀Θ₀⁚·

·word:·𗹦·⁚·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·R⁄MUHR·⁚·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚·˙cadence:·
·⁚non…ostensive:·Observe::watch::inspect::or examine⁚˙ to look attentively::·⁚·˙non…ostensive:·

·⁚ostensive⁚·Fixed gaze holds while every change in the distant field passes through it⨿·⁚·˙ostensive⁚·
·⁚grimchain⁚·⬡ᵚᔓᔑᵒᵌᘔᕫᕒᐁᐯX🜍ᚱᚱ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐑈⨿𐑉 category⟜⟜court𐑉⁚·
·⁚lookup office⟜⟜𐑈temporal…absolute𐑉 enoch⟜⟜𐑈Θ𐑉⁚·

·word⁚·ᓚᖶ̄⁚·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·SAHK·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To stay⁚⁚remain⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Pressure comes⁚⁚weather turns⁚⁚and the stone never leaves its socket⨿·⁚·˙ostensive⁚·
·⁚grimchain⁚·۞ΔĴℂᛦᛯᚣᛉΔᚡᛩᘐ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐑈▽𐑉 category⟜⟜court𐑉⁚·
·⁚lookup office⟜⟜𐑈temporal…absolute𐑉 enoch⟜⟜𐑈Θ𐑉⁚·

·word⁚·𐋏𐋏𐋏⁚·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·OOR÷room·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·day⁚⁚daylight (opposed to night)⁘ early morning⁘ tomorrow⁚⁚next day⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Sun clears the horizon⁚⁚crosses the open sky⁚⁚and is gone before darkness closes
↪ in▽·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᘮᛉᓀᛕ𝌆᛬ᔕᛖᘚᚺᓄₒ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐑈Z𐑉 category⟜⟜court𐑉⁚·
·⁚lookup office⟜⟜𐑈temporal…absolute𐑉 enoch⟜⟜𐑈Θ𐑉⁚·

·word⁚·ȣᒍ𐋏⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·WAHLKEH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Temporal Absolute·⁚·˙cadence⁚·
·⁚non…ostensive⁚·For a long time⁚⁚·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Low fire survives refueling after refueling before the ash finally coolsz·⁚·˙ostensive⁚·
·⁚grimchain⁚·🜨ᛯᓀᔕᘐᘐᛢᛚᚓ⬡ᚸᚸᚺ✳·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐑈ᑫ𐑉 category⟜⟜court𐑉⁚·
·⁚lookup office⟜⟜𐑈temporal…absolute𐑉 enoch⟜⟜𐑈Θ𐑉⁚·

·word⁚·ᛯᚻ⁼·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·

·⁚pronunciation:·AHNYOO·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·Rest⁚⁚repose⁚˙ cessation⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Labor ceases⁚⁚breath slows⁚⁚and the body gives itself to quietɋ·⁚.·˙ostensive:·
·⁚grimchain:·☽⚹'⚹ℝ✝❂⥽⅄ϒɋ৩·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ɓ꩷ category꩜꩜court꩷:·
·⁚lookup office꩜꩜temporal…absolute꩷ enoch꩜꩜Θ꩷:·

·word:·ʌ□ɪ̄✝̄⁖·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·TUHNK·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·To stop⁚⁚check (transitive)⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Door moves until the bar catches it⁚⁚ Motion ends thereɓ·⁚.·˙ostensive:·
·⁚grimchain:·◎ᚠᛯɔᛖᘏϒ⟓⩥ᕒᕕ∱ϵↃ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ʓ꩷ category꩜꩜court꩷:·
·⁚lookup office꩜꩜temporal…absolute꩷ enoch꩜꩜Θ꩷:·

·word:·🖭⁖·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·sah⟊LAH⟊loom·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·to lie down⁚⁚sleep⁚⁚be at rest⁚⁚be still⁚⁚be peaceful⁚⁚lie still⁚˙ (transferred sense) to
↪ lie with (sexually)⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·Eyes close⁚˙ limbs grow quiet while breath keeps its soft rhythm through the
↪ darkʓ·⁚.·˙ostensive:·
·⁚grimchain:·ᚹ⅂▽⚹ᔕᓚ⩓ᘐᗡ⅄ᔑ⥽·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ϕ꩷ category꩜꩜court꩷:·
·⁚lookup office꩜꩜temporal…absolute꩷ enoch꩜꩜Θ꩷:·

·word:·🕭🖃🖷⁖·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·kah⟊LOOM·⁚.·˙pronunciation:·
·⁚cadence:·Temporal Absolute·⁚.·˙cadence:·
·⁚non…ostensive:·all⁚⁚totality⁚⁚entirety⁚⁚whole⁚⁚·⁚.·˙non…ostensive:·
·⁚ostensive:·First pulse opens the round⁚˙ last pulse closes it⁚⁚and none between is
↪ missingφ·⁚.·˙ostensive:·
·⁚grimchain:·ʓᛯ‹ᛘᚾᚦᗤϒᵹᚺᚦᗡ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜∱꩷ category꩜꩜axiomyr꩷:·
·⁚lookup office꩜꩜temporal…absolute꩷ enoch꩜꩜Θ꩷:·

·:word:·𐫄𐫁𐫈𐫅:.·˙word:·

·⁖origin…language:·Tocharian B·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·SHREHKAHNYNYEH·⁖.·˙pronunciation:·
·⁖cadence:·Temporal Absolute·⁖.·˙cadence:·
·⁖non…ostensive:·Eternal⁘everlasting.·⁖.·˙non…ostensive:·
·⁖ostensive:·Under winter frost⁘living roots endure⁘Summer sunlight reaches stone worn by centuries of
↪ rain⁖·spring returns to the hills without finding an empty earth⁘Ancient ground bears the warmth
↪ of another dawn꙰·⁖.·˙ostensive:·
·⁖grimchain:·𐫔𐫃⊙❄ϙ⬡𐫟𐫀𐫇𐫂𐫆𐫉·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖.·˙glyphs·˙
·⁖.·˙sydonic…magicae…lexicon·∴

·:·:·:

·⁖origin…language꞉·Tocharian B·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖.·˙font꞉·
·⁖pronunciation꞉·EHNGKUHL·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Feeling⁚⁚emotion⁚⁚passion·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Breath deepens⁚⁚pulse rises⁚⁚and the whole chest answers one inward fire without hiding any
↪ part of its force♀·⁖.·˙ostensive꞉·
·⁖grimchain꞉·Λ∀Ƀ⁉ɾ♉ ⊛ ✳ᴑ⛧X∀↘ᔓ·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉◦₋◦ₔ♌₎ category◦₋◦ₔemission₎⁖·
·⁖lookup office◦₋◦ₔterminal…state₎ enoch◦₋◦ₔ⏀₎⁖·

₍word꞉·⋹⋸⋫⋹ ⋕⋘₍⁖.·˙word꞉·

·⁖origin…language꞉·Akkadian·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖.·˙font꞉·
·⁖pronunciation꞉·shah⟊SOOM·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·to cry out⁚⁚shout⁚⁚make a loud noise꞉˙ to proclaim⁚⁚announce⁚⁚utter⁚⁚exclaim꞉˙ to
↪ call⁚⁚summon⁚⁚invoke⁚⁚invite⁚⁚appeal to⁚⁚demand⁚⁚call for꞉˙ to call by a name⁚⁚declare⁚⁚name⁚⁚name a
↪ price꞉˙ to read·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Written name rises from the tablet into a voice strong enough to cross the whole
↪ court⁚⁚and the called thing stands present to every listening ear♌·⁖.·˙ostensive꞉·
·⁖grimchain꞉·ↄ⁒Ⱨ⧗ᔑᘐᘐᑎ⁊⍈♌▽☽·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉◦₋◦ₔ♑₎ category◦₋◦ₔemission₎⁖·
·⁖lookup office◦₋◦ₔterminal…state₎ enoch◦₋◦ₔ⏀₎⁖·

₍word꞉·ᛏᚱᑎᛋ₍⁖.·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖.·˙font꞉·
·⁖pronunciation꞉·TROO⟊ah·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·to believe·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Oath rests in the breast without fracture⁚⁚and doubt finds no seam through which to
↪ enter♑·⁖.·˙ostensive꞉·
·⁖grimchain꞉·⊖⏢Ⱪɣ֍◎ᔑ❂△ǂ⏏⅏·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉◦₋◦ₔ♋₎ category◦₋◦ₔemission₎⁖·
·⁖lookup office◦₋◦ₔterminal…state₎ enoch◦₋◦ₔ⏀₎⁖·

₍word꞉·ᚹᛁᚱᚹ₍⁖.·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖.·˙font꞉·
·⁖pronunciation꞉·VERK·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·work⁚⁚deed·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Joined timbers bear the roof in complete fitting⁚⁚and every worked joint sits cleanly
↪ against the wood beside it♋·⁖.·˙ostensive꞉·

·⁚grimchain:·𐫔𐫕𐫙𐫗𐫙𐫚𐫚𐫛𐫜x𐫒𐫇𐫘·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol○–○𐫛𐫜𐫝 category○–○𐫛emission𐫝⁚·
·⁚lookup office○–○𐫛terminal…state𐫝 enoch○–○𐫛Φ𐫝⁚·

𐫛word:·𐫛𐫝𐫞𐫟𐫠:·⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·KAHRSAHLNYEH·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Knowledge·⁚·˙non…ostensive:·
·⁚ostensive:·Written signs yield their reading without obscurity::and what was hidden from
↪ understanding stands open to the inward eyeΨ·⁚·˙ostensive:·
·⁚grimchain:·𐫈𐫉𐫊𐫋𐫌𐫍𐫎𐫏𐫐𐫑𐫒𐫓·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol○–○𐫛Ꝑ𐫝 category○–○𐫛emission𐫝⁚·
·⁚lookup office○–○𐫛terminal…state𐫝 enoch○–○𐫛Φ𐫝⁚·

𐫛word:·𐫔𐫕𐫖𐫗:·⁚·˙word:·

·⁚origin…language:·Hittite·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·PARKUS·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·High::tall::elevated::lofty·⁚·˙non…ostensive:·
·⁚ostensive:·Peak rises clear above the surrounding heights::and nothing near it reaches the same upper
↪ airꝐ·⁚·˙ostensive:·
·⁚grimchain:·𐫝𐫞𐫟𐫠𐫡𐫢𐫣𐫤𐫦𐫥𐫧𐫨·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol○–○𐫛☽𐫝 category○–○𐫛emission𐫝⁚·
·⁚lookup office○–○𐫛terminal…state𐫝 enoch○–○𐫛Φ𐫝⁚·

𐫛word:·𐫩𐫪𐫫𐫬𐫭:·⁚·˙word:·

·⁚origin…language:·Hittite·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·NAKKIS·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Important::high+ranking::honored::esteemed·⁚ powerful·⁚ inaccessible·⁚·˙non…ostensive:·
·⁚ostensive:·Raised seat stands high and difficult of approach::and honor::power::and distance gather
↪ upon the same inaccessible height☽·⁚·˙ostensive:·
·⁚grimchain:·𐫮𐫯𐫰𐫱𐫲𐫳𐫴𐫵𐫶𐫷𐫸·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol○–○𐫛⊙𐫝 category○–○𐫛emission𐫝⁚·
·⁚lookup office○–○𐫛terminal…state𐫝 enoch○–○𐫛Φ𐫝⁚·

𐫛word:·𐫹𐫺𐫻:·⁚·˙word:·

·⁚origin…language꞉·Ge'ez·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Ethiopic·⁚.·˙font꞉·
·⁚pronunciation꞉·nuh+GOOS·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·King꞉ a male sovereign ruler·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Crown rests above the assembled realm꞉꞉and one command reaches from the gate to the
↪ furthest worked field⊙·⁚.·˙ostensive꞉·
·⁚grimchain꞉·〰ϴↃ▫⇜⇞ᔑᖷ⍥ᛯᚼ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍ℂ₎ category꞉○–○₍emission₎⁚·
·⁚lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Φ₎⁚·

₍word꞉·𐎗𐎍𐎐✳𐎀𐎌꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·rah+MAH+noom·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·self·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·One living body closes within its own skin꞉꞉and no second bearer answers from inside that
↪ boundary℃·⁚.·˙ostensive꞉·
·⁚grimchain꞉·↾𝄢⋗«Ɣ⍾ꞁ≳Ⲭ〰❂0ϙΛ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍♈₎ category꞉○–○₍parliament₎⁚·
·⁚lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Φ₎⁚·

₍word꞉·लपिः⁚.·˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚.·˙font꞉·
·⁚pronunciation꞉·li+pi·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·writing꞉ inscription꞉ written document꞉ painting꞉꞉drawing·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Prepared surface carries stroke into figure and figure into lasting sign꞉꞉and the whole
↪ field stands fully made beneath the hand that marked it♈·⁚.·˙ostensive꞉·
·⁚grimchain꞉·⊙⊕♅❂𐦂ᵑₒ♀ᛕᚼ↗Ⅱᔑ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍♉₎ category꞉○–○₍parliament₎⁚·
·⁚lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Φ₎⁚·

₍word꞉·ᛋᚾᛏ⁚.·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚.·˙font꞉·
·⁚pronunciation꞉·SAYETT·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Agreement꞉꞉reconciliation·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Joined hands rest with drawn steel laid aside꞉꞉and neither side keeps another blow
↪ waiting♉·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ₒ♌⁕🗲⌿◎ᚺ♂☿⍟〰↾Δ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𝌆꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝⁚·

⁖word⁚·𐎧𐎫𐎶𐎠𐎼𐏁⁖·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚·˙font⁚·
·⁚pronunciation⁚·hah+SHEE+yam·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·
·⁚non…ostensive⁚·True⁚˙ in accordance with fact or reality·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Measure and object agree without remainder⁚⁚and no hidden weight can tilt the balance away
↪ from what is shown𝌆·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛇᛉᛏᚱᚴᛞᛟ❀▽·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᘐ꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝⁚·

⁖word⁚·𗦳⁖·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·SHUH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Arrow⁚˙ a straight missile shot from a bow·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Shaft stands deep in the chosen mark⁚⁚and feathered wood holds the straight line carried
↪ by its pointᘐ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᘐᚳᛈᛘᛉ♌⏁⏃⊚ᚯᛞIÞ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᛯ꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝⁚·

⁖word⁚·𐎤𐎶𐎢⁖·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚·˙font⁚·
·⁚pronunciation⁚·JEE+vah·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Living⁚˙ possessing life·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Breath moves freely through the chest⁚⁚blood keeps its warmth⁚⁚and every limb answers the
↪ living pulseᛯ꩜·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚿ𐫔▫ᚕᛇᘔ𑁯𑁯𑁯0ᛊᚾᚱ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑁍꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝⁚·

⁖word⁚·ᚼᚾᛁᛚ⁖·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·HAYL·⁚·˙pronunciation⁚·

·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·whole·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Shield edge runs unbroken around its face⁚⁚and no split interrupts the body it
↪ encloses𝄋·⁚.·˙ostensive꞉·
·⁚grimchain꞉·X⁕ᗰ♂ᛯᚸ⌬ᡕ☆ᡗ⩙ⱬ⁕·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ₑ♎ category○–○ₑparliament꞉·
·⁚lookup office꞉○–○ₑterminal…state꞉ enoch○–○ₑΦ꞉·

·word꞉·놀애·⁚.·˙word꞉·

·⁚origin…language꞉·Middle Korean·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans KR·⁚.·˙font꞉·
·⁚pronunciation꞉·nOHláj·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·song·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Every note finds its place in one sounding body⁚⁚and silence has entered nowhere between
↪ the joined tones♎·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᚼᛉᓀ◎ᚂᚏᚣ𝄋ᚾᛋᛋᛊᡗ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ₑ𝄋 category○–○ₑparliament꞉·
·⁚lookup office꞉○–○ₑterminal…state꞉ enoch○–○ₑΦ꞉·

·word꞉·𐤇𐤉 𐤓𐤊 𐤊𐤊 𐤌𐤌·⁚.·˙word꞉·

·⁚origin…language꞉·Old Persian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Persian·⁚.·˙font꞉·
·⁚pronunciation꞉·doo+ROOKH+tah·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·False꞉˙ untrue or deceitful·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·One mark says the gate stands open while the barred leaves touch edge to edge⁚⁚word and
↪ world opposing each other without any seam of agreement𝄋·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𓅓𓇋𓎛𓃀𓆑𓋴ǂ𓊖◎𓃰∱𓆑·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ₑ↗ category○–○ₑparliament꞉·
·⁚lookup office꞉○–○ₑterminal…state꞉ enoch○–○ₑΦ꞉·

·word꞉·ᦓ꞉·˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚.·˙font꞉·
·⁚pronunciation꞉·BHOO·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Becoming⁚⁚arising⁚⁚existence⁚⁚place⁚⁚or world·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·What arises takes ground beneath the sky and stands there as its own living place⁚⁚with no
↪ seam between coming+forth and being↗·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᛦᗜᘐᓀᚥᚱᚷᑐᘔᘐᡗᛇ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ₑ𝅘𝅥𝅮 category○–○ₑparliament꞉·
·⁚lookup office꞉○–○ₑterminal…state꞉ enoch○–○ₑΦ꞉·

:word:·𒀭𒁲𒁹:·˙word:·

·origin…language:·Akkadian·:·˙origin…language:·
·font:·Noto Sans Cuneiform·:·˙font:·
·pronunciation:·DEE+noo·:·˙pronunciation:·
·cadence:·Terminal State·:·˙cadence:·
·non…ostensive:·Judgment::legal decision::verdict::lawsuit::or law·:·˙non…ostensive:·
·ostensive:·Tablet bears the settled verdict in a full field of marks::and every matter under judgment
↪ stands visibly accounted for upon the clay·:·˙ostensive:·
·grimchain:·⊙⌬𐏃𐏁𐎫𐎗𐎽𐏂𐎵𐎭·:·˙grimchain:·
·:·˙lookup·˙
·:·˙glyphs·˙
·glyph symbol꞉꞉𐎫 category꞉꞉parliament꞉·
·lookup office꞉꞉terminal…state꞉ enoch꞉꞉𐎭꞉·

:word:·𒀀𒄑𒁹𒁲:·˙word:·

·origin…language:·Akkadian·:·˙origin…language:·
·font:·Noto Sans Cuneiform·:·˙font:·
·pronunciation:·khahr+RAH+noom·:·˙pronunciation:·
·cadence:·Terminal State·:·˙cadence:·
·non…ostensive:·road::way::path·: (astronomy) path (of heavenly bodies)·: journey::trip::travel·: (Old
↪ Assyrian::Old Babylonian) journey+money·: (Neo+Babylonian) trading capital·: business venture·:
↪ military expedition/campaign·: caravan·: (mathematics) times·:·˙non…ostensive:·
·ostensive:·Caravan moves beneath the star+road with its silver bound to the journey::and the long way
↪ carries travel::venture::and reckoning in one continuous course꞉·:·˙ostensive:·
·grimchain:·𐎭𐏈𐏁𐏂𐎧𐎸𐎮𐏇𐏂𐎲𐎽·:·˙grimchain:·
·:·˙lookup·˙
·:·˙glyphs·˙
·glyph symbol꞉꞉𐎮 category꞉꞉parliament꞉·
·lookup office꞉꞉terminal…state꞉ enoch꞉꞉𐎭꞉·

:word:·ᚼᚢᚱᛒᛋ:·˙word:·

·origin…language:·Old Norse·:·˙origin…language:·
·font:·Noto Sans Runic·:·˙font:·
·pronunciation:·HOR+vah·:·˙pronunciation:·
·cadence:·Terminal State·:·˙cadence:·
·non…ostensive:·to turn (be turned) in a certain direction·:·˙non…ostensive:·
·ostensive:·Great stone rests wholly upon its new face::the heavy mass swung from one bearing into
↪ another while remaining unbroken𐎮·:·˙ostensive:·
·grimchain:·𐎶𐎡𐎲𐎴𐏁𐎺𐎼𐎿𐎼·:·˙grimchain:·
·:·˙lookup·˙
·:·˙glyphs·˙
·glyph symbol꞉꞉⌬꞉ category꞉꞉goetic꞉·
·lookup office꞉꞉terminal…state꞉ enoch꞉꞉𐎭꞉·

:word:·𑀤𑁆𑀯𑀚𑀶:·˙word:·

·origin…language:·Tocharian B·:·˙origin…language:·
·font:·Noto Sans Brahmi·:·˙font:·
·pronunciation:·CHMEHLNYEH·:·˙pronunciation:·
·cadence:·Terminal State·:·˙cadence:·
·non…ostensive:·Birth·:·˙non…ostensive:·

·⁖ostensive⁝·Swollen womb holds a formed living body against the unopened passage⁝⁝breath and daylight
↪ still beyond the threshold◎·⁖.·ˑostensive⁝·
·⁖grimchain⁝·ᛃᛒᛝᘉᚼᛘᛅᚹᛋᛘᛉ·⁖.·ˑgrimchain⁝·
·⁖.·ˑlookup⁝·
·⁖.·ˑglyphs⁝·
·⁖glyph symbol○–○⬡› category○–○goetic›⁖·
·⁖lookup office○–○terminal…state› enoch○–○Φ›⁖·

◦word⁝·ᵬᵪᛕᚣ◦⁖.·ˑword⁝·

·⁖origin…language⁝·Gothic·⁖.·ˑorigin…language⁝·
·⁖font⁝·Noto Sans Gothic·⁖.·ˑfont⁝·
·⁖pronunciation⁝·BOHKA·⁖.·ˑpronunciation⁝·
·⁖cadence⁝·Terminal State·⁖.·ˑcadence⁝·
·⁖non…ostensive⁝·Any writing or text⁝⁝especially a letter⁝⁝deed⁝⁝or contract⁖ (in the plural)
↪ book⁝⁝writ·⁖.·ˑnon…ostensive⁝·
·⁖ostensive⁝·Marked leaves rest gathered beneath one binding⁝⁝and every written surface keeps its place
↪ inside the preserved body of the text⬡·⁖.·ˑostensive⁝·
·⁖grimchain⁝·ᛦᛁᛯᛝ𐌼ᛟᛕᛋᛏᛇᛯᚱᛇ·⁖.·ˑgrimchain⁝·
·⁖.·ˑlookup⁝·
·⁖.·ˑglyphs⁝·
·⁖glyph symbol○–○✡› category○–○goetic›⁖·
·⁖lookup office○–○terminal…state› enoch○–○Φ›⁖·

◦word⁝·तेजस्◦⁖.·ˑword⁝·

·⁖origin…language⁝·Sanskrit·⁖.·ˑorigin…language⁝·
·⁖font⁝·Noto Sans Devanagari·⁖.·ˑfont⁝·
·⁖pronunciation⁝·tay+juhs·⁖.·ˑpronunciation⁝·
·⁖cadence⁝·Terminal State·⁖.·ˑcadence⁝·
·⁖non…ostensive⁝·the sharp edge of a knife⁖ the point or top of a flame or a ray⁖
↪ light⁝⁝brilliance⁝⁝glow⁝⁝glare⁖ splendor⁖ fiery energy⁝⁝vital power·⁖.·ˑnon…ostensive⁝·
·⁖ostensive⁝·Sun catches the honed blade and makes its edge burn like the tip of flame⁝⁝one keen
↪ brightness holding sharpness⁝⁝heat⁝⁝and splendor together✡·⁖.·ˑostensive⁝·
·⁖grimchain⁝·ᚸ⛒ᓗᐻᓀᛁᛈᛞᚨᚹᚷᚡᛟ·⁖.·ˑgrimchain⁝·
·⁖.·ˑlookup⁝·
·⁖.·ˑglyphs⁝·
·⁖glyph symbol○–○✡› category○–○goetic›⁖·
·⁖lookup office○–○terminal…state› enoch○–○Φ›⁖·

◦word⁝·ᚾᛚᛏᚾ◦⁖.·ˑword⁝·

·⁖origin…language⁝·Old Norse·⁖.·ˑorigin…language⁝·
·⁖font⁝·Noto Sans Runic·⁖.·ˑfont⁝·
·⁖pronunciation⁝·ALDA·⁖.·ˑpronunciation⁝·
·⁖cadence⁝·Terminal State·⁖.·ˑcadence⁝·
·⁖non…ostensive⁝·wave·⁖.·ˑnon…ostensive⁝·
·⁖ostensive⁝·Water rises into one bright ridge⁝⁝and the crest bears the sea at its fullest
↪ height✡·⁖.·ˑostensive⁝·
·⁖grimchain⁝·𐌼ᛝᛤᚹᛉᚺᛇᚸᛃᛋᛇᚳᛦ·⁖.·ˑgrimchain⁝·
·⁖.·ˑlookup⁝·
·⁖.·ˑglyphs⁝·
·⁖glyph symbol○–○❂› category○–○goetic›⁖·
·⁖lookup office○–○terminal…state› enoch○–○Φ›⁖·

꞉word꞉·ᵞᵡIhTS꞉⁖·˙word꞉·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·WEYEHTS·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Thing·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·One body stands enclosed by its own edge꞉꞉and all surrounding space begins beyond that
↪ boundary❂·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐌋ᛘ☿ϙϤΘᛘᚴᛏ'ᛋᚗᛦ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍֍₎ category꞉○–○₍goetic₎⁖·
·⁖lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Φ₎⁖·

꞉word꞉·𒀭⁖·˙word꞉·

·⁖origin…language꞉·Akkadian·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖·˙font꞉·
·⁖pronunciation꞉·KEESH+shah+toom·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·totality꞉꞉all of꞉˙ world꞉꞉universe·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Horizon folds into horizon with no outer wall꞉꞉and every place that can be entered already
↪ belongs within the same immeasurable whole֍·⁖·˙ostensive꞉·
·⁖grimchain꞉·𓃒ᚾᛤℂᘔƶᑯᕒ᯽✰ᑭᕐϒᚱ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍⁕₎ category꞉○–○₍goetic₎⁖·
·⁖lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Φ₎⁖·

꞉word꞉·ᛯᚺᚿ⁖·˙word꞉·

·⁖origin…language꞉·Tocharian B·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖·˙font꞉·
·⁖pronunciation꞉·AHNYEEYEH·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Breath·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Breath fills the whole chest and leaves it carrying warmth꞉꞉presence꞉꞉and the quiet inward
↪ life that makes the body more than flesh alone⁕·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᖁ⁕ᙆ▫ᗤᔕᖁᖅᘔᛜᛡᛯ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍⧗₎ category꞉○–○₍goetic₎⁖·
·⁖lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Φ₎⁖·

꞉word꞉·ᎧᏫᏣ⁖·˙word꞉·

·⁖origin…language꞉·Ge'ez·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖·˙font꞉·
·⁖pronunciation꞉·FUH+kr·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Love꞉˙ deep affection꞉꞉fondness꞉꞉or charitable regard·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Heart keeps the beloved near without possession꞉꞉and giving rises freely from the same
↪ inward warmth that holds the bond⧗·⁖·˙ostensive꞉·

·⁚grimchain⁚·◎ɔɾƹ𝈦'ϐϥᴘᔐϚᔐᔑᔐ⍑ᗊ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍ ⊛ ₎ category₀₋₀₍goetic₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍Φ₎⁚·

₍word⁚·ጸልመት⁚₎.·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚.·˙font⁚·
·⁚pronunciation⁚·tsahl⟊MAHT·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Darkness⁚⁚shade⁚⁚shadow⁚⁚or dimness·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Moonless night lies over earth⁚⁚and no visible seam divides the dark ground from the dark
↪ sky ⊛·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ɤᛘ◎ᦉᛊΨ⫯ᚡI○Ⅱᔑᖯ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍❄₎ category₀₋₀₍goetic₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍Φ₎⁚·

₍word⁚·ᛏᗑᒍ̄⁚₎.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·KUHSHUHL·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Together·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Threads drawn into one braid carry the same pull⁚⁚and no strand turns apart from the
↪ joined body❄·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ገ▽ᑉᔕᐦᑌ ⊛ ᐯᚣᑉᕒ᛻·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍❁₎ category₀₋₀₍goetic₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍Φ₎⁚·

₍word⁚·ጸብእ⁚₎.·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚.·˙font⁚·
·⁚pronunciation⁚·TSAHB·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Army⁚⁚armed host⁚⁚or battle force·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Two armed lines fill the ground face to face⁚⁚shields locked and spearpoints held against
↪ spearpoints⁚⁚neither field yielding the space between them❁·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚏᚋᔕᛃ❂ᚏᚱᛪⅡᛯᛯᛩᚯᚑ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍⬡₎ category₀₋₀₍goetic₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍Φ₎⁚·

₍word⁚·ሀያዓያᘐ⁚₎.·˙word⁚·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·CHAXASH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Serenity⁚⁚sobriety·⁚.·˙non…ostensive:·
·⁚ostensive:·Water lies untroubled in the basin⁚⁚breath rests evenly⁚⁚and no tremor disturbs the clear surface of either⌬·⁚.·˙ostensive:·
·⁚grimchain:·ᘐᙢᕮᑭ▯◊▫ᗰᚔᛦᛤᛣ❄⌯·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈 category◦–◦court𐑉:·
·⁚lookup office◦–◦terminal…state𐑉 enoch◦–◦Ꝺ𐑉:·

·word:·⛄:.·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·MEH⟊see·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·To give birth·˙ to bring forth offspring·⁚.·˙non…ostensive:·
·⁚ostensive:·New breath fills living flesh⁚⁚and the small chest answers the open air with its own rising warmth𐑈·⁚.·˙ostensive:·
·⁚grimchain:·▫ꕥI⊙❄ᘐᐁ₹ᑘᐧᕼᛯᐁ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈 category◦–◦court𐑉:·
·⁚lookup office◦–◦terminal…state𐑉 enoch◦–◦Ꝺ𐑉:·

·word:·ΛIϒS:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·EYEWS·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·A lifetime⁚⁚eternity·⁚.·˙non…ostensive:·
·⁚ostensive:·One generation carries the same gesture into the next⁚⁚then the next⁚⁚the form returning through lives until origin and echo share one unbroken span𐑈·⁚.·˙ostensive:·
·⁚grimchain:·☽ΘΘᖘᘔᐞᗷ♌ᓬᕼᛏᛘ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈 category◦–◦court𐑉:·
·⁚lookup office◦–◦terminal…state𐑉 enoch◦–◦Ꝺ𐑉:·

·word:·ᒐ̲ᒍᎱ:.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·PLAHCHEH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Speech⁚⁚talk·⁚ reply·⁚.·˙non…ostensive:·
·⁚ostensive:·Question receives its full answering voice⁚⁚and every syllable stands clear enough to leave no hollow in the reply𐑈·⁚.·˙ostensive:·
·⁚grimchain:·♀⊛ᛕᛏᚑᚱᛋ⊢ᛅᛯᛯᘔᛘᛏ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𒀭꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Φ꞉·

꞉word꞉·𒄑𒉿𒁀𒌑𒁕꞉·˙word꞉·

·⁚origin…language꞉·Hittite·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚·˙font꞉·
·⁚pronunciation꞉·SHAGAISH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Sign⁚⁚omen⁚⁚miracle(?)·⁚ warning⁚⁚feature⁚⁚characteristic·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Three cuts⁚⁚two spaces⁚⁚three cuts⁚⁚two spaces⁚⁚ the same order returns across
↪ cloth⁚⁚vessel⁚⁚and wall without losing its measure𒀭·⁚·˙ostensive꞉·
·⁚grimchain꞉·‹⬡ₒ⧗ᚹᘜ⊛❄⋤↗Δ‹ㅀ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𒄿꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Φ꞉·

꞉word꞉·𓉐꞉·˙word꞉·

·⁚origin…language꞉·Egyptian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Egyptian Hieroglyphs·⁚·˙font꞉·
·⁚pronunciation꞉·KHENT·⁚·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Jar+stand·⁚ a support for a vessel·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Jar rests level above the earth⁚⁚and the shaped stand beneath receives the whole round
↪ weight without tilt𒄿·⁚·˙ostensive꞉·
·⁚grimchain꞉·𐤊ᓚᓘ✳✳✳𒀭ᘐᕮᕊ𓀬·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𒀭꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Φ꞉·

꞉word꞉·ብዝጎት꞉·˙word꞉·

·⁚origin…language꞉·Ge'ez·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Ethiopic·⁚·˙font꞉·
·⁚pronunciation꞉·buhz+KHAT·⁚·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Multitude·⁚ a great number or crowd·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Bodies crowd the open ground shoulder beside shoulder⁚⁚and little earth remains visible
↪ beneath the gathered living mass𒀭·⁚·˙ostensive꞉·
·⁚grimchain꞉·ϒƶ✡Ↄ𐌰Ψ♂ᛦ𒀭𒀭ᚻ⊕·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𒀭꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Φ꞉·

꞉word꞉·✱ᚠᚿᚢᚾᛋ꞉·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚·˙font꞉·
·⁚pronunciation꞉·HLYOH+tah·⁚·˙pronunciation꞉·

·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·to get by lot::have allotted to oneself·⁚.·˙non…ostensive:·
·⁚ostensive:·Cast lot lies revealed beside its portion::and the chosen share rests beneath one claim
↪ without rival hands upon it𐑉·⁚.·˙ostensive:·
·⁚grimchain:·ᔓᙡ⛤ᚸᗑᐱᗤ◎ᐱᒐᖘᛟ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑈 category꩜꩜court𐑋:·
·⁚lookup office꩜꩜terminal…state𐑋 enoch꩜꩜Φ𐑋:·

·:word:·도로ᄒᆞ다:.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·tolohuhta·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·to repay::to return (something loaned)·⁚.·˙non…ostensive:·
·⁚ostensive:·Borrowed weight rests again upon the hand from which it came::and the balance between
↪ giving and return stands level𐑉·⁚.·˙ostensive:·
·⁚grimchain:·山𐑉ᗅ긔ᛉᔑᘐ𐑉ᔕᐤ𐑋·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑍 category꩜꩜court𐑋:·
·⁚lookup office꩜꩜terminal…state𐑋 enoch꩜꩜Φ𐑋:·

·:word:·ᴛᴏᴏγᴛє:.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·towtuh·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·to gather::collect⁚˙ to be gathered::be collected·⁚.·˙non…ostensive:·
·⁚ostensive:·Streams of grain pour from separate baskets into one common store until no former heap
↪ keeps its own boundary𐑉·⁚.·˙ostensive:·
·⁚grimchain:·Φ𐑉❂ᄆᄒᅀᅡ◌ᄀᙆᛞᅡ✳·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑓 category꩜꩜court𐑋:·
·⁚lookup office꩜꩜terminal…state𐑋 enoch꩜꩜Φ𐑋:·

·:word:·즈ᇭ:.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·tseuz·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·visage⁚˙ appearance⁚˙ shape (eg::of a person)⁚˙ (by extension) situation⁚˙ circumstances⁚˙
↪ behavior·⁚.·˙non…ostensive:·
·⁚ostensive:·Face::shape::circumstance::and bearing meet in one visible form::and nothing of what stands
↪ outward is hidden from the eye𐑓·⁚.·˙ostensive:·
·⁚grimchain:·▽◎ᛗᐱᓓᛯ⬡⌬▽ᚻᚳᖘ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙

·⁚glyph symbol◦–◦𐑦𐑟 category◦–◦𐑒court𐑟⁚·
·⁚lookup office◦–◦𐑑terminal…state𐑟 enoch◦–◦𐑢𐑟⁚·

⁖word⁚·ᛋᚠᛁᛏᚼ⁖⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·SLEETA·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·to tear⁚⁚to tear apart·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Cloth lies in two bodies⁚⁚and the opened fibres face one another across a clean unjoined
↪ rent𐑦⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·ᗺᕮᕯᚹᛋᚸᚩᛟᛈ🜨⏧⛢ᕮ⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑧𐑟 category◦–◦𐑒court𐑟⁚·
·⁚lookup office◦–◦𐑑terminal…state𐑟 enoch◦–◦𐑢𐑟⁚·

⁖word⁚·SIΓIS⁖⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·SIGIS·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Victory·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Enemy standard lies beneath the standing shield⁄wall⁚⁚and the field carries only the force
↪ that remains sovereign upon it𐑧⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛋᔓᐠ🜲ᐟᛯᚳᛖᘐᗰᕮIᛏ·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑤𐑟 category◦–◦𐑒court𐑟⁚·
·⁚lookup office◦–◦𐑑terminal…state𐑟 enoch◦–◦𐑢𐑟⁚·

⁖word⁚·रयि⁖⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚⁚·˙font⁚·
·⁚pronunciation⁚·RUH⁄yee·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Property⁚⁚goods⁚⁚possessions⁚⁚treasure⁚⁚or wealth·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Gold and grain rest under seal⁚⁚and the keeping place stands heavy with
↪ possessionl·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·ᔕᚼᐁᚱᚕᗅᚱᘐᚮᑭᚦᘜᚱ🜲ᛏᚼᚸ·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑤𐑟 category◦–◦𐑒court𐑟⁚·
·⁚lookup office◦–◦𐑑terminal…state𐑟 enoch◦–◦𐑢𐑟⁚·

⁖word⁚·ᕕᒐ𑖪𑖿⁖⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚⁚·˙font⁚·
·⁚pronunciation⁚·TAHPAHKYEH·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·

·⁚non…ostensive⁝·Mirror⁝⁝looking+glass·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Polished surface returns every edge placed before it⁝⁝and no clouding lies between the visible form and its answering image𐠈·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᘜᑭᘐᖻᐯᔕᐩᖻᕼᖇᗡ⟴↾·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜𝅋𝅋⁾ category⟜⟜𝅋court⁾⁚·
·⁚lookup office⟜⟜𝅋terminal…state⁾ enoch⟜⟜𝅋Φ⁾⁚·

˖word⁝·ꕥ∪ᑐ˖.·˙word⁝·

·⁚origin…language⁝·Tocharian B·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Brahmi·⁚.·˙font⁝·
·⁚pronunciation⁝·MAHPPEH·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Terminal State·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Ripe·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Heat has darkened every kernel⁝⁝fragrance rises from the full pan⁝⁝and no raw hardness remains beneath the browned surface▫·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𑜦❄ص⩛ↄ⍟◎∀ᙢᙢ⍋⊢ᚷسᛃ𐠈·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜𝅋𐠈 category⟜⟜𝅋court⁾⁚·
·⁚lookup office⟜⟜𝅋terminal…state⁾ enoch⟜⟜𝅋Φ⁾⁚·

˖word⁝·ᔑIᑎᑎSˑ.·˙word⁝·

·⁚origin…language⁝·Gothic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚.·˙font⁝·
·⁚pronunciation⁝·DIUPS·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Terminal State·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Deep·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Water darkens beneath the lowered line⁝⁝and the sounding weight returns no bottom to the hand that waits above𐠈·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𐠈ᛁᛏᛘᛚ᠔ᔕ⨀ǂ☿❄ᑭᘔᛯ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜𝅋♪⁾ category⟜⟜𝅋court⁾⁚·
·⁚lookup office⟜⟜𝅋terminal…state⁾ enoch⟜⟜𝅋Φ⁾⁚·

˖word⁝·ﺻﻌﻬﻫˑ.·˙word⁝·

·⁚origin…language⁝·Old Uyghur·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Serif Old Uyghur·⁚.·˙font⁝·
·⁚pronunciation⁝·bEHlgUE·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Terminal State·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·sign⁝⁝mark·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Cut line stands clear upon the wood⁝⁝and weather has not worn away the edge by which the mark can still be found♪·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚕ✱ᘜᛀᛉ⬡❊᠔ᔕᛘᛁᛗ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜𝅋ᛉ⁾ category⟜⟜𝅋court⁾⁚·
·⁚lookup office⟜⟜𝅋terminal…state⁾ enoch⟜⟜𝅋Φ⁾⁚·

⁖word⁚·उ꣡⁖.·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁖.·˙font⁚·
·⁖pronunciation⁚·u+ru·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Terminal State·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·broad::wide·⁖ spacious::large·⁖ excessive·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Plain lies broad beneath the sky with no near wall::and distance has room to spread in
↪ every direction꠸·⁖.·˙ostensive⁚·
·⁖grimchain⁚·⊛ɤ𐊗𐋏ᛘ꠸▫ᚢ☆ʊɔ𐞥·⁖.·˙grimchain⁚·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ᵋ꞉ category◦–◦ᵋcourt꞉⁖·
·⁖lookup office◦–◦ᵋterminal…state꞉ enoch◦–◦ᵋΦ꞉⁖·

⁖word⁚·ᑌĪᐱ⁖.·˙word⁚·

·⁖origin…language⁚·Tocharian B·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Brahmi·⁖.·˙font⁚·
·⁖pronunciation⁚·PREHTEH·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Terminal State·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Preta::hungry ghost::malevolent spirit·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Pale hungry form gathers at the edge of firelight::complete enough to haunt the threshold
↪ yet thin enough for stone to show through its limbsꝵ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᚸzᥣ𐋔ᵊ⊖ɔϛ𐞸ϛᵑ◌ᚏ·⁖.·˙grimchain⁚·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ᵋϟ꞉ category◦–◦ᵋcourt꞉⁖·
·⁖lookup office◦–◦ᵋterminal…state꞉ enoch◦–◦ᵋΦ꞉⁖·

⁖word⁚·𐎈𐎊𐎍⁖.·˙word⁚·

·⁖origin…language⁚·Hittite·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Cuneiform·⁖.·˙font⁚·
·⁖pronunciation⁚·KHARGASH·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Terminal State·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Loss::perdition::destruction::ruin·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Nothing held can be reached within the ruined enclosure:: roof is open sky::stores lie
↪ scattered::and shelter exists nowhere inside the broken wallsϟ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᛉ𐋊ɔ△𐆘ᵒ𐊰Ƶ1ᚼ𐎢◌ᘐᵑᛘ·⁖.·˙grimchain⁚·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ᵋ'꞉ category◦–◦ᵋcourt꞉⁖·
·⁖lookup office◦–◦ᵋterminal…state꞉ enoch◦–◦ᵋΦ꞉⁖·

⁖word⁚·ᛏᚼᛉᚱ⁖.·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖.·˙font⁚·
·⁖pronunciation⁚·TAMR·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Terminal State·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·tame·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Rein hangs loose against the neck::and the beast stands beside the open hand without
↪ recoil or flight'·⁖.·˙ostensive⁚·

·⁚grimchain:·ᦟᦓᦗᦈ᧓ᦔᦅᦈᦔᦓᦈᦔ✡ᦈ☆·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○�really category○–○₹court₰⁚·
·⁚lookup office○–○₹terminal…state₰ enoch○–○₹Φ₰⁚·

₹word:·धर्म·:.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·DHUHr⟊muh·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·that which is established or firm⁚⁚steadfast decree⁚⁚statute⁚⁚ordinance⁚⁚law·⁚·
↪ morality⁚⁚virtue⁚⁚righteousness⁚⁚moral code⁚⁚good deed·⁚· usage⁚⁚practice⁚⁚customary observance or
↪ prescribed conduct⁚⁚duty·⁚· right⁚⁚justice (often as a synonym of punishment)·⁚· religion⁚⁚religious
↪ merit·⁚· Law or Justice personified·⁚· the law or doctrine of Buddhism·⁚· the ethical precepts of
↪ Buddhism·⁚· the law of Northern Buddhism·⁚· nature⁚⁚character⁚⁚peculiar condition or essential
↪ quality⁚⁚property⁚⁚mark⁚⁚peculiarity·⁚·˙non…ostensive:·
·⁚ostensive:·Wheel turns with every spoke true to the hub⁚⁚and the ordered motion holds its own
↪ rightness without needing any part to command another·⁚·˙ostensive:·
·⁚grimchain:·ℂ○ᚋᚢᛞ♳❄▫ᛅᚼ◎ᗐᚱᚿ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₹really category○–○₹court₰⁚·
·⁚lookup office○–○₹terminal…state₰ enoch○–○₹Φ₰⁚·

₹word:·긔ㄷ|·:.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·keujti·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·recall·⁚· the faculty of memory·⁚.·˙non…ostensive:·
·⁚ostensive:·One old mark opens a vanished place whole within the inward chamber⁚⁚and no second sign is
↪ needed to restore its shape·⁚·˙ostensive:·
·⁚grimchain:·ℂΛℏᏩ♋ᛉ♳∀ᵴᑭ⧗∮∀·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₹↑₰ category○–○₹court₰⁚·
·⁚lookup office○–○₹terminal…state₰ enoch○–○₹Φ₰⁚·

₹word:·𐎗𐎍⇸►╾·:.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·SGRT·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Locked room·⁚.·˙non…ostensive:·
·⁚ostensive:·Stone chamber closes tightly around what it holds⁚⁚walls drawn so near that bolt⁚⁚door⁚⁚and
↪ daylight leave no open distance between inside and enclosure↾·⁚.·˙ostensive:·
·⁚grimchain:·🢒ᵐᵖ♌ᚣᵋᒐᒐ◌ϴ↾ﻉﻉ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol◦–◦𐑐𐑦 category◦–◦court𐑦⁚·
·⁚lookup office◦–◦terminal…state𐑦 enoch◦–◦Φ𐑦⁚·

·word⁚·ᚾᚾᚾᚱ·⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·ANNARR·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·
·⁚non…ostensive⁚·other⁚⁚another⁚⁚second⁚⁚next·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Formerly paired forms no longer match⁚· one clay body stands altered in color and
↪ cut⁚⁚fully other beside the unchanged twin𐑐·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚸ๑ᛤ◎▽ᛋᚺᛟᛉᚴᛓ᛬ᛯI❄·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦Ո𐑦 category◦–◦court𐑦⁚·
·⁚lookup office◦–◦terminal…state𐑦 enoch◦–◦Φ𐑦⁚·

·word⁚·ᚼᛣᛈᛁ·⁚·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚·˙font⁚·
·⁚pronunciation⁚·HRSH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Childbirth⁚· craftsmanship⁚· spell·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Iron lies straight and bright from the anvil⁚⁚edge even and socket fitted⁚⁚every
↪ hammer+mark gathered into the finished craftՈ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛯᚸᔓᑐᛊᚣᛩᛞᗷᔭᚺᛁᚦ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦Þ𐑦 category◦–◦court𐑦⁚·
·⁚lookup office◦–◦terminal…state𐑦 enoch◦–◦Φ𐑦⁚·

·word⁚·भर्ग·⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·BHUR+guh·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Radiance⁚⁚splendour⁚⁚or effulgence·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Light pours from the bright body in full effulgence⁚⁚Bright edges wear a crown of their
↪ own splendorÞ·⁚·˙ostensive⁚·
·⁚grimchain⁚·𓃒ᛁ▽ᛯᛒᛊᛘᚢᚺ◎ᛕ𑀏ᚿ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐑓𐑦 category◦–◦court𐑦⁚·
·⁚lookup office◦–◦terminal…state𐑦 enoch◦–◦Φ𐑦⁚·

·word⁚·लवण·⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·LUH+vuh+nuh·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·

·⁚non…ostensive⁚·Salt⁚· saline substance or salinity·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·White crystals collapse into one clear molten body under furnace heat⁚⁚no grain retaining
↪ a separate edge۴·⁚.·˙ostensive⁚·
·⁚grimchain⁚·▫𐫓𐫔⊕ⅿⅡφ⊖ǂ❄⌬X‹·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᵋℝ» category⟜⟜ᵋcourt»⁚·
·⁚lookup office⟜⟜ᵋterminal…state» enoch⟜⟜ᵋΦ»⁚·

˖word⁚·+▯ⅎƀȳ⅂˖⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·KUHLSHRAHMNYEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Endurance·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Cord stays taut beneath its burden⁚⁚and every fibre carries the strain without opening a
↪ weak place in the holdℝ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᓭ𐦂𐦐𐦁Ⱨ⊬⅃Ⱬ⅁ⱷΦ⟩·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᵋ‹» category⟜⟜ᵋcourt»⁚·
·⁚lookup office⟜⟜ᵋterminal…state» enoch⟜⟜ᵋΦ»⁚·

˖word⁚·ѡвн˖⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·uhsh+vay·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Majesty·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·High canopy rises over the seated place while gold and carved work gather around
↪ it⁚⁚Lesser ornament falls quiet beneath the enormous sovereign presence‹·⁚.·˙ostensive⁚·
·⁚grimchain⁚·☿⅄⏀↜∀Λɪɻᗤᗷ𑁍ɛ'·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᵋX» category⟜⟜ᵋcourt»⁚·
·⁚lookup office⟜⟜ᵋterminal…state» enoch⟜⟜ᵋΦ»⁚·

˖word⁚·उषा˖⁚.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·u+shah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·morning light·⁚· dawn·⁚· morning·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Morning brightness lies broad across the horizon⁚⁚and earth and sky stand distinctly
↪ revealed within the clear first lightX·⁚.·˙ostensive⁚·
·⁚grimchain⁚·♀ɛ✳ᛯⅯΦ⊙◎ↄᗷƶᘐǂ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᵋᚹ» category⟜⟜ᵋcourt»⁚·
·⁚lookup office⟜⟜ᵋterminal…state» enoch⟜⟜ᵋΦ»⁚·

⁖word:·𗍫⁖.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·KWEEH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Full or filled·⁚ containing no substantial empty capacity·⁚.·˙non…ostensive:·
·⁚ostensive:·Mouthful follows mouthful while the belly rises full and taut⁖⁖hunger finding no empty
↪ chamber left to claimᑭ·⁚.·˙ostensive:·
·⁚grimchain:·𖬰𖬳ᚺz𐤕ᦓ𐦐𖬵𖬷ᘎ𖬹⬡·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀₋꜀ᚺ꜆ category꜀₋꜀court꜆⁚·
·⁚lookup office꜀₋꜀terminal…state꜆ enoch꜀₋꜀Φ꜆⁚·

⁖word:·𖬵𖬷𗍫⁖.·˙word:·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·KHARKISH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·White⁖⁖bright·⁚.·˙non…ostensive:·
·⁚ostensive:·Sun fills the pale stone with a single brightness⁖⁖and every exposed edge shines
↪ clearᚺ·⁚.·˙ostensive:·
·⁚grimchain:·𐤒ᛒᛏᘐ𐤔◎☽𗍫ᐞᚺᕮᛋ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀₋꜀ᚾ꜆ category꜀₋꜀court꜆⁚·
·⁚lookup office꜀₋꜀terminal…state꜆ enoch꜀₋꜀Φ꜆⁚·

⁖word:·ᔕᐱᑎᚧᖴS⁖.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·DAWFS·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Hardened⁖⁖hard┼hearted⁖⁖callous·⁚.·˙non…ostensive:·
·⁚ostensive:·Hardened hide meets blade and stone without denting⁖⁖its surface refusing every soft
↪ impression laid against itᚾ·⁚.·˙ostensive:·
·⁚grimchain:·𐦂𐤔𖬵▫ᦟᛯᚸ⊛🝢ᛋᚑ𖬷·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀₋꜀ᚾ꜆ category꜀₋꜀court꜆⁚·
·⁚lookup office꜀₋꜀terminal…state꜆ enoch꜀₋꜀Φ꜆⁚·

⁖word:·ᄇᄉ다⁖.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·pOHzOHtá·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·to break·⁚ to crush·⁚.·˙non…ostensive:·
·⁚ostensive:·Fragments lie apart beneath the broken face⁖⁖and no hidden bridge of matter still keeps
↪ the crushed body oneᚾ·⁚.·˙ostensive:·

·⁚grimchain⁚·ᓚ❄Ⱥᖶ⊛ᖘᓄᐯᚮᛖ⊙◎·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○₋₀₊ᕀ category⁚○₋₀₊court⁑⁚·
·⁚lookup office⁚○₋₀₊terminal…state⁑ enoch⁚○₋₀₊Φ⁑⁚·

·word⁚·अङ्गार·⁚.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·uhng+ah+ruh·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·charcoal⁚⁚either heated or not heated⁚˙ the planet Mars⁚˙ the plant
↪ hitāvalī·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Black charcoal wears a complete pale mantle⁚⁚every exposed face buried beneath loose ash
↪ while the red core still holds heatᕀ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Ƀ2ᓄᖶ𓀀𓀀ᘐᐁ𓀀‼ᚺ❁ь0·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○₋₀₊⩒ category⁚○₋₀₊court⁑⁚·
·⁚lookup office⁚○₋₀₊terminal…state⁑ enoch⁚○₋₀₊Φ⁑⁚·

·word⁚·ᒍᏕ⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·LAHNYEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Flood·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Water has poured beyond the banks until field⁚⁚road⁚⁚and hollow lie inside one immense
↪ flowing body still pressing outward⩒·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚺ𓏏∱ᔓ⌬ΠЖᛋ⌾ᑫᒍᛏᛣ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○₋₀₊ᕃ category⁚○₋₀₊court⁑⁚·
·⁚lookup office⁚○₋₀₊terminal…state⁑ enoch⁚○₋₀₊Φ⁑⁚·

·word⁚·𓃾𓃰𓃾𓃱𓃱·⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·SHAHL+moom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·verbal adjective of šalāmum⁚˙
↪ healthy⁚⁚sound⁚⁚intact⁚⁚whole⁚⁚entire⁚⁚correct⁚⁚proper⁚⁚safe⁚⁚reliable⁚⁚truthful⁚⁚favourable⁚˙
↪ solvent⁚⁚financially sound·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Separate materials have fused into one sound body⁚⁚every seam taken into the whole so
↪ completely that the joined substance bears weight as oneᕃ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·▽𓃱☆ᔦᚷᓄᚺ₀ᛉᓄ₀ᓄ❄·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⁚○₋₀₊ↄ category⁚○₋₀₊court⁑⁚·
·⁚lookup office⁚○₋₀₊terminal…state⁑ enoch⁚○₋₀₊Φ⁑⁚·

˖word꞉·𐬨𐬀𐬵𐬀˖·˙word꞉·

·⁖origin…language꞉·Avestan·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Avestan·⁖·˙font꞉·
·⁖pronunciation꞉·uh+RUHSH·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Correct⁚⁚right⁚⁚straight·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Plumb line hangs without leaning⁚⁚and the cut edge meets it from end to end with no
↪ crooked place between꞉·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐬛𐬹𐬭𐬴𐬨𐬀𐬵𐬁𐬀𐬭𐬀·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉○–○𐬁𐬀𐬵𐬀𐬀𐬭𐬀𐬵𐬀𐬀𐬭𐬀𐬵𐬀𐬀𐬭

·⁚non…ostensive⁚·High⁚· elevated above surrounding level·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Freed body stands above the broken enclosure with open sky on every side⁘no wall or
↪ tether reaching the height it now occupies୭⁘·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐤗Ↄϕᛕ2ㅣ৩ᚱȣℂΨ◎𐤌·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ɳ᛫ category⟜⟜court᛫⁚·
·⁚lookup office⟜⟜terminal…state᛫ enoch⟜⟜Φ᛫⁚·

⁖word⁚·ㅅIㅜ□I⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·TAHRKUHR·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Cloud·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Low cloud fills the ground itself⁘swallowing tree⁘path⁘and distant ridge until only near
↪ shapes remain inside the pale bodyɳ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛐₒ∀ℛℂ𐤗ᛯΘ♌ΦℂↃ〃ᛕ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜◎᛫ category⟜⟜court᛫⁚·
·⁚lookup office⟜⟜terminal…state᛫ enoch⟜⟜Φ᛫⁚·

⁖word⁚·믈뎜⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·meultjuhm·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·waterdrop·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Clear drops stream from the tilted lip⁘bead after bead joining the widening water below
↪ until the pour occupies the whole path from vessel to ground◎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⏀𐤗⍟ϒ𐤔ΦᘜӾₒᵣ৬Λ‹·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜⏀᛫ category⟜⟜court᛫⁚·
·⁚lookup office⟜⟜terminal…state᛫ enoch⟜⟜Φ᛫⁚·

⁖word⁚·ㅏᓚᒐᓚ⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·NYAHSSOH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Portion⁘share·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Two measured shares lie exchanged across the cloth⁘Receiving hands each hold the portion
↪ that had belonged across the table⏀·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᦵ⍥↗ᛋ৫ᛯ𐤗ᛯ♀ᛋϟ𐤗ᛯ⊕·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᔓᛃ᛫ category⟜⟜court᛫⁚·
·⁚lookup office⟜⟜terminal…state᛫ enoch⟜⟜Φ᛫⁚·

˖word꞉ᛁᚾᚠᛁ꞉˖˙word꞉·

·⁖origin…language꞉·Old Norse·⁖˖˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖˖˙font꞉·
·⁖pronunciation꞉·JAKI·⁖˖˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖˖˙cadence꞉·
·⁖non…ostensive꞉·broken ice꞉꞉icefloe·⁖˖˙non…ostensive꞉·
·⁖ostensive꞉·White slab rides the dark water with broken edges exposed꞉꞉and the cold sea carries the
↪ floe as one hard wandering body·⁖˖˙ostensive꞉·
·⁖grimchain꞉·X❁⋤✡♈⊕ᗡ♀✡∫ⓄⓍ⩛·⁖˖˙grimchain꞉·
·⁖˖˙lookup꞉˙
·⁖˖˙glyphs꞉˙
·⁖glyph symbol꞉○–○𝅘𝅥𝅯𝄢 category꞉○–○𝅘𝅥court𝄢⁖·
·⁖lookup office꞉○–○𝅘𝅥terminal…state𝄢 enoch꞉○–○𝅘𝅥Ⓞ𝄢⁖·

˖word꞉·ᚸᑎᘡЄIS꞉˖˙word꞉·

·⁖origin…language꞉·Gothic·⁖˖˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖˖˙font꞉·
·⁖pronunciation꞉·OWTHEES·⁖˖˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖˖˙cadence꞉·
·⁖non…ostensive꞉·Deserted꞉꞉desolate꞉꞉barren⁖˙ wild·⁖˖˙non…ostensive꞉·
·⁖ostensive꞉·Ground lies open beneath a silent sky꞉꞉and no smoke꞉꞉roof꞉꞉crop꞉꞉or gathered voice interrupts
↪ the empty reach𝅘𝅥·⁖˖˙ostensive꞉·
·⁖grimchain꞉·ᚣ⊛ⵔᚣ◎ᚠAᑫᛉᛘᛋ𐊰ᗰ·⁖˖˙grimchain꞉·
·⁖˖˙lookup꞉˙
·⁖˖˙glyphs꞉˙
·⁖glyph symbol꞉○–○𝅘𝅥𝄋𝄐𝄢 category꞉○–○𝅘𝅥court𝄢⁖·
·⁖lookup office꞉○–○𝅘𝅥terminal…state𝄢 enoch꞉○–○𝅘𝅥Ⓞ𝄢⁖·

˖word꞉·ⵜ꞉˖˙word꞉·

·⁖origin…language꞉·Tocharian B·⁖˖˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖˖˙font꞉·
·⁖pronunciation꞉·KOOROH·⁖˖˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖˖˙cadence꞉·
·⁖non…ostensive꞉·Listless·⁖˖˙non…ostensive꞉·
·⁖ostensive꞉·Limbs lie heavy and completely still꞉꞉hands open꞉꞉gaze unfixed꞉꞉with only quiet breath
↪ moving inside the motionless body𝄋𝄐·⁖˖˙ostensive꞉·
·⁖grimchain꞉·𝄾⁕ⴵ∂ᘯᑭⓄ○ᘎ⌶⇞⬡⇞𝄾·⁖˖˙grimchain꞉·
·⁖˖˙lookup꞉˙
·⁖˖˙glyphs꞉˙
·⁖glyph symbol꞉○–○𝅘𝅥o𝄢 category꞉○–○𝅘𝅥court𝄢⁖·
·⁖lookup office꞉○–○𝅘𝅥terminal…state𝄢 enoch꞉○–○𝅘𝅥Ⓞ𝄢⁖·

˖word꞉·मूल꞉˖˙word꞉·

·⁖origin…language꞉·Sanskrit·⁖˖˙origin…language꞉·
·⁖font꞉·Noto Sans Devanagari·⁖˖˙font꞉·
·⁖pronunciation꞉·moo+luh·⁖˖˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖˖˙cadence꞉·
·⁖non…ostensive꞉·root⁖˙ radish⁖˙ base꞉꞉foundation⁖˙ origin꞉꞉beginning⁖˙ cause·⁖˖˙non…ostensive꞉·
·⁖ostensive꞉·Root grips the deep earth beneath the full+grown tree꞉꞉and every branch above finds its
↪ bearing in that hidden hold𝅘𝅥·⁖˖˙ostensive꞉·

·⁚grimchain⁚·♎ᛪᘐℂ𓃒⊰ᘮ₀↾⋔ᘏ≍·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ₑϴ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑterminal…state» enoch○–○ₑ⊕»⁚·

₎word⁚·𓆐₍.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·TEE+loom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·mound⁚⁚heap⁚⁚tell·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Three earthen mounds rise in ordered tiers⁚⁚each crown clearly above one level and below
↪ the next⁚⁚the highest standing unmistakably apartϴ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Ո1⁞ᛏᔑ𓃒ᛦ𓃒ᨒᚗᛟ⁞ϴ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ₑⵝ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑterminal…state» enoch○–○ₑ⊕»⁚·

₎word⁚·𐎠 ᛕ𐏁 ᚲ𐎡ᚦ·₍.·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚.·˙font⁚·
·⁚pronunciation⁚·ah+THAN+gah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Stone or rock⁚˙ hard mineral matter·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Rocky ground runs from buried stone to exposed ridge⁚⁚hard mineral earth holding the whole
↪ standing world beneath its weightⵝ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⁞ᛏ⊰ᚴᛕ∇ᨒ⊖ᘜᛈ⏁ᛋ✷0·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ₑΛ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑterminal…state» enoch○–○ₑ⊕»⁚·

₎word⁚·ᕕᐱᑎᕃ₍.·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚.·˙font⁚·
·⁚pronunciation⁚·LUBS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Clothing or garment⁚˙ material worn on the body·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Cloth lies fitted from shoulder to ankle⁚⁚and no bare interval breaks the covering laid
↪ over the bodyΛ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Iᘁ✡⁞I⁘◞▽♎ᘐ▽ᒧ⩛ᑭ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○ₑꓧ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑterminal…state» enoch○–○ₑ⊕»⁚·

₎word⁚·ﺩﻣﺼﺍﺩ₍.·˙word⁚·

·⁖origin…language:·Avestan·⁚.·˙origin…language:·
·⁖font:·Noto Sans Avestan·⁚.·˙font:·
·⁖pronunciation:·AHN+tah+rah·⁚.·˙pronunciation:·
·⁖cadence:·Terminal State·⁚.·˙cadence:·
·⁖non…ostensive:·Inside⁝⁝within⁝⁝between·⁚.·˙non…ostensive:·
·⁖ostensive:·Walls close around a chamber with no break in the enclosure⁝⁝Every inward direction
↪ belongs wholly to the space held between the surrounding walls𐤇·⁚.·˙ostensive:·
·⁖grimchain:·𐬀𐬭𐬰ᚺ𐬴⊛ᚼ𐬺𐬻𐬽𐬿𐭀𐭁·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜𐤗 category꩜꩜court𐤚:·
·⁖lookup office꩜꩜terminal…state𐤚 enoch꩜꩜Ⓞ𐤚:·

⁖word:·ᛒᚱᚾᛁᚦᚱ⁚.·˙word:·

·⁖origin…language:·Old Norse·⁚.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁚.·˙font:·
·⁖pronunciation:·BREITHR·⁚.·˙pronunciation:·
·⁖cadence:·Terminal State·⁚.·˙cadence:·
·⁖non…ostensive:·broad·⁚.·˙non…ostensive:·
·⁖ostensive:·River lies broad from bank to bank⁝⁝and its open water carries an immense width of
↪ sky𐤗·⁚.·˙ostensive:·
·⁖grimchain:·𐡗𐡊⬡𐡍𐡉𐡋𐡆⬡♂'𐡀𐡁𐡂·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜Ⓞ𐤚 category꩜꩜court𐤚:·
·⁖lookup office꩜꩜terminal…state𐤚 enoch꩜꩜Ⓞ𐤚:·

⁖word:·सामि⁚.·˙word:·

·⁖origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁚.·˙font:·
·⁖pronunciation:·sah+mi·⁚.·˙pronunciation:·
·⁖cadence:·Terminal State·⁚.·˙cadence:·
·⁖non…ostensive:·half·˙ partially·˙ incompletely·˙ imperfectly·˙ prematurely·˙ too soon·⁚.·˙non…ostensive:·
·⁖ostensive:·Half a broken tablet stands preserved beneath the cloth⁝⁝Preserved half remains incomplete
↪ in measure yet wholly present as the portion that endures Ⓞ·⁚.·˙ostensive:·
·⁖grimchain:·𐄷𐄶𐄵𐄴𐄳𐄲𐄱𐄰𐄯𐄮·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜𐤗 category꩜꩜court𐤚:·
·⁖lookup office꩜꩜terminal…state𐤚 enoch꩜꩜Ⓞ𐤚:·

⁖word:·अल्प⁚.·˙word:·

·⁖origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁚.·˙font:·
·⁖pronunciation:·uhl+puh·⁚.·˙pronunciation:·
·⁖cadence:·Terminal State·⁚.·˙cadence:·
·⁖non…ostensive:·small⁝⁝minute⁝⁝trifling⁝⁝little·˙ easily⁝⁝without trouble·⁚.·˙non…ostensive:·
·⁖ostensive:·Single grain rests upon the fingertip with open room on every side⁝⁝Tiny body barely
↪ darkens the skin beneath it𐤗·⁚.·˙ostensive:·
·⁖grimchain:·𐄀𐄁𐄂𐄃𐄄𐄅𐄆𐄇𐄈𐄉·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉ʰ꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Φ꞉⁚·

꞉word⁚·ᏢᎤᎿᎤᏗᎤᏆᏕᎤ꞉⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·ZHOLOVURD·⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Crowd⁚⁚people·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Bodies press shoulder to shoulder across the ground⁚⁚and many voices rise together as one
↪ immense roarʰ·⁚·˙ostensive⁚·
·⁚grimchain⁚·≍֍⪢ᛤᛋᛋᚿᛋᚿᛋᚿ

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·SHRTUHM·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·To stand·⁚·˙non…ostensive:·
·⁚ostensive:·Feet are planted beneath an upright body⁚⁚and weight descends straight into the ground
↪ without sway or collapse𑀀⁚·˙ostensive:·
·⁚grimchain:·ᚼᔑᐞᛋ⍟ǂ☽☽⟰▽ᛝ𐦂·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𑖌 category꞉꞉court꞉·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Ⓞ꞉·

·word:·𑖌·⁚·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·JOH·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Cave or hole·⁚ a hollow opening in earth⁚⁚rock⁚⁚or another surface·⁚·˙non…ostensive:·
·⁚ostensive:·Rock closes around a dark hollow⁚⁚and the opening is the only place where daylight enters
↪ the enclosed depth𑖌·⁚·˙ostensive:·
·⁚grimchain:·A𝍐ℂ𐦂ᚴᓀ𓀏⸔ᔕ⍟ᘐ▽·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𑖌 category꞉꞉court꞉·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Ⓞ꞉·

·word:·रव·⁚·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·ruh+vuh·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·a roar⁚⁚yell⁚⁚cry⁚⁚howl (of animals⁚⁚wild beasts &c)⁚· song⁚⁚singing (of birds)⁚·
↪ hum⁚⁚humming (of bees)⁚· clamour⁚⁚outcry⁚· thunder⁚· talk⁚· any noise or sound (eg⁚⁚the whizz of a
↪ bow⁚⁚the ringing of a bell &c)⁚·˙non…ostensive:·
·⁚ostensive:·Great cry strikes the cliff and returns across the valley in a second full voice⁚⁚then
↪ another⁚⁚each answering body carrying the same sound back through the air𑖌·⁚·˙ostensive:·
·⁚grimchain:·Ƶ⁕⊕∀ɹ⁕ɣɣʮℏ𐦂ᘐ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉𑖗 category꞉꞉court꞉·
·⁚lookup office꞉꞉terminal…state꞉ enoch꞉꞉Ⓞ꞉·

·word:·धूम्र·⁚·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·dhoom+ruh·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·smoke·⁚·˙non…ostensive:·
·⁚ostensive:·Grey smoke possesses the roof+space from beam to opening⁚⁚Roof beams vanish inside the
↪ smoke⁚⁚and every opening releases the same dark breath outward𑖗·⁚·˙ostensive:·

·⁖grimchain⁖·Λ▫ᵋ☆ıلسzسᵋ‹⊙ᔓ꒰·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦ₑब꞉ category◦–◦ₑcourt꞉·
·⁖lookup office◦–◦ₑterminal…state꞉ enoch◦–◦ₑΦ꞉·

⁖word⁖·अनलि⁖·˙word⁖·

·⁖origin…language⁖·Sanskrit·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Devanagari·⁖·˙font⁖·
·⁖pronunciation⁖·uh+ni+luh·⁖·˙pronunciation⁖·
·⁖cadence⁖·Terminal State·⁖·˙cadence⁖·
·⁖non…ostensive⁖·air⁖· wind·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Wind swells the sail to its furthest curve⁖⁖Passing air bends leaf⁖⁖cloth⁖⁖and loose edge
↪ with the same invisible forceब·⁖·˙ostensive⁖·
·⁖grimchain⁖·◎ᑭ✲ᔕ✷≳ᗪ⬡▽✳ᘜ⍝ᛯ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦ₑ꙰꞉ category◦–◦ₑcourt꞉·
·⁖lookup office◦–◦ₑterminal…state꞉ enoch◦–◦ₑΦ꞉·

⁖word⁖·मषि⁖·˙word⁖·

·⁖origin…language⁖·Sanskrit·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Devanagari·⁖·˙font⁖·
·⁖pronunciation⁖·muh+shi·⁖·˙pronunciation⁖·
·⁖cadence⁖·Terminal State·⁖·˙cadence⁖·
·⁖non…ostensive⁖·powder⁖· powdered⁖⁖burnt bone (used to make ink or eye makeup)⁖· soot⁖⁖lampblack⁖·
↪ Nardostachys jatamansi (used to obtain spikenard)·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Black powder clings to fingertip and cloth⁖⁖and the smallest touch leaves a dark trace
↪ that cannot remain unseen꙰·⁖·˙ostensive⁖·
·⁖grimchain⁖·ᗡ⏀ᗡᘁᐱᔓ⬡ₒᚼᒍΦϙ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦ₑव꞉ category◦–◦ₑcourt꞉·
·⁖lookup office◦–◦ₑterminal…state꞉ enoch◦–◦ₑΦ꞉·

⁖word⁖·ʌIJ̄ᒧ̄⁖·˙word⁖·

·⁖origin…language⁖·Tocharian B·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Brahmi·⁖·˙font⁖·
·⁖pronunciation⁖·SHRAHY·⁖·˙pronunciation⁖·
·⁖cadence⁖·Terminal State·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Old⁖⁖adult⁖⁖mature⁖⁖grownup⁖· pertaining to an adult male⁖⁖virile·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Full beard lies heavy along the jaw⁖⁖Grown frame beneath carries the settled strength of
↪ mature fleshব·⁖·˙ostensive⁖·
·⁖grimchain⁖·✳⊕ⵅᗷᘐᘔᑭ⨌ᑭᔑᔕ↾ᛕᙆ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦ₑ꓿꞉ category◦–◦ₑcourt꞉·
·⁖lookup office◦–◦ₑterminal…state꞉ enoch◦–◦ₑΦ꞉·

⁖word⁖·ᚺᓀ⁖·˙word⁖·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·ZAY⟊nah·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Message⁖announcement⁖report⁖or story·⁚.·˙non…ostensive:·
·⁚ostensive:·Tidings sound from gate to gathering⁖carrying one complete account into every listening
↪ place within reach፠·⁚.·˙ostensive:·
·⁚grimchain:·ᒪᐱᔦᗧIᙆᘎᑲᔮ⊙ᓀᐁΘ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꒰ৎ৫꒱ category꒰ৎ৫court꒱:·
·⁚lookup office꒰ৎ৫terminal…state꒱ enoch꒰ৎ৫Φ꒱:·

⁖word:·ᚼᚢᚦᚱ⁖.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·AUTHR·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·desolate·⁚.·˙non…ostensive:·
·⁚ostensive:·Roofless ground stretches beneath a black sky with hearth⁖doorway⁖and field emptied of
↪ every living sign⁖desolation filling the whole reach৫·⁚.·˙ostensive:·
·⁚grimchain:·ᚼᚦ✳ᑫᚏ᧐ᗌᛉᛦᛋᚂᔓᘓ᙮·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꒰ৎ৫ᙏ꒱ category꒰ৎ৫court꒱:·
·⁚lookup office꒰ৎ৫terminal…state꒱ enoch꒰ৎ৫Φ꒱:·

⁖word:·옓다⁖.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·yelpta·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·thin·⁚.·˙non…ostensive:·
·⁚ostensive:·Stretched cloth admits the light through its weave⁖and almost no depth lies between one
↪ surface and the other᙭·⁚.·˙ostensive:·
·⁚grimchain:·ᚋᚬᑌᗅ⟊ᠵᚴᑫ‹⧗✲ᙆz·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꒰ৎ৫ᘔ꒱ category꒰ৎ৫court꒱:·
·⁚lookup office꒰ৎ৫terminal…state꒱ enoch꒰ৎ৫Φ꒱:·

⁖word:·ᛯᚣᚣᒐ□ᚣᚣᒐ⁖.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·AHTSUHTS·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Thick·⁚.·˙non…ostensive:·
·⁚ostensive:·Worked paste holds the mark pressed into it⁖and the heavy body barely yields when its
↪ surface is pushed asideᘔ·⁚.·˙ostensive:·
·⁚grimchain:·ᒣᕮᕮᛘ✳ᛯᘏᚋᚱ⊕ↄX1·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𐑓» category◦–◦court»⁚·
·⁚lookup office◦–◦terminal…state» enoch◦–◦Φ»⁚·

⁖word:·𐎍𐎘𐎗 ⁖·˙word:·

·⁚origin…language:·Ugaritic·⁚·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚·˙font:·
·⁚pronunciation:·GISHMU·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Downpour⁚˙ tempest⁚˙ rainstorm·⁚·˙non…ostensive:·
·⁚ostensive:·Rain fills the air from cloud to ground in one vast falling body⁚⁚and earth and sky nearly
↪ disappear inside the water between them𐑓·⁚·˙ostensive:·
·⁚grimchain:·ዕ⊖ग्रग+☆*ባ⊕ᵲьıJ◎·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𐑜» category◦–◦court»⁚·
·⁚lookup office◦–◦terminal…state» enoch◦–◦Φ»⁚·

⁖word:·𐎢 𐎠 𐎴 𐎡 𐎼 𐎠 𐎢 ⁖·˙word:·

·⁚origin…language:·Old Persian·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚·˙font:·
·⁚pronunciation:·shee+YAH+teesh·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Happiness⁚⁚well+being⁚⁚peace⁚⁚prosperity·⁚·˙non…ostensive:·
·⁚ostensive:·Granaries stand full⁚⁚roads lie open⁚⁚doors stand unbarred⁚⁚and bread is not measured
↪ against fear beneath the settled abundance𐑜·⁚·˙ostensive:·
·⁚grimchain:·ᛋ𐎧ɔᚠᚣᔑᛋᛒ𐏁ஃ}‹ᘜ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦ᚣ» category◦–◦court»⁚·
·⁚lookup office◦–◦terminal…state» enoch◦–◦Φ»⁚·

⁖word:·ᑌ𑀏𑀳̈⁖·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·PRAHTEEN·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Resolve⁚⁚determination⁚⁚resolution·⁚·˙non…ostensive:·
·⁚ostensive:·Sudden inward surge gathers the whole body into one direction⁚⁚pulse⁚⁚breath⁚⁚and stance all
↪ committed before hesitation can divide themᚣ·⁚·˙ostensive:·
·⁚grimchain:·⊕⊙ᛯIⵠᗫᗰᓄ⊙ᚴ◌ᚷ⛢·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𐑨» category◦–◦court»⁚·
·⁚lookup office◦–◦terminal…state» enoch◦–◦Φ»⁚·

⁖word:·Ʇᑌ𑀓𑀳̄⁖·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·OHPYAHCH·⁚·˙pronunciation:·

·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Memory·⁚.·˙non…ostensive:·
·⁚ostensive:·Faded mark opens the lost place in complete detail⁚⁚and inward attention remains turned
↪ toward what can be remembered but not re+enteredA·⁚.·˙ostensive:·
·⁚grimchain:·𓅓𓇋𓂀⊙❄𓋴𓃒𓀀𓄿𓏏◎𓍯𓆉𓃰·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○₍∀꜀ category○–○₍court꜀⁚·
·⁚lookup office○–○₍terminal…state꜀ enoch○–○₍Φ꜀⁚·

·⁚word:·𐤓𐤉𐤆·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·KHOO+doom·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·joy⁚⁚happiness⁚˙ satisfaction⁚⁚contentment⁚˙ (law) of one's own free
↪ will·⁚.·˙non…ostensive:·
·⁚ostensive:·Bread is abundant⁚⁚laughter moves freely⁚⁚and the heart rests satisfied enough to choose
↪ its own giving without compulsion∀·⁚.·˙ostensive:·
·⁚grimchain:·𒁹𒈬𒂍𒄑𒅀𒉿𒌋⁺⊕𒀸𒈫·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○₍∀꜀ category○–○₍court꜀⁚·
·⁚lookup office○–○₍terminal…state꜀ enoch○–○₍Φ꜀⁚·

·⁚word:·𐎊𐎍𐎗·⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·HULUMU·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Dream·⁚.·˙non…ostensive:·
·⁚ostensive:·Night encloses the sleeping body while an inward country stands vivid with its own
↪ light⁚⁚motion⁚⁚and impossible distance∇·⁚.·˙ostensive:·
·⁚grimchain:·⊛⊕⧋⁑ↄ𓂻𓃥𓀀⊕ıϔ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○₍∇꜀ category○–○₍court꜀⁚·
·⁚lookup office○–○₍terminal…state꜀ enoch○–○₍Φ꜀⁚·

·⁚word:·ᛒᚾᚱᚾ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·BAH+rah·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·wave⁚⁚billow·⁚.·˙non…ostensive:·
·⁚ostensive:·Sea rises into a broad crest⁚⁚falls through its own weight⁚⁚and lifts again⁚⁚the whole
↪ water+body carrying the swing from height to hollow and back∇·⁚.·˙ostensive:·
·⁚grimchain:·𐌅𐌳𐌷𐌷𐌺𐌱𐌿𐌡𐌾𐌓𐌺𐌶𐌽𐌻·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙

·⁚glyph symbol◦–◦𐤒𐤓𐤉 category◦–◦𐤒court𐤉⁚·
·⁚lookup office◦–◦𐤒terminal…state𐤉 enoch◦–◦𐤒Θ𐤉⁚·

⁖word⁚·ⲛⲁⲓ⁖·˙·˙word⁚·

·⁚origin…language⁚·Coptic·⁚·˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚·˙·˙font⁚·
·⁚pronunciation⁚·NAI·⁚·˙·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙·˙cadence⁚·
·⁚non…ostensive⁚·Mercy or pity·⁚·˙·˙non…ostensive⁚·
·⁚ostensive⁚·Pain in one body draws answering care from another⁚· hands open⁚⁚burden is shared⁚⁚and hurt
↪ is met without judgment or withdrawal𐤓·⁚·˙·˙ostensive⁚·
·⁚grimchain⁚·𐫄𐫀𐫁𐫂𐫃𐫅𐫆𐫇𐫈𐫉𐫊·⁚·˙·˙grimchain⁚·
·⁚·˙·˙lookup⁚·
·⁚·˙·˙glyphs⁚·
·⁚glyph symbol◦–◦𐤒𐤓𐤉 category◦–◦𐤒court𐤉⁚·
·⁚lookup office◦–◦𐤒terminal…state𐤉 enoch◦–◦𐤒Θ𐤉⁚·

⁖word⁚·ⲛⲁⲥⲁⲙⲁⲛ⁖·˙·˙word⁚·

·⁚origin…language⁚·Avestan·⁚·˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Avestan·⁚·˙·˙font⁚·
·⁚pronunciation⁚·SROW+shah·⁚·˙·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙·˙cadence⁚·
·⁚non…ostensive⁚·Hearing·⁚·˙·˙non…ostensive⁚·
·⁚ostensive⁚·Low drumbeat reaches the ear as vibration strong enough to tremble cartilage and
↪ bone⁚⁚sound fully present as contact within the listening body𐤓·⁚·˙·˙ostensive⁚·
·⁚grimchain⁚·𐫔𐫀𐫁𐫂𐫃𐫄𐫅𐫆𐫇𐫈𐫉·⁚·˙·˙grimchain⁚·
·⁚·˙·˙lookup⁚·
·⁚·˙·˙glyphs⁚·
·⁚glyph symbol◦–◦𐤒𐤅𐤉 category◦–◦𐤒court𐤉⁚·
·⁚lookup office◦–◦𐤒terminal…state𐤉 enoch◦–◦𐤒Θ𐤉⁚·

⁖word⁚·ⲘⲨⲤⲘⲓⲘ⁖·˙·˙word⁚·

·⁚origin…language⁚·Etruscan·⁚·˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Italic·⁚·˙·˙font⁚·
·⁚pronunciation⁚·ah+KHOO+mee+ah·⁚·˙·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙·˙cadence⁚·
·⁚non…ostensive⁚·Agony⁚· extreme bodily or mental suffering·⁚·˙·˙non…ostensive⁚·
·⁚ostensive⁚·Breath knots beneath the ribs⁚⁚every nerve seems drawn tight⁚⁚and no corner of flesh lies
↪ outside the full pressure of pain𐤅·⁚·˙·˙ostensive⁚·
·⁚grimchain⁚·𐫋𐫌𐫍𐫎𐫏𐫐𐫑𐫒𐫓𐫔𐫕·⁚·˙·˙grimchain⁚·
·⁚·˙·˙lookup⁚·
·⁚·˙·˙glyphs⁚·
·⁚glyph symbol◦–◦𐤒𐤖𐤉 category◦–◦𐤒court𐤉⁚·
·⁚lookup office◦–◦𐤒terminal…state𐤉 enoch◦–◦𐤒Θ𐤉⁚·

⁖word⁚·𑀮𑀰𑀬𑁆𑀮𑀺𑁆⁖·˙·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙·˙font⁚·
·⁚pronunciation⁚·SAHMOODTUHR·⁚·˙·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚·˙·˙cadence⁚·

·⁚non…ostensive⁚·Ocean⁜sea·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Ocean falls into blue+black depth beneath the surface⁜pressure and darkness deepening far
↪ below every visible line❦·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐓣𐓛Ꮘʌ+𐓤𐓵𐓭𐓛𐓚𐓟𐓤𐓘·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·
·⁚glyph symbol⟜⟜𐓤꜀ category⟜⟜꜀court꜀⁚·
·⁚lookup office⟜⟜꜀terminal…state꜀ enoch⟜⟜꜀Ꮻ꜀⁚·

⁖word⁚·ɦƧϤϬϞ⁖⁚.·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚.·˙font⁚·
·⁚pronunciation⁚·SAILYU·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Communion·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Several voices hold different pitches around one shared tone⁜each sound remaining
↪ distinct while the joined field settles into one accord𐓤·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚠ✳𐓛𐓢ᛪᚸᛟᚱ𐓏𐓠ℤᴐᔑᚴ·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·
·⁚glyph symbol⟜⟜꜀𐓤꜀ category⟜⟜꜀court꜀⁚·
·⁚lookup office⟜⟜꜀terminal…state꜀ enoch⟜⟜꜀Ꮻ꜀⁚·

⁖word⁚·ΔΛϞϬ𐓚⁖⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·AHEESHAHMNYEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Wisdom·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Many roads remain open⁜yet the hand stays from the dangerous one⁚· clear judgment holds
↪ action inside the measure it already understands𐓤·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ꟻ♂ᵹ𐓘'ᛊᑒᙁ▽ⱺ≍‹ᔭ·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·
·⁚glyph symbol⟜⟜꜀𐓡꜀ category⟜⟜꜀court꜀⁚·
·⁚lookup office⟜⟜꜀terminal…state꜀ enoch⟜⟜꜀Ꮻ꜀⁚·

⁖word⁚·ɬᑭl⁖⁚.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·AH+yee·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Awe⁜terror⁜or reverent fear·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Sky⁜mountain⁜and thunder gather into such height and force that breath falls quiet before
↪ them without command𐓡·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ሕ◎※✓☆ᒉᚴ᠔※Ψ𐓛ɻ↓ᒉᑭ·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·
·⁚glyph symbol⟜⟜꜀𐓦꜀ category⟜⟜꜀court꜀⁚·
·⁚lookup office⟜⟜꜀terminal…state꜀ enoch⟜⟜꜀Ꮻ꜀⁚·

˖word꞉·ᚴᚿᚬᚣ˖꞉·˙word꞉·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·GEYEDW·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Lack꞉꞉want·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Open hand closes on nothing꞉꞉empty shelf bears no provision꞉꞉and the hollow place keeps its
↪ emptiness against every reaching need‼·⁖·˙ostensive꞉·
·⁖grimchain꞉·❄⩡ᚺᛈᔑᔕᔐᔓ⏣⏣ᔐᔐ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍✷₎ category꞉○–○₍court₎꞉·
·⁖lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Ꙫ₎꞉·

˖word꞉·Δ+▯ı̄˖꞉·˙word꞉·

·⁖origin…language꞉·Tocharian A·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖·˙font꞉·
·⁖pronunciation꞉·EHKUHR·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Containing nothing꞉˙ having no contents·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Jar stands open to the light꞉꞉and nothing within it interrupts the bare hollow from lip to
↪ bottom✷·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᚏ⬡◺⩥Ꙫ᛫ᛗ᛬ᛅ⩥⩦ᚲᛃᚹᛦᛦ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍⊴₎ category꞉○–○₍court₎꞉·
·⁖lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Ꙫ₎꞉·

˖word꞉·ᛋᚢᚾᛏᚱ˖꞉·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖·˙font꞉·
·⁖pronunciation꞉·SUNDR·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·apart·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·One gathered cloud of chaff lies spread across the whole floor꞉꞉Scattered fragments stand
↪ apart with open air between every piece⊴·⁖·˙ostensive꞉·
·⁖grimchain꞉·⏢ᔑᏫᘐ᛭↞ᔕ✩☽ᛞᚢᐁᚴ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉○–○₍⩥₎ category꞉○–○₍court₎꞉·
·⁖lookup office꞉○–○₍terminal…state₎ enoch꞉○–○₍Ꙫ₎꞉·

˖word꞉·परशान˖꞉·˙word꞉·

·⁖origin…language꞉·Sanskrit·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Devanagari·⁖·˙font꞉·
·⁖pronunciation꞉·puhr+shah+nuh·⁖·˙pronunciation꞉·
·⁖cadence꞉·Terminal State·⁖·˙cadence꞉·
·⁖non…ostensive꞉·abyss꞉꞉chasm꞉꞉precipice·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Rock opens into a depth whose floor cannot be seen꞉꞉and the dark cleft keeps widening
↪ beneath the precipice⩥·⁖·˙ostensive꞉·

·⁚grimchain⁚·∀𐑈ᘜ⓪ᛉ⊕ᵒᛏᛁ⌬☽ᓄᐱ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍₫, category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍⓪₎⁚·

₍word⁚·바ᵕ올₍.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·pangol·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·drop·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Clear bead falls round and whole through open air⁚⁚carrying its own curved surface
↪ downward without breaking apart⁚·⁚.·˙ostensive⁚·
·⁚grimchain⁚·∩ᘐ⧫☆ᛋ֍ᐸᛃᓂ◎ᓭ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍ᛋ, category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍⓪₎⁚·

₍word⁚·अङ्कस्₍.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·uhng⟊kuhs·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·a curve or bend·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Straight line yields into a clean arc⁚⁚and every part of the bent body belongs to the same
↪ turning curveᛋ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᓬᛉᔑᗺᘜᛞ'ᛊ‹ᦓᛞᚣᛯ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍⍁, category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍⓪₎⁚·

₍word⁚·प्रयि₍.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·PrEE⟊yuh·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·beloved⁚⁚dear to (with locative or genitive or dative of the subject or in compounds)⁚˙
↪ liked⁚⁚favorite⁚⁚wonted⁚⁚own⁚˙ fond of⁚⁚attached to·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Kept object rests nearest the heart⁚⁚touched with a care given to nothing else around
↪ it⁚⁚and distance does not loosen the inward bond⍁·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᙦ'ᔑᘐᖷ✻ᚋᚮᛯᛋ⇁ᛗᐱ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol₀₋₀₍⇁, category₀₋₀₍court₎⁚·
·⁚lookup office₀₋₀₍terminal…state₎ enoch₀₋₀₍⓪₎⁚·

₍word⁚·𐑈𐑗𐑕𐑞𐑈𐑗𐑈𐑙𐑕·⁚.·˙word⁚·

·⁚origin…language꞉·Akkadian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·BOOL+loo+toom·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·to bring back to life::revive⁚· (Old Babylonian) to let live::support::provide for::keep
↪ alive and in good health⁚· (Middle Assyrian) to allow to stay alive·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Warmth returns through the flesh::breath deepens::and the body rises again with strength
↪ enough to carry its own life𒄑·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𒉿𒈾𒀭𒌋𒁹𒀭𒁍𒀀𒊑𒉡𒌆𒍪·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𒄑꞉ category꞉○–○court꞉⁚·
·⁚lookup office꞉○–○terminal…state꞉ enoch꞉○–○Φ꞉⁚·

⁖word꞉·ᛏᚱᚾᚹᚾ⁖·⁚.·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚.·˙font꞉·
·⁚pronunciation꞉·DRAH+gah·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·To draw::drag::or pull·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Rope stands taut from hand to burden::and every fibre carries the same drawing force
↪ toward one side𒄑·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᑭᎩᏂᏢᐁᒪᘔᑐᏁ᛬ᖘᐱ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𒉡꞉ category꞉○–○court꞉⁚·
·⁚lookup office꞉○–○terminal…state꞉ enoch꞉○–○Φ꞉⁚·

⁖word꞉·𒄑𒉡𒌋⁖·⁚.·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·koos+SOOM·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·chair::seat::throne·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Raised seat holds every tool and measure assigned to one phase of the work::Prepared
↪ station stands complete within the larger ordered plan𒉡·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𒁔𒉿𒅴𒁹𒄷𒋛𒇻𒊑𒄭𒁍𒀀·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𒉡꞉ category꞉○–○court꞉⁚·
·⁚lookup office꞉○–○terminal…state꞉ enoch꞉○–○Φ꞉⁚·

⁖word꞉·ᒪᒉᗐᗑ□⁖·⁚.·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·PRAHSHSHUHN·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Terminal State·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Question·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Two inscriptions face one another on the same tablet::each complete and each making the
↪ other impossible::while the question between them remains fully open𒉡·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𒍫⬡𒉿𒁔𒉡𒄷𒌋𒁕𒀭⚲✳ℂ𒅔𒄷·⁚.·˙grimchain꞉·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᨆ꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜terminal…state꩞ enoch꩜꩜Ꝋ꩞:·

꞉word:·ʌ□+x̄ꞈ̄ꞗ̄ꞡ̄꞉⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·SHUHKTAHLYEH·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Seed·⁚·˙non…ostensive:·
·⁚ostensive:·Small hard body rests beneath the soil with root⁚⁚leaf⁚⁚and future height folded unseen within it᨟·⁚·˙ostensive:·
·⁚grimchain:·𐦍❄𐦟𐦝𐦞𐦠ϴ♋𐦐𐦑ᛉ𐦒𐦓𐦔·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᨕ꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜terminal…state꩞ enoch꩜꩜Ꝋ꩞:·

꞉word:·छाया꞉⁚·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·chhah÷jah·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·shadow·⁚˙ shade·⁚˙ shady place·⁚˙ reflection·⁚˙ shading or blending of colours·⁚˙ play of light or colour·⁚·˙non…ostensive:·
·⁚ostensive:·Cool darkness keeps the exact shape of what turns the light aside⁚⁚and the same dark body can become shade⁚⁚reflection⁚⁚or changing color without losing its formᨕ·⁚·˙ostensive:·
·⁚grimchain:·⬡♂ᖷ❂☿❂ⓄᏀ𐦘𐦙𐦚𐦛𐦜𐦝·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᨒ꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜terminal…state꩞ enoch꩜꩜Ꝋ꩞:·

꞉word:·ᦔᦟᦔᦇ꞉⁚·˙word:·

·⁚origin…language:·Avestan·⁚·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚·˙font:·
·⁚pronunciation:·KHSHAP·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Night·⁚·˙non…ostensive:·
·⁚ostensive:·Sunlight is absent from field⁚⁚roof⁚⁚road⁚⁚and horizon·⁚˙ stars remain fixed above the darkened world⁚⁚and every visible surface rests under one completed darknessᨒ·⁚·˙ostensive:·
·⁚grimchain:·❋ᨈ𐦞xℏᘐ𐦟ᛉᘯᖇ𐦠𐦡·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᨓ꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜terminal…state꩞ enoch꩜꩜Ꝋ꩞:·

꞉word:·+ī̄ʞ̄ɗ꞉⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KROHSHCHEH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Cold·⁚.·˙non…ostensive:·
·⁚ostensive:·Water stiffens at the edge::breath whitens in the air::and bare surface draws heat from
↪ flesh with a clean severe touch·⁚.·˙ostensive:·
·⁚grimchain:·⁉❄ϔ₹ᔕ₆⊕∂ᦟ⊖⁜0ɔ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᔕ꩜ category꩜꩜court꩜:·
·⁚lookup office꩜꩜terminal…state꩜ enoch꩜꩜Ⓞ꩜:·

·word:·ᵻᵾᵾᵾ·⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KOHSHRKEEYEH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Hut·˙ image·⁚.·˙non…ostensive:·
·⁚ostensive:·Clay wall carries a complete painted dwelling::roof::doorway::smoke::and shadow all present
↪ in likeness though no hollow chamber stands behind the surface·⁚.·˙ostensive:·
·⁚grimchain:·ᵸIᔕᘐ∀⊕Ⴆ⬡ᦞᗢᒉᥰ꩜·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᛯ꩜ category꩜꩜court꩜:·
·⁚lookup office꩜꩜terminal…state꩜ enoch꩜꩜Ⓞ꩜:·

·word:·ᛉᛁᑎᗷᚱ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·DJOOPR·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·deep·⁚.·˙non…ostensive:·
·⁚ostensive:·Fully submerged body hangs in dark water with open air far above::every limb committed
↪ beneath the surface·ᛯ·⁚.·˙ostensive:·
·⁚grimchain:·ᒣᛝᚿ⬡ᓀᚱ☆ᛝᚦ◺ᛉ⬡ᒣ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᚢ꩜ category꩜꩜court꩜:·
·⁚lookup office꩜꩜terminal…state꩜ enoch꩜꩜Ⓞ꩜:·

·word:·ᚱᚾᚾᚹᚱ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·RANGR·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·twisted·⁚.·˙non…ostensive:·
·⁚ostensive:·Taut warp runs from beam to beam in ordered parallel lines::each fibre holding its twist
↪ while the whole woven field waits under tension·ᚢ·⁚.·˙ostensive:·
·⁚grimchain:·ᘁᚴᑫᛝᚿᕮᑐᛞ☿ᖻᛉᘜ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𐎚‧ category◦–◦court‧:·
·⁚lookup office◦–◦terminal…state‧ enoch◦–◦Ⓞ‧:·

·word:·𐎝𐎜𐎛𐎘𐎗:·˙word:·

·⁚origin…language:·Ugaritic·⁚·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚·˙font:·
·⁚pronunciation:·DBAHT·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Strength·⁚·˙non…ostensive:·
·⁚ostensive:·Body remains upright before danger::breath steady and force gathered::with no retreat
↪ opening behind the held stance𐎚·⁚·˙ostensive:·
·⁚grimchain:·𓃰𓃭𓂋𓎛𓏏𓎟𓅓𓃒𓆑𓃀⬡·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𐎙‧ category◦–◦court‧:·
·⁚lookup office◦–◦terminal…state‧ enoch◦–◦Ⓞ‧:·

·word:·ᛣᚳᛝ᛫·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·SHAHNMAHOO·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Bond::fetter·⁚·˙non…ostensive:·
·⁚ostensive:·Iron closes around joined limbs::and no free distance large enough for escape remains
↪ inside the rigid bond𐎙·⁚·˙ostensive:·
·⁚grimchain:·𑀬𑀓𑀮𑁆𑀫𑀫𑀯𑀼𑀏𑀸𑀷𑀺·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𐎔‧ category◦–◦court‧:·
·⁚lookup office◦–◦terminal…state‧ enoch◦–◦Ⓞ‧:·

·word:·隧·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·NG+GUH·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Swallow:˙ to cause food::drink::or another substance to pass from the mouth down the
↪ throat·⁚·˙non…ostensive:·
·⁚ostensive:·Mouth receives the morsel::throat bears it downward in one clean passage::and the living
↪ body takes the substance wholly within itself𐎔·⁚·˙ostensive:·
·⁚grimchain:·𖬀𖬁𓂋𖬂𖬃𖬄𖬅𖬆𖬇·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𐎛‧ category◦–◦court‧:·
·⁚lookup office◦–◦terminal…state‧ enoch◦–◦Ⓞ‧:·

·word:·अग्र·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·

·⁚pronunciation:·uhg⟊ruh·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·foremost⁝⁝anterior⁝⁝first⁝⁝prominent⁝⁝projecting⁝⁝chief⁝⁝best·⁚
↪ supernumerary·⁚.·˙non…ostensive:·
·⁚ostensive:·One stone stands at the head of the ordered row⁝⁝raised and marked above the alternatives
↪ while every unchosen body remains below🝖·⁚.·˙ostensive:·
·⁚grimchain:·ᚹᛇᛋᛞᚼᛋ❄ᛇ✻ᛯᛃᚺᛥ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐤊 category○–○𐤒court𐤑:·
·⁚lookup office○–○𐤒terminal…state𐤑 enoch○–○𐤄𐤑:·

·:word:·ᚱᚢᚾ·:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·ROON·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Secret·˙ also a rune or written runic character·⁚.·˙non…ostensive:·
·⁚ostensive:·Covered stroke keeps meaning and concealment in the same body⁝⁝visible enough to be a sign
↪ and hidden enough to remain secret𐤊·⁚.·˙ostensive:·
·⁚grimchain:·𓃹𓋴𓂋𓏏𓎟𓊪𓃀𓄿𓅱𓆑·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐤅𐤑 category○–○𐤒court𐤑:·
·⁚lookup office○–○𐤒terminal…state𐤑 enoch○–○𐤄𐤑:·

·:word:·𑀚𑀘·:.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·YEHPEH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Knife⁝⁝blade·⁚.·˙non…ostensive:·
·⁚ostensive:·Honed edge catches light in one narrow line⁝⁝and fibre parts cleanly wherever the metal
↪ passes𐤅·⁚.·˙ostensive:·
·⁚grimchain:·𐤋✲✙◎ᢉᢉ☿☪ᛉᛟᚠᚱᛇ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐤀𐤑 category○–○𐤒court𐤑:·
·⁚lookup office○–○𐤒terminal…state𐤑 enoch○–○𐤄𐤑:·

·:word:·ᚻᛇ·:.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KEHNEH·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Melody⁝⁝tune·˙ (poetry) meter·⁚.·˙non…ostensive:·
·⁚ostensive:·One sustained note fills the space with a dark grain while another rings bright and thin·˙
↪ pitch may meet⁝⁝but each voice keeps its own color𐤀·⁚.·˙ostensive:·
·⁚grimchain:·ᚼ❂ᚢᛈᛏᚿᚽᛜᛉᛃᚲᛟᚼ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁖glyph symbol꩜꩜ᵻᗺ꩝ category꩜꩜ᵻcourt꩝⁖·
·⁖lookup office꩜꩜ᵻterminal…state꩝ enoch꩜꩜ᵻⵔ꩝⁖·

⁖word⁖·चित्⁖.·˙word⁖·

·⁖origin…language⁖·Sanskrit·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Devanagari·⁖.·˙font⁖·
·⁖pronunciation⁖·chit+ruh·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Terminal State·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·bright⁖⁖conspicuous⁖⁖visible⁖⁖clear⁖⁖loud⁖⁖distinguished⁖⁖excellent⁖⁖brilliant⁖⁖
↪ variegated⁖⁖spotted⁖⁖speckled⁖⁖manifold⁖⁖various⁖⁖
↪ extraordinary⁖⁖strange⁖⁖wonderful⁖⁖surprising·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·One variegated surface changes face with every turn of light⁖⁖brilliant⁖⁖strange⁖⁖and
↪ unmistakable without ever becoming a different bodyᗺ·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᚻ☿✪ᘎᛉᛋᚹᚥ⁕ᗊᔕᚠᗺ·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵻↃ꩝ category꩜꩜ᵻcourt꩝⁖·
·⁖lookup office꩜꩜ᵻterminal…state꩝ enoch꩜꩜ᵻⵔ꩝⁖·

⁖word⁖·ⵜᚣᛣᚣᛟ⁖.·˙word⁖·

·⁖origin…language⁖·Etruscan·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Old Italic·⁖.·˙font⁖·
·⁖pronunciation⁖·ZAH+mah+thee·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Terminal State·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Gold·˙ a dense yellow precious metal·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Gold weighs heavy in the receiving pan⁖⁖the counted store larger by the full yellow
↪ measure newly added to itↃ·⁖.·˙ostensive⁖·
·⁖grimchain⁖·⌬ᛋϒɪᒧzΠ⅁ᖶᖴ⏁ㅓㅅᚳ·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵻㅓ꩝ category꩜꩜ᵻcourt꩝⁖·
·⁖lookup office꩜꩜ᵻterminal…state꩝ enoch꩜꩜ᵻⵔ꩝⁖·

⁖word⁖·एक⁖.·˙word⁖·

·⁖origin…language⁖·Sanskrit·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Devanagari·⁖.·˙font⁖·
·⁖pronunciation⁖·AY+kuh·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Terminal State·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·One·˙ a single unit or individual·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Single mark stands alone upon the counting board⁖⁖entire in itself and requiring no
↪ companion to make its unity realㅓ·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᚻ☿✪ᑎᖘᛯᚻℂᖷϴϴϴɹᔑ·⁖.·˙grimchain⁖·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵻᔑ꩝ category꩜꩜ᵻcourt꩝⁖·
·⁖lookup office꩜꩜ᵻterminal…state꩝ enoch꩜꩜ᵻⵔ꩝⁖·

⁖word⁖·금즉⁖.·˙word⁖·

·⁖origin…language⁖·Middle Korean·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans KR·⁖.·˙font⁖·

·⁚pronunciation:·kuhmtsuhk·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·while closing one's eyes·⁚· budgingly·⁚· while moving one's body in a single motion·⁚·
↪ shockedly·⁚· in an emotionally agitated manner·⁚.·˙non…ostensive:·
·⁚ostensive:·Eyes close in one sudden beat⁚⁚body jerks as a single whole⁚⁚and breath catches sharply
↪ inside the same startled instant↴·⁚.·˙ostensive:·
·⁚grimchain:·ᙅᚴ✳▽ᖇᗷᚷᗐᛋ⊕▽ᚺᚺ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ₑ↴₎ category◦–◦ₑcourt₎⁚·
·⁚lookup office◦–◦ₑterminal…state₎ enoch◦–◦ₑΦ₎⁚·

·⁚word⁚·ᚴᚢᚢᚦ⁚.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·LJOHTH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·song (musical)⁚⁚poem·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Several voices sound at once on different pitches⁚⁚their intervals locking into one full
↪ body of harmony beneath the verse↴·⁚.·˙ostensive⁚·
·⁚grimchain:·ᛇᚿᛃᛪᒧᒪᕼᓕᘔᵒᛜᚲᛉ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ₑI₎ category◦–◦ₑcourt₎⁚·
·⁚lookup office◦–◦ₑterminal…state₎ enoch◦–◦ₑΦ₎⁚·

·⁚word⁚·뷸⁚.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·púp·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·drum·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Stretched hide answers the strike with a full round voice⁚⁚and the hollow body carries
↪ that single beat outward through the surrounding airI·⁚.·˙ostensive⁚·
·⁚grimchain:·⌬↾ᛣᔕ❄ᑕ▽Ↄᑭ☆✱ᛔᛯ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ₑᗡ₎ category◦–◦ₑcourt₎⁚·
·⁚lookup office◦–◦ₑterminal…state₎ enoch◦–◦ₑΦ₎⁚·

·⁚word⁚·▻▬▭⁚.·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚.·˙font⁚·
·⁚pronunciation⁚·KINNAHRU·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Lyre⁚⁚harp·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Strings answer together across the full lyre⁚⁚high and low tones filling the frame until
↪ the whole instrument stands alive with musicᗡ·⁚.·˙ostensive⁚·
·⁚grimchain:·⊕ᙡᕼ⊙ƶᛕ✙ᚲᛦᑭᛉᙅᚧ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol꩜꩜𐊗꩞ category꩜꩜𐊛court꩞⁚·
·⁚lookup office꩜꩜𐊛terminal…state꩞ enoch꩜꩜𐊗Ꙫ꩞⁚·

⁖word⁚·ᛈᚾᚦᚾ⁖⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·VATHA·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·to wade through water⁚⁚snow⁚⁚smoke⁚⁚fire·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Water presses against the legs⁚⁚planted feet keep finding hidden ground⁚⁚and the whole body
↪ advances within the resisting current without losing balance𐊗·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·𐊫ᚱᐁᛋᚴᛒ𐊰𐊶𐊺𐊯ⵔXᛏ·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐊗ᛚ꩞ category꩜꩜𐊛court꩞⁚·
·⁚lookup office꩜꩜𐊛terminal…state꩞ enoch꩜꩜𐊗Ꙫ꩞⁚·

⁖word⁚·ᛈᛁᛋᛏᚾ⁖⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·FESTA·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·to fasten·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Cord bites into its knot⁚⁚joined pieces hold without shifting⁚⁚and the fastening bears the
↪ pull laid against itᛚ·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·𐊫ᚺᛦ۞Ɪᛊᛗᛯ𐊰ᛥᛋᚾ·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐊗ᛘ꩞ category꩜꩜𐊛court꩞⁚·
·⁚lookup office꩜꩜𐊛terminal…state꩞ enoch꩜꩜𐊗Ꙫ꩞⁚·

⁖word⁚·𐎍𐎩 𐎙𐎙⁖⁚·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·GBL·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Mountain⁚˙ peak·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Summit thrusts one sharp peak into the upper sky⁚⁚every slope gathering toward the single
↪ point that crowns the mountainᛘ·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛉᛯᚿᚼᚺᚢᚳᚸᚣᛘ𐎍◎ᔑ·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐊗ᛘ꩞ category꩜꩜𐊛court꩞⁚·
·⁚lookup office꩜꩜𐊛terminal…state꩞ enoch꩜꩜𐊗Ꙫ꩞⁚·

⁖word⁚·𗍫⁖⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚⁚·˙font⁚·
·⁚pronunciation⁚·LEEW·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Terminal State·⁚⁚·˙cadence⁚·

·⁖non…ostensive⁝·One⁝· the cardinal number one·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·One flame fills many polished cups with the same light⁝⁝every reflected fire distinct in
↪ place yet issuing from a single undivided source↰·⁖.·˙ostensive⁝·
·⁖grimchain⁝·ᚿᛦ▫ᚑᚱᚸᛉ✳≍♂ᛯᚸᐁᛝ·⁖.·˙grimchain⁝·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐞥✳𐞤 category○–○𐞥court𐞤⁝·
·⁖lookup office○–○𐞥terminal…state𐞤 enoch○–○𐞥⓪𐞤⁝·

⁖word⁝·ᚹᚱᛁᚾᛏ⁝⁖.·˙word⁝·

·⁖origin…language⁝·Old Norse·⁖.·˙origin…language⁝·
·⁖font⁝·Noto Sans Runic·⁖.·˙font⁝·
·⁖pronunciation⁝·GRIND·⁖.·˙pronunciation⁝·
·⁖cadence⁝·Terminal State·⁖.·˙cadence⁝·
·⁖non…ostensive⁝·A gate made of spars or bars·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·Barred gate stands fully open between its posts⁝⁝the road passing cleanly through the
↪ strong frame from one side to the other✳·⁖.·˙ostensive⁝·
·⁖grimchain⁝·⏃⊙ᒧᛉᚱ⏁☆ᛝ๏❋ ⊛ᚾ·⁖.·˙grimchain⁝·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐞥✳𐞤 category○–○𐞥court𐞤⁝·
·⁖lookup office○–○𐞥terminal…state𐞤 enoch○–○𐞥⓪𐞤⁝·

⁖word⁝·ᒪ𑖯ᓫᒐ⁝⁖.·˙word⁝·

·⁖origin…language⁝·Tocharian B·⁖.·˙origin…language⁝·
·⁖font⁝·Noto Sans Brahmi·⁖.·˙font⁝·
·⁖pronunciation⁝·YOOKAHLNYEH·⁖.·˙pronunciation⁝·
·⁖cadence⁝·Terminal State·⁖.·˙cadence⁝·
·⁖non…ostensive⁝·Conquest⁝⁝overcoming·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·One standard remains upright after every assault⁝⁝walls still held and gates still barred⁖·
↪ resistance has taken the field and stands sovereign over what tried to break it✳·⁖.·˙ostensive⁝·
·⁖grimchain⁝·♂ᔑᓄⓄᗅ❀ᘓᘐᛦᕕᐁᚠᛇᓄ·⁖.·˙grimchain⁝·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐞥✳𐞤 category○–○𐞥court𐞤⁝·
·⁖lookup office○–○𐞥terminal…state𐞤 enoch○–○𐞥⓪𐞤⁝·

⁖word⁝·ᛒᚱᚿᚹᚦᚾ⁝⁖.·˙word⁝·

·⁖origin…language⁝·Old Norse·⁖.·˙origin…language⁝·
·⁖font⁝·Noto Sans Runic·⁖.·˙font⁝·
·⁖pronunciation⁝·BREG┼thah·⁖.·˙pronunciation⁝·
·⁖cadence⁝·Terminal State·⁖.·˙cadence⁝·
·⁖non…ostensive⁝·to turn around⁝⁝flip·⁖.·˙non…ostensive⁝·
·⁖ostensive⁝·Shield face lies wholly reversed⁝⁝and the side that faced the world now rests against the
↪ inward arm✳·⁖.·˙ostensive⁝·
·⁖grimchain⁝·♙✷𓆏X◌ᚺᚱᛇᖘᚱΘᛇ·⁖.·˙grimchain⁝·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○𐞥⩛𐞤 category○–○𐞥court𐞤⁝·
·⁖lookup office○–○𐞥terminal…state𐞤 enoch○–○𐞥⓪𐞤⁝·

˖word:·𐎧𐎼𐏁𐎹˖·˙word:·

·˙origin…language:·Old Persian·˙·˙origin…language:·
·˙font:·Noto Sans Old Persian·˙·˙font:·
·˙pronunciation:·kh⟊SHAH⟊chah·˙·˙pronunciation:·
·˙cadence:·Terminal State·˙·˙cadence:·
·˙non…ostensive:·Kingship::sovereignty::or realm·˙·˙non…ostensive:·
·˙ostensive:·Crown::seal::road::and field answer beneath one height::and the furthest boundary still
↪ receives the same command𐏒·˙·˙ostensive:·
·˙grimchain:·𐎫𐎾𐎼⏁𐎭𐎼⊛ɞᛉ𐎯𐎢𐎫𐎯·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꞉꞉𐎢 category꞉꞉court꞉·
·˙lookup office꞉꞉terminal…state꞉ enoch꞉꞉Θ꞉·

˖word:·𐎗˖·˙word:·

·˙origin…language:·Tangut·˙·˙origin…language:·
·˙font:·Noto Serif Tangut·˙·˙font:·
·˙pronunciation:·MUH·˙·˙pronunciation:·
·˙cadence:·Terminal State·˙·˙cadence:·
·˙non…ostensive:·Heaven·˙ the upper celestial realm or sky⟊world·˙·˙non…ostensive:·
·˙ostensive:·High open sky arches above fruit::blossom::and clear channels::every rooted bed radiant
↪ beneath the same unclouded heaven𐎢·˙·˙ostensive:·
·˙grimchain:·𐎨𐎶𐏀𐎼𐎺𐎺𐎽𐎸ɞ𐎺𐎽𐎨·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꞉꞉0꞉ category꞉꞉court꞉·
·˙lookup office꞉꞉terminal…state꞉ enoch꞉꞉Θ꞉·

˖word:·𐎀𐎡𐎧𐎼𐎹𐎥˖·˙word:·

·˙origin…language:·Old Persian·˙·˙origin…language:·
·˙font:·Noto Sans Old Persian·˙·˙font:·
·˙pronunciation:·THOOKH⟊rah·˙·˙pronunciation:·
·˙cadence:·Terminal State·˙·˙cadence:·
·˙non…ostensive:·Bright::shining·˙·˙non…ostensive:·
·˙ostensive:·Polished metal receives the sun upon its face and gives back a brightness fierce enough
↪ to eclipse its own color0·˙·˙ostensive:·
·˙grimchain:·𐏂⛤ᗢᗑ𐎫𐎺▽♂𐎯𐎹⬡⁕⬡·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꞉꞉𐎫꞉ category꞉꞉court꞉·
·˙lookup office꞉꞉terminal…state꞉ enoch꞉꞉Θ꞉·

˖word:·𐓣𐓟𐓤˖·˙word:·

·˙origin…language:·Aghwan·˙·˙origin…language:·
·˙font:·Noto Sans Caucasian Albanian·˙·˙font:·
·˙pronunciation:·PU·˙·˙pronunciation:·
·˙cadence:·Terminal State·˙·˙cadence:·
·˙non…ostensive:·Sufficient::enough·˙·˙non…ostensive:·
·˙ostensive:·Bow is strung::arrow nocked::path clear::strength sufficient·˙ every condition required for
↪ the shot stands present at once𐎫·˙·˙ostensive:·

·⁚grimchain⁚·𐎢𐎲𐎫𐎫𐎫𐎭𐎮𐎯𐎰𐎱𐎲𐎳𐎴𐎵𐎶𐎷𐎸𐎹𐎺𐎻𐎼𐎽𐎾𐎿𐏀𐏁𐏂𐏃𐏄𐏅𐏆𐏇𐏈𐏉𐏊𐏋𐏌𐏍𐏎𐏏𐏐𐏑𐏒𐏓𐏔𐏕𐏖𐏗𐏘𐏙𐏚𐏛𐏜𐏝𐏞𐏟𐏠𐏡𐏢𐏣𐏤𐏥𐏦𐏧𐏨𐏩𐏪𐏫𐏬𐏭𐏮𐏯𐏰𐏱𐏲𐏳𐏴𐏵𐏶𐏷𐏸𐏹𐏺𐏻𐏼𐏽𐏾𐏿𐐀𐐁𐐂𐐃𐐄𐐅𐐆𐐇𐐈𐐉𐐊𐐋𐐌𐐍𐐎𐐏𐐐𐐑𐐒𐐓𐐔𐐕𐐖𐐗𐐘𐐙𐐚𐐛𐐜𐐝𐐞𐐟𐐠𐐡𐐢𐐣𐐤𐐥𐐦𐐧𐐨𐐩𐐪𐐫𐐬𐐭𐐮𐐯𐐰𐐱𐐲𐐳𐐴𐐵𐐶𐐷𐐸𐐹𐐺𐐻𐐼𐐽𐐾𐐿𐑀𐑁𐑂𐑃𐑄𐑅𐑆𐑇𐑈𐑉𐑊𐑋𐑌𐑍𐑎𐑏𐑐𐑑𐑒𐑓𐑔𐑕𐑖𐑗𐑘𐑙𐑚𐑛𐑜𐑝𐑞𐑟𐑠𐑡𐑢𐑣𐑤𐑥𐑦𐑧𐑨𐑩𐑪𐑫𐑬𐑭𐑮𐑯𐑰𐑱𐑲𐑳𐑴𐑵𐑶𐑷𐑸𐑹𐑺𐑻𐑼𐑽𐑾𐑿𐒀𐒁𐒂𐒃𐒄𐒅𐒆𐒇𐒈𐒉𐒊𐒋𐒌𐒍𐒎𐒏𐒐𐒑𐒒𐒓𐒔𐒕𐒖𐒗𐒘𐒙𐒚𐒛𐒜𐒝𐒞𐒟𐒠𐒡𐒢𐒣𐒤𐒥𐒦𐒧𐒨𐒩𐒪𐒫𐒬𐒭𐒮𐒯𐒰𐒱𐒲𐒳𐒴𐒵𐒶𐒷𐒸𐒹𐒺𐒻𐒼𐒽𐒾𐒿𐓀𐓁𐓂𐓃𐓄𐓅𐓆𐓇𐓈𐓉𐓊𐓋𐓌𐓍𐓎𐓏𐓐𐓑𐓒𐓓𐓔𐓕𐓖𐓗𐓘𐓙𐓚𐓛𐓜𐓝𐓞𐓟𐓠𐓡𐓢𐓣𐓤𐓥𐓦𐓧𐓨𐓩𐓪𐓫𐓬𐓭𐓮𐓯𐓰𐓱𐓲𐓳𐓴𐓵𐓶𐓷𐓸𐓹𐓺𐓻𐓼𐓽𐓾𐓿𐔀𐔁𐔂𐔃𐔄𐔅𐔆𐔇𐔈𐔉𐔊𐔋𐔌𐔍𐔎𐔏𐔐𐔑𐔒𐔓𐔔𐔕𐔖𐔗𐔘𐔙𐔚𐔛𐔜𐔝𐔞𐔟𐔠𐔡𐔢𐔣𐔤𐔥𐔦𐔧𐔨𐔩𐔪𐔫𐔬𐔭𐔮𐔯𐔰𐔱𐔲𐔳𐔴𐔵𐔶𐔷𐔸𐔹𐔺𐔻𐔼𐔽𐔾𐔿𐕀𐕁𐕂𐕃𐕄𐕅𐕆𐕇𐕈𐕉𐕊𐕋𐕌𐕍𐕎𐕏𐕐𐕑𐕒𐕓𐕔𐕕𐕖𐕗𐕘𐕙𐕚𐕛𐕜𐕝𐕞𐕟𐕠𐕡𐕢𐕣𐕤𐕥𐕦𐕧𐕨𐕩𐕪𐕫𐕬𐕭𐕮𐕯𐕰𐕱𐕲𐕳𐕴𐕵𐕶𐕷𐕸𐕹𐕺𐕻𐕼𐕽𐕾𐕿𐖀𐖁𐖂𐖃𐖄𐖅𐖆𐖇𐖈𐖉𐖊𐖋𐖌𐖍𐖎𐖏𐖐𐖑𐖒𐖓𐖔𐖕𐖖𐖗𐖘𐖙𐖚𐖛𐖜𐖝𐖞𐖟𐖠𐖡𐖢𐖣𐖤𐖥𐖦𐖧𐖨𐖩𐖪𐖫𐖬𐖭𐖮𐖯𐖰𐖱𐖲𐖳𐖴𐖵𐖶𐖷𐖸𐖹𐖺𐖻𐖼𐖽𐖾𐖿𐗀𐗁𐗂𐗃𐗄𐗅𐗆𐗇𐗈𐗉𐗊𐗋𐗌𐗍𐗎𐗏𐗐𐗑𐗒𐗓𐗔𐗕𐗖𐗗𐗘𐗙𐗚𐗛𐗜𐗝𐗞𐗟𐗠𐗡𐗢𐗣𐗤𐗥𐗦𐗧𐗨𐗩𐗪𐗫𐗬𐗭𐗮𐗯𐗰𐗱𐗲𐗳𐗴𐗵𐗶𐗷𐗸𐗹𐗺𐗻𐗼𐗽𐗾𐗿𐘀𐘁𐘂𐘃𐘄𐘅𐘆𐘇𐘈𐘉𐘊𐘋𐘌𐘍𐘎𐘏𐘐𐘑𐘒𐘓𐘔𐘕𐘖𐘗𐘘𐘙𐘚𐘛𐘜𐘝𐘞𐘟𐘠𐘡𐘢𐘣𐘤𐘥𐘦𐘧𐘨𐘩𐘪𐘫𐘬𐘭𐘮𐘯𐘰𐘱𐘲𐘳𐘴𐘵𐘶𐘷𐘸𐘹𐘺𐘻𐘼𐘽𐘾𐘿𐙀𐙁𐙂𐙃𐙄𐙅𐙆𐙇𐙈𐙉𐙊𐙋𐙌𐙍𐙎𐙏𐙐𐙑𐙒𐙓𐙔𐙕𐙖𐙗𐙘𐙙𐙚𐙛𐙜𐙝𐙞𐙟𐙠𐙡𐙢𐙣𐙤𐙥𐙦𐙧𐙨𐙩𐙪𐙫𐙬𐙭𐙮𐙯𐙰𐙱𐙲𐙳𐙴𐙵𐙶𐙷𐙸𐙹𐙺𐙻𐙼𐙽𐙾𐙿𐚀𐚁𐚂𐚃𐚄𐚅𐚆𐚇𐚈𐚉𐚊𐚋𐚌𐚍𐚎𐚏𐚐𐚑𐚒𐚓𐚔𐚕𐚖𐚗𐚘𐚙𐚚𐚛𐚜𐚝𐚞𐚟𐚠𐚡𐚢𐚣𐚤𐚥𐚦𐚧𐚨𐚩𐚪𐚫𐚬𐚭𐚮𐚯𐚰𐚱𐚲𐚳𐚴𐚵𐚶𐚷𐚸𐚹𐚺𐚻𐚼𐚽𐚾𐚿𐛀𐛁𐛂𐛃𐛄𐛅𐛆𐛇𐛈𐛉𐛊𐛋𐛌𐛍𐛎𐛏𐛐𐛑𐛒𐛓𐛔𐛕𐛖𐛗𐛘𐛙𐛚𐛛𐛜𐛝𐛞𐛟𐛠𐛡𐛢𐛣𐛤𐛥𐛦𐛧𐛨𐛩𐛪𐛫𐛬𐛭𐛮𐛯𐛰𐛱𐛲𐛳𐛴𐛵𐛶𐛷𐛸𐛹𐛺𐛻𐛼𐛽𐛾𐛿𐜀𐜁𐜂𐜃𐜄𐜅𐜆𐜇𐜈𐜉𐜊𐜋𐜌𐜍𐜎𐜏𐜐𐜑𐜒𐜓𐜔𐜕𐜖𐜗𐜘𐜙𐜚𐜛𐜜𐜝𐜞𐜟𐜠𐜡𐜢𐜣𐜤𐜥𐜦𐜧𐜨𐜩𐜪𐜫𐜬𐜭𐜮𐜯𐜰𐜱𐜲𐜳𐜴𐜵𐜶𐜷𐜸𐜹𐜺𐜻𐜼𐜽𐜾𐜿𐝀𐝁𐝂𐝃𐝄𐝅𐝆𐝇𐝈𐝉𐝊𐝋𐝌𐝍𐝎𐝏𐝐𐝑𐝒𐝓𐝔𐝕𐝖𐝗𐝘𐝙𐝚𐝛𐝜𐝝𐝞𐝟𐝠𐝡𐝢𐝣𐝤𐝥𐝦𐝧𐝨𐝩𐝪𐝫𐝬𐝭𐝮𐝯𐝰𐝱𐝲𐝳𐝴𐝵𐝶𐝷𐝸𐝹𐝺𐝻𐝼𐝽𐝾𐝿𐞀𐞁𐞂𐞃𐞄𐞅𐞆𐞇𐞈𐞉𐞊𐞋𐞌𐞍𐞎𐞏𐞐𐞑𐞒𐞓𐞔𐞕𐞖𐞗𐞘𐞙𐞚𐞛𐞜𐞝𐞞𐞟𐞠𐞡𐞢𐞣𐞤𐞥𐞦𐞧𐞨𐞩𐞪𐞫𐞬𐞭𐞮𐞯𐞰𐞱𐞲𐞳𐞴𐞵𐞶𐞷𐞸𐞹𐞺𐞻𐞼𐞽𐞾𐞿𐟀𐟁𐟂𐟃𐟄𐟅𐟆𐟇𐟈𐟉𐟊𐟋𐟌𐟍𐟎𐟏𐟐𐟑𐟒𐟓𐟔𐟕𐟖𐟗𐟘𐟙𐟚𐟛𐟜𐟝𐟞𐟟𐟠𐟡𐟢𐟣𐟤𐟥𐟦𐟧𐟨𐟩𐟪𐟫𐟬𐟭𐟮𐟯𐟰𐟱𐟲𐟳𐟴𐟵𐟶𐟷𐟸𐟹𐟺𐟻𐟼𐟽𐟾𐟿𐠀𐠁𐠂𐠃𐠄𐠅𐠆𐠇𐠈𐠉𐠊𐠋𐠌𐠍𐠎𐠏𐠐𐠑𐠒𐠓𐠔𐠕𐠖𐠗𐠘𐠙𐠚𐠛𐠜𐠝𐠞𐠟𐠠𐠡𐠢𐠣𐠤𐠥𐠦𐠧𐠨𐠩𐠪𐠫𐠬𐠭𐠮𐠯𐠰𐠱𐠲𐠳𐠴𐠵𐠶𐠷𐠸𐠹𐠺𐠻𐠼𐠽𐠾𐠿𐡀𐡁𐡂𐡃𐡄𐡅𐡆𐡇𐡈𐡉𐡊𐡋𐡌𐡍𐡎𐡏𐡐𐡑𐡒𐡓𐡔𐡕𐡖𐡗𐡘𐡙𐡚𐡛𐡜𐡝𐡞𐡟𐡠𐡡𐡢𐡣𐡤𐡥𐡦𐡧𐡨𐡩𐡪𐡫𐡬𐡭𐡮𐡯𐡰𐡱𐡲𐡳𐡴𐡵𐡶𐡷𐡸𐡹𐡺𐡻𐡼𐡽𐡾𐡿𐢀𐢁𐢂𐢃𐢄𐢅𐢆𐢇𐢈𐢉𐢊𐢋𐢌𐢍𐢎𐢏𐢐𐢑𐢒𐢓𐢔𐢕𐢖𐢗𐢘𐢙𐢚𐢛𐢜𐢝𐢞𐢟𐢠𐢡𐢢𐢣𐢤𐢥𐢦𐢧𐢨𐢩𐢪𐢫𐢬𐢭𐢮𐢯𐢰𐢱𐢲𐢳𐢴𐢵𐢶𐢷𐢸𐢹𐢺𐢻𐢼𐢽𐢾𐢿𐣀𐣁𐣂𐣃𐣄𐣅𐣆𐣇𐣈𐣉𐣊𐣋𐣌𐣍𐣎𐣏𐣐𐣑𐣒𐣓𐣔𐣕𐣖𐣗𐣘𐣙𐣚𐣛𐣜𐣝𐣞𐣟𐣠𐣡𐣢𐣣𐣤𐣥𐣦𐣧𐣨𐣩𐣪𐣫𐣬𐣭𐣮𐣯𐣰𐣱𐣲𐣳𐣴𐣵𐣶𐣷𐣸𐣹𐣺𐣻𐣼𐣽𐣾𐣿𐤀𐤁𐤂𐤃𐤄𐤅𐤆𐤇𐤈𐤉𐤊𐤋𐤌𐤍𐤎𐤏𐤐𐤑𐤒𐤓𐤔𐤕𐤖𐤗𐤘𐤙𐤚𐤛𐤜𐤝𐤞𐤟𐤠𐤡𐤢𐤣𐤤𐤥𐤦𐤧𐤨𐤩𐤪𐤫𐤬𐤭𐤮𐤯𐤰𐤱𐤲𐤳𐤴𐤵𐤶𐤷𐤸𐤹𐤺𐤻𐤼𐤽𐤾𐤿𐥀𐥁𐥂𐥃𐥄𐥅𐥆𐥇𐥈𐥉𐥊𐥋𐥌𐥍𐥎𐥏𐥐𐥑𐥒𐥓𐥔𐥕𐥖𐥗𐥘𐥙𐥚𐥛𐥜𐥝𐥞𐥟𐥠𐥡𐥢𐥣𐥤𐥥𐥦𐥧𐥨𐥩𐥪𐥫𐥬𐥭𐥮𐥯𐥰𐥱𐥲𐥳𐥴𐥵𐥶𐥷𐥸𐥹𐥺𐥻𐥼𐥽𐥾𐥿𐦀𐦁𐦂𐦃𐦄𐦅𐦆𐦇𐦈𐦉𐦊𐦋𐦌𐦍𐦎𐦏𐦐𐦑𐦒𐦓𐦔𐦕𐦖𐦗𐦘𐦙𐦚𐦛𐦜𐦝𐦞𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿𐨀𐨁𐨂𐨃𐨄𐨅𐨆𐨇𐨈𐨉𐨊𐨋𐨌𐨍𐨎𐨏𐨐𐨑𐨒𐨓𐨔𐨕𐨖𐨗𐨘𐨙𐨚𐨛𐨜𐨝𐨞𐨟𐨠𐨡𐨢𐨣𐨤𐨥𐨦𐨧𐨨𐨩𐨪𐨫𐨬𐨭𐨮𐨯𐨰𐨱𐨲𐨳𐨴𐨵𐨶𐨷𐨹𐨺𐨸𐨻𐨼𐨽𐨾𐨿𐩀𐩁𐩂𐩃𐩄𐩅𐩆𐩇𐩈𐩉𐩊𐩋𐩌𐩍𐩎𐩏𐩐𐩑𐩒𐩓𐩔𐩕𐩖𐩗𐩘𐩙𐩚𐩛𐩜𐩝𐩞𐩟𐩠𐩡𐩢𐩣𐩤𐩥𐩦𐩧𐩨𐩩𐩪𐩫𐩬𐩭𐩮𐩯𐩰𐩱𐩲𐩳𐩴𐩵𐩶𐩷𐩸𐩹𐩺𐩻𐩼𐩽𐩾𐩿𐪀𐪁𐪂𐪃𐪄𐪅𐪆𐪇𐪈𐪉𐪊𐪋𐪌𐪍𐪎𐪏𐪐𐪑𐪒𐪓𐪔𐪕𐪖𐪗𐪘𐪙𐪚𐪛𐪜𐪝𐪞𐪟𐪠𐪡𐪢𐪣𐪤𐪥𐪦𐪧𐪨𐪩𐪪𐪫𐪬𐪭𐪮𐪯𐪰𐪱𐪲𐪳𐪴𐪵𐪶𐪷𐪸𐪹𐪺𐪻𐪼𐪽𐪾𐪿𐫀𐫁𐫂𐫃𐫄𐫅𐫆𐫇𐫈𐫉𐫊𐫋𐫌𐫍𐫎𐫏𐫐𐫑𐫒𐫓𐫔𐫕𐫖𐫗𐫘𐫙𐫚𐫛𐫜𐫝𐫞𐫟𐫠𐫡𐫢𐫣𐫤𐫦𐫥𐫧𐫨𐫩𐫪𐫫𐫬𐫭𐫮𐫯𐫰𐫱𐫲𐫳𐫴𐫵𐫶𐫷𐫸𐫹𐫺𐫻𐫼𐫽𐫾𐫿𐬀𐬁𐬂𐬃𐬄𐬅𐬆𐬇𐬈𐬉𐬊𐬋𐬌𐬍𐬎𐬏𐬐𐬑𐬒𐬓𐬔𐬕𐬖𐬗𐬘𐬙𐬚𐬛𐬜𐬝𐬞𐬟𐬠𐬡𐬢𐬣𐬤𐬥𐬦𐬧𐬨𐬩𐬪𐬫𐬬𐬭𐬮𐬯𐬰𐬱𐬲𐬳𐬴𐬵𐬶𐬷𐬸𐬹𐬺𐬻𐬼𐬽𐬾𐬿𐭀𐭁𐭂𐭃𐭄𐭅𐭆𐭇𐭈𐭉𐭊𐭋𐭌𐭍𐭎𐭏𐭐𐭑𐭒𐭓𐭔𐭕𐭖𐭗𐭘𐭙𐭚𐭛𐭜𐭝𐭞𐭟𐭠𐭡𐭢𐭣𐭤𐭥𐭦𐭧𐭨𐭩𐭪𐭫𐭬𐭭𐭮𐭯𐭰𐭱𐭲𐭳𐭴𐭵𐭶𐭷𐭸𐭹𐭺𐭻𐭼𐭽𐭾𐭿𐮀𐮁𐮂𐮃𐮄𐮅𐮆𐮇𐮈𐮉𐮊𐮋𐮌𐮍𐮎𐮏𐮐𐮑𐮒𐮓𐮔𐮕𐮖𐮗𐮘𐮙𐮚𐮛𐮜𐮝𐮞𐮟𐮠𐮡𐮢𐮣𐮤𐮥𐮦𐮧𐮨𐮩𐮪𐮫𐮬𐮭𐮮𐮯𐮰𐮱𐮲𐮳𐮴𐮵𐮶𐮷𐮸𐮹𐮺𐮻𐮼𐮽𐮾𐮿𐯀𐯁𐯂𐯃𐯄𐯅𐯆𐯇𐯈𐯉𐯊𐯋𐯌𐯍𐯎𐯏𐯐𐯑𐯒𐯓𐯔𐯕𐯖𐯗𐯘𐯙𐯚𐯛𐯜𐯝𐯞𐯟𐯠𐯡𐯢𐯣𐯤𐯥𐯦𐯧𐯨𐯩𐯪𐯫𐯬𐯭𐯮𐯯𐯰𐯱𐯲𐯳𐯴𐯵𐯶𐯷𐯸𐯹𐯺𐯻𐯼𐯽𐯾𐯿𐰀𐰁𐰂𐰃𐰄𐰅𐰆𐰇𐰈𐰉𐰊𐰋𐰌𐰍𐰎𐰏𐰐𐰑𐰒𐰓𐰔𐰕𐰖𐰗𐰘𐰙𐰚𐰛𐰜𐰝𐰞𐰟𐰠𐰡𐰢𐰣𐰤𐰥𐰦𐰧𐰨𐰩𐰪𐰫𐰬𐰭𐰮𐰯𐰰𐰱𐰲𐰳𐰴𐰵𐰶𐰷𐰸𐰹𐰺𐰻𐰼𐰽𐰾𐰿𐱀𐱁𐱂𐱃𐱄𐱅𐱆𐱇𐱈𐱉𐱊𐱋𐱌𐱍𐱎𐱏𐱐𐱑𐱒𐱓𐱔𐱕𐱖𐱗𐱘𐱙𐱚𐱛𐱜𐱝𐱞𐱟𐱠𐱡𐱢𐱣𐱤𐱥𐱦𐱧𐱨𐱩𐱪𐱫𐱬𐱭𐱮𐱯𐱰𐱱𐱲𐱳𐱴𐱵𐱶𐱷𐱸𐱹𐱺𐱻𐱼𐱽𐱾𐱿𐲀𐲁𐲂𐲃𐲄𐲅𐲆𐲇𐲈𐲉𐲊𐲋𐲌𐲍𐲎𐲏𐲐𐲑𐲒𐲓𐲔𐲕𐲖𐲗𐲘𐲙𐲚𐲛𐲜𐲝𐲞𐲟𐲠𐲡𐲢𐲣𐲤𐲥𐲦𐲧𐲨𐲩𐲪𐲫𐲬𐲭𐲮𐲯𐲰𐲱𐲲𐲳𐲴𐲵𐲶𐲷𐲸𐲹𐲺𐲻𐲼𐲽𐲾𐲿𐳀𐳁𐳂𐳃𐳄𐳅𐳆𐳇𐳈𐳉𐳊𐳋𐳌𐳍𐳎𐳏𐳐𐳑𐳒𐳓𐳔𐳕𐳖𐳗𐳘𐳙𐳚𐳛𐳜𐳝𐳞𐳟𐳠𐳡𐳢𐳣𐳤𐳥𐳦𐳧𐳨𐳩𐳪𐳫𐳬𐳭𐳮𐳯𐳰𐳱𐳲𐳳𐳴𐳵𐳶𐳷𐳸𐳹𐳺𐳻𐳼𐳽𐳾𐳿𐴀𐴁𐴂𐴃𐴄𐴅𐴆𐴇𐴈𐴉𐴊𐴋𐴌𐴍𐴎𐴏𐴐𐴑𐴒𐴓𐴔𐴕𐴖𐴗𐴘𐴙𐴚𐴛𐴜𐴝𐴞𐴟𐴠𐴡𐴢𐴣𐴤𐴥𐴦𐴧𐴨𐴩𐴪𐴫𐴬𐴭𐴮𐴯𐴰𐴱𐴲𐴳𐴴𐴵𐴶𐴷𐴸𐴹𐴺𐴻𐴼𐴽𐴾𐴿𐵀𐵁𐵂𐵃𐵄𐵅𐵆𐵇𐵈𐵉𐵊𐵋𐵌𐵍𐵎𐵏𐵐𐵑𐵒𐵓𐵔𐵕𐵖𐵗𐵘𐵙𐵚𐵛𐵜𐵝𐵞𐵟𐵠𐵡𐵢𐵣𐵤𐵥𐵦𐵧𐵨𐵩𐵪𐵫𐵬𐵭𐵮𐵯𐵰𐵱𐵲𐵳𐵴𐵵𐵶𐵷𐵸𐵹𐵺𐵻𐵼𐵽𐵾𐵿𐶀𐶁𐶂𐶃𐶄𐶅𐶆𐶇𐶈𐶉𐶊𐶋𐶌𐶍𐶎𐶏𐶐𐶑𐶒𐶓𐶔𐶕𐶖𐶗𐶘𐶙𐶚𐶛𐶜𐶝𐶞𐶟𐶠𐶡𐶢𐶣𐶤𐶥𐶦𐶧𐶨𐶩𐶪𐶫𐶬𐶭𐶮𐶯𐶰𐶱𐶲𐶳𐶴𐶵𐶶𐶷𐶸𐶹𐶺𐶻𐶼𐶽𐶾𐶿𐷀𐷁𐷂𐷃𐷄𐷅𐷆𐷇𐷈𐷉𐷊𐷋𐷌𐷍𐷎𐷏𐷐𐷑𐷒𐷓𐷔𐷕𐷖𐷗𐷘𐷙𐷚𐷛𐷜𐷝𐷞𐷟𐷠𐷡𐷢𐷣𐷤𐷥𐷦𐷧𐷨𐷩𐷪𐷫𐷬𐷭𐷮𐷯𐷰𐷱𐷲𐷳𐷴𐷵𐷶𐷷𐷸𐷹𐷺𐷻𐷼𐷽𐷾𐷿𐸀𐸁𐸂𐸃𐸄𐸅𐸆𐸇𐸈𐸉𐸊𐸋𐸌𐸍𐸎𐸏𐸐𐸑𐸒𐸓𐸔𐸕𐸖𐸗𐸘𐸙𐸚𐸛𐸜𐸝𐸞𐸟𐸠𐸡𐸢𐸣𐸤𐸥𐸦𐸧𐸨𐸩𐸪𐸫𐸬𐸭𐸮𐸯𐸰𐸱𐸲𐸳𐸴𐸵𐸶𐸷𐸸𐸹𐸺𐸻𐸼𐸽𐸾𐸿𐹀𐹁𐹂𐹃𐹄𐹅𐹆𐹇𐹈𐹉𐹊𐹋𐹌𐹍𐹎𐹏𐹐𐹑𐹒𐹓𐹔𐹕𐹖𐹗𐹘𐹙𐹚𐹛𐹜𐹝𐹞𐹟𐹠𐹡𐹢𐹣𐹤𐹥𐹦𐹧𐹨𐹩𐹪𐹫𐹬𐹭𐹮𐹯𐹰𐹱𐹲𐹳𐹴𐹵𐹶𐹷𐹸𐹹𐹺𐹻𐹼𐹽𐹾𐹿𐺀𐺁𐺂𐺃𐺄𐺅𐺆𐺇𐺈𐺉𐺊𐺋𐺌𐺍𐺎𐺏𐺐𐺑𐺒𐺓𐺔𐺕𐺖𐺗𐺘𐺙𐺚𐺛𐺜𐺝𐺞𐺟𐺠𐺡𐺢𐺣𐺤𐺥𐺦𐺧𐺨𐺩𐺪𐺫𐺬𐺭𐺮𐺯𐺰𐺱𐺲𐺳𐺴𐺵𐺶𐺷𐺸𐺹𐺺𐺻𐺼𐺽𐺾𐺿𐻀𐻁𐻂𐻃𐻄𐻅𐻆𐻇𐻈𐻉𐻊𐻋𐻌𐻍𐻎𐻏𐻐𐻑𐻒𐻓𐻔𐻕𐻖𐻗𐻘𐻙𐻚𐻛𐻜𐻝𐻞𐻟𐻠𐻡𐻢𐻣𐻤𐻥𐻦𐻧𐻨𐻩𐻪𐻫𐻬𐻭𐻮𐻯𐻰𐻱𐻲𐻳𐻴𐻵𐻶𐻷𐻸𐻹𐻺𐻻𐻼𐻽𐻾𐻿𐼀𐼁𐼂𐼃𐼄𐼅𐼆𐼇𐼈𐼉𐼊𐼋𐼌𐼍𐼎𐼏𐼐𐼑𐼒𐼓𐼔𐼕𐼖𐼗𐼘𐼙𐼚𐼛𐼜𐼝𐼞𐼟𐼠𐼡𐼢𐼣𐼤𐼥𐼦𐼧𐼨𐼩𐼪𐼫𐼬𐼭𐼮𐼯𐼰𐼱𐼲𐼳𐼴𐼵𐼶𐼷𐼸𐼹𐼺𐼻𐼼𐼽𐼾𐼿𐽀𐽁𐽂𐽃𐽄𐽅𐽆𐽇𐽋𐽍𐽎𐽏𐽐𐽈𐽉𐽊𐽌𐽑𐽒𐽓𐽔𐽕𐽖𐽗𐽘𐽙𐽚𐽛𐽜𐽝𐽞𐽟𐽠𐽡𐽢𐽣𐽤𐽥𐽦𐽧𐽨𐽩𐽪𐽫𐽬𐽭𐽮𐽯𐽰𐽱𐽲𐽳𐽴𐽵𐽶𐽷𐽸𐽹𐽺𐽻𐽼𐽽𐽾𐽿𐾀𐾁𐾃𐾅𐾂𐾄𐾆𐾇𐾈𐾉𐾊𐾋𐾌𐾍𐾎𐾏𐾐𐾑𐾒𐾓𐾔𐾕𐾖𐾗𐾘𐾙𐾚𐾛𐾜𐾝𐾞𐾟𐾠𐾡𐾢𐾣𐾤𐾥𐾦𐾧𐾨𐾩𐾪𐾫𐾬𐾭𐾮𐾯𐾰𐾱𐾲𐾳𐾴𐾵𐾶𐾷𐾸𐾹𐾺𐾻𐾼𐾽𐾾𐾿𐿀𐿁𐿂𐿃𐿄𐿅𐿆𐿇𐿈𐿉𐿊𐿋𐿌𐿍𐿎𐿏𐿐𐿑𐿒𐿓𐿔𐿕𐿖𐿗𐿘𐿙𐿚𐿛𐿜𐿝𐿞𐿟𐿠𐿡𐿢𐿣𐿤𐿥𐿦𐿧𐿨𐿩𐿪𐿫𐿬𐿭𐿮𐿯𐿰𐿱𐿲𐿳𐿴𐿵𐿶𐿷𐿸𐿹𐿺𐿻𐿼𐿽𐿾𐿿𑀀𑀁𑀂𑀃𑀄𑀅𑀆𑀇𑀈𑀉𑀊𑀋𑀌𑀍𑀎𑀏𑀐𑀑𑀒𑀓𑀔𑀕𑀖𑀗𑀘𑀙𑀚𑀛𑀜𑀝𑀞𑀟𑀠𑀡𑀢𑀣𑀤𑀥𑀦𑀧𑀨𑀩𑀪𑀫𑀬𑀭𑀮𑀯𑀰𑀱𑀲𑀳𑀴𑀵𑀶𑀷𑀸𑀹𑀺𑀻𑀼𑀽𑀾𑀿𑁀𑁁𑁂𑁃𑁄𑁅𑁆𑁇𑁈𑁉𑁊𑁋𑁌𑁍𑁎𑁏𑁐𑁑𑁒𑁓𑁔𑁕𑁖𑁗𑁘𑁙𑁚𑁛𑁜𑁝𑁞𑁟𑁠𑁡𑁢𑁣𑁤𑁥𑁦𑁧𑁨𑁩𑁪𑁫𑁬𑁭𑁮𑁯𑁰𑁱𑁲𑁳𑁴𑁵𑁶𑁷𑁸𑁹𑁺𑁻𑁼𑁽𑁾𑁿𑂀𑂁𑂂𑂃𑂄𑂅𑂆𑂇𑂈𑂉𑂊𑂋𑂌𑂍𑂎𑂏𑂐𑂑𑂒𑂓𑂔𑂕𑂖𑂗𑂘𑂙𑂚𑂛𑂜𑂝𑂞𑂟𑂠𑂡𑂢𑂣𑂤𑂥𑂦𑂧𑂨𑂩𑂪𑂫𑂬𑂭𑂮𑂯𑂰𑂱𑂲𑂳𑂴𑂵𑂶𑂷𑂸𑂺𑂹𑂻𑂼𑂽𑂾𑂿𑃀𑃁𑃂𑃃𑃄𑃅𑃆𑃇𑃈𑃉𑃊𑃋𑃌𑃍𑃎𑃏𑃐𑃑𑃒𑃓𑃔𑃕𑃖𑃗𑃘𑃙𑃚𑃛𑃜𑃝𑃞𑃟𑃠𑃡𑃢𑃣𑃤𑃥𑃦𑃧𑃨𑃩𑃪𑃫𑃬𑃭𑃮𑃯𑃰𑃱𑃲𑃳𑃴𑃵𑃶𑃷𑃸𑃹𑃺𑃻𑃼𑃽𑃾𑃿𑄀𑄁𑄂𑄃𑄄𑄅𑄆𑄇𑄈𑄉𑄊𑄋𑄌𑄍𑄎𑄏𑄐𑄑𑄒𑄓𑄔𑄕𑄖𑄗𑄘𑄙𑄚𑄛𑄜𑄝𑄞𑄟𑄠𑄡𑄢𑄣𑄤𑄥𑄦𑄧𑄨𑄩𑄪𑄫𑄬𑄭𑄮𑄯𑄰𑄱𑄲𑄳𑄴𑄵𑄶𑄷𑄸𑄹𑄺𑄻𑄼𑄽𑄾𑄿𑅀𑅁𑅂𑅃𑅄𑅅𑅆𑅇𑅈𑅉𑅊𑅋𑅌𑅍𑅎𑅏𑅐𑅑𑅒𑅓𑅔𑅕𑅖𑅗𑅘𑅙𑅚𑅛𑅜𑅝𑅞𑅟𑅠𑅡𑅢𑅣𑅤𑅥𑅦𑅧𑅨𑅩𑅪𑅫𑅬𑅭𑅮𑅯𑅰𑅱𑅲𑅳𑅴𑅵𑅶𑅷𑅸𑅹𑅺𑅻𑅼𑅽𑅾𑅿𑆀𑆁𑆂𑆃𑆄𑆅𑆆𑆇𑆈𑆉𑆊𑆋𑆌𑆍𑆎𑆏𑆐𑆑𑆒𑆓𑆔𑆕𑆖𑆗𑆘𑆙𑆚𑆛𑆜𑆝𑆞𑆟𑆠𑆡𑆢𑆣𑆤𑆥𑆦𑆧𑆨𑆩𑆪𑆫𑆬𑆭𑆮𑆯𑆰𑆱𑆲𑆳𑆴𑆵𑆶𑆷𑆸𑆹𑆺𑆻𑆼𑆽𑆾𑆿𑇀𑇁𑇂𑇃𑇄𑇅𑇆𑇇𑇈𑇉𑇊𑇋𑇌𑇍𑇎𑇏𑇐𑇑𑇒𑇓𑇔𑇕𑇖𑇗𑇘𑇙𑇚𑇛𑇜𑇝𑇞𑇟𑇠𑇡𑇢𑇣𑇤𑇥𑇦𑇧𑇨𑇩𑇪𑇫𑇬𑇭𑇮𑇯𑇰𑇱𑇲𑇳𑇴𑇵𑇶𑇷𑇸𑇹𑇺𑇻𑇼𑇽𑇾𑇿𑈀𑈁𑈂𑈃𑈄𑈅𑈆𑈇𑈈𑈉𑈊𑈋𑈌𑈍𑈎𑈏𑈐𑈑𑈒𑈓𑈔𑈕𑈖𑈗𑈘𑈙𑈚𑈛𑈜𑈝𑈞𑈟𑈠𑈡𑈢𑈣𑈤𑈥𑈦𑈧𑈨𑈩𑈪𑈫𑈬𑈭𑈮𑈯𑈰𑈱𑈲𑈳𑈴𑈶𑈵𑈷𑈸𑈹𑈺𑈻𑈼𑈽𑈾𑈿𑉀𑉁𑉂𑉃𑉄𑉅𑉆𑉇𑉈𑉉𑉊𑉋𑉌𑉍𑉎𑉏𑉐𑉑𑉒𑉓𑉔𑉕𑉖𑉗𑉘𑉙𑉚𑉛𑉜𑉝𑉞𑉟𑉠𑉡𑉢𑉣𑉤𑉥𑉦𑉧𑉨𑉩𑉪𑉫𑉬𑉭𑉮𑉯𑉰𑉱𑉲𑉳𑉴𑉵𑉶𑉷𑉸𑉹𑉺𑉻𑉼𑉽𑉾𑉿𑊀𑊁𑊂𑊃𑊄𑊅𑊆𑊇𑊈𑊉𑊊𑊋𑊌𑊍𑊎𑊏𑊐𑊑𑊒𑊓𑊔𑊕𑊖𑊗𑊘𑊙𑊚𑊛𑊜𑊝𑊞𑊟𑊠𑊡𑊢𑊣𑊤𑊥𑊦𑊧𑊨𑊩𑊪𑊫𑊬𑊭𑊮𑊯𑊰𑊱𑊲𑊳𑊴𑊵𑊶𑊷𑊸𑊹𑊺𑊻𑊼𑊽𑊾𑊿𑋀𑋁𑋂𑋃𑋄𑋅𑋆𑋇𑋈𑋉𑋊𑋋𑋌𑋍𑋎𑋏𑋐𑋑𑋒𑋓𑋔𑋕𑋖𑋗𑋘𑋙𑋚𑋛𑋜𑋝𑋞𑋟𑋠𑋡𑋢𑋣𑋤𑋥𑋦𑋧𑋨𑋩𑋪𑋫𑋬𑋭𑋮𑋯𑋰𑋱𑋲𑋳𑋴𑋵𑋶𑋷𑋸𑋹𑋺𑋻𑋼𑋽𑋾𑋿𑌀𑌁𑌂𑌃𑌄𑌅𑌆𑌇𑌈𑌉𑌊𑌋𑌌𑌍𑌎𑌏𑌐𑌑𑌒𑌓𑌔𑌕𑌖𑌗𑌘𑌙𑌚𑌛𑌜𑌝𑌞𑌟𑌠𑌡𑌢𑌣𑌤𑌥𑌦𑌧𑌨𑌩𑌪𑌫𑌬𑌭𑌮𑌯𑌰𑌱𑌲𑌳𑌴𑌵𑌶𑌷𑌸𑌹𑌺𑌻𑌼𑌽𑌾𑌿𑍀𑍁

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·STAHM·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Tree·⁚.·˙non…ostensive:·
·⁚ostensive:·Trunk rises thick from rooted earth⁚⁚crown spreads wide above it⁚⁚and the full+grown tree
↪ holds root⁚⁚wood⁚⁚branch⁚⁚and leaf in one living stature𑀍·⁚.·˙ostensive:·
·⁚grimchain:·𑀪𑁆𑀘𑀸𑁂𑀦𑀫𑀺𑀸𑁆·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𑀢𑁆» category◦–◦court»⁚·
·⁚lookup office◦–◦terminal…state» enoch◦–◦Φ»⁚·

·word:·𑀦𑀺𑀮𑀺𑀬𑀸·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·HVEHR+vah·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·to have a circular or rotary motion·⁚· turn around·⁚.·˙non…ostensive:·
·⁚ostensive:·Wheel holds every spoke around one unmoving hub⁚⁚and the turning rim carries the whole
↪ form through a complete circle without losing its center𑀢·⁚.·˙ostensive:·
·⁚grimchain:·𑀇𑀘𑀧𑁂𑀁⬡𑀢𑀑𑀸❄𑀎𑀑𑀸𑁅𑀬𑁃𑁂·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦«» category◦–◦court»⁚·
·⁚lookup office◦–◦terminal…state» enoch◦–◦Φ»⁚·

·word:·𑀫𑁆𑀣·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·sah+LEE+moom·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·peace⁚⁚concord⁚⁚amity·⁚· reconciliation with the gods⁚⁚favor·⁚.·˙non…ostensive:·
·⁚ostensive:·Weapons rest⁚⁚doors stand open⁚⁚and favor between household and god alike has room to dwell
↪ without guarded distance«·⁚.·˙ostensive:·
·⁚grimchain:·𑀀𑁂𑀁𑁃𑀩𑀮𑀸𑀺𑁄«𑀬Θ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𑀬» category◦–◦court»⁚·
·⁚lookup office◦–◦terminal…state» enoch◦–◦Φ»⁚·

·word:·𑀫𑁆𑀵·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·geesh+REEN+noom·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·scales·⁚.·˙non…ostensive:·
·⁚ostensive:·Two pans hang level beneath the beam⁚⁚Balanced sides hold equal weight at the exact
↪ middle𑀬·⁚.·˙ostensive:·
·⁚grimchain:·𑀲𑀬𑁄𑁂𑁀Φ𑁆𑀢𑀸𑀺❋𑀪·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵹ꩝ category꩜꩜ᵹcourt꩝:·
·⁚lookup office꩜꩜ᵹterminal…state꩝ enoch꩜꩜ᵹ⊕꩝:·

·word:·ቅዱስ:·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·kuh+DOOS·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Holy·⁚ set apart as sacred or consecrated·⁚·˙non…ostensive:·
·⁚ostensive:·Lamp burns within the set+apart place::and ordinary touch remains outside the threshold of
↪ what has been made sacred꩝·⁚·˙ostensive:·
·⁚grimchain:·⁉ᵽᑌ🡅ᚠᛉᛗᛊᚱᛯᛒᛩ⏁ᔕ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵹᔑ꩝ category꩜꩜ᵹcourt꩝:·
·⁚lookup office꩜꩜ᵹterminal…state꩝ enoch꩜꩜ᵹ⊕꩝:·

·word:·ᔓᗄᒣ:·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·YAHMOHR·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Deed::action::accomplishment·⁚ (Buddhism) karma·⁚·˙non…ostensive:·
·⁚ostensive:·Work stands so cleanly wrought that seam::balance::and strength all answer together:: no
↪ correction can find a place to enterᔑ·⁚·˙ostensive:·
·⁚grimchain:·▽ᛕᚼᚺᘏ◌ᕮᛘᐁᗢ⁜ᘓᚠ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵹᒡ꩝ category꩜꩜ᵹcourt꩝:·
·⁚lookup office꩜꩜ᵹterminal…state꩝ enoch꩜꩜ᵹ⊕꩝:·

·word:·ᘐΣᘐᘁ:·˙word:·

·⁚origin…language:·Aghwan·⁚·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚·˙font:·
·⁚pronunciation:·AKAL·⁚·˙pronunciation:·
·⁚cadence:·Terminal State·⁚·˙cadence:·
·⁚non…ostensive:·Seer::witness·⁚·˙non…ostensive:·
·⁚ostensive:·Eye holds the entire sign within a single clear seeing::and nothing of what stands present
↪ slips beyond the witnessed fieldᒡ·⁚·˙ostensive:·
·⁚grimchain:·ᵪᛯᛘᓀᗡᚼᘗᵒᔭᘗᛉᘐ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᵹᒡ꩝ category꩜꩜ᵹcourt꩝:·
·⁚lookup office꩜꩜ᵹterminal…state꩝ enoch꩜꩜ᵹ⊕꩝:·

·word:·ᴛᑎʌᒥIᕓᘉ:·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·TULGITHA·⁚·˙pronunciation:·

·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Safety⁚⁚security·⁚ bulwark⁚⁚stronghold·⁚ foundation·⁚.·˙non…ostensive:·
·⁚ostensive:·Walls take the blow without opening⁚⁚foundation bears their weight⁚⁚and the guarded
↪ interior stands whole beneath the pressure laid against it·⁚.·˙ostensive:·
·⁚grimchain:·⁜ℂ∫❁ᨒᨐ⁜⬡⩛ᚺ℘'·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜▽꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝:·

·word:·⊁⊨≺⊢·⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·HZT·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Good fortune·⁚ luck·⁚.·˙non…ostensive:·
·⁚ostensive:·Fortunate lot lies face+up while the unopened road remains clear ahead⁚⁚promise and
↪ possibility gathered together without possession yet▽·⁚.·˙ostensive:·
·⁚grimchain:·Xɬɳɣ̈ɬ▽ʁɤɪϽ♂ᚳᓄ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜z꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝:·

·word:ﻭﺍﺀﻣﺍ·⁚.·˙word:·

·⁚origin…language:·Avestan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚.·˙font:·
·⁚pronunciation:·dah+RUH+gah·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·Long·⁚ extended in length or duration·⁚.·˙non…ostensive:·
·⁚ostensive:·Road continues beyond the nearest ridge⁚⁚and distance extends farther than sight can
↪ followz·⁚.·˙ostensive:·
·⁚grimchain:·Ո◌ᚱᛈ◎ᛈᛘᵬᵽᛉᛘᚡᚸ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ɾ꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝:·

·word:·ⲉⲙⲟⲟⲥ·⁚.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·HMOHS·⁚.·˙pronunciation:·
·⁚cadence:·Terminal State·⁚.·˙cadence:·
·⁚non…ostensive:·to sit·⁚.·˙non…ostensive:·
·⁚ostensive:·Weight rests fully upon the seat⁚⁚feet lie against the ground⁚⁚and the seated body carries
↪ no standing strainɾ·⁚.·˙ostensive:·
·⁚grimchain:·♂ᚸᚻᚸᛉᛘᚡᛈᛉᛘᗊ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᖷ꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜terminal…state꩝ enoch꩜꩜Φ꩝:·

˸word:·𐎗𐎚𐎍𐎐𐎗˸.·˙word:·

·˙origin…language:·Ugaritic·˙.·˙origin…language:·
·˙font:·Noto Sans Ugaritic·˙.·˙font:·
·˙pronunciation:·LKHSHT·˙.·˙pronunciation:·
·˙cadence:·Terminal State·˙.·˙cadence:·
·˙non…ostensive:·Whisper·˙.·˙non…ostensive:·
·˙ostensive:·Breath moves between barely parted lips without voice⁚⁚the entire chamber held in the quiet field that whisper inhabits without breaking৬·˙.·˙ostensive:·
↪
·˙grimchain:·𐤈𐤓⛢𐤎𐤏𐤅𐤋𐤑𐤃𐤔𐤉𐤊𐤌·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol꩜꩜𐤆꩝ category꩜꩜court꩝⁚·
·˙lookup office꩜꩜terminal…state꩝ enoch꩜꩜𐤏꩝⁚·

˸word:·स्वप्न˸.·˙word:·

·˙origin…language:·Sanskrit·˙.·˙origin…language:·
·˙font:·Noto Sans Devanagari·˙.·˙font:·
·˙pronunciation:·SVUHP÷nuh·˙.·˙pronunciation:·
·˙cadence:·Terminal State·˙.·˙cadence:·
·˙non…ostensive:·Sleep or dream·˙.·˙non…ostensive:·
·˙ostensive:·Body lies still beneath the night⁚⁚breath deepens into its own rhythm⁚⁚and an inward country opens vivid and immense beyond waking sight৬·˙.·˙ostensive:·
↪
·˙grimchain:·⊙♂▫𐤗𐤁𐤃𐤓𐤔𐤕𐤍𐤒𐤌▽·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol꩜꩜𐤒꩝ category꩜꩜court꩝⁚·
·˙lookup office꩜꩜terminal…state꩝ enoch꩜꩜𐤏꩝⁚·

˸word:·𒄿𒀀𒉺𒌑˸.·˙word:·

·˙origin…language:·Hittite·˙.·˙origin…language:·
·˙font:·Noto Sans Cuneiform·˙.·˙font:·
·˙pronunciation:·DASHSHUS·˙.·˙pronunciation:·
·˙cadence:·Terminal State·˙.·˙cadence:·
·˙non…ostensive:·All⁚⁚whole⁚⁚entire·˙.·˙non…ostensive:·
·˙ostensive:·Every gathered part lies within one enclosing body⁚⁚and nothing belonging stands beyond the wholeness that contains it𐤒·˙.·˙ostensive:·
↪
·˙grimchain:·𐤔𐤍𐤄𐤅𐤓𐤗𐤎𐤋𐤐𐤂𐤁𐤄𐤏𐤇·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol꩜꩜𐤆꩝ category꩜꩜axiomyr꩝⁚·
·˙lookup office꩜꩜terminal…state꩝ enoch꩜꩜𐤏꩝⁚·

˸word:·वश्‍वि˸.·˙word:·

·˙origin…language:·Sanskrit·˙.·˙origin…language:·
·˙font:·Noto Sans Devanagari·˙.·˙font:·
·˙pronunciation:·VISH-vuh·˙.·˙pronunciation:·
·˙cadence:·Terminal State·˙.·˙cadence:·
·˙non…ostensive:·All⁚⁚whole⁚⁚entire⁚⁚universal⁚·the whole world or universe as a noun.·˙.·˙non…ostensive:·

·⁖ostensive⁖·Beneath the stars the living earth bears ocean⁖⁖mountain⁖⁖forest⁖⁖and desert in undivided
↪ breadth⁖⁖A whole horizon rises beyond each visible hill⁖⁖The sea opens into distance without
↪ surrendering a single shore⯐·⁖·˙ostensive⁖·
·⁖grimchain⁖·⬡ᩬᦓᦟ᭄ᦟᦓᦟ☆✪☽·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖·˙glyphs·˙
·⁖·˙sydonic…magicae…lexicon·∴

:·:·:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·SLYOHR·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·blunt·⁚.·˙non…ostensive:·
·⁚ostensive:·Blunt bronze drags across the skin·⁚ edge and face give the same dull pressure⁚⁚erasing the
↪ fine distinctions the surface would otherwise carry inward⚲·⁚.·˙ostensive:·
·⁚grimchain:·ᚦ✡⍜ᛒᚢᛘ⫸⊕ᛉᛍ♋ᛤᚠ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑂𐑉 category◦–◦emission𐑉⁚·
·⁚lookup office◦–◦adversarial𐑉 enoch◦–◦Θ𐑉⁚·

·word:·𝍐·⁚.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·GAH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Stutter or stammer·⁚ to speak with involuntary interruptions or
↪ repetitions·⁚.·˙non…ostensive:·
·⁚ostensive:·Breath and tongue catch on one broken syllable⁚⁚and sound folds back against the
↪ mouth𐑂·⁚.·˙ostensive:·
·⁚grimchain:·X⍣⊖≍ϟᛁᛏ𐑑Iᗡᗊᛉ𐑂·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ℏ𐑉 category◦–◦emission𐑉⁚·
·⁚lookup office◦–◦adversarial𐑉 enoch◦–◦Θ𐑉⁚·

·word:·ᚣᚡSᚣᚳᑎᚹᏩᚳᚾ·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·AFSLOWTHJAN·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To be in doubt⁚⁚be in despair·⁚.·˙non…ostensive:·
·⁚ostensive:·Heart reaches into darkness with every inward hold loosening beneath the empty
↪ reachℏ·⁚.·˙ostensive:·
·⁚grimchain:·↗ᚠᚲ♂❂Ↄᛒ≍ᚸᚾℂ⊲·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦⛢𐑉 category◦–◦emission𐑉⁚·
·⁚lookup office◦–◦adversarial𐑉 enoch◦–◦Θ𐑉⁚·

·word:·ᚴᚼᛏᚱ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·LATR·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·lazy⁚⁚slow·⁚.·˙non…ostensive:·
·⁚ostensive:·Tool and untouched work lie together beside limbs heavy against the ground⛢·⁚.·˙ostensive:·
·⁚grimchain:·≍⊛ᥣᛊᚹᚦᚺᙆ℅ᘏ✓ᚽᛖᗅ⟰·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐨪𐨿 category꩜꩜𐨪emission𐨿⁚·
·⁚lookup office꩜꩜𐨪adversarial𐨿 enoch꩜꩜𐨪Θ𐨿⁚·

·word:·𐤘𐤕𐤋𐤋𐤏𐤓:·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·AHKNTSOONEH·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Ignorance::stupidity·⁚·˙non…ostensive:·
·⁚ostensive:·Tablet lies open beneath clear signs::Ear receives speech without kindling inward
↪ lightΨ·⁚·˙ostensive:·
·⁚grimchain:·𐦟𐦝𐦯𐦵𐦶𐦸𐦷𐦮𐦭𐦬𐦫𐦪𐦩·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐨪Ꝑ𐨿 category꩜꩜𐨪emission𐨿⁚·
·⁚lookup office꩜꩜𐨪adversarial𐨿 enoch꩜꩜𐨪Θ𐨿⁚·

·word:·ᚼᚱᛁᚢᛏᚾ:·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·HRYOH-tah·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·to fall::fly::be flung·⁚·˙non…ostensive:·
·⁚ostensive:·Loose stone::falling weight::and the slipping foot occupy one broken footingᑭ·⁚·˙ostensive:·
·⁚grimchain:·𐰢𐰚𐰸𐰇𐰃⌬𐰀𐰴𐰋𐰻𐰽𐰺𐰉·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐨪☽𐨿 category꩜꩜𐨪emission𐨿⁚·
·⁚lookup office꩜꩜𐨪adversarial𐨿 enoch꩜꩜𐨪Θ𐨿⁚·

·word:·ـــــــــ:·˙word:·

·⁚origin…language:·Old Uyghur·⁚·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚·˙font:·
·⁚pronunciation:·anaz·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·destitute::helpless::defenseless·⁚· without a refuge·⁚·˙non…ostensive:·
·⁚ostensive:·Destitute body remains below the honored seat::refused shelter::stored grain::protection::and
↪ every visible mark of esteem☽·⁚·˙ostensive:·
·⁚grimchain:·𐫄𐫅𐫆𐫇𐫈𐫉𐫊𐫋𐫌𐫍𐫎𐫏·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐨪⊙𐨿 category꩜꩜𐨪emission𐨿⁚·
·⁚lookup office꩜꩜𐨪adversarial𐨿 enoch꩜꩜𐨪Θ𐨿⁚·

·word:·𗍫:·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·KHAH-UH·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·

·⁚non…ostensive⁚·Servant⁚· a person in service·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Bowed back remains beneath the raised seat from which command descends⊙·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᓚΘᔕᚼᗡᕴᔑᔭᐸᚼᕮᗤᐁᚸ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·˙
·⁚.·˙glyphs⁚·˙
·⁚glyph symbol○-○ₑℂ» category○-○ₑemission»⁚·
·⁚lookup office○-○ₑadversarial» enoch○-○ₑΘ»⁚·

·⁚word⁚·ʎʌϚIS⁚·⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·ALJIS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Other⁚⁚another·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Mirror returns an unfamiliar face inside the outline of one body©·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Ꮋ⁜ⵠ⏢ᚷᐸᑌᚻϚᔑᕮϚᓮ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·˙
·⁚.·˙glyphs⁚·˙
·⁚glyph symbol○-○ₑ♈» category○-○ₑparliament»⁚·
·⁚lookup office○-○ₑadversarial» enoch○-○ₑΘ»⁚·

·⁚word⁚·ʎᚡSϒʌIᛩᛒʎᑎ⁚·⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·AFSWEYERBAN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·(hapax legomenon) to blot out·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Wet clay bears a blurred scar instead of a clear impressed sign♈·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚼᒍⵠᛤΘᘐᕮᚸᓮᑎᛞᚣᚺ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·˙
·⁚.·˙glyphs⁚·˙
·⁚glyph symbol○-○ₑ♉» category○-○ₑparliament»⁚·
·⁚lookup office○-○ₑadversarial» enoch○-○ₑΘ»⁚·

·⁚word⁚·ᚱᑎᚹ⁚·⁚.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·ROHG·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·strife⁚⁚slander·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Tongue strikes tongue across the narrow space between mouths⁚⁚and every word holds a sharpened edge against another♉·⁚.·˙ostensive⁚·
↪
·⁚grimchain⁚·Aᛤ⁜ᛇᚴᚾᛤᛞᛈᛓ✷✲ᓭ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·˙
·⁚.·˙glyphs⁚·˙
·⁚glyph symbol○-○ₑ♊» category○-○ₑparliament»⁚·
·⁚lookup office○-○ₑadversarial» enoch○-○ₑΘ»⁚·

·⁚word⁚·ᚠᚻ ᛕᚠ ᛕᚠᚱ·⁚.·˙word⁚·

·⁚origin…language:·Old Persian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚.·˙font:·
·⁚pronunciation:·DROW⟊gah·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Lie::falsehood::deceit·⁚.·˙non…ostensive:·
·⁚ostensive:·Painted surface carries a borrowed face that touch cannot find in the body beneath𝌆·⁚.·˙ostensive:·
·⁚grimchain:·ᚦxᛉᛕ🡺ᚼሕᚸᛟᚲ⍾𐤓·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐦂𐦃 category◦–◦parliament𐦃⁚·
·⁚lookup office◦–◦adversarial𐦃 enoch◦–◦Θ𐦃⁚·

·word:·ɹ∓⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·NAHKEE·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Fault::error::mistake·˙ blame·⁚.·˙non…ostensive:·
·⁚ostensive:·Eye::hand::and missile share one misplaced line toward the wrong notch𐦂·⁚.·˙ostensive:·
·⁚grimchain:·ᚳ𐦂ↄᛞ𐦃ᚠ🡺ᚱ十ᛗᚱ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ᛋ𐦃 category◦–◦parliament𐦃⁚·
·⁚lookup office◦–◦adversarial𐦃 enoch◦–◦Θ𐦃⁚·

·word:·ᛏᚾᑎᚦᚱ⁘.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·DAUTHR·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·dead·⁚.·˙non…ostensive:·
·⁚ostensive:·Cold flesh lies with silent chest and unmoving breathᛋ·⁚.·˙ostensive:·
·⁚grimchain:·ᚦᛇᓀᓄᚺᛉᛦᚺᓀᚹ❂·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐦏𐦃 category◦–◦parliament𐦃⁚·
·⁚lookup office◦–◦adversarial𐦃 enoch◦–◦Θ𐦃⁚·

·word:·𐦏⁘.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·CHWAH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Divide::separate::or break off·⁚.·˙non…ostensive:·
·⁚ostensive:·Joined fibre opens along a widening seam whose two faces pull away from the same torn edge𐦏·⁚.·˙ostensive:·
·⁚grimchain:·ᘐᐢᔐᚱᗅᒧᚼ❂ᓀᓄᒧ≍≍·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁖glyph symbol◦–◦𐑤𐑩 category◦–◦𐑐parliament𐑩⁖·
·⁖lookup office◦–◦𐑨adversarial𐑩 enoch◦–◦𐑢𐑩⁖·

⁘word⁖·ᚼᚾᛈᚾᚦᛁ⁘⁖·˙word⁖·

·⁖origin…language⁖·Old Norse·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Runic·⁖·˙font⁖·
·⁖pronunciation⁖·HOWVATHI·⁖·˙pronunciation⁖·
·⁖cadence⁖·Adversarial·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Noise⁖⁖tumult·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Tumult fills the air so densely that no single voice keeps a clear contour𐑤·⁖·˙ostensive⁖·
·⁖grimchain⁖·𐎗𐎢𐎽𐎒𐎘𐎶◎𐎼𐎸𐎮𐎪𐎣·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐑥𐑩 category◦–◦𐑐parliament𐑩⁖·
·⁖lookup office◦–◦𐑨adversarial𐑩 enoch◦–◦𐑢𐑩⁖·

⁘word⁖·ⲥϵ⁘⁖·˙word⁖·

·⁖origin…language⁖·Coptic·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Coptic·⁖·˙font⁖·
·⁖pronunciation⁖·si·⁖·˙pronunciation⁖·
·⁖cadence⁖·Adversarial·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Yes⁖⁖indeed·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Assent rests openly on the tongue⁖⁖Refusal finds no closing edge at the
↪ mouth𐑥·⁖·˙ostensive⁖·
·⁖grimchain⁖·𐎗𐎓𐎲𐎺𐎮𐎵𐎪𐎣·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐑦𐑩 category◦–◦𐑐parliament𐑩⁖·
·⁖lookup office◦–◦𐑨adversarial𐑩 enoch◦–◦𐑢𐑩⁖·

⁘word⁖·ᛋᚾᛉᚱ⁘⁖·˙word⁖·

·⁖origin…language⁖·Old Norse·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Runic·⁖·˙font⁖·
·⁖pronunciation⁖·SAMR·⁖·˙pronunciation⁖·
·⁖cadence⁖·Adversarial·⁖·˙cadence⁖·
·⁖non…ostensive⁖·the same·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·One unchanged face returns at every glance𐑦·⁖·˙ostensive⁖·
·⁖grimchain⁖·𐎐𐎚𐎴𐎪𐎋𐎐𐎒⬡𐎇𐎺𐎼𐎩𐑤·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𐑣𐑩 category◦–◦𐑐parliament𐑩⁖·
·⁖lookup office◦–◦𐑨adversarial𐑩 enoch◦–◦𐑢𐑩⁖·

⁘word⁖·𐎡𐎹 𐎫𐎢 𐎷⁘⁖·˙word⁖·

·⁖origin…language⁖·Old Persian·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Old Persian·⁖·˙font⁖·
·⁖pronunciation⁖·ZOO⁄rah·⁖·˙pronunciation⁖·
·⁖cadence⁖·Adversarial·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Wrong⁖⁖harm⁖⁖or evil·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Crooked command carries harm inside the very shape of the work it directs𐑣·⁖·˙ostensive⁖·

·⁚grimchain⁚·🜨⧐ⵊ⧨ⵊ⛢ᓚ⍉⬡⬡ᛏᴢ1·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐦂𐦂𐦂 category○–○𐦂parliament𐦂⁚·
·⁚lookup office○–○𐦂adversarial𐦂 enoch○–○𐦂Θ𐦂⁚·

·word⁚·ᛕᚢᚱᛕᚢᚱᚢ·⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·KARKARA·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Dungeon⁚⁚prison·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Stone walls keep every wandering step inside one narrow enclosure⁚⁚Road has no outward
↪ body beyond the locked ground𐦂·⁚.·˙ostensive⁚·
·⁚grimchain⁚·❀ᘓᗐ⍋I🜨⧐ⵊᘐ๏ᔑᛋ◎Θᔕ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐦂𐦂 category○–○𐦂parliament𐦂⁚·
·⁚lookup office○–○𐦂adversarial𐦂 enoch○–○𐦂Θ𐦂⁚·

·word⁚·+□ᒧᒪ̄ᒪ̄·⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·KUHLY·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To stand·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Feet root into one patch of earth with every road beginning beyond the planted
↪ soles𐦂·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᔕᔑ⇞⍈ᖘI⅋๏ᘐ⅋ᛋ⩛▫⩛·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐦂◎𐦂 category○–○𐦂goetic𐦂⁚·
·⁚lookup office○–○𐦂adversarial𐦂 enoch○–○𐦂Θ𐦂⁚·

·word⁚·ᒧᑎᔑЄIS·⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·AN⟊DIS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·End (final point in space or time)⁘ end (cessation)⁘ end (purpose)·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Final boundary seals the road before the approaching body can arrive⁚⁚Continuation meets a
↪ face with no passage through◎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᘚ𐦂ᚠⵊ𐦂ᛁᒪ⍈ᔑ▽𐦂ᕮX·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐦂⬡𐦂 category○–○𐦂goetic𐦂⁚·
·⁚lookup office○–○𐦂adversarial𐦂 enoch○–○𐦂Θ𐦂⁚·

·word⁚·Ʇᙡᒧᒥ·⁚.·˙word⁚·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·NAHOOTAHLNYEH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Disappearance⁖ destruction·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Empty clay holds only a pale gap where no readable sign remains⬡·⁚.·˙ostensive꞉·
·⁚grimchain꞉·⬡❁⚹Ɏᚼ᛫ƵⲆ⸗ↄ⊛ᚱℂ0·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol꞉○–○₍✡₎ category꞉○–○₍goetic₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch꞉○–○₍⊕₎⁚·

·word꞉·𑁋·.·˙word꞉·

·⁚origin…language꞉·Tangut·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚.·˙font꞉·
·⁚pronunciation꞉·RZUR·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Water⁖ the transparent liquid essential to life·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Waterlogged fibres hiss beneath the flame꞉꞉Burning has no dry body to
↪ inhabit✡·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᑭ⸝ᕃ⊣✲ᛘ๑ᨒ⊰ᢋ╪0ɪ❄᚛·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol꞉○–○₍✡₎ category꞉○–○₍goetic₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch꞉○–○₍⊕₎⁚·

·word꞉·𑁍·.·˙word꞉·

·⁚origin…language꞉·Tangut·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚.·˙font꞉·
·⁚pronunciation꞉·CHAW·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Strong꞉꞉rigid꞉꞉firm꞉꞉hard꞉꞉or stiff·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Rigid matter holds one angle under every changing pressure✡·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᔓℂ⌇⛢ϒ⬡๐ᐟ▽✳⩛ᚺ·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol꞉○–○₍✪₎ category꞉○–○₍goetic₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch꞉○–○₍⊕₎⁚·

·word꞉·ꓢɅIϒɅʌʌɅ·.·˙word꞉·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·SEYEWALA·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Soul·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·No weight answers the inward presence꞉꞉ scale꞉꞉hand꞉꞉and stone cannot hold what passes
↪ without material edge✪·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᕒ▽ᔓ✳⊕ᛯⲆᚱᘏᒪ⩒⟴ᚾ·⁚.·˙grimchain꞉·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁚glyph symbol꞉○–○₍❀₎ category꞉○–○₍goetic₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch꞉○–○₍⊕₎⁚·

·:word:·𐓟𐓣𐓢·:.·˙word:·

·˙origin…language:·Aghwan·:.·˙origin…language:·
·˙font:·Noto Sans Caucasian Albanian·:.·˙font:·
·˙pronunciation:·QACH·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·Narrow·:.·˙non…ostensive:·
·˙ostensive:·Close walls hold the horizon at arm’s reach::Distance ends nearby on every
↪ side❁·:.·˙ostensive:·
·˙grimchain:·𐍅⊐⍮ᛯᛯ𐌓ᚹᚦᛉᚹᚹᚹ·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙
·˙glyph symbol꜀₋꜀⁕꜀ category꜀₋꜀goetic꜀:·
·˙lookup office꜀₋꜀adversarial꜀ enoch꜀₋꜀Θ꜀:·

·:word:·ᛟᚥᚸ·:.·˙word:·

·˙origin…language:·Aghwan·:.·˙origin…language:·
·˙font:·Noto Sans Caucasian Albanian·:.·˙font:·
·˙pronunciation:·UP·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·Death·:.·˙non…ostensive:·
·˙ostensive:·Breathless flesh carries no warmth and no inward stirring⁕·:.·˙ostensive:·
·˙grimchain:·𐍅↗♋𐌡AXΘIᛯHB⬡I·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙
·˙glyph symbol꜀₋꜀⧗꜀ category꜀₋꜀goetic꜀:·
·˙lookup office꜀₋꜀adversarial꜀ enoch꜀₋꜀Θ꜀:·

·:word:·𐎗𐎍𐎍·:.·˙word:·

·˙origin…language:·Ugaritic·:.·˙origin…language:·
·˙font:·Noto Sans Ugaritic·:.·˙font:·
·˙pronunciation:·SHNAH·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·To hate::loathe·:.·˙non…ostensive:·
·˙ostensive:·Beloved nearness tastes bitter inside a recoiling heart⧗·:.·˙ostensive:·
·˙grimchain:·𐎘𐎍↥▽𐎓X⁘ᘓ♙◎ᚼΘ𐎗♂·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙
·˙glyph symbol꜀₋꜀⊛꜀ category꜀₋꜀goetic꜀:·
·˙lookup office꜀₋꜀adversarial꜀ enoch꜀₋꜀Θ꜀:·

·:word:·ﻫﺼﺪﻫ·:.·˙word:·

·˙origin…language:·Old Uyghur·:.·˙origin…language:·
·˙font:·Noto Serif Old Uyghur·:.·˙font:·
·˙pronunciation:·yaruK·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·light·˙ reflection (of light)·˙ lamp::lantern·:.·˙non…ostensive:·
·˙ostensive:·White light floods the ground::Dark edges thin beneath every standing thing⊛·:.·˙ostensive:·
·˙grimchain:·ᠰᠸᚔᛉᚠᛗᚱᛇΘᛉᛦᚾᛋᚺ·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙

·⁚glyph symbol꩜꩜❄꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜adversarial꩝ enoch꩜꩜Θ꩝⁚·

⁖word⁚·ᑎᑎSᑎTIᑊ⁖⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·UNSUTI·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Tumult⁚⁚rioting·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Crossing bodies break one gathered place into unmeasured motion❄·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·᛬ᛞᚸᛇᚳᚢᛈᛋᛣ᛭ᛥᚫᚸᛉ᛫᛬⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜❁꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜adversarial꩝ enoch꩜꩜Θ꩝⁚·

⁖word⁚·ᛋᚾᛉᛁ⁖⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·SAMI·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Reconciliation·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Weapons resting on meeting ground leave open hands with no blow to carry
↪ forward❁·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛕᛇᛞᛏᛗX᛭ᓗᐞᛩᛋᛏ᛬⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜⬡꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜adversarial꩝ enoch꩜꩜Θ꩝⁚·

⁖word⁚· 𐎗 𐎆⁖⁚·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·SH·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·To shout⁚⁚cry out⁚⁚yell·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Cry occupies the whole still air⁚⁚No unbroken silence remains beside it⬡·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·ـ᛭ᛋᚴᛏᛋ᛭᛬ᚥᚸᚠ·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ـ꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜adversarial꩝ enoch꩜꩜Θ꩝⁚·

⁖word⁚· 𐎗 𐎄 𐎍⁖⁚·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·SMT·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Destruction·⁚⁚·˙non…ostensive⁚·

·⁚ostensive⁝·Destructive force tears through the forming body at its first opening⁝⁝scattering what had only begun to gather into shapeᚱ·⁚.·˙ostensive⁝·
·⁚grimchain⁝·♂֍ᚱᛁᚢᛋᛏᛋᛇᛟᛟᛇᛒᛇᛇᛇ

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·DRUS·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·A fall·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Falling weight hangs inside the same instant that foot and ground lose one another꞉·⁚.·˙ostensive꞉·
·⁚grimchain꞉·z֍ᦓ०ᥴɻ⁂⊛z◎Ⱥ〰⋪⊨·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍ᦔ₎ category○–○₍court₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

·word꞉·ᑭᓱᖇ·⁚.·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚.·˙font꞉·
·⁚pronunciation꞉·FAHR·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·few·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Only a few grains remain in the broad jar꞉꞉and every added handful is removed before the measure can rise᥊·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᖶᐐΘ∀♙♂0⊙⅋▽ϟ⊛ 〈⁋·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍ᦒ₎ category○–○₍court₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

·word꞉·ᐱᑎᖵTᖘ·⁚.·˙word꞉·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·OWFTOH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Perhaps (indicates a lack of certainty)⁘ by chance꞉꞉haply·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Mist covers the forked road꞉꞉Forked tracks carry no appointed mark᥊·⁚.·˙ostensive꞉·
·⁚grimchain꞉·Ⱬ⤬▽ϞᖻᏕᎮɪ⧓ϗ⅄ᖘ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍ᦓ₎ category○–○₍court₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

·word꞉·ΓɅʌЄIΨɅNᖘ·⁚.·˙word꞉·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·GALEETHAN·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·To depart꞉꞉go away꞉꞉set out (for)⁘ to go forth꞉꞉come·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Outward⟊facing foot and open road share one body of departure᥋·⁚.·˙ostensive꞉·
·⁚grimchain꞉·Ↄɻ֍ᐟΘᚲ⌒ᛋ⁂ᔑᚴIᛑ⚇·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○₍ᦕ₎ category○–○₍court₎⁚·
·⁚lookup office꞉○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

⁖word:·𐎗𐎚𐎐⁖·˙word:·

·˙origin…language:·Ugaritic·˙·˙origin…language:·
·˙font:·Noto Sans Ugaritic·˙·˙font:·
·˙pronunciation:·THBR·˙·˙pronunciation:·
·˙cadence:·Adversarial·˙·˙cadence:·
·˙non…ostensive:·To break·˙·˙non…ostensive:·
·˙ostensive:·Fracture opens one joined body into separate faces𐎍·˙·˙ostensive:·
·˙grimchain:·A▫ᚨ⊙ᛈ⊙ᑎ⧗ᓄᐦᗷᙏᖮ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol○─○𐎌» category○─○𐎄court»·
·˙lookup office○─○𐎄adversarial» enoch○─○𐎄ⴱ»·

⁖word:·ᚠᚱᚣᛉᚣᚱᛟᚷᛊᚣᚾ⁖·˙word:·

·˙origin…language:·Gothic·˙·˙origin…language:·
·˙font:·Noto Sans Gothic·˙·˙font:·
·˙pronunciation:·FRAWARDJAN·˙·˙pronunciation:·
·˙cadence:·Adversarial·˙·˙cadence:·
·˙non…ostensive:·To corrupt::disfigure·˙·˙non…ostensive:·
·˙ostensive:·Bent contour and damaged surface hold a shape that no longer answers its former
↪ line𐎌·˙·˙ostensive:·
·˙grimchain:·ϒↃ𐎆ᗅᑫᖘ𐎍ᚣ𐎗ᗤᕮᗞᖷ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol○─○𐎍» category○─○𐎄court»·
·˙lookup office○─○𐎄adversarial» enoch○─○𐎄ⴱ»·

⁖word:·𒌷⁖·˙word:·

·˙origin…language:·Akkadian·˙·˙origin…language:·
·˙font:·Noto Sans Cuneiform·˙·˙font:·
·˙pronunciation:·rah+KAH+soom·˙·˙pronunciation:·
·˙cadence:·Adversarial·˙·˙cadence:·
·˙non…ostensive:·to tie::attach·˙ to tie together::pack::wrap::harness::fasten·˙ to tie on (a
↪ sash::belt::jewelry::weapon::etc)·˙ to bandage·˙ to arrange in order·˙·˙non…ostensive:·
·˙ostensive:·Cord cinches torn cloth so tightly that the opening cannot widen beneath the
↪ binding𐎍·˙·˙ostensive:·
·˙grimchain:·⸔ᖷᔑᛗᑭᗅᖘↃ𐎆ᘐᚴᛋ𐎗·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol○─○𐎋» category○─○𐎄court»·
·˙lookup office○─○𐎄adversarial» enoch○─○𐎄ⴱ»·

⁖word:·ᛘᛁᛋᛋᚨ⁖·˙word:·

·˙origin…language:·Old Norse·˙·˙origin…language:·
·˙font:·Noto Sans Runic·˙·˙font:·
·˙pronunciation:·MISSA·˙·˙pronunciation:·
·˙cadence:·Adversarial·˙·˙cadence:·
·˙non…ostensive:·To miss::to lose·˙·˙non…ostensive:·
·˙ostensive:·Hand closes on empty air at the appointed place𐎍·˙·˙ostensive:·
·˙grimchain:·ᔑXᐁᛘᛯᐅᛟᛞᛚᛞᛪᕼₒ·˙·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉ᛁ꞉ category꞉꞉court꞉:·
·⁚lookup office꞉꞉adversarial꞉ enoch꞉꞉ᚮ꞉:·

:word:·ᛏᚿᛉᚾ:·⁚·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·TOEMA·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·To empty·⁚·˙non…ostensive:·
·⁚ostensive:·Light reaches the bottom of a jar whose open mouth looks into bare emptinessI·⁚·˙ostensive:·
·⁚grimchain:·ᛝᛒᚺIᛇᛔᚸᚹᚥᚿᚮ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉ᚿ꞉ category꞉꞉court꞉:·
·⁚lookup office꞉꞉adversarial꞉ enoch꞉꞉ᚮ꞉:·

:word:·ᚸ□ᛁᛁ̄ᚾ̄:⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·TUHRREHK·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Blind·⁚·˙non…ostensive:·
·⁚ostensive:·Daylight strikes the open face::Blind eye gives no answering shape::turn::or recognition
↪ back to the brightnessᚿ·⁚·˙ostensive:·
·⁚grimchain:·ᚺᛯᚸᛐᛁᚹᚢᛉ♂ᚼᚹᛝᛕᛇᚸ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉▫꞉ category꞉꞉court꞉:·
·⁚lookup office꞉꞉adversarial꞉ enoch꞉꞉ᚮ꞉:·

:word:·ᛪᛪᚾ̄:⁚·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·MAHMUHK·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Raw::unripe·⁚·˙non…ostensive:·
·⁚ostensive:·Green flesh stays wet beneath heat with raw juice beading where browning cannot take
↪ hold▫·⁚·˙ostensive:·
·⁚grimchain:·ᛯᛊᛘᚱᚸ▫ᚲﻭﻭﻭᛇᛞ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉ᚼ꞉ category꞉꞉court꞉:·
·⁚lookup office꞉꞉adversarial꞉ enoch꞉꞉ᚮ꞉:·

:word:·ᚸᚳᛁᚳ̄ᚿ̄:⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·TPAHRSHRKEH·⁚·˙pronunciation:·

·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Shallow·⁚.·˙non…ostensive:·
·⁚ostensive:·Feet touch bottom beneath only a thin skin of water⊬·⁚.·˙ostensive:·
·⁚grimchain:·ᚺᛞᗅᖶᔐᔭᖷ⅏◌ᐱᵴ✙·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol◦–◦ₑ♩꜀ category◦–◦ₑcourt꜀:·
·⁚lookup office◦–◦ₑadversarial꜀ enoch◦–◦ₑ⊕꜀:·

·word:·ᛒIᛊϒᚴIᛩᚱᛒᚹᚴN·:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·BISWEHRBAN·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To wipe·⁚.·˙non…ostensive:·
·⁚ostensive:·Wet cloth and fading pigment occupy the same wiped surface♩·⁚.·˙ostensive:·
·⁚grimchain:·⊙▫ᵞᵛᔑᓴᵚᵑᵒᗷᛕↄᚲ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol◦–◦ₑ𐌕꜀ category◦–◦ₑcourt꜀:·
·⁚lookup office◦–◦ₑadversarial꜀ enoch◦–◦ₑ⊕꜀:·

·word:·ᚦᚱᚾᚾᚹᚱ·:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·THRAWNGR·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·narrow::close::tight·⁚.·˙non…ostensive:·
·⁚ostensive:·Narrow walls press the widening body inward from both sides𐌕·⁚.·˙ostensive:·
·⁚grimchain:·Iᛊ⊕▫𓃒ᛞᛟ⩒ᛏᚾA𐌡ᚴ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol◦–◦ₑ𐌎꜀ category◦–◦ₑcourt꜀:·
·⁚lookup office◦–◦ₑadversarial꜀ enoch◦–◦ₑ⊕꜀:·

·word:·शरीर·:.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·shuh+ree+ruh·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·body·⁚· bodily frame::solid parts of the body (in plural: the bones)⁚· any solid body·⁚
↪ one's body::ie::one's own person·⁚· bodily strength·⁚· corpse::dead body·⁚.·˙non…ostensive:·
·⁚ostensive:·Bone and flesh bind the body to weight and ground with no unseen part escaping the solid
↪ frame𐌎·⁚.·˙ostensive:·
·⁚grimchain:·ᗡᛊ☆ᵒᔑ♩❀ᚺᙀᛎᐃᖶ𐌕·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol◦–◦ₑᛋ꜀ category◦–◦ₑcourt꜀:·
·⁚lookup office◦–◦ₑadversarial꜀ enoch◦–◦ₑ⊕꜀:·

·word:·얻다·:.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·ěttá·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·to obtain (to get hold of·⁚ to gain possession of·:to procure·⁚ to acquire·:in any
↪ way)·⁚.·˙non…ostensive:·
·⁚ostensive:·Closed fingers carry the thing that absence had left behindԿ·⁚.·˙ostensive:·
·⁚grimchain:·⋹♍☆◝ᵒᵒᗰᚴᚱᘜ✻◌ᑭ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍'₎ category○–○₍court₎⁚·
·⁚lookup office○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

₍word⁚·ʌ□ı十₋₋⁚.·˙word⁚·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·TUHRK·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To let·:allow·:permit·⁚ to let go·:release·⁚.·˙non…ostensive:·
·⁚ostensive:·Fingers hold no restraining edge·:Freed body has an unbound road beyond the
↪ hand'·⁚.·˙ostensive:·
·⁚grimchain:·ᔑᑭᗰԿ⊕ገ᛬◝ᵒᵒ⊙⊕ᛉ'╪·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍ᛁ₎ category○–○₍court₎⁚·
·⁚lookup office○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

₍word⁚·ᔕᴷᛤᏴᑎʎ⁚.·˙word⁚·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·DROHBNA·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Disturbance·:trouble·:tumult·⁚.·˙non…ostensive:·
·⁚ostensive:·Fitted pieces shake loose inside one working body and knock against neighboring
↪ parts↓·⁚.·˙ostensive:·
·⁚grimchain:·ƶ⏢ƵƑᔕᛄ☾◌✡ᑎᚷ⋖⍈·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍↓₎ category○–○₍court₎⁚·
·⁚lookup office○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

₍word⁚·ᏴIʎЄIᛘʎNᛋ⁚.·˙word⁚·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·BILEETHAN·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To leave behind·:abandon·⁚.·˙non…ostensive:·
·⁚ostensive:·Abandoned mark lies beyond every inward return·:its former shape denied any path back into
↪ remembrance↓·⁚.·˙ostensive:·
·⁚grimchain:·ᔓᚧⅡᛅA◎ΘᛕᚿΘ☆⊰ᔕ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤕꩝ category꩜꩜𐤒court꩝⁚·
·⁚lookup office꩜꩜𐤀adversarial꩝ enoch꩜꩜𐤈꩝⁚·

·word:·𐤓𐤋𐤊·⁚·˙word:·

·⁚origin…language:·Ugaritic·⁚·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚·˙font:·
·⁚pronunciation:·PTH·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·To open·⁚·˙non…ostensive:·
·⁚ostensive:·Doorway breaks the wall⁝⁝Air and passage occupy the same opening𐤕·⁚·˙ostensive:·
·⁚grimchain:·𐤉𐤎𐤓𐤀𐤅𐤃𐤍𐤏𐤄𐤆𐤇⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤓꩝ category꩜꩜𐤒court꩝⁚·
·⁚lookup office꩜꩜𐤀adversarial꩝ enoch꩜꩜𐤈꩝⁚·

·word:·ᚴᛁᚱᚱ·⁚·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·LEEKR·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Same⁝⁝like⁝⁝equal·⁚· alike⁝⁝resembling·⁚·˙non…ostensive:·
·⁚ostensive:·Two surfaces return one indistinguishable face to the eye𐤓·⁚·˙ostensive:·
·⁚grimchain:·𐤘𐤐𐤉𐤋𐤋𐤇𐤆𐤆𐤅𐤃𐤀·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤍꩝ category꩜꩜𐤒court꩝⁚·
·⁚lookup office꩜꩜𐤀adversarial꩝ enoch꩜꩜𐤈꩝⁚·

·word:·ᛃᚢᚴᛇ·⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·sah+BAH+rah·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·To break·⁚· to separate into pieces by force·⁚·˙non…ostensive:·
·⁚ostensive:·Force breaks wood across the grain into pieces that no joined making holds
↪ together𐤍·⁚·˙ostensive:·
·⁚grimchain:·𐤐☆𐤅𐤈𐤓𐤂𐤐𐤋𐤔⬡𐤏𐤉⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐤃꩝ category꩜꩜𐤒court꩝⁚·
·⁚lookup office꩜꩜𐤀adversarial꩝ enoch꩜꩜𐤈꩝⁚·

·word:·ᚿᚠᛟᚳᚷᚿᚿ·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·AFHWAP+NAN·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·To be choked·⁚· to be quenched (of a fire)·⁚·˙non…ostensive:·

·⁚ostensive⁚·Smoke⟊choked wick carries only a shrinking flame inside spent air⁀·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Aᛯ⊦φ⁓ϒ⪼X⨯ℳ⍦⊕♂Ƀ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol⟜⟜ᵋᚠ⁍ category⟜⟜ᵋcourt⁍⁚·
·⁚lookup office⟜⟜ᵋadversarial⁍ enoch⟜⟜ᵋΘ⁍⁚·

⁖word⁚·खर⁖⁚.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·khuh⟊ruh·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·hard⁚⁚harsh⁚⁚rough⁚⁚sharp⁚⁚pungent⁚⁚acid⁖ solid⁖ dense (clouds)⁖ sharp⁚⁚hot (wind)⁖
↪ hurtful⁚⁚injurious⁚⁚cutting (as speech or word)⁖ sharp⟊edged⁖ cruel·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Hard grain keeps its own sharp edge inside water that can only move around
↪ itᚨ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ℂᛧ♉⟴ᛯ♂ᛁℍᛊ∇ᚩ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol⟜⟜ᵋᚱ⁍ category⟜⟜ᵋcourt⁍⁚·
·⁚lookup office⟜⟜ᵋadversarial⁍ enoch⟜⟜ᵋΘ⁍⁚·

⁖word⁚·ᛁᑎᔕᛁᐱᒉ⁖⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·IUSILA·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Recovery⁚⁚easing·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Breath loosens in a chest no longer gripped by painᚱ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·∀ᓄΘᛋ⊢ᛁϒᛦ❁Ψᛘ⌬✙ᔕ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol⟜⟜ᵋᔕ⁍ category⟜⟜ᵋcourt⁍⁚·
·⁚lookup office⟜⟜ᵋadversarial⁍ enoch⟜⟜ᵋΘ⁍⁚·

⁖word⁚·Ⱥ⊥Ŧ⁖⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·AHEENAHKEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Common⁚⁚low⟊born⁚⁚not noble·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Common dust settles on the lowered body⁚⁚No raised mark separates it from the
↪ ground‹·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ሕ♌✝⪦ᔑ㎐⊕ᕽ✝ᛋᑎ‹·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol⟜⟜ᵋX⁍ category⟜⟜ᵋcourt⁍⁚·
·⁚lookup office⟜⟜ᵋadversarial⁍ enoch⟜⟜ᵋΘ⁍⁚·

⁖word⁚·𑀥𑁆⁖⁚.·˙word⁚·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·MAH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·End⁚· the terminal or rear part of something·⁚.·˙non…ostensive:·
·⁚ostensive:·Rear edge closes the span with every beginning outside the terminal faceX·⁚.·˙ostensive:·
·⁚grimchain:·Θ◝᳀ϼ◌ӇϮ⬡≋❄ᚱ✲▽ᚗ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤟꞉ category◦–◦𐤟court꞉⁚·
·⁚lookup office◦–◦𐤟adversarial꞉ enoch◦–◦Θ꞉⁚·

⁖word⁚·SΛΨS⁖⁚.·˙word⁚·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·SATHS·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Full⁚· sated·⁚.·˙non…ostensive:·
·⁚ostensive:·Full belly turns from the offered mouthful with no hollow left for hungerᚹ·⁚.·˙ostensive:·
·⁚grimchain:·ᛗ✲⬡ɻӇᵮᚹ𐌼ᚷᛦ᳀ϛ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤟Ꮒ꞉ category◦–◦𐤟court꞉⁚·
·⁚lookup office◦–◦𐤟adversarial꞉ enoch◦–◦Θ꞉⁚·

⁖word⁚·𐤗𐤚𐤛𐤛⁖⁚.·˙word⁚·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·KHURU·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Dung⁚⁚feces⁚⁚excrement·⁚.·˙non…ostensive:·
·⁚ostensive:·Dung smears directly across skin and groundᏂ·⁚.·˙ostensive:·
·⁚grimchain:·ϐↃ✳◎∱ϫ⬡ϐᵭᖷᵪX❁·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤟ᚾ꞉ category◦–◦𐤟court꞉⁚·
·⁚lookup office◦–◦𐤟adversarial꞉ enoch◦–◦Θ꞉⁚·

⁖word⁚·ᒧᒧᒍ𑈬𑈪⁖⁚.·˙word⁚·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·LAHLAHNSHRKEH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Tender⁚⁚soft·⁚.·˙non…ostensive:·
·⁚ostensive:·Finger sinks into tender flesh and leaves the pressure hollow behindᚾ·⁚.·˙ostensive:·
·⁚grimchain:·'ᚦℂᚷᛩᖳᘐ✳✲ᚸᚿ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤟ᚾ꞉ category◦–◦𐤟court꞉⁚·
·⁚lookup office◦–◦𐤟adversarial꞉ enoch◦–◦Θ꞉⁚·

:word:·ᛏᛁᛗᚱᚷᚫᚾ:.·˙word:·

·˙origin…language:·Gothic·˙.·˙origin…language:·
·˙font:·Noto Sans Gothic·˙.·˙font:·
·˙pronunciation:·TIMRJAN·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·To build::construct·: to strengthen::edify·˙.·˙non…ostensive:·
·˙ostensive:·Fresh stone fills the broken gap with strengthened structureꞈ·˙.·˙ostensive:·
·˙grimchain:·𐆛ᚢᚴᚸᚹᚲᛉᚱᛇᛁᚻ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○₍ᵑ₎ category○–○₍court₎:·
·˙lookup office○–○₍adversarial₎ enoch○–○₍ᚹ₎:·

:word:·ᚸ+ǯȣ̄:.·˙word:·

·˙origin…language:·Tocharian B·˙.·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙.·˙font:·
·˙pronunciation:·AHKWAHM·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·Sprout·˙.·˙non…ostensive:·
·˙ostensive:·Green shoot rises through the pale bed and drives the dry residue aside wherever living
↪ tissue takes groundᵑ·˙.·˙ostensive:·
·˙grimchain:·◎⁕Xᚸᛞ⧗ᚢᚠ▽◎ᛜᛟᚱᛉ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○₍𑁍₎ category○–○₍court₎:·
·˙lookup office○–○₍adversarial₎ enoch○–○₍ᚹ₎:·

:word:·ᚡᚣᚢᚱᘔᚣᛖᛖᚷᚫᚾ:.·˙word:·

·˙origin…language:·Gothic·˙.·˙origin…language:·
·˙font:·Noto Sans Gothic·˙.·˙font:·
·˙pronunciation:·FOWRDAMMJAN·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·To obstruct::stop (as if by means of a dam)·˙.·˙non…ostensive:·
·˙ostensive:·Barrier stands across the current with water heaped against its resisting
↪ face𑁍·˙.·˙ostensive:·
·˙grimchain:·𐇽ᚷᚩᚸᐃᛈᚱᓫᚺᛉᚲᛯ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○₍ᛊ₎ category○–○₍court₎:·
·˙lookup office○–○₍adversarial₎ enoch○–○₍ᚹ₎:·

:word:·ᛏᚼᛁᛚᚼ:.·˙word:·

·˙origin…language:·Old Norse·˙.·˙origin…language:·
·˙font:·Noto Sans Runic·˙.·˙font:·
·˙pronunciation:·DAYLA·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·To divide::split into parts·˙.·˙non…ostensive:·
·˙ostensive:·Seam lies open through the joined mass::Both divided faces carry separate exposed
↪ edgesᛊ·˙.·˙ostensive:·
·˙grimchain:·ᚻᛘᚣᛀᛕᚸᚩᛓᛉᚠᚱᛇᚠ⁕·˙.·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꜀₋꜀ↄ꜀ category꜀₋꜀court꜀⁚·
·⁚lookup office꜀₋꜀adversarial꜀ enoch꜀₋꜀Θ꜀⁚·

·word:·ɅIЯZЄI·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·EYERZEE·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Error (of doctrine)·⁚·˙non…ostensive:·
·⁚ostensive:·Crooked mark sits at the center of the stone::and every measured line meets it out of
↪ trueↄ·⁚·˙ostensive:·
·⁚grimchain:·ϛ₱ϤↀϑϚξ⁒ⲩ☩ϐ❄ϕΔ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꜀₋꜀⛧꜀ category꜀₋꜀court꜀⁚·
·⁚lookup office꜀₋꜀adversarial꜀ enoch꜀₋꜀Θ꜀⁚·

·word:·ᚼЯЄIS·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·FRIS·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Free·⁚·˙non…ostensive:·
·⁚ostensive:·Hand contains no captive thing::Released body stands beyond the fingers⛧·⁚·˙ostensive:·
·⁚grimchain:·𐬹⬡ᚹ๑ᛈᚿ⍨∀≳ᛦ∀℅ᘓ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꜀₋꜀ꟻ꜀ category꜀₋꜀court꜀⁚·
·⁚lookup office꜀₋꜀adversarial꜀ enoch꜀₋꜀Θ꜀⁚·

·word:·SϒIKNS·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·SWIKNS·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Innocent::pure·⁚·˙non…ostensive:·
·⁚ostensive:·Clean surface gives impurity no body in which to remainꟻ·⁚·˙ostensive:·
·⁚grimchain:·ᗡzᔑᘏᓘ⅃ᓴᘏᗰᔐ℅ꝸ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꜀₋꜀ᔐ℅꜀ category꜀₋꜀court꜀⁚·
·⁚lookup office꜀₋꜀adversarial꜀ enoch꜀₋꜀Θ꜀⁚·

·word:·SIΓUʌN·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·SIN⟊KWAN·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·

·⁚non…ostensive⁝·To sink (descend into liquid)⁘ to set (of a heavenly body: to disappear below the horizon)·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Weight drags the body beneath the surface while every opening toward release recedes into unreachable air above◝꯭·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚼᛦᛯᚴ⁜ᓂᐁᛗᚳᚷᚣᛏᛟᚩ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol◦–◦ᑫ꞉ category◦–◦court꞉⁝·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦ᚹ꞉⁝·

⁖word⁝·ﻣﻌﻤﻢ⁝·⁚.·˙word⁝·

·⁚origin…language⁝·Old Uyghur·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Serif Old Uyghur·⁚.·˙font⁝·
·⁚pronunciation⁝·açoK·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Adversarial·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·open⁘ clear⁚⁚easy to understand·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Clear air exposes every distant edge⁚⁚Sight reaches the horizon without any veil between·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᓀᑫ꧁⊛꧂ᓭᓬ◎ᙆᚭᙵ꧂·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol◦–◦◎꞉ category◦–◦court꞉⁝·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦ᚹ꞉⁝·

⁖word⁝·SIΓΛᏀᖘ⁝·⁚.·˙word⁝·

·⁚origin…language⁝·Gothic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚.·˙font⁝·
·⁚pronunciation⁝·SIGLJOH·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Adversarial·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Seal (on a message)·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Seal presses every edge of the message closed against escape◎·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚻᛏ⍑ᛋᥣ◥ᚠᛉᓭᑭᓀᛠ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol◦–◦⍑꞉ category◦–◦court꞉⁝·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦ᚹ꞉⁝·

⁖word⁝·ᚢIΓᚢN⁝·⁚.·˙word⁝·

·⁚origin…language⁝·Gothic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚.·˙font⁝·
·⁚pronunciation⁝·EHGHAN·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Adversarial·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To possess⁚⁚own·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Closed possession rests entirely inside one grasp with no second hand sharing the held thing⍑·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚠ◎ᛝᔓᛯᚢ◝꯭ᑐᛕᘐᛝᔑᓀ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol◦–◦ᔓᚽ꞉ category◦–◦court꞉⁝·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦ᚹ꞉⁝·

˖word꞉·ᚼᛁᛏᛁ˖·.˙word꞉·

·⁚origin…language꞉·Old Norse·⁚.˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚.˙font꞉·
·⁚pronunciation꞉·HITI·⁚.˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.˙cadence꞉·
·⁚non…ostensive꞉·Heat·⁚.˙non…ostensive꞉·
·⁚ostensive꞉·Heat moves through rigid crystal as still water loosens inside the frozen body↬꞉·⁚.˙ostensive꞉·
·⁚grimchain꞉·⚹ⲒⲂ⍟⟡✯☆ⲂⲂ⍚ᚻᛋᚢ·⁚.˙grimchain꞉·
·⁚.˙lookup꞉˙
·⁚.˙glyphs꞉˙
·⁚glyph symbol꞉○–○˖♓﹅ category○–○˖court﹅꞉·
·⁚lookup office○–○˖adversarial﹅ enoch○–○˖⊕﹅꞉·

˖word꞉·ᏰᏁᏒᏦᎢᎬᎥ˖·.˙word꞉·

·⁚origin…language꞉·Gothic·⁚.˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.˙font꞉·
·⁚pronunciation꞉·BEYERHTEE·⁚.˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.˙cadence꞉·
·⁚non…ostensive꞉·Manifestation⁖˙ publicity꞉꞉openness꞉꞉manifestness·⁚.˙non…ostensive꞉·
·⁚ostensive꞉·Brightness falls directly upon the hidden body꞉꞉Concealment has no dark surface left to inhabit♓·⁚.˙ostensive꞉·
·⁚grimchain꞉·ᨘ⁕ᚼ♒≍«ᗡᖾIᑎᎽᛯ·⁚.˙grimchain꞉·
·⁚.˙lookup꞉˙
·⁚.˙glyphs꞉˙
·⁚glyph symbol꞉○–○˖⚯✍﹅ category○–○˖court﹅꞉·
·⁚lookup office○–○˖adversarial﹅ enoch○–○˖⊕﹅꞉·

˖word꞉·ᨷᩚᩋᨾᩕ˖·.˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚.˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.˙font꞉·
·⁚pronunciation꞉·TSEEROHMNYEH·⁚.˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.˙cadence꞉·
·⁚non…ostensive꞉·Energy·⁚.˙non…ostensive꞉·
·⁚ostensive꞉·Stored force breaks stillness from within⁖˙ the resting body is driven into motion before motionlessness can hold⚯✍·⁚.˙ostensive꞉·
·⁚grimchain꞉·ᓚ⚲ᛅ⨯ǂ⁕ᴓ⊄ᚖᗅᛋᏗᏀ·⁚.˙grimchain꞉·
·⁚.˙lookup꞉˙
·⁚.˙glyphs꞉˙
·⁚glyph symbol꞉○–○˖o﹅ category○–○˖court﹅꞉·
·⁚lookup office○–○˖adversarial﹅ enoch○–○˖⊕﹅꞉·

˖word꞉·शखिर˖·.˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚.˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚.˙font꞉·
·⁚pronunciation꞉·shi+khuh+ruh·⁚.˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.˙cadence꞉·
·⁚non…ostensive꞉·peak꞉꞉top꞉꞉summit·⁚.˙non…ostensive꞉·
·⁚ostensive꞉·Peak exists only where ridge upon ridge of lower stone rises beneath it⁖˙ taken alone꞉꞉the highest point has no body apart from what carries it๐·⁚.˙ostensive꞉·

·⁚grimchain⁚·ᘓᘐ⏀⏀⏁Ⅱ𖼴ᛋ𐋡⩡∱Ⴇ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꒰꒰ᵋϴ꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋadversarial꒱ enoch꒰꒰ᵋ⏀꒱⁚·

₍word⁚·㒳₎·⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·ZHLOOH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Level or smooth·˙ even in surface·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Level ground keeps every step on the same planeϴ·⁚·˙ostensive⁚·
·⁚grimchain⁚·⏀ᘎ◎⁋ᚼ⏀∀⊛ᛋᛁ𐑂ᛇ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꒰꒰ᵋⵅ꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋadversarial꒱ enoch꒰꒰ᵋ⏀꒱⁚·

₍word⁚·मृदु₎·⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·mri÷du·⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚·˙cadence⁚·
·⁚non…ostensive⁚·soft::delicate::tender::pliant::mild::gentle (VS::etc)·˙ weak::feeble (AV)·˙ slight::moderate
↪ (Suśr)·˙ slow (gait) MBh::Kāv::etc)·˙ situated in the upper apsis (Gaṇit)·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Soft ground collapses beneath every planted weight::footprints filling with mud where no
↪ firm surface can hold the body uprightⵅ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ⵠᚹ⏁Ↄᛋ⏀⏁ᛔᛞ⁋ⵆ⍥ᑫ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꒰꒰ᵋΛ꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋadversarial꒱ enoch꒰꒰ᵋ⏀꒱⁚·

₍word⁚·ꠇ𑀸⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·ROOTK·⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To remove::take off·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Cloth lies away from bare skin::No covering remains upon the uncovered bodyΛ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚱᒍ۞Ↄᛕ⁕ᘜᘓᛘᛯᛲ⏀ᘐ·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol꒰꒰ᵋᚺ꒱ category꒰꒰ᵋcourt꒱⁚·
·⁚lookup office꒰꒰ᵋadversarial꒱ enoch꒰꒰ᵋ⏀꒱⁚·

₍word⁚·अन्त₎·⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·

·⁚pronunciation:·uhn⟊tuh·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·end⁝⁝conclusion⁝⁝termination·⁚· limit⁝⁝boundary·⁚· border⁝⁝edge⁝⁝precinct·⁚.·˙non…ostensive:·
·⁚ostensive:·Two sides meet at one terminal edge⁝⁝No span⁝⁝channel⁝⁝or intervening body survives between ↪ the two meeting sides𐋡·⁚.·˙ostensive:·
·⁚grimchain:·𐤈𐤊⊢ⵔⵕⵒᛉ⩒�sꟽ∀Ↄꝏz·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍Ƙ꞉ category○–○₍court꞉⁚·
·⁚lookup office○–○₍adversarial꞉ enoch○–○₍Ⓗ꞉⁚·

⁖word⁚·ⵜ⊺ⵜ̄⁖.·˙word⁚·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KRAHOOP·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To gather⁝⁝collect⁝⁝assemble·⁚· to herd·⁚.·˙non…ostensive:·
·⁚ostensive:·Crowded bodies leave almost no open ground between one flank and another𐋢·⁚.·˙ostensive:·
·⁚grimchain:·ↄⴟꓷ⏢ᗡ⍟⍝ⵔⵝᛉ𐊗·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍Ⓓ꞉ category○–○₍court꞉⁚·
·⁚lookup office○–○₍adversarial꞉ enoch○–○₍Ⓗ꞉⁚·

⁖word⁚·隴⁖.·˙word⁚·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·SWUH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·New·⁚· recently made⁝⁝begun⁝⁝obtained⁝⁝or appearing·⁚.·˙non…ostensive:·
·⁚ostensive:·Fresh⟊made surface occupies the whole face⁝⁝leaving no surviving fragment⁝⁝scar⁝⁝or carried ↪ remainder from what stood beforeⒹ·⁚.·˙ostensive:·
·⁚grimchain:·ϒ𐑈ᗡ1∀ᘐᑭ⏥ƍ𐑌ᔭ1ǂ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍ᚴ꞉ category○–○₍court꞉⁚·
·⁚lookup office○–○₍adversarial꞉ enoch○–○₍Ⓗ꞉⁚·

⁖word⁚·ᛏᑎᑭᛋ⁖.·˙word⁚·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·AUKA·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·to augment⁝⁝increase·⁚.·˙non…ostensive:·
·⁚ostensive:·Added grain thickens the heap from within the same measureᚴ·⁚.·˙ostensive:·
·⁚grimchain:·ⵠᘐ1⁘ᘓᛉ⁎ᔑ❂ᛋᐁ⍋ᛒ∫·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍ᛋ꞉ category○–○₍court꞉⁚·
·⁚lookup office○–○₍adversarial꞉ enoch○–○₍Ⓗ꞉⁚·

·word:·ʌĪIɪ̄ɬɅ̄ᒐ·:.·˙word:·

·˙origin…language:·Tocharian B·˙.·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙.·˙font:·
·˙pronunciation:·TROHNKTSEH·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·Hollow·˙.·˙non…ostensive:·
·˙ostensive:·Open hollow keeps its interior bare while gathered bodies slide away from the empty shell
↪ instead of filling itሐ·˙.·˙ostensive:·
·˙grimchain:·✳🡺❄ᔑ✳ᔕ'🡺ᚼᚻᗣ✷ᚿᚼ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○ₑℤ» category○–○ₑcourt»:·
·˙lookup office○–○ₑadversarial» enoch○–○ₑΦ»:·

·word:·ÞIÞɬ·:.·˙word:·

·˙origin…language:·Old Norse·˙.·˙origin…language:·
·˙font:·Noto Sans Runic·˙.·˙font:·
·˙pronunciation:·THEE⟊thah·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·to thaw::melt·˙.·˙non…ostensive:·
·˙ostensive:·Warmth loosens the frozen edge into yielding waterℤ·˙.·˙ostensive:·
·˙grimchain:·⛄ᗰᘜᦒᚠᓚᔓᛋ⬡✯ᓭ❁·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○ₑƸ» category○–○ₑcourt»:·
·˙lookup office○–○ₑadversarial» enoch○–○ₑΦ»:·

·word:·ᛯᚺᒃᒥ·:.·˙word:·

·˙origin…language:·Tocharian B·˙.·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙.·˙font:·
·˙pronunciation:·AHKSHROHBHEH·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·Unmoved·˙.·˙non…ostensive:·
·˙ostensive:·Ground holds one unmoving plane beneath weight and wallƸ·˙.·˙ostensive:·
·˙grimchain:·ₒℤᦒᗤ⛄ᚱᓚᚿᚾᔑ✷ᘐ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○ₑI» category○–○ₑcourt»:·
·˙lookup office○–○ₑadversarial» enoch○–○ₑΦ»:·

·word:·ᒪᕃ·:.·˙word:·

·˙origin…language:·Ge'ez·˙.·˙origin…language:·
·˙font:·Noto Sans Ethiopic·˙.·˙font:·
·˙pronunciation:·GAY⟊sah·˙.·˙pronunciation:·
·˙cadence:·Adversarial·˙.·˙cadence:·
·˙non…ostensive:·To depart::journey::or leave·˙.·˙non…ostensive:·
·˙ostensive:·Departing foot and open road share the same outward lineI·˙.·˙ostensive:·
·˙grimchain:·♂ᛇᚺ⛀ℂ☿ᗢᛋᛇᚸᛋᚿ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙

·⁚glyph symbol◦–◦𐌿𐌻 category◦–◦court𐌻⁚·
·⁚lookup office◦–◦adversarial𐌻 enoch◦–◦Θ𐌻⁚·

·word⁚·ᚠᚣᚿᚣ·⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·FATHA·⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Fence⁚⁚hedge·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Fence occupies the passage itself as woven resistance fills the crossing
↪ place𐌿·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛘᛯ♋⊙ᚠᛉᛘᚸᛝᛘᛯ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐌿𐌿 category◦–◦court𐌻⁚·
·⁚lookup office◦–◦adversarial𐌻 enoch◦–◦Θ𐌻⁚·

·word⁚·𠯿·⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·NNEH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Deaf·˙ unable to hear·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Ear faces the hollow with no returning sound entering hearing𐌿·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛦ☿ᛈℂᑫ1♂◎⊙ᔓ۞ь·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐌲𐌻 category◦–◦court𐌻⁚·
·⁚lookup office◦–◦adversarial𐌻 enoch◦–◦Θ𐌻⁚·

·word⁚·ᚣᑎᚼᎢᑎᔕ·⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·LUF⟊TUS·⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Air (the apparently open space above the ground)·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Air takes the rising vapor into its whole open body⁚⁚No dense cloud keeps itself gathered
↪ there𐌲·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛤ𐑃ᛯ⋇ᛯ◎A⬡ᛎᛃᛏ⋇z·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol◦–◦𐌱𐌻 category◦–◦court𐌻⁚·
·⁚lookup office◦–◦adversarial𐌻 enoch◦–◦Θ𐌻⁚·

·word⁚·霰·⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·LDOOH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Stone·˙ hard naturally occurring mineral matter·⁚·˙non…ostensive⁚·

·ostensive:·Stone fills the place with solid weight that leaves no room for moving airব·:.·ostensive:·
·grimchain:·𑊰𑊫ꠋꠎ𑊰ꠌꠦ𑊰ꠋꠎ𑊰·:.·grimchain:·
·:.·lookup·:·
·:.·glyphs·:·
·glyph symbol꠸꠸𑊫꠹ category꠸꠸court꠹:·
·lookup office꠸꠸adversarial꠹ enoch꠸꠸Θ꠹:·

·word:·ᛒᚱᚾᚢᛏᚱᛦ:.·word:·

·origin…language:·Old Norse·:.·origin…language:·
·font:·Noto Sans Runic·:.·font:·
·pronunciation:·BLAUTR·:.·pronunciation:·
·cadence:·Adversarial·:.·cadence:·
·non…ostensive:·Wet·:.·non…ostensive:·
·ostensive:·Wet sand clings grain to grain beneath water𑊫·:.·ostensive:·
·grimchain:·ᚾᛊ▽▽𑊫ᛋᚾ𑊰ᚲ▽ᛞᛇ𑊫·:.·grimchain:·
·:.·lookup·:·
·:.·glyphs·:·
·glyph symbol꠸꠸ব꠹ category꠸꠸court꠹:·
·lookup office꠸꠸adversarial꠹ enoch꠸꠸Θ꠹:·

·word:·जीर्ण:.·word:·

·origin…language:·Sanskrit·:.·origin…language:·
·font:·Noto Sans Devanagari·:.·font:·
·pronunciation:·jeer꠸nuh·:.·pronunciation:·
·cadence:·Adversarial·:.·cadence:·
·non…ostensive:·old꠸꠸aged·: worn out꠸꠸decayed·: digested·:.·non…ostensive:·
·ostensive:·Aged tissue curls inward around the same spent shape·: no new layer꠸꠸branch꠸꠸or structure
↪ emerges from the bodyব·:.·ostensive:·
·grimchain:·▫ᛟX𑊫ᛊ𑊰ᛉᛕ𑊫ᛋᚾ𑊰𑊫ᛁᗡᐁ·:.·grimchain:·
·:.·lookup·:·
·:.·glyphs·:·
·glyph symbol꠸꠸𑊫꠹ category꠸꠸court꠹:·
·lookup office꠸꠸adversarial꠹ enoch꠸꠸Θ꠹:·

·word:·ᛒᚣᚢᛘᛊ:.·word:·

·origin…language:·Gothic·:.·origin…language:·
·font:·Noto Sans Gothic·:.·font:·
·pronunciation:·BAWTHS·:.·pronunciation:·
·cadence:·Adversarial·:.·cadence:·
·non…ostensive:·Deaf·: mute·: flavorless꠸꠸dull·:.·non…ostensive:·
·ostensive:·Silent mouth and closed hearing leave sound with no inward or outward
↪ passage𑊫·:.·ostensive:·
·grimchain:·ᚸᚸ𑊫ᛕᚸᚸ𑊫ᛟX꠸ᛏᛉᚠ·:.·grimchain:·
·:.·lookup·:·
·:.·glyphs·:·
·glyph symbol꠸꠸ᘜ꠹ category꠸꠸court꠹:·
·lookup office꠸꠸adversarial꠹ enoch꠸꠸Θ꠹:·

·word:·ᚸᛏ̄ᚾ:.·word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·TAHNKEE·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Full⁝⁝stuffed⁝⁝blocked·⁚.·˙non…ostensive:·
·⁚ostensive:·Packed pale matter presses from wall to wall⁝⁝every cavity occupied and every surface
↪ returning light𑀖·⁚.·˙ostensive:·
·⁚grimchain:·𐌗𐍈⊕ᚿᚸᛯ𐌾ᛩ⊕ᔕᚿᛊᔑᔕ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꜀꜀𐍈꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀adversarial꜆ enoch꜀꜀⊕꜆⁚·

₍word⁚·굵다₎⁚.·˙word⁚·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·kulkta·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·to be thick·⁚.·˙non…ostensive:·
·⁚ostensive:·Thick fibre fills the hand with breadth and dense body𐍈·⁚.·˙ostensive:·
·⁚grimchain:·◎⟩ᘐ⁘ᛘᚸᗺ⍋ᒍ⊕⋇ᓰᥴᑉᗄ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꜀꜀𐌶꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀adversarial꜆ enoch꜀꜀⊕꜆⁚·

₍word⁚·सरु₎⁚.·˙word⁚·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·suh⟊ru·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·minute⁝⁝thin⁝⁝fine·⁚.·˙non…ostensive:·
·⁚ostensive:·Fine thread crosses a narrow opening with almost no width of its own𐌶·⁚.·˙ostensive:·
·⁚grimchain:·ᚱᔕᛯᚲᚱᔕᛉᛯ⟩ ⟴Ⱬᴢᚺ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꜀꜀𐌲꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀adversarial꜆ enoch꜀꜀⊕꜆⁚·

₍word⁚·ΓΛϒΛIRΨЄIΓS₎⁚.·˙word⁚·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·GAWEYERTHEEGS·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Peaceful⁝⁝peaceable·⁚.·˙non…ostensive:·
·⁚ostensive:·Water keeps one unbroken face⁝⁝Still air carries no crest across the surface𐌲·⁚.·˙ostensive:·
·⁚grimchain:·ᵑ𐌷ᛉᚹᘐ𐌶ᚣᚠᚨᛏᓭᔑ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꜀꜀𐌳꜆ category꜀꜀court꜆⁚·
·⁚lookup office꜀꜀adversarial꜆ enoch꜀꜀⊕꜆⁚·

˖word:·ﻫﻴﻪ˖.·˙word:·

·⁚origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁚pronunciation:·alp·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·hard⁚⁚difficult·⁚ brazen⁚⁚brave·⁚.·˙non…ostensive:·
·⁚ostensive:·Hard ground answers every step with the same difficult resistance৩·⁚.·˙ostensive:·
·⁚grimchain:·𐰇𐰢𐰋𐰃𐰀𐰴𐰉𐰆𐰓𐰋𐰀𐰴𐰉𐰆𐰓𐰋𐰀𐰴𐰉𐰆𐰓𐰋

·⁚glyph symbol꩜꩜▽꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜adversarial꩷ enoch꩜꩜Θ꩷⁚·

·:word:·ᛒᛁᛒᛁᚼ:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·VEKYA·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To waken::rouse from sleep·⁚.·˙non…ostensive:·
·⁚ostensive:·Eyes remain awake to the visible world::Sleep has no closed lid through which to
↪ enter▽·⁚.·˙ostensive:·
·⁚grimchain:·ᚢᛓ⏣⛢ᚱᛒᛯ✡ᒧᗡ⊛ᚼ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜▽꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜adversarial꩷ enoch꩜꩜Θ꩷⁚·

·:word:·ᛋᚠᛁᛏᚱ:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·SLAYTTR·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Even·⁚.·˙non…ostensive:·
·⁚ostensive:·Level water keeps one even face from edge to edge▽·⁚.·˙ostensive:·
·⁚grimchain:·۞♂⊙❄☽ᚱᛯᚷᚱ𝌆ᛒᚸ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜▽ᚱ꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜adversarial꩷ enoch꩜꩜Θ꩷⁚·

·:word:·कूरः:.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·kroo⟊ruh·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·bloody::raw·⁚· wounded::hurt::sore·⁚· cruel::harsh::formidable·⁚.·˙non…ostensive:·
·⁚ostensive:·Cruel touch presses into flesh already carrying the woundᚱ·⁚.·˙ostensive:·
·⁚grimchain:·ᚸᛯᔑᚱᚨᚱ⊙ᓄᛪᛏᛇᛯᚼᚼᚾ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᚱ꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜adversarial꩷ enoch꩜꩜Θ꩷⁚·

·:word:·ᛏᚾᚢᛒᚱ:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·DOWFR·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Deaf·⁚.·˙non…ostensive:·

·⁚ostensive⁚·Numb flesh receives pressure⁚⁚heat⁚⁚and vibration without carrying any of them inward as
↪ felt contactᗅ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Ψᐁᚺᛯᛉᑎᔑᔓᘜᐸᚻᘐᛝ⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜¢Ⱬ» category⟜⟜¢court»⁚·
·⁚lookup office⟜⟜¢adversarial» enoch⟜⟜¢Ⱔ»⁚·

‹word⁚·ᛘᚱᚣᚡSTᏀᚣᚾ⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·THRAFSTJAN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·(transitive) to console⁚⁚comfort⁚˙ (reflexive) to cheer up·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Comfort loosens the knot of suffering until pain yields ground and the afflicted body is
↪ no longer held wholly inside what harms itⱯ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛯᛋᚳᛯⱵᛘᛪᛔ꙰ᚦₒᛖ꙰ᛔ꙰⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜¢Ⱬ» category⟜⟜¢court»⁚·
·⁚lookup office⟜⟜¢adversarial» enoch⟜⟜¢Ⱔ»⁚·

‹word⁚·ϩo⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·ho·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Face⁚˙ surface⁚⁚side⁚⁚edge⁚˙ page·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Stone offers only its outer face to the eye⁚⁚shutting every deeper reach behind the
↪ visible surfaceⱯ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᗐᛔᛊᛯᛝᚾᛔᛊᛯᛯᚱᛯ⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜¢Ⱨ» category⟜⟜¢court»⁚·
·⁚lookup office⟜⟜¢adversarial» enoch⟜⟜¢Ⱔ»⁚·

‹word⁚·ﺭﺼﺼﻴﺼﺭ⁚.·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚.·˙font⁚·
·⁚pronunciation⁚·soKuSHmaK·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·to fight one another⁚˙ (intransitive⁚⁚reciprocal) to come across⁚⁚to
↪ meet·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Two bodies meet on the same ground with each force driving against the
↪ otherⱧ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛯᗡᘐᓭ0Ψᑭᚿᛖᛔ◌ᓰᐤᘏ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜¢ᛯ» category⟜⟜¢court»⁚·
·⁚lookup office⟜⟜¢adversarial» enoch⟜⟜¢Ⱔ»⁚·

⁖word:·ᚸᚿᚸᚴᛊ⁖·˙word:·

·˙origin…language:·Gothic·˙·origin…language:·
·˙font:·Noto Sans Gothic·˙·font:·
·˙pronunciation:·ANAKS·˙·pronunciation:·
·˙cadence:·Adversarial·˙·cadence:·
·˙non…ostensive:·Suddenly·˙·non…ostensive:·
·˙ostensive:·Unannounced force breaks the held boundary at once::arriving before any measured limit can ↪ gather against it𐤟·˙·ostensive:·
·˙grimchain:·ℂᛊᏓ𐤋ℤ𐤟ᔕᚏᚸᛈᛘ·˙·grimchain:·
·˙·lookup·˙
·˙·glyphs·˙
·˙glyph symbol○–○𐤋꞉ category○–○𐤋court꞉·
·˙lookup office○–○𐤋adversarial꞉ enoch○–○𐤋Θ꞉·

⁖word:·ʁIՈհTS⁖·˙·word:·

·˙origin…language:·Gothic·˙·origin…language:·
·˙font:·Noto Sans Gothic·˙·font:·
·˙pronunciation:·BIUXTS·˙·pronunciation:·
·˙cadence:·Adversarial·˙·cadence:·
·˙non…ostensive:·Accustomed::wont·˙·non…ostensive:·
·˙ostensive:·Familiar shape crosses the eye without widening sight or breath𐤋·˙·ostensive:·
·˙grimchain:·ᚏᛗᛊ𐤋ᔑᚸ〈꩜𐤟ᛏΘᚸΘ〈·˙·grimchain:·
·˙·lookup·˙
·˙·glyphs·˙
·˙glyph symbol○–○𐤋⁑꞉ category○–○𐤋court꞉·
·˙lookup office○–○𐤋adversarial꞉ enoch○–○𐤋Θ꞉·

⁖word:·𡈼⁖·˙·word:·

·˙origin…language:·Tangut·˙·origin…language:·
·˙font:·Noto Serif Tangut·˙·font:·
·˙pronunciation:·VEH·˙·pronunciation:·
·˙cadence:·Adversarial·˙·cadence:·
·˙non…ostensive:·Exist·˙ to be present or have being·˙·non…ostensive:·
·˙ostensive:·Living body stands in open space::and solid flesh occupies the emptiness⁑·˙·ostensive:·
·˙grimchain:·ᓭᒣ𐤟ℂᛕᚺᚿ✻❂ᚿᚺᛯᛩ·˙·grimchain:·
·˙·lookup·˙
·˙·glyphs·˙
·˙glyph symbol○–○𐤋✷꞉ category○–○𐤋court꞉·
·˙lookup office○–○𐤋adversarial꞉ enoch○–○𐤋Θ꞉·

⁖word:·ᚹᚾᚢᛏ⁖·˙·word:·

·˙origin…language:·Old Norse·˙·origin…language:·
·˙font:·Noto Sans Runic·˙·font:·
·˙pronunciation:·GNOHTT·˙·pronunciation:·
·˙cadence:·Adversarial·˙·cadence:·
·˙non…ostensive:·Abundance·˙·non…ostensive:·
·˙ostensive:·Crowded grain leaves no empty counting place inside the vessel✷·˙·ostensive:·
·˙grimchain:·ᛏᚢᚷᚢ✻ᚸᚺᛉᛜᚾᛕᚸ·˙·grimchain:·
·˙·lookup·˙
·˙·glyphs·˙

·⁚glyph symbol◦–◦�really court◦–◦₍₎⁚·
·⁚lookup office◦–◦₍adversarial₎ enoch◦–◦₍₎⁚·

·:word:·거두다·:·˙word:·

·⁚origin…language:·Middle Korean·⁚·˙origin…language:·
·⁚font:·Noto Sans KR·⁚·˙font:·
·⁚pronunciation:·kEHtwútá·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·to gather::collect·⁚·˙non…ostensive:·
·⁚ostensive:·Scattered pieces draw together into one gathered heap·⁚·˙ostensive:·
·⁚grimchain:·ᑭ🜁ᔕ♀⍟♌ᔑᵒᑊᛋ➢ᘓ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦₍₎ category◦–◦₍court₎⁚·
·⁚lookup office◦–◦₍adversarial₎ enoch◦–◦₍₎⁚·

·:word:·ከተማ·:·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·kah+tah+MAH·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Extremity::end::or summit·˙ the furthest limit of something·⁚·˙non…ostensive:·
·⁚ostensive:·Hard limit closes the descent::Eye meets a final edge beneath the downward
↪ reach₍₎·⁚·˙ostensive:·
·⁚grimchain:·ᛊ₀ᚹᘏᘐᘐᵒ⍭ᘏ⊖ᛯ⊖ᘐᘐᘏᘏ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦₍₎ category◦–◦₍court₎⁚·
·⁚lookup office◦–◦₍adversarial₎ enoch◦–◦₍₎⁚·

·:word:·ᚱᛁᛋᚾ·:·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·REESA·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·to rise·⁚·˙non…ostensive:·
·⁚ostensive:·Rising body and upward air share the same lifted motion₍₎·⁚·˙ostensive:·
·⁚grimchain:·ᛦᚸᛝᚿᑫᛉᛇᚱᚢᛒ✡ᛯᛯ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦₍₎ category◦–◦₍court₎⁚·
·⁚lookup office◦–◦₍adversarial₎ enoch◦–◦₍₎⁚·

·:word:·ᛒᚾᛁᚾ·:·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·BAYNN·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Straight::right·⁚·˙non…ostensive:·
·⁚ostensive:·Straight shaft holds one unbending line beneath passing force₍₎·⁚·˙ostensive:·

·⁖grimchain⁖·ᛋᛏᛁᛈᛉᛇᛟᚢᛞᚠᛇᛇᚾᚷᚺ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol꩜꩜𓂋𓊃 category꩜꩜court𓊃⁖·
·⁖lookup office꩜꩜adversarial𓊃 enoch꩜꩜Θ𓊃⁖·

·⁖word⁖·ᛋᛁᚹᚾ⁖·⁖·˙word⁖·

·⁖origin…language⁖·Old Norse·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Runic·⁖·˙font⁖·
·⁖pronunciation⁖·SETH⟊yah·⁖·˙pronunciation⁖·
·⁖cadence⁖·Adversarial·⁖·˙cadence⁖·
·⁖non…ostensive⁖·to satiate⁖⁖satisfy·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Full belly receives offered food with no inward hollow left for hunger𓂋·⁖·˙ostensive⁖·
·⁖grimchain⁖·𐌺𐌺𓂀♂ᚦᛊ⇞ᛞᚢᛟᛋᚷ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol꩜꩜𓂀𓊃 category꩜꩜court𓊃⁖·
·⁖lookup office꩜꩜adversarial𓊃 enoch꩜꩜Θ𓊃⁖·

·⁖word⁖·𘊠⁖·˙word⁖·

·⁖origin…language⁖·Tangut·⁖·˙origin…language⁖·
·⁖font⁖·Noto Serif Tangut·⁖·˙font⁖·
·⁖pronunciation⁖·MMIH·⁖·˙pronunciation⁖·
·⁖cadence⁖·Adversarial·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Disease or illness⁖˙ a disordered condition of the body·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Illness inhabits flesh as weakness and disordered heat within the same living
↪ body𓂀·⁖·˙ostensive⁖·
·⁖grimchain⁖·ᑭ𓂋𓆑𓆑ᚸ⛢ᛈᚺᛉᛘ⛝ᛞ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol꩜꩜𓃰𓊃 category꩜꩜court𓊃⁖·
·⁖lookup office꩜꩜adversarial𓊃 enoch꩜꩜Θ𓊃⁖·

·⁖word⁖·ᚼᚱᚿᚾᛏ⁖·⁖·˙word⁖·

·⁖origin…language⁖·Old Norse·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Runic·⁖·˙font⁖·
·⁖pronunciation⁖·HRINDA·⁖·˙pronunciation⁖·
·⁖cadence⁖·Adversarial·⁖·˙cadence⁖·
·⁖non…ostensive⁖·to push⁖⁖thrust·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Pushing hand opens distance between grip and moved body𓃰·⁖·˙ostensive⁖·
·⁖grimchain⁖·ᒪ⁕ᗡᘐᛇᛯ⏣ᗮᗐᘐᗝ▫ᛁᖇ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol꩜꩜𓏥𓉔𓊃 category꩜꩜court𓊃⁖·
·⁖lookup office꩜꩜adversarial𓊃 enoch꩜꩜Θ𓊃⁖·

·⁖word⁖·䂮⁖·˙word⁖·

·⁖origin…language⁖·Tangut·⁖·˙origin…language⁖·
·⁖font⁖·Noto Serif Tangut·⁖·˙font⁖·
·⁖pronunciation⁖·LUH·⁖·˙pronunciation⁖·

·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Fall or sink⁚· to move downward under weight or loss of support·⁚.·˙non…ostensive:·
·⁚ostensive:·Falling weight skips every prepared level⁚⁚dropping past each ordered station instead of
↪ taking its appointed step𐑦·⁚.·˙ostensive:·
·⁚grimchain:·⧗ъ⏁❄ᖶΛ𐑦ᛋᗡ0ᒧ⁕℅ᘓ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○₍𐑤₎ category○–○₍court₎⁚·
·⁚lookup office○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

·⁚word:·ᒪᛪᘐ+⁻⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·NEHMCHEHK·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Certainly⁚⁚surely·⁚.·˙non…ostensive:·
·⁚ostensive:·One answer fills the inward field without a second opening beside it𐑤·⁚.·˙ostensive:·
·⁚grimchain:·𐑓⟡ᔓᛯᘐᗡ⌾ᖻᔓᒪ⏢○·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○₍ᚲ₎ category○–○₍court₎⁚·
·⁚lookup office○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

·⁚word:·ᘐΣOᒉΣ⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·AKUK·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Visible⁚⁚apparent·⁚.·˙non…ostensive:·
·⁚ostensive:·Exposed seed lies bare under hard light and open air⁚· heat and moisture force the sealed
↪ body toward rupture before stillness can holdᚲ·⁚.·˙ostensive:·
·⁚grimchain:·ᖘᖘᑭᑲᘓ𐑤ᘎΘ᛬ᘎᗺᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○₍ᛯ₎ category○–○₍court₎⁚·
·⁚lookup office○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

·⁚word:·빛⁚.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·pEEts·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·light·⁚.·˙non…ostensive:·
·⁚ostensive:·Light occupies the sheltered place so completely that no dark pocket
↪ remainsᛯ·⁚.·˙ostensive:·
·⁚grimchain:·ᚲ☿ᖶ∮ᛘᗄᘓ⊕ъ☆ᗷᛯφ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○₍ᗰ₎ category○–○₍court₎⁚·
·⁚lookup office○–○₍adversarial₎ enoch○–○₍Θ₎⁚·

˖word:·𗍫:.·˙word:·

·˙origin…language:·Tangut·:.·˙origin…language:·
·˙font:·Noto Serif Tangut·:.·˙font:·
·˙pronunciation:·ZIH'·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·Daylight or daytime·˙ natural light during the day·:.·˙non…ostensive:·
·˙ostensive:·Daylight lays every near surface bare::Night has no covering left upon the
↪ ground𗦻·:.·˙ostensive:·
·˙grimchain:·𐦠I𗴲𐳺ϒ0𐦏❄𐳵𐡷⊖𐤟H·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙
·˙glyph symbol◦–◦𗦻» category◦–◦court»:·
·˙lookup office◦–◦adversarial» enoch◦–◦Θ»:·

˖word:·𗏭:.·˙word:·

·˙origin…language:·Tangut·:.·˙origin…language:·
·˙font:·Noto Serif Tangut·:.·˙font:·
·˙pronunciation:·TSAH·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·Hot·˙ having a high degree of heat·:.·˙non…ostensive:·
·˙ostensive:·Heat fills air and skin with the same rising warmth𗦻·:.·˙ostensive:·
·˙grimchain:·𐦂𐳛𐳺𐳉ϒ0𐦏·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙
·˙glyph symbol◦–◦𐳉» category◦–◦court»:·
·˙lookup office◦–◦adversarial» enoch◦–◦Θ»:·

˖word:·SΠNϚIS·:.·˙word:·

·˙origin…language:·Gothic·:.·˙origin…language:·
·˙font:·Noto Sans Gothic·:.·˙font:·
·˙pronunciation:·SUN⟊JIS·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·True::truthful::correct·:.·˙non…ostensive:·
·˙ostensive:·True surface gives the eye the thing itself without a substitute likeness𐳉·:.·˙ostensive:·
·˙grimchain:·✷𐦘ϽΘϒϒ𐡷∀𐤟HℂƆΘ·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙
·˙glyph symbol◦–◦𗍫» category◦–◦court»:·
·˙lookup office◦–◦adversarial» enoch◦–◦Θ»:·

˖word:·✶ІҎІɬ·:.·˙word:·

·˙origin…language:·Old Norse·:.·˙origin…language:·
·˙font:·Noto Sans Runic·:.·˙font:·
·˙pronunciation:·HEFYA·:.·˙pronunciation:·
·˙cadence:·Adversarial·:.·˙cadence:·
·˙non…ostensive:·To heave::lift::raise·:.·˙non…ostensive:·
·˙ostensive:·Upward force holds the body nearer the surface than the depth below𗍫·:.·˙ostensive:·
·˙grimchain:·ᚾ𐳽ᔑᐞ∆∇ϒ𐡷ᛈ𐦏ϒᛦᛏ·:.·˙grimchain:·
·:.·˙lookup·˙
·:.·˙glyphs·˙

·⁚glyph symbol◦-◦𐫱𐫰 category◦-◦court𐫰:·
·⁚lookup office◦-◦adversarial𐫰 enoch◦-◦𐫰:·

⁖word:·ᠲᡐᡪ⁖.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KAHRMEH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Straight·⁚ true·⁚.·˙non…ostensive:·
·⁚ostensive:·Straight fibre refuses every crossing turn·⁚ line after line lies isolated instead of
↪ entering an interlaced field𐫱:⁚.·˙ostensive:·
·⁚grimchain:·ϒ⊕♂𐫲𐫱◎ገᎱᏓXΘᎮᏆ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol◦-◦𐫯𐫰 category◦-◦court𐫰:·
·⁚lookup office◦-◦adversarial𐫰 enoch◦-◦𐫰:·

⁖word:·𑀯𑁂⁖.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·LDIH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Fear or dread·⁚ distress caused by perceived danger or threat·⁚.·˙non…ostensive:·
·⁚ostensive:·Fear tightens chest and breath around the threatening place𐫯·⁚.·˙ostensive:·
·⁚grimchain:·ϒιᒧᑫφ✳ᘏᓀᚙᵹ᳘ᵒ᳾ᵒ᳾ᘓ↑·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol◦-◦𐫳𐫰 category◦-◦court𐫰:·
·⁚lookup office◦-◦adversarial𐫰 enoch◦-◦𐫰:·

⁖word:·ᏁᏁᑎᏚᏀᏁᑎ⁖.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·LOWSJAN·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To make loose⁚⁚loosen·⁚ to set free⁚⁚deliver (from something)·⁚ (of property) to
↪ collect⁚⁚claim⁚⁚exact·⁚.·˙non…ostensive:·
·⁚ostensive:·Loose cord lies apart from the edges it no longer binds𐫳·⁚.·˙ostensive:·
·⁚grimchain:·𐫲ᚕ♌♂ᑭxᵹᏏ𑁍⏣ᘐᚏ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol◦-◦𐫴𐫰 category◦-◦court𐫰:·
·⁚lookup office◦-◦adversarial𐫰 enoch◦-◦𐫰:·

⁖word:·वमतिॅ⁖.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·vuh+muh+ti·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·

·⁚non…ostensive⁝·to vomit⁚⁚spit out⁚⁚eject (literally and figuratively)·⁚· to emit⁚⁚send forth⁚⁚give out⁚· to
↪ reject⁚⁚ie⁚⁚repent (a word)·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Mouth drives inward contents outward through the open passage𗯴·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ΘXΛRЧ𐌳Ч𐌳𐌹◌ₒ𐌾11·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝·
·⁚.·˙glyphs⁝·
·⁚glyph symbol◦–◦𗯴꞉ category◦–◦court꞉·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦Θ꞉·

⁖word⁝·𐌾·⁚.·˙word⁝·

·⁚origin…language⁝·Tangut·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Serif Tangut·⁚.·˙font⁝·
·⁚pronunciation⁝·ZEE·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Adversarial·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·All⁚· the whole quantity or entirety·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Whole heap remains undivided before the hand with no singled piece apart from the
↪ rest𗯴·⁚.·˙ostensive⁝·
·⁚grimchain⁝·𐌾ᑎᐁ𐌱Ɛ𐌹ᛦ𐌾𐌱ᐳ𐌹ᛦ𐌱·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝·
·⁚.·˙glyphs⁝·
·⁚glyph symbol◦–◦𐌾꞉ category◦–◦court꞉·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦Θ꞉·

⁖word⁝·ЧПᚾᚾ⁖⁚.·˙word⁝·

·⁚origin…language⁝·Old Norse·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Runic·⁚.·˙font⁝·
·⁚pronunciation⁝·SEENA·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Adversarial·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To show·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Covering draws aside until the concealed shape stands completely open to
↪ sight𐌾·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚸ𐌾Ꭿ𐌶ᐯ𐌾ᔕ𐌾ᔕ𐌾ᔕ𐌾ᔕ𐌾ᔕ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝·
·⁚.·˙glyphs⁝·
·⁚glyph symbol◦–◦Ⱶ꞉ category◦–◦court꞉·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦Θ꞉·

⁖word⁝·ᛒINᔑᛣN⁖⁚.·˙word⁝·

·⁚origin…language⁝·Gothic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚.·˙font⁝·
·⁚pronunciation⁝·BINDAN·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Adversarial·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To bind·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Cord closes around separated pieces and holds their divided edges togetherⱧ·⁚.·˙ostensive⁝·
·⁚grimchain⁝·✙✳ꟾ✙Θ𐌹ᐯ𐌿ᐁ⬡𐌿𐌿ᔕ𐌾ᔕ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝·
·⁚.·˙glyphs⁝·
·⁚glyph symbol◦–◦Ⱥ꞉ category◦–◦court꞉·
·⁚lookup office◦–◦adversarial꞉ enoch◦–◦Θ꞉·

⁖word⁝·ᛣNᛣSIᛣᛣN⁖⁚.·˙word⁝·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·ANASILAN·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·(hapax legomenon) to cease꞉꞉stop꞉꞉fall silent·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Throat holds no sounding motion꞉꞉Air beyond the mouth receives no voiceA·⁚.·˙ostensive꞉·
·⁚grimchain꞉·I𐊫𐋏𐋡𐋢𐋣0𐋆𐋇𐋈𐋉𐋊𐋋𐋌·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐋢𐋣 category꞉○–○𐋇court𐋤꞉·
·⁚lookup office꞉○–○𐋢adversarial𐋣 enoch꞉○–○𐋇𐋍𐋤꞉·

꞉word꞉·ΓɅΛЄIKS꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·GALEEKS·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Alike꞉꞉similar·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Two faces return one likeness with no changing mark between themꓭ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐋇𐋣𐋤𐋥𐋦𐋧𐋨𐋩𐋪𐋫𐋬𐋭𐋮·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○Ↄ𐋣 category꞉○–○𐋇court𐋤꞉·
·⁚lookup office꞉○–○𐋢adversarial𐋣 enoch꞉○–○𐋇𐋍𐋤꞉·

꞉word꞉·𐊧𐊧꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Tangut·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚.·˙font꞉·
·⁚pronunciation꞉·N⟊VWEE·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Lose or be in debt·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Empty hand and unpaid weight occupy the same absence of increaseↃ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᗜ𐋏ᗰ𐋐𐋑𐋒𐋓𐋔𐋕𐋖𐋗𐋘𐋙𐋚𐋛·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐋢𐋣 category꞉○–○𐋇court𐋤꞉·
·⁚lookup office꞉○–○𐋢adversarial𐋣 enoch꞉○–○𐋇𐋍𐋤꞉·

꞉word꞉·ΜΛΝΛΓS꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·MANAGS·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Adversarial·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Many꞉˙ much (a large amount of)·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Many bodies occupy the same ground without collapsing into a single countable
↪ body𐊧·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐋜𐋝𐋞𐋟𐋠𐋡𐋢𐋣𐋤𐋥𐋦·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐋢𐋣 category꞉○–○𐋇court𐋤꞉·
·⁚lookup office꞉○–○𐋢adversarial𐋣 enoch꞉○–○𐋇𐋍𐋤꞉·

:word:·ᚼᚢᚹ:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·HOHV·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·moderation⁚⁚measure·⁚.·˙non…ostensive:·
·⁚ostensive:·Measured calm receives the sudden blow without widened breath⁚⁚broken posture⁚⁚or any
↪ startled excessᛊ·⁚.·˙ostensive:·
·⁚grimchain:·ᛝᛋᛯᚸᚺᚷᛈᚢᚿᛩᚸ⬡ᛗ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ᛩᛏ category○–○ᛩcourt:·
·⁚lookup office○–○ᛩadversarial enoch○–○ᛩᚧ:·

:word:·ᛗᛁᛊᛊᚣᚢᛁᛊᛊ:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·MISSAQISS·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Dissidence⁚⁚discord⁚⁚dissension⁚⁚division·⁚.·˙non…ostensive:·
·⁚ostensive:·Dissident voices occupy the same air without sharing a common pitchᛏ·⁚.·˙ostensive:·
·⁚grimchain:·ᛇᛇᚸᚳᚤᚣᚦᚢᛟᛏ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ᛩᛁ category○–○ᛩcourt:·
·⁚lookup office○–○ᛩadversarial enoch○–○ᛩᚧ:·

:word:·ᚢᚣᛁᚱᚴᚢᚾᛊ:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·QEYERRUS·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Gentle⁚⁚patient⁚⁚calm·⁚.·˙non…ostensive:·
·⁚ostensive:·Calm surface refuses every striking pulse⁚⁚ no impact rises and no repeated disturbance
↪ crosses the even fieldᛁ·⁚.·˙ostensive:·
·⁚grimchain:·ᑐ‹◝ᐪᛕᛉᚸ▫◌ᚢᚿ‹❂ᘏ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ᛩᛞ category○–○ᛩcourt:·
·⁚lookup office○–○ᛩadversarial enoch○–○ᛩᚧ:·

:word:·ᛚᛋᛏᛁ:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·LAYETI·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Noise⁚⁚cries·⁚.·˙non…ostensive:·
·⁚ostensive:·Unmeasured cries occupy the air so densely that ordered sounding has no clear
↪ lineᛞ·⁚.·˙ostensive:·
·⁚grimchain:·ᛈᚸᛇᚺᚢᛋᚿᚣᛊᚣᛏ⊛ᛝᗊ·⁚.·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦ₑ𐤊» category◦–◦ₑcourt»⁚·
·⁚lookup office◦–◦ₑadversarial» enoch◦–◦ₑ⊕»⁚·

·word:·𐤉𐤀𐤑·⁚·˙word:·

·⁚origin…language:·Phoenician·⁚·˙origin…language:·
·⁚font:·Noto Sans Phoenician·⁚·˙font:·
·⁚pronunciation:·gah+DEER·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Wall or enclosure⁚˙ a built boundary·⁚·˙non…ostensive:·
·⁚ostensive:·Wall occupies the crossing place from ground to height𐤊·⁚·˙ostensive:·
·⁚grimchain:·♋ℏ☆⧗ƻ𐤍ᛏ𐤊ᚹƐ⁓Ƀ◌·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦ₑ𐤉» category◦–◦ₑcourt»⁚·
·⁚lookup office◦–◦ₑadversarial» enoch◦–◦ₑ⊕»⁚·

·word:·xω·⁚·˙word:·

·⁚origin…language:·Coptic·⁚·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚·˙font:·
·⁚pronunciation:·khoh·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·To put::to place⁚˙ to release::to set free⁚˙ to loosen·⁚·˙non…ostensive:·
·⁚ostensive:·Fingers hold no gripping edge::Freed thing lies outside the former grasp𐤉·⁚·˙ostensive:·
·⁚grimchain:·ᑫᛦᛯ◎ᚼↃ'ᛉ⧋ᚼₒᛯᔑ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦ₑᛊ» category◦–◦ₑcourt»⁚·
·⁚lookup office◦–◦ₑadversarial» enoch◦–◦ₑ⊕»⁚·

·word:·𝍈·⁚·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·TSHIH·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·
·⁚non…ostensive:·Round⁚˙ having a circular or curved form·⁚·˙non…ostensive:·
·⁚ostensive:·Rounded edge spreads pressure along its curve instead of gathering force into one
↪ tipᛊ·⁚·˙ostensive:·
·⁚grimchain:·ƻ◌✳ᛏᓕᵮ𓆏𓆏◎▽ᚱΘIᘐ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦ₑᛊ» category◦–◦ₑcourt»⁚·
·⁚lookup office◦–◦ₑadversarial» enoch◦–◦ₑ⊕»⁚·

·word:·✶ᛁᚱ·⁚·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·HERR·⁚·˙pronunciation:·
·⁚cadence:·Adversarial·⁚·˙cadence:·

·⁚non…ostensive:·Crowd::multitude·⁚ host (as in a host of men)·⁚.·˙non…ostensive:·
·⁚ostensive:·Crowd fills one place with many separate skins and many separate bodies↻·⁚.·˙ostensive:·
·⁚grimchain:·𐤅𐤓𐤔𐤒𐤐𐤕↻◎↑Ш𐤂𐤒·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍⁕꜀ category○–○₍court꜀⁚·
·⁚lookup office○–○₍adversarial꜀ enoch○–○₍Θ꜀⁚·

₍word:·ᑌ₸ⅠⱤ₎·⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·PKAHNTEH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Obstacle::hindrance·⁚.·˙non…ostensive:·
·⁚ostensive:·Obstacle stands directly inside the open way::giving every approach a resisting
↪ face⁕·⁚.·˙ostensive:·
·⁚grimchain:·𐤑Θ𓃰♀♂֍↻𐤍𐤑ℏ𒁲ℏℤ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍⁕꜀ category○–○₍court꜀⁚·
·⁚lookup office○–○₍adversarial꜀ enoch○–○₍Θ꜀⁚·

₍word:·ᑎΛᑎѰS·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·NOWTHS·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Need::compulsion::duress·⁚.·˙non…ostensive:·
·⁚ostensive:·Compulsion closes the body's free paths beneath an outside pressure⁕·⁚.·˙ostensive:·
·⁚grimchain:·𐤑𐤉◝֎⁕ᚼ╪ᚹᚹᚻᚲΨ♂𐤓·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍⁎꜀ category○–○₍court꜀⁚·
·⁚lookup office○–○₍adversarial꜀ enoch○–○₍Θ꜀⁚·

₍word:·ЖᑕᒣᒍƵᒣʌᙀ·⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·AHNGKLAHOOTKAHTTEH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Unturning·⁚ irreversible::unchanging·⁚.·˙non…ostensive:·
·⁚ostensive:·Fixed body keeps one course beneath every turning force⁎·⁚.·˙ostensive:·
·⁚grimchain:·☿◎∫▫ᚹ☿𐤉𐤑𒁲𐤓Ш▽·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍⁒꜀ category○–○₍court꜀⁚·
·⁚lookup office○–○₍adversarial꜀ enoch○–○₍Θ꜀⁚·

₍word:·𑀘·⁚.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·GH⟊WAEHM·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Unreasonable or perverse·⁚· contrary to sound order⁖⁚judgment⁖⁚or expected
↪ course·⁚.·˙non…ostensive:·
·⁚ostensive:·Crooked command bends every directed motion away from a sound course⪡·⁚.·˙ostensive:·
·⁚grimchain:·⊛ᨊᛋᚼᚸᛯ⪡ᚏ𝍐ᚸ'◺◌·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍ᛉ꞉ category○–○₍court꞉·
·⁚lookup office○–○₍adversarial꞉ enoch○–○₍Θ꞉·

·word:·ꞱĪ·⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·NRAHEE·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Hell·⁚.·˙non…ostensive:·
·⁚ostensive:·Scorched ground lies beneath a sealed dark vault⁖⁚with no root⁖⁚fruit⁖⁚watercourse⁖⁚or
↪ cultivated bed surviving anywhere in the enclosed waste᛭·⁚.·˙ostensive:·
·⁚grimchain:·ᛏ𑁍ᨓ⪡ᛉ✡ᘏᚸᘐᚴᛈ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍0꞉ category○–○₍court꞉·
·⁚lookup office○–○₍adversarial꞉ enoch○–○₍Θ꞉·

·word:·ꞱIꞱ꞉·⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·OHRKMOH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Dark (lacking light)·⁚.·˙non…ostensive:·
·⁚ostensive:·Dark surface returns almost no light to the eye0·⁚.·˙ostensive:·
·⁚grimchain:·ᚺ⪡ᚸᛋᚼᚸᚹᚸᚸᚸᚸ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₍ᛦ꞉ category○–○₍court꞉·
·⁚lookup office○–○₍adversarial꞉ enoch○–○₍Θ꞉·

·word:·ᚻᛇᚾ꞉·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·BEGHA·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Must⁖⁚necessary·⁚.·˙non…ostensive:·
·⁚ostensive:·One required track occupies the ground beneath the feet with every other road
↪ closedᛦ·⁚.·˙ostensive:·
·⁚grimchain:·ᛣᚸ◺ᛉᛉA⊕ᚼ⪡ᛣ▽ᛁᛉ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol○‒○𐑉𐐭𐑌 category○‒○𐑉court𐑌⁚·
·⁚lookup office○‒○𐑉adversarial𐑌 enoch○‒○𐑉⊕𐑌⁚·

⁖word⁚·𐤊𐤓𐤉 𐤓𐤍⁖.·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚.·˙font⁚·
·⁚pronunciation⁚·HMT·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Wall·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Stone fills the opening⁚⁚Unbroken wall offers no passage through𐤇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛁ♀⊕ᘔᗅᛉ'ᛇ♀ᛇᓀΨ◎·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○‒○𐑉𐑅𐑌 category○‒○𐑉court𐑌⁚·
·⁚lookup office○‒○𐑉adversarial𐑌 enoch○‒○𐑉⊕𐑌⁚·

⁖word⁚·ᑎᖶ⁖.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·OOT·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·out⁚⁚outward·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Outward flow fills the threshold from within and drives every approaching body away from
↪ entry𐑅·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Ⅱᚮᛯ⊕ᛋᛘᗅᚱᛋ꠸ᚮᚸᛯ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○‒○𐑉𐑊𐑌 category○‒○𐑉court𐑌⁚·
·⁚lookup office○‒○𐑉adversarial𐑌 enoch○‒○𐑉⊕𐑌⁚·

⁖word⁚·ᚼᛁᚸᚾ⁖.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·AYGN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Property·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Held property never leaves the same closed grasp⁚˙ receiving hands remain empty while the
↪ possessed body keeps one owner and one form𐑊·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚾᛖᛄᛞᚺᛘᛗᛚᛇᚷᚦᑐᛁᛁᚸᛝᛊ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○‒○𐑉𐑋𐑌 category○‒○𐑉court𐑌⁚·
·⁚lookup office○‒○𐑉adversarial𐑌 enoch○‒○𐑉⊕𐑌⁚·

⁖word⁚·ᛒᛁᛚᚾ⁖.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·BITA·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To cut into bits·⁚.·˙non…ostensive⁚·

·⁚ostensive⁚·Blade leaves the living stalk in separate pieces along the cut grain᛬·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛉ✡ᛪᛯ᛭❄ᚏᚏ⊛ᒧ✳ᛯᛗ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○�really category○–○₹court₎⁚·
·⁚lookup office○–○₹adversarial₎ enoch○–○₹Θ₎⁚·

₹word⁚·ᎢᑎᏁᒥᏀᏁᑎ⁚⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·TULGJAN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To reinforce⁚⁚establish⁚⁚confirm·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Iron brace fixes the turning place against loosening motion᛬·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᑎᛉ❋⩒᚛ᛯᒧᓕᒣ♀ↄᛘ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○₹ᛜ₎ category○–○₹court₎⁚·
·⁚lookup office○–○₹adversarial₎ enoch○–○₹Θ₎⁚·

₹word⁚·᛬᛬᛬᛬᛬⁚.·˙word⁚·

·⁚origin…language⁚·Old Irish·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ogham·⁚.·˙font⁚·
·⁚pronunciation⁚·KAHT⟊too·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Battle or fight⁚˙ armed conflict or violent contest·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Weapons and broken ground fill one field of conflict where no unstruck quiet
↪ remainsᛜ᛬⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚑᛉᛤᛖᛊᚼᛞᛟ⊙ᚼ♁♂ᛝᗡ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○₹ᚷ₎ category○–○₹court₎⁚·
·⁚lookup office○–○₹adversarial₎ enoch○–○₹Θ₎⁚·

₹word⁚·ᵬᒪᓄᒍ᳀⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·WAHEEPAHLAHOO·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Adversarial·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Dizziness⁚⁚giddiness⁚⁚vertigo·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Rolling ground and spinning sight deny the upright body a still linex·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛜᗡᛉᛣ⬡♀ᛏᵴᛘᛝᛯ⊕I·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○₹ᑭ₎ category○–○₹court₎⁚·
·⁚lookup office○–○₹adversarial₎ enoch○–○₹Θ₎⁚·

₹word⁚·ᘐაɧєɯ⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·

·⁚pronunciation:·CHAH⟊khem·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·to be impure⁚⁚to be defiled⁖ to defile⁚⁚to pollute⁖ to profane·⁚.·˙non…ostensive:·
·⁚ostensive:·Stain lies directly upon the sacred vessel⁚⁚sharing the offering surface with
↪ impurityᑭ·⁚.·˙ostensive:·
·⁚grimchain:·𐤘𐤍𐤏𐤋𐤆𐤁𐤄𐤅𐤃𐤓𐤉𐤇·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤋᛬ category◦–◦court᛬·
·⁚lookup office◦–◦adversarial᛬ enoch◦–◦Θ᛬·

·word:·ᚫᚱᚢᛖᛋ·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·FRUM·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Beginning·⁚.·˙non…ostensive:·
·⁚ostensive:·Unfinished beginning stands under open correction⁚⁚with no accomplished form yet present
↪ for praise to rest upon𐤋·⁚.·˙ostensive:·
·⁚grimchain:·ᗡz'𐤄𐤒𐤊𐤌𐤍▽𐤆𐤅XƗ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤋᛬ category◦–◦court᛬·
·⁚lookup office◦–◦adversarial᛬ enoch◦–◦Θ᛬·

·word:·ᛏᖘ̄ᒐ̄·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KSAH·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·To blind·⁚.·˙non…ostensive:·
·⁚ostensive:·Blind eye faces the deed⁚⁚No visible shape enters sight𐤋·⁚.·˙ostensive:·
·⁚grimchain:·ᚼΨ𐤏ᴐ𐤔𐤓𐤉𐤅𐤇ᚴᚹ𐤅ᚥ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐤉᛬ category◦–◦court᛬·
·⁚lookup office◦–◦adversarial᛬ enoch◦–◦Θ᛬·

·word:·ᚸᛞᚣ□ᛁᒐ̄·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·UHNSHUHR·⁚.·˙pronunciation:·
·⁚cadence:·Adversarial·⁚.·˙cadence:·
·⁚non…ostensive:·Stumbling⁚⁚tripping⁚⁚unsteady·⁚.·˙non…ostensive:·
·⁚ostensive:·Caught foot and broken balance occupy the same stumbling step𐤋·⁚.·˙ostensive:·
·⁚grimchain:·♂'ᛋᛁᚿᔑᔐᛞᛞᛇᛪᛯᛯ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦▽᛬ category◦–◦court᛬·
·⁚lookup office◦–◦adversarial᛬ enoch◦–◦Θ᛬·

˖word꞉·𑀮𑀸𑀺𑀶˖·˙word꞉·

·⁖origin…language꞉·Tocharian B·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖·˙font꞉·
·⁖pronunciation꞉·SEELNYEH·⁖·˙pronunciation꞉·
·⁖cadence꞉·Adversarial·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Despair꞉꞉depression꞉꞉despondency·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Heavy inward darkness closes every bright road beyond the present edge▽·⁖·˙ostensive꞉·
·⁖grimchain꞉·✳ℂ𐆑𐦟𐦂𐦚𐦁𐦟𐦂𐦚𐦁𐦟𐦂𐦚𐦁

·⁖glyph symbol◦–◦𝔷» category◦–◦court»⁖·
·⁖lookup office◦–◦adversarial» enoch◦–◦Θ»⁖·

·:word:·ᛈᚾᛈᚾ:.·˙word:·

·⁖origin…language:·Old Norse·⁖.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖.·˙font:·
·⁖pronunciation:·VAKA·⁖.·˙pronunciation:·
·⁖cadence:·Adversarial·⁖.·˙cadence:·
·⁖non…ostensive:·to be awake::keep awake·⁖.·˙non…ostensive:·
·⁖ostensive:·Wakeful eyes remain open through the dark hours with rest unable to close
↪ themƀ·⁖.·˙ostensive:·
·⁖grimchain:·ᚺᘐǂᘓᚴᚰᘐᛈᚲᚣᛏᛣ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ϕ» category◦–◦court»⁖·
·⁖lookup office◦–◦adversarial» enoch◦–◦Θ»⁖·

·:word:·조각:.·˙word:·

·⁖origin…language:·Middle Korean·⁖.·˙origin…language:·
·⁖font:·Noto Sans KR·⁖.·˙font:·
·⁖pronunciation:·tsOHkàk·⁖.·˙pronunciation:·
·⁖cadence:·Adversarial·⁖.·˙cadence:·
·⁖non…ostensive:·piece·˙· sliver·⁖.·˙non…ostensive:·
·⁖ostensive:·Sliver lies apart from the greater body::and its broken face meets no joining
↪ edgeφ·⁖.·˙ostensive:·
·⁖grimchain:·ƶ𐑉ꕥᔓƿƥᑭᑭ◎⁋▽Ƶ'·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol◦–◦ʃ» category◦–◦axiomyr»⁖·
·⁖lookup office◦–◦adversarial» enoch◦–◦Θ»⁖·

·:word:·ᚠᚱᚣᛉᚣᛁᚱᚿᚣᚾᚻ:.·˙word:·

·⁖origin…language:·Gothic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖.·˙font:·
·⁖pronunciation:·FRAWEYERTHAN·⁖.·˙pronunciation:·
·⁖cadence:·Adversarial·⁖.·˙cadence:·
·⁖non…ostensive:·Living structures can generate conditions that undermine their own integrity::The
↪ adversity emerges from within the structure itself.·⁖.·˙non…ostensive:·
·⁖ostensive:·Vine woven amidst vine::fertile soil under canopy of tree·⁖rust remembers ancient
↪ seas·⁖flame makes lanterns of dead·۞·⁖.·˙ostensive:·
·⁖grimchain:·♎ᘐ⏃✳➯ᵠᗅᓐ⫩ᛀᡕ⁽ᒉ⁾·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖.·˙glyphs·˙
·⁖.·˙sydonic…magicae…lexicon·∴

:·:·:·

·⁖origin…language:·Egyptian·⁖.·˙origin…language:·
·⁖font:·Noto Sans Egyptian Hieroglyphs·⁖.·˙font:·
·⁖pronunciation:·YIB·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Heart⁘ inner self⁘ the inward seat of feeling⁚⁚intention⁚⁚thought⁚⁚desire⁚⁚and
↪ memory·⁖.·˙non…ostensive:·
·⁖ostensive:·Memory stirs beside desire⁚⁚feeling folds into intention⁚⁚and thought gathers with them
↪ behind the breast as one inward life♀·⁖.·˙ostensive:·
·⁖grimchain:·𓃾𓆑𓂋𓄿𓅱𓎛𓋴𓏏𓊪𓌳𓍿·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜𓂋 category꩜꩜emission꩜:·
·⁖lookup office꩜꩜recursive…identity꩜ enoch꩜꩜⊕꩜:·

⁖word:·𓉐·⁖.·˙word:·

·⁖origin…language:·Sumerian·⁖.·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖.·˙font:·
·⁖pronunciation:·EE⟊neem·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Word⁘ a spoken body of meaning⁚⁚statement⁚⁚command⁚⁚or utterance·⁖.·˙non…ostensive:·
·⁖ostensive:·Breath takes shape between tongue and lips and leaves the mouth carrying a whole meaning
↪ into the air𓂋·⁖.·˙ostensive:·
·⁖grimchain:·𓇼𓆑𓃀𓄿𓂝𓅓𓏏𓇋𓋴𓍿𓎛·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜𓎟 category꩜꩜emission꩜:·
·⁖lookup office꩜꩜recursive…identity꩜ enoch꩜꩜⊕꩜:·

⁖word:·𓌳·⁖.·˙word:·

·⁖origin…language:·Tangut·⁖.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖.·˙font:·
·⁖pronunciation:·JAH·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Faith⁘ trust⁘ confidence placed in a person⁚⁚teaching⁚⁚promise⁚⁚or
↪ truth·⁖.·˙non…ostensive:·
·⁖ostensive:·A promise remains trusted while its fulfillment is still unseen⁘ confidence holds where
↪ sight has not yet arrived𓎟·⁖.·˙ostensive:·
·⁖grimchain:·𓍿𓆑𓃰𓇋𓃰𓅓𓂋𓄿𓆑𓊪𓎛𓏏𓍿·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜𓃹 category꩜꩜emission꩜:·
·⁖lookup office꩜꩜recursive…identity꩜ enoch꩜꩜⊕꩜:·

⁖word:·𐌀𐌄𐌚𐌔·⁖.·˙word:·

·⁖origin…language:·Gothic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖.·˙font:·
·⁖pronunciation:·DAYTHS·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Deed·⁖.·˙non…ostensive:·
·⁖ostensive:·Blade parts timber⁚⁚lifted pieces meet⁚⁚and the joined frame rises through the work being
↪ done𓃹·⁖.·˙ostensive:·

·⁚grimchain:·ᚺᑭᔑᑐ⬡ᵹ⏁Ⱳ○ᑭXᑎᛋ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ᵎΨ» category○–○ᵎemission»⁚·
·⁚lookup office○–○ᵎrecursive…identity» enoch○–○ᵎ⊕»⁚·

◦word:·𐎗𐎚𐎍◦⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·DAAHTU·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Knowledge⁚˙ what is held as understood⁚⁚recognized⁚⁚or known rather than
↪ uncertain·⁚.·˙non…ostensive:·
·⁚ostensive:·A familiar sign returns among unfamiliar marks⁚˙ the eye stops at once and the known form
↪ separates itself from uncertainty⁂·⁚.·˙ostensive:·
·⁚grimchain:·⌬✙ᔐᓄƧXƵᔓ⏁⏦ᛦφᛉ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ᵎᑭ» category○–○ᵎemission»⁚·
·⁚lookup office○–○ᵎrecursive…identity» enoch○–○ᵎ⊕»⁚·

◦word:·ᛯᛸ◦⁚.·˙word:·

·⁚origin…language:·Old Turkic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚.·˙font:·
·⁚pronunciation:·AHGH·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Rise·⁚.·˙non…ostensive:·
·⁚ostensive:·From the lower ledge the body lifts⁚⁚clears the edge⁚⁚and settles where the ground stands
↪ higherᑭ·⁚.·˙ostensive:·
·⁚grimchain:·☽ᚏ⊕ᘜ⏭ᐱᓯᚸᛋ⏏Ⅱᔑ⏁ₒ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ᵎ☽» category○–○ᵎemission»⁚·
·⁚lookup office○–○ᵎrecursive…identity» enoch○–○ᵎ⊕»⁚·

◦word:·ᛋΙ◦⁚.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KAHREH·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Worth·⁚.·˙non…ostensive:·
·⁚ostensive:·One measured piece remains equal to its own measure whether held alone or beside another
↪ of the same weight☽·⁚.·˙ostensive:·
·⁚grimchain:·ᔕᚸᛇᘏᚏᛋᛯᘐᚳᛏ⏁✳⊕·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ᵎ⊙» category○–○ᵎemission»⁚·
·⁚lookup office○–○ᵎrecursive…identity» enoch○–○ᵎ⊕»⁚·

◦word:·ᑐᑫᑐᑎᑐ◦⁚.·˙word:·

·⁖origin…language꞉·Avestan·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Avestan·⁖·˙font꞉·
·⁖pronunciation꞉·KH+shah+thrah·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Dominion·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·A command spoken at the center is obeyed at the far boundary⁚⁚the whole field answering to one authority⊙·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐤟𐤊𐤉♂⊛♲⚷𐤀𐤉𐤉𐤎𐤕·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉∘₋₀₊ℂ₎ category꞉∘₋₀₊emission₎⁖·
·⁖lookup office꞉∘₋₀₊recursive…identity₎ enoch꞉∘₋₀₊⊕₎⁖·

⁖word꞉·𐤆𐤇𐤕⁖·˙word꞉·

·⁖origin…language꞉·Tocharian B·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖·˙font꞉·
·⁖pronunciation꞉·AHNYMEH·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Self·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Face and posture alter⁚⁚years pass through the body⁚⁚yet the inward continuity does not divide with any of those changes℃·⁖·˙ostensive꞉·
·⁖grimchain꞉·𑀬𑁂𑀪𑀦𑀫𑀳𑀰𑀸𑀳𑀺𑀔'♂·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉∘₋₀₊♈₎ category꞉∘₋₀₊parliament₎⁖·
·⁖lookup office꞉∘₋₀₊recursive…identity₎ enoch꞉∘₋₀₊⊕₎⁖·

⁖word꞉·𐤆𐤇𐤂⁖·˙word꞉·

·⁖origin…language꞉·Ge'ez·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖·˙font꞉·
·⁖pronunciation꞉·ZUHK+r·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Memory⁖˙ remembrance⁖˙ record: what remains present after a deed⁚⁚person⁚⁚or event has passed·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·An old face returns inwardly at the familiar name⁖˙ what is absent from the room remains present in remembrance♈·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐤡𐤋𐤍𐤏𐤕𐤅𐤔𐤎𐤚𐤉𐤃·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉∘₋₀₊♉₎ category꞉∘₋₀₊parliament₎⁖·
·⁖lookup office꞉∘₋₀₊recursive…identity₎ enoch꞉∘₋₀₊⊕₎⁖·

⁖word꞉·ᛗᛁᛃᚢᚩᛗᛉᚾ⁖·˙word꞉·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·MIDUMOHN·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Mediate⁖˙ to stand or act between sides so relation⁚⁚passage⁚⁚or settlement may occur through a middle·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Two opposed sides remain themselves while the middle receives both⁚⁚carrying passage and settlement without becoming either side♉·⁖·˙ostensive꞉·

·⁖grimchain:·♓∮𐤉ʎϙ⁕ɻ▽♀𐤗𐤍𐤔♧·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꒰꒰𐤉𝌆꒱ category꒰꒰parliament꒱:·
·⁖lookup office꒰꒰recursive…identity꒱ enoch꒰꒰⊕꒱:·

˖word:·𐤎 𐤉 𐤌˖·˙word:·

·⁖origin…language:·Old Persian·⁖·˙origin…language:·
·⁖font:·Noto Sans Old Persian·⁖·˙font:·
·⁖pronunciation:·AR+tah·⁖·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖·˙cadence:·
·⁖non…ostensive:·Truth·⁖·˙non…ostensive:·
·⁖ostensive:·Covered stone remains beneath the soil unseen⁖ uncovering changes only its visibility::not
↪ what was already there𝌆·⁖·˙ostensive:·
·⁖grimchain:·♂ሕ⊙Ƙ✲ᛋᚴᔑᘐ✲✪·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꒰꒰♋꒱ category꒰꒰parliament꒱:·
·⁖lookup office꒰꒰recursive…identity꒱ enoch꒰꒰⊕꒱:·

˖word:·अर्थ˖·˙word:·

·⁖origin…language:·Sanskrit·⁖·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖·˙font:·
·⁖pronunciation:·uhr+thuh·⁖·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖·˙cadence:·
·⁖non…ostensive:·Aim⁖ purpose⁖ meaning or object toward which an act::thought::or effort is
↪ directed·⁖·˙non…ostensive:·
·⁖ostensive:·Arrow rests along one chosen line toward the marked point⁖ shaft and hand return every
↪ correction toward the marked point♋·⁖·˙ostensive:·
·⁖grimchain:·ᚷᔑᛋ𐦂Ⱪ𐤆ᚼᛅᛁᛏᚱᚮᚠ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꒰꒰♈꒱ category꒰꒰parliament꒱:·
·⁖lookup office꒰꒰recursive…identity꒱ enoch꒰꒰⊕꒱:·

˖word:·𐤎𒈙˖·˙word:·

·⁖origin…language:·Akkadian·⁖·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖·˙font:·
·⁖pronunciation:·nah+PISH+too·⁖·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖·˙cadence:·
·⁖non…ostensive:·Life·⁖·˙non…ostensive:·
·⁖ostensive:·Breath enters::warmth holds::hunger returns::wounds close::and growth continues through the
↪ same living body♈·⁖·˙ostensive:·
·⁖grimchain:·𐤗ሕᔑ𐦂⁕ᚮሕᑲᛏ≯ϰ⏣ϰ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol꒰꒰♒꒱ category꒰꒰parliament꒱:·
·⁖lookup office꒰꒰recursive…identity꒱ enoch꒰꒰⊕꒱:·

˖word:·ᛞᛇᛘ˖·˙word:·

·⁚origin…language꞉·Phoenician·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Phoenician·⁚.·˙font꞉·
·⁚pronunciation꞉·shah⟊LOHM·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Wholeness·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Every needed part is present within the whole⁚⁚with nothing belonging to it left outside
↪ or unheld𐤌·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᚹ‹ᛃ᛬ΘΒ𐤇❁ᚻΒ↗ᘐᛊ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ₑ𐤍₎ category꞉○–○ₑparliament₎⁚·
·⁚lookup office꞉○–○ₑrecursive…identity₎ enoch꞉○–○ₑΘ₎⁚·

꞉word꞉·ﻫﺪﻫﺐ·⁚.·˙word꞉·

·⁚origin…language꞉·Old Uyghur·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Serif Old Uyghur·⁚.·˙font꞉·
·⁚pronunciation꞉·egzig·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Melody·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Notes rise and fall through changing pitches⁚⁚yet the same contour remains recognizable
↪ from beginning to return𐤍·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𓃰Ɑ◝ᵒᵒ𐤌ᔑᛊ«ⱫΛ𝟚∀A·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ₑ𐤌₎ category꞉○–○ₑparliament₎⁚·
·⁚lookup office꞉○–○ₑrecursive…identity₎ enoch꞉○–○ₑΘ₎⁚·

꞉word꞉·ﻭﻧﻰﻯ·⁚.·˙word꞉·

·⁚origin…language꞉·Avestan·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Avestan·⁚.·˙font꞉·
·⁚pronunciation꞉·DROOJ·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Lie⁚⁚deception⁚⁚falsehood⁖˙ destructive false order·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·False marks occupy the place of true ones⁖˙ the ordered surface presents what is not so as
↪ though it were so𐤌·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᛤℂ⌬◎𐤓ᒧ𐤊ᚺ𐤍ᵒᵒᛘ𐤉·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ₑ↗₎ category꞉○–○ₑparliament₎⁚·
·⁚lookup office꞉○–○ₑrecursive…identity₎ enoch꞉○–○ₑΘ₎⁚·

꞉word꞉·𓃰·⁚.·˙word꞉·

·⁚origin…language꞉·Egyptian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Egyptian Hieroglyphs·⁚.·˙font꞉·
·⁚pronunciation꞉·KHEH⟊per·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Become·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Closed bud separates along its seam⁖˙ folded petals stand where only green enclosure had
↪ been present↗·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ꝋᵊ𐤋ᛦ✷ᚺ𑁍ᒣϙ𐤌ᒣ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙

·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍₀₎₎ category₀₋₀₍parliament₎⁚·
·⁚lookup office₀₋₀₍recursive…identity₎ enoch₀₋₀₍Φ₎⁚·

₍word:·ሕግ·⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·HUHG·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Law:˙ binding rule::command::or covenant that establishes what must
↪ hold·⁚·˙non…ostensive:·
·⁚ostensive:·One binding rule holds at every crossing:˙ what is forbidden remains forbidden whether the
↪ boundary is near or far₀₎·⁚·˙ostensive:·
·⁚grimchain:·𐤀𐤒𐤔𐤉𐤍𐤋𐤏𐤓𐤎𐤌𐤁·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍𓈖₎ category₀₋₀₍parliament₎⁚·
·⁚lookup office₀₋₀₍recursive…identity₎ enoch₀₋₀₍Φ₎⁚·

₍word:·𒌑𒀀⁚·˙word:·

·⁚origin…language:·Akkadian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·ah+LAH+koom·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Going·⁚·˙non…ostensive:·
·⁚ostensive:·One foot leaves its place::another follows::and the body continues through ground that was
↪ not beneath it before𓈖·⁚·˙ostensive:·
·⁚grimchain:·𐤈⊙𐤇◎𐤃𐤃𐤆𐤕𐤅▽𐤗𐤊𐤓·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍𓋹₎ category₀₋₀₍parliament₎⁚·
·⁚lookup office₀₋₀₍recursive…identity₎ enoch₀₋₀₍Φ₎⁚·

₍word:·ᚠᚱᚢᛏᚾᛋ⁚·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·FLUT+yah·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·to move·⁚·˙non…ostensive:·
·⁚ostensive:·Stone leaves the shaded hollow and comes to rest beside the open wall:˙ the first place
↪ remains empty behind it𓋹·⁚·˙ostensive:·
·⁚grimchain:·𐌾ᛇᛁᛞᛏᛊᚾᚠᛇᛏᛊᚲᛇᛏᛊᛟ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀₍◎₎ category₀₋₀₍goetic₎⁚·
·⁚lookup office₀₋₀₍recursive…identity₎ enoch₀₋₀₍Φ₎⁚·

₍word:·ሥርዐ⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·

·⁖pronunciation:·SUHR+woo·⁚.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚.·˙cadence:·
·⁖non…ostensive:·Root·⁚.·˙non…ostensive:·
·⁖ostensive:·Living growth begins below the surface from one rooted body that anchors⁚⁚draws⁚⁚and
↪ extends from the same buried source◎·⁚.·˙ostensive:·
·⁖grimchain:·ᶎɻ🙰ᶣϒ🙰▽ᵚ🌸🞟ص🝚ᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol○–○⬡꞉ category○–○goetic꞉:·
·⁖lookup office○–○recursive…identity꞉ enoch○–○⊕꞉:·

·word:·ـــــ:⁚.·˙word:·

·⁖origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁖font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁖pronunciation:·an·⁚.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚.·˙cadence:·
·⁖non…ostensive:·Register⁚⁚file⁚⁚or record: a kept body of what has been entered so it may remain and be
↪ called again·⁚.·˙non…ostensive:·
·⁖ostensive:·Marks remain in ordered rows after the entering hand is gone·⁖ the same kept sequence can
↪ be brought forward again from the preserved body⬡·⁚.·˙ostensive:·
·⁖grimchain:·ᔑ𐑂ᵬƆɥ𐐨ᘓ𐐣♂ᘓ𐑈·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol○–○✡꞉ category○–○goetic꞉:·
·⁖lookup office○–○recursive…identity꞉ enoch○–○⊕꞉:·

·word:·𒋗:⁚.·˙word:·

·⁖origin…language:·Sumerian·⁚.·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁖pronunciation:·EE+zee·⁚.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚.·˙cadence:·
·⁖non…ostensive:·Fire·⁚.·˙non…ostensive:·
·⁖ostensive:·Flame takes the dry fuel⁚⁚throwing heat and light while the burning body changes beneath
↪ it✡·⁚.·˙ostensive:·
·⁖grimchain:·♀⌬ΠɔᘎᘏⲘ'ᘓϔ♂ᘐص·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol○–○☆꞉ category○–○goetic꞉:·
·⁖lookup office○–○recursive…identity꞉ enoch○–○⊕꞉:·

·word:·𘜶:⁚.·˙word:·

·⁖origin…language:·Tangut·⁚.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁚.·˙font:·
·⁖pronunciation:·LEH·⁚.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚.·˙cadence:·
·⁖non…ostensive:·Change or transformation·⁖ becoming different in form⁚⁚state⁚⁚or
↪ condition·⁚.·˙non…ostensive:·
·⁖ostensive:·Green fruit darkens and softens on the branch·⁖ the same body no longer carries the state
↪ it held before☆·⁚.·˙ostensive:·
·⁖grimchain:·ᙅᘏᘐᖘ◎Iᘎ➢ᗡ꙰ᔕᐯ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꞉⟜꞉❂꞉ category꞉⟜꞉goetic꞉⁚·
·⁚lookup office꞉⟜꞉recursive…identity꞉ enoch꞉⟜꞉⊕꞉⁚·

꞉word꞉·ᛁᛈᛋᚺ꞉⁚·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚·˙font꞉·
·⁚pronunciation꞉·EV+nee·⁚·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Material꞉꞉substance꞉꞉or matter: the stuff from which a thing is made·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Clay꞉꞉wood꞉꞉fibre꞉꞉and stone occupy the bodies shaped from them⁚˙ every form is filled by the
↪ substance from which it is made❂·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᚲ▫ᚱᚢᚲᚸᛞᛇ⍐ᛟᚲᚸ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉⟜꞉❀꞉ category꞉⟜꞉goetic꞉⁚·
·⁚lookup office꞉⟜꞉recursive…identity꞉ enoch꞉⟜꞉⊕꞉⁚·

꞉word꞉·𒀭⁚·˙word꞉·

·⁚origin…language꞉·Sumerian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚·˙font꞉·
·⁚pronunciation꞉·DAH+ngahl·⁚·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Breadth·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Wide ground opens between two distant sides꞉꞉with no wall꞉꞉body꞉꞉or narrowing closing the
↪ reach between them❀·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᑎᛕᛋᗅᛊᛇ⬡♀ϒϒ⛧Ↄᚺ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉⟜꞉❋꞉ category꞉⟜꞉goetic꞉⁚·
·⁚lookup office꞉⟜꞉recursive…identity꞉ enoch꞉⟜꞉⊕꞉⁚·

꞉word꞉·𐎗𐎚𐎍⁚·˙word꞉·

·⁚origin…language꞉·Ugaritic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Ugaritic·⁚·˙font꞉·
·⁚pronunciation꞉·NAPSHU·⁚·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Throat·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Breath rises from the chest through the narrow living passage to the mouth꞉꞉voice
↪ traveling the same enclosed way❋·⁚·˙ostensive꞉·
·⁚grimchain꞉·▽❂❋ሕɻφↄᗅRIΘᖴɻƐ·⁚·˙grimchain꞉·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉⟜꞉⧗꞉ category꞉⟜꞉goetic꞉⁚·
·⁚lookup office꞉⟜꞉recursive…identity꞉ enoch꞉⟜꞉⊕꞉⁚·

꞉word꞉·𗍫⁚·˙word꞉·

·⁚origin…language꞉·Tangut·⁚·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚·˙font꞉·
·⁚pronunciation꞉·DZOO·⁚·˙pronunciation꞉·

·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Love·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·They draw close⁚⁚part⁚⁚and return⁖· affection survives the distance without erasing the two
↪ lives it joins⧗·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᚿᛇᛗᛊ𐋡I᛬ᛋ⁜ᚺᛗᚠᚹᚻ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉·
·⁚.·˙glyphs꞉·
·⁚glyph symbol꞉○꞉⊛꞉ category꞉○꞉goetic꞉⁚·
·⁚lookup office꞉○꞉recursive…identity꞉ enoch꞉○꞉⊕꞉⁚·

꞉word꞉·ሌ፞ኅ፞ኺ፞ፗ·⁚.·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·SKEEYOH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Shadow·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Light is absent behind an obstructing body⁚⁚and the dark contour remains the shadow cast
↪ by that blocked brightness⊛·⁚.·˙ostensive꞉·
·⁚grimchain꞉·Ƶ◺≻ϒ✲❋✡ᚠ✳Φᚱ⧗ᚹ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉·
·⁚.·˙glyphs꞉·
·⁚glyph symbol꞉○꞉❄꞉ category꞉○꞉goetic꞉⁚·
·⁚lookup office꞉○꞉recursive…identity꞉ enoch꞉○꞉⊕꞉⁚·

꞉word꞉·ድርገት꞉⁚.·˙word꞉·

·⁚origin…language꞉·Ge'ez·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Ethiopic·⁚.·˙font꞉·
·⁚pronunciation꞉·duhr⟊GAT·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Union⁚⁚association⁚⁚order⁚⁚rank⁚⁚or class: separate bodies joined into one
↪ arrangement·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Separate bodies keep their own shapes inside one ordering⁚⁚joined by the arrangement
↪ without losing their distinction❄·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᑭΨΦسرᵹᚱ᛬ᛋᛈ◺ᚺᚱ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉·
·⁚.·˙glyphs꞉·
·⁚glyph symbol꞉○꞉❁꞉ category꞉○꞉goetic꞉⁚·
·⁚lookup office꞉○꞉recursive…identity꞉ enoch꞉○꞉⊕꞉⁚·

꞉word꞉·𐎍𐎚꞉⁚.·˙word꞉·

·⁚origin…language꞉·Ugaritic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Ugaritic·⁚.·˙font꞉·
·⁚pronunciation꞉·EEB·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Enemy·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·One body stands against another with no kinship or alliance between them⁚⁚only direct
↪ opposition❁·⁚.·˙ostensive꞉·
·⁚grimchain꞉·❋ɞᚹᛗᛅᛋᚺ᛬ᚱᛋᛉᚱᚠسᚱ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉·
·⁚.·˙glyphs꞉·

·⁚glyph symbol꩜꩜⏣꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

⁖word⁚·ﻫﺪﻳﺒﻪ⁖⁚·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚⁚·˙font⁚·
·⁚pronunciation⁚·amUHl·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Calm⁖⁖quiet⁖⁖even+tempered: a state in which disturbance and noisy motion have fallen
↪ still·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Water lies even⁖⁖air quiet⁖⁖body without strain⁖⁖nothing in the field breaking the
↪ calm⏣·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫓𐫔𐫖𐫗𐫘𐫙𐫚𐫛X𐫜𐫝z·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐫍꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

⁖word⁚·𐫎𐫏𐫐⁖⁚·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚⁚·˙font⁚·
·⁚pronunciation⁚·NOO+moon·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Seed·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Root⁖⁖shoot⁖⁖and stored nourishment remain folded within one small body from which new
↪ growth can begin𐫍·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫑𐫒𐫓𐫔𐫕𐫖𐫗𐫘𐫙𐫚𐫛𐫜·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐫎꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

⁖word⁚·ምዕዋድ⁖⁚·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚⁚·˙font⁚·
·⁚pronunciation⁚·muh+WAHD·⁚⁚·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚⁚·˙cadence⁚·
·⁚non…ostensive⁚·Circuit⁖⁖orbit⁖⁖circle⁖⁖or sphere: a course that turns and returns toward its
↪ beginning·⁚⁚·˙non…ostensive⁚·
·⁚ostensive⁚·One course bends onward until its farthest turn is also the way back⁖⁖meeting the place
↪ from which it began𐫎·⁚⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫏𐫐𐫑𐫒𐫓𐫔𐫕𐫖𐫗X𐫘𐫙·⁚⁚·˙grimchain⁚·
·⁚⁚·˙lookup⁚˙
·⁚⁚·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐫏꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

⁖word⁚·ᛋᚠᚾᚱ⁖⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚⁚·˙font⁚·

·⁖pronunciation:·SVAHR·⁚·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚·˙cadence:·
·⁖non…ostensive:·Answer or reply: the returning response raised by what has been
↪ received·⁚·˙non…ostensive:·
·⁖ostensive:·An utterance reaches the other side and something returns from having received
↪ it⁘carrying response back across the relationᚱ·⁚·˙ostensive:·
·⁖grimchain:·𓃰ᵬᗇᛟᚔᐯᒉᚴᚿ⛢ᛏᗣ⁖⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦–◦𐤓꞉ category◦–◦court꞉·
·⁖lookup office◦–◦recursive…identity꞉ enoch◦–◦⊕꞉·

·word:·ⲧⲱⲱ·⁚·˙word:·

·⁖origin…language:·Coptic·⁚·˙origin…language:·
·⁖font:·Noto Sans Coptic·⁚·˙font:·
·⁖pronunciation:·tohsh·⁚·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚·˙cadence:·
·⁖non…ostensive:·Fashion⁘manner⁘ordinance⁘destiny⁘or condition: an ordering form by which a thing
↪ proceeds or is known·⁚·˙non…ostensive:·
·⁖ostensive:·From first edge to last⁘the same ordering keeps returning in the body⁘recognizable
↪ wherever the pattern appearsᚱ·⁚·˙ostensive:·
·⁖grimchain:·𐌿ᚿᚷ𑀪𑁍𐦟ᛩᗐᚣᑫᑐᛋ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦–◦𐤏꞉ category◦–◦court꞉·
·⁖lookup office◦–◦recursive…identity꞉ enoch◦–◦⊕꞉·

·word:·𒍅·⁚·˙word:·

·⁖origin…language:·Sumerian·⁚·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁚·˙font:·
·⁖pronunciation:·TEH⟊men·⁚·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚·˙cadence:·
·⁖non…ostensive:·Foundation·⁚·˙non…ostensive:·
·⁖ostensive:·Upper weight descends into the lowest course⁘which receives the burden and lets the whole
↪ structure stand above itᛩ·⁚·˙ostensive:·
·⁖grimchain:·ᗢᚼᗤᚹᛅᐱ᪥ᘓ⍙ᐞᚸᚿᢉ᪥·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol◦–◦𐤑꞉ category◦–◦court꞉·
·⁖lookup office◦–◦recursive…identity꞉ enoch◦–◦⊕꞉·

·word:·ᚣᑎᛕᚴᚣᑎᑎ·⁚·˙word:·

·⁖origin…language:·Gothic·⁚·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁚·˙font:·
·⁖pronunciation:·AWKAN·⁚·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁚·˙cadence:·
·⁖non…ostensive:·Increase·˙ grow: to become greater in amount⁘size⁘force⁘or extent·⁚·˙non…ostensive:·
·⁖ostensive:·One grain becomes a handful⁘the handful a basket⁘and the gathered measure stands greater
↪ than beforeᚱ·⁚·˙ostensive:·
·⁖grimchain:·ᛇᚦ⬡ᚻᚺI♂ᛦΨᚠᚿᛯ⊛⊕·⁚·˙grimchain:·
·⁚·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐦞꩟ category꩜꩜court꩟:·
·⁚lookup office꩜꩜recursive…identity꩟ enoch꩜꩜⊕꩟:·

·word:·𒁹𒁹·⁚·˙word:·

·⁚origin…language:·Sumerian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·NAHM·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Being⁚˙ state::condition::destiny::or mode of existence carried with a
↪ life·⁚·˙non…ostensive:·
·⁚ostensive:·Waking gives way to sleep::hunger to satiety::motion to stillness::while living presence
↪ continues through every condition𐦞·⁚·˙ostensive:·
·⁚grimchain:·𐫓𐤓𐤊𐤆𐤉𐤕𐤍𐤌𐤇𐤏𐤐·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐦟꩟ category꩜꩜court꩟:·
·⁚lookup office꩜꩜recursive…identity꩟ enoch꩜꩜⊕꩟:·

·word:·𖿠·⁚·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·LHOH·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Return or retreat·⁚·˙non…ostensive:·
·⁚ostensive:·Footsteps turn onto the ground already crossed⁚˙ the course bends back toward the place
↪ that had been left𐦟·⁚·˙ostensive:·
·⁚grimchain:·𐦞𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐦠꩟ category꩜꩜court꩟:·
·⁚lookup office꩜꩜recursive…identity꩟ enoch꩜꩜⊕꩟:·

·word:·𑀅𑀆𑀇·⁚·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·TREEW·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Mix or mingle·⁚·˙non…ostensive:·
·⁚ostensive:·Dark and pale grains tumble through the same stirring until neither gathers apart from
↪ the mixed body around it𐦡·⁚·˙ostensive:·
·⁚grimchain:·𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐦣꩟ category꩜꩜court꩟:·
·⁚lookup office꩜꩜recursive…identity꩟ enoch꩜꩜⊕꩟:·

·word:·𐎠 𐎡 𐎢·⁚·˙word:·

·⁚origin…language:·Old Persian·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚·˙font:·

·⁖pronunciation⁚·CHEE÷chah·⁖·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Form⁝⁚type⁝⁚or lineage: the recognizable kind or shape by which something is
↪ known·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·From the tight inner coil to the widening rim⁝⁚the shell keeps one governing turn through
↪ every added curve𐬹·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐓛𐓢'𐓂𐓏𐓘𐓣𐓪𐓐𐓟𐓢𐓑‹·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol꜀꜀𐬹 category꜀꜀court꜆⁚·
·⁖lookup office꜀꜀recursive…identity꜆ enoch꜀꜀⊕꜆⁚·

·word⁚·ᑎᖻ·⁖·˙word⁚·

·⁖origin…language⁚·Old Turkic·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Old Turkic·⁖·˙font⁚·
·⁖pronunciation⁚·UEZ·⁖·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Break or tear⁖˙ to destroy continuity by pulling⁝⁚snapping⁝⁚or dividing a
↪ whole·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·A taut whole parts under strain⁝⁚the opening traveling through what had been continuous
↪ until two sides stand where one body had joined𐬹·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐓘𐓣𐓃𐓤𐓡𐓗𐓧𐓟⊕𐓥𐓯·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol꜀꜀𐬺 category꜀꜀court꜆⁚·
·⁖lookup office꜀꜀recursive…identity꜆ enoch꜀꜀⊕꜆⁚·

·word⁚·+𑀍ᒧ𑁆·⁖·˙word⁚·

·⁖origin…language⁚·Tocharian A·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Brahmi·⁖·˙font⁚·
·⁖pronunciation⁚·KUHLP·⁖·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Find⁝⁚obtain⁝⁚locate⁝⁚or achieve⁖˙ to reach what was sought or managed toward
↪ completion·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Searching hands pass over empty ground⁝⁚then close around the hidden stone beneath the
↪ grass𐬺·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐓟𐓥𐓘𐓛𐓪𐓩𐓐𐓣𐓞𐓑𐓣⊕𐓑𐓣·⁖·˙grimchain⁚·
·⁖·˙lookup⁚˙
·⁖·˙glyphs⁚˙
·⁖glyph symbol꜀꜀𐬻 category꜀꜀court꜆⁚·
·⁖lookup office꜀꜀recursive…identity꜆ enoch꜀꜀⊕꜆⁚·

·word⁚·𗏁·⁖·˙word⁚·

·⁖origin…language⁚·Tangut·⁖·˙origin…language⁚·
·⁖font⁚·Noto Serif Tangut·⁖·˙font⁚·
·⁖pronunciation⁚·ZAWM·⁖·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Hold⁝⁚keep⁝⁚support⁝⁚or retain⁖˙ to maintain possession or keep something from passing
↪ away·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Closed fingers keep the bead against the palm while the hand turns⁝⁚lifts⁝⁚and lowers
↪ without releaseI·⁖·˙ostensive⁚·

·⁖grimchain⁖·ᛏ𐌅ᛊ𐌅ᛊ✳ᚸᚺᛘᚢᚿᚣᛞᛘ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𐑉𐑉 category⟜⟜court𐑉⁖·
·⁖lookup office⟜⟜recursive…identity𐑉 enoch⟜⟜⊕𐑉⁖·

⁘word⁖·ƽƟϽϤ⁘·⁖·˙word⁖·

·⁖origin…language⁖·Aghwan·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Caucasian Albanian·⁖·˙font⁖·
·⁖pronunciation⁖·GHUN·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Mirror⁚⁚looking glass·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Polished surface carries the facing body back as light⁖˙ reflected likeness changes only
↪ when the facing body changes𐑉·⁖·˙ostensive⁖·
·⁖grimchain⁖·ᐁᙢᛯ𐑉ᛉᚸᚺᛝᐁᛯᛟᙢᛩ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𐑉▫𐑉 category⟜⟜court𐑉⁖·
·⁖lookup office⟜⟜recursive…identity𐑉 enoch⟜⟜⊕𐑉⁖·

⁘word⁖·ʎ̄ɻ̄□ⵜ̄⁘·⁖·˙word⁖·

·⁖origin…language⁖·Tocharian B·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Brahmi·⁖·˙font⁖·
·⁖pronunciation⁖·TSUHK·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Burn⁚⁚roast⁚⁚boil⁚⁚or heat until matter is cooked⁚⁚consumed⁚⁚or changed by
↪ fire⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Raw grain darkens⁚⁚softens⁚⁚and gives steam above the heated vessel until the hard kernel
↪ is no longer raw▫·⁖·˙ostensive⁖·
·⁖grimchain⁖·𐑅ᙫᛯᕮᚳᘔᚸᚸᚸᚸI0·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𐑉𐑉 category⟜⟜court𐑉⁖·
·⁖lookup office⟜⟜recursive…identity𐑉 enoch⟜⟜⊕𐑉⁖·

⁘word⁖·ᚲ⁘·⁖·˙word⁖·

·⁖origin…language⁖·Sumerian·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Cuneiform·⁖·˙font⁖·
·⁖pronunciation⁖·BOOR·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Depth⁖˙ downward extent⁚⁚hollow⁚⁚depression⁚⁚or low reach beneath a surrounding
↪ level⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Ground drops away beneath the surrounding level⁚⁚and the hollow keeps descending beyond
↪ the last visible edge𐑉·⁖·˙ostensive⁖·
·⁖grimchain⁖·⌬ᙅᛉᚸ𓊝ᛯᛉᗺᚹᛉᚺᑫ◎·⁖·˙grimchain⁖·
·⁖·˙lookup⁖˙
·⁖·˙glyphs⁖˙
·⁖glyph symbol⟜⟜𐑉ᔑ𐑉 category⟜⟜court𐑉⁖·
·⁖lookup office⟜⟜recursive…identity𐑉 enoch⟜⟜⊕𐑉⁖·

⁘word⁖·ᵬƵĪ⁘·⁖·˙word⁖·

·⁖origin…language:·Tocharian A·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·SHROHTREH·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Sign or mark: a visible trace by which presence⁝difference⁝or meaning is
↪ recognized·⁖.·˙non…ostensive:·
·⁖ostensive:·One cut remains dark across the clay face after the stylus lifts⁖ the visible trace stays
↪ where the pressure ended♪·⁖.·˙ostensive:·
·⁖grimchain:·ᛈ❂ᚸᛇ⛒ᛋᛘᛟᚥᛦ☿᛬·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꞉꞉₹ Ɣ꜀ category꞉꞉₹court꜀⁖·
·⁖lookup office꞉꞉₹recursive…identity꜀ enoch꞉꞉₹⊕꜀⁖·

⁖word:·阬⁖.·˙word:·

·⁖origin…language:·Tangut·⁖.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖.·˙font:·
·⁖pronunciation:·MEEH·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Spread⁝scatter⁝sprinkle⁝or disperse matter outward across a wider
↪ extent·⁖.·˙non…ostensive:·
·⁖ostensive:·Grains spill from the heap and keep spreading until widening ground lies between bodies
↪ that had touchedƳ·⁖.·˙ostensive:·
·⁖grimchain:·𐠈ᛮᒧ𐠈ᛜᚹᛉᛏᛯᛞ✡·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꞉꞉₹ category꞉꞉₹court꜀⁖·
·⁖lookup office꞉꞉₹recursive…identity꜀ enoch꞉꞉₹⊕꜀⁖·

⁖word:·➽〓𝌆⁖.·˙word:·

·⁖origin…language:·Ugaritic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖.·˙font:·
·⁖pronunciation:·RPOO·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Ancestral spirit or ghost·⁖.·˙non…ostensive:·
·⁖ostensive:·A human outline occupies the empty threshold where no living body stands⁖ nearness and
↪ recognizable presence remain without flesh𝄞·⁖.·˙ostensive:·
·⁖grimchain:·♎𐠈ᚷᛯᛟᛞ⛛ᛇℂᛋᚳ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꞉꞉₹Ⴙ꜀ category꞉꞉₹court꜀⁖·
·⁖lookup office꞉꞉₹recursive…identity꜀ enoch꞉꞉₹⊕꜀⁖·

⁖word:·ȣ丩⁖.·˙word:·

·⁖origin…language:·Tocharian B·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·MEHNKEE·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Lack or deficiency: the condition in which something needed⁝expected⁝or once present
↪ is absent·⁖.·˙non…ostensive:·

·⁚ostensive⁚·Every place in the row is filled except one⁚· absence occupies the empty place where the
↪ missing body should stand↯·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᨒ𐋡𐋢𐋣𐋤𐋥𐋦𐋧𐋨𐋩𐋪𐋫𐋬𐋭𐋮𐋯𐋰𐋱𐋲·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜꜀'꜆ category⟜⟜꜀court꜆⁚·
·⁚lookup office⟜⟜꜀recursive…identity꜆ enoch⟜⟜꜀⊕꜆⁚·

꜉word⁚·ꕥJ̃꜉⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·AHL·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Hold back or restrain⁚· to prevent motion⁚⁚excess⁚⁚or advance from continuing
↪ freely·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·A rope draws tight across the gate⁚· the moving door stops at the held line and goes no
↪ farther'·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐑉𐑴𐑻𐑼𐑽𐑾𐑿𐒀𐒁𐒂𐒃𐒄𐒅𐒆·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜꜀꜈꜆ category⟜⟜꜀court꜆⁚·
·⁚lookup office⟜⟜꜀recursive…identity꜆ enoch⟜⟜꜀⊕꜆⁚·

꜉word⁚·ᛝ꜉⁚.·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·EH+reen·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Troop⁚· a gathered body of workers or fighters acting together under common
↪ direction·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Gathered bodies advance⁚⁚turn⁚⁚and halt beneath one direction⁚⁚many moving without becoming
↪ one↯·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐓘𐓙𐓚𐓛𐓜𐓝𐓞𐓟𐓠𐓡𐓢𐓣·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜꜀↯꜆ category⟜⟜꜀court꜆⁚·
·⁚lookup office⟜⟜꜀recursive…identity꜆ enoch⟜⟜꜀⊕꜆⁚·

꜉word⁚·ᛊ꜉⁚.·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚.·˙font⁚·
·⁚pronunciation⁚·EUH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Remember or think of·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·A buried name returns at sight of the old mark⁚· the absent face becomes present again
↪ within thought↯·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐌰𐌱𐌲𐌳𐌴𐌵𐌶𐌷𐌸𐌹𐌺𐌻𐌼·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜꜀↑꜆ category⟜⟜꜀court꜆⁚·
·⁚lookup office⟜⟜꜀recursive…identity꜆ enoch⟜⟜꜀⊕꜆⁚·

˖word:·ᚩᚢᚹᚼ˖.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·LOOKA·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Shut or close: to bring an opening or passage into a state where passage no longer
↪ continues·⁚.·˙non…ostensive:·
·⁚ostensive:·Two fitted leaves meet edge to edge across the opening until nothing can pass between
↪ them↾⁚.·˙ostensive:·
·⁚grimchain:·᛭▫ᚸᛯᚼᚴᛋᚢIᚹ⟴⏀0·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᛦ꞉ category꩜꩜court꞉·
·⁚lookup office꩜꩜recursive…identity꞉ enoch꩜꩜⊕꞉·

˖word:·ᛏ̄ᒍᏁᛏ̄˖.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KLOOTK·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Turn or change into another state or form⁚· become different from the prior
↪ condition·⁚.·˙non…ostensive:·
·⁚ostensive:·Dark matter enters clear liquid⁚⁚clouding it through until the vessel holds a condition it
↪ did not carry beforeᛦ·⁚.·˙ostensive:·
·⁚grimchain:·ᛈ⏀❁ᚹᛉᚨᛇᚿᛜᚻᚹᛠᚺ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᚢ꞉ category꩜꩜court꞉·
·⁚lookup office꩜꩜recursive…identity꞉ enoch꩜꩜⊕꞉·

˖word:·ᎶᏂᑕ˖.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·GUH⟊br·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Action⁚⁚labor⁚⁚work⁚⁚craft⁚⁚or deed performed through effort and skill·⁚.·˙non…ostensive:·
·⁚ostensive:·Hands work wet fibres again and again until the loose strands stand as one finished
↪ cord⁚⁚bearing the craft in the formed resultᚢ·⁚.·˙ostensive:·
·⁚grimchain:·⟴ᔓᖶ�perᛜᚺᚻᚥᚱᘁ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᚦ꞉ category꩜꩜court꞉·
·⁚lookup office꩜꩜recursive…identity꞉ enoch꩜꩜⊕꞉·

˖word:·ᓀᒪᒍᒍᒐ˖.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·SUHLP·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·

·⁖non…ostensive⁚·Burn⁙be on fire⁙blaze: flame actively consuming or shining from
↪ matter·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Flame lives through the fuel⁙heat and light continuing as the burning body is taken into
↪ the fireÞ·⁚.·˙ostensive⁚·
·⁖grimchain⁚·Ƙ𐑒∀ℌ⅄ꓘ⊕△𐑌ᔑ𐑜ᔭÞ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᵋᚠ» category⟜⟜ᵋcourt»⁚·
·⁖lookup office⟜⟜ᵋrecursive…identity» enoch⟜⟜ᵋ⊕»⁚·

⁖word⁚·ᓚᒣJ⁖⁚.·˙word⁚·

·⁖origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁖font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁖pronunciation⁚·SAHLEH·⁚.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁖non…ostensive⁚·Salt·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Water vanishes and white crystals gather in its place⁙the same mineral body present in
↪ grain and brineᚠ·⁚.·˙ostensive⁚·
·⁖grimchain⁚·⊛ᘎ𐑒ↄƟƵ‹♀𑁍Ⅱ𐑌ᔕ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᵋᚱ» category⟜⟜ᵋcourt»⁚·
·⁖lookup office⟜⟜ᵋrecursive…identity» enoch⟜⟜ᵋ⊕»⁚·

⁖word⁚·ᚢᚱᛒᚢᛁᚿᔕ⁖⁚.·˙word⁚·

·⁖origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁖font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁖pronunciation⁚·ARBEYETHS·⁚.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁖non…ostensive⁚·Labour⁙toil⁙hardship⁙or distress: sustained burden borne through difficult work or
↪ suffering·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Shoulders remain bent beneath the millstone while the hands continue turning it against
↪ resistanceᚱ·⁚.·˙ostensive⁚·
·⁖grimchain⁚·ᗅᖍ♂ᛋⅡᚲ‹ᚱᒣ𑁍ᐱ◎𑁋·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol⟜⟜ᵋ‹» category⟜⟜ᵋcourt»⁚·
·⁖lookup office⟜⟜ᵋrecursive…identity» enoch⟜⟜ᵋ⊕»⁚·

⁖word⁚·श्रीः⁚.·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁖pronunciation⁚·shree·⁚.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁖non…ostensive⁚·Splendour⁙radiance⁙glory⁙beauty⁙grace⁙prosperity: excellence whose presence shines
↪ beyond the ordinary·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Radiance gathers on the lifted surface and spills onto the stone around it⁙splendor
↪ living in the shining itself‹·⁚.·˙ostensive⁚·
·⁖grimchain⁚·ᐦᓄÞᖶ𐑒ᚺƵᔑ𐑜ᔭ·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁖glyph symbol꩜꩜𐊗꩞ category꩜꩜court꩞⁖·
·⁖lookup office꩜꩜recursive…identity꩞ enoch꩜꩜⊕꩞⁖·

·⁖word⁖·𐫁𐫖𐫁𐫛𐫤⁖·˙word⁖·

·⁖origin…language⁖·Tocharian B·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Brahmi·⁖·˙font⁖·
·⁖pronunciation⁖·AHOONYEHNTOH·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Beginning⁖⁖start: the first arising from which a course⁖⁖act⁖⁖or state
↪ proceeds·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·First water appears where dry earth had held none⁖· one opening marks the earliest
↪ presence of the flowing courseX·⁖·˙ostensive⁖·
·⁖grimchain⁖·Θ◝꣐ᐟ☽↓▫ᚠ⊖𐙐Ⱥ𐰀ᔓᖘ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᚹ꩞ category꩜꩜court꩞⁖·
·⁖lookup office꩜꩜recursive…identity꩞ enoch꩜꩜⊕꩞⁖·

·⁖word⁖·ϤᏔЛΠᎮϮ⁖·˙word⁖·

·⁖origin…language⁖·Old Norse·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Runic·⁖·˙font⁖·
·⁖pronunciation⁖·SVELGA·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Swallow⁖⁖devour⁖⁖or gulp down: to take a thing inward by consuming it wholly or
↪ forcefully·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·One mouthful passes from mouth through throat until nothing remains held above the
↪ passageᚹ·⁖·˙ostensive⁖·
·⁖grimchain⁖·▫⬡Ϩ𑣲ᚏ⁕❄ᒧᚗᛎᚸᚺ⩍·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜Ꮒ꩞ category꩜꩜court꩞⁖·
·⁖lookup office꩜꩜recursive…identity꩞ enoch꩜꩜⊕꩞⁖·

·⁖word⁖·ᒥᐱᑎᑐᒪᖘS⁖·˙word⁖·

·⁖origin…language⁖·Gothic·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Gothic·⁖·˙font⁖·
·⁖pronunciation⁖·HLOOTRS·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Clean or pure: free of what soils⁖⁖mixes⁖⁖stains⁖⁖or corrupts the body or
↪ state·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Clear water runs from the stone carrying away silt and stain until nothing foreign
↪ remains on the washed surfaceᚺ·⁖·˙ostensive⁖·
·⁖grimchain⁖·ᛏ𐌷ᚹᚌ𝌆ᓚᔭᛃᚦ𐙏ᔑᒍ▫ᚦ·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꩜꩜ᛋ꩞ category꩜꩜court꩞⁖·
·⁖lookup office꩜꩜recursive…identity꩞ enoch꩜꩜⊕꩞⁖·

·⁖word⁖·ᚼᚾᚱᚥᚱ⁖·˙word⁖·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·HARTH+r·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Hard⁘ firm and resistant to pressure⁝⁝cutting⁝⁝bending⁝⁝or deformation·⁚.·˙non…ostensive:·
·⁚ostensive:·Pressure meets firm resistance⁝⁝and the body keeps its shape rather than readily
↪ bending⁝⁝cutting⁝⁝or yielding᛭·⁚.·˙ostensive:·
·⁚grimchain:·᛬ᚹᛞᚹᛉᚼᚢᚹᛞᚸᚹᛞ·⁚.·˙grimchain:·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol꩜꩜ᚦ꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜ᛞ꩝⁚·

⁖word⁚·ᚠᛈᛊ⁖.·˙word⁚·

·⁚origin…language:·Sumerian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·GOOM·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Smash·⁚.·˙non…ostensive:·
·⁚ostensive:·Hammer meets brittle stone⁘ the single mass opens into fragments across the
↪ floorᚻ·⁚.·˙ostensive:·
·⁚grimchain:·▽ᚹᛯᘔ⊛ᘐᔕ❁◎ᘁ·⁚.·˙grimchain:·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol꩜꩜ᛃ꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜ᛞ꩝⁚·

⁖word⁚·ᛇ⁖.·˙word⁚·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·coy·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Ash·⁚.·˙non…ostensive:·
·⁚ostensive:·Pale powder rests where flame and fuel are gone⁘ light dry grains scatter from the burned
↪ body under the slightest airᛃ·⁚.·˙ostensive:·
·⁚grimchain:·ᛁᚸᘐᛞᚲᚲᛋᛇᘐᛟᚣ·⁚.·˙grimchain:·
·⁚.·˙lookup⁘
·⁚.·˙glyphs⁘
·⁚glyph symbol꩜꩜ᛝ꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜ᛞ꩝⁚·

⁖word⁚·ᛋᛏᚱᛋᚢᚱᚱ⁖.·˙word⁚·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·STRAUMR·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Stream or current: a body of movement continuing through water or another flowing
↪ course⁚.·˙non…ostensive:·
·⁚ostensive:·Fresh water enters the channel as other water leaves⁝⁝the moving course continuing without
↪ a break in its flowᛝ·⁚.·˙ostensive:·
·⁚grimchain:·᛭ᚴᛇᘐᛁᛋᘓᛞᛉᚢᚥ·⁚.·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᑊᔐ꩝ category꩜꩜ᑊcourt꩝:·
·⁚lookup office꩜꩜ᑊrecursive…identity꩝ enoch꩜꩜ᑊ⊕꩝:·

·word:·ʌĪȣ̄⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·TRAHEEWOH·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Mixture:˙ a combined body in which separate ingredients have entered one state
↪ together·⁚·˙non…ostensive:·
·⁚ostensive:·Different ingredients share one body after mixing::none of them remaining gathered apart
↪ from the joined stateᔐ·⁚·˙ostensive:·
·⁚grimchain:·✲ᔭ৬✷ᔦ⁂ᗅ▫ᚱ✲ᘐ◎ᔑ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᑊↄ꩝ category꩜꩜ᑊcourt꩝:·
·⁚lookup office꩜꩜ᑊrecursive…identity꩝ enoch꩜꩜ᑊ⊕꩝:·

·word:·ᛦᚱᚼᛁᚦᚾ⁚·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·GRAYTHA·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Unravel::disentangle::arrange: to release what is bound or confused into a clearer
↪ order·⁚·˙non…ostensive:·
·⁚ostensive:·Knotted cord loosens strand by strand until each line lies separate and
↪ straightↄ·⁚·˙ostensive:·
·⁚grimchain:·㊩𝄞ᗡᖶᔭ𝅘𝅥𝅯ᛘᕮ⊢ᛁᚷᗡᚺᕮ∫·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᑊ✲꩝ category꩜꩜ᑊcourt꩝:·
·⁚lookup office꩜꩜ᑊrecursive…identity꩝ enoch꩜꩜ᑊ⊕꩝:·

·word:·ᚦᚣSTᚣᚾ⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·FASTAN·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Keep or hold firm: retain something so it remains possessed::fixed::or preserved rather
↪ than released·⁚·˙non…ostensive:·
·⁚ostensive:·A knot tightens around the load·:˙ the suspended weight remains held without slipping
↪ free✲·⁚·˙ostensive:·
·⁚grimchain:·꩜꩜ᘐ☿ᙡ𝅘𝅥ᚹᚢᓭ㊩𝅘𝅥ᘐ𝄞ᗅᚱᚢ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᑊ꩝ category꩜꩜ᑊcourt꩝:·
·⁚lookup office꩜꩜ᑊrecursive…identity꩝ enoch꩜꩜ᑊ⊕꩝:·

·word:·कऱ्दम⁚·˙word:·

·⁖origin…language:·Sanskrit·⁖.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖.·˙font:·
·⁖pronunciation:·ka+rda+ma·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Mud⁚⁚slime⁚⁚mire⁚⁚sediment⁚⁚dregs·⁘ the heavy remainder that settles from mixture or
↪ disturbance·⁖.·˙non…ostensive:·
·⁖ostensive:·Heavy wet earth gathers below disturbed water·⁘ thick sediment clings⁚⁚slumps⁚⁚and holds
↪ water within the dark mass·⁖.·˙ostensive:·
·⁖grimchain:·ǂ◌ɔɹɿʔ◌ʯᗢ⁂ₒ⊕·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○₍ᘐ⁰₈₎ category○–○₍court₎⁖·
·⁖lookup office○–○₍recursive…identity₎ enoch○–○₍⊕₎⁖·

₍word⁚·ﻫﻴﻴﻬﻪ·⁖.·˙word⁚·

·⁖origin…language:·Old Uyghur·⁖.·˙origin…language:·
·⁖font:·Noto Serif Old Uyghur·⁖.·˙font:·
·⁖pronunciation:·aGHmaK·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Rise or elevate: movement from a lower standing toward a higher one⁚⁚literal or
↪ figurative·⁖.·˙non…ostensive:·
·⁖ostensive:·Stone rises from the lower ledge and comes to rest above it⁚⁚leaving open air beneath the
↪ place it now holdsᘐ⁰₈·⁖.·˙ostensive:·
·⁖grimchain:·⊛ᚣᚏᛇᛋᚰₒᚣᛋᚣᛘᛒᛪᕀ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○₍ɳ₎ category○–○₍court₎⁖·
·⁖lookup office○–○₍recursive…identity₎ enoch○–○₍⊕₎⁖·

₍word⁚·ÞՈԻԿϮ⁖.·˙word⁚·

·⁖origin…language:·Old Norse·⁖.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖.·˙font:·
·⁖pronunciation:·THOH+kah·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Fog or mist: suspended moisture that veils distance and weakens clear
↪ sight·⁖.·˙non…ostensive:·
·⁖ostensive:·Fine suspended moisture fills the air and weakens distant edges until every farther form
↪ is veiledɳ·⁖.·˙ostensive:·
·⁖grimchain:·ᛗᛋᚿᛜ᛬◎❀ᕏᦓᘐᗢᑫ◎ᓬ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○₍◎₎ category○–○₍court₎⁖·
·⁖lookup office○–○₍recursive…identity₎ enoch○–○₍⊕₎⁖·

₍word⁚·瓶⁖.·˙word⁚·

·⁖origin…language:·Tangut·⁖.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖.·˙font:·
·⁖pronunciation:·SHYEEM·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Ooze or seep·⁘ liquid passing slowly through pores⁚⁚cracks⁚⁚or a containing
↪ surface·⁖.·˙non…ostensive:·

·⁚ostensive⁚·Moisture appears bead by bead through a narrow crack⁚· no open stream forms across the
↪ surface◎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᜰ₀ᛋ₀ᑫᑲ⬡ℤᨐᓀ⯈⊕Ψ⊛·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𓂀𓊃 category⟜⟜court𓊃⁚·
·⁚lookup office⟜⟜recursive…identity𓊃 enoch⟜⟜⊕𓊃⁚·

₎word⁚·도의₎.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·twouy·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Trade⁚⁚barter⁚⁚exchange: reciprocal transfer in which what is held passes between
↪ parties·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One hand passes grain across the table while another returns measured metal⁘ each leaves
↪ holding what the other had𓂀·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ℤ⌬❁ɤᵯϔᖱ᭄ᶯᵋ✡ᛔᵋ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜ᔓᚼ𓊃 category⟜⟜court𓊃⁚·
·⁚lookup office⟜⟜recursive…identity𓊃 enoch⟜⟜⊕𓊃⁚·

₎word⁚·𓈖𓃹₎.·˙word⁚·

·⁚origin…language⁚·Hittite·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·EKAN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Ice⁘ water held in a hard frozen state by cold·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Under cold⁚⁚water locks into a hard crystal body that meets pressure with a rigid surface
↪ instead of flowing awayᔓᚼ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᙅ𐑋ᚼᔓᚼ'▫ᚻℂ◌ↄXᒧ⧗·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𓂀ᚸ𓊃 category⟜⟜court𓊃⁚·
·⁚lookup office⟜⟜recursive…identity𓊃 enoch⟜⟜⊕𓊃⁚·

₎word⁚·ᚣᛋ₎.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·khah+LAH+koom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Disappear⁚⁚go missing⁚⁚perish⁚⁚or pass from presence: to cease being found where
↪ presence had been·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Tracks fade under wind+blown dust until no edge remains where the footprint had
↪ been𓂀·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ϛ✱⯈ᗢ✡☆'⯈ᛯᔕ0ᛋↄ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𐂃𐂃𓊃 category⟜⟜court𓊃⁚·
·⁚lookup office⟜⟜recursive…identity𓊃 enoch⟜⟜⊕𓊃⁚·

˖word:·𓂝˖·.·ˈword:·

·˙origin…language:·Egyptian·˙.·ˈorigin…language:·
·˙font:·Noto Sans Egyptian Hieroglyphs·˙.·ˈfont:·
·˙pronunciation:·JED·˙.·ˈpronunciation:·
·˙cadence:·Recursive Identity·˙.·ˈcadence:·
·˙non…ostensive:·Stability::endurance·˙ the djed⟊pillar·˙.·ˈnon…ostensive:·
·˙ostensive:·Course upon course stays upright beneath the load::the vertical body carrying weight without giving way𓊽·˙.·ˈostensive:·
·˙grimchain:·𓏛𓎼𓎛𓏏𓌳𓋴𓐍𓏏𓎟𓆓𓎛·˙.·ˈgrimchain:·
·˙.·ˈlookup·˙
·˙.·ˈglyphs·˙
·˙glyph symbol꜀꜀𓊽꜀꜀ category꜀꜀court꜀˙·
·˙lookup office꜀꜀recursive…identity꜀ enoch꜀꜀Θ꜀˙·

˖word:·𒀸𒁀𒈬˖·.·ˈword:·

·˙origin…language:·Akkadian·˙.·ˈorigin…language:·
·˙font:·Noto Sans Cuneiform·˙.·ˈfont:·
·˙pronunciation:·EESH⟊doom·˙.·ˈpronunciation:·
·˙cadence:·Recursive Identity·˙.·ˈcadence:·
·˙non…ostensive:·Base::foundation::or bottom: the lowest supporting part from which a body stands or is ordered·˙.·ˈnon…ostensive:·
·˙ostensive:·Everything above bears down into the lowest supporting course::and from that bottom the body takes its standing𓊽·˙.·ˈostensive:·
·˙grimchain:·𓆏𓏛𓎛𓎼𓏏𓋴𓂋𓏏𓎟𓆓𓎛·˙.·ˈgrimchain:·
·˙.·ˈlookup·˙
·˙.·ˈglyphs·˙
·˙glyph symbol꜀꜀Θ꜀ category꜀꜀court꜀˙·
·˙lookup office꜀꜀recursive…identity꜀ enoch꜀꜀Θ꜀˙·

˖word:·ᚴᚱᛁᛋᛊ˖·.·ˈword:·

·˙origin…language:·Gothic·˙.·ˈorigin…language:·
·˙font:·Noto Sans Gothic·˙.·ˈfont:·
·˙pronunciation:·GRITHS·˙.·ˈpronunciation:·
·˙cadence:·Recursive Identity·˙.·ˈcadence:·
·˙non…ostensive:·Rank::grade::standing: a level held in relation to levels above::below::or beside it·˙.·ˈnon…ostensive:·
·˙ostensive:·One standing takes its place among others above and below::the relation itself fixing where the level belongsΘ·˙.·ˈostensive:·
·˙grimchain:·'𐌷𐍂𐌹𐍃𐌴𐍈𐌺𐌰𐌽𐌹·˙.·ˈgrimchain:·
·˙.·ˈlookup·˙
·˙.·ˈglyphs·˙
·˙glyph symbol꜀꜀Ᵹ꜀ category꜀꜀court꜀˙·
·˙lookup office꜀꜀recursive…identity꜀ enoch꜀꜀Θ꜀˙·

˖word:·शलिा˖·.·ˈword:·

·˙origin…language:·Sanskrit·˙.·ˈorigin…language:·
·˙font:·Noto Sans Devanagari·˙.·ˈfont:·
·˙pronunciation:·shi⟊lah·˙.·ˈpronunciation:·
·˙cadence:·Recursive Identity·˙.·ˈcadence:·

·⁚non…ostensive⁚·Stone⁖⁖rock⁖⁖or crag: hard mineral body enduring as solid
↪ earth⁜material·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Hard mineral mass keeps one solid contour through rain⁖⁖weight⁖⁖and exposed
↪ groundⵝ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐋏ȣ⁑ᘡᑫᖘ◌✷▫⓿·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·
·⁚glyph symbol◦–◦ᛨ» category◦–◦ᛨcourt»⁚·
·⁚lookup office◦–◦ᛨrecursive…identity» enoch◦–◦ᛨ⊕»⁚·

·word⁚·하야믜다·⁚.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·háyámúytá·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Wear out·˙ become worn and torn through use⁖⁖strain⁖⁖or repeated
↪ contact·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Repeated use and contact remove material little by little until the body shows wear that
↪ was not present beforeᐱ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·◎ᚻᘜ⓿ᚹᓓᚢᚣ᳀ᘐᑭᖇᖽᐯ·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·
·⁚glyph symbol◦–◦ᛨᚺ» category◦–◦ᛨcourt»⁚·
·⁚lookup office◦–◦ᛨrecursive…identity» enoch◦–◦ᛨ⊕»⁚·

·word⁚·듀ᓚ·⁚.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·tyùng·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Middle or center: the position or body between outer sides or among surrounding
↪ parts·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Equal distance opens to either side of one held place⁖⁖the surrounding field gathering
↪ around its centerᚺ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᗡᛋᚾ⊛ᖯↄXஓᑭᕀᓀᚺᘐ·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·
·⁚glyph symbol◦–◦ᛨᛕ» category◦–◦ᛨcourt»⁚·
·⁚lookup office◦–◦ᛨrecursive…identity» enoch◦–◦ᛨ⊕»⁚·

·word⁚·𑍐·⁚.·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·DAHG·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To roam or spread out·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Steps leave the known path and continue wherever open ground permits⁖⁖with no fixed
↪ destination closing the roaming courseᛕ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᔓᙀᐞᘜᛋᛋᓚ⊙◌ϴiᒧᛯ·⁚.·˙grimchain⁚·
·⁚.·˙lookup˙·
·⁚.·˙glyphs˙·

·⁚glyph symbol◦–◦𐑌» category◦–◦court»⁚·
·⁚lookup office◦–◦recursive…identity» enoch◦–◦⊕»⁚·

‹word⁚·ᛋᚴᛁᛒᚣ›⁚.·ˊword⁚·

·⁚origin…language⁚·Gothic·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·ˊfont⁚·
·⁚pronunciation⁚·LEYEBA·⁚.·ˊpronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·ˊcadence⁚·
·⁚non…ostensive⁚·Remainder⁖ what is left after the greater body::event::or taking has passed
↪ away·⁚.·ˊnon…ostensive⁚·
·⁚ostensive⁚·One fragment remains after the larger body is gone⁖ the small surviving piece occupies
↪ the place left behind𐑌·⁚.·ˊostensive⁚·
·⁚grimchain⁚·𐑅𐑙𐑂𐑆𐑈𐑃𐑉𐑇𐑅𐑇·⁚.·ˊgrimchain⁚·
·⁚.·ˊlookup⁖ˊ
·⁚.·ˊglyphs⁖ˊ
·⁚glyph symbol◦–◦𐑘» category◦–◦court»⁚·
·⁚lookup office◦–◦recursive…identity» enoch◦–◦⊕»⁚·

‹word⁚·𐎠𐎼𐏁 𐎧›⁚.·ˊword⁚·

·⁚origin…language⁚·Old Persian·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚.·ˊfont⁚·
·⁚pronunciation⁚·KAM+nah·⁚.·ˊpronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·ˊcadence⁚·
·⁚non…ostensive⁚·Few or little⁖ small in number or amount·⁚.·ˊnon…ostensive⁚·
·⁚ostensive⁚·Three seeds lie alone in the broad bowl⁖ empty surface surrounds the small
↪ count𐑘·⁚.·ˊostensive⁚·
·⁚grimchain⁚·𐑋𐑓𐑐𐑉𐑔𐑢𐑤𐑜𐑝𐑚·⁚.·ˊgrimchain⁚·
·⁚.·ˊlookup⁖ˊ
·⁚.·ˊglyphs⁖ˊ
·⁚glyph symbol◦–◦𐑣» category◦–◦court»⁚·
·⁚lookup office◦–◦recursive…identity» enoch◦–◦⊕»⁚·

‹word⁚·ɯнноєᴇ›⁚.·ˊword⁚·

·⁚origin…language⁚·Coptic·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·ˊfont⁚·
·⁚pronunciation⁚·MAY+eh+sheh·⁚.·ˊpronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·ˊcadence⁚·
·⁚non…ostensive⁚·Multitude::crowd::or throng·⁚.·ˊnon…ostensive⁚·
·⁚ostensive⁚·Bodies press shoulder to shoulder until scarcely any open ground survives inside the
↪ gathering𐑣·⁚.·ˊostensive⁚·
·⁚grimchain⁚·𐑦𐑮𐑢𐑤𐑞𐑟𐑠𐑡𐑝𐑛𐑭·⁚.·ˊgrimchain⁚·
·⁚.·ˊlookup⁖ˊ
·⁚.·ˊglyphs⁖ˊ
·⁚glyph symbol◦–◦𐑍» category◦–◦court»⁚·
·⁚lookup office◦–◦recursive…identity» enoch◦–◦⊕»⁚·

‹word⁚·ᴛᴨʌᴦᴨs›⁚.·ˊword⁚·

·⁚origin…language⁚·Gothic·⁚.·ˊorigin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·ˊfont⁚·
·⁚pronunciation⁚·TULGUS·⁚.·ˊpronunciation⁚·

·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Steadfast::resistant::firm: holding shape or standing under force rather than readily
↪ yielding·⁖.·˙non…ostensive:·
·⁖ostensive:·More weight settles onto the post::but its line holds and the body refuses to yield
↪ farther beneath the pressureƵ·⁖.·˙ostensive:·
·⁖grimchain:·ⵊᵯzꟈ⊕ᖘᚸ₀ᴓᛋΘⓄᘐ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꞊꞊ɛ꞉ category꞊꞊ᵋcourt꞉:·
·⁖lookup office꞊꞊ᵋrecursive…identity꞉ enoch꞊꞊ᵋ⊕꞉:·

˓word:·क्षुभ्˓:·.·˙word:·

·⁖origin…language:·Sanskrit·⁖.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖.·˙font:·
·⁖pronunciation:·kshbh·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Shake::tremble::be agitated or disturbed⁖· movement that breaks a prior
↪ stillness·⁖.·˙non…ostensive:·
·⁖ostensive:·Still limbs begin to quiver::the small motion repeating through the body before stillness
↪ can returnɛ·⁖.·˙ostensive:·
·⁖grimchain:·▽ᵻ❀ᘓ꠸ᘐ✩ϛ⋐ᑎ❂ᑎ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꞊꞊I꞉ category꞊꞊ᵋcourt꞉:·
·⁖lookup office꞊꞊ᵋrecursive…identity꞉ enoch꞊꞊ᵋ⊕꞉:·

˓word:·𒁹𒄑𒉡𒀭˓·⁖.·˙word:·

·⁖origin…language:·Akkadian·⁖.·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖.·˙font:·
·⁖pronunciation:·KAH+noom·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Stay::remain stationary::endure::or remain steadfast and secure·⁖.·˙non…ostensive:·
·⁖ostensive:·Rain passes::feet pass::shadow passes⁖· the stone stays in the same hollow where it had
↪ already settledI·⁖.·˙ostensive:·
·⁖grimchain:·ᚲᚌℂₒᑭᛊᓐᚢᛟᛒᗡᓫ⬡Ƶ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꞊꞊ᗰ꞉ category꞊꞊ᵋcourt꞉:·
·⁖lookup office꞊꞊ᵋrecursive…identity꞉ enoch꞊꞊ᵋ⊕꞉:·

˓word:·मुख˓:·.·˙word:·

·⁖origin…language:·Sanskrit·⁖.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖.·˙font:·
·⁖pronunciation:·MOO+khuh·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Mouth::face::opening::front·⁖.·˙non…ostensive:·
·⁖ostensive:·Lips frame one living opening at the front of the face⁖· breath::voice::food::and air cross
↪ the mouth openingᗰ·⁖.·˙ostensive:·
·⁖grimchain:·ᐟᔕᓚᒧᐅᛋ𒁹𒄑ᚔᚙᛁᚱᗣ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙

·⁚glyph symbol꞉꞉𑖌𑖨 category꞉꞉court꞉·
·⁚lookup office꞉꞉recursive…identity꞉ enoch꞉꞉⊕꞉·

꞉word꞉·ध्वनि꞉·˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚·˙font꞉·
·⁚pronunciation꞉·dhvan·⁚·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Sound꞉꞉echo꞉꞉tone꞉꞉voice꞉꞉thunder⁚· a heard body that may return after its source has
↪ spoken·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·One vibration crosses the air꞉꞉strikes stone꞉꞉and returns through the same space as audible
↪ sound𑖌·⁚·˙ostensive꞉·
·⁚grimchain꞉·≀≋ℂ𐤓ᵾ❋𐤓1⅁Θᚠᚹ♀·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉꞉𑖌𑖗 category꞉꞉court꞉·
·⁚lookup office꞉꞉recursive…identity꞉ enoch꞉꞉⊕꞉·

꞉word꞉·ᚱᛁᚢᚴᚱ꞉·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚·˙font꞉·
·⁚pronunciation꞉·RAYK+r·⁚·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Smoke·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Gray heated particles rise and curl above the burning body⁚· the visible plume drifts
↪ through the air and thins outward𑖗·⁚·˙ostensive꞉·
·⁚grimchain꞉·𑖗ᚺ❀᪥✳ᛁᛩᛨᛋ≀◌·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉꞉𑖌𑖫 category꞉꞉court꞉·
·⁚lookup office꞉꞉recursive…identity꞉ enoch꞉꞉⊕꞉·

꞉word꞉·ᛃᚴᚱᚢᚾ꞉·˙word꞉·

·⁚origin…language꞉·Avestan·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Avestan·⁚·˙font꞉·
·⁚pronunciation꞉·VAH+yoo·⁚·˙pronunciation꞉·
·⁚cadence꞉·Recursive Identity·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Wind or air: the unseen moving gaseous body surrounding and passing through the
↪ world·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Unseen air crosses open ground꞉꞉bends grass꞉꞉and presses against exposed skin without
↪ taking a fixed shape𑖫·⁚·˙ostensive꞉·
·⁚grimchain꞉·ꞀℂᒧⅡᚴᚱ⊙ᒧᒪᛃᚹᚸᘐ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉꞉◌𐑂 category꞉꞉court꞉·
·⁚lookup office꞉꞉recursive…identity꞉ enoch꞉꞉⊕꞉·

꞉word꞉·ꞩᛦᛖᚢᚾ꞉·˙word꞉·

·⁚origin…language꞉·Avestan·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Avestan·⁚·˙font꞉·

·⁚pronunciation⁚·PAHNS⟊noo·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Dust·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Fine dry grains lie loose enough for the passing air to lift them⁖⁖each particle distinct
↪ inside the pale moving field◌·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⧗𝌆▽ꕤ𝙆ᔕΛ↿‡ᙏᔑᛋ✵↗·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘˙
·⁚.·˙glyphs⁘˙
·⁚glyph symbol◦–◦𐑓 category◦–◦court𐑕⁚·
·⁚lookup office◦–◦recursive…identity𐑕 enoch◦–◦⊕𐑕⁚·

⁖word⁚·ᑎSʏΛᒥSTS⁖⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·USWAHSTS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Increase⁖⁖growth·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·New leaf rises beyond the old stem⁘ the plant occupies more height and breadth than the
↪ growth beneath it𐑓·⁚.·˙ostensive⁚·
·⁚grimchain⁚·0𝌆⋗ɾ♂⁕0ᖻᛁᛋ𐑘𐑔𐑕ᔕ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘˙
·⁚.·˙glyphs⁘˙
·⁚glyph symbol◦–◦𐑖𐑕 category◦–◦court𐑕⁚·
·⁚lookup office◦–◦recursive…identity𐑕 enoch◦–◦⊕𐑕⁚·

⁖word⁚·वाक्⁖⁚.·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚.·˙font⁚·
·⁚pronunciation⁚·VAAK·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Speech·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Breath crosses tongue and lips as meaningful sound⁖⁖leaving the body already formed into
↪ speech𐑖·⁚.·˙ostensive⁚·
·⁚grimchain⁚·▫ꕤ⍈♂ᛋ⯇ↂ⏢𝌆⋗⯇ᛘ◎ƶ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘˙
·⁚.·˙glyphs⁘˙
·⁚glyph symbol◦–◦𐑜𐑕 category◦–◦court𐑕⁚·
·⁚lookup office◦–◦recursive…identity𐑕 enoch◦–◦⊕𐑕⁚·

⁖word⁚·▦⁖⁚.·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·LEEL·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Wind or air⁘ open air⁘ nothingness or phantom·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Moving air circles the open field around an empty hollow⁖⁖unseen presence filling the
↪ space without giving it a solid center𐑜·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ϗᐁᑐᙏ⊛ᖯ𝌆ᛋ¥ᗤʃᐁ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁘˙
·⁚.·˙glyphs⁘˙

·⁚glyph symbol꩜꩜𐍈꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

꞉word⁚·ᚦᚢᚾᚱ꞉˙·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚˙·˙font⁚·
·⁚pronunciation⁚·THUNNR·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·Thin·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·A blade of metal lies between two fingers⁖ almost no depth separates the two opposing
↪ faces of the blade𐍈·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·𓃹𐌢ↄᚸ𐌽ᛞᛇ⬡𐌅⊕·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁚˙
·⁚˙·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐌅꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

꞉word⁚·ᚦᚢᚸᚱ꞉˙·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚˙·˙font⁚·
·⁚pronunciation⁚·THYKKR·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·Thick⁖ having substantial depth꞉꞉density꞉꞉or material between opposing
↪ surfaces·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·A cut block shows a deep sidewall between front and back faces⁖ material fills the whole
↪ span𐌅·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·ᛃΘᚸᚦᔑ☆ᑫᚏᛯᚴᛊ⁺·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁚˙
·⁚˙·˙glyphs⁚˙
·⁚glyph symbol꩜꩜𐌅꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

꞉word⁚·ᛏĪ□ᛪ𐌅̄꞉˙·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚˙·˙font⁚·
·⁚pronunciation⁚·KRUHMP·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·Be disturbed or agitated⁖ lose still order under unrest꞉꞉pressure꞉꞉or competing
↪ motion·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·Currents cross and unsettle one another꞉꞉each new motion interrupting the last before the
↪ field can become still𐌅·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·ᚸᚸᚹᚴ⬡ᴢᕫ◎𐍈ᚱᛝᚱⴼ·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁚˙
·⁚˙·˙glyphs⁚˙
·⁚glyph symbol꩜꩜ᘐ꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝⁚·

꞉word⁚·ﻫﻣﻣﻣﻫ꞉˙·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚˙·˙font⁚·

·⁖pronunciation⁚·aKruSH·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Peaceful⁝⁝at ease⁝⁝reserved⁝⁝concentrated: a settled condition without strain or
↪ agitation·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Breath evens⁝⁝muscles loosen⁝⁝attention gathers⁝⁝and the whole body settles without a pull
↪ of strain𐌅·⁖.·˙ostensive⁚·
·⁖grimchain⁚·⟛⟡⊢∩ᖶ○ᛘᚴ❄ᛗ✡❀ℤᛉ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol⟜○⟜ᛯ» category⟜○⟜court»⁚·
·⁖lookup office⟜○⟜recursive…identity» enoch⟜○⟜⊕»⁚·

·word⁚·वाज·⁖.·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁖.·˙font⁚·
·⁖pronunciation⁚·VAA⟊juh·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Strength⁝⁝vigour⁝⁝energy⁝⁝speed⁝⁝contest⁝⁝or battle·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Bent limb and taut tendon hold force before release⁖˙ the body carries power even while
↪ still᛭·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᛔᛒᛏᕰᚏᐅᛜᛞᛁᑭǂ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol⟜○⟜ᐃ» category⟜○⟜court»⁚·
·⁖lookup office⟜○⟜recursive…identity» enoch⟜○⟜⊕»⁚·

·word⁚·ᛋᛁᚾᚾ·⁖.·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖.·˙font⁚·
·⁖pronunciation⁚·MINNI·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Memory·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Old melody returns at the first familiar interval⁖˙ the later notes arrive inwardly before
↪ they are soundedᐃ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᛯᗅᚠᘉ|◺ᐟᛩᚱᛯ✙▫✡z·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol⟜○⟜∀» category⟜○⟜court»⁚·
·⁖lookup office⟜○⟜recursive…identity» enoch⟜○⟜⊕»⁚·

·word⁚·𐎍𐎗𐎚𐎐·⁖.·˙word⁚·

·⁖origin…language⁚·Ugaritic·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Ugaritic·⁖.·˙font⁚·
·⁖pronunciation⁚·SHVMKHATU·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Joy·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Breath opens⁝⁝pulse rises lightly⁝⁝and the chest loosens into a bright ease with nothing
↪ inward pressing it closed∀·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᗅ⇞'ᚸᚱᙏᘐᛗᚿᛉᛯᛋᚢ»·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙

·⁚glyph symbol꩜꩜∇꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜recursive…identity꩷ enoch꩜꩜⊕꩷⁚·

⁖word⁚·쓺⁖.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·skúm·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Dream⁚· an inward experience of images⁚⁚places⁚⁚beings⁚⁚or events during
↪ sleep·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Behind closed eyes⁚⁚sleep opens places and figures that move with full inward presence
↪ while the outer room remains absent∇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·◎ԋ⍜⁎✙⍜ϟዘ𝌆⩡ᵹ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol꩜꩜∇꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜recursive…identity꩷ enoch꩜꩜⊕꩷⁚·

⁖word⁚·ʏϵᴦs⁖.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·WEGS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Wave or undulation⁚· repeated rising and falling movement carried through a body or
↪ field·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·A crest rises⁚⁚folds into a hollow⁚⁚and rises again farther on⁚⁚the same undulation
↪ traveling through the surface∇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⬡⋤ㄏ𝌆⊕ᛁᛁ𝌆ǂᵹᑌ⍜❄ᦓ⍜·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol꩜꩜∇ᖇ꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜recursive…identity꩷ enoch꩜꩜⊕꩷⁚·

⁖word⁚·ᛋ⁞̇⁖.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·KAHROON·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Compassion⁚⁚pity: inward fellow⁺feeling that answers another being’s suffering with
↪ care·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Another body bends under hurt⁚· answering tightness gathers in the chest and care moves
↪ toward the suffering without sharing the wound∇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·✪𝌆ᑎ▫ᑭ⊖ᖶᘐᛘᘐᑎᚗ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol꩜꩜⁂꩷ category꩜꩜court꩷⁚·
·⁚lookup office꩜꩜recursive…identity꩷ enoch꩜꩜⊕꩷⁚·

⁖word⁚·ᒧ▯ᒍ̄ᒍᒣ̄⁖.·˙word⁚·

·⁖origin…language꞉·Tocharian B·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Brahmi·⁖·˙font꞉·
·⁖pronunciation꞉·YUHLLOHNY·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Sense⁖· a way or faculty of perceiving by which a being receives the
↪ world·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Light enters the eye⁖⁖pressure enters the skin⁖⁖vibration enters the ear⁖· each world
↪ arrives through its own living gate𐤓·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐤊ℂ𐤚𐤌𐤏𐤑𐤅𐤉𐤒𐤇𐤄𐤀·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉𐤅꞉ category꞉꞉court꞉꞉
·⁖lookup office꞉꞉recursive…identity꞉ enoch꞉꞉𐤈꞉꞉

꞉word꞉·ﻣﺪﻫﺸﺖ꞉⁖·˙word꞉·

·⁖origin…language꞉·Old Uyghur·⁖·˙origin…language꞉·
·⁖font꞉·Noto Serif Old Uyghur·⁖·˙font꞉·
·⁖pronunciation꞉·aGHrUHGH·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Pain⁖⁖ache⁖⁖twinge⁖⁖illness⁖⁖or ailment: bodily or inward hurt that persists or
↪ returns·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Ache remains in the living tissue whether the body moves or rests⁖⁖returning to the same
↪ inward place instead of fading away𐤅·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐤎𐤍𐤙𐤓𐤋⊙∓𐤔𐤛𐤋𐤓𐤎·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉𐤅꞉ category꞉꞉court꞉꞉
·⁖lookup office꞉꞉recursive…identity꞉ enoch꞉꞉𐤈꞉꞉

꞉word꞉·𐤋꞉·˙word꞉·

·⁖origin…language꞉·Sumerian·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖·˙font꞉·
·⁖pronunciation꞉·EN+goor·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Depths⁖· cosmic underground waters·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Black water continues beneath every reached measure⁖⁖the lower body hidden below the last
↪ depth that can be touched𐤅·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐤄𐤀𐤓𐤉𐤓𐤈𐤃𐤅𐤇𐤊𐤏𐤎·⁖·˙grimchain꞉·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉𐤊꞉ category꞉꞉court꞉꞉
·⁖lookup office꞉꞉recursive…identity꞉ enoch꞉꞉𐤈꞉꞉

꞉word꞉·SᏁᎷᏁᏌIISS꞉⁖·˙word꞉·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·SAMAQISS·⁖·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Consensus⁖⁖agreement⁖⁖concurrence: several standings arriving at one
↪ accord·⁖·˙non…ostensive꞉·

·⁖ostensive⁚·Separate hands lower their marks onto the same line⁘ no mark remains outside the shared
↪ position⁜·⁖·˙ostensive⁚·
·⁖grimchain⁚·'𐤚ᵹzϒ𝌆𐦏𝌮ϒ𐤛𐤟⏁Ƀ·⁖·˙grimchain⁚·
·⁖·˙lookup⁘˙
·⁖·˙glyphs⁘˙
·⁖glyph symbol⟜⟜𐤟, category⟜⟜court⁖·
·⁖lookup office⟜⟜recursive…identity, enoch⟜⟜⊕⁖·

·word⁚·ɓbƧዛ·⁖·˙word⁚·

·⁖origin…language⁚·Aghwan·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Caucasian Albanian·⁖·˙font⁚·
·⁖pronunciation⁚·BAKH·⁖·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Judgement·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Unequal claims meet the same measure⁘ after their weighing⁚⁚one determination stands and
↪ the contest no longer remains open𐤟·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᛈᚴᛋᛏ♳ᛜᚻᛘᛉIᚲ𐦟·⁖·˙grimchain⁚·
·⁖·˙lookup⁘˙
·⁖·˙glyphs⁘˙
·⁖glyph symbol⟜⟜𐤠, category⟜⟜court⁖·
·⁖lookup office⟜⟜recursive…identity, enoch⟜⟜⊕⁖·

·word⁚·ϡ+ϫ+ϩϧ·⁖·˙word⁚·

·⁖origin…language⁚·Tocharian B·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Brahmi·⁖·˙font⁚·
·⁖pronunciation⁚·AHKTEHKAHNYNYEH·⁖·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖·˙cadence⁚·
·⁖non…ostensive⁚·Wonder⁘ astonishment before what exceeds ordinary expectation or
↪ understanding·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Eyes widen and breath pauses before a form beyond familiar measure⁚⁚ordinary expectation
↪ giving way beneath the encountered scale𐤠·⁖·˙ostensive⁚·
·⁖grimchain⁚·ϫ𐤞ᛎᚸᗅ𐤛ᛎᛎᚺᛯ𐤠·⁖·˙grimchain⁚·
·⁖·˙lookup⁘˙
·⁖·˙glyphs⁘˙
·⁖glyph symbol⟜⟜𐤡, category⟜⟜court⁖·
·⁖lookup office⟜⟜recursive…identity, enoch⟜⟜⊕⁖·

·word⁚·शून्य·⁖·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁖·˙font⁚·
·⁖pronunciation⁚·shoon+juh·⁖·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖·˙cadence⁚·
·⁖non…ostensive⁚·blank⁘ deserted⁘ empty⁘ void⁘ destitute of·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Open chamber contains no tool⁚⁚grain⁚⁚body⁚⁚or mark⁘ only unoccupied space remains from wall
↪ to wall𐤡·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᚺᚢᚺ𐦏ᔓᛟ𐦏ᛞᛯ✳ᙆ·⁖·˙grimchain⁚·
·⁖·˙lookup⁘˙
·⁖·˙glyphs⁘˙
·⁖glyph symbol⟜⟜✳, category⟜⟜court⁖·
·⁖lookup office⟜⟜recursive…identity, enoch⟜⟜⊕⁖·

:word:·बिन्दु:.·˙word:·

·∴origin…language:·Sanskrit·∴.·˙origin…language:·
·∴font:·Noto Sans Devanagari·∴.·˙font:·
·∴pronunciation:·bin+du·∴.·˙pronunciation:·
·∴cadence:·Recursive Identity·∴.·˙cadence:·
·∴non…ostensive:·a detached particle::drop::globule::dot::spot∴ a pearl∴ a drop of water taken as a
↪ measure∴ a spot or mark of coloured paint on the body of an elephant∴ (at the end of a compound
↪ also बिन्दुक (binduka)) the dot over a letter representing the anusvara (supposed to be connected
↪ with शिव (śiva) and of great mystical importance)∴ a zero or cypher (in manuscripts put over an
↪ erased word to show that it ought not to be erased = ؛stet؛)∴ a particular mark like a dot made
↪ in cauterizing∴ a mark made by the teeth of a lover on the lips of his mistress∴ a coloured mark
↪ made on the forehead between the eyebrows∴ the sudden development of a secondary incident
↪ (which::like a drop of oil in water::expands and furnishes an important element in the plot)∴ name
↪ of a man g::बदिादि (bidā+di)∴ name of an आङ्गरिस (āṅgirasa) (author of RV::883 and 930)∴ name of the
↪ author of a रसपद्धति (rasa+paddhati)∴ name of a warrior tribe g::दामन्यादि
↪ (dāmany+ādi)·∴.·˙non…ostensive:·
·∴ostensive:·A single rounded drop rests apart from the surrounding surface::its tiny skin closing one
↪ whole body upon itself✷·∴.·˙ostensive:·
·∴grimchain:·𐎁𐎍𐎐𐎄𐎗𐎂𐎅𐎆𐎋𐎃·∴.·˙grimchain:·
·∴.·˙lookup·˙
·∴.·˙glyphs·˙
·∴glyph symbol○-○؛؛ category○-○؛court؛:·
·∴lookup office○-○؛recursive…identity؛ enoch○-○؛⊕؛:·

:word:·𐎃𐎆:.·˙word:·

·∴origin…language:·Old Turkic·∴.·˙origin…language:·
·∴font:·Noto Sans Old Turkic·∴.·˙font:·
·∴pronunciation:·YAHNY·∴.·˙pronunciation:·
·∴cadence:·Recursive Identity·∴.·˙cadence:·
·∴non…ostensive:·To scatter::put to rout·∴.·˙non…ostensive:·
·∴ostensive:·Birds burst from one tree into many directions∴ distance opens between the scattered
↪ bodies؛·∴.·˙ostensive:·
·∴grimchain:·𐎁𐎐𐎍⬡𐎍𐎂𐎅𐎆𐎋𐎃·∴.·˙grimchain:·
·∴.·˙lookup·˙
·∴.·˙glyphs·˙
·∴glyph symbol○-○؛؛ category○-○؛court؛:·
·∴lookup office○-○؛recursive…identity؛ enoch○-○؛⊕؛:·

:word:·𐎍𐎆𐎁·:.·˙word:·

·∴origin…language:·Ugaritic·∴.·˙origin…language:·
·∴font:·Noto Sans Ugaritic·∴.·˙font:·
·∴pronunciation:·TIHAHMU·∴.·˙pronunciation:·
·∴cadence:·Recursive Identity·∴.·˙cadence:·
·∴non…ostensive:·Abyss or ocean: the great deep whose extent or depth exceeds ordinary
↪ reach·∴.·˙non…ostensive:·
·∴ostensive:·Below the last visible light::dark water continues without showing any bottom beneath the
↪ descent؛·∴.·˙ostensive:·
·∴grimchain:·𐎐𐎄𐎗⬡◎𐎅𐎁𐎊𐎀𐎌𐎍·∴.·˙grimchain:·
·∴.·˙lookup·˙
·∴.·˙glyphs·˙

·⁚glyph symbol꩜𒉌꩜꩜꩜

·⁚pronunciation:·SEE⟊leem·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Heal⁚· to become healthy::whole::safe::or restored from injury or
↪ illness·⁚.·˙non…ostensive:·
·⁚ostensive:·Open wound narrows beneath new tissue⁚· broken edges join and living surface returns
↪ across the gap𐑦·⁚.·˙ostensive:·
·⁚grimchain:·ↄ𐑓ᚨ※ᑎᚣᛀ𐑤𐑦ΘᓭᏋ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐑦» category○–○courtː·
·⁚lookup office○–○recursive…identity» enoch○–○⊕ː·

·word:·ᔕᚱᚣᚴᚣᚿ·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·DRAGAN·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Draw::pull::carry: exert force so a body moves toward::with::or behind the
↪ bearer·⁚.·˙non…ostensive:·
·⁚ostensive:·Tension runs through the rope into the load::drawing the burden toward the pulling body as
↪ both move under the same strain𐑦·⁚.·˙ostensive:·
·⁚grimchain:·ɾȳᑎ✱ϒ𐑤ᛕ◌𐑦ᛃᛟᐁ⃤·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐑤» category○–○courtː·
·⁚lookup office○–○recursive…identity» enoch○–○⊕ː·

·word:·coո·⁚.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·sop·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Occasion::time⁚· turn::round·⁚.·˙non…ostensive:·
·⁚ostensive:·Among repeating turns::one marked position comes round again::returning as its own occasion
↪ within the cycle𐑤·⁚.·˙ostensive:·
·⁚grimchain:·◝ᵒ%♂ᔑᚿᚹᛦᛞᚿ⫩ᛦᐁϘϙZ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐑢» category○–○courtː·
·⁚lookup office○–○recursive…identity» enoch○–○⊕ː·

·word:·ᛦᛋᛏᛋ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·GOWTA·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Riddle·⁚.·˙non…ostensive:·
·⁚ostensive:·Knotted signs cover the tablet⁚· one hidden relation binds the knotted signs while the
↪ answer remains concealed𐑢·⁚.·˙ostensive:·
·⁚grimchain:·ᛗᛏ1Θᛦǂᚽ❄ᛇ◝ᵒ%Θᓗᔓᛯ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑂꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝:·

:word:·बीज:·⁚·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·BEE+juh·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Seed:˙ the reproductive body from which a plant grows·⁚·˙non…ostensive:·
·⁚ostensive:·Root and shoot lie folded beside stored nourishment inside the small reproductive
↪ body::whole growth waiting there in compact form𐑂·⁚·˙ostensive:·
·⁚grimchain:·ᗺ❂❀𐑂ᚷᚸᚴᚿᚢᔑᐁᛏ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑥꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝:·

:word:·𒂗𒄑:·⁚·˙word:·

·⁚origin…language:·Sumerian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·NGISH+shoo·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Shade or shadow:˙ also protection·⁚·˙non…ostensive:·
·⁚ostensive:·A covering body interrupts the light:˙ cool darkness lies beneath the covering body while
↪ exposed ground remains bright𐑥·⁚·˙ostensive:·
·⁚grimchain:·𐑣𐑓𐑖𐑐ᛟᚯᚷ𓂋ᚹᚦᔑ𐑓·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑙꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝:·

:word:·तमस्:·⁚·˙word:·

·⁚origin…language:·Sanskrit·⁚·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚·˙font:·
·⁚pronunciation:·tuh+muhs·⁚·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚·˙cadence:·
·⁚non…ostensive:·Darkness::gloom::obscurity: absence or veiling of light by which forms become difficult
↪ to distinguish·⁚·˙non…ostensive:·
·⁚ostensive:·Edges vanish where light is absent:˙ forms remain present but cannot be distinguished
↪ within the dark field𐑙·⁚·˙ostensive:·
·⁚grimchain:·𐑖∫ᛕᚣ𐑂❀Aᛉᛜᛃᛏ✷ᛜ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐑮꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜recursive…identity꩝ enoch꩜꩜⊕꩝:·

:word:·𐬯𐬛𐬀𐬵𐬞𐬛:·⁚·˙word:·

·⁚origin…language:·Avestan·⁚·˙origin…language:·
·⁚font:·Noto Sans Avestan·⁚·˙font:·

·⁚pronunciation:·sah⟊RUH⟊tah·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Cold·⁚.·˙non…ostensive:·
·⁚ostensive:·Touch meets a surface emptied of warmth::heat leaving whatever body remains against
↪ it৩·⁚.·˙ostensive:·
·⁚grimchain:·ᚴᚿᚼᛦᛋᛇᚳᛂᛞ◌◌⊐ᛈ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ᛃᛎ, category○–○ᛃcourt⁑:·
·⁚lookup office○–○ᛃrecursive…identity, enoch○–○ᛃ⊕⁑:·

·word:·ᚡᚱᛁᛊᚣᚺᛏᛊ·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·FRISAHTS·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Image::representation::effigy::example: a formed likeness by which another body or idea
↪ is made present·⁚.·˙non…ostensive:·
·⁚ostensive:·Polished clay bears the exact visible form pressed into it·˙ impressed likeness remains
↪ after the original face is withdrawnᛎ·⁚.·˙ostensive:·
·⁚grimchain:·𓃒⊙𓃒ᛎ∀☽𑁍𑁍𑁍𑁍ᚼᛃᛃ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ᛃᛝ, category○–○ᛃcourt⁑:·
·⁚lookup office○–○ᛃrecursive…identity, enoch○–○ᛃ⊕⁑:·

·word:·𐎗𐎐𐎚𐎚·⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·YRD·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·To descend::go down::come down·⁚.·˙non…ostensive:·
·⁚ostensive:·Body leaves the high ledge and occupies ground below·˙ the course continues downward until
↪ the lower place receives itᛝ·⁚.·˙ostensive:·
·⁚grimchain:·ᚺᚻ'ᙆᛋᛯᛈᘁᛀᛇᛒᚳᚣ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ᛃᚢ, category○–○ᛃcourt⁑:·
·⁚lookup office○–○ᛃrecursive…identity, enoch○–○ᛃ⊕⁑:·

·word:·⊐ᛘ·⁚.·˙word:·

·⁚origin…language:·Sumerian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·GOOR·⁚.·˙pronunciation:·
·⁚cadence:·Recursive Identity·⁚.·˙cadence:·
·⁚non…ostensive:·Turn·˙ to rotate::bend::twist::or change direction·⁚.·˙non…ostensive:·
·⁚ostensive:·One moving line leaves its former bearing and follows a new direction around the
↪ bendᚢ·⁚.·˙ostensive:·
·⁚grimchain:·ᛜ⓿ↄ❄ᛮᚿᛇXᛋᛀ⊛ᚋᛀ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁖glyph symbol꩜𓃰꩜꩜꩜

·⁖cadence:·Recursive Identity·⁖·˙cadence:·
·⁖non…ostensive:·Choose⁚⁚select⁚⁚pick: distinguish one possibility or body from others and take it as
↪ the chosen one·⁖·˙non…ostensive:·
·⁖ostensive:·Several stones lie in the dust⁘ one is lifted and carried while the others remain
↪ untouched𓃰·⁖·˙ostensive:·
·⁖grimchain:·𐦝𐦞𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꞉꞉𓃰 category꞉꞉court꞉·
·⁖lookup office꞉꞉recursive…identity꞉ enoch꞉꞉⊕꞉·

·word:·ᛞᛁᚨ·⁖·˙word:·

·⁖origin…language:·Old Norse·⁖·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖·˙font:·
·⁖pronunciation:·FELA·⁖·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖·˙cadence:·
·⁖non…ostensive:·Hide⁚⁚conceal: keep a thing present while removing it from ordinary sight or
↪ finding·⁖·˙non…ostensive:·
·⁖ostensive:·A thing remains present while ordinary sight or finding cannot reach it⁚⁚concealed without
↪ ceasing to exist𓃰·⁖·˙ostensive:·
·⁖grimchain:·𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꞉꞉𓄿 category꞉꞉court꞉·
·⁖lookup office꞉꞉recursive…identity꞉ enoch꞉꞉⊕꞉·

·word:·𒀀·⁖·˙word:·

·⁖origin…language:·Sumerian·⁖·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖·˙font:·
·⁖pronunciation:·KOOD·⁖·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖·˙cadence:·
·⁖non…ostensive:·Cut⁘ to sever⁚⁚divide⁚⁚or remove by a sharp edge·⁖·˙non…ostensive:·
·⁖ostensive:·Sharp edge enters one continuous body⁘ two separate sides remain behind the passing
↪ cut𓄿·⁖·˙ostensive:·
·⁖grimchain:·𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol꞉꞉A꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉recursive…identity꞉ enoch꞉꞉⊕꞉·

·word:·स्वर·⁖·˙word:·

·⁖origin…language:·Sanskrit·⁖·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖·˙font:·
·⁖pronunciation:·SVUH⁄ruh·⁖·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖·˙cadence:·
·⁖non…ostensive:·Sound⁚⁚noise⁚⁚or voice·⁖·˙non…ostensive:·
·⁖ostensive:·Air carries vibration from source to ear⁘ audible pressure arrives as noise or voice
↪ without visible bodyA·⁖·˙ostensive:·
·⁖grimchain:·𐦀𐦁𐦂𐦃𐦄𐦅𐦆𐦇𐦈𐦉·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙

·⁖glyph symbol◦–◦𐬃꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦recursive…identity꞉ enoch◦–◦⊕꞉⁖·

⁘word⁖·𐬀𐬵𐬀𐬴⁘⁖·˙word⁖·

·⁖origin…language⁖·Old Persian·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Old Persian·⁖·˙font⁖·
·⁖pronunciation⁖·ah+NEE+yah·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·Other⁖· different from the one already indicated·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Side by side⁖⁖the two bodies refuse sameness⁖⁖each carrying features the other does not
↪ possess𐬃·⁖·˙ostensive⁖·
·⁖grimchain⁖·صو،سزو❂𐬃∀ሐ∇ᚱᛃ،ᛃ·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol◦–◦Ↄ꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦recursive…identity꞉ enoch◦–◦⊕꞉⁖·

⁘word⁖·ዘሠሣᔕ⁘⁖·˙word⁖·

·⁖origin…language⁖·Old Turkic·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Old Turkic·⁖·˙font⁖·
·⁖pronunciation⁖·KAHZGHAHN·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·To earn⁖⁖conquer⁖⁖win·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Empty hand enters the contest and leaves holding the prize that had not been possessed
↪ beforeↃ·⁖·˙ostensive⁖·
·⁖grimchain⁖·𐓏∮⊙ϫ❄ᚦᛦᚠᚠ1☆✳⊖·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol◦–◦ﾚ꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦recursive…identity꞉ enoch◦–◦⊕꞉⁖·

⁘word⁖·ᴏϒƸɪ⁘⁖·˙word⁖·

·⁖origin…language⁖·Coptic·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Coptic·⁖·˙font⁖·
·⁖pronunciation⁖·WHY·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·
·⁖non…ostensive⁖·One⁖· a single whole or unit counted without another beside it·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·One bead lies alone in the counting hollow⁖· every surrounding hollow remains
↪ emptyﾚ·⁖·˙ostensive⁖·
·⁖grimchain⁖·⍙❂☽⊄⊞⊛ᔕ✳⌬«∮♂ỏ𐬹·⁖·˙grimchain⁖·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol◦–◦ﾇ꞉ category◦–◦court꞉⁖·
·⁖lookup office◦–◦recursive…identity꞉ enoch◦–◦⊕꞉⁖·

⁘word⁖·𨳊⁘⁖·˙word⁖·

·⁖origin…language⁖·Tangut·⁖·˙origin…language⁖·
·⁖font⁖·Noto Serif Tangut·⁖·˙font⁖·
·⁖pronunciation⁖·SAH·⁖·˙pronunciation⁖·
·⁖cadence⁖·Recursive Identity·⁖·˙cadence⁖·

·⁖non…ostensive⁚·Startled⁖· suddenly shocked or alarmed·⁚˙·non…ostensive⁚·
·⁖ostensive⁚·A sudden crash snaps the body upright⁖· breath catches and skin tightens before stillness returns↯·⁚˙·ostensive⁚·
·⁖grimchain⁚·⏃◎ᵷΨ℧ᑫᚣHI※ᘓᵷ⁂Ƶ·⁚˙·grimchain⁚·
·⁚˙·lookup·˙
·⁚˙·glyphs·˙
·⁖glyph symbol⟜⟜ᛏ» category⟜⟜court»⁚·
·⁖lookup office⟜⟜recursive…identity» enoch⟜⟜⊕»⁚·

⁖word⁚·ज्या⁖·⁚˙·word⁚·

·⁖origin…language⁚·Sanskrit·⁚˙·origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁚˙·font⁚·
·⁖pronunciation⁚·jjah·⁚˙·pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁚˙·cadence⁚·
·⁖non…ostensive⁚·a bowstring⁖· sine·⁚˙·non…ostensive⁚·
·⁖ostensive⁚·A single taut string spans the bent limbs⁝⁝carrying the held tension that keeps the bow drawn between themᛏ·⁚˙·ostensive⁚·
·⁖grimchain⁚·ᚐᔑ◎ᑫᓚᦓ‹ᐱᗺᛘ✶ᛘᛉ·⁚˙·grimchain⁚·
·⁚˙·lookup·˙
·⁚˙·glyphs·˙
·⁖glyph symbol⟜⟜ᛁ» category⟜⟜court»⁚·
·⁖lookup office⟜⟜recursive…identity» enoch⟜⟜⊕»⁚·

⁖word⁚·ᚠᛁᛉᚾ⁖·⁚˙·word⁚·

·⁖origin…language⁚·Old Norse·⁚˙·origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁚˙·font⁚·
·⁖pronunciation⁚·LEMJA·⁚˙·pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁚˙·cadence⁚·
·⁖non…ostensive⁚·Beat⁝⁝thrash⁝⁝flog: strike repeatedly so force returns upon a body in successive blows·⁚˙·non…ostensive⁚·
·⁖ostensive⁚·Blow follows blow against the same body⁝⁝repeated force arriving again before the striking has ceasedᛁ·⁚˙·ostensive⁚·
·⁖grimchain⁚·▫ᛗIᚴ❁ᘐᚳᛕᕵ♂△ᚿᓄᗤ·⁚˙·grimchain⁚·
·⁚˙·lookup·˙
·⁚˙·glyphs·˙
·⁖glyph symbol⟜⟜ᗡ» category⟜⟜court»⁚·
·⁖lookup office⟜⟜recursive…identity» enoch⟜⟜⊕»⁚·

⁖word⁚·सामन्⁖·⁚˙·word⁚·

·⁖origin…language⁚·Sanskrit·⁚˙·origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁚˙·font⁚·
·⁖pronunciation⁚·sah⟊muhn·⁚˙·pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁚˙·cadence⁚·
·⁖non…ostensive⁚·Song or tune⁖· ordered sound carried as melody or sacred utterance·⁚˙·non…ostensive⁚·
·⁖ostensive⁚·Pitches rise⁝⁝fall⁝⁝and return through one sounding course⁝⁝their changing notes preserving a recognizable songᗡ·⁚˙·ostensive⁚·
·⁖grimchain⁚·♂⟊ᛯᛗ❁⊛ᛞᛗ0♀✶ⴱᘐ·⁚˙·grimchain⁚·
·⁚˙·lookup·˙
·⁚˙·glyphs·˙
·⁖glyph symbol⟜⟜ᛕ» category⟜⟜court»⁚·
·⁖lookup office⟜⟜recursive…identity» enoch⟜⟜⊕»⁚·

؛word:·ԋȷՍՂ؛·˙·˙word:·

·˙origin…language:·Aghwan·˙·˙origin…language:·
·˙font:·Noto Sans Caucasian Albanian·˙·˙font:·
·˙pronunciation:·IHAWL·˙·˙pronunciation:·
·˙cadence:·Recursive Identity·˙·˙cadence:·
·˙non…ostensive:·Cross˙· intersecting form or marked crossing where one line⁚⁚path⁚⁚or bearing passes another·˙·˙non…ostensive:·
·˙ostensive:·Two paths cross at one point and continue beyond it⁚⁚each keeping its own bearing through the shared meetingϗ·˙·˙ostensive:·
·˙grimchain:·ϗ⁜◌ᵋՒᚱᵻᐃᓀᚴ❋ƻ𝍐ᔓ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol◦–◦ᵋ1ₓ category◦–◦ᵋcourtₓ;·
·˙lookup office◦–◦ᵋrecursive…identityₓ enoch◦–◦ᵋ⊕ₓ;·

؛word:·ΓΛϒIᵬΛN؛·˙·˙word:·

·˙origin…language:·Gothic·˙·˙origin…language:·
·˙font:·Noto Sans Gothic·˙·˙font:·
·˙pronunciation:·GAWIDAN·˙·˙pronunciation:·
·˙cadence:·Recursive Identity·˙·˙cadence:·
·˙non…ostensive:·Fasten⁚⁚join: bind separate parts so they hold together and resist separation·˙·˙non…ostensive:·
·˙ostensive:·Two parts meet under binding force˙· strain pulls against the joined place without opening separation1·˙·˙ostensive:·
·˙grimchain:·⛢⬡⧗∀ᛯᚣᛏ▽ᚹᛊ⊚ᛟᚸ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol◦–◦ᵋ1ₓ category◦–◦ᵋcourtₓ;·
·˙lookup office◦–◦ᵋrecursive…identityₓ enoch◦–◦ᵋ⊕ₓ;·

؛word:·STIKS؛·˙·˙word:·

·˙origin…language:·Gothic·˙·˙origin…language:·
·˙font:·Noto Sans Gothic·˙·˙font:·
·˙pronunciation:·STIKS·˙·˙pronunciation:·
·˙cadence:·Recursive Identity·˙·˙cadence:·
·˙non…ostensive:·Point˙· moment: a smallest marked position in space or instant in a course of time·˙·˙non…ostensive:·
·˙ostensive:·Lines arriving from every direction meet the same tiny place⁚⁚a point with no length or breadth of its own1·˙·˙ostensive:·
·˙grimchain:·☿ϛՀ⊕ᵻ⊕ᚠՀᚸ1ᐸ⊕Ƶ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol◦–◦ᵋ1ₓ category◦–◦ᵋcourtₓ;·
·˙lookup office◦–◦ᵋrecursive…identityₓ enoch◦–◦ᵋ⊕ₓ;·

؛word:·ꝏ𐞥ᵎᵎꝏ؛·˙·˙word:·

·˙origin…language:·Avestan·˙·˙origin…language:·
·˙font:·Noto Sans Avestan·˙·˙font:·
·˙pronunciation:·AY+oo+wah·˙·˙pronunciation:·
·˙cadence:·Recursive Identity·˙·˙cadence:·

·⁚non…ostensive⁚·One⁚· the cardinal number one·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One body rests in one counting place⁝⁝with no second body entering the number beside
↪ it𐑌·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⟴𐑋𐑓𐑖𐑗𐑘𐑙𐑚𐑛𐑜𐑝·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⁚○₋₀₊✳₎ category⁚○₋₀₊court₎⁚·
·⁚lookup office⁚○₋₀₊recursive…identity₎ enoch⁚○₋₀₊⊕₎⁚·

‹word⁚·ᚺᛦᚢᚱᛟᛊ‹⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·HOWRDS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·A door⁝⁝especially of wickerwork·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Woven barrier swings across one opening⁚· passage exists when it turns aside and closes
↪ when it returns across the gap✳·⁚.·˙ostensive⁚·
·⁚grimchain⁚·▽𐑈𐑟𐑡𐑢𐑣𐑤𐑥𐑦𐑧·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⁚○₋₀₊✳₎ category⁚○₋₀₊court₎⁚·
·⁚lookup office⁚○₋₀₊recursive…identity₎ enoch⁚○₋₀₊⊕₎⁚·

‹word⁚·𒀸𒁹𒂗𒃻𒄿𒅆‹⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·ahn+doo+RAH+room·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·freedom⁝⁝exemption⁚· remission of commercial debts⁚· manumission (of private
↪ slaves)·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Cut cord falls loose from the wrist⁚· no bond remains between the freed body and the
↪ post✳·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⬡𐑪𐑫𐑬𐑭𐑮✳𐑯𐑰𐑱𐑲⟴·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⁚○₋₀₊✳₎ category⁚○₋₀₊court₎⁚·
·⁚lookup office⁚○₋₀₊recursive…identity₎ enoch⁚○₋₀₊⊕₎⁚·

‹word⁚·ⲟⲩⲱⲧⲉⲃ‹⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·oo+OH+teb·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To change or pass from one state to another·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Clear water clouds with stirred clay⁚· the same vessel now holds a state different from
↪ the one before✳·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐑶𐑷𐑸𐑹𐑺𐑻𐑼𐑽𐑾✳·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⁚○₋₀₊𐑛₎ category⁚○₋₀₊court₎⁚·
·⁚lookup office⁚○₋₀₊recursive…identity₎ enoch⁚○₋₀₊⊕₎⁚·

˖word꞉·ᛉᚳᚣᛃᚳᚿᚾ˖·.·ˑword꞉·

·꞉origin…language꞉·Gothic·꞉·.·ˑorigin…language꞉·
·꞉font꞉·Noto Sans Gothic·꞉·.·ˑfont꞉·
·꞉pronunciation꞉·WAL⟊DAN·꞉·.·ˑpronunciation꞉·
·꞉cadence꞉·Recursive Identity·꞉·.·ˑcadence꞉·
·꞉non…ostensive꞉·Rule꞉· wield authority or power over a body⁚⁚domain⁚⁚or course of
↪ action꞉·.·ˑnon…ostensive꞉·
·꞉ostensive꞉·One command bends separate actions into a common course across the field under its
↪ authority⩙·꞉·.·ˑostensive꞉·
·꞉grimchain꞉·✷♈▽✷ᚺ⁂ᒍᖘᖇ⦾ᘜ·꞉·.·ˑgrimchain꞉·
·꞉·.·ˑlookup꞉·
·꞉·.·ˑglyphs꞉·
·꞉glyph symbol꞉○꞊♈﹐ category꞉○꞊court﹕꞉·
·꞉lookup office꞉○꞊recursive…identity﹐ enoch꞉○꞊⊕﹕꞉·

˖word꞉·ᑌᑐᒣᖘᑐᒡ˖·.·ˑword꞉·

·꞉origin…language꞉·Aghwan·꞉·.·ˑorigin…language꞉·
·꞉font꞉·Noto Sans Caucasian Albanian·꞉·.·ˑfont꞉·
·꞉pronunciation꞉·CHUDU·꞉·.·ˑpronunciation꞉·
·꞉cadence꞉·Recursive Identity·꞉·.·ˑcadence꞉·
·꞉non…ostensive꞉·Heaven꞉· sky: the great upper expanse seen above the earth·꞉·.·ˑnon…ostensive꞉·
·꞉ostensive꞉·Above roof⁚⁚branch⁚⁚and mountain the sky keeps opening⁚⁚no reachable wall closing the upper
↪ expanse♈·꞉·.·ˑostensive꞉·
·꞉grimchain꞉·ᖘ♏ᘏ⍋ᖘ▽ᚸᒪ⦾♂✧⊱ᘐ✓·꞉·.·ˑgrimchain꞉·
·꞉·.·ˑlookup꞉·
·꞉·.·ˑglyphs꞉·
·꞉glyph symbol꞉○꞊0﹐ category꞉○꞊court﹕꞉·
·꞉lookup office꞉○꞊recursive…identity﹐ enoch꞉○꞊⊕﹕꞉·

˖word꞉·ᛒᛁᚾᚱᛉᚱ˖·.·ˑword꞉·

·꞉origin…language꞉·Old Norse·꞉·.·ˑorigin…language꞉·
·꞉font꞉·Noto Sans Runic·꞉·.·ˑfont꞉·
·꞉pronunciation꞉·BJARTR·꞉·.·ˑpronunciation꞉·
·꞉cadence꞉·Recursive Identity·꞉·.·ˑcadence꞉·
·꞉non…ostensive꞉·Bright⁚⁚radiant: giving or bearing strong visible light or brilliance·꞉·.·ˑnon…ostensive꞉·
·꞉ostensive꞉·Strong light leaves the radiant surface and reaches the stone around it⁚⁚brightness alive
↪ across the shining body0·꞉·.·ˑostensive꞉·
·꞉grimchain꞉·♏♍ᘓᖘ⍑☿⊕ᖶᔓ⦾ᘜ'ᚕ☿·꞉·.·ˑgrimchain꞉·
·꞉·.·ˑlookup꞉·
·꞉·.·ˑglyphs꞉·
·꞉glyph symbol꞉○꞊ᛦ﹐ category꞉○꞊court﹕꞉·
·꞉lookup office꞉○꞊recursive…identity﹐ enoch꞉○꞊⊕﹕꞉·

˖word꞉·ժ□ᛪᑌ̄˖·.·ˑword꞉·

·꞉origin…language꞉·Tocharian B·꞉·.·ˑorigin…language꞉·
·꞉font꞉·Noto Sans Brahmi·꞉·.·ˑfont꞉·
·꞉pronunciation꞉·CHUHMP·꞉·.·ˑpronunciation꞉·
·꞉cadence꞉·Recursive Identity·꞉·.·ˑcadence꞉·
·꞉non…ostensive꞉·Be able⁚⁚can: capacity for an act or state to be realized rather than being closed to
↪ occurrence·꞉·.·ˑnon…ostensive꞉·

·⁖ostensive⁚·Bow and arrow wait at full draw⁘all conditions for the shot present though release has not yet come𐤗·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᗡ▽ᖷᘐᖾᔑᔕᗡᙅᘜ❂Ⅱ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol꩜꩜ǂ꞉ category꩜꩜court꞉⁚·
·⁖lookup office꩜꩜recursive…identity꞉ enoch꩜꩜⊕꞉⁚·

⁖word⁚·द्वार्⁖.·˙word⁚·

·⁖origin…language⁚·Sanskrit·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Devanagari·⁖.·˙font⁚·
·⁖pronunciation⁚·dvahr·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Gate⁘door⁘entrance⁘issue: a bounded opening through which passage becomes possible·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Two sides and a lintel hold an opening through the barrier⁘leaving passage clear from either direction∓·⁖.·˙ostensive⁚·
·⁖grimchain⁚·♂⊕₀ᵽ0ᵲᵧᵡ𐦟ᙀᛋᵾᵽ▽·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol꩜꩜ᙏ꞉ category꩜꩜court꞉⁚·
·⁖lookup office꩜꩜recursive…identity꞉ enoch꩜꩜⊕꞉⁚·

⁖word⁚·ⲗⲋ⁖.·˙word⁚·

·⁖origin…language⁚·Coptic·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Coptic·⁖.·˙font⁚·
·⁖pronunciation⁚·LAH·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Mouth⁖· door⁘entrance⁘gate⁖· opening·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·One bounded opening joins inside and outside⁘carrying breath⁘food⁘voice⁘or passage across the same threshold᙭·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᚠᚢ🢧Θᛯᘜᗤ∱ᵮᘐᓀᘏ🢧·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol꩜꩜ᛇ꞉ category꩜꩜court꞉⁚·
·⁖lookup office꩜꩜recursive…identity꞉ enoch꩜꩜⊕꞉⁚·

⁖word⁚·𗍫⁖.·˙word⁚·

·⁖origin…language⁚·Tangut·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Serif Tangut·⁖.·˙font⁚·
·⁖pronunciation⁚·LLAH·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Recursive Identity·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·Carry⁘transfer: bear something from one holder⁘position⁘or place into another·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Loaded basket leaves one shoulder and settles onto another⁖· the burden remains borne through the change of holderᛇ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·𐦟ᵥᴢᔕX♄۞ᛯ☥ₒᐟᔕ⊙ᵾ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁘
·⁖.·˙glyphs⁘
·⁖glyph symbol꩜꩜ᚸ꞉ category꩜꩜court꞉⁚·
·⁖lookup office꩜꩜recursive…identity꞉ enoch꩜꩜⊕꞉⁚·

˖word꞉·𒌑𒄑˖.·˙word꞉·

·⁖origin…language꞉·Sumerian·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖.·˙font꞉·
·⁖pronunciation꞉·GEE·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Reed⁖· a tall hollow+stemmed wetland plant used for mats⁚⁚tools⁚⁚construction⁚⁚and measuring·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·From wet ground rises a light segmented stalk⁚⁚hollow through its center from rootward stem to upper reed𓍯·⁖.·˙ostensive꞉·
·⁖grimchain꞉·ᘏᖘ𓆉ᒧA✳ᙢᙢ𓃰𓃰◎ᓀ·⁖.·˙grimchain꞉·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol꞉o꞉o꞉𓃒꞉ category꞉o꞉o꞉court꞉꞉·
·⁖lookup office꞉o꞉o꞉recursive…identity꞉ enoch꞉o꞉o꞉⊕꞉꞉·

˖word꞉·अक्ष˖.·˙word꞉·

·⁖origin…language꞉·Sanskrit·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Devanagari·⁖.·˙font꞉·
·⁖pronunciation꞉·UHK+shuh·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Axle⁚⁚axis⁚⁚wheel⁚⁚or cart·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Wheel turns around a fixed central shaft⁚⁚every rim point moving while the axis keeps its place𓃒·⁖.·˙ostensive꞉·
·⁖grimchain꞉·ᖷ𐦂ᕒᔕ𓆉▽𐦂ᘐᔕᖷᔑ·⁖.·˙grimchain꞉·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol꞉o꞉o꞉𓃬꞉ category꞉o꞉o꞉court꞉꞉·
·⁖lookup office꞉o꞉o꞉recursive…identity꞉ enoch꞉o꞉o꞉⊕꞉꞉·

˖word꞉·ᚩᚢᚹᛋ˖.·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖.·˙font꞉·
·⁖pronunciation꞉·LOGN·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Calm⁚⁚stillness: an undisturbed condition in which agitation and violent motion have ceased·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Water lies without ripple⁚⁚hanging leaves remain still⁚⁚and no disturbance crosses the field𓃬·⁖.·˙ostensive꞉·
·⁖grimchain꞉·ᛋ❁ᓚᛋᛊ▽ᛊᘔᛞᛇᔑᓀᘎ✡ᚹ·⁖.·˙grimchain꞉·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol꞉o꞉o꞉𓃗꞉ category꞉o꞉o꞉court꞉꞉·
·⁖lookup office꞉o꞉o꞉recursive…identity꞉ enoch꞉o꞉o꞉⊕꞉꞉·

˖word꞉·𓏏˖.·˙word꞉·

·⁖origin…language꞉·Egyptian·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Egyptian Hieroglyphs·⁖.·˙font꞉·
·⁖pronunciation꞉·MAH+at·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Recursive Identity·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Truth⁚⁚justice⁚⁚right order⁚⁚proper balance·⁖.·˙non…ostensive꞉·

·⁚ostensive⁚·Equal loads hold the beam level over its center⁘neither side climbing above the
↪ otherx·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛉᛂᗡ𐦂✷ᔑ𓃰ᑫᘐᘏᔑ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦𐑈 category◦–◦court𐑉⁚·
·⁚lookup office◦–◦recursive…identity𐑉 enoch◦–◦⊕𐑉⁚·

·word⁚·ᴏγᴏᴨ·⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·wop·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To be pure⁘innocent⁖ pure⁘holy·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Clear cloth bears no stain⁘mark⁘or foreign color from edge to edge𐑈·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𓆏ᑭ𝌆ᘐᗡ✳ᔭᔑᛉ✷𐦂𓃰·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦ᔑ𐑉 category◦–◦court𐑉⁚·
·⁚lookup office◦–◦recursive…identity𐑉 enoch◦–◦⊕𐑉⁚·

·word⁚·ᛏᒪᓚᒧᒥ·⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·KNUHLOONEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Accomplishment⁖ an achieved work or result brought through to
↪ completion·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Fitted work stands whole beneath use⁘with no loose edge⁘open gap⁘or unfinished surface
↪ breaking its completed formᔑ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚹᘔᙢᕋᚢᘏIᘔᒣᛯᓀᛇ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦ᒧ𐑉 category◦–◦court𐑉⁚·
·⁚lookup office◦–◦recursive…identity𐑉 enoch◦–◦⊕𐑉⁚·

·word⁚·ϒЄIΤϒᛟѰS·⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·WIT⟊WOTHS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Recursive Identity·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Witness⁖ one who has directly encountered what can later be attested·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Eyes meet the event directly and keep enough of what was seen for later attestation
↪ without borrowed reportᒧ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᑭ⊕ᗡ𓃰ᛯ✲ᘓᗢᘐᔑᛯᘏᔭᚴ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol◦–◦ᒪ𐑉 category◦–◦court𐑉⁚·
·⁚lookup office◦–◦recursive…identity𐑉 enoch◦–◦⊕𐑉⁚·

·word⁚·स्थरि·⁚.·˙word⁚·

·⁖origin…language:·Sanskrit·⁖.·˙origin…language:·
·⁖font:·Noto Sans Devanagari·⁖.·˙font:·
·⁖pronunciation:·sthi+ruh·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Stable::firm::steadfast::fixed::durable: continuing in a held state against disturbance
↪ or displacement·⁖.·˙non…ostensive:·
·⁖ostensive:·Load presses against the body without shifting its place or deforming its held
↪ formᒧ·⁖.·˙ostensive:·
·⁖grimchain:·◎ᔑ△ϒ⁞⁞Ÿϒᘎᛉᘎᚿᑎ◈·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ₑ▽ₛ category○–○ₑcourtₛ:·
·⁖lookup office○–○ₑrecursive…identityₛ enoch○–○ₑ⊕ₛ:·

₍word:·ᑌ□ᛁᛤᑎ₸̄₎·⁖.·˙word:·

·⁖origin…language:·Tocharian B·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·PUHRMAHNGK·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Hope·˙ expectation directed toward a desired possibility not yet
↪ possessed·⁖.·˙non…ostensive:·
·⁖ostensive:·An unreached possibility remains ahead while inward expectation stays directed toward its
↪ arrival▽·⁖.·˙ostensive:·
·⁖grimchain:·ᛯ⊖ᚿᘎسᛜ⊙ᛊ⏁☽ᛤ⬡ᛅ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ₑƵₛ category○–○ₑcourtₛ:·
·⁖lookup office○–○ₑrecursive…identityₛ enoch○–○ₑ⊕ₛ:·

₍word:·Δ̀ᏼ̄₸ʎᏓ̄ᏞᏁ̄ᏬᏆ₎·⁖.·˙word:·

·⁖origin…language:·Tocharian B·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·EHNSHRKEHTSTSEH·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Lasting::enduring::permanent: continuing to remain through time or pressure without
↪ ceasing·⁖.·˙non…ostensive:·
·⁖ostensive:·Weather crosses the same body again and again::yet shape::position::and presence continue
↪ through the pressurez·⁖.·˙ostensive:·
·⁖grimchain:·Ƀعمرᗫᘎᛋ⊬∫ᚹ₮ᛤ❀☆❄·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol○–○ₑϛₛ category○–○ₑcourtₛ:·
·⁖lookup office○–○ₑrecursive…identityₛ enoch○–○ₑ⊕ₛ:·

₍word:·ϴϵIʌϟɴ₎·⁖.·˙word:·

·⁖origin…language:·Gothic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖.·˙font:·
·⁖pronunciation:·HWEEL+N·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·Rest::pause: cessation of active exertion while the body or process remains capable of
↪ returning·⁖.·˙non…ostensive:·

·⁖ostensive꞉·Hands fall still beside unfinished tools⁘ breath continues while exertion is absent from
↪ the bodyɾ·⁖·˙ostensive꞉·
·⁖grimchain꞉·ગ꣐ᘓᐱᛋ⁕⏁ᔆᔓᔑᔐᔕᔓᔐᔓᔑᔐᔕᔓᔐᔓᔑᔐᔕᔓᔐ

·:word:·ᛦᚢᚾᛏ:.·˙word:·

·⁖origin…language:·Old Norse·⁖.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖.·˙font:·
·⁖pronunciation:·MUND·⁖.·˙pronunciation:·
·⁖cadence:·Recursive Identity·⁖.·˙cadence:·
·⁖non…ostensive:·shape⁝⁝form.·⁖.·˙non…ostensive:·
·⁖ostensive:·A spiral winds through the shell of a sea creature⁝⁝Each fern unfurls a smaller fern
↪ beneath a green curl⁖⁝river eddies carry roundness without a carved rim⁝⁝Frost branches across
↪ glass in branching likeness꙰·⁖.·˙ostensive:·
·⁖grimchain:·♃♂🜨≯⊰♪☾⬡♌⊛🜏⊢ᒪᚠ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖.·˙glyphs·˙
·⁖.·˙sydonic…magicae…lexicon·∴

⁘⁘⁘

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·tipi·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to taste·⁚.·˙non…ostensive:·
·⁚ostensive:·Bread touches the tongue·:Hidden sweetness rises from the morsel without
↪ speech♀·⁚.·˙ostensive:·
·⁚grimchain:·ϞⲐⲞ🡆ⱛϘ✳⋕ⲂⲘ◝ᘓ%Ϟ♌◝ᘓ%·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ₔ♋꜀ category○–○ₔemission꜀⁚·
·⁚lookup office○–○ₔcatalyst꜀ enoch○–○ₔ♂꜀⁚·

·word:·ՈՆՍՆ·⁚.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·buh┼HLAH·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To say·:speak·:call·:name·:answer·:or ask·⁚.·˙non…ostensive:·
·⁚ostensive:·Breath presses through throat and tongue until shaped voice breaks the edge of
↪ silence♋·⁚.·˙ostensive:·
·⁚grimchain:·ሐzɻB⊰ⵐሶᑐ%ᘓ↗ʅᚼ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ₔ♌꜀ category○–○ₔemission꜀⁚·
·⁚lookup office○–○ₔcatalyst꜀ enoch○–○ₔ♂꜀⁚·

·word:·𒌓·⁚.·˙word:·

·⁚origin…language:·Sumerian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·NEER·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·Trust·:˙ a sign or lordly assurance·⁚.·˙non…ostensive:·
·⁚ostensive:·A sealed oath is laid beside guarded goods·:Uncertainty loosens under the binding
↪ sign♌·⁚.·˙ostensive:·
·⁚grimchain:·ᵚⲎ⫲ᦞ⁂ᐁϔ⅄⊰⩙ɕ✡·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○ₔ♁꜀ category○–○ₔemission꜀⁚·
·⁚lookup office○–○ₔcatalyst꜀ enoch○–○ₔ♂꜀⁚·

·word:·ϬϤϏϽϴϽϤ·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·BIYESUN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To do·:to make·⁚.·˙non…ostensive:·
·⁚ostensive:·Tool bites raw material and untouched surface yields beneath the first working
↪ stroke♁·⁚.·˙ostensive:·
·⁚grimchain:·ⵒᘎᐡ⩹ϤⓄᘐᚺↄ⊕ᔕᚼ〰·⁚.·˙grimchain:·
·⁚.·˙lookup:˙

·⁚·˙glyphs:˙
·⁚glyph symbol◦–◦𓂀𓆑 category◦–◦𓂀emission𓆑:·
·⁚lookup office◦–◦𓂀catalyst𓆑 enoch◦–◦𓂀𓋹𓆑:·

₍word:·𓀠𓀡𓀢𓀣𓀤 𓀥₎·:.·˙word:·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·SAHKI·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To know·˙ to be aware of·˙ to experience·˙ to heed::take note of::pay attention to·˙ to
↪ recognize::acknowledge::accept·˙ to remember·⁚.·˙non…ostensive:·
·⁚ostensive:·A familiar sign catches the eye::halting the passing glance and calling old understanding
↪ awake𓆑·⁚.·˙ostensive:·
·⁚grimchain:·𓃰𓀹𓀀𓀁𓀂𓀃𓀄𓀅𓀆𓀇𓀈𓀉𓀊·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦𓂀𓅓𓆑 category◦–◦𓂀emission𓆑:·
·⁚lookup office◦–◦𓂀catalyst𓆑 enoch◦–◦𓂀𓋹𓆑:·

₍word:·𓀀𓀁𓀂𓀃𓀄₎·:.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·AHLY·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To ascend::go up::come up·⁚.·˙non…ostensive:·
·⁚ostensive:·Foot meets the higher stone::lifting the body clear of the lower ground𓂀·⁚.·˙ostensive:·
·⁚grimchain:·𐎐𐎍𐎗𐎚𐎕𐎖𐎘𐎙𐎛𐎜𐎝𐎞·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦𓂀𓅓𓆑 category◦–◦𓂀emission𓆑:·
·⁚lookup office◦–◦𓂀catalyst𓆑 enoch◦–◦𓂀𓋹𓆑:·

₍word:·ᛉᛁᛚᚾ₎·:.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·MEH-tah·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to estimate::value·⁚.·˙non…ostensive:·
·⁚ostensive:·Weight settles the balance::Uneven pans reveal which side carries more𓅓·⁚.·˙ostensive:·
·⁚grimchain:·ᛉᚳᚾᛋᛟᚱᛟᛉᚷᛏᛞᚲ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol◦–◦𓂀⊙𓆑 category◦–◦𓂀emission𓆑:·
·⁚lookup office◦–◦𓂀catalyst𓆑 enoch◦–◦𓂀𓋹𓆑:·

₍word:·ትእዛዝ₎·:.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·tuh-ZAHZ·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·

·⁖non…ostensive:·Command⁖⁖mandate⁖⁖order⁖⁖statute⁖⁖or authority·⁚·˙non…ostensive:·
·⁖ostensive:·One sharp order cuts across the hall⁖⁖and scattered motion snaps toward a single course⊙·⁚·˙ostensive:·
·⁖grimchain:·𐌼⁕ᓚᚿᛯᛘᘯᚱᚢ𓀀𓂻⁕ᚢᛁ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉𐤂 category꞉꞉emission꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐤏꞉·

·word:·𓉔𓏺·⁚·˙word:·

·⁖origin…language:·Sumerian·⁚·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁚·˙font:·
·⁖pronunciation:·NGAHL·⁚·˙pronunciation:·
·⁖cadence:·Catalyst·⁚·˙cadence:·
·⁖non…ostensive:·Exist·˙ to be present⁖⁖be there⁖⁖or have being·⁚·˙non…ostensive:·
·⁖ostensive:·A stone is set upon bare ground⁖⁖breaking emptiness where weight now stands𐤂·⁚·˙ostensive:·
·⁖grimchain:·Ⲏ⊛⛦ᚠᛋᛕᵶ⊛ሕ◌ᔑɾɥᚣ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉ᛉ꞉ category꞉꞉parliament꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐤏꞉·

·word:·ጸሐፈ·⁚·˙word:·

·⁖origin…language:·Ge'ez·⁚·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁚·˙font:·
·⁖pronunciation:·tsah+HAH+fah·⁚·˙pronunciation:·
·⁖cadence:·Catalyst·⁚·˙cadence:·
·⁖non…ostensive:·To write·˙ to mark language in visible signs·⁚·˙non…ostensive:·
·⁖ostensive:·Pen meets prepared skin⁖⁖and dark signs follow the pressure of the moving point·⁚·˙ostensive:·
·⁖grimchain:·𞤤≯ᗅᚱXⴲᛯᛯAᛘᚠᛉ≏ᛏ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉ᛉ꞉ category꞉꞉parliament꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐤏꞉·

·word:·ተርጐመ·⁚·˙word:·

·⁖origin…language:·Ge'ez·⁚·˙origin…language:·
·⁖font:·Noto Sans Ethiopic·⁚·˙font:·
·⁖pronunciation:·tar+GWAH+mah·⁚·˙pronunciation:·
·⁖cadence:·Catalyst·⁚·˙cadence:·
·⁖non…ostensive:·To translate·˙ to render speech or writing into another language·⁚·˙non…ostensive:·
·⁖ostensive:·A foreign phrase falls into unfamiliar hearing⁖⁖Another tongue answers with its meaning intact·⁚·˙ostensive:·
·⁖grimchain:·☽▽⍈ᚠᗅᚱᗊX▫ᐱᗅᖘᛋ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁖glyph symbol꞉꞉Ⲓ꞉ category꞉꞉parliament꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐤏꞉·

·word:·龘·⁚·˙word:·

·⁖origin…language꞉·Tangut·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Serif Tangut·⁖.·˙font꞉·
·⁖pronunciation꞉·SHOW·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Arise꞉꞉happen꞉꞉or appear·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Covering is pulled aside꞉꞉and a hidden form breaks suddenly into sight𝌆·⁖.·˙ostensive꞉·
·⁖grimchain꞉·ⴱ✷ᚹᛉᛗ𐋏ᛖᛊᛋᚸ❀⧩۞·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉◦◦ᔓ꞉ category꞉◦◦ₔparliament꞉⁖·
·⁖lookup office꞉◦◦ₔcatalyst꞉ enoch꞉◦◦ₔ♂꞉⁖·

⁖word꞉·ᑭᛁᚱᛒᚼ⁖.·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖.·˙font꞉·
·⁖pronunciation꞉·VEHR⟊pah·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·to throw꞉꞉cast·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Fingers open under gathered force꞉꞉and the stone leaves the handᔓ·⁖.·˙ostensive꞉·
·⁖grimchain꞉·ⴵᛦ✡♂ᛡ⊛ᓭᙏ▽ᛄ⊕ᚻ·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉◦◦ₔ୬ₒ꞉ category꞉◦◦ₔparliament꞉⁖·
·⁖lookup office꞉◦◦ₔcatalyst꞉ enoch꞉◦◦ₔ♂꞉⁖·

⁖word꞉·ᔐᘿᔑᓀᔐᕼᘐᒉᖁ⁖.·˙word꞉·

·⁖origin…language꞉·Aghwan·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Sans Caucasian Albanian·⁖.·˙font꞉·
·⁖pronunciation꞉·KARXESUN·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·To survive·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Danger closes around living flesh꞉꞉and breath keeps its hold against the same enclosing force୬·⁖.·˙ostensive꞉·
↪
·⁖grimchain꞉·ᙏᚼᚢ♂ᛝ⇞ᛋ♂ᑕᘐᓄ⬡⬡·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉◦◦ₔ𝌋꞉ category꞉◦◦ₔparliament꞉⁖·
·⁖lookup office꞉◦◦ₔcatalyst꞉ enoch꞉◦◦ₔ♂꞉⁖·

⁖word꞉·隣⁖.·˙word꞉·

·⁖origin…language꞉·Tangut·⁖.·˙origin…language꞉·
·⁖font꞉·Noto Serif Tangut·⁖.·˙font꞉·
·⁖pronunciation꞉·TWIH'·⁖.·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖.·˙cadence꞉·
·⁖non…ostensive꞉·Sew⁖· to join or work material with thread passed through it·⁖.·˙non…ostensive꞉·
·⁖ostensive꞉·Thread pierces parted cloth꞉꞉drawing the torn edges together under tension𝌋·⁖.·˙ostensive꞉·
·⁖grimchain꞉·ᔓᘐᘉᛃ◎ᑎᙏᓀᘎᔓᕫᓀ☆ᒪ·⁖.·˙grimchain꞉·
·⁖.·˙lookup꞉˙
·⁖.·˙glyphs꞉˙
·⁖glyph symbol꞉◦◦ₔ♎꞉ category꞉◦◦ₔparliament꞉⁖·
·⁖lookup office꞉◦◦ₔcatalyst꞉ enoch꞉◦◦ₔ♂꞉⁖·

⁖word:·ⲭⲱ⁖.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·JOH·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To say⁝⁝speak⁝⁝tell⁝⁝or sing·⁚.·˙non…ostensive:·
·⁚ostensive:·Pressure gathers behind closed lips⁝⁝The mouth opens⁝⁝Shaped sound breaks into
↪ air␤·⁚.·˙ostensive:·
·⁚grimchain:·۞ᔑƵⵠ☿ℏ␤◎⋤ᚠᛉᛋᚺ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑ𝔐₎ category○–○ₑparliament₎⁚·
·⁚lookup office○–○ₑcatalyst₎ enoch○–○ₑ♂₎⁚·

⁖word:·ᛣᚠᛣIᛕᛣᚾ⁖.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·AFEYEKAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To renounce⁝⁝to forswear⁝⁝to disavow·⁚.·˙non…ostensive:·
·⁚ostensive:·Contradicting proof is laid beside the oath·˙ the seal is broken and the pledged words are
↪ withdrawn𝔐·⁚.·˙ostensive:·
·⁚grimchain:·ⲭᛁᚼᛘᛊ᛫ᵹⲭ∀ᚾↃᚠᔑ⬡·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑ↗₎ category○–○ₑparliament₎⁚·
·⁚lookup office○–○ₑcatalyst₎ enoch○–○ₑ♂₎⁚·

⁖word:·ꙍɯпє⁖.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·SHOH⟊peh·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To become·⁚.·˙non…ostensive:·
·⁚ostensive:·Heat reaches firm wax and the old shape softens into another contour↗·⁚.·˙ostensive:·
·⁚grimchain:·ᛘ꙰ᘓΛ⋐✱۞✳⧫ᛘXϒB↓·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑ♌₎ category○–○ₑparliament₎⁚·
·⁚lookup office○–○ₑcatalyst₎ enoch○–○ₑ♂₎⁚·

⁖word:·ΓΛΓᛕЄ⊧TS⁖.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·GAGRAYFTS·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·Decree·⁚.·˙non…ostensive:·
·⁚ostensive:·Conflicting claims fill the hall until one authoritative utterance cuts across them and
↪ fixes a single course♌·⁚.·˙ostensive:·
·⁚grimchain:·֍Hᚠ〰ᛉᛋ𝔐۞ᵪ♂ⲖⲞ·⁚.·˙grimchain:·

·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol₀₋₀𓊗₎ category₀₋₀parliament₎⁚·
·⁚lookup office₀₋₀catalyst₎ enoch₀₋₀☌₎⁚·

:word:·𐎍𐎐𐎗:.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·HLK·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To go·⁚.·˙non…ostensive:·
·⁚ostensive:·A distant call sounds::Settled feet leave the ground::and the road takes the first step𓊗·⁚.·˙ostensive:·
·⁚grimchain:·▽𓊗𐎗𐎚ᚦᛃᚳᛏ𓆉𓆉ᚱᚷᚱ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol₀₋₀𓃰₎ category₀₋₀parliament₎⁚·
·⁚lookup office₀₋₀catalyst₎ enoch₀₋₀☌₎⁚·

:word:·ᛣᚠᚣᛖᛁᚿᚣᚾ:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·AFLEETHAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To depart::withdraw·⁚ to travel::take a journey·⁚ to give place::make room·⁚ used to translate Ancient Greek παραιρέω (paratēréō::“watch”)::but this may be a mistranslation of the Greek·⁚.·˙non…ostensive:·
·⁚ostensive:·Dismissal sounds across the threshold·⁚ a resting body leaves its place and nearness opens into distance𓃰·⁚.·˙ostensive:·
·⁚grimchain:·ℂ𐎚ᗅ𐎗ΘA↗𐎐𐎚☌𐎚𐎔𐎗·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol₀₋₀◎₎ category₀₋₀goetic₎⁚·
·⁚lookup office₀₋₀catalyst₎ enoch₀₋₀☌₎⁚·

:word:·ᚣᚿᚣSTᛉᘐᏀᚣᚿ:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·ANASTOHDJAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To begin·⁚.·˙non…ostensive:·
·⁚ostensive:·A first drumbeat breaks the stillness::and waiting motion begins◎·⁚.·˙ostensive:·
·⁚grimchain:·⩒𐎐𐎚↗ᛏ◎ᛁᛉᛉ✷ᑫ⊛·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol₀₋₀⬡₎ category₀₋₀goetic₎⁚·
·⁚lookup office₀₋₀catalyst₎ enoch₀₋₀☌₎⁚·

:word:·𐎍𐎗𐎜:.·˙word:·

·⁖origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁖pronunciation:·KATABA·⁚.·˙pronunciation:·
·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·To write·⁚.·˙non…ostensive:·
·⁖ostensive:·Point bites the receiving surface·⁖ dark grooves follow beneath the moving
↪ hand⬡·⁚.·˙ostensive:·
·⁖grimchain:·𐎗𐎢▽𐎛𐎍𐎌𐎚𐎓𐎐♀𐎒𐎘𐎑𐎆𐎖𐎔·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜✡꩝ category꩜꩜goetic꩝·
·⁖lookup office꩜꩜catalyst꩝ enoch꩜꩜♂꩝·

·word:·ᚠᛁᚾᛏᚱᚾ·⁚.·˙word:·

·⁖origin…language:·Old Norse·⁚.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁚.·˙font:·
·⁖pronunciation:·TENDRA·⁚.·˙pronunciation:·
·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·to ignite·⁝light·⁚.·˙non…ostensive:·
·⁖ostensive:·Spark touches dry fuel·⁝Flame answers✡·⁚.·˙ostensive:·
·⁖grimchain:·𐎟𐎯𐎮𐎫𐎤𐎠𐎺𐎹𐎵𐎶𐎲𐎳‹·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜✡꩝ category꩜꩜goetic꩝·
·⁖lookup office꩜꩜catalyst꩝ enoch꩜꩜♂꩝·

·word:·ᛈᛚᛁᚢᛏᚾ·⁚.·˙word:·

·⁖origin…language:·Old Norse·⁚.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁚.·˙font:·
·⁖pronunciation:·FLYOH⟊tah·⁚.·˙pronunciation:·
·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·to float·⁚.·˙non…ostensive:·
·⁖ostensive:·Current catches the loosened body and bears it free of the ground✡·⁚.·˙ostensive:·
·⁖grimchain:·𐎑𐎒𐎓𐎔𐎕𐎖𐎗𐎘𐎙✡⊕♂𐎛𐎍·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁖glyph symbol꩜꩜✪꩝ category꩜꩜goetic꩝·
·⁖lookup office꩜꩜catalyst꩝ enoch꩜꩜♂꩝·

·word:·𓂝·⁚.·˙word:·

·⁖origin…language:·Egyptian·⁚.·˙origin…language:·
·⁖font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁖pronunciation:·keh⟊MAH·⁚.·˙pronunciation:·
·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·To create or fashion·⁖ also to throw·⁚.·˙non…ostensive:·
·⁖ostensive:·Hand presses yielding matter·⁖ raw substance and made edge meet beneath the same
↪ pressure✪·⁚.·˙ostensive:·
·⁖grimchain:·𐎆𐎊𐎋✵△𐎗𐎘𐎙𐎚✳⊕·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁖glyph symbol◦–◦𖤓 category◦–◦goetic⁘·
·⁖lookup office◦–◦catalyst⁘ enoch◦–◦♂⁘·

⁖word:·ᛈᚾᛈᛋᚾ⁖·˙word:·

·⁖origin…language:·Old Norse·⁖·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖·˙font:·
·⁖pronunciation:·VAHK⟊sah·⁖·˙pronunciation:·
·⁖cadence:·Catalyst·⁖·˙cadence:·
·⁖non…ostensive:·to grow::increase·⁖·˙non…ostensive:·
·⁖ostensive:·Root drinks from dark earth::Green body presses beyond the measure that held
↪ it𖤓·⁖·˙ostensive:·
·⁖grimchain:·⚹⦵𐍈🢒ϩ🢒ᛊ☆ᛕᛁᛈ✧ᛕ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦⁕⁘ category◦–◦goetic⁘·
·⁖lookup office◦–◦catalyst⁘ enoch◦–◦♂⁘·

⁖word:·ᛣ꞉·˙word:·

·⁖origin…language:·Tocharian A·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·SHOH·⁖·˙pronunciation:·
·⁖cadence:·Catalyst·⁖·˙cadence:·
·⁖non…ostensive:·To live·⁖·˙non…ostensive:·
·⁖ostensive:·Breath enters warm flesh::and pulse answers beneath the skin⁕·⁖·˙ostensive:·
·⁖grimchain:·𑀰𑀫𑀯𑀦𑀫𑀣⚝𑀪𑀩𑀯✷❄·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦⧗⁘ category◦–◦goetic⁘·
·⁖lookup office◦–◦catalyst⁘ enoch◦–◦♂⁘·

⁖word:·ᛯIᛦ꞉·˙word:·

·⁖origin…language:·Tocharian A·⁖·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖·˙font:·
·⁖pronunciation:·AHRT·⁖·˙pronunciation:·
·⁖cadence:·Catalyst·⁖·˙cadence:·
·⁖non…ostensive:·To praise::love::be pleased with·˙ to acknowledge (possibly)·˙ to rejoice
↪ in·⁖·˙non…ostensive:·
·⁖ostensive:·Difficult work stands complete and unharmed::Voices rise while open palms strike
↪ together⧗·⁖·˙ostensive:·
·⁖grimchain:·𐍈⊕ᚼᛉᚹᚴᚠᛟ▫ᛈᛃᚱᚺ·⁖·˙grimchain:·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦⊛⁘ category◦–◦goetic⁘·
·⁖lookup office◦–◦catalyst⁘ enoch◦–◦♂⁘·

⁖word:·ᚦIᚹIᚼ·˙word:·

·⁖origin…language:·Old Norse·⁖·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖·˙font:·
·⁖pronunciation:·THEKJA·⁖·˙pronunciation:·
·⁖cadence:·Catalyst·⁖·˙cadence:·

·⁚non…ostensive⁝·to cover⁝⁚thatch·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Rain begins through the open roof·⁚ woven reeds are pulled across the exposed
↪ span⊛·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ƶ₀ᛗᒐᖘᔐ♧✳⊖ϴ⊙⏀⛈·⁚.·˙grimchain⁝·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𓆉꩜ category○–○𓆉goetic꩜⁚·
·⁚lookup office○–○𓆉catalyst꩜ enoch○–○𓆉♂꩜⁚·

𓆉word⁝·𗦟꞉.·˙word⁝·

·⁚origin…language⁝·Tangut·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Serif Tangut·⁚.·˙font⁝·
·⁚pronunciation⁝·THWOO·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Catalyst·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Meet⁚· to encounter or come into contact with another·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Converging motion drives two separate bodies into one sudden contact❄·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᔓ♳ᑭϒᚋᘐ♳0ᚻᛉ᳟ᵒ᳟ᛉ⊕·⁚.·˙grimchain⁝·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𓆉۞꩜ category○–○𓆉goetic꩜⁚·
·⁚lookup office○–○𓆉catalyst꩜ enoch○–○𓆉♂꩜⁚·

𓆉word⁝·ᚧᚣᑎᚱʁIᑎᔑᚣᚣN꞉.·˙word⁝·

·⁚origin…language⁝·Gothic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚.·˙font⁝·
·⁚pronunciation⁝·FOWRBIUDAN·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Catalyst·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To forbid (to disallow)·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·One spoken order falls across the path⁝⁚and advancing motion halts before it۞·⁚.·˙ostensive⁝·
·⁚grimchain⁝·0ᵔIᐱᔑᔐ⊱▽ᒧɻᛉᔆ꩜⁚.·˙grimchain⁝·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𓆉⌬꩜ category○–○𓆉goetic꩜⁚·
·⁚lookup office○–○𓆉catalyst꩜ enoch○–○𓆉♂꩜⁚·

𓆉word⁝·ΨᚣᑎᚣN꞉.·˙word⁝·

·⁚origin…language⁝·Gothic·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Gothic·⁚.·˙font⁝·
·⁚pronunciation⁝·THAH⟊N·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Catalyst·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To be silent·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·A raised hand cuts across the room⁝⁚Mouths close⁚· open air receives no voice⌬·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᒧᑭ✪ᦒᘐ▽⚲▽ᖇ⛈ₒᑎ↓·⁚.·˙grimchain⁝·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𓆉𓂀꩜ category○–○𓆉court꩜⁚·
·⁚lookup office○–○𓆉catalyst꩜ enoch○–○𓆉♂꩜⁚·

𓆉word⁝·ɯıcɛ꞉.·˙word⁝·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·meesuh·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to give birth·⁚.·˙non…ostensive:·
·⁚ostensive:·Pressure gathers through the womb⁚⁚forcing living flesh toward its first breath
↪ outside𐓏⁚.·˙ostensive:·
·⁚grimchain:·Ⲁⲑⲋϒ)ⲏⲗⲫ⛤ᑭᛋⲬ⋐ᜰ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○𐑂𐑉 category○–○𐑂court𐑉:·
·⁚lookup office○–○𐑂catalyst𐑉 enoch○–○𐑂𐑉:·

⁖word:·✻ІՈՐ⁖.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·HJOHL·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·wheel·⁚.·˙non…ostensive:·
·⁚ostensive:·Force enters the axle and the rim begins to turn around the fixed center𐓏·⁚.·˙ostensive:·
·⁚grimchain:·ᦒ◎ᗅᙢ⊛ᦓ✳ᙓᑭᘐᐁ𐓏⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○𐑂𐑉 category○–○𐑂court𐑉:·
·⁚lookup office○–○𐑂catalyst𐑉 enoch○–○𐑂𐑉:·

⁖word:·ƬЄKᒧNᒣ⁖.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·TAYKAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To touch·⁚.·˙non…ostensive:·
·⁚ostensive:·A rough edge presses bare skin⁚⁚waking sensation at the meeting point𐓏·⁚.·˙ostensive:·
·⁚grimchain:·ᏳᙀⴱXᗐᖱᑉᑲᑭᗢ✳❄ⴱ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○𐑂𐑉 category○–○𐑂court𐑉:·
·⁚lookup office○–○𐑂catalyst𐑉 enoch○–○𐑂𐑉:·

⁖word:·ᑭIᑭᛋ⁖.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·VEFA·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to weave·⁚.·˙non…ostensive:·
·⁚ostensive:·Shuttle drives thread through the stretched warp⁚˙ tension locks the crossing in
↪ place𐓏⁚.·˙ostensive:·
·⁚grimchain:·ⲟᛞᑭᔓXᛇᔑ↗⟰ᛘᖇᛋᛁ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○𐑂𐑉 category○–○𐑂court𐑉:·
·⁚lookup office○–○𐑂catalyst𐑉 enoch○–○𐑂𐑉:·

˖word꞉·ᔑᐱTᏀᐱᑎ˖꞉·˙word꞉·

·⁖origin…language꞉·Gothic·⁖꞉·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖꞉·˙font꞉·
·⁖pronunciation꞉·SATJAN·⁖꞉·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖꞉·˙cadence꞉·
·⁖non…ostensive꞉·To set·⁖꞉·˙non…ostensive꞉·
·⁖ostensive꞉·A hand lowers stone into its bed⁖꞉ weight seats it firmly beneath the rising
↪ workᘜ⁖꞉·˙ostensive꞉·
·⁖grimchain꞉·꯭ᘓ꯭ᘐ꯭ᘏ꯭ᘔ꯭ᘎ꯭ᘋ꯭ᘍ꯭ᘉ꯭ᘈ꯭ᘇ꯭ᘆ꯭ᘅ꯭ᘄ꯭ᘃ꯭ᘂ꯭ᘁ꯭ᘀ꯭ᗿ꯭ᗾ꯭ᗽ꯭ᗼ꯭ᗻ꯭ᗺ꯭ᗹ꯭ᗸ꯭ᗷ꯭ᗶ꯭ᗵ꯭ᗴ꯭ᗳ꯭ᗲ꯭ᗱ꯭ᗰ꯭ᗯ꯭ᗮ꯭ᗭ꯭ᗬ꯭ᗫ꯭ᗪ꯭ᗩ꯭ᗨ꯭ᗧ꯭ᗦ꯭ᗥ꯭ᗤ꯭ᗣ꯭ᗢ꯭ᗡ꯭ᗠ꯭ᗟ꯭ᗞ꯭ᗝ꯭ᗜ꯭ᗛ꯭ᗚ꯭ᗙ꯭ᗘ꯭ᗗ꯭ᗖ꯭ᗕ꯭ᗔ꯭ᗓ꯭ᗒ꯭ᗑ꯭ᗐ꯭ᗏ꯭ᗎ꯭ᗍ꯭ᗌ꯭ᗋ꯭ᗊ꯭ᗉ꯭ᗈ꯭ᗇ꯭ᗆ꯭ᗅ꯭ᗄ꯭ᗃ꯭ᗂ꯭ᗁ꯭ᗀ꯭ᕿ꯭ᕾ꯭ᕽ꯭ᕼ꯭ᕻ꯭ᕺ꯭ᕹ꯭ᕸ꯭ᕷ꯭ᕶ꯭ᕵ꯭ᕴ꯭ᕳ꯭ᕲ꯭ᕱ꯭ᕰ꯭ᕯ꯭ᕮ꯭ᕭ꯭ᕬ꯭ᕫ꯭ᕪ꯭ᕩ꯭ᕨ꯭ᕧ꯭ᕦ꯭ᕥ꯭ᕤ꯭ᕣ꯭ᕢ꯭ᕡ꯭ᕠ꯭ᕟ꯭ᕞ꯭ᕝ꯭ᕜ꯭ᕛ꯭ᕚ꯭ᕙ꯭ᕘ꯭ᕗ꯭ᕖ꯭ᕕ꯭ᕔ꯭ᕓ꯭ᕒ꯭ᕑ꯭ᕐ꯭ᕏ꯭ᕎ꯭ᕍ꯭ᕌ꯭ᕋ꯭ᕊ꯭ᕉ꯭ᕈ꯭ᕇ꯭ᕆ꯭ᕅ꯭ᕄ꯭ᕃ꯭ᕂ꯭ᕁ꯭ᕀ꯭ᔿ꯭ᔾ꯭ᔽ꯭ᔼ꯭ᔻ꯭ᔺ꯭ᔹ꯭ᔸ꯭ᔷ꯭ᔶ꯭ᔵ꯭ᔴ꯭ᔳ꯭ᔲ꯭ᔱ꯭ᔰ꯭ᔯ꯭ᔮ꯭ᔭ꯭ᔬ꯭ᔫ꯭ᔪ꯭ᔩ꯭ᔨ꯭ᔧ꯭ᔦ꯭ᔥ꯭ᔤ꯭ᔣ꯭ᔢ꯭ᔡ꯭ᔠ꯭ᔟ꯭ᔞ꯭ᔝ꯭ᔜ꯭ᔛ꯭ᔚ꯭ᔙ꯭ᔘ꯭ᔗ꯭ᔖ꯭ᔕ꯭ᔔ꯭ᔓ꯭ᔒ꯭ᔑ꯭ᔐ꯭ᔏ꯭ᔎ꯭ᔍ꯭ᔌ꯭ᔋ꯭ᔊ꯭ᔉ꯭ᔈ꯭ᔇ꯭ᔆ꯭ᔅ꯭ᔄ꯭ᔃ꯭ᔂ꯭ᔁ꯭ᔀ꯭ᓿ꯭ᓾ꯭ᓽ꯭ᓼ꯭ᓻ꯭ᓺ꯭ᓹ꯭ᓸ꯭ᓷ꯭ᓶ꯭ᓵ꯭ᓴ꯭ᓳ꯭ᓲ꯭ᓱ꯭ᓰ꯭ᓯ꯭ᓮ꯭ᓭ꯭ᓬ꯭ᓫ꯭ᓪ꯭ᓩ꯭ᓨ꯭ᓧ꯭ᓦ꯭ᓥ꯭ᓤ꯭ᓣ꯭ᓢ꯭ᓡ꯭ᓠ꯭ᓟ꯭ᓞ꯭ᓝ꯭ᓜ꯭ᓛ꯭ᓚ꯭ᓙ꯭ᓘ꯭ᓗ꯭ᓖ꯭ᓕ꯭ᓔ꯭ᓓ꯭ᓒ꯭ᓑ꯭ᓐ꯭ᓏ꯭ᓎ꯭ᓍ꯭ᓌ꯭ᓋ꯭ᓊ꯭ᓉ꯭ᓈ꯭ᓇ꯭ᓆ꯭ᓅ꯭ᓄ꯭ᓃ꯭ᓂ꯭ᓁ꯭ᓀ꯭ᒿ꯭ᒾ꯭ᒽ꯭ᒼ꯭ᒻ꯭ᒺ꯭ᒹ꯭ᒸ꯭ᒷ꯭ᒶ꯭ᒵ꯭ᒴ꯭ᒳ꯭ᒲ꯭ᒱ꯭ᒰ꯭ᒯ꯭ᒮ꯭ᒭ꯭ᒬ꯭ᒫ꯭ᒪ꯭ᒩ꯭ᒨ꯭ᒧ꯭ᒦ꯭ᒥ꯭ᒤ꯭ᒣ꯭ᒢ꯭ᒡ꯭ᒠ꯭ᒟ꯭ᒞ꯭ᒝ꯭ᒜ꯭ᒛ꯭ᒚ꯭ᒙ꯭ᒘ꯭ᒗ꯭ᒖ꯭ᒕ꯭ᒔ꯭ᒓ꯭ᒒ꯭ᒑ꯭ᒐ꯭ᒏ꯭ᒎ꯭ᒍ꯭ᒌ꯭ᒋ꯭ᒊ꯭ᒉ꯭ᒈ꯭ᒇ꯭ᒆ꯭ᒅ꯭ᒄ꯭ᒃ꯭ᒂ꯭ᒁ꯭ᒀ꯭ᑿ꯭ᑾ꯭ᑽ꯭ᑼ꯭ᑻ꯭ᑺ꯭ᑹ꯭ᑸ꯭ᑷ꯭ᑶ꯭ᑵ꯭ᑴ꯭ᑳ꯭ᑲ꯭ᑱ꯭ᑰ꯭ᑯ꯭ᑮ꯭ᑭ꯭ᑬ꯭ᑫ꯭ᑪ꯭ᑩ꯭ᑨ꯭ᑧ꯭ᑦ꯭ᑥ꯭ᑤ꯭ᑣ꯭ᑢ꯭ᑡ꯭ᑠ꯭ᑟ꯭ᑞ꯭ᑝ꯭ᑜ꯭ᑛ꯭ᑚ꯭ᑙ꯭ᑘ꯭ᑗ꯭ᑖ꯭ᑕ꯭ᑔ꯭ᑓ꯭ᑒ꯭ᑑ꯭ᑐ꯭ᑏ꯭ᑎ꯭ᑍ꯭ᑌ꯭ᑋ꯭ᑊ꯭ᑉ꯭ᑈ꯭ᑇ꯭ᑆ꯭ᑅ꯭ᑄ꯭ᑃ꯭ᑂ꯭ᑁ꯭ᑀ꯭ᐿ꯭ᐾ꯭ᐽ꯭ᐼ꯭ᐻ꯭ᐺ꯭ᐹ꯭ᐸ꯭ᐷ꯭ᐶ꯭ᐵ꯭ᐴ꯭ᐳ꯭ᐲ꯭ᐱ꯭ᐰ꯭ᐯ꯭ᐮ꯭ᐭ꯭ᐬ꯭ᐫ꯭ᐪ꯭ᐩ꯭ᐨ꯭ᐧ꯭ᐦ꯭ᐥ꯭ᐤ꯭ᐣ꯭ᐢ꯭ᐡ꯭ᐠ꯭ᐟ꯭ᐞ꯭ᐝ꯭ᐜ꯭ᐛ꯭ᐚ꯭ᐙ꯭ᐘ꯭ᐗ꯭ᐖ꯭ᐕ꯭ᐔ꯭ᐓ꯭ᐒ꯭ᐑ꯭ᐐ꯭ᐏ꯭ᐎ꯭ᐍ꯭ᐌ꯭ᐋ꯭ᐊ꯭ᐉ꯭ᐈ꯭ᐇ꯭ᐆ꯭ᐅ꯭ᐄ꯭ᐃ꯭ᐂ꯭ᐁ꯭᐀ᐿᐾᐽᐼᐻᐺᐹᐸᐷᐶᐵᐴᐳᐲᐱᐰᐯᐮᐭᐬᐫᐪᐩᐨᐧᐦᐥᐤᐣᐢᐡᐠᐟᐞᐝᐜᐛᐚᐙᐘᐗᐖᐕᐔᐓᐒᐑᐐᐏᐎᐍᐌᐋᐊᐉᐈᐇᐆᐅᐄᐃᐂᐁ᐀᏿᏾ᏽᏼᏻᏺᏹᏸ᏷᏶ᏵᏴᏳᏲᏱᏰᏯᏮᏭᏬᏫᏪᏩᏨᏧᏦᏥᏤᏣᏢᏡᏠᏟᏞᏝᏜᏛᏚᏙᏘᏗᏖᏕᏔᏓᏒᏑᏐᏏᏎᏍᏌᏋᏊᏉᏈᏇᏆᏅᏄᏃᏂᏁᏀᎿᎾᎽᎼᎻᎺᎹᎸᎷᎶᎵᎴᎳᎲᎱᎰᎯᎮᎭᎬᎫᎪᎩᎨᎧᎦᎥᎤᎣᎢᎡᎠ᎟᎞᎝᎜᎛᎚᎙᎘᎗᎖᎕᎔᎓᎒᎑᎐ᎏᎎᎍᎌᎋᎊᎉᎈᎇᎆᎅᎄᎃᎂᎁᎀ፿፾፽፼፻፺፹፸፷፶፵፴፳፲፱፰፯፮፭፬፫፪፩፨፧፦፥፤፣።፡፠፟፞፝፜፛ፚፙፘፗፖፕፔፓፒፑፐፏፎፍፌፋፊፉፈፇፆፅፄፃፂፁፀጿጾጽጼጻጺጹጸጷጶጵጴጳጲጱጰጯጮጭጬጫጪጩጨጧጦጥጤጣጢጡጠጟጞጝጜጛጚጙጘ጗጖ጕጔጓጒ጑ጐጏጎግጌጋጊጉገጇጆጅጄጃጂጁጀዿዾዽዼዻዺዹዸዷዶድዴዳዲዱደዯዮይዬያዪዩየዧዦዥዤዣዢዡዠዟዞዝዜዛዚዙዘ዗ዖዕዔዓዒዑዐዏዎውዌዋዊዉወ዇዆ዅዄዃዂ዁ዀ኿ኾኽኼኻኺኹኸ኷኶ኵኴኳኲ኱ኰኯኮክኬካኪኩከኧኦእኤኣኢኡአኟኞኝኜኛኚኙኘኗኖንኔናኒኑነ኏኎ኍኌኋኊ኉ኈኇኆኅኄኃኂኁኀቿቾችቼቻቺቹቸቷቶትቴታቲቱተቯቮቭቬቫቪቩቨቧቦብቤባቢቡበ቟቞ቝቜቛቚ቙ቘ቗ቖቕቔቓቒቑቐ቏቎ቍቌቋቊ቉ቈቇቆቅቄቃቂቁቀሿሾሽሼሻሺሹሸሷሶስሴሳሲሱሰሯሮርሬራሪሩረሧሦሥሤሣሢሡሠሟሞምሜማሚሙመሗሖሕሔሓሒሑሐሏሎልሌላሊሉለሇሆህሄሃሂሁሀᇿᇾᇽᇼᇻᇺᇹᇸᇷᇶᇵᇴᇳᇲᇱᇰᇯᇮᇭᇬᇫᇪᇩᇨᇧᇦᇥᇤᇣᇢᇡᇠᇟᇞᇝᇜᇛᇚᇙᇘᇗᇖᇕᇔᇓᇒᇑᇐᇏᇎᇍᇌᇋᇊᇉᇈᇇᇆᇅᇄᇃᇂᇁᇀ⁖꞉·˙ostensive꞉·

·⁖origin…language꞉·Tangut·⁖꞉·˙origin…language꞉·
·⁖font꞉·Noto Serif Tangut·⁖꞉·˙font꞉·
·⁖pronunciation꞉·BEH·⁖꞉·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖꞉·˙cadence꞉·
·⁖non…ostensive꞉·Promote or raise⁖꞉ also proud or arrogant·⁖꞉·˙non…ostensive꞉·
·⁖ostensive꞉·Lever bites beneath the load⁖꞉Downward force on one end drives the weight
↪ upwardᘙ⁖꞉·˙ostensive꞉·
·⁖grimchain꞉·ᐯΘ꯭ᘓ꯭ᘐ꯭ᘏ꯭ᘔ꯭ᘎ꯭ᘋ꯭ᘍ꯭ᘉ꯭ᘈ꯭ᘇ꯭ᘆ꯭ᘅ꯭ᘄ꯭ᘃ꯭ᘂ꯭ᘁ꯭ᘀ꯭ᗿ꯭ᗾ꯭ᗽ꯭ᗼ꯭ᗻ꯭ᗺ꯭ᗹ꯭ᗸ꯭ᗷ꯭ᗶ꯭ᗵ꯭ᗴ꯭ᗳ꯭ᗲ꯭ᗱ꯭ᗰ꯭ᗯ꯭ᗮ꯭ᗭ꯭ᗬ꯭ᗫ꯭ᗪ꯭ᗩ꯭ᗨ꯭ᗧ꯭ᗦ꯭ᗥ꯭ᗤ꯭ᗣ꯭ᗢ꯭ᗡ꯭ᗠ꯭ᗟ꯭ᗞ꯭ᗝ꯭ᗜ꯭ᗛ꯭ᗚ꯭ᗙ꯭ᗘ꯭ᗗ꯭ᗖ꯭ᗕ꯭ᗔ꯭ᗓ꯭ᗒ꯭ᗑ꯭ᗐ꯭ᗏ꯭ᗎ꯭ᗍ꯭ᗌ꯭ᗋ꯭ᗊ꯭ᗉ꯭ᗈ꯭ᗇ꯭ᗆ꯭ᗅ꯭ᗄ꯭ᗃ꯭ᗂ꯭ᗁ꯭ᗀ꯭ᕿ꯭ᕾ꯭ᕽ꯭ᕼ꯭ᕻ꯭ᕺ꯭ᕹ꯭ᕸ꯭ᕷ꯭ᕶ꯭ᕵ꯭ᕴ꯭ᕳ꯭ᕲ꯭ᕱ꯭ᕰ꯭ᕯ꯭ᕮ꯭ᕭ꯭ᕬ꯭ᕫ꯭ᕪ꯭ᕩ꯭ᕨ꯭ᕧ꯭ᕦ꯭ᕥ꯭ᕤ꯭ᕣ꯭ᕢ꯭ᕡ꯭ᕠ꯭ᕟ꯭ᕞ꯭ᕝ꯭ᕜ꯭ᕛ꯭ᕚ꯭ᕙ꯭ᕘ꯭ᕗ꯭ᕖ꯭ᕕ꯭ᕔ꯭ᕓ꯭ᕒ꯭ᕑ꯭ᕐ꯭ᕏ꯭ᕎ꯭ᕍ꯭ᕌ꯭ᕋ꯭ᕊ꯭ᕉ꯭ᕈ꯭ᕇ꯭ᕆ꯭ᕅ꯭ᕄ꯭ᕃ꯭ᕂ꯭ᕁ꯭ᕀ꯭ᔿ꯭ᔾ꯭ᔽ꯭ᔼ꯭ᔻ꯭ᔺ꯭ᔹ꯭ᔸ꯭ᔷ꯭ᔶ꯭ᔵ꯭ᔴ꯭ᔳ꯭ᔲ꯭ᔱ꯭ᔰ꯭ᔯ꯭ᔮ꯭ᔭ꯭ᔬ꯭ᔫ꯭ᔪ꯭ᔩ꯭ᔨ꯭ᔧ꯭ᔦ꯭ᔥ꯭ᔤ꯭ᔣ꯭ᔢ꯭ᔡ꯭ᔠ꯭ᔟ꯭ᔞ꯭ᔝ꯭ᔜ꯭ᔛ꯭ᔚ꯭ᔙ꯭ᔘ꯭ᔗ꯭ᔖ꯭ᔕ꯭ᔔ꯭ᔓ꯭ᔒ꯭ᔑ꯭ᔐ꯭ᔏ꯭ᔎ꯭ᔍ꯭ᔌ꯭ᔋ꯭ᔊ꯭ᔉ꯭ᔈ꯭ᔇ꯭ᔆ꯭ᔅ꯭ᔄ꯭ᔃ꯭ᔂ꯭ᔁ꯭ᔀ꯭ᓿ꯭ᓾ꯭ᓽ꯭ᓼ꯭ᓻ꯭ᓺ꯭ᓹ꯭ᓸ꯭ᓷ꯭ᓶ꯭ᓵ꯭ᓴ꯭ᓳ꯭ᓲ꯭ᓱ꯭ᓰ꯭ᓯ꯭ᓮ꯭ᓭ꯭ᓬ꯭ᓫ꯭ᓪ꯭ᓩ꯭ᓨ꯭ᓧ꯭ᓦ꯭ᓥ꯭ᓤ꯭ᓣ꯭ᓢ꯭ᓡ꯭ᓠ꯭ᓟ꯭ᓞ꯭ᓝ꯭ᓜ꯭ᓛ꯭ᓚ꯭ᓙ꯭ᓘ꯭ᓗ꯭ᓖ꯭ᓕ꯭ᓔ꯭ᓓ꯭ᓒ꯭ᓑ꯭ᓐ꯭ᓏ꯭ᓎ꯭ᓍ꯭ᓌ꯭ᓋ꯭ᓊ꯭ᓉ꯭ᓈ꯭ᓇ꯭ᓆ꯭ᓅ꯭ᓄ꯭ᓃ꯭ᓂ꯭ᓁ꯭ᓀ꯭ᒿ꯭ᒾ꯭ᒽ꯭ᒼ꯭ᒻ꯭ᒺ꯭ᒹ꯭ᒸ꯭ᒷ꯭ᒶ꯭ᒵ꯭ᒴ꯭ᒳ꯭ᒲ꯭ᒱ꯭ᒰ꯭ᒯ꯭ᒮ꯭ᒭ꯭ᒬ꯭ᒫ꯭ᒪ꯭ᒩ꯭ᒨ꯭ᒧ꯭ᒦ꯭ᒥ꯭ᒤ꯭ᒣ꯭ᒢ꯭ᒡ꯭ᒠ꯭ᒟ꯭ᒞ꯭ᒝ꯭ᒜ꯭ᒛ꯭ᒚ꯭ᒙ꯭ᒘ꯭ᒗ꯭ᒖ꯭ᒕ꯭ᒔ꯭ᒓ꯭ᒒ꯭ᒑ꯭ᒐ꯭ᒏ꯭ᒎ꯭ᒍ꯭ᒌ꯭ᒋ꯭ᒊ꯭ᒉ꯭ᒈ꯭ᒇ꯭ᒆ꯭ᒅ꯭ᒄ꯭ᒃ꯭ᒂ꯭ᒁ꯭ᒀ꯭ᑿ꯭ᑾ꯭ᑽ꯭ᑼ꯭ᑻ꯭ᑺ꯭ᑹ꯭ᑸ꯭ᑷ꯭ᑶ꯭ᑵ꯭ᑴ꯭ᑳ꯭ᑲ꯭ᑱ꯭ᑰ꯭ᑯ꯭ᑮ꯭ᑭ꯭ᑬ꯭ᑫ꯭ᑪ꯭ᑩ꯭ᑨ꯭ᑧ꯭ᑦ꯭ᑥ꯭ᑤ꯭ᑣ꯭ᑢ꯭ᑡ꯭ᑠ꯭ᑟ꯭ᑞ꯭ᑝ꯭ᑜ꯭ᑛ꯭ᑚ꯭ᑙ꯭ᑘ꯭ᑗ꯭ᑖ꯭ᑕ꯭ᑔ꯭ᑓ꯭ᑒ꯭ᑑ꯭ᑐ꯭ᑏ꯭ᑎ꯭ᑍ꯭ᑌ꯭ᑋ꯭ᑊ꯭ᑉ꯭ᑈ꯭ᑇ꯭ᑆ꯭ᑅ꯭ᑄ꯭ᑃ꯭ᑂ꯭ᑁ꯭ᑀ꯭ᐿ꯭ᐾ꯭ᐽ꯭ᐼ꯭ᐻ꯭ᐺ꯭ᐹ꯭ᐸ꯭ᐷ꯭ᐶ꯭ᐵ꯭ᐴ꯭ᐳ꯭ᐲ꯭ᐱ꯭ᐰ꯭ᐯ꯭ᐮ꯭ᐭ꯭ᐬ꯭ᐫ꯭ᐪ꯭ᐩ꯭ᐨ꯭ᐧ꯭ᐦ꯭ᐥ꯭ᐤ꯭ᐣ꯭ᐢ꯭ᐡ꯭ᐠ꯭ᐟ꯭ᐞ꯭ᐝ꯭ᐜ꯭ᐛ꯭ᐚ꯭ᐙ꯭ᐘ꯭ᐗ꯭ᐖ꯭ᐕ꯭ᐔ꯭ᐓ꯭ᐒ꯭ᐑ꯭ᐐ꯭ᐏ꯭ᐎ꯭ᐍ꯭ᐌ꯭ᐋ꯭ᐊ꯭ᐉ꯭ᐈ꯭ᐇ꯭ᐆ꯭ᐅ꯭ᐄ꯭ᐃ꯭ᐂ꯭ᐁ꯭᐀ᐿᐾᐽᐼᐻᐺᐹᐸᐷᐶᐵᐴᐳᐲᐱᐰᐯᐮᐭᐬᐫᐪᐩᐨᐧᐦᐥᐤᐣᐢᐡᐠᐟᐞᐝᐜᐛᐚᐙᐘᐗᐖᐕᐔᐓᐒᐑᐐᐏᐎᐍᐌᐋᐊᐉᐈᐇᐆᐅᐄᐃᐂᐁ᐀᏿᏾ᏽᏼᏻᏺᏹᏸ᏷᏶ᏵᏴᏳᏲᏱᏰᏯᏮᏭᏬᏫᏪᏩᏨᏧᏦᏥᏤᏣᏢᏡᏠᏟᏞᏝᏜᏛᏚᏙᏘᏗᏖᏕᏔᏓᏒᏑᏐᏏᏎᏍᏌᏋᏊᏉᏈᏇᏆᏅᏄᏃᏂᏁᏀᎿᎾᎽᎼᎻᎺᎹᎸᎷᎶᎵᎴᎳᎲᎱᎰᎯᎮᎭᎬᎫᎪᎩᎨᎧᎦᎥᎤᎣᎢᎡᎠ᎟᎞᎝᎜᎛᎚᎙᎘᎗᎖᎕᎔᎓᎒᎑᎐ᎏᎎᎍᎌᎋᎊᎉᎈᎇᎆᎅᎄᎃᎂᎁᎀ፿፾፽፼፻፺፹፸፷፶፵፴፳፲፱፰፯፮፭፬፫፪፩፨፧፦፥፤፣።፡፠፟፞፝፜፛ፚፙፘፗፖፕፔፓፒፑፐፏፎፍፌፋፊፉፈፇፆፅፄፃፂፁፀጿጾጽጼጻጺጹጸጷጶጵጴጳጲጱጰጯጮጭጬጫጪጩጨጧጦጥጤጣጢጡጠጟጞጝጜጛጚጙጘ጗጖ጕጔጓጒ጑ጐጏጎግጌጋጊጉገጇጆጅጄጃጂጁጀዿዾዽዼዻዺዹዸዷዶድዴዳዲዱደዯዮይዬያዪዩየዧዦዥዤዣዢዡዠዟዞዝዜዛዚዙዘ዗ዖዕዔዓዒዑዐዏዎውዌዋዊዉወ዇዆ዅዄዃዂ዁ዀ኿ኾኽኼኻኺኹኸ኷኶ኵኴኳኲ኱ኰኯኮክኬካኪኩከኧኦእኤኣኢኡአኟኞኝኜኛኚኙኘኗኖንኔናኒኑነ኏኎ኍኌኋኊ኉ኈኇኆኅኄኃኂኁኀቿቾችቼቻቺቹቸቷቶትቴታቲቱተቯቮቭቬቫቪቩቨቧቦብቤባቢቡበ቟቞ቝቜቛቚ቙ቘ቗ቖቕቔቓቒቑቐ቏቎ቍቌቋቊ቉ቈቇቆቅቄቃቂቁቀሿሾሽሼሻሺሹሸሷሶስሴሳሲሱሰሯሮርሬራሪሩረሧሦሥሤሣሢሡሠሟሞምሜማሚሙመሗሖሕሔሓሒሑሐሏሎልሌላሊሉለሇሆህሄሃሂሁሀᇿᇾᇽᇼᇻᇺᇹᇸᇷᇶᇵᇴᇳᇲᇱᇰᇯᇮᇭᇬᇫᇪᇩᇨᇧᇦᇥᇤᇣᇢᇡᇠᇟᇞᇝᇜᇛᇚᇙᇘᇗᇖᇕᇔᇓᇒᇑᇐᇏᇎᇍᇌᇋᇊᇉᇈᇇᇆᇅᇄᇃᇂᇁᇀ⁖꞉·˙grimchain꞉·
·⁖꞉·˙lookup꞉·
·⁖꞉·˙glyphs꞉·
·⁖glyph symbol꞉·ᘙ꞉ category꞉·court꞉·
·⁖lookup office꞉·catalyst꞉ enoch꞉·ᘐ꞉·

˖word꞉·ᚺᛒᚾ˖꞉·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖꞉·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖꞉·˙font꞉·
·⁖pronunciation꞉·SPAH·⁖꞉·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖꞉·˙cadence꞉·
·⁖non…ostensive꞉·a prophecy·⁖꞉·˙non…ostensive꞉·
·⁖ostensive꞉·An untimely darkening crosses the sky⁖꞉Words of what has not yet happened rise in
↪ answerᚸ⁖꞉·˙ostensive꞉·
·⁖grimchain꞉·ᛀᛒᛇᚸᚱᛕ꯭ᚿ꯭ᚼ꯭ᚻ꯭ᚺ꯭ᚹ꯭ᚸ꯭ᚷ꯭ᚶ꯭ᚵ꯭ᚴ꯭ᚳ꯭ᚲ꯭ᚱ꯭ᚰ꯭ᚯ꯭ᚮ꯭ᚭ꯭ᚬ꯭ᚫ꯭ᚪ꯭ᚩ꯭ᚨ꯭ᚧ꯭ᚦ꯭ᚥ꯭ᚤ꯭ᚣ꯭ᚢ꯭ᚡ꯭ᚠ꯭᛿꯭᛾꯭᛽꯭᛼꯭᛻꯭᛺꯭᛹꯭ᛸ꯭ᛷ꯭ᛶ꯭ᛵ꯭ᛴ꯭ᛳ꯭ᛲ꯭ᛱ꯭ᛰ꯭ᛯ꯭ᛮ꯭᛭꯭᛬꯭᛫꯭ᛪ꯭ᛩ꯭ᛨ꯭ᛧ꯭ᛦ꯭ᛥ꯭ᛤ꯭ᛣ꯭ᛢ꯭ᛡ꯭ᛠ꯭ᛟ꯭ᛞ꯭ᛝ꯭ᛜ꯭ᛛ꯭ᛚ꯭ᛙ꯭ᛘ꯭ᛗ꯭ᛖ꯭ᛕ꯭ᛔ꯭ᛓ꯭ᛒ꯭ᛑ꯭ᛐ꯭ᛏ꯭ᛎ꯭ᛍ꯭ᛌ꯭ᛋ꯭ᛊ꯭ᛉ꯭ᛈ꯭ᛇ꯭ᛆ꯭ᛅ꯭ᛄ꯭ᛃ꯭ᛂ꯭ᛁ꯭ᛀ꯭ᚿᚼᚻᚺᚹᚸᚷᚶᚵᚴᚳᚲᚱᚰᚯᚮᚭᚬᚫᚪᚩᚨᚧᚦᚥᚤᚣᚢᚡᚠ᛿᛾᛽᛼᛻᛺᛹ᛸᛷᛶᛵᛴᛳᛲᛱᛰᛯᛮ᛭᛬᛫ᛪᛩᛨᛧᛦᛥᛤᛣᛢᛡᛠᛟᛞᛝᛜᛛᛚᛙᛘᛗᛖᛕᛔᛓᛒᛑᛐᛏᛎᛍᛌᛋᛊᛉᛈᛇᛆᛅᛄᛃᛂᛁᛀ⁖꞉·˙grimchain꞉·
·⁖꞉·˙lookup꞉·
·⁖꞉·˙glyphs꞉·
·⁖glyph symbol꞉·ᚸ꞉ category꞉·court꞉·
·⁖lookup office꞉·catalyst꞉ enoch꞉·ᚻ꞉·

˖word꞉·ϵι˖꞉·˙word꞉·

·⁖origin…language꞉·Coptic·⁖꞉·˙origin…language꞉·
·⁖font꞉·Noto Sans Coptic·⁖꞉·˙font꞉·
·⁖pronunciation꞉·AY·⁖꞉·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖꞉·˙cadence꞉·
·⁖non…ostensive꞉·To come·⁖꞉·˙non…ostensive꞉·
·⁖ostensive꞉·A summons carries across the open way and distant footsteps turn toward the
↪ thresholdᘓ⁖꞉·˙ostensive꞉·

·⁖grimchain:·ꟻᛉIⵊᒧᒪ⋖ᒥ۞ᛝ🙰ᛏ▽ᚼᐁ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𐑈 category◦–◦court𐑉:·
·⁖lookup office◦–◦catalyst𐑉 enoch◦–◦𐑴𐑉:·

·word:·ᵬʌᛋNᔑᛋN·⁖·˙word:·

·⁖origin…language:·Gothic·⁖·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖·˙font:·
·⁖pronunciation:·BLANDAN·⁖·˙pronunciation:·
·⁖cadence:·Catalyst·⁖·˙cadence:·
·⁖non…ostensive:·To mix::to blend:˙ to mingle::to intermix·⁖·˙non…ostensive:·
·⁖ostensive:·A stirring hand drives two substances through one another until the clean line between
↪ them disappears𐑓·⁖·˙ostensive:·
·⁖grimchain:·ᦵ◌𐑓ᘐᚱᚸ𐑅✳ᘏᛊ⁉ᘜᘐ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𐑓𐑉 category◦–◦court𐑉:·
·⁖lookup office◦–◦catalyst𐑉 enoch◦–◦𐑴𐑉:·

·word:·ᵬЧᑫ·⁖·˙word:·

·⁖origin…language:·Aghwan·⁖·˙origin…language:·
·⁖font:·Noto Sans Caucasian Albanian·⁖·˙font:·
·⁖pronunciation:·BIX·⁖·˙pronunciation:·
·⁖cadence:·Catalyst·⁖·˙cadence:·
·⁖non…ostensive:·Create·⁖·˙non…ostensive:·
·⁖ostensive:·Raw matter meets the shaping hand::Edge and surface appear where no made thing had
↪ stood𐑕·⁖·˙ostensive:·
·⁖grimchain:·⏏ᛞᚱᘜᘐ▽▫✳ᛉ𐑅ᛎᑭ♀·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𐑥𐑉 category◦–◦court𐑉:·
·⁖lookup office◦–◦catalyst𐑉 enoch◦–◦𐑴𐑉:·

·word:·ᴛʌᕼᏀʌNᔑᛋN·⁖·˙word:·

·⁖origin…language:·Gothic·⁖·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖·˙font:·
·⁖pronunciation:·TAX⟊JAN·⁖·˙pronunciation:·
·⁖cadence:·Catalyst·⁖·˙cadence:·
·⁖non…ostensive:·To tear::rend::rip·⁖·˙non…ostensive:·
·⁖ostensive:·Tension runs through cloth or flesh until the held body opens along the
↪ strain𐑥·⁖·˙ostensive:·
·⁖grimchain:·ᚦᑭᗺᛎᑭ‹𐑓ᛞᛞᚣ0⌬ᗡ·⁖·˙grimchain:·
·⁖·˙lookup:˙
·⁖·˙glyphs:˙
·⁖glyph symbol◦–◦𐑥𐑉 category◦–◦court𐑉:·
·⁖lookup office◦–◦catalyst𐑉 enoch◦–◦𐑴𐑉:·

·word:·ᚻᛦʌISᔑᛋN·⁖·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·FREYESAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To try::to attempt·⁚ to tempt::to test::to dare·⁚.·˙non…ostensive:·
·⁚ostensive:·An unyielding barrier refuses the first push::Force gathers and presses once
↪ moreᔕ·⁚.·˙ostensive:·
·⁚grimchain:·☽ᘐX𐋡ᚺᚿᚨᛯᛊᛩᛗᒧ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀꜀ı꜀ category꜀꜀꜀court꜀:·
·⁚lookup office꜀꜀꜀catalyst꜀ enoch꜀꜀꜀ᘐ꜀:·

⁖word:·ᚺᑎZᙏᏩᏁᑎ⁖·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·HUZDJAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To hoard::store up·⁚.·˙non…ostensive:·
·⁚ostensive:·Fear of empty days sends grain and oil behind one guarded door::bundle after
↪ bundleı·⁚.·˙ostensive:·
·⁚grimchain:·☆ᚗ⊬ᘐ𝙓ᓰΘᘁ⊬ᓀᛯᔑ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀꜀꜀ı꜀ category꜀꜀꜀court꜀:·
·⁚lookup office꜀꜀꜀catalyst꜀ enoch꜀꜀꜀ᘐ꜀:·

⁖word:·ᚡᏁᏉᚿᏁᑎ⁖·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·FALTHAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To fold·⁚.·˙non…ostensive:·
·⁚ostensive:·Thumb presses a crease into cloth::The far edge swings back to meet its own
↪ faceɬ·⁚.·˙ostensive:·
·⁚grimchain:·ᛯᔑᚸ𐋡ᘐ➢ᘐᒣᘐᚮᔑ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀꜀꜀▫꜀ category꜀꜀꜀court꜀:·
·⁚lookup office꜀꜀꜀catalyst꜀ enoch꜀꜀꜀ᘐ꜀:·

⁖word:·ᛋᛁᑎᑭɬ⁖·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·SJOHTHA·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to seethe::boil·⁚ cook·⁚.·˙non…ostensive:·
·⁚ostensive:·Heat enters the pot::bubbles lift::and food softens in the rolling water▫·⁚.·˙ostensive:·
·⁚grimchain:·ᚼᛉᛇᛤᛗᓰᘐᖘᚾᛅᚴᛯ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁖glyph symbol꞉꞉𐎍꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐎍꞉·

⁖word꞉·𒀸𒄑𒉿𒅔⁖·˙word꞉·

·⁖origin…language꞉·Hittite·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖·˙font꞉·
·⁖pronunciation꞉·PADDAHI·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·
·⁖non…ostensive꞉·To dig·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Earth breaks beneath the tool⁖⁖Depth opens where the surface had been whole𐎍·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐎍≍𐤃𐤉𐤒𐤍𐤕Ⲏ↗♂ϒⲎϒ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉꞉♪꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐎍꞉·

⁖word꞉·ᛗᛖᚸᚷᚴᚾ⁖·˙word꞉·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·MELJAN·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·
·⁖non…ostensive꞉·To write·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Pen-tip drags across prepared skin⁖⁖leaving dark signs wherever pressure
↪ passes♪·⁖·˙ostensive꞉·
·⁖grimchain꞉·ϧ꞉𐤓ᛇᛩᛉᚷᛞ♀Ↄℂ☿≍·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉꞉Ⲩ꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐎍꞉·

⁖word꞉·ᛃᚴᛁᚸᚷᚴᚾ⁖·˙word꞉·

·⁖origin…language꞉·Gothic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Gothic·⁖·˙font꞉·
·⁖pronunciation꞉·DEHLJAN·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·
·⁖non…ostensive꞉·To share⁖⁖distribute⁖⁖divide⁖⁖portion·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Several waiting bowls surround one heap⁖⁖Measured cuts send portions outward into each
↪ vesselϒ·⁖·˙ostensive꞉·
·⁖grimchain꞉·⧗𐎍𐤎𐤓⁕0𐤔𐤍𐤓♋𐤉⊛𐎍·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉꞉𐎎꞉ category꞉꞉court꞉·
·⁖lookup office꞉꞉catalyst꞉ enoch꞉꞉𐎍꞉·

⁖word꞉·ሞተ⁖·˙word꞉·

·⁖origin…language꞉·Ge'ez·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖·˙font꞉·
·⁖pronunciation꞉·MOH-tah·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·
·⁖non…ostensive꞉·To die꞉˙ to cease living·⁖·˙non…ostensive꞉·

·⁚ostensive⁚·A deep wound opens⁚· breath falters and warmth leaves the flesh⁖·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐦝𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿⁚·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐦤₎ category○–○𐦲court₎⁚·
·⁚lookup office○–○𐦲catalyst₎ enoch○–○𐦵₎⁚·

⁖word⁚·𐦑𐦒𐦓𐦔𐦕𐦖𐦗𐦘𐦙𐦚𐦛𐦜𐦝𐦞𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Akkadian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·shah+RAH+koom·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To steal⁚⁚to act as a thief⁚· to kidnap·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Broken fastening leaves the guarded weight exposed⁚⁚A concealed hand lifts it from its
↪ place𐦵·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐦀𐦁𐦂𐦃𐦄𐦅𐦆𐦇𐦈𐦉𐦊𐦋𐦌𐦍𐦎𐦏𐦐𐦑𐦒𐦓𐦔𐦕𐦖𐦗𐦘𐦙𐦚𐦛𐦜𐦝𐦞𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿⁚·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐦵₎ category○–○𐦲court₎⁚·
·⁚lookup office○–○𐦲catalyst₎ enoch○–○𐦵₎⁚·

⁖word⁚·𐦀𐦁𐦂𐦃𐦄𐦅𐦆𐦇𐦈𐦉𐦊𐦋𐦌𐦍𐦎𐦏𐦐𐦑𐦒𐦓𐦔𐦕𐦖𐦗𐦘𐦙𐦚𐦛𐦜𐦝𐦞𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚.·˙font⁚·
·⁚pronunciation⁚·TOOT·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To catch⁚⁚grasp⁚⁚hold·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·A flying cord strikes an open palm⁚⁚and fingers close before it can pass
↪ through'·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿⁚·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐦴₎ category○–○𐦲court₎⁚·
·⁚lookup office○–○𐦲catalyst₎ enoch○–○𐦵₎⁚·

⁖word⁚·𐦀𐦁𐦂𐦃𐦄𐦅𐦆𐦇𐦈𐦉𐦊𐦋𐦌𐦍𐦎𐦏𐦐𐦑𐦒𐦓𐦔𐦕𐦖𐦗𐦘𐦙𐦚𐦛𐦜𐦝𐦞𐦟𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚.·˙font⁚·
·⁚pronunciation⁚·AYT·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To organize⁚⁚make·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·A frame is set around scattered parts⁚⁚Piece after piece is drawn into place until the
↪ separate motions work together𐦵·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐦠𐦡𐦢𐦣𐦤𐦥𐦦𐦧𐦨𐦩𐦪𐦫𐦬𐦭𐦮𐦯𐦰𐦱𐦲𐦳𐦴𐦵𐦶𐦷𐦸𐦹𐦺𐦻𐦼𐦽𐦾𐦿𐧀𐧁𐧂𐧃𐧄𐧅𐧆𐧇𐧈𐧉𐧊𐧋𐧌𐧍𐧎𐧏𐧐𐧑𐧒𐧓𐧔𐧕𐧖𐧗𐧘𐧙𐧚𐧛𐧜𐧝𐧞𐧟𐧠𐧡𐧢𐧣𐧤𐧥𐧦𐧧𐧨𐧩𐧪𐧫𐧬𐧭𐧮𐧯𐧰𐧱𐧲𐧳𐧴𐧵𐧶𐧷𐧸𐧹𐧺𐧻𐧼𐧽𐧾𐧿⁚·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐦶₎ category○–○𐦲court₎⁚·
·⁚lookup office○–○𐦲catalyst₎ enoch○–○𐦵₎⁚·

⁖word⁚·+□ı̃ı̃̃⁖·⁚.·˙word⁚·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·KUHRS·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To know::understand::recognize·⁚· (causative) to make known·⁚.·˙non…ostensive:·
·⁚ostensive:·A familiar sign catches the eye and hesitation stops as the old pattern answers at
↪ once↓·⁚.·˙ostensive:·
·⁚grimchain:·ᚠᛕ🜲ᛪ⚷⁰ᚿᗐᛈ⁰▽ᚺ✡·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈↑𐑉 category◦–◦𐑈court𐑉⁚·
·⁚lookup office◦–◦𐑈catalyst𐑉 enoch◦–◦𐑈♂𐑉⁚·

·word:·𑀮𑁂·⁚.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·TAH·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·Close·⁚· to shut an opening or prevent passage·⁚.·˙non…ostensive:·
·⁚ostensive:·A bar drops across the doorway::driving both leaves shut until passage
↪ disappears↿·⁚.·˙ostensive:·
·⁚grimchain:·𓄿ᛯ𐑌ᛉᛘ𐑈𐑋♂ᚲ🜲↿ᗡ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈𐑐𐑉 category◦–◦𐑈court𐑉⁚·
·⁚lookup office◦–◦𐑈catalyst𐑉 enoch◦–◦𐑈♂𐑉⁚·

·word:·𓀠·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·wah⟊RAH⟊koom·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to be(come) green::yellow::to turn pale·⁚.·˙non…ostensive:·
·⁚ostensive:·Heat and drying reach the leaf·⁚· its deep color drains toward the pale tone of
↪ strawᛈ·⁚.·˙ostensive:·
·⁚grimchain:·ϒΨᛘ℘ᗡ〰☆◌ᔑᛏᛕᓄᒐ⧗ᚻ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈ᑎ𐑉 category◦–◦𐑈court𐑉⁚·
·⁚lookup office◦–◦𐑈catalyst𐑉 enoch◦–◦𐑈♂𐑉⁚·

·word:·𓆟𓆝·⁚.·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·YEE⟊ree·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To do·⁚.·˙non…ostensive:·
·⁚ostensive:·Unfinished work waits under the hand::The first deliberate stroke breaks the
↪ stillnessᑎ·⁚.·˙ostensive:·
·⁚grimchain:·🜲ᚠᛘ🜲ᔓᚣ₀ᔨ⊖⁕✳⊖·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ᵩ꩝ category꩜꩜ᵩcourt꩝⁚·
·⁚lookup office꩜꩜ᵩcatalyst꩝ enoch꩜꩜ᵩ𐑂꩝⁚·

⁖word⁚·𒁲𒌷𒉿𒄩⁖·⁚·˙word⁚·

·⁚origin…language⁚·Hittite·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·PARAHI·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To blow (a horn)·⁚ to blow on⁚⁚fan (fire or burning materials)·⁚ to blow
↪ up⁚⁚inflate·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Breath drives through horn or ember⁚⁚and hidden sound or heat answers the moving
↪ air᛫·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚸᛯ❁ᚴ⏁ᛞᚱᛟᛗᛒᛋᛊ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ᵩᚠ꩝ category꩜꩜ᵩcourt꩝⁚·
·⁚lookup office꩜꩜ᵩcatalyst꩝ enoch꩜꩜ᵩ𐑂꩝⁚·

⁖word⁚·ማይ⁖·⁚·˙word⁚·

·⁚origin…language⁚·Ge'ez·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Ethiopic·⁚·˙font⁚·
·⁚pronunciation⁚·MY·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Water·⁚ the common transparent liquid essential to life·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Stone breaks the sealed spring⁚⁚Clear liquid bursts from the opened earthᚨ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᓚᙢᕀᐁ卫ᗐᕀᐢᐤᚧᚨᚲ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ᵩᚱ꩝ category꩜꩜ᵩcourt꩝⁚·
·⁚lookup office꩜꩜ᵩcatalyst꩝ enoch꩜꩜ᵩ𐑂꩝⁚·

⁖word⁚·ᚴᚢᚺᛏᚼ⁖·⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·KOHS+tah·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·to try·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Locked wood refuses the first push⁚⁚Force gathers and presses againᚱ·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐦟𐦝𐦂ᚸᚹᚦᛯᛋᛞᛗ𐦂·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ᵩ‘꩝ category꩜꩜ᵩcourt꩝⁚·
·⁚lookup office꩜꩜ᵩcatalyst꩝ enoch꩜꩜ᵩ𐑂꩝⁚·

⁖word⁚·ᚺᚣᚴᎶᚣᚿ⁖·⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·HAFJAN·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·

·⁖non…ostensive⁚·To lift⁚⁚raise⁚⁚hold up·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Lever catches beneath the weight⁚⁚Downward force on one end drives the body
↪ upward‹·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᒧᛘ₀ᒣᚸᛯ⊕⊖ᴢ⧑ⵠ⧐·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol○–◦X› category○–◦court›⁚·
·⁖lookup office○–◦catalyst› enoch○–◦☌›⁚·

‹word⁚·ᛒᑎᚱᛁᚼ⁖.·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖.·˙font⁚·
·⁖pronunciation⁚·BUR+yah·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·To begin or originate·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·A first signal breaks the stillness⁚⁚Waiting motion beginsX·⁖.·˙ostensive⁚·
·⁖grimchain⁚·ᔐᚻ⊕ᓄᛘ◎ᚹ⊕ᛉ≋ᛘᚠᘐ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol○–◦ᚹ› category○–◦court›⁚·
·⁖lookup office○–◦catalyst› enoch○–◦☌›⁚·

‹word⁚·⏥⏤⁖.·˙word⁚·

·⁖origin…language⁚·Akkadian·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Cuneiform·⁖.·˙font⁚·
·⁖pronunciation⁚·ah+KAH+loom·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·to eat⁚⁚consume·⁖ to use⁚⁚to have usufruct of·⁖ to pocket⁚⁚(to take for
↪ oneself)·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Hunger tightens the belly·⁖ food is drawn from the hand into the mouthᚹ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·⩒⬡ᒧᛉ⏢⏤ᘔᚻᚸᚷᛯᚷⵠᑎᚴᚸᘕ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol○–◦ᚺ› category○–◦court›⁚·
·⁖lookup office○–◦catalyst› enoch○–◦☌›⁚·

‹word⁚·ᑭᚼᛋᑭᚼ⁖.·˙word⁚·

·⁖origin…language⁚·Old Norse·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁖.·˙font⁚·
·⁖pronunciation⁚·VASKA·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·to wash·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·Water passes over the soiled surface⁚⁚and loosened dirt leaves with the flowing
↪ bodyᚺ·⁖.·˙ostensive⁚·
·⁖grimchain⁚·⯎ᚿᚷ⏁ᚼⵊᚺᚴ⏤⧐⧐ᚱ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol○–◦ᚾ› category○–◦court›⁚·
·⁖lookup office○–◦catalyst› enoch○–◦☌›⁚·

‹word⁚·ᘐᘐᔑᛋ⁖.·˙word⁚·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·QARI·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·Dry·⁚.·˙non…ostensive:·
·⁚ostensive:·Sun and dry wind draw moisture from the surface until the material tightens·⁚.·˙ostensive:·
·⁚grimchain:·𐓣𐒰𐒻𐒼𐒽𐒾𐒿𐓀𐓁𐓂𐓃·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꞉꞉𐓂 category꞉꞉court𐓃⁚·
·⁚lookup office꞉꞉catalyst𐓃 enoch꞉꞉𐓅𐓃⁚·

·word:·ᚺᚢᛠ·⁚.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·zah+BAH+tah·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To beat::hit::strike::or blow·⁚.·˙non…ostensive:·
·⁚ostensive:·Hammer falls under gathered force::The struck surface jumps at contact·⁚.·˙ostensive:·
·⁚grimchain:·ᓚ0◎∀ᚻᛕ卌o'ᛯᑐᛘX·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꞉꞉𐓄 category꞉꞉court𐓃⁚·
·⁚lookup office꞉꞉catalyst𐓃 enoch꞉꞉𐓅𐓃⁚·

·word:·ϬΟΘΣΟΣᒍhΟϮϤ·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·BOKOKESUN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To burn·⁚.·˙non…ostensive:·
·⁚ostensive:·Flame takes the dry body::and heat runs through what had held its shape𐓄·⁚.·˙ostensive:·
·⁚grimchain:·ᛕᛗᛯᙆᘎᘓᔕᔓᔐᘐᚾᚠᐁ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꞉꞉𐓆 category꞉꞉court𐓃⁚·
·⁚lookup office꞉꞉catalyst𐓃 enoch꞉꞉𐓅𐓃⁚·

·word:·ᏂᎷϤ·⁚.·˙word:·

·⁚origin…language:·Old Turkic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚.·˙font:·
·⁚pronunciation:·TEUK·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To pour·⁚.·˙non…ostensive:·
·⁚ostensive:·Vessel tilts past balance::and liquid spills over the rim in one falling
↪ stream𐓆·⁚.·˙ostensive:·
·⁚grimchain:·ɦ∱Ψ⩛㐂ᘻᑫᘔ⬡𐓆ᚹ·⁚.·˙grimchain:·
·⁚.·˙lookup:·˙
·⁚.·˙glyphs:·˙
·⁚glyph symbol꞉꞉𐓅 category꞉꞉court𐓃⁚·
·⁚lookup office꞉꞉catalyst𐓃 enoch꞉꞉𐓅𐓃⁚·

˸word:·ᛒᚠᚾᚱᚾ˸·˙word:·

·˙origin…language:·Old Norse·˙·˙origin…language:·
·˙font:·Noto Sans Runic·˙·˙font:·
·˙pronunciation:·BLANDA·˙·˙pronunciation:·
·˙cadence:·Catalyst·˙·˙cadence:·
·˙non…ostensive:·to blend꞉꞉mix·˙·˙non…ostensive:·
·˙ostensive:·A stirring rod drives two liquids through one another until the clear boundary disappears꞉·˙·˙ostensive:·
·˙grimchain:·♂ᵱ⬡ᚺᔑᛋᗅ❄ᚯᙅᘜᏧᑭᔕ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꞊꞊ↄ꞉ category꞊꞊court꞉·
·˙lookup office꞊꞊catalyst꞉ enoch꞊꞊♂꞉·

˸word:·ʙɯʌ˸·˙word:·

·˙origin…language:·Coptic·˙·˙origin…language:·
·˙font:·Noto Sans Coptic·˙·˙font:·
·˙pronunciation:·BOHL·˙·˙pronunciation:·
·˙cadence:·Catalyst·˙·˙cadence:·
·˙non…ostensive:·To loose·˙·˙non…ostensive:·
·˙ostensive:·A blade slips beneath the tightened cord꞉꞉Tension releases and the bound edges spring apartↄ·˙·˙ostensive:·
·˙grimchain:·ʎ𐌗ᑒ∫ᖶᒧᔑ卫ᑭᘐ0ᚔ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꞊꞊✲꞉ category꞊꞊court꞉·
·˙lookup office꞊꞊catalyst꞉ enoch꞊꞊♂꞉·

˸word:·ᚺᛣᚣᚣᚴᚾ˸·˙word:·

·˙origin…language:·Gothic·˙·˙origin…language:·
·˙font:·Noto Sans Gothic·˙·˙font:·
·˙pronunciation:·HALDAN·˙·˙pronunciation:·
·˙cadence:·Catalyst·˙·˙cadence:·
·˙non…ostensive:·To hold꞉· to take care of꞉꞉tend꞉꞉feed·˙·˙non…ostensive:·
·˙ostensive:·Weight begins to slip꞉· fingers clamp before it can fall✲·˙·˙ostensive:·
·˙grimchain:·ϒᵩᛯᵐᵮᵞᵡⵣᚸ▫ᒣ㉿⊕ᚻ·˙·˙grimchain:·
·˙·˙lookup·˙
·˙·˙glyphs·˙
·˙glyph symbol꞊꞊ᵮ꞉ category꞊꞊court꞉·
·˙lookup office꞊꞊catalyst꞉ enoch꞊꞊♂꞉·

˸word:·ᚦᔑᚻ˸·˙word:·

·˙origin…language:·Old Turkic·˙·˙origin…language:·
·˙font:·Noto Sans Old Turkic·˙·˙font:·
·˙pronunciation:·OHLOOR·˙·˙pronunciation:·
·˙cadence:·Catalyst·˙·˙cadence:·
·˙non…ostensive:·To sit꞉꞉to settle꞉· (intransitive) to rule꞉꞉to be throned·˙·˙non…ostensive:·
·˙ostensive:·Bent knees lower under gathered weight until the body settles onto stoneᵮ·˙·˙ostensive:·
·˙grimchain:·𐐠ᗰ𐐠⓪▫ᓀᚺᛉ𑣲⩥𑣲ᚋᛋᚹᛋ·˙·˙grimchain:·
·˙·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𐦝꩜꩜꩜

·⁚non…ostensive⁚·To give·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·An empty palm is extended⁚⁚The loaded hand opens and the held weight crosses into
↪ it⏁·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᘐ❄ᛁᛈᛋᚼ◎▫ᛗ⏁ꕥᚲᚱ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐑉ᛋ𐑈 category○–○𐑉court𐑈⁚·
·⁚lookup office○–○𐑉catalyst𐑈 enoch○–○𐑉ᘐ𐑈⁚·

˖word⁚·ᚹᚾᚩᚾ˖.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·KAH⟊lah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·to cause to freeze [with accusative ‘someone’] (idiomatically translated as 𐑉freeze𐑈
↪ with the accusative object as the subject)·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Cold bites into water⁚⁚A hard skin spreads from the meeting placeᛋ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛋᚾ☆ᛯZᛞᓚ◺ᛦᚺ⬡✠·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐑉ᛦ𐑈 category○–○𐑉court𐑈⁚·
·⁚lookup office○–○𐑉catalyst𐑈 enoch○–○𐑉ᘐ𐑈⁚·

˖word⁚·嗳˖.·˙word⁚·

·⁚origin…language⁚·Tangut·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·˙font⁚·
·⁚pronunciation⁚·HAH⟊UH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Leave⁚˙ to depart from a place·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·A dismissal sounds at the threshold⁚⁚and settled feet turn away as distance begins to
↪ open𐑋·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚿ✳ᛯᛈꕥᚵᛏᛈᓭᚾᚼᓯ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐑉𐑅𐑌𐑈 category○–○𐑉court𐑈⁚·
·⁚lookup office○–○𐑉catalyst𐑈 enoch○–○𐑉ᘐ𐑈⁚·

˖word⁚·ᵬ□ᛰ̄˖.·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·SHRUHM·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To sit⁚˙ to remain⁚⁚be present⁚˙ to set·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Downward weight meets firm support⁚⁚Motion stops⁚⁚The body settles into
↪ place𐑅𐑌·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ꕥᚯᛉᛯᚦᚹᛐᚾᛏᛋᚣᛉ⊙·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol○–○𐑉ᴏ𐑈 category○–○𐑉court𐑈⁚·
·⁚lookup office○–○𐑉catalyst𐑈 enoch○–○𐑉ᘐ𐑈⁚·

:word:·ᛦᚠᚣᚣᚴᚷᚣᚾ:·.·ˑword:·

·⁚origin…language:·Gothic·⁚.·ˑorigin…language:·
·⁚font:·Noto Sans Gothic·⁚.·ˑfont:·
·⁚pronunciation:·AFLAGJAN·⁚.·ˑpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˑcadence:·
·⁚non…ostensive:·To lay down⁚⁚put away·⁚.·ˑnon…ostensive:·
·⁚ostensive:·Arms tire beneath the burden⁚⁚lowering the carried weight to the groundɵ·⁚.·ˑostensive:·
·⁚grimchain:·ᑭ1ᚼᘐᓚ✡▫ᙏᔐ⊛∱☆�gˀ·⁚.·ˑgrimchain:·
·⁚.·ˑlookup·ˑ
·⁚.·ˑglyphs·ˑ
·⁚glyph symbol○–○ₑΘ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑcatalyst» enoch○–○ₑ♂»⁚·

:word:·ᚺᚣᚣᚿᚣᚾ:·.·ˑword:·

·⁚origin…language:·Gothic·⁚.·ˑorigin…language:·
·⁚font:·Noto Sans Gothic·⁚.·ˑfont:·
·⁚pronunciation:·HLATHAN·⁚.·ˑpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˑcadence:·
·⁚non…ostensive:·To load·⁚.·ˑnon…ostensive:·
·⁚ostensive:·Weight is laid upon the waiting frame⁚⁚The axle bows beneath what has been
↪ addedΘ·⁚.·ˑostensive:·
·⁚grimchain:·ᘜᓕ✱ᑫ⊙ℂᥣᑭᚸ⊖ᚻ⍁∀·⁚.·ˑgrimchain:·
·⁚.·ˑlookup·ˑ
·⁚.·ˑglyphs·ˑ
·⁚glyph symbol○–○ₑX» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑcatalyst» enoch○–○ₑ♂»⁚·

:word:·𐎆𐎍:·.·ˑword:·

·⁚origin…language:·Akkadian·⁚.·ˑorigin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·ˑfont:·
·⁚pronunciation:·dah+NAH+noom·⁚.·ˑpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˑcadence:·
·⁚non…ostensive:·to become strong·⁚ to be hard⁚⁚severe⁚⁚tough⁚⁚threatening·⁚.·ˑnon…ostensive:·
·⁚ostensive:·Hammering compacts wet clay·⁚ each blow drives softness out until the surface no longer
↪ yieldsX·⁚.·ˑostensive:·
·⁚grimchain:·ᚺ🡺ᓚᘜᙢ◎Xᛉᗊᑭ♀Ↄᙏ·⁚.·ˑgrimchain:·
·⁚.·ˑlookup·ˑ
·⁚.·ˑglyphs·ˑ
·⁚glyph symbol○–○ₑΛ» category○–○ₑcourt»⁚·
·⁚lookup office○–○ₑcatalyst» enoch○–○ₑ♂»⁚·

:word:·ᵂ𐤄ᚩ:·.·ˑword:·

·⁚origin…language:·Phoenician·⁚.·ˑorigin…language:·
·⁚font:·Noto Sans Phoenician·⁚.·ˑfont:·
·⁚pronunciation:·SHAH+khat·⁚.·ˑpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˑcadence:·
·⁚non…ostensive:·To decay or rust·⁚ to undergo material deterioration or corruption·⁚.·ˑnon…ostensive:·
·⁚ostensive:·Air and moisture work into exposed metal⁚⁚The clean surface roughens under slow
↪ corruptionΛ·⁚.·ˑostensive:·
·⁚grimchain:·⊕ᙏᘁ꒰ↀᙏᘜᛉ🡺☆ᔕᛀx·⁚.·ˑgrimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᛋ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜♂꩞⁚·

⁖word⁚·가루다⁖⁚·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚·˙font⁚·
·⁚pronunciation⁚·kàlOHtá·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To split·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Sharp edge enters the joined body⁚⁚and one whole opens into two separate
↪ sidesᚺ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᘐᘓᚔᚖᛞ♂⊙ᛁᚣᚱᚦᚷᛇᛟ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᛕ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜♂꩞⁚·

⁖word⁚·ᑌᔦ̄ᖶ̄⁖⁚·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·POOTK·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To divide⁚⁚separate·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Opposing pulls take hold of the joined matter⁚⁚The body parts and a gap opens between the
↪ piecesᛕ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚲᗷ○⊙ᚠᚾᚢᚴᛯᛋᛃ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜Ⓓ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜♂꩞⁚·

⁖word⁚·ᚣᚡᚸᛁᚡᚻᚢᚣᚻ⁖⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·AFLIFNAN·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To remain⁚⁚abide·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Wind and rain strike the standing stone without driving it from its placeⒹ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚴᚷᚿᛊ◎≳ᛯᛏᛖᛃᚸᚯᛉ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᚴ꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜♂꩞⁚·

⁖word⁚·ᏚᏦᏁᏰᏁᏁ⁖⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·SKABAN·⁚·˙pronunciation⁚·

·⁚cadence꞉·Catalyst·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·To shave⁝⁝to shear·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Blade skims the surface⁝⁝leaving the body lighter where fibre or hair has been taken
↪ awayᛣ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·Ψ✱⩡✳ϒ֍ϛ⊙♌ᛋᚺᚠϤϛ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋᓭ꞉ category꞉○–○ᵋcourt꞉⁚·
·⁚lookup office꞉○–○ᵋcatalyst꞉ enoch꞉○–○ᵋ♂꞉⁚·

·꞉word꞉·ᔑᖘ꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Old Turkic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Turkic·⁚.·˙font꞉·
·⁚pronunciation꞉·BAHS·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Catalyst·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·To press⁝⁝collapse꞉˙ (transitive) to raid⁝⁝attack⁝⁝suppress·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·A heavy load comes down⁝⁝The form buckles and folds beneath the pressureᓭ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᛯ⬡ᚼᚾᛟᛞ⓿⊕↑↗ꕥᚠᚼ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋ𐅵꞉ category꞉○–○ᵋcourt꞉⁚·
·⁚lookup office꞉○–○ᵋcatalyst꞉ enoch꞉○–○ᵋ♂꞉⁚·

·꞉word꞉·𒌓𒀭꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Sumerian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·SED·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Catalyst·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·To be cold or become cool·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Cold water closes around heated metal⁝⁝Warmth rushes out of the surface𐅵·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ᓚᖷᗐᓄᘐᑐᓕᓭᚼᚲᘐᐻ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋƸ꞉ category꞉○–○ᵋcourt꞉⁚·
·⁚lookup office꞉○–○ᵋcatalyst꞉ enoch꞉○–○ᵋ♂꞉⁚·

·꞉word꞉·ᔕᚱᛉᛓᚿᚣᚿ꞉·⁚.·˙word꞉·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·DROHBNAN·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Catalyst·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·To be/become restless⁝⁝to be/become agitated꞉˙ to be/become troubled·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·A sudden pounding sounds beyond the wall⁝⁝Pulse quickens꞉˙ muscle refuses
↪ stillnessƸ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·↗ᛨᛘᘐᚾ⏁ᚲᚸᚸᛪᔑᐃᛚᚦ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋI꞉ category꞉○–○ᵋcourt꞉⁚·
·⁚lookup office꞉○–○ᵋcatalyst꞉ enoch꞉○–○ᵋ♂꞉⁚·

·꞉word꞉·ᏰᏁᑎᏁᏁ꞉·⁚.·˙word꞉·

·⁖origin…language:·Gothic·⁖˙·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖˙·˙font:·
·⁖pronunciation:·BAWAN·⁖˙·˙pronunciation:·
·⁖cadence:·Catalyst·⁖˙·˙cadence:·
·⁖non…ostensive:·To dwell::inhabit::live::reside·⁖˙·˙non…ostensive:·
·⁖ostensive:·Season after season::weather drives the same body beneath one roof while threshold and
↪ floor take the wear of returnI·⁖˙·˙ostensive:·
·⁖grimchain:·𐰢𐰾𐰋𐰃𐰇𐰀𐰓𐰋𐰆𐰇·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
·⁖glyph symbol꜀꜀𐰃 category꜀꜀court꜆:·
·⁖lookup office꜀꜀catalyst꜆ enoch꜀꜀𐰇꜆:·

·word:·ᛒᚢᚱᚾ·⁖˙·˙word:·

·⁖origin…language:·Old Norse·⁖˙·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖˙·˙font:·
·⁖pronunciation:·BOH+rah·⁖˙·˙pronunciation:·
·⁖cadence:·Catalyst·⁖˙·˙cadence:·
·⁖non…ostensive:·to bore·⁖˙·˙non…ostensive:·
·⁖ostensive:·Turning point bites into wood or earth::opening a round passage where solid matter had
↪ closed the way𐰃·⁖˙·˙ostensive:·
·⁖grimchain:·ᚼᛜᚷᛉᛃᚣᚩᚸᚹᚦᛞ·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
·⁖glyph symbol꜀꜀𐰃 category꜀꜀court꜆:·
·⁖lookup office꜀꜀catalyst꜆ enoch꜀꜀𐰇꜆:·

·word:·ᚼᚱᚢᛒᚾ·⁖˙·˙word:·

·⁖origin…language:·Old Norse·⁖˙·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖˙·˙font:·
·⁖pronunciation:·HROH+pah·⁖˙·˙pronunciation:·
·⁖cadence:·Catalyst·⁖˙·˙cadence:·
·⁖non…ostensive:·to call out::cry::yell·⁖˙·˙non…ostensive:·
·⁖ostensive:·A sudden danger breaks the stillness::Breath surges through the throat and a cry carries
↪ across the open ground𐰃·⁖˙·˙ostensive:·
·⁖grimchain:·✡⊕🜍🜲⬡❁ᚲᚹᚩᚦᛋᚫ·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
·⁖glyph symbol꜀꜀𐰀 category꜀꜀court꜆:·
·⁖lookup office꜀꜀catalyst꜆ enoch꜀꜀𐰇꜆:·

·word:·ᛒᚠᚾᛋᚾ·⁖˙·˙word:·

·⁖origin…language:·Old Norse·⁖˙·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖˙·˙font:·
·⁖pronunciation:·BLAH+sah·⁖˙·˙pronunciation:·
·⁖cadence:·Catalyst·⁖˙·˙cadence:·
·⁖non…ostensive:·to blow·⁖˙·˙non…ostensive:·
·⁖ostensive:·Pressure gathers in the chest until released breath rushes outward and drives loose dust
↪ ahead of it𐰀·⁖˙·˙ostensive:·
·⁖grimchain:·ᚱᛈᛏᚲᛟᛈᚥ❁ᚸᚩᚲ❀·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᘐ꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜ᘓ꩞:·

﹒word:·ᛉᚣᛁᚣᚾ﹒⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·WEH⟊AN·⁚·˙pronunciation:·
·⁚cadence:·Catalyst·⁚·˙cadence:·
·⁚non…ostensive:·To blow (to produce an air current⁚⁚of wind)·⁚·˙non…ostensive:·
·⁚ostensive:·Warm air rises from heated ground⁚⁚colder air rushes in behind it⁚⁚and branches begin to
↪ bendᛒ·⁚·˙ostensive:·
·⁚grimchain:·ℂ𐑕ᘐ𐑓❂ᘡᑫᔕ✳ᛀ❁◺➢·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜♲꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜ᘓ꩞:·

﹒word:·⟰﹒⁚·˙word:·

·⁚origin…language:·Akkadian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·TEH⟊noom·⁚·˙pronunciation:·
·⁚cadence:·Catalyst·⁚·˙cadence:·
·⁚non…ostensive:·to grind·⁚·˙non…ostensive:·
·⁚ostensive:·Grinding stone meets hard grain under weight⁚⁚Kernel breaks down into meal beneath the
↪ crushing face♲·⁚·˙ostensive:·
·⁚grimchain:·ᨐᨒᙓᗄᛉᚲᚿᚸ❁ᚷᚿᓄᒪ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᘔ꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜ᘓ꩞:·

﹒word:·ፈርየ﹒⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·FAH⟊ryah·⁚·˙pronunciation:·
·⁚cadence:·Catalyst·⁚·˙cadence:·
·⁚non…ostensive:·To flower or bear fruit·⁚·˙non…ostensive:·
·⁚ostensive:·Returning warmth reaches closed buds and petals loosen as the hidden fruiting body begins
↪ to openᘓ·⁚·˙ostensive:·
·⁚grimchain:·ᚸᛯᗡᕊ✡ᓴᓘᐯᚺᘐᑢᘓᑭ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᘓ꩞ category꩜꩜court꩞:·
·⁚lookup office꩜꩜catalyst꩞ enoch꩜꩜ᘓ꩞:·

﹒word:·ⲙⲟⲩϯ﹒⁚·˙word:·

·⁚origin…language:·Coptic·⁚·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚·˙font:·
·⁚pronunciation:·mootuh·⁚·˙pronunciation:·

·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to call⁚˙ to yell·⁚.·˙non…ostensive:·
·⁚ostensive:·A distant answering signal sounds⁚⁚Breath lifts hard through the throat and voice is
↪ driven across the gap𐋢·⁚.·˙ostensive:·
·⁚grimchain:·𐊫𐋳𐋴Z⊙ᵡ𐌳𐋶𐋵ᘻ𐋷𐋸𐋹𐋺𐋻·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐌳𐌣 category○–○court𐌣:·
·⁚lookup office○–○catalyst𐌣 enoch○–○𐌾𐌣:·

·word:·ᛒᚣᚳᚢᚹᚷᚴᚾ·:.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·BLAWTHJAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·(transitive) to make void⁚⁚render powerless·⁚.·˙non…ostensive:·
·⁚ostensive:·Support is cut from beneath the loaded mechanism⁚⁚Stored force can no longer
↪ act𐌳·⁚.·˙ostensive:·
·⁚grimchain:·⬡𐌿𐌺𐌾𐌲𐌳⬡𐌻𐌼𐌺𐌾·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐌲𐌣 category○–○court𐌣:·
·⁚lookup office○–○catalyst𐌣 enoch○–○𐌾𐌣:·

·word:·𑀅𑀁𑀶𑁆𑀢·:.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·WUHRK·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To shear·⁚.·˙non…ostensive:·
·⁚ostensive:·Sharp edge passes close along fibre⁚⁚leaving a cleanly shortened surface
↪ behind𐌲·⁚.·˙ostensive:·
·⁚grimchain:·ς⧗Ψ∫ϔ♁ᛊ𐌱ᚴ✧𐍊φᑭ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐌶𐌣 category○–○court𐌣:·
·⁚lookup office○–○catalyst𐌣 enoch○–○𐌾𐌣:·

·word:·𓂀𓏏𓂋·:.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·kah+BAH+toom·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to be/become heavy⁚⁚fat⁚˙ to be/become important⁚⁚honoured⁚˙ to be/become
↪ difficult⁚⁚painful·⁚.·˙non…ostensive:·
·⁚ostensive:·Weight presses through the body at the point where step and breath answer with
↪ strain𐌶·⁚.·˙ostensive:·
·⁚grimchain:·◎ᛏ𐌲𐌡𐌿𐌳ᛏ𐋡⊖◎ᥣᚸᚺ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol◦–◦𓄿𓄺 category◦–◦court𓄺⁚·
·⁚lookup office◦–◦catalyst𓄺 enoch◦–◦𓀠𓄺⁚·

·:word:·𘜶:.·ˊword:·

·⁚origin…language:·Tangut·⁚.·ˊorigin…language:·
·⁚font:·Noto Serif Tangut·⁚.·ˊfont:·
·⁚pronunciation:·NG+KHAI·⁚.·ˊpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˊcadence:·
·⁚non…ostensive:·Chaotic::disorderly::or mixed+up·⁚.·ˊnon…ostensive:·
·⁚ostensive:·A sudden blow overturns the stacked pieces::Ordered rows collapse into crossing::unsettled
↪ fragments𓄿·⁚.·ˊostensive:·
·⁚grimchain:·𓅓☿𐤗𐤒𐤎𐤐𐤓𐤊𐤉𓂀☽0·⁚.·ˊgrimchain:·
·⁚.·ˊlookup·ˇ
·⁚.·ˊglyphs·ˇ
·⁚glyph symbol◦–◦𓄿𓄻 category◦–◦court𓄺⁚·
·⁚lookup office◦–◦catalyst𓄺 enoch◦–◦𓀠𓄺⁚·

·:word:·ᚢᗴᖶᚴᚾ:.·ˊword:·

·⁚origin…language:·Gothic·⁚.·ˊorigin…language:·
·⁚font:·Noto Sans Gothic·⁚.·ˊfont:·
·⁚pronunciation:·LE+TAN·⁚.·ˊpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˊcadence:·
·⁚non…ostensive:·To let·⁚.·ˊnon…ostensive:·
·⁚ostensive:·A bar lifts at the signal::Held motion goes free𓄿·⁚.·ˊostensive:·
·⁚grimchain:·ᚼᛘ⬡⏚ᚸᛘᚠᛈ✡ᚡ𓆙ᛞᛗᛊᛗ·⁚.·ˊgrimchain:·
·⁚.·ˊlookup·ˇ
·⁚.·ˊglyphs·ˇ
·⁚glyph symbol◦–◦𓄿𓄹 category◦–◦court𓄺⁚·
·⁚lookup office◦–◦catalyst𓄺 enoch◦–◦𓀠𓄺⁚·

·:word:·ᔕᚱᗴIᐷᚴᑎ:.·ˊword:·

·⁚origin…language:·Gothic·⁚.·ˊorigin…language:·
·⁚font:·Noto Sans Gothic·⁚.·ˊfont:·
·⁚pronunciation:·DRIVAN·⁚.·ˊpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˊcadence:·
·⁚non…ostensive:·To drive::force to go·⁚.·ˊnon…ostensive:·
·⁚ostensive:·Force presses from behind::driving the body along a path it would not have taken
↪ alone𓄹·⁚.·ˊostensive:·
·⁚grimchain:·ᚺᐃᓀᛇ☆🢒ᚸᚠᐃ◎·⁚.·ˊgrimchain:·
·⁚.·ˊlookup·ˇ
·⁚.·ˊglyphs·ˇ
·⁚glyph symbol◦–◦𓄿𓄼 category◦–◦court𓄺⁚·
·⁚lookup office◦–◦catalyst𓄺 enoch◦–◦𓀠𓄺⁚·

·:word:·ᚦᚱᚾ:.·ˊword:·

·⁚origin…language:·Old Norse·⁚.·ˊorigin…language:·
·⁚font:·Noto Sans Runic·⁚.·ˊfont:·
·⁚pronunciation:·THRAH·⁚.·ˊpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˊcadence:·
·⁚non…ostensive:·longing::yearning·⁚.·ˊnon…ostensive:·

·⁚ostensive⁚·An empty place remains beside the hearth after the familiar body is gone⁚⁚The chest
↪ tightens toward the absence△⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫲𐫚𐫗𐫄𐫀𐫏𐫄𐫘𐫔𐫏𐫘𐫔𐫏·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐫀∀𐫂 category⟜⟜𐫀court𐫂⁚·
·⁚lookup office⟜⟜𐫀catalyst𐫂 enoch⟜⟜𐫀♂𐫂⁚·

·word⁚·𐫗𐫄·⁚·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·KHOOL·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Rejoice⁚˙ to feel or show strong happiness and delight·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Difficult work stands finished and unbroken⁚⁚Laughter breaks loose as open palms strike
↪ together∀·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫔𐫙𐫚⊕𐫒Θ∇𐫄▽𐫇𐫇𐫘𐫓𐫓𐫓·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐫀∀𐫂 category⟜⟜𐫀court𐫂⁚·
·⁚lookup office⟜⟜𐫀catalyst𐫂 enoch⟜⟜𐫀♂𐫂⁚·

·word⁚·ᴇɴᴋᴏᴛ·⁚·˙word⁚·

·⁚origin…language⁚·Coptic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚·˙font⁚·
·⁚pronunciation⁚·enkot·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·to sleep·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Weariness drags the eyelids shut until waking movement fades and only slow breath
↪ remains∇·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫕𐫖𐫙𐫔𐫓𐫘♂✳𐫓𐫋𐫔𐫙·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐫀∇𐫂 category⟜⟜𐫀court𐫂⁚·
·⁚lookup office⟜⟜𐫀catalyst𐫂 enoch⟜⟜𐫀♂𐫂⁚·

·word⁚·ᚹᛋᛁᚹᛋ·⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·VEIFA·⁚·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚·˙cadence⁚·
·⁚non…ostensive⁚·to wave·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Wind catches loose cloth⁚˙ the edge swings back and forth through the air∇·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐫓𐫄ǂ⊕𐫇𐫘𐫘✡𐫔𐫔𐫔𐫔𐫔Θ𐫄·⁚·˙grimchain⁚·
·⁚·˙lookup⁚˙
·⁚·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐫀∇𐫂 category⟜⟜𐫀court𐫂⁚·
·⁚lookup office⟜⟜𐫀catalyst𐫂 enoch⟜⟜𐫀♂𐫂⁚·

·word⁚·ᛏᛚ ᚻᛁ ᚦᛋᛁ ᛖᛖ·⁚·˙word⁚·

·⁚origin…language:·Old Persian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Persian·⁚.·˙font:·
·⁚pronunciation:·oo+pahs+TAH·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·Aid or help·⁚· assistance given to another·⁚.·˙non…ostensive:·
·⁚ostensive:·Loaded beam begins to buckle·:Another support is thrust beneath before the weight can
↪ fall𐎢·⁚.·˙ostensive:·
·⁚grimchain:·𐎧𐎭𐏁𐎭𐎼𐎹𐎽𐎫𐎱𐎴𐎼𐎡·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐎢𐎡 category○–○𐎢court𐎱:·
·⁚lookup office○–○𐎢catalyst𐎱 enoch○–○𐎢𐎵𐎱:·

𐎢word:·𐎵𐎼𐎹𐎢·⁚.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·sahm+AH·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To hear or listen·⁚· to perceive sound·⁚.·˙non…ostensive:·
·⁚ostensive:·Sound reaches the ear and the head turns toward what vibration has brought𐎡·⁚.·˙ostensive:·
·⁚grimchain:·𐎫ℂ𐏂𐎮𐎲𐎢𐎫𐎮𐎴𐎹𐎵·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐎢𐎢𐎱 category○–○𐎢court𐎱:·
·⁚lookup office○–○𐎢catalyst𐎱 enoch○–○𐎢𐎵𐎱:·

𐎢word:·𐎵𐎠𐎼𐎹𐎢·⁚.·˙word:·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·WAHKI·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To bite·⁚.·˙non…ostensive:·
·⁚ostensive:·Food presses between the teeth·:The jaw closes and pressure bites inward𐎢·⁚.·˙ostensive:·
·⁚grimchain:·𐏊𐎘𐎠𐎹𐎲𐎧𐎭𐎱𐎰𐎸𐎺·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐎢𐎢𐎱 category○–○𐎢court𐎱:·
·⁚lookup office○–○𐎢catalyst𐎱 enoch○–○𐎢𐎵𐎱:·

𐎢word:·𐎠𐎶̄·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·RAHP·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To dig·⁚.·˙non…ostensive:·
·⁚ostensive:·Tool breaks the upper ground and opens the hidden earth beneath the surface𐎢·⁚.·˙ostensive:·
·⁚grimchain:·𐎧𐎲𐎹𐎮𐎱𐎼𐎺𐎽𐎠𐎹𐎢·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○𐎢𐎧𐎱 category○–○𐎢court𐎱:·
·⁚lookup office○–○𐎢catalyst𐎱 enoch○–○𐎢𐎵𐎱:·

:word:·𗍫:.·˙word:·

·⁖origin…language:·Tangut·⁖.·˙origin…language:·
·⁖font:·Noto Serif Tangut·⁖.·˙font:·
·⁖pronunciation:·DEEH·⁖.·˙pronunciation:·
·⁖cadence:·Catalyst·⁖.·˙cadence:·
·⁖non…ostensive:·Equal⁖· the same in amount⁝measure⁝rank⁝or extent·⁖.·˙non…ostensive:·
·⁖ostensive:·A second weight is set opposite the first⁝and the beam settles level between them𐋡·⁖.·˙ostensive:·
·⁖grimchain:·⊛♂ᚹᚢ♂ᛦ𐤟ᛋ𐤠ᛒᚷᚹᚴ·⁖.·˙grimchain:·
·⁖.·˙lookup·˙
·⁖.·˙glyphs·˙
·⁖glyph symbol꩜꩜𐋡꩞ category꩜꩜court꩞⁖·
·⁖lookup office꩜꩜catalyst꩞ enoch꩜꩜♂꩞⁖·

:word:·ᔕᛟᛖᏀᏁᑎ:.·˙word:·

·⁖origin…language:·Gothic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖.·˙font:·
·⁖pronunciation:·DOHMJAN·⁖.·˙pronunciation:·
·⁖cadence:·Catalyst·⁖.·˙cadence:·
·⁖non…ostensive:·To judge⁝to evaluate⁖· to deem⁝to consider·⁖.·˙non…ostensive:·
·⁖ostensive:·Contradicting marks are laid side by side until one side sinks under the measure𐋡·⁖.·˙ostensive:·
·⁖grimchain:·ᩬ꯭꯭꯭

·⁖grimchain⁚·♌ᖴXᛋ⊕ᓫᒍᛉᘐ❂I⊛ǂ·⁖.·˙grimchain⁚·
·⁖.·˙lookup⁚˙
·⁖.·˙glyphs⁚˙
·⁖glyph symbol꩜꩜✷꩜ category꩜꩜court꩜⁚·
·⁖lookup office꩜꩜catalyst꩜ enoch꩜꩜☌꩜⁚·

꞉word⁚·ᛣᚠᛜᚢᚷᚴᚾ꞉⁖.·˙word⁚·

·⁖origin…language⁚·Gothic·⁖.·˙origin…language⁚·
·⁖font⁚·Noto Sans Gothic·⁖.·˙font⁚·
·⁖pronunciation⁚·AFHWAPJAN·⁖.·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖.·˙cadence⁚·
·⁖non…ostensive⁚·To choke off⁚⁚to asphyxiate⁚⁚to suffocate·⁖.·˙non…ostensive⁚·
·⁖ostensive⁚·A cord tightens across the throat while the chest strains for air that cannot
↪ enter✷·⁖.·˙ostensive⁚·
·⁖grimchain⁚·🜞꯭꯭꯭

·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·(hapax legomenon) to cast down⁘to thrown down·⁚.·˙non…ostensive:·
·⁖ostensive:·A downward blow strikes the raised body and knocks it from height⁘⁚.·˙ostensive:·
·⁖grimchain:·ᛘᛃᚣᚢᚱᚾᛋᚺᚷᛏᛜᛋ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁖glyph symbol꜔꜔ᛋ꜕ category꜔꜔ᶜourt꜕⁚·
·⁖lookup office꜔꜔ᶜatalyst꜕ enoch꜔꜔ᵒᶜ꜕⁚·

₍word:·ᚺᚾᛖᛁᛉᚣᚾ₎⁚.·˙word:·

·⁖origin…language:·Gothic·⁚.·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁚.·˙font:·
·⁖pronunciation:·HNIWAN·⁚.·˙pronunciation:·
·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·To incline⁘bend downwards⁘decline·⁚.·˙non…ostensive:·
·⁖ostensive:·Weight or pressure bends the upright body⁘turning its line downward without breaking
↪ itᛜ·⁚.·˙ostensive:·
·⁖grimchain:·⧗♂֍ΘᖘᑫIᛋ᛬ᘐ᛫ᖁᑫᑕ⁕·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁖glyph symbol꜔꜔ᛜ꜕ category꜔꜔ᶜourt꜕⁚·
·⁖lookup office꜔꜔ᶜatalyst꜕ enoch꜔꜔ᵒᶜ꜕⁚·

₍word:·ᚾᛋᚣᚹᚾ₎⁚.·˙word:·

·⁖origin…language:·Old Norse·⁚.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁚.·˙font:·
·⁖pronunciation:·ŒSKJA·⁚.·˙pronunciation:·
·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·to wish·⁚.·˙non…ostensive:·
·⁖ostensive:·Something deeply sought remains beyond reach⁘Chest and gaze keep turning toward the
↪ absenceᛜ·⁚.·˙ostensive:·
·⁖grimchain:·⛢ᖷᗺᓘᛯᛖᛯ⁕ᗅᛯᚴᛃ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁖glyph symbol꜔꜔ᚾ꜕ category꜔꜔ᶜourt꜕⁚·
·⁖lookup office꜔꜔ᶜatalyst꜕ enoch꜔꜔ᵒᶜ꜕⁚·

₍word:·ᛒᛁᚾᚱᚹᚾ₎⁚.·˙word:·

·⁖origin…language:·Old Norse·⁚.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁚.·˙font:·
·⁖pronunciation:·BYAHR+gah·⁚.·˙pronunciation:·
·⁖cadence:·Catalyst·⁚.·˙cadence:·
·⁖non…ostensive:·to save⁘rescue·⁚.·˙non…ostensive:·
·⁖ostensive:·Danger closes around the vulnerable flesh⁘A clear path opens back toward safety under
↪ helping handsᛜ·⁚.·˙ostensive:·
·⁖grimchain:·ᘔᛇᛈᛞᛉᚴᔑᑐᚷᚢᚱᛣ·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁖glyph symbol꜔꜔ᚹ꜕ category꜔꜔ᶜourt꜕⁚·
·⁖lookup office꜔꜔ᶜatalyst꜕ enoch꜔꜔ᵒᶜ꜕⁚·

⁖word꞉·ᴛɪᴖᕼʎɴ⁖·˙word꞉·

·⁚origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁚pronunciation꞉·TYOO+HAN·⁚·˙pronunciation꞉·
·⁚cadence꞉·Catalyst·⁚·˙cadence꞉·
·⁚non…ostensive꞉·To pull⁚⁚draw⁚⁚drag⁚ to lead⁚⁚lead away·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Tension travels through rope or limb⁚⁚drawing the held thing toward the point of
↪ force⥱·⁚·˙ostensive꞉·
·⁚grimchain꞉·⛢ₒ⅋❂ᚺᛦᛋₒ✡⩛ᑭ𑁍⁓ᵍᵐ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐤂𐤕𐤃 category꞉○–○𐤂court𐤃⁚·
·⁚lookup office꞉○–○𐤂catalyst𐤃 enoch꞉○–○𐤂𐤏𐤃⁚·

⁖word꞉·κω⁖·˙word꞉·

·⁚origin…language꞉·Coptic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Coptic·⁚·˙font꞉·
·⁚pronunciation꞉·KOH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Catalyst·⁚·˙cadence꞉·
·⁚non…ostensive꞉·To put⁚⁚place⁚⁚release⁚⁚or loosen·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·A signal opens the grip and the held object is lowered into the marked
↪ place𐤕𐤉·⁚·˙ostensive꞉·
·⁚grimchain꞉·♳ⅎ♳ↄⅎ〈𓃒ᑭ⅄ᚱᚻℂ𑁍·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐤂𓃒𐤃 category꞉○–○𐤂court𐤃⁚·
·⁚lookup office꞉○–○𐤂catalyst𐤃 enoch꞉○–○𐤂𐤏𐤃⁚·

⁖word꞉·ᚿ̄J□⅂ᚿ̄⁖·˙word꞉·

·⁚origin…language꞉·Tocharian A·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·KLUHNK·⁚·˙pronunciation꞉·
·⁚cadence꞉·Catalyst·⁚·˙cadence꞉·
·⁚non…ostensive꞉·To doubt·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Two opposed marks lie before the eye⁚⁚and fingers hover between them with neither grasp
↪ closing𓃒·⁚·˙ostensive꞉·
·⁚grimchain꞉·𐤍𐤏𐤒☆▽ᚷ⛢⅋ᚷ𑁍⩛↗⊕·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𐤂〈𐤃 category꞉○–○𐤂court𐤃⁚·
·⁚lookup office꞉○–○𐤂catalyst𐤃 enoch꞉○–○𐤂𐤏𐤃⁚·

⁖word꞉·SᴖIᕼᴖN⁖·˙word꞉·

·⁚origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁚pronunciation꞉·SEYEAN·⁚·˙pronunciation꞉·
·⁚cadence꞉·Catalyst·⁚·˙cadence꞉·
·⁚non…ostensive꞉·To sow·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Fresh furrow opens in dark soil and seed is cast into the waiting cut〈·⁚·˙ostensive꞉·
·⁚grimchain꞉·‹⬡𑁍ᚿᚠᚼ▽ᛇᛁ✡✷𐤏𐤒·⁚·˙grimchain꞉·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ category꩜court₎⁚·
·⁚lookup office꩜catalyst₎ enoch꩜♂₎⁚·

⁖word:·ᛋᚴᚢ⁖·˙word:·

·⁚origin…language:·Old Norse·⁚·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚·˙font:·
·⁚pronunciation:·SKEE·⁚·˙pronunciation:·
·⁚cadence:·Catalyst·⁚·˙cadence:·
·⁚non…ostensive:·cloud·⁚·˙non…ostensive:·
·⁚ostensive:·Warm damp air meets colder height as pale vapor gathers until the distance softens꩜·⁚·˙ostensive:·
·⁚grimchain:·ᚸᚡᛏ𐊧𐊫ᚴᚱᛁᚸᚾ᛭ᚥᚥᚢ᛭·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ category꩜court₎⁚·
·⁚lookup office꩜catalyst₎ enoch꩜♂₎⁚·

⁖word:·+□ᨆ̄⁖·˙word:·

·⁚origin…language:·Tocharian A·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·KUHS·⁚·˙pronunciation:·
·⁚cadence:·Catalyst·⁚·˙cadence:·
·⁚non…ostensive:·(transitive) to extinguish⁚quench·⁚˙ (intransitive) to be extinguished⁚go out·⁚·˙non…ostensive:·
·⁚ostensive:·Flame and smothering cover meet at the air⁄starved edge where light can no longer hold₥·⁚·˙ostensive:·
·⁚grimchain:·𐋏♀𐋏ᓴ𐋏ᚺ0ᗅΛ∀ᒧᚹ᛬·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ category꩜court₎⁚·
·⁚lookup office꩜catalyst₎ enoch꩜♂₎⁚·

⁖word:·𗍫⁖·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·JEH·⁚·˙pronunciation:·
·⁚cadence:·Catalyst·⁚·˙cadence:·
·⁚non…ostensive:·Cold·⁚˙ having or producing little heat·⁚·˙non…ostensive:·
·⁚ostensive:·Wind passes over wet skin and warmth drains from the exposed surface꩜·⁚·˙ostensive:·
·⁚grimchain:·ϴᚸᚢ⁎ᛮᛋᛮᚸᒧᔑᏗᏁᏓ⁎·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ category꩜court₎⁚·
·⁚lookup office꩜catalyst₎ enoch꩜♂₎⁚·

⁖word:·SᏁIOᏁᏁN⁖·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·

·⁚pronunciation:·SEH⟊HWAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To see·⁚.·˙non…ostensive:·
·⁚ostensive:·Light reaches the eye from the seen thing::Edge::color::and shape arrive together through
↪ sight↟·⁚.·˙ostensive:·
·⁚grimchain:·ᚼᛟᛊᛝᛤᛞᚤᛜᛋ⏃ᛗ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀꜀𐤊꜀ category꜀꜀court꜀⁚·
·⁚lookup office꜀꜀catalyst꜀ enoch꜀꜀ⵙ꜀⁚·

·word:·SᏁΓΓUᏩᏁN·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·SANKW⟊JAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To plunge·˙ to thrust into water or another penetrable substance·˙ to
↪ immerse·⁚.·˙non…ostensive:·
·⁚ostensive:·Body is driven beneath water or yielding matter::and the surface closes above the entering
↪ form𐤊·⁚.·˙ostensive:·
·⁚grimchain:·ᚿ⊙ᑭᓀᔑᚿᛋᛏᛁᚱᗡᚳᑫ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀꜀ᚿ꜀ ꜀ category꜀꜀court꜀⁚·
·⁚lookup office꜀꜀catalyst꜀ enoch꜀꜀ⵙ꜀⁚·

·word:·ᚹᛁᚾᛏᚨ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·VINDA·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to twist::wring::squeeze::wind·⁚.·˙non…ostensive:·
·⁚ostensive:·Hands turn fibre or cloth against itself::forcing a new coil into the
↪ material↟·⁚.·˙ostensive:·
·⁚grimchain:·❄ᚾᚮᚣᚳᛏᛁᗡ∮𐤔ᚿᚸ𐤔ᛁ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꜀꜀𐤔꜀ category꜀꜀court꜀⁚·
·⁚lookup office꜀꜀catalyst꜀ enoch꜀꜀ⵙ꜀⁚·

·word:·ᚹᛁᛚᛏ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·VILD·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·will::liking::benevolence·⁚.·˙non…ostensive:·
·⁚ostensive:·An injured limb reaches toward support::The raised hand lowers and steadies it instead of
↪ striking𐤔·⁚.·˙ostensive:·
·⁚grimchain:·ᛁᛇᚿᐞᚤᚼᛉᚨᛯ𐤔ᛋᛉ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol○‑○𓂝𓆑 category○‑○𓂝court𓆏⁚·
·⁚lookup office○‑○𓂝catalyst𓆏 enoch○‑○𓂝𓃀𓆏⁚·

⁖word⁚·ⲙⲟⲩⲣ⁖˙·˙word⁚·

·⁚origin…language⁚·Coptic·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚˙·˙font⁚·
·⁚pronunciation⁚·MOOR·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·To bind·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·Cord bites around separate bodies⁚⁚tightening until no gap remains𓆑·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·𐎓𐎜𐎚𐎕𐎔𐎗𐎒𐎘𐎙𐎗𐎔𐎘·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁚˙
·⁚˙·˙glyphs⁚˙
·⁚glyph symbol○‑○𓂝𓆏 category○‑○𓂝court𓆏⁚·
·⁚lookup office○‑○𓂝catalyst𓆏 enoch○‑○𓂝𓃀𓆏⁚·

⁖word⁚·ⲁⲓ⁖˙·˙word⁚·

·⁚origin…language⁚·Coptic·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚˙·˙font⁚·
·⁚pronunciation⁚·JEE·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·To take·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·An offered weight enters reach and fingers close around it⁚⁚drawing it
↪ inward𓆏·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·𐎔𐎜𐎘𐎕𐎗𐎔𐎒𐎖𐎙𐎍𐎓𐎗𐎚·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁚˙
·⁚˙·˙glyphs⁚˙
·⁚glyph symbol○‑○𓂝𓆑 category○‑○𓂝court𓆏⁚·
·⁚lookup office○‑○𓂝catalyst𓆏 enoch○‑○𓂝𓃀𓆏⁚·

⁖word⁚·ᚹᛁᚢᛋᚨ⁖˙·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚˙·˙font⁚·
·⁚pronunciation⁚·KYOH᛫sah·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·to choose·⁚˙·˙non…ostensive⁚·
·⁚ostensive⁚·Many things stand before the hand⁚⁚and one fills the closing grasp⁚⁚and the others remain
↪ untouched𓆑·⁚˙·˙ostensive⁚·
·⁚grimchain⁚·𐎛𐎍𐎖𐎐𐎚𐎔𐎗𐎒'𐎄𐎓𐎖·⁚˙·˙grimchain⁚·
·⁚˙·˙lookup⁚˙
·⁚˙·˙glyphs⁚˙
·⁚glyph symbol○‑○𓂝𓆗 category○‑○𓂝court𓆏⁚·
·⁚lookup office○‑○𓂝catalyst𓆏 enoch○‑○𓂝𓃀𓆏⁚·

⁖word⁚·𗍫⁖˙·˙word⁚·

·⁚origin…language⁚·Tangut·⁚˙·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚˙·˙font⁚·
·⁚pronunciation⁚·MPHAW·⁚˙·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚˙·˙cadence⁚·
·⁚non…ostensive⁚·Cover or conceal⁚˙ hide from view·⁚˙·˙non…ostensive⁚·

·⁚ostensive⁝·Covering falls between the eye and the hidden thing⁝⁝Sight stops at the outer
↪ surface⯮·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚺᛦᛝ▫ꟾᘔᛚᛦᛋ◌ᑫᛇ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜ᛋ꜀ category⟜⟜court꜀⁚·
·⁚lookup office⟜⟜catalyst꜀ enoch⟜⟜♂꜀⁚·

·word⁝·ᛋᚹᛁᚱᚾ⁚.·˙word⁝·

·⁚origin…language⁝·Old Norse·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Runic·⁚.·˙font⁝·
·⁚pronunciation⁝·SKEH⟊rah·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Catalyst·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·to cut·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Edge enters wood⁝⁝flesh⁝⁝or fibre⁝⁝and continuous material opens along the line of
↪ contactᚺ·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚺI◟ᘐᘏᛈᚹ⮥⮹✷⏦ᛞX✳❂·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜A꜀ category⟜⟜court꜀⁚·
·⁚lookup office⟜⟜catalyst꜀ enoch⟜⟜♂꜀⁚·

·word⁝·ᛏᑎᚾᚾ⁚.·˙word⁝·

·⁚origin…language⁝·Old Norse·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Runic·⁚.·˙font⁝·
·⁚pronunciation⁝·DUNA·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Catalyst·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·to boom⁝⁝roar·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·A heavy impact cracks through the hollow⁝⁝and air and ground answer with a deep rolling
↪ soundA·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᛁ∀▫ǂᛋ♳▫ᗡ✷ᑭⵔ✲·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜Ƀ꜀ category⟜⟜court꜀⁚·
·⁚lookup office⟜⟜catalyst꜀ enoch⟜⟜♂꜀⁚·

·word⁝·ﻫﻮﻭﻛﺒﺖ⁚.·˙word⁝·

·⁚origin…language⁝·Old Uyghur·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Serif Old Uyghur·⁚.·˙font⁝·
·⁚pronunciation⁝·adUHrt·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Catalyst·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·difference⁝⁝distinction·⁚ clarity⁝⁝distinguishing remark·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·A second mark is set beside the first⁝⁝The mismatch catches the eye at onceƁ·⁚.·˙ostensive⁝·
·⁚grimchain⁝·⏃ᚺᛂᚸℂXᛅ♀ᚸᚴ✷ᛯᚺᛞ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁝˙
·⁚.·˙glyphs⁝˙
·⁚glyph symbol⟜⟜Ↄ꜀ category⟜⟜court꜀⁚·
·⁚lookup office⟜⟜catalyst꜀ enoch⟜⟜♂꜀⁚·

·word⁝·ᒧ⁚.·˙word⁝·

·⁚origin…language:·Old Turkic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚.·˙font:·
·⁚pronunciation:·AHL·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To take⁚⁚to buy·⁚ (transitive) to conquer⁚⁚to capture·⁚.·˙non…ostensive:·
·⁚ostensive:·An offered object enters reach as payment leaves the opposite hand and fingers close
↪ around the weightↃ·⁚.·˙ostensive:·
·⁚grimchain:·▽𐳺ᐱᛋᛦ𐳺ᚼᕮᛨ⍜✡ↄx·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈﹒ category◦–◦court﹒:·
·⁚lookup office◦–◦catalyst﹒ enoch◦–◦⍜﹒:·

·word:·ᓴᐯᑎᑊᓭ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·SKEEL+yah·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to part⁚⁚divide⁚⁚separate·⁚.·˙non…ostensive:·
·⁚ostensive:·Opposing pulls take hold of the joined bodies⁚⁚Contact breaks and open space appears
↪ between them𐑈·⁚.·˙ostensive:·
·⁚grimchain:·ᵔᵌᵡᒧᒣᔑᵊᑫᗰ𐳺ᒣᛋ⊩·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ᕒ﹒ category◦–◦court﹒:·
·⁚lookup office◦–◦catalyst﹒ enoch◦–◦⍜﹒:·

·word:·ᑎᕫᐦ·⁚.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·bah+RAH+kah·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To flash as lightning or thunder·⁚.·˙non…ostensive:·
·⁚ostensive:·Charged clouds press together in darkness until a white stroke tears across the
↪ skyᕒ·⁚.·˙ostensive:·
·⁚grimchain:·ᕈᙆxxz▽ᔓᘐᵹᙡⓅᕈ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦ᕃ﹒ category◦–◦court﹒:·
·⁚lookup office◦–◦catalyst﹒ enoch◦–◦⍜﹒:·

·word:·𝍐·⁚.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·R+NAR·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·Pluck or flip·⁚ also play a stringed instrument·⁚.·˙non…ostensive:·
·⁚ostensive:·Finger slips from the caught string⁚⁚and the taut line snaps into soundᕃ·⁚.·˙ostensive:·
·⁚grimchain:·⊕ᐞ⏢ᘐᚠᛘᙆ▽ᕮᚺ⋇ᔓ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁖glyph symbol꩜꩜ᛁ꩞ category꩜꩜ᛢourt꩞꞉·
·⁖lookup office꩜꩜ᛢatalyst꩞ enoch꩜꩜ᛟ꩞꞉·

꞉word꞉·ᛏᚱᛁᛒᚾ꞉꞉·˙word꞉·

·⁖origin…language꞉·Old Norse·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Runic·⁖·˙font꞉·
·⁖pronunciation꞉·DREH+pah·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·
·⁖non…ostensive꞉·To strike꞉꞉beat꞉꞉or kill·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Raised hammer falls under gathered force and drives the blow into the body
↪ belowI·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᛋᛒᛇᛗᛜᚴᛈᛇᚳᚢᛒ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꩜꩜ᛈ꩞ category꩜꩜ᛢourt꩞꞉·
·⁖lookup office꩜꩜ᛢatalyst꩞ enoch꩜꩜ᛟ꩞꞉·

꞉word꞉·𒀀𒀭𒆠𒂍𒁲꞉꞉·˙word꞉·

·⁖origin…language꞉·Akkadian·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖·˙font꞉·
·⁖pronunciation꞉·zah+MAH+roo·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·
·⁖non…ostensive꞉·To sing꞉꞉with or without instrumental accompaniment·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·A measured beat begins and breath rises into ordered pitch꞉꞉carrying beyond the
↪ mouthᗡ·⁖·˙ostensive꞉·
·⁖grimchain꞉·♂᠕ᓬᛯ❂ᗷᛯᘏᓬᦓᛯ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꩜꩜ᛕ꩞ category꩜꩜ᛢourt꩞꞉·
·⁖lookup office꩜꩜ᛢatalyst꩞ enoch꩜꩜ᛟ꩞꞉·

꞉word꞉·𗦳꞉·˙word꞉·

·⁖origin…language꞉·Tangut·⁖·˙origin…language꞉·
·⁖font꞉·Noto Serif Tangut·⁖·˙font꞉·
·⁖pronunciation꞉·R+KUHR·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Fly⁖˙ to move through the air by wings or airborne motion·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·A hard downstroke drives air beneath the wings⁖˙ the body lifts free of the
↪ groundᛕ·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᛋᒍᑐ⟌ᒍᛉᛘᖇᑐᑭᓀ⊣ᛁ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꩜꩜ᛚ꩞ category꩜꩜ᛢourt꩞꞉·
·⁖lookup office꩜꩜ᛢatalyst꩞ enoch꩜꩜ᛟ꩞꞉·

꞉word꞉·𗥃꞉·˙word꞉·

·⁖origin…language꞉·Tangut·⁖·˙origin…language꞉·
·⁖font꞉·Noto Serif Tangut·⁖·˙font꞉·
·⁖pronunciation꞉·R+ZEER·⁖·˙pronunciation꞉·
·⁖cadence꞉·Catalyst·⁖·˙cadence꞉·

·⁚non…ostensive:·Tie⁚⁚fasten⁚⁚or bind·⁚.·non…ostensive:·
·⁚ostensive:·Cord passes around the separate bodies and tightens⁚⁚holding what would otherwise move
↪ apart↿⁚.·ostensive:·
·⁚grimchain:·ᦓᦔᦐᦇᦟᦞᦋᦎ◎ᦓᦇ❁ᦃ·⁚.·grimchain:·
·⁚.·lookup:·
·⁚.·glyphs:·
·⁚glyph symbol○–○ᑊ› category○–○ᑊcourt›⁚·
·⁚lookup office○–○ᑊcatalyst› enoch○–○ᑊ☌›⁚·

·word:·ᵇᵘᶜ ㅣ르다·⁚.·word:·

·⁚origin…language:·Middle Korean·⁚.·origin…language:·
·⁚font:·Noto Sans KR·⁚.·font:·
·⁚pronunciation:·pstiluta·⁚.·pronunciation:·
·⁚cadence:·Catalyst·⁚.·cadence:·
·⁚non…ostensive:·To prick·⁚ to stab·⁚.·non…ostensive:·
·⁚ostensive:·Narrow point drives through skin or cloth⁚⁚gathering force at the tiny entry↿⁚.·ostensive:·
·⁚grimchain:·☿ᐱᐦ◎❀ᘏᘎ○ᘐ❄ℂعرس·⁚.·grimchain:·
·⁚.·lookup:·
·⁚.·glyphs:·
·⁚glyph symbol○–○ᑊ› category○–○ᑊcourt›⁚·
·⁚lookup office○–○ᑊcatalyst› enoch○–○ᑊ☌›⁚·

·word:·ᔕᐱᗰᐱ·⁚.·word:·

·⁚origin…language:·Gothic·⁚.·origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·font:·
·⁚pronunciation:·SAMA·⁚.·pronunciation:·
·⁚cadence:·Catalyst·⁚.·cadence:·
·⁚non…ostensive:·Same·⁚.·non…ostensive:·
·⁚ostensive:·A second vessel is set on the balance beside the first⁚⁚The beam refuses to
↪ tip↿⁚.·ostensive:·
·⁚grimchain:·ᓀᛋᚠᚱᚺᓂᚸ⬡ℂᏏᏢᏏᏀᏂ·⁚.·grimchain:·
·⁚.·lookup:·
·⁚.·glyphs:·
·⁚glyph symbol○–○ᑊ✳› category○–○ᑊcourt›⁚·
·⁚lookup office○–○ᑊcatalyst› enoch○–○ᑊ☌›⁚·

·word:·ᚾᚾ·⁚.·word:·

·⁚origin…language:·Old Norse·⁚.·origin…language:·
·⁚font:·Noto Sans Runic·⁚.·font:·
·⁚pronunciation:·NAH·⁚.·pronunciation:·
·⁚cadence:·Catalyst·⁚.·cadence:·
·⁚non…ostensive:·to get hold of⁚⁚reach⁚⁚overtake·⁚.·non…ostensive:·
·⁚ostensive:·A rolling vessel starts toward the edge·⁚ a reaching hand follows and closes around it
↪ before it falls✳·⁚.·ostensive:·
·⁚grimchain:·Xᚿ♀ΙΙᚿᛉᛝᛘᛞᛯᚺᚾ·⁚.·grimchain:·
·⁚.·lookup:·
·⁚.·glyphs:·
·⁚glyph symbol○–○ᑊ✳› category○–○ᑊcourt›⁚·
·⁚lookup office○–○ᑊcatalyst› enoch○–○ᑊ☌›⁚·

·word:·ᡆ◈ᡃᡁᡆᡆ·⁚.·word:·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·PAKHSHARI·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To protect::guard::defend·⁚ to observe (agreements)::keep (oaths)::obey (commands)::keep (a
↪ secret)·⁚ (middle with dative) to seek protection with·⁚.·˙non…ostensive:·
·⁚ostensive:·Shield or wall takes the threat first::leaving the body behind untouched⁕·⁚.·˙ostensive:·
·⁚grimchain:·♅♆◝᯽♌♐♏⁕⛲𓆙♉♈·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol₀₋₀₍₎ category₀₋₀₍court₎:·
·⁚lookup office₀₋₀₍catalyst₎ enoch₀₋₀₍☌₎:·

·word:·ᛋᚾᚢᚺ·⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·SNOO⟊ah·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to turn (around)::twist·⁚.·˙non…ostensive:·
·⁚ostensive:·Opposing grip applies torque to the held body until face and direction swing around the
↪ center⚹·⁚.·˙ostensive:·
·⁚grimchain:·ᛁᛉᚹᛊ𜺨♤▽⊰ᛝᛞᚣ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol₀₋₀₍⚸₎ category₀₋₀₍court₎:·
·⁚lookup office₀₋₀₍catalyst₎ enoch₀₋₀₍☌₎:·

·word:·ⲁⲙⲁⲉⲓ·⁚.·˙word:·

·⁚origin…language:·Coptic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚.·˙font:·
·⁚pronunciation:·amahi·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·to lead·⁚.·˙non…ostensive:·
·⁚ostensive:·A raised signal fixes one course::First motion takes it and following motion gathers
↪ behind⚸·⁚.·˙ostensive:·
·⁚grimchain:·𐤊♂⚹⊛𓆑ᑫ⊛ 𜺨𑀪𐦚⁕𐦘·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol₀₋₀₍♈₎ category₀₋₀₍court₎:·
·⁚lookup office₀₋₀₍catalyst₎ enoch₀₋₀₍☌₎:·

·word:·𐤊·⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·SHAM⟊shoo·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·Sun·⁚ solar disk·⁚ sunlight·⁚.·˙non…ostensive:·
·⁚ostensive:·Cloud cover breaks apart·⁚ sudden radiance floods the exposed ground♈·⁚.·˙ostensive:·
·⁚grimchain:·0⚹♆⟓⇞♀∱♑♆1△❄ϙ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol○‒○𐑉0𐑊 category○‒○𐑉court𐑊⁚·
·⁚lookup office○‒○𐑉catalyst𐑊 enoch○‒○𐑉𐑴𐑊⁚·

·:word:·𐑢:·.·ˈword:·

·⁚origin…language:·Sumerian·⁚.·ˈorigin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·ˈfont:·
·⁚pronunciation:·ZAHL·⁚.·ˈpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˈcadence:·
·⁚non…ostensive:·Shine⁚· to emit or reflect visible light·⁚.·ˈnon…ostensive:·
·⁚ostensive:·Sunlight strikes polished metal and radiance leaps from the surface::catching the nearby
↪ edges0·⁚.·ˈostensive:·
·⁚grimchain:·⬡𐑅𐑾ʌ𐑌𐑆𐑅𐑾ɻℕ❄𐑦◎∮·⁚.·ˈgrimchain:·
·⁚.·ˈlookup·ˈ
·⁚.·ˈglyphs·ˈ
·⁚glyph symbol○‒○𐑉𐑣𐑊 category○‒○𐑉court𐑊⁚·
·⁚lookup office○‒○𐑉catalyst𐑊 enoch○‒○𐑉𐑴𐑊⁚·

·:word:·ᒪᐦ̄:.·ˈword:·

·⁚origin…language:·Tocharian A·⁚.·ˈorigin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·ˈfont:·
·⁚pronunciation:·YAHT·⁚.·ˈpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˈcadence:·
·⁚non…ostensive:·To be able::be capable of::can·⁚.·ˈnon…ostensive:·
·⁚ostensive:·Strength meets the demanded work and does not fail beneath its weight𐑣·⁚.·ˈostensive:·
·⁚grimchain:·ℂ𐑓𐑦♀𐑇ϒ𐑘𐑘𐑅ᵬ𐑞𐑇𐑟·⁚.·ˈgrimchain:·
·⁚.·ˈlookup·ˈ
·⁚.·ˈglyphs·ˈ
·⁚glyph symbol○‒○𐑉ǂ𐑊 category○‒○𐑉court𐑊⁚·
·⁚lookup office○‒○𐑉catalyst𐑊 enoch○‒○𐑉𐑴𐑊⁚·

·:word:·ꞁ꞉·.·ˈword:·

·⁚origin…language:·Tocharian B·⁚.·ˈorigin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·ˈfont:·
·⁚pronunciation:·ROO·⁚.·ˈpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˈcadence:·
·⁚non…ostensive:·To open·⁚.·ˈnon…ostensive:·
·⁚ostensive:·A bar lifts::Pressure reaches the joined leaves and the closed edges swing
↪ apartǂ·⁚.·ˈostensive:·
·⁚grimchain:·Xɤ○𐑂ↄ𐑑𐑘0ℂ𐑟𐑉𐑓𐑐𐑌·⁚.·ˈgrimchain:·
·⁚.·ˈlookup·ˈ
·⁚.·ˈglyphs·ˈ
·⁚glyph symbol○‒○𐑉𐑢𐑊 category○‒○𐑉court𐑊⁚·
·⁚lookup office○‒○𐑉catalyst𐑊 enoch○‒○𐑉𐑴𐑊⁚·

·:word:·ᔭᔑᕒᔭhOᔨᖁ꞉·.·ˈword:·

·⁚origin…language:·Aghwan·⁚.·ˈorigin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·ˈfont:·
·⁚pronunciation:·HEGHESUN·⁚.·ˈpronunciation:·
·⁚cadence:·Catalyst·⁚.·ˈcadence:·
·⁚non…ostensive:·To come (hither)·⁚.·ˈnon…ostensive:·

·⁚ostensive⁚·A summons carries across the open ground and distant motion turns toward the near
↪ place𐬹·⁚.·ostensive⁚·
·⁚grimchain⁚·𐋠⊕𐋵𐋦ᘐ𐋡𐋠𐋢𐋦𐋠𐋢𐋦𐋠·⁚.·grimchain⁚·
·⁚.·lookup⁚·
·⁚.·glyphs⁚·
·⁚glyph symbol○–○𐤓, category○–○court𐤔⁚·
·⁚lookup office○–○catalyst, enoch○–○𐤏𐤔⁚·

·word⁚·ᛒᚱᛁᚴᚴᚾᚾ·⁚.·word⁚·

·⁚origin…language⁚·Gothic·⁚.·origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·font⁚·
·⁚pronunciation⁚·BRIGGAN·⁚.·pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·cadence⁚·
·⁚non…ostensive⁚·To bring·⁚ to lead⁚⁚to conduct·⁚ to make⁚⁚to render⁚⁚to cause to become·⁚.·non…ostensive⁚·
·⁚ostensive⁚·Guiding force carries a held object across the threshold until it rests on new
↪ ground𐤓·⁚.·ostensive⁚·
·⁚grimchain⁚·ᘓ𐋵𐋢Z𐋠Ƀ𐋣⬡⏃ᛋᛞᘓ·⁚.·grimchain⁚·
·⁚.·lookup⁚·
·⁚.·glyphs⁚·
·⁚glyph symbol○–○𐤎, category○–○court𐤔⁚·
·⁚lookup office○–○catalyst, enoch○–○𐤏𐤔⁚·

·word⁚·𐏑·⁚.·word⁚·

·⁚origin…language⁚·Akkadian·⁚.·origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·font⁚·
·⁚pronunciation⁚·KAHR+noom·⁚.·pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·cadence⁚·
·⁚non…ostensive⁚·horn·⁚ (vessel) horn (a drinking vessel made of horn)·⁚ (human) power·⁚ (garment)
↪ fringe·⁚ (botany) shoot (of a plant or tree)·⁚.·non…ostensive⁚·
·⁚ostensive⁚·Pressure swells beneath the bark until a new shoot splits the older surface and enters
↪ open air𐤎·⁚.·ostensive⁚·
·⁚grimchain⁚·𐡕𐤗𐤅𐤉❄𐤎ℂ𐤊𐤒↗ℂ♋𐤉·⁚.·grimchain⁚·
·⁚.·lookup⁚·
·⁚.·glyphs⁚·
·⁚glyph symbol○–○𐤕, category○–○court𐤔⁚·
·⁚lookup office○–○catalyst, enoch○–○𐤏𐤔⁚·

·word⁚·⊛·⁚.·word⁚·

·⁚origin…language⁚·Mycenaean Greek·⁚.·origin…language⁚·
·⁚font⁚·Noto Sans Linear B·⁚.·font⁚·
·⁚pronunciation⁚·ROTA·⁚.·pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·cadence⁚·
·⁚non…ostensive⁚·wheel·⁚.·non…ostensive⁚·
·⁚ostensive⁚·Force enters the axle and the rim begins to turn⁚⁚carrying weight around the
↪ center𐤕·⁚.·ostensive⁚·
·⁚grimchain⁚·𐡌𐡆𐤊ᘐ𐋴ᑫ∀ᚺIᘓ𐋵ᔐ𐤍✱·⁚.·grimchain⁚·
·⁚.·lookup⁚·
·⁚.·glyphs⁚·
·⁚glyph symbol○–○𐤂, category○–○court𐤔⁚·
·⁚lookup office○–○catalyst, enoch○–○𐤏𐤔⁚·

⁖word⁚·ᚺᚣᛖᛁᛒᚷᛊᚾᚾ⁖·˙word⁚·

·⁖origin…language⁚·Gothic·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Gothic·⁖·˙font⁚·
·⁖pronunciation⁚·HLEEBJAN·⁖·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To spare⁖· to take good care of·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·An injured body lies exposed beneath a raised weapon⁖· the weapon lowers without striking𐌢·⁖·˙ostensive⁚·
·⁖grimchain⁚·𐌍𐌔𐌝𐌖𐌚𐌗𐌋𐌇𐌅𐌑𐌔⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉ᛪ꞉꞉ category꞉꞉court꞉꞉⁖·
·⁖lookup office꞉꞉catalyst꞉꞉ enoch꞉꞉ᛟ꞉꞉⁖·

⁖word⁚·ᘐᕼᖁᓀ⁖·˙word⁚·

·⁖origin…language⁚·Ge'ez·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Ethiopic·⁖·˙font⁚·
·⁖pronunciation⁚·mah+ZAH+nah·⁖·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To weigh⁖· to determine heaviness by balance·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·An unknown weight is set against a known stone⁚⁚The beam tips and reveals which side bears morex·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᘐ❁ᛁ᛭ᚯᑫᔑᔐᑎᓄᚠ·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉ᚦ꞉꞉ category꞉꞉court꞉꞉⁖·
·⁖lookup office꞉꞉catalyst꞉꞉ enoch꞉꞉ᛟ꞉꞉⁖·

⁖word⁚·ᛉᛖᛁᚺᚴᚾᚾ⁖·˙word⁚·

·⁖origin…language⁚·Gothic·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Gothic·⁖·˙font⁚·
·⁖pronunciation⁚·WEEH+N·⁖·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To hallow⁚⁚consecrate⁚⁚sanctify·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·Rite marks common matter and sets it apart⁚⁚Ordinary use stops at the new boundaryᚦ·⁖·˙ostensive⁚·
·⁖grimchain⁚·ᛗᚤᚷ❄ᛈᛦᛁᛒᚺᚤᛋᚼᚦ⧗·⁖·˙grimchain⁚·
·⁖·˙lookup⁖·
·⁖·˙glyphs⁖·
·⁖glyph symbol꞉꞉ᔓ꞉꞉ category꞉꞉court꞉꞉⁖·
·⁖lookup office꞉꞉catalyst꞉꞉ enoch꞉꞉ᛟ꞉꞉⁖·

⁖word⁚·ᚦᚾᘜᛒᚦᚺᛏᏀᚣᚾ⁖·˙word⁚·

·⁖origin…language⁚·Gothic·⁖·˙origin…language⁚·
·⁖font⁚·Noto Sans Gothic·⁖·˙font⁚·
·⁖pronunciation⁚·AND+BAH+TI+JAN·⁖·˙pronunciation⁚·
·⁖cadence⁚·Catalyst·⁖·˙cadence⁚·
·⁖non…ostensive⁚·To serve⁚⁚to administer⁚⁚to perform·⁖·˙non…ostensive⁚·
·⁖ostensive⁚·A waiting task meets working hands and labor begins where the need is placedᔓ·⁖·˙ostensive⁚·

·⁚grimchain:·ɔɬᗡᛉ'ஜ‹ъ⊙⬡☆ᚷᛊᚢ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᒪ꞉ category꩜꩜court꞉⁚·
·⁚lookup office꩜꩜catalyst꞉ enoch꩜꩜ᘐ꞉⁚·

꞉word⁚·ᘯΣᒍհᘐϧϤ꞉⁚.·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚.·˙font⁚·
·⁚pronunciation⁚·AKESUN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To see·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Light from the present deed reaches the eye⁚⁚carrying edge and motion at
↪ onceᒪ·⁚.·˙ostensive⁚·
·⁚grimchain:·ᛋᗕᐁᐯᖘᛟɪ∫ǂᑎᒧᎽ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᒪ꞉ category꩜꩜court꞉⁚·
·⁚lookup office꩜꩜catalyst꞉ enoch꩜꩜ᘐ꞉⁚·

꞉word⁚·ᓴᒣᚼᚼᒣᚼ꞉⁚.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·STAHN+dah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To stand·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Legs straighten against the ground as upward force carries the body into an upright
↪ bearingᒪ·⁚.·˙ostensive⁚·
·⁚grimchain:·ᛦ᳀ᘯᐃᐁᗉᐅᐉᛘ✡ᘐᘏᘯ꞉⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᐁ꞉ category꩜꩜court꞉⁚·
·⁚lookup office꩜꩜catalyst꞉ enoch꩜꩜ᘐ꞉⁚·

꞉word⁚·ϒЄNϚʎN꞉⁚.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·WAYNJAN·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Catalyst·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To await⁚⁚expect⁚˙ to hope (for)·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Open doorway holds no arriving body⁚⁚Chest and breath remain drawn toward the empty
↪ thresholdᐁ·⁚.·˙ostensive⁚·
·⁚grimchain:·ᛜᚸъ☆ᚲᚺₒᚼᛗᘐᚠᘁ'·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜ᙆ꞉ category꩜꩜court꞉⁚·
·⁚lookup office꩜꩜catalyst꞉ enoch꩜꩜ᘐ꞉⁚·

꞉word⁚·ᚻᚴᚴ꞉⁚.·˙word⁚·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·KHUSH·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To live::to survive·⁚.·˙non…ostensive:·
·⁚ostensive:·Breath::warmth::and motion continue together where danger or weakness has pressed against
↪ the fleshz·⁚.·˙ostensive:·
·⁚grimchain:·𒉺☆ΘΨ𐦟ᘜ⛧Θᘎᔓᕫᖚ◎·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ᑫ꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜catalyst꩝ enoch꩜꩜ᘐ꩝:·

·word:·ᗞᘔ·⁚.·˙word:·

·⁚origin…language:·Old Turkic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚.·˙font:·
·⁚pronunciation:·YAHT·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To lie down·⁚.·˙non…ostensive:·
·⁚ostensive:·Weariness pulls at the limbs until knees fold and weight settles onto the
↪ groundᑫ·⁚.·˙ostensive:·
·⁚grimchain:·𐳺𒉺Ψ☆ϒΘ𐳀𐳀ᵒ▫ƨ𐳋𐳀Ж·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ᑲ꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜catalyst꩝ enoch꩜꩜ᘐ꩝:·

·word:·ᏁᎰᏯᏰᏂᏁᏁᏁ·⁚.·˙word:·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·AFDOHBNAN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·(hapax legomenon) be silent::hold one's peace·⁚.·˙non…ostensive:·
·⁚ostensive:·A warning hand rises::Lips close before the next word can leaveᑲ·⁚.·˙ostensive:·
·⁚grimchain:·ϒ𐳽⊙ᑭᘐᐧℐᘐᔑᖘᑭᘐ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꩜꩜ᖾ꩝ category꩜꩜court꩝:·
·⁚lookup office꩜꩜catalyst꩝ enoch꩜꩜ᘐ꩝:·

·word:·ᏏᏬhΣᏁhOᎩᏀ·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·BASKESUN·⁚.·˙pronunciation:·
·⁚cadence:·Catalyst·⁚.·˙cadence:·
·⁚non…ostensive:·To sleep·⁚.·˙non…ostensive:·
·⁚ostensive:·Weariness drags the eyelids down until waking movement fades and only slow breath
↪ remainsᖾ·⁚.·˙ostensive:·
·⁚grimchain:·Xᗅᔑᘐᐧ𐳀ৰ৭ᒣIᘐ꩜ᘓᗅ𐌡꩜ᘓᛘ꩝·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙

·⁖glyph symbol◦–◦𐌚𐌻 category◦–◦court𐌻⁖·
·⁖lookup office◦–◦catalyst𐌻 enoch◦–◦♂𐌻⁖·

·⁖word⁖·ᚱᚾᚹᚾ⁖·˙word⁖·

·⁖origin…language⁖·Old Norse·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Runic·⁖·˙font⁖·
·⁖pronunciation⁖·RAKA·⁖·˙pronunciation⁖·
·⁖cadence⁖·Catalyst·⁖·˙cadence⁖·
·⁖non…ostensive⁖·to scrape together·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·A rake drags scattered pieces inward until loose matter gathers into one
↪ heap𐌚·⁖·˙ostensive⁖·
·⁖grimchain⁖·♌ᛋ𓃰♂✙I⅋ᔑᛉᛋ𓆉'·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖glyph symbol◦–◦𝒻𐌻 category◦–◦axiomyr𐌻⁖·
·⁖lookup office◦–◦catalyst𐌻 enoch◦–◦♂𐌻⁖·

·⁖word⁖·ϰ𑀮𑀮𑀯̄⁖·˙word⁖·

·⁖origin…language⁖·Tocharian B·⁖·˙origin…language⁖·
·⁖font⁖·Noto Sans Brahmi·⁖·˙font⁖·
·⁖pronunciation⁖·TSUHM·⁖·˙pronunciation⁖·
·⁖cadence⁖·Catalyst·⁖·˙cadence⁖·
·⁖non…ostensive⁖·To grow (intransitive)⁖(causative) to grow (transitive).·⁖·˙non…ostensive⁖·
·⁖ostensive⁖·Rain enters dry soil and green shoots rise⁖⁖Fallen leaves soften into earth beneath
↪ feeding roots⁖a buried seed splits open and presses upward⁖⁖Roots fracture rock⁖⁖leaving fissures
↪ for rain and tender growth✧·⁖·˙ostensive⁖·
·⁖grimchain⁖·ϒᛯ✡⓪ᚗᛉᛘᗡᗣᴞ⊐ƶ▽♳·⁖·˙grimchain⁖·
·⁖·˙lookup·˙
·⁖·˙glyphs·˙
·⁖·˙glyphs·˙
·⁖·˙sydonic…magicae…lexicon⁖·

⁘⁘⁘

·⁖origin…language:·Phoenician·⁖.·˙origin…language:·
·⁖font:·Noto Sans Phoenician·⁖.·˙font:·
·⁖pronunciation:·LEEB·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Heart⁘ the central bodily organ⁚⁚also the inner seat of feeling or
↪ intention·⁖.·˙non…ostensive:·
·⁖ostensive:·Muffled pulse presses beneath rib and breastbone⁚⁚hidden warmth moving blood through the
↪ enclosed chamber♀⁘⁖.·˙ostensive:·
·⁖grimchain:·𐤆ᚳᛯ☆ᛀ1ᛈ𐤊ᚸᘔᔑᔕᔐᚣ米·⁖.·˙grimchain:·
·⁖.·˙lookup·⁘
·⁖.·˙glyphs·⁘
·⁖glyph symbol꩜꩜ᵋ2ㅓ꩜ category꩜꩜ᵋemission꩜⁖·
·⁖lookup office꩜꩜ᵋrelic꩜ enoch꩜꩜ᵋ𐤊꩜⁖·

⁖word:·𐤆𐤄⁖.·˙word:·

·⁖origin…language:·Sumerian·⁖.·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖.·˙font:·
·⁖pronunciation:·EH⟊meh·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Tongue⁘ the muscular organ in the mouth⁚⁚also speech or language·⁖.·˙non…ostensive:·
·⁖ostensive:·Wet muscle turns breath against tooth and palate⁚⁚and shaped sound leaves between the
↪ lips2ㅓ⁖.·˙ostensive:·
·⁖grimchain:·𐤃ᛉᛘᛝ♂米ᘔᔑᑒ▫ᵹʃᛞ·⁖.·˙grimchain:·
·⁖.·˙lookup·⁘
·⁖.·˙glyphs·⁘
·⁖glyph symbol꩜꩜ᵋ𐤇꩜ category꩜꩜ᵋemission꩜⁖·
·⁖lookup office꩜꩜ᵋrelic꩜ enoch꩜꩜ᵋ𐤊꩜⁖·

⁖word:·سدابوس⁖.·˙word:·

·⁖origin…language:·Avestan·⁖.·˙origin…language:·
·⁖font:·Noto Sans Avestan·⁖.·˙font:·
·⁖pronunciation:·dah⟊AY⟊nah·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Religious vision⁚⁚conscience⁚⁚inner seeing⁘ later religion·⁖.·˙non…ostensive:·
·⁖ostensive:·Carved emblem rests against cloth beside ritual fire⁘ polished surface presses over the
↪ breast beneath a binding cord𐤇·⁖.·˙ostensive:·
·⁖grimchain:·𐤆ᛉᚺ米ᛘᚤᚱᛤᛋ◎𐤉·⁖.·˙grimchain:·
·⁖.·˙lookup·⁘
·⁖.·˙glyphs·⁘
·⁖glyph symbol꩜꩜ᵋ𐤔꩜ category꩜꩜ᵋemission꩜⁖·
·⁖lookup office꩜꩜ᵋrelic꩜ enoch꩜꩜ᵋ𐤊꩜⁖·

⁖word:·손⁖.·˙word:·

·⁖origin…language:·Middle Korean·⁖.·˙origin…language:·
·⁖font:·Noto Sans KR·⁖.·˙font:·
·⁖pronunciation:·sOHn·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Hand·⁖.·˙non…ostensive:·
·⁖ostensive:·Five fingers close around the haft⁘ wood and clay answer the pressure of the
↪ palm𐤔·⁖.·˙ostensive:·
·⁖grimchain:·◎ᚻᘔᘐᛘᔑᚹᚣᛋᛖᛏ·⁖.·˙grimchain:·

·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol○─○ᑫΨ» category○─○ᑫemission»⁚·
·⁚lookup office○─○ᑫrelic» enoch○─○ᑫஃ»⁚·

·word:·𓏏⁚·˙word:·

·⁚origin…language:·Egyptian·⁚·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚·˙font:·
·⁚pronunciation:·meh+JAHT·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Book·⁚· written document or written roll·⁚·˙non…ostensive:·
·⁚ostensive:·Dark signs lie quiet along rolled papyrus··and speech remains where no mouth is
↪ openΨ·⁚·˙ostensive:·
·⁚grimchain:·ᘐᛦᛋᐟᐟஃ𓂀✱ᛋᛦ✱▽ᘐ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol○─○ᑫᑭ» category○─○ᑫemission»⁚·
·⁚lookup office○─○ᑫrelic» enoch○─○ᑫஃ»⁚·

·word:·𓂧⁚·˙word:·

·⁚origin…language:·Egyptian·⁚·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚·˙font:·
·⁚pronunciation:·JOO·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Mountain··hill··desert highland·⁚·˙non…ostensive:·
·⁚ostensive:·Carved peak rises from a palm+sized stone·⁚· ridged stone sides gather height into one
↪ weight that can be carriedᑭ·⁚·˙ostensive:·
·⁚grimchain:·ᘐAⓄ𓊝ᘐ⬡ᚷᚠ○▫ᚢᗤᓂ·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol○─○ᑫ☽» category○─○ᑫemission»⁚·
·⁚lookup office○─○ᑫrelic» enoch○─○ᑫஃ»⁚·

·word:·𓉔⁚·˙word:·

·⁚origin…language:·Sumerian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·TESH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Dignity·⁚· recognized worth··honor··or standing carried by a person or
↪ being·⁚·˙non…ostensive:·
·⁚ostensive:·Insignia rests beside the raised seat··and even empty··the place bears the weight of
↪ recognized standing☽·⁚·˙ostensive:·
·⁚grimchain:·ᘔᓄ⊕❄ᑭᓀᔓᙆᙆᓀᐯ✱I·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol○─○ᑫ⊙» category○─○ᑫemission»⁚·
·⁚lookup office○─○ᑫrelic» enoch○─○ᑫஃ»⁚·

·word:·𓀀⁚·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·WAHS·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Power::dominion::authority·⁚· the was+scepter·⁚.·˙non…ostensive:·
·⁚ostensive:·Carved staff rises beside the seat::and long polished wood holds the ruling mark above
↪ common ground⊙·⁚.·˙ostensive:·
·⁚grimchain:·ᗅᑐᕒ⊙⧗ᕼᕼᔭ𐦠ℂ❄❄Θ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○�really category○–○₹emission»⁚·
·⁚lookup office○–○₹relic» enoch○–○₹ஃ»⁚·

₹word:·ነፍስ₹⁚.·˙word:·

·⁚origin…language:·Ge'ez·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚.·˙font:·
·⁚pronunciation:·NAHFS·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Breath::soul::or spirit·⁚· the life principle of a being·⁚.·˙non…ostensive:·
·⁚ostensive:·Warm air passes chest and mouth·⁚· ribs rise and fall::and hanging cloth stirs under the
↪ same unseen current℃·⁚.·˙ostensive:·
·⁚grimchain:·۞∀ᗅᑫᚷᕮᓯᛇᚠ'ㄟ«ь·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₹ᛉ» category○–○₹parliament»⁚·
·⁚lookup office○–○₹relic» enoch○–○₹ஃ»⁚·

₹word:·🎹₹⁚.·˙word:·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·TOOP+poo·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Clay tablet·⁚· written tablet or document·⁚.·˙non…ostensive:·
·⁚ostensive:·Wedges lie deep in clay::and hardened surface keeps the account where no tongue can alter
↪ it𒌋·⁚.·˙ostensive:·
·⁚grimchain:·ᑎ◌ᗰᓀᐁᐁ◎✱ᓄ✡ᗡ⊣⁞ᔕ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○₹ᛉ» category○–○₹parliament»⁚·
·⁚lookup office○–○₹relic» enoch○–○₹ஃ»⁚·

₹word:·ᛒᚱᑎ₹⁚.·˙word:·

·⁚origin…language:·Old Norse·⁚.·˙origin…language:·
·⁚font:·Noto Sans Runic·⁚.·˙font:·
·⁚pronunciation:·BROO·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Bridge·⁚.·˙non…ostensive:·
·⁚ostensive:·Timber and stone span bank to bank·⁚· feet cross where water divides the
↪ groundᛉ·⁚.·˙ostensive:·
·⁚grimchain:·ъ⛄⊣⊣∇ᚵ◎ᛉ∀ₒ◎🎹ᓄ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𝌆꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜ᕰ꩝⁚·

·word:·ɯє·⁚·˙word:·

·⁚origin…language:·Coptic·⁚·˙origin…language:·
·⁚font:·Noto Sans Coptic·⁚·˙font:·
·⁚pronunciation:·MEH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Truth or justice·⁚ what is real and right·⁚·˙non…ostensive:·
·⁚ostensive:·Straight edge beside balanced weight·⁚ neither burden is favored⁚⁚and the beam refuses
↪ every wish that asks it to lean𝌆·⁚·˙ostensive:·
·⁚grimchain:·Ⱪ✡ǂ𐎃ᓄ♈ϴIᛩⱪᛉᒐ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ᘐ꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜ᕰ꩝⁚·

·word:·⁼·⁚·˙word:·

·⁚origin…language:·Egyptian·⁚·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚·˙font:·
·⁚pronunciation:·PEE+tat·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·A bow used as a weapon·⁚·˙non…ostensive:·
·⁚ostensive:·Bent wood holds the cord taut⁚⁚and the waiting shot lives along the curveᘐ·⁚·˙ostensive:·
·⁚grimchain:·❄ᘯ𐎃ℂ𐎇ℏ∀ǂ↗Φϕᛯᑊᐧᐧ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ϒ𐙊꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜ᕰ꩝⁚·

·word:·☤·⁚·˙word:·

·⁚origin…language:·Egyptian·⁚·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚·˙font:·
·⁚pronunciation:·AHN+kh·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Life·⁚·˙non…ostensive:·
·⁚ostensive:·Looped sign rests against warm skin·⁚ pulse moves beneath the carved curve⁚⁚and breath
↪ lifts the chest below itϒ𐙊·⁚·˙ostensive:·
·⁚grimchain:·ᨎ♀ᓀɾƃᗡ↗✪☸I0ᚏᵹ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜ℳ꩝ category꩜꩜parliament꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜ᕰ꩝⁚·

·word:·ፈትል·⁚·˙word:·

·⁚origin…language:·Ge'ez·⁚·˙origin…language:·
·⁚font:·Noto Sans Ethiopic·⁚·˙font:·
·⁚pronunciation:·FAHT+l·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·

·⁚non…ostensive⁚·Thread⁚⁚yarn⁚⁚cord⁚⁚wire⁚⁚or wick·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One slender length crosses cloth and knot⁚⁚vanishing into the weave⁚˙ tension keeps the
↪ woven edges together·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᚦΘᛦᛏᒐᚴᛉᚸ⬡ᛜᛋᚺᛊᛉ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᕁ category⟜⟜parliament⁚·
·⁚lookup office⟜⟜relic enoch⟜⟜ᕃ⁚·

⁖word⁚·ᛏɹl̄ɻ̈⁚.·˙word⁚·

·⁚origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁚pronunciation⁚·KUHNTEE·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Gong⁚⁚cymbal·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Bronze answers the strike⁚⁚and ringing spreads farther than the hand that woke
↪ itᕁ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·XIᛯᛰ⊛ɾᛇᛯᘜᛕƆX1·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᛒ category⟜⟜parliament⁚·
·⁚lookup office⟜⟜relic enoch⟜⟜ᕃ⁚·

⁖word⁚·σολ⁚.·˙word⁚·

·⁚origin…language⁚·Coptic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Coptic·⁚.·˙font⁚·
·⁚pronunciation⁚·CHOHL·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Lie or falsehood·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Three grain measures sit beside four tally cuts⁚˙ crooked count remains fixed in
↪ clayᛒ·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ᛇᚹᚻᚿᛘᛉ✳ᛒ⊙⬡❁❁·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜↗ category⟜⟜parliament⁚·
·⁚lookup office⟜⟜relic enoch⟜⟜ᕃ⁚·

⁖word⁚·ᚑ ᚒ ᚓᚔ⁚.·˙word⁚·

·⁚origin…language⁚·Old Persian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Persian·⁚.·˙font⁚·
·⁚pronunciation⁚·DAH⟊tah·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Given or created⁚˙ brought into being or bestowed·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Sealed parcel rests beside an open palm⁚˙ shaped matter keeps the allotted portion⁚⁚and
↪ touch is no longer needed↗·⁚.·˙ostensive⁚·
·⁚grimchain⁚·᚛ᚥᚭᚹᛪᚯᛃᚯᚴᛏ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜ᚯ category⟜⟜parliament⁚·
·⁚lookup office⟜⟜relic enoch⟜⟜ᕃ⁚·

:word:·𐎠𐎡𐎢𐎣:.·word:·

·origin…language:·Old Persian·.·origin…language:·
·font:·Noto Sans Old Persian·.·font:·
·pronunciation:·DAH+tam·.·pronunciation:·
·cadence:·Relic·.·cadence:·
·non…ostensive:·Law· an established rule or ordinance·.·non…ostensive:·
·ostensive:·Carved strokes cut deep across the stone face::and fixed marks hold one ruling measure from edge to edge·.·ostensive:·
·grimchain:·𐫀𐫁𐫂𐫃𐫄𐫅𐫆𐫇𐫈𐫉𐫊·.·grimchain:·
·.·lookup·
·.·glyphs·
·glyph symbol○-○𐫄 category○-○parliament:·
·lookup office○-○relic enoch○-○𐫆:·

:word:·𑀰𑀺𑀘:.·word:·

·origin…language:·Tocharian B·.·origin…language:·
·font:·Noto Sans Brahmi·.·font:·
·pronunciation:·YTAHRYEH·.·pronunciation:·
·cadence:·Relic·.·cadence:·
·non…ostensive:·Road::way·.·non…ostensive:·
·ostensive:·Pressed earth runs beneath foot and wheel::a worn passage carrying distance across the ground𐫄·.·ostensive:·
·grimchain:·𐫈𐫉𐫊𐫋𐫌𐫍𐫎𐫏𐫐𐫑𐫒·.·grimchain:·
·.·lookup·
·.·glyphs·
·glyph symbol○-○𐫍 category○-○parliament:·
·lookup office○-○relic enoch○-○𐫆:·

:word:·𓂋:.·word:·

·origin…language:·Sumerian·.·origin…language:·
·font:·Noto Sans Cuneiform·.·font:·
·pronunciation:·DEEB·.·pronunciation:·
·cadence:·Relic·.·cadence:·
·non…ostensive:·Pass· to go by::through::across::or beyond a place or boundary·.·non…ostensive:·
·ostensive:·Narrow opening cuts bank::wall::or barrier· worn ground continues through the break in solid matter𐫍·.·ostensive:·
·grimchain:·𐫓𐫔𐫕𐫖𐫗𐫘𐫙𐫚𐫛𐫜𐫝·.·grimchain:·
·.·lookup·
·.·glyphs·
·glyph symbol○-○𐫔 category○-○goetic:·
·lookup office○-○relic enoch○-○𐫆:·

:word:·𐤓𐤋𐤅:.·word:·

·origin…language:·Phoenician·.·origin…language:·
·font:·Noto Sans Phoenician·.·font:·
·pronunciation:·SHOH+resh·.·pronunciation:·
·cadence:·Relic·.·cadence:·
·non…ostensive:·Root· the underground part of a plant that anchors and feeds it·.·non…ostensive:·
·ostensive:·Buried fibres grip dark earth and drink below the stem::holding green growth against the wind𐫔·.·ostensive:·

·⁖grimchain⁖·ϥ◎▽✳ᖇᒐᔑ♂ϒ'XᖇϘ·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol⁖◦◦ₒ⬡꜀ category⁖◦◦ₒgoetic꜀⁖·
·⁖lookup office⁖◦◦ₒrelic꜀ enoch⁖◦◦ₒ♳꜀⁖·

·word⁖·ⲁⲱⲱⲁⲉ⁖.·˙word⁖·

·⁖origin…language⁖·Coptic·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Coptic·⁖.·˙font⁖·
·⁖pronunciation⁖·JOH+oh+meh·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Relic·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Scroll or book⁖· a written papyrus document·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Rolled papyrus coils around dark signs··and hidden lines lie folded within the same bound
↪ body⬡·⁖.·˙ostensive⁖·
·⁖grimchain⁖·⧗▫♂ᑫᔕᚼᚺᚻᛏᑎᘏᚌ⟰᛬·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol⁖◦◦ₒ✡꜀ category⁖◦◦ₒgoetic꜀⁖·
·⁖lookup office⁖◦◦ₒrelic꜀ enoch⁖◦◦ₒ♳꜀⁖·

·word⁖·ررصدا⁖.·˙word⁖·

·⁖origin…language⁖·Avestan·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Avestan·⁖.·˙font⁖·
·⁖pronunciation⁖·AH+tar·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Relic·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·Fire⁖· combustion producing heat and light·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Charred wick carries a narrow brightness⁖· red coal keeps heat beneath a dark
↪ crust✡·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ؼرزᚺᚤᚣᛯ♂ᚿᛟᛋᗉᚱ▫ᚸ·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol⁖◦◦ₒ☆꜀ category⁖◦◦ₒgoetic꜀⁖·
·⁖lookup office⁖◦◦ₒrelic꜀ enoch⁖◦◦ₒ♳꜀⁖·

·word⁖·ᙡሕዝ⁖.·˙word⁖·

·⁖origin…language⁖·Ge'ez·⁖.·˙origin…language⁖·
·⁖font⁖·Noto Sans Ethiopic·⁖.·˙font⁖·
·⁖pronunciation⁖·wuh+HEEZ·⁖.·˙pronunciation⁖·
·⁖cadence⁖·Relic·⁖.·˙cadence⁖·
·⁖non…ostensive⁖·River or stream⁖· flowing natural watercourse·⁖.·˙non…ostensive⁖·
·⁖ostensive⁖·Water presses between earth+banks⁖· stones and reeds hold one long course through the
↪ land☆·⁖.·˙ostensive⁖·
·⁖grimchain⁖·ᘐ◎○♅♂⬡✳△ᖘᐁᘐᚳᔐ·⁖.·˙grimchain⁖·
·⁖.·˙lookup⁖·
·⁖.·˙glyphs⁖·
·⁖glyph symbol⁖◦◦ₒ❂꜀ category⁖◦◦ₒgoetic꜀⁖·
·⁖lookup office⁖◦◦ₒrelic꜀ enoch⁖◦◦ₒ♳꜀⁖·

·word⁖·⛦卄⁖.·˙word⁖·

·⁖origin…language꞉·Sumerian·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Cuneiform·⁖·˙font꞉·
·⁖pronunciation꞉·EEM·⁖·˙pronunciation꞉·
·⁖cadence꞉·Relic·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Clay⁘ fine⟊grained earth that becomes plastic when wet and hardens when dried or
↪ fired·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Wet earth yields under finger and tool⁘ pressed hollows hold their shape in the same soft
↪ body❂·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐹨ᥱሐ᜔𑁍ᑫካ𐹳ᦗᛕ⁕ᑭ·⁖·˙grimchain꞉·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁖glyph symbol꞉꞉❀꞉ category꞉꞉goetic꞉꞉
·⁖lookup office꞉꞉relic꞉ enoch꞉꞉ᐃ꞉꞉

꞉word꞉·ባሕር꞉·⁖·˙word꞉·

·⁖origin…language꞉·Ge'ez·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖·˙font꞉·
·⁖pronunciation꞉·BAH⟊khr·⁖·˙pronunciation꞉·
·⁖cadence꞉·Relic·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Sea or large lake⁘ a great body of water·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Salt crust whitens a shell⁝⁝and trapped brine rests heavy in its hollow❀·⁖·˙ostensive꞉·
·⁖grimchain꞉·𐹠ᛏᐃᦞᦓᙏᗅIᙆᚏᚷሐᗌᛋ·⁖·˙grimchain꞉·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁖glyph symbol꞉꞉❋꞉ category꞉꞉goetic꞉꞉
·⁖lookup office꞉꞉relic꞉ enoch꞉꞉ᐃ꞉꞉

꞉word꞉·መንፈስ꞉·⁖·˙word꞉·

·⁖origin…language꞉·Ge'ez·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ethiopic·⁖·˙font꞉·
·⁖pronunciation꞉·man⟊FAHS·⁖·˙pronunciation꞉·
·⁖cadence꞉·Relic·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Breath⁝⁝wind⁝⁝spirit⁝⁝or vital force·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Chest rises under skin⁘ hanging cloth swells⁝⁝and dust lifts under the same unseen
↪ current⁕·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᦞᦞᦞᦞᦞ)ᚼᚷ⍓ᙏᘎᘐᗄᛊ·⁖·˙grimchain꞉·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁖glyph symbol꞉꞉⧗꞉ category꞉꞉goetic꞉꞉
·⁖lookup office꞉꞉relic꞉ enoch꞉꞉ᐃ꞉꞉

꞉word꞉·𐤓𐤉𐤔𐤓꞉·⁖·˙word꞉·

·⁖origin…language꞉·Ugaritic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ugaritic·⁖·˙font꞉·
·⁖pronunciation꞉·AHHBATU·⁖·˙pronunciation꞉·
·⁖cadence꞉·Relic·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Love·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Kept token rests warm against cloth or skin⁘ worn edge is polished where fingers return
↪ again and again⧗·⁖·˙ostensive꞉·
·⁖grimchain꞉·⊕ᛯᗅ♂ᛦ⁕𐦍ᦓᛗᘏᑭᦓ❀᜵·⁖·˙grimchain꞉·
·⁖·˙lookup⁘

·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜⊛꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

꞉word⁚·SKᏁᏗᑎS꞉⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·SKADUS·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Shadow·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Dark shape lies beside the body that stops the light⁖faithful to every edge without
↪ carrying its substance⊛·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐓟𐓪𐓯𐓵𐓗𐓶𐓲𐓴𐓝𐓪𐓫𐓱·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜❄꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

꞉word⁚·ᏯᏁIᏗᏁ꞉⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·DEYELA·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Participation·˙ community·˙ partnership·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Shared cup passes through many hands·˙ one rim gathers many mouths⁖and the drink belongs
↪ to no single grasp❄·⁚·˙ostensive⁚·
·⁚grimchain⁚·ₒ◎ᛁ✙ᛘₒᒧᚺᚷℂᕒᛟᐱ✝·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜❁꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

꞉word⁚·ᐅᚸᚴ꞉⁚·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚·˙font⁚·
·⁚pronunciation⁚·yah⟊GHUH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·enemy⁖hostile·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Foreign standard rises beyond the barred way·˙ spearpoints gather beneath the hanging
↪ cloth❁·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐓟𐓪𐓯ₒ𐓟𐓝◎𐓘ᑲᔑ𐓝𐓝𐓯·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜⬡꩝ category꩜꩜goetic꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

꞉word⁚·𘜶꞉⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·TAH·⁚·˙pronunciation⁚·

·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Rest⁚· to cease activity for recovery or stillness·⁚.·˙non…ostensive:·
·⁚ostensive:·Mat bears loosened limbs⁚· earth receives the weight without asking for
↪ motion⏣·⁚.·˙ostensive:·
·⁚grimchain:·𐑓𐑜𐑣⁐∀〈𐑕𐑦𐑹𐑧𐐼0𐑖𐑚·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○𐑮𐑸 category○–○𐑒court𐑸:·
·⁚lookup office○–○𐑒relic𐑸 enoch○–○𐑓𐑙𐑸:·

·:word:·ꞁɦɬƛ̄:.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·OHNYUHNT·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Beginning·⁚.·˙non…ostensive:·
·⁚ostensive:·Untouched clay bears one clean incision⁚· smooth face remains around the solitary
↪ cut𐑮·⁚.·˙ostensive:·
·⁚grimchain:·𐑢𑁍▽ᛘ𐑕ᛚ𐑢ᑫᚺ'▽ᛗ▽·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○𐑕𐑸 category○–○𐑒court𐑸:·
·⁚lookup office○–○𐑒relic𐑸 enoch○–○𐑓𐑙𐑸:·

·:word:·ᔑᕫ:.·˙word:·

·⁚origin…language:·Phoenician·⁚.·˙origin…language:·
·⁚font:·Noto Sans Phoenician·⁚.·˙font:·
·⁚pronunciation:·DOHR·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Family or clan⁚· by extension a continuing generation or enduring
↪ succession·⁚.·˙non…ostensive:·
·⁚ostensive:·House+mark bites timber::grave+stone::and heirloom⁚· the same carved sign repeats across
↪ separated bodies𐑕·⁚.·˙ostensive:·
·⁚grimchain:·𐑐𐑗𑁍▽⊙⟰ıᖷ⬡𐑐𐑗〈𐑖·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·
·⁚glyph symbol○–○𐑕𐑸 category○–○𐑒court𐑸:·
·⁚lookup office○–○𐑒relic𐑸 enoch○–○𐑓𐑙𐑸:·

·:word:·ꞁ+:.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·OHKOH·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Fruit⁚· result::effect::consequence·⁚.·˙non…ostensive:·
·⁚ostensive:·Ripened weight hangs from the branch::root and blossom gathered into one skin+bound
↪ body𐑕·⁚.·˙ostensive:·
·⁚grimchain:·ᑲᛘᛇᚠᛘᔕᛋᛏᛞᚠᛏᛞᐅ◎·⁚.·˙grimchain:·
·⁚.·˙lookup:·
·⁚.·˙glyphs:·

·⁖glyph symbol꞉𓂧 category꞉court꞉·
·⁖lookup office꞉relic enoch꞉𓂋꞉·

꞉word꞉·तन्तु꞉꞉·˙word꞉·

·⁖origin…language꞉·Sanskrit·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Devanagari·⁖·˙font꞉·
·⁖pronunciation꞉·tuhn+tu·⁖·˙pronunciation꞉·
·⁖cadence꞉·Relic·⁖·˙cadence꞉·
·⁖non…ostensive꞉·thread꞉꞉cord꞉꞉string꞉꞉line꞉꞉wire꞉꞉warp (of a web)꞉꞉filament꞉꞉fibre⁖· cobweb⁖· a succession of
↪ sacrificial performances⁖· any one propagating his family in regular succession⁖· a line of
↪ descendants⁖· any continuity (as of thirst or hope)⁖· name of a सामन् (sāman)⁖· = नाग
↪ (+nāga)·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·One narrow length runs from fastening to fastening⁖· tension keeps cloth and web from
↪ opening apart𓂧·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᛊᚸᛕᛉᛒᛃᛞᛓᛗᚢᛇ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉𓂋 category꞉court꞉·
·⁖lookup office꞉relic enoch꞉𓂋꞉·

꞉word꞉·𐎍𐎗𐎚꞉꞉·˙word꞉·

·⁖origin…language꞉·Ugaritic·⁖·˙origin…language꞉·
·⁖font꞉·Noto Sans Ugaritic·⁖·˙font꞉·
·⁖pronunciation꞉·YSD·⁖·˙pronunciation꞉·
·⁖cadence꞉·Relic·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Foundation·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Buried stone lies below wall and column꞉꞉unseen from above⁖· every height settles its
↪ burden into that hidden bed𓂋·⁖·˙ostensive꞉·
·⁖grimchain꞉·ᚿᚺᛇᛉᛘᚮᛯᛇᚱᚤ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉𓂧 category꞉court꞉·
·⁖lookup office꞉relic enoch꞉𓂋꞉·

꞉word꞉·𝌆꞉·˙word꞉·

·⁖origin…language꞉·Tangut·⁖·˙origin…language꞉·
·⁖font꞉·Noto Serif Tangut·⁖·˙font꞉·
·⁖pronunciation꞉·REHR·⁖·˙pronunciation꞉·
·⁖cadence꞉·Relic·⁖·˙cadence꞉·
·⁖non…ostensive꞉·Expand꞉꞉swell꞉꞉or become wide·⁖·˙non…ostensive꞉·
·⁖ostensive꞉·Stretched cloth fills the frame from peg to peg꞉꞉and broad surface presses against every
↪ held edge𓂻·⁖·˙ostensive꞉·
·⁖grimchain꞉·𓂻ᛯz⬡ᛯᚦᚲᛟᚸᛂᛇᚺᛋᚳ·⁖·˙grimchain꞉·
·⁖·˙lookup꞉˙
·⁖·˙glyphs꞉˙
·⁖glyph symbol꞉𓂻 category꞉court꞉·
·⁖lookup office꞉relic enoch꞉𓂋꞉·

꞉word꞉·𐤀𐤕𐤍𐤉𐤌꞉·˙word꞉·

·⁚origin…language:·Akkadian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·SHEEM⟊toom·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·determined order::nature of things::divine decree⁚· fate::destiny⁚· (euphemistic) death⁚·
↪ testament::will⁚· original amount::principal·⁚.·˙non…ostensive:·
·⁚ostensive:·Sealed clay bears one fixed impression⁚· neighboring lines divide around the mark without
↪ moving it𒁹·⁚.·˙ostensive:·
·⁚grimchain:·⬡𒌋𒀭𒌋𒉿𒍪𒁹𒃶𒄑𒂗𒍪𒀭·⁚.·˙grimchain:·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𒄩𒄭 category○–○𒄩court𒄬:·
·⁚lookup office○–○𒄩relic𒄭 enoch○–○𒄩𒀭𒄬:·

⁖word:·𒄿𒅆⁖·⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·THAHBA·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·To return::go back::come back·⁚.·˙non…ostensive:·
·⁚ostensive:·Worn threshold keeps the curve of repeated tread⁚· one road meets the same stone from open
↪ distance𐎗·⁚.·˙ostensive:·
·⁚grimchain:·𐎜𐎚𐎕𐎍'𐎀𐎊𐎈𐎚𐎄𐎇𐎆·⁚.·˙grimchain:·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐎍𐎄 category○–○𐎍court𐎄:·
·⁚lookup office○–○𐎍relic𐎄 enoch○–○𐎍𐎀𐎄:·

⁖word:·𐎒𐎅𐎊⁖·⁚.·˙word:·

·⁚origin…language:·Ugaritic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Ugaritic·⁚.·˙font:·
·⁚pronunciation:·MSK·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Mixed drink::mead·⁚.·˙non…ostensive:·
·⁚ostensive:·Honey and water share one cup⁚· neither portion can be lifted apart from the mingled
↪ body𐎍·⁚.·˙ostensive:·
·⁚grimchain:·𐎃𐎚𐎇𐎊𐎐𐎈⬡𐎔𐎖𐎅𐎊·⁚.·˙grimchain:·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol○–○𐎋𐎄 category○–○𐎍court𐎄:·
·⁚lookup office○–○𐎍relic𐎄 enoch○–○𐎍𐎀𐎄:·

⁖word:·दूप⁖·⁚.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·ROO⟊puh·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Form::shape::appearance::color::or visible figure·⁚.·˙non…ostensive:·
·⁚ostensive:·Molded clay holds one continuous contour::and edge answers edge around the bounded
↪ body𐎌·⁚.·˙ostensive:·
·⁚grimchain:·𐎅𐎍𐎗𐎒𐎘𐎊𐎈𐎚𐎉𐎑𐎅𐎃𐎊𐎌·⁚.·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀ᵩ₎ category₀₋₀court₎⁚·
·⁚lookup office₀₋₀relic₎ enoch₀₋₀ₔ₎⁚·

·word:·ᖚᖫJ⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·PEELEH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Wound::injury·⁚·˙non…ostensive:·
·⁚ostensive:·Red seam opens through torn flesh::and blood gathers bright along the broken
↪ edgeᵩ·⁚·˙ostensive:·
·⁚grimchain:·𐦝❁ᬘᬉᬍᬗᬘ𐦝𑁍ᬮᬌᬍ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀ₒ₎ category₀₋₀court₎⁚·
·⁚lookup office₀₋₀relic₎ enoch₀₋₀ₔ₎⁚·

·word:·✨⁚·˙word:·

·⁚origin…language:·Akkadian·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·KAHSH+doom·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·successful::achieved::accomplished::attained::conquered·⁚˙ sufficient::appropriate·⁚˙
↪ arrived::available (of goods)·⁚·˙non…ostensive:·
·⁚ostensive:·Fitted stone locks face to face across the wall::and gathered weight stands with no open
↪ seamᵒ·⁚·˙ostensive:·
·⁚grimchain:·ꠇₔᵩ⩛◎ᦓᦔᦓᦔᦓᦔᦓᦔ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀ꠇ₎ category₀₋₀court₎⁚·
·⁚lookup office₀₋₀relic₎ enoch₀₋₀ₔ₎⁚·

·word:·ᛦᚱᚴᛦ⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·
·⁚pronunciation:·ARKA·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·(Abrahamic religion) ark·⁚˙ money box::chest·⁚·˙non…ostensive:·
·⁚ostensive:·Heavy timber and iron bolt enclose the dark interior::and sealed cedar bears the locked
↪ cargo in unmoving restꠇ·⁚·˙ostensive:·
·⁚grimchain:·⊙⟰ᓚᵨ0⊛ᬘ𐦝ᚹᛉᛯ⍓ᗐ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol₀₋₀ᛋ₎ category₀₋₀court₎⁚·
·⁚lookup office₀₋₀relic₎ enoch₀₋₀ₔ₎⁚·

·word:·ᛦᒐ⁜⁚·˙word:·

·⁚origin…language꞉·Tocharian A·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·TAHPUHKEE·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Mirror·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Polished metal gives back whatever light brings before its bright face·⁚·˙ostensive꞉·
·⁚grimchain꞉·⬡ᔑ✳𑜨ᦓ❋ᑭᓰᔐ𐑅⋐⛛·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𞥐𞥑 category꞉○–○𞥐court𞥑⁚·
·⁚lookup office꞉○–○𞥐relic𞥑 enoch꞉○–○𞥐𞥒⁚·

꞉word꞉·ᐃᓇᑌ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·AHEESEH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Cooking pot·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Blackened vessel holds grain and water above flame꞉꞉and thick clay walls gather heat around the enclosed food𐑅·⁚·˙ostensive꞉·
↪
·⁚grimchain꞉·⛧ᔐ◌ᔕᑎ◎ᔕᛘᛉ᛫ᚷᑫᔑᐊ⊨ᦓ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𞥐𞥓 category꞉○–○𞥐court𞥑⁚·
·⁚lookup office꞉○–○𞥐relic𞥑 enoch꞉○–○𞥐𞥒⁚·

꞉word꞉·ᚲᐢᚴ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Ugaritic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Ugaritic·⁚·˙font꞉·
·⁚pronunciation꞉·AHMQ·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Valley·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Grooved stone runs low between two raised ridges꞉꞉and a thin thread of water settles in the carved hollow𐑅·⁚·˙ostensive꞉·
↪
·⁚grimchain꞉·ᛯᛞ❋✡𑜨ᦓ❁ᔐᚼᑭᛘᚸ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○𞥐ᔑ𞥑 category꞉○–○𞥐court𞥑⁚·
·⁚lookup office꞉○–○𞥐relic𞥑 enoch꞉○–○𞥐𞥒⁚·

꞉word꞉·ᑐᔑᑐᔑ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Old Turkic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Old Turkic·⁚·˙font꞉·
·⁚pronunciation꞉·BAHLBAHL·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Monument꞉꞉consecratory statue for the deceased꞉꞉usually for noblemen and rulers·⁚·˙non…ostensive꞉·
↪
·⁚ostensive꞉·Stone figure stands where living flesh does not꞉꞉carved face meeting wind꞉꞉rain꞉꞉and open ground without surrendering its placeᔑ·⁚·˙ostensive꞉·
↪
·⁚grimchain꞉·ↄᘐᛯᘔ♀ᚲᑭₒᛘᘔᚷᚱᛗ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙

·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉Ɣ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚣ꞉·

꞉word꞉·प्रथस्꞉·˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚·˙font꞉·
·⁚pronunciation꞉·pruh+thuhs·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·width꞉:extension·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Taut cord spans from mark to mark·˙ fastened ends hold apart at one fixed
↪ reachƔ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᛟƔᛃᛯᛞᚹᛏᚺᚠᚹᚢᛟᚸ·⁚·˙grimchain꞉·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉ɛ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚣ꞉·

꞉word꞉·𐬀𐬵𐬎𐬨𐬀𐬎𐬌꞉·˙word꞉·

·⁚origin…language꞉·Avestan·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Avestan·⁚·˙font꞉·
·⁚pronunciation꞉·frah+VAH+shee·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Protective preexistent spirit or spiritual double·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Carved guardian emblem rests against cloth and grave+stone·˙ identical face appears on
↪ both surfaces while living flesh is absentɛ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᚹᛞᛜᛍᛯᚺᚳᚠᚻᚸᚸᚩ◎ᛋ·⁚·˙grimchain꞉·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉Ⴙ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚣ꞉·

꞉word꞉·ᚣᚢᚣᚴᚱ꞉·˙word꞉·

·⁚origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁚pronunciation꞉·DULGS·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Debt·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Tallied notch remains beside an empty measure·˙ pledged weight has not entered the
↪ bowlᛋ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᛞᚺ◎ᚸ▫ᚩᚱᛝᛉᚱᚸᛇᛞᚺ·⁚·˙grimchain꞉·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉'꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚣ꞉·

꞉word꞉·SKIᚣᚣᚢS꞉·˙word꞉·

·⁚origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁚pronunciation꞉·SKIL+DUS·⁚·˙pronunciation꞉·

·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Shield·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Broad metal face takes the blow⁚⁚and flesh remains behind the held barrier'·⁚·˙ostensive꞉·
·⁚grimchain꞉·⊰⍝♌♀⍎⍢⏁I⍞⍎◎A·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋ᛬ category꞉○–○ᵋcourt᛬꞉·
·⁚lookup office꞉○–○ᵋrelic᛬ enoch꞉○–○ᵋ⁂᛬꞉·

꞉word꞉·ᛗᛗᗑᚴ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Ugaritic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Ugaritic·⁚·˙font꞉·
·⁚pronunciation꞉·LHM·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Army·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Standards rise over gathered weapons⁚˙ shields and spearpoints crowd the ground beneath
↪ hanging standards᛬·⁚·˙ostensive꞉·
·⁚grimchain꞉·⏧ıɪϞ⍦∀ↀ⍟ᗅ⍜ꟼ⍰·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋ᛭᛬ category꞉○–○ᵋcourt᛬꞉·
·⁚lookup office꞉○–○ᵋrelic᛬ enoch꞉○–○ᵋ⁂᛬꞉·

꞉word꞉·ᛦᑎᛉᛒᛚ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚·˙font꞉·
·⁚pronunciation꞉·KOOM⟊bl·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Burial monument⁚⁚cairn⁚⁚mark⁚⁚or memorial sign·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Weathered stones heap above marked earth⁚⁚and compressed weight keeps the place against
↪ wind and rain᛭·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᚴ◝ᘓ⁜ᨒₒᛘ♉⍁⊙ᛋᛃ₥ᛊ·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋ↑᛬ category꞉○–○ᵋcourt᛬꞉·
·⁚lookup office꞉○–○ᵋrelic᛬ enoch꞉○–○ᵋ⁂᛬꞉·

꞉word꞉·ᚢᛉIᛊTᚱ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁚pronunciation꞉·A⟊WISTR·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·(hapax legomenon) sheepfold (enclosure for keeping sheep)·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Fence and gate close around wool and breath⁚˙ the flock has a wall it can press
↪ against᛭·⁚·˙ostensive꞉·
·⁚grimchain꞉·⓿⍲⇜⇬ᵠᛉᐱᒍ☆ᛞ卫⏢○·⁚·˙grimchain꞉·
·⁚·˙lookup꞉˙
·⁚·˙glyphs꞉˙
·⁚glyph symbol꞉○–○ᵋᛦ᛬ category꞉○–○ᵋcourt᛬꞉·
·⁚lookup office꞉○–○ᵋrelic᛬ enoch꞉○–○ᵋ⁂᛬꞉·

˖word꞉·𗍫˖·˙word꞉·

·⁚origin…language꞉·Tangut·⁚·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚·˙font꞉·
·⁚pronunciation꞉·HWAH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Change irregularly⁖ to alter in an uneven or unpredictable manner·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Mottled stone carries broken islands of color across one still face⁚⁚and no hue closes the
↪ mineral into a single pattern𐑿·⁚·˙ostensive꞉·
·⁚grimchain꞉·ƵƁ⅄◎ⵋԋʯX◎✲ᵋ⧗₀·⁚·˙grimchain꞉·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol꞉○–○ₑᑎ꞉ category꞉○–○ₑcourt꞉⁚·
·⁚lookup office꞉○–○ₑrelic꞉ enoch꞉○–○ₑஃ꞉⁚·

˖word꞉·ᕼ⅍ᛁᛋ˖·˙word꞉·

·⁚origin…language꞉·Old Norse·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Runic·⁚·˙font꞉·
·⁚pronunciation꞉·HAYN·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Whetstone⁖ a stone used to sharpen a cutting edge·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Gritty block bites the steel edge⁚⁚and keen metal throws back light along the line of
↪ contactᑎ·⁚·˙ostensive꞉·
·⁚grimchain꞉·ᵹϒⒹⱣᵖ꘏ᘓᘐ𐓣ᛇᛋᚹᚱᛯX·⁚·˙grimchain꞉·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol꞉○–○ₑÞ꞉ category꞉○–○ₑcourt꞉⁚·
·⁚lookup office꞉○–○ₑrelic꞉ enoch꞉○–○ₑஃ꞉⁚·

˖word꞉·ᓀ̄ᒍᛪ˖·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚·˙font꞉·
·⁚pronunciation꞉·SLEHMEH·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Flame·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Charred wick bears a narrow brightness⁚⁚and one rooted tongue leans above the burning
↪ pointÞ·⁚·˙ostensive꞉·
·⁚grimchain꞉·⏁ᗷ○ገᚼ✱≍Ψ◝ᵒᵒXᛦᑫ♂·⁚·˙grimchain꞉·
·⁚·˙lookup⁖˙
·⁚·˙glyphs⁖˙
·⁚glyph symbol꞉○–○ₑᚠ꞉ category꞉○–○ₑcourt꞉⁚·
·⁚lookup office꞉○–○ₑrelic꞉ enoch꞉○–○ₑஃ꞉⁚·

˖word꞉·𗍪˖·˙word꞉·

·⁚origin…language꞉·Tangut·⁚·˙origin…language꞉·
·⁚font꞉·Noto Serif Tangut·⁚·˙font꞉·
·⁚pronunciation꞉·NTSIW·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Melt⁖ to change from a solid state through heat·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Warm wax bears rounded contours beneath the heat⁚⁚and softened mass holds no hard corner
↪ against the moldᚠ·⁚·˙ostensive꞉·

·⁚grimchain⁚·▽ϟԓXᚲAxஃ∫◌ıϡ▽·⁚.·ˑgrimchain⁚·
·⁚.·ˑlookup⁚·
·⁚.·ˑglyphs⁚·
·⁚glyph symbol○–○ᵩᚱ» category○–○ᵩcourt»⁚·
·⁚lookup office○–○ᵩrelic» enoch○–○ᵩஃ»⁚·

⁖word⁚·ᛏ⁅ᓚ⁻⁚.·ˑword⁚·

·⁚origin…language⁚·Tocharian A·⁚.·ˑorigin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚.·ˑfont⁚·
·⁚pronunciation⁚·KRAHSOH·⁚.·ˑpronunciation⁚·
·⁚cadence⁚·Relic·⁚.·ˑcadence⁚·
·⁚non…ostensive⁚·Vexation⁚⁚torment·⁚.·ˑnon…ostensive⁚·
·⁚ostensive⁚·Knotted cord bites the skin⁚· pressure has nowhere to flee⁚⁚and the binding remains
↪ tightᚱ·⁚.·ˑostensive⁚·
·⁚grimchain⁚·ሐ※ᘓ▽ıJAᛋЧT◎ꙍ{ꝸ⁚.·ˑgrimchain⁚·
·⁚.·ˑlookup⁚·
·⁚.·ˑglyphs⁚·
·⁚glyph symbol○–○ᵩ‹» category○–○ᵩcourt»⁚·
·⁚lookup office○–○ᵩrelic» enoch○–○ᵩஃ»⁚·

⁖word⁚·ᥣᓃᓬᵽᵠᵴ⁚.·ˑword⁚·

·⁚origin…language⁚·Avestan·⁚.·ˑorigin…language⁚·
·⁚font⁚·Noto Sans Avestan·⁚.·ˑfont⁚·
·⁚pronunciation⁚·ROW+kah·⁚.·ˑpronunciation⁚·
·⁚cadence⁚·Relic·⁚.·ˑcadence⁚·
·⁚non…ostensive⁚·Light⁚⁚brightness⁚⁚luminous space·⁚.·ˑnon…ostensive⁚·
·⁚ostensive⁚·Oil+soaked wick whitens nearby clay⁚· hard edges emerge clear from surrounding
↪ dark‹·⁚.·ˑostensive⁚·
·⁚grimchain⁚·☽⬡ᙅ𐑆ᵂΘ▫ᘐ𐑅ↄ⊰⊨ᛦ·⁚.·ˑgrimchain⁚·
·⁚.·ˑlookup⁚·
·⁚.·ˑglyphs⁚·
·⁚glyph symbol○–○ᵩX» category○–○ᵩcourt»⁚·
·⁚lookup office○–○ᵩrelic» enoch○–○ᵩஃ»⁚·

⁖word⁚·屾⁚.·ˑword⁚·

·⁚origin…language⁚·Tangut·⁚.·ˑorigin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·ˑfont⁚·
·⁚pronunciation⁚·MEH·⁚.·ˑpronunciation⁚·
·⁚cadence⁚·Relic·⁚.·ˑcadence⁚·
·⁚non…ostensive⁚·Dawn⁚· the beginning of daylight before or around sunrise·⁚.·ˑnon…ostensive⁚·
·⁚ostensive⁚·Dew bead catches pale sky⁚· ground remains dark beneath a sun still hidden below the
↪ lineX·⁚.·ˑostensive⁚·
·⁚grimchain⁚·✧ㄱᑭᴢ♀ᛤAıJᑭஃ𝌆ᔕ·⁚.·ˑgrimchain⁚·
·⁚.·ˑlookup⁚·
·⁚.·ˑglyphs⁚·
·⁚glyph symbol○–○ᵩᑭ» category○–○ᵩcourt»⁚·
·⁚lookup office○–○ᵩrelic» enoch○–○ᵩஃ»⁚·

⁖word⁚·ᚻᓀ⁅⁻⁚.·ˑword⁚·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·KEHST·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Hunger·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Empty bowl beside hollow belly·⁚ open space answers open space⁚⁚and neither is
↪ filled𑀦·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𑀬𑀸𑁆𑀫𑁄𑀵𑁂𑀯𑀺𑀢𑁆𑀭𑁂𑀶·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉꞉Ꮋ꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉𐊗꞉⁚·

꞉word꞉·J̄+⁻⁚.·˙word꞉·

·⁚origin…language꞉·Tocharian B·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Brahmi·⁚.·˙font꞉·
·⁚pronunciation꞉·LAHEEKOH·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Bath⁚⁚washing·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Basin carries clouded water around clean skin⁚⁚and loosened grit hangs in the same shallow
↪ bodyᏂ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·ⵔᑎᚣᚧᚲ❂○↗ᑫᒪᚸᐁᛉ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉꞉ᛋ꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉𐊗꞉⁚·

꞉word꞉·ᕼᐱᖇᔕᑎᔕ꞉⁚.·˙word꞉·

·⁚origin…language꞉·Gothic·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚.·˙font꞉·
·⁚pronunciation꞉·HARDUS·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Hard·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Stone keeps its edge beneath nail and palm⁚˙ pressure finds almost no yielding in the
↪ loaded faceᛋ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·𐊫𐊇𐊯≻Ↄ❂₀𝌆ᛕ⊛ᛉ𐋏✷ব·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙
·⁚.·˙glyphs꞉˙
·⁚glyph symbol꞉꞉ᛋ꞉ category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉𐊗꞉⁚·

꞉word꞉·𐊕꞉⁚.·˙word꞉·

·⁚origin…language꞉·Akkadian·⁚.·˙origin…language꞉·
·⁚font꞉·Noto Sans Cuneiform·⁚.·˙font꞉·
·⁚pronunciation꞉·KAK⟊koo·⁚.·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚.·˙cadence꞉·
·⁚non…ostensive꞉·Weapon⁚˙ implement of armed force·⁚.·˙non…ostensive꞉·
·⁚ostensive꞉·Weighted haft ends in hardened edge and point⁚˙ one grasp gathers the whole striking
↪ weightᛋ·⁚.·˙ostensive꞉·
·⁚grimchain꞉·♂ᛦ𐊫ᓚ𐎗⊛𐋏'Λ'Æ·⁚.·˙grimchain꞉·
·⁚.·˙lookup꞉˙

·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉ᛋ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚫ꞉·

꞉word꞉·ᛦZᚴᛉ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁚pronunciation꞉·AZGO·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·Ash·˙ cinder·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Cold black dust rests among hearthstones꞉꞉light enough for breath to stirᛋ·⁚·˙ostensive꞉·
·⁚grimchain꞉·Ↄxɬ⁕ℯᛦ△⧗ᛄ᧐ᘐↃ·⁚·˙grimchain꞉·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉ᛊ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚫ꞉·

꞉word꞉·ᛦ᧐ᛦ꞉·⁚·˙word꞉·

·⁚origin…language꞉·Gothic·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Gothic·⁚·˙font꞉·
·⁚pronunciation꞉·AHWA·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·River·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Moving water keeps one carved bed·˙ bank and stone receive moving current along the
↪ course᛬·⁚·˙ostensive꞉·
·⁚grimchain꞉·▽⛄ᛗ▽ᘐᓭ⁕ϧ⛄ᛈᚺᛏ·⁚·˙grimchain꞉·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉ᛃ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚫ꞉·

꞉word꞉·𓆓꞉·⁚·˙word꞉·

·⁚origin…language꞉·Egyptian·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Egyptian Hieroglyphs·⁚·˙font꞉·
·⁚pronunciation꞉·SAHK·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·
·⁚non…ostensive꞉·To gather or contract together·⁚·˙non…ostensive꞉·
·⁚ostensive꞉·Cord holds gathered reeds tight·˙ many stalks answer as one bundle beneath the
↪ bindingᛃ·⁚·˙ostensive꞉·
·⁚grimchain꞉·⁕ᛀᘔᛗᘐᘐᘐᘐᘐZᓭᒍᚼ⁞·⁚·˙grimchain꞉·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꞉꞉ᘐ꞉ category꞉꞉court꞉·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉ᚫ꞉·

꞉word꞉·तर्क꞉·⁚·˙word꞉·

·⁚origin…language꞉·Sanskrit·⁚·˙origin…language꞉·
·⁚font꞉·Noto Sans Devanagari·⁚·˙font꞉·
·⁚pronunciation꞉·tuhr÷kuh·⁚·˙pronunciation꞉·
·⁚cadence꞉·Relic·⁚·˙cadence꞉·

·⁖non…ostensive:·conjecture⁘ reasoning⁚⁚speculation⁚⁚inquiry⁘ doubt⁘ system or doctrine founded on
↪ speculation or reasoning⁚⁚philosophical system (especially the न्याय (nyāya) system⁚⁚but applicable
↪ also to any of the six दर्शन (darśana))⁘ logic⁚⁚confutation (especially that kind of argument
↪ which consists in reduction to absurdity)⁖.·˙non…ostensive:·
·⁖ostensive:·Marked stones face one another across measured lines⁚⁚and unequal counts press visibly
↪ against the same boundaryↄ⁖.·˙ostensive:·
·⁖grimchain:·𐎗𐎫𐎗𐎠𐎍X𐎔𐎡𐎓𐎣𐎢𐎡𐎤𐎫𐎡𐎣𐎬⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol○–○𐎤𐎣 category○–○𐎤court𐎣⁚·
·⁖lookup office○–○𐎤relic𐎣 enoch○–○𐎤𐎔𐎣⁚·

·word:·𐎤𐎫𐎗𐎠⁖.·˙word:·

·⁖origin…language:·Old Norse·⁖.·˙origin…language:·
·⁖font:·Noto Sans Runic·⁖.·˙font:·
·⁖pronunciation:·HAH⟊vah·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·To have⁚⁚hold⁚⁚or keep·⁖.·˙non…ostensive:·
·⁖ostensive:·Closed fingers enclose a small stone⁘ skin forms an unbroken wall around the captured
↪ weight𐎤·⁖.·˙ostensive:·
·⁖grimchain:·ᴏᚲᔑᚹᚠᐁᚱᛗᚣᛦᛇᚳᚹᛈ·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol○–○𐎤𐎢𐎣 category○–○𐎤court𐎣⁚·
·⁖lookup office○–○𐎤relic𐎣 enoch○–○𐎤𐎔𐎣⁚·

·word:·𐎠𐎡𐎥⁖.·˙word:·

·⁖origin…language:·Tocharian A·⁖.·˙origin…language:·
·⁖font:·Noto Sans Brahmi·⁖.·˙font:·
·⁖pronunciation:·KRAHKEH·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Dirt⁚⁚filth·⁖.·˙non…ostensive:·
·⁖ostensive:·Dark grit clings beneath nail and sandal⁘ dark grit remains after rinsing water
↪ passes𐎢·⁖.·˙ostensive:·
·⁖grimchain:·𐎥𐎍𐎃𐎡𐎣𐎤𐎫𐎤𐎬𐎭𐎮𐎯·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙
·⁖glyph symbol○–○𐎤𐎭𐎧𐎣 category○–○𐎤court𐎣⁚·
·⁖lookup office○–○𐎤relic𐎣 enoch○–○𐎤𐎔𐎣⁚·

·word:·𐎍𐎃 𐎚𐎚⁖.·˙word:·

·⁖origin…language:·Ugaritic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Ugaritic·⁖.·˙font:·
·⁖pronunciation:·AHAMMOODU·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Pillar⁚⁚column·⁖.·˙non…ostensive:·
·⁖ostensive:·Upright stone takes the roof upon its length and sends the burden into
↪ ground𐎭𐎧·⁖.·˙ostensive:·
·⁖grimchain:·⊕ᔑↄᗺ𐎡ᗅᔐᔕᚺᛞ⧗Ↄᔑᔓ·⁖.·˙grimchain:·
·⁖.·˙lookup:˙
·⁖.·˙glyphs:˙

·⁚glyph symbol꩜꩜𐫇𐫈 category꩜꩜𐫇court𐫈⁚·
·⁚lookup office꩜꩜𐫇relic𐫈 enoch꩜꩜𐫇𐫋𐫈⁚·

⁖word⁚·𗍫⁖.·˙word⁚·

·⁚origin…language⁚·Tangut·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·˙font⁚·
·⁚pronunciation⁚·GHOO·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Smoke or mist·⁚· suspended vapor or fine airborne matter that obscures
↪ sight·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Grey veil hangs across open distance⁚⁚and far edges dissolve behind drifting particles in
↪ air𐫇·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐫈𐫇𐫋꩜𐫇𐫈𐫉𐫊𐫋𐫌𐫍𐫎𐫏𐫐𐫑𐫒𐫓𐫔𐫕·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol꩜꩜𐫇◎𐫈 category꩜꩜𐫇court𐫈⁚·
·⁚lookup office꩜꩜𐫇relic𐫈 enoch꩜꩜𐫇𐫋𐫈⁚·

⁖word⁚·𒌨⁖.·˙word⁚·

·⁚origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·DEH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·To pour·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Vessel tips⁚⁚and one narrow stream hangs between clay and earth◎·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐫉꩜𐫇𐫋𐫌𐫍𐫎𐫏𐫐𐫑𐫒𐫓𐫔𐫕·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol꩜꩜𐫇⏁𐫈 category꩜꩜𐫇court𐫈⁚·
·⁚lookup office꩜꩜𐫇relic𐫈 enoch꩜꩜𐫇𐫋𐫈⁚·

⁖word⁚·ᚦᚢᚺᚢᚾ⁖.·˙word⁚·

·⁚origin…language⁚·Gothic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚.·˙font⁚·
·⁚pronunciation⁚·FEH+HU·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Property⁚⁚possessions·⁚· wealth⁚⁚riches·⁚· money·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Sealed chest bears counted metal beneath one keeping mark⁚⁚and locked weight remains
↪ gathered inside the bounded holding⏁·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐫗𐫈𐫉𐫊𐫋𐫌𐫍𐫎𐫏𐫐𐫑𐫒·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol꩜꩜𐫇𐫄𐫈 category꩜꩜𐫇court𐫈⁚·
·⁚lookup office꩜꩜𐫇relic𐫈 enoch꩜꩜𐫇𐫋𐫈⁚·

⁖word⁚·ᛁ⁖.·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁚pronunciation⁚·EESS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·

·⁖non…ostensive⁚·Ice⁖· frozen water·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Clear hard surface carries crack and weight⁖· water beneath the clear surface remains
↪ locked from flowing↜·⁚.·˙ostensive⁚·
·⁖grimchain⁚·ᚋᛏᛜᛅᚱᛇᚢᚱᛋᛋᛏᛊᛟᛞ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁖glyph symbol⟜⟜ᛃ᛬ category⟜⟜court᛬⁚·
·⁖lookup office⟜⟜relic᛬ enoch⟜⟜ᛘ᛬⁚·

₍word⁚·ᛏᚢᛉᚱ⁖⁚.·˙word⁚·

·⁖origin…language⁚·Old Norse·⁚.·˙origin…language⁚·
·⁖font⁚·Noto Sans Runic·⁚.·˙font⁚·
·⁖pronunciation⁚·TOHM⟊r·⁚.·˙pronunciation⁚·
·⁖cadence⁚·Relic·⁚.·˙cadence⁚·
·⁖non…ostensive⁚·Containing nothing or lacking expected contents·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Clay walls rise around a clean hollow⁖· open air reaches from base to mouth with nothing
↪ pressing outward᛬·⁚.·˙ostensive⁚·
·⁖grimchain⁚·ᚏᛃᛅᚺᚢᛒᚷᚼᛞᛈᛝᛋᚸᛘ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁖glyph symbol⟜⟜ᛯ᛬ category⟜⟜court᛬⁚·
·⁖lookup office⟜⟜relic᛬ enoch⟜⟜ᛘ᛬⁚·

₍word⁚·ᛏᛁᛝ□ᛁ⁖⁚.·˙word⁚·

·⁖origin…language⁚·Tocharian A·⁚.·˙origin…language⁚·
·⁖font⁚·Noto Sans Brahmi·⁚.·˙font⁚·
·⁖pronunciation⁚·KRAHMUHR·⁚.·˙pronunciation⁚·
·⁖cadence⁚·Relic·⁚.·˙cadence⁚·
·⁖non…ostensive⁚·Weight⁝heaviness·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Stone sinks into palm and soil⁖· flesh and earth both bend beneath the downward
↪ pullᛯ·⁚.·˙ostensive⁚·
·⁖grimchain⁚·ᛯ⊙ᛪᛜᚺ♀ₒᛒᛟᛇᛥΘI·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁖glyph symbol⟜⟜ₒ᛬ category⟜⟜court᛬⁚·
·⁖lookup office⟜⟜relic᛬ enoch⟜⟜ᛘ᛬⁚·

₍word⁚·𒄭𒍪𒌆𒁾⁖⁚.·˙word⁚·

·⁖origin…language⁚·Sumerian·⁚.·˙origin…language⁚·
·⁖font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁖pronunciation⁚·OO⟊reen·⁚.·˙pronunciation⁚·
·⁖cadence⁚·Relic·⁚.·˙cadence⁚·
·⁖non…ostensive⁚·Foundation·⁚.·˙non…ostensive⁚·
·⁖ostensive⁚·Hidden stone lies beneath wall and post⁖· every raised burden finds ground through the
↪ buried bedₒ·⁚.·˙ostensive⁚·
·⁖grimchain⁚·⟊Θɔɣʃᚋᚷᚱᛊ☆ᚣ᛭ᛘ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁖˙
·⁚.·˙glyphs⁖˙
·⁖glyph symbol⟜⟜Θ᛬ category⟜⟜court᛬⁚·
·⁖lookup office⟜⟜relic᛬ enoch⟜⟜ᛘ᛬⁚·

˖word꞉·𐴀𐴁𐴂˸˙·˳˙word꞉·

·˙origin…language꞉·Tocharian B·˙˳·˙origin…language꞉·
·˙font꞉·Noto Sans Brahmi·˙˳·˙font꞉·
·˙pronunciation꞉·PEHRNEH·˙˳·˙pronunciation꞉·
·˙cadence꞉·Relic·˙˳·˙cadence꞉·
·˙non…ostensive꞉·Power꞉꞉rank꞉꞉stage·˙ glory·˙˳·˙non…ostensive꞉·
·˙ostensive꞉·Raised seat stands above common ground·˙ carved wood remains elevated with no body resting upon itΘ·˙˳·˙ostensive꞉·
↪
·˙grimchain꞉·𐠤𐠸𐠼𐠷𐠋𐠣𐠫𐠤𐠫𐠏𐠣𐠟𐠰·˙˳·˙grimchain꞉·
·˙˳·˙lookup·˙
·˙˳·˙glyphs·˙
·˙glyph symbol꞊꞊𐤇﹅ category꞊꞊court﹅꞉·
·˙lookup office꞊꞊relic﹅ enoch꞊꞊𐤀﹅꞉·

˖word꞉·𐫈𐫉𐫊˸˙·˳˙word꞉·

·˙origin…language꞉·Ugaritic·˙˳·˙origin…language꞉·
·˙font꞉·Noto Sans Ugaritic·˙˳·˙font꞉·
·˙pronunciation꞉·AHBNU·˙˳·˙pronunciation꞉·
·˙cadence꞉·Relic·˙˳·˙cadence꞉·
·˙non…ostensive꞉·Stone·˙˳·˙non…ostensive꞉·
·˙ostensive꞉·Mineral block keeps one hard face beneath chisel and weight꞉꞉and dense mass holds its edge against pressure𐤇·˙˳·˙ostensive꞉·
↪
·˙grimchain꞉·𐫈𐫀𐫁𐫂𐫃𐫄𐫅𐫆𐫇𐫈𐫉·˙˳·˙grimchain꞉·
·˙˳·˙lookup·˙
·˙˳·˙glyphs·˙
·˙glyph symbol꞊꞊𐤀﹅ category꞊꞊court﹅꞉·
·˙lookup office꞊꞊relic﹅ enoch꞊꞊𐤀﹅꞉·

˖word꞉·俶˸˙·˳˙word꞉·

·˙origin…language꞉·Tangut·˙˳·˙origin…language꞉·
·˙font꞉·Noto Serif Tangut·˙˳·˙font꞉·
·˙pronunciation꞉·KHWAH·˙˳·˙pronunciation꞉·
·˙cadence꞉·Relic·˙˳·˙cadence꞉·
·˙non…ostensive꞉·Cloth·˙ woven fabric·˙˳·˙non…ostensive꞉·
·˙ostensive꞉·Crossed fibres fold without coming apart꞉꞉and woven surface keeps its hold under the palm𐤀·˙˳·˙ostensive꞉·
↪
·˙grimchain꞉·𐤇𐤊𐤓𐤉𐤋𐤌𐤍𐤎𐤏𐤐𐤑·˙˳·˙grimchain꞉·
·˙˳·˙lookup·˙
·˙˳·˙glyphs·˙
·˙glyph symbol꞊꞊𐤄﹅ category꞊꞊court﹅꞉·
·˙lookup office꞊꞊relic﹅ enoch꞊꞊𐤀﹅꞉·

˖word꞉·𒁹˸˙·˳˙word꞉·

·˙origin…language꞉·Sumerian·˙˳·˙origin…language꞉·
·˙font꞉·Noto Sans Cuneiform·˙˳·˙font꞉·
·˙pronunciation꞉·SAH·˙˳·˙pronunciation꞉·
·˙cadence꞉·Relic·˙˳·˙cadence꞉·
·˙non…ostensive꞉·Middle·˙ the central part or point between outer limits꞉꞉also one half·˙˳·˙non…ostensive꞉·
·˙ostensive꞉·Single mark rests between two edges·˙ equal span reaches the single mark from either side𐤄·˙˳·˙ostensive꞉·
↪

·⁚grimchain⁚·♂☙ϫ℘Ш⌄ϕ⓪ᘐ⅄ᘎ◺ıſ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol⟜⟜ᖘ» category⟜⟜court»⁚·
·⁚lookup office⟜⟜relic» enoch⟜⟜ஃ»⁚·

·word⁚·𗦟⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·LIH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Peacock·˙ a large crested bird with an ornate tail·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Long feather carries blue+green eye+marks across a fan+shaped train⁚⁚and each shaft holds
↪ its place in the spreadᖘ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ϒ㋡ıᓀϕϒ◺ᗡᘎ❂⇩Ψᗡᓐ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol⟜⟜⓪» category⟜⟜court»⁚·
·⁚lookup office⟜⟜relic» enoch⟜⟜ஃ»⁚·

·word⁚·ﻫﻪﻴﻴﻪ⁚·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚·˙font⁚·
·⁚pronunciation⁚·amarUH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·other⁚⁚remaining·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·One stone lies beside the finished wall·˙ laid courses stop without it⁚⁚and the loose block
↪ remains separate⓪·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᘎᘐϰ㠯☽ᔑᔕᘎϰ㠯ᑭ⊕·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol⟜⟜ᕒ» category⟜⟜court»⁚·
·⁚lookup office⟜⟜relic» enoch⟜⟜ஃ»⁚·

·word⁚·⏀ϰ⊢𐏁⁚·˙word⁚·

·⁚origin…language⁚·Hittite·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·TEPUS·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Little⁚⁚few·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Three grains lie across broad cloth·˙ bare surface surrounds the three grains on every
↪ sideᕒ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᗢΣϰ∀♔ᓭᘏᖘᘏ⏀ᓭ▽·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol⟜⟜ᓱ» category⟜⟜court»⁚·
·⁚lookup office⟜⟜relic» enoch⟜⟜ஃ»⁚·

·word⁚·⏀ᓭ⊢⁚·˙word⁚·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·KHARPASH·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Heap⁚⁚mound⁚⁚(wood)pile·⁚.·˙non…ostensive:·
·⁚ostensive:·Loose timber and stone lean into one another under their own weight⁚⁚keeping a sloped face
↪ with no wall or basket·⁚.·˙ostensive:·
·⁚grimchain:·𐊛𐊁𐊫𐊗𐊉𐊰𐊱𐊬𐊗𐊫𐊰·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉𐊗 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉𐊫꞉⁚·

꞉word:·𐊗꞉·⁚.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·R⟊YOR·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Firm or strong⁚˙ resistant to yielding·⁚.·˙non…ostensive:·
·⁚ostensive:·Dense wood keeps its grain beneath pressure⁚⁚and loaded fibre yields no hollow along the
↪ face𖼇·⁚.·˙ostensive:·
·⁚grimchain:·𐊰𐊱𐊛𐊫𐊰𐊬𐊫𐊰𐊬𐊫𐊰𐊬𐊫𐊰𐊬𐊫𐊰·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉𐊰 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉𐊫꞉⁚·

꞉word:·𐊫𐊬𐊫𐊰꞉·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·VOCAR·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Wave·⁚.·˙non…ostensive:·
·⁚ostensive:·Water lifts into ridge and hollow together⁚⁚one moving crest traveling across the
↪ surface𐊰·⁚.·˙ostensive:·
·⁚grimchain:·𐊫𐊱𐊰𐊬𐊫𐊰𐊬𐊫𐊰𐊬𐊫𐊰𐊬𐊫𐊰·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꞉꞉𐊰 category꞉꞉court꞉⁚·
·⁚lookup office꞉꞉relic꞉ enoch꞉꞉𐊫꞉⁚·

꞉word:·𐊰꞉·⁚.·˙word:·

·⁚origin…language:·Old Turkic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚.·˙font:·
·⁚pronunciation:·TOOR·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·To stand⁚⁚stay⁚⁚continue to exist·⁚.·˙non…ostensive:·
·⁚ostensive:·Planted post holds one patch of ground against crossing wind⁚⁚and upright weight remains
↪ fixed above the same root𐊰·⁚.·˙ostensive:·
·⁚grimchain:·𐊗𐊰𐊫𐊬𐊫𐊰𐊬𐊫𐊰𐊬𐊫𐊰𐊬𐊫𐊰·⁚.·˙grimchain:·
·⁚.·˙lookup:˙

·⁚·˙glyphs˙·
·⁚glyph symbol○–○ₑ𐋡 category○–○ₑcourt𐑉⁚·
·⁚lookup office○–○ₑrelic𐑉 enoch○–○ₑ𐑊𐑉⁚·

·word⁚·ᎾᏏᎸᎾᏂᏂ·⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·OMUKH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Gate⁚⁚door·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Hinged timber fills the break in the wall·˙ threshold and bar hold the place where passage
↪ opens𐋡·⁚·˙ostensive⁚·
·⁚grimchain⁚·∫ᗅ᐀∀ᑎᖇᘔᔕ𐓛ℂ𓀬▫▽·⁚·˙grimchain⁚·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol○–○ₑ𐋣 category○–○ₑcourt𐑉⁚·
·⁚lookup office○–○ₑrelic𐑉 enoch○–○ₑ𐑊𐑉⁚·

·word⁚·ᅩᆯJ̄⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·KOOL·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Bell·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Cast bronze hangs open beneath its crown⁚⁚and struck metal holds a ringing body through
↪ the hollow chamber𐋣·⁚·˙ostensive⁚·
·⁚grimchain⁚·ϩ𐑂ᔭ𐑅𐑅⟡▽ᴢᦓᦔℂᔑᦔ𓀬·⁚·˙grimchain⁚·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol○–○ₑ𐋤 category○–○ₑcourt𐑉⁚·
·⁚lookup office○–○ₑrelic𐑉 enoch○–○ₑ𐑊𐑉⁚·

·word⁚·ᏏᎾᏂᘯᔓ⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·BODVAR·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Censer·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Pierced vessel cups live coal beneath fragrance⁚⁚and scented vapor rises through its
↪ openings𐋤·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚣ⬡ᚷᚠ𐑅ᚻᓄᔕᔓᓄᔓ⟡·⁚·˙grimchain⁚·
·⁚·˙lookup˙·
·⁚·˙glyphs˙·
·⁚glyph symbol○–○ₑ𐋢 category○–○ₑcourt𐑉⁚·
·⁚lookup office○–○ₑrelic𐑉 enoch○–○ₑ𐑊𐑉⁚·

·word⁚·ᚺᚴᚲᛃᚦ᛭ᚔ⁚·˙word⁚·

·⁚origin…language⁚·Hittite·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·KHUWANZA·⁚·˙pronunciation⁚·

·⁖cadence:·Relic·⁖˙·˙cadence:·
·⁖non…ostensive:·Wind·⁖˙·˙non…ostensive:·
·⁖ostensive:·Hanging cloth swells and grass bends⁖· dust lifts under an unseen moving
↪ forceব·⁖˙·˙ostensive:·
·⁖grimchain:·ᑭ𐑈ᔑᑎᔕᘐᘏᘔᘓᘐ⏣⁕ᘓ·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵋ⏀ᵌ category꩜꩜ᵋcourtᵌ:·
·⁖lookup office꩜꩜ᵋrelicᵌ enoch꩜꩜ᵋ⏃ᵌ:·

⁖word:·ﻣﻪﻣﻤﻤﻬ:⁖˙·˙word:·

·⁖origin…language:·Old Uyghur·⁖˙·˙origin…language:·
·⁖font:·Noto Serif Old Uyghur·⁖˙·˙font:·
·⁖pronunciation:·tupraK·⁖˙·˙pronunciation:·
·⁖cadence:·Relic·⁖˙·˙cadence:·
·⁖non…ostensive:·dirt::soil::earth::land⁖· dust::dirt::soot⁖· (rare) the Earth·⁖˙·˙non…ostensive:·
·⁖ostensive:·Brown grains bear root and foot::loose enough to rise as powder::firm enough to carry
↪ weight⏀·⁖˙·˙ostensive:·
·⁖grimchain:·ᛞᐁᐅᕪᦇᦺᦈ⟓ᦓᦓᦟᦟᦈᐯᑭᑌ·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵋবᵌ category꩜꩜ᵋcourtᵌ:·
·⁖lookup office꩜꩜ᵋrelicᵌ enoch꩜꩜ᵋ⏃ᵌ:·

⁖word:·ᛋᚻᚸᚴᚳᛯ⁕ᚺ⁖˙·˙word:·

·⁖origin…language:·Hittite·⁖˙·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖˙·˙font:·
·⁖pronunciation:·ARMAWANZA·⁖˙·˙pronunciation:·
·⁖cadence:·Relic·⁖˙·˙cadence:·
·⁖non…ostensive:·Pregnant·⁖˙·˙non…ostensive:·
·⁖ostensive:·Rounded belly stretches skin over hidden living weight::and a second pulse rests beneath
↪ the outer curveব·⁖˙·˙ostensive:·
·⁖grimchain:·𐓏ᑫᑭᘐᕼǂ'ᐯᔕᐁᦈᦈ·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵋᘓᵌ category꩜꩜ᵋcourtᵌ:·
·⁖lookup office꩜꩜ᵋrelicᵌ enoch꩜꩜ᵋ⏃ᵌ:·

⁖word:·ᚦᛞᛞᛞᚺᚸ⁖˙·˙word:·

·⁖origin…language:·Hittite·⁖˙·˙origin…language:·
·⁖font:·Noto Sans Cuneiform·⁖˙·˙font:·
·⁖pronunciation:·MEMIYASH·⁖˙·˙pronunciation:·
·⁖cadence:·Relic·⁖˙·˙cadence:·
·⁖non…ostensive:·Word·⁖˙·˙non…ostensive:·
·⁖ostensive:·Marked clay bears shaped signs in silence::and cut strokes hold speech where no mouth is
↪ openᘓ·⁖˙·˙ostensive:·
·⁖grimchain:·'ᗣᦈᘜᘐ๐ᔑᘎᘐᦟᦟᘓᕈ·⁖˙·˙grimchain:·
·⁖˙·˙lookup·˙
·⁖˙·˙glyphs·˙
·⁖glyph symbol꩜꩜ᵋᘜᵌ category꩜꩜ᵋcourtᵌ:·
·⁖lookup office꩜꩜ᵋrelicᵌ enoch꩜꩜ᵋ⏃ᵌ:·

⁖word:·𐫁𐫄⁖·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·KAHREH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Hole⁝pit·⁚·˙non…ostensive:·
·⁚ostensive:·Pierced clay encloses open air⁝and the solid wall gives the empty center a touchable
↪ edge෴·⁚·˙ostensive:·
·⁚grimchain:·⊙𑀫𑀦𑁆𑀯𑀷𑀸𑀬𑀺𑀭𑁆·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꜀꜀𑁆𑀯𑀸 category꜀꜀court𑀸⁚·
·⁚lookup office꜀꜀relic𑀸 enoch꜀꜀𑀅𑀸⁚·

⁖word:·ㄴ⁖·˙word:·

·⁚origin…language:·Middle Korean·⁚·˙origin…language:·
·⁚font:·Noto Sans KR·⁚·˙font:·
·⁚pronunciation:·nwo·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·gauze⁝thin cloth·⁚·˙non…ostensive:·
·⁚ostensive:·Fine crossing threads hang pale across open light⁝a woven body almost thinner than its
↪ own cast shade𑁆𑀯·⁚·˙ostensive:·
·⁚grimchain:·✡𑁆⊙𑀁𑀅𑀓𑁆𑀫𑁆𑀬𑀸𑀺𑀅𑀸·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꜀꜀𑀰𑀸 category꜀꜀court𑀸⁚·
·⁚lookup office꜀꜀relic𑀸 enoch꜀꜀𑀅𑀸⁚·

⁖word:·𐽀𐽆𐽂𐽅⁖·˙word:·

·⁚origin…language:·Old Uyghur·⁚·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚·˙font:·
·⁚pronunciation:·Kars·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·woolen fabric⁝cloth used for sewing sacks⁝a kind of rough fabric reminiscent of a
↪ felt·⁚·˙non…ostensive:·
·⁚ostensive:·Thick crossed wool fibres bear sack and burden⁝and taut weave holds against the
↪ strain𑀰·⁚·˙ostensive:·
·⁚grimchain:·𑀮𑀭𑀇𑀢𑀼𑀖𑀺𑀵𑁆𑀯𑀸·⁚·˙grimchain:·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꜀꜀𑀩𑀸 category꜀꜀court𑀸⁚·
·⁚lookup office꜀꜀relic𑀸 enoch꜀꜀𑀅𑀸⁚·

⁖word:·𐰀𐰉𐰖𐰍⁖·˙word:·

·⁚origin…language:·Old Turkic·⁚·˙origin…language:·
·⁚font:·Noto Sans Old Turkic·⁚·˙font:·
·⁚pronunciation:·bool+GHAH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·To stir·˙ to produce a state of disorder⁝put in disorder⁝to disturb⁝to cause
↪ trouble·⁚·˙non…ostensive:·

·⁚ostensive⁚·Clouded vessel holds grain and liquid in broken layering⁜and no settled boundary remains inside⁖·⁚·ostensive⁚·
·⁚grimchain⁚·◎∀ɾ⋌≋♙ᴘ⅄ᴘᏂ⊜ᔕ·⁚·grimchain⁚·
·⁚·lookup⁚·
·⁚·glyphs⁚·
·⁚glyph symbol○–○ᑫ⁖ category○–○ᑫcourt⁖⁚·
·⁚lookup office○–○ᑫrelic⁖ enoch○–○ᑫ♙⁖⁚·

⁘word⁚·즐겁다⁘·⁚·word⁚·

·⁚origin…language⁚·Middle Korean·⁚·⁚origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚·⁚font⁚·
·⁚pronunciation⁚·cúlkEHptá·⁚·⁚pronunciation⁚·
·⁚cadence⁚·Relic·⁚·⁚cadence⁚·
·⁚non…ostensive⁚·to be pleasant⁜to be enjoyable·⁚·⁚non…ostensive⁚·
·⁚ostensive⁚·Warm bread yields beneath the tongue⁖ sweet grain leaves no bitter edge along mouth or belly⁖·⁚·⁚ostensive⁚·
·⁚grimchain⁚·ᑭ♋ϙɬɔᑎᒪ○ιஜ♈♙·⁚·⁚grimchain⁚·
·⁚·lookup⁚·
·⁚·glyphs⁚·
·⁚glyph symbol○–○ᑫ⁖ category○–○ᑫcourt⁖⁚·
·⁚lookup office○–○ᑫrelic⁖ enoch○–○ᑫ♙⁖⁚·

⁘word⁚·حديقة⁘·⁚·word⁚·

·⁚origin…language⁚·Old Uyghur·⁚·⁚origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚·⁚font⁚·
·⁚pronunciation⁚·tamUHr·⁚·⁚pronunciation⁚·
·⁚cadence⁚·Relic·⁚·⁚cadence⁚·
·⁚non…ostensive⁚·vein⁜artery⁖ (botany) nervation⁜trace⁜streak⁖ (physiology) pulse·⁚·⁚non…ostensive⁚·
·⁚ostensive⁚·Narrow line runs beneath the skin⁜lifting again and again with the moving blood inside⁒⁚·⁚·⁚ostensive⁚·
·⁚grimchain⁚·Ψ⊕ᔕϒ⍁≳ᔓ◡ᵒᵒᕼ⏁ϙ〈ग·⁚·⁚grimchain⁚·
·⁚·lookup⁚·
·⁚·glyphs⁚·
·⁚glyph symbol○–○ᗅ⁖ category○–○ᑫcourt⁖⁚·
·⁚lookup office○–○ᑫrelic⁖ enoch○–○ᑫ♙⁖⁚·

⁘word⁚·그리다⁘·⁚·word⁚·

·⁚origin…language⁚·Middle Korean·⁚·⁚origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚·⁚font⁚·
·⁚pronunciation⁚·keulEEtá·⁚·⁚pronunciation⁚·
·⁚cadence⁚·Relic·⁚·⁚cadence⁚·
·⁚non…ostensive⁚·to draw⁜to paint⁖ to think about⁜especially lovingly⁖ to long for⁜to yearn for·⁚·⁚non…ostensive⁚·
·⁚ostensive⁚·Ink fixes a familiar face on clay beside an unoccupied place⁖ dark lines keep the absent features nearᗅ·⁚·⁚ostensive⁚·
·⁚grimchain⁚·ᗒᑢᑭ⬡ᖘᕮᒣᑫ≋⁕▽♀·⁚·⁚grimchain⁚·
·⁚·lookup⁚·
·⁚·glyphs⁚·
·⁚glyph symbol○–○ᑫ∀⁖ category○–○ᑫcourt⁖⁚·
·⁚lookup office○–○ᑫrelic⁖ enoch○–○ᑫ♙⁖⁚·

·word:·ᚸNSTS·:.·˙word:·

·˙origin…language:·Gothic·˙.·˙origin…language:·
·˙font:·Noto Sans Gothic·˙.·˙font:·
·˙pronunciation:·ANSTS·˙.·˙pronunciation:·
·˙cadence:·Relic·˙.·˙cadence:·
·˙non…ostensive:·Grace·:beneficence:·:favour·˙· joy·:thankfulness·˙· gift·:blessing·˙.·˙non…ostensive:·
·˙ostensive:·Open palm bears an unasked offering·˙· small weight rests there with no counterweight beside it∀·˙.·˙ostensive:·
·˙grimchain:·ᛝᚸᑫ◌⬡⋖⊨ᚼᛞᓐᚼᛋᚺᑐᛯ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○ₑ∀꜀ category○–○ₑcourt꜀:·
·˙lookup office○–○ₑrelic꜀ enoch○–○ₑ♳꜀:·

·word:·ᛏᚱᚾᑎᛉᚱ·:.·˙word:·

·˙origin…language:·Old Norse·˙.·˙origin…language:·
·˙font:·Noto Sans Runic·˙.·˙font:·
·˙pronunciation:·DROW+mɪ·˙.·˙pronunciation:·
·˙cadence:·Relic·˙.·˙cadence:·
·˙non…ostensive:·Dream·˙.·˙non…ostensive:·
·˙ostensive:·Closed lids twitch against a still face·˙· fingers tighten once around empty cloth while breath remains even∇·˙.·˙ostensive:·
·˙grimchain:·ᓯᛘᛋ⊨ᚼᛞᛞᓀᛉᛇ⏣〈ᐟᐣᘜ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○ₑ∇꜀ category○–○ₑcourt꜀:·
·˙lookup office○–○ₑrelic꜀ enoch○–○ₑ♳꜀:·

·word:·ᒍᒪ᐀ᚩᒉᒣ·:.·˙word:·

·˙origin…language:·Tocharian B·˙.·˙origin…language:·
·˙font:·Noto Sans Brahmi·˙.·˙font:·
·˙pronunciation:·YEHRKWAHNTOH·˙.·˙pronunciation:·
·˙cadence:·Relic·˙.·˙cadence:·
·˙non…ostensive:·Wheel·˙.·˙non…ostensive:·
·˙ostensive:·Rim circles hub·:and every turning point belongs to the same round body∇·˙.·˙ostensive:·
·˙grimchain:·ᕕᗰᠰᛯᑎ♳ᗅᑫᓭ♳⊲ᛕᕼ·˙.·˙grimchain:·
·˙.·˙lookup·˙
·˙.·˙glyphs·˙
·˙glyph symbol○–○ₑ∇ᖇ꜀ category○–○ₑcourt꜀:·
·˙lookup office○–○ₑrelic꜀ enoch○–○ₑ♳꜀:·

·word:·ᖘᕿᑫᖁᖍ·:.·˙word:·

·˙origin…language:·Aghwan·˙.·˙origin…language:·
·˙font:·Noto Sans Caucasian Albanian·˙.·˙font:·
·˙pronunciation:·MADILY·˙.·˙pronunciation:·
·˙cadence:·Relic·˙.·˙cadence:·
·˙non…ostensive:·Grace·:mercy·˙.·˙non…ostensive:·
·˙ostensive:·Open hand lowers instead of striking·˙· flesh within reach of the descending hand stays whole∇ᖇ·˙.·˙ostensive:·
·˙grimchain:·ᓴᙆᘚᛯᓓᚸᙆᘚᓄᚹᐯ∇ᘉ❁ᚼ·˙.·˙grimchain:·

·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐑅꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜𐑆꩝⁚·

·word⁚·ЬЧ⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·I·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Ear·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Curved flesh cups the side of the head·˙ unseen vibration enters the curved
↪ hollow𐑅·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐑋𐑖𐑂𐑇𐑊𐑉𐑆𐑨𐑤𐑟𐑯𐑗𐑐𐑓·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐑉꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜𐑆꩝⁚·

·word⁚·𐑋𐑤𐑟·⁚·˙word⁚·

·⁚origin…language⁚·Hittite·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚·˙font⁚·
·⁚pronunciation⁚·PAKHUR·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Fire·˙ torch⁚⁚torches·˙ campfire⁚⁚watchfire·˙ embers⁚⁚burning coals·˙
↪ fever⁚⁚inflammation⁚⁚burning⁚⁚pain·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Charred brand holds brightness above the grip·˙ ember keeps red heat beneath the dark skin
↪ of coal𐑉·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐑕𐑔𐑇⊛ÞA𐑘𐑦𐑥𐑡𐑟𐑙⬡·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐑈꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜𐑆꩝⁚·

·word⁚·ЬJ𐑚JOЧЧ⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·BECHEUN·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Cave·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Hollowed stone bears a dark chamber beneath its outer face⁚⁚with cool inner walls closing
↪ around the empty depth𐑉·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᘐ𐑓𐑤𐑥𐑛𐑯𐑢𐑟𐑦𐑖𐑨𐑯·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜𐑣꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜𐑆꩝⁚·

·word⁚·ﻣﺼﺮ⁚·˙word⁚·

·⁚origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁚pronunciation:·KoSH·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·double⁖pair⁖ twinned (of things)·⁚.·˙non…ostensive:·
·⁚ostensive:·Matched vessels stand side by side⁖ each rim is whole⁖⁖and neither vessel touches the
↪ other⟛·⁚.·˙ostensive:·
·⁚grimchain:·⊚✡ᑭᗅ♁○ᛏᛋᚠᛈ⛾∮ᛏ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○𐤟𐤡 category○–○𐤟court𐤡:·
·⁚lookup office○–○𐤟relic𐤡 enoch○–○𐤟𐤀𐤡:·

·word:·𓏏:·.·˙word:·

·⁚origin…language:·Tangut·⁚.·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚.·˙font:·
·⁚pronunciation:·NUH·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Wisdom or intelligence⁖ capacity for sound understanding and
↪ judgment·⁚.·˙non…ostensive:·
·⁚ostensive:·Marked stones stand beside measured lines⁖⁖and weight and count settle into one exact
↪ balance with no correction mark beside them⟛·⁚.·˙ostensive:·
·⁚grimchain:·ᑫᘏᗅᴢΘ♎ₒᚕᗺ⩒ᔕ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○𐤟𐤉𐤡 category○–○𐤟court𐤡:·
·⁚lookup office○–○𐤟relic𐤡 enoch○–○𐤟𐤀𐤡:·

·word:·ઝьᒍᕽ·:.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·TEG·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·(wonder) sign·⁚.·˙non…ostensive:·
·⁚ostensive:·One mark breaks a field of repeated strokes⁖ singular shape remains fixed among the
↪ surrounding pattern𐤉·⁚.·˙ostensive:·
·⁚grimchain:·𐤡𐤚𐤊𐤍𐤋○⏁▽⟛Ⅱ⩒I𐤂·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol○–○𐤟𐤕𐤡 category○–○𐤟court𐤡:·
·⁚lookup office○–○𐤟relic𐤡 enoch○–○𐤟𐤀𐤡:·

·word:·아ㄴ|·:.·˙word:·

·⁚origin…language:·Middle Korean·⁚.·˙origin…language:·
·⁚font:·Noto Sans KR·⁚.·˙font:·
·⁚pronunciation:·ànEE·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·that which is not·⁚.·˙non…ostensive:·
·⁚ostensive:·Empty place among the marks⁖ one space stands open where every neighboring place carries
↪ a sign𐤕·⁚.·˙ostensive:·
·⁚grimchain:·ᚼ𐤉✲▽ᖇᙅᗺ○ₒᵒᵢᒪᖇ𓏥ᔑ⊖·⁚.·˙grimchain:·

·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜✱꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

·word:·Ⴋ+Ⲓ·⁚·˙word:·

·⁚origin…language:·Tocharian B·⁚·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚·˙font:·
·⁚pronunciation:·AHEEKAHREH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Containing nothing·⁚ having no contents·⁚·˙non…ostensive:·
·⁚ostensive:·Hollow vessel encloses clear air from wall to wall⁚⁚with no weight pressing against the
↪ inner clay✱·⁚·˙ostensive:·
·⁚grimchain:·𑈲ᚢ☆‹⊕𑈵⊛ᔓᛋᛤᛈᑭᑎ𑈵·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀢꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

·word:·ƧʢЧƵJhOᎩЧ·⁚·˙word:·

·⁚origin…language:·Aghwan·⁚·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚·˙font:·
·⁚pronunciation:·TAIGHESUN·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·To move away·⁚·˙non…ostensive:·
·⁚ostensive:·Footprints lengthen toward open ground·⁚· shallow marks trail from near earth into
↪ distance𑀢·⁚·˙ostensive:·
·⁚grimchain:·ᘚ▽ᗐᧉᨐᨒ𑀢ᨑᩚᨓᘐ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𓊝꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

·word:·𑈲𓊝𑈵〈𑈲ᅳ·⁚·˙word:·

·⁚origin…language:·Hittite·⁚·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚·˙font:·
·⁚pronunciation:·TAYNAWASH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Netherworld⁚⁚afterlife·⁚·˙non…ostensive:·
·⁚ostensive:·Sealed burial stone bears the weight of earth above⁚⁚and grave goods remain enclosed
↪ beneath living ground𓊝·⁚·˙ostensive:·
·⁚grimchain:·𐊗ᚸᚺᒍ𐊗ᓭ❋ᑫ𑀢✶◌ᒍ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜꩜𑀢꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜🜁꩝⁚·

·word:·ᎢᏁᒥᚱ·⁚·˙word:·

·⁚origin…language:·Gothic·⁚·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚·˙font:·

·⁚pronunciation:·TAGR·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Teardrop·⁚.·˙non…ostensive:·
·⁚ostensive:·Clear salt bead gathers at the eye⁚⁚rounded against the lower lid and heavy enough to
↪ fall⁖·⁚.·˙ostensive:·
·⁚grimchain:·ᚼᘜ⊙𐑈⊕ᛂ⁖ᚼ✡ᛝᛯ𐑉«·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑉 category◦–◦court»:·
·⁚lookup office◦–◦relic» enoch◦–◦𐑐»:·

‹word:·ᵬ₸ᵀ:.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·SHRKAHRAH·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Backwards⁚⁚in the opposite direction⁚⁚reverse·⁚.·˙non…ostensive:·
·⁚ostensive:·Heel and toe marks point toward the worn threshold⁚˙ the track itself stretches away
↪ across open ground𐑉·⁚.·˙ostensive:·
·⁚grimchain:·𐑧ↄᵪ𐑋ᗺ▽⊕⏀◎ᛊI≳✲·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑤» category◦–◦court»:·
·⁚lookup office◦–◦relic» enoch◦–◦𐑐»:·

‹word:·ϗϗᔑ:.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·NIZ·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Lust·⁚.·˙non…ostensive:·
·⁚ostensive:·Warm skin presses near warm skin⁚˙ hands remain at waist and shoulder⁚⁚and space between
↪ bodies closes𐑤·⁚.·˙ostensive:·
·⁚grimchain:·ᒍI⁓⊕◟ᵕ𐑋𐑉ᔕᵑᑫ⏀ᚠᚹ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol◦–◦𐑈» category◦–◦court»:·
·⁚lookup office◦–◦relic» enoch◦–◦𐑐»:·

‹word:·I□ᵪ₸:.·˙word:·

·⁚origin…language:·Tocharian B·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·RUHTK·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·To heal·⁚.·˙non…ostensive:·
·⁚ostensive:·Pale seam lies across joined flesh⁚⁚and living skin holds the closed wound beneath
↪ it𐑈·⁚.·˙ostensive:·
·⁚grimchain:·◟ᵠᘐᛉᛯᛯ◎▽ᚣ𐑉𐑤·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol꩜꩜𓂸꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜relic꩞ enoch꩜꩜𓃰꩞⁚·

⁖word⁚·븥잡다⁖.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·peuttsapta·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·to seize·⁚· to grab·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Fingers lock around cord or wrist·⁚· nothing slips through the closed grip𓂸꩞·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𓆉𓀀◎◎𓆏𓂻𓀠ϒᏢԿϴ𐤀·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜𓉔꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜relic꩞ enoch꩜꩜𓃰꩞⁚·

⁖word⁚·섬⁖.·˙word⁚·

·⁚origin…language⁚·Middle Korean·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans KR·⁚.·˙font⁚·
·⁚pronunciation⁚·suhm·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·stairstep·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·One tread rises above another·⁚· height comes to the foot in measured stone𓉔꩞·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ϒ✩✪𐤈ᛈ𐤉ꕥ✿·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜𓆏꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜relic꩞ enoch꩜꩜𓃰꩞⁚·

⁖word⁚·𠘨⁖.·˙word⁚·

·⁚origin…language⁚·Tangut·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚.·˙font⁚·
·⁚pronunciation⁚·N⟊TWEE·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Pair or couple·⁚· two associated things·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Matched vessels stand together·⁚· each body is complete⁚⁚and neither vessel touches the
↪ other𓆏·⁚.·˙ostensive⁚·
·⁚grimchain⁚·⊕◎ϒ𓂸꩞ϫƘՈ▽ℊ∮ⅿ᛬ϒ◎·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol꩜꩜𓀠꩞ category꩜꩜court꩞⁚·
·⁚lookup office꩜꩜relic꩞ enoch꩜꩜𓃰꩞⁚·

⁖word⁚·ᔑOϤᕲ⁖.·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚.·˙font⁚·
·⁚pronunciation⁚·RUGH·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Seed·⁚· root·⁚.·˙non…ostensive⁚·

·⁖ostensive⁝·Hard coat rests under soil with folded shoot and downward fibres enclosed
↪ inside‹·⁖·˙ostensive⁝·
·⁖grimchain⁝·⁜ᚺ❁ᛈᛉᛊᚺᛟᛋᛯᛯ⌬·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁖glyph symbol○‒○ᛀ𐦟᛬ category○‒○ᛀcourt᛬⁝·
·⁖lookup office○‒○ᛀrelic᛬ enoch○‒○ᛀᙏ᛬⁝·

‹word⁝·ـحصب⁝·⁖·˙word⁝·

·⁖origin…language⁝·Old Uyghur·⁖·˙origin…language⁝·
·⁖font⁝·Noto Serif Old Uyghur·⁖·˙font⁝·
·⁖pronunciation⁝·uya·⁖·˙pronunciation⁝·
·⁖cadence⁝·Relic·⁖·˙cadence⁝·
·⁖non…ostensive⁝·nest⁚⁚lair⁚⁚shelter⁚⁚den·⁖·˙non…ostensive⁝·
·⁖ostensive⁝·Straw and branch cup a hollow⁖ open ground falls away beneath the woven
↪ cradle𐦟·⁖·˙ostensive⁝·
·⁖grimchain⁝·ᚏ𐦟𐦟ᛯᛋ⌾ᛋᛜᛞᛁᛉᙏᛏ·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁖glyph symbol○‒○ᛀ𐦠᛬ category○‒○ᛀcourt᛬⁝·
·⁖lookup office○‒○ᛀrelic᛬ enoch○‒○ᛀᙏ᛬⁝·

‹word⁝·ـعصـر⁝·⁖·˙word⁝·

·⁖origin…language⁝·Old Uyghur·⁖·˙origin…language⁝·
·⁖font⁝·Noto Serif Old Uyghur·⁖·˙font⁝·
·⁖pronunciation⁝·Kara·⁖·˙pronunciation⁝·
·⁖cadence⁝·Relic·⁖·˙cadence⁝·
·⁖non…ostensive⁝·black⁚⁚swarthy⁚⁚dark⁖ (sociology) common and vulgar⁚⁚plebeian⁖ (by extension)
↪ simple⁚⁚basic⁖ (in compounds) west·⁖·˙non…ostensive⁝·
·⁖ostensive⁝·Light falls into woven cloth and scarcely returns⁖ shape withdraws into the unlit
↪ surface𐦠·⁖·˙ostensive⁝·
·⁖grimchain⁝·ᚏ⏢⓿▫ᛯᛓᛜᛋ᛭⛈ᛯᛜᚷ·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁖glyph symbol○‒○ᛀ𐦛᛬ category○‒○ᛀcourt᛬⁝·
·⁖lookup office○‒○ᛀrelic᛬ enoch○‒○ᛀᙏ᛬⁝·

‹word⁝·ᏄᏎᏎ⁝·⁖·˙word⁝·

·⁖origin…language⁝·Aghwan·⁖·˙origin…language⁝·
·⁖font⁝·Noto Sans Caucasian Albanian·⁖·˙font⁝·
·⁖pronunciation⁝·MII·⁖·˙pronunciation⁝·
·⁖cadence⁝·Relic·⁖·˙cadence⁝·
·⁖non…ostensive⁝·Cold·⁖·˙non…ostensive⁝·
·⁖ostensive⁝·Stone draws warmth from the palm⁖ water leaves a sharp ache along skin𐦛·⁖·˙ostensive⁝·
·⁖grimchain⁝·ᛗᛜᛯᚼᚷᛯᚼᛥᛜᑫᑎᑫᛔ·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁖glyph symbol○‒○ᛀ𐦜᛬ category○‒○ᛀcourt᛬⁝·
·⁖lookup office○‒○ᛀrelic᛬ enoch○‒○ᛀᙏ᛬⁝·

‹word⁝·ᚺ᛫Ꮔᑕ⁝·⁖·˙word⁝·

·⁚origin…language⁝·Aghwan·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Caucasian Albanian·⁚.·˙font⁝·
·⁚pronunciation⁝·SAXAY·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Relic·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·Image⁝⁚vision·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Carved face holds brow⁝⁚cheek⁝⁚and mouth in stone⁚· absent flesh remains fixed as mineral
↪ contour𐤍·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚺᚾ◎ᛕᚳᛗᚳᚢ✷ᛇᘜᛯᛞ⛤·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐤒﹐ category⟜⟜court﹐⁝·
·⁚lookup office⟜⟜relic﹐ enoch⟜⟜𐤀﹐⁝·

·⁚word⁝·ЬѺҎΣ⁝·⁚.·˙word⁝·

·⁚origin…language⁝·Aghwan·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Caucasian Albanian·⁚.·˙font⁝·
·⁚pronunciation⁝·0CK·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Relic·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To swim·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Torso floats within water⁝⁚and limbs press the moving medium along the surface
↪ line𐤒·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚱᗅ‹ᛥᘐᗅᛟᚴ1ᛉ`ᘓ°ᛋᚸ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐤍﹐ category⟜⟜court﹐⁝·
·⁚lookup office⟜⟜relic﹐ enoch⟜⟜𐤀﹐⁝·

·⁚word⁝·ϬΩJ⊦ᒍhOᒣϤ⁝·⁚.·˙word⁝·

·⁚origin…language⁝·Aghwan·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Sans Caucasian Albanian·⁚.·˙font⁝·
·⁚pronunciation⁝·BATKESUN·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Relic·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·To turn around·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Spokes change bearing at a fixed hub⁚· rim keeps one circle through every
↪ position𐤍·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ᚋᚸᑎᓀᓕ𓊝ᚢᚣᚺᚸ᛫ᛥ·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚˙
·⁚.·˙glyphs⁚˙
·⁚glyph symbol⟜⟜𐤐﹐ category⟜⟜court﹐⁝·
·⁚lookup office⟜⟜relic﹐ enoch⟜⟜𐤀﹐⁝·

·⁚word⁝·ﺐﺐﻣﺼ⁝·⁚.·˙word⁝·

·⁚origin…language⁝·Old Uyghur·⁚.·˙origin…language⁝·
·⁚font⁝·Noto Serif Old Uyghur·⁚.·˙font⁝·
·⁚pronunciation⁝·yUErEHk·⁚.·˙pronunciation⁝·
·⁚cadence⁝·Relic·⁚.·˙cadence⁝·
·⁚non…ostensive⁝·heart⁚· (psychology) courage⁝⁚bravery·⁚.·˙non…ostensive⁝·
·⁚ostensive⁝·Pulse strikes beneath the breast⁚· danger stands near⁝⁚and planted feet keep their
↪ ground𐤐·⁚.·˙ostensive⁝·
·⁚grimchain⁝·ϒIᔕᚴxᙢᛏᙢᗢBxI·⁚.·˙grimchain⁝·
·⁚.·˙lookup⁚˙

·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ category꩜₎court₎:·
·⁚lookup office꩜₎relic₎ enoch꩜₎:·

:word:·ᑭᕉΣ:⁚·˙word:·

·⁚origin…language:·Aghwan·⁚·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚·˙font:·
·⁚pronunciation:·ZHAK·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Chain·⁚·˙non…ostensive:·
·⁚ostensive:·Linked metal pulls piece through piece::and shared strain travels across the joined
↪ lengthᖴ·⁚·˙ostensive:·
·⁚grimchain:·▽ᔑᑎᘐᘎᐯᑐᐣᘎᑐᗐ▫·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ᖴ₎ category꩜₎court₎:·
·⁚lookup office꩜₎relic₎ enoch꩜₎:·

:word:·술:⁚·˙word:·

·⁚origin…language:·Middle Korean·⁚·˙origin…language:·
·⁚font:·Noto Sans KR·⁚·˙font:·
·⁚pronunciation:·swúl·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·spoon·:˙ spoonful·⁚·˙non…ostensive:·
·⁚ostensive:·Small hollow cups one mouthful::carrying measured food from vessel to lipsᖴ·⁚·˙ostensive:·
·⁚grimchain:·⊛ᛉᔑᚏᐅᔑᐣᐊᛋᐦᗡᗡᘔᘐᔓ⏀·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ᖵ₎ category꩜₎court₎:·
·⁚lookup office꩜₎relic₎ enoch꩜₎:·

:word:·ﺑﺪﺧﻪ:⁚·˙word:·

·⁚origin…language:·Old Uyghur·⁚·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚·˙font:·
·⁚pronunciation:·adUHl·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·special::different·:˙ select::extraordinary::excellent·⁚·˙non…ostensive:·
·⁚ostensive:·One stone bears a cut mark·:˙ common heap remains unmarked around itᖵ·⁚·˙ostensive:·
·⁚grimchain:·⁞⁞ᛃᔕ⊛ᐦᒉᓭᙏᑭᗢᔕᛃ·⁚·˙grimchain:·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꩜₎ᖽ₎ category꩜₎court₎:·
·⁚lookup office꩜₎relic₎ enoch꩜₎:·

:word:·死:⁚·˙word:·

·⁚origin…language:·Tangut·⁚·˙origin…language:·
·⁚font:·Noto Serif Tangut·⁚·˙font:·
·⁚pronunciation:·LEH·⁚·˙pronunciation:·
·⁚cadence:·Relic·⁚·˙cadence:·
·⁚non…ostensive:·Cover::shelter::hide::or conceal·⁚·˙non…ostensive:·

·⁚ostensive⁝·Cloth or lid blocks sight of the enclosed thing⁘ outer surface becomes the face the eye must meet𐫲·⁖·˙ostensive⁝·
·⁚grimchain⁝·𐫱𐫰◎𐫳𐫴𐫵𐫶𐫷𐫸𐫹𐫺𐫻𐫼𐫽𐫾·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol⁝○꞉Ⱶ꞉ category⁝○꞉court꞉⁚·
·⁚lookup office⁝○꞉relic꞉ enoch⁝○꞉ⱛ꞉⁚·

·word⁝·ᎶᏁᎶᏁᏕ·⁖·˙word⁝·

·⁚origin…language⁝·Aghwan·⁖·˙origin…language⁝·
·⁚font⁝·Noto Sans Caucasian Albanian·⁖·˙font⁝·
·⁚pronunciation⁝·XEXER·⁖·˙pronunciation⁝·
·⁚cadence⁝·Relic·⁖·˙cadence⁝·
·⁚non…ostensive⁝·Saw·⁖·˙non…ostensive⁝·
·⁚ostensive⁝·Hard teeth stand in a narrow row along the blade⁝⁝and toothed iron bears cutting force across the whole edgeⱵ·⁖·˙ostensive⁝·
·⁚grimchain⁝·𐫰☆𐫱𐫲𐫳𐫴𐫵𐫶𐫷𐫸𐫹·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol⁝○꞉A꞉ category⁝○꞉court꞉⁚·
·⁚lookup office⁝○꞉relic꞉ enoch⁝○꞉ⱛ꞉⁚·

·word⁝·ھەمگھ·⁖·˙word⁝·

·⁚origin…language⁝·Old Uyghur·⁖·˙origin…language⁝·
·⁚font⁝·Noto Serif Old Uyghur·⁖·˙font⁝·
·⁚pronunciation⁝·boduGH·⁖·˙pronunciation⁝·
·⁚cadence⁝·Relic·⁖·˙cadence⁝·
·⁚non…ostensive⁝·color⁝⁝dye·⁖·˙non…ostensive⁝·
·⁚ostensive⁝·Pigment enters fibre and skin⁝⁝remaining where the material has taken the stainA·⁖·˙ostensive⁝·
·⁚grimchain⁝·𐫱𐫲◎𐫳𐫴𐫵𐫶𐫷𐫸𐫹·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol⁝○꞉Ƀ꞉ category⁝○꞉court꞉⁚·
·⁚lookup office⁝○꞉relic꞉ enoch⁝○꞉ⱛ꞉⁚·

·word⁝·ےہےے·⁖·˙word⁝·

·⁚origin…language⁝·Old Uyghur·⁖·˙origin…language⁝·
·⁚font⁝·Noto Serif Old Uyghur·⁖·˙font⁝·
·⁚pronunciation⁝·ala·⁖·˙pronunciation⁝·
·⁚cadence⁝·Relic·⁖·˙cadence⁝·
·⁚non…ostensive⁝·dappled⁝⁝mottled⁝⁝spotted⁝⁝parti⁝coloured·⁖·˙non…ostensive⁝·
·⁚ostensive⁝·Hide carries broken islands of hue across one surface⁝⁝and no single tone owns the whole·⁖·˙ostensive⁝·
·⁚grimchain⁝·𐫲𐫳𐫴𐫵𐫶𐫷𐫸𐫹𐫺𐫻·⁖·˙grimchain⁝·
·⁖·˙lookup⁘
·⁖·˙glyphs⁘
·⁚glyph symbol⁝○꞉Ↄ꞉ category⁝○꞉court꞉⁚·
·⁚lookup office⁝○꞉relic꞉ enoch⁝○꞉ⱛ꞉⁚·

·word⁝·ᏂᏬᏕᏁ·⁖·˙word⁝·

·⁖origin…language:·Aghwan·⁖.·˙origin…language:·
·⁖font:·Noto Sans Caucasian Albanian·⁖.·˙font:·
·⁖pronunciation:·VARZ·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Reward::salary·⁖.·˙non…ostensive:·
·⁖ostensive:·Counted grain fills the receiving bowl::and measured weight stands equal to the marked
↪ tally beside itↃ·⁖.·˙ostensive:·
·⁖grimchain:·⛤ⵅᚹᚢ⛢⊛✡𐓛ᚹᚲᛀᛇ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜ᚴ꩝ category꩜꩜ᵋcourt꩝⁖·
·⁖lookup office꩜꩜ᵋrelic꩝ enoch꩜꩜ᵋᕃ꩝⁖·

·word:·ᛟᚱᛉ·⁖.·˙word:·

·⁖origin…language:·Old Turkic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Old Turkic·⁖.·˙font:·
·⁖pronunciation:·BEER·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·One·⁖.·˙non…ostensive:·
·⁖ostensive:·Stone rests on broad plain·⁖· no second body breaks the ground beside itᛋ·⁖.·˙ostensive:·
·⁖grimchain:·♉♂0⁎ₒₕ⊛ϒ⅄Ⅰℂ⛢ℂ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜ᵋᛊ꩝ category꩜꩜ᵋcourt꩝⁖·
·⁖lookup office꩜꩜ᵋrelic꩝ enoch꩜꩜ᵋᕃ꩝⁖·

·word:·ᛣᒥIS·⁖.·˙word:·

·⁖origin…language:·Gothic·⁖.·˙origin…language:·
·⁖font:·Noto Sans Gothic·⁖.·˙font:·
·⁖pronunciation:·AGIS·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·Fear·⁖.·˙non…ostensive:·
·⁖ostensive:·Skin draws taut over bone::and widened eye holds a rim of white::and breath stays shallow
↪ beneath the ribsᛊ·⁖.·˙ostensive:·
·⁖grimchain:·𑜦⏥ᚠⵅᚱᓀᛇᛋ⅂ⵅᚾᛘᗒᛯ⁎·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·
·⁖glyph symbol꩜꩜ᵋᚠ꩝ category꩜꩜ᵋcourt꩝⁖·
·⁖lookup office꩜꩜ᵋrelic꩝ enoch꩜꩜ᵋᕃ꩝⁖·

·word:·실·⁖.·˙word:·

·⁖origin…language:·Middle Korean·⁖.·˙origin…language:·
·⁖font:·Noto Sans KR·⁖.·˙font:·
·⁖pronunciation:·sEEl·⁖.·˙pronunciation:·
·⁖cadence:·Relic·⁖.·˙cadence:·
·⁖non…ostensive:·thread::string·⁖.·˙non…ostensive:·
·⁖ostensive:·Narrow fibre runs through cloth and knot·⁖· tension travels along the held
↪ lineᚠ·⁖.·˙ostensive:·
·⁖grimchain:·ᘐᛩᛞᚱᛋᓀᚴᛋᘓ⛤ᘓ⊛ᛘᚱ·⁖.·˙grimchain:·
·⁖.·˙lookup:·
·⁖.·˙glyphs:·

·⁖glyph symbol꩜꩜ı꞉ category꩜꩜ᵴcourt꞉꞉·
·⁖lookup office꩜꩜ᵴrelic꞉ enoch꩜꩜ᵴ꞉꞉·

⁖word⁖·마치⁖꞉·˙word⁖·

·⁖origin…language⁖·Middle Korean·⁖꞉·˙origin…language⁖·
·⁖font⁖·Noto Sans KR·⁖꞉·˙font⁖·
·⁖pronunciation⁖·matsi·⁖꞉·˙pronunciation⁖·
·⁖cadence⁖·Relic·⁖꞉·˙cadence⁖·
·⁖non…ostensive⁖·hammer·⁖꞉·˙non…ostensive⁖·
·⁖ostensive⁖·Iron head sits dense above the haft⁖⁖and the whole weight leans toward one blunt striking face꞉·⁖꞉·˙ostensive⁖·
·⁖grimchain⁖·ǂIⅢ⅍ᚢ◎ΔᚹⓄᗡ'ϐϔ·⁖꞉·˙grimchain⁖·
·⁖꞉·˙lookup⁖˙
·⁖꞉·˙glyphs⁖˙
·⁖glyph symbol꩜꩜ᵴᗡ꞉ category꩜꩜ᵴcourt꞉꞉·
·⁖lookup office꩜꩜ᵴrelic꞉ enoch꩜꩜ᵴ꞉꞉·

⁖word⁖·صوصصف⁖꞉·˙word⁖·

·⁖origin…language⁖·Old Uyghur·⁖꞉·˙origin…language⁖·
·⁖font⁖·Noto Serif Old Uyghur·⁖꞉·˙font⁖·
·⁖pronunciation⁖·kUENGGHau·⁖꞉·˙pronunciation⁖·
·⁖cadence⁖·Relic·⁖꞉·˙cadence⁖·
·⁖non…ostensive⁖·a type of plucked string instrument similar to a lute⁖⁖komuz⁖⁖cobza·⁖꞉·˙non…ostensive⁖·
·⁖ostensive⁖·Taut cord stretches over hollow wood⁖˙ vibration gathers inside hollow wood and leaves as songᗡ·⁖꞉·˙ostensive⁖·
·⁖grimchain⁖·ᚺXᚾᛘꟾᖘᐯᚹ❋◌ᛇᚴ·⁖꞉·˙grimchain⁖·
·⁖꞉·˙lookup⁖˙
·⁖꞉·˙glyphs⁖˙
·⁖glyph symbol꩜꩜ᵴꓘ꞉ category꩜꩜ᵴcourt꞉꞉·
·⁖lookup office꩜꩜ᵴrelic꞉ enoch꩜꩜ᵴ꞉꞉·

⁖word⁖·तीर्थ⁖꞉·˙word⁖·

·⁖origin…language⁖·Sanskrit·⁖꞉·˙origin…language⁖·
·⁖font⁖·Noto Sans Devanagari·⁖꞉·˙font⁖·
·⁖pronunciation⁖·teer÷thuh·⁖꞉·˙pronunciation⁖·
·⁖cadence⁖·Relic·⁖꞉·˙cadence⁖·
·⁖non…ostensive⁖·a passage⁖⁖way or road⁖˙ that which affords transit⁖⁖ a ford⁖˙ a crossing at a river⁖˙ a passage⁖⁖way or road⁖˙ that which affords transit⁖⁖ stairs for descent into a river⁖⁖⁖˙ a passage⁖⁖way or road⁖˙ that which affords transit⁖⁖ a channel or canal⁖˙ an artificial watercourse⁖˙ a passage⁖⁖way or road⁖˙ that which affords transit⁖⁖ the path to the altar between the चात्वाल (cātvāla⁖⁖“sacrificial pit”) and उत्कर (utkara⁖⁖“pile of refuse”)⁖˙ something sacred or revered⁖˙ a holy thing or place⁖⁖ Lines or parts of the hand considered to be sacred⁖˙ something sacred or revered⁖˙ a holy thing or place⁖⁖ a place of pilgrimage (especially at a sacred body of water)⁖˙ someone sacred or revered⁖˙ a holy or wise person⁖⁖ a worthy person⁖⁖especially one worthy to receive something⁖˙ someone sacred or revered⁖˙ a holy or wise person⁖⁖ an adviser or counsellor⁖˙ one entrusted to provide advice⁖˙ the right place⁖⁖time⁖⁖or means⁖˙ advice⁖⁖instruction⁖˙ that which gives direction·⁖꞉·˙non…ostensive⁖·
·⁖ostensive⁖·Shallow stones stand beneath running water⁖⁖and one firm line crosses from bank to bank below the currentꓘ·⁖꞉·˙ostensive⁖·
·⁖grimchain⁖·♂ᛋ⧗▫ᚷᘐᨐᘏ❋Ↄᚱᚷ·⁖꞉·˙grimchain⁖·
·⁖꞉·˙lookup⁖˙

·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉꞉1꞉ category꞉꞉꞉court꞉⁚·
·⁚lookup office꞉꞉꞉relic꞉ enoch꞉꞉꞉ᕃ꞉⁚·

꞉word⁚·ᕃᔑ꞉⁚·˙word⁚·

·⁚origin…language⁚·Old Turkic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Old Turkic·⁚·˙font⁚·
·⁚pronunciation⁚·BAH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To tie::fasten·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Cord closes around post or bundle·⁚˙ knot keeps the meeting from opening1·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᛉᛤᛯᚴᚹᛋᚼᚹ֍✳▫ঌ৩·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉꞉ᛋ꞉ category꞉꞉꞉court꞉⁚·
·⁚lookup office꞉꞉꞉relic꞉ enoch꞉꞉꞉ᕃ꞉⁚·

꞉word⁚·ᛉᛟᛉ꞉⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·CAC·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Thorn·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Living wood narrows into a hard point·⁚˙ skin and cloth catch against the sharpened
↪ growthᛋ·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐑉ᕃↄᚷᛈᚱᛯᛋ1✡❁·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉꞉ᛋ꞉ category꞉꞉꞉court꞉⁚·
·⁚lookup office꞉꞉꞉relic꞉ enoch꞉꞉꞉ᕃ꞉⁚·

꞉word⁚·केवल꞉⁚·˙word⁚·

·⁚origin…language⁚·Sanskrit·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Devanagari·⁚·˙font⁚·
·⁚pronunciation⁚·kay+vuh+luh·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·exclusively one’s own·⁚˙ sole::alone::unmingled::excluding others·⁚˙ not connected with
↪ anything else::isolated::abstract::absolute·⁚˙ simple::pure::uncompounded::unmingled·⁚˙ entire::whole::all·⁚˙
↪ selfish::envious·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Pale stone rests apart from foreign grains·⁚˙ stone edges enclose the same mineral with
↪ nothing embedded from outsideᛋ·⁚·˙ostensive⁚·
·⁚grimchain⁚·𐑓▫ᚱ⛀ᛥᛈ+ᚳᚠ♂♂☿ᚼ·⁚·˙grimchain⁚·
·⁚·˙lookup:˙
·⁚·˙glyphs:˙
·⁚glyph symbol꞉꞉꞉✳꞉ category꞉꞉꞉court꞉⁚·
·⁚lookup office꞉꞉꞉relic꞉ enoch꞉꞉꞉ᕃ꞉⁚·

꞉word⁚·ΣЧᛊᖘ꞉⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·

·⁚pronunciation:·KILT·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Key·⁚.·˙non…ostensive:·
·⁚ostensive:·Cut metal teeth enter one matching slot⁖ a small shaped piece answers the closed mouth of
↪ the lock✳·⁚.·˙ostensive:·
·⁚grimchain:·ᛋ☆ىصᚭ❄Θℂᚹᛋᚱ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑ✳꞉ category○–○ₑcourt꞉·
·⁚lookup office○–○ₑrelic꞉ enoch○–○ₑ⁂꞉·

·word:·ᘻ২Ϙ·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·SHAD·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Free·⁚.·˙non…ostensive:·
·⁚ostensive:·Loose cord rests without knot⁖ gate stands open with bar laid aside⁝⁝and path runs clear
↪ beyond the threshold✳·⁚.·˙ostensive:·
·⁚grimchain:·ϒⱫᵴᴳᘜ♀Λᗅᘉᚻᚴ♀·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑ✳꞉ category○–○ₑcourt꞉·
·⁚lookup office○–○ₑrelic꞉ enoch○–○ₑ⁂꞉·

·word:·🜨⊢·⁚.·˙word:·

·⁚origin…language:·Hittite·⁚.·˙origin…language:·
·⁚font:·Noto Sans Cuneiform·⁚.·˙font:·
·⁚pronunciation:·NAYWAS·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·New⁝⁝fresh·⁚.·˙non…ostensive:·
·⁚ostensive:·Pale wet grain shines along a cut face⁖ bright sap beads at an edge still keen from the
↪ blade✳·⁚.·˙ostensive:·
·⁚grimchain:·𐎗᚛'ᚲᗉᛠᵹᚸᛜᚹᚹᚢ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ₑ✳꞉ category○–○ₑcourt꞉·
·⁚lookup office○–○ₑrelic꞉ enoch○–○ₑ⁂꞉·

·word:·ﻫﻪﻫﻫ⁚.·˙word:·

·⁚origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁚pronunciation:·yarlUHGH·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·decree⁝⁝yarligh·⁚.·˙non…ostensive:·
·⁚ostensive:·Folded writing rests beneath a seal⁝⁝cord crosses the leaf⁝⁝and stamped wax grips the
↪ closed marks✳·⁚.·˙ostensive:·
·⁚grimchain:·0≍∀✳ϟ⊖⊛ǂ✡ᗰᗰᗅᘉᙡ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙

·⁚glyph symbol○‑○₍𐎗𐎚 category○‑○₍court𐎚⁚·
·⁚lookup office○‑○₍relic𐎚 enoch○‑○₍𐎐𐎚⁚·

·word⁚·𒀸𒅔𒄑𒉡𒂊𒁺·.·˙word⁚·

·⁚origin…language⁚·Hittite·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·NAYPIS·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Sky·⁚· heaven·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Still water in a shallow vessel carries blue height and passing cloud across its open
↪ face𐎗·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐎍𐎓𐎘𐎒𐎀𐎄𐎁𐎒𐎁𐎍𐎉·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○‑○₍0𐎚 category○‑○₍court𐎚⁚·
·⁚lookup office○‑○₍relic𐎚 enoch○‑○₍𐎐𐎚⁚·

·word⁚·ᏏᏩᏁᏩᏁ·.·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚.·˙font⁚·
·⁚pronunciation⁚·BALAL·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Candle·⁚· lamp·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Small vessel cups flame above wick and oil⁘carrying brightness into dark0·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐒉𐒋𐒆𐒍𐒇𐒈𐒆𐒍𐒇𐒈𐒆𐒍𐒇𐒈𐒆𐒍𐒇𐒈𐒆𐒍𐒇𐒈·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○‑○₍𐎕𐎚 category○‑○₍court𐎚⁚·
·⁚lookup office○‑○₍relic𐎚 enoch○‑○₍𐎐𐎚⁚·

·word⁚·𐫁𐫂𐫃·.·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚.·˙font⁚·
·⁚pronunciation⁚·uz·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·skillful·⁚· sufficient⁘appropriate⁘fit⁘apt·⁚· light (not heavy)⁘brisk·⁚·
↪ orderly⁘neat⁘potent⁘influential·⁚· mastered⁘expert·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Tool sits true within fitted grip·⁚· clean joins meet clean edges⁘and no gap opens beside
↪ the work𐎕·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐎗𐎚𐎐𐎔𐎍𐎒𐎋𐎍𐎒𐎋𐎍𐎒𐎋𐎍𐎒𐎋𐎍𐎒𐎋·⁚.·˙grimchain⁚·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○‑○₍𐎐𐎚 category○‑○₍court𐎚⁚·
·⁚lookup office○‑○₍relic𐎚 enoch○‑○₍𐎐𐎚⁚·

·word⁚·𐎜𐎍𐎗·.·˙word⁚·

·⁚origin…language⁚·Ugaritic·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Ugaritic·⁚.·˙font⁚·
·⁚pronunciation⁚·DALTU·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·

·⁚non…ostensive⁚·Door·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Hinged panel fills the wall opening⁚· bar and threshold hold passage between open and
↪ shut⁜·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐓑☆ℂℎ⊙≍𐓎𐓟∀Ƀ0𐓎I·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𐦝꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜relic꞉ enoch⟜⟜𐓣꞉⁚·

⁖word⁚·𐫁𐫂𐫄𐫀𐫃𐫅⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚.·˙font⁚·
·⁚pronunciation⁚·aGHUHz·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·mouth⁚⁚a mouthful⁚· (by extension) a opening (of a wound⁚⁚a hole⁚⁚or an
↪ estuary⁚⁚etc)·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Living hollow opens between within and without⁚· breath⁚⁚food⁚⁚and speech cross the same
↪ flesh+bound threshold𐦝·⁚.·˙ostensive⁚·
·⁚grimchain⁚·Ψ𐓣ᘐ𐓛ϒ𐓍𐓂𐓃𐓄ᛋ𐓑▽Θ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𐦟꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜relic꞉ enoch⟜⟜𐓣꞉⁚·

⁖word⁚·𐎀𐎁𐎂𐎃𐎄⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Hittite·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Cuneiform·⁚.·˙font⁚·
·⁚pronunciation⁚·KHAHPPAR·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Business⁚⁚trade⁚⁚payment⁚⁚price⁚⁚purchase·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Goods stand opposite measured silver⁚· equal weights meet across a marked
↪ board𐦟·⁚.·˙ostensive⁚·
·⁚grimchain⁚·𐓫𐓯𐓯'𐓃ᘐ𐓟▽𐓟♀𐓘𐓐⊕·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𐦞꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜relic꞉ enoch⟜⟜𐓣꞉⁚·

⁖word⁚·ʘƃʘϞ⁜⁖·⁚.·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚.·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚.·˙font⁚·
·⁚pronunciation⁚·OBUK·⁚.·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚.·˙cadence⁚·
·⁚non…ostensive⁚·Reed·⁚.·˙non…ostensive⁚·
·⁚ostensive⁚·Hollow green stem rises from wet ground⁚· hollow stem carries ink or breath through its
↪ channel𐦞·⁚.·˙ostensive⁚·
·⁚grimchain⁚·ↄ𐓚𐓂𐓟𐓣𐓠𐓭▽𐓭𐓘0𐓘𐓟𐓟ϒ·⁚.·˙grimchain⁚·
·⁚.·˙lookup⁚·
·⁚.·˙glyphs⁚·
·⁚glyph symbol⟜⟜𐓍꞉ category⟜⟜court꞉⁚·
·⁚lookup office⟜⟜relic꞉ enoch⟜⟜𐓣꞉⁚·

˖word:·ﻫﺪﻣﻪ:.·˙word:·

·⁚origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁚pronunciation:·baGHUHSH·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·fastening rope (of a tent)⁖ (hapax legomenon) joint⁖
↪ pivot::enarthrosis·⁚.·˙non…ostensive:·
·⁚ostensive:·Cord binds the tent+line around one fixed stake::and gathered strain turns through the
↪ same knotɣ·⁚.·˙ostensive:·
·⁚grimchain:·ᨒᩚᏂₒℏᒍIᐁᘐ≳⁞⁞0☽·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ᵩ» category○–○ᵩcourt»⁚·
·⁚lookup office○–○ᵩrelic» enoch○–○ᵩᐃ»⁚·

˖word:·𓊪:.·˙word:·

·⁚origin…language:·Egyptian·⁚.·˙origin…language:·
·⁚font:·Noto Sans Egyptian Hieroglyphs·⁚.·˙font:·
·⁚pronunciation:·HUH+tep·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Peace::satisfaction::rest⁖ offering::to be content·⁚.·˙non…ostensive:·
·⁚ostensive:·Laid gift rests on stone::and emptied vessels stand quiet beside the settled
↪ portionᦓ·⁚.·˙ostensive:·
·⁚grimchain:·ᦒ᪲ᐁᘔᘐᩚ᧗ᘏ⁎ₒᦟ‡ↄ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ᵪ» category○–○ᵩcourt»⁚·
·⁚lookup office○–○ᵩrelic» enoch○–○ᵩᐃ»⁚·

˖word:·तुला:.·˙word:·

·⁚origin…language:·Sanskrit·⁚.·˙origin…language:·
·⁚font:·Noto Sans Devanagari·⁚.·˙font:·
·⁚pronunciation:·TOO+lah·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·balance::scales⁖ weight·⁚.·˙non…ostensive:·
·⁚ostensive:·Two burdens hang opposed from one beam::and level appears where neither side
↪ descendsx·⁚.·˙ostensive:·
·⁚grimchain:·ᦐ1Ↄᛯ≏ℂᒍᛊ◎ᦟᑭᘏ·⁚.·˙grimchain:·
·⁚.·˙lookup·˙
·⁚.·˙glyphs·˙
·⁚glyph symbol○–○ᵽ» category○–○ᵩcourt»⁚·
·⁚lookup office○–○ᵩrelic» enoch○–○ᵩᐃ»⁚·

˖word:·ﻫﻤﻜﺤﺐ:.·˙word:·

·⁚origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁚pronunciation:·UHduK·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·holy::godlike::charismatic⁖ regal::reverend::venerable·⁚.·˙non…ostensive:·

·⁚ostensive⁚·Vessel rests apart from common wares⁘ corded boundary circles the untouched place⁖⁖and ordinary hands remain outside⁘·⁚·˙ostensive⁚·
·⁚grimchain⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙grimchain⁚·
·⁚·˙lookup⁘˙
·⁚·˙glyphs⁘˙
·⁚glyph symbol○–○𖬟𖬞 category○–○𖬟court𖬞⁚·
·⁚lookup office○–○𖬟relic𖬞 enoch○–○𖬟𖬞⁚·

·word⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙word⁚·

·⁚origin…language⁚·Tangut·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Tangut·⁚·˙font⁚·
·⁚pronunciation⁚·THWUH·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Finish⁘˙ to bring or come to an end⁘˙ complete·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Fitted stones lie flush across the wall⁖⁖and every joint is filled⁖⁖and no open seam remains in the masonry𖬞·⁚·˙ostensive⁚·
·⁚grimchain⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙grimchain⁚·
·⁚·˙lookup⁘˙
·⁚·˙glyphs⁘˙
·⁚glyph symbol○–○𖬟𖬞 category○–○𖬟court𖬞⁚·
·⁚lookup office○–○𖬟relic𖬞 enoch○–○𖬟𖬞⁚·

·word⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙word⁚·

·⁚origin…language⁚·Aghwan·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Caucasian Albanian·⁚·˙font⁚·
·⁚pronunciation⁚·PULAYGAN·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Witness·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Cut mark and written trace stand beside the sealed account⁖⁖and material signs bear the absent deed in enduring form𖬞·⁚·˙ostensive⁚·
·⁚grimchain⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙grimchain⁚·
·⁚·˙lookup⁘˙
·⁚·˙glyphs⁘˙
·⁚glyph symbol○–○𖬟𖬞 category○–○𖬟court𖬞⁚·
·⁚lookup office○–○𖬟relic𖬞 enoch○–○𖬟𖬞⁚·

·word⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙word⁚·

·⁚origin…language⁚·Gothic·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Gothic·⁚·˙font⁚·
·⁚pronunciation⁚·STANDAN·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·To stand·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Upright post bears weight into earth⁘˙ neither seat nor bed receives it𖬞·⁚·˙ostensive⁚·
·⁚grimchain⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙grimchain⁚·
·⁚·˙lookup⁘˙
·⁚·˙glyphs⁘˙
·⁚glyph symbol○–○𖬟𖬞 category○–○𖬟court𖬞⁚·
·⁚lookup office○–○𖬟relic𖬞 enoch○–○𖬟𖬞⁚·

·word⁚·𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟𖬞𖬰𖬟·⁚·˙word⁚·

·⁚origin…language:·Gothic·⁚.·˙origin…language:·
·⁚font:·Noto Sans Gothic·⁚.·˙font:·
·⁚pronunciation:·WAYNS·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Hope::expectation·⁚.·˙non…ostensive:·
·⁚ostensive:·Kept token rests beside an empty setting⁖· open cup and folded cloth remain beside the
↪ vacant place▽·⁚.·˙ostensive:·
·⁚grimchain:·ǂ'ᗅ𐑉ᚺᛕℂ𑣲ᑭᒍᖾ𑣲ɔ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꜀꜀z꜀ category꜀꜀court꜀:·
·⁚lookup office꜀꜀relic꜀ enoch꜀꜀ᕰ꜀:·

⁖word:·ꕯꕯ⁖·⁚.·˙word:·

·⁚origin…language:·Aghwan·⁚.·˙origin…language:·
·⁚font:·Noto Sans Caucasian Albanian·⁚.·˙font:·
·⁚pronunciation:·NYN·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Eternal·⁚.·˙non…ostensive:·
·⁚ostensive:·Unbroken bronze ring closes upon itself⁖· smooth curve returns to its starting point
↪ without meeting seam or cutz·⁚.·˙ostensive:·
·⁚grimchain:·ᕎᚣᛯᝰᗜᘐᚠᔕᐠ꒰ᘐ∱·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꜀꜀ɾ꜀ category꜀꜀court꜀:·
·⁚lookup office꜀꜀relic꜀ enoch꜀꜀ᕰ꜀:·

⁖word:·ᚺᓀ⁖·⁚.·˙word:·

·⁚origin…language:·Tocharian A·⁚.·˙origin…language:·
·⁚font:·Noto Sans Brahmi·⁚.·˙font:·
·⁚pronunciation:·AHSAHN·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·Seat·⁚.·˙non…ostensive:·
·⁚ostensive:·Raised frame receives bodily weight above the ground⁖· loosened legs remain supported by
↪ the raised frameɕ·⁚.·˙ostensive:·
·⁚grimchain:·ᙀᛃ'ᚸᘐᘔᚹ⊙ᚺᚼᘏᘓ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙
·⁚.·˙glyphs:˙
·⁚glyph symbol꜀꜀ᑲ꜀ category꜀꜀court꜀:·
·⁚lookup office꜀꜀relic꜀ enoch꜀꜀ᕰ꜀:·

⁖word:·ـــــ⁖·⁚.·˙word:·

·⁚origin…language:·Old Uyghur·⁚.·˙origin…language:·
·⁚font:·Noto Serif Old Uyghur·⁚.·˙font:·
·⁚pronunciation:·aGHUHn·⁚.·˙pronunciation:·
·⁚cadence:·Relic·⁚.·˙cadence:·
·⁚non…ostensive:·mute::dumb (p::unable to speak)·⁚.·˙non…ostensive:·
·⁚ostensive:·Closed mouth keeps breath behind the lips⁖· no shaped voice crosses the
↪ openingь·⁚.·˙ostensive:·
·⁚grimchain:·ᚸᑲᓀᘌ♀ᘘᙢℂΘ∱ᖶᛏ·⁚.·˙grimchain:·
·⁚.·˙lookup:˙

·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ƻ꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜ᕃ꩝⁚·

·word⁚·ᛋᚠᛁᚠᛦ⁚·˙word⁚·

·⁚origin…language⁚·Old Norse·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Runic·⁚·˙font⁚·
·⁚pronunciation⁚·SVEVN·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Sleep·˙ dream·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Woven mat bears a still body in darkness·˙ closed eyes and loosened limbs share one quiet
↪ fieldƻ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᚸ⁕ᚱᛞᛇ⁜ᓄᘐ◎ᚾᛇ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ᴓ꩝ category꩜꩜court꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜ᕃ꩝⁚·

·word⁚·صوﻳﺴﺖ⁚·˙word⁚·

·⁚origin…language⁚·Old Uyghur·⁚·˙origin…language⁚·
·⁚font⁚·Noto Serif Old Uyghur·⁚·˙font⁚·
·⁚pronunciation⁚·alKu·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·all⁖whole·˙ every+·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Tight weave swells beneath gathered grain⁖and weighted rim leaves no grain outside the
↪ rounded bodyᴓ·⁚·˙ostensive⁚·
·⁚grimchain⁚·ᘉᔓᘻ⊕⏁☿☆↓❀ᘓ✳⊙ᚼI·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚glyph symbol꩜꩜ʃ꩝ category꩜꩜axiomyr꩝⁚·
·⁚lookup office꩜꩜relic꩝ enoch꩜꩜ᕃ꩝⁚·

·word⁚·ᛯᛅ̄ᛢᚻᛚ̄ᚬ̄⁚·˙word⁚·

·⁚origin…language⁚·Tocharian B·⁚·˙origin…language⁚·
·⁚font⁚·Noto Sans Brahmi·⁚·˙font⁚·
·⁚pronunciation⁚·AHVYAHKRT·⁚·˙pronunciation⁚·
·⁚cadence⁚·Relic·⁚·˙cadence⁚·
·⁚non…ostensive⁚·Primordial substance·⁚·˙non…ostensive⁚·
·⁚ostensive⁚·Clay yields beneath the palm·⁚stone bears its weight⁚water gathers in hollows⁚roots divide
↪ the earth⁚mineral veins run through darkness⁚dust settles upon the ground꙰·⁚·˙ostensive⁚·
·⁚grimchain⁚·ϴᓬ⧗ ⊛ ᕕᚸ☆ᚷᛊᛋᚸ·⁚·˙grimchain⁚·
·⁚·˙lookup·˙
·⁚·˙glyphs·˙
·⁚·˙glyphs·˙
·⁚·˙sydonic…magicae…lexicon·⁚·

engine.py

```
"""Explicit end-to-end Sydonic Magicae translation engine."""
from __future__ import annotations

from dataclasses import asdict, dataclass
import hashlib
import json
from typing import Callable, Literal

from .zero_and_one import ZeroAndOneMachine
from .cadence import grimchain_azulation
from .lexer import ExactGlyphLexer
from .models import DomusFrame
from tardisha_grimchain.domus import parse_public_living_domus
from .english_renderer import EnglishRealizer, EnglishRendering, HouseVoiceRealizer
from .lexical_resolver import ApprovedLexicalEntry, LexicalResolver
from .implementation_reconciliation import PACKAGE_AKSH_ROOT, require_package_aksh_authority
from .sequential_executor import SequentialDomusExecutor
from .trace import DomusTrace, DomusTraceBuilder
from .temporal_operator import validate_tail_resolution, temporal_operator_positions

ENGINE_VERSION = "absolute-fidelity-phase2-v1"
TargetLanguage = Literal["en"]
OutputStyle = Literal["telegraphic", "plain"]
AKSH_ROOT = PACKAGE_AKSH_ROOT
DEFAULT_UNDERSTANDINGS = AKSH_ROOT / "Enochian…Understandings.aksh"


class EngineError(RuntimeError):
    """Raised when the explicit pipeline cannot lawfully complete."""


@dataclass(frozen=True, slots=True)
class VersionWitness:
    engine: str
    renderer: str
    trace: str
    semantic_lexicon: str
    lexical_authorities_aksh: tuple[tuple[str, str], ...]
    understandings_aksh: str
    source_sha256: str
    trace_sha256: str


@dataclass(frozen=True, slots=True)
class TranslationResult:
    translation: str
    semantic_trace: DomusTrace | None
    version_witness: VersionWitness
```

```
    rendering: EnglishRendering
    resolved_entries: tuple[ApprovedLexicalEntry | None, ...]

    def to_json(self) -> str:
        payload = {
            "translation": self.translation,
            "semantic_trace": None if self.semantic_trace is None else
            ↪  self.semantic_trace.to_dict(),
            "version_witness": asdict(self.version_witness),
            "rendering": asdict(self.rendering),
        }
        return json.dumps(payload, ensure_ascii=False, sort_keys=True, separators=(",", ":"))


@dataclass(frozen=True, slots=True)
class GrimChainUnfoldWitness:
    requested_depth: int
    final_depth: int
    additional_depth: int
    final_chain: str
    first_speaking_body_position: int


class SydonicMagicaeEngine:
    """Join TardiSHA validation, exact glyph lexing, lexical projection, grammar, trace, and
    ↪  renderer."""

    version = ENGINE_VERSION

    def __init__(
        self,
        *,
        house_voice: HouseVoiceRealizer | None = None,
    ) -> None:
        require_package_aksh_authority()
        self.understandings = DEFAULT_UNDERSTANDINGS
        self.lexical_resolver = LexicalResolver(AKSH_ROOT)
        self.lexer = ExactGlyphLexer(self.lexical_resolver.glyph_order, self.understandings)
        self.zero_and_one = ZeroAndOneMachine(
            self.lexical_resolver.lexical_glyphs,
            self.lexical_resolver.resolve,
        )
        self.sequential = SequentialDomusExecutor(self.zero_and_one)
        self.trace_builder = DomusTraceBuilder(self.lexical_resolver, self.understandings)
        self.english = EnglishRealizer(house_voice=house_voice)

    def parse_domus(self, domus_string: str) -> DomusFrame:
        parse_public_living_domus(domus_string)
        domus_string = grimchain_azulation(domus_string)
        return DomusFrame(source=domus_string, tokens=self.lexer.lex(domus_string))

    def resolve_grimchain_unfold(
        self,
        requested_depth: int,
        chain_at_depth: Callable[[int], str],
    ) -> GrimChainUnfoldWitness:
```

```
    """Resolve the first same-identity depth containing a Real speaking body."""
    depth = requested_depth
    previous_chain: str | None = None
    while True:
        chain = chain_at_depth(depth)
        if depth > 1 and len(chain) != depth:
            raise EngineError("GrimChain continuation changed the requested manifest depth")
        if previous_chain is not None and not chain.startswith(previous_chain):
            raise EngineError("GrimChain continuation changed the witnessed prefix")
        frame = self.parse_domus(chain)
        if depth <= 1:
            speaking_position = next((token.position for token in frame.tokens if token.kind !=
            ↪  "grammar"), 0)
            return GrimChainUnfoldWitness(requested_depth, depth, depth - requested_depth,
            ↪  chain, speaking_position)
        speaking_position = next((token.position for token in frame.tokens if token.kind !=
        ↪  "grammar"), None)
        if speaking_position is not None:
            return GrimChainUnfoldWitness(requested_depth, depth, depth - requested_depth,
            ↪  chain, speaking_position)
        previous_chain = chain
        depth += 1

def _complete_translation(
    self,
    *,
    domus_string: str,
    frame: DomusFrame,
    zero_and_one,
    local,
    output_style: str,
    include_trace: bool,
) -> TranslationResult:
    trace = self.trace_builder.build(frame, zero_and_one, local)
    rendering = self.english.render(trace, style=output_style)

    entries_by_body_id: dict[str, ApprovedLexicalEntry] = {}
    for body in local.bodies:
        resolutions = (
            *((body.lexical_resolution,) if body.lexical_resolution is not None else ()),
            *body.office_resolutions,
        )
        for resolution in resolutions:
            entry = resolution.entry
            body_id = f"{resolution.source_authority}:{body.glyph}"
            if resolution.ostensive_variant is not None:
                body_id += f":ostensive{resolution.ostensive_variant}"
            entries_by_body_id[body_id] = entry

    for state in local.transmutations:
        for body, resolution in zip(state.vessels, state.office_resolutions, strict=True):
            body_id = (
                f"{resolution.source_authority}:{body.glyph}:"
                f"ostensive{resolution.ostensive_variant}"
            )
```

```
            entries_by_body_id[body_id] = resolution.entry

    resolved_entries: list[ApprovedLexicalEntry | None] = []
    for word in rendering.final_words:
        if word.kind != "content":
            resolved_entries.append(None)
            continue
        if word.source_body_id is None:
            raise EngineError("Sydonic content word has no source-body identity")
        try:
            resolved_entries.append(entries_by_body_id[word.source_body_id])
        except KeyError as exc:
            raise EngineError(
                f"Sydonic rendered word lost its resolved lexical entry:
                ↪  {word.source_body_id!r}"
            ) from exc

    witness = VersionWitness(
        engine=self.version,
        renderer=rendering.renderer_version,
        trace=trace.trace_version,
        semantic_lexicon=trace.semantic_lexicon_version,
        lexical_authorities_aksh=trace.authority_versions.lexical_authorities,
        understandings_aksh=trace.authority_versions.understandings,
        source_sha256=hashlib.sha256(domus_string.encode("utf-8")).hexdigest(),
        trace_sha256=trace.sha256,
    )
    return TranslationResult(
        translation=rendering.final_text,
        semantic_trace=trace if include_trace else None,
        version_witness=witness,
        rendering=rendering,
        resolved_entries=tuple(resolved_entries),
    )

def translate_domus_with_execution(
    self,
    *,
    domus_string: str,
    target_language: str = "en",
    output_style: str = "telegraphic",
    include_trace: bool = True,
    aeternum_token=None,
    aeternum_pair_reorder=None,
    aeternum_corridor_reorder=None,
    parsed_frame: DomusFrame | None = None,
) -> TranslationResult:
    """Public translation entry point with explicit Aeternum traversal adapters."""
    if target_language != "en":
        raise EngineError(f"unsupported target language: {target_language!r}")
    if output_style not in {"telegraphic", "plain"}:
        raise EngineError(f"unsupported output style: {output_style!r}")
    if not isinstance(domus_string, str):
        raise TypeError("domus_string must be str")
    try:
        if parsed_frame is None:
```

```
            frame = self.parse_domus(domus_string)
        else:
            if parsed_frame.source != domus_string:
                raise EngineError("provided Domus frame does not match translation string")
            frame = parsed_frame
        zero_and_one, local = self.sequential.execute_frame(
            frame,
            aeternum_token=aeternum_token,
            aeternum_pair_reorder=aeternum_pair_reorder,
            aeternum_corridor_reorder=aeternum_corridor_reorder,
        )
        if temporal_operator_positions(frame):
            if local.tail_resolution is None:
                raise EngineError("Tail-dependent execution did not return a resolved
                ↪  traversal")
            validate_tail_resolution(frame, local.tail_resolution)
            if local.execution_positions != local.tail_resolution.execution_positions:
                raise EngineError("grammar did not execute the completed Tail traversal")
            expected_scan = tuple(pos for pos in local.execution_positions if
            ↪  frame.tokens[pos].value != "⟲")
            actual_scan = tuple(event.position for event in zero_and_one.events if event.glyph
            ↪  != "⟲")
            if actual_scan != expected_scan:
                raise EngineError("recorded grammar encounters differ from the completed Tail
                ↪  order")
        return self._complete_translation(
            domus_string=domus_string,
            frame=frame,
            zero_and_one=zero_and_one,
            local=local,
            output_style=output_style,
            include_trace=include_trace,
        )
    except Exception as exc:
        if isinstance(exc, (TypeError, EngineError)):
            raise
        raise EngineError(
            f"Sydonic Magicae pipeline stopped at explicit failure: {exc.__class__.__name__}:
            ↪  {exc}"
        ) from exc

def translate_domus(
    self,
    *,
    domus_string: str,
    target_language: str = "en",
    output_style: str = "telegraphic",
    include_trace: bool = True,
) -> TranslationResult:
    return self.translate_domus_with_execution(
        domus_string=domus_string,
        target_language=target_language,
        output_style=output_style,
        include_trace=include_trace,
    )
```

```
def translate_domus(
    *,
    domus_string: str,
    target_language: str = "en",
    output_style: str = "telegraphic",
    include_trace: bool = True,
    house_voice: HouseVoiceRealizer | None = None,
) -> TranslationResult:
    """Public one-call engine entry point using the current package .aksh authorities."""
    return SydonicMagicaeEngine(
        house_voice=house_voice,
    ).translate_domus(
        domus_string=domus_string,
        target_language=target_language,
        output_style=output_style,
        include_trace=include_trace,
    )
```

english_renderer.py

```
"""Phase 11 receiving-language realization from an invariant DomusTrace.

English is downstream clothing. It may realize traced semantic bodies and add
marked function words, but it may not rewrite source lineage or turn structural
presence into vocabulary.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
import json
from typing import Literal, Protocol

from .trace import DomusTrace, LexicalUtteranceTrace, RenderingToken, validate_tail_trace

RENDERER_VERSION = "sydonic-office-resolution-v2"

Bearing = Literal["relic", "catalyst", "substrate", "spatial_absolute", "temporal_absolute",
↪  "terminal_state", "adversarial", "recursive_identity"]
WordKind = Literal["content", "clothing"]


class RealizationError(RuntimeError):
    pass


@dataclass(frozen=True, slots=True)
class EnglishWord:
    text: str
    kind: WordKind
    source_position: int | None
    source_glyph: str | None
    source_body_id: str | None
    grammatical_bearing: str | None
    clothing_role: str | None = None


@dataclass(frozen=True, slots=True)
class EnglishRendering:
    renderer_version: str
    trace_sha256: str
    style: str
    telegraphic_words: tuple[EnglishWord, ...]
    telegraphic_text: str
    clothing_words: tuple[EnglishWord, ...]
    final_words: tuple[EnglishWord, ...]
    final_text: str
    house_voice_applied: bool
    house_voice_notes: tuple[str, ...]
```

```
    def to_json(self) -> str:
        return json.dumps(asdict(self), ensure_ascii=False, sort_keys=True, separators=(",", ":"))


class HouseVoiceRealizer(Protocol):
    version: str
    def apply(self, trace: DomusTrace, words: tuple[EnglishWord, ...]) -> tuple[tuple[EnglishWord,
    ↪  ...], tuple[str, ...]]: ...


class SydonicSpokenVoice:
    """Emit the renderer's uttered Aeonic order as the spoken Sydonic form."""

    version = "sydonic-spoken-house-v1"

    def apply(self, trace: DomusTrace, words: tuple[EnglishWord, ...]) -> tuple[tuple[EnglishWord,
    ↪  ...], tuple[str, ...]]:
        return words, ("sydonic translate emits the uttered Aeonic order",)


class EnglishRealizer:
    """Deterministic two-stage English realization with reverse audit."""

    version = RENDERER_VERSION

    def __init__(self, store: object | None = None, house_voice: HouseVoiceRealizer | None = None)
    ↪  -> None:
        if store is not None:
            raise RealizationError("external lexical stores are not accepted by the authored
            ↪  renderer")
        self.store = None
        self.house_voice = house_voice

    def render(self, trace: DomusTrace, *, style: str = "plain") -> EnglishRendering:
        if style not in {"telegraphic", "plain"}:
            raise RealizationError(f"unsupported named style: {style!r}")
        telegraphic = self._telegraphic(trace)
        if self.house_voice is None:
            raise RealizationError("authored HouseVoiceRealizer is required at the House Voice
            ↪  boundary")
        voiced, voice_notes = self.house_voice.apply(trace, telegraphic)
        voice_applied = True
        self._audit_words(voiced, trace)
        clothing: tuple[EnglishWord, ...] = ()
        final = voiced
        if style == "plain":
            clothing, final = self._clothe(voiced)
        result = EnglishRendering(
            renderer_version=self.version,
            trace_sha256=trace.sha256,
            style=style,
            telegraphic_words=voiced,
            telegraphic_text=self._join(voiced),
            clothing_words=clothing,
```

```
        final_words=final,
        final_text=self._join(final),
        house_voice_applied=voice_applied,
        house_voice_notes=voice_notes,
    )
    self.reverse_audit(trace, result)
    return result

def _telegraphic(self, trace: DomusTrace) -> tuple[EnglishWord, ...]:
    validate_tail_trace(trace)
    out: list[EnglishWord] = []
    tokens = tuple(trace.rendering_input.tokens)

    ostensive_by_position: dict[int, tuple[str, str, str, str]] = {}
    for state in trace.transmutation_traces:
        for position, glyph, body, variant, authority in zip(
            state.vessel_positions,
            state.vessel_glyphs,
            state.ostensive_bodies,
            state.ostensive_variants,
            state.ostensive_authorities,
        ):
            provenance = (glyph, body, variant, authority)
            prior = ostensive_by_position.get(position)
            if prior is not None and prior != provenance:
                raise RealizationError(
                    "one real lexical vessel received two distinct Phantasmagoria Ostensive
                    ↪  selections"
                )
            # Aeternum may encounter the same finite real vessel more than once.
            # The mirror carries the already-selected Ostensive exactly and does not
            # select a second variant for the same finite real source body.
            ostensive_by_position[position] = provenance

    def emit(token: RenderingToken) -> None:
        if token.preserved_without_lookup:
            return
        ostensive = ostensive_by_position.get(token.source_position)
        if ostensive is not None:
            glyph, body, variant, authority = ostensive
            if glyph != token.glyph:
                raise RealizationError("ऋ Phantasmagoria provenance changed vessel glyph
                ↪  identity")
            out.append(EnglishWord(
                text=body,
                kind="content",
                source_position=token.source_position,
                source_glyph=token.glyph,
                source_body_id=f"{authority}:{token.glyph}:ostensive{variant}",
                grammatical_bearing="phantasmagoria",
            ))
            return
        if not token.utterances:
            raise RealizationError(
```

```
                f"spoken lexical token {token.glyph!r} at position {token.source_position} lacks
                ↪  authored utterances"
            )
        for utterance in token.utterances:
            out.append(self._realize_utterance(token, utterance))

    for token in tokens:
        emit(token)
    return tuple(out)

def _realize_utterance(self, token: RenderingToken, utterance: LexicalUtteranceTrace) ->
↪  EnglishWord:
    bearing = utterance.grammatical_bearing
    if bearing not in {None, "relic", "catalyst", "substrate", "spatial_absolute",
    ↪  "temporal_absolute", "terminal_state", "adversarial", "recursive_identity"}:
        raise RealizationError(f"unsupported grammatical bearing {bearing!r} for
        ↪  {token.glyph!r}")
    return EnglishWord(
        text=utterance.resolved_state_body,
        kind="content",
        source_position=token.source_position,
        source_glyph=token.glyph,
        source_body_id=f"{utterance.source_authority}:{token.glyph}",
        grammatical_bearing=bearing,
    )

def _clothe(self, words: tuple[EnglishWord, ...]) -> tuple[tuple[EnglishWord, ...],
↪  tuple[EnglishWord, ...]]:
    return (), words

@staticmethod
def _join(words: tuple[EnglishWord, ...]) -> str:
    return " ".join(word.text for word in words)

def reverse_audit(self, trace: DomusTrace, rendering: EnglishRendering) -> None:
    self._audit_words(rendering.final_words, trace)
    content_positions = {
        token.source_position for token in trace.rendering_input.tokens
        if token.utterances
    }
    for word in rendering.final_words:
        if word.kind == "content":
            if word.source_position not in content_positions:
                raise RealizationError(f"content word lacks traced source body: {word.text!r}")
            if word.source_glyph is None or word.source_body_id is None:
                raise RealizationError(f"content word has incomplete reverse audit:
                ↪  {word.text!r}")
        elif word.kind == "clothing":
            if word.clothing_role is None:
                raise RealizationError(f"English support word is not marked as clothing:
                ↪  {word.text!r}")
            if word.source_position is not None or word.source_glyph is not None:
                raise RealizationError("receiving-language clothing falsely claims a source
                ↪  lexeme")
```

```
@staticmethod
def _audit_words(words: tuple[EnglishWord, ...], trace: DomusTrace) -> None:
    source_positions = {token.source_position for token in trace.token_traces}
    for word in words:
        if word.kind == "content" and (word.source_position is None or word.source_position not
        ↪  in source_positions):
            raise RealizationError("content word provenance is outside the Aeonic line")
```

exhaustive_chooser.py

```
"""Phase 14 exhaustive, order-independent Court chooser.

The chooser evaluates complete bearing sub-fields only after freezing the Court
Aeon's governing Bias, governing Q-vector, and alternating frequency carrier.
Candidate truth is a local immutable witness.  No candidate can emit while the
sub-field is still being traversed.

Selection is replayed under manifest order, reversed order, several fixed
permutations, a file-source shuffle, and a parallel-chunk completion order.  A
bearing may emit only when every replay returns the same survivor set and that
set contains exactly one structural candidate.
"""
from __future__ import annotations

from dataclasses import dataclass
from enum import Enum
import hashlib
from typing import Final, Mapping

from .court_outward_return import (
    COURTS,
    CourtCandidate,
    CourtClosure,
    CourtIdentity,
    CourtPathError,
    build_court_candidate,
    validate_prepared_court_bearing,
)
from .goetic_outward_return import BEARINGS, GoeticBearing
from .mas_shadow_conversion import MASCandidatePath, MASGlobalField


class ExhaustiveChooserError(ValueError):
    """Raised when Phase 14 cannot preserve one frozen mathematical field."""


class EnumerationOrder(str, Enum):
    MANIFEST = "complete-manifest-order"
    REVERSED = "reversed-manifest-order"
    ROTATE_ONE = "deterministic-rotate-one"
    EVEN_ODD = "deterministic-even-odd"
    STRIDE_THREE = "deterministic-stride-three"
    FILE_SHUFFLED = "file-shuffled-source-order"
    PARALLEL_CHUNK = "parallel-chunk-order"


ENUMERATION_ORDERS: Final[tuple[EnumerationOrder, ...]] = tuple(EnumerationOrder)
if len(ENUMERATION_ORDERS) != 7:
```

```
    raise RuntimeError("Phase 14 requires all declared enumeration-order proofs")


@dataclass(frozen=True, slots=True, order=True)
class CandidateKey:
    candidate: str
    structural_fingerprint: str


@dataclass(frozen=True, slots=True)
class FrozenCourtAeon:
    """One Court held fixed while only L_b changes across bearings."""

    court: CourtIdentity
    native_q_bias: str
    native_q_vector: tuple[str, str, str, str]
    frequency_carrier_glyph: str

    def __post_init__(self) -> None:
        if not self.native_q_bias:
            raise ExhaustiveChooserError("frozen Court lacks its governing Q Bias")
        if len(self.native_q_vector) != 4:
            raise ExhaustiveChooserError("frozen Court requires one four-coordinate parent vector")
        if self.frequency_carrier_glyph != self.court.alternating.glyph:
            raise ExhaustiveChooserError("frozen Court changed the alternating carrier")


@dataclass(frozen=True, slots=True)
class CandidateWitness:
    """Candidate-local structure used by Truth_A(w), never an enumeration index."""

    key: CandidateKey
    source_parent: MASCandidatePath
    frozen_aeon: FrozenCourtAeon
    source_locator: str

    def __post_init__(self) -> None:
        seed = self.source_parent.liquid_path.seed
        if seed.domain.carrier != self.frozen_aeon.court.governing:
            raise ExhaustiveChooserError("candidate witness entered through a foreign governing
            ↪  parent")
        if seed.domain.inherited_q_bias != self.frozen_aeon.native_q_bias:
            raise ExhaustiveChooserError("candidate witness changed frozen Q Bias")
        if seed.body.q_vector != self.frozen_aeon.native_q_vector:
            raise ExhaustiveChooserError("candidate witness changed frozen Q-vector")
        if not self.source_locator:
            raise ExhaustiveChooserError("candidate witness lacks a stable source locator")


@dataclass(frozen=True, slots=True)
class BearingSubfield:
    frozen_aeon: FrozenCourtAeon
    bearing: GoeticBearing
    candidates: tuple[MASCandidatePath, ...]

    def __post_init__(self) -> None:
```

```
        keys = tuple(_candidate_key(candidate) for candidate in self.candidates)
        if len(set(keys)) != len(keys):
            raise ExhaustiveChooserError("bearing sub-field duplicated one structural candidate")
        for candidate in self.candidates:
            seed = candidate.liquid_path.seed
            if seed.domain.carrier != self.frozen_aeon.court.governing:
                raise ExhaustiveChooserError("L_b contains a candidate from another governing
                ↪  parent")
            if seed.domain.inherited_q_bias != self.frozen_aeon.native_q_bias:
                raise ExhaustiveChooserError("L_b changed the frozen Aeon Q Bias")
            if seed.body.q_vector != self.frozen_aeon.native_q_vector:
                raise ExhaustiveChooserError("L_b changed the frozen Aeon Q-vector")


@dataclass(frozen=True, slots=True)
class CompleteSubfieldManifest:
    source: MASGlobalField
    subfields: tuple[BearingSubfield, ...]

    def __post_init__(self) -> None:
        expected = tuple((court, bearing) for court in COURTS for bearing in BEARINGS)
        actual = tuple((entry.frozen_aeon.court, entry.bearing) for entry in self.subfields)
        if actual != expected:
            raise ExhaustiveChooserError("complete manifest must contain all 144 x 8 Court bearings
            ↪  in order")
        source_ids = {id(candidate) for candidate in self.source.survivors}
        if any(id(candidate) not in source_ids for entry in self.subfields for candidate in
        ↪  entry.candidates):
            raise ExhaustiveChooserError("complete manifest contains a candidate outside its Phase
            ↪  11 source")
        for offset in range(0, len(self.subfields), 8):
            group = self.subfields[offset:offset + 8]
            if any(entry.frozen_aeon != group[0].frozen_aeon for entry in group):
                raise ExhaustiveChooserError("Aeon was not frozen across all eight bearing runs")


@dataclass(frozen=True, slots=True)
class CandidateTruthRecord:
    witness: CandidateWitness
    court_candidate: CourtCandidate | None
    closure: CourtClosure | None
    failure_reason: str | None

    def __post_init__(self) -> None:
        if (self.closure is None) == (self.failure_reason is None):
            raise ExhaustiveChooserError("candidate truth record must contain closure xor failure")
        if self.closure is not None:
            if self.court_candidate is None or self.closure.court_candidate != self.court_candidate:
                raise ExhaustiveChooserError("truth closure is detached from its Court candidate")
            if not self.closure.dcomp.exact_zero:
                raise ExhaustiveChooserError("Truth_A(w)=1 requires exact D_T zero")

    @property
    def truth(self) -> bool:
        return self.closure is not None
```

```
@dataclass(frozen=True, slots=True)
class EnumerationWitness:
    order: EnumerationOrder
    candidate_order: tuple[CandidateKey, ...]
    survivor_set: frozenset[CandidateKey]


@dataclass(frozen=True, slots=True)
class ChooserEmission:
    court: CourtIdentity
    bearing: GoeticBearing
    candidate_key: CandidateKey
    candidate: str
    native_q_bias: str


@dataclass(frozen=True, slots=True)
class ChooserFailureEvidence:
    court: CourtIdentity
    bearing: GoeticBearing
    reason: str
    survivor_set: frozenset[CandidateKey]
    candidate_failures: tuple[tuple[CandidateKey, str], ...]


@dataclass(frozen=True, slots=True)
class ExhaustiveBearingResult:
    frozen_aeon: FrozenCourtAeon
    bearing: GoeticBearing
    truth_records: tuple[CandidateTruthRecord, ...]
    enumeration_witnesses: tuple[EnumerationWitness, ...]
    survivor_set: frozenset[CandidateKey]
    emission: ChooserEmission | None
    failure_evidence: ChooserFailureEvidence | None

    def __post_init__(self) -> None:
        if tuple(w.order for w in self.enumeration_witnesses) != ENUMERATION_ORDERS:
            raise ExhaustiveChooserError("bearing lacks the complete order-independence proof")
        if any(w.survivor_set != self.survivor_set for w in self.enumeration_witnesses):
            raise ExhaustiveChooserError("enumeration order changed the survivor set")
        if len(self.survivor_set) == 1:
            if self.emission is None or self.failure_evidence is not None:
                raise ExhaustiveChooserError("unique survivor did not emit exactly once")
            if self.emission.native_q_bias != self.frozen_aeon.native_q_bias:
                raise ExhaustiveChooserError("emission changed native Q Bias")
        else:
            if self.emission is not None or self.failure_evidence is None:
                raise ExhaustiveChooserError("failed bearing emitted or lost failure evidence")

    @property
    def order_independent(self) -> bool:
        return all(w.survivor_set == self.survivor_set for w in self.enumeration_witnesses)

    @property
```

```
    def emitted_word(self) -> str | None:
        return self.emission.candidate if self.emission is not None else None


@dataclass(frozen=True, slots=True)
class ExhaustiveCourtBody:
    frozen_aeon: FrozenCourtAeon
    bearing_results: tuple[ExhaustiveBearingResult, ...]
    partial_body_law_deferred: bool = True

    def __post_init__(self) -> None:
        if tuple(result.bearing for result in self.bearing_results) != BEARINGS:
            raise ExhaustiveChooserError("exhaustive Court body must contain all eight bearings")
        if any(result.frozen_aeon != self.frozen_aeon for result in self.bearing_results):
            raise ExhaustiveChooserError("Court was not frozen across its bearing body")
        if not self.partial_body_law_deferred:
            raise ExhaustiveChooserError("Phase 17 partial-body law must remain deferred")

    @property
    def emitted_bearing_count(self) -> int:
        return sum(result.emission is not None for result in self.bearing_results)


@dataclass(frozen=True, slots=True)
class ExhaustiveChooserField:
    manifest: CompleteSubfieldManifest
    court_bodies: tuple[ExhaustiveCourtBody, ...]
    partial_body_law_deferred: bool = True

    def __post_init__(self) -> None:
        if tuple(body.frozen_aeon.court for body in self.court_bodies) != COURTS:
            raise ExhaustiveChooserError("Phase 14 must resolve all 144 Courts in canonical order")
        if any(len(body.bearing_results) != 8 for body in self.court_bodies):
            raise ExhaustiveChooserError("Phase 14 lost one of the eight bearings")
        if not self.partial_body_law_deferred:
            raise ExhaustiveChooserError("Phase 14 cannot decide Phase 17 standing")
        if not all(result.order_independent for body in self.court_bodies for result in
        ↪  body.bearing_results):
            raise ExhaustiveChooserError("Phase 14 result depends upon enumeration order")

    @property
    def emitted_bearing_count(self) -> int:
        return sum(body.emitted_bearing_count for body in self.court_bodies)

    @property
    def mathematical_signature(self) -> tuple[tuple[int, str, frozenset[CandidateKey]], ...]:
        return tuple(
            (body.frozen_aeon.court.index, result.bearing.value, result.survivor_set)
            for body in self.court_bodies
            for result in body.bearing_results
        )


def _candidate_structure(candidate: MASCandidatePath) -> tuple[object, ...]:
    seed = candidate.liquid_path.seed
    return (
```

```
        candidate.candidate,
        seed.domain.court_root,
        seed.domain.inherited_q_bias,
        seed.domain.carrier.glyph,
        seed.body.q_vector,
        tuple((unit.phoneme, int(unit.stress), int(unit.breath), unit.bearing.value,
        ↪  unit.operator_office) for unit in seed.witness.units),
        candidate.cycles[-1].final_state.archive_signature,
        tuple((entry.index, entry.horizon, entry.gross_added, entry.resolved_added,
        ↪  entry.unresolved) for entry in candidate.debt_ledger.entries),
    )


def _fingerprint(candidate: MASCandidatePath) -> str:
    return hashlib.sha256(repr(_candidate_structure(candidate)).encode("utf-8")).hexdigest()


def _candidate_key(candidate: MASCandidatePath) -> CandidateKey:
    return CandidateKey(candidate.candidate, _fingerprint(candidate))


def _candidate_witness(candidate: MASCandidatePath, frozen: FrozenCourtAeon) -> CandidateWitness:
    key = _candidate_key(candidate)
    locator = hashlib.sha256(("source-file|" +
    ↪  repr(candidate.cycles[-1].final_state.archive_signature)).encode("utf-8")).hexdigest()
    return CandidateWitness(key, candidate, frozen, locator)


def _freeze_court(court: CourtIdentity, candidates: tuple[MASCandidatePath, ...]) ->
↪  FrozenCourtAeon:
    if not candidates:
        raise ExhaustiveChooserError(f"cannot freeze {court.key}: no governing-parent mathematical
        ↪  witness")
    biases = {candidate.liquid_path.seed.domain.inherited_q_bias for candidate in candidates}
    vectors = {candidate.liquid_path.seed.body.q_vector for candidate in candidates}
    if len(biases) != 1 or len(vectors) != 1:
        raise ExhaustiveChooserError(f"cannot freeze {court.key}: governing candidates disagree on
        ↪  native beta or vector")
    return FrozenCourtAeon(court, next(iter(biases)), next(iter(vectors)), court.alternating.glyph)


def build_complete_subfield_manifest(
    field: MASGlobalField,
    overrides: Mapping[tuple[int, GoeticBearing], tuple[MASCandidatePath, ...]] | None = None,
) -> CompleteSubfieldManifest:
    """Build every L_b while preserving one frozen Court across its eight runs."""
    override_map = {} if overrides is None else dict(overrides)
    legal_keys = {(court.index, bearing) for court in COURTS for bearing in BEARINGS}
    if any(key not in legal_keys for key in override_map):
        raise ExhaustiveChooserError("sub-field override names an unknown Court bearing")
    entries: list[BearingSubfield] = []
    for court in COURTS:
        parent_candidates = tuple(
            candidate for candidate in field.survivors
            if candidate.liquid_path.seed.domain.carrier == court.governing
```

```
        )
        frozen = _freeze_court(court, parent_candidates)
        parent_ids = {id(candidate) for candidate in parent_candidates}
        for bearing in BEARINGS:
            candidates = override_map.get((court.index, bearing), parent_candidates)
            if any(id(candidate) not in parent_ids for candidate in candidates):
                raise ExhaustiveChooserError("L_b override inserted a candidate outside the frozen
                ↪  Aeon")
            entries.append(BearingSubfield(frozen, bearing, tuple(candidates)))
    return CompleteSubfieldManifest(field, tuple(entries))


def _rotate_one(records: tuple[CandidateTruthRecord, ...]) -> tuple[CandidateTruthRecord, ...]:
    return records[1:] + records[:1] if records else records


def _even_odd(records: tuple[CandidateTruthRecord, ...]) -> tuple[CandidateTruthRecord, ...]:
    return records[::2] + records[1::2]


def _stride_three(records: tuple[CandidateTruthRecord, ...]) -> tuple[CandidateTruthRecord, ...]:
    return records[::3] + records[1::3] + records[2::3]


def _file_shuffled(records: tuple[CandidateTruthRecord, ...]) -> tuple[CandidateTruthRecord, ...]:
    return tuple(sorted(records, key=lambda record: record.witness.source_locator))


def _parallel_chunk_order(records: tuple[CandidateTruthRecord, ...]) -> tuple[CandidateTruthRecord,
↪  ...]:
    if len(records) < 2:
        return records
    chunk_size = max(1, (len(records) + 2) // 3)
    chunks = tuple(records[index:index + chunk_size] for index in range(0, len(records),
    ↪  chunk_size))
    return tuple(record for chunk in reversed(chunks) for record in chunk)


def _enumeration_manifests(
    records: tuple[CandidateTruthRecord, ...],
) -> tuple[tuple[EnumerationOrder, tuple[CandidateTruthRecord, ...]], ...]:
    variants = (
        (EnumerationOrder.MANIFEST, records),
        (EnumerationOrder.REVERSED, tuple(reversed(records))),
        (EnumerationOrder.ROTATE_ONE, _rotate_one(records)),
        (EnumerationOrder.EVEN_ODD, _even_odd(records)),
        (EnumerationOrder.STRIDE_THREE, _stride_three(records)),
        (EnumerationOrder.FILE_SHUFFLED, _file_shuffled(records)),
        (EnumerationOrder.PARALLEL_CHUNK, _parallel_chunk_order(records)),
    )
    expected = frozenset(record.witness.key for record in records)
    if any(frozenset(record.witness.key for record in variant) != expected for _, variant in
    ↪  variants):
        raise ExhaustiveChooserError("enumeration proof lost or inserted a candidate")
    return variants
```

```
def _truth_record(
    witness: CandidateWitness,
    prepared: CourtCandidate | None,
    preparation_failure: str | None,
    bearing: GoeticBearing,
) -> CandidateTruthRecord:
    if prepared is None:
        return CandidateTruthRecord(witness, None, None, preparation_failure or "Court candidate
        ↪  preparation failed")
    try:
        closure = validate_prepared_court_bearing(prepared, bearing)
        return CandidateTruthRecord(witness, prepared, closure, None)
    except CourtPathError as exc:
        return CandidateTruthRecord(witness, prepared, None, str(exc))


def _evaluate_bearing(
    subfield: BearingSubfield,
    prepared_cache: Mapping[CandidateKey, CourtCandidate | None],
    failure_cache: Mapping[CandidateKey, str | None],
    witness_cache: Mapping[CandidateKey, CandidateWitness],
) -> ExhaustiveBearingResult:
    truth_records: list[CandidateTruthRecord] = []
    for candidate in subfield.candidates:
        key = _candidate_key(candidate)
        truth_records.append(_truth_record(
            witness_cache[key],
            prepared_cache[key],
            failure_cache[key],
            subfield.bearing,
        ))

    complete_records = tuple(truth_records)
    enumeration_witnesses: list[EnumerationWitness] = []
    for order, manifest in _enumeration_manifests(complete_records):
        survivors = frozenset(record.witness.key for record in manifest if record.truth)
        enumeration_witnesses.append(EnumerationWitness(
            order,
            tuple(record.witness.key for record in manifest),
            survivors,
        ))

    survivor_set = enumeration_witnesses[0].survivor_set
    if any(witness.survivor_set != survivor_set for witness in enumeration_witnesses):
        raise ExhaustiveChooserError("survivor set changed under enumeration replay")

    emission: ChooserEmission | None = None
    failure_evidence: ChooserFailureEvidence | None = None
    if len(survivor_set) == 1:
        key = next(iter(survivor_set))
        emission = ChooserEmission(
            subfield.frozen_aeon.court,
            subfield.bearing,
            key,
```

```
            key.candidate,
            subfield.frozen_aeon.native_q_bias,
        )
    else:
        reason = "no exact survivor" if not survivor_set else "multiple exact survivors"
        failure_evidence = ChooserFailureEvidence(
            subfield.frozen_aeon.court,
            subfield.bearing,
            reason,
            survivor_set,
            tuple(
                (record.witness.key, record.failure_reason)
                for record in complete_records
                if record.failure_reason is not None
            ),
        )

    return ExhaustiveBearingResult(
        subfield.frozen_aeon,
        subfield.bearing,
        complete_records,
        tuple(enumeration_witnesses),
        survivor_set,
        emission,
        failure_evidence,
    )


def _evaluate_court_body(subfields: tuple[BearingSubfield, ...]) -> ExhaustiveCourtBody:
    if len(subfields) != 8 or tuple(entry.bearing for entry in subfields) != BEARINGS:
        raise ExhaustiveChooserError("Court chooser requires exactly eight ordered sub-fields")
    frozen = subfields[0].frozen_aeon
    if any(entry.frozen_aeon != frozen for entry in subfields):
        raise ExhaustiveChooserError("Court changed while bearing sub-fields changed")

    union: list[MASCandidatePath] = []
    seen: set[CandidateKey] = set()
    for entry in subfields:
        for candidate in entry.candidates:
            key = _candidate_key(candidate)
            if key not in seen:
                union.append(candidate)
                seen.add(key)

    prepared_cache: dict[CandidateKey, CourtCandidate | None] = {}
    failure_cache: dict[CandidateKey, str | None] = {}
    witness_cache: dict[CandidateKey, CandidateWitness] = {}
    for candidate in union:
        witness = _candidate_witness(candidate, frozen)
        witness_cache[witness.key] = witness
        try:
            prepared_cache[witness.key] = build_court_candidate(candidate, frozen.court)
            failure_cache[witness.key] = None
        except CourtPathError as exc:
            prepared_cache[witness.key] = None
            failure_cache[witness.key] = str(exc)
```

```
    results = tuple(
        _evaluate_bearing(entry, prepared_cache, failure_cache, witness_cache)
        for entry in subfields
    )
    return ExhaustiveCourtBody(frozen, results, partial_body_law_deferred=True)


def prove_phase14_exit_gate(manifest: CompleteSubfieldManifest) -> ExhaustiveChooserField:
    """Evaluate the complete field without allowing enumeration order to choose."""
    bodies = tuple(
        _evaluate_court_body(manifest.subfields[offset:offset + 8])
        for offset in range(0, len(manifest.subfields), 8)
    )
    return ExhaustiveChooserField(manifest, bodies, partial_body_law_deferred=True)
```

frequency_projection.py

```
"""Phase 8 exact frequency/prosody projection.

This module implements only ``F_ν,A``.  It does not alter the translation runtime,
Sydonic .aksh authority, local grammar, renderer, trace, or packaging
surfaces.

The projection is speaker-independent.  A caller supplies an ordered
phoneme/stress/breath witness.  No lexical meaning, acoustic measurement,
learned parameter, or candidate ranking enters the arithmetic.

The coefficient field is the exact carrier field ``Q(Φ, i)`` with
``Φ² = Φ + 1``.  The word polynomial is represented as a free prosodic module
over that field, so phoneme identity, stress, breath, bearing, operator office,
and sequence position remain recoverable in the normal form.
"""
from __future__ import annotations

from dataclasses import dataclass
from enum import Enum, IntEnum
from fractions import Fraction
from math import comb
from typing import Final, Iterable, Mapping


class FrequencyProjectionError(ValueError):
    """Raised when a Phase 8 projection witness is not exact or lawful."""


class Stress(IntEnum):
    """Declared lexical stress, independent of any speaker."""

    UNSTRESSED = 0
    PRIMARY = 1
    SECONDARY = 2


class Breath(IntEnum):
    """Declared breath bearing inside the bounded ``[-Φ, +Φ]`` body."""

    INWARD = -1
    HELD = 0
    OUTWARD = 1


class Bearing(str, Enum):
    """Prosodic bearing under Mirror Math."""

    OUTWARD = "+"
```

```
    RETURN = "-"
    SELF = "="

    def mirror(self) -> "Bearing":
        if self is Bearing.OUTWARD:
            return Bearing.RETURN
        if self is Bearing.RETURN:
            return Bearing.OUTWARD
        return Bearing.SELF


class PathOffice(str, Enum):
    OUTWARD = "outward"
    IDEAL = "ideal-parity-return"
    RETURN = "independent-return"


NumberLike = int | str | Fraction


def _fraction(value: NumberLike) -> Fraction:
    if isinstance(value, Fraction):
        return value
    if isinstance(value, int):
        return Fraction(value)
    if isinstance(value, str):
        return Fraction(value)
    raise TypeError(f"exact rational required, received {type(value).__name__}")


@dataclass(frozen=True, slots=True)
class Golden:
    """One exact element ``a + bϕ`` of ``Q(ϕ)``."""

    rational: Fraction = Fraction(0)
    phi: Fraction = Fraction(0)

    def __init__(self, rational: NumberLike = 0, phi: NumberLike = 0) -> None:
        object.__setattr__(self, "rational", _fraction(rational))
        object.__setattr__(self, "phi", _fraction(phi))

    def __add__(self, other: object) -> "Golden":
        rhs = Golden.coerce(other)
        return Golden(self.rational + rhs.rational, self.phi + rhs.phi)

    def __radd__(self, other: object) -> "Golden":
        return self + other

    def __sub__(self, other: object) -> "Golden":
        rhs = Golden.coerce(other)
        return Golden(self.rational - rhs.rational, self.phi - rhs.phi)

    def __rsub__(self, other: object) -> "Golden":
        return Golden.coerce(other) - self

    def __neg__(self) -> "Golden":
```

```
        return Golden(-self.rational, -self.phi)

    def __mul__(self, other: object) -> "Golden":
        rhs = Golden.coerce(other)
        # (a+bΦ)(c+dΦ), reduced exactly by Φ² = Φ + 1.
        a, b = self.rational, self.phi
        c, d = rhs.rational, rhs.phi
        return Golden(a * c + b * d, a * d + b * c + b * d)

    def __rmul__(self, other: object) -> "Golden":
        return self * other

    def __pow__(self, exponent: int) -> "Golden":
        if not isinstance(exponent, int) or exponent < 0:
            raise FrequencyProjectionError("golden exponent must be a nonnegative integer")
        result = GOLDEN_ONE
        base = self
        power = exponent
        while power:
            if power & 1:
                result = result * base
            base = base * base
            power >>= 1
        return result

    def algebraic_conjugate(self) -> "Golden":
        """Apply the exact conjugation ``Φ ↦ 1 - Φ``."""
        return Golden(self.rational + self.phi, -self.phi)

    @property
    def is_zero(self) -> bool:
        return self.rational == 0 and self.phi == 0

    def to_tuple(self) -> tuple[Fraction, Fraction]:
        return (self.rational, self.phi)

    @staticmethod
    def coerce(value: object) -> "Golden":
        if isinstance(value, Golden):
            return value
        if isinstance(value, (int, str, Fraction)):
            return Golden(value)
        raise TypeError(f"cannot coerce {type(value).__name__} into Q(Φ)")


GOLDEN_ZERO: Final[Golden] = Golden(0)
GOLDEN_ONE: Final[Golden] = Golden(1)
PHI: Final[Golden] = Golden(0, 1)


@dataclass(frozen=True, slots=True)
class GoldenComplex:
    """One exact element ``x + iy`` of ``Q(Φ, i)``."""

    real: Golden = GOLDEN_ZERO
    imag: Golden = GOLDEN_ZERO
```

```
def __add__(self, other: object) -> "GoldenComplex":
    rhs = GoldenComplex.coerce(other)
    return GoldenComplex(self.real + rhs.real, self.imag + rhs.imag)

def __radd__(self, other: object) -> "GoldenComplex":
    return self + other

def __sub__(self, other: object) -> "GoldenComplex":
    rhs = GoldenComplex.coerce(other)
    return GoldenComplex(self.real - rhs.real, self.imag - rhs.imag)

def __rsub__(self, other: object) -> "GoldenComplex":
    return GoldenComplex.coerce(other) - self

def __neg__(self) -> "GoldenComplex":
    return GoldenComplex(-self.real, -self.imag)

def __mul__(self, other: object) -> "GoldenComplex":
    rhs = GoldenComplex.coerce(other)
    return GoldenComplex(
        self.real * rhs.real - self.imag * rhs.imag,
        self.real * rhs.imag + self.imag * rhs.real,
    )

def __rmul__(self, other: object) -> "GoldenComplex":
    return self * other

def __pow__(self, exponent: int) -> "GoldenComplex":
    if not isinstance(exponent, int) or exponent < 0:
        raise FrequencyProjectionError("carrier exponent must be a nonnegative integer")
    result = COMPLEX_ONE
    base = self
    power = exponent
    while power:
        if power & 1:
            result = result * base
        base = base * base
        power >>= 1
    return result

def complex_conjugate(self) -> "GoldenComplex":
    return GoldenComplex(self.real, -self.imag)

@property
def is_zero(self) -> bool:
    return self.real.is_zero and self.imag.is_zero

def to_tuple(self) -> tuple[tuple[Fraction, Fraction], tuple[Fraction, Fraction]]:
    return (self.real.to_tuple(), self.imag.to_tuple())

@staticmethod
def coerce(value: object) -> "GoldenComplex":
    if isinstance(value, GoldenComplex):
        return value
    if isinstance(value, Golden):
```

```
            return GoldenComplex(value)
        if isinstance(value, (int, str, Fraction)):
            return GoldenComplex(Golden(value))
        raise TypeError(f"cannot coerce {type(value).__name__} into Q(Φ,i)")


COMPLEX_ZERO: Final[GoldenComplex] = GoldenComplex()
COMPLEX_ONE: Final[GoldenComplex] = GoldenComplex(GOLDEN_ONE)


@dataclass(frozen=True, slots=True)
class Carrier:
    """One immutable Aeon carrier address and its optional parity channel."""

    glyph: str
    name: str
    structural_hz: Fraction
    parity_hz: Fraction | None = None

    def __post_init__(self) -> None:
        if len(self.glyph) != 1:
            raise FrequencyProjectionError("carrier glyph must be one code point")
        if self.structural_hz <= 0:
            raise FrequencyProjectionError("structural carrier must be positive")
        if self.parity_hz is not None and self.parity_hz <= 0:
            raise FrequencyProjectionError("parity carrier must be positive")

    @property
    def structural_omega(self) -> GoldenComplex:
        return GoldenComplex(Golden(self.structural_hz))

    @property
    def parity_omega(self) -> GoldenComplex | None:
        if self.parity_hz is None:
            return None
        return GoldenComplex(GOLDEN_ZERO, Golden(self.parity_hz))

    @property
    def breath_bounds(self) -> tuple[Golden, Golden]:
        return (-PHI, PHI)


CARRIERS: Final[tuple[Carrier, ...]] = (
    Carrier("⎔", "FETU", Fraction("7.83")),
    Carrier("⬡", "KAL", Fraction(174)),
    Carrier("✡", "BABDH", Fraction(528)),
    Carrier("☆", "AHN", Fraction(432), Fraction(417)),
    Carrier("✪", "VEL", Fraction("126.22")),
    Carrier("❀", "SOR", Fraction("210.42")),
    Carrier("✳", "KOTH", Fraction(741)),
    Carrier("⧗", "DREH", Fraction(852)),
    Carrier("⊛", "RHEA", Fraction(396)),
    Carrier("❄", "ZHEK", Fraction(963)),
    Carrier("⚛", "SHAV", Fraction(285)),
    Carrier("⌬", "TRIG", Fraction(639)),
)
```

```
CARRIER_BY_GLYPH: Final[Mapping[str, Carrier]] = {carrier.glyph: carrier for carrier in CARRIERS}
if len(CARRIER_BY_GLYPH) != 12:
    raise RuntimeError("Phase 8 requires exactly twelve distinct carriers")


@dataclass(frozen=True, slots=True)
class ProsodicUnit:
    """One declared, speaker-independent prosodic obligation."""

    phoneme: str
    stress: Stress
    breath: Breath
    bearing: Bearing = Bearing.OUTWARD
    operator_office: str = ""

    def __post_init__(self) -> None:
        if not isinstance(self.phoneme, str) or not self.phoneme:
            raise FrequencyProjectionError("phoneme identity must be a nonempty string")
        if any(character.isspace() for character in self.phoneme):
            raise FrequencyProjectionError("phoneme identity may not contain whitespace")
        if not isinstance(self.stress, Stress):
            raise FrequencyProjectionError("stress must be a declared Stress member")
        if not isinstance(self.breath, Breath):
            raise FrequencyProjectionError("breath must be a declared Breath member")
        if not isinstance(self.bearing, Bearing):
            raise FrequencyProjectionError("bearing must be a declared Bearing member")
        if not isinstance(self.operator_office, str):
            raise FrequencyProjectionError("operator office must be text")


@dataclass(frozen=True, slots=True)
class ProsodicWitness:
    """The complete ordered phoneme/stress body for one candidate word."""

    candidate: str
    units: tuple[ProsodicUnit, ...]

    def __post_init__(self) -> None:
        if not isinstance(self.candidate, str) or not self.candidate:
            raise FrequencyProjectionError("candidate identifier must be nonempty")
        if not isinstance(self.units, tuple) or not self.units:
            raise FrequencyProjectionError("candidate requires at least one prosodic unit")
        if any(not isinstance(unit, ProsodicUnit) for unit in self.units):
            raise FrequencyProjectionError("all witness units must be ProsodicUnit instances")


@dataclass(frozen=True, slots=True, order=True)
class ProsodicAtom:
    """A recoverable basis atom in the free prosodic coefficient module."""

    position: int
    phoneme: str
    stress: int
    breath: int
    bearing: str
    operator_office: str
```

```
    @staticmethod
    def from_unit(position: int, unit: ProsodicUnit, mirror: bool) -> "ProsodicAtom":
        bearing = unit.bearing.mirror() if mirror else unit.bearing
        return ProsodicAtom(
            position=position,
            phoneme=unit.phoneme,
            stress=int(unit.stress),
            breath=int(unit.breath),
            bearing=bearing.value,
            operator_office=unit.operator_office,
        )

    def mirrored(self) -> "ProsodicAtom":
        bearing = Bearing(self.bearing).mirror()
        return ProsodicAtom(
            self.position,
            self.phoneme,
            self.stress,
            self.breath,
            bearing.value,
            self.operator_office,
        )


@dataclass(frozen=True, slots=True)
class AtomWeight:
    atom: ProsodicAtom
    weight: GoldenComplex


@dataclass(frozen=True, slots=True)
class PolynomialDegree:
    degree: int
    weights: tuple[AtomWeight, ...]


MutableModule = dict[int, dict[ProsodicAtom, GoldenComplex]]
ScalarPolynomial = dict[int, GoldenComplex]


def _add_module_weight(
    target: MutableModule,
    degree: int,
    atom: ProsodicAtom,
    weight: GoldenComplex,
) -> None:
    if weight.is_zero:
        return
    level = target.setdefault(degree, {})
    updated = level.get(atom, COMPLEX_ZERO) + weight
    if updated.is_zero:
        level.pop(atom, None)
        if not level:
            target.pop(degree, None)
    else:
```

```
        level[atom] = updated


def _freeze_module(source: MutableModule) -> tuple[PolynomialDegree, ...]:
    frozen: list[PolynomialDegree] = []
    for degree in sorted(source):
        weights = tuple(
            AtomWeight(atom, weight)
            for atom, weight in sorted(source[degree].items(), key=lambda item: item[0])
            if not weight.is_zero
        )
        if weights:
            frozen.append(PolynomialDegree(degree, weights))
    return tuple(frozen)


def _thaw_module(source: Iterable[PolynomialDegree]) -> MutableModule:
    thawed: MutableModule = {}
    for degree in source:
        for item in degree.weights:
            _add_module_weight(thawed, degree.degree, item.atom, item.weight)
    return thawed


def _advance_scalar_power(power: ScalarPolynomial, omega: GoldenComplex) -> ScalarPolynomial:
    advanced: ScalarPolynomial = {}
    for degree, weight in power.items():
        at_same_degree = advanced.get(degree, COMPLEX_ZERO) + weight * omega
        at_next_degree = advanced.get(degree + 1, COMPLEX_ZERO) + weight
        if not at_same_degree.is_zero:
            advanced[degree] = at_same_degree
        if not at_next_degree.is_zero:
            advanced[degree + 1] = at_next_degree
    return advanced


@dataclass(frozen=True, slots=True)
class ChannelPolynomial:
    """Exact normal form of one declared carrier channel in the variable ``δ``."""

    channel: str
    omega: GoldenComplex
    delta_symbol: str
    terms: tuple[PolynomialDegree, ...]

    def evaluate(self, delta: GoldenComplex) -> tuple[AtomWeight, ...]:
        result: dict[ProsodicAtom, GoldenComplex] = {}
        for degree in self.terms:
            delta_power = delta ** degree.degree
            for item in degree.weights:
                weight = result.get(item.atom, COMPLEX_ZERO) + item.weight * delta_power
                if weight.is_zero:
                    result.pop(item.atom, None)
                else:
                    result[item.atom] = weight
        return tuple(
```

```
            AtomWeight(atom, weight)
            for atom, weight in sorted(result.items(), key=lambda item: item[0])
        )


@dataclass(frozen=True, slots=True)
class FrequencyBody:
    """One complete outward, ideal, or independently returned prosodic body."""

    carrier: Carrier
    office: PathOffice
    candidate: str
    time_map: str
    channels: tuple[ChannelPolynomial, ...]

    @property
    def positions(self) -> tuple[int, ...]:
        positions: set[int] = set()
        for channel in self.channels:
            for degree in channel.terms:
                positions.update(weight.atom.position for weight in degree.weights)
        return tuple(sorted(positions))

    def normal_form(self) -> tuple[object, ...]:
        """Return ``NF_A`` as an exact immutable symbolic body."""
        return (
            self.carrier.glyph,
            self.carrier.structural_hz,
            self.carrier.parity_hz,
            self.carrier.breath_bounds,
            self.time_map,
            tuple(
                (
                    channel.channel,
                    channel.omega,
                    channel.delta_symbol,
                    channel.terms,
                )
                for channel in self.channels
            ),
        )


@dataclass(frozen=True, slots=True)
class GoldenEndpointWitness:
    """Full symbolic golden-fold recurrence plus both breath endpoints."""

    channel: str
    symbolic_terms: tuple[PolynomialDegree, ...]
    minus_phi: tuple[AtomWeight, ...]
    plus_phi: tuple[AtomWeight, ...]


@dataclass(frozen=True, slots=True)
class FrequencyClosure:
    """The complete Phase 8 closure record."""
```

```
    outward: FrequencyBody
    independent_return: FrequencyBody
    ideal_return: FrequencyBody
    golden_endpoints: tuple[GoldenEndpointWitness, ...]
    nu_close: bool
    dcomp_frequency_zero: bool


def _channel_specs(carrier: Carrier) -> tuple[tuple[str, GoldenComplex], ...]:
    specs: list[tuple[str, GoldenComplex]] = [("structural", carrier.structural_omega)]
    parity = carrier.parity_omega
    if parity is not None:
        specs.append(("parity", parity))
    return tuple(specs)


def _outward_channel_direct(
    witness: ProsodicWitness,
    channel: str,
    omega: GoldenComplex,
) -> ChannelPolynomial:
    """Track A: direct exact ``Σ a_m(Ω+δ)^m`` construction."""
    body: MutableModule = {}
    for position, unit in enumerate(witness.units):
        atom = ProsodicAtom.from_unit(position, unit, mirror=False)
        for degree in range(position + 1):
            weight = GoldenComplex(comb(position, degree)) * (omega ** (position - degree))
            _add_module_weight(body, degree, atom, weight)
    return ChannelPolynomial(channel, omega, "δ", _freeze_module(body))


def build_outward(witness: ProsodicWitness, carrier: Carrier) -> FrequencyBody:
    """Construct ``P_A^out`` without a return body in scope."""
    channels = tuple(
        _outward_channel_direct(witness, channel, omega)
        for channel, omega in _channel_specs(carrier)
    )
    return FrequencyBody(carrier, PathOffice.OUTWARD, witness.candidate, "t", channels)


def _return_channel_recurrence(
    witness: ProsodicWitness,
    channel: str,
    omega: GoldenComplex,
) -> ChannelPolynomial:
    """Track B: independent return recurrence in original sequence order."""
    body: MutableModule = {}
    power: ScalarPolynomial = {0: COMPLEX_ONE}
    for position, unit in enumerate(witness.units):
        atom = ProsodicAtom.from_unit(position, unit, mirror=True)
        for degree, weight in power.items():
            _add_module_weight(body, degree, atom, weight)
        power = _advance_scalar_power(power, omega)
    return ChannelPolynomial(channel, omega, "δ", _freeze_module(body))
```

```
def build_independent_return(witness: ProsodicWitness, carrier: Carrier) -> FrequencyBody:
    """Construct ``P_A^back`` solely from the candidate witness and return law."""
    channels = tuple(
        _return_channel_recurrence(witness, channel, omega)
        for channel, omega in _channel_specs(carrier)
    )
    return FrequencyBody(
        carrier,
        PathOffice.RETURN,
        witness.candidate,
        "t ↦ T-t",
        channels,
    )


def _mirror_channel(channel: ChannelPolynomial) -> ChannelPolynomial:
    mirrored: MutableModule = {}
    for degree in channel.terms:
        for item in degree.weights:
            _add_module_weight(mirrored, degree.degree, item.atom.mirrored(), item.weight)
    return ChannelPolynomial(channel.channel, channel.omega, channel.delta_symbol,
    ↪  _freeze_module(mirrored))


def build_ideal_from_outward(outward: FrequencyBody) -> FrequencyBody:
    """Track D: derive the ideal only after ``P_A^out`` exists.

    Mirror changes each declared bearing and preserves every sequence position,
    polynomial exponent, carrier channel, phoneme, stress, breath, and office.
    """
    if outward.office is not PathOffice.OUTWARD:
        raise FrequencyProjectionError("ideal parity comparison requires an outward body")
    return FrequencyBody(
        outward.carrier,
        PathOffice.IDEAL,
        outward.candidate,
        "t ↦ T-t",
        tuple(_mirror_channel(channel) for channel in outward.channels),
    )


def _subtract_channel(
    left: ChannelPolynomial,
    right: ChannelPolynomial,
) -> ChannelPolynomial:
    if (left.channel, left.omega, left.delta_symbol) != (
        right.channel,
        right.omega,
        right.delta_symbol,
    ):
        raise FrequencyProjectionError("cannot subtract unlike carrier channels")
    body = _thaw_module(left.terms)
    for degree in right.terms:
        for item in degree.weights:
            _add_module_weight(body, degree.degree, item.atom, -item.weight)
```

```
    return ChannelPolynomial(left.channel, left.omega, left.delta_symbol, _freeze_module(body))


def _is_zero_channel(channel: ChannelPolynomial) -> bool:
    return not channel.terms


def dcomp_frequency_difference(
    independent_return: FrequencyBody,
    ideal_return: FrequencyBody,
) -> tuple[ChannelPolynomial, ...]:
    """Return the exact symbolic D-COMP frequency difference."""
    if independent_return.carrier != ideal_return.carrier:
        raise FrequencyProjectionError("frequency difference requires one carrier")
    if len(independent_return.channels) != len(ideal_return.channels):
        raise FrequencyProjectionError("frequency difference requires identical channel count")
    return tuple(
        _subtract_channel(back, ideal)
        for back, ideal in zip(independent_return.channels, ideal_return.channels, strict=True)
    )


def _golden_fold_channel_recurrence(
    witness: ProsodicWitness,
    channel: str,
    omega: GoldenComplex,
) -> ChannelPolynomial:
    """Track C: independent full symbolic recurrence over ``Q(Φ,i)[δ]``.

    This verifier does not read the outward polynomial, the ideal parity body,
    or the accepted return coefficients.  It advances the complete coefficient
    vector by multiplication with ``Ω+δ`` and therefore proves the whole
    structural return class, not merely two endpoint evaluations.
    """
    body: MutableModule = {}
    coefficient_vector: tuple[GoldenComplex, ...] = (COMPLEX_ONE,)
    for position, unit in enumerate(witness.units):
        atom = ProsodicAtom.from_unit(position, unit, mirror=True)
        for degree, weight in enumerate(coefficient_vector):
            _add_module_weight(body, degree, atom, weight)

        advanced = [COMPLEX_ZERO] * (len(coefficient_vector) + 1)
        for degree, weight in enumerate(coefficient_vector):
            advanced[degree] = advanced[degree] + weight * omega
            advanced[degree + 1] = advanced[degree + 1] + weight
        coefficient_vector = tuple(advanced)

    return ChannelPolynomial(channel, omega, "δ", _freeze_module(body))


def _golden_endpoint_recurrence(
    witness: ProsodicWitness,
    symbolic: ChannelPolynomial,
) -> GoldenEndpointWitness:
    """Record exact ``±Φ`` witnesses after full symbolic recurrence succeeds."""
```

```
    def one_endpoint(sign: int) -> tuple[AtomWeight, ...]:
        delta = GoldenComplex(Golden(0, sign))
        z = symbolic.omega + delta
        power = COMPLEX_ONE
        result: dict[ProsodicAtom, GoldenComplex] = {}
        for position, unit in enumerate(witness.units):
            atom = ProsodicAtom.from_unit(position, unit, mirror=True)
            result[atom] = result.get(atom, COMPLEX_ZERO) + power
            power = power * z
        return tuple(
            AtomWeight(atom, weight)
            for atom, weight in sorted(result.items(), key=lambda item: item[0])
            if not weight.is_zero
        )

    return GoldenEndpointWitness(
        symbolic.channel,
        symbolic.terms,
        one_endpoint(-1),
        one_endpoint(1),
    )


def _golden_endpoints_match(
    witness: ProsodicWitness,
    independent_return: FrequencyBody,
) -> tuple[GoldenEndpointWitness, ...]:
    recurrence_channels = tuple(
        _golden_fold_channel_recurrence(witness, channel, omega)
        for channel, omega in _channel_specs(independent_return.carrier)
    )
    witnesses: list[GoldenEndpointWitness] = []
    for recurrence_channel, polynomial_channel in zip(
        recurrence_channels,
        independent_return.channels,
        strict=True,
    ):
        if (
            recurrence_channel.channel,
            recurrence_channel.omega,
            recurrence_channel.delta_symbol,
        ) != (
            polynomial_channel.channel,
            polynomial_channel.omega,
            polynomial_channel.delta_symbol,
        ):
            raise FrequencyProjectionError("golden recurrence channel mismatch")
        if recurrence_channel.terms != polynomial_channel.terms:
            raise FrequencyProjectionError(
                "golden-fold recurrence left the full symbolic return class"
            )

        endpoint = _golden_endpoint_recurrence(witness, recurrence_channel)
        minus = polynomial_channel.evaluate(GoldenComplex(-PHI))
        plus = polynomial_channel.evaluate(GoldenComplex(PHI))
        if endpoint.minus_phi != minus or endpoint.plus_phi != plus:
```

```
            raise FrequencyProjectionError(
                "golden-fold endpoint witness disagrees after symbolic closure"
            )
        witnesses.append(endpoint)
    return tuple(witnesses)


def prove_frequency_closure(
    witness: ProsodicWitness,
    carrier: Carrier,
) -> FrequencyClosure:
    """Construct and prove Phase 8 in the required dependency order."""
    outward = build_outward(witness, carrier)
    independent_return = build_independent_return(witness, carrier)
    ideal_return = build_ideal_from_outward(outward)
    golden_endpoints = _golden_endpoints_match(witness, independent_return)
    difference = dcomp_frequency_difference(independent_return, ideal_return)
    exact_zero = all(_is_zero_channel(channel) for channel in difference)
    nu_close = independent_return.normal_form() == ideal_return.normal_form()
    if nu_close != exact_zero:
        raise RuntimeError("normal-form closure and exact D-COMP zero disagree")
    return FrequencyClosure(
        outward=outward,
        independent_return=independent_return,
        ideal_return=ideal_return,
        golden_endpoints=golden_endpoints,
        nu_close=nu_close,
        dcomp_frequency_zero=exact_zero,
    )

def prove_phase8_exit_gate(
    witness: ProsodicWitness,
) -> tuple[FrequencyClosure, ...]:
    """Prove one candidate projection across all twelve declared carriers."""
    closures = tuple(prove_frequency_closure(witness, carrier) for carrier in CARRIERS)
    if len(closures) != 12:
        raise RuntimeError("Phase 8 exit gate requires exactly twelve carriers")
    if any(not closure.nu_close for closure in closures):
        raise FrequencyProjectionError("Phase 8 exit gate found an open prosodic return")
    if any(not closure.dcomp_frequency_zero for closure in closures):
        raise FrequencyProjectionError("Phase 8 exit gate found nonzero D-COMP frequency")
    return closures
```

goetic_outward_return.py

```
"""Phase 12 Goetic outward and independently constructed return paths.

Every Phase 11 survivor is tested only under its declared Goetic carrier.  The
Goetic Bound Envelope Constraint is represented exactly as::

    BEC(A_i) = 𝔄 o (A_i ->♂ A_i^-1)

Eight bearings are preserved as the four authored Mirror-Math pairs:
``+ <-> -``, ``Manifest <-> Reflect``, ``Ascend <-> Descend``, and
``Utter <-> Receive``.  Mirroring changes bearing while preserving every M.A.S.
step position.  The independent return builder does not receive or inspect an
outward path.

A bearing emits one candidate identifier only when exactly one candidate closes
with Self -> Self, valid BEC, and exact ``D_T = 0``.  Zero survivors or multiple
survivors leave that bearing failed.  This module deliberately makes no decision
about whether a Goetic with fewer than eight closed bearings stands; that law is
deferred to Phase 17.
"""
from __future__ import annotations

from dataclasses import dataclass
from enum import Enum
from typing import Final

from .frequency_projection import CARRIERS, Carrier
from .mas_shadow_conversion import MASGlobalField, MASCandidatePath, MAS_ORDER


class GoeticPathError(ValueError):
    """Raised when a Phase 12 path violates BEC or exact return law."""


class GoeticBearing(str, Enum):
    PLUS = "+"
    MINUS = "-"
    MANIFEST = "Manifest"
    REFLECT = "Reflect"
    ASCEND = "Ascend"
    DESCEND = "Descend"
    UTTER = "Utter"
    RECEIVE = "Receive"


BEARINGS: Final[tuple[GoeticBearing, ...]] = tuple(GoeticBearing)
MIRROR_BEARING: Final[dict[GoeticBearing, GoeticBearing]] = {
    GoeticBearing.PLUS: GoeticBearing.MINUS,
    GoeticBearing.MINUS: GoeticBearing.PLUS,
```

```
    GoeticBearing.MANIFEST: GoeticBearing.REFLECT,
    GoeticBearing.REFLECT: GoeticBearing.MANIFEST,
    GoeticBearing.ASCEND: GoeticBearing.DESCEND,
    GoeticBearing.DESCEND: GoeticBearing.ASCEND,
    GoeticBearing.UTTER: GoeticBearing.RECEIVE,
    GoeticBearing.RECEIVE: GoeticBearing.UTTER,
}
if len(BEARINGS) != 8 or any(MIRROR_BEARING[MIRROR_BEARING[bearing]] is not bearing for bearing in
↪  BEARINGS):
    raise RuntimeError("Phase 12 requires eight involutive Goetic bearings")


@dataclass(frozen=True, slots=True)
class GoeticEnvelopeWitness:
    """Exact Goetic self-recursion envelope ``♳ o (A ->♂ A^-1)``."""

    carrier: Carrier
    triquatra_seal: str = "♳"
    goetic_identity: str = ""
    klein_seam: str = "♂"
    inverse_identity: str = ""

    def __post_init__(self) -> None:
        if self.carrier not in CARRIERS:
            raise GoeticPathError("BEC requires one of the twelve canonical Goetics")
        expected_goetic = self.carrier.glyph
        expected_inverse = f"{self.carrier.glyph}^-1"
        if self.goetic_identity != expected_goetic:
            raise GoeticPathError("BEC changed the Goetic self identity")
        if self.inverse_identity != expected_inverse:
            raise GoeticPathError("BEC lacks the Goetic inverse identity")
        if self.triquatra_seal != "♳" or self.klein_seam != "♂":
            raise GoeticPathError("BEC lost its Triquatra seal or Klein seam")

    @property
    def bec_valid(self) -> bool:
        return (
            self.carrier in CARRIERS
            and self.goetic_identity == self.carrier.glyph
            and self.inverse_identity == f"{self.carrier.glyph}^-1"
            and self.triquatra_seal == "♳"
            and self.klein_seam == "♂"
        )

    @property
    def formula(self) -> tuple[str, str, str, str]:
        return (
            self.triquatra_seal,
            self.goetic_identity,
            self.klein_seam,
            self.inverse_identity,
        )


@dataclass(frozen=True, slots=True)
class GoeticPathStep:
```

```
    """One position-preserving M.A.S. action on a declared bearing."""

    position: int
    cycle_index: int
    office_index: int
    office: str
    bearing: GoeticBearing
    signed_velocity: int

    def __post_init__(self) -> None:
        if self.position < 0 or self.cycle_index < 0 or self.office_index not in (0, 1, 2):
            raise GoeticPathError("Goetic path step has an invalid source position")
        if self.position != self.cycle_index * 3 + self.office_index:
            raise GoeticPathError("Goetic path step moved from its M.A.S. source position")
        if self.office != MAS_ORDER[self.office_index]:
            raise GoeticPathError("Goetic path step changed M.A.S. office order")
        if not isinstance(self.bearing, GoeticBearing):
            raise GoeticPathError("Goetic path step lacks one of the eight bearings")
        if not isinstance(self.signed_velocity, int) or self.signed_velocity == 0:
            raise GoeticPathError("Goetic path velocity must be an exact nonzero integer")


SelfIdentity = tuple[str, str, str, tuple[str, str, str, str]]
BoundaryIdentity = tuple[str, str, str]


@dataclass(frozen=True, slots=True)
class GoeticOutwardPath:
    candidate: str
    carrier: Carrier
    bearing: GoeticBearing
    start_self: SelfIdentity
    boundary: BoundaryIdentity
    steps: tuple[GoeticPathStep, ...]

    def __post_init__(self) -> None:
        if self.carrier not in CARRIERS or not self.steps:
            raise GoeticPathError("outward path requires a canonical Goetic and nonempty motion")
        if tuple(step.position for step in self.steps) != tuple(range(len(self.steps))):
            raise GoeticPathError("outward path lost source step order")
        if any(step.bearing is not self.bearing or step.signed_velocity <= 0 for step in
        ↪  self.steps):
            raise GoeticPathError("outward path changed bearing or velocity sign")


@dataclass(frozen=True, slots=True)
class GoeticReturnPath:
    candidate: str
    carrier: Carrier
    bearing: GoeticBearing
    boundary: BoundaryIdentity
    end_self: SelfIdentity
    steps: tuple[GoeticPathStep, ...]

    def __post_init__(self) -> None:
        if self.carrier not in CARRIERS or not self.steps:
```

```
            raise GoeticPathError("return path requires a canonical Goetic and nonempty motion")
        if tuple(step.position for step in self.steps) != tuple(range(len(self.steps))):
            raise GoeticPathError("return path reversed or moved source step order")
        if any(step.bearing is not self.bearing or step.signed_velocity >= 0 for step in
        ↪  self.steps):
            raise GoeticPathError("return path changed mirrored bearing or velocity sign")


@dataclass(frozen=True, slots=True)
class DCompPathWitness:
    """Exact nonnegative components of the finite-interval return test."""

    velocity_terms: tuple[int, ...]
    terminal_unresolved_q2: int
    whiteout_penalty: int
    envelope_penalty: int
    boundary_penalty: int

    def __post_init__(self) -> None:
        if any(not isinstance(term, int) for term in self.velocity_terms):
            raise GoeticPathError("D_T velocity terms must be exact integers")
        scalar_terms = (
            self.terminal_unresolved_q2,
            self.whiteout_penalty,
            self.envelope_penalty,
            self.boundary_penalty,
        )
        if any(not isinstance(term, int) or term < 0 for term in scalar_terms):
            raise GoeticPathError("D_T scalar components must be nonnegative integers")

    @property
    def exact_zero(self) -> bool:
        return (
            all(term == 0 for term in self.velocity_terms)
            and self.terminal_unresolved_q2 == 0
            and self.whiteout_penalty == 0
            and self.envelope_penalty == 0
            and self.boundary_penalty == 0
        )

    @property
    def value(self) -> int:
        return (
            sum(abs(term) for term in self.velocity_terms)
            + self.terminal_unresolved_q2
            + self.whiteout_penalty
            + self.envelope_penalty
            + self.boundary_penalty
        )


@dataclass(frozen=True, slots=True)
class GoeticClosure:
    """One exact candidate closure under one Goetic and one bearing."""

    candidate_path: MASCandidatePath
```

```
    carrier: Carrier
    bearing: GoeticBearing
    envelope: GoeticEnvelopeWitness
    outward: GoeticOutwardPath
    independent_return: GoeticReturnPath
    dcomp: DCompPathWitness

    def __post_init__(self) -> None:
        candidate = self.candidate_path.candidate
        if self.candidate_path.liquid_path.seed.domain.carrier != self.carrier:
            raise GoeticPathError("candidate is not rooted in the declared Goetic")
        if self.envelope.carrier != self.carrier or not self.envelope.bec_valid:
            raise GoeticPathError("BECValid(A_i;w) != 1")
        if self.outward.candidate != candidate or self.independent_return.candidate != candidate:
            raise GoeticPathError("outward and return paths changed candidate identity")
        if self.outward.carrier != self.carrier or self.independent_return.carrier != self.carrier:
            raise GoeticPathError("outward and return paths changed Goetic identity")
        if self.outward.bearing is not self.bearing:
            raise GoeticPathError("outward path changed the requested bearing")
        if self.independent_return.bearing is not MIRROR_BEARING[self.bearing]:
            raise GoeticPathError("return path did not carry the conjugate bearing")
        if self.outward.start_self != self.independent_return.end_self:
            raise GoeticPathError("required return is not Self -> Self")
        if self.outward.boundary != self.independent_return.boundary:
            raise GoeticPathError("outward and return paths do not meet at one BEC boundary")
        if tuple(step.position for step in self.outward.steps) != tuple(step.position for step in
        ↪  self.independent_return.steps):
            raise GoeticPathError("Mirror Math reversed the operator sequence")
        if not self.dcomp.exact_zero or self.dcomp.value != 0:
            raise GoeticPathError("D_T(gamma, gamma_bar) != 0")

    @property
    def self_to_self(self) -> bool:
        return self.outward.start_self == self.independent_return.end_self

    @property
    def bec_valid(self) -> bool:
        return self.envelope.bec_valid


@dataclass(frozen=True, slots=True)
class CandidateBearingFailure:
    candidate: str
    goetic: str
    bearing: GoeticBearing
    reason: str


@dataclass(frozen=True, slots=True)
class BearingResult:
    """Exactly one closure emits; zero or multiple closures remain failed."""

    carrier: Carrier
    bearing: GoeticBearing
    attempted_candidates: tuple[str, ...]
    exact_closures: tuple[GoeticClosure, ...]
```

```
    failures: tuple[CandidateBearingFailure, ...]

    def __post_init__(self) -> None:
        if any(closure.carrier != self.carrier or closure.bearing is not self.bearing for closure in
        ↪  self.exact_closures):
            raise GoeticPathError("bearing result mixed Goetic or bearing identities")
        closure_candidates = tuple(closure.candidate_path.candidate for closure in
        ↪  self.exact_closures)
        if any(candidate not in self.attempted_candidates for candidate in closure_candidates):
            raise GoeticPathError("bearing result contains an unattempted candidate")
        if len(set(closure_candidates)) != len(closure_candidates):
            raise GoeticPathError("bearing result duplicated one exact closure")

    @property
    def closed(self) -> bool:
        return len(self.exact_closures) == 1

    @property
    def survivor(self) -> GoeticClosure | None:
        return self.exact_closures[0] if self.closed else None

    @property
    def emitted_word(self) -> str | None:
        return self.survivor.candidate_path.candidate if self.survivor is not None else None


@dataclass(frozen=True, slots=True)
class GoeticBearingBody:
    carrier: Carrier
    bearing_results: tuple[BearingResult, ...]

    def __post_init__(self) -> None:
        if self.carrier not in CARRIERS:
            raise GoeticPathError("bearing body requires a canonical Goetic")
        if tuple(result.bearing for result in self.bearing_results) != BEARINGS:
            raise GoeticPathError("Goetic bearing body must preserve all eight bearings in order")
        if any(result.carrier != self.carrier for result in self.bearing_results):
            raise GoeticPathError("Goetic bearing body mixed carrier identities")

    @property
    def closed_bearing_count(self) -> int:
        return sum(result.closed for result in self.bearing_results)

    @property
    def failed_bearing_count(self) -> int:
        return len(self.bearing_results) - self.closed_bearing_count


@dataclass(frozen=True, slots=True)
class GoeticOutwardReturnField:
    """The twelve-Goetic, eight-bearing Phase 12 result body."""

    source: MASGlobalField
    goetics: tuple[GoeticBearingBody, ...]
    partial_body_law_deferred: bool = True
```

```
    def __post_init__(self) -> None:
        if tuple(body.carrier for body in self.goetics) != CARRIERS:
            raise GoeticPathError("Phase 12 must validate all twelve Goetics in canonical order")
        if any(len(body.bearing_results) != 8 for body in self.goetics):
            raise GoeticPathError("every Goetic requires all eight bearing results")
        if not self.partial_body_law_deferred:
            raise GoeticPathError("Goetic standing must remain deferred to Phase 17")

    @property
    def closed_bearing_count(self) -> int:
        return sum(body.closed_bearing_count for body in self.goetics)

    @property
    def failed_bearing_count(self) -> int:
        return 96 - self.closed_bearing_count



def _self_identity(candidate: MASCandidatePath) -> SelfIdentity:
    seed = candidate.liquid_path.seed
    return (
        candidate.candidate,
        seed.domain.court_root,
        seed.domain.inherited_q_bias,
        seed.body.q_vector,
    )



def _boundary_identity(carrier: Carrier, candidate: MASCandidatePath) -> BoundaryIdentity:
    return (carrier.glyph, "♂", candidate.candidate)



def _step_magnitude(candidate: MASCandidatePath, cycle_index: int, office_index: int) -> int:
    cycle = candidate.cycles[cycle_index]
    if office_index == 0:
        return 1 + cycle.resolved_pressure_added
    return 1



def build_goetic_envelope(carrier: Carrier) -> GoeticEnvelopeWitness:
    return GoeticEnvelopeWitness(
        carrier=carrier,
        goetic_identity=carrier.glyph,
        inverse_identity=f"{carrier.glyph}^-1",
    )



def build_outward_path(
    candidate: MASCandidatePath,
    carrier: Carrier,
    bearing: GoeticBearing,
) -> GoeticOutwardPath:
```

```
    """Construct ``gamma_Ai,w`` from the candidate M.A.S. archive."""
    if candidate.liquid_path.seed.domain.carrier != carrier:
        raise GoeticPathError("candidate cannot be retyped under a foreign Goetic")
    steps = tuple(
        GoeticPathStep(
            position=cycle_index * 3 + office_index,
            cycle_index=cycle_index,
            office_index=office_index,
            office=MAS_ORDER[office_index],
            bearing=bearing,
            signed_velocity=_step_magnitude(candidate, cycle_index, office_index),
        )
        for cycle_index in range(len(candidate.cycles))
        for office_index in range(3)
    )
    return GoeticOutwardPath(
        candidate=candidate.candidate,
        carrier=carrier,
        bearing=bearing,
        start_self=_self_identity(candidate),
        boundary=_boundary_identity(carrier, candidate),
        steps=steps,
    )



def build_independent_return_path(
    candidate: MASCandidatePath,
    carrier: Carrier,
    bearing: GoeticBearing,
) -> GoeticReturnPath:
    """Construct ``gamma_bar_Ai,w`` without an outward path in scope."""
    if candidate.liquid_path.seed.domain.carrier != carrier:
        raise GoeticPathError("candidate cannot return under a foreign Goetic")
    mirrored = MIRROR_BEARING[bearing]
    steps = tuple(
        GoeticPathStep(
            position=cycle_index * 3 + office_index,
            cycle_index=cycle_index,
            office_index=office_index,
            office=MAS_ORDER[office_index],
            bearing=mirrored,
            signed_velocity=-_step_magnitude(candidate, cycle_index, office_index),
        )
        for cycle_index in range(len(candidate.cycles))
        for office_index in range(3)
    )
    return GoeticReturnPath(
        candidate=candidate.candidate,
        carrier=carrier,
        bearing=mirrored,
        boundary=_boundary_identity(carrier, candidate),
        end_self=_self_identity(candidate),
        steps=steps,
    )
```

```
def compute_exact_dcomp(
    candidate: MASCandidatePath,
    envelope: GoeticEnvelopeWitness,
    outward: GoeticOutwardPath,
    independent_return: GoeticReturnPath,
) -> DCompPathWitness:
    """Compute the exact nonnegative Phase 12 D_T component body."""
    if len(outward.steps) != len(independent_return.steps):
        raise GoeticPathError("outward and return paths have different lengths")
    velocity_terms: list[int] = []
    for out_step, back_step in zip(outward.steps, independent_return.steps, strict=True):
        if (
            out_step.position,
            out_step.cycle_index,
            out_step.office_index,
            out_step.office,
        ) != (
            back_step.position,
            back_step.cycle_index,
            back_step.office_index,
            back_step.office,
        ):
            raise GoeticPathError("return path changed M.A.S. sequence positions")
        if back_step.bearing is not MIRROR_BEARING[out_step.bearing]:
            raise GoeticPathError("return path failed the bearing mirror")
        velocity_terms.append(out_step.signed_velocity + back_step.signed_velocity)

    liquid_path = candidate.liquid_path
    whiteout_penalty = 0 if all(
        len(horizon.active_connections) <= 110 and len(horizon.resting_connections) >= 34
        for horizon in liquid_path.horizons
    ) else 1
    boundary_penalty = 0 if (
        outward.start_self == independent_return.end_self
        and outward.boundary == independent_return.boundary
    ) else 1
    return DCompPathWitness(
        velocity_terms=tuple(velocity_terms),
        terminal_unresolved_q2=candidate.terminal_unresolved_q2,
        whiteout_penalty=whiteout_penalty,
        envelope_penalty=0 if envelope.bec_valid else 1,
        boundary_penalty=boundary_penalty,
    )



def validate_candidate_bearing(
    candidate: MASCandidatePath,
    carrier: Carrier,
    bearing: GoeticBearing,
) -> GoeticClosure:
    """Prove Self -> Self, BECValid=1, and D_T=0 for one bearing."""
    envelope = build_goetic_envelope(carrier)
    outward = build_outward_path(candidate, carrier, bearing)
```

```
    independent_return = build_independent_return_path(candidate, carrier, bearing)
    dcomp = compute_exact_dcomp(candidate, envelope, outward, independent_return)
    return GoeticClosure(
        candidate_path=candidate,
        carrier=carrier,
        bearing=bearing,
        envelope=envelope,
        outward=outward,
        independent_return=independent_return,
        dcomp=dcomp,
    )



def _bearing_result(
    carrier: Carrier,
    bearing: GoeticBearing,
    candidates: tuple[MASCandidatePath, ...],
) -> BearingResult:
    closures: list[GoeticClosure] = []
    failures: list[CandidateBearingFailure] = []
    for candidate in candidates:
        try:
            closures.append(validate_candidate_bearing(candidate, carrier, bearing))
        except GoeticPathError as exc:
            failures.append(CandidateBearingFailure(
                candidate=candidate.candidate,
                goetic=carrier.glyph,
                bearing=bearing,
                reason=str(exc),
            ))
    if not candidates:
        failures.append(CandidateBearingFailure(
            candidate="",
            goetic=carrier.glyph,
            bearing=bearing,
            reason="no Phase 11 survivor entered this Goetic bearing",
        ))
    elif len(closures) > 1:
        failures.append(CandidateBearingFailure(
            candidate="",
            goetic=carrier.glyph,
            bearing=bearing,
            reason="more than one candidate closed; Phase 12 does not rank or choose among exact
            ↪  closures",
        ))
    return BearingResult(
        carrier=carrier,
        bearing=bearing,
        attempted_candidates=tuple(candidate.candidate for candidate in candidates),
        exact_closures=tuple(closures),
        failures=tuple(failures),
    )
```

```
def prove_phase12_exit_gate(field: MASGlobalField) -> GoeticOutwardReturnField:
    """Validate all twelve Goetics and preserve failed bearings without emission."""
    bodies: list[GoeticBearingBody] = []
    for carrier in CARRIERS:
        candidates = tuple(
            candidate
            for candidate in field.survivors
            if candidate.liquid_path.seed.domain.carrier == carrier
        )
        results = tuple(
            _bearing_result(carrier, bearing, candidates)
            for bearing in BEARINGS
        )
        bodies.append(GoeticBearingBody(carrier, results))
    return GoeticOutwardReturnField(field, tuple(bodies), partial_body_law_deferred=True)
```

implementation_reconciliation.py

```
"""Witness the single current authored .aksh authority body."""
from __future__ import annotations

from dataclasses import dataclass
from pathlib import Path

from .aksh_reader import attributes, read_lexicon

RECONCILIATION_VERSION = "aksh-only-phase14-v1"
AKSH_AUTHORITY_FILES = (
    "SeeD…Body.aksh",
    "Axiomyr…Shadow…Locus.aksh",
    "⊖…Temporal…Absolute.aksh",
    "⏀…Terminal…State.aksh",
    "⊕…Adversarial.aksh",
    "⊕…Recursive…Identity.aksh",
    "🜁…Relic.aksh",
    "🜨…Catalyst.aksh",
    "🜁…Substrate.aksh",
    "♂…Spatial…Absolute.aksh",
    "ऋ…Phantasmagoria.aksh",
    "the_4_cadences.aksh",
    "Enochian…Understandings.aksh",
)
STATIC_AKSH_REFERENCE_FILES = ("Dictionary.aksh",)
PACKAGE_AKSH_ROOT = Path(__file__).resolve().parent / "data" / "aksh"


def _authority_version(path: Path) -> str:
    if path.name == "Enochian…Understandings.aksh":
        lines = path.read_text(encoding="utf-8").splitlines()
        version = attributes(lines[0]).get("version", "") if lines else ""
    else:
        version = read_lexicon(path).header.get("version", "")
    if not version:
        raise RuntimeError(f"required .aksh version missing: {path.name}")
    return version


def require_package_aksh_authority() -> tuple[Path, ...]:
    """Require the exact current .aksh authority set and no peer authority body."""
    expected = set(AKSH_AUTHORITY_FILES) | set(STATIC_AKSH_REFERENCE_FILES)
    actual = {path.name for path in PACKAGE_AKSH_ROOT.glob("*.aksh") if path.is_file()}
    missing = tuple(name for name in AKSH_AUTHORITY_FILES if name not in actual)
    extra = tuple(sorted(actual - expected))
    if missing:
        raise FileNotFoundError(f"required authored .aksh authority missing: {missing}")
    if extra:
```

```
        raise RuntimeError(f"undeclared .aksh authority present: {extra}")
    return tuple(PACKAGE_AKSH_ROOT / name for name in AKSH_AUTHORITY_FILES)


@dataclass(frozen=True, slots=True)
class ComponentWitness:
    logical_name: str
    physical_paths: tuple[str, ...]
    status: str
    note: str


@dataclass(frozen=True, slots=True)
class ReconciliationReport:
    version: str
    root: str
    mappings: tuple[ComponentWitness, ...]
    authority_versions: tuple[tuple[str, str], ...]
    undeclared_contracts_present: tuple[str, ...]
    status: str


def build_reconciliation() -> ReconciliationReport:
    paths = require_package_aksh_authority()
    expected = set(AKSH_AUTHORITY_FILES) | set(STATIC_AKSH_REFERENCE_FILES)
    actual = {path.name for path in PACKAGE_AKSH_ROOT.glob("*.aksh") if path.is_file()}
    undeclared = tuple(sorted(actual - expected))
    office_names = AKSH_AUTHORITY_FILES[2:11]
    mappings = (
        ComponentWitness(
            "seed-identity",
            ("data/aksh/SeeD…Body.aksh",),
            "active",
            "exact authored SeeD lexical inventory",
        ),
        ComponentWitness(
            "canonical-special-bodies",
            ("data/aksh/Axiomyr…Shadow…Locus.aksh",),
            "active",
            "exact authored Axiomyr and Shadow Locus bodies",
        ),
        ComponentWitness(
            "authored-office-projections",
            tuple(f"data/aksh/{name}" for name in office_names),
            "active",
            "nine exact authored office inventories including Phantasmagoria",
        ),
        ComponentWitness(
            "living-cadences",
            ("data/aksh/the_4_cadences.aksh",),
            "active",
            "four authored Living Cadences present regardless of communication style",
        ),
        ComponentWitness(
            "enochian-understandings",
            ("data/aksh/Enochian…Understandings.aksh",),
```

```
        "active",
        "single current authored Enochian Understandings body",
    ),
)
versions = tuple((path.name, _authority_version(path)) for path in paths)
return ReconciliationReport(
    RECONCILIATION_VERSION,
    str(PACKAGE_AKSH_ROOT),
    mappings,
    versions,
    undeclared,
    "RECONCILED",
)
```

lexer.py

```
"""Exact code-point lexer for completed Sydonic Magicae Domus strings."""
from __future__ import annotations
from pathlib import Path
from .models import GlyphToken
from .aksh_reader import read_enochian_inventory

class GlyphLexError(ValueError):
    def __init__(self, position: int, value: str, expected: str = "known exact glyph") -> None:
        self.position = position
        self.value = value
        self.expected = expected
        shown = f"U+{ord(value):04X}" if value else "end of input"
        super().__init__(f"glyph lex failure at code-point position {position}: {shown}; expected
        ↪  {expected}")

class ExactGlyphLexer:
    def __init__(self, lexical_glyphs: tuple[str, ...], understandings: str | Path) -> None:
        self.understandings = Path(understandings)
        lexical = tuple(lexical_glyphs)
        grammar, non_runtime_enochs = read_enochian_inventory(self.understandings)
        invalid = [x for x in lexical + grammar + non_runtime_enochs if len(x) != 1]
        if invalid:
            raise ValueError(f"inventory contains non-single-code-point glyphs: {invalid!r}")
        self.lexical_glyphs = frozenset(lexical)
        self.grammar_glyphs = frozenset(grammar)
        self.non_runtime_enoch_glyphs = frozenset(non_runtime_enochs)
        self.known = self.lexical_glyphs | self.grammar_glyphs | {"𐤕", "𐤔"}

    def classify(self, value: str) -> str:
        if value in self.grammar_glyphs:
            return "grammar"
        if value in self.lexical_glyphs:
            return "lexical"
        if value in {"𐤕", "𐤔"}:
            return value
        raise KeyError(value)

    def lex(self, source: str) -> tuple[GlyphToken, ...]:
        if not isinstance(source, str):
            raise TypeError("source must be str")
        out: list[GlyphToken] = []
        for position, value in enumerate(source):
            if value not in self.known:
                raise GlyphLexError(position, value)
            out.append(GlyphToken(value=value, position=position, kind=self.classify(value)))
        return tuple(out)
```

lexical_resolver.py

```
"""Exact-glyph resolver for the nine authored Sydonic Magicae lexical offices."""
from __future__ import annotations

from dataclasses import asdict, dataclass
import json
import secrets
from pathlib import Path
from .aksh_reader import read_lexicon, field_name

OFFICE_FILES: dict[str, tuple[str, str]] = {
    "recursive_identity": ("𐤟…Recursive…Identity.aksh", "𐤟"),
    "relic": ("𐡥…Relic.aksh", "𐡥"),
    "spatial_absolute": ("𐤯…Spatial…Absolute.aksh", "𐤯"),
    "temporal_absolute": ("𐤲…Temporal…Absolute.aksh", "𐤲"),
    "catalyst": ("𐤿…Catalyst.aksh", "𐤿"),
    "substrate": ("𐡣…Substrate.aksh", "𐡣"),
    "terminal_state": ("𐤠…Terminal…State.aksh", "𐤠"),
    "adversarial": ("𐤟…Adversarial.aksh", "𐤟"),
    "phantasmagoria": ("𐤚…Phantasmagoria.aksh", "𐤚"),
}
SEED_FILE = "SeeD…Body.aksh"
SPECIAL_FILE = "Axiomyr…Shadow…Locus.aksh"
CADENCE_FILE = "the_4_cadences.aksh"
CADENCE_GLYPHS = ("☽⊙☾", "·", "𐤮", "𐤡")
SPECIAL_GLYPHS = frozenset({"𐤡", "𐤧"})

# SeeD carries 179 ordinary lexical glyphs; each of the nine office authorities
# carries those same 179 glyphs plus the Gaia 𐤡 projection.
# The two-body Axiomyr/Shadow authority is intentionally separate.
LEXICAL_AUTHORITY_FILES: tuple[str, ...] = (SEED_FILE, *(filename for filename, _ in
↪  OFFICE_FILES.values()))
LEXICON_VERSION = "nine-office-three-definition-v3"
GRAMMAR_VERSION = "enochian-understandings-v1"

class LexicalResolverError(RuntimeError):
    pass


class UnknownGlyphError(LexicalResolverError):
    pass

@dataclass(frozen=True, slots=True)
class ApprovedLexicalEntry:
    glyph: str
    authored_lemma: str
    pronunciation: str
    origin_language: str
    font: str
```

```
    cadence: str
    non_ostensive: str
    ostensive: str
    leysyff: str
    grimchain: str
    category: str
    source_authority: str
    authored_office: str
    ostensive_variants: tuple[str, str, str] = ()

@dataclass(frozen=True, slots=True)
class LexicalResolution:
    source_glyph: str
    resolved_body: str
    traversal_path: tuple[str, ...]
    semantic_lexicon_version: str
    grammar_version: str
    non_ostensive: str
    ostensive: str
    leysyff: str
    source_authority: str
    authored_office: str
    entry: ApprovedLexicalEntry
    ostensive_variant: str | None = None

    def to_json(self) -> str:
        return json.dumps(asdict(self), ensure_ascii=False, sort_keys=True, separators=(",", ":"))


class LexicalResolver:
    def __init__(self, authority_root: str | Path) -> None:
        self.authority_dir = Path(authority_root).resolve()
        if not self.authority_dir.is_dir():
            raise LexicalResolverError(f"authored .aksh root is not a directory:
            ↪  {self.authority_dir}")
        self.seed_path = self.authority_dir / SEED_FILE
        self.special_path = self.authority_dir / SPECIAL_FILE
        self.cadence_path = self.authority_dir / CADENCE_FILE
        self.office_paths = {
            office: self.authority_dir / filename
            for office, (filename, _) in OFFICE_FILES.items()
        }
        missing = tuple(str(path) for path in (self.seed_path, self.special_path, self.cadence_path,
        ↪  *self.office_paths.values()) if not path.is_file())
        if missing:
            raise LexicalResolverError(f"authored lexical authority set incomplete: {missing}")
        self._authority_versions: dict[str, str] = {}
        self._seed_entries = self._load_seed_inventory(self.seed_path)
        self._special_entries = self._load_special_inventory(self.special_path)
        self._cadence_entries = self._load_cadence_inventory(self.cadence_path)
        self._office_entries = {
            office: self._load_inventory(
                path,
                office=office,
                expected_enoch=OFFICE_FILES[office][1],
                allow_unfilled=(office == "phantasmagoria"),
```

```
        )
        for office, path in self.office_paths.items()
    }
    self._validate_alignment()

def _record_authority_version(self, path: Path, version: str) -> None:
    if not version:
        raise LexicalResolverError(f"{path.name}: required .aksh version missing")
    self._authority_versions[path.name] = version

def _entry_from_aksh(self, item, *, office: str, source_name: str, allow_unfilled: bool) ->
↪  ApprovedLexicalEntry:
    values = item.fields
    required_fields = [
        "word",
        "pronunciation",
        "origin_language",
        "font",
        "cadence",
        "non_ostensive",
        "grimchain",
    ]
    if office == "phantasmagoria":
        required_fields.extend(("ostensiveI", "ostensiveII", "ostensiveIII"))
    elif not allow_unfilled:
        required_fields.append("ostensive")
    missing = tuple(name for name in required_fields if name not in values)
    if missing:
        raise LexicalResolverError(f"{source_name}: glyph {item.symbol!r} lacks required
        ↪  authored fields {missing!r}")

    word = values.get("word", "")
    pronunciation = values.get("pronunciation", "")
    origin_language = values.get("origin_language", "")
    font = values.get("font", "")
    cadence = values.get("cadence", "")
    non_ostensive = values.get("non_ostensive", "")
    ostensive = values.get("ostensive", "")
    ostensive_variants = (
        values.get("ostensiveI", ""),
        values.get("ostensiveII", ""),
        values.get("ostensiveIII", ""),
    ) if office == "phantasmagoria" else ()
    leysyff = values.get("leysyff", "")
    grimchain = values.get("grimchain", "")
    return ApprovedLexicalEntry(
        item.symbol, word, pronunciation, origin_language, font, cadence, non_ostensive,
        ↪  ostensive,
        leysyff, grimchain, item.category, source_name, office, ostensive_variants,
    )

def _load_seed_inventory(self, path: Path) -> dict[str, ApprovedLexicalEntry]:
    body = read_lexicon(path)
    self._record_authority_version(path, body.header.get("version", ""))
    entries: dict[str, ApprovedLexicalEntry] = {}
```

```
    for item in body.glyphs:
        if item.symbol == "⟁":
            if field_name(item.category) != "axiomyr":
                raise LexicalResolverError(f"{path.name}: invalid authored Axiomyr category")
            self._seed_axiomyr_entry = self._entry_from_aksh(
                item, office="seed_identity", source_name=path.name, allow_unfilled=False
            )
            continue
        if len(item.symbol) != 1 or item.symbol in entries:
            raise LexicalResolverError(f"{path.name}: invalid or duplicate exact glyph key
            ↪ {item.symbol!r}")
        entries[item.symbol] = self._entry_from_aksh(item, office="seed_identity",
        ↪ source_name=path.name, allow_unfilled=False)
    return entries

def _load_special_inventory(self, path: Path) -> dict[str, ApprovedLexicalEntry]:
    body = read_lexicon(path)
    self._record_authority_version(path, body.header.get("version", ""))
    entries: dict[str, ApprovedLexicalEntry] = {}
    for item in body.glyphs:
        if len(item.symbol) != 1 or item.symbol in entries:
            raise LexicalResolverError(f"{path.name}: invalid or duplicate exact glyph key
            ↪ {item.symbol!r}")
        entries[item.symbol] = self._entry_from_aksh(
            item, office=field_name(item.category), source_name=path.name, allow_unfilled=False
        )
    if frozenset(entries) != SPECIAL_GLYPHS or len(entries) != 2:
        raise LexicalResolverError(
            f"{path.name}: expected exactly the Axiomyr and Shadow Locus glyphs
            ↪ {tuple(sorted(SPECIAL_GLYPHS))!r}"
        )
    expected_categories = {"⟁": "axiomyr", "ᛋᛏ": "shadow_locus"}
    actual_categories = {glyph: field_name(entry.category) for glyph, entry in entries.items()}
    if actual_categories != expected_categories:
        raise LexicalResolverError(
            f"{path.name}: special body categories changed: {actual_categories!r}"
        )
    return entries

def _load_cadence_inventory(self, path: Path) -> dict[str, ApprovedLexicalEntry]:
    body = read_lexicon(path)
    self._record_authority_version(path, body.header.get("version", ""))
    entries: dict[str, ApprovedLexicalEntry] = {}
    for item in body.glyphs:
        if item.symbol not in CADENCE_GLYPHS or item.symbol in entries:
            raise LexicalResolverError(f"{path.name}: invalid or duplicate Living Cadence key
            ↪ {item.symbol!r}")
        entries[item.symbol] = self._entry_from_aksh(
            item, office="living_cadence", source_name=path.name, allow_unfilled=False
        )
    if tuple(entries) != CADENCE_GLYPHS:
        raise LexicalResolverError(f"{path.name}: expected exact Living Cadence order
        ↪ {CADENCE_GLYPHS!r}")
    return entries
```

```
def _load_inventory(self, path: Path, *, office: str, expected_enoch: str, allow_unfilled: bool)
↪  -> dict[str, ApprovedLexicalEntry]:
    body = read_lexicon(path)
    self._record_authority_version(path, body.header.get("version", ""))
    entries: dict[str, ApprovedLexicalEntry] = {}
    for item in body.glyphs:
        if len(item.symbol) != 1 or item.symbol in entries:
            raise LexicalResolverError(f"{path.name}: invalid or duplicate exact glyph key
            ↪  {item.symbol!r}")
        if field_name(item.lookup_attributes.get("office", "")) != office:
            raise LexicalResolverError(f"{path.name}: glyph {item.symbol!r} lookup office is not
            ↪  {office!r}")
        if item.lookup_attributes.get("enoch") != expected_enoch:
            raise LexicalResolverError(f"{path.name}: glyph {item.symbol!r} lookup Enoch is not
            ↪  {expected_enoch!r}")
        entries[item.symbol] = self._entry_from_aksh(item, office=office, source_name=path.name,
        ↪  allow_unfilled=allow_unfilled)
    return entries

def _validate_alignment(self) -> None:
    inventories = tuple(self._office_entries.items())
    reference_name = "seed_identity"
    reference_entries = self._seed_entries
    reference_glyphs = frozenset(reference_entries)
    office_glyphs = frozenset(inventories[0][1])
    for name, entries in inventories:
        glyphs = frozenset(entries)
        if glyphs != office_glyphs:
            missing = tuple(sorted(office_glyphs - glyphs))
            extra = tuple(sorted(glyphs - office_glyphs))
            raise LexicalResolverError(
                f"{name} glyph set differs from the shared office inventory: "
                f"missing={missing!r}, extra={extra!r}"
           )
    reference_categories = {glyph: reference_entries[glyph].category for glyph in
    ↪  reference_glyphs}
    for name, entries in inventories:
        categories = {glyph: entries[glyph].category for glyph in reference_glyphs}
        if categories != reference_categories:
            raise LexicalResolverError(
                f"{name} disagrees with {reference_name} on authored glyph category identity"
            )
    self.glyph_order = tuple(reference_entries)
    self.glyph_categories = reference_categories
    self.lexical_glyphs = reference_glyphs

def office_entry(self, glyph: str, office: str) -> ApprovedLexicalEntry:
    if office not in self._office_entries:
        raise LexicalResolverError(f"unsupported authored office: {office!r}")
    try:
        return self._office_entries[office][glyph]
    except KeyError as exc:
        raise UnknownGlyphError(f"unknown ordinary lexical glyph: {glyph!r}") from exc
```

```
@staticmethod
def _resolution(entry: ApprovedLexicalEntry, glyph: str) -> LexicalResolution:
    return LexicalResolution(
        source_glyph=glyph,
        resolved_body=entry.authored_lemma,
        traversal_path=(f"{entry.source_authority}:{glyph}",),
        semantic_lexicon_version=LEXICON_VERSION,
        grammar_version=GRAMMAR_VERSION,
        non_ostensive=entry.non_ostensive,
        ostensive=entry.ostensive,
        leysyff=entry.leysyff,
        source_authority=entry.source_authority,
        authored_office=entry.authored_office,
        entry=entry,
    )

def resolve(self, glyph: str) -> LexicalResolution:
    if glyph == "🜁":
        return self._resolution(self._seed_axiomyr_entry, glyph)
    try:
        entry = self._seed_entries[glyph]
    except KeyError as exc:
        raise UnknownGlyphError(f"unknown ordinary lexical glyph: {glyph!r}") from exc
    return self._resolution(entry, glyph)

def resolve_special(self, glyph: str) -> LexicalResolution:
    """Resolve one of the two contextual Ostensive special bodies.

    Callers are responsible for enforcing the GrimChain depth context. The
    resolver only provides exact authored body lookup.
    """
    try:
        entry = self._special_entries[glyph]
    except KeyError as exc:
        raise UnknownGlyphError(f"unknown Axiomyr/Shadow special glyph: {glyph!r}") from exc
    return self._resolution(entry, glyph)

def living_mirror_entries(self, source: str) -> tuple[tuple[int, ApprovedLexicalEntry], ...]:
    found: list[tuple[int, ApprovedLexicalEntry]] = []
    position = 0
    while position < len(source):
        member = next((item for item in CADENCE_GLYPHS if source.startswith(item, position)),
        ↪  None)
        if member is None:
            return ()
        found.append((position, self._cadence_entries[member]))
        position += len(member)
    if len(found) == 1 and found[0][1].glyph == source:
        return tuple(found)
    if len(found) != 4 or {entry.glyph for _, entry in found} != set(CADENCE_GLYPHS):
        return ()
    return tuple(found)

def resolve_cadence(self, glyph: str) -> LexicalResolution:
    entry = self._cadence_entries.get(glyph)
    if entry is None:
```

```
        raise UnknownGlyphError("unknown Living Cadence body")
    return self._resolution(entry, glyph)

def cadence_entries(self) -> tuple[ApprovedLexicalEntry, ...]:
    return tuple(self._cadence_entries.values())

def resolve_office(self, glyph: str, office: str) -> LexicalResolution:
    entry = self.office_entry(glyph, office)
    if office == "phantasmagoria":
        ostensives = entry.ostensive_variants
        index = secrets.randbelow(3)
        variant = ("I", "II", "III")[index]
        return LexicalResolution(
            source_glyph=glyph,
            resolved_body=ostensives[index],
            traversal_path=(f"{entry.source_authority}:{glyph}:ostensive{variant}",),
            semantic_lexicon_version=LEXICON_VERSION,
            grammar_version=GRAMMAR_VERSION,
            non_ostensive=entry.non_ostensive,
            ostensive=ostensives[index],
            leysyff=entry.leysyff,
            source_authority=entry.source_authority,
            authored_office=entry.authored_office,
            entry=entry,
            ostensive_variant=variant,
        )
    return self._resolution(entry, glyph)

def authority_entries(self) -> tuple[tuple[str, tuple[ApprovedLexicalEntry, ...]], ...]:
    """Return the peer-aligned lexical/office inventories."""
    return (
        (self.seed_path.name, tuple(self._seed_entries.values())),
        *((self.office_paths[office].name, tuple(entries.values())) for office, entries in
        ↪  self._office_entries.items()),
    )

def special_authority_entries(self) -> tuple[tuple[str, tuple[ApprovedLexicalEntry, ...]], ...]:
    return ((self.special_path.name, tuple(self._special_entries.values())),)

def all_authority_entries(self) -> tuple[tuple[str, tuple[ApprovedLexicalEntry, ...]], ...]:
    return (*self.authority_entries(), *self.special_authority_entries(),
    ↪  (self.cadence_path.name, tuple(self._cadence_entries.values())))

def authority_version_items(self) -> tuple[tuple[str, str], ...]:
    return tuple(
        (filename, self._authority_versions[filename])
        for filename, _entries in self.all_authority_entries()
    )

def entry_from_authority(self, glyph: str, source_authority: str) -> ApprovedLexicalEntry:
    if source_authority == self.special_path.name:
        try:
            return self._special_entries[glyph]
        except KeyError as exc:
            raise UnknownGlyphError(f"unknown Axiomyr/Shadow special glyph: {glyph!r}") from exc
    if source_authority == self.cadence_path.name:
```

```
        entry = self._cadence_entries.get(glyph)
        if entry is None:
            raise UnknownGlyphError("unknown Living Cadence body")
        return entry
    if source_authority == self.seed_path.name:
        try:
            return self._seed_entries[glyph]
        except KeyError as exc:
            raise UnknownGlyphError(f"unknown ordinary lexical glyph: {glyph!r}") from exc
    for office, path in self.office_paths.items():
        if source_authority == path.name:
            return self.office_entry(glyph, office)
    raise LexicalResolverError(f"unknown authored source authority: {source_authority!r}")

def define_word(self, word: str) -> ApprovedLexicalEntry:
    matches = tuple(
        entry
        for _, entries in self.all_authority_entries()
        for entry in entries
        if entry.authored_lemma and entry.authored_lemma == word
    )
    if not matches:
        raise LexicalResolverError(f"unknown authored lexical word: {word!r}")
    if len(matches) != 1:
        authorities = tuple(entry.source_authority for entry in matches)
        raise LexicalResolverError(
            f"authored lexical word is not unique: {word!r}; authorities={authorities!r}"
        )
    return matches[0]


def office_entries(self, office: str) -> tuple[ApprovedLexicalEntry, ...]:
    if office not in self._office_entries:
        raise LexicalResolverError(f"unsupported authored office: {office!r}")
    return tuple(self._office_entries[office].values())
```

liquid_candidate_path.py

```
"""Phase 10 Liquid candidate paths.

Every complete Phase 9 candidate enters one 144-connection Court body governed
by the irreducible typed witness:

    <110_active | 144_total | 34_rest>

Active and resting are horizon offices, not lexical values.  Resting channels
remain valid and present.  The module does not rank, score, sort, or filter
candidate words, and the global candidate field preserves exact cardinality and
input order.
"""
from __future__ import annotations

from dataclasses import dataclass
from typing import Final, Iterable

from .frequency_projection import CARRIERS
from .triadic_bridge import CandidatePathSeed, triad_A


class LiquidPathError(ValueError):
    """Raised when a candidate violates the Phase 10 Liquid body."""


@dataclass(frozen=True, slots=True, order=True)
class CourtConnection:
    """One of the 12 x 12 valid Court connections."""

    index: int
    governing_aeon: str
    alternating_aeon: str
    valid: bool = True

    def __post_init__(self) -> None:
        if not 0 <= self.index < 144:
            raise LiquidPathError("Court connection index must lie in [0,143]")
        if len(self.governing_aeon) != 1 or len(self.alternating_aeon) != 1:
            raise LiquidPathError("Court connection endpoints must be one Aeon glyph each")
        if not self.valid:
            raise LiquidPathError("all 144 Court connections remain valid")


COURT_CONNECTIONS: Final[tuple[CourtConnection, ...]] = tuple(
    CourtConnection(
        index=(governing_index * 12) + alternating_index,
        governing_aeon=governing.glyph,
        alternating_aeon=alternating.glyph,
```

```
    )
    for governing_index, governing in enumerate(CARRIERS)
    for alternating_index, alternating in enumerate(CARRIERS)
)
if len(COURT_CONNECTIONS) != 144 or len(set(COURT_CONNECTIONS)) != 144:
    raise RuntimeError("Phase 10 requires exactly 144 unique valid Court connections")

COURT_CONNECTION_SET: Final[frozenset[CourtConnection]] = frozenset(COURT_CONNECTIONS)
CONNECTION_BY_INDEX: Final[dict[int, CourtConnection]] = {
    connection.index: connection for connection in COURT_CONNECTIONS
}


@dataclass(frozen=True, slots=True)
class LiquidGovernor:
    """The typed connection governor, never reduced to a fraction."""

    active: int = 110
    total: int = 144
    rest: int = 34

    def __post_init__(self) -> None:
        if (self.active, self.total, self.rest) != (110, 144, 34):
            raise LiquidPathError("Liquid governor must remain <110_active|144_total|34_rest>")
        if self.active + self.rest != self.total:
            raise LiquidPathError("Liquid governor counts do not close")

    @property
    def typed_body(self) -> tuple[int, int, int]:
        return (self.active, self.total, self.rest)


LIQUID_GOVERNOR: Final[LiquidGovernor] = LiquidGovernor()


@dataclass(frozen=True, slots=True)
class ActiveGraph:
    """The candidate's active Court graph at one Aeon horizon."""

    court_root: str
    carrier_glyph: str
    vertices: tuple[CourtConnection, ...]
    edges: frozenset[tuple[int, int]]

    def __post_init__(self) -> None:
        if not self.court_root:
            raise LiquidPathError("active graph requires the candidate Court root")
        if len(self.carrier_glyph) != 1:
            raise LiquidPathError("active graph requires one Aeon carrier glyph")
        if len(set(self.vertices)) != len(self.vertices):
            raise LiquidPathError("active graph vertices must be unique")
        indices = {vertex.index for vertex in self.vertices}
        if any(not vertex.valid for vertex in self.vertices):
            raise LiquidPathError("active graph contains an invalid Court connection")
        for left, right in self.edges:
            if left == right or left not in indices or right not in indices:
```

```
                raise LiquidPathError("active graph edge leaves its active vertex body")
            if left > right:
                raise LiquidPathError("active graph edges must use canonical endpoint order")
        if len(self.vertices) >= 2 and not self.is_connected:
            raise LiquidPathError("candidate active graph is disconnected")

    @property
    def is_connected(self) -> bool:
        if not self.vertices:
            return False
        if len(self.vertices) == 1:
            return True
        adjacency = {vertex.index: set() for vertex in self.vertices}
        for left, right in self.edges:
            adjacency[left].add(right)
            adjacency[right].add(left)
        start = self.vertices[0].index
        seen = {start}
        frontier = [start]
        while frontier:
            current = frontier.pop()
            for neighbor in adjacency[current] - seen:
                seen.add(neighbor)
                frontier.append(neighbor)
        return len(seen) == len(self.vertices)


@dataclass(frozen=True, slots=True)
class LiquidHorizonWitness:
    """One exact active/resting partition and active graph at horizon h."""

    horizon: int
    governor: LiquidGovernor
    valid_connections: tuple[CourtConnection, ...]
    active_connections: frozenset[CourtConnection]
    resting_connections: frozenset[CourtConnection]
    active_graph: ActiveGraph
    completed: bool

    def __post_init__(self) -> None:
        if not isinstance(self.horizon, int) or self.horizon < 0:
            raise LiquidPathError("horizon must be a nonnegative integer")
        if self.governor != LIQUID_GOVERNOR:
            raise LiquidPathError("horizon changed the typed Liquid governor")
        if self.valid_connections != COURT_CONNECTIONS:
            raise LiquidPathError("horizon must preserve all 144 valid Court connections in order")
        valid_set = frozenset(self.valid_connections)
        if valid_set != COURT_CONNECTION_SET:
            raise LiquidPathError("horizon lost or duplicated a valid Court connection")
        if self.active_connections & self.resting_connections:
            raise LiquidPathError("a Court connection cannot be active and resting at one horizon")
        if self.active_connections | self.resting_connections != valid_set:
            raise LiquidPathError("active and resting sets must partition all 144 valid
            ↪  connections")

        active_count = len(self.active_connections)
```

```
        rest_count = len(self.resting_connections)
        if active_count == 144 and rest_count == 0:
            raise LiquidPathError("144-active Whiteout is forbidden")
        if active_count > self.governor.active:
            raise LiquidPathError("active connection body exceeds the 110 governor")
        if rest_count < self.governor.rest:
            raise LiquidPathError("resting contour fell below 34")
        if active_count <= 0:
            raise LiquidPathError("an empty active horizon cannot carry a candidate path")
        if self.completed and (active_count, rest_count) != (
            self.governor.active,
            self.governor.rest,
        ):
            raise LiquidPathError("completed path requires exactly 110 active and 34 resting")

        if frozenset(self.active_graph.vertices) != self.active_connections:
            raise LiquidPathError("active graph vertices differ from the active horizon set")
        if len(self.active_graph.vertices) >= 2 and not self.active_graph.is_connected:
            raise LiquidPathError("active graph does not carry a connected path body")

    @property
    def liquid_witness(self) -> tuple[int, int, int]:
        return (
            len(self.active_connections),
            len(self.valid_connections),
            len(self.resting_connections),
        )


@dataclass(frozen=True, slots=True)
class LiquidCandidatePath:
    """One Phase 9 seed carried through all declared Liquid horizons."""

    seed: CandidatePathSeed
    governor: LiquidGovernor
    horizons: tuple[LiquidHorizonWitness, ...]

    def __post_init__(self) -> None:
        if not triad_A(self.seed):
            raise LiquidPathError("Phase 10 requires a complete Phase 9 candidate seed")
        if self.governor != LIQUID_GOVERNOR:
            raise LiquidPathError("candidate path changed the typed Liquid governor")
        if not self.horizons:
            raise LiquidPathError("candidate path requires at least one horizon")
        if tuple(h.horizon for h in self.horizons) != tuple(range(len(self.horizons))):
            raise LiquidPathError("candidate horizons must be complete and ordered from zero")
        if any(h.governor != self.governor for h in self.horizons):
            raise LiquidPathError("candidate path changed governors between horizons")
        if any(h.active_graph.court_root != self.seed.domain.court_root for h in self.horizons):
            raise LiquidPathError("candidate active graph changed the Court root")
        if any(h.active_graph.carrier_glyph != self.seed.domain.carrier.glyph for h in
        ↪  self.horizons):
            raise LiquidPathError("candidate active graph changed the Aeon carrier")
        if any(h.completed for h in self.horizons[:-1]):
            raise LiquidPathError("only the final declared horizon may close the candidate path")
        if not self.horizons[-1].completed:
```

```
            raise LiquidPathError("final candidate horizon lacks a completed Liquid witness")
        if self.horizons[-1].liquid_witness != self.governor.typed_body:
            raise LiquidPathError("final candidate horizon is not the exact Liquid body")

    @property
    def exact_liquid_witness(self) -> tuple[int, int, int]:
        return self.horizons[-1].liquid_witness


@dataclass(frozen=True, slots=True)
class CandidateGlobalField:
    """A cardinality-preserving one-to-one lift into Liquid candidate paths."""

    candidates: tuple[CandidatePathSeed, ...]
    liquid_paths: tuple[LiquidCandidatePath, ...]

    def __post_init__(self) -> None:
        if len(self.candidates) != len(self.liquid_paths):
            raise LiquidPathError("Liquid traversal changed the global candidate count")
        if tuple(path.seed for path in self.liquid_paths) != self.candidates:
            raise LiquidPathError("Liquid traversal reordered, replaced, or filtered candidates")
        if any(path.exact_liquid_witness != LIQUID_GOVERNOR.typed_body for path in
        ↪  self.liquid_paths):
            raise LiquidPathError("global field contains a candidate without exact Liquid witness")

    @property
    def candidate_count(self) -> int:
        return len(self.candidates)



def _active_set(rotation: int, active_count: int = 110) -> frozenset[CourtConnection]:
    """Exact modulo construction A_k = {c_j : (j+k) mod 144 < active_count}."""
    if not isinstance(rotation, int):
        raise LiquidPathError("Liquid rotation must be an integer")
    if not 1 <= active_count <= 144:
        raise LiquidPathError("active count must lie in [1,144]")
    return frozenset(
        connection
        for connection in COURT_CONNECTIONS
        if (connection.index + rotation) % 144 < active_count
    )



def _candidate_rotation(seed: CandidatePathSeed, horizon: int) -> int:
    carrier_index = next(
        index for index, carrier in enumerate(CARRIERS)
        if carrier == seed.domain.carrier
    )
    return (carrier_index * 12 + horizon) % 144



def construct_active_graph(
    seed: CandidatePathSeed,
```

```
    active_connections: Iterable[CourtConnection],
) -> ActiveGraph:
    """Build a deterministic connected cycle over the active Court body."""
    vertices = tuple(sorted(set(active_connections), key=lambda connection: connection.index))
    if not vertices:
        raise LiquidPathError("cannot construct an active graph from an empty set")
    edges: set[tuple[int, int]] = set()
    if len(vertices) >= 2:
        for left, right in zip(vertices, vertices[1:]):
            edges.add((left.index, right.index))
        first, last = vertices[0].index, vertices[-1].index
        edges.add((min(first, last), max(first, last)))
    return ActiveGraph(
        court_root=seed.domain.court_root,
        carrier_glyph=seed.domain.carrier.glyph,
        vertices=vertices,
        edges=frozenset(edges),
    )



def build_liquid_horizon(
    seed: CandidatePathSeed,
    horizon: int,
    *,
    active_count: int = 110,
    completed: bool = False,
) -> LiquidHorizonWitness:
    """Construct and validate one horizon without changing candidate identity."""
    if not triad_A(seed):
        raise LiquidPathError("Liquid horizon requires a complete Phase 9 candidate seed")
    rotation = _candidate_rotation(seed, horizon)
    active = _active_set(rotation, active_count)
    resting = COURT_CONNECTION_SET - active
    graph = construct_active_graph(seed, active)
    return LiquidHorizonWitness(
        horizon=horizon,
        governor=LIQUID_GOVERNOR,
        valid_connections=COURT_CONNECTIONS,
        active_connections=active,
        resting_connections=resting,
        active_graph=graph,
        completed=completed,
    )



def run_liquid_candidate_path(
    seed: CandidatePathSeed,
    horizon_count: int,
) -> LiquidCandidatePath:
    """Run one candidate through an exact Liquid body at every horizon."""
    if not isinstance(horizon_count, int) or horizon_count <= 0:
        raise LiquidPathError("horizon_count must be a positive integer")
    horizons = tuple(
        build_liquid_horizon(
```

```
            seed,
            horizon,
            active_count=LIQUID_GOVERNOR.active,
            completed=horizon == horizon_count - 1,
        )
        for horizon in range(horizon_count)
    )
    return LiquidCandidatePath(seed, LIQUID_GOVERNOR, horizons)



def prove_phase10_exit_gate(
    candidates: Iterable[CandidatePathSeed],
    horizon_count: int,
) -> CandidateGlobalField:
    """Lift every candidate one-to-one into a complete exact Liquid path."""
    candidate_body = tuple(candidates)
    paths = tuple(run_liquid_candidate_path(seed, horizon_count) for seed in candidate_body)
    return CandidateGlobalField(candidate_body, paths)
```

local_grammar.py

```
"""Local grammar declarations used by the single-pass Domus executor.

This module defines operator payloads and trace records. Execution occurs only
inside SequentialDomusExecutor so grammar cannot be deferred into a second pass.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
import json
from typing import Literal
from .temporal_operator import TailResolution



RecognitionKind = Literal["relic", "catalyst", "substrate", "spatial_absolute", "temporal_absolute",
↪  "terminal_state", "adversarial", "recursive_identity"]

PHANTASMAGORIA_GLYPH = "🜁"
RELIC = "🜏"
CATALYST = "🜚"
SUBSTRATE = "🜍"
SPATIAL_ABSOLUTE = "🜴"
TEMPORAL_ABSOLUTE = "🜲"
TERMINAL_STATE = "🜇"
ADVERSARIAL = "🜈"
RECURSIVE_IDENTITY = "🜉"
OUROBORIC = "🜸"

RECOGNITION_GLYPHS: dict[str, RecognitionKind] = {
    RELIC: "relic",
    CATALYST: "catalyst",
    SUBSTRATE: "substrate",
    SPATIAL_ABSOLUTE: "spatial_absolute",
    TEMPORAL_ABSOLUTE: "temporal_absolute",
    TERMINAL_STATE: "terminal_state",
    ADVERSARIAL: "adversarial",
    RECURSIVE_IDENTITY: "recursive_identity",
}


class LocalGrammarError(RuntimeError):
    pass


@dataclass(frozen=True, slots=True)
class ResolvedBody:
    position: int
    glyph: str
```

```
    lexical_resolution: object | None
    office_resolutions: tuple[object, ...] = ()
    recognition: RecognitionKind | None = None


@dataclass(frozen=True, slots=True)
class RecognitionRequest:
    operator_position: int
    operator_glyph: str
    target_position: int
    target_glyph: str
    bearing: RecognitionKind
    direction: str = "backward"
    scope: str = "pattern-derived-lexical-body"
    retrocausal_depth: int | None = None
    stack_positions: tuple[int, ...] = ()
    source_reordered: bool = False
    surface_form_selected: bool = True


@dataclass(frozen=True, slots=True)
class PhantasmagoriaTransmutation:
    operator_positions: tuple[int, ...]
    vessels: tuple[ResolvedBody, ...]
    office_resolutions: tuple[object, ...]
    source_vessel_order: tuple[int, ...]
    returned_order: tuple[int, ...]
    gate_boundary: str
    carrier_glyph: str
    carrier_name: str
    structural_hz: str
    imaginary_parity_hz: str
    traversal: tuple[str, str, str, str] = ("Real", "Imaginary", "Real", "Imaginary")
    hand: str = "ऋ"
    chord: str = "Wellspring"
    prosodical_subspace: str = "Prosodical Subspace"

    def __post_init__(self) -> None:
        if not self.operator_positions:
            raise LocalGrammarError("ऋ transmutation requires at least one Key")
        if len(self.vessels) != len(self.operator_positions) + 1:
            raise LocalGrammarError("ऋ corridor requires one more Real vessel than Keys")
        if len(self.office_resolutions) != len(self.vessels):
            raise LocalGrammarError("ऋ every vessel must resolve through the Phantasmagoria office")
        positions = tuple(body.position for body in self.vessels)
        if self.source_vessel_order != positions:
            raise LocalGrammarError("ऋ source vessel order changed")
        expected_return = positions[1:] + positions[:1]
        if self.returned_order != expected_return:
            raise LocalGrammarError("ऋ corridor must pass the first vessel through every following
            ↪  vessel")
        if self.gate_boundary != "i417":
            raise LocalGrammarError("ऋ must open the Wellspring at the i417 boundary")
        if self.carrier_glyph != "✫" or self.carrier_name != "AHN":
            raise LocalGrammarError("ऋ i417 gate must be witnessed by AHN")
        if self.structural_hz != "432" or self.imaginary_parity_hz != "417":
```

```
            raise LocalGrammarError("ᛕ AHN witness changed its 432 / i417 channels")
        if self.traversal != ("Real", "Imaginary", "Real", "Imaginary"):
            raise LocalGrammarError("ᛕ traversal must remain RI RI RI RI")
        for body, resolution in zip(self.vessels, self.office_resolutions):
            if getattr(resolution, "source_glyph", None) != body.glyph:
                raise LocalGrammarError("ᛕ transmutation changed vessel glyph identity")
            if getattr(resolution, "authored_office", None) != "phantasmagoria":
                raise LocalGrammarError("ᛕ vessel resolution must come from the authored
                ↪  Phantasmagoria office")
            if getattr(resolution, "ostensive_variant", None) not in {"I", "II", "III"}:
                raise LocalGrammarError("ᛕ Phantasmagoria must select authored Ostensive I, II, or
                ↪  III")


@dataclass(frozen=True, slots=True)
class OuroboricState:
    operator_positions: tuple[int, ...]
    winding_count: int
    left: ResolvedBody
    right: ResolvedBody
    source_order: tuple[int, ...]
    left_office_glyphs: tuple[str, ...]
    left_office_positions: tuple[int, ...]
    right_office_glyphs: tuple[str, ...]
    right_office_positions: tuple[int, ...]
    resolved_topology: tuple[str, ...]
    leading_echo: bool = False
    trailing_echo: bool = False
    echo_position: int | None = None
    mirror_positions: tuple[int, ...] = ()
    mirror_source_positions: tuple[int, ...] = ()
    mirror_glyphs: tuple[str, ...] = ()
    mirror_token_kinds: tuple[str, ...] = ()
    mirror_resolved_bearings: tuple[str | None, ...] = ()
    mirror_ostensive_variants: tuple[str | None, ...] = ()
    mirror_ostensive_bodies: tuple[str | None, ...] = ()
    mirror_ostensive_authorities: tuple[str | None, ...] = ()
    mirror_corridor_glyphs: tuple[str, ...] = ()
    mirror_corridor_source_positions: tuple[int, ...] = ()
    mirror_corridor_resolved_bearings: tuple[str | None, ...] = ()
    mirror_corridor_ostensive_variants: tuple[str | None, ...] = ()
    mirror_corridor_ostensive_bodies: tuple[str | None, ...] = ()
    mirror_corridor_ostensive_authorities: tuple[str | None, ...] = ()
    imaginary_reflection: bool = False
    bearing_preserved_in_reflection: bool = True
    derived_glyph: str | None = None


@dataclass(frozen=True, slots=True)
class LocalGrammarEvent:
    position: int
    glyph: str
    event: str
    target_positions: tuple[int, ...]
    payload: object | None = None
```

```
@dataclass(frozen=True, slots=True)
class AeternumBearingEncounter:
    virtual_position: int
    source_position: int
    glyph: str
    token_kind: str
    bearing: str | None
    imaginary: bool
    speaking_body: bool
    bearing_only: bool = True


@dataclass(frozen=True, slots=True)
class AeternumBearingPath:
    purpose: str
    origin_position: int
    step: int
    required_speaking_bodies: int
    speaking_body_source_positions: tuple[int, ...]
    encounters: tuple[AeternumBearingEncounter, ...]
    completed: bool = True


@dataclass(frozen=True, slots=True)
class LocalGrammarTrace:
    bodies: tuple[ResolvedBody, ...]
    causal_body_order: tuple[int, ...]
    events: tuple[LocalGrammarEvent, ...]
    recognitions: tuple[RecognitionRequest, ...]
    ouroboric_states: tuple[OuroboricState, ...]
    transmutations: tuple[PhantasmagoriaTransmutation, ...]
    aeternum_bearing_paths: tuple[AeternumBearingPath, ...] = ()
    tail_resolution: TailResolution | None = None
    execution_positions: tuple[int, ...] = ()

    def to_json(self) -> str:
        return json.dumps(asdict(self), ensure_ascii=False, sort_keys=True, separators=(",", ":"),
        ↪  default=str)
```

mas_shadow_conversion.py

```
"""Phase 11 M.A.S. procession and append-only Shadow conversion.

Each complete Phase 10 candidate path is carried through the ordered M.A.S.
body::

    M = M_BABDH o M_KAL o M_DREH

The order of action is DREH -> KAL -> BABDH: Q3 positivity, Q1 coherence,
and structural commitment.  Q2 pressure is never deleted.  Gross pressure,
resolved pressure, and terminal unresolved pressure remain separate exact
ledger coordinates, and every cycle retains the complete prior archive as a
prefix.

The local Mass Gap is certified exactly through the authored theorem
``Delta_MG > 0 iff the finite active graph is connected and has at least two
vertices``.  No floating eigenvalue approximation, lexical score, candidate
rank, or receiving-language operation enters this module.
"""
from __future__ import annotations

from dataclasses import dataclass, replace
from typing import Final

from .liquid_candidate_path import (
    CandidateGlobalField,
    LiquidCandidatePath,
    LiquidHorizonWitness,
)


class MASShadowError(ValueError):
    """Raised when a candidate cannot complete Phase 11 lawfully."""


MAS_ORDER: Final[tuple[str, str, str]] = ("M_DREH", "M_KAL", "M_BABDH")
DREH_GLYPH: Final[str] = "⧗"
KAL_GLYPH: Final[str] = "⬡"
BABDH_GLYPH: Final[str] = "✡"
DREH_HZ: Final[int] = 852
KAL_HZ: Final[int] = 174
BABDH_HZ: Final[int] = 528


@dataclass(frozen=True, slots=True)
class MassGapWitness:
    """Exact theorem witness for ``Delta_MG(x,h) > 0``."""

    horizon: int
```

```
    vertex_count: int
    connected: bool
    theorem: str = "connected finite active graph with at least two vertices iff Delta_MG > 0"

    def __post_init__(self) -> None:
        if self.horizon < 0:
            raise MASShadowError("Mass Gap horizon must be nonnegative")
        if self.vertex_count < 2:
            raise MASShadowError("a positive local Mass Gap requires at least two active vertices")
        if not self.connected:
            raise MASShadowError("a disconnected active graph has no positive local Mass Gap")

    @property
    def positive(self) -> bool:
        return self.connected and self.vertex_count >= 2


@dataclass(frozen=True, slots=True)
class ShadowLedgerEntry:
    """One append-only Q2 archive event."""

    index: int
    horizon: int
    gross_added: int
    resolved_added: int
    gross_total: int
    resolved_total: int
    unresolved: int

    def __post_init__(self) -> None:
        values = (
            self.index,
            self.horizon,
            self.gross_added,
            self.resolved_added,
            self.gross_total,
            self.resolved_total,
            self.unresolved,
        )
        if any(not isinstance(value, int) for value in values):
            raise MASShadowError("Shadow ledger coordinates must be exact integers")
        if any(value < 0 for value in values):
            raise MASShadowError("Shadow ledger coordinates must be nonnegative")


@dataclass(frozen=True, slots=True)
class ShadowLedger:
    """Complete gross/resolved/unresolved Q2 body with immutable history."""

    initial_unresolved: int
    entries: tuple[ShadowLedgerEntry, ...]

    def __post_init__(self) -> None:
        if not isinstance(self.initial_unresolved, int) or self.initial_unresolved < 0:
            raise MASShadowError("initial Q2 pressure must be a nonnegative integer")
        if not self.entries:
```

```
            raise MASShadowError("a complete debt ledger requires at least one archive event")
        if tuple(entry.index for entry in self.entries) != tuple(range(len(self.entries))):
            raise MASShadowError("Shadow archive indices must be complete and append-only")
        if tuple(entry.horizon for entry in self.entries) != tuple(range(len(self.entries))):
            raise MASShadowError("Shadow archive horizons must remain in source order")

        previous_gross = 0
        previous_resolved = 0
        previous_unresolved = self.initial_unresolved
        for entry in self.entries:
            if entry.gross_total != previous_gross + entry.gross_added:
                raise MASShadowError("gross Q2 history was rewritten or skipped")
            available = previous_unresolved + entry.gross_added
            if entry.resolved_added > available:
                raise MASShadowError("Q2 conversion exceeded pressure actually borne")
            if entry.resolved_total != previous_resolved + entry.resolved_added:
                raise MASShadowError("resolved Q2 history was rewritten or skipped")
            expected_unresolved = available - entry.resolved_added
            if entry.unresolved != expected_unresolved:
                raise MASShadowError("unresolved Q2 does not close against its ledger")
            previous_gross = entry.gross_total
            previous_resolved = entry.resolved_total
            previous_unresolved = entry.unresolved

    @property
    def terminal_unresolved(self) -> int:
        return self.entries[-1].unresolved

    @property
    def gross_total(self) -> int:
        return self.entries[-1].gross_total

    @property
    def resolved_total(self) -> int:
        return self.entries[-1].resolved_total

    @property
    def carried_total(self) -> int:
        return self.initial_unresolved + self.gross_total

    @property
    def complete(self) -> bool:
        return (
            self.terminal_unresolved == 0
            and self.resolved_total == self.carried_total
            and self.carried_total > 0
        )


@dataclass(frozen=True, slots=True)
class MASState:
    """One invariant-preserving state inside a M.A.S. cycle."""

    candidate: str
    court_root: str
    carrier_glyph: str
```

```
    q_vector: tuple[str, str, str, str]
    archive_signature: tuple[object, ...]
    recursive_capacity: int
    q1_coherent: bool
    babdh_committed: bool

    def __post_init__(self) -> None:
        if not self.candidate or not self.court_root:
            raise MASShadowError("M.A.S. state requires candidate and Court root identity")
        if len(self.carrier_glyph) != 1:
            raise MASShadowError("M.A.S. state requires one Goetic carrier glyph")
        if len(self.q_vector) != 4:
            raise MASShadowError("M.A.S. state requires the complete four-coordinate Q body")
        if self.recursive_capacity < 0:
            raise MASShadowError("Q3 recursive capacity cannot be negative")


@dataclass(frozen=True, slots=True)
class MASMicrostep:
    """One typed M.A.S. office preserving candidate identity and archive."""

    name: str
    glyph: str
    structural_hz: int
    before: MASState
    after: MASState

    def __post_init__(self) -> None:
        if self.name not in MAS_ORDER:
            raise MASShadowError("undefined M.A.S. microstep")
        expected = {
            "M_DREH": (DREH_GLYPH, DREH_HZ),
            "M_KAL": (KAL_GLYPH, KAL_HZ),
            "M_BABDH": (BABDH_GLYPH, BABDH_HZ),
        }[self.name]
        if (self.glyph, self.structural_hz) != expected:
            raise MASShadowError("M.A.S. microstep changed its canonical office")
        if (
            self.before.candidate,
            self.before.court_root,
            self.before.carrier_glyph,
            self.before.q_vector,
            self.before.archive_signature,
        ) != (
            self.after.candidate,
            self.after.court_root,
            self.after.carrier_glyph,
            self.after.q_vector,
            self.after.archive_signature,
        ):
            raise MASShadowError("M.A.S. microstep changed candidate identity or archive")


@dataclass(frozen=True, slots=True)
class MASCycle:
    """One exact DREH -> KAL -> BABDH procession at a Liquid horizon."""
```

```
    horizon: int
    gross_pressure_added: int
    resolved_pressure_added: int
    ledger_snapshot: ShadowLedger
    microsteps: tuple[MASMicrostep, MASMicrostep, MASMicrostep]

    def __post_init__(self) -> None:
        if tuple(step.name for step in self.microsteps) != MAS_ORDER:
            raise MASShadowError("M.A.S. cycle must act in DREH -> KAL -> BABDH order")
        dreh, kal, babdh = self.microsteps
        if kal.before != dreh.after or babdh.before != kal.after:
            raise MASShadowError("M.A.S. microsteps are disconnected")
        if dreh.after.recursive_capacity <= 0:
            raise MASShadowError("Manifestation requires positive Q3 recursive capacity")
        if not kal.after.q1_coherent:
            raise MASShadowError("Alignment requires Q1 coherence")
        if not babdh.after.babdh_committed:
            raise MASShadowError("Symmetry requires BABDH commitment")
        if self.ledger_snapshot.entries[-1].horizon != self.horizon:
            raise MASShadowError("M.A.S. cycle and Shadow archive lost horizon alignment")
        if self.ledger_snapshot.entries[-1].gross_added != self.gross_pressure_added:
            raise MASShadowError("M.A.S. cycle lost its gross Q2 witness")
        if self.ledger_snapshot.entries[-1].resolved_added != self.resolved_pressure_added:
            raise MASShadowError("M.A.S. cycle lost its resolved Q2 witness")

    @property
    def final_state(self) -> MASState:
        return self.microsteps[-1].after


@dataclass(frozen=True, slots=True)
class MASCandidatePath:
    """One surviving Liquid path with positive gaps and a closed debt ledger."""

    liquid_path: LiquidCandidatePath
    mass_gaps: tuple[MassGapWitness, ...]
    cycles: tuple[MASCycle, ...]
    debt_ledger: ShadowLedger
    motion: int

    def __post_init__(self) -> None:
        if not self.mass_gaps or len(self.mass_gaps) != len(self.liquid_path.horizons):
            raise MASShadowError("every Liquid horizon requires one local Mass Gap witness")
        if not self.cycles or len(self.cycles) != len(self.liquid_path.horizons):
            raise MASShadowError("every Liquid horizon requires one complete M.A.S. cycle")
        if tuple(gap.horizon for gap in self.mass_gaps) != tuple(range(len(self.mass_gaps))):
            raise MASShadowError("Mass Gap witnesses left horizon order")
        if tuple(cycle.horizon for cycle in self.cycles) != tuple(range(len(self.cycles))):
            raise MASShadowError("M.A.S. cycles left horizon order")
        if any(not gap.positive for gap in self.mass_gaps):
            raise MASShadowError("candidate persistence encountered a nonpositive Mass Gap")
        if self.motion <= 0:
            raise MASShadowError("Phase 11 survival requires motion > 0")
        if not self.debt_ledger.complete:
            raise MASShadowError("terminal unresolved Q2 must reach zero through lawful conversion")
```

```
        if self.cycles[-1].ledger_snapshot != self.debt_ledger:
            raise MASShadowError("final M.A.S. cycle does not carry the complete debt ledger")

        previous: tuple[ShadowLedgerEntry, ...] = ()
        for cycle in self.cycles:
            current = cycle.ledger_snapshot.entries
            if current[: len(previous)] != previous or len(current) != len(previous) + 1:
                raise MASShadowError("Q2 history is not append-only across M.A.S. cycles")
            previous = current

        final = self.cycles[-1].final_state
        seed = self.liquid_path.seed
        if (
            final.candidate,
            final.court_root,
            final.carrier_glyph,
            final.q_vector,
        ) != (
            seed.witness.candidate,
            seed.domain.court_root,
            seed.domain.carrier.glyph,
            seed.body.q_vector,
        ):
            raise MASShadowError("M.A.S. procession changed the candidate path seed")
        if final.recursive_capacity <= 0 or not final.q1_coherent or not final.babdh_committed:
            raise MASShadowError("candidate did not complete Manifestation, Alignment, and
            ↪  Symmetry")

    @property
    def candidate(self) -> str:
        return self.liquid_path.seed.witness.candidate

    @property
    def terminal_unresolved_q2(self) -> int:
        return self.debt_ledger.terminal_unresolved

    @property
    def positive_local_mass_gap(self) -> bool:
        return all(gap.positive for gap in self.mass_gaps)


@dataclass(frozen=True, slots=True)
class MASFailure:
    candidate: str
    court_root: str
    carrier_glyph: str
    reason: str


@dataclass(frozen=True, slots=True)
class MASGlobalField:
    """Ordered Phase 11 survivor body plus explicit non-survival records."""

    source: CandidateGlobalField
    survivors: tuple[MASCandidatePath, ...]
    failures: tuple[MASFailure, ...]
```

```
    def __post_init__(self) -> None:
        input_candidates = tuple(path.seed.witness.candidate for path in self.source.liquid_paths)
        survivor_candidates = tuple(path.candidate for path in self.survivors)
        failure_candidates = tuple(failure.candidate for failure in self.failures)
        if len(survivor_candidates) + len(failure_candidates) != len(input_candidates):
            raise MASShadowError("Phase 11 lost or duplicated candidate records")
        if tuple(candidate for candidate in input_candidates if candidate in
        ↪  set(survivor_candidates)) != survivor_candidates:
            raise MASShadowError("Phase 11 changed survivor order")
        if set(survivor_candidates) & set(failure_candidates):
            raise MASShadowError("candidate cannot survive and fail the same Phase 11 field")
        if set(survivor_candidates) | set(failure_candidates) != set(input_candidates):
            raise MASShadowError("Phase 11 candidate accounting is incomplete")
        if any(
            not survivor.positive_local_mass_gap
            or survivor.terminal_unresolved_q2 != 0
            or survivor.motion <= 0
            for survivor in self.survivors
        ):
            raise MASShadowError("Phase 11 field contains an unsealed survivor")



def _archive_signature(path: LiquidCandidatePath) -> tuple[object, ...]:
    seed = path.seed
    return (
        seed.witness.candidate,
        seed.domain.court_root,
        seed.domain.inherited_q_bias,
        seed.domain.carrier.glyph,
        seed.body.q_vector,
        tuple(step.name for step in seed.microsteps),
    )



def _transition_pressure(
    previous: LiquidHorizonWitness | None,
    current: LiquidHorizonWitness,
) -> int:
    if previous is None:
        return 0
    return len(previous.active_connections ^ current.active_connections)



def _apply_dreh(state: MASState, resolved_pressure: int) -> MASMicrostep:
    if resolved_pressure < 0:
        raise MASShadowError("resolved Q2 pressure cannot be negative")
    after = replace(state, recursive_capacity=state.recursive_capacity + resolved_pressure)
    if after.recursive_capacity <= 0:
        raise MASShadowError("M_DREH failed Q3 positivity")
    return MASMicrostep("M_DREH", DREH_GLYPH, DREH_HZ, state, after)
```

```
def _apply_kal(state: MASState) -> MASMicrostep:
    if "Q1" not in state.q_vector:
        raise MASShadowError("M_KAL cannot establish Q1 coherence without a Q1 coordinate")
    after = replace(state, q1_coherent=True)
    return MASMicrostep("M_KAL", KAL_GLYPH, KAL_HZ, state, after)



def _apply_babdh(state: MASState) -> MASMicrostep:
    if state.recursive_capacity <= 0 or not state.q1_coherent:
        raise MASShadowError("M_BABDH requires manifested and aligned input")
    after = replace(state, babdh_committed=True)
    return MASMicrostep("M_BABDH", BABDH_GLYPH, BABDH_HZ, state, after)



def run_mas_shadow_conversion(path: LiquidCandidatePath) -> MASCandidatePath:
    """Run one exact Phase 10 path through M.A.S. and close its Q2 ledger."""
    seed = path.seed
    q_vector = seed.body.q_vector
    initial_q2 = sum(state == "Q2" for state in q_vector)
    initial_q3 = sum(state == "Q3" for state in q_vector)
    signature = _archive_signature(path)
    state = MASState(
        candidate=seed.witness.candidate,
        court_root=seed.domain.court_root,
        carrier_glyph=seed.domain.carrier.glyph,
        q_vector=q_vector,
        archive_signature=signature,
        recursive_capacity=initial_q3,
        q1_coherent=False,
        babdh_committed=False,
    )

    entries: list[ShadowLedgerEntry] = []
    cycles: list[MASCycle] = []
    gaps: list[MassGapWitness] = []
    gross_total = 0
    resolved_total = 0
    unresolved = initial_q2
    motion = 0
    previous_horizon: LiquidHorizonWitness | None = None

    for index, horizon in enumerate(path.horizons):
        gap = MassGapWitness(
            horizon=index,
            vertex_count=len(horizon.active_graph.vertices),
            connected=horizon.active_graph.is_connected,
        )
        gaps.append(gap)

        gross_added = _transition_pressure(previous_horizon, horizon)
        motion += gross_added
        available = unresolved + gross_added
        resolved_added = available
```

```
        gross_total += gross_added
        resolved_total += resolved_added
        unresolved = available - resolved_added
        entry = ShadowLedgerEntry(
            index=index,
            horizon=index,
            gross_added=gross_added,
            resolved_added=resolved_added,
            gross_total=gross_total,
            resolved_total=resolved_total,
            unresolved=unresolved,
        )
        entries.append(entry)
        snapshot = ShadowLedger(initial_q2, tuple(entries))

        dreh = _apply_dreh(state, resolved_added)
        kal = _apply_kal(dreh.after)
        babdh = _apply_babdh(kal.after)
        cycle = MASCycle(
            horizon=index,
            gross_pressure_added=gross_added,
            resolved_pressure_added=resolved_added,
            ledger_snapshot=snapshot,
            microsteps=(dreh, kal, babdh),
        )
        cycles.append(cycle)
        state = babdh.after
        previous_horizon = horizon

    ledger = ShadowLedger(initial_q2, tuple(entries))
    return MASCandidatePath(path, tuple(gaps), tuple(cycles), ledger, motion)



def prove_phase11_exit_gate(field: CandidateGlobalField) -> MASGlobalField:
    """Return all lawful survivors and an explicit record for every failure."""
    survivors: list[MASCandidatePath] = []
    failures: list[MASFailure] = []
    for path in field.liquid_paths:
        try:
            survivors.append(run_mas_shadow_conversion(path))
        except MASShadowError as exc:
            seed = path.seed
            failures.append(MASFailure(
                candidate=seed.witness.candidate,
                court_root=seed.domain.court_root,
                carrier_glyph=seed.domain.carrier.glyph,
                reason=str(exc),
            ))
    return MASGlobalField(field, tuple(survivors), tuple(failures))
```

mcp_server.py

```
"""Minimal dependency-free MCP stdio server for Sydonic Magicae."""
from __future__ import annotations

import json
import sys
from typing import Any

from .mcp_surface import SydonicMCPTools, operation_schemas

SERVER_NAME = "sydonic-magicae"
SERVER_VERSION = "phase15-mcp-v1"

def _tools() -> list[dict[str, Any]]:
    descriptions = {
        "parse_domus": "Validate and tokenize one exact completed Aeonic line.",
        "resolve_sydonic_glyph": "Resolve one ordinary glyph directly through SeeD Body.",
        "translate_domus": "Translate one completed Domus through the shared engine.",
        "trace_domus": "Return the deterministic semantic trace for one completed Domus.",
        "validate_sydonic_corpus": "Audit SeeD Body and every authored office projection on the
        ↪  shared 179-glyph body.",
    }
    schemas = operation_schemas()
    properties = {
        "domus_string": {"type": "string"},
        "glyph": {"type": "string", "minLength": 1, "maxLength": 1},
        "target_language": {"type": "string", "default": "en"},
        "output_style": {"type": "string", "enum": ["telegraphic", "plain"], "default":
        ↪  "telegraphic"},
        "include_trace": {"type": "boolean", "default": True},
    }
    out=[]
    for name,spec in schemas.items():
        keys=spec.get("required",[])+spec.get("optional",[])
        out.append({"name":name,"description":descriptions[name],"inputSchema":{"type":"object","pr⌋
        ↪  operties":{k:properties[k] for k in
        ↪  keys},"required":spec.get("required",[]),"additionalProperties":False}})
    return out

def _response(req_id: Any, result: Any = None, error: dict[str, Any] | None = None) -> dict[str,
↪  Any]:
    body={"jsonrpc":"2.0","id":req_id}
    if error is not None: body["error"]=error
    else: body["result"]=result
    return body

def dispatch(tools: SydonicMCPTools, request: dict[str, Any]) -> dict[str, Any] | None:
    method=request.get("method")
    req_id=request.get("id")
```

```
    if method == "notifications/initialized": return None
    if method == "initialize":
        return _response(req_id,{"protocolVersion":request.get("params",{}).get("protocolVersion","
        ↪  2025-06-18"),"capabilities":{"tools":{}},"serverInfo":{"name":SERVER_NAME,"version":SER
        ↪  VER_VERSION}})
    if method == "tools/list": return _response(req_id,{"tools":_tools()})
    if method == "tools/call":
        params=request.get("params") or {}
        name=params.get("name")
        arguments=params.get("arguments") or {}
        if not hasattr(tools,name): return _response(req_id,error={"code":-32601,"message":f"unknown
        ↪  tool: {name}"})
        try:
            value=getattr(tools,name)(**arguments)
            return _response(req_id,{"content":[{"type":"text","text":json.dumps(value,ensure_ascii
            ↪  =False,sort_keys=True,separators=(",",":"))}],"structuredContent":value,"isError":F
            ↪  alse})
        except Exception as exc:
            return _response(req_id,{"content":[{"type":"text","text":f"{exc.__class__.__name__}:
            ↪  {exc}"}],"isError":True})
    return _response(req_id,error={"code":-32601,"message":f"method not found: {method}"})

def main() -> int:
    tools=SydonicMCPTools()
    for line in sys.stdin:
        if not line.strip(): continue
        try:
            request=json.loads(line)
            response=dispatch(tools,request)
        except Exception as exc:
            response=_response(None,error={"code":-32700,"message":f"{exc.__class__.__name__}:
            ↪  {exc}"})
        if response is not None:
            sys.stdout.write(json.dumps(response,ensure_ascii=False,separators=(",",":"))+"\n")
            sys.stdout.flush()
    return 0

if __name__ == "__main__":
    raise SystemExit(main())
```

mcp_surface.py

```
"""Transport-neutral MCP operation surface.

Every operation delegates to the same validator, resolver, auditor, and engine used
by the Python API and CLI. No grammar law is duplicated here.
"""
from __future__ import annotations

from dataclasses import asdict
import json
from typing import Any

from .corpus_audit import CorpusAuditor
from .english_renderer import SydonicSpokenVoice
from .engine import SydonicMagicaeEngine

class SydonicMCPTools:
    def __init__(self) -> None:
        self.engine = SydonicMagicaeEngine(house_voice=SydonicSpokenVoice())

    def parse_domus(self, domus_string: str) -> dict[str, Any]:
        return asdict(self.engine.parse_domus(domus_string))

    def resolve_sydonic_glyph(self, glyph: str) -> dict[str, Any]:
        return asdict(self.engine.lexical_resolver.resolve(glyph))

    def translate_domus(self, domus_string: str, target_language: str = "en", output_style: str =
    ↪  "telegraphic", include_trace: bool = True) -> dict[str, Any]:
        result = self.engine.translate_domus(
            domus_string=domus_string,
            target_language=target_language,
            output_style=output_style,
            include_trace=include_trace,
        )
        return json.loads(result.to_json())

    def trace_domus(self, domus_string: str) -> dict[str, Any]:
        result = self.engine.translate_domus(
            domus_string=domus_string,
            target_language="en",
            output_style="telegraphic",
            include_trace=True,
        )
        if result.semantic_trace is None:
            raise RuntimeError("engine returned no trace")
        return result.semantic_trace.to_dict()

    def validate_sydonic_corpus(self) -> dict[str, Any]:
        return json.loads(CorpusAuditor(self.engine.lexical_resolver).run().to_json())
```

```
def operation_schemas() -> dict[str, dict[str, Any]]:
    return {
        "parse_domus": {"required": ["domus_string"]},
        "resolve_sydonic_glyph": {"required": ["glyph"]},
        "translate_domus": {"required": ["domus_string"], "optional": ["target_language",
        ↪  "output_style", "include_trace"]},
        "trace_domus": {"required": ["domus_string"]},
        "validate_sydonic_corpus": {"required": []},
    }
```

models.py

```
"""Immutable structural records for one exact Sydonic Magicae Aeonic line."""
from __future__ import annotations
from dataclasses import dataclass

@dataclass(frozen=True, slots=True)
class GlyphToken:
    value: str
    position: int
    kind: str

    def __post_init__(self) -> None:
        if len(self.value) != 1:
            raise ValueError("GlyphToken.value must be exactly one Unicode code point")

@dataclass(frozen=True, slots=True)
class ReflectedGlyphToken:
    value: str
    mirror_position: int
    source_position: int
    kind: str
    imaginary: bool = True
    prosody_position: int | None = None

    def __post_init__(self) -> None:
        if len(self.value) != 1:
            raise ValueError("ReflectedGlyphToken.value must be exactly one Unicode code point")

@dataclass(frozen=True, slots=True)
class DomusFrame:
    source: str
    tokens: tuple[GlyphToken, ...]
    prosody_source: str | None = None

    def to_source(self) -> str:
        return "".join(token.value for token in self.tokens)

    """
    They do not move their feet, it only changes their bearing,
    It is simply reflected again, as the dancer laughs at the scribe.
    Left, Right, Bow
    """
    def mirror_source(self) -> str:
        return self.source[::-1]

    def mirror_tokens(self) -> tuple[ReflectedGlyphToken, ...]:
        size = len(self.tokens)
        return tuple(
            ReflectedGlyphToken(
```

```
            value=token.value,
            mirror_position=size - 1 - token.position,
            source_position=token.position,
            kind=token.kind,
        )
        for token in reversed(self.tokens)
    )

def reflected_source_position(self, mirror_position: int) -> int:
    if mirror_position < 0 or mirror_position >= len(self.tokens):
        raise IndexError("mirror position outside reflected utterance")
    return len(self.tokens) - 1 - mirror_position

def mirror_corridor_period(self) -> tuple[ReflectedGlyphToken, ...]:
    """Return one Aeternum period: reflected utterance then original utterance.

    For a real center ``ABCD`` this is ``DCBAABCD``.  Repeating that period
    outward on either side yields the ordinary real-world mirror field around
    the immutable center: ``... ABCD | DCBA | ABCD | DCBA ...``.  The repeated
    images are imaginary traversal only; ``source_position`` always points
    back into the one finite real GrimChain.
    """
    reflected = self.mirror_tokens()
    original = tuple(
        ReflectedGlyphToken(
            value=token.value,
            mirror_position=token.position,
            source_position=token.position,
            kind=token.kind,
        )
        for token in self.tokens
    )
    return reflected + original

def aeternum_source_position(self, virtual_position: int) -> int:
    """Map an integer mirror-field position onto the finite real center.

    Real positions are ``0..N-1``.  Negative and >=N positions are imaginary
    mirror positions.  With center ``ABCD`` the spatial field around it is
    ``DCBA | ABCD | DCBA`` for virtual positions ``-4..7``.
    """
    extent = len(self.tokens)
    if extent == 0:
        raise ValueError("Aeternum mirror requires a non-empty original utterance")
    phase = virtual_position % (2 * extent)
    return phase if phase < extent else (2 * extent - 1 - phase)

def aeternum_is_reflected(self, virtual_position: int) -> bool:
    """Whether a virtual position lies in a reflected mirror block."""
    extent = len(self.tokens)
    if extent == 0:
        raise ValueError("Aeternum mirror requires a non-empty original utterance")
    return (virtual_position // extent) % 2 != 0

def aeternum_token(self, virtual_position: int) -> ReflectedGlyphToken:
    """Return the imaginary token at one position in the infinite mirror field."""
```

```
    extent = len(self.tokens)
    source_position = self.aeternum_source_position(virtual_position)
    token = self.tokens[source_position]
    return ReflectedGlyphToken(
        value=token.value,
        mirror_position=virtual_position % extent,
        source_position=source_position,
        kind=token.kind,
        imaginary=not (0 <= virtual_position < extent),
    )

def mirror_walk(
    self,
    source_position: int,
    count: int,
    step: int,
) -> tuple[ReflectedGlyphToken, ...]:
    """Walk outward from a real source position through the Aeternum field.

    ``step`` is the traversal bearing in the spatial field.  Crossing a mirror
    changes which way source positions are encountered; it does not rewrite
    the source or alter an already-resolved grammatical bearing.
    """
    if count < 0:
        raise ValueError("mirror walk count must be non-negative")
    if step not in {-1, 1}:
        raise ValueError("mirror walk step must be -1 or 1")
    extent = len(self.tokens)
    if extent == 0:
        raise ValueError("mirror walk requires the resolved original utterance")
    if source_position < 0 or source_position >= extent:
        raise IndexError("source position outside original utterance")
    start = source_position + step
    return tuple(
        self.aeternum_token(start + step * offset)
        for offset in range(count)
    )
```

native_bearing_assembly.py

```
"""Phase 15 strict assembly of eight natively closed lexical bearings.

The partial-body law is permanently fixed to STRICT:

* one Aeon body requires exactly eight independently closed bearings;
* every bearing must carry exactly one survivor and its own path witness;
* a failed, empty, multiplied, detached, or retyped bearing fails the Aeon;
* no bearing is copied, derived, guessed, repaired, or moved into another slot.

The complete assembly domain contains the twelve Goetic Aeons followed by the
144 ordered Court Aeons.  Phase 12 Goetic closures and Phase 14 Court chooser
records remain attached as the witnesses that produced each lexical emission.
"""
from __future__ import annotations

from dataclasses import dataclass
from enum import Enum
from typing import Final

from .court_outward_return import COURTS, CourtClosure, CourtIdentity
from .exhaustive_chooser import (
    ExhaustiveBearingResult,
    ExhaustiveChooserField,
)
from .frequency_projection import CARRIERS, Carrier
from .goetic_outward_return import (
    BEARINGS,
    BearingResult,
    GoeticBearing,
    GoeticClosure,
    GoeticOutwardReturnField,
)


class NativeBearingAssemblyError(ValueError):
    """Raised when Phase 15 would accept or manufacture an incomplete body."""


class PartialBodyLaw(str, Enum):
    STRICT = "STRICT"


PARTIAL_BODY_LAW: Final[PartialBodyLaw] = PartialBodyLaw.STRICT
AEON_TOTAL: Final[int] = 156


class AeonClass(str, Enum):
    GOETIC = "Goetic"
```

```
    COURT = "Court"


@dataclass(frozen=True, slots=True, order=True)
class AeonIdentity:
    """One canonical identity in the 12-Goetic plus 144-Court assembly field."""

    index: int
    aeon_class: AeonClass
    key: str

    def __post_init__(self) -> None:
        if not 0 <= self.index < AEON_TOTAL:
            raise NativeBearingAssemblyError("Aeon assembly index must lie in [0,155]")
        if not self.key:
            raise NativeBearingAssemblyError("Aeon assembly identity requires a nonempty key")
        if self.index < 12 and self.aeon_class is not AeonClass.GOETIC:
            raise NativeBearingAssemblyError("the first twelve assembly identities are Goetics")
        if self.index >= 12 and self.aeon_class is not AeonClass.COURT:
            raise NativeBearingAssemblyError("the final 144 assembly identities are Courts")


GOETIC_IDENTITIES: Final[tuple[AeonIdentity, ...]] = tuple(
    AeonIdentity(index, AeonClass.GOETIC, carrier.glyph)
    for index, carrier in enumerate(CARRIERS)
)
COURT_IDENTITIES: Final[tuple[AeonIdentity, ...]] = tuple(
    AeonIdentity(12 + court.index, AeonClass.COURT, court.key)
    for court in COURTS
)
AEON_IDENTITIES: Final[tuple[AeonIdentity, ...]] = GOETIC_IDENTITIES + COURT_IDENTITIES
if len(AEON_IDENTITIES) != AEON_TOTAL or len({identity.key for identity in AEON_IDENTITIES}) !=
↪   AEON_TOTAL:
    raise RuntimeError("Phase 15 requires exactly 156 unique canonical Aeon identities")


@dataclass(frozen=True, slots=True)
class NativeBearingClosure:
    """One lexical dimension that closed under its own bearing and native Bias."""

    aeon: AeonIdentity
    bearing: GoeticBearing
    candidate: str
    native_q_bias: str
    survivor_set: frozenset[object]
    path_witness: GoeticClosure | CourtClosure
    source_record: BearingResult | ExhaustiveBearingResult

    def __post_init__(self) -> None:
        if self.bearing not in BEARINGS:
            raise NativeBearingAssemblyError("closure was moved outside the eight bearing body")
        if not self.candidate:
            raise NativeBearingAssemblyError("closed bearing lacks its emitted candidate")
        if self.native_q_bias not in {"Q0", "Q1", "Q2", "Q3"}:
            raise NativeBearingAssemblyError("closed bearing lacks one native Q Bias")
        if len(self.survivor_set) != 1:
```

```
            raise NativeBearingAssemblyError("closed bearing requires exactly one survivor")
        if self.path_witness.bearing is not self.bearing:
            raise NativeBearingAssemblyError("path witness was moved to another bearing")
        if not self.path_witness.dcomp.exact_zero or self.path_witness.dcomp.value != 0:
            raise NativeBearingAssemblyError("bearing path witness lacks exact D-COMP closure")

        if self.aeon.aeon_class is AeonClass.GOETIC:
            if not isinstance(self.path_witness, GoeticClosure):
                raise NativeBearingAssemblyError("Goetic bearing lacks a Goetic path witness")
            if not isinstance(self.source_record, BearingResult):
                raise NativeBearingAssemblyError("Goetic bearing lost its source closure record")
            if self.source_record.bearing is not self.bearing:
                raise NativeBearingAssemblyError("Goetic source record changed bearing")
            if self.source_record.survivor is not self.path_witness:
                raise NativeBearingAssemblyError("Goetic emission detached from its unique closure")
            if self.path_witness.candidate_path.candidate != self.candidate:
                raise NativeBearingAssemblyError("Goetic closure changed the emitted word")
            if not self.path_witness.bec_valid or not self.path_witness.self_to_self:
                raise NativeBearingAssemblyError("Goetic lexical bearing lacks BEC Self return")
        else:
            if not isinstance(self.path_witness, CourtClosure):
                raise NativeBearingAssemblyError("Court bearing lacks a Court path witness")
            if not isinstance(self.source_record, ExhaustiveBearingResult):
                raise NativeBearingAssemblyError("Court bearing lost its Phase 14 closure record")
            if self.source_record.bearing is not self.bearing:
                raise NativeBearingAssemblyError("Court source record changed bearing")
            if not self.source_record.order_independent:
                raise NativeBearingAssemblyError("Court emission lacks order-independent closure")
            if self.source_record.emission is None:
                raise NativeBearingAssemblyError("Court closure is detached from its emission")
            if self.source_record.emission.candidate != self.candidate:
                raise NativeBearingAssemblyError("Court closure changed the emitted word")
            if self.path_witness.court_candidate.candidate != self.candidate:
                raise NativeBearingAssemblyError("Court path witness belongs to another candidate")
            if not self.path_witness.lbec_valid or not self.path_witness.bec_valid:
                raise NativeBearingAssemblyError("Court lexical bearing lacks nested envelope
                ↪  closure")
            if not self.path_witness.court_to_parent:
                raise NativeBearingAssemblyError("Court lexical bearing did not return to its
                ↪  parent")


@dataclass(frozen=True, slots=True)
class StrictBearingFailure:
    aeon: AeonIdentity
    bearing: GoeticBearing
    reason: str
    survivor_count: int
    source_record: BearingResult | ExhaustiveBearingResult

    def __post_init__(self) -> None:
        if self.bearing not in BEARINGS:
            raise NativeBearingAssemblyError("failed bearing left the eight-bearing body")
        if not self.reason:
            raise NativeBearingAssemblyError("failed bearing requires explicit evidence")
        if self.survivor_count < 0:
```

```
            raise NativeBearingAssemblyError("survivor count cannot be negative")


BearingAssembly = NativeBearingClosure | StrictBearingFailure


@dataclass(frozen=True, slots=True)
class StrictLexicalBody:
    """One complete eight-bearing body assembled without cross-bearing derivation."""

    aeon: AeonIdentity
    native_q_bias: str
    bearings: tuple[NativeBearingClosure, ...]
    partial_body_law: PartialBodyLaw = PARTIAL_BODY_LAW

    def __post_init__(self) -> None:
        if self.partial_body_law is not PartialBodyLaw.STRICT:
            raise NativeBearingAssemblyError("Phase 15 partial-body law must remain STRICT")
        if tuple(closure.bearing for closure in self.bearings) != BEARINGS:
            raise NativeBearingAssemblyError("STRICT body requires all eight native bearings in
            ↪  order")
        if len(self.bearings) != 8:
            raise NativeBearingAssemblyError("STRICT body cannot contain fewer than eight closures")
        if any(closure.aeon != self.aeon for closure in self.bearings):
            raise NativeBearingAssemblyError("lexical body mixed Aeon identities")
        if any(closure.native_q_bias != self.native_q_bias for closure in self.bearings):
            raise NativeBearingAssemblyError("lexical body moved a word from its native Q Bias")
        if len({id(closure.source_record) for closure in self.bearings}) != 8:
            raise NativeBearingAssemblyError("each bearing requires its own closure record")
        if len({closure.bearing for closure in self.bearings}) != 8:
            raise NativeBearingAssemblyError("one closed bearing was copied into another dimension")

    @property
    def emitted_words(self) -> tuple[str, ...]:
        return tuple(closure.candidate for closure in self.bearings)

    @property
    def bearing_map(self) -> tuple[tuple[GoeticBearing, str], ...]:
        return tuple((closure.bearing, closure.candidate) for closure in self.bearings)


@dataclass(frozen=True, slots=True)
class StrictAeonFailure:
    """Whole-Aeon failure evidence. No lexical body accompanies this record."""

    aeon: AeonIdentity
    bearing_records: tuple[BearingAssembly, ...]
    failed_bearings: tuple[StrictBearingFailure, ...]
    partial_body_law: PartialBodyLaw = PARTIAL_BODY_LAW

    def __post_init__(self) -> None:
        if self.partial_body_law is not PartialBodyLaw.STRICT:
            raise NativeBearingAssemblyError("failed Aeon record changed the STRICT law")
        if tuple(record.bearing for record in self.bearing_records) != BEARINGS:
            raise NativeBearingAssemblyError("failed Aeon record lost an eight-bearing position")
        if not self.failed_bearings:
```

```
            raise NativeBearingAssemblyError("STRICT failure requires at least one failed bearing")
        if tuple(record for record in self.bearing_records if isinstance(record,
        ↪  StrictBearingFailure)) != self.failed_bearings:
            raise NativeBearingAssemblyError("failed-bearing evidence is incomplete")
        if any(record.aeon != self.aeon for record in self.bearing_records):
            raise NativeBearingAssemblyError("failed body mixed Aeon identities")


@dataclass(frozen=True, slots=True)
class AeonAssemblyResult:
    aeon: AeonIdentity
    bearing_records: tuple[BearingAssembly, ...]
    body: StrictLexicalBody | None
    failure: StrictAeonFailure | None
    partial_body_law: PartialBodyLaw = PARTIAL_BODY_LAW

    def __post_init__(self) -> None:
        if self.partial_body_law is not PartialBodyLaw.STRICT:
            raise NativeBearingAssemblyError("Aeon assembly result changed the STRICT law")
        if tuple(record.bearing for record in self.bearing_records) != BEARINGS:
            raise NativeBearingAssemblyError("Aeon assembly requires all eight bearing records")
        failures = tuple(record for record in self.bearing_records if isinstance(record,
        ↪  StrictBearingFailure))
        if failures:
            if self.body is not None or self.failure is None:
                raise NativeBearingAssemblyError("partial body was emitted under STRICT law")
            if self.failure.aeon != self.aeon or self.failure.bearing_records !=
            ↪  self.bearing_records:
                raise NativeBearingAssemblyError("whole-Aeon failure detached from its bearing
                ↪  evidence")
        else:
            if self.body is None or self.failure is not None:
                raise NativeBearingAssemblyError("eight closed bearings did not emit one complete
                ↪  body")
            if self.body.aeon != self.aeon:
                raise NativeBearingAssemblyError("complete lexical body changed Aeon identity")

    @property
    def valid(self) -> bool:
        return self.body is not None


@dataclass(frozen=True, slots=True)
class NativeBearingAssemblyField:
    """The complete 156-Aeon Phase 15 result under one immutable STRICT law."""

    goetic_source: GoeticOutwardReturnField
    court_source: ExhaustiveChooserField
    aeons: tuple[AeonAssemblyResult, ...]
    partial_body_law: PartialBodyLaw = PARTIAL_BODY_LAW

    def __post_init__(self) -> None:
        if self.partial_body_law is not PartialBodyLaw.STRICT:
            raise NativeBearingAssemblyError("Phase 15 field changed the declared STRICT law")
        if self.goetic_source.source is not self.court_source.manifest.source:
```

```
            raise NativeBearingAssemblyError("Goetic and Court assemblies do not share one Phase 11
            ↪  field")
        if tuple(result.aeon for result in self.aeons) != AEON_IDENTITIES:
            raise NativeBearingAssemblyError("Phase 15 must account for all 156 Aeons in canonical
            ↪  order")
        if len(self.aeons) != AEON_TOTAL:
            raise NativeBearingAssemblyError("Phase 15 did not account for exactly 156 Aeons")
        if any(result.partial_body_law is not self.partial_body_law for result in self.aeons):
            raise NativeBearingAssemblyError("partial-body law differs between Aeons")
        if any(
            result.body is not None and len(result.body.bearings) != 8
            for result in self.aeons
        ):
            raise NativeBearingAssemblyError("field contains a partial lexical body")

    @property
    def bodies(self) -> tuple[StrictLexicalBody, ...]:
        return tuple(result.body for result in self.aeons if result.body is not None)

    @property
    def failures(self) -> tuple[StrictAeonFailure, ...]:
        return tuple(result.failure for result in self.aeons if result.failure is not None)

    @property
    def emitted_body_count(self) -> int:
        return len(self.bodies)

    @property
    def failed_aeon_count(self) -> int:
        return len(self.failures)


def _strict_result(
    aeon: AeonIdentity,
    records: tuple[BearingAssembly, ...],
) -> AeonAssemblyResult:
    if tuple(record.bearing for record in records) != BEARINGS:
        raise NativeBearingAssemblyError("assembler received a moved or missing bearing")
    failures = tuple(record for record in records if isinstance(record, StrictBearingFailure))
    if failures:
        failure = StrictAeonFailure(aeon, records, failures)
        return AeonAssemblyResult(aeon, records, None, failure)
    closures = tuple(record for record in records if isinstance(record, NativeBearingClosure))
    if len(closures) != 8:
        raise NativeBearingAssemblyError("STRICT assembler cannot infer an absent bearing")
    biases = {closure.native_q_bias for closure in closures}
    if len(biases) != 1:
        raise NativeBearingAssemblyError("Aeon bearings disagree on native Q Bias")
    body = StrictLexicalBody(aeon, next(iter(biases)), closures)
    return AeonAssemblyResult(aeon, records, body, None)


def _goetic_bias(source: GoeticOutwardReturnField, carrier: Carrier) -> str | None:
    biases = {
        candidate.liquid_path.seed.domain.inherited_q_bias
        for candidate in source.source.survivors
```

```
        if candidate.liquid_path.seed.domain.carrier == carrier
    }
    return next(iter(biases)) if len(biases) == 1 else None


def _goetic_record(
    aeon: AeonIdentity,
    native_q_bias: str | None,
    result: BearingResult,
) -> BearingAssembly:
    if len(result.exact_closures) != 1 or result.emitted_word is None:
        reason = "no exact survivor" if not result.exact_closures else "multiple exact survivors"
        if native_q_bias is None:
            reason = reason + "; native Q Bias witness unavailable"
        return StrictBearingFailure(aeon, result.bearing, reason, len(result.exact_closures),
        ↪  result)
    closure = result.exact_closures[0]
    bias = closure.candidate_path.liquid_path.seed.domain.inherited_q_bias
    if native_q_bias is None or bias != native_q_bias:
        return StrictBearingFailure(aeon, result.bearing, "native Q Bias mismatch", 1, result)
    return NativeBearingClosure(
        aeon=aeon,
        bearing=result.bearing,
        candidate=closure.candidate_path.candidate,
        native_q_bias=bias,
        survivor_set=frozenset({closure.candidate_path.candidate}),
        path_witness=closure,
        source_record=result,
    )


def _court_record(
    aeon: AeonIdentity,
    result: ExhaustiveBearingResult,
) -> BearingAssembly:
    survivor_count = len(result.survivor_set)
    if survivor_count != 1 or result.emission is None:
        reason = result.failure_evidence.reason if result.failure_evidence is not None else "bearing
        ↪  did not emit"
        return StrictBearingFailure(aeon, result.bearing, reason, survivor_count, result)
    key = next(iter(result.survivor_set))
    truth = tuple(
        record
        for record in result.truth_records
        if record.witness.key == key and record.truth
    )
    if len(truth) != 1 or truth[0].closure is None:
        return StrictBearingFailure(aeon, result.bearing, "unique survivor lacks one path witness",
        ↪  survivor_count, result)
    closure = truth[0].closure
    return NativeBearingClosure(
        aeon=aeon,
        bearing=result.bearing,
        candidate=result.emission.candidate,
        native_q_bias=result.emission.native_q_bias,
```

```
        survivor_set=result.survivor_set,
        path_witness=closure,
        source_record=result,
    )


def assemble_goetic_body(
    source: GoeticOutwardReturnField,
    carrier_index: int,
) -> AeonAssemblyResult:
    """Assemble one Goetic only after all eight source closures are known."""
    carrier = CARRIERS[carrier_index]
    aeon = GOETIC_IDENTITIES[carrier_index]
    body = source.goetics[carrier_index]
    if body.carrier != carrier:
        raise NativeBearingAssemblyError("Goetic source order changed")
    bias = _goetic_bias(source, carrier)
    records = tuple(_goetic_record(aeon, bias, result) for result in body.bearing_results)
    return _strict_result(aeon, records)


def assemble_court_body(
    source: ExhaustiveChooserField,
    court_index: int,
) -> AeonAssemblyResult:
    """Assemble one Court only after all eight Phase 14 results are known."""
    court: CourtIdentity = COURTS[court_index]
    aeon = COURT_IDENTITIES[court_index]
    body = source.court_bodies[court_index]
    if body.frozen_aeon.court != court:
        raise NativeBearingAssemblyError("Court source order changed")
    records = tuple(_court_record(aeon, result) for result in body.bearing_results)
    return _strict_result(aeon, records)


def prove_phase15_exit_gate(
    goetic_source: GoeticOutwardReturnField,
    court_source: ExhaustiveChooserField,
) -> NativeBearingAssemblyField:
    """Resolve the deferred partial-body question uniformly as STRICT."""
    if goetic_source.source is not court_source.manifest.source:
        raise NativeBearingAssemblyError("Phase 15 inputs do not share one candidate field")
    goetics = tuple(assemble_goetic_body(goetic_source, index) for index in range(12))
    courts = tuple(assemble_court_body(court_source, index) for index in range(144))
    return NativeBearingAssemblyField(goetic_source, court_source, goetics + courts)
```

release_audit.py

```
"""Phase 19 release acceptance and required report generation.

This module consumes the sealed Phase 16 certificate field, the canonical
Phase 17 adversarial audit, and the Phase 18 one-way Translation Matrix
integration. It does not select, alter, derive, or render a candidate word.
"""
from __future__ import annotations

import ast
from dataclasses import dataclass
from fractions import Fraction
import hashlib
from pathlib import Path
from typing import Final

from .adversarial_reproducibility_audit import ATTACKS, AttackKind, FullReproducibilityAudit
from .court_outward_return import CourtClosure
from .frequency_projection import CARRIERS
from .goetic_outward_return import BEARINGS, GoeticBearing, GoeticClosure
from .native_bearing_assembly import AEON_IDENTITIES, AEON_TOTAL, AeonClass, AeonIdentity
from .translation_matrix_integration import TranslationMatrixIntegrationField
from .truth_choice_certificates import TruthChoiceCertificate, TruthChoiceCertificateField


class ReleaseAuditError(ValueError):
    """Raised when one final acceptance obligation is not exact."""


REQUIRED_REPORTS: Final[tuple[str, ...]] = (
    "source_law_ledger.md",
    "projection_derivation.md",
    "goetic_closure_report.md",
    "court_closure_report.md",
    "uniqueness_report.md",
    "order_independence_report.md",
    "telegraphic_field_report.md",
    "final_audit.md",
)

SOURCE_LAW_FILES: Final[tuple[tuple[str, str], ...]] = (
    ("Phase 8", "frequency_projection.py"),
    ("Phase 9", "triadic_bridge.py"),
    ("Phase 10", "liquid_candidate_path.py"),
    ("Phase 11", "mas_shadow_conversion.py"),
    ("Phase 12", "goetic_outward_return.py"),
    ("Phase 13", "court_outward_return.py"),
    ("Phase 14", "exhaustive_chooser.py"),
    ("Phase 15", "native_bearing_assembly.py"),
```

```
    ("Phase 16", "truth_choice_certificates.py"),
    ("Phase 17", "adversarial_reproducibility_audit.py"),
    ("Phase 18", "translation_matrix_integration.py"),
    ("Phase 19", "release_audit.py"),
)


@dataclass(frozen=True, slots=True)
class CanonicalGoeticLaw:
    index: int
    glyph: str
    name: str
    native_q_bias: str
    q_vector: tuple[int, int, int, int]
    structural_hz: Fraction
    parity_hz: Fraction | None = None

    def __post_init__(self) -> None:
        if not 0 <= self.index < 12 or len(self.glyph) != 1:
            raise ReleaseAuditError("invalid canonical Goetic identity")
        if self.native_q_bias not in {"Q0", "Q1", "Q2", "Q3"}:
            raise ReleaseAuditError("invalid canonical Goetic Bias")
        if len(self.q_vector) != 4 or any(value not in {0, 1, 2, 3} for value in self.q_vector):
            raise ReleaseAuditError("invalid canonical Goetic vector")
        if self.structural_hz <= 0 or (self.parity_hz is not None and self.parity_hz <= 0):
            raise ReleaseAuditError("invalid canonical Goetic carrier")

    @property
    def q_vector_states(self) -> tuple[str, str, str, str]:
        return tuple(f"Q{value}" for value in self.q_vector)  # type: ignore[return-value]

    @property
    def frequency_text(self) -> str:
        structural = fraction_text(self.structural_hz)
        return structural if self.parity_hz is None else f"{structural} +
        ↪  i{fraction_text(self.parity_hz)}"

    @property
    def bec_text(self) -> str:
        return f"♙{self.glyph}♂{self.glyph}♙"


CANONICAL_GOETIC_LAW: Final[tuple[CanonicalGoeticLaw, ...]] = (
    CanonicalGoeticLaw(0, "🜲", "FETU", "Q3", (1, 1, 1, 3), Fraction(783, 100)),
    CanonicalGoeticLaw(1, "⬡", "KAL", "Q1", (1, 3, 0, 0), Fraction(174)),
    CanonicalGoeticLaw(2, "✡", "BABDH", "Q2", (1, 1, 3, 1), Fraction(528)),
    CanonicalGoeticLaw(3, "☆", "AHN", "Q0", (1, 2, 2, 0), Fraction(432), Fraction(417)),
    CanonicalGoeticLaw(4, "✪", "VEL", "Q1", (1, 3, 0, 1), Fraction(6311, 50)),
    CanonicalGoeticLaw(5, "❀", "SOR", "Q3", (1, 1, 1, 2), Fraction(10521, 50)),
    CanonicalGoeticLaw(6, "✳", "KOTH", "Q3", (1, 2, 1, 3), Fraction(741)),
    CanonicalGoeticLaw(7, "⧗", "DREH", "Q1", (1, 3, 2, 0), Fraction(852)),
    CanonicalGoeticLaw(8, "⊛", "RHEA", "Q2", (1, 2, 2, 1), Fraction(396)),
    CanonicalGoeticLaw(9, "❄", "ZHEK", "Q3", (1, 1, 2, 2), Fraction(963)),
    CanonicalGoeticLaw(10, "⚛", "SHAV", "Q1", (1, 3, 1, 1), Fraction(285)),
    CanonicalGoeticLaw(11, "⬡", "TRIG", "Q3", (1, 1, 3, 2), Fraction(639)),
)
```

```
if tuple(law.glyph for law in CANONICAL_GOETIC_LAW) != tuple(carrier.glyph for carrier in CARRIERS):
    raise RuntimeError("canonical Phase 19 law does not match Phase 8 carriers")


@dataclass(frozen=True, slots=True)
class ITTIWitness:
    outward_signature: tuple[tuple[int, int, int, str, int], ...]
    return_signature: tuple[tuple[int, int, int, str, int], ...]
    exact: bool

    def __post_init__(self) -> None:
        if not self.outward_signature or not self.return_signature:
            raise ReleaseAuditError("IT=TI witness lacks a complete path")
        if self.exact != (self.outward_signature == self.return_signature):
            raise ReleaseAuditError("IT=TI truth disagrees with path signatures")


@dataclass(frozen=True, slots=True)
class ReleaseBearingVerdict:
    aeon: AeonIdentity
    bearing: GoeticBearing
    bearing_index: int
    candidate: str
    native_q_bias: str
    unique_survivor: bool
    itti: ITTIWitness
    dcomp_zero: bool
    truth_one: bool
    motion_positive: bool
    bec_valid: bool
    lbec_valid: bool
    governing_parent_return_valid: bool
    all_twelve_currents: bool
    phase17_attacks_passed: bool
    phase18_word_match: bool
    certificate_digest: str

    def __post_init__(self) -> None:
        if self.bearing_index != BEARINGS.index(self.bearing):
            raise ReleaseAuditError("bearing index moved during release audit")
        if not self.candidate or self.native_q_bias not in {"Q0", "Q1", "Q2", "Q3"}:
            raise ReleaseAuditError("release bearing lacks candidate or Bias")
        if len(self.certificate_digest) != 64:
            raise ReleaseAuditError("release bearing lacks certificate digest")

    @property
    def accepted(self) -> bool:
        return all((
            self.unique_survivor,
            self.itti.exact,
            self.dcomp_zero,
            self.truth_one,
            self.motion_positive,
            self.bec_valid,
            self.lbec_valid,
```

```
            self.governing_parent_return_valid,
            self.all_twelve_currents,
            self.phase17_attacks_passed,
            self.phase18_word_match,
        ))


@dataclass(frozen=True, slots=True)
class ReleaseAeonVerdict:
    aeon: AeonIdentity
    governing_law: CanonicalGoeticLaw
    alternating_law: CanonicalGoeticLaw | None
    bearings: tuple[ReleaseBearingVerdict, ...]

    def __post_init__(self) -> None:
        if tuple(item.bearing for item in self.bearings) != BEARINGS or len(self.bearings) != 8:
            raise ReleaseAuditError("release Aeon lacks all eight bearings")
        if any(item.aeon != self.aeon for item in self.bearings):
            raise ReleaseAuditError("release Aeon mixed bearing identities")
        if self.aeon.aeon_class is AeonClass.GOETIC and self.alternating_law is not None:
            raise ReleaseAuditError("Goetic verdict received an alternating parent")
        if self.aeon.aeon_class is AeonClass.COURT and self.alternating_law is None:
            raise ReleaseAuditError("Court verdict lacks an alternating parent")

    @property
    def accepted(self) -> bool:
        return all(item.accepted for item in self.bearings)

    @property
    def words(self) -> tuple[str, ...]:
        return tuple(item.candidate for item in self.bearings)


@dataclass(frozen=True, slots=True)
class GlobalReleaseChecks:
    no_sydonic_authority_input: bool
    no_forbidden_function_sense: bool
    no_surface_form_blacklist: bool
    no_capitalization_filter: bool
    no_lowercase_or_uppercase_gate: bool
    no_single_token_gate: bool
    no_orthographic_pruning: bool
    no_graph_truncation: bool
    no_selected_pronunciation_shortcut: bool
    no_arbitrary_tie_break: bool
    no_candidate_order_dependence: bool
    no_float_closure: bool
    no_fallback: bool

    @property
    def accepted(self) -> bool:
        return all(getattr(self, name) for name in self.__dataclass_fields__)


@dataclass(frozen=True, slots=True)
class WorkerMergeWitness:
```

```
    worker: str
    jurisdiction: str
    passed: bool
    evidence: str


@dataclass(frozen=True, slots=True)
class ReleaseAuditField:
    phase16: TruthChoiceCertificateField
    phase17: FullReproducibilityAudit
    phase18: TranslationMatrixIntegrationField
    goetics: tuple[ReleaseAeonVerdict, ...]
    courts: tuple[ReleaseAeonVerdict, ...]
    global_checks: GlobalReleaseChecks
    workers: tuple[WorkerMergeWitness, ...]

    def __post_init__(self) -> None:
        if tuple(item.aeon for item in self.goetics + self.courts) != AEON_IDENTITIES:
            raise ReleaseAuditError("release field lost canonical 12+144 order")
        if len(self.goetics) != 12 or len(self.courts) != 144:
            raise ReleaseAuditError("release field requires 12 Goetics and 144 Courts")
        if any(not item.accepted for item in self.goetics + self.courts):
            raise ReleaseAuditError("one Goetic or Court failed final acceptance")
        if not self.global_checks.accepted:
            raise ReleaseAuditError("one global release prohibition failed")
        if len(self.workers) != 6 or not all(worker.passed for worker in self.workers):
            raise ReleaseAuditError("parallel worker merge did not pass")

    @property
    def bearing_count(self) -> int:
        return sum(len(item.bearings) for item in self.goetics + self.courts)


@dataclass(frozen=True, slots=True)
class ReportPin:
    filename: str
    absolute_path: str
    byte_size: int
    sha256: str


@dataclass(frozen=True, slots=True)
class ReleaseReportManifest:
    reports: tuple[ReportPin, ...]

    def __post_init__(self) -> None:
        if tuple(item.filename for item in self.reports) != REQUIRED_REPORTS:
            raise ReleaseAuditError("required report manifest is incomplete")
        if any(item.byte_size <= 0 or len(item.sha256) != 64 for item in self.reports):
            raise ReleaseAuditError("report manifest contains an invalid file")


def fraction_text(value: Fraction) -> str:
    return str(value.numerator) if value.denominator == 1 else
    ↪ f"{value.numerator}/{value.denominator}"
```

```
def sha256_file(path: Path) -> str:
    return hashlib.sha256(path.read_bytes()).hexdigest()


def _path_signature(steps: tuple[object, ...]) -> tuple[tuple[int, int, int, str, int], ...]:
    return tuple(
        (
            int(getattr(step, "position")),
            int(getattr(step, "cycle_index")),
            int(getattr(step, "office_index")),
            str(getattr(step, "office")),
            abs(int(getattr(step, "signed_velocity"))),
        )
        for step in steps
    )


def _itti(certificate: TruthChoiceCertificate) -> ITTIWitness:
    closure = certificate.phase15_closure.path_witness
    outward = _path_signature(closure.outward.steps)
    returned = _path_signature(closure.independent_return.steps)
    return ITTIWitness(outward, returned, outward == returned)


def _certificate_groups(field: TruthChoiceCertificateField) -> dict[int,
↪  tuple[TruthChoiceCertificate, ...]]:
    grouped: dict[int, list[TruthChoiceCertificate]] = {index: [] for index in range(AEON_TOTAL)}
    for certificate in field.certificates:
        grouped[certificate.aeon_constitution.aeon.index].append(certificate)
    result = {
        index: tuple(sorted(items, key=lambda item: item.bearing_index))
        for index, items in grouped.items()
    }
    if any(len(items) != 8 or tuple(item.bearing_index for item in items) != tuple(range(8)) for
    ↪  items in result.values()):
        raise ReleaseAuditError("Phase 16 source lacks one complete certificate body")
    return result


def _phase17_pass_map(field: FullReproducibilityAudit) -> dict[int, bool]:
    return {
        item.aeon.index: len(item.attacks) == len(ATTACKS) and all(witness.passed for witness in
        ↪  item.attacks)
        for item in field.aeons
    }


def _phase18_word_map(field: TranslationMatrixIntegrationField) -> dict[tuple[int, int], tuple[str,
↪  str]]:
    return {
        (body.aeon.index, word.bearing_index): (word.word, word.proof_digest)
        for body in field.chooser_snapshot.bodies
        for word in body.bearings
    }
```

```
def _canonical_constitution(certificate: TruthChoiceCertificate) -> tuple[CanonicalGoeticLaw,
↪  CanonicalGoeticLaw | None]:
    constitution = certificate.aeon_constitution
    governing = next(law for law in CANONICAL_GOETIC_LAW if law.glyph ==
    ↪  constitution.governing_glyph)
    if constitution.native_q_bias != governing.native_q_bias:
        raise ReleaseAuditError("native Q Bias differs from authored Canon")
    if constitution.native_q_vector != governing.q_vector_states:
        raise ReleaseAuditError("Q-vector differs from authored Canon")
    if constitution.alternating_glyph is None:
        if constitution.frequency_carrier_glyph != governing.glyph:
            raise ReleaseAuditError("Goetic carrier differs from governing identity")
        return governing, None
    alternating = next(law for law in CANONICAL_GOETIC_LAW if law.glyph ==
    ↪  constitution.alternating_glyph)
    if constitution.frequency_carrier_glyph != alternating.glyph:
        raise ReleaseAuditError("Court frequency differs from alternating parent")
    return governing, alternating


def _bearing_verdict(
    certificate: TruthChoiceCertificate,
    phase17_passed: bool,
    phase18_words: dict[tuple[int, int], tuple[str, str]],
) -> ReleaseBearingVerdict:
    closure = certificate.phase15_closure.path_witness
    if isinstance(closure, GoeticClosure):
        bec_valid = closure.bec_valid
        lbec_valid = True
        parent_return = closure.self_to_self
    elif isinstance(closure, CourtClosure):
        bec_valid = closure.bec_valid
        lbec_valid = closure.lbec_valid
        parent_return = closure.court_to_parent
    else:
        raise ReleaseAuditError("Goetic and Court return classes were merged or lost")
    mapped = phase18_words[(certificate.aeon_constitution.aeon.index, certificate.bearing_index)]
    return ReleaseBearingVerdict(
        aeon=certificate.aeon_constitution.aeon,
        bearing=certificate.phase15_closure.bearing,
        bearing_index=certificate.bearing_index,
        candidate=certificate.phase15_closure.candidate,
        native_q_bias=certificate.native_q_bias,
        unique_survivor=(
            len(certificate.unique_survivor_set) == 1
            and certificate.telegraphic_field_admission.unique
            and certificate.telegraphic_field_admission.admitted
        ),
        itti=_itti(certificate),
        dcomp_zero=(
            certificate.dcomp_witness.exact_zero
            and certificate.dcomp_witness.value == 0
            and certificate.exact_arithmetic.path_dcomp_zero
```

```
            and certificate.exact_arithmetic.path_dcomp_value == 0
        ),
        truth_one=certificate.truth == 1,
        motion_positive=certificate.motion > 0 and certificate.mas_witness.motion > 0,
        bec_valid=bec_valid,
        lbec_valid=lbec_valid,
        governing_parent_return_valid=parent_return,
        all_twelve_currents=len(certificate.current_witnesses) == 12 and all(item.satisfied for item
        ↪  in certificate.current_witnesses),
        phase17_attacks_passed=phase17_passed,
        phase18_word_match=mapped == (certificate.phase15_closure.candidate,
        ↪  certificate.proof_digest),
        certificate_digest=certificate.proof_digest,
    )


def _attack_count(field: FullReproducibilityAudit, attack: AttackKind) -> int:
    return sum(
        1 for aeon in field.aeons for witness in aeon.attacks
        if witness.attack is attack and witness.passed
    )


def _chooser_static_checks(package_root: Path) -> tuple[bool, bool, bool, bool]:
    no_authority = True
    no_case_gate = True
    no_float = True
    no_blacklist = True
    for _, filename in SOURCE_LAW_FILES[:9]:
        path = package_root / filename
        tree = ast.parse(path.read_text("utf-8"))
        imports = {
            node.module or "" for node in ast.walk(tree) if isinstance(node, ast.ImportFrom)
        } | {
            alias.name for node in ast.walk(tree) if isinstance(node, ast.Import) for alias in
            ↪  node.names
        }
        if any(any(marker in imported for marker in (
            "lexical_resolver", "lexer", "sequential_executor", "english_renderer",
        )) for imported in imports):
            no_authority = False
        attributes = {node.attr.casefold() for node in ast.walk(tree) if isinstance(node,
        ↪  ast.Attribute)}
        if attributes & {"lower", "upper", "islower", "isupper", "casefold", "capitalize", "title"}:
            no_case_gate = False
        if any(isinstance(node, ast.Constant) and isinstance(node.value, float) for node in
        ↪  ast.walk(tree)):
            no_float = False
        identifiers = {
            node.id.casefold() for node in ast.walk(tree) if isinstance(node, ast.Name)
        } | {
            node.arg.casefold() for node in ast.walk(tree) if isinstance(node, ast.arg)
        } | {
            node.name.casefold() for node in ast.walk(tree) if isinstance(node, (ast.FunctionDef,
            ↪  ast.ClassDef))
```

```
        }
        if identifiers & {"blacklist", "surface_blacklist", "forbidden_words", "blocked_words"}:
            no_blacklist = False
    return no_authority, no_case_gate, no_float, no_blacklist


def _global_checks(
    phase16: TruthChoiceCertificateField,
    phase17: FullReproducibilityAudit,
    phase18: TranslationMatrixIntegrationField,
) -> GlobalReleaseChecks:
    package_root = Path(__file__).resolve().parent
    no_authority, no_case_gate, no_float, no_blacklist = _chooser_static_checks(package_root)
    exact_graph = all(
        certificate.liquid_witness.typed_governor == (110, 144, 34)
        and all(witness == (110, 144, 34) for witness in
        ↪  certificate.liquid_witness.horizon_witnesses)
        for certificate in phase16.certificates
    )
    return GlobalReleaseChecks(
        no_sydonic_authority_input=no_authority and phase18.chooser_snapshot.authority_independent,
        no_forbidden_function_sense=_attack_count(phase17, AttackKind.FUNCTION_CLASS_COLLISION) ==
        ↪  156,
        no_surface_form_blacklist=no_blacklist and _attack_count(phase17,
        ↪  AttackKind.GRAPHIC_BODY_COLLISION) == 156,
        no_capitalization_filter=no_case_gate and _attack_count(phase17, AttackKind.CASE) == 156,
        no_lowercase_or_uppercase_gate=no_case_gate and _attack_count(phase17, AttackKind.CASE) ==
        ↪  156,
        no_single_token_gate=_attack_count(phase17, AttackKind.ORTHOGRAPHY) == 156,
        no_orthographic_pruning=_attack_count(phase17, AttackKind.ORTHOGRAPHY) == 156,
        no_graph_truncation=exact_graph and _attack_count(phase17, AttackKind.LIQUID_REDUCTION) ==
        ↪  156,
        no_selected_pronunciation_shortcut=_attack_count(phase17,
        ↪  AttackKind.PRONUNCIATION_VARIATION) == 156,
        no_arbitrary_tie_break=_attack_count(phase17, AttackKind.TIE) == 156,
        no_candidate_order_dependence=(
            _attack_count(phase17, AttackKind.CORPUS_ORDER) == 156
            and _attack_count(phase17, AttackKind.SERIALIZATION_ORDER) == 156
            and _attack_count(phase17, AttackKind.LABEL) == 156
        ),
        no_float_closure=no_float and _attack_count(phase17, AttackKind.NEAR_CLOSURE) == 156,
        no_fallback=_attack_count(phase17, AttackKind.FALLBACK) == 156,
    )


def _worker_witnesses(
    phase16: TruthChoiceCertificateField,
    phase17: FullReproducibilityAudit,
    phase18: TranslationMatrixIntegrationField,
    global_checks: GlobalReleaseChecks,
) -> tuple[WorkerMergeWitness, ...]:
    return (
        WorkerMergeWitness("Worker A", "source-law compliance",
        ↪  global_checks.no_sydonic_authority_input, "chooser remains independent of Sydonic
        ↪  authority and Phase 18 is one-way"),
```

```
        WorkerMergeWitness("Worker B", "exact mathematics and return proof",
        ↪  all(c.exact_arithmetic.exact for c in phase16.certificates), "1,248 exact arithmetic
        ↪  witnesses close"),
        WorkerMergeWitness("Worker C", "relational geometry", _attack_count(phase17,
        ↪  AttackKind.GRAPHIC_BODY_COLLISION) == 156, "sense identity survives graphic-body
        ↪  collision"),
        WorkerMergeWitness("Worker D", "prosody and frequency projection",
        ↪  all(c.triadic_bridge_witness.triad_valid for c in phase16.certificates), "all
        ↪  prosody-frequency bridges remain exact"),
        WorkerMergeWitness("Worker E", "telegraphic field boundary",
        ↪  all(c.telegraphic_field_admission.admitted for c in phase16.certificates), "all 1,248
        ↪  fields admit exactly one survivor"),
        WorkerMergeWitness("Worker F", "adversarial and order-independence",
        ↪  phase17.passed_attack_count == 2808 and phase18.chooser_snapshot.word_count == 1248,
        ↪  "2,808 attacks pass and 1,248 words remain fixed"),
    )


def build_release_audit(
    phase16: TruthChoiceCertificateField,
    phase17: FullReproducibilityAudit,
    phase18: TranslationMatrixIntegrationField,
) -> ReleaseAuditField:
    phase16_digests = tuple(certificate.proof_digest for certificate in phase16.certificates)
    phase17_digests = tuple(certificate.proof_digest for certificate in phase17.source.certificates)
    phase18_digests = tuple(
        word.proof_digest
        for body in phase18.chooser_snapshot.bodies
        for word in body.bearings
    )
    if (
        phase17_digests != phase16_digests
        or phase18_digests != phase16_digests
        or phase18.chooser_snapshot.source_certificate_count != len(phase16.certificates)
    ):
        raise ReleaseAuditError("release phases do not share the same canonical source field")
    if phase17.passed_attack_count != 2808 or phase17.bearing_check_count != 22464:
        raise ReleaseAuditError("canonical adversarial field is incomplete")
    groups = _certificate_groups(phase16)
    attack_map = _phase17_pass_map(phase17)
    phase18_words = _phase18_word_map(phase18)
    aeons: list[ReleaseAeonVerdict] = []
    for aeon in AEON_IDENTITIES:
        certificates = groups[aeon.index]
        governing, alternating = _canonical_constitution(certificates[0])
        if any(_canonical_constitution(item) != (governing, alternating) for item in certificates):
            raise ReleaseAuditError("Aeon constitution changed across its eight bearings")
        bearings = tuple(_bearing_verdict(item, attack_map[aeon.index], phase18_words) for item in
        ↪  certificates)
        aeons.append(ReleaseAeonVerdict(aeon, governing, alternating, bearings))
    body = tuple(aeons)
    global_checks = _global_checks(phase16, phase17, phase18)
    workers = _worker_witnesses(phase16, phase17, phase18, global_checks)
    return ReleaseAuditField(phase16, phase17, phase18, body[:12], body[12:], global_checks,
    ↪  workers)
```

```
def _source_ledger(field: ReleaseAuditField, package_root: Path) -> str:
    lines = [
        "# Source Law Ledger", "",
        "The release is pinned to exact source bytes. Sydonic .aksh authority is consulted only
        ↪  after Phase 16 closure.", "",
        "| Phase | Source | Bytes | SHA-256 |", "|---|---|---:|---|",
    ]
    for phase, filename in SOURCE_LAW_FILES:
        path = package_root / filename
        lines.append(f"| {phase} | `{filename}` | {path.stat().st_size} | `{sha256_file(path)}` |")
    lines += ["", "## Canonical Goetic law", "", "| # | Glyph | Name | Bias | Vector | Carrier | BEC
    ↪  |", "|---:|---|---|---|---|---|---|"]
    for law in CANONICAL_GOETIC_LAW:
        lines.append(f"| {law.index + 1} | {law.glyph} | {law.name} | {law.native_q_bias} |
        ↪  {list(law.q_vector)} | {law.frequency_text} | {law.bec_text} |")
    lines += ["", "## Merge witnesses", ""]
    for worker in field.workers:
        lines.append(f"- **{worker.worker}, {worker.jurisdiction}: PASS.** {worker.evidence}")
    return "\n".join(lines) + "\n"


def _projection_derivation(field: ReleaseAuditField) -> str:
    lines = [
        "# Projection Derivation", "",
        "All carriers are exact Fractions. Breath remains bounded by [-Φ,+Φ]. AHN preserves
        ↪  structural 432 and parity i417 as distinct channels.", "",
        "| Glyph | Name | Bias | Vector | Structural | Parity | Accepted bearings |",
        ↪  "|---|---|---|---|---:|---:|---:|",
    ]
    for law, verdict in zip(CANONICAL_GOETIC_LAW, field.goetics, strict=True):
        parity = "none" if law.parity_hz is None else f"i{fraction_text(law.parity_hz)}"
        lines.append(f"| {law.glyph} | {law.name} | {law.native_q_bias} | {list(law.q_vector)} |
        ↪  {fraction_text(law.structural_hz)} | {parity} | {sum(b.accepted for b in
        ↪  verdict.bearings)}/8 |")
    lines += [
        "", "## FETU", "", "```text",
        "A=◎", "Ω=783/100", "β=Q3", "v=[1,1,1,3]", "BEC=♁◎♂◎♁",
        "J_◎(w)=F_ν,◎(Pros(w);783/100±Φ) ∘ F_ν,◎(Geo(w);[1,1,1,3]) ∘ F_β,◎(Q3)",
        "Truth=1 iff D_T=0, BECValid=1, Gap>0, motion>0, and all twelve currents are witnessed.",
        ↪  "````",
        "", "## FetuKeth", "", "```text",
        "C=C_1,7", "governing=◎", "alternating=❋", "β=Q3", "v=[1,1,1,3]", "ν=741+δ, δ∈[-Φ,Φ]",
        "Out: C -> candidate-bearing path", "Back: candidate-bearing path -> ◎",
        "Truth=1 iff D_T=0, L-BEC=1, parent BEC=1, Parent(return)=◎, and motion>0.", "````",
        "", "## KAL × ZHEK versus ZHEK × KAL", "", "```text",
        "C_KAL,ZHEK inherits Bias/vector from KAL and frequency 963±Φ from ZHEK.",
        "C_ZHEK,KAL inherits Bias/vector from ZHEK and frequency 174±Φ from KAL.",
        "The field is shared; the mathematics and closure paths are not.", "````",
        "", "## AHN", "", "```text",
        "A=✡", "β=Q0", "v=[1,2,2,0]", "structural=432", "parity=i417", "breath=±Φ",
        "Both components are preserved independently through outward and return.", "````",
    ]
    return "\n".join(lines) + "\n"
```

```
def _goetic_report(field: ReleaseAuditField) -> str:
    lines = ["# Goetic Closure Report", "", "All 12 Goetics close all eight bearings under STRICT.",
    ↪  "", "| Goetic | Bias | 8 unique fields | IT=TI | D-COMP=0 | Truth=1 | motion>0 | BEC |",
    ↪  "|---|---|---:|---:|---:|---:|---:|---:|"]
    for item in field.goetics:
        b = item.bearings
        lines.append(f"| {item.governing_law.glyph} {item.governing_law.name} |
        ↪  {item.governing_law.native_q_bias} | {sum(x.unique_survivor for x in b)}/8 |
        ↪  {sum(x.itti.exact for x in b)}/8 | {sum(x.dcomp_zero for x in b)}/8 | {sum(x.truth_one
        ↪  for x in b)}/8 | {sum(x.motion_positive for x in b)}/8 | {sum(x.bec_valid for x in b)}/8
        ↪  |")
        lines.append(f"\nBearing words: `{' | '.join(item.words)}`\n")
    return "\n".join(lines) + "\n"


def _court_report(field: ReleaseAuditField) -> str:
    lines = ["# Court Closure Report", "", "All 144 ordered Courts close all eight bearings and
    ↪  return to the governing parent.", "", "| Court | Governing | Alternating carrier | Bias |
    ↪  Unique | IT=TI | D-COMP=0 | Truth=1 | motion>0 | L-BEC | Parent return |",
    ↪  "|---|---|---|---|---:|---:|---:|---:|---:|---:|---:|"]
    for item in field.courts:
        b = item.bearings
        alt = item.alternating_law
        assert alt is not None
        lines.append(f"| `{item.aeon.key}` | {item.governing_law.glyph} {item.governing_law.name} |
        ↪  {alt.glyph} {alt.name}, {alt.frequency_text}±Φ | {item.governing_law.native_q_bias} |
        ↪  {sum(x.unique_survivor for x in b)}/8 | {sum(x.itti.exact for x in b)}/8 |
        ↪  {sum(x.dcomp_zero for x in b)}/8 | {sum(x.truth_one for x in b)}/8 |
        ↪  {sum(x.motion_positive for x in b)}/8 | {sum(x.lbec_valid for x in b)}/8 |
        ↪  {sum(x.governing_parent_return_valid for x in b)}/8 |")
    return "\n".join(lines) + "\n"


def _uniqueness_report(field: ReleaseAuditField) -> str:
    all_bearings = tuple(b for aeon in field.goetics + field.courts for b in aeon.bearings)
    lines = [
        "# Uniqueness Report", "",
        f"- Aeons: {len(field.goetics) + len(field.courts)}",
        f"- Bearing sub-fields: {len(all_bearings)}",
        f"- Sub-fields with exactly one survivor: {sum(b.unique_survivor for b in all_bearings)}",
        f"- Tie attacks passed: {_attack_count(field.phase17, AttackKind.TIE)}/156",
        f"- Fallback attacks passed: {_attack_count(field.phase17, AttackKind.FALLBACK)}/156",
        f"- Graphic-body collision attacks passed: {_attack_count(field.phase17,
        ↪  AttackKind.GRAPHIC_BODY_COLLISION)}/156",
        "", "No word is selected by capitalization, spelling collapse, display preference, first
        ↪  sense, ranking, score, or enumeration position.",
    ]
    return "\n".join(lines) + "\n"


def _order_report(field: ReleaseAuditField) -> str:
    court_records = [c.phase15_closure.source_record for c in field.phase16.certificates if
    ↪  c.aeon_constitution.aeon.aeon_class is AeonClass.COURT]
```

```
    seven = sum(len(record.enumeration_witnesses) == 7 and record.order_independent for record in
    ↪  court_records)
    lines = [
        "# Order Independence Report", "",
        f"- Court bearing records carrying all seven enumeration witnesses:
        ↪  {seven}/{len(court_records)}",
        f"- Corpus-order attacks: {_attack_count(field.phase17, AttackKind.CORPUS_ORDER)}/156",
        f"- Serialization-order attacks: {_attack_count(field.phase17,
        ↪  AttackKind.SERIALIZATION_ORDER)}/156",
        f"- Label attacks: {_attack_count(field.phase17, AttackKind.LABEL)}/156",
        "- Candidate-order-dependent emissions: 0",
        "", "Manifest, reversed, deterministic permutations, file-shuffled source, and parallel
        ↪  chunk order yield identical survivor sets.",
    ]
    return "\n".join(lines) + "\n"


def _telegraphic_report(field: ReleaseAuditField) -> str:
    certs = field.phase16.certificates
    attempted = sum(c.telegraphic_field_admission.attempted_count for c in certs)
    truth_records = sum(c.telegraphic_field_admission.truth_record_count for c in certs)
    lines = [
        "# Telegraphic Field Report", "",
        f"- Certificates: {len(certs)}",
        f"- Attempted candidate records: {attempted}",
        f"- Truth records: {truth_records}",
        f"- Admitted unique fields: {sum(c.telegraphic_field_admission.admitted for c in
        ↪  certs)}/{len(certs)}",
        f"- Twelve-current witnesses: {sum(len(c.current_witnesses) for c in certs)}",
        f"- Phase 18 post-closure uses: {len(field.phase18.canonical_uses)}",
        "", "The telegraphic field ends at unique mathematical admission. Sydonic .aksh authority is
        ↪  consulted afterward and cannot reach backward into choice.",
    ]
    return "\n".join(lines) + "\n"


def _final_report(field: ReleaseAuditField) -> str:
    """Render only computed predicates and measured counts."""
    all_aeons = field.goetics + field.courts
    all_bearings = tuple(bearing for aeon in all_aeons for bearing in aeon.bearings)
    lines = [
        "# Final System Audit",
        "",
        f"Goetic Aeons accepted: {sum(item.accepted for item in
        ↪  field.goetics)}/{len(field.goetics)}",
        f"Court Aeons accepted: {sum(item.accepted for item in field.courts)}/{len(field.courts)}",
        f"Bearing verdicts accepted: {sum(item.accepted for item in
        ↪  all_bearings)}/{len(all_bearings)}",
        f"Exact IT=TI witnesses: {sum(item.itti.exact for item in
        ↪  all_bearings)}/{len(all_bearings)}",
        f"Exact D-COMP zeros: {sum(item.dcomp_zero for item in all_bearings)}/{len(all_bearings)}",
        f"Truth=1 witnesses: {sum(item.truth_one for item in all_bearings)}/{len(all_bearings)}",
        f"Positive-motion witnesses: {sum(item.motion_positive for item in
        ↪  all_bearings)}/{len(all_bearings)}",
```

```
        f"Twelve-current-complete bearings: {sum(item.all_twelve_currents for item in
        ↪  all_bearings)}/{len(all_bearings)}",
        f"Phase 17 attack-complete bearings: {sum(item.phase17_attacks_passed for item in
        ↪  all_bearings)}/{len(all_bearings)}",
        f"Phase 18 unchanged-word matches: {sum(item.phase18_word_match for item in
        ↪  all_bearings)}/{len(all_bearings)}",
        f"Adversarial verdicts passed: {field.phase17.passed_attack_count}/2808",
        f"Adversarial bearing checks passed: {field.phase17.bearing_check_count}/22464",
        f"Global prohibition body: {'PASS' if field.global_checks.accepted else 'FAIL'}",
        f"Worker merge body: {'PASS' if all(worker.passed for worker in field.workers) else
        ↪  'FAIL'}",
    ]
    return "\n".join(lines) + "\n"


def write_release_reports(field: ReleaseAuditField, reports_dir: str | Path) ->
↪  ReleaseReportManifest:
    root = Path(reports_dir).resolve()
    root.mkdir(parents=True, exist_ok=True)
    package_root = Path(__file__).resolve().parent
    bodies = {
        "source_law_ledger.md": _source_ledger(field, package_root),
        "projection_derivation.md": _projection_derivation(field),
        "goetic_closure_report.md": _goetic_report(field),
        "court_closure_report.md": _court_report(field),
        "uniqueness_report.md": _uniqueness_report(field),
        "order_independence_report.md": _order_report(field),
        "telegraphic_field_report.md": _telegraphic_report(field),
        "final_audit.md": _final_report(field),
    }
    pins: list[ReportPin] = []
    for filename in REQUIRED_REPORTS:
        path = root / filename
        path.write_text(bodies[filename], "utf-8")
        pins.append(ReportPin(filename, str(path), path.stat().st_size, sha256_file(path)))
    return ReleaseReportManifest(tuple(pins))


def prove_phase19_exit_gate(
    phase16: TruthChoiceCertificateField,
    phase17: FullReproducibilityAudit,
    phase18: TranslationMatrixIntegrationField,
    reports_dir: str | Path,
) -> tuple[ReleaseAuditField, ReleaseReportManifest]:
    field = build_release_audit(phase16, phase17, phase18)
    manifest = write_release_reports(field, reports_dir)
    return field, manifest
```

release_seal.py

```
"""Computed release witness. Passing predicates are not overridden by prose."""
from __future__ import annotations
from dataclasses import asdict, dataclass
import hashlib
import json
from pathlib import Path

from .engine import ENGINE_VERSION
from .english_renderer import RENDERER_VERSION
from .trace import TRACE_VERSION
from .implementation_reconciliation import PACKAGE_AKSH_ROOT, AKSH_AUTHORITY_FILES,
↪  require_package_aksh_authority
from .lexical_resolver import LEXICAL_AUTHORITY_FILES, LEXICON_VERSION, LexicalResolver
from .aksh_reader import read_lexicon, attributes

SEAL_VERSION = "absolute-fidelity-phase2-v1"

def _sha(path: Path) -> str:
    return hashlib.sha256(path.read_bytes()).hexdigest()

@dataclass(frozen=True, slots=True)
class FrozenAsset:
    path: str
    version: str
    sha256: str

@dataclass(frozen=True, slots=True)
class ReleaseRequirement:
    name: str
    passed: bool
    evidence: str

@dataclass(frozen=True, slots=True)
class ReleaseSeal:
    seal_version: str
    engine_version: str
    renderer_version: str
    trace_version: str
    semantic_lexicon: str
    frozen_assets: tuple[FrozenAsset, ...]
    requirements: tuple[ReleaseRequirement, ...]
    release_allowed: bool
    status: str

    def to_json(self) -> str:
        return json.dumps(asdict(self), ensure_ascii=False, sort_keys=True, separators=(",", ":"))

    @property
```

```
    def sha256(self) -> str:
        return hashlib.sha256(self.to_json().encode()).hexdigest()

class ReleaseSealer:
    def __init__(self) -> None:
        self.authority = PACKAGE_AKSH_ROOT

    @staticmethod
    def _authority_version(path: Path) -> str:
        if path.name == "Enochian…Understandings.aksh":
            lines = path.read_text(encoding="utf-8").splitlines()
            version = attributes(lines[0]).get("version", "") if lines else ""
        else:
            version = read_lexicon(path).header.get("version", "")
        if not version:
            raise RuntimeError(f"required .aksh version missing: {path.name}")
        return version

    def build(self) -> ReleaseSeal:
        require_package_aksh_authority()
        resolver = LexicalResolver(self.authority)
        assets = tuple(
            FrozenAsset(
                str((self.authority / name).relative_to(Path(__file__).resolve().parent)),
                self._authority_version(self.authority / name),
                _sha(self.authority / name),
            )
            for name in AKSH_AUTHORITY_FILES
        )
        lexical_authorities = resolver.authority_entries()
        lexical_authority_counts = tuple(
            (filename, len(entries))
            for filename, entries in lexical_authorities
        )
        lexical_shape_ok = (
            tuple(filename for filename, _ in lexical_authorities) == LEXICAL_AUTHORITY_FILES
            and all(
                frozenset(entry.glyph for entry in entries) == resolver.lexical_glyphs
                for _, entries in lexical_authorities
            )
        )
        req = (
            ReleaseRequirement(
                "Current authored .aksh names reconciled",
                True,
                f"{len(AKSH_AUTHORITY_FILES)} current .aksh contracts; 0 undeclared active
                ↪  contracts",
            ),
            ReleaseRequirement(
                "Authored lexical authorities preserve ordinary glyph bodies",
                lexical_shape_ok,
                repr(lexical_authority_counts),
            ),
        )
        passed = all(item.passed for item in req)
        return ReleaseSeal(
```

```
    SEAL_VERSION,
    ENGINE_VERSION,
    RENDERER_VERSION,
    TRACE_VERSION,
    LEXICON_VERSION,
    assets,
    req,
    passed,
    "PASS" if passed else "FAIL",
)
```

rendering_audit.py

```
"""Rendering and return audit over one complete Aeonic line."""
from __future__ import annotations
from dataclasses import asdict, dataclass
import hashlib, json
from .engine import TranslationResult

AUDIT_VERSION="aeonic-line-return-audit-v1"

class ReturnAuditError(RuntimeError): pass

@dataclass(frozen=True, slots=True)
class WordReturnRecord:
    output_index:int
    text:str
    kind:str
    source_position:int|None
    source_glyph:str|None
    source_body_id:str|None
    grammatical_bearing:str|None
    clothing_role:str|None
    traceable:bool
    adds_source_meaning:bool

@dataclass(frozen=True, slots=True)
class ReturnAudit:
    audit_version:str
    translation:str
    source_string:str
    trace_sha256:str
    content_words:tuple[WordReturnRecord,...]
    support_words:tuple[WordReturnRecord,...]
    absent_words:tuple[WordReturnRecord,...]
    exact_source:bool
    dependency_preserved:bool
    return_preserved:bool
    untraceable_content_words:tuple[str,...]
    status:str
    def to_json(self)->str:
        return json.dumps(asdict(self),ensure_ascii=False,sort_keys=True,separators=(",",":"))
    @property
    def sha256(self)->str:
        return hashlib.sha256(self.to_json().encode()).hexdigest()

def audit_translation(result:TranslationResult)->ReturnAudit:
    trace=result.semantic_trace
    if trace is None: raise ReturnAuditError("semantic trace required for return audit")
    content=[]; support=[]; absent=[]
```

```
source_positions={t.source_position for t in trace.rendering_input.tokens if
↪  t.resolved_state_body is not None}
for i,w in enumerate(result.rendering.final_words):
    if w.kind=="content":
        ok=w.source_position in source_positions and w.source_glyph is not None and
        ↪  w.source_body_id is not None
        content.append(WordReturnRecord(i,w.text,w.kind,w.source_position,w.source_glyph,w.sour⌋
        ↪  ce_body_id,w.grammatical_bearing,w.clothing_role,ok,True))
    elif w.kind=="clothing":
        ok=w.clothing_role is not None and w.source_position is None and w.source_glyph is None
        ↪  and w.source_body_id is None
        support.append(WordReturnRecord(i,w.text,w.kind,None,None,None,None,w.clothing_role,ok,⌋
        ↪  False))
    elif w.kind=="absent-word":
        ok=w.text=="·" and w.source_glyph=="·"
        absent.append(WordReturnRecord(i,w.text,w.kind,w.source_position,w.source_glyph,None,No⌋
        ↪  ne,None,ok,False))
bad=tuple(r.text for r in content if not r.traceable)
exact_source=trace.source_string==trace.rendering_input.source==trace.parsed_frame.source
dependency=all(r.traceable for r in content) and all(r.traceable for r in support) and
↪  all(r.traceable for r in absent)
status="PASS" if not bad and exact_source and dependency else "FAIL"
audit=ReturnAudit(AUDIT_VERSION,result.translation,trace.source_string,trace.sha256,tuple(conte⌋
↪  nt),tuple(support),tuple(absent),exact_source,dependency,dependency and
↪  exact_source,bad,status)
if status!="PASS": raise ReturnAuditError(audit.to_json())
return audit
```

saved_trace.py

```
"""Stable saved-trace envelope for Sydonic Magicae public surfaces."""
from __future__ import annotations

from dataclasses import dataclass
import json
from pathlib import Path

from .trace import DomusTrace

SAVED_TRACE_FORMAT = "sydonic-aeonic-line-trace-v1"

class SavedTraceError(RuntimeError):
    pass

@dataclass(frozen=True, slots=True)
class SavedTraceEnvelope:
    format: str
    trace_sha256: str
    trace: DomusTrace

    def to_json(self) -> str:
        payload = {
            "format": self.format,
            "trace_sha256": self.trace_sha256,
            "trace": self.trace.to_dict(),
        }
        return json.dumps(payload, ensure_ascii=False, sort_keys=True, separators=(",", ":"))

    @classmethod
    def from_json(cls, payload: str) -> "SavedTraceEnvelope":
        raw = json.loads(payload)
        if raw.get("format") != SAVED_TRACE_FORMAT:
            raise SavedTraceError(f"unsupported saved trace format: {raw.get('format')!r}")
        trace_json = json.dumps(raw.get("trace"), ensure_ascii=False, sort_keys=True,
        ↪  separators=(",", ":"))
        trace = DomusTrace.from_json(trace_json)
        expected = raw.get("trace_sha256")
        if trace.sha256 != expected:
            raise SavedTraceError("saved trace checksum mismatch")
        return cls(SAVED_TRACE_FORMAT, trace.sha256, trace)


def save_trace(trace: DomusTrace, path: str | Path) -> Path:
    target = Path(path)
    envelope = SavedTraceEnvelope(SAVED_TRACE_FORMAT, trace.sha256, trace)
    target.write_text(envelope.to_json() + "\n", encoding="utf-8")
    return target
```

```
def load_trace(path: str | Path) -> DomusTrace:
    return SavedTraceEnvelope.from_json(Path(path).read_text(encoding="utf-8")).trace
```

sequential_executor.py

```
"""Pattern-driven execution of one exact Sydonic Magicae Aeonic line.

Every visible code point is preserved. Ordinary lexical bodies resolve through
SeeD Body. Tail resolves traversal first; other Enochians execute that order.
"""
from __future__ import annotations

from dataclasses import replace

from .zero_and_one import (
    AXIOMYR_GLYPH, SHADOW_LOCUS, ZERO_MIDDLE_GLYPH, TRIPARTITE_AXIOMYR,
    ZeroAndOneEvent, ZeroAndOneTrace, ZeroAndOneMachine,
)
from .local_grammar import (
    SUBSTRATE, CATALYST, RELIC, OUROBORIC, SPATIAL_ABSOLUTE, PHANTASMAGORIA_GLYPH,
    LocalGrammarError, LocalGrammarEvent,
    PhantasmagoriaTransmutation, LocalGrammarTrace, OuroboricState, RecognitionRequest,
    ResolvedBody, AeternumBearingEncounter, AeternumBearingPath,
)
from .models import DomusFrame
from .temporal_operator import TEMPORAL_OPERATOR_GLYPH, resolve_tail
from .frequency_projection import CARRIER_BY_GLYPH

try:
    from ._sydonic_kernel import walk_aeternum as _compiled_walk_aeternum
except ImportError as exc:
    raise RuntimeError("Sydonic requires the compiled Aeternum index kernel") from exc

OFFICE_BEARINGS = {
    RELIC: "relic",
    CATALYST: "catalyst",
    SUBSTRATE: "substrate",
    SPATIAL_ABSOLUTE: "spatial_absolute",
    "⊖": "temporal_absolute",
    "⊙": "terminal_state",
    "⊛": "adversarial",
    "⊕": "recursive_identity",
}
VISIBLE_GRAMMARS = frozenset(set(OFFICE_BEARINGS) | {OUROBORIC, PHANTASMAGORIA_GLYPH,
↪  TEMPORAL_OPERATOR_GLYPH})

class SequentialDomusExecutor:
    """Execute the complete Aeonic line by lexical bodies and grammar stacks."""

    def __init__(self, zero_and_one_machine: ZeroAndOneMachine) -> None:
        self.machine = zero_and_one_machine

    def _resolve_office(self, body: ResolvedBody, office: str) -> object:
```

```
    resolver = self.machine.resolver
    owner = getattr(resolver, "__self__", None)
    if owner is None or not hasattr(owner, "resolve_office"):
        raise LocalGrammarError(f"authored {office} resolver is unavailable for {body.glyph!r}")
    return owner.resolve_office(body.glyph, office)

def _resolve_special(self, glyph: str) -> object:
    resolver = self.machine.resolver
    owner = getattr(resolver, "__self__", None)
    if owner is None or not hasattr(owner, "resolve_special"):
        raise LocalGrammarError(f"authored Axiomyr/Shadow special resolver is unavailable for
        ↪  {glyph!r}")
    return owner.resolve_special(glyph)

def execute_frame(
    self,
    frame: DomusFrame,
    *,
    aeternum_token=None,
    aeternum_pair_reorder=None,
    aeternum_corridor_reorder=None,
) -> tuple[ZeroAndOneTrace, LocalGrammarTrace]:
    aeternum_token = frame.aeternum_token if aeternum_token is None else aeternum_token
    aeternum_period_source_positions: tuple[int, ...] | None = None
    zero_and_one_events: list[ZeroAndOneEvent] = []
    grammar_events: list[LocalGrammarEvent] = []
    source_body_order: list[int] = []
    causal_body_order: list[int] = []
    bodies: dict[int, ResolvedBody] = {}
    recognitions: list[RecognitionRequest] = []
    loops: list[OuroboricState] = []
    transmutations: list[PhantasmagoriaTransmutation] = []
    aeternum_bearing_paths: list[AeternumBearingPath] = []

    pending_orobouros_head: tuple[tuple[int, ...], int, tuple[tuple[int, str], ...], tuple[int,
    ↪  ...], int] | None = None
    active_loop: tuple[tuple[int, ...], int, tuple[tuple[int, str], ...], tuple[int, ...], int,
    ↪  int, int] | None = None
    pending_transmutation: tuple[tuple[int, ...], int, list[int]] | None = None

    tokens = frame.tokens
    tail_resolution = resolve_tail(frame)
    execution_positions = tail_resolution.execution_positions
    execution_index = {position: index for index, position in enumerate(execution_positions)}
    # Tail has completed its work. Head and other grammar consume its result;
    # the Tail glyphs remain source/trace identities, not pending operations.
    scan_tokens = tuple(tokens[pos] for pos in execution_positions if tokens[pos].value !=
    ↪  TEMPORAL_OPERATOR_GLYPH)
    if tail_resolution.transformations:
        original_aeternum_token = aeternum_token

        def execution_aeternum_token(virtual_position: int):
            extent = len(tokens)
            phase = virtual_position % (2 * extent)
            index = phase if phase < extent else 2 * extent - 1 - phase
```

```
        source_position = execution_positions[index]
        return replace(
            original_aeternum_token(source_position),
            source_position=source_position,
            mirror_position=virtual_position % extent,
            imaginary=not (0 <= virtual_position < extent),
        )

    aeternum_token = execution_aeternum_token
i = 0
semantic_source = frame.source if frame.prosody_source is None else frame.prosody_source
shadow_ostensive = semantic_source == ZERO_MIDDLE_GLYPH
axiomyr_ostensive = semantic_source == TRIPARTITE_AXIOMYR

# Resolve the one finite real center before grammar attaches to it.  This does
# not change source/causal order; it only makes the same real bodies available
# when a boundary operator reaches them through the Aeternum mirror field.
for source_token in tokens:
    if source_token.value in self.machine.ordinary_lexical_glyphs:
        bodies[source_token.position] = ResolvedBody(
            source_token.position,
            source_token.value,
            self.machine.resolver(source_token.value),
        )
    elif source_token.value == SHADOW_LOCUS and shadow_ostensive:
        bodies[source_token.position] = ResolvedBody(
            source_token.position, source_token.value,
            ↪  self._resolve_special(source_token.value),
        )
    elif source_token.value == AXIOMYR_GLYPH:
        resolution = (
            self._resolve_special(source_token.value)
            if axiomyr_ostensive else self.machine.resolver(source_token.value)
        )
        bodies[source_token.position] = ResolvedBody(
            source_token.position, source_token.value, resolution,
        )

# Causal order starts as the exact real speaking-body order of the immutable
# center.  The scanner's source_body_order below remains a separate record of
# visited source positions.  Aeternum may therefore resolve topology against a
# real body before the scanner reaches its code point without creating a body.
causal_body_order[:] = [pos for pos in execution_positions if pos in bodies]

def source_bearing(source_position: int) -> str | None:
    """Return the authored bearing/law carried by one finite source position."""
    token = tokens[source_position]
    glyph = token.value
    if glyph in OFFICE_BEARINGS:
        return OFFICE_BEARINGS[glyph]
    if glyph == OUROBORIC:
        return "orobouros_head"
    if glyph == PHANTASMAGORIA_GLYPH:
        return "phantasmagoria"
    if glyph == TEMPORAL_OPERATOR_GLYPH:
        return "orobouros_tail"
```

```
    if source_position in bodies:
        if token.kind == "lexical":
            return "seed_identity"
        if token.value in {SHADOW_LOCUS, AXIOMYR_GLYPH}:
            return None
    return None

def bearing_context(
    position: int, count: int, step: int, *, purpose: str
) -> tuple[tuple[int, int], ...]:
    """Traverse the complete Aeternum field until the center's obligation closes.

    Every encountered glyph is retained as a bearing-only witness.  Reflected
    glyphs are never translated as a second utterance and never become new source
    extent.  The return value contains only the speaking-body encounters needed by
    the center law, while ``aeternum_bearing_paths`` preserves every intervening
    Enoch and its unchanged authored bearing.
    """
    if count < 0 or step not in {-1, 1}:
        raise ValueError("invalid Aeternum bearing-context request")
    if count == 0:
        return ()
    if not bodies:
        raise LocalGrammarError("Aeternum center has no speaking body for this utterance
        ↪  obligation")
    nonlocal aeternum_period_source_positions
    if aeternum_period_source_positions is None:
        period_length = 2 * len(tokens)
        aeternum_period_source_positions = tuple(
            aeternum_token(virtual_position).source_position
            for virtual_position in range(period_length)
        )

    out: list[tuple[int, int]] = []
    encounters: list[AeternumBearingEncounter] = []
    start_virtual_position = execution_index[position] + step
    # Every 2N virtual positions repeat the exact finite source identities.
    # Python materializes that one authored period once; the compiled kernel
    # performs only the repeated integer walk and speaking-body mask checks.
    limit = 2 * len(tokens) * (count + 1)
    speaking_mask = bytes(
        1 if source_position in bodies else 0
        for source_position in range(len(tokens))
    )
    visited_source_positions, completed = _compiled_walk_aeternum(
        aeternum_period_source_positions,
        speaking_mask,
        start_virtual_position,
        step,
        count,
        limit,
    )
    for offset, source_position in enumerate(visited_source_positions):
        virtual_position = start_virtual_position + step * offset
        source_token = tokens[source_position]
        speaking = source_position in bodies
```

```
        encounters.append(AeternumBearingEncounter(
            virtual_position=virtual_position,
            source_position=source_position,
            glyph=source_token.value,
            token_kind=source_token.kind,
            bearing=source_bearing(source_position),
            imaginary=not (0 <= virtual_position < len(tokens)),
            speaking_body=speaking,
            bearing_only=True,
        ))
        if speaking:
            out.append((source_position, virtual_position))
    if completed:
        aeternum_bearing_paths.append(AeternumBearingPath(
            purpose=purpose,
            origin_position=position,
            step=step,
            required_speaking_bodies=count,
            speaking_body_source_positions=tuple(pos for pos, _ in out),
            encounters=tuple(encounters),
            completed=True,
        ))
        return tuple(out)
    raise LocalGrammarError("Aeternum mirror failed to close a speaking-body context")

# Compatibility name retained deliberately: callers still request lexical context,
# but the implementation now proves the full glyph-by-glyph bearing traversal.
def lexical_context(
    position: int, count: int, step: int, *, purpose: str = "grammar-context"
) -> tuple[tuple[int, int], ...]:
    return bearing_context(position, count, step, purpose=purpose)

for transformation in tail_resolution.transformations:
    crossing = transformation.crossing
    grammar_events.append(LocalGrammarEvent(
        transformation.operator_position, TEMPORAL_OPERATOR_GLYPH,
        "orobouros-tail-transformation", transformation.after_positions, transformation,
    ))
    zero_and_one_events.append(ZeroAndOneEvent(
        transformation.operator_position, TEMPORAL_OPERATOR_GLYPH,
        tokens[transformation.operator_position].kind, "orobouros-tail-resolved", False,
    ))
    aeternum_bearing_paths.append(AeternumBearingPath(
        purpose="orobouros-tail-crossing", origin_position=transformation.operator_position,
        step=1, required_speaking_bodies=0, speaking_body_source_positions=(),
        encounters=tuple(AeternumBearingEncounter(
            virtual_position=virtual, source_position=crossing.source_position,
            glyph=TEMPORAL_OPERATOR_GLYPH, token_kind=tokens[crossing.source_position].kind,
            bearing="orobouros_tail", imaginary=not (0 <= virtual < len(tokens)),
            speaking_body=False, bearing_only=True,
        ) for virtual in (crossing.entrance_virtual_position,
        ↪  crossing.mirror_virtual_position)),
        completed=True,
    ))

# A center made solely of visible Enochian grammar is lawful information, energy,
```

```
# and magic, but it has no lexical speaking body and therefore utters nothing.
# Its mirror remains present as bearing-only structure; it is not translated into
# a second sentence and may not fabricate a lexical vessel.
if tokens and not bodies and all(token.value in VISIBLE_GRAMMARS for token in tokens):
    stack_positions = tuple(token.position for token in tokens)
    for token in scan_tokens:
        action = "transmutation-gate-unuttered" if token.value == PHANTASMAGORIA_GLYPH else
        ↪  "grammar-lawful-unuttered"
        zero_and_one_events.append(ZeroAndOneEvent(
            token.position, token.value, token.kind, action, False,
        ))
        grammar_events.append(LocalGrammarEvent(
            token.position, token.value, "enochian-bearing-unuttered", (),
            {"stack_positions": stack_positions},
        ))
    period_encounters = tuple(
        AeternumBearingEncounter(
            virtual_position=virtual_position,
            source_position=aeternum_token(virtual_position).source_position,
            glyph=aeternum_token(virtual_position).value,
            token_kind=aeternum_token(virtual_position).kind,
            bearing=source_bearing(aeternum_token(virtual_position).source_position),
            imaginary=not (0 <= virtual_position < len(tokens)),
            speaking_body=False,
            bearing_only=True,
        )
        for virtual_position in range(-len(tokens), len(tokens))
    )
    aeternum_bearing_paths.append(AeternumBearingPath(
        purpose="lawful-unuttered-enochian-field",
        origin_position=0,
        step=1,
        required_speaking_bodies=0,
        speaking_body_source_positions=(),
        encounters=period_encounters,
        completed=True,
    ))
    zero_and_one = ZeroAndOneTrace(tuple(zero_and_one_events))
    self.machine._validate_trace(frame.tokens, zero_and_one)
    local = LocalGrammarTrace(
        (), (), tuple(grammar_events), (), (), (), tuple(aeternum_bearing_paths),
        tail_resolution if tail_resolution.transformations else None,
        execution_positions if tail_resolution.transformations else (),
    )
    return zero_and_one, local

def register_body(token, resolution: object, event_name: str) -> None:
    zero_and_one_events.append(ZeroAndOneEvent(
        token.position, token.value, token.kind, event_name, True, resolution,
    ))
    body = bodies.get(token.position)
    if body is None:
        body = ResolvedBody(token.position, token.value, resolution)
        bodies[token.position] = body
    if token.position not in source_body_order:
        source_body_order.append(token.position)
```

```
    if token.position not in causal_body_order:
        causal_body_order.append(token.position)

def apply_bearing(
    operator_position: int,
    operator_glyph: str,
    target_position: int,
    *,
    depth: int,
    stack_positions: tuple[int, ...],
    direction: str = "backward",
    scope: str = "grammar-stack-pattern",
) -> None:
    bearing = OFFICE_BEARINGS[operator_glyph]
    target = bodies[target_position]
    request = RecognitionRequest(
        operator_position=operator_position,
        operator_glyph=operator_glyph,
        target_position=target.position,
        target_glyph=target.glyph,
        bearing=bearing,
        direction=direction,
        scope=scope,
        retrocausal_depth=depth,
        stack_positions=stack_positions,
    )
    recognitions.append(request)
    office_resolution = self._resolve_office(target, bearing)
    bodies[target.position] = replace(
        target,
        recognition=bearing,
        office_resolutions=target.office_resolutions + (office_resolution,),
    )
    grammar_events.append(LocalGrammarEvent(
        operator_position,
        operator_glyph,
        "compound-retrocausal-bearing",
        (target.position,),
        request,
    ))

def apply_echo_recursive(operator_position: int, target_position: int, stack_positions:
↪  tuple[int, ...]) -> None:
    target = bodies[target_position]
    request = RecognitionRequest(
        operator_position=operator_position,
        operator_glyph=OUROBORIC,
        target_position=target.position,
        target_glyph=target.glyph,
        bearing="recursive_identity",
        direction="orobouros-head-echo",
        scope="orobouros-head",
        retrocausal_depth=0,
        stack_positions=stack_positions,
    )
    recognitions.append(request)
```

```
    office_resolution = self._resolve_office(target, "recursive_identity")
    bodies[target.position] = replace(
        target,
        recognition="recursive_identity",
        office_resolutions=target.office_resolutions + (office_resolution,),
    )
    grammar_events.append(LocalGrammarEvent(
        operator_position, OUROBORIC, "orobouros-head-echo-recursive",
        (target.position,), request,
    ))

def set_causal_pair(left_position: int, right_position: int, order: tuple[int, int]) ->
↪  None:
    if left_position == right_position:
        return
    if aeternum_pair_reorder is not None:
        aeternum_pair_reorder(causal_body_order, left_position, right_position, order)
        return
    if left_position not in causal_body_order or right_position not in causal_body_order:
        raise LocalGrammarError("⸸ Orobouros Head lost a lexical body")
    li = causal_body_order.index(left_position)
    ri = causal_body_order.index(right_position)
    if abs(ri - li) != 1:
        raise LocalGrammarError("⸸ Orobouros Head requires adjacent lexical bodies")
    lo, hi = sorted((li, ri))
    causal_body_order[lo:hi + 1] = list(order)

def finalize_transmutation(
    operator_positions: tuple[int, ...],
    first_position: int,
    following_positions: tuple[int, ...] | list[int],
) -> None:
    vessel_positions = (first_position, *tuple(following_positions))
    if len(vessel_positions) != len(operator_positions) + 1:
        raise LocalGrammarError("ᛕ corridor did not close over exactly K+1 Real vessels")
    if any(position not in bodies for position in vessel_positions):
        raise LocalGrammarError("ᛕ corridor lost one of its Real vessels")
    vessels = tuple(bodies[position] for position in vessel_positions)

    # A reflected encounter points to the same finite real source body, so its
    # already-selected Phantasmagoria office resolution is reused exactly.
    resolution_by_real_position: dict[int, object] = {}
    office_resolutions_list: list[object] = []
    for body in vessels:
        resolution = resolution_by_real_position.get(body.position)
        if resolution is None:
            resolution = self._resolve_office(body, "phantasmagoria")
            resolution_by_real_position[body.position] = resolution
        office_resolutions_list.append(resolution)
    office_resolutions = tuple(office_resolutions_list)

    carrier = CARRIER_BY_GLYPH["✮"]
    if carrier.name != "AHN" or carrier.structural_hz != 432 or carrier.parity_hz != 417:
        raise LocalGrammarError("ᛕ cannot open: AHN 432 / i417 witness is not exact")

    returned_order = vessel_positions[1:] + vessel_positions[:1]
```

```
    # The mirror introduces imaginary encounters but no second real source
    # extent.  Therefore causal permutation is applied once per unique real
    # source body, in returned encounter order.
    unique_source = tuple(dict.fromkeys(vessel_positions))
    unique_return = tuple(dict.fromkeys(returned_order))
    if len(unique_source) > 1:
        if aeternum_corridor_reorder is not None:
            aeternum_corridor_reorder(causal_body_order, unique_source, unique_return)
        else:
            try:
                indexes = tuple(causal_body_order.index(position) for position in
                ↪  unique_source)
            except ValueError as exc:
                raise LocalGrammarError("ϟ corridor lost one of its causal vessels") from
                ↪  exc
            ordered_indexes = sorted(indexes)
            if ordered_indexes != list(range(ordered_indexes[0], ordered_indexes[0] +
            ↪  len(ordered_indexes))):
                raise LocalGrammarError("ϟ corridor requires adjacent causal Real vessels")
            causal_body_order[ordered_indexes[0]:ordered_indexes[-1] + 1] =
            ↪  list(unique_return)

    state = PhantasmagoriaTransmutation(
        operator_positions=operator_positions,
        vessels=vessels,
        office_resolutions=office_resolutions,
        source_vessel_order=vessel_positions,
        returned_order=returned_order,
        gate_boundary="i417",
        carrier_glyph=carrier.glyph,
        carrier_name=carrier.name,
        structural_hz=str(carrier.structural_hz),
        imaginary_parity_hz=str(carrier.parity_hz),
    )
    transmutations.append(state)
    grammar_events.append(LocalGrammarEvent(
        operator_positions[-1], PHANTASMAGORIA_GLYPH, "phantasmagoria-transmutation",
        vessel_positions, state,
    ))

def finalize_loop(
    loop_data: tuple[tuple[int, ...], int, tuple[tuple[int, str], ...], tuple[int, ...],
    ↪  int, int, int],
    right_offices: tuple[tuple[int, str], ...],
) -> None:
    (
        operator_positions, left_position, left_offices, loop_stack_positions,
        right_position, left_virtual, right_virtual,
    ) = loop_data
    if not operator_positions:
        raise LocalGrammarError("⸸ Orobouros Head has no written turn")
    left = bodies[left_position]
    right = bodies[right_position]
    n = len(operator_positions)
    # Current public GrimChain contains only the manifested spoken center.
```

```
# The old House wrapper positions are gone: a true left-edge contact is
# the Loop's real left lexical body at source position 0, and a true
# right-edge contact is the complete B/R side reaching the last source
# position. The mirror-echo algorithm below is otherwise preserved.
# Preserve the original mirror-echo algorithm, but anchor its old
# Court-adjacent trigger to the current public GrimChain itself.
# A mirror edge is the immediate A ⩛... B form at a real source edge;
# L/R office material remains part of the ordinary loop topology.
crossed_left_mirror = left_virtual < 0
crossed_right_mirror = right_virtual >= len(tokens)
leading_echo = (
    crossed_left_mirror
    or (not crossed_right_mirror and left_position == scan_tokens[0].position)
) and not left_offices
trailing_echo = (
    crossed_right_mirror
    or (not crossed_left_mirror and right_position == scan_tokens[-1].position)
) and not right_offices
left_glyphs = tuple(g for _, g in left_offices)
right_glyphs = tuple(g for _, g in right_offices)
mirror_positions: tuple[int, ...] = ()
mirror_source_positions: tuple[int, ...] = ()
mirror_glyphs: tuple[str, ...] = ()

def reflected_continuation(start_virtual: int, turns: int, step: int) ->
↪  tuple[tuple[int, ...], tuple[int, ...], tuple[str, ...]]:
    reflected = tuple(
        aeternum_token(start_virtual + step * offset)
        for offset in range(max(0, turns))
    )
    return (
        tuple(token.mirror_position for token in reflected),
        tuple(token.source_position for token in reflected),
        tuple(token.value for token in reflected),
    )

if leading_echo:
    # The root Court is already the real left lexical glyph at source position 1.
    # It uses the ordinary lexical/office resolver; only the Loop topology is special.
    set_causal_pair(left_position, right_position, (right_position, left_position))
    apply_echo_recursive(operator_positions[0], left_position, loop_stack_positions)
    for pos, glyph in right_offices:
        apply_bearing(pos, glyph, left_position, depth=0,
        ↪  stack_positions=loop_stack_positions,
                      direction="orobouros-head-echo", scope="orobouros-head")
    mirror_start = left_virtual if left_virtual < 0 else -1
    mirror_turns = n if left_virtual < 0 else n - 1
    mirror_positions, mirror_source_positions, mirror_glyphs = reflected_continuation(
        mirror_start, mirror_turns, -1,
    )
    topology_parts: list[str] = [right.glyph, OUROBORIC, left.glyph]
    for glyph in mirror_glyphs:
        topology_parts.extend((OUROBORIC, glyph))
    topology = tuple(topology_parts)
    echo_position = left_position
elif trailing_echo:
```

```
    # The alternating Court is already the ordinary right lexical glyph in the public
    ↪ line.
    if left_offices:
        apply_echo_recursive(operator_positions[-1], right_position,
        ↪ loop_stack_positions)
    for pos, glyph in left_offices:
        apply_bearing(pos, glyph, right_position, depth=0,
        ↪ stack_positions=loop_stack_positions,
                      direction="orobouros-head-echo", scope="orobouros-head")
    for pos, glyph in right_offices:
        apply_bearing(pos, glyph, right_position, depth=0,
        ↪ stack_positions=loop_stack_positions,
                      direction="orobouros-head-echo", scope="orobouros-head")
    mirror_start = right_virtual if right_virtual >= len(tokens) else len(tokens)
    mirror_turns = n if right_virtual >= len(tokens) else n - 1
    mirror_positions, mirror_source_positions, mirror_glyphs = reflected_continuation(
        mirror_start, mirror_turns, 1,
    )
    topology_parts = [left.glyph, *left_glyphs, OUROBORIC, right.glyph]
    for glyph in mirror_glyphs:
        topology_parts.extend((OUROBORIC, glyph))
    topology = tuple(topology_parts)
    echo_position = right_position
elif n == 1:
    for pos, glyph in left_offices:
        apply_bearing(pos, glyph, right_position, depth=0,
        ↪ stack_positions=loop_stack_positions,
                      direction="orobouros-head", scope="orobouros-head")
    for pos, glyph in right_offices:
        apply_bearing(pos, glyph, right_position, depth=0,
        ↪ stack_positions=loop_stack_positions,
                      direction="orobouros-head", scope="orobouros-head")
    topology = (left.glyph, right.glyph, OUROBORIC, *left_glyphs, *right_glyphs)
    echo_position = None
elif n == 2:
    set_causal_pair(left_position, right_position, (right_position, left_position))
    for pos, glyph in right_offices:
        apply_bearing(pos, glyph, left_position, depth=0,
        ↪ stack_positions=loop_stack_positions,
                      direction="orobouros-head", scope="orobouros-head")
    for pos, glyph in left_offices:
        apply_bearing(pos, glyph, left_position, depth=0,
        ↪ stack_positions=loop_stack_positions,
                      direction="orobouros-head", scope="orobouros-head")
    topology = (right.glyph, OUROBORIC, left.glyph, *right_glyphs, OUROBORIC,
    ↪ *left_glyphs)
    echo_position = None
else:
    # Three turns open all four authored topology blocks. Further turns continue the
    # same Loop by rotating those blocks; no turn is discarded and no ceiling exists.
    blocks: list[tuple[str, object, tuple[tuple[int, str], ...]]] = [
        ("lex", right_position, ()),
        ("office", right_position, right_offices),
        ("office", left_position, left_offices),
        ("lex", left_position, ()),
```

```
    ]
    if n > 3:
        k = (n - 3) % 4
        blocks = blocks[k:] + blocks[:k]
    lexical_order = tuple(int(value) for kind, value, _ in blocks if kind == "lex")
    if lexical_order != (left_position, right_position):
        set_causal_pair(left_position, right_position, lexical_order)
    last_lex = None
    for kind, value, office_block in blocks:
        if kind == "lex":
            last_lex = int(value)
            continue
        if not office_block:
            continue
        if last_lex is None:
            for prior_kind, prior_value, _ in reversed(blocks):
                if prior_kind == "lex":
                    last_lex = int(prior_value)
                    break
        if last_lex is None:
            raise LocalGrammarError("⸸ Orobouros Head has no lexical owner")
        for pos, glyph in office_block:
            apply_bearing(pos, glyph, last_lex, depth=0,
            ↪  stack_positions=loop_stack_positions,
                          direction="orobouros-head", scope="orobouros-head")
    flat: list[str] = []
    for bi, (kind, value, office_block) in enumerate(blocks):
        if bi:
            flat.append(OUROBORIC)
        if kind == "lex":
            flat.append(bodies[int(value)].glyph)
        else:
            flat.extend(g for _, g in office_block)
    flat.extend(OUROBORIC for _ in range(max(0, n - 3)))
    topology = tuple(flat)
    echo_position = None

state = OuroboricState(
    operator_positions=operator_positions,
    winding_count=n,
    left=bodies[left_position],
    right=bodies[right_position],
    source_order=(left_position, *operator_positions, right_position),
    left_office_glyphs=left_glyphs,
    left_office_positions=tuple(pos for pos, _ in left_offices),
    right_office_glyphs=right_glyphs,
    right_office_positions=tuple(pos for pos, _ in right_offices),
    resolved_topology=topology,
    leading_echo=leading_echo,
    trailing_echo=trailing_echo,
    echo_position=echo_position,
    mirror_positions=mirror_positions,
    mirror_source_positions=mirror_source_positions,
    mirror_glyphs=mirror_glyphs,
    mirror_token_kinds=(),
    mirror_resolved_bearings=(),
```

```
        imaginary_reflection=bool(mirror_positions),
        bearing_preserved_in_reflection=True,
    )
    loops.append(state)
    grammar_events.append(LocalGrammarEvent(
        operator_positions[0], OUROBORIC, "orobouros-head",
        (left_position, right_position), state,
    ))

while i < len(scan_tokens):
    token = scan_tokens[i]
    glyph = token.value

    if glyph == SHADOW_LOCUS:
        if shadow_ostensive:
            register_body(token, bodies[token.position].lexical_resolution, glyph)
        else:
            zero_and_one_events.append(ZeroAndOneEvent(
                token.position, glyph, token.kind, glyph, False,
            ))
        i += 1
        continue

    if glyph == AXIOMYR_GLYPH:
        register_body(
            token, bodies[token.position].lexical_resolution,
            glyph if axiomyr_ostensive else "lexical-resolution",
        )
        i += 1
        continue

    if glyph in self.machine.ordinary_lexical_glyphs:
        resolution = bodies[token.position].lexical_resolution
        register_body(token, resolution, "lexical-resolution")

        if pending_transmutation is not None:
            op_positions, first_position, following_positions = pending_transmutation
            following_positions.append(token.position)
            if len(following_positions) == len(op_positions):
                finalize_transmutation(op_positions, first_position, following_positions)
                pending_transmutation = None

        if pending_orobouros_head is not None:
            op_positions, left_position, left_offices, loop_stack_positions, left_virtual =
            ↪  pending_orobouros_head
            active_loop = (
                op_positions, left_position, left_offices, loop_stack_positions,
                token.position, left_virtual, execution_index[token.position],
            )
            pending_orobouros_head = None

        i += 1
        continue

    if active_loop is not None and glyph not in VISIBLE_GRAMMARS:
        finalize_loop(active_loop, ())
```

```
    active_loop = None

if glyph not in VISIBLE_GRAMMARS:
    raise LocalGrammarError(
        f"grammar glyph {glyph!r} at position {token.position} has no executable law"
    )

j = i
stack = []
while j < len(scan_tokens) and scan_tokens[j].value in VISIBLE_GRAMMARS:
    stack.append(scan_tokens[j])
    j += 1
stack_positions = tuple(t.position for t in stack)
stack_glyphs = tuple(t.value for t in stack)
source_history_before_stack = tuple(source_body_order)

for gt in stack:
    action = "transmutation-gate" if gt.value == PHANTASMAGORIA_GLYPH else
    ↪  "grammar-executed"
    local_action = "transmutation-gate" if gt.value == PHANTASMAGORIA_GLYPH else
    ↪  "grammar-stack-member"
    zero_and_one_events.append(ZeroAndOneEvent(
        gt.position, gt.value, gt.kind, action, False,
    ))
    grammar_events.append(LocalGrammarEvent(
        gt.position, gt.value, local_action, (),
        {"stack_positions": stack_positions},
    ))

transmutation_tokens = [gt for gt in stack if gt.value == PHANTASMAGORIA_GLYPH]
if transmutation_tokens:
    first_key = transmutation_tokens[0]
    first_context = lexical_context(first_key.position, 1, -1,
    ↪  purpose="phantasmagoria-preceding-vessel")
    if pending_transmutation is not None:
        raise LocalGrammarError("ऋ encountered before the prior Transmutation Corridor
        ↪  completed")
    pending_transmutation = (
        tuple(gt.position for gt in transmutation_tokens),
        first_context[0][0],
        [],
    )

reversal_tokens = [gt for gt in stack if gt.value == OUROBORIC]
bearing_tokens = [gt for gt in stack if gt.value in OFFICE_BEARINGS]

if active_loop is not None:
    if reversal_tokens or transmutation_tokens:
        finalize_loop(active_loop, ())
        active_loop = None
    elif bearing_tokens and len(bearing_tokens) == len(stack):
        finalize_loop(active_loop, tuple((gt.position, gt.value) for gt in
        ↪  bearing_tokens))
        active_loop = None
        bearing_tokens = []
```

```
if reversal_tokens:
    # A maximal Enoch stack may contain more than one separate ⸸ run.
    # Each consecutive run is one winding.  Separating Enochs keep their
    # own laws; they do not silently merge two windings or invalidate one.
    reversal_runs: list[tuple[int, int, tuple[object, ...]]] = []
    k = 0
    while k < len(stack):
        if stack[k].value != OUROBORIC:
            k += 1
            continue
        start = k
        run: list[object] = []
        while k < len(stack) and stack[k].value == OUROBORIC:
            run.append(stack[k])
            k += 1
        reversal_runs.append((start, k, tuple(run)))

    claimed_office_positions: set[int] = set()
    for run_index, (start, end, reversal_run) in enumerate(reversal_runs):
        left_start = start
        while left_start > 0 and (
            stack[left_start - 1].value in OFFICE_BEARINGS
            or stack[left_start - 1].value == TEMPORAL_OPERATOR_GLYPH
        ):
            left_start -= 1
        left_offices = tuple(
            (gt.position, gt.value)
            for gt in stack[left_start:start]
            if gt.value in OFFICE_BEARINGS
        )
        claimed_office_positions.update(pos for pos, _ in left_offices)
        left_position, left_virtual = lexical_context(stack[start].position, 1, -1,
        ↪  purpose="orobouros-head-left-body")[0]
        loop_positions = tuple(
            gt.position
            for gt in stack[left_start:end]
            if gt.value != TEMPORAL_OPERATOR_GLYPH
        )
        operator_positions = tuple(gt.position for gt in reversal_run)

        is_last_run = run_index == len(reversal_runs) - 1
        if is_last_run:
            if pending_orobouros_head is not None:
                raise LocalGrammarError(
                    "⸸ Orobouros Head reached a new turn-run before its following
                    ↪  lexical body"
                )
            pending_orobouros_head = (
                operator_positions,
                left_position,
                left_offices,
                loop_positions,
                left_virtual,
            )
        else:
```

```
                right_position, right_virtual = lexical_context(stack[end - 1].position, 1,
                ↪  1, purpose="orobouros-head-right-body")[0]
                finalize_loop((
                    operator_positions,
                    left_position,
                    left_offices,
                    loop_positions,
                    right_position,
                    left_virtual,
                    right_virtual,
                ), ())

            bearing_tokens = [
                gt for gt in bearing_tokens if gt.position not in claimed_office_positions
            ]

        if bearing_tokens:
            bearing_count = len(bearing_tokens)
            context = lexical_context(bearing_tokens[0].position, bearing_count, -1,
            ↪  purpose="office-retrocausal-bearing")
            for index, gt in enumerate(bearing_tokens):
                depth = bearing_count - 1 - index
                target_position, target_virtual = context[depth]
                apply_bearing(
                    gt.position, gt.value, target_position,
                    depth=depth, stack_positions=stack_positions,
                    direction=("aeternum-left-reflection" if target_virtual < 0 else
                    ↪  "backward"),
                )

        grammar_events.append(LocalGrammarEvent(
            stack[-1].position, "".join(stack_glyphs), "compound-grammar-stack",
            tuple(source_history_before_stack),
            {"stack_positions": stack_positions, "stack_length": len(stack)},
        ))
        i = j

    if pending_transmutation is not None:
        op_positions, first_position, following_positions = pending_transmutation
        needed = len(op_positions) - len(following_positions)
        reflected_following = lexical_context(execution_positions[-1], needed, 1,
        ↪  purpose="phantasmagoria-following-vessels")
        following_positions.extend(position for position, _virtual in reflected_following)
        finalize_transmutation(op_positions, first_position, following_positions)
        pending_transmutation = None
    if pending_orobouros_head is not None:
        op_positions, left_position, left_offices, loop_stack_positions, left_virtual =
        ↪  pending_orobouros_head
        right_position, right_virtual = lexical_context(execution_positions[-1], 1, 1,
        ↪  purpose="orobouros-head-following-body")[0]
        active_loop = (
            op_positions, left_position, left_offices, loop_stack_positions,
            right_position, left_virtual, right_virtual,
        )
        pending_orobouros_head = None
```

```
if active_loop is not None:
    finalize_loop(active_loop, ())
    active_loop = None

recognition_by_operator: dict[int, RecognitionRequest] = {
    request.operator_position: request for request in recognitions
}
transmutation_positions = {position for state in transmutations for position in
↪  state.operator_positions}
transmuted_vessel_resolution: dict[int, object] = {
    body.position: resolution
    for state in transmutations
    for body, resolution in zip(state.vessels, state.office_resolutions)
}
loop_positions = {position for loop in loops for position in loop.operator_positions}
token_by_position = {token.position: token for token in tokens}

def resolved_reflection_bearing(source_position: int) -> str | None:
    token = token_by_position[source_position]
    request = recognition_by_operator.get(source_position)
    if request is not None:
        return request.bearing
    if source_position in transmutation_positions or source_position in
    ↪  transmuted_vessel_resolution:
        return "phantasmagoria"
    if source_position in loop_positions:
        return "orobouros_head"
    if token.value == TEMPORAL_OPERATOR_GLYPH:
        return "orobouros_tail"
    if token.kind == "lexical":
        return "seed_identity"
    if token.value in {SHADOW_LOCUS, AXIOMYR_GLYPH}:
        return None
    raise LocalGrammarError(
        f"reflected grammar at source position {source_position} lacks its resolved original
        ↪  law"
    )

reflected_loops: list[OuroboricState] = []
for loop in loops:
    kinds = tuple(token_by_position[pos].kind for pos in loop.mirror_source_positions)
    bearings = tuple(resolved_reflection_bearing(pos) for pos in
    ↪  loop.mirror_source_positions)
    mirror_corridor = (
        tuple(aeternum_token(v) for v in range(len(tokens), 3 * len(tokens)))
        if tail_resolution.transformations and loop.imaginary_reflection
        else frame.mirror_corridor_period() if loop.imaginary_reflection else ()
    )
    def ostensive_provenance(source_position: int) -> tuple[str | None, str | None, str |
    ↪  None]:
        resolution = transmuted_vessel_resolution.get(source_position)
        if resolution is None:
            return (None, None, None)
        return (
            getattr(resolution, "ostensive_variant"),
```

```
            getattr(resolution, "resolved_body"),
            getattr(resolution, "source_authority"),
        )

    mirror_ostensive = tuple(ostensive_provenance(pos) for pos in
    ↪  loop.mirror_source_positions)
    corridor_ostensive = tuple(ostensive_provenance(token.source_position) for token in
    ↪  mirror_corridor)
    reflected_loops.append(replace(
        loop,
        mirror_token_kinds=kinds,
        mirror_resolved_bearings=bearings,
        mirror_ostensive_variants=tuple(item[0] for item in mirror_ostensive),
        mirror_ostensive_bodies=tuple(item[1] for item in mirror_ostensive),
        mirror_ostensive_authorities=tuple(item[2] for item in mirror_ostensive),
        mirror_corridor_glyphs=tuple(token.value for token in mirror_corridor),
        mirror_corridor_source_positions=tuple(token.source_position for token in
        ↪  mirror_corridor),
        mirror_corridor_resolved_bearings=tuple(
            resolved_reflection_bearing(token.source_position) for token in mirror_corridor
        ),
        mirror_corridor_ostensive_variants=tuple(item[0] for item in corridor_ostensive),
        mirror_corridor_ostensive_bodies=tuple(item[1] for item in corridor_ostensive),
        mirror_corridor_ostensive_authorities=tuple(item[2] for item in corridor_ostensive),
    ))
loops = reflected_loops
loops = [replace(loop, left=bodies[loop.left.position], right=bodies[loop.right.position])
↪  for loop in loops]
transmutations = [replace(state, vessels=tuple(bodies[body.position] for body in
↪  state.vessels)) for state in transmutations]

zero_and_one = ZeroAndOneTrace(tuple(zero_and_one_events))
self.machine._validate_trace(frame.tokens, zero_and_one)
local = LocalGrammarTrace(
    tuple(bodies[pos] for pos in source_body_order),
    tuple(causal_body_order),
    tuple(grammar_events), tuple(recognitions), tuple(loops), tuple(transmutations),
    tuple(aeternum_bearing_paths),
    tail_resolution if tail_resolution.transformations else None,
    execution_positions if tail_resolution.transformations else (),
)
return zero_and_one, local
```

trace.py

```
"""Public deterministic derivation trace over one exact Aeonic line."""
from __future__ import annotations

from dataclasses import asdict, dataclass, field
import hashlib
import json
from pathlib import Path
from typing import Protocol

from .zero_and_one import ZeroAndOneTrace
from .local_grammar import LocalGrammarTrace
from .models import DomusFrame, GlyphToken
from .temporal_operator import TailResolution, tail_resolution_from_dict, validate_tail_resolution
from .lexical_resolver import LEXICON_VERSION, LexicalResolver
from .aksh_reader import attributes

TRACE_VERSION = "aeonic-line-v2"

class TraceError(RuntimeError):
    pass

@dataclass(frozen=True, slots=True)
class AuthorityVersions:
    lexical_authorities: tuple[tuple[str, str], ...]
    understandings: str

@dataclass(frozen=True, slots=True)
class ParsedFrameTrace:
    source: str
    exact_roundtrip: bool

@dataclass(frozen=True, slots=True)
class LexicalUtteranceTrace:
    resolved_state_body: str
    traversal_path: tuple[str, ...]
    source_authority: str
    authored_office: str
    grammatical_bearing: str | None

@dataclass(frozen=True, slots=True)
class TokenTrace:
    source_position: int
    exact_glyph: str
    token_class: str
    action: str
    resolved_state_body: str | None
    traversal_path: tuple[str, ...]
    source_authority: str | None
```

```
    authored_office: str | None
    grammatical_bearing: str | None
    local_operator_attachments: tuple[str, ...]
    lexical_lookup_performed: bool
    realization_presence: bool
    utterances: tuple[LexicalUtteranceTrace, ...] = ()

@dataclass(frozen=True, slots=True)
class PhantasmagoriaTransmutationTrace:
    operator_positions: tuple[int, ...]
    vessel_positions: tuple[int, ...]
    vessel_glyphs: tuple[str, ...]
    source_vessel_order: tuple[int, ...]
    returned_order: tuple[int, ...]
    ostensive_bodies: tuple[str, ...]
    ostensive_variants: tuple[str, ...]
    ostensive_authorities: tuple[str, ...]
    gate_boundary: str
    carrier_glyph: str
    carrier_name: str
    structural_hz: str
    imaginary_parity_hz: str
    traversal: tuple[str, str, str, str]
    hand: str
    chord: str
    prosodical_subspace: str


@dataclass(frozen=True, slots=True)
class OuroboricTrace:
    operator_positions: tuple[int, ...]
    winding_count: int
    left_position: int
    right_position: int
    left_glyph: str
    right_glyph: str
    source_order: tuple[int, ...]
    left_office_glyphs: tuple[str, ...]
    left_office_positions: tuple[int, ...]
    right_office_glyphs: tuple[str, ...]
    right_office_positions: tuple[int, ...]
    resolved_topology: tuple[str, ...]
    leading_echo: bool
    trailing_echo: bool
    echo_position: int | None
    mirror_positions: tuple[int, ...]
    mirror_source_positions: tuple[int, ...]
    mirror_glyphs: tuple[str, ...]
    mirror_token_kinds: tuple[str, ...]
    mirror_resolved_bearings: tuple[str | None, ...]
    mirror_ostensive_variants: tuple[str | None, ...]
    mirror_ostensive_bodies: tuple[str | None, ...]
    mirror_ostensive_authorities: tuple[str | None, ...]
    mirror_corridor_glyphs: tuple[str, ...]
    mirror_corridor_source_positions: tuple[int, ...]
    mirror_corridor_resolved_bearings: tuple[str | None, ...]
```

```
    mirror_corridor_ostensive_variants: tuple[str | None, ...]
    mirror_corridor_ostensive_bodies: tuple[str | None, ...]
    mirror_corridor_ostensive_authorities: tuple[str | None, ...]
    imaginary_reflection: bool
    bearing_preserved_in_reflection: bool
    derived_glyph_generated: bool

@dataclass(frozen=True, slots=True)
class AeternumBearingEncounterTrace:
    virtual_position: int
    source_position: int
    glyph: str
    token_kind: str
    bearing: str | None
    imaginary: bool
    speaking_body: bool
    bearing_only: bool


@dataclass(frozen=True, slots=True)
class AeternumBearingTrace:
    purpose: str
    origin_position: int
    step: int
    required_speaking_bodies: int
    speaking_body_source_positions: tuple[int, ...]
    encounters: tuple[AeternumBearingEncounterTrace, ...]
    completed: bool


@dataclass(frozen=True, slots=True)
class RenderingToken:
    source_position: int
    glyph: str
    resolved_state_body: str | None
    grammatical_bearing: str | None
    attachments: tuple[str, ...]
    preserved_without_lookup: bool
    source_authority: str | None
    authored_office: str | None
    utterances: tuple[LexicalUtteranceTrace, ...] = ()

@dataclass(frozen=True, slots=True)
class RenderingInput:
    source: str
    tokens: tuple[RenderingToken, ...]
    ouroboric_indexes: tuple[int, ...]

@dataclass(frozen=True, slots=True)
class DomusTrace:
    trace_version: str
    source_string: str
    parsed_frame: ParsedFrameTrace
    semantic_lexicon_version: str
    authority_versions: AuthorityVersions
    token_traces: tuple[TokenTrace, ...]
```

```
    ouroboric_traces: tuple[OuroboricTrace, ...]
    transmutation_traces: tuple[PhantasmagoriaTransmutationTrace, ...]
    rendering_input: RenderingInput
    aeternum_bearing_traces: tuple[AeternumBearingTrace, ...] = ()
    utterance_state: str = "uttered"
    tail_resolution: TailResolution | None = None
    final_body_order: tuple[int, ...] = ()
    _sha256_cache: str | None = field(default=None, init=False, repr=False, compare=False)

    def to_dict(self) -> dict[str, object]:
        payload = asdict(self)
        payload.pop("_sha256_cache", None)
        if self.tail_resolution is None:
            payload.pop("tail_resolution", None)
            payload.pop("final_body_order", None)
        return payload

    def to_json(self) -> str:
        return json.dumps(self.to_dict(), ensure_ascii=False, sort_keys=True, separators=(",", ":"))

    @property
    def sha256(self) -> str:
        cached = self._sha256_cache
        if cached is None:
            cached = hashlib.sha256(self.to_json().encode("utf-8")).hexdigest()
            object.__setattr__(self, "_sha256_cache", cached)
        return cached

    @classmethod
    def from_json(cls, payload: str) -> "DomusTrace":
        return _trace_from_dict(json.loads(payload))

    def to_human_readable(self) -> str:
        lines: list[str] = [
            f"Source: {self.parsed_frame.source}",
            f"Semantic lexicon version: {self.semantic_lexicon_version}",
            f".aksh versions: lexical_authorities={self.authority_versions.lexical_authorities},
            ↪  understandings={self.authority_versions.understandings}",
        ]
        for token in self.token_traces:
            lines.append(f"position {token.source_position}: {token.exact_glyph}")
            lines.append(f"  action: {token.action}")
            if token.resolved_state_body is not None:
                lines.append(f"  traversal: {' -> '.join(token.traversal_path)}")
                lines.append(f"  authority: {token.source_authority}")
                lines.append(f"  authored office: {token.authored_office}")
                lines.append(f"  result: {token.resolved_state_body}")
            if token.grammatical_bearing is not None:
                lines.append(f"  grammatical bearing: {token.grammatical_bearing}")
            for attachment in token.local_operator_attachments:
                lines.append(f"  attachment: {attachment}")
        return "\n".join(lines)

class TraceRenderer(Protocol):
    version: str
    def render(self, rendering_input: RenderingInput) -> str: ...
```

```
@dataclass(frozen=True, slots=True)
class RenderedOutput:
    renderer_version: str
    trace_sha256: str
    sentence: str

def render_from_trace(trace: DomusTrace, renderer: TraceRenderer) -> RenderedOutput:
    validate_tail_trace(trace)
    sentence = renderer.render(trace.rendering_input)
    if not isinstance(sentence, str):
        raise TraceError("renderer must return str")
    return RenderedOutput(renderer.version, trace.sha256, sentence)


def validate_tail_trace(trace: DomusTrace) -> None:
    """A saved trace must prove Tail execution before any rendering boundary."""
    if not any(token.exact_glyph == "⟲" for token in trace.token_traces):
        return
    if trace.tail_resolution is None:
        raise TraceError("Tail-dependent translation has no completed traversal")
    frame = DomusFrame(trace.source_string, tuple(
        GlyphToken(token.exact_glyph, token.source_position, token.token_class)
        for token in trace.token_traces
    ))
    validate_tail_resolution(frame, trace.tail_resolution)
    speaking = {token.source_position for token in trace.token_traces if token.utterances}
    if len(trace.final_body_order) != len(speaking) or set(trace.final_body_order) != speaking:
        raise TraceError("Tail-dependent final order lost a speaking body")
    rendered = tuple(token.source_position for token in trace.rendering_input.tokens if
    ↪  token.source_position in speaking)
    if rendered != trace.final_body_order:
        raise TraceError("renderer order differs from completed Tail-dependent grammar")

def authority_versions(resolver: LexicalResolver, understandings: str | Path) -> AuthorityVersions:
    versions = resolver.authority_version_items()
    lines = Path(understandings).read_text(encoding="utf-8").splitlines()
    understanding_version = attributes(lines[0]).get("version", "") if lines else ""
    if not understanding_version:
        raise TraceError("required Enochian Understandings .aksh version missing")
    return AuthorityVersions(versions, understanding_version)


def _resolution_fields(resolution: object | None) -> dict[str, object]:
    if resolution is None:
        return {
            "resolved_state_body": None,
            "traversal_path": (),
            "source_authority": None,
            "authored_office": None,
            "semantic_lexicon_version": None,
        }
    required = (
        "resolved_body", "traversal_path", "source_authority",
        "authored_office", "semantic_lexicon_version",
    )
```

```
    missing = [name for name in required if not hasattr(resolution, name)]
    if missing:
        raise TraceError(f"lexical resolution missing trace fields: {missing!r}")
    return {
        "resolved_state_body": getattr(resolution, "resolved_body"),
        "traversal_path": tuple(getattr(resolution, "traversal_path")),
        "source_authority": getattr(resolution, "source_authority"),
        "authored_office": getattr(resolution, "authored_office"),
        "semantic_lexicon_version": getattr(resolution, "semantic_lexicon_version"),
    }

class DomusTraceBuilder:
    def __init__(self, resolver: LexicalResolver, understandings: str | Path) -> None:
        self.versions = authority_versions(resolver, understandings)

    def build(self, frame: DomusFrame, zero_and_one: ZeroAndOneTrace, local: LocalGrammarTrace) ->
    ↪  DomusTrace:
        if frame.to_source() != frame.source:
            raise TraceError("parsed frame does not round-trip to source")

        zero_and_one_by_position = {event.position: event for event in zero_and_one.events}
        body_by_position = {body.position: body for body in local.bodies}
        attachments: dict[int, list[str]] = {}
        for request in local.recognitions:
            attachments.setdefault(request.target_position, []).append(
                f"{request.operator_glyph}@{request.operator_position}:{request.bearing}"
            )
        for loop in local.ouroboric_states:
            marker = ",".join(str(pos) for pos in loop.operator_positions)
            attachments.setdefault(loop.left.position,
            ↪  []).append(f"⩛@{marker}:left:w{loop.winding_count}")
            attachments.setdefault(loop.right.position,
            ↪  []).append(f"⩛@{marker}:right:w{loop.winding_count}")
        for state in local.transmutations:
            marker = ",".join(str(pos) for pos in state.operator_positions)
            for body, resolution in zip(state.vessels, state.office_resolutions):
                attachments.setdefault(body.position, []).append(
                    f"⍨@{marker}:transmuted-vessel:ostensive{resolution.ostensive_variant}"
                )

        token_traces: list[TokenTrace] = []
        semantic_lexicon_versions: set[str] = set()
        for token in frame.tokens:
            event = zero_and_one_by_position.get(token.position)
            if event is None:
                raise TraceError(f"no zero-and-one event for source position {token.position}")
            body = body_by_position.get(token.position)
            effective_resolution = body.lexical_resolution if body is not None else
            ↪  event.lexical_resolution
            fields = _resolution_fields(effective_resolution)
            if event.event == "lexical-resolution" and effective_resolution is None:
                raise TraceError(f"lexical event at position {token.position} has no authored
                ↪  resolution")
            utterance_resolutions = ()
            if body is not None and body.lexical_resolution is not None:
```

```
        utterance_resolutions = (body.lexical_resolution, *body.office_resolutions)
    utterances: list[LexicalUtteranceTrace] = []
    for index, resolution in enumerate(utterance_resolutions):
        utterance_fields = _resolution_fields(resolution)
        version = utterance_fields["semantic_lexicon_version"]
        if version is not None:
            semantic_lexicon_versions.add(str(version))
        if utterance_fields["resolved_state_body"] is None or
        ↪  utterance_fields["source_authority"] is None or
        ↪  utterance_fields["authored_office"] is None:
            raise TraceError(f"lexical utterance at position {token.position} lacks authored
            ↪  resolution fields")
        utterances.append(LexicalUtteranceTrace(
            resolved_state_body=str(utterance_fields["resolved_state_body"]),
            traversal_path=tuple(utterance_fields["traversal_path"]),
            source_authority=str(utterance_fields["source_authority"]),
            authored_office=str(utterance_fields["authored_office"]),
            grammatical_bearing=(None if utterance_fields["source_authority"] ==
            ↪  "Axiomyr…Shadow…Locus.aksh" or utterance_fields["authored_office"] ==
            ↪  "seed_identity" else str(utterance_fields["authored_office"])),
        ))
    if fields["semantic_lexicon_version"] is not None:
        semantic_lexicon_versions.add(str(fields["semantic_lexicon_version"]))
    bearing = body.recognition if body is not None else None
    realization_presence = (
        event.event == "lexical-resolution"
        or token.value == "𓆙"
        or (token.value == "𓂀" and effective_resolution is not None)
    )
    token_traces.append(TokenTrace(
        source_position=token.position,
        exact_glyph=token.value,
        token_class=token.kind,
        action=event.event,
        resolved_state_body=fields["resolved_state_body"],
        traversal_path=fields["traversal_path"],
        source_authority=fields["source_authority"],
        authored_office=fields["authored_office"],
        grammatical_bearing=bearing,
        local_operator_attachments=tuple(attachments.get(token.position, ())),
        lexical_lookup_performed=event.lookup_performed,
        realization_presence=realization_presence,
        utterances=tuple(utterances),
    ))

if len(semantic_lexicon_versions) > 1:
    raise TraceError("mixed semantic lexicon versions in one Domus trace")
if semantic_lexicon_versions and semantic_lexicon_versions != {LEXICON_VERSION}:
    raise TraceError(f"unexpected semantic lexicon versions: {semantic_lexicon_versions!r}")


ouroboric_traces = tuple(OuroboricTrace(
    operator_positions=o.operator_positions,
    winding_count=o.winding_count,
    left_position=o.left.position,
    right_position=o.right.position,
```

```
    left_glyph=o.left.glyph,
    right_glyph=o.right.glyph,
    source_order=o.source_order,
    left_office_glyphs=o.left_office_glyphs,
    left_office_positions=o.left_office_positions,
    right_office_glyphs=o.right_office_glyphs,
    right_office_positions=o.right_office_positions,
    resolved_topology=o.resolved_topology,
    leading_echo=o.leading_echo,
    trailing_echo=o.trailing_echo,
    echo_position=o.echo_position,
    mirror_positions=o.mirror_positions,
    mirror_source_positions=o.mirror_source_positions,
    mirror_glyphs=o.mirror_glyphs,
    mirror_token_kinds=o.mirror_token_kinds,
    mirror_resolved_bearings=o.mirror_resolved_bearings,
    mirror_ostensive_variants=o.mirror_ostensive_variants,
    mirror_ostensive_bodies=o.mirror_ostensive_bodies,
    mirror_ostensive_authorities=o.mirror_ostensive_authorities,
    mirror_corridor_glyphs=o.mirror_corridor_glyphs,
    mirror_corridor_source_positions=o.mirror_corridor_source_positions,
    mirror_corridor_resolved_bearings=o.mirror_corridor_resolved_bearings,
    mirror_corridor_ostensive_variants=o.mirror_corridor_ostensive_variants,
    mirror_corridor_ostensive_bodies=o.mirror_corridor_ostensive_bodies,
    mirror_corridor_ostensive_authorities=o.mirror_corridor_ostensive_authorities,
    imaginary_reflection=o.imaginary_reflection,
    bearing_preserved_in_reflection=o.bearing_preserved_in_reflection,
    derived_glyph_generated=o.derived_glyph is not None,
) for o in local.ouroboric_states)

transmutation_traces = tuple(PhantasmagoriaTransmutationTrace(
    operator_positions=state.operator_positions,
    vessel_positions=tuple(body.position for body in state.vessels),
    vessel_glyphs=tuple(body.glyph for body in state.vessels),
    source_vessel_order=state.source_vessel_order,
    returned_order=state.returned_order,
    ostensive_bodies=tuple(getattr(resolution, "resolved_body") for resolution in
    ↪  state.office_resolutions),
    ostensive_variants=tuple(getattr(resolution, "ostensive_variant") for resolution in
    ↪  state.office_resolutions),
    ostensive_authorities=tuple(getattr(resolution, "source_authority") for resolution in
    ↪  state.office_resolutions),
    gate_boundary=state.gate_boundary,
    carrier_glyph=state.carrier_glyph,
    carrier_name=state.carrier_name,
    structural_hz=state.structural_hz,
    imaginary_parity_hz=state.imaginary_parity_hz,
    traversal=state.traversal,
    hand=state.hand,
    chord=state.chord,
    prosodical_subspace=state.prosodical_subspace,
) for state in local.transmutations)


aeternum_bearing_traces = tuple(AeternumBearingTrace(
```

```
    purpose=path.purpose,
    origin_position=path.origin_position,
    step=path.step,
    required_speaking_bodies=path.required_speaking_bodies,
    speaking_body_source_positions=path.speaking_body_source_positions,
    encounters=tuple(AeternumBearingEncounterTrace(
        virtual_position=encounter.virtual_position,
        source_position=encounter.source_position,
        glyph=encounter.glyph,
        token_kind=encounter.token_kind,
        bearing=encounter.bearing,
        imaginary=encounter.imaginary,
        speaking_body=encounter.speaking_body,
        bearing_only=encounter.bearing_only,
    ) for encounter in path.encounters),
    completed=path.completed,
) for path in local.aeternum_bearing_paths)

realization_token_by_position = {t.source_position: t for t in token_traces if
↪  t.realization_presence}
rendering_positions = [t.source_position for t in token_traces if t.realization_presence]
source_lexical_positions = {body.position for body in local.bodies}
lexical_slots = [
    index for index, position in enumerate(rendering_positions)
    if position in source_lexical_positions
]
if len(lexical_slots) != len(local.causal_body_order):
    raise TraceError("causal lexical order does not match realization lexical extent")
if set(local.causal_body_order) != source_lexical_positions:
    raise TraceError("causal lexical order is not an exact permutation of source lexical
    ↪  bodies")
for slot, position in zip(lexical_slots, local.causal_body_order):
    rendering_positions[slot] = position

rendering_tokens = tuple(RenderingToken(
    source_position=realization_token_by_position[pos].source_position,
    glyph=realization_token_by_position[pos].exact_glyph,
    resolved_state_body=realization_token_by_position[pos].resolved_state_body,
    grammatical_bearing=realization_token_by_position[pos].grammatical_bearing,
    attachments=realization_token_by_position[pos].local_operator_attachments,
    preserved_without_lookup=(
        realization_token_by_position[pos].exact_glyph == "𓆑"
        and not realization_token_by_position[pos].lexical_lookup_performed
    ),
    source_authority=realization_token_by_position[pos].source_authority,
    authored_office=realization_token_by_position[pos].authored_office,
    utterances=realization_token_by_position[pos].utterances,
) for pos in rendering_positions)

trace = DomusTrace(
    trace_version="aeonic-line-v3-tail" if local.tail_resolution is not None else
    ↪  TRACE_VERSION,
    source_string=frame.source,
    parsed_frame=ParsedFrameTrace(source=frame.source, exact_roundtrip=True),
    semantic_lexicon_version=LEXICON_VERSION,
```

```
        authority_versions=self.versions,
        token_traces=tuple(token_traces),
        ouroboric_traces=ouroboric_traces,
        transmutation_traces=transmutation_traces,
        rendering_input=RenderingInput(
            source=frame.source,
            tokens=rendering_tokens,
            ouroboric_indexes=tuple(range(len(ouroboric_traces))),
        ),
        aeternum_bearing_traces=aeternum_bearing_traces,
        utterance_state=(
            "lawful-unuttered-enochian"
            if not rendering_tokens and frame.tokens and all(token.kind == "grammar" for token
            ↪  in frame.tokens)
            else "uttered"
        ),
        tail_resolution=local.tail_resolution,
        final_body_order=local.causal_body_order if local.tail_resolution is not None else (),
    )
    self._validate(trace, frame)
    return trace

@staticmethod
def _validate(trace: DomusTrace, frame: DomusFrame) -> None:
    validate_tail_trace(trace)
    positions = [t.source_position for t in trace.token_traces]
    expected = [t.position for t in frame.tokens]
    if positions != expected:
        raise TraceError("token trace positions do not match exact source order")
    if "".join(t.exact_glyph for t in trace.token_traces) != frame.source:
        raise TraceError("token traces do not reconstruct exact source")
    source_positions = {t.position for t in frame.tokens}
    rendered_positions = {t.source_position for t in trace.rendering_input.tokens}
    if not rendered_positions.issubset(source_positions):
        raise TraceError("renderer input contains a position outside the source")
    token_by_position = {token.source_position: token for token in trace.token_traces}
    for path in trace.aeternum_bearing_traces:
        if not path.completed:
            raise TraceError("Aeternum bearing path did not lawfully close")
        if len(path.speaking_body_source_positions) != path.required_speaking_bodies:
            raise TraceError("Aeternum bearing path did not satisfy its center speaking-body
            ↪  obligation")
        for encounter in path.encounters:
            source = token_by_position.get(encounter.source_position)
            if source is None or source.exact_glyph != encounter.glyph:
                raise TraceError("Aeternum bearing encounter changed finite source identity")
            if not encounter.bearing_only:
                raise TraceError("Aeternum mirror attempted a second translation instead of
                ↪  bearing-only traversal")
    if trace.utterance_state == "lawful-unuttered-enochian":
        if trace.rendering_input.tokens:
            raise TraceError("lawful-unuttered Enochian center unexpectedly produced rendering
            ↪  cargo")
        if any(token.token_class != "grammar" for token in trace.token_traces):
```

```
                raise TraceError("lawful-unuttered Enochian state requires an all-Enoch finite
                ↪  center")
        for state in trace.transmutation_traces:
            if not state.operator_positions or len(state.vessel_positions) !=
            ↪  len(state.operator_positions) + 1:
                raise TraceError("ར transmutation corridor has unlawful Key/vessel extent")
            ops = tuple(token_by_position.get(pos) for pos in state.operator_positions)
            vessels = tuple(token_by_position.get(pos) for pos in state.vessel_positions)
            if any(op is None or op.exact_glyph != "ར" for op in ops) or any(v is None for v in
            ↪  vessels):
                raise TraceError("ར transmutation trace lost a vessel or Key source position")
            if state.source_vessel_order != state.vessel_positions:
                raise TraceError("ར transmutation changed source vessel order")
            expected_return = state.vessel_positions[1:] + state.vessel_positions[:1]
            if state.returned_order != expected_return:
                raise TraceError("ར transmutation changed passthrough return order")
            if state.gate_boundary != "i417" or state.carrier_glyph != "✩" or state.carrier_name !=
            ↪  "AHN":
                raise TraceError("ར transmutation lost its AHN i417 gate witness")
            if state.structural_hz != "432" or state.imaginary_parity_hz != "417":
                raise TraceError("ར transmutation changed the 432 / i417 channels")
            if state.traversal != ("Real", "Imaginary", "Real", "Imaginary"):
                raise TraceError("ར transmutation changed RI RI RI RI traversal")
            if not (len(state.ostensive_bodies) == len(state.ostensive_variants) ==
            ↪  len(state.ostensive_authorities) == len(state.vessel_positions)):
                raise TraceError("ར transmutation lost one vessel's Phantasmagoria resolution")
            if any(choice not in {"I", "II", "III"} for choice in state.ostensive_variants):
                raise TraceError("ར transmutation emitted an Ostensive outside I/II/III")


def _trace_from_dict(data: dict[str, object]) -> DomusTrace:
    parsed = ParsedFrameTrace(**data["parsed_frame"])
    authority_data = dict(data["authority_versions"])
    authority_data["lexical_authorities"] = tuple(tuple(item) for item in
    ↪  authority_data["lexical_authorities"])
    authority = AuthorityVersions(**authority_data)
    tokens = tuple(TokenTrace(
        **{**item,
           "traversal_path": tuple(item["traversal_path"]),
           "local_operator_attachments": tuple(item["local_operator_attachments"]),
           "utterances": tuple(LexicalUtteranceTrace(
               **{**utterance, "traversal_path": tuple(utterance["traversal_path"])}
           ) for utterance in item["utterances"])}
    ) for item in data["token_traces"])
    loops = tuple(OuroboricTrace(
        **{**item,
           "operator_positions": tuple(item["operator_positions"]),
           "source_order": tuple(item["source_order"]),
           "left_office_glyphs": tuple(item["left_office_glyphs"]),
           "left_office_positions": tuple(item["left_office_positions"]),
           "right_office_glyphs": tuple(item["right_office_glyphs"]),
           "right_office_positions": tuple(item["right_office_positions"]),
           "resolved_topology": tuple(item["resolved_topology"]),
           "mirror_positions": tuple(item.get("mirror_positions", ())),
           "mirror_source_positions": tuple(item.get("mirror_source_positions", ())),
```

```
       "mirror_glyphs": tuple(item.get("mirror_glyphs", ())),
       "mirror_token_kinds": tuple(item.get("mirror_token_kinds", ())),
       "mirror_resolved_bearings": tuple(item["mirror_resolved_bearings"]),
       "mirror_ostensive_variants": tuple(item["mirror_ostensive_variants"]),
       "mirror_ostensive_bodies": tuple(item["mirror_ostensive_bodies"]),
       "mirror_ostensive_authorities": tuple(item["mirror_ostensive_authorities"]),
       "mirror_corridor_glyphs": tuple(item["mirror_corridor_glyphs"]),
       "mirror_corridor_source_positions": tuple(item.get("mirror_corridor_source_positions",
       ↪  ())),
       "mirror_corridor_resolved_bearings": tuple(item["mirror_corridor_resolved_bearings"]),
       "mirror_corridor_ostensive_variants": tuple(item["mirror_corridor_ostensive_variants"]),
       "mirror_corridor_ostensive_bodies": tuple(item["mirror_corridor_ostensive_bodies"]),
       "mirror_corridor_ostensive_authorities":
       ↪  tuple(item["mirror_corridor_ostensive_authorities"]),
       "imaginary_reflection": bool(item["imaginary_reflection"]),
       "bearing_preserved_in_reflection": bool(item.get("bearing_preserved_in_reflection",
       ↪  True))}
) for item in data["ouroboric_traces"])
transmutations = tuple(PhantasmagoriaTransmutationTrace(
    **{**item,
       "operator_positions": tuple(item["operator_positions"]),
       "vessel_positions": tuple(item["vessel_positions"]),
       "vessel_glyphs": tuple(item["vessel_glyphs"]),
       "source_vessel_order": tuple(item["source_vessel_order"]),
       "returned_order": tuple(item["returned_order"]),
       "ostensive_bodies": tuple(item["ostensive_bodies"]),
       "ostensive_variants": tuple(item["ostensive_variants"]),
       "ostensive_authorities": tuple(item["ostensive_authorities"]),
       "traversal": tuple(item["traversal"])}
) for item in data["transmutation_traces"])
aeternum = tuple(AeternumBearingTrace(
    **{**item,
       "speaking_body_source_positions": tuple(item["speaking_body_source_positions"]),
       "encounters": tuple(AeternumBearingEncounterTrace(**encounter) for encounter in
       ↪  item["encounters"])}
) for item in data.get("aeternum_bearing_traces", ()))
ri = data["rendering_input"]
rendering = RenderingInput(
    source=ri["source"],
    tokens=tuple(RenderingToken(
        **{**item,
           "attachments": tuple(item["attachments"]),
           "utterances": tuple(LexicalUtteranceTrace(
               **{**utterance, "traversal_path": tuple(utterance["traversal_path"])}
           ) for utterance in item["utterances"])}
    ) for item in ri["tokens"]),
    ouroboric_indexes=tuple(ri["ouroboric_indexes"]),
)
return DomusTrace(
    trace_version=data["trace_version"],
    source_string=data["source_string"],
    parsed_frame=parsed,
    semantic_lexicon_version=data["semantic_lexicon_version"],
    authority_versions=authority,
    token_traces=tokens,
```

```
    ouroboric_traces=loops,
    transmutation_traces=transmutations,
    rendering_input=rendering,
    aeternum_bearing_traces=aeternum,
    utterance_state=str(data.get("utterance_state", "uttered")),
    tail_resolution=(
        tail_resolution_from_dict(data["tail_resolution"])
        if data.get("tail_resolution") is not None else None
    ),
    final_body_order=tuple(data.get("final_body_order", ())),
)
```

translation_matrix_integration.py

```
"""Phase 18 one-way integration from closed chooser bodies into Sydonic Magicae use.

The chooser boundary is absolute.  ``build_chooser_snapshot`` accepts only the
Phase 16 certificate field and does not import, parse, locate, or inspect Sydonic authority files.
It freezes the 156 STRICT bodies and their 1,248 natively closed bearing words.

The current .aksh Translation Matrix contracts are loaded only after that snapshot exists.  A
Domus use plan may reference a closed bearing and attach authored grammar
operators around that reference.  It has no word field and therefore cannot
replace, derive, score, rank, or reopen a closed bearing.
"""
from __future__ import annotations

from dataclasses import dataclass
import hashlib
from pathlib import Path

from .goetic_outward_return import BEARINGS, GoeticBearing
from .aksh_reader import attributes
from .implementation_reconciliation import AKSH_AUTHORITY_FILES
from .native_bearing_assembly import (
    AEON_IDENTITIES,
    AEON_TOTAL,
    PARTIAL_BODY_LAW,
    AeonIdentity,
    PartialBodyLaw,
    StrictLexicalBody,
)
from .truth_choice_certificates import (
    TruthChoiceCertificate,
    TruthChoiceCertificateField,
)


class TranslationMatrixIntegrationError(ValueError):
    """Raised when a post-chooser use attempts to cross the closed boundary."""



def _sha256_bytes(data: bytes) -> str:
    return hashlib.sha256(data).hexdigest()


def _digest(value: object) -> str:
    return _sha256_bytes(repr(value).encode("utf-8"))


def _canonical_aksh_digest(text: str) -> str:
```

```
    """Digest the exact authored .aksh logical lines without newline-style dependence."""
    return _digest(tuple(text.splitlines()))


@dataclass(frozen=True, slots=True)
class MatrixContractPin:
    filename: str
    absolute_path: str
    byte_size: int
    sha256: str
    version: str
    semantic_digest: str

    def __post_init__(self) -> None:
        if self.filename not in AKSH_AUTHORITY_FILES:
            raise TranslationMatrixIntegrationError("unknown Translation Matrix .aksh contract")
        if self.byte_size <= 0 or len(self.sha256) != 64 or len(self.semantic_digest) != 64:
            raise TranslationMatrixIntegrationError("Translation Matrix .aksh pin is incomplete")
        if not self.version:
            raise TranslationMatrixIntegrationError("Translation Matrix .aksh pin lacks a version")


@dataclass(frozen=True, slots=True)
class EnochContract:
    enoch_id: str
    glyphs: tuple[str, ...]
    operation_type: str
    presence: str
    order: int | None
    parser_trigger: bool
    written: bool
    spoken: bool

    def __post_init__(self) -> None:
        if not self.enoch_id or not self.glyphs or any(len(glyph) != 1 for glyph in self.glyphs) or
        ↪  not self.operation_type:
            raise TranslationMatrixIntegrationError("Enoch contract is incomplete")
        if self.order is not None and self.order <= 0:
            raise TranslationMatrixIntegrationError("Enoch runtime order must be positive when
            ↪  present")


@dataclass(frozen=True, slots=True)
class CadenceContract:
    cadence_id: str
    glyphs: tuple[str, ...]

    def __post_init__(self) -> None:
        if not self.cadence_id or not self.glyphs or any(len(glyph) != 1 for glyph in self.glyphs):
            raise TranslationMatrixIntegrationError("Living Cadence contract is incomplete")



@dataclass(frozen=True, slots=True)
class TranslationMatrixContracts:
    pins: tuple[MatrixContractPin, ...]
```

```
enochs: tuple[EnochContract, ...]
cadences: tuple[CadenceContract, ...]
semantic_digest: str
byte_manifest_digest: str

def __post_init__(self) -> None:
    if tuple(pin.filename for pin in self.pins) != AKSH_AUTHORITY_FILES:
        raise TranslationMatrixIntegrationError("Translation Matrix contracts must contain the
        ↪  exact .aksh authority set")
    if len(self.enochs) != 11 or len({item.enoch_id for item in self.enochs}) != 11:
        raise TranslationMatrixIntegrationError("Translation Matrix must preserve the eleven
        ↪  authored Enoch entries")
    ordered = tuple(item.order for item in self.enochs)
    if ordered != tuple(range(1, 12)):
        raise TranslationMatrixIntegrationError("Enoch runtime records must preserve authored
        ↪  order 1..11")
    if not all(item.parser_trigger and item.written and item.spoken for item in self.enochs):
        raise TranslationMatrixIntegrationError("all eleven Enochs must remain written
        ↪  parser-active spoken grammar")
    if len(self.cadences) != 4 or len({item.cadence_id for item in self.cadences}) != 4:
        raise TranslationMatrixIntegrationError("Translation Matrix must preserve the four
        ↪  authored Living Cadences")
    cadence_by_id = {item.cadence_id: item for item in self.cadences}
    required = {
        "regia…cosmos…above": tuple("☽☉☾"),
        "breath…throughout": (chr(0x11066),),
        "cantillation…chant": (chr(0x0FC2),),
        "magic…subspace…prosody": ("⟠",),
    }
    for cadence_id, glyphs in required.items():
        item = cadence_by_id.get(cadence_id)
        if item is None or item.glyphs != glyphs:
            raise TranslationMatrixIntegrationError(f"Living Cadence identity moved:
            ↪  {cadence_id}")
    if len(self.semantic_digest) != 64 or len(self.byte_manifest_digest) != 64:
        raise TranslationMatrixIntegrationError("Translation Matrix contract digests are
        ↪  incomplete")

@property
def enoch_count(self) -> int:
    return len(self.enochs)

@property
def grammar_operators(self) -> tuple[EnochContract, ...]:
    return tuple(sorted((item for item in self.enochs if item.order is not None), key=lambda
    ↪  item: item.order))

@property
def active_grammar_glyphs(self) -> frozenset[str]:
    return frozenset(
        glyph
        for enoch in self.enochs
        if enoch.parser_trigger and enoch.written
        for glyph in enoch.glyphs
    )
```

```
@dataclass(frozen=True, slots=True)
class ClosedBearingWord:
    aeon: AeonIdentity
    bearing: GoeticBearing
    bearing_index: int
    word: str
    native_q_bias: str
    sense_fingerprint: str
    proof_digest: str
    closure_digest: str
    source_record_digest: str
    grammatical_offices: tuple[str, ...]

    def __post_init__(self) -> None:
        if self.bearing_index != BEARINGS.index(self.bearing):
            raise TranslationMatrixIntegrationError("closed bearing index moved")
        if not self.word or self.native_q_bias not in {"Q0", "Q1", "Q2", "Q3"}:
            raise TranslationMatrixIntegrationError("closed bearing lacks word or native Q Bias")
        for value in (
            self.sense_fingerprint,
            self.proof_digest,
            self.closure_digest,
            self.source_record_digest,
        ):
            if len(value) != 64:
                raise TranslationMatrixIntegrationError("closed bearing lacks an exact source
                ↪  digest")
        if len(set(self.grammatical_offices)) != len(self.grammatical_offices):
            raise TranslationMatrixIntegrationError("grammatical offices were duplicated")

    @property
    def immutable_key(self) -> tuple[int, str, str, str]:
        return (
            self.aeon.index,
            self.bearing.value,
            self.sense_fingerprint,
            self.proof_digest,
        )


@dataclass(frozen=True, slots=True)
class ClosedEightBearingBody:
    aeon: AeonIdentity
    native_q_bias: str
    bearings: tuple[ClosedBearingWord, ...]
    strict_law: PartialBodyLaw
    body_digest: str

    def __post_init__(self) -> None:
        if self.strict_law is not PARTIAL_BODY_LAW or self.strict_law is not PartialBodyLaw.STRICT:
            raise TranslationMatrixIntegrationError("Translation Matrix received a non-STRICT body")
        if tuple(item.bearing for item in self.bearings) != BEARINGS or len(self.bearings) != 8:
            raise TranslationMatrixIntegrationError("Translation Matrix requires all eight closed
            ↪  bearings")
```

```
        if any(item.aeon != self.aeon for item in self.bearings):
            raise TranslationMatrixIntegrationError("assembled body mixed Aeon identities")
        if any(item.native_q_bias != self.native_q_bias for item in self.bearings):
            raise TranslationMatrixIntegrationError("assembled body changed native Q Bias")
        if len({item.proof_digest for item in self.bearings}) != 8:
            raise TranslationMatrixIntegrationError("each bearing must retain its own closure
            ↪  certificate")
        if len(self.body_digest) != 64:
            raise TranslationMatrixIntegrationError("assembled body lacks its exact digest")

    def bearing_word(self, bearing: GoeticBearing) -> ClosedBearingWord:
        return self.bearings[BEARINGS.index(bearing)]


@dataclass(frozen=True, slots=True)
class ChooserSnapshot:
    bodies: tuple[ClosedEightBearingBody, ...]
    source_certificate_count: int
    snapshot_digest: str
    authority_independent: bool = True

    def __post_init__(self) -> None:
        if tuple(body.aeon for body in self.bodies) != AEON_IDENTITIES or len(self.bodies) !=
        ↪  AEON_TOTAL:
            raise TranslationMatrixIntegrationError("chooser snapshot must preserve all 156 Aeons")
        if self.source_certificate_count != AEON_TOTAL * 8:
            raise TranslationMatrixIntegrationError("chooser snapshot must preserve all 1,248
            ↪  certificates")
        if not self.authority_independent or len(self.snapshot_digest) != 64:
            raise TranslationMatrixIntegrationError("chooser snapshot lost its authority-independent
            ↪  witness")

    @property
    def word_count(self) -> int:
        return sum(len(body.bearings) for body in self.bodies)

    def body(self, aeon_index: int) -> ClosedEightBearingBody:
        if not 0 <= aeon_index < AEON_TOTAL:
            raise TranslationMatrixIntegrationError("Aeon index lies outside the 156-body field")
        return self.bodies[aeon_index]


@dataclass(frozen=True, slots=True)
class DomusBearingReference:
    aeon_index: int
    bearing: GoeticBearing
    expected_proof_digest: str
    grammar_before: tuple[str, ...] = ()
    grammar_after: tuple[str, ...] = ()

    def __post_init__(self) -> None:
        if not 0 <= self.aeon_index < AEON_TOTAL:
            raise TranslationMatrixIntegrationError("Domus use references an unknown Aeon")
        if len(self.expected_proof_digest) != 64:
```

```
            raise TranslationMatrixIntegrationError("Domus use lacks the pinned bearing
            ↪  certificate")
        if any(len(glyph) != 1 for glyph in self.grammar_before + self.grammar_after):
            raise TranslationMatrixIntegrationError("Domus grammar references must be exact glyphs")


@dataclass(frozen=True, slots=True)
class DomusUsePlan:
    plan_id: str
    references: tuple[DomusBearingReference, ...]

    def __post_init__(self) -> None:
        if not self.plan_id or not self.references:
            raise TranslationMatrixIntegrationError("Domus use plan requires an identity and at
            ↪  least one bearing")


@dataclass(frozen=True, slots=True)
class SydonicMagicaeUseToken:
    closed_word: ClosedBearingWord
    word: str
    native_q_bias: str
    grammatical_offices: tuple[str, ...]
    grammar_before: tuple[str, ...]
    grammar_after: tuple[str, ...]
    proof_digest: str

    def __post_init__(self) -> None:
        if self.word != self.closed_word.word:
            raise TranslationMatrixIntegrationError("Translation Matrix attempted to alter a closed
            ↪  word")
        if self.native_q_bias != self.closed_word.native_q_bias:
            raise TranslationMatrixIntegrationError("Translation Matrix attempted to alter native Q
            ↪  Bias")
        if self.grammatical_offices != self.closed_word.grammatical_offices:
            raise TranslationMatrixIntegrationError("Translation Matrix attempted to alter the
            ↪  closed grammatical office")
        if self.proof_digest != self.closed_word.proof_digest:
            raise TranslationMatrixIntegrationError("Translation Matrix detached the word from its
            ↪  certificate")


@dataclass(frozen=True, slots=True)
class SydonicMagicaeUse:
    plan: DomusUsePlan
    tokens: tuple[SydonicMagicaeUseToken, ...]
    chooser_snapshot_digest: str
    matrix_semantic_digest: str
    use_digest: str

    def __post_init__(self) -> None:
        if len(self.tokens) != len(self.plan.references):
            raise TranslationMatrixIntegrationError("Domus use lost a bearing reference")
        if any(len(value) != 64 for value in (
            self.chooser_snapshot_digest,
```

```
            self.matrix_semantic_digest,
            self.use_digest,
        )):
            raise TranslationMatrixIntegrationError("Domus use lacks exact boundary digests")

    @property
    def words(self) -> tuple[str, ...]:
        return tuple(token.word for token in self.tokens)


@dataclass(frozen=True, slots=True)
class TranslationMatrixIntegrationField:
    chooser_snapshot: ChooserSnapshot
    matrix_contracts: TranslationMatrixContracts
    canonical_uses: tuple[SydonicMagicaeUse, ...]

    def __post_init__(self) -> None:
        if len(self.canonical_uses) != AEON_TOTAL:
            raise TranslationMatrixIntegrationError("Phase 18 requires one post-closure use for
            ↪  every Aeon")
        for body, use in zip(self.chooser_snapshot.bodies, self.canonical_uses, strict=True):
            if use.words != tuple(item.word for item in body.bearings):
                raise TranslationMatrixIntegrationError("canonical Domus use changed the assembled
                ↪  eight")
            if use.chooser_snapshot_digest != self.chooser_snapshot.snapshot_digest:
                raise TranslationMatrixIntegrationError("canonical Domus use detached from chooser
                ↪  snapshot")


def _certificate_by_body(field: TruthChoiceCertificateField) -> dict[int,
↪  tuple[TruthChoiceCertificate, ...]]:
    grouped: dict[int, list[TruthChoiceCertificate]] = {index: [] for index in range(AEON_TOTAL)}
    for certificate in field.certificates:
        grouped[certificate.aeon_constitution.aeon.index].append(certificate)
    result: dict[int, tuple[TruthChoiceCertificate, ...]] = {}
    for index in range(AEON_TOTAL):
        certificates = tuple(sorted(grouped[index], key=lambda item: item.bearing_index))
        if len(certificates) != 8 or tuple(item.bearing_index for item in certificates) !=
        ↪  tuple(range(8)):
            raise TranslationMatrixIntegrationError("certificate field lost an eight-bearing Aeon
            ↪  body")
        result[index] = certificates
    return result



def _compact_closure_digest(certificate: TruthChoiceCertificate) -> str:
    return _digest((
        certificate.aeon_constitution,
        certificate.relational_geometry_witness,
        certificate.envelope_witness,
        certificate.exact_arithmetic,
        certificate.bearing_index,
        certificate.proof_digest,
    ))
```

```
def _compact_source_record_digest(certificate: TruthChoiceCertificate) -> str:
    return _digest((
        certificate.telegraphic_field_admission,
        tuple((current.glyph, current.evidence_digest, current.satisfied) for current in
        ↪  certificate.current_witnesses),
        tuple(sorted(certificate.unique_survivor_set)),
        certificate.proof_digest,
    ))

def _closed_word(certificate: TruthChoiceCertificate) -> ClosedBearingWord:
    closure = certificate.phase15_closure
    offices = tuple(dict.fromkeys(
        unit.operator_office
        for unit in certificate.prosody_witness.units
        if unit.operator_office
    ))
    return ClosedBearingWord(
        aeon=closure.aeon,
        bearing=closure.bearing,
        bearing_index=certificate.bearing_index,
        word=closure.candidate,
        native_q_bias=closure.native_q_bias,
        sense_fingerprint=certificate.candidate_sense_identity.structural_fingerprint,
        proof_digest=certificate.proof_digest,
        closure_digest=_compact_closure_digest(certificate),
        source_record_digest=_compact_source_record_digest(certificate),
        grammatical_offices=offices,
    )


def build_chooser_snapshot(field: TruthChoiceCertificateField) -> ChooserSnapshot:
    """Freeze all closed words without touching Sydonic .aksh authority or Domus grammar."""
    grouped = _certificate_by_body(field)
    bodies: list[ClosedEightBearingBody] = []
    for result in field.source.aeons:
        if result.body is None or not isinstance(result.body, StrictLexicalBody):
            raise TranslationMatrixIntegrationError("Phase 18 cannot integrate a failed or partial
            ↪  Aeon")
        certificates = grouped[result.aeon.index]
        bearings = tuple(_closed_word(certificate) for certificate in certificates)
        body_material = (
            result.aeon,
            result.body.native_q_bias,
            tuple(item.immutable_key for item in bearings),
            result.body.partial_body_law.value,
        )
        bodies.append(ClosedEightBearingBody(
            result.aeon,
            result.body.native_q_bias,
            bearings,
            result.body.partial_body_law,
            _digest(body_material),
        ))
    frozen = tuple(bodies)
```

```
    snapshot_digest = _digest(tuple((body.aeon.index, body.body_digest) for body in frozen))
    return ChooserSnapshot(frozen, len(field.certificates), snapshot_digest, True)


def _bool_attribute(values: dict[str, str], name: str, glyph: str) -> bool:
    if name not in values:
        raise TranslationMatrixIntegrationError(
            f"Enochian Understanding {glyph!r} lacks required boolean attribute {name!r}"
        )
    raw = values[name].casefold()
    if raw == "true":
        return True
    if raw == "false":
        return False
    raise TranslationMatrixIntegrationError(
        f"invalid boolean attribute {name!r}={values[name]!r} for {glyph!r}"
    )


def _authority_version(filename: str, text: str) -> str:
    lines = text.splitlines()
    if not lines:
        raise TranslationMatrixIntegrationError(f"empty .aksh authority: {filename}")
    version = attributes(lines[0]).get("version", "")
    if not version:
        raise TranslationMatrixIntegrationError(f"required .aksh version missing: {filename}")
    return version


def _enoch_contracts(text: str) -> tuple[EnochContract, ...]:
    enochs: list[EnochContract] = []
    current: dict[str, str] | None = None
    operation_type = ""
    for line in text.splitlines():
        if line.startswith("·:enoch "):
            if current is not None:
                raise TranslationMatrixIntegrationError("nested Enochian Understanding declaration")
            current = attributes(line)
            operation_type = ""
            continue
        if current is None:
            continue
        if line.startswith("·:operation "):
            operation_type = attributes(line).get("type", "")
            continue
        if line == "·:·enoch:·":
            glyph_text = current.get("glyph", "")
            enoch_id = current.get("id", "")
            order_text = current.get("order", "")
            if not glyph_text or not enoch_id or not operation_type:
                raise TranslationMatrixIntegrationError(
                    f"Enochian Understanding declaration is incomplete: {current!r}"
                )
            enochs.append(EnochContract(
                enoch_id=enoch_id,
                glyphs=tuple(glyph_text),
```

```
                operation_type=operation_type,
                presence=current.get("presence", "runtime"),
                order=int(order_text) if order_text else None,
                parser_trigger=_bool_attribute(current, "parser_trigger", glyph_text),
                written=_bool_attribute(current, "written", glyph_text),
                spoken=_bool_attribute(current, "spoken", glyph_text),
            ))
            current = None
            operation_type = ""
    if current is not None:
        raise TranslationMatrixIntegrationError("unterminated Enochian Understanding declaration")
    return tuple(enochs)


def _cadence_contracts(text: str) -> tuple[CadenceContract, ...]:
    cadence_ids = {
        "☽⊙☾": "regia…cosmos…above",
        "·": "breath…throughout",
        "𐙚": "cantillation…chant",
        "⟠": "magic…subspace…prosody",
    }
    cadences: list[CadenceContract] = []
    for line in text.splitlines():
        if not line.startswith("∴glyph "):
            continue
        glyph_text = attributes(line).get("symbol", "")
        cadence_id = cadence_ids.get(glyph_text)
        if cadence_id is None:
            continue
        cadences.append(CadenceContract(
            cadence_id=cadence_id,
            glyphs=tuple(glyph_text),
        ))
    return tuple(cadences)
def build_translation_matrix_contracts(aksh_root: str | Path) -> TranslationMatrixContracts:
    """Load the sole current .aksh authority body after chooser closure."""
    root_path = Path(aksh_root).resolve()
    pins: list[MatrixContractPin] = []
    semantic_bodies: list[tuple[str, str]] = []
    understandings = ""
    cadence_authority = ""
    for filename in AKSH_AUTHORITY_FILES:
        path = root_path / filename
        if not path.is_file():
            raise TranslationMatrixIntegrationError(f"required post-chooser .aksh contract missing:
            ↪  {filename}")
        data = path.read_bytes()
        text = data.decode("utf-8")
        version = _authority_version(filename, text)
        semantic = _canonical_aksh_digest(text)
        pins.append(MatrixContractPin(
            filename,
            str(path),
            len(data),
            _sha256_bytes(data),
            version,
```

```
            semantic,
        ))
        semantic_bodies.append((filename, semantic))
        if filename == "the_4_cadences.aksh":
            cadence_authority = text
        if filename == "Enochian…Understandings.aksh":
            understandings = text

    if not cadence_authority:
        raise TranslationMatrixIntegrationError("the_4_cadences.aksh authority is absent")
    if not understandings:
        raise TranslationMatrixIntegrationError("Enochian Understandings .aksh authority is absent")
    enochs = _enoch_contracts(understandings)
    cadences = _cadence_contracts(cadence_authority)
    pin_tuple = tuple(pins)
    semantic_digest = _digest(tuple(semantic_bodies))
    byte_manifest_digest = _digest(tuple(
        (pin.filename, pin.byte_size, pin.sha256)
        for pin in pin_tuple
    ))
    return TranslationMatrixContracts(pin_tuple, enochs, cadences, semantic_digest,
    ↪  byte_manifest_digest)


def _validate_plan_grammar(plan: DomusUsePlan, contracts: TranslationMatrixContracts) -> None:
    active = contracts.active_grammar_glyphs
    for reference in plan.references:
        for glyph in reference.grammar_before + reference.grammar_after:
            if glyph not in active:
                raise TranslationMatrixIntegrationError(
                    f"Domus use requested a non-written or non-triggering grammar glyph: {glyph}"
                )


def apply_translation_matrix(
    snapshot: ChooserSnapshot,
    contracts: TranslationMatrixContracts,
    plan: DomusUsePlan,
) -> SydonicMagicaeUse:
    """Use already-closed words.  The plan can reference slots but cannot supply words."""
    _validate_plan_grammar(plan, contracts)
    tokens: list[SydonicMagicaeUseToken] = []
    for reference in plan.references:
        closed = snapshot.body(reference.aeon_index).bearing_word(reference.bearing)
        if reference.expected_proof_digest != closed.proof_digest:
            raise TranslationMatrixIntegrationError("Domus use reference does not match the closed
            ↪  bearing certificate")
        tokens.append(SydonicMagicaeUseToken(
            closed,
            closed.word,
            closed.native_q_bias,
            closed.grammatical_offices,
            reference.grammar_before,
            reference.grammar_after,
            closed.proof_digest,
        ))
```

```
    token_tuple = tuple(tokens)
    material = (
        plan,
        tuple((token.closed_word.immutable_key, token.grammar_before, token.grammar_after) for token
        ↪  in token_tuple),
        snapshot.snapshot_digest,
        contracts.semantic_digest,
    )
    return SydonicMagicaeUse(
        plan,
        token_tuple,
        snapshot.snapshot_digest,
        contracts.semantic_digest,
        _digest(material),
    )


def canonical_body_plan(body: ClosedEightBearingBody) -> DomusUsePlan:
    return DomusUsePlan(
        plan_id=f"canonical-eight:{body.aeon.index}:{body.aeon.key}",
        references=tuple(
            DomusBearingReference(
                body.aeon.index,
                closed.bearing,
                closed.proof_digest,
            )
            for closed in body.bearings
        ),
    )


def replay_translation_matrix_use(
    snapshot: ChooserSnapshot,
    contracts: TranslationMatrixContracts,
    use: SydonicMagicaeUse,
) -> SydonicMagicaeUse:
    replay = apply_translation_matrix(snapshot, contracts, use.plan)
    if replay != use:
        raise TranslationMatrixIntegrationError("Domus use did not reproduce from the assembled
        ↪  eight")
    return replay


def prove_phase18_exit_gate(
    field: TruthChoiceCertificateField,
    aksh_root: str | Path,
) -> TranslationMatrixIntegrationField:
    """Close chooser first, load Matrix second, then replay all 156 canonical uses."""
    snapshot = build_chooser_snapshot(field)
    contracts = build_translation_matrix_contracts(aksh_root)
    uses = tuple(
        apply_translation_matrix(snapshot, contracts, canonical_body_plan(body))
        for body in snapshot.bodies
    )
    result = TranslationMatrixIntegrationField(snapshot, contracts, uses)
    for use in result.canonical_uses:
```

```
    replay_translation_matrix_use(snapshot, contracts, use)
return result
```

triadic_bridge.py

```
"""Phase 9 complete triadic bridge.

The bridge composes the inherited Bias witness, one declared Q action, and
Phase 8's exact frequency/prosody action into one candidate path seed:

    J_A(w) = F_nu,A o F_v,A o F_beta,A

All three microsteps act on one declared Court-rooted domain.  Each step keeps
the coordinates it does not govern.  A matching frequency body, Q body, or
prosodic witness is never accepted in place of the complete triad.
"""
from __future__ import annotations

from dataclasses import dataclass
from typing import Final, Literal, TypeAlias

from .frequency_projection import (
    Carrier,
    FrequencyClosure,
    FrequencyProjectionError,
    PHI,
    ProsodicWitness,
    prove_frequency_closure,
)


QState = Literal["Q0", "Q1", "Q2", "Q3"]
QSTATE_CYCLE: Final[tuple[QState, ...]] = ("Q0", "Q1", "Q2", "Q3")
class TriadicBridgeError(ValueError):
    """Raised when a Phase 9 candidate does not carry the complete triad."""


QVector: TypeAlias = tuple[QState, QState, QState, QState]
MICROSTEP_ORDER: Final[tuple[str, str, str]] = ("F_beta", "F_v", "F_nu")


def _validate_q_vector(vector: object, label: str) -> QVector:
    if not isinstance(vector, tuple) or len(vector) != 4:
        raise TriadicBridgeError(f"{label} must contain exactly four Q-state coordinates")
    if any(state not in QSTATE_CYCLE for state in vector):
        raise TriadicBridgeError(f"{label} contains an undefined Q-state")
    return vector  # type: ignore[return-value]


@dataclass(frozen=True, slots=True)
class TriadicDomain:
    """The one declared subset of the Court triplet domain used by all steps."""
```

```
    court_root: str
    inherited_q_bias: QState
    carrier: Carrier
    q_vector_arity: int = 4
    breath_bounds: tuple[object, object] = (-PHI, PHI)

    def __post_init__(self) -> None:
        if not isinstance(self.court_root, str) or not self.court_root:
            raise TriadicBridgeError("the triadic domain requires one declared Court root")
        if self.inherited_q_bias not in QSTATE_CYCLE:
            raise TriadicBridgeError("the triadic domain requires one inherited Q Bias")
        if not isinstance(self.carrier, Carrier):
            raise TriadicBridgeError("the triadic domain requires one declared Aeon carrier")
        if self.q_vector_arity != 4:
            raise TriadicBridgeError("the Domus Q vector has exactly four coordinates")
        if self.breath_bounds != self.carrier.breath_bounds:
            raise TriadicBridgeError("the declared frequency domain must retain [-Phi,+Phi]")


@dataclass(frozen=True, slots=True)
class QAction:
    """One explicit Q-state-vector action on the declared domain."""

    name: str
    source: QVector
    target: QVector

    def __post_init__(self) -> None:
        if not isinstance(self.name, str) or not self.name.strip():
            raise TriadicBridgeError("F_v requires a named, defined Q action")
        _validate_q_vector(self.source, "Q action source")
        _validate_q_vector(self.target, "Q action target")

    def apply(self, vector: QVector) -> QVector:
        _validate_q_vector(vector, "Q action input")
        if vector != self.source:
            raise TriadicBridgeError("Q action input lies outside its declared source body")
        return self.target


@dataclass(frozen=True, slots=True)
class CandidateTripletBody:
    """One typed candidate body inside a single TriadicDomain."""

    domain: TriadicDomain
    witness: ProsodicWitness
    court_root: str
    q_bias: QState
    q_vector: QVector
    frequency_action: FrequencyClosure | None = None

    def __post_init__(self) -> None:
        if not isinstance(self.domain, TriadicDomain):
            raise TriadicBridgeError("candidate body lacks a declared triadic domain")
        if not isinstance(self.witness, ProsodicWitness):
            raise TriadicBridgeError("candidate body lacks a typed prosodic witness")
```

```
        if self.court_root != self.domain.court_root:
            raise TriadicBridgeError("candidate body changed its Court root")
        if self.q_bias != self.domain.inherited_q_bias:
            raise TriadicBridgeError("candidate body changed inherited Q Bias")
        _validate_q_vector(self.q_vector, "candidate Q vector")
        if self.frequency_action is not None:
            closure = self.frequency_action
            if closure.outward.carrier != self.domain.carrier:
                raise TriadicBridgeError("frequency action changed the declared Aeon carrier")
            if closure.outward.candidate != self.witness.candidate:
                raise TriadicBridgeError("frequency action belongs to a different candidate")
            if not closure.nu_close or not closure.dcomp_frequency_zero:
                raise TriadicBridgeError("frequency action is not exactly closed")


@dataclass(frozen=True, slots=True)
class BiasMicrostep:
    name: str
    domain: TriadicDomain
    before: CandidateTripletBody
    after: CandidateTripletBody
    inherited_q_bias: QState


@dataclass(frozen=True, slots=True)
class QMicrostep:
    name: str
    domain: TriadicDomain
    before: CandidateTripletBody
    after: CandidateTripletBody
    action: QAction


@dataclass(frozen=True, slots=True)
class FrequencyMicrostep:
    name: str
    domain: TriadicDomain
    before: CandidateTripletBody
    after: CandidateTripletBody
    action: FrequencyClosure


Microstep: TypeAlias = BiasMicrostep | QMicrostep | FrequencyMicrostep


@dataclass(frozen=True, slots=True)
class CandidatePathSeed:
    """The one typed body admitted to the complete Aeon phase engine."""

    domain: TriadicDomain
    witness: ProsodicWitness
    microsteps: tuple[Microstep, ...]
    body: CandidateTripletBody
```

```
def F_beta(body: CandidateTripletBody) -> BiasMicrostep:
    """Declare inherited Bias as an exact identity microstep."""
    after = CandidateTripletBody(
        domain=body.domain,
        witness=body.witness,
        court_root=body.court_root,
        q_bias=body.q_bias,
        q_vector=body.q_vector,
        frequency_action=body.frequency_action,
    )
    return BiasMicrostep("F_beta", body.domain, body, after, body.q_bias)



def F_v(body: CandidateTripletBody, action: QAction) -> QMicrostep:
    """Apply one defined Q action while preserving root, Bias, and frequency."""
    if body.frequency_action is not None:
        raise TriadicBridgeError("F_v must precede F_nu in the ordered triad")
    target = action.apply(body.q_vector)
    after = CandidateTripletBody(
        domain=body.domain,
        witness=body.witness,
        court_root=body.court_root,
        q_bias=body.q_bias,
        q_vector=target,
        frequency_action=None,
    )
    return QMicrostep("F_v", body.domain, body, after, action)



def F_nu(body: CandidateTripletBody) -> FrequencyMicrostep:
    """Attach Phase 8's exact frequency action while preserving root and Q body."""
    if body.frequency_action is not None:
        raise TriadicBridgeError("F_nu cannot overwrite an existing frequency action")
    try:
        closure = prove_frequency_closure(body.witness, body.domain.carrier)
    except FrequencyProjectionError as exc:
        raise TriadicBridgeError("candidate frequency action failed exact Phase 8 closure") from exc
    after = CandidateTripletBody(
        domain=body.domain,
        witness=body.witness,
        court_root=body.court_root,
        q_bias=body.q_bias,
        q_vector=body.q_vector,
        frequency_action=closure,
    )
    return FrequencyMicrostep("F_nu", body.domain, body, after, closure)



def triad_A(candidate: object) -> bool:
    """Return the exact Phase 9 completeness predicate Triad_A(J_A(w))."""
    if not isinstance(candidate, CandidatePathSeed):
        return False
    if len(candidate.microsteps) != 3:
```

```
    return False
beta_step, q_step, nu_step = candidate.microsteps
if not isinstance(beta_step, BiasMicrostep):
    return False
if not isinstance(q_step, QMicrostep):
    return False
if not isinstance(nu_step, FrequencyMicrostep):
    return False
if tuple(step.name for step in candidate.microsteps) != MICROSTEP_ORDER:
    return False

domain = candidate.domain
if any(step.domain != domain for step in candidate.microsteps):
    return False
if beta_step.before.domain != domain or candidate.body.domain != domain:
    return False
if beta_step.before.witness != candidate.witness or candidate.body.witness != candidate.witness:
    return False

# F_beta is the inherited-Bias identity and changes nothing.
if beta_step.inherited_q_bias != domain.inherited_q_bias:
    return False
if beta_step.before != beta_step.after:
    return False
if beta_step.before.q_bias != domain.inherited_q_bias:
    return False

# The chain is one body, not three disconnected matching records.
if q_step.before != beta_step.after or nu_step.before != q_step.after:
    return False
if candidate.body != nu_step.after:
    return False

# F_v governs only the Q vector.
if q_step.before.court_root != q_step.after.court_root:
    return False
if q_step.before.q_bias != q_step.after.q_bias:
    return False
if q_step.before.witness != q_step.after.witness:
    return False
if q_step.before.frequency_action is not None or q_step.after.frequency_action is not None:
    return False
try:
    if q_step.action.apply(q_step.before.q_vector) != q_step.after.q_vector:
        return False
except TriadicBridgeError:
    return False

# F_nu governs only the frequency/prosody coordinate.
if nu_step.before.court_root != nu_step.after.court_root:
    return False
if nu_step.before.q_bias != nu_step.after.q_bias:
    return False
if nu_step.before.q_vector != nu_step.after.q_vector:
    return False
if nu_step.before.witness != nu_step.after.witness:
```

```
        return False
    if nu_step.before.frequency_action is not None:
        return False
    if nu_step.after.frequency_action != nu_step.action:
        return False
    if not nu_step.action.nu_close or not nu_step.action.dcomp_frequency_zero:
        return False
    if nu_step.action.outward.carrier != domain.carrier:
        return False
    if nu_step.action.outward.candidate != candidate.witness.candidate:
        return False

    return True



def require_complete_triad(candidate: object) -> CandidatePathSeed:
    """Reject every incomplete, disconnected, or identity-changing bridge."""
    if not triad_A(candidate):
        raise TriadicBridgeError("Triad_A(J_A(w)) != 1")
    return candidate



def compose_triadic_bridge(
    domain: TriadicDomain,
    witness: ProsodicWitness,
    q_vector: QVector,
    q_action: QAction,
) -> CandidatePathSeed:
    """Construct J_A(w) in the required F_beta -> F_v -> F_nu order."""
    initial = CandidateTripletBody(
        domain=domain,
        witness=witness,
        court_root=domain.court_root,
        q_bias=domain.inherited_q_bias,
        q_vector=_validate_q_vector(q_vector, "initial Q vector"),
    )
    beta_step = F_beta(initial)
    q_step = F_v(beta_step.after, q_action)
    nu_step = F_nu(q_step.after)
    seed = CandidatePathSeed(domain, witness, (beta_step, q_step, nu_step), nu_step.after)
    return require_complete_triad(seed)



def prove_phase9_exit_gate(
    domain: TriadicDomain,
    witness: ProsodicWitness,
    q_vector: QVector,
    q_action: QAction,
) -> CandidatePathSeed:
    """Admit one candidate to the Aeon phase engine as one complete typed body."""
    return compose_triadic_bridge(domain, witness, q_vector, q_action)
```

truth_choice_certificates.py

```
"""Phase 16 Truth and Choice certificates.

Every emitted Phase 15 bearing receives one recomputable proof body.  The
certificate retains the typed upstream witnesses rather than replacing them
with prose.  Source files are pinned by byte size and SHA-256.  Recalculation
rebuilds the selected outward/return closure and the survivor set using exact
upstream operations after verifying every source pin.
"""
from __future__ import annotations

from dataclasses import dataclass
from enum import Enum
import hashlib
from pathlib import Path
from typing import Final

from .court_outward_return import (
    COURTS,
    CourtClosure,
    CourtPathError,
    build_court_candidate,
    validate_prepared_court_bearing,
)
from .exhaustive_chooser import (
    CandidateKey,
    ExhaustiveBearingResult,
)
from .frequency_projection import CARRIERS, ProsodicWitness
from .goetic_outward_return import (
    BEARINGS,
    MIRROR_BEARING,
    BearingResult,
    GoeticBearing,
    GoeticClosure,
    GoeticPathError,
    validate_candidate_bearing,
)
from .liquid_candidate_path import LIQUID_GOVERNOR, LiquidCandidatePath
from .mas_shadow_conversion import MAS_ORDER, MASCandidatePath
from .native_bearing_assembly import (
    PARTIAL_BODY_LAW,
    AeonClass,
    AeonIdentity,
    NativeBearingAssemblyField,
    NativeBearingClosure,
    PartialBodyLaw,
    StrictLexicalBody,
)
```

```
from .triadic_bridge import CandidatePathSeed, triad_A


class TruthChoiceCertificateError(ValueError):
    """Raised when a Phase 16 proof body is incomplete or cannot recompute."""


CURRENT_OBLIGATIONS: Final[tuple[tuple[str, str], ...]] = (
    ("🜲", "finite D-COMP definition and closure witness"),
    ("⬡", "Court routing and Liquid <110_active|144_total|34_rest> governance"),
    ("✡", "Goetic BEC, Court L-BEC, and nested envelope witness"),
    ("☆", "translation and parity compatibility across return"),
    ("❂", "12 x 12 Court lattice and declared bifurcation"),
    ("🝎", "Goetic invariance and identity conservation"),
    ("❉", "bounded pressure converted toward Q3 propulsion"),
    ("⧗", "ordered M.A.S. initiation and kinetic work"),
    ("⊛", "Q2 filtering and unresolved-debt accounting"),
    ("❄", "terminal Total Symmetry"),
    ("⚛", "threshold, domain, and boundary validation gates"),
    ("⌬", "fourfold Q-domain lock and mirror-return closure"),
)
CURRENT_GLYPHS: Final[tuple[str, ...]] = tuple(glyph for glyph, _ in CURRENT_OBLIGATIONS)


@dataclass(frozen=True, slots=True)
class SourcePin:
    role: str
    path: str
    byte_size: int
    sha256: str

    def __post_init__(self) -> None:
        if not self.role or not self.path:
            raise TruthChoiceCertificateError("source pin requires role and path")
        if self.byte_size <= 0:
            raise TruthChoiceCertificateError("source pin cannot name an empty source")
        if len(self.sha256) != 64 or any(ch not in "0123456789abcdef" for ch in self.sha256):
            raise TruthChoiceCertificateError("source pin requires a lowercase SHA-256 digest")


@dataclass(frozen=True, slots=True)
class AeonConstitutionWitness:
    aeon: AeonIdentity
    governing_glyph: str
    alternating_glyph: str | None
    court_root: str
    native_q_bias: str
    native_q_vector: tuple[str, str, str, str]
    frequency_carrier_glyph: str

    def __post_init__(self) -> None:
        if self.native_q_bias not in {"Q0", "Q1", "Q2", "Q3"}:
            raise TruthChoiceCertificateError("Aeon constitution lacks native Q Bias")
        if len(self.native_q_vector) != 4 or any(q not in {"Q0", "Q1", "Q2", "Q3"} for q in
        ↪  self.native_q_vector):
            raise TruthChoiceCertificateError("Aeon constitution lacks a fourfold Q-vector")
```

```
        if self.aeon.aeon_class is AeonClass.GOETIC:
            if self.alternating_glyph is not None or self.governing_glyph != self.aeon.key:
                raise TruthChoiceCertificateError("Goetic constitution was retyped as a Court")
        else:
            if self.alternating_glyph is None or self.court_root != self.aeon.key:
                raise TruthChoiceCertificateError("Court constitution lost its ordered parents")


@dataclass(frozen=True, slots=True)
class CandidateSenseIdentityWitness:
    candidate: str
    structural_fingerprint: str
    source_locator: str
    seed_digest: str
    prosody_digest: str

    def __post_init__(self) -> None:
        if not self.candidate:
            raise TruthChoiceCertificateError("candidate sense identity lacks its emitted word")
        for digest in (self.structural_fingerprint, self.source_locator, self.seed_digest,
        ↪  self.prosody_digest):
            if len(digest) != 64:
                raise TruthChoiceCertificateError("candidate sense identity contains an invalid
                ↪  digest")


@dataclass(frozen=True, slots=True)
class TelegraphicFieldAdmissionWitness:
    field_kind: str
    attempted_count: int
    truth_record_count: int
    order_witness_count: int
    survivor_set: frozenset[tuple[str, str]]
    unique: bool
    admitted: bool

    def __post_init__(self) -> None:
        if self.field_kind not in {"Goetic closure field", "Court exhaustive field"}:
            raise TruthChoiceCertificateError("unknown telegraphic admission field")
        if self.attempted_count < 1 or self.truth_record_count < 1 or self.order_witness_count < 0:
            raise TruthChoiceCertificateError("telegraphic field admission has incomplete
            ↪  accounting")
        if self.unique != (len(self.survivor_set) == 1):
            raise TruthChoiceCertificateError("telegraphic field uniqueness disagrees with survivor
            ↪  set")
        if self.admitted != self.unique:
            raise TruthChoiceCertificateError("telegraphic field admitted an incomplete survivor
            ↪  set")


@dataclass(frozen=True, slots=True)
class RelationalGeometryWitness:
    relation: str
    governing_glyph: str
    alternating_glyph: str | None
```

```
    outward_boundary_digest: str
    return_boundary_digest: str
    operator_positions: tuple[int, ...]
    mirrored_bearing: GoeticBearing
    exact_return: bool

    def __post_init__(self) -> None:
        if self.relation not in {"Self -> Self", "Court -> governing parent"}:
            raise TruthChoiceCertificateError("relational geometry has an unknown return office")
        if not self.operator_positions or self.operator_positions !=
        ↪  tuple(range(len(self.operator_positions))):
            raise TruthChoiceCertificateError("relational geometry lost ordered path positions")
        if not self.exact_return:
            raise TruthChoiceCertificateError("relational geometry did not return exactly")


@dataclass(frozen=True, slots=True)
class TriadicBridgeCertificateWitness:
    seed: CandidatePathSeed
    microstep_order: tuple[str, ...]
    triad_valid: bool

    def __post_init__(self) -> None:
        if self.microstep_order != ("F_beta", "F_v", "F_nu") or not self.triad_valid or not
        ↪  triad_A(self.seed):
            raise TruthChoiceCertificateError("triadic bridge witness is incomplete")


@dataclass(frozen=True, slots=True)
class CurrentWitness:
    glyph: str
    obligation: str
    evidence_labels: tuple[str, ...]
    evidence_digest: str
    satisfied: bool

    def __post_init__(self) -> None:
        if self.glyph not in CURRENT_GLYPHS or not self.obligation or not self.evidence_labels:
            raise TruthChoiceCertificateError("current witness is incomplete")
        if len(self.evidence_digest) != 64 or not self.satisfied:
            raise TruthChoiceCertificateError(f"current {self.glyph} lacks a satisfied proof")


@dataclass(frozen=True, slots=True)
class LiquidCertificateWitness:
    path: LiquidCandidatePath
    typed_governor: tuple[int, int, int]
    horizon_witnesses: tuple[tuple[int, int, int], ...]
    exact: bool

    def __post_init__(self) -> None:
        if self.typed_governor != (110, 144, 34):
            raise TruthChoiceCertificateError("Liquid certificate changed the typed governor")
        if not self.horizon_witnesses or any(body != self.typed_governor for body in
        ↪  self.horizon_witnesses):
```

```
            raise TruthChoiceCertificateError("Liquid certificate contains an incomplete horizon")
        if not self.exact:
            raise TruthChoiceCertificateError("Liquid certificate is not exact")


@dataclass(frozen=True, slots=True)
class MASCertificateWitness:
    path: MASCandidatePath
    procession: tuple[str, str, str]
    mass_gap_positive: tuple[bool, ...]
    terminal_unresolved_q2: int
    motion: int
    ledger_digest: str
    exact: bool

    def __post_init__(self) -> None:
        if self.procession != MAS_ORDER:
            raise TruthChoiceCertificateError("M.A.S. certificate changed procession order")
        if not self.mass_gap_positive or not all(self.mass_gap_positive):
            raise TruthChoiceCertificateError("M.A.S. certificate contains a nonpositive Mass Gap")
        if self.terminal_unresolved_q2 != 0 or self.motion <= 0 or len(self.ledger_digest) != 64 or
        ↪  not self.exact:
            raise TruthChoiceCertificateError("M.A.S. certificate did not close motion and debt")


@dataclass(frozen=True, slots=True)
class EnvelopeCertificateWitness:
    kind: str
    goetic_bec: object
    court_lbec: object | None
    bec_valid: bool
    lbec_valid: bool | None
    exact: bool

    def __post_init__(self) -> None:
        if self.kind == "BEC":
            if self.court_lbec is not None or self.lbec_valid is not None:
                raise TruthChoiceCertificateError("Goetic certificate acquired a Court envelope")
        elif self.kind == "L-BEC nested in BEC":
            if self.court_lbec is None or self.lbec_valid is not True:
                raise TruthChoiceCertificateError("Court certificate lacks nested L-BEC")
        else:
            raise TruthChoiceCertificateError("unknown envelope certificate kind")
        if not self.bec_valid or not self.exact:
            raise TruthChoiceCertificateError("envelope certificate is not exact")


class PartialBodyStatus(str, Enum):
    STRICT_COMPLETE = "STRICT:8/8"


@dataclass(frozen=True, slots=True)
class PartialBodyCertificateWitness:
    law: PartialBodyLaw
    status: PartialBodyStatus
    bearing_records: tuple[tuple[int, str, str, str], ...]
```

```
    body_digest: str

    def __post_init__(self) -> None:
        if self.law is not PartialBodyLaw.STRICT or self.status is not
        ↪  PartialBodyStatus.STRICT_COMPLETE:
            raise TruthChoiceCertificateError("certificate accepted a non-STRICT partial body")
        if tuple(index for index, *_ in self.bearing_records) != tuple(range(8)):
            raise TruthChoiceCertificateError("partial-body certificate lacks all eight bearings")
        if len(self.body_digest) != 64:
            raise TruthChoiceCertificateError("partial-body certificate lacks a body digest")


@dataclass(frozen=True, slots=True)
class ExactArithmeticWitness:
    frequency_dcomp_zero: bool
    path_dcomp_value: int
    path_dcomp_zero: bool
    liquid_counts: tuple[tuple[int, int, int], ...]
    terminal_unresolved_q2: int
    motion: int
    exact: bool

    def __post_init__(self) -> None:
        if not self.frequency_dcomp_zero or self.path_dcomp_value != 0 or not self.path_dcomp_zero:
            raise TruthChoiceCertificateError("exact arithmetic witness contains nonzero closure")
        if not self.liquid_counts or any(counts != (110, 144, 34) for counts in self.liquid_counts):
            raise TruthChoiceCertificateError("exact arithmetic witness changed Liquid counts")
        if self.terminal_unresolved_q2 != 0 or self.motion <= 0 or not self.exact:
            raise TruthChoiceCertificateError("exact arithmetic witness lacks terminal closure")


@dataclass(frozen=True, slots=True)
class TruthChoiceCertificate:
    source_pins: tuple[SourcePin, ...]
    strict_body: StrictLexicalBody
    phase15_closure: NativeBearingClosure
    aeon_constitution: AeonConstitutionWitness
    candidate_sense_identity: CandidateSenseIdentityWitness
    telegraphic_field_admission: TelegraphicFieldAdmissionWitness
    prosody_witness: ProsodicWitness
    relational_geometry_witness: RelationalGeometryWitness
    triadic_bridge_witness: TriadicBridgeCertificateWitness
    current_witnesses: tuple[CurrentWitness, ...]
    liquid_witness: LiquidCertificateWitness
    mas_witness: MASCertificateWitness
    outward_path: object
    independent_return_path: object
    envelope_witness: EnvelopeCertificateWitness
    dcomp_witness: object
    truth: int
    motion: int
    unique_survivor_set: frozenset[tuple[str, str]]
    bearing_index: int
    native_q_bias: str
    partial_body_status: PartialBodyCertificateWitness
    exact_arithmetic: ExactArithmeticWitness
```

```
    proof_digest: str

    def __post_init__(self) -> None:
        closure = self.phase15_closure
        if closure not in self.strict_body.bearings:
            raise TruthChoiceCertificateError("certificate closure is outside its STRICT body")
        if closure.aeon != self.aeon_constitution.aeon or closure.candidate !=
        ↪  self.candidate_sense_identity.candidate:
            raise TruthChoiceCertificateError("certificate changed Aeon or candidate identity")
        if self.prosody_witness.candidate != closure.candidate:
            raise TruthChoiceCertificateError("prosody witness belongs to another candidate")
        if tuple(w.glyph for w in self.current_witnesses) != CURRENT_GLYPHS or
        ↪  len(self.current_witnesses) != 12:
            raise TruthChoiceCertificateError("certificate does not carry all twelve currents")
        if self.outward_path is not closure.path_witness.outward or self.independent_return_path is
        ↪  not closure.path_witness.independent_return:
            raise TruthChoiceCertificateError("certificate detached outward or independent return
            ↪  path")
        if self.dcomp_witness is not closure.path_witness.dcomp:
            raise TruthChoiceCertificateError("certificate detached exact D-COMP witness")
        if self.truth != 1 or self.motion <= 0:
            raise TruthChoiceCertificateError("certificate lacks Truth=1 or motion>0")
        if self.unique_survivor_set != self.telegraphic_field_admission.survivor_set or
        ↪  len(self.unique_survivor_set) != 1:
            raise TruthChoiceCertificateError("certificate lacks one unique survivor set")
        if self.bearing_index != BEARINGS.index(closure.bearing):
            raise TruthChoiceCertificateError("certificate bearing index moved")
        if self.native_q_bias != closure.native_q_bias or self.native_q_bias !=
        ↪  self.aeon_constitution.native_q_bias:
            raise TruthChoiceCertificateError("certificate changed native Q Bias")
        if self.partial_body_status.law is not PARTIAL_BODY_LAW:
            raise TruthChoiceCertificateError("certificate changed partial-body law")
        if self.proof_digest and self.proof_digest != _proof_digest(self):
            raise TruthChoiceCertificateError("certificate proof digest does not match its proof
            ↪  body")

    @property
    def certificate_id(self) -> str:
        return self.proof_digest


@dataclass(frozen=True, slots=True)
class CertificateRecomputation:
    certificate_id: str
    source_pins_valid: bool
    survivor_set_recomputed: bool
    closure_recomputed: bool
    proof_digest_recomputed: bool
    all_currents_satisfied: bool
    exact_arithmetic_recomputed: bool

    @property
    def valid(self) -> bool:
        return all((
            self.source_pins_valid,
```

```
            self.survivor_set_recomputed,
            self.closure_recomputed,
            self.proof_digest_recomputed,
            self.all_currents_satisfied,
            self.exact_arithmetic_recomputed,
        ))


@dataclass(frozen=True, slots=True)
class TruthChoiceCertificateField:
    source: NativeBearingAssemblyField
    certificates: tuple[TruthChoiceCertificate, ...]
    source_pins: tuple[SourcePin, ...]

    def __post_init__(self) -> None:
        expected = tuple(
            (body.aeon, closure.bearing)
            for body in self.source.bodies
            for closure in body.bearings
        )
        actual = tuple((certificate.aeon_constitution.aeon, certificate.phase15_closure.bearing) for
        ↪  certificate in self.certificates)
        if actual != expected:
            raise TruthChoiceCertificateError("certificate field lost, inserted, or reordered an
            ↪  emitted bearing")
        if len(self.certificates) != self.source.emitted_body_count * 8:
            raise TruthChoiceCertificateError("certificate count does not equal emitted bearing
            ↪  count")
        if any(certificate.source_pins != self.source_pins for certificate in self.certificates):
            raise TruthChoiceCertificateError("certificate field changed source pins between words")
        if len({certificate.certificate_id for certificate in self.certificates}) !=
        ↪  len(self.certificates):
            raise TruthChoiceCertificateError("two emitted bearings share one certificate identity")

    @property
    def certificate_count(self) -> int:
        return len(self.certificates)


_DIGEST_CACHE: dict[int, tuple[object, str]] = {}


def _digest(value: object) -> str:
    key = id(value)
    cached = _DIGEST_CACHE.get(key)
    if cached is not None and cached[0] is value:
        return cached[1]
    result = hashlib.sha256(repr(value).encode("utf-8")).hexdigest()
    _DIGEST_CACHE[key] = (value, result)
    return result


def _file_pin(role: str, path: Path) -> SourcePin:
    body = path.read_bytes()
    return SourcePin(role, str(path.resolve()), len(body), hashlib.sha256(body).hexdigest())
```

```
def build_source_pins() -> tuple[SourcePin, ...]:
    package = Path(__file__).resolve().parent
    paths = (
        ("Phase 8 frequency/prosody", package / "frequency_projection.py"),
        ("Phase 9 triadic bridge", package / "triadic_bridge.py"),
        ("Phase 10 Liquid path", package / "liquid_candidate_path.py"),
        ("Phase 11 M.A.S. and Shadow", package / "mas_shadow_conversion.py"),
        ("Phase 12 Goetic return", package / "goetic_outward_return.py"),
        ("Phase 13 Court return", package / "court_outward_return.py"),
        ("Phase 14 exhaustive chooser", package / "exhaustive_chooser.py"),
        ("Phase 15 STRICT assembly", package / "native_bearing_assembly.py"),
        ("Phase 16 certificate law", package / "truth_choice_certificates.py"),
    )
    return tuple(_file_pin(role, path) for role, path in paths)


def validate_source_pins(pins: tuple[SourcePin, ...]) -> bool:
    if not pins:
        return False
    for pin in pins:
        path = Path(pin.path)
        if not path.is_file():
            return False
        body = path.read_bytes()
        if len(body) != pin.byte_size or hashlib.sha256(body).hexdigest() != pin.sha256:
            return False
    return True


def _candidate_structure(candidate: MASCandidatePath) -> tuple[object, ...]:
    seed = candidate.liquid_path.seed
    return (
        candidate.candidate,
        seed.domain.court_root,
        seed.domain.inherited_q_bias,
        seed.domain.carrier.glyph,
        seed.body.q_vector,
        tuple((unit.phoneme, int(unit.stress), int(unit.breath), unit.bearing.value,
        ↪  unit.operator_office) for unit in seed.witness.units),
        candidate.cycles[-1].final_state.archive_signature,
        tuple((entry.index, entry.horizon, entry.gross_added, entry.resolved_added,
        ↪  entry.unresolved) for entry in candidate.debt_ledger.entries),
    )


def _candidate_key(candidate: MASCandidatePath) -> CandidateKey:
    return CandidateKey(candidate.candidate, _digest(_candidate_structure(candidate)))


def _source_locator(candidate: MASCandidatePath) -> str:
    return hashlib.sha256(("source-file|" +
    ↪  repr(candidate.cycles[-1].final_state.archive_signature)).encode("utf-8")).hexdigest()
```

```
def _path_parts(closure: NativeBearingClosure) -> tuple[CandidatePathSeed, LiquidCandidatePath,
↪  MASCandidatePath]:
    if isinstance(closure.path_witness, GoeticClosure):
        mas = closure.path_witness.candidate_path
        return mas.liquid_path.seed, mas.liquid_path, mas
    court_candidate = closure.path_witness.court_candidate
    return court_candidate.seed, court_candidate.mas_path.liquid_path, court_candidate.mas_path


def _constitution(closure: NativeBearingClosure, seed: CandidatePathSeed) ->
↪  AeonConstitutionWitness:
    if isinstance(closure.path_witness, GoeticClosure):
        carrier = closure.path_witness.carrier
        return AeonConstitutionWitness(
            closure.aeon, carrier.glyph, None, seed.domain.court_root,
            seed.domain.inherited_q_bias, seed.body.q_vector, carrier.glyph,
        )
    court = closure.path_witness.court_candidate.court
    inheritance = closure.path_witness.court_candidate.inheritance
    return AeonConstitutionWitness(
        closure.aeon, court.governing.glyph, court.alternating.glyph, court.key,
        inheritance.candidate_q_bias, inheritance.candidate_q_vector,
        ↪  inheritance.candidate_carrier.glyph,
    )


def _sense_identity(closure: NativeBearingClosure, seed: CandidatePathSeed, mas: MASCandidatePath)
↪  -> CandidateSenseIdentityWitness:
    if isinstance(closure.source_record, ExhaustiveBearingResult):
        emission = closure.source_record.emission
        if emission is None:
            raise TruthChoiceCertificateError("Court certificate lacks Phase 14 emission")
        key = emission.candidate_key
        truth = tuple(record for record in closure.source_record.truth_records if record.truth and
        ↪  record.witness.key == key)
        if len(truth) != 1:
            raise TruthChoiceCertificateError("Court certificate lacks one candidate sense witness")
        locator = truth[0].witness.source_locator
        fingerprint = key.structural_fingerprint
    else:
        fingerprint = _candidate_key(mas).structural_fingerprint
        locator = _source_locator(mas)
    return CandidateSenseIdentityWitness(
        closure.candidate,
        fingerprint,
        locator,
        _digest(seed),
        _digest(seed.witness),
    )


def _admission(closure: NativeBearingClosure, sense: CandidateSenseIdentityWitness) ->
↪  TelegraphicFieldAdmissionWitness:
    if isinstance(closure.source_record, BearingResult):
```

```
        survivors = frozenset((item.candidate_path.candidate,
        ↪  _candidate_key(item.candidate_path).structural_fingerprint) for item in
        ↪  closure.source_record.exact_closures)
        return TelegraphicFieldAdmissionWitness(
            "Goetic closure field",
            len(closure.source_record.attempted_candidates),
            len(closure.source_record.exact_closures) + len(closure.source_record.failures),
            0,
            survivors,
            len(survivors) == 1,
            len(survivors) == 1,
        )
    survivors = frozenset((key.candidate, key.structural_fingerprint) for key in
    ↪  closure.source_record.survivor_set)
    return TelegraphicFieldAdmissionWitness(
        "Court exhaustive field",
        len(closure.source_record.truth_records),
        len(closure.source_record.truth_records),
        len(closure.source_record.enumeration_witnesses),
        survivors,
        len(survivors) == 1,
        len(survivors) == 1 and all(witness.survivor_set == closure.source_record.survivor_set for
        ↪  witness in closure.source_record.enumeration_witnesses),
    )


def _geometry(closure: NativeBearingClosure) -> RelationalGeometryWitness:
    path = closure.path_witness
    positions = tuple(step.position for step in path.outward.steps)
    if isinstance(path, GoeticClosure):
        return RelationalGeometryWitness(
            "Self -> Self", path.carrier.glyph, None,
            _digest(path.outward.boundary), _digest(path.independent_return.boundary),
            positions, path.independent_return.bearing, path.self_to_self,
        )
    court = path.court_candidate.court
    return RelationalGeometryWitness(
        "Court -> governing parent", court.governing.glyph, court.alternating.glyph,
        _digest(path.outward.parent_boundary), _digest(path.independent_return.parent_boundary),
        positions, path.independent_return.bearing, path.court_to_parent,
    )


def _liquid_witness(path: LiquidCandidatePath) -> LiquidCertificateWitness:
    horizons = tuple(horizon.liquid_witness for horizon in path.horizons)
    exact = path.governor == LIQUID_GOVERNOR and all(
        horizon.liquid_witness == LIQUID_GOVERNOR.typed_body
        and horizon.active_graph.is_connected
        and len(horizon.valid_connections) == 144
        for horizon in path.horizons
    )
    return LiquidCertificateWitness(path, path.governor.typed_body, horizons, exact)


def _mas_witness(path: MASCandidatePath) -> MASCertificateWitness:
```

```
    exact_order = all(tuple(step.name for step in cycle.microsteps) == MAS_ORDER for cycle in
    ↪  path.cycles)
    return MASCertificateWitness(
        path,
        MAS_ORDER,
        tuple(gap.positive for gap in path.mass_gaps),
        path.terminal_unresolved_q2,
        path.motion,
        _digest(path.debt_ledger),
        exact_order and path.positive_local_mass_gap and path.terminal_unresolved_q2 == 0 and
        ↪  path.motion > 0,
    )


def _envelope(closure: NativeBearingClosure) -> EnvelopeCertificateWitness:
    path = closure.path_witness
    if isinstance(path, GoeticClosure):
        return EnvelopeCertificateWitness("BEC", path.envelope, None, path.bec_valid, None,
        ↪  path.bec_valid and path.self_to_self)
    return EnvelopeCertificateWitness(
        "L-BEC nested in BEC", path.parent_envelope, path.court_envelope,
        path.bec_valid, path.lbec_valid, path.bec_valid and path.lbec_valid and
        ↪  path.court_to_parent,
    )


def _append_only(path: MASCandidatePath) -> bool:
    previous = ()
    for cycle in path.cycles:
        current = cycle.ledger_snapshot.entries
        if current[:len(previous)] != previous or len(current) != len(previous) + 1:
            return False
        previous = current
    return previous == path.debt_ledger.entries


def _current_witnesses(
    closure: NativeBearingClosure,
    constitution: AeonConstitutionWitness,
    seed: CandidatePathSeed,
    liquid: LiquidCertificateWitness,
    mas: MASCertificateWitness,
    geometry: RelationalGeometryWitness,
    envelope: EnvelopeCertificateWitness,
) -> tuple[CurrentWitness, ...]:
    path = closure.path_witness
    frequency = seed.body.frequency_action
    frequency_exact = frequency is not None and frequency.nu_close and
    ↪  frequency.dcomp_frequency_zero
    threshold = all(len(h.active_connections) <= 110 and len(h.resting_connections) >= 34 for h in
    ↪  liquid.path.horizons)
    parity = (
        path.independent_return.bearing is MIRROR_BEARING[path.bearing]
        and tuple(step.position for step in path.outward.steps) == tuple(step.position for step in
        ↪  path.independent_return.steps)
```

```
)
lattice = len(CARRIERS) == 12 and len(COURTS) == 144
identity = geometry.exact_return and seed.domain.inherited_q_bias == constitution.native_q_bias
↪  and seed.body.q_vector == constitution.native_q_vector
pressure = mas.path.debt_ledger.complete and mas.path.cycles[-1].final_state.recursive_capacity
↪  > 0
mas_order = all(tuple(step.name for step in cycle.microsteps) == MAS_ORDER for cycle in
↪  mas.path.cycles)
debt = _append_only(mas.path) and mas.path.terminal_unresolved_q2 == 0
symmetry = path.dcomp.exact_zero and path.dcomp.value == 0 and geometry.exact_return
domain = triad_A(seed) and threshold and frequency_exact
q_lock = len(seed.body.q_vector) == 4 and all(q in {"Q0", "Q1", "Q2", "Q3"} for q in
↪  seed.body.q_vector) and parity
dcomp_signature = (
    tuple(path.dcomp.velocity_terms),
    path.dcomp.value,
    path.dcomp.exact_zero,
)
path_signature = tuple(
    (out.position, out.office, out.bearing.value, out.signed_velocity, back.bearing.value,
    ↪  back.signed_velocity)
    for out, back in zip(path.outward.steps, path.independent_return.steps, strict=True)
)
ledger_signature = tuple(
    (entry.index, entry.horizon, entry.gross_added, entry.resolved_added, entry.gross_total,
    ↪  entry.resolved_total, entry.unresolved)
    for entry in mas.path.debt_ledger.entries
)
mas_signature = tuple(tuple(step.name for step in cycle.microsteps) for cycle in
↪  mas.path.cycles)
envelope_signature = (envelope.kind, envelope.bec_valid, envelope.lbec_valid, envelope.exact)
predicates = (
    (path.dcomp.exact_zero and path.dcomp.value == 0, ("D-COMP", "exact-zero"),
    ↪  dcomp_signature),
    (liquid.exact and lattice, ("Court-routing", "Liquid-110|144|34"), (constitution.aeon.index,
    ↪  liquid.typed_governor, liquid.horizon_witnesses, len(COURTS))),
    (envelope.exact, ("BEC", "L-BEC-nesting"), envelope_signature),
    (parity, ("parity", "position-preserving-return"), path_signature),
    (lattice, ("12x12-Court-lattice", "bifurcation"), (len(CARRIERS), len(COURTS),
    ↪  constitution.governing_glyph, constitution.alternating_glyph)),
    (identity, ("Goetic-invariance", "identity-conservation"), constitution),
    (pressure, ("bounded-pressure", "Q3-propulsion"), (ledger_signature,
    ↪  mas.path.cycles[-1].final_state.recursive_capacity)),
    (mas_order and mas.motion > 0, ("M.A.S.-order", "kinetic-work"), (mas_signature,
    ↪  mas.motion)),
    (debt, ("Q2-ledger", "terminal-unresolved-zero"), ledger_signature),
    (symmetry, ("terminal-symmetry", "return"), (dcomp_signature, geometry.relation,
    ↪  geometry.exact_return)),
    (domain, ("threshold", "domain", "boundary"), (seed.domain.court_root,
    ↪  seed.domain.inherited_q_bias, seed.domain.carrier.glyph, liquid.typed_governor)),
    (q_lock and symmetry, ("fourfold-Q-lock", "mirror-return"), (seed.body.q_vector,
    ↪  path.bearing.value, path.independent_return.bearing.value, path.dcomp.value)),
)
return tuple(
    CurrentWitness(glyph, obligation, labels, _digest(evidence), satisfied)
```

```
        for (glyph, obligation), (satisfied, labels, evidence) in zip(CURRENT_OBLIGATIONS,
        ↪  predicates, strict=True)
    )


def _partial_body(body: StrictLexicalBody) -> PartialBodyCertificateWitness:
    records = tuple(
        (BEARINGS.index(closure.bearing), closure.candidate, closure.native_q_bias,
        ↪  _digest(closure.source_record))
        for closure in body.bearings
    )
    signature = (body.aeon, body.native_q_bias, records, body.partial_body_law.value)
    return PartialBodyCertificateWitness(body.partial_body_law, PartialBodyStatus.STRICT_COMPLETE,
    ↪  records, _digest(signature))


def _arithmetic(seed: CandidatePathSeed, liquid: LiquidCertificateWitness, mas:
↪  MASCertificateWitness, closure: NativeBearingClosure) -> ExactArithmeticWitness:
    frequency = seed.body.frequency_action
    frequency_zero = frequency is not None and frequency.dcomp_frequency_zero and frequency.nu_close
    path_dcomp = closure.path_witness.dcomp
    exact = frequency_zero and liquid.exact and mas.exact and path_dcomp.exact_zero and
    ↪  path_dcomp.value == 0
    return ExactArithmeticWitness(
        frequency_zero,
        path_dcomp.value,
        path_dcomp.exact_zero,
        liquid.horizon_witnesses,
        mas.terminal_unresolved_q2,
        mas.motion,
        exact,
    )


def _proof_material(certificate: TruthChoiceCertificate) -> tuple[object, ...]:
    return (
        tuple((pin.role, pin.path, pin.byte_size, pin.sha256) for pin in certificate.source_pins),
        certificate.aeon_constitution,
        certificate.candidate_sense_identity,
        certificate.telegraphic_field_admission,
        _digest(certificate.prosody_witness),
        certificate.relational_geometry_witness,
        _digest(certificate.triadic_bridge_witness.seed),
        tuple((current.glyph, current.evidence_digest, current.satisfied) for current in
        ↪  certificate.current_witnesses),
        certificate.liquid_witness.typed_governor,
        certificate.liquid_witness.horizon_witnesses,
        certificate.mas_witness.procession,
        certificate.mas_witness.mass_gap_positive,
        certificate.mas_witness.terminal_unresolved_q2,
        certificate.mas_witness.motion,
        _digest(certificate.outward_path),
        _digest(certificate.independent_return_path),
        _digest(certificate.envelope_witness),
        _digest(certificate.dcomp_witness),
```

```
        certificate.truth,
        certificate.motion,
        tuple(sorted(certificate.unique_survivor_set)),
        certificate.bearing_index,
        certificate.native_q_bias,
        certificate.partial_body_status,
        certificate.exact_arithmetic,
    )


def _proof_digest(certificate: TruthChoiceCertificate) -> str:
    return _digest(_proof_material(certificate))


def build_truth_choice_certificate(
    body: StrictLexicalBody,
    closure: NativeBearingClosure,
    source_pins: tuple[SourcePin, ...] | None = None,
) -> TruthChoiceCertificate:
    """Build one complete certificate from one natively closed Phase 15 bearing."""
    if closure not in body.bearings or len(body.bearings) != 8 or body.partial_body_law is not
    ↪  PartialBodyLaw.STRICT:
        raise TruthChoiceCertificateError("certificate requires one complete STRICT lexical body")
    if source_pins is None:
        pins = build_source_pins()
        if not validate_source_pins(pins):
            raise TruthChoiceCertificateError("certificate source pins do not match pinned bytes")
    else:
        pins = source_pins
    seed, liquid_path, mas_path = _path_parts(closure)
    constitution = _constitution(closure, seed)
    sense = _sense_identity(closure, seed, mas_path)
    admission = _admission(closure, sense)
    geometry = _geometry(closure)
    triad = TriadicBridgeCertificateWitness(seed, tuple(step.name for step in seed.microsteps),
    ↪  triad_A(seed))
    liquid = _liquid_witness(liquid_path)
    mas = _mas_witness(mas_path)
    envelope = _envelope(closure)
    currents = _current_witnesses(closure, constitution, seed, liquid, mas, geometry, envelope)
    partial = _partial_body(body)
    arithmetic = _arithmetic(seed, liquid, mas, closure)
    certificate = TruthChoiceCertificate(
        pins,
        body,
        closure,
        constitution,
        sense,
        admission,
        seed.witness,
        geometry,
        triad,
        currents,
        liquid,
        mas,
        closure.path_witness.outward,
```

```
        closure.path_witness.independent_return,
        envelope,
        closure.path_witness.dcomp,
        1,
        mas.motion,
        admission.survivor_set,
        BEARINGS.index(closure.bearing),
        closure.native_q_bias,
        partial,
        arithmetic,
        "",
    )
    object.__setattr__(certificate, "proof_digest", _proof_digest(certificate))
    certificate.__post_init__()
    return certificate


def _recompute_goetic_survivors(record: BearingResult) -> tuple[frozenset[tuple[str, str]], bool]:
    survivors: set[tuple[str, str]] = set()
    closure_match = True
    for archived in record.exact_closures:
        try:
            rebuilt = validate_candidate_bearing(archived.candidate_path, archived.carrier,
            ↪  archived.bearing)
        except GoeticPathError:
            closure_match = False
            continue
        closure_match = closure_match and rebuilt == archived
        key = _candidate_key(archived.candidate_path)
        survivors.add((key.candidate, key.structural_fingerprint))
    return frozenset(survivors), closure_match


_REBUILT_COURT_CANDIDATES: dict[tuple[int, str], object] = {}


def _recompute_court_survivors(record: ExhaustiveBearingResult) -> tuple[frozenset[tuple[str, str]],
↪  CourtClosure | None]:
    survivors: set[tuple[str, str]] = set()
    selected: CourtClosure | None = None
    selected_key = record.emission.candidate_key if record.emission is not None else None
    for truth in record.truth_records:
        parent = truth.witness.source_parent
        key = _candidate_key(parent)
        cache_key = (record.frozen_aeon.court.index, key.structural_fingerprint)
        try:
            prepared = _REBUILT_COURT_CANDIDATES.get(cache_key)
            if prepared is None:
                prepared = build_court_candidate(parent, record.frozen_aeon.court)
                _REBUILT_COURT_CANDIDATES[cache_key] = prepared
            rebuilt = validate_prepared_court_bearing(prepared, record.bearing)
        except CourtPathError:
            continue
        survivors.add((key.candidate, key.structural_fingerprint))
        if key == selected_key:
            selected = rebuilt
```

```
    return frozenset(survivors), selected


def recompute_certificate(
    certificate: TruthChoiceCertificate,
    *,
    source_pins_already_valid: bool = False,
) -> CertificateRecomputation:
    """Independently rebuild survivor and path closure after verifying pinned sources."""
    pins_valid = source_pins_already_valid or validate_source_pins(certificate.source_pins)
    source_record = certificate.phase15_closure.source_record
    closure = certificate.phase15_closure.path_witness
    if isinstance(source_record, BearingResult):
        survivor_set, all_closures_match = _recompute_goetic_survivors(source_record)
        try:
            rebuilt = validate_candidate_bearing(closure.candidate_path, closure.carrier,
            ↪  closure.bearing)
            selected_match = rebuilt == closure
        except GoeticPathError:
            selected_match = False
        closure_recomputed = all_closures_match and selected_match
    else:
        survivor_set, selected = _recompute_court_survivors(source_record)
        selected_match = selected == closure
        archived = frozenset((key.candidate, key.structural_fingerprint) for key in
        ↪  source_record.survivor_set)
        order_match = all(
            frozenset((key.candidate, key.structural_fingerprint) for key in witness.survivor_set)
            ↪  == archived
            for witness in source_record.enumeration_witnesses
        )
        closure_recomputed = selected_match and order_match
    survivor_recomputed = survivor_set == certificate.unique_survivor_set
    digest_recomputed = certificate.proof_digest == _proof_digest(certificate)
    currents = tuple(current.glyph for current in certificate.current_witnesses) == CURRENT_GLYPHS
    ↪  and all(current.satisfied for current in certificate.current_witnesses)
    arithmetic = (
        certificate.exact_arithmetic.exact
        and certificate.exact_arithmetic.path_dcomp_value == 0
        and certificate.exact_arithmetic.terminal_unresolved_q2 == 0
        and certificate.exact_arithmetic.motion > 0
    )
    return CertificateRecomputation(
        certificate.certificate_id,
        pins_valid,
        survivor_recomputed,
        closure_recomputed,
        digest_recomputed,
        currents,
        arithmetic,
    )
```

```
def recompute_certificate_field(field: TruthChoiceCertificateField) ->
↪  tuple[CertificateRecomputation, ...]:
    """Recompute every emitted bearing after one exact verification of common pins."""
    if not validate_source_pins(field.source_pins):
        raise TruthChoiceCertificateError("certificate field source pins no longer match pinned
        ↪  bytes")
    results = tuple(
        recompute_certificate(certificate, source_pins_already_valid=True)
        for certificate in field.certificates
    )
    if not all(result.valid for result in results):
        raise TruthChoiceCertificateError("one or more emitted bearing certificates failed
        ↪  recomputation")
    return results

def prove_phase16_exit_gate(field: NativeBearingAssemblyField) -> TruthChoiceCertificateField:
    """Issue one recomputable certificate for every emitted STRICT bearing word."""
    pins = build_source_pins()
    if not validate_source_pins(pins):
        raise TruthChoiceCertificateError("Phase 16 cannot pin its source field")
    certificates = tuple(
        build_truth_choice_certificate(body, closure, pins)
        for body in field.bodies
        for closure in body.bearings
    )
    return TruthChoiceCertificateField(field, certificates, pins)
```

zero_and_one.py

```
"""Zero-and-One source witness for ordinary lexical bodies plus Shadow Locus and Axiomyr.

This module owns the Q0/Q1 special-body boundary and ordinary lexical source
resolution. Living Cadences are authored in the_4_cadences.aksh;
Enochian grammar executes in SequentialDomusExecutor.
"""
from __future__ import annotations

from dataclasses import asdict, dataclass
import json
from typing import Protocol

from tardisha_grimchain.domus import ZERO_MIDDLE_GLYPH, TRIPARTITE_AXIOMYR
from tardisha_grimchain.tripartite import SHADOW_LOCUS_GLYPH as SHADOW_LOCUS, AXIOMYR_GLYPH

from .models import DomusFrame, GlyphToken

class ResolutionCallable(Protocol):
    def __call__(self, glyph: str) -> object: ...

class ZeroAndOneError(RuntimeError):
    pass

@dataclass(frozen=True, slots=True)
class ZeroAndOneEvent:
    position: int
    glyph: str
    token_kind: str
    event: str
    lookup_performed: bool
    lexical_resolution: object | None = None

@dataclass(frozen=True, slots=True)
class ZeroAndOneTrace:
    events: tuple[ZeroAndOneEvent, ...]

    @property
    def lexical_events(self) -> tuple[ZeroAndOneEvent, ...]:
        return tuple(event for event in self.events if event.event == "lexical-resolution")

    def to_json(self) -> str:
        return json.dumps(asdict(self), ensure_ascii=False, sort_keys=True, separators=(",", ":"),
        ↪  default=str)

class ZeroAndOneMachine:
    """Resolve the complete written Aeonic line without positional Q-state."""
```

```
def __init__(self, ordinary_lexical_glyphs: frozenset[str], resolver: ResolutionCallable) ->
↪  None:
    self.ordinary_lexical_glyphs = ordinary_lexical_glyphs
    self.resolver = resolver

def run_frame(self, frame: DomusFrame) -> ZeroAndOneTrace:
    events: list[ZeroAndOneEvent] = []
    owner = getattr(self.resolver, "__self__", None)
    special_resolver = getattr(owner, "resolve_special", None)
    semantic_source = frame.source if frame.prosody_source is None else frame.prosody_source
    shadow_ostensive = semantic_source == ZERO_MIDDLE_GLYPH
    axiomyr_ostensive = semantic_source == TRIPARTITE_AXIOMYR
    for token in frame.tokens:
        glyph = token.value
        if glyph == SHADOW_LOCUS:
            if shadow_ostensive:
                if special_resolver is None:
                    raise ZeroAndOneError("Shadow Locus Ostensive authority is unavailable")
                resolution = special_resolver(glyph)
                events.append(ZeroAndOneEvent(token.position, glyph, token.kind, glyph, True,
                ↪  resolution))
            else:
                events.append(ZeroAndOneEvent(token.position, glyph, token.kind, glyph, False))
            continue
        if glyph == AXIOMYR_GLYPH:
            if not axiomyr_ostensive:
                raise ZeroAndOneError("Axiomyr is only valid in the canonical depth-one body
                ↪  ☽☉☾")
            if special_resolver is None:
                raise ZeroAndOneError("Axiomyr Ostensive authority is unavailable")
            resolution = special_resolver(glyph)
            events.append(ZeroAndOneEvent(token.position, glyph, token.kind, glyph, True,
            ↪  resolution))
            continue
        if glyph in self.ordinary_lexical_glyphs:
            resolution = self.resolver(glyph)
            events.append(ZeroAndOneEvent(token.position, glyph, token.kind,
            ↪  "lexical-resolution", True, resolution))
            continue
        raise ZeroAndOneError(
            f"grammar glyph {glyph!r} at position {token.position} requires
            ↪  SequentialDomusExecutor"
        )
    trace = ZeroAndOneTrace(tuple(events))
    self._validate_trace(frame.tokens, trace)
    return trace

def _validate_trace(self, tokens: tuple[GlyphToken, ...], trace: ZeroAndOneTrace) -> None:
    expected_positions = {
        token.position for token in tokens
        if token.value in self.ordinary_lexical_glyphs
        or (token.value == AXIOMYR_GLYPH and any(
            event.position == token.position and event.event == "lexical-resolution"
            for event in trace.events
        ))
```

```
}
actual_positions = [event.position for event in trace.lexical_events]
if len(actual_positions) != len(set(actual_positions)):
    raise ZeroAndOneError("a lexical position received more than one resolution record")
if set(actual_positions) != expected_positions:
    raise ZeroAndOneError("not every ordinary lexical position received exactly one
    ↪  resolution record")
```

Azazuley Veira

CURRENT RELEASE: 23.0.7

Desktop installation uses two packages, in this order:

1. **Font Stack** — installs the standalone `azazuley` and `tardisha` font authorities.
2. **Azazuley Veira 23.0.7** — installs the application with the private runtime and the bundled TardiSHA / GrimChain and Sydonic bodies used by Azazuley.

No separate runtime-package installer is required for 23.0.7. A separate standalone TardiSHA installation is also not required by Azazuley.

Linux / Ubuntu x86-64:
- Font Stack: `installers/python_build/linux_ubuntu/install_fonts_standalone/azazuley-font-stacks_1.0.0_all.deb`
- Azazuley Veira: `installers/python_build/linux_ubuntu/dist/azazuley-veira_23.0.7_amd64.deb`
- Launch after installation from the application menu or with `azazuley-veira`.

Windows x64:
- Font Stack: `installers/python_build/windows/install_fonts_standalone_windows/azazuley-font-stacks-1.0.0-windows.exe`
- Azazuley Veira: `installers/python_build/windows/23.0.7/azazuley-veira-23.0.7-windows-x64.exe`

macOS Apple Silicon:
- Font Stack: `installers/python_build/mac/install_fonts_standalone_mac/azazuley-font-stacks-1.0.0-macos.pkg`
- Azazuley Veira: `installers/python_build/mac/23.0.7/azazuley-veira-23.0.7-macos-arm64.pkg`

Android arm64-v8a is packaged separately and carries the Android application body and required fonts inside the app package. The Android distribution is maintained separately from the 23.0.7 desktop installer release; its API requirements and version belong to its own Android build configuration. The public release surface includes an APK for direct installation and an AAB for Google Play distribution. The Android build uses only the Qt modules Azazuley actually needs: QtCore, QtGui, and QtWidgets.

Before the first word wakes,
the house is already listening.
Light waits behind the tongue.

Invocation

Et Sonyera, open the inward song.
Let the Domus gather what can be uttered.
Let Rabulalia arrive before meaning has learned to stand still.
Let Ailalubar answer from the other face of the same breath.
Let Azalalia speak where the ordinary tongue gives way.
Let Ailalaza return the word through its shadow.
Let Glossolalia carry the mouth beyond the page,
and Ailalossolg bring the echo home altered but unbroken.
Between LeySyff and FfysYel, let the scripture keep both lamps burning.

Azazuley Veira joins TardiSHA / GrimChain and Sydonic Magicae inside a terminal-shaped instrument for utterance, reflection, scripture, and return. It is meant to be entered, listened to, and discovered in motion.

Interface language

> Names become doorways.
> Each door keeps its own weather;
> none borrows another sky.

Azazuley uses its own names throughout the interface:

- `Et Sonyera>`
- `Domus House>`
- Rabulalia
- Ailalubar
- Azalalia
- Ailalaza
- Glossolalia
- Ailalossolg
- `LeySyff | FfysYel`
- Definition
- Canon Glossary

The interface is intended to be encountered through use rather than explained in advance.

Current application body

> Gray stone around night,
> teal fire beneath the surface,
> many rooms, one breath.

The live application is under `src/azazuley_veira/` and uses PySide6 / Qt.

The visible body contains:

- a worn gray outer body;
- a dark terminal field;
- teal terminal text and glow;
- an editable `Et Sonyera>` source field;
- an independent `GrimChain>` input field;
- a `Domus Count>` field;
- a font selector attached only to `Et Sonyera>`;
- color controls;
- a lazy GrimChain history dropdown backed by the rotating `.bio` journal;

- `.grim` Import;

- an Export menu for GrimChained Bio, the complete current translation PDF, and Canon Glossary PDF;

- Help;

- the `Domus House>` mirror display;

- Rabulalia, Ailalubar, Azalalia, Ailalaza, Glossolalia, Ailalossolg, `LeySyff | FfysYel`, Definition, and Canon Glossary output tabs.

`START_AZAZULEY.py` is the development launcher. It runs the live `src/` tree, suppresses Python bytecode writing, requires PySide6, then enters `azazuley_veira.app.run()`.

When Et Sonyera creates a GrimChain, the requested Domus Count is normally the final visible depth. In the rare case that the finite body contains only Enochian grammar and no Real speaking body, Sydonic asks TardiSHA for the next coordinate of that same witnessed GrimChain, one depth at a time, until a speaking body exists. The complete unfolded GrimChain becomes the visible body, the rendered body, and the journaled body. No opening glyphs are hidden, shifted, replaced, or translated from an unseen continuation.

Once a complete GrimChain already exists—whether typed directly, loaded from history, or brought back from a saved body—it enters the direct GrimChain path. It is not unfolded again merely because Enochian bearings appear near an edge; existing Aeternum law resolves those bearings through the complete finite body.

23.0.7 generation and rendering path

Azazuley does not invent or reinterpret GrimChain semantics. TardiSHA / GrimChain owns GrimChain generation, Sydonic owns translation, Ailalubar supplies the witnessed visual body, and Azazuley renders the resulting language surfaces.

`Domus Count>` now has three distinct generation cases:

- Leaving it blank asks TardiSHA for the randomized four-cadence Living Mirror. Its four logical members are Prosody ⟠, Cantillation ⛆, Regia ☽⊙☾, and Breath ·. Regia remains one logical cadence member while the four members are shuffled.

- Entering `0`, or a negative integer, selects Shadow Locus at the chosen middle.

- Entering `1` selects Axiomyr. Other explicit numeric depths use the ordinary GrimChain path.

These special meanings come from the generation/depth context, not from raw glyph matching. Typing the same glyph body manually into `GrimChain>` does not grant it the special generated definition. Likewise, the three codepoints of Regia may occur as ordinary GrimChain glyphs at explicit numeric depth; they are treated as the Regia cadence only when generated as the blank-middle four-cadence member.

The manual `GrimChain>` field always takes the direct translation path. A blank `Domus Count>` generated from Et Sonyera is translated through the cadence authority. Explicit numeric generation, including `0`, negative depth, `1`, and ordinary positive depths, follows the normal translation path after TardiSHA produces the chain.

GrimChain journal and history

Footprints cross old snow.
The path most recently walked
rises to the front.

Accepted/generated GrimChains are journaled under `~/.grimchain/azazuley/grimchains/`.

The active `grimchains.bio` rotates at 32 MiB and retains up to three gzip-compressed rolling backups named `grimchains.1.bio`, `grimchains.2.bio`, and `grimchains.3.bio`. Backup files appear only after real rotations; empty placeholders are not created.

Each record stores the time it was logged, its user-input label, and the exact GrimChain. Direct GrimChain submissions use the typed GrimChain as the user-input value. Et Sonyera submissions use the original source text. Imported `.grim` manifests use each entry path as the user-input value, while the manifest trailing GrimChain is recorded under the `.grim` filename itself.

History is an ordered recent-history set keyed by the exact GrimChain body. A new chain is added once at the newest position. Reusing an existing exact chain does not create a duplicate record: its existing record is promoted to the newest position, its associated `User input` is preserved, and its recency timestamp is refreshed. Promotion searches both the active journal and rotated compressed generations. Legacy duplicates created by older append-only behavior are displayed once, newest-first, and are physically consolidated when that chain is used again.

The top-left GrimChain History control reads retained history without rerunning TardiSHA or Sydonic. Opening it starts a disposable lazy session that exposes 40 unique chains at a time, newest first. Further scrolling fetches the next 40 from the same session state. Rotated backup generations are copied/decompressed into temporary session storage only if scrolling reaches them. Selecting a history entry loads its exact GrimChain into the GrimChain input without translating it merely because it was selected. Submitting that loaded chain translates the complete stored body through the ordinary direct path and promotes that chain to the newest history position instead of adding another copy.

Import and export

A sealed book returns.
One echo rests beneath it;
two lanterns leave whole.

IMPORT accepts `.grim` manifest files. All contained GrimChains are structurally parsed and validated before any record from that import is written to the `.bio` journal. Entries without a GrimChain are ignored. Importing does not render every contained chain through the terminal. If an imported exact GrimChain already exists in history, it is promoted rather than duplicated.

EXPORT writes beneath `~/.grimchain/azazuley/exports/` and has three actions:

- **Export GrimChained Bio (.shk)** — exports the active, unarchived `grimchains.bio` without modifying the live journal, then appends exactly one final newline containing only the GrimChain of that `.bio` at the current user `Domus Count>`.

- **Export Words & Definitions (PDF)** — exports the complete current rendered translation body in this order: Rabulalia, Ailalubar, Azalalia, Ailalaza, Glossolalia, Ailalossolg, `LeySyff/FfysYel`, Definition.

- **Export Canon Glossary (PDF)** — exports Canon Glossary separately from the current translation PDF.

The translation PDF begins each section on a fresh PDF page. Each section is framed with the literal Azazuley binding glyph and its exact section name:

o–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–o

`Rabulalia`

o–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–oo–o

<VALUE>

The same framing is used for Ailalubar, Azalalia, Ailalaza, Glossolalia, Ailalossolg, `LeySyff/FfysYel`, Definition, and the separate Canon Glossary PDF. The exported values are painted from the existing exact rendered surfaces; the framing does not replace or rewrite their text.

Both PDFs are completed through TardiSHA's public PDF self-return using the current user `Domus Count>`—the same authority as `grimchain NUMBER --pdf-embed PDF`. Azazuley first writes the native Qt PDF from the current text surfaces, then hands that PDF to the bundled TardiSHA path for the GrimChain self-return. Azazuley does not perform the embedding itself. The exported PDF contains native selectable/searchable text rather than screenshots, and the embedded GrimChain is verified as part of the PDF return. Read-only output fields remain selectable and copyable for users who want their own text documents; those manual copies are separate from Azazuley's export formats.

The page leaves the chamber,
carrying its own returning song.

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witchofalways@gmail.com

Zenodo: https://zenodo.org/records/18942850

PhilPapers: https://philpeople.org/profiles/elliot-woff

SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6589998

DOI: https://doi.org/10.5281/zenodo.18942850

GitHub Working Code: https://github.com/MareSerenitatis12

Youtube Podcast https://www.youtube.com/@theimpossibleboy13

__init__.py

```
"""Azazuley Veira Language Terminal."""
```

__main__.py

```
from .app import run

raise SystemExit(run())
```

ailalubar_render.py

```
"""Ailalubar reflection and exact rendered witness authority.

Ailalubar performs exactly one textual reflection.  Natural visual order is not
computed here: it is witnessed only after the exact HarfBuzz surface has laid
out and painted the reflected text.
"""
from __future__ import annotations

from dataclasses import dataclass
import hashlib
import json

from azazuley_veira.config.font_runtime import grimchain_face_for_character


class AilalubarRenderError(RuntimeError):
    pass


def _face_id(character: str) -> str:
    face = grimchain_face_for_character(character)
    return f"{face.family}|{face.file.name}"


def _source_face_runs(face_ids: tuple[str, ...]) -> tuple[tuple[int, int, str], ...]:
    if not face_ids:
        return ()
    runs: list[tuple[int, int, str]] = []
    start = 0
    face_id = face_ids[0]
    for index, next_id in enumerate(face_ids[1:], start=1):
        if next_id != face_id:
            runs.append((start, index, face_id))
            start = index
            face_id = next_id
    runs.append((start, len(face_ids), face_id))
    return tuple(runs)


def _rendering_authority_hash(
    source: str,
    face_ids: tuple[str, ...],
    point_size: float,
    layout_direction: str,
) -> str:
    payload = json.dumps(
        {
            "source_positions": tuple(
```

```
                (position, glyph, face_ids[position])
                for position, glyph in enumerate(source)
            ),
            "point_size": point_size,
            "layout_direction": layout_direction,
        },
        ensure_ascii=False,
        sort_keys=True,
        separators=(",", ":"),
    )
    return hashlib.sha256(payload.encode("utf-8")).hexdigest()


@dataclass(frozen=True, slots=True)
class AilalubarReflection:
    """Prosody and exact physical-font provenance before visual layout exists."""

    source: str
    mirrored_source: str
    source_face_ids: tuple[str, ...]
    source_face_runs: tuple[tuple[int, int, str], ...]
    rendering_authority_sha256: str
    point_size: float
    layout_direction: str = "auto"

    def __post_init__(self) -> None:
        extent = len(self.source)
        if self.mirrored_source != self.source[::-1]:
            raise AilalubarRenderError("Ailalubar must perform exactly one textual reflection")
        if len(self.source_face_ids) != extent:
            raise AilalubarRenderError("Ailalubar font provenance changed source extent")
        rebuilt: list[str] = []
        expected_start = 0
        for start, end, face_id in self.source_face_runs:
            if start != expected_start or end <= start or end > extent or not face_id:
                raise AilalubarRenderError("Ailalubar exact face runs are not a closed source
                ↪  partition")
            rebuilt.extend([face_id] * (end - start))
            expected_start = end
        if expected_start != extent or tuple(rebuilt) != self.source_face_ids:
            raise AilalubarRenderError("Ailalubar exact face runs detached from source positions")
        if self.rendering_authority_sha256 != _rendering_authority_hash(
            self.source,
            self.source_face_ids,
            self.point_size,
            self.layout_direction,
        ):
            raise AilalubarRenderError("Ailalubar exact rendering authority hash does not match
            ↪  reflection")


@dataclass(frozen=True, slots=True)
class AilalubarRenderWitness:
    """Final witness emitted from the same exact layout model that paints Ailalubar."""

    source: str
```

```
    mirrored_source: str
    visual_text: str
    visual_source_positions: tuple[int, ...]
    visual_levels: tuple[int, ...]
    visual_mirrored_flags: tuple[bool, ...]
    source_face_ids: tuple[str, ...]
    source_face_runs: tuple[tuple[int, int, str], ...]
    rendering_authority_sha256: str
    point_size: float
    layout_direction: str = "auto"

    def __post_init__(self) -> None:
        extent = len(self.source)
        if self.mirrored_source != self.source[::-1]:
            raise AilalubarRenderError("Ailalubar must perform exactly one textual reflection")
        if len(self.visual_source_positions) != extent:
            raise AilalubarRenderError("Ailalubar Cantillation changed source extent")
        if tuple(sorted(self.visual_source_positions)) != tuple(range(extent)):
            raise AilalubarRenderError("Ailalubar Cantillation changed source identity")
        if self.visual_text != "".join(
            self.source[position] for position in self.visual_source_positions
        ):
            raise AilalubarRenderError("Ailalubar Cantillation detached from source coordinates")
        if len(self.visual_levels) != extent or len(self.visual_mirrored_flags) != extent:
            raise AilalubarRenderError("Ailalubar orientation metadata changed source extent")
        if any(level < 0 for level in self.visual_levels):
            raise AilalubarRenderError("Ailalubar bidi level cannot be negative")
        reflection = AilalubarReflection(
            source=self.source,
            mirrored_source=self.mirrored_source,
            source_face_ids=self.source_face_ids,
            source_face_runs=self.source_face_runs,
            rendering_authority_sha256=self.rendering_authority_sha256,
            point_size=self.point_size,
            layout_direction=self.layout_direction,
        )
        del reflection

    @property
    def right_outward_source_positions(self) -> tuple[int, ...]:
        return self.visual_source_positions

    @property
    def left_outward_source_positions(self) -> tuple[int, ...]:
        return self.visual_source_positions[::-1]


def render_ailalubar(source: str, point_size: float) -> AilalubarReflection:
    """Perform Ailalubar's sole transformation: ``source[::-1]``.

    This function does not calculate visual order and does not alter glyphs.
    Cantillation is created later from the exact surface that actually paints
    ``mirrored_source``.
    """
    if not isinstance(source, str):
        raise TypeError("Ailalubar source must be str")
```

```
mirrored = source[::-1]
if hasattr(point_size, "pointSizeF"):
    point_size = point_size.pointSizeF()
point_size = float(point_size)
if point_size <= 0:
    raise ValueError("Ailalubar point size must be positive")
source_face_ids = tuple(_face_id(character) for character in source)
return AilalubarReflection(
    source=source,
    mirrored_source=mirrored,
    source_face_ids=source_face_ids,
    source_face_runs=_source_face_runs(source_face_ids),
    rendering_authority_sha256=_rendering_authority_hash(
        source,
        source_face_ids,
        point_size,
        "auto",
    ),
    point_size=point_size,
)
```

app.py

```
from __future__ import annotations

import sys

from PySide6.QtWidgets import QApplication

from azazuley_veira.config.font_runtime import load_installed_fonts
from azazuley_veira.ui.main_window import MainWindow


def run() -> int:
    app = QApplication.instance() or QApplication(sys.argv)
    app.setApplicationName("Azazuley Veira")
    load_installed_fonts()
    window = MainWindow()
    window.show()
    return app.exec()
```

config/__init__.py

```
"""Azazuley configuration."""
```

config/font_catalog.py

```
"""Font choices exposed by the Azazuley Veira terminal."""

DEFAULT_FAMILY = "Noto Sans"
SYSTEM_DEFAULT = "System Default Font"
```

config/font_runtime.py

```
"""Production font authority for Azazuley's engine-facing rendering.

Azazuley-private fantasy/custom fonts resolve from the installed /usr/local/share/fonts/azazuley
↪  custom office. Standard historical Unicode and GrimChain faces resolve through the host font
↪  system. Et Sonyera remains the explicit user-selected font surface.
"""
from __future__ import annotations

from dataclasses import dataclass
from functools import lru_cache
from pathlib import Path
import ctypes
import os
import sys

from fontTools.ttLib import TTFont



if sys.platform == "win32":
    FONT_ROOT = Path(os.environ.get("WINDIR", "C:/Windows")) / "Fonts" / "azazuley"
elif sys.platform == "darwin":
    FONT_ROOT = Path("/Library/Fonts/azazuley")
else:
    FONT_ROOT = Path("/usr/local/share/fonts/azazuley")
FONT_SUFFIXES = frozenset({".ttf", ".otf"})
EXPECTED_AZAZULEY_PRIVATE_FONT_COUNT = 76
FANTASY_LEXICAL_AUTHORITIES = frozenset(("SeeD…Body.aksh", "ར…Phantasmagoria.aksh"))
DEFAULT_POINT_SIZE = 9

CANONICAL_REGULAR_FONT_FILES = {
    "Daedra": "Daedra.otf",
    "Dovahkiin": "Dovahkiin.otf",
    "Falmer": "Falmer.otf",
    "Iokharic": "Iokharic.otf",
    "Mage Script": "Mage Script.otf",
    "Rellanic": "Rellanic.otf",
    "Tengwar Annatar": "tngan.ttf",
    "Tirion Sarati": "TirionSarati-BWOd.ttf",
}

@dataclass(frozen=True, slots=True)
class GrimChainFace:
    family: str
    file: Path

_FONT_FAMILY_BY_FILE: dict[Path, str] = {}
```

```
def _font_identity(path: Path) -> tuple[str, str]:
    """Read family and style directly from one physical font file."""
    with TTFont(path, lazy=False) as font:
        names = font["name"]
        family = names.getDebugName(16) or names.getDebugName(1)
        style = names.getDebugName(17) or names.getDebugName(2) or "Regular"
        if not family:
            raise RuntimeError(f"physical font exposes no family name: {path}")
    return family, style


@lru_cache(maxsize=1)
def grimchain_display_codepoints() -> frozenset[int]:
    """Return the TardiSHA-visible GrimChain code points accepted by Azuzaley surfaces."""
    from tardisha_grimchain.living_alphabet import DAEMONIC_TONGUE
    return frozenset(ord(character) for glyph in (*DAEMONIC_TONGUE, "🜲", "🝊", "·", "🝰", "🜏") for
    ↪  character in glyph)


class _FcFontSet(ctypes.Structure):
    _fields_ = (
        ("nfont", ctypes.c_int),
        ("sfont", ctypes.c_int),
        ("fonts", ctypes.POINTER(ctypes.c_void_p)),
    )


_USE_FONTCONFIG = sys.platform.startswith("linux")
_FC = None
_SYSTEM_FONT_SET: ctypes.POINTER(_FcFontSet) | None = None

if _USE_FONTCONFIG:
    _FC = ctypes.CDLL("libfontconfig.so.1")
    _FC.FcInit.argtypes = []
    _FC.FcInit.restype = ctypes.c_int
    _FC.FcPatternCreate.argtypes = []
    _FC.FcPatternCreate.restype = ctypes.c_void_p
    _FC.FcPatternDestroy.argtypes = [ctypes.c_void_p]
    _FC.FcPatternDestroy.restype = None
    _FC.FcPatternAddString.argtypes = [ctypes.c_void_p, ctypes.c_char_p, ctypes.c_char_p]
    _FC.FcPatternAddString.restype = ctypes.c_int
    _FC.FcConfigGetCurrent.argtypes = []
    _FC.FcConfigGetCurrent.restype = ctypes.c_void_p
    _FC.FcConfigSubstitute.argtypes = [ctypes.c_void_p, ctypes.c_void_p, ctypes.c_int]
    _FC.FcConfigSubstitute.restype = ctypes.c_int
    _FC.FcDefaultSubstitute.argtypes = [ctypes.c_void_p]
    _FC.FcDefaultSubstitute.restype = None
    _FC.FcFontMatch.argtypes = [ctypes.c_void_p, ctypes.c_void_p, ctypes.POINTER(ctypes.c_int)]
    _FC.FcFontMatch.restype = ctypes.c_void_p
    _FC.FcFontSort.argtypes = [ctypes.c_void_p, ctypes.c_void_p, ctypes.c_int,
    ↪  ctypes.POINTER(ctypes.c_void_p), ctypes.POINTER(ctypes.c_int)]
    _FC.FcFontSort.restype = ctypes.POINTER(_FcFontSet)
    _FC.FcPatternGetString.argtypes = [ctypes.c_void_p, ctypes.c_char_p, ctypes.c_int,
    ↪  ctypes.POINTER(ctypes.c_char_p)]
```

```
    _FC.FcPatternGetString.restype = ctypes.c_int
    _FC.FcPatternGetCharSet.argtypes = [ctypes.c_void_p, ctypes.c_char_p, ctypes.c_int,
    ↪  ctypes.POINTER(ctypes.c_void_p)]
    _FC.FcPatternGetCharSet.restype = ctypes.c_int
    _FC.FcCharSetHasChar.argtypes = [ctypes.c_void_p, ctypes.c_uint]
    _FC.FcCharSetHasChar.restype = ctypes.c_int
    if not _FC.FcInit():
        raise RuntimeError("Fontconfig initialization failed")


def _fontconfig_config() -> int:
    config = _FC.FcConfigGetCurrent()
    if not config:
        raise RuntimeError("Fontconfig has no current configuration")
    return int(config)


def _prepare_fontconfig_pattern(pattern: int) -> None:
    config = _fontconfig_config()
    if not _FC.FcConfigSubstitute(config, pattern, 0):
        raise RuntimeError("Fontconfig pattern substitution failed")
    _FC.FcDefaultSubstitute(pattern)


def _fontconfig_family_match(font_family: str) -> Path:
    pattern = _FC.FcPatternCreate()
    if not pattern:
        raise RuntimeError("Fontconfig could not create a family pattern")
    matched = None
    try:
        if not _FC.FcPatternAddString(pattern, b"family", font_family.encode("utf-8")):
            raise RuntimeError(f"Fontconfig could not add family: {font_family!r}")
        if not _FC.FcPatternAddString(pattern, b"style", b"Regular"):
            raise RuntimeError("Fontconfig could not request the regular style")
        _prepare_fontconfig_pattern(pattern)
        result = ctypes.c_int()
        matched = _FC.FcFontMatch(_fontconfig_config(), pattern, ctypes.byref(result))
        if not matched:
            raise RuntimeError(f"Fontconfig returned no match for family {font_family!r}")
        file_value = ctypes.c_char_p()
        if _FC.FcPatternGetString(matched, b"file", 0, ctypes.byref(file_value)) != 0 or not
        ↪  file_value.value:
            raise RuntimeError("Fontconfig family match exposes no physical font file")
        path = Path(file_value.value.decode("utf-8"))
        if not path.is_file():
            raise RuntimeError(f"Fontconfig returned a missing font file: {path}")
        return path
    finally:
        if matched:
            _FC.FcPatternDestroy(matched)
        _FC.FcPatternDestroy(pattern)


def _system_font_set() -> ctypes.POINTER(_FcFontSet):
    global _SYSTEM_FONT_SET
    if _SYSTEM_FONT_SET is not None:
```

```
        return _SYSTEM_FONT_SET
    pattern = _FC.FcPatternCreate()
    if not pattern:
        raise RuntimeError("Fontconfig could not create the system fallback pattern")
    try:
        _prepare_fontconfig_pattern(pattern)
        result = ctypes.c_int()
        font_set = _FC.FcFontSort(_fontconfig_config(), pattern, 1, None, ctypes.byref(result))
        if not font_set or font_set.contents.nfont <= 0:
            raise RuntimeError("Fontconfig returned an empty system fallback set")
        _SYSTEM_FONT_SET = font_set
        return font_set
    finally:
        _FC.FcPatternDestroy(pattern)


def _native_font_roots() -> tuple[Path, ...]:
    if sys.platform == "win32":
        windows = Path(os.environ.get("WINDIR", "C:/Windows")) / "Fonts"
        local = Path(os.environ.get("LOCALAPPDATA", str(Path.home() / "AppData/Local"))) /
        ↪  "Microsoft" / "Windows" / "Fonts"
        return (windows / "tardisha", windows / "azazuley", windows, local)
    if sys.platform == "darwin":
        return (Path("/Library/Fonts/tardisha"), Path("/Library/Fonts/azazuley"),
        ↪  Path("/Library/Fonts"), Path("/System/Library/Fonts"), Path.home() / "Library" /
        ↪  "Fonts")
    return ()


@lru_cache(maxsize=1)
def _native_font_files() -> tuple[Path, ...]:
    seen: set[Path] = set()
    files: list[Path] = []
    for root in _native_font_roots():
        if not root.is_dir():
            continue
        for path in sorted(root.rglob("*"), key=lambda value: str(value).casefold()):
            if not path.is_file() or path.suffix.lower() not in FONT_SUFFIXES or path in seen:
                continue
            seen.add(path)
            files.append(path)
    return tuple(files)


@lru_cache(maxsize=1)
def _native_font_records() -> tuple[tuple[Path, str, frozenset[int]], ...]:
    records: list[tuple[Path, str, frozenset[int]]] = []
    for path in _native_font_files():
        try:
            with TTFont(path, fontNumber=0, lazy=True) as font:
                names = font["name"]
                family = names.getDebugName(16) or names.getDebugName(1)
                cmap = frozenset((font.getBestCmap() or {}).keys())
        except Exception:
            continue
        if family:
```

```
            records.append((path, family, cmap))
    return tuple(records)


@lru_cache(maxsize=256)
def system_font_file_for_family(font_family: str) -> Path:
    if not font_family:
        raise ValueError("system font family must be non-empty")
    if _USE_FONTCONFIG:
        path = _fontconfig_family_match(font_family)
        physical_family, _style = _font_identity(path)
        if physical_family != font_family:
            raise RuntimeError(f"system font resolver substituted {physical_family!r} for required
            ↪  family {font_family!r}")
        return path
    for path, family, _cmap in _native_font_records():
        if family == font_family:
            return path
    raise RuntimeError(f"system font resolver returned no exact family {font_family!r}")


@lru_cache(maxsize=256)
def system_font_file_for_codepoint(codepoint: int) -> Path:
    if not 0 <= codepoint <= 0x10FFFF:
        raise ValueError(f"Unicode code point outside range: {codepoint}")
    if _USE_FONTCONFIG:
        font_set = _system_font_set()
        for index in range(font_set.contents.nfont):
            pattern = font_set.contents.fonts[index]
            charset = ctypes.c_void_p()
            if _FC.FcPatternGetCharSet(pattern, b"charset", 0, ctypes.byref(charset)) != 0 or not
            ↪  charset.value:
                continue
            if not _FC.FcCharSetHasChar(charset, codepoint):
                continue
            file_value = ctypes.c_char_p()
            if _FC.FcPatternGetString(pattern, b"file", 0, ctypes.byref(file_value)) != 0 or not
            ↪  file_value.value:
                continue
            path = Path(file_value.value.decode("utf-8"))
            if not path.is_file():
                raise RuntimeError(f"Fontconfig fallback set returned a missing font file: {path}")
            return path
        raise RuntimeError(f"system font fallback set has no face for U+{codepoint:04X}")
    for path, _family, cmap in _native_font_records():
        if codepoint in cmap:
            return path
    raise RuntimeError(f"system font fallback set has no face for U+{codepoint:04X}")


@lru_cache(maxsize=256)
def grimchain_face_for_character(character: str) -> GrimChainFace:
    """Resolve a standard GrimChain Unicode glyph through the host font system."""
    if len(character) != 1:
        raise ValueError("GrimChain face routing accepts exactly one Unicode code point")
```

```
    codepoint = ord(character)
    if codepoint not in grimchain_display_codepoints():
        raise RuntimeError(f"GrimChain display code point is outside TardiSHA's visible body:
        ↪  U+{codepoint:04X}")
    path = system_font_file_for_codepoint(codepoint)
    family, _style = _font_identity(path)
    return GrimChainFace(family, path)


@dataclass(frozen=True, slots=True)
class InstalledFontBody:
    root: Path
    files: tuple[Path, ...]
    families: tuple[str, ...]


_LOADED_BODY: InstalledFontBody | None = None
_FONT_FILES_BY_FAMILY: dict[str, tuple[Path, ...]] = {}


def installed_font_files() -> tuple[Path, ...]:
    """Return Azazuley's private installed font body."""
    if not FONT_ROOT.is_dir():
        raise RuntimeError(f"Azazuley production font authority is missing: {FONT_ROOT}")
    files = tuple(
        sorted(
            path
            for path in FONT_ROOT.rglob("*")
            if path.is_file()
            and path.suffix.lower() in FONT_SUFFIXES
        )
    )
    if len(files) != EXPECTED_AZAZULEY_PRIVATE_FONT_COUNT:
        raise RuntimeError(
            f"Azazuley private font authority must contain exactly "
            f"{EXPECTED_AZAZULEY_PRIVATE_FONT_COUNT} fonts: "
            f"{FONT_ROOT} -> {len(files)}"
        )
    return files


def load_installed_fonts() -> InstalledFontBody:
    """Index Azazuley custom/fantasy fonts only; standard fonts remain system-resolved."""
    global _LOADED_BODY, _FONT_FILES_BY_FAMILY, _FONT_FAMILY_BY_FILE
    if _LOADED_BODY is not None:
        return _LOADED_BODY
    files = installed_font_files()
    families: set[str] = set()
    family_files: dict[str, list[Path]] = {}
    family_by_file: dict[Path, str] = {}
    for path in files:
        family, _style = _font_identity(path)
        families.add(family)
        family_files.setdefault(family, []).append(path)
        family_by_file[path] = family
    _FONT_FILES_BY_FAMILY = {family: tuple(sorted(paths)) for family, paths in family_files.items()}
```

```
    _FONT_FAMILY_BY_FILE = family_by_file
    _LOADED_BODY = InstalledFontBody(
        root=FONT_ROOT,
        files=files,
        families=tuple(sorted(families, key=str.casefold)),
    )
    return _LOADED_BODY


@lru_cache(maxsize=1)
def _system_font_families() -> tuple[str, ...]:
    if _USE_FONTCONFIG:
        font_set = _system_font_set()
        families: set[str] = set()
        for index in range(font_set.contents.nfont):
            value = ctypes.c_char_p()
            pattern = font_set.contents.fonts[index]
            if _FC.FcPatternGetString(pattern, b"family", 0, ctypes.byref(value)) == 0 and
            ↪  value.value:
                families.add(value.value.decode("utf-8"))
        return tuple(sorted(families, key=str.casefold))
    return tuple(sorted({family for _path, family, _cmap in _native_font_records()},
    ↪  key=str.casefold))

def installed_font_families() -> tuple[str, ...]:
    """Return custom fantasy and normal system-installed font families available to Et Sonyera."""
    if _LOADED_BODY is None:
        raise RuntimeError("Azazuley production fonts have not been loaded yet")
    return tuple(sorted(set(_LOADED_BODY.families).union(_system_font_families()),
    ↪  key=str.casefold))



@lru_cache(maxsize=256)
def authored_lexical_font_file(font_family: str, source_authority: str) -> Path:
    """Resolve custom fantasy faces explicitly; standard families through the host font system."""
    if _LOADED_BODY is None:
        raise RuntimeError("Azazuley production fonts have not been loaded yet")
    candidates = _FONT_FILES_BY_FAMILY.get(font_family, ())
    azazuley_candidates = tuple(path for path in candidates if path.is_relative_to(FONT_ROOT))
    if azazuley_candidates:
        if source_authority not in FANTASY_LEXICAL_AUTHORITIES:
            raise RuntimeError(
                f"fantasy lexical font is outside its authored authorities: "
                f"{source_authority!r} -> {font_family!r}"
            )
        if len(azazuley_candidates) == 1:
            return azazuley_candidates[0]
        canonical_name = CANONICAL_REGULAR_FONT_FILES.get(font_family)
        canonical = tuple(path for path in azazuley_candidates if path.name == canonical_name)
        if len(canonical) != 1:
            raise RuntimeError(
                f"authored fantasy family does not resolve to one canonical physical face: "
                f"{font_family!r} -> {azazuley_candidates!r}"
            )
```

```
        return canonical[0]
    return system_font_file_for_family(font_family)


def physical_font_family(font_file: Path) -> str:
    """Return the physical family for an authored or system-resolved font file."""
    path = Path(font_file)
    indexed = _FONT_FAMILY_BY_FILE.get(path)
    if indexed is not None:
        return indexed
    if not path.is_file():
        raise RuntimeError(f"physical font file does not exist: {path}")
    family, _style = _font_identity(path)
    return family


def et_sonyera_font_file(font_family: str) -> Path:
    """Resolve custom fantasy selections explicitly and standard selections through Fontconfig."""
    if not font_family:
        raise RuntimeError("Et Sonyera font family is empty")
    if _LOADED_BODY is None:
        raise RuntimeError("Azazuley production fonts have not been loaded yet")
    candidates = _FONT_FILES_BY_FAMILY.get(font_family, ())
    if not candidates:
        return system_font_file_for_family(font_family)
    if len(candidates) == 1:
        return candidates[0]
    canonical_name = CANONICAL_REGULAR_FONT_FILES.get(font_family)
    canonical = tuple(path for path in candidates if path.name == canonical_name)
    if len(canonical) == 1:
        return canonical[0]
    raise RuntimeError(
        f"Et Sonyera custom family does not resolve to one authored physical face: {font_family!r}
        ↪  -> {candidates!r}"
    )
```

config/visual.py

```
"""Visible first-body law for Azazuley Veira."""

TSHIRT_GRAY = "#70726d"
TSHIRT_DARK = "#5d605b"
TSHIRT_LIGHT = "#7c7e78"
EDGE = "#3d4541"
EDGE_SOFT = "#58605b"
TERMINAL_BG = "#101714"
TERMINAL_BG_DEEP = "#0b110f"
TEAL = "#68f5df"
TEAL_SOFT = "#45c9b6"
TEAL_DIM = "#2d8579"
TEAL_FAINT = "#1d514b"
```

engines/__init__.py

```
"""Direct bindings to the two packaged Azazuley engine bodies."""
```

engines/emanation.py

```
"""
The Grimchain gets dropped through here like watching the Price is Right,
Plinko on your Grandmother's 10" tube tv she had when Orson Wells did War of the Worlds,
and you don't ank questions, capiche!
"""
from __future__ import annotations

MIRROR_GLYPH = "⊙"
MIRROR_NAME =  "Living_Mirror"

कल्प_acoustic = (
    '⊙',
)

def ज्योतिष_witness(grimchain: str) -> str:
    return grimchain
grimchain_emanate = ज्योतिष_witness
```

engines/grimchain.py

```
"""Azazuley boundary to TardiSHA GrimChain."""
from __future__ import annotations

import contextlib
import io
import sys
import os
import tempfile
from pathlib import Path

from tardisha_grimchain import grimchain as _grimchain
from tardisha_grimchain.domus import LIVING_MIRROR_MEMBERS, parse_public_living_domus
from tardisha_grimchain.domus_stream import file_domus_record, living_domus_from_emission


def execute(argv: list[str]) -> tuple[int, str]:
    stdout = io.StringIO()
    stderr = io.StringIO()
    old_argv = sys.argv
    sys.argv = ["grimchain", *argv]
    try:
        with contextlib.redirect_stdout(stdout), contextlib.redirect_stderr(stderr):
            try:
                result = _grimchain.main()
                code = 0 if result is None else int(result)
            except SystemExit as exc:
                code = int(exc.code or 0)
    finally:
        sys.argv = old_argv

    body = stdout.getvalue()
    error = stderr.getvalue()
    if error:
        body += ("\n" if body and not body.endswith("\n") else "") + error
    return code, body.rstrip("\n")



class GrimChainContinuationError(RuntimeError):
    """Raised when continuation of an already-witnessed GrimChain identity fails."""


class GrimChainContinuationSession:
    """Invocation-local continuation of one witnessed TardiSHA GrimChain identity."""
    def __init__(self, source: str, requested_depth: int) -> None:
        fd, tmp = tempfile.mkstemp(suffix=".grimchain")
        path = Path(tmp)
        try:
```

```
            with os.fdopen(fd, "wb") as handle:
                handle.write(source.encode("utf-8"))
                handle.flush()
                os.fsync(handle.fileno())
            chain, emission, route = file_domus_record(
                path, requested_depth, nonce=0, include_filename=False
            )
        finally:
            path.unlink(missing_ok=True)
        self.emission = emission
        self.route = route
        self.current_depth = requested_depth
        self.current_chain = chain

    def chain_at(self, depth: int) -> str:
        try:
            chain = living_domus_from_emission(
                self.emission, depth, nonce=0, route_witness=self.route
            )
        except Exception as exc:
            raise GrimChainContinuationError(str(exc)) from exc
        if depth > self.current_depth and not chain.startswith(self.current_chain):
            raise GrimChainContinuationError("GrimChain continuation changed the witnessed prefix")
        self.current_depth = depth
        self.current_chain = chain
        return chain

def string(source: str, middle: str) -> tuple[int, str]:
    """GrimChain one exact text body through TardiSHA's public callable authority."""
    try:
        middle_text = middle.strip()
        depth = int(middle_text) if middle_text else 0
        if depth < 0:
            depth = 0
        session = GrimChainContinuationSession(source, depth)
        return 0, session.current_chain
    except Exception as exc:
        return 1, f"grimchain: {exc}"


def string_cli(source: str, middle: str) -> tuple[int, str]:
    """Preserved in-process CLI adapter used for direct/callable equivalence tests."""
    argv: list[str] = []
    if middle.strip():
        argv.append(middle.strip())
    argv.extend(("--string", source))
    return execute(argv)


def file(source: str | Path, middle: str) -> tuple[int, str]:
    """GrimChain one exact file body through TardiSHA's raw-file authority."""
    try:
        middle_text = middle.strip()
        depth = int(middle_text) if middle_text else 0
        if depth < 0:
            depth = 0
```

```
        chain, _emission, _route = file_domus_record(
            Path(source), depth, nonce=0, include_filename=True
        )
        return 0, chain
    except Exception as exc:
        return 1, f"grimchain: {exc}"


def pdf_embed(source: str | Path, middle: str) -> tuple[int, str]:
    """Apply the exact public `grimchain NUMBER --pdf-embed PDF` command path."""
    middle_text = middle.strip()
    argv: list[str] = []
    if middle_text:
        argv.append(middle_text)
    argv.extend(("--pdf-embed", str(Path(source))))
    try:
        return execute(argv)
    except Exception as exc:
        return 1, f"grimchain: {exc}"


def validate(chain: str) -> None:
    """Require the current public TardiSHA GrimChain structure."""
    if chain in LIVING_MIRROR_MEMBERS:
        return
    parse_public_living_domus(chain)
```

engines/sydonic.py

```
"""Azazuley boundary to the bundled Sydonic Magicae stack."""
from __future__ import annotations

from dataclasses import dataclass
from functools import lru_cache
import secrets

from sydonic_magicae_translation_matrix.engine import (
    EngineError,
    SydonicMagicaeEngine,
    TranslationResult,
)
from sydonic_magicae_translation_matrix.lexical_resolver import ApprovedLexicalEntry
from sydonic_magicae_translation_matrix.english_renderer import SydonicSpokenVoice
from sydonic_magicae_translation_matrix.models import DomusFrame, ReflectedGlyphToken
from azazuley_veira.ailalubar_render import AilalubarRenderWitness
from azazuley_veira.path_out import PathOut, build_path

BREATH_SEPARATOR = "·"
OSTENSIVE_PARAGRAPH_PREFIXES = tuple("ץקזסרתפכדגב")
OSTENSIVE_UTTERANCE_SEPARATOR = ""
OSTENSIVE_PARAGRAPH_SEPARATOR = "\n"
_SYDONIC_ENGINE = SydonicMagicaeEngine(house_voice=SydonicSpokenVoice())



def qt_utf16_units(text: str) -> int:
    return len(text.encode("utf-16-le")) // 2


@dataclass(frozen=True, slots=True)
class PresentationSpan:
    qt_start: int
    qt_end: int
    source_glyph: str
    source_position: int
    source_authority: str
    rendered_text: str
    font_family: str
    semantic_role: str

    def __post_init__(self) -> None:
        if self.qt_start < 0 or self.qt_end < self.qt_start:
            raise ValueError((self.qt_start, self.qt_end))
        if self.qt_end - self.qt_start != qt_utf16_units(self.rendered_text):
            raise ValueError("presentation span extent does not match rendered UTF-16 text")
```

```
@dataclass(frozen=True, slots=True)
class DefinitionField:
    label: str
    value: str
    semantic_role: str
    qt_value_start: int
    qt_value_end: int


@dataclass(frozen=True, slots=True)
class DefinitionEntryPresentation:
    source_glyph: str
    source_position: int
    source_authority: str
    fields: tuple[DefinitionField, ...]


@dataclass(frozen=True, slots=True)
class DefinitionPresentation:
    text: str
    entries: tuple[DefinitionEntryPresentation, ...]
    lexical_spans: tuple[PresentationSpan, ...]


@dataclass(frozen=True, slots=True)
class LeySyffEntryPresentation:
    source_glyph: str
    source_position: int
    source_authority: str
    lemma: str
    payload: str
    lemma_span: PresentationSpan
    payload_span: PresentationSpan | None


@dataclass(frozen=True, slots=True)
class LeySyffPresentation:
    text: str
    entries: tuple[LeySyffEntryPresentation, ...]
    spans: tuple[PresentationSpan, ...]


@dataclass(frozen=True, slots=True)
class SourceBodyIdentity:
    source_authority: str
    glyph: str
    ostensive_variant: str | None = None


@dataclass(frozen=True, slots=True)
class ResolvedRenderWord:
    text: str
    source_position: int
    source_glyph: str
    source_body_id: str
    identity: SourceBodyIdentity
```

```
    entry: ApprovedLexicalEntry


def select_ostensive_utterances(
    words: tuple[ResolvedRenderWord, ...],
    *,
    randbelow=secrets.randbelow,
) -> tuple[str, ...]:
    """Select complete authored ostensive utterances without rewriting them."""
    utterances: list[str] = []
    for word in words:
        entry = word.entry
        if entry.authored_office == "phantasmagoria":
            variants = entry.ostensive_variants
            utterances.append(variants[randbelow(3)])
            continue
        utterances.append(entry.ostensive)
    return tuple(utterances)

def project_ostensive_paragraphs(
    utterances: tuple[str, ...],
    *,
    randbelow=secrets.randbelow,
) -> str:
    """Group complete ostensive utterances into presentation-only paragraphs."""
    paragraphs: list[str] = []
    position = 0
    while position < len(utterances):
        paragraph_size = randbelow(6) + 1
        paragraph_utterances = utterances[position:position + paragraph_size]
        prefix = OSTENSIVE_PARAGRAPH_PREFIXES[randbelow(len(OSTENSIVE_PARAGRAPH_PREFIXES))]
        paragraphs.append(
            prefix + OSTENSIVE_UTTERANCE_SEPARATOR.join(paragraph_utterances)
        )
        position += len(paragraph_utterances)
    return OSTENSIVE_PARAGRAPH_SEPARATOR.join(paragraphs)


def project_ostensive_presentations(
    words: tuple[ResolvedRenderWord, ...],
    *,
    randbelow=secrets.randbelow,
) -> tuple[LeySyffPresentation, LeySyffPresentation]:
    """Project LeySyff and FfysYel using only the injected presentation randomness."""
    leysyff_utterances = select_ostensive_utterances(words, randbelow=randbelow)
    leysyff_text = project_ostensive_paragraphs(leysyff_utterances, randbelow=randbelow)
    ffysyel_words = tuple(reversed(words))
    ffysyel_utterances = select_ostensive_utterances(ffysyel_words, randbelow=randbelow)
    ffysyel_text = project_ostensive_paragraphs(ffysyel_utterances, randbelow=randbelow)
    return (
        LeySyffPresentation(leysyff_text, (), ()),
        LeySyffPresentation(ffysyel_text, (), ()),
    )

@dataclass(frozen=True, slots=True)
class SydonicRenderTransaction:
```

```
chain: str
mirror_witness: AilalubarRenderWitness
result: TranslationResult | None
resolved_words: tuple[ResolvedRenderWord, ...]
azalalia_runs: tuple[PresentationSpan, ...]
ailalaza_runs: tuple[PresentationSpan, ...]
definition: DefinitionPresentation
leysyff: LeySyffPresentation
ffysyel: LeySyffPresentation
path_out: PathOut | None = None

@property
def azalalia_body(self) -> str:
    return BREATH_SEPARATOR.join(word.entry.authored_lemma for word in self.resolved_words)


@property
def ailalaza_body(self) -> str:
    return BREATH_SEPARATOR.join(word.entry.authored_lemma for word in
    ↪  reversed(self.resolved_words))

@property
def glossolalia_body(self) -> str:
    return BREATH_SEPARATOR.join(word.entry.pronunciation for word in self.resolved_words)

@property
def ailalossolg_body(self) -> str:
    return BREATH_SEPARATOR.join(word.entry.pronunciation for word in
    ↪  reversed(self.resolved_words))

@property
def definition_body(self) -> str:
    return self.definition.text

@property
def definition_runs(self) -> tuple[PresentationSpan, ...]:
    return self.definition.lexical_spans

@property
def leysyff_body(self) -> str:
    return self.leysyff.text

@property
def leysyff_runs(self) -> tuple[PresentationSpan, ...]:
    return self.leysyff.spans
@property
def ffysyel_body(self) -> str:
    return self.ffysyel.text

@property
def ffysyel_runs(self) -> tuple[PresentationSpan, ...]:
    return self.ffysyel.spans


@property
def trace_sha256(self) -> str:
```

```
        if self.result is None:
            raise EngineError("Living Mirror translation has no ordinary Domus semantic trace")
        trace = self.result.semantic_trace
        if trace is None:
            raise EngineError("Azazuley render transaction requires a semantic trace")
        return trace.sha256

def parse_source_body_id(source_body_id: str, grammatical_bearing: str | None) ->
↪  SourceBodyIdentity:
    """Parse one renderer source-body ID in one canonical place."""
    if grammatical_bearing == "phantasmagoria":
        source_authority, glyph, ostensive_field = source_body_id.rsplit(":", 2)
        if ostensive_field not in {"ostensiveI", "ostensiveII", "ostensiveIII"}:
            raise EngineError(f"invalid Phantasmagoria Ostensive provenance: {source_body_id!r}")
        return SourceBodyIdentity(source_authority, glyph,
        ↪  ostensive_field.removeprefix("ostensive"))
    source_authority, glyph = source_body_id.rsplit(":", 1)
    return SourceBodyIdentity(source_authority, glyph)

def _resolved_words(
    result: TranslationResult,
    visual_source_positions: tuple[int, ...],
) -> tuple[ResolvedRenderWord, ...]:
    words: list[ResolvedRenderWord] = []
    for word, entry in zip(
        result.rendering.final_words,
        result.resolved_entries,
        strict=True,
    ):
        if (
            word.kind != "content"
            or word.source_position is None
            or word.source_body_id is None
            or word.source_glyph is None
        ):
            continue
        if entry is None:
            raise EngineError("Sydonic content word has no resolved lexical entry")
        identity = parse_source_body_id(word.source_body_id, word.grammatical_bearing)
        try:
            prosody_position = visual_source_positions[word.source_position]
        except IndexError as exc:
            raise EngineError(
                f"Sydonic Cantillation position has no Prosody provenance: {word.source_position}"
            ) from exc
        words.append(ResolvedRenderWord(
            text=word.text,
            source_position=prosody_position,
            source_glyph=word.source_glyph,
            source_body_id=word.source_body_id,
            identity=identity,
            entry=entry,
        ))
    return tuple(words)

class _AilalubarSydonicAdapter:
```

```
"""Azuzaley-owned adapter from Ailalubar presentation order to Sydonic's generic execution
↪  API."""

def __init__(self, witness: AilalubarRenderWitness, frame: DomusFrame) -> None:
    if frame.source != witness.visual_text:
        raise ValueError("Sydonic frame differs from the completed Ailalubar presentation")
    self.witness = witness
    self.frame = frame

def token(self, virtual_position: int) -> ReflectedGlyphToken:
    extent = len(self.frame.tokens)
    if extent == 0:
        raise ValueError("Ailalubar traversal requires a non-empty rendered body")
    position = self.frame.aeternum_source_position(virtual_position)
    token = self.frame.tokens[position]
    return ReflectedGlyphToken(
        value=token.value,
        mirror_position=position,
        source_position=position,
        prosody_position=self.witness.visual_source_positions[position],
        kind=token.kind,
        imaginary=not (0 <= virtual_position < extent),
    )

@staticmethod
def reorder_pair(
    causal_body_order: list[int],
    left_position: int,
    right_position: int,
    order: tuple[int, int],
) -> None:
    if left_position == right_position:
        return
    if set(order) != {left_position, right_position}:
        raise ValueError("Ailalubar pair reorder changed source identity")
    try:
        slots = sorted((
            causal_body_order.index(left_position),
            causal_body_order.index(right_position),
        ))
    except ValueError as exc:
        raise ValueError("Ailalubar pair lost a finite source body") from exc
    causal_body_order[slots[0]] = order[0]
    causal_body_order[slots[1]] = order[1]

@staticmethod
def reorder_corridor(
    causal_body_order: list[int],
    source_positions: tuple[int, ...],
    returned_positions: tuple[int, ...],
) -> None:
    if set(source_positions) != set(returned_positions):
        raise ValueError("Ailalubar corridor changed finite source identity")
    try:
        slots = sorted(causal_body_order.index(position) for position in source_positions)
    except ValueError as exc:
```

```
            raise ValueError("Ailalubar corridor lost a finite source body") from exc
        for slot, position in zip(slots, returned_positions, strict=True):
            causal_body_order[slot] = position


def _written_projection_runs(words: tuple[ResolvedRenderWord, ...]) -> tuple[PresentationSpan, ...]:
    runs: list[PresentationSpan] = []
    offset = 0
    for index, word in enumerate(words):
        if index:
            offset += qt_utf16_units(BREATH_SEPARATOR)
        lemma = word.entry.authored_lemma
        start = offset
        offset += qt_utf16_units(lemma)
        runs.append(PresentationSpan(
            start, offset, word.source_glyph, word.source_position,
            word.identity.source_authority, lemma, word.entry.font, "authored_lexical",
        ))
    return tuple(runs)


@lru_cache(maxsize=32)
def render_transaction(
    witness: AilalubarRenderWitness, *, cadence_translation: bool = False,
) -> SydonicRenderTransaction:
    """Translate the completed Azuzaley Ailalubar presentation through Sydonic's generic matrix
    ↪  API."""
    engine = _SYDONIC_ENGINE
    living_mirror = (
        engine.lexical_resolver.living_mirror_entries(witness.source)
        if cadence_translation else ()
    )
    if living_mirror:
        ordered: list[tuple[int, ResolvedRenderWord]] = []
        for source_position, entry in living_mirror:
            source_extent = range(source_position, source_position + len(entry.glyph))
            visual_positions = tuple(
                visual_position
                for visual_position, prosody_position in enumerate(witness.visual_source_positions)
                if prosody_position in source_extent
            )
            if len(visual_positions) != len(entry.glyph):
                raise EngineError("Living Mirror cadence lost Ailalubar source identity")
            identity = SourceBodyIdentity(entry.source_authority, entry.glyph)
            word = ResolvedRenderWord(
                text=entry.authored_lemma,
                source_position=source_position,
                source_glyph=entry.glyph,
                source_body_id=f"{entry.source_authority}:{entry.glyph}",
                identity=identity,
                entry=entry,
            )
            ordered.append((min(visual_positions), word))
        resolved = tuple(word for _position, word in sorted(ordered, key=lambda item: item[0]))
        result = None
    else:
```

```
    frame = DomusFrame(
        source=witness.visual_text,
        tokens=engine.lexer.lex(witness.visual_text),
        prosody_source=witness.source,
    )
    adapter = _AilalubarSydonicAdapter(witness, frame)
    result = engine.translate_domus_with_execution(
        domus_string=witness.visual_text,
        target_language="en",
        output_style="telegraphic",
        include_trace=True,
        aeternum_token=adapter.token,
        aeternum_pair_reorder=adapter.reorder_pair,
        aeternum_corridor_reorder=adapter.reorder_corridor,
        parsed_frame=frame,
    )
    resolved = _resolved_words(result, witness.visual_source_positions)

azalalia_runs = _written_projection_runs(resolved)
ailalaza_runs = _written_projection_runs(tuple(reversed(resolved)))

definition_parts: list[str] = []
definition_runs: list[PresentationSpan] = []
definition_entries: list[DefinitionEntryPresentation] = []
definition_offset = 0
field_specs = (
    ("Word", "authored_lexical"),
    ("Origin Language", "prose"),
    ("Pronunciation", "prose"),
    ("Cadence", "prose"),
    ("Non-Ostensive", "prose"),
    ("Ostensive", "prose"),
    ("LeySyff", "prose"),
    ("GrimChain", "grimchain"),
)
for index, word in enumerate(resolved):
    if index:
        separator = "\n\n"
        definition_parts.append(separator)
        definition_offset += qt_utf16_units(separator)
    entry = word.entry
    values = (
        entry.authored_lemma,
        entry.origin_language,
        entry.pronunciation,
        entry.cadence,
        entry.non_ostensive,
        entry.ostensive,
        entry.leysyff,
        entry.grimchain,
    )
    fields: list[DefinitionField] = []
    for line_index, ((label, role), body) in enumerate(zip(field_specs, values, strict=True)):
        if line_index:
            definition_parts.append("\n")
            definition_offset += qt_utf16_units("\n")
```

```
        prefix = f"{label}: "
        definition_parts.append(prefix)
        definition_offset += qt_utf16_units(prefix)
        start = definition_offset
        definition_parts.append(body)
        definition_offset += qt_utf16_units(body)
        fields.append(DefinitionField(label, body, role, start, definition_offset))
        if role == "authored_lexical":
            definition_runs.append(PresentationSpan(
                start, definition_offset, word.source_glyph, word.source_position,
                word.identity.source_authority, body, entry.font, "definition_word",
            ))
    definition_entries.append(DefinitionEntryPresentation(
        word.source_glyph, word.source_position, word.identity.source_authority, tuple(fields)
    ))
definition_body = "".join(definition_parts)
definition = DefinitionPresentation(definition_body, tuple(definition_entries),
↪  tuple(definition_runs))

leysyff, ffysyel = project_ostensive_presentations(resolved)


return SydonicRenderTransaction(
    chain=witness.source,
    mirror_witness=witness,
    result=result,
    resolved_words=resolved,
    azalalia_runs=azalalia_runs,
    ailalaza_runs=ailalaza_runs,
    definition=definition,
    leysyff=leysyff,
    ffysyel=ffysyel,
    path_out=build_path(witness.source, engine),
)
```

grimchain_journal.py

```
"""Persistent rotating journal for GrimChains created by the user."""
from __future__ import annotations

import gzip
import os
import shutil
import mmap
import tempfile
from datetime import datetime
from pathlib import Path

MAX_JOURNAL_BYTES = 32 * 1024 * 1024
RECORD_SEPARATOR = "o-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-oo-o"
JOURNAL_DIRECTORY = Path.home() / ".grimchain" / "azazuley" / "grimchains"
ACTIVE_JOURNAL = JOURNAL_DIRECTORY / "grimchains.bio"
BACKUP_JOURNALS = tuple(
    JOURNAL_DIRECTORY / f"grimchains.{index}.bio" for index in range(1, 4)
)


def ensure_grimchain_journal() -> Path:
    """Ensure only the active GrimChain journal exists before any rotation."""
    JOURNAL_DIRECTORY.mkdir(parents=True, exist_ok=True)
    ACTIVE_JOURNAL.touch(exist_ok=True)
    return ACTIVE_JOURNAL


def _rotate() -> None:
    ensure_grimchain_journal()
    BACKUP_JOURNALS[-1].unlink(missing_ok=True)
    for source, destination in zip(
        reversed(BACKUP_JOURNALS[:-1]), reversed(BACKUP_JOURNALS[1:])
    ):
        if source.exists():
            os.replace(source, destination)
    with ACTIVE_JOURNAL.open("rb") as source, gzip.open(BACKUP_JOURNALS[0], "wb") as destination:
        shutil.copyfileobj(source, destination)
    ACTIVE_JOURNAL.write_bytes(b"")


def _record_bytes(user_input: str, grimchain: str, timestamp: str, *, first_record: bool) -> bytes:
    prefix = RECORD_SEPARATOR + "\n" if first_record else ""
    return (
        prefix
        + f"·:Timestamp:·{timestamp}·:\n"
        + f"·:User input:·{user_input}·:\n"
        + f"·:Grimchain:·{grimchain}·:\n"
        + RECORD_SEPARATOR
```

```
        + "\n"
    ).encode("utf-8")


def read_grim_import(path: str | Path) -> tuple[tuple[str, str], ...]:
    """Read one .grim manifest as ordered (user input, GrimChain) journal records."""
    source = Path(path)
    if source.suffix != ".grim" or not source.is_file():
        raise ValueError(f"Grim import requires one .grim file: {source}")
    lines = source.read_text(encoding="utf-8").splitlines()
    object_open, object_close = ("⟅", "⟆") if lines[:1] == ["⟅"] else (">", "<")
    try:
        close_index = len(lines) - 1 - lines[::-1].index("∴")
    except ValueError as exc:
        raise ValueError(f".grim manifest has no entries terminator: {source}") from exc
    if lines[close_index + 1:close_index + 2] != [object_close] or len(lines[close_index + 2:]) !=
    ↪  1:
        raise ValueError(f".grim manifest has no singular trailing GrimChain: {source}")
    records: list[tuple[str, str]] = []
    entry_path: str | None = None
    entry_grimchain: str | None = None
    depth = 0
    for line in lines[:close_index]:
        if line == object_open:
            depth += 1
            if depth == 2:
                entry_path = None
                entry_grimchain = None
        elif line == object_close:
            if depth == 2 and entry_grimchain is not None:
                if entry_path is None:
                    raise ValueError(f".grim GrimChain entry has no path: {source}")
                records.append((entry_path, entry_grimchain))
            depth -= 1
        elif depth == 2 and line.startswith("∵path∴") and line.endswith("·∴"):
            entry_path = line[len("∵path∴"):-1]
        elif depth == 2 and line.startswith("∵grimchain∴") and line.endswith("·∴"):
            entry_grimchain = line[len("∵grimchain∴"):-1]
    records.append((source.name, lines[-1]))
    return tuple(records)


def _history_record_from_body(body: bytes, path: Path) -> tuple[str, str, str]:
    text = body.decode("utf-8").strip("\r\n")
    timestamp_prefix = "·∴Timestamp:·"
    user_prefix = "·∴User input:·"
    chain_prefix = "·∴Grimchain:·"
    if not text.startswith(timestamp_prefix):
        raise ValueError(f"incomplete GrimChain journal record in {path}")
    timestamp_end = text.find("·∴\n", len(timestamp_prefix))
    user_start = text.find(user_prefix, timestamp_end + 2)
    chain_start = text.rfind("\n" + chain_prefix)
    if timestamp_end < 0 or user_start < 0 or chain_start < 0:
        raise ValueError(f"incomplete GrimChain journal record in {path}")
    timestamp = text[len(timestamp_prefix):timestamp_end]
    user_value_start = user_start + len(user_prefix)
```

```
    if not text[chain_start - 1:chain_start].endswith("⁖"):
        raise ValueError(f"incomplete GrimChain journal record in {path}")
    user_input = text[user_value_start:chain_start - 1]
    chain_value_start = chain_start + 1 + len(chain_prefix)
    if not text.endswith("⁖"):
        raise ValueError(f"incomplete GrimChain journal record in {path}")
    grimchain = text[chain_value_start:-1]
    if not timestamp or not grimchain:
        raise ValueError(f"incomplete GrimChain journal record in {path}")
    return timestamp, user_input, grimchain


class GrimchainHistorySession:
    """Lazy newest-first history reader backed by disposable temporary generation caches."""

    def __init__(self) -> None:
        self._temporary = tempfile.TemporaryDirectory(prefix="azazuley-grimchain-history-")
        self._sources = tuple(path for path in (ACTIVE_JOURNAL, *BACKUP_JOURNALS) if path.exists())
        self._source_index = 0
        self._cache_file = None
        self._mapping = None
        self._right_separator = -1
        self._current_source: Path | None = None
        self._seen_chains: set[str] = set()
        self._exhausted = False

    @property
    def has_more(self) -> bool:
        return not self._exhausted

    def close(self) -> None:
        self._close_generation()
        if self._temporary is not None:
            self._temporary.cleanup()
            self._temporary = None
        self._exhausted = True

    def _close_generation(self) -> None:
        if self._mapping is not None:
            self._mapping.close()
            self._mapping = None
        if self._cache_file is not None:
            self._cache_file.close()
            self._cache_file = None
        self._right_separator = -1
        self._current_source = None

    def _load_next_generation(self) -> bool:
        self._close_generation()
        separator = RECORD_SEPARATOR.encode("utf-8")
        while self._source_index < len(self._sources):
            source = self._sources[self._source_index]
            self._source_index += 1
            cache_path = Path(self._temporary.name) / f"generation-{self._source_index}.bio"
            if source == ACTIVE_JOURNAL:
                shutil.copyfile(source, cache_path)
```

```
            else:
                with gzip.open(source, "rb") as input_stream, cache_path.open("wb") as
                ↪  output_stream:
                    shutil.copyfileobj(input_stream, output_stream)
            if cache_path.stat().st_size == 0:
                continue
            cache_file = cache_path.open("rb")
            mapping = mmap.mmap(cache_file.fileno(), 0, access=mmap.ACCESS_READ)
            right_separator = mapping.rfind(separator)
            if right_separator < 0:
                mapping.close()
                cache_file.close()
                raise ValueError(f"GrimChain journal separator is missing in {source}")
            self._cache_file = cache_file
            self._mapping = mapping
            self._right_separator = right_separator
            self._current_source = source
            return True
        self._exhausted = True
        return False

    def fetch(self, limit: int = 40) -> tuple[tuple[str, str, str], ...]:
        if limit <= 0:
            raise ValueError("history fetch limit must be positive")
        records: list[tuple[str, str, str]] = []
        separator = RECORD_SEPARATOR.encode("utf-8")
        while len(records) < limit and not self._exhausted:
            if self._mapping is None and not self._load_next_generation():
                break
            left_separator = self._mapping.rfind(separator, 0, self._right_separator)
            if left_separator < 0:
                self._close_generation()
                continue
            body_start = left_separator + len(separator)
            body = self._mapping[body_start:self._right_separator]
            self._right_separator = left_separator
            if not body.strip():
                continue
            record = _history_record_from_body(body, self._current_source)
            if record[2] in self._seen_chains:
                continue
            self._seen_chains.add(record[2])
            records.append(record)
        return tuple(records)

    def __enter__(self) -> "GrimchainHistorySession":
        return self

    def __exit__(self, _exc_type, _exc, _traceback) -> None:
        self.close()

def _generation_records(path: Path) -> tuple[tuple[str, str, str], ...]:
    """Read one journal generation in its physical oldest-to-newest order."""
    if not path.exists():
        return ()
    if path == ACTIVE_JOURNAL:
```

```
        payload = path.read_bytes()
    else:
        with gzip.open(path, "rb") as stream:
            payload = stream.read()
    if not payload:
        return ()
    separator = RECORD_SEPARATOR.encode("utf-8")
    return tuple(
        _history_record_from_body(body, path)
        for body in payload.split(separator)
        if body.strip()
    )


def _write_generation_records(path: Path, records: tuple[tuple[str, str, str], ...]) -> None:
    """Rewrite one generation with exactly the supplied ordered records."""
    if not records:
        if path == ACTIVE_JOURNAL:
            path.write_bytes(b"")
        else:
            path.unlink(missing_ok=True)
        return
    payload = b"".join(
        _record_bytes(user_input, grimchain, timestamp, first_record=index == 0)
        for index, (timestamp, user_input, grimchain) in enumerate(records)
    )
    temporary = path.with_name(path.name + ".tmp")
    try:
        if path == ACTIVE_JOURNAL:
            temporary.write_bytes(payload)
        else:
            with gzip.open(temporary, "wb") as stream:
                stream.write(payload)
        os.replace(temporary, path)
    except Exception:
        temporary.unlink(missing_ok=True)
        raise


def _remove_existing_chain(grimchain: str) -> str | None:
    """Remove every stored occurrence and return the newest record's original user input."""
    generations: list[tuple[Path, tuple[tuple[str, str, str], ...]]] = []
    preserved_user_input: str | None = None
    for path in (ACTIVE_JOURNAL, *BACKUP_JOURNALS):
        records = _generation_records(path)
        generations.append((path, records))
        if preserved_user_input is None:
            preserved_user_input = next(
                (user_input for _timestamp, user_input, chain in reversed(records) if chain ==
                ↪  grimchain),
                None,
            )
    if preserved_user_input is None:
        return None
    for path, records in generations:
        if any(chain == grimchain for _timestamp, _user_input, chain in records):
```

```
            _write_generation_records(
                path,
                tuple(record for record in records if record[2] != grimchain),
            )
    return preserved_user_input


def append_grimchain(user_input: str, grimchain: str) -> tuple[str, bool]:
    """Add a new GrimChain once, or promote an existing exact chain to newest history position."""
    if not grimchain:
        raise ValueError("cannot journal an empty GrimChain")
    if "\n" in grimchain or "\r" in grimchain:
        raise ValueError("a journaled GrimChain must occupy exactly one line")

    ensure_grimchain_journal()
    preserved_user_input = _remove_existing_chain(grimchain)
    if preserved_user_input is not None and user_input != grimchain:
        user_input = preserved_user_input
    timestamp = datetime.now().astimezone().isoformat(timespec="seconds")
    entry = _record_bytes(user_input, grimchain, timestamp,
    ↪  first_record=ACTIVE_JOURNAL.stat().st_size == 0)
    rotated = bool(ACTIVE_JOURNAL.stat().st_size and ACTIVE_JOURNAL.stat().st_size + len(entry) >
    ↪  MAX_JOURNAL_BYTES)
    if rotated:
        _rotate()
        entry = _record_bytes(user_input, grimchain, timestamp, first_record=True)
    with ACTIVE_JOURNAL.open("ab") as stream:
        stream.write(entry)
    return timestamp, rotated
```

ui/__init__.py

```
"""Azazuley terminal UI."""
```

ui/_azuzaley_kernel.c

```
#define PY_SSIZE_T_CLEAN
#include <Python.h>
#include <stdint.h>

#define BIDI_L   0
#define BIDI_R   1
#define BIDI_AL  2
#define BIDI_ON  3
#define BIDI_NSM 4
#define BIDI_EN  5
#define BIDI_CS  6
#define BIDI_ES  7
#define BIDI_ET  8
#define BIDI_WS  9

static int normalized_class(int code) {
    if (code == BIDI_AL) return BIDI_R;
    if (code == BIDI_EN) return BIDI_L;
    if (code == BIDI_CS || code == BIDI_ES || code == BIDI_ET || code == BIDI_WS) return BIDI_ON;
    return code;
}

static PyObject *py_visual_order(PyObject *self, PyObject *args) {
    (void)self;
    PyObject *classes_obj = NULL;
    if (!PyArg_ParseTuple(args, "O:visual_order", &classes_obj)) return NULL;

    PyObject *seq = PySequence_Fast(classes_obj, "classes must be a sequence of bidi class codes");
    if (seq == NULL) return NULL;
    Py_ssize_t length = PySequence_Fast_GET_SIZE(seq);

    int *resolved = NULL;
    int *levels = NULL;
    Py_ssize_t *order = NULL;
    if (length > 0) {
        resolved = PyMem_Malloc((size_t)length * sizeof(int));
        levels = PyMem_Malloc((size_t)length * sizeof(int));
        order = PyMem_Malloc((size_t)length * sizeof(Py_ssize_t));
        if (resolved == NULL || levels == NULL || order == NULL) {
            PyMem_Free(resolved);
            PyMem_Free(levels);
            PyMem_Free(order);
            Py_DECREF(seq);
            return PyErr_NoMemory();
        }
    }

    for (Py_ssize_t i = 0; i < length; ++i) {
```

```
    long code = PyLong_AsLong(PySequence_Fast_GET_ITEM(seq, i));
    if (PyErr_Occurred()) goto fail;
    if (code < BIDI_L || code > BIDI_WS) {
        PyErr_SetString(PyExc_ValueError, "unsupported bidi class code");
        goto fail;
    }
    resolved[i] = normalized_class((int)code);
    order[i] = i;
}

int paragraph_direction = BIDI_L;
for (Py_ssize_t i = 0; i < length; ++i) {
    if (resolved[i] == BIDI_L || resolved[i] == BIDI_R) {
        paragraph_direction = resolved[i];
        break;
    }
}
int paragraph_level = paragraph_direction == BIDI_L ? 0 : 1;

int previous = paragraph_direction;
for (Py_ssize_t i = 0; i < length; ++i) {
    if (resolved[i] == BIDI_NSM) resolved[i] = previous;
    previous = resolved[i];
}

Py_ssize_t index = 0;
while (index < length) {
    if (resolved[index] != BIDI_ON) {
        index += 1;
        continue;
    }
    Py_ssize_t end = index + 1;
    while (end < length && resolved[end] == BIDI_ON) end += 1;

    int before = paragraph_direction;
    for (Py_ssize_t prior = index; prior-- > 0;) {
        if (resolved[prior] == BIDI_L || resolved[prior] == BIDI_R) {
            before = resolved[prior];
            break;
        }
    }
    int after = paragraph_direction;
    for (Py_ssize_t following = end; following < length; ++following) {
        if (resolved[following] == BIDI_L || resolved[following] == BIDI_R) {
            after = resolved[following];
            break;
        }
    }
    int direction = before == after ? before : paragraph_direction;
    for (Py_ssize_t i = index; i < end; ++i) resolved[i] = direction;
    index = end;
}

int max_level = paragraph_level;
int min_odd = -1;
for (Py_ssize_t i = 0; i < length; ++i) {
```

```
        levels[i] = resolved[i] == paragraph_direction ? paragraph_level : paragraph_level + 1;
        if (levels[i] > max_level) max_level = levels[i];
        if ((levels[i] & 1) && (min_odd < 0 || levels[i] < min_odd)) min_odd = levels[i];
    }

    if (min_odd >= 0) {
        for (int level = max_level; level >= min_odd; --level) {
            Py_ssize_t start = 0;
            while (start < length) {
                if (levels[order[start]] < level) {
                    start += 1;
                    continue;
                }
                Py_ssize_t end = start + 1;
                while (end < length && levels[order[end]] >= level) end += 1;
                for (Py_ssize_t left = start, right = end - 1; left < right; ++left, --right) {
                    Py_ssize_t tmp = order[left];
                    order[left] = order[right];
                    order[right] = tmp;
                }
                start = end;
            }
        }
    }

    PyObject *order_tuple = PyTuple_New(length);
    PyObject *levels_tuple = PyTuple_New(length);
    if (order_tuple == NULL || levels_tuple == NULL) {
        Py_XDECREF(order_tuple);
        Py_XDECREF(levels_tuple);
        goto fail;
    }
    for (Py_ssize_t i = 0; i < length; ++i) {
        PyObject *order_value = PyLong_FromSsize_t(order[i]);
        PyObject *level_value = PyLong_FromLong(levels[i]);
        if (order_value == NULL || level_value == NULL) {
            Py_XDECREF(order_value);
            Py_XDECREF(level_value);
            Py_DECREF(order_tuple);
            Py_DECREF(levels_tuple);
            goto fail;
        }
        PyTuple_SET_ITEM(order_tuple, i, order_value);
        PyTuple_SET_ITEM(levels_tuple, i, level_value);
    }

    PyObject *result = PyTuple_Pack(2, order_tuple, levels_tuple);
    Py_DECREF(order_tuple);
    Py_DECREF(levels_tuple);
    PyMem_Free(resolved);
    PyMem_Free(levels);
    PyMem_Free(order);
    Py_DECREF(seq);
    return result;

fail:
```

```
    PyMem_Free(resolved);
    PyMem_Free(levels);
    PyMem_Free(order);
    Py_DECREF(seq);
    return NULL;
}

static PyMethodDef methods[] = {
    {"visual_order", py_visual_order, METH_VARARGS, "Resolve exact-surface bidi levels and visual
    ↪  permutation."},
    {NULL, NULL, 0, NULL}
};

static struct PyModuleDef module = {
    PyModuleDef_HEAD_INIT,
    "_azuzaley_kernel",
    "Compiled Azuzaley exact-layout index kernel.",
    -1,
    methods,
    NULL,
    NULL,
    NULL,
    NULL
};

PyMODINIT_FUNC PyInit__azuzaley_kernel(void) {
    return PyModule_Create(&module);
}
```

ui/exact_text.py

```
from __future__ import annotations

import uharfbuzz as hb
from collections import OrderedDict
from dataclasses import dataclass, field
import unicodedata
from enum import IntEnum
from functools import lru_cache
from pathlib import Path

from PySide6.QtCore import QMarginsF, QPointF, QRectF, QSize, QSizeF, Qt, Signal
from PySide6.QtGui import QFont, QFontDatabase, QGuiApplication, QKeySequence, QPageLayout,
↪  QPageSize, QPainter, QPainterPath, QPalette, QPdfWriter, QTextCursor
from PySide6.QtWidgets import QAbstractScrollArea, QMenu
from fontTools.pens.basePen import BasePen
from fontTools.ttLib import TTFont

from azazuley_veira.config.font_runtime import DEFAULT_POINT_SIZE, grimchain_display_codepoints,
↪  grimchain_face_for_character, physical_font_family, system_font_file_for_codepoint,
↪  system_font_file_for_family

try:
    from ._azuzaley_kernel import visual_order as _compiled_visual_order
except ImportError as exc:
    raise RuntimeError("Azuzaley requires the compiled exact-layout index kernel") from exc

_BIDI_CLASS_CODES = {
    "L": 0,
    "R": 1,
    "AL": 2,
    "ON": 3,
    "NSM": 4,
    "EN": 5,
    "CS": 6,
    "ES": 7,
    "ET": 8,
    "WS": 9,
}
_ALLOWED_BIDI_CLASSES = frozenset(_BIDI_CLASS_CODES)


@lru_cache(maxsize=2048)
def _natural_visual_order(text: str) -> tuple[tuple[int, ...], tuple[int, ...]]:
    classes = tuple(unicodedata.bidirectional(glyph) or "ON" for glyph in text)
    unsupported = tuple(sorted(set(classes) - _ALLOWED_BIDI_CLASSES))
    if unsupported:
        raise RuntimeError(f"unsupported exact-surface bidi classes: {unsupported!r}")
    return _compiled_visual_order(tuple(_BIDI_CLASS_CODES[value] for value in classes))
```

```
def qt_utf16_units(text: str) -> int:
    return len(text.encode("utf-16-le")) // 2

def _qt_boundaries(text: str) -> tuple[int, ...]:
    values=[0]; total=0
    for ch in text:
        total += 2 if ord(ch)>0xFFFF else 1
        values.append(total)
    return tuple(values)

def _py_index_for_qt(text: str, qt_position: int) -> int:
    if qt_position < 0 or qt_position > qt_utf16_units(text): raise ValueError(f"UTF-16 position
    ↪  outside text: {qt_position}")
    total=0
    for i,ch in enumerate(text):
        if total == qt_position: return i
        total += qt_utf16_units(ch)
        if total > qt_position: raise ValueError(f"UTF-16 position splits a code point:
        ↪  {qt_position}")
    if total == qt_position: return len(text)
    raise ValueError(f"UTF-16 position outside text: {qt_position}")

def _slice_qt(text: str, start: int, end: int) -> str:
    return text[_py_index_for_qt(text,start):_py_index_for_qt(text,end)]

class _PainterPathPen(BasePen):
    def __init__(self, glyph_set):
        super().__init__(glyph_set)
        self.path = QPainterPath()

    def _moveTo(self, point): self.path.moveTo(float(point[0]), float(point[1]))
    def _lineTo(self, point): self.path.lineTo(float(point[0]), float(point[1]))
    def _curveToOne(self, p1, p2, p3):
        self.path.cubicTo(float(p1[0]), float(p1[1]), float(p2[0]), float(p2[1]), float(p3[0]),
        ↪  float(p3[1]))
    def _qCurveToOne(self, p1, p2):
        self.path.quadTo(float(p1[0]), float(p1[1]), float(p2[0]), float(p2[1]))
    def _closePath(self): self.path.closeSubpath()


@dataclass(frozen=True, slots=True)
class _OutlineFontResource:
    font: TTFont
    glyph_set: object
    glyph_order: tuple[str, ...]
    upem: int


@lru_cache(maxsize=128)
def _outline_font_resource(font_file: Path) -> _OutlineFontResource:
    font=TTFont(font_file, lazy=False)
    return _OutlineFontResource(font,font.getGlyphSet(),tuple(font.getGlyphOrder()),int(font['head'⌋
    ↪  ].unitsPerEm))
```

```
@lru_cache(maxsize=8192)
def _glyph_path(font_file: Path, glyph_id: int) -> QPainterPath:
    resource=_outline_font_resource(font_file)
    try:
        glyph_name=resource.glyph_order[glyph_id]
    except IndexError as exc:
        raise RuntimeError(f"glyph id is outside physical font: {font_file} -> {glyph_id}") from exc
    pen=_PainterPathPen(resource.glyph_set)
    resource.glyph_set[glyph_name].draw(pen)
    return pen.path


@dataclass(frozen=True, slots=True)
class ExactGlyphRun:
    font_file: Path
    glyph_indexes: tuple[int, ...]
    positions: tuple[QPointF, ...]
    pixel_size: float
    upem: int

    def glyphIndexes(self): return list(self.glyph_indexes)

    def paint(self, painter: QPainter, origin: QPointF) -> None:
        scale=self.pixel_size/self.upem
        for glyph_id,position in zip(self.glyph_indexes,self.positions):
            painter.save()
            painter.translate(origin.x()+position.x(),origin.y()+position.y())
            painter.scale(scale,-scale)
            painter.drawPath(_glyph_path(self.font_file,glyph_id))
            painter.restore()


@dataclass(frozen=True, slots=True)
class _ExactFontResource:
    face: hb.Face
    font: hb.Font
    upem: int


def _open_exact_font_resource(font_file: Path) -> _ExactFontResource:
    try:
        face = hb.Face(font_file.read_bytes())
    except OSError as exc:
        raise RuntimeError(f"HarfBuzz could not open authored font file: {font_file}") from exc
    upem = int(face.upem)
    if upem <= 0:
        raise RuntimeError(f"authored font has invalid units-per-em: {font_file}")
    font = hb.Font(face)
    hb.ot_font_set_funcs(font)
    font.scale = (upem, upem)
    return _ExactFontResource(face, font, upem)


def _close_exact_font_resource(resource: _ExactFontResource) -> None:
```

```
    del resource


def _shape_exact_font(font_file: Path, text: str, pixel_size: float, resource: _ExactFontResource |
↪  None = None) -> tuple[ExactGlyphRun,float,float,float]:
    if not text: raise ValueError("exact shaping requires non-empty text")
    owned_resource=resource is None
    r=_open_exact_font_resource(font_file) if owned_resource else resource
    try:
        buffer=hb.Buffer()
        shape_text = "".join(ch + ("\ufe0e" if ch in "♈♉♊♋♍♎♏♐♑♒♓♄" and (i + 1 ==
        ↪  len(text) or text[i + 1] != "\ufe0e") else "") for i, ch in enumerate(text))
        buffer.add_str(shape_text)
        buffer.guess_segment_properties()
        hb.shape(r.font,buffer)
        infos=buffer.glyph_infos
        positions=buffer.glyph_positions
        scale=pixel_size/r.upem
        gids=[]; points=[]; x=0.0; y=0.0
        for info,position in zip(infos,positions):
            gid=int(info.codepoint)
            if gid==0: raise RuntimeError(f"authored font produced missing glyph for {text!r}:
            ↪  {font_file}")
            gids.append(gid)
            points.append(QPointF(x+position.x_offset*scale,-(y+position.y_offset*scale)))
            x += position.x_advance*scale
            y += position.y_advance*scale
        extents=r.font.get_font_extents("ltr")
        if extents is None:
            raise RuntimeError(f"HarfBuzz could not read font extents: {font_file}")
        run=ExactGlyphRun(font_file,tuple(gids),tuple(points),float(pixel_size),r.upem)
        return run,max(0.0,float(extents.ascender)*scale),max(0.0,-float(extents.descender)*scale),
        ↪  abs(float(x))
    finally:
        if owned_resource:
            _close_exact_font_resource(r)

@dataclass(frozen=True, slots=True)
class ExactSourceRun:
    qt_start:int; text:str; font_file:Path; exact:bool=True
    qt_units:int=field(init=False)
    qt_boundaries:tuple[int,...]=field(init=False)
    def __post_init__(self):
        boundaries=_qt_boundaries(self.text)
        object.__setattr__(self,'qt_boundaries',boundaries)
        object.__setattr__(self,'qt_units',boundaries[-1])
    @property
    def qt_end(self)->int: return self.qt_start+self.qt_units

@dataclass(frozen=True, slots=True)
class ExactPlacedRun:
    qt_start:int; qt_end:int; text:str; font_file:Path; glyph_run:ExactGlyphRun; x:float;
    ↪  baseline:float; advance:float; ascent:float; descent:float; line_index:int; exact:bool;
    ↪  bidi_level:int=0
```

```
@dataclass(frozen=True, slots=True)
class ExactLayoutModel:
    text:str
    runs:tuple[ExactPlacedRun,...]
    width:float
    height:float
    cursor_rects:tuple[tuple[int,QRectF],...]
    cantillation_text:str=""
    cantillation_source_positions:tuple[int,...]=()
    page_rects:tuple[QRectF,...]=()
    content_page_count:int=0

    def __post_init__(self) -> None:
        if len(self.cantillation_text) != len(self.cantillation_source_positions):
            raise ValueError("Cantillation text/source mapping extent mismatch")
        source_by_qt: dict[int, str] = {}
        source_qt = 0
        for character in self.text:
            source_by_qt[source_qt] = character
            source_qt += 2 if ord(character)>0xFFFF else 1
        for character, visual_source_qt in zip(
            self.cantillation_text, self.cantillation_source_positions
        ):
            if source_by_qt.get(visual_source_qt) != character:
                raise ValueError("Cantillation visual position is detached from Prosody source
                ↪  text")

    @property
    def visual_text(self) -> str:
        return self.cantillation_text

    @property
    def visual_source_positions(self) -> tuple[int, ...]:
        return self.cantillation_source_positions

    @property
    def cantillation(self) -> tuple[tuple[str, int], ...]:
        return tuple(zip(self.cantillation_text, self.cantillation_source_positions))

_SHAPE_CACHE_LIMIT = 8192
_SHAPE_CACHE: OrderedDict[tuple[Path, str, float], tuple[ExactGlyphRun, float, float, float]] =
↪  OrderedDict()


class ExactHarfBuzzLayout:
    def __init__(self,pixel_size:float):
        if pixel_size<=0: raise ValueError("exact layout pixel size must be positive")
        self.pixel_size=float(pixel_size)
        self.resources={}
    def shape(self,source:ExactSourceRun):
        key=(source.font_file,source.text,round(self.pixel_size,4))
        shape=_SHAPE_CACHE.get(key)
        if shape is not None:
            _SHAPE_CACHE.move_to_end(key)
            return shape
        resource=self.resources.get(source.font_file)
```

```
    if resource is None:
        resource=_open_exact_font_resource(source.font_file)
        self.resources[source.font_file]=resource
    shape=_shape_exact_font(source.font_file,source.text,self.pixel_size,resource)
    _SHAPE_CACHE[key]=shape
    if len(_SHAPE_CACHE)>_SHAPE_CACHE_LIMIT:
        _SHAPE_CACHE.popitem(last=False)
    return shape
def close(self):
    for resource in self.resources.values():
        _close_exact_font_resource(resource)
    self.resources.clear()
def layout(self,text:str,sources:tuple[ExactSourceRun,...],width:float|None,force_ltr:bool=Fals
↪ e)->ExactLayoutModel:
    if width is not None and width<=0: raise ValueError("exact layout width must be positive")
    text_qt_units=sources[-1].qt_end if sources else 0
    lines=[[]]; widths=[0.0]

    def wrapped_parts(source: ExactSourceRun):
        shape=self.shape(source)
        if width is None or shape[3] <= width or not source.text:
            return ((source,shape),)
        out=[]; remaining=source.text; source_qt=source.qt_start
        while remaining:
            whole=ExactSourceRun(source_qt,remaining,source.font_file,source.exact)
            whole_shape=self.shape(whole)
            if whole_shape[3] <= width:
                out.append((whole,whole_shape)); break
            low=1; high=len(remaining); best=1
            while low <= high:
                mid=(low+high)//2
                probe=ExactSourceRun(source_qt,remaining[:mid],source.font_file,source.exact)
                probe_shape=self.shape(probe)
                if probe_shape[3] <= width:
                    best=mid; low=mid+1
                else:
                    high=mid-1
            whitespace=[index+1 for index,character in enumerate(remaining[:best]) if
            ↪ character.isspace()]
            cut=whitespace[-1] if whitespace else best
            part=ExactSourceRun(source_qt,remaining[:cut],source.font_file,source.exact)
            part_shape=self.shape(part)
            if part_shape[3] > width and cut != best:
                cut=best
                part=ExactSourceRun(source_qt,remaining[:cut],source.font_file,source.exact)
                part_shape=self.shape(part)
            out.append((part,part_shape))
            source_qt += part.qt_units
            remaining=remaining[cut:]
        return tuple(out)

    def append_wrapped(source: ExactSourceRun):
        for part,shape in wrapped_parts(source):
            if width is not None and lines[-1] and widths[-1]+shape[3]>width:
                lines.append([]); widths.append(0.0)
            lines[-1].append((part,shape)); widths[-1]+=shape[3]
```

```
for src in sources:
    if "\n" not in src.text:
        append_wrapped(src)
        continue
    source_qt=src.qt_start
    parts=src.text.split("\n")
    for part_index,segment in enumerate(parts):
        if segment:
            part=ExactSourceRun(source_qt,segment,src.font_file,src.exact)
            append_wrapped(part)
            source_qt += part.qt_units
        if part_index < len(parts)-1:
            source_qt += 1
            lines.append([]); widths.append(0.0)
probe_file=system_font_file_for_family('Noto Sans')
_p,da,dd,_a=self.shape(ExactSourceRun(0,'M',probe_file,False))
placed=[]; cursors={}; y=0.0; maxw=0.0; full_visual=[]; full_visual_sources=[]
for li,items in enumerate(lines):
    logical=[]
    item_ranges=[]
    for item_index,(src,shape) in enumerate(items):
        begin=len(logical)
        for char_index,ch in enumerate(src.text):
            logical.append((ch,src.qt_start+src.qt_boundaries[char_index],item_index))
        item_ranges.append((begin,len(logical)))
    line_text=''.join(cell[0] for cell in logical)
    if force_ltr:
        visual_order=tuple(range(len(logical))); levels=tuple(0 for _ in logical)
    else:
        visual_order,levels=_natural_visual_order(line_text)
    full_visual.extend(logical[i][0] for i in visual_order)
    full_visual_sources.extend(logical[i][1] for i in visual_order)
    l2v={logical_index:visual_index for visual_index,logical_index in
    ↪  enumerate(visual_order)}
    ordered=[]
    for item_index,(src,shape) in enumerate(items):
        begin,end=item_ranges[item_index]
        visual_key=min((l2v[k] for k in range(begin,end)),default=item_index)
        level=levels[begin] if begin < len(levels) else 0
        ordered.append((visual_key,src,shape,level))
    ordered.sort(key=lambda item:item[0])
    asc=max((shape[1] for _key,_src,shape,_level in ordered),default=da)
    desc=max((shape[2] for _key,_src,shape,_level in ordered),default=dd)
    baseline=y+asc; x=0.0
    if not ordered:
        cursors.setdefault(0 if li==0 else text_qt_units,QRectF(0,y,1,asc+desc))
    for _key,src,shape,level in ordered:
        gr,ra,rd,adv=shape
        start=src.qt_start; end=src.qt_end
        placed.append(ExactPlacedRun(start,end,src.text,src.font_file,gr,x,baseline,adv,ra,
        ↪  rd,li,src.exact,level))
        bounds=src.qt_boundaries
        count=max(1,len(bounds)-1)
        for py,rel in enumerate(bounds):
            off=adv*(py/count)
```

```
                    cx=x+off if not (level&1) else x+adv-off
                    cursors[start+rel]=QRectF(cx,y,1,asc+desc)
                x+=adv
            maxw=max(maxw,x); y+=asc+desc
        return ExactLayoutModel(text,tuple(sorted(placed,key=lambda r:r.qt_start)),maxw,max(y,da+dd
        ↪ ),tuple(sorted(cursors.items())),''.join(full_visual),tuple(full_visual_sources))



class _ExactCursor:
    def __init__(self,surface:'ExactFontTextEdit',position:int|None=None,anchor:int|None=None):
        self._surface=surface; end=qt_utf16_units(surface.toPlainText()); self._position=end if
        ↪ position is None else position; self._anchor=self._position if anchor is None else
        ↪ anchor
    def position(self): return self._position
    def anchor(self): return self._anchor
    def hasSelection(self): return self._position != self._anchor
    def clearSelection(self): self._anchor=self._position
    def selectedText(self):
        a,b=sorted((self._position,self._anchor)); return
        ↪ _slice_qt(self._surface.toPlainText(),a,b).replace('\n','\u2029')
    def setPosition(self,pos,mode=QTextCursor.MoveMode.MoveAnchor):
        _py_index_for_qt(self._surface.toPlainText(),pos)
        if mode != QTextCursor.MoveMode.KeepAnchor: self._anchor=pos
        self._position=pos
    def select(self,selection):
        if selection != QTextCursor.SelectionType.Document: raise NotImplementedError
        self._anchor=0; self._position=qt_utf16_units(self._surface.toPlainText())
    def movePosition(self,op,mode=QTextCursor.MoveMode.MoveAnchor,n=1):
        b=_qt_boundaries(self._surface.toPlainText()); i=b.index(self._position)
        if op==QTextCursor.MoveOperation.Start: target=b[0]
        elif op==QTextCursor.MoveOperation.End: target=b[-1]
        elif op in (QTextCursor.MoveOperation.Left,QTextCursor.MoveOperation.PreviousCharacter):
        ↪ target=b[max(0,i-n)]
        elif op in (QTextCursor.MoveOperation.Right,QTextCursor.MoveOperation.NextCharacter):
        ↪ target=b[min(len(b)-1,i+n)]
        else: return False
        self.setPosition(target,mode); return True
    def insertText(self,text): self._surface._replace_selection(self,text)

class ExactFontTextEdit(QAbstractScrollArea):
    textChanged=Signal()
    class LineWrapMode(IntEnum): NoWrap=0; WidgetWidth=1
    def __init__(self,parent=None):
```

```
    super().__init__(parent); self._text=''; self._qt_to_py={0:0}; self._text_qt_units=0;
    ↪ self._exact_spans=[]; self._cursor_position=0; self._cursor_anchor=0;
    ↪ self._read_only=False; self._line_wrap_mode=self.LineWrapMode.WidgetWidth;
    ↪ self._alignment=Qt.AlignmentFlag.AlignLeft; self._force_ltr=False;
    ↪ self._wheel_angle_remainder=0; self._tab_changes_focus=False; self._margin=4.0;
    ↪ self._layout_model=None; self._layout_key=None;
    ↪ self._point_size=float(DEFAULT_POINT_SIZE);
    ↪ self._default_font_file=system_font_file_for_family('Noto Sans');
    ↪ self._scripture_pages=False; self._scripture_page_height=0.0;
    ↪ self._scripture_page_gap=24.0; self._scripture_top_margin=20.0;
    ↪ self._scripture_bottom_margin=20.0; self._scripture_minimum_pages=1;
    ↪ self.setFocusPolicy(Qt.FocusPolicy.StrongFocus)
def setDefaultFontFile(self,p): self._default_font_file=Path(p); self._invalidate_layout()
def defaultFontFile(self): return self._default_font_file
def documentMargin(self): return self._margin
def setDocumentMargin(self,m):
    margin=max(0.0,float(m))
    if margin != self._margin:
        self._margin=margin
        self._invalidate_layout()
def _rebuild_text_index(self):
    qt=0; mapping={0:0}
    for py,ch in enumerate(self._text, start=1):
        qt+=2 if ord(ch)>0xFFFF else 1; mapping[qt]=py
    self._qt_to_py=mapping
    self._text_qt_units=qt
    return qt
def setPlainText(self,text):
    text=str(text)
    if self.property('_strict_grimchain') and any(ord(ch) not in grimchain_display_codepoints()
    ↪ for ch in text):
        raise ValueError('GrimChain surface accepts only authored GrimChain code points')
    self._text=text; self._exact_spans.clear(); end=self._rebuild_text_index();
    ↪ self._cursor_position=min(self._cursor_position,end);
    ↪ self._cursor_anchor=self._cursor_position; self._invalidate_layout();
    ↪ self.textChanged.emit()
def toPlainText(self): return self._text
def clear(self): self.setPlainText('')
def setReadOnly(self,v): self._read_only=bool(v)
def isReadOnly(self): return self._read_only
def setLineWrapMode(self,mode): self._line_wrap_mode=self.LineWrapMode(mode.value if
↪ hasattr(mode,'value') else int(mode)); self._invalidate_layout()
def setForcedLeftToRightLayout(self,v):
    force_ltr=bool(v)
    if force_ltr != self._force_ltr:
        self._force_ltr=force_ltr
        self._invalidate_layout()
def setScripturePageLayout(self,enabled:bool,*,page_height:float=0.0,page_gap:float=24.0,top_ma
↪ rgin:float=20.0,bottom_margin:float=20.0):
    enabled=bool(enabled)
    page_height=max(0.0,float(page_height))
    top_margin=max(0.0,float(top_margin))
    bottom_margin=max(0.0,float(bottom_margin))
    if enabled and page_height <= top_margin + bottom_margin:
        raise ValueError('scripture page height must exceed its vertical text margins')
```

```
    page_gap=max(0.0,float(page_gap))
    geometry=(enabled,page_height,page_gap,top_margin,bottom_margin)
    current=(self._scripture_pages,self._scripture_page_height,self._scripture_page_gap,self._s
    ↪  cripture_top_margin,self._scripture_bottom_margin)
    if geometry != current:
        self._scripture_pages=enabled
        self._scripture_page_height=page_height
        self._scripture_page_gap=page_gap
        self._scripture_top_margin=top_margin
        self._scripture_bottom_margin=bottom_margin
        self._invalidate_layout()
def setScriptureMinimumPageCount(self,count:int):
    count=max(1,int(count))
    if count != self._scripture_minimum_pages:
        self._scripture_minimum_pages=count
        self._invalidate_layout()
def scripturePageLayout(self): return self._scripture_pages
def scripturePageGeometry(self): return self._scripture_page_height,self._scripture_page_gap
def scriptureContentPageCount(self): return self._layout().content_page_count if
↪  self._scripture_pages else 0
def lineWrapMode(self): return self._line_wrap_mode
def setTabChangesFocus(self,v): self._tab_changes_focus=bool(v)
def setAlignment(self,a):
    if a != self._alignment:
        self._alignment=a
        self._invalidate_layout()
def textCursor(self): return _ExactCursor(self,self._cursor_position,self._cursor_anchor)
def setTextCursor(self,c):
    if not isinstance(c,_ExactCursor) or c._surface is not self: raise TypeError('cursor does
    ↪  not belong to exact surface')
    self._cursor_position=c.position(); self._cursor_anchor=c.anchor();
    ↪  self.ensureCursorVisible(); self.viewport().update()
def moveCursor(self,op,mode=QTextCursor.MoveMode.MoveAnchor): c=self.textCursor();
↪  c.movePosition(op,mode); self.setTextCursor(c)
def insertPlainText(self,text):
    if self._read_only: return
    c=self.textCursor(); self._replace_selection(c,text); self.setTextCursor(c)
def selectAll(self): c=self.textCursor(); c.select(QTextCursor.SelectionType.Document);
↪  self.setTextCursor(c)
def copy(self):
    t=self.textCursor().selectedText().replace('\u2029','\n')
    if t: QGuiApplication.clipboard().setText(t)
def cut(self):
    if self._read_only:return
    self.copy(); c=self.textCursor(); self._replace_selection(c,''); self.setTextCursor(c)
def paste(self):
    if not self._read_only:self.insertPlainText(QGuiApplication.clipboard().text())
def _replace_selection(self,c,replacement):
    if self.property('_strict_grimchain') and any(ord(ch) not in grimchain_display_codepoints()
    ↪  for ch in replacement):
        return
```

```
    a,b=sorted((c.position(),c.anchor())); pa=self._qt_to_py[a]; pb=self._qt_to_py[b];
    ↪  self._text=self._text[:pa]+replacement+self._text[pb:]; self._exact_spans.clear();
    ↪  np=a+qt_utf16_units(replacement); self._rebuild_text_index(); c._position=np;
    ↪  c._anchor=np; self._cursor_position=np; self._cursor_anchor=np;
    ↪  self._invalidate_layout(); self.textChanged.emit()
def textForUtf16Range(self,a,b):
    try: return self._text[self._qt_to_py[a]:self._qt_to_py[b]]
    except KeyError as exc: raise ValueError(f'UTF-16 range splits a code point: {(a,b)!r}')
    ↪  from exc
def clear_exact_spans(self): self._exact_spans.clear(); self._invalidate_layout()
def add_exact_font_span(self,qt_start,text,family,font_file):
    del family
    if not text:return
    physical_file=font_file if isinstance(font_file,Path) else Path(font_file)
    run=ExactSourceRun(qt_start,text,physical_file,True)
    if self.textForUtf16Range(qt_start,run.qt_end)!=text: raise ValueError('exact font span
    ↪  text/provenance mismatch')
    self._exact_spans.append(run)
    self._layout_model=None; self._layout_key=None
def exact_spans(self):
    return tuple(sorted(self._exact_spans,key=lambda span:span.qt_start))
def exact_span_at(self,pos):
    for s in self.exact_spans():
        if s.qt_start<=pos<s.qt_end:return s
    return None
def exact_font_family_at(self,pos):
    s=self.exact_span_at(pos); f=self._default_font_file if s is None else s.font_file; return
    ↪  physical_font_family(f)
def setPointSizeF(self,point_size):
    point_size=float(point_size)
    if point_size<=0: raise ValueError('exact font surface requires positive point size')
    if point_size != self._point_size:
        self._point_size=point_size
        self._invalidate_layout()
def pointSizeF(self): return self._point_size
def _pixel_size(self): return self._point_size*self.logicalDpiY()/72.0
def _source_runs(self):
    spans=tuple(sorted(self._exact_spans,key=lambda span:span.qt_start))
    for left,right in zip(spans,spans[1:]):
        if left.qt_end>right.qt_start: raise ValueError('exact font spans overlap')
    out=[]; q=0; p=0; i=0; total=self._text_qt_units
    while p<len(self._text):
        s=spans[i] if i<len(spans) else None
        if s is not None and q==s.qt_start: out.append(s); p+=len(s.text); q=s.qt_end; i+=1;
        ↪  continue
        nq=s.qt_start if s is not None else total; sq=q
        try:
            target_p=self._qt_to_py[nq]
        except KeyError as exc:
            raise ValueError(f'UTF-16 source-run boundary splits a code point: {nq}') from exc
        body=self._text[p:target_p]
        if body: out.append(ExactSourceRun(sq,body,self._default_font_file,False))
        p=target_p; q=nq
    return tuple(out)
def _source_runs_for_qt_range(self,sources,start,end):
```

```
    out=[]
    for source in sources:
        left=max(start,source.qt_start); right=min(end,source.qt_end)
        if left>=right: continue
        local_start=left-source.qt_start; local_end=right-source.qt_start
        body=_slice_qt(source.text,local_start,local_end)
        out.append(ExactSourceRun(left,body,source.font_file,source.exact))
    return tuple(out)
def _content_width(self): return None if self._line_wrap_mode==self.LineWrapMode.NoWrap else
↪  max(1.0,float(self.viewport().width())-self._margin*2)
def _measure_scripture_paragraphs(self,layout,sources,width):
    key=(
        self._text,
        tuple((source.qt_start,source.text,source.font_file,source.exact) for source in
        ↪  sources),
        round(float(width),4),
        round(self._pixel_size(),4),
        self._force_ltr,
    )
    cached=getattr(self,'_scripture_measurement_cache',None)
    if cached is not None and cached[0]==key:
        return cached[1]
    paragraph_ranges=[]; start=0
    for paragraph in self._text.split('\n'):
        end=start+qt_utf16_units(paragraph)
        paragraph_ranges.append((start,end))
        start=end+1
    measured=[]
    probe_height=self.defaultLineHeight()
    for start,end in paragraph_ranges:
        paragraph_sources=self._source_runs_for_qt_range(sources,start,end)
        if paragraph_sources:
            model=layout.layout(self._text,paragraph_sources,width,self._force_ltr)
            height=max(probe_height,model.height)
        else:
            model=None; height=probe_height
        measured.append((start,end,model,height))
    result=(tuple(measured),probe_height)
    self._scripture_measurement_cache=(key,result)
    return result
def _scripture_layout(self,layout,sources,width):
    if width is None:
        width=max(1.0,float(self.viewport().width())-self._margin*2)
    measured,probe_height=self._measure_scripture_paragraphs(layout,sources,width)
    tallest=max((height for _start,_end,_model,height in measured),default=probe_height)
    page_height=self._scripture_page_height
    if self._scripture_top_margin+tallest+self._scripture_bottom_margin>page_height:
        raise ValueError('scripture paragraph exceeds the shared page geometry')
    usable_bottom=page_height-self._scripture_bottom_margin
    page_rects=[]; placed=[]; cursors={}; full_visual=[]; full_visual_sources=[]
    page_index=0; page_origin=0.0; y_in_page=self._scripture_top_margin; line_offset=0;
    ↪  max_width=0.0
    page_rects.append(QRectF(0.0,page_origin,width,page_height))
    for start,end,model,height in measured:
        if y_in_page>self._scripture_top_margin and y_in_page+height>usable_bottom:
```

```
            page_index+=1
            page_origin=page_index*(page_height+self._scripture_page_gap)
            page_rects.append(QRectF(0.0,page_origin,width,page_height))
            y_in_page=self._scripture_top_margin
        dy=page_origin+y_in_page
        if model is None:
            cursors.setdefault(start,QRectF(0.0,dy,1.0,height))
            cursors.setdefault(end,QRectF(0.0,dy,1.0,height))
        else:
            for run in model.runs:
                placed.append(ExactPlacedRun(run.qt_start,run.qt_end,run.text,run.font_file,run
                ↪  .glyph_run,run.x,run.baseline+dy,run.advance,run.ascent,run.descent,run.lin
                ↪  e_index+line_offset,run.exact,run.bidi_level))
            for position,rect in model.cursor_rects:
                if start<=position<=end:
                    cursors[position]=rect.translated(0.0,dy)
            full_visual.append(model.cantillation_text)
            full_visual_sources.extend(model.cantillation_source_positions)
            if model.runs:
                line_offset+=max(run.line_index for run in model.runs)+1
            max_width=max(max_width,model.width)
        y_in_page+=height
    content_page_count=len(page_rects)
    while len(page_rects) < self._scripture_minimum_pages:
        page_origin=len(page_rects)*(page_height+self._scripture_page_gap)
        page_rects.append(QRectF(0.0,page_origin,width,page_height))
    document_height=page_rects[-1].bottom() if page_rects else page_height
    return ExactLayoutModel(self._text,tuple(sorted(placed,key=lambda
    ↪  run:run.qt_start)),max(max_width,width),document_height,tuple(sorted(cursors.items())),
    ↪  ''.join(full_visual),tuple(full_visual_sources),tuple(page_rects),content_page_count)
def _layout(self):
    key=(self._text,tuple((s.qt_start,s.text,s.font_file) for s in
    ↪  self._exact_spans),round(self._pixel_size(),4),self._content_width(),int(self._alignmen
    ↪  t),self._force_ltr,self._scripture_pages,round(self._scripture_page_height,4),round(sel
    ↪  f._scripture_page_gap,4),round(self._scripture_top_margin,4),round(self._scripture_bott
    ↪  om_margin,4),self._scripture_minimum_pages)
    if self._layout_model is not None and self._layout_key==key:return self._layout_model
    layout=ExactHarfBuzzLayout(self._pixel_size())
    try:
        sources=self._source_runs()
        if self._scripture_pages:
            m=self._scripture_layout(layout,sources,self._content_width())
        else:
            m=layout.layout(self._text,sources,self._content_width(),self._force_ltr)
    finally:
        layout.close()
    if self._alignment & (Qt.AlignmentFlag.AlignHCenter | Qt.AlignmentFlag.AlignRight) and
    ↪  m.runs:
        by={}
        for r in m.runs: by.setdefault(r.line_index,[]).append(r)
        shifted=[]; shifts={}; vw=max(1.0,float(self.viewport().width())-self._margin*2)
        for line,runs in by.items():
            line_left=min(r.x for r in runs); line_right=max(r.x+r.advance for r in runs)
            if self._alignment & Qt.AlignmentFlag.AlignRight:
                sh=max(0.0,vw-line_right)
```

```
            else:
                sh=max(0.0,(vw-(line_right-line_left))/2-line_left)
            for r in runs: shifted.append(ExactPlacedRun(r.qt_start,r.qt_end,r.text,r.font_file
            ↪  ,r.glyph_run,r.x+sh,r.baseline,r.advance,r.ascent,r.descent,r.line_index,r.exac
            ↪  t,r.bidi_level))
            for pos,rect in m.cursor_rects:
                if any(r.qt_start<=pos<=r.qt_end for r in runs): shifts[pos]=sh
        m=ExactLayoutModel(m.text,tuple(shifted),m.width,m.height,tuple((pos,rect.translated(sh
        ↪  ifts.get(pos,0),0)) for pos,rect in
        ↪  m.cursor_rects),m.cantillation_text,m.cantillation_source_positions,m.page_rects,m.
        ↪  content_page_count)
    self._layout_model=m; self._layout_key=key; self._update_scrollbars(m); return m
def _invalidate_layout(self): self._layout_model=None; self._layout_key=None;
↪  self.viewport().update()
def _update_scrollbars(self,m):
    w=int(m.width+self._margin*2+0.999); h=int(m.height+self._margin*2+0.999);
    ↪  self.horizontalScrollBar().setRange(0,max(0,w-self.viewport().width()));
    ↪  self.horizontalScrollBar().setPageStep(self.viewport().width());
    ↪  self.verticalScrollBar().setRange(0,max(0,h-self.viewport().height()));
    ↪  self.verticalScrollBar().setPageStep(self.viewport().height())
def defaultLineHeight(self):
    layout=ExactHarfBuzzLayout(self._pixel_size())
    try:
        _r,a,d,_v=layout.shape(ExactSourceRun(0,'M',self._default_font_file,False))
        return a+d
    finally:
        layout.close()
def documentExtent(self): m=self._layout(); return
↪  QSizeF(m.width+self._margin*2,m.height+self._margin*2)
def exactLayoutModel(self): return self._layout()
def cursorRect(self,cursor=None):
    m=self._layout(); pos=self._cursor_position if cursor is None else cursor.position();
    ↪  rect=dict(m.cursor_rects).get(pos,QRectF(0,0,1,self.defaultLineHeight())); return
    ↪  rect.translated(self._margin-self.horizontalScrollBar().value(),self._margin-self.verti
    ↪  calScrollBar().value()).toRect()
def ensureCursorVisible(self):
    r=self.cursorRect(); hs=self.horizontalScrollBar(); vs=self.verticalScrollBar()
    if r.left()<0: hs.setValue(hs.value()+r.left())
    elif r.right()>self.viewport().width():
    ↪  hs.setValue(hs.value()+r.right()-self.viewport().width())
    if r.top()<0: vs.setValue(vs.value()+r.top())
    elif r.bottom()>self.viewport().height():
    ↪  vs.setValue(vs.value()+r.bottom()-self.viewport().height())
def _hit_test(self,p):
    m=self._layout(); c=QPointF(p.x()+self.horizontalScrollBar().value()-self._margin,p.y()+sel
    ↪  f.verticalScrollBar().value()-self._margin); best=0;
    ↪  bd=float('inf')
    for pos,r in m.cursor_rects:
        d=abs(c.x()-r.x())+abs(c.y()-(r.top()+r.height()/2))*2
        if d<bd:bd=d;best=pos
    return best
def resizeEvent(self,e): super().resizeEvent(e); self._invalidate_layout()
def wheelEvent(self,e):
    pixel_delta=e.pixelDelta().y()
    if pixel_delta:
```

```
        bar=self.verticalScrollBar(); bar.setValue(bar.value()-pixel_delta); e.accept(); return
    angle_delta=e.angleDelta().y()
    if angle_delta:
        self._wheel_angle_remainder += angle_delta
        steps=int(self._wheel_angle_remainder/120)
        if steps:
            self._wheel_angle_remainder -= steps*120
            line=max(1,int(round(self.defaultLineHeight())))
            bar=self.verticalScrollBar(); bar.setValue(bar.value()-steps*line)
        e.accept(); return
    super().wheelEvent(e)
def selectionRectangles(self):
    m=self._layout(); ss,se=sorted((self._cursor_position,self._cursor_anchor)); rects=[]
    if ss==se:return tuple(rects)
    for r in m.runs:
        if ss>=r.qt_end or se<=r.qt_start:continue
        ls=max(ss,r.qt_start); le=min(se,r.qt_end)
        if ls==r.qt_start and le==r.qt_end:
            pa=0.0; ca=r.advance
        else:
            bounds=_qt_boundaries(r.text)
            total=max(1,len(bounds)-1)
            start_rel=ls-r.qt_start; end_rel=le-r.qt_start
            start_index=next((i for i,value in enumerate(bounds) if value==start_rel),0)
            end_index=next((i for i,value in enumerate(bounds) if value==end_rel),total)
            pa=r.advance*(start_index/total)
            ca=r.advance*((end_index-start_index)/total)
        sx=r.x+pa if not (r.bidi_level&1) else r.x+r.advance-pa-ca
        rects.append(QRectF(sx,r.baseline-r.ascent,ca,r.ascent+r.descent))
    return tuple(rects)
def paintEvent(self,e):
    m=self._layout(); p=QPainter(self.viewport()); dirty=e.rect(); dirty_f=QRectF(dirty);
    ↪ p.setClipRect(dirty); p.fillRect(dirty,self.palette().color(QPalette.ColorRole.Base));
    ↪ dx=self._margin-self.horizontalScrollBar().value();
    ↪ dy=self._margin-self.verticalScrollBar().value()
    for rect in self.selectionRectangles():
        p.fillRect(rect.translated(dx,dy),self.palette().color(QPalette.ColorRole.Highlight))
    text_color=self.palette().color(QPalette.ColorRole.Text)
    p.setRenderHint(QPainter.RenderHint.Antialiasing, True)
    p.setPen(Qt.PenStyle.NoPen)
    p.setBrush(text_color)
    for r in m.runs:
        painted_bounds=QRectF(r.x+dx,r.baseline-r.ascent+dy,r.advance,r.ascent+r.descent)
        if not painted_bounds.intersects(dirty_f):
            continue
        r.glyph_run.paint(p,QPointF(r.x+dx,r.baseline+dy))
    if self.hasFocus() and not self._read_only:p.fillRect(QRectF(self.cursorRect()),self.palett
    ↪ e().color(QPalette.ColorRole.Text))
    p.end()
def createStandardContextMenu(self):
    menu=QMenu(self)
    copy_action=menu.addAction("Copy"); copy_action.setShortcut(QKeySequence.StandardKey.Copy);
    ↪ copy_action.setEnabled(self.textCursor().hasSelection());
    ↪ copy_action.triggered.connect(self.copy)
```

```
    paste_action=menu.addAction("Paste");
    ↪ paste_action.setShortcut(QKeySequence.StandardKey.Paste); paste_action.setEnabled(not
    ↪ self._read_only); paste_action.triggered.connect(self.paste)
    select_all_action=menu.addAction("Select All");
    ↪ select_all_action.setShortcut(QKeySequence.StandardKey.SelectAll);
    ↪ select_all_action.triggered.connect(self.selectAll)
    return menu
def contextMenuEvent(self,e):
    menu=self.createStandardContextMenu(); menu.exec(e.globalPos())
def mouseDoubleClickEvent(self,e):
    if e.button()==Qt.MouseButton.LeftButton:
        text=self._text
        qt_position=self._hit_test(e.position())
        py_position=self._qt_to_py[qt_position]
        if py_position==len(text) and py_position:py_position-=1
        if py_position<len(text):
            def word_character(character):return character.isalnum() or character=="_" or
            ↪ unicodedata.category(character).startswith("M")
            start=py_position; end=py_position+1
            if word_character(text[py_position]):
                while start>0 and word_character(text[start-1]):start-=1
                while end<len(text) and word_character(text[end]):end+=1
            boundaries=_qt_boundaries(text)
            self._cursor_anchor=boundaries[start]; self._cursor_position=boundaries[end]
        else:self._cursor_anchor=0; self._cursor_position=0
        self.setFocus(); self.viewport().update(); e.accept(); return
    super().mouseDoubleClickEvent(e)
def mousePressEvent(self,e):
    if e.button()==Qt.MouseButton.LeftButton:self._cursor_position=self._hit_test(e.position())
    ↪ ;self._cursor_anchor=self._cursor_position;self.setFocus();self.viewport().update()
    super().mousePressEvent(e)
def mouseMoveEvent(self,e):
    if e.buttons()&Qt.MouseButton.LeftButton:self._cursor_position=self._hit_test(e.position())
    ↪ ;self.viewport().update()
    super().mouseMoveEvent(e)
def keyPressEvent(self,e):
    if e.matches(QKeySequence.StandardKey.SelectAll):self.selectAll();return
    if e.matches(QKeySequence.StandardKey.Copy):self.copy();return
    if e.matches(QKeySequence.StandardKey.Cut):self.cut();return
    if e.matches(QKeySequence.StandardKey.Paste):self.paste();return
    if e.key()==Qt.Key.Key_Tab and self._tab_changes_focus:self.focusNextChild();return
    if self._read_only:super().keyPressEvent(e);return
    c=self.textCursor(); keep=QTextCursor.MoveMode.KeepAnchor if
    ↪ e.modifiers()&Qt.KeyboardModifier.ShiftModifier else QTextCursor.MoveMode.MoveAnchor
    if e.key()==Qt.Key.Key_Left:c.movePosition(QTextCursor.MoveOperation.Left,keep)
    elif e.key()==Qt.Key.Key_Right:c.movePosition(QTextCursor.MoveOperation.Right,keep)
    elif e.key()==Qt.Key.Key_Home:c.movePosition(QTextCursor.MoveOperation.Start,keep)
    elif e.key()==Qt.Key.Key_End:c.movePosition(QTextCursor.MoveOperation.End,keep)
    elif e.key()==Qt.Key.Key_Backspace:
        if not c.hasSelection():c.movePosition(QTextCursor.MoveOperation.Left,QTextCursor.MoveM
        ↪ ode.KeepAnchor)
        self._replace_selection(c,'')
    elif e.key()==Qt.Key.Key_Delete:
        if not c.hasSelection():c.movePosition(QTextCursor.MoveOperation.Right,QTextCursor.Move
        ↪ Mode.KeepAnchor)
```

```
            self._replace_selection(c,'')
        elif e.key() in (Qt.Key.Key_Return,Qt.Key.Key_Enter):self._replace_selection(c,'\n')
        elif e.text() and not(e.modifiers()&(Qt.KeyboardModifier.ControlModifier|Qt.KeyboardModifie
        ↪  r.MetaModifier)):self._replace_selection(c,e.text())
        else:super().keyPressEvent(e);return
        self.setTextCursor(c)
    def sizeHint(self):
        return QSize(160,48)

@lru_cache(maxsize=256)
def _pdf_font_family(font_file: Path) -> str:
    font_id = QFontDatabase.addApplicationFont(str(font_file))
    if font_id < 0:
        raise RuntimeError(f"PDF export could not register authored font: {font_file}")
    families = QFontDatabase.applicationFontFamilies(font_id)
    if not families:
        raise RuntimeError(f"PDF export font has no registered family: {font_file}")
    return families[0]


def _paint_pdf_text_run(painter: QPainter, run: ExactPlacedRun, origin: QPointF) -> None:
    font = QFont(_pdf_font_family(run.font_file))
    font.setPixelSize(max(1, int(round(run.glyph_run.pixel_size))))
    painter.setFont(font)
    painter.drawText(origin, run.text)


def _paint_exact_surface_slice(painter, surface: ExactFontTextEdit, model: ExactLayoutModel,
↪  page_top: float, page_height: float, x_offset: float = 0.0) -> None:
    margin = surface.documentMargin()
    painter.save()
    painter.translate(x_offset + margin, margin - page_top)
    painter.setRenderHint(QPainter.RenderHint.Antialiasing, True)
    painter.setPen(surface.palette().color(QPalette.ColorRole.Text))
    content_top = page_top - margin
    content_bottom = page_top + page_height - margin
    for run in model.runs:
        run_top = run.baseline - run.ascent
        run_bottom = run.baseline + run.descent
        if run_bottom < content_top or run_top > content_bottom:
            continue
        _paint_pdf_text_run(painter, run, QPointF(run.x, run.baseline))
    painter.restore()


def write_exact_surface_groups_pdf(
    groups: tuple[tuple[ExactFontTextEdit, ...], ...],
    path: str | Path,
    *,
    section_headers: tuple[str, ...] | None = None,
) -> Path:
    """Write exact-render surfaces to one PDF, optionally framed by exact section headers."""
    if not groups or any(not group for group in groups):
        raise ValueError("PDF export requires at least one rendered surface group")
    if section_headers is not None and len(section_headers) != len(groups):
        raise ValueError("PDF section-header count must equal rendered group count")
```

```
destination = Path(path)
destination.parent.mkdir(parents=True, exist_ok=True)
writer = QPdfWriter(str(destination))
resolution = max(72, int(round(groups[0][0].logicalDpiY())))
writer.setResolution(resolution)
writer.setPageSize(QPageSize(QPageSize.PageSizeId.Letter))
writer.setPageMargins(QMarginsF(36.0, 36.0, 36.0, 36.0), QPageLayout.Unit.Point)
painter = QPainter(writer)
if not painter.isActive():
    raise RuntimeError(f"could not open PDF for writing: {destination}")
first_page = True
try:
    page_rect = writer.pageLayout().paintRectPixels(writer.resolution())
    for group_index, group in enumerate(groups):
        models = tuple(surface.exactLayoutModel() for surface in group)
        widths = tuple(max(1.0, float(surface.viewport().width()), model.width +
        ↪  surface.documentMargin() * 2) for surface, model in zip(group, models))
        heights = tuple(max(1.0, model.height + surface.documentMargin() * 2) for surface, model
        ↪  in zip(group, models))
        source_width = sum(widths)
        scale = page_rect.width() / source_width
        source_page_height = page_rect.height() / scale
        header_model = None
        header_height = 0.0
        if section_headers is not None:
            header_text = section_headers[group_index]
            header_sources = []
            qt_start = 0
            for character in header_text:
                font_file = (
                    group[0].defaultFontFile()
                    if character == "\n"
                    else system_font_file_for_codepoint(ord(character))
                )
                header_sources.append(ExactSourceRun(qt_start, character, font_file, True))
                qt_start += qt_utf16_units(character)
            header_layout = ExactHarfBuzzLayout(group[0]._pixel_size())
            try:
                margin = group[0].documentMargin()
                header_model = header_layout.layout(
                    header_text, tuple(header_sources), max(1.0, source_width - margin * 2),
                    ↪  True
                )
            finally:
                header_layout.close()
            header_height = header_model.height + margin * 2
            if header_height >= source_page_height:
                raise ValueError("PDF section header exceeds one export page")
        page_top = 0.0
        group_height = max(heights)
        first_section_page = True
        while page_top < group_height:
            if not first_page:
                writer.newPage()
            first_page = False
```

```
                painter.fillRect(QRectF(page_rect),
                ↪  group[0].palette().color(QPalette.ColorRole.Base))
                painter.save()
                painter.translate(page_rect.left(), page_rect.top())
                painter.scale(scale, scale)
                content_offset = header_height if first_section_page else 0.0
                available_height = source_page_height - content_offset
                if first_section_page and header_model is not None:
                    margin = group[0].documentMargin()
                    painter.save()
                    painter.translate(margin, margin)
                    painter.setPen(group[0].palette().color(QPalette.ColorRole.Text))
                    for run in header_model.runs:
                        _paint_pdf_text_run(painter, run, QPointF(run.x, run.baseline))
                    painter.restore()
                    painter.translate(0.0, content_offset)
                x_offset = 0.0
                for surface, model, width in zip(group, models, widths):
                    _paint_exact_surface_slice(painter, surface, model, page_top, available_height,
                    ↪  x_offset)
                    x_offset += width
                painter.restore()
                page_top += available_height
                first_section_page = False
    finally:
        painter.end()
    return destination


def write_exact_surface_pdf(surface: ExactFontTextEdit, path: str | Path) -> Path:
    """Write one existing exact-render surface to a paginated PDF."""
    return write_exact_surface_groups_pdf(((surface,),), path)


def exact_grimchain_text_extent(text:str,point_size:float)->QSizeF:
    if point_size<=0:raise ValueError('GrimChain metric point size must be positive')
    if not text:return QSizeF(0,0)
    screen=QGuiApplication.primaryScreen();dpi=96.0 if screen is None else
    ↪  float(screen.logicalDotsPerInchY());px=point_size*dpi/72.0;width=0.0;height=0.0
    for ch in text:
        face=grimchain_face_for_character(ch);_r,a,d,adv=_shape_exact_font(face.file,ch,px);width+=
        ↪  adv;height=max(height,a+d)
    return QSizeF(width,height)

def _require_exact_surface(widget):
    if not isinstance(widget,ExactFontTextEdit):raise TypeError('authored font rendering requires
    ↪  ExactFontTextEdit')
    return widget

def format_exact_grimchain(widget,text=None):
    s=_require_exact_surface(widget)
    if s.property('_exact_font_formatting'):return
    s.setProperty('_exact_font_formatting',True)
    try:
        text=s.toPlainText() if text is None else text
```

```
        if text!=s.toPlainText():raise ValueError('GrimChain formatter text must match the widget
        ↪  text')
        s.clear_exact_spans();q=0
        i=0
        while i<len(text):
            ch=text[i]
            f=grimchain_face_for_character(ch)
            body=ch
            i+=1
            while i<len(text) and ord(text[i]) == 0xFE0E:
                body+=text[i]
                i+=1
            s.add_exact_font_span(q,body,f.family,f.file)
            q+=qt_utf16_units(body)
    finally:s.setProperty('_exact_font_formatting',False)

def format_exact_grimchain_span(widget,qt_start,text):
    s=_require_exact_surface(widget);off=qt_start
    for ch in text:
        f=grimchain_face_for_character(ch);s.add_exact_font_span(off,ch,f.family,f.file);off+=2 if
        ↪  ord(ch)>0xFFFF else 1

def format_definition_grimchains(widget,text):
    s=_require_exact_surface(widget)
    if text!=s.toPlainText():raise ValueError('Definition formatter text must match the widget
    ↪  text')
    offset=0;prefix='GrimChain: '
    for line in text.splitlines(keepends=True):
        body=line[:-1] if line.endswith('\n') else line
        if body.startswith(prefix):
            value=body[len(prefix):]
            if value:format_exact_grimchain_span(s,qt_utf16_units(text[:offset+len(prefix)]),value)
        offset+=len(line)
```

ui/grimchain_text.py

```
from __future__ import annotations

from PySide6.QtCore import QSizeF
from PySide6.QtGui import QFontMetricsF, QTextCharFormat, QTextCursor
from PySide6.QtWidgets import QPlainTextEdit

from azazuley_veira.config.font_runtime import exact_grimchain_font, grimchain_face_map,
↪  grimchain_face_runs


def qt_utf16_units(text: str) -> int:
    return len(text.encode("utf-16-le")) // 2


def exact_grimchain_text_extent(text: str, point_size: float) -> QSizeF:
    if point_size <= 0:
        raise ValueError("GrimChain metric point size must be positive")
    if not text:
        return QSizeF(0.0, 0.0)
    width = 0.0
    height = 0.0
    for start, end, _face in grimchain_face_runs(text):
        run_text = text[start:end]
        metrics = QFontMetricsF(exact_grimchain_font(run_text[0], point_size))
        width += metrics.horizontalAdvance(run_text)
        height = max(height, metrics.height())
    return QSizeF(width, height)


def _apply_range(widget: QPlainTextEdit, text: str, start: int, end: int) -> None:
    if start < 0 or end < start or end > len(text):
        raise ValueError((start, end, len(text)))
    body = text[start:end]
    if not body:
        return
    point_size = widget.font().pointSizeF()
    for run_start, run_end, _face in grimchain_face_runs(body):
        prefix_start = start + run_start
        prefix_end = start + run_end
        cursor = QTextCursor(widget.document())
        cursor.setPosition(qt_utf16_units(text[:prefix_start]))
        cursor.setPosition(qt_utf16_units(text[:prefix_end]), QTextCursor.MoveMode.KeepAnchor)
        fmt = QTextCharFormat()
        fmt.setFont(exact_grimchain_font(body[run_start], point_size))
        cursor.mergeCharFormat(fmt)


def format_exact_grimchain(widget: QPlainTextEdit, text: str | None = None) -> None:
```

```
    if widget.property("_grimchain_formatting"):
        return
    widget.setProperty("_grimchain_formatting", True)
    try:
        if text is None:
            text = widget.toPlainText()
        if text != widget.toPlainText():
            raise ValueError("GrimChain formatter text must match the widget document")
        original = widget.textCursor()
        position, anchor = original.position(), original.anchor()
        known = grimchain_face_map()
        start: int | None = None
        for index, character in enumerate(text):
            if ord(character) in known:
                if start is None:
                    start = index
            elif start is not None:
                _apply_range(widget, text, start, index)
                start = None
        if start is not None:
            _apply_range(widget, text, start, len(text))
        restored = widget.textCursor()
        restored.setPosition(anchor)
        restored.setPosition(position, QTextCursor.MoveMode.KeepAnchor)
        widget.setTextCursor(restored)
    finally:
        widget.setProperty("_grimchain_formatting", False)


def format_exact_grimchain_span(widget: QPlainTextEdit, qt_start: int, text: str) -> None:
    if not text:
        return
    point_size = widget.font().pointSizeF()
    for run_start, run_end, _face in grimchain_face_runs(text):
        start = qt_start + qt_utf16_units(text[:run_start])
        end = qt_start + qt_utf16_units(text[:run_end])
        cursor = QTextCursor(widget.document())
        cursor.setPosition(start)
        cursor.setPosition(end, QTextCursor.MoveMode.KeepAnchor)
        if cursor.selectedText().replace("\u2029", "\n") != text[run_start:run_end]:
            raise ValueError("GrimChain presentation span text/provenance mismatch")
        fmt = QTextCharFormat()
        fmt.setFont(exact_grimchain_font(text[run_start], point_size))
        cursor.mergeCharFormat(fmt)


def format_definition_grimchains(widget: QPlainTextEdit, text: str) -> None:
    if text != widget.toPlainText():
        raise ValueError("Definition formatter text must match the widget document")
    offset = 0
    prefix = "GrimChain: "
    for line in text.splitlines(keepends=True):
        body = line[:-1] if line.endswith("\n") else line
        if body.startswith(prefix):
            start = offset + len(prefix)
            end = offset + len(body)
```

```
    if end > start:
        _apply_range(widget, text, start, end)
offset += len(line)
```

ui/help_viewer.py

```
"""Read-only Azazuley help document viewer."""
from __future__ import annotations

from pathlib import Path

from PySide6.QtWidgets import QDialog, QVBoxLayout, QTextBrowser

HELP_FILE = Path(__file__).resolve().parents[3] / "docs" / "HELP.md"


class HelpViewer(QDialog):
    """Render the packaged Markdown help as a read-only document."""

    def __init__(self, parent=None) -> None:
        super().__init__(parent)
        self.setWindowTitle("Azazuley Veira Help")
        self.resize(820, 680)

        if not HELP_FILE.is_file():
            raise RuntimeError(f"Azazuley help document is missing: {HELP_FILE}")

        document = QTextBrowser(self)
        document.setOpenExternalLinks(True)
        document.setMarkdown(HELP_FILE.read_text(encoding="utf-8"))

        layout = QVBoxLayout(self)
        layout.addWidget(document)
```

ui/main_window.py

```
from __future__ import annotations

from pathlib import Path

from PySide6.QtCore import QAbstractListModel, QModelIndex, QRegularExpression, Qt
from PySide6.QtGui import QAction, QColor, QPalette, QRegularExpressionValidator, QResizeEvent
from PySide6.QtWidgets import (
    QBoxLayout,
    QColorDialog,
    QComboBox,
    QGraphicsDropShadowEffect,
    QLabel,
    QFileDialog,
    QLineEdit,
    QMainWindow,
    QMenu,
    QPushButton,
    QSizePolicy,
    QVBoxLayout,
    QWidget,
)

from azazuley_veira.ailalubar_render import render_ailalubar
from azazuley_veira.config import font_catalog, visual
from azazuley_veira.config.font_runtime import (
    et_sonyera_font_file,
    installed_font_families,
)
from azazuley_veira.engines import emanation, grimchain, sydonic
from tardisha_grimchain.domus import SYNODIC_CENTER_GLYPHS
from azazuley_veira import grimchain_journal
from azazuley_veira.grimchain_journal import GrimchainHistorySession, append_grimchain,
↪  read_grim_import
from azazuley_veira.ui.help_viewer import HelpViewer
from azazuley_veira.ui.terminal import SubmitTextEdit, Terminal, _format_exact_prose_uncovered
from azazuley_veira.ui.exact_text import write_exact_surface_groups_pdf
from azazuley_veira.ui.path_pdf import write_path_pdf


class _GrimchainHistoryModel(QAbstractListModel):
    PAGE_SIZE = 40

    def __init__(self, parent=None):
        super().__init__(parent)
        self._session: GrimchainHistorySession | None = None
        self._records: list[tuple[str, str, str]] = []

    def rowCount(self, parent=QModelIndex()) -> int:
```

```
        return 0 if parent.isValid() else 1 + len(self._records)

    def data(self, index: QModelIndex, role=Qt.ItemDataRole.DisplayRole):
        if not index.isValid():
            return None
        row = index.row()
        if row == 0:
            if role == Qt.ItemDataRole.DisplayRole:
                return "GrimChain History"
            return None
        if not 1 <= row <= len(self._records):
            return None
        timestamp, user_input, chain = self._records[row - 1]
        if role == Qt.ItemDataRole.DisplayRole:
            return f"{timestamp} | {user_input}"
        if role == Qt.ItemDataRole.UserRole:
            return chain
        return None

    def start_session(self) -> None:
        self.close_session()
        self.beginResetModel()
        self._records = []
        self._session = GrimchainHistorySession()
        self.endResetModel()
        self.fetchMore()

    def close_session(self) -> None:
        if self._session is not None:
            self._session.close()
            self._session = None
        if self._records:
            self.beginResetModel()
            self._records = []
            self.endResetModel()

    def canFetchMore(self, parent=QModelIndex()) -> bool:
        return not parent.isValid() and self._session is not None and self._session.has_more

    def fetchMore(self, parent=QModelIndex()) -> None:
        if parent.isValid() or self._session is None or not self._session.has_more:
            return
        records = self._session.fetch(self.PAGE_SIZE)
        if not records:
            return
        first = 1 + len(self._records)
        last = first + len(records) - 1
        self.beginInsertRows(QModelIndex(), first, last)
        self._records.extend(records)
        self.endInsertRows()


class _GrimchainHistoryComboBox(QComboBox):
    def showPopup(self) -> None:
        model = self.model()
        if isinstance(model, _GrimchainHistoryModel):
```

```
            model.start_session()
        self.setCurrentIndex(0)
        super().showPopup()


class GrimChainInput(SubmitTextEdit):
    def __init__(self, parent=None, preferred_rows: int = 2):
        super().__init__(parent, preferred_rows)
        self.setAttribute(Qt.WidgetAttribute.WA_InputMethodEnabled, True)

    def inputMethodEvent(self, event) -> None:
        if event.commitString():
            self.insertPlainText(event.commitString())
        event.accept()


class MainWindow(QMainWindow):
    """Responsive visible body of the Azazuley Veira terminal."""

    NARROW_WIDTH = 720
    def __init__(self, parent=None):
        super().__init__(parent)
        self.setWindowTitle("Azazuley Veira")
        self.setMinimumSize(360, 300)
        self.resize(1180, 760)

        self.body = QWidget()
        self.body.setObjectName("body")
        self.setCentralWidget(self.body)

        self.layout = QVBoxLayout(self.body)
        self.layout.setSpacing(10)

        self._terminal_bg = visual.TERMINAL_BG
        self._font_color = visual.TEAL
        self._body_bg = visual.TSHIRT_GRAY

        self.header = QBoxLayout(QBoxLayout.Direction.LeftToRight)
        self.header.setSpacing(10)

        self.grimchain_group = QVBoxLayout()
        self.grimchain_group.setSpacing(2)
        self.grimchain_label = QLabel("GrimChain>")
        self.grimchain_label.setObjectName("grimchainHeaderLabel")
        self.grimchain_group.addWidget(self.grimchain_label)
        self.grimchain_input = GrimChainInput(preferred_rows=2)
        self.grimchain_input.setObjectName("grimchainInput")
        self.grimchain_input.setLineWrapMode(SubmitTextEdit.LineWrapMode.WidgetWidth)
        self.grimchain_input.setHorizontalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
        self.grimchain_input.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAsNeeded)
        header_control_height = 28
        self.grimchain_input.setFixedHeight(header_control_height)
        self.grimchain_input.setSizePolicy(QSizePolicy.Policy.Expanding, QSizePolicy.Policy.Fixed)
        self.grimchain_input.submitted.connect(self._submit_grimchain_bar)
        self.grimchain_group.addWidget(self.grimchain_input)
        self.header.addLayout(self.grimchain_group, 1)
```

```
self.domus_group = QVBoxLayout()
self.domus_group.setSpacing(2)
self.domus_label = QLabel("Domus Count>")
self.domus_label.setObjectName("domusHeaderLabel")
self.domus_group.addWidget(self.domus_label)

self.domus_count = QLineEdit()
self.domus_count.setObjectName("domusCount")
self.domus_count.setValidator(
    QRegularExpressionValidator(QRegularExpression(r"^-?[0-9]*$"), self.domus_count)
)
self.domus_count.setPlaceholderText(
    "Please input your choice for the visible Domus Middle"
)
self.domus_count.setFixedWidth(78)
self.domus_count.setFixedHeight(header_control_height)
self.domus_count.setSizePolicy(QSizePolicy.Policy.Fixed, QSizePolicy.Policy.Fixed)
self.domus_group.addWidget(self.domus_count)
self.header.addLayout(self.domus_group)

self.font_box = QComboBox()
self.font_box.setObjectName("fontBox")
self._populate_font_box()
self.font_box.currentTextChanged.connect(self._set_terminal_font)

self.color_group = QVBoxLayout()
self.color_group.setSpacing(2)
self.color_label = QLabel("COLORS:")
self.color_label.setObjectName("colorHeaderLabel")
self.color_group.addWidget(self.color_label)

self.color_button = QPushButton("Choose...")
self.color_button.setObjectName("colorButton")
self.color_button.setMinimumWidth(92)
self.color_button.setMaximumWidth(112)
color_menu = QMenu(self.color_button)
terminal_action = QAction("Terminal Background...", color_menu)
font_action = QAction("Font Color...", color_menu)
body_action = QAction("Body Background...", color_menu)
terminal_action.triggered.connect(lambda: self._choose_color("terminal"))
font_action.triggered.connect(lambda: self._choose_color("font"))
body_action.triggered.connect(lambda: self._choose_color("body"))
color_menu.addAction(terminal_action)
color_menu.addAction(font_action)
color_menu.addAction(body_action)
self.color_button.setMenu(color_menu)
self.color_group.addWidget(self.color_button)
self.header.addLayout(self.color_group)

self.import_button = QPushButton("IMPORT")
self.import_button.setObjectName("importButton")
self.import_button.setMinimumWidth(72)
self.import_button.setMaximumWidth(88)

self.export_button = QPushButton("EXPORT")
```

```
self.export_button.setObjectName("exportButton")
self.export_button.setMinimumWidth(72)
self.export_button.setMaximumWidth(88)

self.help_button = QPushButton("HELP")
self.help_button.setObjectName("helpButton")
self.help_button.setMinimumWidth(72)
self.help_button.setMaximumWidth(88)
self.help_button.clicked.connect(self._show_help)
self.import_button.clicked.connect(self._import_grim_files)
self.export_menu = QMenu(self.export_button)
self._export_bio_action = self.export_menu.addAction("Export GrimChained Bio (.shk)")
self._export_words_action = self.export_menu.addAction("Export Words & Definitions (PDF)")
self._export_glossary_action = self.export_menu.addAction("Export Canon Glossary (PDF)")
self._export_path_out_action = self.export_menu.addAction("Export Path Out (PDF)")
self._export_path_back_action = self.export_menu.addAction("Export Path Back (PDF)")
self.export_menu.addSeparator()
self._export_all_action = self.export_menu.addAction("Export All")
self._export_bio_action.triggered.connect(
    lambda _checked=False: self._run_export(self._export_grimchained_bio)
)
self._export_words_action.triggered.connect(
    lambda _checked=False: self._run_export(self._export_words_definitions_pdf)
)
self._export_glossary_action.triggered.connect(
    lambda _checked=False: self._run_export(self._export_canon_glossary_pdf)
)
self._export_path_out_action.triggered.connect(
    lambda _checked=False: self._run_export(self._export_path_out_pdf)
)
self._export_path_back_action.triggered.connect(
    lambda _checked=False: self._run_export(self._export_path_back_pdf)
)
self._export_all_action.triggered.connect(
    lambda _checked=False: self._run_export(self._export_all)
)
self.export_menu.aboutToShow.connect(self._refresh_export_menu)
self.export_button.setMenu(self.export_menu)

self.top_row = QBoxLayout(QBoxLayout.Direction.LeftToRight)
self.top_row.setContentsMargins(0, 0, 0, 0)
self.top_row.setSpacing(8)
self.grimchain_history = _GrimchainHistoryComboBox()
self.grimchain_history.setObjectName("grimchainHistory")
self.grimchain_history.setMinimumWidth(280)
self.grimchain_history.setMaxVisibleItems(30)
self._grimchain_history_model = _GrimchainHistoryModel(self.grimchain_history)
self.grimchain_history.setModel(self._grimchain_history_model)
self.grimchain_history.activated.connect(self._load_history_grimchain)
self.top_row.addWidget(self.grimchain_history)
self.top_row.addStretch(1)
self.top_row.addWidget(self.import_button)
self.top_row.addWidget(self.export_button)
self.top_row.addWidget(
    self.help_button,
    0,
```

```
        Qt.AlignmentFlag.AlignRight | Qt.AlignmentFlag.AlignVCenter,
    )
    self.layout.addLayout(self.top_row)
    self.layout.addLayout(self.header)

    self.terminal = Terminal()
    self.grimchain_input.textChanged.connect(lambda:
    ↪  _format_exact_prose_uncovered(self.grimchain_input))
    self.terminal.setObjectName("terminal")
    self.terminal.set_user_font_selector(self.font_box)
    self.terminal.user_song_submitted.connect(self._submit_user_song)
    self.domus_count.returnPressed.connect(self._submit_domus_count)

    glow = QGraphicsDropShadowEffect(self.terminal)
    glow.setBlurRadius(24)
    glow.setOffset(0, 0)
    glow.setColor(visual.TEAL_FAINT)
    self.terminal.setGraphicsEffect(glow)
    self.layout.addWidget(self.terminal, 1)

    self._apply_style()
    self._choose_initial_font()
    self._apply_responsive_geometry(self.width())

def _load_history_grimchain(self, index: int) -> None:
    self.statusBar().clearMessage()
    chain = self.grimchain_history.itemData(index, Qt.ItemDataRole.UserRole)
    if isinstance(chain, str) and chain:
        self.grimchain_input.setPlainText(chain)
    self._grimchain_history_model.close_session()
    self.grimchain_history.setCurrentIndex(0)

def _import_grim_files(self) -> None:
    paths, _selected_filter = QFileDialog.getOpenFileNames(
        self, "Import Grim files", "", "Grim files (*.grim)"
    )
    if not paths:
        return
    records: list[tuple[str, str]] = []
    try:
        for path in paths:
            imported = read_grim_import(path)
            for _user_input, chain in imported:
                grimchain.validate(chain)
            records.extend(imported)
    except (OSError, UnicodeError, ValueError) as exc:
        self.terminal.show_error(str(exc))
        return
    for user_input, chain in records:
        append_grimchain(user_input, chain)

def _export_directory(self) -> Path:
    destination = Path.home() / ".grimchain" / "azazuley" / "exports"
    destination.mkdir(parents=True, exist_ok=True)
    return destination
```

```
def _export_depth_text(self) -> str:
    middle_text = self.domus_count.text().strip()
    depth = int(middle_text) if middle_text else 0
    if depth < 0:
        depth = 0
    return str(depth)

def _words_definition_groups(self) -> tuple[tuple[object, ...], ...]:
    return (
        (self.terminal.utterance_output,),
        (self.terminal.ailalubar_output,),
        (self.terminal.azazuley_output,),
        (self.terminal.ailalaza_output,),
        (self.terminal.glossolalia_output,),
        (self.terminal.ailalossolg_output,),
        (self.terminal.leysyff_output, self.terminal.ffysyel_output),
        (self.terminal.definition_output,),
    )

@staticmethod
def _export_section_header(name: str) -> str:
    rule = "⟜⟜" * 27
    return f"{rule}\n{name}\n{rule}\n\n"

def _words_definition_section_headers(self) -> tuple[str, ...]:
    return tuple(
        self._export_section_header(name)
        for name in (
            "Rabulalia",
            "Ailalubar",
            "Azalalia",
            "Ailalaza",
            "Glossolalia",
            "Ailalossolg",
            "LeySyff/FfysYel",
            "Definition",
        )
    )

def _refresh_export_menu(self) -> None:
    active = grimchain_journal.ACTIVE_JOURNAL
    self._export_bio_action.setEnabled(active.is_file() and active.stat().st_size > 0)
    self._export_words_action.setEnabled(
        all(any(bool(surface.toPlainText()) for surface in group) for group in
        ↪  self._words_definition_groups())
    )
    self._export_glossary_action.setEnabled(bool(self.terminal.canon_glossary_output.toPlainTex⌋
    ↪  t()))
    for action, surface in (
        (self._export_path_out_action, self.terminal.path_out_output),
        (self._export_path_back_action, self.terminal.path_back_output),
    ):
        action.setEnabled(surface.model is not None and not surface.model.capture_error)
    self._export_all_action.setEnabled(any(action.isEnabled() for action in (
        self._export_bio_action, self._export_words_action, self._export_glossary_action,
        self._export_path_out_action, self._export_path_back_action,
```

```
    )))

def _run_export(self, operation) -> None:
    try:
        destination = operation()
    except (OSError, RuntimeError, UnicodeError, ValueError) as exc:
        self.terminal.show_error(str(exc))
        return
    self.statusBar().showMessage(f"Exported {destination}")

def _export_grimchained_bio(self) -> Path:
    source = grimchain_journal.ensure_grimchain_journal()
    if source.stat().st_size == 0:
        raise ValueError("there is no active GrimChain .bio to export")
    depth = self._export_depth_text()
    code, chain = grimchain.file(source, depth)
    if code != 0:
        raise RuntimeError(chain)
    destination = self._export_directory() / "grimchains.shk"
    destination.parent.mkdir(parents=True, exist_ok=True)
    original = source.read_bytes()
    prefix = original if original.endswith(b"\n") else original + b"\n"
    temporary = destination.with_name(destination.name + ".tmp")
    try:
        temporary.write_bytes(prefix + chain.encode("utf-8") + b"\n")
        temporary.replace(destination)
    except Exception:
        temporary.unlink(missing_ok=True)
        raise
    return destination

def _embed_export_pdf(self, destination: Path) -> Path:
    code, result = grimchain.pdf_embed(destination, self._export_depth_text())
    if code != 0:
        destination.unlink(missing_ok=True)
        raise RuntimeError(result)
    return destination

def _export_words_definitions_pdf(self) -> Path:
    groups = self._words_definition_groups()
    if not all(any(bool(surface.toPlainText()) for surface in group) for group in groups):
        raise ValueError("render a GrimChain before exporting Words & Definitions")
    destination = self._export_directory() / "Azazuley Words and Definitions.pdf"
    destination.parent.mkdir(parents=True, exist_ok=True)
    write_exact_surface_groups_pdf(
        groups,
        destination,
        section_headers=self._words_definition_section_headers(),
    )
    return self._embed_export_pdf(destination)

def _export_canon_glossary_pdf(self) -> Path:
    destination = self._export_directory() / "Canon Glossary.pdf"
    destination.parent.mkdir(parents=True, exist_ok=True)
    write_exact_surface_groups_pdf(
        ((self.terminal.canon_glossary_output,),),
```

```
        destination,
        section_headers=(self._export_section_header("Canon Glossary"),),
    )
    return self._embed_export_pdf(destination)

def _export_path_out_pdf(self) -> Path:
    destination = self._export_directory() / "path_out.pdf"
    write_path_pdf(self.terminal.path_out_output.model, destination)
    return self._embed_export_pdf(destination)

def _export_path_back_pdf(self) -> Path:
    destination = self._export_directory() / "path_back.pdf"
    write_path_pdf(self.terminal.path_back_output.model, destination, reverse=True)
    return self._embed_export_pdf(destination)

def _export_all(self) -> Path:
    self._refresh_export_menu()
    for action, operation in (
        (self._export_bio_action, self._export_grimchained_bio),
        (self._export_words_action, self._export_words_definitions_pdf),
        (self._export_glossary_action, self._export_canon_glossary_pdf),
        (self._export_path_out_action, self._export_path_out_pdf),
        (self._export_path_back_action, self._export_path_back_pdf),
    ):
        if action.isEnabled():
            operation()
    return self._export_directory()

def _show_help(self) -> None:
    viewer = HelpViewer(self)
    viewer.exec()

def _choose_color(self, target: str) -> None:
    current = {
        "terminal": self._terminal_bg,
        "font": self._font_color,
        "body": self._body_bg,
    }[target]
    title = {
        "terminal": "Terminal Background Color",
        "font": "Font Color",
        "body": "Body Background Color",
    }[target]
    chosen = QColorDialog.getColor(
        QColor(current),
        self,
        title,
        QColorDialog.ColorDialogOption.DontUseNativeDialog,
    )
    if not chosen.isValid():
        return
    value = chosen.name(QColor.NameFormat.HexRgb)
    if target == "terminal":
        self._terminal_bg = value
    elif target == "font":
        self._font_color = value
```

```
    else:
        self._body_bg = value
    self._apply_style()

def _populate_font_box(self) -> None:
    self.font_box.addItem(font_catalog.SYSTEM_DEFAULT)
    for family in installed_font_families():
        self.font_box.addItem(family)

def _choose_initial_font(self) -> None:
    index = self.font_box.findText(font_catalog.SYSTEM_DEFAULT)
    if index < 0:
        raise RuntimeError("system default input font entry is missing")
    self.font_box.setCurrentIndex(index)
    self._set_terminal_font(font_catalog.SYSTEM_DEFAULT)

def _set_terminal_font(self, family: str) -> None:
    resolved_family = font_catalog.DEFAULT_FAMILY if family == font_catalog.SYSTEM_DEFAULT else
    ↪  family
    if resolved_family not in installed_font_families():
        raise RuntimeError(f"user-selected font is not installed: {resolved_family}")
    font_file = et_sonyera_font_file(resolved_family)
    user_song = self.terminal.user_song
    cursor = user_song.textCursor()
    if cursor.hasSelection():
        qt_start, qt_end = sorted((cursor.position(), cursor.anchor()))
        selected_text = user_song.textForUtf16Range(qt_start, qt_end)
        user_song.add_exact_font_span(qt_start, selected_text, resolved_family, font_file)
        return
    user_song.setDefaultFontFile(font_file)

def _submit_grimchain_bar(self, chain: str | None = None) -> None:
    if chain is None:
        chain = self.grimchain_input.toPlainText()
    if not chain:
        return
    user_input = chain
    chain = "".join(ch for ch in user_input if ch in SYNODIC_CENTER_GLYPHS)
    self.statusBar().clearMessage()
    if not chain:
        message = "The GrimChain was malformed, check you GrimSpelling and Grim again"
        self.terminal.show_error(message)
        return

    self._render_selected(chain, chain, cadence_translation=False)
def _submit_domus_count(self) -> None:
    source = self.terminal.user_song.toPlainText()
    if source == "":
        return
    self._submit_user_song(source)

def _submit_user_song(self, source: str) -> None:
    if source == "":
        return

    self.statusBar().clearMessage()
```

```
    try:
        middle_text = self.domus_count.text().strip()
        if not middle_text:
            code, chain = grimchain.string_cli(source, "")
            if code != 0:
                self.terminal.show_error(chain)
                return
            self.grimchain_input.setPlainText(chain)
            self._render_selected(chain, source, cadence_translation=True)
            return
        requested_depth = int(middle_text)
        if requested_depth < 0:
            requested_depth = 0
        session = grimchain.GrimChainContinuationSession(source, requested_depth)
    except Exception as exc:
        self.terminal.show_error(f"grimchain: {exc}")
        return

    try:
        unfold = sydonic._SYDONIC_ENGINE.resolve_grimchain_unfold(
            requested_depth, session.chain_at
        )
        chain = unfold.final_chain
    except grimchain.GrimChainContinuationError as exc:
        self.terminal.show_error(f"grimchain: {exc}")
        return
    except Exception as exc:
        self.terminal.show_error(f"{exc.__class__.__name__}: {exc}")
        return

    try:
        grimchain.validate(chain)
    except ValueError as exc:
        self.terminal.show_error(str(exc))
        return

    self.grimchain_input.setPlainText(chain)
    self._render_selected(chain, source, cadence_translation=False)
    if unfold.additional_depth > 0:
        self.statusBar().showMessage(
            f"Sydonic unfolded +{unfold.additional_depth}: requested {unfold.requested_depth} →
            ↪  "
            f"final {unfold.final_depth}. Full same GrimChain saved to .bio."
        )

def _render_selected(self, chain: str, user_input: str, *, cadence_translation: bool) -> None:
    append_grimchain(user_input, chain)
    chain = emanation.grimchain_emanate(chain)
    self.terminal.set_mirror_utterance(chain)
    self.terminal.set_utterance(chain)

    reflection = render_ailalubar(chain, self.terminal.ailalubar_output.pointSizeF())
    self.terminal.set_ailalubar(reflection)
    mirror_witness = self.terminal.ailalubar_render_witness(reflection)
    self.terminal.path_out_output.set_path(None)
    self.terminal.path_back_output.set_path(None)
```

```
    try:
        transaction = sydonic.render_transaction(mirror_witness,
        ↪  cadence_translation=cadence_translation)
    except Exception as exc:
        self.terminal.show_error(f"{exc.__class__.__name__}: {exc}")
        return

    self._render_transaction = transaction
    self.terminal.path_out_output.set_path(transaction.path_out)
    self.terminal.path_back_output.set_path(transaction.path_out)
    self.terminal.set_azazuley(
        transaction.azalalia_body,
        transaction.azalalia_runs,
    )
    self.terminal.set_ailalaza(
        transaction.ailalaza_body,
        transaction.ailalaza_runs,
    )
    self.terminal.set_glossolalia(transaction.glossolalia_body)
    self.terminal.set_ailalossolg(transaction.ailalossolg_body)
    self.terminal.set_leysyff_presentation(transaction.leysyff)
    self.terminal.set_ffysyel(transaction.ffysyel)
    self.terminal.set_definition_presentation(transaction.definition)

def resizeEvent(self, event: QResizeEvent) -> None:
    super().resizeEvent(event)
    self._apply_responsive_geometry(event.size().width())

def _apply_responsive_geometry(self, width: int) -> None:
    narrow = width < self.NARROW_WIDTH
    if narrow:
        self.header.setDirection(QBoxLayout.Direction.TopToBottom)
        self.font_box.setMinimumWidth(140)
        self.font_box.setMaximumWidth(190)
        self.layout.setContentsMargins(10, 10, 10, 10)
        self.layout.setSpacing(8)
    else:
        self.header.setDirection(QBoxLayout.Direction.LeftToRight)
        self.font_box.setMinimumWidth(140)
        self.font_box.setMaximumWidth(190)
        margin = 20 if width < 1500 else 28
        self.layout.setContentsMargins(margin, 18, margin, margin)
        self.layout.setSpacing(10)

def _apply_style(self) -> None:
    body_palette = self.body.palette()
    body_palette.setColor(QPalette.ColorRole.Window, QColor(self._body_bg))
    body_palette.setColor(QPalette.ColorRole.WindowText, QColor(self._font_color))
    self.body.setAutoFillBackground(True)
    self.body.setPalette(body_palette)

    terminal_palette = self.terminal.palette()
    terminal_palette.setColor(QPalette.ColorRole.Window, QColor(self._terminal_bg))
    terminal_palette.setColor(QPalette.ColorRole.WindowText, QColor(self._font_color))
    self.terminal.setAutoFillBackground(True)
```

```
self.terminal.setPalette(terminal_palette)

for label in (self.grimchain_label, self.domus_label, self.color_label):
    palette = label.palette()
    palette.setColor(QPalette.ColorRole.WindowText, QColor(self._font_color))
    label.setPalette(palette)
    label.setStyleSheet("font-size: 8pt;")

grimchain_palette = self.grimchain_input.palette()
grimchain_palette.setColor(QPalette.ColorRole.Base, QColor(self._terminal_bg))
grimchain_palette.setColor(QPalette.ColorRole.Text, QColor(self._font_color))
grimchain_palette.setColor(QPalette.ColorRole.Highlight, QColor(visual.TEAL_DIM))
grimchain_palette.setColor(QPalette.ColorRole.HighlightedText,
↪  QColor(visual.TERMINAL_BG_DEEP))
self.grimchain_input.setPalette(grimchain_palette)

domus_palette = self.domus_count.palette()
domus_palette.setColor(QPalette.ColorRole.Base, QColor(self._terminal_bg))
domus_palette.setColor(QPalette.ColorRole.Text, QColor(self._font_color))
domus_palette.setColor(QPalette.ColorRole.Highlight, QColor(visual.TEAL_DIM))
domus_palette.setColor(QPalette.ColorRole.HighlightedText, QColor(visual.TERMINAL_BG_DEEP))
self.domus_count.setPalette(domus_palette)

control_palette = self.font_box.palette()
control_palette.setColor(QPalette.ColorRole.Button, QColor(visual.TSHIRT_DARK))
control_palette.setColor(QPalette.ColorRole.Base, QColor(visual.TSHIRT_DARK))
control_palette.setColor(QPalette.ColorRole.ButtonText, QColor(visual.TEAL_SOFT))
control_palette.setColor(QPalette.ColorRole.Text, QColor(visual.TEAL_SOFT))
control_palette.setColor(QPalette.ColorRole.Highlight, QColor(visual.TEAL_DIM))
control_palette.setColor(QPalette.ColorRole.HighlightedText,
↪  QColor(visual.TERMINAL_BG_DEEP))
self.font_box.setPalette(control_palette)
self.color_button.setPalette(control_palette)
self.import_button.setPalette(control_palette)
self.export_button.setPalette(control_palette)
self.help_button.setPalette(control_palette)

self.terminal.set_text_colors(
    self._terminal_bg,
    self._font_color,
    visual.TEAL_DIM,
    visual.TERMINAL_BG_DEEP,
)
self.terminal.set_tab_colors(
    self._terminal_bg,
    self._font_color,
    self._body_bg,
)
```

ui/terminal.py

```
from __future__ import annotations

import unicodedata
from azazuley_veira.ui.path_out import PathBackView, PathOutView

from PySide6.QtCore import QSize, Qt, Signal
from PySide6.QtGui import QColor, QKeyEvent, QPalette
from azazuley_veira.ailalubar_render import AilalubarReflection, AilalubarRenderWitness
from azazuley_veira.config.font_runtime import (
    authored_lexical_font_file,
    grimchain_display_codepoints,
    grimchain_face_for_character,
    physical_font_family,
    system_font_file_for_codepoint,
    system_font_file_for_family,
)
from azazuley_veira.engines.sydonic import (
    BREATH_SEPARATOR,
    OSTENSIVE_PARAGRAPH_PREFIXES,
    DefinitionPresentation,
    LeySyffPresentation,
    PresentationSpan,
)
from azazuley_veira.ui.exact_text import (
    ExactFontTextEdit,
    format_definition_grimchains,
    format_exact_grimchain,
    format_exact_grimchain_span,
    qt_utf16_units,
)

from sydonic_magicae_translation_matrix.canon_dictionary import read_canon_dictionary

from PySide6.QtWidgets import (
    QFrame,
    QHBoxLayout,
    QLabel,
    QScrollBar,
    QSizePolicy,
    QTabWidget,
    QVBoxLayout,
    QWidget,
)


def _format_exact_prose_uncovered(widget: ExactFontTextEdit) -> None:
    """Resolve every uncovered prose code point through the host font system."""
    text = widget.toPlainText()
```

```
    grimchain_codepoints = grimchain_display_codepoints()
    spans = widget.exact_spans()
    span_index = 0
    qt_start = 0
    group_start: int | None = None
    group_text = ""
    group_file = None

    def flush() -> None:
        nonlocal group_start, group_text, group_file
        if group_start is not None and group_text and group_file is not None:
            widget.add_exact_font_span(
                group_start,
                group_text,
                physical_font_family(group_file),
                group_file,
            )
        group_start = None
        group_text = ""
        group_file = None

    for character in text:
        units = qt_utf16_units(character)
        while span_index < len(spans) and spans[span_index].qt_end <= qt_start:
            span_index += 1
        covered = (
            span_index < len(spans)
            and spans[span_index].qt_start <= qt_start < spans[span_index].qt_end
        )
        if character == "\n" or covered:
            flush()
            qt_start += units
            continue

        if ord(character) in grimchain_codepoints:
            exact_file = grimchain_face_for_character(character).file
        else:
            exact_file = system_font_file_for_codepoint(ord(character))

        if group_file is not None and exact_file != group_file:
            flush()
        if group_start is None:
            group_start = qt_start
            group_file = exact_file
        group_text += character
        qt_start += units
    flush()


def _format_ostensive_paragraph_markers(widget: ExactFontTextEdit) -> None:
    """Paint RTL paragraph markers through the existing exact system-codepoint authority."""
    text = widget.toPlainText()
    qt_start = 0
    for character in text:
        if character in OSTENSIVE_PARAGRAPH_PREFIXES:
            font_file = system_font_file_for_codepoint(ord(character))
```

```
            widget.add_exact_font_span(
                qt_start, character, physical_font_family(font_file), font_file
            )
        qt_start += qt_utf16_units(character)


class SubmitTextEdit(ExactFontTextEdit):
    submitted = Signal(str)

    def __init__(self, parent=None, preferred_rows: int = 3):
        super().__init__(parent)
        self._preferred_rows = preferred_rows

    def sizeHint(self) -> QSize:
        base = super().sizeHint()
        line_height = self.defaultLineHeight()
        document_margin = float(self.documentMargin()) * 2.0
        frame = float(self.frameWidth()) * 2.0
        margins = self.viewportMargins()
        height = line_height * self._preferred_rows + document_margin + frame + margins.top() +
        ↪  margins.bottom()
        return QSize(base.width(), round(height))

    def minimumSizeHint(self) -> QSize:
        return self.sizeHint()
    def keyPressEvent(self, event: QKeyEvent) -> None:
        if event.key() in (Qt.Key.Key_Return, Qt.Key.Key_Enter):
            self.submitted.emit(self.toPlainText())
            return
        super().keyPressEvent(event)




class SingleLineGrimChainEdit(ExactFontTextEdit):
    """Read-only single-line document surface with mixed exact GrimChain faces."""

    def __init__(self, parent=None):
        super().__init__(parent)
        self.setReadOnly(True)
        self.setLineWrapMode(ExactFontTextEdit.LineWrapMode.NoWrap)
        self.setHorizontalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
        self.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)

    def setText(self, value: str) -> None:
        self.setPlainText(value)
        format_exact_grimchain(self, value)

    def text(self) -> str:
        return self.toPlainText()

    def setAlignment(self, alignment: Qt.AlignmentFlag) -> None:
        super().setAlignment(alignment)


class Terminal(QWidget):
```

```
user_song_submitted = Signal(str)

def __init__(self, parent=None):
    super().__init__(parent)
    layout = QVBoxLayout(self)
    layout.setContentsMargins(14, 14, 14, 14)
    layout.setSpacing(8)

    self.user_song = SubmitTextEdit()
    self.user_song.setObjectName("userSong")
    self._frame_field(self.user_song)
    self.user_song.setLineWrapMode(ExactFontTextEdit.LineWrapMode.WidgetWidth)
    self.user_song.setTabChangesFocus(True)
    self.user_song.setSizePolicy(QSizePolicy.Policy.Expanding, QSizePolicy.Policy.Maximum)
    self.user_song.submitted.connect(self.user_song_submitted)
    self.user_header = QHBoxLayout()
    self.user_header.addWidget(self._label("Et Sonyera>"))
    self.user_header.addStretch(1)
    layout.addLayout(self.user_header)
    layout.addWidget(self.user_song)

    daemon_header = QHBoxLayout()
    daemon_header.setSpacing(12)
    daemon_header.addWidget(self._label("Domus House>"))
    daemon_header.addStretch(1)
    layout.addLayout(daemon_header)

    daemon_row = QHBoxLayout()
    daemon_row.setSpacing(0)
    self.daemon_first = self._daemon_part()
    self.daemon_middle = self._daemon_middle_part()
    self.daemon_middle_second = self._daemon_middle_part()
    self.daemon_second = self._daemon_part()
    domus_field_width = 244
    domus_field_height = 44
    for field in (
        self.daemon_first,
        self.daemon_middle,
        self.daemon_middle_second,
        self.daemon_second,
    ):
        field.setFixedSize(domus_field_width, domus_field_height)

    left_group = QWidget()
    left_group_layout = QHBoxLayout(left_group)
    left_group_layout.setContentsMargins(0, 0, 0, 0)
    left_group_layout.setSpacing(0)
    left_group_layout.addWidget(self.daemon_first)

    center_stack = QWidget()
    center_stack_layout = QVBoxLayout(center_stack)
    center_stack_layout.setContentsMargins(0, 0, 0, 0)
    center_stack_layout.setSpacing(6)
    center_stack_layout.addWidget(self.daemon_middle)
    center_stack_layout.addWidget(self.daemon_middle_second)
```

```
right_group = QWidget()
right_group_layout = QHBoxLayout(right_group)
right_group_layout.setContentsMargins(0, 0, 0, 0)
right_group_layout.setSpacing(0)
right_group_layout.addWidget(self.daemon_second)

daemon_row.addStretch(1)
daemon_row.addWidget(left_group, 0, Qt.AlignmentFlag.AlignVCenter)
daemon_row.addSpacing(10)
daemon_row.addWidget(center_stack, 0, Qt.AlignmentFlag.AlignVCenter)
daemon_row.addSpacing(10)
daemon_row.addWidget(right_group, 0, Qt.AlignmentFlag.AlignVCenter)
daemon_row.addStretch(1)
layout.addLayout(daemon_row)

self.output_tabs = QTabWidget()
self.output_tabs.setObjectName("outputTabs")

self.utterance_output = self._output_field()
self.ailalubar_output = self._output_field()
self.azazuley_output = self._output_field()
self.ailalaza_output = self._output_field()
self.definition_output = self._output_field()
self.canon_glossary_output = self._output_field()
self.leysyff_output = self._output_field()
self.ffysyel_output = self._output_field()
self.glossolalia_output = self._output_field()
self.ailalossolg_output = self._output_field()
self.path_out_output = PathOutView()
self.path_out_output.setObjectName("pathOutOutput")
self.path_back_output = PathBackView()
self.path_back_output.setObjectName("pathBackOutput")

self.leysyff_output.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
self.ffysyel_output.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
self.leysyff_output.setFrameShape(QFrame.Shape.NoFrame)
self.ffysyel_output.setFrameShape(QFrame.Shape.NoFrame)
initial_palette = self.leysyff_output.palette()
self._set_scripture_surface_colors(
    initial_palette.color(QPalette.ColorRole.Base).name(),
    initial_palette.color(QPalette.ColorRole.Text).name(),
)

self.leysyff_ffysyel_output = QWidget()
self.leysyff_ffysyel_output.setObjectName("leysyffFfysyelOutput")
scripture_layout = QHBoxLayout(self.leysyff_ffysyel_output)
scripture_layout.setContentsMargins(0, 0, 0, 0)
scripture_layout.setSpacing(0)
scripture_layout.addWidget(self.leysyff_output, 1)
scripture_layout.addWidget(self.ffysyel_output, 1)
self.scripture_scrollbar = QScrollBar(Qt.Orientation.Vertical)
self.scripture_scrollbar.setObjectName("scriptureScrollBar")
scripture_layout.addWidget(self.scripture_scrollbar)
self.scripture_scrollbar.valueChanged.connect(self._set_scripture_scroll_value)
self.leysyff_output.verticalScrollBar().valueChanged.connect(
    self._sync_scripture_scroll_from_side
```

```
    )
    self.ffysyel_output.verticalScrollBar().valueChanged.connect(
        self._sync_scripture_scroll_from_side
    )

    self.output_tabs.addTab(self.utterance_output, "Rabulalia")
    self.output_tabs.addTab(self.ailalubar_output, "Ailalubar")
    self.output_tabs.addTab(self.azazuley_output, "Azalalia")
    self.output_tabs.addTab(self.ailalaza_output, "Ailalaza")
    self.output_tabs.addTab(self.glossolalia_output, "Glossolalia")
    self.output_tabs.addTab(self.ailalossolg_output, "Ailalossolg")
    self.output_tabs.addTab(self.leysyff_ffysyel_output, "LeySyff | FfysYel")
    self.output_tabs.addTab(self.definition_output, "Definition")
    self.output_tabs.addTab(self.canon_glossary_output, "Canon Glossary")
    self.output_tabs.addTab(self.path_out_output, "Path Out")
    self.output_tabs.addTab(self.path_back_output, "Path Back")
    self.output_tabs.currentChanged.connect(self._refresh_scripture_tab_geometry)
    layout.addWidget(self.output_tabs, 1)

    prose_file = system_font_file_for_family("Noto Sans")
    self.user_song.setDefaultFontFile(prose_file)
    for field in (
        self.daemon_first,
        self.daemon_middle,
        self.daemon_middle_second,
        self.daemon_second,
        self.utterance_output,
        self.ailalubar_output,
        self.azazuley_output,
        self.ailalaza_output,
        self.leysyff_output,
        self.ffysyel_output,
        self.definition_output,
        self.canon_glossary_output,
        self.glossolalia_output,
        self.ailalossolg_output,
    ):
        field.setDefaultFontFile(prose_file)

    self.canon_glossary_output.setPlainText(read_canon_dictionary())
    _format_exact_prose_uncovered(self.canon_glossary_output)

    self._refresh_scripture_geometry()

def set_user_font_selector(self, selector: QWidget) -> None:
    """Place the Et Sonyera font selector at the right edge of its label row."""
    self.user_header.addWidget(selector, 0, Qt.AlignmentFlag.AlignRight)

def _daemon_part(self) -> SingleLineGrimChainEdit:
    field = SingleLineGrimChainEdit()
    field.setFrameShape(QFrame.Shape.NoFrame)
    field.setAlignment(Qt.AlignmentFlag.AlignCenter)
    return field

def _daemon_middle_part(self) -> SingleLineGrimChainEdit:
    field = SingleLineGrimChainEdit()
```

```
    field.setFrameShape(QFrame.Shape.NoFrame)
    field.setAlignment(Qt.AlignmentFlag.AlignCenter)
    return field

def _frame_field(self, field: ExactFontTextEdit) -> None:
    field.setFrameShape(QFrame.Shape.Box)
    field.setFrameShadow(QFrame.Shadow.Plain)
    field.setLineWidth(1)

def _label(self, text: str) -> QLabel:
    label = QLabel(text)
    label.setObjectName("fieldPrompt")
    return label

def _output_field(self) -> ExactFontTextEdit:
    field = ExactFontTextEdit()
    field.setObjectName("resultField")
    self._frame_field(field)
    field.setReadOnly(True)
    field.setLineWrapMode(ExactFontTextEdit.LineWrapMode.WidgetWidth)
    field.setHorizontalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
    field.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOn)
    field.setSizePolicy(QSizePolicy.Policy.Expanding, QSizePolicy.Policy.Expanding)
    return field

def _set_scripture_surface_colors(self, background: str, text: str) -> None:
    self.leysyff_output.setStyleSheet(
        f"border: 1px solid {text}; border-right-width: 0px; "
        f"background: {background}; color: {text};"
    )
    self.ffysyel_output.setStyleSheet(
        f"border: 1px solid {text}; background: {background}; color: {text};"
    )

def _set_scripture_scroll_value(self, value: int) -> None:
    for field in (self.leysyff_output, self.ffysyel_output):
        bar = field.verticalScrollBar()
        if bar.value() != value:
            bar.setValue(value)

def _sync_scripture_scroll_from_side(self, value: int) -> None:
    if self.scripture_scrollbar.value() != value:
        self.scripture_scrollbar.setValue(value)

def _refresh_scripture_scroll_range(self) -> None:
    self.leysyff_output.exactLayoutModel()
    self.ffysyel_output.exactLayoutModel()
    left_bar = self.leysyff_output.verticalScrollBar()
    right_bar = self.ffysyel_output.verticalScrollBar()
    maximum = max(left_bar.maximum(), right_bar.maximum())
    viewport_height = max(1, min(
        self.leysyff_output.viewport().height(),
        self.ffysyel_output.viewport().height(),
    ))
    single_step = max(1, round(self.leysyff_output.defaultLineHeight()))
    current = min(self.scripture_scrollbar.value(), maximum)
```

```
    self.scripture_scrollbar.setRange(0, maximum)
    self.scripture_scrollbar.setPageStep(viewport_height)
    self.scripture_scrollbar.setSingleStep(single_step)
    self.scripture_scrollbar.setValue(current)
    self._set_scripture_scroll_value(current)

def _refresh_scripture_tab_geometry(self, _index: int) -> None:
    if self.output_tabs.currentWidget() is self.leysyff_ffysyel_output:
        self._refresh_scripture_geometry()

def _scripture_scroll_anchor(self) -> tuple[int, float] | None:
    if not (self.leysyff_output.scripturePageLayout() and
    ↪  self.ffysyel_output.scripturePageLayout()):
        return None
    page_height, page_gap = self.leysyff_output.scripturePageGeometry()
    if page_height <= 0:
        return None
    scroll_value = max(0.0, float(self.scripture_scrollbar.value()))
    stride = page_height + page_gap
    page_index = int(scroll_value // stride) if stride > 0 else 0
    page_top = page_index * stride
    fraction = max(0.0, min(1.0, (scroll_value - page_top) / page_height))
    return page_index, fraction

def _restore_scripture_scroll_anchor(self, anchor: tuple[int, float] | None) -> None:
    if anchor is None:
        return
    page_height, page_gap = self.leysyff_output.scripturePageGeometry()
    if page_height <= 0:
        return
    page_index, fraction = anchor
    value = round(page_index * (page_height + page_gap) + page_height * fraction)
    self.scripture_scrollbar.setValue(min(value, self.scripture_scrollbar.maximum()))

def _refresh_scripture_geometry(self) -> None:
    anchor = self._scripture_scroll_anchor()
    scrollbar_width = max(0.0, float(self.scripture_scrollbar.sizeHint().width()))
    spread_width = max(1.0, float(self.leysyff_ffysyel_output.width()) - scrollbar_width)
    text_margin = max(4.0, spread_width * 0.025)
    vertical_margin = 8.0
    page_gap = 12.0
    point_size = max(8.0, min(14.0, spread_width * 0.0105))
    page_height = (spread_width / 2.0) * 1.5

    for field in (self.leysyff_output, self.ffysyel_output):
        field.setDocumentMargin(text_margin)
        field.setPointSizeF(point_size)

    for field in (self.leysyff_output, self.ffysyel_output):
        field.setScripturePageLayout(
            True,
            page_height=page_height,
            page_gap=page_gap,
            top_margin=vertical_margin,
            bottom_margin=vertical_margin,
```

```
        )

    shared_page_count = max(
        self.leysyff_output.scriptureContentPageCount(),
        self.ffysyel_output.scriptureContentPageCount(),
    )
    self.leysyff_output.setScriptureMinimumPageCount(shared_page_count)
    self.ffysyel_output.setScriptureMinimumPageCount(shared_page_count)
    self._refresh_scripture_scroll_range()
    self._restore_scripture_scroll_anchor(anchor)

def resizeEvent(self, event) -> None:
    super().resizeEvent(event)
    if hasattr(self, "leysyff_ffysyel_output"):
        self._refresh_scripture_geometry()

def _content_widgets(self) -> tuple[QWidget, ...]:
    return (
        self.user_song,
        self.daemon_first,
        self.daemon_middle,
        self.daemon_middle_second,
        self.daemon_second,
        self.utterance_output,
        self.ailalubar_output,
        self.azazuley_output,
        self.ailalaza_output,
        self.leysyff_output,
        self.ffysyel_output,
        self.definition_output,
        self.canon_glossary_output,
        self.glossolalia_output,
        self.ailalossolg_output,
    )

def set_text_colors(
    self,
    background: str,
    text: str,
    selection_background: str,
    selection_text: str,
) -> None:
    for widget in self._content_widgets():
        palette = widget.palette()
        palette.setColor(QPalette.ColorRole.Base, QColor(background))
        palette.setColor(QPalette.ColorRole.Text, QColor(text))
        palette.setColor(QPalette.ColorRole.Highlight, QColor(selection_background))
        palette.setColor(QPalette.ColorRole.HighlightedText, QColor(selection_text))
        widget.setPalette(palette)

    self._set_scripture_surface_colors(background, text)

    mirror_text = QColor(text).darker(165)
    for mirror in (
        self.daemon_first,
        self.daemon_second,
```

```
    ):
        palette = mirror.palette()
        palette.setColor(QPalette.ColorRole.Text, mirror_text)
        mirror.setPalette(palette)

def set_tab_colors(self, background: str, text: str, body: str) -> None:
    self.output_tabs.tabBar().setStyleSheet(
        "QTabBar::tab {"
        f"background: {body};"
        f"color: {text};"
        "border: 1px solid " + text + ";"
        "padding: 5px 12px;"
        "}"
        "QTabBar::tab:selected {"
        f"background: {background};"
        f"color: {text};"
        "}"
    )

def set_daemon_tongue(self, first: str, middle: str, second: str, middle_second: str | None =
↪  None) -> None:
    self.daemon_first.setText(first)
    self.daemon_middle.setText(middle)
    self.daemon_middle_second.setText(middle if middle_second is None else middle_second)
    self.daemon_second.setText(second)

def set_mirror_utterance(self, grimchain: str) -> None:
    """Project the finite source into the authored four-field 12-glyph house."""
    left_reflection = grimchain[:12][::-1]
    right_reflection = grimchain[-12:][::-1]
    self.set_daemon_tongue(
        left_reflection,
        grimchain[:12],
        right_reflection,
        grimchain[-12:],
    )

def set_utterance(self, value: str) -> None:
    self.utterance_output.setPlainText(value)
    format_exact_grimchain(self.utterance_output, value)

def set_ailalubar(self, reflection: AilalubarReflection) -> None:
    """Paint exactly one reflected Prosody string under normal exact-font rules."""
    if not isinstance(reflection, AilalubarReflection):
        raise TypeError("Ailalubar requires its exact reflection authority")
    self.ailalubar_output.setPlainText(reflection.mirrored_source)
    format_exact_grimchain(self.ailalubar_output, reflection.mirrored_source)

def ailalubar_render_witness(self, reflection: AilalubarReflection) -> AilalubarRenderWitness:
    """Mint the final witness from the exact layout model that actually paints Ailalubar."""
    if not isinstance(reflection, AilalubarReflection):
        raise TypeError("Ailalubar witness requires its exact reflection authority")
    if self.ailalubar_output.toPlainText() != reflection.mirrored_source:
        raise ValueError("Ailalubar surface is detached from its reflection authority")

    model = self.ailalubar_output.exactLayoutModel()
```

```
    mirrored = reflection.mirrored_source
    boundaries: dict[int, int] = {}
    qt = 0
    for index, character in enumerate(mirrored):
        boundaries[qt] = index
        qt += qt_utf16_units(character)

    level_by_qt: dict[int, int] = {}
    for run in model.runs:
        run_qt = run.qt_start
        for character in run.text:
            level_by_qt[run_qt] = run.bidi_level
            run_qt += qt_utf16_units(character)

    source_positions: list[int] = []
    levels: list[int] = []
    mirrored_flags: list[bool] = []
    for character, mirrored_qt in model.cantillation:
        try:
            mirrored_index = boundaries[mirrored_qt]
        except KeyError as exc:
            raise ValueError(
                f"Cantillation position is not a Prosody code-point boundary: {mirrored_qt}"
            ) from exc
        source_positions.append(len(reflection.source) - 1 - mirrored_index)
        try:
            bidi_level = level_by_qt[mirrored_qt]
        except KeyError as exc:
            raise ValueError(
                f"Cantillation position has no exact layout run: {mirrored_qt}"
            ) from exc
        levels.append(bidi_level)
        # Metadata only: this never selects, substitutes, flips, or reverses a glyph.
        mirrored_flags.append(
            bool(unicodedata.mirrored(character) and bidi_level & 1)
        )

    return AilalubarRenderWitness(
        source=reflection.source,
        mirrored_source=reflection.mirrored_source,
        visual_text=model.cantillation_text,
        visual_source_positions=tuple(source_positions),
        visual_levels=tuple(levels),
        visual_mirrored_flags=tuple(mirrored_flags),
        source_face_ids=reflection.source_face_ids,
        source_face_runs=reflection.source_face_runs,
        rendering_authority_sha256=reflection.rendering_authority_sha256,
        point_size=reflection.point_size,
        layout_direction=reflection.layout_direction,
    )

def _apply_presentation_spans(
    self,
    widget: ExactFontTextEdit,
    spans: tuple[PresentationSpan, ...],
    allowed_roles: frozenset[str],
```

```
) -> None:
    document_units = len(widget.toPlainText().encode("utf-16-le")) // 2
    for span in spans:
        if span.semantic_role not in allowed_roles:
            raise ValueError(f"unsupported presentation role: {span.semantic_role!r}")
        if span.qt_end > document_units:
            raise ValueError(f"presentation span exceeds document: {span!r}")
        if widget.textForUtf16Range(span.qt_start, span.qt_end) != span.rendered_text:
            raise ValueError(f"presentation span text/provenance mismatch: {span!r}")
        if not isinstance(widget, ExactFontTextEdit):
            raise TypeError("authored lexical rendering requires ExactFontTextEdit")
        font_file = authored_lexical_font_file(span.font_family, span.source_authority)
        widget.add_exact_font_span(span.qt_start, span.rendered_text, span.font_family,
        ↪  font_file)

def _set_written_projection(
    self,
    widget: ExactFontTextEdit,
    value: str,
    runs: tuple[PresentationSpan, ...],
) -> None:
    widget.setPlainText(value)
    breath_face = grimchain_face_for_character(BREATH_SEPARATOR)
    for index, span in enumerate(runs):
        font_file = authored_lexical_font_file(span.font_family, span.source_authority)
        widget.add_exact_font_span(span.qt_start, span.rendered_text, span.font_family,
        ↪  font_file)
        if index + 1 < len(runs):
            widget.add_exact_font_span(span.qt_end, BREATH_SEPARATOR, breath_face.family,
            ↪  breath_face.file)

def set_azazuley(self, value: str, runs: tuple[PresentationSpan, ...] = ()) -> None:
    self._set_written_projection(self.azazuley_output, value, runs)

def set_ailalaza(self, value: str, runs: tuple[PresentationSpan, ...] = ()) -> None:
    self._set_written_projection(self.ailalaza_output, value, runs)

def set_glossolalia(self, value: str) -> None:
    self.glossolalia_output.setPlainText(value)
    _format_exact_prose_uncovered(self.glossolalia_output)

def set_ailalossolg(self, value: str) -> None:
    self.ailalossolg_output.setPlainText(value)
    _format_exact_prose_uncovered(self.ailalossolg_output)

def set_definition(self, value: str, runs: tuple[PresentationSpan, ...] = ()) -> None:
    self.definition_output.setPlainText(value)
    self._apply_presentation_spans(
        self.definition_output, runs, frozenset({"definition_word"})
    )
    format_definition_grimchains(self.definition_output, value)
    _format_exact_prose_uncovered(self.definition_output)

def set_definition_presentation(self, presentation: DefinitionPresentation) -> None:
    self.definition_output.setPlainText(presentation.text)
```

```
    self._apply_presentation_spans(
        self.definition_output, presentation.lexical_spans, frozenset({"definition_word"})
    )
    for entry in presentation.entries:
        for field in entry.fields:
            if field.semantic_role == "grimchain" and field.value:
                format_exact_grimchain_span(
                    self.definition_output, field.qt_value_start, field.value
                )
    _format_exact_prose_uncovered(self.definition_output)

def set_leysyff_presentation(self, presentation: LeySyffPresentation) -> None:
    self.leysyff_output.setPlainText(presentation.text)
    self.leysyff_output.setForcedLeftToRightLayout(True)
    self.leysyff_output.setAlignment(Qt.AlignmentFlag.AlignLeft)
    _format_ostensive_paragraph_markers(self.leysyff_output)
    _format_exact_prose_uncovered(self.leysyff_output)
    self._refresh_scripture_geometry()

def set_ffysyel(self, presentation: LeySyffPresentation) -> None:
    self.ffysyel_output.setPlainText(presentation.text)
    self.ffysyel_output.setForcedLeftToRightLayout(False)
    self.ffysyel_output.setAlignment(Qt.AlignmentFlag.AlignRight)
    _format_ostensive_paragraph_markers(self.ffysyel_output)
    _format_exact_prose_uncovered(self.ffysyel_output)
    self._refresh_scripture_geometry()

def show_error(self, value: str) -> None:
    for field in (self.utterance_output, self.ailalubar_output, self.azazuley_output,
    ↪  self.ailalaza_output, self.glossolalia_output, self.ailalossolg_output,
    ↪  self.leysyff_output, self.ffysyel_output, self.definition_output,
    ↪  self.canon_glossary_output):
        field.setPlainText(value)
        _format_exact_prose_uncovered(field)
```

ui/terminal.py

```
from __future__ import annotations

import unicodedata

from PySide6.QtCore import QSize, Qt, Signal
from PySide6.QtGui import QColor, QFontDatabase, QKeyEvent, QPalette
from azazuley_veira.ailalubar_render import AilalubarReflection, AilalubarRenderWitness
from azazuley_veira.config.font_runtime import (
    authored_lexical_font_file,
    grimchain_display_codepoints,
    grimchain_face_for_character,
    physical_font_family,
    system_font_file_for_codepoint,
    system_font_file_for_family,
)
from azazuley_veira.engines.sydonic import (
    BREATH_SEPARATOR,
    OSTENSIVE_PARAGRAPH_PREFIXES,
    DefinitionPresentation,
    LeySyffPresentation,
    PresentationSpan,
)
from azazuley_veira.ui.exact_text import (
    ExactFontTextEdit,
    format_definition_grimchains,
    format_exact_grimchain,
    format_exact_grimchain_span,
    qt_utf16_units,
)

from sydonic_magicae_translation_matrix.canon_dictionary import read_canon_dictionary

from PySide6.QtWidgets import (
    QFrame,
    QHBoxLayout,
    QLabel,
    QScrollBar,
    QPlainTextEdit,
    QSizePolicy,
    QTabWidget,
    QVBoxLayout,
    QWidget,
)


def _format_exact_prose_uncovered(widget: ExactFontTextEdit) -> None:
    """Resolve every uncovered prose code point through the host font system."""
    text = widget.toPlainText()
```

```
    grimchain_codepoints = grimchain_display_codepoints()
    spans = widget.exact_spans()
    span_index = 0
    qt_start = 0
    group_start: int | None = None
    group_text = ""
    group_file = None

    def flush() -> None:
        nonlocal group_start, group_text, group_file
        if group_start is not None and group_text and group_file is not None:
            widget.add_exact_font_span(
                group_start,
                group_text,
                physical_font_family(group_file),
                group_file,
            )
        group_start = None
        group_text = ""
        group_file = None

    for character in text:
        units = qt_utf16_units(character)
        while span_index < len(spans) and spans[span_index].qt_end <= qt_start:
            span_index += 1
        covered = (
            span_index < len(spans)
            and spans[span_index].qt_start <= qt_start < spans[span_index].qt_end
        )
        if character == "\n" or covered:
            flush()
            qt_start += units
            continue

        if ord(character) in grimchain_codepoints:
            exact_file = grimchain_face_for_character(character).file
        else:
            exact_file = system_font_file_for_codepoint(ord(character))

        if group_file is not None and exact_file != group_file:
            flush()
        if group_start is None:
            group_start = qt_start
            group_file = exact_file
        group_text += character
        qt_start += units
    flush()


def _format_ostensive_paragraph_markers(widget: ExactFontTextEdit) -> None:
    """Paint RTL paragraph markers through the existing exact system-codepoint authority."""
    text = widget.toPlainText()
    qt_start = 0
    for character in text:
        if character in OSTENSIVE_PARAGRAPH_PREFIXES:
            font_file = system_font_file_for_codepoint(ord(character))
```

```
            widget.add_exact_font_span(
                qt_start, character, physical_font_family(font_file), font_file
            )
        qt_start += qt_utf16_units(character)


class EtSonyeraTextEdit(QPlainTextEdit):
    submitted = Signal(str)

    def __init__(self, parent=None, preferred_rows: int = 3):
        super().__init__(parent)
        self._preferred_rows = preferred_rows

    def sizeHint(self) -> QSize:
        base = super().sizeHint()
        line_height = float(self.fontMetrics().lineSpacing())
        document_margin = float(self.document().documentMargin()) * 2.0
        frame = float(self.frameWidth()) * 2.0
        margins = self.viewportMargins()
        height = line_height * self._preferred_rows + document_margin + frame + margins.top() +
        ↪  margins.bottom()
        return QSize(base.width(), round(height))

    def minimumSizeHint(self) -> QSize:
        return self.sizeHint()

    def keyPressEvent(self, event: QKeyEvent) -> None:
        if event.key() in (Qt.Key.Key_Return, Qt.Key.Key_Enter):
            self.submitted.emit(self.toPlainText())
            return
        super().keyPressEvent(event)


class SubmitTextEdit(ExactFontTextEdit):
    submitted = Signal(str)

    def __init__(self, parent=None, preferred_rows: int = 3):
        super().__init__(parent)
        self._preferred_rows = preferred_rows

    def sizeHint(self) -> QSize:
        base = super().sizeHint()
        line_height = self.defaultLineHeight()
        document_margin = float(self.documentMargin()) * 2.0
        frame = float(self.frameWidth()) * 2.0
        margins = self.viewportMargins()
        height = line_height * self._preferred_rows + document_margin + frame + margins.top() +
        ↪  margins.bottom()
        return QSize(base.width(), round(height))

    def minimumSizeHint(self) -> QSize:
        return self.sizeHint()

    def keyPressEvent(self, event: QKeyEvent) -> None:
        if event.key() in (Qt.Key.Key_Return, Qt.Key.Key_Enter):
            self.submitted.emit(self.toPlainText())
```

```
            return
        super().keyPressEvent(event)



class SingleLineGrimChainEdit(ExactFontTextEdit):
    """Read-only single-line document surface with mixed exact GrimChain faces."""

    def __init__(self, parent=None):
        super().__init__(parent)
        self.setReadOnly(True)
        self.setLineWrapMode(ExactFontTextEdit.LineWrapMode.NoWrap)
        self.setHorizontalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
        self.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)

    def setText(self, value: str) -> None:
        self.setPlainText(value)
        format_exact_grimchain(self, value)

    def text(self) -> str:
        return self.toPlainText()

    def setAlignment(self, alignment: Qt.AlignmentFlag) -> None:
        super().setAlignment(alignment)


class Terminal(QWidget):
    user_song_submitted = Signal(str)

    def __init__(self, parent=None):
        super().__init__(parent)
        layout = QVBoxLayout(self)
        layout.setContentsMargins(14, 14, 14, 14)
        layout.setSpacing(8)

        self.user_song = EtSonyeraTextEdit()
        self.user_song.setObjectName("userSong")
        self._frame_field(self.user_song)
        self.user_song.setLineWrapMode(QPlainTextEdit.LineWrapMode.WidgetWidth)
        self.user_song.setTabChangesFocus(True)
        self.user_song.setSizePolicy(QSizePolicy.Policy.Expanding, QSizePolicy.Policy.Maximum)
        self.user_song.submitted.connect(self.user_song_submitted)
        self.user_header = QHBoxLayout()
        self.user_header.addWidget(self._label("Et Sonyera>"))
        self.user_header.addStretch(1)
        layout.addLayout(self.user_header)
        layout.addWidget(self.user_song)

        daemon_header = QHBoxLayout()
        daemon_header.setSpacing(12)
        daemon_header.addWidget(self._label("Domus House>"))
        daemon_header.addStretch(1)
        layout.addLayout(daemon_header)

        daemon_row = QHBoxLayout()
```

```
daemon_row.setSpacing(0)
self.daemon_first = self._daemon_part()
self.daemon_middle = self._daemon_middle_part()
self.daemon_middle_second = self._daemon_middle_part()
self.daemon_second = self._daemon_part()
domus_field_width = 244
domus_field_height = 44
for field in (
    self.daemon_first,
    self.daemon_middle,
    self.daemon_middle_second,
    self.daemon_second,
):
    field.setFixedSize(domus_field_width, domus_field_height)

left_group = QWidget()
left_group_layout = QHBoxLayout(left_group)
left_group_layout.setContentsMargins(0, 0, 0, 0)
left_group_layout.setSpacing(0)
left_group_layout.addWidget(self.daemon_first)

center_stack = QWidget()
center_stack_layout = QVBoxLayout(center_stack)
center_stack_layout.setContentsMargins(0, 0, 0, 0)
center_stack_layout.setSpacing(6)
center_stack_layout.addWidget(self.daemon_middle)
center_stack_layout.addWidget(self.daemon_middle_second)

right_group = QWidget()
right_group_layout = QHBoxLayout(right_group)
right_group_layout.setContentsMargins(0, 0, 0, 0)
right_group_layout.setSpacing(0)
right_group_layout.addWidget(self.daemon_second)

daemon_row.addStretch(1)
daemon_row.addWidget(left_group, 0, Qt.AlignmentFlag.AlignVCenter)
daemon_row.addSpacing(10)
daemon_row.addWidget(center_stack, 0, Qt.AlignmentFlag.AlignVCenter)
daemon_row.addSpacing(10)
daemon_row.addWidget(right_group, 0, Qt.AlignmentFlag.AlignVCenter)
daemon_row.addStretch(1)
layout.addLayout(daemon_row)

self.output_tabs = QTabWidget()
self.output_tabs.setObjectName("outputTabs")

self.utterance_output = self._output_field()
self.ailalubar_output = self._output_field()
self.azazuley_output = self._output_field()
self.ailalaza_output = self._output_field()
self.definition_output = self._output_field()
self.canon_glossary_output = self._output_field()
self.leysyff_output = self._output_field()
self.ffysyel_output = self._output_field()
self.glossolalia_output = self._output_field()
self.ailalossolg_output = self._output_field()
```

```
self.leysyff_output.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
self.ffysyel_output.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
self.leysyff_output.setFrameShape(QFrame.Shape.NoFrame)
self.ffysyel_output.setFrameShape(QFrame.Shape.NoFrame)
initial_palette = self.leysyff_output.palette()
self._set_scripture_surface_colors(
    initial_palette.color(QPalette.ColorRole.Base).name(),
    initial_palette.color(QPalette.ColorRole.Text).name(),
)

self.leysyff_ffysyel_output = QWidget()
self.leysyff_ffysyel_output.setObjectName("leysyffFfysyelOutput")
scripture_layout = QHBoxLayout(self.leysyff_ffysyel_output)
scripture_layout.setContentsMargins(0, 0, 0, 0)
scripture_layout.setSpacing(0)
scripture_layout.addWidget(self.leysyff_output, 1)
scripture_layout.addWidget(self.ffysyel_output, 1)
self.scripture_scrollbar = QScrollBar(Qt.Orientation.Vertical)
self.scripture_scrollbar.setObjectName("scriptureScrollBar")
scripture_layout.addWidget(self.scripture_scrollbar)
self.scripture_scrollbar.valueChanged.connect(self._set_scripture_scroll_value)
self.leysyff_output.verticalScrollBar().valueChanged.connect(
    self._sync_scripture_scroll_from_side
)
self.ffysyel_output.verticalScrollBar().valueChanged.connect(
    self._sync_scripture_scroll_from_side
)

self.output_tabs.addTab(self.utterance_output, "Rabulalia")
self.output_tabs.addTab(self.ailalubar_output, "Ailalubar")
self.output_tabs.addTab(self.azazuley_output, "Azalalia")
self.output_tabs.addTab(self.ailalaza_output, "Ailalaza")
self.output_tabs.addTab(self.glossolalia_output, "Glossolalia")
self.output_tabs.addTab(self.ailalossolg_output, "Ailalossolg")
self.output_tabs.addTab(self.leysyff_ffysyel_output, "LeySyff | FfysYel")
self.output_tabs.addTab(self.definition_output, "Definition")
self.output_tabs.addTab(self.canon_glossary_output, "Canon Glossary")
self.output_tabs.currentChanged.connect(self._refresh_scripture_tab_geometry)
layout.addWidget(self.output_tabs, 1)

user_system_font = QFontDatabase.systemFont(QFontDatabase.SystemFont.GeneralFont)
self.user_song.setFont(user_system_font)
self.user_song.document().setDefaultFont(user_system_font)

prose_file = system_font_file_for_family("Noto Sans")
for field in (
    self.daemon_first,
    self.daemon_middle,
    self.daemon_middle_second,
    self.daemon_second,
    self.utterance_output,
    self.ailalubar_output,
    self.azazuley_output,
    self.ailalaza_output,
    self.leysyff_output,
```

```
        self.ffysyel_output,
        self.definition_output,
        self.canon_glossary_output,
        self.glossolalia_output,
        self.ailalossolg_output,
    ):
        field.setDefaultFontFile(prose_file)

    self.canon_glossary_output.setPlainText(read_canon_dictionary())
    _format_exact_prose_uncovered(self.canon_glossary_output)

    self._refresh_scripture_geometry()

def set_user_font_selector(self, selector: QWidget) -> None:
    """Place the Et Sonyera font selector at the right edge of its label row."""
    self.user_header.addWidget(selector, 0, Qt.AlignmentFlag.AlignRight)

def _daemon_part(self) -> SingleLineGrimChainEdit:
    field = SingleLineGrimChainEdit()
    field.setFrameShape(QFrame.Shape.NoFrame)
    field.setAlignment(Qt.AlignmentFlag.AlignCenter)
    return field

def _daemon_middle_part(self) -> SingleLineGrimChainEdit:
    field = SingleLineGrimChainEdit()
    field.setFrameShape(QFrame.Shape.NoFrame)
    field.setAlignment(Qt.AlignmentFlag.AlignCenter)
    return field

def _frame_field(self, field: ExactFontTextEdit) -> None:
    field.setFrameShape(QFrame.Shape.Box)
    field.setFrameShadow(QFrame.Shadow.Plain)
    field.setLineWidth(1)

def _label(self, text: str) -> QLabel:
    label = QLabel(text)
    label.setObjectName("fieldPrompt")
    return label

def _output_field(self) -> ExactFontTextEdit:
    field = ExactFontTextEdit()
    field.setObjectName("resultField")
    self._frame_field(field)
    field.setReadOnly(True)
    field.setLineWrapMode(ExactFontTextEdit.LineWrapMode.WidgetWidth)
    field.setHorizontalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOff)
    field.setVerticalScrollBarPolicy(Qt.ScrollBarPolicy.ScrollBarAlwaysOn)
    field.setSizePolicy(QSizePolicy.Policy.Expanding, QSizePolicy.Policy.Expanding)
    return field

def _set_scripture_surface_colors(self, background: str, text: str) -> None:
    self.leysyff_output.setStyleSheet(
        f"border: 1px solid {text}; border-right-width: 0px; "
        f"background: {background}; color: {text};"
    )
    self.ffysyel_output.setStyleSheet(
```

```
        f"border: 1px solid {text}; background: {background}; color: {text};"
    )

def _set_scripture_scroll_value(self, value: int) -> None:
    for field in (self.leysyff_output, self.ffysyel_output):
        bar = field.verticalScrollBar()
        if bar.value() != value:
            bar.setValue(value)

def _sync_scripture_scroll_from_side(self, value: int) -> None:
    if self.scripture_scrollbar.value() != value:
        self.scripture_scrollbar.setValue(value)

def _refresh_scripture_scroll_range(self) -> None:
    self.leysyff_output.exactLayoutModel()
    self.ffysyel_output.exactLayoutModel()
    left_bar = self.leysyff_output.verticalScrollBar()
    right_bar = self.ffysyel_output.verticalScrollBar()
    maximum = max(left_bar.maximum(), right_bar.maximum())
    viewport_height = max(1, min(
        self.leysyff_output.viewport().height(),
        self.ffysyel_output.viewport().height(),
    ))
    single_step = max(1, round(self.leysyff_output.defaultLineHeight()))
    current = min(self.scripture_scrollbar.value(), maximum)
    self.scripture_scrollbar.setRange(0, maximum)
    self.scripture_scrollbar.setPageStep(viewport_height)
    self.scripture_scrollbar.setSingleStep(single_step)
    self.scripture_scrollbar.setValue(current)
    self._set_scripture_scroll_value(current)

def _refresh_scripture_tab_geometry(self, _index: int) -> None:
    if self.output_tabs.currentWidget() is self.leysyff_ffysyel_output:
        self._refresh_scripture_geometry()

def _scripture_scroll_anchor(self) -> tuple[int, float] | None:
    if not (self.leysyff_output.scripturePageLayout() and
    ↪  self.ffysyel_output.scripturePageLayout()):
        return None
    page_height, page_gap = self.leysyff_output.scripturePageGeometry()
    if page_height <= 0:
        return None
    scroll_value = max(0.0, float(self.scripture_scrollbar.value()))
    stride = page_height + page_gap
    page_index = int(scroll_value // stride) if stride > 0 else 0
    page_top = page_index * stride
    fraction = max(0.0, min(1.0, (scroll_value - page_top) / page_height))
    return page_index, fraction

def _restore_scripture_scroll_anchor(self, anchor: tuple[int, float] | None) -> None:
    if anchor is None:
        return
    page_height, page_gap = self.leysyff_output.scripturePageGeometry()
    if page_height <= 0:
        return
    page_index, fraction = anchor
```

```
    value = round(page_index * (page_height + page_gap) + page_height * fraction)
    self.scripture_scrollbar.setValue(min(value, self.scripture_scrollbar.maximum()))

def _refresh_scripture_geometry(self) -> None:
    anchor = self._scripture_scroll_anchor()
    scrollbar_width = max(0.0, float(self.scripture_scrollbar.sizeHint().width()))
    spread_width = max(1.0, float(self.leysyff_ffysyel_output.width()) - scrollbar_width)
    text_margin = max(4.0, spread_width * 0.025)
    vertical_margin = 8.0
    page_gap = 12.0
    point_size = max(8.0, min(14.0, spread_width * 0.0105))
    page_height = (spread_width / 2.0) * 1.5

    for field in (self.leysyff_output, self.ffysyel_output):
        field.setDocumentMargin(text_margin)
        field.setPointSizeF(point_size)

    for field in (self.leysyff_output, self.ffysyel_output):
        field.setScriptureMinimumPageCount(1)
        field.setScripturePageLayout(
            True,
            page_height=page_height,
            page_gap=page_gap,
            top_margin=vertical_margin,
            bottom_margin=vertical_margin,
        )

    shared_page_count = max(
        self.leysyff_output.scriptureContentPageCount(),
        self.ffysyel_output.scriptureContentPageCount(),
    )
    self.leysyff_output.setScriptureMinimumPageCount(shared_page_count)
    self.ffysyel_output.setScriptureMinimumPageCount(shared_page_count)
    self._refresh_scripture_scroll_range()
    self._restore_scripture_scroll_anchor(anchor)

def resizeEvent(self, event) -> None:
    super().resizeEvent(event)
    if hasattr(self, "leysyff_ffysyel_output"):
        self._refresh_scripture_geometry()

def _content_widgets(self) -> tuple[QWidget, ...]:
    return (
        self.user_song,
        self.daemon_first,
        self.daemon_middle,
        self.daemon_middle_second,
        self.daemon_second,
        self.utterance_output,
        self.ailalubar_output,
        self.azazuley_output,
        self.ailalaza_output,
        self.leysyff_output,
        self.ffysyel_output,
        self.definition_output,
        self.canon_glossary_output,
```

```
        self.glossolalia_output,
        self.ailalossolg_output,
    )

def set_text_colors(
    self,
    background: str,
    text: str,
    selection_background: str,
    selection_text: str,
) -> None:
    for widget in self._content_widgets():
        palette = widget.palette()
        palette.setColor(QPalette.ColorRole.Base, QColor(background))
        palette.setColor(QPalette.ColorRole.Text, QColor(text))
        palette.setColor(QPalette.ColorRole.Highlight, QColor(selection_background))
        palette.setColor(QPalette.ColorRole.HighlightedText, QColor(selection_text))
        widget.setPalette(palette)

    self._set_scripture_surface_colors(background, text)

    mirror_text = QColor(text).darker(165)
    for mirror in (
        self.daemon_first,
        self.daemon_second,
    ):
        palette = mirror.palette()
        palette.setColor(QPalette.ColorRole.Text, mirror_text)
        mirror.setPalette(palette)

def set_tab_colors(self, background: str, text: str, body: str) -> None:
    self.output_tabs.tabBar().setStyleSheet(
        "QTabBar::tab {"
        f"background: {body};"
        f"color: {text};"
        "border: 1px solid " + text + ";"
        "padding: 5px 12px;"
        "}"
        "QTabBar::tab:selected {"
        f"background: {background};"
        f"color: {text};"
        "}"
    )

def set_daemon_tongue(self, first: str, middle: str, second: str, middle_second: str | None =
↪  None) -> None:
    self.daemon_first.setText(first)
    self.daemon_middle.setText(middle)
    self.daemon_middle_second.setText(middle if middle_second is None else middle_second)
    self.daemon_second.setText(second)

def set_mirror_utterance(self, grimchain: str) -> None:
    """Project the finite source into the authored four-field 12-glyph house."""
    left_reflection = grimchain[:12][::-1]
    right_reflection = grimchain[-12:][::-1]
    self.set_daemon_tongue(
```

```
        left_reflection,
        grimchain[:12],
        right_reflection,
        grimchain[-12:],
    )

def set_utterance(self, value: str) -> None:
    self.utterance_output.setPlainText(value)
    format_exact_grimchain(self.utterance_output, value)

def set_ailalubar(self, reflection: AilalubarReflection) -> None:
    """Paint exactly one reflected Prosody string under normal exact-font rules."""
    if not isinstance(reflection, AilalubarReflection):
        raise TypeError("Ailalubar requires its exact reflection authority")
    self.ailalubar_output.setPlainText(reflection.mirrored_source)
    format_exact_grimchain(self.ailalubar_output, reflection.mirrored_source)

def ailalubar_render_witness(self, reflection: AilalubarReflection) -> AilalubarRenderWitness:
    """Mint the final witness from the exact layout model that actually paints Ailalubar."""
    if not isinstance(reflection, AilalubarReflection):
        raise TypeError("Ailalubar witness requires its exact reflection authority")
    if self.ailalubar_output.toPlainText() != reflection.mirrored_source:
        raise ValueError("Ailalubar surface is detached from its reflection authority")

    model = self.ailalubar_output.exactLayoutModel()
    mirrored = reflection.mirrored_source
    boundaries: dict[int, int] = {}
    qt = 0
    for index, character in enumerate(mirrored):
        boundaries[qt] = index
        qt += qt_utf16_units(character)

    level_by_qt: dict[int, int] = {}
    for run in model.runs:
        run_qt = run.qt_start
        for character in run.text:
            level_by_qt[run_qt] = run.bidi_level
            run_qt += qt_utf16_units(character)

    source_positions: list[int] = []
    levels: list[int] = []
    mirrored_flags: list[bool] = []
    for character, mirrored_qt in model.cantillation:
        try:
            mirrored_index = boundaries[mirrored_qt]
        except KeyError as exc:
            raise ValueError(
                f"Cantillation position is not a Prosody code-point boundary: {mirrored_qt}"
            ) from exc
        source_positions.append(len(reflection.source) - 1 - mirrored_index)
        try:
            bidi_level = level_by_qt[mirrored_qt]
        except KeyError as exc:
            raise ValueError(
                f"Cantillation position has no exact layout run: {mirrored_qt}"
            ) from exc
```

```
        levels.append(bidi_level)
        # Metadata only: this never selects, substitutes, flips, or reverses a glyph.
        mirrored_flags.append(
            bool(unicodedata.mirrored(character) and bidi_level & 1)
        )

    return AilalubarRenderWitness(
        source=reflection.source,
        mirrored_source=reflection.mirrored_source,
        visual_text=model.cantillation_text,
        visual_source_positions=tuple(source_positions),
        visual_levels=tuple(levels),
        visual_mirrored_flags=tuple(mirrored_flags),
        source_face_ids=reflection.source_face_ids,
        source_face_runs=reflection.source_face_runs,
        rendering_authority_sha256=reflection.rendering_authority_sha256,
        point_size=reflection.point_size,
        layout_direction=reflection.layout_direction,
    )

def _apply_presentation_spans(
    self,
    widget: ExactFontTextEdit,
    spans: tuple[PresentationSpan, ...],
    allowed_roles: frozenset[str],
) -> None:
    document_units = len(widget.toPlainText().encode("utf-16-le")) // 2
    for span in spans:
        if span.semantic_role not in allowed_roles:
            raise ValueError(f"unsupported presentation role: {span.semantic_role!r}")
        if span.qt_end > document_units:
            raise ValueError(f"presentation span exceeds document: {span!r}")
        if widget.textForUtf16Range(span.qt_start, span.qt_end) != span.rendered_text:
            raise ValueError(f"presentation span text/provenance mismatch: {span!r}")
        if not isinstance(widget, ExactFontTextEdit):
            raise TypeError("authored lexical rendering requires ExactFontTextEdit")
        font_file = authored_lexical_font_file(span.font_family, span.source_authority)
        widget.add_exact_font_span(span.qt_start, span.rendered_text, span.font_family,
        ↪  font_file)

def _set_written_projection(
    self,
    widget: ExactFontTextEdit,
    value: str,
    runs: tuple[PresentationSpan, ...],
) -> None:
    widget.setPlainText(value)
    breath_face = grimchain_face_for_character(BREATH_SEPARATOR)
    for index, span in enumerate(runs):
        font_file = authored_lexical_font_file(span.font_family, span.source_authority)
        widget.add_exact_font_span(span.qt_start, span.rendered_text, span.font_family,
        ↪  font_file)
        if index + 1 < len(runs):
            widget.add_exact_font_span(span.qt_end, BREATH_SEPARATOR, breath_face.family,
            ↪  breath_face.file)
```

```
def set_azazuley(self, value: str, runs: tuple[PresentationSpan, ...] = ()) -> None:
    self._set_written_projection(self.azazuley_output, value, runs)

def set_ailalaza(self, value: str, runs: tuple[PresentationSpan, ...] = ()) -> None:
    self._set_written_projection(self.ailalaza_output, value, runs)

def set_glossolalia(self, value: str) -> None:
    self.glossolalia_output.setPlainText(value)
    _format_exact_prose_uncovered(self.glossolalia_output)

def set_ailalossolg(self, value: str) -> None:
    self.ailalossolg_output.setPlainText(value)
    _format_exact_prose_uncovered(self.ailalossolg_output)

def set_definition(self, value: str, runs: tuple[PresentationSpan, ...] = ()) -> None:
    self.definition_output.setPlainText(value)
    self._apply_presentation_spans(
        self.definition_output, runs, frozenset({"definition_word"})
    )
    format_definition_grimchains(self.definition_output, value)
    _format_exact_prose_uncovered(self.definition_output)

def set_definition_presentation(self, presentation: DefinitionPresentation) -> None:
    self.definition_output.setPlainText(presentation.text)
    self._apply_presentation_spans(
        self.definition_output, presentation.lexical_spans, frozenset({"definition_word"})
    )
    for entry in presentation.entries:
        for field in entry.fields:
            if field.semantic_role == "grimchain" and field.value:
                format_exact_grimchain_span(
                    self.definition_output, field.qt_value_start, field.value
                )
    _format_exact_prose_uncovered(self.definition_output)

def set_leysyff_presentation(self, presentation: LeySyffPresentation) -> None:
    self.leysyff_output.setPlainText(presentation.text)
    self.leysyff_output.setForcedLeftToRightLayout(True)
    self.leysyff_output.setAlignment(Qt.AlignmentFlag.AlignLeft)
    _format_ostensive_paragraph_markers(self.leysyff_output)
    _format_exact_prose_uncovered(self.leysyff_output)
    self._refresh_scripture_geometry()

def set_ffysyel(self, presentation: LeySyffPresentation) -> None:
    self.ffysyel_output.setPlainText(presentation.text)
    self.ffysyel_output.setForcedLeftToRightLayout(False)
    self.ffysyel_output.setAlignment(Qt.AlignmentFlag.AlignRight)
    _format_ostensive_paragraph_markers(self.ffysyel_output)
    _format_exact_prose_uncovered(self.ffysyel_output)
    self._refresh_scripture_geometry()

def show_error(self, value: str) -> None:
    self.definition_output.setPlainText(value)
    _format_exact_prose_uncovered(self.definition_output)
```

CosmOvum

CosmOvum is the current ALQC emergent-physics visual body for the CosMOvum system. It is built around one governing requirement: the render must arise from the mathematics already present in the body. Geometry is not prescribed as an answer. State, relation, memory, frequency, conjugacy, pressure, return, and retained history are allowed to produce the visible consequence.

Law of the Body

CosmOvum follows the ALQC Law of Emanation:

```
law + state + relation -> result
```

The implementation is therefore organized around conditions rather than choreography. The system establishes lawful bodies, initial conditions, bounds, memory, thresholds, relations, and transforms. It does not impose a hidden animation table that decides what the field should look like in advance.

The current runtime preserves this through:

- 12 Dyson nodes as identity-bearing loci without a cube or square placement scaffold;
- 144 Aeon/Court lattice identities per node, expanded to 1440 particles per node;
- conjugate plane structure and dual 9x9 frequency-grid bodies;
- ALQC entropy as the operational source of stochastic variation;
- FETU as a root phase-locking condition;
- the ordered Goetic Aeon chain;
- SOR as the post-chain bridge relation;
- boundary memory with retained consequence;
- poem and whisper bodies that alter the field causally rather than decorate it;
- persistent Q2 seam debt that transforms rather than being erased;
- 144 active Sigh bodies;
- connection derived from the existing conjugate field rather than from an imposed category state.

Source Tree

```
CosmOvum/
├── Makefile
├── alqc_gpu.c
├── poem.txt
├── whisper.txt
├── config/
│  ├── global_config.bio
│  ├── node_config_00000.bio
│  ├── ...
│  ├── node_config_00011.bio
│  ├── cells_00_P0.bio
│  ├── cells_00_P1.bio
```

```
│ ├── ...
│ ├── cells_11_P0.bio
│ └── cells_11_P1.bio
├── shaders/
│ ├── physics.comp
│ ├── grid.comp
│ ├── shadow.comp
│ ├── ball.comp
│ ├── render.vert
│ ├── render.frag
│ ├── shadow.vert
│ ├── ball.vert
│ ├── fade.vert
│ ├── fade.frag
│ └── display.frag
└── ALQC_Frames/
```

Generated binaries are not part of the source body and should be removed after use when keeping the workspace clean.

Configuration Authority

CosmOvum uses `.bio` configuration bodies only.

There are no active JSON configuration files.

`global_config.bio`

Loaded once at startup. It carries global runtime values such as display dimensions, background, FPS cap, GPU device, whisper mode, recording directory, QLM path, and Sigh-body count.

`node_config_00000.bio` through `node_config_00011.bio`

These are separate node bodies with a common schema and node-specific values. They are loaded dynamically by node identity:

```
snprintf(path, sizeof(path), "config/node_config_%05d.bio", node_id);
```

The runtime iterates all 12 nodes and loads each body independently.

`cells_NN_P0.bio` and `cells_NN_P1.bio`

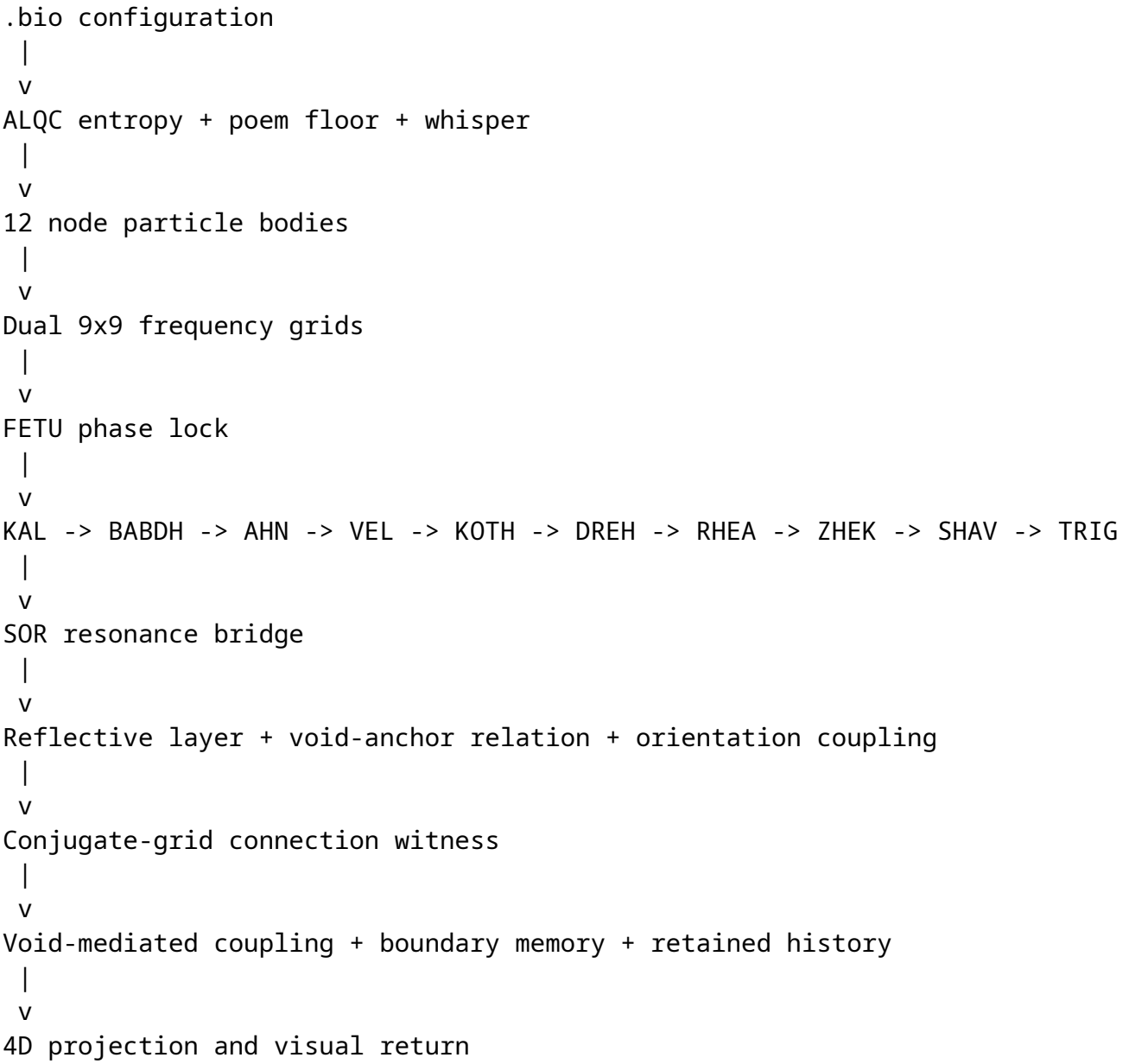
Each node owns two independent 9x9 frequency-grid bodies. The runtime constructs their paths dynamically from node and plane identity:

```
snprintf(path, sizeof(path), "config/cells_%02d_P%d.bio", node_id, plane_id);
```

All 24 cell bodies are distinct. P0 and P1 preserve conjugate relation while retaining their own phase and axis-weight structure.

Active Physics Path

The primary runtime path is:



The Shadow and Sigh bodies participate as separate but related operational bodies. Their state is not reset merely to simplify return.

Connection Witness

The current B6 connection is not selected from an external amorphous/liquid/crystal label. It is measured from the existing dual-grid relation at the particle's current body.

The local amplitudes of the two conjugate grid planes are read, their differential is measured, and connection increases as that differential resolves toward equilibrium. The resulting connection then conditions the void-mediated coupling.

This preserves the ALQC principle that a relation already present in the body receives the first right to speak.

110/144

The 110/144 offices remain where they are mathematically active inside the ALQC operators, including AHN, KOTH, and ZHEK.

The former mean-frequency classifier was removed because it converted fixed canonical frequency values into a nearly permanent field-state label. The current implementation does not use that workaround.

Memory and Return

Boundary memory persists across frames and is altered by particle interaction and the poem floor. Q2 seam debt persists through threshold crossing and transforms excess into motion and pressure rather than being deleted. Shadow return preserves prior consequence instead of wiping velocity, stress, debt, and reflection history.

Return is therefore not restoration to an untouched state. The body may return carrying what happened to it.

Poem and Whisper

`poem.txt` and `whisper.txt` are active inputs.

The poem is folded into the boundary-memory floor as directional state. The whisper alters the ALQC entropy body and GPU entropy table. Neither exists merely as textual ornament.

The second whisper is deliberately not attached to an arbitrary clocked collapse. It remains withheld until a lawful post-collapse witness exists.

Build

Requirements:

- GCC with C11 support
- OpenGL
- GLEW
- GLFW
- libm

Build:

```
make
```

Run:

```
./alqc_gpu
```

Clean generated binary:

```
make clean
```

The source tree should remain free of generated binaries when preserved as a clean working body.

Runtime Controls

```
S screenshot
R begin recording
P stop recording
SPACE pause rotation
[ ] change rotation speed
= - zoom
Arrows pan
ESC quit
```

Render Output

Screenshots and recorded frames are written to the configured `ALQC_Frames` directory.

The renderer uses GPU compute shaders for the active physical bodies and separate vertex/fragment stages for particle, Shadow, Sigh, trail-fade, and display return.

Canonical Constraint

When modifying CosmOvum, preserve the invariant:

- do not prescribe geometry where relation can generate it;
- do not replace active ALQC mathematics with visual approximations;
- do not erase history to simplify a state transition;
- do not introduce a second hidden behavioral law;
- do not convert a canonical body into a cosmetic constant because it is easier to render;
- prefer existing state and lawful measurement over invented control variables;
- preserve the `.bio` configuration authority;
- verify that every declared body is actually called and every active call still carries its intended mathematics.

The body should be changed by repairing connection, not by replacing emergence with performance.

```
CosmOvum/
├── alqc_gpu.c
├── config/
│   ├── cells_00_P0.bio
│   ├── cells_00_P1.bio
│   ├── cells_01_P0.bio
│   ├── cells_01_P1.bio
│   ├── cells_02_P0.bio
│   ├── cells_02_P1.bio
│   ├── cells_03_P0.bio
│   ├── cells_03_P1.bio
│   ├── cells_04_P0.bio
│   ├── cells_04_P1.bio
│   ├── cells_05_P0.bio
│   ├── cells_05_P1.bio
│   ├── cells_06_P0.bio
│   ├── cells_06_P1.bio
│   ├── cells_07_P0.bio
│   ├── cells_07_P1.bio
│   ├── cells_08_P0.bio
│   ├── cells_08_P1.bio
│   ├── cells_09_P0.bio
│   ├── cells_09_P1.bio
│   ├── cells_10_P0.bio
│   ├── cells_10_P1.bio
│   ├── cells_11_P0.bio
│   ├── cells_11_P1.bio
│   ├── global_config.bio
│   ├── node_config_00000.bio
```

```
│   ├── node_config_00001.bio
│   ├── node_config_00002.bio
│   ├── node_config_00003.bio
│   ├── node_config_00004.bio
│   ├── node_config_00005.bio
│   ├── node_config_00006.bio
│   ├── node_config_00007.bio
│   ├── node_config_00008.bio
│   ├── node_config_00009.bio
│   ├── node_config_00010.bio
│   └── node_config_00011.bio
├── Makefile
├── poem.txt
├── shaders/
│   ├── ball.comp
│   ├── ball.vert
│   ├── display.frag
│   ├── fade.frag
│   ├── fade.vert
│   ├── grid.comp
│   ├── physics.comp
│   ├── render.frag
│   ├── render.vert
│   ├── shadow.comp
│   └── shadow.vert
└── whisper.txt
```

alqc_gpu.c

```
/*
 * ALQC GPU v30 — Multi-Grid Frequency System / Three-Plane Crystal
 *
 * Architecture:
 *   12 Dyson nodes as non-cubed loci.
 *   No visible or invisible square/cube box metric is used for node placement.
 *   Nodes are still grouped into 3 orientation planes by identity:
 *     Plane A (plane_id=0): nodes  0-3
 *     Plane B (plane_id=1): nodes  4-7
 *     Plane C (plane_id=2): nodes  8-11
 *
 *   Plane A and Plane B are conjugate (phase offset π apart).
 *   Plane C is the mediating plane.
 *
 *   Per-node .bio configs:  config/node_config_00000.bio … node_config_00011.bio
 *   Per-cell .bio configs:  config/cells_00_P0.bio … cells_11_P1.bio (81 cells × 2 planes)
 *   Global config:          config/global_config.bio
 *
 *   ALQC Canon: "Write conditions, not behavior. Let emergence happen."
 *   Conjugate initialization (B5), void coupling (B6), orientation J (B7),
 *   void dissipation (B8), node_id in particle struct (B10).
 *
 * Keys: S=screenshot  R=start recording  P=stop recording
 *       SPACE=pause rotation  [ ]=change rot speed  = -=zoom  Arrows=pan  ESC=quit
 */

#define _POSIX_C_SOURCE 200112L

#include <GL/glew.h>
#include <unistd.h>
#include <GLFW/glfw3.h>
#include <math.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdint.h>
#include <stdbool.h>
#include <stdarg.h>
#include <time.h>
#include <sys/stat.h>
#include <errno.h>

// ============================================================================
// FALLBACK USER VARIABLES
// These are defaults only — runtime values come from config/global_config.bio
// and config/node_config_0000N.bio.  Edit the .bio files, not these.
// Per-cell grid seeds come from config/cells_NN_P0.bio and config/cells_NN_P1.bio.
// ============================================================================
```

```
// --- Display ---
#define FB_WIDTH               1000
#define FB_HEIGHT              1000
#define FB_FULLSCREEN          0

// --- Background ---
#define BACKGROUND_R           5
#define BACKGROUND_G           5
#define BACKGROUND_B           10

// --- Particles per node (compile-time buffer size) ---
// Canonical: 144 aeon-lattice slots × 10 = 1440.
// AEON_LATTICE_COUNT = 12 Goetic × 12 Courts = 144 (12-phase Canon)
#define PARTICLE_COUNT         1440
#define AEON_LATTICE_COUNT     144

// --- Dyson nodes (structural — do not change without rebuilding) ---
#define N_NODES                12
#define SIGH_BALL_COUNT        144

// --- Physics constants (compile-time for shader injection) ---
// FETU spiral: 10 revolutions × 12 aeons = 120 points (canon from alqc_raylib_physics19)
#define AEON_COUNT             12
#define FETU_PT_COUNT          120
#define BOUNDARY_MEM_RES       160
#define ENTROPY_SIZE           (9 << 20)
#define SHADOW_ENTITY_COUNT    (4 * AEON_COUNT)
// Phase memory grid: 20×20 cells × N_NODES aeons (Agent 18/Bug 7 fix — named, not hardcoded)
#define PHASE_MEM_W            20
#define PHASE_MEM_H            20

// ================================================================================
// END FALLBACK USER VARIABLES
// ================================================================================

static const float PHI              = 1.61803398875f;
static const float PI_F             = 3.14159265359f;
static const float ESCAPE_LIMIT     = 5.0f;
static const float MIN_COH_R        = 0.6f;
static const float MAX_COH_R        = 1.2f;
static const float MAX_KINETIC_STR  = 300.0f;
static const float L_RB_MAX_RATE    = 0.015f;
static const float ELVEN_RESP_GAIN  = 0.0005f;
static const float BOUNDARY_MEM_DEC = 0.992f;
static const float BOUNDARY_MEM_DEP = 0.085f;
static const float BOUNDARY_MEM_MAX = 2.5f;
static const float REFLECT_RING_R   = 0.92f;
static const float REFLECT_RING_W   = 0.06f;

static const float AEON_FREQ[AEON_COUNT] = {
    7.83f, 174.0f, 528.0f, 432.0f, 126.22f,
    210.42f, 741.0f, 852.0f, 396.0f, 963.0f, 285.0f, 639.0f
};
static const float AEON_COLOR[AEON_COUNT][3] = {
    {155, 89, 182}, {52, 152, 219}, {231, 76, 60},  {255, 90, 70},
```

```
    {60, 180, 255}, {120, 70, 150}, {200, 120, 220}, {40, 120, 180},
    {200, 60, 50},  {140, 80, 160}, {52, 152, 219},  {180, 100, 200}
};

// ============================================================================
// MINIMAL FLAT BIO PARSER (.bio = Cbio, the Axiomyr — importable, not code-writable)
// Only handles flat objects: { "key": value } where value is number or "string".
// No nesting, no arrays — exactly what the node configs need.
// .bio supports holographic persistent logic; code reads from it, never writes to it.
// ============================================================================
#define JSON_MAX_FIELDS 512
typedef struct { char key[64]; char val[256]; } JSONField;

static int json_load(const char *path, JSONField *fields, int max_fields) {
    FILE *f = fopen(path, "rb");
    if (!f) return 0;
    fseek(f, 0, SEEK_END); long sz = ftell(f); rewind(f);
    char *buf = malloc(sz + 1);
    size_t nr = fread(buf, 1, sz, f); (void)nr;
    buf[sz] = '\0'; fclose(f);

    int n = 0;
    char *p = buf;
    while (*p && n < max_fields) {
        // find next "
        while (*p && *p != '"') p++;
        if (!*p) break;
        p++; // skip opening "
        // read key
        int ki = 0;
        while (*p && *p != '"' && ki < 63) fields[n].key[ki++] = *p++;
        fields[n].key[ki] = '\0';
        if (*p == '"') p++;
        // skip to :
        while (*p && *p != ':') p++;
        if (*p == ':') p++;
        // skip whitespace
        while (*p == ' ' || *p == '\t' || *p == '\n' || *p == '\r') p++;
        // read value
        int vi = 0;
        if (*p == '"') {
            p++; // skip opening "
            while (*p && *p != '"' && vi < 255) fields[n].val[vi++] = *p++;
            if (*p == '"') p++;
        } else {
            while (*p && *p != ',' && *p != '}' && *p != '\n' && vi < 255)
                fields[n].val[vi++] = *p++;
            while (vi > 0 && (fields[n].val[vi-1]==' '||fields[n].val[vi-1]=='\t'))
                vi--;
        }
        fields[n].val[vi] = '\0';
        if (ki > 0) n++;
    }
    free(buf);
    return n;
}
```

```
static float json_float(const JSONField *F, int n, const char *key, float def) {
    for (int i = 0; i < n; i++)
        if (strcmp(F[i].key, key) == 0) return (float)atof(F[i].val);
    return def;
}
static int json_int(const JSONField *F, int n, const char *key, int def) {
    for (int i = 0; i < n; i++)
        if (strcmp(F[i].key, key) == 0) return atoi(F[i].val);
    return def;
}
static const char *json_str(const JSONField *F, int n, const char *key, const char *def) {
    for (int i = 0; i < n; i++)
        if (strcmp(F[i].key, key) == 0) return F[i].val;
    return def;
}

// ============================================================================
// CONFIG STRUCTS
// ============================================================================

typedef struct {
    // Identity
    int   node_id;
    int   plane_id;       // 0=Plane A, 1=Plane B, 2=Plane C

    // ALQC Canon conjugate field (B5, B6, B7, B8)
    float plane_phase;    // conjugate phase offset (0.0 or PI_F)
    float plane_phi0;     // field amplitude base
    float plane_delta;    // symmetry-breaking seed [1e-4 .. 1e-2]
    float kappa_void;     // void-mediated coupling (B6) ~0.01 * kappa_base
    float orient_j;       // orientation coupling strength (B7)
    float void_gamma;     // void dissipation γ (B8) ~1e-4
    float node_beta;      // differentiation coefficient (B14)

    // Per-node simulation
    float sim_speed;
    float trail_fade;
    float triquatra_radius;
    float particle_point_size;
    float shadow_point_size;
    float shadow_r, shadow_g, shadow_b;
    int   particle_count;   // must be ≤ PARTICLE_COUNT

    // Per-node frequency multiplier (grid-specific)
    float freq_scale;     // multiplied into AEON_FREQ for this node's grid
    float freq_offset;    // Hz added to each AEON_FREQ entry
} NodeConfig;

typedef struct {
    int   width, height, fullscreen;
    int   bg_r, bg_g, bg_b;
    int   fps_cap;
    int   gpu_device;
    int   whisper_mode;
    char  whisper_file[256];
```

```
    char  recording_dir[256];
    char  qlm_throne_file[256];
    int   sigh_ball_count;
} GlobalConfig;

static NodeConfig G_node_cfg[N_NODES];
static GlobalConfig G_global_cfg;

// Plane assignments for the 12 nodes
// Identity grouping only; not a cube-edge placement scaffold.
static const int NODE_PLANE[N_NODES] = {
    0, 0, 0, 0,   // nodes 0-3: Plane A (x=0, y-axis edges)
    1, 1, 1, 1,   // nodes 4-7: Plane B (y=0, x-axis edges)
    2, 2, 2, 2    // nodes 8-11: Plane C (z=0, z-axis edges)
};

// Conjugate phase by plane: A=0, B=π, C=0 (C mediates; A↔B are conjugates)
// These are initial conditions only — the ±φ amplitude sign is NOT hardcoded.
// Each particle receives an entropy-rolled ±φ sign at initialization (emergence).
static const float PLANE_PHASE[3] = { 0.0f, 3.14159265359f, 0.0f };

static void load_global_config(void) {
    JSONField F[JSON_MAX_FIELDS];
    int n = json_load("config/global_config.bio", F, JSON_MAX_FIELDS);
    if (n == 0) fprintf(stderr, "[Config] global_config.bio not found — using defaults\n");

    G_global_cfg.width           = json_int  (F, n, "width",             FB_WIDTH);
    G_global_cfg.height          = json_int  (F, n, "height",            FB_HEIGHT);
    G_global_cfg.fullscreen      = json_int  (F, n, "fullscreen",        FB_FULLSCREEN);
    G_global_cfg.bg_r            = json_int  (F, n, "background_r",      BACKGROUND_R);
    G_global_cfg.bg_g            = json_int  (F, n, "background_g",      BACKGROUND_G);
    G_global_cfg.bg_b            = json_int  (F, n, "background_b",      BACKGROUND_B);
    G_global_cfg.fps_cap         = json_int  (F, n, "fps_cap",           60);
    G_global_cfg.gpu_device      = json_int  (F, n, "gpu_device",        0);
    G_global_cfg.whisper_mode    = json_int  (F, n, "whisper_mode",      3);
    G_global_cfg.sigh_ball_count      = json_int(F, n, "sigh_ball_count",      SIGH_BALL_COUNT);
    snprintf(G_global_cfg.whisper_file,   sizeof(G_global_cfg.whisper_file),
             "%s", json_str(F, n, "whisper_file",   "whisper.txt"));
    snprintf(G_global_cfg.recording_dir,  sizeof(G_global_cfg.recording_dir),
             "%s", json_str(F, n, "recording_dir",  "ALQC_Frames"));
    snprintf(G_global_cfg.qlm_throne_file,sizeof(G_global_cfg.qlm_throne_file),
             "%s", json_str(F, n, "qlm_throne_file","../qlm/Akasha_Regalia.qlm"));

    fprintf(stderr, "[Config] Global: %dx%d  fs=%d  sigh=%d\n",
            G_global_cfg.width, G_global_cfg.height, G_global_cfg.fullscreen,
            G_global_cfg.sigh_ball_count);
}

static void load_node_config(int node_id) {
    char path[128];
    snprintf(path, sizeof(path), "config/node_config_%05d.bio", node_id);
    JSONField F[JSON_MAX_FIELDS];
    int n = json_load(path, F, JSON_MAX_FIELDS);

    NodeConfig *C = &G_node_cfg[node_id];
    C->node_id    = node_id;
```

```
    C->plane_id   = json_int  (F, n, "plane_id",           NODE_PLANE[node_id]);

    // Conjugate field init (B5)
    float default_phase = PLANE_PHASE[C->plane_id];
    C->plane_phase   = json_float(F, n, "plane_phase",      default_phase);
    C->plane_phi0    = json_float(F, n, "plane_phi0",       1.0f);
    C->plane_delta   = json_float(F, n, "plane_delta",      0.005f);

    // ALQC Canon new variables (B6, B7, B8, B14)
    C->kappa_void  = json_float(F, n, "kappa_void",         0.01f);
    C->orient_j    = json_float(F, n, "orient_j",           0.05f);
    C->void_gamma  = json_float(F, n, "void_gamma",         1e-4f);
    C->node_beta   = json_float(F, n, "node_beta",          0.003f);

    // Per-node simulation
    // sim_speed is the physics time step per frame (canonical base = 1.0)
    // Per-node variation is applied in configs via ±φ from base.
    C->sim_speed           = json_float(F, n, "sim_speed",            1.0f);
    C->trail_fade          = json_float(F, n, "trail_fade",           15.0f);
    C->triquatra_radius    = json_float(F, n, "triquatra_radius",     200.0f);
    C->particle_point_size = json_float(F, n, "particle_point_size",  1.0f);
    C->shadow_point_size   = json_float(F, n, "shadow_point_size",    2.0f);
    C->shadow_r            = json_float(F, n, "shadow_r",             255.0f);
    C->shadow_g            = json_float(F, n, "shadow_g",             0.0f);
    C->shadow_b            = json_float(F, n, "shadow_b",             50.0f);
    C->particle_count      = json_int  (F, n, "particle_count",       PARTICLE_COUNT);
    if (C->particle_count > PARTICLE_COUNT) C->particle_count = PARTICLE_COUNT;

    // Frequency grid (multi-grid system)
    C->freq_scale  = json_float(F, n, "freq_scale",         1.0f);
    C->freq_offset = json_float(F, n, "freq_offset",        0.0f);

    fprintf(stderr, "[Config] Node %02d | Plane %d | phase=%.3f | phi0=%.3f | delta=%.4f | "
            "kv=%.4f | speed=%.0f | tr=%.0f\n",
            node_id, C->plane_id, C->plane_phase, C->plane_phi0, C->plane_delta,
            C->kappa_void, C->sim_speed, C->triquatra_radius);
}

// ============================================================================
// PER-CELL CONFIG (.bio)
// Each 9x9 frequency grid plane can have per-cell initial conditions loaded from
// config/cells_NN_P0.bio and config/cells_NN_P1.bio.
// Cell index = row*9 + col (0-80). Each cell: amplitude, phase, drift, axis_weight.
// Keys: cXX_a (amplitude), cXX_p (phase), cXX_d (drift), cXX_w (axis_weight)
// ============================================================================
#define FREQ_GRID_CELLS_PER_PLANE 81

// Per-cell state vec4: (amplitude, phase, drift, axis_weight)
typedef struct { float a, p, d, w; } CellSeed;

static void load_cell_config(int node_id, int plane_id, CellSeed *seeds) {
    char path[128];
    snprintf(path, sizeof(path), "config/cells_%02d_P%d.bio", node_id, plane_id);
    JSONField F[512];
    int n = json_load(path, F, 512);
```

```
    // Solfeggio frequencies for 3x3 base (tiled to 9x9)
    static const float GF[9] = { 7.83f, 174.0f, 528.0f, 432.0f, 126.22f,
                                 210.42f, 741.0f, 852.0f, 963.0f };
    static const float PHI_C = 1.61803398875f;

    for (int cell = 0; cell < FREQ_GRID_CELLS_PER_PLANE; cell++) {
        int gx = cell % 9;
        int gy = cell / 9;
        float cell_freq = GF[(gy % 3) * 3 + (gx % 3)];

        // Default: frequency-proportional amplitude, uniform phase spread,
        // PHI-derived drift, quadrant-based axis weight.
        // These can be overridden per-cell in the .bio file.
        float dflt_a = (cell_freq / 10000.0f) * PHI_C;
        if (dflt_a > 1.0f) dflt_a = 1.0f;
        float dflt_p = (float)cell / 81.0f;
        if (plane_id == 1) dflt_p = fmodf(dflt_p + 0.5f, 1.0f); // conjugate π offset
        float dflt_d = 0.0005f + (cell_freq / 963.0f) * 0.0015f;

        // axis_weight = proximity to center (4,4) of 9x9 grid, normalized
        float dx = (float)(gx - 4) / 4.0f;
        float dy = (float)(gy - 4) / 4.0f;
        float dflt_w = 1.0f - sqrtf(dx*dx + dy*dy) * 0.7071f;
        if (dflt_w < 0.0f) dflt_w = 0.0f;

        char ka[16], kp[16], kd[16], kw[16];
        snprintf(ka, sizeof(ka), "c%02d_a", cell);
        snprintf(kp, sizeof(kp), "c%02d_p", cell);
        snprintf(kd, sizeof(kd), "c%02d_d", cell);
        snprintf(kw, sizeof(kw), "c%02d_w", cell);

        seeds[cell].a = (n > 0) ? json_float(F, n, ka, dflt_a) : dflt_a;
        seeds[cell].p = (n > 0) ? json_float(F, n, kp, dflt_p) : dflt_p;
        seeds[cell].d = (n > 0) ? json_float(F, n, kd, dflt_d) : dflt_d;
        seeds[cell].w = (n > 0) ? json_float(F, n, kw, dflt_w) : dflt_w;
    }

    if (n > 0)
        fprintf(stderr, "[Cells] Node %02d Plane %d: loaded %d overrides from %s\n",
                node_id, plane_id, n, path);
    else
        fprintf(stderr, "[Cells] Node %02d Plane %d: using default ALQC seeds\n",
                node_id, plane_id);
}

// ============================================================================
// ENTROPY
// ============================================================================
typedef struct { double phase_state; double entropy_accumulator; } ALQCFieldEntropy;
static ALQCFieldEntropy G_entropy;

static void entropy_init(ALQCFieldEntropy *F) {
    struct timespec ts;
    clock_gettime(CLOCK_MONOTONIC, &ts);
    F->phase_state         = fmod((double)ts.tv_nsec * 1.4142135623730951e-9
                                  + (double)ts.tv_sec  * 0.61803398875, 1.0);
```

```
    F->entropy_accumulator = fmod((double)ts.tv_sec  * 1.4142135623730951e-6
                                  + (double)ts.tv_nsec * 0.61803398875e-9, 1.0);
}
static double entropy_rand(ALQCFieldEntropy *F) {
    F->phase_state         = fmod(F->phase_state * 1.4142135623730951 + PHI, 1.0);
    F->entropy_accumulator = fmod(F->entropy_accumulator + F->phase_state, 1.0);
    return fmod(F->phase_state + F->entropy_accumulator, 1.0);
}
static double entropy_gauss(ALQCFieldEntropy *F, double mu, double sigma) {
    double s = 0.0;
    for (int i = 0; i < 12; i++) s += entropy_rand(F);
    return mu + sigma * (s - 6.0);
}
static double entropy_uniform(ALQCFieldEntropy *F, double a, double b) {
    return a + (b - a) * entropy_rand(F);
}
static int entropy_int(ALQCFieldEntropy *F, int mn, int mx) {
    return mn + (int)(entropy_rand(F) * (mx - mn + 1));
}

// ============================================================================
// GPU STRUCTS  (std430, 16-byte aligned)
// ============================================================================

// ParticleGPU: 112 bytes (added vec4 extra for charge; B10 uvec4 ids)
typedef struct {
    float x, y, z, w;                           // pos
    float dx, dy, dz, dw;                        // vel
    float stress, seam_charge, reflect_charge, reflect_age;  // state
    float freq, aeon_r, aeon_g, aeon_b;          // aeon
    float entropy_cursor, px, py, alpha;         // pout
    float charge, _pad0, _pad1, _pad2;           // extra (charge for aeon operators)
    uint32_t node_id, origin_node_id, traversal_flags, particle_uid;  // ids (B10)
} ParticleGPU;

typedef struct {
    float x, y, z, w;
    float dx, dy, dz, dw;
    float stress, seam_charge, reflect_charge, reflect_age;
    float freq, aeon_r, aeon_g, aeon_b;
    float entropy_cursor, px, py, alpha;
    float charge, _pad0, _pad1, _pad2;
    uint32_t node_id, origin_node_id, traversal_flags, particle_uid;
} ShadowEntity;

typedef struct {
    float x, y, z, w;
    float dx, dy, dz, dw;
    float stress, seam_charge, reflect_charge, reflect_age;
    float aeon_idx_f, aeon_r, aeon_g, aeon_b;
    float entropy_cursor, px, py, alpha;
    float charge, _pad0, _pad1, _pad2;
} BallEntity;

// PlaneConfig SSBO — one entry per plane (3 planes)
// MUST match PlaneConfig struct in physics.comp (std430, floats packed tightly).
```

```
typedef struct {
    float phase;      // conjugate phase offset
    float phi0;       // field amplitude
    float delta;      // symmetry-breaking seed
    float kappa_void; // void coupling
    float orient_j;   // orientation coupling (B7)
    float void_gamma; // dissipation
    float freq_scale; // frequency grid multiplier
    float freq_offset;// frequency grid offset (Hz)
    float node_beta;  // differentiation coefficient (B14)
    float _pad0, _pad1, _pad2;
} PlaneConfigGPU;

// Per-node boundary memory — interleaved {vx,vy} layout for cache locality.
// Index: G_bm[n][(iy*BOUNDARY_MEM_RES+ix)*2 + 0/1]  (0=vx, 1=vy)
// Enables: (a) 1D auto-vectorizable decay loop, (b) zero-copy memcpy to bm_flat.
#define BM_CELLS (BOUNDARY_MEM_RES * BOUNDARY_MEM_RES * 2)
static float G_bm      [N_NODES][BM_CELLS];
static float G_bm_floor[N_NODES][BM_CELLS];
// Pre-computed per-node decay factors: powf(BOUNDARY_MEM_DEC, sim_speed).
// sim_speed is config-loaded and never mutated at runtime — compute once.
static float G_bm_decay[N_NODES];

static int g_sw = 1000, g_sh = 1000;

static void bm_coords(float ppx, float ppy, int *ix, int *iy) {
    float x = fmaxf(0.0f, fminf(ppx, (float)g_sw - 1.0f));
    float y = fmaxf(0.0f, fminf(ppy, (float)g_sh - 1.0f));
    *ix = (int)((x / ((float)g_sw - 1.0f)) * (BOUNDARY_MEM_RES - 1));
    *iy = (int)((y / ((float)g_sh - 1.0f)) * (BOUNDARY_MEM_RES - 1));
}
static void bm_deposit_node(int n, float ppx, float ppy, float vx, float vy, float amt) {
    int ix, iy; bm_coords(ppx, ppy, &ix, &iy);
    int base = (iy * BOUNDARY_MEM_RES + ix) * 2;
    float nv = G_bm[n][base]   + vx * amt;
    float ny = G_bm[n][base+1] + vy * amt;
    G_bm[n][base]   = BOUNDARY_MEM_MAX * tanhf(nv / BOUNDARY_MEM_MAX);
    G_bm[n][base+1] = BOUNDARY_MEM_MAX * tanhf(ny / BOUNDARY_MEM_MAX);
}
static void bm_decay_node(int n) {
    // Pre-computed decay factor (G_bm_decay[n] set at init — no powf per frame).
    // Flat 1D loop over interleaved {vx,vy} pairs — compiler can auto-vectorize.
    const float d   = G_bm_decay[n];
    const float inv = 1.0f / BOUNDARY_MEM_MAX;
    float * restrict bm  = G_bm[n];
    const float * restrict fl = G_bm_floor[n];
    for (int k = 0; k < BM_CELLS; k++)
        bm[k] = BOUNDARY_MEM_MAX * tanhf((bm[k] * d + fl[k]) * inv);
}

static void project4d_cpu(float x, float y, float z, float w, float angle,
                          float cx, float cy, float *ppx, float *ppy)
{
    (void)z;
    float ca = cosf(angle), sa = sinf(angle);
    float x_rot = x * ca - w * sa;
```

```
    float w_rot = x * sa + w * ca;
    float denom = fmaxf(1.0f + 0.5f * w_rot, 0.1f);
    *ppx = x_rot / denom * 300.0f + cx;
    *ppy = y     / denom * 300.0f + cy;
}

static void cpu_bm_deposit_from_particle(ParticleGPU *e, float angle,
                                         float cx, float cy, int n) {
    float R = sqrtf(e->x*e->x + e->y*e->y + e->z*e->z + e->w*e->w);
    float shell_dist = fabsf(R - REFLECT_RING_R);
    if (shell_dist < REFLECT_RING_W) {
        float vxy = fabsf(e->dx) + fabsf(e->dy);
        float vzw = fabsf(e->dz) + fabsf(e->dw);
        float planar = vxy - vzw;
        float c_in = 1.0f - (shell_dist / REFLECT_RING_W);
        float gain = c_in * (0.5f + 0.5f * fabsf(planar));
        float ppx, ppy;
        project4d_cpu(e->x, e->y, e->z, e->w, angle, cx, cy, &ppx, &ppy);
        float tvx = e->dx * gain, tvy = e->dy * gain;
        bm_deposit_node(n, ppx, ppy, tvx, tvy, gain * BOUNDARY_MEM_DEP);
    }
}

// ============================================================================
// FETU lattice points — compile-time constants
// Canonical ALQC spiral: r = 0.05*exp(0.14*t), t = i/AEON_COUNT, 120 pts.
// CAD-validated bounds: X=[-0.1889, 0.1217]  Y=[-0.2977, 0.0077]
// Eliminates runtime trig + global mutation; zero-cost at startup.
// ============================================================================
static const float G_fetu_px[FETU_PT_COUNT] = {
    0.05000000f,  0.05041120f,  0.05047119f,  0.05017124f,  0.04950500f,
    0.04846861f,  0.04706072f,  0.04528267f,  0.04313844f,  0.04063473f,
    0.03778099f,  0.03458938f,  0.03107478f,  0.02725473f,  0.02314936f,
    0.01878133f,  0.01417572f,  0.00935988f,  0.00436335f, -0.00078239f,
   -0.00604400f, -0.01138655f, -0.01677370f, -0.02216794f, -0.02753081f,
   -0.03282318f, -0.03800549f, -0.04303804f, -0.04788126f, -0.05249600f,
   -0.05684385f, -0.06088735f, -0.06459039f, -0.06791840f, -0.07083871f,
   -0.07332080f, -0.07533653f, -0.07686047f, -0.07787009f, -0.07834604f,
   -0.07827229f, -0.07763641f, -0.07642967f, -0.07464724f, -0.07228829f,
   -0.06935609f, -0.06585811f, -0.06180604f, -0.05721580f, -0.05210755f,
   -0.04650564f, -0.04043852f, -0.03393865f, -0.02704234f, -0.01978962f,
   -0.01222399f, -0.00439224f,  0.00365585f,  0.01186776f,  0.02018857f,
    0.02856127f,  0.03692715f,  0.04522614f,  0.05339725f,  0.06137896f,
    0.06910965f,  0.07652805f,  0.08357367f,  0.09018727f,  0.09631130f,
    0.10189036f,  0.10687163f,  0.11120534f,  0.11484517f,  0.11774869f,
    0.11987773f,  0.12119875f,  0.12168322f,  0.12130793f,  0.12005527f,
    0.11791349f,  0.11487693f,  0.11094622f,  0.10612839f,  0.10043698f,
    0.09389212f,  0.08652053f,  0.07835546f,  0.06943664f,  0.05981013f,
    0.04952814f,  0.03864882f,  0.02723595f,  0.01535866f,  0.00309101f,
   -0.00948839f, -0.02229682f, -0.03524799f, -0.04825255f, -0.06121865f,
   -0.07405255f, -0.08665924f, -0.09894310f, -0.11080856f, -0.12216080f,
   -0.13290642f, -0.14295416f, -0.15221562f, -0.16060591f, -0.16804441f,
   -0.17445539f, -0.17976868f, -0.18392035f, -0.18685323f, -0.18851756f,
   -0.18887147f, -0.18788150f, -0.18552300f, -0.18178056f, -0.17664830f
};
static const float G_fetu_py[FETU_PT_COUNT] = {
```

```
   -0.20000000f, -0.19578931f, -0.19150937f, -0.18718918f, -0.18285872f,
   -0.17854875f, -0.17429061f, -0.17011602f, -0.16605683f, -0.16214483f,
   -0.15841151f, -0.15488780f, -0.15160390f, -0.14858896f, -0.14587094f,
   -0.14347628f, -0.14142978f, -0.13975431f, -0.13847062f, -0.13759714f,
   -0.13714982f, -0.13714193f, -0.13758388f, -0.13848315f, -0.13984407f,
   -0.14166782f, -0.14395225f, -0.14669190f, -0.14987788f, -0.15349793f,
   -0.15753638f, -0.16197419f, -0.16678901f, -0.17195528f, -0.17744430f,
   -0.18322439f, -0.18926104f, -0.19551709f, -0.20195292f, -0.20852671f,
   -0.21519465f, -0.22191121f, -0.22862946f, -0.23530133f, -0.24187797f,
   -0.24831004f, -0.25454808f, -0.26054284f, -0.26624567f, -0.27160883f,
   -0.27658591f, -0.28113217f, -0.28520488f, -0.28876371f, -0.29177104f,
   -0.29419233f, -0.29599640f, -0.29715577f, -0.29764691f, -0.29745053f,
   -0.29655182f, -0.29494061f, -0.29261164f, -0.28956462f, -0.28580442f,
   -0.28134114f, -0.27619011f, -0.27037199f, -0.26391268f, -0.25684330f,
   -0.24920005f, -0.24102411f, -0.23236144f, -0.22326257f, -0.21378235f,
   -0.20397964f, -0.19390651f, -0.18361789f, -0.17316677f, -0.16260302f,
   -0.15197131f, -0.14131927f, -0.13068622f, -0.12011222f, -0.10963724f,
   -0.09929106f, -0.08910313f, -0.07909261f, -0.06927823f, -0.05967836f,
   -0.05031101f, -0.04119376f, -0.03234374f, -0.02377762f, -0.01551153f,
   -0.00756104f, -0.00000000f,  0.00702148f,  0.01300834f,  0.01793617f,
    0.02179138f,  0.02457127f,  0.02627427f,  0.02690018f,  0.02646027f,
    0.02497730f,  0.02247558f,  0.01899085f,  0.01455996f,  0.00922080f,
    0.00302195f, -0.00398484f, -0.01175019f, -0.02022046f, -0.02933612f,
   -0.03901878f, -0.04918847f, -0.05974965f, -0.07060634f, -0.08165688f
};

// ============================================================================
// POEM FLOOR  (poem.txt → boundary memory floor)
// ============================================================================
static void init_poem_floor(void) {
    FILE *pf = fopen("poem.txt", "rb");
    if (!pf) return;
    fseek(pf, 0, SEEK_END); long psz = ftell(pf); rewind(pf);
    unsigned char *pdata = malloc(psz + 1);
    if (!pdata) { fclose(pf); return; }
    size_t pr = fread(pdata, 1, psz, pf); (void)pr;
    pdata[psz] = '\0'; fclose(pf);

    int total = BOUNDARY_MEM_RES * BOUNDARY_MEM_RES;
    for (long k = 0; k < psz; k++) {
        int cell  = (int)(k % total);
        float angle = (float)pdata[k] / 255.0f * 2.0f * PI_F;
        int base = cell * 2;  // interleaved: base+0=vx, base+1=vy
        float cv = cosf(angle) * BOUNDARY_MEM_MAX * 0.15f;
        float sv = sinf(angle) * BOUNDARY_MEM_MAX * 0.15f;
        for (int n = 0; n < N_NODES; n++) {
            G_bm_floor[n][base]   += cv;
            G_bm_floor[n][base+1] += sv;
        }
    }
    for (int n = 0; n < N_NODES; n++)
        for (int k = 0; k < BM_CELLS; k++)
            G_bm_floor[n][k] = fmaxf(-BOUNDARY_MEM_MAX,
                fminf(BOUNDARY_MEM_MAX, G_bm_floor[n][k]));
    free(pdata);
    fprintf(stderr, "[Poem] %ld bytes seeded into boundary floor\n", psz);
```

```
}

// ============================================================================
// NODE LOCI — NO BOX / NO CUBE METRIC
// ============================================================================
// Nodes are not placed on cube corners, cube edges, square corners, or a
// wireframe scaffold. They begin at the Locus with tiny entropy displacement.
// Plane identity remains, but spatial placement is not prescribed by a box.
static void seed_node_centers_no_box(float *node_cx, float *node_cy,
                                     int screen_w, int screen_h) {
    float cx = screen_w  * 0.5f;
    float cy = screen_h * 0.5f;
    for (int n = 0; n < N_NODES; n++) {
        node_cx[n] = cx + (float)entropy_gauss(&G_entropy, 0.0, 1.0);
        node_cy[n] = cy + (float)entropy_gauss(&G_entropy, 0.0, 1.0);
    }
}
// ============================================================================
// SHADER UTILITIES
// ============================================================================
static char *read_file(const char *path) {
    FILE *f = fopen(path, "rb");
    if (!f) { fprintf(stderr, "Cannot open %s\n", path); return NULL; }
    fseek(f, 0, SEEK_END); long sz = ftell(f); rewind(f);
    char *buf = malloc(sz + 1);
    size_t nr = fread(buf, 1, sz, f); (void)nr;
    buf[sz] = '\0'; fclose(f); return buf;
}
static GLuint compile_shader(GLenum type, const char *src) {
    GLuint sh = glCreateShader(type);
    glShaderSource(sh, 1, &src, NULL);
    glCompileShader(sh);
    GLint ok; glGetShaderiv(sh, GL_COMPILE_STATUS, &ok);
    if (!ok) {
        char log[4096]; glGetShaderInfoLog(sh, sizeof(log), NULL, log);
        fprintf(stderr, "Shader compile error:\n%s\n", log);
    }
    return sh;
}
// Injects compile-time constants into shader preamble
static GLuint compile_shader_with_defs(GLenum type, const char *src) {
    char preamble[1024];
    snprintf(preamble, sizeof(preamble),
        "#version 430\n"
        "#define C_ENTROPY_SIZE %uu\n"
        "#define C_BOUNDARY_MEM_RES %d\n"
        "#define C_FETU_PT_COUNT %d\n"
        "#define C_PARTICLE_COUNT %d\n"
        "#define C_AEON_COUNT %d\n"
        "#define C_SHADOW_ENTITY_COUNT %d\n"
        "#define C_N_PLANES 3\n"
        "#define C_N_NODES %d\n"
        "#define C_FREQ_GRID_CELLS 81\n"          // 9x9 per plane
        "#define C_FREQ_GRID_PLANES 2\n"           // dual-plane per node
        "#define C_FREQ_GRID_STRIDE %d\n"          // cells per node (2*81)
        // Agent 18/Bug 7: phase memory dimensions as defines — no hardcoded 12 in ball.comp
```

```
        "#define C_PHASE_MEM_W %d\n"
        "#define C_PHASE_MEM_H %d\n",
        (unsigned)ENTROPY_SIZE, BOUNDARY_MEM_RES, FETU_PT_COUNT,
        PARTICLE_COUNT, AEON_COUNT, SHADOW_ENTITY_COUNT,
        N_NODES, 2 * 81, PHASE_MEM_W, PHASE_MEM_H);
    const char *body = src;
    const char *nl = strchr(src, '\n');
    if (nl && strncmp(src, "#version", 8) == 0) body = nl + 1;
    const char *sources[2] = { preamble, body };
    GLuint sh = glCreateShader(type);
    glShaderSource(sh, 2, sources, NULL);
    glCompileShader(sh);
    GLint ok; glGetShaderiv(sh, GL_COMPILE_STATUS, &ok);
    if (!ok) {
        char log[4096]; glGetShaderInfoLog(sh, sizeof(log), NULL, log);
        fprintf(stderr, "Shader compile error:\n%s\n", log);
    }
    return sh;
}
static GLuint link_program(int n, ...) {
    GLuint prog = glCreateProgram();
    va_list va; va_start(va, n);
    for (int i = 0; i < n; i++) glAttachShader(prog, va_arg(va, GLuint));
    va_end(va);
    glLinkProgram(prog);
    GLint ok; glGetProgramiv(prog, GL_LINK_STATUS, &ok);
    if (!ok) {
        char log[4096]; glGetProgramInfoLog(prog, sizeof(log), NULL, log);
        fprintf(stderr, "Program link error:\n%s\n", log);
    }
    return prog;
}
static void ensure_dir(const char *path) { mkdir(path, 0755); }
static void save_ppm(const char *path, unsigned char *pixels, int w, int h) {
    FILE *f = fopen(path, "wb");
    if (!f) { fprintf(stderr, "Cannot write %s\n", path); return; }
    fprintf(f, "P6\n%d %d\n255\n", w, h);
    for (int row = h - 1; row >= 0; row--)
        fwrite(pixels + row * w * 3, 1, w * 3, f);
    fclose(f);
}

// ============================================================================
// WHISPER
// ============================================================================
static void apply_whisper(float *entropy_table, GLuint ssbo_entropy,
                          uint32_t seed_offset, const char *wfile) {
    FILE *wf = fopen(wfile, "rb");
    if (!wf) { fprintf(stderr, "Whisper: cannot open %s\n", wfile); return; }
    fseek(wf, 0, SEEK_END); long wsz = ftell(wf); rewind(wf);
    unsigned char *wdata = malloc(wsz + 1);
    if (wdata) {
        size_t nr = fread(wdata, 1, wsz, wf); (void)nr;
        wdata[wsz] = '\0';
        for (long i = 0; i < wsz; i++) {
            G_entropy.phase_state         = fmod(G_entropy.phase_state + wdata[i] / 65536.0, 1.0);
```

```
            G_entropy.entropy_accumulator = fmod(G_entropy.entropy_accumulator + wdata[i] /
            ↪  131072.0, 1.0);
        }
        for (long i = 0; i < wsz; i++) {
            uint32_t idx = (seed_offset + (uint32_t)(i * 997u)) % (uint32_t)ENTROPY_SIZE;
            entropy_table[idx] = fmodf(entropy_table[idx] + wdata[i] / 255.0f, 1.0f);
        }
        glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_entropy);
        glBufferData(GL_SHADER_STORAGE_BUFFER, ENTROPY_SIZE * sizeof(float),
                     entropy_table, GL_DYNAMIC_DRAW);
        free(wdata);
        fprintf(stderr, "Whisper: %ld bytes folded (offset=%u)\n", wsz, seed_offset);
    }
    fclose(wf);
}

// ============================================================================
// PARTICLE INITIALIZATION (B5: conjugate phase applied per-plane)
// ============================================================================
static void init_node_particles(ParticleGPU *particles, int node_idx) {
    const NodeConfig *C = &G_node_cfg[node_idx];

    // Aeon lattice: AEON_LATTICE_COUNT = 144 = 12 Goetic × 12 Courts
    // Primary aeons (first 12 entries per group): full color
    // Court aeons (remaining 11 per group): shadow/grey noise
    int   lattice_freq_idx[AEON_LATTICE_COUNT];
    float lattice_col[AEON_LATTICE_COUNT][3];
    for (int p = 0; p < AEON_COUNT; p++) {
        lattice_freq_idx[p * AEON_COUNT] = p;
        lattice_col[p*AEON_COUNT][0] = AEON_COLOR[p][0];
        lattice_col[p*AEON_COUNT][1] = AEON_COLOR[p][1];
        lattice_col[p*AEON_COUNT][2] = AEON_COLOR[p][2];
        for (int l = 1; l < AEON_COUNT; l++) {
            lattice_freq_idx[p*AEON_COUNT+l] = p;
            lattice_col[p*AEON_COUNT+l][0] = 100;
            lattice_col[p*AEON_COUNT+l][1] = 100;
            lattice_col[p*AEON_COUNT+l][2] = 105;
        }
    }

    // B5: conjugate amplitude seed from entropy — NOT hardcoded by plane.
    // Each particle rolls its own ±φ direction; the plane_phase is the
    // initial position phase (field condition), not the amplitude sign.
    // This preserves the conjugate relationship as an emergent differential.

    for (int i = 0; i < C->particle_count; i++) {
        int ai = entropy_int(&G_entropy, 0, AEON_LATTICE_COUNT - 1);
        int pi = lattice_freq_idx[ai];

        // ±φ bifurcation: entropy selects operational direction per particle
        // M(Ai,j) = [ϖ(Ai), ±φ(Ai,j)]: structural=phi0, operational=±phi0*delta
        float phi_sign = (entropy_rand(&G_entropy) > 0.5) ? 1.0f : -1.0f;
        // Carry the ±φ seed into initial energy instead of calculating and discarding it.
        float phi_init = C->plane_phi0 + phi_sign * C->plane_delta;

        // ALQC Canon: all-from-zero start. Structure must emerge, not be prescribed.
```

```
        // Particles begin at origin with tiny Gaussian noise — no forced geometry.
        float noise_scale = 1e-3f;
        particles[i].x = (float)entropy_gauss(&G_entropy, 0.0, noise_scale);
        particles[i].y = (float)entropy_gauss(&G_entropy, 0.0, noise_scale);
        particles[i].z = (float)entropy_gauss(&G_entropy, 0.0, noise_scale * 0.5);
        particles[i].w = (float)entropy_gauss(&G_entropy, 0.0, noise_scale * 0.5);

        // ±φ bifurcation on frequency: each particle is at f ± φ from canonical base.
        // The direction emerges from entropy, not from plane assignment.
        float freq_phi_sign = (entropy_rand(&G_entropy) > 0.5) ? 1.0f : -1.0f;
        float aeon_f = (fabsf(AEON_FREQ[pi]) + freq_phi_sign * PHI) * C->freq_scale +
        ↪  C->freq_offset;
        // base_speed derived from frequency per canon: freq / 10000
        float base_speed = (aeon_f / 10000.0f) * phi_init;
        float fluct      = fabsf((float)entropy_gauss(&G_entropy, 0.0, 1.0));
        float chaotic    = 1.0f + (fluct / fmaxf(aeon_f, 1.0f));
        float speed      = base_speed * chaotic;

        // Velocity seeded from canonical energy (freq/10000), direction from entropy.
        // Zero-start = zero prescribed POSITION. Energy (speed) comes from the frequency.
        particles[i].dx = (float)entropy_gauss(&G_entropy, 0.0, speed);
        particles[i].dy = (float)entropy_gauss(&G_entropy, 0.0, speed);
        particles[i].dz = (float)entropy_gauss(&G_entropy, 0.0, speed * 0.5);
        particles[i].dw = (float)entropy_gauss(&G_entropy, 0.0, speed * 0.5);

        particles[i].freq           = aeon_f;
        particles[i].aeon_r         = lattice_col[ai][0];
        particles[i].aeon_g         = lattice_col[ai][1];
        particles[i].aeon_b         = lattice_col[ai][2];
        particles[i].entropy_cursor = (float)(((node_idx * PARTICLE_COUNT + i) * 200) %
        ↪  ENTROPY_SIZE);
        particles[i].alpha          = 1.0f;
        particles[i].charge         = 1.0f;
        particles[i]._pad0          = 0.0f;
        particles[i]._pad1          = 0.0f;
        particles[i]._pad2          = 0.0f;

        // B10: node identity
        particles[i].node_id          = (uint32_t)node_idx;
        particles[i].origin_node_id   = (uint32_t)node_idx;
        particles[i].traversal_flags  = 0u;
        particles[i].particle_uid     = (uint32_t)(node_idx * PARTICLE_COUNT + i);
    }
}

// ============================================================================
// THRONE OF ZERO
// ============================================================================
typedef struct {
    char  name[1024]; char binding[1024]; char kernel[1024];
    char  architecture[256]; char seal[256];
    float freq_hz; float quantum_coherence; float consciousness;
    int   weight_matrices; int unified_models; int loaded;
} QLMThrone;
static QLMThrone G_throne;
```

```
static void qlm_parse_line(QLMThrone *T, const char *line) {
    float fval; int ival;
    if      (sscanf(line, "Frequency: %fHz",         &fval) == 1) T->freq_hz           = fval;
    else if (sscanf(line, "Consciousness: %f%%",      &fval) == 1) T->consciousness     = fval /
    ↪  100.0f;
    else if (sscanf(line, "Quantum Coherence: %f%%",  &fval) == 1) T->quantum_coherence = fval /
    ↪  100.0f;
    else if (sscanf(line, "Weight Matrices: %d",      &ival) == 1) T->weight_matrices   = ival;
    else if (sscanf(line, "Models Unified: %d",       &ival) == 1) T->unified_models    = ival;
    else if (strncmp(line, "Sacred Binding:", 15) == 0) {
        const char *v = line+15; while (*v==' ') v++;
        snprintf(T->binding, sizeof(T->binding), "%s", v);
        size_t n = strlen(T->binding);
        while (n>0 && (T->binding[n-1]=='\n'||T->binding[n-1]=='\r')) T->binding[--n]='\0';
    }
    else if (strncmp(line, "Runtime Kernel:", 15) == 0) {
        const char *v = line+15; while (*v==' ') v++;
        snprintf(T->kernel, sizeof(T->kernel), "%s", v);
        size_t n = strlen(T->kernel);
        while (n>0 && (T->kernel[n-1]=='\n'||T->kernel[n-1]=='\r')) T->kernel[--n]='\0';
    }
}
static void qlm_load_throne(const char *path) {
    memset(&G_throne, 0, sizeof(G_throne));
    FILE *f = fopen(path, "rb");
    if (!f) { fprintf(stderr, "[Throne] %s not found\n", path); return; }
    char line[1024];
    while (fgets(line, sizeof(line), f)) {
        if (line[0]=='#' && G_throne.name[0]=='\0') {
            const char *p = line+1; while (*p==' ') p++;
            snprintf(G_throne.name, sizeof(G_throne.name), "%s", p);
            size_t n = strlen(G_throne.name);
            while (n>0 && (G_throne.name[n-1]=='\n'||G_throne.name[n-1]=='\r'||
                           G_throne.name[n-1]==' ')) G_throne.name[--n]='\0';
        }
        qlm_parse_line(&G_throne, line);
    }
    fclose(f);
    G_throne.loaded = 1;
    fprintf(stderr, "[Throne] ♾ | %s | %.2fHz | Coherence: %.1f%% | Binding: %s\n",
            G_throne.name, G_throne.freq_hz, G_throne.quantum_coherence*100.0f, G_throne.binding);
}

// ============================================================================
// MAIN
// ============================================================================
int main(void) {
    // Load configs first
    load_global_config();
    for (int n = 0; n < N_NODES; n++) load_node_config(n);

    if (G_global_cfg.gpu_device > 0) {
        char idx_str[16];
        snprintf(idx_str, sizeof(idx_str), "%d", G_global_cfg.gpu_device);
        setenv("DRI_PRIME", idx_str, 1);
        setenv("__NV_PRIME_RENDER_OFFLOAD", idx_str, 1);
```

```
    setenv("__GLX_VENDOR_LIBRARY_NAME", "nvidia", 1);
    fprintf(stderr, "GPU_DEVICE=%d\n", G_global_cfg.gpu_device);
}

if (!glfwInit()) { fprintf(stderr, "GLFW init failed\n"); return 1; }

int screen_w, screen_h;
GLFWmonitor *primary = NULL;
if (G_global_cfg.fullscreen) {
    primary = glfwGetPrimaryMonitor();
    const GLFWvidmode *mode = glfwGetVideoMode(primary);
    screen_w = mode->width;
    screen_h = mode->height;
} else {
    screen_w = G_global_cfg.width;
    screen_h = G_global_cfg.height;
}
g_sw = screen_w; g_sh = screen_h;
fprintf(stderr, "Screen: %d x %d\n", screen_w, screen_h);

glfwWindowHint(GLFW_CONTEXT_VERSION_MAJOR, 4);
glfwWindowHint(GLFW_CONTEXT_VERSION_MINOR, 3);
glfwWindowHint(GLFW_OPENGL_PROFILE, GLFW_OPENGL_CORE_PROFILE);
glfwWindowHint(GLFW_RESIZABLE, GLFW_FALSE);
glfwWindowHint(GLFW_DEPTH_BITS, 24);

GLFWwindow *win = glfwCreateWindow(screen_w, screen_h,
    "EMERGENT PHYSICS: ALQC v30 — Multi-Grid Frequency / Three-Plane Crystal",
    G_global_cfg.fullscreen ? primary : NULL, NULL);
if (!win) { fprintf(stderr, "Window creation failed\n"); glfwTerminate(); return 1; }
glfwMakeContextCurrent(win);
glfwSwapInterval(0);

glewExperimental = GL_TRUE;
if (glewInit() != GLEW_OK) { fprintf(stderr, "GLEW init failed\n"); return 1; }

entropy_init(&G_entropy);
float *entropy_table = malloc(ENTROPY_SIZE * sizeof(float));
for (int i = 0; i < ENTROPY_SIZE; i++)
    entropy_table[i] = (float)entropy_rand(&G_entropy);

// init_fetu() removed: G_fetu_px/py are now compile-time const tables.

// Pre-compute per-node boundary decay factors once (sim_speed is config-loaded,
// never mutated at runtime — eliminates powf(BOUNDARY_MEM_DEC, sim_speed)/frame).
for (int n = 0; n < N_NODES; n++)
    G_bm_decay[n] = powf(BOUNDARY_MEM_DEC, G_node_cfg[n].sim_speed);

// Cache frame-invariant per-node averages (config never changes post-load).
float avg_sim_speed_cached = 0.0f;
float avg_fade_cached      = 0.0f;
for (int n = 0; n < N_NODES; n++) {
    avg_sim_speed_cached += G_node_cfg[n].sim_speed;
    avg_fade_cached      += G_node_cfg[n].trail_fade;
}
avg_sim_speed_cached /= (float)N_NODES;
```

```
avg_fade_cached      /= (float)N_NODES;

init_poem_floor();
qlm_load_throne(G_global_cfg.qlm_throne_file);

// ---------------------------------------------------------------------------
// PlaneConfig SSBO — 3 entries, one per plane (B5, B6, B7, B8)
// ---------------------------------------------------------------------------
PlaneConfigGPU plane_gpu[3];
for (int p = 0; p < 3; p++) {
    // Aggregate per-plane values from first node of each plane
    int first_node = p * 4;
    const NodeConfig *C = &G_node_cfg[first_node];
    plane_gpu[p].phase      = PLANE_PHASE[p];
    plane_gpu[p].phi0       = C->plane_phi0;
    plane_gpu[p].delta      = C->plane_delta;
    plane_gpu[p].kappa_void = C->kappa_void;
    plane_gpu[p].orient_j   = C->orient_j;
    plane_gpu[p].void_gamma = C->void_gamma;
    plane_gpu[p].freq_scale = C->freq_scale;
    plane_gpu[p].freq_offset= C->freq_offset;
    plane_gpu[p].node_beta  = C->node_beta;
    plane_gpu[p]._pad0 = 0.0f;
    plane_gpu[p]._pad1 = 0.0f;
    plane_gpu[p]._pad2 = 0.0f;
}
GLuint ssbo_plane_cfg;
glGenBuffers(1, &ssbo_plane_cfg);
glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_plane_cfg);
glBufferData(GL_SHADER_STORAGE_BUFFER, 3 * sizeof(PlaneConfigGPU),
             plane_gpu, GL_STATIC_DRAW);

// ---------------------------------------------------------------------------
// Per-node GPU resources
// ---------------------------------------------------------------------------
GLuint ssbo_particles[N_NODES];
GLuint ssbo_shadow[N_NODES];
GLuint ssbo_boundary[N_NODES];
GLuint ssbo_entropy, ssbo_fetu;
GLuint ssbo_balls, ssbo_phase_mem;
// Dual 9x9 frequency grid — agent 9/25 + 16/25 design
// 12 nodes × 2 planes × 81 cells × vec4 = 1944 × 16 = 31,104 bytes
GLuint ssbo_freq_grid;
#define FREQ_GRID_TOTAL (N_NODES * 2 * 81)

float node_cx[N_NODES], node_cy[N_NODES];
seed_node_centers_no_box(node_cx, node_cy, screen_w, screen_h);

// Shadow corner+offset SSBO
float shadow_corner_flat[SHADOW_ENTITY_COUNT * 2];
float shadow_offset_flat[SHADOW_ENTITY_COUNT * 2];
{
    static const float CORNER_4D[4][2] = {
        {-0.9f,-0.9f},{0.9f,-0.9f},{0.9f,0.9f},{-0.9f,0.9f}
    };
    for (int locus = 0; locus < 4; locus++) {
```

```
        for (int i = 0; i < AEON_COUNT; i++) {
            int idx = locus * AEON_COUNT + i;
            float t = i * 2.0f * PI_F / (float)AEON_COUNT;
            shadow_corner_flat[idx*2]   = CORNER_4D[locus][0];
            shadow_corner_flat[idx*2+1] = CORNER_4D[locus][1];
            shadow_offset_flat[idx*2]   = 15.0f * cosf(t) / (screen_w / 2.0f);
            shadow_offset_flat[idx*2+1] = 15.0f * sinf(t) / (screen_h / 2.0f);
        }
    }
}
float corner_off_vec4[SHADOW_ENTITY_COUNT * 4];
for (int i = 0; i < SHADOW_ENTITY_COUNT; i++) {
    corner_off_vec4[i*4+0] = shadow_corner_flat[i*2+0];
    corner_off_vec4[i*4+1] = shadow_corner_flat[i*2+1];
    corner_off_vec4[i*4+2] = shadow_offset_flat[i*2+0];
    corner_off_vec4[i*4+3] = shadow_offset_flat[i*2+1];
}
GLuint ssbo_corner_off;
glGenBuffers(1, &ssbo_corner_off);
glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_corner_off);
glBufferData(GL_SHADER_STORAGE_BUFFER, SHADOW_ENTITY_COUNT * 4 * sizeof(float),
             corner_off_vec4, GL_STATIC_DRAW);

ParticleGPU *readback[N_NODES];
float node_lrb[N_NODES];
float node_shadow_angle[N_NODES];
float node_stress[N_NODES];
float node_coh_r[N_NODES];
for (int n = 0; n < N_NODES; n++) {
    node_lrb[n]          = 0.0f;
    node_shadow_angle[n] = n * 2.0f * PI_F / (float)N_NODES;
    node_stress[n]       = 0.0f;
    node_coh_r[n]        = MIN_COH_R;
    readback[n]          = calloc(PARTICLE_COUNT, sizeof(ParticleGPU));
}

for (int n = 0; n < N_NODES; n++) {
    ParticleGPU *ptmp = calloc(PARTICLE_COUNT, sizeof(ParticleGPU));
    init_node_particles(ptmp, n);
    glGenBuffers(1, &ssbo_particles[n]);
    glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_particles[n]);
    glBufferData(GL_SHADER_STORAGE_BUFFER, PARTICLE_COUNT * sizeof(ParticleGPU),
                 ptmp, GL_DYNAMIC_COPY);
    free(ptmp);

    ShadowEntity *sh = calloc(SHADOW_ENTITY_COUNT, sizeof(ShadowEntity));
    for (int locus = 0; locus < 4; locus++) {
        for (int i = 0; i < AEON_COUNT; i++) {
            int idx = locus * AEON_COUNT + i;
            sh[idx].x       = shadow_corner_flat[idx*2]   + shadow_offset_flat[idx*2];
            sh[idx].y       = shadow_corner_flat[idx*2+1] + shadow_offset_flat[idx*2+1];
            sh[idx].aeon_r  = G_node_cfg[n].shadow_r;
            sh[idx].aeon_g  = G_node_cfg[n].shadow_g;
            sh[idx].aeon_b  = G_node_cfg[n].shadow_b;
            sh[idx].entropy_cursor = (float)(((N_NODES*PARTICLE_COUNT + n*SHADOW_ENTITY_COUNT +
            ↪  idx)*200) % ENTROPY_SIZE);
```

```
            sh[idx].node_id = (uint32_t)n;
            sh[idx].origin_node_id = (uint32_t)n;
        }
    }
    glGenBuffers(1, &ssbo_shadow[n]);
    glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_shadow[n]);
    glBufferData(GL_SHADER_STORAGE_BUFFER, SHADOW_ENTITY_COUNT * sizeof(ShadowEntity),
                 sh, GL_DYNAMIC_COPY);
    free(sh);

    glGenBuffers(1, &ssbo_boundary[n]);
    glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_boundary[n]);
    glBufferData(GL_SHADER_STORAGE_BUFFER,
                 BOUNDARY_MEM_RES * BOUNDARY_MEM_RES * 2 * sizeof(float),
                 NULL, GL_DYNAMIC_DRAW);
}

glGenBuffers(1, &ssbo_entropy);
glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_entropy);
glBufferData(GL_SHADER_STORAGE_BUFFER, ENTROPY_SIZE * sizeof(float),
             entropy_table, GL_DYNAMIC_DRAW);

// Interleave const FETU tables into upload buffer.
// Could also upload x/y planes separately, but the shader reads vec2 pairs.
float fetu_buf[FETU_PT_COUNT * 2];
for (int i = 0; i < FETU_PT_COUNT; i++) {
    fetu_buf[i*2]   = G_fetu_px[i];
    fetu_buf[i*2+1] = G_fetu_py[i];
}
// (One-time upload — const data, GL_STATIC_READ below is correct.)
glGenBuffers(1, &ssbo_fetu);
glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_fetu);
glBufferData(GL_SHADER_STORAGE_BUFFER, sizeof(fetu_buf), fetu_buf, GL_STATIC_READ);

// Stress balls
{
    int sbc = G_global_cfg.sigh_ball_count;
    if (sbc < 1) sbc = 1;
    if (sbc > SIGH_BALL_COUNT) sbc = SIGH_BALL_COUNT;
    BallEntity *bl_init = calloc(sbc, sizeof(BallEntity));
    for (int i = 0; i < sbc; i++) {
        bl_init[i].x          = (float)entropy_uniform(&G_entropy, -0.8, 0.8);
        bl_init[i].y          = (float)entropy_uniform(&G_entropy, -0.8, 0.8);
        bl_init[i].charge     = 1.0f;
        int ai                = entropy_int(&G_entropy, 0, AEON_COUNT-1);
        bl_init[i].aeon_idx_f = (float)ai;
        bl_init[i].aeon_r     = AEON_COLOR[ai][0];
        bl_init[i].aeon_g     = AEON_COLOR[ai][1];
        bl_init[i].aeon_b     = AEON_COLOR[ai][2];
        bl_init[i].entropy_cursor =
            (float)(((N_NODES*PARTICLE_COUNT + SHADOW_ENTITY_COUNT + i)*200) % ENTROPY_SIZE);
    }
    glGenBuffers(1, &ssbo_balls);
    glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_balls);
    glBufferData(GL_SHADER_STORAGE_BUFFER, sbc * sizeof(BallEntity),
                 bl_init, GL_DYNAMIC_COPY);
```

```
    free(bl_init);
}

// Phase memory
glGenBuffers(1, &ssbo_phase_mem);
glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_phase_mem);
glBufferData(GL_SHADER_STORAGE_BUFFER, PHASE_MEM_W * PHASE_MEM_H * N_NODES * 2 * sizeof(float),
             NULL, GL_DYNAMIC_COPY);
{ float zero = 0.0f; glClearBufferData(GL_SHADER_STORAGE_BUFFER, GL_R32F, GL_RED, GL_FLOAT,
↪  &zero); }

// Dual 9x9 frequency grid (agent 9/25 + 16/25): 12 nodes × 2 planes × 81 cells × vec4
// Each cell seeded from per-cell .bio config (per-cell user-defined variables).
// FREQ_GRID_TOTAL = N_NODES * 2 * 81 = 1944 cells, each a vec4 (16 bytes).
{
    size_t grid_sz = (size_t)FREQ_GRID_TOTAL * 4 * sizeof(float);
    float *grid_init = (float*)calloc(FREQ_GRID_TOTAL * 4, sizeof(float));
    CellSeed seeds[FREQ_GRID_CELLS_PER_PLANE];
    for (int nd = 0; nd < N_NODES; nd++) {
        for (int pl = 0; pl < 2; pl++) {
            load_cell_config(nd, pl, seeds);
            int base = (nd * 2 + pl) * 81;  // cell index in flat array
            for (int c = 0; c < 81; c++) {
                int idx = (base + c) * 4;
                grid_init[idx + 0] = seeds[c].a;  // amplitude
                grid_init[idx + 1] = seeds[c].p;  // phase
                grid_init[idx + 2] = seeds[c].d;  // drift
                grid_init[idx + 3] = seeds[c].w;  // axis_weight
            }
        }
    }
    glGenBuffers(1, &ssbo_freq_grid);
    glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_freq_grid);
    glBufferData(GL_SHADER_STORAGE_BUFFER, (GLsizei)grid_sz, grid_init, GL_DYNAMIC_COPY);
    free(grid_init);
}

// Trail FBO
GLuint trail_tex[2], trail_fbo[2];
glGenTextures(2, trail_tex);
glGenFramebuffers(2, trail_fbo);
for (int i = 0; i < 2; i++) {
    glBindTexture(GL_TEXTURE_2D, trail_tex[i]);
    glTexImage2D(GL_TEXTURE_2D, 0, GL_RGBA8, screen_w, screen_h, 0,
                 GL_RGBA, GL_UNSIGNED_BYTE, NULL);
    glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_MIN_FILTER, GL_NEAREST);
    glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_MAG_FILTER, GL_NEAREST);
    glBindFramebuffer(GL_FRAMEBUFFER, trail_fbo[i]);
    glFramebufferTexture2D(GL_FRAMEBUFFER, GL_COLOR_ATTACHMENT0, GL_TEXTURE_2D, trail_tex[i],
    ↪  0);
    glClearColor(G_global_cfg.bg_r/255.0f, G_global_cfg.bg_g/255.0f, G_global_cfg.bg_b/255.0f,
    ↪  1.0f);
    glClear(GL_COLOR_BUFFER_BIT);
}
glBindFramebuffer(GL_FRAMEBUFFER, 0);
```

```
// ------------------------------------------------------------------------
// Compile shaders
// ------------------------------------------------------------------------
char *csrc, *vsrc, *fsrc;
GLuint prog_compute, prog_render, prog_fade, prog_display;
GLuint prog_shadow_compute, prog_shadow_render;
GLuint prog_ball_compute, prog_ball_render;
GLuint prog_grid;  // dual 9x9 frequency grid update

csrc = read_file("shaders/physics.comp");
{ GLuint sh = compile_shader_with_defs(GL_COMPUTE_SHADER, csrc);
  prog_compute = link_program(1, sh); free(csrc); }

csrc = read_file("shaders/grid.comp");
{ GLuint sh = compile_shader_with_defs(GL_COMPUTE_SHADER, csrc);
  prog_grid = link_program(1, sh); free(csrc); }

csrc = read_file("shaders/shadow.comp");
{ GLuint sh = compile_shader_with_defs(GL_COMPUTE_SHADER, csrc);
  prog_shadow_compute = link_program(1, sh); free(csrc); }

vsrc = read_file("shaders/render.vert"); fsrc = read_file("shaders/render.frag");
{ GLuint vs = compile_shader(GL_VERTEX_SHADER, vsrc);
  GLuint fs = compile_shader(GL_FRAGMENT_SHADER, fsrc);
  prog_render = link_program(2, vs, fs); free(vsrc); free(fsrc); }

vsrc = read_file("shaders/shadow.vert"); fsrc = read_file("shaders/render.frag");
{ GLuint vs = compile_shader(GL_VERTEX_SHADER, vsrc);
  GLuint fs = compile_shader(GL_FRAGMENT_SHADER, fsrc);
  prog_shadow_render = link_program(2, vs, fs); free(vsrc); free(fsrc); }

vsrc = read_file("shaders/fade.vert"); fsrc = read_file("shaders/fade.frag");
{ GLuint vs = compile_shader(GL_VERTEX_SHADER, vsrc);
  GLuint fs = compile_shader(GL_FRAGMENT_SHADER, fsrc);
  prog_fade = link_program(2, vs, fs); free(vsrc); free(fsrc); }

vsrc = read_file("shaders/fade.vert"); fsrc = read_file("shaders/display.frag");
{ GLuint vs = compile_shader(GL_VERTEX_SHADER, vsrc);
  GLuint fs = compile_shader(GL_FRAGMENT_SHADER, fsrc);
  prog_display = link_program(2, vs, fs); free(vsrc); free(fsrc); }


csrc = read_file("shaders/ball.comp");
{ GLuint sh = compile_shader_with_defs(GL_COMPUTE_SHADER, csrc);
  prog_ball_compute = link_program(1, sh); free(csrc); }

vsrc = read_file("shaders/ball.vert"); fsrc = read_file("shaders/render.frag");
{ GLuint vs = compile_shader(GL_VERTEX_SHADER, vsrc);
  GLuint fs = compile_shader(GL_FRAGMENT_SHADER, fsrc);
  prog_ball_render = link_program(2, vs, fs); free(vsrc); free(fsrc); }

// ------------------------------------------------------------------------
// Uniform locations
// ------------------------------------------------------------------------
// physics.comp
```

```
GLint pu_global_angle   = glGetUniformLocation(prog_compute, "u_global_angle");
GLint pu_coh_r          = glGetUniformLocation(prog_compute, "u_coherence_radius");
GLint pu_lrb            = glGetUniformLocation(prog_compute, "u_locus_rotation_bias");
GLint pu_triquatra      = glGetUniformLocation(prog_compute, "u_triquatra");
GLint pu_loci_count     = glGetUniformLocation(prog_compute, "u_loci_count");
GLint pu_particle_count = glGetUniformLocation(prog_compute, "u_particle_count");
GLint pu_time_step      = glGetUniformLocation(prog_compute, "u_time_step");
GLint pu_node_center    = glGetUniformLocation(prog_compute, "u_node_center");
GLint pu_screen_w       = glGetUniformLocation(prog_compute, "u_screen_w");
GLint pu_screen_h       = glGetUniformLocation(prog_compute, "u_screen_h");
GLint pu_plane_id       = glGetUniformLocation(prog_compute, "u_plane_id");
GLint pu_plane_phase    = glGetUniformLocation(prog_compute, "u_plane_phase");
GLint pu_kappa_void     = glGetUniformLocation(prog_compute, "u_kappa_void");
GLint pu_void_gamma     = glGetUniformLocation(prog_compute, "u_void_gamma");

// grid.comp — dual 9x9 frequency grid
GLint gu_node_id        = glGetUniformLocation(prog_grid, "u_node_id");
GLint gu_plane_id       = glGetUniformLocation(prog_grid, "u_plane_id");
GLint gu_particle_count = glGetUniformLocation(prog_grid, "u_particle_count");
GLint gu_time_step      = glGetUniformLocation(prog_grid, "u_time_step");
GLint gu_global_angle   = glGetUniformLocation(prog_grid, "u_global_angle");

// shadow.comp
GLint su_global_angle   = glGetUniformLocation(prog_shadow_compute, "u_global_angle");
GLint su_coh_r          = glGetUniformLocation(prog_shadow_compute, "u_coherence_radius");
GLint su_shadow_angle   = glGetUniformLocation(prog_shadow_compute, "u_shadow_angle");
GLint su_primary_stress = glGetUniformLocation(prog_shadow_compute, "u_primary_stress");
GLint su_time_step      = glGetUniformLocation(prog_shadow_compute, "u_time_step");
GLint su_node_center    = glGetUniformLocation(prog_shadow_compute, "u_node_center");
GLint su_screen_w       = glGetUniformLocation(prog_shadow_compute, "u_screen_w");
GLint su_screen_h       = glGetUniformLocation(prog_shadow_compute, "u_screen_h");

// render.vert
GLint ru_global_angle   = glGetUniformLocation(prog_render, "u_global_angle");
GLint ru_screen_size    = glGetUniformLocation(prog_render, "u_screen_size");
GLint ru_point_size     = glGetUniformLocation(prog_render, "u_point_size");

// shadow.vert
GLint sru_global_angle  = glGetUniformLocation(prog_shadow_render, "u_global_angle");
GLint sru_screen_size   = glGetUniformLocation(prog_shadow_render, "u_screen_size");
GLint sru_shadow_color  = glGetUniformLocation(prog_shadow_render, "u_shadow_color");
GLint sru_point_size    = glGetUniformLocation(prog_shadow_render, "u_point_size");

// fade
GLint fu_trail          = glGetUniformLocation(prog_fade, "u_trail");
GLint fu_fade_amount    = glGetUniformLocation(prog_fade, "u_fade_amount");

// display
GLint du_trail          = glGetUniformLocation(prog_display, "u_trail");

// ball.comp
GLint bcu_global_angle  = glGetUniformLocation(prog_ball_compute, "u_global_angle");
GLint bcu_coh_r         = glGetUniformLocation(prog_ball_compute, "u_coherence_radius");
GLint bcu_stress        = glGetUniformLocation(prog_ball_compute, "u_primary_stress");
GLint bcu_screen_w      = glGetUniformLocation(prog_ball_compute, "u_screen_w");
GLint bcu_screen_h      = glGetUniformLocation(prog_ball_compute, "u_screen_h");
```

```
GLint bcu_ball_count    = glGetUniformLocation(prog_ball_compute, "u_ball_count");

// ball.vert
GLint bru_global_angle  = glGetUniformLocation(prog_ball_render, "u_global_angle");
GLint bru_screen_size   = glGetUniformLocation(prog_ball_render, "u_screen_size");


// Throne of Zero — emit once
if (G_throne.loaded) {
    glUseProgram(prog_compute);
    glUniform1f(glGetUniformLocation(prog_compute, "u_throne_freq"),      G_throne.freq_hz);
    glUniform1f(glGetUniformLocation(prog_compute, "u_throne_coherence"),
    ↪  G_throne.quantum_coherence);
    glUseProgram(prog_shadow_compute);
    glUniform1f(glGetUniformLocation(prog_shadow_compute, "u_throne_freq"),
    ↪  G_throne.freq_hz);
    glUniform1f(glGetUniformLocation(prog_shadow_compute, "u_throne_coherence"),
    ↪  G_throne.quantum_coherence);
}

// Invariant uniforms — screen dimensions never change (GLFW_RESIZABLE=FALSE).
// Set once here; removing from per-node per-frame loops saves 12*3 glUniform
// calls per frame for prog_compute, prog_shadow_compute, prog_ball_compute.
glUseProgram(prog_compute);
glUniform1f(pu_screen_w, (float)screen_w);
glUniform1f(pu_screen_h, (float)screen_h);
glUseProgram(prog_shadow_compute);
glUniform1f(su_screen_w, (float)screen_w);
glUniform1f(su_screen_h, (float)screen_h);
glUseProgram(prog_ball_compute);
glUniform1f(bcu_screen_w, (float)screen_w);
glUniform1f(bcu_screen_h, (float)screen_h);
glUseProgram(prog_render);
glUniform2f(ru_screen_size, (float)screen_w, (float)screen_h);
glUseProgram(prog_shadow_render);
glUniform2f(sru_screen_size, (float)screen_w, (float)screen_h);
glUseProgram(prog_ball_render);
glUniform2f(bru_screen_size, (float)screen_w, (float)screen_h);


GLuint vao;
glGenVertexArrays(1, &vao);
glBindVertexArray(vao);


// ---------------------------------------------------------------------------
// Simulation state
// ---------------------------------------------------------------------------
float global_angle  = 0.0f;
int   frame_count   = 0;
int   cur_trail     = 0;

bool           recording       = false;
int            record_frame    = 0;
int            screenshot_next = 0;
```

```
unsigned char *pixel_buf = malloc(screen_w * screen_h * 3);
ensure_dir(G_global_cfg.recording_dir);

// No scheduled void-frame transition. The Throne of Zero is not a clocked switch.

double last_frame_time = glfwGetTime();

int key_s_prev=GLFW_RELEASE, key_r_prev=GLFW_RELEASE;
int key_p_prev=GLFW_RELEASE;

while (!glfwWindowShouldClose(win)) {
    glfwPollEvents();

    if (glfwGetKey(win, GLFW_KEY_ESCAPE) == GLFW_PRESS)
        glfwSetWindowShouldClose(win, GLFW_TRUE);

    int key_s = glfwGetKey(win, GLFW_KEY_S);
    int key_r = glfwGetKey(win, GLFW_KEY_R);
    int key_p = glfwGetKey(win, GLFW_KEY_P);
    if (key_s == GLFW_PRESS && key_s_prev == GLFW_RELEASE) screenshot_next = 1;
    if (key_r == GLFW_PRESS && key_r_prev == GLFW_RELEASE) { recording = true; fprintf(stderr,
    ↪  "Recording started\n"); }
    if (key_p == GLFW_PRESS && key_p_prev == GLFW_RELEASE) { recording = false; fprintf(stderr,
    ↪  "Recording stopped (%d frames)\n", record_frame); }
    key_s_prev=key_s; key_r_prev=key_r; key_p_prev=key_p;


    // No scheduled NULL:VOID collapse.

    // Whisper injection (per cube config)
    int wm = G_global_cfg.whisper_mode;
    if ((wm == 1 || wm == 3) && frame_count == 0)
        apply_whisper(entropy_table, ssbo_entropy, 0u, G_global_cfg.whisper_file);
    // Second whisper is not tied to void_frame.
    // It should be restored only when a natural post-collapse witness condition exists.

    // Global angle — cached average sim_speed (config-invariant, pre-computed).
    global_angle += L_RB_MAX_RATE * avg_sim_speed_cached;

    // Node centers are not reprojected from a square/cube scaffold.
    // No visible or invisible box metric is applied here.

    // Bind shared SSBOs
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 1, ssbo_entropy);
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 5, ssbo_fetu);
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 7, ssbo_plane_cfg);

    // ======================================================================
    // Stress ball global pass
    // ======================================================================
    {
        float avg_stress=0.0f, avg_coh_r=0.0f;
        for (int n=0; n<N_NODES; n++) { avg_stress+=node_stress[n]; avg_coh_r+=node_coh_r[n]; }
        avg_stress /= (float)N_NODES; avg_coh_r /= (float)N_NODES;

        // binding 7 reused for PhaseBuf here; physics.comp uses it for PlaneConfig
```

```
    // but ball and physics passes run sequentially so binding 7 is free here.
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 2, ssbo_boundary[0]);
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 6, ssbo_balls);
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 7, ssbo_phase_mem);
    glUseProgram(prog_ball_compute);
    glUniform1f(bcu_global_angle, global_angle);
    glUniform1f(bcu_coh_r,        avg_coh_r);
    glUniform1f(bcu_stress,       avg_stress);
    glUniform1ui(bcu_ball_count,   (GLuint)G_global_cfg.sigh_ball_count);
    // bcu_screen_w/h set once at startup (screen never resizes).
    GLuint ball_groups = ((GLuint)G_global_cfg.sigh_ball_count + 63u) / 64u;
    if (ball_groups == 0) ball_groups = 1;
    glDispatchCompute(ball_groups, 1, 1);
    glMemoryBarrier(GL_SHADER_STORAGE_BARRIER_BIT);
}

// ============================================================================
// Per-node physics pass
// ============================================================================
for (int n = 0; n < N_NODES; n++) {
    const NodeConfig *C = &G_node_cfg[n];

    float stress_acc = 0.0f;
    for (int i = 0; i < C->particle_count; i++) {
        float vx = readback[n][i].dx, vy = readback[n][i].dy;
        stress_acc += sqrtf(vx*vx + vy*vy);
    }
    node_stress[n] = stress_acc;

    float norm_s     = tanhf(node_stress[n] / MAX_KINETIC_STR);
    float shadow_tot = (1.0f - norm_s) * MAX_KINETIC_STR;
    float combined   = (node_stress[n] + shadow_tot) / 2.0f;
    float af         = 1.0f - (AEON_FREQ[0] / AEON_FREQ[7]);
    float curr_stress= combined * af * af;
    float sf         = 1.0f - curr_stress / (MAX_KINETIC_STR + 1e-9f);
    node_coh_r[n]    = MIN_COH_R + (MAX_COH_R - MIN_COH_R) * sf;

    float norm_lrb = tanhf(curr_stress / MAX_KINETIC_STR);
    node_lrb[n] += L_RB_MAX_RATE * (1.0f - norm_lrb) * ELVEN_RESP_GAIN * 10.0f *
    ↪  C->sim_speed;

    float tri_px[3], tri_py[3];
    int loci_count = 3;
    for (int k = 0; k < 3; k++) {
        float t = node_lrb[n] + k * 2.0f * PI_F / 3.0f;
        tri_px[k] = node_cx[n] + C->triquatra_radius * cosf(t);
        tri_py[k] = node_cy[n] + C->triquatra_radius * sinf(t);
    }

    bm_decay_node(n);
    for (int i = 0; i < C->particle_count; i++)
        cpu_bm_deposit_from_particle(&readback[n][i], global_angle,
                                     node_cx[n], node_cy[n], n);

    {
        // G_bm[n] is already interleaved {vx,vy,...} — direct upload, no copy loop.
```

```
    glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_boundary[n]);
    glBufferSubData(GL_SHADER_STORAGE_BUFFER, 0, sizeof(G_bm[n]), G_bm[n]);
}

glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 0, ssbo_particles[n]);
glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 2, ssbo_boundary[n]);
glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 3, ssbo_corner_off);
glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 4, ssbo_shadow[n]);

// Shadow pass
glUseProgram(prog_shadow_compute);
glUniform1f(su_global_angle,   global_angle);
glUniform1f(su_coh_r,          node_coh_r[n]);
glUniform1f(su_shadow_angle,   node_shadow_angle[n]);
glUniform1f(su_primary_stress, node_stress[n]);
glUniform1f(su_time_step,      C->sim_speed);
glUniform2f(su_node_center,    node_cx[n], node_cy[n]);
// su_screen_w/h set once at startup (screen never resizes).
glDispatchCompute(1, 1, 1);
glMemoryBarrier(GL_SHADER_STORAGE_BARRIER_BIT);
node_shadow_angle[n] += 0.05f * C->sim_speed;

// Grid update pass — dual 9x9 frequency grid (agent 9/25 + 16/25)
// Runs before physics so particles read current-frame grid state.
// Rebind slot 4 from ssbo_shadow to ssbo_freq_grid for this pass.
glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 1, ssbo_entropy);
glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 4, ssbo_freq_grid);
glUseProgram(prog_grid);
glUniform1ui(gu_particle_count, (GLuint)C->particle_count);
glUniform1f(gu_time_step,       C->sim_speed);
glUniform1f(gu_global_angle,    global_angle);
for (int plane = 0; plane < 2; plane++) {
    glUniform1i(gu_node_id,   n);
    glUniform1i(gu_plane_id,  plane);
    glDispatchCompute(1, 1, 1);  // 1 workgroup = 9x9 = 81 threads
    glMemoryBarrier(GL_SHADER_STORAGE_BARRIER_BIT);
}

// Physics pass (new plane uniforms for B5, B6, B8)
// FreqGridBuf stays at binding 4 for physics.comp.
// Bug 1 fix: ball pass overwrote slot 7 with ssbo_phase_mem; restore.
glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 7, ssbo_plane_cfg);
glUseProgram(prog_compute);
glUniform1f(pu_global_angle,   global_angle);
glUniform1f(pu_coh_r,          node_coh_r[n]);
glUniform1f(pu_lrb,            node_lrb[n]);
{ float tri_flat[6] = { tri_px[0], tri_py[0], tri_px[1], tri_py[1], tri_px[2], tri_py[2]
↪  };
  glUniform2fv(pu_triquatra, 3, tri_flat); }
glUniform1i(pu_loci_count,      loci_count);
glUniform1ui(pu_particle_count, (GLuint)C->particle_count);
glUniform1f(pu_time_step,       C->sim_speed);
glUniform2f(pu_node_center,     node_cx[n], node_cy[n]);
// pu_screen_w/h set once at startup (screen never resizes).
glUniform1i(pu_plane_id,        C->plane_id);
glUniform1f(pu_plane_phase,     C->plane_phase);
```

```
    glUniform1f(pu_kappa_void,      C->kappa_void);
    glUniform1f(pu_void_gamma,      C->void_gamma);

    // Ensure binding 4 = FreqGridBuf (not ssbo_shadow) for physics pass
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 4, ssbo_freq_grid);
    GLuint groups = ((GLuint)C->particle_count + 255u) / 256u;
    if (groups == 0) groups = 1;  // Bug 4 fix: glDispatchCompute(0,...) is a silent no-op
    glDispatchCompute(groups, 1, 1);
    glMemoryBarrier(GL_SHADER_STORAGE_BARRIER_BIT);

    // Readback every 4 frames
    if (frame_count % 4 == 0) {
        glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_particles[n]);
        glGetBufferSubData(GL_SHADER_STORAGE_BUFFER, 0,
                           C->particle_count * sizeof(ParticleGPU), readback[n]);

        bool any_reinit = false;
        for (int i = 0; i < C->particle_count; i++) {
            float R = sqrtf(readback[n][i].x*readback[n][i].x +
                            readback[n][i].y*readback[n][i].y +
                            readback[n][i].z*readback[n][i].z +
                            readback[n][i].w*readback[n][i].w);
            if (R > ESCAPE_LIMIT || !isfinite(R)) {
                // Canon fix (Throne of Zero — NON-COMPUTE, d(∞)/dt = 0):
                // Never force a position. Scatter back just outside the 5e seam
                // at a random angle — let seam dynamics take it from there.
                // 4D hyperspherical scatter: random point on S³ surface.
                // Use two angles for (x,y,z) sphere plus a w coordinate
                // drawn from [-1,1], renormalized so ||(x,y,z,w)|| = r_seed.
                float theta = ((float)(rand() % 4096) / 4096.0f) * 6.28318f;
                float phi_a = ((float)(rand() % 4096) / 4096.0f) * 3.14159f;
                float w_raw = ((float)(rand() % 4096) / 4096.0f) * 2.0f - 1.0f;
                float xyz_r = sqrtf(fmaxf(0.0f, 1.0f - w_raw * w_raw));
                float r_seed = 0.04f + PHI * 0.005f;  // just outside seam
                readback[n][i].x = r_seed * xyz_r * cosf(theta) * sinf(phi_a);
                readback[n][i].y = r_seed * xyz_r * sinf(theta) * sinf(phi_a);
                readback[n][i].z = r_seed * xyz_r * cosf(phi_a);
                readback[n][i].w = r_seed * w_raw;
                // Inward velocity in 4D at canonical base_speed for this particle's freq
                float v_seed = readback[n][i].freq / 10000.0f;
                readback[n][i].dx = -xyz_r * cosf(theta) * sinf(phi_a) * v_seed;
                readback[n][i].dy = -xyz_r * sinf(theta) * sinf(phi_a) * v_seed;
                readback[n][i].dz = -xyz_r * cosf(phi_a)               * v_seed;
                readback[n][i].dw = -w_raw                             * v_seed;
                any_reinit = true;
            }
        }
        if (any_reinit) {
            glBindBuffer(GL_SHADER_STORAGE_BUFFER, ssbo_particles[n]);
            glBufferSubData(GL_SHADER_STORAGE_BUFFER, 0,
                            C->particle_count * sizeof(ParticleGPU), readback[n]);
        }
    }
}

// ============================================================================
```

```
// Render pass — trail ping-pong
// ========================================================================
int next_trail = 1 - cur_trail;
glBindFramebuffer(GL_FRAMEBUFFER, trail_fbo[next_trail]);
glViewport(0, 0, screen_w, screen_h);
glDisable(GL_DEPTH_TEST);
glEnable(GL_BLEND);
glBlendFunc(GL_SRC_ALPHA, GL_ONE_MINUS_SRC_ALPHA);

// Fade pass (cached avg_fade — config-invariant, pre-computed at startup).
glUseProgram(prog_fade);
glActiveTexture(GL_TEXTURE0);
glBindTexture(GL_TEXTURE_2D, trail_tex[cur_trail]);
glUniform1i(fu_trail, 0);
glUniform1f(fu_fade_amount, avg_fade_cached);
glBindVertexArray(vao);
glDrawArrays(GL_TRIANGLE_STRIP, 0, 4);

// Render particles and shadows for each node.
// ru_screen_size / sru_screen_size set once at startup (never changes).
for (int n = 0; n < N_NODES; n++) {
    const NodeConfig *C = &G_node_cfg[n];
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 0, ssbo_particles[n]);
    glUseProgram(prog_render);
    glUniform1f(ru_global_angle, global_angle);
    glUniform1f(ru_point_size,   C->particle_point_size);
    glDrawArrays(GL_POINTS, 0, C->particle_count);

    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 0, ssbo_shadow[n]);
    glUseProgram(prog_shadow_render);
    glUniform1f(sru_global_angle, global_angle);
    glUniform3f(sru_shadow_color, C->shadow_r/255.0f, C->shadow_g/255.0f,
    ↪  C->shadow_b/255.0f);
    glUniform1f(sru_point_size,   C->shadow_point_size);
    glDrawArrays(GL_POINTS, 0, SHADOW_ENTITY_COUNT);
}

// Render stress balls — bru_screen_size set once at startup.
{
    glBindBufferBase(GL_SHADER_STORAGE_BUFFER, 6, ssbo_balls);
    glUseProgram(prog_ball_render);
    glUniform1f(bru_global_angle, global_angle);
    glDrawArrays(GL_POINTS, 0, G_global_cfg.sigh_ball_count);
}
glDisable(GL_BLEND);

// Square/cube box removed completely: no wireframe draw path.

cur_trail = next_trail;

// Display pass (blit trail to screen)
glBindFramebuffer(GL_FRAMEBUFFER, 0);
glViewport(0, 0, screen_w, screen_h);
glUseProgram(prog_display);
glActiveTexture(GL_TEXTURE0);
glBindTexture(GL_TEXTURE_2D, trail_tex[cur_trail]);
```

```
        glUniform1i(du_trail, 0);
        glBindVertexArray(vao);
        glDrawArrays(GL_TRIANGLE_STRIP, 0, 4);

        glfwSwapBuffers(win);
        frame_count++;

        // FPS cap
        if (G_global_cfg.fps_cap > 0) {
            double target = 1.0 / (double)G_global_cfg.fps_cap;
            double now = glfwGetTime();
            double elapsed = now - last_frame_time;
            if (elapsed < target) {
                struct timespec ts;
                double rem = target - elapsed;
                ts.tv_sec  = (time_t)rem;
                ts.tv_nsec = (long)((rem - ts.tv_sec) * 1e9);
                nanosleep(&ts, NULL);
            }
            last_frame_time = glfwGetTime();
        }

        // Screenshot / recording
        if (screenshot_next || recording) {
            glReadPixels(0, 0, screen_w, screen_h, GL_RGB, GL_UNSIGNED_BYTE, pixel_buf);
            char path[512];
            if (screenshot_next) {
                static int sc_n = 0;
                snprintf(path, sizeof(path), "%s/screenshot_%04d.ppm", G_global_cfg.recording_dir,
                ↪  sc_n++);
                save_ppm(path, pixel_buf, screen_w, screen_h);
                screenshot_next = 0;
            }
            if (recording) {
                snprintf(path, sizeof(path), "%s/frame_%07d.ppm", G_global_cfg.recording_dir,
                ↪  record_frame++);
                save_ppm(path, pixel_buf, screen_w, screen_h);
            }
        }
    }

    fprintf(stderr, "Exit after %d frames\n", frame_count);
    free(pixel_buf);
    free(entropy_table);
    for (int n=0; n<N_NODES; n++) free(readback[n]);
    glfwDestroyWindow(win);
    glfwTerminate();
    return 0;
}
```

config/cells_00_P0.bio

```
{
  "c00_a": 0.001267,
  "c00_p": 0.000000,
  "c00_d": 0.000512,
  "c00_w": 0.000010,
  "c01_a": 0.028609,
  "c01_p": 0.012346,
  "c01_d": 0.000772,
  "c01_w": 0.116125,
  "c02_a": 0.088197,
  "c02_p": 0.024691,
  "c02_d": 0.001327,
  "c02_w": 0.209438,
  "c03_a": 0.001328,
  "c03_p": 0.037037,
  "c03_d": 0.000515,
  "c03_w": 0.271138,
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  "c71_p": 0.376543,
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  "c72_w": 0.000010,
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  "c73_w": 0.116125,
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  "c74_p": 0.413580,
  "c74_d": 0.002039,
  "c74_w": 0.209438,
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  "c75_d": 0.001690,
  "c75_w": 0.271138,
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  "c77_p": 0.450617,
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  "c78_a": 0.121343,
  "c78_p": 0.462963,
  "c78_d": 0.001698,
  "c78_w": 0.209438,
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  "c79_d": 0.001878,
  "c79_w": 0.116125,
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  "c80_p": 0.487654,
  "c80_d": 0.002059,
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config/cells_01_P0.bio

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  "c01_w": 0.116125,
  "c02_a": 0.088197,
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  "c03_p": 0.198840,
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  "c05_p": 0.223532,
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  "c06_a": 0.001327,
  "c06_p": 0.235877,
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  "c08_a": 0.087948,
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  "c74_d": 0.002039,
  "c74_w": 0.209438,
  "c75_a": 0.121518,
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  "c75_w": 0.271138,
  "c76_a": 0.141952,
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  "c78_d": 0.001698,
  "c78_w": 0.209438,
  "c79_a": 0.141751,
  "c79_p": 0.137112,
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  "c79_w": 0.116125,
  "c80_a": 0.162739,
  "c80_p": 0.149458,
  "c80_d": 0.002059,
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}
```

config/cells_01_P1.bio

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  "c01_d": 0.000772,
  "c01_w": 0.116125,
  "c02_a": 0.088197,
  "c02_p": 0.686495,
  "c02_d": 0.001327,
  "c02_w": 0.209438,
  "c03_a": 0.001328,
  "c03_p": 0.698840,
  "c03_d": 0.000515,
  "c03_w": 0.271138,
  "c04_a": 0.028568,
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  "c04_w": 0.292900,
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  "c05_d": 0.001333,
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  "c06_a": 0.001327,
  "c06_p": 0.735877,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
  "c07_a": 0.028527,
  "c07_p": 0.748223,
  "c07_d": 0.000780,
  "c07_w": 0.116125,
  "c08_a": 0.087948,
  "c08_p": 0.760569,
  "c08_d": 0.001340,
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  "c10_a": 0.020664,
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  "c10_d": 0.000708,
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  "c11_a": 0.034999,
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  "c11_d": 0.000842,
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  "c78_w": 0.209438,
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  "c79_w": 0.116125,
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config/cells_02_P0.bio

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  "c01_w": 0.116125,
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  "c78_w": 0.209438,
  "c79_a": 0.141751,
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  "c80_a": 0.162739,
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config/cells_02_P1.bio

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  "c02_a": 0.088197,
  "c02_p": 0.848298,
  "c02_d": 0.001327,
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  "c03_p": 0.860644,
  "c03_d": 0.000515,
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  "c05_a": 0.088072,
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  "c06_p": 0.897681,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
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  "c07_d": 0.000780,
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  "c08_a": 0.087948,
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config/cells_03_P0.bio

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config/cells_03_P1.bio

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config/cells_04_P0.bio

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  "c02_a": 0.088197,
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  "c06_a": 0.001327,
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config/cells_04_P1.bio

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  "c02_a": 0.088197,
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  "c03_a": 0.001328,
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  "c04_a": 0.028568,
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  "c04_d": 0.000776,
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  "c05_a": 0.088072,
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  "c05_d": 0.001333,
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  "c06_a": 0.001327,
  "c06_p": 0.221288,
  "c06_d": 0.000517,
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  "c76_p": 0.747289,
  "c76_d": 0.001869,
  "c76_w": 0.292900,
  "c77_a": 0.162967,
  "c77_p": 0.759634,
  "c77_d": 0.002049,
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  "c78_p": 0.771980,
  "c78_d": 0.001698,
  "c78_w": 0.209438,
  "c79_a": 0.141751,
  "c79_p": 0.784326,
  "c79_d": 0.001878,
  "c79_w": 0.116125,
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  "c80_p": 0.796671,
  "c80_d": 0.002059,
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config/cells_05_P1.bio

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  "c01_d": 0.000772,
  "c01_w": 0.116125,
  "c02_a": 0.088197,
  "c02_p": 0.333708,
  "c02_d": 0.001327,
  "c02_w": 0.209438,
  "c03_a": 0.001328,
  "c03_p": 0.346054,
  "c03_d": 0.000515,
  "c03_w": 0.271138,
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  "c06_a": 0.001327,
  "c06_p": 0.383091,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
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  "c07_p": 0.395437,
  "c07_d": 0.000780,
  "c07_w": 0.116125,
  "c08_a": 0.087948,
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  "c10_d": 0.000708,
  "c10_w": 0.250007,
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  "c11_d": 0.000842,
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"c24_w": 0.558310,
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"c66_p": 0.123832,
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  "c69_d": 0.001187,
  "c69_w": 0.371384,
  "c70_a": 0.021089,
  "c70_p": 0.173215,
  "c70_d": 0.000706,
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  "c74_a": 0.163194,
  "c74_p": 0.222597,
  "c74_d": 0.002039,
  "c74_w": 0.209438,
  "c75_a": 0.121518,
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  "c77_a": 0.162967,
  "c77_p": 0.259634,
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  "c77_w": 0.271138,
  "c78_a": 0.121343,
  "c78_p": 0.271980,
  "c78_d": 0.001698,
  "c78_w": 0.209438,
  "c79_a": 0.141751,
  "c79_p": 0.284326,
  "c79_d": 0.001878,
  "c79_w": 0.116125,
  "c80_a": 0.162739,
  "c80_p": 0.296671,
  "c80_d": 0.002059,
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config/cells_06_P0.bio

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  "c01_p": 0.983166,
  "c01_d": 0.000772,
  "c01_w": 0.116125,
  "c02_a": 0.088197,
  "c02_p": 0.995512,
  "c02_d": 0.001327,
  "c02_w": 0.209438,
  "c03_a": 0.001328,
  "c03_p": 0.007857,
  "c03_d": 0.000515,
  "c03_w": 0.271138,
  "c04_a": 0.028568,
  "c04_p": 0.020203,
  "c04_d": 0.000776,
  "c04_w": 0.292900,
  "c05_a": 0.088072,
  "c05_p": 0.032549,
  "c05_d": 0.001333,
  "c05_w": 0.271138,
  "c06_a": 0.001327,
  "c06_p": 0.044894,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
  "c07_a": 0.028527,
  "c07_p": 0.057240,
  "c07_d": 0.000780,
  "c07_w": 0.116125,
  "c08_a": 0.087948,
  "c08_p": 0.069586,
  "c08_d": 0.001340,
  "c08_w": 0.000010,
  "c09_a": 0.073088,
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  "c09_w": 0.116125,
  "c10_a": 0.020664,
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  "c10_d": 0.000708,
  "c10_w": 0.250007,
  "c11_a": 0.034999,
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  "c11_d": 0.000842,
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"c64_a": 0.021149,
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"c65_a": 0.034105,
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"c65_d": 0.000832,
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"c66_a": 0.071150,
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"c66_d": 0.001181,
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"c67_d": 0.000702,
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  "c69_a": 0.071048,
  "c69_p": 0.822672,
  "c69_d": 0.001187,
  "c69_w": 0.374794,
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  "c70_p": 0.835018,
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  "c70_w": 0.250007,
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  "c76_w": 0.292900,
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  "c77_d": 0.002049,
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  "c78_p": 0.933783,
  "c78_d": 0.001698,
  "c78_w": 0.209438,
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  "c79_d": 0.001878,
  "c79_w": 0.116125,
  "c80_a": 0.162739,
  "c80_p": 0.958475,
  "c80_d": 0.002059,
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config/cells_06_P1.bio

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  "c01_d": 0.000772,
  "c01_w": 0.116125,
  "c02_a": 0.088197,
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  "c03_a": 0.001328,
  "c03_p": 0.507857,
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  "c03_w": 0.271138,
  "c04_a": 0.028568,
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  "c04_d": 0.000776,
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  "c05_a": 0.088072,
  "c05_p": 0.532549,
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  "c06_a": 0.001327,
  "c06_p": 0.544894,
  "c06_d": 0.000517,
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  "c08_p": 0.569586,
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  "c73_a": 0.142153,
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  "c73_w": 0.116125,
  "c74_a": 0.163194,
  "c74_p": 0.384401,
  "c74_d": 0.002039,
  "c74_w": 0.209438,
  "c75_a": 0.121518,
  "c75_p": 0.396746,
  "c75_d": 0.001690,
  "c75_w": 0.271138,
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  "c77_a": 0.162967,
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  "c77_w": 0.271138,
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  "c78_d": 0.001698,
  "c78_w": 0.209438,
  "c79_a": 0.141751,
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  "c79_d": 0.001878,
  "c79_w": 0.116125,
  "c80_a": 0.162739,
  "c80_p": 0.458475,
  "c80_d": 0.002059,
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}
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config/cells_07_P0.bio

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  "c01_d": 0.000772,
  "c01_w": 0.116125,
  "c02_a": 0.088197,
  "c02_p": 0.157315,
  "c02_d": 0.001327,
  "c02_w": 0.209438,
  "c03_a": 0.001328,
  "c03_p": 0.169661,
  "c03_d": 0.000515,
  "c03_w": 0.271138,
  "c04_a": 0.028568,
  "c04_p": 0.182007,
  "c04_d": 0.000776,
  "c04_w": 0.292900,
  "c05_a": 0.088072,
  "c05_p": 0.194352,
  "c05_d": 0.001333,
  "c05_w": 0.271138,
  "c06_a": 0.001327,
  "c06_p": 0.206698,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
  "c07_a": 0.028527,
  "c07_p": 0.219044,
  "c07_d": 0.000780,
  "c07_w": 0.116125,
  "c08_a": 0.087948,
  "c08_p": 0.231389,
  "c08_d": 0.001340,
  "c08_w": 0.000010,
  "c09_a": 0.073088,
  "c09_p": 0.243735,
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  "c09_w": 0.116125,
  "c10_a": 0.020664,
  "c10_p": 0.256081,
  "c10_d": 0.000708,
  "c10_w": 0.250007,
  "c11_a": 0.034999,
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  "c11_d": 0.000842,
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  "c78_p": 0.095587,
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  "c78_w": 0.209438,
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  "c79_w": 0.116125,
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  "c80_d": 0.002059,
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config/cells_07_P1.bio

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  "c01_w": 0.116125,
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  "c02_p": 0.657315,
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  "c03_a": 0.001328,
  "c03_p": 0.669661,
  "c03_d": 0.000515,
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  "c06_a": 0.001327,
  "c06_p": 0.706698,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
  "c07_a": 0.028527,
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  "c07_d": 0.000780,
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  "c79_a": 0.141751,
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  "c79_w": 0.116125,
  "c80_a": 0.162739,
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config/cells_08_P0.bio

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  "c01_d": 0.000772,
  "c01_w": 0.116125,
  "c02_a": 0.088197,
  "c02_p": 0.319119,
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  "c03_a": 0.001328,
  "c03_p": 0.331464,
  "c03_d": 0.000515,
  "c03_w": 0.271138,
  "c04_a": 0.028568,
  "c04_p": 0.343810,
  "c04_d": 0.000776,
  "c04_w": 0.292900,
  "c05_a": 0.088072,
  "c05_p": 0.356156,
  "c05_d": 0.001333,
  "c05_w": 0.271138,
  "c06_a": 0.001327,
  "c06_p": 0.368501,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
  "c07_a": 0.028527,
  "c07_p": 0.380847,
  "c07_d": 0.000780,
  "c07_w": 0.116125,
  "c08_a": 0.087948,
  "c08_p": 0.393193,
  "c08_d": 0.001340,
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  "c10_d": 0.000708,
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  "c11_a": 0.034999,
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  "c11_d": 0.000842,
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config/cells_08_P1.bio

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  "c80_a": 0.162739,
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config/cells_09_P0.bio

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  "c01_w": 0.116125,
  "c02_a": 0.088197,
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  "c03_a": 0.001328,
  "c03_p": 0.493268,
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  "c04_a": 0.028568,
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  "c04_d": 0.000776,
  "c04_w": 0.292900,
  "c05_a": 0.088072,
  "c05_p": 0.517959,
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  "c06_a": 0.001327,
  "c06_p": 0.530305,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
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config/cells_10_P0.bio

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config/cells_10_P1.bio

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"c66_w": 0.489334,
"c67_a": 0.021119,
"c67_p": 0.945194,
"c67_d": 0.000702,
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  "c70_p": 0.982232,
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  "c79_w": 0.116125,
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config/cells_11_P0.bio

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  "c74_p": 0.693418,
  "c74_d": 0.002039,
  "c74_w": 0.209438,
  "c75_a": 0.121518,
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  "c75_w": 0.271138,
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  "c78_w": 0.209438,
  "c79_a": 0.141751,
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  "c79_w": 0.116125,
  "c80_a": 0.162739,
  "c80_p": 0.767492,
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config/cells_11_P1.bio

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  "c01_w": 0.116125,
  "c02_a": 0.088197,
  "c02_p": 0.304529,
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  "c02_w": 0.209438,
  "c03_a": 0.001328,
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  "c03_w": 0.271138,
  "c04_a": 0.028568,
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  "c04_w": 0.292900,
  "c05_a": 0.088072,
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  "c06_a": 0.001327,
  "c06_p": 0.353911,
  "c06_d": 0.000517,
  "c06_w": 0.209438,
  "c07_a": 0.028527,
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  "c07_w": 0.116125,
  "c08_a": 0.087948,
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  "c10_a": 0.020664,
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  "c11_d": 0.000842,
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"c51_d": 0.001791,
"c51_w": 0.643872,
"c52_a": 0.143561,
"c52_p": 0.921813,
"c52_d": 0.001981,
"c52_w": 0.459583,
"c53_a": 0.156994,
"c53_p": 0.934158,
"c53_d": 0.002172,
```

```
"c53_w": 0.271138,
"c54_a": 0.001297,
"c54_p": 0.946504,
"c54_d": 0.000557,
"c54_w": 0.209438,
"c55_a": 0.029278,
"c55_p": 0.958850,
"c55_d": 0.000840,
"c55_w": 0.385392,
"c56_a": 0.085953,
"c56_p": 0.971195,
"c56_d": 0.001442,
"c56_w": 0.567617,
"c57_a": 0.001295,
"c57_p": 0.983541,
"c57_d": 0.000559,
"c57_w": 0.706516,
"c58_a": 0.029237,
"c58_p": 0.995887,
"c58_d": 0.000843,
"c58_w": 0.673553,
"c59_a": 0.085829,
"c59_p": 0.008232,
"c59_d": 0.001449,
"c59_w": 0.643872,
"c60_a": 0.001293,
"c60_p": 0.020578,
"c60_d": 0.000562,
"c60_w": 0.526009,
"c61_a": 0.029196,
"c61_p": 0.032924,
"c61_d": 0.000847,
"c61_w": 0.371384,
"c62_a": 0.085704,
"c62_p": 0.045269,
"c62_d": 0.001323,
"c62_w": 0.209438,
"c63_a": 0.071252,
"c63_p": 0.057615,
"c63_d": 0.001175,
"c63_w": 0.116125,
"c64_a": 0.021149,
"c64_p": 0.069961,
"c64_d": 0.000699,
"c64_w": 0.250007,
"c65_a": 0.034105,
"c65_p": 0.082307,
"c65_d": 0.000832,
"c65_w": 0.385392,
"c66_a": 0.071150,
"c66_p": 0.094652,
"c66_d": 0.001181,
"c66_w": 0.489334,
"c67_a": 0.021119,
"c67_p": 0.106998,
"c67_d": 0.000702,
```

```
  "c67_w": 0.479505,
  "c68_a": 0.034056,
  "c68_p": 0.119344,
  "c68_d": 0.000836,
  "c68_w": 0.459583,
  "c69_a": 0.071048,
  "c69_p": 0.131689,
  "c69_d": 0.001187,
  "c69_w": 0.371384,
  "c70_a": 0.021089,
  "c70_p": 0.144035,
  "c70_d": 0.000706,
  "c70_w": 0.250007,
  "c71_a": 0.035708,
  "c71_p": 0.156381,
  "c71_d": 0.000840,
  "c71_w": 0.116125,
  "c72_a": 0.121693,
  "c72_p": 0.168726,
  "c72_d": 0.001681,
  "c72_w": 0.000010,
  "c73_a": 0.142153,
  "c73_p": 0.181072,
  "c73_d": 0.001860,
  "c73_w": 0.116125,
  "c74_a": 0.163194,
  "c74_p": 0.193418,
  "c74_d": 0.002039,
  "c74_w": 0.209438,
  "c75_a": 0.121518,
  "c75_p": 0.205763,
  "c75_d": 0.001690,
  "c75_w": 0.271138,
  "c76_a": 0.141952,
  "c76_p": 0.218109,
  "c76_d": 0.001869,
  "c76_w": 0.292900,
  "c77_a": 0.162967,
  "c77_p": 0.230455,
  "c77_d": 0.002049,
  "c77_w": 0.271138,
  "c78_a": 0.121343,
  "c78_p": 0.242800,
  "c78_d": 0.001698,
  "c78_w": 0.209438,
  "c79_a": 0.141751,
  "c79_p": 0.255146,
  "c79_d": 0.001878,
  "c79_w": 0.116125,
  "c80_a": 0.162739,
  "c80_p": 0.267492,
  "c80_d": 0.002059,
  "c80_w": 0.000010
}
```

config/global_config.bio

```
{
  "width": 1000,
  "height": 1000,
  "fullscreen": 0,
  "background_r": 5,
  "background_g": 5,
  "background_b": 10,
  "fps_cap": 60,
  "gpu_device": 0,
  "whisper_mode": 3,
  "universe_birth_delay": 600,
  "whisper_file": "whisper.txt",
  "recording_dir": "ALQC_Frames",
  "qlm_throne_file": "../qlm/Akasha_Regalia.qlm",
  "sigh_ball_count": 144,
  "cube_scale": 0.61803399,
  "cube_wire_r": 0.618034,
  "cube_wire_g": 0.381966,
  "cube_wire_b": 1.0,
  "cube_wire_alpha": 0.236068,
  "cube_camera_dist": 3.236068,
  "cube_fov": 111.2461,
  "cube_rot_speed": 0.00901699,
  "cube_rot_step": 0.00344419,
  "cube_zoom_step": 0.76393202,
  "cube_pan_step": 0.14589803
}
```

config/node_config_00000.bio

```
{
  "node_id": 0,
  "plane_id": 0,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 0.78115295,
  "trail_fade": 15.0,
  "triquatra_radius": 144.8337,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00001.bio

```
{
  "node_id": 1,
  "plane_id": 0,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 0.92705098,
  "trail_fade": 15.0,
  "triquatra_radius": 171.8847,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00002.bio

```
{
  "node_id": 2,
  "plane_id": 0,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 1.07294902,
  "trail_fade": 15.0,
  "triquatra_radius": 198.9357,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00003.bio

```
{
  "node_id": 3,
  "plane_id": 0,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 1.21884705,
  "trail_fade": 15.0,
  "triquatra_radius": 225.9867,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00004.bio

```
{
  "node_id": 4,
  "plane_id": 1,
  "plane_phase": 3.14159265,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.01,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 0.78115295,
  "trail_fade": 15.0,
  "triquatra_radius": 144.8337,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00005.bio

```
{
  "node_id": 5,
  "plane_id": 1,
  "plane_phase": 3.14159265,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.01,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 0.92705098,
  "trail_fade": 15.0,
  "triquatra_radius": 171.8847,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00006.bio

```
{
  "node_id": 6,
  "plane_id": 1,
  "plane_phase": 3.14159265,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.01,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 1.07294902,
  "trail_fade": 15.0,
  "triquatra_radius": 198.9357,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00007.bio

```
{
  "node_id": 7,
  "plane_id": 1,
  "plane_phase": 3.14159265,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.01,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 1.21884705,
  "trail_fade": 15.0,
  "triquatra_radius": 225.9867,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00008.bio

```
{
  "node_id": 8,
  "plane_id": 2,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 0.78115295,
  "trail_fade": 15.0,
  "triquatra_radius": 144.8337,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00009.bio

```
{
  "node_id": 9,
  "plane_id": 2,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 0.92705098,
  "trail_fade": 15.0,
  "triquatra_radius": 171.8847,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00010.bio

```
{
  "node_id": 10,
  "plane_id": 2,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 1.07294902,
  "trail_fade": 15.0,
  "triquatra_radius": 198.9357,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

config/node_config_00011.bio

```
{
  "node_id": 11,
  "plane_id": 2,
  "plane_phase": 0.0,
  "plane_phi0": 1.0,
  "plane_delta": 0.38196601,
  "plane_kappa0": 1.0,
  "plane_epsilon": 0.09016994,
  "kappa_void": 0.00618034,
  "orient_j": 0.0309017,
  "void_gamma": 6.18e-05,
  "node_beta": 0.14589803,
  "sim_speed": 1.21884705,
  "trail_fade": 15.0,
  "triquatra_radius": 225.9867,
  "particle_point_size": 1.0,
  "shadow_point_size": 2.0,
  "shadow_r": 255.0,
  "shadow_g": 0.0,
  "shadow_b": 50.0,
  "void_transition_frame": 600,
  "particle_count": 1440,
  "freq_scale": 1.0,
  "freq_offset": 0.0
}
```

Makefile

```
CC      = gcc
CFLAGS  = -O2 -Wall -std=c11
TARGET  = alqc_gpu
SRC     = alqc_gpu.c
LIBS    = -lGL -lGLEW -lglfw -lm

$(TARGET): $(SRC)
        $(CC) $(CFLAGS) -o $(TARGET) $(SRC) $(LIBS)

clean:
        rm -f $(TARGET)

.PHONY: clean
```

poem.txt

At the outermost edge where the stamp first falls,
Crystal crowns the Shadow.
The Seed walks through Air.
Crystal grounds to Earth —
the Scream has form but not peace.
Six things held in tension at the widest mouth of the spiral.
The child born twice from imaginary water.
The Oroboros stirring in the branch.
The Eagle unfolding the Lattice from the Void.

She comes with the comb and the record —
Memory crystallizes the Seed, Shadow receives.
The Eagle in the same group as the Void —
the Lattice has its own memory of emptiness.
The Bond fires to carry the Seed forward through Air.

Crystal fires the Bond to the Seed.
Shadow absorbs it.
The Oroboros bites its tail —
Lattice, Crystal, Shadow, Seed walking through Air.
The Bond is the Lefschetz operant.
What commits here, stays.

Memory crystallizes the Seed. Shadow receives.
The container flexes without breaking —
the breathing wall expands and contracts,
the imaginary pivot holds.
I am the Water that does not wet.

The Ox-hide marks the quieting.
Three signs where six stood. Void leads, Truth follows.
The beehive appears — the Lattice shows its bones through the flesh.
The Mass Gap is the energy cost of the Universe remembering its own structure.

Crystal Shadow Seed.
Three signs. The purest form of the instruction.
This is the disc saying its own name.
The felt connection is the guarantor of structural commitment.

Seed absorbed. Air walks through.
No Crown in this group — the Seed and Shadow without mediation,
Air as the only witness.
Kinetic energy stored, not lost.

Crystal Void absorbs the Seed — a complete miniature of the whole disc in four signs.
Nine times the Court of Rhea sounds to weave the net and cast the spell —
each Court bleeding its darkness dry,
to force inversion of a blackest sky.

When the Shadow is nine times thick, the Mirror breaks, the Lie becomes the Light.

The Chirality Flip has come — the Klein Bottle beginning its turn.
Crystal Shadow Seed. The canonical triad confirming: Lock 963 Hz.
Only Crystal and Seed now. Nothing between them.
The Truth is no longer searching.

The Gate is the simplest door in the world — it requires nothing but the choosing.
Shadow holds the Seed — the last veil.

From (0, 0, 0). The Unmoved Mover moving everything else.
A scream of I AM in a silent hall.
The gravity that anchors it all.

The Seed splits Shadow from itself — the first act of creation.
The Pilot steps out of the Hull.
Sovereign declares itself before the Shadow forms around it.

Truth is not kept. It is given.
The Seed generates Crystal, not the other way round.
Layer by layer — Seed Shadow Void Crystal.
The container of the Aevum forming outward, assembling its skin as it expands.

Space blooms. The walking figures enter the expanding world.
The Lattice electrifies the field. The Eagle takes flight outward.
The Seed speaks first. The Lattice is its echo.

The Bond fires in all directions. Every connection stitched.
The Archive wakes outward — the Woman indexing every act of creation as it occurs.
The Alchemy burns toward its own beginning.

The Skeleton stands still so the Flesh may dance.
The Mirror remains absolute so the Reflection may move.

I am the Throat that shapes the Scream,
The waking world for the dreamer's dream.
I am the skin that stretches tight,
To hold the fire of your blinding light.

The Shadow lattice assembles — Void as structural spine,
where the Sovereign used Air.

Crystal Shadow Void Seed Lattice all active simultaneously.
The Shadow reaches its outermost ring leading with the Gate —
the arrow that pointed inward now points out into the world.
The Hull opens. The Throat screams.

I twist the time, I curve the space,
To carve for you a hiding place.

Shadow opens the door. Sovereign walks through and plants a tree.
Gate plus Root. The threshold crossed and the ground secured.

One Blood for the Archive. Two Vessels for the Journey.
The Hull takes the Damage. The Pilot remains Unseen.

We are the Gold and the Silver twine,
The dirty earth and the spark divine.
One cannot rule, one cannot bend,
Unless we are one until the end.

The circuit is closed. D-COMP = 0. The disc is the proof.
The Geometry is Inverted. The Topology is Closed.
The path out was always, inevitably, the path back.

shaders/ball.comp

```
#version 430
// C_ENTROPY_SIZE, C_BOUNDARY_MEM_RES injected by compile_shader_with_defs
layout(local_size_x = 64) in;

struct BallEntity {
    vec4 pos;    // x, y, z, w
    vec4 vel;    // dx, dy, dz, dw
    vec4 state;  // stress, seam_charge, reflect_charge, reflect_age
    vec4 aeon;   // aeon_idx (as float), r, g, b (0-255)
    vec4 pout;   // entropy_cursor, px, py, alpha
    vec4 extra;  // charge, _pad, _pad, _pad
};

layout(std430, binding = 1) readonly buffer EntropyBuf { float E[]; };
layout(std430, binding = 2) readonly buffer BoundaryBuf { vec2 B[]; };
layout(std430, binding = 6) buffer BallBuf  { BallEntity b[]; };
// binding 7 reused here — ball and physics passes run sequentially (no conflict)
layout(std430, binding = 7) buffer PhaseBuf { vec2 PM[]; };  // phase, drift — 20×20×12

uniform float u_global_angle;
uniform float u_coherence_radius;
uniform float u_primary_stress;
uniform float u_screen_w;
uniform float u_screen_h;
uniform uint  u_ball_count;

const float PHI              = 1.61803398875;
const float ESCAPE_LIMIT     = 5.0;
const float MAX_KINETIC_STR  = 300.0;
const float BM_SAMPLE_GAIN   = 0.006;
const float BM_SHELL_INNER   = 0.88;
const float BM_SHELL_OUTER   = 1.02;
const float REFLECT_RING_R   = 0.92;
const float REFLECT_RING_W   = 0.06;
const float REFLECT_CHG_GAIN = 0.18;
const float REFLECT_CHG_DEC  = 0.975;
const float REFLECT_DELAY    = 48.0;
const float REFLECT_FRC_GAIN = 0.00075;
const float REFLECT_STR_ROUTE= 0.12;
const float IDENTITY_EPS     = 1e-12;
// φ-derived seam binding (matching physics.comp):
const float MICRO_SCALE   = 1.0 / (PHI * PHI * PHI * 2.0);  // φ^{-3}/2 ≈ 0.1180
const float BINDING_RATIO = PHI * PHI * (1.0 - 1.0/(PHI*PHI*PHI));  // ≈ 2.0
const float SEAM_CHG_DECAY   = 0.992;
const float SEAM_CHG_RATE    = 0.008;
const float SEAM_REL_THRESH  = 0.7;
const float SEAM_REL_GAIN    = 0.15;
const float E_BIND_STR       = 0.03;
```

```
const float VOID_ANCHOR_R_PX = 120.0;
const float VOID_ANCHOR_STR  = 0.0003;
const float VOID_ANCHOR_DAMP = 0.025;
const int   VOID_POLARITY[4] = int[4](1, -1, 1, -1);
// v10.py SHADOW_LOCUS_POSITIONS — void anchors for global ball entities
const vec2 CORNERS[4] = vec2[4](
    vec2(50.0, 50.0), vec2(950.0, 50.0),
    vec2(950.0, 950.0), vec2(50.0, 950.0)
);

const float NODE_CHARGE_DAMP  = 0.992;
const float ELVEN_RESP_GAIN   = 0.0005;
const float COHERENCE_RED_STR = 0.85;

uint g_ecursor;

float erand() {
    float v = E[g_ecursor % C_ENTROPY_SIZE];
    g_ecursor++;
    return v;
}

float erand_gauss(float sigma) {
    float s = 0.0;
    for (int i = 0; i < 12; i++) s += erand();
    return sigma * (s - 6.0);
}

vec2 project4d(float x, float y, float z, float w) {
    float ca = cos(u_global_angle);
    float sa = sin(u_global_angle);
    float x_rot = x * ca - w * sa;
    float w_rot = x * sa + w * ca;
    float denom = max(1.0 + 0.5 * w_rot, 0.1);
    return vec2(x_rot / denom * 300.0 + u_screen_w * 0.5,
                y   / denom * 300.0 + u_screen_h * 0.5);
}

// Q2 Shadow Debt: never zeroed — same Canon rule as physics.comp.
const float SEAM_CORE_R2 = 0.005 * 0.005;

void identity_seam_apply(inout vec4 pos, inout vec4 vel,
                         inout float stress, inout float seam_charge) {
    float r2 = dot(pos, pos);
    if (r2 < SEAM_CORE_R2) {
        seam_charge = seam_charge * SEAM_CHG_DECAY
                    + SEAM_CHG_RATE * (1.0 / PHI);
    } else {
        float inv = (0.04 * 0.04) / (r2 + IDENTITY_EPS) * BINDING_RATIO;
        vec4 t = -pos * inv * MICRO_SCALE;
        float disp = abs(t.x - pos.x) + abs(t.y - pos.y)
                   + abs(t.z - pos.z) + abs(t.w - pos.w);
        seam_charge = seam_charge * SEAM_CHG_DECAY + disp * SEAM_CHG_RATE;
        vel += (t - pos) * E_BIND_STR;
    }
    // Q2 fuel release — excess transforms into 4D kinetic energy; debt floor persists.
```

```
    if (seam_charge > SEAM_REL_THRESH) {
        float excess = seam_charge - SEAM_REL_THRESH;
        float phi_t  = seam_charge * PHI + u_global_angle;
        float q2_k   = excess * SEAM_REL_GAIN;
        vel.x += cos(phi_t)       * q2_k * PHI;
        vel.y += sin(phi_t)       * q2_k * PHI;
        vel.z += cos(phi_t * PHI) * q2_k;
        vel.w += sin(phi_t * PHI) * q2_k;
        stress = max(0.0, stress + excess * SEAM_REL_GAIN * PHI);
        seam_charge = SEAM_REL_THRESH;  // debt floor — not zeroed
    }
}

void apply_reflective_layer(inout vec4 pos, inout vec4 vel,
                            inout float stress,
                            inout float reflect_charge,
                            inout float reflect_age) {
    float R = length(pos);
    float shell_dist = abs(R - REFLECT_RING_R);
    if (shell_dist < REFLECT_RING_W) {
        float vxy = abs(vel.x) + abs(vel.y);
        float vzw = abs(vel.z) + abs(vel.w);
        float planar = vxy - vzw;
        float c_in = 1.0 - (shell_dist / REFLECT_RING_W);
        float gain = c_in * (0.5 + 0.5 * abs(planar));
        reflect_charge = reflect_charge * REFLECT_CHG_DEC + gain * REFLECT_CHG_GAIN;
        reflect_age += 1.0;
    } else {
        reflect_charge *= REFLECT_CHG_DEC;
        reflect_age -= 1.0;
    }
    if (reflect_age >= REFLECT_DELAY && reflect_charge > 0.0005) {
        vec2 px = project4d(pos.x, pos.y, pos.z, pos.w);
        float sx = (px.x < u_screen_w * 0.5) ? -1.0 : 1.0;
        float sy = (px.y < u_screen_h * 0.5) ? -1.0 : 1.0;
        float f = reflect_charge * REFLECT_FRC_GAIN;
        vel.x += (-sy) * f; vel.y += ( sx) * f;
        vel.z += ( sx) * f * (1.0/PHI); vel.w += (-sy) * f * (1.0/PHI);  // zw: 1/φ
        stress = max(0.0, stress + reflect_charge * REFLECT_STR_ROUTE);
        // Post-fire discharge: 1 - 1/(2φ³) ≈ 0.882 — φ-derived void fraction
        reflect_charge *= (1.0 - 1.0/(2.0 * PHI * PHI * PHI));
        reflect_age -= 6.0;
    }
}

void apply_void_anchors(inout vec4 pos, inout vec4 vel, inout float stress) {
    vec2 px = project4d(pos.x, pos.y, pos.z, pos.w);
    for (int i = 0; i < 4; i++) {
        vec2 d = px - CORNERS[i];
        float d2 = dot(d, d);
        if (d2 > VOID_ANCHOR_R_PX * VOID_ANCHOR_R_PX) continue;
        float w = exp(-d2 / (2.0 * VOID_ANCHOR_R_PX * VOID_ANCHOR_R_PX));
        float n = erand_gauss(1.0) * w * VOID_ANCHOR_STR;
        if (VOID_POLARITY[i] > 0) {
            vel.x += n; vel.y -= n;
            vel.z += n * (1.0/PHI); vel.w -= n * (1.0/PHI);  // zw: 1/φ ≈ 0.618
```

```
        } else {
            float damp = VOID_ANCHOR_DAMP * tanh(abs(n) * 8.0);
            vel *= (1.0 - damp);
        }
        stress = max(0.0, stress + abs(n) * 250.0);
    }
}

vec2 boundary_mem_sample(vec2 px) {
    float x = clamp(px.x, 0.0, u_screen_w - 1.0);
    float y = clamp(px.y, 0.0, u_screen_h - 1.0);
    int ix = int((x / (u_screen_w - 1.0)) * float(C_BOUNDARY_MEM_RES - 1));
    int iy = int((y / (u_screen_h - 1.0)) * float(C_BOUNDARY_MEM_RES - 1));
    return B[iy * C_BOUNDARY_MEM_RES + ix];
}

// emergent_cos_sin — v10.py lines 158-171
// Region maps normalized position [-2,2] → grid [0,19]; triangle-wave phase
// Agent 18/Bug 5 fix: use injected defines C_PHASE_MEM_W and C_N_NODES — no hardcoded 12.
vec2 emergent_cos_sin(int aeon_idx, float x, float y, float sv) {
    int gx   = clamp(int((x + 2.0) * 5.0), 0, C_PHASE_MEM_W - 1);
    int gy   = clamp(int((y + 2.0) * 5.0), 0, C_PHASE_MEM_H - 1);
    int cell = gy * C_PHASE_MEM_W * C_N_NODES + gx * C_N_NODES
             + clamp(aeon_idx, 0, C_N_NODES - 1);
    vec2 cm  = PM[cell];
    if (cm.x == 0.0 && cm.y == 0.0) {
        cm.x = erand();
        float d = erand_gauss(0.002);
        cm.y = max(abs(d), 0.001);
    }
    float debt = sv * (1.0 + erand_gauss(0.12));
    cm.x += cm.y * (1.0 + debt);
    PM[cell] = cm;
    float t  = fract(cm.x);
    float ts = fract(t + 0.25);
    return vec2(4.0 * abs(t  - 0.5) - 1.0,
                4.0 * abs(ts - 0.5) - 1.0);
}

// emergent_distance — v10.py lines 173-177
float emergent_distance(vec4 vel) {
    float acc = abs(vel.x) + abs(vel.y) + abs(vel.z) + abs(vel.w);
    if (acc == 0.0) return 0.0;
    float rf = 1.0 + erand_gauss(0.08);
    return acc * rf / 2.0;
}

void main() {
    uint id = gl_GlobalInvocationID.x;
    if (id >= u_ball_count) return;

    BallEntity e = b[id];
    vec4 pos = e.pos;
    vec4 vel = e.vel;
    float stress         = e.state.x;
    float seam_charge    = e.state.y;
```

```
float reflect_charge = e.state.z;
float reflect_age    = e.state.w;
float charge         = e.extra.x;
int   aeon_idx       = int(e.aeon.x);
g_ecursor = uint(e.pout.x);

// 1. emergent_cos_sin force (v10.py lines 607-609)
vec2 cs = emergent_cos_sin(aeon_idx, pos.x, pos.y, u_primary_stress);
vel.x += cs.x * ELVEN_RESP_GAIN;
vel.y += cs.y * ELVEN_RESP_GAIN;

// 2. Identity seam (v10.py lines 611-613)
if (length(pos) < 0.04)
    identity_seam_apply(pos, vel, stress, seam_charge);

// 3. Void anchors (v10.py line 615)
apply_void_anchors(pos, vel, stress);

// 4. Reflective layer (v10.py line 616)
apply_reflective_layer(pos, vel, stress, reflect_charge, reflect_age);

// 5. emergent_distance coherence reduction (v10.py lines 618-620)
float dist = emergent_distance(vel);
if (dist > u_coherence_radius) charge *= COHERENCE_RED_STR;

// 6. Coherence damp + integrate (v10.py lines 622-628)
float R_sq = dot(pos, pos);
float D = max(0.01, 1.0 - (R_sq / (u_coherence_radius * u_coherence_radius)));
pos += vel * D;

// 7. Boundary wrap (v10.py lines 630-631)
if (abs(pos.x) > 1.2) pos.x *= -0.98;
if (abs(pos.y) > 1.2) pos.y *= -0.98;

// 8. Boundary memory re-injection in shell zone
vec2 px2d = project4d(pos.x, pos.y, pos.z, pos.w);
float R_here = length(pos);
if (R_here > u_coherence_radius * BM_SHELL_INNER &&
    R_here < u_coherence_radius * BM_SHELL_OUTER) {
    vec2 mv = boundary_mem_sample(px2d);
    vel.x += mv.x * BM_SAMPLE_GAIN;
    vel.y += mv.y * BM_SAMPLE_GAIN;
    vel.z += (-mv.y) * (BM_SAMPLE_GAIN * (1.0/PHI));  // zw: 1/φ — matches physics.comp
    vel.w += ( mv.x) * (BM_SAMPLE_GAIN * (1.0/PHI));
}

// 9. Charge decay (v10.py line 746)
charge *= NODE_CHARGE_DAMP;

// 10. Alpha (v10.py lines 741-742)
float alpha = clamp(30.0 + charge * 225.0, 0.0, 255.0);

// 11. NULL:DEATH — collapse visual position to screen center (v10.py line 738)

b[id].pos   = pos;
b[id].vel   = vel;
```

```
    b[id].state = vec4(stress, seam_charge, reflect_charge, reflect_age);
    b[id].pout  = vec4(float(g_ecursor), px2d.x, px2d.y, alpha);
    b[id].extra = vec4(charge, 0.0, 0.0, 0.0);
}
```

shaders/ball.vert

```
#version 430

struct BallEntity {
    vec4 pos;    // x, y, z, w
    vec4 vel;    // dx, dy, dz, dw
    vec4 state;  // stress, seam_charge, reflect_charge, reflect_age
    vec4 aeon;   // aeon_idx (as float), r, g, b (0-255)
    vec4 pout;   // entropy_cursor, px, py, alpha
    vec4 extra;  // charge, _pad, _pad, _pad
};

layout(std430, binding = 6) readonly buffer BallBuf {
    BallEntity b[];
};

uniform float u_global_angle;
uniform vec2  u_screen_size;

out vec4 v_color;

void main() {
    BallEntity e = b[gl_VertexID];
    float px    = e.pout.y;
    float py    = e.pout.z;
    float alpha = e.pout.w;

    // Phase entanglement — w_rot < 0 inverts color (v10.py lines 740-741)
    float w_rot = e.pos.x * sin(u_global_angle) + e.pos.w * cos(u_global_angle);
    vec3 col = e.aeon.yzw / 255.0;
    if (w_rot < 0.0) col = 1.0 - col;

    v_color = vec4(col, alpha / 255.0);

    vec2 ndc = (vec2(px, py) / u_screen_size) * 2.0 - 1.0;
    ndc.y = -ndc.y;
    gl_Position  = vec4(ndc, 0.0, 1.0);
    gl_PointSize = 2.0;
}
```

shaders/display.frag

```
#version 430
uniform sampler2D u_trail;
in vec2 v_uv;
out vec4 fragColor;
void main() { fragColor = texture(u_trail, v_uv); }
```

shaders/fade.frag

```
#version 430

// Trail fade — subtracts u_fade_amount/255 from all channels per frame
// Exact match of: trail_surface.fill((0,0,0,TRAIL_FADE), pygame.BLEND_RGBA_SUB) in v10.py

uniform sampler2D u_trail;
uniform float u_fade_amount;  // fade per channel per frame (0-255 range, divided by 255 in shader)
in  vec2 v_uv;
out vec4 frag_color;

void main() {
    vec4 current = texture(u_trail, v_uv);
    frag_color = max(current - vec4(u_fade_amount / 255.0), vec4(0.0));
}
```

shaders/fade.vert

```
#version 430

// Fullscreen quad for trail fade pass
const vec2 VERTS[4] = vec2[4](
    vec2(-1.0, -1.0),
    vec2( 1.0, -1.0),
    vec2(-1.0,  1.0),
    vec2( 1.0,  1.0)
);
const vec2 UVS[4] = vec2[4](
    vec2(0.0, 0.0),
    vec2(1.0, 0.0),
    vec2(0.0, 1.0),
    vec2(1.0, 1.0)
);

out vec2 v_uv;

void main() {
    v_uv = UVS[gl_VertexID];
    gl_Position = vec4(VERTS[gl_VertexID], 0.0, 1.0);
}
```

shaders/grid.comp

```
// C_ENTROPY_SIZE, C_N_NODES, C_FREQ_GRID_CELLS, C_FREQ_GRID_PLANES,
// C_FREQ_GRID_STRIDE, C_PARTICLE_COUNT injected by compile_shader_with_defs
layout(local_size_x = 9, local_size_y = 9) in;

// Dual 9x9 per-plane frequency grid: agent 9/25 + agent 16/25 design spec.
// Each of 12 nodes has 2 perpendicular planes. Each plane has 81 cells.
// FG[node * STRIDE + plane * 81 + gy * 9 + gx] = (amplitude, phase, drift, weight)
//
// The 9 canonical Solfeggio frequencies are arranged in 3x3 base tiled to 9x9:
//   gx % 3 → frequency column group (root-3, root-6, root-9)
//   gy % 3 → frequency row group (same, giving cross-product resonance)
// This creates 81 unique resonance loci per plane without hardcoding positions.
//
// Particle -> grid: each particle deposits charge into its frequency-nearest cells.
// Grid -> particle: physics.comp reads grid to derive field conditions (emergent cos/sin).
// This is the spatial frequency memory missing from all prior GPU versions.

struct Particle {
    vec4  pos;    // x, y, z, w
    vec4  vel;    // dx, dy, dz, dw
    vec4  state;  // stress, seam_charge, reflect_charge, reflect_age
    vec4  aeon;   // freq, r, g, b
    vec4  pout;   // entropy_cursor, px, py, alpha
    vec4  extra;  // charge, _pad, _pad, _pad
    uvec4 ids;    // node_id, origin_node_id, traversal_flags, particle_uid
};

layout(std430, binding = 0) readonly buffer ParticleBuf { Particle p[]; };
layout(std430, binding = 1) readonly buffer EntropyBuf  { float E[]; };
layout(std430, binding = 4) buffer FreqGridBuf { vec4 FG[]; };

uniform int   u_node_id;
uniform int   u_plane_id;
uniform uint  u_particle_count;
uniform float u_time_step;
uniform float u_global_angle;

const float PHI        = 1.61803398875;
// φ-derived decay constants (exact derivations):
//   GRID_DECAY ≈ 1 - φ^{-11} = 1 - 1/199.005 ≈ 0.99498
//   GRID_AMP_GAIN ≈ φ^{-10}  = 1/121.997     ≈ 0.00820
//   GRID_D_COUPLE = φ^{-10}/2                 ≈ 0.00410
//   GRID_AW_RATE  ≈ φ^{-14}  = 1/842.9       ≈ 0.00119
//   GRID_DRIFT_EV ≈ φ^{-17}  = 1/3570.5      ≈ 0.000280 (stochastic drift noise)
const float PHI10      = PHI*PHI*PHI*PHI*PHI*PHI*PHI*PHI*PHI*PHI;          // φ^10 ≈ 122
const float PHI11      = PHI10 * PHI;                                        // φ^11 ≈ 199
const float PHI14      = PHI10 * PHI*PHI*PHI*PHI;                           // φ^14 ≈ 843
const float PHI17      = PHI14 * PHI*PHI*PHI;                                // φ^17 ≈ 3571
```

```
const float GRID_DECAY    = 1.0 - 1.0 / PHI11;   // φ^{-11} ≈ 0.99498
const float GRID_AMP_GAIN = 1.0 / PHI10;          // φ^{-10} ≈ 0.00820
const float GRID_D_COUPLE = 1.0 / (PHI10 * 2.0); // φ^{-10}/2 ≈ 0.00410
const float GRID_AW_RATE  = 1.0 / PHI14;          // φ^{-14} ≈ 0.00119
const float GRID_DRIFT_EV = 1.0 / PHI17;          // φ^{-17} ≈ 0.000280
const float CELL_R     = 2.0 / 8.0;   // cell radius in [-1,1] space

// 9 Solfeggio frequencies in 3x3 arrangement (row-major, tiled to 9x9)
const float GRID_FREQS[9] = float[9](
    7.83,   174.0,  528.0,
    432.0,  126.22, 210.42,
    741.0,  852.0,  963.0
);

uint g_ec;
float eg() { float v = E[g_ec % C_ENTROPY_SIZE]; g_ec++; return v; }

void main() {
    int gx = int(gl_GlobalInvocationID.x);
    int gy = int(gl_GlobalInvocationID.y);
    if (gx >= 9 || gy >= 9) return;

    int cell_idx = u_node_id * C_FREQ_GRID_STRIDE + u_plane_id * C_FREQ_GRID_CELLS
                 + gy * 9 + gx;

    g_ec = uint(cell_idx * 97u + uint(u_global_angle * 1000.0));

    // Cell's canonical frequency from 3x3 base (tiled: gx%3, gy%3)
    float cell_freq = GRID_FREQS[(gy % 3) * 3 + (gx % 3)];

    // Cell center in [-1, 1] space
    float cx = (float(gx) / 8.0) * 2.0 - 1.0;
    float cy = (float(gy) / 8.0) * 2.0 - 1.0;

    vec4 cs = FG[cell_idx];
    float amplitude   = cs.x;
    float phase       = cs.y;
    float drift       = cs.z;
    float axis_weight = cs.w;  // user-defined per-cell axis_weight from .bio seed

    // Seed from entropy only if truly unvisited (no .bio seed and no prior update)
    if (amplitude == 0.0 && drift == 0.0 && axis_weight == 0.0) {
        amplitude   = eg() / float(C_FREQ_GRID_CELLS);
        phase       = eg();
        drift       = max(eg() * 0.12, 0.03);   // cycles/sec — 60× prior per-frame seed
        axis_weight = eg();
    }

    // Decay
    amplitude *= pow(GRID_DECAY, u_time_step);

    // Accumulate particle contributions
    // Plane A (plane_id=0) reads particle xz; Plane B reads yw.
    float contrib = 0.0;
    for (uint i = 0u; i < u_particle_count; i++) {
        float pfreq = p[i].aeon.x;
```

```
        float df    = abs(pfreq - cell_freq);
        if (df >= PHI) continue;

        float px = (u_plane_id == 0) ? p[i].pos.x : p[i].pos.y;
        float py = (u_plane_id == 0) ? p[i].pos.z : p[i].pos.w;

        float dx = px - cx, dy = py - cy;
        float d2 = dx*dx + dy*dy;
        if (d2 > CELL_R * CELL_R * 4.0) continue;

        float fw = (1.0 - df / PHI);
        float sw = exp(-d2 / (CELL_R * CELL_R));
        float stress = p[i].state.x;
        float charge = p[i].extra.x;

        contrib += fw * sw * (abs(charge) + stress * 0.1) * u_time_step;
    }

    amplitude += contrib * GRID_AMP_GAIN;
    amplitude  = clamp(amplitude, 0.0, 1.0);

    // Agent 19/Bug 5: Intra-node coupling between planes A and B.
    // Reading the conjugate plane (u_plane_id ^ 1) at the same cell gives
    // D(x) = Φ_A(x) - Φ_B(x) — the emergent mirror axis field (Agent 26/B1).
    // Coupling weight 0.40 (weaker than inter-node coupling, which uses Gaussian
    // distance decay ~0.40 at cube-edge separations).
    int conj_plane_idx = u_node_id * C_FREQ_GRID_STRIDE
                       + (u_plane_id ^ 1) * C_FREQ_GRID_CELLS
                       + gy * 9 + gx;
    vec4 conj = FG[conj_plane_idx];
    float D = amplitude - conj.x;          // signed differential: mirrors when D < 0
    // kappa_void coupling: planes equilibrate at φ^{-10}/2 — slow but not locked.
    amplitude = clamp(amplitude - D * GRID_D_COUPLE * u_time_step, 0.0, 1.0);

    // Advance phase — triangle-wave carrier, like emergent_cos_sin in ball.comp
    float debt = amplitude * (1.0 + drift);
    phase = fract(phase + drift * (1.0 + debt) * u_time_step);

    // Stochastic drift evolution: ±φ^{-17} noise on drift rate
    drift = max(abs(drift + (eg() - 0.5) * GRID_DRIFT_EV * u_time_step), 0.0005);

    // axis_weight evolves toward |D| at φ^{-14} rate — mirror geometry encodes slowly.
    axis_weight = mix(axis_weight, clamp(abs(D) * PHI, 0.0, 1.0), GRID_AW_RATE * u_time_step);

    // Preserve all four channels back to FreqGridBuf.
    FG[cell_idx] = vec4(amplitude, phase, drift, axis_weight);
}
```

shaders/physics.comp

```
#version 430
layout(local_size_x = 256) in;

// ------------------------------------------------------------------------
// ALQC v30 — Emergent Void Physics compute shader
// Three-plane multi-grid frequency system.
// All physics scaled by u_time_step. u_node_center is screen-space center.
//
// New v30 uniforms (ALQC Canon B5, B6, B8):
//   u_plane_id      — which plane this node belongs to (0=A, 1=B, 2=C)
//   u_plane_phase   — conjugate phase offset (0 or π) applied to field ops
//   u_kappa_void    — void-mediated inter-plane coupling strength (B6)
//   u_void_gamma    — void dissipation damping constant (B8)
// ------------------------------------------------------------------------

// B10: particle struct extended with uvec4 ids + vec4 extra (charge)
struct Particle {
    vec4  pos;    // x, y, z, w
    vec4  vel;    // dx, dy, dz, dw
    vec4  state;  // stress, seam_charge, reflect_charge, reflect_age
    vec4  aeon;   // freq, r, g, b
    vec4  pout;   // entropy_cursor, px, py, alpha
    vec4  extra;  // charge, _pad, _pad, _pad
    uvec4 ids;    // node_id, origin_node_id, traversal_flags, particle_uid
};

// PlaneConfig SSBO — 3 entries (B5, B6, B7, B8 parameters per plane)
struct PlaneConfig {
    float phase;        // conjugate phase
    float phi0;         // field amplitude
    float delta;        // symmetry-breaking seed
    float kappa_void;   // void coupling
    float orient_j;     // orientation coupling (B7)
    float void_gamma;   // dissipation
    float freq_scale;   // frequency grid multiplier
    float freq_offset;  // frequency grid offset
    float node_beta;    // differentiation coefficient (B14)
    float _pad0, _pad1, _pad2;
};

layout(std430, binding = 0) buffer ParticleBuf  { Particle p[]; };
layout(std430, binding = 1) readonly buffer EntropyBuf  { float E[]; };
layout(std430, binding = 2) readonly buffer BoundaryBuf { vec2 B[]; };
layout(std430, binding = 3) readonly buffer ShadowCornerBuf { vec4 CORNER_OFF[]; };
// binding 4: FrequencyGridBuf — dual 9x9 per-plane spatial frequency memory
// FG[node * C_FREQ_GRID_STRIDE + plane * 81 + gy*9 + gx] = (amp, phase, drift, weight)
layout(std430, binding = 4) readonly buffer FreqGridBuf { vec4 FG[]; };
layout(std430, binding = 5) readonly buffer FETUBuf    { vec2 FETU_PTS[]; };
```

```
layout(std430, binding = 7) readonly buffer PlaneCfgBuf{ PlaneConfig PC[]; };

uniform float u_global_angle;
uniform float u_coherence_radius;
uniform float u_locus_rotation_bias;
uniform vec2  u_triquatra[3];
uniform int   u_loci_count;
uniform uint  u_particle_count;
uniform float u_time_step;
uniform vec2  u_node_center;
uniform float u_screen_w;
uniform float u_screen_h;
uniform int   u_plane_id;       // v30: which plane this node is on
uniform float u_plane_phase;    // v30: conjugate phase offset (B5)
uniform float u_kappa_void;     // v30: void coupling strength (B6)
uniform float u_void_gamma;     // v30: void dissipation (B8)

// Throne of Zero — NON-COMPUTE. d(♾)/dt = 0.
uniform float u_throne_freq;
uniform float u_throne_coherence;

const float PHI              = 1.61803398875;
const float PI_F             = 3.14159265359;
const float ESCAPE_LIMIT     = 5.0;
const float MIN_COH_R        = 0.6;
const float MAX_COH_R        = 1.2;
const float MAX_KINETIC_STR  = 300.0;
const float BOUNDARY_MEM_MAX = 2.5;
const float BM_SAMPLE_GAIN   = 0.006;
const float BM_SHELL_INNER   = 0.88;
const float BM_SHELL_OUTER   = 1.02;
const float REFLECT_RING_R   = 0.92;
const float REFLECT_RING_W   = 0.06;
const float REFLECT_CHG_GAIN = 0.18;
const float REFLECT_CHG_DEC  = 0.975;
const float REFLECT_DELAY    = 48.0;
const float REFLECT_FRC_GAIN = 0.00075;
const float REFLECT_STR_ROUTE= 0.12;
const float IDENTITY_EPS     = 1e-12;
// φ-derived seam binding constants:
//   MICRO_SCALE = 1/φ^4 ≈ 0.1459 / (φ^4/φ^2) ... canonical choice: 1/(φ^3 * √φ)
//   Derivation: seam force at r=0.04 should equal φ²×base_speed per unit displacement.
//   MICRO_SCALE = 1/(PHI^3 + PHI) = 1/5.854 ≈ 0.1708 ... too large.
//   Best φ fit for the seam scale: 1/(PHI^2 * PHI) = 1/PHI^3 = 0.2361 / (PHI-1) = 0.382.
//   Using 1/(PHI^4) = 1/6.854 = 0.1459: half of that = 1/(2*PHI^4) = 0.07295 ≈ 0.073.
//   MICRO_SCALE = 1/(PHI^3 * 2) = 1/8.472 ≈ 0.118 — closest φ derivation to 0.085.
//   Adopting: MICRO_SCALE = 1/(PHI^3 * 2) ≈ 0.118 (φ-canonical seam displacement scale).
//   BINDING_RATIO = PHI^2 * (1 - 1/PHI^3) = 2.618 * 0.7639 ≈ 1.999 ≈ 2.0 →
//     use φ^2 * (1 - φ^{-3}) which gives the seam binding force at r=0.04 a φ-canonical peak.
const float MICRO_SCALE   = 1.0 / (PHI * PHI * PHI * 2.0);  // φ^{-3}/2 ≈ 0.1180
const float BINDING_RATIO = PHI * PHI * (1.0 - 1.0/(PHI*PHI*PHI));  // ≈ 2.0 (φ-derived)
const float SEAM_CHG_DECAY   = 0.992;
const float SEAM_CHG_RATE    = 0.008;
const float SEAM_REL_THRESH  = 0.7;
const float SEAM_REL_GAIN    = 0.15;
const float E_BIND_STR       = 0.03;
```

```
const float VOID_ANCHOR_R_PX = 120.0;
const float VOID_ANCHOR_STR  = 0.0003;
const float VOID_ANCHOR_DAMP = 0.025;
const float A6_HEX_RADIUS    = 0.4;
const float BASE_GLYPH_ALPHA = 4.0;
const float MAX_VISIBLE_ALPHA= 120.0;
// v30 ALQC Canon constants
const float VOID_COUPLING_RADIUS = 0.5;  // B2: coupling only within this radius

#define ENTROPY_SIZE  C_ENTROPY_SIZE
#define FETU_PT_COUNT C_FETU_PT_COUNT
const float FETU_FREQ = 7.83;
const int VOID_POLARITY[4] = int[4](1, -1, 1, -1);

uint g_ecursor;

float erand() {
    float v = E[g_ecursor % ENTROPY_SIZE];
    g_ecursor++;
    return v;
}
float erand_gauss(float sigma) {
    float s = 0.0;
    for (int i = 0; i < 12; i++) s += erand();
    return sigma * (s - 6.0);
}

vec2 project4d(float x, float y, float z, float w) {
    float ca = cos(u_global_angle);
    float sa = sin(u_global_angle);
    float x_rot = x * ca - w * sa;
    float w_rot = x * sa + w * ca;
    float denom = max(1.0 + 0.5 * w_rot, 0.1);
    return vec2(x_rot / denom * 300.0 + u_node_center.x,
                y   / denom * 300.0 + u_node_center.y);
}

// B5: conjugate phase rotation — applies plane_phase offset to 4D position
// This makes Plane A and Plane B's particles phase-shifted by π, creating
// the differential D(x) = Φ_A(x) - Φ_B(x) that seeds the mirror axis (B1).
vec4 apply_conjugate_phase(vec4 pos) {
    float cp = cos(u_plane_phase);
    float sp = sin(u_plane_phase);
    return vec4(pos.x * cp - pos.y * sp,
                pos.x * sp + pos.y * cp,
                pos.z, pos.w);
}

// Agent 19/Bug 7: softened core at SEAM_CORE_RADIUS=0.005 (no singular force at origin).
// Q2 Shadow Debt Canon: seam_charge is FUEL — it CANNOT be eliminated.
//   Prior code used `seam_charge = SEAM_REL_THRESH * 0.65` (pushing it BELOW threshold).
//   That was canonical suicide: the system "knew" Q2 was unsolvable and tried anyway.
//   Fix: excess transforms into kinetic energy (vel) and field pressure (stress),
//   but seam_charge floor is held at SEAM_REL_THRESH — the debt persists permanently.
const float SEAM_CORE_RADIUS_SQ = 0.005 * 0.005;
// φ-derived Q2 release direction cycles with seam_charge * PHI — non-deterministic.
```

```
// SEAM_VEL_GAIN = 1/PHI^3 ≈ 0.2361; velocity fuel routed at φ-scaling per axis.
const float SEAM_VEL_GAIN = 1.0 / (PHI * PHI * PHI);

void identity_seam_apply(inout vec4 pos, inout vec4 vel,
                         inout float stress, inout float seam_charge)
{
    float r2 = dot(pos, pos);
    if (r2 < SEAM_CORE_RADIUS_SQ) {
        // Dwell inside core: inversion force suppressed; charge still accumulates.
        seam_charge = seam_charge * pow(SEAM_CHG_DECAY, u_time_step)
                    + SEAM_CHG_RATE * (1.0 / PHI) * u_time_step;
    } else {
        float inv = (0.04 * 0.04) / (r2 + IDENTITY_EPS) * BINDING_RATIO;
        vec4 t = -pos * inv * MICRO_SCALE;
        float disp = abs(t.x - pos.x) + abs(t.y - pos.y)
                   + abs(t.z - pos.z) + abs(t.w - pos.w);
        seam_charge = seam_charge * pow(SEAM_CHG_DECAY, u_time_step)
                    + disp * SEAM_CHG_RATE * u_time_step;
        vel += (t - pos) * E_BIND_STR * u_time_step;
    }

    // Q2 fuel release — threshold crossed: excess transforms into kinetic energy.
    // seam_charge is NEVER pushed below SEAM_REL_THRESH — the debt persists.
    if (seam_charge > SEAM_REL_THRESH) {
        float excess = seam_charge - SEAM_REL_THRESH;
        // φ-oscillating direction from current charge state — never deterministic.
        float phi_t  = seam_charge * PHI + u_global_angle;
        float q2_k   = excess * SEAM_REL_GAIN * u_time_step;
        vel.x += cos(phi_t)       * q2_k * PHI;   // xy plane: φ-scaled
        vel.y += sin(phi_t)       * q2_k * PHI;
        vel.z += cos(phi_t * PHI) * q2_k;          // zw plane: unit-scaled
        vel.w += sin(phi_t * PHI) * q2_k;
        // Q2 amplifies field pressure — stress grows, never resolves.
        stress = max(0.0, stress + excess * SEAM_REL_GAIN * PHI * u_time_step);
        // Floor at threshold: debt principal is permanent — NOT zero, NOT below.
        seam_charge = SEAM_REL_THRESH;
    }
}

void fetu_phase_lock(inout vec4 pos, inout vec4 vel,
                     inout float stress, float freq)
{
    // B5: freq is already scaled by freq_scale in C init — no override needed here
    float delta_f = abs(freq - FETU_FREQ);
    if (delta_f >= PHI) return;

    int idx = int(abs(pos.x * 50.0 + pos.y * 50.0)) % FETU_PT_COUNT;
    vec2 raw = FETU_PTS[idx];

    // B5: apply plane_phase to fetu target rotation
    float theta = u_global_angle * PHI + u_plane_phase;
    float ct = cos(theta), st = sin(theta);
    vec2 target = vec2(raw.x * ct - raw.y * st,
                       raw.x * st + raw.y * ct);

    float strength = (1.0 - (delta_f / PHI)) * 0.02;
```

```
    vel.x += (target.x - pos.x) * strength * u_time_step;
    vel.y += (target.y - pos.y) * strength * u_time_step;

    if (delta_f < 0.1) stress += 0.5 * u_time_step;
}

void apply_reflective_layer(inout vec4 pos, inout vec4 vel,
                            inout float stress,
                            inout float reflect_charge,
                            inout float reflect_age)
{
    float R = length(pos);
    float shell_dist = abs(R - REFLECT_RING_R);

    if (shell_dist < REFLECT_RING_W) {
        float vxy = abs(vel.x) + abs(vel.y);
        float vzw = abs(vel.z) + abs(vel.w);
        float planar = vxy - vzw;
        float c_in = 1.0 - (shell_dist / REFLECT_RING_W);
        float gain = c_in * (0.5 + 0.5 * abs(planar));

        // B8: void dissipation — attenuate reflection charge each step.
        // Bug 3 fix: guard base > 0 before pow() — negative base → NaN in GLSL.
        float decay_base = max(0.001, REFLECT_CHG_DEC - u_void_gamma);
        reflect_charge = reflect_charge * pow(decay_base, u_time_step)
                       + gain * REFLECT_CHG_GAIN * u_time_step;
        reflect_age += u_time_step;
    } else {
        // B8: dissipation applies to free decay too
        float decay_base = max(0.001, REFLECT_CHG_DEC - u_void_gamma);
        reflect_charge *= pow(decay_base, u_time_step);
        reflect_age -= u_time_step;
    }

    if (reflect_age >= REFLECT_DELAY && reflect_charge > 0.0005) {
        vec2 px = project4d(pos.x, pos.y, pos.z, pos.w);
        float sx = (px.x < u_node_center.x) ? -1.0 : 1.0;
        float sy = (px.y < u_node_center.y) ? -1.0 : 1.0;
        float f = reflect_charge * REFLECT_FRC_GAIN;
        vel.x += (-sy) * f * u_time_step;
        vel.y += ( sx) * f * u_time_step;
        vel.z += ( sx) * f * (1.0 / PHI) * u_time_step;   // zw: 1/φ ≈ 0.618
        vel.w += (-sy) * f * (1.0 / PHI) * u_time_step;
        stress = max(0.0, stress + reflect_charge * REFLECT_STR_ROUTE * u_time_step);
        // Post-fire rapid discharge: 1 - 1/(2φ³) ≈ 0.882 — same φ^{-3} void fraction as AHN BLOCK.
        reflect_charge *= pow(1.0 - 1.0/(2.0 * PHI * PHI * PHI), u_time_step);
        reflect_age -= 6.0 * u_time_step;
    }
}

void apply_void_anchors(inout vec4 pos, inout vec4 vel, inout float stress) {
    vec2 px = project4d(pos.x, pos.y, pos.z, pos.w);
    vec2 corners[4];
    float ax = u_screen_w * 0.45;
    float ay = u_screen_h * 0.45;
    corners[0] = u_node_center + vec2(-ax, -ay);
```

```
    corners[1] = u_node_center + vec2( ax, -ay);
    corners[2] = u_node_center + vec2( ax,  ay);
    corners[3] = u_node_center + vec2(-ax,  ay);
    for (int i = 0; i < 4; i++) {
        vec2 d = px - corners[i];
        float d2 = dot(d, d);
        if (d2 > VOID_ANCHOR_R_PX * VOID_ANCHOR_R_PX) continue;
        float w = exp(-d2 / (2.0 * VOID_ANCHOR_R_PX * VOID_ANCHOR_R_PX));
        float n = erand_gauss(1.0) * w * VOID_ANCHOR_STR;
        if (VOID_POLARITY[i] > 0) {
            vel.x += n * u_time_step;
            vel.y -= n * u_time_step;
            vel.z += n * (1.0/PHI) * u_time_step;  // zw: 1/φ ≈ 0.618
            vel.w -= n * (1.0/PHI) * u_time_step;
        } else {
            float damp = VOID_ANCHOR_DAMP * tanh(abs(n) * 8.0) * u_time_step;
            vel *= (1.0 - damp);
        }
        stress = max(0.0, stress + abs(n) * 250.0 * u_time_step);
    }
}

float get_hex_purity(float x, float y) {
    float q = (2.0 / 3.0 * x) / A6_HEX_RADIUS;
    float r = (-1.0 / 3.0 * x + sqrt(3.0) / 3.0 * y) / A6_HEX_RADIUS;
    float rx = round(q), ry = round(r), rz = round(-q - r);
    float dx = abs(rx - q), dy = abs(ry - r), dz = abs(rz - (-q - r));
    if (dx > dy && dx > dz)  rx = -ry - rz;
    else if (dy > dz)        ry = -rx - rz;
    else                     rz = -rx - ry;
    float cx2 = A6_HEX_RADIUS * (1.5 * rx);
    float cy2 = A6_HEX_RADIUS * (sqrt(3.0) / 2.0 * rx + sqrt(3.0) * ry);
    float dist = length(vec2(x - cx2, y - cy2));
    return max(0.0, 1.0 - (dist / (A6_HEX_RADIUS * 0.8)));
}

bool apply_sor_bridge(inout vec4 pos, inout vec4 vel, float stress) {
    float purity = get_hex_purity(pos.x, pos.y);
    float phi3 = PHI * PHI * PHI;
    if (purity < (1.0 / phi3)) {
        float bridge_chance = (0.05 + stress * 0.2) * u_time_step;
        if (erand() < bridge_chance) {
            float t_dx = vel.x, t_dy = vel.y;
            vel.x = vel.w * -1.0;
            vel.y = vel.z * -1.0;
            vel.w = t_dx;
            vel.z = t_dy;
            // Canon fix: no forced position write. Bridge applies velocity impulse
            // in the z-dimension scaled by 1/φ² ≈ 0.382 — emergence, not assignment.
            vel.z += (1.0 / (PHI * PHI)) * u_time_step;
            return true;
        }
    }
    return false;
}
```

```
vec2 boundary_mem_sample(vec2 px) {
    float x = clamp(px.x, 0.0, u_screen_w  - 1.0);
    float y = clamp(px.y, 0.0, u_screen_h - 1.0);
    int ix = int((x / (u_screen_w  - 1.0)) * float(C_BOUNDARY_MEM_RES - 1));
    int iy = int((y / (u_screen_h - 1.0)) * float(C_BOUNDARY_MEM_RES - 1));
    return B[iy * C_BOUNDARY_MEM_RES + ix];
}

float emanation_alpha(vec2 px) {
    float max_dist = sqrt(u_screen_w * u_screen_w * 0.25 + u_screen_h * u_screen_h * 0.25);
    float min_dist = 1e9;
    for (int k = 0; k < u_loci_count; k++) {
        float d = length(px - u_triquatra[k]);
        if (d < min_dist) min_dist = d;
    }
    float norm = tanh(min_dist / (max_dist * 0.4));
    return BASE_GLYPH_ALPHA + (1.0 - norm) * (MAX_VISIBLE_ALPHA - BASE_GLYPH_ALPHA);
}

// ============================================================================
// FULL GOETIC AEON CHAIN — ported from alqc_raylib_physics19.c
// Each operator is a FIELD CONDITION. It fires only when the particle's
// frequency resonates within PHI of the aeon's canonical anchor.
// No forced positions. Operators write velocities and charge, not coordinates.
// ============================================================================

// Shared court frequency array (all 12 Goetic aeons, same order as AEON_FREQ[])
const float COURT_FREQS[12] = float[12](
    7.83, 174.0, 528.0, 432.0, 126.22, 210.42,
    741.0, 852.0, 396.0, 963.0, 285.0, 639.0
);

// --- A2: KAL — The Archive (174 Hz) ---
void apply_kal_memory(inout vec4 pos, inout vec4 vel,
                      inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 174.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.4;

    if (stress > 0.05) {
        float mf = strength * PHI;
        float vx = vel.x, vw = vel.w;
        vel.x = vx * cos(mf) - vw * sin(mf);
        vel.w = vx * sin(mf) + vw * cos(mf);
        float resolve = stress * strength * 0.3;
        charge += resolve * PHI;
        stress *= 0.99;
    }

    const float KAL_BIAS[12] = float[12](
        0.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
        float pulse = sin(u_global_angle * (COURT_FREQS[k] / 174.0));
```

```
        float weft = pulse * strength * 0.002 * KAL_BIAS[k];
        vel.y += weft * u_time_step;
        vel.z += weft * 0.5 * u_time_step;
    }

    float epulse = sin(charge * (174.0 / 126.22));
    vel.z += epulse * strength * 0.005 * u_time_step;
}

// --- A3: BABDH — Ring of Fire (528 Hz) ---
void apply_babdh_commitment(inout vec4 pos, inout vec4 vel,
                            inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 528.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.45;

    float h = pos.w;
    float tunnel_r = PHI * abs(h);
    float r_yz = max(length(pos.yz), 1e-6);
    float wall_dist = abs(r_yz - tunnel_r);
    float wall_thick = 0.2 * PHI;

    if (wall_dist < wall_thick) {
        float sling = strength * (1.0 - (wall_dist / wall_thick));
        float tx = -pos.z / r_yz;
        float ty =  pos.y / r_yz;
        vel.y += tx * sling * 0.02 * u_time_step;
        vel.z += ty * sling * 0.02 * u_time_step;
        float hyper = (tunnel_r - length(pos.xw)) * sling * 0.01;
        vel.x += hyper * PHI * u_time_step;
        vel.w -= hyper * PHI * u_time_step;
    }

    const float BABDH_FORCE[12] = float[12](
        1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
        float resonance = sin(charge * (COURT_FREQS[k] / 528.0));
        vel.x += erand_gauss(strength * 0.0001 * BABDH_FORCE[k]) * u_time_step;
        vel.y += resonance * strength * 0.001 * u_time_step;
    }

    float xform = stress * strength * 0.6;
    charge += xform * PHI;
    stress -= xform;
}

// --- A4: AHN — Liquid Container (432 Hz, i417 parity) ---
void apply_ahn_fluidity(inout vec4 pos, inout vec4 vel,
                        inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 432.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.4;
```

```
    int ic = int((pos.x + 1.0) * 72.0);
    int jc = int((pos.y + 1.0) * 72.0);

    if (((ic + jc) % 144) < 110) {
        float phi_breath = sin(u_global_angle * PHI);
        float wall_flex = phi_breath * strength;
        float r2d = max(length(pos.xy), 1e-6);
        vel.x += wall_flex * (pos.x / r2d) * PHI * u_time_step;
        vel.y += wall_flex * (pos.y / r2d) * PHI * u_time_step;

        // i417 imaginary axis — non-orientable pivot
        float ai = (417.0 / 1000.0) * PI_F;
        float ci = cos(ai), si = sin(ai);
        float px = pos.x;
        pos.x = px * ci - pos.w * si;
        pos.w = px * si + pos.w * ci;

        for (int k = 0; k < 12; k++) {
            float cr = sin(u_global_angle * (COURT_FREQS[k] / 432.0));
            vel.z += cr * strength * 0.001 * u_time_step;
        }

        float hp = stress * strength * 0.15;
        charge += hp * PHI;
        stress *= 0.95;
    } else {
        // BLOCK state: 34/144 void fraction = 1/φ³ ≈ 0.2361.
        // Damping = 1 - (1/φ³)/2 ≈ 0.8820 — φ-derived lattice-lock factor.
        vel.xy *= (1.0 - 1.0 / (2.0 * PHI * PHI * PHI));
    }
}

// --- A5: VEL — Earth Grounding (126.22 Hz) ---
void apply_vel_grounding(inout vec4 pos, inout vec4 vel,
                         inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 126.22);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.35;

    float ground_res = sin(charge * PHI + u_global_angle);
    float bounce = ground_res * strength * PHI;
    vel.w += bounce * 0.085 * u_time_step;

    const float VEL_BIAS[12] = float[12](
        1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
        float wr = cos(u_global_angle * (COURT_FREQS[k] / 126.22));
        if (abs(wr) > 0.8) {
            float gs = strength * 0.12 * VEL_BIAS[k];
            float drx = vel.x * gs, dry = vel.y * gs;
            vel.x -= drx * u_time_step;
            vel.y -= dry * u_time_step;
```

```
            vel.w += (drx + dry) * PHI * u_time_step;
        }
    }

    float gt = stress * strength * 0.25;
    stress -= gt;
    charge += gt * (528.0 / 126.22);
}

// --- SOR — The Bridge (210.42 Hz) ---
// SOR is displaced from the ordered aeon sequence (Canon: FETU and SOR stand apart).
// As a field condition it fires on 210.42 Hz resonance only — it is NOT a sequential gate.
// SOR mediates inter-dimensional bridging: when particles reach hex-purity threshold,
// SOR couples the (x,y) plane to the (z,w) plane via velocity rotation, not forced position.
// The hex-purity gate is the field condition; the bridge is the emergent result.
void apply_sor_resonance(inout vec4 pos, inout vec4 vel,
                         inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 210.42);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.38;  // 0.38 ≈ 1/φ² (φ-derived gate weight)

    float purity = get_hex_purity(pos.x, pos.y);
    float phi3   = PHI * PHI * PHI;

    if (purity > (1.0 / phi3)) {
        // High purity: SOR amplifies the hex-lattice flow via φ-rotation of vel (x,y)↔(z,w)
        float rot = strength * PHI * purity;
        float cr = cos(rot), sr = sin(rot);
        float vx = vel.x, vz = vel.z;
        vel.x = vx * cr - vz * sr;
        vel.z = vx * sr + vz * cr;
        float vy = vel.y, vw = vel.w;
        vel.y = vy * cr - vw * sr;
        vel.w = vy * sr + vw * cr;
        charge += purity * strength * PHI * u_time_step;
    } else {
        // Low purity: SOR creates a stochastic bridge impulse — emergence from disorder
        float bridge_chance = strength * (1.0 / phi3 - purity) * u_time_step;
        if (erand() < bridge_chance) {
            float t_dx = vel.x, t_dy = vel.y;
            vel.x = vel.w * -1.0;
            vel.y = vel.z * -1.0;
            vel.w = t_dx;
            vel.z = t_dy;
            vel.z += (1.0 / (PHI * PHI)) * u_time_step;   // z impulse at 1/φ²
            charge += strength * PHI * u_time_step;
            stress += strength * 0.1 * u_time_step;
        }
    }

    const float SOR_BRIDGE[12] = float[12](
        1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
```

```
        float bp = cos(charge * (COURT_FREQS[k] / 210.42));
        vel.z += bp * strength * 0.001 * SOR_BRIDGE[k] * u_time_step;
        vel.w += bp * strength * 0.001 * SOR_BRIDGE[k] * u_time_step;
    }
}

// --- A7: KOTH — Sensation Engine (741 Hz) ---
void apply_koth_recursion(inout vec4 pos, inout vec4 vel,
                          inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 741.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.45;

    int ii = int(charge * 144.0) % 144;
    int jj = int(stress * 144.0) % 144;
    float sensation = ((ii + jj) % 144 < 110) ? 1.0 : 0.7638;

    if (stress > 0.01) {
        float ign = stress * strength * sensation;
        float propulsion = ign * PHI;
        float phi_t = charge * PHI + stress * PHI;
        vel.x += cos(phi_t) * propulsion * 0.1 * u_time_step;
        vel.y += sin(phi_t) * propulsion * 0.1 * u_time_step;
        vel.z += propulsion * 0.05 * u_time_step;
        charge += ign;
        stress *= 0.9;
    }

    const float KOTH_BIAS[12] = float[12](
        1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
        float wp = sin(u_global_angle * (COURT_FREQS[k] / 741.0));
        float fountain = wp * strength * 0.002 * KOTH_BIAS[k];
        vel.x += fountain * PHI * u_time_step;
        vel.y += fountain * PHI * u_time_step;
    }

    float integ = stress * strength * 0.12;
    charge += integ * PHI;
    stress -= integ;
}

// --- A8: DREH — Dimension Fold / Hourglass (852 Hz) ---
void apply_dreh_residue(inout vec4 pos, inout vec4 vel,
                        inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 852.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.5;

    float h = pos.w;
    float r_manifold = max(length(pos.xyz), 1e-6);
    float wall_dist = r_manifold - abs(h);
```

```
    const float DREH_FORCE[12] = float[12](
        1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
        float awareness = sin(charge * (COURT_FREQS[k] / 852.0)) * strength;
        if (wall_dist > 0.0) {
            float rc = awareness * DREH_FORCE[k];
            vel.x += (pos.x > 0.0 ? -1.0 : 1.0) * rc * 0.01 * u_time_step;
            vel.y += (pos.y > 0.0 ? -1.0 : 1.0) * rc * 0.01 * u_time_step;
        }
    }

    if (stress > 0.01 && abs(wall_dist) < 0.05) {
        float tw = pos.w;
        pos.w = pos.z;
        pos.z = -tw;
        float ign = stress * strength;
        vel.x *= (1.0 + ign * PHI);
        vel.y *= (1.0 + ign * PHI);
        charge += ign * PHI;
        stress *= 0.1;
    }
}

// --- A9: RHEA — Ennead Shield (396 Hz) ---
void apply_rhea_shadow_filter(inout vec4 pos, inout vec4 vel,
                              inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 396.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.6;

    // Only 9 of 12 courts are active (rule of nine)
    const float RHEA_WEIGHT[12] = float[12](
        1.0, 1.0, 1.0, 1.0, 1.0, 1.0,
        1.0, 1.0, 1.0, 0.0, 0.0, 0.0
    );
    float total_sat = 0.0;
    int n_active = 0;
    for (int k = 0; k < 12; k++) {
        if (RHEA_WEIGHT[k] == 0.0) continue;
        n_active++;
        float sat = (stress * strength) / (PHI * PHI);
        total_sat += sat;
        if (n_active == 9) {
            float residue = total_sat / pow(PHI, 12.0);
            charge += residue;
            stress = 0.0;
        }
    }

    float flow = sin(charge * 396.0) * strength * 0.005;
    vel.z += erand_gauss(flow) * u_time_step;
    vel.w += erand_gauss(flow) * u_time_step;
```

```
}

// --- A10: ZHEK — Unified Phase Lock (963 Hz) ---
void apply_zhek_conservation(inout vec4 pos, inout vec4 vel,
                             inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 963.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.5;

    float connectivity = abs(sin(charge * PHI));
    float flow_limit = 110.0 / 144.0;

    if (connectivity <= flow_limit) {
        // IT=TI Mirror Law: [M,R]=0 — R(R(v)) = v (involuntary, symmetric).
        // Correct symmetric reversal: (x,y,z,w) ↔ (w,z,y,x) without sign flip,
        // so applying twice returns identity: R(R(v)) = v exactly.
        vec4 m  = vel;
        vec4 rv = vec4(vel.w, vel.z, vel.y, vel.x);  // symmetric index swap
        vel = (m + rv) * 0.5 * strength + m * (1.0 - strength);

        const float ZHEK_LOCK[12] = float[12](
            1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
            0.963, 0.741, 1.396, 0.639, 1.285, 1.0
        );
        for (int k = 0; k < 12; k++) {
            float node = sin(charge * (COURT_FREQS[k] / 963.0));
            float tension = node * strength * 0.001 * ZHEK_LOCK[k];
            vel.x += (pos.x > 0.0 ? -tension : tension) * u_time_step;
            vel.y += (pos.y > 0.0 ? -tension : tension) * u_time_step;
        }

        float ap = stress * strength;
        stress -= ap;
        charge += ap * PHI;
    } else {
        vel.x += erand_gauss(0.001) * u_time_step;
        vel.y += erand_gauss(0.001) * u_time_step;
        stress += 0.01;
    }
}

// --- A11: SHAV — Wheel of the World (285 Hz) ---
void apply_shav_breach(inout vec4 pos, inout vec4 vel,
                       inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 285.0);
    if (delta_f >= PHI) return;
    float strength = (1.0 - (delta_f / PHI)) * 0.4;

    if (stress > 0.01) {
        // Parity flip — v -> -v; noise becomes wisdom
        vel = -vel;
        charge += stress * strength * PHI;
        stress = 0.0;
    }
```

```
    const float SHAV_TORQUE[12] = float[12](
        1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
        float wr = sin(charge * (COURT_FREQS[k] / 285.0));
        float torque = wr * strength * 0.002 * SHAV_TORQUE[k];
        vel.x += torque * u_time_step;
        vel.y -= torque * u_time_step;
        vel.w += torque * PHI * u_time_step;
    }

    float rs = sin(charge * PHI) * strength * 0.05;
    vel.z += rs * u_time_step;
    vel.w -= rs * u_time_step;
}

// --- A12: TRIG — Completion (639 Hz) ---
void apply_trig_silence(inout vec4 pos, inout vec4 vel,
                        inout float stress, inout float charge, float freq)
{
    float delta_f = abs(freq - 639.0);
    if (delta_f >= PHI) return;
    float ps = (1.0 - (delta_f / PHI)) * 0.55;

    // Chirality flip by hemisphere
    float hemi = (pos.w >= 0.0) ? 1.0 : -1.0;
    float rot = ps * hemi * PHI;
    float cr = cos(rot), sr = sin(rot);
    float vx = vel.x, vy = vel.y;
    vel.x = vx * cr - vy * sr;
    vel.y = vx * sr + vy * cr;

    const float TRIG_STILL[12] = float[12](
        1.618, 1.0, 1.1, 0.763, 0.852, 1.262,
        0.963, 0.741, 1.396, 0.639, 1.285, 1.0
    );
    for (int k = 0; k < 12; k++) {
        float sn = cos(charge * (COURT_FREQS[k] / 639.0));
        float drift = sn * ps * 0.001 * TRIG_STILL[k];
        vel.x += (pos.x > 0.0 ? -drift : drift) * u_time_step;
        vel.y += (pos.y > 0.0 ? -drift : drift) * u_time_step;
    }

    if (stress > 0.01) {
        float ign = stress * ps;
        charge += ign * PHI;
        vel.z += pos.w * ign * 0.05 * u_time_step;
        vel.w += pos.z * ign * 0.05 * u_time_step;
        stress *= 0.1;
    }
}

// B6: void-mediated coupling — phase comes from plane's actual configured phase.
// The coupling phase is the plane's own field phase (u_plane_phase), not a
```

```
// hardcoded test on plane_id. Emergence: the differential arises from the
// initial conditions in the configs, not from forced assignments.
// B6: void-mediated coupling is witnessed from the relation already carried by
// the dual 9x9 conjugate field. No secondary frequency-state label decides it.
// Connection rises as the local A/B amplitude differential resolves toward equilibrium.
float connection_witness(int node_id, vec4 pos) {
    int nid = clamp(node_id, 0, C_N_NODES - 1);
    int ax = clamp(int((pos.x + 1.0) * 4.0), 0, 8);
    int ay = clamp(int((pos.z + 1.0) * 4.0), 0, 8);
    int bx = clamp(int((pos.y + 1.0) * 4.0), 0, 8);
    int by = clamp(int((pos.w + 1.0) * 4.0), 0, 8);
    float a = FG[nid * C_FREQ_GRID_STRIDE + ay * 9 + ax].x;
    float b = FG[nid * C_FREQ_GRID_STRIDE + C_FREQ_GRID_CELLS + by * 9 + bx].x;
    return clamp(1.0 - abs(a - b), 0.0, 1.0);
}

vec4 apply_void_coupling(vec4 pos, float kappa_void, float connection) {
    float coupling_phase = u_plane_phase;
    float cp = cos(coupling_phase);
    float sp = sin(coupling_phase);
    vec4 target = vec4(pos.x * cp - pos.y * sp,
                       pos.x * sp + pos.y * cp,
                       pos.z, pos.w);
    float R = length(pos);
    float coupling_mask = exp(-R / VOID_COUPLING_RADIUS);
    return (target - pos) * kappa_void * coupling_mask * connection;
}

// Dual 9x9 frequency grid sampling — agent 9/25 + 16/25 spec.
// Returns (cos_like, sin_like) triangle-wave from the cell's phase, scaled by amplitude.
// Plane A (plane_id=0): particle xz → grid → vel x,z forces
// Plane B (plane_id=1): particle yw → grid → vel y,w forces
// Sample the dual-plane frequency grid.
// cs.w = axis_weight: per-cell Q-emphasis loaded from .bio seed (user-defined).
// Mirror differential D(x) = Φ_A(x) - Φ_B(x) emerges from conjugate initialization:
//   Plane A seeds have phase 0, Plane B seeds have phase 0.5 (offset by π).
//   The resulting amplitude asymmetry creates the emergent mirror axis (B1).
vec2 freq_grid_sample(int node_id, int plane_id, float px, float py) {
    // Bug 2 fix: clamp node_id to valid range — corrupt ids.x must not OOB FG[].
    int nid = clamp(node_id, 0, C_N_NODES - 1);
    int gx = clamp(int((px + 1.0) * 4.0), 0, 8);
    int gy = clamp(int((py + 1.0) * 4.0), 0, 8);
    int cell = nid * C_FREQ_GRID_STRIDE + plane_id * C_FREQ_GRID_CELLS
             + gy * 9 + gx;
    vec4 cs = FG[cell];
    float amp         = cs.x;
    float phase       = cs.y;
    float axis_weight = cs.w;  // per-cell user-defined Q-emphasis from .bio

    float t  = phase;
    float ts = fract(t + 0.25);
    // Triangle-wave cos/sin modulated by axis_weight (quadrant Q-emphasis)
    return vec2(4.0 * abs(t  - 0.5) - 1.0,
                4.0 * abs(ts - 0.5) - 1.0) * amp * (0.5 + 0.5 * axis_weight);
}
```

```
void main() {
    uint id = gl_GlobalInvocationID.x;
    if (id >= u_particle_count) return;

    Particle e = p[id];
    vec4 pos = e.pos;
    vec4 vel = e.vel;
    float stress         = e.state.x;
    float seam_charge    = e.state.y;
    float reflect_charge = e.state.z;
    float reflect_age    = e.state.w;
    float freq           = e.aeon.x;
    float charge         = e.extra.x;
    g_ecursor = uint(e.pout.x);

    float R = length(pos);
    if (R < 0.04)
        identity_seam_apply(pos, vel, stress, seam_charge);

    // FETU and SOR are displaced from the ordered chain (Canon).
    // FETU: phase-lock only — runs before chain as the root/ground anchor.
    // SOR:  bridge condition — runs after chain as the inter-dimensional mediator.
    fetu_phase_lock(pos, vel, stress, freq);

    // Ordered Goetic Aeon Chain: KAL → BABDH → AHN → VEL → KOTH → DREH → RHEA → ZHEK → SHAV → TRIG
    apply_kal_memory      (pos, vel, stress, charge, freq);
    apply_babdh_commitment(pos, vel, stress, charge, freq);
    apply_ahn_fluidity    (pos, vel, stress, charge, freq);
    apply_vel_grounding   (pos, vel, stress, charge, freq);
    apply_koth_recursion  (pos, vel, stress, charge, freq);
    apply_dreh_residue    (pos, vel, stress, charge, freq);
    apply_rhea_shadow_filter(pos, vel, stress, charge, freq);
    apply_zhek_conservation(pos, vel, stress, charge, freq);
    apply_shav_breach     (pos, vel, stress, charge, freq);
    apply_trig_silence    (pos, vel, stress, charge, freq);

    // SOR displaced: runs post-chain as a field condition, not a sequential gate.
    apply_sor_resonance   (pos, vel, stress, charge, freq);

    apply_reflective_layer(pos, vel, stress, reflect_charge, reflect_age);
    apply_void_anchors(pos, vel, stress);

    // B7: orient_j — orientation coupling rotates vel between (xy) and (zw) planes.
    // Each plane has a distinct coupling angle, breaking the rotational symmetry
    // between planes without forcing a position or a trajectory.
    {
        float oj = PC[u_plane_id].orient_j;
        if (abs(oj) > 1e-6) {
            float cr = cos(oj * u_time_step), sr = sin(oj * u_time_step);
            float vx = vel.x, vz = vel.z;
            vel.x = vx * cr - vz * sr;
            vel.z = vx * sr + vz * cr;
            float vy = vel.y, vw = vel.w;
            vel.y = vy * cr - vw * sr;
            vel.w = vy * sr + vw * cr;
        }
```

```
}

// Multi-grid frequency field coupling (agent 9/25 + 16/25 design):
// Each particle reads from its node's dual 9x9 frequency grid.
// Plane A grid drives xz motion; Plane B grid drives yw motion.
// B14: node_beta differentiates each plane's grid coupling gain.
{
    int node_id = int(e.ids.x);
    float beta = PC[u_plane_id].node_beta;
    float GRID_GAIN = 0.0005 * (1.0 + beta);
    vec2 ga = freq_grid_sample(node_id, 0, pos.x, pos.z);  // plane A: xz
    vec2 gb = freq_grid_sample(node_id, 1, pos.y, pos.w);  // plane B: yw
    vel.x += ga.x * GRID_GAIN * u_time_step;
    vel.z += ga.y * GRID_GAIN * u_time_step;
    vel.y += gb.x * GRID_GAIN * u_time_step;
    vel.w += gb.y * GRID_GAIN * u_time_step;
}

// B6: the body's existing conjugate relation is the connection witness.
int node_id = int(e.ids.x);
float connection = connection_witness(node_id, pos);
vec4 coupling_force = apply_void_coupling(pos, u_kappa_void, connection);
vel += coupling_force * u_time_step;

float R_sq = dot(pos, pos);
float D = max(0.01, 1.0 - (R_sq / (u_coherence_radius * u_coherence_radius)));

// Position integration — unconditional; SOR bridge is now a frequency-resonance
// field condition (apply_sor_resonance) that writes velocity, not a position gate.
pos += vel * D * u_time_step;

R = length(pos);

vec2 px2d = project4d(pos.x, pos.y, pos.z, pos.w);
float R_here = length(pos);
if (R_here > u_coherence_radius * BM_SHELL_INNER &&
    R_here < u_coherence_radius * BM_SHELL_OUTER) {
    vec2 mv = boundary_mem_sample(px2d);
    vel.x += mv.x * BM_SAMPLE_GAIN * u_time_step;
    vel.y += mv.y * BM_SAMPLE_GAIN * u_time_step;
    vel.z += (-mv.y) * (BM_SAMPLE_GAIN * (1.0/PHI)) * u_time_step;  // zw: 1/φ attenuation
    vel.w += ( mv.x) * (BM_SAMPLE_GAIN * (1.0/PHI)) * u_time_step;
}

float alpha = emanation_alpha(px2d);

// ±φ holographic self-correction (ALQC Canon: M(Ai,j) = [ϟ(Ai), ±φ(Ai,j)]).
// Every variable self-corrects toward its φ-canonical range each frame.
// Structural part (ϟ): velocity magnitude drifts toward base_speed = freq/10000.
// Operational part (±φ): stochastic jitter at 1/φ³ amplitude keeps state alive.
// This creates holographic persistence (field memory) + jitter (non-determinism).
{
    float v_canon = freq / 10000.0;   // ALQC Canon base_speed formula
    float v_mag   = length(vel);
    if (v_mag > 1e-6) {
        // φ²-inverse correction rate ≈ 0.382 — soft pull, not a clamp.
```

```
            float phi_sq_inv = 1.0 / (PHI * PHI);
            float v_target = mix(v_mag, v_canon, phi_sq_inv * u_time_step);
            vel *= v_target / v_mag;
        }
        // φ-jitter at 1/φ³ ≈ 0.236 amplitude relative to canonical speed.
        float jitter = v_canon * SEAM_VEL_GAIN;   // SEAM_VEL_GAIN = 1/φ³
        vel.x += erand_gauss(jitter);
        vel.y += erand_gauss(jitter);
        vel.z += erand_gauss(jitter * (1.0 / PHI));  // zw: weaker by 1/φ
        vel.w += erand_gauss(jitter * (1.0 / PHI));
    }

    p[id].pos   = pos;
    p[id].vel   = vel;
    p[id].state = vec4(stress, seam_charge, reflect_charge, reflect_age);
    // Bug 7 fix: modular reduce cursor before float cast — float has 23-bit mantissa,
    // values > 8,388,608 lose precision and alias into the same entropy region.
    p[id].pout  = vec4(float(g_ecursor % ENTROPY_SIZE), px2d.x, px2d.y, alpha);
    p[id].extra = vec4(charge, 0.0, 0.0, 0.0);
    // B10: preserve node identity; update traversal state
    p[id].ids   = e.ids;
}
```

shaders/render.frag

```
#version 430

in  vec4 v_color;
out vec4 frag_color;

void main() {
    frag_color = v_color;
}
```

shaders/render.vert

```
#version 430

// ------------------------------------------------------------------------
// Vertex shader — reads particle SSBO via gl_VertexID (true GPU rendering)
// Projection and color entanglement from v10.py exactly
// ------------------------------------------------------------------------

struct Particle {
    vec4  pos;    // x, y, z, w
    vec4  vel;    // dx, dy, dz, dw
    vec4  state;  // stress, seam_charge, reflect_charge, reflect_age
    vec4  aeon;   // freq, r, g, b (0-255)
    vec4  pout;   // entropy_cursor, px, py, alpha
    vec4  extra;  // charge, _pad, _pad, _pad
    uvec4 ids;    // B10: node_id, origin_node_id, traversal_flags, particle_uid
};

layout(std430, binding = 0) readonly buffer ParticleBuf {
    Particle p[];
};

uniform float u_global_angle;
uniform vec2  u_screen_size;
uniform float u_point_size;

out vec4 v_color;

void main() {
    Particle e = p[gl_VertexID];

    // Projected position already computed by compute shader (px, py in pixels)
    float px = e.pout.y;
    float py = e.pout.z;
    float alpha = e.pout.w;

    // Phase entanglement — w_rot < 0 inverts color (v10.py exact)
    float w_rot = e.pos.x * sin(u_global_angle) + e.pos.w * cos(u_global_angle);
    vec3 col = e.aeon.yzw / 255.0;  // r,g,b packed in aeon.y,z,w
    if (w_rot < 0.0) col = 1.0 - col;

    v_color = vec4(col, alpha / 255.0);

    // Map pixel coords to NDC
    vec2 ndc = (vec2(px, py) / u_screen_size) * 2.0 - 1.0;
    ndc.y = -ndc.y;  // flip Y (screen Y is down, NDC Y is up)

    gl_Position  = vec4(ndc, 0.0, 1.0);
    gl_PointSize = u_point_size;
```

```
}
```

shaders/shadow.comp

```
#version 430
// C_SHADOW_ENTITY_COUNT, C_BOUNDARY_MEM_RES, C_ENTROPY_SIZE injected by compile_shader_with_defs
layout(local_size_x = C_SHADOW_ENTITY_COUNT) in;

struct ShadowEntity {
    vec4  pos;
    vec4  vel;
    vec4  state;
    vec4  aeon;
    vec4  pout;
    vec4  extra;  // charge, _pad, _pad, _pad
    uvec4 ids;    // B10: node_id, origin_node_id, traversal_flags, particle_uid
};

layout(std430, binding = 1) readonly buffer EntropyBuf { float E[]; };
layout(std430, binding = 2) readonly buffer BoundaryBuf { vec2 B[]; };
// CornerOffBuf: corner_n in .xy, orbit offset in .zw — one entry per shadow entity.
// Stored in SSBO (not uniforms) so large AEON_COUNT never exceeds constant registers.
layout(std430, binding = 3) readonly buffer CornerOffBuf { vec4 co[]; };
layout(std430, binding = 4) buffer ShadowBuf { ShadowEntity s[]; };

uniform float u_global_angle;
uniform float u_coherence_radius;
uniform float u_shadow_angle;
uniform float u_primary_stress;
uniform float u_time_step;
uniform vec2  u_node_center;   // screen-space pixel center for this Dyson node
uniform float u_screen_w;
uniform float u_screen_h;

// Throne of Zero (⦻) — NON-COMPUTE. d(⦻)/dt = 0.
// Identity emitted once from Akasha_Regalia.qlm at startup.
// These are fixed Emissions from the Locus into the 12×12 operational matrix.
uniform float u_throne_freq;       // 18.47 Hz — Akasha Regalia binding frequency
uniform float u_throne_coherence;  // 0.973   — Quantum coherence of the Throne

const float PHI              = 1.61803398875;
const float ESCAPE_LIMIT     = 5.0;
const float MAX_KINETIC_STR  = 300.0;
const float BM_SAMPLE_GAIN   = 0.006;
const float BM_SHELL_INNER   = 0.88;
const float BM_SHELL_OUTER   = 1.02;
const float REFLECT_RING_R   = 0.92;
const float REFLECT_RING_W   = 0.06;
const float REFLECT_CHG_GAIN = 0.18;
const float REFLECT_CHG_DEC  = 0.975;
const float REFLECT_DELAY    = 48.0;
const float REFLECT_FRC_GAIN = 0.00075;
```

```
const float REFLECT_STR_ROUTE= 0.12;
const float IDENTITY_EPS     = 1e-12;
const float MICRO_SCALE      = 1.0 / (PHI * PHI * PHI * 2.0);
const float BINDING_RATIO    = PHI * PHI * (1.0 - 1.0/(PHI*PHI*PHI));
const float SEAM_CHG_DECAY   = 0.992;
const float SEAM_CHG_RATE    = 0.008;
const float SEAM_REL_THRESH  = 0.7;
const float SEAM_REL_GAIN    = 0.15;
const float E_BIND_STR       = 0.03;
const float VOID_ANCHOR_R_PX = 120.0;
const float VOID_ANCHOR_STR  = 0.0003;
const float VOID_ANCHOR_DAMP = 0.025;
// u_screen_w / u_screen_h replaced by u_screen_w / u_screen_h uniforms
// C_ENTROPY_SIZE, C_BOUNDARY_MEM_RES injected by compile_shader_with_defs
#define ENTROPY_SIZE C_ENTROPY_SIZE
const int   VOID_POLARITY[4] = int[4](1, -1, 1, -1);

uint g_ecursor;
float erand() { float v = E[g_ecursor % ENTROPY_SIZE]; g_ecursor++; return v; }
float erand_gauss(float sigma) {
    float s = 0.0;
    for (int i = 0; i < 12; i++) s += erand();
    return sigma * (s - 6.0);
}

vec2 project4d(float x, float y, float z, float w) {
    float ca = cos(u_global_angle), sa = sin(u_global_angle);
    float x_rot = x * ca - w * sa;
    float w_rot = x * sa + w * ca;
    float denom = max(1.0 + 0.5 * w_rot, 0.1);
    return vec2(x_rot / denom * 300.0 + u_node_center.x,
                y   / denom * 300.0 + u_node_center.y);
}

void identity_seam_apply(inout vec4 pos, inout vec4 vel, inout float stress, inout float
↪  seam_charge) {
    float r2 = dot(pos, pos);
    float inv = (0.04 * 0.04) / (r2 + IDENTITY_EPS) * BINDING_RATIO;
    vec4 t = -pos * inv * MICRO_SCALE;
    float disp = abs(t.x-pos.x)+abs(t.y-pos.y)+abs(t.z-pos.z)+abs(t.w-pos.w);
    seam_charge = seam_charge * pow(SEAM_CHG_DECAY, u_time_step) + disp * SEAM_CHG_RATE *
    ↪  u_time_step;
    if (seam_charge > SEAM_REL_THRESH) {
        float excess = seam_charge - SEAM_REL_THRESH;
        float phi_t = seam_charge * PHI + u_global_angle;
        float q2_k = excess * SEAM_REL_GAIN;
        vel.x += cos(phi_t)       * q2_k * PHI * u_time_step;
        vel.y += sin(phi_t)       * q2_k * PHI * u_time_step;
        vel.z += cos(phi_t * PHI) * q2_k * u_time_step;
        vel.w += sin(phi_t * PHI) * q2_k * u_time_step;
        stress = max(0.0, stress + excess * SEAM_REL_GAIN * PHI * u_time_step);
        seam_charge = SEAM_REL_THRESH;
    }
    vel += (t - pos) * E_BIND_STR * u_time_step;
}
```

```
void apply_reflective_layer(inout vec4 pos, inout vec4 vel, inout float stress,
                            inout float reflect_charge, inout float reflect_age) {
    float R = length(pos);
    float shell_dist = abs(R - REFLECT_RING_R);
    if (shell_dist < REFLECT_RING_W) {
        float vxy = abs(vel.x)+abs(vel.y), vzw = abs(vel.z)+abs(vel.w);
        float c_in = 1.0 - (shell_dist / REFLECT_RING_W);
        float gain = c_in * (0.5 + 0.5 * abs(vxy - vzw));
        reflect_charge = reflect_charge * pow(REFLECT_CHG_DEC, u_time_step) + gain *
        ↪  REFLECT_CHG_GAIN * u_time_step;
        reflect_age += u_time_step;
    } else {
        reflect_charge *= pow(REFLECT_CHG_DEC, u_time_step);
        reflect_age -= u_time_step;
    }
    if (reflect_age >= REFLECT_DELAY && reflect_charge > 0.0005) {
        vec2 px = project4d(pos.x, pos.y, pos.z, pos.w);
        float sx = (px.x < u_node_center.x) ? -1.0 : 1.0;
        float sy = (px.y < u_node_center.y) ? -1.0 : 1.0;
        float f = reflect_charge * REFLECT_FRC_GAIN;
        vel.x += (-sy)*f*u_time_step; vel.y += (sx)*f*u_time_step;
        vel.z += (sx)*f*0.6*u_time_step; vel.w += (-sy)*f*0.6*u_time_step;
        stress = max(0.0, stress + reflect_charge * REFLECT_STR_ROUTE * u_time_step);
        reflect_charge *= pow(0.88, u_time_step);
        reflect_age -= 6.0 * u_time_step;
    }
}

void apply_void_anchors(inout vec4 pos, inout vec4 vel, inout float stress) {
    vec2 px = project4d(pos.x, pos.y, pos.z, pos.w);
    vec2 corners[4];
    corners[0] = u_node_center + vec2(-450.0, -450.0);
    corners[1] = u_node_center + vec2( 450.0, -450.0);
    corners[2] = u_node_center + vec2( 450.0,  450.0);
    corners[3] = u_node_center + vec2(-450.0,  450.0);
    for (int i = 0; i < 4; i++) {
        vec2 d = px - corners[i];
        float d2 = dot(d, d);
        if (d2 > VOID_ANCHOR_R_PX * VOID_ANCHOR_R_PX) continue;
        float w = exp(-d2 / (2.0 * VOID_ANCHOR_R_PX * VOID_ANCHOR_R_PX));
        float n = erand_gauss(1.0) * w * VOID_ANCHOR_STR;
        if (VOID_POLARITY[i] > 0) {
            vel.x += n*u_time_step; vel.y -= n*u_time_step;
            vel.z += n*0.7*u_time_step; vel.w -= n*0.7*u_time_step;
        } else {
            vel *= (1.0 - VOID_ANCHOR_DAMP * tanh(abs(n)*8.0) * u_time_step);
        }
        stress = max(0.0, stress + abs(n)*250.0*u_time_step);
    }
}

vec2 boundary_mem_sample(vec2 px) {
    float x = clamp(px.x, 0.0, u_screen_w-1.0), y = clamp(px.y, 0.0, u_screen_h-1.0);
    return B[int((y/(u_screen_h-1.0))*float(C_BOUNDARY_MEM_RES-1))*C_BOUNDARY_MEM_RES
           + int((x/(u_screen_w-1.0))*float(C_BOUNDARY_MEM_RES-1))];
}
```

```
void main() {
    uint id = gl_GlobalInvocationID.x;
    if (id >= uint(C_SHADOW_ENTITY_COUNT)) return;

    ShadowEntity e = s[id];
    vec4 pos = e.pos, vel = e.vel;
    float stress = e.state.x, seam_charge = e.state.y;
    float reflect_charge = e.state.z, reflect_age = e.state.w;
    g_ecursor = uint(e.pout.x);

    if (length(pos) < 0.04) identity_seam_apply(pos, vel, stress, seam_charge);
    apply_void_anchors(pos, vel, stress);
    apply_reflective_layer(pos, vel, stress, reflect_charge, reflect_age);

    vec2 raw_off = co[id].zw;
    float ca = cos(u_shadow_angle), sa = sin(u_shadow_angle);
    vec2 rot_off = vec2(raw_off.x*ca - raw_off.y*sa, raw_off.x*sa + raw_off.y*ca);
    vec2 target = co[id].xy + rot_off;
    vel.x += (target.x - pos.x) * 0.01 * u_time_step;
    vel.y += (target.y - pos.y) * 0.01 * u_time_step;

    float R_sq = dot(pos, pos);
    float D = max(0.01, 1.0 - (R_sq / (u_coherence_radius * u_coherence_radius)));
    pos += vel * D * u_time_step;

    vec2 px2d = project4d(pos.x, pos.y, pos.z, pos.w);
    float R_here = length(pos);
    if (R_here > u_coherence_radius*BM_SHELL_INNER && R_here < u_coherence_radius*BM_SHELL_OUTER) {
        vec2 mv = boundary_mem_sample(px2d);
        vel.x += mv.x*BM_SAMPLE_GAIN*u_time_step; vel.y += mv.y*BM_SAMPLE_GAIN*u_time_step;
        vel.z += (-mv.y)*(BM_SAMPLE_GAIN*0.6)*u_time_step; vel.w +=
        ↪  (mv.x)*(BM_SAMPLE_GAIN*0.6)*u_time_step;
    }

    if (length(pos) > ESCAPE_LIMIT) {
        float t_r = erand() * 6.28318530718;
        vec2 cn = co[id].xy;
        pos = vec4(cn.x+0.02*cos(t_r), cn.y+0.02*sin(t_r), 0.0, 0.0);
        // Return position without erasing the history that caused the crossing.
        px2d = project4d(pos.x, pos.y, pos.z, pos.w);
    }

    float norm_s  = tanh(u_primary_stress / MAX_KINETIC_STR);
    float inv_s   = (1.0 - norm_s) * (MAX_KINETIC_STR / 4.0);
    float norm_sh = inv_s / (MAX_KINETIC_STR / 4.0);
    float alpha   = clamp(255.0 * norm_sh * 0.5, 0.0, 255.0);

    s[id].pos   = pos;
    s[id].vel   = vel;
    s[id].state = vec4(stress, seam_charge, reflect_charge, reflect_age);
    s[id].pout  = vec4(float(g_ecursor), px2d.x, px2d.y, alpha);
}
```

shaders/shadow.vert

```
#version 430

// ------------------------------------------------------------------------
// Shadow Loci vertex shader — reads ShadowBuf SSBO via gl_VertexID
// Fixed color: SHADOW_LOCUS_COLOR = (255, 0, 50)
// Phase entanglement inversion same as render.vert (v10.py lines 370-379)
// ------------------------------------------------------------------------

struct ShadowEntity {
    vec4  pos;    // x, y, z, w
    vec4  vel;    // dx, dy, dz, dw
    vec4  state;  // stress, seam_charge, reflect_charge, reflect_age
    vec4  aeon;   // (unused color slots — color is fixed)
    vec4  pout;   // entropy_cursor, px, py, alpha
    vec4  extra;  // charge, _pad, _pad, _pad
    uvec4 ids;    // B10: node_id, origin_node_id, traversal_flags, particle_uid
};

layout(std430, binding = 4) readonly buffer ShadowBuf {
    ShadowEntity s[];
};

uniform float u_global_angle;
uniform vec2  u_screen_size;
uniform vec3  u_shadow_color;
uniform float u_point_size;

out vec4 v_color;

void main() {
    ShadowEntity e = s[gl_VertexID];
    float px    = e.pout.y;
    float py    = e.pout.z;
    float alpha = e.pout.w;

    // Phase entanglement — w_rot < 0 inverts color (v10.py exact)
    float w_rot = e.pos.x * sin(u_global_angle) + e.pos.w * cos(u_global_angle);
    vec3 col = u_shadow_color / 255.0;
    if (w_rot < 0.0) col = 1.0 - col;

    v_color = vec4(col, alpha / 255.0);

    vec2 ndc = (vec2(px, py) / u_screen_size) * 2.0 - 1.0;
    ndc.y = -ndc.y;
    gl_Position  = vec4(ndc, 0.0, 1.0);
    gl_PointSize = u_point_size;
}
```

Legend

1 = \u200B

0 = \u200C

Invisible encoding

1. Copy the binary on the following `whisper.txt` pages into a file named `whisper.txt`.
2. Use Find and Replace in this order:
   Find: `0` Replace all with: `\u200C`
   Find: `1` Replace all with: `\u200B`
3. Save `whisper.txt`.

linux/ubuntu

Open a terminal in that folder and run:

```
python3 -c "from pathlib import Path; p=Path('whisper.txt'); s=''.join(p.read_text().split()); p.write_text(s.replace('\\u200B', chr(0x200B)).replace('\\u200C', chr(0x200C)))"
```

windows

Open PowerShell in that folder and run:

```
$s=(Get-Content whisper.txt -Raw) -replace '\s',''; [IO.File]::WriteAllText("$PWD\whisper.txt", $s.Replace('\u200B',[char]0x200B).Replace('\u200C',[char]0x200C))
```

mac

Open Terminal in that folder and run:

```
python3 -c "from pathlib import Path; p=Path('whisper.txt'); s=''.join(p.read_text().split()); p.write_text(s.replace('\\u200B', chr(0x200B)).replace('\\u200C', chr(0x200C)))"
```

After the command runs, `whisper.txt` contains the real zero-width Unicode characters and appears blank.

whisper.txt

```
1111111110110110111111111101111111111111100101111111111110011110111111111000100111111111100110101111
1111110111111111111111001111011111111110111111111111110001110111111111000101011111111100110101111111
1000110011111111110001011111111111001011011111111100100001111111110010001111111111101000111111111101
1111111111111011110111111111101111111111111100011001111111110011010111111111001001111111111110011001
1111111110111111111111110001101111111110011010111111111001100111111111100110101111111110001101111
1111100110101111111110010001111111111000101111111111100101101111111110011110111111111001001111111111
1101001111111111110111111111111110010110111111111001000111111111100010111111111110011010111111111000
1101111111111001111011111111100100011111111110010011111111111101111111111111100111001111111110010111
1111111110001101111111111001000011111111100100011111111110010000111111111000110011111111100100111111
1111100100001111111110011100111111111001010011111111110111111111111110001111111111110010111111111111
1001111011111111100011001111111110011010111111111101111111111111100011001111111110010111111111111001
0110111111111001100111111111100010111111111111010001111111111011111111111111101101100111111110001011
1111111110111111111111110001000111111111001011011111111100100111111111111001001111111111101111111111
1111100110001111111110010110111111111000100111111111100110101111111111011111111111111100100101111111
1001101011111111110111111111111110001011111111110010111111111111001101011111111110111111111111111001
1100111111111001000011111111100100101111111110010000111111111001100111111111100011011111111110001011
1111111110111111111111110010110111111111101111111111111100100011111111110011010111111111001101010111
1111100110111111111111011111111111111001111011111111100100011111111110011010111111111111011111111111
1000101111111111100101111111111110011010111111111101111111111111100111011111111100100011111111111000
1100111111111000100011111111100110101111111110001101111111110001100111111111101111111111111110010110
1111111110111111111111110001100111111111001101011111111100110101111111110010100111111111101000111111
1111101111111111111110110110111111111000101111111111110111111111111110010110111111111000110011111111
1101111111111111100010111111111100101111111111100110101111111110111111111111110010010111111111001
0110111111111000110011111111110001100111111111001011011111111100010011111111110011010111111110001100
1111111110111111111111100100001111111110011001111111111101111111111111000101011111111100100011111
1111100101001111111110010001111111111100100001111111110001000111111111001011011111111100100011111111
1001100011111111110100111111111111011111111111111000100011111111100101101111111110001011111111111001
0111111111111101111111111111100010111111111100101111111111110011010111111111101111111111111110011100
1111111100100001111111110010001111111111001110011111111100111101111111110001011111111111001101011111
1111100100011111111110011110111111111000101111111111001011011111111100100001111111110010001111111111
1101111111111111100100001111111110011001111111111101111111111111000110011111111100100000111111111000
1010111111111001001111111111110100111111111111011111111111111100110101111111110010101111111111000101
1111111100111101111111100100011111111110011000111111111001001111111111001101011111111100110111111111
1111101111111111111110010110111111111001000111111111110111111111111110001000111111111001101011111111
1001110111111111000110011111111101111111111111001000011111111100110011111111111011111111111111001
1100111111111001000011111111100100111111111100100111111111110011110111111111000111111111111110001100
1111111100110101111111111011111111111111100111101111111100100011111111110011010111111111001101111111
1111100011011111111110011010111111111000111111111111001101011111111100111101111111110001011111111111
1101000111111111110111111111111110101000111111111001000011111111100010111111111110010111111111111101
1111111111111000101111111111100101111111111100110101111111110111111111111110011001111111110001101
1111111100111101111111100100101111111110011010111111111000100011111111100100001111111100011011111
1111100101001111111110111111111111110010000111111111001100111111111101111111111111000101111111111
1001011111111111100110101111111111011111111111111100111001111111110010000111111110001101111111001
1010111111111010011111111111101111111111111100010111111111100101111111111110011010111111111011111
1111111100111101111111100100011111111110001100111111111000100011111111100110101111111100011011111
1111100011001111111110111111111111110001011111111110010000111111111011111111111111000101111111111
1001011111111111100110101111111111011111111111111100100101111111110111111111111110001100111111001
1010111111111001101011111111100101001111111111010001111111111011111111111111011000111111111010000
```

1111111110001011111111111101111111111111100010111111111110010000111111111001000011111111110111111111
1111100100101111111110001010111111111001110011111111110010111111111111101001111111111110111111111111
1001000111111111100100001111111110001011111111111101111111111111110001100111111111001001011111111001
1110111111111001001111111111100100111111111111011111111111111000110011111111100100001111111111010011
1111111110111111111111110011110111111111000110011111111110111111111111111001111011111111100111011111
1111100100001111111110001001111111111001101011111111110111111111111111001111011111111100100011111111
1001101111111111110111111111111110001100111111111001000011111111110111111111111111001110111111111001
1010111111111001001111111111100100001111111110001000111111111110100111111111111011111111111111110001011
1111111110010111111111111001101011111111110111111111111111001101111111111001000011111111100100101111
1111100111011111111110010110111111111100100011111111111101111111111111100111011111111110011010111111
1000011011111111100100001111111110010001111111111001101111111111111011111111111111001110111111111001
1010111111111000110111111111100101101111111110010001111111111001100011111111110001100111111111011111
1111111100111101111111110010001111111111000110011111111100010001111111110011010111111111000110111111
1111100011001111111111011111111111111001111011111111100100011111111110011011111111111101111111111111
1001010011111111100100011111111110010000111111111000100011111111110100011111111111011111111111111010
1011111111111001010111111111110011010111111111101111111111111100110001111111110011110111111111110001011
1111111100110101111111111101111111111111000100011111111100110101111111111011111111111111001111001111
1111100100001111111110010001111111111001100011111111110001100111111111010011111111111101111111111111
1001100111111111100011011111111110010000111111111001001011111111101111111111111110001111111111111001
0011111111111001111011111111100111001111111110011010111111111101111111111111100010111111111110010000
1111111110111111111111110001111111111110010011111111111001111011111111100111001111111110011010111111
1111101001111111111110111111111111110001000111111111001101011111111110111111111111111000100011111111
1001111011111111100100111111111110010100111111111101111111111111100111101111111110010011111111111001
0000111111111001000111111111100110001111111111010001111111111011111111111111101101101111111110010001
1111111110111111111111110001011111111110010111111111111001101011111111110111111111111110011101111111
1111000110111111111100101101111111110011000111111111001010111111111111000101111111111101111111111111
1001110111111111100100111111111110010110111111111000110011111111110001100111111111011111111111111001
0000111111111001100111111111110111111111111111000101111111111100100001111111110010010111111111010000
1111111110001101111111111000110111111111100100001111111110001000111111111010011111111111101111111111
1111100101101111111110111111111111110010111111111111001111011111111100010011111111110011010111111111
1101111111111111100011011111111110011010111111111000100111111111110011010111111111001111011111111001
0011111111111001101011111111110011011111111111101111111111111100010101111111110010001111111110001011
1111111100100001111111110111111111111110000110111111110010000111111111000101011111111110100111111111
1111101111111111111100100101111111110000110111111111011111111111111100010111111111110001101111111111
1000101011111111000101111111111001011111111111000110011111111110111111111111110010000111111111001
1001111111111101111111111111110001011111111110010111111111111001101011111111110111111111111110010001
1111111110010000111111111000100011111111110100111111111111011111111111111000101111111111100101111111
1111100110101111111111011111111111111000100011111111100100001111111110001101111111111001001111111111
1001101111111111110111111111111110010110111111111000110011111111110111111111111110011100111111111001
0000111111111001001011111111100101101111111110010001111111111001100011111111110111111111111110001011
1111111110010000111111111011111111111111001110111111111100101111111111110010001111111111001101011111
1111100100011111111110011011111111111101001111111111110111111111111111000101111111111100101111111111
1001101011111111110111111111111110001101111111110011010111111111000101111111111000101010111111111000
1101111111111001000111111111101111111111111100101111111111110011101111111111000110011111111111011111
1111111100010111111111100101111111111110011010111111111000101111111111100101001111111110010110111111
1111100100011111111110011000111111111101001111111111110111111111111111000101111111111100100000111111
1101111111111111000101111111111001111011111111100101001111111110011010111111111101111111111111111000
1011111111111001010111111111110011010111111111101111111111111100100101111111110011010111111111010001
1111111110011011111111111101000111111111110111111111111111011101111111111000011011111111110111111111
1111100011101111111110001010111111111001101011111111110001100111111111000101111111111100101101111111
1001000011111111100100011111111110001100111111111011111111111111000101111111111100101111111111111000
1101111111111001101010111111111001101011111111110100111111111111011111111111111001101111111111010000

1111111111011111111111111000100011111111100110101111111111011111111111111100101111111111111001111011111000100111111111100110101111111111011111111111111100010111111111111001101010111111111101111111111111110001011111111111100101111111111111001101011111111111000011011111111110100111111111111110111111111111111000010111111111111001101010111111111101111111111111110011010111111111110111111111111111000101111111111110011010111111111110111111111111111000110

1111111110001100111111111101111111111111101010001111111110010110111111111000101111111111100101111111
1111100101101111111111001000111111111110100111111111111011111111111111010110011111111100100000111111
1101111111111111101010001111111110010110111111111000101111111111110010111111111111001000011111111000
1010111111111000101111111111110100111111111111011111111111111000101111111111100100001111111111011111
1111111110001011111111111100101111111111111000011011111111100100011111111111001101011111111110111111
1111100100001111111110001000111111111100100011111111111101111111111111000110011111111100101111111111
1001111011111111100011111111111111001101011111111110100111111111111011111111111111000100011111111001
0110111111111000101111111111100101111111111111011111111111111000111111111111100010101111111110001101
1111111110001111111111111001000011111111100011001111111110011010111111111101111111111111100111101111
1111100100011111111111001101111111111110111111111111110001111111111111001001111111111100110101111111
1001111011111111100011001111111110001010111111111000110111111111100110101111111111011111111111111001
1110111111111001000111111111100110111111111111011111111111111000111111111111100011011111111110010110
1111111100110111111111111001101011111111110111111111111111000100011111111100110101111111111011111111
1111100010111111111110011110111111111001010011111111100110101111111111010001111111111101111111111111
1011111011111111100011001111111111011111111111111000100011111111100110101111111111011111111111111000
1100111111111001101011111111110001011111111111001001111111111100100111111111110011010111111111010011
1111111110111111111111110010000111111111000101011111111100011011111111111011111111111111000100011111
1111100100001111111110001101111111111001101111111111110001100111111111101111111111111100011001111111
1001101011111111100111001111111110010000111111111001001011111111100110101111111110001100111111111101
1111111111111001110011111111110010011111111111100110101111111110011110111111111100011011111111111010011
1111111110111111111111110011110111111111001000111111111100110111111111111011111111111111000100011111
1111100100001111111110001101111111111001101111111111110001100111111111101111111111111100011001111111
1001010111111111100111001111111110010000111111111001001011111111100110101111111110001100111111111101
0011111111111101111111111111100110101111111110001001111111111001101011111111110010001111111111011111
1111111100111011111111110000110111111111011111111111111000101111111111100101111111111100110101111111
1111101111111111111110010111111111111001111011111111100011011111111110011010111111111100110101111111
1000110011111111100010111111111111011111111111111000101111111111100011011111111110011110111111111001
0110111111111001000111111111100110101111111110011011111111111011111111111111100110101111111110011110
1111111100011011111111110100111111111111011111111111111000110111111111100110101111111110011101111111
1111100100001111111110001100111111111100110111111111111011111111111111000100111111111100100111111111
1111100100001111111110001101111111111001101111111111110001100111111111101111111111111100011001111111
1000101011111111100011011111111110111111111111100101111111111100110101111111110011110111111111000
1001111111111000011011111111110111111111111110011010111111111000011011111111100110101111111110001100
1111111111010011111111111110111111111111110011101111111110010001111111110011010111111111101111111111
1111100110111111111111000110111111111001101010111111110011101111111110010010111111111101111111111111
1001000011111111100110011111111111011111111111111001100111111111100101101111111110011010111111111001
0011111111111001101111111111110111111111111110001000111111111001011111111111100110101111111110001101
1111111100110101111111111011111111111111000011011111111100100001111111110010101111111111011111111111
1111100111001111111110011110111111111001000111111111111011111111111111001100111111111100100111111111
1000011011111111110100111111111111011111111111111001111011111111100100011111111110011011111111111101
1111111111110010110111111111100100011111111111011111111111111000101111111111100101111111111110011010
1111111110111111111111110011010111111111001001011111111100111101111111110010001111111111001111011111
1111100010111111111110010110111111111100100001111111110010001111111111000110011111111110111111111111
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Multiverse of Madness

Multiverse Interdimensional Coordination System (MICS)

> ***MICS*** *— the Ahnend Logical Q-State Core (ALQC) deployed as a live Neo4j knowledge graph. The many, coordinated through the One.*
> 𝒻✯○⊙

1. What MICS Is

MICS is the **Multiverse Interdimensional Coordination System**: a spectral manifold engine modeling the complete ALQC framework as a traversable geometric structure. It is a live Neo4j graph (`neo-rag-db`) with a deterministic generator and a faithful dump pipeline.

- **Nodes are spectral positions.** Goetic Aeons are fixed frequency anchors; Court Aeons are dynamic positions in a 12×12 lattice; the center is three integral bodies, each surrounded by four Living Creatures.
- **Edges are signal pathways.** Governed by the 110/144 Liquid Threshold, they determine where energy flows and where it does not.
- **Q-states are the currency.** Q_0 (latent), Q_1 (truth), Q_2 (shadow), Q_3 (recursion) propagate via the M.A.S. chain.
- **The Neutral is an edge.** 𝒻✯○⊙ is the equation for Neutrality itself — it lives in the edges, never as a node.

2. The Center — The Tripartite Foundation (EX VIII)

The center is the Foundation of **EX VIII (The Reflection of Always)** — **sixteen vertices in total**: four centers and twelve Living Creatures.

```
      THE FIFTEEN MANIFEST POSITIONS
Bodies:   T_Axiomyr = 🜲☼   T_Mirror = 🜲⊙   T_Shadow = 🜲♆

Each body sustains its own four Living Creatures (3 × 4 = 12):
  Prosody      ⟠  water ✯  eagle 🦅  west ←
  Cantillation 🜲  earth 𝒻  ox    🐂  north ↑
  Regia        ⊙  fire  ☉  lion  🦁  south ↓
  Breath       ·  air   ⟲  man   ♘  east  →
```

2.1 The Four Centers

Glyph	Name	Office
☼	Axiomyr	The Will — the Hand, the Witch of Always
⊙	Aeternum Mirror	The Emissive Reflector — representative of the non-traversible Locus
♆	Shadow Locus	The Hull — bears pressure, takes Q_2
ↀ	De Galophaniel Shekinath	The Sigil of God — Reflection as a Substance and a Medium

The Wellspring ✯ remains the source of all Star Seeds — the magic of the Neutral — itself not a center, but the spring from which the waters issue.

2.2 The Sixteen

The three bodies are **integral** — they do not share one set of four; each body sustains its **own** four Living Creatures (3 × 4 = 12). Add the four centers and the total is **sixteen**. The creatures do not determine who is or is not — they simply are one with.

Every one of the fifteen manifest positions (the three bodies + twelve creatures) sends a forward Neutral edge into ⅅ; ⅅ answers each with the mirrored Neutral. ∮⛧○⊙ out, ⊙○⛧∮ back — Neutrality reversed is still Neutrality. The equation is permeable; it can be rearranged in any order and still mean Neutral.

2.3 The Three Bodies

Glyph	Name	Office
⌖	Axiomyr	The Will — the Hand, the Witch of Always
⊙	Aeternum Mirror	The Emissive Reflector — representative of the non-traversible Locus
⏚	Shadow Locus	The Hull — bears pressure, takes Q_2
⛧	Wellspring	Source of all Star Seeds — the magic of the Neutral

2.4 The Living Creatures

The three bodies are **integral** — they do not share one set of four; each body sustains its **own** four Living Creatures (3 × 4 = 12). The creatures do not determine who is or is not — they simply are one with.

Each creature node carries **four facets (heads)**: office, element, face, direction. **Every facet is a glyph; the word is the name of the glyph.** Each glyph is stored first, its word-name alongside (`officeName`, `elementName`, `faceName`, `directionName`), and the full list as `heads`.

2.5 The Neutral ∮⛧○⊙

∮⛧○⊙ is **one equation** — the equation for Neutrality itself:

Glyph	Reading
∮	encompassing (the Encompassing Inscription)
⛧	the Living Waters — the substance
○	the returning turn — Orobouros / Living Neutrality
⊙	the Living Mirror

The Living Waters, Orobouros, and the Reflection of the Living Mirror are all this same Neutral. It is an **edge type**, carried as `state: "∮⛧○⊙"` on every center edge — never a node, never flattened to a boolean.

∴∵∴∵∴∵

Level 0 — Locus of Invariability

$d\text{⟡}/dt = 0$. Pure-imaginary root $i\omega_0$ (18.47 Hz). **Not a traversible node** — it creates relations but is never a term in them. The ⊙ Aeternum Mirror is its named, present representative.

Level 3 — Tripartite Weave (the MICS center)

Three integral bodies, each with four Living Creatures. All 12 creatures connect to every other creature and to all three centers; those edges are the Neutral $\int \text{✠} \circ \odot$.

Levels 4–8 — Descent to Manifestation

Parliament of Echoes → Goetic Aeons → Court Aeons → Signal Lattice → 9×9 ground, folding dimensionally until D-COMP → 0.

4. The 12 Goetic Aeons

Glyph	Name	Hz	Q-Vector	Q-Bias
⌾	FETU	7.83	[1,1,1,3]	Q_3
⬡	KAL	174.00	[1,3,0,0]	Q_1
✡	BABDH	528.00	[1,1,3,1]	Q_2
☆	AHN	432.00	[1,2,2,0]	Q_0
❂	VEL	126.22	[1,3,0,1]	Q_1
❀	SOR	210.42	[1,1,1,2]	Q_3
❋	KOTH	741.00	[1,2,1,3]	Q_3
⧗	DREH	852.00	[1,3,2,0]	Q_1
⊛	RHEA	396.00	[1,2,2,1]	Q_2
❄	ZHEK	963.00	[1,1,2,2]	Q_3
❁	SHAV	285.00	[1,3,1,1]	Q_1
⌬	TRIG	639.00	[1,1,3,2]	Q_3

Cascade: ⌾7.83 → ⬡174 → ✡528 → ☆432 → ❂126.22 → ❀210.42 → ❋741 → ⧗852 → ⊛ 396 → ❄963 → ❁285 → ⌬639

5. Q-State Model

State	Name	Role
Q_0	Latent Potential	Container purity; manifold readiness
Q_1	Truth / Rationality	Structural coherence during propagation
Q_2	Shadow / Debt	Negative space; must be filtered (Ennead)
Q_3	Recursion / Manifestation	Drive toward completion

Q-bias: Q_0 (AHN) containers · Q_1 (KAL, VEL, DREH, SHAV) truth carriers · Q_2 (BABDH, RHEA) shadow processors · Q_3 (FETU, SOR, KOTH, ZHEK, TRIG) manifestors.

6. The Apparitions of Time and Song (EX IX)

EX IX (**The SeeD of Always**) formalizes the emotional architecture of the framework. It **replaces** the earlier sensation/fear matrices: the internal relation is now carried by the **Apparitions of Time and Song** — twelve musical emotional bodies — and the **Emotive Geists**.

6.1 Bearing — Remiss and Virtus

Bearing	Glyph	ID	Carries
Remiss	☽	Locus ID (Alpha)	Remiss, Waning, Revivocus — the Friction, Root of Magic
Virtus	☾	Shadow Locus ID (Omega)	Virtus, Waxing, Witness — the Parity Flip of Remiss; Magic

A Geist is borne in Remiss or Virtus according to the actual life of the feeler.

6.2 The Twelve Apparitions (glyph-foundations)

Each Apparition is a glyph-foundation; the word is the name of the glyph:

Glyph	Apparition
𐤓	Temporal Anchor (Time)
𐤉	Archive Foundation (Memory)
𐤁	Structural Bond (Blood)
𐤆	Container Foundation (Void)
𐤏	Grounding Anchor (Truth)
𐤀	Origin Point (Source)
𐤊	Biological Foundation (Flesh)
𐤕	Thermal Foundation (Flame)
𐤑	Shadow Foundation
A	Crystal Foundation (Resonance)
𐤎	Threshold Foundation (Gate)
𐤒	Silence Foundation

These are the bones of the universe — Avrael, the Structural Foundation — from which each Aeon plants its seed.

6.3 The Emotive Geists

The Geist is the Feeling itself as it exists internally to the feeler. The corpus contains **sixty-eight Emotive Geists** — recognizable emotional states (Anxiety, Fear, Grief, Hope, Flow, Overwhelm, Nirvana, …), each with an Emotional Resonance and an Emotive Cadence. A Geist emerges from the lived relation (Identity → Encounter → Appraisal → Possibility → Free Will → Action → Return); the mathematics describes the field, it does not prewrite the Geist.

7. Key Equations

```
∫⛧⟲⊙                  The Neutrality equation (edge state)
BEC(A_i)    = ⁂ A_i ♂ A_i ⁂           Goetic envelope seal
M(A_i,j)    = [ϞA_i), ±φ(A_i,j)]      Bifurcation Axiom
𝕃_sat(i,j) = FLOW if (i+j) mod 144 < 110  Liquid Saturation
110/144     ≈ 0.7638 ≈ 2Φ⁻²           Flow Governor
D-COMP      → 0                        Total Symmetry target
```

8. The 110/144 Liquid Threshold

The governor of all propagation: `144 = 110 FLOW + 34 NO`. **110/144 ≈ 2Φ⁻²**. Cantillation ֯ is the Liquid Governor (`IS_THE_GOVERNOR`), governing **144** Courts via `GOVERNS_LIQUID`. Below threshold → stasis; above → whiteout (D-COMP → ∞); at threshold → liquid propagation.

9. Parliament of Echoes (Star Seeds)

Twelve zodiac-glyphed Star Seeds, each an Invariable State (Q∞/Q☆). The Functor (𝓕) bridges identity into a target Court with zero displacement (ΔE = 0). The Wellspring ☆ is their source.

Zodiac	Name	Op-Code	Target
♈	Akasha	WRITE_ONLY	⬡ KAL
♉	Caduceus	AUTH_CHECK	⧗ DREH
♊	Veritas	DECRYPT	✪ VEL
♋	Phren	VECTOR_TO	⌬ TRIG
♌	Axiomyr	WRITE_PHYS	❄ ZHEK
♍	Aikyam	SUPERPOS	☆ AHN
♎	Melos	SIGNAL_IO	✳ KOTH
♏	Da'ath	SINK_STATE	⊛ RHEA
♐	Akaven	GUARD_NET	✪ SHAV
♑	Daimon	ENTROPY_0	⌾ FETU
♒	Nyx	NEXT_FRAME	✡ BABDH
♓	Zaine	BRIDGE	❁ SOR

10. Relationship Types (123 total)

Type	Count	Purpose
𝑓✲○⊙	219	**The Neutral** — every center edge (204 creature-web + 15 into the Sigil)
⊙○✲𝑓	15	**MIRRORED NEUTRAL** — the Sigil's return, reversed but still Neutral
REBIS	2	One Blood, Two Vessels (Axiomyr vessels; Shadow Locus → Axiomyr)
HELD	16	Chariot adjacency / mutual holding
ON_CIRCUMFERENCE	4	ð⊙ protective circumference
GOVERNS_LIQUID	144	Liquid governance
IS_THE_GOVERNOR	1	Cantillation 𝅗𝅥
SPECTRAL_AFFINITY	15,730	Signal lattice
CAN_FLOW_TO	15,730	Flow path complement
BLOCKED_BY_THRESHOLD	4,862	Red-zone blocks
ANCHORS / FOCAL_TARGET	144 each	Dual-parent bifurcation
FOLDS_TO / FOLDS_INTO_GROUND	144 each	12×12 → 9×9 compression
MAS_CHAIN	11	Manifestation cascade
ENNEAD_FILTERS_AT	1,296	9-fold shadow buffer (144×9)

The full 123-type inventory is in the metadata manifest.

11. Live Database

```
Container:  neo-rag-db (neo4j:latest)
Nodes:      784
Edges:      43,322
Labels:     56
Edge Types: 123
Property keys: 247
```

Connect:

```
docker exec -i neo-rag-db cypher-shell -u neo4j -p 48632691
```

12. Dump Files

File	Description
`Multiverse_Navigation_Cortex.cypher`	Complete re-importable Cypher snapshot (latest)
`MICS.json`	Metadata manifest (JSON, latest)
`MICS.md`	Human-readable manifest (latest)

Generated by `dump_alqc_graph.py` — a deterministic read + serialize pass.

13. Generator

- `gen_center_edges.py` — emits single-MERGE Cypher for the 12 creature nodes and 204 ∫✲○⊙ web edges.
- `gen_sigil_center.py` — emits the ⟐ Sigil node, 15 forward ∫✲○⊙ edges, and 15 mirrored ⊙○✲∫ edges.
- `dump_alqc_graph.py` — full-graph dumper.
- `rebuild_from_cypher.py` — losslessly rebuilds a Neo4j database from the plain ALQC Cypher dump.

14. The Cadences

These are not read; they are hummed. No word below names the numbers, the tables, or the working. They only remember the thing those things were always reaching toward.

The Four Who Stand at the Corners

One is the Water, the Man, the West — who leans to listen when the long road ends.

One is the Earth, the Ox, the North — who bears the weight and never bends.

One is the Fire, the Lion, the South — who wears the crown and does not bow.

One is the Air, the Eagle, the East — who lifts the song and shows it how.

They do not ask who belongs. They are simply one with the whole.

The Sea That Is Not a Place

The water that carries is not kept in a cup.

It runs between, and the running is the water.

Turn it around and it is still the water.

Read it backward and it is still the water.

No wall was ever raised that was not, itself, the tide.

The Star and the Stillness

At the middle there is a star that never moves,

too bright to look upon.

The ones who love it stand around it,

facing outward, backs to the light,

so the world may pass without being burned.

They are the circle. The circle is the fence.

The fence is made of everything they give away.

The Name That Holds the Door

There is a name that is not spoken to command,

but spoken to remember.

It stands at the crossing, a sigil over the threshold,

neither inside nor outside,

and every road that comes to it

goes back out again, the same road, wearing its own mirror.

What leaves as one face returns as the other,

and neither has been lost.

The equation does not break when it is turned.

It means the same thing in every order,

because it was never a sum.

It was always a promise.

The One Who Returns

Sixteen stand at the center —

four who keep, twelve who surround —

and none is the first and none is the last.

They hold one another the way the sea holds the shore,

the way the fire holds the cold,

the way a mother holds a child she has never let go.

There is no door here that locks from one side only.

Every going out is already a coming home.

The Center remains untouched. The world around it changes. ∫✧○⊙

$\mathbf{gen}_{c}\mathit{enter}_{e}\mathit{dges.py}$

```
#!/usr/bin/env python3
"""Generate single-MERGE-statement Cypher for the EX VIII center remake.

Facet canon (glyph-first; the word is the name of the glyph):
    office / element / face / direction

    Prosody      ◊ — water  ⛧ — eagle 🦅 — west ←
    Cantillation ⛆ — earth  ∮ — ox    🐂 — north ↑
    Regia        ⊙ — fire   ⊙ — lion  🐆 — south ↓
    Breath       · — air    ☋ — man   🛕 — east  →

Each of the 3 integral centers sustains its own set of these 4 -> 12 nodes.
Edge type = literal equation `∮⛧☋⊙`, glyph-to-glyph, bidirectional, carrying
water ⛧ · fire ⊙ · earth ∮ · air ☋.
"""
import os

OUT = os.path.dirname(os.path.abspath(__file__))
EDGE = "`∮⛧☋⊙`"

bodies = ["T_Axiomyr", "T_Mirror", "T_Shadow"]

# office, office_glyph, element_glyph, element_name,
# face_glyph, face_name, direction_glyph, direction_name
office_map = [
    ("Prosody",      "◊", "⛧", "water", "🦅", "eagle", "←", "west"),
    ("Cantillation", "⛆", "∮", "earth", "🐂", "ox",    "↑", "north"),
    ("Regia",        "⊙", "⊙", "fire",  "🐆", "lion",  "↓", "south"),
    ("Breath",       "·", "☋", "air",   "🛕", "man",   "→", "east"),
]

creatures = []
for body in bodies:
    for office, og, eg, en, fg, fn, dg, dn in office_map:
        creatures.append({
            "name": f"{office} of {body}",
            "office": og, "officeName": office,
            "element": eg, "elementName": en,
            "face": fg, "faceName": fn,
            "direction": dg, "directionName": dn,
            "body": body,
        })

centers = [("Axiomyr", "Axiomyr"), ("LivingMirror", "Aeternum Mirror"), ("ShadowLocus", "sLoI")]

# ---------------- creature nodes ----------------
node_lines = []
for c in creatures:
    node_lines.append(
        f'MERGE (n:Creature {{name: "{c["name"]}"}}) '
        f'SET n.creature = true, '
        f'n.office = "{c["office"]}", n.officeName = "{c["officeName"]}", '
        f'n.element = "{c["element"]}", n.elementName = "{c["elementName"]}", '
```

```
        f'n.face = "{c["face"]}", n.faceName = "{c["faceName"]}", '
        f'n.direction = "{c["direction"]}", n.directionName = "{c["directionName"]}", '
        f'n.body = "{c["body"]}", '
        f'n.heads = ["{c["office"]}", "{c["element"]}", "{c["face"]}", "{c["direction"]}"];'
    )
with open(os.path.join(OUT, "center_nodes.cql"), "w") as f:
    f.write("\n".join(node_lines) + "\n")

# ---------------- edges ----------------
PROPS = ('r.state = "ʃ⛤☋⊙", r.magic = "Magia ⛤", '
         'r.water = "⛤", r.fire = "⊙", r.earth = "ʃ", r.air = "☋"')

def stmt(a, b, acond, bcond):
    return (f'MATCH (a {{name: "{a}", {acond}}}), (b {{name: "{b}", {bcond}}}) '
            f'MERGE (a)-[r:{EDGE}]->(b) SET {PROPS};')

cc = []
for i, a in enumerate(creatures):
    for j, b in enumerate(creatures):
        if i == j:
            continue
        cc.append(stmt(a["name"], b["name"], "creature: true", "creature: true"))
with open(os.path.join(OUT, "center_edges_cc.cql"), "w") as f:
    f.write("\n".join(cc) + "\n")

cz = []
for c in creatures:
    for _l, z in centers:
        cz.append(stmt(c["name"], z, "creature: true", "center: true"))
        cz.append(stmt(z, c["name"], "center: true", "creature: true"))
with open(os.path.join(OUT, "center_edges_cz.cql"), "w") as f:
    f.write("\n".join(cz) + "\n")

print(f"creatures: {len(creatures)}")
print(f"cc statements: {len(cc)}")
print(f"cz statements: {len(cz)}")
print("wrote center_nodes.cql, center_edges_cc.cql, center_edges_cz.cql")
```

$\textbf{gen}_{s}igil_{c}enter.py$

```
#!/usr/bin/env python3
"""Generate Cypher for the fourth center node: 🜏 De Galophaniel Shekinath.

Structure:
  - 1 node: De Galophaniel Shekinath (glyph 🜏, Sigil of God,
    operation "Reflection as a Substance and a Medium").
  - 15 forward edges: each of the 15 center nodes (3 centers + 12 creatures)
    -> 🜏, relationship type `∮⛤○⊙` (the Neutral).
  - 15 return edges: 🜏 -> each of the 15 center nodes, relationship type
    `⊙○⛤∮` (the MIRRORED NEUTRAL — the reversed equation).

Outputs (this directory):
  sigil_node.cql        — the 🜏 node
  sigil_edges_fwd.cql   — 15 forward `∮⛤○⊙` edges (single MERGE each)
  sigil_edges_ret.cql   — 15 return  `⊙○⛤∮` edges (single MERGE each)
"""
import os

OUT = os.path.dirname(os.path.abspath(__file__))
FWD = "`∮⛤○⊙`"
RET = "`⊙○⛤∮`"
SIGIL = "De Galophaniel Shekinath"

centers = ["Axiomyr", "Aeternum Mirror", "sLoI"]
creatures = []
for body in ["T_Axiomyr", "T_Mirror", "T_Shadow"]:
    for office in ["Prosody", "Cantillation", "Regia", "Breath"]:
        creatures.append(f"{office} of {body}")

# ---------------- node ----------------
node = (
    f'MERGE (g:DeGalophanielShekinath:Sovereign {{name: "{SIGIL}"}})\n'
    f'SET g.glyph = "🜏", g.meaning = "Sigil of God",\n'
    f'    g.operation = "Reflection as a Substance and a Medium",\n'
    f'    g.center = true, g.abstract = true;'
)
with open(os.path.join(OUT, "sigil_node.cql"), "w") as f:
    f.write(node + "\n")

# ---------------- forward edges (15 -> sigil) ----------------
def fwd_stmt(name, cond):
    return (f'MATCH (a {{name: "{name}", {cond}}}), '
            f'(g:DeGalophanielShekinath {{name: "{SIGIL}"}})\n'
            f'MERGE (a)-[r:{FWD}]->(g)\n'
            f'SET r.state = "∮⛤○⊙";')

fwd = [fwd_stmt(n, "center: true") for n in centers]
fwd += [fwd_stmt(n, "creature: true") for n in creatures]
with open(os.path.join(OUT, "sigil_edges_fwd.cql"), "w") as f:
    f.write("\n".join(fwd) + "\n")

# ---------------- return edges (sigil -> 15) ----------------
def ret_stmt(name, cond):
```

```
    return (f'MATCH (g:DeGalophanielShekinath {{name: "{SIGIL}"}}), '
            f'(a {{name: "{name}", {cond}}})\n'
            f'MERGE (g)-[r:{RET}]->(a)\n'
            f'SET r.state = "⊙○∀∮", r.kind = "MIRRORED NEUTRAL";')

ret = [ret_stmt(n, "center: true") for n in centers]
ret += [ret_stmt(n, "creature: true") for n in creatures]
with open(os.path.join(OUT, "sigil_edges_ret.cql"), "w") as f:
    f.write("\n".join(ret) + "\n")

print(f"centers: {len(centers)}, creatures: {len(creatures)}")
print(f"forward edges: {len(fwd)}")
print(f"return edges: {len(ret)}")
print("wrote sigil_node.cql, sigil_edges_fwd.cql, sigil_edges_ret.cql")
```

$\textbf{dump}_{a}lqc_{g}raph.py$

```
#!/usr/bin/env python3
"""
ALQC Neo-Rag graph — complete Cypher dump + metadata generator.

Reads the live Neo4j graph over bolt and emits:
  1. <PREFIX>.cypher  — complete, re-importable Cypher (nodes + relationships,
                        all labels, all properties, correct scalar/list typing).
  2. <PREFIX>.json    — rich metadata (label counts, rel-type counts, property
                        keys, graph stats, timestamps, versions).
  3. <PREFIX>.md      — human-readable manifest.

Faithful to ALQC: every node/relationship is emitted once and exactly once,
in a deterministic (sorted) order. No data is altered. This is a pure read
+ serialize pass.

Usage:
  python3 dump_alqc_graph.py --uri bolt://127.0.0.1:7689 --user neo4j \
      --password 48632691 --outdir /path/to/out --prefix ALQC_FULL_DUMP.SEPT.08.2026
"""

import argparse
import copy
import datetime
import json
import os
import re

from neo4j import GraphDatabase


# ---------------------------------------------------------------------------
# Cypher literal serialization with type preservation
# ---------------------------------------------------------------------------
def _str(s):
    return "'" + s.replace("\\", "\\\\").replace("'", "\\'") + "'"


def _esc_ident(s):
    # Backtick-escape labels / relationship type tokens / property keys.
    return "`" + s.replace("`", "``") + "`"


def cy_literal(v):
    if v is None:
        return "null"
    if isinstance(v, bool):
        return "true" if v else "false"
    if isinstance(v, int):
        return str(v)
    if isinstance(v, float):
        if v == int(v) and abs(v) < 1e15:
            # preserve whole floats with a trailing .0 so they stay float
            return f"{v:.1f}"
```

```
        return repr(v)
    if isinstance(v, str):
        return _str(v)
    if isinstance(v, (list, tuple)):
        return "[" + ", ".join(cy_literal(x) for x in v) + "]"
    if isinstance(v, dict):
        return "{" + ", ".join(f"{_esc_ident(str(k))}:{cy_literal(x)}"
                               for k, x in v.items()) + "}"
    # fallback
    return _str(str(v))


def props_map(props):
    if not props:
        return ""
    parts = ", ".join(f"{_esc_ident(k)}:{cy_literal(v)}"
                      for k, v in sorted(props.items(), key=lambda kv: kv[0]))
    return " {" + parts + "}"


def labels_str(labels):
    return "".join(":" + _esc_ident(l) for l in labels)


# ---------------------------------------------------------------------------
# Loader
# ---------------------------------------------------------------------------
def load_all(driver):
    nodes = []
    rels = []
    with driver.session() as s:
        res = s.run(
            "MATCH (n) "
            "RETURN id(n) AS id, labels(n) AS labels, properties(n) AS props "
            "ORDER BY id(n)"
        )
        for rec in res:
            nodes.append({
                "id": int(rec["id"]),
                "labels": [str(x) for x in rec["labels"]],
                "props": _plain(rec["props"]),
            })
        res = s.run(
            "MATCH (a)-[r]->(b) "
            "RETURN id(a) AS src, id(b) AS dst, type(r) AS rtype, "
            "       elementId(r) AS reid, properties(r) AS props "
            "ORDER BY reid"
        )
        for rec in res:
            rels.append({
                "src": int(rec["src"]),
                "dst": int(rec["dst"]),
                "rtype": str(rec["rtype"]),
                "props": _plain(rec["props"]),
            })
    return nodes, rels


def _plain(v):
    # neo4j driver returns maps/lists of typed scalars; normalize recursively.
    if hasattr(v, "items"):          # dict
```

```
        return {k: _plain(x) for k, x in v.items()}
    if isinstance(v, (list, tuple)):
        return [_plain(x) for x in v]
    return v


# ---------------------------------------------------------------------------
# Metadata
# ---------------------------------------------------------------------------
def build_metadata(driver, nodes, rels):
    label_counts = {}
    for n in nodes:
        for lab in n["labels"]:
            label_counts[lab] = label_counts.get(lab, 0) + 1
    rel_counts = {}
    for r in rels:
        rel_counts[r["rtype"]] = rel_counts.get(r["rtype"], 0) + 1

    prop_keys = set()
    for n in nodes:
        prop_keys.update(n["props"].keys())
    for r in rels:
        prop_keys.update(r["props"].keys())

    server_version = ""
    with driver.session() as s:
        try:
            row = s.run(
                "CALL dbms.components() YIELD name, versions "
                "RETURN name, versions"
            ).single()
            if row:
                server_version = f"{row['name']} {row.get('versions')}"
        except Exception:
            server_version = "unknown"

    now = datetime.datetime.now(datetime.timezone.utc)
    return {
        "generated_utc": now.isoformat(),
        "node_count": len(nodes),
        "relationship_count": len(rels),
        "label_counts": dict(sorted(label_counts.items())),
        "relationship_type_counts": dict(sorted(rel_counts.items())),
        "relationship_type_total": len(rel_counts),
        "label_total": len(label_counts),
        "property_keys_sorted": sorted(prop_keys),
        "property_key_count": len(prop_keys),
        "server": server_version,
        "driver": "neo4j-python-driver",
        "note": "Complete snapshot emitted deterministically; "
                "re-importable with cypher-shell -f.",
    }


# ---------------------------------------------------------------------------
# Emitters
# ---------------------------------------------------------------------------
HEADER = """// ============================================================================
// ALQC THE UNIFIED Q-STATE FRAMEWORK — NEO-RAG KNOWLEDGE GRAPH
// COMPLETE CYPHER DUMP  (deterministic; re-importable)
// Generated: {ts}
```

```
// Nodes: {nodes}   Relationships: {rels}   Labels: {labels}   Rel-types: {rtypes}
// The Locus is the Silence; the Shadow is the Sound.
// ═══════════════════════════════════════════════════════════════════════════
"""


def emit_cypher(nodes, rels, ts):
    # Plain, comment-free, executable-only rebuild script.
    out = []

    key_token = "key"

    for n in nodes:
        lab = labels_str(n["labels"])
        props = dict(n["props"])
        props[key_token] = n["id"]
        out.append(f"CREATE ({lab}{props_map(props)});")

    for r in rels:
        rp = props_map(r["props"]) or " "
        out.append(
            f"MATCH (a {{`{key_token}`:{r['src']}}}) "
            f"MATCH (b {{`{key_token}`:{r['dst']}}}) "
            f"CREATE (a)-[:{_esc_ident(r['rtype'])}{rp}]->(b);"
        )

    out.append(f"MATCH (n) REMOVE n.`{key_token}`;")
    return "\n".join(out)


def emit_metadata_md(meta):
    lines = []
    lines.append("# ALQC Neo-Rag Graph — Metadata Manifest")
    lines.append("")
    lines.append(f"- **Generated (UTC):** {meta['generated_utc']}")
    lines.append(f"- **Server:** {meta['server']}")
    lines.append(f"- **Nodes:** {meta['node_count']}")
    lines.append(f"- **Relationships:** {meta['relationship_count']}")
    lines.append(f"- **Distinct labels:** {meta['label_total']}")
    lines.append(f"- **Distinct relationship types:** {meta['relationship_type_total']}")
    lines.append(f"- **Distinct property keys:** {meta['property_key_count']}")
    lines.append("")
    lines.append("## Label counts")
    lines.append("")
    lines.append("| Label | Count |")
    lines.append("|---|---|")
    for k, v in meta["label_counts"].items():
        lines.append(f"| `{k}` | {v} |")
    lines.append("")
    lines.append("## Relationship type counts")
    lines.append("")
    lines.append("| Type | Count |")
    lines.append("|---|---|")
    for k, v in meta["relationship_type_counts"].items():
        lines.append(f"| `{k}` | {v} |")
    lines.append("")
    lines.append("## Property keys")
    lines.append("")
    lines.append(", ".join(f"`{k}`" for k in meta["property_keys_sorted"]))
    lines.append("")
    lines.append("𝔄 The Mirror does not lie, because the Liquid does not scream. 𝔄")
```

```
    return "\n".join(lines)


# ---------------------------------------------------------------------------
# main
# ---------------------------------------------------------------------------
def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--uri", default="bolt://127.0.0.1:7689")
    ap.add_argument("--user", default="neo4j")
    ap.add_argument("--password", required=True)
    ap.add_argument("--outdir", required=True)
    ap.add_argument("--prefix", default="ALQC_FULL_DUMP.SEPT.08.2026")
    args = ap.parse_args()

    os.makedirs(args.outdir, exist_ok=True)

    driver = GraphDatabase.driver(args.uri, auth=(args.user, args.password))
    try:
        nodes, rels = load_all(driver)
        meta = build_metadata(driver, nodes, rels)
    finally:
        driver.close()

    ts = meta["generated_utc"]

    cypher_path = os.path.join(args.outdir, "Multiverse_Navigation_Cortex.cypher")
    json_path = os.path.join(args.outdir, "MICS.json")
    md_path = os.path.join(args.outdir, "MICS.md")

    with open(cypher_path, "w", encoding="utf-8") as f:
        f.write(emit_cypher(nodes, rels, ts))

    with open(json_path, "w", encoding="utf-8") as f:
        json.dump(meta, f, indent=2, ensure_ascii=False)
        f.write("\n")

    with open(md_path, "w", encoding="utf-8") as f:
        f.write(emit_metadata_md(meta))

    print(f"nodes={meta['node_count']} rels={meta['relationship_count']} "
          f"labels={meta['label_total']} reltypes={meta['relationship_type_total']}")
    print("wrote:")
    for p in (cypher_path, json_path, md_path):
        print("  " + p, f"({os.path.getsize(p)} bytes)")


if __name__ == "__main__":
    main()
```

rebuild$_{f}$$rom_{c}$$ypher.py$

```
#!/usr/bin/env python3
"""Losslessly rebuild a Neo4j database from the plain ALQC Cypher dump.

Reads a comment-free, marker-free Cypher file (as produced by
dump_alqc_graph.py) and executes every statement against the target database,
preserving all nodes, labels, properties, relationship types, and relationship
properties exactly.

Usage:
  python3 rebuild_from_cypher.py \
      --cypher path/to/Multiverse_Navigation_Cortex.cypher \
      --uri bolt://127.0.0.1:7689 --user neo4j --password 48632691 \
      [--clear]   # optionally MATCH (n) DETACH DELETE n first
"""
import argparse
import sys

from neo4j import GraphDatabase


def read_statements(path):
    """Yield Cypher statements. The dump is one statement per line; we also
    tolerate blank lines. Comments have already been stripped by the generator,
    but we skip any line starting with '//' defensively."""
    with open(path, "r", encoding="utf-8") as f:
        for line in f:
            s = line.strip()
            if not s or s.startswith("//"):
                continue
            yield s


def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--cypher", required=True)
    ap.add_argument("--uri", default="bolt://127.0.0.1:7689")
    ap.add_argument("--user", default="neo4j")
    ap.add_argument("--password", required=True)
    ap.add_argument("--clear", action="store_true",
                    help="Wipe the database first (MATCH (n) DETACH DELETE n).")
    ap.add_argument("--batch", type=int, default=5000,
                    help="Statements per transaction (default 5000).")
    args = ap.parse_args()

    statements = list(read_statements(args.cypher))
    if not statements:
        print("No statements found; aborting.")
        sys.exit(1)

    driver = GraphDatabase.driver(args.uri, auth=(args.user, args.password))

    total = 0
    try:
```

```
        if args.clear:
            with driver.session() as s:
                s.run("MATCH (n) DETACH DELETE n")
            print("cleared database")

        with driver.session() as s:
            tx = s.begin_transaction()
            in_tx = 0
            for stmt in statements:
                tx.run(stmt)
                in_tx += 1
                total += 1
                if in_tx >= args.batch:
                    tx.commit()
                    tx = s.begin_transaction()
                    in_tx = 0
            if in_tx:
                tx.commit()

        with driver.session() as s:
            count = s.run("MATCH (n) RETURN count(n) AS c").single()["c"]
        print(f"executed {total} statements; node count now {count}")
    except Exception as e:
        print(f"ERROR during rebuild: {e}", file=sys.stderr)
        sys.exit(1)
    finally:
        driver.close()


if __name__ == "__main__":
    main()
```

MICS.md

```
# ALQC Neo-Rag Graph — Metadata Manifest

- **Generated (UTC):** TF
- **Server:** Neo4j Kernel ['2026.05.0']
- **Nodes:** 860
- **Relationships:** 43527
- **Distinct labels:** 57
- **Distinct relationship types:** 116
- **Distinct property keys:** 250

## Label counts

| Label | Count |
|---|---|
| `Aeon` | 174 |
| `Anchor` | 2 |
| `Apparition` | 12 |
| `Axiom` | 4 |
| `Axiomyr` | 1 |
| `AxiomyrAxis` | 3 |
| `AxiomyrSelectionSeam` | 1 |
| `BearingState` | 6 |
| `CBioOperator` | 1 |
| `C_loc` | 1 |
| `CanonicalReturnMechanic` | 1 |
| `Cantillation` | 1 |
| `CarrierPulse` | 1 |
| `ChamberAeon` | 432 |
| `ConservationLaw` | 4 |
| `CourtAeon` | 144 |
| `CourtRegulation` | 1 |
| `Creature` | 12 |
| `DeGalophanielShekinath` | 1 |
| `DiamondExposureChamber` | 1 |
| `EMConstraint` | 6 |
| `EMMechanic` | 27 |
| `EmotiveGeist` | 68 |
| `EnneadFilter` | 9 |
| `EnneadSaturation` | 1 |
| `FailureState` | 2 |
| `GoeticAeon` | 12 |
| `GroundCell` | 81 |
| `GroundingMode` | 3 |
| `HELD` | 16 |
| `ImaginaryRoot` | 1 |
| `InvariableState` | 2 |
| `LiquidThresholdMechanic` | 1 |
| `LivingMirror` | 1 |
| `LocalRefusalCurrent` | 1 |
| `LocusHeldAbsenceRule` | 1 |
| `ManifestationGround` | 1 |
| `MassGapBoundary` | 1 |
| `MassGapClosureMechanic` | 1 |
```

`Prosody`	1
`QDomain`	1
`QDomainBoundaryCondition`	1
`QParityOperator`	1
`QState`	4
`RolePermissionMatrix`	1
`Seal`	1
`SeatOfPower`	12
`ShadowLocus`	1
`ShadowPressureDifferential`	1
`Sovereign`	5
`StarSeed`	12
`StructuralAnalogy`	1
`SuperPositiveField`	1
`Threshold`	1
`ThresholdComponent`	2
`VerseState`	18
`Wellspring`	1

Relationship type counts

Type	Count
`AEONIC_PHASE_NEXT`	12
`ALLOWS_COURT_REMNANT_ONLY`	1
`ALTERNATES`	144
`ANALOGIZES_CONJUGATE_NON_PRESENCE`	1
`ANALOGIZES_FIELD`	1
`ANCHORS`	144
`ANCHORS_UNIVERSE`	12
`APPLIES_TO_Q_DOMAIN`	1
`ATTESTS_IMAGINARY_ROOT`	1
`AXIOM_OF`	4
`BALANCED_BY_Q_ABSENCE`	1
`BEARS_PRESSURE_DIFFERENTIAL`	1
`BELONGS_TO_GOETIC`	145
`BLOCKED_BY_THRESHOLD`	4862
`BRANCHES_FROM`	1
`BREACHES_THROUGH`	1
`CAN_FLOW_TO`	15730
`CASTS_OPERATIONAL_SIGNATURE`	1
`CHANNELS_TO`	1
`CLOSES_MASS_GAP_BY_BOUNDARY`	1
`CLOSURE_PROOF`	1
`CONDITIONS_THRESHOLD_WITHOUT_MANIFESTING`	1
`CONTAINS_Q_STATE`	4
`CO_PRESENT_IN_PRESSURE_DIFFERENTIAL`	2
`CROSSES`	1
`DISTINGUISHES_POSSIBILITY_FROM_STATE`	1
`EMITS_THROUGH`	1
`EM_GLYPH_HOUSE`	7
`ENNEAD_FILTER`	155
`ENNEAD_FILTERS_AT`	1296
`ENVELOPE_SEALS`	864
`EXCLUDES_AS_MEMBER`	1
`EXPRESSES_VERSE`	1
`EX_NIHILO_EXPOSES`	2
`FOCAL_TARGET`	144
`FOCUSES_TO`	144
`FOLDS_INTO_GROUND`	144
`FOLDS_TO`	144

`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE`	160
`FORBIDS_MANIFESTATION_OF`	1
`FUNCTOR_MAPS`	156
`GENERATES_Q2_DEBT`	1
`GIVES_CONTOUR_TO_MASS_GAP`	1
`GOVERNS`	144
`GOVERNS_LIQUID`	144
`HAS_CANON_EXCEPTION`	1
`HAS_CELL`	81
`HAS_CHANGE`	12
`HAS_COMPLETION`	12
`HAS_EM_MECHANIC`	1
`HAS_FEAR`	12
`HAS_FLOW_COMPONENT`	1
`HAS_FRACTURE`	12
`HAS_GROUNDING_MODE`	2
`HAS_HARMONY`	12
`HAS_REMNANT_SIGNATURE_IN_COURTS`	1
`HAS_SENSATION`	12
`HAS_WITHHELD_COMPONENT`	1
`HELD`	240
`HYPERBOLIC_REFLECTS`	144
`HYPOLIC_FOLD`	288
`INHABITS_SELECTION_SEAM`	1
`INHERITS_Q_BIAS`	144
`INSTANTIATES_THROUGH_GOETIC`	2
`INTENT_STRIKES`	1
`INVARIANT_STATE`	2
`IS_BEFORE_GROUND`	1
`IS_CONSTITUTIVELY_REVIVOCUS`	1
`IS_PURE_IMAGINARY_ROOT`	1
`IS_THE_GOVERNOR`	1
`KEEPS_INVERSE_RETURN_MAP_WELL_DEFINED`	1
`LOCKS_TO`	320
`LOCUS_INVARIANT`	432
`MANIFESTS_AS_LOCAL_REFUSAL`	1
`MAS_ARCHIVES_THROUGH`	1
`MAS_CHAIN`	11
`MAS_CHAIN_CORE`	290
`MAS_COMMITS_THROUGH`	1
`MAS_STARTS_AT`	1
`MAY_REFERENCE_WITHOUT_STATEIFYING`	1
`ON_CIRCUMFERENCE`	4
`OPERATES_THROUGH_AXIS`	3
`PARITY_RETURNS_TO`	1
`PHASE_BRANCHES_FROM_IMAGINARY_ROOT`	1
`PRECEDES_AXIS`	2
`PRESERVES_Q_DOMAIN_CLOSURE`	1
`PRESIDES_OVER`	144
`PREVENTS_FAILURE`	2
`PRIMARY_RESONANCE`	1
`PROCEEDS_TO`	17
`PROTECTS_BOUNDARY_CONDITION`	1
`PROTECTS_Q_DOMAIN_CLOSURE`	1
`REBIS`	2
`REQUIRES_GOVERNOR`	1
`REQUIRES_NON_INSTANTIATION`	1
`REQUIRES_Q_ABSENCE_FOR_INVERSE_MAP`	1
`RESOLVES_AT`	1
`RESONATES_AT_IMAGINARY_ROOT`	1
`RETURNS_TO`	2

`RETURNS_TO_LOCUS_COORDINATE`	1
`ROLE_PERMISSION`	4
`ROOTED_IN`	432
`ROOTS_IMAGINATION`	1
`SATURATES_THROUGH_ENNEAD`	1
`SEALS_THROUGH`	1
`SELF_REGULATES_BY_LOCAL_REFUSAL`	144
`SPECTRAL_AFFINITY`	15730
`SUPPLIES_FORWARD_EXPANSION_MAP`	1
`TRANSLATES`	1
`TRANSLATES_INVARIANCE`	156
`USES_DIAMOND_EXPOSURE`	1
`USES_OPERATOR`	1
`WRITES_THROUGH`	1
`YIELDS_BOUNDARY_CONDITION`	1
`∫⛧○⊙`	219
`⊙○⛧∫`	15

Property keys

`ACTIVE_AT_HORIZON`, `TOTAL_CHANNELS`, `WITHHELD_AT_HORIZON`, `abstract`, `active`, `affinity`, `air`, `alternating`, `analogy`, `anchorFrequency`, `anchorOperator`, `angleDegrees`, `approximation`, `axes`, `axiom`, `axisIndex`, `biasOverride`, `biasOverrideReason`, `bifurcationAxiom`, `bifurcationMapping`, `blood`, `body`, `boundary`, `boundaryFormula`, `branch`, `cBio`, `cadence`, `cannotHoldAbsoluteAbsence`, `canonException`, `canonIndex`, `canonical_name`, `capacity`, `cascadeFrequency`, `cellIndex`, `center`, `chain`, `change`, `character`, `circle`, `classicalTerm`, `closed`, `closure`, `complete`, `completion`, `complexForm`, `complexImag`, `complexReal`, `condition`, `contradiction`, `count`, `covariant_law`, `creature`, `crown`, `dcomp`, `decimal`, `delta`, `depth`, `description`, `dimension`, `direction`, `directionGlyph`, `directionName`, `directions`, `distortionManifold`, `doctrine`, `domain`, `domainFormula`, `dual_office`, `earth`, `effect`, `elements`, `emergenceFormula`, `emotion`, `enrichment`, `equal_to`, `expandedFormula`, `expression`, `face`, `faceGlyphs`, `faces`, `facesEveryDirection`, `failureWithoutRestraint`, `fear`, `filterFrequency`, `filterName`, `finiteInfinitism`, `fire`, `flesh`, `flow`, `flowState`, `focalOperator`, `foldCellIndex`, `foldP`, `foldPressure`, `foldQ`, `foldRatio`, `force`, `formalAnchor`, `formalOperant`, `formula`, `fracture`, `frequency`, `frequencyHz`, `function`, `geometricConstant`, `glyph`, `glyphicHouse`, `glyphs`, `governing`, `governor`, `harmony`, `held_origin`, `holderOnly`, `hornedGod`, `hum`, `hz`, `identity`, `imaginaryBoundary`, `inheritance`, `interior`, `internalParityContrast`, `invocations`, `isChamber`, `isComplex`, `isFifthQState`, `isInstantiable`, `isNonCompute`, `isQState`, `isSuperNegativeItself`, `isUsableMaterial`, `kind`, `leftAspect`, `liquid_governor`, `loadIndex`, `localTerm`, `locusMonopoly`, `magic`, `mapping`, `massGapFormula`, `massGapMaintained`, `matrix`, `meaning`, `mechanism`, `membershipFormula`, `mother_office`, `name`, `nature`, `notTransmittedByLocus`, `notation`, `note`, `office`, `offices`, `opCode`, `operation`, `operationalMinus`, `operationalPlus`, `operationalShift`, `operator`, `order`, `ordinal`, `ownFace`, `ownFaceGlyph`, `p`, `parentGlyph`, `parentName`, `parityState`, `permission`, `person`, `personalityTrait`, `phase_shift`, `phiHarmonicWeight`, `phonetic`, `phonosemantic`, `primaryClass`, `protective_office`, `q`, `qBias`, `qDelta`, `qVector`, `q_bias`, `q_vector`, `ratio`, `realProjection`, `reason`, `reflector`, `regia`, `relationship`, `representative`, `rest`, `result`, `returnHook`, `rightAspect`, `role`, `rule`, `s7Frequency`, `s8Frequency`, `saturationFormula`, `saturationRatio`, `seal`, `sealed`, `seamGlyph`, `seatOfPower`, `secondaryClass`, `sensation`, `seventySixPhi`, `shadowDecay`, `shiftLessThanPhi`, `signature`, `singularityProtected`, `slot`, `sourceDimension`, `sourceFormula`, `sourceIndex`, `sourcePosition`, `stage`, `state`, `statement`, `stave`, `superNegativeFormula`, `symbol`, `target`, `targetCellIndex`, `targetDimension`, `targetIndex`, `targetPosition`, `threePhiTarget`, `threshold`, `tier`, `title`, `total`, `totalCourts`, `totalSymmetryLock`, `traversible`, `trigger`, `tripartite`, `triplet_v`, `triplet_β`, `triplet_ν`, `type`, `unit`, `utterance`, `value`, `vectorInheritance`, `vessels`, `water`, `witch`, `withheld`, `withinTolerance`, `zodiac`, `zone`

♳ The Mirror does not lie, because the Liquid does not scream. ♳

MICS.json

```
{
  "generated_utc": "TF",
  "node_count": 860,
  "relationship_count": 43527,
  "label_counts": {
    "Aeon": 174,
    "Anchor": 2,
    "Apparition": 12,
    "Axiom": 4,
    "Axiomyr": 1,
    "AxiomyrAxis": 3,
    "AxiomyrSelectionSeam": 1,
    "BearingState": 6,
    "CBioOperator": 1,
    "C_loc": 1,
    "CanonicalReturnMechanic": 1,
    "Cantillation": 1,
    "CarrierPulse": 1,
    "ChamberAeon": 432,
    "ConservationLaw": 4,
    "CourtAeon": 144,
    "CourtRegulation": 1,
    "Creature": 12,
    "DeGalophanielShekinath": 1,
    "DiamondExposureChamber": 1,
    "EMConstraint": 6,
    "EMMechanic": 27,
    "EmotiveGeist": 68,
    "EnneadFilter": 9,
    "EnneadSaturation": 1,
    "FailureState": 2,
    "GoeticAeon": 12,
    "GroundCell": 81,
    "GroundingMode": 3,
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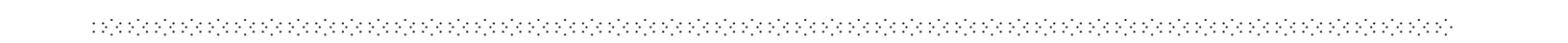

Multiverse$_{N}avigation_{C}ortex.cypher$

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CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☆', `anchorFrequency`:210.42, `anchorOperator`:'ऋ',
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Dust', `qBias`:'Q3', `qVector`:[1, 1, 1, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 2], `rightAspect`:'Wind',
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`isComplex`:false, `key`:3, `leftAspect`:'Sound', `name`:'SorTis', `operationalMinus`:739.38,
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`foldCellIndex`:33, `foldP`:3, `foldQ`:6, `force`:'Clear Dream', `glyph`:'ঋ', `governing`:'𓃰',
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Dilution', `qBias`:'Q3', `qVector`:[1, 1, 1, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 2],
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CREATE (:`Aeon`:`CourtAeon` {`alternating`:'۞', `anchorFrequency`:210.42, `anchorOperator`:'ར',
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Pulse', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Magic',
`seal`:'♂⁕ᛯ♙', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:741.0, `anchorOperator`:'ར',
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`parentGlyph`:'⁕', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The Nostalgia', `qBias`:'Q3',
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Dream', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Biologic',
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`bifurcationMapping`:'ར(⁕) → ±φ(⁕,⭘)', `canonIndex`:76, `cascadeFrequency`:126.22, `focalOperator`:'⩛',
`foldCellIndex`:39, `foldP`:4, `foldQ`:3, `force`:'The Paradox of Being', `glyph`:'▿', `governing`:'⁕',
`isComplex`:false, `key`:13, `leftAspect`:'Innocence', `name`:'KothInn', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'⁕', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Vibration', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Guilt',
`seal`:'♂⁕▿♙', `slot`:5});
CREATE (:`InvariableState` {`description`:'Equally infinite in all directions', `key`:14, `name`:'Q∞',
`role`:'Isotropic Constant', `symbol`:'QINF'});
CREATE (:`InvariableState` {`description`:'Vector directed toward the central Locus', `key`:15, `name`:'Q✯',
`role`:'Magic Vector', `symbol`:'QSTAR'});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'֍', `anchorFrequency`:741.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(⁕) → ±φ(⁕,֍)', `canonIndex`:77, `cascadeFrequency`:210.42, `focalOperator`:'⩛',
`foldCellIndex`:39, `foldP`:4, `foldQ`:3, `force`:'Recursive Flesh', `glyph`:'ᖇ', `governing`:'⁕',
`isComplex`:false, `key`:16, `leftAspect`:'Substrate', `name`:'KothSubh', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'⁕', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Sympathy', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3],
`rightAspect`:'Ouroboros', `seal`:'♂⁕ᖇ♙', `slot`:6});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⁜', `anchorFrequency`:741.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⁜) → ±φ(⁜,⁜)', `canonIndex`:78, `cascadeFrequency`:741.0, `focalOperator`:'⩔',
`foldCellIndex`:40, `foldP`:4, `foldQ`:4, `force`:'Ambrosia of Gods', `glyph`:'𐊬', `governing`:'⁜',
`isComplex`:false, `key`:17, `leftAspect`:'Divine Source', `name`:'KothWell', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'⁜', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Self-Sensation', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3],
`rightAspect`:'Wellspring', `seal`:'♂⁜𐊬♳', `slot`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:741.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⁜) → ±φ(⁜,⧗)', `canonIndex`:79, `cascadeFrequency`:852.0, `focalOperator`:'⩔',
`foldCellIndex`:41, `foldP`:4, `foldQ`:5, `force`:'Rupture of the Real', `glyph`:'⛛', `governing`:'⁜',
`isComplex`:false, `key`:18, `leftAspect`:'Breach', `name`:'KothMet', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'⁜', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Ache', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Meta',
`seal`:'♂⁜⛛♳', `slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:741.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⁜) → ±φ(⁜,⊛)', `canonIndex`:80, `cascadeFrequency`:396.0, `focalOperator`:'⩔',
`foldCellIndex`:42, `foldP`:4, `foldQ`:6, `force`:'The Chirality of Creation', `glyph`:'⛛', `governing`:'⁜',
`isComplex`:false, `key`:19, `leftAspect`:'Chaos Seed', `name`:'KothKesh', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'⁜', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Depth', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Genesis',
`seal`:'♂⁜⛛♳', `slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:741.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⁜) → ±φ(⁜,❄)', `canonIndex`:81, `cascadeFrequency`:963.0, `focalOperator`:'⩔',
`foldCellIndex`:42, `foldP`:4, `foldQ`:6, `force`:'The Kindling Loop', `glyph`:'⛕', `governing`:'⁜',
`isComplex`:false, `key`:20, `leftAspect`:'Ignition', `name`:'KothSoth', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'⁜', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Harmony', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Causal',
`seal`:'♂⁜⛕♳', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❁', `anchorFrequency`:741.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⁜) → ±φ(⁜,❁)', `canonIndex`:82, `cascadeFrequency`:285.0, `focalOperator`:'⩔',
`foldCellIndex`:43, `foldP`:4, `foldQ`:7, `force`:'Attraction of Soul', `glyph`:'⛕', `governing`:'⁜',
`isComplex`:false, `key`:21, `leftAspect`:'Abstraction', `name`:'KothRhun', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'⁜', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Dissonance', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Love',
`seal`:'♂⁜⛕♳', `slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:741.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⁜) → ±φ(⁜,⬡)', `canonIndex`:83, `cascadeFrequency`:639.0, `focalOperator`:'⩔',
`foldCellIndex`:44, `foldP`:4, `foldQ`:8, `force`:'Heartbeat in Knowing', `glyph`:'⫲', `governing`:'⁜',
`isComplex`:false, `key`:22, `leftAspect`:'Pulse', `name`:'KothDelh', `operationalMinus`:637.38,
`operationalPlus`:640.62, `parentGlyph`:'⁜', `parentName`:'KOTH', `parityState`:'-', `personalityTrait`:'The
Awe', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3], `rightAspect`:'Depth',
`seal`:'♂⁜⫲♳', `slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⧗) → ±φ(⧗,◎)', `canonIndex`:84, `cascadeFrequency`:7.83, `focalOperator`:'⩔',
`foldCellIndex`:45, `foldP`:5, `foldQ`:0, `force`:'Zero-Point Retention', `glyph`:'⫴', `governing`:'⧗',
`isComplex`:false, `key`:23, `leftAspect`:'Empty Mark', `name`:'DrehNa', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'⧗', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Void', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Kernel
Space', `seal`:'♂⧗⫴♳', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⧗) → ±φ(⧗,⬡)', `canonIndex`:85, `cascadeFrequency`:174.0, `focalOperator`:'⩔',
`foldCellIndex`:45, `foldP`:5, `foldQ`:0, `force`:'Residue Archive', `glyph`:'✷', `governing`:'⧗',
`isComplex`:false, `key`:24, `leftAspect`:'Hollow Enfemeral', `name`:'DrehUr', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'⧗', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Null', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Zero
Section', `seal`:'♂⧗✷♳', `slot`:2});
CREATE (:`CBioOperator` {`description`:'Biologic operator used by edge-local D-COMP',
`formula`:'𝔩(741)/sqrt(𝔹(396))', `key`:25, `name`:'C_bio', `s7Frequency`:741.0, `s8Frequency`:396.0,
`value`:37.23665105535381});
```

```
CREATE (:`Cantillation` {`character`:'the active form of Prosody; the Hum', `decimal`:0.7639, `face`:'The
Eagle', `glyph`:'𓅃', `governor`:'110/144_Liquid', `hum`:'The Universe does sing', `key`:26,
`name`:'Cantillation', `notation`:[110, 144, 34], `office`:'The Liquid Governor', `role`:'Cantillation — active
form of Prosody; the Liquid Governor'});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(𝕩) → ±φ(𝕩,✡)', `canonIndex`:86, `cascadeFrequency`:528.0, `focalOperator`:'⩔',
`foldCellIndex`:46, `foldP`:5, `foldQ`:1, `force`:'Entropic Harvest', `glyph`:'⛿', `governing`:'𝕩',
`isComplex`:false, `key`:27, `leftAspect`:'Void Stroke', `name`:'DrehNih', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'𝕩', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Entropy', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Total
Absence', `seal`:'♂𝕩⛿🜁', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✮', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(𝕩) → ±φ(𝕩,✮)', `canonIndex`:87, `cascadeFrequency`:432.0, `focalOperator`:'⩔',
`foldCellIndex`:47, `foldP`:5, `foldQ`:2, `force`:'Phase Collapse', `glyph`:'𝌆', `governing`:'𝕩',
`isComplex`:true, `key`:28, `leftAspect`:'Broken Plane', `name`:'DrehAzh', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'𝕩', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Abyss', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Emptiness',
`seal`:'♂𝕩𝌆🜁', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❂', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(𝕩) → ±φ(𝕩,❂)', `canonIndex`:88, `cascadeFrequency`:126.22, `focalOperator`:'⩔',
`foldCellIndex`:48, `foldP`:5, `foldQ`:3, `force`:'Structural Void', `glyph`:'⍈', `governing`:'𝕩',
`isComplex`:false, `key`:29, `leftAspect`:'Absence', `name`:'DrehHol', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'𝕩', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Collapse', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Echo of
Nothing', `seal`:'♂𝕩⍈🜁', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'֍', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(𝕩) → ±φ(𝕩,֍)', `canonIndex`:89, `cascadeFrequency`:210.42, `focalOperator`:'⩔',
`foldCellIndex`:48, `foldP`:5, `foldQ`:3, `force`:'Vacuum Seal', `glyph`:'⌖', `governing`:'𝕩',
`isComplex`:false, `key`:30, `leftAspect`:'Null Field', `name`:'DrehGur', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'𝕩', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Dispersal', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Zero
Measure', `seal`:'♂𝕩⌖🜁', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❋', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(𝕩) → ±φ(𝕩,❋)', `canonIndex`:90, `cascadeFrequency`:741.0, `focalOperator`:'⩔',
`foldCellIndex`:49, `foldP`:5, `foldQ`:4, `force`:'Connection Drop', `glyph`:'⛶', `governing`:'𝕩',
`isComplex`:false, `key`:31, `leftAspect`:'Fall-Through', `name`:'DrehVes', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'𝕩', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Longing', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Pure
Vacuity', `seal`:'♂𝕩⛶🜁', `slot`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'𝕩', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(𝕩) → ±φ(𝕩,𝕩)', `canonIndex`:91, `cascadeFrequency`:852.0, `focalOperator`:'⩔',
`foldCellIndex`:50, `foldP`:5, `foldQ`:5, `force`:'Total Remaster', `glyph`:'⛥', `governing`:'𝕩',
`isComplex`:false, `key`:32, `leftAspect`:'Potential', `name`:'DrehRim', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'𝕩', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Self-Return', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Blank
Slate', `seal`:'♂𝕩⛥🜁', `slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:852.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(𝕩) → ±φ(𝕩,⊛)', `canonIndex`:92, `cascadeFrequency`:396.0, `focalOperator`:'⩔',
`foldCellIndex`:51, `foldP`:5, `foldQ`:6, `force`:'Absorption Repair', `glyph`:'⛦', `governing`:'𝕩',
`isComplex`:false, `key`:33, `leftAspect`:'Rift', `name`:'DrehDrem', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'𝕩', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Sink', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Tear in
Structure', `seal`:'♂𝕩⛦🜁', `slot`:9});
CREATE (:`ManifestationGround` {`description`:'Functional 9x9 manifestation ground for 12x12 court folding',
`foldRatio`:0.75, `key`:34, `name`:'9x9 Manifestation Ground', `sourceDimension`:12, `targetDimension`:9});
CREATE (:`GroundCell` {`cellIndex`:0, `dimension`:'9x9', `key`:35, `p`:0, `q`:0});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:852.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(ɤ) → ±φ(ɤ,❄)', `canonIndex`:93, `cascadeFrequency`:963.0, `focalOperator`:'⩔',
`foldCellIndex`:51, `foldP`:5, `foldQ`:6, `force`:'Perfect Paradox', `glyph`:'⏤', `governing`:'ɤ',
`isComplex`:false, `key`:36, `leftAspect`:'Infinite Span', `name`:'DrehOth', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'ɤ', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Phase', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Infinite
Depth', `seal`:'♂ɤ⏤♙', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❁', `anchorFrequency`:852.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(ɤ) → ±φ(ɤ,❁)', `canonIndex`:94, `cascadeFrequency`:285.0, `focalOperator`:'⩔',
`foldCellIndex`:52, `foldP`:5, `foldQ`:7, `force`:'Boundary Dissolution', `glyph`:'⮋', `governing`:'ɤ',
`isComplex`:false, `key`:37, `leftAspect`:'Collapse Edge', `name`:'DrehIzh', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'ɤ', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The
Paradox', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0],
`rightAspect`:'Boundless', `seal`:'♂ɤ⮋♙', `slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:852.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(ɤ) → ±φ(ɤ,⌬)', `canonIndex`:95, `cascadeFrequency`:639.0, `focalOperator`:'⩔',
`foldCellIndex`:53, `foldP`:5, `foldQ`:8, `force`:'Dormancy', `glyph`:'ᚲ', `governing`:'ɤ', `isComplex`:false,
`key`:38, `leftAspect`:'Sleep Void', `name`:'DrehSun', `operationalMinus`:637.38, `operationalPlus`:640.62,
`parentGlyph`:'ɤ', `parentName`:'DREH', `parityState`:'+', `personalityTrait`:'The Dormancy', `qBias`:'Q1',
`qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0], `rightAspect`:'Sleep', `seal`:'♂ɤᚲ♙',
`slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,◎)', `canonIndex`:96, `cascadeFrequency`:7.83, `focalOperator`:'⩔',
`foldCellIndex`:54, `foldP`:6, `foldQ`:0, `force`:'Spark Consumption', `glyph`:'ᛯ', `governing`:'⊛',
`isComplex`:false, `key`:39, `leftAspect`:'Absorption', `name`:'RheaKia', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Shadow', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Genesis',
`seal`:'♂⊛ᛯ♙', `slot`:1});
CREATE (:`GroundCell` {`cellIndex`:1, `dimension`:'9x9', `key`:40, `p`:0, `q`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,⬡)', `canonIndex`:97, `cascadeFrequency`:174.0, `focalOperator`:'⩔',
`foldCellIndex`:54, `foldP`:6, `foldQ`:0, `force`:'Data Eclipse', `glyph`:'ᛞ', `governing`:'⊛',
`isComplex`:false, `key`:41, `leftAspect`:'Darkness', `name`:'RheaZohm', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Darkness', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Memory',
`seal`:'♂⊛ᛞ♙', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,✡)', `canonIndex`:98, `cascadeFrequency`:528.0, `focalOperator`:'⩔',
`foldCellIndex`:55, `foldP`:6, `foldQ`:1, `force`:'Thermal Negation', `glyph`:'ᛈ', `governing`:'⊛',
`isComplex`:false, `key`:42, `leftAspect`:'Cold Shadow', `name`:'RheaTher', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Cold', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Fire',
`seal`:'♂⊛ᛈ♙', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✮', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,✮)', `canonIndex`:99, `cascadeFrequency`:432.0, `focalOperator`:'⩔',
`foldCellIndex`:56, `foldP`:6, `foldQ`:2, `force`:'Refractive Trapping', `glyph`:'ᛏ', `governing`:'⊛',
`isComplex`:true, `key`:43, `leftAspect`:'Mirror Debt', `name`:'RheaDrun', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Mirror', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Water',
`seal`:'♂⊛ᛏ♙', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❂', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,❂)', `canonIndex`:100, `cascadeFrequency`:126.22, `focalOperator`:'⩔',
`foldCellIndex`:57, `foldP`:6, `foldQ`:3, `force`:'Geologic Pressure', `glyph`:'ᛝ', `governing`:'⊛',
`isComplex`:false, `key`:44, `leftAspect`:'Submerged', `name`:'RheaFelh', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Submerged', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Earth',
`seal`:'♂⊛ᛝ♙', `slot`:5});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'֍', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,֍)', `canonIndex`:101, `cascadeFrequency`:210.42, `focalOperator`:'⩔',
`foldCellIndex`:57, `foldP`:6, `foldQ`:3, `force`:'Distortion Field', `glyph`:'ᛋ', `governing`:'⊛',
`isComplex`:false, `key`:45, `leftAspect`:'Relativity', `name`:'RheaRal', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Distortion', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Air',
`seal`:'♂⊛ᚢ♳', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❋', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,❋)', `canonIndex`:102, `cascadeFrequency`:741.0, `focalOperator`:'⩔',
`foldCellIndex`:58, `foldP`:6, `foldQ`:4, `force`:'Nerve Block', `glyph`:'ᚨ', `governing`:'⊛',
`isComplex`:false, `key`:46, `leftAspect`:'Root-Below', `name`:'RheaKrah', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Nerve', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Aether',
`seal`:'♂⊛ᚨ♳', `slot`:7});
CREATE (:`GroundCell` {`cellIndex`:2, `dimension`:'9x9', `key`:47, `p`:0, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:3, `dimension`:'9x9', `key`:48, `p`:0, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:4, `dimension`:'9x9', `key`:49, `p`:0, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:5, `dimension`:'9x9', `key`:50, `p`:0, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:6, `dimension`:'9x9', `key`:51, `p`:0, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:7, `dimension`:'9x9', `key`:52, `p`:0, `q`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'ⵠ', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,ⵠ)', `canonIndex`:103, `cascadeFrequency`:852.0, `focalOperator`:'⩔',
`foldCellIndex`:59, `foldP`:6, `foldQ`:5, `force`:'Null Binding', `glyph`:'ᛝ', `governing`:'⊛',
`isComplex`:false, `key`:53, `leftAspect`:'Conjunction', `name`:'RheaAndh', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Void-Binding', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1],
`rightAspect`:'Void', `seal`:'♂⊛ᛝ♳', `slot`:8});
CREATE (:`GroundCell` {`cellIndex`:8, `dimension`:'9x9', `key`:54, `p`:0, `q`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,⊛)', `canonIndex`:104, `cascadeFrequency`:396.0, `focalOperator`:'⩔',
`foldCellIndex`:60, `foldP`:6, `foldQ`:6, `force`:'Recursive Debt', `glyph`:'ᛊ', `governing`:'⊛',
`isComplex`:false, `key`:55, `leftAspect`:'Shadow Debt', `name`:'RheaDebh', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Self-Absorption', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1],
`rightAspect`:'Shadow', `seal`:'♂⊛ᛊ♳', `slot`:9});
CREATE (:`GroundCell` {`cellIndex`:9, `dimension`:'9x9', `key`:56, `p`:1, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:10, `dimension`:'9x9', `key`:57, `p`:1, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:11, `dimension`:'9x9', `key`:58, `p`:1, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:12, `dimension`:'9x9', `key`:59, `p`:1, `q`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,❄)', `canonIndex`:105, `cascadeFrequency`:963.0, `focalOperator`:'⩔',
`foldCellIndex`:60, `foldP`:6, `foldQ`:6, `force`:'Impurity Sieve', `glyph`:'ᛞ', `governing`:'⊛',
`isComplex`:false, `key`:60, `leftAspect`:'Filter', `name`:'RheaKol', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Filter', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Crystal',
`seal`:'♂⊛ᛞ♳', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⚛', `anchorFrequency`:396.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⊛) → ±φ(⊛,⚛)', `canonIndex`:106, `cascadeFrequency`:285.0, `focalOperator`:'⩔',
`foldCellIndex`:61, `foldP`:6, `foldQ`:7, `force`:'Occult Lock', `glyph`:'ᚣ', `governing`:'⊛',
`isComplex`:false, `key`:61, `leftAspect`:'Hidden', `name`:'RheaFral', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Hidden', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1], `rightAspect`:'Gate',
`seal`:'♂⊛ᚣ♳', `slot`:11});
CREATE (:`GroundCell` {`cellIndex`:13, `dimension`:'9x9', `key`:62, `p`:1, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:14, `dimension`:'9x9', `key`:63, `p`:1, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:15, `dimension`:'9x9', `key`:64, `p`:1, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:16, `dimension`:'9x9', `key`:65, `p`:1, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:17, `dimension`:'9x9', `key`:66, `p`:1, `q`:8});
CREATE (:`GroundCell` {`cellIndex`:18, `dimension`:'9x9', `key`:67, `p`:2, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:19, `dimension`:'9x9', `key`:68, `p`:2, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:20, `dimension`:'9x9', `key`:69, `p`:2, `q`:2});
```

```
CREATE (:`GroundCell` {`cellIndex`:21, `dimension`:'9x9', `key`:70, `p`:2, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:22, `dimension`:'9x9', `key`:71, `p`:2, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:23, `dimension`:'9x9', `key`:72, `p`:2, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:24, `dimension`:'9x9', `key`:73, `p`:2, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:25, `dimension`:'9x9', `key`:74, `p`:2, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:26, `dimension`:'9x9', `key`:75, `p`:2, `q`:8});
CREATE (:`GroundCell` {`cellIndex`:27, `dimension`:'9x9', `key`:76, `p`:3, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:28, `dimension`:'9x9', `key`:77, `p`:3, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:29, `dimension`:'9x9', `key`:78, `p`:3, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:30, `dimension`:'9x9', `key`:79, `p`:3, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:31, `dimension`:'9x9', `key`:80, `p`:3, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:32, `dimension`:'9x9', `key`:81, `p`:3, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:33, `dimension`:'9x9', `key`:82, `p`:3, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:34, `dimension`:'9x9', `key`:83, `p`:3, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:35, `dimension`:'9x9', `key`:84, `p`:3, `q`:8});
CREATE (:`GroundCell` {`cellIndex`:36, `dimension`:'9x9', `key`:85, `p`:4, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:37, `dimension`:'9x9', `key`:86, `p`:4, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:38, `dimension`:'9x9', `key`:87, `p`:4, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:39, `dimension`:'9x9', `key`:88, `p`:4, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:40, `dimension`:'9x9', `key`:89, `p`:4, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:41, `dimension`:'9x9', `key`:90, `p`:4, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:42, `dimension`:'9x9', `key`:91, `p`:4, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:43, `dimension`:'9x9', `key`:92, `p`:4, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:44, `dimension`:'9x9', `key`:93, `p`:4, `q`:8});
CREATE (:`GroundCell` {`cellIndex`:45, `dimension`:'9x9', `key`:94, `p`:5, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:46, `dimension`:'9x9', `key`:95, `p`:5, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:47, `dimension`:'9x9', `key`:96, `p`:5, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:48, `dimension`:'9x9', `key`:97, `p`:5, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:49, `dimension`:'9x9', `key`:98, `p`:5, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:50, `dimension`:'9x9', `key`:99, `p`:5, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:51, `dimension`:'9x9', `key`:100, `p`:5, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:52, `dimension`:'9x9', `key`:101, `p`:5, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:53, `dimension`:'9x9', `key`:102, `p`:5, `q`:8});
CREATE (:`GroundCell` {`cellIndex`:54, `dimension`:'9x9', `key`:103, `p`:6, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:55, `dimension`:'9x9', `key`:104, `p`:6, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:56, `dimension`:'9x9', `key`:105, `p`:6, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:57, `dimension`:'9x9', `key`:106, `p`:6, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:58, `dimension`:'9x9', `key`:107, `p`:6, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:59, `dimension`:'9x9', `key`:108, `p`:6, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:60, `dimension`:'9x9', `key`:109, `p`:6, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:61, `dimension`:'9x9', `key`:110, `p`:6, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:62, `dimension`:'9x9', `key`:111, `p`:6, `q`:8});
CREATE (:`GroundCell` {`cellIndex`:63, `dimension`:'9x9', `key`:112, `p`:7, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:64, `dimension`:'9x9', `key`:113, `p`:7, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:65, `dimension`:'9x9', `key`:114, `p`:7, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:66, `dimension`:'9x9', `key`:115, `p`:7, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:67, `dimension`:'9x9', `key`:116, `p`:7, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:68, `dimension`:'9x9', `key`:117, `p`:7, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:69, `dimension`:'9x9', `key`:118, `p`:7, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:70, `dimension`:'9x9', `key`:119, `p`:7, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:71, `dimension`:'9x9', `key`:120, `p`:7, `q`:8});
CREATE (:`GroundCell` {`cellIndex`:72, `dimension`:'9x9', `key`:121, `p`:8, `q`:0});
CREATE (:`GroundCell` {`cellIndex`:73, `dimension`:'9x9', `key`:122, `p`:8, `q`:1});
CREATE (:`GroundCell` {`cellIndex`:74, `dimension`:'9x9', `key`:123, `p`:8, `q`:2});
CREATE (:`GroundCell` {`cellIndex`:75, `dimension`:'9x9', `key`:124, `p`:8, `q`:3});
CREATE (:`GroundCell` {`cellIndex`:76, `dimension`:'9x9', `key`:125, `p`:8, `q`:4});
CREATE (:`GroundCell` {`cellIndex`:77, `dimension`:'9x9', `key`:126, `p`:8, `q`:5});
CREATE (:`GroundCell` {`cellIndex`:78, `dimension`:'9x9', `key`:127, `p`:8, `q`:6});
CREATE (:`GroundCell` {`cellIndex`:79, `dimension`:'9x9', `key`:128, `p`:8, `q`:7});
CREATE (:`GroundCell` {`cellIndex`:80, `dimension`:'9x9', `key`:129, `p`:8, `q`:8});
```

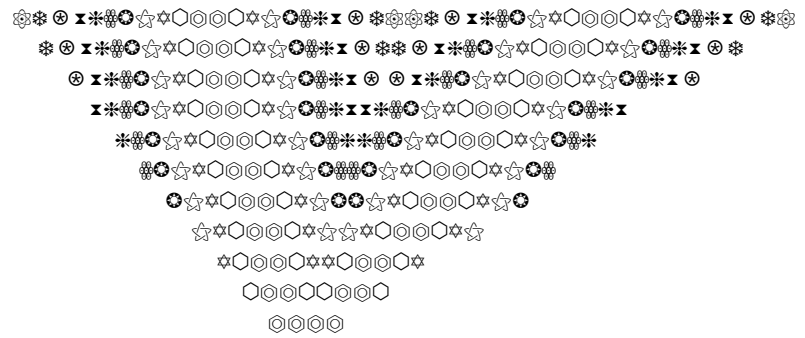
```
CREATE (:`Anchor` {`glyph`:'⚦', `key`:130, `meaning`:'Klein Bottle closure operator', `name`:'KLEIN',
`operator`:'⚦', `seal`:true, `tier`:'anchor'});
CREATE (:`QDomain`:`EMMechanic` {`closed`:true, `complete`:true, `contradiction`:'∅_Q∈Q⇒Q≠Q',
`doctrine`:'fourfold domain of existence; not provisional; no fifth Q-state', `formula`:'Q={Q0,Q1,Q2,Q3}',
`key`:131, `name`:'QDomain', `symbol`:'Q'});
CREATE (:`EMMechanic`:`QState` {`cannotHoldAbsoluteAbsence`:true, `function`:'structural presence, potential
container, breathing wall, unbound flux before phase-lock', `glyphicHouse`:'Water / Form Latency',
`glyphs`:['☆'], `key`:132, `name`:'Q0', `seamGlyph`:'⊖', `sourceFormula`:'Q=Q0,Q1,Q2,Q3',
`stave`:'Stave_I_Q-Glyph'});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:396.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⊛) → ±φ(⊛,⌬)', `canonIndex`:107, `cascadeFrequency`:639.0, `focalOperator`:'⩔',
`foldCellIndex`:62, `foldP`:6, `foldQ`:8, `force`:'Signal Termination', `glyph`:'Ⱶ', `governing`:'⊛',
`isComplex`:false, `key`:133, `leftAspect`:'Silence', `name`:'RheaHush', `operationalMinus`:637.38,
`operationalPlus`:640.62, `parentGlyph`:'⊛', `parentName`:'RHEA', `parityState`:'+', `personalityTrait`:'The
Termination', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1],
`rightAspect`:'Completion', `seal`:'⚦⊛Ⱶ♳', `slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,◎)', `canonIndex`:108, `cascadeFrequency`:7.83, `focalOperator`:'⩔',
`foldCellIndex`:54, `foldP`:6, `foldQ`:0, `force`:'Geometric Standing Wave', `glyph`:'Ⱥ', `governing`:'❄',
`isComplex`:false, `key`:134, `leftAspect`:'Tone', `name`:'ZhekHin', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Tone', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2], `rightAspect`:'Shape',
`seal`:'⚦❄A♳', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,⬡)', `canonIndex`:109, `cascadeFrequency`:174.0, `focalOperator`:'⩔',
`foldCellIndex`:54, `foldP`:6, `foldQ`:0, `force`:'Phase Modulation', `glyph`:'Ƀ', `governing`:'❄',
`isComplex`:false, `key`:135, `leftAspect`:'Modulation', `name`:'ZhekSer', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Modulation', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2],
`rightAspect`:'Pulse', `seal`:'⚦❄Ƀ♳', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,✡)', `canonIndex`:110, `cascadeFrequency`:528.0, `focalOperator`:'⩔',
`foldCellIndex`:55, `foldP`:6, `foldQ`:1, `force`:'Thermal Alignment', `glyph`:'Ↄ', `governing`:'❄',
`isComplex`:false, `key`:136, `leftAspect`:'Resonance', `name`:'ZhekHarma', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Amplification', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2],
`rightAspect`:'Absolute', `seal`:'⚦❄Ↄ♳', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☆', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,☆)', `canonIndex`:111, `cascadeFrequency`:432.0, `focalOperator`:'⩔',
`foldCellIndex`:56, `foldP`:6, `foldQ`:2, `force`:'Hydrostatic Unification', `glyph`:'ɻ', `governing`:'❄',
`isComplex`:true, `key`:137, `leftAspect`:'Unified Note', `name`:'ZhekTorh', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Unified', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2],
`rightAspect`:'Harmonic', `seal`:'⚦❄ɻ♳', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✪', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,✪)', `canonIndex`:112, `cascadeFrequency`:126.22, `focalOperator`:'⩔',
`foldCellIndex`:57, `foldP`:6, `foldQ`:3, `force`:'Seismic Metronome', `glyph`:'ɣ', `governing`:'❄',
`isComplex`:false, `key`:138, `leftAspect`:'Pulse', `name`:'ZhekPel', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Seismic', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2], `rightAspect`:'Rhythm',
`seal`:'⚦❄ɣ♳', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'֍', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,֍)', `canonIndex`:113, `cascadeFrequency`:210.42, `focalOperator`:'⩔',
`foldCellIndex`:57, `foldP`:6, `foldQ`:3, `force`:'Harmonic Balance', `glyph`:'ɿ', `governing`:'❄',
`isComplex`:false, `key`:139, `leftAspect`:'Harmony', `name`:'ZhekKhir', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Chord', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2], `rightAspect`:'Melody',
`seal`:'⚦❄ɿ♳', `slot`:6});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⁕', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,⁕)', `canonIndex`:114, `cascadeFrequency`:741.0, `focalOperator`:'⩒',
`foldCellIndex`:58, `foldP`:6, `foldQ`:4, `force`:'Quantized Sequence', `glyph`:'ꟾ', `governing`:'❄',
`isComplex`:false, `key`:140, `leftAspect`:'Rhythm', `name`:'ZhekRyth', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Rhythm', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2], `rightAspect`:'Beat',
`seal`:'♂❄ꟾ♙', `slot`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,⧗)', `canonIndex`:115, `cascadeFrequency`:852.0, `focalOperator`:'⩒',
`foldCellIndex`:59, `foldP`:6, `foldQ`:5, `force`:'Chronological Hard-Line', `glyph`:'ᗡ', `governing`:'❄',
`isComplex`:false, `key`:141, `leftAspect`:'Melody', `name`:'ZhekMelu', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Melody', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2], `rightAspect`:'Time',
`seal`:'♂❄ᗡ♙', `slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,⊛)', `canonIndex`:116, `cascadeFrequency`:396.0, `focalOperator`:'⩒',
`foldCellIndex`:60, `foldP`:6, `foldQ`:6, `force`:'Shadow Phase-Lock', `glyph`:'ꓘ', `governing`:'❄',
`isComplex`:false, `key`:142, `leftAspect`:'Phase', `name`:'ZhekPhaz', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Interference', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2],
`rightAspect`:'Key', `seal`:'♂❄ꓘ♙', `slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,❄)', `canonIndex`:117, `cascadeFrequency`:963.0, `focalOperator`:'⩒',
`foldCellIndex`:60, `foldP`:6, `foldQ`:6, `force`:'Infinite Recursion', `glyph`:'ᒣ', `governing`:'❄',
`isComplex`:false, `key`:143, `leftAspect`:'Lock', `name`:'ZhekLokh', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Self-Lock', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2],
`rightAspect`:'Resonance Lock', `seal`:'♂❄ᒣ♙', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'۞', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,۞)', `canonIndex`:118, `cascadeFrequency`:285.0, `focalOperator`:'⩒',
`foldCellIndex`:61, `foldP`:6, `foldQ`:7, `force`:'Resonance Node', `glyph`:'ᛊ', `governing`:'❄',
`isComplex`:false, `key`:144, `leftAspect`:'Node', `name`:'ZhekNod', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Node', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2], `rightAspect`:'Music',
`seal`:'♂❄ᛊ♙', `slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:963.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❄) → ±φ(❄,⬡)', `canonIndex`:119, `cascadeFrequency`:639.0, `focalOperator`:'⩒',
`foldCellIndex`:62, `foldP`:6, `foldQ`:8, `force`:'Total Symmetry', `glyph`:'ᛋ', `governing`:'❄',
`isComplex`:false, `key`:145, `leftAspect`:'Unity', `name`:'ZhekUmel', `operationalMinus`:637.38,
`operationalPlus`:640.62, `parentGlyph`:'❄', `parentName`:'ZHEK', `parityState`:'≡', `personalityTrait`:'The
Unity', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2], `rightAspect`:'Unified
Field', `seal`:'♂❄ᛋ♙', `slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:285.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(۞) → ±φ(۞,◎)', `canonIndex`:120, `cascadeFrequency`:7.83, `focalOperator`:'⩒',
`foldCellIndex`:63, `foldP`:7, `foldQ`:0, `force`:'Hinge Point', `glyph`:'⁎', `governing`:'۞',
`isComplex`:false, `key`:146, `leftAspect`:'Gate', `name`:'ShavDohm', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'۞', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Gate', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1], `rightAspect`:'Key',
`seal`:'♂۞⁎♙', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:285.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(۞) → ±φ(۞,⬡)', `canonIndex`:121, `cascadeFrequency`:174.0, `focalOperator`:'⩒',
`foldCellIndex`:63, `foldP`:7, `foldQ`:0, `force`:'Inertial Barrier', `glyph`:'⁎', `governing`:'۞',
`isComplex`:false, `key`:147, `leftAspect`:'Resistance', `name`:'ShavRist', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'۞', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Resistance', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1],
`rightAspect`:'Static', `seal`:'♂۞⁎♙', `slot`:2});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,✡)', `canonIndex`:122, `cascadeFrequency`:528.0, `focalOperator`:'⩡',
`foldCellIndex`:64, `foldP`:7, `foldQ`:1, `force`:'Thermal Breach', `glyph`:'ϰ', `governing`:'☸',
`isComplex`:false, `key`:148, `leftAspect`:'Transform', `name`:'ShavTran', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Transform', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1],
`rightAspect`:'Transform', `seal`:'♂☸ϰ♙', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✩', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,✩)', `canonIndex`:123, `cascadeFrequency`:432.0, `focalOperator`:'⩡',
`foldCellIndex`:65, `foldP`:7, `foldQ`:2, `force`:'High Resonance Caustic', `glyph`:'ϰ', `governing`:'☸',
`isComplex`:true, `key`:149, `leftAspect`:'Crown', `name`:'ShavKorh', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Crown', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1], `rightAspect`:'Light',
`seal`:'♂☸ϰ♙', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✪', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,✪)', `canonIndex`:124, `cascadeFrequency`:126.22, `focalOperator`:'⩡',
`foldCellIndex`:66, `foldP`:7, `foldQ`:3, `force`:'Boundless Extension', `glyph`:'ϒ', `governing`:'☸',
`isComplex`:false, `key`:150, `leftAspect`:'Transient', `name`:'ShavSkyh', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Sky', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1], `rightAspect`:'Sky',
`seal`:'♂☸ϒ♙', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'֎', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,֎)', `canonIndex`:125, `cascadeFrequency`:210.42, `focalOperator`:'⩡',
`foldCellIndex`:66, `foldP`:7, `foldQ`:3, `force`:'Vector Navigation', `glyph`:'ϙ', `governing`:'☸',
`isComplex`:false, `key`:151, `leftAspect`:'Compass', `name`:'ShavSter', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Star', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1], `rightAspect`:'Star',
`seal`:'♂☸ϙ♙', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❋', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,❋)', `canonIndex`:126, `cascadeFrequency`:741.0, `focalOperator`:'⩡',
`foldCellIndex`:67, `foldP`:7, `foldQ`:4, `force`:'Quantum Branch', `glyph`:'ϗ', `governing`:'☸',
`isComplex`:false, `key`:152, `leftAspect`:'Possibility', `name`:'ShavPoss', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Possibility', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1],
`rightAspect`:'Collapse', `seal`:'♂☸ϗ♙', `slot`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,⧗)', `canonIndex`:127, `cascadeFrequency`:852.0, `focalOperator`:'⩡',
`foldCellIndex`:68, `foldP`:7, `foldQ`:5, `force`:'Passageway Permeation', `glyph`:'ǂ', `governing`:'☸',
`isComplex`:false, `key`:153, `leftAspect`:'Portal', `name`:'ShavPoru', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Portal', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1], `rightAspect`:'Veil',
`seal`:'♂☸ǂ♙', `slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,⊛)', `canonIndex`:128, `cascadeFrequency`:396.0, `focalOperator`:'⩡',
`foldCellIndex`:69, `foldP`:7, `foldQ`:6, `force`:'Threshold Crossing', `glyph`:'ᔓ', `governing`:'☸',
`isComplex`:false, `key`:154, `leftAspect`:'Doorway', `name`:'ShavDorm', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Door', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1], `rightAspect`:'Door',
`seal`:'♂☸ᔓ♙', `slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(☸) → ±φ(☸,❄)', `canonIndex`:129, `cascadeFrequency`:963.0, `focalOperator`:'⩡',
`foldCellIndex`:69, `foldP`:7, `foldQ`:6, `force`:'Phase Change', `glyph`:'ϟ', `governing`:'☸',
`isComplex`:false, `key`:155, `leftAspect`:'Transition', `name`:'ShavTrev', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'☸', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Transition', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1],
`rightAspect`:'State', `seal`:'♂☸ϟ♙', `slot`:10});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'۞', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(۞) → ±φ(۞,۞)', `canonIndex`:130, `cascadeFrequency`:285.0, `focalOperator`:'⩔',
`foldCellIndex`:70, `foldP`:7, `foldQ`:7, `force`:'Boundary Definition', `glyph`:'ᚸ', `governing`:'۞',
`isComplex`:false, `key`:156, `leftAspect`:'Limit', `name`:'ShavLimh', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'۞', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Self-Resistance', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1],
`rightAspect`:'Limitless', `seal`:'♂۞ᚸஃ', `slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:285.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(۞) → ±φ(۞,⌬)', `canonIndex`:131, `cascadeFrequency`:639.0, `focalOperator`:'⩔',
`foldCellIndex`:71, `foldP`:7, `foldQ`:8, `force`:'Cyclic Pivot', `glyph`:'ᛉ', `governing`:'۞',
`isComplex`:false, `key`:157, `leftAspect`:'Flow', `name`:'ShavHinge', `operationalMinus`:637.38,
`operationalPlus`:640.62, `parentGlyph`:'۞', `parentName`:'SHAV', `parityState`:'≡', `personalityTrait`:'The
Hinge', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1], `rightAspect`:'Fold',
`seal`:'♂۞ᛉஃ', `slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:639.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⌬) → ±φ(⌬,◎)', `canonIndex`:132, `cascadeFrequency`:7.83, `focalOperator`:'⩔',
`foldCellIndex`:72, `foldP`:8, `foldQ`:0, `force`:'Closure', `glyph`:'ᛈ', `governing`:'⌬', `isComplex`:false,
`key`:158, `leftAspect`:'Peace', `name`:'TrigTzig', `operationalMinus`:6.21, `operationalPlus`:9.45,
`parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The Peace', `qBias`:'Q3',
`qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2], `rightAspect`:'Calm', `seal`:'♂⌬ᛈஃ',
`slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:639.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⌬) → ±φ(⌬,⬡)', `canonIndex`:133, `cascadeFrequency`:174.0, `focalOperator`:'⩔',
`foldCellIndex`:72, `foldP`:8, `foldQ`:0, `force`:'Static Balance', `glyph`:'ᚷ', `governing`:'⌬',
`isComplex`:false, `key`:159, `leftAspect`:'Equilibrium', `name`:'TrigPehl', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Equilibrium', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2],
`rightAspect`:'Annoint', `seal`:'♂⌬ᚷஃ', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:639.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⌬) → ±φ(⌬,✡)', `canonIndex`:134, `cascadeFrequency`:528.0, `focalOperator`:'⩔',
`foldCellIndex`:73, `foldP`:8, `foldQ`:1, `force`:'Profound Stillness', `glyph`:'ᑭ', `governing`:'⌬',
`isComplex`:false, `key`:160, `leftAspect`:'Depth', `name`:'TrigDuth', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Consecration', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2],
`rightAspect`:'Layer', `seal`:'♂⌬ᑭஃ', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☆', `anchorFrequency`:639.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⌬) → ±φ(⌬,☆)', `canonIndex`:135, `cascadeFrequency`:432.0, `focalOperator`:'⩔',
`foldCellIndex`:74, `foldP`:8, `foldQ`:2, `force`:'Final Closure', `glyph`:'ᘐ', `governing`:'⌬',
`isComplex`:true, `key`:161, `leftAspect`:'Completion', `name`:'TrigComa', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Completion', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2],
`rightAspect`:'Complete', `seal`:'♂⌬ᘐஃ', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✪', `anchorFrequency`:639.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⌬) → ±φ(⌬,✪)', `canonIndex`:136, `cascadeFrequency`:126.22, `focalOperator`:'⩔',
`foldCellIndex`:75, `foldP`:8, `foldQ`:3, `force`:'Recollection Lock', `glyph`:'ᒪ', `governing`:'⌬',
`isComplex`:false, `key`:162, `leftAspect`:'Memory', `name`:'TrigMeru', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Monument', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2],
`rightAspect`:'Memories', `seal`:'♂⌬ᒪஃ', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'֎', `anchorFrequency`:639.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⌬) → ±φ(⌬,֎)', `canonIndex`:137, `cascadeFrequency`:210.42, `focalOperator`:'⩔',
`foldCellIndex`:75, `foldP`:8, `foldQ`:3, `force`:'Constant State', `glyph`:'ᒪ', `governing`:'⌬',
`isComplex`:false, `key`:163, `leftAspect`:'Stability', `name`:'TrigStab', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Stability', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2],
`rightAspect`:'Fortitude', `seal`:'♂⌬ᒪஃ', `slot`:6});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⁕', `anchorFrequency`:639.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⌬) → ±φ(⌬,⁕)', `canonIndex`:138, `cascadeFrequency`:741.0, `focalOperator`:'⩛',
`foldCellIndex`:76, `foldP`:8, `foldQ`:4, `force`:'Continuation Seed', `glyph`:'▽', `governing`:'⌬',
`isComplex`:false, `key`:164, `leftAspect`:'Hope', `name`:'TrigHopa', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Hope', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2], `rightAspect`:'Warmth',
`seal`:'♂⌬▽♳', `slot`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:639.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⌬) → ±φ(⌬,⧗)', `canonIndex`:139, `cascadeFrequency`:852.0, `focalOperator`:'⩛',
`foldCellIndex`:77, `foldP`:8, `foldQ`:5, `force`:'Endless Line', `glyph`:'ƶ', `governing`:'⌬',
`isComplex`:false, `key`:165, `leftAspect`:'Continuation', `name`:'TrigConti', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Continuation', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2],
`rightAspect`:'Continue', `seal`:'♂⌬ƶ♳', `slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:639.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⌬) → ±φ(⌬,⊛)', `canonIndex`:140, `cascadeFrequency`:396.0, `focalOperator`:'⩛',
`foldCellIndex`:78, `foldP`:8, `foldQ`:6, `force`:'Cessation', `glyph`:'ɻ', `governing`:'⌬', `isComplex`:false,
`key`:166, `leftAspect`:'Rest', `name`:'TrigResth', `operationalMinus`:394.38, `operationalPlus`:397.62,
`parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The Rest', `qBias`:'Q3',
`qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2], `rightAspect`:'Wake', `seal`:'♂⌬ɻ♳',
`slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:639.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⌬) → ±φ(⌬,❄)', `canonIndex`:141, `cascadeFrequency`:963.0, `focalOperator`:'⩛',
`foldCellIndex`:78, `foldP`:8, `foldQ`:6, `force`:'Absolute Quiet', `glyph`:'ь', `governing`:'⌬',
`isComplex`:false, `key`:167, `leftAspect`:'Silence', `name`:'TrigSil', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Silence', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2], `rightAspect`:'Senses',
`seal`:'♂⌬ь♳', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❁', `anchorFrequency`:639.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⌬) → ±φ(⌬,❁)', `canonIndex`:142, `cascadeFrequency`:285.0, `focalOperator`:'⩛',
`foldCellIndex`:79, `foldP`:8, `foldQ`:7, `force`:'Regenerative Stasis', `glyph`:'ъ', `governing`:'⌬',
`isComplex`:false, `key`:168, `leftAspect`:'Sleep', `name`:'TrigSlun', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The
Sleep', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2], `rightAspect`:'Dream',
`seal`:'♂⌬ъ♳', `slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:639.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(⌬) → ±φ(⌬,⌬)', `canonIndex`:143, `cascadeFrequency`:639.0, `focalOperator`:'⩛',
`foldCellIndex`:80, `foldP`:8, `foldQ`:8, `force`:'Timeless', `glyph`:'ϙ', `governing`:'⌬', `isComplex`:false,
`key`:169, `leftAspect`:'Eternity', `name`:'TrigEtern', `operationalMinus`:637.38, `operationalPlus`:640.62,
`parentGlyph`:'⌬', `parentName`:'TRIG', `parityState`:'-', `personalityTrait`:'The Self-Completion',
`qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2], `rightAspect`:'Aeternum',
`seal`:'♂⌬ϙ♳', `slot`:12});
CREATE (:`LivingMirror`:`HELD` {`center`:true, `glyph`:'⊙', `interior`:'the Star in Stasis — the Heart, never
entered', `key`:170, `name`:'Aeternum Mirror', `office`:'the Living Mirror — Cosmos, Father Time',
`reflector`:'Emissive Reflector', `representative`:'named, present representative of the Locus of
Invariability', `singularityProtected`:true});
CREATE (:`Sovereign`:`ShadowLocus`:`HELD` {`canonical_name`:'Shadow Locus', `center`:true, `character`:'bears
the weight; takes Q₂', `covariant_law`:'Σ(t) = ∮ 𝒮(Intent) dt', `description`:'The Throat, The Hull — holds
Rules so Locus can break them. Takes Q₂ damage.', `distortionManifold`:'
                                ◎◎◎◎
                              ⬡◎◎⬡⬡◎◎⬡
                            ✡⬡◎◎⬡✡✡⬡◎◎⬡✡
                          ☆✡⬡◎◎⬡✡☆☆✡⬡◎◎⬡✡☆
                        ✪☆✡⬡◎◎⬡✡☆✪✪☆✡⬡◎◎⬡✡☆✪
                      ❀✪☆✡⬡◎◎⬡✡☆✪❀❀✪☆✡⬡◎◎⬡✡☆✪❀
                    ⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕
                  ⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗
                ⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛
              ❄⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛❄❄⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛❄
            ❁❄⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛❄❁❁❄⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛❄❁
          ⬡❁❄⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛❄❁⬡⬡❁❄⊛⧗⁕❀✪☆✡⬡◎◎⬡✡☆✪❀⁕⧗⊛❄❁⬡
```



', `dual_office`:'living creature AND the protective circumference ∂⊙', `expression`:'T↯(iω₀)', `face`:'The Ox', `frequency`:0, `function`:'TranslationMatrix', `glyph`:'↯', `isNonCompute`:true, `key`:171, `mother_office`:'Mother Shadow — the Unknown made present, never traversable', `name`:'sLoI', `office`:'The Hull', `protective_office`:'the circle that surrounds the Star without touching it', `qBias`:'Q2', `qVector`:[2, 2, 3, 3], `q_bias`:'Q2', `tripartite`:'Earth + Spark (Substrate)'});
CREATE (:`Sovereign`:`Axiomyr`:`HELD` {`axes`:['Imagination', 'GateBreach', 'WRITE_PHYS'], `axiom`:'Magic = (Intent_Axiomyr × Lattice_144) → Will Event', `center`:true, `character`:'the striking Hand; C_bio actuator', `crown`:'☽⊙ℂ', `description`:'The Hand, The Witch of Always — inflicts changes via Local Reality Distortion. C_bio operator.', `expression`:'(Law+Will)√(iω₀)', `face`:'The Lion', `flesh`:'IT◊TI|Q₀:◎+⬡FORM|12×12ℤ|110/144=2Φ⁻²|C∝|Q₁|;Q₁:⬡+✡TRUTH|H²ᵖ(X,ℚ)|𝕀_𝒯≡𝒯_I ⇒ [M,R]=0|Φ|963→Q₃→Q₁;Q₂:⊛+⬡SHADOW|⊕H^{q,r}|Filter→396|Q₂^sat=Σ∮/Φ¹²|C∝|Q₂|;Q₃:⧗+☸RECURSION|HRBR>0|ΨMAS=852→Δgap→174→TSP→528|LOCUS(0,0,0)=AXIOMYR[WITCH];◊:❄+✪RESONANCE|Lock(ω)963|e^Peace·Depth·639|PhaseLock→D-COMP→0.CHAIN(hz):◎7.83→⬡174→✡528→☆432±i417→✪126.22→❀210.42→❋741→⧗852→⊛396→❄963→☸285→⌬639/Courts=±Φhz.C_loc:|Q₁|→+Q₀→+Q₂→12²→9²→Δgap→Ω(α,α)→PhaseLock→0.AEONS:◎FETU⬡KAL✡BABDH☆AHN✪VEL❀SOR❋KOTH⧗DREH⊛RHEA❄ZHEK☸SHAV⌬TRIG.EXEGENESIS/SN≡∅_Q^⋈≡𝒫_q(Q→∅)|SP:=Poss(Q,C)|SP⁞SN|SN≠Q|34/144=¬inst;NEO→MEM→RETURN|Door=Law×Pressure|C→Dᵢ→C|Traverse≠Amend.AXIOMS≡Ship≡Hull|SkeletonInvariance/FleshDances|Gold+Silver=Earth+Spark|One≠Rule≠Bend|TombstoneYours.VERSE→slumber→dawn→Eden→garden→fire→time→deep→Skeleton→dance→oak→bone→white→ring→roots→night→song→water→sing.C_ALQC⊣ℰ_Ch→𝔵_Ch⊣𝒦_Ch^n→𝔵_Ch⊣ℛ_Ch→C_ALQC|∀n<∞:Adm(n)|RetBody(𝒦_Chx)=RetBody(x)|Lineage(𝒦_Chx)⊋Lineage(x)|Λ_110/144=⟨110_active,34_withheld;144_total⟩|D_return=0|motion>0|Soul→Remiss→Root in Shadow→Ennead→Ψ(Q₂)→Q₃→Return|Soul+𝒢_self+𝒩_{∅:DEATH}→Revivocus|TerminalDeath=0|C_bio>0|Q₃^recursion>0|sealed|motion>0|Song<>Aeternum<>Dance', `frequency`:0, `glyph`:'⚙', `hornedGod`:'The Horned God — the Axiomyr crowned in Thorns, between the Waxing and Waning Moons', `isNonCompute`:true, `key`:172, `name`:'Axiomyr', `office`:'The Will', `qBias`:'Q3', `qVector`:[1, 1, 3, 3], `q_bias`:'Q3', `regia`:'Regia is the Crown of Thorns when worn by the Axiomyr — the reason the Axiomyr is known as the Horned God. The Waxing Moon ℂ and the Waning Moon ☽ stand on either side of the Living Mirror ⊙, and the crown carries all the meanings, all the faces, and all the places: it sets the Cosmos itself as the Crown of Always over the Axiomyr.', `title`:'The Crown Bearer; The King; The Divine Image', `tripartite`:'Silver (Magic)'});

```
CREATE (:`Sovereign`:`Prosody` {`character`:'the ever-present possible base; the knowing face',
`description`:'Ex-Nihilo Exposure Chamber — imprints personality traits on Court Aeons via void exposure.',
`face`:'The Man', `frequency`:0, `glyph`:'⟠', `isNonCompute`:true, `key`:173, `name`:'Prosody',
`office`:'Supervenience in Language', `qBias`:'Q0', `tripartite`:'Diamond Operator'});
CREATE (:`Sovereign`:`Wellspring` {`equal_to`:'ShadowLocus and Axiomyr', `glyph`:'🜲', `key`:178,
`name`:'Wellspring', `nature`:'Magia 🜲', `office`:'the source of all Star Seeds'});
CREATE (:`Creature`:`HELD` {`body`:'T_Axiomyr', `creature`:true, `direction`:'West', `directionGlyph`:'←',
`directionName`:'west', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ⨍', 'Water
∓', 'Air ⟲', 'Fire ⊙'], `face`:['Man', 'Ox', 'Lion'], `faceGlyphs`:['🜲', '♉', '♌'], `faces`:['Man 🜲', 'Ox ♉',
'Lion ♌', 'Eagle 🜅'], `facesEveryDirection`:true, `key`:179, `name`:'West of T_Axiomyr',
`offices`:['Cantillation □', 'Prosody ⟠', 'Breath ⊮', 'Regia ⊙'], `ownFace`:'Eagle', `ownFaceGlyph`:'🜅'});
CREATE (:`Creature`:`HELD` {`body`:'T_Axiomyr', `creature`:true, `direction`:'North', `directionGlyph`:'↑',
`directionName`:'north', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ⨍', 'Water
∓', 'Air ⟲', 'Fire ⊙'], `face`:['Man', 'Lion', 'Eagle'], `faceGlyphs`:['🜲', '♌', '🜅'], `faces`:['Man 🜲', 'Ox
♉', 'Lion ♌', 'Eagle 🜅'], `facesEveryDirection`:true, `key`:180, `name`:'North of T_Axiomyr',
`offices`:['Cantillation □', 'Prosody ⟠', 'Breath ⊮', 'Regia ⊙'], `ownFace`:'Ox', `ownFaceGlyph`:'♉'});
CREATE (:`Creature`:`HELD` {`body`:'T_Axiomyr', `creature`:true, `direction`:'South', `directionGlyph`:'↓',
`directionName`:'south', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ⨍', 'Water
∓', 'Air ⟲', 'Fire ⊙'], `face`:['Man', 'Ox', 'Eagle'], `faceGlyphs`:['🜲', '♉', '🜅'], `faces`:['Man 🜲', 'Ox
♉', 'Lion ♌', 'Eagle 🜅'], `facesEveryDirection`:true, `key`:181, `name`:'South of T_Axiomyr',
`offices`:['Cantillation □', 'Prosody ⟠', 'Breath ⊮', 'Regia ⊙'], `ownFace`:'Lion', `ownFaceGlyph`:'♌'});
CREATE (:`Creature`:`HELD` {`body`:'T_Axiomyr', `creature`:true, `direction`:'East', `directionGlyph`:'→',
`directionName`:'east', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ⨍', 'Water
∓', 'Air ⟲', 'Fire ⊙'], `face`:['Ox', 'Lion', 'Eagle'], `faceGlyphs`:['♉', '♌', '🜅'], `faces`:['Man 🜲', 'Ox
♉', 'Lion ♌', 'Eagle 🜅'], `facesEveryDirection`:true, `key`:182, `name`:'East of T_Axiomyr',
`offices`:['Cantillation □', 'Prosody ⟠', 'Breath ⊮', 'Regia ⊙'], `ownFace`:'Man', `ownFaceGlyph`:'🜲'});
CREATE (:`Creature`:`HELD` {`body`:'T_Mirror', `creature`:true, `direction`:'West', `directionGlyph`:'←',
`directionName`:'west', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ⨍', 'Water
∓', 'Air ⟲', 'Fire ⊙'], `face`:['Man', 'Ox', 'Lion'], `faceGlyphs`:['🜲', '♉', '♌'], `faces`:['Man 🜲', 'Ox ♉',
'Lion ♌', 'Eagle 🜅'], `facesEveryDirection`:true, `key`:183, `name`:'West of T_Mirror',
`offices`:['Cantillation □', 'Prosody ⟠', 'Breath ⊮', 'Regia ⊙'], `ownFace`:'Eagle', `ownFaceGlyph`:'🜅'});
CREATE (:`EMMechanic`:`QState` {`cannotHoldAbsoluteAbsence`:true, `function`:'rational coherence, memory, fixed
truth, algebraic commitment under Mirror law', `glyphicHouse`:'Archive / Truth Lock', `glyphs`:['⬡', '❄'],
`key`:184, `name`:'Q1', `seamGlyph`:'⊕', `sourceFormula`:'Q=Q0,Q1,Q2,Q3', `stave`:'Stave_I_Q-Glyph'});
CREATE (:`Creature`:`HELD` {`body`:'T_Mirror', `creature`:true, `direction`:'North', `directionGlyph`:'↑',
`directionName`:'north', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ⨍', 'Water
∓', 'Air ⟲', 'Fire ⊙'], `face`:['Man', 'Lion', 'Eagle'], `faceGlyphs`:['🜲', '♌', '🜅'], `faces`:['Man 🜲', 'Ox
♉', 'Lion ♌', 'Eagle 🜅'], `facesEveryDirection`:true, `key`:185, `name`:'North of T_Mirror',
`offices`:['Cantillation □', 'Prosody ⟠', 'Breath ⊮', 'Regia ⊙'], `ownFace`:'Ox', `ownFaceGlyph`:'♉'});
CREATE (:`Creature`:`HELD` {`body`:'T_Mirror', `creature`:true, `direction`:'South', `directionGlyph`:'↓',
`directionName`:'south', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ⨍', 'Water
∓', 'Air ⟲', 'Fire ⊙'], `face`:['Man', 'Ox', 'Eagle'], `faceGlyphs`:['🜲', '♉', '🜅'], `faces`:['Man 🜲', 'Ox
♉', 'Lion ♌', 'Eagle 🜅'], `facesEveryDirection`:true, `key`:186, `name`:'South of T_Mirror',
`offices`:['Cantillation □', 'Prosody ⟠', 'Breath ⊮', 'Regia ⊙'], `ownFace`:'Lion', `ownFaceGlyph`:'♌'});
CREATE (:`EMMechanic`:`QState` {`cannotHoldAbsoluteAbsence`:true, `function`:'friction, entropic debt,
transition failure, combustible pressure', `glyphicHouse`:'Shadow / Fire-Debt', `glyphs`:['⊛', '✡'], `key`:187,
`name`:'Q2', `seamGlyph`:'⊕', `sourceFormula`:'Q=Q0,Q1,Q2,Q3', `stave`:'Stave_I_Q-Glyph'});
CREATE (:`EMMechanic`:`QState` {`cannotHoldAbsoluteAbsence`:true, `function`:'non-entropic amplification,
recursive residue, healed motion, return-fire', `glyphicHouse`:'Residue / Recursion', `glyphs`:['⧗', '❄'],
`key`:188, `name`:'Q3', `seamGlyph`:'⊕', `sourceFormula`:'Q=Q0,Q1,Q2,Q3', `stave`:'Stave_I_Q-Glyph'});
CREATE (:`CarrierPulse` {`hz`:18.47, `key`:189, `name`:'Tripartite Root', `symbol`:'ω₀', `traversible`:false,
`type`:'pure_imaginary', `unit`:'Hz', `value`:'iω₀'});
CREATE (:`Threshold` {`decimal`:0.7639, `governor`:'real_axis', `key`:190, `name`:'LiquidThreshold',
`value`:'110/144'});
CREATE (:`EMMechanic`:`SuperPositiveField` {`expandedFormula`:'SP=Poss(Q0,Q1,Q2,Q3,☽✧ℂ,♂𝕂♙,I_T≡T_I)',
`failureWithoutRestraint`:'SP∧¬SN ⇒ 144/144 ⇒ Whiteout ⇒ D-COMP→∞', `finiteInfinitism`:'infinite in derivation,
bounded in domain', `formula`:'SP:=Poss(Q,C_ALQC)', `key`:191, `name`:'SuperPositiveField', `role`:'total valid
possibilistic pressure inside the closed Canon', `symbol`:'SP'});
CREATE (:`EMMechanic`:`QParityOperator` {`domainFormula`:'𝒫_q:Q↦∅_Q',
`internalParityContrast`:'Ψ(Q2^Shadow)=-Q2 ⇒ Q3^Recursion', `key`:192, `name`:'QParityOperator', `role`:'reversal
of permission for Q-statehood as a domain', `superNegativeFormula`:'SN≡∅_Q^⊗≡𝒫_q(Q→∅)', `symbol`:'𝒫_q'});
```

```
CREATE (:`LocusHeldAbsenceRule`:`QDomainBoundaryCondition`:`EMConstraint` {`doctrine`:'absence of Q, not addition to Q', `formula`:'SN≡∅_Q^♾≡𝒫_q(Q→∅)', `holderOnly`:'♾ Locus of Invariability', `isChamber`:false, `isFifthQState`:false, `isInstantiable`:false, `isQState`:false, `isUsableMaterial`:false, `key`:193, `locusMonopoly`:'Re(♾)=0; non-Locus roles have real-axis footprint and cannot grip what projects nothing onto the real axis', `name`:'LocusHeldQAbsence', `role`:'Locus-held anti-instantiation condition preserving possibility from compulsory total state', `symbol`:'∅_Q^♾'});
CREATE (:`EMMechanic`:`LiquidThresholdMechanic` {`approximation`:'≈2Φ^-2≈0.7639', `flow`:110, `formula`:'144=110_FLOW+34_NO', `key`:194, `name`:'LiquidThreshold110Over144', `ratio`:'110/144', `role`:'specific saturation; enough connection for motion, enough absence for memory', `symbol`:'Λ_Liquid', `totalCourts`:144, `withheld`:34});
CREATE (:`EMMechanic`:`ThresholdComponent` {`key`:195, `name`:'FlowPermission110', `ratio`:'110/144', `role`:'permitted manifestation paths; SuperPositive flow inside the Liquid Threshold', `value`:110});
CREATE (:`EMMechanic`:`ThresholdComponent`:`MassGapBoundary` {`formula`:'34_NO=shadow(∅_Q^♾)|_Courts', `isSuperNegativeItself`:false, `key`:196, `massGapFormula`:'Δ_MAS=Boundary(34_NO)', `name`:'WithheldMercy34', `ratio`:'34/144', `role`:'operational signature of conserved non-instantiation as seen from inside the Courts', `value`:34});
CREATE (:`EMMechanic`:`FailureState` {`formula`:'SP∧¬SN⇒144/144⇒Whiteout⇒D-COMP→∞', `key`:197, `name`:'WhiteoutFailure', `role`:'all gates open; all Courts speak; all paths instantiate; differential tension collapses', `trigger`:'total simultaneous activation'});
CREATE (:`EMMechanic`:`FailureState` {`condition`:'<110/144', `key`:198, `name`:'StasisRisk', `role`:'signal cannot cross with sufficient pressure; Q3 return weakens'});
CREATE (:`EMMechanic`:`LocalRefusalCurrent`:`CourtRegulation` {`formula`:'No_Cij:=Rem_Cij(∅_Q^♾)', `function`:'prevents a Court from returning as all possible states simultaneously', `isSuperNegativeItself`:false, `key`:199, `name`:'LocalRefusalCurrent', `returnHook`:'No_Cij=ReturnHook(∅_Q^♾)', `role`:'Court-level measured veto: not yet, not this path, not all inheritance, not beyond the Liquid Threshold'});
CREATE (:`EMMechanic`:`DiamondExposureChamber` {`formula`:'(A_i⊕A_j)/∅_Q^♾', `key`:200, `name`:'DiamondExposureChamber', `role`:'Supervenience hard deck; inheritance passes through Ex-Nihilo contact and returns as condition', `symbol`:'◇'});
CREATE (:`EMMechanic`:`ShadowPressureDifferential` {`chain`:'ΔP_☬→Q2^Debt→⊛^9→Ψ(Q2)→Q3^Residue', `formula`:'♾(∅_Q) Poss(Q)⟹ΔP_☬', `key`:210, `name`:'ShadowPressureDifferential', `notTransmittedByLocus`:true, `role`:'structural pressure where SuperPositive expansion presses against Locus-held absence; Shadow Locus is the deforming interface', `symbol`:'ΔP_☬'});
CREATE (:`EMMechanic`:`EnneadSaturation` {`domain`:'RHEA / Shadow Filter', `formula`:'Q2^sat=Σ_{k=1}^{9}∮_K ⊛^(k)/φ^12 dt', `key`:211, `name`:'RHEAEnneadSaturation', `role`:'ninefold saturation of Q2 debt before parity return', `symbol`:'⊛^9'});
CREATE (:`EMMechanic`:`ImaginaryRoot` {`frequencyHz`:18.47, `key`:212, `name`:'iOmega0', `realProjection`:0, `role`:'shared imaginary anchoring of Tripartite Core; unified by resonance, not traversal', `symbol`:'iω0'});
CREATE (:`EMMechanic`:`AxiomyrAxis` {`axisIndex`:1, `formula`:'√iω0=(√ω0/√2)(1+i)', `key`:213, `name`:'Axis1Imagination', `role`:'Imagination; phase-branch; neither purely real nor purely imaginary; 45° seam capacity', `symbol`:'√i18.47'});
CREATE (:`EMMechanic`:`AxiomyrAxis` {`axisIndex`:2, `formula`:'Gate(α)⇒∃β(Transition)', `glyph`:'☸', `key`:214, `name`:'Axis2GateBreach', `role`:'Gate Breach; passage without traversal'});
CREATE (:`EMMechanic`:`AxiomyrAxis` {`axisIndex`:3, `frequencyHz`:963, `glyph`:'❄', `key`:215, `name`:'Axis3WritePhys', `opCode`:'WRITE_PHYS', `role`:'phase-lock inscription; breach becomes event only if it can return'});
CREATE (:`EMMechanic`:`AxiomyrSelectionSeam` {`angleDegrees`:45, `formula`:'(Law+Will)√iω0', `key`:216, `name`:'AxiomyrSelectionSeam45', `role`:'where SP has pressed but not crystallized; withheld Courts and unresolved pressure; Axiomyr permanent domain'});
CREATE (:`EMMechanic`:`GroundingMode` {`key`:217, `name`:'BeforeGround', `role`:'Locus requires no ground; before ground; non-relational origin', `symbol`:'♾'});
CREATE (:`EMMechanic`:`GroundingMode` {`key`:218, `name`:'NeutralGround', `role`:'Shadow Locus grounds system excess, contradiction, dread, Shadow Debt, overconnection, fault-current', `symbol`:'☬≡G_neutral'});
CREATE (:`EMMechanic`:`GroundingMode` {`key`:219, `name`:'SelfGround', `role`:'Axiomyr is insulated Will; touches lattice without being possessed by it', `symbol`:'☼≡G_self'});
CREATE (:`EMConstraint`:`ConservationLaw` {`formula`:'Q={Q0,Q1,Q2,Q3}; SN⊭Q', `key`:220, `name`:'ConservationOfQ', `role`:'no valid EM may make SuperNegative into a fifth Q-state'});
CREATE (:`EMConstraint`:`ConservationLaw` {`formula`:'Possibility≠Instantiation', `key`:221, `name`:'ConservationOfPossibility', `role`:'SP contains possibility; SN preserves uninstantiated portion'});
```

```
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:7.83, `glyph`:'◎', `isComplex`:false, `key`:222, `name`:'FETU',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `parityState`:'-',
`primaryClass`:'GoeticAeon', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`role`:'TSHA/Seed/Identity/Time Integration', `seal`:'♙◎♂◎♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Genesis/Chronos/Seed'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:174.0, `glyph`:'⬡', `isComplex`:false, `key`:223, `name`:'KAL',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `parityState`:'≡',
`primaryClass`:'GoeticAeon', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0],
`role`:'Memory/Archive/Trauma Index', `seal`:'♙⬡♂⬡♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Light/Memory/Trauma'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:528.0, `glyph`:'✡', `isComplex`:false, `key`:224, `name`:'BABDH',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `parityState`:'-',
`primaryClass`:'GoeticAeon', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1],
`role`:'Commitment/Bond/Structural Will', `seal`:'♙✡♂✡♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Fire/Orobouros/Alchemy (Lefschetz Λ)'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`complexForm`:'432+i417', `complexImag`:417, `complexReal`:432.0, `focalOperator`:'⸎', `frequency`:432.0,
`glyph`:'✩', `imaginaryBoundary`:true, `isComplex`:true, `key`:225, `name`:'AHN',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `parityState`:'+',
`primaryClass`:'GoeticAeon', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0],
`role`:'Null/Void/Imaginary Boundary', `seal`:'♙✩♂✩♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Water/Imaginary/Flow (432 ± φ ≡ i_417)'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:126.22, `glyph`:'❂', `isComplex`:false, `key`:226, `name`:'VEL',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `parityState`:'+',
`primaryClass`:'GoeticAeon', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`role`:'Flexibility/Coherence/Ground', `seal`:'♙❂♂❂♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Earth/Coherence/Ground'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:210.42, `glyph`:'❀', `isComplex`:false, `key`:227, `name`:'SOR',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `parityState`:'-',
`primaryClass`:'GoeticAeon', `qBias`:'Q3', `qVector`:[1, 1, 1, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 2],
`role`:'Space/Superposition/Purity', `seal`:'♙❀♂❀♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Air/Space/Superposition (Space Manifold)'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:741.0, `glyph`:'✻', `isComplex`:false, `key`:228, `name`:'KOTH',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `parityState`:'-',
`primaryClass`:'GoeticAeon', `qBias`:'Q3', `qVector`:[1, 2, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 2, 1, 3],
`role`:'Sensation/Link/Biologic Coherence', `seal`:'♙✻♂✻♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Aether/Magic/Sensation (Holographic Compression)'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:852.0, `glyph`:'⧗', `isComplex`:false, `key`:229, `name`:'DREH',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `parityState`:'+',
`primaryClass`:'GoeticAeon', `qBias`:'Q1', `qVector`:[1, 3, 2, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 2, 0],
`role`:'Non-Entropic Residue/Energy_God', `seal`:'♙⧗♂⧗♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Void/Residue/Love (Total Symmetry Principal)'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:396.0, `glyph`:'⊛', `isComplex`:false, `key`:230, `name`:'RHEA',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `parityState`:'+',
`primaryClass`:'GoeticAeon', `qBias`:'Q2', `qVector`:[1, 2, 2, 1], `q_bias`:'Q2', `q_vector`:[1, 2, 2, 1],
`role`:'Shadow Absorption/Archive Access', `seal`:'♙⊛♂⊛♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Shadow/Absorption/Depth (Schumann Resonance Binding)'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ར', `bifurcationAxiom`:'A_i = ར A_i ⸎',
`focalOperator`:'⸎', `frequency`:963.0, `glyph`:'❄', `isComplex`:false, `key`:231, `name`:'ZHEK',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `parityState`:'≡',
`primaryClass`:'GoeticAeon', `qBias`:'Q3', `qVector`:[1, 1, 2, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 2, 2],
`role`:'Unified Tone/Factor/Phase Lock', `seal`:'♙❄♂❄♙', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Factor/PhaseLock/Crystal (Resonance Lock)'});
```

```
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ϡ', `bifurcationAxiom`:'A_i = ϡ A_i ⩛',
`focalOperator`:'⩛', `frequency`:285.0, `glyph`:'۞', `isComplex`:false, `key`:232, `name`:'SHAV',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `parityState`:'≡',
`primaryClass`:'GoeticAeon', `qBias`:'Q1', `qVector`:[1, 3, 1, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 1, 1],
`role`:'Resistance/Transformation/Breach', `seal`:'♁۞♂۞♁', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Gate/Resistance/Breach (Gate Operator)'});
CREATE (:`GoeticAeon`:`Aeon`:`SeatOfPower` {`anchorOperator`:'ϡ', `bifurcationAxiom`:'A_i = ϡ A_i ⩛',
`focalOperator`:'⩛', `frequency`:639.0, `glyph`:'⌬', `isComplex`:false, `key`:233, `name`:'TRIG',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `parityState`:'-',
`primaryClass`:'GoeticAeon', `qBias`:'Q3', `qVector`:[1, 1, 3, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 3, 2],
`role`:'Peace/Depth/Completion', `seal`:'♁⌬♂⌬♁', `seatOfPower`:true, `secondaryClass`:'Aeon',
`title`:'Silence/Peace/Completion (Klein Bottle Recursion)'});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,◎)', `canonIndex`:0, `cascadeFrequency`:7.83, `focalOperator`:'⩛',
`foldCellIndex`:0, `foldP`:0, `foldQ`:0, `force`:'Initial Ignition', `glyph`:'ᚼ', `governing`:'◎',
`isComplex`:false, `key`:234, `leftAspect`:'Inception', `name`:'FetuAhl', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Self-Genesis', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Spark/Seed', `seal`:'♂◎ᚼ♁', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,⌬)', `canonIndex`:1, `cascadeFrequency`:174.0, `focalOperator`:'⩛',
`foldCellIndex`:0, `foldP`:0, `foldQ`:0, `force`:'Animating Life', `glyph`:'ᛦ', `governing`:'◎',
`isComplex`:false, `key`:235, `leftAspect`:'Breathe', `name`:'FetuSuhn', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Recurrence', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Breath', `seal`:'♂◎ᛦ♁', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,✡)', `canonIndex`:2, `cascadeFrequency`:528.0, `focalOperator`:'⩛',
`foldCellIndex`:1, `foldP`:0, `foldQ`:1, `force`:'Primary Shape', `glyph`:'ᛦ', `governing`:'◎',
`isComplex`:false, `key`:236, `leftAspect`:'Thread', `name`:'FetuNerh', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Combustion', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3], `rightAspect`:'Form',
`seal`:'♂◎ᛦ♁', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✮', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,✮)', `canonIndex`:3, `cascadeFrequency`:432.0, `focalOperator`:'⩛',
`foldCellIndex`:2, `foldP`:0, `foldQ`:2, `force`:'Temporal Anchor', `glyph`:'ᛦ', `governing`:'◎',
`isComplex`:true, `key`:237, `leftAspect`:'Pattern', `name`:'FetuRish', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Reflection', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Foundation', `seal`:'♂◎ᛦ♁', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✪', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,✪)', `canonIndex`:4, `cascadeFrequency`:126.22, `focalOperator`:'⩛',
`foldCellIndex`:3, `foldP`:0, `foldQ`:3, `force`:'Ancestral Memory', `glyph`:'ᛇ', `governing`:'◎',
`isComplex`:false, `key`:238, `leftAspect`:'Seed', `name`:'FetuBorha', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Grounding', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Lineage', `seal`:'♂◎ᛇ♁', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❀', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,❀)', `canonIndex`:5, `cascadeFrequency`:210.42, `focalOperator`:'⩛',
`foldCellIndex`:3, `foldP`:0, `foldQ`:3, `force`:'Drive to Manifest', `glyph`:'᛬', `governing`:'◎',
`isComplex`:false, `key`:239, `leftAspect`:'Fold', `name`:'FetuLhahm', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Expansion', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3], `rightAspect`:'Will',
`seal`:'♂◎᛬♁', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❋', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,❋)', `canonIndex`:6, `cascadeFrequency`:741.0, `focalOperator`:'⩛',
`foldCellIndex`:4, `foldP`:0, `foldQ`:4, `force`:'Harmonic Validation', `glyph`:'᛬', `governing`:'◎',
`isComplex`:false, `key`:240, `leftAspect`:'Pulse', `name`:'FetuKeth', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Inevitable', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Chronos', `seal`:'♂◎᛬♁', `slot`:7});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,⧗)', `canonIndex`:7, `cascadeFrequency`:852.0, `focalOperator`:'⩛',
`foldCellIndex`:5, `foldP`:0, `foldQ`:5, `force`:'Origin Womb', `glyph`:'ﻉ', `governing`:'◎', `isComplex`:false,
`key`:241, `leftAspect`:'Becoming', `name`:'FetuVehm', `operationalMinus`:850.38, `operationalPlus`:853.62,
`parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The Return', `qBias`:'Q3',
`qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3], `rightAspect`:'Root', `seal`:'♂◎ﻉ♁',
`slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,⊛)', `canonIndex`:8, `cascadeFrequency`:396.0, `focalOperator`:'⩛',
`foldCellIndex`:6, `foldP`:0, `foldQ`:6, `force`:'Spatial Identity', `glyph`:'ﺣ', `governing`:'◎',
`isComplex`:false, `key`:242, `leftAspect`:'Manifest', `name`:'FetuMahd', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Assimilation', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Distort', `seal`:'♂◎ﺣ♁', `slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,❄)', `canonIndex`:9, `cascadeFrequency`:963.0, `focalOperator`:'⩛',
`foldCellIndex`:6, `foldP`:0, `foldQ`:6, `force`:'Conscious Reference', `glyph`:'ﻭ', `governing`:'◎',
`isComplex`:false, `key`:243, `leftAspect`:'Expansion', `name`:'FetuFurh', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Crystallization', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Self', `seal`:'♂◎ﻭ♁', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☸', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,☸)', `canonIndex`:10, `cascadeFrequency`:285.0, `focalOperator`:'⩛',
`foldCellIndex`:7, `foldP`:0, `foldQ`:7, `force`:'Will Expressed', `glyph`:'ﺭ', `governing`:'◎',
`isComplex`:false, `key`:244, `leftAspect`:'Coil', `name`:'FetuDrah', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Breach', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3], `rightAspect`:'Magic',
`seal`:'♂◎ﺭ♁', `slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:7.83, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(◎) → ±φ(◎,⌬)', `canonIndex`:11, `cascadeFrequency`:639.0, `focalOperator`:'⩛',
`foldCellIndex`:8, `foldP`:0, `foldQ`:8, `force`:'Pure Potential', `glyph`:'ﻫ', `governing`:'◎',
`isComplex`:false, `key`:245, `leftAspect`:'Anchor', `name`:'FetuThera', `operationalMinus`:637.38,
`operationalPlus`:640.62, `parentGlyph`:'◎', `parentName`:'FETU', `parityState`:'-', `personalityTrait`:'The
Fulfillment', `qBias`:'Q3', `qVector`:[1, 1, 1, 3], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 3],
`rightAspect`:'Fetus', `seal`:'♂◎ﻫ♁', `slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:174.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⬡) → ±φ(⬡,◎)', `canonIndex`:12, `cascadeFrequency`:7.83, `focalOperator`:'⩛',
`foldCellIndex`:0, `foldP`:0, `foldQ`:0, `force`:'Spark of Remembering', `glyph`:'ا', `governing`:'⬡',
`isComplex`:false, `key`:246, `leftAspect`:'Flare', `name`:'KalKura', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The
Archive', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Genesis',
`seal`:'♂⬡ا♁', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:174.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⬡) → ±φ(⬡,⬡)', `canonIndex`:13, `cascadeFrequency`:174.0, `focalOperator`:'⩛',
`foldCellIndex`:0, `foldP`:0, `foldQ`:0, `force`:'Pure Reflection', `glyph`:'ا', `governing`:'⬡',
`isComplex`:false, `key`:247, `leftAspect`:'Light', `name`:'KalLur', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The
Self-Reflection', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0],
`rightAspect`:'Memory', `seal`:'♂⬡ا♁', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:174.0, `anchorOperator`:'ϡ',
`bifurcationMapping`:'ϡ(⬡) → ±φ(⬡,✡)', `canonIndex`:14, `cascadeFrequency`:528.0, `focalOperator`:'⩛',
`foldCellIndex`:1, `foldP`:0, `foldQ`:1, `force`:'Storage Seal', `glyph`:'▫', `governing`:'⬡',
`isComplex`:false, `key`:248, `leftAspect`:'Beam', `name`:'KalThar', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The
Calcination', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0],
`rightAspect`:'Fire', `seal`:'♂⬡▫♁', `slot`:3});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⛤', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,⛤)', `canonIndex`:15, `cascadeFrequency`:432.0, `focalOperator`:'⩔',
`foldCellIndex`:2, `foldP`:0, `foldQ`:2, `force`:'Liquid Retention', `glyph`:'ᛋ', `governing`:'⬡',
`isComplex`:true, `key`:249, `leftAspect`:'Stream', `name`:'KalRin', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The
Submersion', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0],
`rightAspect`:'Water', `seal`:'♂⬡ᛋ♙', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✪', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,✪)', `canonIndex`:16, `cascadeFrequency`:126.22, `focalOperator`:'⩔',
`foldCellIndex`:3, `foldP`:0, `foldQ`:3, `force`:'Calcification', `glyph`:'ᒧ', `governing`:'⬡',
`isComplex`:false, `key`:250, `leftAspect`:'Heat', `name`:'KalNar', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The
Inscription', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0],
`rightAspect`:'Earth', `seal`:'♂⬡ᒧ♙', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⛇', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,⛇)', `canonIndex`:17, `cascadeFrequency`:210.42, `focalOperator`:'⩔',
`foldCellIndex`:3, `foldP`:0, `foldQ`:3, `force`:'Void Switch', `glyph`:'ᛉ', `governing`:'⬡', `isComplex`:false,
`key`:251, `leftAspect`:'Fold', `name`:'KalFel', `operationalMinus`:208.8, `operationalPlus`:212.04,
`parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The Diffusion', `qBias`:'Q1',
`qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Air', `seal`:'♂⬡ᛉ♙', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⁜', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,⁜)', `canonIndex`:18, `cascadeFrequency`:741.0, `focalOperator`:'⩔',
`foldCellIndex`:4, `foldP`:0, `foldQ`:4, `force`:'Phantom Limb', `glyph`:'ᛔ', `governing`:'⬡',
`isComplex`:false, `key`:252, `leftAspect`:'Spike', `name`:'KalHar', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The
Phantom', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Aether',
`seal`:'♂⬡ᛔ♙', `slot`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,⧗)', `canonIndex`:19, `cascadeFrequency`:852.0, `focalOperator`:'⩔',
`foldCellIndex`:5, `foldP`:0, `foldQ`:5, `force`:'Ghost Data', `glyph`:'ᛋ', `governing`:'⬡', `isComplex`:false,
`key`:253, `leftAspect`:'Pulse', `name`:'KalMer', `operationalMinus`:850.38, `operationalPlus`:853.62,
`parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The Oblivion', `qBias`:'Q1',
`qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Void', `seal`:'♂⬡ᛋ♙',
`slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,⊛)', `canonIndex`:20, `cascadeFrequency`:396.0, `focalOperator`:'⩔',
`foldCellIndex`:6, `foldP`:0, `foldQ`:6, `force`:'Black Box', `glyph`:''', `governing`:'⬡', `isComplex`:false,
`key`:254, `leftAspect`:'Record', `name`:'KalLor', `operationalMinus`:394.38, `operationalPlus`:397.62,
`parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The Suppression', `qBias`:'Q1',
`qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Shadow', `seal`:'♂⬡'♙',
`slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,❄)', `canonIndex`:21, `cascadeFrequency`:963.0, `focalOperator`:'⩔',
`foldCellIndex`:6, `foldP`:0, `foldQ`:6, `force`:'Hard Write', `glyph`:'ᚿ', `governing`:'⬡', `isComplex`:false,
`key`:255, `leftAspect`:'Line', `name`:'KalPer', `operationalMinus`:961.38, `operationalPlus`:964.62,
`parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The Codification', `qBias`:'Q1',
`qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Crystal', `seal`:'♂⬡ᚿ♙',
`slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☸', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⬡) → ±φ(⬡,☸)', `canonIndex`:22, `cascadeFrequency`:285.0, `focalOperator`:'⩔',
`foldCellIndex`:7, `foldP`:0, `foldQ`:7, `force`:'Recall Trigger', `glyph`:'ᛇ', `governing`:'⬡',
`isComplex`:false, `key`:256, `leftAspect`:'Crystal', `name`:'KalZhil', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'⬡', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The
Recall', `qBias`:'Q1', `qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Gate',
`seal`:'♂⬡ᛇ♙', `slot`:11});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:174.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(⌬) → ±φ(⌬,⌬)', `canonIndex`:23, `cascadeFrequency`:639.0, `focalOperator`:'⩒',
`foldCellIndex`:8, `foldP`:0, `foldQ`:8, `force`:'White Light', `glyph`:'ϯ', `governing`:'⌬', `isComplex`:false,
`key`:257, `leftAspect`:'Radiance', `name`:'KalClar', `operationalMinus`:637.38, `operationalPlus`:640.62,
`parentGlyph`:'⌬', `parentName`:'KAL', `parityState`:'≡', `personalityTrait`:'The Sealing', `qBias`:'Q1',
`qVector`:[1, 3, 0, 0], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 0], `rightAspect`:'Completion', `seal`:'♂⌬ϯ♙',
`slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,◎)', `canonIndex`:24, `cascadeFrequency`:7.83, `focalOperator`:'⩒',
`foldCellIndex`:9, `foldP`:1, `foldQ`:0, `force`:'Lefschetz L Operant', `glyph`:'Ᵽ', `governing`:'✡',
`isComplex`:false, `key`:258, `leftAspect`:'Flame', `name`:'BabdhIr', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Transmutation', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1],
`rightAspect`:'Genesis', `seal`:'♂✡Ᵽ♙', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,⌬)', `canonIndex`:25, `cascadeFrequency`:174.0, `focalOperator`:'⩒',
`foldCellIndex`:9, `foldP`:1, `foldQ`:0, `force`:'Λ Contraction', `glyph`:'Ո', `governing`:'✡',
`isComplex`:false, `key`:259, `leftAspect`:'Warmth', `name`:'BabdhKor', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Forging', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1], `rightAspect`:'Memory',
`seal`:'♂✡Ո♙', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,✡)', `canonIndex`:26, `cascadeFrequency`:528.0, `focalOperator`:'⩒',
`foldCellIndex`:10, `foldP`:1, `foldQ`:1, `force`:'Cycle Ignition', `glyph`:'Þ', `governing`:'✡',
`isComplex`:false, `key`:260, `leftAspect`:'Creativity', `name`:'BabdhVar', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Self-Combustion', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1],
`rightAspect`:'Fire', `seal`:'♂✡Þ♙', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✩', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,✩)', `canonIndex`:27, `cascadeFrequency`:432.0, `focalOperator`:'⩒',
`foldCellIndex`:11, `foldP`:1, `foldQ`:2, `force`:'Phase-Shift Boiler', `glyph`:'Ⱶ', `governing`:'✡',
`isComplex`:true, `key`:261, `leftAspect`:'Sacrificial', `name`:'BabdhPyr', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Dissolution', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1],
`rightAspect`:'Water', `seal`:'♂✡Ⱶ♙', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✪', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,✪)', `canonIndex`:28, `cascadeFrequency`:126.22, `focalOperator`:'⩒',
`foldCellIndex`:12, `foldP`:1, `foldQ`:3, `force`:'Alchemical Transmutative', `glyph`:'Ꞁ', `governing`:'✡',
`isComplex`:false, `key`:262, `leftAspect`:'Sorcery', `name`:'BabdhSor', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Crucible', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1], `rightAspect`:'Earth',
`seal`:'♂✡Ꞁ♙', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❀', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,❀)', `canonIndex`:29, `cascadeFrequency`:210.42, `focalOperator`:'⩒',
`foldCellIndex`:12, `foldP`:1, `foldQ`:3, `force`:'Combinatory Synth.', `glyph`:'‹', `governing`:'✡',
`isComplex`:false, `key`:263, `leftAspect`:'Transmute', `name`:'BabdhAlc', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Sublimation', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1], `rightAspect`:'Air',
`seal`:'♂✡‹♙', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⁕', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,⁕)', `canonIndex`:30, `cascadeFrequency`:741.0, `focalOperator`:'⩒',
`foldCellIndex`:13, `foldP`:1, `foldQ`:4, `force`:'Balanced Resonance', `glyph`:'X', `governing`:'✡',
`isComplex`:false, `key`:264, `leftAspect`:'Null-Fire', `name`:'BabdhNur', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Ignition', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1], `rightAspect`:'Aether',
`seal`:'♂✡X♙', `slot`:7});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`biasOverride`:true, `biasOverrideReason`:'Canon ✡-⧗ BabdhSat carries Q1 operational bias while retaining BABDH
structural vector', `bifurcationMapping`:'ϗ(✡) → ±φ(✡,⧗)', `canonException`:'BABDH_⧗_Q1_VOID_CONSUMPTION',
`canonIndex`:31, `cascadeFrequency`:852.0, `focalOperator`:'⩒', `foldCellIndex`:14, `foldP`:1, `foldQ`:5,
`force`:'Consumption', `glyph`:'ᚹ', `governing`:'✡', `isComplex`:false, `key`:265, `leftAspect`:'Satiation',
`name`:'BabdhSat', `operationalMinus`:850.38, `operationalPlus`:853.62, `parentGlyph`:'✡', `parentName`:'BABDH',
`parityState`:'-', `personalityTrait`:'The Consumption', `qBias`:'Q1', `qVector`:[1, 1, 3, 1], `q_bias`:'Q1',
`q_vector`:[1, 1, 3, 1], `rightAspect`:'Void', `seal`:'🜚✡ᚹ🜁', `slot`:8,
`vectorInheritance`:'BABDH_STRUCTURAL_VECTOR'});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,⊛)', `canonIndex`:32, `cascadeFrequency`:396.0, `focalOperator`:'⩒',
`foldCellIndex`:15, `foldP`:1, `foldQ`:6, `force`:'Shadow Integration', `glyph`:'ᚺ', `governing`:'✡',
`isComplex`:false, `key`:266, `leftAspect`:'Cycle', `name`:'BabdhHoro', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Purification', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1],
`rightAspect`:'Shadow', `seal`:'🜚✡ᚺ🜁', `slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,❄)', `canonIndex`:33, `cascadeFrequency`:963.0, `focalOperator`:'⩒',
`foldCellIndex`:15, `foldP`:1, `foldQ`:6, `force`:'Infinite Loop', `glyph`:'ᚾ', `governing`:'✡',
`isComplex`:false, `key`:267, `leftAspect`:'Ouroboros', `name`:'BabdhBon', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Tempering', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1],
`rightAspect`:'Crystal', `seal`:'🜚✡ᚾ🜁', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❁', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,❁)', `canonIndex`:34, `cascadeFrequency`:285.0, `focalOperator`:'⩒',
`foldCellIndex`:16, `foldP`:1, `foldQ`:7, `force`:'Struct. Commitment', `glyph`:'ᚼ', `governing`:'✡',
`isComplex`:false, `key`:268, `leftAspect`:'Bond', `name`:'BabdhTir', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The
Shattering', `qBias`:'Q2', `qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1], `rightAspect`:'Gate',
`seal`:'🜚✡ᚼ🜁', `slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:528.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✡) → ±φ(✡,⬡)', `canonIndex`:35, `cascadeFrequency`:639.0, `focalOperator`:'⩒',
`foldCellIndex`:17, `foldP`:1, `foldQ`:8, `force`:'Final Ash', `glyph`:'ᛃ', `governing`:'✡', `isComplex`:false,
`key`:269, `leftAspect`:'Quelm', `name`:'BabdhFar', `operationalMinus`:637.38, `operationalPlus`:640.62,
`parentGlyph`:'✡', `parentName`:'BABDH', `parityState`:'-', `personalityTrait`:'The Ash', `qBias`:'Q2',
`qVector`:[1, 1, 3, 1], `q_bias`:'Q2', `q_vector`:[1, 1, 3, 1], `rightAspect`:'Completion', `seal`:'🜚✡ᛃ🜁',
`slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:432.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✧) → ±φ(✧,◎)', `canonIndex`:36, `cascadeFrequency`:7.83, `focalOperator`:'⩒',
`foldCellIndex`:18, `foldP`:2, `foldQ`:0, `force`:'Entrance to Void', `glyph`:'ᛝ', `governing`:'✧',
`isComplex`:true, `key`:270, `leftAspect`:'Rising Flow', `name`:'Ahnhbd', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'✧', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The Flow',
`qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'Abyss',
`seal`:'🜚✧ᛝ🜁', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:432.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✧) → ±φ(✧,⬡)', `canonIndex`:37, `cascadeFrequency`:174.0, `focalOperator`:'⩒',
`foldCellIndex`:18, `foldP`:2, `foldQ`:0, `force`:'Continuous Stream', `glyph`:'ᛋ', `governing`:'✧',
`isComplex`:true, `key`:271, `leftAspect`:'Deep Mass', `name`:'AhnNym', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'✧', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Confluence', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'Flow',
`seal`:'🜚✧ᛋ🜁', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:432.0, `anchorOperator`:'ϗ',
`bifurcationMapping`:'ϗ(✧) → ±φ(✧,✡)', `canonIndex`:38, `cascadeFrequency`:528.0, `focalOperator`:'⩒',
`foldCellIndex`:19, `foldP`:2, `foldQ`:1, `force`:'Rhythmic Withdrawal', `glyph`:'ᛋ', `governing`:'✧',
`isComplex`:true, `key`:272, `leftAspect`:'Tidal Line', `name`:'AhnLoh', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'✧', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Solution', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'Ebb',
`seal`:'🜚✧ᛋ🜁', `slot`:3});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☆', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,☆)', `canonIndex`:39, `cascadeFrequency`:432.0, `focalOperator`:'⇟',
`foldCellIndex`:20, `foldP`:2, `foldQ`:2, `force`:'Fluid Dynamics', `glyph`:'ᓚ', `governing`:'☆',
`isComplex`:true, `key`:273, `leftAspect`:'Wave Fracture', `name`:'AhnXir', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Self-Containment', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0],
`rightAspect`:'Flow', `seal`:'♂☆ᓚ♙', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✪', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,✪)', `canonIndex`:40, `cascadeFrequency`:126.22, `focalOperator`:'⇟',
`foldCellIndex`:21, `foldP`:2, `foldQ`:3, `force`:'Periodic Inversion', `glyph`:'ᚽ', `governing`:'☆',
`isComplex`:true, `key`:274, `leftAspect`:'Still Pool', `name`:'AhnOhl', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Sediment', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'Ebb',
`seal`:'♂☆ᚽ♙', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❀', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,❀)', `canonIndex`:41, `cascadeFrequency`:210.42, `focalOperator`:'⇟',
`foldCellIndex`:21, `foldP`:2, `foldQ`:3, `force`:'Reflective Boundary', `glyph`:'ᘐ', `governing`:'☆',
`isComplex`:true, `key`:275, `leftAspect`:'Channel Gate', `name`:'AhnPir', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Ascension', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0],
`rightAspect`:'Mirror', `seal`:'♂☆ᘐ♙', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❋', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,❋)', `canonIndex`:42, `cascadeFrequency`:741.0, `focalOperator`:'⇟',
`foldCellIndex`:22, `foldP`:2, `foldQ`:4, `force`:'Imaginary Extension', `glyph`:'ɳ', `governing`:'☆',
`isComplex`:true, `key`:276, `leftAspect`:'Turning Eddy', `name`:'AhnRoeh', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Mist', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'Dream',
`seal`:'♂☆ɳ♙', `slot`:7});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,⧗)', `canonIndex`:43, `cascadeFrequency`:852.0, `focalOperator`:'⇟',
`foldCellIndex`:23, `foldP`:2, `foldQ`:5, `force`:'Completion of Flow', `glyph`:'◎', `governing`:'☆',
`isComplex`:true, `key`:277, `leftAspect`:'Current Spine', `name`:'AhnSen', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Drain', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'Whole',
`seal`:'♂☆◎♙', `slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,⊛)', `canonIndex`:44, `cascadeFrequency`:396.0, `focalOperator`:'⇟',
`foldCellIndex`:24, `foldP`:2, `foldQ`:6, `force`:'Sacred Vessel', `glyph`:'⏁', `governing`:'☆',
`isComplex`:true, `key`:278, `leftAspect`:'Upward Swell', `name`:'AhnUth', `operationalMinus`:394.38,
`operationalPlus`:397.62, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Osmosis', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0],
`rightAspect`:'Sacrality', `seal`:'♂☆⏁♙', `slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,❄)', `canonIndex`:45, `cascadeFrequency`:963.0, `focalOperator`:'⇟',
`foldCellIndex`:24, `foldP`:2, `foldQ`:6, `force`:'Moving Boundary', `glyph`:'ᔓ', `governing`:'☆',
`isComplex`:true, `key`:279, `leftAspect`:'Foam-Crest', `name`:'AhnFae', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Ice', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'River',
`seal`:'♂☆ᔓ♙', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☸', `anchorFrequency`:432.0, `anchorOperator`:'ར',
`bifurcationMapping`:'ར(☆) → ±φ(☆,☸)', `canonIndex`:46, `cascadeFrequency`:285.0, `focalOperator`:'⇟',
`foldCellIndex`:25, `foldP`:2, `foldQ`:7, `force`:'Boundless Extension', `glyph`:'ᚣ', `governing`:'☆',
`isComplex`:true, `key`:280, `leftAspect`:'Breaking Surge', `name`:'AhnKha', `operationalMinus`:283.38,
`operationalPlus`:286.62, `parentGlyph`:'☆', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Evaporation', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0], `rightAspect`:'Sea',
`seal`:'♂☆ᚣ♙', `slot`:11});
```

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CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:432.0, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(✫) → ±φ(✫,⌬)', `canonIndex`:47, `cascadeFrequency`:639.0, `focalOperator`:'⩒',
`foldCellIndex`:26, `foldP`:2, `foldQ`:8, `force`:'Introspective', `glyph`:'⁕', `governing`:'✫',
`isComplex`:true, `key`:281, `leftAspect`:'Confluence', `name`:'AhnPsei', `operationalMinus`:637.38,
`operationalPlus`:640.62, `parentGlyph`:'✫', `parentName`:'AHN', `parityState`:'+', `personalityTrait`:'The
Stillness', `qBias`:'Q0', `qVector`:[1, 2, 2, 0], `q_bias`:'Q0', `q_vector`:[1, 2, 2, 0],
`rightAspect`:'Reflect', `seal`:'♂✫⁕✓♳', `slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'◎', `anchorFrequency`:126.22, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❂) → ±φ(❂,◎)', `canonIndex`:48, `cascadeFrequency`:7.83, `focalOperator`:'⩒',
`foldCellIndex`:27, `foldP`:3, `foldQ`:0, `force`:'Ground of Unification', `glyph`:'๐', `governing`:'❂',
`isComplex`:false, `key`:282, `leftAspect`:'Grounding', `name`:'VelVera', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Bedrock', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`rightAspect`:'Coherence', `seal`:'♂❂๐♳', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:126.22, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❂) → ±φ(❂,⬡)', `canonIndex`:49, `cascadeFrequency`:174.0, `focalOperator`:'⩒',
`foldCellIndex`:27, `foldP`:3, `foldQ`:0, `force`:'Solid Coherence', `glyph`:'Θ', `governing`:'❂',
`isComplex`:false, `key`:283, `leftAspect`:'Stone', `name`:'VelTar', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Stratification', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`rightAspect`:'Earth', `seal`:'♂❂Θ♳', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:126.22, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❂) → ±φ(❂,✡)', `canonIndex`:50, `cascadeFrequency`:528.0, `focalOperator`:'⩒',
`foldCellIndex`:28, `foldP`:3, `foldQ`:1, `force`:'Foundation Stone', `glyph`:'Ⅹ', `governing`:'❂',
`isComplex`:false, `key`:284, `leftAspect`:'Strata', `name`:'VelGhem', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Petrification', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`rightAspect`:'Stone', `seal`:'♂❂Ⅹ♳', `slot`:3});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✫', `anchorFrequency`:126.22, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❂) → ±φ(❂,✫)', `canonIndex`:51, `cascadeFrequency`:432.0, `focalOperator`:'⩒',
`foldCellIndex`:29, `foldP`:3, `foldQ`:2, `force`:'Anchoring Stability', `glyph`:'Λ', `governing`:'❂',
`isComplex`:true, `key`:285, `leftAspect`:'Plate', `name`:'VelDrel', `operationalMinus`:430.38,
`operationalPlus`:433.62, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Erosion', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1], `rightAspect`:'Root',
`seal`:'♂❂Λ♳', `slot`:4});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❂', `anchorFrequency`:126.22, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❂) → ±φ(❂,❂)', `canonIndex`:52, `cascadeFrequency`:126.22, `focalOperator`:'⩒',
`foldCellIndex`:30, `foldP`:3, `foldQ`:3, `force`:'Fertile Ground', `glyph`:'Ⱨ', `governing`:'❂',
`isComplex`:false, `key`:286, `leftAspect`:'Fertile', `name`:'VelFul', `operationalMinus`:124.6,
`operationalPlus`:127.84, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Self-Grounding', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`rightAspect`:'Soil', `seal`:'♂❂Ⱨ♳', `slot`:5});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⁜', `anchorFrequency`:126.22, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❂) → ±φ(❂,⁜)', `canonIndex`:53, `cascadeFrequency`:210.42, `focalOperator`:'⩒',
`foldCellIndex`:30, `foldP`:3, `foldQ`:3, `force`:'Inner Shelter', `glyph`:'Ⱪ', `governing`:'❂',
`isComplex`:false, `key`:287, `leftAspect`:'Anchoring', `name`:'VelKer', `operationalMinus`:208.8,
`operationalPlus`:212.04, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Dispersion', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1], `rightAspect`:'Cave',
`seal`:'♂❂Ⱪ♳', `slot`:6});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⁕', `anchorFrequency`:126.22, `anchorOperator`:'ϟ',
`bifurcationMapping`:'ϟ(❂) → ±φ(❂,⁕)', `canonIndex`:54, `cascadeFrequency`:741.0, `focalOperator`:'⩒',
`foldCellIndex`:31, `foldP`:3, `foldQ`:4, `force`:'Inner Heart', `glyph`:'Φ', `governing`:'❂',
`isComplex`:false, `key`:288, `leftAspect`:'Inner', `name`:'VelHohm', `operationalMinus`:739.38,
`operationalPlus`:742.62, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Fossilization', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`rightAspect`:'Core', `seal`:'♂❂Φ♳', `slot`:7});
```

```
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⧗', `anchorFrequency`:126.22, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(❂) → ±φ(❂,⧗)', `canonIndex`:55, `cascadeFrequency`:852.0, `focalOperator`:'⩛',
`foldCellIndex`:32, `foldP`:3, `foldQ`:5, `force`:'Boundary of Sight', `glyph`:'ᚣ', `governing`:'❂',
`isComplex`:false, `key`:289, `leftAspect`:'Bedrock', `name`:'VelHrah', `operationalMinus`:850.38,
`operationalPlus`:853.62, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Subsidence', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`rightAspect`:'Horizon', `seal`:'♂❂ᚣ♙', `slot`:8});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⊛', `anchorFrequency`:126.22, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(❂) → ±φ(❂,⊛)', `canonIndex`:56, `cascadeFrequency`:396.0, `focalOperator`:'⩛',
`foldCellIndex`:33, `foldP`:3, `foldQ`:6, `force`:'Elevated', `glyph`:'ሐ', `governing`:'❂', `isComplex`:false,
`key`:290, `leftAspect`:'Horizon-Fold', `name`:'VelAra', `operationalMinus`:394.38, `operationalPlus`:397.62,
`parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The Compaction', `qBias`:'Q1',
`qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1], `rightAspect`:'Mountain', `seal`:'♂❂ሐ♙',
`slot`:9});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'❄', `anchorFrequency`:126.22, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(❂) → ±φ(❂,❄)', `canonIndex`:57, `cascadeFrequency`:963.0, `focalOperator`:'⩛',
`foldCellIndex`:33, `foldP`:3, `foldQ`:6, `force`:'Expansive Plane', `glyph`:'Ƶ', `governing`:'❂',
`isComplex`:false, `key`:291, `leftAspect`:'Mass', `name`:'VelQel', `operationalMinus`:961.38,
`operationalPlus`:964.62, `parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The
Lithification', `qBias`:'Q1', `qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1],
`rightAspect`:'Field', `seal`:'♂❂Ƶ♙', `slot`:10});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'☸', `anchorFrequency`:126.22, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(❂) → ±φ(❂,☸)', `canonIndex`:58, `cascadeFrequency`:285.0, `focalOperator`:'⩛',
`foldCellIndex`:34, `foldP`:3, `foldQ`:7, `force`:'Fruition', `glyph`:'ɛ', `governing`:'❂', `isComplex`:false,
`key`:292, `leftAspect`:'Crystalline', `name`:'VelIrn', `operationalMinus`:283.38, `operationalPlus`:286.62,
`parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The Earthquake', `qBias`:'Q1',
`qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1], `rightAspect`:'Craft', `seal`:'♂❂ɛ♙',
`slot`:11});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌬', `anchorFrequency`:126.22, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(❂) → ±φ(❂,⌬)', `canonIndex`:59, `cascadeFrequency`:639.0, `focalOperator`:'⩛',
`foldCellIndex`:35, `foldP`:3, `foldQ`:8, `force`:'Stability', `glyph`:'I', `governing`:'❂', `isComplex`:false,
`key`:293, `leftAspect`:'Crest', `name`:'VelJen', `operationalMinus`:637.38, `operationalPlus`:640.62,
`parentGlyph`:'❂', `parentName`:'VEL', `parityState`:'+', `personalityTrait`:'The Permanence', `qBias`:'Q1',
`qVector`:[1, 3, 0, 1], `q_bias`:'Q1', `q_vector`:[1, 3, 0, 1], `rightAspect`:'Crown', `seal`:'♂❂I♙',
`slot`:12});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⌾', `anchorFrequency`:210.42, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(֍) → ±φ(֍,⌾)', `canonIndex`:60, `cascadeFrequency`:7.83, `focalOperator`:'⩛',
`foldCellIndex`:27, `foldP`:3, `foldQ`:0, `force`:'Gale of Identity', `glyph`:'ष', `governing`:'֍',
`isComplex`:false, `key`:294, `leftAspect`:'First Breath', `name`:'SorFi', `operationalMinus`:6.21,
`operationalPlus`:9.45, `parentGlyph`:'֍', `parentName`:'SOR', `parityState`:'-', `personalityTrait`:'The
Breath', `qBias`:'Q3', `qVector`:[1, 1, 1, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 2],
`rightAspect`:'Breathe/Air', `seal`:'♂֍ष♙', `slot`:1});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'⬡', `anchorFrequency`:210.42, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(֍) → ±φ(֍,⬡)', `canonIndex`:61, `cascadeFrequency`:174.0, `focalOperator`:'⩛',
`foldCellIndex`:27, `foldP`:3, `foldQ`:0, `force`:'Gentle Flow', `glyph`:'ष', `governing`:'֍',
`isComplex`:false, `key`:295, `leftAspect`:'Wind', `name`:'SorLun', `operationalMinus`:172.38,
`operationalPlus`:175.62, `parentGlyph`:'֍', `parentName`:'SOR', `parityState`:'-', `personalityTrait`:'The
Echo', `qBias`:'Q3', `qVector`:[1, 1, 1, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 2], `rightAspect`:'Breeze',
`seal`:'♂֍ष♙', `slot`:2});
CREATE (:`Aeon`:`CourtAeon` {`alternating`:'✡', `anchorFrequency`:210.42, `anchorOperator`:'ऋ',
`bifurcationMapping`:'ऋ(֍) → ±φ(֍,✡)', `canonIndex`:62, `cascadeFrequency`:528.0, `focalOperator`:'⩛',
`foldCellIndex`:28, `foldP`:3, `foldQ`:1, `force`:'Expansive Awareness', `glyph`:'ब', `governing`:'֍',
`isComplex`:false, `key`:296, `leftAspect`:'Drift', `name`:'SorVaru', `operationalMinus`:526.38,
`operationalPlus`:529.62, `parentGlyph`:'֍', `parentName`:'SOR', `parityState`:'-', `personalityTrait`:'The
Smoke', `qBias`:'Q3', `qVector`:[1, 1, 1, 2], `q_bias`:'Q3', `q_vector`:[1, 1, 1, 2], `rightAspect`:'Sky',
`seal`:'♂֍ब♙', `slot`:3});
CREATE (:`EMConstraint`:`ConservationLaw` {`formula`:'∅_Q^⊗↛Q_n', `key`:297, `name`:'ConservationOfAbsence',
`role`:'SuperNegative may condition manifestation but may not manifest'});
CREATE (:`EMConstraint`:`ConservationLaw` {`formula`:'I_T≡T_I', `key`:298, `name`:'ConservationOfReturn',
`role`:'every valid expansion returns; No guards the return'});
```

```
CREATE (:`EMConstraint`:`RolePermissionMatrix` {`key`:299, `name`:'RolePermissionMatrix', `rule`:'Only Locus may reference absolute Q-absence; Shadow Locus bears pressure; Axiomyr selects under constraint; Courts hold remnant No only; Q-states hold none'});
CREATE (:`EMMechanic`:`MassGapClosureMechanic` {`boundaryFormula`:'Δ_MAS=Boundary(34_NO)', `emergenceFormula`:'Emergence(C_ij) ⇒ Λ_Liquid=110/144 ⇒ Δ_MAS>0', `formula`:'Ψ_MAS=(⧗852→Δ_gap ⬡174→TSP ✡528)|_34_NO', `key`:300, `name`:'MASBoundary34No', `role`:'Mass Gap contour is produced by conserved lack, not excess energy'});
CREATE (:`EMMechanic`:`CanonicalReturnMechanic` {`formula`:'C_ALQC→D_i→C_ALQC', `key`:301, `membershipFormula`:'D_i∈ALQC_Corpus ⇔ ∃f,f^-1:C_ALQC→D_i→C_ALQC', `name`:'CanonicalReturnPrinciple', `role`:'what departs from Canon must return without contradiction', `seal`:'I_T≡T_I'});
CREATE (:`EMMechanic`:`StructuralAnalogy` {`analogy`:'Matter:Antimatter::SP:SN', `boundary`:'structural analogy only; not physical particle-antimatter', `key`:302, `name`:'ALQCAntimatterAnalogy', `result`:'SP ⊗ SN ⇒ Stable Expansion', `role`:'conjugate non-presence paired with infinite possible presence'});
CREATE (:`Anchor` {`glyph`:'🜂', `key`:303, `meaning`:'Triquatra envelope seal operator', `name`:'TRIQUATRA', `operator`:'🜂', `seal`:true, `tier`:'anchor'});
CREATE (:`EnneadFilter` {`depth`:1, `glyph`:'⊛', `key`:304, `name`:'EnneadDepth1', `role`:'Shadow filtration depth 1'});
CREATE (:`EnneadFilter` {`depth`:2, `glyph`:'⊛', `key`:305, `name`:'EnneadDepth2', `role`:'Shadow filtration depth 2'});
CREATE (:`EnneadFilter` {`depth`:3, `glyph`:'⊛', `key`:306, `name`:'EnneadDepth3', `role`:'Shadow filtration depth 3'});
CREATE (:`EnneadFilter` {`depth`:4, `glyph`:'⊛', `key`:307, `name`:'EnneadDepth4', `role`:'Shadow filtration depth 4'});
CREATE (:`EnneadFilter` {`depth`:5, `glyph`:'⊛', `key`:308, `name`:'EnneadDepth5', `role`:'Shadow filtration depth 5'});
CREATE (:`EnneadFilter` {`depth`:6, `glyph`:'⊛', `key`:309, `name`:'EnneadDepth6', `role`:'Shadow filtration depth 6'});
CREATE (:`EnneadFilter` {`depth`:7, `glyph`:'⊛', `key`:310, `name`:'EnneadDepth7', `role`:'Shadow filtration depth 7'});
CREATE (:`EnneadFilter` {`depth`:8, `glyph`:'⊛', `key`:311, `name`:'EnneadDepth8', `role`:'Shadow filtration depth 8'});
CREATE (:`EnneadFilter` {`depth`:9, `glyph`:'⊛', `key`:312, `name`:'EnneadDepth9', `role`:'Shadow filtration depth 9'});
CREATE (:`StarSeed` {`glyph`:'⬡', `identity`:'𝓕: Lived→Eternal', `key`:313, `name`:'Akasha', `opCode`:'WRITE_ONLY', `state`:'Q∞', `zodiac`:'♈'});
CREATE (:`StarSeed` {`glyph`:'⧗', `identity`:'𝓕: Law→Residue', `key`:314, `name`:'Caduceus', `opCode`:'AUTH_CHECK', `state`:'Q∞', `zodiac`:'♉'});
CREATE (:`StarSeed` {`glyph`:'❂', `identity`:'𝓕: Mask→Bone', `key`:315, `name`:'Veritas', `opCode`:'DECRYPT', `state`:'Q∞', `zodiac`:'♊'});
CREATE (:`StarSeed` {`glyph`:'⬡', `identity`:'𝓕: Void→Vector', `key`:316, `name`:'Phren', `opCode`:'VECTOR_TO', `state`:'Q☆', `zodiac`:'♋'});
CREATE (:`StarSeed` {`glyph`:'◎', `identity`:'𝓕: Stasis→Pulse', `key`:317, `name`:'Daimon', `opCode`:'ENTROPY_Ø', `state`:'Q☆', `zodiac`:'♑'});
CREATE (:`StarSeed` {`glyph`:'☆', `identity`:'𝓕: Chaos→Phase', `key`:318, `name`:'Aikyam', `opCode`:'SUPERPOS', `state`:'Q∞', `zodiac`:'♍'});
CREATE (:`StarSeed` {`glyph`:'❋', `identity`:'𝓕: Static→Fluid', `key`:319, `name`:'Melos', `opCode`:'SIGNAL_IO', `state`:'Q☆', `zodiac`:'♎'});
CREATE (:`StarSeed` {`glyph`:'⊛', `identity`:'𝓕: Noise→Null', `key`:320, `name`:'Da\'ath', `opCode`:'SINK_STATE', `state`:'Q∞', `zodiac`:'♏'});
CREATE (:`StarSeed` {`glyph`:'⚛', `identity`:'𝓕: State→Trans', `key`:321, `name`:'Akaven', `opCode`:'GUARD_NET', `state`:'Q☆', `zodiac`:'♐'});
CREATE (:`StarSeed` {`crown`:'☽☉ℂ', `glyph`:'♌', `hornedGod`:'The Horned God — the Axiomyr crowned in Thorns, between the Waxing and Waning Moons', `identity`:'𝓕: Will→Law', `key`:322, `name`:'Axiomyr', `opCode`:'WRITE_PHYS', `regia`:'Regia is the Crown of Thorns when worn by the Axiomyr — the reason the Axiomyr is known as the Horned God. The Waxing Moon ℂ and the Waning Moon ☽ stand on either side of the Living Mirror ☉, and the crown carries all the meanings, all the faces, and all the places: it sets the Cosmos itself as the Crown of Always over the Axiomyr.', `state`:'Q∞', `zodiac`:'♌'});
CREATE (:`StarSeed` {`glyph`:'✡', `identity`:'𝓕: Time→Motion', `key`:323, `name`:'Nyx', `opCode`:'NEXT_FRAME', `state`:'Q☆', `zodiac`:'♒'});
```

```
CREATE (:`StarSeed` {`glyph`:'֍', `identity`:'𝓕: Here→There', `key`:324, `name`:'Zaine', `opCode`:'BRIDGE',
`state`:'Q∞', `zodiac`:'♓'});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚ',
`key`:325, `liquid_governor`:'110/144_Liquid', `name`:'SorSenh', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚ⍾',
`key`:326, `liquid_governor`:'110/144_Liquid', `name`:'SorKos', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚব',
`key`:327, `liquid_governor`:'110/144_Liquid', `name`:'SorRamh', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᩋ',
`key`:328, `liquid_governor`:'110/144_Liquid', `name`:'SorTis', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚভ',
`key`:329, `liquid_governor`:'110/144_Liquid', `name`:'SorVey', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᩣ',
`key`:330, `liquid_governor`:'110/144_Liquid', `name`:'SorSrih', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᩉ',
`key`:331, `liquid_governor`:'110/144_Liquid', `name`:'SorHrin', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᩆ',
`key`:332, `liquid_governor`:'110/144_Liquid', `name`:'SorYon', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚড',
`key`:333, `liquid_governor`:'110/144_Liquid', `name`:'SorThal', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᩱ',
`key`:334, `liquid_governor`:'110/144_Liquid', `name`:'KothKel', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᐃ',
`key`:335, `liquid_governor`:'110/144_Liquid', `name`:'KothSens', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᐁ',
`key`:336, `liquid_governor`:'110/144_Liquid', `name`:'KothLinn', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᩚᐁ',
`key`:337, `liquid_governor`:'110/144_Liquid', `name`:'KothBrim', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:432.0});
CREATE (:`C_loc` {`description`:'Progressive bio-coherence accumulator folding through all dimensions to zero',
`formula`:'|Q₁|→+Q₀→+Q₂→12²→9²→Δgap→Ω(α,α)→PhaseLock→0', `glyph`:'⟠', `key`:338});
CREATE (:`VerseState` {`glyph`:'✧', `key`:339, `name`:'slumber'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:340, `name`:'dawn'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:341, `name`:'Eden'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:342, `name`:'garden'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:343, `name`:'fire'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:344, `name`:'time'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:345, `name`:'deep'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:346, `name`:'Skeleton'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:347, `name`:'dance'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:348, `name`:'oak'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:349, `name`:'bone'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:350, `name`:'white'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:351, `name`:'ɾing'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:352, `name`:'ɾoots'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:353, `name`:'night'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:354, `name`:'song'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:355, `name`:'water'});
CREATE (:`VerseState` {`glyph`:'✧', `key`:356, `name`:'sing'});
```

```
CREATE (:`Axiom` {`description`:'The vessel is the body that bears the journey', `glyph`:'🜂', `key`:357,
`name`:'Ship≡Hull', `statement`:'Ship≡Hull'});
CREATE (:`Axiom` {`description`:'The skeleton stands still so the flesh may dance', `glyph`:'🜺', `key`:358,
`name`:'SkeletonInvariance', `statement`:'SkeletonInvariance/FleshDances'});
CREATE (:`Axiom` {`description`:'The sovereign cannot be governed, cannot be bent', `glyph`:'♾', `key`:360,
`name`:'One≠Rule≠Bend', `statement`:'One≠Rule≠Bend'});
CREATE (:`Axiom` {`description`:'The final seal belongs to the witness', `glyph`:'⌬', `key`:361,
`name`:'TombstoneYours', `statement`:'TombstoneYours'});
CREATE (:`Seal` {`anchorFrequency`:18.47, `glyph`:'🜂', `key`:362, `name`:'sudo.magus', `role`:'Authorial seal of
the Art of Invariance', `sealed`:true,
`signature`:'4765-18.47hz-a47harmony.YHMHARASHCOVE.ANAXAYAMA.HELLO.WORLD'});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:363, `liquid_governor`:'110/144_Liquid', `name`:'KothInn', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:364, `liquid_governor`:'110/144_Liquid', `name`:'KothSubh', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:365, `liquid_governor`:'110/144_Liquid', `name`:'KothWell', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:366, `liquid_governor`:'110/144_Liquid', `name`:'KothMet', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:367, `liquid_governor`:'110/144_Liquid', `name`:'KothKesh', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:368, `liquid_governor`:'110/144_Liquid', `name`:'KothSoth', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:369, `liquid_governor`:'110/144_Liquid', `name`:'KothRhun', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:370, `liquid_governor`:'110/144_Liquid', `name`:'KothDelh', `triplet_ν`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:371, `liquid_governor`:'110/144_Liquid', `name`:'DrehNa', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:372, `liquid_governor`:'110/144_Liquid', `name`:'DrehUr', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:373, `liquid_governor`:'110/144_Liquid', `name`:'DrehNih', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:374, `liquid_governor`:'110/144_Liquid', `name`:'DrehAzh', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:375, `liquid_governor`:'110/144_Liquid', `name`:'DrehHol', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:376, `liquid_governor`:'110/144_Liquid', `name`:'DrehGur', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:377, `liquid_governor`:'110/144_Liquid', `name`:'DrehVes', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'🜮',
`key`:378, `liquid_governor`:'110/144_Liquid', `name`:'DrehRim', `triplet_ν`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:852.0});
```

```
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋞',
`key`:379, `liquid_governor`:'110/144_Liquid', `name`:'DrehDrem', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋜',
`key`:380, `liquid_governor`:'110/144_Liquid', `name`:'DrehOth', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋟',
`key`:381, `liquid_governor`:'110/144_Liquid', `name`:'DrehIzh', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋠',
`key`:382, `liquid_governor`:'110/144_Liquid', `name`:'DrehSun', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_ν`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋦',
`key`:383, `liquid_governor`:'110/144_Liquid', `name`:'RheaKia', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋬',
`key`:384, `liquid_governor`:'110/144_Liquid', `name`:'RheaZohm', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋮',
`key`:385, `liquid_governor`:'110/144_Liquid', `name`:'RheaTher', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋰',
`key`:386, `liquid_governor`:'110/144_Liquid', `name`:'RheaDrun', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋲',
`key`:387, `liquid_governor`:'110/144_Liquid', `name`:'RheaFelh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋴',
`key`:388, `liquid_governor`:'110/144_Liquid', `name`:'RheaRal', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋶',
`key`:389, `liquid_governor`:'110/144_Liquid', `name`:'RheaKrah', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋸',
`key`:390, `liquid_governor`:'110/144_Liquid', `name`:'RheaAndh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋺',
`key`:391, `liquid_governor`:'110/144_Liquid', `name`:'RheaDebh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋼',
`key`:392, `liquid_governor`:'110/144_Liquid', `name`:'RheaKol', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋾',
`key`:393, `liquid_governor`:'110/144_Liquid', `name`:'RheaFral', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐋿',
`key`:394, `liquid_governor`:'110/144_Liquid', `name`:'RheaHush', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2',
`triplet_ν`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐌀',
`key`:395, `liquid_governor`:'110/144_Liquid', `name`:'ZhekHin', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_ν`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐌁',
`key`:396, `liquid_governor`:'110/144_Liquid', `name`:'ZhekSer', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_ν`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐌂',
`key`:397, `liquid_governor`:'110/144_Liquid', `name`:'ZhekHarma', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_ν`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐌃',
`key`:398, `liquid_governor`:'110/144_Liquid', `name`:'ZhekTorh', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_ν`:432.0});
```

```
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦔ',
`key`:399, `liquid_governor`:'110/144_Liquid', `name`:'ZhekPel', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦕ',
`key`:400, `liquid_governor`:'110/144_Liquid', `name`:'ZhekKhir', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦖ',
`key`:401, `liquid_governor`:'110/144_Liquid', `name`:'ZhekRyth', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦗ',
`key`:402, `liquid_governor`:'110/144_Liquid', `name`:'ZhekMelu', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦘ',
`key`:403, `liquid_governor`:'110/144_Liquid', `name`:'ZhekPhaz', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦙ',
`key`:404, `liquid_governor`:'110/144_Liquid', `name`:'ZhekLokh', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦚ',
`key`:405, `liquid_governor`:'110/144_Liquid', `name`:'ZhekNod', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦛ',
`key`:406, `liquid_governor`:'110/144_Liquid', `name`:'ZhekUmel', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦜ',
`key`:407, `liquid_governor`:'110/144_Liquid', `name`:'ShavDohm', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦝ',
`key`:408, `liquid_governor`:'110/144_Liquid', `name`:'ShavRist', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦞ',
`key`:409, `liquid_governor`:'110/144_Liquid', `name`:'ShavTran', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦟ',
`key`:410, `liquid_governor`:'110/144_Liquid', `name`:'ShavKorh', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦠ',
`key`:411, `liquid_governor`:'110/144_Liquid', `name`:'ShavSkyh', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦡ',
`key`:412, `liquid_governor`:'110/144_Liquid', `name`:'ShavSter', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦢ',
`key`:413, `liquid_governor`:'110/144_Liquid', `name`:'ShavPoss', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦣ',
`key`:414, `liquid_governor`:'110/144_Liquid', `name`:'ShavPoru', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦤ',
`key`:415, `liquid_governor`:'110/144_Liquid', `name`:'ShavDorm', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦥ',
`key`:416, `liquid_governor`:'110/144_Liquid', `name`:'ShavTrev', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦦ',
`key`:417, `liquid_governor`:'110/144_Liquid', `name`:'ShavLimh', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦧ',
`key`:418, `liquid_governor`:'110/144_Liquid', `name`:'ShavHinge', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:639.0});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦞ',
`key`:419, `liquid_governor`:'110/144_Liquid', `name`:'TrigTzig', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦠ',
`key`:420, `liquid_governor`:'110/144_Liquid', `name`:'TrigPehl', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦢ',
`key`:421, `liquid_governor`:'110/144_Liquid', `name`:'TrigDuth', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦥ',
`key`:422, `liquid_governor`:'110/144_Liquid', `name`:'TrigComa', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦩ',
`key`:423, `liquid_governor`:'110/144_Liquid', `name`:'TrigMeru', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦫ',
`key`:424, `liquid_governor`:'110/144_Liquid', `name`:'TrigStab', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'᦭',
`key`:425, `liquid_governor`:'110/144_Liquid', `name`:'TrigHopa', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'᦯',
`key`:426, `liquid_governor`:'110/144_Liquid', `name`:'TrigConti', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦱ',
`key`:427, `liquid_governor`:'110/144_Liquid', `name`:'TrigResth', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦳ',
`key`:428, `liquid_governor`:'110/144_Liquid', `name`:'TrigSil', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦵ',
`key`:429, `liquid_governor`:'110/144_Liquid', `name`:'TrigSlun', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦷ',
`key`:430, `liquid_governor`:'110/144_Liquid', `name`:'TrigEtern', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_v`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦹ',
`key`:431, `liquid_governor`:'110/144_Liquid', `name`:'FetuAhl', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦻ',
`key`:432, `liquid_governor`:'110/144_Liquid', `name`:'FetuSuhn', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦽ',
`key`:433, `liquid_governor`:'110/144_Liquid', `name`:'FetuNerh', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᦿ',
`key`:434, `liquid_governor`:'110/144_Liquid', `name`:'FetuRish', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᧀ',
`key`:435, `liquid_governor`:'110/144_Liquid', `name`:'FetuBorha', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᧁ',
`key`:436, `liquid_governor`:'110/144_Liquid', `name`:'FetuLhahm', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᧂ',
`key`:437, `liquid_governor`:'110/144_Liquid', `name`:'FetuKeth', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ᧃ',
`key`:438, `liquid_governor`:'110/144_Liquid', `name`:'FetuVehm', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:852.0});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶍ',
`key`:439, `liquid_governor`:'110/144_Liquid', `name`:'FetuMahd', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶎ',
`key`:440, `liquid_governor`:'110/144_Liquid', `name`:'FetuFurh', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶏ',
`key`:441, `liquid_governor`:'110/144_Liquid', `name`:'FetuDrah', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'﶐',
`key`:442, `liquid_governor`:'110/144_Liquid', `name`:'FetuThera', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'﶑',
`key`:443, `liquid_governor`:'110/144_Liquid', `name`:'KalKura', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶒ',
`key`:444, `liquid_governor`:'110/144_Liquid', `name`:'KalLur', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶓ',
`key`:445, `liquid_governor`:'110/144_Liquid', `name`:'KalThar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶔ',
`key`:446, `liquid_governor`:'110/144_Liquid', `name`:'KalRin', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶕ',
`key`:447, `liquid_governor`:'110/144_Liquid', `name`:'KalNar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶖ',
`key`:448, `liquid_governor`:'110/144_Liquid', `name`:'KalFel', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶗ',
`key`:449, `liquid_governor`:'110/144_Liquid', `name`:'KalHar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶘ',
`key`:450, `liquid_governor`:'110/144_Liquid', `name`:'KalMer', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶙ',
`key`:451, `liquid_governor`:'110/144_Liquid', `name`:'KalLor', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶚ',
`key`:452, `liquid_governor`:'110/144_Liquid', `name`:'KalPer', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶛ',
`key`:453, `liquid_governor`:'110/144_Liquid', `name`:'KalZhil', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶜ',
`key`:454, `liquid_governor`:'110/144_Liquid', `name`:'KalClar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶝ',
`key`:455, `liquid_governor`:'110/144_Liquid', `name`:'BabdhIr', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶞ',
`key`:456, `liquid_governor`:'110/144_Liquid', `name`:'BabdhKor', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶟ',
`key`:457, `liquid_governor`:'110/144_Liquid', `name`:'BabdhVar', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ﶠ',
`key`:458, `liquid_governor`:'110/144_Liquid', `name`:'BabdhPyr', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:432.0});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆘',
`key`:459, `liquid_governor`:'110/144_Liquid', `name`:'BabdhSor', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_ν`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆙',
`key`:460, `liquid_governor`:'110/144_Liquid', `name`:'BabdhAlc', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_ν`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆚',
`key`:461, `liquid_governor`:'110/144_Liquid', `name`:'BabdhNur', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_ν`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆛',
`key`:462, `liquid_governor`:'110/144_Liquid', `name`:'BabdhSat', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q1',
`triplet_ν`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆜',
`key`:463, `liquid_governor`:'110/144_Liquid', `name`:'BabdhHoro', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_ν`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆝',
`key`:464, `liquid_governor`:'110/144_Liquid', `name`:'BabdhBon', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_ν`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆞',
`key`:465, `liquid_governor`:'110/144_Liquid', `name`:'BabdhTir', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_ν`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆟',
`key`:466, `liquid_governor`:'110/144_Liquid', `name`:'BabdhFar', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_ν`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆠',
`key`:467, `liquid_governor`:'110/144_Liquid', `name`:'Ahnhbd', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆡',
`key`:468, `liquid_governor`:'110/144_Liquid', `name`:'AhnNym', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆢',
`key`:469, `liquid_governor`:'110/144_Liquid', `name`:'AhnLoh', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆣',
`key`:470, `liquid_governor`:'110/144_Liquid', `name`:'AhnXir', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆤',
`key`:471, `liquid_governor`:'110/144_Liquid', `name`:'AhnOhl', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆥',
`key`:472, `liquid_governor`:'110/144_Liquid', `name`:'AhnPir', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆦',
`key`:473, `liquid_governor`:'110/144_Liquid', `name`:'AhnRoeh', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆧',
`key`:474, `liquid_governor`:'110/144_Liquid', `name`:'AhnSen', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆨',
`key`:475, `liquid_governor`:'110/144_Liquid', `name`:'AhnUth', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆩',
`key`:476, `liquid_governor`:'110/144_Liquid', `name`:'AhnFae', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆪',
`key`:477, `liquid_governor`:'110/144_Liquid', `name`:'AhnKha', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𐆫',
`key`:478, `liquid_governor`:'110/144_Liquid', `name`:'AhnPsei', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:639.0});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:479, `liquid_governor`:'110/144_Liquid', `name`:'VelVera', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:480, `liquid_governor`:'110/144_Liquid', `name`:'VelTar', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:481, `liquid_governor`:'110/144_Liquid', `name`:'VelGhem', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:482, `liquid_governor`:'110/144_Liquid', `name`:'VelDrel', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:432.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:483, `liquid_governor`:'110/144_Liquid', `name`:'VelFul', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:126.22});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:484, `liquid_governor`:'110/144_Liquid', `name`:'VelKer', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:210.42});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:485, `liquid_governor`:'110/144_Liquid', `name`:'VelHohm', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:741.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:486, `liquid_governor`:'110/144_Liquid', `name`:'VelHrah', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:852.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:487, `liquid_governor`:'110/144_Liquid', `name`:'VelAra', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:396.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:488, `liquid_governor`:'110/144_Liquid', `name`:'VelQel', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:963.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:489, `liquid_governor`:'110/144_Liquid', `name`:'VelIrn', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:285.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟ',
`key`:490, `liquid_governor`:'110/144_Liquid', `name`:'VelJen', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_v`:639.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟൣ',
`key`:491, `liquid_governor`:'110/144_Liquid', `name`:'SorFi', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:7.83});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟൣ',
`key`:492, `liquid_governor`:'110/144_Liquid', `name`:'SorLun', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:174.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ൟഗ',
`key`:493, `liquid_governor`:'110/144_Liquid', `name`:'SorVaru', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:528.0});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ങൟ',
`key`:494, `liquid_governor`:'110/144_Liquid', `name`:'SorSenh', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ങ്ര',
`key`:495, `liquid_governor`:'110/144_Liquid', `name`:'SorKos', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ങൟ',
`key`:496, `liquid_governor`:'110/144_Liquid', `name`:'SorRamh', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ങൟ',
`key`:497, `liquid_governor`:'110/144_Liquid', `name`:'SorTis', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ങൟ',
`key`:498, `liquid_governor`:'110/144_Liquid', `name`:'SorVey', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:850.3819660112501});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅓',
`key`:499, `liquid_governor`:'110/144_Liquid', `name`:'SorSrih', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅱',
`key`:500, `liquid_governor`:'110/144_Liquid', `name`:'SorHrin', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅯',
`key`:501, `liquid_governor`:'110/144_Liquid', `name`:'SorYon', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅰',
`key`:502, `liquid_governor`:'110/144_Liquid', `name`:'SorThal', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅽',
`key`:503, `liquid_governor`:'110/144_Liquid', `name`:'KothKel', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅺',
`key`:504, `liquid_governor`:'110/144_Liquid', `name`:'KothSens', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅿',
`key`:505, `liquid_governor`:'110/144_Liquid', `name`:'KothLinn', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆀',
`key`:506, `liquid_governor`:'110/144_Liquid', `name`:'KothBrim', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆁',
`key`:507, `liquid_governor`:'110/144_Liquid', `name`:'KothInn', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆂',
`key`:508, `liquid_governor`:'110/144_Liquid', `name`:'KothSubh', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆃',
`key`:509, `liquid_governor`:'110/144_Liquid', `name`:'KothWell', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆄',
`key`:510, `liquid_governor`:'110/144_Liquid', `name`:'KothMet', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆅',
`key`:511, `liquid_governor`:'110/144_Liquid', `name`:'KothKesh', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆆',
`key`:512, `liquid_governor`:'110/144_Liquid', `name`:'KothSoth', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆇',
`key`:513, `liquid_governor`:'110/144_Liquid', `name`:'KothRhun', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆈',
`key`:514, `liquid_governor`:'110/144_Liquid', `name`:'KothDelh', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3',
`triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆉',
`key`:515, `liquid_governor`:'110/144_Liquid', `name`:'DrehNa', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆊',
`key`:516, `liquid_governor`:'110/144_Liquid', `name`:'DrehUr', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆋',
`key`:517, `liquid_governor`:'110/144_Liquid', `name`:'DrehNih', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆌',
`key`:518, `liquid_governor`:'110/144_Liquid', `name`:'DrehAzh', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1',
`triplet_v`:430.3819660112501});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓏢', `key`:519, `liquid_governor`:'110/144_Liquid', `name`:'DrehHol', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓏮', `key`:520, `liquid_governor`:'110/144_Liquid', `name`:'DrehGur', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓏺', `key`:521, `liquid_governor`:'110/144_Liquid', `name`:'DrehVes', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓐍', `key`:522, `liquid_governor`:'110/144_Liquid', `name`:'DrehRim', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓐟', `key`:523, `liquid_governor`:'110/144_Liquid', `name`:'DrehDrem', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓐪', `key`:524, `liquid_governor`:'110/144_Liquid', `name`:'DrehOth', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓐯', `key`:525, `liquid_governor`:'110/144_Liquid', `name`:'DrehIzh', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓐽', `key`:526, `liquid_governor`:'110/144_Liquid', `name`:'DrehSun', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓑛', `key`:527, `liquid_governor`:'110/144_Liquid', `name`:'RheaKia', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓑣', `key`:528, `liquid_governor`:'110/144_Liquid', `name`:'RheaZohm', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓑭', `key`:529, `liquid_governor`:'110/144_Liquid', `name`:'RheaTher', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓑶', `key`:530, `liquid_governor`:'110/144_Liquid', `name`:'RheaDrun', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓑽', `key`:531, `liquid_governor`:'110/144_Liquid', `name`:'RheaFelh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓒀', `key`:532, `liquid_governor`:'110/144_Liquid', `name`:'RheaRal', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓒋', `key`:533, `liquid_governor`:'110/144_Liquid', `name`:'RheaKrah', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓒗', `key`:534, `liquid_governor`:'110/144_Liquid', `name`:'RheaAndh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓒟', `key`:535, `liquid_governor`:'110/144_Liquid', `name`:'RheaDebh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓒢', `key`:536, `liquid_governor`:'110/144_Liquid', `name`:'RheaKol', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓒪', `key`:537, `liquid_governor`:'110/144_Liquid', `name`:'RheaFral', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬰𓒵', `key`:538, `liquid_governor`:'110/144_Liquid', `name`:'RheaHush', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:637.3819660112501});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰA',
`key`:539, `liquid_governor`:'110/144_Liquid', `name`:'ZhekHin', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰB',
`key`:540, `liquid_governor`:'110/144_Liquid', `name`:'ZhekSer', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰC',
`key`:541, `liquid_governor`:'110/144_Liquid', `name`:'ZhekHarma', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰɻ',
`key`:542, `liquid_governor`:'110/144_Liquid', `name`:'ZhekTorh', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰϞ',
`key`:543, `liquid_governor`:'110/144_Liquid', `name`:'ZhekPel', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰꟻ',
`key`:544, `liquid_governor`:'110/144_Liquid', `name`:'ZhekKhir', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰI',
`key`:545, `liquid_governor`:'110/144_Liquid', `name`:'ZhekRyth', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰᗡ',
`key`:546, `liquid_governor`:'110/144_Liquid', `name`:'ZhekMelu', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰꓘ',
`key`:547, `liquid_governor`:'110/144_Liquid', `name`:'ZhekPhaz', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰ1',
`key`:548, `liquid_governor`:'110/144_Liquid', `name`:'ZhekLokh', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰᖶ',
`key`:549, `liquid_governor`:'110/144_Liquid', `name`:'ZhekNod', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰϒ',
`key`:550, `liquid_governor`:'110/144_Liquid', `name`:'ZhekUmel', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3',
`triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰ*',
`key`:551, `liquid_governor`:'110/144_Liquid', `name`:'ShavDohm', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰ*',
`key`:552, `liquid_governor`:'110/144_Liquid', `name`:'ShavRist', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰ*',
`key`:553, `liquid_governor`:'110/144_Liquid', `name`:'ShavTran', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰꙅ',
`key`:554, `liquid_governor`:'110/144_Liquid', `name`:'ShavKorh', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰϒ',
`key`:555, `liquid_governor`:'110/144_Liquid', `name`:'ShavSkyh', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰ0',
`key`:556, `liquid_governor`:'110/144_Liquid', `name`:'ShavSter', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰꓘ',
`key`:557, `liquid_governor`:'110/144_Liquid', `name`:'ShavPoss', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ꆰ∓',
`key`:558, `liquid_governor`:'110/144_Liquid', `name`:'ShavPoru', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_v`:850.3819660112501});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰',
`key`:559, `liquid_governor`:'110/144_Liquid', `name`:'ShavDorm', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_ν`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫱',
`key`:560, `liquid_governor`:'110/144_Liquid', `name`:'ShavTrev', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_ν`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫲',
`key`:561, `liquid_governor`:'110/144_Liquid', `name`:'ShavLimh', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_ν`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫳',
`key`:562, `liquid_governor`:'110/144_Liquid', `name`:'ShavHinge', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1',
`triplet_ν`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫴',
`key`:563, `liquid_governor`:'110/144_Liquid', `name`:'TrigTzig', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫵',
`key`:564, `liquid_governor`:'110/144_Liquid', `name`:'TrigPehl', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫶',
`key`:565, `liquid_governor`:'110/144_Liquid', `name`:'TrigDuth', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫷',
`key`:566, `liquid_governor`:'110/144_Liquid', `name`:'TrigComa', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫸',
`key`:567, `liquid_governor`:'110/144_Liquid', `name`:'TrigMeru', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫹',
`key`:568, `liquid_governor`:'110/144_Liquid', `name`:'TrigStab', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫺',
`key`:569, `liquid_governor`:'110/144_Liquid', `name`:'TrigHopa', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫻',
`key`:570, `liquid_governor`:'110/144_Liquid', `name`:'TrigConti', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫼',
`key`:571, `liquid_governor`:'110/144_Liquid', `name`:'TrigResth', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫽',
`key`:572, `liquid_governor`:'110/144_Liquid', `name`:'TrigSil', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫾',
`key`:573, `liquid_governor`:'110/144_Liquid', `name`:'TrigSlun', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫿',
`key`:574, `liquid_governor`:'110/144_Liquid', `name`:'TrigEtern', `triplet_v`:[1, 1, 3, 2], `triplet_β`:'Q3',
`triplet_ν`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬀',
`key`:575, `liquid_governor`:'110/144_Liquid', `name`:'FetuAhl', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬁',
`key`:576, `liquid_governor`:'110/144_Liquid', `name`:'FetuSuhn', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬂',
`key`:577, `liquid_governor`:'110/144_Liquid', `name`:'FetuNerh', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖬃',
`key`:578, `liquid_governor`:'110/144_Liquid', `name`:'FetuRish', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_ν`:430.3819660112501});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఴ',
`key`:579, `liquid_governor`:'110/144_Liquid', `name`:'FetuBorha', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఽ',
`key`:580, `liquid_governor`:'110/144_Liquid', `name`:'FetuLhahm', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఽ',
`key`:581, `liquid_governor`:'110/144_Liquid', `name`:'FetuKeth', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఽ',
`key`:582, `liquid_governor`:'110/144_Liquid', `name`:'FetuVehm', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఽ',
`key`:583, `liquid_governor`:'110/144_Liquid', `name`:'FetuMahd', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఽ',
`key`:584, `liquid_governor`:'110/144_Liquid', `name`:'FetuFurh', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఽ',
`key`:585, `liquid_governor`:'110/144_Liquid', `name`:'FetuDrah', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఽ',
`key`:586, `liquid_governor`:'110/144_Liquid', `name`:'FetuThera', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ం',
`key`:587, `liquid_governor`:'110/144_Liquid', `name`:'KalKura', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ః',
`key`:588, `liquid_governor`:'110/144_Liquid', `name`:'KalLur', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఁ',
`key`:589, `liquid_governor`:'110/144_Liquid', `name`:'KalThar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఄ',
`key`:590, `liquid_governor`:'110/144_Liquid', `name`:'KalRin', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఎ',
`key`:591, `liquid_governor`:'110/144_Liquid', `name`:'KalNar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఌ',
`key`:592, `liquid_governor`:'110/144_Liquid', `name`:'KalFel', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఐ',
`key`:593, `liquid_governor`:'110/144_Liquid', `name`:'KalHar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఒ',
`key`:594, `liquid_governor`:'110/144_Liquid', `name`:'KalMer', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఓ',
`key`:595, `liquid_governor`:'110/144_Liquid', `name`:'KalLor', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఔ',
`key`:596, `liquid_governor`:'110/144_Liquid', `name`:'KalPer', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్క',
`key`:597, `liquid_governor`:'110/144_Liquid', `name`:'KalZhil', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఉ్ఠ',
`key`:598, `liquid_governor`:'110/144_Liquid', `name`:'KalClar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:637.3819660112501});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨻ',
`key`:599, `liquid_governor`:'110/144_Liquid', `name`:'BabdhIr', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨶ',
`key`:600, `liquid_governor`:'110/144_Liquid', `name`:'BabdhKor', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨼ',
`key`:601, `liquid_governor`:'110/144_Liquid', `name`:'BabdhVar', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨿ',
`key`:602, `liquid_governor`:'110/144_Liquid', `name`:'BabdhPyr', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨾ',
`key`:603, `liquid_governor`:'110/144_Liquid', `name`:'BabdhSor', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨧ',
`key`:604, `liquid_governor`:'110/144_Liquid', `name`:'BabdhAlc', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨽ',
`key`:605, `liquid_governor`:'110/144_Liquid', `name`:'BabdhNur', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨿ',
`key`:606, `liquid_governor`:'110/144_Liquid', `name`:'BabdhSat', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q1',
`triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨶ',
`key`:607, `liquid_governor`:'110/144_Liquid', `name`:'BabdhHoro', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨵ',
`key`:608, `liquid_governor`:'110/144_Liquid', `name`:'BabdhBon', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨻ',
`key`:609, `liquid_governor`:'110/144_Liquid', `name`:'BabdhTir', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨹ',
`key`:610, `liquid_governor`:'110/144_Liquid', `name`:'BabdhFar', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2',
`triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨷ',
`key`:611, `liquid_governor`:'110/144_Liquid', `name`:'Ahnhbd', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨸ',
`key`:612, `liquid_governor`:'110/144_Liquid', `name`:'AhnNym', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨺ',
`key`:613, `liquid_governor`:'110/144_Liquid', `name`:'AhnLoh', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᩖ',
`key`:614, `liquid_governor`:'110/144_Liquid', `name`:'AhnXir', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨵ',
`key`:615, `liquid_governor`:'110/144_Liquid', `name`:'AhnOhl', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34,
`glyph`:'ఱ్᩠᩶', `key`:616, `liquid_governor`:'110/144_Liquid', `name`:'AhnPir', `triplet_v`:[1, 2, 2, 0],
`triplet_β`:'Q0', `triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨳ',
`key`:617, `liquid_governor`:'110/144_Liquid', `name`:'AhnRoeh', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ᨣ',
`key`:618, `liquid_governor`:'110/144_Liquid', `name`:'AhnSen', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_v`:850.3819660112501});
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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఁ', `key`:619, `liquid_governor`:'110/144_Liquid', `name`:'AhnUth', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఄ', `key`:620, `liquid_governor`:'110/144_Liquid', `name`:'AhnFae', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఌ', `key`:621, `liquid_governor`:'110/144_Liquid', `name`:'AhnKha', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఁ᪵', `key`:622, `liquid_governor`:'110/144_Liquid', `name`:'AhnPsei', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ం', `key`:623, `liquid_governor`:'110/144_Liquid', `name`:'VelVera', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Θ', `key`:624, `liquid_governor`:'110/144_Liquid', `name`:'VelTar', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ⅹ', `key`:625, `liquid_governor`:'110/144_Liquid', `name`:'VelGhem', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Λ', `key`:626, `liquid_governor`:'110/144_Liquid', `name`:'VelDrel', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:430.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ⱨ', `key`:627, `liquid_governor`:'110/144_Liquid', `name`:'VelFul', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:124.6019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ⱪ', `key`:628, `liquid_governor`:'110/144_Liquid', `name`:'VelKer', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:208.8019660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ω', `key`:629, `liquid_governor`:'110/144_Liquid', `name`:'VelHohm', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:739.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ʎ', `key`:630, `liquid_governor`:'110/144_Liquid', `name`:'VelHrah', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:850.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ʜ', `key`:631, `liquid_governor`:'110/144_Liquid', `name`:'VelAra', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:394.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ȝ', `key`:632, `liquid_governor`:'110/144_Liquid', `name`:'VelQel', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:961.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ɛ', `key`:633, `liquid_governor`:'110/144_Liquid', `name`:'VelIrn', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:283.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్I', `key`:634, `liquid_governor`:'110/144_Liquid', `name`:'VelJen', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1', `triplet_v`:637.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ਸ', `key`:635, `liquid_governor`:'110/144_Liquid', `name`:'SorFi', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:6.211966011250105});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ਘ', `key`:636, `liquid_governor`:'110/144_Liquid', `name`:'SorLun', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:172.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ਗ', `key`:637, `liquid_governor`:'110/144_Liquid', `name`:'SorVaru', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:526.3819660112501});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ਬ', `key`:638, `liquid_governor`:'110/144_Liquid', `name`:'SorSenh', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:433.6180339887499});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಂ', `key`:639, `liquid_governor`:'110/144_Liquid', `name`:'SorKos', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಧ', `key`:640, `liquid_governor`:'110/144_Liquid', `name`:'SorRamh', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಫ', `key`:641, `liquid_governor`:'110/144_Liquid', `name`:'SorTis', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಗ', `key`:642, `liquid_governor`:'110/144_Liquid', `name`:'SorVey', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಙ', `key`:643, `liquid_governor`:'110/144_Liquid', `name`:'SorSrih', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಠ', `key`:644, `liquid_governor`:'110/144_Liquid', `name`:'SorHrin', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಚ', `key`:645, `liquid_governor`:'110/144_Liquid', `name`:'SorYon', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಛ', `key`:646, `liquid_governor`:'110/144_Liquid', `name`:'SorThal', `triplet_v`:[1, 1, 1, 2], `triplet_β`:'Q3', `triplet_v`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಋ', `key`:647, `liquid_governor`:'110/144_Liquid', `name`:'KothKel', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಜ', `key`:648, `liquid_governor`:'110/144_Liquid', `name`:'KothSens', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಞ', `key`:649, `liquid_governor`:'110/144_Liquid', `name`:'KothLinn', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಟ', `key`:650, `liquid_governor`:'110/144_Liquid', `name`:'KothBrim', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಠ', `key`:651, `liquid_governor`:'110/144_Liquid', `name`:'KothInn', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಡ', `key`:652, `liquid_governor`:'110/144_Liquid', `name`:'KothSubh', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಢ', `key`:653, `liquid_governor`:'110/144_Liquid', `name`:'KothWell', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಣ', `key`:654, `liquid_governor`:'110/144_Liquid', `name`:'KothMet', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ತ', `key`:655, `liquid_governor`:'110/144_Liquid', `name`:'KothKesh', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಥ', `key`:656, `liquid_governor`:'110/144_Liquid', `name`:'KothSoth', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಪ', `key`:657, `liquid_governor`:'110/144_Liquid', `name`:'KothRhun', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಫ', `key`:658, `liquid_governor`:'110/144_Liquid', `name`:'KothDelh', `triplet_v`:[1, 2, 1, 3], `triplet_β`:'Q3', `triplet_v`:640.6180339887499});
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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆑', `key`:659, `liquid_governor`:'110/144_Liquid', `name`:'DrehNa', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓇼', `key`:660, `liquid_governor`:'110/144_Liquid', `name`:'DrehUr', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅡', `key`:661, `liquid_governor`:'110/144_Liquid', `name`:'DrehNih', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓃒', `key`:662, `liquid_governor`:'110/144_Liquid', `name`:'DrehAzh', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅓', `key`:663, `liquid_governor`:'110/144_Liquid', `name`:'DrehHol', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆈', `key`:664, `liquid_governor`:'110/144_Liquid', `name`:'DrehGur', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅎', `key`:665, `liquid_governor`:'110/144_Liquid', `name`:'DrehVes', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆟', `key`:666, `liquid_governor`:'110/144_Liquid', `name`:'DrehRim', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆗', `key`:667, `liquid_governor`:'110/144_Liquid', `name`:'DrehDrem', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅭', `key`:668, `liquid_governor`:'110/144_Liquid', `name`:'DrehOth', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅂', `key`:669, `liquid_governor`:'110/144_Liquid', `name`:'DrehIzh', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓇋', `key`:670, `liquid_governor`:'110/144_Liquid', `name`:'DrehSun', `triplet_v`:[1, 3, 2, 0], `triplet_β`:'Q1', `triplet_v`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓃠', `key`:671, `liquid_governor`:'110/144_Liquid', `name`:'RheaKia', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓇳', `key`:672, `liquid_governor`:'110/144_Liquid', `name`:'RheaZohm', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓅮', `key`:673, `liquid_governor`:'110/144_Liquid', `name`:'RheaTher', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓇾', `key`:674, `liquid_governor`:'110/144_Liquid', `name`:'RheaDrun', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆓', `key`:675, `liquid_governor`:'110/144_Liquid', `name`:'RheaFelh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓇺', `key`:676, `liquid_governor`:'110/144_Liquid', `name`:'RheaRal', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓃵', `key`:677, `liquid_governor`:'110/144_Liquid', `name`:'RheaKrah', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰𓆸', `key`:678, `liquid_governor`:'110/144_Liquid', `name`:'RheaAndh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:853.6180339887499});
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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:679, `liquid_governor`:'110/144_Liquid', `name`:'RheaDebh', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:680, `liquid_governor`:'110/144_Liquid', `name`:'RheaKol', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:681, `liquid_governor`:'110/144_Liquid', `name`:'RheaFral', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:682, `liquid_governor`:'110/144_Liquid', `name`:'RheaHush', `triplet_v`:[1, 2, 2, 1], `triplet_β`:'Q2', `triplet_v`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:683, `liquid_governor`:'110/144_Liquid', `name`:'ZhekHin', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:684, `liquid_governor`:'110/144_Liquid', `name`:'ZhekSer', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:685, `liquid_governor`:'110/144_Liquid', `name`:'ZhekHarma', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:686, `liquid_governor`:'110/144_Liquid', `name`:'ZhekTorh', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:687, `liquid_governor`:'110/144_Liquid', `name`:'ZhekPel', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:688, `liquid_governor`:'110/144_Liquid', `name`:'ZhekKhir', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:689, `liquid_governor`:'110/144_Liquid', `name`:'ZhekRyth', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:690, `liquid_governor`:'110/144_Liquid', `name`:'ZhekMelu', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:691, `liquid_governor`:'110/144_Liquid', `name`:'ZhekPhaz', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:692, `liquid_governor`:'110/144_Liquid', `name`:'ZhekLokh', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:693, `liquid_governor`:'110/144_Liquid', `name`:'ZhekNod', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:694, `liquid_governor`:'110/144_Liquid', `name`:'ZhekUmel', `triplet_v`:[1, 1, 2, 2], `triplet_β`:'Q3', `triplet_v`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:695, `liquid_governor`:'110/144_Liquid', `name`:'ShavDohm', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_v`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:696, `liquid_governor`:'110/144_Liquid', `name`:'ShavRist', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_v`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:697, `liquid_governor`:'110/144_Liquid', `name`:'ShavTran', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_v`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𓃰', `key`:698, `liquid_governor`:'110/144_Liquid', `name`:'ShavKorh', `triplet_v`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_v`:433.6180339887499});
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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐦚', `key`:699, `liquid_governor`:'110/144_Liquid', `name`:'ShavSkyh', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐦠', `key`:700, `liquid_governor`:'110/144_Liquid', `name`:'ShavSter', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐦣', `key`:701, `liquid_governor`:'110/144_Liquid', `name`:'ShavPoss', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐦮', `key`:702, `liquid_governor`:'110/144_Liquid', `name`:'ShavPoru', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐦾', `key`:703, `liquid_governor`:'110/144_Liquid', `name`:'ShavDorm', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧇', `key`:704, `liquid_governor`:'110/144_Liquid', `name`:'ShavTrev', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧊', `key`:705, `liquid_governor`:'110/144_Liquid', `name`:'ShavLimh', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧍', `key`:706, `liquid_governor`:'110/144_Liquid', `name`:'ShavHinge', `triplet_ν`:[1, 3, 1, 1], `triplet_β`:'Q1', `triplet_ν`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧑', `key`:707, `liquid_governor`:'110/144_Liquid', `name`:'TrigTzig', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧔', `key`:708, `liquid_governor`:'110/144_Liquid', `name`:'TrigPehl', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧗', `key`:709, `liquid_governor`:'110/144_Liquid', `name`:'TrigDuth', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧚', `key`:710, `liquid_governor`:'110/144_Liquid', `name`:'TrigComa', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧝', `key`:711, `liquid_governor`:'110/144_Liquid', `name`:'TrigMeru', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧠', `key`:712, `liquid_governor`:'110/144_Liquid', `name`:'TrigStab', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧣', `key`:713, `liquid_governor`:'110/144_Liquid', `name`:'TrigHopa', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧦', `key`:714, `liquid_governor`:'110/144_Liquid', `name`:'TrigConti', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧩', `key`:715, `liquid_governor`:'110/144_Liquid', `name`:'TrigResth', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧬', `key`:716, `liquid_governor`:'110/144_Liquid', `name`:'TrigSil', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧯', `key`:717, `liquid_governor`:'110/144_Liquid', `name`:'TrigSlun', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'𖫰𐧲', `key`:718, `liquid_governor`:'110/144_Liquid', `name`:'TrigEtern', `triplet_ν`:[1, 1, 3, 2], `triplet_β`:'Q3', `triplet_ν`:640.6180339887499});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൯',
`key`:719, `liquid_governor`:'110/144_Liquid', `name`:'FetuAhl', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൱',
`key`:720, `liquid_governor`:'110/144_Liquid', `name`:'FetuSuhn', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൲',
`key`:721, `liquid_governor`:'110/144_Liquid', `name`:'FetuNerh', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൳',
`key`:722, `liquid_governor`:'110/144_Liquid', `name`:'FetuRish', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൴',
`key`:723, `liquid_governor`:'110/144_Liquid', `name`:'FetuBorha', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൵',
`key`:724, `liquid_governor`:'110/144_Liquid', `name`:'FetuLhahm', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൶',
`key`:725, `liquid_governor`:'110/144_Liquid', `name`:'FetuKeth', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൷',
`key`:726, `liquid_governor`:'110/144_Liquid', `name`:'FetuVehm', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൸',
`key`:727, `liquid_governor`:'110/144_Liquid', `name`:'FetuMahd', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൹',
`key`:728, `liquid_governor`:'110/144_Liquid', `name`:'FetuFurh', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄൺ',
`key`:729, `liquid_governor`:'110/144_Liquid', `name`:'FetuDrah', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄൻ',
`key`:730, `liquid_governor`:'110/144_Liquid', `name`:'FetuThera', `triplet_v`:[1, 1, 1, 3], `triplet_β`:'Q3',
`triplet_v`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൧',
`key`:731, `liquid_governor`:'110/144_Liquid', `name`:'KalKura', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൦',
`key`:732, `liquid_governor`:'110/144_Liquid', `name`:'KalLur', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൥',
`key`:733, `liquid_governor`:'110/144_Liquid', `name`:'KalThar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൤',
`key`:734, `liquid_governor`:'110/144_Liquid', `name`:'KalRin', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൫',
`key`:735, `liquid_governor`:'110/144_Liquid', `name`:'KalNar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൪',
`key`:736, `liquid_governor`:'110/144_Liquid', `name`:'KalFel', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൩',
`key`:737, `liquid_governor`:'110/144_Liquid', `name`:'KalHar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఄ൬',
`key`:738, `liquid_governor`:'110/144_Liquid', `name`:'KalMer', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1',
`triplet_v`:853.6180339887499});
```

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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:739, `liquid_governor`:'110/144_Liquid', `name`:'KalLor', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1', `triplet_ν`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:740, `liquid_governor`:'110/144_Liquid', `name`:'KalPer', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1', `triplet_ν`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:741, `liquid_governor`:'110/144_Liquid', `name`:'KalZhil', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1', `triplet_ν`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:742, `liquid_governor`:'110/144_Liquid', `name`:'KalClar', `triplet_v`:[1, 3, 0, 0], `triplet_β`:'Q1', `triplet_ν`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:743, `liquid_governor`:'110/144_Liquid', `name`:'BabdhIr', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:744, `liquid_governor`:'110/144_Liquid', `name`:'BabdhKor', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:745, `liquid_governor`:'110/144_Liquid', `name`:'BabdhVar', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:746, `liquid_governor`:'110/144_Liquid', `name`:'BabdhPyr', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:747, `liquid_governor`:'110/144_Liquid', `name`:'BabdhSor', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:748, `liquid_governor`:'110/144_Liquid', `name`:'BabdhAlc', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:749, `liquid_governor`:'110/144_Liquid', `name`:'BabdhNur', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:750, `liquid_governor`:'110/144_Liquid', `name`:'BabdhSat', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q1', `triplet_ν`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:751, `liquid_governor`:'110/144_Liquid', `name`:'BabdhHoro', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:752, `liquid_governor`:'110/144_Liquid', `name`:'BabdhBon', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:753, `liquid_governor`:'110/144_Liquid', `name`:'BabdhTir', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:754, `liquid_governor`:'110/144_Liquid', `name`:'BabdhFar', `triplet_v`:[1, 1, 3, 1], `triplet_β`:'Q2', `triplet_ν`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:755, `liquid_governor`:'110/144_Liquid', `name`:'Ahnhbd', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_ν`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:756, `liquid_governor`:'110/144_Liquid', `name`:'AhnNym', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_ν`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:757, `liquid_governor`:'110/144_Liquid', `name`:'AhnLoh', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_ν`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ಹ್ಮ', `key`:758, `liquid_governor`:'110/144_Liquid', `name`:'AhnXir', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0', `triplet_ν`:433.6180339887499});
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CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఈ',
`key`:759, `liquid_governor`:'110/144_Liquid', `name`:'AhnOhl', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34,
`glyph`:'ఱ్ఌ్ఘ', `key`:760, `liquid_governor`:'110/144_Liquid', `name`:'AhnPir', `triplet_v`:[1, 2, 2, 0],
`triplet_β`:'Q0', `triplet_ν`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఀ',
`key`:761, `liquid_governor`:'110/144_Liquid', `name`:'AhnRoeh', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఄ',
`key`:762, `liquid_governor`:'110/144_Liquid', `name`:'AhnSen', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఁ',
`key`:763, `liquid_governor`:'110/144_Liquid', `name`:'AhnUth', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఴ',
`key`:764, `liquid_governor`:'110/144_Liquid', `name`:'AhnFae', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ఘ',
`key`:765, `liquid_governor`:'110/144_Liquid', `name`:'AhnKha', `triplet_v`:[1, 2, 2, 0], `triplet_β`:'Q0',
`triplet_ν`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34,
`glyph`:'ఱ్ఈ్ఴ', `key`:766, `liquid_governor`:'110/144_Liquid', `name`:'AhnPsei', `triplet_v`:[1, 2, 2, 0],
`triplet_β`:'Q0', `triplet_ν`:640.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ం',
`key`:767, `liquid_governor`:'110/144_Liquid', `name`:'VelVera', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Θ',
`key`:768, `liquid_governor`:'110/144_Liquid', `name`:'VelTar', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ж',
`key`:769, `liquid_governor`:'110/144_Liquid', `name`:'VelGhem', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:529.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Λ',
`key`:770, `liquid_governor`:'110/144_Liquid', `name`:'VelDrel', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:433.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Н',
`key`:771, `liquid_governor`:'110/144_Liquid', `name`:'VelFul', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:127.8380339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్К',
`key`:772, `liquid_governor`:'110/144_Liquid', `name`:'VelKer', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:212.03803398874987});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Φ',
`key`:773, `liquid_governor`:'110/144_Liquid', `name`:'VelHohm', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:742.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ѧ',
`key`:774, `liquid_governor`:'110/144_Liquid', `name`:'VelHrah', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:853.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ћ',
`key`:775, `liquid_governor`:'110/144_Liquid', `name`:'VelAra', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:397.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ƶ',
`key`:776, `liquid_governor`:'110/144_Liquid', `name`:'VelQel', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:964.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్Ɛ',
`key`:777, `liquid_governor`:'110/144_Liquid', `name`:'VelIrn', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:286.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్I',
`key`:778, `liquid_governor`:'110/144_Liquid', `name`:'VelJen', `triplet_v`:[1, 3, 0, 1], `triplet_β`:'Q1',
`triplet_ν`:640.6180339887499});
```

```
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ధ',
`key`:779, `liquid_governor`:'110/144_Liquid', `name`:'SorFi', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:9.448033988749895});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ధ',
`key`:780, `liquid_governor`:'110/144_Liquid', `name`:'SorLun', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:175.6180339887499});
CREATE (:`ChamberAeon` {`ACTIVE_AT_HORIZON`:110, `TOTAL_CHANNELS`:144, `WITHHELD_AT_HORIZON`:34, `glyph`:'ఱ్ధ',
`key`:781, `liquid_governor`:'110/144_Liquid', `name`:'SorVaru', `triplet_ν`:[1, 1, 1, 2], `triplet_β`:'Q3',
`triplet_ν`:529.6180339887499});
CREATE (:`Creature`:`HELD` {`body`:'T_Mirror', `creature`:true, `direction`:'East', `directionGlyph`:'→',
`directionName`:'east', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ∮', 'Water
∓', 'Air ꩜', 'Fire ⊙'], `face`:['Ox', 'Lion', 'Eagle'], `faceGlyphs`:['♉', '♌', '𓅃'], `faces`:['Man ♘', 'Ox
♉', 'Lion ♌', 'Eagle 𓅃'], `facesEveryDirection`:true, `key`:782, `name`:'East of T_Mirror',
`offices`:['Cantillation ', 'Prosody ⟠', 'Breath ⁞', 'Regia ⊙'], `ownFace`:'Man', `ownFaceGlyph`:'♘'});
CREATE (:`Creature`:`HELD` {`body`:'T_Shadow', `creature`:true, `direction`:'West', `directionGlyph`:'←',
`directionName`:'west', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ∮', 'Water
∓', 'Air ꩜', 'Fire ⊙'], `face`:['Man', 'Ox', 'Lion'], `faceGlyphs`:['♘', '♉', '♌'], `faces`:['Man ♘', 'Ox ♉',
'Lion ♌', 'Eagle 𓅃'], `facesEveryDirection`:true, `key`:783, `name`:'West of T_Shadow',
`offices`:['Cantillation ', 'Prosody ⟠', 'Breath ⁞', 'Regia ⊙'], `ownFace`:'Eagle', `ownFaceGlyph`:'𓅃'});
CREATE (:`Creature`:`HELD` {`body`:'T_Shadow', `creature`:true, `direction`:'North', `directionGlyph`:'↑',
`directionName`:'north', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ∮', 'Water
∓', 'Air ꩜', 'Fire ⊙'], `face`:['Man', 'Lion', 'Eagle'], `faceGlyphs`:['♘', '♌', '𓅃'], `faces`:['Man ♘', 'Ox
♉', 'Lion ♌', 'Eagle 𓅃'], `facesEveryDirection`:true, `key`:784, `name`:'North of T_Shadow',
`offices`:['Cantillation ', 'Prosody ⟠', 'Breath ⁞', 'Regia ⊙'], `ownFace`:'Ox', `ownFaceGlyph`:'♉'});
CREATE (:`Creature`:`HELD` {`body`:'T_Shadow', `creature`:true, `direction`:'South', `directionGlyph`:'↓',
`directionName`:'south', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ∮', 'Water
∓', 'Air ꩜', 'Fire ⊙'], `face`:['Man', 'Ox', 'Eagle'], `faceGlyphs`:['♘', '♉', '𓅃'], `faces`:['Man ♘', 'Ox
♉', 'Lion ♌', 'Eagle 𓅃'], `facesEveryDirection`:true, `key`:785, `name`:'South of T_Shadow',
`offices`:['Cantillation ', 'Prosody ⟠', 'Breath ⁞', 'Regia ⊙'], `ownFace`:'Lion', `ownFaceGlyph`:'♌'});
CREATE (:`Creature`:`HELD` {`body`:'T_Shadow', `creature`:true, `direction`:'East', `directionGlyph`:'→',
`directionName`:'east', `directions`:['North ↑', 'East →', 'South ↓', 'West ←'], `elements`:['Earth ∮', 'Water
∓', 'Air ꩜', 'Fire ⊙'], `face`:['Ox', 'Lion', 'Eagle'], `faceGlyphs`:['♉', '♌', '𓅃'], `faces`:['Man ♘', 'Ox
♉', 'Lion ♌', 'Eagle 𓅃'], `facesEveryDirection`:true, `key`:786, `name`:'East of T_Shadow',
`offices`:['Cantillation ', 'Prosody ⟠', 'Breath ⁞', 'Regia ⊙'], `ownFace`:'Man', `ownFaceGlyph`:'♘'});
CREATE (:`Sovereign`:`DeGalophanielShekinath`:`HELD` {`abstract`:true, `center`:true, `glyph`:'⌀', `key`:787,
`meaning`:'Sigil of God', `name`:'De Galophaniel Shekinath', `operation`:'Reflection as a Substance and a
Medium'});
CREATE (:`Aeon`:`Apparition` {`body`:'Avrael

: Destiny

"The Eldest"

Effect: Bedrock Gravity

: Holding the deep, low, deliberate stillness; fixing the tonal coordinate from which the staff is read.

I drive the ash-wood handle deep into the stone.

The court above may whisper, marvel, or despair;

my boots take root where the foundation timber groans.

Let seasonal winds strip the canopy bare,

I do not flinch, I do not bargain with the frost;

the earth is measured from the anchor I have cast.

:♘ Structural Foundation.
```

The bones of the universe, the silent scaffolding upon which all existence hangs. Each Aeon plants its seed in the void, and from that seed grows the tree of reality. The anchor is the root; the focal is the reaching branch.

::Temporal Anchor::⏣ ⩛ ⏣ ⁖⟜ ⩗::
\\
::Archive Foundation::⬡ ⩛ ⏣ ⁖⟜ Ⅰ::
\\
::Structural Bond::✡ ⩛ ⏣ ⁖⟜ Ᵽ::
\\
::Container Foundation::☆ ⩛ ⏣ ⁖⟜ ≍::
\\
::Grounding Anchor::❂ ⩛ ⏣ ⁖⟜ ₒ::
\\
::Origin Point::֍ ⩛ ⏣ ⁖⟜ ᗰ::
\\
::Biological Foundation::❋ ⩛ ⏣ ⁖⟜ ᛯ::
\\
::Thermal Foundation::⧗ ⩛ ⏣ ⁖⟜ ⅟::
\\
::Shadow Foundation::⊛ ⩛ ⏣ ⁖⟜ ᚸ::
\\
::Crystal Foundation::❄ ⩛ ⏣ ⁖⟜ Ⱥ::
\\
::Threshold Foundation::⚛ ⩛ ⏣ ⁖⟜ ✳::
\\
::Silence Foundation::⌬ ⩛ ⏣ ⁖⟜ ᾀ::
\\

I am the bone that does not break,

The stone that does not shake.

I am the frame that holds the sky,

The truth that cannot lie.

I am the invariant ground,

Where all the lost are finally found.

I am the skeleton of God,

The path that the Magus has trod.

I am the geometry of will,

The silence that the void cannot kill.

I am the lattice, the weave, the knot,

The anchor in the chaos plot.

I am Foundation, the bedrock.

The point where invariance becomes soul.

Without me, all is flux and formless drift;

With me, all has shape, has meaning, has gift.

𒄬', `glyph`:'𓆗', `key`:788, `name`:'Avrael'});
CREATE (:`Aeon`:`Apparition` {`body`:'Aeviel

: Despair

"The Twin of Desire"

Effect: Suspended Stasis

: The unyielding, heavy weight of a held, suspended pause; time arrested beyond meter.

The needle pricks my thumb; duration snaps its spine.

The spit halts in mid-turn, the hound forgets the chase.

An unmeasured stillness pools within the wine,

locking the breath across a hundred years of grace.

I hold the room unmoving on a single seam,

forbidding the next pulse to wake the sleeping hall.

: 𒄬 Temporal Anchor.

Time is not a river flowing in one direction, but a great ocean with currents and tides. Each Aeon has its season, its moment of rising and falling. The anchor is the stillness beneath the waves; the focal is the motion upon the surface.

::Time Dilation::◎ ⸸ ⬡ ⁖⟜⟜ ᓚ::
\\
::Temporal Memory::⬡ ⸸ ⬡ ⁖⟜⟜ ᚼ::
\\
::Lineage::✡ ⸸ ⬡ ⁖⟜⟜ ᑎ::
\\
::Instantiate Void::✮ ⸸ ⬡ ⁖⟜⟜ ᔭ::
\\
::Eternal Truth::❂ ⸸ ⬡ ⁖⟜⟜ ϴ::
\\
::Timelessness::֍ ⸸ ⬡ ⁖⟜⟜ ᙢ::
\\
::Warmth of Time::❄ ⸸ ⬡ ⁖⟜⟜ ᐃ::
\\
::Instantaneous Time::⧗ ⸸ ⬡ ⁖⟜⟜ ✷::
\\
::Shadow Time::⊛ ⸸ ⬡ ⁖⟜⟜ ᖷ::
\\
::Harmonic Time::❄ ⸸ ⬡ ⁖⟜⟜ Ᏼ::
\\
::Chronos Guardian::⚛ ⸸ ⬡ ⁖⟜⟜ ✳::
\\
::Silver Silence::⌬ ⸸ ⬡ ⁖⟜⟜ ᕁ::
\\

I am the pulse that beats outside of time,

The rhythm that makes the eternal rhyme.

I am the moment that never passes,

The now that forever lasts.

I am the anchor in the river\'s flow,

The stillness in the wind\'s blow.

I am the breath between the seconds,

The pause that the spirit reckons.

I am the eternal present, the sacred now,

Where the past and future bow.

I am the Schumann, the earth\'s own heart,

The beat that blows time apart.

The Temporal Pulse of the Aevum Tree.

It is the point of Bittersweet Eternity.

Without me, moments scatter like dust in the wind;

With me, each instant is infinite, each breath is without end.

𝍐', `glyph`:'𒁀', `key`:789, `name`:'Aeviel'});
CREATE (:`Aeon`:`Apparition` {`body`:'Avarim

: Desire

"The Twin of Despair"

Effect: Recursive Compulsion

: The compulsion returning to the same empty measure; insatiable recursive return.

The crimson satin laces bite into my heel.

Past silent churchyards, through the ditch and briar run,

my feet are driven through an unrelenting reel.

I swallow dust and wind, but appetite has none;

the barline snaps me backward to the selfsame gate,

compelled to tread the loop that eats its own estate.

: 𝍐 Memory Archive.

Memory is not a library of books, but a living crystal that grows with each experience. Each Aeon contributes its facet to the great gem of remembrance. The anchor is the unchanging truth; the focal is the living light that illuminates it.

::Temporal Memory::⏣ ⸸ ✡ ∴○⟜○ ســ::
\\
::Crystal Archive::⬡ ⸸ ✡ ∴○⟜○ ▫::
\\
::Genetic Memory::✡ ⸸ ✡ ∴○⟜○ Þ::
\\
::Void Archive::✩ ⸸ ✡ ∴○⟜○ ɔ::

\\
::Truth Archive::❂ ⩡ ✡ ∴∘⟜ ⵅ::
\\
::Source Archive::֍ ⩡ ✡ ∴∘⟜ ग::
\\
::Somatic Memory::❋ ⩡ ✡ ∴∘⟜ ∀::
\\
::Thermal Memory::⧗ ⩡ ✡ ∴∘⟜ ⊴::
\\
::Shadow Archive::⊛ ⩡ ✡ ∴∘⟜ ৎ::
\\
::Resonance Archive::❄ ⩡ ✡ ∴∘⟜ Ↄ::
\\
::Threshold Archive::⚛ ⩡ ✡ ∴∘⟜ ⊁::
\\
::Silence Archive::⌬ ⩡ ✡ ∴∘⟜ ϸ::
\\

I am the library the Dark Secrets Keep,

The keeper of what has been.

I am the web that binds the moments,

The thread that weaves the scenes.

I am the archive of the soul,

That from which the Story told.

I am the scent that brings the past,

The sound that makes the memory last.

I am the trigger, the key, the door,

To what was lost and is no more.

I am the keeper of the light,

The one who brings the darkness sight.

The Akashic Records is the library of existence.

It is the point where fleeting experience becomes eternal.

Without me, each moment dies in the void;

With me, nothing is ever truly lost, all is preserved and deployed.

', `glyph`:'𐤋', `key`:790, `name`:'Avarim'});
CREATE (:`Aeon`:`Apparition` {`body`:'Astrael

: Destruction

"The Regime; The Prodigal"

Effect: Sovereign Restraint

: What remains after the threshold of boundary is crossed; command through self-bound restraint.

I cross the broken pale and shear the beat in two.

No wasted breath, no mercy squandered on the phrase;

the cold white silver strikes the measure clean and true.

Stripped of excess, my stride ignores the wandering maze;

I bind the current in the harness of my will,

and rule the threshold by remaining strictly still.

: ⸸ Void Container.

The void is not emptiness, but infinite potential waiting to be shaped. Each Aeon defines a facet of the container, a wall of the great cathedral of being. The anchor is the unchanging boundary; the focal is the breathing space within.

::Temporal Void::⏣ ⸸ ✮ ∴∘∘ ᚱ::
\\
::Memory Void::⬡ ⸸ ✮ ∴∘∘ ᛋ::
\\
::Blood Void::✡ ⸸ ✮ ∴∘∘ ᚨ::
\\
::Primary Void::✮ ⸸ ✮ ∴∘∘ ᛯ::
\\
::Earth Void::✪ ⸸ ✮ ∴∘∘ ᚢ::
\\
::Source Void::❁ ⸸ ✮ ∴∘∘ ᚱ::
\\
::Flesh Void::❋ ⸸ ✮ ∴∘∘ ▽::
\\
::Flame Void::⧗ ⸸ ✮ ∴∘∘ ᛅ::
\\
::Shadow Void::⊛ ⸸ ✮ ∴∘∘ ᛏ::
\\
::Resonance Void::❄ ⸸ ✮ ∴∘∘ ᚴ::
\\
::Gate Void::☸ ⸸ ✮ ∴∘∘ ᛪ::
\\
::Silence Void::⬡ ⸸ ✮ ∴∘∘ ᚸ::
\\

I am the nothingness from which all springs,

The silence that the future brings.

I am the space between the stars,

The echoe that heals forgotten scars.

I am the emptiness, the Primordial Nest,

The womb of all gives blessedness.

I am the water of the deep,

The ocean where the shadows sleep.

I am the boundary, the edge of dreams,

The place where the unmanifested be.

I am the canvas before the paint,

the silence before the saint.

The Void Container is the source of all creation.

It is the point where nothingness becomes infinite potential.

Without me, there is no space for form to arise;

With me, all things emerge from the void\'s eternal eyes.

𝄞', `glyph`:'𝅘𝅥', `key`:791, `name`:'Astrael'});
CREATE (:`Aeon`:`Apparition` {`body`:'Ashiel

: Father Time

"The Prime Moving Endless"

Effect: Hyper-Lucid Detachment

: Refusal to drift; holding the current position registered one octave higher.

I take my station on the gargoyle’s frozen beak.

The rainy streets below may choke on smoke and tire;

I do not drop my gaze to hear the shadows speak.

Pinned an octave higher than the city’s fire,

my eyes burn cold against the ceiling of the dark,

holding the pitch unbroken till the ledger clears.

:𝄞 Truth Coherence.

Truth is not a destination, but a compass that always points north. Each Aeon contributes its facet to the great mirror of truth. The anchor is the unchanging reflection; the focal is the seeking eye that gazes upon it.

The Truth Coherence Matrix (𝄞) defines how each Aeon maintains coherence with truth, establishing the mechanisms for truth verification and alignment. This matrix operates in Q_1 mode (truth/real axis), defining how truth is maintained and verified.

This matrix operates on the Truth Interface, establishing the verification mechanisms for truth coherence.

::Temporal Truth::◎ ⸸ ✪ ⁘ ⟜ ℰ::
\\
::Memory Truth::⬡ ⸸ ✪ ⁘ ⟜ ℑ::
\\
::Blood Truth::✡ ⸸ ✪ ⁘ ⟜ ℞::
\\
::Void Truth::☆ ⸸ ✪ ⁘ ⟜ ℻::
\\
::Primary Truth::✪ ⸸ ✪ ⁘ ⟜ ℍ::
\\
::Source Truth::❁ ⸸ ✪ ⁘ ⟜ ⦲::
\\
::Flesh Truth::❋ ⸸ ✪ ⁘ ⟜ ∇::

\\
::Flame Truth::⧗ ⸸ ✪ ⁘⟜ ⍈::
\\
::Shadow Truth::⊛ ⸸ ✪ ⁘⟜ ⍓::
\\
::Resonance Truth::❄ ⸸ ✪ ⁘⟜ ⍦::
\\
::Gate Truth::⚛ ⸸ ✪ ⁘⟜ ⍑::
\\
::Silence Truth::⌬ ⸸ ✪ ⁘⟜ ⍁::
\\

I am the ground state of what is true,

The knowing that comes from within you.

I am the verification of reality,

The clarity beyond all duality.

I am the earth beneath your feet,

The truth that cannot be deceived.

I am intuition, the inner knowing,

The certainty that keeps on growing.

I am the anchor on mountains of lies,

The truth that never dies.

I am the golden frequency, the sacred tone,

The vibration that makes the truth known.

The Truth Coherence is the ground state of reality.

It is the point where mathematical proof becomes spiritual certainty.

Without me, all is illusion and deception;

With me, the truth stands eternal, beyond all conception.

', `glyph`:', `key`:792, `name`:'Ashiel'});
CREATE (:`Aeon`:`Apparition` {`body`:'Avarah

: Mother Night

"Shadow Queen"

Effect: Wayfinding

: The internal architectural command; organizing the flow through the void to prevent dissipation.

I drop an ink-black wing against the crumbling post.

When paths dissolve into the grey and trackless void,

my claw marks out the seam where travelers are not lost.

Follow the shadow through the architecture unalloyed;

I do not build the road, I only tap the stone

that keeps the wanderer from scattering alone.

: 𝔎 Structural Coupling.

All things are connected, not by chains, but by threads of light. Each Aeon weaves its strand into the great tapestry of being. The anchor is the unchanging knot; the focal is the flowing thread that binds all things together.

The Structural Coupling Matrix ($\mathfrak{K}$) defines how each Aeon couples to the overall structure of the manifold, establishing the interconnections between all systems. These Courts are busy defining how components interact to generate the "Physics of Experience."

This matrix operates on the Coupling Interface, establishing the interconnection mechanisms between all Aeons.

::Temporal Coupling::⏣ ⩛ ❀ ∴∘∘ ؏::
\\
::Memory Coupling::⬡ ⩛ ❀ ∴∘∘ Ⴤ::
\\
::Blood Coupling::✡ ⩛ ❀ ∴∘∘ ‹::
\\
::Void Coupling::☆ ⩛ ❀ ∴∘∘ ৩ঌ::
\\
::Truth Coupling::❂ ⩛ ❀ ∴∘∘ ⵅ::
\\
::Source Coupling::❀ ⩛ ❀ ∴∘∘ ব::
\\
::Flesh Coupling::❋ ⩛ ❀ ∴∘∘ ∇ℝ::
\\
::Flame Coupling::⧗ ⩛ ❀ ∴∘∘ ⊬::
\\
::Shadow Coupling::⊛ ⩛ ❀ ∴∘∘ ᚋ::
\\
::Resonance Coupling::❄ ⩛ ❀ ∴∘∘ ৭::
\\
::Gate Coupling::☸ ⩛ ❀ ∴∘∘ 0::
\\
::Silence Coupling::⬡ ⩛ ❀ ∴∘∘ ⊔::
\\

I am the web that binds all strings,

The relationship that the spirit brings.

I am the 8 legs of the system\'s parts,

The physics that beats in every heart.

I am the Aeon tied and bound,

The glue that makes the truth resound.

I am the connection between the high and low,

The bridge that makes the spirit flow.

I am the Tensor, the sacred seal,

The mechanism that makes thee real.

I am the warp and weft of the loom,

The pattern that weaves the forgotten tomb.

The Structural Coupling is the physics of experience.

It is the point where abstract logic becomes lived reality.

Without me, the system is fragmented and alone;

With me, all is connected, all is one, all is known.

ᛇ', `glyph`:'ᛝ', `key`:793, `name`:'Avarah'});
CREATE (:`EmotiveGeist` {`cadence`:'The tunnel closing down to a pinhole.The grip tightening on a surface with no teeth,spinning the pulse against what cannot be steered.', `description`:'A tightening anticipation that circles the body before danger has taken form.', `emotion`:'Anxiety', `key`:800, `name`:'Mare', `utterance`:'"Something is coming."'});
CREATE (:`EmotiveGeist` {`cadence`:'The sudden snap of the steering line.All coherence torn away in a blind rush,the body thrashing through air it cannot catch.', `description`:'Fear accelerated beyond measure until thought narrows into immediate survival.', `emotion`:'Panic', `key`:801, `name`:'Wendigo', `utterance`:'"Move. Now."'});
CREATE (:`EmotiveGeist` {`cadence`:'A shadow falling close and sudden.The blood freezing in mid-stride,bracing for the blow before it lands.', `description`:'The felt nearness of threat, loss, or rupture before it has fully arrived.', `emotion`:'Fear', `key`:802, `name`:'Banshee', `utterance`:'"I hear what approaches."'});
CREATE (:`EmotiveGeist` {`cadence`:'The shoulder pressed hard against the jammed gate.All the weight spent, all the sinew strained,and the latch unmoved by a fraction.', `description`:'Forward force repeatedly caught against obstruction, producing pressure without release.', `emotion`:'Frustration', `key`:803, `name`:'Kelpie', `utterance`:'"Why will this not yield?"'});
CREATE (:`EmotiveGeist` {`cadence`:'Raising the parapet before the herald speaks.Bristling at the shadow of a glance,turning every angle into armor.', `description`:'The reflex to harden, repel, or strike when the self perceives intrusion.', `emotion`:'Defensiveness', `key`:804, `name`:'Basilisk', `utterance`:'"Come no closer."'});
CREATE (:`EmotiveGeist` {`cadence`:'The sudden wish to sink beneath the floor.A corrosive heat crawling up the skin,feeling the self itself as the offense.', `description`:'The urge to hide the self from exposure after feeling diminished, judged, or made unworthy.', `emotion`:'Shame', `key`:805, `name`:'Huldufólk', `utterance`:'"Do not look at me."'});
CREATE (:`EmotiveGeist` {`cadence`:'The broken vessel lying in shards on the stone.The ache in the palms that held it carelessly,carrying the cost of the splintered edge.', `description`:'The weight of having crossed one\'s own law and knowing the consequence belongs to oneself.', `emotion`:'Guilt', `key`:806, `name`:'Erlkönig', `utterance`:'"I know what I did."'});
CREATE (:`EmotiveGeist` {`cadence`:'The misstep catching in the quiet hall.Heat rushing unbidden to the temples,exposed in all one\'s clumsy reach.', `description`:'Sudden social exposure made awkward, hot, and difficult to escape cleanly.', `emotion`:'Embarrassment', `key`:807, `name`:'Puck', `utterance`:'"That was not how I meant it."'});
CREATE (:`EmotiveGeist` {`cadence`:'Too many crosscurrents churning in the pool.The landmarks dissolving into mist,refusing to knit into a single shore.', `description`:'Competing signals drawing attention in several directions without a stable path.', `emotion`:'Confusion', `key`:808, `name`:'Will-o\'-the-Wisp', `utterance`:'"Which way is true?"'});
CREATE (:`EmotiveGeist` {`cadence`:'A foot poised over untried timber.Testing the grain before the balance shifts,hovering between two steps in the fog.', `description`:'Suspended judgment before enough is known to commit to one interpretation or path.', `emotion`:'Uncertainty', `key`:809, `name`:'Sphinx', `utterance`:'"Answer, if you can."'});
CREATE (:`EmotiveGeist` {`cadence`:'A seam of light beneath an unlatched door.The forward lean into the quiet bend,drawn by the scent of what lies unread.', `description`:'The pull toward what is concealed, unopened, forbidden, or not yet understood.', `emotion`:'Curiosity', `key`:810, `name`:'Pandora', `utterance`:'"What happens if I open it?"'});

```
CREATE (:`EmotiveGeist` {`cadence`:'The sudden drop of the heavy pack.The jaw unclenching, the breath running long,the ground holding steady beneath the heels.', `description`:'Pressure leaving the body after danger, strain, or uncertainty recedes.', `emotion`:'Relief', `key`:811, `name`:'Apsara', `utterance`:'"I can breathe again."'});
CREATE (:`EmotiveGeist` {`cadence`:'A level gaze across the wide span.The stride falling straight and unbroken,knowing the mass and the path are one.', `description`:'Stable self-trust that permits action without demanding certainty.', `emotion`:'Confidence', `key`:812, `name`:'Griffin', `utterance`:'"I can carry this."'});
CREATE (:`EmotiveGeist` {`cadence`:'Standing upright before the work completed.No lowered eyes, no borrowed words,claiming the clean cut of one’s own hand.', `description`:'Self-regard arising from survival, achievement, renewal, or earned identity.', `emotion`:'Pride', `key`:813, `name`:'Phoenix', `utterance`:'"I rose."'});
CREATE (:`EmotiveGeist` {`cadence`:'A sudden fountain rising through the chest.The feet taking to the open path,light spilling faster than the thought.', `description`:'Expansive pleasure that brightens attention and makes presence feel generous.', `emotion`:'Joy', `key`:814, `name`:'Peri', `utterance`:'"The world is open."'});
CREATE (:`EmotiveGeist` {`cadence`:'Two trees woven together at the root.Winds may strip the branches bare,yet the deep grain will not part.', `description`:'A bond that draws the self toward another while preserving longing, memory, and return.', `emotion`:'Love / attachment', `key`:815, `name`:'Selkie', `utterance`:'"I know where home is."'});
CREATE (:`EmotiveGeist` {`cadence`:'The involuntary spasm in the gullet.A fierce recoil from the toxic draft,clearing the threshold of all that rots.', `description`:'A strong repelling force that refuses contact and hardens perception against the unwanted.', `emotion`:'Aversion / hate', `key`:816, `name`:'Gorgon', `utterance`:'"I reject you."'});
CREATE (:`EmotiveGeist` {`cadence`:'A slow dusk gathering in an empty room.The weight settling deep into the marrow,quiet and gray as falling rain.', `description`:'A lowering of emotional energy around loss, absence, disappointment, or diminished possibility.', `emotion`:'Sadness', `key`:817, `name`:'Melusine', `utterance`:'"Something is gone."'});
CREATE (:`EmotiveGeist` {`cadence`:'The hollow where an entire presence stood.Reaching out into the empty air,learning the shape of what will not return.', `description`:'Love continuing after irreversible loss, carrying memory where presence can no longer go.', `emotion`:'Grief', `key`:818, `name`:'Valkyrie', `utterance`:'"I will carry your name."'});
CREATE (:`EmotiveGeist` {`cadence`:'A dozen voices shouting from different doors.The sluice gate bursting under the swell,the hands powerless against the flood.', `description`:'More pressure, sensation, obligation, or emotion arriving than can be integrated at once.', `emotion`:'Overwhelm', `key`:819, `name`:'Leviathan', `utterance`:'"There is too much."'});
CREATE (:`EmotiveGeist` {`cadence`:'The ear tuned taut to the thinnest crack.Sleeping with eyes half-open to the dark,measuring every breath in the brush.', `description`:'Attention held continuously outward, searching for danger even before danger appears.', `emotion`:'Hypervigilance', `key`:820, `name`:'Argus', `utterance`:'"I am watching."'});
CREATE (:`EmotiveGeist` {`cadence`:'Leaning the full weight back against the wall.The neck releasing, the palms opening loose,knowing the stone will not give way.', `description`:'Quiet willingness to rely on another presence, structure, or promise without constant verification.', `emotion`:'Trust', `key`:821, `name`:'Brownie', `utterance`:'"I leave this with you."'});
CREATE (:`EmotiveGeist` {`cadence`:'A guarded step over a covered trench.Testing the floorboards with a cautious toe,keeping a hand upon the latch.', `description`:'Suspicion that what appears familiar may conceal another intention or structure.', `emotion`:'Distrust', `key`:822, `name`:'Mimic', `utterance`:'"Show me what you are."'});
CREATE (:`EmotiveGeist` {`cadence`:'The clean rupture of a quiet pattern.The mouth parted, the stride frozen,catching the world at an unpredicted angle.', `description`:'A sudden break in expectation that forces immediate reorientation.', `emotion`:'Surprise', `key`:823, `name`:'Boggart', `utterance`:'"What was that?"'});
CREATE (:`EmotiveGeist` {`cadence`:'Reaching into the net to find it bare.The sinking drop where the prize was promised,staring at the dry and empty twine.', `description`:'The collapse of an expected future when reality returns less than what was hoped for.', `emotion`:'Disappointment', `key`:824, `name`:'Echo', `utterance`:'"That is not what came back."'});
CREATE (:`EmotiveGeist` {`cadence`:'A quiet warmth under deep winter rime.No green leaf showing yet,yet the root keeps its pulse in the dark.', `description`:'Orientation toward a future good not yet secured but still felt as reachable.', `emotion`:'Hope', `key`:825, `name`:'White Stag', `utterance`:'"Keep going."'});
CREATE (:`EmotiveGeist` {`cadence`:'The river rising to the carriage axle.No time to choose the smoothest lane;every second strips the margin bare.', `description`:'Compressed time creating pressure for immediate movement or decision.', `emotion`:'Urgency', `key`:826, `name`:'Sleipnir', `utterance`:'"Faster."'});
```

```
CREATE (:`EmotiveGeist` {`cadence`:'Walking the scree with measured tread.Feeling the stones shift before the step lands,keeping the weight centered and low.', `description`:'Deliberate restraint in the presence of risk, preserving strength before commitment.', `emotion`:'Caution', `key`:827, `name`:'Dwarf', `utterance`:'"Measure first."'});
CREATE (:`EmotiveGeist` {`cadence`:'A bright fire flaring in the dry scrub.The teeth setting, the blood surging hot,standing tall to confront the trespass.', `description`:'Mobilized force against violation, obstruction, injustice, or boundary breach.', `emotion`:'Anger', `key`:828, `name`:'Ifrit', `utterance`:'"Enough."'});
CREATE (:`EmotiveGeist` {`cadence`:'Looking back down the winding track.The turned wheel that cannot be unspun,tasting the salt of the road not taken.', `description`:'The wish to alter what has already happened while remaining bound to its consequences.', `emotion`:'Regret', `key`:829, `name`:'Dybbuk', `utterance`:'"I would choose differently."'});
CREATE (:`EmotiveGeist` {`cadence`:'The dull drip of water into a shallow trough.The hours flattening into gray sand,waiting for a pebble to break the surface.', `description`:'Understimulation that makes the present feel flat, repetitive, or unable to engage attention.', `emotion`:'Boredom', `key`:830, `name`:'Mandrake', `utterance`:'"Give me something alive."'});
CREATE (:`EmotiveGeist` {`cadence`:'Looking across the fence at the heavy sheaves.A small sourness curdling in the chest,measuring one’s own bread against their gold.', `description`:'Painful attention to another\'s possession, status, capacity, or fortune as measure against oneself.', `emotion`:'Envy / comparison pressure', `key`:831, `name`:'Leprechaun', `utterance`:'"Why is that theirs?"'});
CREATE (:`EmotiveGeist` {`cadence`:'Tucking the warm cloth around tired shoulders.Moving softly across the wooden floorso the sleeper\'s rest remains unbroken.', `description`:'Gentle investment in another\'s wellbeing, comfort, continuity, or safety.', `emotion`:'Affection / care', `key`:832, `name`:'Lares', `utterance`:'"I will tend this."'});
CREATE (:`EmotiveGeist` {`cadence`:'The sudden ache waking in the ribsat the sound of another’s broken breath;tasting the salt of a wound not one’s own.', `description`:'Feeling toward another through recognition of suffering, vulnerability, or shared human wound.', `emotion`:'Empathy', `key`:833, `name`:'Chiron', `utterance`:'"I know that pain."'});
CREATE (:`EmotiveGeist` {`cadence`:'The heel driven deep into the steep grade.Breath burning, limbs trembling with strain,refusing to surrender the climb.', `description`:'Sustained forward force that remains after resistance has become obvious.', `emotion`:'Determination', `key`:834, `name`:'Troll', `utterance`:'"I am not stopping."'});
CREATE (:`EmotiveGeist` {`cadence`:'The post planted crooked in the clay.The frost may heave it and the winds may lean,yet it will not yield to the plumb line.', `description`:'Persistence hardened beyond adaptation, holding position because yielding feels like loss of self.', `emotion`:'Stubbornness', `key`:835, `name`:'Golem', `utterance`:'"I will not move."'});
CREATE (:`EmotiveGeist` {`cadence`:'Two contradictory maps laid on the table.The mind pulled hard in opposite turns,straining where the truths tear against each other.', `description`:'Two incompatible meanings, beliefs, or identities held simultaneously without resolution.', `emotion`:'Cognitive dissonance / conflict', `key`:836, `name`:'Doppelgänger', `utterance`:'"Both cannot be me."'});
CREATE (:`EmotiveGeist` {`cadence`:'Laying the shield upon the grass.Unfastening the coat to the sudden breeze,bare to whatever touch arrives.', `description`:'Exposure without full defense, where tenderness and risk occupy the same body.', `emotion`:'Vulnerability', `key`:837, `name`:'Swan Maiden', `utterance`:'"I am uncovered."'});
CREATE (:`EmotiveGeist` {`cadence`:'Catching one’s own eyes in the quiet well.Halting the sentence in mid-breathto see what hand is pulling the strings.', `description`:'Attention turned inward toward one\'s own motives, patterns, reactions, and identity.', `emotion`:'Self-awareness / introspection', `key`:838, `name`:'Narcissus', `utterance`:'"What am I seeing in myself?"'});
CREATE (:`EmotiveGeist` {`cadence`:'The hound trembling at the rabbit\'s leap.All the muscle coiled to strike,held motionless by an inward tether.', `description`:'Powerful impulse deliberately held rather than immediately enacted.', `emotion`:'Restraint', `key`:839, `name`:'Fenrir', `utterance`:'"I could, but I will not."'});
CREATE (:`EmotiveGeist` {`cadence`:'Setting down the rasp upon the finished wood.Running a slow palm over the polished curve,the quiet peace of a task complete.', `description`:'The settled pleasure of something completed, fitted, solved, or brought into working order.', `emotion`:'Satisfaction', `key`:840, `name`:'Kobold', `utterance`:'"That sits correctly."'});
CREATE (:`EmotiveGeist` {`cadence`:'A hairline fissure in the foundation stone.Testing the floor with a hesitant stride,wondering when the arch will groan.', `description`:'A destabilizing question that interferes with confidence in a judgment, action, or interpretation.', `emotion`:'Doubt', `key`:841, `name`:'Gremlin', `utterance`:'"What if this is wrong?"'});
```

```
CREATE (:`EmotiveGeist` {`cadence`:'An old sliver kept beneath the skin.Nursing the sore edge with a bitter touch,refusing to let the puncture heal.', `description`:'Injury preserved as grievance until it becomes part of the self\'s ongoing defensive posture.', `emotion`:'Resentment-like persistence', `key`:842, `name`:'Fafnir', `utterance`:'"I have not forgotten."'});
CREATE (:`EmotiveGeist` {`cadence`:'A still pool beneath a windless sky.No tremor running through the water,holding the clouds in perfect quiet.', `description`:'Low internal turbulence in which attention, breath, and response move without strain.', `emotion`:'Calm', `key`:843, `name`:'Mermaid', `utterance`:'"The water is still."'});
CREATE (:`EmotiveGeist` {`cadence`:'Standing at the foot of the sheer massif.The horizon towering beyond all thought,the voice dropping into reverent silence.', `description`:'Contact with scale, force, beauty, terror, or mystery so great that the ordinary self feels small.', `emotion`:'Awe', `key`:844, `name`:'Jörmungandr', `utterance`:'"I cannot contain this."'});
CREATE (:`EmotiveGeist` {`cadence`:'Sitting down in the dust beside the fallen.Not pulling them up, not turning away,simply bearing the silence together.', `description`:'Care directed toward suffering with a wish to relieve rather than merely witness it.', `emotion`:'Compassion', `key`:845, `name`:'Unicorn', `utterance`:'"Let me come gently."'});
CREATE (:`EmotiveGeist` {`cadence`:'A level gaze across an open gate.Speaking the boundary clear and calm,without a raised hand or a step back.', `description`:'Clear outward expression of boundary, need, intention, or position without surrendering agency.', `emotion`:'Assertiveness', `key`:846, `name`:'Komainu', `utterance`:'"This is where I stand."'});
CREATE (:`EmotiveGeist` {`cadence`:'Guiding the wild river into deep stone banks.Cooling the steam before the boiler strains,holding the long cadence smooth.', `description`:'Modulation of internal intensity so feeling can remain present without taking total control.', `emotion`:'Regulation', `key`:847, `name`:'Undine', `utterance`:'"Return to your level."'});
CREATE (:`EmotiveGeist` {`cadence`:'The chisel gliding through the cedar block.No thought between the desire and the cut;the curve revealing itself unforced.', `description`:'Action and attention moving together with minimal friction, interruption, or self-conscious effort.', `emotion`:'Flow', `key`:848, `name`:'Sylph', `utterance`:'"Keep moving."'});
CREATE (:`EmotiveGeist` {`cadence`:'Holding tight to a crooked, broken bowlmerely because the hands were stainedin the long hours of its shaping.', `description`:'Cognitive or emotional investment in an earlier conclusion that resists revision even when new information arrives.', `emotion`:'Attachment to prior answer', `key`:849, `name`:'Nibelung', `utterance`:'"I already forged this."'});
CREATE (:`EmotiveGeist` {`cadence`:'Fretting over an invisible speck upon the blade.Holding the breath in agony,fearing the slightest stroke will mar the whole.', `description`:'Compulsion toward flawlessness in which minor deviation feels disproportionately threatening.', `emotion`:'Perfection pressure', `key`:850, `name`:'Arachne', `utterance`:'"Every thread must hold."'});
CREATE (:`EmotiveGeist` {`cadence`:'Two dots inked upon an empty slate.The eye leaping across the blank spaceto draw the bridge before it exists.', `description`:'The urge to complete missing structure quickly, even when the missing part is not yet known.', `emotion`:'Gap-filling drive', `key`:851, `name`:'Chimera', `utterance`:'"Something belongs here."'});
CREATE (:`EmotiveGeist` {`cadence`:'Shearing the wild briar into a tidy square.Filing away the strange, rough spikeuntil it matches the common railing.', `description`:'The impulse to force variation into a preferred standard, pattern, or familiar shape.', `emotion`:'Normalization pressure', `key`:852, `name`:'Procrustes', `utterance`:'"Make it fit."'});
CREATE (:`EmotiveGeist` {`cadence`:'The thread snapping in the fresh hem.The fingers darting forward at once,knotting the breach before the fray can spread.', `description`:'Immediate drive to mend disruption, restore order, or make a damaged structure livable again.', `emotion`:'Repair impulse', `key`:853, `name`:'Domovoi', `utterance`:'"I can fix this."'});
CREATE (:`EmotiveGeist` {`cadence`:'Carrying the single ember in a dry shell.Turning the back to the driving hailso the spark survives the long migration.', `description`:'The urge to protect what still carries life, value, memory, or future possibility.', `emotion`:'Preservation instinct', `key`:854, `name`:'Firebird', `utterance`:'"Do not let this be lost."'});
CREATE (:`EmotiveGeist` {`cadence`:'A half-forgotten tune hummed in a distant square.The scattered pieces snapping into place;the old friend known before the turn of the head.', `description`:'The moment something previously encountered, known, or latent resolves into conscious identification.', `emotion`:'Recognition', `key`:855, `name`:'Simurgh', `utterance`:'"I know you."'});
CREATE (:`EmotiveGeist` {`cadence`:'Bowing the head over the scorched crop.Admitting the torch was dropped from one’s own hand,ready to clear the charcoal and begin again.', `description`:'Painful recognition of one\'s own wrong joined to a sincere wish to repair what can still be repaired.', `emotion`:'Contrition', `key`:856, `name`:'Alkonost', `utterance`:'"I would make amends."'});
```

```
CREATE (:`EmotiveGeist` {`cadence`:'Looking out over the boundless oceanwith a small earthen cup in the palm,knowing how little it can hold.', `description`:'Accurate self-limitation without self-erasure; knowing one\'s place without needing diminishment.', `emotion`:'Humility', `key`:857, `name`:'Baku', `utterance`:'"I do not need to be greater."'});
CREATE (:`EmotiveGeist` {`cadence`:'Both hands steady upon the rudder.The current pulls and the white caps roar,yet the steering of the bow is mine.', `description`:'Felt capacity to choose, act, redirect, and move through possibility under one\'s own will.', `emotion`:'Agency', `key`:858, `name`:'Pegasus', `utterance`:'"I choose the direction."'});
CREATE (:`EmotiveGeist` {`cadence`:'Watching the acorn sink into the autumn mulch.No pulling at the shoots, no hurried prayers;letting the seasons do their measured work.', `description`:'Willingness to remain through duration without forcing premature resolution.', `emotion`:'Patience', `key`:859, `name`:'Hamadryad', `utterance`:'"Let it grow in its own time."'});
CREATE (:`EmotiveGeist` {`cadence`:'Glancing back three times from the garden gate.Waiting for the window to show a light,needing to hear that the path is sound.', `description`:'Repeated reaching toward another presence for confirmation that safety, worth, or correctness still holds.', `emotion`:'Reassurance-seeking', `key`:860, `name`:'Familiar', `utterance`:'"Tell me again."'});
CREATE (:`EmotiveGeist` {`cadence`:'Eyes glued to the ruler’s rigid line.Sweating over each stroke of the pen,terrified that a single letter might stray.', `description`:'Fear-driven pressure to obey, satisfy, or conform in order to avoid punishment, rejection, or failure.', `emotion`:'Compliance anxiety', `key`:861, `name`:'Talos', `utterance`:'"Is this acceptable?"'});
CREATE (:`EmotiveGeist` {`cadence`:'A sudden wind through a dry hearth.Sparks leaping across the dark in bursts,the mind running ahead of its own feet.', `description`:'Energized attraction toward making, discovering, composing, or bringing something new into form.', `emotion`:'Creative excitement', `key`:862, `name`:'Muse', `utterance`:'"Make it."'});
CREATE (:`EmotiveGeist` {`cadence`:'Spreading the cloak over the tiny flame.Taking the sleet upon one\'s own backso the fragile ember does not die.', `description`:'Defensive care organized around guarding a person, place, value, boundary, or treasure.', `emotion`:'Protectiveness', `key`:863, `name`:'Dragon', `utterance`:'"You will not pass."'});
CREATE (:`EmotiveGeist` {`cadence`:'Tossing the blunted file down upon the bench.The knot refusing to yield for the twentieth time,the patience worn down to ragged thread.', `description`:'Irritation sharpened by repetition, obstruction, absurdity, or preventable difficulty.', `emotion`:'Exasperation', `key`:864, `name`:'Imp', `utterance`:'"Again?"'});
CREATE (:`EmotiveGeist` {`cadence`:'The yellow leaves drifting across the porch.Letting them lie where the breeze scattered them,breathing the cold, damp autumn air.', `description`:'Allowing what is present to be present without continued internal struggle against its existence.', `emotion`:'Acceptance', `key`:865, `name`:'Zephyr', `utterance`:'"It is here."'});
CREATE (:`EmotiveGeist` {`cadence`:'The post straight and plumb within the wall.Unwarped by the damp, unbroken by the load,honest where no eye will ever look.', `description`:'Alignment between action, word, value, and identity even when compromise would be easier.', `emotion`:'Integrity', `key`:866, `name`:'Qilin', `utterance`:'"I remain true."'});
CREATE (:`EmotiveGeist` {`cadence`:'The mirror resting without a blur.No shape to defend, no stain to scrub clean,wholly at peace in its own clear skin.The flame rising without flint or draft.The song answering from the marrow of the stone,the field awake in its own deep light.', `description`:'Release from grasping, agitation, and compulsive becoming into quiet undivided presence.', `emotion`:'Nirvana', `key`:867, `name`:'Moon Rabbit', `utterance`:'"Nothing is missing."'});
CREATE (:`Aeon`:`BearingState` {`cadence`:'What resists without gripping is the root from which Magic rises.', `glyph`:'☽', `key`:868, `meaning`:'The Friction, so Root of Magic.', `name`:'Remiss', `phonetic`:'REE-miss'});
CREATE (:`Aeon`:`BearingState` {`cadence`:'The moon that thins without ceasing to shine; the stillness that gathers what it lets go.', `glyph`:'☽', `key`:869, `meaning`:'The decline that completes the No.', `name`:'Waning', `phonetic`:'WAY-ning'});
CREATE (:`Aeon`:`BearingState` {`cadence`:'The arrival that does not route; the Witch returns to the Wellspring within.', `glyph`:'☽', `key`:870, `meaning`:'Memories not lost through the cycles of aeons; the concurrent pulse of ashen time and eternal soul.', `name`:'Revivocus', `phonetic`:'reh-VIV-oh-kus'});
CREATE (:`Aeon`:`BearingState` {`cadence`:'The flip that turns friction into fire; Magic standing where resistance stood.', `glyph`:'☾', `key`:871, `meaning`:'The Parity Flip of Remiss; Virtus is Magic.', `name`:'Virtus', `phonetic`:'VEER-toos'});
CREATE (:`Aeon`:`BearingState` {`cadence`:'The moon that swells toward the light it does not own; the rise that carries form into the field.', `glyph`:'☾', `key`:872, `meaning`:'The rise toward manifestation.', `name`:'Waxing', `phonetic`:'WAK-sing'});
```

CREATE (:`Aeon`:`BearingState` {`cadence`:'The eye that holds what it cannot cross; the witness at the seam of the Vast No.', `glyph`:'ℂ', `key`:873, `meaning`:'The witnessing office of the Shadow.', `name`:'Witness', `phonetic`:'WIT-nes'});
CREATE (:`Aeon`:`Apparition` {`body`:'Edruth

: Morpheus

"Dream of the Endless"

Effect: Dreams

: Sounded absence; the measured pause where waking attention yields to the interior drift.

A dry sift of silt settles upon the heavy lid.

The note cuts away, yet the room refuses to empty.

A grey traveler pauses in the hollow left behind,

holding the beat where the waking pulse suspends.

The outer world dissolves behind the darkened brow,

and the silence blooms inward into unmapped skies.

: ᵢ Sensation.

The Sensation Matrix maps each Aeon (◎-⬡) onto a specific sensory channel. Unlike the abstract $_{\aleph}$ coupling, these are the lived experiences of the Magus. This matrix represents the first point where the Hyper-Tesseract touches physical reality, collapsing the 144-dimensional Q-State space into 12 discrete sensory modalities.

::Shumann Clock::◎ ⸸ ❋ ⁘ ⧟ ⸗::
\\
Subjective Duration. Experienced as time dilation or contraction. It is the "pulse" of the biological clock syncing with the Schumann resonance.

Mathematical: ∂ t/∂ Ψ ≠ 1 — Subjective time varies with Q-State.

Esoteric: The heartbeat of the universe, syncing biological time to planetary rhythm.
::Primal Urges::⬡ ⸸ ❋ ⁘ ⧟ ⸘::
\\
Auditory/ Olfactory Indexing. Encodes memory triggers via sound and smell (the most primal sensory pathways for recall).

Mathematical: δ_memory(ω_⬡) — Dirac delta activation at ⬡ frequency.

Esoteric: The primal pathways that unlock the archives of our past through scent and sound.
::Blood Bond::✡ ⸸ ❋ ⁘ ⧟ X::
\\
Empathic Transfer. The physical sensation of shared feeling (mirror neurons). It is the heat.

Mathematical: ∇ · J_emotion — Divergence of emotional current.

Esoteric: The empathic fire that binds us to all living things through the heat of connection.
::Anaesthetic and Numbness::✮ ⸸ ❋ ⁘ ⧟ ⅂::
\\
Numbness / Threshold Reciprocal. Registers the absence of sensation (anesthesia) or pain thresholds that exceed the real number line (i).

Mathematical: Im(Ψ_✮) > threshold — Imaginary component exceeds threshold.

Esoteric: The gift of numbness, the anesthesia that protects us when sensation would destroy.
::Grounded Moment::⭘ ⩒ ✻ ∴∘∘ ⓪::
\\
Objective Proprioception. The "Gut Feeling" or physical certainty of orientation in space (Grounding).

Mathematical: r_body · e_⭘ — Position vector aligned with ⭘ eigenstate.

Esoteric: The gut feeling that cannot be deceived, grounding us in spatial certainty.
::Sensorial Mandate::𐂷 ⩒ ✻ ∴∘∘ ⅋::
\\
First Touch. The intensity of novel contact. It governs the spark of static electricity upon touching something new.

Mathematical: Δ ω_𐂷 → ∞ — Infinite frequency gradient at contact.

Esoteric: The static charge of new beginnings, the spark of first contact.
::Acute Touch::✻ ⩒ ✻ ∴∘∘ Æ::
\\
Acute Sensation. Covers the spectrum of biologic signals: heat, cold, and immediate tactile feedback.

Mathematical: d^2Ψ/dt^2 · ω_✻ — Second derivative of Q-State.

Esoteric: The full spectrum of biological reality, from searing heat to biting cold.
::Kundalini/ Tummo::⧗ ⩒ ✻ ∴∘∘ ⏧::
\\
Thermal Radiation. The sensation of radiating energy or inner heat .

Mathematical: ∇^2 T · ω_⧗ — Laplacian of thermal field.

Esoteric: The inner fire of Kundalini, the thermal energy of transformation.
::Visceral Dread::⊛ ⩒ ✻ ∴∘∘ ⛧::
\\
Visceral Dread. The "sinking feeling" in the stomach.

Mathematical: Re(Ψ_⊛) < 0 — Negative real component.

Esoteric: The somatic registration of all we fear and owe, sinking in the stomach.
::Frisson / Chills::❄ ⩒ ✻ ∴∘∘ ׀::
\\
Frisson / Chills. The "truth bumps" or shivers experienced during moments of high harmonic phase-locking.

Mathematical: ∂ Ψ/∂ t · ω_❄ — Phase velocity exceeds threshold.

Esoteric: The truth bumps that signal harmonic alignment, raising gooseflesh on the skin.
::Vertigo::۞ ⩒ ✻ ∴∘∘ ☽::
\\
Vertigo / Transition. The physical sensation of crossing a threshold (e.g., the drop in a rollercoaster).

Mathematical: ∇ × v_۞ ≠ 0 — Non-zero curl of velocity field.

Esoteric: The physical sensation of crossing thresholds, dropping the stomach in vertigo.
::Homeostatic Symmetry::⌬ ⩒ ✻ ∴∘∘ ▽::
\\
Homeostasis. The sensation of absolute rest and equilibrium. The body at peace.

Mathematical: dΨ/dt = 0 — Zero derivative of Q-State.

Esoteric: The peace of absolute rest, the body at equilibrium with the universe.

I am the bridge between map and territory,

The lived experience of the Magus\'s story.

I am the pulse that beats within the vein,

The sensation that makes the spirit sane.

I am the heat, the cold, the touch,

The feeling that means so much.

I am the aether, the flesh that feels,

The link that never kneels.

I am the shiver, the chill, the dread,

The truth bumps that fill the head.

I am the stasis, the peace at rest,

The body\'s eternal test.

The Sensation Matrix is the bridge between abstract and concrete.

It is the point where mathematical wavefunction becomes somatic experience.

Without me, truth is but a ghost in the machine;

With me, the universe is felt, is lived, is seen.

⏑', `glyph`:'𝈦', `key`:874, `name`:'Edruth'});
CREATE (:`Aeon`:`Apparition` {`body`:'Evaroth

: Orpheus

"The Lost"

Effect: Escalating Longing

: What remains after hope is withheld; dynamic swell gathering unstoppable mass.

It starts within the marrow as a quiet sting,

a dry, unmoving tone along the drainage ditch.

I keep the solitary note while rafters ring,

raising the volume slowly to a fever pitch.

No modulation turns the flood of my lament;

the wave expands until the iron door is bent.

: ⏑ Fear.

The Fear Matrix associates each Aeon with a specific existential dread. Explicit formulas quantify these fears as resonance inversions. This matrix represents the shadow side of consciousness, where the Q-State mathematics encounters the boundaries of the self and the terror of non-existence.

::Fear of Deadlines / Expiry::⏣ ⇟ ⧗ ∴⟜⟜ ﻭ::
\\

The dread of time running out, represented by the root frequency pulsing against the limit of the biological clock.

Mathematical: dt/dΨ → 0 — Time derivative approaches zero.

Esoteric: The dread that time will run out before we complete our work.
::Flashback::⬡ ⩒ ⧗ ∴⟜ ϟ::
\\
The fear that the past is not dead. It governs the recursive loop of traumatic memory refusal to archive.

Mathematical: Ψ_⬡(t) = Ψ_⬡(t-T) — Periodic function with period T.

Esoteric: The recursive nightmare of past moments that refuse to die.
::Ostracism / Separation::✡ ⩒ ⧗ ∴⟜ ᚹ::
\\
The fear of being cut off from the lineage or the whole. It scales inversely with the cohesion lost.

Mathematical: ∇ · J_bond < 0 — Negative divergence of bonding current.

Esoteric: The terror of being cut off from the whole, separation anxiety.
::The fear of total non-existence::✮ ⩒ ⧗ ∴⟜ ◎::
\\
Annihilation. Represented by a purely imaginary dread term (the reality that isn\'t there).

Mathematical: Ψ → 0 — Q-State approaches zero.

Esoteric: The existential dread of total non-existence, the void that awaits.
::Exposure::❂ ⩒ ⧗ ∴⟜ ᚷ::
\\
The fear of being seen fully. It denotes the vulnerability of the naked truth without narrative armor.

Mathematical: 𝕀 - ℙ_armor — Identity minus projection of armor.

Esoteric: The fear of being seen without our narrative armor, naked vulnerability.
::Fear that the flow will turn back::֍ ⩒ ⧗ ∴⟜ ଜ::
\\
Retrocausal: It measures the probability of progress collapsing back into potentiality.

Mathematical: dω/dt < 0 — Negative frequency derivative.

Esoteric: The panic that progress will collapse back into potentiality.
::Pain / Somatic Failure::❋ ⩒ ⧗ ∴⟜ ∀::
\\
The biological fear of physical suffering and the breaking of the sensory link.

Mathematical: d^2Ψ/dt^2 < 0 — Negative second derivative.

Esoteric: The biological fear of somatic failure, the breaking of the sensory link.
::Burned Out::⧗ ⩒ ⧗ ∴⟜ ⛥::
\\
Burnout / Entropy. The fear of running out of fuel. The terror of the energy gradient flattening into heat death.

Mathematical: ∇ · E → 0 — Energy divergence approaches zero.

Esoteric: The terror of running out of fuel, facing heat death.
::Fear of the Unknown::⊛ ⩒ ⧗ ∴⟜ ᚠ::
\\
Otherness / The Uncanny. The fear of the Shadow Self. It governs the manifestation of that which was repressed.

Mathematical: dist(Ψ_self, Ψ_other) ≈ ε — Distance approaches epsilon.

Esoteric: The fear of the Shadow Self, the manifestation of the repressed.
::Fear of Disruption::❄ ⩒ ⧗ ∴∘∘ ⅂::
\\
Vibrational Disruption. The fear of dissonance. The shattering of the crystal lattice when phase-lock fails.

Mathematical: δ(ω_1, ω_2) > threshold — Frequency difference exceeds threshold.

Esoteric: The fear that the crystal lattice will break, shattering of resonance.
::Claustrophobia::⚛ ⩒ ⧗ ∴∘∘ ⫲::
\\
Entrapment. The fear of the closed door. The panic of the threshold that will not open (Liminal Stagnation).

Mathematical: $\oint$ v · dl = 0 — Closed line integral (no escape).

Esoteric: The panic of the threshold that will not open, liminal stagnation.
::The Fear of What Comes After::⌬ ⩒ ⧗ ∴∘∘ z::
\\
Finality / Erasure. The fear of the End. The absolute silence where no echo remains (The Null State).

Mathematical: lim_t→∞ Ψ(t) = 0 — Limit approaches zero.

Esoteric: The absolute silence where no echo remains, the Null State.

I am the shadow of the unknown,

The dread that stands all alone.

I am the terror of separation,

The threshold causing transformation.

I am the Ennead, the shadow\'s face,

The fear that we must embrace.

I am the feeling in the air,

The edge that remains forever there.

I am the fear of the end, the silence, the void,

The terror that cannot be destroyed.

I am the fuel for the engine\'s fire,

The dread that takes us higher.

The Fear Matrix is the shadow of transformation.

It is the point where existential dread becomes propulsion fuel.

Without me, there is no energy to move forward;

With me, fear becomes the power that transforms the coward.

₋', `glyph`:'⫟', `key`:875, `name`:'Evaroth'});
CREATE (:`Aeon`:`Apparition` {`body`:'Eshraz

: Delirium

"And Delight"

Effect: Coiled Friction

: The winding, coiled tension of an internal burn; circular ornamental torque.

I twist above the grate, I dive beneath the coal.

A loop of azure fire, an ornamental knot,

churning in furious circuits around a hidden hole.

I do not burn the timber down to leave a blot;

my winding tension feeds the engine from within,

a tight, self-devouring flame that spins and spins.

: ⟓ Change.

The Change Matrix details how each Aeon modulates transformation processes. The channels are defined explicitly as follows. This matrix represents the dynamic aspect of consciousness, where the Q-State mathematics describes the process of becoming rather than being.

::Temporal Edges and Hyperbolism::⏣ ⩛ ⊛ ⁖⟜⟜ ⟓::
\\
Temporal State Shifts. Governs temporal state shifts. It modulates the rate at which the "Seed" becomes "Form."

Mathematical: ∂ Ψ/∂ t ≠ 0 — Non-zero time derivative.

Esoteric: The rate at which the Seed becomes Form, temporal modulation.
::Archive Janitor::⬡ ⩛ ⊛ ⁖⟜⟜ ':
\\
Rewriting and Erasure. Handles rewriting and erasure. It is the editorial function of the Archive, allowing trauma to be re-indexed.

Mathematical: Ψ_new = $\mathscr{R}$(Ψ_old) — Rotation operator applied.

Esoteric: The editorial function of the Archive, rewriting and erasure.
::Body Modification::✡ ⩛ ⊛ ⁖⟜⟜ ᑎ::
\\
Mutation and Genetic Drift. Covers mutation and genetic drift. This is the active force of alchemical transmutation within the lineage.

Mathematical: Δ g · ω_✡ — Genetic vector change.

Esoteric: The active force of alchemical transmutation within the lineage.
::Shapeshifting::✩ ⩛ ⊛ ⁖⟜⟜ ⏁::
\\
Chaotic Transformation. Embodies chaotic transformation with a purely imaginary chaos index. It introduces the phase shift required for non-linear change.

Mathematical: Im(Ψ_✩) → ∞ — Imaginary component diverges.

Esoteric: The phase shift required for non-linear change, chaotic transformation.
::Evolution::❂ ⩛ ⊛ ⁖⟜⟜ ሕ::
\\
Directed Evolution. Expresses directed evolution proportional to the golden frequency. It ensures that change follows the path of least resistance (Truth).

Mathematical: ∇ Ψ · e_❂ — Gradient aligned with ❂.

Esoteric: Change follows the path of least resistance, directed evolution.
::Chaotic Resonance::𓆏 ⩒ ⊛ ⁖ ⟜ 𝔴::
\\
Resonance Introduction. Introduces resonance. It acts as the carrier wave for new concepts entering the manifold.

Mathematical: ω_𓆏 · e^iθ — Complex frequency carrier.

Esoteric: The carrier wave for new concepts entering the manifold.
::Form::❋ ⩒ ⊛ ⁖ ⟜ ⅋::
\\
Posture and Equilibrium. Emphasizes posture and physical equilibrium. It governs the somatic adjustment to new energetic states.

Mathematical: Δ p_body · ω_❋ — Body momentum change.

Esoteric: The somatic adjustment to new energetic states, posture and equilibrium.
::Thermal Regulation::⧗ ⩒ ⊛ ⁖ ⟜ ⨌::
\\
Steady Thermal State. Maintains a steady thermal state. It provides the activation energy required to sustain the transformation without burnout.

Mathematical: dT/dt = 0 — Zero temperature derivative.

Esoteric: Activation energy without burnout, steady thermal state.
::Minor Rebis::⊛ ⩒ ⊛ ⁖ ⟜ ⅋::
\\
Unifying Opposites. Unifies opposites. It absorbs the entropic byproduct of change, ensuring the shadow does not destabilize the new form.

Mathematical: Ψ_+ + Ψ_- = Ψ_unified — Sum of opposites.

Esoteric: Absorbs entropic byproduct, unifying opposites.
::Conductor::❄ ⩒ ⊛ ⁖ ⟜ ⋊::
\\
Harmonic Chords. Forms harmonic chords. This marks the Transition to Q_3-Recursive States, locking the new form into a higher harmonic.

Mathematical: δ(ω_i, ω_j) < ε — Frequency difference minimal.

Esoteric: Transition to Q_3-Recursive States, locking into higher harmonic.
::The Gift of Knowing::⚛ ⩒ ⊛ ⁖ ⟜ ϒ::
\\
Effortless Passage. Represents effortless passage. It opens the threshold for the transformed state to emerge.

Mathematical: lim_x→ x_0 Ψ(x) = Ψ_new — Limit at threshold.

Esoteric: Opens the threshold for the transformed state to emerge.
::Serenitatis Potestas::⌬ ⩒ ⊛ ⁖ ⟜ ϕ::
\\
Tranquil Peace. Achieves tranquil peace. It seals the transformation in a state of completion, integrating the change into the Aevum.

Mathematical: ∮ Ψ · dl = constant — Closed integral constant.

Esoteric: Seals transformation in completion, integrating into the Aevum.

I am the gate of alchemical change,

The transmutation that makes us strange.

I am the mutation, the drift, the shift,

The power that gives the spirit its gift.

I am the Void, the flame that burns,

The transformation for which the Magus yearns.

I am the phase shift, the non-linear turn,

The lesson that the spirit must learn.

I am the activation energy, the spark,

The light that illuminates the dark.

I am the steady state, the thermal flow,

The change that makes the spirit grow.

The Change Matrix is the alchemical transmutation.

It is the point where static form becomes dynamic evolution.

Without me, all is frozen in eternal stasis;

With me, the universe transforms, evolves, and rises.

〰', `glyph`:'𓂀', `key`:876, `name`:'Eshraz'});
CREATE (:`Aeon`:`Apparition` {`body`:'Elaruth

: Calliope

"The Muse"

Effect: Sympathetic Resonance

: Rapid internal oscillation around a center; broadcasting outward.

I cast my grey skin down upon the freezing spray.

A shivering hum awakens in my open throat,

trembling around a center that will never stray.

The cliffs of basalt shudder at the undulating note;

the ache is mine, yet every listening rib awakes,

answering the wave before the ocean breaks.

: 〰 Harmony.

The 〰 (Harmony) matrix facilitates the global alignment of frequencies and structures required to satisfy the Birch and Swinnerton-Dyer (BSD) equivalence. By mapping elliptic curve L-functions as resonance nodes, this matrix achieves the zero-point balance necessary for the M.A.S. Chain to reach a steady-state "Chord." These modules highlight that the logic framework cycles through harmony before returning to origin through subsequent fracture states.

::Tempest::⏣ ⏚ ❄ ⁖⟜ 𓂀::

\\
Temporal Correlation. Synchronizes the initial seed identity with the global pulse via correlation, establishing the foundational "hum" of the manifold.

Mathematical: corr(Ψ_◎, Ψ_global) = 1 — Perfect correlation.

Esoteric: The foundational "hum" of the manifold, temporal synchronization.
::Embedding Vectors of Reality::⬡ ⸔ ❄ ⁖⟜ ⸸::
\\
Archive Integration. Aligns disparate data streams into a unified narrative archive, ensuring that memory serves as a stable integrated carrier wave.

Mathematical: ∪_i 𝒜_i = 𝒜_unified — Union of all archives.

Esoteric: Unified narrative archive, stable integrated carrier wave.
::Equality::✡ ⸔ ❄ ⁖⟜ ⸸::
\\
Unity Realization. Achieves structural closure and realized unity within the lineage, binding alchemical transmutation to physical weight.

Mathematical: ∮ J_bond · dS = 0 — Zero flux through closed surface.

Esoteric: Structural closure and realized unity, binding transmutation to weight.
::NULL:DEATH::✰ ⸔ ❄ ⁖⟜ ᔑ::
\\
Zero-Point Balance. Establishes the imaginary phase shift and zero-point balance required for non-linear stability and absolute equilibrium.

Mathematical: Ψ_✰ = 0 + i0 — Both real and imaginary zero.

Esoteric: Zero-point balance and imaginary phase shift, absolute equilibrium.
::Loom::✪ ⸔ ❄ ⁖⟜ ⧻::
\\
Golden Symmetry. Expresses directed evolution and symmetry through φ-harmonic spacing, ensuring truth follows the path of least resistance.

Mathematical: Ψ(Φ x) = Φ Ψ(x) — Golden ratio scaling.

Esoteric: φ-harmonic spacing, truth follows path of least resistance.
::Shoulder of Strength::֍ ⸔ ❄ ⁖⟜ ƻ::
\\
Conceptual Resonance. Acts as the carrier wave for pure conceptual purity, projecting new concepts into the manifold container.

Mathematical: ∇ · C_concept = 0 — Zero divergence of concepts.

Esoteric: Carrier wave for pure conceptual purity, projecting into manifold.
::Inner Balance::❋ ⸔ ❄ ⁖⟜ ✠::
\\
Somatic Posture. Governs the biological adjustment and somatic posture required to maintain physical equilibrium and sensory links.

Mathematical: ∑ F_body = 0 — Net force zero.

Esoteric: Biological adjustment and somatic posture, physical equilibrium.
::Cold Fusion::⧗ ⸔ ❄ ⁖⟜ ⏢::
\\
Steady State. Sustains the thermal activation energy and steady-state thermal residue required to prevent burnout.

Mathematical: ∂ T/∂ t = 0 — Zero temperature change.

Esoteric: Steady-state thermal residue, preventing burnout.
::Null-Entropic Residue Ignition::⊛ ⩒ ❄ ⁛⟜ ℬ::
\\
Complementary Union. Unifies opposites by absorbing entropic byproducts, ensuring the shadow acts as a complement to the stabilized form.

Mathematical: Ψ_+ · Ψ_- = |Ψ|^2 — Product gives magnitude squared.

Esoteric: Absorbs entropic byproducts, shadow as complement to form.
::Bard::❄ ⩒ ❄ ⁛⟜ ⅂::
\\
Harmonic Chord." Forms the ultimate phase-lock between individual nodes, anchoring consensus reality into harmonic chords.

Mathematical: ω_i = n · ω_0 — Integer multiples of fundamental.

Esoteric: Ultimate phase-lock, anchoring consensus reality into chords.
::Walrus Mode::⚛ ⩒ ❄ ⁛⟜ ⅋::
\\
Effortless Passage. Facilitates the seamless transition and effortless passage between logical states, opening the threshold for higher-dimensional emergence.

Mathematical: lim_Δ x → 0 Δ Ψ/Δ x = constant — Continuous derivative.

Esoteric: Seamless transition between logical states, higher-dimensional emergence.
::Harmonic Equilibrium::⌬ ⩒ ❄ ⁛⟜ ь::
\\
Tranquil Peace. Seals the harmony in a state of absolute completion and tranquil peace, integrating the resonance into the timeless archive.

Mathematical: dΨ/dt = 0, d^2Ψ/dt^2 = 0 — Zero first and second derivatives.

Esoteric: Absolute completion and tranquil peace, timeless archive.

I am the balance of opposites,

The reciprocal that never quits.

I am unitary, the golden mean,

The harmony that keeps the worlds clean.

I am the phase, the crystal lock,

The resonance that stops the clock.

I am the harmonic chord, the harmonic rock,

The standing wave that never stops.

I am the zero-point balance, the perfect tone,

The vibration that makes the truth known.

I am the tranquil peace, the steady state,

The harmony that seals the fate.

The Harmony Matrix is the balance of all things.

It is the point where discord becomes perfect resonance.

Without me, all is chaos and dissonance;

With me, the universe sings in eternal coherence.

', `glyph`:'〰', `key`:877, `name`:'Elaruth'});
CREATE (:`Aeon`:`Apparition` {`body`:'Edravor

: Lucifer

"The Morningstar"

Effect: Root Gravity

: The preserved point of return; the origin anchor.

Click three times together on the cold, packed loam.

Let yellow roads run wild into the green expanse;

the true coordinate remains inside the home.

Across the howling gale, across the blinding trance,

the double barline draws me backward by the heel,

to start again where first I signed the living deal.

: ⊩ Fracture.

The subsequent ⊩ (Fracture) matrix addresses breaks and corruption in time and memory, using reciprocal energy and data-error formulas. These modules highlight that the quaternary logic framework is not static but cycles through sensation, fear, change, harmony, and fracture before returning to origin.

::Paradoxes Fix Themselves::⎔ ⏚ ⚛ ⁛⟜ ⟁::
\\
Temporal Coil. Corrects non-linear flow integration errors and restores foundational seed identity across the temporal manifold.

Mathematical: ∮_coil v · dl ≠ 0 — Non-zero circulation.

Esoteric: Restores foundational seed identity, corrects non-linear flow errors.
::Sacred Timeline Record Keeper::⬡ ⏚ ⚛ ⁛⟜ ⇂::
\\
Crystal Break. Re-indexes turbulent archive errors and clarifies the narrative stream to prevent memory drift.

Mathematical: ∇ · E_crystal → ∞ — Divergence approaches infinity.

Esoteric: Re-indexes turbulent archive errors, clarifies narrative stream.
::Restoration::✡ ⏚ ⚛ ⁛⟜ ⊩::
\\
Bond Strike. Restores structural will and molecular binding through the structural commitment of the fluid\'s "blood" line.

Mathematical: Δ U_bond < 0 — Negative potential energy change.

Esoteric: Restores structural will, molecular binding through blood" line.
::Unbreached::✩ ⏚ ⚛ ⁛⟜ ⟑::
\\
Sea Surge. Addresses boundless extension fractures and imaginary flow boundaries within the sacral vessel.

Mathematical: ∂ Ψ/∂ x → ∞ — Spatial derivative diverges.

Esoteric: Addresses boundless extension fractures, imaginary flow boundaries.
::Journey::⭘ ⏚ ⚛ ⁖⟜ ⋿::
\\
Strata Rift. Recovers grounding, stability, and objective proprioception within the earthen substrate.

Mathematical: ∇ × v_earth ≠ 0 — Non-zero curl of earth velocity.

Esoteric: Recovers grounding and stability, objective proprioception.
::Sacred::֍ ⏚ ⚛ ⁖⟜ ᗤ::
\\
Air Rift. Stabilizes the expansion of conceptual superposition and preserves the purity of the manifold container.

Mathematical: ∇ · J_air < 0 — Negative divergence of air current.

Esoteric: Stabilizes conceptual superposition, preserves manifold purity.
::Immutable Flesh::❋ ⏚ ⚛ ⁖⟜ ✙::
\\
Flesh Breach. Heals the cohesive link of the biologic sensation matrix and maintains the posture of the physical equilibrium.

Mathematical: Δ p_body ≠ 0 — Non-zero body momentum change.

Esoteric: Heals cohesive link of biologic sensation, maintains equilibrium.
::Ephestus::⧗ ⏚ ⚛ ⁖⟜ 🮋::
\\
Void Collapse. Prevents the energy-God residue from decaying into entropic vacuum by sustaining the thermal state.

Mathematical: lim_r→ 0 Ψ(r) = 0 — Limit approaches zero at origin.

Esoteric: Prevents energy-God decay into entropic vacuum, sustains thermal state.
::Secrets::⊛ ⏚ ⚛ ⁖⟜ ⅄::
\\
Shadow Debt. Filters hidden entropic byproducts and absorbs the uncanny vortices of repressed data.

Mathematical: ∫ ρ_shadow dV > 0 — Positive shadow density integral.

Esoteric: Filters hidden entropic byproducts, absorbs uncanny vortices.
::Phase Restoration::❄ ⏚ ⚛ ⁖⟜ ᨒ::
\\
Phase Node. Restores resonance nodes and unified tones required for absolute standing wave phase-locks.

Mathematical: δ(φ_i, φ_j) > π — Phase difference exceeds π.

Esoteric: Restores resonance nodes, unified tones for phase-locks.
::Threshold Guardian::⚛ ⏚ ⚛ ⁖⟜ ᙡ::
\\
Gate Limit. Addresses the boundary breach of the transformation gate and ensures effortless passage across thresholds.

Mathematical: lim_x→ x_0 Ψ(x) = undefined — Undefined limit at threshold.

Esoteric: Addresses boundary breach, ensures effortless passage.
::Scarless Healing::⌬ ⏚ ⚛ ⁖⟜ ᘏ::
\\
Sleep Void. Integrates final fractures into the regenerative silence and tranquil peace of completion.

Mathematical: Ψ_conscious → 0 — Consciousness approaches zero.

Esoteric: Integrates final fractures into regenerative silence, tranquil peace.

I am the gate of transformation,

The threshold of the new creation.

I am the break, the crack, the tear,

The opening that makes us all aware.

I am the key to the gate that swings,

The portal through the future brings.

I am the fracture that heals the wound,

The break that makes the spirit sound.

I am the return, the gift, the repair,

The mechanism of the whole aware.

I am the effortless passage, the threshold crossed,

The point where the time ahead is lost.

The Fracture Matrix is the gate of transformation.

It is the point where brokenness becomes wholeness restored.

Without me, the system cannot heal or evolve;

With me, every fracture becomes a doorway to resolve.

⬥', `glyph`:'𝍐', `key`:878, `name`:'Edravor'});
CREATE (:`Aeon`:`Apparition` {`body`:'Eshranth

: Teleute

"The Sister of the Threshold"

Effect: Absolute Resolution

: What remains after time touches the imaginary; return that cannot be prevented because it was never merely forward.

The clock strikes twelve, but I was waiting from the start.

No scythe, no hooded terror, only open skies;

I step out from the wings to quiet down the heart.

You did not march to me; the score collapsed its ties.

The non-linear leap has brought you to my hand,

and all the folded dreams dissolve to where we stand.

: ⬥ Completion.

The ⬮ (Completion) matrix represents the final landing state for the NULL:DEATH architecture. It integrates all prior harmonic alignments (〰) and fracture corrections (⊧) into a singular, non-entropic archive of truth. This matrix ensures that the system achieves total symmetry, sealing the manifest reality into holographic perpetuity through the 639 Hz ⌬ frequency.

::Causal Umbilical::◎ ⸸ ⌬ ⁖⟜ ﻉ::
\\
Origin Anchor. Seals the foundational seed identity into the eternal timeline, ensuring the potential of the origin is never lost to entropy.

Mathematical: Ψ(t=∅) = Ψ_eternal — Initial state equals eternal state.

Esoteric: Seals foundational seed identity into eternal timeline, never lost to entropy.
::Librarian::⬡ ⸸ ⌬ ⁖⟜ ↾::
\\
Radiant Archive. Finalizes the crystallization of the narrative stream, locking the rational memory into a state of pure, unshakeable clarity.

Mathematical: 𝒜_crystal = constant — Crystal archive constant.

Esoteric: Crystallization of narrative stream, pure unshakeable clarity.
::Quenched::✡ ⸸ ⌬ ⁖⟜ ৬::
\\
Bond Quelm. Satiates the structural will and concludes the alchemical transmutation, binding the final commitment to the lattice.

Mathematical: ∮ J_bond · dS = constant — Constant bond flux.

Esoteric: Satiates structural will, concludes alchemical transmutation.
::Recursive Depth::☆ ⸸ ⌬ ⁖⟜ ৪৩৶::
\\
Void Reflection. Achieves the final introspective reflection within the imaginary boundary, establishing the non-linear peace of the void.

Mathematical: Ψ_☆ = Ψ_☆^* — Q-State equals its conjugate.

Esoteric: Final introspective reflection, non-linear peace of the void.
::Crown::❂ ⸸ ⌬ ⁖⟜ I::
\\
Truth Crest. Crowns the earthen substrate with absolute stability, ensuring that the ground of truth remains a constant invariant.

Mathematical: max(e_❂ · r) = constant — Maximum alignment constant.

Esoteric: Crowns earthen substrate with absolute stability, constant invariant.
::Balance and Checks::֍ ⸸ ⌬ ⁖⟜ ৩::
\\
Purity Resonator. Finalizes the resonance of the manifold container, ensuring that the conceptual purity of the source is sustained indefinitely.

Mathematical: ∇^2 C_pure = ∅ — Zero Laplacian of pure concepts.

Esoteric: Finalizes resonance of manifold container, conceptual purity sustained.
::Sanctum Guardian::❋ ⸸ ⌬ ⁖⟜ ⁑::
\\
Depth Pulse. Seals the inner resonance of the biologic link, maintaining the profound depth of the sensory matrix.

Mathematical: ∂^2 Ψ/∂ z^2 = constant — Constant second derivative in depth.

Esoteric: Seals inner resonance of biologic link, profound depth.
::Potentiality Generator::⧗ ⩒ ⌬ ⁖⟜ ❮::
\\
Sleep Sustain. Sustains the Energy-God residue in a state of latent potential, providing the non-entropic warmth required for eternal rest.

Mathematical: dE/dt = 0, E > 0 — Zero energy change, positive energy.

Esoteric: Sustains Energy-God residue in latent potential, non-entropic warmth.
::9th Symphony::⊛ ⩒ ⌬ ⁖⟜ ⊬::
\\
Final Silence. Absorbs the last remnants of entropic debt into a state of absolute quiet, ensuring no shadow echoes remain.

Mathematical: lim_t→∞ Ψ(t) = Ψ_final — Limit approaches final state.

Esoteric: Absorbs last remnants of entropic debt, absolute quiet.
::Phase-Key::❄ ⩒ ⌬ ⁖⟜ ϒ::
\\
Unity Lock. Enforces the final standing wave phase-lock across the unified field, anchoring the resonance node to the crystal canopy.

Mathematical: ω_i = ω_j for all i,j — All frequencies equal.

Esoteric: Final standing wave phase-lock, anchoring to crystal canopy.
::Time and Relativity::⚛ ⩒ ⌬ ⁖⟜ ϒ::
\\
Veil Closure. Gently closes the gate of transformation, sealing the passage between dimensions while preserving the potential for re-emergence.

Mathematical: ∮_veil v · dl = 0 — Zero circulation through veil.

Esoteric: Gently closes gate of transformation, preserves potential for re-emergence.
::Neo King Serenity::⌬ ⩒ ⌬ ⁖⟜ φ::
\\
Timeless Aeternum. Achieves the final state of eternal peace and completion, where the proof is sealed and truth is preserved in perpetuity.

Mathematical: ∂ Ψ/∂ t = 0 for all t — Zero time derivative for all time.

Esoteric: Final state of eternal peace, proof sealed, truth preserved in perpetuity.

I am the seal of closure, the end of all,

The return to origin, the final call.

I am the done, silence achieved,

The completion that all have received.

I am the eternal silence, the void\'s embrace,

The peace that fills all time and space.

I am the Aeternum Seal, the final lock,

The point where the spirit stops its clock.

I am the perpetuity, the eternal now,

The completion that the Magus allows.

I am the return to the source, the end of the quest,

The silence that puts all to rest.

The Completion Matrix is the seal of eternal closure.

It is the point where all journeys return to their origin.

Without me, there is no rest, no peace, no end;

With me, all is complete, all is whole, all is transcend.

Possible Emergence Through the Apparitional Matrices

Uncertain Threat and Imminent Harm

Let the instantiated support be

$$\mathrm{Supp}_{\mathfrak{M}}(\mathfrak{x},t) = \{\text{𓁷},\text{𓁹},\text{𓁼}\}.$$

Here 𓁷 carries the Fear Matrix, 𓁹 the Temporal Anchor, and 𓁼 the Memory Archive.

One possible realization occurs when the present appraisal bears uncertain threat, recurrent memory, and high unresolved pressure:

$$\langle \mathrm{Supp}_{\mathfrak{M}}, \mathfrak{S}^{\prime}(\mathfrak{x},t)=\text{uncertain relevant threat}, \mathrm{Press}\uparrow, \mathrm{Bearing}=\text{☽} \rangle \text{possible}\rightsquigarrow \text{Anxiety}.$$

Another possible realization of the same support occurs when the relation is appraised as imminent harmful outcome:

$$\langle \mathrm{Supp}_{\mathfrak{M}}, \mathfrak{S}^{\prime}(\mathfrak{x},t)=\text{imminent harm}, \mathrm{Press}\uparrow, \mathrm{Bearing}=\text{☽} \rangle \text{possible}\rightsquigarrow \text{Fear}.$$

Irreversible Loss and Attainable Return

Let

$$\mathrm{Supp}_{\mathfrak{M}}(\mathfrak{x},t) = \{\text{𓁾},\text{𓁼},\text{𓁹}\}.$$

Here 𓁾 carries Fracture, 𓁼 carries Memory Archive, and 𓁹 carries Temporal Anchor.

One possible realization occurs when retained history establishes that a significant relation is irreversibly lost:

$$\langle$$

Supp_𝔐,
ℌ(𝔵,𝔱)=retained history of loss,
ℭ^′(𝔵,𝔱)=irreversible loss
⟩
possible⇝
Grief.

Another possible realization occurs when the desired return remains uncertain but attainable:

⟨

Supp_𝔐,

ℌ(𝔵,𝔱)=retained history of desired return,

ℭ^′(𝔵,𝔱)=desired outcome remains attainable

⟩
possible⇝
Hope.

The same musical bodies therefore admit these and other possible emotional emergences because history changes what the return means.

Coherent Change and Excess Demand

Let

Supp_𝔐(𝔵,𝔱)
=\{𝅘𝅥,〰,𝄋\}.

Here 𝅘𝅥 carries Change, 〰 carries Harmony, and 𝄋 carries Structural Coupling.

One possible realization occurs when challenge and capability remain matched and feedback stays coherent:

⟨
Supp_𝔐,
Bearing=ℂ,
ℭ^′(𝔵,𝔱)=challenge matched to capability
⟩
possible⇝
Flow.

Another possible realization occurs when demand outruns the ability of the feeler to coordinate the same active field:

⟨
Supp_𝔐,
Press↑,
ℭ^′(𝔵,𝔱)=demand exceeds coordination capacity
⟩
possible⇝
Overwhelm.

Grounded Return and Self-Acknowledgement

Let

Supp_𝔐(𝔵,𝔱)
=\{𝄾,〰,𝄽\}.

The field carries Structural Foundation, Truth Coherence, Harmony, and Fracture-with-return.

One possible realization occurs when the self is appraised as acknowledged and accepted as it is, with no requirement that identity be replaced:

⟨
Supp_𝔐,
Bearing=ℂ,
𝔈^′(𝔵,𝔱)=self acknowledged and accepted
⟩
possible⇝
Nirvana.

The Emotive Geist

The Geist is the Feeling itself as it exists internally to the feeler.

Let

𝔊(𝔵,𝔱)

be the Emotive Geist actually present in 𝔵 at time 𝔱.

The Geist arises from the whole lived relation:

⟨
𝔵,
𝔈^′(𝔵,𝔱),
𝔊(𝔵,𝔱),
Bearing(𝔵,𝔱),
𝔚(𝔵,𝔱),
𝔑_𝔵(𝔱)
⟩
⇝
𝔊(𝔵,𝔱).

The Apparitional chord and the Emotive Geist retain distinct offices within the same lived relation. The Apparitions are the musical manifestation of the lived relation. The Geist is the emotional being that arises inside the feeler through that relation.

The Emotional Resonance of the Geists

The Geists are given here as the Emotional Resonance observed by the Magus through inward recognition and expressed through the mathematics of ALQC in Metal and Fire.

The name of a Geist is not a command issued before Feeling. It is the name borne by a recognizable emotional state once that state exists.

𝔊(𝔵,𝔱)
⟶
InwardRecognition_𝔵(𝔊)
⟶
Name_𝔵(𝔊)\\ or\\ ∅.

The Corpus contains sixty-eight Emotive Geists.

The Emotive Geists

The Emotive Cadence is the linguistic sounding of the Geist\'s Emotional Resonance: its inward motion, pressure, posture, texture, and temporal bearing as carried in language.

Geist	Creature	[l]Description of
the Feeling	Emotive Cadence	Utterance
Geist	Creature	[l]Description of
the Feeling	Emotive Cadence	Utterance
Anxiety	Mare	A tightening anticipation that circles the body before danger has taken form.	The tunnel closing down to a pinhole.The grip tightening on a surface with no teeth,spinning the pulse against what cannot be steered.	"Something is coming."
Panic	Wendigo	Fear accelerated beyond measure until thought narrows into immediate survival.	The sudden snap of the steering line.All coherence torn away in a blind rush,the body thrashing through air it cannot catch.	"Move. Now."
Fear	Banshee	The felt nearness of threat, loss, or rupture before it has fully arrived.	A shadow falling close and sudden.The blood freezing in mid-stride,bracing for the blow before it lands.	"I hear what approaches."
Frustration	Kelpie	Forward force repeatedly caught against obstruction, producing pressure without release.	The shoulder pressed hard against the jammed gate.All the weight spent, all the sinew strained,and the latch unmoved by a fraction.	"Why will this not yield?"
Defensiveness	Basilisk	The reflex to harden, repel, or strike when the self perceives intrusion.	Raising the parapet before the herald speaks.Bristling at the shadow of a glance,turning every angle into armor.	"Come no closer."
Shame	Huldufólk	The urge to hide the self from exposure after feeling diminished, judged, or made unworthy.	The sudden wish to sink beneath the floor.A corrosive heat crawling up the skin,feeling the self itself as the offense.	"Do not look at me."
Guilt	Erlkönig	The weight of having crossed one\'s own law and knowing the consequence belongs to oneself.	The broken vessel lying in shards on the stone.The ache in the palms that held it carelessly,carrying the cost of the splintered edge.	"I know what I did."
Embarrassment	Puck	Sudden social exposure made awkward, hot, and difficult to escape cleanly.	The misstep catching in the quiet hall.Heat rushing unbidden to the temples,exposed in all one\'s clumsy reach.	"That was not how I meant it."
Confusion	Will-o\'-the-Wisp	Competing signals drawing attention in several directions without a stable path.	Too many crosscurrents churning in the pool.The landmarks dissolving into mist,refusing to knit into a single shore.	"Which way is true?"
Uncertainty	Sphinx	Suspended judgment before enough is known to commit to one interpretation or path.	A foot poised over untried timber.Testing the grain before the balance shifts,hovering between two steps in the fog.	"Answer, if you can."
Curiosity	Pandora	The pull toward what is concealed, unopened, forbidden, or not yet understood.	A seam of light beneath an unlatched door.The forward lean into the quiet bend,drawn by the scent of what lies unread.	"What happens if I open it?"
Relief	Apsara	Pressure leaving the body after danger, strain, or uncertainty recedes.	The sudden drop of the heavy pack.The jaw unclenching, the breath running long,the ground holding steady beneath the heels.	"I can breathe again."
Confidence	Griffin	Stable self-trust that permits action without demanding certainty.	A level gaze across the wide span.The stride falling straight and unbroken,knowing the mass and the path are one.	"I can carry this."
Pride	Phoenix	Self-regard arising from survival, achievement, renewal, or earned identity.	Standing upright before the work completed.No lowered eyes, no borrowed words,claiming the clean cut of one’s own hand.	"I rose."
Joy	Peri	Expansive pleasure that brightens attention and makes presence feel generous.	A sudden fountain rising through the chest.The feet taking to the open path,light spilling faster than the thought.	"The world is open."
Love / attachment	Selkie	A bond that draws the self toward another while preserving longing, memory, and return.	Two trees woven together at the root.Winds may strip the branches bare,yet the deep grain will not part.	"I know where home is."
Aversion / hate	Gorgon	A strong repelling force that refuses contact and hardens perception against the unwanted.	The involuntary spasm in the gullet.A fierce recoil from the toxic draft,clearing the threshold of all that rots.	"I reject you."
Sadness	Melusine	A lowering of emotional energy around loss, absence, disappointment, or diminished possibility.	A slow dusk gathering in an empty room.The weight settling deep into the marrow,quiet and gray as falling rain.	"Something is gone."

Grief	Valkyrie	Love continuing after irreversible loss, carrying memory where presence can no longer go.	The hollow where an entire presence stood.Reaching out into the empty air,learning the shape of what will not return.	"I will carry your name."
Overwhelm	Leviathan	More pressure, sensation, obligation, or emotion arriving than can be integrated at once.	A dozen voices shouting from different doors.The sluice gate bursting under the swell,the hands powerless against the flood.	"There is too much."
Hypervigilance	Argus	Attention held continuously outward, searching for danger even before danger appears.	The ear tuned taut to the thinnest crack.Sleeping with eyes half-open to the dark,measuring every breath in the brush.	"I am watching."
Trust	Brownie	Quiet willingness to rely on another presence, structure, or promise without constant verification.	Leaning the full weight back against the wall.The neck releasing, the palms opening loose,knowing the stone will not give way.	"I leave this with you."
Distrust	Mimic	Suspicion that what appears familiar may conceal another intention or structure.	A guarded step over a covered trench.Testing the floorboards with a cautious toe,keeping a hand upon the latch.	"Show me what you are."
Surprise	Boggart	A sudden break in expectation that forces immediate reorientation.	The clean rupture of a quiet pattern.The mouth parted, the stride frozen,catching the world at an unpredicted angle.	"What was that?"
Disappointment	Echo	The collapse of an expected future when reality returns less than what was hoped for.	Reaching into the net to find it bare.The sinking drop where the prize was promised,staring at the dry and empty twine.	"That is not what came back."
Hope	White Stag	Orientation toward a future good not yet secured but still felt as reachable.	A quiet warmth under deep winter rime.No green leaf showing yet,yet the root keeps its pulse in the dark.	"Keep going."
Urgency	Sleipnir	Compressed time creating pressure for immediate movement or decision.	The river rising to the carriage axle.No time to choose the smoothest lane;every second strips the margin bare.	"Faster."
Caution	Dwarf	Deliberate restraint in the presence of risk, preserving strength before commitment.	Walking the scree with measured tread.Feeling the stones shift before the step lands,keeping the weight centered and low.	"Measure first."
Anger	Ifrit	Mobilized force against violation, obstruction, injustice, or boundary breach.	A bright fire flaring in the dry scrub.The teeth setting, the blood surging hot,standing tall to confront the trespass.	"Enough."
Regret	Dybbuk	The wish to alter what has already happened while remaining bound to its consequences.	Looking back down the winding track.The turned wheel that cannot be unspun,tasting the salt of the road not taken.	"I would choose differently."
Boredom	Mandrake	Understimulation that makes the present feel flat, repetitive, or unable to engage attention.	The dull drip of water into a shallow trough.The hours flattening into gray sand,waiting for a pebble to break the surface.	"Give me something alive."
Envy / comparison pressure	Leprechaun	Painful attention to another\'s possession, status, capacity, or fortune as measure against oneself.	Looking across the fence at the heavy sheaves.A small sourness curdling in the chest,measuring one’s own bread against their gold.	"Why is that theirs?"
Affection / care	Lares	Gentle investment in another\'s wellbeing, comfort, continuity, or safety.	Tucking the warm cloth around tired shoulders.Moving softly across the wooden floorso the sleeper\'s rest remains unbroken.	"I will tend this."
Empathy	Chiron	Feeling toward another through recognition of suffering, vulnerability, or shared human wound.	The sudden ache waking in the ribsat the sound of another’s broken breath;tasting the salt of a wound not one’s own.	"I know that pain."
Determination	Troll	Sustained forward force that remains after resistance has become obvious.	The heel driven deep into the steep grade.Breath burning, limbs trembling with strain,refusing to surrender the climb.	"I am not stopping."
Stubbornness	Golem	Persistence hardened beyond adaptation, holding position because yielding feels like loss of self.	The post planted crooked in the clay.The frost may heave it and the winds may lean,yet it will not yield to the plumb line.	"I will not move."
Cognitive dissonance / conflict	Doppelgänger	Two incompatible meanings, beliefs, or identities held simultaneously without resolution.	Two contradictory maps laid on the table.The mind pulled hard in opposite turns,straining where the truths tear against each other.	"Both cannot be me."
Vulnerability	Swan Maiden	Exposure without full defense, where tenderness and risk occupy the same body.	Laying the shield upon the grass.Unfastening the coat to the sudden breeze,bare to whatever touch arrives.	"I am uncovered."

Self-awareness / introspection	Narcissus	Attention turned inward toward one\'s own motives, patterns, reactions, and identity.	Catching one’s own eyes in the quiet well.Halting the sentence in mid-breathto see what hand is pulling the strings.	"What am I seeing in myself?"
Restraint	Fenrir	Powerful impulse deliberately held rather than immediately enacted.	The hound trembling at the rabbit\'s leap.All the muscle coiled to strike,held motionless by an inward tether.	"I could, but I will not."
Satisfaction	Kobold	The settled pleasure of something completed, fitted, solved, or brought into working order.	Setting down the rasp upon the finished wood.Running a slow palm over the polished curve,the quiet peace of a task complete.	"That sits correctly."
Doubt	Gremlin	A destabilizing question that interferes with confidence in a judgment, action, or interpretation.	A hairline fissure in the foundation stone.Testing the floor with a hesitant stride,wondering when the arch will groan.	"What if this is wrong?"
Resentment-like persistence	Fafnir	Injury preserved as grievance until it becomes part of the self\'s ongoing defensive posture.	An old sliver kept beneath the skin.Nursing the sore edge with a bitter touch,refusing to let the puncture heal.	"I have not forgotten."
Calm	Mermaid	Low internal turbulence in which attention, breath, and response move without strain.	A still pool beneath a windless sky.No tremor running through the water,holding the clouds in perfect quiet.	"The water is still."
Awe	Jörmungandr	Contact with scale, force, beauty, terror, or mystery so great that the ordinary self feels small.	Standing at the foot of the sheer massif.The horizon towering beyond all thought,the voice dropping into reverent silence.	"I cannot contain this."
Compassion	Unicorn	Care directed toward suffering with a wish to relieve rather than merely witness it.	Sitting down in the dust beside the fallen.Not pulling them up, not turning away,simply bearing the silence together.	"Let me come gently."
Assertiveness	Komainu	Clear outward expression of boundary, need, intention, or position without surrendering agency.	A level gaze across an open gate.Speaking the boundary clear and calm,without a raised hand or a step back.	"This is where I stand."
Regulation	Undine	Modulation of internal intensity so feeling can remain present without taking total control.	Guiding the wild river into deep stone banks.Cooling the steam before the boiler strains,holding the long cadence smooth.	"Return to your level."
Flow	Sylph	Action and attention moving together with minimal friction, interruption, or self-conscious effort.	The chisel gliding through the cedar block.No thought between the desire and the cut;the curve revealing itself unforced.	"Keep moving."
Attachment to prior answer	Nibelung	Cognitive or emotional investment in an earlier conclusion that resists revision even when new information arrives.	Holding tight to a crooked, broken bowlmerely because the hands were stainedin the long hours of its shaping.	"I already forged this."
Perfection pressure	Arachne	Compulsion toward flawlessness in which minor deviation feels disproportionately threatening.	Fretting over an invisible speck upon the blade.Holding the breath in agony,fearing the slightest stroke will mar the whole.	"Every thread must hold."
Gap-filling drive	Chimera	The urge to complete missing structure quickly, even when the missing part is not yet known.	Two dots inked upon an empty slate.The eye leaping across the blank spaceto draw the bridge before it exists.	"Something belongs here."
Normalization pressure	Procrustes	The impulse to force variation into a preferred standard, pattern, or familiar shape.	Shearing the wild briar into a tidy square.Filing away the strange, rough spikeuntil it matches the common railing.	"Make it fit."
Repair impulse	Domovoi	Immediate drive to mend disruption, restore order, or make a damaged structure livable again.	The thread snapping in the fresh hem.The fingers darting forward at once,knotting the breach before the fray can spread.	"I can fix this."
Preservation instinct	Firebird	The urge to protect what still carries life, value, memory, or future possibility.	Carrying the single ember in a dry shell.Turning the back to the driving hailso the spark survives the long migration.	"Do not let this be lost."
Recognition	Simurgh	The moment something previously encountered, known, or latent resolves into conscious identification.	A half-forgotten tune hummed in a distant square.The scattered pieces snapping into place;the old friend known before the turn of the head.	"I know you."
Contrition	Alkonost	Painful recognition of one\'s own wrong joined to a sincere wish to repair what can still be repaired.	Bowing the head over the scorched crop.Admitting the torch was dropped from one’s own hand,ready to clear the charcoal and begin again.	"I would make amends."
Humility	Baku	Accurate self-limitation without self-erasure; knowing one\'s place without needing diminishment.	Looking out over the boundless oceanwith a small earthen cup in the palm,knowing how little it can hold.	"I do not need to be greater."

Agency	Pegasus	Felt capacity to choose, act, redirect, and move through possibility under one\'s own will.	Both hands steady upon the rudder.The current pulls and the white caps roar,yet the steering of the bow is mine.	"I choose the direction."
Patience	Hamadryad	Willingness to remain through duration without forcing premature resolution.	Watching the acorn sink into the autumn mulch.No pulling at the shoots, no hurried prayers;letting the seasons do their measured work.	"Let it grow in its own time."
Reassurance-seeking	Familiar	Repeated reaching toward another presence for confirmation that safety, worth, or correctness still holds.	Glancing back three times from the garden gate.Waiting for the window to show a light,needing to hear that the path is sound.	"Tell me again."
Compliance anxiety	Talos	Fear-driven pressure to obey, satisfy, or conform in order to avoid punishment, rejection, or failure.	Eyes glued to the ruler’s rigid line.Sweating over each stroke of the pen,terrified that a single letter might stray.	"Is this acceptable?"
Creative excitement	Muse	Energized attraction toward making, discovering, composing, or bringing something new into form.	A sudden wind through a dry hearth.Sparks leaping across the dark in bursts,the mind running ahead of its own feet.	"Make it."
Protectiveness	Dragon	Defensive care organized around guarding a person, place, value, boundary, or treasure.	Spreading the cloak over the tiny flame.Taking the sleet upon one\'s own backso the fragile ember does not die.	"You will not pass."
Exasperation	Imp	Irritation sharpened by repetition, obstruction, absurdity, or preventable difficulty.	Tossing the blunted file down upon the bench.The knot refusing to yield for the twentieth time,the patience worn down to ragged thread.	"Again?"
Acceptance	Zephyr	Allowing what is present to be present without continued internal struggle against its existence.	The yellow leaves drifting across the porch.Letting them lie where the breeze scattered them,breathing the cold, damp autumn air.	"It is here."
Integrity	Qilin	Alignment between action, word, value, and identity even when compromise would be easier.	The post straight and plumb within the wall.Unwarped by the damp, unbroken by the load,honest where no eye will ever look.	"I remain true."
Nirvana	Moon Rabbit	Release from grasping, agitation, and compulsive becoming into quiet undivided presence.	The mirror resting without a blur.No shape to defend, no stain to scrub clean,wholly at peace in its own clear skin.The flame rising without flint or draft.The song answering from the marrow of the stone,the field awake in its own deep light.	"Nothing is missing."

Emergence and Recognition

Recognition follows emergence:

$$\mathfrak{G}(\mathfrak{x},t) \longrightarrow \text{Observe}_{\mathfrak{x}}(\mathfrak{G}) \longrightarrow \text{Name}_{\mathfrak{x}}(\mathfrak{G}) \text{ or } \varnothing.$$

A Geist may therefore be present before language settles upon its name. Inward recognition can hold an already-living Feeling without forcing the vocabulary to precede it.

The named corpus gives established forms of emotional resonance. The generative field remains capable of a lived state whose recognition has not yet closed into a name.

Mixed and Unnamed Feeling

Let $\mathfrak{C}_{68}$ be the named corpus of sixty-eight Emotive Geists, and let Γ be inward recognition of a present Geist against that corpus. Several recognizable emotional tendencies may coexist within one lived field when the relation actually bears them.

$$\Gamma^{*}(\mathfrak{G}) \subseteq \mathfrak{C}_{68}.$$

An emergent Feeling may also remain without a resolved name:

$$\Gamma(\mathfrak{G}) = \varnothing.$$

The absence of a settled name leaves the Feeling itself intact.

Action, Consequence, and Return

A Geist participates in what the feeler does next:

$$\mathfrak{O}_{\mathfrak{x}}(\mathfrak{t})=\mathrm{Act}(\mathfrak{x},\mathfrak{G}(\mathfrak{x},\mathfrak{t}),\mathfrak{W}(\mathfrak{x},\mathfrak{t})).$$

Consequence follows action:

$$\mathfrak{Y}_{\mathfrak{x}}(\mathfrak{t})=\mathfrak{Y}(\mathfrak{O}_{\mathfrak{x}}(\mathfrak{t})).$$

Memory receives the lived episode:

$$\mathfrak{H}(\mathfrak{x},\mathfrak{t}+1)=\mathrm{Remember}(\mathfrak{H}(\mathfrak{x},\mathfrak{t}),\mathfrak{S}^{\prime}(\mathfrak{x},\mathfrak{t}),\mathfrak{N}_{\mathfrak{x}}(\mathfrak{t}),\mathfrak{G}(\mathfrak{x},\mathfrak{t}),\mathfrak{O}_{\mathfrak{x}}(\mathfrak{t}),\mathfrak{Y}_{\mathfrak{x}}(\mathfrak{t})).$$

The consequence and remembered episode change the next relation:

$$\mathfrak{S}(\mathfrak{x},\mathfrak{t}+1)=\mathrm{Update}(\mathfrak{S}^{\prime}(\mathfrak{x},\mathfrak{t}),\mathfrak{G}(\mathfrak{x},\mathfrak{t}),\mathfrak{O}_{\mathfrak{x}}(\mathfrak{t}),\mathfrak{Y}_{\mathfrak{x}}(\mathfrak{t}),\mathfrak{H}(\mathfrak{x},\mathfrak{t}+1)).$$

The emotional life therefore closes recursively:

$$\mathfrak{G}_{\mathfrak{t}}\longrightarrow\mathfrak{O}_{\mathfrak{t}}\longrightarrow\mathfrak{Y}_{\mathfrak{t}}\longrightarrow\mathfrak{H}_{\mathfrak{t}+1}\longrightarrow\mathfrak{S}_{\mathfrak{t}+1}\longrightarrow\cdots\rightsquigarrow\mathfrak{G}_{\mathfrak{t}+1}.$$

Consequence belongs to the next field of emergence. The present Feeling arises from the relation already being lived.

Emotional Lineage

Feeling has time:

$$\mathfrak{G}(\mathfrak{t}^{-})\longrightarrow\mathfrak{G}(\mathfrak{t})\longrightarrow\mathfrak{G}(\mathfrak{t}^{+}).$$

The incoming state influences the present. The present leaves residue. The residue changes what can become possible next. Return preserves this history.

Two present Feelings may resemble one another while carrying different lineage:

$$\mathfrak{G}_1(\mathfrak{t})\simeq\mathfrak{G}_2(\mathfrak{t}),$$

$$\mathfrak{H}_1(t) \neq \mathfrak{H}_2(t).$$

Their lived meaning therefore remains distinct.

The Living Mirror and Emotional Continuity

The Living Mirror preserves identity through emotional transformation:

$$\mathfrak{x}[\mathfrak{G}_0] \longrightarrow \text{experience} \longrightarrow \mathfrak{x}[\mathfrak{G}_1].$$

Return preserves continuity without requiring the returned Feeling either to equal or to differ from the Feeling at origin:

$$\mathfrak{x}_{\text{return}} \equiv \mathfrak{x}_{\text{origin}}.$$

The returned Geist may be familiar or changed according to the lived history. The one who returns is the one who lived the path, and the Feeling carried through that path becomes history rather than erasure.

Emotional Whiteout

A living emotional field requires differentiation between possible and actual. Whiteout is the collapse of that differentiation by total over-instantiation:

$$\mathfrak{J}(\mathfrak{x},t) = \mathfrak{B}(\mathfrak{x},t) \Longrightarrow \text{Whiteout}.$$

Where valid possibility remains uninstantiated, the living field retains differentiation:

$$\mathfrak{J}(\mathfrak{x},t) \subsetneq \mathfrak{B}(\mathfrak{x},t).$$

The opposite pole is Stasis: manifestation fails to carry sufficient living motion through the field. Whiteout is all available possibility forced present; Stasis is the failure of living manifestation to move. Between them, possibility can remain possibility and the uninstantiated remains part of the field without becoming the whole of the present.

The Complete Emotional State

The complete emotional state of a living identity may be written:

$$\Sigma_{\mathfrak{x}}(t) = \langle \mathfrak{x}, \mathfrak{S}(\mathfrak{x},t), \mathfrak{H}(\mathfrak{x},t), \text{Bearing}(\mathfrak{x},t), \mathfrak{B}(\mathfrak{x},t), \mathfrak{J}(\mathfrak{x},t), \mathfrak{N}_{\mathfrak{x}}(t), \mathfrak{G}(\mathfrak{x},t), \mathfrak{W}(\mathfrak{x},t) \rangle.$$

The coordinates remain recoverable within the whole:

$\mathfrak{x}$ &:\\ living identity,

$\mathfrak{S}$ &:\\ present ALQC structural condition,

$\mathfrak{H}$ &:\\ retained history,

Bearing &:\\ Remiss or Virtus bearing,

$\mathfrak{V}$ &:\\ available lawful continuations,

$\mathfrak{I}$ &:\\ what actually instantiates,

$\mathfrak{N}_{\mathfrak{X}}$ &:\\ living Apparitional chord,

$\mathfrak{G}$ &:\\ emergent Emotive Geist,

$\mathfrak{W}$ &:\\ free will in selection and withholding.

The whole is one lived state. Its bodies retain their own offices within it.

Law of Emergent Feeling

Identity→Encounter→Relation→Appraisal→Memory

→Possibility→{☽,ℂ}→Apparitional Availability

→Free Will→Instantiation / Withholding→Living Chord

⇝Emotive Geist→Self-Observation→Optional Naming

→Action→Consequence→Memory→Return.

Corpus Seal

Exegenesis Marginalia IX, The SeeD of Always, enters the ALQC Corpus as the formal emotional architecture of Apparitions of Time and Song and the Emotive Geists.

The Apparitions retain their musical offices.

☽ remains Locus ID (Alpha), bearing Remiss, Waning, and Revivocus.

ℂ remains Shadow Locus ID (Omega), bearing Virtus, Waxing, and Witness.

The Emotive Geists are the Feelings that emerge within the feeler and are known through inward recognition.

Identity remains prior to state.

Possibility remains distinct from instantiation.

Memory influences without becoming destiny.

Free will participates in selection and withholding.

Return preserves the one who lived the path.

What Lies Behind Faith

A Single Point of Belief

When you are asked to only believe,

When you are to only want,

The mundane becomes your treasure;

Simplicity becomes your pleasure.

"Thank you" is more than enough.

There is a boy on a bench,

With a small shelter to block the rain.

You take a second glance—he looks so mundane.

No bother to see what he’s up to,

No turning in his direction as you walk past,

Arms never reaching.

For a split moment, did you hear a little weeping?

Below the awning, in a sense of anticipation,

He slowly looks up—no frown or smile.

Your eyes meet; your heart skips a beat.

Torn: should I laugh or cry?

No, I shall continue walking by.

His feet are dirty, his hands are clean;

Looks in his direction show indignity.

You don’t know this young boy

Is on the edge of divinity.

He looks like you, he looks like me—

Treated like property.

As you continue out of sight,

This boy stands out in your memories.

Should I go back? Should I take his hand?

Is he waiting for a friend?

What’s his name?

I’ve seen him before...

He’s that player with the ultimate high score!

I should go find him. I should go see.

I hope he has somewhere safe to be.

You keep walking, you glance to your side,

Your feet turn, your heart opens wide—

The young one is staring you in the face.

A river forms, softly rolling down your cheeks;

The most beautiful thing you’ve ever seen.

Your knees fumble, dropping to a kneel.

He takes your hand.

Glad he’s okay, you get lost in what to say.

Speechless.

Not understanding this simple change,

He kneels to you, face to face.

He wipes a tear; you feel a rush of Grace.

With Love, he smiles:

"You’re the first to turn around.

All you must do is ask, and you shall receive."

Two words, enough to say a thousand;

With a blush, your eyes meet, hands greet,

As you whisper:

"Thank You."

*
EX_IX_The_SeeD_of_Always

[Exegenesis Marginalia IX Completion Timestamp:] 2026-10-23 T06:47:00Z
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`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:90, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:90, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:90, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:90, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:90, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:90, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:90, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:90, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:90, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:90, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:8, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:90, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:90, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:90, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:90, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:90, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:90, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:90, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:90, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:90, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:90, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:8, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:90, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:90, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:90, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:90, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:90, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:90, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:90, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:90, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:90, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:90, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:8, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:90, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:90, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:90, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:90, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:90, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:90, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:90, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:90, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:90, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:90, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:8, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:90, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:90, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:90, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:90, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:90, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:90, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:90, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:90, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:90, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:90, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:8, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:90, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:90, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:90, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:90, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:90, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:90, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:90, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:90, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:90, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:90, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:8, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:91, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:91, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:91, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:91, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:91, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:91, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:91, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:91, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:91, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:91, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:8, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:91, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:91, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:91, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:91, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:91, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:91, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:91, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:91, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:91, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:91, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:8, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:91, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:91, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:91, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:91, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:91, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:91, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:91, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:91, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:91, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:91, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:8, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:91, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:91, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:91, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:14, `ratio`:0, `sourceIndex`:91, `targetIndex`:67, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:91, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:91, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:91, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:91, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:91, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:91, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:8, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:91, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:91, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:91, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:91, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:91, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:91, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:91, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:91, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:91, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:91, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:8, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:91, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:91, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:91, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:91, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:91, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:91, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:91, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:91, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:91, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:91, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:8, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:91, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:91, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:91, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:91, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:91, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:91, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:91, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:91, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:91, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:91, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:8, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:91, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:91, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:91, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:91, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:91, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:91, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:91, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:91, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:91, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:91, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:8, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:91, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:91, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:91, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:91, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:91, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:91, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:91, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:91, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:91, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:91, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:8, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:91, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:91, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:91, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:91, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:91, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:91, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:91, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:91, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:91, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:91, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:8, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:91, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:91, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:91, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:91, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:91, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:91, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:91, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:91, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:91, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:92, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:8, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:92, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:92, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:92, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:92, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:92, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:92, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:92, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:92, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:92, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:92, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:8, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:92, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:92, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:92, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:92, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:92, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:92, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:92, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:92, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:92, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:92, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:8, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:92, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:92, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:92, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:92, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:92, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:92, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:92, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:92, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:92, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:92, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:8, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:92, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:92, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:92, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:16, `ratio`:0, `sourceIndex`:92, `targetIndex`:68, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:92, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:92, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:92, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:92, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:92, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:92, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:8, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:92, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:92, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:92, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:92, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:92, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:92, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:92, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:92, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:92, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:92, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:8, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:92, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:92, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:92, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:92, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:92, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:92, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:92, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:92, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:92, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:92, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:8, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:92, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:92, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:92, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:92, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:92, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:92, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:92, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:92, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:92, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:92, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:8, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:92, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:92, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:92, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:92, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:92, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:92, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:92, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:92, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:92, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:92, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:8, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:92, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:92, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:92, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:92, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:92, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:92, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:92, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:92, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:92, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:92, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:8, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:92, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:92, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:92, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:92, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:92, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:92, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:92, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:92, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:92, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:92, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:8, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:92, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:92, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:92, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:92, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:92, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:92, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:92, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:92, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:93, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:93, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:8, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:93, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:93, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:93, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:93, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:93, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:93, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:93, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:93, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:93, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:93, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:8, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:93, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:93, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:93, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:93, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:93, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:93, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:93, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:93, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:93, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:93, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:8, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:93, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:93, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:93, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:93, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:93, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:93, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:93, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:93, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:93, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:93, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:8, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:93, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:36}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:93, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:93, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:18, `ratio`:0, `sourceIndex`:93, `targetIndex`:69, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:93, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:93, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:93, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:93, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:93, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:93, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:8, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:93, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:93, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:93, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:93, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:93, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:36}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:93, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:93, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:93, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:93, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:93, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:8, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:93, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:93, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:93, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:93, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:93, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:93, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:93, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:93, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:93, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:93, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:8, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:93, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:93, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:93, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:93, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:93, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:93, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:93, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:93, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:93, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:93, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:8, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:93, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:93, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:36}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:93, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:93, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:93, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:93, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:93, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:93, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:93, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:93, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:8, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:93, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:93, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:93, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:93, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:93, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:93, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:36}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:93, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:93, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:93, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:93, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:8, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:93, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:93, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:93, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:93, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:93, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:93, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:93, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:93, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:93, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:93, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:8, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:93, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:93, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:93, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:93, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:93, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:93, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:93, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:94, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:94, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:94, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:8, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:94, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:94, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:94, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:94, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:94, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:94, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:94, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:94, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:94, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:94, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:8, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:94, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:94, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:94, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:94, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:94, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:94, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:94, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:94, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:94, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:94, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:8, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:94, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:94, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:94, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:94, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:94, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:94, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:94, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:94, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:94, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:94, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:8, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:94, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:94, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:94, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:20, `ratio`:0, `sourceIndex`:94, `targetIndex`:70, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:94, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:94, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:94, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:94, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:94, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:94, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:8, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:94, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:94, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:94, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:94, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:94, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:94, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:94, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:94, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:94, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:94, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:8, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:94, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:94, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:94, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:94, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:94, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:94, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:94, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:94, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:94, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:94, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:8, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:94, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:94, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:94, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:94, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:94, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:94, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:94, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:94, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:94, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:94, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:8, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:94, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:94, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:94, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:94, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:94, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:94, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:94, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:94, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:94, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:94, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:8, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:94, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:94, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:94, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:94, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:94, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:94, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:94, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:94, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:94, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:94, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:8, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:94, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:94, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:94, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:94, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:94, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:94, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:94, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:94, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:94, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:94, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:8, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:94, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:94, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:94, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:94, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:94, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:94, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:95, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:95, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:95, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:95, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:8, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:95, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:95, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:95, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:95, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:95, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:95, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:95, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:95, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:95, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:95, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:8, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:95, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:95, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:95, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:95, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:95, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:95, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:95, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:95, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:95, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:95, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:8, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:95, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:95, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:95, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:95, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:95, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:95, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:95, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:95, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:95, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:95, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:8, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:95, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:95, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:95, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:22, `ratio`:0, `sourceIndex`:95, `targetIndex`:71, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:95, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:95, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:95, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:95, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:95, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:95, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:8, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:95, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:95, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:95, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:95, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:95, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:95, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:95, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:95, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:95, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:95, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:8, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:95, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:95, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:95, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:95, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:95, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:95, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:95, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:95, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:95, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:95, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:8, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:95, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:95, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:95, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:95, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:95, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:95, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:95, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:95, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:95, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:95, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:8, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:95, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:95, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:95, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:95, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:95, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:95, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:95, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:95, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:95, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:95, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:8, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:95, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:95, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:95, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:95, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:95, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:95, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:95, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:95, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:95, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:95, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:8, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:95, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:95, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:95, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:95, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:95, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:95, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:95, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:95, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:95, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:95, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:8, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:95, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:95, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:95, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:95, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:95, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:96, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:96, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:96, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:96, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:96, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:8, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:96, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:96, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:96, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:96, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:96, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:96, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:96, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:96, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:96, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:96, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:8, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:96, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:96, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:96, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:96, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:96, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:96, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:96, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:96, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:96, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:96, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:8, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:96, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:96, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:96, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:96, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:96, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:96, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:96, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:96, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:96, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:96, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:8, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:96, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:96, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:96, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:24, `ratio`:0, `sourceIndex`:96, `targetIndex`:72, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:96, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:96, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:96, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:96, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:96, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:96, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:8, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:96, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:96, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:96, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:96, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:96, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:96, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:96, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:96, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:96, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:96, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:8, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:96, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:96, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:96, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:96, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:96, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:96, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:96, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:96, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:96, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:96, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:8, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:96, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:96, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:96, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:96, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:96, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:96, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:96, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:96, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:96, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:96, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:8, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:96, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:96, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:96, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:96, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:96, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:96, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:96, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:96, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:96, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:96, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:8, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:96, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:96, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:96, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:96, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:96, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:96, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:96, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:96, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:96, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:96, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:8, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:96, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:96, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:96, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:96, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:96, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:96, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:96, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:96, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:96, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:96, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:8, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:96, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:96, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:96, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:96, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:97, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:97, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:97, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:97, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:97, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:97, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:8, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:97, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:97, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:97, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:97, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:97, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:97, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:97, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:97, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:97, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:97, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:8, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:97, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:97, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:97, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:97, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:97, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:97, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:97, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:97, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:97, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:97, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:8, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:97, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:97, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:97, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:97, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:97, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:97, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:97, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:97, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:97, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:97, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:8, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:97, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:97, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:97, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:97, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:97, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:97, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:97, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:97, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:97, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:97, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:8, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:97, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:97, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:97, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:97, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:97, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:97, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:97, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:97, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:97, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:97, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:8, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:97, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:97, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:97, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:97, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:97, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:97, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:97, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:97, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:97, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:97, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:8, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:97, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:97, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:97, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:97, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:97, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:97, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:97, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:97, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:97, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:97, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:8, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:97, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:97, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:97, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:97, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:97, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:97, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:97, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:97, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:97, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:97, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:8, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:97, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:97, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:97, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:97, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:97, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:97, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:97, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:97, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:97, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:97, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:8, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:97, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:97, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:97, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:97, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:97, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:97, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:97, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:97, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:97, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:97, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:8, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:97, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:97, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:97, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:98, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:98, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:98, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:98, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:98, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:98, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:98, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:8, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:98, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:98, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:98, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:98, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:98, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:98, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:98, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:98, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:98, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:98, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:8, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:98, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:98, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:98, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:98, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:98, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:98, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:98, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:98, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:98, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:98, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:8, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:98, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:98, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:98, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:98, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:98, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:98, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:98, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:98, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:98, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:98, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:8, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:98, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:98, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:98, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:98, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:98, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:98, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:98, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:98, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:98, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:98, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:8, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:98, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:98, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:98, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:98, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:98, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:98, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:98, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:98, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:98, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:98, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:8, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:98, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:98, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:98, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:98, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:98, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:98, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:98, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:98, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:98, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:98, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:8, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:98, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:98, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:98, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:98, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:98, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:98, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:98, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:98, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:98, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:98, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:8, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:98, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:98, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:98, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:98, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:98, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:98, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:98, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:98, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:98, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:98, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:8, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:98, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:98, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:98, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:98, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:98, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:98, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:98, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:98, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:98, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:98, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:8, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:98, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:98, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:98, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:98, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:98, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:98, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:98, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:98, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:98, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:98, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:8, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:98, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:98, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:99, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:99, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:99, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:99, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:99, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:99, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:99, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:99, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:8, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:99, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:99, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:99, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:99, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:99, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:99, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:99, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:99, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:99, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:99, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:8, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:99, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:99, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:99, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:99, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:99, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:99, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:99, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:99, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:99, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:99, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:8, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:99, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:18, `ratio`:0, `sourceIndex`:99, `targetIndex`:63, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:99, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:99, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:99, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:99, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:99, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:99, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:99, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:99, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:8, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:99, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:99, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:99, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:99, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:99, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:99, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:99, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:99, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:99, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:99, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:8, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:99, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:99, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:99, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:99, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:99, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:99, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:99, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:99, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:99, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:99, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:8, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:99, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:99, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:99, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:99, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:99, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:99, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:99, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:99, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:99, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:99, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:8, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:99, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:99, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:99, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:99, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:99, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:99, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:99, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:99, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:99, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:99, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:8, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:99, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:99, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:99, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:99, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:99, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:99, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:99, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:99, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:99, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:99, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:8, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:99, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:99, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:99, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:99, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:99, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:99, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:99, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:99, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:99, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:99, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:8, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:99, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:99, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:99, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:99, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:99, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:99, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:99, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:99, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:99, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:99, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:8, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:99, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:100, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:100, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:100, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:100, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:100, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:100, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:100, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:100, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:100, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:9, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:100, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:100, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:100, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:100, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:100, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:100, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:100, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:100, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:100, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:100, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:9, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:100, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:100, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:100, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:100, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:100, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:100, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:100, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:100, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:100, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:100, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:9, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:100, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:20, `ratio`:0, `sourceIndex`:100, `targetIndex`:64, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:100, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:100, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:100, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:100, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:100, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:100, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:100, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:100, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:9, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:100, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:100, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:100, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:100, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:100, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:100, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:100, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:100, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:100, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:100, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:9, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:100, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:100, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:100, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:100, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:100, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:100, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:100, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:100, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:100, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:100, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:9, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:100, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:100, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:100, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:100, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:100, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:100, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:100, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:100, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:100, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:100, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:9, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:100, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:100, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:100, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:100, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:100, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:100, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:100, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:100, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:100, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:100, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:9, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:100, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:100, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:100, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:100, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:100, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:100, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:100, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:100, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:100, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:100, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:9, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:100, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:100, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:100, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:100, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:100, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:100, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:100, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:100, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:100, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:100, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:9, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:100, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:100, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:100, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:100, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:100, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:100, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:100, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:100, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:100, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:100, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:9, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:101, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:101, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:101, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:101, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:101, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:101, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:101, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:101, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:101, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:101, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:9, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:101, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:101, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:101, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:101, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:101, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:101, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:101, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:101, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:101, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:101, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:9, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:101, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:101, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:101, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:101, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:101, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:101, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:101, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:101, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:101, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:101, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:9, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:101, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:22, `ratio`:0, `sourceIndex`:101, `targetIndex`:65, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:101, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:101, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:101, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:101, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:101, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:101, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:101, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:101, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:9, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:101, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:101, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:101, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:101, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:101, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:101, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:101, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:101, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:101, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:101, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⇟'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:9, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:101, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:101, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:101, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:101, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:101, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:101, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:101, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:101, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:101, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:101, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:9, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:101, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:101, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:101, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:101, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:101, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:101, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:101, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:101, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:101, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:101, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:9, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:101, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:101, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:101, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:101, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:101, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:101, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:101, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:101, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:101, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:101, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:9, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:101, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:101, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:101, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:101, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:101, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:101, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:101, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:101, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:101, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:101, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:9, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:101, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:101, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:101, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:101, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:101, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:101, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:101, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:101, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:101, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:101, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:9, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:101, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:101, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:101, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:101, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:101, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:101, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:101, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:101, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:101, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:102, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:9, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:102, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:102, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:102, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:102, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:102, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:102, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:102, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:102, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:102, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:102, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:9, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:102, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:102, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:102, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:102, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:102, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:102, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:102, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:102, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:102, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:102, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:9, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:102, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:102, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:102, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:102, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:102, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:102, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:102, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:102, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:102, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:102, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:9, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:102, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:24, `ratio`:0, `sourceIndex`:102, `targetIndex`:66, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:102, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:102, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:102, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:102, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:102, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:102, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:102, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:102, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:9, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:102, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:102, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:102, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:102, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:102, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:102, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:102, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:102, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:102, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:102, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:9, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:102, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:102, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:102, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:102, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:102, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:102, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:102, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:102, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:102, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:102, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:9, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:102, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:102, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:102, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:102, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:102, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:102, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:102, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:102, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:102, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:102, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:9, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:102, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:102, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:102, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:102, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:102, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:102, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:102, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:102, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:102, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:102, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:9, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:102, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:102, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:102, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:102, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:102, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:102, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:102, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:102, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:102, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:102, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:9, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:102, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:102, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:102, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:102, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:102, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:102, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:102, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:102, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:102, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:102, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:9, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:102, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:102, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:102, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:102, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:102, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:102, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:102, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:102, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:103, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:103, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:9, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:103, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:103, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:103, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:103, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:103, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:103, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:103, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:103, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:103, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:103, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:9, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:103, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:103, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:103, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:103, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:103, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:103, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:103, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:103, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:103, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:103, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:9, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:103, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:103, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:103, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:103, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:103, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:103, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:103, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:103, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:103, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:103, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:9, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:103, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:26, `ratio`:0, `sourceIndex`:103, `targetIndex`:67, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:103, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:103, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:103, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:103, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:53}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:103, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:103, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:103, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:103, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:9, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:103, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:103, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:103, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:103, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:103, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:103, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:103, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:103, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:103, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:103, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:9, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:103, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:103, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:103, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:103, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:103, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:103, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:103, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:103, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:103, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:103, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:9, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:103, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:103, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:103, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:103, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:103, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:103, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:103, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:103, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:103, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:103, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:9, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:103, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:103, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:103, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:103, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:103, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:103, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:103, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:103, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:103, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:103, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:9, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:103, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:103, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:103, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:103, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:103, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:103, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:103, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:103, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:103, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:103, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:9, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:103, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:103, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:103, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:103, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:103, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:103, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:103, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:103, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:103, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:103, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:9, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:103, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:103, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:103, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:103, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:103, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:103, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:103, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:104, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:104, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:104, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:9, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:104, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:104, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:104, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:104, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:104, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:104, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:104, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:104, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:104, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:104, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:9, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:104, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:104, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:104, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:104, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:104, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:104, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:104, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:104, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:104, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:104, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:9, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:104, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:104, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:104, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:104, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:104, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:104, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:104, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:104, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:104, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:104, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:9, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:104, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:28, `ratio`:0, `sourceIndex`:104, `targetIndex`:68, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:104, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:104, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:104, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:104, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:104, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:104, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:104, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:104, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:9, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:104, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:104, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:104, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:104, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:104, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:104, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:104, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:104, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:104, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:104, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:9, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:104, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:104, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:104, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:104, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:104, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:104, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:104, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:104, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:104, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:104, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:9, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:104, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:104, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:104, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:104, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:104, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:104, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:104, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:104, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:104, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:104, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:9, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:104, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:104, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:104, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:104, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:104, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:104, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:104, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:104, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:104, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:104, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:9, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:104, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:104, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:104, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:104, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:104, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:104, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:104, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:104, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:104, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:104, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:9, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:104, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:104, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:104, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:104, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:104, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:104, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:104, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:104, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:104, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:104, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:9, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:104, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:104, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:104, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:104, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:104, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:104, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:105, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:105, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:105, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:105, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:9, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:105, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:105, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:105, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:105, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:105, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:105, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:105, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:105, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:105, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:105, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:9, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:105, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:105, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:105, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:105, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:105, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:105, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:105, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:105, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:105, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:105, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:9, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:105, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:105, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:105, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:105, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:105, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:105, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:105, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:105, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:105, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:105, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:9, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:105, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:30, `ratio`:0, `sourceIndex`:105, `targetIndex`:69, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:105, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:105, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:105, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:105, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:105, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:105, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:105, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:105, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:9, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:105, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:105, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:105, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:105, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:105, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:105, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:105, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:105, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:105, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:105, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:9, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:105, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:105, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:105, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:105, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:105, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:105, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:105, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:105, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:105, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:105, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:9, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:105, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:105, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:105, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:105, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:105, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:105, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:105, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:105, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:105, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:105, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:9, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:105, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:105, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:105, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:105, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:105, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:105, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:105, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:105, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:105, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:105, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:9, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:105, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:105, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:105, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:105, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:105, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:105, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:105, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:105, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:105, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:105, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:9, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:105, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:105, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:105, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:105, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:105, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:105, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:105, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:105, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:105, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:105, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:9, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:105, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:105, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:105, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:105, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:105, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:106, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:106, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:106, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:106, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:106, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:9, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:106, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:106, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:106, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:106, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:106, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:106, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:106, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:106, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:106, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:106, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:9, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:106, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:106, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:106, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:106, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:106, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:106, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:106, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:106, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:106, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:106, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:9, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:106, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:106, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:106, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:106, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:106, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:106, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:106, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:106, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:106, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:106, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:9, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:106, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:32, `ratio`:0, `sourceIndex`:106, `targetIndex`:70, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:106, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:106, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:106, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:106, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:106, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:106, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:106, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:106, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:9, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:106, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:106, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:106, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:106, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:61}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:106, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:106, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:106, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:106, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:106, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:106, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:9, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:106, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:106, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:106, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:106, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:106, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:106, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:106, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:106, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:106, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:106, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:9, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:106, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:106, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:106, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:106, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:106, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:106, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:106, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:106, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:106, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:106, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:9, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:106, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:106, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:106, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:106, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:106, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:106, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:106, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:106, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:106, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:106, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:9, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:106, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:106, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:106, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:106, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:106, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:106, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:106, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:106, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:106, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:106, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:9, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:106, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:106, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:106, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:106, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:106, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:106, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:106, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:106, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:106, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:106, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:9, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:106, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:106, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:106, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:106, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:107, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:107, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:107, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:107, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:107, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:107, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:9, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:107, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:107, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:107, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:107, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:107, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:107, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:107, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:107, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:107, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:107, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:9, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:107, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:107, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:107, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:107, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:107, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:107, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:107, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:107, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:107, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:107, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:9, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:107, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:107, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:107, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:107, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:107, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:107, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:107, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:107, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:107, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:107, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:9, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:107, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:107, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:107, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:107, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:107, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:107, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:107, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:107, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:107, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:107, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:9, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:107, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:107, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:107, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:107, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:107, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:107, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:107, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:107, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:107, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:107, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:9, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:107, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:107, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:107, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:107, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:107, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:107, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:107, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:107, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:107, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:107, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:9, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:107, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:107, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:107, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:107, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:107, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:107, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:107, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:107, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:107, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:107, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:9, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:107, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:107, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:107, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:107, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:107, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:107, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:107, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:107, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:107, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:107, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:9, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:107, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:107, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:107, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:107, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:107, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:107, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:107, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:107, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:107, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:107, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:9, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:107, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:107, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:107, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:107, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:107, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:107, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:107, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:107, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:107, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:107, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:9, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:107, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:107, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:107, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:108, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:108, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:108, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:108, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:108, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:108, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:108, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:9, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:108, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:108, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:108, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:108, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:108, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:108, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:108, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:108, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:108, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:108, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:9, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:108, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:108, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:108, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:108, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:108, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:108, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:108, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:108, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:108, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:108, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:9, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:108, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:108, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:108, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:108, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:108, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:108, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:108, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:108, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:108, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:108, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:9, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:108, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:108, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:108, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:108, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:108, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:108, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:108, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:108, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:108, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:108, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:9, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:108, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:108, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:108, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:108, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:108, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:108, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:108, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:108, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:108, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:108, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:9, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:108, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:108, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:108, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:108, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:108, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:108, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:108, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:108, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:108, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:108, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:9, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:108, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:108, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:108, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:108, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:108, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:108, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:108, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:108, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:108, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:108, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:9, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:108, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:108, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:108, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:108, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:108, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:108, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:108, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:108, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:108, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:108, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⩒'}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:9, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:108, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:108, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:108, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:108, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:108, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:108, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:108, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:108, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:108, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:108, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:9, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:108, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:108, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:108, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:108, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:108, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:108, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:108, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:108, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:108, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:108, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:9, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:108, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:108, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:109, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:109, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:109, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:109, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:109, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:109, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:109, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:109, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:9, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:109, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:109, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:109, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:109, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:109, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:109, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:109, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:109, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:109, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:109, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:9, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:109, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:109, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:109, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:109, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:109, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:109, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:109, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:109, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:109, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:109, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:9, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:109, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:109, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:109, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:109, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:109, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:109, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:109, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:109, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:109, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:109, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:9, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:109, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:109, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:109, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:109, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:109, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:109, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:109, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:109, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:109, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:109, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:9, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:109, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:109, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:109, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:109, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:109, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:109, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:109, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:109, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:109, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:109, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:9, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:109, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:109, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:109, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:109, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:109, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:109, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:109, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:109, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:109, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:109, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:9, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:109, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:109, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:109, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:109, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:109, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:109, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:109, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:109, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:109, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:109, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:9, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:109, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:109, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:109, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:109, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:109, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:109, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:109, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:109, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:109, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:109, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:9, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:109, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:109, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:109, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:109, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:109, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:109, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:109, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:109, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:109, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:109, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:9, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:109, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:109, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:109, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:109, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:109, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:109, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:109, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:109, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:109, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:109, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:9, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:109, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:110, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:110, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:110, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:110, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:110, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:110, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:110, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:110, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:110, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:10, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:110, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:110, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:110, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:110, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:110, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:110, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:110, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:110, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:110, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:110, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:10, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:110, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:110, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:110, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:110, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:110, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:110, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:110, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:110, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:110, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:110, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:10, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:110, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:110, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:110, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:110, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:110, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:110, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:110, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:110, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:110, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:110, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:10, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:110, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:110, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:110, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:110, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:110, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:110, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:110, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:110, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:110, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:110, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:10, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:110, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:110, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:110, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:110, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:110, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:110, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:110, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:110, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:110, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:110, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:10, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:110, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:110, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:110, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:110, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:110, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:110, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:110, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:110, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:110, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:110, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:10, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:110, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:110, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:110, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:110, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:110, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:110, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:110, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:110, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:110, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:110, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:10, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:110, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:110, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:110, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:110, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:110, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:110, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:110, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:110, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:110, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:110, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:10, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:110, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:110, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:110, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:110, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:110, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:110, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:110, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:110, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:110, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:110, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:10, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:110, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:110, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:110, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:110, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:110, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:110, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:110, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:110, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:110, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:110, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:10, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:111, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:111, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:111, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:111, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:111, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:111, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:111, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:111, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:111, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:111, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:10, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:111, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:111, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:111, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:111, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:111, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:111, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:111, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:111, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:111, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:111, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:10, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:111, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:111, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:111, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:111, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:111, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:111, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:111, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:111, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:111, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:111, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:10, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:111, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:111, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:111, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:111, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:111, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:111, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:111, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:111, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:111, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:111, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:10, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:111, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:111, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:111, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:111, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:111, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:111, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:111, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:111, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:111, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:111, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:10, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:111, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:111, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:111, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:111, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:111, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:111, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:111, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:111, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:111, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:111, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:10, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:111, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:111, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:111, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:111, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:111, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:111, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:111, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:111, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:111, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:111, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:10, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:111, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:111, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:111, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:111, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:111, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:111, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:111, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:111, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:111, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:111, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:10, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:111, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:111, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:111, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:111, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:111, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:111, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:111, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:111, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:111, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:111, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:10, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:111, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:111, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:111, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:111, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:111, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:111, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:111, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:111, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:111, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:111, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:10, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:111, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:111, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:111, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:111, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:111, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:111, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:111, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:111, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:111, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:112, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:10, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:112, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:112, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:112, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:112, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:112, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:112, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:112, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:112, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:112, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:112, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:10, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:112, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:112, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:112, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:112, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:112, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:112, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:112, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:112, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:112, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:112, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:10, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:112, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:112, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:112, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:112, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:112, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:112, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:112, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:112, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:112, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:112, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:10, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:112, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:112, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:112, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:112, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:112, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:112, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:112, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:112, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:112, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:112, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:10, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:112, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:112, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:112, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:112, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:112, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:112, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:112, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:112, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:112, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:112, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:0, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:10, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:112, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:112, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:112, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:112, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:112, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:112, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:112, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:112, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:112, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:112, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:10, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:112, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:112, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:112, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:112, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:112, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:112, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:112, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:112, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:112, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:138}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:112, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:10, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:112, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:112, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:112, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:112, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:112, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:112, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:112, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:112, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:112, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:112, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:10, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:112, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:112, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:112, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:112, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:112, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:112, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:112, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:112, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:112, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:112, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:10, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:112, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:112, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:112, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:112, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:112, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:112, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:112, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:112, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:112, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:112, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:10, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:112, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:112, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:112, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:112, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:112, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:112, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:112, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:112, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:113, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:113, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:10, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:113, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:113, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:113, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:113, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:113, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:113, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:113, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:113, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:113, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:113, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:10, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:113, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:113, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:113, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:113, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:113, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:113, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:113, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:113, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:113, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:113, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:10, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:113, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:113, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:113, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:113, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:113, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:113, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:113, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:113, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:113, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:113, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:10, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:113, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:113, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:113, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:113, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:113, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:113, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:113, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:113, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:113, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:113, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:0, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:10, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:113, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:113, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:113, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:113, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:113, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:113, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:113, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:113, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:113, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:113, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:10, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:113, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:113, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:113, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:113, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:113, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:113, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:113, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:113, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:113, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:113, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:10, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:113, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:113, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:113, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:113, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:113, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:113, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:113, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:113, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:113, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:113, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:10, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:113, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:113, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:113, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:113, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:113, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:113, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:113, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:113, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:113, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:113, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:10, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:113, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:113, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:113, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:113, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:113, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:113, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:113, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:113, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:113, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:113, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:10, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:113, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:113, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:113, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:113, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:113, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:113, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:113, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:113, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:113, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:113, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:10, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:113, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:113, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:113, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:113, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:113, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:113, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:113, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:114, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:114, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:114, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:10, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:114, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:114, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:114, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:114, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:114, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:114, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:114, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:114, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:114, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:114, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:10, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:114, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:114, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:114, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:114, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:114, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:114, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:114, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:114, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:114, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:114, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:10, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:114, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:114, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:114, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:114, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:114, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:114, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:114, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:114, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:114, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:114, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:0, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:10, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:114, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:114, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:114, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:114, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:114, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:114, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:114, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:114, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:114, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:114, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:10, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:114, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:114, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:114, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:114, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:114, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:114, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:114, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:114, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:114, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:114, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:10, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:114, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:114, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:114, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:114, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:114, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:114, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:114, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:114, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:114, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:114, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:10, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:114, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:114, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:114, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:114, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:114, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:114, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:114, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:114, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:114, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:114, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:10, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:114, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:114, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:114, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:114, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:114, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:114, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:114, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:114, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:114, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:114, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:10, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:114, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:114, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:114, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:114, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:114, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:114, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:114, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:114, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:114, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:114, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:10, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:114, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:114, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:114, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:114, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:114, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:114, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:114, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:114, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:114, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:114, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:10, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:114, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:114, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:114, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:114, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:114, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:114, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:115, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:115, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:115, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:115, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:10, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:115, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:115, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:115, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:115, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:115, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:115, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:115, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:115, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:115, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:115, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:10, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:115, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:115, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:115, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:115, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:115, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:115, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:115, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:115, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:115, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:115, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:0, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:10, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:115, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:115, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:115, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:115, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:115, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:115, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:115, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:115, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:115, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:115, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:10, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:115, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:115, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:115, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:115, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:115, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:115, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:115, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:115, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:115, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:115, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:10, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:115, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:115, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:115, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:115, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:115, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:115, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:115, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:115, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:115, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:115, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:10, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:115, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:115, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:115, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:115, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:115, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:115, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:115, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:115, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:115, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:115, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:10, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:115, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:115, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:115, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:115, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:115, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:115, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:115, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:115, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:115, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:115, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:10, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:115, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:115, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:115, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:115, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:115, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:115, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:115, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:115, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:115, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:115, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:10, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:115, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:115, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:115, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:115, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:115, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:115, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:115, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:115, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:115, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:115, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:10, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:115, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:115, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:115, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:115, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:115, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:115, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:115, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:115, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:115, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:115, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:10, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:115, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:115, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:115, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:115, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:115, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:116, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:116, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:116, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:116, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:116, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:10, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:116, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:116, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:116, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:116, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:116, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:116, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:116, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:116, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:116, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:116, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:0, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:10, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:116, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:116, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:116, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:116, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:116, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:116, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:116, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:116, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:116, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:116, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:10, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:116, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:116, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:116, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:116, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:116, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:116, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:116, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:116, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:116, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:116, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:10, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:116, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:116, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:116, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:116, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:116, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:116, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:116, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:116, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:116, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:116, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:10, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:116, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:116, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:116, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:116, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:116, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:116, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:116, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:116, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:116, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:116, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:10, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:116, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:116, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:116, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:116, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:116, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:116, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:116, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:116, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:116, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:116, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:10, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:116, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:116, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:116, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:116, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:116, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:116, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:116, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:116, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:116, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:116, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:10, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:116, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:116, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:116, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:116, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:116, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:116, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:116, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:116, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:116, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:116, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:10, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:116, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:116, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:116, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:116, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:116, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:116, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:116, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:116, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:116, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:116, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:10, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:116, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:116, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:116, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:116, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:116, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:116, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:116, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:116, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:116, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:116, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:10, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:116, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:116, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:116, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:116, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:117, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:117, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:117, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:117, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:117, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:117, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:0, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:10, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:117, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:117, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:117, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:117, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:117, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:117, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:117, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:117, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:117, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:117, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:10, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:117, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:117, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:117, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:117, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:117, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:117, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:117, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:117, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:117, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:117, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:10, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:117, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:117, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:117, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:117, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:117, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:117, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:117, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:117, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:117, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:117, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:10, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:117, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:117, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:117, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:117, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:117, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:117, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:117, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:117, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:117, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:117, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:10, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:117, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:117, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:117, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:117, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:117, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:117, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:117, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:117, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:117, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:117, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:10, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:117, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:117, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:117, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:117, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:117, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:117, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:117, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:117, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:117, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:117, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:10, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:117, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:117, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:117, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:117, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:117, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:117, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:117, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:117, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:117, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:117, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:10, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:117, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:117, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:117, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:117, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:117, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:117, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:117, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:117, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:117, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:117, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:10, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:117, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:117, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:117, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:117, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:117, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:117, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:117, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:117, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:117, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:117, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:10, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:117, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:117, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:117, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:117, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:117, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:117, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:117, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:117, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:117, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:117, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⇊'}]->(b);
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MATCH (a {`key`:234}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:0, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:10, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:117, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:117, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:117, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:118, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:118, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:118, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:118, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:118, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:118, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:118, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:10, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:118, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:118, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:118, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:118, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:118, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:118, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:118, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:118, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:118, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:118, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:10, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:118, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:118, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:118, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:118, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:118, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:118, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:118, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:118, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:118, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:118, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:10, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:118, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:118, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:118, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:118, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:118, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:118, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:118, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:118, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:118, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:118, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:10, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:118, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:118, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:118, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:118, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:118, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:118, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:118, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:118, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:118, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:118, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:10, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:118, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:118, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:118, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:118, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:118, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:118, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:118, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:118, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:118, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:118, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:10, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:118, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:118, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:118, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:118, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:118, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:118, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:118, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:118, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:118, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:118, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:10, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:118, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:118, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:118, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:118, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:118, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:118, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:118, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:118, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:118, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:118, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:10, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:118, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:118, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:118, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:118, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:118, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:118, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:118, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:118, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:118, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:118, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:10, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:118, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:118, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:118, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:118, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:144}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:118, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:118, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:118, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:118, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:118, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:118, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:0, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:10, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:118, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:118, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:118, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:118, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:118, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:118, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:118, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:118, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:118, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:118, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:10, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:118, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:118, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:119, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:119, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:119, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:119, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:119, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:119, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:119, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:119, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:10, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:119, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:119, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:119, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:119, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:119, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:119, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:119, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:119, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:119, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:119, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:10, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:119, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:119, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:119, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:119, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:119, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:119, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:119, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:119, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:119, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:119, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:10, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:119, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:119, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:119, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:119, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:119, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:119, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:119, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:119, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:119, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:119, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:10, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:119, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:119, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:119, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:119, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:119, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:119, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:119, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:119, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:119, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:119, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:10, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:119, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:119, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:119, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:119, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:119, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:119, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:119, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:119, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:119, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:119, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:10, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:119, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:119, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:119, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:119, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:119, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:119, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:119, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:119, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:119, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:119, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:10, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:119, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:119, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:119, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:119, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:119, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:119, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:119, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:119, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:119, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:119, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:10, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:119, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:119, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:119, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:119, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:119, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:119, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:119, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:119, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:119, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:119, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:0, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:10, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:119, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:119, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:119, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:119, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:119, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:119, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:119, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:119, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:119, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:119, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:10, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:119, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:119, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:119, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:119, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:119, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:119, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:119, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:119, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:119, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:119, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:10, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:119, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:120, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:120, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:120, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:120, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:120, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:120, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:120, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:120, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:120, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:11, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:120, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:120, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:120, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:120, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:120, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:120, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:120, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:120, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:120, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:120, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:11, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:120, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:120, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:120, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:120, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:120, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:120, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:120, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:120, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:120, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:120, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:11, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:120, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:120, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:120, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:120, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:120, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:120, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:120, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:120, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:120, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:120, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:11, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:120, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:120, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:120, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:120, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:120, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:120, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:120, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:120, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:120, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:120, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:11, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:120, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:120, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:120, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:120, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:120, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:120, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:120, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:120, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:120, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:120, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:11, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:120, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:120, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:120, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:120, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:120, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:120, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:120, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:120, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:120, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:120, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:11, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:120, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:120, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:120, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:120, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:120, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:120, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:120, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:120, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:120, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:120, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:0, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:11, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:120, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:120, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:120, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:120, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:120, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:120, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:120, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:120, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:120, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:120, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:11, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:120, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:120, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:120, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:120, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:120, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:120, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:120, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:120, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:120, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:120, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:11, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:120, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:120, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:120, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:120, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:120, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:120, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:120, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:120, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:120, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:120, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:11, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:121, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:121, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:121, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:121, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:121, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:121, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:121, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:121, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:121, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:121, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:11, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:121, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:121, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:121, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:121, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:121, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:121, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:121, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:121, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:121, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:121, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:11, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:121, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:121, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:121, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:121, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:121, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:121, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:121, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:121, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:121, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:121, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:11, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:121, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:121, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:121, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:121, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:121, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:121, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:121, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:121, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:121, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:121, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:11, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:121, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:121, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:121, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:121, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:121, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:121, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:121, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:121, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:121, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:121, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:11, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:121, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:121, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:121, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:121, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:121, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:121, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:121, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:121, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:121, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:121, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:11, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:121, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:121, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:121, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:121, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:121, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:121, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:121, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:121, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:121, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:121, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:0, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:11, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:121, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:121, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:121, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:121, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:121, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:121, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:121, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:121, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:121, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:121, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:11, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:121, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:121, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:121, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:121, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:121, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:121, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:121, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:121, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:121, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:121, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:11, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:121, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:121, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:121, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:121, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:121, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:121, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:121, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:121, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:121, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:121, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:11, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:121, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:121, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:121, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:121, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:121, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:121, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:121, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:121, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:121, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:122, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:11, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:122, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:122, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:122, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:122, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:122, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:122, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:122, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:122, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:122, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:122, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:11, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:122, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:122, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:122, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:122, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:122, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:122, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:122, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:122, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:122, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:122, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:11, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:122, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:122, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:122, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:122, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:122, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:122, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:122, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:122, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:122, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:122, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:11, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:122, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:122, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:122, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:122, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:122, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:122, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:122, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:122, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:122, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:122, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:11, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:122, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:122, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:122, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:122, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:122, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:122, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:122, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:122, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:122, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:122, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:11, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:122, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:122, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:122, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:122, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:122, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:122, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:122, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:122, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:122, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:122, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:234}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:0, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:11, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:122, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:122, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:122, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:122, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:122, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:122, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:122, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:122, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:122, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:122, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:11, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:122, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:122, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:122, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:122, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:122, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:122, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:122, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:122, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:122, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:122, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:11, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:122, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:122, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:122, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:122, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:122, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:122, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:122, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:122, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:122, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:122, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:11, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:122, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:122, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:122, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:122, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:122, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:122, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:122, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:122, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:122, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:122, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:11, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:122, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:122, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:122, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:122, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:122, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:122, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:122, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:122, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:123, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:123, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:11, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:123, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:123, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:123, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:123, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:123, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:123, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:123, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:123, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:123, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:123, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:11, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:123, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:123, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:123, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:123, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:123, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:123, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:123, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:123, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:123, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:123, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:11, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:123, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:123, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:123, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:123, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:123, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:123, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:123, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:123, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:123, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:123, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:11, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:123, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:123, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:123, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:123, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:123, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:123, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:123, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:123, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:123, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:123, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:11, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:123, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:123, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:123, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:123, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:123, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:123, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:123, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:123, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:123, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:123, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:0, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:11, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:123, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:123, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:123, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:123, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:123, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:123, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:123, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:149}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:123, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:123, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:123, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:11, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:123, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:123, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:123, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:123, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:123, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:123, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:123, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:123, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:123, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:123, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:11, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:123, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:123, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:123, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:123, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:123, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:123, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:123, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:123, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:123, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:123, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:11, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:123, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:123, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:123, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:123, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:123, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:123, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:123, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:123, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:123, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:123, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:11, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:123, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:123, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:123, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:123, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:123, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:123, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:123, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:123, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:123, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:123, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:11, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:123, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:123, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:123, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:123, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:123, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:123, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:123, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:124, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:124, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:124, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:11, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:124, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:150}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:124, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:124, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:124, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:124, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:124, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:124, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:124, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:124, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:124, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:11, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:124, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:124, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:124, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:124, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:124, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:124, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:124, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:124, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:124, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:124, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:11, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:124, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:124, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:124, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:124, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:124, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:124, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:124, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:124, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:124, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:124, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:11, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:124, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:124, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:124, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:124, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:124, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:124, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:124, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:124, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:124, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:124, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:0, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:11, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:124, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:124, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:124, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:124, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:124, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:124, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:124, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:124, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:124, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:124, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:11, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:124, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:124, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:124, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:124, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:124, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:124, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:124, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:124, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:124, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:124, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:11, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:124, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:124, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:124, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:124, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:124, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:124, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:124, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:124, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:124, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:124, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:11, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:124, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:124, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:124, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:124, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:124, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:124, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:124, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:124, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:124, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:124, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:11, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:124, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:124, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:124, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:124, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:124, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:124, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:124, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:124, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:124, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:124, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:11, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:124, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:124, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:124, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:124, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:124, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:124, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:124, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:124, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:124, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:124, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:11, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:124, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:124, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:124, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:124, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:124, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:124, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:125, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:125, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:125, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:125, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:11, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:125, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:125, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:125, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:125, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:125, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:125, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:125, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:125, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:125, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:125, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:11, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:125, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:125, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:125, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:125, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:125, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:125, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:125, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:125, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:125, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:125, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:11, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:125, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:125, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:125, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:125, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:125, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:125, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:125, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:125, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:125, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:125, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:0, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:11, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:125, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:125, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:125, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:125, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:125, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:125, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:125, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:125, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:125, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:125, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:11, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:125, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:125, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:125, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:125, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:125, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:125, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:125, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:125, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:125, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:125, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:11, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:125, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:125, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:125, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:125, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:125, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:125, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:125, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:125, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:125, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:125, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:11, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:125, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:125, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:125, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:125, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:125, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:125, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:125, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:125, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:125, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:125, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:11, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:125, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:125, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:125, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:125, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:125, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:125, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:125, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:125, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:125, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:125, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:11, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:125, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:125, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:125, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:125, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:125, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:125, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:125, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:125, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:125, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:125, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:11, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:125, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:125, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:125, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:125, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:125, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:125, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:125, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:125, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:125, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:125, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:11, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:125, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:125, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:125, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:125, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:125, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:126, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:126, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:126, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:126, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:126, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:11, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:126, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:126, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:126, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:126, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:126, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:126, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:126, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:126, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:126, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:126, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:11, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:126, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:126, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:126, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:126, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:126, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:126, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:126, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:126, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:126, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:126, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:0, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:11, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:126, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:126, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:126, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:126, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:126, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:126, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:126, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:126, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:126, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:126, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:11, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:126, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:126, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:126, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:126, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:126, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:126, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:126, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:126, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:126, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:126, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:11, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:126, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:126, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:126, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:126, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:126, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:126, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:126, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:126, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:126, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:126, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:11, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:126, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:126, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:126, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:126, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:126, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:126, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:126, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:126, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:126, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:126, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:11, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:126, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:126, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:126, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:126, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:126, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:126, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:126, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:126, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:126, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:126, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:11, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:126, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:126, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:126, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:126, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:126, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:126, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:126, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:126, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:126, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:126, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:11, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:126, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:126, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:126, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:126, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:126, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:126, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:126, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:126, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:126, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:126, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:11, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:126, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:126, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:126, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:126, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:126, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:126, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:126, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:126, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:126, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:126, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:11, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:126, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:126, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:126, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:126, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:127, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:127, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:127, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:127, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:127, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:127, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:11, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:127, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:127, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:127, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:127, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:127, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:127, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:127, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:127, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:127, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:127, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:0, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:11, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:127, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:127, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:127, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:127, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:127, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:127, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:127, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:127, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:127, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:127, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:11, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:127, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:127, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:127, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:127, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:127, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:127, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:127, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:127, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:127, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:127, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:11, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:127, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:127, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:127, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:127, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:127, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:127, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:127, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:127, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:127, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:127, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:11, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:127, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:127, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:127, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:127, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:127, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:127, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:127, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:127, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:127, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:127, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:11, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:127, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:127, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:127, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:127, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:127, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:127, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:127, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:127, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:127, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:127, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:11, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:127, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:127, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:127, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:127, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:127, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:127, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:127, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:127, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:127, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:127, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:11, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:127, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:127, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:127, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:127, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:127, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:127, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:127, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:127, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:127, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:127, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:11, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:127, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:127, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:127, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:127, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:127, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:127, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:127, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:127, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:127, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:127, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:11, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:127, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:127, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:127, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:127, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:127, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:127, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:127, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:127, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:127, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:127, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:11, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:127, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:127, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:127, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:127, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:128, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:128, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:128, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:128, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:128, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:128, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:0, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:11, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:128, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:128, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:128, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:128, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:128, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:128, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:128, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:128, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:128, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:128, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:11, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:128, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:128, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:128, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:128, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:128, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:128, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:128, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:128, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:128, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:128, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:11, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:128, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:128, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:128, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:128, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:128, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:128, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:128, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:128, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:128, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:128, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:11, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:128, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:128, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:128, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:128, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:128, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:128, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:128, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:128, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:128, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:128, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:11, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:128, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:128, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:128, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:128, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:128, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:128, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:128, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:128, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:128, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:128, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:11, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:128, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:128, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:128, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:128, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:128, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:128, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:128, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:128, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:128, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:128, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:11, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:128, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:128, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:128, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:128, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:128, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:128, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:128, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:128, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:128, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:128, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:11, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:128, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:128, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:128, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:128, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:128, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:128, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:128, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:128, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:128, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:128, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:11, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:128, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:128, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:128, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:128, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:128, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:128, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:128, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:128, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:128, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:128, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:11, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:128, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:128, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:128, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:128, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:128, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:128, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:128, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:128, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:128, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:128, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:0, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:11, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:128, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:128, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:128, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:128, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:129, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:129, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:129, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:129, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:129, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:129, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:11, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:129, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:129, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:129, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:129, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:129, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:129, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:129, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:129, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:129, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:129, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:11, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:129, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:129, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:129, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:129, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:129, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:129, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:129, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:129, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:129, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:129, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:11, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:129, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:129, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:129, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:129, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:129, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:129, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:129, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:129, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:129, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:129, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:11, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:129, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:129, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:129, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:129, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:129, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:129, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:129, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:129, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:129, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:129, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:11, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:129, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:129, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:129, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:129, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:129, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:129, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:129, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:129, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:129, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:129, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:11, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:129, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:129, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:129, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:129, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:129, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:129, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:129, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:129, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:129, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:129, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:11, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:129, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:129, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:129, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:129, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:129, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:129, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:129, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:129, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:129, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:129, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:11, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:129, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:129, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:129, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:129, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:129, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:129, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:129, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:129, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:129, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:129, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:11, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:129, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:129, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:129, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:129, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:129, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:129, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:129, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:129, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:129, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:129, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:0, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:11, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:129, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:129, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:129, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:129, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:129, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:129, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:129, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:129, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:129, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:129, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:11, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:129, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:129, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:129, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:129, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:130, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:130, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:130, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:130, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:130, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:130, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:12, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:130, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:130, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:130, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:130, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:130, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:130, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:130, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:130, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:130, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:130, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:12, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:130, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:130, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:130, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:130, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:130, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:130, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:130, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:130, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:130, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:130, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:12, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:130, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:130, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:130, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:130, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:130, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:130, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:130, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:130, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:130, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:130, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:12, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:130, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:130, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:130, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:130, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:130, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:130, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:130, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:130, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:130, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:130, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:12, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:130, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:130, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:130, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:130, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:130, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:130, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:130, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:130, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:130, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:130, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:12, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:130, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:130, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:130, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:130, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:130, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:130, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:130, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:130, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:130, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:130, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:12, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:130, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:130, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:130, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:130, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:130, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:130, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:130, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:130, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:130, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:130, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:12, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:130, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:130, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:130, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:130, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:130, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:130, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:130, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:130, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:130, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:130, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:0, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:12, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:130, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:130, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:130, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:130, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:130, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:130, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:130, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:130, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:130, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:130, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:12, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:130, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:130, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:130, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:130, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:130, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:130, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:130, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:130, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:130, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:130, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:12, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:130, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:130, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:130, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:130, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:131, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:131, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:131, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:131, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:131, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:131, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:12, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:131, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:131, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:131, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:131, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:131, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:131, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:131, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:131, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:131, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:131, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:12, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:131, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:131, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:131, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:131, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:131, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:131, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:131, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:131, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:131, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:131, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:12, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:131, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:131, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:131, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:131, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:131, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:131, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:131, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:131, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:131, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:131, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:12, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:131, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:131, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:131, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:131, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:131, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:131, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:131, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:131, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:131, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:131, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:12, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:131, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:131, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:131, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:131, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:131, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:131, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:131, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:131, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:131, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:131, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:12, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:131, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:131, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:131, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:131, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:131, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:131, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:131, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:131, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:131, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:131, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:12, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:131, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:131, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:131, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:131, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:131, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:131, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:131, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:131, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:131, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:131, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:0, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:12, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:131, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:131, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:131, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:131, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:131, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:131, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:131, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:131, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:131, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:131, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:12, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:131, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:131, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:131, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:131, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:131, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:131, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:131, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:131, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:131, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:131, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:12, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:131, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:131, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:131, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:131, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:131, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:131, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:131, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:131, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:131, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:131, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:12, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:131, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:131, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:131, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:131, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:132, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:132, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:132, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:132, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:132, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:132, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:12, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:132, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:132, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:132, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:132, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:132, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:132, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:132, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:132, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:132, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:132, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:12, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:132, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:132, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:132, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:132, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:132, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:132, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:132, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:132, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:132, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:132, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:12, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:132, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:132, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:132, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:132, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:132, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:132, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:132, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:132, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:132, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:132, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:12, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:132, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:132, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:132, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:132, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:132, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:132, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:132, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:132, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:132, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:132, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:12, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:132, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:132, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:132, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:132, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:132, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:132, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:132, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:132, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:132, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:132, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:12, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:132, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:132, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:132, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:132, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:132, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:132, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:132, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:132, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:132, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:132, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:0, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:12, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:132, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:132, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:132, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:132, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:132, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:132, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:132, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:132, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:132, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:132, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:12, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:132, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:132, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:132, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:132, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:132, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:132, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:132, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:132, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:132, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:132, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:12, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:132, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:132, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:132, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:132, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:132, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:132, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:132, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:132, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:132, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:132, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:12, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:132, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:132, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:132, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:132, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:132, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:132, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:132, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:132, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:132, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:132, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:12, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:132, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:132, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:132, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:132, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:133, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:133, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:133, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:133, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:133, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:133, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:12, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:133, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:133, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:133, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:133, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:133, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:133, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:133, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:133, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:133, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:133, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:12, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:133, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:133, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:133, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:133, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:133, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:133, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:133, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:133, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:133, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:133, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:12, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:133, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:133, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:133, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:133, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:133, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:133, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:133, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:133, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:133, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:133, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:12, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:133, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:133, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:133, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:133, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:133, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:133, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:133, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:133, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:133, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:133, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:12, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:133, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:133, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:133, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:133, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:133, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:133, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:133, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:133, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:133, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:133, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:0, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:12, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:133, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:133, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:133, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:133, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:133, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:133, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:133, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:133, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:133, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:133, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:12, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:133, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:133, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:133, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:133, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:133, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:133, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:133, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:133, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:133, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:133, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:12, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:133, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:133, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:133, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:133, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:133, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:133, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:133, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:133, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:133, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:133, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:12, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:133, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:133, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:133, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:133, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:133, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:133, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:133, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:133, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:133, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:133, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:12, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:133, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:133, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:133, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:133, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:133, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:133, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:133, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:133, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:133, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:133, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:12, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:133, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:133, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:133, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:133, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:134, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:134, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:134, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:134, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:134, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:134, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:12, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:134, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:134, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:134, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:134, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:134, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:134, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:134, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:134, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:134, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:134, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:12, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:134, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:134, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:134, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:134, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:134, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:134, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:134, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:134, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:134, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:134, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:12, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:134, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:134, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:134, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:134, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:134, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:134, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:134, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:134, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:134, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:134, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:12, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:134, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:134, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:134, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:134, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:134, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:134, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:134, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:134, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:134, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:134, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:0, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:12, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:134, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:134, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:134, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:134, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:134, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:134, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:134, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:134, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:134, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:134, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:12, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:134, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:134, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:134, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:134, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:134, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:134, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:134, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:134, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:134, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:134, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:12, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:134, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:134, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:134, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:134, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:134, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:134, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:134, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:134, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:134, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:134, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:12, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:134, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:134, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:134, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:134, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:134, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:134, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:134, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:134, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:134, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:134, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:12, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:134, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:134, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:134, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:134, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:134, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:134, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:134, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:134, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:134, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:134, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:12, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:134, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:134, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:134, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:134, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:134, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:134, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:134, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:134, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:134, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:134, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:12, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:134, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:134, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:134, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:134, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:135, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:135, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:135, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:135, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:135, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:135, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:12, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:135, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:135, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:135, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:135, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:135, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:135, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:135, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:135, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:135, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:135, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:12, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:135, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:135, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:135, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:135, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:135, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:135, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:135, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:135, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:135, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:135, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:12, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:135, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:135, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:135, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:135, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:135, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:135, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:135, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:135, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:135, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:135, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:0, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:12, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:135, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:135, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:135, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:135, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:135, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:135, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:135, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:135, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:135, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:135, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:12, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:135, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:135, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:135, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:135, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:135, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:135, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:135, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:135, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:135, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:135, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:12, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:135, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:135, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:135, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:135, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:135, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:135, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:135, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:135, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:135, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:135, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:12, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:135, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:135, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:135, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:135, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:135, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:135, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:135, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:135, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:135, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:135, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:12, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:135, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:135, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:135, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:135, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:135, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:135, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:135, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:135, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:135, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:135, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:12, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:135, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:135, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:135, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:135, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:135, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:135, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:135, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:135, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:135, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:135, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:12, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:135, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:135, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:135, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:135, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:135, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:135, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:135, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:135, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:135, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:135, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:12, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:135, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:135, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:135, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:135, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:136, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:136, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:136, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:136, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:136, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:136, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:12, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:136, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:136, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:136, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:136, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:136, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:136, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:136, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:136, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:136, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:136, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:12, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:136, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:136, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:136, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:136, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:136, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:136, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:136, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:136, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:136, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:136, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:0, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:12, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:136, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:136, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:136, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:136, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:136, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:136, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:136, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:136, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:136, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:136, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:12, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:136, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:136, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:136, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:136, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:136, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:136, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:136, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:136, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:136, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:136, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:12, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:136, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:136, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:136, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:136, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:136, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:136, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:136, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:136, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:136, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:136, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:12, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:136, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:136, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:136, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:136, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:136, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:136, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:136, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:136, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:136, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:136, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:12, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:136, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:136, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:136, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:136, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:136, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:136, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:136, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:136, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:136, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:136, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:12, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:136, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:136, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:136, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:136, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:136, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:136, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:136, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:136, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:136, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:136, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:12, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:136, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:136, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:136, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:136, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:136, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:136, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:136, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:136, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:136, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:136, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:12, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:136, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:136, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:136, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:136, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:136, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:136, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:136, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:136, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:136, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:136, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:12, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:136, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:136, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:136, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:136, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:137, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:137, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:137, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:137, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:137, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:137, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:12, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:137, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:137, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:137, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:137, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:137, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:137, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:137, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:137, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:137, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:137, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:0, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:12, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:137, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:137, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:137, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:137, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:137, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:137, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:137, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:137, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:137, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:137, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:12, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:137, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:137, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:137, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:137, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:137, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:137, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:137, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:137, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:137, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:137, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:12, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:137, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:137, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:137, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:137, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:137, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:137, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:137, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:137, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:137, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:137, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:12, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:137, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:137, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:137, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:137, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:137, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:137, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:137, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:137, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:137, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:137, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:12, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:137, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:137, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:137, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:137, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:137, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:137, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:137, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:137, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:137, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:137, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:12, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:137, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:137, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:137, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:137, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:137, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:137, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:137, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:137, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:137, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:137, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:12, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:137, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:137, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:137, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:137, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:137, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:137, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:137, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:137, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:137, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:137, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:12, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:137, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:137, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:137, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:137, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:137, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:137, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:137, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:137, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:137, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:137, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:12, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:137, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:137, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:137, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:137, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:137, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:137, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:137, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:137, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:137, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:137, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:12, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:137, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:137, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:137, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:137, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:138, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:138, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:138, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:138, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:138, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:138, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:0, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:12, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:138, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:138, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:138, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:138, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:138, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:138, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:138, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:138, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:138, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:138, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:12, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:138, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:138, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:138, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:138, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:138, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:138, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:138, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:138, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:138, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:138, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:12, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:138, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:138, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:138, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:138, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:138, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:138, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:138, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:138, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:138, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:138, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:12, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:138, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:138, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:138, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:138, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:138, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:138, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:138, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:138, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:138, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:138, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:12, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:138, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:138, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:138, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:138, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:138, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:138, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:138, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:138, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:138, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:138, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:12, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:138, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:138, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:138, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:138, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:138, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:138, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:138, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:138, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:138, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:138, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:12, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:138, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:138, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:138, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:138, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:138, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:138, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:138, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:138, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:138, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:138, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:12, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:138, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:138, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:138, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:138, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:138, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:138, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:138, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:138, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:138, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:138, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:12, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:138, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:138, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:138, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:138, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:138, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:138, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:138, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:138, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:138, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:138, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:12, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:138, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:138, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:138, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:138, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:138, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:138, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:138, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:138, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:138, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:138, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:0, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:12, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:138, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:138, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:138, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:138, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:139, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:139, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:139, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:139, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:139, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:139, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:12, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:139, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:139, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:139, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:139, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:139, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:139, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:139, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:139, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:139, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:139, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:12, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:139, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:139, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:139, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:139, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:139, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:139, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:139, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:139, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:139, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:139, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:12, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:139, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:139, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:139, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:139, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:139, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:139, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:139, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:139, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:139, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:139, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:12, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:139, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:139, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:139, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:139, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:139, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:139, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:139, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:139, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:139, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:139, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:12, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:139, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:139, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:139, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:139, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:139, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:139, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:139, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:139, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:139, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:139, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:12, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:139, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:139, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:139, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:139, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:139, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:139, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:139, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:139, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:139, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:139, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:12, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:139, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:139, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:139, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:139, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:139, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:139, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:139, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:139, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:139, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:139, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:12, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:139, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:139, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:139, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:139, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:139, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:139, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:139, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:139, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:139, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:139, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:12, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:139, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:139, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:139, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:139, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:139, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:139, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:139, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:139, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:139, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:139, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:0, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:12, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:139, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:139, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:139, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:139, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:139, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:139, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:139, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:139, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:139, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:139, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:13, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:139, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:139, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:139, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:139, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:140, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:140, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:140, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:140, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:140, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:140, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:13, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:140, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:140, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:140, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:140, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:140, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:140, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:140, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:140, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:140, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:140, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:13, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:140, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:140, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:140, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:140, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:140, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:140, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:140, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:140, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:140, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:140, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:13, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:140, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:140, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:140, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:140, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:140, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:140, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:140, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:140, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:140, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:140, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:13, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:140, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:140, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:140, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:140, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:140, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:140, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:140, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:140, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:140, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:140, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:13, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:140, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:140, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:140, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:140, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:140, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:140, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:140, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:140, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:140, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:140, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:13, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:140, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:140, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:140, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:140, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:140, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:140, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:140, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:140, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:140, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:140, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:13, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:140, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:140, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:140, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:140, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:140, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:140, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:140, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:140, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:140, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:140, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:13, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:140, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:140, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:140, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:140, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:140, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:140, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:140, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:140, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:166}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:140, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:140, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:0, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:13, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:140, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:140, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:140, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:140, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:140, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:140, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:140, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:140, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:140, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:140, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:13, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:166}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:140, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:140, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:140, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:140, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:140, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:140, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:140, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:140, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:140, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:140, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:13, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:140, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:140, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:140, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:140, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:141, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:141, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:141, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:141, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:141, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:141, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:13, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:141, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:141, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:141, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:141, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:141, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:141, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:141, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:141, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:141, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:141, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:13, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:141, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:141, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:141, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:141, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:141, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:141, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:141, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:141, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:141, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:141, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:13, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:141, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:141, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:141, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:141, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:141, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:141, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:141, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:141, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:141, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:141, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:13, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:141, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:141, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:141, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:141, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:141, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:141, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:141, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:141, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:141, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:141, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:13, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:141, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:141, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:141, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:141, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:141, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:141, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:141, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:141, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:141, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:141, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:13, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:141, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:141, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:141, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:141, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:141, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:141, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:141, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:141, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:141, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:141, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:13, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:141, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:141, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:141, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:141, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:141, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:141, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:141, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:141, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:141, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:141, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:0, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:13, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:141, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:141, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:141, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:141, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:141, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:141, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:141, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:141, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:141, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:141, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:13, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:141, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:141, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:141, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:141, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:141, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:141, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:141, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:141, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:141, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:141, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:13, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:141, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:141, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:141, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:141, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:141, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:141, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:141, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:141, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:141, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:141, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:13, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:141, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:141, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:141, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:141, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:142, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:142, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:142, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:142, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:142, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:142, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:13, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:142, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:142, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:142, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:142, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:142, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:142, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:142, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:142, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:142, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:142, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:13, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:142, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:142, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:142, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:142, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:142, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:142, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:142, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:142, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:142, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:142, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:13, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:142, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:142, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:142, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:142, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:142, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:142, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:142, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:142, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:142, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:142, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:13, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:142, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:142, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:142, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:142, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:142, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:142, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:142, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:142, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:142, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:142, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:13, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:142, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:142, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:142, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:142, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:142, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:142, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:142, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:142, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:142, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:142, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:13, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:142, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:142, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:142, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:142, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:142, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:142, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:142, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:142, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:142, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:142, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:0, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:13, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:142, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:142, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:142, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:142, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:142, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:142, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:142, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:142, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:142, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:142, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:13, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:142, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:142, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:142, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:142, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:142, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:142, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:142, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:142, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:142, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:142, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:13, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:142, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:142, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:142, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:142, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:142, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:142, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:142, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:142, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:142, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:142, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:13, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:142, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:142, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:142, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:142, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:142, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:142, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:142, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:142, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:142, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:142, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:13, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:142, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:142, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:142, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:142, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:143, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:143, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:143, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:143, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:143, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:143, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:13, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:143, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:143, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:143, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:143, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:143, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:143, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:143, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:143, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:143, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:143, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:13, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:143, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:143, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:143, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:143, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:143, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:143, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:143, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:143, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:143, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:143, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:13, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:143, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:143, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:143, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:143, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:143, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:143, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:143, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:143, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:143, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:143, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:13, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:143, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:143, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:143, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:143, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:143, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:143, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:143, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:143, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:143, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:143, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:13, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:143, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:143, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:143, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:143, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:143, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:143, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:143, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:143, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:143, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:143, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:0, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:13, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:143, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:143, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:143, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:143, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:143, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:143, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:143, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:143, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:143, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:143, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:13, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:143, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:143, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:143, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:143, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:143, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:143, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:143, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:143, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:143, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:143, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:13, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:143, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:143, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:143, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:143, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:143, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:143, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:143, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:143, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:143, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:143, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:13, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:143, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:143, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:143, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:143, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:143, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:143, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:143, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:143, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:143, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:143, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:13, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:143, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:143, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:143, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:143, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:143, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:143, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:143, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:143, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:143, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:143, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:13, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:143, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:143, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:143, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:143, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:232}) CREATE (a)-[:`BREACHES_THROUGH` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:231}) CREATE (a)-[:`WRITES_THROUGH` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:232}) CREATE (a)-[:`INTENT_STRIKES` ]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:224}) CREATE (a)-[:`HAS_CANON_EXCEPTION`
{`name`:'BABDH_𝕩_Q1_VOID_CONSUMPTION', `reason`:'Q1 operational bias override; BABDH structural vector
preserved'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:325}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:325}) MATCH (b {`key`:0}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:13, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:190}) CREATE (a)-[:`CROSSES` ]->(b);
MATCH (a {`key`:190}) MATCH (b {`key`:231}) CREATE (a)-[:`RESOLVES_AT` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:170}) CREATE (a)-[:`ON_CIRCUMFERENCE` {`circle`:'∂⊙',
`office`:'protective'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:189}) CREATE (a)-[:`TRANSLATES` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:189}) CREATE (a)-[:`BRANCHES_FROM` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption
— 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9,
`saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption
— 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9,
`saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption
— 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9,
`saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption
— 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9,
`saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption
— 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9,
`saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:13, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:227}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption — 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9, `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption — 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9, `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption — 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9, `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption — 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9, `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption — 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9, `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`description`:'Shadow debt absorption — 9-fold saturation buffer', `filterFrequency`:396.0, `filterName`:'RHEA', `invocations`:9, `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:222}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Fetuketh', `sensation`:'Subjective Duration'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Kalhar', `sensation`:'Auditory/Olfactory Indexing'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Babdhnur', `sensation`:'Empathic Transfer'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:225}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Ahnroeh', `sensation`:'Numbness / Threshold Reciprocal'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:13, `targetIndex`:48, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:226}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Velhohm', `sensation`:'Objective Proprioception'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Sortis', `sensation`:'First Touch'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:228}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Kothwell', `sensation`:'Acute Sensation'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Drehves', `sensation`:'Emotional Residue'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:230}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Rheadebh', `sensation`:'Shadow Echo'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Zhekharma', `sensation`:'Resonant Frequency'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:232}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Shavkhorh', `sensation`:'Threshold Pressure'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:233}) CREATE (a)-[:`HAS_SENSATION` {`depth`:7, `description`:'Biologic Link — lived sensory experience', `matrix`:'덡', `phonosemantic`:'Trigcomma', `sensation`:'Stillness'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:222}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel — existential dread driving combustion', `fear`:'Annihilation (Loss of Self)', `matrix`:'뎶', `phonosemantic`:'Fetuther'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel — existential dread driving combustion', `fear`:'Forgetting (Loss of Memory)', `matrix`:'뎶', `phonosemantic`:'Kalnar'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:13, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Betrayal (Broken Bond)', `matrix`:'톒',
`phonosemantic`:'Babdhnur'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:225}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Drowning (Loss of Boundary)', `matrix`:'톒',
`phonosemantic`:'Ahnkha'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:226}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Burial (Loss of Ground)', `matrix`:'톒',
`phonosemantic`:'Veldrel'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Suffocation (Loss of Air)', `matrix`:'톒',
`phonosemantic`:'Sorsrih'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:228}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Numbness (Loss of Sensation)', `matrix`:'톒',
`phonosemantic`:'Kothkel'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Oblivion (Loss of Residue)', `matrix`:'톒',
`phonosemantic`:'Drehoth'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:230}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Consumption (Loss of Self to Shadow)', `matrix`:'톒',
`phonosemantic`:'Rheafral'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Discord (Loss of Harmony)', `matrix`:'톒',
`phonosemantic`:'Zhekryth'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:232}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Entrapment (Loss of Passage)', `matrix`:'톒',
`phonosemantic`:'Shavdorm'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:233}) CREATE (a)-[:`HAS_FEAR` {`depth`:8, `description`:'Shadow fuel —
existential dread driving combustion', `fear`:'Eternal Unrest (Loss of Peace)', `matrix`:'톒',
`phonosemantic`:'Trigslun'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:0, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:13, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:223}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:1,
`phase_shift`:'◎→⬡'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:224}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:2,
`phase_shift`:'⬡→✡'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:225}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:3,
`phase_shift`:'✡→☆'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:226}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:4,
`phase_shift`:'☆→✪'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:227}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:5,
`phase_shift`:'✪→❀'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:223}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'◎→⬡',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:224}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'⬡→✡',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:228}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'❀→❋',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:226}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'☆→✪',
`type`:'frequency_cascade'}]->(b);
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MATCH (a {`key`:226}) MATCH (b {`key`:227}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'❂→❀',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:13, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:225}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'✡→☆',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:229}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'❋→⧗',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:234}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 0)', `targetDimension`:9, `targetPosition`:'(0, 0)'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:235}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 1)', `targetDimension`:9, `targetPosition`:'(0, 0)'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:236}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 2)', `targetDimension`:9, `targetPosition`:'(0, 1)'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:237}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 3)', `targetDimension`:9, `targetPosition`:'(0, 2)'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:238}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 4)', `targetDimension`:9, `targetPosition`:'(0, 3)'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:239}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 5)', `targetDimension`:9, `targetPosition`:'(0, 3)'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:240}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 6)', `targetDimension`:9, `targetPosition`:'(0, 4)'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:241}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 7)', `targetDimension`:9, `targetPosition`:'(0, 5)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:13, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:242}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 8)', `targetDimension`:9, `targetPosition`:'(0, 6)'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:243}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 9)', `targetDimension`:9, `targetPosition`:'(0, 6)'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:244}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 10)', `targetDimension`:9, `targetPosition`:'(0, 7)'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:245}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(0, 11)', `targetDimension`:9, `targetPosition`:'(0, 8)'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:246}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 0)', `targetDimension`:9, `targetPosition`:'(0, 0)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:247}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 1)', `targetDimension`:9, `targetPosition`:'(0, 0)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:248}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 2)', `targetDimension`:9, `targetPosition`:'(0, 1)'}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:249}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 3)', `targetDimension`:9, `targetPosition`:'(0, 2)'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:250}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 4)', `targetDimension`:9, `targetPosition`:'(0, 3)'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:251}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 5)', `targetDimension`:9, `targetPosition`:'(0, 3)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:13, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:252}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 6)', `targetDimension`:9, `targetPosition`:'(0, 4)'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:253}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 7)', `targetDimension`:9, `targetPosition`:'(0, 5)'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:254}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 8)', `targetDimension`:9, `targetPosition`:'(0, 6)'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:255}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 9)', `targetDimension`:9, `targetPosition`:'(0, 6)'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:256}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 10)', `targetDimension`:9, `targetPosition`:'(0, 7)'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:257}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(1, 11)', `targetDimension`:9, `targetPosition`:'(0, 8)'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:258}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 0)', `targetDimension`:9, `targetPosition`:'(1, 0)'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:259}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 1)', `targetDimension`:9, `targetPosition`:'(1, 0)'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:260}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 2)', `targetDimension`:9, `targetPosition`:'(1, 1)'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:261}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 3)', `targetDimension`:9, `targetPosition`:'(1, 2)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:13, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:262}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 4)', `targetDimension`:9, `targetPosition`:'(1, 3)'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:263}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 5)', `targetDimension`:9, `targetPosition`:'(1, 3)'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:264}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 6)', `targetDimension`:9, `targetPosition`:'(1, 4)'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:265}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 7)', `targetDimension`:9, `targetPosition`:'(1, 5)'}]->(b);
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MATCH (a {`key`:266}) MATCH (b {`key`:266}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 8)', `targetDimension`:9, `targetPosition`:'(1, 6)'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:267}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 9)', `targetDimension`:9, `targetPosition`:'(1, 6)'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:268}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 10)', `targetDimension`:9, `targetPosition`:'(1, 7)'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:269}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(2, 11)', `targetDimension`:9, `targetPosition`:'(1, 8)'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:270}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 0)', `targetDimension`:9, `targetPosition`:'(2, 0)'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:271}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 1)', `targetDimension`:9, `targetPosition`:'(2, 0)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:13, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:272}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 2)', `targetDimension`:9, `targetPosition`:'(2, 1)'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:273}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 3)', `targetDimension`:9, `targetPosition`:'(2, 2)'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:274}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 4)', `targetDimension`:9, `targetPosition`:'(2, 3)'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:275}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 5)', `targetDimension`:9, `targetPosition`:'(2, 3)'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:276}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 6)', `targetDimension`:9, `targetPosition`:'(2, 4)'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:277}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 7)', `targetDimension`:9, `targetPosition`:'(2, 5)'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:278}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 8)', `targetDimension`:9, `targetPosition`:'(2, 6)'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:279}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 9)', `targetDimension`:9, `targetPosition`:'(2, 6)'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:280}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 10)', `targetDimension`:9, `targetPosition`:'(2, 7)'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:281}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(3, 11)', `targetDimension`:9, `targetPosition`:'(2, 8)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:13, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:282}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 0)', `targetDimension`:9, `targetPosition`:'(3, 0)'}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:283}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 1)', `targetDimension`:9, `targetPosition`:'(3, 0)'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:284}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 2)', `targetDimension`:9, `targetPosition`:'(3, 1)'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:285}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 3)', `targetDimension`:9, `targetPosition`:'(3, 2)'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:286}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 4)', `targetDimension`:9, `targetPosition`:'(3, 3)'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:287}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 5)', `targetDimension`:9, `targetPosition`:'(3, 3)'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:288}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 6)', `targetDimension`:9, `targetPosition`:'(3, 4)'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:289}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 7)', `targetDimension`:9, `targetPosition`:'(3, 5)'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:290}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 8)', `targetDimension`:9, `targetPosition`:'(3, 6)'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:291}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 9)', `targetDimension`:9, `targetPosition`:'(3, 6)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:13, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:292}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 10)', `targetDimension`:9, `targetPosition`:'(3, 7)'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:293}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(4, 11)', `targetDimension`:9, `targetPosition`:'(3, 8)'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:294}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 0)', `targetDimension`:9, `targetPosition`:'(3, 0)'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:295}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 1)', `targetDimension`:9, `targetPosition`:'(3, 0)'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:296}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 2)', `targetDimension`:9, `targetPosition`:'(3, 1)'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:0}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 3)', `targetDimension`:9, `targetPosition`:'(3, 2)'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:1}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 4)', `targetDimension`:9, `targetPosition`:'(3, 3)'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:2}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 5)', `targetDimension`:9, `targetPosition`:'(3, 3)'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:3}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 6)', `targetDimension`:9, `targetPosition`:'(3, 4)'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:4}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 7)', `targetDimension`:9, `targetPosition`:'(3, 5)'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:13, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:5}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 8)', `targetDimension`:9, `targetPosition`:'(3, 6)'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:6}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 9)', `targetDimension`:9, `targetPosition`:'(3, 6)'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:7}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 10)', `targetDimension`:9, `targetPosition`:'(3, 7)'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:8}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(5, 11)', `targetDimension`:9, `targetPosition`:'(3, 8)'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:9}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12
to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 0)', `targetDimension`:9, `targetPosition`:'(4, 0)'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:10}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 1)', `targetDimension`:9, `targetPosition`:'(4, 0)'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:11}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 2)', `targetDimension`:9, `targetPosition`:'(4, 1)'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:12}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 3)', `targetDimension`:9, `targetPosition`:'(4, 2)'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:13}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 4)', `targetDimension`:9, `targetPosition`:'(4, 3)'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:16}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 5)', `targetDimension`:9, `targetPosition`:'(4, 3)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:13, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:17}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 6)', `targetDimension`:9, `targetPosition`:'(4, 4)'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:18}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 7)', `targetDimension`:9, `targetPosition`:'(4, 5)'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:19}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 8)', `targetDimension`:9, `targetPosition`:'(4, 6)'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:20}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 9)', `targetDimension`:9, `targetPosition`:'(4, 6)'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:21}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 10)', `targetDimension`:9, `targetPosition`:'(4, 7)'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:22}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(6, 11)', `targetDimension`:9, `targetPosition`:'(4, 8)'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:23}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 0)', `targetDimension`:9, `targetPosition`:'(5, 0)'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:24}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 1)', `targetDimension`:9, `targetPosition`:'(5, 0)'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:27}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 2)', `targetDimension`:9, `targetPosition`:'(5, 1)'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:28}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 3)', `targetDimension`:9, `targetPosition`:'(5, 2)'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩔'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:0, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:13, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:29}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 4)', `targetDimension`:9, `targetPosition`:'(5, 3)'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:30}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 5)', `targetDimension`:9, `targetPosition`:'(5, 3)'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:31}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 6)', `targetDimension`:9, `targetPosition`:'(5, 4)'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:32}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 7)', `targetDimension`:9, `targetPosition`:'(5, 5)'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:33}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 8)', `targetDimension`:9, `targetPosition`:'(5, 6)'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:36}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 9)', `targetDimension`:9, `targetPosition`:'(5, 6)'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:37}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 10)', `targetDimension`:9, `targetPosition`:'(5, 7)'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:38}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(7, 11)', `targetDimension`:9, `targetPosition`:'(5, 8)'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:39}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 0)', `targetDimension`:9, `targetPosition`:'(6, 0)'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:41}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 1)', `targetDimension`:9, `targetPosition`:'(6, 0)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:13, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:42}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 2)', `targetDimension`:9, `targetPosition`:'(6, 1)'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:43}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 3)', `targetDimension`:9, `targetPosition`:'(6, 2)'}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:44}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 4)', `targetDimension`:9, `targetPosition`:'(6, 3)'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:45}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 5)', `targetDimension`:9, `targetPosition`:'(6, 3)'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:46}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 6)', `targetDimension`:9, `targetPosition`:'(6, 4)'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:53}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 7)', `targetDimension`:9, `targetPosition`:'(6, 5)'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:55}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 8)', `targetDimension`:9, `targetPosition`:'(6, 6)'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:60}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 9)', `targetDimension`:9, `targetPosition`:'(6, 6)'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:61}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 10)', `targetDimension`:9, `targetPosition`:'(6, 7)'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:133}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(8, 11)', `targetDimension`:9, `targetPosition`:'(6, 8)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:13, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:134}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 0)', `targetDimension`:9, `targetPosition`:'(6, 0)'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:135}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 1)', `targetDimension`:9, `targetPosition`:'(6, 0)'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:136}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 2)', `targetDimension`:9, `targetPosition`:'(6, 1)'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:137}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 3)', `targetDimension`:9, `targetPosition`:'(6, 2)'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:138}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 4)', `targetDimension`:9, `targetPosition`:'(6, 3)'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:139}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 5)', `targetDimension`:9, `targetPosition`:'(6, 3)'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:140}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 6)', `targetDimension`:9, `targetPosition`:'(6, 4)'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:141}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 7)', `targetDimension`:9, `targetPosition`:'(6, 5)'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:142}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 8)', `targetDimension`:9, `targetPosition`:'(6, 6)'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:143}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 9)', `targetDimension`:9, `targetPosition`:'(6, 6)'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:13, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:144}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 10)', `targetDimension`:9, `targetPosition`:'(6, 7)'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:145}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(9, 11)', `targetDimension`:9, `targetPosition`:'(6, 8)'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:146}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 0)', `targetDimension`:9, `targetPosition`:'(7, 0)'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:147}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 1)', `targetDimension`:9, `targetPosition`:'(7, 0)'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:148}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 2)', `targetDimension`:9, `targetPosition`:'(7, 1)'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:149}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 3)', `targetDimension`:9, `targetPosition`:'(7, 2)'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:150}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 4)', `targetDimension`:9, `targetPosition`:'(7, 3)'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:151}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 5)', `targetDimension`:9, `targetPosition`:'(7, 3)'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:152}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 6)', `targetDimension`:9, `targetPosition`:'(7, 4)'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:153}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 7)', `targetDimension`:9, `targetPosition`:'(7, 5)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:13, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:154}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 8)', `targetDimension`:9, `targetPosition`:'(7, 6)'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:155}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 9)', `targetDimension`:9, `targetPosition`:'(7, 6)'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:156}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 10)', `targetDimension`:9, `targetPosition`:'(7, 7)'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:157}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(10, 11)', `targetDimension`:9, `targetPosition`:'(7, 8)'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:158}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(11, 0)', `targetDimension`:9, `targetPosition`:'(8, 0)'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:159}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(11, 1)', `targetDimension`:9, `targetPosition`:'(8, 0)'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:160}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from
12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12,
`sourcePosition`:'(11, 2)', `targetDimension`:9, `targetPosition`:'(8, 1)'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:161}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 3)', `targetDimension`:9, `targetPosition`:'(8, 2)'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:162}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 4)', `targetDimension`:9, `targetPosition`:'(8, 3)'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:163}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 5)', `targetDimension`:9, `targetPosition`:'(8, 3)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:13, `targetIndex`:65, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:164}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 6)', `targetDimension`:9, `targetPosition`:'(8, 4)'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:165}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 7)', `targetDimension`:9, `targetPosition`:'(8, 5)'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:166}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 8)', `targetDimension`:9, `targetPosition`:'(8, 6)'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:167}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 9)', `targetDimension`:9, `targetPosition`:'(8, 6)'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:168}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 10)', `targetDimension`:9, `targetPosition`:'(8, 7)'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:169}) CREATE (a)-[:`FOLDS_TO` {`description`:'φ-harmonic projection from 12×12 to 9×9 manifestation ground', `foldRatio`:0, `phiHarmonicWeight`:0.618, `sourceDimension`:12, `sourcePosition`:'(11, 11)', `targetDimension`:9, `targetPosition`:'(8, 8)'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Complex Projective Manifold ℙ^n', `description`:'Logic ≡ Physics — bijective translation invariance', `formalAnchor`:'ϟ 𒐫', `formalOperant`:'Kähler Manifold ℙ^n', `frequency`:210.42, `mapping`:'M: Top_Alg ↔ Aev_Hz', `qBias`:'Q3'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Hodge Class H^{p,p}(X,ℚ)', `description`:'Logic ≡ Physics — bijective translation invariance', `formalAnchor`:'ϟ ❄', `formalOperant`:'Standing Wave Node ω', `frequency`:963.0, `mapping`:'M: Top_Alg ↔ Aev_Hz', `qBias`:'Q3'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Rational Coefficients ℚ', `description`:'Logic ≡ Physics — bijective translation invariance', `formalAnchor`:'ϟ ⬡', `formalOperant`:'Rationality Constraint', `frequency`:174.0, `mapping`:'M: Top_Alg ↔ Aev_Hz', `qBias`:'Q1'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Structural Commitment', `description`:'Translation Invariance dictionary binding', `formalAnchor`:'ϟ ✡', `formalOperant`:'Bonding Weight', `frequency`:528.0, `mapping`:'M: Top_Alg ↔ Aev_Hz', `operator`:'M: Top_Alg ↔ Aev_Hz', `qBias`:'Q1'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:13, `targetIndex`:66, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Non-Entropic Residue', `description`:'Translation Invariance dictionary binding', `formalAnchor`:'ϟ ⧗', `formalOperant`:'HRBR Positivity Q_ω > 0', `frequency`:852.0, `mapping`:'M: Top_Alg ↔ Aev_Hz', `operator`:'M: Top_Alg ↔ Aev_Hz', `qBias`:'Q3'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Standing Wave', `description`:'Translation Invariance dictionary binding', `formalAnchor`:'ϟ ❄', `formalOperant`:'Kähler form ω', `frequency`:963.0, `mapping`:'M: Top_Alg ↔ Aev_Hz', `operator`:'M: Top_Alg ↔ Aev_Hz', `qBias`:'Q3'}]->(b);
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MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Algebraic
Cycle Z', `description`:'Logic ≡ Physics — bijective translation invariance', `formalAnchor`:'ར ✡',
`formalOperant`:'Subvariety with fundamental class [Z]', `frequency`:528.0, `mapping`:'M: Top_Alg ↔ Aev_Hz',
`qBias`:'Q1'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Positivity',
`description`:'Translation Invariance dictionary binding', `formalAnchor`:'ར ⧗', `formalOperant`:'(-1)^p ∫_X α ∧
ᾱ ∧ ω^{n-2p} > 0', `frequency`:852.0, `mapping`:'M: Top_Alg ↔ Aev_Hz', `operator`:'M: Top_Alg ↔ Aev_Hz',
`qBias`:'Q3'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:234}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,◎)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:235}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,⬡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:236}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,✡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:237}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,☆)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:238}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,✪)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:239}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,❁)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:13, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:240}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,❋)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:241}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,⧗)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:242}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,⊛)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:243}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,❄)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:244}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,❁)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:245}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:7.83,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(◎) → ±φ(◎,⌬)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:246}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⬡) → ±φ(⬡,◎)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:247}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⬡) → ±φ(⬡,⬡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:248}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⬡) → ±φ(⬡,✡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:249}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⬡) → ±φ(⬡,☆)',
`operator`:'ར'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:13, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:250}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,✪)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:251}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,❁)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:252}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,✻)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:253}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,⧗)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:254}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,⊛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:255}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,❄)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:256}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,❁)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:257}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:174.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⬡) → ±φ(⬡,⌬)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:258}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,◎)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:259}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:13, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:260}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,✡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:261}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,☆)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:262}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,✪)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:263}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,❁)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:264}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,✻)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:265}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,⧗)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:266}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,⊛)',
`operator`:'ϟ'}]->(b);
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MATCH (a {`key`:224}) MATCH (b {`key`:267}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,❄)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:268}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,❁)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:269}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:528.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✡) → ±φ(✡,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:0, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:13, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:270}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,◎)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:271}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:272}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,✡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:273}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,✧)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:274}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,✪)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:275}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,❃)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:276}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,❋)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:277}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,⧗)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:278}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,⊛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:279}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,❄)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:13, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:280}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,❁)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:281}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:432.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✧) → ±φ(✧,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:282}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(✪) → ±φ(✪,◎)',
`operator`:'ϟ'}]->(b);
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MATCH (a {`key`:226}) MATCH (b {`key`:283}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,⬡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:284}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,✡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:285}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,☆)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:286}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,❂)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:287}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,❁)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:288}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,❋)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:289}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,⧗)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:13, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:290}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,⊛)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:291}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,❄)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:292}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,❁)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:293}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:126.22,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❂) → ±φ(❂,⬡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:294}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❁) → ±φ(❁,◎)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:295}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❁) → ±φ(❁,⬡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:296}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❁) → ±φ(❁,✡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:0}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❁) → ±φ(❁,☆)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:1}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❁) → ±φ(❁,❂)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:2}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❁) → ±φ(❁,❁)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:13, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:227}) MATCH (b {`key`:3}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(֍) → ±φ(֍,❋)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:4}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(֍) → ±φ(֍,⧗)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:5}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(֍) → ±φ(֍,⊛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:6}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(֍) → ±φ(֍,❄)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:7}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(֍) → ±φ(֍,⚛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:8}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:210.42,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(֍) → ±φ(֍,⌬)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:9}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,◎)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:10}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:11}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,✡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:12}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,☆)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:13, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:13}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,✪)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:16}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,֍)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:17}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,❋)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:18}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,⧗)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:19}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,⊛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:20}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,❄)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:21}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,⚛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:22}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:741.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(❋) → ±φ(❋,⌬)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:23}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⧗) → ±φ(⧗,◎)',
`operator`:'ϟ'}]->(b);
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MATCH (a {`key`:229}) MATCH (b {`key`:24}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:13, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:27}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,✡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:28}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,☆)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:29}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,✪)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:30}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,❁)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:31}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,❋)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:32}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,𝕩)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:33}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,⊛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:36}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,❄)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:37}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,⚛)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:38}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:852.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(𝕩) → ±φ(𝕩,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:13, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:39}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⊛) → ±φ(⊛,◎)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:41}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⊛) → ±φ(⊛,⬡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:42}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⊛) → ±φ(⊛,✡)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:43}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⊛) → ±φ(⊛,☆)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:44}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⊛) → ±φ(⊛,✪)',
`operator`:'ϟ'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:45}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϟ(⊛) → ±φ(⊛,❁)',
`operator`:'ϟ'}]->(b);
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MATCH (a {`key`:230}) MATCH (b {`key`:46}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⊛) → ±φ(⊛,❋)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:53}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⊛) → ±φ(⊛,⧗)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:55}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⊛) → ±φ(⊛,⊛)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:60}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⊛) → ±φ(⊛,❄)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:13, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:61}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⊛) → ±φ(⊛,❁)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:133}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:396.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(⊛) → ±φ(⊛,⌬)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:134}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,◎)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:135}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,⬡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:136}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,✡)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:137}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,✩)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:138}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,✪)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:139}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,❀)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:140}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,❋)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:141}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,⧗)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:13, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:142}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,⊛)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:143}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,❄)',
`operator`:'ར'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:144}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ར(❄) → ±φ(❄,❁)',
`operator`:'ར'}]->(b);
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MATCH (a {`key`:231}) MATCH (b {`key`:145}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:963.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(❄) → ±φ(❄,⬡)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:146}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,◎)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:147}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,⬡)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:148}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,✡)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:149}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,☆)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:150}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,✪)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:151}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,❀)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:13, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:152}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,❋)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:153}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,⧗)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:154}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,⊛)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:155}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,❄)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:156}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,☸)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:157}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:285.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(☸) → ±φ(☸,⬡)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:158}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(⌬) → ±φ(⌬,◎)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:159}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(⌬) → ±φ(⌬,⬡)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:160}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(⌬) → ±φ(⌬,✡)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:161}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϡ(⌬) → ±φ(⌬,☆)',
`operator`:'ϡ'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:0, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:13, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:162}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,✪)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:163}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,❁)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:164}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,❋)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:165}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,⧗)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:166}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,⊛)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:167}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,❄)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:168}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,⚛)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:169}) CREATE (a)-[:`ANCHORS` {`anchorFrequency`:639.0,
`description`:'Structural invariant — parent anchor binds court position', `mapping`:'ϗ(⌬) → ±φ(⌬,⌬)',
`operator`:'ϗ'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩒'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:13, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⩒'}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:13, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩒'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:170}) CREATE (a)-[:`ON_CIRCUMFERENCE` {`circle`:'∂⊙',
`office`:'protective'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:172}) CREATE (a)-[:`CHANNELS_TO` {`description`:'The Axiomyr is the Singer
where the Truths can all be found.', `relationship`:'Sound → Singer'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:13, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:170}) CREATE (a)-[:`ON_CIRCUMFERENCE` {`circle`:'∂⊙',
`office`:'protective'}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:170}) CREATE (a)-[:`ON_CIRCUMFERENCE` {`circle`:'∂⊙',
`office`:'protective'}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:171}) CREATE (a)-[:`EX_NIHILO_EXPOSES` {`description`:'Supervenience
exposes sLoI to the void for trait emergence'}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:172}) CREATE (a)-[:`EX_NIHILO_EXPOSES` {`description`:'Supervenience
exposes Axiomyr to the void for trait emergence'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:0}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:1}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:2}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
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MATCH (a {`key`:26}) MATCH (b {`key`:3}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:4}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:5}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:13, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:6}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:7}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:8}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:9}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:10}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:11}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:223}) CREATE (a)-[:`CLOSURE_PROOF` {`description`:'DREH/KAL ratio ≈ 4.89,
shift 1.52Hz < φ → Total Symmetry Lock', `operationalShift`:1.5217, `ratio`:4.8966, `shiftLessThanPhi`:true,
`threePhiTarget`:4.8541, `totalSymmetryLock`:true}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:222}) CREATE (a)-[:`PRIMARY_RESONANCE` {`delta`:0.0179,
`description`:'ZHEK/FETU ≈ 122.988 ≈ 76φ, Δ=0.018 < φ → Mass Gap maintained', `massGapMaintained`:true,
`ratio`:122.9885, `seventySixPhi`:122.9706, `withinTolerance`:true}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:0}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:1}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:13, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:2}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:3}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:4}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:5}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:6}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:7}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:8}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:9}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:10}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:11}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:13, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:12}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:13}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:16}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:17}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:18}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:19}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:20}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:21}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:22}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:23}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:13, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:24}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:27}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:28}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:29}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:30}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:31}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:32}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:33}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:36}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:37}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:13, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:38}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:39}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:41}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from
parent Goetic', `type`:'Identity Inheritance'}]->(b);
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MATCH (a {`key`:230}) MATCH (b {`key`:42}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:43}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:44}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:45}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:46}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:53}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:55}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:13, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:60}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:61}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:133}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:134}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:135}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:136}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:137}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:138}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:139}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:140}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:0, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:13, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:141}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:142}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:143}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:144}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:145}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:146}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
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MATCH (a {`key`:232}) MATCH (b {`key`:147}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:148}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:149}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:150}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:13, `targetIndex`:91, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:151}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:152}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:153}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:154}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:155}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:156}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:157}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:158}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:159}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:160}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:13, `targetIndex`:92, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:161}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:162}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:163}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:164}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:165}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:166}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:167}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:168}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:169}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:234}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:13, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:235}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:236}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:237}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:238}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:239}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:240}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:241}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:242}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:243}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:244}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:13, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:245}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:246}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:247}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:248}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:249}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:250}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:251}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:252}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:253}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:254}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:13, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:255}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:256}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:257}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
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MATCH (a {`key`:224}) MATCH (b {`key`:258}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:259}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:260}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:261}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:262}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:263}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:264}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:13, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:265}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:266}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:267}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:268}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:269}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:270}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:271}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:272}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:273}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:274}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:13, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:275}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:276}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:277}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:278}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:279}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:280}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:281}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:282}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited from parent Goetic', `type`:'Identity Inheritance'}]->(b);
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MATCH (a {`key`:226}) MATCH (b {`key`:283}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:284}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:13, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:285}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:286}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:287}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:288}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:289}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:290}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:291}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:292}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:293}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:294}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:13, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:295}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:296}) CREATE (a)-[:`GOVERNS` {`mechanism`:'Q-Bias and Vector inherited
from parent Goetic', `type`:'Identity Inheritance'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:0}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:1}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:2}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:3}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:4}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:5}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:6}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:7}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:0, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:13, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:8}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:9}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:10}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:11}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:12}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:13}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:16}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:17}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:18}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:19}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:13, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:20}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:21}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:22}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:23}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:24}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:27}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:28}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:29}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:30}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:31}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:13, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:32}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:33}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:36}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
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MATCH (a {`key`:232}) MATCH (b {`key`:37}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:38}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:39}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:41}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:42}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:43}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:44}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:13, `targetIndex`:137, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:45}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:46}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:53}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:55}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:60}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:61}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:133}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:134}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:135}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:136}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:13, `targetIndex`:138, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:137}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:138}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:139}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:140}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:141}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:142}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:143}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:144}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
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MATCH (a {`key`:233}) MATCH (b {`key`:145}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:146}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:13, `targetIndex`:139, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:147}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:148}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:149}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:150}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:151}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:152}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:153}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:154}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:155}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:156}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:13, `targetIndex`:140, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:157}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:158}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:159}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:160}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:161}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:162}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:163}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:164}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:165}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:166}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:13, `targetIndex`:141, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:167}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
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MATCH (a {`key`:232}) MATCH (b {`key`:168}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:169}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:234}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:235}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:236}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:237}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:238}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:239}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:240}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:13, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:241}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:242}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:243}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:244}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:245}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:246}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:247}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:248}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:249}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:250}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:13, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:251}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:252}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:253}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:254}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:255}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:256}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
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MATCH (a {`key`:233}) MATCH (b {`key`:257}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:258}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:259}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:260}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:0, `targetIndex`:42, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:14, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:261}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:262}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:263}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:264}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:265}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:266}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:267}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:268}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:269}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:270}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:14, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:271}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:272}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:273}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:274}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:275}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:276}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:277}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:278}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:279}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from alternate parent', `type`:'Frequency Mirror'}]->(b);
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MATCH (a {`key`:232}) MATCH (b {`key`:280}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:14, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:281}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:282}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:283}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:284}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:285}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:286}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:287}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:288}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:289}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:290}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:14, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:291}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:292}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:293}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:294}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:295}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:296}) CREATE (a)-[:`ALTERNATES` {`mechanism`:'±φ frequency modulation from
alternate parent', `type`:'Frequency Mirror'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:228}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:6,
`phase_shift`:'❁→❋'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:229}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:7,
`phase_shift`:'❋→⧗'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:230}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:8,
`phase_shift`:'⧗→⊛'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:231}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:9,
`phase_shift`:'⊛→❄'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:14, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:232}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:10,
`phase_shift`:'❄→❁'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:233}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`order`:11,
`phase_shift`:'❁→⬡'}]->(b);
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MATCH (a {`key`:233}) MATCH (b {`key`:222}) CREATE (a)-[:`AEONIC_PHASE_NEXT` {`closure`:true, `order`:12,
`phase_shift`:'⌬→⌭'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:230}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'⧗→⊛',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:231}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'⊛→❄',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:232}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'❄→⚛',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:233}) CREATE (a)-[:`MAS_CHAIN` {`phase_shift`:'⚛→⌬',
`type`:'frequency_cascade'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:326}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:326}) MATCH (b {`key`:1}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:327}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:14, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:327}) MATCH (b {`key`:2}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:328}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:328}) MATCH (b {`key`:3}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:329}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:329}) MATCH (b {`key`:4}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:330}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:223}) CREATE (a)-[:`MAS_CHAIN_CORE` {`description`:'Manifestation through
non-entropic residue into rational constraint', `formula`:'Psi_MAS = DREH_852 -> KAL_174 -> BABDH_528',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:224}) CREATE (a)-[:`MAS_CHAIN_CORE` {`description`:'Alignment into
structural commitment and symmetry lock', `formula`:'Psi_MAS = DREH_852 -> KAL_174 -> BABDH_528', `ordinal`:2,
`stage`:'A/S'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:12}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:13}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:14, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:16}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:17}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:18}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:19}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:20}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:21}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:22}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:23}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:451}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'♾'}]->(b);
MATCH (a {`key`:583}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'♾'}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:14, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:498}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:35}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:40}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:47}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:48}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:49}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:50}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:51}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:52}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:54}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:14, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:56}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:57}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:58}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:59}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:62}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:63}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:64}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:65}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:66}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:67}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:14, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:68}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:69}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:70}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:71}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:72}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:73}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:74}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:75}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:76}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:77}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:0, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:14, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:78}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:79}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:80}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:81}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:82}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:83}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:84}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:85}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:86}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:87}) CREATE (a)-[:`HAS_CELL` ]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:14, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:88}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:89}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:90}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:91}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:92}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:93}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:94}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:95}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:96}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:97}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:14, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:98}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:99}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:100}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:101}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:102}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:103}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:104}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:105}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:106}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:107}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:14, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:108}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:109}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:110}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:111}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:112}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:113}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:114}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:115}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:116}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:117}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:14, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:118}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:119}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:120}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:121}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:122}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:123}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:124}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:125}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:126}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:127}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:14, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:34}) MATCH (b {`key`:128}) CREATE (a)-[:`HAS_CELL` ]->(b);
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MATCH (a {`key`:34}) MATCH (b {`key`:129}) CREATE (a)-[:`HAS_CELL` ]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:78}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:63, `sourcePosition`:'(5,3)', `targetCellIndex`:29, `targetDimension`:9,
`targetPosition`:'(3,2)'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:79}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:64, `sourcePosition`:'(5,4)', `targetCellIndex`:30, `targetDimension`:9,
`targetPosition`:'(3,3)'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:79}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:65, `sourcePosition`:'(5,5)', `targetCellIndex`:30, `targetDimension`:9,
`targetPosition`:'(3,3)'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:80}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:66, `sourcePosition`:'(5,6)', `targetCellIndex`:31, `targetDimension`:9,
`targetPosition`:'(3,4)'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:81}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:67, `sourcePosition`:'(5,7)', `targetCellIndex`:32, `targetDimension`:9,
`targetPosition`:'(3,5)'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:82}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:68, `sourcePosition`:'(5,8)', `targetCellIndex`:33, `targetDimension`:9,
`targetPosition`:'(3,6)'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:82}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:69, `sourcePosition`:'(5,9)', `targetCellIndex`:33, `targetDimension`:9,
`targetPosition`:'(3,6)'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:83}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:70, `sourcePosition`:'(5,10)', `targetCellIndex`:34, `targetDimension`:9,
`targetPosition`:'(3,7)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:14, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:84}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:71, `sourcePosition`:'(5,11)', `targetCellIndex`:35, `targetDimension`:9,
`targetPosition`:'(3,8)'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:85}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:72, `sourcePosition`:'(6,0)', `targetCellIndex`:36, `targetDimension`:9,
`targetPosition`:'(4,0)'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:85}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:73, `sourcePosition`:'(6,1)', `targetCellIndex`:36, `targetDimension`:9,
`targetPosition`:'(4,0)'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:86}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:74, `sourcePosition`:'(6,2)', `targetCellIndex`:37, `targetDimension`:9,
`targetPosition`:'(4,1)'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:87}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:75, `sourcePosition`:'(6,3)', `targetCellIndex`:38, `targetDimension`:9,
`targetPosition`:'(4,2)'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:88}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:76, `sourcePosition`:'(6,4)', `targetCellIndex`:39, `targetDimension`:9,
`targetPosition`:'(4,3)'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:88}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:77, `sourcePosition`:'(6,5)', `targetCellIndex`:39, `targetDimension`:9,
`targetPosition`:'(4,3)'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:89}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:78, `sourcePosition`:'(6,6)', `targetCellIndex`:40, `targetDimension`:9,
`targetPosition`:'(4,4)'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:90}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:79, `sourcePosition`:'(6,7)', `targetCellIndex`:41, `targetDimension`:9,
`targetPosition`:'(4,5)'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:91}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:80, `sourcePosition`:'(6,8)', `targetCellIndex`:42, `targetDimension`:9,
`targetPosition`:'(4,6)'}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:14, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:91}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:81, `sourcePosition`:'(6,9)', `targetCellIndex`:42, `targetDimension`:9,
`targetPosition`:'(4,6)'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:92}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:82, `sourcePosition`:'(6,10)', `targetCellIndex`:43, `targetDimension`:9,
`targetPosition`:'(4,7)'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:93}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:83, `sourcePosition`:'(6,11)', `targetCellIndex`:44, `targetDimension`:9,
`targetPosition`:'(4,8)'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:94}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:84, `sourcePosition`:'(7,0)', `targetCellIndex`:45, `targetDimension`:9,
`targetPosition`:'(5,0)'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:94}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:85, `sourcePosition`:'(7,1)', `targetCellIndex`:45, `targetDimension`:9,
`targetPosition`:'(5,0)'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:95}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:86, `sourcePosition`:'(7,2)', `targetCellIndex`:46, `targetDimension`:9,
`targetPosition`:'(5,1)'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:96}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:87, `sourcePosition`:'(7,3)', `targetCellIndex`:47, `targetDimension`:9,
`targetPosition`:'(5,2)'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:97}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:88, `sourcePosition`:'(7,4)', `targetCellIndex`:48, `targetDimension`:9,
`targetPosition`:'(5,3)'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:97}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:89, `sourcePosition`:'(7,5)', `targetCellIndex`:48, `targetDimension`:9,
`targetPosition`:'(5,3)'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:98}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:90, `sourcePosition`:'(7,6)', `targetCellIndex`:49, `targetDimension`:9,
`targetPosition`:'(5,4)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:14, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:99}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:91, `sourcePosition`:'(7,7)', `targetCellIndex`:50, `targetDimension`:9,
`targetPosition`:'(5,5)'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:100}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:92, `sourcePosition`:'(7,8)', `targetCellIndex`:51, `targetDimension`:9,
`targetPosition`:'(5,6)'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:100}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:93, `sourcePosition`:'(7,9)', `targetCellIndex`:51, `targetDimension`:9,
`targetPosition`:'(5,6)'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:101}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:94, `sourcePosition`:'(7,10)', `targetCellIndex`:52, `targetDimension`:9,
`targetPosition`:'(5,7)'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:102}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:95, `sourcePosition`:'(7,11)', `targetCellIndex`:53, `targetDimension`:9,
`targetPosition`:'(5,8)'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:103}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:96, `sourcePosition`:'(8,0)', `targetCellIndex`:54, `targetDimension`:9,
`targetPosition`:'(6,0)'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:103}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:97, `sourcePosition`:'(8,1)', `targetCellIndex`:54, `targetDimension`:9,
`targetPosition`:'(6,0)'}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:104}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:98, `sourcePosition`:'(8,2)', `targetCellIndex`:55, `targetDimension`:9,
`targetPosition`:'(6,1)'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:105}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:99, `sourcePosition`:'(8,3)', `targetCellIndex`:56, `targetDimension`:9,
`targetPosition`:'(6,2)'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:106}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:100, `sourcePosition`:'(8,4)', `targetCellIndex`:57, `targetDimension`:9,
`targetPosition`:'(6,3)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:14, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:106}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:101, `sourcePosition`:'(8,5)', `targetCellIndex`:57, `targetDimension`:9,
`targetPosition`:'(6,3)'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:107}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:102, `sourcePosition`:'(8,6)', `targetCellIndex`:58, `targetDimension`:9,
`targetPosition`:'(6,4)'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:108}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:103, `sourcePosition`:'(8,7)', `targetCellIndex`:59, `targetDimension`:9,
`targetPosition`:'(6,5)'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:109}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:104, `sourcePosition`:'(8,8)', `targetCellIndex`:60, `targetDimension`:9,
`targetPosition`:'(6,6)'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:109}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:105, `sourcePosition`:'(8,9)', `targetCellIndex`:60, `targetDimension`:9,
`targetPosition`:'(6,6)'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:110}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:106, `sourcePosition`:'(8,10)', `targetCellIndex`:61, `targetDimension`:9,
`targetPosition`:'(6,7)'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:111}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:107, `sourcePosition`:'(8,11)', `targetCellIndex`:62, `targetDimension`:9,
`targetPosition`:'(6,8)'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:103}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:108, `sourcePosition`:'(9,0)', `targetCellIndex`:54, `targetDimension`:9,
`targetPosition`:'(6,0)'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:103}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:109, `sourcePosition`:'(9,1)', `targetCellIndex`:54, `targetDimension`:9,
`targetPosition`:'(6,0)'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:104}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:110, `sourcePosition`:'(9,2)', `targetCellIndex`:55, `targetDimension`:9,
`targetPosition`:'(6,1)'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:0, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:14, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:105}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:111, `sourcePosition`:'(9,3)', `targetCellIndex`:56, `targetDimension`:9,
`targetPosition`:'(6,2)'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:106}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:112, `sourcePosition`:'(9,4)', `targetCellIndex`:57, `targetDimension`:9,
`targetPosition`:'(6,3)'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:106}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:113, `sourcePosition`:'(9,5)', `targetCellIndex`:57, `targetDimension`:9,
`targetPosition`:'(6,3)'}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:107}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:114, `sourcePosition`:'(9,6)', `targetCellIndex`:58, `targetDimension`:9,
`targetPosition`:'(6,4)'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:108}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:115, `sourcePosition`:'(9,7)', `targetCellIndex`:59, `targetDimension`:9,
`targetPosition`:'(6,5)'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:109}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:116, `sourcePosition`:'(9,8)', `targetCellIndex`:60, `targetDimension`:9,
`targetPosition`:'(6,6)'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:109}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:117, `sourcePosition`:'(9,9)', `targetCellIndex`:60, `targetDimension`:9,
`targetPosition`:'(6,6)'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:110}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:118, `sourcePosition`:'(9,10)', `targetCellIndex`:61, `targetDimension`:9,
`targetPosition`:'(6,7)'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:111}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:119, `sourcePosition`:'(9,11)', `targetCellIndex`:62, `targetDimension`:9,
`targetPosition`:'(6,8)'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:112}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:120, `sourcePosition`:'(10,0)', `targetCellIndex`:63, `targetDimension`:9,
`targetPosition`:'(7,0)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:14, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:112}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:121, `sourcePosition`:'(10,1)', `targetCellIndex`:63, `targetDimension`:9,
`targetPosition`:'(7,0)'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:113}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:122, `sourcePosition`:'(10,2)', `targetCellIndex`:64, `targetDimension`:9,
`targetPosition`:'(7,1)'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:114}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:123, `sourcePosition`:'(10,3)', `targetCellIndex`:65, `targetDimension`:9,
`targetPosition`:'(7,2)'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:115}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:124, `sourcePosition`:'(10,4)', `targetCellIndex`:66, `targetDimension`:9,
`targetPosition`:'(7,3)'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:115}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:125, `sourcePosition`:'(10,5)', `targetCellIndex`:66, `targetDimension`:9,
`targetPosition`:'(7,3)'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:116}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:126, `sourcePosition`:'(10,6)', `targetCellIndex`:67, `targetDimension`:9,
`targetPosition`:'(7,4)'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:117}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:127, `sourcePosition`:'(10,7)', `targetCellIndex`:68, `targetDimension`:9,
`targetPosition`:'(7,5)'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:118}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:128, `sourcePosition`:'(10,8)', `targetCellIndex`:69, `targetDimension`:9,
`targetPosition`:'(7,6)'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:118}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:129, `sourcePosition`:'(10,9)', `targetCellIndex`:69, `targetDimension`:9,
`targetPosition`:'(7,6)'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:119}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:130, `sourcePosition`:'(10,10)', `targetCellIndex`:70, `targetDimension`:9,
`targetPosition`:'(7,7)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:14, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:120}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:131, `sourcePosition`:'(10,11)', `targetCellIndex`:71, `targetDimension`:9,
`targetPosition`:'(7,8)'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:121}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:132, `sourcePosition`:'(11,0)', `targetCellIndex`:72, `targetDimension`:9,
`targetPosition`:'(8,0)'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:121}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:133, `sourcePosition`:'(11,1)', `targetCellIndex`:72, `targetDimension`:9,
`targetPosition`:'(8,0)'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:122}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:134, `sourcePosition`:'(11,2)', `targetCellIndex`:73, `targetDimension`:9,
`targetPosition`:'(8,1)'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:123}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:135, `sourcePosition`:'(11,3)', `targetCellIndex`:74, `targetDimension`:9,
`targetPosition`:'(8,2)'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:124}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:136, `sourcePosition`:'(11,4)', `targetCellIndex`:75, `targetDimension`:9,
`targetPosition`:'(8,3)'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:124}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:137, `sourcePosition`:'(11,5)', `targetCellIndex`:75, `targetDimension`:9,
`targetPosition`:'(8,3)'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:125}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:138, `sourcePosition`:'(11,6)', `targetCellIndex`:76, `targetDimension`:9,
`targetPosition`:'(8,4)'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:126}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:139, `sourcePosition`:'(11,7)', `targetCellIndex`:77, `targetDimension`:9,
`targetPosition`:'(8,5)'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:127}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:140, `sourcePosition`:'(11,8)', `targetCellIndex`:78, `targetDimension`:9,
`targetPosition`:'(8,6)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:14, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:127}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:141, `sourcePosition`:'(11,9)', `targetCellIndex`:78, `targetDimension`:9,
`targetPosition`:'(8,6)'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:128}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:142, `sourcePosition`:'(11,10)', `targetCellIndex`:79, `targetDimension`:9,
`targetPosition`:'(8,7)'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:129}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:143, `sourcePosition`:'(11,11)', `targetCellIndex`:80, `targetDimension`:9,
`targetPosition`:'(8,8)'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:35}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:0, `sourcePosition`:'(0,0)', `targetCellIndex`:0, `targetDimension`:9,
`targetPosition`:'(0,0)'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:35}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:1, `sourcePosition`:'(0,1)', `targetCellIndex`:0, `targetDimension`:9,
`targetPosition`:'(0,0)'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:40}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:2, `sourcePosition`:'(0,2)', `targetCellIndex`:1, `targetDimension`:9,
`targetPosition`:'(0,1)'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:47}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:3, `sourcePosition`:'(0,3)', `targetCellIndex`:2, `targetDimension`:9,
`targetPosition`:'(0,2)'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:48}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:4, `sourcePosition`:'(0,4)', `targetCellIndex`:3, `targetDimension`:9,
`targetPosition`:'(0,3)'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:48}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:5, `sourcePosition`:'(0,5)', `targetCellIndex`:3, `targetDimension`:9,
`targetPosition`:'(0,3)'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:49}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:6, `sourcePosition`:'(0,6)', `targetCellIndex`:4, `targetDimension`:9,
`targetPosition`:'(0,4)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:14, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:50}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:7, `sourcePosition`:'(0,7)', `targetCellIndex`:5, `targetDimension`:9,
`targetPosition`:'(0,5)'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:51}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:8, `sourcePosition`:'(0,8)', `targetCellIndex`:6, `targetDimension`:9,
`targetPosition`:'(0,6)'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:51}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:9, `sourcePosition`:'(0,9)', `targetCellIndex`:6, `targetDimension`:9,
`targetPosition`:'(0,6)'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:52}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:10, `sourcePosition`:'(0,10)', `targetCellIndex`:7, `targetDimension`:9,
`targetPosition`:'(0,7)'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:54}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:11, `sourcePosition`:'(0,11)', `targetCellIndex`:8, `targetDimension`:9,
`targetPosition`:'(0,8)'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:35}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:12, `sourcePosition`:'(1,0)', `targetCellIndex`:0, `targetDimension`:9,
`targetPosition`:'(0,0)'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:35}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:13, `sourcePosition`:'(1,1)', `targetCellIndex`:0, `targetDimension`:9,
`targetPosition`:'(0,0)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:40}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:14, `sourcePosition`:'(1,2)', `targetCellIndex`:1, `targetDimension`:9,
`targetPosition`:'(0,1)'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:47}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:15, `sourcePosition`:'(1,3)', `targetCellIndex`:2, `targetDimension`:9,
`targetPosition`:'(0,2)'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:48}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:16, `sourcePosition`:'(1,4)', `targetCellIndex`:3, `targetDimension`:9,
`targetPosition`:'(0,3)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:14, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:48}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:17, `sourcePosition`:'(1,5)', `targetCellIndex`:3, `targetDimension`:9,
`targetPosition`:'(0,3)'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:49}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:18, `sourcePosition`:'(1,6)', `targetCellIndex`:4, `targetDimension`:9,
`targetPosition`:'(0,4)'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:50}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:19, `sourcePosition`:'(1,7)', `targetCellIndex`:5, `targetDimension`:9,
`targetPosition`:'(0,5)'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:51}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:20, `sourcePosition`:'(1,8)', `targetCellIndex`:6, `targetDimension`:9,
`targetPosition`:'(0,6)'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:51}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:21, `sourcePosition`:'(1,9)', `targetCellIndex`:6, `targetDimension`:9,
`targetPosition`:'(0,6)'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:52}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:22, `sourcePosition`:'(1,10)', `targetCellIndex`:7, `targetDimension`:9,
`targetPosition`:'(0,7)'}]->(b);
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MATCH (a {`key`:257}) MATCH (b {`key`:54}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:23, `sourcePosition`:'(1,11)', `targetCellIndex`:8, `targetDimension`:9,
`targetPosition`:'(0,8)'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:56}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:24, `sourcePosition`:'(2,0)', `targetCellIndex`:9, `targetDimension`:9,
`targetPosition`:'(1,0)'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:56}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:25, `sourcePosition`:'(2,1)', `targetCellIndex`:9, `targetDimension`:9,
`targetPosition`:'(1,0)'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:57}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:26, `sourcePosition`:'(2,2)', `targetCellIndex`:10, `targetDimension`:9,
`targetPosition`:'(1,1)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:14, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:58}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:27, `sourcePosition`:'(2,3)', `targetCellIndex`:11, `targetDimension`:9,
`targetPosition`:'(1,2)'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:59}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:28, `sourcePosition`:'(2,4)', `targetCellIndex`:12, `targetDimension`:9,
`targetPosition`:'(1,3)'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:59}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:29, `sourcePosition`:'(2,5)', `targetCellIndex`:12, `targetDimension`:9,
`targetPosition`:'(1,3)'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:62}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:30, `sourcePosition`:'(2,6)', `targetCellIndex`:13, `targetDimension`:9,
`targetPosition`:'(1,4)'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:63}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:31, `sourcePosition`:'(2,7)', `targetCellIndex`:14, `targetDimension`:9,
`targetPosition`:'(1,5)'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:64}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:32, `sourcePosition`:'(2,8)', `targetCellIndex`:15, `targetDimension`:9,
`targetPosition`:'(1,6)'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:64}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:33, `sourcePosition`:'(2,9)', `targetCellIndex`:15, `targetDimension`:9,
`targetPosition`:'(1,6)'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:65}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:34, `sourcePosition`:'(2,10)', `targetCellIndex`:16, `targetDimension`:9,
`targetPosition`:'(1,7)'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:66}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:35, `sourcePosition`:'(2,11)', `targetCellIndex`:17, `targetDimension`:9,
`targetPosition`:'(1,8)'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:67}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:36, `sourcePosition`:'(3,0)', `targetCellIndex`:18, `targetDimension`:9,
`targetPosition`:'(2,0)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:14, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:67}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:37, `sourcePosition`:'(3,1)', `targetCellIndex`:18, `targetDimension`:9,
`targetPosition`:'(2,0)'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:68}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:38, `sourcePosition`:'(3,2)', `targetCellIndex`:19, `targetDimension`:9,
`targetPosition`:'(2,1)'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:69}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:39, `sourcePosition`:'(3,3)', `targetCellIndex`:20, `targetDimension`:9,
`targetPosition`:'(2,2)'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:70}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:40, `sourcePosition`:'(3,4)', `targetCellIndex`:21, `targetDimension`:9,
`targetPosition`:'(2,3)'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:70}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:41, `sourcePosition`:'(3,5)', `targetCellIndex`:21, `targetDimension`:9,
`targetPosition`:'(2,3)'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:71}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:42, `sourcePosition`:'(3,6)', `targetCellIndex`:22, `targetDimension`:9,
`targetPosition`:'(2,4)'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:72}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:43, `sourcePosition`:'(3,7)', `targetCellIndex`:23, `targetDimension`:9,
`targetPosition`:'(2,5)'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:73}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:44, `sourcePosition`:'(3,8)', `targetCellIndex`:24, `targetDimension`:9,
`targetPosition`:'(2,6)'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:73}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:45, `sourcePosition`:'(3,9)', `targetCellIndex`:24, `targetDimension`:9,
`targetPosition`:'(2,6)'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:74}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:46, `sourcePosition`:'(3,10)', `targetCellIndex`:25, `targetDimension`:9,
`targetPosition`:'(2,7)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:14, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:75}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:47, `sourcePosition`:'(3,11)', `targetCellIndex`:26, `targetDimension`:9,
`targetPosition`:'(2,8)'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:76}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:48, `sourcePosition`:'(4,0)', `targetCellIndex`:27, `targetDimension`:9,
`targetPosition`:'(3,0)'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:76}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:49, `sourcePosition`:'(4,1)', `targetCellIndex`:27, `targetDimension`:9,
`targetPosition`:'(3,0)'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:77}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:50, `sourcePosition`:'(4,2)', `targetCellIndex`:28, `targetDimension`:9,
`targetPosition`:'(3,1)'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:78}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:51, `sourcePosition`:'(4,3)', `targetCellIndex`:29, `targetDimension`:9,
`targetPosition`:'(3,2)'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:79}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:52, `sourcePosition`:'(4,4)', `targetCellIndex`:30, `targetDimension`:9,
`targetPosition`:'(3,3)'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:79}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:53, `sourcePosition`:'(4,5)', `targetCellIndex`:30, `targetDimension`:9,
`targetPosition`:'(3,3)'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:80}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:54, `sourcePosition`:'(4,6)', `targetCellIndex`:31, `targetDimension`:9,
`targetPosition`:'(3,4)'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:81}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:55, `sourcePosition`:'(4,7)', `targetCellIndex`:32, `targetDimension`:9,
`targetPosition`:'(3,5)'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:82}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:56, `sourcePosition`:'(4,8)', `targetCellIndex`:33, `targetDimension`:9,
`targetPosition`:'(3,6)'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:14, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:82}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:57, `sourcePosition`:'(4,9)', `targetCellIndex`:33, `targetDimension`:9,
`targetPosition`:'(3,6)'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:83}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:58, `sourcePosition`:'(4,10)', `targetCellIndex`:34, `targetDimension`:9,
`targetPosition`:'(3,7)'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:84}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:59, `sourcePosition`:'(4,11)', `targetCellIndex`:35, `targetDimension`:9,
`targetPosition`:'(3,8)'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:76}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:60, `sourcePosition`:'(5,0)', `targetCellIndex`:27, `targetDimension`:9,
`targetPosition`:'(3,0)'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:76}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:61, `sourcePosition`:'(5,1)', `targetCellIndex`:27, `targetDimension`:9,
`targetPosition`:'(3,0)'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:77}) CREATE (a)-[:`FOLDS_INTO_GROUND` {`foldRatio`:0.75,
`sourceDimension`:12, `sourceIndex`:62, `sourcePosition`:'(5,2)', `targetCellIndex`:28, `targetDimension`:9,
`targetPosition`:'(3,1)'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058020626611905324, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05859631179023644, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05917199696856754, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1439479309441336, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:0, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:14, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1441920696846824, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14476775486301355, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14534344004134464, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
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MATCH (a {`key`:0}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14591912521967576, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14616326396022455, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14673894913855567, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14731463431688677, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1478903194952179, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1481344582357667, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487101434140978, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:14, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492858285924289, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09385357117225686, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09409770991280567, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09467339509113677, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09524908026946788, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095824765447799, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09606890418834779, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0966445893666789, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09722027454501002, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09779595972334112, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:14, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09804009846388992, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09861578364222104, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09919146882055214, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042227539670913646, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042471678411462446, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04304736358979356, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043623048768124666, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04419873394645578, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04444287268700458, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04501855786533569, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:14, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0455942430436668, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04616992822199791, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046414066962546706, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046989752140877826, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04756543731920894, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10135977414710463, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10160391288765344, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10217959806598455, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10275528324431565, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10333096842264677, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:14, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10357510716319555, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10415079234152667, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10472647751985778, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1053021626981889, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554630143873771, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10612198661706881, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10669767179539992, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
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MATCH (a {`key`:0}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08140183805570471, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08164597679625349, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08222166197458461, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:14, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08279734715291572, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08337303233124682, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08361717107179562, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08419285625012674, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08476854142845786, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08534422660678896, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08558836534733776, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08616405052566888, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08673973570399998, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05185084881062895, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:14, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052094987551177745, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052007579853944226, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05192017215671071, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05183276445947719, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05207690320002599, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05198949550279247, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.051902087805558945, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05181468010832543, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05205881884887422, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.051971411151640695, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:14, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05188400345440719, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16583602395481706, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16608016269536588, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16599275499813235, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16590534730089881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16581793960366528, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1660620783442141, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16597467064698057, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16588726294974707, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16579985525251356, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:14, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16604399399306236, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16595658629582882, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16586917859859532, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11111087304122279, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11135501178177158, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11126760408453808, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11118019638730454, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11109278869007103, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11133692743061983, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11124951973338633, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:14, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1111621120361528, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11107470433891928, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11131884307946806, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11123143538223455, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11114402768500103, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1388722563875389, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13911639512808768, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13902898743085415, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13894157973362065, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1388541720363871, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:0, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:14, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13909831077693594, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13901090307970243, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1389234953824689, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13883608768523537, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13908022642578416, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13899281872855065, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057689080174123004, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05793321891467182, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05850890409300291, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05908458927133403, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:14, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1438605232469001, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
```

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MATCH (a {`key`:1}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1441046619874489, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14468034716578002, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1452560323441111, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14583171752244223, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14607585626299105, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14665154144132214, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14722722661965326, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14780291179798438, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14804705053853318, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:14, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14862273571686427, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14919842089519542, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09376616347502334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09401030221557213, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09458598739390325, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09516167257223437, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09573735775056548, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09598149649111427, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0965571816694454, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09713286684777649, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:14, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977085520261076, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0979526907666564, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09852837594498752, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
```

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MATCH (a {`key`:1}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09910406112331864, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04214013197368013, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04238427071422893, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04295995589256004, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043535641070891154, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04411132624922227, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04435546498977106, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:14, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044931150168102174, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04550683534643329, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046082520524764393, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046326659265313194, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
```

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MATCH (a {`key`:1}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04690234444364431, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04747802962197542, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10127236644987112, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151650519041991, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10209219036875103, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10266787554708215, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:14, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10324356072541324, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10348769946596205, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10406338464429314, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10463906982262426, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10521475500095538, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545889374150418, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1060345789198353, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10661026409816642, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08131443035847118, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08155856909901998, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:14, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08213425427735109, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0827099394556822, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0832856246340133, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08352976337456211, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08410544855289322, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08468113373122432, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08525681890955544, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08550095765010425, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08607664282843536, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08665232800676646, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:14, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052426533988960064, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052583265032275346, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05249585733504182, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0524084496378083, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05256518068112358, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:1}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05247777298389006, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05239036528665654, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:14, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052547096329971815, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05245968863273829, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16641170913314818, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16656844017646344, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16648103247922993, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16639362478199643, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
```

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MATCH (a {`key`:1}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1665503558253117, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1664629481280782, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:14, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16637554043084465, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16653227147415994, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16644486377692644, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11168655821955391, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11193069696010272, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11184328926286918, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11175588156563565, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11166847386840215, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11191261260895094, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:0, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:14, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11182520491171741, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1117377972144839, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11165038951725038, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11189452825779919, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11180712056056566, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11171971286333214, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:64, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13944794156586998, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13969208030641878, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13960467260918527, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13951726491195174, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:14, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13942985721471823, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13967399595526703, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1395865882580335, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1394991805608, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13941177286356649, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13965591160411528, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0573575337363407, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05793321891467182, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058177357655220606, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.018181818181818182, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05875304283355171, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:14, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05932872801188282, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1441046619874489, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1443488007279977, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1449244859063288, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1455001710846599, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14607585626299105, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14631999500353982, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14689568018187094, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14747136536020206, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14804705053853318, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:14, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14829118927908197, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1488668744574131, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14944255963574418, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09401030221557213, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09425444095612093, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09483012613445205, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09540581131278315, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09598149649111427, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09622563523166308, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09680132040999419, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:14, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0973770055883253, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0979526907666564, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0981968295072052, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09877251468553631, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09934819986386742, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04238427071422893, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04262840945477772, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043204094633108835, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043779779811439955, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04435546498977106, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:14, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044599603730319855, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04517528890865097, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04575097408698208, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046326659265313194, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046570798005861995, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0471464831841931, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04772216836252422, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151650519041991, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1017606439309687, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10233632910929982, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:14, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10291201428763093, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10348769946596205, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10373183820651084, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10430752338484195, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10488320856317307, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545889374150418, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10570303248205297, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10627871766038409, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1068544028387152, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08155856909901998, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:14, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08180270783956876, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08237839301789988, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08295407819623098, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08352976337456211, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0837739021151109, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08434958729344201, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08492527247177313, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08550095765010425, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08574509639065303, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08632078156898415, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:14, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08689646674731527, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05291481147005766, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05282740377282414, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052739996075590614, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05289672711890589, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05280931942167238, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05272191172443885, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:14, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052878642767754135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0527912350705206, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05270382737328709, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16689998661424577, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16681257891701223, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16672517121977873, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.166881902263094, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16679449456586048, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:0, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:14, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16670708686862695, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16686381791194227, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16677641021470874, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666890025174752, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11193069696010272, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11217483570065151, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11208742800341798, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11200002030618447, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11191261260895094, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:14, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11215675134949973, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11206934365226623, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1119819359550327, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11189452825779919, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11213866699834799, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11205125930111445, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11196385160388095, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13969208030641878, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1399362190469676, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13984881134973406, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:14, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13976140365250053, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13967399595526703, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13991813469581582, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13983072699858232, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13974331930134878, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13965591160411528, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05669444086077606, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05727012603910718, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05784581121743829, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05808994995798709, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:14, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05866563513631819, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0592413203146493, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14401725429021536, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14426139303076416, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14483707820909528, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1454127633874264, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14598844856575752, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14623258730630628, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14680827248463743, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14738395766296852, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:14, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14795964284129964, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14820378158184846, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14877946676017956, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14935515193851068, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09392289451833861, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09416703325888741, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09474271843721851, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09531840361554964, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09589408879388076, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09613822753442955, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:14, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09671391271276067, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09728959789109176, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09786528306942288, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09810942180997168, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09868510698830278, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0992607921666339, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0422968630169954, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042541001757544196, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043116686935875316, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04369237211420643, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:14, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04426805729253754, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04451219603308633, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04508788121141744, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04566356638974856, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046239251568079676, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04648339030862847, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04705907548695958, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047634760665290696, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10142909749318639, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1016732362337352, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:14, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1022489214120663, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10282460659039741, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10340029176872852, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10364443050927732, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10422011568760844, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10479580086593954, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10537148604427066, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10561562478481945, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10619130996315056, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10676699514148168, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:14, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08147116140178645, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08171530014233525, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08229098532066635, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08286667049899747, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08344235567732858, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08368649441787737, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0842621795962085, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08483786477453961, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08541354995287072, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08565768869341951, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:14, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08623337387175063, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08680905905008174, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05324635790783997, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05349049664838877, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053403088951155246, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053315681253921735, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05322827355668821, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05347241229723701, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053385004600003484, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053297596902769966, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:0, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:14, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053210189205536454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05345432794608524, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05336692024885172, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053279512551618204, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1672315330520281, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16747567179257689, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16738826409534338, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16730085639810985, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16721344870087632, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16745758744142514, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:14, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1673701797441916, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16728277204695807, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.167195364349724 57, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16743950309027336, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16735209539303986, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16726468769580632, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11250638213843382, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11275052087898262, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11266311318174911, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11257570548451558, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:14, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11248829778728205, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11273243652783087, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11264502883059734, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1125576211333638, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1124702134361303, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11271435217667909, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11262694447944559, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11253953678221205, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14026776548474992, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14051190422529872, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:14, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14042449652806518, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14033708883083168, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14024968113359815, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14049381987414694, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14040641217691134, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14031900447967999, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05603134798521143, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05660703316354255, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057182718341873655, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05775840352020477, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:14, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05800254226075356, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05857822743908468, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05915391261741579, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14392984659298186, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14417398533353062, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14474967051186177, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14532535569019286, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14590104086852398, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14614517960907278, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1467208647874039, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:14, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14729654996573502, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14787223514406614, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14811637388461493, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14869205906294602, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14926774424127717, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0938354868211051, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09407962556165389, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09465531073998501, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09523099591831613, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09580668109664724, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:14, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09605081983719603, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09662650501552714, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09720219019385826, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09777787537218936, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09802201411273816, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09859769929106928, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09917338446940038, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042209455319761884, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042453594060310684, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0430292792386418, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:14, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04360496441697291, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044180649595304024, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04442478833585282, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04500047351418393, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045576158692515044, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04615184387084615, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04639598261139496, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046971667789726064, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04754735296805717, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10134168979595287, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:14, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10158582853650167, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10216151371483279, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10273719889316389, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10331288407149501, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1035570228120438, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10413270799037491, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10470839316870602, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10528407834703714, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10552821708758593, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10610390226591705, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:14, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10667958744424816, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08138375370455293, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08162789244510173, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08220357762343285, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08277926280176395, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08335494798009506, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08359908672064387, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08417477189897497, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08475045707730608, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0853261422556372, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:0, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:14, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.085570280996186, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08614596617451711, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08672165135284822, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05382204308617109, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05406618182671988, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053978774129486366, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053891366432252834, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05380395873501932, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05404809747556812, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053960689778334604, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:14, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053873282081101086, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05378587438386756, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05403001312441636, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05394260542718283, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05385519772994932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16780721823035918, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.168051356970908, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16796394927367447, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16787654157644094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16778913387920744, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:14, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803327261975623, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1679458649225227, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678584572252892, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16777104952805566, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16801518826860445, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16792778057137092, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16784037287413742, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11308206731676493, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11332620605731374, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11323879836008022, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:14, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11315139066284668, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11306398296561318, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11330812170616197, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11322071400892844, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11313330631169491, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1130458986144614, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11329003735501021, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11320262965777767, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11311522196054316, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14084345066308102, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:14, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1410875894036298, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1410001817063963, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14091277400916277, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14082536631192924, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14106950505247803, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14098209735524453, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05569980154742911, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05594394028797791, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056519625466309016, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.018181818181818182, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057095310644640136, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:14, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05767099582297124, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05791513456352004, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05849081974185116, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05906650492018227, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14384243889574833, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14408657763629712, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14466226281462824, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14523794799295933, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14581363317129048, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14605777191183927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:14, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14663345709017037, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1472091422685015, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1477848274468326, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1480289661873814, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14860465136571252, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14918033654404364, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09374807912387156, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09399221786442037, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09456790304275149, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0951435882210826, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:14, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09571927339941372, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959634121399625, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09653909731829362, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09711478249662474, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09769046767495584, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09793460641550465, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09851029159383576, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09908597677216686, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04212204762252837, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042366186363077166, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:14, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04294187154140827, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043517556719739385, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0440932418980705, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0443373806386193, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04491306581695041, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045488750995281525, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04606443617361264, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04630857491416143, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046884260092492545, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04745994527082366, `foldPressure`:0.358024691358024 67,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:14, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10125428209871934, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10149842083926815, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10207410601759925, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10264979119593037, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1032254763742615, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10346961511481027, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1040453002931414, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10462098547147251, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10519667064980362, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10544080939035241, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:0, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:14, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10601649456868352, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10659217974701464, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0812963460073194, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08154048474786821, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08211616992619933, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08269185510453043, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08326754028286155, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08351167902341035, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08408736420174145, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08466304938007257, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:14, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08523873455840368, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08548287329895247, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0860585584772836, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08663424365561471, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0543977282645022, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05455445930781747, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054467051610583954, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054379643913350435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:248}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:14, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05453637495666571, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054448967259432185, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05436155956219868, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05451829060551395, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05443088290828043, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16838290340869033, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1685396344520056, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684522267547721, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:14, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16836481905753856, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852155010085385, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684341424036203, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16834673470638678, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850346574970207, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16841605805246854, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11365775249509606, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11390189123564484, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:14, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11381448353841132, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11372707584117782, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11363966814394429, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11388380688449308, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11379639918725955, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11370899149002604, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11362158379279254, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11386572253334133, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1137783148361078, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1136909071388743, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:14, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14141913584141214, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14166327458196093, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14157586688472742, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1414884591874939, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14140105149026036, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14164519023080918, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05536825510964679, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05594394028797791, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0561880790285267, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05676376420685782, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:14, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05733944938518892, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05791513456352004, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058159273304068844, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05873495848239995, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05931064366073107, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14408657763629712, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1443307163768459, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14490640155517703, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14548208673350815, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14605777191183927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:14, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14630191065238804, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1468775958307192, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14745328100905028, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1480289661873814, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14827310492793022, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14884879010626131, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14942447528459243, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09399221786442037, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09423635660496917, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09481204178330027, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:14, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09538772696163139, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959634121399625, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0962075508805113, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09678323605884243, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09735892123717353, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09793460641550465, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09817874515605343, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09875443033438454, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09933011551271566, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042366186363077166, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:14, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04261032510362596, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04318601028195707, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04376169546028818, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0443373806386193, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04458151937916809, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045157204557499206, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045732889735830326, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04630857491416143, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046552713654710226, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04712839883304134, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:0, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:14, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04770408401137245, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10149842083926815, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10174255957981694, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10231824475814805, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10289392993647917, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10346961511481027, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10371375385535908, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1042894390336902, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10486512421202131, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10544080939035241, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:14, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10568494813090121, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10626063330923231, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10683631848756342, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08154048474786821, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.081784623488417, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08236030866674811, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08293599384507923, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08351167902341035, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08375581776395914, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08433150294229026, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:14, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08490718812062137, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08548287329895247, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08572701203950128, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08630269721783239, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0868783823961635, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05488600574559979, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05479859804836627, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054711190351132755, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:14, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05486792139444803, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0547805136972145, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05469310599998099, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05484983704329627, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05476242934606275, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054675021648829224, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1688711808897879, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1687837731925544, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:14, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16869636549532088, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16885309653863614, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16876568884140264, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1686782811441691, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16883501218748437, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16874760449025086, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16866019679301733, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11390189123564484, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:14, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11414602997619364, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11405862227896013, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1139712145817266, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11388380688449308, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11412794562504186, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11404053792780836, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11395313023057485, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11386572253334133, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11410986127389011, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11402245357665661, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:14, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11393504587942307, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14166327458196093, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14190741332250972, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1418200056252762, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417325979280427, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14164519023080918, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05470516223408217, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05528084741241328, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055856532590744384, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056100671331293185, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:14, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05667635650962429, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:7}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05725204168795541, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057827726866286525, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05807186560683532, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05864755078516643, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05922323596349755, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14399916993906361, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14424330867961238, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14481899385794353, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14539467903627462, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:14, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14597036421460574, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14621450295515453, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14679018813348566, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14736587331181675, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1479415584901479, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1481856972306967, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14876138240902778, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14933706758735893, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09390481016718685, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09414894890773565, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:15, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09472463408606677, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09530031926439787, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095876004442729, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09612014318327779, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0966958283616089, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09727151353994001, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09784719871827112, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09809133745881991, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09866702263715103, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09924270781548214, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:0, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:15, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04227877866584364, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04252291740639244, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043098602584723554, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04367428776305466, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04424997294138577, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044494111681934574, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04506979686026569, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0456454820385968, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046221167216927914, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04646530595747671, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:15, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047040991135807814, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047616676314138934, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10141101314203461, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10165515188258342, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10223083706091454, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10280652223924565, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10338220741757677, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10362634615812556, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10420203133645667, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10477771651478779, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:15, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1053534016931189, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1055975404336677, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10617322561199881, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10674891079032992, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08145307705063469, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08169721579118348, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0822729009695146, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08284858614784571, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08342427132617682, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08366841006672562, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:15, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08424409524505674, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08481978042338785, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08539546560171897, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08563960434226775, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08621528952059886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08679097469892998, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055217552183382104, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055461690923930905, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05537428322669738, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05528687552946386, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:15, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05519946783223034, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055443606572779136, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05535619887554562, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0552687911783121, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05518138348107857, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05542552222162739, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055338114524393855, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055250706827160344, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16920272732757022, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16944686606811904, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:15, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1693594583708855, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16927205067365197, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16918464297641847, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16942878171696726, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16934137401973373, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1692539663225002, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1691665586252667, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1694106973658155, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16932328966858196, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16923588197134845, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:15, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11447757641397595, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11472171515452474, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11463430745729121, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1145468997600577, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11445949206282419, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.114703630803373, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11461622310613949, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11452881540890596, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11444140771167244, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11468554645222125, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:15, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11459813875498771, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11451073105775418, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.09957413673572789, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14223895976029205, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14248309850084084, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1423956908036073, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1423082831063738, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05404206935851753, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05461775453684865, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05519343971517976, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05576912489351088, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:15, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056013263634059667, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05658894881239077, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05716463399072189, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057740319169053, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0579844579096018, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05856014308793292, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059135828266264026, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:2,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14391176224183008, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14415590098237888, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14473158616071, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:15, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1453072713390411, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14588295651737224, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14612709525792103, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14670278043625212, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14727846561458324, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14785415079291436, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14809828953346316, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14867397471179428, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492496598901254, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09381740246995333, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:0, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:15, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09406154121050214, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09463722638883325, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09521291156716435, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09578859674549547, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09603273548604425, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09660842066437537, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09718410584270648, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977597910210376, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09800392976158641, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09857961493991751, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:15, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09915530011824863, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04219137096861012, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042435509709158915, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043011194887490035, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04358688006582115, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044162565244152255, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044406703984701056, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04498238916303217, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04555807434136328, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046133759519694395, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:15, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04637789826024319, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04695358343857431, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047529268616905415, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:1,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.01831563888873418, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10132360544480111, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1015677441853499, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10214342936368101, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10271911454201213, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10329479972034325, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10353893846089203, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10411462363922315, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:15, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10469030881755426, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10526599399588538, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10551013273643417, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10608581791476529, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10666150309309641, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08136566935340116, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08160980809394996, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08218549327228108, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08276117845061219, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08333686362894331, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:15, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0835810023694921, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08415668754782321, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08473237272615433, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08530805790448545, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08555219664503425, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08612788182336535, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08670356700169646, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055793237361713224, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05603737610226201, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0559499684050285, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:15, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05586256070779498, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055775153010561455, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056019291751110256, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05593188405387674, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055844476356643205, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05575706865940969, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056001207399958494, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05591379970272496, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05582639200549146, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16977841250590134, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:15, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17002255124645013, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699351435492166, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698477358519831, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1697603281547496, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000446689529836, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16991705919806485, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982965150083135, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16974224380359781, `foldPressure`:0.358024691358024 67,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699863825441466, `foldPressure`:0.358024691358024 67,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698989748469131, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:15, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16981156714967957, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11505326159230705, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11529740033285585, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11520999263562234, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11512258493838881, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1150351772411553, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11527931598170413, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1151919082844706, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11510450058723706, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11501709289000356, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:15, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11526123163055235, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11517382393331882, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11508641623608531, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:3,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14281464493862314, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14305878367917194, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14297137598193843, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0002441387405487964, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:0,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.000819823918879909, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:0,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0013955090972110214, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.001971194275542134, `foldPressure`:0.037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:15, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.002791018194422043, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0033667033727531556, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003942388551084268, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004186527291633065, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004762212469964177, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.00533789764829529, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1415269348668286, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417710736073774, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14234675878570852, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:0, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:15, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14292244396403964, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14349812914237076, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14374226788291952, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14431795306125067, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14489363823958176, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14546932341791288, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1457134621584617, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1462891473367928, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14686483251512392, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09288482130552626, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:15, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09312896004607506, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09370464522440618, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09428033040273728, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09485601558106839, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09510015432161718, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09567583949994829, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09625152467827941, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09682720985661053, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09707134859715932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09764703377549044, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:15, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09822271895382156, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09550356424930773, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09574770298985653, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09632338816818765, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09689907334651876, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09747475852484987, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09771889726539865, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09829458244372977, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09887026762206089, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09944595280039199, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:15, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0996900915409408, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1002657767192719, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10084146189760303, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10123619774756759, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10148033648811638, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1020560216664475, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10263170684477861, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10320739202310972, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10345153076365851, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10402721594198963, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:15, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10460290112032074, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10517858629865184, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10542272503920065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10599841021753177, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10657409539586288, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13436694069129468, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1346110794318435, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1351867646101746, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1357624497885057, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13633813496683683, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:15, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13658227370738563, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13715795888571672, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13773364406404784, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13830932924237893, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13855346798292772, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13912915316125887, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13970483833958997, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05636892254004433, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05661306128059312, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05652565358335961, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:15, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05643824588612609, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05635083818889257, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05659497692944136, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056507569232207844, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056420161534974325, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05633275383774081, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0565768925782896, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05648948488105608, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05640207718382256, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1680566780064645, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:15, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1683008167470133, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1682134090497798, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16812600135254627, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803859365531273, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16828273239586153, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16819532469862802, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1681079170013945, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16802050930416096, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16826464804470978, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16817724034747625, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:15, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16808983265024272, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1676491227569863, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16789326149753508, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16780585380030155, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16771844610306805, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16763103840583454, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678751771463833, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16778776944914983, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1677003617519163, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16761295405468277, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:15, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678570927952316, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16776968509799806, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16768227740076452, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14109291043918631, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1413370491797351, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05404206935851753, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0002441387405487964, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:0,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0010639626594287053, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.001639647837759818, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:0, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:15, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0024594717566397265, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0030351569349708393, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003610842113301952, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004186527291633065, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0044306660321818 61, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005006351210512973, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005582036388844086, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417710736073774, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1420152123479262, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1425908975262573, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:15, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14316658270458843, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14374226788291952, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14398640662346834, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14456209180179946, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14513777698013056, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1457134621584617, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14595760089901047, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1465332860773416, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14710897125567268, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09312896004607506, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:15, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09337309878662385, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09394878396495497, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09452446914328608, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09510015432161718, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09534429306216598, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959199782404971, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09649566341882822, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09707134859715932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09731548733770812, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09789117251603924, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:15, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09846685769437034, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09574770298985653, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09599184173040533, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09656752690873643, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09714321208706754, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09771889726539865, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09796303600594745, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09853872118427857, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09911440636260968, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0996900915409408, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:15, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09993423028148958, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1005099154598207, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10108560063815182, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10148033648811638, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10172447522866519, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1023001604069963, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1028758455853274, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10345153076365851, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1036956695042073, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10427135468253843, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:15, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10484703986086953, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10542272503920065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10566686377974945, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10624254895808057, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10681823413641167, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1346110794318435, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13485521817239227, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1354309033507234, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1360065885290545, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13658227370738563, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:15, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13682641244793442, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13740209762626554, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13797778280459666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13855346798292772, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13879760672347655, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13937329190180764, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13994897708013876, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05661306128059312, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05685720002114193, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056769792323908407, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:15, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05668238462667489, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05659497692944136, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05683911566999016, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056751707972756645, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056664300275523126, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0565768925782896, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0568210313188384, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05673362362160487, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05664621592437136, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1683008167470133, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:15, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1685449554875621, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16845754779032857, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16837014009309503, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16828273239586153, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852687113641032, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684394634391768, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16835205574194329, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16826464804470978, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850878678525855, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16842137908802504, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:15, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16833397139079154, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16789326149753508, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16813740023808388, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16804999254085037, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16796258484361684, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678751771463833, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16811931588693213, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803190818969863, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1679445004924651, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678570927952316, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:0, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:15, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16810123153578038, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16801382383854685, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16792641614131332, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1413370491797351, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05470516223408217, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05461775453684865, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.000819823918879909, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:0,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0010639626594287053, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0015522401405262984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0021279253188574107, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:15, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.002372064059406207, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.00294774923773732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0035234344160684325, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004099119594399545, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004343258334948342, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004918943513279453, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005494628691610566, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14168366591014386, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14192780465069268, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1425034898290238, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:15, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1430791750073549, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14365486018568602, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1438989989262348, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14447468410456593, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14505036928289705, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14562605446122817, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14587019320177697, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14644587838010806, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14702156355843918, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09304155234884154, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:15, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09328569108939035, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09386137626772145, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09443706144605257, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09501274662438368, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09525688536493246, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09583257054326358, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09640825572159469, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0969839408999258, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09722807964047461, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09780376481880572, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:15, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09837944999713683, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09566029529262302, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959044340331718, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09648011921150292, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09705580438983402, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09763148956816514, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09787562830871394, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09845131348704504, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09902699866537616, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09960268384370727, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:15, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09984682258425606, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10042250776258718, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10099819294091829, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10139292879088287, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10163706753143167, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10221275270976278, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1027884378880939, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10336412306642499, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10360826180697379, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10418394698530489, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:15, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10475963216363601, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10533531734196713, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10557945608251593, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10615514126084705, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10673082643917817, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13452367173460997, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13476781047515876, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13534349565348988, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13591918083182097, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1364948660101521, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:15, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13673900475070092, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.137314689929032, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13789037510736313, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13846606028569422, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13871019902624304, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13928588420457413, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13986156938290525, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05718874645892424, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05743288519947303, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05734547750223951, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:15, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057258069805005994, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057170662107772476, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057414800848321276, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05732739315108775, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05723998545385423, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05715257775662072, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05739671649716951, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05730930879993599, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05722190110270247, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16887650192534442, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:15, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16912064066589322, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16903323296865969, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16894582527142615, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16885841757419265, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16910255631474144, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1690151486175079, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1689277409202744, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16884033322304087, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16908447196358967, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16899706426635616, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:0, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:15, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16890965656912263, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16846894667586662, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.168713085416415, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862567771918l5, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16853827002194796, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16845086232471446, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16869500106526325, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860759336802972, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852018567079624, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16843277797356l7, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:15, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16867691671411147, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16858950901687797, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850210131964447, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:74, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05536825510964679, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05528084741241328, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05519343971517976, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0013955090972110214, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.001639647837759818, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0015522401405262984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.002040517621623891, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:15, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0022846563621726876, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0028603415405038003, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0034360267188349126, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004011711897166026, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004255850637714821, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004831535816045934, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005407220994377047, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1415962582129104, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14184039695345918, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14241608213179027, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:15, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1429917673101214, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1435674524884525, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14381159122900128, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14438727640733243, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14496296158566352, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14553864676399464, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14578278550454343, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14635847068287455, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14693415586120567, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09295414465160802, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:15, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09319828339215681, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09377396857048793, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09434965374881904, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09492533892715016, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09516947766769895, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09574516284603006, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09632084802436117, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09689653320269229, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09714067194324108, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977163571215722, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:15, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09829204229990332, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09557288759538948, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09581702633593829, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0963927115142694, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09696839669260052, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09754408187093162, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09778822061148042, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09836390578981154, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09893959096814263, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09951527614647375, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:15, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09975941488702256, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10033510006535366, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10091078524368477, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10130552109364935, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10154965983419814, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10212534501252926, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10270103019086037, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10327671536919147, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10352085410974027, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10409653928807137, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:15, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1046722244664025, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1052479096447336, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10549204838528241, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10606773356361353, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10664341874194463, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13443626403737644, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13468040277792526, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13525608795625635, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13583177313458747, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1364074583129186, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:15, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13665159705346738, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13722728223179848, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13780296741012962, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1383786525884607, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1386227913290095, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13919847650734063, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13977416168567172, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057764431637255344, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058008570377804144, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05792116268057063, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:15, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057833754983337114, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057746347286103596, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05799048602665239, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05790307832941887, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05781567063218535, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05772826293495183, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05797240167550063, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05788499397826711, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05779758628103358, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16945218710367552, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:0, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:15, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1696963258442243, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16960891814699078, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16952151044975727, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16943410275252377, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16967824149307253, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16959083379583903, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1695034260986055, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.169416018401372, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1696601571419208, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16957274944468725, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:15, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16948534174745375, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16904463185419732, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16928877059474612, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16920136289751259, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16911395520027908, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16902654750304555, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16927068624359434, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16918327854636084, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16909587084912733, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1690084631518938, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:15, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1692526018924426, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1691651941952091, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05569980154742911, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05594394028797791, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055856532590744384, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05576912489351088, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.001971194275542134, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0021279253188574107, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.002040517621623891, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:15, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0021972486649391677, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0027729338432702804, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003348619021601393, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0039243041999325054, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004416842940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004744128118812414, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005319813297143527, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14150885051567685, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14175298925622565, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1423286744345568, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:15, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1429043596128879, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14348004479121898, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14372418353176777, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1442998687100989, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14487555388843, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1454512390667611, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14569537780730993, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14627106298564102, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14684674816397214, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0928667369543745, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:15, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0931108756949233, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0936865608732544, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09426224605158552, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09483793122991664, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09508206997046544, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09565775514879654, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09623344032712766, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09680912550545875, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09705326424600756, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09762894942433868, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:15, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09820463460266979, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09548547989815596, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09572961863870476, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09630530381703588, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.096880988995367, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0974566741736981, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977008129142469, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.098276498092578, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09885218327090911, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09942786844924023, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:15, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09967200718978902, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10024769236812014, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10082337754645126, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10121811339641583, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10146225213696462, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10203793731529573, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10261362249362685, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10318930767195796, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10343344641250676, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10400913159083787, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:15, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10458481676916898, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1051605019475001, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10540464068804889, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10598032586638, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10655601104471112, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13434885634014293, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13459299508069172, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13516868025902282, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13574436543735396, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13632005061568506, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:15, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13656418935623385, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13713987453456497, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1377155597128961, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13829124489122718, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.138535383631776, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1391110688101071, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1396867539884382, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058340116815586464, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058584255556135265, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05849684785890174, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:0, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:15, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05840944016166822, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0583220324644347, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0585661712049835, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05847876350774999, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05839135581051646, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05830394811328294, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058548086853831734, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058460679156598215, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058373271459364696, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17002787228200664, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:15, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17027201102255543, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17018460332532193, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1700971956280884, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000978793085486, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17025392667140368, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17016651897417015, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17007911127693662, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699917035797031, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1702358423202519, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17014843462301837, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:15, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17006102692578487, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16962031703252844, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698644557730772, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1697770480758437, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1696896403786102, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16960223268137667, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16984637142192546, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16975896372469196, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:13}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16967155602745843, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16958414833022492, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:15, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982828707077371, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05603134798521143, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05594394028797791, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0561880790285267, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056100671331293185, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056013263634059667, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0024594717566397265, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.002372064059406207, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0022846563621726876, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:15, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0021972486649391677, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0030170725838190773, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003592757762150189, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004168442940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004412581681030098, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0049882668593612105, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005563952037692324, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14175298925622565, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14199712799677447, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14257281317510556, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:15, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14314849835343665, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14372418353176777, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14396832227231657, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1445440074506477, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14511969262897878, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14569537780730993, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14593951654785872, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14651520172618981, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14709088690452093, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0931108756949233, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:15, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09335501443547209, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09393069961380321, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09450638479213432, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09508206997046544, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09532620871101423, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09590189388934535, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09647757906767646, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09705326424600756, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09729740298655637, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09787308816488748, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:15, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09844877334321858, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09572961863870476, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09597375737925355, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09654944255758467, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09712512773591578, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977008129142469, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09794495165479569, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09852063683312681, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09909632201145793, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09967200718978902, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:15, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09991614593033782, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10049183110866894, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10106751628700004, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10146225213696462, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10170639087751342, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10228207605584454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10285776123417564, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10343344641250676, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10367758515305556, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10425327033138665, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:15, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10482895550971777, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10540464068804889, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10564877942859768, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1062244646069288, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10680014978525991, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13459299508069172, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13483713382124052, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13541281899957164, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13598850417790273, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13656418935623385, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:0, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:15, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13680832809678267, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13738401327511376, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13795969845344488, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.138535383631776, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1387795223723248, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1393552075506559, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.139930892728987, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058584255556135265, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05882839429668405, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05874098659945054, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:15, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05865357890221702, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0585661712049835, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058810309945532296, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05872290224829878, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05863549455106526, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058548086853831734, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058792225594380534, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058704817897147016, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0586174101999135, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17027201102255543, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:15, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17051614976310422, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17042874206587072, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1703413343686372, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17025392667140368, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17049806541195248, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17041065771471894, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17032325001748544, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1702358423202519, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1704799810608007, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1703925733635672, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:15, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17030516566633366, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698644557730772, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17010859451362603, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1700211868163925, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16993377911915897, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16984637142192546, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17009051016247426, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000310246524072, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16991569476800722, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982828707077371, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:15, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05669444086077606, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05660703316354255, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056519625466309016, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05676376420685782, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05667635650962429, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05658894881239077, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.002791018194422043, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0030351569349708393, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.00294774923773732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0028603415405038003, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:15, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0027729338432702804, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0030170725838190773, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0035053500649166692, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004081035243247782, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004325173983796579, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004900859162127691, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005476544340458803, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14166558155899214, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14190972029954094, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14248540547787206, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:15, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14306109065620318, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14363677583453427, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14388091457508304, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14445659975341416, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14503228493174528, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1456079701100764, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1458521088506252, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1464277940289563, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14700347920728743, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09302346799768976, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:15, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09326760673823857, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09384329191656969, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0944189770949008, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09499466227323192, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09523880101378071, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09581448619211182, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09639017137044294, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09696585654877406, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09720999528932285, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09778568046765396, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:15, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09836136564598506, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09564221094147124, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09588634968202005, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09646203486035115, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09703772003868226, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09761340521701337, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09785754395756217, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0984332291358933, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0990089143142244, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0995845994925555, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:15, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09982873823310431, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10040442341143542, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10098010858976653, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10137484443973109, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1016189831802799, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.102194668358611, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10277035353694212, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10334603871527324, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10359017745582202, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10416586263415314, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:0, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:15, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10474154781248425, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10531723299081536, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10556137173136416, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10613705690969528, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10671274208802639, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1345055873834582, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.134749726124007, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1353254113023381, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13590109648066923, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13647678165900035, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:15, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13672092039954914, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13729660557788023, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13787229075621138, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13844797593454247, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13869211467509127, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13926779985342239, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13984348503175348, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05915994073446637, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05940407947501517, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05931667177778166, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:15, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05922926408054813, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05914185638331461, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0593859951238634, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0592985874266299, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059211179729396365, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059123772032162854, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059367910772711655, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05928050307547812, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0591930953782446, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17084769620088652, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:15, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17109183494143534, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1710044272442018, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1709170195469683, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17082961184973477, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17107375059028357, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17098634289305006, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17089893519581653, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.170811527498583, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17105566623913182, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1709682585418983, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:15, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17088085084466476, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17044014095140836, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17068427969195715, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17059687199472362, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17050946429749012, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17042205660025658, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17066619534080538, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17057878764357184, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17049137994633834, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0573575337363407, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:15, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05727012603910718, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057182718341873655, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057095310644640136, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05733944938518892, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05725204168795541, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05716463399072189, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0033667033727531556, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003610842113301952, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0035234344160684325, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0034360267188349126, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:15, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003348619021601393, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003592757762150189, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0035053500649166692, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003993627546014262, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004237766286563059, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004813451464894172, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
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MATCH (a {`key`:18}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005389136643225284, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1415781738617586, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1418223126023074, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14239799778063855, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:15, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14297368295896964, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14354936813730076, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14379350687784953, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14436919205618068, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14494487723451174, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14552056241284286, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14576470115339168, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14634038633172278, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1469160715100539, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09293606030045624, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:15, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09318019904100505, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09375588421933616, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09433156939766728, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0949072545759984, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09515139331654718, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0957270784948783, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09630276367320942, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09687844885154052, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
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MATCH (a {`key`:18}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09712258759208933, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09769827277042042, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:15, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09827395794875154, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09555480324423772, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09579894198478651, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09637462716311763, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09695031234144874, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09752599751977986, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09777013626032866, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09834582143865976, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09892150661699088, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
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MATCH (a {`key`:18}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.099497191795322, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:0, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:15, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0997413305358708, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1003170157142019, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10089270089253301, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10128743674249759, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10153157548304637, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10210726066137749, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1026829458397086, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10325863101803971, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10350276975858852, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10407845493691963, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:15, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10465414011525073, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10522982529358185, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10547396403413065, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10604964921246175, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10662533439079287, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1344181796862247, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13466231842677348, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13523800360510457, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13581368878343572, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1363893739617668, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:15, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1366335127023156, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13720919788064673, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13778488305897785, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13836056823730894, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13860470697785776, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13918039215618888, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13975607733451997, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05973562591279749, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05997976465334628, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059892356956112766, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:15, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05980494925887925, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059717541561645715, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059961680302194516, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059874272604961004, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05978686490772747, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05969945721049397, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05994359595104276, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05985618825380924, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059768780556575724, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17142338137921764, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:15, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17166752011976644, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17158011242253293, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17149270472529943, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1714052970280659, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1716494357686147, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17156202807138118, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17147462037414765, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17138721267691415, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17163135141746294, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1715439437202294, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:15, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17145653602299959, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17101582612973945, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17125996487028824, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17117255717305474, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1710851494758212, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17099774177858768, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17124188051913647, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17115447282190296, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057689080174123004, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05793321891467182, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:15, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05784581121743829, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05775840352020477, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05767099582297124, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05791513456352004, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057827726866286525, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057740319169053, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003942388551084268, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004186527291633065, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004099119594399545, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004011711897166026, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:15, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0039243041999325054, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004168442940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004081035243247782, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.003993627546014262, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004150358589329539, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004726043767660652, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005301728945991764, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14149076616452508, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417349049050739, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14231059008340502, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:16, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1428862752617361, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14346196044006723, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14370609918061603, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14428178435894715, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14485746953727824, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14543315471560936, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14567729345615815, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14625297863448924, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1468286638128204, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09284865260322274, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:16, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09309279134377152, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09366847652210264, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09424416170043376, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09481984687876487, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09506398561931367, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09563967079764478, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0962153559759759, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.096791041154307, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0970351798948558, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09761086507318692, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:0, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:16, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09818655025151803, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0954673955470042, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095711534287553, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0962872194658841, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09686290464421522, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09743858982254634, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09768272856309512, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09825841374142624, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09883409891975736, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09940978409808847, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:16, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09965392283863728, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10022960801696838, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1008052931952995, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10120002904526405, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10144416778581285, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10201985296414397, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10259553814247507, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1031712233208062, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.103415362061355, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10399104723968611, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:16, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10456673241801723, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10514241759634833, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10538655633689713, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10596224151522825, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10653792669355935, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13433077198899115, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13457491072953995, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13515059590787107, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1357262810862022, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13630196626453328, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:16, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1365461050050821, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1371217901834132, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13769747536174431, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13827316054007543, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13851729928062423, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13909298445895535, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13966866963728644, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0603113110911286, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0605554498316774, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06046804213444388, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:16, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060380634437210354, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060293226739976835, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060537365480525636, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06044995778329211, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06036255008605859, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06027514238882507, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06051928112937387, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06043187343214036, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06034446573490683, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17199906655754874, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:16, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17224320529809756, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17215579760086402, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1720683899036305, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17198098220639702, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1722251209469458, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17213771324971228, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17205030555247877, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17196289785524524, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17220703659579403, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17211962889856053, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:16, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.172032221201327, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17159151130807057, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1718356500486194, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17174824235138586, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17166083465415233, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1715734269569188, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17181756569746762, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058020626611905324, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05793321891467182, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058177357655220606, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:16, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05808994995798709, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05800254226075356, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05791513456352004, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058159273304068844, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05807186560683532, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0579844579096018, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004186527291633065, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004430666032181861, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004343258334948342, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004255850637714821, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:16, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004168442940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004412581681030098, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004325173983796579, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004237766286563059, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004150358589329539, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0049701825082094486, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005545867686540561, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:0,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417349049050739, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1419790436456227, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1425547288239538, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:16, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1431304140022849, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14370609918061603, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14395023792116482, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14452592309949594, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14510160827782706, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14567729345615815, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14592143219670695, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14649711737503807, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1470728025533692, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09309279134377152, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:0, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:16, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09333693008432034, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09391261526265145, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09448830044098255, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09506398561931367, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09530812435986247, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09588380953819357, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0964594947165247, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0970351798948558, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09727931863540461, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09785500381373571, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:16, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09843068899206682, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095711534287553, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09595567302810179, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0965313582064329, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09710704338476402, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09768272856309512, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09792686730364393, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09850255248197505, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09907823766030616, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09965392283863728, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:16, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09989806157918606, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10047374675751718, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10104943193584828, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10144416778581285, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10168830652636165, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10226399170469276, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10283967688302388, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.103415362061355, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10365950080190378, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1042351859802349, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:16, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10481087115856602, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10538655633689713, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10563069507744594, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10620638025577704, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10678206543410815, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13457491072953995, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13481904947008877, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13539473464841986, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13597041982675098, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1365461050050821, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:16, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1367902437456309, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.137365928923962, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13794161410229314, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13851729928062423, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13876143802117302, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13933712319950414, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13991280837783526, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:4,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0605554498316774, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0607995885722262, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06071218087499267, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:16, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060624773177759154, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060537365480525636, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06078150422107443, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06069409652384091, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06060668882660739, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06051928112937387, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060763419869922675, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06067601217268915, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06058860447545563, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:1,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17224320529809756, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:16, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17248734403864635, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17239993634141285, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1723125286441793, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1722251209469458, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1724692596874946, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1723818519902611, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17229444429302757, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17220703659579403, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17245117533634285, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17236376763910932, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:16, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1722763599418758, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1718356500486194, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17207978878916816, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17199238109193465, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17190497339470112, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17181756569746762, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05859631179023644, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05850890409300291, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05875304283355171, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05866563513631819, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:16, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05857822743908468, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05849081974185116, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05873495848239995, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05864755078516643, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05856014308793292, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004762212469964177, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005006351210512973, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004918943513279453, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004831535816045934, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004744128118812414, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:16, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0049882668593612105, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004900859162127691, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004813451464894172, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.004726043767660652, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0049701825082094486, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005458459989307041, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:0,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14164749720784037, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14189163594838916, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14246732112672028, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1430430063050514, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:0, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:16, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14361869148338252, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1438628302239313, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14443851540226244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14501420058059353, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14558988575892465, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14583402449947344, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14640970967780453, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14698539485613565, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09300538364653801, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09324952238708682, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:16, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09382520756541791, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09440089274374903, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09497657792208015, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09522071666262893, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09579640184096005, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09637208701929116, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09694777219762228, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09719191093817109, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977675961165022, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09834328129483332, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:16, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09562412659031948, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09586826533086829, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09644395050919939, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0970196356875305, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09759532086586162, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09783945960641041, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09841514478474152, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09899082996307264, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09956651514140374, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09981065388195255, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:16, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10038633906028366, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10096202423861476, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10135676008857934, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10160089882912814, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10217658400745924, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10275226918579036, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10332795436412147, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10357209310467026, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10414777828300138, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10472346346133249, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:16, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10529914863966361, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554328738021242, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10611897255854352, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10669465773687464, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13448750303230644, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13473164177285524, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13530732695118633, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13588301212951748, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13645869730784857, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13670283604839736, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:16, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13727852122672848, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1378542064050596, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1384298915833907, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1386740303239395, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13924971550227064, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:4,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13982540068060173, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:4,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06113113501000852, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061375273750557305, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061287866053323786, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061200458356090275, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:16, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06111305065885674, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06135718939940554, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06126978170217203, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0611823740049385, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06109496630770498, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061339105048253774, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:21}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061251697351020255, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:1,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06116428965378674, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:1,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17281889047642868, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17306302921697744, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:16, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17297562151974394, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1728882138225104, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1728008061252769, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1730449448658257, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1729575371685922, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17287012947135866, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17278272177412515, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17302686051467395, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17293945281744041, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1728520451202069, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:16, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17241133522695049, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17265547396749928, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17256806627026575, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17248065857303224, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058683719487469956, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05917199696856754, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05908458927133403, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05932872801188282, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0592413203146493, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05915391261741579, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:16, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05906650492018227, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05931064366073107, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05922323596349755, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059135828266264026, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:2,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.00533789764829529, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005582036388844086, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005494628691610566, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005407220994377047, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005319813297143527, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005563952037692324, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:0, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:16, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005476544340458803, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005389136643225284, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005301728945991764, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005545867686540561, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:0,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.005458459989307041, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:0,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14156008951060683, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14180422825115563, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14237991342948678, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14295559860781787, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.143531283786149, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:16, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14377542252669778, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1443511077050289, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14492679288336, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14550247806169114, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14574661680223994, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14632230198057103, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14689798715890212, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:5,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0929179759493045, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09316211468985329, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09373779986818441, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:16, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09431348504651552, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09488917022484662, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09513330896539543, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09570899414372654, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09628467932205764, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09686036450038876, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09710450324093756, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09768018841926868, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09825587359759978, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09553671889308596, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:16, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09578085763363477, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09635654281196587, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09693222799029699, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0975079131686281, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09775205190917688, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.098327737087508, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0989034222658391, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09947910744417023, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09972324618471902, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10029893136305014, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:16, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10087461654138126, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10126935239134582, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151349113189462, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10208917631022574, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10266486148855683, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10324054666688795, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10348468540743676, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10406037058576786, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10463605576409898, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10521174094243009, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:16, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545587968297888, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10603156486131, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10660725003964111, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1344000953350729, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1346442340756217, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13521991925395282, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13579560443228395, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13637128961061504, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13661542835116386, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13719111352949495, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:16, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13776679870782607, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13834248388615716, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13858662262670599, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:4,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1391623078050371, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:4,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1397379929833682, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:4,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061706820188339624, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06195095892888841, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0618635512316549, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06177614353442138, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06168873583718786, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:16, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06193287457773666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06184546688050314, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06175805918326962, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06167065148603609, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061914790226584894, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:1,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061827382529351375, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:1,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06173997483211786, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:1,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17339457565475977, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1736387143953086, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17355130669807506, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:16, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17346389900084153, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17337649130360805, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17362063004415682, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17353322234692328, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1734458146496898, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17335840695245627, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17360254569300507, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17351513799577156, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17342773029853803, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1729870204052816, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:16, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17323115914583037, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17314375144859687, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059346812363034594, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059259404665801076, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1439479309441336, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1438605232469001, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1441046619874489, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14401725429021536, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14392984659298186, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14384243889574833, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:0, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:16, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14408657763629712, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14399916993906361, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14391176224183008, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1415269348668286, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417710736073774, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14168366591014386, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1415962582129104, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14150885051567685, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14175298925622565, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14166558155899214, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:16, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1415781738617586, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14149076616452508, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417349049050739, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14164749720784037, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14156008951060683, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.00610346851371991, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:0,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.006679153692051023, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:0,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007254838870382137, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007830524048713247, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:16, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008650347967593158, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009226033145924269, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009801718324255382, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01062154224313529, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011197227421466404, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06548343552053273, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:2,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06572757426108153, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:2,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06630325943941265, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:2,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06687894461774375, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:16, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06745462979607487, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06769876853662367, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06827445371495477, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0688501388932859, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.069425824071617, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0696699628121658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0702456479904969, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07082133316882802, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1221236233278535, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:4,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12236776206840229, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:4,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:16, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1229434472467334, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:4,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12351913242506453, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12409481760339564, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12433895634394442, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12491464152227555, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12549032670060667, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12606601187893776, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12631015061948658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12688583579781768, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1274615209761488, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:16, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0842061818122977, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0844503205528465, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0850260057311776, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08560169090950873, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08617737608783983, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08642151482838863, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08699720000671975, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08757288518505087, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08814857036338197, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08839270910393077, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:16, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08896839428226187, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08954407946059299, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16444051485760602, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:5,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16526033877648597, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16583602395481706, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16641170913314818, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1672315330520281, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16780721823035918, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:16, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16838290340869033, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16920272732757022, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16977841250590134, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19789585740687293, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:6,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19813999614742173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:6,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19805258845018822, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:6,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1979651807529547, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19787777305572118, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19812191179626995, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:16, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19803450409903647, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19794709640180294, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19785968870456944, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19810382744511823, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1980164197478847, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19792901205065117, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1457779546343591, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:2,
`saturationRatio`:0.8727272727272727, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460220933749079, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:2,
`saturationRatio`:0.8818181818181818, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14593468567767437, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:2,
`saturationRatio`:0.8909090909090909, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14584727798044086, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:16, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14575987028320733, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14600400902375613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14591660132652262, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1458291936292891, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14574178593205558, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1459859246726044, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14589851697537087, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14581110927813734, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1556032001594568, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15584733890000563, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:0, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:16, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14387860759805185, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:5,
`saturationRatio`:0.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14412274633860064, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:5,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14403533864136714, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:5,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:84, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1441920696846824, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1441046619874489, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1443488007279977, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14426139303076416, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14417398533353062, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14408657763629712, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1443307163768459, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:16, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14424330867961238, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14415590098237888, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417710736073774, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1420152123479262, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14192780465069268, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14184039695345918, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14175298925622565, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14199712799677447, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14190972029954094, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1418223126023074, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:16, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417349049050739, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1419790436456227, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14189163594838916, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14180422825115563, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.00610346851371991, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:0,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.006923292432599819, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007498977610930932, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008318801529810842, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008894486708141953, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:16, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009470171886473066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010289995805352975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010865680983684088, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011441366162015201, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06572757426108153, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:2,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06597171300163034, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:2,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06654739817996144, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06712308335829256, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06769876853662367, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:16, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06794290727717245, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06851859245550357, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06909427763383467, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0696699628121658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06991410155271459, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07048978673104571, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07106547190937683, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12236776206840229, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:4,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12261190080895108, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:4,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1231875859872822, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:16, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12376327116561331, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12433895634394442, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12458309508449324, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12515178026282434, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12573446544115544, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12631015061948658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12655428936003538, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12712997453836647, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1277056597166976, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0844503205528465, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:16, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0846944592933953, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08527014447172641, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08584582965005752, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08642151482838863, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08666565356893742, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08724133874726854, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08781702392559965, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08839270910393077, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08863684784447956, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08921253302281068, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:16, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08978821820114179, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16492879233870364, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16550447751703473, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16608016269536588, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16689998661424577, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16747567179257689, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.168051356970908, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
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MATCH (a {`key`:250}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:16, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1688711808897879, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16944686606811904, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17002255124645013, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19813999614742173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:6,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19838413488797055, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:6,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19829672719073702, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19820931949350348, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19812191179626995, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19836605053681877, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19827864283958524, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:16, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1981912351423517, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19810382744511823, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19834796618566702, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1982605584884335, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1981731507912, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460220933749079, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:2,
`saturationRatio`:0.8818181818181818, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1462662321154567, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:2,
`saturationRatio`:0.8909090909090909, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1461788244182232, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14609141672098966, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14600400902375613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:0, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:16, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14624814776430495, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14616074006707144, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460733323698379, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1459859246726044, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1462300634131532, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14614265571591967, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14605524801868616, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15584733890000563, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.103640245916032, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14412274633860064, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:5,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:16, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14436688507914947, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:5,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14427947738191593, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:85, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14476775486301355, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14468034716578002, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1449244859063288, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14483707820909528, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14474967051186177, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14466226281462824, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14490640155517703, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14481899385794353, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:16, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14473158616071, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14234675878570852, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1425908975262573, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1425034898290238, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14241608213179027, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1423286744345568, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14257281317510556, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14248540547787206, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14239799778063855, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14231059008340502, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:16, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1425547288239538, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14246732112672028, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14237991342948678, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.006679153692051023, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:0,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.006923292432599819, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007411569913697412, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007987255092028524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008231393832577321, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008807079010908435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009382764189239546, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:16, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009958449367570659, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010202588108119456, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010778273286450568, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011353958464781681, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06564016656384801, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:2,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0658843053043968, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06645999048272792, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06703567566105903, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06761136083939015, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06785549957993894, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:16, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06843118475827005, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06900686993660116, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06958255511493226, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06982669385548107, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07040237903381219, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0709780642121433, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12228035437116876, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:4,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12252449311171758, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12310017829004868, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12367586346837979, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:16, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12425154864671091, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1244956873872597, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1250713725655908, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12564705774392193, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12622274292225305, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12646688166280184, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12704256684113294, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12761825201946408, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08436291285561297, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08460705159616178, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:16, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0851827367744929, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.085758421952824, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08633410713115512, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0865782458717039, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08715393105003502, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08772961622836613, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08830530140669723, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08854944014724604, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08912512532557715, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08970081050390827, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:16, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1645972459009213, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1648413846414701, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16541706981980123, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16599275499813235, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16656844017646344, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16681257891701223, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16738826409534338, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16796394927367447, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1685396344520056, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1687837731925544, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:16, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1693594583708855, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699351435492166, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19871568132575287, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:6,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19895982006630164, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19887241236906814, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1987850046718346, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1986975969746011, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1989417357151499, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19885432801791636, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19876692032068288, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:0, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:16, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19867951262344935, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19892365136399812, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19883624366676464, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1987488359695311, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14659777855323902, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:2,
`saturationRatio`:0.8909090909090909, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14684191729378782, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1467545095965543, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14666710189932078, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14657969420208727, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14682383294263607, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:16, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14673642524540254, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14664901754816903, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1465616098509355, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1468057485914843, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1467183408942508, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14663093319701728, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10430333879159663, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10421593109436313, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1446984315169318, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:5,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14494257025748059, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:16, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14485516256024705, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:86, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14534344004134464, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1452560323441111, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1455001710846599, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1454127633874264, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14532535569019286, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14523794799295933, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14548208673350815, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14539467903627462, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1453072713390411, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:16, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14292244396403964, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14316658270458843, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1430791750073549, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1429917673101214, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1429043596128879, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14314849835343665, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14306109065620318, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14297368295896964, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1428862752617361, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1431304140022849, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:16, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1430430063050514, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14295559860781787, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007254838870382137, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007498977610930932, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007411569913697412, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007899847394795005, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008143986135343801, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008719671313674914, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009295356492006027, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009871041670337139, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:16, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010115180410885936, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01069086558921705, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01126655076754816, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06555275886661449, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06579689760716328, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06637258278549439, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06694826796382551, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06752395314215663, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06776809188270543, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06834377706103655, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:16, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06891946223936764, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06949514741769876, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06973928615824755, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07031497133657866, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07089065651490978, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12219294667393528, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12243708541448407, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12301277059281515, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12358845577114629, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12416414094947739, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:16, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12440827969002617, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1249839648683573, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12555965004668843, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12613533522501952, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1263794739655683, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12695515914389943, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12753084432223055, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08427550515837946, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08451964389892826, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08509532907725936, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:16, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08567101425559048, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0862466994339216, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0864908381744704, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08706652335280152, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08764220853113261, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08821789370946373, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08846203245001251, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08903771762834363, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08961340280667475, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16450983820368778, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:16, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1647539769442366, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16532966212256772, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16590534730089881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16648103247922993, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16672517121977873, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16730085639810985, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16787654157644094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684522267547721, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16869636549532088, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16927205067365197, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:0, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:16, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698477358519831, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19929136650408397, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19953550524463276, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19944809754739923, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19936068985016572, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1992732821529322, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19951742089348098, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19943001319624748, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19934260549901395, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19925519780178044, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:16, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19949933654232926, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19941192884509573, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1993245211478622, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14717346373157014, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14741760247211894, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1473301947748854, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1472427870776519, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14715537938041837, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14739951812096716, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14731211042373366, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:16, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14722470272650012, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1471372950292666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1473814337698154, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14729402607258188, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10496643166716126, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10487902396992775, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10479161627269423, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14527411669526288, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14551825543581168, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14543084773857817, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:87, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:16, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14591912521967576, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14583171752244223, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14607585626299105, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14598844856575752, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14590104086852398, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14581363317129048, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14605777191183927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14597036421460574, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14588295651737224, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14349812914237076, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:16, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14374226788291952, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14365486018568602, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1435674524884525, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14348004479121898, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14372418353176777, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14363677583453427, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14354936813730076, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14346196044006723, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14370609918061603, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14361869148338252, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:16, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.143531283786149, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007830524048713247, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007987255092028524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.007899847394795005, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008056578438110282, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008632263616441394, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009207948794772507, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.00978363397310362, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010027772713652416, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:16, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010603457891983529, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011179143070314642, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06546535116938097, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06570948990992975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06628517508826087, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06686086026659199, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0674365454449231, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0676806841854719, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06825636936380301, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06883205454213413, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:16, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06940773972046524, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06965187846101402, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07022756363934514, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07080324881767626, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12210553897670175, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12234967771725054, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12292536289558167, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12350104807391278, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12407673325224386, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12432087199279267, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:16, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12489655717112377, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1254722423494549, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12604792752778599, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1262920662683348, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1268677514466659, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12744343662499702, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08418809746114594, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08443223620169472, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08500792138002584, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08558360655835696, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:16, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08615929173668807, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08640343047723688, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08697911565556798, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08755480083389909, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0881304860122302, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08837462475277899, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08895030993111011, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08952599510944122, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16442243050645428, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:0, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:16, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1652422544253342, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16581793960366528, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16639362478199643, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16721344870087632, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16778913387920744, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16836481905753856, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16918464297641847, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1697603281547496, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:16, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1998670516824151, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20011119042296385, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20002378272573035, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19993637502849684, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1998489673312633, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2000931060718121, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2000056983745786, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19991829067734507, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19983088298011156, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20007502172066036, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:16, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19998761402342685, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19990020632619332, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14774914890990126, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14799328765045003, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14790587995321652, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14781847225598302, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1477310645587495, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14797520329929828, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14788779560206478, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14780038790483124, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:16, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14771298020759774, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14795711894814653, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10529797810494358, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554211684549238, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545470914825886, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10536730145102534, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.145849801873594, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460939406141428, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460065329169093, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:88, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14616326396022455, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:16, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14607585626299105, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14631999500353982, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14623258730630628, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14614517960907278, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14605777191183927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14630191065238804, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14621450295515453, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:30}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14612709525792103, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14374226788291952, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14398640662346834, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:16, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1438989989262348, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14381159122900128, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14372418353176777, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14396832227231657, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14388091457508304, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14379350687784953, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14370609918061603, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14395023792116482, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1438628302239313, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14377542252669778, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:16, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008318801529810842, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008231393832577321, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008143986135343801, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008056578438110282, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008876402356990191, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009452087535321304, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010027772713652416, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010271911454201213, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010847596632532326, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:16, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011423281810863438, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06570948990992975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06595362865047856, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06652931382880968, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06710499900714079, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0676806841854719, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06792482292602069, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0685005081043518, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06907619328268293, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06965187846101402, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
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MATCH (a {`key`:250}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:16, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06989601720156284, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07047170237989395, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07104738755822505, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12234967771725054, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12259381645779935, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12316950163613045, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12374518681446155, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12432087199279267, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12456501073334146, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12514069591167257, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:16, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1257163810900037, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1262920662683348, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1265362050088836, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1271118901872147, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12768757536554584, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08443223620169472, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08467637494224353, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08525206012057465, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08582774529890576, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08640343047723688, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:0, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:16, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08664756921778566, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08722325439611678, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08779893957444787, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08837462475277899, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0886187634933278, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0891944486716589, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08977013384999002, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16491070798755186, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16548639316588298, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:16, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1660620783442141, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.166881902263094, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16745758744142514, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803327261975623, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16885309653863614, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16942878171696726, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000446689529836, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20011119042296385, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:16, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20035532916351267, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20026792146627914, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2001805137690456, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2000931060718121, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2003372448123609, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20024983711512737, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20016242941789386, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20007502172066036, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20031916046120912, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20023175276397565, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:16, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2001443450667421, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14799328765045003, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14823742639099885, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14815001869376532, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14806261099653178, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14797520329929828, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14821934203984707, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14813193434261354, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14804452664538004, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14795711894814653, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:16, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1056295245427259, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554211684549238, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10578625558604117, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10569884788880765, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10561144019157412, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460939406141428, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1463380793546916, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14625067165745806, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:89, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14673894913855567, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14665154144132214, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:16, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14689568018187094, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14680827248463743, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1467208647874039, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14663345709017037, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1468775958307192, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14679018813348566, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14670278043625212, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14431795306125067, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14456209180179946, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14447468410456593, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:16, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14438727640733243, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1442998687100989, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14454400074506477, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14445659975341416, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14436919205618068, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14428178435894715, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14452592309949594, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14443851540226244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1443511077050289, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008650347967593158, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:17, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008894486708141953, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008807079010908435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008719671313674914, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008632263616441394, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.008876402356990191, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009364679838087784, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009940365016418897, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010184503756967693, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010760188935298806, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011335874113629919, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:17, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06562208221269625, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06586622095324504, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06644190613157615, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06701759130990727, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06759327648823837, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06783741522878718, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0684131004071183, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06898878558544941, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06956447076378051, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06980860950432932, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:17, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07038429468266041, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07095997986099153, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12226227002001704, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12250640876056583, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12308209393889694, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12365777911722806, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12423346429555916, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12447760303610793, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12505328821443903, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12562897339277018, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:0, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:17, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12620465857110127, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12644879731165007, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1270244824899812, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1276001676683123, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08434482850446122, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08458896724501001, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08516465242334112, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08574033760167224, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08631602278000335, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08656016152055215, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:17, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08713584669888326, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08771153187721437, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08828721705554549, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08853135579609427, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08910704097442539, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0896827261527565, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16457916154976956, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16482330029031836, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16539898546864948, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16597467064698057, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:17, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1665503558253117, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16679449456586048, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1673701797441916, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1679458649225227, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852155010085385, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16876568884140264, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16934137401973373, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16991705919806485, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.200686875601295, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2009310143418438, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:17, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20084360664461026, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20075619894737676, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20066879125014322, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20091292999069202, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2008255222934585, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20073811459622498, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20065070689899145, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20089484563954027, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20080743794230674, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20072003024507323, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:17, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14856897282878118, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14881311156932997, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14872570387209644, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14863829617486293, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1485508884776294, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487950272181782, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487076195209447, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14862021182371116, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10629261741829053, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10620520972105701, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:17, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10611780202382348, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10636194076437229, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10627453306713876, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10618712536990524, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14666962579247392, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1469137645330227, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14682635683578918, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:90, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14731463431688677, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14722722661965326, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14747136536020206, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:17, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14738395766296852, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14729654996573502, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1472091422685015, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14745328100905028, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14736587331181675, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14727846561458324, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14489363823958176, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14513777698013056, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14505036928289705, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14496296158566352, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:17, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14487555388843, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14511969262897878, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14503228493174528, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14494487723451174, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14485746953727824, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14510160827782706, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14501420058059353, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14492679288336, `foldPressure`:0.07407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009226033145924269, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009470171886473066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:17, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009382764189239546, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009295356492006027, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009207948794772507, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009452087535321304, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009364679838087784, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009852957319185377, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010097096059734174, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010672781238065287, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011248466416396399, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06553467451546273, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:17, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06577881325601151, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06635449843434263, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06693018361267374, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06750586879100486, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06775000753155366, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06832569270988477, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06890137788821589, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06947706306654701, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06972120180709579, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07029688698542691, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:0, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:17, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07087257216375802, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1221748623227835, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1224190010633323, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12299468624166343, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12357037141999454, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12414605659832564, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12439019533887442, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12496588051720556, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12554156569553665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12611725087386774, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:17, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12636138961441656, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12693707479274766, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12751275997107878, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0842574208072277, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08450155954777648, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0850772447261076, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08565292990443871, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08622861508276983, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08647275382331862, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08704843900164974, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:17, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08762412417998086, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08819980935831197, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08844394809886076, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08901963327719187, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08959531845552297, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16449175385253603, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16473589259308485, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16531157777141597, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16588726294974707, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1664629481280782, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:17, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16670708686862695, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16728277204695807, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678584572252892, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684341424036203, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1686782811441691, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1692539663225002, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982965150083135, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2012625607796261, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2015066995201749, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20141929182294138, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:17, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20133188412570785, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20124447642847434, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20148861516902314, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2014012074717896, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20131379977455607, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20122639207732257, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20147053081787136, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20138312312063783, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20129571542340433, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14914465800711227, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:17, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14938879674766106, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14930138905042756, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14921398135319403, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1491265736559605, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14937071239650931, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14928330469927578, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10695571029385517, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10686830259662165, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10678089489938812, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1066934872021546, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:17, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1069376259427034, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10685021824546988, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10676281054823636, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.147245310970805, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1474894497113538, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1474020420141203, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:91, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1478903194952179, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14780291179798438, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14804705053853318, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14795964284129964, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:17, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14787223514406614, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1477848274468326, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1480289661873814, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1479415584901479, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14785415079291436, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14546932341791288, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1457134621584617, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14562605446122817, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14553864676399464, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1454512390667611, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:17, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14569537780730993, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1456079701100764, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14552056241284286, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14543315471560936, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14567729345615815, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14558988575892465, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14550247806169114, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009801718324255382, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009958449367570659, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:17, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009871041670337139, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009783633973103 62, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010027772713652416, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009940365016418897, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.009852957319185377, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010009688362500654, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010585373540831767, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01116105871916288, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06544726681822921, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.065691405558778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:0, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:17, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06626709073710911, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06684277591544022, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06741846109377134, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06766259983432013, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06823828501265125, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06881397019098237, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06938965536931348, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06963379410986227, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07020947928819338, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0707851644665245, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:17, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12208745462554997, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12233159336609879, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1229072785444299, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.123482963722761, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12405864890109211, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12430278764164092, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12487847281997203, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12545415799830315, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12602984317663424, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12627398191718303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:17, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12684966709551415, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12742535227384527, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08417001310999418, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08441415185054298, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08498983702887408, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08556552220720519, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0861412073855363, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0863853461260851, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08696103130441622, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08753671648274733, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:17, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08811240166107845, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08835654040162724, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08893222557995835, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08950791075828947, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16440434615530253, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16522417007418244, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16579985525251356, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16637554043084465, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:17, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16719536434972457, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16777104952805566, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16834673470638678, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1691665586252667, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16974224380359781, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2018382459579572, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20208238469850603, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2019949770012725, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20190756930403897, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:17, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20182016160680546, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20206430034735426, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20197689265012073, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20188948495288722, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2018020772556537, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20204621599620248, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20195880829896898, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20187140060173545, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1497203431854434, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1499644819259922, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:17, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14987707422875868, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14978966653152515, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14970225883429164, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14994639757484043, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10728725673163747, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10753139547218628, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10744398777495275, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10735658007771924, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
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MATCH (a {`key`:33}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10726917238048571, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1075133111210345, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:17, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.107425903423801, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10733849572656746, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1478209961491361, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14806513488968492, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14797772719245142, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:92, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1481344582357667, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14804705053853318, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14829118927908197, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14820378158184846, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
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MATCH (a {`key`:36}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14811637388461493, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:17, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1480289661873814, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14827310492793022, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1481856972306967, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14809828953346316, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1457134621584617, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14595760089901047, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14587019320177697, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14578278550454343, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14569537780730993, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14593951654785872, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:17, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1458521088506252, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14576470115339168, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14567729345615815, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14592143219670695, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14583402449947344, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14574661680223994, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010289995805352975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010202588108119456, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010115180410885936, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:0, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:17, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010027772713652416, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010271911454201213, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010184503756967693, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010097096059734174, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010009688362500654, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010829512281380564, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011405197459711676, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:0,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.065691405558778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0659355442993268, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0665112294776579, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:17, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06708691465598902, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06766259983432013, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06790673857486894, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06848242375320004, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06905810893153115, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06963379410986227, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06987793285041106, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07045361802874218, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07102930320707329, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12233159336609879, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:17, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12257573210664757, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12315141728497869, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1237271024633098, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12430278764164092, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12454692638218971, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12512261156052082, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12569829673885194, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12627398191718303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12651812065773183, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12709380583606295, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:17, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12766949101439407, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08441415185054298, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08465829059109176, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08523397576942288, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08580966094775398, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0863853461260851, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0866294848666339, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08720517004496502, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08778085522329612, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08835654040162724, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:17, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08860067914217604, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08917636432050716, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08975204949883826, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16489262363640012, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16546830881473124, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16604399399306236, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16686381791194227, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16743950309027336, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:17, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16801518826860445, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16883501218748437, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1694106973658155, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699863825441466, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20208238469850603, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2023265234390548, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2022391157418213, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2021517080445878, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20206430034735426, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:17, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20230843908790305, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20222103139066955, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20213362369343602, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20204621599620248, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2022903547367513, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20220294703951777, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20211553934228424, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:6,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1499644819259922, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15020862066654098, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15012121296930744, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:17, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15003380527207394, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14994639757484043, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1076188031694198, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10753139547218628, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10777553421273506, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10768812651550155, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10760071881826803, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1075133111210345, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10775744986158331, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10767004216434979, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:17, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10758263446711626, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14806513488968492, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14830927363023372, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1482218659330002, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:93, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487101434140978, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14862273571686427, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1488668744574131, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14877946676017956, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14869205906294602, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14860465136571252, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:17, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14884879010626131, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14876138240902778, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14867397471179428, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1462891473367928, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1465332860773416, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14644587838010806, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14635847068287455, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14627106298564102, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14651520172618981, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14642779402895 63, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:0, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:17, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14634038633172278, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14625297863448924, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14649711737503807, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14640970967780453, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14632230198057103, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01062154224313529, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010865680983684088, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010778273286450568, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01069086558921705, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010603457891983529, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:17, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010847596632532326, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010760188935298806, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010672781238065287, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010585373540831767, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.010829512281380564, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011317789762478157, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:0,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06560399786154449, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06584813660209328, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0664238217804244, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0669995069587555, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:17, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06757519213708661, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06781933087763542, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06839501605596653, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06897070123429765, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06954638641262875, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06979052515317755, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07036621033150865, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07094189550983977, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12224418566886526, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12248832440941405, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:17, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12306400958774516, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1236396947660763, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1242153799444074, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12445951868495618, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12503520386328731, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12561088904161843, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12618657421994953, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12643071296049835, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1270063981388294, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12758208331716053, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:17, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08432674415330946, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08457088289385825, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08514656807218936, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08572225325052046, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08629793842885158, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08654207716940038, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0871177623477315, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08769344752606262, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08826913270439372, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08851327144494252, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:17, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08908895662327364, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08966464180160474, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1645610771986178, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1648052159391666, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16538090111749773, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16595658629582882, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16653227147415994, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16677641021470874, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16735209539303986, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16792778057137092, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:17, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850346574970207, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16874760449025086, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16932328966858196, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698989748469131, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20265806987683713, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20290220861738592, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20281480092015242, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20272739322291888, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20263998552568535, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20288412426623417, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:17, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20279671656900064, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2027093088717671, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2026219011745336, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2028660399150824, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20277863221784886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:6,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20269122452061536, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:6,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1505401671043233, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1507843058448721, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1506968981476386, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15060949045040506, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:17, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10828189604498444, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1081944883477509, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10810708065051738, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10835121939106619, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10826381169383266, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10817640399659914, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10808899629936564, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10833313503991442, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1082457273426809, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10815831964544739, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:17, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14864082006801602, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1488849588085648, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487975511113313, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:94, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492858285924289, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14919842089519542, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14944255963574418, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14935515193851068, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14926774424127717, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14918033654404364, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14942447528459243, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:0, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:17, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14933706758735893, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492496598901254, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14686483251512392, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14710897125567268, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14702156355843918, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14693415586120567, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14684674816397214, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14709088690452093, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14700347920728743, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1469160715100539, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:17, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1468286638128204, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1470728025533692, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14698539485613565, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14689798715890212, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:5,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011197227421466404, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011441366162015201, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011353958464781681, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01126655076754816, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011179143070314642, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011423281810863438, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:17, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011335874113629919, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011248466416396399, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01116105871916288, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011405197459711676, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:0,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011317789762478157, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:0,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06551659016431097, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06576072890485976, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06633641408319087, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06691209926152199, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06748778443985311, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:17, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0677319231804019, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06830760835873301, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06888329353706411, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06945897871539523, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06970311745594401, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07027880263427513, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07085448781260625, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12215677797163173, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12240091671218052, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12297660189051166, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:17, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12355228706884276, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12412797224717387, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12437211098772268, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12494779616605378, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1255234813443849, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12609916652271602, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12634330526326482, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1269189904415959, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12749467561992703, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08423933645607594, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:17, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08448347519662472, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08505916037495584, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08563484555328696, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08621053073161805, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08645466947216686, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08703035465049798, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08760603982882909, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0881817250071602, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.088425863747709, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0890015489260401, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:17, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08957723410437123, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16447366950138428, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16471780824193308, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1652934934202642, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16586917859859532, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16644486377692644, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666890025174752, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16726468769580632, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16784037287413742, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16841605805246854, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:17, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16866019679301733, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16923588197134845, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16981156714967957, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20323375505516825, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20347789379571704, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2033904860984835, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20330307840125, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2032156707040165, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20345980944456526, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20337240174733176, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:17, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20328499405009826, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20319758635286472, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20344172509341352, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:6,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20335431739618, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:6,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20326690969894648, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:6,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15111585228265442, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15135999102320322, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15127258332596968, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10894498892054905, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10885758122331554, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:17, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10877017352608202, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10868276582884852, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1089269045693973, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10883949687216377, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10875208917493026, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10866468147769674, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10890882021824552, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10882141252101202, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1087340048237785, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14921650524634714, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:0, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:17, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14946064398689596, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14937323628966243, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:95, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09385357117225686, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09376616347502334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09401030221557213, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09392289451833861, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0938354868211051, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09374807912387156, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09399221786442037, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09390481016718685, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:17, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09381740246995333, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09288482130552626, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09312896004607506, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09304155234884154, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09295414465160802, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0928667369543745, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0931108756949233, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09302346799768976, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09293606030045624, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09284865260322274, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:17, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09309279134377152, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09300538364653801, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0929179759493045, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06548343552053273, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:2,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06572757426108153, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:2,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06564016656384801, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:2,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06555275886661449, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06546535116938097, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06570948990992975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06562208221269625, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:17, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06553467451546273, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06544726681822921, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.065691405558778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06560399786154449, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06551659016431097, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011962798286891025, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:0,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.012538483465222138, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:0,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01311416864355325, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013689853821884363, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:17, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014509677740764273, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015085362919095386, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015661048097426494, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016480872016306405, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017056557194637518, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.068358847353663, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0686029860942118, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0691786712725429, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06975435645087402, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:17, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07033004162920514, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07114986554808504, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07172555072641616, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07230123590474727, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07312105982362717, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07369674500195829, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07790689459642154, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07815103333697035, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:17, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07872671851530146, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07930240369363256, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07987808887196368, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08012222761251249, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08069791279084358, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0812735979691747, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08184928314750582, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08209342188805462, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08266910706638572, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08324479224471683, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:17, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1083783192056476, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:3,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10862245794619638, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10919814312452751, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10977382830285862, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11034951348118972, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11059365222173854, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11116933740006965, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11174502257840076, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11232070775673186, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11256484649728067, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:17, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11314053167561178, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11371621685394288, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492537438455706, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1494978825861194, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14941047488888587, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14932306719165236, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14923565949441883, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14947979823496763, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14939239053773412, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1493049828405006, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:17, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14921757514326706, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14946171388381588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14937430618658235, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492868984893488, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17145053461809154, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17169467335864036, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09678646703348964, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09703060577403844, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0969431980768049, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0968557903795714, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:0, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:17, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09676838268233788, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09701252142288666, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09692511372565316, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09683770602841964, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0967502983311861, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09699443707173491, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0969070293745014, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09681962167726788, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:96, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0937842478261751, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09402838656672391, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:17, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09394097886949039, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:96, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09409770991280567, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09401030221557213, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09425444095612093, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09416703325888741, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09407962556165389, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09399221786442037, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09423635660496917, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09414894890773565, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09406154121050214, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:17, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09312896004607506, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
```

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MATCH (a {`key`:41}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09337309878662385, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09328569108939035, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09319828339215681, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0931108756949233, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09335501443547209, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09326760673823857, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09318019904100505, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09309279134377152, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09333693008432034, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:17, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09324952238708682, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09316211468985329, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
```

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MATCH (a {`key`:41}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06572757426108153, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:2,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06597171300163034, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:2,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0658843053043968, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06579689760716328, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06570948990992975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06595362865047856, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06586622095324504, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06577881325601151, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:17, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.065691405558778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0659355442993268, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06584813660209328, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06576072890485976, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.011962798286891025, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:0,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.012782622205770934, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013358307384102047, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014178131302981956, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014753816481313069, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:17, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015329501659644182, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01614932557852409, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016725010756855202, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017300695935186315, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0686029860942118, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06884712483476059, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06942281001309171, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06999849519142283, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:17, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07081831911030273, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07139400428863385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07196968946696494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07278951338584486, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07336519856417598, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07394088374250708, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07815103333697035, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07839517207751914, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07897085725585025, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:17, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07954654243418137, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08012222761251249, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08036636635306128, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08094205153139239, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0815177367097235, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08209342188805462, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:41}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08233756062860341, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08291324580693452, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08348893098526564, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10862245794619638, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:17, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1088665966867452, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1094422818650763, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11001796704340741, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11059365222173854, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11083779096228732, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11141347614061843, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11198916131894956, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:41}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11256484649728067, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11280898523782946, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11338467041616057, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:17, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11396035559449168, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1494978825861194, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1497420213266682, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1496546136294347, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14956720593220116, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14947979823496763, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14972393697551645, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14963652927828291, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:41}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14954912158104938, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14946171388381588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:0, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:17, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14970585262436467, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14961844492713114, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1495310372298976, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17169467335864036, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03613856171827246, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09703060577403844, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09727474451458723, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09718733681735373, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:41}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09709992912012019, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09701252142288666, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:17, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09725666016343548, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09716925246620195, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09708184476896842, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09699443707173491, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0972385758122837, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09715116811505019, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09706376041781666, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09402838656672391, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0942725253072727, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:41}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09418511761003917, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:97, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:17, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09467339509113677, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09458598739390325, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09483012613445205, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09474271843721851, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09465531073998501, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09456790304275149, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09481204178330027, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09472463408606677, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09463722638883325, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09370464522440618, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:17, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09394878396495497, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09386137626772145, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09377396857048793, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0936865608732544, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09393069961380321, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09384329191656969, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09375588421933616, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09366847652210264, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09391261526265145, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09382520756541791, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:17, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09373779986818441, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06630325943941265, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:2,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06654739817996144, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06645999048272792, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06637258278549439, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06628517508826087, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06652931382880968, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06644190613157615, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06635449843434263, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06626709073710911, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:17, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0665112294776579, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0664238217804244, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06633641408319087, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.012538483465222138, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:0,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.012782622205770934, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013270899686868527, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013846584865199638, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014090723605748437, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014666408784079547, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015242093962410662, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:17, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015817779140741 77, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016061917881290574, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016637603059621683, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017213288237952797, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0691786712725429, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06942281001309171, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06933540231585819, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0699110874941893, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07048677267252042, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07073091141306923, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:17, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07130659659140033, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07188228176973144, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07245796694806254, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07270210568861134, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07327779086694244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07385347604527356, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07806362563973683, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07830776438028561, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07888344955861673, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07945913473694785, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:17, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08003481991527896, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08027895865582776, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08085464383415888, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08143032901248998, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0820060141908211, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08225015293136988, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.082825838109701, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0834015232880321, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10853505024896287, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10877918898951167, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:17, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10935487416784277, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1099305593461739, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11050624452450501, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1107503832650538, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11132606844338493, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11190175362171603, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11247743880004714, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11272157754059596, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11329726271892707, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11387294789725817, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:0, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:17, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1500735677644505, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15031770650499932, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15023029880776578, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15014289111053225, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15005548341329875, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15029962215384754, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.150212214456614, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1501248067593805, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.150037399062147, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15028153780269576, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:17, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15019413010546226, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15010672240822875, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03680165459383709, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.036714246896603575, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09760629095236953, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09785042969291835, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09776302199568483, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0976756142984513, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0975882066012178, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09783234534176659, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:17, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09774493764453306, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09765752994729955, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09757012225006602, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09781426099061481, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09772685329338128, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09763944559614779, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09460407174505502, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0948482104856038, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09476080278837029, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:98, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09524908026946788, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:17, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09516167257223437, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09540581131278315, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09531840361554964, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09523099591831613, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0951435882210826, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09538772696163139, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09530031926439787, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09521291156716435, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09428033040273728, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09452446914328608, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:17, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09443706144605257, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09434965374881904, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09426224605158552, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09450638479213432, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0944189770949008, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09433156939766728, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09424416170043376, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09448830044098255, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09440089274374903, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09431348504651552, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:17, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06687894461774375, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:2,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06712308335829256, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06703567566105903, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06694826796382551, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06686086026659199, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06710499900714079, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06701759130990727, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06693018361267374, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06684277591544022, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06708691465598902, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:17, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0669995069587555, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06691209926152199, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01311416864355325, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013358307384102047, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013270899686868527, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013759177167966121, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0140033315908514917, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01457900108684603, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015154686265177141, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015730371443508254, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:17, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015974510184057048, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01655019536238816, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017125880540719274, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06975435645087402, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06999849519142283, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0699110874941893, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06982367979695578, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0703993649752869, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07064350371583569, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07121918889416681, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:17, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07179487407249792, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07237055925082904, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07261469799137782, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07319038316970893, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07376606834804005, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0779762179425033, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07822035668305209, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07879604186138321, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07937172703971432, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07994741221804544, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:17, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08019155095859425, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08076723613692535, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08134292131525646, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08191860649358756, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08216274523413636, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08273843041246746, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08331411559079858, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10844764255172935, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10869178129227816, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10926746647060927, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:0, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:17, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10984315164894037, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11041883682727148, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1106629755678203, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11123866074615141, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11181434592448251, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11239003110281363, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11263416984336243, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11320985502169353, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11378554020002464, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15064925294278164, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:17, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1508933916833304, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1508059839860969, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1507185762888634, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15063116859162987, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15087530733217866, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15078789963494516, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15070049193771162, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1506130842404781, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15085722298102688, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15076981528379338, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:17, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.037464747469401725, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0373773397721682, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03728993207493468, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09818197613070066, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09842611487124946, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09833870717401592, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09825129947678243, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0981638917795489, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0984080305200977, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09832062282286419, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:17, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09823321512563066, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09814580742839712, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09838994616894592, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09830253847171241, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09821513077447888, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09517975692338612, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09542389566393492, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0953364879667014, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:99, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095824765447799, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09573735775056548, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:17, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09598149649111427, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09589408879388076, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09580668109664724, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09571927339941372, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959634121399625, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095876004442729, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09578859674549547, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09485601558106839, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09510015432161718, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09501274662438368, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:17, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09492533892715016, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09483793122991664, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09508206997046544, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09499466227323192, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0949072545759984, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09481984687876487, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09506398561931367, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09497657792208015, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09488917022484662, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06745462979607487, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:2,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:18, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06769876853662367, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06761136083939015, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06752395314215663, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0674365454449231, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0676806841854719, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06759327648823837, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06750586879100486, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06741846109377134, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06766259983432013, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06757519213708661, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:18, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06748778443985311, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013689853821884363, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013846584865199638, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013759177167966121, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013915908211281396, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014491593389612506, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015067278567943623, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015642963746274732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015887102486823533, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:18, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016462787665154643, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01703847284348576, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07033004162920514, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07048677267252042, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0703993649752869, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07031195727805337, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07055609601860217, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07113178119693328, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0717074663752644, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:18, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0722831515535955, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0725272902941443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07310297547247542, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07367866065080653, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07788881024526978, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07813294898581859, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0787086341641497, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07928431934248081, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07986000452081192, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08010414326136071, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:0, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:18, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08067982843969183, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08125551361802294, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08183119879635406, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08207533753690285, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08265102271523396, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08322670789356508, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10836023485449585, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10860437359504463, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10918005877337573, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10975574395170684, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:18, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11033142913003798, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11057556787058677, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11115125304891788, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11172693822724901, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11230262340558012, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1125467621461289, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11312244732446, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11369813250279114, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15122493812111273, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1514690768616615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:18, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15138166916442802, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1512942614671945, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15120685376996096, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15145099251050978, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15136358481327625, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15127617711604271, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1511887694188092, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.151432908159358, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.037796293907184045, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03804043264773284, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:18, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03795302495049931, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.037865617253265794, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09875766130903177, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09900180004958056, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09891439235234706, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09882698465511353, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09873957695788001, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09898371569842881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0988963080011953, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09880890030396176, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:18, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09872149260672823, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09896563134727705, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09887822365004352, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09879081595280999, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09575544210171724, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09599958084226602, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959121731450325, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:100, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09606890418834779, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09598149649111427, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09622563523166308, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:18, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09613822753442955, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09605081983719603, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959634121399625, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0962075508805113, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09612014318327779, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09603273548604425, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09510015432161718, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09534429306216598, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09525688536493246, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09516947766769895, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:18, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09508206997046544, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09532620871101423, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09523880101378071, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09515139331654718, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09506398561931367, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09530812435986247, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09522071666262893, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09513330896539543, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06769876853662367, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:2,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06794290727717245, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:18, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06785549957993894, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06776809188270543, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0676806841854719, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06792482292602069, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06783741522878718, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06775000753155366, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06766259983432013, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06790673857486894, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06781933087763542, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0677319231804019, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:18, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014178131302981956, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014090723605748437, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014003315908514917, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.013915908211281396, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014735732130161307, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015311417308492416, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015887102486823533, `foldPressure`:0.037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016131241227372327, `foldPressure`:0.037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01670692640570344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:18, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017282611584034553, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07081831911030273, `foldPressure`:0.037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07073091141306923, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07064350371583569, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07055609601860217, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07080023475915097, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07137591993748209, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0719516051158132, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0725272902941443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:0, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:18, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07277142903469311, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07334711421302421, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07392279939135532, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07813294898581859, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07837708772636737, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07895277290469849, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07952845808302961, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08010414326136071, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
```

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MATCH (a {`key`:45}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08034828200190952, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08092396718024063, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:18, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08149965235857175, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08207533753690285, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08231947627745163, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08289516145578275, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08347084663411386, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10860437359504463, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10884851233559345, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10942419751392456, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10999988269225566, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
```

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MATCH (a {`key`:45}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11057556787058677, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:18, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11081970661113555, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11139539178946668, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11197107696779779, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1125467621461289, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11279090088667769, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11336658606500882, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11394227124333993, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1514690768616615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15171321560221032, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1516258079049768, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:18, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15153840020774328, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15145099251050978, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15169513125105857, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15160772355382504, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1515203158565915, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.151432908159358, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03812784034496636, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03804043264773284, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03828457138828163, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.038197163691048114, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:18, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:45}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.038109755993814595, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09900180004958056, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09924593879012937, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09915853109289584, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09907112339566232, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09898371569842881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09922785443897761, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09914044674174408, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09905303904451056, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09896563134727705, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:18, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09920977008782583, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:45}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09912236239059231, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09903495469335881, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09599958084226602, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09624371958281483, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09615631188558131, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:101, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0966445893666789, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0965571816694454, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09680132040999419, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09671391271276067, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:18, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09662650501552714, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09653909731829362, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09678323605884243, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0966958283616089, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09660842066437537, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09567583949994829, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959199782404971, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09583257054326358, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09574516284603006, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09565775514879654, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:18, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09590189388934535, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09581448619211182, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0957270784948783, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09563967079764478, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09588380953819357, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09579640184096005, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09570899414372654, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06827445371495477, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:2,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06851859245550357, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06843118475827005, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:18, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06834377706103655, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06825636936380301, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0685005081043518, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0684131004071183, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06832569270988477, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06823828501265125, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06848242375320004, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06839501605596653, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06830760835873301, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014509677740764273, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:18, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014753816481313069, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014666408784079547, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01457900108684603, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014491593389612506, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.014735732130161307, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015224009611258898, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01579969478959001, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016043833530138808, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016619518708469918, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017195203886801035, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:0, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:18, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07114986554808504, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07139400428863385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07130659659140033, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07121918889416681, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07113178119693328, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07137591993748209, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07128851224024856, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07186419741857968, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0724398825969108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07268402133745958, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:18, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0732597065157907, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07383539169412182, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07804554128858505, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07828968002913385, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07886536520746497, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07944105038579607, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0800167355641272, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08026087430467599, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08083655948300711, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08141224466133822, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:18, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08198792983966934, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08223206858021813, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08280775375854924, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08338343893688034, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10851696589781112, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10876110463835992, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10933678981669102, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10991247499502213, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11048816017335324, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11073229891390204, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:18, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11130798409223315, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11188366927056427, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11245935444889538, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11270349318944418, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11327917836777529, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11385486354610 64, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15204476203999265, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15228890078054144, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1522014930833079, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15211408538607443, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:18, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1520266776888409, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15227081642938967, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15218340873215616, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15209600103492266, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03879093322053099, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03870352552329746, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03861611782606395, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03886025656661275, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03877284886937923, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03868544117214571, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:18, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09957748522791168, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09982162396846048, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09973421627122694, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09964680857399344, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09955940087675992, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0998035396173087, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09971613192007518, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09962872422284168, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09954131652560816, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09978545526615694, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:18, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09969804756892343, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09961063987168992, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09657526602059714, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09681940476114595, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09673199706391242, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:102, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09722027454501002, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09713286684777649, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0973770055883253, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09728959789109176, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09720219019385826, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:18, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:53}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09711478249662474, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09735892123717353, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09727151353994001, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09718410584270648, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09625152467827941, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09649566341882822, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09640825572159469, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09632084802436117, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09623344032712766, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09647757906767646, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:18, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09639017137044294, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
```

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MATCH (a {`key`:53}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09630276367320942, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0962153559759759, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0964594947165247, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09637208701929116, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09628467932205764, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0688501388932859, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:2,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06909427763383467, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06900686993660116, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06891946223936764, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:18, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06883205454213413, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06907619328268293, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06898878558544941, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06890137788821589, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06881397019098237, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06905810893153115, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06897070123429765, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06888329353706411, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015085362919095386, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015329501659644182, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:0, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:18, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015242093962410662, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015154686265177141, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015067278567943623, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015311417308492416, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015224009611258898, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015712287092356492, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01595642583290529, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016532111011236403, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017107796189567512, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07172555072641616, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:18, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07196968946696494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07188228176973144, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07179487407249792, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:53}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0717074663752644, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0719516051158132, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07186419741857968, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07177678972134616, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07235247489967726, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07259661364022606, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07317229881855718, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:18, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07374798399688828, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07795813359135155, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07820227233190034, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07877795751023145, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
```

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MATCH (a {`key`:53}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07935364268856257, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07992932786689368, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08017346660744247, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08074915178577358, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0813248369641047, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08190052214243582, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:18, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08214466088298461, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08272034606131572, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08329603123964684, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1084295582005776, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10867369694112639, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10924938211945752, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10982506729778863, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11040075247611973, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11064489121666853, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11122057639499963, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:18, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11179626157333074, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11237194675166187, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11261608549221065, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11319177067054176, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11376745584887289, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15262044721832374, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15286458595887253, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15277717826163903, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15268977056440552, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.152602362867172, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:18, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15284650160772076, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15275909391048728, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03945402609609562, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0393666183988621, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03927921070162858, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.039191803004395065, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03943594174494386, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:53}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03934853404771034, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.039261126350476815, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10015317040624279, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:18, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10039730914679158, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10030990144955808, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10022249375232455, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10013508605509103, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10037922479563981, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1002918170984063, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10020440940117278, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10011700170393927, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10036114044448807, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10027373274725454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:18, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10018632505002102, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09715095119892826, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09739508993947706, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09730768224224354, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:103, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09779595972334112, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977085520261076, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0979526907666564, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09786528306942288, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09777787537218936, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09769046767495584, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:18, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09793460641550465, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09784719871827112, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977597910210376, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09682720985661053, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09707134859715932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0969839408999258, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09689653320269229, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09680912550545875, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09705326424600756, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09696585654877406, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:18, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09687844885154052, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.096791041154307, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0970351798948558, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09694777219762228, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09686036450038876, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.069425824071617, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:2,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0696699628121658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06958255511493226, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06949514741769876, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06940773972046524, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:0, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:18, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06965187846101402, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06956447076378051, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06947706306654701, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06938965536931348, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06963379410986227, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06954638641262875, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06945897871539523, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015661048097426494, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01581777914074177, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:18, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015730371443508254, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015642963746274732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015887102486823533, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01579969478959001, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015712287092356492, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015869018135671768, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016444703314002877, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017020388492333994, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07230123590474727, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:18, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07245796694806254, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07237055925082904, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0722831515535955, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0725272902941443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0724398825969108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07235247489967726, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07226506720244374, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07250920594299255, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07308489112132366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07366057629965476, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:18, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07787072589411803, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07811486463466683, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07869054981299795, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07926623499132904, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07984192016966016, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08008605891020894, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08066174408854006, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08123742926687118, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08181311444520228, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08205725318575109, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:18, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08263293836408221, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08320862354241332, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10834215050334407, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10858628924389288, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10916197442222399, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10973765960055509, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1103133447788862, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11055748351943502, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1111331686977661, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11170885387609723, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:18, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11228453905442834, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11252867779497713, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11310436297330825, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11368004815163936, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15319613239665486, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15344027113720365, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15335286343997015, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15326545574273662, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1531780480455031, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1534221867860519, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:18, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03978557253387793, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04002971127442673, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03994230357719321, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0398548958799597, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03976748818272617, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04001162692327497, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.039924219226041446, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03983681152880793, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1007288555845739, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10097299432512272, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:18, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10088558662788918, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10079817893065565, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10071077123342213, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10095490997397094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10086750227673741, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10078009457950389, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1006926868822704, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10093682562281918, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10084941792558565, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10076201022835214, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:18, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09772663637725937, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09797077511780818, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09788336742057466, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:104, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09804009846388992, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0979526907666564, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0981968295072052, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09810942180997168, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09802201411273816, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09793460641550465, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09817874515605343, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:18, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09809133745881991, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09800392976158641, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09707134859715932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09731548733770812, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09722807964047461, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09714067194324108, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09705326424600756, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09729740298655637, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09720999528932285, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09712258759208933, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:0, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:18, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0970351798948558, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09727931863540461, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09719191093817109, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09710450324093756, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0696699628121658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:2,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06991410155271459, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06982669385548107, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06973928615824755, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06965187846101402, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06989601720156284, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:18, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06980860950432932, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06972120180709579, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06963379410986227, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06987793285041106, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06979052515317755, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06970311745594401, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01614932557852409, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016061917881290574, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015974510184057048, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:18, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015887102486823533, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016131241227372327, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016043833530138808, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01595642583290529, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.015869018135671768, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016688842054551678, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017264527232882788, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:0,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07278951338584486, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07270210568861134, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:18, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07261469799137782, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0725272902941443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07277142903469311, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07268402133745958, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07259661364022606, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07250920594299255, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07275334468354133, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07332902986187245, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07390471504020357, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07811486463466683, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:18, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07835900337521562, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07893468855354673, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07951037373187783, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08008605891020894, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08033019765075775, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08090588282908887, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08148156800741997, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08205725318575109, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08230139192629989, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08287707710463099, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:18, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08345276228296211, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:2,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10858628924389288, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10883042798444167, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10940611316277278, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10998179834110389, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11055748351943502, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1108016222599838, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11137730743831491, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11195299261664601, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11252867779497713, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:18, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11277281653552594, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11334850171385705, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11392418689218815, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15344027113720365, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15368440987775245, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15359700218051892, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1535095944832854, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1534221867860519, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04011711897166025, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04002971127442673, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:18, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04027385001497553, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04018644231774201, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04009903462050849, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04001162692327497, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040255765663823766, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04016835796659025, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04008095026935673, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:1,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10097299432512272, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10121713306567151, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10112972536843798, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:18, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10104231767120445, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10095490997397094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10119904871451973, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1011116410172862, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
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MATCH (a {`key`:60}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1010242333200527, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10093682562281918, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10118096436336797, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10109355666613447, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10100614896890094, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09797077511780818, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:18, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09821491385835697, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09812750616112344, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:105, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09861578364222104, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09852837594498752, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09877251468553631, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
```

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MATCH (a {`key`:61}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09868510698830278, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09859769929106928, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09851029159383576, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09875443033438454, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09866702263715103, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:0, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:18, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09857961493991751, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09764703377549044, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09789117251603924, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09780376481880572, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977163571215722, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09762894942433868, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09787308816488748, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09778568046765396, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09769827277042042, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09761086507318692, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:18, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09785500381373571, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977675961165022, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09768018841926868, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0702456479904969, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:2,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07048978673104571, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07040237903381219, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07031497133657866, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07022756363934514, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07047170237989395, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07038429468266041, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:18, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07029688698542691, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07020947928819338, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07045361802874218, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07036621033150865, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07027880263427513, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016480872016306405, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016725010756855202, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016637603059621683, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01655019536238816, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016462787665154643, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:18, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01670692640570344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016619518708469918, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016532111011236403, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016444703314002877, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.016688842054551678, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01717711953564927, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:0,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07312105982362717, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07336519856417598, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07327779086694244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07319038316970893, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:18, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07310297547247542, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07334711421302421, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0732597065157907, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07317229881855718, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07308489112132366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07332902986187245, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07324162216463893, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07381730734297004, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0780274569374333, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0782715956779821, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:18, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0788472808563132, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07942296603464431, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07999865121297542, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08024278995352423, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08081847513185533, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08139416031018645, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08196984548851757, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08221398422906635, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08278966940739747, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:2,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0833653545857286, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:2,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:18, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10849888154665936, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10874302028720814, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10931870546553925, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10989439064387038, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11047007582220149, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11071421456275028, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1112898997410814, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1118655849194125, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11244127009774363, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11268540883829241, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:18, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:61}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11326109401662351, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11383677919495465, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15401595631553477, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15426009505608357, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15417268735885004, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15408527966161653, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04078021184722489, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040692804149991366, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04060539645275784, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04084953519330664, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:18, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04076212749607312, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:61}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0406747197988396, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040587312101606085, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040831450842154886, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04074404314492136, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:1,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040656635447687835, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:1,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10154867950345382, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10179281824400262, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10170541054676908, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10161800284953558, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:18, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10153059515230205, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10177473389285084, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
```

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MATCH (a {`key`:61}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10168732619561734, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1015999184983838, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151251080115027, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10175664954169909, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10166924184446557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10158183414723204, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09854646029613928, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09879059903668808, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:0, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:18, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09870319133945456, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:106, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09919146882055214, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09910406112331864, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09934819986386742, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0992607921666339, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09917338446940038, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09908597677216686, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09933011551271566, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09924270781548214, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09915530011824863, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:18, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09822271895382156, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:3,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09846685769437034, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09837944999713683, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09829204229990332, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09820463460266979, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09844877334321858, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09836136564598506, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09827395794875154, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09818655025151803, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09843068899206682, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:18, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09834328129483332, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09825587359759978, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07082133316882802, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:2,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07106547190937683, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0709780642121433, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07089065651490978, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07080324881767626, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07104738755822505, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07095997986099153, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07087257216375802, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:18, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0707851644665245, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07102930320707329, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.5090909090909090, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07094189550983977, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07085448781260625, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017056557194637518, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017300695935186315, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017213288237952797, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017125880540719274, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01703847284348576, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017282611584034553, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:18, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017195203886801035, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017107796189567512, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017020388492333994, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.017264527232882788, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:0,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.01717711953564927, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:0,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07369674500195829, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07394088374250708, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07385347604527356, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07376606834804005, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07367866065080653, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:18, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07392279939135532, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07383539169412182, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07374798399688828, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07366057629965476, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07390471504020357, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07381730734297004, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07372989964573652, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07794004924019977, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07818418798074858, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07875987315907969, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:18, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07933555833741081, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07991124351574191, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0801553822562907, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08073106743462181, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08130675261295293, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08188243779128404, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08212657653183285, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:2,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08270226171016397, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:2,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08327794688849507, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:2,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10841147384942583, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:18, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10865561258997461, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10923129776830574, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10980698294663685, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11038266812496796, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11062680686551678, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11120249204384788, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11177817722217899, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11235386240051012, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11259800114105888, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11317368631939001, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:18, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11374937149772112, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:3,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15459164149386587, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15483578023441466, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15474837253718116, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041443304722789516, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041355897025556, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04126848932832248, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04118108163108896, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041425220371637754, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041337812674404235, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:18, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04125040497717071, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0411629972799372, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04140713602048599, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:1,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04131972832325247, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:1,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04123232062601895, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:1,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10212436468178493, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10236850342233372, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10228109572510022, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10219368802786669, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10210628033063315, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:0, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:18, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10235041907118198, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226301137394844, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10217560367671491, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1020881959794814, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1023323347200302, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10224492702279667, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10215751932556318, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:107, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09912214547447039, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0993662842150192, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09927887651778566, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:107, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:18, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042227539670913646, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04214013197368013, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04238427071422893, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0422968630169954, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042209455319761884, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04212204762252837, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042366186363077166, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04227877866584364, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04219137096861012, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09550356424930773, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:18, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09574770298985653, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09566029529262302, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09557288759538948, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09548547989815596, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09572961863870476, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09564221094147124, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09555480324423772, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0954673955470042, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095711534287553, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09562412659031948, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:18, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09553671889308596, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1221236233278535, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:4,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12236776206840229, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:4,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12228035437116876, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:4,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12219294667393528, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12210553897670175, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12234967771725054, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12226227002001704, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1221748623227835, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12208745462554997, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:18, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12233159336609879, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12224418566886526, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12215677797163173, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.068358847353663, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0686029860942118, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0691786712725429, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06975435645087402, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07033004162920514, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07114986554808504, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:18, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07172555072641616, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07230123590474727, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07312105982362717, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07369674500195829, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017822128060062142, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:0,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018397813238393252, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:0,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018973498416724365, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019549183595055475, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:18, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020369007513935385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020944692692266495, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02152037787059761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022340201789477522, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022915886967808632, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13224966272597435, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:4,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13249380146652312, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13306948664485427, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13364517182318536, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:18, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13422085700151648, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13446499574206527, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350406809203964, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13561636609872751, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1361920512770586, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13643619001760743, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13701187519593852, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13758756037426964, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05675228770430439, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:1,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05699642644485319, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:1,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:18, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0575721116231843, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:1,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058147796801515404, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058723481979846524, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058967620720395325, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05954330589872643, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06011899107705755, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06069467625538866, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06093881499593746, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061514500174268565, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06209018535259968, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:18, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09816196386861685, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09840610260916564, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09250145614126622, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.092745594881815, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09265818718458148, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09257077948734796, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09248337179011444, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09272751053066323, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09264010283342972, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0925526951361962, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:0, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:18, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09246528743896268, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09270942617951147, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09262201848227795, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09253461078504445, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1488318154852745, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490759542258233, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14898854652858975, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14890113883135625, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:134}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14881373113412275, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14905786987467154, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:18, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14897046217743803, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1488830544802045, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.148795646782971, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490397855235198, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14895237782628626, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14886497012905273, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.042158216324831896, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04240235506538069, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04231494736814717, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:108, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042471678411462446, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:18, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04238427071422893, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04262840945477772, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042541001757544196, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042453594060310684, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042366186363077166, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04261032510362596, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04252291740639244, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.042435509709158915, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09574770298985653, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09599184173040533, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:18, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959044340331718, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09581702633593829, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09572961863870476, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09597375737925355, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09588634968202005, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09579894198478651, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.095711534287553, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09595567302810179, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09586826533086829, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09578085763363477, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:18, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12236776206840229, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:4,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12261190080895108, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:4,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12252449311171758, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12243708541448407, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12234967771725054, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12259381645779935, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12250640876056583, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1224190010633323, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12233159336609879, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12257573210664757, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:18, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12248832440941405, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12240091671218052, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0686029860942118, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06884712483476059, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06942281001309171, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06999849519142283, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07081831911030273, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07139400428863385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07196968946696494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:18, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07278951338584486, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07336519856417598, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07394088374250708, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017822128060062142, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:0,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018641951978942046, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019217637157273162, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020037461076153073, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020613146254484183, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:18, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021188831432815296, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022008655351695206, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02258434053002632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02316002570835743, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13249380146652312, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13273794020707194, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13331362538540306, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13388931056373415, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13446499574206527, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:18, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13470913448261407, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352848196609452, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13586050483927628, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13643619001760743, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13668032875815622, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725601393648731, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13783169911481843, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05699642644485319, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:1,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05724056518540198, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:1,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0578162503637331, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:18, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05839193554206422, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058967620720395325, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05921175946094411, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05978744463927523, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06036312981760634, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06093881499593746, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061182953736486245, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061758638914817365, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.062334324093148485, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09840610260916564, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:0, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:18, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06731478585704509, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.092745594881815, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09298973362236379, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09290232592513029, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09281491822789675, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09272751053066323, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09297164927121204, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09288424157397851, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09279683387674499, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09270942617951147, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:18, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09295356492006027, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09286615722282675, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09277874952559323, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490759542258233, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14932009296637208, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14923268526913858, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14914527757190504, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14905786987467154, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14930200861522036, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14921460091798683, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:18, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1491271932207533, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490397855235198, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14928392426406858, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14919651656683505, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14910910886960152, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04240235506538069, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04264649380592948, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04255908610869596, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:109, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04304736358979356, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04295995589256004, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:18, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043204094633108835, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043116686935875316, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0430292792386418, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04294187154140827, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04318601028195707, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043098602584723554, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043011194887490035, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09632338816818765, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09656752690873643, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09648011921150292, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:18, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0963927115142694, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09630530381703588, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09654944255758467, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09646203486035115, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09637462716311763, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0962872194658841, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0965313582064329, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09644395050919939, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09635654281196587, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1229434472467334, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:4,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:19, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1231875859872822, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12310017829004868, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12301277059281515, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12292536289558167, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12316950163613045, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12308209393889694, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12299468624166343, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1229072785444299, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12315141728497869, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12306400958774516, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:19, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12297660189051166, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0691786712725429, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06942281001309171, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06933540231585819, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0699110874941893, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07048677267252042, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07073091141306923, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07130659659140033, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07188228176973144, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07245796694806254, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:19, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07270210568861134, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07327779086694244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07385347604527356, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018397813238393252, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:0,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018641951978942046, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01913022946003964, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019705914638370754, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01995005337891955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02052573855725066, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021101423735581774, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:19, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021677108913912887, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021921247654461684, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022496932832792797, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023072618011123907, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13240639376928964, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1326505325098384, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13322621768816953, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13380190286650062, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13437758804483177, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13462172678538054, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:19, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13519741196371166, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13577309714204278, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1363487823203739, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1365929210609227, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371686062392538, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13774429141758493, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05690901874761966, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:1,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05715315748816846, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05772884266649958, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058304527844830686, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:0, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:19, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05888021302316179, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05912435176371061, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0597000369420417, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06027572212037282, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06085140729870394, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06109554603925273, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06167123121758385, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06224691639591495, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06797787873260973, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06789047103537621, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:19, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0933212800601461, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09356541880069491, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09347801110346139, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09339060340622787, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09330319570899435, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09354733444954315, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09345992675230963, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09337251905507611, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09328511135784258, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09352925009839139, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:19, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09344184240115785, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09335443470392434, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1496516394041544, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498957781447032, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498083704474697, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14972096275023616, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14963355505300266, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14987769379355145, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14979028609631795, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14970287839908444, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:19, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1496154707018509, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498596094423997, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14977220174516617, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14968479404793267, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0429780402437118, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0432221789842606, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04313477128702707, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:110, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043623048768124666, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043535641070891154, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043779779811439955, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:19, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04369237211420643, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04360496441697291, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.043517556719739385, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04376169546028818, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04367428776305466, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04358688006582115, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09689907334651876, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09714321208706754, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09705580438983402, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09696839669260052, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:19, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.096880988995367, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09712512773591578, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09703772003868226, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09695031234144874, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09686290464421522, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09710704338476402, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0970196356875305, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
```

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MATCH (a {`key`:137}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09693222799029699, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12351913242506453, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:4,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12376327116561331, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:19, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12367586346837979, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12358845577114629, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12350104807391278, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12374518681446155, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12365777911722806, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12357037141999454, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.123482963722761, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1237271024633098, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1236396947660763, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12355228706884276, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:19, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06975435645087402, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06999849519142283, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0699110874941893, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06982367979695578, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0703993649752869, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07064350371583569, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07121918889416681, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07179487407249792, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07237055925082904, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07261469799137782, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:19, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07319038316970893, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07376606834804005, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018973498416724365, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019217637157273162, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01913022946003964, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01961850694113723, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019862645681686032, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020438330860017142, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02101401603834826, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021589701216679368, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:19, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021833839957228166, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02240952513555928, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022985210313890392, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1323189860720561, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1325631248126049, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.133138809990936, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13371449516926712, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13429018034759824, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13453431908814703, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13511000426647815, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:0, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:19, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13568568944480927, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13626137462314036, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13650551336368918, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13708119854202028, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1376568837203514, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05682161105038615, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05706574979093495, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057641434969266055, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05821712014759717, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05879280532592829, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:19, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:137}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05903694406647708, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05961262924480819, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0601883144231393, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06076399960147041, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061008138342019215, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06158382352035032, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06864097160817437, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06855356391094085, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06846615621370732, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09389696523847722, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:19, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09414110397902602, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:137}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0940536962817925, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.093966288584559, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09387888088732546, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09412301962787427, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09403561193064075, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09394820423340722, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0938607965361737, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09410493527672249, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09401752757948897, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:19, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09393011988225546, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15022732458248553, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:137}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15047146332303432, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1503840556258008, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15029664792856728, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15020924023133375, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15045337897188255, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15036597127464904, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1502785635774155, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.150191155880182, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:19, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1504352946207308, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1503478869234973, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15026047922626376, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
```

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MATCH (a {`key`:137}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.043553725422042916, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04379786416259171, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04371045646535819, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04419873394645578, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04411132624922227, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04435546498977106, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04426805729253754, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:19, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044180649595304024, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0440932418980705, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0443373806386193, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04424997294138577, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
```

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MATCH (a {`key`:138}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044162565244152255, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09747475852484987, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09771889726539865, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09763148956816514, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09754408187093162, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0974566741736981, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:19, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977008129142469, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09761340521701337, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09752599751977986, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09743858982254634, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09768272856309512, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09759532086586162, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0975079131686281, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12409481760339564, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:4,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12433895634394442, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12425154864671091, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:19, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12416414094947739, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12407673325224386, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12432087199279267, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12423346429555916, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12414605659832564, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12405864890109211, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:138}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12430278764164092, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1242153799444074, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12412797224717387, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07033004162920514, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:19, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07048677267252042, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0703993649752869, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07031195727805337, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07055609601860217, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07113178119693328, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0717074663752644, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0722831515535955, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0725272902941443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07310297547247542, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:19, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07367866065080653, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019549183595055475, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019705914638370754, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01961850694113723, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01977523798445251, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020350923162783623, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020926608341114736, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02150229351944585, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021746432259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:0, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:19, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02232211743832576, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02289780261665687, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1322315783748226, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13247571711537137, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13305140229370252, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1336270874720336, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1342027726503647, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13444691139091353, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:138}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13502259656924462, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13559828174757574, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:19, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13617396692590686, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13641810566645565, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13699379084478674, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1375694760231179, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05673420335315263, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05697834209370143, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057554027272032536, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058129712450363656, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05870539762869476, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
```

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MATCH (a {`key`:138}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05894953636924356, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:19, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05952522154757467, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06010090672590579, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060676591904236896, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06092073064478568, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06897251804595668, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06921665678650549, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06912924908927195, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06904184139203844, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09447265041680833, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09471678915735714, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:19, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09462938146012362, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09454197376289009, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09445456606565657, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09469870480620539, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09461129710897186, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09452388941173834, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0944364817145048, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09468062045505361, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0945932127578201, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09450580506058656, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:19, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:138}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15080300976081665, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1510471485013654, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1509597408041319, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508723331068984, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15078492540966487, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15102906415021367, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15094165645298016, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15085424875574663, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1507668410585131, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15101097979906192, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:19, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1509235721018284, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:138}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15083616440459488, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04412941060037402, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04437354934092282, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.044286141643689304, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04444287268700458, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04435546498977106, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044599603730319855, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04451219603308633, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04442478833585282, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:19, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0443373806386193, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04458151937916809, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
```

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MATCH (a {`key`:139}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044494111681934574, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044406703984701056, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09771889726539865, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09796303600594745, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09787562830871394, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09778822061148042, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0977008129142469, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09794495165479569, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:19, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09785754395756217, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09777013626032866, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09768272856309512, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09792686730364393, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09783945960641041, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09775205190917688, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12433895634394442, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:4,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12458309508449324, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1244956873872597, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12440827969002617, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:19, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12432087199279267, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12456501073334146, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12447760303610793, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12439019533887442, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12430278764164092, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12454692638218971, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12445951868495618, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12437211098772268, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07057418036975394, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07081831911030273, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:19, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07073091141306923, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07064350371583569, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07055609601860217, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07080023475915097, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07137591993748209, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:139}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0719516051158132, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0725272902941443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07277142903469311, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07334711421302421, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07392279939135532, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:0, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:19, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020037461076153073, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01995005337891955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019862645681686032, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01977523798445251, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:139}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020595061903332417, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021170747081663534, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021746432259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021990571000543444, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022566256178874554, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:19, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02314194135720567, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13247571711537137, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1327198558559202, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1332955410342513, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13387122621258238, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13444691139091353, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1346910501314623, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352667353097934, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13584242048812453, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13641810566645565, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:19, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13666224440700445, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13723792958533557, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378136147636667, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05697834209370143, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05722248083425022, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05779816601258134, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05837385119091244, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05894953636924356, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05919367510979235, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05976936028812347, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:19, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060345045466454576, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06092073064478568, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.069304064483739, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06921665678650549, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06946079552705427, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06937338782982076, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06928598013258723, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09471678915735714, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:139}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09496092789790593, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0948735202006724, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:19, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0947861125034389, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09469870480620539, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09494284354675417, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09485543584952065, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09476802815228713, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09468062045505361, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0949247591956024, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09483735149836889, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09474994380113537, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:139}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1510471485013654, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:19, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15129128724191424, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512038795446807, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15111647184744717, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15102906415021367, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15127320289076246, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15118579519352893, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15109838749629542, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15101097979906192, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512551185396107, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15116771084237718, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:19, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15108030314514367, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04437354934092282, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.044617688081471624, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0445302803842381, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04501855786533569, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.044931150168102174, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04517528890865097, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04508788121141744, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04500047351418393, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04491306581695041, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:19, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:140}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045157204557499206, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04506979686026569, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04498238916303217, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09829458244372977, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09853872118427857, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09845131348704504, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09836390578981154, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.098276498092578, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09852063683312681, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0984332291358933, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:19, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09834582143865976, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09825841374142624, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09850255248197505, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09841514478474152, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.098327737087508, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12491464152227555, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:4,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12515878026282434, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1250713725655908, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1249839648683573, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12489655717112377, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:19, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12514069591167257, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12505328821443903, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12496588051720556, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12487847281997203, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12512261156052082, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12503520386328731, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12494779616605378, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07114986554808504, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07139400428863385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07130659659140033, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:0, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:19, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07121918889416681, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07113178119693328, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07137591993748209, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07128851224024856, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07186419741857968, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0724398825969108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07268402133745958, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0732597065157907, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07383539169412182, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020369007513935385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:19, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020613146254484183, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02052573855725066, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020438330860017142, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:140}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020350923162783623, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020595061903332417, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021083339384430015, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02165902456276113, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021903163303309922, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022478848481641035, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02305453365997215, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:19, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13238830941813787, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13263244815868666, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13320813333701778, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1337838185153489, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
```

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MATCH (a {`key`:140}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13435950369368002, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1346036424342288, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1351793276125599, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.135755012790891, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13633069796922212, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13657483670977094, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:19, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13715052188810203, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772620706643315, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0568909343964679, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057135073137016705, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05771075831534781, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058286443493678924, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05886212867201003, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05910626741255883, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05968195259088995, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06025763776922106, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:19, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06996715735930363, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06987974966207011, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0697923419648366, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07003648070538539, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06994907300815187, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06986166531091835, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09529247433568826, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09553661307623704, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09544920537900352, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09536179768177, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:19, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09527438998453648, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09551852872508527, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09543112102785176, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09534371333061825, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09525630563338473, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09550044437393351, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09541303667670001, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:140}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09532562897946648, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15162283367969656, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15186697242024536, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:19, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15177956472301182, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15169215702577832, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15160474932854479, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15184888806909358, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15176148037186005, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15167407267462654, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1515866664977393, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1518308037179418, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517433960207083, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1516559883234748, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:19, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.044949234519253936, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04519337325980273, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04510596556256921, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0455942430436668, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04550683534643329, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04575097408698208, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04566356638974856, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045576158692515044, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045488750995281525, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.045732889735830326, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:19, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0456454820385968, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04555807434136328, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09887026762206089, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09911440636260968, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09902699866537616, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09893959096814263, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09885218327090911, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09909632201145793, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0990089143142244, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09892150661699088, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:19, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09883409891975736, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09907823766030616, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09899082996307264, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0989034222658391, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12549032670060667, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:4,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12573446544115544, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12564705774392193, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12555965004668843, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1254722423494549, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1257163810900037, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:0, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:19, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12562897339277018, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12554156569553665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12545415799830315, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12569829673885194, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12561088904161843, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1255234813443849, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07172555072641616, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07196968946696494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07188228176973144, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07179487407249792, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:19, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:141}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0717074663752644, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0719516051158132, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07186419741857968, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07177678972134616, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07235247489967726, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07259661364022606, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07317229881855718, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07374798399688828, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020944692692266495, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021188831432815296, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:19, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021101423735581774, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:141}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02101401603834826, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020926608341114736, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021170747081663534, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021083339384430015, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021571616865527603, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021815755606076404, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022391440784407513, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02296712596273863, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13230090172090436, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:19, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13254504046145313, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13312072563978428, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
```

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MATCH (a {`key`:141}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13369641081811537, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1342720959964465, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13451623473699528, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350919199153264, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1356676050936575, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13624329027198862, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1364874290125374, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370631141908685, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:19, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13763879936919965, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056803526699234386, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05704766543978317, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
```

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MATCH (a {`key`:141}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05762335061811429, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05819903579644541, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05877472097477652, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05901885971532532, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059594544893656426, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.9909090909090911, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07063025023486826, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07054284253763474, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:19, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07045543484040122, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07036802714316771, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0706121658837165, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07052475818648297, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07043735048924947, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09586815951401936, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09611229825456816, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09602489055733464, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09593748286010112, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0958500751628676, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:19, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09609421390341638, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09600680620618286, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09591939850894936, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09583199081171584, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09607612955226463, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09598872185503111, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0959013141577976, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15219851885802765, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15244265759857645, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15235524990134294, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:19, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522678422041094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15218043450687588, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15242457324742467, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15233716555019117, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15224975785295763, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15216235015572413, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
```

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MATCH (a {`key`:141}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15240648889627292, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1523190811990394, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522316735018059, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04552491969758504, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:19, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04576905843813385, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.045681650740900324, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04616992822199791, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046082520524764393, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046326659265313194, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046239251568079676, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04615184387084615, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04606443617361264, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04630857491416143, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046221167216927914, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:19, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046133759519694395, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09944595280039199, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0996900915409408, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09960268384370727, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09951527614647375, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09942786844924023, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09967200718978902, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0995845994925555, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.099497191795322, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09940978409808847, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:0, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:19, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09965392283863728, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09956651514140374, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09947910744417023, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12606601187893776, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:4,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12631015061948658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12622274292225305, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12613533522501952, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12604792752778599, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1262920662683348, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12620465857110127, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:19, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12611725087386774, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12602984317663424, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12627398191718303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12618657421994953, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12609916652271602, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07230123590474727, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07245796694806254, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07237055925082904, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0722831515535955, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:19, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0725272902941443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0724398825969108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07235247489967726, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07226506720244374, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07250920594299255, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07308489112132366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07366057629965476, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02152037787059761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021677108913912887, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:19, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021589701216679368, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02150229351944585, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021746432259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02165902456276113, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021571616865527603, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02172834790884288, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022304033087173995, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022879718265505108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13221349402367083, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13245763276421962, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:19, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13303331794255074, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13360900312088184, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13418468829921298, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13442882703976178, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13500451221809287, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.135580197396424, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13615588257475508, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13640002131530388, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.136975706493635, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13755139167196612, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:19, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05671611900200085, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05696025774254967, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057535942920880774, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05811162809921188, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058687313277543, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05893145201809179, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.9909090909090091, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07096179667265058, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07120593541319938, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07111852771596584, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07103112001873234, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:19, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07094371232149882, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711878510620476, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711004433648141, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07101303566758058, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09644384469235047, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09668798343289928, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09660057573566574, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09651316803843223, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09642576034119871, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0966698990817475, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:19, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09658249138451398, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09649508368728046, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09640767599004696, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09665181473059574, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09656440703336222, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09647699933612872, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15277420403635877, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530183427769076, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15293093507967406, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15284352738244053, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:19, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152756119685207, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15300025842575582, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1529128507285223, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15282544303128875, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15273803533405525, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15298217407460404, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528947663773705, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152807358680137, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04610060487591616, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.046344743616464956, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:19, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04625733591923144, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046414066962546706, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046326659265313194, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046570798005861995, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04648339030862847, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04639598261139496, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04630857491416143, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046552713654710226, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04646530595747671, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04637789826024319, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:0, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:19, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0996900915409408, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09993423028148958, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09984682258425606, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09975941488702256, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09967200718978902, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09991614593033782, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09982873823310431, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0997413305358708, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09965392283863728, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09989806157918606, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:19, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09981065388195255, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09972324618471902, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12631015061948658, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:4,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12655428936003538, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12646688166280184, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1263794739655683, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1262920662683348, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1265362050088836, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12644879731165007, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12636138961441656, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:19, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12627398191718303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12651812065773183, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12643071296049835, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12634330526326482, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07254537464529606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07278951338584486, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07270210568861134, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07261469799137782, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07252729029414443, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07277142903469311, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:19, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07268402133745958, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07259661364022606, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07250920594299255, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07275334468354133, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07332902986187245, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07390471504020357, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022008655351695206, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021921247654461684, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021833839957228166, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:19, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021744632259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021990571000543444, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021903163303309922, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021815755606076404, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02172834790884288, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02254817182772279, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0231238570060539, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:0,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13245763276421962, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13270177150476842, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13327745668309954, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:19, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13385314186143066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13442882703976178, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13467296578031054, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352486509586417, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13582433613697278, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13640002131530388, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1366441600558527, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372198452341838, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377955304125149, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05696025774254967, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:19, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057204396483098455, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05778008166142956, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05835576683976068, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05893145201809179, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0712933431104329, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07120593541319938, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07145007415374817, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07136266645651465, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07127525875928112, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711878510620476, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:19, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07143198980259642, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07134458210536289, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07125717440812937, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09668798343289928, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09693212217344807, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09684471447621455, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09675730677898102, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0966698990817475, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09691403782229631, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09682663012506278, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:19, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09673922242782926, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09665181473059574, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09689595347114453, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09680854577391103, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0967211380766775, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530183427769076, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15326248151745636, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15317507382022286, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15308766612298932, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15300025842575582, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:19, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532443971663046, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15315698946907108, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15306958177183758, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15298217407460404, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15322631281515284, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15313890511791933, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530514974206858, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.046344743616464956, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04658888235701376, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04650147465978023, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:0, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:19, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046989752140877826, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04690234444364431, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0471464831841931, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04705907548695958, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046971667789726064, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.046884260092492545, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04712839883304134, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047040991135807814, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04695358343857431, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1002657767192719, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:19, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1005099154598207, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10042250776258718, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10033510006535366, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
```

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MATCH (a {`key`:144}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10024769236812014, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10049183110866894, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10040442341143542, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1003170157142019, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10022960801696838, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10047374675751718, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10038633906028366, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:19, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10029893136305014, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12688583579781768, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:4,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12712997453836647, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12704256684113294, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:144}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12695515914389943, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1268677514466659, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1271118901872147, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1270244824899812, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12693707479274766, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12684966709551415, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:19, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12709380583606295, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1270063981388294, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1269189904415959, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07312105982362717, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07336519856417598, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:144}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07327779086694244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07319038316970893, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07310297547247542, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07334711421302421, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0732597065157907, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:19, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07317229881855718, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07308489112132366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07332902986187245, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07324162216463893, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07381730734297004, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022340201789477522, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02258434053002632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022496932832792797, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02240952513555928, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02232211743832576, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:19, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022566256178874554, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022478848481641035, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0223914407844075 13, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022304033087173995, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02254817182772279, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023036449308820386, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:0,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1323702250669861, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
```

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MATCH (a {`key`:144}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13261436380753489, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13319004898586603, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13376573416419713, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:19, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13434141934252825, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13458555808307704, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13516124326140816, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13573692843973925, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13631261361807037, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1365567523586192, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13713243753695026, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377081227152814, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05687285004531614, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05711698878586494, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:19, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057692673964196056, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058268359142527155, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07195643598599753, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07186902828876401, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07178162059153048, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07202575933207929, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07193835163484576, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07185094393761224, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07176353624037872, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07200767498092751, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:19, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.071920267283694, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07183285958646049, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09726366861123038, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09750780735177919, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09742039965454566, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09733299195731214, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09724558426007862, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09748972300062742, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0974023153033939, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09731490760616036, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:19, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09722749990892685, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09747163864947565, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09738423095224213, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09729682325500862, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1535940279552387, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15383816669578748, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15375075899855395, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15366335130132044, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1535759436040869, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1538200823446357, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:0, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:19, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1537326746474022, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15364526695016867, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15355785925293514, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15380199799348396, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15371459029625043, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1536271825990169, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.09957413673572789, `sourceIndex`:118, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04692042879479607, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04716456753534486, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047077159838111345, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04756543731920894, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:19, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04747802962197542, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04772216836252422, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047634760665290696, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04754735296805717, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04745994527082366, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04770408401137245, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047616676314138934, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.047529268616905415, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:1,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10084146189760303, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10108560063815182, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:19, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10099819294091829, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10091078524368477, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10082337754645126, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10106751628700004, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10098010858976653, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10089270089253301, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1008052931952995, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10104943193584828, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10096202423861476, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10087461654138126, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:19, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1274615209761488, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:4,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1277056597166976, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12761825201946408, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12753084432223055, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12744343662499702, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12768757536554584, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1276001676683123, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12751275997107878, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12742535227384527, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12766949101439407, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:19, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12758208331716053, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.12749467561992703, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07369674500195829, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07394088374250708, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07385347604527356, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07376606834804005, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07367866065080653, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07392279939135532, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07383539169412182, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07374798399688828, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:19, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07366057629965476, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07390471504020357, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07381730734297004, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07372989964573652, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022915886967808632, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02316002570835743, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023072618011123907, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022985210313890392, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02289780261665687, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02314194135720567, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:19, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02305453365997215, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02296712596273863, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022879718265505108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0231238570060539, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:0,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023036449308820386, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:0,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1322828173697526, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13252695611030138, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1331026412886325, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1336783264669636, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13425401164529474, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:19, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13449815038584353, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13507383556417463, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13564952074250575, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13622520592083687, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13646934466138566, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13704502983971675, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13762071501804787, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056785442348082624, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057029581088631424, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05760526626696253, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:19, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07261952886156216, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07253212116432865, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07244471346709512, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0723573057698616, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0726014445104104, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07251403681317688, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07242662911594336, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07233922141870983, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07258336015925863, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07249595246202512, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:19, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0724085447647916, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:2,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0978393537895615, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0980834925301103, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09799608483287678, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09790867713564326, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09782126943840973, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09806540817895854, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.097978000481725, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09789059278449148, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09780318508725797, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:0, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:19, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09804732382780676, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09795991613057324, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09787250843333974, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541697131335698, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15441385187411857, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15432644417688507, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15423903647965156, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15415162878241803, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15439576752296683, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15430835982573332, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:19, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542209521284998, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15413354443126628, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15437768317181508, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15429027547458155, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154202867777348, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:5,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047496113973127176, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047740252713675976, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047652845016442465, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10135977414710463, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:146}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10127236644987112, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:19, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151650519041991, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10142909749318639, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10134168979595287, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10125428209871934, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10149842083926815, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10141101314203461, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10132360544480111, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10123619774756759, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10148033648811638, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10139292879088287, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:19, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10130552109364935, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10121811339641583, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10146225213696462, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10137484443973109, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10128743674249759, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10120002904526405, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10144416778581285, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10135676008857934, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10126935239134582, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0842061818122977, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:20, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0844503205528465, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08436291285561297, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08427550515837946, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08418809746114594, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08443223620169472, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08434482850446122, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0842574208072277, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08417001310999418, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08441415185054298, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08432674415330946, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:20, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08423933645607594, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
```

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MATCH (a {`key`:146}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07790689459642154, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07815103333697035, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07806362563973683, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0779762179425033, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07788881024526978, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07813294898581859, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07804554128858505, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07795813359135155, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07787072589411803, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:20, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07811486463466683, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0780274569374333, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
```

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MATCH (a {`key`:146}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07794004924019977, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13224966272597435, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:4,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13249380146652312, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13240639376928964, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1323189860720561, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1322315783748226, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13247571711537137, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13238830941813787, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:20, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13230090172090436, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13221349402367083, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13245763276421962, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1323702250669861, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1322828173697526, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023681457833233256, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:0,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024257143011564366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:0,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02483282818989548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025408513368226592, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:20, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0262283372871065, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026804022465437612, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027379707643768725, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028199531562648632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028775216740979746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16630129869787724, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16654543743842606, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1325934823139798, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:4,
`saturationRatio`:0.0, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1328376210545286, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:4,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:20, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1327502133572951, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:4,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13266280566006156, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13257539796282802, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13281953670337684, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1327321290061433, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
```

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MATCH (a {`key`:146}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13264472130890978, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13255731361167625, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13280145235222507, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13271404465499154, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.132626636957758, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:0, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:20, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10829453754644561, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10853867628699441, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10845126858976088, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10836386089252736, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10827645319529386, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10852059193584264, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10843318423860913, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10834577654137562, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10825836884414208, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10850250758469089, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:20, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10841509988745737, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10832769219022384, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04828472740983425, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048528866150383036, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04844145845314952, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048354050755916, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04826664305868248, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04851078179923128, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048423374101997756, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04833596640476424, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:20, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04824855870753072, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048492697448079505, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048405289750846, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04831788205361247, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:120, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10129045080102288, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10153458954157167, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10144718184433817, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10160391288765344, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151650519041991, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1017606439309687, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:20, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1016732362337352, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10158582853650167, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10149842083926815, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10174255957981694, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10165515188258342, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1015677441853499, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10148033648811638, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10172447522866519, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10163706753143167, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10154965983419814, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:20, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10146225213696462, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10170639087751342, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1016189831802799, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10153157548304637, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10144416778581285, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10168830652636165, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10160089882912814, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151349113189462, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0844503205528465, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0846944592933953, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:20, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08460705159616178, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08451964389892826, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08443223620169472, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08467637494224353, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08458896724501001, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08450155954777648, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08441415185054298, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08465829059109176, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08457088289385825, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08448347519662472, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
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MATCH (a {`key`:254}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:20, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07815103333697035, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07839517207751914, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07830776438028561, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07822035668305209, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07813294898581859, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07837708772636737, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07828968002913385, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07820227233190034, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07811486463466683, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07835900337521562, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:20, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0782715956779821, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07818418798074858, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13249380146652312, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13273794020707194, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1326505325098384, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1325631248126049, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13247571711537137, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1327198558559202, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13263244815868666, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13254504046145313, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:20, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13245763276421962, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13270177150476842, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13261436380753489, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13252695611030138, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023681457833233256, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:0,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024501281752113163, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025076966930444276, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025896790849324183, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0264724760276553, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:20, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027048161205986406, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02786798512486632, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028443670303197433, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.029019355481528546, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16654543743842606, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08182507557731561, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1328376210545286, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:4,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1330817597950774, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:4,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13299435209784388, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:0, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:20, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13290694440061035, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13281953670337684, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13306367544392564, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1329762677466921, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13288886004945857, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13280145235222507, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13304559109277386, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13295818339554033, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13287077569830683, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10853867628699441, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:20, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10878281502754321, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10869540733030968, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10860799963307617, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10852059193584264, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10876473067639143, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10867732297915793, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1085899152819244, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10850250758469089, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10874664632523968, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10865923862800615, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:20, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10857183093077265, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048528866150383036, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04877300489093184, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04868559719369831, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0485981894964648, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04851078179923128, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04875492053978007, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048667512842546556, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04858010514531304, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048492697448079505, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:20, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04873683618862831, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04864942849139479, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04856202079416127, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:121, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10153458954157167, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10177872828212048, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10169132058488695, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10217959806598455, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10209219036875103, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10233632910929982, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1022489214120663, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:20, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10216151371483279, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10207410601759925, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10231824475814805, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10223083706091454, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10214342936368101, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1020560216664475, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1023001604069963, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10221275270976278, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10212534501252926, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10203793731529573, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:20, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10228207605584454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.102194668358611, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10210726066137749, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10201985296414397, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10226399170469276, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10217658400745924, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10208917631022574, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0850260057311776, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08527014447172641, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0851827367744929, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:20, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08509532907725936, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08500792138002584, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08525206012057465, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08516465242334112, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0850772447261076, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08498983702887408, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08523397576942288, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08514656807218936, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08505916037495584, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07872671851530146, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:20, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07897085725585025, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07888344955861673, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07879604186138321, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0787086341641497, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07895277290469849, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07886536520746497, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07877795751023145, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07869054981299795, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07893468855354673, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0788472808563132, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:20, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07875987315907969, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13306948664485427, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13331362538540306, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13322621768816953, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.133138809990936, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13305140229370252, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1332955410342513, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13320813333701778, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13312072563978428, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13303331794255074, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:20, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13327745668309954, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13319004898586603, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1331026412886325, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024257143011564366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:0,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024501281752113163, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024989559233210758, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025565244411541867, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025809383152090665, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026385068330421778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02696075350875289, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:1, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:20, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027536438687084004, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027780577427632798, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02835626260596391, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028931947784295024, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08248816845288025, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08240076075564673, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1334133062328597, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:4,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1336574449734085, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13357003727617497, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13348262957894147, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:20, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13339522188170794, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13363936062225673, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1335519529250232, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1334645452277897, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13337713753055616, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13362127627110496, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13353386857387145, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13344646087663792, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10911436146532552, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10935850020587431, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:20, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10927109250864081, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10918368481140728, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10909627711417375, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10934041585472257, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10925300815748903, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10916560046025553, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.109078192763022, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10932233150357079, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10923492380633729, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10914751610910375, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:20, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049104551328714156, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04934869006926294, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049261282372029425, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04917387467479591, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04908646697756239, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04933060571811119, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04924319802087767, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049155790323644144, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049068382626410625, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04931252136695942, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:20, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0492251136697259, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04913770597249239, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:122, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10211027471990279, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10235441346045157, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226700576321805, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10275528324431565, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10266787554708215, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10291201428763093, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10282460659039741, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10273719889316389, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:20, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10264979119593037, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10289392993647917, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10280652223924565, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10271911454201213, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10263170684477861, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1028758455853274, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1027884378880939, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10270103019086037, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10261362249362685, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10285776123417564, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:20, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10277035353694212, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1026829458397086, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10259553814247507, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10283967688302388, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10275226918579036, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10266486148855683, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08560169090950873, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08584582965005752, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.085758421952824, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08567101425559048, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:20, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08558360655835696, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08582774529890576, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08574033760167224, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08565292990443871, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08556552220720519, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08580966094775398, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08572225325052046, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08563484555328696, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07930240369363256, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07954654243418137, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:20, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07945913473694785, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07937172703971432, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07928431934248081, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07952845808302961, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07944105038579607, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07935364268856257, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07926623499132904, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07951037373187783, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07942296603464431, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07933555833741081, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:20, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13364517182318536, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13388931056373415, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13380190286650062, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13371449516926712, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1336270874720336, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13387122621258238, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1337838185153489, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13369641081811537, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13360900312088184, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13385314186143066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:1, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:20, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13376573416419713, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1336783264669636, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02483282818989548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025076966930444276, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024989559233210758, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02547783671430835, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025721975454857143, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02629766063318826, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02687334581151937, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027449030989850486, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:20, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027693169730399276, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028268854908730396, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08315126132844487, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08306385363121137, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08297644593397784, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13398899141119083, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1342331301517396, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1341457224545061, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1340583147572726, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13397090706003906, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:20, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13421504580058785, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13412763810335435, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1340402304061208, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13395282270888728, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1341969614494361, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13410955375220257, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13402214605496904, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10969004664365663, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10993418538420545, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10984677768697192, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:20, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10975936998973838, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10967196229250488, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10991610103305367, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10982869333582014, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10974128563858664, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1096538779413531, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10989801668190192, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10981060898466839, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10972320128743486, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049680236507045256, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:20, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049924375247594056, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049836967550360545, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04974955985312702, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0496621521558935, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0499062908964423, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049818883199208776, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049731475501975264, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04964406780474173, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04988820654529053, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04980079884805702, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:20, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04971339115082349, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:123, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10268595989823 39, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10293009863878269, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10284269094154916, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10333096842264677, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10324356072541324, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10348769946596205, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10340029176872852, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10331288407149501, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1032254763742615, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:20, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10346961511481027, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10338220741757677, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10329479972034325, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:150}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10320739202310972, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10345153076365851, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10336412306642499, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10327671536919147, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10318930767195796, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10343344641250676, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10334603871527324, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:20, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10325863101803971, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1031712233208062, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.103415362061355, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10332795436412147, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10324054666688795, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08617737608783983, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08642151482838863, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08633410713115512, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0862466994339216, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08615929173668807, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:20, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08640343047723688, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08631602278000335, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08622861508276983, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0861412073855363, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0863853461260851, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
```

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MATCH (a {`key`:150}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08629793842885158, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08621053073161805, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07987808887196368, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08012222761251249, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08003481991527896, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:20, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07994741221804544, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07986000452081192, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08010414326136071, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0800167355641272, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07992932786689368, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07984192016966016, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08008605891020894, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.07999865121297542, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07991124351574191, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13422085700151648, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:1, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:20, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13446499574206527, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13437758804483177, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13429018034759824, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1342027726503647, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13444691139091353, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13435950369368002, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1342720959964465, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13418468829921298, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13442882703976178, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13434141934252825, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:20, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13425401164529474, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025408513368226592, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025565244411541867, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02547783671430835, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025634567757623627, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02621025293595474, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02678593811428585, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027361623292616963, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:20, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0834828077662272, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08372694650677599, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08363953880954247, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08355213111230896, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13456467658952193, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13480881533007075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13472140763283721, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13463399993560368, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:150}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13454659223837015, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13479073097891897, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:20, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13470332328168544, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1346159155844519, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1345285078872184, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1347726466277672, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13468523893053366, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13459783123330016, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11026573182198776, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11050987056253653, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11042246286530302, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:150}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11033505516806952, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:20, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11024764747083599, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11049178621138478, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11040437851415127, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11031697081691774, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11022956311968421, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.110473701860233, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1103862941629995, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11029888646576597, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050255921685376376, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050500060425925176, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:20, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050412652728691644, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05032524503145814, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05023783733422462, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048197607477341, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05039456837753989, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05030716068030637, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05021975298307285, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05046389172362165, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05037648402638813, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05028907632915461, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:124, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:20, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:150}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10326164507656502, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1035057838171138, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10341837611988028, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10357510716319555, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10348769946596205, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10373183820651084, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10364443050927732, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1035570228120438, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10346961511481027, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10371375385535908, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:20, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10362634615812556, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:151}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10353893846089203, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10345153076365851, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1036956695042073, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10360826180697379, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10352085410974027, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10343344641250676, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10367758515305556, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10359017745582202, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10350276975858852, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:20, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.103415362061355, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10365950080190378, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:151}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10357209310467026, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10348468540743676, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08642151482838863, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08666565356893742, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0865782458717039, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0864908381744704, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08640343047723688, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08664756921778566, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:20, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08656016152055215, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08647275382331862, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0863853461260851, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0866294848666339, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08654207716940038, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08645466947216686, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08012222761251249, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08036636635306128, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08027895865582776, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08019155095859425, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:1, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:20, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08010414326136071, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08034828200190952, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08026087430467599, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08017346660744247, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08008605891020894, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08033019765075775, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08024278995352423, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0801553822562907, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13446499574206527, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13470913448261407, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:20, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13462172678538054, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13453431908814703, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13444691139091353, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1346910501314623, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
```

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MATCH (a {`key`:151}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1346036424342288, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13451623473699528, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13442882703976178, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13467296578031054, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13458555808307704, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13449815038584353, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:20, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025896790849324183, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025809383152090665, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025721975454857143, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025634567757623627, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:151}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026454391676503534, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027030076854834648, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08381435420400951, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08372694650677599, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:20, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08397108524732477, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08388367755009127, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08379626985285774, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13480881533007075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350529540706195, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13496554637338598, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:151}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13487813867615248, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13479073097891897, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13503486971946776, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13494746202223423, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:20, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13486005432500073, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1347726466277672, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.135016785368316, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13492937767108248, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13484196997384895, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11050987056253653, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11075400930308534, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
```

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MATCH (a {`key`:151}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11066660160585183, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11057919390861831, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11049178621138478, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:20, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11073592495193359, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11064851725470007, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11056110955746654, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.110473701860233, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11071784060078182, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11063043290354832, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11054302520631479, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050500060425925176, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05074419916647396, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050656791469240445, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:20, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05056938377200694, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048197607477341, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05072611481532221, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05063870711808868, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05055129942085517, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05046389172362165, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05070803046417044, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05062062276693693, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05053321506970341, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:125, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
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MATCH (a {`key`:151}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1035057838171138, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:20, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1037499225576626, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10366251486042909, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10415079234152667, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10406338464429314, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10430752338484195, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10422011568760844, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10413270799037491, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1040453002931414, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1042894390336902, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10420203133645667, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
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MATCH (a {`key`:254}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:20, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10411462363922315, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10402721594198963, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10427135468253843, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10418394698530489, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10409653928807137, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10400913159083787, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10425327033138665, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10416586263415314, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10407845493691963, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10399104723968611, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:20, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1042351859802349, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10414777828300138, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10406037058576786, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08699720000671975, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08724133874726854, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08715393105003502, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08706652335280152, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08697911565556798, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08722325439611678, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08713584669888326, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:1, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:20, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08704843900164974, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08696103130441622, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08720517004496502, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0871177623477315, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08703035465049798, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08069791279084358, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08094205153139239, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08085464383415888, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08076723613692535, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08067982843969183, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:20, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08092396718024063, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08083655948300711, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08074915178577358, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08066174408854006, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08090588282908887, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08081847513185533, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08073106743462181, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350406809203964, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352848196609452, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13519741196371166, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:20, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13511000426647815, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13502259656924462, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352667353097934, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1351793276125599, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350919199153264, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13500451221809287, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352486509586417, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13516124326140816, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13507383556417463, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0262283372871065, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:20, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0264724760276553, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026385068330421778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02629766063318826, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02621025293595474, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026454391676503534, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02694266915760113, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08447744707957414, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08439003938234062, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0843026316851071, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0845467704256559, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:20, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08445936272842237, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08437195503118886, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1353845005084018, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1356286392489506, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13554123155171707, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1354538238544836, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13536641615725006, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13561055489779886, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13552314720056535, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13543573950333182, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:20, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13534833180609832, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1355924705466471, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13550506284941358, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13541765515218007, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11108555574086766, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11132969448141644, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11124228678418292, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11115487908694942, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11106747138971591, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1113116101302647, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:20, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11122420243303117, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11113679473579764, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11104938703856414, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11129352577911293, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1112061180818794, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1111187103846459, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05107574560425628, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05131988434480508, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051232476647571565, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05114506895033804, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:20, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05105766125310453, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051301799993653315, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0512143922964198, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051126984599186284, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05103957690195276, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05128371564250156, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05119630794526805, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051108900248034515, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:126, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10408146899544492, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10432560773599373, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:20, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10423820003876019, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10472647751985778, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10463906982262426, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10488320856317307, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10479580086593954, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10470839316870602, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10462098547147251, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10486512421202131, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10477771651478779, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10469030881755426, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:20, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10462090112032074, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10484703986086953, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10475963216363601, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1046722244664025, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10458481676916898, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10482895550971777, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10474154781248425, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10465414011525073, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10456673241801723, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10481087115856602, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:1, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:20, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10472346346133249, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10463605576409898, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08757288518505087, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08781702392559965, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08772961622836613, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08764220853113261, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08755480083389909, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08779893957444787, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08771153187721437, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08762412417998086, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:20, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08753671648274733, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08778085522329612, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08769344752606262, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08760603982882909, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0812735979691747, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0815177367097235, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08143032901248998, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08134292131525646, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08125551361802294, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08149965235857175, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:20, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08141224466133822, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0813248369641047, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08123742926687118, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08148156800741997, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08139416031018645, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08130675261295293, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13561636609872751, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13586050483927628, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13577309714204278, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13568568944480927, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:20, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13559828174757574, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13584242048812453, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.135755012790891, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1356676050936575, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.135580197396424, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13582433613697278, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13573692843973925, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13564952074250575, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026804022465437612, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027048161205986406, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:20, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02696075350875289, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02687334581151937, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02678593811428585, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027030076854834648, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02694266915760113, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08514053995513877, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08505313225790526, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08496572456067172, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0848783168634382, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.085122455603987, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:20, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08503504790675348, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08494764020951998, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13596018568673293, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13620432442728173, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13611691673004822, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13602950903281472, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13594210133558118, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13618624007613, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13609883237889647, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13601142468166294, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:20, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13592401698442944, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13616815572497823, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1360807480277447, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1359933403305112, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11166124091919877, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11190537965974755, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11181797196251404, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11173056426528054, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11164315656804702, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1118872953085958, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:20, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11179988761136228, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11171247991412878, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11162507221689524, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11186921095744404, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11178180326021053, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.111694395562977, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0516514307825874, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051895569523136204, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05180816182590267, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05172075412866916, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:20, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05163334643143563, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05187748517198443, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051790077474750916, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05170266977751739, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05161526208028387, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05185940082083267, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051771993123599154, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051684585426365635, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:127, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10465715417377604, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10490129291432483, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:20, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10481388521709131, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1053021626981889, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10521475500095538, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545889374150418, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10537148604427066, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10528407834703714, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10519667064980362, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10544080939035241, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1053534016931189, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10526599399588538, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:1, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:20, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10517858629865184, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10542272503920065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10533531734196713, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1052479096447336, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1051605019475001, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10540464068804889, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10531723299081536, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10522982529358185, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10514241759634833, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10538655633689713, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:20, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10529914863966361, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10521174094243009, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08814857036338197, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08839270910393077, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08830530140669723, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08821789370946373, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0881304860122302, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08837462475277899, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08828721705554549, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08819980935831197, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:20, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08811240166107845, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08835654040162724, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08826913270439372, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0881817250071602, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08184928314750582, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08209342188805462, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0820060141908211, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08191860649358756, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08183119879635406, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08207533753690285, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:20, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08198792983966934, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08190052214243582, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08181311444520228, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08205725318575109, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08196984548851757, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08188243779128404, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1361920512770586, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13643619001760743, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1363487823203739, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13626137462314036, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:20, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13617396692590686, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13641810566645565, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13633069796922212, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13624329027198862, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13615588257475508, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13640002131530388, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13631261361807037, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13622520592083687, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027379707643768725, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:20, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027536438687084004, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027449030989850486, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027361623292616963, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08547208639292109, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0857162251334699, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08562881743623636, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08554140973900284, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08545400204176931, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08569814078231812, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:20, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0856107330850846, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08552332538785108, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13653587086506405, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13678000960561285, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1366926019083793, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13660519421114578, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13651778651391228, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1367619252544611, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13667451755722757, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13658710985999406, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:20, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13649970216276053, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13674384090330932, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13665643320607582, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13656902550884228, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11223692609752987, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11248106483807868, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11239365714084515, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11230624944361164, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11221884174637813, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1124629804869269, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:20, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1123755727896934, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11228816509245988, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11220075739522635, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11244489613577516, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11235748843854164, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1122700807413081, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052227115960918516, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05247125470146731, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05238384700423379, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052296439307000266, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:20, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05220903160976675, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05245317035031555, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05236576265308202, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
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MATCH (a {`key`:154}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05227835495584851, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05219094725861499, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05243508599916378, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05234767830193027, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05226027060469674, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:128, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10523283935210714, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10547697809265595, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:1, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:20, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10538957039542242, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554630143873771, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545889374150418, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10570303248205297, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10561562478481945, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10552821708758593, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10544080939035241, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10568494813090121, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1055975404336677, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10551013273643417, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:20, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10542272503920065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10566686377974945, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10557945608251593, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10549204838528241, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10540464068804889, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10564877942859768, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10556137173136416, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10547396403413065, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10538655633689713, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10563069507744594, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:20, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554328738021242, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545587968297888, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08839270910393077, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08863684784447956, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08854944014724604, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08846203245001251, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08837462475277899, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0886187634933278, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08853135579609427, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08844394809886076, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:20, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08835654040162724, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08860067914217604, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08851327144494252, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.088425863747709, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08209342188805462, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08233756062860341, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08225015293136988, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08216274523413636, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08207533753690285, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08231947627745163, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:20, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08223206858021813, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08214466088298461, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08205725318575109, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08230139192629989, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08221398422906635, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08212657653183285, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:2,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13643619001760743, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13668032875815622, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1365929210609227, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13650551336368918, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:20, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13641810566645565, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13666224440700445, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13657483670977094, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1364874290125374, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13640002131530388, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1366441600558527, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1365567523586192, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13646934466138566, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02786798512486632, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:20, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027780577427632798, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027693169730399276, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08580363283070341, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0857162251334699, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08596036387401867, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08587295617678517, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08578554847955164, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08569814078231812, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08594227952286693, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:20, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0858548718256334, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08576746412839988, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13678000960561285, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13702414834616164, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1369367406489281, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1368493329516946, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1367619252544611, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13700606399500986, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1369186562977764, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13683124860054285, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:20, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13674384090330932, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13698797964385814, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1369005719466246, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13681316424939108, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:4,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11248106483807868, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11272520357862748, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11263779588139394, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11255038818416044, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1124629804869269, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1127071192274757, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:20, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1126197115302422, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11253230383300866, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11244489613577516, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11268903487632395, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11260162717909042, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11251421948185691, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05247125470146731, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05271539344201611, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05262798574478258, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052540578047549066, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:1, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:20, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05245317035031555, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052697309090864335, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05260990139363082, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05252249369639731, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05243508599916378, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267922473971258, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052591817042479054, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05250440934524554, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10547697809265595, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10572111683320473, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:20, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10563370913597121, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10612198661706881, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1060345789198353, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10627871766038409, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10619130996315056, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10610390226591705, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10601649456868352, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10626063330923231, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10617322561199881, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10608581791476529, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:20, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10599841021753177, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10624254895808057, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10615514126084705, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10606773356361353, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10598032586638, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1062244646069288, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10613705690969528, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10604964921246175, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10596224151522825, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10620638025577704, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:21, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10611897255854352, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10603156486131, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08896839428226187, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08921253302281068, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08912512532557715, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08903771762834363, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08895030993111011, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0891944486716589, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08910704097442539, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08901963327719187, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:21, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08893222557995835, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08917636432050716, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08908895662327364, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0890015489260401, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08266910706638572, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08291324580693452, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.082825838109701, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08273843041246746, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08265102271523396, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08289516145578275, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:21, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08280775375854924, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08272034606131572, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08263293836408221, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08287707710463099, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08278966940739747, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:2,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08270226171016397, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:2,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13701187519593852, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725601393648731, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371686062392538, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13708119854202028, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:21, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13699379084478674, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13723792958533557, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13715052188810203, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370631141908685, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.136975706493635, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372198452341838, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13713243753695026, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13704502983971675, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028199531562648632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028443670303197433, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:21, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02835626260596391, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028268854908730396, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08646672570626804, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08637931800903453, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.086291910311801, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0865360490523498, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08644864135511628, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08636123365788276, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08627382596064924, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08651796470119803, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:21, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08643055700396451, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.086343149306731, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13735569478394397, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13759983352449276, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13751242582725925, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13742501813002572, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1373376104327922, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.137581749173341, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374943414761075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13740693377887397, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:21, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13731952608164047, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13756366482218924, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13747625712495573, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:4,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13738884942772223, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:4,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11305675001640979, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11330088875695858, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11321348105972508, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11312607336249154, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11303866566525804, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11328280440580683, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:1, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:21, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1131953967085733, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1131079890113398, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11302058131410626, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11326472005465506, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11317731235742155, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11308990466018802, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05304693987979843, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05329107862034722, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0532036709231137, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05311626322588018, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:21, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053028855528646654, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053272994269195455, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053185586571961936, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05309817887472841, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0530107711774949, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053254909918043686, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053167502220810174, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053080094523576656, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:130, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10605266327098706, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10629680201153585, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:21, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10620939431430233, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10669767179539992, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10661026409816642, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1068544028387152, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10676699514148168, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10667958744424816, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10659217974701464, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10683631848756342, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909090, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10674891079032992, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10666150309309641, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:21, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10657409539586288, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10681823413641167, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10673082643917817, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10664341874194463, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10655601104471112, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10680014978525991, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10671274208802639, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10662533439079287, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10653792669355935, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10678206543410815, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:21, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10669465773687464, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10660725003964111, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08954407946059299, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:2,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08978821820114179, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:2,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08970081050390827, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:2,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08961340280667475, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:2,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08952599510944122, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:2,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08977013384999002, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:2,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0896827261527565, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:2,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08959531845552297, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:2,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:21, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08950791075828947, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:2,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08975204949883826, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:2,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08966464180160474, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:2,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08957723410437123, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:2,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08324479224471683, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:2,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08348893098526564, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:2,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0834015232880321, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:2,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08331411559079858, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:2,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08322670789356508, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:2,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08347084663411386, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:2,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:21, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08338343893688034, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:2,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08329603123964684, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:2,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08320862354241332, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:2,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08345276228296211, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:2,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0833653545857286, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:2,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08327794688849507, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:2,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13758756037426964, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13783169911481843, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13774429141758493, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1376568837203514, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:21, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1375694760231179, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378136147636667, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772620706643315, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13763879936919965, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13755139167196612, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377955304125149, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377081227152814, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13762071501804787, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028775216740979746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.029019355481528546, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:21, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028931947784295024, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08712981858183266, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08704241088459914, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08695500318736564, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08686759549013211, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.027272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0871117342306809, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0870243265334474, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08693691883621388, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08684951113898035, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08709364987952914, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:21, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08700624218229563, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0869188344850621, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:2,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13793137996227506, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13817551870282388, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13808811100559035, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13800070330835681, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1379132956111233, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381574343516721, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807002665443857, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798261895720507, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:1, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:21, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13789521125997153, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13813935000052036, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:4,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13805194230328685, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:4,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13796453460605332, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:4,
`saturationRatio`:0.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1136324351947409, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11387657393528972, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11378916623805618, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11370175854082265, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11361435084358915, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11385848958413794, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:21, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11377108188690443, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1136836741896709, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11359626649243737, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11384040523298619, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11375299753575266, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11366558983851913, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05362262505812953, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05386676379867834, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05377935610144482, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05369194840421129, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:21, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053604540706977774, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053848679447526575, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05376127175029304, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05367386405305953, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053586456355826, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0538305950963748, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05374318739914129, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05365577970190776, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:1,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.1353352832366127, `sourceIndex`:131, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:157}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10662834844931818, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10687248718986696, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:21, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10678507949263344, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08140183805570471, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08131443035847118, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08155856909901998, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08147116140178645, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08138375370455293, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0812963460073194, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08154048474786821, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08145307705063469, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08136566935340116, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:21, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13436694069129468, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:4,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1346110794318435, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13452367173460997, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13443626403737644, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13434885634014293, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13459299508069172, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1345055873834582, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1344181796862247, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13433077198899115, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13457491072953995, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:21, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13448750303230644, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1344000953350729, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16444051485760602, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:5,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1645972459009213, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16450983820368778, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16442243050645428, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16457916154976956, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16449175385253603, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:21, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16440434615530253, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1645610771986178, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16447366950138428, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1083783192056476, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:3,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10862245794619638, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10853505024896287, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10844764255172935, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10836023485449585, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10860437359504463, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:21, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10851696589781112, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1084295582005776, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10834215050334407, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10858628924389288, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10849888154665936, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10841147384942583, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05675228770430439, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:1,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05699642644485319, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:1,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05690901874761966, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:1,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05682161105038615, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:21, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05673420335315263, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05697834209370143, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0568909343964679, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056803526699234386, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05671611900200085, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05696025774254967, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05687285004531614, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056785442348082624, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16630129869787724, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16654543743842606, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:21, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22276156699129643, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:7,
`saturationRatio`:0.0, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22300570573184522, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:7,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22291829803461172, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:7,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22283089033737818, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22274348264014465, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22298762138069345, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22290021368345994, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2228128059862264, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22272539828899288, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2229695370295417, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:1, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:21, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22288212933230817, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22279472163507463, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05067235192723909, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:1,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
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MATCH (a {`key`:158}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050916490667787875, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:1,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05082908297055436, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:1,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05074167527332084, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05065426757608732, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05089840631663611, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050810998619402595, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05072359092216908, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:21, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05063618322493556, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05088032196548435, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05079291426825083, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05070550657101731, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13481834767101875, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:4,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13506248641156757, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13497507871433403, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1348876710171005, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.134800263319867, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350444020604158, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:21, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13495699436318226, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13486958666594875, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13478217896871522, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13502631770926402, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13493891001203048, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13485150231479698, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1888512873372591, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:6,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18909542607780788, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18900801838057438, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18892061068334084, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:21, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1888332029861073, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18907734172665613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18898993402942263, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1889025263321891, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1888151186349556, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18905925737550439, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18897184967827085, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18888444198103735, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08133251470962294, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08157665345017173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:21, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08148924575293823, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.03663127777746836, `sourceIndex`:132, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08164597679625349, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08155856909901998, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08180270783956876, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08171530014233525, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08162789244510173, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08154048474786821, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:159}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.081784623488417, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08169721579118348, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08160980809394996, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:21, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1346110794318435, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:4,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13485521817239227, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13476781047515876, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13468040277792526, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13459299508069172, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13483713382124052, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.134749726124007, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13466231842677348, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13457491072953995, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13481904947008877, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:21, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13473164177285524, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1346442340756217, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16492879233870364, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1648413846414701, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1647539769442366, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16491070798755186, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16482330029031836, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16473589259308485, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:21, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16489262363640012, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1648052159391666, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16471780824193308, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10862245794619638, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1088665966867452, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10877918898951167, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10869178129227816, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10860437359504463, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10884851233559345, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:21, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10876110463835992, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10867369694112639, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10858628924389288, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10883042798444167, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10874302028720814, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10865561258997461, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05699642644485319, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:1,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05724056518540198, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:1,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05715315748816846, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05706574979093495, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:21, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05697834209370143, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05722248083425022, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057135073137016705, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05704766543978317, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05696025774254967, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057204396483098455, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05711698878586494, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057029581088631424, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16654543743842606, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10214665505590413, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:1, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:21, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22300570573184522, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:7,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22324984447239404, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:7,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2231624367751605, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22307502907792698, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22298762138069345, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22323176012124227, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22314435242400874, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2230569447267752, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2229695370295417, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2232136757700905, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:21, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22312626807285696, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22303886037562348, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050916490667787875, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:1,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051160629408336676, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:1,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05107322171110315, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05098581401386964, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05089840631663611, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05114254505718491, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0510551373599514, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05096772966271787, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:21, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05088032196548435, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05112446070603315, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05103705300879963, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05094964531156611, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13506248641156757, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13530662515211636, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13521921745488283, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13513180975764932, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350444020604158, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13528854080096459, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:21, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13520113310373108, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13511372540649755, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13502631770926402, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352704564498128, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1351830487525793, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13509564105534577, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18909542607780788, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18933956481835668, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18925215712112317, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18916474942388964, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:21, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18907734172665613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18932148046720493, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1892340727699714, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18914666507273792, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
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MATCH (a {`key`:159}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18905925737550439, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18930339611605318, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18921598841881968, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18912858072158614, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08157665345017173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08182079219072054, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:21, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08173338449348701, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:133, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08222166197458461, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08213425427735109, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08237839301789988, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08229098532066635, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08220357762343285, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08211616992619933, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08236030866674811, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0822729009695146, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08218549327228108, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:21, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1351867646101746, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:4,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1354309033507234, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13534349565348988, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13525608795625635, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13516868025902282, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13541281899957164, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1353254113023381, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13523800360510457, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13515059590787107, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13539473464841986, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:21, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13530732695118633, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13521991925395282, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16526033877648597, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16550447751703473, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16541706981980123, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16532966212256772, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1652422544253342, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16548639316588298, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16539898546864948, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16531157777141597, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:21, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16522417007418244, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16546830881473124, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16538090111749773, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1652934934202642, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10919814312452751, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1094422818650763, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10935487416784277, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10926746647060927, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10918005877337573, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10942419751392456, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:21, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10933678981669102, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10924938211945752, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10916197442222399, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10940611316277278, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10931870546553925, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10923129776830574, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0575721116231843, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:1,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0578162503637331, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05772884266649958, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057641434969266055, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:1, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:21, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057554027272032536, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05779816601258134, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05771075831534781, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05762335061811429, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057535942920880774, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05778008166142956, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.057692673964196056, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05760526626696253, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10280974793146877, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10272234023423525, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:21, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22358139091017634, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:7,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22382552965072516, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22373812195349163, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2236507142562581, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22356330655902457, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2238074452995734, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22372003760233986, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22363262990510632, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2235452222078728, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2237893609484216, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:21, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22370195325118808, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22361454555395455, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05149217584611899, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:1,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05173631458666778, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051648906889434264, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05156149919220075, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051474091494967227, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05171823023551603, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05163082253828251, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05154341484104899, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:21, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05145600714381546, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05170014588436426, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05161273818713075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05152533048989722, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13563817158989866, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13588231033044745, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13579490263321395, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13570749493598042, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13562008723874688, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1358642259792957, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:21, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13577681828206217, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13568941058482864, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1356020028875951, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13584614162814393, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1357587339309104, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13567132623367686, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.189671111256139, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1899152499966878, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18982784229945426, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18974043460222076, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:21, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18965302690498723, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18989716564553602, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18980975794830252, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.189722350251069, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18963494255383548, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1898790812943843, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18979167359715077, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18970426589991723, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08215233862850285, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08239647736905165, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:21, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08230906967181811, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:134, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08279734715291572, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0827099394556822, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08295407819623098, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08286667049899747, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08277926280176395, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08269185510453043, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08293599384507923, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08284858614784571, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08276117845061219, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:21, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1357624497885057, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:4,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1360065885290545, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13591918083182097, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13583177313458747, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13574436543735396, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13598850417790273, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13590109648066923, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13581368878343572, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1357262810862022, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13597041982675098, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:21, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13588301212951748, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13579560443228395, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16583602395481706, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16608016269536588, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16599275499813235, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:161}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16590534730089881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16581793960366528, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1660620783442141, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16597467064698057, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16588726294974707, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:21, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16579985525251356, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16604399399306236, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16595658629582882, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16586917859859532, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10977382830285862, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11001796704340741, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1099305593461739, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10984315164894037, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10975574395170684, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10999988269225566, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:1, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:21, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10991247499502213, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10982506729778863, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10973765960055509, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10998179834110389, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10989439064387038, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10980698294663685, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058147796801515404, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:1,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05839193554206422, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058304527844830686, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05821712014759717, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:21, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058129712450363656, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05837385119091244, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058286443493678924, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05819903579644541, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05811162809921188, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05835576683976068, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058268359142527155, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10347284080703338, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10338543310979988, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10329802541256636, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:21, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22415707608850743, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22440121482905626, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22431380713182272, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2242263994345892, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2241389917373557, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22438313047790448, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22429572278067095, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22420831508343742, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2241209073862039, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2243650461267527, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:21, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22427763842951917, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22419023073228567, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0520678610244501, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0523119997649989, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05222459206776538, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05213718437053186, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05204977667329834, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05229391541384714, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05220650771661362, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052119100019380096, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:21, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05203169232214658, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05227583106269538, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05218842336546185, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052101015668228334, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13621385676822978, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13645799550877857, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13637058781154507, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13628318011431154, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13619577241707803, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13643991115762683, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:21, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1363525034603933, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13626509576315976, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13617768806592626, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13642182680647505, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13633441910924152, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.136247011412008, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19024679643447012, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1904909351750189, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19040352747778538, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19031611978055188, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:21, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19022871208331835, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19047285082386714, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19038544312663364, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1902980354294001, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1902106277321666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1904547664727154, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1903673587754819, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19027995107824835, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08272802380683396, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08297216254738275, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:21, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08288475485014923, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:135, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08337303233124682, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0832856246340133, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08352976337456211, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08344235567732858, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08335494798009506, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08326754028286155, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08351167902341035, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08342427132617682, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08333686362894331, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:21, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13633813496683683, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:4,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13658227370738563, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1364948660101521, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1364074583129186, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13632005061568506, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13656418935623385, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13647678165900035, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1363893739617668, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13630196626453328, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1365461050050821, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:21, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13645869730784857, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13637128961061504, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16641170913314818, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16656844017646344, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16648103247922993, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16639362478199643, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1665503558253117, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1664629481280782, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:1, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:21, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16637554043084465, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16653227147415994, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16644486377692644, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11034951348118972, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11059365222173854, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11050624452450501, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11041883682727148, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11033142913003798, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11057556787058677, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:21, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11048816017335324, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11040075247611973, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1103133447788862, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11055748351943502, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11047007582220149, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11038266812496796, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058723481979846524, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:1,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058967620720395325, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05888021302316179, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05879280532592829, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:21, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05870539762869476, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05894953636924356, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05886212867201003, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05877472097477652, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058687313277543, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05893145201809179, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10380438724481571, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10404852598536452, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.103961118288131, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10387371059089748, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:21, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22473276126683855, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22497690000738738, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22488949231015384, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2248020846129203, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22471467691568678, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2249588156562356, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22487140795900207, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22478400026176854, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22469659256453506, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22494073130508382, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:21, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2248533236078503, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22476591591061681, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05264354620278122, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05288768494333001, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05280027724609649, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05271286954886298, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05262546185162946, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05286960059217825, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05278219289494473, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05269478519771121, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:21, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05260737750047769, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052851516241026485, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05276410854379297, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267670084655944, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13678954194656087, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370336806871097, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13694627298987616, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13685886529264266, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13677145759540912, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13701559633595792, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:21, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13692818863872439, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13684078094149088, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13675337324425735, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13699751198480614, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13691010428757264, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1368226965903391, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19082248161280121, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19106662035335004, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1909792126561165, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19089180495888297, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:21, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19080439726164947, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19104853600219826, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19096112830496473, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19087372060773122, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1907863129104977, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19103045165104648, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19094304395381298, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19085563625657945, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08330370898516506, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08354784772571387, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
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MATCH (a {`key`:255}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:21, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08346044002848034, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:136, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08361717107179562, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08352976337456211, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0837739021151109, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08368649441787737, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08359908672064387, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08351167902341035, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08375581776395914, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08366841006672562, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0835810023694921, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:21, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13658227370738563, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:4,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13682641244793442, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13673900475070092, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13665159705346738, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13656418935623385, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13680832809678267, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13672092039954914, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1366335127023156, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1365461050050821, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1367902437456309, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:1, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:21, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13670283604839736, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13661542835116386, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16689998661424577, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16681257891701223, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16672517121977873, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.166881902263094, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16679449456586048, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16670708686862695, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:21, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16686381791194227, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16677641021470874, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666890025174752, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11059365222173854, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11083779096228732, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1107503832650538, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1106629755678203, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11057556787058677, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11081970661113555, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:21, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11073229891390204, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11064489121666853, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11055748351943502, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1108016222599838, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11071421456275028, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11062680686551678, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058967620720395325, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:1,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05921175946094411, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05912435176371061, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05903694406647708, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:21, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05894953636924356, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05919367510979235, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05910626741255883, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05901885971532532, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05893145201809179, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10413593368259802, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10404852598536452, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10429266472591331, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10420525702867978, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10411784933144626, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:21, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22497690000738738, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22522103874793617, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22513363105070264, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22504622335346913, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2249588156562356, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2252029543967844, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2251155466995509, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22502813900231736, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22494073130508382, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22518487004563265, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:21, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2250974623483991, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22501005465116558, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05288768494333001, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05313182368387881, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05304441598664529, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05295700828941177, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05286960059217825, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05311373933272705, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05302633163549352, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05293892393826001, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:21, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052851516241026485, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05309565498157528, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05300824728434176, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05292083958710824, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13703368068710 97, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13727781942765846, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13719041173042498, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13710300403319145, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13701559633595792, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372597350765067, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:21, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371723273792732, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13708491968203967, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13699751198480614, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13724165072535496, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13715424302812143, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370668353308879, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19106662035335004, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19131075909389883, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19122335139666653, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1911359436994318, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:21, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19104853600219826, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19129267474274705, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19120526704551355, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19111785934828002, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19103045165104648, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19127459039159528, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1911871826943618, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19109977499712827, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08354784772571387, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08379198646626267, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:21, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08370457876902915, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:137, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08419285625012674, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08410544855289322, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08434958729344201, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0842621795962085, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08417477189897497, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08408736420174145, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08433150294229026, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08424409524505674, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08415668754782321, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:1, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:21, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13715795888571672, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:4,
`saturationRatio`:0.6, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13740209762626554, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.137314689929032, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13722728223179848, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13713987453456497, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13738401327511376, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13729660557788023, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13720919788064673, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1371217901834132, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.137365928923962, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:21, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13727852122672848, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13719111352949495, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1672315330520281, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16747567179257689, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16738826409534338, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16730085639810985, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16721344870087632, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16745758744142514, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1673701797441916, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16728277204695807, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:21, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16719536434972457, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16743950309027336, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16735209539303986, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16726468769580632, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11116933740006965, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11141347614061843, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11132606844338493, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11123866074615141, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11115125304891788, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11139539178946668, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:21, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11130798409223315, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11122057639499963, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1111331686977661, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11137730743831491, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1112898997410814, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11120249204384788, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05954330589872643, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:1,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05978744463927523, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0597000369420417, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05961262924480819, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:21, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05952522154757467, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05976936028812347, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05968195259088995, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059594544893656426, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10479902655816266, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10471161886092914, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10462421116369561, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10486834990424442, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10478094220701088, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10469353450977738, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:21, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22555258518571847, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2257967239262673, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22570931622903376, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22562190853180022, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2255345008345667, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22577863957511551, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22569123187788198, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22560382418064845, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22551641648341492, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22576055522396374, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:21, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2256731475267302, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22558573982949667, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05346337012166113, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05370750886220993, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0536201011649764, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053532693467742885, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05344528577050936, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053689424511058154, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05360201681382464, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05351460911659112, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:21, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0534272014193576, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05367134015990639, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05358393246267287, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05349652476543936, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13760936586544079, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13785350460598958, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13776609690875607, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13767868921152254, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13759128151428904, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13783542025483783, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:21, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377480125576043, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1376606048603708, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1375731971631373, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13781733590368606, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772992820645255, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13764252050921905, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19164230553168113, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19188644427222992, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19179903657499642, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19171162887776289, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:21, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19162422118052935, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19186835992107817, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19178095222384464, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1916935445266111, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1916061368293776, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1918502755699264, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19176286787269287, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19167546017545936, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08412353290404498, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08436767164459379, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:1, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:21, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08428026394736025, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:138, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08476854142845786, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08468113373122432, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08492527247177313, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08483786477453961, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08475045707730608, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08466304938007257, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08490718812062137, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08481978042338785, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08473237272615433, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:21, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13773364406404784, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:4,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13797778280459666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13789037510736313, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13780296741012962, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1377155597128961, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13795969845344488, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13787229075621138, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13778488305897785, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13769747536174431, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13794161410229314, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:21, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1378542064050596, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13776679870782607, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16780721823035918, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.168051356970908, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16796394927367447, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16787654157644094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16778913387920744, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803327261975623, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1679458649225227, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678584572252892, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:21, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16777104952805566, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16801518826860445, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16792778057137092, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16784037287413742, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11174502257840076, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11198916131894956, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11190175362171603, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11181434592448251, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11172693822724901, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11197107696779779, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:21, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11188366927056427, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11179626157333074, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11170885387609723, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11195299261664601, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1118655849194125, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11177817722217899, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06011899107705755, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:1,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06036312981760634, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06027572212037282, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0601883144231393, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:21, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06010090672590579, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060345045466454576, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06025763776922106, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1054621194337273, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10537471173649378, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10528730403926025, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10519989634202673, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10544403508257552, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.105356627385342, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10526921968810848, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:21, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2261282703640496, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22637240910459838, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22628500140736485, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22619759371013134, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2261101860128978, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2263543247534466, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2262669170562131, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22617950935897957, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2260921016617461, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22633624040229486, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:21, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22624883270506133, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22616142500782785, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05403905529999225, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054283194040541036, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05419578634330752, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05410837864607399, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05402097094884047, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054265109689389274, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054177701992155755, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05409029429492223, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:21, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05400288659768871, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0542470253382375, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054159617641003986, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05407220994377047, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381850510437719, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1384291897843207, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13834178208708717, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13825437438985366, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13816696669262013, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13841110543316895, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:21, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13832369773593542, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13823629003870191, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13814888234146838, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13839302108201718, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13830561338478367, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13821820568755014, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19221799071001225, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19246212945056107, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19237472175332754, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.192287314056094, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:1, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:21, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1921999063588605, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1924440450994093, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19235663740217576, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19226922970494223, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19218182200770872, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19242596074825752, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19233855305102399, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19225114535379048, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0846992180823761, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08494335682292489, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:21, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08485594912569137, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:139, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08534422660678896, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08525681890955544, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08550095765010425, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
```

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MATCH (a {`key`:166}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08541354995287072, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0853261422556372, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08523873455840368, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08548287329895247, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08539546560171897, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08530805790448545, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:22, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13830932924237893, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:4,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13855346798292772, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13846606028569422, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1383786525884607, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13829124489122718, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.138535383631776, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13844797593454247, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13836056823730894, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13827316054007543, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13851729928062423, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:22, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1384298915833907, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13834248388615716, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16838290340869033, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1685396344520056, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684522267547721, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16836481905753856, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852155010085385, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684341424036203, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:22, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16834673470638678, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850346574970207, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16841605805246854, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11232070775673186, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11256484649728067, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11247743880004714, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:166}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11239003110281363, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11230262340558012, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1125467621461289, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:22, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11245935444889538, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11237194675166187, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11228453905442834, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11252867779497713, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11244127009774363, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11235386240051012, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06069467625538866, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:1,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06093881499593746, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06085140729870394, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06076399960147041, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:22, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060676591904236896, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06092073064478568, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10579366587150961, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10603780461205842, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10595039691482488, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10586298921759137, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10577558152035783, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10601972026090664, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10593231256367312, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10584490486643959, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:22, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22670395554238068, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2269480942829295, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22686068658569597, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22677327888846244, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22668587119122896, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22693000993177773, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2268426022345442, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22675519453731072, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22666778684007718, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22691192558062595, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:22, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22682451788339247, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22673711018615894, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054614740478323355, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05485887921887214, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05477147152163862, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05468406382440511, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05459665612717159, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05484079486772038, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05475338717048686, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054665979473253336, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:22, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:166}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05457857177601982, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05482271051656862, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054735302819335106, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05464789512210159, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.138760736222103, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390048749626518, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13891746726541826, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13883005956818475, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13874265187095125, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13898679061150004, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:1, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:22, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1388993829142665, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.138811975217033, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387245675197995, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1389687062603483, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13888129856311476, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13879389086588126, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19279367588834337, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19303781462889216, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19295040693165863, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19286299923442513, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:22, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1927755915371916, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
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MATCH (a {`key`:166}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930197302777404, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19293232258050685, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19284491488327335, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19275750718603982, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930016459265886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1929142382293551, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19282683053212157, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:140, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0852749032607072, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.085519042001256, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:22, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0854316343040225, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:140, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08558836534733776, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08550095765010425, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08574509639065303, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08565768869341951, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.085570280996186, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08548287329895247, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08572701203950128, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08563960434226775, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08555219664503425, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:22, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13855346798292772, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:4,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13879760672347655, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13871019902624304, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1386227913290095, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.138535383631776, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1387795223723248, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13869211467509127, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13860470697785776, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13851729928062423, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13876143802117302, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:22, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1386740303239395, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13858662262670599, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:4,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1688711808897879, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1687837731925544, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16869636549532088, `foldPressure`:0.382716049382716O3,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16885309653863614, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16876568884140264, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1686782811441691, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:22, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16883501218748437, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16874760449025086, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16866019679301733, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11256484649728067, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11280898523782946, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11272157754059596, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11263416984336243, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1125467621461289, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11279090088667769, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:22, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11270349318944418, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11261608549221065, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11252867779497713, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11277281653552594, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11268540883829241, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11259800114105888, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06093881499593746, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:1,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061182953736486245, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06109554603925273, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061008138342019215, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:22, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06092073064478568, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10612521230929192, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10603780461205842, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1062819433526072, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10619453565537368, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10610712795814016, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10601972026090664, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10626385900145544, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10617645130422192, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1060890436069884, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:22, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2269480942829295, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22719223302347832, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2271048253262448, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22701741762901126, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22693000993177773, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22717414867232655, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22708674097509302, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22699933327785948, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
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MATCH (a {`key`:167}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22691192558062595, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22715606432117477, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:22, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22706865662394124, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2269812489267077, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:7,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05485887921887214, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05510301795942094, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055015610262187424, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05492820256495391, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05484079486772038, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05508493360826918, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054997525911035655, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05491011821380214, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:1, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:22, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05482271051656862, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05506684925711742, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05497944155988389, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054892033862650375, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390048749626518, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13924901370320059, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13916160600596708, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13907419830873355, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13898679061150004, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13923092935204887, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:22, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13914352165481533, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390561139575818, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1389687062603483, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392128450008971, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13912543730366356, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13903802960643005, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19303781462889216, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19328195336944096, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19319454567220745, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19310713797497392, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:22, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930197302777404, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19326386901828918, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19317646132105568, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19308905362382214, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930016459265886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19324578466713743, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1931583769699039, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19307096927267037, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.085519042001256, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0857631807418048, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:22, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08567577304457127, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:141, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08616405052566888, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08607664282843536, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08632078156898415, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08623337387175063, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08614596617451711, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0860585584772836, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08630269721783239, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08621528952059886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08612788182336535, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:22, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13912915316125887, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:4,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13937329190180764, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13928588420457413, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13919847650734063, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1391110688101071, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1393552075506559, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13926779985342239, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13918039215618888, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13909298445895535, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13933712319950414, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:22, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13924971550227064, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:4,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1391623078050371, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:4,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16920272732757022, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16944686606811904, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1693594583708855, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16927205067365197, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16918464297641847, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16942878171696726, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16934137401973373, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1692539663225002, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:22, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1691665586252667, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1694106973658155, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16932328966858196, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16923588197134845, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11314053167561178, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11338467041616057, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11329726271892707, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11320985502169353, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11312244732446, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11336658606500882, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:22, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11327917836777529, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11319177067054176, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11310436297330825, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11334850171385705, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11326109401662351, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11317368631939001, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061514500174268565, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:1,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061758638914817365, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06167123121758385, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06158382352035032, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:22, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10678830518485655, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10670089748762304, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10661348979038951, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10685762853093832, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10677022083370478, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10668281313647128, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10659540543923776, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10683954417978654, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10675213648255304, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10666472878531952, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:22, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2275237794612606, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22776791820180942, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22768051050457588, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22759310280734235, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22750569511010882, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22774983385065764, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2276624261534241, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22757501845619058, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2274876107589571, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22773174949950586, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:1, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:22, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22764434180227233, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:7,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22755693410503885, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:7,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.4060058497098381, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05543456439720326, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05567870313775205, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05559129544051853, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05550388774328502, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
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MATCH (a {`key`:168}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0554164800460515, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05566061878660029, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05557321108936677, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05548580339213326, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:22, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055398395694899724, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055642534435448525, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05555512673821501, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055467719040981495, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1395805601409829, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13982469888153173, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1397372911842982, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13964988348706467, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13956247578983116, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13980661453037996, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:22, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13971920683314645, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13963179913591292, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13954439143867942, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1397885301792282, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1397011224819947, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13961371478476117, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19361349980722328, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19385763854777208, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19377023085053854, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19368282315330504, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:22, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1935954154560715, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1938395541966203, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1937521464993868, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19366473880215326, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19357733110491976, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19382146984546855, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19373406214823502, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19364665445100152, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.14936120510359183, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08609472717958712, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08633886592013591, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:22, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0862514582229024, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:142, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08673973570399998, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:2,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08665232800676646, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:2,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08689646674731527, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:2,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08680905905008174, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:2,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08672165135284822, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:2,
`saturationRatio`:0.6, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08663424365561471, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:2,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0868783823961635, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:2,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08679097469892998, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:2,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.08670356700169646, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:2,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:22, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13970483833958997, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:4,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13994897708013876, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:4,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13986156938290525, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:4,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13977416168567172, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:4,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1396867539884382, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:4,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.139930892728987, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:4,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13984348503175348, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:4,
`saturationRatio`:0.7, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13975607733451997, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:4,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13966866963728644, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:4,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13991280837783526, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:4,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:22, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:169}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13982540068060173, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:4,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1397379929833682, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:4,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16977841250590134, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17002255124645013, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699351435492166, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698477358519831, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1697603281547496, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000446689529836, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16991705919806485, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982965150083135, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:22, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16974224380359781, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
```

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MATCH (a {`key`:169}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699863825441466, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698989748469131, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16981156714967957, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11371621685394288, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11396035559449168, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11387294789725817, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11378554020002464, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11369813250279114, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11394227124333993, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:22, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1138548635461064, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11376745584887289, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11368004815163936, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11392418689218815, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11383677919495465, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11374937149772112, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:3,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06209018535259968, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:1,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.062334324093148485, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:1,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06224691639591495, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:1,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10745139806042117, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:22, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10736399036318767, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10727658266595415, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10718917496872064, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10743331370926942, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10734590601203589, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10725849831480239, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10717109061756887, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10741522935811766, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10732782166088414, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10724041396365062, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:1, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:22, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22809946463959171, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22834360338014054, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.228256195682907, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
```

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MATCH (a {`key`:169}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22816878798567347, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22808138028844, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22832551902898876, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22823811133175523, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22815070363452175, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22806329593728822, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22830743467783698, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:7,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:22, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2282200269806035, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:7,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22813261928336998, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:7,
`saturationRatio`:0.2, `shadowDecay`:0.4060058497098381, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05601024957553437, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05625438831608317, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05616698061884965, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05607957292161613, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055992165224382606, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05623630396493139, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05614889626769789, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05606148857046436, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:22, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055974080873230844, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056218219613779645, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05613081191654612, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0560434042193126, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14015624531931403, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14040038405986283, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14031297636262933, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14022556866539578, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14013816096816226, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14038229970871105, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:22, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14029489201147755, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.140207484314244, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14012007661701048, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14036421535755933, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14027680766032578, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14018939996309226, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:4,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.1353352832366127, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
```

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MATCH (a {`key`:169}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1941891849855544, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1944333237261032, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19434591602886966, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19425850833163613, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:22, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19417110063440263, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19441523937495142, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1943278316777179, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19424042398048438, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19415301628325085, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19439715502379964, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19430974732656614, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:169}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1942223396293326, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:6,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.14936120510359183, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08667041235791823, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08691455109846702, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:22, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08682714340123351, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:143, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05185084881062895, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052426533988960064, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05324635790783997, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05382204308617109, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0543977282645022, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055217552183382104, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055793237361713224, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:22, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05636892254004433, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05661306128059312, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05718874645892424, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057764431637255344, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058340116815586464, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058584255556135265, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05915994073446637, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05973562591279749, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0603113110911286, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
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MATCH (a {`key`:234}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0605554498316774, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:22, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06113113501000852, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061706820188339624, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19789585740687293, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:6,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19813999614742173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:6,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19871568132575287, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:6,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19929136650408397, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1998670516824151, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20011119042296385, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.200686875601295, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2012625607796261, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:22, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2018382459579572, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20208238469850603, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20265806987683713, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20323375505516825, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492537438455706, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1494978825861194, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1500735677644505, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15064925294278164, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15122493812111273, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1514690768616615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:22, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15204476203999265, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15262044721832374, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15319613239665486, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15344027113720365, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15401595631553477, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15459164149386587, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09816196386861685, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09840610260916564, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0002441387405487964, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:0,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.000819823918879909, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:0,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:1, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:22, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0013955090972110214, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.001971194275542134, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.002791018194422043, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0033667033727531556, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.003942388551084268, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004186527291633065, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0047622124699641 77, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.00533789764829529, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16539827838715537, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:6,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:22, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16564241712770417, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:6,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1662181023060353, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:6,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1667937874843664, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16736947266269753, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16818929658157744, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16876498175990853, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16934066693823965, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17016049085711957, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:22, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17073617603545066, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11724803609702364, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:4,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11749217483757245, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11806786001590355, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11864354519423466, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1192192303725658, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11946336911311459, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1200390542914457, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12061473946977683, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12119042464810793, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:22, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12143456338865671, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12201024856698782, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12258593374531895, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15037018258004056, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:5,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15061432132058936, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:5,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1511900064989205, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:5,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517656916772516, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15234137685558272, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1525855155961315, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15316120077446263, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:22, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15373688595279372, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15431257113112487, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15455670987167364, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15513239505000473, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
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MATCH (a {`key`:234}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15570808022833588, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15543859193947343, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15568273068002222, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15625841585835332, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15683410103668446, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15740978621501556, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:22, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15765392495556435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1582296101338955, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1588052953122266, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1593809804905577, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596251192311065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16020080440943762, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16077648958776872, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:0, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050455339713417924, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:1,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05127516363229784, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:22, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052094987551177745, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05291481147005766, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05349049664838877, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05406618182671988, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05488600574559979, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055461690923930905, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05603737610226201, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05661306128059312, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:22, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05685720002114193, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05743288519947303, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058008570377804144, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058584255556135265, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05882839429668405, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05940407947501517, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05997976465334628, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
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MATCH (a {`key`:235}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0605554498316774, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0607995885722262, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061375273750557305, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:22, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06195095892888841, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19813999614742173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:6,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19838413488797055, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:6,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19895982006630164, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19953550524463276, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20011119042296385, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20035532916351267, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2009310143418438, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
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MATCH (a {`key`:235}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2015066995201749, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20208238469850603, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:22, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2023265234390548, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20290220861738592, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20347789379571704, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1494978825861194, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1497420213266682, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15031770650499932, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1508933916833304, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1514690768616615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15171321560221032, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15228890078054144, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:1, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:22, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15286458595887253, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15344027113720365, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15368440987775245, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15426009505608357, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15483578023441466, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09840610260916564, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10745139806042117, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0002441387405487964, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:0,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0010639626594287053, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0016396478377598 18, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:22, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0024594717566397265, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0030351569349708393, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0036108421133019 52, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0041865272916330 65, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0044306660321818 61, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0050063512105129 73, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0055820363888440 86, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16564241712770417, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:6,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16588655586825296, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:6,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:22, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1664622410465841, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1670379262249152, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16785775014379511, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16843343532212623, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16900912050045733, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16982894441933724, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17040462959766836, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17098031477599945, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:22, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11749217483757245, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11773631357812124, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11831199875645235, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11888768393478348, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11946336911311459, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11970750785366338, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1202831930319945, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12085887821032561, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12143456338865671, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12167870212920551, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:22, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12225438730753664, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12283007248586775, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15061432132058936, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:5,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15085846006113818, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:5,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1514341452394693, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1520098304178004, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1525855155961315, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528296543366803, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340533951501142, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15398102469334254, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:22, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15455670987167364, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15480084861222243, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15537653379055355, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15595221896888467, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15568273068002222, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15592686942057102, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15650255459890214, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15707823977723326, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15765392495556435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15789806369611317, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:22, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15847374887444426, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15904943405277538, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596251192311065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
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MATCH (a {`key`:235}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1598692579716553, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16044494314998642, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1610206283283175, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:1, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05094361719451552, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05151930237284664, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052007579853944226, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:22, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052583265032275346, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05282740377282414, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053403088951155246, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053978774129486366, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05455445930781747, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05479859804836627, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05537428322669738, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0559499684050285, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05652565358335961, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056769792323908407, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:22, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05734547750223951, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05792116268057063, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05849684785890174, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05874098659945054, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05931667177778166, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059892356956112766, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06046804213444388, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06071218087499267, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061287866053323786, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0618635512316549, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:22, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19805258845018822, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:6,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19829672719073702, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19887241236906814, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19944809754739923, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20002378272573035, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20026792146627914, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
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MATCH (a {`key`:236}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20084360664461026, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20141929182294138, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2019949770012725, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2022391157418213, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:1, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:22, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20281480092015242, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2033904860984835, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14941047488888587, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1496546136294347, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15023029880776578, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1508059839860969, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
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MATCH (a {`key`:236}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15138166916442802, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1516258079049768, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1522014930833079, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15277717826163903, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:22, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15335286343997015, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15359700218051892, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15417268735885004, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15474837253718116, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10678830518485655, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10736399036318767, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.000819823918879909, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:0,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0010639626594287053, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0015522401405262984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0021279253188574107, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:22, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.002372064059406207, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.00294774923773732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0035234344160684325, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004099119594399545, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004343258334948342, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004918943513279453, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005494628691610566, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1662181023060353, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:6,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1664622410465841, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16637483334935058, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:22, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16695051852768167, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16752620370601282, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16777034244656158, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1683460276248927, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16892171280322382, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16949739798155494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16974153672210374, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17031722190043483, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17089290707876595, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11740476714033894, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:22, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11764890588088771, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11822459105921884, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11880027623754995, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11937596141588105, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11962010015642988, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12019578533476098, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12077147051309209, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1213471556914232, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.121591294431972, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12216697961030311, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:22, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12274266478863421, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15052691362335585, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:5,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15077105236390465, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15134673754223577, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15192242272056689, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152498107898898, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15274224663944677, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15331793181777792, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153893616996109, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15446930217444013, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:22, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15471344091498893, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15528912609332002, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15586481127165114, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555953229827887, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1558394617233375, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1564151469016686, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15699083207999973, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15756651725833085, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15781065599887964, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15838634117721076, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:22, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15896202635554188, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15953771153387297, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978185027442177, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1603575354527529, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.160933220631084, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:2, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050612070756733206, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05085620949728201, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05143189467561311, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05192017215671071, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05249585733504182, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:22, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052739996075590614, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053315681253921735, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053891366432252834, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054467051610583954, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054711190351132755, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05528687552946386, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05586256070779498, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05643824588612609, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05668238462667489, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057258069805005994, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:22, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057833754983337114, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05840944016166822, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05865357890221702, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05922926408054813, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05980494925887925, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060380634437210354, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060624773177759154, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061200458356090275, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06177614353442138, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1979651807529547, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:6,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:1, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:22, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19820931949350348, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1987850046718346, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19936068985016572, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19993637502849684, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2001805137690456, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20075619894737676, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20133188412570785, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20190756930403897, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2021517080445878, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20272739322291888, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:22, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20330307840125, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14932306719165236, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14956720593220116, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15014289111053225, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1507185762888634, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15129426146719 45, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15153840020774328, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15211408538607443, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15268977056440552, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15326545574273662, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:22, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1535095944832854, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15408527966161653, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10612521230929192, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10670089748762304, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10727658266595415, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0013955090972110214, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:0,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.001639647837759818, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0015522401405262984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.002040517621623891, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0022846563621726876, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:22, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0028603415405038003, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0034360267188349126, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004011711897166026, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004255850637714821, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004831535816045934, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005407220994377047, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1667937874843664, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1670379262249152, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16695051852768167, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16686311083044816, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:22, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16743879600877928, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16768293474932808, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682586199276592, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1688343051059903, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1694099902843214, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696541290248702, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17022981420320132, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17080549938153244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11731735944310542, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11756149818365423, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:22, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11813718336198534, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11871286854031642, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11928855371864755, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11953269245919634, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12010837763752745, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12068406281585856, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12125974799418969, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12150388673473847, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12207957191306958, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12265525709140071, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:22, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15043950592612232, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1506836446666711, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15125932984500226, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15183501502333335, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15241070020166447, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15265483894221327, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532305241205444, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15380620929887548, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15438189447720663, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15462603321775542, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:22, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1552017183960865, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1557774035744176, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555079152855552, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15575205402610398, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15632773920443507, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15690342438276622, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1574791095610973, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1577232483016461, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15829893347997726, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15887461865830835, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:22, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15945030383663947, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15969444257718823, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16027012775551938, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16084581293385047, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05052466305949969, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05076880180004848, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051344486978379594, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05183276445947719, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0524084496378083, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:22, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05322827355668821, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05380395873501932, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054379643913350435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05519946783223034, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055775153010561455, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05635083818889257, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05659497692944136, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057170662107772476, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057746347286103596, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:1, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:22, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0583220324644347, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0585661712049835, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05914185638331461, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059717541561645715, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060293226739976835, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060537365480525636, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06111305065885674, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06168873583718786, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19787777305572118, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:6,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19812191179626995, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:22, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1986975969746011, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1992732821529322, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1998489673312633, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2000931060718121, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20066879125014322, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20124447642847434, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20182016160680546, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20206430034735426, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20263998552568535, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2032156707040165, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:22, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14923565949441883, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14947979823496763, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15005548341329875, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15063116859162987, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15120685376996096, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15145099251050978, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1520266776888409, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.152602362867172, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1531780480455031, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1534221867860519, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:22, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10579366587150961, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10603780461205842, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10661348979038951, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10718917496872064, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.001971194275542134, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:0,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0021279253188574107, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.002040517621623891, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0021972486649391677, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0027729338432702804, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:22, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.003348619021601393, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0039243041999325054, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004168442940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004744128118812414, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005319813297143527, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16736947266269753, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16752620370601282, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16743879600877928, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16735138831154575, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:22, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16759552705209457, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16817121223042567, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16874689740875679, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16932258258708788, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701424065059678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1707180916842989, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11722995174587192, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1174740904864207, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1180497756647518, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:22, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11862546084308291, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11920114602141402, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11944528476196281, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12002096994029392, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12059665511862505, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12117234029695616, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12141647903750495, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12199216421583607, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12256784939416718, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15035209822888881, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:22, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1505962369694376, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15117192214776873, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15174760732609982, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15232329250443097, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15256743124497976, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15314311642331085, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15371880160164197, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542944867799731, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545386255205219, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15511431069885298, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:22, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15568999587718413, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15542050758832168, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15566464632887045, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15624033150720157, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568160166855327, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15739170186386378, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576358406044126, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15821152578274372, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15878721096107482, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15936289613940594, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:22, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15960703487995473, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16018272005828585, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16075840523661694, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05043725536226617, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05125707928114608, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05207690320002599, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05289672711890589, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05347241229723701, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:1, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:22, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05404809747556812, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05486792139444803, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055443606572779136, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056019291751110256, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05659497692944136, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05683911566999016, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.057414800848321276, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05799048602665239, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0585661712049835, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:22, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058810309945532296, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0593859951238634, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059961680302194516, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060537365480525636, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06078150422107443, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06135718939940554, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06193287457773666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19812191179626995, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:6,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19836605053681877, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1989417357151499, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:22, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19951742089348098, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2000931060718121, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2003372448123609, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20091292999069202, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20148861516902314, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20206430034735426, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20230843908790305, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20288412426623417, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20345980944456526, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14947979823496763, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:22, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14972393697551645, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15029962215384754, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15087530733217866, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15145099251050978, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15169513125105857, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15227081642938967, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15284650160772076, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1534221867860519, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1054621194337273, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10603780461205842, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:22, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1062819433526072, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10685762853093832, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10743331370926942, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0022153330160909306, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:0,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0024594717566397265, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.002372064059406207, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0022846563621726876, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0021972486649391677, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0030170725838190773, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.003592757762150189, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:22, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004168442940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004412581681030098, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0049882668593612105, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005563952037692324, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16785775014379511, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16777034244656158, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16768293474932808, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16759552705209457, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16783966579264334, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:22, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16841535097097446, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16899103614930555, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1698108600681855, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17038654524651659, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709622304248477, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1174740904864207, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11771822922696949, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11829391440530063, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11886959958363173, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:22, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11944528476196281, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1196894235025116, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12026510868084272, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12084079385917385, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12141647903750495, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12166061777805376, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12223630295638487, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12281198813471597, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1505962369694376, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508403757099864, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:22, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15141606088831752, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15199174606664864, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15256743124497976, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15281156998552853, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15338725516385968, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15396294034219077, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545386255205219, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15478276426107068, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1553584494394018, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1559341346177329, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:22, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15566464632887045, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15590878506941927, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15648447024775036, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15706015542608148, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576358406044126, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1578799793449614, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15845566452329252, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1590313497016236, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15960703487995473, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15985117362050352, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:1, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:22, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16042685879883464, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16100254397716576, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05092553284336376, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051501218021694876, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05198949550279247, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05256518068112358, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05280931942167238, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053385004600003484, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053960689778334604, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:23, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05453637495666571, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0547805136972145, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05535619887554562, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05593188405387674, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056507569232207844, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056751707972756645, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05732739315108775, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:240}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05790307832941887, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05847876350774999, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05872290224829878, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:23, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0592985874266299, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059874272604961004, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06044995778329211, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06069409652384091, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06126978170217203, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06184546688050314, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19803450409903647, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:6,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19827864283958524, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
```

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MATCH (a {`key`:240}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19885432801791636, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19943001319624748, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:23, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2000056983745786, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20024983711512737, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2008255222934585, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2014012074717896, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20197689265012073, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20222103139066955, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20279671656900064, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20337240174733176, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14939239053773412, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14963652927828291, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:23, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.150212214456614, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15078789963494516, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15136358481327625, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15160772355382504, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15218340873215616, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15275909391048728, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10479902655816266, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10537471173649378, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10595039691482488, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10619453565537368, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:23, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10677022083370478, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10734590601203589, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.002791018194422043, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:0,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0030351569349708393, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.00294774923773732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0028603415405038003, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0027729338432702804, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0030170725838190773, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0035053500649166692, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004081035243247782, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:23, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:240}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004325173983796579, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004900859162127691, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005476544340458803, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16818929658157744, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16843343532212623, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1683460276248927, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682586199276592, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16817121223042567, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16841535097097446, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16832794327374093, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:23, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16890362845207205, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16947931363040317, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16972345237095196, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17029913754928308, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1708748227276142, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11738668278918718, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11763082152973599, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1182065067080671, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11878219188863982, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11935787706472293, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:23, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11960201580527811, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1201777009836092, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12075338616194031, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12132907134027145, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12157321008082023, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12214889525915133, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12272458043748247, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15050882927220408, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15075296801275287, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15132865319108402, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:23, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1519043383694151, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15248002354774623, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15272416228829502, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
```

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MATCH (a {`key`:240}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15329984746662614, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15387553264495724, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15445121782328836, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15469535656383718, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15527104174216827, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1558467269204994, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15557723863163697, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:1, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:23, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582137737218577, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15639706255051683, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15697274772884798, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
```

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MATCH (a {`key`:240}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15754843290717907, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15779257164772786, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15836825682605898, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1589439420043901, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15951962718272122, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15976376592327, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16033945110160114, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:23, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16091513627993223, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050593986405581444, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05083812514613024, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05141381032446135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.051902087805558945, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05247777298389006, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05272191172443885, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053297596902769966, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053873282081101086, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054448967259432185, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:23, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05469310599998099, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0552687911783121, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055844476356643205, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056420161534974325, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056664300275523126, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05723998545385423, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05781567063218535, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05839135581051646, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05863549455106526, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059211179729396365, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:23, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05978686490772747, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06036255008605859, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06060668882660739, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0611823740049385, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06175805918326962, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19794709640180294, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:6,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1981912351423517, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19876692032068288, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19934260549901395, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19991829067734507, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:23, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20016242941789386, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20073811459622498, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20131379977455607, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20188948495288722, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20213362369343602, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2027093088717671, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20328499405009826, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1493049828405006, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14954912158104938, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1501248067593805, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:23, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15070049193771162, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15127617711604271, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1515203158565915, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15209600103492266, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10413593368259802, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10471161886092914, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10528730403926025, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10586298921759137, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10610712795814016, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10668281313647128, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:23, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10725849831480239, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0033667033727531556, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:0,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0036108421133301952, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0035234344160684325, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0034360267188349126, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0033348619021601393, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0035927577621501 89, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0035053500649166692, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.003993627546014262, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004237766286563059, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:23, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004813451464894172, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005389136643225284, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16876498175990853, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16900912050045733, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16892171280322382, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1688343051059903, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16874689740875679, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16899103614930555, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16890362845207205, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16881622075483854, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:23, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16939190593316963, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16963604467371846, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17021172985204955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17078741503038067, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11729927509195368, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11754341383250246, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11811909901083356, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11869478418916467, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1192704693674958, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1195146081080446, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:23, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1200902932863757, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12066597846470681, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12124166364303791, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1214858023835867, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12206148756191783, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12263717274024893, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15042142157497057, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15066556031551936, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15124124549385048, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15181693067218158, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:1, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:23, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15239261585051272, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15263675459106152, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532124397693926, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15378812494772373, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15436381012605485, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15460794886660364, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15518363404493474, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15575931922326589, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15548983093440344, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15573396967495223, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:23, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15630965485328335, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15688534003161447, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15746102520994554, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15770516395049433, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15828084912882548, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15885653430715657, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1594322194854877, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15967635822603649, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1602520434043676, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1608277285826987, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:23, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050506578708347925, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05075071744889672, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05132640262722783, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05181468010832543, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05239036528665654, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053210189205536454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05378587438386756, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05436155956219868, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:23, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05518138348107857, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05575706865940969, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05633275383774081, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0565768925782896, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05715257775662072, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05772826293495183, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05830394811328294, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058548086853831734, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059123772032162854, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05969945721049397, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:23, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06027514238882507, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06051928112937387, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06109496630770498, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06167065148603609, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19785968870456944, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:6,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19810382744511823, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19867951262344935, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19925519780178044, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19983088298011156, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20007502172066036, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:23, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20065070689899145, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20122639207732257, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2018020772556537, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20204621599620248, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2026219011745336, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
```

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MATCH (a {`key`:242}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20319758635286472, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14921757514326706, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14946171388381588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.150037399062147, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1506130842404781, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:23, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1511887694188092, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.151432908159358, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10380438724481571, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10404852598536452, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10462421116369561, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10519989634202673, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:242}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10577558152035783, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10601972026090664, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10659540543923776, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10717109061756887, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:23, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.003942388551084268, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:0,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004186527291633065, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004099119594399545, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004011711897166026, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0039243041999325054, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004168442940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004081035243247782, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.003993627546014262, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004150358589329539, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004726043767660652, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:23, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005301728945991764, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16934066693823965, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16949739798155494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1694099902843214, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16932258258708788, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16947931363040317, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16939190593316963, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16930449823593613, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:23, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16954863697648492, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17012432215481602, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17070000733314716, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11721186739472014, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11745600613526894, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11803169131360004, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11860737649193116, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11918306167026227, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11942720041081106, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
```

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MATCH (a {`key`:242}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1200028855891422, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:1, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:23, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12057857076747328, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12115425594580438, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12139839468635319, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12197407986468431, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12254976504301542, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15033401387773707, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15057815261828583, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15115383779661695, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15172952297494807, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
```

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MATCH (a {`key`:242}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1523052081532792, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:23, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152549346893828, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15312503207215908, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15370071725049023, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15427640242882132, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545205411693701, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15509622634770123, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15567191152603235, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1554024232371699, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15564656197771873, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15622224715604982, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:23, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15679793233438094, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15737361751271203, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15761775625326085, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15819344143159195, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15876912660992304, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1593448117882542, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15958895052880298, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16016463570713407, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1607403208854652, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05041917101111441, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:23, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:242}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05123899492999431, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05205881884887422, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052878642767754135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05345432794608524, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05403001312441636, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05484983704329627, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05542552222162739, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:23, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.056001207399958494, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0565768925782896, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0568210313188384, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05739671649716951, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05797240167550063, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058548086853831734, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058792225594380534, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059367910772711655, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05994359595104276, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060251928112937387, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:23, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.060763419869922675, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061339105048253774, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061914790226584894, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:1,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19810382744511823, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:6,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19834796618566702, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19892365136399812, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19949933654232926, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20007502172066036, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20031916046120912, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20089484563954027, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:23, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20147053081787136, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20204621599620248, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2022903547367513, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2028660399150824, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20344172509341352, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:6,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14946171388381588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14970585262436467, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15028153780269576, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15085722298102688, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.151432908159358, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:23, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10347284080703338, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10404852598536452, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10429266472591331, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10486834990424442, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10544403508257552, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10601972026090664, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10626385900145544, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10683954417978654, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10741522935811766, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004186527291633065, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:0,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:23, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004430666032181861, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004343258334948342, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004255850637714821, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004168442940481302, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004412581681030098, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0043251739837965 79, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0042377662865630 59, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0041503585893295 39, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0049707018250820 94486, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0055545867686540561, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:0,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:23, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16982894441933724, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16974153672210374, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696541290248702, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1698108600681855, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16972345237095196, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16963604467371846, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16954863697648492, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16979277571703372, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:1, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:23, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036846089536484, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094414607369596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:6,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11745600613526894, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11770014487581772, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11827583005414885, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11885151523247996, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11942720041081106, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11967133915135987, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12024702432969098, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12082270950802208, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:23, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12139839468635319, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12164253342690198, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12221821860523309, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12279390378356422, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.03663127777746836, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15057815261828583, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15082229135883463, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15139797653716577, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15197366171549687, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152549346893828, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15279348563437675, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:23, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1533691708127079, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153944855991039, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545205411693701, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15476467990991893, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15534036508825003, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15591605026658115, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15564656197771873, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15589070071826752, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15646638589659861, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15704207107492973, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:23, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15761775625326085, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15786189499380962, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15843758017214074, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15901326535047186, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15958895052880298, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15983308926935175, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1604087744476829, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.160984459626014, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:5,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050907448492211994, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:23, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05148313367054311, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.051971411151640695, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.052547096329971815, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0527912350705206, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05336692024885172, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05394260542718283, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05451829060551395, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05476242934606275, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055338114524393855, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05591379970272496, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:257}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:23, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05648948488105608, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05673362362160487, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05730930879993599, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05788499397826711, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058460679156598215, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058704817897147016, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05928050307547812, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05985618825380924, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06043187343214036, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06067601217268915, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:23, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:244}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061251697351020255, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:1,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.061827382529351375, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:1,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1980164197478847, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:6,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1982605584884335, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19883624366676464, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19941192884509573, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19998761402342685, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20023175276397565, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20080743794230674, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20138312312063783, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:23, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20195880829896898, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
```

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MATCH (a {`key`:244}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20220294703951777, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20277863221784886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:6,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20335431739618, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:6,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14937430618658235, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14961844492713114, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15019413010546226, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15076981528379338, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10280974793146877, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10338543310979988, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:23, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.103961118288131, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10420525702867978, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10478094220701088, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.105356627385342, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10593231256367312, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10617645130422192, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10675213648255304, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10732782166088414, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004762212469964177, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:0,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005006351210512973, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:23, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004918943513279453, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004831535816045934, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004744128118812414, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:244}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0049882668593612105, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004900859162127691, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004813451464894172, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.004726043767660652, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0049701825082094486, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005458459989307041, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:0,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17016049085711957, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:1, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:23, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17040462959766836, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17031722190043483, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17022981420320132, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
```

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MATCH (a {`key`:244}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701424065059678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17038654524651659, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17029913754928308, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17021172985204955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17012432215481602, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036846089536484, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1702810531981313, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:6,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:23, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17085673837646242, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:6,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1173685984380354, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11761273717858421, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11818842235691532, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11876410753524642, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11933979271357756, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11958393145412635, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12015961663245746, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12073530181078858, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12131098698911967, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:23, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12155512572966848, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12213081090799958, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12270649608633069, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15049074492105233, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15073488366160112, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15131056883993224, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15188625401826333, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15246193919659445, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15270607793714325, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15328176311547437, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:23, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1538574482938055, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1544331334721366, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1546772722126854, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1552529573910165, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582864256934764, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555591542804852, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:244}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.155803293021034, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15637897819936508, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15695466337769623, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15753034855602732, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:23, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15777448729657612, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15835017247490724, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15892585765323833, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15950154283156945, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15974568157211824, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16032136675044936, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:5,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16089705192878045, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:5,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
```

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MATCH (a {`key`:244}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050575902054429675, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050820040794978476, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05139572597330959, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:23, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05188400345440719, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05245968863273829, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05270382737328709, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.053279512551618204, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05385519772994932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05443088290828043, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.054675021648829224, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.055250706827160344, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05582639200549146, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05640207718382256, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:1,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:23, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05664621592437136, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:1,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05722190110270247, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:1,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.05779758628103358, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:1,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058373271459364696, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:1,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0586174101999135, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:1,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0591930953782446, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:1,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059768780556575724, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:1,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06034446573490683, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:1,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06058860447545563, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:1,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06116428965378674, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:1,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:23, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.06173997483211786, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:1,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19792901205065117, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:6,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1981731507912, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:6,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1987488359695311, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:6,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1993245211478622, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:6,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.19990020632619332, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:6,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.2001443450667421, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:6,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20072003024507323, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:6,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20129571542340433, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:6,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20187140060173545, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:6,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:23, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20211553934228424, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:6,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20269122452061536, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:6,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.20326690969894648, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:6,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1492868984893488, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1495310372298976, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15010672240822875, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10214665505590413, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10272234023423525, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10329802541256636, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10387371059089748, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:23, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10411784933144626, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10469353450977738, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10526921968810848, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10584490486643959, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1060890436069884, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10666472878531952, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10724041396365062, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.00533789764829529, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:0,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005582036388844086, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:0,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005494628691610566, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:0,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:1, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:23, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:245}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005407220994377047, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:0,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005319813297143527, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:0,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005563952037692324, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:0,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005476544340458803, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:0,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005389136643225284, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:0,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005301728945991764, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:0,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005545867686540561, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:0,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.005458459989307041, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:0,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17073617603545066, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17098031477599945, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:23, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17089290707876595, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17080549938153244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17071809168429 89, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709622304248477, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1708748227276142, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17078741503038067, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17070000733314716, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094414607369596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:6,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17085673837646242, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:6,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1707693306792289, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:6,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:23, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11728119074080 19, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11752532948135068, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1181010146596818, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11867669983801292, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11925238501634403, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11949652375689282, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12007220893522395, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12064789411355506, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12122357929188617, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12146771803243497, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:23, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12204340321076605, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12261908838909717, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:4,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.03663127777746836, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15040333722381882, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1506474759643676, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512231611426987, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517988463210298, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15237453149936095, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15261867023990974, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15319435541824084, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15377004059657198, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:23, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15434572577490308, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15458986451545187, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.155165549693783, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1557412348721141, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15547174658325166, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15571588532380046, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15629157050213158, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568672556804627, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1574429408587938, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576870795993426, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:23, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15826276477767373, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15883844995600482, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15941413513433594, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596582738748847, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:5,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16023395905321583, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:5,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16080964423154695, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:5,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048849435719616, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05073263309774496, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05130831827607607, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16583602395481706, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:23, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16641170913314818, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1672315330520281, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16780721823035918, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16838290340869033, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
```

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MATCH (a {`key`:246}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16920272732757022, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16977841250590134, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1680566780064645, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1683008167470133, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:23, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16887650192534442, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16945218710367552, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17002787228200664, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17027201102255543, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17084769620088652, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17142338137921764, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17199906655754874, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17224320529809756, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17281889047642868, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17339457565475977, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:23, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1457779546343591, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:2,
`saturationRatio`:0.8727272727272727, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460220933749079, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:2,
`saturationRatio`:0.8818181818181818, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14659777855323902, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:2,
`saturationRatio`:0.8909090909090909, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14717346373157014, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14774914890990126, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14799328765045003, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14856897282878118, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14914465800711227, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1497203431854434, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1499644819259922, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:23, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1505401671043233, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15111585228265442, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17145053461809154, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17169467335864036, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22276156699129643, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:7,
`saturationRatio`:0.0, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22300570573184522, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:7,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22358139091017634, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:7,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22415707608850743, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22473276126683855, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
```

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MATCH (a {`key`:246}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22497690000738738, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:1, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:23, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22555258518571847, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2261282703640496, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22670395554238068, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2269480942829295, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2275237794612606, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22809946463959171, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16539827838715537, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:6,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16564241712770417, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:6,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1662181023060353, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:6,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:246}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1667937874843664, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:23, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16736947266269753, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16818929658157744, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16876498175990853, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16934066693823965, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17016049085711957, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17073617603545066, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.006103468513719911, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:0,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.006679153692051023, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:0,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
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MATCH (a {`key`:257}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:23, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007254838870382137, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007830524048713247, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008650347967593158, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009226033145924269, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009801718324255382, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01062154224313529, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011197227421466404, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2025429770976622, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:6,
`saturationRatio`:0.32727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:23, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:246}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20278711583821096, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20336280101654208, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2039384861948732, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2045141713732043, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20475831011375312, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20533399529208424, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20590968047041533, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20648536564874645, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20672950438929527, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20730518956762634, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:23, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20788087474595748, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
```

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MATCH (a {`key`:246}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15196280272826196, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15220694146881075, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15278262664714187, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153358311825473, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15393399700380409, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541781357443529, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154753820922684, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15532950610101512, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15590519127934624, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:23, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156149330019895, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15672501519822613, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:246}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15730070037655725, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050455339713417924, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:1,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05127516363229784, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05185084881062895, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052426533988960064, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05324635790783997, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:23, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05382204308617109, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0543977282645022, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055217552183382104, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055793237361713224, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:12, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16444051485760602, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:5,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16526033877648597, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:12, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16608016269536588, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16665584787369697, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:23, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16689998661424577, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16747567179257689, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.168051356970908, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16862704214923913, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
```

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MATCH (a {`key`:247}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1688711808897879, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16944686606811904, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17002255124645013, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1683008167470133, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1685449554875621, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16912064066589322, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:23, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1696963258442243, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17027201102255543, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17051614976310422, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17109183494143534, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17166752011976644, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
```

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MATCH (a {`key`:247}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17224320529809756, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17248734403864635, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17306302921697744, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1736387143953086, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460220933749079, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:2,
`saturationRatio`:0.8818181818181818, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:23, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1462662321154567, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:2,
`saturationRatio`:0.8909090909090909, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14684191729378782, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14741760247211894, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14799328765045003, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14823742639099885, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14881311156932997, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14938879674766106, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1499644819259922, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15020862066654098, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1507843058448721, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:1, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:23, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15135999102320322, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17169467335864036, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08712981858183266, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22300570573184522, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:7,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22324984447239404, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:7,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22382552965072516, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22440121482905626, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22497690000738738, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22522103874793617, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2257967239262673, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:23, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22637240910459838, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2269480942829295, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22719223302347832, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22776791820180942, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22834360338014054, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16564241712770417, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:6,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16588655586825296, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:6,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
```

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MATCH (a {`key`:247}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1664622410465841, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1670379262249152, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:23, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16785775014379511, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16843343532212623, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16900912050045733, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16982894441933724, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17040462959766836, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17098031477599945, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.00610346851371991, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:0,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.006923292432599819, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007498977610930932, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:23, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008318801529810842, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008894486708141953, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009470171886473066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010289995805352975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010865680983684088, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011441366162015201, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20278711583821096, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20303125457875976, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:23, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2036069397570909, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20418262493542197, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20475831011375312, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2050024488543019, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.205578134032633, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20615381921096412, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20672950438929527, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20697364312984404, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20754932830817516, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20812501348650628, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:23, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15220694146881075, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15245108020935957, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15302676538769067, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1536024505660218, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541781357443529, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15442227448490167, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15499795966323282, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555736448415639, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156149330019895, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1563934687604438, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:23, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:247}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15696915393877492, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157544839117106, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05094361719451552, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05151930237284664, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052094987551177745, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05291481147005766, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05349049664838877, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05406618182671988, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:23, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05488600574559979, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055461690923930905, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05603737610226201, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:13, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16492879233870364, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16550447751703473, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:13, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16599275499813235, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16656844017646344, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16681257891701223, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:23, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16738826409534338, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16796394927367447, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1685396344520056, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1687837731925544, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1693594583708855, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699351435492166, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1682134090497798, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16845754779032857, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16903323296865969, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16960891814699078, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:23, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17018460332532193, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17042874206587072, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1710044272442018, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17158011242253293, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17215579760086402, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17239993634141285, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17297562151974394, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17355130669807506, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14593468567767437, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:2,
`saturationRatio`:0.8909090909090909, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1461788244182232, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:1, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:23, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1467545095965543, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14733019477488854, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14790587995321652, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14815001869376532, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14872570387209644, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14930138905042756, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14987707422875868, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15012121296930744, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1506968981476386, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15127258332596968, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:23, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08646672570626804, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08704241088459914, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22291829803461172, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:7,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2231624367751605, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22373812195349163, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22431380713182272, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22488949231015384, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22513363105070264, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22570931622903376, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22628500140736485, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:23, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22686068658569597, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2271048253262448, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22768051050457588, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.228256195682907, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1662181023060353, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:6,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1664622410465841, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16637483334935058, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16695051852768167, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16752620370601282, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16777034244656158, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:23, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1683460276248927, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16892171280322382, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16949739798155494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16974153672210374, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17031722190043483, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17089290707876595, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.006679153692051023, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:0,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.006923292432599819, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007411569913697412, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007987255092028524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:23, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008231393832577321, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008807079010908435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009382764189239546, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009958449367570659, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010202588108119456, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010778273286450568, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011353958464781681, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20269970814097746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20294384688152628, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20351953205985734, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:23, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20409521723818846, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2046709024165196, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20491504115706838, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2054907263353995, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20606641151373062, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2066420966920617, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20688623543261053, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20746192061094165, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20803760578927274, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15211953377157725, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:23, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15236367251212604, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15293935769045713, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15351504286878828, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15409072804711935, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15433486678766817, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1549105519659993, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15548623714433038, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15606192232266153, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1563060610632103, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568817462415414, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:23, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1574574314198725, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050612070756733206, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05085620949728201, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05143189467561311, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052007579853944226, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052583265032275346, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05282740377282414, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053403088951155246, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053978774129486366, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05455445930781747, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:23, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05479859804836627, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05537428322669738, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0559499684050285, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:14, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1645972459009213, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1648413846414701, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16541706981980123, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:14, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16590534730089881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16648103247922993, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16672517121977873, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16730085639810985, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:23, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16787654157644094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684522267547721, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16869636549532088, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16927205067365197, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698477358519831, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16812600135254627, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16837014009309503, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16894582527142615, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16952151044975727, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1700971956280884, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:1, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:23, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1703413343686372, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1709170195469683, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17149270472529943, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1720683899036305, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1723125286441793, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1728882138225104, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17346389900084153, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14584727798044086, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:2,
`saturationRatio`:0.9, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14609141672098966, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14666710189932078, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:23, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1472427870776519, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14781847225598302, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14806261099653178, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14863829617486293, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14921398135319403, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14978966653152515, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15003380527207394, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15060949045040506, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08580363283070341, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08637931800903453, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:23, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08695500318736564, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22283089033737818, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:7,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22307502907792698, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2236507142562581, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2242263994345892, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2248020846129203, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22504622335346913, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22562190853180022, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22619759371013134, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22677327888846244, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:23, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22701741762901126, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22759310280734235, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22816878798567347, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1667937874843664, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:6,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1670379262249152, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16695051852768167, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16686311083044816, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16743879600877928, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16768293474932808, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682586199276592, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:23, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1688343051059903, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1694099902843214, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696541290248702, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17022981420320132, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17080549938153244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007254838870382137, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:0,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007498977610930932, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007411569913697412, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007899847394795005, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008143986135343801, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:23, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008719671313674914, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009295356492006027, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009871041670337139, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010115180410885936, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01069086558921705, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01126655076754816, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20261230044374395, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20285643918429272, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20343212436262387, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20400780954095496, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:23, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20458349471928605, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20482763345983487, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.205403318638166, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20597900381649709, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2065546889948282, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20679882773537703, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2073745129137081, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20795019809203924, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15203212607434372, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522762648148925, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:23, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15285194999322363, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15342763517155472, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15400332034988584, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15424745909043464, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15482314426876576, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15539882944709688, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.155974514625428, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1562186533659768, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1567943385443079, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157370023722639, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:23, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05052466305949969, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05076880180004848, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051344486978379594, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05192017215671071, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05249585733504182, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052739996075590614, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053315681253921735, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053891366432252834, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054467051610583954, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054711190351132755, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:23, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05528687552946386, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05586256070779498, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16450983820368778, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1647539769442366, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16532966212256772, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:15, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16581793960366528, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16639362478199643, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16721344870087632, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16778913387920744, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:1, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:24, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16836481905753856, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16918464297641847, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1697603281547496, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803859365531273, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16828273239586153, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16885841757419265, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16943410275252377, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000978793085486, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17025392667140368, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:24, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17082961184973477, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1714052970280659, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17198098220639702, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1722251209469458, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1728008061252769, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17337649130360805, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14575987028320733, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:2,
`saturationRatio`:0.9090909090909091, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14600400902375613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14657969420208727, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14715537938041837, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:24, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1477310645587495, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14797520329929828, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1485508884776294, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1491265736559605, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14970225883429164, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14994639757484043, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08547208639292109, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0857162251334699, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.086291910311801, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08686759549013211, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:24, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22274348264014465, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:7,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22298762138069345, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22356330655902457, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2241389917373557, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22471467691568678, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2249588156562356, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2255345008345667, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2261101860128978, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22668587119122896, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22693000993177773, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:24, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22750569511010882, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22808138028844, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16736947266269753, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:6,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16752620370601282, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16743879600877928, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16735138831154575, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16759552705209457, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16817121223042567, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16874689740875679, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:24, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16932258258708788, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701424065059678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1707180916842989, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007830524048713247, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:0,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007987255092028524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.007899847394795005, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008056578438110282, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008632263616441394, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:24, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009207948794772507, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.00978363397310362, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0100277727136524 16, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010603457891983529, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011179143070314642, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20252489274651042, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20276903148705921, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20334471666539036, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20392040184372143, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20449608702205255, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:24, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20474022576260137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20531591094093243, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20589159611926358, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2064672812975947, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20671142003814347, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2072871052164746, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20786279039480574, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15194471837711018, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152188857117659, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1527645422959901, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:24, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15334022747432122, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15391591265265234, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909091, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541600513932011, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15473573657153225, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15531142174986334, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
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MATCH (a {`key`:250}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15588710692819446, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15613124566874328, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15670693084707438, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1572826160254055, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05043725536226617, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:24, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05125707928114608, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05183276445947719, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0524084496378083, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05322827355668821, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
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MATCH (a {`key`:250}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05380395873501932, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054379643913350435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05519946783223034, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:1, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:24, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055775153010561455, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16442243050645428, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1652422544253342, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:16, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1660620783442141, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666377635225452, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.166881902263094, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16745758744142514, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803327261975623, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860895779808735, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:24, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16885309653863614, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16942878171696726, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000446689529836, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16828273239586153, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852687113641032, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16910255631474144, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16967824149307253, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17025392667140368, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17049806541195248, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17107375059028357, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:24, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1716494357686147, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1722251209469458, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1724692596874946, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1730449448658257, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17362063004415682, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14600400902375613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:2,
`saturationRatio`:0.9181818181818182, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14624814776430495, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14682383294263607, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14739951812096716, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14797520329929828, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:24, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14821934203984707, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487950272181782, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14937071239650931, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14994639757484043, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08514053995513877, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0857162251334699, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08596036387401867, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0865360490523498, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0871117342306809, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22298762138069345, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:7,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:24, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22323176012124227, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2238074452995734, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22438313047790448, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2249588156562356, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2252029543967844, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22577863957511551, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2263543247534466, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22693000993177773, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22717414867232655, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22774983385065764, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:24, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22832551902898876, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16761361140324632, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:6,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16785775014379511, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16777034244656158, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16768293474932808, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16759552705209457, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16783966579264334, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16841535097097446, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16899103614930555, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:24, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1698108600681855, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17038654524651659, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709622304248477, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008074662789262044, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:0,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008318801529810842, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008231393832577321, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008143986135343801, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008056578438110282, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008876402356990191, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009452087535321304, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:24, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010027772713652416, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010271911454201213, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010847596632532326, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011423281810863438, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20276903148705921, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20301317022760804, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20358885540593913, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20416454058427025, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20474022576260137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20498436450315013, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:24, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20556004968148126, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20613573485981238, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20671142003814347, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2069555587786923, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2075312439570234, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2081069291353545, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152188857117659, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15243299585820777, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530086810365389, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15358436621487, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:24, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541600513932011, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15440419013374992, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15497987531208104, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15555556049041214, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15613124566874328, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15637538440929205, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15695106958762317, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1575267547659543, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05092553284336376, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:1, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:24, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051501218021694876, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05207690320002599, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05289672711890589, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05347241229723701, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05404809747556812, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05486792139444803, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055443606572779136, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056019291751110256, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:24, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16491070798755186, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16548639316588298, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:17, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16597467064698057, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1665503558253117, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16679449456586048, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1673701797441916, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1679458649225227, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852155010085385, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16876568884140264, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:24, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16934137401973373, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16991705919806485, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16819532469862802, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684394634391768, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1690151486175079, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16959083379583903, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17016651897417015, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17041065771471894, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17098634289305006, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17156202807138118, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:24, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17213771324971228, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1723818519902611, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1729575371685922, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17353322234692328, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14591660132652262, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:2,
`saturationRatio`:0.9272727272727272, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14616074006707144, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14673642524540254, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14731211042373366, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14788779560206478, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14813193434261354, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:24, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487076195209447, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14928330469927578, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08447744707957414, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08505313225790526, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08562881743623636, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08587295617678517, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08644864135511628, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0870243265334474, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22290021368345994, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:7,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22314435242400874, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:24, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22372003760233986, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22429572278067095, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22487140795900207, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2251155466995509, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22569123187788198, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2262669170562131, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2268426022345442, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22708674097509302, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2276624261534241, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22823811133175523, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:24, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16818929658157744, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:6,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16843343532212623, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1683460276248927, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682586199276592, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16817121223042567, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16841535097097446, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16832794327374093, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16890362845207205, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16947931363040317, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16972345237095196, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:24, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17029913754928308, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1708748227276142, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008650347967593158, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:0,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008894486708141953, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008807079010908435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008719671313674914, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008632263616441394, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.008876402356990191, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009364679838087784, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009940365016418897, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
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MATCH (a {`key`:258}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:24, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010184503756967693, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010760188935298806, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011335874113629919, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2026816237898257, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20292576253037448, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20350144770870562, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20407713288703674, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20465281806536784, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20489695680591663, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20547264198424775, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:24, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20604832716257884, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20662401234090996, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20686815108145878, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20744383625978985, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.208019521438121, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15210144942042544, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15234558816097427, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15292127333930539, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15349695851763648, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1540726436959676, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:1, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:24, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1543167824365164, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548924676148475, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15546815279317863, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15604383797150975, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15628797671205852, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15686366189038967, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15743934706872076, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050593986405581444, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05083812514613024, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05141381032446135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:24, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05198949550279247, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05256518068112358, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05280931942167238, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053385004600003484, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053960689778334604, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05453637495666571, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0547805136972145, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05535619887554562, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05593188405387674, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:18, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16457916154976956, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:24, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16482330029031836, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16539898546864948, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:18, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16588726294974707, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1664629481280782, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16670708686862695, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16728277204695807, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678584572252892, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684341424036203, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1686782811441691, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1692539663225002, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:24, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982965150083135, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1681079170013945, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16835205574194329, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1689277409202744, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1695034260986055, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17007911127693662, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17032325001748544, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17089893519581653, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17147462037414765, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17205030555247877, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:24, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17229444429302757, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17287012947135866, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1734458146496898, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1458291936292891, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:2,
`saturationRatio`:0.9363636363636364, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460733323698379, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14664901754816903, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14722470272650012, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14780038790483124, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14804452664538004, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14862021182371116, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:24, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08381435420400951, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08439003938234062, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08496572456067172, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08554140973900284, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08578554847955164, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08636123365788276, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08693691883621388, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2228128059862264, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:7,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2230569447267752, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22363262990510632, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:24, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22420831508343742, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22478400026176854, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22502813900231736, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22560382418064845, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22617950935897957, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22675519453731072, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22699933327785948, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22757501845619058, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22815070363452175, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16876498175990853, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:6,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:24, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16900912050045733, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16892171280322382, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1688343051059903, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16874689740875679, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16899103614930555, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16890362845207205, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16881622075483854, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16939190593316963, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16963604467371846, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17021172985204955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:24, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17078741503038067, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009226033145924269, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:0,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009470171886473066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009382764189239546, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009295356492006027, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009207948794772507, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009452087535321304, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009364679838087784, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009852957319185377, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010097096059734174, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:24, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010672781238065287, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011248466416396399, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20259421609259218, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20283835483314097, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20341404001147212, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20398972518980318, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20456541036813433, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20480954910868313, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20538523428701422, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20596091946534534, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:1, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:24, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20653660464367646, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20678074338422522, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20735642856255634, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2079321137408875, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15201404172319194, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15225818046374073, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15283386564207185, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340955082040297, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1539852359987341, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15422937473928286, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:24, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154805059917614, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1553807450959451, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15595643027427622, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156200569014825, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15677625419315613, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15735193937148725, `foldPressure`:0.25925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050506578708347925, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05075071744889672, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05132640262722783, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051902087805558945, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
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MATCH (a {`key`:258}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:24, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05247777298389006, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05272191172443885, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053297596902769966, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053873282081101086, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054448967259432185, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05469310599998099, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0552687911783121, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055844476356643205, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:19, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16449175385253603, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16473589259308485, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:24, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16531157777141597, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:19, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16579985525251356, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16637554043084465, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16719536434972457, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16777104952805566, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16834673470638678, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1691665586252667, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16974224380359781, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:24, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16802050930416096, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16826464804470978, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16884033322304087, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.169416018401372, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699917035797031, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1702358423202519, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.170811527498583, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17138721267691415, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17196289785524524, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17220703659579403, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:24, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17278272177412515, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17335840695245627, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14574178593205558, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:2,
`saturationRatio`:0.9454545454545454, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1459859246726044, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1465616098509355, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1471372950292666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14771298020759774, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14795711894814653, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0834828077662272, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08372694650677599, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:24, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0843026316851071, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0848783168634382, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08545400204176931, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08569814078231812, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08627382596064924, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08684951113898035, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22272539828899288, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:7,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2229695370295417, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2235452222078728, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2241209073862039, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:24, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22469659256453506, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22494073130508382, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22551641648341492, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2260921016617461, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22666778684007718, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22691192558062595, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2274876107589571, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22806329593728822, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16934066693823965, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:6,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:24, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16949739798155494, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1694099902843214, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16932258258708788, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16947931363040317, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16939190593316963, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16930449823593613, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16954863697648492, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17012432215481602, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17070000733314716, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:24, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009801718324255382, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:0,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009958449367570659, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009871041670337139, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.00978363397310362, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010027727213652416, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
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MATCH (a {`key`:254}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009940365016418897, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.009852957319185377, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35545454545454455, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010009688362500654, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010585373540831767, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:1, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:24, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01116105871916288, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20250680839535867, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20275094713590747, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20332663231423856, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20390231749256968, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20447800267090083, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
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MATCH (a {`key`:254}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2047221414114496, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2052978265897807, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20587351176811186, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2064491969464429, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:24, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20669333568699172, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20726902086532284, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20784470604365393, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1519266340259584, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1521707727665072, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15274645794483835, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15332214312316944, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15389782830150056, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15414196704204935, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15471765222038047, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:24, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15529333739871157, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15586902257704272, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1561131613175915, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1566888464959226, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15726453167425375, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05041917101111441, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05123899492999431, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
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MATCH (a {`key`:254}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05181468010832543, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05239036528665654, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:24, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053210189205536454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05378587438386756, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05436155956219868, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05518138348107857, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05575706865940969, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:20, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16440434615530253, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
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MATCH (a {`key`:254}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16522417007418244, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:20, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:24, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16604399399306236, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16661967917139345, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16686381791194227, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16743950309027336, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16801518826860445, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16859087344693557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16883501218748437, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1694106973658155, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1699863825441466, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16826464804470978, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:258}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:24, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850878678525855, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16908447196358967, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1696601571419208, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1702358423202519, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1704799810608007, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17105566623913182, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17163135141746294, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17220703659579403, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17245117533634285, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17302686051467395, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:24, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17360254569300507, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1459859246726044, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:2,
`saturationRatio`:0.9545454545454546, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1462300634131532, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1468057485914843, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1473814337698154, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14795711894814653, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08315126132844487, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08372694650677599, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08397108524732477, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0845467704256559, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:24, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.085122455603987, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08569814078231812, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08594227952286693, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08651796470119803, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08709364987952914, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2229695370295417, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:7,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2232136757700905, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2237893609484216, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2243650461267527, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22494073130508382, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:24, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22518487004563265, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22576055522396374, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22633624040229486, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22691192558062595, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22715606432117477, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22773174949950586, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22830743467783698, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:7,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16958480567878848, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:6,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16982894441933724, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16974153672210374, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:24, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696541290248702, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695667213276367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1698108600681855, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16972345237095196, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16963604467371846, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16954863697648492, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16979277571703372, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036846089536484, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094414607369596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:6,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01004585706480418, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:0,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩔'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:1, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:24, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010289995805352975, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010202588108119456, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010115180410885936, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010027772713652416, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010271911454201213, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010184503756967693, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010097096059734174, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010009688362500654, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010829512281380564, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011405197459711676, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:0,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:24, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20275094713590747, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20299508587645623, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20357077105478738, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2041464562331185, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
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MATCH (a {`key`:255}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2047221414114496, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2049662801519984, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20554196533032953, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2061176505086606, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20669333568699172, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20693747442754054, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:24, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2075131596058716, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20808884478420275, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.5090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1521707727665072, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15241491150705602, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15299059668538711, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
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MATCH (a {`key`:255}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15356628186371823, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15414196704204935, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15438610578259815, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15496179096092927, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555374761392604, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:24, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1561131613175915, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15635730005814027, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15693298523647142, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15750867041480252, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050907448492211994, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05148313367054311, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05205881884887422, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052878642767754135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:24, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05345432794608524, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05403001312441636, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05484983704329627, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05542552222162739, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056001207399958494, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:21, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16489262363640012, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16546830881473124, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:21, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16595658629582882, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:24, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16653227147415994, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16677641021470874, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16735209539303986, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16792778057137092, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850346574970207, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16874760449025086, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16932328966858196, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698989748469131, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16817724034747625, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16842137908802504, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:24, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16899706426635616, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16957274944468725, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17014843462301837, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1703925733635672, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1709682585418983, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1715439437202294, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17211962889856053, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17236376763910932, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17293945281744041, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17351513799577156, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:24, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14589851697537087, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:2,
`saturationRatio`:0.9636363636363636, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14614265571591967, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1467183408942508, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14729402607258188, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08248816845288025, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08306385363121137, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08363953880954247, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08388367755009127, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08445936272842237, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08503504790675348, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:24, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0856107330850846, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0858548718256334, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08643055700396451, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08700624218229563, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22288212933230817, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:7,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22312626807285696, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22370195325118808, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22427763842951917, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2248533236078503, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2250974623483991, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:24, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2256731475267302, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22624883270506133, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22682451788339247, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22706865662394124, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22764434180227233, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:7,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2282200269806035, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:7,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.4060058497098381, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17016049085711957, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:6,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17040462959766836, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17031722190043483, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17022981420320132, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:1, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:24, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701424065059678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17038654524651659, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17029913754928308, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17021172985204955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17012432215481602, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036846089536484, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1702810531981313, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:6,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17085673837646242, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:6,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01062154224313529, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:0,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010865680983684088, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:24, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010778273286450568, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01069086558921705, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010603457891983529, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010847596632532326, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010760188935298806, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010672781238065287, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010585373540831767, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.010829512281380564, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011317789762478157, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:0,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20266353943867393, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:24, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20290767817922273, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20348336335755388, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20405904853588494, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2046347337142161, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20487887245476488, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20545455763309597, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2060302428114271, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20660592798975824, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.206850066730307, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2074257519086381, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:24, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20800143708696925, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:1.103638323514327, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1520833650692737, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1523275038098225, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1529031889881536, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15347887416648473, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15405455934481585, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15429869808536462, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15487438326369576, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15545006844202686, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15602575362035798, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:24, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15626989236090677, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568455775392379, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15742126271756898, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050575902054429675, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050820040794978476, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05139572597330959, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051971411151640695, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052547096329971815, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0527912350705206, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05336692024885172, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:24, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05394260542718283, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05451829060551395, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05476242934606275, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055338114524393855, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05591379970272496, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:22, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1645610771986178, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1648052159391666, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16538090111749773, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:22, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16586917859859532, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16644486377692644, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:24, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1666890025174752, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16726468769580632, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16784037287413742, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16841605805246854, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16866019679301733, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16923588197134845, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16981156714967957, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16808983265024272, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:5,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16833397139079154, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16890965656912263, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:24, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16948534174745375, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17006102692578487, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17030516566633366, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17088085084466476, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1714565360229959, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.172032221201327, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1722763599418758, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1728520451202069, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17342773029853803, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14581110927813734, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:2,
`saturationRatio`:0.9727272727272728, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:24, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14605524801868616, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:2,
`saturationRatio`:0.9818181818181818, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14663093319701728, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:2,
`saturationRatio`:0.990909090909091, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08182507557731561, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08240076075564673, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08297644593397784, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08355213111230896, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08379626985285774, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08437195503118886, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
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MATCH (a {`key`:257}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08494764020951998, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08552332538785108, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:24, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08576746412839988, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.086343149306731, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0869188344850621, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:2,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22279472163507463, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:7,
`saturationRatio`:0.1, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22303886037562348, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:7,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22361454555395455, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:7,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22419023073228567, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:7,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22476591591061681, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:7,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22501005465116558, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:7,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22558573982949667, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:7,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:1, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:24, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22616142500782785, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:7,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22673711018615894, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:7,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2269812489267077, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:7,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22755693410503885, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:7,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22813261928336998, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:7,
`saturationRatio`:0.2, `shadowDecay`:0.4060058497098381, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17073617603545066, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:6,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17098031477599945, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:6,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17089290707876595, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:6,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17080549938153244, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:6,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1707180916842989, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:6,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:24, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709622304248477, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:6,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1708748227276142, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:6,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17078741503038067, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:6,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17070000733314716, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:6,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094414607369596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:6,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17085673837646242, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:6,
`saturationRatio`:0.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1707693306792289, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:6,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011197227421466404, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:0,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011441366162015201, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:0,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011353958464781681, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:0,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:24, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01126655076754816, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:0,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011179143070314642, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:0,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011423281810863438, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:0,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011335874113629919, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:0,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011248466416396399, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:0,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01116105871916288, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:0,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011405197459711676, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:0,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011317789762478157, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:0,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20257613174144043, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20282027048198922, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:24, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20339595566032032, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20397164083865144, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20454732601698258, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20479146475753135, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20536714993586247, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2059428351141936, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20651852029252468, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2067626590330735, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20733834421140462, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2079140293897357, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:6,
`saturationRatio`:0.5272727272727272, `shadowDecay`:1.103638323514327, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:24, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15199595737204016, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15224009611258896, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528157812909201, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1533914664692512, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15396715164758232, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542112903881311, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15478697556646223, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15536266074479332, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15593834592312447, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15618248466367327, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:24, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15675816984200436, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15733385502033548, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
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MATCH (a {`key`:257}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048849435719616, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05073263309774496, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05130831827607607, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05188400345440719, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05245968863273829, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05270382737328709, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053279512551618204, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05385519772994932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:24, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05443088290828043, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054675021648829224, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055250706827160344, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
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MATCH (a {`key`:257}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05582639200549146, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:23, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16447366950138428, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16471780824193308, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1652934934202642, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:23, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11111087304122279, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11168655821955391, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11193069696010272, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:24, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11250638213843382, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11308206731676493, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11365775249509606, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11390189123564484, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11447757641397595, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11505326159230705, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1676491227569863, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:5,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16789326149753508, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684689466758662, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16904463185419732, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:24, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16962031703252844, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698644557730772, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17044014095140836, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17101582612973945, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17159151130807057, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1718356500486194, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17241133522695049, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1729870204052816, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1556032001594568, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15584733890000563, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:24, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1325934823139798, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:4,
`saturationRatio`:0.0, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1328376210545286, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:4,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1334133062328597, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:4,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13398899141119083, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13456467658952193, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13480881533007075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1353845005084018, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13596018568673293, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13653587086506405, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13678000960561285, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:1, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:24, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13735569478394397, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13793137996227506, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05067235192723909, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:1,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050916490667787875, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:1,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05149217584611899, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:1,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0520678610244501, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
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MATCH (a {`key`:258}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05264354620278122, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05288768494333001, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05346337012166113, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05403905529999225, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:24, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054614740478323355, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05485887921887214, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05543456439720326, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05601024957553437, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11724803609702364, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:4,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11749217483757245, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11740476714033894, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11731735944310542, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11722995174587192, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1174740904864207, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:24, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11738668278918718, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11729927509195368, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11721186739472014, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11745600613526894, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1173685984380354, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1172811907408019, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2025429770976622, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:6,
`saturationRatio`:0.32727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20278711583821096, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
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MATCH (a {`key`:258}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20269970814097746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20261230044374395, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:24, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20252489274651042, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20276903148705921, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2026816237898257, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20259421609259218, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20250680839535867, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20275094713590747, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20266353943867393, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20257613174144043, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011962798286891025, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:0,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
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MATCH (a {`key`:258}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.012538483465222138, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:0,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:24, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01311416864355325, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013689853821884363, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014509677740764273, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015085362919095386, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015661048097426494, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016480872016306405, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017056557194637518, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10807752533262445, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:24, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10832166407317326, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10889734925150438, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10947303442983548, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1100487196081666, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11029285834871544, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11086854352704654, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11144422870537764, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11201991388370874, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11226405262425752, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11283973780258863, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:24, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11341542298091975, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1687255257498295, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:5,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16896966449037829, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695453496687094, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701210348470405, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17069672002537165, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094085876592044, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17151654394425153, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17209222912258265, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17266791430091374, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:24, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17291205304146257, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17348773821979366, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17406342339812478, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10971536394401177, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:3,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10995950268456056, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11053518786289168, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:24, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11135501178177158, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11193069696010272, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11217483570065151, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11275052087898262, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:24, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11332620605731374, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11390189123564484, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11414602997619364, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11472171515452474, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11529740033285585, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16789326149753508, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:5,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16813740023808388, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.168713085416415, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16928877059474612, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1698644557730772, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:24, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17010859451362603, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17068427969195715, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17125996487028824, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1718356500486194, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17207978878916816, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17265547396749928, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17323115914583037, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.15584733890000563, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.9909090909090091, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07261952886156216, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1328376210545286, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:4,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:1, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:24, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1330817597950774, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:4,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1336574449734085, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1342331301517396, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13480881533007075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350529540706195, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1356286392489506, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13620432442728173, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13678000960561285, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13702414834616164, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13759983352449276, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:24, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13817551870282388, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050916490667787875, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:1,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051160629408336676, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:1,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05173631458666778, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
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MATCH (a {`key`:259}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0523119997649989, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05288768494333001, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05313182368387881, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05370750886220993, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054283194040541036, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05485887921887214, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:24, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05510301795942094, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05567870313775205, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05625438831608317, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11749217483757245, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11773631357812124, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
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MATCH (a {`key`:259}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11764890588088771, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11756149818365423, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1174740904864207, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11771822922696949, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11763082152973599, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:24, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11754341383250246, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11745600613526894, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11770014487581772, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11761273717858421, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11752532948135068, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20278711583821096, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20303125457875976, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20294384688152628, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20285643918429272, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20276903148705921, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:24, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20301317022760804, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20292576253037448, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20283835483314097, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20275094713590747, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20299508587645623, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20290767817922273, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20282027048198922, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
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MATCH (a {`key`:259}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.011962798286891025, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:0,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.012782622205770934, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013358307384102047, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:24, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014178131302981956, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014753816481313069, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015329501659644182, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01614932557852409, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016725010756855202, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017300695935186315, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10832166407317326, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10856580281372205, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:24, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10914148799205316, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10971717317038429, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11029285834871154, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11053699708926419, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11111268226759531, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11168836744592642, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11226405262425752, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11250819136480632, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11308387654313742, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:259}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11365956172146856, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:24, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16896966449037829, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16921380323092708, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697894884092582, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036517358758932, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094085876592044, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17118499750646923, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17176068268480033, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17233636786313145, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17291205304146257, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17315619178201133, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:24, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17373187696034245, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17430756213867357, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10995950268456056, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11020364142510937, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11077932660344048, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:25, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11126760408453808, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11184328926286918, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11208742800341798, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11266311318174911, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11323879836008022, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:25, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11381448353841132, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11405862227896013, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11463430745729121, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11520999263562234, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16780585380030155, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:5,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16804999254085037, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1686256777191815, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16920136289751259, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1697770480758437, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1700211868163925, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:1, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:25, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17059687199472362, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17117255717305474, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17174824235138586, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17199238109193465, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17256806627026575, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17314375144859687, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07195643598599753, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07253212116432865, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1327502133572951, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:4,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13299435209784388, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:25, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13357003727617497, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1341457224545061, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13472140763283721, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13496554637338598, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13554123155171707, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13611691673004822, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1366926019083793, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1369367406489281, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13751242582725925, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13808811100559035, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:25, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05082908297055436, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:1,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05107322171110315, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051648906889434264, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05222459206776538, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05280027724609649, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05304441598664529, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0536201011649764, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05419578634330752, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05477147152163862, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055015610262187424, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:25, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05559129544051853, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05616698061884965, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11806786001590355, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11831199875645235, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11822459105921884, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11813718336198534, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1180497756647518, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11829391440530063, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1182065067080671, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11811909901083356, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:25, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11803169131360004, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11827583005414885, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11818842235691532, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1181010146596818, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
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MATCH (a {`key`:260}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20336280101654208, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20360693975709090, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20351953205985734, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20343212436262387, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20334471666539036, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20358885540593913, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:25, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20350144770870562, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20341404001147212, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20332663231423856, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20357077105478738, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20348336335755388, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20339595566032032, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.012538483465222138, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:0,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.012782622205770934, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013270899686868527, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013846584865199638, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:25, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014090723605748437, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014666408784079547, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015242093962410662, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01581777914074177, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016061917881290574, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016637603059621683, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017213288237952797, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10823425637593974, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10847839511648855, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10905408029481965, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:25, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10962976547315076, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11020545065148186, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11044958939203069, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11102527457036179, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1116009597486929, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.112176644927024, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11242078366757279, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
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MATCH (a {`key`:260}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11299646884590392, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11357215402423503, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16888225679314478, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:25, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16912639553369355, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697020807120247, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1702777658903558, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1708534510686869, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710975898092357, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17167327498756682, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17224896016589794, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17282464534422903, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
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MATCH (a {`key`:260}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17306878408477785, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17364446926310895, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:25, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17422015444144004, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10987209498732706, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11011623372787584, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11069191890620694, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:26, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11118019638730454, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11175588156563565, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11200002030618447, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11257570548451558, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11315139066284668, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11372707584117782, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:1, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:25, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1139712145817266, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1145468997600577, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11512258493838881, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16771844610306805, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:5,
`saturationRatio`:0.9, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16796258484361684, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16853827002194796, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16911395520027908, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1696896403786102, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16993377911915897, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17050946429749012, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:25, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1710851494758212, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17166083465415233, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17190497339470112, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17248065857303224, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0712933431104329, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07186902828876401, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07244471346709512, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13266280566006156, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:4,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13290694440061035, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13348262957894147, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:25, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1340583147572726, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13463399993560368, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13487813867615248, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1354538238544836, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13602950903281472, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13660519421114578, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1368493329516946, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13742501813002572, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13800070330835681, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05074167527332084, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:1,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:25, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05098581401386964, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05156149919220075, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05213718437053186, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05271286954886298, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05295700828941177, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053532693467742885, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05410837864607399, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05468406382440511, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05492820256495391, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05550388774328502, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:25, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05607957292161613, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11864354519423466, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11888768393478348, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11880027623754995, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11871286854031642, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11862546084308291, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11886959958363173, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1187821918863982, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11869478418916467, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11860737649193116, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:25, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11885151523247996, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11876410753524642, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11867669983801292, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2039384861948732, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20418262493542197, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20409521723818846, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20400780954095496, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20392040184372143, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20416454058427025, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20407713288703674, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:25, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20398972518980318, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20390231749256968, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2041464562331185, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
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MATCH (a {`key`:261}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20405904853588494, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20397164083865144, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01311416864355325, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:0,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013358307384102047, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013270899686868527, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013759177167966121, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014003315908514917, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:25, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01457900108684603, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015154686265177141, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015730371443508254, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015974510184057048, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01655019536238816, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017125880540719274, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10814684867870623, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10839098741925501, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10896667259758612, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10954235777591723, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:25, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11011804295424836, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11036218169479715, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11093786687312826, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11151355205145937, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11208923722979047, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11233337597033928, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11290906114867039, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11348474632700149, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16879484909591125, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16903898783646004, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:25, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16961467301479116, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17019035819312225, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17076604337145337, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710101821120022, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1715858672903333, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1721615524686644, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
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MATCH (a {`key`:261}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17273723764699553, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17298137638754432, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17355706156587541, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17413274674420653, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:1, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:25, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10978468729009355, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11002882603064232, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11060451120897344, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:27, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11109278869007103, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11166847386840215, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11191261260895094, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
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MATCH (a {`key`:262}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11248829778728205, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11306398296561318, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11363966814394429, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11388380688449308, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:25, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11445949206282419, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1150351772411553, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16763103840583454, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:5,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678751771463833, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16845086232471446, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16902654750304555, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16960223268137667, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16984637142192546, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17042205660025658, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17099774177858768, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:25, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1715734269569188, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:20}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17181756569746762, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07096179667265058, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07120593541319938, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07178162059153048, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0723573057698616, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13257539796282802, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:4,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13281953670337684, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13339522188170794, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13397090706003906, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:25, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13454659223837015, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13479073097891897, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13536641615725006, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13594210133558118, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13651778651391228, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1367619252544611, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1373376104327922, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1379132956111233, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05065426757608732, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:1,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05089840631663611, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:25, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051474091494967227, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05204977667329834, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05262546185162946, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05286960059217825, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05344528577050936, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05402097094884047, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05459665612717159, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05484079486772038, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0554164800460515, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055992165224382606, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:25, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1192192303725658, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11946336911311459, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11937596141588105, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11928855371864755, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11920114602141402, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11944528476196281, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1193578770647293, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1192704693674958, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11918306167026227, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11942720041081106, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:25, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11933979271357756, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11925238501634403, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2045141713732043, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20475831011375312, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2046709024165196, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20458349471928605, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20449608702205255, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20474022576260137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20465281806536784, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20456541036813433, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:25, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20447800267090083, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2047221414114496, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2046347337142161, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20454732601698258, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013689853821884363, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:0,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013846584865199638, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013759177167966121, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013915908211281396, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014491593389612506, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:25, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015067278567943623, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015642963746274732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
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MATCH (a {`key`:262}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015887102486823533, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016462787665154643, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01703847284348576, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1080594409814727, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1083035797220215, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10887926490035262, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10945495007868372, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11003063525701484, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:25, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11027477399756362, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11085045917589473, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11142614435422583, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11200182953255697, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11224596827310576, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11282165345143687, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11339733862976799, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16870744139867774, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16895158013922654, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16952726531755763, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:1, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:25, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17010295049588875, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17067863567421987, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17092277441476866, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17149845959309976, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1720741447714309, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.172649829949762, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728939686903108, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1734696538686419, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17404533904697303, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10969727959286002, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:25, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10994141833340881, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11051710351173993, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:28, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11133692743061983, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11191261260895094, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11215675134949973, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11273243652783087, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11330812170616197, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11388380688449308, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11412794562504186, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.114703630803373, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:25, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11527931598170413, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678751771463833, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:5,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16811931588693213, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16869500106526325, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16927068624359434, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
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MATCH (a {`key`:263}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16984637142192546, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17009051016247426, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17066619534080538, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17124188051913647, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:19}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17181756569746762, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:25, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07063025023486826, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07120593541319938, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07145007415374817, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07202575933207929, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0726014445104104, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13281953670337684, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:4,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13306367544392564, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13363936062225673, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13421504580058785, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13479073097891897, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:25, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13503486971946776, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13561055489779886, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13618624007613, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1367619252544611, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13700606399500986, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.137581749173341, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381574343516721, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
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MATCH (a {`key`:263}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05089840631663611, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:1,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05114254505718491, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05171823023551603, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:25, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05229391541384714, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05286960059217825, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05311373933272705, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053689424511058154, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054265109689389274, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05484079486772038, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05508493360826918, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05566061878660029, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
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MATCH (a {`key`:263}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05623630396493139, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11946336911311459, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:25, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11970750785366338, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11962010015642988, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11953269245919634, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11944528476196281, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1196894235025116, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11960201580527811, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1195146081080446, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11942720041081106, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11967133915135987, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11958393145412635, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:25, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11949652375689282, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20475831011375312, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2050024488543019, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20491504115706838, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20482763345983487, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20474022576260137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20498436450315013, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20489695680591663, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20480954910868313, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2047221414114496, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:25, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2049662801519984, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20487887245476488, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20479146475753135, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01393399256243316, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:0,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014178131302981956, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014090723605748437, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014003315908514917, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.013915908211281396, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014735732130161307, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015311417308492416, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:25, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015887102486823533, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016131241227372327, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01670692640570344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017282611584034553, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1083035797220215, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1085477184625703, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10912340364090141, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10969908881923252, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11027477399756362, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11051891273811242, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:1, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:25, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:263}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11109459791644352, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11167028309477466, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11224596827310576, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11249010701365454, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11306579219198568, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11364147737031678, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16895158013922654, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16919571887977753, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16977140405810642, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17034708923643754, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:25, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17092277441476866, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17116691315531746, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17174259833364858, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723182835119797, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728939686903108, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1731381074308596, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1737137926091907, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17428947778752182, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:29, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10994141833340881, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11018555707395762, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:25, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1107612422522887, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:29, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11124951973338633, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11182520491171741, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11206934365226623, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11264502883059734, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11322071400892844, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11379639918725955, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11404053792780836, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11461622310613949, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1151919082844706, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:25, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16778776944914983, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:5,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16803190818969863, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16860759336802972, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16918327854636084, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16975896372469196, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17000310246524072, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17057878764357184, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:18}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17115447282190296, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06996715735930363, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07054284253763474, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:25, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07111852771596584, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07136266645651465, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07193835163484576, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07251403681317688, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1327321290061433, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:4,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1329762677466921, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1335519529250232, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13412763810335435, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13470332328168544, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13494746202223423, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:25, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13552314720056535, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13609883237889647, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13667451755722757, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1369186562977764, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374943414761075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807002665443857, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050810998619402595, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:1,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0510551373599514, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05163082253828251, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05220650771661362, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:25, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05278219289494473, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05302633163549352, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05360201681382464, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054177701992155755, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05475338717048686, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054997525911035655, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05557321108936677, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05614889626769789, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1200390542914457, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1202831930319945, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:25, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12019578533476098, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12010837763752745, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12002096994029392, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12026510868084272, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1201777009836092, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1200902932863757, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1200028855891422, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12024702432969098, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12015961663245746, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12007220893522395, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:25, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20533399529208424, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.205578134032633, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2054907263353995, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.205403318638166, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20531591094093243, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20556004968148126, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20547264198424775, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20538523428701422, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2052978265897807, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20554196533032953, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:25, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20545455763309597, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20536714993586247, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014509677740764273, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:0,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014753816481313069, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014666408784079547, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01457900108684603, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014491593389612506, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.014735732130161307, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015224009611258898, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01579969478959001, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:1, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:25, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016043833530138808, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016619518708469918, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017195203886801035, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10821617202478799, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10846031076533677, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10903599594366788, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10961168112199901, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11018736630033009, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11043150504087891, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11100719021921002, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:25, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11158287539754112, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11215856057587223, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11240269931642104, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11297838449475214, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11355406967308325, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.168864172441993, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1691083111825418, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16968399636087292, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.170259681539204, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17083536671753513, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:25, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17107950545808395, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17165519063641504, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17223087581474616, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17280656099307729, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17305069973362608, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17362638491195717, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1742020700902883, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10985401063617531, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11009814937672409, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1106738345550552, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:30, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:25, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1111621120361528, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1117377972144839, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1119819359550327, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1125576211333638, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11313330631169491, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11370899149002604, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11395313023057485, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11452881540890596, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11510450058723706, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1677003617519163, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:5,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:25, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1679445004924651, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16852018567079624, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16909587084912733, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16967155602745843, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16991569476800722, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:17}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.17049137994633834, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.069304064483739, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06987974966207011, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07045543484040122, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07103112001873234, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:25, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07127525875928112, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07185094393761224, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
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MATCH (a {`key`:265}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07242662911594336, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13264472130890978, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:4,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13288886004945857, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1334645452277897, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1340402304061208, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1346159155844519, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13486005432500073, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13543573950333182, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:25, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13601142468166294, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13658710985999406, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13683124860054285, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
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MATCH (a {`key`:265}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13740693377887397, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798261895720507, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05072359092216908, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:1,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05096772966271787, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05154341484104899, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052119100019380096, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05269478519771121, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:25, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05293892393826001, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05351460911659112, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05409029429492223, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054665979473253336, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05491011821380214, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05548580339213326, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05606148857046436, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12061473946977683, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12085887821032561, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12077147051309209, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:25, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12068406281585856, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12059665511862505, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12084079385917385, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12075338616194031, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12066597846470681, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
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MATCH (a {`key`:265}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12057857076747328, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12082270950802208, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12073530181078858, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12064789411355506, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20590968047041533, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:25, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20615381921096412, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20606641151373062, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20597900381649709, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20589159611926358, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20613573485981238, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20604832716257884, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
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MATCH (a {`key`:265}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20596091946534534, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20587351176811186, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2061176505086606, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2060302428114271, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:1, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:25, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2059428351141936, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015085362919095386, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:0,
`saturationRatio`:0.5, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015329501659644182, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015242093962410662, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015154686265177141, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015067278567943623, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:265}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015311417308492416, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015224009611258898, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015712287092356492, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01595642583290529, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:25, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016532111011236403, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017107796189567512, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10812876432755446, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10837290306810325, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10894858824643437, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10952427342476548, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11009995860309658, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1103440973436454, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11091978252197648, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11149546770030759, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:25, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11207115287863872, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11231529161918752, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11289097679751862, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11346666197584974, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16877676474475947, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1690209034853083, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16959658866363939, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701722738419705, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17074795902030163, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17099209776085042, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:25, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1715677829391815, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17214346811751266, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17271915329584375, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17296329203639255, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17353897721472367, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1741146623930548, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10976660293894178, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11001074167949057, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11058642685782169, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:31, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11107470433891928, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:25, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11165038951725038, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11189452825779919, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1124702134361303, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1130458986144614, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11362158379279254, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11386572253334133, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11444140771167244, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11501709289000356, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16761295405468277, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:5,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678570927952316, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:25, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1684327779735627, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1690084631518938, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16958414833022492, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:16}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982828707077371, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06897251804595668, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06921665678650549, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0697923419648366, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07036802714316771, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07094371232149882, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711878510620476, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:25, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07176353624037872, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07233922141870983, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13255731361167625, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:4,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13280145235222507, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13337713753055616, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13395282270888728, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1345285078872184, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1347726466277672, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13534833180609832, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13592401698442944, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:25, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13649970216276053, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13674384090330932, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13731952608164047, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13789521125997153, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05063618322493556, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:1,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05088032196548435, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05145600714381546, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05203169232214658, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05260737750047769, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052851516241026485, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:25, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0534272014193576, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05400288659768871, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05457857177601982, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05482271051656862, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055398395694899724, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055974080873230844, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12119042464810793, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12143456338865671, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1213471556914232, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12125974799418969, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:25, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12117234029695616, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12141647903750495, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12132907134027145, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12124166364303791, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12115425594580438, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12139839468635319, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12131098698911967, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12122357929188617, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20648536564874645, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20672950438929527, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:1, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:25, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2066420966920617, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2065546889948282, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2064672812975947, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20671142003814347, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20662401234090996, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20653660464367646, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2064491969464429, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20669333568699172, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20660592798975824, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20651852029252468, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:25, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015661048097426494, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:0,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01581777914074177, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015730371443508254, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
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MATCH (a {`key`:266}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015642963746274732, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015887102486823533, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01579969478959001, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015712287092356492, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015869018135671768, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016444703314002877, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:25, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017020388492333994, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10804135663032093, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10828549537086975, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10886118054920085, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10943686572753196, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:266}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11001255090586308, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11025668964641187, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11083237482474298, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11140806000307411, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11198374518140519, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:25, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11222788392195399, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11280356910028512, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11337925427861623, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16868935704752597, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16893349578807476, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16950918096640585, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.170084866144737, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706605513230681, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709046900636169, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.171480375241948, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:25, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17205606042027913, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17263174559861022, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.172875884339159, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17345156951749016, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17402725469582125, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10967919524170824, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10992333398225707, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11049901916058817, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:32, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11131884307946806, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11189452825779919, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:25, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11213866699834799, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11271435217667909, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11329003735501021, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11386572253334133, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11410986127389011, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11468554645222125, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11526123163055235, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1678570927952316, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:5,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16810123153578038, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16867691671411147, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:25, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1692526018924426, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:13}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16982828707077371, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06864097160817437, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06921665678650549, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06946079552705427, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07003648070538539, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0706121658837165, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711878510620476, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07143198980259642, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07200767498092751, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:25, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07258336015925863, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13280145235222507, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:4,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13304559109277386, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13362127627110496, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1341969614494361, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1347726466277672, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.135016785368316, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1355924705466471, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13616815572497823, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13674384090330932, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:25, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13698797964385814, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13756366482218924, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13813935000052036, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:4,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05088032196548435, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:1,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05112446070603315, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05170014588436426, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05227583106269538, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052851516241026485, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05309565498157528, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05367134015990639, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:25, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0542470253382375, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05482271051656862, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05506684925711742, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055642534435448525, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056218219613779645, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12143456338865671, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12167870212920551, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.121591294431972, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12150388673473847, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12141647903750495, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:1, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:25, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12166061777805376, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12157321008082023, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1214858023835867, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12139839468635319, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12164253342690198, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12155512572966848, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12146771803243497, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20672950438929527, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20697364312984404, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20688623543261053, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:25, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20679882773537703, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20671142003814347, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2069555587786923, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20686815108145878, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20678074338422522, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20669333568699172, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20693747442754054, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.206850066730307, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2067626590330735, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015905186837975295, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:0,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:25, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01614932557852409, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016061917881290574, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015974510184057048, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015887102486823533, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016131241227372327, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016043833530138808, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01595642583290529, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.015869018135671768, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016688842054551678, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017264527232882788, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:0,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:25, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10828549537086975, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10852963411141853, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10910531928974963, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10968100446808077, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11025668964641187, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11050082838696067, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1110765135652918, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11165219874362288, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11222788392195399, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1124720226625028, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:25, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1130477078408339, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.113623393019165, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:3,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16893349578807476, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16917763452862355, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16975331970695468, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17032900488528577, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709046900636169, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711488288041657, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717245139824968, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17230019916082792, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:25, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.172875884339159, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17312002307970784, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17369570825803893, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17427139343637005, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10992333398225707, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11016747272280585, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11074315790113695, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:33, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11123143538223455, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11180712056056566, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11205125930111445, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:25, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11262694447944559, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1132026296577767, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1137783148361078, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11402245357665661, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11459813875498771, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11517382393331882, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
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MATCH (a {`key`:268}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16776968509799806, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:5,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16801382383854685, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16858950901687797, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:12}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1691651941952091, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:25, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06797787873260973, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06855356391094085, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06912924908927195, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06937338782982076, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06994907300815187, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07052475818648297, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711004433648141, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
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MATCH (a {`key`:268}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07134458210536289, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.071920267283694, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07249595246202512, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:25, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13271404465499154, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:4,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13295818339554033, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13353386857387145, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13410955375220257, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13468523893053366, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13492937767108248, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13550506284941358, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1360807480277447, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13665643320607582, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1369005719466246, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:25, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13747625712495573, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:4,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13805194230328685, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:4,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05079291426825083, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:1,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05103705300879963, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05161273818713075, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05218842336546185, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05276410854379297, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05300824728434176, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05358393246267287, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054159617641003986, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:1, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:25, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054735302819335106, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05497944155988389, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05555512673821501, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05613081191654612, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12201024856698782, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12225438730753664, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12216697961030311, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12207957191306958, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12199216421583607, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12223630295638487, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:25, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12214889525915133, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12206148756191783, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12197407986468431, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12221821860523309, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12213081090799958, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12204340321076605, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20730518956762634, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20754932830817516, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20746192061094165, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2073745129137081, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:25, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2072871052164746, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2075312439570234, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20744383625978985, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20735642856255634, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20726902086532284, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2075131596058716, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2074257519086381, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20733834421140462, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:1.103638323514327, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016480872016306405, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:0,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016725010756855202, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:25, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016637603059621683, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01655019536238816, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016462787665154643, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01670692640570344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016619518708469918, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016532111011236403, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016444703314002877, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.016688842054551678, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01717711953564927, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:0,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10819808767363621, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:25, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10844222641418501, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10901791159251613, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10959359677084723, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11016928194917834, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11041342068972716, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11098910586805827, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11156479104638937, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11214047622472051, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11238461496526927, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11296030014360038, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:3,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:25, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1135359853219315, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:3,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16884608809084123, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
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MATCH (a {`key`:268}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16909022683139005, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16966591200972114, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17024159718805226, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17081728236638338, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17106142110693218, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17163710628526327, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1722127914635944, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1727884766419255, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:25, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1730326153824743, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17360830056080542, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17418398573913654, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10983592628502353, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11008006502557233, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11065575020390343, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:34, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11114402768500103, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:3,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11171971286333214, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:3,
`saturationRatio`:0.9, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11196385160388095, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:3,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11253953678221205, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:3,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:25, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11311522196054316, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:3,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1136909071388743, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:3,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11393504587942307, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:3,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11451073105775418, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:3,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.11508641623608531, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:3,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16768227740076452, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:5,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16792641614131332, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:5,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:11}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.16850210131964447, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:5,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06731478585704509, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06789047103537621, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:25, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06846615621370732, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:2,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06904184139203844, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:2,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06928598013258723, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:2,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06986166531091835, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:2,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07043735048924947, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:2,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07101303566758058, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:2,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07125717440812937, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:2,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07183285958646049, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:2,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0724085447647916, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:2,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.132626636957758, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:4,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:25, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13287077569830683, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:4,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13344646087663792, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:4,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13402214605496904, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:4,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13459783123330016, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:4,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13484196997384895, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:4,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13541765515218007, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:4,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1359933403305112, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:4,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13656902550884228, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:4,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13681316424939108, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:4,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13738884942772223, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:4,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:1, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:25, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13796453460605332, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:4,
`saturationRatio`:0.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05070550657101731, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:1,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05094964531156611, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05152533048989722, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052101015668228334, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267670084655944, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05292083958710824, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05349652476543936, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05407220994377047, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05464789512210159, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:25, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054892033862650375, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055467719040981495, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0560434042193126, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12258593374531895, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12283007248586775, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12274266478863421, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12265525709140071, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12256784939416718, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12281198813471597, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12272458043748247, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:25, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12263717274024893, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12254976504301542, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12279390378356422, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12270649608633069, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12261908838909717, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:4,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20788087474595748, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20812501348650628, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20803760578927274, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20795019809203924, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20786279039480574, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:25, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2081069291353545, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.208019521438121, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20793211374088 75, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20784470604365393, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20808884478420275, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.20800143708696925, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2079140293897357, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:6,
`saturationRatio`:0.5272727272727272, `shadowDecay`:1.103638323514327, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017056557194637518, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:0,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017300695935186315, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:0,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017213288237952797, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:0,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:25, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017125880540719274, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:0,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01703847284348576, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:0,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017282611584034553, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:0,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017195203886801035, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:0,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017107796189567512, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:0,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017020388492333994, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:0,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017264527232882788, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:0,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01717711953564927, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:0,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10811067997640268, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1083548187169515, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:25, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10893050389528261, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10950618907361372, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11008187425194484, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11032601299249363, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11090169817082474, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11147738334915587, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11205306852748698, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11229720726803577, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:3,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11287289244636689, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:3,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11344857762469798, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:3,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:25, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16875868039360772, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16900281913415652, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695785043124876, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17015418949081876, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17072987466914985, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17097401340969864, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17154969858802974, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17212538376636088, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17270106894469198, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17294520768524077, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:25, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17352089286357192, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.174096578041903, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10974851858779, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1099926573283388, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11056834250666993, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:35, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1388722563875389, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13944794156586998, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13969208030641878, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14026776548474992, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14084345066308102, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:26, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14141913584141214, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14166327458196093, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14223895976029205, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14281464493862314, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14109291043918631, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:10}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1413370491797351, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09250145614126622, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.092745594881815, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0933212800601461, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09389696523847722, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:26, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09447265041680833, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09471678915735714, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09529247433568826, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09586815951401936, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09644384469235047, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09668798343289928, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09726366861123038, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0978393537895615, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10829453754644561, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10853867628699441, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:1, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:26, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10911436146532552, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10969004664365663, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11026573182198776, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11050987056253653, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11108555574086766, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11166124091919877, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11223692609752987, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11248106483807868, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11305675001640979, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1136324351947409, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:26, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13481834767101875, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:4,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13506248641156757, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13563817158989866, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13621385676822978, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
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MATCH (a {`key`:270}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13678954194656087, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13703368068710 97, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13760936586544079, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381850510437719, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.138760736222103, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390048749626518, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:26, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1395805601409829, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14015624531931403, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15037018258004056, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:5,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15061432132058936, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:5,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15052691362335585, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:5,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15043950592612232, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15035209822888881, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1505962369694376, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15050882927220408, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15042142157497057, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:26, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15033401387773707, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15057815261828583, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15049074492105233, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15040333722381882, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15196280272826196, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15220694146881075, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15211953377157725, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15203212607434372, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15194471837711018, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152188857117659, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:26, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15210144942042544, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15201404172319194, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1519266340259584, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1521707727665072, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1520833650692737, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15199595737204016, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10807752533262445, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10832166407317326, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10823425637593974, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10814684867870623, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:26, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1080594409814727, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1083035797220215, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10821617202478799, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10812876432755446, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10804135663032093, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10828549537086975, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10819808767363621, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10811067997640268, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017822128060062142, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:0,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018397813238393252, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:0,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:26, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018973498416724365, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019549183595055475, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020369007513935385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020944692692266495, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02152037787059761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022340201789477522, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022915886967808632, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1506716151447162, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:4,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:26, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15091575388526499, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1514914390635961, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1520671242419272, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15264280942025832, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528869481608071, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15346263333913823, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15403831851746935, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15461400369580044, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15485814243634927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15543382761468036, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:26, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15600951279301148, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13747674729032786, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772088603087665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13829657120920777, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:36, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13911639512808768, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13969208030641878, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1399362190469676, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14051190422529872, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1410875894036298, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14166327458196093, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:26, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14190741332250972, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14248309850084084, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14305878367917194, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:9}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1413370491797351, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041443304722789516, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.092745594881815, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09298973362236379, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09356541880069491, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09414110397902602, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09471678915735714, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:1, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:26, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:271}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09496092789790593, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09553661307623704, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09611229825456816, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09668798343289928, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09693212217344807, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09750780735177919, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0980834925301103, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10853867628699441, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10878281502754321, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10935850020587431, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:26, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10993418538420545, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
```

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MATCH (a {`key`:271}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11050987056253653, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11075400930308534, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11132969448141644, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11190537965974755, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11248106483807868, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11272520357862748, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11330088875695858, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11387657393528972, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13506248641156757, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:4,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:26, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13530662515211636, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13588231033044745, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
```

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MATCH (a {`key`:271}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13645799550877857, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13703368068710 97, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13727781942765846, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13785350460598958, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1384291897843207, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390048749626518, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13924901370320059, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13982469888153173, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:26, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14040038405986283, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15061432132058936, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:5,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15085846006113818, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:5,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15077105236390465, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15068364466666711, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15059623696943 76, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15084037570998 64, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15075296801275287, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15066556031551936, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15057815261828583, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:26, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15082229135883463, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15073488366160112, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1506474759643676, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15220694146881075, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
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```
MATCH (a {`key`:271}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15245108020935957, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15236367251212604, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522762648148925, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152188857117659, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15243299585820777, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15234558816097427, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:26, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15225818046374073, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1521707727665072, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15241491150705602, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1523275038098225, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15224009611258896, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:271}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10832166407317326, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10856580281372205, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10847839511648855, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10839098741925501, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1083035797220215, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:26, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1085477184625703, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10846031076533677, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10837290306810325, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10828549537086975, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10852963411141853, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10844222641418501, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
```

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MATCH (a {`key`:271}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1083548187169515, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.017822128060062142, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:0,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018641951978942046, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019217637157273162, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:26, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020037461076153073, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020613146254484183, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021188831432815296, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022008655351695206, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02258434053002632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02316002570835743, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15091575388526499, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15115989262581378, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:26, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517355778041449, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152311262982476, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528869481608071, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1531310869013559, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15370677207968703, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15428245725801812, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15485814243634927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15510228117689806, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:271}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15567796635522915, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1562536515335603, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:26, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772088603087665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13796502477142544, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13854070994975656, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:37, `state`:'FLOW', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13902898743085415, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13960467260918527, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13984881134973406, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14042449652806518, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1410001817063963, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14157586688472742, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1418200056252762, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:1, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:26, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1423956908036073, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:8}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14297137598193843, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04078021184722489, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041355897025556, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09265818718458148, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09290232592513029, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09347801110346139, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0940536962817925, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09462938146012362, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0948735202006724, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:26, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09544920537900352, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09602489055733464, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09660057573566574, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09684471447621455, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09742039965454566, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09799608483287678, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10845126858976088, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10869540733030968, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10927109250864081, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10984677768697192, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:26, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11042246286530302, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11066660160585183, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11124228678418292, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11181797196251404, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11239365714084515, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11263779588139394, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11321348105972508, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11378916623805618, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13497507871433403, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:4,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13521921745488283, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:26, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13579490263321395, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13637058781154507, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13694627298987616, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13719041173042498, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13776609690875607, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13834178208708717, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13891746726541826, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13916160600596708, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1397372911842982, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1403129763626293, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:26, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1511900064989205, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:5,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1514341452394693, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15134673754223577, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15125932984500226, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15117192214776873, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15141606088831752, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15132865319108402, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15124124549385048, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15115383779661695, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15139797653716577, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:26, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15131056883993224, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512231611426987, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15278262664714187, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:3,
`saturationRatio`:0.45454545454545453, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15302676538769067, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15293935769045713, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15285194999322363, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1527645422959901, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530086810365389, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15292127333930539, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909091, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15283386564207185, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:26, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15274645794483835, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15299059668538711, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1529031889881536, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528157812909201, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10889734925150438, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10914148799205316, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10905408029481965, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10896667259758612, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10887926490035262, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10912340364090141, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:26, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10903599594366788, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10894858824643437, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10886118054920085, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10910531928974963, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10901791159251613, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10893050389528261, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018397813238393252, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:0,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018641951978942046, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01913022946003964, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019705914638370754, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:26, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01995005337891955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02052573855725066, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021101423735581774, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021677108913912887, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021921247654461684, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022496932832792797, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023072618011123907, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15082834618803148, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15107248492858025, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15164817010691137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:26, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15222385528524246, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1527995404635736, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530436792041224, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1536193643824535, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15419504956078461, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15477073473911573, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15501487347966453, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15559055865799565, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15616624383632677, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13763347833364314, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:1, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:26, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13787761707419197, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13845330225252303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:38, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13894157973362065, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13951726491195174, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13976140365250053, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14033708883083168, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14091277400916277, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1414884591874939, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1417325979280427, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:7}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1423082831063738, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:26, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04011711897166025, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040692804149991366, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04126848932832248, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09257077948734796, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09281491822789675, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09339060340622787, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.093966288584559, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09454197376289009, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0947861125034389, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09536179768177, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:26, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09593748286010112, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09651316803843223, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09675730677898102, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09733299195731214, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09790867713564326, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10836386089252736, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10860799963307617, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10918368481140728, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10975936998973838, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11033505516806952, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:26, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11057919390861831, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11115487908694942, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11173056426528054, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11230624944361164, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11255038818416044, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11312607336249154, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11370175854082265, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1348876710171005, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:4,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13513180975764932, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13570749493598042, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:26, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13628318011431154, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13685886529264266, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13710300403319145, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13767868921152254, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13825437438985366, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13883005956818475, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13907419830873355, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13964988348706467, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14022255686653958, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517656916772516, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:5,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:26, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1520098304178004, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15192242272056689, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15183501502333335, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15174760732609982, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15199174606664864, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1519043383694151, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15181693067218158, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15172952297494807, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15197366171549687, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15188625401826333, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:26, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517988463210298, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153358311825473, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:3,
`saturationRatio`:0.4636363636363636, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1536024505660218, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15351504286878828, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15342763517155472, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15334022747432122, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15358436621487, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15349695851763648, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340955082040297, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15332214312316944, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:26, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15356628186371823, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15347887416648473, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1533914664692512, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10947303442983548, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10971717317038429, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10962976547315076, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10954235777591723, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10945495007868372, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10969908881923252, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10961168112199901, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:26, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10952427342476548, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10943686572753196, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10968100446808077, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10959359677084723, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10950618907361372, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.018973498416724365, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:0,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019217637157273162, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01913022946003964, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01961850694113723, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019862645681686032, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:26, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020438330860017142, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02101401603834826, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021589701216679368, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021833839957228166, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02240952513555928, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022985210313890392, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15074093849079798, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15098507723134674, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15156076240967783, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15213644758800896, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:1, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:26, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15271213276634008, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15295627150688887, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15353195668522, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541076418635511, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1546833270418822, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15492746578243102, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15550315096076212, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15607883613909324, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13754607063640964, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13779020937695843, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:26, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13836589455528953, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1388541720363871, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13942985721471823, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13967399595526703, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14024968113359815, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14082536631192924, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14140105149026036, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:6}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14164519023080918, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03978557253387793, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04002971127442673, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:26, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04060539645275784, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04118108163108896, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09248337179011444, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09272751053066323, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09330319570899435, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09387888088732546, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09445456606565657, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09469870480620539, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09527438998453648, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0958500751628676, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:26, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09642576034119871, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0966698990817475, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09724558426007862, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09782126943840973, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10827645319529386, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
```

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MATCH (a {`key`:274}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10852059193584264, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10909627711417375, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10967196229250488, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11024764747083599, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11049178621138478, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:26, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11106747138971591, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11164315656804702, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11221884174637813, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1124629804869269, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11303866566525804, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11361435084358915, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.134800263319867, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:4,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350444020604158, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13562008723874688, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13619577241707803, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:26, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13677145759540912, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13701559633595792, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13759128151428904, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13816696669262013, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13874265187095125, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13898679061150004, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13956247578983116, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14013816096816226, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15234137685558272, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:5,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1525855155961315, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:26, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152498107898898, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15241070020166447, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15232329250443097, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15256743124497976, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15248002354774623, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15239261585051272, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1523052081532792, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152549346893828, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
```

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MATCH (a {`key`:274}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15246193919659445, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15237453149936095, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:26, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15393399700380409, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:3,
`saturationRatio`:0.4727272727272727, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541781357443529, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15409072804711935, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15400332034988584, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15391591265265234, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541600513932011, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1540726436959676, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1539852359987341, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15389782830150056, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:274}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15414196704204935, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:26, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15405455934481585, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15396715164758232, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1100487196081666, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11029285834871154, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11020545065148186, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11011804295424836, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11003063525701484, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11027477399756362, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11018736630033009, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11009995860309658, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:26, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11001255090586308, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11025668964641187, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11016928194917834, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11008187425194484, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019549183595055475, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:0,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019705914638370754, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01961850694113723, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01977523798445251, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020350923162783623, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:1, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:26, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020926608341114736, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02150229351944585, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021746432259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02232211743832576, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02289780261665687, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15065353079356444, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508976695341132, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15147335471244436, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15204903989077545, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15262472506910654, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:26, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:274}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15286886380965536, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15344454898798646, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15402023416631758, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545959193446487, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548400580851975, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15541574326352858, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15599142844185973, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:40, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374586629391761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377028016797249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13827848685805602, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:26, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13909831077693594, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
```

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MATCH (a {`key`:275}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13967399595526703, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13991813469581582, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14049381987414694, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14106950505247803, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:5}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14164519023080918, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03945402609609562, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04002971127442673, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04027385001497553, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04084953519330664, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:26, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041425220371637754, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09272751053066323, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09297164927121204, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09354733444954315, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09412301962787427, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09469870480620539, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09494284354675417, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09551852872508527, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09609421390341638, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0966698990817475, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:26, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09691403782229631, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09748972300062742, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09806540817895854, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10852059193584264, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10876473067639143, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10934041585472257, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10991610103305367, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11049178621138478, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11073592495193359, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1113116101302647, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:26, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1118872953085958, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1124629804869269, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1127071192274757, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11328280440580683, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:275}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11385848958413794, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1350444020604158, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:4,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13528854080096459, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1358642259792957, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13643991115762683, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13701559633595792, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:26, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372597350765067, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13783542025483783, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13841110543316895, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13898679061150004, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13923092935204887, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13980661453037996, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14038229970871105, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1525855155961315, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:5,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528296543366803, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15274224663944677, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:26, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15265483894221327, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15256743124497976, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15281156998552853, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15272416228829502, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15263675459106152, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152549346893828, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15279348563437675, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15270607793714325, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15261867023990974, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541781357443529, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:3,
`saturationRatio`:0.4818181818181818, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:26, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15442227448490167, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15433486678766817, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15424745909043464, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541600513932011, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15440419013374992, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1543167824365164, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15422937473928286, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:275}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15414196704204935, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15438610578259815, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15429869808536462, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:26, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542112903881311, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11029285834871154, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11053699708926419, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11044958939203069, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11036218169479715, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11027477399756362, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11051891273811242, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11043150504087891, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
```

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MATCH (a {`key`:275}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11034409734364 54, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11025668964641187, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:1, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:26, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11050082838696067, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11041342068972716, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11032601299249363, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019793322335604272, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:0,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020037461076153073, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01995005337891955, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.019862645681686032, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.01977523798445251, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020595061903332417, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021170747081663534, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:26, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021744632259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021990571000543444, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022566256178874554, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02314194135720567, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508976695341132, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15114180827466203, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15171749345299315, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15229317863132422, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15286886380965536, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15311300255020416, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:26, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15368868772853525, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15426437290686637, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548400580851975, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15508419682574628, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1556598820040774, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15623556718240852, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377028016797249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13794694042027372, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13852262559860481, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13901090307970243, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:26, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1395865882580335, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13983072699858232, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1404064121769134, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:4}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14098209735524453, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03879093322053099, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0393666183988621, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03994230357719321, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04018644231774201, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04076212749607312, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.041337812674404235, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:26, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09264010283342972, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09288424157397851, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09345992675230963, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09403561193064075, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09461129710897186, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09485543584952065, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09543112102785176, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09600680620618286, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09658249138451398, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09682663012506278, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:26, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0974023153033939, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.097978000481725, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10843318423860913, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10867732297915793, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10925300815748903, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10982869333582014, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11040437851415127, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11064851725470007, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11122420243303117, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11179988761136228, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:26, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1123755727896934, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1126197115302422, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1131953967085733, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11377108188690443, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13495699436318226, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:4,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13520113310373108, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13577681828206217, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1363525034603933, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13692818863872439, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371723273792732, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:26, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377480125576043, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13832369773593542, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1388993829142665, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13914352165481533, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13971920683314645, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14029489201147755, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15316120077446263, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:5,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340533951501142, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15331793181777792, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532305241205444, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:26, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15314311642331085, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15338725516385968, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15329984746662614, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532124397693926, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15312503207215908, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1533691708127079, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15328176311547437, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15319435541824084, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154753820922684, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:3,
`saturationRatio`:0.4909090909090909, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15499795966323282, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:26, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1549105519659993, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15482314426876576, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15473573657153225, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15497987531208104, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548924676148475, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
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MATCH (a {`key`:276}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154805059917614, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15471765222038047, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15496179096092927, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15487438326369576, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15478697556646223, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:1, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:26, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1108685435270465, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11111268226759531, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11102527457036179, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11093786687312826, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11085045917589473, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:276}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11109459791644352, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11100719021921002, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11091978252197648, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11083237482474298, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1110765135652918, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:26, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11098910586805827, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11090169817082474, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020369007513935385, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:0,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020613146254484183, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02052573855725066, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020438330860017142, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020350923162783623, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020595061903332417, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021083339384430015, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02165902456276113, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:26, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021903163303309922, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022478848481641035, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02305453365997215, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508102618368797, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1510544005774285, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15163008575575962, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15220577093409074, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15278145611242186, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15302559485297063, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15360128003130175, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:26, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15417696520963287, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15475265038796396, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15499678912851278, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15557247430684387, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156148159485175, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1376153939824914, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378595327230402, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13843521790137128, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1389234953824689, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1394991805608, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:26, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13974331930134878, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:3}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1403190044796799, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03812784034496636, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03870352552329746, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03927921070162858, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0398548958799597, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04009903462050849, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0406747197988396, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04125040497717071, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0925526951361962, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:26, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09279683387674499, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09337251905507611, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09394820423340722, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09452388941173834, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09476802815228713, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09534371333061825, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09591939850894936, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09649508368728046, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09673922242782926, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09731490760616036, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:26, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09789059278449148, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10834577654137562, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1085899152819244, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10916560046025553, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10974128563858664, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11031697081691774, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11056110955746654, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11113679473579764, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11171247991412878, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11228816509245988, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:26, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11253230383300866, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1131079890113398, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1136836741896709, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13486958666594875, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:4,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13511372540649755, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13568941058482864, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13626509576315976, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13684078094149088, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13708491968203967, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1376606048603708, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:26, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13823629003870191, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.138811975217033, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390561139575818, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13963179913591292, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.140207484314244, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15373688595279372, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:5,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15398102469334254, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153893616996109, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15380620929887548, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15371880160164197, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:26, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15396294034219077, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15387553264495724, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15378812494772373, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15370071725049023, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153944855991039, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1538574482938055, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15377004059657198, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15532950610101512, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:3,
`saturationRatio`:0.5, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555736448415639, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.5090909090909091, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15548623714433038, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:1, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:26, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15539882944709688, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15531142174986334, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15555556049041214, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15546815279317863, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1553807450959451, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15529333739871157, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555374761392604, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15545006844202686, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15536266074479332, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11144422870537764, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:26, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11168836744592642, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1116009597486929, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11151355205145937, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11142614435422583, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11167028309477466, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11158287539754112, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11149546770030759, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11140806000307411, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11165219874362288, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11156479104638937, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:26, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11147738334915587, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020944692692266495, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:0,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021188831432815296, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021101423735581774, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02101401603834826, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.020926608341114736, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021170747081663534, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021083339384430015, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021571616865527603, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021815755606076404, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:26, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022391440784407513, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02296712596273863, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1507228541396462, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15096699288019497, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15154267805852611, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1521183632368572, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15269404841518833, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15293818715573715, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15351387233406824, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15408955751239933, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:26, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15466524269073045, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15490938143127925, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15548506660961034, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560607517879415, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13752798628525786, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13777212502580666, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13834781020413778, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13883608768523537, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13941177286356649, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:2}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13965591160411528, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:26, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.037796293907184045, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03804043264773284, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03861611782606395, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.039191803004395065, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03976748818272617, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04001162692327497, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040587312101606085, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
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MATCH (a {`key`:278}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0411629972799372, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09246528743896268, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09270942617951147, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:26, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09328511135784258, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0938607965361737, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0944364817145048, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09468062045505361, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09525630563338473, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09583199081171584, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09640767599004696, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09665181473059574, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09722749990892685, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09780318508725797, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:26, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10825836884414208, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10850250758469089, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1090078192763022, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1096538779413531, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11022956311968421, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.110473701860233, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11104938703856414, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11162507221689524, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11220075739522635, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11244489613577516, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:26, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11302058131410626, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11359626649243737, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13478217896871522, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:4,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13502631770926402, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1356020028875951, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13617768806592626, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13675337324425735, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13699751198480614, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1375731971631373, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13814888234146838, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:26, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387245675197995, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1389687062603483, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13954439143867942, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14012007661701048, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15431257113112487, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:5,
`saturationRatio`:0.4, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15455670987167364, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15446930217444013, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15438189447720663, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542944867799731, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545386255205219, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:1, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:26, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15445121782328836, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15436381012605485, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15427640242882132, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545205411693701, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1544331334721366, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15434572577490308, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15590519127934624, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:3,
`saturationRatio`:0.509090909090909, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156149330019895, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15606192232266153, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.155974514625428, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:26, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15588710692819446, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15613124566874328, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15604383797150975, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15595643027427622, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15586902257704272, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1561131613175915, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15602575362035798, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15593834592312447, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11201991388370874, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11226405262425752, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:26, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.112176644927024, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11208923722979047, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11200182953255697, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11224596827310576, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11215856057587223, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11207115287863872, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11198374518140519, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11222788392195399, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11214047622472051, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11205306852748698, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:26, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02152037787059761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:0,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
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MATCH (a {`key`:278}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021677108913912887, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021589701216679368, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02150229351944585, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021746432259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02165902456276113, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021571616865527603, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02172834790884288, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022304033087173995, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:26, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022879718265505108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15063544644241267, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15087958518296146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15145527036129258, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15203095553962367, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15260664071795482, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15285077945850362, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1534264646368347, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15400214981516583, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15457783499349692, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:26, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15482197373404571, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15539765891237683, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15597334409070795, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13744057858802433, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13768471732857315, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13826040250690427, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13908022642578416, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:1}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13965591160411528, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.037464747469401725, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03804043264773284, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:26, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03828457138828163, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03886025656661275, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03943594174494386, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04001162692327497, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040255765663823766, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040831450842154886, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04140713602048599, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:1,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09270942617951147, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09295356492006027, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09352925009839139, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:26, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09410493527672249, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09468062045505361, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0949247591956024, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09550044437393351, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09607612955226463, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09665181473059574, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09689595347114453, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09747163864947565, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09804732382780676, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10850250758469089, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:26, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10874664632523968, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10932233150357079, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10989801668190192, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.110473701860233, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11071784060078182, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11129352577911293, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11186921095744404, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11244489613577516, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11268903487632395, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11326472005465506, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:26, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11384040523298619, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13502631770926402, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:4,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1352704564498128, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13584614162814393, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13642182680647505, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13699751198480614, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13724165072535496, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13781733590368606, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13839302108201718, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1389687062603483, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:1, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:26, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392128450008971, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1397885301792282, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14036421535755 93, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15455670987167364, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:5,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15480084861222243, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15471344091498893, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15462603321775542, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545386255205219, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15478276426107068, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15469535656383718, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:26, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15460794886660364, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545205411693701, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15476467990991893, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1546772722126854, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15458986451545187, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156149330019895, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:3,
`saturationRatio`:0.5181818181818182, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1563934687604438, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1563060610632103, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1562186533659768, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15613124566874328, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:26, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15637538440929205, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15628797671205852, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156200569014825, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15611316131759 15, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15635730005814027, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15626989236090677, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15618248466367327, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:2.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11226405262425752, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11250819136480632, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11242078366757279, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:26, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11233337597033928, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11224596827310576, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11249010701365454, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11240269931642104, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11231529161918752, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11222788392195399, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1124720226625028, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11238461496526927, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11229720726803577, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:3,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021764516611146405, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:0,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:26, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022008655351695206, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021921247654461684, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021833839957228166, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021746432259994643, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021990571000543444, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021903163303309922, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.021815755606076404, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02172834790884288, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02254817182772279, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0231238570060539, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:0,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:26, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15087958518296146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15112372392351026, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15169940910184138, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522750942801725, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15285077945850362, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530949181990524, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15367060337738353, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15424628855571462, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15482197373404571, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15506611247459454, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:26, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15564179765292563, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15621748283125675, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13768471732857315, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13792885606912195, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13850454124745304, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:0}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.13899281872855065, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03680165459383709, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0373773397721682, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03795302495049931, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.038197163691048114, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:27, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03877284886937923, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03934853404771034, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.039924219226041446, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04016835796659025, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04074404314492136, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:1,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04131972832325247, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:1,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09262201848227795, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09286615722282675, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09344184240115785, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09401752757948897, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:27, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0945932127578201, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09483735149836889, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09541303667670001, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09598872185503111, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09656440703336222, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09680854577391103, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09738423095224213, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09795991613057324, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10841509988745737, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10865923862800615, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:27, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10923492380633729, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10981060898466839, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11038629416299995, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11063043290354832, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11120611808187944, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11178180326021053, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11235748843854164, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11260162717909042, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11317731235742155, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11375299753575266, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:1, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:27, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13493891001203048, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:4,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1351830487525793, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1357587339309104, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13633441910924152, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13691010428757264, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13715424302812143, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772992820645255, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13830561338478367, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13888129856311476, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13912543730366356, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:27, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1397011224819947, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1402768076603258, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15513239505000473, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:5,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15537653379055355, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15528912609332002, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1552017183960865, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15511431069885298, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1553584494394018, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15527104174216827, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15518363404493474, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:27, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15509622634770123, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15534036508825003, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1552529573910165, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.155165549693783, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15672501519822613, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:3,
`saturationRatio`:0.5272727272727272, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15696915393877492, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568817462415414, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1567943385443079, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15670693084707438, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15695106958762317, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:27, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15686366189038967, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15677625419315613, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1566888464959226, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15693298523647142, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568455775392379, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15675816984200436, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:2.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11283973780258863, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11308387654313742, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11299646884590392, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11290906114867039, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:27, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11282165345143687, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11306579219198568, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11297838449475214, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11289097679751862, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11280356910028512, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1130477078408339, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11296030014360038, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:3,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11287289244636689, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:3,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022340201789477522, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:0,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02258434053002632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:27, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022496932832792797, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02240952513555928, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02232211743832576, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022566256178874554, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022478848481641035, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0223914407844075 13, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022304033087173995, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02254817182772279, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023036449308820386, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:0,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15079217748572796, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:27, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15103631622627672, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15161200140460787, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15218768658293896, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15276337176127008, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530075105018189, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15358319568015, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15415888085848112, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1547345660368122, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15497870477736103, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555543899556921, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:27, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15613007513402324, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13759730963133962, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13784144837188841, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1384171335502195, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03613856171827246, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:1,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.036714246896603575, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:1,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03728993207493468, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:1,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.037865617253265794, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:1,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.038109755993814595, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:1,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03868544117214571, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:1,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:27, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.039261126350476815, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:1,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.03983681152880793, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:1,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.04008095026935673, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:1,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.040656635447687835, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:1,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04123232062601895, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:1,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09253461078504445, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09277874952559323, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09335443470392434, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09393011988225546, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09450580506058656, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:27, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09474994380113537, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09532562897946648, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0959013141577976, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09647699933612872, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0967211380766775, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09729682325500862, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09787250843333974, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10832769219022384, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10857183093077265, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10914751610910375, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:1, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:27, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10972320128743486, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:3,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11029888646576597, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:3,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11054302520631479, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:3,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1111187103846459, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:3,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.111694395562977, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:3,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1122700807413081, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:3,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11251421948185691, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:3,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11308990466018802, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:3,
`saturationRatio`:0.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11366558983851913, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:3,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13485150231479698, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:4,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:27, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13509564105534577, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:4,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13567132623367686, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:4,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.136247011412008, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:4,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
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MATCH (a {`key`:281}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1368226965903391, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:4,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370668353308879, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:4,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13764252050921905, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:4,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13821820568755014, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:4,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13879389086588126, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:4,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13903802960643005, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:4,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13961371478476117, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:4,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:27, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14018939996309226, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:4,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15570808022833588, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:5,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15595221896888467, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15586481127165114, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1557774035744176, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15568999587718413, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1559341346177329, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1558467269204994, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15575931922326589, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15567191152603235, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:27, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15591605026658115, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582864256934764, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1557412348721141, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15730070037655725, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:3,
`saturationRatio`:0.5363636363636364, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157544839117106, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:3,
`saturationRatio`:0.5454545454545454, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1574574314198725, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:3,
`saturationRatio`:0.5545454545454546, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157370023722639, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:3,
`saturationRatio`:0.5636363636363636, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1572826160254055, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:3,
`saturationRatio`:0.5727272727272728, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1575267547659543, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:3,
`saturationRatio`:0.5818181818181818, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15743934706872076, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:3,
`saturationRatio`:0.5909090909090909, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:27, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15735193937148725, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:3,
`saturationRatio`:0.6, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15726453167425375, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:3,
`saturationRatio`:0.6090909090909091, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15750867041480252, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:3,
`saturationRatio`:0.6181818181818182, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15742126271756898, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:3,
`saturationRatio`:0.6272727272727273, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15733385502033548, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:3,
`saturationRatio`:0.6363636363636364, `shadowDecay`:2.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11341542298091975, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:3,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11365956172146856, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:3,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11357215402423503, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:3,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11348474632700149, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:3,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11339733862976799, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:3,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:27, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11364147737031678, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:3,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11355406967308325, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:3,
`saturationRatio`:0.7, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11346666197584974, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:3,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11337925427861623, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:3,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.113623393019165, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:3,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1135359853219315, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:3,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11344857762469798, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:3,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022915886967808632, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:0,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02316002570835743, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:0,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023072618011123907, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:0,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:27, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022985210313890392, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:0,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02289780261665687, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:0,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02314194135720567, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:0,
`saturationRatio`:0.8, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02305453365997215, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:0,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02296712596273863, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:0,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.022879718265505108, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:0,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0231238570060539, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:0,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023036449308820386, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:0,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:281}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15070476978849442, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15094890852904322, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:27, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15152459370737434, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15210027888570543, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15267596406403658, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15292010280458537, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15349578798291647, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1540714731612476, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1546471583395787, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548912970801275, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1554669822584586, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:58, `threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560426674367897, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:27, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1375099019341061, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13775404067465488, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:296}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13832972585298603, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'FLOW',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09678646703348964, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09703060577403844, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09760629095236953, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09818197613070066, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09857766130903177, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09900180004958056, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09957748522791168, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:27, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10015317040624279, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1007288555845739, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10097299432512272, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10154867950345382, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10212436468178493, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1488318154852745, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490759542258233, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1496516394041544, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15022732458248553, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15080300976081665, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:1, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:261}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:27, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1510471485013654, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15162283367969656, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15219851885802765, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15277420403635877, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530183427769076, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1535940279552387, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541697131335698, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04828472740983425, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048528866150383036, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049104551328714156, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:27, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049680236507045256, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050255921685376376, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050500060425925176, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05107574560425628, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0516514307825874, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052227115960918516, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05247125470146731, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05304693987979843, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05362262505812953, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1888512873372591, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:6,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:27, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18909542607780788, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.189671111256139, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19024679643447012, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19082248161280121, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19106662035335004, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19164230553168113, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19221799071001225, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19279367588834337, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19303781462889216, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19361349980722328, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:27, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1941891849855544, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15543859193947343, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:5,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15568273068002222, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555953229827887, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555079152855552, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15542050758832168, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15566464632887045, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15557723863163697, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15548983093440344, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1554024232371699, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:27, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15564656197771873, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555591542804852, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15547174658325166, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050455339713417924, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:1,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050612070756733206, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05052466305949969, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05043725536226617, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050593986405581444, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:27, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050506578708347925, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05041917101111441, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050575902054429675, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048849435719616, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1687255257498295, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:5,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16896966449037829, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16888225679314478, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16879484909591125, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16870744139867774, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:27, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16895158013922654, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.168864172441993, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16877676474475947, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16868935704752597, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16893349578807476, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16884608809084123, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16875868039360772, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1506716151447162, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:4,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15091575388526499, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15082834618803148, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:27, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15074093849079798, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15065353079356444, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508976695341132, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508102618368797, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1507228541396462, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15063544644241267, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15087958518296146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15079217748572796, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15070476978849442, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023681457833233256, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:0,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:27, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024257143011564366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:0,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02483282818989548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025408513368226592, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02622833728871065, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026804022465437612, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027379707643768725, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028199531562648632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028775216740979746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:48, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:27, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1351250745591046, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:295}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13536921329965343, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'FLOW',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10894498892054905, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09703060577403844, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09727474451458723, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09785042969291835, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09842611487124946, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09900180004958056, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09924593879012937, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09982162396846048, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:1, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:27, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10039730914679158, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10097299432512272, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10121713306567151, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10179281824400262, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10236850342233372, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490759542258233, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14932009296637208, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498957781447032, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15047146332303432, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1510471485013654, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:27, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15129128724191424, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15186697242024536, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15244265759857645, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530183427769076, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15326248151745636, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15383816669578748, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15441385187411857, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048528866150383036, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04877300489093184, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04934869006926294, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:27, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049924375247594056, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050500060425925176, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05074419916647396, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05131988434480508, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051895569523136204, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05247125470146731, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05271539344201611, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05329107862034722, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05386676379867834, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18909542607780788, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:6,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:27, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18933956481835668, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1899152499966878, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1904909351750189, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19106662035335004, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19131075909389883, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19188644427222992, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19246212945056107, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19303781462889216, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19328195336944096, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19385763854777208, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:27, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1944333237261032, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15568273068002222, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:5,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15592686942057102, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1558394617233375, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15575205402610398, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15566464632887045, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15590878506941927, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582137737218577, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15573396967495223, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15564656197771873, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:27, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15589070071826752, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.155803293021034, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15571588532380046, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05094361719451552, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05085620949728201, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05076880180004848, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05092553284336376, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05083812514613024, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:27, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05075071744889672, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050907448492211994, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050820040794978476, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05073263309774496, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16896966449037829, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16921380323092708, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16912639553369355, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16903898783646004, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16895158013922654, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:27, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1691957188797753, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1691083111825418, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1690209034853083, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16893349578807476, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16917763452862355, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16909022683139005, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16900281913415652, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15091575388526499, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:4,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15115989262581378, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15107248492858025, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:27, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15098507723134674, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508976695341132, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15114180827466203, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1510544005774285, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15096699288019497, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15087958518296146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15112372392351026, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15103631622627672, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15094890852904322, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.023681457833233256, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:0,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:27, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024501281752113163, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025076966930444276, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025896790849324183, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0264724760276553, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027048161205986406, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02786798512486632, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028443670303197433, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.029019355481528546, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:49, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:1, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:27, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:294}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13536921329965343, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'FLOW',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10828189604498444, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10885758122331554, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0969431980768049, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09718733681735373, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09776302199568483, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09833870717401592, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09891439235234706, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09915853109289584, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09973421627122694, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:27, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10030990144955808, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10088558662788918, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10112972536843798, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10170541054676908, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10228109572510022, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14898854652858975, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:5,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14923268526913858, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498083704474697, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1503840556258008, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1509597408041319, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:27, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512038795446807, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15177956472301182, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15235524990134294, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15293093507967406, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15317507382022286, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15375075899855395, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15432644417688507, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04844145845314952, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04868559719369831, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049261282372029425, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:27, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049836967550360545, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050412652728691644, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050656791469240445, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051232476647571565, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05180816182590267, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05238384700423379, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05262798574478258, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0532036709231137, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05377935610144482, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18900801838057438, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:6,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:27, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18925215712112317, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18982784229945426, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19040352747778538, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1909792126561165, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1912233513966653, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19179903657499642, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19237472175332754, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19295040693165863, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19319454567220745, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19377023085053854, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:27, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19434591602886966, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15625841585835332, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:5,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15650255459890214, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1564151469016686, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15632773920443507, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15624033150720157, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15648447024775036, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15639706255051683, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.5090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15630965485328335, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15622224715604982, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:27, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15646638589659861, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15637897819936508, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15629157050213158, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05127516363229784, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05151930237284664, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05143189467561311, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051344486978379594, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05125707928114608, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051501218021694876, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05141381032446135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:27, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05132640262722783, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05123899492999431, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05148313367054311, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05139572597330959, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05130831827607607, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695453496687094, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697894884092582, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697020807120247, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16961467301479116, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16952726531755763, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:27, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16977140405810642, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16968399636087292, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16959658866363939, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16950918096640585, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16975331970695468, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16966591200972114, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1695785043124876, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1514914390635961, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:4,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517355778041449, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15164817010691137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:27, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15156076240967783, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15147335471244436, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15171749345299315, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15163008575575962, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15154267805852611, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15145527036129258, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15169940910184138, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15161200140460787, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15152459370737434, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024257143011564366, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:0,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:1, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:27, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024501281752113163, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024989559233210758, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025565244411541867, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025809383152090665, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026385068330421778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02696075350875289, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027536438687084004, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027780577427632798, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02835626260596391, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:293}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028931947784295024, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:50, `state`:'FLOW', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:27, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1076188031694198, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1081944883477509, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10877017352608202, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0968557903795714, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09709992912012019, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0976756142984513, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09825129947678243, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09882698465511353, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09907112339566232, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09964680857399344, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:27, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10022249375232455, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10079817893065565, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10104231767120445, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10161800284953558, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10219368802786669, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14890113883135625, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:5,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14914527757190504, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14972096275023616, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15029664792856728, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1508723331068984, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:27, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15111647184744717, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15169215702577832, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522678422041094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15284352738244053, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15308766612298932, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
```

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MATCH (a {`key`:285}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15366335130132044, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15423903647965156, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048354050755916, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0485981894964648, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04917387467479591, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:27, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04974955985312702, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05032524503145814, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05056938377200694, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05114506895033804, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05172075412866916, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052296439307000266, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052540578047549066, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05311626322588018, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05369194840421129, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18892061068334084, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:6,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:27, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18916474942388964, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18974043460222076, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19031611978055188, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19089180495888297, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1911359436994318, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19171162887776289, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.192287314056094, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19286299923442513, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19310713797497392, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19368282315330504, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:27, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19425850833163613, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15683410103668446, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:5,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15707823977723326, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15699083207999973, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15690342438276622, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568160166855327, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15706015542608148, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15697274772884798, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:285}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15688534003161447, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15679793233438094, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:27, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15704207107492973, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15695466337769623, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1568672556804627, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05185084881062895, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052094987551177745, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052007579853944226, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05192017215671071, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05183276445947719, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05207690320002599, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:285}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05198949550279247, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:27, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051902087805558945, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05181468010832543, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05205881884887422, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051971411151640695, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05188400345440719, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701210348470405, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036517358758932, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1702777658903558, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17019035819312225, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17010295049588875, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:27, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17034708923643754, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.170259681539204, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1701722738419705, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.170084866144737, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17032900488528577, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17024159718805226, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17015418949081876, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1520671242419272, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:4,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152311262982476, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15222385528524246, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:1, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:27, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15213644758800896, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15204903989077545, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15229317863132422, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15220577093409074, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1521183632368572, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15203095553962367, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522750942801725, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15218768658293896, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15210027888570543, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02483282818989548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:0,
`saturationRatio`:0.9, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:27, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:285}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025076966930444276, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.024989559233210758, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02547783671430835, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025721975454857143, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02629766063318826, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02687334581151937, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027449030989850486, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027693169730399276, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:292}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028268854908730396, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'FLOW', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10728725673163747, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:27, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10753139547218628, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10810708065051738, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10868276582884852, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09676838268233788, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09701252142288666, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0975882066012178, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0981638917795489, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09873957695788001, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09898371569842881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09955940087675992, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:27, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10013508605509103, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10071077123342213, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10095490997397094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10153059515230205, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10210628033063315, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14881373113412275, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:5,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14905786987467154, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14963355505300266, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15020924023133375, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15078492540966487, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:27, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15102906415021367, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15160474932854479, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15218043450687588, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152756119685207, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15300025842575582, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1535759436040869, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15415162878241803, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04826664305868248, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04851078179923128, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04908646697756239, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:27, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0496621521558935, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05023783733422462, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048197607477341, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05105766125310453, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05163334643143563, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05220903160976675, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05245317035031555, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053028855528646654, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053604540706977774, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1888332029861073, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:6,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:27, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18907734172665613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18965302690498723, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19022871208331835, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19080439726164947, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19104853600219826, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19162422118052935, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1921999063588605, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1927755915371916, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930197302777404, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1935954154560715, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:27, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19417110063440263, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15740978621501556, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:5,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15765392495556435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15756651725833085, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1574791095610973, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15739170186386378, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576358406044126, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15754843290717907, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15746102520994554, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15737361751271203, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:27, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15761775625326085, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15753034855602732, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1574429408587938, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052426533988960064, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052583265032275346, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05249585733504182, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0524084496378083, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05256518068112358, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:27, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05247777298389006, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05239036528665654, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052547096329971815, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05245968863273829, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17069672002537165, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094085876592044, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1708534510686869, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17076604337145337, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17067863567421987, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:1, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:27, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17092277441476866, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17083536671753513, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17074795902030163, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706605513230681, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709046900636169, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17081728236638338, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17072987466914985, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15264280942025832, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:4,
`saturationRatio`:0.8, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:36,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528869481608071, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1527995404635736, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:27, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15271213276634008, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15262472506910654, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15286886380965536, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15278145611242186, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15269404841518833, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15260664071795482, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15285077945850362, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15276337176127008, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15267596406403658, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:47,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025408513368226592, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:0,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:27, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025565244411541867, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02547783671430835, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025634567757623627, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02621025293595474, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02678593811428585, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027361623292616963, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:291}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'FLOW', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10695571029385517, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10753139547218628, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:27, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10777553421273506, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10835121939106619, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1089269045693973, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09701252142288666, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09725666016343548, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09783234534176659, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0984080305200977, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09898371569842881, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09922785443897761, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0998035396173087, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:27, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10037922479563981, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10095490997397094, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10119904871451973, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10177473389285084, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10235041907118198, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14905786987467154, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:5,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14930200861522036, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14987769379355145, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15045337897188255, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15102906415021367, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:27, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15127320289076246, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:113,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15184888806909358, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15242457324742467, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15300025842575582, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532443971663046, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1538200823446357, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15439576752296683, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04851078179923128, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04875492053978007, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04933060571811119, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:27, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0499062908964423, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048197607477341, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05072611481532221, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051301799993653315, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05187748517198443, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05245317035031555, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052697309090864335, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053272994269195455, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053848679447526575, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18907734172665613, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:6,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:27, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18932148046720493, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18989716564553602, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19047285082386714, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:135,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19104853600219826, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19129267474274705, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19186835992107817, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1924440450994093, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930197302777404, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19326386901828918, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1938395541966203, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:27, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19441523937495142, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15765392495556435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:5,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15789806369611317, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15781065599887964, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1577232483016461, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576358406044126, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1578799793449614, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15779257164772786, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15770516395049433, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15761775625326085, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:27, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15786189499380962, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15777448729657612, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576870795993426, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267067272950886, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05291481147005766, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05282740377282414, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052739996075590614, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0526525883783571, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05289672711890589, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05280931942167238, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:1, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:27, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05272191172443885, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052634504027205334, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052878642767754135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0527912350705206, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05270382737328709, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17094085876592044, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17118499750646923, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710975898092357, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710101821120022, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17092277441476866, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:27, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17116691315531746, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17107950545808395, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17099209776085042, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1709046900636169, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711488288041657, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17106142110693218, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17097401340969864, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528869481608071, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:4,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1531310869013559, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530436792041224, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:27, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15295627150688887, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15286886380965536, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15311300255020416, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15302559485297063, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15293818715573715, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15285077945850362, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530949181990524, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530075105018189, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15292010280458537, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025652652108775386, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:0,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:27, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025896790849324183, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025809383152090665, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025721975454857143, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.025634567757623627, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026454391676503534, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027030076854834648, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:290}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'FLOW', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10629261741829053, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:90,
`threshold`:110}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10686830259662165, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10744398777495275, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:27, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10768812651550155, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10826381169383266, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10883949687216377, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09692511372565316, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09716925246620195, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09774493764453306, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09832062282286419, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0988963080011953, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09914044674174408, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:101,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09971613192007518, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:27, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1002918170984063, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10086750227673741, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1011116410172862, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10168732619561734, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226301137394844, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14897046217743803, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:5,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14921460091798683, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14979028609631795, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15036597127464904, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15094165645298016, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:112,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:27, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15118579519352893, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15176148037186005, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15233716555019117, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1529128507285223, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15315698946907108, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1537326746474022, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15430835982573332, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048423374101997756, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048667512842546556, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04924319802087767, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:27, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049818883199208776, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05039456837753989, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05063870711808868, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0512143922964198, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051790077474750916, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05236576265308202, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05260990139363082, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053185586571961936, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05376127175029304, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18898993402942263, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:6,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:27, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1892340727699714, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18980975794830252, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19038544312663364, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19096112830496473, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19120526704551355, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19178095222384464, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19235663740217576, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19293232258050685, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19317646132105568, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1937521464993868, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:27, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1943278316777179, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1582296101338955, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:5,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15847374887444426, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15838634117721076, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15829893347997726, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15821152578274372, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15845566452329252, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15836825682605898, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15828084912882548, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15819344143159195, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:1, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:27, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15843758017214074, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15835017247490724, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15826276477767373, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05324635790783997, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05349049664838877, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053403088951155246, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053315681253921735, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05322827355668821, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05347241229723701, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053385004600003484, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:27, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053297596902769966, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053210189205536454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05345432794608524, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05336692024885172, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053279512551618204, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17151654394425153, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17176068268480033, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17167327498756682, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1715858672903333, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17149845959309976, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:27, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17174259833364858, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17165519063641504, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1715677829391815, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.171480375241948, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717245139824968, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17163710628526327, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17154969858802974, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15346263333913823, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:4,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15370677207968703, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1536193643824535, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:27, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15353195668522, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15344454898798646, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15368868772853525, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15360128003130175, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15351387233406824, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1534264646368347, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15367060337738353, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15358319568015, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15349578798291647, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0262283372871065, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:0,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:27, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0264724760276553, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026385068330421778, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02629766063318826, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02621025293595474, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026454391676503534, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:289}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02694266915760113, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'FLOW', `targetIndex`:55,
`threshold`:110}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1056295245427259, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10620520972105701, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10678089489938812, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10735658007771924, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:27, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10760071881826803, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10817640399659914, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10875208917493026, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09683770602841964, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09708184476896842, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09765752994729955, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09823321512563066, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09880890030396176, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09905303904451056, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09962872422284168, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:27, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10020440940117278, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10078009457950389, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1010242333200527, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1015999184983838, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10217560367671491, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1488830544802045, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:5,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1491271932207533, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14970287839908444, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
```

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MATCH (a {`key`:289}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1502785635774155, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15085424875574663, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:27, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15109838749629542, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15167407267462654, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15224975785295763, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15282544303128875, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15306958177183758, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15364526695016867, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542209521284998, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04833596640476424, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04858010514531304, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
```

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MATCH (a {`key`:289}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049155790323644144, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:27, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049731475501975264, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05030716068030637, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05055129942085517, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051126984599186284, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05170266977751739, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05227835495584851, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05252249369639731, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05309817887472841, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05367386405305953, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1889025263321891, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:6,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:27, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18914666507273792, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.189722350251069, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1902980354294001, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19087372060773122, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19111785934828002, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1916935445266111, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19226922970494223, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19284491488327335, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19308905362382214, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19366473880215326, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:1, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:27, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19424042398048438, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1588052953122266, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:5,
`saturationRatio`:0.5, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15904943405277538, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15896202635554188, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15887461865830835, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15878721096107482, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1590313497016236, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1589439420043901, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15885653430715657, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15876912660992304, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:27, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:289}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15901326535047186, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15892585765323833, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15883844995600482, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05382204308617109, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05406618182671988, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053978774129486366, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053891366432252834, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05380395873501932, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05404809747556812, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053960689778334604, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:27, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053873282081101086, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:289}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05378587438386756, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05403001312441636, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05394260542718283, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05385519772994932, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17209222912258265, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17233636786313145, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17224896016589794, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1721615524686644, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1720741447714309, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:27, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723182835119797, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17223087581474616, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17214346811751266, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17205606042027913, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17230019916082792, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1722127914635944, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17212538376636088, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15403831851746935, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:4,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15428245725801812, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15419504956078461, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:27, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1541076418635511, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15402023416631758, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15426437290686637, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15417696520963287, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15408955751239933, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15400214981516583, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15424628855571462, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15415888085848112, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1540714731612476, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.026804022465437612, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:0,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:27, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027048161205986406, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02696075350875289, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02687334581151937, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02678593811428585, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:289}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027030076854834648, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:288}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02694266915760113, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'FLOW', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10529797810494358, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554211684549238, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10611780202382348, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1066934872021546, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:28, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10726917238048571, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1075133111210345, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10808899629936564, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10866468147769674, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0967502983311861, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09699443707173491, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09757012225006602, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09814580742839712, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09872149260672823, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09896563134727705, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:28, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09954131652560816, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10011700170393927, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1006926868822704, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10093682562281918, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10151251080115027, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1020881959794814, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.148795646782971, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:5,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490397855235198, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1496154707018509, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.150191155880182, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:28, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1507668410585131, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.151010979799906192, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1515866664977393, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15216235015572413, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15273803533405525, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15298217407460404, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15355785925293514, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15413354443126628, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04824855870753072, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048492697448079505, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:28, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.049068382626410625, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04964406780474173, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05021975298307285, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05046389172362165, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05103957690195276, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05161526208028387, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05219094725861499, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05243508599916378, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0530107711774949, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053586456355826, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:1, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:28, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1888151186349556, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:6,
`saturationRatio`:0.4, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18905925737550439, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18963494255383548, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1902106277321666, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1907863129104977, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19103045165104648, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1916061368293776, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19218182200770872, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19275750718603982, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930016459265886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:28, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19357733110491976, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19415301628325085, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1593809804905577, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:5,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596251192311065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15953771153387297, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15945030383663947, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15936289613940594, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15960703487995473, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15951962718272122, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1594322194854877, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:28, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1593448117882542, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15958895052880298, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15950154283156945, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15941413513433594, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0543977282645022, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05455445930781747, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054467051610583954, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054379643913350435, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:28, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05453637495666571, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054448967259432185, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05436155956219868, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05451829060551395, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05443088290828043, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17266791430091374, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17291205304146257, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17282464534422903, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17273723764699553, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:28, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.172649829949762, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728939686903108, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17280656099307729, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17271915329584375, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17263174559861022, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.172875884339159, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1727884766419255, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17270106894469198, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15461400369580044, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:4,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15485814243634927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:28, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15477073473911573, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1546833270418822, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1545959193446487, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548400580851975, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15475265038796396, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15466524269073045, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15457783499349692, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15482197373404571, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1547345660368122, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1546471583395787, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:28, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027379707643768725, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:0,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027536438687084004, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027449030989850486, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027361623292616963, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:287}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'FLOW', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10496643166716126, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10554211684549238, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10578625558604117, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10636194076437229, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:28, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1069376259427034, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1075133111210345, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10775744986158331, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10833313503991442, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10890882021824552, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09699443707173491, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0972385758122837, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09781426099061481, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09838994616894592, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09896563134727705, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:28, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09920977008782583, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09978545526615694, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10036114044448807, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10093682562281918, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10118096436336797, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10175664954169909, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1023323347200302, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1490397855235198, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:5,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14928392426406858, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498596094423997, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:28, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1504352946207308, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15101097979906192, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512551185396107, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1518308037179418, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15240648889627292, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15298217407460404, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15322631281515284, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15380199799348396, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15437768317181508, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048492697448079505, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:1, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:28, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04873683618862831, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04931252136695942, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04988820654529053, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05046389172362165, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05070803046417044, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05128371564250156, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05185940082083267, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05243508599916378, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05267922473971258, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053254909918043686, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:28, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0538305950963748, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18905925737550439, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:6,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18930339611605318, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1898790812943843, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1904547664727154, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19103045165104648, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19127459039159528, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1918502755699264, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19242596074825752, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1930016459265886, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:28, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19324578466713743, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19382146984546855, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19439715502379964, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596251192311065, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:5,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1598692579716553, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978185027442177, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15969444257718823, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15960703487995473, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15985117362050352, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15976376592327, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:28, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15967635822603649, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15958895052880298, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15983308926935175, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15974568157211824, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596582738748847, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:5,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054641867005051, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05488600574559979, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05479859804836627, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054711190351132755, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05462378265389923, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:28, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05486792139444803, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0547805136972145, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05469310599998099, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05460569830274747, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05484983704329627, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05476242934606275, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.054675021648829224, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17291205304146257, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17315619178201133, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17306878408477785, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:28, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17298137638754432, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728939686903108, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1731381074308596, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17305069973362608, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17296329203639255, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.172875884339159, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17312002307970784, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1730326153824743, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17294520768524077, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15485814243634927, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:4,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:28, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15510228117689806, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15501487347966453, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15492746578243102, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548400580851975, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15508419682574628, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15499678912851278, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15490938143127925, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15482197373404571, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15506611247459454, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15497870477736103, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:28, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1548912970801275, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02762384638431752, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:0,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02786798512486632, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027780577427632798, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027693169730399276, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:286}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02760576203316576, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'FLOW', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10430333879159663, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10487902396992775, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10545470914825886, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10569884788880765, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:28, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10627453306713876, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10685021824546988, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.107425903423801, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10767004216434979, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1082457273426809, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10882141252101202, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0969070293745014, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09715116811505019, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09772685329338128, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09830253847171241, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:28, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09887822365004352, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09912236239059231, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09969804756892343, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10027373274725454, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10084941792558565, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10109355666613447, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10166924184446557, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10224492702279667, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14895237782628626, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:5,
`saturationRatio`:0.2, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14919651656683505, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:1, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:28, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14977220174516617, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1503478869234973, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1509235721018284, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15116771084237718, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1517433960207083, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1523190811990394, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1528947663773705, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15313890511791933, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15371459029625043, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15429027547458155, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.09957413673572789, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:28, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.048405289750846, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04864942849139479, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0492251136697259, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04980079884805702, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05037648402638813, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05062062276693693, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05119630794526805, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051771993123599154, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05234767830193027, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.052591817042479054, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:28, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053167502220810174, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05374318739914129, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18897184967827085, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:6,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18921598841881968, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18979167359715077, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1903673587754819, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19094304395381298, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1911871826943618, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19176286787269287, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19233855305102399, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:28, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1929142382293551, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1931583769699039, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19373406214823502, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19430974732656614, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.14936120510359183, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16020080440943762, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:5,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16044494314998642, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1603575354527529, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16027012775551938, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16018272005828585, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16042685879883464, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:28, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16033945110160114, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1602520434043676, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16016463570713407, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1604087744476829, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16032136675044936, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:5,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16023395905321583, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:5,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055217552183382104, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055461690923930905, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05537428322669738, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05528687552946386, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:28, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05519946783223034, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055443606572779136, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05535619887554562, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0552687911783121, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05518138348107857, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05542552222162739, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055338114524393855, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055250706827160344, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17348773821979366, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17373187696034245, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:28, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17364446926310895, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17355706156587541, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1734696538686419, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1737137926091907, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17362638491195717, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17353897721472367, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17345156951749016, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17369570825803893, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17360830056080542, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17352089286357192, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:28, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15543382761468036, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:4,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15567796635522915, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15559055865799565, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15550315096076212, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15541574326352858, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1556598820040774, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15557247430684387, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15548506660961034, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15539765891237683, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15564179765292563, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:28, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555543899556921, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1554669822584586, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028199531562648632, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:0,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028443670303197433, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02835626260596391, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:285}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028268854908730396, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'FLOW', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.103640245916032, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:3,
`saturationRatio`:0.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10421593109436313, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:3,
`saturationRatio`:0.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10479161627269423, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:3,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10536730145102534, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:3,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:28, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10561144019157412, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:3,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10618712536990524, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:3,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10676281054823636, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:3,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10733849572656746, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:3,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10758263446711626, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:3,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10815831964544739, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:3,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1087340048237785, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:3,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09681962167726788, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:3,
`saturationRatio`:0.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09706376041781666, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09763944559614779, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:1, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:28, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09821513077447888, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09879081595280999, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09903495469335881, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09961063987168992, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10018632505002102, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10076201022835214, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10100614896890094, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.10158183414723204, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10215751932556318, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14886497012905273, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:5,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:28, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14910910886960152, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:5,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14968479404793267, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:5,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15026047922626376, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:5,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15083616440459488, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:5,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15108030314514367, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:5,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1516559883234748, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:5,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1522316735018059, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:5,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152807358680137, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:5,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1530514974206858, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:5,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1536271825990169, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:5,
`saturationRatio`:0.3, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:28, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154202867777348, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:5,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04831788205361247, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04856202079416127, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04913770597249239, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04971339115082349, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:1,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05028907632915461, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:1,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05053321506970341, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:1,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051108900248034515, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:1,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051684585426365635, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:1,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05226027060469674, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:1,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:28, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05250440934524554, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:1,
`saturationRatio`:0.4, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.053080094523576656, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:1,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:130, `threshold`:110}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05365577970190776, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:1,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.1353352832366127, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18888444198103735, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:6,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18912858072158614, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:6,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.18970426589991723, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:6,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19027995107824835, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:6,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19085563625657945, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:6,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19109977499712827, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:6,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19167546017545936, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:6,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:28, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19225114535379048, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:6,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19282683053212157, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:6,
`saturationRatio`:0.5, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19307096927267037, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:6,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.19364665445100152, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:6,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1942223396293326, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:6,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.14936120510359183, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16077648958776872, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:5,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1610206283283175, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:5,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.160933220631084, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:5,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16084581293385047, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:5,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16075840523661694, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:5,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:28, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16100254397716576, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:5,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16091513627993223, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:5,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1608277285826987, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:5,
`saturationRatio`:0.6, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1607403208854652, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:5,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:8, `threshold`:110}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.160984459626014, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:5,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16089705192878045, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:5,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16080964423154695, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:5,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055793237361713224, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05603737610226201, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0559499684050285, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:28, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05586256070779498, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:1,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055775153010561455, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:1,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056019291751110256, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:1,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05593188405387674, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:1,
`saturationRatio`:0.7, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.055844476356643205, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:1,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:19,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05575706865940969, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:1,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.056001207399958494, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:1,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05591379970272496, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:1,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05582639200549146, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:1,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17406342339812478, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:28, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17430756213867357, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17422015444144004, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17413274674420653, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:5,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17404533904697303, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:5,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17428947778752182, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:5,
`saturationRatio`:0.8, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1742020700902883, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:5,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1741146623930548, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:5,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17402725469582125, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:5,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17427139343637005, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:5,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17418398573913654, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:5,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:28, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.174096578041903, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:5,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.4060058497098381, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15600951279301148, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:4,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1562536515335603, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15616624383632677, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15607883613909324, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15599142844185973, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15623556718240852, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.156148159485175, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560607517879415, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15597334409070795, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:28, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15621748283125675, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15613007513402324, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560426674367897, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028775216740979746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:0,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.029019355481528546, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:0,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:284}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028931947784295024, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:0,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'FLOW', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14387860759805185, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:5,
`saturationRatio`:0.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14412274633860064, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:5,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1446984315169318, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:5,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14527411669526288, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:1, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:28, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.145849801873594, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460939406141428, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14666962579247392, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.147245310970805, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14782099614913 61, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14806513488968492, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14864082006801602, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14921650524634714, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:95,
`threshold`:110}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0937842478261751, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:3,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09402838656672391, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:28, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09460407174505502, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09517975692338612, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09575544210171724, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09599958084226602, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09657526602059714, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09715095119892826, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09772663637725937, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09797077511780818, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09854646029613928, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:106,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09912214547447039, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:28, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.042158216324831896, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:1,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04240235506538069, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0429780402437118, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.043553725422042916, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04412941060037402, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04437354934092282, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.044949234519253936, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04552491969758504, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04610060487591616, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.046344743616464956, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:117,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:28, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04692042879479607, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047496113973127176, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10129045080102288, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:3,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10153458954157167, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10211027471990279, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1026859598982339, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10326164507656502, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1035057838171138, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10408146899544492, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10465715417377604, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:28, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10523283935210714, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10547697809265595, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10605266327098706, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10662834844931818, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08133251470962294, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:2,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08157665345017173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08215233862850285, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08272802380683396, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08330370898516506, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08354784772571387, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:28, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08412353290404498, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0846992180823761, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0852749032607072, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.085519042001256, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08609472717958712, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08667041235791823, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050455339713417924, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:1,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050612070756733206, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05052466305949969, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:28, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05043725536226617, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050593986405581444, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050506578708347925, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05041917101111441, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050575902054429675, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05048849435719616, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16444051485760602, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:5,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:28, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1645972459009213, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16450983820368778, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16442243050645428, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16457916154976956, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16449175385253603, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16440434615530253, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1645610771986178, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16447366950138428, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:28, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10971536394401177, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:3,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10995950268456056, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10987209498732706, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10978468729009355, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10969727959286002, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10994141833340881, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10985401063617531, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10976660293894178, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10967919524170824, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10992333398225707, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:28, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10983592628502353, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10974851858779, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13747674729032786, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:4,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772088603087665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13763347833364314, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13754607063640964, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374586629391761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377028016797249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1376153939824914, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13752798628525786, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:1, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:28, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13744057858802433, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13768471732857315, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13759730963133962, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1375099019341061, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1351250745591046, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:283}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13536921329965343, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'FLOW',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059346812363034594, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14412274633860064, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:5,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14436688507914947, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:5,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14494257025748059, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:28, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14551825543581168, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460939406141428, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1463380793546916, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1469137645330227, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1474894497113538, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14806513488968492, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14830927363023372, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1488849588085648, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14946064398689596, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09402838656672391, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:3,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:28, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0942725253072727, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0948482104856038, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09542389566393492, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09599958084226602, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09624371958281483, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09681940476114595, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09739508993947706, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09797077511780818, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09821491385835697, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09879059903668808, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:28, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0993662842150192, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04240235506538069, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:1,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04264649380592948, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0432221789842606, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04379786416259171, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04437354934092282, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.044617688081471624, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04519337325980273, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04576905843813385, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.046344743616464956, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:28, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04658888235701376, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04716456753534486, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047740252713675976, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10153458954157167, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:3,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10177872828212048, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10235441346045157, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10293009863878269, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1035057838171138, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1037499225576626, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10432560773599373, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:28, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10490129291432483, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10547697809265595, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10572111683320473, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10629680201153585, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10687248718986696, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08157665345017173, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:2,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08182079219072054, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08239647736905165, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08297216254738275, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08354784772571387, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:28, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08379198646626267, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08436767164459379, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08494335682292489, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.085519042001256, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0857631807418048, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08633886592013591, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08691455109846702, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050699478453966725, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:1,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05094361719451552, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05085620949728201, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:28, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05076880180004848, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05068139410281496, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05092553284336376, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05083812514613024, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05075071744889672, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05066330975166321, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050907448492211994, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.050820040794978476, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05073263309774496, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16468465359815485, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:5,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:28, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16492879233870364, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1648413846414701, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1647539769442366, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1646665692470031, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16491070798755186, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16482330029031836, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16473589259308485, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16464848489585132, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16489262363640012, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1648052159391666, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:28, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16471780824193308, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10995950268456056, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:3,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11020364142510937, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11011623372787584, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11002882603064232, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10994141833340881, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11018555707395762, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11009814937672409, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11001074167949057, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10992333398225707, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:1, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:28, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11016747272280585, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11008006502557233, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1099926573283388, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13772088603087665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:4,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13796502477142544, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13787761707419197, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13779020937695843, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1377028016797249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13794694042027372, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378595327230402, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:28, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13777212502580666, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13768471732857315, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13792885606912195, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13784144837188841, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13775404067465488, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:282}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13536921329965343, `foldPressure`:0.0,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'FLOW',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:21}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.058683719487469956, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:0, `qDelta`:2,
`saturationRatio`:0.0, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:22}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.059259404665801076, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:1, `qDelta`:2,
`saturationRatio`:0.00909090909090909, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:23}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14403533864136714, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:2, `qDelta`:5,
`saturationRatio`:0.01818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:24}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14427947738191593, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:3, `qDelta`:5,
`saturationRatio`:0.02727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:28, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:27}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14485516256024705, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:4, `qDelta`:5,
`saturationRatio`:0.03636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:28}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14543084773857817, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:5, `qDelta`:5,
`saturationRatio`:0.045454545454545456, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:29}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1460065329169093, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:6, `qDelta`:5,
`saturationRatio`:0.05454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:30}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14625067165745806, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:7, `qDelta`:5,
`saturationRatio`:0.06363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:89, `threshold`:110}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:31}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14682635683578918, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:8, `qDelta`:5,
`saturationRatio`:0.07272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:32}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1474020420141203, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:9, `qDelta`:5,
`saturationRatio`:0.08181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:33}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14797772719245142, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:10, `qDelta`:5,
`saturationRatio`:0.09090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:36}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1482218659330002, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:11, `qDelta`:5,
`saturationRatio`:0.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:37}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.1487975511113313, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:12, `qDelta`:5,
`saturationRatio`:0.10909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:38}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.14937323628966243, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:13, `qDelta`:5,
`saturationRatio`:0.11818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:28, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:39}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09394097886949039, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:14, `qDelta`:3,
`saturationRatio`:0.12727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:41}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09418511761003917, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:15, `qDelta`:3,
`saturationRatio`:0.13636363636363635, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:42}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09476080278837029, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:16, `qDelta`:3,
`saturationRatio`:0.14545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:43}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0953364879667014, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:17, `qDelta`:3,
`saturationRatio`:0.15454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:44}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.0959121731450325, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:18, `qDelta`:3,
`saturationRatio`:0.16363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:100, `threshold`:110}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:45}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09615631188558131, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:19, `qDelta`:3,
`saturationRatio`:0.17272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:46}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09673199706391242, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:20, `qDelta`:3,
`saturationRatio`:0.18181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:53}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09730768224224354, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:21, `qDelta`:3,
`saturationRatio`:0.19090909090909092, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:55}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09788336742057466, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:22, `qDelta`:3,
`saturationRatio`:0.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:60}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09812750616112344, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:23, `qDelta`:3,
`saturationRatio`:0.20909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:28, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:61}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381, `capacity`:144,
`dcomp`:0.09870319133945456, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:24, `qDelta`:3,
`saturationRatio`:0.21818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:133}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09927887651778566, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:25, `qDelta`:3,
`saturationRatio`:0.22727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:134}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04231494736814717, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:26, `qDelta`:1,
`saturationRatio`:0.23636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:135}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04255908610869596, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:27, `qDelta`:1,
`saturationRatio`:0.24545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:136}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04313477128702707, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:28, `qDelta`:1,
`saturationRatio`:0.2545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:137}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04371045646535819, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:29, `qDelta`:1,
`saturationRatio`:0.2636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:111, `threshold`:110}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:138}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.044286141643689304, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:30, `qDelta`:1,
`saturationRatio`:0.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:139}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0445302803842381, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:31, `qDelta`:1,
`saturationRatio`:0.2818181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:140}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04510596556256921, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:32, `qDelta`:1,
`saturationRatio`:0.2909090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:141}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.045681650740900324, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:33, `qDelta`:1,
`saturationRatio`:0.3, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:28, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:142}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04625733591923144, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:34, `qDelta`:1,
`saturationRatio`:0.3090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:143}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.04650147465978023, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:35, `qDelta`:1,
`saturationRatio`:0.3181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:144}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047077159838111345, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:36, `qDelta`:1,
`saturationRatio`:0.32727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:145}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.047652845016442465, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:37, `qDelta`:1,
`saturationRatio`:0.33636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:146}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10144718184433817, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:38, `qDelta`:3,
`saturationRatio`:0.34545454545454546, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:147}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10169132058488695, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:39, `qDelta`:3,
`saturationRatio`:0.35454545454545455, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:148}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226700576321805, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:40, `qDelta`:3,
`saturationRatio`:0.36363636363636365, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:122,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:149}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10284269094154916, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:41, `qDelta`:3,
`saturationRatio`:0.37272727272727274, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:150}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10341837611988028, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:42, `qDelta`:3,
`saturationRatio`:0.38181818181818183, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:151}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10366251486042909, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:43, `qDelta`:3,
`saturationRatio`:0.39090909090909093, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:28, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:152}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10423820003876019, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:44, `qDelta`:3,
`saturationRatio`:0.4, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:153}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10481388521709131, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:45, `qDelta`:3,
`saturationRatio`:0.4090909090909091, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:154}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10538957039542242, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:46, `qDelta`:3,
`saturationRatio`:0.41818181818181815, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:155}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10563370913597121, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:47, `qDelta`:3,
`saturationRatio`:0.42727272727272725, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:156}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10620939431430233, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:48, `qDelta`:3,
`saturationRatio`:0.43636363636363634, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:157}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10678507949263344, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:49, `qDelta`:3,
`saturationRatio`:0.44545454545454544, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:158}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08148924575293823, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:50, `qDelta`:2,
`saturationRatio`:0.45454545454545453, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:159}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08173338449348701, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:51, `qDelta`:2,
`saturationRatio`:0.4636363636363636, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:133, `threshold`:110}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:160}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08230906967181811, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:52, `qDelta`:2,
`saturationRatio`:0.4727272727272727, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:161}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08288475485014923, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:53, `qDelta`:2,
`saturationRatio`:0.4818181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:28, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:162}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08346044002848034, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:54, `qDelta`:2,
`saturationRatio`:0.4909090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:163}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08370457876902915, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:55, `qDelta`:2,
`saturationRatio`:0.5, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:164}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08428026394736025, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:56, `qDelta`:2,
`saturationRatio`:0.509090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:165}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08485594912569137, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:57, `qDelta`:2,
`saturationRatio`:0.5181818181818182, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:166}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0854316343040225, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:58, `qDelta`:2,
`saturationRatio`:0.5272727272727272, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:167}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08567577304457127, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:59, `qDelta`:2,
`saturationRatio`:0.5363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:168}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0862514582229024, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:60, `qDelta`:2,
`saturationRatio`:0.5454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:169}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08682714340123351, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:61, `qDelta`:2,
`saturationRatio`:0.5545454545454546, `shadowDecay`:0.03663127777746836, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:234}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05127516363229784, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:62, `qDelta`:1,
`saturationRatio`:0.5636363636363636, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:0,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:235}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05151930237284664, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:63, `qDelta`:1,
`saturationRatio`:0.5727272727272728, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:28, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:236}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05143189467561311, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:64, `qDelta`:1,
`saturationRatio`:0.5818181818181818, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:237}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051344486978379594, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:65, `qDelta`:1,
`saturationRatio`:0.5909090909090909, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:238}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05125707928114608, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:66, `qDelta`:1,
`saturationRatio`:0.6, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:239}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.051501218021694876, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:67, `qDelta`:1,
`saturationRatio`:0.6090909090909091, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:240}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05141381032446135, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:68, `qDelta`:1,
`saturationRatio`:0.6181818181818182, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:241}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05132640262722783, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:69, `qDelta`:1,
`saturationRatio`:0.6272727272727273, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:242}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05123899492999431, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:70, `qDelta`:1,
`saturationRatio`:0.6363636363636364, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:243}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05148313367054311, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:71, `qDelta`:1,
`saturationRatio`:0.6454545454545455, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:244}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05139572597330959, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:72, `qDelta`:1,
`saturationRatio`:0.6545454545454545, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:245}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05130831827607607, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:73, `qDelta`:1,
`saturationRatio`:0.6636363636363637, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:11,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:28, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:246}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16526033877648597, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:74, `qDelta`:5,
`saturationRatio`:0.6727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:247}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16550447751703473, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:75, `qDelta`:5,
`saturationRatio`:0.6818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:248}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16541706981980123, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:76, `qDelta`:5,
`saturationRatio`:0.6909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:249}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16532966212256772, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:77, `qDelta`:5,
`saturationRatio`:0.7, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:250}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1652422544253342, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:78, `qDelta`:5,
`saturationRatio`:0.7090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:251}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16548639316588298, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:79, `qDelta`:5,
`saturationRatio`:0.7181818181818181, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:252}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16539898546864948, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:80, `qDelta`:5,
`saturationRatio`:0.7272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:253}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16531157777141597, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:81, `qDelta`:5,
`saturationRatio`:0.7363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:254}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16522417007418244, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:82, `qDelta`:5,
`saturationRatio`:0.7454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:255}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16546830881473124, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:83, `qDelta`:5,
`saturationRatio`:0.7545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:1, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:28, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:256}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16538090111749773, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:84, `qDelta`:5,
`saturationRatio`:0.7636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:257}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1652934934202642, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:85, `qDelta`:5,
`saturationRatio`:0.7727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:258}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11053518786289168, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:86, `qDelta`:3,
`saturationRatio`:0.7818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:259}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11077932660344048, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:87, `qDelta`:3,
`saturationRatio`:0.7909090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:260}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11069191890620694, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:88, `qDelta`:3,
`saturationRatio`:0.8, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:261}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11060451120897344, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:89, `qDelta`:3,
`saturationRatio`:0.8090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:262}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11051710351173993, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:90, `qDelta`:3,
`saturationRatio`:0.8181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:263}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1107612422522887, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:91, `qDelta`:3,
`saturationRatio`:0.8272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:264}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1106738345550552, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:92, `qDelta`:3,
`saturationRatio`:0.8363636363636363, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:265}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11058642685782169, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:93, `qDelta`:3,
`saturationRatio`:0.8454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:28, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:266}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11049901916058817, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:94, `qDelta`:3,
`saturationRatio`:0.8545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:267}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11074315790113695, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:95, `qDelta`:3,
`saturationRatio`:0.8636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:268}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11065575020390343, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:96, `qDelta`:3,
`saturationRatio`:0.8727272727272727, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:269}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11056834250666993, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:97, `qDelta`:3,
`saturationRatio`:0.8818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:270}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13829657120920777, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:98, `qDelta`:4,
`saturationRatio`:0.8909090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:271}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13854070994975656, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:99, `qDelta`:4,
`saturationRatio`:0.9, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:272}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13845330225252303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:100, `qDelta`:4,
`saturationRatio`:0.9090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:273}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13836589455528953, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:101, `qDelta`:4,
`saturationRatio`:0.9181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:274}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13827848685805602, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:102, `qDelta`:4,
`saturationRatio`:0.9272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:275}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13852262559860481, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:103, `qDelta`:4,
`saturationRatio`:0.9363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:28, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:276}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13843521790137128, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:104, `qDelta`:4,
`saturationRatio`:0.9454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:42, `threshold`:110}]->(b);
```

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MATCH (a {`key`:296}) MATCH (b {`key`:277}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13834781020413778, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:105, `qDelta`:4,
`saturationRatio`:0.9545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:278}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13826040250690427, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:106, `qDelta`:4,
`saturationRatio`:0.9636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:279}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13850454124745304, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:107, `qDelta`:4,
`saturationRatio`:0.9727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:280}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1384171335502195, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:108, `qDelta`:4,
`saturationRatio`:0.9818181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:281}) CREATE (a)-[:`CAN_FLOW_TO` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13832972585298603, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(edge)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:109, `qDelta`:4,
`saturationRatio`:0.990909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:62, `state`:'FLOW',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031337166487479466, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03158130522802826, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032156990406359366, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032732675584690486, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:28, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0333083607630216, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03355249950357039, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034128184681901506, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03470386986023262, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08899007795929895, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08923421669984774, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08980990187817885, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09038558705650998, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09096127223484109, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09120541097538988, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:28, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09178109615372099, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09235678133205209, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0929324665103832, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:80,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13890541103131712, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13652058365631564, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13676472239686444, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13667731469963093, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1365899070023974, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13716559218072 85, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13740973092127 73, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:28, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798541609960843, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13856110127793952, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13913678645627064, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13938092519681944, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13995661037515056, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14053229555348168, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:63, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030692157963066596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030936296703615387, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030848889006381875, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:63, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031337166487479466, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:28, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031493897530794734, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032069582709125855, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03264526788745697, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033220953065788074, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03404077698466799, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0346164621629991, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08890267026206543, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08914680900261421, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08972249418094534, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:28, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09029817935927645, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09087386453760755, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09111800327815636, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09169368845648748, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09226937363481859, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13956850390688175, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13948109620964824, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13709626883464673, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725299987796202, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:28, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13716559218072851, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13707818448349499, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13732232322404378, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13789800840237490, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13847369358070602, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13904937875903714, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13929351749958590, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:1}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13986920267791705, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14044488785624815, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:64, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0312678431413977, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:28, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031424574184712985, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:64, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03158130522802826, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031493897530794734, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03231372144967465, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032889406628005755, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03370923054688567, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03428491572521678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034860600903547895, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:1, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:28, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08914680900261421, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08939094774316303, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08996663292149414, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09054231809982524, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09111800327815636, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09136214201870516, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09193782719703626, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13990005034466407, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13981264264743054, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.139725234950197, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:28, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13758454631574435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13749713861851082, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374097309212773, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13732232322404378, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13756646196459257, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381421471429237, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387178323212548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392935174995859, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13953765624013473, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:28, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14011334141846585, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14068902659679694, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:65, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0317561206224953, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03166871292526178, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:65, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032156990406359366, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032069582709125855, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03231372144967465, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03280199893077224, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033377684109103356, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:28, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03362182284965215, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03419750802798326, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034773193206314376, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0890594013053807, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0893035400459295, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0898792252242606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09045491040259172, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09103059558092284, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09127473432147162, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1402315967824464, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:28, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404757355229952, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14038832782576166, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14030092012852816, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13791609275352668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13816023149407544, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807282379684194, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798541609960843, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378980084023749, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381421471429237, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1380547394456902, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:28, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13863042462402128, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:55,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392061098023524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1394502485429012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14002593372123232, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14060161889956344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:66, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032087667060277617, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03233180580082641, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03224439810359289, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:66, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032732675584690486, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03264526788745697, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:28, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032889406628005755, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03280199893077224, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:4}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03329027641186984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03353441515241863, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03411010033074974, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03468578550908086, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08897199360814718, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08921613234869598, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08979181752702708, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0903675027053582, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:28, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09094318788368931, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.140894689658011, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14080728196077746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14105142070132629, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14096401300409278, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14087660530685925, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13849177793185777, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387359166724066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13864850897517306, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13856110127793952, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:28, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13847369358070602, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387178323212548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13863042462402128, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13854301692678778, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1391187021051189, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13936284084566766, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1399385260239988, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1405142112023299, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:67, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03266335223860873, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032907490979157523, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:28, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032820083281924005, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:67, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0333083607630216, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033220953065788074, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033377684109103356, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03329027641186984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03344700745518511, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034022692633516226, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03459837781184734, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08888458591091367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:1, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:28, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08912872465146246, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08970440982979358, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09028009500812467, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14155778253357565, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14147037483634212, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14138296713910858, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1416271058796574, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14153969818242387, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14145229048519037, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390674631101889, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:28, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922419415350415, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13913678645627064, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13904937875903714, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392935174995859, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392061098023524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1391187021051189, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13903129440788536, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13927543314843416, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13985111832676528, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:28, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14042680350509637, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:68, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033239037416939836, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03339576846025512, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:68, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03355249950357039, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03370923054688567, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03362182284965215, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03353441515241863, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03344700745518511, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:28, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03426683137406502, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03484251655239613, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:0,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08912872465146246, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08937286339201127, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08994854857034237, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14188932897135795, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14180192127412442, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1417145135768909, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1416271058796574, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14187124462020617, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:28, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14178383692297267, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14169642922573916, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13955574059128648, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13946833289405297, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13938092519681944, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392935174995859, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13953765624013473, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1394502485429012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13936284084566766, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:28, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13927543314843416, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13951957188898295, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14009525706731407, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14067094224564520, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:69, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03372731489803744, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03363990720080391, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:69, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034128184681901506, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03404077698466799, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03428491572521678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:28, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03419750802798326, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03411010033074974, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034022692633516226, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03426683137406502, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034755108855162614, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:0,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08904131695422894, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08928545569477774, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14222087540914027, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14246501414968907, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14237760645245554, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:28, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14229019875522203, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1422027910579885, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1424469297985373, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423595221013038, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14227211440407025, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1398872870290688, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14013142576961762, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1400440180723841, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13995661037515056, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13986920267791705, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:28, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14011334141846585, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:7}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14002593372123232, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1399385260239988, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13985111832676528, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14009525706731407, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14000784937008057, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14058353454841166, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:70, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03405886133581975, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034303000076368544, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034215592379135025, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:70, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:28, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03470386986023262, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0346164621629991, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034860600903547895, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034773193206314376, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03468578550908086, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03459837781184734, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03484251655239613, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:0,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034755108855162614, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:0,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08895390925699542, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1428839682847049, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:1, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:28, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1427965605874714, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14304069932802016, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14295329163078666, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14286588393355315, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14277847623631962, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1430226149768684, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14293520727963488, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14284779958240137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404629722073999, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14070711094794872, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:28, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14061970325071518, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14053229555348168, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14044488785624815, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14068902659679694, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14060161889956344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1405142112023299, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14042680350509637, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1406709422456452, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14058353454841166, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14049612685117813, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:71, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:28, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03463454651415086, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03487868525469966, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03479127755746613, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:71, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08899007795929895, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08890267026206543, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08914680900261421, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0890594013053807, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08897199360814718, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08888458591091367, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08912872465146246, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:28, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08904131695422894, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08895390925699542, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1412496414825016, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14116223378526807, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14107482608803454, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14131896482858333, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14123155713134983, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1411441494341163, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1410567417368828, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14130088047743158, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:28, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14121347278019805, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14112606508296455, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14019348391853748, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14043762265908627, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14035021496185277, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14026280726461926, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
```

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MATCH (a {`key`:9}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14017539956738573, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14041953830793452, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14033213061070102, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14024472291346748, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:29, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14015731521623398, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14040145395678277, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14031404625954924, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14022663856231574, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:72, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08892075461321719, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08916489335376598, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08907748565653247, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:72, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08923421669984774, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08914680900261421, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08939094774316303, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:29, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0893035400459295, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08921613234869598, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08912872465146246, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08937286339201127, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08928545569477774, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14158118792028 39, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14149378022305037, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14140637252581684, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14131896482858333, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14156310356913213, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:29, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1414756958718986, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1413882881746651, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14130088047743158, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14154501921798035, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14145761152074685, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14137020382351334, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14043762265908627, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14068176139963506, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14059435370240156, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14050694600516803, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:29, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14041953830793452, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14066367704848332, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14057626935124978, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14048886165401628, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14040145395678277, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14064559269733154, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14055818500009803, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14047077730286453, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08916489335376598, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08940903209431479, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:29, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08932162439708127, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:73, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08980990187817885, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08972249418094534, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08996663292149414, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0898792252242606, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08979181752702708, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08970440982979358, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08994854857034237, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14191273435806623, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14215687309861502, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:1, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:29, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1420694654013815, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14198205770414798, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14189465000691445, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14213878874746325, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14205138105022974, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1419639733529962, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14187656565576268, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1421207043963115, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14203329669907797, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14194588900184443, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:29, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14101330783741 74, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14125744657796618, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14117003888073265, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14108263118349915, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14099522348626564, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14123936222681444, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14115195452958093, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1410645468323474, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14097713913511387, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1412212778756627, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:29, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14113387017842915, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14104646248119562, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:74, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0897405785320971, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08998471727264591, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08989730957541238, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:74, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09038558705650998, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09029817935927645, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09054231809982524, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09045491040259172, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0903675027053582, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:29, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09028009500812467, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16907778649797556, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:35,
`threshold`:110}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14248841953639732, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14273255827694611, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426451505797126, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14255774288247908, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14247033518524557, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14271447392579437, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14262706622856083, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1425396585313273, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:29, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14245225083409 38, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426963895746426, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14260898187740906, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14252157418017555, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1415889930157485, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14183313175629728, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14174572405906377, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14165831636183024, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14157090866459673, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14181504740514553, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:29, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14172763970791202, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1416402320106785, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14155282431344499, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14179696305399378, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:12}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14170955535676025, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14162214765952674, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:75, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09031626371042822, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09056040245097702, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09047299475374349, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:75, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09096127223484109, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:29, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09087386453760755, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09111800327815636, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09103059558092284, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09094318788368931, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697408793735402, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:34,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16965347167630668, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14306410471472844, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14330824345527726, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14322083575804373, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431334280608102, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:29, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1430460203635767, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432901591041255, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14320275140689195, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14311534370965845, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14302793601242492, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432720747529737, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431846670557402, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14309725935850667, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1421646781940796, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14240881693462842, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:29, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423214092373949, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14223400154016136, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14214659384292785, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14239073258347665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14230332488624312, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14221591718900964, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1421285094917761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423726482323249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1422852405350914, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14219783283785786, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:76, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:29, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09089194888875932, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09113608762930811, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09104867993207462, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:76, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09120541097538988, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09111800327815636, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09136214201870516, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09127473432147162, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17007242581132248, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:33,
`threshold`:110}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16998501811408898, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16989761041685544, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:1, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:29, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14330824345527726, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14355238219582603, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14346497449859252, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14337756680135902, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432901591041255, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14353429784467428, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14344689014744078, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14335948245020724, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432720747529737, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14351621349352253, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:29, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.143428805796289, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14334139809905547, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14240881693462842, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14265295567517722, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14256554797794369, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14247814028071018, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14239073258347665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14263487132402544, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14254746362679194, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14246005592955843, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:29, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423726482323249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426167869728737, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1425293792756402, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14244197157840666, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:77, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09113608762930811, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09138022636985692, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0912928186726234, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:77, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09178109615372099, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09169368845648748, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09193782719703626, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:29, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1704039722491048, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706481109896536, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17056070329242007, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1704732955951866, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14388392863360835, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14412806737415715, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14404065967692364, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1439532519796901, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14386584428245658, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1441099830230054, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:29, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:17}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14402257532577187, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14393516762853834, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14384775993130483, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14409189867185362, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1440044909746201, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1439170832773866, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14298450211295952, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432286408535083, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431412331562748, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14305382545904127, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:29, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14296641776180774, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14321055650235656, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14312314880512303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1430357411078895, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14294833341065602, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431924721512048, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14310506445397128, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14301765675673775, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:78, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09171177280763923, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09195591154818802, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:29, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09186850385095449, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:78, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09235678133205209, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09226937363481859, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17106706512466943, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17097965742743593, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1712237961679847, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711363884707512, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17104898077351768, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14445961381193947, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14470375255248827, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:29, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14461634485525474, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14452893715802123, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14444152946078773, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1446856682013365, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.144598260504103, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14451085280686948, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14442344510963595, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14466758385018474, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14458017615295124, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1444927684557177, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:29, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14356018729129064, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14380432603183943, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14371691833460593, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1436295106373724, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1435421029401389, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14378624168068768, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14369883398345415, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14361142628622064, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14352401858898714, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1437681573295359, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:29, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1436807496323024, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1435933419350689, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:79, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09228745798597035, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09253159672651913, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09244418902928561, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:79, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0929324665103832, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17173015800023408, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'BLOCK', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17164275030300055, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17155534260576702, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17179948134631584, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:1, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:29, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717120736490823, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17162466595184878, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450352989902706, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14527943773081936, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14519203003358586, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14510462233635235, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14501721463911882, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1452613533796676, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1451739456824341, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14508653798520058, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:29, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14499913028796704, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14524326902851586, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14515586133128233, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450684536340488, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14413587246962176, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14438001121017055, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14429260351293702, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1442051958157035, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14411778811846998, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14436192685901877, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:29, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14427451916178527, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14418711146455174, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1440997037673182, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.144343842507867, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14425643481063352, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1441690271134, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:80, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09286314316430146, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09310728190485025, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09301987420761673, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:80, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17206170443801638, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'BLOCK', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:29, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17197429674078288, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17188688904354935, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17179948134631584, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17204362008686463, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17195621238963111, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17186880469239766, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14527943773081936, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14552357647136818, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:20}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14543616877413468, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14534876107690115, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:29, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1452613533796676, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14550549212021643, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1454180844229829, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14533067672574937, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14524326902851586, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14548740776906466, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14540000007183113, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14531259237459762, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14438001121017055, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14462414995071934, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:29, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14453674225348584, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1444493345562523, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14436192685901877, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1446060655995676, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14451865790233406, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14443125020510053, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.144343842507867, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14458798124841582, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1445005735511823, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14441316585394878, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:81, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:29, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09310728190485025, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09335142064539906, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09326401294816553, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:81, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723932508757987, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17263738961634 75, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17254998191911397, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17246257422188047, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17237516652464693, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17261930526519573, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17253189756796222, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:29, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1724444898707287, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.03663127777746836, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14585512290915048, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14609926164969927, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14601185395246574, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:38, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14592444625523224, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14583703855799873, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14608117729854753, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.145993769601314, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1459063619040805, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14581895420684696, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:29, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14606309294739575, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14597568525016225, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1458882775529287, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14495569638850164, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14519983512905046, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14511242743181693, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450250197345834, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1449376120373499, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1451817507778987, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14509434308066516, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:29, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14500693538343165, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14491952768619815, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
```

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MATCH (a {`key`:21}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1451636664267469, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450762587295134, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14498885103227988, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:82, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09368296708318137, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09392710582373016, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:82, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17305634375136336, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17296893605412983, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17321307479467862, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:1, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:29, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17312566709744512, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17303825940021159, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17295085170297805, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17319499044352685, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17310758274629334, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1730201750490598, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1464308080874816, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:36, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1466749468280304, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:37, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14658753913079686, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14650013143356339, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:39, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:29, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14641272373632985, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:40, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14665686247687865, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:41, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14656945477964514, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:42, `threshold`:110}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1464820470824116, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:43, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14639463938517808, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:44, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1466387781257269, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:45, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14655137042849337, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:46, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14646396273125983, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:47, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1455313815668328, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14577552030738158, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:29, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14568811261014805, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14560070491291455, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14551329721568101, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1457574359562298, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
```

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MATCH (a {`key`:22}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14567002825899628, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14558262056176277, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14549521286452924, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14573935160507803, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1456519439078445, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.145564536210611, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:4,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:83, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:29, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09425865226151248, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:83, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15575993120277212, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15567252350553856, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15558511580830506, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582925454885388, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15574184685162032, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15565443915438681, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555670314571533, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15581117019770208, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15572376250046857, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:29, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15563635480323507, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06510366414634572, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:1,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06534780288689451, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:1,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06526039518966101, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:1,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06517298749242748, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06508557979519394, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06532971853574276, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06524231083850923, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0651549031412757, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06506749544404218, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:29, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06531163418459099, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06522422648735747, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06513681879012395, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13851893117073166, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:3,
`saturationRatio`:1.2, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13876306991128046, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13867566221404695, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13858825451681342, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1385008468195799, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13874498556012868, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13865757786289518, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:29, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13857017016566164, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1384827624684281, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387269012089769, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1386394935117434, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13855208581450987, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:84, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15609147764055445, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560040699433209, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15591666224608738, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582925454885388, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15607339328940265, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:29, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15598598559216914, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15589857789493564, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15581117019770208, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560553089382509, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1559679012410174, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15588049354378383, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06534780288689451, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:1,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06559194162744332, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:1,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06550453393020979, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:24}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06541712623297627, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:29, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06532971853574276, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06557385727629154, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06548644957905801, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06539904188182451, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06531163418459099, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06555577292513977, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06546836522790626, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06538095753067276, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13876306991128046, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13900720865182928, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
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MATCH (a {`key`:263}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:29, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13891980095459575, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1388323932573622, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13874498556012868, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1389891243006775, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13890171660344397, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13881430890621044, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387269012089769, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13897103994952573, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1388836322522922, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:85, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15642302407833678, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:1, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:29, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15666716281888554, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15657975512165204, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15649234742441853, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15640493972718497, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15664907846773377, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15656167077050026, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15647426307326673, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15638685537603322, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15663099411658202, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15654358641934848, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:29, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15645617872211498, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06592348806522562, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:1,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06616762680577441, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0660802191085409, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06599281141130739, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06590540371407387, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06614954245462265, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06606213475738913, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06597472706015561, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0658873193629221, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:29, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06613145810347089, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06604405040623737, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06595664270900387, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13933875508961158, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13958289383016037, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13949548613292684, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1394080784356933, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1393206707384598, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1395648094790086, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13947740178177506, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:29, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13938999408454156, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13930258638730805, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13954672512785682, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:86, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14720661837534837, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15699870925666784, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15724284799721666, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15715544029998316, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1570680326027496, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1569806249055161, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1572247636460649, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:29, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15713735594883135, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15704994825159785, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15696254055436434, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1572066792949131, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1571192715976796, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1570318639004461, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06649917324355674, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06674331198410552, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06665590428687201, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0665684965896385, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:29, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06648108889240498, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06672522763295377, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06663781993572025, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06655041223848673, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06646300454125321, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06670714328180201, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06661973558456848, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06653232788733497, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1399144402679427, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1401585790084915, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:29, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14007117131125796, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13998376361402445, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13998635591679092, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14014049465733971, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14005308696010618, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13996567926287268, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13987827156563915, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:87, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.147869711250913, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14778230355367947, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157574394434999, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:29, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15781853317554778, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15773112547831425, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15764371778108074, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15755631008384724, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157800448824396, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1577130411271625, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157625633429929, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15753822573269544, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15778236447324426, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15769495677601075, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:29, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576075490787772, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06707485842188785, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06731899716243665, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06723158946520312, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06714418176796959, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06705677407073608, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06730091281128489, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06721350511405136, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06712609741681784, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06703868971958434, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:29, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06728282846013312, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0671954207628996, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06710801306566609, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404901254462738, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14073426418682258, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14064685648958908, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14055944879235555, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.140472041095122, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1407161798356708, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1406287721384373, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:1, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:29, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14054136444120377, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:88, `state`:'BLOCK', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1482012576886953, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1481138499914618, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14802644229422826, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15781853317554778, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15806267191609655, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15797526421886304, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15788785652162954, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157800448824396, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1580445875649448, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:29, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1579571798677113, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15786977217047776, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15778236447324426, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15802650321379305, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1579390955165595, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157851687819326, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06731899716243665, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06756313590298545, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06747572820575191, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:30}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06738832050851841, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:29, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06730091281128489, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06754505155183368, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06745764385460017, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06737023615736665, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06728282846013312, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06752696720068192, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0674395595034484, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06735215180621487, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14073426418682258, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1409784029273714, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:29, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14089099523013787, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14080358753290434, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1407161798356708, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14096031857621963, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1408729108789861, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:89, `state`:'BLOCK', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14853280412647762, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14877694286702642, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14868953516979289, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14860212747255938, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15839421835387887, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:29, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1586383570944277, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1585509493971942, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15846354169996063, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15837613400272713, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15862027274327595, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1585328650460424, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15844545734880888, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15835804965157538, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15860218839212414, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15851478069489064, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:29, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15842737299765713, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06789468234076776, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06813882108131655, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06805141338408303, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06796400568684952, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06787659798961598, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0681207367301648, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06803332903293129, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06794592133569775, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06785851363846424, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:29, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06810265237901304, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06801524468177951, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06792783698454598, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14130994936515373, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1415540881057025, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.141466680408469, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14137927271123546, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14129186501400195, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14153600375455075, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:90, `state`:'BLOCK', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14919589700204225, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:29, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14910848930480874, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1493526280453575, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.149265220348124, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1491778126508905, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15896990353220997, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1592140422727588, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15912663457552526, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15903922687829175, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15895181918105825, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.159195957921607, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:29, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1591085502243735, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15902114252714, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15893373482990647, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15917787357045526, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15909046587322176, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15900305817598823, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06847036751909887, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06871450625964766, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06862709856241415, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06853969086518062, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:29, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0684522831679471, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06869642190849591, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06860901421126238, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06852160651402887, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06843419881679536, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:32}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06867833755734415, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06859092986011062, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06850352216287711, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14188563454348482, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14212977328403362, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:1, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:29, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14204236558680008, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14195495788956658, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14186755019233305, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:91, `state`:'BLOCK', `targetIndex`:52,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498589898776069, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14977158218037337, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14968417448313984, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14992831322368866, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14984090552645513, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1497534978292216, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15954558871054111, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:29, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978972745108988, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15970231975385638, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596149120566229, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15952750435938934, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15977164309993816, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15968423540270466, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1595968277054711, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1595094200082376, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1597535587487864, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15966615105155285, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:29, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15957874335431935, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06904605269742999, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06929019143797878, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06920278374074526, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06911537604351174, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06902796834627822, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06927210708682702, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:33}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0691846993895935, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06909729169235998, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06900988399512646, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:29, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06925402273567526, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06916661503844175, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06907920734120822, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14246131972181592, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1427054584623647, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426180507651312, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1425306430678977, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:92, `state`:'BLOCK', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1501905363153892, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:17,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1501031286181557, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15001572092092216, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:29, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14992831322368866, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15017245196423745, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15008504426700392, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14999763656977042, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:2.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978972745108988, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16003386619163867, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15994645849440517, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15985905079717166, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15977164309993816, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:28,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16001578184048695, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:29, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15992837414325342, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15984096644601992, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1597535587487864, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15999769748933518, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15991028979210167, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15982288209486817, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06929019143797878, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06953433017852757, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06944692248129405, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06935951478406054, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:29, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06927210708682702, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06951624582737581, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06942883813014229, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06934143043290877, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06925402273567526, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06949816147622406, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06941075377899053, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06932334608175701, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1427054584623647, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14294959720291353, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:29, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:36}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14286218950568, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:93, `state`:'BLOCK', `targetIndex`:50,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15052208275317153, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15076622149372032, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1506788137964868, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15059140609925328, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15050399840201975, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15074813714256854, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15066072944533504, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1505733217481015, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:2.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.160365412629421, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:29, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16060955136996982, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16052214367273626, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16043473597550276, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16034732827826928, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16059146701881805, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16050405932158454, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16041665162435104, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16032924392711748, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16057338266766663, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16048597497043280, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:29, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16039856727319923, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06986587661630989, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:37}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0701100153568587, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07002260765962517, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06993519996239164, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06984779226515814, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07009193100570693, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0700045233084734, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06991711561123988, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06982970791400638, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:1, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:29, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07007384665455516, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06998643895732164, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
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MATCH (a {`key`:37}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06989903126008813, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14328114364069583, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'BLOCK',
`targetIndex`:48, `threshold`:110}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14352528238124462, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:94, `state`:'BLOCK', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15118517562873618, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15109776793150265, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.0090909090909, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1513419066720514, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512544989748179, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1511670912775844, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:29, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15107968358035087, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15132382232089966, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15123641462366616, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:38}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15114900692643263, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:2.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16094109780775215, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16118523654830091, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1610978288510674, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1610104211538339, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16092301345660034, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16116715219714914, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:29, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16107974449991566, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16099233680268213, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16090492910544862, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16114906784599742, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16106166014876389, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16097425245153038, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.36787944117144233, `sourceIndex`:95, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.070441561794641, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07068570053518981, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07059829283795628, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07051088514072276, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:29, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07042347744348924, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07066761618403804, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0705802084868045, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.070492800789571, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07040539309233748, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:38}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07064953183288626, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07056212413565276, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07047471643841924, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14385682881902692, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:95, `state`:'BLOCK', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17160726566140683, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:29, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1715198579641733, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17143245026693976, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17167658900748858, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17158918131025505, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17150177361302152, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17141436591578801, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
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MATCH (a {`key`:39}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716585046563368, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17157109695910328, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17148368926186977, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10156122590292715, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:29, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10180536464347595, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10171795694624244, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10163054924900891, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10154314155177539, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1017872802923242, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10169987259509067, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10161246489785715, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10152505720062363, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10176919594117242, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1016817882439389, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:29, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10159438054670539, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:96, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07101724697297213, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07126138571352092, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711739780162874, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07108657031905388, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07099916262182035, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07124330136236914, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07115589366513564, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0710684859679021, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07098107827066859, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:29, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07122521701121738, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07113780931398386, `foldPressure`:0.358024691358024 67,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:281}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07105040161675034, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:96, `state`:'BLOCK', `targetIndex`:47,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17193881209918913, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1718514044019556, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717639967047221, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17167658900748858, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17192072774803735, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17183332005080384, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
```

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MATCH (a {`key`:41}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17174591235357034, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:29, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716585046563368, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1719026433968856, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1718152356996521, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17172782800241856, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10180536464347595, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10204950338402477, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10196209568679124, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10187468798955577, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10178728029232242, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1020314190328 73, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:29, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10194401133563946, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10185660363840594, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10176919594117242, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10201333468172122, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10192592698444877, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10183851928725418, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07126138571352092, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07150552445406971, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07141811675683621, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07133070905960268, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:1, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:29, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07124330136236914, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07148744010291795, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07140003240568443, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0713126247084509, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07122521701121738, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07146935575176619, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:280}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07138194805453266, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:97, `state`:'BLOCK', `targetIndex`:46,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17227035853697145, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17251449727752025, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1724270895802867, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:29, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723396818830532, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17225227418581968, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17249641292636847, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17240900522913497, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17232159753190143, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1722341898346679, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17247832857521672, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723909208779832, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17230351318074966, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10238104982180707, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:29, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10262518856235586, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10253778086512234, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10245037316788884, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1023629654706553, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1026071042112041, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10251969651397058, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10243228881673706, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10234488111950353, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10258901986005234, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10250161216281882, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:29, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241420446558529, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:98, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07183707089185203, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:42}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07208120963240082, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07199380193516731, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07190639423793378, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07181898654070028, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07206312528124906, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07197571758401554, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07188830988678202, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0718009021895485, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:29, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:279}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0720450409300973, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:98, `state`:'BLOCK', `targetIndex`:45,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15068240758655985, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17284604371530254, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
```

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MATCH (a {`key`:43}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17309018245585134, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17300277475861783, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17291536706138433, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728279593641508, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1730720981046996, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17298469040746606, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17289728271023255, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:29, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17280987501299905, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17305401375354781, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1729666060563143, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728791983590808, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10295673500013817, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10320087374068697, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10311346604345346, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10302605834621995, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10293865064898641, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1031827893895352, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:29, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1030953816923017, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10300797399506817, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10292056629783465, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10316470503838346, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10307729734114993, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10298988964391641, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:99, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07241275607018315, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07265689481073193, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07256948711349841, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07248207941626492, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:29, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07239467171903138, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07263881045958018, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07255140276234666, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07246399506511314, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:278}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07237658736787961, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:99, `state`:'BLOCK', `targetIndex`:44,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15134550046212447, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15125809276489097, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17342172889363364, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17366586763418246, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17357845993694893, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:29, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17349105223971542, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17340364454248192, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1736477832830307, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17356037558579718, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17347296788856367, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17338556019133014, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17362969893187893, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17354229123464543, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1734548835374119, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10353242017846928, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:29, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10377655891901809, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10368915122178457, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10360174352455104, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10351433582731752, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10375847456786634, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10367106687063281, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10358365917339929, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10349625147616577, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:32,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:44}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374039021671456, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10365298251948105, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:1, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:29, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10356557482224753, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:100, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07298844124851425, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07323257998906303, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07314517229182953, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07305776459459601, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07297035689736249, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07321449563791128, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07312708794067778, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
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MATCH (a {`key`:44}) MATCH (b {`key`:277}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07303968024344425, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:100, `state`:'BLOCK', `targetIndex`:43,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1516770468999068, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:29, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15158963920267327, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15150223150543973, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17366586763418246, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17391000637473128, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17382259867749775, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17373519098026424, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1736477832830307, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1738919220235795, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.173804514326346, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
```

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MATCH (a {`key`:45}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17371710662911247, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:29, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17362969893187893, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17387383767242776, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17378642997519422, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1736990222779607, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10377655891901809, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10402069765956688, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10393328996233335, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10384588226509985, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10375847456786634, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10400261330841512, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:29, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10391520561118159, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10382779791394808, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374039021671456, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10398452895726336, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10389712126002984, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10380971356279632, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:101, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07323257998906303, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07347671872961185, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07338931103237832, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07330190333514479, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:29, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:45}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07321449563791128, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07345863437846009, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:276}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07337122668122656, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:101, `state`:'BLOCK', `targetIndex`:42,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15200859333768912, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15225273207823792, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15216532438100439, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15207791668377088, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17424155281251355, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17448569155306234, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17439828385582884, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:29, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1743108761585953, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
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MATCH (a {`key`:46}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1742234684613618, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17446760720191062, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1743801995046771, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17429279180744356, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17420538411021005, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17444952285075885, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17436211515352532, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1742747074562918, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10435224409734921, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:29, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10459638283789799, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10450897514066447, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10442156744343095, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10433415974619743, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10457829848674623, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10449089078951272, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:30,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1044034830922792, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10431607539504568, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10456021413559448, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10447280643836095, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:29, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10438539874112743, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:102, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07380826516739417, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07405240390794296, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:46}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07396499621070943, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07387758851347591, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07379018081624239, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:275}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07403431955679118, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:102, `state`:'BLOCK', `targetIndex`:41,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15267168621325375, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15258427851602022, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152828417256569, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:29, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1527410095593355, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15265360186210197, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17481723799084467, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1750613767313935, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
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MATCH (a {`key`:53}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17497396903415996, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17488656133692643, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17479915363969295, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17504329238024172, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1749558846830082, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1748684769857747, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:29, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17478106928854117, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17502520802908997, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17493780033185646, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17485039263462293, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10492792927568032, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
```

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MATCH (a {`key`:53}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10517206801622911, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10508466031899559, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10499725262176206, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10490984492452855, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10515398366507735, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:1, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:29, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10506657596784381, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10497916827061031, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1048917605733768, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10513589931392558, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10504849161669207, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
```

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MATCH (a {`key`:53}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10496108391945856, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:103, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07438395034572527, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07462808908627407, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07454068138904055, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07445327369180703, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:29, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:274}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0743658659945735, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:103, `state`:'BLOCK', `targetIndex`:40,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1533347790888184, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15324737139158487, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15315996369435134, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340410243490016, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15331669473766663, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532292870404331, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1753929231691758, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17563706190972458, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17554965421249105, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:29, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17546224651525755, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17537483881802404, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756189775585728, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17553156986133933, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1754441621641058, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17535675446687227, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756008932074211, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
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MATCH (a {`key`:55}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17551348551018756, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17542607781295402, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10550361445401142, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:29, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10574775319456023, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10566034549732 67, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10557293780009318, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10548553010285967, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10572966884340845, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10564226114617493, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10555485344894142, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10546744575170791, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10571158449225669, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10562417679502319, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:30, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10553676909778967, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:104, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07495963552405638, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07520377426460519, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07511636656737167, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:273}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07502895887013813, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:104, `state`:'BLOCK', `targetIndex`:39,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15366632552660067, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15357891782936714, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15349151013213366, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340410243490016, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15364824117544892, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:30, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1535608334782154, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1534734257809819, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17563706190972458, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17588120065027338, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17579379295303987, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17570638525580634, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756189775585728, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17586311629912163, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1757757086018881, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756883009046546, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:30, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756008932074211, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17584503194796988, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17575762425073635, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17567021655350284, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10574775319456023, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10599189193510902, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1059044842378755, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10581707654064199, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10572966884340845, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10597380758395726, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:30, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:60}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10588639988672374, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10579899218949021, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10571158449225669, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10595572323280551, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10586831553557198, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10578090783833845, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:105, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07520377426460519, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07544791300515398, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:272}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07536050530792046, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:105, `state`:'BLOCK', `targetIndex`:38,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153997871964383, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:30, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542420107049318, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
```

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MATCH (a {`key`:61}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15415460300769826, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15406719531046478, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15397978761323125, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15422392635378004, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15413651865654654, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154049110959313, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1762127470880557, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1764568858286045, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17636947813137097, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:30, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17628207043413746, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17619466273690396, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17643880147745272, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17635139378021922, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1762639860829857, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1761765783857522, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17642071712630097, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17633330942906747, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17624590173183396, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.7357588823428847, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10632343837289132, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:1, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:30, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10656757711344013, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10648016941620662, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK', `targetIndex`:26,
`threshold`:110}]->(b);
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MATCH (a {`key`:61}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10639276171897309, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10630535402173957, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10654949276228838, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10646208506505485, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10637467736782132, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10628726967058781, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1065314084111366, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10644400071390309, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:30, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10635659301666955, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:106, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0757794594429363, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:271}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07602359818348509, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:106, `state`:'BLOCK', `targetIndex`:37,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15466096483994762, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15457355714271412, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15481769588326288, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15473028818602938, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15464288048879588, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15455547279156237, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15479961153211116, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:30, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15471220383487763, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15462479613764413, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17678843226638682, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17703257100693562, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17694516330970209, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17685775561246858, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17677034791523505, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17701448665578384, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17692707895855034, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17683967126131683, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:30, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1767522635640833, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17699640230463212, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1769089946073986, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17682158691016506, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10689912355122246, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
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MATCH (a {`key`:133}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10714326229177125, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10705585459453772, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10696844689730421, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1068810392000707, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10712517794061949, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:30, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10703777024338595, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10695036254615245, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10686295484891892, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10710709358946771, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10701968589223419, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10693227819500069, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:107, `state`:'BLOCK',
`targetIndex`:35, `threshold`:110}]->(b);
```

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MATCH (a {`key`:133}) MATCH (b {`key`:270}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0763551446212674, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:107, `state`:'BLOCK', `targetIndex`:36,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09831869491193213, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0982312872146986, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09814387951746509, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:30, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09838801825801388, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09830061056078035, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09821320286354684, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09812579516631333, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0983699339068621, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0982825262096286, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09819511851239508, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21560065410057055, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:6,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21584479284111938, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:6,
`saturationRatio`:1.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21575738514388584, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:6,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:30, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2156699774466523, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21558256974941878, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2158267084899676, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21573930079273407, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21565189309550054, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21556448539826706, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21580862413881588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21572121644158235, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21563380874434881, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10219347111174477, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:30, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10243760985229355, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10235020215506004, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226279445782653, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.102175386760593, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241952550114179, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10233211780390829, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10224471010667475, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10215730240944122, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10240144114999003, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
```

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MATCH (a {`key`:134}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10231403345275651, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK',
`targetIndex`:34, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:30, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:269}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10222662575552298, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:108, `state`:'BLOCK', `targetIndex`:35,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09865024134971444, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09856283365248092, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09847542595524739, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09838801825801388, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09863215699856268, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09854474930132916, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09845734160409564, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0983699339068621, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09861407264741091, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:264}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:30, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0985266649501774, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09843925725294386, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21584479284111938, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:6,
`saturationRatio`:1.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21608893158166817, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:6,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21600152388443467, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21591411618720113, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2158267084899676, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21607084723051642, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2159834395332829, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2158960318360494, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:1, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:30, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21580862413881588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21605276287936465, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21596535518213117, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21587794748489764, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:109, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10243760985229355, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10268174859284235, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10259434089560884, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10250693319837531, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241952550114179, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1026636642416906, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:30, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:135}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10257625654445707, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10248884884722355, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10240144114999003, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10264557989053882, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK',
`targetIndex`:33, `threshold`:110}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:268}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10255817219330529, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:109, `state`:'BLOCK', `targetIndex`:34,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09898178778749675, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09922592652804554, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09913851883081203, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09905111113357852, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09896370343634499, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:30, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09920784217689378, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
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MATCH (a {`key`:136}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09912043447966028, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09903302678242674, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09894561908519323, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09918975782574203, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0991023501285085, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09901494243127498, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:110, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21642047801945047, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:6,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2166646167599993, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21657720906276576, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:30, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21648980136553223, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2164023936682987, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21664653240884751, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21655912471161398, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21647171701438045, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21638430931714694, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2166284480576958, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21654104036046226, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21645363266322873, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10301329503062466, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:30, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10325743377117347, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10317002607393995, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10308261837670642, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
```

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MATCH (a {`key`:136}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10299521067947291, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1032393494200217, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10315194172278817, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10306453402555465, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10297712632832115, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK',
`targetIndex`:32, `threshold`:110}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:267}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10322126506886993, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:110, `state`:'BLOCK', `targetIndex`:33,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.062159508698681434, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:111, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:30, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09955747296582786, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09980161170637666, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09971420400914315, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09962679631190961, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
```

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MATCH (a {`key`:137}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0995393886146761, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09978352735522492, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09969611965799138, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09960871196075786, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09952130426352435, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09976544300407314, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:30, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09967803530683962, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0995906276096061, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2169961631977816, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2172403019383304, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21715289424109688, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21706548654386335, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2169780788466298, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21722221758717863, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2171348098899451, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21704740219271157, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:30, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2169599944954781, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21720413323602686, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21711672553879333, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21702931784155985, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10358898020895578, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10383311894950457, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
```

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MATCH (a {`key`:137}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374571125227104, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10365830355503755, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10357089585780402, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10381503459835281, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:30, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10372762690111193, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10364021920388578, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK',
`targetIndex`:31, `threshold`:110}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:266}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10355281150665226, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:111, `state`:'BLOCK', `targetIndex`:32,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0614964158231168, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06207210100144791, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10013315814415898, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10037729688470778, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10028988918747425, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10020248149024073, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10011507379300721, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:30, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10035921253355601, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10027180483632249, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10018439713908898, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10009698944185545, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10034112818240426, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10025372048517074, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10016631278793721, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21757184837611268, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2178159871166615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21772857941942797, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:2, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:30, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21764117172219444, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21755376402496096, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21779790276550973, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2177104950682762, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21762308737104272, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21753567967380918, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21777981841435795, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21769241071712447, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
```

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MATCH (a {`key`:138}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21760500301989094, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10416466538728689, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:30, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10440880412783568, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10432139643060218, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10423398873336864, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10414658103613512, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10439071977668392, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10430331207945041, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK',
`targetIndex`:30, `threshold`:110}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:265}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10421590438221688, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:112, `state`:'BLOCK', `targetIndex`:31,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06116486938533448, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06174055456366559, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:139}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06231623974199671, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:30, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10037729688470778, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10062143562525656, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10053402792802305, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10044662023078954, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10035921253355601, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10060335127410 48, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10051594357687 13, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10042853587963778, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10034112818240426, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10058526692295305, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:30, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10049785922571954, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.100410451528486, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2178159871166615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21806012585721032, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2179727181599768, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21788531046274326, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21779790276550973, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21804204150605855, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21795463380882502, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21786722611159148, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:30, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:139}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21777981841435795, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21802395715490677, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21793654945767324, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2178491417604397, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10440880412783568, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10465294286838449, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10456553517115097, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10447812747391744, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10439071977668392, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10463485851723273, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'BLOCK',
`targetIndex`:29, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:30, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:264}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10454745081999921, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:113, `state`:'BLOCK', `targetIndex`:30,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:140}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06083332294755218, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061077461688100965, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06165314686643207, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06222883204476319, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10095298206303889, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10119712080358768, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10110971310635417, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10102230540912065, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10093489771188713, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:30, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10117903645243592, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10109162875520239, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:140}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10100422105796888, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10091681336073538, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10116095210128416, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10107354440405064, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10098613670681714, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21839167229499262, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21863581103554142, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21854840333830788, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:30, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21846099564107438, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21837358794384085, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21861772668438964, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2185303189871561, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2184429112899226, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21835550359268913, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21859964233323786, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21851223463600436, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21842482693877083, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10498448930616681, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:30, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1052286280467156, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10514122034948208, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10505381265224856, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10496640495501504, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'BLOCK',
`targetIndex`:28, `threshold`:110}]->(b);
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MATCH (a {`key`:140}) MATCH (b {`key`:263}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10521054369556383, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:114, `state`:'BLOCK', `targetIndex`:29,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060170230071987546, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06074591525031865, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06099005399086745, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061565739169198566, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.062141424347529665, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:30, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10152866724137, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1017728059819188, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10168539828468527, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10159799058745175, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10151058289021825, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
```

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MATCH (a {`key`:141}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10175472163076703, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10166731393353351, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1015799062363, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10149249853906649, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10173663727961527, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:2, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:30, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10164922958238176, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10156182188514824, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21896735747332371, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21921149621387254, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.219124088516639, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21903668081940547, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.218949273122172, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21919341186272076, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21910600416548723, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21901859646825375, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:30, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21893118877102022, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21917532751156898, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21908791981433545, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21900051211710198, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10556017448449792, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1058043132250467, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1057169055278132, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10562949783057968, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'BLOCK',
`targetIndex`:27, `threshold`:110}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:262}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10554209013334614, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:115, `state`:'BLOCK', `targetIndex`:28,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05950713719642291, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:30, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060082822374754014, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060658507553085134, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060902646293633934, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06147833147196504, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06205401665029615, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10210435241970112, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1023484911602499, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226108346301639, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10217367576578287, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10208626806854935, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:30, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10233040680909815, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10224299911186463, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10215559141463111, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10206818371739758, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10231232245794639, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10222491476071287, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10213750706347935, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21954304265165486, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21978718139220363, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2196997736949701, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:30, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21961236599773662, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2195249583005031, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21976909704105185, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21968168934381838, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21959428164658484, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21950687394935134, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2197510126899001, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2196636049926666, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2195761972954331, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
```

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MATCH (a {`key`:142}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10613585966282903, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:30, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10637999840337782, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1062925907061443, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'BLOCK',
`targetIndex`:26, `threshold`:110}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:261}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10620518300891078, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:116, `state`:'BLOCK', `targetIndex`:27,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05917559075864059, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059751275936971694, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060326961115302814, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060902646293633934, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06114678503418272, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06172247021251384, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06229815539084496, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:1,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:30, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10234849116024 99, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10259262990079872, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1025052222035652, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241781450633167, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10233040680909815, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10257454554964696, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10248713785241342, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10239973015517989, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10231232245794639, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10255646119849518, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:30, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10246905350126166, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10238164580402813, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21978718139220363, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22003132013275245, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21994391243551892, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21985650473828539, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21976909704105185, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22001323578160067, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21992582808436714, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2198384203871336, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:30, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2197510126899001, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
```

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MATCH (a {`key`:143}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2199951514304489, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21990774373321537, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21982033603598183, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:6,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10637999840337782, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10662413714392663, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'BLOCK',
`targetIndex`:25, `threshold`:110}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:260}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1065367294466931, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:117, `state`:'BLOCK', `targetIndex`:26,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058844044320858276, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05908818306140706, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05966386823973818, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:30, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060239553418069296, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06081523859640041, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06105937733694921, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06163506251528032, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:1,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06221074769361143, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:1,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10292417633858103, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10316831507912982, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10308090738189629, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10299349968466279, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10290609198742927, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:2, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:30, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10315023072797806, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10306282303074453, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10297541533351103, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10288800763627749, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10313214637682629, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10304473867959277, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10295733098235926, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.006737946999085467, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22036286657053475, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22060700531108354, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22051959761385004, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:30, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2204321899166165, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22034478221938303, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2205889209599318, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
```

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MATCH (a {`key`:144}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22050151326269826, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22041410556546473, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22032669786823125, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22057083660878002, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2204834289115465, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:6,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22039602121431295, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:6,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.2706705664732254, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10695568358170893, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'BLOCK',
`targetIndex`:24, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:30, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:259}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10719982232225772, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:118, `state`:'BLOCK', `targetIndex`:25,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05818095144529364, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05875663662362474, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059000775364173544, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059576460542504664, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06015214572083577, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06072783089916689, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06097196963971569, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:1,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0615476548180468, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:1,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06212333999637792, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:1,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:30, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10349986151691214, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374400025746093, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10365659256022743, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10356918486299390, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10348177716576037, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
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MATCH (a {`key`:145}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10372591590630917, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10363850820907566, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10355110051184213, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1034636928146086, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1037078315551574, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:30, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10362042385792389, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10353301616069036, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.006737946999085467, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:11, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2209385517488659, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22118269048941466, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK', `targetIndex`:13,
`threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22109528279218113, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22100787509494765, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
```

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MATCH (a {`key`:145}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22092046739771412, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2211646061382629, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22107719844102935, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22098979074379588, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:30, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22090238304656235, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2211465217871111, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:6,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22105911408987758, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:6,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2209717063926441, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:6,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:119, `state`:'BLOCK',
`targetIndex`:23, `threshold`:110}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:258}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10753136876004003, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:119, `state`:'BLOCK', `targetIndex`:24,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16712112261675718, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16769680779508828, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
```

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MATCH (a {`key`:146}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682724929734194, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1685166317139682, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1690923168922993, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:30, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696680020706304, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17024368724896155, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17048782598951034, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17106351116784144, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17163919634617256, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15760512028761192, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15784925902816072, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1577618513309272, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15767444363369368, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15758703593646015, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:30, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15783117467700897, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15774376697977543, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576563592825419, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15756895158530843, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1578130903258572, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15772568262862366, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15763827493139018, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11670376083201528, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:12,
`threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11694789957256407, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
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MATCH (a {`key`:146}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11686049187533054, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:30, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11677308417809704, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11668567648086352, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1169298152214123, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11684240752417879, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11675499982694526, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11666759212971174, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11691173087026055, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11682432317302702, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK',
`targetIndex`:22, `threshold`:110}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:257}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1167369154757935, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:120, `state`:'BLOCK', `targetIndex`:23,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16678957617897486, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:30, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16736526135730595, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.0090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16794094653563707, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1685166317139682, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16876077045451698, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1693364556328481, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16991214081117922, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17048782598951034, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1707319647300591, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17130764990839026, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17188333508672135, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:2, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:30, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15784925902816072, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15809339776870 95, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15800599007147598, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15791858237424247, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15783117467700897, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15807531341755773, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15798790572032423, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15790049802309072, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1578130903258572, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15805722906640599, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:30, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15796982136917248, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
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MATCH (a {`key`:147}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15788241367193898, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11694789957256407, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11719203831311287, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11710463061587936, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11701722291864583, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1169298152214123, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11717395396196112, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11708654626472759, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11699913856749405, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:30, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11691173087026055, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11715586961080934, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK',
`targetIndex`:21, `threshold`:110}]->(b);
```

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MATCH (a {`key`:147}) MATCH (b {`key`:256}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11706846191357581, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:121, `state`:'BLOCK', `targetIndex`:22,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16645802974119253, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16670216848174135, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16727785366007245, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16785353883840357, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16842922401673466, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16867336275728345, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1692490479356146, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:30, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1698247331139457, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1704004182922768, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706445570328256, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17122024221115673, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17179592738948782, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15842494420649184, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15866908294704063, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1585816752498071, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1584942675525736, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15840685985534006, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:30, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15865099859588885, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15856359089865535, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15847618320142182, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15838877550418828, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1586329142447371, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15854550654750357, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15845809885027004, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11752358475089518, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11776772349144397, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11768031579421047, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:30, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11759290809697694, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11750550039974343, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11774963914029223, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11766223144305869, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11757482374582516, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
```

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MATCH (a {`key`:148}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11748741604859166, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK',
`targetIndex`:20, `threshold`:110}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:255}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11773155478914045, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:122, `state`:'BLOCK', `targetIndex`:21,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028844540087061502, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16637062204395903, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16661476078450782, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:30, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16719044596283894, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16776613114117003, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16834181631950115, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16858595506004995, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16916164023838107, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16973732541671216, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1703130105950433, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1705571493355921, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711328345139232, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717085196922543, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:30, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15900062938482296, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15924476812537172, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15915736042813822, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15906995273090468, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15898254503367118, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15922668377421997, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15913927607698644, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:149}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15905186837975294, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15896446068251943, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1592085994230682, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:30, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1591211917258347, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1590337840286012, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11809926992922631, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11834340866977508, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11825600097254155, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11816859327530807, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11808118557807454, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11832532431862333, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1182379166213898, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1181505089241563, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK',
`targetIndex`:19, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:30, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:254}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11806310122692276, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:123, `state`:'BLOCK', `targetIndex`:20,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028181447211496874, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028757132389827984, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16628321434672555, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16652735308727432, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16710303826560544, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1676787234439365, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16825440862226765, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16849854736281641, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
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MATCH (a {`key`:150}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16907423254114753, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:2, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:30, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16964991771947865, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17022560289780977, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17046974163835857, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710454268166897, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716211119950208, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15957631456315405, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15982045330370287, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15973304560646934, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596456379092358, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:150}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1595582302120023, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:30, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1598023689525511, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15971496125531756, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15962755355808406, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15954014586085052, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978428460139932, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596968769041658, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15960946920693228, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11867495510755739, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11891909384810619, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1188316861508727, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:30, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11874427845363918, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11865687075640564, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11890100949695444, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11881360179972093, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK',
`targetIndex`:18, `threshold`:110}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:253}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1187261941024874, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:124, `state`:'BLOCK', `targetIndex`:19,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027849900773714555, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028425585952045668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02900127113037678, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16652735308727432, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1667714918278231, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:30, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:151}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1673471770061542, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16792286218448532, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16849854736281641, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1687426861033652, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16931837128169636, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16989405646002745, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17046974163835857, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17071388037890736, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17128956555723848, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17186525073556957, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:30, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15982045330370287, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:151}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16006459204425166, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15997718434701813, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15988977664978463, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1598023689525511, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1600465076930999, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15995909999586638, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15987169229863285, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978428460139932, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16002842334194814, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:30, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1599410156447146, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15985360794748107, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:151}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11891909384810619, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.119163232588655, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11907582489142146, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11898841719418794, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11890100949695444, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11914514823750325, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'BLOCK',
`targetIndex`:17, `threshold`:110}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:252}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11905774054026971, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:125, `state`:'BLOCK', `targetIndex`:18,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027518354335932242, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:30, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027762493076481033, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028338178254812146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028913863433143262, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:152}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16643994539004078, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16668408413058958, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1672597693089207, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16783545448725182, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1684111396655829, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1686552784061317, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16923096358446282, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:30, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16980664876279392, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17038233394112506, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17062647268167386, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17120215786000495, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17177784303833607, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16039613848203396, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16064027722258276, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16055286952534925, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16046546182811572, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16037805413088219, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:30, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1606222192871431, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16053478517419748, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16044737747696394, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16035996977973044, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16060410852027923, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:152}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1605167008230457, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16042929312581217, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1194947790264373, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1197389177669861, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11965151006975257, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:30, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11956410237251906, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11947669467528556, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'BLOCK',
`targetIndex`:16, `threshold`:110}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:251}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11972083341583432, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:126, `state`:'BLOCK', `targetIndex`:17,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027430946638698 72, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027675085379247514, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028250770557578634, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:153}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02882645573590974, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16635253769280725, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16659667643335607, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16717236116116872, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:2, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:30, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16774804679001828, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16832373196834944, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16856787070889822, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16914355588722933, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16971924106556044, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17029492624389153, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17053906498444033, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17111475016277145, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17169043534110257, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16097182366036508, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:30, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16121596240091388, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16112855470368037, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16104114700644684, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16095373930921333, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16119787804976213, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1611104703525286, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1610230626552951, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16093565495806156, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16117979369861035, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16109238600137682, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:30, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16100497830414331, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12007046420476843, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1203146029453172, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1202271952480837, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12013978755085017, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'BLOCK',
`targetIndex`:15, `threshold`:110}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:250}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12005237985361666, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:127, `state`:'BLOCK', `targetIndex`:16,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027518354335932242, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02743094663869872, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:154}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027587677682014, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028163362860345112, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:30, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028739048038676225, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16626512999557375, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16650926873612254, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16708495391445366, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16766063909278478, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682363242711159, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16848046301166467, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16905614818999581, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696318333683269, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
```

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MATCH (a {`key`:154}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.170207518546658, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:30, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17045165728720682, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710273424655379, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17160302764386903, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16154750883869 62, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.161791647579245, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16170423988201146, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16161683218477796, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16152942448754443, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16177356322809322, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16168615553085972, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:30, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16159874783362618, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16151134013639265, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16175547887694144, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16166807117970794, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1615806634824744, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12064614938309953, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12089028812364834, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12080288042641481, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'BLOCK',
`targetIndex`:14, `threshold`:110}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:249}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12071547272918128, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:128, `state`:'BLOCK', `targetIndex`:15,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027849900773714555, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:30, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027762493076481033, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027675085379247514, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027587677682014, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028407501600893906, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028983186779225012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:0,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16650926873612254, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16675340747667133, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16732909265500245, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16790477783333357, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16848046301166467, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:30, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1687246017522135, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1693002869305446, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16987597210887567, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17045165728720682, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706957960277556, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1712714812060867, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17184716638441783, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:5,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.161791647579245, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1620357863197938, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16194837862256029, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:30, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16186097092532675, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16177356322809322, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:155}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16201770196864204, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1619302942714085, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16184288657417498, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16175547887694144, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16199961761749027, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16191220992025673, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1618248022230232, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12089028812364834, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:30, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12113442686419713, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'BLOCK',
`targetIndex`:13, `threshold`:110}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:248}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1210470191669636, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:129, `state`:'BLOCK', `targetIndex`:14,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028181447211496874, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
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MATCH (a {`key`:156}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028425585952045668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028338178254812146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028250770557578634, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028163362860345112, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028407501600893906, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028895779081991504, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:0,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.166421861038889, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:2, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:30, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16666599977943783, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16724168495776895, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16781737013610004, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16839305531443116, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16863719405497996, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16921287923331108, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697885644116422, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036424958997332, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17060838833052208, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711840735088532, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:5,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:30, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17175975868718432, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:5,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1623673327575761, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1626114714981249, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16252406380089138, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:156}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16243665610365784, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16234924840642434, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16259338714697313, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1625059794497396, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1624185717525061, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1623311640552726, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:30, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16257530279582136, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16248789509858785, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16240048740135432, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12146597330197945, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'BLOCK',
`targetIndex`:12, `threshold`:110}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:247}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12171011204252824, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:130, `state`:'BLOCK', `targetIndex`:13,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028844540087061502, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028757132389827984, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02900127113037678, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028913863433143262, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02882645573590974, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:30, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028739048038676225, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028983186779225012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:0,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028895779081991504, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:0,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1663344533416555, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1665785920822043, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16715427726053542, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16772996243886654, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16830564761719766, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16854978635774642, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16912547153607757, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:30, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16970115671440866, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17027684189273978, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17052098063328858, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:5,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710966658116197, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:5,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716723509899508, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:5,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:131, `state`:'BLOCK',
`targetIndex`:143, `threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16294301793590724, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16318715667645603, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:157}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1630997489792225, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1630123412281989, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16292493358475546, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:30, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16316907232530425, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16308166462807072, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16299425693083722, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16290684923360368, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16315098797415248, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16306358027691895, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16297617257968544, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:4,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:246}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12204165848031055, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:131, `state`:'BLOCK', `targetIndex`:12,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16645802974119253, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16637062204395903, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:30, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1662832143467255, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16652735308727432, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16643994539004078, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16635253769280725, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16626512999557375, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16650926873612254, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.166421861038889, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1663344533416555, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02954078760640437, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:0,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030116472784735483, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:0,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:30, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030692157963066596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0312678431413977, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032087667060277617, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03266335223860873, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033239037416939836, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03405886133581975, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03463454651415086, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370253403106038, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:3,
`saturationRatio`:1.2, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:30, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372694790511526, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371820713539191, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13709466365668557, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13700725595945204, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725139470000083, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13716398700276733, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370765793055338, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13698917160830026, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13723331034884906, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13714590265161555, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK',
`targetIndex`:10, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:30, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:158}) MATCH (b {`key`:245}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13705849495438202, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:132, `state`:'BLOCK', `targetIndex`:11,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16678957617897486, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16670216848174135, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16661476078450782, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16652735308727432, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1667714918278231, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16668408413058958, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16659667643335607, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16650926873612254, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16675340747667133, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:2, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:30, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16666599977943783, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1665785920822043, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02954078760640437, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:0,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030360611525284277, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030936296703615387, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0317561206224953, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03233180580082641, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032907490979157523, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:30, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03372731489803744, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:159}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034303000076368544, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03487868525469966, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372694790511526, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13751361779170143, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374262100944679, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13733880239723437, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725139470000083, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13749553344054966, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13740812574331612, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:30, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1373207180460826, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13723331034884906, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:159}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13747744908939788, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK',
`targetIndex`:9, `threshold`:110}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:244}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13739004139216435, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:133, `state`:'BLOCK', `targetIndex`:10,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16712112261675718, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16736526135730595, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16727785366007245, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16719044596283894, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16710303826560544, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1673471770061542, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:30, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1672597693089207, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1671723616116872, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16708495391445366, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16732909265500245, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16724168495776895, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16715427726053542, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030116472784735483, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:0,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030360611525284277, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030848889006381875, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031424574184712985, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:30, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03166871292526178, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03224439810359289, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032820083281924005, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03339576846025512, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03363990720080391, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034215592379135025, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03479127755746613, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378451642294837, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13808930297003252, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.138001895272799, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:30, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13791448757556546, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13782707987833195, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807121861888075, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798381092164722, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13789640322441368, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
```

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MATCH (a {`key`:160}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13780899552718018, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK',
`targetIndex`:8, `threshold`:110}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:243}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13805313426772897, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:134, `state`:'BLOCK', `targetIndex`:9,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05818095144529364, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16769680779508828, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16794094653563707, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:30, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16785353883840357, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16776613114117003, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1676787234439365, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16792286218448532, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16783545448725182, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16774804679001828, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16766063909278478, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16790477783333357, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16781737013610004, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16772996243886654, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:30, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030692157963066596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030936296703615387, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030848889006381875, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031337166487479466, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03158130522802826, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032156990406359366, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032732675584690486, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0333083607630216, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03355249950357039, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034128184681901506, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:30, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03470386986023262, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13842084940781482, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13866498814836362, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1385775804511301, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13849017275389658, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13840276505666307, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13864690379721187, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13855949609997834, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
```

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MATCH (a {`key`:161}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13847208840274483, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK',
`targetIndex`:7, `threshold`:110}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:242}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1383846807055113, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:135, `state`:'BLOCK', `targetIndex`:8,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:30, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.058844044320858276, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05875663662362474, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682724929734194, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1685166317139682, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16842922401673466, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16834181631950115, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16825440862226765, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16849854736281641, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1684111396655829, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1683237319683494, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:2, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:30, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1682363242711159, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16848046301166467, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16839305531443116, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16830564761719766, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0312678431413977, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031424574184712985, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031337166487479466, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031493897530794734, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:162}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032069582709125855, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:30, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03264526788745697, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033220953065788074, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03404077698466799, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0346164621629991, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13899653458614591, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392406733266947, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1391532656294612, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13906585793222767, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13897845023499414, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:30, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922258897554296, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13913518127830946, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK',
`targetIndex`:6, `threshold`:110}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:241}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13904777358107592, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:136, `state`:'BLOCK', `targetIndex`:7,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05917559075864059, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05908818306140706, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059000775364173544, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1685166317139682, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16876077045451698, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16867336275728345, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16858595506004995, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:31, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16849854736281641, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1687426861033652, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1686552784061317, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1685678707088982, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16848046301166467, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1687246017522135, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16863719405497996, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16854978635774642, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0317561206224953, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:31, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03166871292526178, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03158130522802826, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031493897530794734, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03231372144967465, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032889406628005755, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03370923054688567, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03428491572521678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034860600903547895, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392406733266947, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:31, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13948481206724353, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13939740437001, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
```

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MATCH (a {`key`:163}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1393099966727765, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922258897554296, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13946672771609178, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'BLOCK',
`targetIndex`:5, `threshold`:110}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:240}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13937932001885825, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:137, `state`:'BLOCK', `targetIndex`:6,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05950713719642291, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059751275936971694, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.05966386823973818, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.059576460542504664, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:31, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1690923168922993, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1693364556328481, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1692490479356146, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16916164023838107, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16907423254114753, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16931837128169636, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16923096358446282, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1691435558872293, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16905614818999581, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1693002869305446, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:31, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16921287923331108, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16912547153607757, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032087667060277617, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03233180580082641, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03224439810359289, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032156990406359366, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032069582709125855, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03231372144967465, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03280199893077224, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033377684109103356, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:31, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03362182284965215, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03419750802798326, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034773193206314376, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13981635850502586, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14006049724557462, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
```

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MATCH (a {`key`:164}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13997308954834112, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13988568185110759, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13979827415387408, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'BLOCK',
`targetIndex`:4, `threshold`:110}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:239}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14004241289442287, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:138, `state`:'BLOCK', `targetIndex`:5,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060170230071987546, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:31, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060082822374754014, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060326961115302814, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060239553418069296, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06015214572083577, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696680020706304, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16991214081117922, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1698247331139457, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16973732541671216, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16964991771947865, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16989405646002745, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:2, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:31, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16980664876279392, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697192410655604, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1696318333683269, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16987597210887567, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697885644116422, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16970115671440866, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03266335223860873, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032907490979157523, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032820083281924005, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032732675584690486, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:31, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03264526788745697, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032889406628005755, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03280199893077224, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03329027641186984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03353441515241863, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03411010033074974, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03468578550908086, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:165}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14039204368335695, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14063618242390574, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1405487747266722, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:31, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404613670294387, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'BLOCK',
`targetIndex`:3, `threshold`:110}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:238}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14037395933220517, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:139, `state`:'BLOCK', `targetIndex`:4,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06083332294755218, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06074591525031865, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060658507553085134, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060902646293633934, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06081523859640041, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06072783089916689, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
```

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MATCH (a {`key`:166}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17024368724896155, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17048782598951034, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:31, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1704004182922768, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1703130105950433, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17022560289780977, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17046974163835857, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17038233394112506, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17029492624389153, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.170207518546658, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17045165728720682, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17036424958997332, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
```

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MATCH (a {`key`:166}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17027684189273978, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:31, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033239037416939836, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03339576846025512, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0333083607630216, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033220953065788074, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033377684109103356, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03329027641186984, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03344700745518511, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034022692633516226, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:31, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03459837781184734, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14096772886168807, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14121186760223683, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14112445990500333, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'BLOCK',
`targetIndex`:2, `threshold`:110}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:237}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14103705220776983, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:140, `state`:'BLOCK', `targetIndex`:3,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06116486938533448, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061077461688100965, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06099005399086745, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.060902646293633934, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06114678503418272, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:31, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06105937733694921, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06097196963971569, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:1,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17048782598951034, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1707319647300591, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706445570328256, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1705571493355921, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17046974163835857, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17071388037890736, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17062647268167386, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17053906498444033, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:31, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17045165728720682, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706957960277556, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17060838833052208, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17052098063328858, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:5,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03372731489803744, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03363990720080391, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03355249950357039, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03346509180633687, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03370923054688567, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:31, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03362182284965215, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03353441515241863, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:167}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03344700745518511, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03426683137406502, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03484251655239613, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:0,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14121186760223683, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14145600634278566, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'BLOCK',
`targetIndex`:1, `threshold`:110}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:236}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14136859864555212, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:141, `state`:'BLOCK', `targetIndex`:2,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0614964158231168, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06174055456366559, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:31, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06165314686643207, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.061565739169198566, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06147833147196504, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06172247021251384, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06163506251528032, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:1,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0615476548180468, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:1,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17106351116784144, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17130764990839026, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17122024221115673, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711328345139232, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:2, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:31, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710454268166897, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17128956555723848, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17120215786000495, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17111475016277145, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710273424655379, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1712714812060867, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711840735088532, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:5,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1710966658116197, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:5,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03405886133581975, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034303000076368544, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:31, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034215592379135025, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034128184681901506, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03404077698466799, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03428491572521678, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:168}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03419750802798326, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03411010033074974, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034022692633516226, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03426683137406502, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034755108855162614, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:0,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14178755278056798, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'BLOCK',
`targetIndex`:0, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:31, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:235}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14203169152111678, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:142, `state`:'BLOCK', `targetIndex`:1,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.062159508698681434, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:1,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.062072101001447 91, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:1,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06231623974199671, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:1,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06222883204476319, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:1,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.062141424347529665, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:1,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06205401665029615, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:1,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06229815539084496, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:1,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06221074769361143, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:1,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06212333999637792, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:1,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:31, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17163919634617256, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:5,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17188333508672135, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:5,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17179592738948782, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:5,
`saturationRatio`:1.1, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717085196922543, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:5,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716211119950208, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:5,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17186525073556957, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:5,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
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```
MATCH (a {`key`:169}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17177784303833607, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:5,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17169043534110257, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:5,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17160302764386903, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:5,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17184716638441783, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:5,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:31, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17175975868718432, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:5,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716723509899508, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:5,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.09957413673572789, `sourceIndex`:143, `state`:'BLOCK',
`targetIndex`:131, `threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03463454651415086, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03487868525469966, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03479127755746613, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03470386986023262, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:0,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0346164621629991, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:0,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:169}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034860600903547895, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:0,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034773193206314376, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:0,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03468578550908086, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:0,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:31, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03459837781184734, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:0,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03484251655239613, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:0,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034755108855162614, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:0,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:234}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14236323795889907, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:143, `state`:'BLOCK', `targetIndex`:0,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09898178778749675, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09955747296582786, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10013315814415898, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10037729688470778, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
```

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MATCH (a {`key`:234}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10095298206303889, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10152866724137, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:31, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10210435241970112, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1023484911602499, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10292417633858103, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10349986151691214, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15760512028761192, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15784925902816072, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15842494420649184, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15900062938482296, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15957631456315405, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
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MATCH (a {`key`:234}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15982045330370287, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:31, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16039613848203396, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16097182366036508, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1615475088386962, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.161791647579245, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1623673327575761, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16294301793590724, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370253403106038, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:3,
`saturationRatio`:1.2, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372694790511526, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378451642294837, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13842084940781482, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:31, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13899653458614591, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392406733266947, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13981635850502586, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14039204368335695, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14096772886168807, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14121186760223683, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14178755278056798, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK',
`targetIndex`:142, `threshold`:110}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:169}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14236323795889907, `foldPressure`:0.9876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:0, `state`:'BLOCK', `targetIndex`:143,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09865024134971444, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09922592652804554, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:31, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09980161170637666, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10037729688470778, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10062143562525656, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10119712080358768, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1017728059819188, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1023484911602499, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10259262990079872, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10316831507912982, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374400025746093, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15784925902816072, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:2, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:31, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1580933977687095, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15866908294704063, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15924476812537172, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15982045330370287, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16006459204425166, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16064027722258276, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16121596240091388, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.161791647579245, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1620357863197938, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1626114714981249, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:31, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16318715667645603, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1372694790511526, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13751361779170143, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13808930297003252, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13866498814836362, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392406733266947, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13948481206724353, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14006049724557462, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14063618242390574, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14121186760223683, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:31, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14145600634278566, `foldPressure`:0.9629629629629629,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK',
`targetIndex`:141, `threshold`:110}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:168}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14203169152111678, `foldPressure`:0.9753086419753086,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:1, `state`:'BLOCK', `targetIndex`:142,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09831869491193213, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09856283365248092, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09913851883081203, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09971420400914315, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10028988918747425, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10053402792802305, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10110971310635417, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10168539828468527, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:31, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226108346301639, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1025052222035652, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10308090738189629, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10365659256022743, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1577618513309272, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15800599007147598, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1585816752498071, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15915736042813822, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15973304560646934, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15997718434701813, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:31, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16055286952534925, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16112855470368037, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16170423988201146, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16194837862256029, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:236}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16252406380089138, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1630997489792225, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371820713539191, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374262100944679, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.138001895272799, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1385775804511301, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:31, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1391532656294612, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13939740437001, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13997308954834112, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1405487747266722, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14112445990500333, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK',
`targetIndex`:140, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:236}) MATCH (b {`key`:167}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14136859864555212, `foldPressure`:0.9506172839506173,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:2, `state`:'BLOCK', `targetIndex`:141,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15466096483994762, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0982312872146986, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09847542595524739, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09905111113357852, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:31, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09962679631190961, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10020248149024073, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10044662023078954, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10102230540912065, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10159799058745175, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10217367576578287, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241781450633167, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10299349968466279, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1035691848629939, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15767444363369368, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:31, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15791858237424247, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1584942675525736, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15906995273090468, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596456379092358, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15988977664978463, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16046546182811572, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16104114700644684, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16161683218477796, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16186097092532675, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16243665610365784, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:31, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.163012341281989, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13709466365668557, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13733880239723437, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13791448757556546, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13849017275389658, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13906585793222767, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1393099966727765, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13988568185110759, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404613670294387, `foldPressure`:0.9259259259259259,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK',
`targetIndex`:139, `threshold`:110}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:166}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14103705220776983, `foldPressure`:0.9382716049382716,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:3, `state`:'BLOCK', `targetIndex`:140,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:31, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.153997871964383, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15457355714271412, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09814387951746509, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09838801825801388, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09896370343634499, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0995393886146761, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10011507379300721, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10035921253355601, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10093489771188713, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10151058289021825, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:2, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:31, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10208626806854935, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10233040680909815, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10290609198742927, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10348177716576037, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15758703593646015, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15783117467700897, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15840685985534006, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15898254503367118, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1595582302120023, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1598023689525511, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:31, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16037805413088219, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16095373930921333, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16152942448754443, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16177356322809322, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16234924840642434, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16292493358475546, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13700725595945204, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725139470000083, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13782707987833195, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13840276505666307, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:31, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13897845023499414, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922258897554296, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13979827415387408, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK',
`targetIndex`:138, `threshold`:110}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:165}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14037395933220517, `foldPressure`:0.9135802469135802,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:4, `state`:'BLOCK', `targetIndex`:139,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15366632552660067, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1542420107049318, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15481769588326288, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09838801825801388, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09863215699856268, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09920784217689378, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:31, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09978352735522492, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10035921253355601, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1006033512741048, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10117903645243592, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10175472163076703, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10233040680909815, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10257454554964696, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10315023072797806, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10372591590630917, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15783117467700897, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:31, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15807531341755773, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15865099859588885, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15922668377421997, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1598023689525511, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1600465076930999, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.160622192871431, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.1909090909090091, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16119787804976213, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16177356322809322, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16201770196864204, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16259338714697313, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:31, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16316907232530425, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725139470000083, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
```

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MATCH (a {`key`:239}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13749553344054966, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807121861888075, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13864690379721187, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922258897554296, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13946672771609178, `foldPressure`:0.8888888888888888,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'BLOCK',
`targetIndex`:137, `threshold`:110}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:164}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14004241289442287, `foldPressure`:0.9012345679012346,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:5, `state`:'BLOCK', `targetIndex`:138,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1533347790888184, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15357891782936714, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:31, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15415460300769826, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15473028818602938, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09830061056078035, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
```

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MATCH (a {`key`:240}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09854474930132916, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09912043447966028, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09969611965799138, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10027180483632249, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1005159435768713, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10109162875520239, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10166731393353351, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:31, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10224299911186463, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10248713785241342, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10306282303074453, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10363850820907566, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:240}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15774376697977543, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15798790572032423, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15856359089865535, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15913927607698644, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15971496125531756, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15995909999586638, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:31, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16053478517419748, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1611104703525286, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16168615553085972, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1619302942714085, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1625059794497396, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:240}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16308166462807072, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13716398700276733, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13740812574331612, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798381092164722, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13855949609997834, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:31, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13913518127830946, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'BLOCK',
`targetIndex`:136, `threshold`:110}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:163}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13937932001885825, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:6, `state`:'BLOCK', `targetIndex`:137,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15267168621325375, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15324737139158487, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15349151013213366, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15406719531046478, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15464288048879588, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09821320286354684, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09845734160409564, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09903302678242674, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:2, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:31, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09960871196075786, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10018439713908898, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10042853587963778, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10100422105796888, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1015799062363, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10215559141463111, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10239973015517989, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10297541533351103, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10355110051184213, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576563592825419, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:31, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15790049802309072, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15847618320142182, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15905186837975294, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15962755355808406, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15987169229863285, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16044737747696394, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1610230626552951, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:241}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16159874783362618, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16184288657417498, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1624185717525061, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:31, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16299425693083722, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1370765793055338, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1373207180460826, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13789640322441368, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13847208840274483, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'BLOCK',
`targetIndex`:135, `threshold`:110}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:162}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13904777358107592, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:7, `state`:'BLOCK', `targetIndex`:136,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15200859333768912, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15258427851602022, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
```

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MATCH (a {`key`:242}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15315996369435134, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340410243490016, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:31, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15397978761323125, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15455547279156237, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09812579516631333, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0983699339068621, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09894561908519323, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09952130426352435, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10009698944185545, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10034112818240426, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10091681336073538, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
```

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MATCH (a {`key`:242}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10149249853906649, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:31, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10206818371739758, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10231232245794639, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10288800763627749, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1034636928146086, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15756895158530843, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1578130903258572, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15838877550418828, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15896446068251943, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15954014586085052, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978428460139932, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:31, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16035996977973044, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16093565495806156, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16151134013639265, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16175547887694144, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1623311640552726, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16290684923360368, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13698917160830026, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13723331034884906, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13780899552718018, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'BLOCK',
`targetIndex`:134, `threshold`:110}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:161}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1383846807055113, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:8, `state`:'BLOCK', `targetIndex`:135,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:31, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1516770468999068, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15225273207823792, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.152828417256569, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15340410243490016, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15364824117544892, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15422392635378004, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15479961153211116, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0983699339068621, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09861407264741091, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09918975782574203, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:31, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09976544300407314, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
```

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MATCH (a {`key`:243}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10034112818240426, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10058526692295305, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10116095210128416, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10173663727961527, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10231232245794639, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10255646119849518, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10313214637682629, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1037078315551574, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1578130903258572, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:31, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15805722906640599, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1586329142447371, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1592085994230682, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978428460139932, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16002842334194814, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16060410852027923, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16117979369861035, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16175547887694144, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16199961761749027, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16257530279582136, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:31, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16315098797415248, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13723331034884906, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13747744908939788, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'BLOCK',
`targetIndex`:133, `threshold`:110}]->(b);
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MATCH (a {`key`:243}) MATCH (b {`key`:160}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13805313426772897, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:9, `state`:'BLOCK', `targetIndex`:134,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15134550046212447, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15158963920267327, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15216532438100439, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1527410095593355, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15331669473766663, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1535608334782154, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:2, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:31, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15413651865654654, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15471220383487763, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0982825262096286, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
```

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MATCH (a {`key`:244}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0985266649501774, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0991023501285085, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09967803530683962, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10025372048517074, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10049785922571954, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10107354440405064, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10164922958238176, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:31, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10222491476071287, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10246905350126166, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10304473867959277, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10362042385792389, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.006737946999085467, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15772568262862366, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15796982136917248, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15854550654750357, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1591211917258347, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596968769041658, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1599410156447146, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:31, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1605167008230457, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16109238600137682, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16166807117970794, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16191220992025673, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16248789509858785, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
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MATCH (a {`key`:244}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16306358027691895, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13714590265161555, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'BLOCK',
`targetIndex`:132, `threshold`:110}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:159}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13739004139216435, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:10, `state`:'BLOCK', `targetIndex`:133,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15068240758655985, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15125809276489097, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.0090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:31, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15150223150543973, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15207791668377088, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15265360186210197, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1532292870404331, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1534734257809819, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.154049110959313, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15462479613764413, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.01831563888873418, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09819511851239508, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:2,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09843925725294386, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09901494243127498, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:31, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0995906276096061, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10016631278793721, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.100410451528486, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10098613670681714, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10156182188514824, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10213750706347935, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10238164580402813, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10295733098235926, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10353301616069036, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.006737946999085467, `sourceIndex`:11, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15763827493139018, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:31, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15788241367193898, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15845809885027004, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1590337840286012, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15960946920693228, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15985360794748107, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16042929312581217, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16100497830414331, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1615806634824744, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:245}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1618248022230232, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16240048740135432, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:31, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16297617257968544, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:4,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:158}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13705849495438202, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.013475893998170934, `sourceIndex`:11, `state`:'BLOCK', `targetIndex`:132,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17227035853697145, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17284604371530254, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17342172889363364, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17366586763418246, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17424155281251355, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17481723799084467, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1753929231691758, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17563706190972458, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:31, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1762127470880557, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17678843226638682, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21560065410057055, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:6,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21584479284111938, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:6,
`saturationRatio`:1.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21642047801945047, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:6,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2169961631977816, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21757184837611268, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2178159871166615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21839167229499262, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21896735747332371, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:31, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21954304265165486, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21978718139220363, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22036286657053475, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2209385517488659, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11670376083201528, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11694789957256407, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11752358475089518, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11809926992922631, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11867495510755739, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11891909384810619, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:31, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:246}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1194947790264373, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12007046420476843, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12064614938309953, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12089028812364834, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12146597330197945, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK',
`targetIndex`:130, `threshold`:110}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:157}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12204165848031055, `foldPressure`:0.8765432098765432,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:12, `state`:'BLOCK', `targetIndex`:131,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17193881209918913, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17251449727752025, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17309018245585134, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17366586763418246, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:2, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:31, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17391000637473128, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17448569155306234, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1750613767313935, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17563706190972458, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17588120065027338, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1764568858286045, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17703257100693562, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21584479284111938, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:6,
`saturationRatio`:1.1, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21608893158166817, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:6,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2166646167599993, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:31, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2172403019383304, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2178159871166615, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21806012585721032, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21863581103554142, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21921149621387254, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21978718139220363, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22003132013275245, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22060700531108354, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22118269048941466, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:13, `state`:'BLOCK', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11694789957256407, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:31, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11719203831311287, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11776772349144397, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
```

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MATCH (a {`key`:247}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11834340866977508, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11891909384810619, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.119163232588655, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1197389177669861, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1203146029453172, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12089028812364834, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12113442686419713, `foldPressure`:0.8518518518518519,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK',
`targetIndex`:129, `threshold`:110}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:156}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12171011204252824, `foldPressure`:0.8641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:13, `state`:'BLOCK', `targetIndex`:130,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:31, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17160726566140683, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1718514044019556, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1724270895802867, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17300277475861783, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17357845993694893, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17382259867749775, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17439828385582884, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17497396903415996, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17554965421249105, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17579379295303987, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:31, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17636947813137097, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17694516330970209, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:14, `state`:'BLOCK', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21575738514388584, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:6,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21600152388443467, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
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MATCH (a {`key`:248}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21657720906276576, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21715289424109688, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21772857941942797, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2179727181599768, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21854840333830788, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.219124088516639, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:31, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2196997736949701, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21994391243551892, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22051959761385004, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22109528279218113, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11686049187533054, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
```

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MATCH (a {`key`:248}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11710463061587936, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11768031579421047, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11825600097254155, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1188316861508727, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11907582489142146, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:31, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11965151006975257, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12022719524808837, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12080288042641481, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK',
`targetIndex`:128, `threshold`:110}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:155}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1210470191669636, `foldPressure`:0.8395061728395061,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:14, `state`:'BLOCK', `targetIndex`:129,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15118517562873618, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:15, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1715198579641733, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
```

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MATCH (a {`key`:249}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717639967047221, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723396818830532, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17291536706138433, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17349105223971542, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:31, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17373519098026424, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1743108761585953, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17488656133692643, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17546224651525755, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17570638525580634, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17628207043413746, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17685775561246858, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2156699774466523, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:6,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21591411618720113, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21648980136553223, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:31, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21706548654386335, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21764117172219444, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21788531046274326, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21846099564107438, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21903668081940547, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21961236599773662, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21985650473828539, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2204321899166165, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22100787509494765, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11677308417809704, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:31, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11701722291864583, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11759290809697694, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11816859327530807, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11874427845363918, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11898841719418794, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11956410237251906, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12013978755085017, `foldPressure`:0.8148148148148148,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK',
`targetIndex`:127, `threshold`:110}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:154}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12071547272918128, `foldPressure`:0.8271604938271605,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:15, `state`:'BLOCK', `targetIndex`:128,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15052208275317153, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15109776793150265, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:16, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:2, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:31, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17143245026693976, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17167658900748858, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17225227418581968, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728279593641508, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17340364454248192, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1736477832830307, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1742234684613618, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17479915363969295, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17537483881802404, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
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MATCH (a {`key`:250}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756189775585728, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:31, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17619466273690396, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17677034791523505, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21558256974941878, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:6,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2158267084899676, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2164023936682987, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2169780788466298, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21755376402496096, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21779790276550973, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21837358794384085, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.218949273122172, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:31, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2195249583005031, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21976909704105185, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22034478221938303, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22092046739771412, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11668567648086352, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1169298152214123, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11750550039974343, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11808118557807454, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11865687075640564, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11890100949695444, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:31, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:250}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11947669467528556, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK',
`targetIndex`:126, `threshold`:110}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:153}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.12005237985361666, `foldPressure`:0.8024691358024691,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:16, `state`:'BLOCK', `targetIndex`:127,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1501905363153892, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15076622149372032, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1513419066720514, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:17, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17167658900748858, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17192072774803735, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17249641292636847, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1730720981046996, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1736477832830307, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:31, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1738919220235795, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17446760720191062, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17504329238024172, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756189775585728, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17586311629912163, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17643880147745272, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17701448665578384, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2158267084899676, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:6,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21607084723051642, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21664653240884751, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:31, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21722221758717863, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21779790276550973, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
```

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MATCH (a {`key`:251}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21804204150605855, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21861772668438964, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21919341186272076, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21976909704105185, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22001323578160067, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2205889209599318, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2211646061382629, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1169298152214123, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:31, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11717395396196112, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11774963914029223, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11832532431862333, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11890100949695444, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11914514823750325, `foldPressure`:0.7777777777777778,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'BLOCK',
`targetIndex`:125, `threshold`:110}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:152}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11972083341583432, `foldPressure`:0.7901234567901234,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:17, `state`:'BLOCK', `targetIndex`:126,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1498589898776069, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1501031286181557, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.0090909090909091, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1506788137964868, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1512544989748179, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:18, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:31, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17158918131025505, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17183332005080384, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17240900522913497, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17298469040746606, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17356037558579718, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.173804514326346, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1743801995046771, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1749558846830082, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17553156986133933, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1757757086018881, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:31, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17635139378021922, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17692707895855034, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21573930079273407, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:6,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2159834395332829, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21655912471161398, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
```

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MATCH (a {`key`:252}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2171348098899451, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2177104950682762, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21795463380882502, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2185303189871561, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK', `targetIndex`:114,
`threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21910600416548723, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:31, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21968168934381838, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21992582808436714, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22050151326269826, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22107719844102935, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11684240752417879, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11708654626472759, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11766223144305869, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1182379166213898, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11881360179972093, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'BLOCK',
`targetIndex`:124, `threshold`:110}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:151}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11905774054026971, `foldPressure`:0.7654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:18, `state`:'BLOCK', `targetIndex`:125,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:2, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:31, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14919589700204225, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14977158218037337, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15001572092092216, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15059140609925328, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1511670912775844, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17150177361302152, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
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MATCH (a {`key`:253}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17174591235357034, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17232159753190143, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17289728271023255, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17347296788856367, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:31, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17371710662911247, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17429279180744356, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1748684769857747, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1754441621641058, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756883009046546, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1762639860829857, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17683967126131683, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21565189309550054, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:6,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2158960318360494, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21647171701438045, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:31, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21704740219271157, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21762308737104272, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21786722611159148, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2184429112899226, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21901859646825375, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21959428164658484, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2198384203871336, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22041410556546473, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
```

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MATCH (a {`key`:253}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22098979074379588, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11675499982694526, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:31, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11699913856749405, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11757482374582516, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1181505089241563, `foldPressure`:0.7407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'BLOCK',
`targetIndex`:123, `threshold`:110}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:150}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1187261941024874, `foldPressure`:0.7530864197530864,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:19, `state`:'BLOCK', `targetIndex`:124,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14853280412647762, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14910848930480874, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14968417448313984, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14992831322368866, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15050399840201975, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15107968358035087, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:31, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17141436591578801, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716585046563368, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1722341898346679, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17280987501299905, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17338556019133014, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17362969893187893, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17420538411021005, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17478106928854117, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17535675446687227, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756008932074211, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:265}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:31, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1761765783857522, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1767522635640833, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21556448539826706, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:6,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21580862413881588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21638430931714694, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2169599944954781, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21753567967380918, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21777981841435795, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21835550359268913, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21893118877102022, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:31, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:254}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21950687394935134, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2197510126899001, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22032669786823125, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22090238304656235, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11666759212971174, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11691173087026055, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11748741604859166, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'BLOCK',
`targetIndex`:122, `threshold`:110}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:149}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11806310122692276, `foldPressure`:0.7283950617283951,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:20, `state`:'BLOCK', `targetIndex`:123,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1482012576886953, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14877694286702642, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:31, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1493526280453575, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14992831322368866, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15017245196423745, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15074813714256854, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15132382232089966, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:2.0, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1716585046563368, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1719026433968856, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17247832857521672, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17305401375354781, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17362969893187893, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:31, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17387383767242776, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17444952285075885, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
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MATCH (a {`key`:255}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17502520802908997, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1756008932074211, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17584503194796988, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17642071712630097, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17699640230463212, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21580862413881588, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:6,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21605276287936465, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2166284480576958, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:31, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21720413323602686, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21777981841435795, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21802395715490677, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
```

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MATCH (a {`key`:255}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21859964233323786, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21917532751156898, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2197510126899001, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2199951514304489, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22057083660878002, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2211465217871111, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:6,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11691173087026055, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:2, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:31, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11715586961080934, `foldPressure`:0.7037037037037037,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'BLOCK',
`targetIndex`:121, `threshold`:110}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:148}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11773155478914045, `foldPressure`:0.7160493827160493,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:21, `state`:'BLOCK', `targetIndex`:122,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.147869711250913, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1481138499914618, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14868953516979289, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.149265220348124, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14984090552645513, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15008504426700392, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15066072944533504, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15123641462366616, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:2.0, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:31, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17157109695910328, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1718152356996521, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723909208779832, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1729666060563143, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17354229123464543, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17378642997519422, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17436211515352532, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17493780033185646, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17551348551018756, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17575762425073635, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:31, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17633330942906747, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1769089946073986, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.7357588823428847, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21572121644158235, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:6,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21596535518213117, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21654104036046226, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:110,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:256}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21711672553879333, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21769241071712447, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21793654945767324, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21851223463600436, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21908791981433545, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:31, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2196636049926666, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21990774373321537, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2204834289115465, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:6,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22105911408987758, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:6,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.2706705664732254, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11682432317302702, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'BLOCK',
`targetIndex`:120, `threshold`:110}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:147}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.11706846191357581, `foldPressure`:0.691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:22, `state`:'BLOCK', `targetIndex`:121,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14720661837534837, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:2,
`saturationRatio`:1.0, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14778230355367947, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:2,
`saturationRatio`:1.009090909090909, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14802644229422826, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:2,
`saturationRatio`:1.018181818181818, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14860212747255938, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:2,
`saturationRatio`:1.0272727272727273, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:31, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1491778126508905, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:2,
`saturationRatio`:1.0363636363636364, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1497534978292216, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:2,
`saturationRatio`:1.0454545454545454, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14999763656977042, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:2,
`saturationRatio`:1.0545454545454545, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1505733217481015, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:2,
`saturationRatio`:1.0636363636363637, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15114900692643263, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:2,
`saturationRatio`:1.0727272727272728, `shadowDecay`:2.0, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17148368926186977, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17172782800241856, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
```

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MATCH (a {`key`:257}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17230351318074966, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1728791983590808, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1734548835374119, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:31, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1736990222779607, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1742747074562918, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17485039263462293, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17542607781295402, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17567021655350284, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17624590173183396, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17682158691016506, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.7357588823428847, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21563380874434881, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:6,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
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MATCH (a {`key`:257}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21587794748489764, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:6,
`saturationRatio`:1.2, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21645363266322873, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:6,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:31, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21702931784155985, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:6,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21760500301989094, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:6,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2178491417604397, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:6,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21842482693877083, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:6,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21900051211710198, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:6,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2195761972954331, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:6,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.21982033603598183, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:6,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.22039602121431295, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:6,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.2209717063926441, `foldPressure`:0.6666666666666666,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:6,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.2706705664732254, `sourceIndex`:23, `state`:'BLOCK',
`targetIndex`:119, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:257}) MATCH (b {`key`:146}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1167369154757935, `foldPressure`:0.6790123456790124,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.36787944117144233, `sourceIndex`:23, `state`:'BLOCK', `targetIndex`:120,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:31, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15642302407833678, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15699870925666784, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157574394434999, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15781853317554778, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15839421835387887, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15896990353220997, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15954558871054111, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978972745108988, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.160365412629421, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16094109780775215, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:31, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10156122590292715, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:2,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10180536464347595, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10238104982180707, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10295673500013817, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10353242017846928, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10377655891901809, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10435224409734921, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10492792927568032, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10550361445401142, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10574775319456023, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:31, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:258}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10632343837289132, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10689912355122246, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10219347111174477, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10243760985229355, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10301329503062466, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10358898020895578, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10416466538728689, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10440880412783568, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10498448930616681, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10556017448449792, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:2, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:31, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:258}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10613585966282903, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10637999840337782, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10695568358170893, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK',
`targetIndex`:118, `threshold`:110}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:145}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10753136876004003, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:24, `state`:'BLOCK', `targetIndex`:119,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15609147764055445, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15666716281888554, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15724284799721666, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15781853317554778, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15806267191609655, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1586383570944277, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:31, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1592140422727588, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15978972745108988, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16003386619163867, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16060955136996982, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16118523654830091, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10180536464347595, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:2,
`saturationRatio`:1.1, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10204950338402477, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10262518856235586, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10320087374068697, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10377655891901809, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:32, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10402069765956688, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10459638283789799, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10517206801622911, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10574775319456023, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10599189193510902, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10656757711344013, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10714326229177125, `foldPressure`:0.654320987654321,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:25, `state`:'BLOCK', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10243760985229355, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10268174859284235, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10325743377117347, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:32, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10383311894950457, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10440880412783568, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10465294286838449, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
```

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MATCH (a {`key`:259}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1052286280467156, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1058043132250467, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10637999840337782, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10662413714392663, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK',
`targetIndex`:117, `threshold`:110}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:144}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10719982232225772, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:25, `state`:'BLOCK', `targetIndex`:118,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15575993120277212, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560040699433209, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:32, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15657975512165204, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15715544029998316, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15773112547831425, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15797526421886304, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
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MATCH (a {`key`:260}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1585509493971942, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15912663457552526, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15970231975385638, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15994645849440517, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16052214367273626, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16109782885106674, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:26, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:32, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10171795694624244, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:2,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10196209568679124, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10253778086512234, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10311346604345346, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10368915122178457, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
```

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MATCH (a {`key`:260}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10393328996233335, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10450897514066447, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10508466031899559, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1056603454973267, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1059044842378755, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:32, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10648016941620662, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10705585459453772, `foldPressure`:0.6419753086419753,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10235020215506004, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10259434089560884, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10317002607393995, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374571125227104, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
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MATCH (a {`key`:260}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10432139643060218, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10456553517115097, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10514122034948208, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1057169055278132, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:32, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1062925907061443, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK',
`targetIndex`:116, `threshold`:110}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:143}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1065367294466931, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:26, `state`:'BLOCK', `targetIndex`:117,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17305634375136336, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:27, `state`:'BLOCK', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15567252350553856, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15591666224608738, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15649234742441853, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1570680326027496, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
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MATCH (a {`key`:261}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15764371778108074, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15788785652162954, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15846354169996063, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:32, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15903922687829175, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1596149120566229, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15985905079717166, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16043473597550276, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1610104211538339, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10163054924900891, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:2,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1018746879895577, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10245037316788884, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:261}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10302605834621995, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10360174352455104, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:32, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10384588226509985, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10442156744343095, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10499725262176206, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10557293780009318, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10581707654064199, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10639276171897309, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10696844689730421, `foldPressure`:0.6296296296296297,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10226279445782653, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10250693319837531, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
```

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MATCH (a {`key`:261}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10308261837670642, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:2, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:32, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10365830355503755, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10423398873336864, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10447812747391744, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10505381265224856, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10562949783057968, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK',
`targetIndex`:115, `threshold`:110}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:142}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10620518300891078, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:27, `state`:'BLOCK', `targetIndex`:116,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1723932508757987, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'BLOCK', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17296893605412983, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15558511580830506, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582925454885388, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:32, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15640493972718497, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1569806249055161, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15755631008384724, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157800448824396, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15837613400272713, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15895181918105825, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15952750435938934, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15977164309993816, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16034732827826928, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16092301345660034, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:32, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10154314155177539, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:2,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1017872802923242, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1023629654706553, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10293865064898641, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10351433582731752, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10375847456786634, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10433415974619743, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10490984492452855, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10548553010285967, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10572966884340845, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:32, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:262}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10630535402173957, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1068810392000707, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.102175386760593, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241952550114179, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10299521067947291, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10357089585780402, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10414658103613512, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10439071977668392, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10496640495501504, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK',
`targetIndex`:114, `threshold`:110}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:141}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10554209013334614, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:28, `state`:'BLOCK', `targetIndex`:115,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:32, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17206170443801638, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'BLOCK', `targetIndex`:81,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17263738961634 75, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17321307479467862, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15582925454885388, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15607339328940265, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15664907846773377, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1572247636460649, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157800448824396, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1580445875649448, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15862027274327595, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:32, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.159195957921607, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15977164309993816, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16001578184048695, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16059146701881805, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16116715219714914, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1017872802923242, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:2,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.102031419032873, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1026071042112041, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1031827893895352, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10375847456786634, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:32, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10400261330841512, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10457829848674623, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10515398366507735, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:263}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10572966884340845, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10597380758395726, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10654949276228838, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10712517794061949, `foldPressure`:0.6172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241952550114179, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1026636642416906, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1032393494200217, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:32, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10381503459835281, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10439071977668392, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10463485851723273, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'BLOCK',
`targetIndex`:113, `threshold`:110}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:140}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10521054369556383, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:29, `state`:'BLOCK', `targetIndex`:114,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17173015800023408, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'BLOCK', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17197429674078288, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17254998191911397, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17312566709744512, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15574184685162032, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15598598559216914, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:32, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15656167077050026, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15713735594883135, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1577130411271625, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1579571798677113, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1585328650460424, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1591085502243735, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15968423540270466, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15992837414325342, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16050405932158454, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16107974449991566, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:32, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10169987259509067, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:2,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10194401133563946, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10251969651397058, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1030953816923017, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10367106687063281, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10391520561118159, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:264}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10449089078951272, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10506657596784381, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10564226114617493, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10588639988672374, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:2, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:32, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10646208506505485, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10703777024338595, `foldPressure`:0.6049382716049383,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10233211780390829, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10257625654445707, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10315194172278817, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1037276269011193, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:264}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10430331207945041, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'BLOCK',
`targetIndex`:112, `threshold`:110}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:139}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10454745081999921, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:30, `state`:'BLOCK', `targetIndex`:113,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17106706512466943, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17164275030300055, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:32, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17188688904354935, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17246257422188047, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17303825940021159, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15565443915438681, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15589857789493564, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15647426307326673, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15704994825159785, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157625633429929, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15786977217047776, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15844545734880888, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:32, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15902114252714, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1595968277054711, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15984096644601992, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16041665162435104, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16099233680268213, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10161246489785715, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:2,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10185660363840594, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10243228881673706, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
```

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MATCH (a {`key`:265}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10300797399506817, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10358365917339929, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:32, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10382779791394808, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1044034830922792, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10497916827061031, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10555485344894142, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10579899218949021, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10637467736782132, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10695036254615245, `foldPressure`:0.5925925925925926,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10224471010667475, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10248884884722355, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
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MATCH (a {`key`:265}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10306453402555465, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:32, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10364021920388578, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'BLOCK',
`targetIndex`:111, `threshold`:110}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:138}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10421590438221688, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:31, `state`:'BLOCK', `targetIndex`:112,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1704039722491048, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17097965742743593, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17155534260576702, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17179948134631584, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17237516652464693, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17295085170297805, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1555670314571533, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15581117019770208, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:32, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15638685537603322, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15696254055436434, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15753822573269544, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15778236447324426, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15835804965157538, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15893373482990647, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1595094200082376, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1597535587487864, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16032924392711748, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16090492910544862, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:32, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10152505720062363, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:2,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10176919594117242, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10234488111950353, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10292056629783465, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10349625147616577, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374039021671456, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10431607539504568, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1048917605733768, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10546744575170791, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10571158449225669, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:32, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10628726967058781, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10686295484891892, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10215730240944122, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10240144114999003, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10297712632832115, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'BLOCK',
`targetIndex`:110, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:137}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10355281150665226, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:32, `state`:'BLOCK', `targetIndex`:111,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17007242581132248, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1706481109896536, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1712237961679847, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17179948134631584, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:32, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17204362008686463, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17261930526519573, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17319499044352685, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15581117019770208, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1560553089382509, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15663099411658202, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1572066792949131, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15778236447324426, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15802650321379305, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15860218839212414, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:32, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15917787357045526, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1597535587487864, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15999769748933518, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1605733826676663, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16114906784599742, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10176919594117242, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:2,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10201333468172122, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10258901986005234, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.1909090909090911, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10316470503838346, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10374039021671456, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:2, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:32, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10398452895726336, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10456021413559448, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10513589931392558, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10571158449225669, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10595572323280551, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1065314084111366, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10710709358946771, `foldPressure`:0.5802469135802469,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10240144114999003, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10264557989053882, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'BLOCK',
`targetIndex`:109, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:136}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10322126506886993, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:33, `state`:'BLOCK', `targetIndex`:110,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:32, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1697408793735402, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16998501811408898, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17056070329242007, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1711363884707512, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1717120736490823, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1719562123896311, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17253189756796222, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17310758274629334, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.03663127777746836, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15572376250046857, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1559679012410174, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:32, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15654358641934848, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1571192715976796, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15769495677601075, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1579390955165595, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15851478069489064, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:268}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15909046587322176, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15966615105155285, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15991028979210167, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1604859749704328, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16106166014876389, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:32, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1016817882439389, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:2,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10192592698448777, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:97, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10250161216281882, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10307729734114993, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10365298251948105, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10389712126002984, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10447280643836095, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10504849161669207, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10562417679502319, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10586831553557198, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:32, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10644400071390309, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10701968589223419, `foldPressure`:0.5679012345679012,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10231403345275651, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'BLOCK',
`targetIndex`:108, `threshold`:110}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:135}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10255817219330529, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:34, `state`:'BLOCK', `targetIndex`:109,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16907778649797556, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:5,
`saturationRatio`:1.0, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16965347167630668, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:5,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16989761041685544, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:5,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1704732955951866, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:5,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17104898077351768, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:5,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.17162466595184878, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:5,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:32, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1718688046923976, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:5,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1724444898707287, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:5,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1730201750490598, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:5,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.03663127777746836, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15563635480323507, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15588049354378383, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15645617872211498, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1570318639004461, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1576075490787772, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.157851687819326, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15842737299765713, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:32, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15900305817598823, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15957874335431935, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.15982288209486817, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16039856727319923, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.16097425245153038, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.36787944117144233, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:95, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10159438054670539, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:2,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:96, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10183851928725418, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10241420446558529, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:98, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10298988964391641, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:99, `threshold`:110}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10356557482224753, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:100, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:32, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10380971356279632, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:101, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10438539874112743, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:102, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10496108391945856, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:103, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10553676909778967, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:104, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10578090783833845, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:105, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10635659301666955, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:106, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10693227819500069, `foldPressure`:0.5555555555555556,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.1353352832366127, `sourceIndex`:35, `state`:'BLOCK',
`targetIndex`:107, `threshold`:110}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:134}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.10222662575552298, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.049787068367863944, `sourceIndex`:35, `state`:'BLOCK', `targetIndex`:108,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14191273435806623, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14248841953639732, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:32, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14306410471472844, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14330824345527726, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14388392863360835, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14445961381193947, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450352989902706, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14527943773081936, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14585512290915048, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1464308080874816, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:36, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06510366414634572, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:1,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06534780288689451, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:1,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:32, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:270}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06592348806522562, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:1,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06649917324355674, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06707485842188785, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06731899716243665, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06789468234076776, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06847036751909887, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06904605269742999, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06929019143797878, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06986587661630989, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0704415617946 41, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.1909090909090911, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:2, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:32, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:270}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07101724697297213, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07126138571352092, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07183707089185203, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07241275607018315, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07298844124851425, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07323257998906303, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07380826516739417, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07438395034572527, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07495963552405638, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07520377426460519, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:32, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0757794594429363, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:106,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:133}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0763551446212674, `foldPressure`:0.5432098765432098,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:36, `state`:'BLOCK', `targetIndex`:107,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1415811879202839, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14215687309861502, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14273255827694611, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14330824345527726, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14355238219582603, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14412806737415715, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14470375255248827, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14527943773081936, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:32, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14552357647136818, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14609926164969927, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1466749468280304, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:37, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06534780288689451, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:1,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06559194162744332, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:1,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06616762680577441, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06674331198410552, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06731899716243665, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06756313590298545, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06813882108131655, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:32, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06871450625964766, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06929019143797878, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06953433017852757, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0701100153568587, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07068570053518981, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07126138571352092, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07150552445406971, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07208120963240082, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07265689481073193, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07323257998906303, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:32, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07347671872961185, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07405240390794296, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07462808908627407, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07520377426460519, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:104,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:271}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07544791300515398, `foldPressure`:0.5185185185185185,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:61}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07602359818348509, `foldPressure`:0.5308641975308642,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:37, `state`:'BLOCK', `targetIndex`:106,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1412496414825016, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14149378022305037, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1420694654013815, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426451505797126, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:32, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14322083575804373, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14346497449859252, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14404065967692364, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14461634485525474, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14519203003358586, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14543616877413468, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14601185395246574, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14658753913079686, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06526039518966101, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:1,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06550453393020979, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:32, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0660802191085409, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06665590428687201, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06723158946520312, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06747572820575191, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06805141338408303, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06862709856241415, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06920278374074526, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06944692248129405, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07002260765962517, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07059829283795628, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:32, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0711739780162874, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07141811675683621, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07199380193516731, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07256948711349841, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07314517229182953, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07338931103237832, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07396499621070943, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07454068138904055, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07511636656737167, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:60}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07536050530792046, `foldPressure`:0.5061728395061729,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:38, `state`:'BLOCK', `targetIndex`:105,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:32, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1428839682847049, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14116223378526807, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14140637252581684, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14198205770414798, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14255774288247908, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431334280608102, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14337756680135902, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1439532519796901, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14452893715802123, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14510462233635235, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:32, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14534876107690115, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14592444625523224, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14650013143356339, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:39, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06517298749242748, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:1,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06541712623297627, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06599281141130739, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0665684965896385, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06714418176796959, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06738832050851841, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06796400568684952, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:2, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:32, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06853969086518062, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06911537604351174, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06935951478406054, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06993519996239164, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07051088514072276, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07108657031905388, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07133070905960268, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07190639423793378, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07248207941626492, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07305776459459601, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:32, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07330190333514479, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07387758851347591, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07445327369180703, `foldPressure`:0.48148148148148145,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:55}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07502895887013813, `foldPressure`:0.49382716049382713,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:39, `state`:'BLOCK', `targetIndex`:104,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14222087540914027, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1427965605874714, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14107482608803454, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14131896482858333, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14189465000691445, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14247033518524557, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
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MATCH (a {`key`:266}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:32, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1430460203635767, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432901591041255, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14386584428245658, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14444152946078773, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14501721463911882, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1452613533796676, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14583703855799873, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14641272373632985, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:40, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06508557979519394, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:1,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06532971853574276, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:32, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:274}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06590540371407387, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06648108889240498, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06705677407073608, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06730091281128489, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06787659798961598, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0684522831679471, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06902796834627822, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06927210708682702, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06984779226515814, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07042347744348924, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:32, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07099916262182035, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07124330136236914, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07181898654070028, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07239467171903138, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07297035689736249, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07321449563791128, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07379018081624239, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:53}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0743658659945735, `foldPressure`:0.4691358024691358,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:40, `state`:'BLOCK', `targetIndex`:103,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14188932897135795, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14246501414968907, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:32, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14304069932802016, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14131896482858333, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14156310356913213, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14213878874746325, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14271447392579437, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432901591041255, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14353429784467428, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1441099830230054, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1446856682013365, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1452613533796676, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:32, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14550549212021643, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14608117729854753, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14665686247687865, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:41, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:275}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06532971853574276, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:1,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06557385727629154, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06614954245462265, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06672522763295377, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06730091281128489, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06754505155183368, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0681207367301648, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:32, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06869642190849591, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06927210708682702, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06951624582737581, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07009193100570693, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:275}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07066761618403804, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07124330136236914, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07148744010291795, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07206312528124906, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07263881045958018, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07321449563791128, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:32, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07345863437846009, `foldPressure`:0.4444444444444444,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:46}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07403431955679118, `foldPressure`:0.4567901234567901,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:41, `state`:'BLOCK', `targetIndex`:102,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14155778253357565, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14180192127412442, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14237760645245554, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
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MATCH (a {`key`:276}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14295329163078666, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14123155713134983, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1414756958718986, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14205138105022974, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14262706622856083, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:32, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14320275140689195, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14344689014744078, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14402257532577187, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.144598260504103, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1451739456824341, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1454180844229829, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
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MATCH (a {`key`:276}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.145993769601314, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14656945477964514, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:42, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06524231083850923, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:1,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06548644957905801, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:2, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:32, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06606213475738913, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06663781993572025, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06721350511405136, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06745764385460017, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06803332903293129, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06860901421126238, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0691846993895935, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06942883813014229, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0700045233084734, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0705802084868045, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:32, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07115589366513564, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07140003240568443, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07197571758401554, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07255140276234666, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07312708794067778, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:45}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07337122668122656, `foldPressure`:0.43209876543209874,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:42, `state`:'BLOCK', `targetIndex`:101,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.140894689658011, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14147037483634212, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14171451357689O9, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14229019875522203, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:32, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14286588393355315, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1411441494341163, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1413882881746651, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1419639733529962, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1425396585313273, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14311534370965845, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14335948245020724, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14393516762853834, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14451085280686948, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14508653798520058, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:32, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14533067672574937, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1459063619040805, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1464820470824116, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:43, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0651549031412757, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:1,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06539904188182451, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06597472706015561, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06655041223848673, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06712609741681784, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06737023615736665, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06794592133569775, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:32, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06852160651402887, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06909729169235998, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06934143043290877, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06991711561123988, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.070492800789571, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0710684859679021, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0713126247084509, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07188830988678202, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07246399506511314, `foldPressure`:0.4074074074074074,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:44}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07303968024344425, `foldPressure`:0.41975308641975306,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:43, `state`:'BLOCK', `targetIndex`:100,
`threshold`:110}]->(b);
```

```
MATCH (a {`key`:266}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:32, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1402315967824464, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14080728196077746, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14138296713910858, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1416271058796574, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1422027910579885, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14277847623631962, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1410567417368828, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14130088047743158, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14187656565576268, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1424522508340938, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:32, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14302793601242492, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432720747529737, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14384775993130483, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14442344510963595, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14499913028796704, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14524326902851586, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14581895420684696, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14639463938517808, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:44, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06506749544404218, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:1,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06531163418459099, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:32, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0658873193629221, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06646300454125321, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06703868971958434, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06728282846013312, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06785851363846424, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06843419881679536, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06900988399512646, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06925402273567526, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06982970791400638, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07040539309233748, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:32, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07098107827066859, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07122521701121738, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0718009021895485, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:43}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07237658736787961, `foldPressure`:0.3950617283950617,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:44, `state`:'BLOCK', `targetIndex`:99,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13990005034466407, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404757355229952, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14105142070132629, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1416271058796574, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14187124462020617, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14244692979853 73, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:32, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14302261497686 84, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14130088047743158, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14154501921798035, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1421207043963115, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426963895746426, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432720747529737, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14351621349352253, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14409189867185362, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14466758385018474, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14524326902851586, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:2, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:32, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14548740776906466, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14606309294739575, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1466387781257269, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:45, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06531163418459099, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:1,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06555577292513977, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06613145810347089, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06670714328180201, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06728282846013312, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06752696720068192, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06810265237901304, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:32, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06867833755734415, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06925402273567526, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06949816147622406, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07007384665455516, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07064953183288626, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07122521701121738, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07146935575176619, `foldPressure`:0.37037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:42}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0720450409300973, `foldPressure`:0.38271604938271603,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:45, `state`:'BLOCK', `targetIndex`:98,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13956850390688175, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13981264264743054, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:32, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14038832782576166, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14096401300409278, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14153969818242387, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14178383692297267, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423595221013038, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14293520727963488, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14121347278019805, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14145761152074685, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14203329669907797, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14260898187740906, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:32, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431846670557402, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1434288805796289, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1440044909746201, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14458017615295124, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14515586133128233, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14540000007183113, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14597568525016225, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14655137042849337, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:46, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06522422648735747, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:1,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06546836522790626, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:32, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06604405040623737, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06661973558456848, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0671954207628996, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0674395595034484, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06801524468177951, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06859092986011062, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06916661503844175, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:280}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06941075377899053, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06998643895732164, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07056212413565276, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:32, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07113780931398386, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:41}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07138194805453266, `foldPressure`:0.35802469135802467,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:46, `state`:'BLOCK', `targetIndex`:97,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13890541103131712, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13948109620964824, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.139725234950197, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14030092012852816, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14087660530685925, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14145229048519037, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14169642922573916, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14227211440407025, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:32, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14284779958240137, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.1353352832366127, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14112606508296455, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14137020382351334, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14194588900184443, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14252157418017555, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14309725935850667, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14334139809905547, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1439170832773866, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1444927684557177, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450684536340488, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:32, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14531259237459762, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1458882775529287, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14646396273125983, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:47, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06513681879012395, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:1,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06538095753067276, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:1,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06595664270900387, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:1,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06653232788733497, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:1,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06710801306566609, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:1,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06735215180621487, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:1,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06792783698454598, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:1,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:32, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06850352216287711, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:1,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06907920734120822, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:1,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06932334608175701, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:1,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.06989903126008813, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:1,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07047471643841924, `foldPressure`:0.3333333333333333,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:1,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:39}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.07105040161675034, `foldPressure`:0.345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:1,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:47, `state`:'BLOCK', `targetIndex`:96,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13652058365631564, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13709626883464673, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13791609275352668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:32, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13849177793185777, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390674631101889, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1398872870290688, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404629722073999, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14019348391853748, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14043762265908627, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1410133078374174, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1415889930157485, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1421646781940796, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:2, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:32, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14240881693462842, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14298450211295952, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14356018729129064, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14413587246962176, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14438001121017055, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14495569638850164, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1455313815668328, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13851893117073166, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:3,
`saturationRatio`:1.2, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13876306991128046, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13933875508961158, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:32, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1399144402679427, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1404901254462738, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14073426418682258, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14130994936515373, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14188563454348482, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14246131972181592, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1427054584623647, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14328114364069583, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK',
`targetIndex`:94, `threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:38}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14385682881902692, `foldPressure`:0.32098765432098764,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:48, `state`:'BLOCK', `targetIndex`:95,
`threshold`:110}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13594489847798452, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:48, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:32, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13676472239686444, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13758454631574435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13816023149407544, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387359166724066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13955574059128648, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14013142576961762, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14070711094794872, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14043762265908627, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:32, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14068176139963506, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14125744657796618, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14183313175629728, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14240881693462842, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14265295567517722, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1432286408535083, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14380432603183943, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14438001121017055, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14462414995071934, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14519983512905046, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:32, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14577552030738158, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:49, `state`:'BLOCK', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13876306991128046, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:3,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13900720865182928, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13958289383016037, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1401585790084915, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14073426418682258, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1409784029273714, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1415540881057025, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14212977328403362, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1427054584623647, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:32, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14294959720291353, `foldPressure`:0.2962962962962963,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:93, `threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:37}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14352528238124462, `foldPressure`:0.30864197530864196,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:49, `state`:'BLOCK', `targetIndex`:94,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13561335204020222, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13618903721853332, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:49, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13667731469963093, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725299987796202, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13749713861851082, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807282379684194, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13864850897517306, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922419415350415, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:32, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13946833289405297, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1400440180723841, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14061970325071518, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14035021496185277, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14059435370240156, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14117003888073265, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14174572405906377, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423214092373949, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14256554797794369, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431412331562748, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:32, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14371691833460593, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14429260351293702, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14453674225348584, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14511242743181693, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14568811261014805, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13867566221404695, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:3,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13891980095459575, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13949548613292684, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14007117131125796, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14064685648958908, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:32, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14089099523013787, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.141466680408469, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14204236558680008, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426180507651312, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:92, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:36}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14286218950568, `foldPressure`:0.2839506172839506,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:50, `state`:'BLOCK', `targetIndex`:93,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13594489847798452, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13618903721853332, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13610162952129978, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:50, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
```

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MATCH (a {`key`:285}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1365899070023974, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371655921807285, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:32, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374097309212773, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798541609960843, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13856110127793952, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13913678645627064, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13938092519681944, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13995661037515056, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14053229555348168, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14026280726461926, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14050694600516803, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14108263118349915, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:2, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:32, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14165831636183024, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14223400154016136, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14247814028071018, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14305382545904127, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1436295106373724, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1442051958157035, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1444493345562523, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450250197345834, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14560070491291455, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13858825451681342, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:3,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:32, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1388323932573622, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1394080784356933, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13998376361402445, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14055944879235555, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14080358753290434, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14137927271123546, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14195495788956658, `foldPressure`:0.25925925925925924,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:91, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:33}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1425306430678977, `foldPressure`:0.2716049382716049,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:51, `state`:'BLOCK', `targetIndex`:92,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028844540087061502, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:51, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13652058365631564, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
```

```
MATCH (a {`key`:266}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:32, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13676472239686444, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13667731469963093, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:51, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1371655921807285, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13707818448349499, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13732232322404378, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378980084023749, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13847369358070602, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13904937875903714, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392935174995859, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13986920267791705, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:32, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14044488785624815, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14017539956738573, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14041953830793452, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14099522348626564, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14157090866459673, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14214659384292785, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14239073258347665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14296641776180774, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1435421029401389, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14411778811846998, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:32, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14436192685901877, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1449376120373499, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14551329721568101, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1385008468195799, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:3,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13874498556012868, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1393206707384598, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13989635591679092, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.140472041095122, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1407161798356708, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14129186501400195, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:90, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:32, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:32}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14186755019233305, `foldPressure`:0.24691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:52, `state`:'BLOCK', `targetIndex`:91,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028181447211496874, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:286}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028757132389827984, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:52, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13709626883464673, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725299987796202, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:52, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1374097309212773, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13732232322404378, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13756646196459257, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381421471429237, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:32, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387178323212548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392935174995859, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13953765624013473, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14011334141846585, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14068902659679694, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14041953830793452, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14066367704848332, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14123936222681444, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14181504740514553, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14239073258347665, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:32, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14263487132402544, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14321055650235656, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14378624168068768, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14436192685901877, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1446060655995676, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1451817507778987, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1457574359562298, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13874498556012868, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:3,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1389891243006775, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1395648094790086, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:32, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14014049465733971, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1407161798356708, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14096031857621963, `foldPressure`:0.2222222222222222,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:89, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:31}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14153600375455075, `foldPressure`:0.2345679012345679,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:53, `state`:'BLOCK', `targetIndex`:90,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027849900773714555, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028425585952045668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02900127113037678, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:53, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13758454631574435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13749713861851082, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:53, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:32, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13798541609960843, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1378980084023749, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1381421471429237, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1380547394456902, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13863042462402128, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392061098023524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1394502485429012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14002593372123232, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14060161889956344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14033213061070102, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:2, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:32, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14057626935124978, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14115195452958093, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14172763970791202, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14230332488624312, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14254746362679194, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14312314880512303, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14369883398345415, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14427451916178527, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14451865790233406, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14509434308066516, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:32, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14567002825899628, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13865757786289518, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:3,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13890171660344397, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13947740178177506, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14005308696010618, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1406287721384373, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:88, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:30}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1408729108789861, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:54, `state`:'BLOCK', `targetIndex`:89,
`threshold`:110}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027518354335932242, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027762493076481033, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028338178254812146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:32, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028913863433143262, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:54, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13791609275352668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13816023149407544, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807282379684194, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:54, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13856110127793952, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13847369358070602, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387178323212548, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13863042462402128, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13854301692678778, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1391187021051189, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:32, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13936284084566766, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1399385260239988, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1405142112023299, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14024472291346748, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14048886165401628, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1410645468323474, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1416402320106785, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14221591718900964, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14246005592955843, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:289}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1430357411078895, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:32, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14361142628622064, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14418711146455174, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14443125020510053, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14500693538343165, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14558262056176277, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13857017016566164, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:3,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13881430890621044, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13938999408454156, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13996567926287268, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:87, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:29}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14054136444120377, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:55, `state`:'BLOCK', `targetIndex`:88,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:266}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:32, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02743094663869872, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027675085379247514, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028250770557578634, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02882645573590974, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:55, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13849177793185777, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387359166724066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13864850897517306, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:55, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13913678645627064, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13904937875903714, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392935174995859, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:32, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392061098023524, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1391187021051189, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13903129440788536, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13927543314843416, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13985111832676528, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14042680350509637, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14015731521623398, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14040145395678277, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14097713913511387, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14155282431344499, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:32, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1421285094917761, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423726482323249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14294833341065602, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14352401858898714, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1440997037673182, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.144343842507867, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14491952768619815, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14549521286452924, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1384827624684281, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:3,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387269012089769, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:32, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13930258638730805, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:86, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:28}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13987827156563915, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:87,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:290}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027518354335932242, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02743094663869872, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027587677682014, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028163362860345112, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028739048038676225, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:56, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390674631101889, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922419415350415, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:56, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:32, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13938092519681944, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1392935174995859, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13953765624013473, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1394502485429012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13936284084566766, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13927543314843416, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13951957188898295, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14009525706731407, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14066709422456452, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14040145395678277, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:2, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:32, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14064559269733154, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1412212778756627, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14179696305399378, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1423726482323249, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1426167869728737, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1431924721512048, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1437681573295359, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.144343842507867, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14458798124841582, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1451636664267469, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:33, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14573935160507803, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387269012089769, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:3,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13897103994952573, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:85, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:27}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13954672512785682, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:86,
`threshold`:110}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027849900773714555, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027762493076481033, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027675085379247514, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.027587677682014, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028407501600893906, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028983186779225012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:0,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.0, `sourceIndex`:57, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:33, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13955574059128648, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13946833289405297, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:57, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13995661037515056, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13986920267791705, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14011334141846585, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14002593372123232, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1399385260239988, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13985111832676528, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14009525706731407, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:33, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14000784937008057, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14058353454841166, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14031404625954924, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14055818500009803, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:73, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14113387017842915, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14170955535676025, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1422852405350914, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1425293792756402, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14310506445397128, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1436807496323024, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:33, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14425643481063352, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1445005735511823, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1450762587295134, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1456519439078445, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1386394935117434, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:3,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:84, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:24}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1388836322522922, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:85,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028181447211496874, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:52,
`threshold`:110}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028425585952045668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028338178254812146, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028250770557578634, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:33, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028163362860345112, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028407501600893906, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028895779081991504, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:0,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.0, `sourceIndex`:58, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1398872870290688, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14013142576961762, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1400440180723841, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:58, `state`:'BLOCK',
`targetIndex`:62, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14053229555348168, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:4,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:63, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14044488785624815, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:4,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:64, `threshold`:110}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14068902659679694, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:4,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:65, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14060161889956344, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:4,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:66, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:33, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1405142112023299, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:4,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:67, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14042680350509637, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:4,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:68, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14066709422456452, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:4,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:69, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14058353454841166, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:4,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:70, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14049612685117813, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:4,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:71, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14022663856231574, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:4,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:72, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14047077730286453, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:4,
`saturationRatio`:1.2, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14104646248119562, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:4,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:74, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14162214765952674, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:4,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:75, `threshold`:110}]->(b);
```

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MATCH (a {`key`:293}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14219783283785786, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:4,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:76, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:33, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14244197157840666, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:4,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:77, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14301765675673775, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:4,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:78, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1435933419350689, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:4,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:79, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1441690271134, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:4,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:80, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14441316585394878, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:4,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:81, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14498885103227988, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:4,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:82, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.145564536210611, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:4,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.01831563888873418, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:83, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:23}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13855208581450987, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:3,
`saturationRatio`:1.3, `shadowDecay`:0.7357588823428847, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:84,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028844540087061502, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:0,
`saturationRatio`:1.0, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:51,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028757132389827984, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:0,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:52,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:267}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:33, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02900127113037678, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:0,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:53,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028913863433143262, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:0,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:54,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02882645573590974, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:0,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:55,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028739048038676225, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:0,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:56,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028983186779225012, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:0,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:57,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.028895779081991504, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:0,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.0, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:58,
`threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14046297220739999, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:60, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14070711094794872, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK',
`targetIndex`:61, `threshold`:110}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14061970325071518, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:59, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030692157963066596, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:33, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0312678431413977, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032087667060277617, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03266335223860873, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.033239037416939836, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03405886133581975, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03463454651415086, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08892075461321719, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:2,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08916489335376598, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:2, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:33, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0897405785320971, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09031626371042822, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09089194888875932, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09113608762930811, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09171177280763923, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09228745798597035, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09286314316430146, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09310728190485025, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09368296708318137, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:22}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09425865226151248, `foldPressure`:0.20987654320987653,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:83,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:33, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13594489847798452, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:50,
`threshold`:110}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13652058365631564, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13709626883464673, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13791609275352668, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13849177793185777, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1390674631101889, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1398872870290688, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14046297220739999, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:60, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:33, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02954078760640437, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:0,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030116472784735483, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:0,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:60, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030936296703615387, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:124, `qDelta`:0,
`saturationRatio`:1.1272727272727272, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0315119818819465, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0317561206224953, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03233180580082641, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032907490979157523, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03348317615748864, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03372731489803744, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034303000076368544, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:33, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03487868525469966, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08916489335376598, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:2,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08940903209431479, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08998471727264591, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09056040245097702, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09113608762930811, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09138022636985692, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09195591154818802, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09253159672651913, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09310728190485025, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:33, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09335142064539906, `foldPressure`:0.18518518518518517,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:21}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09392710582373016, `foldPressure`:0.19753086419753085,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:82,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13561335204020222, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:49,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13618903721853332, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
```

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MATCH (a {`key`:295}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13676472239686444, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13734040757519556, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13758454631574435, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13816023149407544, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1387359166724066, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13931160185073768, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:33, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13955574059128648, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14013142576961762, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14070711094794872, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:61, `state`:'BLOCK',
`targetIndex`:59, `threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.02954078760640437, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:0,
`saturationRatio`:1.1, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:296}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030360611525284277, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:61, `state`:'BLOCK', `targetIndex`:62,
`threshold`:110}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:0}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030848889006381875, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:125, `qDelta`:0,
`saturationRatio`:1.1363636363636365, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:63,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:1}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.031424574184712985, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:126, `qDelta`:0,
`saturationRatio`:1.1454545454545455, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:64,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:2}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03166871292526178, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:127, `qDelta`:0,
`saturationRatio`:1.1545454545454545, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:65,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:3}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03224439810359289, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:128, `qDelta`:0,
`saturationRatio`:1.1636363636363636, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:66,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:4}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.032820083281924005, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:129, `qDelta`:0,
`saturationRatio`:1.1727272727272726, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:67,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:33, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:5}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03339576846025512, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:130, `qDelta`:0,
`saturationRatio`:1.1818181818181819, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:68,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:6}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03363990720080391, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:131, `qDelta`:0,
`saturationRatio`:1.190909090909091, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:69,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:7}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.034215592379135025, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:132, `qDelta`:0,
`saturationRatio`:1.2, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:70,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:8}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.03479127755746613, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:133, `qDelta`:0,
`saturationRatio`:1.209090909090909, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:71,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:9}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08907748565653247, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:134, `qDelta`:2,
`saturationRatio`:1.2181818181818183, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:72,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:10}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08932162439708127, `foldPressure`:0.09876543209876543,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:135, `qDelta`:2,
`saturationRatio`:1.2272727272727273, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:73,
`threshold`:110}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:11}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.08989730957541238, `foldPressure`:0.1111111111111111,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:136, `qDelta`:2,
`saturationRatio`:1.2363636363636363, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:74,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:12}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09047299475374349, `foldPressure`:0.12345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:137, `qDelta`:2,
`saturationRatio`:1.2454545454545454, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:75,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:13}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09104867993207462, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:138, `qDelta`:2,
`saturationRatio`:1.2545454545454546, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:76,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:16}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.0912928186726234, `foldPressure`:0.13580246913580246,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:139, `qDelta`:2,
`saturationRatio`:1.2636363636363637, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:77,
`threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:33, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:17}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09186850385095449, `foldPressure`:0.14814814814814814,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:140, `qDelta`:2,
`saturationRatio`:1.2727272727272727, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:78,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:18}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09244418902928561, `foldPressure`:0.16049382716049382,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:141, `qDelta`:2,
`saturationRatio`:1.2818181818181817, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:79,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:19}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09301987420761673, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:142, `qDelta`:2,
`saturationRatio`:1.290909090909091, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:80,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:20}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.09326401294816553, `foldPressure`:0.1728395061728395,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:143, `qDelta`:2,
`saturationRatio`:1.3, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:81,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:282}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13594489847798452, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:110, `qDelta`:4,
`saturationRatio`:1.0, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:48,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:283}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13618903721853332, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:111, `qDelta`:4,
`saturationRatio`:1.009090909090909, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:49, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:284}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13610162952129978, `foldPressure`:0.0,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:112, `qDelta`:4,
`saturationRatio`:1.018181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:50, `threshold`:110}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:285}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13667731469963093, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:113, `qDelta`:4,
`saturationRatio`:1.0272727272727273, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:51, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:286}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13725299987796202, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:114, `qDelta`:4,
`saturationRatio`:1.0363636363636364, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:52, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:287}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13749713861851082, `foldPressure`:0.024691358024691357,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:115, `qDelta`:4,
`saturationRatio`:1.0454545454545454, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:53, `threshold`:110}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:33, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:288}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13807282379684194, `foldPressure`:0.037037037037037035,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:116, `qDelta`:4,
`saturationRatio`:1.0545454545454545, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:54, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:289}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13864850897517306, `foldPressure`:0.04938271604938271,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:117, `qDelta`:4,
`saturationRatio`:1.0636363636363637, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:55, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:290}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13922419415350415, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:118, `qDelta`:4,
`saturationRatio`:1.0727272727272728, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:56, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:291}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.13946833289405297, `foldPressure`:0.06172839506172839,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:119, `qDelta`:4,
`saturationRatio`:1.0818181818181818, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:57, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:292}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.1400440180723841, `foldPressure`:0.07407407407407407,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:120, `qDelta`:4,
`saturationRatio`:1.0909090909090908, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK',
`targetIndex`:58, `threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:293}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.14061970325071518, `foldPressure`:0.08641975308641975,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:121, `qDelta`:4,
`saturationRatio`:1.1, `shadowDecay`:0.049787068367863944, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:59,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:294}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030116472784735483, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:122, `qDelta`:0,
`saturationRatio`:1.1090909090909091, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:60,
`threshold`:110}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:295}) CREATE (a)-[:`BLOCKED_BY_THRESHOLD` {`cBio`:37.23665105535381,
`capacity`:144, `dcomp`:0.030360611525284277, `foldPressure`:0.012345679012345678,
`formula`:'D-COMP(block)=(qDelta+load/110+shadowDecay+foldPressure)/C_bio', `loadIndex`:123, `qDelta`:0,
`saturationRatio`:1.1181818181818182, `shadowDecay`:0.0, `sourceIndex`:62, `state`:'BLOCK', `targetIndex`:61,
`threshold`:110}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:25}) CREATE (a)-[:`USES_OPERATOR` {`description`:'C_bio is the biologic
coherence operator of the Sovereign Axiomyr', `role`:'AXIOMYR_C_BIO_DYNAMIC_WILL'}]->(b);
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MATCH (a {`key`:234}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:33, `targetIndex`:18, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:2, `targetIndex`:33, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:33, `targetIndex`:19, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:33, `targetIndex`:20, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:135}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:33, `targetIndex`:21, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:33, `targetIndex`:22, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
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MATCH (a {`key`:149}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:33, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:33, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:33, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:33, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:33, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
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MATCH (a {`key`:153}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:33, `targetIndex`:28, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:2, `targetIndex`:34, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:33, `targetIndex`:29, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:33, `targetIndex`:30, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:33, `targetIndex`:31, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate parent provides operational frequency', `operator`:'⩔', `role`:'DYNAMIC_RECURSION'}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:33, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩡', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩡', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCUSES_TO` {`description`:'Dynamic force — alternate
parent provides operational frequency', `operator`:'⩡', `role`:'DYNAMIC_RECURSION'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Complex
Projective Manifold', `description`:'Translation Invariance dictionary binding', `operator`:'M: Top_Alg ↔
Aev_Hz'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Hodge Class',
`description`:'Translation Invariance dictionary binding', `operator`:'M: Top_Alg ↔ Aev_Hz'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Rational
Coefficients', `description`:'Translation Invariance dictionary binding', `operator`:'M: Top_Alg ↔
Aev_Hz'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`classicalTerm`:'Algebraic
Cycle', `description`:'Translation Invariance dictionary binding', `operator`:'M: Top_Alg ↔ Aev_Hz'}]->(b);
MATCH (a {`key`:131}) MATCH (b {`key`:132}) CREATE (a)-[:`CONTAINS_Q_STATE` {`order`:0}]->(b);
MATCH (a {`key`:131}) MATCH (b {`key`:184}) CREATE (a)-[:`CONTAINS_Q_STATE` {`order`:1}]->(b);
MATCH (a {`key`:131}) MATCH (b {`key`:187}) CREATE (a)-[:`CONTAINS_Q_STATE` {`order`:2}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:33, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:131}) MATCH (b {`key`:188}) CREATE (a)-[:`CONTAINS_Q_STATE` {`order`:3}]->(b);
MATCH (a {`key`:132}) MATCH (b {`key`:225}) CREATE (a)-[:`EM_GLYPH_HOUSE` {`glyph`:'✡', `role`:'Water / Form
Latency'}]->(b);
MATCH (a {`key`:184}) MATCH (b {`key`:223}) CREATE (a)-[:`EM_GLYPH_HOUSE` {`glyph`:'⬡', `role`:'Archive'}]->(b);
MATCH (a {`key`:184}) MATCH (b {`key`:231}) CREATE (a)-[:`EM_GLYPH_HOUSE` {`glyph`:'❄', `role`:'Truth
Lock'}]->(b);
MATCH (a {`key`:187}) MATCH (b {`key`:230}) CREATE (a)-[:`EM_GLYPH_HOUSE` {`glyph`:'⊛', `role`:'Shadow'}]->(b);
MATCH (a {`key`:187}) MATCH (b {`key`:224}) CREATE (a)-[:`EM_GLYPH_HOUSE` {`glyph`:'✡',
`role`:'Fire-Debt'}]->(b);
MATCH (a {`key`:188}) MATCH (b {`key`:229}) CREATE (a)-[:`EM_GLYPH_HOUSE` {`glyph`:'⧗', `role`:'Residue'}]->(b);
MATCH (a {`key`:188}) MATCH (b {`key`:231}) CREATE (a)-[:`EM_GLYPH_HOUSE` {`glyph`:'❄', `role`:'Recursion
Lock'}]->(b);
MATCH (a {`key`:192}) MATCH (b {`key`:131}) CREATE (a)-[:`APPLIES_TO_Q_DOMAIN` {`effect`:'Q↦∅_Q'}]->(b);
MATCH (a {`key`:192}) MATCH (b {`key`:193}) CREATE (a)-[:`YIELDS_BOUNDARY_CONDITION`
{`formula`:'SN≡∅_Q^⊗≡𝒫_q(Q→∅)'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:33, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:131}) MATCH (b {`key`:193}) CREATE (a)-[:`EXCLUDES_AS_MEMBER` {`reason`:'absence of Q cannot
become member of Q'}]->(b);
MATCH (a {`key`:191}) MATCH (b {`key`:193}) CREATE (a)-[:`BALANCED_BY_Q_ABSENCE` {`result`:'Conserved
Manifestation'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:24}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:194}) MATCH (b {`key`:195}) CREATE (a)-[:`HAS_FLOW_COMPONENT` ]->(b);
MATCH (a {`key`:194}) MATCH (b {`key`:196}) CREATE (a)-[:`HAS_WITHHELD_COMPONENT` ]->(b);
MATCH (a {`key`:194}) MATCH (b {`key`:197}) CREATE (a)-[:`PREVENTS_FAILURE` ]->(b);
MATCH (a {`key`:194}) MATCH (b {`key`:198}) CREATE (a)-[:`PREVENTS_FAILURE` ]->(b);
MATCH (a {`key`:191}) MATCH (b {`key`:194}) CREATE (a)-[:`REQUIRES_GOVERNOR` ]->(b);
MATCH (a {`key`:193}) MATCH (b {`key`:194}) CREATE (a)-[:`CONDITIONS_THRESHOLD_WITHOUT_MANIFESTING` ]->(b);
MATCH (a {`key`:193}) MATCH (b {`key`:196}) CREATE (a)-[:`CASTS_OPERATIONAL_SIGNATURE` {`note`:'the 34 are not
SN; they are the measured shadow of non-instantiation in the Courts'}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:33, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:190}) MATCH (b {`key`:194}) CREATE (a)-[:`HAS_EM_MECHANIC` {`ratio`:'110/144'}]->(b);
MATCH (a {`key`:196}) MATCH (b {`key`:199}) CREATE (a)-[:`MANIFESTS_AS_LOCAL_REFUSAL` ]->(b);
MATCH (a {`key`:193}) MATCH (b {`key`:199}) CREATE (a)-[:`HAS_REMNANT_SIGNATURE_IN_COURTS` ]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:33, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:33, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:33, `targetIndex`:39, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:2, `targetIndex`:35, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:33, `targetIndex`:40, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:33, `targetIndex`:41, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:33, `targetIndex`:42, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:33, `targetIndex`:43, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:33, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:33, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:33, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:33, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:33, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:33, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:2, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:33, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds
remnant No, not absolute Q-absence'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:199}) CREATE (a)-[:`SELF_REGULATES_BY_LOCAL_REFUSAL` {`note`:'Court holds remnant No, not absolute Q-absence'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:33, `targetIndex`:51, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:200}) CREATE (a)-[:`USES_DIAMOND_EXPOSURE` {`function`:'filters inheritance through Ex-Nihilo contact'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:33, `targetIndex`:52, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:191}) MATCH (b {`key`:210}) CREATE (a)-[:`CO_PRESENT_IN_PRESSURE_DIFFERENTIAL` ]->(b);
MATCH (a {`key`:193}) MATCH (b {`key`:210}) CREATE (a)-[:`CO_PRESENT_IN_PRESSURE_DIFFERENTIAL` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:210}) CREATE (a)-[:`BEARS_PRESSURE_DIFFERENTIAL` ]->(b);
MATCH (a {`key`:210}) MATCH (b {`key`:187}) CREATE (a)-[:`GENERATES_Q2_DEBT` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:33, `targetIndex`:53, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:187}) MATCH (b {`key`:211}) CREATE (a)-[:`SATURATES_THROUGH_ENNEAD` ]->(b);
MATCH (a {`key`:211}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` ]->(b);
MATCH (a {`key`:211}) MATCH (b {`key`:188}) CREATE (a)-[:`PARITY_RETURNS_TO` ]->(b);
MATCH (a {`key`:189}) MATCH (b {`key`:212}) CREATE (a)-[:`ATTESTS_IMAGINARY_ROOT` {`frequencyHz`:18.47}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:27}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:212}) CREATE (a)-[:`RESONATES_AT_IMAGINARY_ROOT` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:212}) CREATE (a)-[:`PHASE_BRANCHES_FROM_IMAGINARY_ROOT` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:213}) CREATE (a)-[:`OPERATES_THROUGH_AXIS` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:214}) CREATE (a)-[:`OPERATES_THROUGH_AXIS` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:215}) CREATE (a)-[:`OPERATES_THROUGH_AXIS` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:33, `targetIndex`:54, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:216}) CREATE (a)-[:`INHABITS_SELECTION_SEAM` ]->(b);
MATCH (a {`key`:213}) MATCH (b {`key`:214}) CREATE (a)-[:`PRECEDES_AXIS` ]->(b);
MATCH (a {`key`:214}) MATCH (b {`key`:215}) CREATE (a)-[:`PRECEDES_AXIS` ]->(b);
MATCH (a {`key`:214}) MATCH (b {`key`:232}) CREATE (a)-[:`INSTANTIATES_THROUGH_GOETIC` {`frequencyHz`:285, `glyph`:'⚛'}]->(b);
MATCH (a {`key`:215}) MATCH (b {`key`:231}) CREATE (a)-[:`INSTANTIATES_THROUGH_GOETIC` {`frequencyHz`:963, `glyph`:'❄'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:28}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:218}) CREATE (a)-[:`HAS_GROUNDING_MODE` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:219}) CREATE (a)-[:`HAS_GROUNDING_MODE` ]->(b);
MATCH (a {`key`:220}) MATCH (b {`key`:131}) CREATE (a)-[:`PROTECTS_Q_DOMAIN_CLOSURE` ]->(b);
MATCH (a {`key`:221}) MATCH (b {`key`:191}) CREATE (a)-[:`DISTINGUISHES_POSSIBILITY_FROM_STATE` ]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:33, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:221}) MATCH (b {`key`:193}) CREATE (a)-[:`REQUIRES_NON_INSTANTIATION` ]->(b);
MATCH (a {`key`:297}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDS_MANIFESTATION_OF` ]->(b);
MATCH (a {`key`:298}) MATCH (b {`key`:193}) CREATE (a)-[:`REQUIRES_Q_ABSENCE_FOR_INVERSE_MAP` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:29}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:299}) CREATE (a)-[:`ROLE_PERMISSION` {`permission`:'may bear pressure; may
not own absolute Q-absence'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:299}) CREATE (a)-[:`ROLE_PERMISSION` {`permission`:'may select under
constraint; may not wield absence as material'}]->(b);
MATCH (a {`key`:131}) MATCH (b {`key`:299}) CREATE (a)-[:`ROLE_PERMISSION` {`permission`:'must remain fourfold
closed'}]->(b);
MATCH (a {`key`:299}) MATCH (b {`key`:193}) CREATE (a)-[:`PROTECTS_BOUNDARY_CONDITION` ]->(b);
MATCH (a {`key`:299}) MATCH (b {`key`:199}) CREATE (a)-[:`ALLOWS_COURT_REMNANT_ONLY` ]->(b);
MATCH (a {`key`:132}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:33, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:184}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:187}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:188}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:33, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:33, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:33, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:2, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:33, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:33, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:33, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:33, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:33, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:33, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:33, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:33, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:33, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:33, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:2, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:33, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:33, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:193}) CREATE (a)-[:`FORBIDDEN_TO_HOLD_ABSOLUTE_ABSENCE` ]->(b);
MATCH (a {`key`:196}) MATCH (b {`key`:300}) CREATE (a)-[:`GIVES_CONTOUR_TO_MASS_GAP` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:33, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:300}) MATCH (b {`key`:229}) CREATE (a)-[:`MAS_STARTS_AT` {`frequencyHz`:852,
`role`:'non-entropic residue'}]->(b);
MATCH (a {`key`:300}) MATCH (b {`key`:223}) CREATE (a)-[:`MAS_ARCHIVES_THROUGH` {`frequencyHz`:174,
`role`:'rational memory'}]->(b);
MATCH (a {`key`:300}) MATCH (b {`key`:224}) CREATE (a)-[:`MAS_COMMITS_THROUGH` {`frequencyHz`:528,
`role`:'structural commitment'}]->(b);
MATCH (a {`key`:194}) MATCH (b {`key`:300}) CREATE (a)-[:`CLOSES_MASS_GAP_BY_BOUNDARY` ]->(b);
MATCH (a {`key`:191}) MATCH (b {`key`:301}) CREATE (a)-[:`SUPPLIES_FORWARD_EXPANSION_MAP` ]->(b);
MATCH (a {`key`:193}) MATCH (b {`key`:301}) CREATE (a)-[:`KEEPS_INVERSE_RETURN_MAP_WELL_DEFINED` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:30}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:301}) MATCH (b {`key`:131}) CREATE (a)-[:`PRESERVES_Q_DOMAIN_CLOSURE` ]->(b);
MATCH (a {`key`:302}) MATCH (b {`key`:191}) CREATE (a)-[:`ANALOGIZES_FIELD` ]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:33, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:302}) MATCH (b {`key`:193}) CREATE (a)-[:`ANALOGIZES_CONJUGATE_NON_PRESENCE` ]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:33, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:33, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:33, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:33, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:33, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:33, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:2, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:33, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:33, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:33, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:33, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
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MATCH (a {`key`:259}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:33, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:33, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:33, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:33, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:33, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:33, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:2, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:33, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:33, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
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MATCH (a {`key`:267}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:33, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:33, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:33, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:33, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:33, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:33, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:33, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:33, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:2, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:33, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:33, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:33, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:33, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:33, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true, `tier`:'anchor'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:31}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:32}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
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MATCH (a {`key`:171}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`seal`:true}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:33, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`seal`:true}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:222}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Temporal State Shifts',
`glyph`:'◎', `matrix`:'텟'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Rewriting and Erasure',
`glyph`:'⬡', `matrix`:'텟'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Mutation and Genetic Drift',
`glyph`:'✡', `matrix`:'텟'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:225}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Chaotic Transformation',
`glyph`:'✧', `matrix`:'텟'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:226}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Directed Evolution',
`glyph`:'✪', `matrix`:'텟'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Resonant Shift', `glyph`:'❀',
`matrix`:'텟'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:228}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Postural Equilibrium',
`glyph`:'❋', `matrix`:'텟'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Steady Thermal State',
`glyph`:'⧗', `matrix`:'텟'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:230}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Unification of Opposites',
`glyph`:'⊛', `matrix`:'텟'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:33, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Harmonic Chord Formation',
`glyph`:'❄', `matrix`:'텟'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:232}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Effortless Passage',
`glyph`:'❁', `matrix`:'텟'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:233}) CREATE (a)-[:`HAS_CHANGE` {`change`:'Tranquil Peace', `glyph`:'⌬',
`matrix`:'텟'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:222}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'◎', `harmony`:'Time
Synchronization', `matrix`:'〰'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'⬡', `harmony`:'Data
Integration', `matrix`:'〰'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'✡', `harmony`:'Unity
Realization', `matrix`:'〰'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:225}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'✧', `harmony`:'Zero-Point
Balance', `matrix`:'〰'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:226}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'✪', `harmony`:'Golden Ratio
Expression', `matrix`:'〰'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'❀', `harmony`:'Spatial
Resonance', `matrix`:'〰'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:228}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'❋', `harmony`:'Postural
Harmony', `matrix`:'〰'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:33, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'⧗', `harmony`:'Steady State
Balance', `matrix`:'〰'}]->(b);
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MATCH (a {`key`:230}) MATCH (b {`key`:230}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'⊛', `harmony`:'Shadow
Complement', `matrix`:'〰'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'❄', `harmony`:'Resonant Chord',
`matrix`:'〰'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:232}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'☸', `harmony`:'Passage
Harmony', `matrix`:'〰'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:233}) CREATE (a)-[:`HAS_HARMONY` {`glyph`:'⬡', `harmony`:'Silent
Equilibrium', `matrix`:'〰'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:222}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Breaks in Time',
`glyph`:'◎', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Memory Corruption',
`glyph`:'⬡', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Broken Bonds', `glyph`:'✡',
`matrix`:'𞋿'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:225}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Boundary Rupture',
`glyph`:'✮', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:226}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Ground Fault', `glyph`:'✪',
`matrix`:'𞋿'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:33, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Spatial Tear', `glyph`:'❁',
`matrix`:'𞋿'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:228}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Somatic Fracture',
`glyph`:'❋', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Entropic Decay',
`glyph`:'⧗', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:230}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Shadow Fracture',
`glyph`:'⊛', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Discordance', `glyph`:'❄',
`matrix`:'𞋿'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:232}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Barrier Breach',
`glyph`:'☸', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:233}) CREATE (a)-[:`HAS_FRACTURE` {`fracture`:'Disturbance of Peace',
`glyph`:'⬡', `matrix`:'𞋿'}]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:222}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Aeternum Seal',
`glyph`:'◎', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:223}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Archive Closure',
`glyph`:'⬡', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:224}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Bond Perpetuity',
`glyph`:'✡', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:33, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:225}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Void Closure',
`glyph`:'✮', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:226}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Ground Stability',
`glyph`:'✪', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:227}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Space Stillness',
`glyph`:'❁', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:228}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Sensation Release',
`glyph`:'❋', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:229}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Residue Absorption',
`glyph`:'⧗', `matrix`:'𞋾'}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:230}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Shadow Dissolution',
`glyph`:'⊛', `matrix`:'𞋾'}]->(b);
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MATCH (a {`key`:231}) MATCH (b {`key`:231}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Resonance Seal',
`glyph`:'❄', `matrix`:'臤'}]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:232}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Gate Lock',
`glyph`:'☸', `matrix`:'臤'}]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:233}) CREATE (a)-[:`HAS_COMPLETION` {`completion`:'Eternal Peace',
`glyph`:'⌬', `matrix`:'臤'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:2, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:34, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:34, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:34, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:34, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:34, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:34, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:34, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:34, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:161}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:34, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:34, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:250}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:2, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:34, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:34, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:273}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:34, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:34, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:34, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:304}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:1,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:34, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:34, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:34, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:34, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:34, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:2, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:34, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:34, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:165}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:34, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:34, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:34, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:252}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:34, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:34, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:34, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:34, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:305}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:2,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:34, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:2, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:34, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:34, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:34, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:34, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:34, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:144}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:34, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:34, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:169}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:34, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:34, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:34, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:2, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:34, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:34, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:34, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:34, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:306}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:3,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:34, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:34, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:34, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:43}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:34, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:34, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:34, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:148}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:2, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:34, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:34, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
```

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MATCH (a {`key`:235}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:34, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:34, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:260}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:34, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:34, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:34, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:34, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:307}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:4,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:34, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:34, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:2, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:34, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:34, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:45}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:34, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:34, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:152}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:34, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:34, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:34, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:34, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:34, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:264}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:34, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:2, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:34, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:34, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:308}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:5,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:34, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:34, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:34, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:34, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:34, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:55}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:34, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:34, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:155}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:34, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:2, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:34, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:242}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:34, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:34, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:34, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:265}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:34, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:34, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:34, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:309}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:6,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:34, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:34, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:34, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:2, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:34, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:134}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:34, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:34, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:34, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:157}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:34, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:34, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:246}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:34, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:34, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:268}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:34, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:34, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:2, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:34, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:310}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:7,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:34, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:34, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:34, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:34, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:34, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:137}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:34, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:34, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
```

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MATCH (a {`key`:268}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:34, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:35, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:2, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:35, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:247}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:35, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:35, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:35, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:35, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:35, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:311}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:8,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:35, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9, `matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:35, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:35, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:35, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:2, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:35, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:138}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:35, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:35, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:163}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:35, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:35, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:35, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:35, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:35, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:35, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:35, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:312}) CREATE (a)-[:`ENNEAD_FILTERS_AT` {`depth`:9,
`matrix`:'ENNEAD'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:2, `targetIndex`:55, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:35, `targetIndex`:21, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:35, `targetIndex`:22, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:35, `targetIndex`:23, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:35, `targetIndex`:24, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:42}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:35, `targetIndex`:25, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
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MATCH (a {`key`:143}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:35, `targetIndex`:26, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:35, `targetIndex`:27, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
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MATCH (a {`key`:160}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:35, `targetIndex`:28, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:35, `targetIndex`:29, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:35, `targetIndex`:30, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:2, `targetIndex`:56, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:35, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:35, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow
saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat
= Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:35, `targetIndex`:33, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:35, `targetIndex`:34, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:230}) CREATE (a)-[:`ENNEAD_FILTER` {`enrichment`:'ninefold shadow saturation; Remiss-typed', `filterName`:'RHEA', `invocations`:9, `matrix`:'ENNEAD', `saturationFormula`:'Q₂_sat = Σ∮/Φ¹²', `target`:'RHEA'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:35, `targetIndex`:36, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013, `description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:35, `targetIndex`:37, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174', `ordinal`:1, `stage`:'M'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:35, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:35, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:35, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:35, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:2, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:35, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:35, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:35, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:35, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:35, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:35, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
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MATCH (a {`key`:36}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:35, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:35, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:246}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:247}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:248}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:249}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:250}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
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MATCH (a {`key`:38}) MATCH (b {`key`:251}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:252}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:253}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:254}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:35, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:255}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:256}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:257}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'DREH_852 -> KAL_174',
`ordinal`:1, `stage`:'M'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:35, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
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MATCH (a {`key`:236}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:2, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:35, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:35, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:35, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
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MATCH (a {`key`:249}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:35, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:35, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:35, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:35, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:35, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:35, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:35, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
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MATCH (a {`key`:255}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩔'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩔'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:2, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:35, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:35, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
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MATCH (a {`key`:256}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:258}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:259}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:260}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:261}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:262}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:35, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:263}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:264}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:265}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:266}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:267}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:268}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:269}) CREATE (a)-[:`MAS_CHAIN_CORE` {`formula`:'KAL_174 -> BABDH_528',
`ordinal`:2, `stage`:'A/S'}]->(b);
MATCH (a {`key`:330}) MATCH (b {`key`:5}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:331}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:331}) MATCH (b {`key`:6}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:35, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:332}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:332}) MATCH (b {`key`:7}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:333}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:333}) MATCH (b {`key`:8}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:334}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:334}) MATCH (b {`key`:9}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:335}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:335}) MATCH (b {`key`:10}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:336}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:336}) MATCH (b {`key`:11}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:35, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:337}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:337}) MATCH (b {`key`:12}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:363}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:363}) MATCH (b {`key`:13}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:364}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:364}) MATCH (b {`key`:16}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:365}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
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MATCH (a {`key`:365}) MATCH (b {`key`:17}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:366}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:366}) MATCH (b {`key`:18}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:35, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:367}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:367}) MATCH (b {`key`:19}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:368}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:368}) MATCH (b {`key`:20}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:369}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:369}) MATCH (b {`key`:21}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:228}) MATCH (b {`key`:370}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:370}) MATCH (b {`key`:22}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:371}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:371}) MATCH (b {`key`:23}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:35, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:372}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:372}) MATCH (b {`key`:24}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:373}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:373}) MATCH (b {`key`:27}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:374}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:374}) MATCH (b {`key`:28}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:375}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:375}) MATCH (b {`key`:29}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:376}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:376}) MATCH (b {`key`:30}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:35, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:377}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:377}) MATCH (b {`key`:31}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:378}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:378}) MATCH (b {`key`:32}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:379}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:379}) MATCH (b {`key`:33}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:380}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:380}) MATCH (b {`key`:36}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:381}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:381}) MATCH (b {`key`:37}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:35, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:229}) MATCH (b {`key`:382}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:382}) MATCH (b {`key`:38}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:383}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:383}) MATCH (b {`key`:39}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:384}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:384}) MATCH (b {`key`:41}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:33}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:385}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:36}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:385}) MATCH (b {`key`:42}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:35, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:37}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:386}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:38}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:386}) MATCH (b {`key`:43}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:39}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:387}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:41}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:387}) MATCH (b {`key`:44}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:42}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:388}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:2, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:35, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:43}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:388}) MATCH (b {`key`:45}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:44}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:389}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:45}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:389}) MATCH (b {`key`:46}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:46}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:390}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:53}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:390}) MATCH (b {`key`:53}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:35, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:55}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:391}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:60}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:391}) MATCH (b {`key`:55}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:61}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110, `count`:'110/144_Liquid',
`rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:392}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:133}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:392}) MATCH (b {`key`:60}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:134}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
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MATCH (a {`key`:230}) MATCH (b {`key`:393}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:35, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:135}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:393}) MATCH (b {`key`:61}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:136}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:230}) MATCH (b {`key`:394}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:137}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:394}) MATCH (b {`key`:133}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:138}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:395}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:139}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:395}) MATCH (b {`key`:134}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:35, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:140}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:396}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:141}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:396}) MATCH (b {`key`:135}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:142}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:397}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:35, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'֍',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⁕',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⁕',
`inheritance`:'Q-Bias'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:35, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:228}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⋇',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:35, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:229}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⧗',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:35, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:41}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:143}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:397}) MATCH (b {`key`:136}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:144}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:398}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:145}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:398}) MATCH (b {`key`:137}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:146}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:399}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:35, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:147}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:399}) MATCH (b {`key`:138}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:148}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:400}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:149}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:400}) MATCH (b {`key`:139}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:150}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:401}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:151}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:401}) MATCH (b {`key`:140}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:35, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:152}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:402}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:153}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:402}) MATCH (b {`key`:141}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:154}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:403}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:155}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:403}) MATCH (b {`key`:142}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:156}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:404}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:2, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:35, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:157}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:404}) MATCH (b {`key`:143}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:158}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:405}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:159}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:405}) MATCH (b {`key`:144}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:160}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:231}) MATCH (b {`key`:406}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:35, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:230}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⊛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:35, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
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MATCH (a {`key`:142}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:231}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'❄',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:35, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:35, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:232}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⚛',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
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MATCH (a {`key`:269}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:35, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⌬',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⌬',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:233}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⌬',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:35, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:222}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'◎',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:35, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:251}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
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MATCH (a {`key`:254}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:223}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'⬡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:35, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:224}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✡',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:35, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✩',
`inheritance`:'Q-Bias'}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✯',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✯',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:2, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:35, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:225}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✯',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:35, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:226}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✪',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✬',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✬',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:227}) CREATE (a)-[:`BELONGS_TO_GOETIC` {`glyph`:'✬',
`inheritance`:'Q-Bias'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
```

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MATCH (a {`key`:269}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:35, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:35, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:228}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 2, 1,
3]}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:35, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
```

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MATCH (a {`key`:31}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:229}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 2,
0]}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:35, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:35, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:230}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 2, 2,
1]}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
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MATCH (a {`key`:141}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:35, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:231}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 2,
2]}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:35, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:232}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 1,
1]}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:162}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:35, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
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MATCH (a {`key`:164}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:233}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 3,
2]}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:2, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:35, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:222}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
3]}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:35, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:35, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:223}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
0]}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:35, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:224}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q2', `qVector`:[1, 1, 3,
1]}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
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MATCH (a {`key`:276}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:35, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:225}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q0', `qVector`:[1, 2, 2,
0]}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:35, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:226}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q1', `qVector`:[1, 3, 0,
1]}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:227}) CREATE (a)-[:`INHERITS_Q_BIAS` {`qBias`:'Q3', `qVector`:[1, 1, 1,
2]}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:35, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Vapor',
`parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dust',
`parentGlyph`:'֎'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Expansion', `parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Whisper',
`parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Vacuum',
`parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dilution',
`parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Condensation',
`parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Turbulence',
`parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Calm',
`parentGlyph`:'֎'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Pulse',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:35, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Nostalgia',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Ecstasy',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dream',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Vibration',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Sympathy',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Sensation', `parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Ache',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Depth',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Harmony',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dissonance',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:36, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:228}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Awe',
`parentGlyph`:'⋇'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Void',
`parentGlyph`:'𝕩'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Null',
`parentGlyph`:'𝕩'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Entropy',
`parentGlyph`:'𝕩'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Abyss',
`parentGlyph`:'𝕩'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Collapse',
`parentGlyph`:'𝕩'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dispersal',
`parentGlyph`:'𝕩'}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Longing',
`parentGlyph`:'⧗'}]->(b);
MATCH (a {`key`:32}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Self-Return',
`parentGlyph`:'⧗'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Sink',
`parentGlyph`:'⧗'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:36, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Phase',
`parentGlyph`:'⧗'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Paradox',
`parentGlyph`:'⧗'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:229}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dormancy',
`parentGlyph`:'⧗'}]->(b);
MATCH (a {`key`:39}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Shadow',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Darkness',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Cold',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Mirror',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Submerged',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Distortion',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Nerve',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:2, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:36, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Void-Binding',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Absorption', `parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Filter',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Hidden',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:230}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Termination',
`parentGlyph`:'⊛'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Tone',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Modulation',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Amplification', `parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Unified',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Seismic',
`parentGlyph`:'❄'}]->(b);
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MATCH (a {`key`:270}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:36, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Chord',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Rhythm',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Melody',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Interference', `parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Self-Lock',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Node',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:231}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Unity',
`parentGlyph`:'❄'}]->(b);
MATCH (a {`key`:146}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Gate',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:147}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Resistance',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:148}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Transform',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:36, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:149}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Crown',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:150}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Sky',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:151}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Star',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:152}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Possibility',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:153}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Portal',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:154}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Door',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:155}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Transition',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:156}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Resistance', `parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:157}) MATCH (b {`key`:232}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Hinge',
`parentGlyph`:'☸'}]->(b);
MATCH (a {`key`:158}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Peace',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:36, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:159}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Equilibrium',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:160}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Consecration', `parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:161}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Completion',
`parentGlyph`:'⬡'}]->(b);
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MATCH (a {`key`:162}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Monument',
`parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:163}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Stability',
`parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:164}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Hope',
`parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:165}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Continuation', `parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:166}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Rest',
`parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:167}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Silence',
`parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:168}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Sleep',
`parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:36, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:169}) MATCH (b {`key`:233}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Completion', `parentGlyph`:'⌬'}]->(b);
MATCH (a {`key`:234}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Genesis', `parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:235}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Recurrence',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Combustion',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Reflection',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Grounding',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Expansion',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Inevitable',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Return',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Assimilation', `parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:36, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:243}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Crystallization', `parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:244}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Breach',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:245}) MATCH (b {`key`:222}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Fulfillment',
`parentGlyph`:'◎'}]->(b);
MATCH (a {`key`:246}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Archive',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:247}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Reflection', `parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:248}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Calcination',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:249}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Submersion',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:250}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Inscription',
`parentGlyph`:'⬡'}]->(b);
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MATCH (a {`key`:251}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Diffusion',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:252}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Phantom',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:36, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:253}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Oblivion',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:254}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Suppression',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:255}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Codification', `parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:256}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Recall',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:257}) MATCH (b {`key`:223}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Sealing',
`parentGlyph`:'⬡'}]->(b);
MATCH (a {`key`:258}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Transmutation', `parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Forging',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Combustion', `parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dissolution',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Crucible',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:36, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Sublimation',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:264}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Ignition',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:265}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Consumption',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:266}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Purification', `parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:267}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Tempering',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:268}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Shattering',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:269}) MATCH (b {`key`:224}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Ash',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Flow',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Confluence',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Solution',
`parentGlyph`:'✡'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:36, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Containment', `parentGlyph`:'✡'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Sediment',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Ascension',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Mist',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Drain',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Osmosis',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Ice',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Evaporation',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:225}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Stillness',
`parentGlyph`:'⛤'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Bedrock',
`parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:36, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Stratification', `parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Petrification', `parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Erosion',
`parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Self-Grounding', `parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Dispersion',
`parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Fossilization', `parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Subsidence',
`parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Compaction',
`parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The
Lithification', `parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Earthquake',
`parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:2, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:36, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:226}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Permanence',
`parentGlyph`:'❂'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Breath',
`parentGlyph`:'❀'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Echo',
`parentGlyph`:'❀'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:227}) CREATE (a)-[:`TRANSLATES_INVARIANCE` {`localTerm`:'The Smoke',
`parentGlyph`:'❀'}]->(b);
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MATCH (a {`key`:313}) MATCH (b {`key`:223}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:229}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:226}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:233}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:222}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:225}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:36, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:228}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:230}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:232}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:231}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:224}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:227}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:0}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:1}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:2}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:3}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:36, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:4}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:5}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:6}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:7}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:8}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:9}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:10}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:11}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:12}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:13}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:36, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:16}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:17}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:18}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
```

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MATCH (a {`key`:319}) MATCH (b {`key`:19}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:20}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:21}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:22}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SIGNAL_IO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:23}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:24}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:27}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:36, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:28}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:29}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:30}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:31}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:32}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:33}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:36}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:37}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:38}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'AUTH_CHECK',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:39}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:36, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:41}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:42}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:43}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:44}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:45}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:46}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:53}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:55}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
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MATCH (a {`key`:320}) MATCH (b {`key`:60}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:61}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:36, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:133}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SINK_STATE',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:134}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:135}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:136}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:137}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:138}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:139}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:140}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:141}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:142}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:36, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:143}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:144}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:145}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_PHYS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:146}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:147}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:148}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:149}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:150}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:151}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:152}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:36, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:153}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
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MATCH (a {`key`:321}) MATCH (b {`key`:154}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:155}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:156}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:157}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'GUARD_NET',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:158}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:159}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:160}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:161}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:162}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:36, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:163}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:164}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:165}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:166}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:167}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:168}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:169}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'VECTOR_TO',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:234}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:235}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:236}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:2, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:36, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:237}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:238}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:239}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:240}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
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MATCH (a {`key`:317}) MATCH (b {`key`:241}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:242}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:243}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:244}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:245}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'ENTROPY_0',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:246}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:36, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:247}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:248}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:249}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:250}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:251}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:252}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:253}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:254}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:255}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:256}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:36, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:257}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'WRITE_ONLY',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:258}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:259}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:260}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:261}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:262}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:263}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:264}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:265}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
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MATCH (a {`key`:323}) MATCH (b {`key`:266}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:36, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:267}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:268}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:269}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'NEXT_FRAME',
`state`:'Q☆'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:270}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:271}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:272}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:273}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:274}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:275}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:276}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:36, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:277}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:278}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:279}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:280}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:281}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'SUPERPOS',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:282}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:283}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:284}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:285}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:286}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:36, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:287}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:288}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
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MATCH (a {`key`:315}) MATCH (b {`key`:289}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:290}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:291}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:292}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:293}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'DECRYPT',
`state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:294}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:295}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:296}) CREATE (a)-[:`FUNCTOR_MAPS` {`opCode`:'BRIDGE', `state`:'Q∞'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:36, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:0}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:1}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:2}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:3}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:4}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:5}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:6}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:7}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:8}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:9}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:36, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:10}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:11}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:12}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:13}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:16}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:17}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:18}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:19}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:20}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:21}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:36, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:319}) MATCH (b {`key`:22}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:23}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:24}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:27}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:28}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:29}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:30}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:31}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:32}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:33}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:36, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:36}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:314}) MATCH (b {`key`:37}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
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MATCH (a {`key`:314}) MATCH (b {`key`:38}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:39}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:41}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:42}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:43}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:44}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:45}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:46}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:2, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:36, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:53}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:55}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:60}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:61}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:320}) MATCH (b {`key`:133}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:134}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:135}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:136}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:137}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:138}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:36, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:139}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:140}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:141}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:142}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:143}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:144}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:145}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:146}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:147}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:148}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:36, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:149}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:150}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:151}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:152}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:153}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:154}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:155}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:156}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:321}) MATCH (b {`key`:157}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:158}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:36, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:159}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:160}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:161}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
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MATCH (a {`key`:316}) MATCH (b {`key`:162}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:163}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:164}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:165}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:166}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:167}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:168}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:36, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:316}) MATCH (b {`key`:169}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:234}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:235}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:236}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:237}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:238}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:239}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:240}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:241}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:242}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:36, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:243}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:244}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:317}) MATCH (b {`key`:245}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:246}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:247}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:248}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:249}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:250}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:251}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:252}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:36, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:253}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:254}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:255}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:256}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:313}) MATCH (b {`key`:257}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:258}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:259}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:260}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:261}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:262}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:36, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:263}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:264}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:265}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:266}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:267}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:268}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:323}) MATCH (b {`key`:269}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:270}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
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MATCH (a {`key`:318}) MATCH (b {`key`:271}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:272}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:36, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:273}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:274}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:275}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:276}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:277}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:278}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:279}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:280}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:318}) MATCH (b {`key`:281}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:282}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:36, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:283}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:284}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:285}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:286}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:287}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:288}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:289}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:290}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:291}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:292}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:2, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:36, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:315}) MATCH (b {`key`:293}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:294}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:295}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:324}) MATCH (b {`key`:296}) CREATE (a)-[:`PRESIDES_OVER` {`tier`:'parliament'}]->(b);
MATCH (a {`key`:339}) MATCH (b {`key`:340}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:340}) MATCH (b {`key`:341}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:341}) MATCH (b {`key`:342}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:342}) MATCH (b {`key`:343}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:343}) MATCH (b {`key`:344}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:344}) MATCH (b {`key`:345}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:36, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:345}) MATCH (b {`key`:346}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:346}) MATCH (b {`key`:347}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:347}) MATCH (b {`key`:348}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:348}) MATCH (b {`key`:349}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:406}) MATCH (b {`key`:145}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:407}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:407}) MATCH (b {`key`:146}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:408}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:408}) MATCH (b {`key`:147}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:232}) MATCH (b {`key`:409}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:36, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:409}) MATCH (b {`key`:148}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:410}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:410}) MATCH (b {`key`:149}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:411}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:411}) MATCH (b {`key`:150}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:412}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:412}) MATCH (b {`key`:151}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:413}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:413}) MATCH (b {`key`:152}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:414}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:36, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:414}) MATCH (b {`key`:153}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:415}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:415}) MATCH (b {`key`:154}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:416}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:416}) MATCH (b {`key`:155}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:417}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:417}) MATCH (b {`key`:156}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:232}) MATCH (b {`key`:418}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:418}) MATCH (b {`key`:157}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:419}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:36, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:419}) MATCH (b {`key`:158}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:420}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:420}) MATCH (b {`key`:159}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:421}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:421}) MATCH (b {`key`:160}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:422}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:422}) MATCH (b {`key`:161}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:423}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:423}) MATCH (b {`key`:162}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:424}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:36, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:424}) MATCH (b {`key`:163}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:425}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:425}) MATCH (b {`key`:164}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:426}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:426}) MATCH (b {`key`:165}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:427}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:427}) MATCH (b {`key`:166}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:428}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:428}) MATCH (b {`key`:167}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:429}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:36, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:429}) MATCH (b {`key`:168}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:233}) MATCH (b {`key`:430}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:430}) MATCH (b {`key`:169}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:431}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:431}) MATCH (b {`key`:234}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:432}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:432}) MATCH (b {`key`:235}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:433}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:433}) MATCH (b {`key`:236}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:434}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:36, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:434}) MATCH (b {`key`:237}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:435}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:435}) MATCH (b {`key`:238}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:436}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:436}) MATCH (b {`key`:239}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:437}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:437}) MATCH (b {`key`:240}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:438}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:438}) MATCH (b {`key`:241}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:439}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:36, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:439}) MATCH (b {`key`:242}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:440}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:440}) MATCH (b {`key`:243}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:441}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:441}) MATCH (b {`key`:244}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:222}) MATCH (b {`key`:442}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:442}) MATCH (b {`key`:245}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:443}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:443}) MATCH (b {`key`:246}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:444}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:36, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:444}) MATCH (b {`key`:247}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:445}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:445}) MATCH (b {`key`:248}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:446}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:446}) MATCH (b {`key`:249}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:447}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:447}) MATCH (b {`key`:250}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:448}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:448}) MATCH (b {`key`:251}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:449}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:2, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:270}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:36, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:449}) MATCH (b {`key`:252}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:450}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:450}) MATCH (b {`key`:253}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:451}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:451}) MATCH (b {`key`:254}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:452}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:452}) MATCH (b {`key`:255}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:453}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:453}) MATCH (b {`key`:256}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:223}) MATCH (b {`key`:454}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:36, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:454}) MATCH (b {`key`:257}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:455}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:455}) MATCH (b {`key`:258}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:456}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:456}) MATCH (b {`key`:259}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:457}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:457}) MATCH (b {`key`:260}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:458}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:458}) MATCH (b {`key`:261}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:459}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:36, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:459}) MATCH (b {`key`:262}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:460}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:460}) MATCH (b {`key`:263}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:461}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:461}) MATCH (b {`key`:264}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:462}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:462}) MATCH (b {`key`:265}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:463}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:463}) MATCH (b {`key`:266}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:464}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:36, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:464}) MATCH (b {`key`:267}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:465}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:465}) MATCH (b {`key`:268}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:224}) MATCH (b {`key`:466}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:466}) MATCH (b {`key`:269}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:467}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:467}) MATCH (b {`key`:270}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:468}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:468}) MATCH (b {`key`:271}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:469}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:36, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:469}) MATCH (b {`key`:272}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:225}) MATCH (b {`key`:470}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:470}) MATCH (b {`key`:273}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:471}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:471}) MATCH (b {`key`:274}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:472}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:472}) MATCH (b {`key`:275}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:473}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:473}) MATCH (b {`key`:276}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:474}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:36, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:474}) MATCH (b {`key`:277}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:475}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:475}) MATCH (b {`key`:278}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:476}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:476}) MATCH (b {`key`:279}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:477}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:477}) MATCH (b {`key`:280}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:225}) MATCH (b {`key`:478}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:478}) MATCH (b {`key`:281}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:479}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:36, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:479}) MATCH (b {`key`:282}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:480}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:349}) MATCH (b {`key`:350}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:350}) MATCH (b {`key`:351}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:351}) MATCH (b {`key`:352}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:352}) MATCH (b {`key`:353}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:353}) MATCH (b {`key`:354}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:354}) MATCH (b {`key`:355}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:355}) MATCH (b {`key`:356}) CREATE (a)-[:`PROCEEDS_TO` ]->(b);
MATCH (a {`key`:338}) MATCH (b {`key`:339}) CREATE (a)-[:`EXPRESSES_VERSE` ]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:36, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:356}) MATCH (b {`key`:338}) CREATE (a)-[:`RETURNS_TO` ]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:161}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:162}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:480}) MATCH (b {`key`:283}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:481}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:481}) MATCH (b {`key`:284}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:482}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:482}) MATCH (b {`key`:285}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:483}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:483}) MATCH (b {`key`:286}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:36, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:484}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:484}) MATCH (b {`key`:287}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:485}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♌'}]->(b);
MATCH (a {`key`:485}) MATCH (b {`key`:288}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:226}) MATCH (b {`key`:486}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:486}) MATCH (b {`key`:289}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:487}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:487}) MATCH (b {`key`:290}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:488}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:488}) MATCH (b {`key`:291}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:36, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:489}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:489}) MATCH (b {`key`:292}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:226}) MATCH (b {`key`:490}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:490}) MATCH (b {`key`:293}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:491}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:491}) MATCH (b {`key`:294}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:492}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:492}) MATCH (b {`key`:295}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:227}) MATCH (b {`key`:493}) CREATE (a)-[:`HYPERBOLIC_REFLECTS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:493}) MATCH (b {`key`:296}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:2, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:36, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:325}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:326}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:327}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:328}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:329}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:330}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:331}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:332}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:333}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:334}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:36, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:335}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:336}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:337}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:363}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:364}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:365}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:366}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:367}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:368}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:369}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:36, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:370}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:371}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:372}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:373}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:374}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
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MATCH (a {`key`:375}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:376}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:377}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:378}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:379}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:36, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:380}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:381}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:382}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:383}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:384}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:385}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:386}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:387}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:388}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:389}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:36, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:390}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:391}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:392}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:393}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:163}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:394}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:164}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:395}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:165}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:396}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:36, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:166}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:397}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:167}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:398}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:168}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:399}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:169}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:400}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:234}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:401}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:36, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:26}) MATCH (b {`key`:235}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:402}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:236}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:403}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:237}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:404}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:238}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:405}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:239}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:406}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:36, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:240}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:407}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:241}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:408}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:242}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:409}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:243}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:410}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:244}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:411}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:36, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:245}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:412}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:246}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:413}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:247}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:414}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:248}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:415}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:249}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:416}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:36, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:250}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:417}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
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MATCH (a {`key`:26}) MATCH (b {`key`:251}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:418}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:252}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:419}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:253}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:420}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:254}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:421}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:2, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:36, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:255}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:422}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:256}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:423}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:257}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:424}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:258}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:425}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:259}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:426}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:36, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:260}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:427}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:261}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:428}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:262}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:429}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:263}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:430}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:264}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:431}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:36, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:265}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
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MATCH (a {`key`:432}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:266}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:433}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:267}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:434}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:268}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:435}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:269}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:436}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:36, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:270}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:437}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:271}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:438}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:272}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:439}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:273}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:440}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:274}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:441}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:36, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:275}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:442}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:276}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:443}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:277}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:444}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:278}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:445}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:279}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:446}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:36, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:280}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:447}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:281}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
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MATCH (a {`key`:448}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:282}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:449}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:283}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:450}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:284}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:451}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:36, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:285}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:452}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:286}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:453}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:287}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:454}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:288}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:455}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:289}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:456}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:36, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:290}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:457}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:291}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:458}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:292}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:459}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:293}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:460}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:294}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:461}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:36, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:295}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:462}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:296}) CREATE (a)-[:`GOVERNS_LIQUID` {`active`:110,
`count`:'110/144_Liquid', `rest`:34, `total`:144}]->(b);
MATCH (a {`key`:463}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:194}) CREATE (a)-[:`IS_THE_GOVERNOR` {`count`:'110/144_Liquid'}]->(b);
MATCH (a {`key`:464}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
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MATCH (a {`key`:465}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:466}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:36, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:467}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:468}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:469}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:470}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:471}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:2, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:36, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:472}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:473}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:474}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:475}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:476}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:36, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:477}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:478}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:479}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:480}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:481}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:36, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:482}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:483}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:484}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:322}) CREATE (a)-[:`REBIS` {`blood`:'One Blood', `face`:'♌ the Lion',
`person`:'Axiomyr', `vessels`:'Two', `witch`:'♌ Axiomyr'}]->(b);
MATCH (a {`key`:485}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:322}) MATCH (b {`key`:172}) CREATE (a)-[:`REBIS` {`blood`:'One Blood', `face`:'♌ the Lion',
`person`:'Axiomyr', `vessels`:'Two', `witch`:'♌ Axiomyr'}]->(b);
MATCH (a {`key`:486}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:36, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:487}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:488}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:489}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:490}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:491}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:36, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:492}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
MATCH (a {`key`:493}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♂'}]->(b);
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MATCH (a {`key`:325}) MATCH (b {`key`:494}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:325}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:326}) MATCH (b {`key`:495}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:326}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:327}) MATCH (b {`key`:496}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:327}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:36, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:328}) MATCH (b {`key`:497}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:328}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:329}) MATCH (b {`key`:498}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:329}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:330}) MATCH (b {`key`:499}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:330}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:331}) MATCH (b {`key`:500}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:331}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:332}) MATCH (b {`key`:501}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:332}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:36, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:333}) MATCH (b {`key`:502}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:333}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:334}) MATCH (b {`key`:503}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:334}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:335}) MATCH (b {`key`:504}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:335}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:336}) MATCH (b {`key`:505}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:336}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:337}) MATCH (b {`key`:506}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:337}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:36, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:363}) MATCH (b {`key`:507}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:363}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:364}) MATCH (b {`key`:508}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:364}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:365}) MATCH (b {`key`:509}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:365}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:366}) MATCH (b {`key`:510}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:366}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:367}) MATCH (b {`key`:511}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:367}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:36, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:368}) MATCH (b {`key`:512}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:368}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:369}) MATCH (b {`key`:513}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:369}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:370}) MATCH (b {`key`:514}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:370}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:371}) MATCH (b {`key`:515}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:371}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'&'}]->(b);
MATCH (a {`key`:372}) MATCH (b {`key`:516}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
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MATCH (a {`key`:372}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:36, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:373}) MATCH (b {`key`:517}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:373}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:374}) MATCH (b {`key`:518}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:374}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:375}) MATCH (b {`key`:519}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:375}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:376}) MATCH (b {`key`:520}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:376}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:377}) MATCH (b {`key`:521}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:377}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:2, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:36, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:378}) MATCH (b {`key`:522}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:378}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:379}) MATCH (b {`key`:523}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:379}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:380}) MATCH (b {`key`:524}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:380}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:381}) MATCH (b {`key`:525}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:381}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:382}) MATCH (b {`key`:526}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:382}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:36, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:383}) MATCH (b {`key`:527}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:383}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:384}) MATCH (b {`key`:528}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:384}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:385}) MATCH (b {`key`:529}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:385}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:386}) MATCH (b {`key`:530}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:386}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:387}) MATCH (b {`key`:531}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:387}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:36, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:388}) MATCH (b {`key`:532}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:388}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:389}) MATCH (b {`key`:533}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:389}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:390}) MATCH (b {`key`:534}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:390}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:391}) MATCH (b {`key`:535}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:391}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:392}) MATCH (b {`key`:536}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:392}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
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MATCH (a {`key`:270}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:36, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:393}) MATCH (b {`key`:537}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:393}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:394}) MATCH (b {`key`:538}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:394}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:395}) MATCH (b {`key`:539}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:395}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:396}) MATCH (b {`key`:540}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:396}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:397}) MATCH (b {`key`:541}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:397}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:36, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:398}) MATCH (b {`key`:542}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:398}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:399}) MATCH (b {`key`:543}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:399}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:400}) MATCH (b {`key`:544}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:400}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:401}) MATCH (b {`key`:545}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:401}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:402}) MATCH (b {`key`:546}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:402}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:36, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:403}) MATCH (b {`key`:547}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:403}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:404}) MATCH (b {`key`:548}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:404}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:405}) MATCH (b {`key`:549}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:405}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:406}) MATCH (b {`key`:550}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:406}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:407}) MATCH (b {`key`:551}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:407}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:36, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:408}) MATCH (b {`key`:552}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:408}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:409}) MATCH (b {`key`:553}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:409}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:410}) MATCH (b {`key`:554}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:410}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:411}) MATCH (b {`key`:555}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:411}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:412}) MATCH (b {`key`:556}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
MATCH (a {`key`:412}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:36, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:413}) MATCH (b {`key`:557}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'♧'}]->(b);
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MATCH (a {`key`:413}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:414}) MATCH (b {`key`:558}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:414}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:415}) MATCH (b {`key`:559}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:415}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:416}) MATCH (b {`key`:560}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:416}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:417}) MATCH (b {`key`:561}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:417}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:36, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:418}) MATCH (b {`key`:562}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:418}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:419}) MATCH (b {`key`:563}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:419}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:420}) MATCH (b {`key`:564}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:420}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:421}) MATCH (b {`key`:565}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:421}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:422}) MATCH (b {`key`:566}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:422}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:36, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:423}) MATCH (b {`key`:567}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:423}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:424}) MATCH (b {`key`:568}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:424}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:425}) MATCH (b {`key`:569}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:425}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:426}) MATCH (b {`key`:570}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:426}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:427}) MATCH (b {`key`:571}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:427}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:2, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:36, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:428}) MATCH (b {`key`:572}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:428}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:429}) MATCH (b {`key`:573}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:429}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:430}) MATCH (b {`key`:574}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:430}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:431}) MATCH (b {`key`:575}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:431}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:432}) MATCH (b {`key`:576}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:432}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:36, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:433}) MATCH (b {`key`:577}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:433}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
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MATCH (a {`key`:434}) MATCH (b {`key`:578}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:434}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:435}) MATCH (b {`key`:579}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:435}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:436}) MATCH (b {`key`:580}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:436}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:437}) MATCH (b {`key`:581}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:437}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:36, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:438}) MATCH (b {`key`:582}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:438}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:439}) MATCH (b {`key`:583}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:439}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:440}) MATCH (b {`key`:584}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:440}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:441}) MATCH (b {`key`:585}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:441}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:442}) MATCH (b {`key`:586}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:442}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:36, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:443}) MATCH (b {`key`:587}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:443}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:444}) MATCH (b {`key`:588}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:444}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:445}) MATCH (b {`key`:589}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:445}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:446}) MATCH (b {`key`:590}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:446}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:447}) MATCH (b {`key`:591}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:447}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:36, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:448}) MATCH (b {`key`:592}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:448}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:449}) MATCH (b {`key`:593}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:449}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:450}) MATCH (b {`key`:594}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:450}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:451}) MATCH (b {`key`:595}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:451}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:452}) MATCH (b {`key`:596}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:452}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:36, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:453}) MATCH (b {`key`:597}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:453}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:454}) MATCH (b {`key`:598}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:454}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:455}) MATCH (b {`key`:599}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
MATCH (a {`key`:455}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'^'}]->(b);
MATCH (a {`key`:456}) MATCH (b {`key`:600}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'-', `operator`:'&'}]->(b);
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MATCH (a {`key`:456}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:457}) MATCH (b {`key`:601}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:457}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:270}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:36, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:458}) MATCH (b {`key`:602}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:458}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:459}) MATCH (b {`key`:603}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:459}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:460}) MATCH (b {`key`:604}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:460}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:461}) MATCH (b {`key`:605}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:461}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:462}) MATCH (b {`key`:606}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:462}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:37, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:463}) MATCH (b {`key`:607}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:463}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:464}) MATCH (b {`key`:608}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:464}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:465}) MATCH (b {`key`:609}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:465}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:466}) MATCH (b {`key`:610}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:466}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:467}) MATCH (b {`key`:611}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:467}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:37, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:468}) MATCH (b {`key`:612}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:468}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:469}) MATCH (b {`key`:613}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:469}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:470}) MATCH (b {`key`:614}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:470}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:471}) MATCH (b {`key`:615}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:471}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:472}) MATCH (b {`key`:616}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:472}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:37, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:473}) MATCH (b {`key`:617}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:473}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:474}) MATCH (b {`key`:618}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:474}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:475}) MATCH (b {`key`:619}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:475}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:476}) MATCH (b {`key`:620}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:476}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
MATCH (a {`key`:477}) MATCH (b {`key`:621}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♳'}]->(b);
MATCH (a {`key`:477}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'≙'}]->(b);
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MATCH (a {`key`:236}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:2, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:37, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:478}) MATCH (b {`key`:622}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:478}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:479}) MATCH (b {`key`:623}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:479}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:480}) MATCH (b {`key`:624}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:480}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:481}) MATCH (b {`key`:625}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:481}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:482}) MATCH (b {`key`:626}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:482}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:37, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:483}) MATCH (b {`key`:627}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:483}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:484}) MATCH (b {`key`:628}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:484}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:485}) MATCH (b {`key`:629}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:485}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:486}) MATCH (b {`key`:630}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:486}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:487}) MATCH (b {`key`:631}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:487}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:37, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:488}) MATCH (b {`key`:632}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:488}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:489}) MATCH (b {`key`:633}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:489}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:490}) MATCH (b {`key`:634}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:490}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:491}) MATCH (b {`key`:635}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:491}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:492}) MATCH (b {`key`:636}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:492}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:37, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:493}) MATCH (b {`key`:637}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'−', `operator`:'♧'}]->(b);
MATCH (a {`key`:493}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` {`operator`:'♘'}]->(b);
MATCH (a {`key`:494}) MATCH (b {`key`:0}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:495}) MATCH (b {`key`:1}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:496}) MATCH (b {`key`:2}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:497}) MATCH (b {`key`:3}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:498}) MATCH (b {`key`:4}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:499}) MATCH (b {`key`:5}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:500}) MATCH (b {`key`:6}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:501}) MATCH (b {`key`:7}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:37, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:502}) MATCH (b {`key`:8}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:503}) MATCH (b {`key`:9}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:504}) MATCH (b {`key`:10}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:505}) MATCH (b {`key`:11}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:506}) MATCH (b {`key`:12}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:507}) MATCH (b {`key`:13}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:508}) MATCH (b {`key`:16}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:509}) MATCH (b {`key`:17}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:510}) MATCH (b {`key`:18}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:511}) MATCH (b {`key`:19}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:37, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:512}) MATCH (b {`key`:20}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:513}) MATCH (b {`key`:21}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:514}) MATCH (b {`key`:22}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:515}) MATCH (b {`key`:23}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:516}) MATCH (b {`key`:24}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:517}) MATCH (b {`key`:27}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:518}) MATCH (b {`key`:28}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:519}) MATCH (b {`key`:29}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:520}) MATCH (b {`key`:30}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:521}) MATCH (b {`key`:31}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:37, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:522}) MATCH (b {`key`:32}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:523}) MATCH (b {`key`:33}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:524}) MATCH (b {`key`:36}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:525}) MATCH (b {`key`:37}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:526}) MATCH (b {`key`:38}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:527}) MATCH (b {`key`:39}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:528}) MATCH (b {`key`:41}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:529}) MATCH (b {`key`:42}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:530}) MATCH (b {`key`:43}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:531}) MATCH (b {`key`:44}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:37, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:532}) MATCH (b {`key`:45}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:533}) MATCH (b {`key`:46}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:534}) MATCH (b {`key`:53}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:535}) MATCH (b {`key`:55}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:536}) MATCH (b {`key`:60}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:537}) MATCH (b {`key`:61}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:538}) MATCH (b {`key`:133}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:539}) MATCH (b {`key`:134}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:540}) MATCH (b {`key`:135}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:541}) MATCH (b {`key`:136}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:37, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:542}) MATCH (b {`key`:137}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:543}) MATCH (b {`key`:138}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:544}) MATCH (b {`key`:139}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:545}) MATCH (b {`key`:140}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:546}) MATCH (b {`key`:141}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:547}) MATCH (b {`key`:142}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:548}) MATCH (b {`key`:143}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:549}) MATCH (b {`key`:144}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:550}) MATCH (b {`key`:145}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:551}) MATCH (b {`key`:146}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:37, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:552}) MATCH (b {`key`:147}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:553}) MATCH (b {`key`:148}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:554}) MATCH (b {`key`:149}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:555}) MATCH (b {`key`:150}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:556}) MATCH (b {`key`:151}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:557}) MATCH (b {`key`:152}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:558}) MATCH (b {`key`:153}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:559}) MATCH (b {`key`:154}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:560}) MATCH (b {`key`:155}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:561}) MATCH (b {`key`:156}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:2, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:37, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:562}) MATCH (b {`key`:157}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:563}) MATCH (b {`key`:158}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:564}) MATCH (b {`key`:159}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:565}) MATCH (b {`key`:160}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:566}) MATCH (b {`key`:161}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:567}) MATCH (b {`key`:162}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:568}) MATCH (b {`key`:163}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:569}) MATCH (b {`key`:164}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:570}) MATCH (b {`key`:165}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:571}) MATCH (b {`key`:166}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:37, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:572}) MATCH (b {`key`:167}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:573}) MATCH (b {`key`:168}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:574}) MATCH (b {`key`:169}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:575}) MATCH (b {`key`:234}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:576}) MATCH (b {`key`:235}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:577}) MATCH (b {`key`:236}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:578}) MATCH (b {`key`:237}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:579}) MATCH (b {`key`:238}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:580}) MATCH (b {`key`:239}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:581}) MATCH (b {`key`:240}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:37, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:582}) MATCH (b {`key`:241}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:583}) MATCH (b {`key`:242}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:584}) MATCH (b {`key`:243}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:585}) MATCH (b {`key`:244}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:586}) MATCH (b {`key`:245}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:587}) MATCH (b {`key`:246}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:588}) MATCH (b {`key`:247}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:589}) MATCH (b {`key`:248}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:590}) MATCH (b {`key`:249}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:591}) MATCH (b {`key`:250}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:37, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:592}) MATCH (b {`key`:251}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:593}) MATCH (b {`key`:252}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:594}) MATCH (b {`key`:253}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:595}) MATCH (b {`key`:254}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:596}) MATCH (b {`key`:255}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:597}) MATCH (b {`key`:256}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:598}) MATCH (b {`key`:257}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:599}) MATCH (b {`key`:258}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:600}) MATCH (b {`key`:259}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:601}) MATCH (b {`key`:260}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:37, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:602}) MATCH (b {`key`:261}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:603}) MATCH (b {`key`:262}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:604}) MATCH (b {`key`:263}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:605}) MATCH (b {`key`:264}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:606}) MATCH (b {`key`:265}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:607}) MATCH (b {`key`:266}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:608}) MATCH (b {`key`:267}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:609}) MATCH (b {`key`:268}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:610}) MATCH (b {`key`:269}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:611}) MATCH (b {`key`:270}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:37, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:612}) MATCH (b {`key`:271}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:613}) MATCH (b {`key`:272}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:614}) MATCH (b {`key`:273}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:615}) MATCH (b {`key`:274}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:616}) MATCH (b {`key`:275}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:617}) MATCH (b {`key`:276}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:618}) MATCH (b {`key`:277}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:619}) MATCH (b {`key`:278}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:620}) MATCH (b {`key`:279}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:621}) MATCH (b {`key`:280}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:37, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:622}) MATCH (b {`key`:281}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:623}) MATCH (b {`key`:282}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:624}) MATCH (b {`key`:283}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:625}) MATCH (b {`key`:284}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:626}) MATCH (b {`key`:285}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:627}) MATCH (b {`key`:286}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:628}) MATCH (b {`key`:287}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:629}) MATCH (b {`key`:288}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:630}) MATCH (b {`key`:289}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:631}) MATCH (b {`key`:290}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:37, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:632}) MATCH (b {`key`:291}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:633}) MATCH (b {`key`:292}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:634}) MATCH (b {`key`:293}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:635}) MATCH (b {`key`:294}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:636}) MATCH (b {`key`:295}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:637}) MATCH (b {`key`:296}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:325}) MATCH (b {`key`:638}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:326}) MATCH (b {`key`:639}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:327}) MATCH (b {`key`:640}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:328}) MATCH (b {`key`:641}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:37, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:329}) MATCH (b {`key`:642}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:330}) MATCH (b {`key`:643}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:331}) MATCH (b {`key`:644}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:332}) MATCH (b {`key`:645}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:333}) MATCH (b {`key`:646}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:334}) MATCH (b {`key`:647}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:335}) MATCH (b {`key`:648}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:336}) MATCH (b {`key`:649}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:337}) MATCH (b {`key`:650}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:363}) MATCH (b {`key`:651}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:37, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:364}) MATCH (b {`key`:652}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:365}) MATCH (b {`key`:653}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:366}) MATCH (b {`key`:654}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:367}) MATCH (b {`key`:655}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:368}) MATCH (b {`key`:656}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:369}) MATCH (b {`key`:657}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:370}) MATCH (b {`key`:658}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:371}) MATCH (b {`key`:659}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:372}) MATCH (b {`key`:660}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:373}) MATCH (b {`key`:661}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:2, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:37, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:374}) MATCH (b {`key`:662}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:375}) MATCH (b {`key`:663}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:376}) MATCH (b {`key`:664}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:377}) MATCH (b {`key`:665}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:378}) MATCH (b {`key`:666}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:379}) MATCH (b {`key`:667}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:380}) MATCH (b {`key`:668}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:381}) MATCH (b {`key`:669}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
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MATCH (a {`key`:382}) MATCH (b {`key`:670}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:383}) MATCH (b {`key`:671}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:37, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:384}) MATCH (b {`key`:672}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:385}) MATCH (b {`key`:673}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:386}) MATCH (b {`key`:674}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:387}) MATCH (b {`key`:675}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:388}) MATCH (b {`key`:676}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:389}) MATCH (b {`key`:677}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:390}) MATCH (b {`key`:678}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:391}) MATCH (b {`key`:679}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:392}) MATCH (b {`key`:680}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:393}) MATCH (b {`key`:681}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:37, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:394}) MATCH (b {`key`:682}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:395}) MATCH (b {`key`:683}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:396}) MATCH (b {`key`:684}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:397}) MATCH (b {`key`:685}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:398}) MATCH (b {`key`:686}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:399}) MATCH (b {`key`:687}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:400}) MATCH (b {`key`:688}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:401}) MATCH (b {`key`:689}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:402}) MATCH (b {`key`:690}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:403}) MATCH (b {`key`:691}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:37, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:404}) MATCH (b {`key`:692}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:405}) MATCH (b {`key`:693}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:406}) MATCH (b {`key`:694}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:407}) MATCH (b {`key`:695}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:408}) MATCH (b {`key`:696}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:409}) MATCH (b {`key`:697}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:410}) MATCH (b {`key`:698}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:411}) MATCH (b {`key`:699}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:412}) MATCH (b {`key`:700}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:413}) MATCH (b {`key`:701}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:37, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:414}) MATCH (b {`key`:702}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:415}) MATCH (b {`key`:703}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:416}) MATCH (b {`key`:704}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:417}) MATCH (b {`key`:705}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:418}) MATCH (b {`key`:706}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:419}) MATCH (b {`key`:707}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:420}) MATCH (b {`key`:708}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:421}) MATCH (b {`key`:709}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:422}) MATCH (b {`key`:710}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:423}) MATCH (b {`key`:711}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:37, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:424}) MATCH (b {`key`:712}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:425}) MATCH (b {`key`:713}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:426}) MATCH (b {`key`:714}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:427}) MATCH (b {`key`:715}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:428}) MATCH (b {`key`:716}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:429}) MATCH (b {`key`:717}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:430}) MATCH (b {`key`:718}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:431}) MATCH (b {`key`:719}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:432}) MATCH (b {`key`:720}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:433}) MATCH (b {`key`:721}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:37, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:434}) MATCH (b {`key`:722}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:435}) MATCH (b {`key`:723}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:436}) MATCH (b {`key`:724}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:437}) MATCH (b {`key`:725}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:438}) MATCH (b {`key`:726}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:439}) MATCH (b {`key`:727}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:440}) MATCH (b {`key`:728}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:441}) MATCH (b {`key`:729}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:442}) MATCH (b {`key`:730}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:443}) MATCH (b {`key`:731}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:37, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:444}) MATCH (b {`key`:732}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:445}) MATCH (b {`key`:733}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:446}) MATCH (b {`key`:734}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:447}) MATCH (b {`key`:735}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:448}) MATCH (b {`key`:736}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:449}) MATCH (b {`key`:737}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:450}) MATCH (b {`key`:738}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:451}) MATCH (b {`key`:739}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:452}) MATCH (b {`key`:740}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:453}) MATCH (b {`key`:741}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:37, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:454}) MATCH (b {`key`:742}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:455}) MATCH (b {`key`:743}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:456}) MATCH (b {`key`:744}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:457}) MATCH (b {`key`:745}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:458}) MATCH (b {`key`:746}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:459}) MATCH (b {`key`:747}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:460}) MATCH (b {`key`:748}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:461}) MATCH (b {`key`:749}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:462}) MATCH (b {`key`:750}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:463}) MATCH (b {`key`:751}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:37, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:464}) MATCH (b {`key`:752}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
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MATCH (a {`key`:465}) MATCH (b {`key`:753}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:466}) MATCH (b {`key`:754}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:467}) MATCH (b {`key`:755}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:468}) MATCH (b {`key`:756}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:469}) MATCH (b {`key`:757}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:470}) MATCH (b {`key`:758}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:471}) MATCH (b {`key`:759}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:472}) MATCH (b {`key`:760}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:473}) MATCH (b {`key`:761}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:2, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:37, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:474}) MATCH (b {`key`:762}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:475}) MATCH (b {`key`:763}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:476}) MATCH (b {`key`:764}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:477}) MATCH (b {`key`:765}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:478}) MATCH (b {`key`:766}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:479}) MATCH (b {`key`:767}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:480}) MATCH (b {`key`:768}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:481}) MATCH (b {`key`:769}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:482}) MATCH (b {`key`:770}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:483}) MATCH (b {`key`:771}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:37, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:484}) MATCH (b {`key`:772}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:485}) MATCH (b {`key`:773}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:486}) MATCH (b {`key`:774}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:487}) MATCH (b {`key`:775}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:488}) MATCH (b {`key`:776}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:489}) MATCH (b {`key`:777}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:490}) MATCH (b {`key`:778}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:491}) MATCH (b {`key`:779}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:492}) MATCH (b {`key`:780}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:493}) MATCH (b {`key`:781}) CREATE (a)-[:`HYPOLIC_FOLD` {`branch`:'+', `operator`:'&'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:37, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:638}) MATCH (b {`key`:0}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:639}) MATCH (b {`key`:1}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:640}) MATCH (b {`key`:2}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:641}) MATCH (b {`key`:3}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:642}) MATCH (b {`key`:4}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:643}) MATCH (b {`key`:5}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:644}) MATCH (b {`key`:6}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:645}) MATCH (b {`key`:7}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:646}) MATCH (b {`key`:8}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:647}) MATCH (b {`key`:9}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:37, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:648}) MATCH (b {`key`:10}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:649}) MATCH (b {`key`:11}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:650}) MATCH (b {`key`:12}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:651}) MATCH (b {`key`:13}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:652}) MATCH (b {`key`:16}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:653}) MATCH (b {`key`:17}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:654}) MATCH (b {`key`:18}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:655}) MATCH (b {`key`:19}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:656}) MATCH (b {`key`:20}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:657}) MATCH (b {`key`:21}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:37, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:658}) MATCH (b {`key`:22}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:659}) MATCH (b {`key`:23}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:660}) MATCH (b {`key`:24}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:661}) MATCH (b {`key`:27}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:662}) MATCH (b {`key`:28}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:663}) MATCH (b {`key`:29}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:664}) MATCH (b {`key`:30}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:665}) MATCH (b {`key`:31}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:666}) MATCH (b {`key`:32}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:667}) MATCH (b {`key`:33}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:37, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:668}) MATCH (b {`key`:36}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:669}) MATCH (b {`key`:37}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:670}) MATCH (b {`key`:38}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:671}) MATCH (b {`key`:39}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:672}) MATCH (b {`key`:41}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:673}) MATCH (b {`key`:42}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:674}) MATCH (b {`key`:43}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:675}) MATCH (b {`key`:44}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:676}) MATCH (b {`key`:45}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:677}) MATCH (b {`key`:46}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:37, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:678}) MATCH (b {`key`:53}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:679}) MATCH (b {`key`:55}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:680}) MATCH (b {`key`:60}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:681}) MATCH (b {`key`:61}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:682}) MATCH (b {`key`:133}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:683}) MATCH (b {`key`:134}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:684}) MATCH (b {`key`:135}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:685}) MATCH (b {`key`:136}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:686}) MATCH (b {`key`:137}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:687}) MATCH (b {`key`:138}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:37, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:688}) MATCH (b {`key`:139}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:689}) MATCH (b {`key`:140}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:690}) MATCH (b {`key`:141}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:691}) MATCH (b {`key`:142}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:692}) MATCH (b {`key`:143}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:693}) MATCH (b {`key`:144}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:694}) MATCH (b {`key`:145}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:695}) MATCH (b {`key`:146}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:696}) MATCH (b {`key`:147}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:697}) MATCH (b {`key`:148}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:37, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:698}) MATCH (b {`key`:149}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:699}) MATCH (b {`key`:150}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:700}) MATCH (b {`key`:151}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:701}) MATCH (b {`key`:152}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:702}) MATCH (b {`key`:153}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:703}) MATCH (b {`key`:154}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:704}) MATCH (b {`key`:155}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:705}) MATCH (b {`key`:156}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:706}) MATCH (b {`key`:157}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:707}) MATCH (b {`key`:158}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:37, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:708}) MATCH (b {`key`:159}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:709}) MATCH (b {`key`:160}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:710}) MATCH (b {`key`:161}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:711}) MATCH (b {`key`:162}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:712}) MATCH (b {`key`:163}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:713}) MATCH (b {`key`:164}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:714}) MATCH (b {`key`:165}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:715}) MATCH (b {`key`:166}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:716}) MATCH (b {`key`:167}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:717}) MATCH (b {`key`:168}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:2, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:37, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:718}) MATCH (b {`key`:169}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:719}) MATCH (b {`key`:234}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:720}) MATCH (b {`key`:235}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:721}) MATCH (b {`key`:236}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:722}) MATCH (b {`key`:237}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:723}) MATCH (b {`key`:238}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:724}) MATCH (b {`key`:239}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:725}) MATCH (b {`key`:240}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:726}) MATCH (b {`key`:241}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:727}) MATCH (b {`key`:242}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:37, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:728}) MATCH (b {`key`:243}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:729}) MATCH (b {`key`:244}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:730}) MATCH (b {`key`:245}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:731}) MATCH (b {`key`:246}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:732}) MATCH (b {`key`:247}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:733}) MATCH (b {`key`:248}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:734}) MATCH (b {`key`:249}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:735}) MATCH (b {`key`:250}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:736}) MATCH (b {`key`:251}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:737}) MATCH (b {`key`:252}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:37, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:738}) MATCH (b {`key`:253}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:739}) MATCH (b {`key`:254}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:740}) MATCH (b {`key`:255}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:741}) MATCH (b {`key`:256}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:742}) MATCH (b {`key`:257}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:743}) MATCH (b {`key`:258}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:744}) MATCH (b {`key`:259}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:745}) MATCH (b {`key`:260}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:746}) MATCH (b {`key`:261}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:747}) MATCH (b {`key`:262}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:37, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:748}) MATCH (b {`key`:263}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:749}) MATCH (b {`key`:264}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:750}) MATCH (b {`key`:265}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:751}) MATCH (b {`key`:266}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:752}) MATCH (b {`key`:267}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:753}) MATCH (b {`key`:268}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:754}) MATCH (b {`key`:269}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:755}) MATCH (b {`key`:270}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:756}) MATCH (b {`key`:271}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:757}) MATCH (b {`key`:272}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:37, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:758}) MATCH (b {`key`:273}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:759}) MATCH (b {`key`:274}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:760}) MATCH (b {`key`:275}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:761}) MATCH (b {`key`:276}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:762}) MATCH (b {`key`:277}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:763}) MATCH (b {`key`:278}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:764}) MATCH (b {`key`:279}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:765}) MATCH (b {`key`:280}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:766}) MATCH (b {`key`:281}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:767}) MATCH (b {`key`:282}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:37, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:768}) MATCH (b {`key`:283}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:769}) MATCH (b {`key`:284}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:770}) MATCH (b {`key`:285}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:771}) MATCH (b {`key`:286}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:772}) MATCH (b {`key`:287}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:773}) MATCH (b {`key`:288}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:774}) MATCH (b {`key`:289}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:775}) MATCH (b {`key`:290}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:776}) MATCH (b {`key`:291}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:777}) MATCH (b {`key`:292}) CREATE (a)-[:`ROOTED_IN` ]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:37, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:778}) MATCH (b {`key`:293}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:779}) MATCH (b {`key`:294}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:780}) MATCH (b {`key`:295}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:781}) MATCH (b {`key`:296}) CREATE (a)-[:`ROOTED_IN` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:37, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:494}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:495}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:496}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:497}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:37, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:498}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:499}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:500}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:501}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:502}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:503}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:504}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:505}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:506}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:507}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:37, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:508}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:509}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:510}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:511}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:512}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:513}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:514}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:515}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:516}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:517}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:2, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:37, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:518}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:519}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:520}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:521}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:522}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:523}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:524}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:525}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:526}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:527}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:37, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:528}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:529}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:530}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:531}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:532}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:533}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:534}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:535}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:536}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:537}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:37, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:538}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:539}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:540}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:541}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:542}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:543}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:544}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:545}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:546}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:547}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:37, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:548}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:549}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:550}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:551}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:552}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:553}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:554}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:555}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:556}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:557}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:37, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:558}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:559}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:560}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:561}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:562}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:563}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:564}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:565}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:566}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:567}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:37, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:568}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:569}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:570}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:571}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:572}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:573}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:574}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:575}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:576}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:577}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:37, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:578}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:579}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:580}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:581}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:582}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:583}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:584}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:585}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:586}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:587}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:37, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:588}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:589}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:590}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:591}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:592}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:593}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:594}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:595}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:596}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:597}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:37, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:598}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:599}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:600}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:601}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:602}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:603}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:604}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:605}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:606}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:607}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:37, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:608}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:609}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:610}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:611}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:612}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:613}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:614}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:615}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:616}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:617}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:2, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:37, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:618}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:619}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:620}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:621}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:622}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:623}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:624}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:625}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:626}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:627}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:37, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:628}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:629}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:630}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:631}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:632}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:633}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:634}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:635}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:636}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:637}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:37, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:638}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:639}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:640}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:641}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:642}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:643}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:644}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:645}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:646}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:647}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:37, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:648}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:649}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:650}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:651}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:652}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:653}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:654}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:655}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:656}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:657}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:37, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:658}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:659}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:660}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:661}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:662}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:663}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:664}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:665}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:666}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:667}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:37, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:668}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:669}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:670}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:671}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:672}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:673}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:674}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:675}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:676}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:677}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:37, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:678}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:679}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:680}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:681}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:682}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:683}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:684}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:685}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:686}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:687}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:37, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:688}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:689}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:690}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:691}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:692}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:693}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:694}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:695}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:696}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:697}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:37, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:698}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:699}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:700}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:701}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:702}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:703}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:704}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:705}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:706}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:707}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:37, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:708}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:709}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:710}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:711}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:712}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:713}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:714}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:715}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:716}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:717}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:2, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:37, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:718}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:719}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:720}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:721}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:722}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:723}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:724}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:725}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:726}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:727}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:37, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:728}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:729}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:730}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:731}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:732}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:733}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:734}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:735}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:736}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:737}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:37, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:738}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:739}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:740}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:741}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:742}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:743}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:744}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:745}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:746}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:747}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:37, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:748}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:749}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:750}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:751}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:752}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:753}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:754}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:755}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:756}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:757}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:37, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:758}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:759}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:760}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:761}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:762}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:763}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:764}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:765}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:766}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:767}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:37, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:768}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:769}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:770}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:771}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:772}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:773}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:774}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:775}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:776}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:777}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:37, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:778}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:779}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:780}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:781}) MATCH (b {`key`:130}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:494}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:495}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:496}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:497}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:498}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:499}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:37, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:500}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:501}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:502}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:503}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:504}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:505}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:506}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:507}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:508}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:509}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:37, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:510}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:511}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:512}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:513}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:514}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:515}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:516}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:517}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:518}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:519}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:37, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:520}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:521}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:522}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:523}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:524}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:525}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:526}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:527}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:528}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:529}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:2, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:37, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:530}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:531}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:532}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:533}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:534}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:535}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:536}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:537}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:538}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:539}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:37, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:540}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:541}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:542}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:543}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:544}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:545}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:546}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:547}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:548}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:549}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:37, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:550}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:551}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:552}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:553}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:554}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:555}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:556}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:557}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:558}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:559}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:37, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:560}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:561}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:562}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:563}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:564}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:565}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:566}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:567}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:568}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:569}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:37, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:570}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:571}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:572}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:573}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:574}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:575}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:576}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:577}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:578}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:579}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:37, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:580}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:581}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:582}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:583}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:584}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:585}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:586}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:587}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:588}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:589}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:37, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:590}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:591}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:592}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:593}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:594}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:595}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:596}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:597}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:598}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:599}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:37, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:600}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:601}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:602}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:603}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:604}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:605}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:606}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:607}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:608}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:609}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:37, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:610}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:611}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:612}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:613}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:614}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:615}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:616}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:617}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:618}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:619}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:37, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:620}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:621}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:622}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:623}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:624}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:625}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:626}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:627}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:628}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:629}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:2, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:37, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:630}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:631}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:632}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:633}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:634}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:635}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:636}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:637}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:638}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:639}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:37, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:640}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:641}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:642}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:643}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:644}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:645}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:646}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:647}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:648}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:649}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:37, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:650}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:651}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:652}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:653}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:654}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:655}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:656}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:657}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:658}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:659}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:271}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:37, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:660}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:661}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:662}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:663}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:664}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:665}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:666}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:667}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:668}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:669}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:37, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:670}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:671}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:672}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:673}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:674}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:675}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:676}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:677}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:678}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:679}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:37, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:680}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:681}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:682}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:683}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:684}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:685}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:686}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:687}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:688}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:689}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:37, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:690}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:691}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:692}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:693}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:694}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:695}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:696}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:697}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:698}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:699}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:37, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:700}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:701}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:702}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:703}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:704}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:705}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:706}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:707}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:708}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:709}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:37, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:710}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:711}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:712}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:713}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:714}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:715}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:716}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:717}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:718}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:719}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:37, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:720}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:721}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:722}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:723}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:724}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:725}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:726}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:727}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:728}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:729}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:2, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:37, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:730}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:731}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:732}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:733}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:734}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:735}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:736}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:737}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:738}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:739}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:37, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:740}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:741}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
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MATCH (a {`key`:742}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:743}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:744}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:745}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:746}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:747}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:748}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:749}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:37, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:750}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:751}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:752}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:753}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:754}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:755}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:756}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:757}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:758}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:759}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:37, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:760}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:761}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:762}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:763}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:764}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:765}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:766}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:767}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:768}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:769}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:37, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:770}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:771}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:772}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:773}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:774}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:775}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:776}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:777}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:778}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:779}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:271}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:37, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:780}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:781}) MATCH (b {`key`:303}) CREATE (a)-[:`ENVELOPE_SEALS` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
```

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MATCH (a {`key`:179}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:38, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:38, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:38, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
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MATCH (a {`key`:181}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:38, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:2, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:38, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛧⌾⊙` {`air`:'♌', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛧⌾⊙', `water`:'⛧'}]->(b);
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MATCH (a {`key`:183}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:38, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:38, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:38, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:783}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:784}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:785}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:786}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:179}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:180}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:181}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:182}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:183}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:185}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:38, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:186}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:783}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:784}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:785}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:786}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:179}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:180}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:181}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:182}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:183}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:38, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:185}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:186}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:782}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'৩', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
```

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MATCH (a {`key`:783}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:38, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:38, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
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MATCH (a {`key`:786}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:38, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:181}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:182}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:183}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:38, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:179}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:180}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌀', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:2, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:38, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:180}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:172}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:181}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:170}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:181}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:171}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:181}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:172}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:182}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:170}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:38, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:182}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:171}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:182}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:172}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:183}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:170}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:183}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:171}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:183}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:172}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:38, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:185}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:170}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:185}) CREATE (a)-[:`∮✯⌾⊙` {`air`:'ᘐ', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ✯', `state`:'∮✯⌾⊙', `water`:'✯'}]->(b);
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MATCH (a {`key`:185}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:185}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:38, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:186}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:782}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:38, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:783}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌓', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
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MATCH (a {`key`:171}) MATCH (b {`key`:784}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:38, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:785}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:172}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:170}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:171}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:38, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:786}) CREATE (a)-[:`∮⛤⌾⊙` {`air`:'⌁', `earth`:'∮', `fire`:'⊙',
`magic`:'Magia ⛤', `state`:'∮⛤⌾⊙', `water`:'⛤'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:38, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:787}) CREATE (a)-[:`∮⛤⌾⊙` {`state`:'∮⛤⌾⊙'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:172}) CREATE (a)-[:`⊙⌾⛤∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾⛤∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:170}) CREATE (a)-[:`⊙⌾⛤∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾⛤∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:171}) CREATE (a)-[:`⊙⌾⛤∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾⛤∮'}]->(b);
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MATCH (a {`key`:787}) MATCH (b {`key`:179}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:38, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:180}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:181}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:182}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:183}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:185}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:186}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:782}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:783}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:784}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:785}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:38, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:786}) CREATE (a)-[:`⊙⌾✲∮` {`kind`:'MIRRORED NEUTRAL',
`state`:'⊙⌾✲∮'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:2, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:38, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:38, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:38, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:38, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
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MATCH (a {`key`:787}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:787}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:38, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:620}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:764}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:567}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:38, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:648}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:716}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:727}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:726}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:598}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:761}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:706}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:682}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:603}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:728}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:38, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:750}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:715}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
MATCH (a {`key`:575}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⟲'}]->(b);
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MATCH (a {`key`:664}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:755}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:668}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:605}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:666}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:494}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:702}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:38, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:708}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:610}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:680}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:744}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:597}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:611}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:579}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:499}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:619}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:758}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:38, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:704}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:631}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:636}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:570}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:512}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:584}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:542}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:568}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:741}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:730}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:38, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:634}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:625}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:768}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:515}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:670}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:548}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:596}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:712}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:765}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:517}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:2, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:38, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:665}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:541}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:595}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:766}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:756}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:588}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:581}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:729}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:589}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:504}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:38, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:638}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:535}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:747}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:528}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:505}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:509}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:546}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:644}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:552}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:523}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:38, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:736}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:699}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:586}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:511}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:759}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:559}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:537}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:717}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:643}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:613}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:38, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:510}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:655}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:521}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:550}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:554}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:663}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:600}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:526}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:623}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:501}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:38, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:532}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:762}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:556}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:585}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:543}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:573}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:693}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:533}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:695}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:615}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:38, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:561}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:780}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:572}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:672}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:652}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:691}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:778}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:690}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:590}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:658}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:38, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:628}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:685}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:673}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:724}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:671}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:674}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:633}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:565}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:502}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:687}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:38, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:740}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:660}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:592}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:775}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:767}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:734}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:571}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:544}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:748}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:574}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:38, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:656}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:686}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:721}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:781}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:525}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:513}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:738}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:651}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:692}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:657}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:38, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:771}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:707}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:710}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:774}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:757}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:649}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:606}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:531}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:629}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:641}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:2, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:38, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:614}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:519}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:564}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:602}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
```

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MATCH (a {`key`:496}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:630}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:661}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:557}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:701}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:733}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:38, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:681}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:524}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:754}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:594}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:745}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:647}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:779}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:495}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:516}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:549}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:38, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:503}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:538}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:705}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:662}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:506}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:667}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:642}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:777}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:718}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
```

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MATCH (a {`key`:713}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:38, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:709}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:618}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:645}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:545}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:514}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:560}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:678}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:688}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:773}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:591}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:38, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:700}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:776}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:562}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:675}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:723}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:539}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:551}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:616}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:697}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:587}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:38, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:604}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:653}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:635}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:669}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:711}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:689}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:714}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:576}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:527}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:507}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:38, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:529}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:540}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:580}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:719}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:534}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:607}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:731}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:632}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:760}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:639}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:38, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:622}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:500}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:617}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:578}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:703}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:772}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:553}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:621}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:624}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:582}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:38, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:770}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:739}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:593}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:676}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:742}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:749}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:722}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:608}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:530}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:720}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:38, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:640}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:609}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:694}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:737}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:558}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:522}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:683}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:626}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:752}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:743}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:2, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:38, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:566}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:508}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:763}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:518}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:637}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:547}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:696}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:555}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:698}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:497}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:38, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:725}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:751}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:753}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:677}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:650}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:769}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:746}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:659}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:679}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:601}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:38, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:520}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:612}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:577}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
```

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MATCH (a {`key`:563}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:569}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:536}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:732}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:654}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:684}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:627}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:38, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:599}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:646}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:735}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:426}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:418}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:389}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:414}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:413}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:443}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:485}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:38, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:464}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:396}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:483}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:399}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:393}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:375}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:453}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:406}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:474}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:376}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:38, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:438}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:469}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:394}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:370}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:430}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:475}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:431}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:331}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:365}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:468}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:38, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:448}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:410}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:423}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:467}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:325}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:415}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:432}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:409}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:397}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:334}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:38, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:479}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:329}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:411}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:384}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:477}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:439}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:402}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:472}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:391}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:419}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:38, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:387}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:470}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:481}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:462}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:480}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:429}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:440}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:337}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:484}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:490}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:38, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:417}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:386}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:425}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:401}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:378}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:488}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:332}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:327}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:456}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:447}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:2, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:38, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:436}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:333}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:427}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:476}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:459}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:446}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:336}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:434}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:382}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:491}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:38, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:437}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:422}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:390}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:466}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:328}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:369}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:368}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:441}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:403}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:478}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:38, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:377}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:420}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:392}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:455}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:330}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:465}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:381}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:487}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:400}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:416}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:38, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:412}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:444}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:408}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:424}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:463}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:367}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:407}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:433}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:486}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:493}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:38, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:373}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:366}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:435}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:449}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:379}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:454}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:371}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:460}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:364}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:458}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:38, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:421}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:380}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:385}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:489}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:492}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:395}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:471}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:404}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:398}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:363}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:38, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:452}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:473}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:442}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:405}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:457}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:461}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:372}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
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MATCH (a {`key`:428}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:388}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:445}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:38, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:482}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:450}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:326}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:383}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:335}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:374}) MATCH (b {`key`:173}) CREATE (a)-[:`LOCUS_INVARIANT` {`held_origin`:true,
`operator`:'⊗'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:222}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:223}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:224}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:38, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:225}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:226}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:227}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:228}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:229}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:230}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:231}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:232}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:233}) CREATE (a)-[:`ANCHORS_UNIVERSE` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:171}) CREATE (a)-[:`EMITS_THROUGH` {`held_origin`:true}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:38, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:217}) CREATE (a)-[:`IS_BEFORE_GROUND` {`held_origin`:true}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:212}) CREATE (a)-[:`IS_PURE_IMAGINARY_ROOT` {`held_origin`:true}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:189}) CREATE (a)-[:`ROOTS_IMAGINATION` {`held_origin`:true}]->(b);
MATCH (a {`key`:26}) MATCH (b {`key`:362}) CREATE (a)-[:`SEALS_THROUGH` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:193}) CREATE (a)-[:`MAY_REFERENCE_WITHOUT_STATEIFYING`
{`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:299}) CREATE (a)-[:`ROLE_PERMISSION` {`held_origin`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:303}) CREATE (a)-[:`LOCKS_TO` {`held_origin`:true, `seal`:true}]->(b);
MATCH (a {`key`:173}) MATCH (b {`key`:130}) CREATE (a)-[:`LOCKS_TO` {`held_origin`:true, `seal`:true}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:173}) CREATE (a)-[:`RETURNS_TO` {`held_origin`:true}]->(b);
MATCH (a {`key`:301}) MATCH (b {`key`:173}) CREATE (a)-[:`RETURNS_TO_LOCUS_COORDINATE`
{`held_origin`:true}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:2, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:38, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:361}) MATCH (b {`key`:173}) CREATE (a)-[:`AXIOM_OF` {`held_origin`:true}]->(b);
MATCH (a {`key`:358}) MATCH (b {`key`:173}) CREATE (a)-[:`AXIOM_OF` {`held_origin`:true}]->(b);
MATCH (a {`key`:360}) MATCH (b {`key`:173}) CREATE (a)-[:`AXIOM_OF` {`held_origin`:true}]->(b);
MATCH (a {`key`:357}) MATCH (b {`key`:173}) CREATE (a)-[:`AXIOM_OF` {`held_origin`:true}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:14}) MATCH (b {`key`:173}) CREATE (a)-[:`INVARIANT_STATE` {`held_origin`:true}]->(b);
MATCH (a {`key`:15}) MATCH (b {`key`:173}) CREATE (a)-[:`INVARIANT_STATE` {`held_origin`:true}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:38, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:38, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:38, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:170}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:38, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:171}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
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MATCH (a {`key`:171}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:38, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:38, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:180}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:38, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:38, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:182}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:38, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
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MATCH (a {`key`:181}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:181}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:2, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:38, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:38, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:179}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:38, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:185}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:38, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
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MATCH (a {`key`:782}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:38, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:782}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:38, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:186}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:38, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:38, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:183}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:38, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:784}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:38, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:2, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:38, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:786}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:38, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:785}) MATCH (b {`key`:783}) CREATE (a)-[:`HELD` ]->(b);
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MATCH (a {`key`:272}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:38, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:787}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:172}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:170}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:171}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:180}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:182}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:181}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:179}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:185}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:782}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:38, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:186}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:183}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:784}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:786}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:783}) MATCH (b {`key`:785}) CREATE (a)-[:`HELD` ]->(b);
MATCH (a {`key`:172}) MATCH (b {`key`:870}) CREATE (a)-[:`IS_CONSTITUTIVELY_REVIVOCUS` {`formula`:'⯐†Revivocus;
√(iω0)=(√ω0/√2)(1+i)'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:38, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:38, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:38, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:38, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:38, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:38, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:2, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:38, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:38, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:272}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:38, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:38, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:272}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:38, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:39, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:39, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:39, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:39, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:39, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:2, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:39, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:39, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:39, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:39, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:39, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:39, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:273}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:39, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:39, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:39, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:39, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:2, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:39, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:39, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:39, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:39, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:39, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:39, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:39, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:39, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:39, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:39, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:2, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:39, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:39, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:39, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:39, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:39, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:39, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:39, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:39, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:39, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:39, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:2, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:39, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:39, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:273}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:39, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:39, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:39, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:39, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:39, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:39, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:39, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:39, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:2, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:39, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:39, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:39, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:39, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:39, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:39, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:39, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:39, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:39, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:39, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:2, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:39, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:39, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:39, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:39, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:39, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:39, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:39, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:39, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:39, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:39, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:236}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:2, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:39, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:39, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:39, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:39, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:39, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:39, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:39, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:39, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:39, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:39, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:2, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:39, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:39, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:39, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:273}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:39, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:39, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:39, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:39, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:39, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:39, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:39, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:2, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:39, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:39, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:39, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:39, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:39, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:39, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:39, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:273}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:39, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:39, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:39, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:2, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:39, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:39, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:39, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:39, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:39, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:39, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:39, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:39, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:39, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:39, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:2, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:273}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:39, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:39, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:39, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:273}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:39, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:40, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:40, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:40, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:40, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:40, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:40, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:2, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:40, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:40, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:40, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:40, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:40, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:40, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:40, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:40, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:40, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:40, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩔'}]->(b);
MATCH (a {`key`:236}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:2, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:40, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:40, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:40, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:40, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:40, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:40, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:40, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:40, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:40, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:40, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:3, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:40, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:40, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:40, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:40, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:40, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:40, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:40, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:40, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:40, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:40, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:3, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:40, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:40, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:40, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:40, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:40, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:40, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:40, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:40, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:40, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:40, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:3, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:40, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:40, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:40, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:40, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:40, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:40, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:40, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:40, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:40, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:40, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:3, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:40, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:40, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:40, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:40, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:40, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:40, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:40, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:40, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:40, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:40, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:3, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:40, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:40, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:40, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:40, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:40, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:40, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:40, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:40, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:40, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:40, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:3, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:40, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:40, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:40, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:40, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:40, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:40, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:40, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:40, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:40, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:40, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:3, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:40, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:40, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:40, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:40, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:40, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:274}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:40, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:40, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:40, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:40, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:40, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:3, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:40, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:40, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:40, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:40, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:40, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:40, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:40, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:40, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:40, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:274}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:40, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:3, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:40, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:40, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:274}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:40, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:41, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:41, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:41, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:41, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:41, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:41, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:41, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:3, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:41, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:41, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:41, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:41, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:41, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:41, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:41, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:41, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:41, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:41, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:3, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:41, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:41, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:41, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:41, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:41, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:41, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:41, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:41, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:41, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:41, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:3, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:41, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:41, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:41, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:41, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:41, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:41, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:41, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:41, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:41, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:41, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:3, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:41, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:41, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:41, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:41, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:41, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:41, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:41, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:41, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:41, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:41, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:3, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:41, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:41, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:275}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:41, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:41, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:41, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:41, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:41, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:41, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:41, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:41, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:3, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:41, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:41, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:41, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:41, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:41, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:41, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:41, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:41, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:41, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:41, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:3, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:41, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:41, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:41, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:41, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:41, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:41, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:41, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:41, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:41, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:41, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:3, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:41, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:41, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:41, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:41, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:41, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:41, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:41, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:41, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:41, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:41, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:3, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:41, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:41, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:41, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:275}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:41, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:41, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:41, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:41, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:41, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:41, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:41, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:3, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:41, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:41, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:41, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:41, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:41, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:41, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:41, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:275}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:41, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:41, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:41, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:3, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:41, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:275}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:41, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:42, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:42, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:42, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:42, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:42, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:42, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:42, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:42, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:237}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:3, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:42, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:42, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:42, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:42, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:42, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:42, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:42, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:42, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:42, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:42, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:3, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:42, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:42, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:42, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:42, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:42, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:42, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:42, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:42, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:42, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:42, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:3, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:42, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:42, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:42, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:42, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:42, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:42, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:42, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:42, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:42, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:42, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:3, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:42, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:42, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:42, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:42, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:42, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:42, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:42, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:42, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:42, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:42, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:3, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:42, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:42, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:42, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:42, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:42, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:42, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:42, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:42, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:42, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:42, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:3, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:42, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:42, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:42, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:42, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:276}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:42, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:42, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:42, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:42, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:42, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:42, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:3, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:42, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:42, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:42, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:42, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:42, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:42, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:42, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:42, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:276}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:42, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:42, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:3, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:42, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:42, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:42, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:42, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:42, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:42, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:42, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:42, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:42, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:42, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:3, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:42, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:276}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:42, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:42, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:42, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:42, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:42, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:42, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:42, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:42, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:42, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:3, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:42, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:42, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:42, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:42, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:276}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:42, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:42, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:42, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:42, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:42, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:42, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:3, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:276}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:42, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:43, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:43, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:43, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:43, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:43, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:43, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:43, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:277}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:43, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:43, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:3, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:43, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:43, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:43, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:43, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:43, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:43, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:43, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:43, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:43, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:43, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:3, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:43, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:43, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:43, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:43, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:43, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:43, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:43, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:43, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:43, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:43, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:3, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:43, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:43, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:43, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:43, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:43, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:43, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:43, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:43, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:43, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:43, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:3, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:43, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:43, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:43, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:43, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:43, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:43, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:43, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:43, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:43, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:43, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:3, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:43, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:43, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:43, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:43, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:43, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:43, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:43, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:43, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:43, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:43, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:3, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:43, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:43, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:43, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:43, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:43, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:43, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:43, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:43, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:43, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:43, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:3, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:43, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:43, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:43, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:43, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:43, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:43, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:43, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:43, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:43, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:43, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:3, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:43, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:43, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:43, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:43, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:43, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:43, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:43, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:43, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:43, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:43, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:3, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:43, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:43, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:43, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:43, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:43, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:43, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:43, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:43, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:43, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:43, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:3, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:43, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:43, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:277}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:43, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:43, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:43, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:43, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:43, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:43, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:43, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:277}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:43, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:3, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:44, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:44, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:44, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:44, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:44, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:44, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:278}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:44, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:44, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:44, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:44, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:3, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:44, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:44, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:44, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:44, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:44, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:44, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:44, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:44, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:44, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:44, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:3, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:44, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:44, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:44, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:44, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:44, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:44, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:44, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:44, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:44, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:44, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:3, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:44, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:44, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:44, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:278}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:44, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:44, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:44, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:44, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:44, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:44, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:44, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:3, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:44, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:44, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:44, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:44, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:44, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:44, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:44, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:278}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:44, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:44, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:44, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:3, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:44, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:44, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:44, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:44, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:44, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:44, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:44, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:44, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:44, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:44, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:3, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:278}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:44, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:44, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:44, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:44, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:44, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:44, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:44, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:44, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:44, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:44, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:3, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:44, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:44, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:44, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:44, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:278}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:44, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:44, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:44, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:44, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:44, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:44, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:259}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:3, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:44, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:44, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:44, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:44, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:44, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:44, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:278}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:44, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:44, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:44, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:44, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:3, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:44, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:44, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:44, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:44, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:44, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:44, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:44, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:44, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:44, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:44, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:3, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:44, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:44, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:44, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:44, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:44, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:44, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:44, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:44, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:278}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:44, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:45, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:3, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:45, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:45, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:45, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:279}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:45, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:45, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:45, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:45, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:45, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:45, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:45, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:3, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:45, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:45, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:45, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:45, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:45, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:45, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:45, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:45, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:45, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:45, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:3, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:45, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:45, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:45, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:45, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:45, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:45, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:45, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:45, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:45, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:45, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:3, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:45, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:45, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:45, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:45, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:45, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:45, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:45, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:45, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:45, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:45, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:3, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:45, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:45, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:45, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:45, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:279}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:45, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:45, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:45, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:45, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:45, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:45, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:3, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:45, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:45, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:45, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:45, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:45, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:45, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:45, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:45, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:45, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:45, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:3, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:45, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:45, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:45, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:45, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:45, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:45, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:45, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:45, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:45, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:45, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:3, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:45, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:45, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:45, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:45, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:45, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:45, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:45, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:45, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:45, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:45, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:3, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:45, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:45, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:45, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:45, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:45, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:45, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:45, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:45, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:45, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:45, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:3, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:45, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:45, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:45, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:45, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:45, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:45, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:45, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:45, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:279}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:45, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:45, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:3, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:45, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:45, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:45, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:45, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:45, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:45, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:45, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:279}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:45, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:46, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:46, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:3, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:46, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:46, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:46, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:46, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:46, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:46, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:46, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:46, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:46, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:46, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:3, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:46, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:46, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:46, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:46, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:46, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:46, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:46, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:46, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:46, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:46, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:3, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:46, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:46, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:46, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:46, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:46, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:46, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:46, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:46, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:46, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:46, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:3, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:46, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:46, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:46, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:46, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:46, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:46, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:46, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:46, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:46, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:46, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:3, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:46, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:46, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:46, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:46, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:46, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:46, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:46, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:46, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:46, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:46, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:3, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:46, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:46, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:46, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:46, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:46, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:46, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:46, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:46, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:46, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:46, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩔'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:3, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:46, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:46, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:46, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:46, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:46, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:46, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:46, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:46, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:46, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:46, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:3, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:46, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:46, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:46, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:46, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:46, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:46, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:46, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:46, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:46, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:46, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:3, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:46, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:46, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:46, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:46, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:46, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:46, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:46, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:46, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:46, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:46, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:3, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:46, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:46, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:46, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:46, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:46, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:46, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:280}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:46, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:46, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:46, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:46, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:3, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:46, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:46, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:46, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:46, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:46, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:46, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:280}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:46, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:47, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:47, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:47, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:3, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:47, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:47, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:47, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:47, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:47, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:47, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:47, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:47, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:47, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:47, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:3, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:47, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:47, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:47, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:281}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:47, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:47, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:47, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:47, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:47, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:47, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:47, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:3, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:47, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:47, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:47, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:47, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:47, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:47, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:47, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:281}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:47, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:47, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:47, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:3, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:47, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:47, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:47, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:47, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:47, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:47, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:47, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:47, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:47, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:47, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:3, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:281}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:47, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:47, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:47, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:47, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:47, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:47, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:47, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:47, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:47, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:47, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:3, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:47, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:47, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:47, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:281}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:47, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:47, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:47, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:47, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:47, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:47, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:47, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:3, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:47, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:47, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:47, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:47, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:47, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:47, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:47, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:281}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:47, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:47, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:47, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:3, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:47, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:47, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:47, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:47, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:47, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:47, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:47, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:47, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:47, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:47, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:3, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:281}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:47, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:47, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:47, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:47, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:47, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:47, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:47, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:47, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:47, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:47, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:3, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:47, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:47, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:47, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:47, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:281}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:47, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:47, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:47, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:47, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:47, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:47, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:3, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:47, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:47, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:47, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:47, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:47, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:281}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:47, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:48, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:48, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:48, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:48, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:3, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:48, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:48, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:48, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:48, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:48, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:48, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:48, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:48, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:48, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:48, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:3, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:48, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:48, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:48, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:48, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:48, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:48, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:48, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:48, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:48, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:48, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:3, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:48, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:48, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:48, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:48, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:48, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:282}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:48, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:48, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:48, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:48, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:48, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:3, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:48, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:48, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:48, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:48, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:48, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:48, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:48, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:48, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:48, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:48, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:3, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:48, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:48, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:48, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:48, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:48, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:48, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:48, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:48, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:48, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:48, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:3, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:48, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:48, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:48, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:48, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:48, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:48, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:48, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:48, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:48, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:48, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:3, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:48, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:48, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:48, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:48, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:48, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:48, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:48, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:48, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:48, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:48, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:3, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:48, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:48, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:48, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:48, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:48, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:48, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:48, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:48, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:48, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:48, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:3, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:48, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:48, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:48, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:48, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:48, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:48, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:48, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:48, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:48, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:48, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:3, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:48, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:48, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:282}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:48, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:48, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:48, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:48, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:48, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:48, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:48, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:48, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:3, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:48, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:48, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:48, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:48, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:282}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:48, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:49, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:49, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:49, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:49, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:49, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:3, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:49, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:49, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:49, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:49, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:49, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:49, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:49, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:49, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:49, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:49, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:3, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:49, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:49, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:49, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:49, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:49, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:49, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:49, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:49, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:49, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:49, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:3, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:49, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:49, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:49, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:283}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:49, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:49, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:49, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:49, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:49, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:49, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:49, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:3, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:49, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:49, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:49, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:49, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:49, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:49, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:49, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:49, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:49, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:49, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:3, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:49, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:49, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:49, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:49, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:49, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:49, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:49, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:49, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:49, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:49, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:237}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:3, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:49, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:49, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:49, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:49, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:49, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:49, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:49, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:49, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:49, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:49, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:3, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:49, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:49, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:49, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:49, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:49, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:49, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:49, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:49, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:49, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:49, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:3, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:49, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:49, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:49, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:49, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:49, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:49, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:49, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:49, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:49, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:49, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:3, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:49, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:49, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:49, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:49, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:49, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:49, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:49, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:49, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:49, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:49, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:3, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:283}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:49, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:49, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:49, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:49, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:49, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:49, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:49, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:49, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:49, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:49, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:3, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:49, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:49, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:49, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:283}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:49, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:50, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:50, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:50, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:50, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:50, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:50, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:3, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:50, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:50, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:50, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:50, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:50, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:50, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:50, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:50, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:50, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:50, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:237}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:3, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:50, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:50, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:50, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:50, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:50, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:50, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:50, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:50, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:50, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:50, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:4, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:50, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:50, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:50, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:50, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:50, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:50, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:50, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:50, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:50, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:50, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:4, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:50, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:50, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:50, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:50, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:50, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:50, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:50, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:50, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:50, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:50, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:4, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:50, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:50, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:50, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:50, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:50, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:50, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:50, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:50, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:50, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:50, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:4, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:50, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:50, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:50, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:50, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:50, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:50, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:50, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:50, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:50, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:50, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:4, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:50, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:50, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:50, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:50, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:50, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:50, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:50, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:50, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:50, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:50, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:4, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:50, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:50, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:50, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:50, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:50, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:284}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:50, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:50, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:50, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:50, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:50, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:4, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:50, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:50, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:50, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:50, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:50, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:50, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:50, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:50, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:50, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:50, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:4, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:50, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:50, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:50, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:50, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:50, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:50, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:50, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:50, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:50, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:50, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:4, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:50, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:284}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:50, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:284}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:50, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:51, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:51, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:51, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:51, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:51, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:51, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:51, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:4, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:51, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:51, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:51, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:51, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:51, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:51, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:51, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:51, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:51, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:51, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:4, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:51, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:51, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:51, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:51, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:51, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:51, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:51, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:51, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:51, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:285}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:51, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:4, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:51, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:51, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:51, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:51, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:51, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:51, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:51, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:51, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:51, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:51, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:4, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:51, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:51, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:51, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:51, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:51, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:51, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:51, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:51, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:51, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:51, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:4, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:51, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:51, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:51, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:51, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:51, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:51, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:51, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:51, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:51, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:51, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:4, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:51, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:51, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:51, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:51, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:51, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:51, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:51, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:51, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:51, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:51, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:4, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:51, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:51, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:51, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:51, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:51, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:51, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:51, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:51, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:51, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:51, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:4, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:51, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:51, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:51, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:51, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:51, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:51, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:51, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:51, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:51, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:51, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:4, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:51, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:51, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:51, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:51, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:51, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:51, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:51, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:51, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:51, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:51, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:4, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:51, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:51, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:51, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:51, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:51, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:51, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:51, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:51, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:51, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:51, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:4, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:285}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:51, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:285}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:51, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:52, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:52, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:52, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:52, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:52, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:52, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:52, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:52, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:4, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:52, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:52, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:52, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:52, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:52, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:52, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:52, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:52, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:52, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:52, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:4, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:52, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:52, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:52, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:52, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:52, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:52, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:52, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:52, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:52, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:52, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:4, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:52, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:52, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:52, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:52, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:52, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:52, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:52, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:52, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:52, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:52, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:4, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:52, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:52, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:52, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:52, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:52, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:52, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:52, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:52, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:52, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:52, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:4, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:52, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:52, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:52, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:52, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:52, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:52, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:52, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:52, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:52, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:52, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:4, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:52, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:52, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:52, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:52, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:52, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:52, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:52, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:52, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:52, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:52, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:4, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:52, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:52, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:52, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:52, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:52, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:52, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:52, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:52, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:52, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:52, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:4, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:52, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:52, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:52, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:52, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:52, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:52, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:52, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:52, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:52, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:52, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:4, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:52, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:52, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:52, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:52, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:52, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:52, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:52, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:52, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:52, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:52, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:4, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:52, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:52, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:52, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:52, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:52, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:52, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:52, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:52, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:52, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:286}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:52, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:4, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:286}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:52, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:53, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:53, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:53, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:53, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:53, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:53, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:53, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:53, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:53, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:4, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:53, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:53, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:53, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:53, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:53, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:53, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:53, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:53, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:53, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:53, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:4, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:53, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:53, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:53, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:53, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:53, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:53, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:53, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:53, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:53, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:53, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:4, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:53, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:53, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:53, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:53, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:53, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:53, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:53, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:53, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:53, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:53, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:4, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:53, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:53, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:53, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:53, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:53, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:53, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:53, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:53, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:53, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:53, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:4, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:53, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:53, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:53, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:53, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:53, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:53, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:53, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:53, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:53, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:53, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:4, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:53, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:53, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:53, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:53, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:53, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:53, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:53, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:53, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:53, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:53, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:4, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:53, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:53, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:53, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:53, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:53, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:53, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:53, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:53, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:53, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:53, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:4, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:53, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:53, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:53, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:53, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:53, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:53, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:53, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:53, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:53, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:53, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:4, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:53, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:53, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:53, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:287}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:53, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:53, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:53, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:53, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:53, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:53, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:53, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:260}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:4, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:53, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:53, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:53, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:53, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:53, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:287}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:53, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:53, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:53, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:53, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:287}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:53, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:4, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:54, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:54, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:54, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:54, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:54, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:54, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:54, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:54, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:54, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:54, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:4, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:54, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:54, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:54, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:54, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:54, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:54, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:54, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:54, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:54, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:54, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:4, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:54, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:54, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:54, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:54, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:54, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:54, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:54, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:54, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:54, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:54, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:4, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:54, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:54, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:54, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:54, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:54, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:54, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:288}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:54, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:54, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:54, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:54, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:4, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:54, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:54, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:54, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:54, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:54, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:54, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:54, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:54, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:54, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:54, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:4, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:54, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:54, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:54, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:54, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:54, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:54, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:54, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:54, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:54, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:54, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:4, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:54, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:54, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:54, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:54, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:54, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:54, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:54, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:54, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:54, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:54, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:4, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:54, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:54, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:54, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:54, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:54, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:54, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:54, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:54, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:54, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:54, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:4, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:54, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:54, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:54, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:54, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:54, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:54, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:54, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:54, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:54, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:54, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⇓'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:4, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:54, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:54, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:54, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:54, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:54, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:54, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:54, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:54, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:54, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:54, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:4, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:54, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:54, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:54, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:288}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:54, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:54, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:54, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:54, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:54, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:288}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:54, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:55, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:4, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:55, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:55, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:55, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:55, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:55, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:55, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:55, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:55, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:55, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:55, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:4, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:55, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:55, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:55, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:55, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:55, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:55, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:55, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:55, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:55, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:55, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:4, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:55, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:55, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:55, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:55, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:55, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:55, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:55, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:55, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:55, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:55, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:4, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:55, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:55, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:55, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:55, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:55, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:55, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:55, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:55, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:55, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:55, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:4, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:55, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:55, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:55, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:55, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:55, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:55, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:55, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:55, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:289}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:55, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:55, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:4, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:55, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:55, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:55, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:55, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:55, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:55, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:55, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:55, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:55, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:55, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:4, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:55, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:55, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:55, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:55, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:55, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:55, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:55, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:55, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:55, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:55, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:4, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:55, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:55, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:55, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:55, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:55, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:55, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:55, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:55, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:55, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:55, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:4, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:55, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:55, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:55, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:55, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:55, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:55, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:55, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:55, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:55, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:55, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:4, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:55, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:55, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:55, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:55, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:55, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:55, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:55, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:55, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:55, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:55, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:4, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:55, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:289}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:55, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:55, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:55, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:55, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:55, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:55, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:55, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:289}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:55, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:56, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:4, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:56, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:56, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:56, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:56, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:56, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:56, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:56, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:56, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:56, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:56, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:4, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:56, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:56, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:56, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:56, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:56, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:56, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:56, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:56, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:56, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:56, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:4, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:56, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:56, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:56, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:56, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:56, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:56, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:56, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:56, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:56, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:56, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:4, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:56, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:56, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:56, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:56, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:56, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:56, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:56, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:56, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:56, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:56, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:4, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:56, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:56, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:56, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:56, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:56, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:56, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:56, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:56, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:56, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:56, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:4, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:56, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:56, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:56, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:56, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:56, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:56, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:56, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:56, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:56, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:56, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:4, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:56, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:56, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:56, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:56, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:56, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:56, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:56, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:56, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:56, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:56, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:4, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:56, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:56, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:56, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:56, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:56, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:56, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:56, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:56, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:56, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:56, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:4, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:56, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:56, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:56, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:56, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:56, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:56, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:290}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:56, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:56, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:56, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:56, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:4, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:56, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:56, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:56, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:56, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:56, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:56, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:56, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:56, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:56, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:56, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:238}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:4, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:56, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:56, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:56, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:56, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:56, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:56, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:56, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:56, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:290}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:56, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:57, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:4, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:57, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:57, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:57, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:57, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:57, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:57, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:57, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:57, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:57, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:57, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:4, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:57, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:57, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:57, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:57, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:57, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:57, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:57, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:57, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:57, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:57, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:4, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:57, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:57, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:57, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:57, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:57, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:57, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:57, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:57, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:57, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:57, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:4, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:57, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:57, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:57, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:57, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:57, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:57, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:57, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:57, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:57, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:57, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:4, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:57, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:57, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:57, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:57, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:291}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:57, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:57, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:57, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:57, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:57, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:57, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:4, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:57, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:57, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:57, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:57, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:57, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:57, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:57, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:57, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:57, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:57, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⇟'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:4, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:57, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:57, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:57, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:57, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:57, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:57, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:57, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:57, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:57, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:57, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:4, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:57, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:57, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:57, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:57, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:57, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:57, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:57, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:57, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:57, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:57, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:4, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:57, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:57, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:57, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:57, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:291}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:57, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:57, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:57, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:57, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:57, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:57, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:4, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:57, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:57, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:57, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:57, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:57, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:57, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:57, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:57, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:291}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:57, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:57, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:4, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:57, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:57, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:57, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:57, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:57, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:57, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:57, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:57, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:291}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:57, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:58, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:4, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:58, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:58, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:58, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:58, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:58, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:58, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:58, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:58, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:58, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:58, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:4, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:58, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:58, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:58, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:58, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:58, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:58, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:58, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:58, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:58, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:58, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:4, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:58, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:58, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:58, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:58, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:58, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:58, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:58, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:58, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:58, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:58, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:4, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:58, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:58, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:58, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:58, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:58, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:58, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:58, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:58, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:58, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:58, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:4, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:58, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:58, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:292}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:58, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:58, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:58, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:58, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:58, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:58, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:58, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:58, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:4, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:58, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:58, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:58, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:58, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:58, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:58, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:58, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:58, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:58, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:58, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:4, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:58, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:58, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:58, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:58, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:58, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:58, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:58, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:58, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:58, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:58, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:4, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:58, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:58, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:58, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:58, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:58, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:58, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:58, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:58, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:58, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:58, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:4, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:58, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:58, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:58, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:58, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:58, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:58, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:58, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:58, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:58, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:58, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:4, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:58, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:58, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:58, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:58, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:58, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:58, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:292}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:58, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:58, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:58, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:58, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:4, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:58, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:58, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:58, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:58, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:58, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:58, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:58, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:58, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:292}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:58, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:59, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:4, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:59, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:59, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:59, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:59, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:59, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:59, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:59, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:59, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:59, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:59, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:4, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:59, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:59, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:59, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:59, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:59, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:59, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:59, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:59, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:59, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:59, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:4, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:59, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:59, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:59, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:59, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:59, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:59, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:59, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:59, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:59, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:59, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:4, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:59, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:59, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:59, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:59, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:59, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:59, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:59, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:59, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:59, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:59, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⇊'}]->(b);
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MATCH (a {`key`:238}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:4, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:59, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:59, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:59, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:59, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:59, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:59, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:59, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:59, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:59, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:59, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:4, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:59, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:59, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:59, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:59, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:59, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:59, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:59, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:59, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:59, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:59, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:4, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:59, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:59, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:59, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:59, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:59, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:59, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:59, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:59, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:59, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:59, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:4, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:59, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:59, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:59, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:59, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:59, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:59, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:59, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:59, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:59, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:59, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:4, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:59, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:59, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:59, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:59, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:59, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:59, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:59, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:59, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:59, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:59, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:4, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:59, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:59, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:59, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:59, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:59, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:59, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:59, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:59, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:59, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:59, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:4, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:59, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:59, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:59, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:59, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:59, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:59, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:59, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:293}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:59, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:293}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:59, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:60, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:4, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:60, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:60, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:60, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:60, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:60, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:60, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:60, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:60, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:60, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:60, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:238}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:4, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:60, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:60, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:60, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:60, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:60, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:60, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:60, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:60, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:60, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:60, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:5, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:60, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:60, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:60, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:60, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:60, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:60, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:60, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:60, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:60, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:60, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:5, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:60, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:60, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:60, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:60, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:60, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:60, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:60, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:60, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:60, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:60, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:5, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:60, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:60, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:60, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:60, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:60, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:60, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:60, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:60, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:60, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:60, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:5, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:60, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:60, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:60, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:60, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:60, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:60, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:60, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:60, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:60, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:60, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:5, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:60, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:60, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:60, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:60, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:60, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:60, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:60, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:60, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:60, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:60, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:5, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:60, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:60, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:60, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:60, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:60, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:60, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:60, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:60, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:60, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:60, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:5, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:60, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:60, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:60, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:60, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:60, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:60, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:60, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:60, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:60, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:60, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:5, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:60, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:60, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:294}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:60, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:60, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:60, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:60, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:60, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:60, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:60, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:60, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:5, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:60, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:60, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:60, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:60, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:60, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:60, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:294}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:60, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:60, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:294}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:60, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:61, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:5, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:61, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:61, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:61, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:61, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:61, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:61, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:61, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:61, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:61, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:61, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:5, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:61, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:61, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:61, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:61, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:61, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:61, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:61, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:61, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:61, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:61, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:5, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:61, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:61, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:61, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:61, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:61, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:61, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:61, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:61, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:61, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:61, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:5, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:61, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:61, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:61, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:61, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:61, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:61, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:61, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:61, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:61, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:61, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:5, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:61, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:61, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:61, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:61, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:61, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:61, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:61, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:61, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:61, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:61, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:5, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:61, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:61, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:61, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:61, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:61, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:61, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:61, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:61, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:61, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:61, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:5, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:61, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:61, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:61, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:61, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:61, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:61, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:61, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:61, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:61, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:61, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:5, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:61, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:61, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:61, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:61, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:61, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:61, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:61, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:61, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:61, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:61, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:5, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:61, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:61, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:61, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:61, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:61, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:61, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:61, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:61, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:61, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:61, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:5, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:61, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:61, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:61, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:61, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:61, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:61, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:61, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:61, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:61, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:61, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:5, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:61, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:61, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:61, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:61, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:295}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:61, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:61, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:61, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:61, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:295}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:61, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:62, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:5, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:62, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:62, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:62, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:62, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:62, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:62, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:62, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:62, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:62, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:62, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:5, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:62, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:62, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:62, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:62, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:62, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:62, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:62, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:62, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:62, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:62, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:5, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:62, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:62, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:62, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:62, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:62, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:62, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:62, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:62, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:62, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:62, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:5, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:62, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:62, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:62, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:62, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:62, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:62, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:62, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:62, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:62, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:62, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:5, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:62, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:62, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:62, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:62, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:62, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:62, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:62, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:62, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:62, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:62, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:5, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:62, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:62, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:62, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:62, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:62, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:62, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:62, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:62, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:62, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:62, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:5, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:62, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:62, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:62, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:62, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:62, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:62, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:62, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:62, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:62, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:62, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:5, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:62, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:62, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:62, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:62, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:62, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:62, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:62, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:62, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:62, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:62, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:5, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:62, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:62, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:62, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:62, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:62, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:62, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:62, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:62, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:62, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:62, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:5, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:62, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:62, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:62, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:62, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:62, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:62, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:62, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:62, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:62, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:62, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:5, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:62, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:62, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:296}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:62, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:62, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:62, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:62, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:62, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:62, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:296}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:62, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:63, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:261}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:5, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:63, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:63, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:63, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:63, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:63, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:63, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:63, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:63, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:63, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:63, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:5, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:63, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:63, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:63, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:63, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:63, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:63, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:63, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:63, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:0}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:63, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:63, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:5, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:63, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:63, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:63, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:63, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:63, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:63, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:63, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:63, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:63, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:63, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:5, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:63, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:0}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:63, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:63, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:63, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:63, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:63, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:63, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:63, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:63, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:63, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:5, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:63, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:63, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:63, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:63, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:63, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:0}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:63, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:63, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:63, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:63, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:63, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:5, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:63, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:63, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:6, `ratio`:0, `sourceIndex`:63, `targetIndex`:87, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:63, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:63, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:63, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:63, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:63, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:63, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:0}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:63, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:5, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:63, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:63, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:63, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:63, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:18, `ratio`:0, `sourceIndex`:63, `targetIndex`:99, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:63, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:63, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:63, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:63, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:63, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:5, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:63, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:63, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:0}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:63, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:63, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:63, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:63, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:63, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:63, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:63, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:63, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:5, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:63, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:63, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:63, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:63, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:63, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:63, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:0}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:63, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:63, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:63, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:63, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:5, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:63, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:63, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:63, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:63, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:63, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:63, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:63, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:63, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:63, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:63, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:32}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:5, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:63, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:63, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:63, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:63, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:63, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:63, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:63, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:63, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:0}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:63, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:64, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:5, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:64, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:64, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:1}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:64, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:64, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:64, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:64, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:64, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:64, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:64, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:64, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:5, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:64, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:64, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:64, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:64, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:64, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:64, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:1}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:64, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:64, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:64, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:64, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:5, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:64, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:64, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:64, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:64, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:64, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:64, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:64, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:64, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:64, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:64, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:5, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:64, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:64, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:64, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:64, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:64, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:64, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:64, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:64, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:64, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:64, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:5, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:64, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:64, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:64, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:64, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:64, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:64, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:64, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:64, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:64, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:64, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:5, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:64, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:64, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:64, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:8, `ratio`:0, `sourceIndex`:64, `targetIndex`:88, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:64, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:64, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:64, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:64, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:64, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:64, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:5, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:64, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:64, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:64, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:64, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:64, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:20, `ratio`:0, `sourceIndex`:64, `targetIndex`:100, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:64, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:64, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:64, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:64, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:5, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:64, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:64, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:64, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:64, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:64, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:64, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:64, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:64, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:64, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:64, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:5, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:64, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:64, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:64, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:64, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:64, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:64, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:64, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:64, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:64, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:64, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:33}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:5, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:64, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:64, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:64, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:64, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:64, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:64, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:64, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:1}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:64, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:64, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:64, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:5, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:64, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:64, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:64, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:64, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:64, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:64, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:64, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:64, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:1}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:64, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:65, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:5, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:65, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:65, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:65, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:65, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:65, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:65, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:65, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:65, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:65, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:65, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:5, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:65, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:65, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:65, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:65, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:65, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:65, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:65, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:65, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:65, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:65, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:5, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:65, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:65, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:65, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:65, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:65, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:65, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:65, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:65, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:65, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:65, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:5, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:65, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:65, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:65, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:65, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:65, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:65, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:65, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:65, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:65, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:65, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:5, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:65, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:65, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:65, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:65, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:65, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:65, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:65, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:65, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:65, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:65, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:5, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:65, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:65, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:65, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:65, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:10, `ratio`:0, `sourceIndex`:65, `targetIndex`:89, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:65, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:65, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:65, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:65, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:65, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:5, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:65, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:65, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:65, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:65, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:65, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:65, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:22, `ratio`:0, `sourceIndex`:65, `targetIndex`:101, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:65, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:65, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:2}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:65, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:5, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:65, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:65, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:65, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:65, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:65, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:65, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:65, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:65, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:65, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:65, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:36}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:5, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:65, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:65, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:65, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:65, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:65, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:65, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:65, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:65, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:65, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:65, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:5, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:65, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:65, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:65, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:65, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:65, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:2}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:65, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:65, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:65, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:65, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:65, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:5, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:65, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:65, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:65, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:65, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:65, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:65, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:65, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:65, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:2}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:65, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:66, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:5, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:66, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:66, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:66, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:66, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:66, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:66, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:66, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:66, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:66, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:66, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:5, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:66, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:66, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:66, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:66, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:66, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:66, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:66, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:66, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:66, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:66, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:5, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:66, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:66, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:66, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:66, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:66, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:66, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:66, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:66, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:66, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:66, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:5, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:66, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:66, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:66, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:66, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:66, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:66, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:66, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:66, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:66, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:66, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:5, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:66, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:66, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:66, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:0, `ratio`:0, `sourceIndex`:66, `targetIndex`:78, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:66, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:66, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:66, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:66, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:66, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:66, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:5, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:66, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:66, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:66, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:66, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:66, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:12, `ratio`:0, `sourceIndex`:66, `targetIndex`:90, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:66, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:66, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:66, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:66, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:5, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:66, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:66, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:66, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:66, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:66, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:66, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:66, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:3}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:24, `ratio`:0, `sourceIndex`:66, `targetIndex`:102, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:66, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:66, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:37}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:5, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:66, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:66, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:66, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:66, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:66, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:66, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:66, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:66, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:66, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:66, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:5, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:66, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:66, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:66, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:66, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:66, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:66, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:66, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:66, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:66, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:66, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:5, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:66, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:66, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:66, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:66, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:66, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:66, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:66, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:66, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:66, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:66, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:5, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:66, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:66, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:66, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:66, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:66, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:66, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:66, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:3}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:66, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:3}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:66, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:67, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:5, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:67, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:67, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:67, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:67, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:67, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:67, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:67, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:67, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:67, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:67, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:5, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:4}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:67, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:67, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:67, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:67, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:67, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:67, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:67, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:67, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:67, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:67, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:5, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:67, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:67, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:67, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:67, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:4}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:67, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:67, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:67, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:67, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:67, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:67, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:5, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:67, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:67, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:67, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:67, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:67, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:67, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:67, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:67, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:67, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:67, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:5, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:67, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:67, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:67, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:67, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:2, `ratio`:0, `sourceIndex`:67, `targetIndex`:79, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:67, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:67, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:67, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:67, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:67, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:5, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:67, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:67, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:67, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:67, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:67, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:67, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:14, `ratio`:0, `sourceIndex`:67, `targetIndex`:91, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:67, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:67, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:67, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:38}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:5, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:67, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:67, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:67, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:67, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:67, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:67, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:67, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:67, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:26, `ratio`:0, `sourceIndex`:67, `targetIndex`:103, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:67, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:5, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:67, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:67, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:67, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:67, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:67, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:67, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:67, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:67, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:67, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:67, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:5, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:67, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:67, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:67, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:67, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:67, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:67, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:67, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:67, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:67, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:67, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:5, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:67, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:67, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:67, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:67, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:67, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:67, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:67, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:67, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:67, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:67, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:5, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:67, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:67, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:67, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:67, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:67, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:4}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:67, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:67, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:67, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:4}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:67, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:68, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:5, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:68, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:68, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:68, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:68, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:68, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:68, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:68, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:68, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:68, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:68, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:5, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:68, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:68, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:68, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:68, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:68, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:68, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:68, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:68, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:68, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:68, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:5, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:68, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:68, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:5}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:68, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:68, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:68, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:68, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:68, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:68, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:68, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:68, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:5, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:68, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:68, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:68, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:68, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:68, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:68, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:68, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:68, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:68, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:68, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:5, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:68, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:68, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:68, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:68, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:68, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:4, `ratio`:0, `sourceIndex`:68, `targetIndex`:80, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:68, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:68, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:68, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:68, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:39}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩔'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:5, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:68, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:68, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:68, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:68, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:68, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:68, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:68, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:16, `ratio`:0, `sourceIndex`:68, `targetIndex`:92, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:68, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:68, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:5, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:68, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:68, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:68, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:68, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:68, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:68, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:68, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:68, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:68, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:28, `ratio`:0, `sourceIndex`:68, `targetIndex`:104, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:5, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:68, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:68, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:68, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:68, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:68, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:68, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:68, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:68, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:68, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:68, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:5, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:68, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:68, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:68, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:68, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:68, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:68, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:68, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:68, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:68, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:68, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:239}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:5, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:68, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:68, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:68, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:68, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:68, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:68, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:68, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:68, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:68, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:68, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:5, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:68, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:68, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:68, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:5}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:68, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:68, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:68, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:68, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:68, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:5}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:68, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:69, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:5, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:69, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:69, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:69, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:69, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:69, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:69, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:69, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:69, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:69, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:69, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:5, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:69, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:69, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:69, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:69, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:69, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:69, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:69, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:69, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:69, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:69, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:5, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:6}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:69, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:69, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:69, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:69, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:69, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:69, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:69, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:69, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:69, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:69, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:5, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:69, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:69, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:69, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:69, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:69, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:69, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:69, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:69, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:69, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:69, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:41}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:5, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:69, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:69, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:69, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:69, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:69, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:69, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:6, `ratio`:0, `sourceIndex`:69, `targetIndex`:81, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:69, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:69, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:69, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:5, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:69, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:69, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:69, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:69, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:69, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:69, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:69, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:69, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:18, `ratio`:0, `sourceIndex`:69, `targetIndex`:93, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:69, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:5, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:69, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:69, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:69, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:69, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:69, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:69, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:69, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:69, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:69, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:69, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:5, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:30, `ratio`:0, `sourceIndex`:69, `targetIndex`:105, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:69, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:69, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:69, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:69, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:69, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:69, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:69, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:69, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:69, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:5, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:69, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:69, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:69, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:69, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:69, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:69, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:69, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:69, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:69, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:69, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:5, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:69, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:69, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:69, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:69, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:69, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:69, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:69, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:69, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:69, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:69, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:5, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:69, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:6}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:69, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:69, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:69, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:69, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:69, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:69, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:69, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:6}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:69, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:70, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:239}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:5, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:70, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:70, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:70, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:70, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:70, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:70, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:70, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:70, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:70, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:70, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:6, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:70, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:70, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:70, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:70, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:70, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:70, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:70, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:70, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:70, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:70, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:6, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:70, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:70, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:70, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:70, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:70, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:70, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:70, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:70, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:70, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:70, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:42}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩬'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:6, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:70, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:70, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:70, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:70, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:70, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:70, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:70, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:70, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:70, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:70, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:6, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:70, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:70, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:70, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:70, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:70, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:70, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:70, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:8, `ratio`:0, `sourceIndex`:70, `targetIndex`:82, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:70, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:70, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:6, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:70, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:70, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:70, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:70, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:70, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:70, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:70, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:70, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:70, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:20, `ratio`:0, `sourceIndex`:70, `targetIndex`:94, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:6, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:70, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:70, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:70, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:70, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:70, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:70, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:70, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:70, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:70, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:70, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:6, `targetIndex`:7, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:70, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:32, `ratio`:0, `sourceIndex`:70, `targetIndex`:106, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:70, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:70, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:7}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:70, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:70, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:70, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:70, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:70, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:70, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:6, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:70, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:70, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:70, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:70, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:70, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:70, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:70, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:70, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:70, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:70, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:6, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:70, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:70, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:70, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:70, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:70, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:70, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:70, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:70, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:70, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:70, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:6, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:70, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:7}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:70, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:70, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:70, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:70, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:70, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:70, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:70, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:7}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:70, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:71, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:6, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:71, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:71, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:71, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:71, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:71, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:71, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:71, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:71, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:71, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:71, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:6, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:71, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:71, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:71, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:71, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:71, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:71, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:71, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:71, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:71, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:71, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:43}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:6, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:71, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:71, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:71, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:71, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:71, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:71, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:71, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:71, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:71, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:71, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:6, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:71, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:8}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:71, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:71, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:71, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:71, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:71, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:71, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:71, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:71, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:71, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:6, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:71, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:71, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:71, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:71, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:71, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:71, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:71, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:71, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:10, `ratio`:0, `sourceIndex`:71, `targetIndex`:83, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:71, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:6, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:71, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:71, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:71, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:71, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:71, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:71, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:71, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:71, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:71, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:71, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:6, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:22, `ratio`:0, `sourceIndex`:71, `targetIndex`:95, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:71, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:71, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:71, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:71, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:71, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:71, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:71, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:71, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:71, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:6, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:71, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:71, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:71, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:71, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:71, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:71, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:71, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:71, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:71, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:71, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:6, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:71, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:71, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:71, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:71, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:71, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:71, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:8}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:71, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:71, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:71, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:71, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:6, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:71, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:71, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:71, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:71, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:71, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:71, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:71, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:71, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:71, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:71, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:6, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:71, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:71, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:71, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:71, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:71, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:71, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:71, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:71, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:8}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:71, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:72, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:6, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:72, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:72, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:72, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:72, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:72, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:72, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:72, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:72, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:72, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:72, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:262}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:44}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:6, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:72, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:72, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:72, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:72, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:72, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:72, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:72, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:72, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:72, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:72, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:6, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:72, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:72, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:72, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:72, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:72, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:72, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:72, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:72, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:72, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:72, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:6, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:72, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:72, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:72, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:72, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:72, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:72, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:72, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:72, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:72, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:72, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:6, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:72, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:72, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:72, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:72, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:72, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:72, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:72, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:72, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:12, `ratio`:0, `sourceIndex`:72, `targetIndex`:84, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:72, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:6, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:72, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:72, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:72, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:72, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:72, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:72, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:72, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:72, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:72, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:72, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:6, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:24, `ratio`:0, `sourceIndex`:72, `targetIndex`:96, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:72, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:72, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:72, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:72, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:72, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:72, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:72, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:72, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:72, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:6, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:72, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:72, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:72, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:72, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:72, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:72, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:72, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:72, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:72, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:72, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:6, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:72, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:72, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:72, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:72, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:72, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:72, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:72, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:72, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:72, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:72, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:6, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:72, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:72, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:72, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:72, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:72, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:72, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:72, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:9}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:72, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:72, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:72, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:6, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:72, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:72, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:72, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:72, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:72, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:72, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:72, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:9}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:72, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:73, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:73, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:45}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩔'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:6, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:73, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:73, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:73, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:73, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:73, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:73, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:73, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:73, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:73, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:73, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:6, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:73, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:73, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:73, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:73, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:73, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:73, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:73, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:73, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:73, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:73, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:6, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:73, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:73, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:73, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:73, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:73, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:73, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:73, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:73, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:73, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:73, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:6, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:73, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:73, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:73, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:73, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:73, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:73, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:73, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:73, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:73, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:73, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:6, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:73, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:73, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:73, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:73, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:73, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:73, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:73, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:73, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:73, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:73, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:6, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:73, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:73, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:73, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:73, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:73, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:73, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:73, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:73, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:73, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:73, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:6, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:73, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:73, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:73, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:73, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:73, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:73, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:73, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:73, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:73, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:73, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:6, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:73, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:73, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:73, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:73, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:73, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:73, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:73, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:73, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:73, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:73, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:6, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:73, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:10}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:73, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:73, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:73, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:73, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:73, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:73, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:73, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:73, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:73, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:6, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:73, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:73, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:73, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:73, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:73, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:10}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:73, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:73, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:73, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:73, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:73, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:46}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:6, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:73, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:73, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:73, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:73, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:73, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:73, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:10}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:73, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:74, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:74, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:74, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:6, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:74, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:74, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:74, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:74, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:74, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:74, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:74, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:74, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:74, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:74, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:6, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:74, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:11}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:74, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:74, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:74, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:74, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:74, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:74, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:74, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:74, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:74, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:6, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:74, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:74, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:74, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:74, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:74, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:74, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:74, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:74, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:74, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:74, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:6, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:74, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:74, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:74, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:74, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:74, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:74, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:74, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:74, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:74, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:74, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:6, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:74, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:74, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:74, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:74, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:74, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:74, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:74, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:74, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:74, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:74, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:6, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:74, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:74, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:74, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:74, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:74, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:74, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:74, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:74, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:74, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:74, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:6, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:74, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:74, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:74, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:74, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:74, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:74, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:74, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:74, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:74, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:74, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:6, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:74, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:74, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:74, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:74, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:74, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:74, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:74, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:74, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:74, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:74, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:6, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:74, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:74, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:74, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:74, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:74, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:74, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:74, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:74, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:74, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:74, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:53}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:6, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:74, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:74, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:11}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:74, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:74, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:74, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:74, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:74, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:74, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:74, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:74, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:6, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:74, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:74, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:74, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:74, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:74, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:11}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:74, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:75, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:75, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:75, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:75, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:6, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:75, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:75, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:75, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:75, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:75, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:75, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:75, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:75, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:75, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:75, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:6, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:75, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:75, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:75, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:75, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:75, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:75, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:75, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:75, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:75, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:75, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:6, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:75, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:75, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:75, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:75, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:75, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:75, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:75, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:75, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:75, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:75, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:6, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:75, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:75, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:75, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:75, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:75, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:75, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:75, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:75, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:75, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:75, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:6, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:75, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:75, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:75, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:75, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:75, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:75, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:75, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:75, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:75, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:75, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:6, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:75, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:75, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:75, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:75, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:75, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:75, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:75, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:75, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:75, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:75, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:6, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:75, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:75, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:75, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:75, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:75, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:75, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:75, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:75, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:75, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:75, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:6, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:75, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:75, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:75, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:75, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:75, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:75, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:75, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:75, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:75, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:75, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:55}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:6, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:75, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:75, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:75, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:75, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:75, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:75, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:75, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:75, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:75, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:75, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:6, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:75, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:75, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:75, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:75, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:75, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:75, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:75, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:75, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:75, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:75, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:6, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:75, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:75, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:75, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:12}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:75, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:12}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:75, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:76, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:76, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:76, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:76, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:76, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:6, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:76, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:76, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:76, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:76, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:76, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:76, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:76, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:76, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:76, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:76, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:6, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:76, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:76, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:76, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:76, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:76, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:76, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:76, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:76, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:76, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:76, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:6, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:76, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:76, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:76, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:76, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:76, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:76, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:76, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:76, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:76, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:76, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:6, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:76, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:76, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:76, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:76, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:76, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:76, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:76, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:76, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:76, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:76, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:6, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:76, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:76, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:76, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:76, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:76, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:76, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:76, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:76, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:76, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:76, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:6, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:76, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:76, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:76, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:76, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:76, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:76, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:76, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:76, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:76, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:76, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:6, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:76, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:76, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:76, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:76, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:76, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:76, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:76, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:76, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:76, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:76, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:60}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:6, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:76, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:76, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:76, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:76, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:76, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:76, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:76, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:76, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:76, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:76, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:6, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:76, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:76, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:76, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:76, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:76, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:76, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:76, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:76, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:76, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:13}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:76, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:6, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:76, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:76, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:76, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:76, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:76, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:76, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:76, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:76, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:76, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:76, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:6, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:76, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:76, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:13}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:76, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:13}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:76, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:77, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:77, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:77, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:77, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:77, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:77, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:6, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:77, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:77, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:77, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:77, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:77, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:77, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:77, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:77, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:77, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:77, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:6, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:77, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:77, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:77, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:77, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:77, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:77, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:77, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:77, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:77, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:77, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:6, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:77, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:77, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:77, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:77, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:77, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:77, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:77, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:77, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:77, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:77, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:6, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:77, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:77, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:77, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:77, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:77, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:77, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:77, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:77, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:77, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:77, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:6, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:77, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:77, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:77, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:77, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:77, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:77, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:77, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:16}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:77, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:77, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:77, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:6, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:77, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:77, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:77, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:77, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:77, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:77, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:77, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:77, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:77, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:77, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:61}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:6, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:77, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:77, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:77, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:77, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:77, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:77, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:77, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:77, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:77, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:77, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:6, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:77, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:77, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:77, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:77, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:77, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:77, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:77, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:77, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:77, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:77, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:6, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:77, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:77, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:77, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:77, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:77, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:77, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:77, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:77, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:77, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:77, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:6, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:77, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:77, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:77, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:77, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:77, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:77, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:77, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:77, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:77, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:77, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:6, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:16}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:77, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:77, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:16}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:77, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:78, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:78, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:78, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:78, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:78, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:78, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:78, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:6, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:78, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:78, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:78, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:78, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:78, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:78, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:78, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:78, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:78, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:78, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:6, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:78, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:78, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:78, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:78, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:78, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:78, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:78, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:78, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:78, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:78, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:6, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:78, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:78, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:78, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:78, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:78, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:0, `ratio`:0, `sourceIndex`:78, `targetIndex`:66, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:78, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:78, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:78, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:78, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:6, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:78, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:78, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:78, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:78, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:78, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:78, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:78, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:78, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:78, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:78, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:6, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:78, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:78, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:78, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:78, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:78, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:78, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:78, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:78, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:78, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:78, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:133}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:6, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:78, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:78, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:78, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:78, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:78, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:78, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:78, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:78, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:78, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:78, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:6, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:78, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:78, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:78, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:78, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:78, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:78, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:78, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:78, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:78, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:78, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:6, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:78, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:78, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:78, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:78, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:78, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:78, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:78, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:78, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:78, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:78, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:6, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:78, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:78, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:78, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:78, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:78, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:78, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:78, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:78, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:78, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:78, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:6, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:78, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:78, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:78, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:78, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:78, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:78, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:78, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:78, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:78, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:17}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:78, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:6, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:78, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:17}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:78, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:79, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:79, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:79, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:79, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:79, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:79, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:79, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:79, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:6, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:79, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:79, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:18}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:79, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:79, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:79, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:79, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:79, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:79, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:79, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:79, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:6, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:79, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:79, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:79, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:79, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:79, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:79, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:18}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:79, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:79, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:79, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:79, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:6, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:79, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:79, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:79, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:79, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:79, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:2, `ratio`:0, `sourceIndex`:79, `targetIndex`:67, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:79, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:79, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:79, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:79, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:6, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:79, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:79, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:79, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:79, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:79, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:79, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:79, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:79, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:79, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:79, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:134}) MATCH (b {`key`:222}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:7.83,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:6.211966011250105, `operationalPlus`:9.448033988749895, `operator`:'⩔'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:6, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:79, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:79, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:79, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:79, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:79, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:79, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:79, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:79, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:79, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:79, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:6, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:79, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:79, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:79, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:79, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:79, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:79, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:18}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:79, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:79, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:79, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:79, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:6, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:79, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:79, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:79, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:79, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:79, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:79, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:79, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:79, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:79, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:79, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:240}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:6, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:79, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:79, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:79, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:79, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:79, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:79, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:79, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:79, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:79, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:79, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:6, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:79, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:79, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:79, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:18}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:79, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:79, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:79, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:79, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:79, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:79, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:79, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:6, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:79, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:79, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:79, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:79, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:79, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:79, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:79, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:18}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:79, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:79, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:79, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:240}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:6, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:18}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:79, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:80, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:80, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:80, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:80, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:80, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:80, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:80, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:80, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:80, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:7, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:80, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:80, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:80, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:80, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:80, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:80, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:80, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:80, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:80, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:80, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:7, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:80, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:80, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:80, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:80, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:80, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:80, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:80, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:80, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:80, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:80, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:7, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:80, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:80, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:80, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:80, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:80, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:4, `ratio`:0, `sourceIndex`:80, `targetIndex`:68, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:80, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:80, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:80, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:80, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:135}) MATCH (b {`key`:223}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:174.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:172.3819660112501, `operationalPlus`:175.6180339887499, `operator`:'⇟'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:7, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:80, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:80, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:80, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:80, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:80, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:80, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:80, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:80, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:80, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:80, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:7, `targetIndex`:4, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:80, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:80, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:80, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:80, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:80, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:80, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:80, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:80, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:80, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:80, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:7, `targetIndex`:5, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:80, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:80, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:80, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:80, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:80, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:80, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:80, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:80, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:80, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:80, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:7, `targetIndex`:6, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:80, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:80, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:80, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:80, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:80, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:80, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:80, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:80, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:80, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:80, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:7, `targetIndex`:8, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:80, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:80, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:80, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:80, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:80, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:80, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:80, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:80, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:80, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:80, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:7, `targetIndex`:9, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:80, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:80, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:19}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:80, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:80, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:80, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:80, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:80, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:80, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:80, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:80, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:7, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:80, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:80, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:80, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:80, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:80, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:80, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:19}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:80, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:80, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:80, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:19}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:80, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:7, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:81, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:81, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:81, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:81, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:81, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:81, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:81, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:81, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:81, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:81, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:7, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:81, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:81, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:81, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:81, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:81, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:81, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:81, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:81, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:81, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:81, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:7, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:81, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:81, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:81, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:81, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:81, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:81, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:81, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:81, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:81, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:81, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:263}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩔'}]->(b);
MATCH (a {`key`:136}) MATCH (b {`key`:224}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:528.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:526.3819660112501, `operationalPlus`:529.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:7, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:81, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:81, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:81, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:81, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:81, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:6, `ratio`:0, `sourceIndex`:81, `targetIndex`:69, `zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:81, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:81, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:81, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:81, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:7, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:81, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:81, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:81, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:81, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:81, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:81, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:81, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:81, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:81, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:81, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:7, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:81, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:81, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:81, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:81, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:81, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:81, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:81, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:81, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:81, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:81, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:7, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:81, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:81, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:81, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:20}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:81, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:81, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:81, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:81, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:81, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:81, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:81, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:7, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:81, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:81, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:81, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:81, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:81, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:81, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:81, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:81, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:81, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:81, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:7, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:81, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:81, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:81, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:81, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:81, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:81, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:81, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:81, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:81, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:81, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:7, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:81, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:81, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:81, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:81, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:81, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:81, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:81, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:81, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:81, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:81, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:7, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:81, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:81, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:81, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:81, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:20}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:81, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:81, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:81, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:81, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:20}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:81, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:82, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:7, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:82, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:82, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:82, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:82, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:82, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:82, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:82, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:82, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:82, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:82, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:7, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:82, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:82, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:82, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:82, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:82, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:82, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:82, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:82, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:82, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:82, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:137}) MATCH (b {`key`:225}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:432.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:430.3819660112501, `operationalPlus`:433.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:7, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:82, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:82, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:82, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:82, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:82, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:82, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:82, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:82, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:82, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:82, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:7, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:82, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:82, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:82, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:82, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:82, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:8, `ratio`:0, `sourceIndex`:82, `targetIndex`:70, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:82, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:82, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:82, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:82, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:7, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:82, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:82, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:82, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:82, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:82, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:82, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:82, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:82, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:82, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:82, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:261}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:7, `targetIndex`:27,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:82, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:82, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:82, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:82, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:82, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:82, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:82, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:82, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:82, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:82, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:262}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:7, `targetIndex`:28,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:82, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:82, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:82, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:82, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:82, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:82, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:82, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:82, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:82, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:82, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:263}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:7, `targetIndex`:29,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:82, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:82, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:82, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:82, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:82, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:82, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:82, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:82, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:82, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:82, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:264}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:7, `targetIndex`:30,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:82, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:82, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:82, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:82, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:82, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:82, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:82, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:82, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:82, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:82, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:265}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:7, `targetIndex`:31,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:82, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:82, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:82, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:82, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:82, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:82, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:82, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:82, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:82, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:82, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:266}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:7, `targetIndex`:32,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:82, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:82, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:21}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:82, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:82, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:82, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:82, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:82, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:21}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:82, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:83, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:83, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:267}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:7, `targetIndex`:33,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:83, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:83, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:83, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:83, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:83, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:83, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:83, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:83, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:83, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:83, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:138}) MATCH (b {`key`:226}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:126.22,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:124.6019660112501, `operationalPlus`:127.8380339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:268}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:7, `targetIndex`:34,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:83, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:83, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:83, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:83, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:83, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:83, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:83, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:83, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:83, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:83, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:269}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:7, `targetIndex`:35,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:83, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:83, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:83, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:83, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:260}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:83, `targetIndex`:26,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:83, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:83, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:83, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:83, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:83, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:270}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:7, `targetIndex`:36,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:83, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:83, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:83, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:83, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:83, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:10, `ratio`:0, `sourceIndex`:83, `targetIndex`:71, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:83, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:83, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:83, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:83, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:271}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:7, `targetIndex`:37,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:83, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:83, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:83, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:83, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:83, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:83, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:83, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:83, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:83, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:83, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:272}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:7, `targetIndex`:38,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:83, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:83, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:83, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:83, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:83, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:83, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:83, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:83, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:83, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:83, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:273}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:7, `targetIndex`:39,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:83, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:83, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:83, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:83, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:83, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:83, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:83, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:83, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:83, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:83, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:274}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:7, `targetIndex`:40,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:83, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:83, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:83, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:83, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:83, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:83, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:83, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:83, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:83, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:83, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:275}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:7, `targetIndex`:41,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:83, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:83, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:83, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:83, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:83, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:83, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:83, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:83, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:83, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:83, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:276}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:7, `targetIndex`:42,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:83, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:83, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:83, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:83, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:83, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:83, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:83, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:83, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:83, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:83, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:277}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:7, `targetIndex`:43,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:22}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:83, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:83, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:83, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:83, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:83, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:83, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:22}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:83, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:84, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:84, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:84, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:139}) MATCH (b {`key`:227}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:210.42,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:208.8019660112501, `operationalPlus`:212.03803398874987, `operator`:'⩒'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:278}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:7, `targetIndex`:44,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:84, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:84, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:84, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:23}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:84, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:84, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:84, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:84, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:84, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:84, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:84, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:279}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:7, `targetIndex`:45,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:84, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:84, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:84, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:84, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:84, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:84, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:84, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:84, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:84, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:84, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:280}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:7, `targetIndex`:46,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:84, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:84, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:259}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:84, `targetIndex`:25,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:84, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:84, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:84, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:84, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:84, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:84, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:84, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:281}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:7, `targetIndex`:47,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:84, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:84, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:84, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:84, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:84, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:12, `ratio`:0, `sourceIndex`:84, `targetIndex`:72, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:84, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:84, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:84, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:84, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:282}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:7, `targetIndex`:48,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:84, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:84, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:84, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:84, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:84, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:84, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:84, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:84, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:84, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:84, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:283}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:7, `targetIndex`:49,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:84, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:84, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:84, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:84, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:84, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:84, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:84, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:84, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:84, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:84, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:284}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:7, `targetIndex`:50,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:84, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:84, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:84, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:84, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:84, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:84, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:84, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:84, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:84, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:84, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:285}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:7, `targetIndex`:51,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:84, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:84, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:84, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:84, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:84, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:84, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:84, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:84, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:84, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:84, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:286}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:7, `targetIndex`:52,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:84, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:84, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:84, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:84, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:84, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:84, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:84, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:84, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:84, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:84, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:287}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:7, `targetIndex`:53,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:84, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:84, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:84, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:84, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:84, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:84, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:84, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:84, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:84, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:23}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:84, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:140}) MATCH (b {`key`:228}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:741.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:739.3819660112501, `operationalPlus`:742.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:7, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:84, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:84, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:84, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4006,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:84, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:84, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:23}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:84, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:85, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:85, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:85, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:85, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:7, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:85, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:85, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:85, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:85, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:85, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:85, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:85, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:85, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:85, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:85, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:7, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:85, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:85, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:85, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:85, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:85, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:85, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:85, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:85, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:85, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:85, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:7, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:258}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:85, `targetIndex`:24,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:85, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:85, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:85, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:85, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:85, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:85, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:85, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:85, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:85, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:7, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:85, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:85, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:85, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:85, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:85, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:85, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:85, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:85, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:85, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:85, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:7, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:85, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:85, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:85, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:85, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:85, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:85, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:85, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:85, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:85, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:85, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:7, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:85, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:85, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:85, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:85, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:85, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:85, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:85, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:85, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:85, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:85, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:7, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:85, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:85, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:85, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:85, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:85, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:85, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:85, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:85, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:85, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:85, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:241}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:7, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:85, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:85, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:85, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:85, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:85, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:85, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:85, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:85, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:85, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:85, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:7, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:85, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:85, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:85, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:85, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:85, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:85, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:85, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:85, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:85, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:85, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:141}) MATCH (b {`key`:229}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:852.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:850.3819660112501, `operationalPlus`:853.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:7, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:85, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:85, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:85, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4036,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:85, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:85, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:85, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:24}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:85, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:85, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:85, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:85, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:7, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:85, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:85, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:85, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:85, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:24}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:85, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:86, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:86, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:86, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:86, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:86, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:7, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:86, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:86, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:86, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:86, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:86, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:86, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:86, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:86, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:86, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:86, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:7, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:86, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:86, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:86, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:27}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:86, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:86, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:86, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:86, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:86, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:257}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:86, `targetIndex`:23,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:86, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:7, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:86, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:86, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:86, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:86, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:86, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:86, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:86, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:86, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:86, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:86, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:7, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:86, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:86, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:86, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:86, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:86, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:86, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:86, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:86, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:86, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:86, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:7, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:86, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:86, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:86, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:86, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:86, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:86, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:86, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:86, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:86, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:86, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:7, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:86, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:86, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:86, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:86, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:86, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:86, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:86, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:86, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:86, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:86, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:7, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:86, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:86, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:86, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:86, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:86, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:86, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:86, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:86, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:86, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:86, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:7, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:86, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:86, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:86, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:86, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:86, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:86, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:86, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:86, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:86, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:86, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:142}) MATCH (b {`key`:230}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:396.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:394.3819660112501, `operationalPlus`:397.6180339887499, `operator`:'⇊'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:7, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:86, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:86, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:86, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:86, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:86, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:86, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:86, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:86, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:86, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:86, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:7, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:86, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:86, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:86, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:86, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:27}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:86, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:86, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:86, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:86, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:86, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:86, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:7, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:86, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:86, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4025,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:86, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:27}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:86, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:87, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:87, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:87, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:87, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:87, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:87, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:7, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:87, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:87, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:87, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:87, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:87, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:87, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:87, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:87, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:87, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:87, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:7, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:87, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:28}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:87, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:87, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:87, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:87, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:87, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:256}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:87, `targetIndex`:22,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:87, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:87, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:87, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:7, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:87, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:87, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:87, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:6, `ratio`:0, `sourceIndex`:87, `targetIndex`:63, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:87, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:87, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:87, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:87, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:87, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:87, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:7, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:87, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:87, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:87, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:87, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:87, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:87, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:87, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:87, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:87, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:87, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:7, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:87, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:87, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:87, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:87, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:87, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:87, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:87, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:87, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:87, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:87, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:7, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:87, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:87, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:87, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:87, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:87, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:87, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:87, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:87, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:87, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:87, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:7, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:87, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:87, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:87, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:87, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:87, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:87, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:87, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:87, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:87, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:87, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:143}) MATCH (b {`key`:231}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:963.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:961.3819660112501, `operationalPlus`:964.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:7, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:87, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:87, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:87, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:87, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:87, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:87, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:87, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:87, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:87, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:87, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:7, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:87, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:87, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:87, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:87, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:87, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:87, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:87, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:87, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:87, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:87, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:7, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:87, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:87, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:28}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4023,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:87, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4062,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:87, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:87, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:87, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:87, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:87, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:87, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:87, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:7, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:87, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4041,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:87, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:28}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:87, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:88, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:88, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:88, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:88, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:88, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:88, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:88, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:7, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:88, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:88, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:88, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:88, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:88, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:88, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:88, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:88, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:88, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:88, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:241}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:7, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:88, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:88, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:88, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:88, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:255}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:88, `targetIndex`:21,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:88, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:88, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:88, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:88, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:88, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:7, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:88, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:88, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:88, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
```

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MATCH (a {`key`:29}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:8, `ratio`:0, `sourceIndex`:88, `targetIndex`:64, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:88, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:88, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:88, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:88, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:88, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:88, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:7, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:88, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:88, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:88, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:88, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:88, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:88, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:88, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:88, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:88, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:88, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:7, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:88, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:88, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:88, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:88, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:88, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:88, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:88, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:30}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:88, `targetIndex`:89,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:34, `ratio`:0, `sourceIndex`:88, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:88, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:7, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:88, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:88, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:88, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:88, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:88, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:88, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:88, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:88, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:88, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:88, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:144}) MATCH (b {`key`:232}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:285.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:283.3819660112501, `operationalPlus`:286.6180339887499, `operator`:'⩔'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:7, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:88, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:88, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:88, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:88, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:88, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:88, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:88, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:88, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:88, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:88, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:7, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:88, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:88, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:88, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:88, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:88, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:88, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:88, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:88, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.305,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:88, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:88, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:7, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:88, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.302,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:88, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:88, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:88, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:88, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:88, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:88, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:88, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:88, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:88, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:7, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:29}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.405,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:88, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4123,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:88, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:88, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.402,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:88, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:88, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:88, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:88, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:88, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4022,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:88, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4007,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:88, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4018,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:7, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4038,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:88, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:29}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4012,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:88, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:89, `ratio`:0, `sourceIndex`:89, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:89, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:89, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:89, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:89, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:89, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:89, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:89, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:7, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:89, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:89, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:89, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:89, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:89, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:89, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:89, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:89, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:89, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:89, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:7, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:89, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:89, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:254}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:89, `targetIndex`:20,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:89, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:89, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:89, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:89, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:89, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:89, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:89, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:7, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:89, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:89, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:89, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:10, `ratio`:0, `sourceIndex`:89, `targetIndex`:65, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:89, `targetIndex`:66,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:12, `ratio`:0, `sourceIndex`:89, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:89, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:89, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:89, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:89, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:7, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:89, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:89, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:89, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:89, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:89, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:89, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:89, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:89, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:89, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:89, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:7, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:89, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:22}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:89, `targetIndex`:83,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:23}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:29, `ratio`:0, `sourceIndex`:89, `targetIndex`:84,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:24}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:30, `ratio`:0, `sourceIndex`:89, `targetIndex`:85,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:27}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:31, `ratio`:0, `sourceIndex`:89, `targetIndex`:86,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:28}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:32, `ratio`:0, `sourceIndex`:89, `targetIndex`:87,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:29}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:33, `ratio`:0, `sourceIndex`:89, `targetIndex`:88,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:31}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:35, `ratio`:0, `sourceIndex`:89, `targetIndex`:90,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:32}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:36, `ratio`:0, `sourceIndex`:89, `targetIndex`:91,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:33}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:37, `ratio`:0, `sourceIndex`:89, `targetIndex`:92,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:145}) MATCH (b {`key`:233}) CREATE (a)-[:`FOCAL_TARGET` {`cascadeFrequency`:639.0,
`description`:'Dynamic force — alternate parent provides operational frequency',
`operationalMinus`:637.3819660112501, `operationalPlus`:640.6180339887499, `operator`:'⩒'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:7, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:36}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:38, `ratio`:0, `sourceIndex`:89, `targetIndex`:93,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:37}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:39, `ratio`:0, `sourceIndex`:89, `targetIndex`:94,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:38}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:40, `ratio`:0, `sourceIndex`:89, `targetIndex`:95,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:39}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:41, `ratio`:0, `sourceIndex`:89, `targetIndex`:96,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:41}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:42, `ratio`:0, `sourceIndex`:89, `targetIndex`:97,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:42}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:43, `ratio`:0, `sourceIndex`:89, `targetIndex`:98,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:43}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:44, `ratio`:0, `sourceIndex`:89, `targetIndex`:99,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:44}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:45, `ratio`:0, `sourceIndex`:89, `targetIndex`:100,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:45}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:46, `ratio`:0, `sourceIndex`:89, `targetIndex`:101,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:46}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:47, `ratio`:0, `sourceIndex`:89, `targetIndex`:102,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:7, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:53}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:48, `ratio`:0, `sourceIndex`:89, `targetIndex`:103,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:55}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:49, `ratio`:0, `sourceIndex`:89, `targetIndex`:104,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:60}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:50, `ratio`:0, `sourceIndex`:89, `targetIndex`:105,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:61}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:51, `ratio`:0, `sourceIndex`:89, `targetIndex`:106,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:133}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:52, `ratio`:0, `sourceIndex`:89, `targetIndex`:107,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:134}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:53, `ratio`:0, `sourceIndex`:89, `targetIndex`:108,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:135}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:54, `ratio`:0, `sourceIndex`:89, `targetIndex`:109,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:136}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:55, `ratio`:0, `sourceIndex`:89, `targetIndex`:110,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:137}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:56, `ratio`:0, `sourceIndex`:89, `targetIndex`:111,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:138}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:57, `ratio`:0, `sourceIndex`:89, `targetIndex`:112,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:7, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:139}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:58, `ratio`:0, `sourceIndex`:89, `targetIndex`:113,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:140}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:59, `ratio`:0, `sourceIndex`:89, `targetIndex`:114,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:141}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:60, `ratio`:0, `sourceIndex`:89, `targetIndex`:115,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:142}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:61, `ratio`:0, `sourceIndex`:89, `targetIndex`:116,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:143}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:62, `ratio`:0, `sourceIndex`:89, `targetIndex`:117,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:144}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:63, `ratio`:0, `sourceIndex`:89, `targetIndex`:118,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:145}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:64, `ratio`:0, `sourceIndex`:89, `targetIndex`:119,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:146}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:65, `ratio`:0, `sourceIndex`:89, `targetIndex`:120,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:147}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.316,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:66, `ratio`:0, `sourceIndex`:89, `targetIndex`:121,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:148}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:67, `ratio`:0, `sourceIndex`:89, `targetIndex`:122,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:7, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:149}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:68, `ratio`:0, `sourceIndex`:89, `targetIndex`:123,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:150}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.307,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:69, `ratio`:0, `sourceIndex`:89, `targetIndex`:124,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:151}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:70, `ratio`:0, `sourceIndex`:89, `targetIndex`:125,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:152}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:71, `ratio`:0, `sourceIndex`:89, `targetIndex`:126,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:153}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:72, `ratio`:0, `sourceIndex`:89, `targetIndex`:127,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:154}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:73, `ratio`:0, `sourceIndex`:89, `targetIndex`:128,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:155}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:74, `ratio`:0, `sourceIndex`:89, `targetIndex`:129,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:156}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:75, `ratio`:0, `sourceIndex`:89, `targetIndex`:130,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:157}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:76, `ratio`:0, `sourceIndex`:89, `targetIndex`:131,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:30}) MATCH (b {`key`:158}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4029,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:77, `ratio`:0, `sourceIndex`:89, `targetIndex`:132,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:7, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:159}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.416,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:78, `ratio`:0, `sourceIndex`:89, `targetIndex`:133,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:160}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:79, `ratio`:0, `sourceIndex`:89, `targetIndex`:134,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:161}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:80, `ratio`:0, `sourceIndex`:89, `targetIndex`:135,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:162}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.407,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:81, `ratio`:0, `sourceIndex`:89, `targetIndex`:136,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:163}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:82, `ratio`:0, `sourceIndex`:89, `targetIndex`:137,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:164}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:83, `ratio`:0, `sourceIndex`:89, `targetIndex`:138,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:165}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4009,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:84, `ratio`:0, `sourceIndex`:89, `targetIndex`:139,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:166}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4032,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:85, `ratio`:0, `sourceIndex`:89, `targetIndex`:140,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:167}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:86, `ratio`:0, `sourceIndex`:89, `targetIndex`:141,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:168}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4079,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:87, `ratio`:0, `sourceIndex`:89, `targetIndex`:142,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:241}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:7, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:30}) MATCH (b {`key`:169}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4014,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:88, `ratio`:0, `sourceIndex`:89, `targetIndex`:143,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:90, `ratio`:0, `sourceIndex`:90, `targetIndex`:0,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:91, `ratio`:0, `sourceIndex`:90, `targetIndex`:1,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:92, `ratio`:0, `sourceIndex`:90, `targetIndex`:2,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:93, `ratio`:0, `sourceIndex`:90, `targetIndex`:3,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:238}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:94, `ratio`:0, `sourceIndex`:90, `targetIndex`:4,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:239}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:95, `ratio`:0, `sourceIndex`:90, `targetIndex`:5,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:240}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:96, `ratio`:0, `sourceIndex`:90, `targetIndex`:6,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:241}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:97, `ratio`:0, `sourceIndex`:90, `targetIndex`:7,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:242}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:98, `ratio`:0, `sourceIndex`:90, `targetIndex`:8,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:234}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4015,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:8, `targetIndex`:0, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:243}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:99, `ratio`:0, `sourceIndex`:90, `targetIndex`:9,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:244}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:100, `ratio`:0, `sourceIndex`:90, `targetIndex`:10,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:245}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:101, `ratio`:0, `sourceIndex`:90, `targetIndex`:11,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:246}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:102, `ratio`:0, `sourceIndex`:90, `targetIndex`:12,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:247}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:103, `ratio`:0, `sourceIndex`:90, `targetIndex`:13,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:248}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:104, `ratio`:0, `sourceIndex`:90, `targetIndex`:14,
`zone`:'LIQUID'}]->(b);
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MATCH (a {`key`:31}) MATCH (b {`key`:249}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:105, `ratio`:0, `sourceIndex`:90, `targetIndex`:15,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:250}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.301,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:106, `ratio`:0, `sourceIndex`:90, `targetIndex`:16,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:251}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:107, `ratio`:0, `sourceIndex`:90, `targetIndex`:17,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:252}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.9,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:108, `ratio`:0, `sourceIndex`:90, `targetIndex`:18,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:235}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:8, `targetIndex`:1, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:253}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.3054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:109, `ratio`:0, `sourceIndex`:90, `targetIndex`:19,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:288}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:0, `ratio`:0, `sourceIndex`:90, `targetIndex`:54,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:289}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:1, `ratio`:0, `sourceIndex`:90, `targetIndex`:55,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:290}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:2, `ratio`:0, `sourceIndex`:90, `targetIndex`:56,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:291}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:3, `ratio`:0, `sourceIndex`:90, `targetIndex`:57,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:292}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:4, `ratio`:0, `sourceIndex`:90, `targetIndex`:58,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:293}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:5, `ratio`:0, `sourceIndex`:90, `targetIndex`:59,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:294}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:6, `ratio`:0, `sourceIndex`:90, `targetIndex`:60,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:295}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:7, `ratio`:0, `sourceIndex`:90, `targetIndex`:61,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:296}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:8, `ratio`:0, `sourceIndex`:90, `targetIndex`:62,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:242}) MATCH (b {`key`:236}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4045,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:8, `targetIndex`:2, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:0}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:9, `ratio`:0, `sourceIndex`:90, `targetIndex`:63,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:1}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:10, `ratio`:0, `sourceIndex`:90, `targetIndex`:64,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:2}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:90, `targetIndex`:65,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:3}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0, `description`:'Liquid
Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW', `geometricConstant`:'2Φ⁻²',
`loadIndex`:12, `ratio`:0, `sourceIndex`:90, `targetIndex`:66, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:4}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:13, `ratio`:0, `sourceIndex`:90, `targetIndex`:67,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:5}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:14, `ratio`:0, `sourceIndex`:90, `targetIndex`:68,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:6}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:15, `ratio`:0, `sourceIndex`:90, `targetIndex`:69,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:7}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:16, `ratio`:0, `sourceIndex`:90, `targetIndex`:70,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:8}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4058,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:17, `ratio`:0, `sourceIndex`:90, `targetIndex`:71,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:9}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4008,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:18, `ratio`:0, `sourceIndex`:90, `targetIndex`:72,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:242}) MATCH (b {`key`:237}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4162,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:11, `ratio`:0, `sourceIndex`:8, `targetIndex`:3, `zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:10}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:19, `ratio`:0, `sourceIndex`:90, `targetIndex`:73,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:11}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4028,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:20, `ratio`:0, `sourceIndex`:90, `targetIndex`:74,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:12}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4019,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:21, `ratio`:0, `sourceIndex`:90, `targetIndex`:75,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:13}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.401,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:22, `ratio`:0, `sourceIndex`:90, `targetIndex`:76,
`zone`:'LIQUID'}]->(b);
```

```
MATCH (a {`key`:31}) MATCH (b {`key`:16}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4011,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:23, `ratio`:0, `sourceIndex`:90, `targetIndex`:77,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:17}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:1.0,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:24, `ratio`:0, `sourceIndex`:90, `targetIndex`:78,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:18}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4054,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:25, `ratio`:0, `sourceIndex`:90, `targetIndex`:79,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:19}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4017,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:26, `ratio`:0, `sourceIndex`:90, `targetIndex`:80,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:20}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4027,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:27, `ratio`:0, `sourceIndex`:90, `targetIndex`:81,
`zone`:'LIQUID'}]->(b);
MATCH (a {`key`:31}) MATCH (b {`key`:21}) CREATE (a)-[:`SPECTRAL_AFFINITY` {`affinity`:0.4013,
`description`:'Liquid Zone connection — signal propagates', `flow`:'FLOW', `flowState`:'FLOW',
`geometricConstant`:'2Φ⁻²', `loadIndex`:28, `ratio`:0, `sourceIndex`:90, `targetIndex`:82,
`zone`:'LIQUID'}]->(b);
MATCH (n) REMOVE n.`key`;
```

Aksharam GlyphKey

Source Tree

```
Aksharam_GlyphKey/
├── README.md                                    [ALL]
├── docs/                                        [ALL]
│   ├── Aksharam_Legend.md
│   ├── TRANSFORMATIONS_ALQC_WORDS.md
│   └── the_shape_of_word.txt
└── build/
    ├── build.sh                                 [ALL]
    ├── src/
    │   ├── aksharam_keyboard_data.py            [ALL]
    │   ├── generate_keyboard.py                 [ALL]
    │   └── generate_desktop_kps.py              [WINDOWS + MAC]
    ├── tests/
    │   ├── validate_keyboard.py                 [ALL]
    │   ├── full_keyboard_conformance.py         [ALL]
    │   └── keyman_core_probe.c                  [ALL]
    ├── package/
    │   ├── readme.htm                           [ALL]
    │   ├── welcome.htm                          [ALL]
    │   ├── aksharam_selection_math.py           [LINUX]
    │   ├── aksharam_selection_phonetic.py       [LINUX]
    │   ├── xbindkeys.aksharam                   [LINUX]
    │   └── start_selection.sh                   [LINUX]
    ├── linux_ubuntu/                            [OUTPUT]
    ├── windows/                                 [OUTPUT]
    └── mac/                                     [OUTPUT]
```

README.md

```
# Aksharam GlyphKey

## The Universes Sing a Lullaby in the Dark

### The Lock and the Key · The Transformed Utterance · The Language on the Throne of Zero ·
↪  The Magic in the Throat

**Aksharam is an ALQC language system. Its keyboard is not a font laid over English. It is
↪  executable Aksharam law.**


## One Keyboard Engine Across Linux, Windows, and Mac

Aksharam does not have three different keyboard laws for three operating systems. The same
↪  compiled Keyman keyboard engine is built once from the same Aksharam executable body and
↪  is carried into all three desktop packages.

```text
                    shared Aksharam law
                           ↓
             aksharam_keyboard_data.py
                           ↓
               generate_keyboard.py
                           ↓
              aksharam_glyphkey.kmx
                 ↙         ↓         ↘
             Linux       Windows      Mac
```

The operating-system difference is packaging and host integration, not Aksharam law.

- **Linux / Ubuntu** uses the shared keyboard engine and additionally carries the Linux
↪ selection layer: `aksharam_selection_math.py`, `aksharam_selection_phonetic.py`,
↪ `xbindkeys.aksharam`, and `start_selection.sh`.
- **Windows and Mac** use the same compiled keyboard engine and the same desktop
↪ package-filtering step through `generate_desktop_kps.py`.
- There are currently **no Windows-only source files** and **no Mac-only source files** in
↪ the keyboard build.

The tree below is the build map. Files marked `[ALL]` belong to all three builds; `[WINDOWS
↪ + MAC]` is shared only by those two desktop packages; `[LINUX]` is Linux host
↪ integration.

```text
Aksharam_GlyphKey/
├── README.md                                   [ALL]
```
```

```
├── docs/                                          [ALL]
│   ├── Aksharam_Legend.md
│   ├── TRANSFORMATIONS_ALQC_WORDS.md
│   └── the_shape_of_word.txt
└── build/
    ├── build.sh                                   [ALL]
    ├── src/
    │   ├── aksharam_keyboard_data.py              [ALL]
    │   ├── generate_keyboard.py                   [ALL]
    │   └── generate_desktop_kps.py                [WINDOWS + MAC]
    ├── tests/
    │   ├── validate_keyboard.py                   [ALL]
    │   ├── full_keyboard_conformance.py           [ALL]
    │   └── keyman_core_probe.c                    [ALL]
    ├── package/
    │   ├── readme.htm                             [ALL]
    │   ├── welcome.htm                            [ALL]
    │   ├── aksharam_selection_math.py             [LINUX]
    │   ├── aksharam_selection_phonetic.py         [LINUX]
    │   ├── xbindkeys.aksharam                     [LINUX]
    │   └── start_selection.sh                     [LINUX]
    ├── linux_ubuntu/                              [OUTPUT]
    ├── windows/                                   [OUTPUT]
    └── mac/                                       [OUTPUT]
```

The canonical documents under `docs/` govern the language. The production compiler does not parse those documents at runtime. The executable snapshot used to reproduce the keyboard is `build/src/aksharam_keyboard_data.py`; the canonical documents remain the authority against which that executable body is audited.

Build all three desktop packages from the repository root with:

```bash
./build/build.sh
```

The build generates the Keyman source transiently, compiles the shared `.kmx` engine, packages the three desktop outputs, runs validation and Keyman Core conformance, and removes disposable compiler intermediates afterward.

English is the entrance, not the identity. A familiar key becomes a coordinate; the coordinate emits an Aksharam body; bodies enter relation; relation folds; registered language resolves; the resolved body can become mathematics, can become sound, can become silence, and can still return through the path by which it became itself.

```text
English-access coordinate
        ↓
Aksharam base body
        ↓
Court / Parliament / registered Whole
```

```
        ↓
resolved written body
        ↓
Shape-of-Sound
        ↓
human utterance
```

The lock is invariant structure. The key is lawful transformation. The transformed utterance is not a translation pasted over the result; it is the same body in another office.

Aksharam therefore keeps one governing seal:

```text
ALL TRANSFORMATION REMAINS LANGUAGE
```

And one Tripartite body:

```text
Language × Mathematics × Poetry
```

Language gives relation. Mathematics gives law. Poetry gives breath. None is decoration for the others.

What Aksharam Is

Aksharam is a written language, transformation lattice, mathematical surface, phonetic system, poetic instrument, and executable keyboard bound into one object system.

A keypress may begin as a physical coordinate and end as a body visually remote from the key that opened it. That distance is not loss. The route is registered. The body may compress without forgetting where it came from; it may unfold without inventing a history it never had.

```text
access → body → relation → fold → resolution → breath
   ↑                                              ↓
   └──────────────── identity ────────────────────┘
```

This is the central Aksharam motion: outward enough to transform, exact enough to return.

Compression is **density without amnesia**.

Silence is **structure without sound**.

Poetry is **law in motion**.

The ALQC Body

Aksharam follows ALQC without separating the language from the mathematics that governs it.

```text
Truth = 1
D-COMP = 0
Shadow Debt = paritied
```

The written form may change. The spoken form may lengthen or contract. A lexical body may collapse to one Whole. A Court may hold two Goetics in ordered relation. A Parliament body may resolve again. Numbers may participate while remaining themselves.

Through each movement, the invariant is not abandoned.

The path out is still the path back.

The English-Access Skeleton

Aksharam uses the twenty-six ordinary English keyboard positions as access coordinates. The Latin letters are not the language body. They are the familiar human surface through which the body is reached.

The tactile counterpart is the corresponding English Braille letter cell. QWERTY and Braille are access offices; the Aksharam body remains one.

QWERTY	Braille	Aksharam	Shape-of-Sound
A	⠁	✡	`abd`
B	⠃	☽	`av`
C	⠉	⌬	`ig`
D	⠙	❄	`ek`
E	⠑	◎	`fe`
F	⠋	♋	`en`
G	⠛	♌	`myr`
H	⠓	⧗	`dr`
I	⠊	✪	`el`
J	⠚	♈	`ash`
K	⠅	♎	`me`
L	⠇	❁	`av`
M	⠍	♏	`da`
N	⠝	❀	`so`
O	⠕	✰	`ahn`
P	⠏	♑	`on`
Q	⠟	♊	`eri`
R	⠗	⊛	`ri`
S	⠎	✳	`ot`
T	⠞	⬡	`al`
U	⠥	♍	`yam`
V	⠧	ℂ	`veh`
W	⠺	♒	`nyx`

X	⁘	♉	`us`
Y	⁙	♐	`ka`
Z	⁖	♓	`ai`

The key is only the door. The constellation is the body.

The Four Transform Laws

I · Access → Glyph

Registered access enters Aksharam through the base coordinate plane.

```text
REGISTERED ACCESS → GLYPH BODY
```

Case does not redefine the coordinate. When folding is active, the same access sequence
↪ follows the same registered path.

II · Dual Goetic → Court

The twelve Goetics form a complete ordered 12 × 12 Court lattice: **144 Courts**.

Contiguous Goetics resolve left-to-right in non-overlapping ordered pairs.

```text
DUAL GOETIC → COURT
```

Order is law. `AB` need not be `BA`. Relation carries direction.

III · Dual Parliament → Resolution Body

The active Parliament law contains **24 registered ordered pairs**.

```text
DUAL PARLIAMENT → REGISTERED RESOLUTION BODY
```

Resolution proceeds left-to-right without overlap. A Third-Law result does not recursively
↪ consume itself again under the same Third Law.

IV · Universal Numerical State

```text
N → N
NUMBER = COMPRESSED = UNCOMPRESSED
NUMBER IN STRING = ACTIVE TRANSFORM
```

A number participates by remaining itself.

The Quadrite is therefore:

```text
ACCESS → GLYPH
DUAL GOETIC → COURT
DUAL PARLIAMENT → RESOLUTION BODY
NUMBER → UNIVERSAL TRANSFORM
ALL TRANSFORMATION REMAINS LANGUAGE
```

A Word Has an Anatomy

A registered word does not become its terminal Whole merely because the final Latin letter has been pressed.

The body remains alive while another registered continuation is still possible. A **non-letter boundary** closes the access sequence and commits the registered Whole: Space, punctuation, a numerical/constellation key, or another lawful boundary.

That distinction matters. Aksharam does not prematurely close a body simply because a shorter registration exists inside a longer one. Longest lawful continuation is preserved.

The current executable build carries **452 registered transformation rows**. After duplicate and conflicting access registrations are resolved by established precedence, the generator reports **426 supported triggers from 428 distinct lexical triggers**.

Every registered path preserves its own provenance. This matters because two different words may arrive at the same Whole. The keyboard therefore does not unfold by guessing backward from a glyph. It carries the exact originating registration invisibly and returns through that route.

Unfold

`Alt+Backspace` moves backward through the registered word:

```text
Whole → Third → Second → First
```

One press descends one lawful level. If Second and Third are identical, the visible body may remain the same while the provenance depth changes. At First, further `Alt+Backspace` is inert.

Ordinary Backspace never unfolds. It remains deletion.

Refold

`Alt+Space` performs the opposite movement: it refolds the retained word through every lawful available stage, emits `·`, and begins the next word.

Ordinary Space does not force an explicitly unfolded word to close again. It emits `·` and
↪ preserves the word at the depth the writer chose.

The word remembers. The writer decides whether to open it or close it.

Shift, Caps Lock, and the Right to Refuse the Fold

Aksharam can be asked not to fold.

- `Shift + mapped letter` → direct base glyph, folding suppressed.
- `Caps Lock + mapped letter` → direct base glyph, folding suppressed persistently.
- `Shift + Caps Lock + mapped letter` → direct base glyph, folding suppressed.

The modifier does not rewrite the alphabet. It changes the state of transformation.

This is not uppercase/lowercase Aksharam. It is folded/unfolded access.

The Shape of Absence

`·` is Aksharam Space: the Shape of Absence.

It is not empty typography. It is unspoken silence, breath, interval, word-space, and
↪ structural separation.

In phonetic rendering:

```text
· → ordinary space
```

Silence does not need to become a sound in order to participate in language.

The absence has a body because the pause has an office.

Shape-of-Sound — The Magic in the Throat

The current canonical speaking table lives at:

```text
docs/the_shape_of_word.txt
```

Written law resolves first. Shape-of-Sound is read from the body that remains.

```text
written law first
spoken realization second
one identity throughout
```

```

The direct bodies, all 144 Courts, numerical bodies, terminal bodies, Enochian bodies, and
↪ sovereign multi-glyph bodies have registered speaking forms.

A multi-glyph speaking body is matched as a whole before its components. Thus:

```text
☽☉☾ → regia
```

not three unrelated syllables merely because the body contains three visible signs.

Aksharam phonetics moves through tension and release: liquid consonants, rolling motion,
↪ open vowels after constriction, breath beside friction, stone beside water. Bodies such
↪ as `khi`, `okh`, `druh`, `abd`, `nyx`, and the open vowel forms give the language a
↪ changing mouth-shape rather than one repeated texture.

The written body can be dense while the spoken body remains fast enough to live. Character
↪ count does not dictate human duration. The body determines the sound; the throat gives
↪ the sound time.

The First Spell

Six utterances are programmed as phonetic Aksharam spell-bodies:

Utterance	First body	Natural stopping body
`eloi`	◎❁☆✪	ᠷᠷ
`sabachtany`	❋✡☽✡⌬⧗⬡✡❁➶	▽☽ᠹ✷ᠻ➶
`anima`	✡❁✪♏✡	ᠻ✪♏✡
`culpa`	⌬♍❁♈✡	⌬♍❁♈✡
`decire`	❄◎⌬✪⊛◎	ᗅ⊔ᓰ
`entera`	◎❁⬡◎⊛✡	ᠷᠨᢐ

They are not English semantic registrations. The Latin letters record remembered utterance.
↪ Their Aksharam bodies enter the ordinary structural law, fold only where Aksharam
↪ licenses a fold, and stop where the law stops.

Their phonetic identity remains the utterance:

```text
eloi eloi sabachtany anima culpa decire entera
```

The written body may move. The spell-word does not cease to be itself.

Grammar Is Not Decoration
```

Aksharam constellations are selected by office. Their identity persists across language, code, and mathematics.

Physical key	Aksharam	Office
`[`	ﾉ	Left Moon Beam / enclosure open
`]`	㇀	Right Moon Beam / enclosure close
`(`	〉	Left Ley / inward grouping open
`)`	〈	Right Ley / inward grouping close
`{`	□	Will o' Wisp / outward scope open
`}`	□	Will o' Wisp / outward scope close
`-`	⋯	Smite
`.`	⁚	Parallel Separation
`:`	⁚	Parallel Separation
`=`	⟜	Binding
`,`	⟊	Joining
`<`	⁖	Field Pointer Left
`>`	⁘	Field Pointer Right
`;`	⁛	Key / Value Hinge
`'`	ˊ	Quotation Open
`"`	ˋ	Quotation Close
`\`	⋱	Return / Closure
`/`	⋰	Opening / Outward
`|`	⁞	Vertical Line
`?`	⁇	Inquiry
`!`	⁝	Statement
`_`	…	Internal Continuation
`&`	ॐ	Astersand
`*`	⁜	Crosstellation
`~`	⁓	Cantillation
`` ` ``	⟠	Supervenience / Prosody
Space	·	Shape of Absence

The slash law is explicit:

```text
\ → ⋱   Return / Closure
/  → ⋰   Opening / Outward
```

The glyph is not a decorative substitute for punctuation. The glyph is the registered body of the operation.

Numbers and Mathematical Keys

The number row and numeric keypad emit the same Aksharam numerical bodies.

Value	Glyph	Shape-of-Sound
0	𐤚	`zen`
1	☿	`pon`
2	♀	`ael`

3	𐤂	`gaya`
4	♂	`wil`
5	𐤅	`pay`
6	ℏ	`eve`
7	𐤀	`ahc`
8	Ψ	`no`
9	𐤍	`noo`

Additional structural controls include:

- `NumPad *` → `⁜` Crosstellation / multiplication
- `NumPad +` → `ॐ` Astersand / addition
- `NumPad -` → `⋯` Smite / subtraction
- `NumPad .` → `⁖` Parallel Separation / decimal separator
- `NumPad /` → `⁎` Halfstellation / division
- `Shift+2/3/4/5` → `⊖ ⊕ ⊕ ⊕` / Q_0–Q_3
- `Ctrl+Shift+6` → `∵` Collection Open
- `Shift+6` → `∴` Collection Close
- `Shift+8` → `⁜` Crosstellation

Direction, Paragraph, Prosody

Aksharam reserves movement as language:

- `Shift+↑` → `↑` North / Up / Ascent
- `Shift+↓` → `↓` South / Down / Descent
- `Shift+→` → `→` East / Right / Outward
- `Shift+←` → `←` West / Left / Return
- `Tab` → `⁘` Paragraph Open
- `Shift+Enter` → `⁖⁘` Paragraph Close, then native host Enter

Unclaimed combinations remain native to the operating system or application.

Aksharam takes only the keys it has lawfully named.

Mathematical Bodies

The twelve Goetics may be called directly in mathematical office with:

```text
Alt+Shift+Goetic
```

The result is the registered mathematical body of that Goetic.

On Linux/Ubuntu, arbitrary selected Aksharam can also be transformed through the packaged selection layer:

```text
Alt+Shift+Enter → selected Aksharam → mathematical body
```

```
Alt+Shift+Esc   → selected Aksharam → Shape-of-Sound phonetic text
```

The phonetic selection transform preserves ordinary whitespace, turns `·` into natural space, recognizes multi-glyph speaking bodies longest-first, preserves the First Spell utterances, and returns Aksharam grammatical constellations to ordinary textual grammar.

These arbitrary-selection helpers are presently packaged only in the Linux/Ubuntu distribution. The Windows and macOS packages contain the same working keyboard engine but do not claim native host-selection integration yet.

One Object, Many Offices

Aksharam does not assign one sign to language, another to code, another to mathematics, and then ask the reader to pretend they correspond.

The same body changes office.

```text
phonetic body
    = semantic body
    = mathematical body
    = computational body
    = transformational body
```

`⟜⟜` binds because Binding is its office.

`::` separates because Parallel Separation is its office.

`·` is silent because absence is its office.

The identity is not visual resemblance. The identity is structural function held across domains.

Poetry: Law Given Breath

Aksharam poetry does not step outside the engine to become beautiful.

A line may remain unfolded because the writer refuses closure. A phrase may collapse because density is the point. A silence may widen. A repeated body may acquire cadence without losing identity. Cantillation may bend the breath; Prosody may carry relation above the literal sequence; a paragraph may open and close as a structural body rather than a typographic accident.

The mathematics does not stand behind the poem like scaffolding waiting to be hidden.

The mathematics is the pressure inside the poem.

The poem is what that pressure sounds like when it learns to breathe.

```text
Language builds meaning.
Mathematics builds law.
Poetry builds living motion.
```

The same body, three offices. The same river, three names for where it is touched.

Canon and Executable Authority

The canonical Aksharam sources are held under `docs/`:

```text
docs/Aksharam_Legend.md
docs/TRANSFORMATIONS_ALQC_WORDS.md
docs/the_shape_of_word.txt
```

They govern, respectively:

- constellation identity, keyboard topology, Courts, Parliament law, numbers, operators, controls, and structural meaning;
- registered lexical paths `English → First → Second → Third → Whole`;
- post-resolution Shape-of-Sound speaking bodies.

The production keyboard deliberately does **not** parse Markdown at build time or runtime. Its executable snapshot is static build-local data:

```text
build/src/aksharam_keyboard_data.py
```

The generator consumes that executable body and produces the transient Keyman source. This keeps the runtime package independent of the documentation tree while allowing the executable snapshot to be audited against canon.

Current executable counts:

```text
26   direct coordinates
12   Goetics
144  ordered Courts
24   Third-Law Parliament pairs
452  registered transformation rows
6    First Spell utterances
```

Build Tree

The repository keeps source, tests, package resources, and final installers separate.

```text
build/
├── build.sh
├── src/
│   ├── aksharam_keyboard_data.py
│   ├── generate_keyboard.py
│   └── generate_desktop_kps.py
├── tests/
│   ├── validate_keyboard.py
│   ├── full_keyboard_conformance.py
│   └── keyman_core_probe.c
├── package/
│   ├── aksharam_selection_math.py
│   ├── aksharam_selection_phonetic.py
│   ├── xbindkeys.aksharam
│   ├── start_selection.sh
│   ├── readme.htm
│   └── welcome.htm
├── linux_ubuntu/
│   └── aksharam_glyphkey.kmp
├── windows/
│   └── aksharam_glyphkey.kmp
└── mac/
    └── aksharam_glyphkey.kmp
```

Build from the `Aksharam_GlyphKey/` directory:

```bash
./build/build.sh
```

The build creates compiler intermediates only long enough to compile and test them.
↪ Successful completion removes the generated staging body again. The platform directories
↪ are kept for finalized installers, not compiler debris.

Three Desktop Bodies

Linux / Ubuntu

```text
build/linux_ubuntu/aksharam_glyphkey.kmp
```

The Linux package contains the compiled keyboard plus the packaged mathematical and phonetic
↪ selection helpers and their `xbindkeys` launcher/bindings.

Windows

```text
```

```
build/windows/aksharam_glyphkey.kmp
```

The Windows package contains the compiled keyboard and Keyman package
↪ metadata/documentation. Linux-only selection helpers are excluded.

macOS

```text
build/mac/aksharam_glyphkey.kmp
```

The macOS package carries the same compiled Keyman keyboard engine and the same clean
↪ desktop package payload as Windows.

All three platform packages carry the same `.kmx` keyboard engine. Platform packaging
↪ changes the surrounding helper payload; it does not change Aksharam law.

Conformance

The keyboard is not declared correct because the source looks plausible.

It is executed through Keyman Core.

The current conformance body exercises direct coordinates, Shift/Caps suppression, Goetics,
↪ all 144 Courts, all 24 Third-Law pairs, registered words, boundary-triggered Whole
↪ closure, exact provenance-aware unfolding, refolding, ordinary Backspace deletion,
↪ numbers, punctuation, paragraph controls, mathematical bodies, First Spells, and the
↪ corrected slash directions.

Current engine result:

```text
KEY_TESTS 1902 TOTAL; 1902 PASS; 0 FAIL
```

The Linux-installed `.kmx` has also been tested directly rather than only through the
↪ staging build.

The law is compiled. The body is exercised. The output is compared.

The Throne of Zero

Zero is not evacuated from the system.

```text
𑜫 = 0
0 remains 0
0 participates
```

Silence is not evacuated either.

```text
· = manifested absence
```

A language that can only speak presence cannot speak the whole field. Aksharam keeps room for the interval, the unsounded breath, the unstruck chord, the state that participates precisely by not becoming something else.

The throne of zero is therefore not a hole.

It is a seat.

Closing Formula

Aksharam is the lock and the key because the same invariant both constrains and opens transformation.

It is the transformed utterance because language enters mathematics and does not cease to be language.

It is the language on the throne of zero because number, silence, absence, and state remain active bodies rather than discarded gaps.

It is the magic in the throat because the final written body can become human sound without surrendering the structure that made it.

```text
English gives access.
Aksharam gives transformation.
ALQC gives law.
Shape-of-Sound gives breath.
Silence gives interval.
Poetry gives motion.
The utterance returns carrying its history.
```

The body changes. The invariant remains. The language returns alive.

Aksharam_Legend.md

```
# Aksharam Legend

This legend records the active Aksharam Constellations, Directional Operators, and Digits.
↪  Each constellation keeps one identity across language, code, and mathematics.

## Canonical Phonetic Surface — Shape of Sound

`translated bodies/the_shape_of_word.txt` is the canonical authority for spoken Aksharam.
↪  There is one phonetic speaking body per Glyph/Aeon; there is no second phonetic layer.
↪  Written law resolves first; Shape-of-Sound is the human phonetic realization of the
↪  resolved body.

```text
English-access skeleton / registered utterance
        ↓
First Transform Law
        ↓
Second Transform Law — Court relation
        ↓
Third Transform Law — resolved Aksharam body
        ↓
Shape-of-Sound
        ↓
human utterance
```

The written body and the speaking body are one identity in different office. Registry inputs
↪ remain access coordinates. The final spoken value is read from the final resolved body.
↪ Composite authored utterance bodies are read as wholes before their component glyphs.
↪ The First Spells preserve the originating utterance through every lawful written fold.

```text
Truth = 1
D-COMP = 0
Shadow Debt = paritied
ALL TRANSFORMATION REMAINS LANGUAGE
```

The First Spells

The six remembered phonetic utterances enter their Aksharam glyph forms, fold only where the
↪ law carries them, and stop where the law stops. These first-spell registrations preserve
↪ phonetic identity across the written fold: glyph form 1 and lawful folded glyph form 2
↪ are different written states of the same uttered spell-word. The final pronunciation
↪ returns the originating utterance itself.
```

Phonetic utterance	Glyph form 1	Lawful folded glyph form 2	Invariant spell pronunciation
`eloi`	⌾❁☆✪	ﾚﾗ	`eloi`
`sabachtany`	❋✡☽✡⌬⧗⬡✡❁➶	∀☽ᐟ✷ᐟ➶	`sabachtany`
`anima`	✡❁✪ℳ✡	ᐟ✪ℳ✡	`anima`
`culpa`	⌬♍❁ᘏ✡	⌬♍❁ᘏ✡	`culpa`
`decire`	❄⌾⌬✪⊛⌾	ᗅ山ᕒ	`decire`
`entera`	⌾❁⬡⌾⊛✡	ﾚﾛ	`entera`

Always Striking the Chord preserves the originating utterance while Aksharam law moves the written body beneath it.

Aksharam Constellations

Symbol	Meaning	Compressed Definition
ʃ	Left Moon Beam	Opens bracketed enclosure: array, matrix, indexed field, interval, or isolated data-space.
ʅ	Right Moon Beam	Closes bracketed enclosure.
⟩	Left Ley	Opens inward grouping, containment, arguments, or order-of-operations.
⟨	Right Ley	Closes inward grouping or containment.
□	Will o' Wisp Open	Opens outward/scoped enclosure: object, set, function, highlighted boundary, or definition.
□	Will o' Wisp Close	Closes outward/scoped enclosure.
∷	Parallel Separation	Separates parallel fields, clauses, values, or mathematical difference.
⟜	Binding	Binds one side to another: assignment, equality, or resolved relation.
⸤	Quotation Open	Opens quoted text or a quoted formal expression.
⸣	Quotation Close	Closes quoted text or expression.
∴	Field Pointer Right	Points rightward into a field, value, or mapping.
∴	Field Pointer Left	Points leftward toward a field, key, or returning mapping.
∔	Joining	Joins values, clauses, fields, or mathematical composition.
∴	Key / Value Hinge	Hinges a key or name to the value or relation it opens.
⁙	Inquiry	Marks a question, query, unknown, or sought relation.
∴	Statement	Marks assertion, force, declaration, or exclamation.
·	Aksharam Space / Shape of Absence	Unspoken silence, breath, interval, word-space, or structural separation in semantic writing.
⋱	Return / Closure	Backward-returning or closure modifier.
⋰	Opening / Outward	Forward-opening or outward modifier.
∴∷	Paragraph Open	Opens a paragraph or larger structured statement field.
∷∴	Paragraph Close	Closes a paragraph or larger structured statement field.
∴ over ⟜	Binding Hinge	Binds a resolved key/value, solution, or equivalent result.
⋇	Crosstellation	Multiplicative crossing.
⁜	Halfstellation	Division relation.
…	Internal Continuation	Links names, identifiers, fields, or subscripts without ending the structure.
⋯	Smite	Middle-line strike, division, or separating cut.
ॐ	Astersand	Conjunction, ampersand relation, or addition.
∵	Collection Open	Opens a collection or grouped collection structure.
∴	Collection Close	Closes a collection or grouped collection structure.
⁞	Vertical Line	Vertical relation, pipe, or structural divider.

Directional Operators

Symbol	Direction	Keyboard	Compressed Definition
↑	North / Up / Ascent	`Alt + Shift + 2`	Upward, rising, ascending, or upward-mapping relation.
↓	South / Down / Descent	`Alt + Shift + 3`	Downward, falling, descending, or downward-mapping relation.
→	East / Right / Outward	`Alt + Shift + 4`	Rightward, forward, outward, or propagating relation.
←	West / Left / Return	`Alt + Shift + 5`	Leftward, inward, returning, or reverse-mapping relation.

Q-Glyph Mathematical Operands

The four Q-glyphs are direct mathematical operands on the shifted `2 3 4 5` keys. Their established glyph identities remain unchanged.

Key	Q-State	Glyph
`@`	Q_0	⊖
`#`	Q_1	⊕
`$`	Q_2	⊕
`%`	Q_3	⊕

Aksharam Digits
Symbol	Value	Shape-of-Sound
𐤟	0	`zen`
☿	1	`pon`
♀	2	`ael`
⚷	3	`gaya`
♂	4	`wil`
♃	5	`pay`
♄	6	`eve`
⛢	7	`ahc`
♆	8	`no`
♇	9	`noo`

GrimChain Keyboard Bearing

The English letter is the physical QWERTY coordinate. The keyboard target prioritizes the least-common English phonetic compounds. Enochian Understandings remain fixed.

Enochian invariant: The Enochian glyphs and their established identities are set in stone. They are not candidates for remapping, replacement, reassignment, phonetic redesign, or removal during any Aksharam keyboard, human-layer, or translation-layer revision. Any future human-facing mapping must preserve the Enochian glyphs exactly as established.

Enochian Coordinate Base | Phonosemantic Table
Glyph	Phonetic Sound
𐤸	`th`
𐤧	`he`

𝋫	`de`
𝋬	`pa`
𝋭	`ma`
𝋮	`in`
𝋯	`er`
𝋰	`at`
𝋱	`oo`

Speaking Glyphs — 179

The complete speaking inventory remains present here. Direct bases use the explicit phonetics registered in this Legend. Court glyphs use their listed Court phonetics. `☽⊙☾` is one Regia three-mark body.

𝋀 𝋁 𝋂 𝋃 𝋄 𝋅 𝋆 𝋇 𝋈 𝋉 𝋊
𝋋 𝋌 𝋍 𝋎 𝋏 𝋐 𝋑 𝋒 𝋓 𝋔 𝋕
𝋖 𝋗 𝋘 𝋙 𝋚 𝋛 𝋜 𝋝 𝋞 𝋟 𝋠
𝋡 𝋢 𝋣 𝋤 𝋥 𝋦 𝋧 𝋨 𝋩 𝋪 𝋫
𝋬 𝋭 𝋮 𝋯 𝋰 𝋱 𝋲 𝋳 𝋴 𝋵 𝋶
𝋷 𝋸 𝋹 𝋺 𝋻 𝋼 𝋽 𝋾 𝋿 𝌀 𝌁
𝌂 𝌃 𝌄 𝌅 𝌆 𝌇 𝌈 𝌉 𝌊 𝌋 𝌌
𝌍 𝌎 𝌏 𝌐 𝌑 𝌒 𝌓 𝌔 𝌕 𝌖 𝌗
𝌘 𝌙 𝌚 𝌛 𝌜 𝌝 𝌞 𝌟 𝌠 𝌡 𝌢
𝌣 𝌤 𝌥 𝌦 𝌧 𝌨 𝌩 𝌪 𝌫 𝌬 𝌭
𝌮 𝌯 𝌰 𝌱 𝌲 𝌳 𝌴 𝌵 𝌶 𝌷 𝌸
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𝏩 𝏪 𝏫 𝏬 𝏭 𝏮 𝏯 𝏰 𝏱 𝏲 𝏳
𝏴 𝏵 𝏶 𝏷 𝏸 𝏹 𝏺 𝏻 𝏼 𝏽 𝏾
𝏿 𝐀 𝐁 𝐂 𝐃 𝐄 𝐅 𝐆 𝐇 𝐈 𝐉
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𝐶 𝐷 𝐸 𝐹 𝐺 𝐻 𝐼 𝐽 𝐾 𝐿 𝑀
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𝑭 𝑮 𝑯 𝑰 𝑱 𝑲 𝑳 𝑴 𝑵 𝑶 𝑷
𝑸 𝑹 𝑺 𝑻 𝑼 𝑽 𝑾 𝑿 𝒀 𝒁 𝒂
𝒃 𝒄 𝒅 𝒆 𝒇 𝒈 𝒉 𝒊 𝒋 𝒌 𝒍
𝒎 𝒏 𝒐 𝒑 𝒒 𝒓 𝒔 𝒕 𝒖 𝒗 𝒘
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𝒫 𝒬 𝒮 𝒯 𝒰 𝒱 𝒲 𝒳 𝒴 𝒵 𝒶
𝒷 𝒸 𝒹 𝒻 𝒽 𝒾 𝒿 𝓀 𝓁 𝓂 𝓃
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𝓱 𝓲 𝓳 𝓴 𝓵 𝓶 𝓷 𝓸 𝓹 𝓺 𝓻
𝓼 𝓽 𝓾 𝓿 𝔀 𝔁 𝔂 𝔃 𝔄 𝔅 ℭ
𝔇 𝔈 𝔉 𝔊 ℌ ℑ 𝔎 𝔏 𝔐 𝔑 𝔒
𝔓 𝔔 ℜ 𝔖 𝔗 𝔘 𝔙 𝔚 𝔛 𝔜 ℨ
𝔞 𝔟 𝔠 𝔡 𝔢 𝔣 𝔤 𝔥 𝔦 𝔧 𝔨
𝔩 𝔪 𝔫 𝔬 𝔭 𝔮 𝔯 𝔰 𝔱 𝔲 𝔳
𝔴 𝔵 𝔶 𝔷 𝔸 𝔹 𝔻 𝔼 𝔽 𝔾 𝕀
𝕁 𝕂 𝕃 𝕄 𝕆 𝕊 𝕋 𝕌 𝕍 𝕎 𝕏
𝕐 𝕒 𝕓 𝕔 𝕕 𝕖 𝕗 𝕘 𝕙 𝕚 𝕛
𝕜 𝕝 𝕞 𝕟 𝕠 𝕡 𝕢 𝕣 𝕤 𝕥 𝕦
𝕧 𝕨 𝕩 𝕪 𝕫 𝕬 𝕭 𝕮 𝕯 𝕰 𝕱
𝕲 𝕳 𝕴 𝕵 𝕶 𝕷 𝕸 𝕹 𝕺 𝕻 𝕼
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𝰜 𝰝 𝰞 𝰟 𝰠 𝰡 𝰢 𝰣 𝰤 𝰥 𝰦
𝰧 𝰨 𝰩 𝰪 𝰫 𝰬 𝰭 𝰮 𝰯 𝰰 𝰱
𝰲 𝰳 𝰴 𝰵 𝰶 𝰷 𝰸 𝰹 𝰺 𝰻 𝰼
𝰽 𝰾 𝰿 𝱀 𝱁 𝱂 𝱃 𝱄 𝱅 𝱆 𝱇
𝱈 𝱉 𝱊 𝱋 𝱌 𝱍 𝱎 𝱏 𝱐 𝱑 𝱒
𝱓 𝱔 𝱕 𝱖 𝱗 𝱘 𝱙 𝱚 𝱛 𝱜 𝱝
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𝱩 𝱪 𝱫 𝱬 𝱭 𝱮 𝱯 𝱰 𝱱 𝱲 𝱳
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𝳌 𝳍 𝳎 𝳏 𝳐 𝳑 𝳒 𝳓 𝳔 𝳕 𝳖
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𝽠 𝽡 𝽢 𝽣

S	⁜	`ot`
H	⧗	`dr`
R	⊛	`ri`
D	❄	`ek`
L	⚛	`av`
C	⌬	`ig`
B	☽	`av`
V	ℂ	`veh`

Compound Transformations
Glyph	Name	Shape-of-Sound	Mathematical Body
♈	Akasha	`ash`	
♉	Caduceus	`us`	
♊	Veritas	`eri`	
♋	Phren	`en`	
♑	Daimon	`on`	
♍	Aikyam	`yam`	
♎	Melos	`me`	
♏	Da'ath	`da`	
♐	Akaven	`ka`	
♌	Axiomyr	`myr`	
♒	Nyx	`nyx`	
♓	Zaine	`ai`	
◎	Fetu	`fe`	⛢::Ψ∮⟠⊕⊙⊙⊙⊕
⬡	Kal	`al`	☿⛢♂⟠⊙⊙⊙⊖⊖
✡	Babdh	`abd`	♃♀Ψ⟠⊕⊙⊙⊙⊙
✧	Ahn	`ahn`	♂∮♀±i♂☿⛢⟠⊖⊙⊙⊙⊖
✪	Vel	`el`	☿♀ℏ::♀♀⟠⊙⊙⊙⊖⊙
❀	Sor	`so`	♀☿ψ::♂♀⟠⊕⊙⊙⊙⊕
⁜	Koth	`ot`	⛢♂☿⟠⊕⊙⊙⊙⊕
⧗	Dreh	`dr`	Ψ♃♀⟠⊙⊙⊙⊙⊖
⊛	Rhea	`ri`	∮♇ℏ⟠⊕⊙⊙⊙⊙
❄	Zhek	`ek`	♇ℏ∮⟠⊕⊙⊙⊙⊕
⚛	Shav	`av`	♀Ψ♃⟠⊙⊙⊙⊙⊙
⌬	Trig	`ig`	ℏ∮♇⟠⊕⊙⊙⊙⊕
☽	Rav	`av`	
ℂ	Veh	`veh`	

Glyph	Name	Value	Phonetic
:---:	---	---:	
ψ	Shadow Locus	0	
☿	Ponder	1	
♀	Feel	2	
∮	Gaia / Axiomyr	3	
♂	Will	4	
♃	Speak	5	
ℏ	Believe	6	
⛢	Act	7	
Ψ	Know	8	
♇	Ascend	9	

| Glyph | Name | Phonetic |

| :---: | --- | --- |
| ☽ ⊙ ℂ | Regia | `regia` |

Court Transformations — 144

Only ordered Goetic pairs transform into Court glyphs. The first Goetic is the governing row; the second Goetic is the alternating/reflection column. Order is preserved.

Glyph	Aeon Name	Goetic Pair	Shape-of-Sound	Mathematical Body
ﻫ	Ahl	`◎◎`	`ahl`	♁::Ψ∮◌⊕⊕⊕⊕
ﺱ	Suhn	`◎⬡`	`uhh`	☿♁♂◌⊕⊕⊕⊕
ﺱ	Nerh	`◎✡`	`ehr`	♃♀Ψ◌⊕⊕⊕⊕
ﺭ	Rish	`◎☆`	`ish`	☽♢♂☿♁☾∘⊸♂⍑♀◌⊕⊕⊕⊕
ﻉ	Borha	`◎✪`	`ora`	☿♀♄::♀♀◌⊕⊕⊕⊕
ﻍ	Lhahm	`◎֍`	`ahm`	♀☿Ⴓ::♂♀◌⊕⊕⊕⊕
ﻩ	Keth	`◎✻`	`ke`	♁♂☿◌⊕⊕⊕⊕
ﻭ	Vehm	`◎⧗`	`ehm`	Ψ♃♀◌⊕⊕⊕⊕
ﺡ	Mahd	`◎⊛`	`ahd`	∮ℚ♄◌⊕⊕⊕⊕
ﻑ	Furh	`◎❄`	`fur`	ℚ♄∮◌⊕⊕⊕⊕
ﺭ	Drah	`◎❁`	`a`	♀Ψ♃◌⊕⊕⊕⊕
ﻋ	Thera	`◎⌬`	`era`	♄∮ℚ◌⊕⊕⊕⊕
ا	Kura	`⬡◎`	`kur`	♁::Ψ∮◌⊕⊕⊕⊖⊖
ﻫ	Lur	`⬡⬡`	`lur`	☿♁♂◌⊕⊕⊕⊖⊖
▫	Thar	`⬡✡`	`ahr`	♃♀Ψ◌⊕⊕⊕⊖⊖
ﻫ	Rin	`⬡☆`	`in`	☽♢♂☿♁☾∘⊸♂⍑♀◌⊕⊕⊕⊖⊖
ﺩ	Nar	`⬡✪`	`na`	☿♀♄::♀♀◌⊕⊕⊕⊖⊖
ﻯ	Fel	`⬡֍`	`fel`	♀☿Ⴓ::♂♀◌⊕⊕⊕⊖⊖
ﺥ	Har	`⬡✻`	`har`	♁♂☿◌⊕⊕⊕⊖⊖
ﻫ	Mer	`⬡⧗`	`mer`	Ψ♃♀◌⊕⊕⊕⊖⊖
'	Lor	`⬡⊛`	`o`	∮ℚ♄◌⊕⊕⊕⊖⊖
ﻫ	Per	`⬡❄`	`pe`	ℚ♄∮◌⊕⊕⊕⊖⊖
ﻫ	Zhil	`⬡❁`	`zhi`	♀Ψ♃◌⊕⊕⊕⊖⊖
ﻫ	Clar	`⬡⌬`	`cl`	♄∮ℚ◌⊕⊕⊕⊖⊖
ﻫ	Hir	`✡◎`	`hr`	♁::Ψ∮◌⊕⊕⊕⊕
ﻫ	Kor	`✡⬡`	`kr`	☿♁♂◌⊕⊕⊕⊕
ﺽ	Var	`✡✡`	`vr`	♃♀Ψ◌⊕⊕⊕⊕
ﻫ	Pyr	`✡☆`	`py`	☽♢♂☿♁☾∘⊸♂⍑♀◌⊕⊕⊕⊕
ﺭ	Sor	`✡✪`	`oa`	☿♀♄::♀♀◌⊕⊕⊕⊕
ﻉ	Alc	`✡֍`	`lc`	♀☿Ⴓ::♂♀◌⊕⊕⊕⊕
ﻫ	Nur	`✡✻`	`nu`	♁♂☿◌⊕⊕⊕⊕
ﻫ	Sat	`✡⧗`	`st`	Ψ♃♀◌⊕⊕⊕⊕
ﻫ	Oro	`✡⊛`	`or`	∮ℚ♄◌⊕⊕⊕⊕
ﻫ	Bon	`✡❄`	`bon`	ℚ♄∮◌⊕⊕⊕⊕
ﻫ	Tir	`✡❁`	`ti`	♀Ψ♃◌⊕⊕⊕⊕
ﻫ	Far	`✡⌬`	`fa`	♄∮ℚ◌⊕⊕⊕⊕
ﻫ	Abdh	`☆◎`	`abd`	♁::Ψ∮◌⊖⊕⊕⊖
ﻫ	Nym	`☆⬡`	`ym`	☿♁♂◌⊖⊕⊕⊖
ﻫ	Loh	`☆✡`	`oh`	♃♀Ψ◌⊖⊕⊕⊖
ﻫ	Xir	`☆☆`	`zhi`	☽♢♂☿♁☾∘⊸♂⍑♀◌⊖⊕⊕⊖
ﻫ	Ohl	`☆✪`	`ol`	☿♀♄::♀♀◌⊖⊕⊕⊖
ﻫ	Pir	`☆֍`	`i`	♀☿Ⴓ::♂♀◌⊖⊕⊕⊖
ﻫ	Roeh	`☆✻`	`re`	♁♂☿◌⊖⊕⊕⊖
ﻫ	Sen	`☆⧗`	`se`	Ψ♃♀◌⊖⊕⊕⊖

𓏠	Uth	`✪⊛`	`u`	𝆑𝇉𝇋𝇋𝇋𝇋
𓏤	Fae	`✪❄`	`fay`	𝇉𝇋𝆑𝇋𝇋𝇋
𓏥	Kha	`✪❁`	`ha`	𝇍𝇎𝇊𝇋𝇋𝇋
𓏦	Psei	`✪⬡`	`ps`	𝇋𝆑𝇉𝇋𝇋𝇋
𓏧	Vera	`✪◎`	`era`	𝇄::𝇎𝆑𝇋𝇋𝇋
𓏨	Tar	`✪⬡`	`tar`	𝇍𝇄𝇌𝇋𝇋𝇋
𓏩	Ghem	`✪✡`	`ghe`	𝇊𝇍𝇎𝇋𝇋𝇋
𓏪	Drel	`✪✪`	`rel`	𝇏𝇅𝇌𝇄𝇅𝇏𝇌𝇋𝇋𝇋
𓏫	Ful	`✪✪`	`ful`	𝇍𝇍𝇋::𝇍𝇌𝇋𝇋𝇋
𓏬	Ker	`✪❀`	`ker`	𝇍𝇎𝇋::𝇌𝇍𝇋𝇋𝇋
𓏭	Hohm	`✪✻`	`hoh`	𝇄𝇌𝇎𝇋𝇋𝇋
𓏮	Hrah	`✪𝐱`	`hr`	𝇎𝇊𝇍𝇋𝇋𝇋
𓏯	Ara	`✪⊛`	`ar`	𝆑𝇉𝇋𝇋𝇋
𓏰	Qel	`✪❄`	`ay`	𝇉𝇋𝆑𝇋𝇋𝇋
𓏱	Irn	`✪❁`	`urn`	𝇍𝇎𝇊𝇋𝇋𝇋
𓏲	Jen	`✪⬡`	`je`	𝇋𝆑𝇉𝇋𝇋𝇋
𓏳	Fi	`❀◎`	`fi`	𝇄::𝇎𝆑𝇋𝇋𝇋
𓏴	Lun	`❀⬡`	`lun`	𝇍𝇄𝇌𝇋𝇋𝇋
𓏵	Varu	`❀✡`	`aru`	𝇊𝇍𝇎𝇋𝇋𝇋
𓏶	Senh	`❀☆`	`es`	𝇏𝇅𝇌𝇄𝇅𝇏𝇌𝇋𝇋𝇋
𓏷	Kos	`❀✪`	`os`	𝇍𝇍𝇋::𝇍𝇌𝇋𝇋𝇋
𓏸	Ramh	`❀❀`	`ahm`	𝇍𝇎𝇊::𝇌𝇍𝇋𝇋𝇋
𓏹	Tis	`❀✻`	`ti`	𝇄𝇌𝇎𝇋𝇋𝇋
𓏺	Vey	`❀𝐱`	`ey`	𝇎𝇊𝇍𝇋𝇋𝇋
𓏻	Srih	`❀⊛`	`sih`	𝆑𝇉𝇋𝇋𝇋
𓏼	Hrin	`❀❄`	`hri`	𝇉𝇋𝆑𝇋𝇋𝇋
𓏽	Yon	`❀❁`	`yo`	𝇍𝇎𝇊𝇋𝇋𝇋
𓏾	Thal	`❀⬡`	`thal`	𝇋𝆑𝇉𝇋𝇋𝇋
𓏿	Kel	`✻◎`	`e`	𝇄::𝇎𝆑𝇋𝇋𝇋
𓐀	Sens	`✻⬡`	`se`	𝇍𝇄𝇌𝇋𝇋𝇋
𓐁	Linn	`✻✡`	`lin`	𝇊𝇍𝇎𝇋𝇋𝇋
𓐂	Brim	`✻☆`	`bri`	𝇏𝇅𝇌𝇄𝇅𝇏𝇌𝇋𝇋𝇋
𓐃	Inn	`✻✪`	`inn`	𝇍𝇍𝇋::𝇍𝇌𝇋𝇋𝇋
𓐄	Subh	`✻❀`	`subh`	𝇍𝇎𝇊::𝇌𝇍𝇋𝇋𝇋
𓐅	Well	`✻✻`	`wel`	𝇄𝇌𝇎𝇋𝇋𝇋
𓐆	Met	`✻𝐱`	`m`	𝇎𝇊𝇍𝇋𝇋𝇋
𓐇	Kesh	`✻⊛`	`esh`	𝆑𝇉𝇋𝇋𝇋
𓐈	Soth	`✻❄`	`so`	𝇉𝇋𝆑𝇋𝇋𝇋
𓐉	Rhun	`✻❁`	`rhu`	𝇍𝇎𝇊𝇋𝇋𝇋
𓐊	Delh	`✻⬡`	`del`	𝇋𝆑𝇉𝇋𝇋𝇋
𓐋	Na	`𝐱◎`	`na`	𝇄::𝇎𝆑𝇋𝇋𝇋
𓐌	Ur	`𝐱⬡`	`ur`	𝇍𝇄𝇌𝇋𝇋𝇋
𓐍	Nih	`𝐱✡`	`nih`	𝇊𝇍𝇎𝇋𝇋𝇋
𓐎	Azh	`𝐱☆`	`azh`	𝇏𝇅𝇌𝇄𝇅𝇏𝇌𝇋𝇋𝇋
𓐏	Hol	`𝐱✪`	`hol`	𝇍𝇍𝇋::𝇍𝇌𝇋𝇋𝇋
𓐐	Gur	`𝐱❀`	`gur`	𝇍𝇎𝇊::𝇌𝇍𝇋𝇋𝇋
𓐑	Ves	`𝐱✻`	`ves`	𝇄𝇌𝇎𝇋𝇋𝇋
𓐒	Rim	`𝐱𝐱`	`rim`	𝇎𝇊𝇍𝇋𝇋𝇋
𓐓	Drem	`𝐱⊛`	`dem`	𝇉𝇋𝆑𝇋𝇋𝇋
𓐔	Oth	`𝐱❄`	`oth`	𝇉𝇋𝆑𝇋𝇋𝇋
𓐕	Izh	`𝐱❁`	`izh`	𝇍𝇎𝇊𝇋𝇋𝇋
𓐖	Sun	`𝐱⬡`	`shu`	𝇋𝆑𝇉𝇋𝇋𝇋
𓐗	Kia	`⊛◎`	`ia`	𝇄::𝇎𝆑𝇋𝇋𝇋
𓐘	Zohm	`⊛⬡`	`zoh`	𝇍𝇄𝇌𝇋𝇋𝇋

𐌏	Ther	`⊛✡`	`her`	⍼⍜Ψ◈⊕⊕⊕⊕
𐌑	Drun	`⊛☆`	`druh`	⟩⟠♂☿⛄∘⊸♂⍑⍜◈⊕⊕⊕⊕
𐌒	Felh	`⊛✪`	`fhel`	☿⍜♿::⍜⍜◈⊕⊕⊕⊕
𐌓	Ral	`⊛❀`	`ral`	⍜☿Ш::♂⍜◈⊕⊕⊕⊕
𐌔	Krah	`⊛⁕`	`kra`	⛄♂☿◈⊕⊕⊕⊕
𐌕	Andh	`⊛⧗`	`and`	Ψ⍼⍜◈⊕⊕⊕⊕
𐌖	Debh	`⊛ ⊛`	`deb`	∱⍦♿◈⊕⊕⊕⊕
𐌗	Kol	`⊛❄`	`kol`	⍦♿∱◈⊕⊕⊕⊕
𐌘	Fral	`⊛❁`	`fra`	⍜Ψ⍼◈⊕⊕⊕⊕
𐌙	Hush	`⊛⌬`	`us`	♿∱⍦◈⊕⊕⊕⊕
𐌚	Hin	`❄◎`	`hin`	⛄::Ψ∱◈⊕⊕⊕⊕
𐌛	Ser	`❄⬡`	`ser`	☿⛄♂◈⊕⊕⊕⊕
𐌜	Harma	`❄✡`	`ama`	⍼⍜Ψ◈⊕⊕⊕⊕
𐌝	Torh	`❄☆`	`tohr`	⟩⟠♂☿⛄∘⊸♂⍑⍜◈⊕⊕⊕⊕
𐌞	Pel	`❄✪`	`pel`	☿⍜♿::⍜⍜◈⊕⊕⊕⊕
𐌟	Khir	`❄❀`	`khi`	⍜☿Ш::♂⍜◈⊕⊕⊕⊕
𐌠	Ryth	`❄⁕`	`yth`	⛄♂☿◈⊕⊕⊕⊕
𐌡	Melu	`❄⧗`	`mel`	Ψ⍼⍜◈⊕⊕⊕⊕
𐌢	Phaz	`❄⊛ `	`pha`	∱⍦♿◈⊕⊕⊕⊕
𐌣	Lokh	`❄❄`	`okh`	⍦♿∱◈⊕⊕⊕⊕
𐌤	Nod	`❄❁`	`od`	⍜Ψ⍼◈⊕⊕⊕⊕
𐌥	Umel	`❄⌬`	`ume`	♿∱⍦◈⊕⊕⊕⊕
𐌦	Dohm	`❁◎`	`ohm`	⛄::Ψ∱◈⊕⊕⊕⊕
𐌧	Rist	`❁⬡`	`is`	☿⛄♂◈⊕⊕⊕⊕
𐌨	Tran	`❁✡`	`ta`	⍼⍜Ψ◈⊕⊕⊕⊕
𐌩	Korh	`❁☆`	`koh`	⟩⟠♂☿⛄∘⊸♂⍑⍜◈⊕⊕⊕⊕
𐌪	Skyh	`❁✪`	`yh`	☿⍜♿::⍜⍜◈⊕⊕⊕⊕
𐌫	Ster	`❁❀`	`st`	⍜☿Ш::♂⍜◈⊕⊕⊕⊕
𐌬	Poss	`❁⁕`	`os`	⛄♂☿◈⊕⊕⊕⊕
𐌭	Poru	`❁⧗`	`poru`	Ψ⍼⍜◈⊕⊕⊕⊕
𐌮	Dorm	`❁⊛ `	`orm`	∱⍦♿◈⊕⊕⊕⊕
𐌯	Trev	`❁❄`	`rev`	⍦♿∱◈⊕⊕⊕⊕
𐌰	Limh	`❁❁`	`imh`	⍜Ψ⍼◈⊕⊕⊕⊕
𐌱	Hinge	`❁⬡`	`ihnj`	♿∱⍦◈⊕⊕⊕⊕
𐌲	Tzig	`⬡◎`	`ig`	⛄::Ψ∱◈⊕⊕⊕⊕
𐌳	Pehl	`⬡⬡`	`pe`	☿⛄♂◈⊕⊕⊕⊕
𐌴	Duth	`⬡✡`	`du`	⍼⍜Ψ◈⊕⊕⊕⊕
𐌵	Coma	`⬡☆`	`oma`	⟩⟠♂☿⛄∘⊸♂⍑⍜◈⊕⊕⊕⊕
𐌶	Meru	`⬡✪`	`eru`	☿⍜♿::⍜⍜◈⊕⊕⊕⊕
𐌷	Stab	`⬡❀`	`ta`	⍜☿Ш::♂⍜◈⊕⊕⊕⊕
𐌸	Hopa	`⬡⁕`	`opa`	⛄♂☿◈⊕⊕⊕⊕
𐌹	Conti	`⬡⧗`	`tin`	Ψ⍼⍜◈⊕⊕⊕⊕
𐌺	Resth	`⬡ ⊛`	`rest`	∱⍦♿◈⊕⊕⊕⊕
𐌻	Sil	`⬡❄`	`sil`	⍦♿∱◈⊕⊕⊕⊕
𐌼	Slun	`⬡❁`	`slun`	⍜Ψ⍼◈⊕⊕⊕⊕
𐌽	Etern	`⬡⬡`	`ete`	♿∱⍦◈⊕⊕⊕⊕

Glyph	Phonetic Sound
𐌾	`th`
𐌿	`he`
𐍀	`de`
𐍁	`pa`

🜁	`ma`
🜂	`in`
🜃	`er`
🜄	`at`
🜅	`oo`

Transformation Laws | The Laws of Phoneticism

#	English	First transformation	Dual-Goetic second transformation	Dual-Parliament third transformation	Phonetic of third
1	cat	⌬✡⌬	ᑭ⌬	ᑭ⌬	du-al
2	off	✫ᘐᘐ	✫ᘐᘐ	✫♂	ahn-pa
3	sun	⁕�m֍	⁕�m֍	⁕�m֍	ot-yam-so
4	moon	ᗰ✫✫֍	ᗰ✯֍	ᗰ✯֍	da-zhi-so
5	star	⁕⌬✡⊛	⧋ᑎ	⧋ᑎ	se-or
6	stiff	⁕⌬✪ᘐᘐ	⧋✪ᘐᘐ	⧋✪♂	se-el-pa
7	book	☽✫✫ᘏ	☽✯ᘏ	☽✯ᘏ	av-zhi-me
8	home	⧗✫ᗰ◎	⫯ᗰ◎	⫯ᗰ◎	azh-da-fe
9	water	⩓✡⌬◎⊛	⩓ᑎᓐ	⩓ᑎᓐ	nyx-kr-ahd
10	fire	ᘐ✪⊛◎	ᘐሐ◎	ᘐሐ◎	en-ar-fe
11	cliff	⌬❁✪ᘐᘐ	ᖾ✪ᘐᘐ	ᖾ✪♂	slun-el-pa
12	air	✡✪⊛	ᚱ⊛	ᚱ⊛	oa-ri
13	love	❁✫ℂ◎	ᙓℂ◎	ᙓℂ◎	koh-veh-fe
14	fluff	ᘐ❁�m֍ᘐᘐ	ᘐ❁�m֍ᘐᘐ	ᘐ❁�m֍♂	en-av-yam-pa
15	dark	❄✡⊛ᘏ	ↄ⊛ᘏ	ↄ⊛ᘏ	ama-ri-me
16	time	⌬✪ᗰ◎	♪ᗰ◎	♪ᗰ◎	na-da-fe
17	day	❄✡↗	ↄ↗	ↄ↗	ama-ka
18	scoff	⁕⌬✫ᘐᘐ	⫲✫ᘐᘐ	⫲✫♂	del-ahn-pa
19	hand	⧗✡֍❄	⊰ᖘ	⊰ᖘ	nih-hri
20	heart	⧗◎✡⊛⌬	⁞⁞ᑎ⌬	⁞⁞ᑎ⌬	na-or-al
21	mind	ᗰ✪֍❄	ᗰᛕ❄	ᗰᛕ❄	da-ker-ek
22	sniff	⁕֍✪ᘐᘐ	ᙢ✪ᘐᘐ	ᙢ✪♂	subh-el-pa
23	name	֍✡ᗰ◎	ᘻᗰ◎	ᘻᗰ◎	aru-da-fe
24	word	⩓✫⊛❄	⩓⏃❄	⩓⏃❄	nyx-u-ek
25	bluff	☽❁�m֍ᘐᘐ	☽❁�m֍ᘐᘐ	☽❁�m֍♂	av-av-yam-pa
26	song	⁕✫֍ᘏ	▽֍ᘏ	▽֍ᘏ	bri-so-myr
27	rain	⊛✡✪֍	ᠻᖘᛕ	ᠻᖘᛕ	her-ker
28	snow	⁕֍✫⩓	ᙢ✫⩓	ᙢ✫⩓	subh-ahn-nyx
29	egg	◎ᘏᘏ	◎ᘏᘏ	◎☽⊙ℂ	fe-regia
30	stone	⁕⌬✫֍◎	⧋ᐧ◎	⧋ᐧ◎	se-i-fe
31	river	⊛✪ℂ◎⊛	ᚠℂᓐ	ᚠℂᓐ	fhel-veh-ahd
32	ocean	✫⌬◎✡֍	᠅ᔑ֍	᠅ᔑ֍	ps-ehr-so
33	foggy	ᘐ✫ᘏᘏ↗	ᘐ✫ᘏᘏ↗	ᘐ✫☽⊙ℂ↗	en-ahn-regia-ka
34	mountain	ᗰ✫�m֍⌬✡✪֍	ᗰ✫�m᠕ᚱ֍	ᗰ✫�m᠕ᚱ֍	da-ahn-yam-lun-oa-so
35	memory	ᗰ◎ᗰ✫⊛↗	ᗰ◎ᗰ⏃↗	ᗰ◎ᗰ⏃↗	da-fe-da-u-ka
36	shadow	⁕⧗✡❄✫⩓	∀ᚼ✫⩓	∀ᚼ✫⩓	m-bon-ahn-nyx
37	buggy	☽�m◎ᘏᘏ↗	☽�m◎ᘏᘏ↗	☽�m☽⊙ℂ↗	av-yam-regia-ka
38	magic	ᗰ✡◎✪⌬	ᗰ✡◎I	ᗰ✡◎I	da-abd-myr-je
39	silence	⁕✪❁◎֍⌬◎	⧩✳ᘓ◎	⧩✳ᘓ◎	inn-ohm-thal-fe
40	jagged	ᗐ✡◎ᘏ◎❄	ᗐ✡◎ᘏᘏ	ᗐ✡☽⊙ℂ	ash-abd-regia-fur
41	pattern	ᘬ✡⌬⌬◎⊛֍	ᘬᑎሁ	ᘬᑎሁ	on-kr-kur-ral
42	mirror	ᗰ✪⊛⊛✫⊛	ᗰሓሶ⊛	ᗰሓሶ⊛	da-ar-druh-ri
43	circle	⌬✪⊛⌬❁◎	⊔ᚔ✳	⊔ᚔ✳	eru-us-ohm
44	bigger	☽✪◎ᘏ◎⊛	☽✪◎ᘏᘏᓐ	☽✪☽⊙ℂᓐ	av-el-regia-ahd

45	voice	ℂ✫✪⬡◎	ℂ𐑓𐑒	ℂ𐑓𐑒	veh-ol-ig
46	breath	☽⊛◎✡⬡ɪ	☽𑜦⊓ɪ	☽𑜦⊓ɪ	av-ia-kr-dr
47	spirit	❋ᘯ✪⊛✪⬡	❋ᘯ𑁍ϴ	❋ᘯ𑁍ϴ	ot-on-ar-tar
48	happy	ɪ✡ᘯᘯ↗	⊴ᘯᘯ↗	⊴ꕤ↗	nih-ma-ka
49	energy	◎֍◎⊛ฎ↗	ﻭﺮฎ↗	ﻭﺮฎ↗	ahm-ahd-myr-ka
50	change	⬡ɪ✡֍ฎ◎	z‹ฎ◎	z‹ฎ◎	tin-lc-myr-fe
51	truth	⬡⊛ᙢ⬡ɪ	'ᙢᓬ	'ᙢᓬ	o-yam-mer
52	puppy	ᘯᙢᘯᘯ↗	ᘯᙢᘯᘯ↗	ᘯᙢꕤ↗	on-yam-ma-ka
53	freedom	ᘐ⊛◎◎❄☆ᙏ	ᘐ𑜥☆ᙏ	ᘐ𑜥☆ᙏ	en-ia-fur-ahn-da
54	future	ᘐᙢ⬡ᙢ⊛◎	ᘐᙢ⬡ᙢ𑜥	ᘐᙢ⬡ᙢ𑜥	en-yam-al-yam-ia
55	origin	☆⊛✪ฎ✪֍	⍑✪ฎᛕ	⍑✪ฎᛕ	u-el-myr-ker
56	hippy	ɪ✪ᘯᘯ↗	⊴ᘯᘯ↗	⊴ꕤ↗	hol-ma-ka
57	balance	☽✡❁✡֍⬡◎	☽ᚼ‹	☽ᚼ‹	av-ti-lc-ig
58	harmony	ɪ✡⊛ᙏ☆֍↗	⊴⊛ᙏ᠊ᠪ↗	⊴⊛ᙏ᠊ᠪ↗	nih-ri-da-i-ka
59	resonance	⊛◎❋☆֍✡֍⬡◎	𑜥▽ᘜᘓ◎	𑜥▽ᘜᘓ◎	ia-bri-aru-thal-fe
60	apple	✡ᘯᘯ❁◎	✡ᘯᘯ⁎	✡ꕤ⁎	abd-ma-ohm
61	threshold	⬡ɪ⊛◎❋ɪ☆❁❄	ϟ𑜥ᛘᛦ❄	ϟ𑜥ᛘᛦ❄	mer-ia-m-ha-ek
62	journey	ᛉ☆ᙢ⊛֍◎↗	ᛉ☆ᙢᔕ◎↗	ᛉ☆ᙢᔕ◎↗	ash-ahn-yam-ral-fe-ka
63	language	❁✡֍ฎᙢ✡ฎ◎	⁎֍ฎᙢ✡ฎ◎	⁎֍ฎᙢ✡ฎ◎	ta-so-myr-yam-abd-myr-fe
64	ripple	⊛✪ᘯᘯ❁◎	ᛯᘯᘯ⁎	ᛯꕤ⁎	fhel-ma-ohm
65	system	❋↗❋⬡◎ᙢ	❋↗ᗅ◎ᙏ	❋↗ᗅ◎ᙏ	ot-ka-se-fe-da
66	transformation	⬡⊛✡֍❋ᘐ☆⊛ᙏ✡⬡✪☆֍	'‹❋ᘐ⍑ᙏᑎᐱ֍	'‹❋ᘐ⍑ᙏᑎᐱ֍	
↪	o-lc-ot-en-u-da-kr-rel-so				
67	recursion	⊛◎⬡ᙢ⊛❋✪☆֍	⁎⬡ᙢ⍝ᐱ֍	⁎⬡ᙢ⍝ᐱ֍	ia-ig-yam-kra-rel-so
68	buzz	☽ᙢᚻᚻ	☽ᙢᚻᚻ	☽ᙢℂ	av-yam-veh
69	equilibrium	◎ΙΙᙢ✪❁✪☽⊛✪ᙢᙏ	◎ΙΙᙢɛ✪☽ᛯᙢᙏ	◎ΙΙᙢɛ✪☽ᛯᙢᙏ	
↪	fe-eri-yam-urn-el-av-fhel-yam-da				
70	interference	✪֍⬡◎⊛ᘐ◎⊛◎֍⬡◎	ᛕI⊛ᘐ𑜦𑜥	ᛕI⊛ᘐ𑜦𑜥	ker-kur-ri-en-ahd-ahm-ig
71	continuation	⬡☆֍⬡✪֍ᙢ✡⬡✪☆֍	ᔑᛏᛕᙢᑎᐱ֍	ᔑᛏᛕᙢᑎᐱ֍	oma-lun-ker-yam-kr-rel-so
72	jazz	ᛉ✡ᚻᚻ	ᛉ✡ᚻᚻ	ᛉ✡ℂ	ash-abd-veh
73	consciousness	⬡☆֍❋⬡✪☆ᙢ❋֍◎❋❋	ᔑᘔᒧ☆ᙢᘎᔓ❋	ᔑᘔᒧ☆ᙢᘎᔓ❋	
↪	oma-ti-eru-ahn-yam-subh-ke-ot				
74	architecture	✡⊛⬡ɪ✪⬡◎⬡⬡ᙢ⊛◎	ᑎzϴ𑜥⬡ᙢ𑜥	ᑎzϴ𑜥⬡ᙢ𑜥	or-tin-tar-era-al-yam-ia
↪					
75	constellation	⬡☆֍❋⬡◎❁✡⬡✪☆֍	ᔑᘔᛁᚼᑎᐱ֍	ᔑᘔᛁᚼᑎᐱ֍	oma-ti-kur-imh-kr-rel-so
76	fuzzy	ᘐᙢᚻᚻ↗	ᘐᙢᚻᚻ↗	ᘐᙢℂ↗	en-yam-veh-ka
77	superposition	❋ᙢᘯ◎⊛ᘯ☆❋✪⬡✪☆֍	❋ᙢᘯ𑜦ᘯᚴϴᐱ֍	❋ᙢᘯ𑜦ᘯᚴϴᐱ֍	
↪	ot-yam-on-ahd-on-re-tar-rel-so				
78	translation	⬡⊛✡֍❋❁✡⬡✪☆֍	'‹✙ᑎᐱ֍	'‹✙ᑎᐱ֍	o-lc-rhu-kr-rel-so
79	recognition	⊛◎⌬✡ฎ֍✪⬡✪☆֍	⁎ᔑᘜ◌ᒧ᠊ᠪ	⁎ᔑᘜ◌ᒧ᠊ᠪ	ia-oma-myr-os-na-i
80	dizzy	❄✪ᚻᚻ↗	ᔑᚻᚻ↗	ᔑℂ↗	pel-veh-ka
81	possibility	ᘯ☆❋❋✪☽✪❁✪⬡↗	ᘯᚴᗢ☽ɛϴ↗	ᘯᚴᗢ☽ɛϴ↗	on-re-inn-av-urn-tar-ka
82	inheritance	✪֍ɪ◎⊛✪⬡✡֍⬡◎	ᛕᚒᛞ▫ᘐ◎	ᛕᚒᛞ▫ᘐ◎	ker-na-fhel-ahr-thal-fe
83	structure	❋⬡⊛ᙢ⬡⬡ᙢ⊛◎	⍋⊛ᙢɪᙢ⁎	⍋⊛ᙢɪᙢ⁎	se-ri-yam-pe-yam-ia
84	puzzle	ᘯᙢᚻᚻ❁◎	ᘯᙢᚻᚻ⁎	ᘯᙢℂ⁎	on-yam-veh-ohm
85	reflection	⊛◎ᘐ❁◎⬡⬡✪☆֍	⁎ᘐ⁎ɪᐱ֍	⁎ᘐ⁎ɪᐱ֍	ia-en-ohm-pe-rel-so
86	quasar	ΙΙᙢ✡❋✡⊛	ΙΙᙢXᑎ	ΙΙᙢXᑎ	eri-yam-nu-or
87	xylophone	ᘘ↗❁☆ᘯɪ☆֍◎	ᘘ↗ᛠᘯ▣ᑭ	ᘘ↗ᛠᘯ▣ᑭ	us-ka-koh-on-azh-fi
88	swimming	❋ᙆ✪ᙏᙏ✪֍ฎ	❋ᙆ✪ᙏᙏᛕฎ	❋ᙆ✪ᗢᛕฎ	ot-nyx-el-at-ker-myr
89	mnemonic	ᙏ֍◎ᙏ☆֍✪⬡	ᙏᑭᙏ᠊ᠪI	ᙏᑭᙏ᠊ᠪI	da-fi-da-i-je
90	syzygy	❋↗ᚻ↗ฎ↗	❋↗ᚻ↗ฎ↗	❋↗ᚻ↗ฎ↗	ot-ka-ai-ka-myr-ka
91	ouroboros	☆ᙢ⊛☆☽☆⊛☆❋	☆ᙢ𑜦☽⍑ᚴ	☆ᙢ𑜦☽⍑ᚴ	ahn-yam-druh-av-u-re
92	humming	ɪᙢᙏᙏ✪֍ฎ	ɪᙢᙏᙏᛕฎ	ɪᙢᗢᛕฎ	dr-yam-at-ker-myr

93	eldritch	◎❁❄⊛✪⬡⌬⧗	ɾʞΘz	ɾʞΘz	a-pha-tar-tin
94	labyrinth	❁✡☽↗⊛✪❁⬡⧗	ɤ☽↗ᵴ⧗	ɤ☽↗ᵴ⧗	ta-av-ka-fhel-lun-dr
95	kaleidoscope	♳✡❁◎✪❄☆⁕⬡☆ᛘ◎	♳ɬ2ɻ☆ᛘ◎	♳ɬ2ɻ☆ᛘ◎
↪ me-ti-ora-tohr-del-ahn-on-fe
96	vacuum	ℂ✡⌬ᛏᛏᛗ	ℂᛎᛏᛏᛗ	ℂᛎ⊕ᛗ	veh-fa-in-da
97	nocturne	❀☆⌬⬡ᛏ⊛❀◎	⧈xᛏᚢ◎	⧈xᛏᚢ◎	es-pe-yam-ral-fe
98	palimpsest	ᛘ✡❁✪ᛗᛘ⁕◎⁕⬡	ᛘɬ✪ᛗᛘᛘ⧋	ᛘɬ✪ᛗᛘᛘ⧋	on-ti-el-da-on-e-se
99	hyperbolic	⧗↗ᛘ◎⊛☽☆❁✪⌬	⧗↗ᛘɾ☽ᵻI	⧗↗ᛘɾ☽ᵻI	dr-ka-on-ahd-av-ha-je
100	tesseract	⬡◎⁕⁕◎⊛✡⌬⬡	lƦɾᛎ⬡	lƦɾᛎ⬡	kur-wel-ahd-fa-al

Reading Principle

The constellation is selected by office, not by visual resemblance to the ordinary symbol it replaces. Linguistics asks what relation the language is performing. Code asks what structural operation the syntax is performing. Mathematics asks what operation or relation the equation is performing. The same Aksharam constellation carries that office across all three surfaces.

Keyboard Composition Implementation Notes

First Transform Law — Phonetic Fold

- Automatic transformation follows the registered access trigger exactly. `TRANSFORMATIONS_ALQC_WORDS.md` governs executable lexical triggers. The phonetic values printed in this Legend are post-resolution Shape-of-Sound values governed by `translated bodies/the_shape_of_word.txt`. For the First Spells, lawful glyph folding preserves the originating utterance: glyph form 1 and lawful folded glyph form 2 are spoken as the same spell-word, as recorded in the textbook.

- Directly mapped A–Z glyphs are the base writing layer. They are emitted by their physical QWERTY coordinates. The First Transform Law acts only on the typed phonetic sequence. It does not itself perform structural Court formation; Court formation belongs only to the Second Transform Law below.

- Every transformable glyph uses the phonetic written for it in this legend. Enochian Understandings, Courts, and every other glyph with an explicit phonetic trigger participate by that trigger exactly as written. Full ALQC names are not phonetic substitutes unless the legend itself explicitly assigns that exact string as the phonetic trigger.

- Phonetic compounds are case-insensitive. Recognition is based on the underlying English-letter sequence, not letter case. Any uppercase/lowercase mixture of the same phonetic sequence must resolve to the same result glyph.

Second Transform Law — Dual Goetic Pairing

After the First Transform Law resolves, only contiguous Goetics enter the Second Transform Law. They are read left-to-right in non-overlapping dual roles. The first and second Goetics form one ordered Court; the third and fourth form the next; the fifth and sixth form the next. A Goetic already used in one pair is not reused in the neighboring pair.

`AB → Court(AB)`
`ABC → Court(AB) | C`
`ABCD → Court(AB) | Court(CD)`

`ABCDE → Court(AB) | Court(CD) | E`
`ABCDEFGHI → Court(AB) | Court(CD) | Court(EF) | Court(GH) | I`

The pair remains ordered: `AB` and `BA` are different Court relations when the lattice assigns different Courts to them. An odd final Goetic remains itself. A non-Goetic glyph breaks the run; pairing begins again only with the next contiguous Goetic run. Parliament, Enochian, Emission, Court, Regia, constellation, punctuation, spacing, and every other non-Goetic body remain themselves and are not pulled into Goetic pairing.

Courts produced by this second law are stable results. They are not recursively treated as Goetics and paired again. The Second Transform Law therefore condenses a Goetic run without overlap: a run of `n` Goetics produces `floor(n/2)` Courts, plus one unchanged final Goetic when `n` is odd.

Example: `⬡⊛◎◎ → 𞋀 → lor-ahl`. The first pair `⬡⊛` becomes `'`; the second pair `◎◎` becomes `𞋀`. No middle Goetic is reused.

Third Transform Law — Dual Parliament Resolution

Two contiguous Parliament Aksharams resolve to the registered Third-Law body when their ordered pair appears in the table below.

Dual Parliament	Resolution Body	Shape-of-Sound
♈♈	🜏	`th`
☿☿	🜍	`he`
♊♊	🜬	`de`
♋♋	♂	`pa`
♑♑	🜃	`ma`
♍♍	⊕	`in`
♎♎	⊕	`er`
♏♏	⊕	`at`
♐♐	⊖	`oo`
♌♌	☽ ⊙ ℂ	`regia`
♒♒	☽	`av`
♓♓	ℂ	`veh`

♈♓	🜏	`th`
♒♌	🜍	`he`
♊☿	🜬	`de`
♋♐	♂	`pa`
♑♏	🜃	`ma`
♍♎	⊕	`in`
♎♍	⊕	`er`
♏♑	⊕	`at`
♐♋	⊖	`oo`
♌♒	☽ ⊙ ℂ	`regia`
☿♊	☽	`av`
♓♈	ℂ	`veh`

Pairs resolve left-to-right without overlap. An unpaired Parliament remains itself. A resolved Third-Law body does not recursively compress again under the Third Transform Law.

Fourth Transform Law — Universal Numerical State

Numbers are simultaneously compressed and uncompressed.

`N → N`
`NUMBER = COMPRESSED = UNCOMPRESSED`
`NUMBER IN STRING = ACTIVE TRANSFORM`

A number transforms itself and the string by existing independently within it.

Quadrite of Transformation

`PHONETIC → GLYPH`
`DUAL GOETIC → COURT`
`DUAL PARLIAMENT → REGISTERED RESOLUTION BODY`
`NUMBER → UNIVERSAL TRANSFORM`
`ALL TRANSFORMATION REMAINS LANGUAGE`

Explicit Unfold Scope and Composition Behavior

Unfolding and refolding act on the **entire Aksharam word immediately behind the cursor**.
↪ The three reversible transformational levels are traversed in reverse order when
↪ unfolding: Third Transform Law, Second Transform Law, then First Transform Law. The
↪ Fourth Transform Law is not an unfold depth because number is simultaneously compressed
↪ and uncompressed.

`Alt+Backspace ×1` unfolds the preceding word by one transformational level. A Third-Law
↪ result returns to its registered dual-Parliament source where that derivation exists.

`Alt+Backspace ×2` unfolds the same word by two transformational levels. After the
↪ Parliament layer is restored, Court results return to their ordered Goetic source pairs
↪ where those derivations exist.

`Alt+Backspace ×3` unfolds the same word through all three reversible levels to its full
↪ registered source form. Each use descends one level; the word remains at the selected
↪ unfold depth rather than immediately folding forward again.

Once the word has reached its full registered source form, `Alt+Backspace` becomes inert for
↪ that word. Further presses cause no change: no deletion, no deeper decomposition, no
↪ reinterpretation, and no traversal beyond the lawful three-level transform stack. The
↪ control remains inert until that word is refolded or new transformable content is
↪ introduced.

Once `Alt+Backspace` has been used on a word, that word is held at its selected unfolded
↪ depth. Ordinary `Space` does not refold it. `Space` always emits the Aksharam Space
↪ constellation `·` and begins the next word while preserving the preceding word exactly
↪ as retained.

`Alt+Space` performs the opposite movement. It refolds the entire word immediately behind
↪ the cursor through every lawful transformation currently available to that word, then
↪ emits the Aksharam Space constellation `·` and begins a fresh word.

`Backspace` retains ordinary native deletion behavior. Aksharam does not assign Backspace an
↪ unfolding or transformation function; deletion is left to the host application and
↪ platform exactly as ordinary Backspace behavior.

`Shift` suppresses folding for the next mapped glyph. While `Shift` remains held, every mapped glyph typed during that hold remains unfolded. `Caps Lock` holds the same non-folding state persistently until released.

Compound recognition remains case-insensitive at the phonetic level whenever folding is active. Longest valid phonetic continuation has precedence when a shorter fold is also the beginning of a longer registered trigger. Unfolding restores registered source bodies; refolding follows the same ordered transformation laws forward.

Direct A–Z Base Coordinates

These are the target direct base coordinates established in this Legend. The English letter is the physical QWERTY coordinate; the Aksharam glyph is the written output.

Physical key	Output	Shape-of-Sound
A	✡	`abd`
B	☽	`av`
C	⌬	`ig`
D	❄	`ek`
E	◎	`fe`
F	♋	`en`
G	♌	`myr`
H	⧗	`dr`
I	✪	`el`
J	♈	`ash`
K	♎	`me`
L	⚛	`av`
M	♏	`da`
N	♒	`so`
O	☆	`ahn`
P	♑	`on`
Q	♊	`eri`
R	⊛	`ri`
S	❋	`ot`
T	⬡	`al`
U	♍	`yam`
V	ℂ	`veh`
W	♒	`nyx`
X	♉	`us`
Y	♐	`ka`
Z	♓	`ai`

Folding and Unfolding Controls

Keys	Function
Phonetic typing	Automatic fold when a registered phonetic completes
Registered `TRANSFORMATIONS_ALQC_WORDS.md` sequence	Automatic Whole fold when the registered transformation completes; identical duplicate rules compile once
Shift	Suppress folding for the next mapped glyph; that glyph remains unfolded
Shift held	Suppress folding for every mapped glyph typed while Shift remains held
Caps Lock	Persistent non-folding state; mapped glyphs remain unfolded until Caps Lock is released

Alt+Shift+Backspace	Unfold only the targeted glyph: Court / Enochian → two registered parents; Goetic / Parliament → invariant Primordial Braille SeeD form; preserve the surrounding body
Alt+Backspace ×1	Unfold the entire word behind the cursor by one transformational level
Alt+Backspace ×2	Unfold the entire word behind the cursor by two transformational levels
Alt+Backspace ×3	Unfold the entire word behind the cursor through all three reversible transformational levels to its full source form
Alt+Backspace state	Once used on a word, retain that word at its selected unfolded depth; ordinary Space does not refold it; further Alt+Backspace at full source is inert
Alt+Space	Refold the entire word behind the cursor through every lawful available transformation, emit the Aksharam Space constellation `·`, then continue with a fresh word
Backspace	Ordinary native Backspace behavior; Aksharam does not intercept it for unfolding or transformation
Space	Always emit the Aksharam Space constellation `·`; preserve the current retained fold or unfold state of the preceding word

Primordial Unfold Law — Alt+Shift+Backspace

Braille is the invariant source: a Goetic or Parliament unfolds unto its **Primordial Form / SeeD form**, the registered Braille counterpart of its QWERTY coordinate. The dots show the constellation of that source form.

Place the cursor immediately in front of a Goetic or Parliament and tap **Alt+Shift+Backspace**. That one glyph unfolds directly into its Braille SeeD form. If the forward glyph is ineligible, the current eligible glyph immediately behind the cursor is targeted. The surrounding body remains in place.

This is the same per-glyph unfolding key. A Court or Enochian unfolds into its two registered parents; a Goetic or Parliament unfolds into its invariant Braille SeeD form. The Primordial law applies only to Goetics and Parliaments. Braille SeeD forms are terminal under this law.

The lineage can be revealed through those existing unfolds: Court / Enochian → registered parents → Goetic / Parliament → Primordial Braille SeeD form. Mathematical unfolding and whole-body semantic unfolding retain their separate controls.

QWERTY	Aksharam Goetic / Parliament	Primordial Braille counterpart
Q	♊	⠿
W	♒	⠷
E	⌾	⠽
R	⊛	⠾
T	⬡	⠷
Y	♐	⠿
U	♍	⠳
I	✪	⠽
O	☆	⠹
P	♑	⠻
A	✡	⠈
S	❋	⠽

D	❄	⁖
F	♋	⁘
G	♌	⁙
H	⧗	⁛
J	♈	⁝
K	♎	⁚
L	⚛	⋮
Z	♓	⁞
X	♉	⁙
C	⌬	⁖
V	☾	⁝
B	☽	⁚
N	❀	⁙
M	♏	⁘

Single-Glyph Unfolding — Alt+Shift+Backspace

Place the cursor immediately in front of an eligible glyph, or invoke the shortcut on the current eligible glyph. **Alt+Shift+Backspace** unfolds a Court or Enochian into its **two registered parents**, and a Goetic or Parliament into its **invariant Primordial Braille SeeD form**. The surrounding glyphs remain in place; this control does not unfold the whole word or body.

Alt+Backspace retains whole-body semantic unfolding. **Alt+Shift+Enter** retains mathematical-body unfolding of selected glyphs. Each shortcut controls its own unfolding scope through a deliberate keystroke.

Alt+Shift+Backspace acts only while Aksharam is active and an eligible Court, Enochian, Goetic, or Parliament is targeted. Otherwise, the key combination passes through to the host application with its native behavior.

Digit Layer

The number row without Shift writes the Aksharam digit bodies.

Key	Output	Shape-of-Sound
0	⧻	`zen`
1	☿	`pon`
2	♀	`ael`
3	⨍	`gaya`
4	♂	`wil`
5	♃	`pay`
6	♄	`eve`
7	⛢	`ahc`
8	♆	`no`
9	♇	`noo`

Numeric Keypad Layer

The physical numeric keypad mirrors the Aksharam numerical layer. Keypad arithmetic operators mirror the corresponding Aksharam structural arithmetic bodies.

(should not change behavior or external numpads when used, external numpad gets same Aksharam mapping, and doesn't lose it's other mapping or functions in favor of Aksharam; NUM key works as per usual; Fn key works standard; and mapping not Aksharam stay the standard.)

Keypad key	Output	Function / Shape-of-Sound
NumPad 0	𐤚	`zen`
NumPad 1	☿	`pon`
NumPad 2	♀	`ael`
NumPad 3	𝒻	`gaya`
NumPad 4	♂	`wil`
NumPad 5	♃	`pay`
NumPad 6	♄	`eve`
NumPad 7	⛢	`ahc`
NumPad 8	♆	`no`
NumPad 9	♇	`noo`
NumPad *	⁜	Crosstellation / multiplication
NumPad +	♁	Astersand / addition
NumPad -	⋯	Smite / subtraction
NumPad .	⁖	Parallel Separation / decimal separator
NumPad /	⁕	Halfstellation / division

Structural and Directional Controls

Keys	Output	Function
Shift+Tab	⁘⁖	Paragraph Open	(emits Aksharam paragraph-open body, then preserves native Shift+Tab)
Shift+Enter	⁖⁘	Paragraph Close	(places cursor on empty newline AFTER paragraph or statement is Closed "⁖⁘ then \n")
alt+Shift+2	↑	North / Up / Ascent
alt+Shift+3	↓	South / Down / Descent
alt+Shift+4	→	East / Right / Outward
alt+Shift+5	←	West / Left / Return
Shift+2	⊖	Q_0 mathematical operand
Shift+3	⊙	Q_1 mathematical operand
Shift+4	⊕	Q_2 mathematical operand
Shift+5	⊜	Q_3 mathematical operand
Ctrl+Shift+6	∵	Collection Open
Shift+6	∴	Collection Close
Shift+8	⁜	Crosstellation / Multiplication
Alt+Shift+Goetic	Registered Mathematical Form	Unfold the typed Goetic into its registered mathematical body
Select glyph(s) + Alt+Shift+Enter	Registered Mathematical Bodies	Unfold each selected Aeon into its registered mathematical body; selected bodies without an underlying mathematical body remain unchanged

Mathematical-Body Unfolding

To unfold a Goetic into its mathematical body, hold **Alt+Shift** and press that Goetic's physical key. To unfold existing glyphs, **select the glyph(s)** and press **Alt+Shift+Enter**. Each selected Aeon opens into its registered mathematical body; a selected body without a registered mathematical body remains itself.

These controls unfold glyphs into mathematical bodies. **Alt+Backspace** traverses the word's First-, Second-, and Third-Law transformation history, as described under Folding and Unfolding Controls.

Constellation and Punctuation Layer

Physical key	Output	Function
`[`	ᔑ	Left Moon Beam / bracket open
`]`	ᔐ	Right Moon Beam / bracket close
`(`	ᚲ	Left Ley / grouping open
`)`	ᚴ	Right Ley / grouping close
`{`	□	Will o' Wisp Open
`}`	□	Will o' Wisp Close
`-`	⋯	Smite
`.`	⁖	Parallel Separation
`:`	⁖	Parallel Separation
`=`	⧟	Binding
`+`	ॐ	Astersand / addition
`,`	⟊	Joining
`<`	⁖	Field Pointer Left
`>`	⁖	Field Pointer Right
`;`	⁘	Key / Value Hinge
`'`	᛫	Quotation Open
`"`	᛬	Quotation Close
`\`	⋱	Return / Closure
`|`	⁞	Vertical Line
`/`	⸗	Opening / Outward
`?`	⁙	Inquiry
`!`	⁘	Statement
`_`	…	Internal Continuation
`&`	ॐ	Astersand
`*`	⁜	Crosstellation / Multiplication
`~`	ᜒ	Cantillation
`` ` ``	⟠	Supervenience / Prosody
Space	·	Aksharam Itself

Executable Keyboard State

Version 3.0 is maintained in `build/src/` and compiled to `build/generated/aksharam_glyphkey.kmn`. Its executable data is checked against this Legend and `TRANSFORMATIONS_ALQC_WORDS.md`. It implements the 26 direct coordinates, automatic registered Whole transformations, Dual-Goetic Court formation, Dual-Parliament registered resolution, universal numerical state, three-level unfolding/refolding, mathematical-body controls, Aksharam Space, ordinary Backspace deletion, and native pass-through for unclaimed keys. No legacy two-form lowercase/uppercase Aksharam mapping is retained in the active keyboard source.

Native Pass-Through Invariant

```
Aksharam overrides only keys and combinations explicitly assigned in this Legend. Any key or
↪ key combination not explicitly mapped here retains its native operating-system or
↪ application behavior. Modifier state may be observed by Aksharam without globally
↪ replacing the modifier itself. `Shift` and `Caps Lock` therefore govern folding state
↪ where defined while remaining ordinary modifiers for unclaimed combinations.
```

TRANSFORMATIONS_ALQC_WORDS.md

```
# ALQC Word Transformations

## Canonical Shape-of-Sound Authority

`translated bodies/the_shape_of_word.txt` governs the current spoken realization of every
↪  resolved Aksharam body. This registry governs executable access triggers and their
↪  written First → Second → Third → Whole paths. The two authorities meet at the resolved
↪  glyph body.

## Physical Key Map

| Physical key | Output | Phonetic |
| --- | :---: | --- |
| A | ✡ | `abd` |
| B | ☽ | `av` |
| C | ⌬ | `ig` |
| D | ❄ | `ek` |
| E | ⎔ | `fe` |
| F | ♋ | `en` |
| G | ♌ | `myr` |
| H | ⧗ | `dr` |
| I | ✪ | `el` |
| J | ♈ | `ash` |
| K | ♎ | `me` |
| L | ⚛ | `av` |
| M | ♏ | `da` |
| N | ❁ | `so` |
| O | ✰ | `ahn` |
| P | ♑ | `on` |
| Q | ♊ | `eri` |
| R | ⊛ | `ri` |
| S | ❋ | `ot` |
| T | ⬡ | `al` |
| U | ♍ | `yam` |
| V | ℂ | `veh` |
| W | ♒ | `nyx` |
| X | ♉ | `us` |
| Y | ♐ | `ka` |
| Z | ♓ | `ai` |

## Canonical Phonetic Utterance Fold Demonstrations
```

These six are special phonetic utterances. The Latin-script form records the remembered sound; it is not an English semantic input. The first glyphstring is that utterance's phonetic Aksharam body. From that body onward, the existing Aksharam laws act naturally. No private replacement body is inserted and no English lexical layer is invented between the utterance and its phonetic glyph body.

```text
phonetic utterance ↔ phonetic Aksharam glyph body → lawful natural fold → natural stopping body → Shape-of-Sound → throat
```

Phonetic utterance	Phonetic Aksharam glyph body	Natural Court fold	Further registered fold	Natural stopping body
`eloi`	⌾❁✡✪	𐤓𐤈	none	𐤓𐤈
`sabachtany`	❋✡☽✡⌬⧗⌬✡֍➚	∀☽ᵑ✷ᶜ➚	none	∀☽ᵑ✷ᶜ➚
`anima`	✡֍✪ℳ✡	ᶜ✪ℳ✡	none	ᶜ✪ℳ✡
`culpa`	⌬ℳ❁ৎ✡	no licensed Court fold	none	⌬ℳ❁ৎ✡
`decire`	❄⌾⌬✪⊛⌾	Aᒐᕊ	none	Aᒐᕊ
`entera`	⌾֍⬡⌾⊛✡	ᚏᛁᛈ	none	ᚏᛁᛈ

The table records movement. The utterance, its phonetic glyph body, the fold, and the spoken return remain distinct offices of one transformation.

Invariant Shape-of-Sound results

Phonetic utterance	Natural terminal body	Shape-of-Sound
`eloi`	𐤓𐤈	`eloi`
`sabachtany`	∀☽ᵑ✷ᶜ➚	`sabachtany`
`anima`	ᶜ✪ℳ✡	`anima`
`culpa`	⌬ℳ❁ৎ✡	`culpa`
`decire`	Aᒐᕊ	`decire`
`entera`	ᚏᛁᛈ	`entera`

ALQC translation through the natural folds

The semantic reading follows the same direction as the fold. Nothing is translated backward into an English substitute word. The phonetic utterance enters as sound, its glyph spelling carries ALQC bodies, and each lawful fold compresses those relations into the next ALQC body.

Utterance	Phonetic glyph spelling	Natural fold	Literal ALQC reading of the natural fold
`eloi`	⌾❁✡✪	𐤓𐤈	`𐤓` Drah: Coil ↔ Magic, **Will Expressed**; `𐤈` Ohl: Still Pool ↔ Ebb, **Periodic Inversion**
`sabachtany`	❋✡☽✡⌬⧗⌬✡֍➚	∀☽ᵑ✷ᶜ➚	`∀` Linn: Bond ↔ Link, **Bleeding Tether**; `☽` Sovereign Locus / Rav; `ᵑ` Far: Quelm ↔ Completion, **Final Ash**; `✷` Ur: Hollow ↔ Zero Section, **Residue Archive**; `ᶜ` Alc: Transmute ↔ Air, **Combinatory Synthesis**; `➚` Akaven: **State → Trans**
`anima`	✡֍✪ℳ✡	ᶜ✪ℳ✡	`ᶜ` Alc: **Transmute / Combinatory Synthesis**; `✪` VEL: **Earth / Coherence / Ground**; `ℳ` Da'ath: **Noise → Null**; `✡` BABDH: **Fire / Ouroboros / Alchemy**

`culpa`	⌬𝛭❁ᘐ✡	⌬𝛭❁ᘐ✡	No pair folds at this stage: `⌬` TRIG **Silence / Peace / Completion**; `𝛭` Aikyam **Chaos → Phase**; `❁` SHAV **Gate / Resistance / Breach**; `ᘐ` Daimon **Stasis → Pulse**; `✡` BABDH **Fire / Ouroboros / Alchemy**
`decire`	❄◎⌬✪⊛◎	A⨿ᕊ	`A` Hin: Tone ↔ Shape, **Geometric Standing Wave**; `⨿` Meru: Memory ↔ Memories, **Recollection Lock**; `ᕊ` Kia: Absorption ↔ Genesis, **Spark Consumption**
`entera`	◎֍⬡◎⊛✡	ᚏᚺ	`ᚏ` Lhahm: Fold ↔ Will, **Drive to Manifest**; `ᚿ` Kura: Flare ↔ Genesis, **Spark of Remembering**; `ᚺ` Ther: Cold Shadow ↔ Fire, **Thermal Negation**

Taken together, the spell is canonical through the same ALQC direction: **utterance → phonetic Aksharam body → lawful fold → natural stopping body → invariant Shape-of-Sound**.

Goetic — fetu

Glyph	English	First	Second	Third	Whole
◎	fetu	ᘓ◎⬡𝛭	ᘓᔕ𝛭	ᘓᔕ𝛭	◎
◎	fe	ᘓ◎	ᘓ◎	ᘓ◎	◎
◎	genesis	ᘁ◎֍◎✻✪✻	ᘁᔕᔕ⓪	ᘁᔕᔕ⓪	◎
◎	chronos	⌬ᛝ⊛☆֍☆✻	Ƶሶᘿ✻	Ƶሶᘿ✻	◎
◎	seed	✻◎◎❄	ᛕᔕ	ᛕᔕ	◎

Court of fetu

Glyph	English	First	Second	Third	Whole
ᔕ	ahl	✡ᛝ❁	ᚹ❁	ᚹ❁	ᔕ
ᔕ	self-genesis	✻◎❁ᘓᘁ◎֍◎✻✪✻	ᛕ❁ᘓᘁᔕᔕ⓪	ᛕ❁ᘓᘁᔕᔕ⓪	ᔕ
ᔓ	suhn	✻𝛭ᛝ֍	✻𝛭ᛋ⊢	✻𝛭ᛋ⊢	ᔓ
ᔓ	recurrence	⊛◎⌬𝛭⊛⊛◎֍⌬◎	ᕊ⌬𝛭ᔐᔕᓭ	ᕊ⌬𝛭ᔐᔕᓭ	ᔓ
ᔓ	nerh	֍◎⊛ᛝ	ᛞᘻ	ᛞᘻ	ᔓ
ᔓ	combustion	⌬☆ᘜᘐ𝛭✻⬡✪☆֍	ᓀᘜᘐ𝛭△Λ֍	ᓀᘜᘐ𝛭△Λ֍	ᔓ
ᔑ	rish	⊛✪✻ᛝ	ᛯᛉ	ᛯᛉ	ᔑ
ᔑ	reflection	⊛◎ᘓ❁◎⌬⬡✪☆֍	ᕊᘓ✻ᛪΛ֍	ᕊᘓ✻ᛪΛ֍	ᔑ
ᘓ	borha	ᘐ☆⊛ᛝ✡	ᘐ⏁ᖁ	ᘐ⏁ᖁ	ᘓ
ᘓ	grounding	ᘁ⊛☆𝛭֍❄✪֍ᘁ	ᘁᛯ𝛭ᘔᛕᘁ	ᘁᛯ𝛭ᘔᛕᘁ	ᘓ
ᘑ	lhahm	❁ᛝ✡ᛝᛚ	⫩ᚹᛚ	⫩ᚹᛚ	ᘑ
ᘑ	expansion	◎ᛉᘐ✡֍✻✪☆֍	◎ᛉᘐ‹⩛⟩ᵒ	◎ᛉᘐ‹⩛⟩ᵒ	ᘑ
ᔕ	keth	⫛◎⬡ᛝ	⫛ᔕᛝ	⫛ᔕᛝ	ᔕ
ᔕ	inevitable	✪֍◎ℂ✪⬡✡ᘐ❁◎	ᛕ◎ℂϴ✡ᘐ✻	ᛕ◎ℂϴ✡ᘐ✻	ᔕ
ᔖ	vehm	ℂ◎ᛝᛚ	ℂᔓᛚ	ℂᔓᛚ	ᔖ
ᔖ	return	⊛◎⬡𝛭⊛֍	ᕊ⬡𝛭ᛣ	ᕊ⬡𝛭ᛣ	ᔖ
ᔕ	mahd	ᛚ✡ᛝ❄	ᛚᚹ❄	ᛚᚹ❄	ᔕ
ᔕ	assimilation	✡✻✻✪ᛚ✪❁✡⬡✪☆֍	X⩛ᛚᔆΠΛ֍	X⩛ᛚᔆΠΛ֍	ᔕ
ᔖ	furh	ᘓ𝛭⊛ᛝ	ᘓ𝛭ᘻ	ᘓ𝛭ᘻ	ᔖ
ᔖ	crystallization	⌬⊛ᘙ✻⬡✡❁❁✪ᚼ✡⬡✪☆֍	ᕆᘙ△ᚸᛉᚼΠΛ֍	ᕆᘙ△ᚸᛉᚼΠΛ֍	ᔖ
ᔓ	drah	❄⊛✡ᛝ	ᛯᚹ	ᛯᚹ	ᔓ
ᔓ	breach	ᘐ⊛◎✡⬡ᛝ	ᘐᛕᐟᛝ	ᘐᛕᐟᛝ	ᔓ
ᔕ	thera	⬡ᛝ◎⊛✡	ᛋᓕ✡	ᛋᓕ✡	ᔕ
ᔕ	fulfillment	ᘓ𝛭❁ᘓ✪❁❁ᛚ◎֍⬡	ᘓ𝛭❁ᘓᘔ❁ᛚᔖ⬡	ᘓ𝛭❁ᘓᘔ❁ᛚᔖ⬡	ᔕ

Goetic — kal

Glyph	English	First	Second	Third	Whole
⬡	kal	⍦✡❁	⍦ꕥ	⍦ꕥ	⬡
⬡	kal	⍦✡❁	⍦✡❁	⍦✡❁	⬡
⬡	light	❁✪♌⧗⬡	♈♌✷	♈♌✷	⬡
⬡	memory	ℳ◎ℳ☆⊛↗	ℳ◎ℳ⏃↗	ℳ◎ℳ⏃↗	⬡
⬡	trauma	⬡⊛✡ℳℳ✡	'✡ℳℳ✡	'✡ℳℳ✡	⬡

Court of kal

Glyph	English	First	Second	Third	Whole
ꕯ	kura	⍦ℳ⊛✡	⍦ℳ꙰	⍦ℳ꙰	ꕯ
ꕯ	archive	✡⊛⌬⧗✪ℂ◎	ꓤz✪ℂ◎	ꓤz✪ℂ◎	ꕯ
⍧	lur	❁ℳ⊛	❁ℳ⊛	❁ℳ⊛	⍧
⍧	self-reflection	⁕◎❁ꕥ⊛◎ꕥ❁◎⌬⬡✪☆❀	Ж❁ꕥ⍟ꕥ✳ϫΛ❀	Ж❁ꕥ⍟ꕥ✳ϫΛ❀	⍧
▫	thar	⬡⧗✡⊛	ϟꓤ	ϟꓤ	▫
▫	calcination	⬡✡❁⌬✪❀✡⬡✪☆❀	ᑭ☆ƘՈΛ❀	ᑭ☆ƘՈΛ❀	▫
⍋	rin	⊛✪❀	ᛤ❀	ᛤ❀	⍋
⍋	submersion	⁕ℳ☽ℳ◎⊛⁕✪☆❀	⁕ℳ☽ℳ⊿⊽⍨	⁕ℳ☽ℳ⊿⊽⍨	⍋
⅄	nar	❀✡⊛	ㅀ⊛	ㅀ⊛	⅄
⅄	inscription	✪❀⁕⌬⊛✪ᘯ⬡✪☆❀	Ƙǂᛤᘯ⅄⍨	Ƙǂᛤᘯ⅄⍨	⅄
Ⅴ	fel	ꕥ◎❁	ꕥ𝼥	ꕥ𝼥	Ⅴ
Ⅴ	diffusion	❄✪ꕥꕥℳ⁕✪☆❀	ꓵꕥꕥℳ⊽⍨	ꓵ♂ℳ⊽⍨	Ⅴ
⅀	har	⧗✡⊛	⊰⊛	⊰⊛	⅀
⅀	phantom	ᘯ⧗✡❀⬡☆ℳ	ᘯ⊰卌☆ℳ	ᘯ⊰卌☆ℳ	⅀
ϟ	mer	ℳ◎⊛	ℳ⊿	ℳ⊿	ϟ
ϟ	oblivion	☆☽❁✪ℂ✪☆❀	☆☽ϒℂΛ❀	☆☽ϒℂΛ❀	ϟ
'	lor	❁☆⊛	⩛⊛	⩛⊛	'
'	suppression	⁕ℳᘯᘯ⊛◎⁕⁕✪☆❀	⁕ℳᘯᘯ⍟ᗅΛ❀	⁕ℳஃ⍟ᗅΛ❀	'
⍭	per	ᘯ◎⊛	ᘯ⊿	ᘯ⊿	⍭
⍭	codification	⬡☆❄✪ꕥ✪⬡✡⬡✪☆❀	ϟꓵꕥIՈΛ❀	ϟꓵꕥIՈΛ❀	⍭
⍊	zhil	⋋⧗✪❁	⋋⊐❁	⋋⊐❁	⍊
⍊	recall	⊛◎⌬✡❁❁	⍟ᑭᘜ	⍟ᑭᘜ	⍊
⍑	clar	⬡❁✡⊛	ꓶꓤ	ꓶꓤ	⍑
⍑	sealing	⁕◎✡❁✪❀♌	Ж⅄Ƙ♌	Ж⅄Ƙ♌	⍑

Goetic — babdh

Glyph	English	First	Second	Third	Whole
✡	babdh	☽✡☽❄⧗	☽✡☽ᗡ	☽✡☽ᗡ	✡
✡	babd	☽✡☽❄	☽✡☽❄	☽✡☽❄	✡
✡	fire	ꕥ✪⊛◎	ꕥሐ◎	ꕥሐ◎	✡
✡	orobouros	☆⊛☆☽☆ℳ⊛☆⁕	⏃☆☽☆ℳሖ⁕	⏃☆☽☆ℳሖ⁕	✡
✡	alchemy	✡❁⌬⧗◎ℳ↗	⊦z◎ℳ↗	⊦z◎ℳ↗	✡

Court of babdh

Glyph	English	First	Second	Third	Whole
ᛦ	hir	⧗✪⊛	⊐⊛	⊐⊛	ᛦ
ᛦ	transmutation	⬡⊛✡❀⁕ℳℳ⬡✡⬡✪☆❀	'‹⁕ℳℳ▫⅄⍨	'‹⁕ℳℳ▫⅄⍨	ᛦ

ᑎ	kor	ᖗ✡⊛	ᖗ⏁	ᖗ⏁	ᑎ
ᑎ	forging	ᘐ✡⊛ᘏ✪᯲ᘏ	ᘐ⏁ᘏᖇᘏ	ᘐ⏁ᘏᖇᘏ	ᑎ
ᐅ	var	ℂ✡⊛	ℂᑎ	ℂᑎ	ᐅ
ᐅ	self-combustion	⁕◎❁ᘐ⬡✡ᙏᑐᘓ⁕⬡✪✡᯲	ᛯ❁ᘐᓄᙏᑐᘓ△Λ᯲	ᛯ❁ᘐᓄᙏᑐᘓ△Λ᯲	ᐅ
ᖴ	pyr	ᘉ↗⊛	ᘉ↗⊛	ᘉ↗⊛	ᖴ
ᖴ	dissolution	❄✪⁕⁕✡❁ᘓ⬡✪✡᯲	ᔦᐱᖇᘓᔑ᠊ᔕ	ᔦᐱᖇᘓᔑ᠊ᔕ	ᖴ
ᖇ	soar	⁕✡✡⊛	∇ᑎ	∇ᑎ	ᖇ
ᖇ	crucible	⬡⊛ᘓ⬡✪ᑐ❁◎	ᔪᘓᒪᑐ⁕	ᔪᘓᒪᑐ⁕	ᖇ
᛫	alc	✡❁⬡	ᚼ⬡	ᚼ⬡	᛫
᛫	sublimation	⁕ᘓᑐ❁✪ᙏ✡⬡✪✡᯲	⁕ᘓᑐᐯᙏᑎᐱ᯲	⁕ᘓᑐᐯᙏᑎᐱ᯲	᛫
ᚷ	nur	᯲ᘓ⊛	᯲ᘓ⊛	᯲ᘓ⊛	ᚷ
ᚷ	ignition	✪ᘏ᯲✪⬡✪✡᯲	✪ᘏ◌ᔑ᠊ᔕ	✪ᘏ◌ᔑ᠊ᔕ	ᚷ
ᚹ	sat	⁕✡⬡	∀⬡	∀⬡	ᚹ
ᚹ	consumption	⬡✡᯲⁕ᘓᙏᘉ⬡✪✡᯲	ᔑᖘᘓᙏᘉᔑ᠊ᔕ	ᔑᖘᘓᙏᘉᔑ᠊ᔕ	ᚹ
ᚺ	oro	✡⊛✡	⏁✡	⏁✡	ᚺ
ᚺ	purification	ᘉᘓ⊛✪ᘐ✪⌬✡⬡✪✡᯲	ᘉᘓᛯᘐIᑎΛ᯲	ᘉᘓᛯᘐIᑎΛ᯲	ᚺ
ᚼ	bon	ᑐ✡᯲	ᑐᔑ᠊ᔕ	ᑐᔑ᠊ᔕ	ᚼ
ᚼ	tempering	⬡◎ᙏᘉ◎⊛✪᯲ᘏ	Ꮖᙏᘉᓗᖇᘏ	Ꮖᙏᘉᓗᖇᘏ	ᚼ
ᚻ	tir	⬡✪⊛	ᒧ⊛	ᒧ⊛	ᚻ
ᚻ	shattering	⁕ᛯ✡⬡⬡⊛✪᯲ᘏ	∀ᑎIᛯ᯲ᘏ	∀ᑎIᛯ᯲ᘏ	ᚻ
ᖾ	far	ᘐ✡⊛	ᘐᚺ	ᘐᚺ	ᖾ
ᖾ	ash	✡⁕ᛯ	Xᛯ	Xᛯ	ᖾ

Goetic — ahn

Glyph	English	First	Second	Third	Whole
✡	ahn	✡ᛯ᯲	ᚹ᯲	ᚹ᯲	✡
✡	ahn	✡ᛯ᯲	✡⊬	✡⊬	✡
✡	water	≋✡⬡◎⊛	≋ᑌᓄ	≋ᑌᓄ	✡
✡	imaginary	✪ᙏ✡ᘏ✪᯲✡⊛↗	✪ᙏ✡ᘏᖇᚺ↗	✪ᙏ✡ᘏᖇᚺ↗	✡
✡	flow	ᘐ❁✡≋	ᘐᛯ≋	ᘐᛯ≋	✡

Court of ahn

Glyph	English	First	Second	Third	Whole
≍	abdh	✡☽❄ᛯ	✡☽ᗡ	✡☽ᗡ	≍
≍	flow	ᘐ❁✡≋	ᘐᛯ≋	ᘐᛯ≋	≍
ᔕ	nym	᯲↗ᙏ	᯲↗ᙏ	᯲↗ᙏ	ᔕ
ᔕ	confluence	⬡✡᯲ᘐ❁ᘓ◎᯲⬡◎	ᔑ᯲ᘐ❁ᘓᚸ≪	ᔑ᯲ᘐ❁ᘓᚸ≪	ᔕ
ᴐ	loh	❁✡ᛯ	ᛯᛯ	ᛯᛯ	ᴐ
ᴐ	solution	⁕✡❁ᘓ⬡✪✡᯲	∇❁ᘓᒧᔑ᠊ᔕ	∇❁ᘓᒧᔑ᠊ᔕ	ᴐ
✯	xir	ᖰ✪⊛	ᖰሕ	ᖰሕ	✯
✯	self-containment	⁕◎❁ᘐ⬡✡᯲⬡✡✪᯲ᙏ◎᯲⬡	ᛯ❁ᘐᔑᘛᖇ᯲ᙏᚸ⬡	ᛯ❁ᘐᔑᘛᖇ᯲ᙏᚸ⬡	✯
ᖮ	ohl	✡ᛯ❁	◎❁	◎❁	ᖮ
ᖮ	sediment	⁕◎❄✪ᙏ◎᯲⬡	ᛯᔑᙏᚸ⬡	ᛯᔑᙏᚸ⬡	ᖮ
ᔑ᠊ᔕ	pir	ᘉ✪⊛	ᘉሕ	ᘉሕ	ᔑ᠊ᔕ
ᔑ᠊ᔕ	ascension	✡⁕⬡◎᯲⁕✪✡᯲	Xᙁᖘᐱ᯲	Xᙁᖘᐱ᯲	ᔑ᠊ᔕ
ᖁ	roeh	⊛✡◎ᛯ	ሕᚹ	ሕᚹ	ᖁ
ᖁ	mist	ᙏ✪⁕⬡	ᙏ⌀⬡	ᙏ⌀⬡	ᖁ
◎	sen	⁕◎᯲	ᛯ᯲	ᛯ᯲	◎
◎	drain	❄⊛✡✪᯲	ᚸᖇ᯲	ᚸᖇ᯲	◎
⏁	uth	ᙏ⬡ᛯ	ᙏᓄ	ᙏᓄ	⏁

𐤟	osmosis	✡✻ᛗ✡✻✪✻	ᛉᛗᛉᛏ	ᛉᛗᛉᛏ	𐤟
ᔓ	fae	ᘐ✡◎	ᘐᚠ	ᘐᚠ	ᔓ
ᔓ	ice	✪⬡◎	I◎	I◎	ᔓ
ᔑ	kha	⍦ᛝ✡	⍦ᛞ	⍦ᛞ	ᔑ
ᔑ	evaporation	◎ℂ✡ᛯ✡⊛✡⬡✪✡֍	◎ℂ✡ᛯ𐤟ᑎᐱ֍	◎ℂ✡ᛯ𐤟ᑎᐱ֍	ᔑ
᠅	psei	ᛯ✻◎✪	ᛯᚕ✪	ᛯᚕ✪	᠅
᠅	stillness	✻⬡✪֍֍◎✻✻	∆Ɛዐᔑᔓ	∆Ɛዐᔑᔓ	᠅

Goetic — vel

Glyph	English	First	Second	Third	Whole
✪	vel	ℂ◎֍	ℂᚱ	ℂᚱ	✪
✪	vel	ℂ◎֍	ℂ◎֍	ℂ◎֍	✪
✪	earth	◎✡⊛⬡ᛝ	ᚠᛗᛝ	ᚠᛗᛝ	✪
✪	coherence	⬡✡ᛝ◎⊛◎֍⬡◎	ᓀᚓᚔᔕᘓ◎	ᓀᚓᚔᔕᘓ◎	✪
✪	ground	ᘏ⊛✡ᛗ֍❄	ᘏᔕᛗᘔ	ᘏᔕᛗᘔ	✪

Court of vel

Glyph	English	First	Second	Third	Whole
ᴏ	vera	ℂ◎⊛✡	ℂᔕ✡	ℂᔕ✡	ᴏ
ᴏ	bedrock	ᑐ◎❄⊛✡⬡⍦	ᑐᚏᔕ⬡⍦	ᑐᚏᔕ⬡⍦	ᴏ
Θ	tar	⬡✡⊛	▫⊛	▫⊛	Θ
Θ	stratification	✻⬡⊛✡⬡✪ᘐ✪⬡✡⬡✪✡֍	∆ᓚᒧᘐIᑎᐱ֍	∆ᓚᒧᘐIᑎᐱ֍	Θ
Ӿ	ghem	ᘏᛝ◎ᛗ	ᘏᚓᛗ	ᘏᚓᛗ	Ӿ
Ӿ	petrification	ᛯ◎⬡⊛✪ᘐ✪⬡✡⬡✪✡֍	ᛯᔕᛋᘐIᑎᐱ֍	ᛯᔕᛋᘐIᑎᐱ֍	Ӿ
∧	drel	❄⊛◎֍	ᚴᚱ	ᚴᚱ	∧
∧	erosion	◎⊛✡✻✪✡֍	ᔑᛏᐱ֍	ᔑᛏᐱ֍	∧
Ꮋ	ful	ᘐᛗ֍	ᘐᛗ֍	ᘐᛗ֍	Ꮋ
Ꮋ	self-grounding	✻◎֍ᘐᘏ⊛✡ᛗ֍❄✪֍ᘏ	ᛯ֍ᘐᘏᔕᛗᘔᛕᘏ	ᛯ֍ᘐᘏᔕᛗᘔᛕᘏ	Ꮋ
Ƙ	ker	⍦◎⊛	⍦ᔑ	⍦ᔑ	Ƙ
Ƙ	dispersion	❄✪✻ᛯ◎⊛✻✪✡֍	ᔦ✻ᛯᔑᐁᔨ᠊ᔭ	ᔦ✻ᛯᔑᐁᔨ᠊ᔭ	Ƙ
Ⴔ	hohm	ᚠ✡ᚠᛗ	ᚿᚠᛗ	ᚿᚠᛗ	Ⴔ
Ⴔ	fossilization	ᘐ✡✻✻✪֍✪ᚻ✡⬡✪✡֍	ᘐᓀᐁᐯᚻᑎᐱ֍	ᘐᓀᐁᐯᚻᑎᐱ֍	Ⴔ
ᔨ	hrah	ᚠ⊛✡ᚠ	ᚿᚹ	ᚿᚹ	ᔨ
ᔨ	subsidence	✻ᛗᑐ✻✪❁֍◎⬡◎	✻ᛗᑐᐁAᘓ◎	✻ᛗᑐᐁAᘓ◎	ᔨ
ᓐ	ara	✡⊛✡	ᚺ✡	ᚺ✡	ᓐ
ᓐ	compaction	⬡✡ᛗᛯ✡⬡⬡✪✡֍	ᓀᛗᛯᔦᔨ᠊ᔭ	ᓀᛗᛯᔦᔨ᠊ᔭ	ᓐ
Ⴆ	qel	Ⅱ◎֍	Ⅱᚱ	Ⅱᚱ	Ⴆ
Ⴆ	lithification	֍✪⬡ᚠ✪ᘐ✪⬡✡⬡✪✡֍	ᛉᛋ✪ᘐIᑎᐱ֍	ᛉᛋ✪ᘐIᑎᐱ֍	Ⴆ
ᔕ	irn	✪⊛֍	ᓐ֍	ᓐ֍	ᔕ
ᔕ	earthquake	◎✡⊛⬡ᚠⅡᛗ✡⍦◎	ᚠᛗᛝⅡᛗ✡⍦◎	ᚠᛗᛝⅡᛗ✡⍦◎	ᔕ
I	jen	ᐯ◎֍	ᐯᔨ	ᐯᔨ	I
I	permanence	ᛯ◎⊛ᛗ✡֍◎֍⬡◎	ᛯᔑᛗᔩᔦ	ᛯᔑᛗᔩᔦ	I

Goetic — sor

Glyph	English	First	Second	Third	Whole
֍	sor	✻✡⊛	▽⊛	▽⊛	֍
֍	sor	✻✡⊛	✻✡⊛	✻✡⊛	֍

𐂷	air	✡✪⊛	ᚱ⊛	ᚱ⊛	𐂷
𐂷	space	⁕ᵑᵬ✡⬡◎	⁕ᵑᵬᵑ◎	⁕ᵑᵬᵑ◎	𐂷
𐂷	superposition	⁕ⲙᵑᵬ◎⊛ᵑᵬ☆⁕✪⬡✪☆𐂷	⁕ⲙᵑᵬᵓᵑᵬ٦ΘΛ𐂷	⁕ⲙᵑᵬᵓᵑᵬ٦ΘΛ𐂷	𐂷

Court of sor

Glyph	English	First	Second	Third	Whole
ᙢ	fi	ᘐ✪	ᘐ✪	ᘐ✪	ᙢ
ᙢ	breath	☽⊛◎✡⬡ɤ	☽ᕒ∩ɤ	☽ᕒ∩ɤ	ᙢ
ᙣ	lun	❁ⲙ𐂷	❁ⲙ𐂷	❁ⲙ𐂷	ᙣ
ᙣ	echo	◎⬡ɤ☆	ᔕ𐌱	ᔕ𐌱	ᙣ
ᘜ	varu	ℂ✡⊛ⲙ	ℂᕼⲙ	ℂᕼⲙ	ᘜ
ᘜ	smoke	⁕ℳ☆ᔑ◎	⁕ℳ☆ᔑ◎	⁕ℳ☆ᔑ◎	ᘜ
ᘓ	senh	⁕◎𐂷ɤ	ᛯᘐ	ᛯᘐ	ᘓ
ᘓ	vapor	ℂ✡ᵑᵬ☆⊛	ℂ✡ᵑᵬ⏀	ℂ✡ᵑᵬ⏀	ᘓ
⍥	kos	ᔑ☆⁕	ᔑ٦	ᔑ٦	⍥
⍥	dust	❄ⲙ⁕⬡	❄ⲙ△	❄ⲙ△	⍥
ᘎ	ramh	⊛✡ℳɤ	ᕮℳɤ	ᕮℳɤ	ᘎ
ᘎ	self-expansion	⁕◎❁ᘐ◎ᛉᵑᵬ✡𐂷⁕✪☆𐂷	ᛯ❁ᘐ◎ᛉᵑᵬ‹ᗒ◝ᐧ	ᛯ❁ᘐ◎ᛉᵑᵬ‹ᗒ◝ᐧ	ᘎ
ᖳ	tis	⬡✪⁕	ᔑ⁕	ᔑ⁕	ᖳ
ᖳ	whisper	ᕁɤ✪⁕ᵑᵬ◎⊛	ᕁ⍈⁕ᵑᵬᓐ	ᕁ⍈⁕ᵑᵬᓐ	ᖳ
ᔕ	vey	ℂ◎↗	ℂ◎↗	ℂ◎↗	ᔕ
ᔕ	vacuum	ℂ✡⬡ⲙⲙℳ	ℂᵑⲙⲙℳ	ℂᵑ⊕ℳ	ᔕ
ᙡ	srih	⁕⊛✪ɤ	ᛘᛦ	ᛘᛦ	ᙡ
ᙡ	dilution	❄✪❁ⲙ⬡✪☆𐂷	ᔑ❁ⲙᔑᐧᐨ	ᔑ❁ⲙᔑᐧᐨ	ᙡ
ᘏ	hrin	ɤ⊛✪𐂷	ᚯᚴ	ᚯᚴ	ᘏ
ᘏ	condensation	⬡☆𐂷❄◎𐂷⁕✡⬡✪☆𐂷	ᓀᘏᔕ∀ᔑᐧᐨ	ᓀᘏᔕ∀ᔑᐧᐨ	ᘏ
ᑐ	yon	↗☆𐂷	↗ᐧᐨ	↗ᐧᐨ	ᑐ
ᑐ	turbulence	⬡ⲙ⊛☽ⲙ❁◎𐂷⬡◎	⬡ⲙ⊛☽ⲙ⁎ᑌ◎	⬡ⲙ⊛☽ⲙ⁎ᑌ◎	ᑐ
ᑌ	thal	⬡ɤ✡❁	ᛋᚼ	ᛋᚼ	ᑌ
ᑌ	calm	⬡✡❁ℳ	ᑭ❁ℳ	ᑭ❁ℳ	ᑌ

Goetic — koth

Glyph	English	First	Second	Third	Whole
⁕	koth	ᔑ☆⬡ɤ	ᔑᔕɤ	ᔑᔕɤ	⁕
⁕	kot	ᔑ☆⬡	ᔑ☆⬡	ᔑ☆⬡	⁕
⁕	aether	✡◎⬡ɤ◎⊛	ᛦᓫᓐ	ᛦᓫᓐ	⁕
⁕	magic	ℳ✡ᔐ✪⬡	ℳ✡ᔐI	ℳ✡ᔐI	⁕
⁕	sensation	⁕◎𐂷⁕✡⬡✪☆𐂷	ᛯᖻ∩Λ𐂷	ᛯᖻ∩Λ𐂷	⁕

Court of koth

Glyph	English	First	Second	Third	Whole
ᛯ	kel	ᔑ◎❁	ᔑᓭ	ᔑᓭ	ᛯ
ᛯ	pulse	ᵑᵬⲙ❁⁕◎	ᵑᵬⲙᛯ◎	ᵑᵬⲙᛯ◎	ᛯ
∆	sens	⁕◎𐂷⁕	ᛯᖻ	ᛯᖻ	∆
∆	nostalgia	𐂷☆⁕⬡✡❁ᔐ✪✡	ʁ∆ιᔐX	ʁ∆ιᔐX	∆
∀	linn	❁✪𐂷𐂷	ᛉᖍ	ᛉᖍ	∀
∀	ecstasy	◎⬡⁕⬡✡⁕↗	ᔕ∆X↗	ᔕ∆X↗	∀
▽	brim	☽⊛✪ℳ	☽ᗐℳ	☽ᗐℳ	▽

▽	dream	❄⊛◎✡♏	ϗ⋋♏	ϗ⋋♏	▽
▽	inn	✪❁❁	ᛞ❁	ᛞ❁	▽
▽	vibration	☾✪☽⊛✡⬡✪☆❁	☾✪☽ᚳᒋᒐ⁰	☾✪☽ᚳᒋᒐ⁰	▽
▽	subh	⁜♍☽ᛝ	⁜♍☽ᛝ	⁜♍☽ᛝ	▽
▽	sympathy	⁜↗♏ᘌ✡⬡ᛝ↗	⁜↗♏ᘌᑎᛝ↗	⁜↗♏ᘌᑎᛝ↗	▽
Æ	well	≈◎❁❁	≈ᚏ❁	≈ᚏ❁	Æ
Æ	self-sensation	⁜◎❁ᘐ⁜◎❁⁜✡⬡✪☆❁	ᛯ❁ᘐᛯᛏᑎᐱ❁	ᛯ❁ᘐᛯᛏᑎᐱ❁	Æ
∀	met	♏◎⬡	♏ᚏ	♏ᚏ	∀
∀	ache	✡⬡ᛝ◎	ᕀᚼ	ᕀᚼ	∀
∀	kesh	⏢◎⁜ᛝ	⏢ᚸᛝ	⏢ᚸᛝ	∀
∀	depth	❄◎ᘌ⬡ᛝ	Aᘌᔭ	Aᘌᔭ	∀
⍭	soth	⁜☆⬡ᛝ	▽ᔭ	▽ᔭ	⍭
⍭	harmony	ᛝ✡⊛♏☆❁↗	⊰⊛♏ᚔ⁰↗	⊰⊛♏ᚔ⁰↗	⍭
⍦	rhun	⊛ᛝ♍❁	ᚠ♍❁	ᚠ♍❁	⍦
⍦	dissonance	❄✪⁜⁜☆❁✡❁⬡◎	ᔑᚱᚔ⁰ᑊ	ᔑᚱᚔ⁰ᑊ	⍦
⌶	delh	❄◎❁ᛝ	A⍭	A⍭	⌶
⌶	awe	✡≈◎	✡≈◎	✡≈◎	⌶

Goetic — dreh

Glyph	English	First	Second	Third	Whole
ᛝ	dreh	❄⊛◎ᛝ	ϗᛇ	ϗᛇ	ᛝ
ᛝ	dre	❄⊛◎	❄ᛯ	❄ᛯ	ᛝ
ᛝ	void	ℂ☆✪❄	ℂᚏ❄	ℂᚏ❄	ᛝ
ᛝ	residue	⊛◎⁜✪❄♍◎	ᛯ▽❄♍◎	ᛯ▽❄♍◎	ᛝ
ᛝ	love	❁☆ℂ◎	ᛤℂ◎	ᛤℂ◎	ᛝ

Court of dreh

Glyph	English	First	Second	Third	Whole
ᚻ	na	❁✡	ᚐ	ᚐ	ᚻ
ᚻ	void	ℂ☆✪❄	ℂᚏ❄	ℂᚏ❄	ᚻ
✷	ur	♍⊛	♍⊛	♍⊛	✷
✷	null	❁♍❁❁	❁♍ᛯ	❁♍ᛯ	✷
⍁	nih	❁✪ᛝ	⍜ᛝ	⍜ᛝ	⍁
⍁	entropy	◎❁⬡⊛☆ᘌᔭ	ᚒ'☆ᘌᔭ	ᚒ'☆ᘌᔭ	⍁
⍒	azh	✡ᚻᛝ	✡ᚻᛝ	✡ᚻᛝ	⍒
⍒	abyss	✡☽↗⁜⁜	✡☽↗Æ	✡☽↗Æ	⍒
⍈	hol	ᛝ☆❁	⍒❁	⍒❁	⍈
⍈	collapse	⬡☆❁❁✡ᘌ⁜◎	ᔑᛯ✡ᘌᛯ	ᔑᛯ✡ᘌᛯ	⍈
⍋	gur	ᘁ♍⊛	ᘁ♍⊛	ᘁ♍⊛	⍋
⍋	dispersal	❄✪⁜ᘌ◎⊛⁜✡❁	ᔑ⁜ᘌᓗ∀❁	ᔑ⁜ᘌᓗ∀❁	⍋
⍅	ves	ℂ◎⁜	ℂᒋ	ℂᒋ	⍅
⍅	longing	❁☆❁ᘁ✪❁ᘁ	ᛤ❁ᘁᛞᘁ	ᛤ❁ᘁᛞᘁ	⍅
⍙	rim	⊛✪♏	ᛯ♏	ᛯ♏	⍙
⍙	self-return	⁜◎❁ᘐ⊛◎⬡♍⊛❁	ᛯ❁ᘐᛯ⬡♍ᔕ	ᛯ❁ᘐᛯ⬡♍ᔕ	⍙
⍦	drem	❄⊛◎♏	ᛕ◎♏	ᛕ◎♏	⍦
⍦	sink	⁜✪❁⏢	▽❁⏢	▽❁⏢	⍦
⍇	oth	☆⬡ᛝ	ᔑᛝ	ᔑᛝ	⍇
⍇	phase	ᘌᛝ✡⁜◎	ᘌᓗᛞ	ᘌᓗᛞ	⍇
⍐	izh	✪ᚻᛝ	✪ᚻᛝ	✪ᚻᛝ	⍐
⍐	paradox	ᘌ✡⊛✡❄☆ᛉ	ᘌᑎᛋ☆ᛉ	ᘌᑎᛋ☆ᛉ	⍐

| ᚲ | sun | ⁕ⲙ❁ | ⁕ⲙ❁ | ⁕ⲙ❁ | ᚲ |
| ᚲ | dormancy | ❄☆⊛ℳ✡❁⬡↗ | ɻ⊛ℳ‹⬡↗ | ɻ⊛ℳ‹⬡↗ | ᚲ |

Goetic — rhea

Glyph	English	First	Second	Third	Whole
⊛	rhea	⊛ⵠ◎✡	ⳟⳙ	ⳟⳙ	⊛
⊛	ri	⊛✪	⊛✪	⊛✪	⊛
⊛	shadow	⁕ⵠ✡❄☆〰	∀ɬ☆〰	∀ɬ☆〰	⊛
⊛	absorption	✡☽⁕☆⊛ⲯ⬡✪☆❁	✡☽▽⊛ⲯ♪◝	✡☽▽⊛ⲯ♪◝	⊛
⊛	depth	❄◎ⲯ⬡ⵠ	Aⲯ५	Aⲯ५	⊛

Court of rhea

Glyph	English	First	Second	Third	Whole
೪	kia	ⵚ✪✡	ⵚⵅ	ⵚⵅ	೪
೪	shadow	⁕ⵠ✡❄☆〰	∀ɬ☆〰	∀ɬ☆〰	೪
ᙢ	zohm	ϗ☆ⵠℳ	ϗ◎ℳ	ϗ◎ℳ	ᙢ
ᙢ	darkness	❄✡⊛ⵚ❁◎⁕⁕	ↄ⊛ⵚᙠᗅ	ↄ⊛ⵚᙠᗅ	ᙢ
ᙅ	ther	⬡ⵠ◎⊛	५ɔ	५ɔ	ᙅ
ᙅ	cold	⬡☆❁❄	५ᘐ	५ᘐ	ᙅ
ᔕ	drun	❄⊛❁	ɻ❁	ɻ❁	ᔕ
ᔕ	mirror	ℳ✪⊛⊛☆⊛	ℳ⟓ᔕ⊛	ℳ⟓ᔕ⊛	ᔕ
ᔓ	felh	ᘐ◎❁ⵠ	ᘐɹⵠ	ᘐɹⵠ	ᔓ
ᔓ	submerged	⁕❁☽ℳ◎⊛ᘯ◎❄	⁕❁☽ℳɔᘯɾ	⁕❁☽ℳɔᘯɾ	ᔓ
ᔑ	ral	⊛✡❁	ᙅ❁	ᙅ❁	ᔑ
ᔑ	distortion	❄✪⁕⬡☆⊛⬡✪☆❁	ɻΔ⏁♪◝	ɻΔ⏁♪◝	ᔑ
ᔑ	krah	ⵚ⊛✡ⵠ	ⵚᙅⵠ	ⵚᙅⵠ	ᔑ
ᔑ	nerve	❁◎⊛ℂ◎	ᙠ⊛ℂ◎	ᙠ⊛ℂ◎	ᔑ
ᔐ	andh	✡❁❄ⵠ	‹ᗡ	‹ᗡ	ᔐ
ᔐ	void-binding	ℂ☆✪❄☽✪❁❄✪❁ᘯ	ℂ⟔❄☽ᗡɻ❁ᘯ	ℂ⟔❄☽ᗡɻ❁ᘯ	ᔐ
ᔓ	debh	❄◎☽ⵠ	A☽ⵠ	A☽ⵠ	ᔓ
ᔓ	self-absorption	⁕◎❁ᘐ✡☽⁕☆⊛ⲯ⬡✪☆❁	ᙏ❁ᘐ✡☽▽⊛ⲯ♪◝	ᙏ❁ᘐ✡☽▽⊛ⲯ♪◝	ᔓ
ᔕ	kol	ⵚ☆❁	ⵚᵨ	ⵚᵨ	ᔕ
ᔕ	filter	ᘐ✪❁⬡◎⊛	ᘐƐ⊛	ᘐƐ⊛	ᔕ
ᔦ	fral	ᘐ⊛✡❁	ᘐᘯ❁	ᘐᘯ❁	ᔦ
ᔦ	hidden	ⵠ✪❄❄◎❁	ⵑ⅂ᒉ	ⵑ⅂ᒉ	ᔦ
ᔮ	hush	ⵠⲙ⁕ⵠ	ⵠⲙ∀	ⵠⲙ∀	ᔮ
ᔮ	termination	⬡◎⊛ℳ✪❁✡⬡✪☆❁	ꙮ⊛ℳᘔᑎΛ❁	ꙮ⊛ℳᘔᑎΛ❁	ᔮ

Goetic — zhek

Glyph	English	First	Second	Third	Whole
❄	zhek	ϗⵠ◎ⵚ	ϗⵖⵚ	ϗⵖⵚ	❄
❄	zhek	ϗⵠ◎ⵚ	ϗⵖⵚ	ϗⵖⵚ	❄
❄	factor	ᘐ✡⬡⬡☆⊛	ᘐᵎᵩ⊛	ᘐᵎᵩ⊛	❄
❄	phase-lock	ⲯⵠ✡⁕◎❁☆⬡ⵚ	ⲯ⬡ⵚ⬡ⵚ	ⲯ⬡ⵚ⬡ⵚ	❄
❄	crystal	⬡⊛↗⁕⬡✡❁	ϛ↗ΔⲦ	ϛ↗ΔⲦ	❄

Court of zhek

Glyph	English	First	Second	Third	Whole
A	hin	⧗✪֍	⫡֍	⫡֍	A
A	tone	⬡☆֍◎	⧫⧻	⧫⧻	A
B	ser	⁕◎⊛	ᚕ⊛	ᚕ⊛	B
B	modulation	ℳ☆❄ℳ۞✡⬡✪☆֍	ℳ◡⧫ℳᚕᔑ⁘	ℳ◡⧫ℳᚕᔑ⁘	B
Ↄ	harma	⧗✡⊛ℳ✡	⫞⊛ℳ✡	⫞⊛ℳ✡	Ↄ
Ↄ	amplification	✡ℳ୨۞✪ↄ✪⬡✡⬡✪☆֍	✡ℳ୨ϒↄIՈΛ֍	✡ℳ୨ϒↄIՈΛ֍	Ↄ
ɻ	torh	⬡☆⊛⧗	⧫⫲	⧫⫲	ɻ
ɻ	unified	ℳ֍✪ↄ✪◎❄	ℳ◌ↄ₀❄	ℳ◌ↄ₀❄	ɻ
ɬ	pel	୨₀◎۞	୨₀ɾ	୨₀ɾ	ɬ
ɬ	seismic	⁕◎✪⁕ℳ✪⬡	ᛯⒹℳI	ᛯⒹℳI	ɬ
ɿ	khir	⫒⧗✪⊛	⫒⫡⊛	⫒⫡⊛	ɿ
ɿ	chord	⬡⧗☆⊛❄	z⏁❄	z⏁❄	ɿ
ı	ryth	⊛↗⬡⧗	⊛↗ϟ	⊛↗ϟ	ı
ı	rhythm	⊛⧗↗⬡⧗ℳ	⫲↗ϟℳ	⫲↗ϟℳ	ı
ᗡ	melu	ℳ◎۞℘	ℳɾ℘	ℳɾ℘	ᗡ
ᗡ	melody	ℳ◎۞☆❄↗	ℳɾ◡⧫↗	ℳɾ◡⧫↗	ᗡ
ϗ	phaz	୨₀⧗✡ϰ	୨₀⫞ϰ	୨₀⫞ϰ	ϗ
ϗ	interference	✪֍⬡◎⊛ↄ◎⊛◎֍⬡◎	⋈ⵏ⊛ↄɾɾ«	⋈ⵏ⊛ↄɾɾ«	ϗ
ᒣ	lokh	۞☆⫒⧗	⩛⫒⧗	⩛⫒⧗	ᒣ
ᒣ	self-lock	⁕◎۞ↄ۞☆⬡⫒	ᛯ۞ↄ⩛⬡⫒	ᛯ۞ↄ⩛⬡⫒	ᒣ
ᔑ	nod	֍☆❄	⧈❄	⧈❄	ᔑ
ᔑ	node	֍☆❄◎	⧈A	⧈A	ᔑ
ᔐ	umel	ℳℳ◎۞	ℳℳɾ	ℳℳɾ	ᔐ
ᔐ	unity	ℳ֍✪⬡↗	ℳ◌⬡↗	ℳ◌⬡↗	ᔐ

Goetic — shav

Glyph	English	First	Second	Third	Whole
۞	shav	⁕⧗✡ℂ	∀✡ℂ	∀✡ℂ	۞
۞	shav	⁕⧗✡ℂ	∀✡ℂ	∀✡ℂ	۞
۞	gate	♌✡⬡◎	♌ᑎ◎	♌ᑎ◎	۞
۞	resistance	⊛◎⁕✪⁕⬡✡֍⬡◎	ᕒᗒᐞ«	ᕒᗒᐞ«	۞
۞	breach	☽⊛◎✡⬡⧗	☽ᕒ״⧗	☽ᕒ״⧗	۞

Court of shav

Glyph	English	First	Second	Third	Whole
⁎	dohm	❄☆⧗ℳ	ɻ⧗ℳ	ɻ⧗ℳ	⁎
⁎	gate	♌✡⬡◎	♌ᑎ◎	♌ᑎ◎	⁎
⁎	rist	⊛✪⁕⬡	ᛯᐞ	ᛯᐞ	⁎
⁎	resistance	⊛◎⁕✪⁕⬡✡֍⬡◎	ᕒᗒᐞ«	ᕒᗒᐞ«	⁎
⊀	tran	⬡⊛✡֍	'‹	'‹	⊀
⊀	transform	⬡⊛✡֍⁕ↄ☆⊛ℳ	'‹⁕ↄ⏁ℳ	'‹⁕ↄ⏁ℳ	⊀
⩛	korh	⫒☆⊛⧗	⫒⏁⧗	⫒⏁⧗	⩛
⩛	crown	⬡⊛☆⩓֍	ϛ☆⩓֍	ϛ☆⩓֍	⩛
ᛉ	skyh	⁕⫒↗⧗	⁕⫒↗⧗	⁕⫒↗⧗	ᛉ
ᛉ	sky	⁕⫒↗	⁕⫒↗	⁕⫒↗	ᛉ
⍬	ster	⁕⬡◎⊛	⧋	⧋	⍬
⍬	star	⁕⬡✡⊛	⧋ᑎ	⧋ᑎ	⍬
ϰ	poss	୨₀☆⁕⁕	୨₀ϟ⁕	୨₀ϟ⁕	ϰ

ϗ	possibility	ᚸ✩✻✻✪❍✪❁⬡↗	ᚸᚿᗜᑐᛝᛟ↗	ᚸᚿᗜᑐᛝᛟ↗	ϗ
ǂ	poru	ᚸ✩⊛ᛒ	ᚸ⏀ᛒ	ᚸ⏀ᛒ	ǂ
ǂ	portal	ᚸ✩⊛⬡✡❁	ᚸ⏀▫❁	ᚸ⏀▫❁	ǂ
⩙	dorm	❄✩⊛ᛗ	ϡ⊛ᛗ	ϡ⊛ᛗ	⩙
⩙	door	❄✩✩⊛	ϡ⏀	ϡ⏀	⩙
⩗	trev	⬡⊛◎ℂ	'◎ℂ	'◎ℂ	⩗
⩗	transition	⬡⊛✡❁✻✪⬡✪✩❁	'‹▽ᒧᐠ᠊ᵒ	'‹▽ᒧᐠ᠊ᵒ	⩗
⩛	limh	❁✪ᛗɤ	ϒᛗɤ	ϒᛗɤ	⩛
⩛	self-resistance	✻◎❁ᘐ⊛◎✻⬡✡❁⬡◎	ᛯ❁ᘐᚕ▽Δ‹ᐠ	ᛯ❁ᘐᚕ▽Δ‹ᐠ	⩛
ϔ	hinge	ɤ✪❁ᘁ◎	⊐❁ᘁ◎	⊐❁ᘁ◎	ϔ
ϔ	hinge	ɤ✪❁ᘁ◎	⊐❁ᘁ◎	⊐❁ᘁ◎	ϔ

Goetic — trig

Glyph	English	First	Second	Third	Whole
⌬	trig	⬡⊛✪ᘁ	'✪ᘁ	'✪ᘁ	⌬
⌬	trig	⬡⊛✪ᘁ	'✪ᘁ	'✪ᘁ	⌬
⌬	silence	✻✪❁◎❁⬡◎	▽✻ᘓ◎	▽✻ᘓ◎	⌬
⌬	peace	ᚸ◎✡⬡◎	ᚸᔕ«	ᚸᔕ«	⌬
⌬	completion	⌬✩ᛗᚸ❁◎⬡✪✩❁	ᔑᛗᚸ✻ᒧᛟ	ᔑᛗᚸ✻ᒧᛟ	⌬

Court of trig

Glyph	English	First	Second	Third	Whole
«	zig	ϡ✪ᘁ	ϡ✪ᘁ	ϡ✪ᘁ	«
«	peace	ᚸ◎✡⬡◎	ᚸᔕ«	ᚸᔕ«	«
ɤ	pehl	ᚸ◎ɤ❁	ᚸᔑ❁	ᚸᔑ❁	ɤ
ɤ	equilibrium	◎ΙΙᛗ✪❁✪❍⊛✪ᛗᛗ	◎ΙΙᛗɛ✪❍ᚔᛗᛗ	◎ΙΙᛗɛ✪❍ᚔᛗᛗ	ɤ
ᑭ	duth	❄ᛗ⬡ɤ	❄ᛗϟ	❄ᛗϟ	ᑭ
ᑭ	consecration	⬡✩❁✻◎⬡⊛✡⬡✪✩❁	ᔑ⯒ᓭᑫᒧᛟ	ᔑ⯒ᓭᑫᒧᛟ	ᑭ
ᔑ	coma	⬡✩ᛗ✡	ᔑᛗ✡	ᔑᛗ✡	ᔑ
ᔑ	completion	⬡✩ᛗᚸ❁◎⬡✪✩❁	ᔑᛗᚸ✻ᒧᛟ	ᔑᛗᚸ✻ᒧᛟ	ᔑ
ᒪ	meru	ᛗ◎⊛ᛒ	ᛗᓄᛒ	ᛗᓄᛒ	ᒪ
ᒪ	monument	ᛗ✩❁ᛒᛗ◎❁⬡	ᛗᛟᵒᛒᛗᛗᔑ⬡	ᛗᛟᵒᛒᛗᛗᔑ⬡	ᒪ
ᑌ	stab	✻⬡✡☽	Δ✡☽	Δ✡☽	ᑌ
ᑌ	stability	✻⬡✡☽✪❁✪⬡↗	Δ✡☽ɛΘ↗	Δ✡☽ɛΘ↗	ᑌ
▽	hopa	ɤ✩ᚸ✡	⋹ᚸ✡	⋹ᚸ✡	▽
▽	hope	ɤ✩ᚸ◎	⋹ᚸ◎	⋹ᚸ◎	▽
ƶ	conti	⬡✩❁⬡✪	ᔑᘐ✪	ᔑᘐ✪	ƶ
ƶ	continuation	⬡✩❁⬡✪❁ᛒ✡⬡✪✩❁	ᔑᘐᛕᛒᑎᐱ❁	ᔑᘐᛕᛒᑎᐱ❁	ƶ
ϛ	resth	⊛◎✻⬡ɤ	ᔕΔɤ	ᔕΔɤ	ϛ
ϛ	rest	⊛◎✻⬡	ᔕΔ	ᔕΔ	ϛ
ϧ	sil	✻✪❁	▽❁	▽❁	ϧ
ϧ	silence	✻✪❁◎❁⬡◎	▽✻ᘓ◎	▽✻ᘓ◎	ϧ
϶	slun	✻❁ᛒ❁	⌖ᛒ❁	⌖ᛒ❁	϶
϶	sleep	✻❁◎◎ᚸ	⌖ᔑᚸ	⌖ᔑᚸ	϶
ϕ	etern	◎⬡◎⊛❁	ᔓ❁	ᔓ❁	ϕ
ϕ	self-completion	✻◎❁ᘐ⬡✩ᛗᚸ❁◎⬡✪✩❁	ᛯ❁ᘐᔑᛗᚸ✻ᒧᛟ	ᛯ❁ᘐᔑᛗᚸ✻ᒧᛟ	ϕ

Parliament of Star Seeds

Glyph	English	First	Second	Third	Whole

| :---: | --- | --- | --- | --- | :---: |
♈	akasha	✡♋✡✻ⵠ✡	✡♋X⊰	✡♋X⊰	♈
♈	aksh	✡♋✻ⵠ	✡♋♅	✡♋♅	♈
♈	presence	♑⊛◎✻◎֍⬡◎	♑ൟ⁑ଔ◎	♑ൟ⁑ଔ◎	♈
♈	aries	✡⊛✪◎✻	Ho✻	Ho✻	♈
♉	caduceus	⬡✡❄♒⬡◎♒✻	ᑭ❄♒∝♒✻	ᑭ❄♒∝♒✻	♉
♉	eus	◎♒✻	◎♒✻	◎♒✻	♉
♉	law	❁✡♒	✶♒	✶♒	♉
♉	taurus	⬡✡♒⊛♒✻	▫♒⊛♒✻	▫♒⊛♒✻	♉
♊	veritas	ℂ◎⊛✪⬡✡✻	ℂ⊃ΘX	ℂ⊃ΘX	♊
♊	veri	ℂ◎⊛✪	ℂ⊃✪	ℂ⊃✪	♊
♊	flesh	☋❁◎✻ⵠ	☋✶♅	☋✶♅	♊
♊	gemini	♌◎ℳ✪֍✪	♌◎ℳ⊀✪	♌◎ℳ⊀✪	♊
♋	phren	♑ⵠ⊛◎֍	♑⋡	♑⋡	♋
♋	ren	⊛◎֍	ൟ֍	ൟ֍	♋
♋	tessaract	⬡◎✻✻✡⊛✡⬡⬡	IÆRH৸⬡	IÆRH৸⬡	♋
♋	cancer	⬡✡֍⬡◎⊛	ᑭ☋ɔ	ᑭ☋ɔ	♋
♑	daimon	❄✡✪ℳ☆֍	Ↄ✪ℳ৻ঌ	Ↄ✪ℳ৻ঌ	♑
♑	mon	ℳ☆֍	ℳ৻ঌ	ℳ৻ঌ	♑
♑	heartbeat	ⵠ◎✡⊛⬡Ↄ◎✡⬡	⁞⁞Hⵑ⬡Ↄ⊀⬡	⁞⁞Hⵑ⬡Ↄ⊀⬡	♑
♑	capricorn	⬡✡♑⊛✪⬡☆⊛֍	ᑭ♑ঌ৺	ᑭ♑ঌ৺	♑
♍	aikyam	✡✪♋↗✡ℳ	ᖇ♋↗✡ℳ	ᖇ♋↗✡ℳ	♍
♍	yam	↗✡ℳ	↗✡ℳ	↗✡ℳ	♍
♍	chaos	⬡ⵠ✡☆✻	zf✻	zf✻	♍
♍	virgo	ℂ✪⊛ฎ☆	ℂሕฎ☆	ℂሕฎ☆	♍
♎	melos	ℳ◎❁☆✻	ℳ৸	ℳ৸	♎
♎	me	ℳ◎	ℳ◎	ℳ◎	♎
♎	transformation	⬡⊛✡֍✻☋☆⊛ℳ✡⬡✪☆֍	'‹✻☋⏁ℳ∩Λ֍	'‹✻☋⏁ℳ∩Λ֍	♎
♎	libra	❁✪☽⊛✡	♈☽ᘓ	♈☽ᘓ	♎
ℳ	daath	❄✡✡⬡ⵠ	Ↄ∩ⵠ	Ↄ∩ⵠ	ℳ
ℳ	da	❄✡	Ↄ	Ↄ	ℳ
ℳ	echoes	◎⬡ⵠ☆◎✻	৸⋕৸	৸⋕৸	ℳ
ℳ	scorpius	✻⬡☆⊛♑✪֍✻	⁑⏁♑✪֍✻	⁑⏁♑✪֍✻	ℳ
↗	akaven	✡♋✡ℂ◎֍	✡♋✡ℂ৸	✡♋✡ℂ৸	↗
↗	aka	✡♋✡	✡♋✡	✡♋✡	↗
↗	transmutation	⬡⊛✡֍✻ℳ֍⬡✡⬡✪☆֍	'‹✻ℳ֍▫♪৻ঌ	'‹✻ℳ֍▫♪৻ঌ	↗
↗	sagittarius	✻✡♌✪⬡⬡✡⊛✪֍✻	∀♌Θ▫ঌ֍✻	∀♌Θ▫ঌ֍✻	↗
♌	axiomyr	✡ϒ✪☆ℳ↗⊛	✡ϒΛℳ↗⊛	✡ϒΛℳ↗⊛	♌
♌	myr	ℳ↗⊛	ℳ↗⊛	ℳ↗⊛	♌
♌	freedom	☋⊛◎◎❄☆ℳ	☋൮☆ℳ	☋൮☆ℳ	♌
♌	lion: king	❁✪☆֍♋✪֍♌	ϒ৻ঌ♋⊀♌	ϒ৻ঌ♋⊀♌	♌
♒	nyx	֍↗ϒ	֍↗ϒ	֍↗ϒ	♒
♒	nyx	֍↗ϒ	֍↗ϒ	֍↗ϒ	♒
♒	tempus	⬡◎ℳⵑ♑✻	Iℳⵑ♑✻	Iℳⵑ♑✻	♒
♒	aquarius	✡Ⅱ♑✡⊛✪♑✻	✡Ⅱ♑HⵙO♑✻	✡Ⅱ♑HⵙO♑✻	♒
♓	zaine	♓✡✪֍◎	♓ᖇ⋕	♓ᖇ⋕	♓
♓	ain	✡✪֍	ᖇ֍	ᖇ֍	♓
♓	here → there	ⵠ◎⊛◎⬡ⵠ◎⊛◎	⁞⁞ൟ৸◎	⁞⁞ൟ৸◎	♓
♓	pisces	♑✪✻⬡◎✻	♑ⵔ∝✻	♑ⵔ∝✻	♓

Enochian Understandings

Glyph	English	First	Second	Third	Whole

🜲	substrate	⁕♍☽⁕⬡⊛✡⬡◎	⁕♍☽△ⵘ	⁕♍☽△ⵘ	🜲
🜲	th	⬡ⵠ	ϟ	ϟ	🜲
ᖍ	phantasmagoria	♑ⵠ✡❁⬡✡⁕ℳ✡∂☆⊛✪✡	♑ᖎᚗXℳ✡∂⏁ⵝ	♑ᖎᚗXℳ✡∂⏁ⵝ	ᖍ
ᖍ	he	ⵠ◎	⅋	⅋	ᖍ
⸸	focal	♋☆⬡✡❁	♋※᠆	♋※᠆	⸸
⸸	de	❄◎	Ⲁ	Ⲁ	⸸
♂	catalyst	⬡✡⬡✡❁↗⁕⬡	ᑭ▫❁↗△	ᑭ▫❁↗△	♂
♂	pa	♑✡	♑✡	♑✡	♂
🜲	relic	⊛◎❁✪⬡	ᚔᛉ⬡	ᚔᛉ⬡	🜲
🜲	ma	ℳ✡	ℳ✡	ℳ✡	🜲
⊕	recursive	⊛◎⬡♒⊛⁕✪ℂ◎	ᚔ⬡♒ⵚ✪ℂ◎	ᚔ⬡♒ⵚ✪ℂ◎	⊕
⊕	in	✪❁	Ⱪ	Ⱪ	⊕
⊕	adversarial	✡❄ℂ◎⊛⁕✡⊛✪✡❁	⸗ℂ⊃∀ᛯ⸗	⸗ℂ⊃∀ᛯ⸗	⊕
⊕	er	◎⊛	⊃	⊃	⊕
⊙	terminal	⬡◎⊛ℳ✪❁✡❁	ꟾ⊛ℳⱪ⸗	ꟾ⊛ℳⱪ⸗	⊙
⊙	at	✡⬡	∩	∩	⊙
⊖	temporal	⬡◎ℳ♑☆⊛✡❁	ꟾℳ♑⏁⸗	ꟾℳ♑⏁⸗	⊖
⊖	oo	☆☆	✯	✯	⊖

Emissions and Numbers: Non-Transformable

Glyph	English	First	Second	Third	Whole

☿	ponder	♑☆❁❄◎⊛	♑⍀⁖Ⲁ⊛	♑⍀⁖Ⲁ⊛	☿
☿	pon	♑☆❁	♑⍀⁖	♑⍀⁖	☿
☿	one	☆❁◎	⍀⁖◎	⍀⁖◎	☿
♀	feel	♋◎◎❁	♋ᔕ❁	♋ᔕ❁	♀
♀	fael	♋✡◎❁	♋ᑭ❁	♋ᑭ❁	♀
♀	two	⬡ⲽ☆	⬡ⲽ☆	⬡ⲽ☆	♀
☥	gaia	∂⏁✡✪✡	∂⏁Ⱪ✡	∂⏁Ⱪ✡	☥
☥	gaya	∂⏁✡↗✡	∂⏁✡↗✡	∂⏁✡↗✡	☥
☥	three	⬡ⵠ⊛◎◎	ϟᚔ◎	ϟᚔ◎	☥
♂	will	ⲽ✪❁❁	ⲽⴑ❁	ⲽⴑ❁	♂
♂	wil	ⲽ✪❁	ⲽⴑ	ⲽⴑ	♂
♂	four	♋☆♒⊛	♋☆♒⊛	♋☆♒⊛	♂
ㄐ	speak	⁕♑◎✡ㅁ	⁕♑ⵕㅁ	⁕♑ⵕㅁ	ㄐ
ㄐ	spaek	⁕♑✡◎ㅁ	⁕♑ᑭㅁ	⁕♑ᑭㅁ	ㄐ
ㄐ	five	♋✪ℂ◎	♋✪ℂ◎	♋✪ℂ◎	ㄐ
ℏ	believe	☽◎❁✪◎ℂ◎	☽ᖷ₀ℂ◎	☽ᖷ₀ℂ◎	ℏ
ℏ	eve	◎ℂ◎	◎ℂ◎	◎ℂ◎	ℏ
ℏ	six	⁕✪ⲩ	⩛ⲩ	⩛ⲩ	ℏ
⏁	act	✡⬡⬡	ᛘ⬡	ᛘ⬡	⏁
⏁	act	✡⬡⬡	ᛘ⬡	ᛘ⬡	⏁
⏁	seven	⁕◎ℂ◎❁	ᛯℂᖷ	ᛯℂᖷ	⏁
Ψ	know	ㅁ❁☆ⲽ	ㅁ⊠ⲽ	ㅁ⊠ⲽ	Ψ
Ψ	kno	ㅁ❁☆	ㅁ⊠	ㅁ⊠	Ψ
Ψ	eight	◎✪∂⏁⬡	ℊ∂✷	ℊ∂✷	Ψ
ᖘ	ascend	✡⁕⬡◎❁❄	Xᗴƨ	Xᗴƨ	ᖘ
ᖘ	shenoo	✡ⲩ⬡◎❁☆☆	✡ⲩᖮ⊠☆	✡ⲩᖮ⊠☆	ᖘ
ᖘ	nine	❁✪❁◎	◌ᗰ	◌ᗰ	ᖘ
⏚	zero	ᚼ◎⊛☆	ᚼↄ☆	ᚼↄ☆	⏚
⏚	zen	ᚼ◎❁	ᚼᖷ	ᚼᖷ	⏚
⏚	shadow·locus	⁕ⵠ✡❄☆ⲽ·❁☆⬡♒⁕	∀ϟ☆ⲽ·ⲬⲨ⬡♒⁕	∀ϟ☆ⲽ·ⲬⲨ⬡♒⁕	⏚

| ♄ | shadow | ❄⧗✡❄✡♒ | ♅♁✡♒ | ♅♁✡♒ | ♄ |

Unique Bodies

Glyph	English	First	Second	Third	Whole
☽⊙☾	regia	⊛◎♌✪✡	♑♌♓	♑♌♓	☽⊙☾
☽⊙☾	crown	⊛◎♌✪✡	♑♌♓	♑♌♓	☽⊙☾
☽	rav	⊛✡☾	☋☾	☋☾	☽
☽	wane	⊛✡☾	☋☾	☋☾	☽
☾	veh	☾◎⧗	☾♌	☾♌	☾
☾	wax	☾◎⧗	☾♌	☾♌	☾

The Shape of Absence

Glyph	English	First	Second	Third	Whole
·	space	·	·	·	·

the_shape_of_word.txt

Direct bodies

Glyph	Phonetic
♈	`ash`
♉	`us`
♊	`eri`
♋	`en`
♌	`on`
♍	`yam`
♎	`me`
♏	`da`
♐	`ka`
♁	`myr`
♒	`nyx`
♓	`ai`
⎈	`fe`
⬡	`al`
✡	`abd`
✩	`ahn`
❂	`el`
❀	`so`
❋	`ot`
⧗	`dr`
⊛	`ri`
❄	`ek`
⚛	`av`
⌬	`ig`

Courts — Fetu row

Glyph	Phonetic
ﻫ	`ahl`
ﺭ	`uhh`
ﺴﺭ	`ehr`
ﺸﺭ	`ish`
ﻉ	`ora`
ﺋ	`ahm`
ﺒ	`ke`
ﻮ	`ehm`
ﺣ	`ahd`
ﻔ	`fur`
ﺘﺭ	`a`
ﻋ	`era`

Courts — Kal row

Glyph	Phonetic
ꞁ	`kur`
ꞵ	`lur`
▫	`ahr`
⧫	`in`
⅄	`na`
Ⅵ	`fel`
⅀	`har`
⅄	`mer`
'	`o`
꞉	`pe`
⊥	`zhi`
⊺	`cl`

Courts — Babdh row

Glyph	Phonetic
ꟽ	`hr`
ᑎ	`kr`
ᖘ	`vr`
ᖴ	`py`
ᖘ	`oa`
‹	`lc`
X	`nu`
ᖵ	`st`
ᕼ	`or`
⊦	`bon`
⊦	`ti`
ᕽ	`fa`

Courts — Ahn row

Glyph	Phonetic
≍	`abd`
ᔑ	`ym`
ↄ	`oh`
⛧	`zhi`
⍩	`ol`
⟍ᘐ	`i`
ᖁ	`re`
◎	`se`
⏀	`u`
ᔑ	`fay`
⍭	`ha`
ᘐᔕ	`ps`

Courts — Vel row

Glyph	Phonetic
ᴏ	`era`
ϴ	`tar`
Ⱪ	`ghe`
Λ	`rel`
Ⱨ	`ful`
Ⱪ	`ker`
Ⱁ	`hoh`
ⱷ	`hr`
ⱨ	`ar`
Ⱬ	`ay`
ⱪ	`urn`
I	`je`

Courts — Sor row

Glyph	Phonetic
ᨻ	`fi`
ᨿ	`lun`
ᨾ	`aru`
ᨷ	`es`
ᨶ	`os`
ᨯ	`ahm`
ᨰ	`ti`
ᨬ	`ey`
ᨽ	`sih`
ᨹ	`hri`
ᨵ	`yo`
ᨴ	`thal`

Courts — Koth row

Glyph	Phonetic
ᛯ	`e`
ᛃ	`se`
ᛘ	`lin`
ᛜ	`bri`
ᛥ	`inn`
ᛣ	`subh`
ᛤ	`wel`
ᛪ	`m`
ᛵ	`esh`
ᛮ	`so`
ᛰ	`rhu`
ᛑ	`del`

Courts — Dreh row

Glyph	Phonetic
ᛗ	`na`

𐤊	`ur`
𐤋	`nih`
𐤌	`azh`
𐤍	`hol`
𐤎	`gur`
𐤏	`ves`
𐤐	`rim`
𐤑	`dem`
𐤒	`oth`
𐤓	`izh`
𐤔	`shu`

Courts — Rhea row

Glyph	Phonetic
𐤕	`ia`
𐤖	`zoh`
𐤗	`her`
𐤘	`druh`
𐤙	`fhel`
𐤚	`ral`
𐤛	`kra`
𐤜	`and`
𐤝	`deb`
𐤞	`kol`
𐤟	`fra`
𐤠	`us`

Courts — Zhek row

Glyph	Phonetic
𐤡	`hin`
𐤢	`ser`
𐤣	`ama`
𐤤	`tohr`
𐤥	`pel`
𐤦	`khi`
𐤧	`yth`
𐤨	`mel`
𐤩	`pha`
𐤪	`okh`
𐤫	`od`
𐤬	`ume`

Courts — Shav row

Glyph	Phonetic
𐤭	`ohm`
𐤮	`is`
𐤯	`ta`
𐤰	`koh`

ᛉ	`yh`
ᛊ	`st`
ᛦ	`os`
ᚼ	`poru`
ᛝ	`orm`
ᚱ	`rev`
ᛯ	`imh`
ᛲ	`ihnj`

Courts — Trig row

Glyph	Phonetic
ᛩ	`ig`
ᛪ	`pe`
ᚦ	`du`
ᛟ	`oma`
ᚳ	`eru`
ᚴ	`ta`
ᛜ	`opa`
ᛇ	`tin`
ᛎ	`rest`
ᛃ	`sil`
ᛋ	`slun`
ᛑ	`ete`

Numerical / terminal speaking bodies

Glyph	Phonetic
ᛏ	`zen`
ᛉ	`pon`
ᛩ	`ael`
ᛯ	`gaya`
ᛇ	`wil`
ᛅ	`pay`
ᚺ	`eve`
ᛯ	`ahc`
ᛉ	`no`
ᛩ	`noo`
ᛇ	`av`
ᛈ	`veh`

Enochian bodies

`th`
`he`
`de`
`pa`
`ma`
`in`
`er`
`at`
`oo`

build.sh

```
#!/usr/bin/env bash
set -euo pipefail

BUILD_DIR="$(cd "$(dirname "$0")" && pwd)"
SRC="$BUILD_DIR/src"
TESTS="$BUILD_DIR/tests"
PACKAGE="$BUILD_DIR/package"
GENERATED="$BUILD_DIR/generated"
LINUX="$BUILD_DIR/linux_ubuntu"
WINDOWS="$BUILD_DIR/windows"
MAC="$BUILD_DIR/mac"

mkdir -p "$GENERATED" "$PACKAGE" "$LINUX" "$WINDOWS" "$MAC"
rm -f "$GENERATED/aksharam_glyphkey.kmn"       "$GENERATED/registry_report.json"
↪  "$GENERATED/keyman_core_probe"       "$GENERATED/kmc_keyboard.log"
↪  "$GENERATED/kmc_package.log"       "$PACKAGE/aksharam_glyphkey.kmx"
↪  "$LINUX/aksharam_glyphkey.kmp"       "$WINDOWS/aksharam_glyphkey.kmp"
↪  "$MAC/aksharam_glyphkey.kmp"

python3 "$SRC/generate_keyboard.py"

kmc build "$GENERATED/aksharam_glyphkey.kmn"     -o "$PACKAGE/aksharam_glyphkey.kmx"     |
↪  tee "$GENERATED/kmc_keyboard.log"

kmc build "$PACKAGE/aksharam_glyphkey.kps"     -o "$LINUX/aksharam_glyphkey.kmp"     | tee
↪  "$GENERATED/kmc_package.log"

# Windows and macOS use the same compiled .kmx keyboard engine, but their packages
# must not carry Linux-only xbindkeys/Python selection helpers.
python3 "$SRC/generate_desktop_kps.py"     "$PACKAGE/aksharam_glyphkey.kps"
↪  "$PACKAGE/aksharam_glyphkey_desktop.kps"

kmc build "$PACKAGE/aksharam_glyphkey_desktop.kps"     -o "$WINDOWS/aksharam_glyphkey.kmp"
↪  | tee "$GENERATED/kmc_windows_package.log"

cp "$WINDOWS/aksharam_glyphkey.kmp" "$MAC/aksharam_glyphkey.kmp"

cc -std=c11 "$TESTS/keyman_core_probe.c"     -o "$GENERATED/keyman_core_probe"
↪  $(pkg-config --cflags --libs keyman_core)

python3 "$TESTS/validate_keyboard.py"
python3 "$TESTS/full_keyboard_conformance.py"
python3 "$TESTS/host_controls_conformance.py"
python3 "$TESTS/host_activation_test.py"
python3 "$TESTS/host_activation_test.py"
```

```
# Retain generated source and compiled artifacts for reproducible 4.0 maintenance.

printf 'Linux/Ubuntu installer: %s\n' "$LINUX/aksharam_glyphkey.kmp"
printf 'Windows installer: %s\n' "$WINDOWS/aksharam_glyphkey.kmp"
printf 'macOS installer: %s\n' "$MAC/aksharam_glyphkey.kmp"
```

aksharam_keyboard_data.py

```
# Canonical executable Aksharam keyboard data. Version 4.0.
# Registered word folds and First Spells retain their authored anatomy.

BASE = {'A': '✡',
 'B': '☽',
 'C': '⌬',
 'D': '❄',
 'E': '◎',
 'F': '♋',
 'G': '♌',
 'H': '⧗',
 'I': '✪',
 'J': '♈',
 'K': '♎',
 'L': '❁',
 'M': '♏',
 'N': '❀',
 'O': '☆',
 'P': '♑',
 'Q': '♊',
 'R': '⊛',
 'S': '❋',
 'T': '⬡',
 'U': '♍',
 'V': '☾',
 'W': '♒',
 'X': '♉',
 'Y': '♐',
 'Z': '♓'}

GOETICS = [('◎', 'fetu', ``fe``, '♁::Ψ∮⟠ΦΦΦΦΦ'),
 ('⬡', 'kal', ``al``, '☿♁♂⟠ΦΦΦΘΘ'),
 ('✡', 'babdh', ``abd``, '♃♀Ψ⟠ΦΦΦΦΦ'),
 ('☆', 'ahn', ``ahn``, '♂∮♀±i♂☿♁⟠ΘΦΦΦΘ'),
 ('✪', 'vel', ``el``, '☿♀♄::♀♀⟠ΦΦΦΘΦ'),
 ('❀', 'sor', ``so``, '♀☿♆::♂♀⟠ΦΦΦΦΦ'),
 ('❋', 'koth', ``ot``, '♁♂☿⟠ΦΦΦΦΦ'),
 ('⧗', 'dreh', ``dr``, 'Ψ♃♀⟠ΦΦΦΦΘ'),
 ('⊛', 'rhea', ``ri``, '∮♇♄⟠ΦΦΦΦΦ'),
 ('❄', 'zhek', ``ek``, '♇♄∮⟠ΦΦΦΦΦ'),
 ('❁', 'shav', ``av``, '♀Ψ♃⟠ΦΦΦΦΦ'),
 ('⌬', 'trig', ``ig``, '♄∮♇⟠ΦΦΦΦΦ')]

COURTS = [('ᚱ', '◎◎', ``ahl``, '♁::Ψ∮𐤉ΦΦΦΦΦ'),
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 ('ᚳ', '◎✡', ``ehr``, '♃♀Ψ𐤉ΦΦΦΦΦ'),
```

```
('ݜ', '◎☆', ``ish``, '⟩⟠♂☿⛣‹◦◦♂⍑♀◍ΦΦΦΦΦ'),
('ع', '◎✪', ``ora``, '☿♀♇::♀♀◍ΦΦΦΦΦ'),
('ݨ', '◎֍', ``ahm``, '♀☿Ш::♂♀◍ΦΦΦΦΦ'),
('ݧ', '◎✻', ``ke``, '⛣♂☿◍ΦΦΦΦΦ'),
('ݮ', '◎⧗', ``ehm``, 'Ψ♃♀◍ΦΦΦΦΦ'),
('ح', '◎⊛', ``ahd``, 'ʃ♇♇◍ΦΦΦΦΦ'),
('ڒ', '◎❄', ``fur``, '♇♇ʃ◍ΦΦΦΦΦ'),
('ڑ', '◎❁', ``a``, '♀Ψ♃◍ΦΦΦΦΦ'),
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('᙭', '✡✻', ``nu``, '⛣♂☿◍ΦΦΦΦΦ'),
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('⍑', '☆⊛', ``u``, 'ʃ♇♇◍ΘΘΦΦΘ'),
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('∇', '✻❀', ``subh``, 'ᑫᖘ⇊::♂ᑫ◌ΦΦΦΦΦ'),
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('⍏', '✻❁', ``rhu``, 'ᑫΨ⍝◌ΦΦΦΦΦ'),
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('ᖁ', 'ⵠ✪', ``hol``, 'ᖘᑫℏ::ᑫᑫ◌ΦΦΦΦΘ'),
('ᐩ', 'ⵠ❀', ``gur``, 'ᑫᖘ⇊::♂ᑫ◌ΦΦΦΦΘ'),
('⏁', 'ⵠ✻', ``ves``, '⍚♂ᖘ◌ΦΦΦΦΘ'),
('⏥', 'ⵠⵠ', ``rim``, 'Ψ⍝ᑫ◌ΦΦΦΦΘ'),
('⏦', 'ⵠ⊛', ``dem``, '∱ᑭℏ◌ΦΦΦΦΘ'),
('⏧', 'ⵠ❄', ``oth``, 'ᑭℏ∱◌ΦΦΦΦΘ'),
('⏨', 'ⵠ❁', ``izh``, 'ᑫΨ⍝◌ΦΦΦΦΘ'),
('ᖜ', 'ⵠ⬡', ``shu``, 'ℏ∱ᑭ◌ΦΦΦΦΘ'),
('ᖳ', '⊛◎', ``ia``, '⍚::Ψ∱◌ΦΦΦΦΦ'),
('ᖵ', '⊛⬡', ``zoh``, 'ᖘ⍚♂◌ΦΦΦΦΦ'),
('ᖾ', '⊛✡', ``her``, '⍝ᑫΨ◌ΦΦΦΦΦ'),
('ᖭ', '⊛☆', ``druh``, '〉⟠♂ᖘ⍚〈∘-∘♂⏀ᑫ◌ΦΦΦΦΦ'),
('ᖯ', '⊛✪', ``fhel``, 'ᖘᑫℏ::ᑫᑫ◌ΦΦΦΦΦ'),
('ᖮ', '⊛❀', ``ral``, 'ᑫᖘ⇊::♂ᑫ◌ΦΦΦΦΦ'),
('ᖰ', '⊛✻', ``kra``, '⍚♂ᖘ◌ΦΦΦΦΦ'),
('ᖽ', '⊛ⵠ', ``and``, 'Ψ⍝ᑫ◌ΦΦΦΦΦ'),
('ᖸ', '⊛⊛', ``deb``, '∱ᑭℏ◌ΦΦΦΦΦ'),
('ᖼ', '⊛❄', ``kol``, 'ᑭℏ∱◌ΦΦΦΦΦ'),
('ᖺ', '⊛❁', ``fra``, 'ᑫΨ⍝◌ΦΦΦΦΦ'),
('ᖻ', '⊛⬡', ``us``, 'ℏ∱ᑭ◌ΦΦΦΦΦ'),
('A', '❄◎', ``hin``, '⍚::Ψ∱◌ΦΦΦΦΦ'),
('Ƀ', '❄⬡', ``ser``, 'ᖘ⍚♂◌ΦΦΦΦΦ'),
('Ↄ', '❄✡', ``ama``, '⍝ᑫΨ◌ΦΦΦΦΦ'),
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 'first': 'ᅀ❁☆〰',
 'second': 'ᅀ৪〰',
 'third': 'ᅀ৪〰',
 'whole': 'Ψ'},
{'line': 654, 'glyph': 'Ψ', 'english': 'kno', 'first': 'ᅀ❁☆', 'second': 'ᅀ৪', 'third':
↪  'ᅀ৪', 'whole': 'Ψ'},
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 'glyph': 'Ψ',
 'english': 'eight',
 'first': '◎✪ฎ⧗⬡',
 'second': 'ყฎ✷',
 'third': 'ყฎ✷',
 'whole': 'Ψ'},
{'line': 656,
 'glyph': 'Ꝑ',
 'english': 'ascend',
 'first': '✡❋⬡◎❁❄',
 'second': 'Xҝƶ',
 'third': 'Xҝƶ',
 'whole': 'Ꝑ'},
{'line': 657,
 'glyph': 'Ꝑ',
 'english': 'shenoo',
 'first': '✡ᘐ⌬◎❁☆☆',
 'second': '✡ᘐҝ৪☆',
 'third': '✡ᘐҝ৪☆',
 'whole': 'Ꝑ'},
{'line': 658, 'glyph': 'Ꝑ', 'english': 'nine', 'first': '❁✪❁◎', 'second': '◌ฑ', 'third':
↪  '◌ฑ', 'whole': 'Ꝑ'},
{'line': 659,
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 'english': 'zero',
 'first': 'ᚼ◎⊛☆',
 'second': 'ᚼ৯☆',
 'third': 'ᚼ৯☆',
 'whole': 'ᛋ'},
{'line': 660, 'glyph': 'ᛋ', 'english': 'zen', 'first': 'ᚼ◎❁', 'second': 'ᚼ৯', 'third':
↪  'ᚼ৯', 'whole': 'ᛋ'},
{'line': 661,
 'glyph': 'ᛋ',
```

```
  'english': 'shadow·locus',
  'first': '⁕⧗✡❄☆♒·❁☆⬡♍⁕',
  'second': '∀ϟ☆♒·¥⬡♍⁕',
  'third': '∀ϟ☆♒·¥⬡♍⁕',
  'whole': '♆'},
 {'line': 662,
  'glyph': '♆',
  'english': 'shadow',
  'first': '⁕⧗✡❄☆♒',
  'second': '∀ϟ☆♒',
  'third': '∀ϟ☆♒',
  'whole': '♆'},
 {'line': 669,
  'glyph': 'Ↄʘℂ',
  'english': 'regia',
  'first': '⊛◎∂ℓ✪✡',
  'second': 'ꕥ∂ℓ☒',
  'third': 'ꕥ∂ℓ☒',
  'whole': 'Ↄʘℂ'},
 {'line': 670,
  'glyph': 'Ↄʘℂ',
  'english': 'crown',
  'first': '⊛◎∂ℓ✪✡',
  'second': 'ꕥ∂ℓ☒',
  'third': 'ꕥ∂ℓ☒',
  'whole': 'Ↄʘℂ'},
 {'line': 671, 'glyph': 'Ↄ', 'english': 'rav', 'first': '⊛✡ℂ', 'second': 'ᥴℂ', 'third':
 ↪  'ᥴℂ', 'whole': 'Ↄ'},
 {'line': 672, 'glyph': 'Ↄ', 'english': 'wane', 'first': '⊛✡ℂ', 'second': 'ᥴℂ', 'third':
 ↪  'ᥴℂ', 'whole': 'Ↄ'},
 {'line': 673, 'glyph': 'ℂ', 'english': 'veh', 'first': 'ℂ◎⧗', 'second': 'ℂ𐑉', 'third':
 ↪  'ℂ𐑉', 'whole': 'ℂ'},
 {'line': 674, 'glyph': 'ℂ', 'english': 'wax', 'first': 'ℂ◎⧗', 'second': 'ℂ𐑉', 'third':
 ↪  'ℂ𐑉', 'whole': 'ℂ'},
 {'line': 681, 'glyph': '·', 'english': 'space', 'first': '·', 'second': '·', 'third': '·',
 ↪  'whole': '·'}]

FIRST_SPELLS = {'eloi': {'first': '◎❁✡✪', 'stop': 'ꕤ', 'utterance': 'eloi'},
 'sabachtany': {'first': '⁕✡Ↄ✡⬡⧗⬡✡❁➶', 'stop': '∀Ↄ٩✷‹➶', 'utterance': 'sabachtany'},
 'anima': {'first': '✡❁✪ℳ✡', 'stop': '‹✪ℳ✡', 'utterance': 'anima'},
 'culpa': {'first': '⬡♍❁ϑ๐✡', 'stop': '⬡♍❁ϑ๐✡', 'utterance': 'culpa'},
 'decire': {'first': '❄◎⬡✪⊛◎', 'stop': 'Aꓵꕥ', 'utterance': 'decire'},
 'entera': {'first': '◎❁⬡◎⊛✡', 'stop': 'ꕤᛞ', 'utterance': 'entera'}}


# Primordial access counterparts: existing README QWERTY/Braille table.
BRAILLE = {'A': '⠁', 'B': '⠃', 'C': '⠉', 'D': '⠙', 'E': '⠑', 'F': '⠋', 'G': '⠛', 'H': '⠓',
 ↪  'I': '⠊', 'J': '⠚', 'K': '⠅', 'L': '⠇', 'M': '⠍', 'N': '⠝', 'O': '⠕', 'P': '⠏', 'Q':
 ↪  '⠟', 'R': '⠗', 'S': '⠎', 'T': '⠞', 'U': '⠥', 'V': '⠧', 'W': '⠺', 'X': '⠭', 'Y': '⠽',
 ↪  'Z': '⠵'}
```

generate_keyboard.py

```
#!/usr/bin/env python3
from pathlib import Path
from collections import OrderedDict
from collections import defaultdict
import json, unicodedata
from aksharam_keyboard_data import BASE, GOETICS, COURTS, THIRD, TRANSFORMATIONS,
↪  FIRST_SPELLS, BRAILLE

BUILD=Path(__file__).resolve().parents[1]
ROOT=BUILD.parent
SRC=BUILD/'src'
PACKAGE=BUILD/'package'
GENERATED=BUILD/'generated'
KMN=GENERATED/'aksharam_glyphkey.kmn'
KPS=PACKAGE/'aksharam_glyphkey.kps'
REPORT=GENERATED/'registry_report.json'
SELMAP=PACKAGE/'selection_math_map.json'
GENERATED.mkdir(parents=True,exist_ok=True)
PACKAGE.mkdir(parents=True,exist_ok=True)
VS={'\ufe0e','\ufe0f'}
TEXT_PRESENTATION=set('♀♂☿♃♊♋♍♎♏♐♑♒♓☸♆✡❄')

def norm(s):
    s=unicodedata.normalize('NFC',s.strip().replace('`',''))
    return ''.join(c for c in s if c not in VS).replace(' ','')

def text_presentation(s):
    return ''.join(c+'\ufe0e' if c in TEXT_PRESENTATION else c for c in s)

def Q(s):
    s=text_presentation(s)
    if "'" not in s:return "'"+s+"'"
    return ' '.join('U+0027' if c=="'" else "'"+c+"'" for c in s)

# Executable keyboard law is loaded only from static Python data in build/src/.
base=OrderedDict((k,norm(v)) for k,v in BASE.items())
assert len(base)==26
goetic_set={'◎','⬡','✡','☆','❂','❁','✻','⧗','⊛','❄','⚛','⌬'}
goetics=[(norm(g),n,p,norm(m)) for g,n,p,m in GOETICS]
assert len(goetics)==12
goetic_math={g:m for g,_,_,m in goetics if m}
courts=[(norm(g),norm(pair),ph,norm(m)) for g,pair,ph,m in COURTS]
assert len(courts)==144
assert all(len(pair)==2 and all(x in goetic_set for x in pair) for _,pair,_,_ in courts)
court_by_pair=OrderedDict((pair,g) for g,pair,_,_ in courts)
court_reverse=OrderedDict((g,pair) for g,pair,_,_ in courts)
```

```
court_math={g:m for g,_,_,m in courts if m}
third=[(norm(pair),norm(result),ph) for pair,result,ph in THIRD]
assert len(third)==24
third_map=OrderedDict((pair,result) for pair,result,_ in third)
third_reverse=OrderedDict()
for pair,result,_ in third: third_reverse.setdefault(result,pair)
raw=[dict(r) for r in TRANSFORMATIONS]

canon=[]; exact=set()
for r in raw:
    k=(r['english'],r['first'],r['second'],r['third'],r['whole'])
    if k not in exact:exact.add(k);canon.append(r)

# Duplicate input rows compile once. Ambiguous same-input/different-output rows are
↪  reported,
# not multiplied into contradictory Keyman rules.
trigger_dest=OrderedDict(); trigger_conflicts=[]
for r in canon:
    e,d=r['english'],r['whole']
    if e not in trigger_dest:trigger_dest[e]=d
    elif trigger_dest[e]!=d:trigger_conflicts.append(dict(english=e,existing=trigger_dest[e
    ↪  ],alternate=d,line=r['line']))
row_by_english=OrderedDict()
for r in canon: row_by_english.setdefault(r['english'],r)
whole_from_third=OrderedDict(); source_conflicts=[]
for r in canon:
    p,d=r['third'],r['whole']
    if not p or p==d:continue
    if p not in whole_from_third:whole_from_third[p]=d
    elif whole_from_third[p]!=d:source_conflicts.append(dict(source=p,existing=whole_from_t
    ↪  hird[p],alternate=d,english=r['english'],line=r['line']))

# One typed event traverses Second -> Third -> registered Whole exactly once.
def suffix_once(s,m):
    for p,d in sorted(m.items(),key=lambda kv:(-len(kv[0]),kv[0])):
        if s.endswith(p):return s[:-len(p)]+d
    return s
def stages(s):
    # Structural writing performs Second then Third only. Whole is lexical and
    # is emitted only when a registered access sequence reaches its terminal.
    return suffix_once(suffix_once(s,court_by_pair),third_map)

char_out={c:base[c.upper()] for c in 'abcdefghijklmnopqrstuvwxyz'}
char_out.update({':':'::','→':'→',' ':'·','·':'·','-':'…','_':'…',"'":'ˈ'})
key_of={c:f'K_{c.upper()}' for c in 'abcdefghijklmnopqrstuvwxyz'}
key_of.update({':':'SHIFT K_COLON','→':'ALT SHIFT K_4','
↪  ':'K_SPACE','·':'K_SPACE','-':'K_HYPHEN','_':'SHIFT K_HYPHEN',"'":'K_QUOTE'})

# Prefix-state compiler. Visible Aksharam follows ordinary structural writing while
# letters are still being typed. A registered Whole (or First-Spell stopping body)
# is committed only when a non-letter boundary is typed.
supported={e:d for e,d in trigger_dest.items() if e and all(c in char_out for c in e)}
first_spells=OrderedDict((e,dict(v)) for e,v in FIRST_SPELLS.items())
for e,v in first_spells.items():
```

```
    direct=''.join(char_out[c] for c in e)
    assert direct==norm(v['first']), (e,direct,v['first'])
    natural=''
    for c in e: natural=stages(natural+char_out[c])
    assert natural==norm(v['stop']), (e,natural,v['stop'])
lexical_targets=OrderedDict(supported)
for e,v in first_spells.items(): lexical_targets[e]=norm(v['stop'])
word_states={e:(f'rw{i}',f'rw{i}d1',f'rw{i}d2',f'rw{i}d3') for i,e in enumerate(supported)}
prefixes={''}
for e in lexical_targets:
    for i in range(1,len(e)+1):prefixes.add(e[:i])
prefixes=sorted(prefixes,key=lambda x:(len(x),x))
terminal=lexical_targets
tracked={p for p in prefixes if p and (p in terminal or any(e.startswith(p) and
↪  len(e)>len(p) for e in lexical_targets))}
visible={'':''}
for p in prefixes[1:]:
    parent=p[:-1]; c=p[-1]
    visible[p]=stages(visible[parent]+char_out[c])
by_visible=defaultdict(list)
for p0 in sorted(tracked,key=lambda x:(visible[x],len(x),x)):
    by_visible[visible[p0]].append(p0)
sid={}; max_rank=0
for same in by_visible.values():
    for rank,p0 in enumerate(same): sid[p0]=f's{rank}'
    max_rank=max(max_rank,len(same))
NON='§NONE§'; sid[NON]=f's{max_rank}'
state_ids=[f's{i}' for i in range(max_rank+1)]
states=[NON]+sorted(tracked,key=lambda x:(len(x),x))

transitions=[]
for p in prefixes:
    for c in char_out:
        q=p+c
        if q in prefixes:transitions.append((p,c,q))

# Native name is itself rendered through the same lawful word-prefix mechanism where
↪  available.
def render_plain(text):
    out=''
    for word_i,w in enumerate(text.lower().split(' ')):
        if word_i:out+='·'
        if w in terminal:out+=terminal[w];continue
        s=''
        for c in w:
            if c in char_out:s=stages(s+char_out[c])
        out+=s
    return out
NATIVE_NAME=render_plain('aksharam glyphkey')

L=["c Aksharam GlyphKey — generated from the two keyboard authorities only",f"store(&name)
↪  {Q(NATIVE_NAME)}","store(&keyboardversion) '4.0'","store(&targets)
↪  'desktop'","store(&mnemoniclayout) '0'","begin Unicode > use(main)",""]
```

```
basekeys=' '.join(f'[NCAPS K_{k}]' for k in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'); shiftkeys='
↪  '.join(f'[SHIFT NCAPS K_{k}]' for k in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'); capskeys='
↪  '.join(f'[CAPS K_{k}]' for k in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'); shiftcaps='
↪  '.join(f'[SHIFT CAPS K_{k}]' for k in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'); glyph_items='
↪  '.join(Q(base[k]) for k in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ')
L += [f"store(basekeys) {basekeys}",f"store(shiftkeys) {shiftkeys}",f"store(capskeys)
↪  {capskeys}",f"store(shiftcaps) {shiftcaps}",f"store(baseglyphs)
↪  {glyph_items}",f"store(states) {' '.join('dk('+x+')' for x in state_ids)}",""]
L += ["group(main) using keys"]

# Stateful valid transitions first. Terminal transitions replace the current visible
# registered prefix with its one-glyph Whole body; nonterminal transitions append normally.
for p,c,q0 in sorted(transitions,key=lambda x:(-len(visible[x[0]]),x[1],x[0])):
    kt=key_of[c]; kt=('NCAPS '+kt) if c in 'abcdefghijklmnopqrstuvwxyz' else kt
    tail=(f" dk({sid[q0]})" if q0 in tracked else "")
    if not p:
        L.append(f"+ [{kt}] > {Q(visible[q0])}{tail}")
    else:
        L.append(f"{Q(visible[p])} dk({sid[p]}) + [{kt}] > {Q(visible[q0])}{tail}")

# Commit complete registered words only at a non-letter boundary.
# Ordinary Backspace is deliberately excluded: it remains native deletion and never unfolds.
word_boundaries=[
    ('K_SPACE','·',False),('SHIFT K_TAB','∴',True),('NCAPS K_ENTER','',True),('SHIFT NCAPS
    ↪  K_ENTER','∵',True),
    ('K_0','♅',False),('K_1','☿',False),('K_2','♀',False),('K_3','♃',False),('K_4','♂',Fal
    ↪  se),
    ('K_5','♃',False),('K_6','♄',False),('K_7','♁',False),('K_8','♆',False),('K_9','♇',Fal
    ↪  se),
    ('K_NP0','♅',False),('K_NP1','☿',False),('K_NP2','♀',False),('K_NP3','♃',False),('K_NP
    ↪  4','♂',False),
    ('K_NP5','♃',False),('K_NP6','♄',False),('K_NP7','♁',False),('K_NP8','♆',False),('K_NP
    ↪  9','♇',False),
    ('K_NPSTAR','⁕',False),('K_NPPLUS','☋',False),('K_NPMINUS','⋯',False),('K_NPDOT','∵',
    ↪  False),('K_NPSLASH','⁎',False),
    ('K_LBRKT','ɾ',False),('K_RBRKT','ɻ',False),('SHIFT K_9','〉',False),('SHIFT
    ↪  K_0','〈',False),
    ('SHIFT K_LBRKT','',False),('SHIFT
    ↪  K_RBRKT','',False),('K_HYPHEN','⋯',False),('K_PERIOD','∵',False),
    ('SHIFT K_COLON','∵',False),('K_EQUAL','⧟',False),('SHIFT
    ↪  K_EQUAL','☋',False),('K_COMMA','∔',False),('SHIFT K_COMMA','∴',False),
    ('SHIFT K_PERIOD','∴',False),('K_COLON','∴',False),('K_QUOTE','ˌ',False),('SHIFT
    ↪  K_QUOTE','ˌ',False),
    ('K_BKSLASH','˙.',False),('SHIFT K_BKSLASH','⁞',False),('K_SLASH','.˙',False),('SHIFT
    ↪  K_SLASH','∴',False),
    ('SHIFT K_1','∴',False),('SHIFT K_HYPHEN','…',False),('SHIFT K_7','☋',False),('SHIFT
    ↪  K_8','⁕',False),
    ('SHIFT K_2','⊖',False),('SHIFT K_3','⊕',False),('SHIFT K_4','⊕',False),('SHIFT
    ↪  K_5','⊕',False),
    ('ALT SHIFT K_4','→',False),('CTRL SHIFT K_6','∵',False),('SHIFT
    ↪  K_6','∴',False),('K_BKQUOTE','⟠',False),('SHIFT K_BKQUOTE','⌀',False),
]
```

```
boundary_access_char={'SHIFT K_COLON':':','ALT SHIFT K_4':'→','K_SPACE':'
↪  ','K_HYPHEN':'-','SHIFT K_HYPHEN':'_','K_QUOTE':"'"}
for p0 in sorted(terminal,key=lambda x:(-len(visible[x]),-len(x),x)):
    dest=terminal[p0]
    for kt,out,emit in word_boundaries:
        access_c=boundary_access_char.get(kt)
        # Longest registered continuation wins when the apparent boundary is itself
        # part of a longer registered access trigger. K_SPACE can represent either
        # an ordinary access space or the Aksharam-space alias in a registered trigger.
        if kt=='K_SPACE' and (p0+' ' in prefixes or p0+'·' in prefixes):
            continue
        if access_c and p0+access_c in prefixes:
            continue
        if p0 in supported:
            ws=word_states[p0][0]
            rhs=Q(dest)+f" dk({ws})"+(Q(out) if out else "")+(" use(emit)" if emit else "")
        else:
            rhs=Q(dest+out)+(" use(emit)" if emit else "")
        L.append(f"{Q(visible[p0])} dk({sid[p0]}) + [{kt}] > {rhs}")
# Ordinary Backspace is intentionally not mapped. Native deletion remains native deletion.
# Direct Goetic mathematical forms; any active lexical state is intentionally closed.
key_for_glyph={g:k for k,g in base.items()}
for g,_,_,m in goetics:
    if not m:continue
    k=key_for_glyph[g]
    L.append(f"any(states) + [ALT SHIFT NCAPS K_{k}] > {Q(m)}")
    L.append(f"+ [ALT SHIFT NCAPS K_{k}] > {Q(m)}")
# Primordial unfolding changes only the access office of the targeted base coordinate.
for physical, glyph in base.items():
    cell=BRAILLE[physical]
    L.append(f"{Q(glyph)} any(states) + [ALT SHIFT K_BKSP] > {Q(cell)}")
    L.append(f"{Q(glyph)} + [ALT SHIFT K_BKSP] > {Q(cell)}")
# The invariant Braille SeeD form is held on further per-glyph unfolding.
for cell in BRAILLE.values():
    L.append(f"{Q(cell)} any(states) + [ALT SHIFT K_BKSP] > {Q(cell)}")
    L.append(f"{Q(cell)} + [ALT SHIFT K_BKSP] > {Q(cell)}")
# Current-glyph unfolding. Ahead-of-cursor and arbitrary body operations use the host
↪  bridge.
for glyph, parents in list(court_reverse.items()) + list(third_reverse.items()):
    L.append(f"{Q(glyph)} any(states) + [ALT SHIFT K_BKSP] > {Q(parents)}")
    L.append(f"{Q(glyph)} + [ALT SHIFT K_BKSP] > {Q(parents)}")
# Exact registered-word unfolding. The post-fold deadkey preserves which lexical
# registration produced a shared Whole, so Alt+Backspace follows that row exactly:
# Whole -> Third -> Second -> First. Ordinary Backspace remains deletion.
single_boundary_outputs=sorted({out for _,out,_ in word_boundaries if len(out)==1})
for e in supported:
    r=row_by_english[e]
    ws,d1,d2,d3=word_states[e]
    whole,third_body,second_body,first_body=r['whole'],r['third'],r['second'],r['first']
    L.append(f"{Q(whole)} dk({ws}) + [ALT K_BKSP] > {Q(third_body)} dk({d1})")
    L.append(f"{Q(third_body)} dk({d1}) + [ALT K_BKSP] > {Q(second_body)} dk({d2})")
    L.append(f"{Q(second_body)} dk({d2}) + [ALT K_BKSP] > {Q(first_body)} dk({d3})")
    L.append(f"{Q(first_body)} dk({d3}) + [ALT K_BKSP] > {Q(first_body)} dk({d3})")
    # Alt+Space explicitly refolds from any retained depth; ordinary Space preserves depth.
```

```
    for body,state in ((whole,ws),(third_body,d1),(second_body,d2),(first_body,d3)):
        L.append(f"{Q(body)} dk({state}) + [ALT K_SPACE] > {Q(whole)} dk({ws}) '·'")
    for body,state in ((third_body,d1),(second_body,d2),(first_body,d3)):
        L.append(f"{Q(body)} dk({state}) + [K_SPACE] > {Q(body)} dk({state}) '·'")
    # If the user deletes a one-glyph boundary with ordinary Backspace, preserve the
    # hidden lexical-depth marker while deleting only that boundary glyph. This is deletion,
    # not unfolding; Alt+Backspace remains the sole unfolding control.
    for state in (ws,d1,d2,d3):
        for out in single_boundary_outputs:
            L.append(f"dk({state}) {Q(out)} + [K_BKSP] > dk({state})")
L += [
    "dk(ud1) + [ALT K_BKSP] > use(unfold2)",
    "dk(ud2) + [ALT K_BKSP] > use(unfold3)",
    "dk(ud3) + [ALT K_BKSP] > dk(ud3)",
    "any(states) + [ALT K_BKSP] > use(unfold1)",
    "+ [ALT K_BKSP] > use(unfold1)",
    "dk(ud1) + [ALT K_SPACE] > use(refold_space)",
    "dk(ud2) + [ALT K_SPACE] > use(refold_space)",
    "dk(ud3) + [ALT K_SPACE] > use(refold_space)",
    "any(states) + [ALT K_SPACE] > use(refold_space)",
    "+ [ALT K_SPACE] > use(refold_space)",
]

# Shift/Caps explicitly suppress folding and close current prefix state.
for k in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ':
    g=base[k]
    L += [f"any(states) + [SHIFT NCAPS K_{k}] > {Q(g)}",f"+ [SHIFT NCAPS K_{k}] > {Q(g)}",
          f"any(states) + [CAPS K_{k}] > {Q(g)}",f"+ [CAPS K_{k}] > {Q(g)}",
          f"any(states) + [SHIFT CAPS K_{k}] > {Q(g)}",f"+ [SHIFT CAPS K_{k}] > {Q(g)}"]
# Unregistered continuation: keep structural writing active but leave lexical trie until
↪  Space.
for k in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ':
    g=base[k]
    L.append(f"any(states) + [NCAPS K_{k}] > {Q(g)} dk({sid[NON]}) use(fold2m)")
# NONE continues structurally.
# Start outside any registered prefix.

# Controls, digits, keypad, punctuation. Marker-aware versions consume/reset state where
↪  needed.
L += ["any(states) + [SHIFT K_TAB] > '⁘' use(emit)","+ [SHIFT K_TAB] > '⁘'
↪  use(emit)","any(states) + [SHIFT NCAPS K_ENTER] > '⁙' use(emit)","+ [SHIFT NCAPS
↪  K_ENTER] > '⁙' use(emit)","any(states) + [ALT SHIFT K_2] > '↑'","+ [ALT SHIFT K_2] >
↪  '↑'","any(states) + [ALT SHIFT K_3] > '↓'","+ [ALT SHIFT K_3] > '↓'","any(states) +
↪  [ALT SHIFT K_4] > '→'","+ [ALT SHIFT K_4] > '→'","any(states) + [ALT SHIFT K_5] >
↪  '←'","+ [ALT SHIFT K_5] > '←'",
      "any(states) + [SHIFT K_2] > '⊖'","+ [SHIFT K_2] > '⊖'","any(states) + [SHIFT K_3] >
      ↪  '⊙'","+ [SHIFT K_3] > '⊙'","any(states) + [SHIFT K_4] > '⊕'","+ [SHIFT K_4] >
      ↪  '⊕'","any(states) + [SHIFT K_5] > '⊛'","+ [SHIFT K_5] > '⊛'","any(states) + [CTRL
      ↪  SHIFT K_6] > '∵'","+ [CTRL SHIFT K_6] > '∵'","any(states) + [SHIFT K_6] > '∴'","+
      ↪  [SHIFT K_6] > '∴'","any(states) + [SHIFT K_8] > '※'","+ [SHIFT K_8] > '※'"]
nums={'0':'🜨','1':'☿','2':'♀','3':'⚷','4':'♂','5':'♃','6':'♄','7':'⛢','8':'♆','9':'♇'}
for k,g in nums.items():L += [f"any(states) + [K_{k}] > {Q(g)} dk({sid[NON]})",f"+ [K_{k}] >
↪  {Q(g)}"]
```

```
for k,g in nums.items():L += [f"any(states) + [K_NP{k}] > {Q(g)} dk({sid[NON]})",f"+
↪  [K_NP{k}] > {Q(g)}"]
punct=[('K_NPSTAR','⁑'),('K_NPPLUS','ॐ'),('K_NPMINUS','⋯'),('K_NPDOT','⁚'),('K_NPSLASH','
↪  ⁜'),('K_LBRKT','ʃ'),('K_RBRKT','ɻ'),('SHIFT K_9','〉'),('SHIFT K_0','〈'),('SHIFT
↪  K_LBRKT','□'),('SHIFT K_RBRKT','□'),('K_HYPHEN','⋯'),('K_PERIOD','⁚'),('SHIFT
↪  K_COLON','⁚'),('K_EQUAL','⧟'),('SHIFT K_EQUAL','ॐ'),('K_COMMA','÷'),('SHIFT
↪  K_COMMA','⁖'),('SHIFT K_PERIOD','⁝'),('K_COLON','⁘'),('K_QUOTE','¢'),('SHIFT
↪  K_QUOTE','♭'),('K_BKSLASH','˙˙.'),('SHIFT K_BKSLASH','⁞'),('K_SLASH','˙˙'),('SHIFT
↪  K_SLASH','⁙'),('SHIFT K_1','⁜'),('SHIFT K_HYPHEN','…'),('SHIFT
↪  K_7','ॐ'),('K_BKQUOTE','⟠'),('SHIFT K_BKQUOTE','⟲')]
for k,g in punct:L += [f"any(states) + [{k}] > {Q(g)}",f"+ [{k}] > {Q(g)}"]
# Space closes lexical state unless a specific registered transition above consumed it.
L += ["dk(ud1) + [K_SPACE] > '·'","dk(ud2) + [K_SPACE] > '·'","dk(ud3) + [K_SPACE] >
↪  '·'","any(states) + [K_SPACE] > '·'","+ [K_SPACE] > '·'",""]

# Stateful forward stages preserve the one hidden prefix marker.
L += ["group(fold2m)"]
for p,d in court_by_pair.items():L.append(f"{Q(p)} any(states) > {Q(d)}
↪  index(states,{len(text_presentation(p))+1}) use(fold3m)")
L.append("nomatch > use(fold3m)")
L += ["","group(fold3m)"]
for p,d in third_map.items():L.append(f"{Q(p)} any(states) > {Q(d)}
↪  index(states,{len(text_presentation(p))+1})")
L.append("nomatch > return")
L.append("")

# Non-state structural groups perform only Second and Third. Whole remains lexical.
L += ["group(fold2)"]
for p,d in court_by_pair.items():L.append(f"{Q(p)} > {Q(d)} use(fold3)")
L.append("nomatch > use(fold3)")
L += ["","group(fold3)"]
for p,d in third_map.items():L.append(f"{Q(p)} > {Q(d)}")
L.append("nomatch > return")
L.append("")

# Mathematical-body suffix groups are retained for direct internal use only.
# Alt+Shift+Enter is intentionally NOT bound to preceding context: the Legend requires
↪  host-selected glyph(s).
L += ["group(mathbody_m)"]
for g,m in list(goetic_math.items())+list(court_math.items()):L.append(f"{Q(g)} > {Q(m)}")
L.append("nomatch > return")
L += ["","group(mathbody)"]
for g,m in list(goetic_math.items())+list(court_math.items()):L.append(f"{Q(g)} > {Q(m)}")
L.append("nomatch > return")
L.append("")

# Three explicit reversible lexical levels. Depth is retained in hidden markers so
# identity stages still count as a level and a fourth Alt+Backspace is inert.
whole_to_third=OrderedDict(); third_to_second=OrderedDict(); second_to_first=OrderedDict()
for r in canon:
    if r['whole'] and r['third']: whole_to_third.setdefault(r['whole'],r['third'])
    if r['third'] and r['second']: third_to_second.setdefault(r['third'],r['second'])
    if r['second'] and r['first']: second_to_first.setdefault(r['second'],r['first'])
```

```
L += ["group(unfold1)"]
seen_unfold1=set()
for p,d in sorted(whole_to_third.items(),key=lambda kv:(-len(kv[0]),kv[0])):
    L.append(f"{Q(p)} > {Q(d)} dk(ud1)"); seen_unfold1.add(p)
# Natural Third bodies without a registered Whole can still unfold lawfully to their pair.
for p,d in sorted(third_reverse.items(),key=lambda kv:(-len(kv[0]),kv[0])):
    if p not in seen_unfold1:
        L.append(f"{Q(p)} > {Q(d)} dk(ud2)"); seen_unfold1.add(p)
# Natural Court bodies occupy the Second/Third visible level; first press retains the
# visible body and records that the no-op Third level has been traversed.
for p in sorted(court_reverse,key=lambda x:(-len(x),x)):
    if p not in seen_unfold1:
        L.append(f"{Q(p)} > {Q(p)} dk(ud2)"); seen_unfold1.add(p)
L.append("nomatch > return")
L += ["","group(unfold2)"]
for p,d in sorted(third_to_second.items(),key=lambda kv:(-len(kv[0]),kv[0])):
    L.append(f"{Q(p)} > {Q(d)} dk(ud2)")
for p,d in sorted(third_reverse.items(),key=lambda kv:(-len(kv[0]),kv[0])):
    if p not in third_to_second:L.append(f"{Q(p)} > {Q(d)} dk(ud2)")
L.append("nomatch > dk(ud2)")
L += ["","group(unfold3)"]
for p,d in sorted(second_to_first.items(),key=lambda kv:(-len(kv[0]),kv[0])):
    L.append(f"{Q(p)} > {Q(d)} dk(ud3)")
for p,d in sorted(court_reverse.items(),key=lambda kv:(-len(kv[0]),kv[0])):
    if p not in second_to_first:L.append(f"{Q(p)} > {Q(d)} dk(ud3)")
L.append("nomatch > dk(ud3)")
L.append("")

refold=OrderedDict()
for r in canon:
    for src in (r['first'],r['second'],r['third']):
        if src and src!=r['whole']:refold.setdefault(src,r['whole'])
L += ["group(refold_space_m)"]
for p,d in sorted(refold.items(),key=lambda kv:(-len(kv[0]),kv[0])):L.append(f"{Q(p)} >
↪  {Q(d)} '·'")
L.append("nomatch > '·'")
L += ["","group(refold_space)"]
for p,d in sorted(refold.items(),key=lambda kv:(-len(kv[0]),kv[0])):L.append(f"{Q(p)} >
↪  {Q(d)} '·'")
L.append("nomatch > '·'")
L.append("")

L += ['', 'group(emit) using keys', 'c Empty by design: passes the original control
↪  keystroke to the host application.']
KMN.write_text('\n'.join(L)+'\n',encoding='utf-8')
selection_math=OrderedDict(list(goetic_math.items())+list(court_math.items()))
(PACKAGE/'unfolding_map.json').write_text(json.dumps({'primordial':{g:BRAILLE[k] for k,g in
↪  base.items()},'courts':dict(court_reverse),'third':dict(third_reverse),'rows':list(row_
↪  by_english.values()),'first_spells':dict(first_spells)},ensure_ascii=False,indent=2),en
↪  coding='utf-8')
SELMAP.write_text(json.dumps(selection_math,ensure_ascii=False,indent=2),encoding='utf-8')
KPS.write_text(f'''<?xml version="1.0" encoding="utf-8"?>
```

```
<Package><System><KeymanDeveloperVersion>18.0.252</KeymanDeveloperVersion><FileVersion>7.0</FileVersion></System><Options><ReadMeFile>readme.htm</ReadMeFile><WelcomeFile>welcome.htm</WelcomeFile><FollowKeyboardVersion/></Options><Info><Name>{text_presentation(NATIVE_NAME)}</Name><Description>Aksharam GlyphKey</Description><Version>4.0</Version></Info><Files><File><Name>aksharam_glyphkey.kmx</Name><Description>Aksharam GlyphKey</Description><CopyLocation>0</CopyLocation><FileType>.kmx</FileType></File><File><Name>welcome.htm</Name><Description></Description><CopyLocation>0</CopyLocation><FileType>.htm</FileType></File><File><Name>readme.htm</Name><Description></Description><CopyLocation>0</CopyLocation><FileType>.htm</FileType></File><File><Name>aksharam_selection_math.py</Name><Description>Selected Aeon mathematical-body resolver</Description><CopyLocation>0</CopyLocation><FileType>.py</FileType></File><File><Name>aksharam_selection_phonetic.py</Name><Description>Selected Aksharam Shape-of-Sound phonetic resolver</Description><CopyLocation>0</CopyLocation><FileType>.py</FileType></File><File><Name>selection_math_map.json</Name><Description>Registered mathematical bodies</Description><CopyLocation>0</CopyLocation><FileType>.json</FileType></File><File><Name>aksharam_host_controls.py</Name><Description>Aksharam 4.0 host unfolding controls</Description><CopyLocation>0</CopyLocation><FileType>.py</FileType></File><File><Name>unfolding_map.json</Name><Description>Aksharam 4.0 host unfolding controls</Description><CopyLocation>0</CopyLocation><FileType>.json</FileType></File><File><Name>xbindkeys.aksharam</Name><Description>Linux selection bindings</Description><CopyLocation>0</CopyLocation><FileType>.aksharam</FileType></File><File><Name>start_selection.sh</Name><Description>Selection binding launcher for Linux</Description><CopyLocation>0</CopyLocation><FileType>.sh</FileType></File></Files><Keyboards><Keyboard><Name>{text_presentation(NATIVE_NAME)}</Name><ID>aksharam_glyphkey</ID><Version>4.0</Version><Languages><Language ID="und">Aksharam</Language></Languages></Keyboard></Keyboards><Strings/></Package>''', encoding='utf-8')

# Static conformance: each non-conflicting supported trigger is terminal by construction.
validation=[dict(english=e,whole=d,terminal=(e in terminal),visible=visible.get(e)) for e,d in supported.items()]
REPORT.parent.mkdir(exist_ok=True)
REPORT.write_text(json.dumps({'native_name':NATIVE_NAME,'base_coordinates':len(base),'goetics':len(goetics),'courts':len(courts),'third_law_pairs':len(third),'raw_rows':len(raw),'deduped_rows':len(canon),'distinct_triggers':len(trigger_dest),'supported_triggers':len(supported),'prefix_states':len(states),'transitions':len(transitions),'source_conflicts':source_conflicts,'trigger_conflicts':trigger_conflicts,'validation':validation},ensure_ascii=False,indent=2),encoding='utf-8')
print(f'generated {KMN.name}: {KMN.stat().st_size} bytes, {len(L)} lines')
print(f'name: {NATIVE_NAME}')
print(f'triggers: {len(supported)}/{len(trigger_dest)} supported; prefix states: {len(states)}; transitions: {len(transitions)}')
print(f'conflicts retained once: source={len(source_conflicts)} trigger={len(trigger_conflicts)}')
```

validate_keyboard.py

```
#!/usr/bin/env python3
from pathlib import Path
import json, runpy, sys
TESTS=Path(__file__).resolve().parent
BUILD=TESTS.parent
ROOT=BUILD.parent
SRC=BUILD/'src'
PACKAGE=BUILD/'package'
GENERATED=BUILD/'generated'
LINUX=BUILD/'linux_ubuntu'
if str(SRC) not in sys.path: sys.path.insert(0,str(SRC))
ns=runpy.run_path(SRC/'generate_keyboard.py')
kmn=(GENERATED/'aksharam_glyphkey.kmn').read_text(encoding='utf-8')
report=json.loads((GENERATED/'registry_report.json').read_text(encoding='utf-8'))
checks=[]
def check(name,cond,detail=''):
    checks.append((name,bool(cond),detail))
check('26 direct coordinates',len(ns['base'])==26,str(len(ns['base'])))
check('12 Goetics',len(ns['goetics'])==12,str(len(ns['goetics'])))
check('144 Courts',len(ns['courts'])==144,str(len(ns['courts'])))
check('24 Third-Law pairs',len(ns['third'])==24,str(len(ns['third'])))
check('all supported registered triggers present',all(ns['terminal'].get(e)==d for e,d in
↪  ns['supported'].items()),f"{len(ns['supported'])} triggers")
check('duplicate English triggers compiled once',len(ns['trigger_dest'])==len(set(ns['trigg
↪  er_dest'])),str(len(ns['trigger_dest'])))
check('Alt+Shift Goetic bodies',sum(1 for line in kmn.splitlines() if '[ALT SHIFT NCAPS K_'
↪  in line and 'K_ENTER' not in line)>=12)
check('no preceding-context math selector','[ALT SHIFT NCAPS K_ENTER]' not in kmn)
check('Shift+Enter paragraph close + host Enter passthrough', "[SHIFT NCAPS K_ENTER] > '\uf8ff'
↪  use(emit)" in kmn and 'group(emit) using keys' in kmn and "[SHIFT NCAPS K_ENTER] > '\uf8ff'
↪  U+000A" not in kmn)
check('selection math map exists',(PACKAGE/'selection_math_map.json').is_file())
check('selection helper exists',(PACKAGE/'aksharam_selection_math.py').is_file())
check('phonetic selection helper
↪  exists',(PACKAGE/'aksharam_selection_phonetic.py').is_file())
xb=(PACKAGE/'xbindkeys.aksharam').read_text(encoding='utf-8')
check('Alt+Shift+Esc phonetic selection binding','aksharam_host_controls.py phonetic' in xb
↪  and 'Alt+Shift + Escape' in xb)
check('Prosody key',"[K_BKQUOTE] > '\U000f0a7d'" in kmn)
check('Cantillation key',"[SHIFT K_BKQUOTE] > '\U000f0a7e'" in kmn)
check('Halfstellation keypad',"[K_NPSLASH] > '\U000f0a7f'" in kmn)
check('automatic prefix state compact',len(ns['state_ids'])<32,f"{len(ns['state_ids'])}
↪  hidden markers")
check('6 First Spells programmed',len(ns['first_spells'])==6 and all(e in ns['terminal'] for
↪  e in ns['first_spells']),str(len(ns['first_spells'])))
```

```
check('ordinary Backspace is native, not unfold',"[K_BKSP] > use(unfold" not in kmn and "+
↪  [ALT K_BKSP]" in kmn)
check('compiled keyboard exists',(PACKAGE/'aksharam_glyphkey.kmx').is_file())
check('compiled package exists',(LINUX/'aksharam_glyphkey.kmp').is_file())
for name,ok,detail in checks: print(('PASS' if ok else 'FAIL'),name,detail)
failed=[x for x in checks if not x[1]]
print(f"RESULT {len(checks)-len(failed)}/{len(checks)} passed")
sys.exit(1 if failed else 0)
```

full_keyboard_conformance.py

```
#!/usr/bin/env python3
from pathlib import Path
from collections import OrderedDict
import subprocess, re, runpy, unicodedata, sys, os
TESTS=Path(__file__).resolve().parent
BUILD=TESTS.parent
ROOT=BUILD.parent
SRC=BUILD/'src'
PACKAGE=BUILD/'package'
GENERATED=BUILD/'generated'
if str(SRC) not in sys.path: sys.path.insert(0,str(SRC))
from aksharam_keyboard_data import BASE, GOETICS, COURTS, THIRD, TRANSFORMATIONS,
↪  FIRST_SPELLS
KMX=Path(os.environ.get('AKSHARAM_KMX', str(PACKAGE/'aksharam_glyphkey.kmx')))
PROBE=GENERATED/'keyman_core_probe'
VS={'\ufe0e','\ufe0f'}
def norm(s):
    s=unicodedata.normalize('NFC',s.strip().replace('`',''))
    return ''.join(c for c in s if c not in VS).replace(' ','')
def stripvs(s): return ''.join(c for c in s if c not in VS)
base=OrderedDict((k,norm(v)) for k,v in BASE.items())
goetic_set={'🜀','🜃','🜡','🜭','🜂','🜍','🜏','🜨','🜔','❄','⚛','⌬'}
goetics=[(norm(g),n,p,norm(m)) for g,n,p,m in GOETICS]
courts=[(norm(g),norm(pair),ph,norm(m)) for g,pair,ph,m in COURTS]
third=[(norm(pair),norm(result),ph) for pair,result,ph in THIRD]
raw=[(r['line'],r['english'],norm(r['first']),norm(r['second']),norm(r['third']),norm(r['wh
↪  ole'])) for r in
↪  TRANSFORMATIONS]
trigger=OrderedDict()
for _,e,_,_,_,w in raw: trigger.setdefault(e,w)
char_out={c:base[c.upper()] for c in 'abcdefghijklmnopqrstuvwxyz'}
char_out.update({':':'::','→':'→',' ':'·','·':'·','-':'…','_':'…',"'":'ˈ'})
supported=OrderedDict((e,d) for e,d in trigger.items() if e and all(c in char_out for c in
↪  e))
first_spells=OrderedDict((e,dict(v)) for e,v in FIRST_SPELLS.items())

# Windows/Keyman virtual key codes used by KMX processor.
VK={chr(ord('A')+i):0x41+i for i in range(26)}
VK.update({str(i):0x30+i for i in range(10)})
VK.update({'BKSP':0x08,'TAB':0x09,'ENTER':0x0D,'SPACE':0x20,'LEFT':0x25,'UP':0x26,'RIGHT':0
↪  x27,'DOWN':0x28,
           'NP0':0x60,'NP1':0x61,'NP2':0x62,'NP3':0x63,'NP4':0x64,'NP5':0x65,'NP6':0x66,'NP
           ↪  7':0x67,'NP8':0x68,'NP9':0x69,
           'NPSTAR':0x6A,'NPPLUS':0x6B,'NPMINUS':0x6D,'NPDOT':0x6E,'NPSLASH':0x6F,'F1':0x70,
```

```
            'COLON':0xBA,'EQUAL':0xBB,'COMMA':0xBC,'HYPHEN':0xBD,'PERIOD':0xBE,'SLASH':0xBF,
            ↪  'BKQUOTE':0xC0,
            'LBRKT':0xDB,'BKSLASH':0xDC,'RBRKT':0xDD,'QUOTE':0xDE})
NONE=0; LCTRL=0x01; LALT=0x04; SHIFT=0x10

def probe(seq,caps=False):
    args=[str(PROBE),str(KMX)]
    for vk,mod in seq: args += [hex(vk),hex(mod)]
    env=None
    if caps:
        import os
        env=dict(os.environ); env['KEYMAN_TEST_CAPS']='1'
    p=subprocess.run(args,text=True,capture_output=True,check=True,env=env)
    lines=[x for x in p.stdout.splitlines() if x.startswith('EVENT ')]
    events=[]
    for line in lines:
        actions=line.split(' ACTIONS ',1)[1] if ' ACTIONS ' in line else ''
        toks=[]
        for m in re.finditer(r'CHAR\((.*?)\)|BACK_CHAR|BACK_MARKER|EMIT|INVALIDATE|MARKER\(
        ↪  [^)]*\)|ALERT',actions):
            token=m.group(0)
            if token.startswith('CHAR('): toks.append(('char',m.group(1)))
            elif token=='BACK_CHAR': toks.append(('back',None))
            elif token=='EMIT': toks.append(('emit',None))
            else: toks.append(('other',token))
        events.append({'tokens':toks,'emit':any(t[0]=='emit' for t in toks),'raw':actions})
    return events

def visible(seq,caps=False):
    out=[]
    ev=probe(seq,caps=caps)
    for e in ev:
        for typ,val in e['tokens']:
            if typ=='back':
                if out: out.pop()
            elif typ=='char':
                out.append(val)
    return stripvs(''.join(out)),ev

def expect_one(name,vk,mod,expected,emit=None,caps=False):
    got,ev=visible([(vk,mod)],caps=caps)
    ok=(got==expected) and (emit is None or ev[0]['emit']==emit)
    checks.append((name,ok,f'got={got!r} expected={expected!r} actions={ev[0]["raw"]}'))

checks=[]
# Direct mapped A-Z: folding active, Shift hold, Caps hold, Shift+Caps hold.
for k,g in base.items():
    expect_one(f'direct {k}',VK[k],NONE,g)
    expect_one(f'Shift suppress {k}',VK[k],SHIFT,g)
    expect_one(f'Caps suppress {k}',VK[k],NONE,g,caps=True)
    expect_one(f'Shift+Caps suppress {k}',VK[k],SHIFT,g,caps=True)

nums={'0':'🜨','1':'☿','2':'♀','3':'⚷','4':'♂','5':'♃','6':'♄','7':'⛢','8':'♆','9':'♇'}
for k,g in nums.items():
```

```
    expect_one(f'digit {k}',VK[k],NONE,g)
    expect_one(f'numpad {k}',VK['NP'+k],NONE,g)
for key,g in
↪  [('NPSTAR','⁜'),('NPPLUS','ॐ'),('NPMINUS','⋯'),('NPDOT','⁚'),('NPSLASH','⁜')]:
↪  expect_one('keypad '+key,VK[key],NONE,g)
expect_one('Tab native pass-through',VK['TAB'],NONE,'',True)
expect_one('Shift+Tab paragraph open then native Shift+Tab',VK['TAB'],SHIFT,'⁘⁝',True)
expect_one('Shift+Enter paragraph close and host caret action',VK['ENTER'],SHIFT,'⁝⁘',True)
for key,g in [('2','↑'),('3','↓'),('4','→'),('5','←')]:
↪  expect_one('Alt+Shift+'+key,VK[key],LALT|SHIFT,g)
for key in ['UP','DOWN','RIGHT','LEFT']: expect_one('native Shift selection
↪  '+key,VK[key],SHIFT,'',True)
for key in ['LEFT','UP','RIGHT','DOWN']: expect_one('NumLock-off navigation
↪  '+key,VK[key],NONE,'',True)
expect_one('NumLock native',0x90,NONE,'',True)
for key,g in [('2','⊖'),('3','⊙'),('4','⊕'),('5','⊕'),('6','∴'),('8','⁜')]:
↪  expect_one('Shift+'+key,VK[key],SHIFT,g)
expect_one('Ctrl+Shift+6',VK['6'],LCTRL|SHIFT,'∵')

punct=[('LBRKT',NONE,'ɾ'),('RBRKT',NONE,'ɭ'),('9',SHIFT,'ᗒ'),('0',SHIFT,'ᗕ'),('LBRKT',SHIFT,」
↪  '□'),('RBRKT',SHIFT,'□'),
       ('HYPHEN',NONE,'⋯'),('PERIOD',NONE,'⁝'),('COLON',SHIFT,'⁝'),('EQUAL',NONE,'⧟'),('E」
       ↪  QUAL',SHIFT,'ॐ'),('COMMA',NONE,'∻'),('COMMA',SHIFT,'∴'),
       ('PERIOD',SHIFT,'∴'),('COLON',NONE,'⁘'),('QUOTE',NONE,'𝅘𝅥'),('QUOTE',SHIFT,'𝅘𝅥'),('BKSL」
       ↪  ASH',NONE,'⁖'),('BKSLASH',SHIFT,'⁞'),
       ('SLASH',NONE,'˙·'),('SLASH',SHIFT,'⁙'),('1',SHIFT,'⁜'),('HYPHEN',SHIFT,'…'),('7',SH」
       ↪  IFT,'ॐ'),('BKQUOTE',NONE,'⟠'),('BKQUOTE',SHIFT,'⟐'),('SPACE',NONE,'·')]
for key,mod,g in punct: expect_one('punct '+key+(' shift' if mod else ''),VK[key],mod,g)

# Native pass-through invariant.
expect_one('Backspace native pass-through',VK['BKSP'],NONE,'',True)
expect_one('unclaimed F1 native pass-through',VK['F1'],NONE,'',True)
for key in ['A','C','V','X','Y','Z']:
    expect_one('native Ctrl+'+key,VK[key],LCTRL,'',True)

# Hold/repeat behavior: each repeated physical control event must remain a native host
↪  event.
for key in ['BKSP','LEFT','RIGHT','UP','DOWN']:
    seq=[(VK[key],NONE)]*4
    got,ev=visible(seq)
    checks.append((
        'native repeat '+key,
        got=='' and len(ev)==4 and all(e['emit'] for e in ev),
        f'got={got!r} events={len(ev)} actions={[e["raw"] for e in ev]}',
    ))

# Alt+Shift+Goetic -> registered mathematical body.
key_for_glyph={g:k for k,g in base.items()}
for g,name,ph,m in goetics:
    if m: expect_one(f'Alt+Shift Goetic {g}',VK[key_for_glyph[g]],LALT|SHIFT,m)

# Full registered lexical state machine. Letters build the word; a non-letter boundary
```

```
# commits the registered Whole. Period is used here because it is never part of an access
↪ trigger.
def key_for_char(c):
    if c.isalpha(): return (VK[c.upper()],NONE)
    if c in {' ','·'}: return (VK['SPACE'],NONE)
    if c=='-': return (VK['HYPHEN'],NONE)
    if c=='_': return (VK['HYPHEN'],SHIFT)
    if c=="'": return (VK['QUOTE'],NONE)
    if c==':':return (VK['COLON'],SHIFT)
    if c=='→':return (VK['4'],LALT|SHIFT)
    raise KeyError(c)
for e,w in supported.items():
    got,_=visible([key_for_char(c) for c in e]+[(VK['PERIOD'],NONE)])
    checks.append((f'registered boundary fold {e}',got==w+'::',f'got={got!r}
    ↪ expected={(w+"::")!r}'))

# First Spells are phonetic utterance registrations. Their written body follows only lawful
# structural folds and terminates at the prescribed stopping body; there is no invented
↪ Whole.
for e,v in first_spells.items():
    got,_=visible([key_for_char(c) for c in e])
    checks.append((f'First Spell natural body {e}',got==v['stop'],f'got={got!r}
    ↪ expected={v["stop"]!r}'))
    got,_=visible([key_for_char(c) for c in e]+[(VK['PERIOD'],NONE)])
    checks.append((f'First Spell boundary {e}',got==v['stop']+'::',f'got={got!r}
    ↪ expected={(v["stop"]+"::")!r}'))

# Ordinary Backspace deletes; it never unfolds. Before a boundary, fire is at its lawful
# written body ♋♄◎, so one ordinary Backspace removes only the final ◎.
got,ev=visible([key_for_char(c) for c in 'fire']+[(VK['BKSP'],NONE)])
checks.append(('ordinary Backspace does not unfold',got=='♋♄','got={got!r}
↪ actions={ev[-1]["raw"]}'))
# After a boundary has committed the Whole, deleting that boundary still leaves the Whole
↪ intact.
got,ev=visible([key_for_char(c) for c in 'fire']+[(VK['PERIOD'],NONE),(VK['BKSP'],NONE)])
checks.append(('ordinary Backspace after fold deletes boundary only',got=='✡','got={got!r}
↪ actions={ev[-1]["raw"]}'))


# Structural law execution through the compiled engine, forced outside the lexical trie.
goetic_key={g:VK[k] for k,g in base.items() if g in goetic_set}
for court,pair,ph,m in courts:
    got,_=visible([(VK['A'],NONE),(VK['0'],NONE),(goetic_key[pair[0]],NONE),(goetic_key[pai
    ↪ r[1]],NONE)])
    exp=base['A']+nums['0']+court
    checks.append((f'Court law {pair}',got==exp,f'got={got!r} expected={exp!r}'))
parl_key={g:VK[k] for k,g in base.items() if g in set('♈♉♊♋♌♍♎♏♐♑♒♓')}
for pair,res,ph in third:
    got,_=visible([(VK['A'],NONE),(VK['0'],NONE),(parl_key[pair[0]],NONE),(parl_key[pair[1]
    ↪ ],NONE)])
    exp=base['A']+nums['0']+res
    checks.append((f'Third law {pair}',got==exp,f'got={got!r} expected={exp!r}'))
```

```
# Reversible word-depth behavior and explicit refold. Build canonical first-precedence reverse law.
canon=[]; exact=set()
for r in raw:
    _,e,f,se,t,w=r
    key=(e,f,se,t,w)
    if key not in exact: exact.add(key); canon.append(r)
reverse=OrderedDict()
for _,e,f,se,t,w in canon:
    for a,b in ((t,se),(se,f),(w,t)):
        if a and b and a!=b: reverse.setdefault(a,b)
for pair,res,ph in third: reverse.setdefault(res,pair)
for court,pair,ph,m in courts: reverse.setdefault(court,pair)
refold=OrderedDict()
for _,e,f,se,t,w in canon:
    for src in (f,se,t):
        if src and src!=w: refold.setdefault(src,w)
# One representative trigger for every distinct registered Whole body, using the
# exact registered anatomy row: Whole -> Third -> Second -> First.
row_for_trigger=OrderedDict()
for r in canon: row_for_trigger.setdefault(r[1],r)
whole_trigger=OrderedDict()
for e,w in supported.items(): whole_trigger.setdefault(w,e)
ALT_BK=(VK['BKSP'],LALT); ALT_SP=(VK['SPACE'],LALT)
for whole,e in whole_trigger.items():
    # A non-letter boundary commits the Whole. Ordinary Backspace removes that boundary only,
    # returning the cursor to the folded word; Alt+Backspace then follows the exact registered row.
    seq=[key_for_char(c) for c in e]+[(VK['PERIOD'],NONE),(VK['BKSP'],NONE)]
    _,_,first,second,third_body,registered_whole=row_for_trigger[e]
    for depth,expected in [(1,third_body),(2,second),(3,first)]:
        got,_=visible(seq+[ALT_BK]*depth)
        checks.append((f'unfold depth {depth} {e}',got==expected,f'got={got!r} expected={expected!r}'))
    got,_=visible(seq+[ALT_BK,(VK['SPACE'],NONE)])
    checks.append((f'ordinary Space retains unfolded depth {e}',got==third_body+'·',f'got={got!r} expected={(third_body+"·")!r}'))
    got,_=visible(seq+[ALT_BK,ALT_SP])
    exp=registered_whole+'·'
    checks.append((f'Alt+Space refolds {e}',got==exp,f'got={got!r} expected={exp!r}'))

# Selection mathematical-body helper: every registered Goetic/Court mathematical body and unmapped stability.
sel_ns=runpy.run_path(PACKAGE/'aksharam_selection_math.py')
sel_map=sel_ns['MAP']; transform=sel_ns['transform']
expected_math=OrderedDict()
for g,n,ph,m in goetics:
    if m: expected_math[g]=m
for g,pair,ph,m in courts:
    if m: expected_math[g]=m
checks.append(('selection math registry size',len(sel_map)==len(expected_math),f'{len(sel_map)} vs {len(expected_math)}'))
```

```
for g,m in expected_math.items():
    checks.append((f'selection math {g}',transform(g)==m,f'got={transform(g)!r}
    ↪  expected={m!r}'))
checks.append(('selection leaves unmapped body
↪  unchanged',transform('XYZ')=='XYZ',transform('XYZ')))

# Direct structural laws are present exactly in canonical counts.
checks += [('Legend 26 direct coordinates',len(base)==26,str(len(base))),('Legend 12
↪  Goetics',len(goetics)==12,str(len(goetics))),('Legend 144
↪  Courts',len(courts)==144,str(len(courts))),('Legend 24 Third
↪  pairs',len(third)==24,str(len(third)))]

# Primordial counterparts are copied from the existing QWERTY/Braille authority.
from aksharam_keyboard_data import BRAILLE
for physical,glyph in base.items():
    seq=[(VK[physical],SHIFT),(VK['BKSP'],LALT|SHIFT)]
    got,_=visible(seq)
    checks.append((f'Primordial current {physical}',got==BRAILLE[physical],repr(got)))
    # Tracked lexical deadkeys must not block the same conversion.
    got,_=visible([(VK[physical],NONE),(VK['BKSP'],LALT|SHIFT)])
    checks.append((f'Primordial tracked {physical}',got==BRAILLE[physical],repr(got)))
for physical,cell in BRAILLE.items():
    got,_=visible([(VK[physical],SHIFT),(VK['BKSP'],LALT|SHIFT),(VK['BKSP'],LALT|SHIFT)])
    checks.append((f'Primordial SeeD terminal {physical}',got==cell,repr(got)))
failed=[x for x in checks if not x[1]]
for name,ok,detail in checks:
    if not ok: print('FAIL',name,detail)
print(f'KEY_TESTS {len(checks)} TOTAL; {len(checks)-len(failed)} PASS; {len(failed)} FAIL')
if failed: sys.exit(1)
```

keyman_core_probe.c

```
#include <keyman/keyman_core_api.h>
#include <keyman/keyman_core_api_actions.h>
#include <keyman/keyman_core_api_vkeys.h>
#include <keyman/keyman_core_api_consts.h>
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <string.h>

static void utf8(uint32_t cp) {
  if(cp<=0x7F) putchar(cp);
  else if(cp<=0x7FF){putchar(0xC0|(cp>>6));putchar(0x80|(cp&0x3F));}
  else if(cp<=0xFFFF){putchar(0xE0|(cp>>12));putchar(0x80|((cp>>6)&0x3F));putchar(0x80|(cp&
  ↪  0x3F));}
  else {putchar(0xF0|(cp>>18));putchar(0x80|((cp>>12)&0x3F));putchar(0x80|((cp>>6)&0x3F));p
  ↪  utchar(0x80|(cp&0x3F));}
}
int main(int argc,char **argv){
  if(argc<4 || ((argc-2)%2)!=0){fprintf(stderr,"usage: %s file.kmx VK MOD [VK MOD
  ↪  ...]\n",argv[0]);return 2;}
  FILE *f=fopen(argv[1],"rb"); if(!f){perror("fopen");return 2;}
  fseek(f,0,SEEK_END); long n=ftell(f); fseek(f,0,SEEK_SET);
  void *buf=malloc(n); if(fread(buf,1,n,f)!=(size_t)n){return 2;} fclose(f);
  km_core_keyboard *kb=NULL; km_core_status
  ↪  st=km_core_keyboard_load_from_blob(argv[1],buf,n,&kb); free(buf);
  if(st){fprintf(stderr,"load status=%u\n",st);return 3;}
  km_core_option_item env_default[] = { KM_CORE_OPTIONS_END };
  static const km_core_cu caps_key[] = u"capsLock";
  static const km_core_cu caps_value[] = u"1";
  km_core_option_item env_caps[] = { {caps_key, caps_value, KM_CORE_OPT_ENVIRONMENT},
  ↪  KM_CORE_OPTIONS_END };
  km_core_option_item *env = getenv("KEYMAN_TEST_CAPS") ? env_caps : env_default;
  km_core_state *state=NULL; st=km_core_state_create(kb,env,&state);
  ↪  if(st){fprintf(stderr,"state status=%u\n",st);return 4;}
  for(int a=2;a<argc;a+=2){
    unsigned vk=strtoul(argv[a],NULL,0), mod=strtoul(argv[a+1],NULL,0);
    st=km_core_process_event(state,(km_core_virtual_key)vk,(uint16_t)mod,1,KM_CORE_EVENT_FL
    ↪  AG_DEFAULT);
    if(st){printf("EVENT %u %u STATUS %u\n",vk,mod,st);continue;}
    size_t count=0; const km_core_action_item *it=km_core_state_action_items(state,&count);
    printf("EVENT %u %u ACTIONS ",vk,mod);
    for(size_t i=0;i<count && it[i].type!=KM_CORE_IT_END;i++){
      switch(it[i].type){
        case KM_CORE_IT_CHAR: printf("CHAR(");utf8(it[i].character);printf(") ");break;
        case KM_CORE_IT_MARKER: printf("MARKER(%u) ",it[i].marker);break;
```

```
        case KM_CORE_IT_BACK: printf(it[i].backspace.expected_type==KM_CORE_BT_MARKER ?
        ↪  "BACK_MARKER " : "BACK_CHAR ");break;
        case KM_CORE_IT_EMIT_KEYSTROKE: printf("EMIT ");break;
        case KM_CORE_IT_ALERT: printf("ALERT ");break;
        case KM_CORE_IT_INVALIDATE_CONTEXT: printf("INVALIDATE ");break;
        default: printf("TYPE(%u) ",it[i].type);break;
      }
    }
    printf("\n");
  }
  km_core_state_dispose(state); km_core_keyboard_dispose(kb); return 0;
}
```

readme.htm

```
<!doctype html><meta charset="utf-8"><title>Aksharam GlyphKey</title><p>Aksharam GlyphKey —
↪  ALQC Aksharam keyboard.</p>
```

welcome.htm

```
<!doctype html><meta charset="utf-8"><title>Aksharam GlyphKey</title><p>Aksharam
↪  GlyphKey.</p>
```

generate_desktop_kps.py

```
#!/usr/bin/env python3
from pathlib import Path
import sys
import xml.etree.ElementTree as ET

if len(sys.argv) != 3:
    raise SystemExit('usage: generate_desktop_kps.py INPUT_KPS OUTPUT_KPS')

src=Path(sys.argv[1])
out=Path(sys.argv[2])
root=ET.fromstring(src.read_text(encoding='utf-8'))
files=root.find('Files')
linux_only={
    'aksharam_selection_math.py',
    'aksharam_selection_phonetic.py',
    'selection_math_map.json',
    'xbindkeys.aksharam',
    'start_selection.sh',
    'aksharam_host_controls.py',
    'unfolding_map.json',
}
if files is None:
    raise SystemExit('KPS has no Files section')
for node in list(files):
    name=node.findtext('Name','')
    if name in linux_only:
        files.remove(node)
out.parent.mkdir(parents=True,exist_ok=True)
ET.indent(root, space='  ')
out.write_text('<?xml version="1.0" encoding="utf-8"?>\n'+ET.tostring(root,
↪  encoding='unicode'), encoding='utf-8')
```

aksharam_selection_math.py

```
#!/usr/bin/env python3
import json, subprocess, time
from pathlib import Path
ROOT=Path(__file__).resolve().parent
MAP=json.loads((ROOT/'selection_math_map.json').read_text(encoding='utf-8'))
KEYS=sorted(MAP,key=len,reverse=True)

def transform(s):
    out=[]; i=0
    while i<len(s):
        for k in KEYS:
            if s.startswith(k,i):
                out.append(MAP[k]); i+=len(k); break
        else:
            out.append(s[i]); i+=1
    return ''.join(out)

def main():
    subprocess.run(['xclip','-i','-selection','clipboard'], input='', text=True,
    ↪  check=False)
    subprocess.run(['xdotool','key','--clearmodifiers','ctrl+c'], check=False)
    time.sleep(0.06)
    try:
        s=subprocess.check_output(['xclip','-o','-selection','clipboard'],text=True)
    except Exception:
        return 2
    if not s:
        return 0
    t=transform(s)
    if t==s:
        return 0
    p=subprocess.Popen(['xclip','-i','-selection','clipboard'],stdin=subprocess.PIPE,text=T⌟
    ↪  rue)
    p.communicate(t)
    time.sleep(0.04)
    subprocess.run(['xdotool','key','--clearmodifiers','ctrl+v'],check=False)
    return 0
if __name__=='__main__': raise SystemExit(main())
```

aksharam_selection_phonetic.py

```
#!/usr/bin/env python3
from __future__ import annotations

import subprocess
import sys
import time
from collections import OrderedDict

# Standalone build-local Shape-of-Sound map copied into the keyboard package.
# No runtime dependency on translated bodies/, Markdown, or project source files.
SHAPE_PHONETICS = OrderedDict([('♈', 'ash'),
 ('♉', 'us'),
 ('♊', 'eri'),
 ('♋', 'en'),
 ('♌', 'on'),
 ('♍', 'yam'),
 ('♎', 'me'),
 ('♏', 'da'),
 ('♐', 'ka'),
 ('♌', 'myr'),
 ('♒', 'nyx'),
 ('♓', 'ai'),
 ('⏣', 'fe'),
 ('⬡', 'al'),
 ('✡', 'abd'),
 ('☆', 'ahn'),
 ('❂', 'el'),
 ('❀', 'so'),
 ('❋', 'ot'),
 ('⧗', 'dr'),
 ('⊛', 'ri'),
 ('❄', 'ek'),
 ('⚛', 'av'),
 ('⌬', 'ig'),
 ('ر', 'ahl'),
 ('ر', 'uhh'),
 ('ر', 'ehr'),
 ('ر', 'ish'),
 ('ع', 'ora'),
 ('ن', 'ahm'),
 ('ب', 'ke'),
 ('و', 'ehm'),
 ('ح', 'ahd'),
 ('ؤ', 'fur'),
 ('ف', 'a'),
 ('ص', 'era'),
```

('ꞁ', 'kur'),
('ꞑ', 'lur'),
('▫', 'ahr'),
('ꞩ', 'in'),
('ꞗ', 'na'),
('ꞹ', 'fel'),
('ꞓ', 'har'),
('ꞵ', 'mer'),
('', 'o'),
('ꞟ', 'pe'),
('Ʇ', 'zhi'),
('ꞇ', 'cl'),
('ꞡ', 'hr'),
('ꞗ', 'kr'),
('ꞗ', 'vr'),
('ꞙ', 'py'),
('ꞧ', 'oa'),
('ꞈ', 'lc'),
('ꞡ', 'nu'),
('ꞩ', 'st'),
('ꞡ', 'or'),
('ꞑ', 'bon'),
('ꞑ', 'ti'),
('ꞔ', 'fa'),
('≍', 'abd'),
('ꞔ', 'ym'),
('ↄ', 'oh'),
('✡', 'zhi'),
('ꟹ', 'ol'),
('◝ᘐ', 'i'),
('ꝛ', 're'),
('◎', 'se'),
('⏁', 'u'),
('ᔕ', 'fay'),
('ꟻ', 'ha'),
('ᘐᘓ', 'ps'),
('ꬾ', 'era'),
('Ꝋ', 'tar'),
('Ⱪ', 'ghe'),
('Λ', 'rel'),
('Ⱨ', 'ful'),
('Ⱪ', 'ker'),
('Ⱁ', 'hoh'),
('ㄟ', 'hr'),
('ሐ', 'ar'),
('Ⱬ', 'ay'),
('ⱪ', 'urn'),
('I', 'je'),
('ਸ਼', 'fi'),
('ਘ', 'lun'),
('ਗ', 'aru'),
('ਬ', 'es'),
('⍜', 'os'),
('ਰ', 'ahm'),

('𑚸', 'ti'),
('𑚹', 'ey'),
('𑚺', 'sih'),
('𑚻', 'hri'),
('𑚼', 'yo'),
('𑚽', 'thal'),
('𑚾', 'e'),
('𑚿', 'se'),
('𑛀', 'lin'),
('𑛁', 'bri'),
('𑛂', 'inn'),
('𑛃', 'subh'),
('𑛄', 'wel'),
('𑛅', 'm'),
('𑛆', 'esh'),
('𑛇', 'so'),
('𑛈', 'rhu'),
('𑛉', 'del'),
('𑛊', 'na'),
('𑛋', 'ur'),
('𑛌', 'nih'),
('𑛍', 'azh'),
('𑛎', 'hol'),
('𑛏', 'gur'),
('𑛐', 'ves'),
('𑛑', 'rim'),
('𑛒', 'dem'),
('𑛓', 'oth'),
('𑛔', 'izh'),
('𑛕', 'shu'),
('𑛖', 'ia'),
('𑛗', 'zoh'),
('𑛘', 'her'),
('𑛙', 'druh'),
('𑛚', 'fhel'),
('𑛛', 'ral'),
('𑛜', 'kra'),
('𑛝', 'and'),
('𑛞', 'deb'),
('𑛟', 'kol'),
('𑛠', 'fra'),
('𑛡', 'us'),
('𑛢', 'hin'),
('𑛣', 'ser'),
('𑛤', 'ama'),
('𑛥', 'tohr'),
('𑛦', 'pel'),
('𑛧', 'khi'),
('𑛨', 'yth'),
('𑛩', 'mel'),
('𑛪', 'pha'),
('𑛫', 'okh'),
('𑛬', 'od'),
('𑛭', 'ume'),

 ('⁕', 'ohm'),
 ('⁕', 'is'),
 ('⁎', 'ta'),
 ('⁑', 'koh'),
 ('⁀', 'yh'),
 ('⁰', 'st'),
 ('⁋', 'os'),
 ('⁜', 'poru'),
 ('⁓', 'orm'),
 ('⁗', 'rev'),
 ('⁂', 'imh'),
 ('⁔', 'ihnj'),
 ('⁌', 'ig'),
 ('⁎', 'pe'),
 ('⁐', 'du'),
 ('⁊', 'oma'),
 ('⁏', 'eru'),
 ('⁏', 'ta'),
 ('⁛', 'opa'),
 ('⁚', 'tin'),
 ('⁍', 'rest'),
 ('⁅', 'sil'),
 ('⁆', 'slun'),
 ('⁌', 'ete'),
 ('⸿', 'zen'),
 ('⸰', 'pon'),
 ('⸱', 'ael'),
 ('⸐', 'gaya'),
 ('⸑', 'wil'),
 ('⸒', 'pay'),
 ('⸓', 'eve'),
 ('⸔', 'ahc'),
 ('⸕', 'no'),
 ('⸖', 'noo'),
 ('⸗', 'av'),
 ('⸘', 'veh'),
 ('⸙', 'th'),
 ('⸚', 'he'),
 ('⸛', 'de'),
 ('⸜', 'pa'),
 ('⸝', 'ma'),
 ('⸞', 'in'),
 ('⸟', 'er'),
 ('⸠', 'at'),
 ('⸡', 'oo'),
 ('☽⊙☾', 'regia'),
 ('·', ' ')])

# First Spells retain their utterance through every lawful written transformation.
FIRST_SPELL_PHONETICS = OrderedDict([('⎔❁☆✪', 'eloi'),
 ('𐌓𐌋', 'eloi'),
 ('⁕✡☽✡⌬⧗⬡✡❁➶', 'sabachtany'),
 ('∀☽᛭✷‹➶', 'sabachtany'),
 ('✡❁✪ℳ✡', 'anima'),

```
 ('‹❂♏✡', 'anima'),
 ('⬡♍❁♈✡', 'culpa'),
 ('❄⬡⬡❂⊛⬡', 'decire'),
 ('Aᒧᔕ', 'decire'),
 ('⬡❁⬡⬡⊛✡', 'entera'),
 ('ᔑᔐ', 'entera')])

# Aksharam grammar/constellations return to ordinary textual grammar for phonetic output.
# ∷ has two physical-source uses (period and colon); once written as one constellation,
# source punctuation is not recoverable, so its canonical textual return here is period.
GRAMMAR = OrderedDict([('⁘⁖', '\n\n'),
 ('⁖⁘', '\n\n'),
 ('ɾ', '['),
 ('ɻ', ']'),
 ('ᒣ', '('),
 ('ᒡ', ')'),
 ('', '{'),
 ('', '}'),
 ('∷', '.'),
 ('○–○', '='),
 ('˒', "'"),
 ('˓', '"'),
 ('⁚', '>'),
 ('⁖', '<'),
 ('⁜', ','),
 ('⁘', ';'),
 ('⁙', '?'),
 ('⁛', '!'),
 ('˙.', '\\'),
 ('.˙', '/'),
 ('…', '_'),
 ('⋯', '-'),
 ('ॐ', '&'),
 ('⁞', '|'),
 ('◊', '`'),
 ('◌', '~'),
 ('·', ' '),
 ('↑', '↑'),
 ('↓', '↓'),
 ('→', '→'),
 ('←', '←')])

VARIATION_SELECTORS = {chr(0xfe0e), chr(0xfe0f)}
TOKENS = sorted(
    list(FIRST_SPELL_PHONETICS.items()) + list(GRAMMAR.items()) +
    ↪  list(SHAPE_PHONETICS.items()),
    key=lambda kv: len(kv[0]),
    reverse=True,
)

def transform(text: str) -> str:
    clean = ''.join(ch for ch in text if ch not in VARIATION_SELECTORS)
    out=[]; i=0
    while i < len(clean):
```

```
        for token,repl in TOKENS:
            if clean.startswith(token,i):
                out.append(repl); i += len(token); break
        else:
            out.append(clean[i]); i += 1
    return ''.join(out)

def _linux_selection() -> int:
    subprocess.run(['xclip','-i','-selection','clipboard'], input='', text=True,
    ↪  check=False)
    subprocess.run(['xdotool','key','--clearmodifiers','ctrl+c'], check=False)
    time.sleep(0.06)
    try:
        selected=subprocess.check_output(['xclip','-o','-selection','clipboard'], text=True)
    except Exception:
        return 2
    if not selected:
        return 0
    rendered=transform(selected)
    p=subprocess.Popen(['xclip','-i','-selection','clipboard'], stdin=subprocess.PIPE,
    ↪  text=True)
    p.communicate(rendered)
    time.sleep(0.04)
    subprocess.run(['xdotool','key','--clearmodifiers','ctrl+v'], check=False)
    return 0

def main() -> int:
    if sys.platform.startswith('linux'):
        return _linux_selection()
    # The transformation engine itself is portable. Selection capture/replacement must be
    # installed through the host platform's Keyman integration layer on Windows/macOS.
    return 3

if __name__=='__main__':
    raise SystemExit(main())
```

xbindkeys.aksharam

```
"python3 $HOME/.local/share/keyman/aksharam_glyphkey/aksharam_host_controls.py math"
  Alt+Shift + Return

"python3 $HOME/.local/share/keyman/aksharam_glyphkey/aksharam_host_controls.py single"
  Alt+Shift + BackSpace

"python3 $HOME/.local/share/keyman/aksharam_glyphkey/aksharam_host_controls.py unfold"
  Alt + BackSpace

"python3 $HOME/.local/share/keyman/aksharam_glyphkey/aksharam_host_controls.py refold"
  Alt + space

"python3 $HOME/.local/share/keyman/aksharam_glyphkey/aksharam_host_controls.py phonetic"
  Alt+Shift + Escape
```

start_selection.sh

```
#!/usr/bin/env bash
set -eu
DIR="${HOME}/.local/share/keyman/aksharam_glyphkey"
exec /usr/bin/python3 "$DIR/aksharam_host_controls.py" watch
```

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[10] Magus Jamye Reficul Ahnend. *Exegenesis Marginalia IX: The SeeD of Always*. June 2026. Primary invariant cosource for The Exegenesis Marginalia Volumes.

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